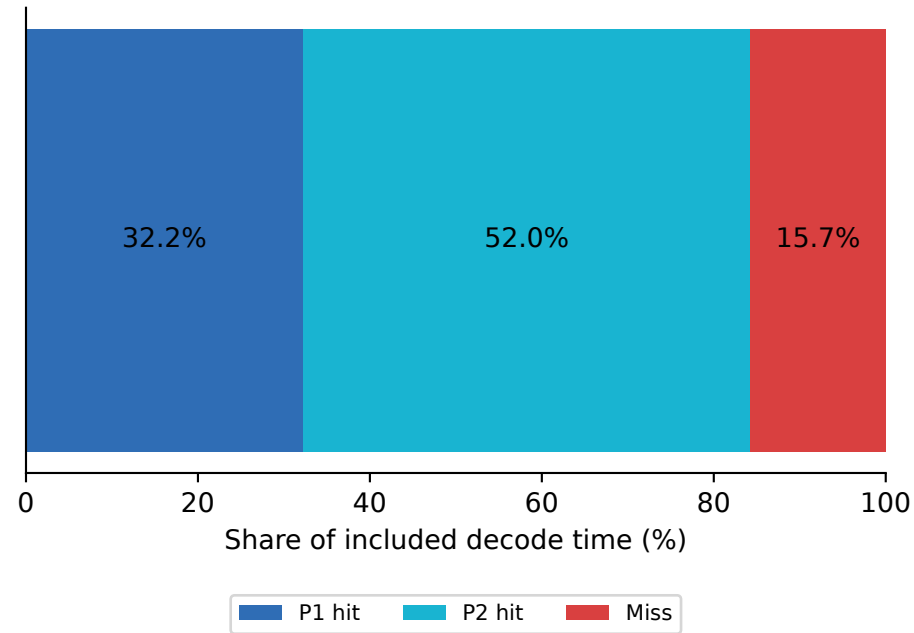
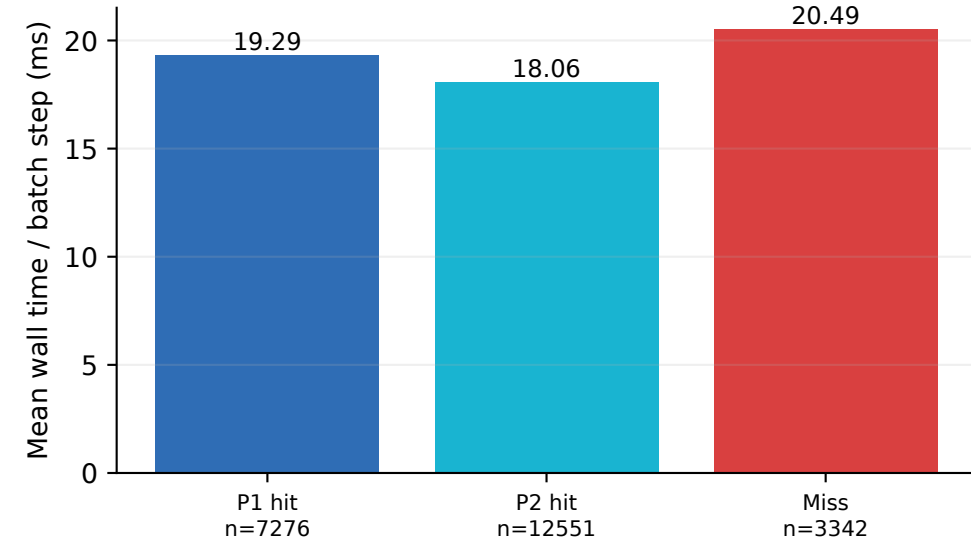


llama2_b1_final_duet_fast_r0
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=0/0
Last pass: B=1, T=0.7, N=480 | AL*=2.365, hit=85.7%, TPS*=125.79

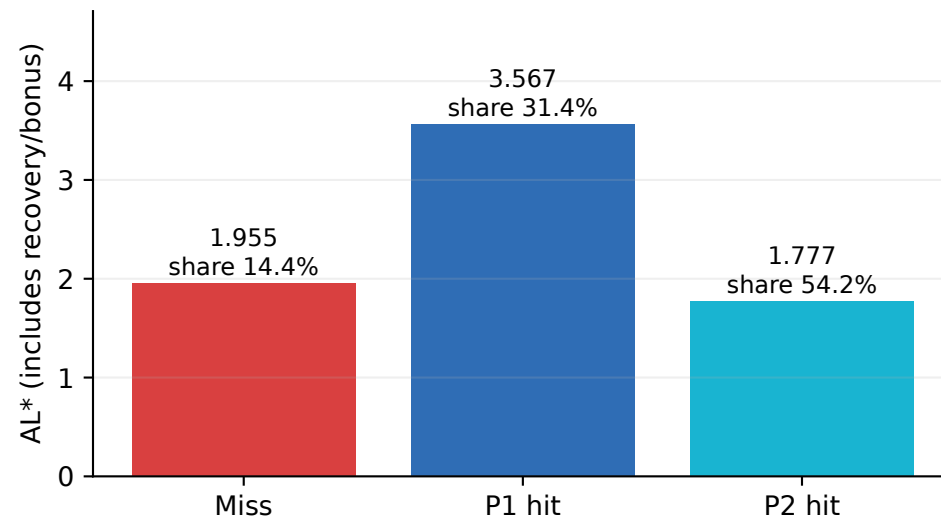
Measured wall time by batch cache status



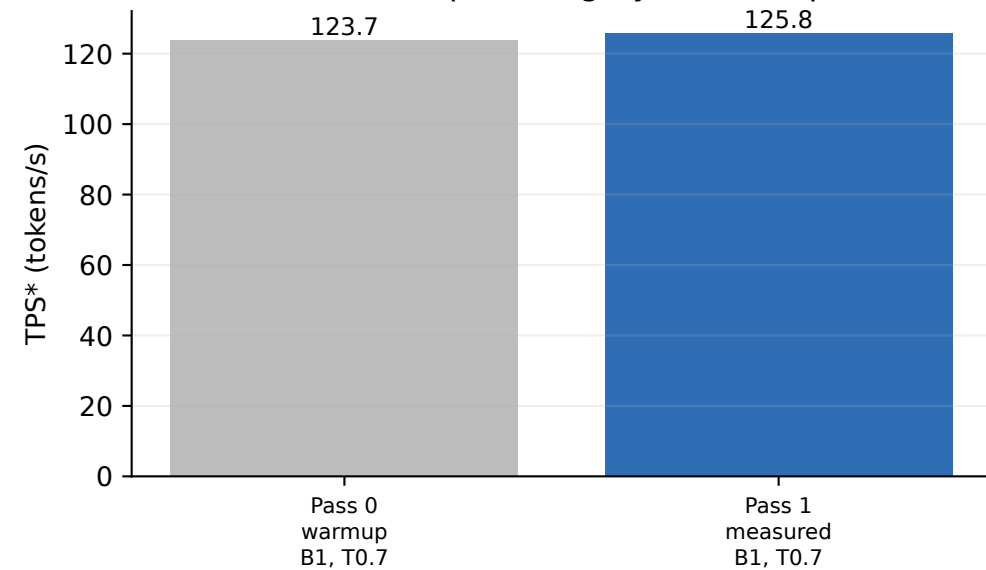
Boundary-free steps, including slow steps



Conditional AL* and source share



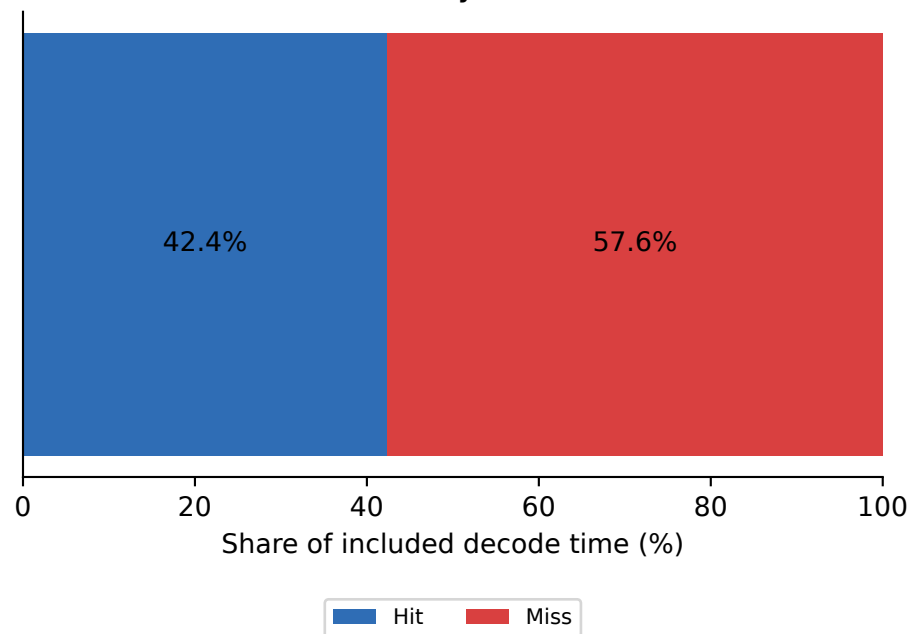
All saved passes; gray = warmup



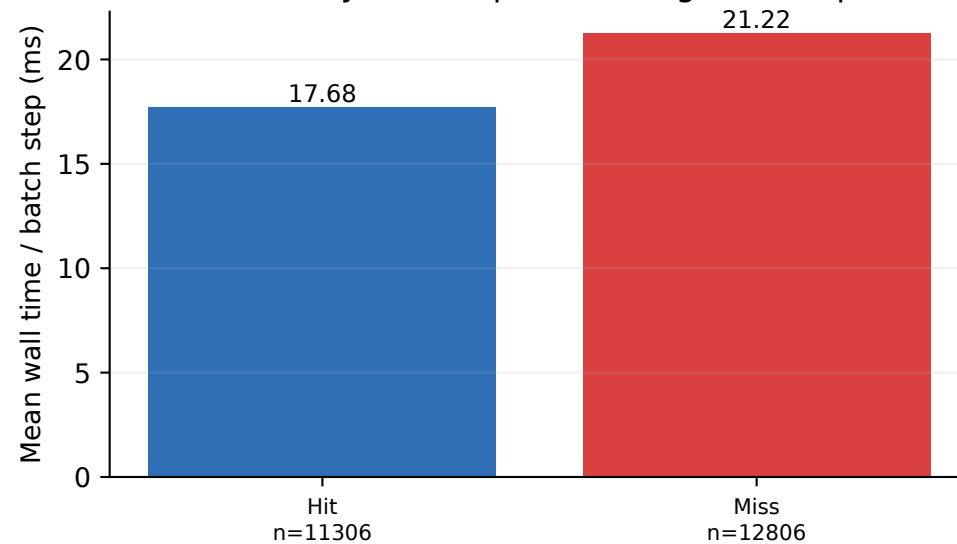
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_final_ssd_al_r0
SSD: K=6, F=5
Last pass: B=1, T=0.7, N=480 | AL*=2.262, hit=47.2%, TPS*=115.63

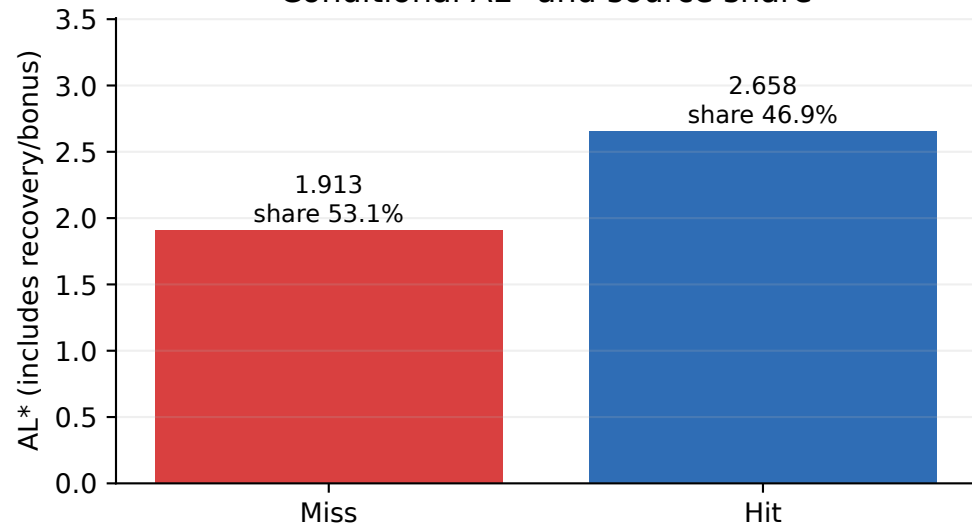
Measured wall time by batch cache status



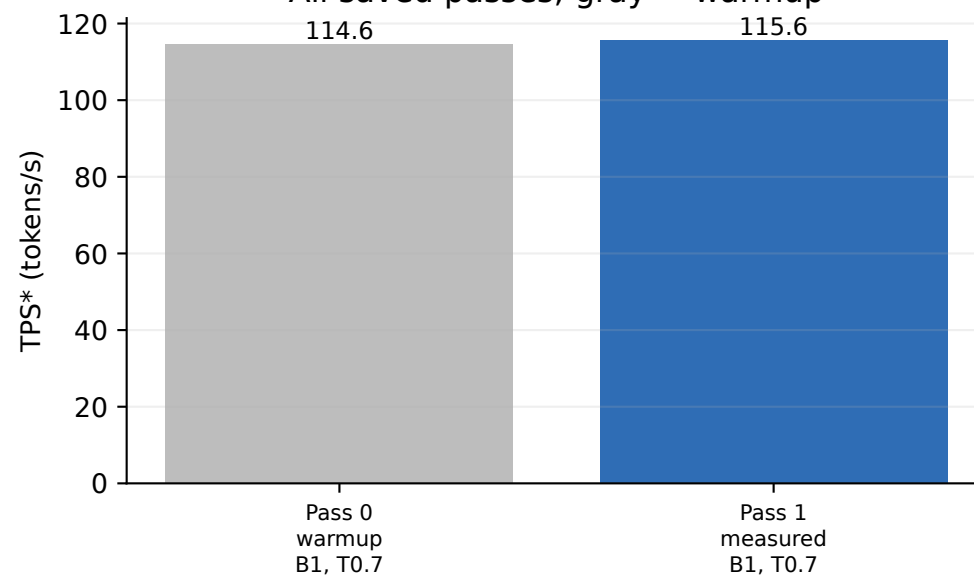
Boundary-free steps, including slow steps



Conditional AL* and source share



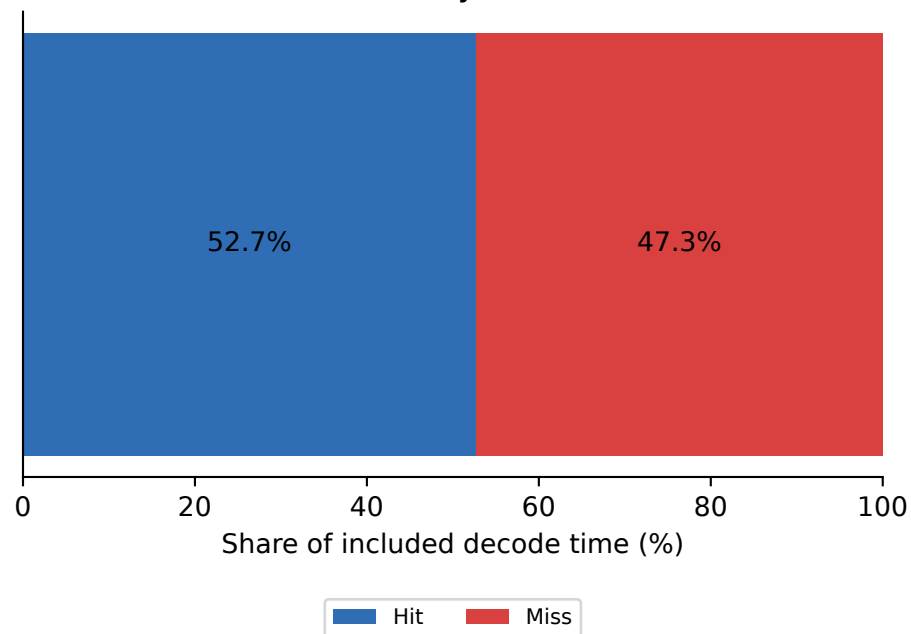
All saved passes; gray = warmup



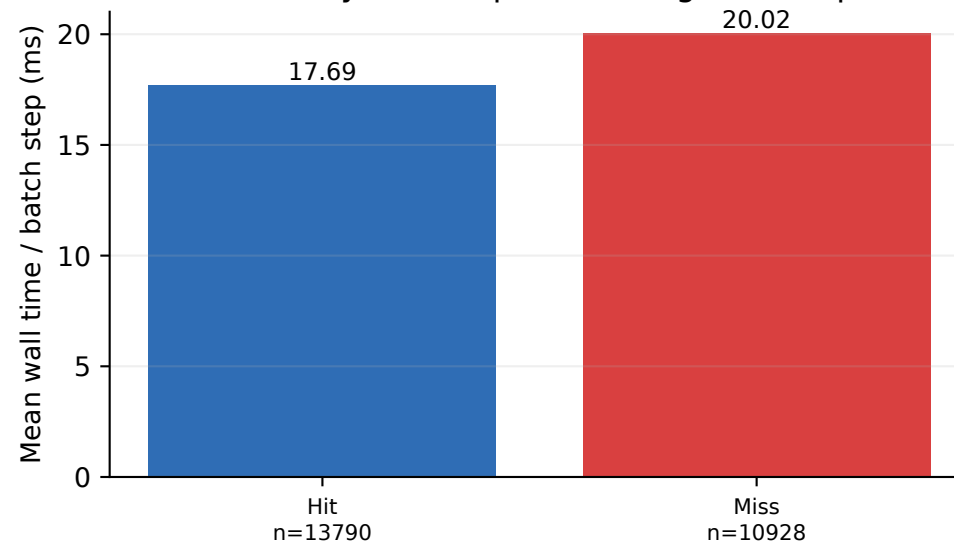
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_final_ssd_fast_r0
SSD: K=4, F=7
Last pass: B=1, T=0.7, N=480 | AL*=2.185, hit=56.0%, TPS*=116.73

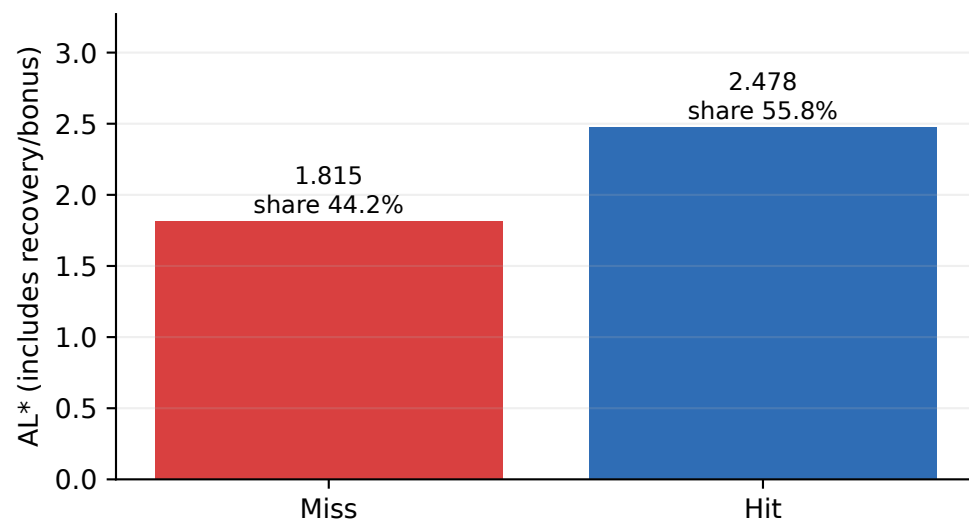
Measured wall time by batch cache status



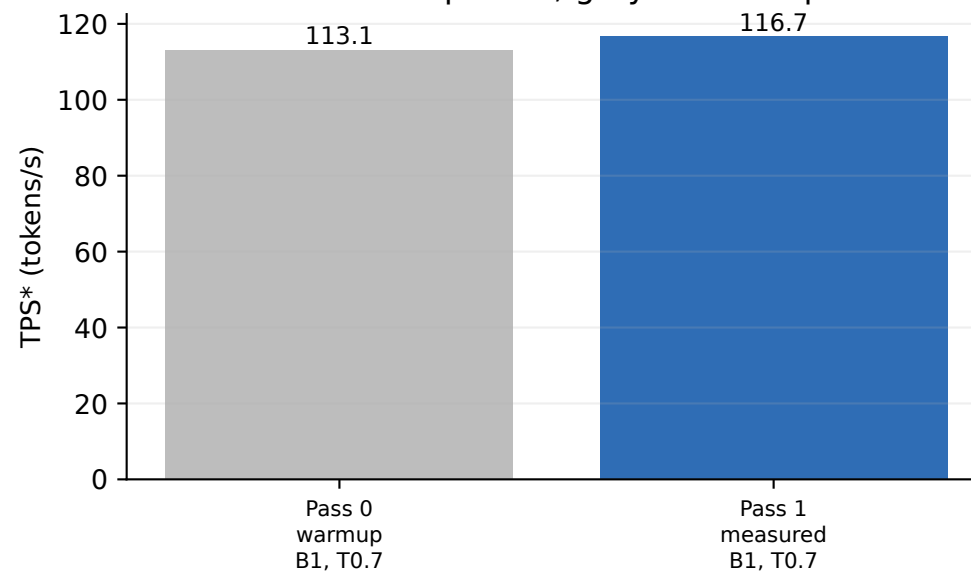
Boundary-free steps, including slow steps



Conditional AL* and source share



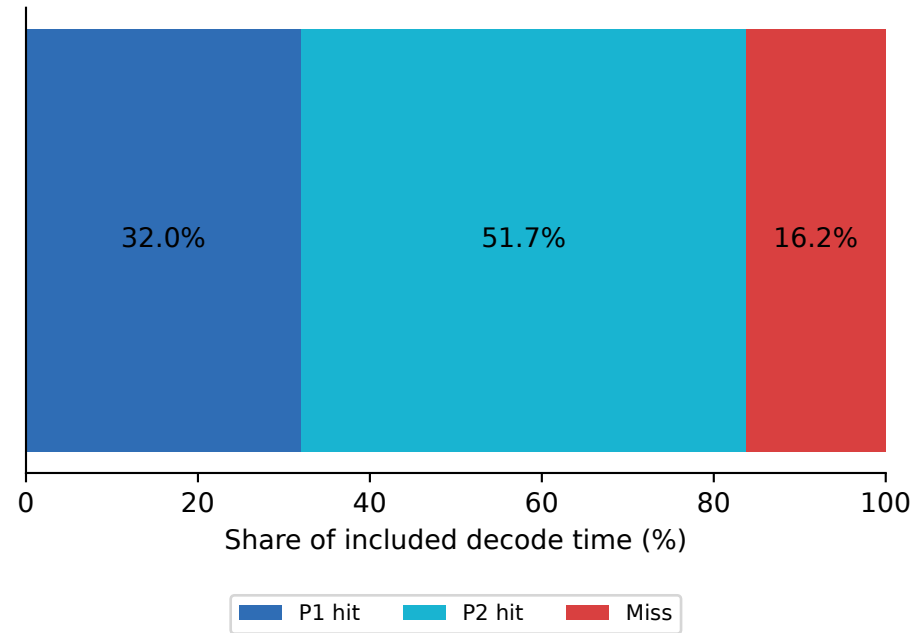
All saved passes; gray = warmup



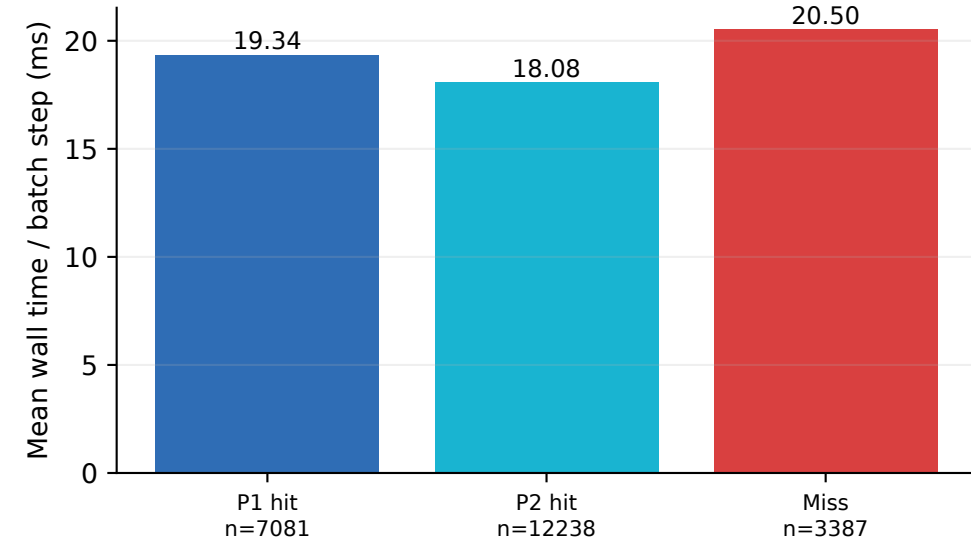
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_final_duet_fast_r1
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=0/0
Last pass: B=1, T=0.7, N=480 | AL*=2.366, hit=85.1%, TPS*=125.62

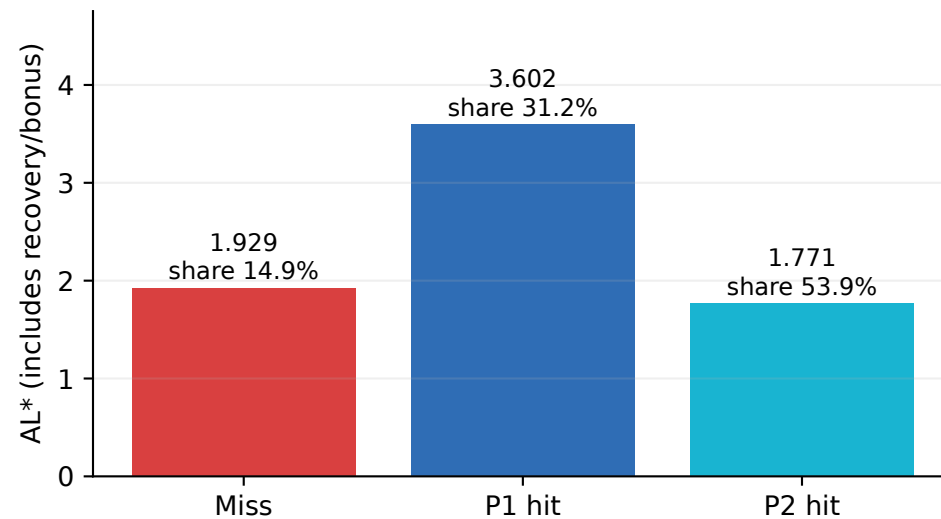
Measured wall time by batch cache status



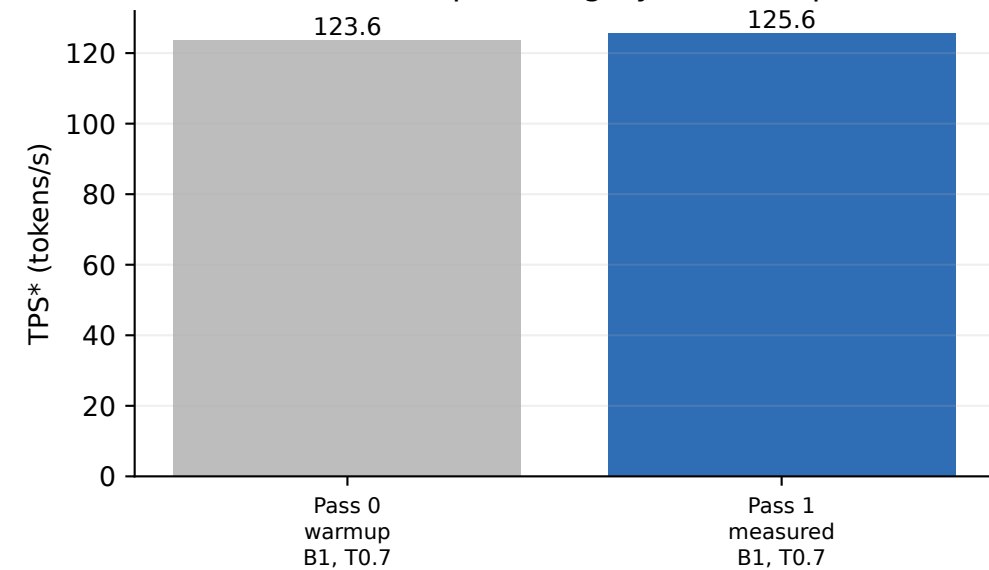
Boundary-free steps, including slow steps



Conditional AL* and source share



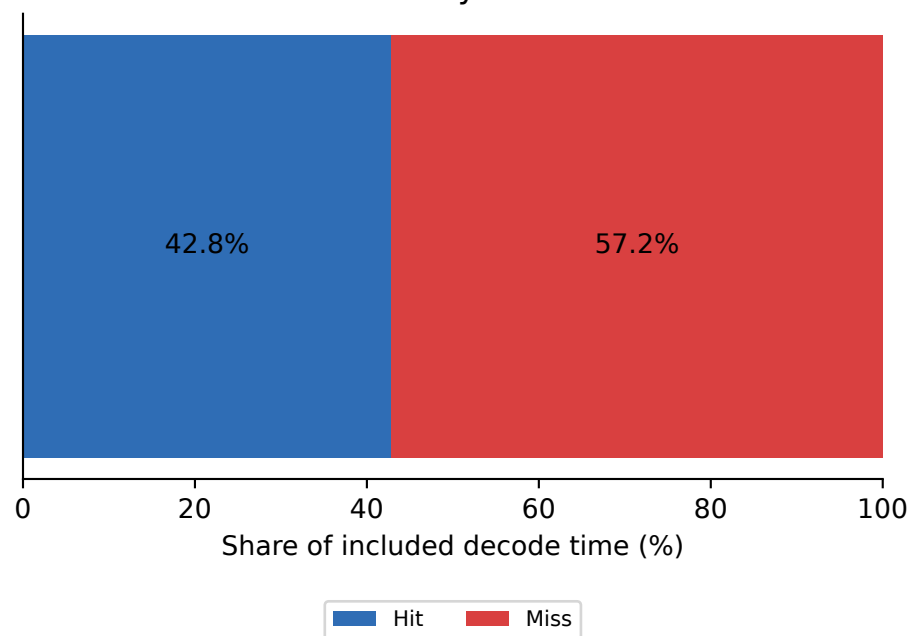
All saved passes; gray = warmup



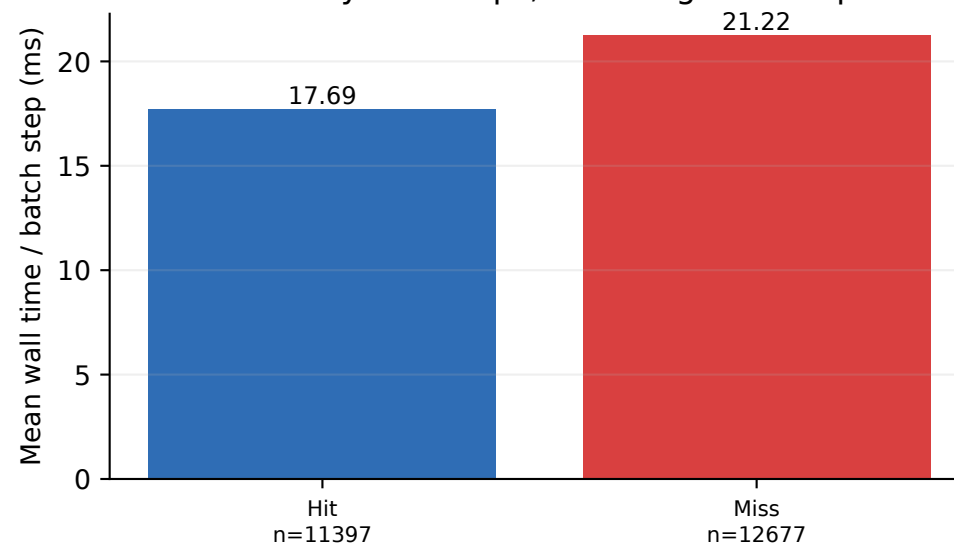
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_final_ssd_al_r1
SSD: K=6, F=5
Last pass: B=1, T=0.7, N=480 | AL*=2.257, hit=47.6%, TPS*=115.44

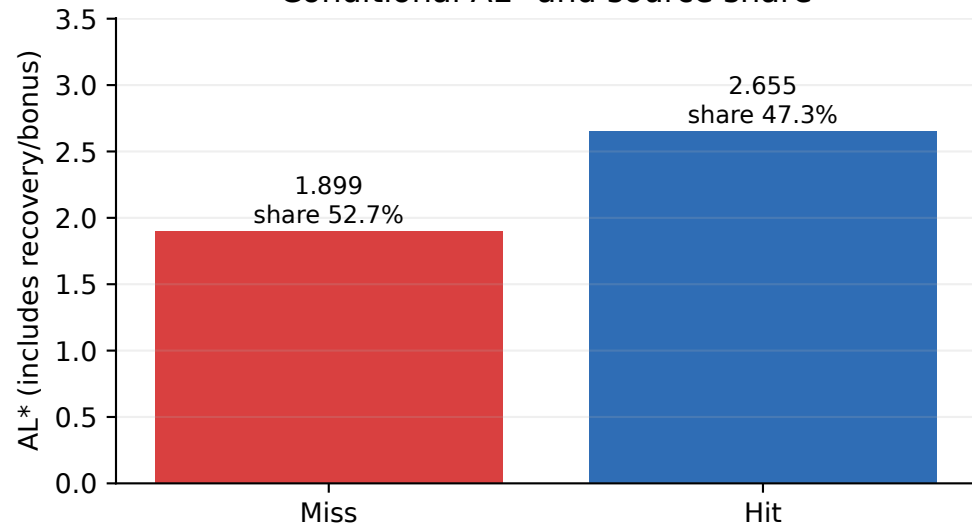
Measured wall time by batch cache status



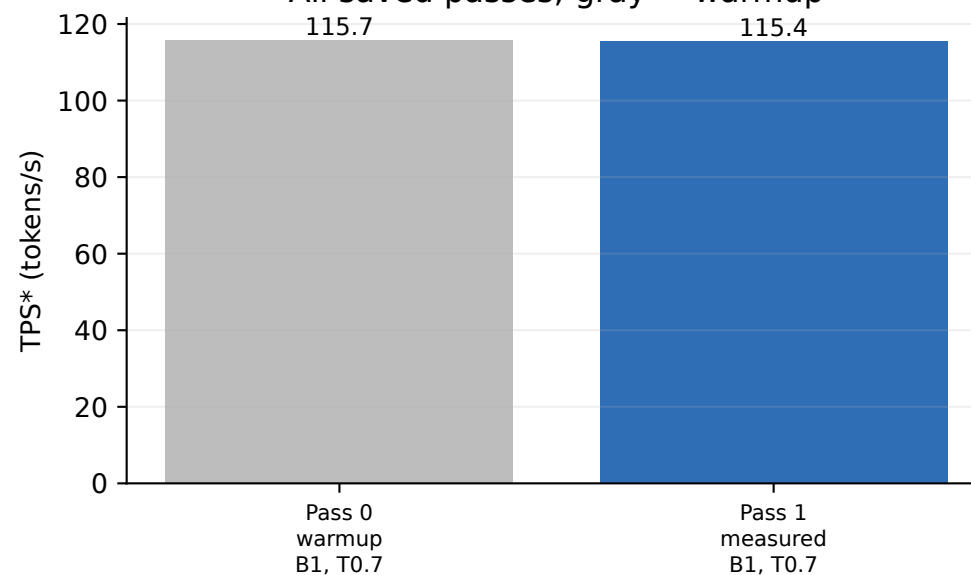
Boundary-free steps, including slow steps



Conditional AL* and source share



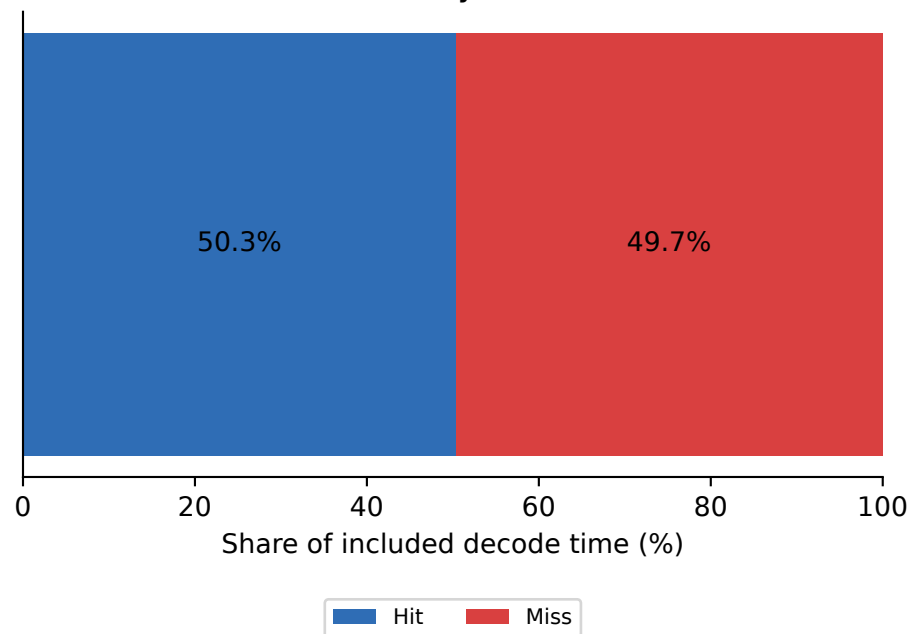
All saved passes; gray = warmup



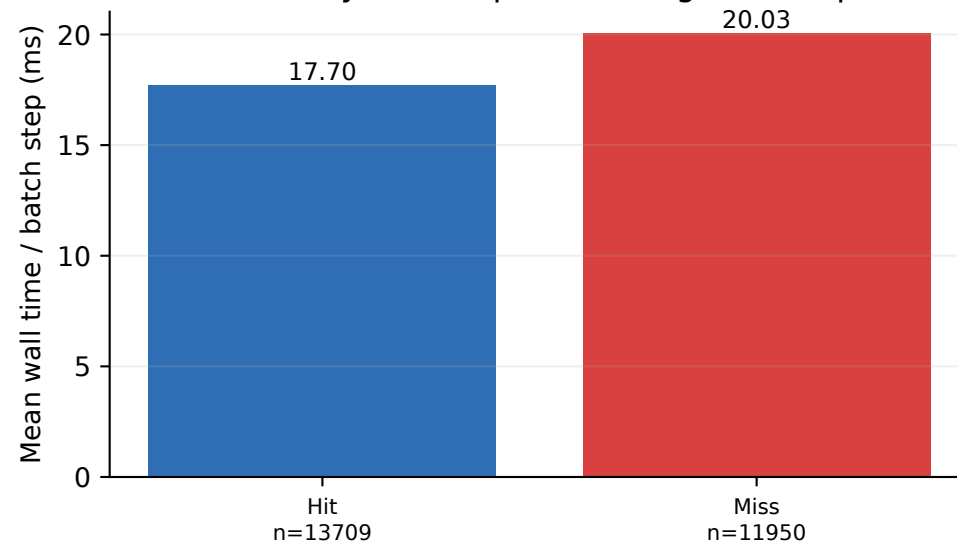
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_final_ssd_fast_r1
SSD: K=4, F=7
Last pass: B=1, T=0.7, N=480 | AL*=2.102, hit=53.7%, TPS*=111.91

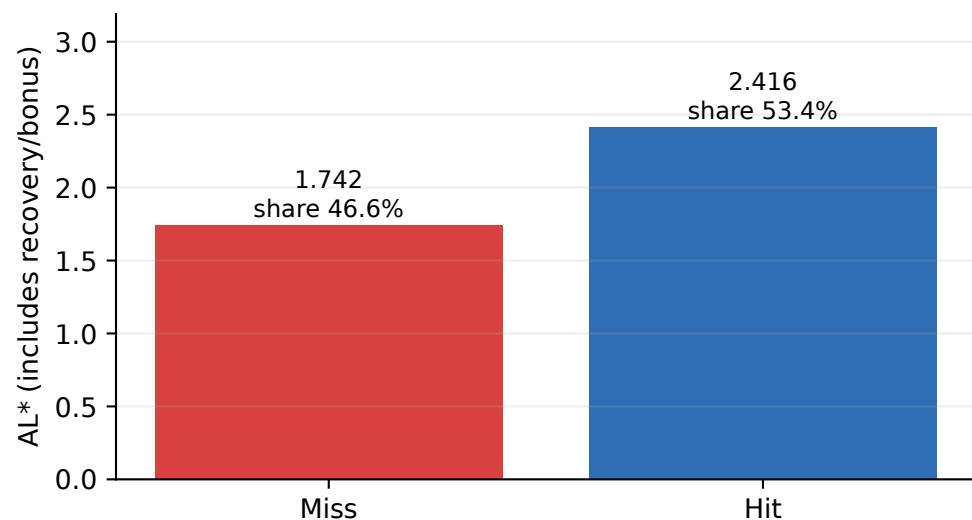
Measured wall time by batch cache status



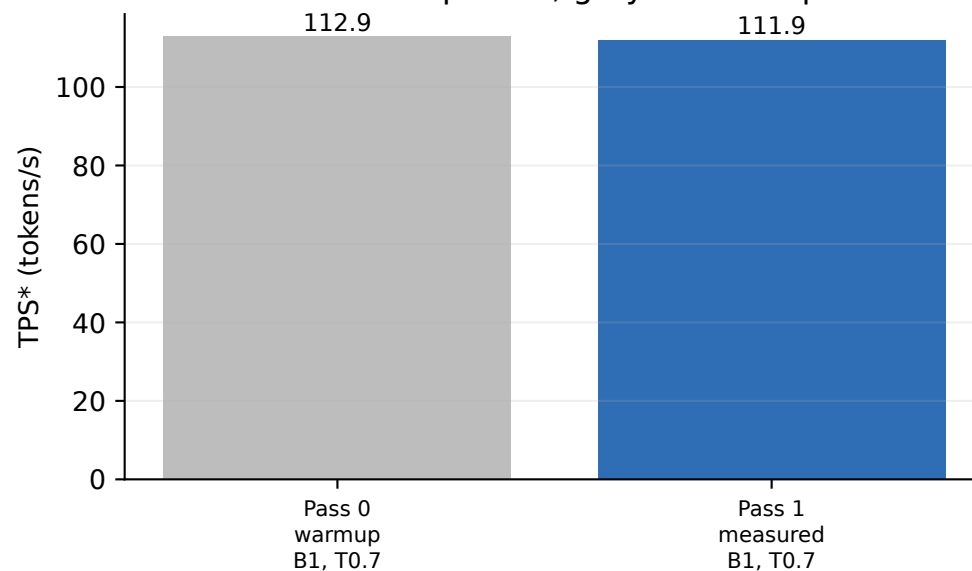
Boundary-free steps, including slow steps



Conditional AL* and source share



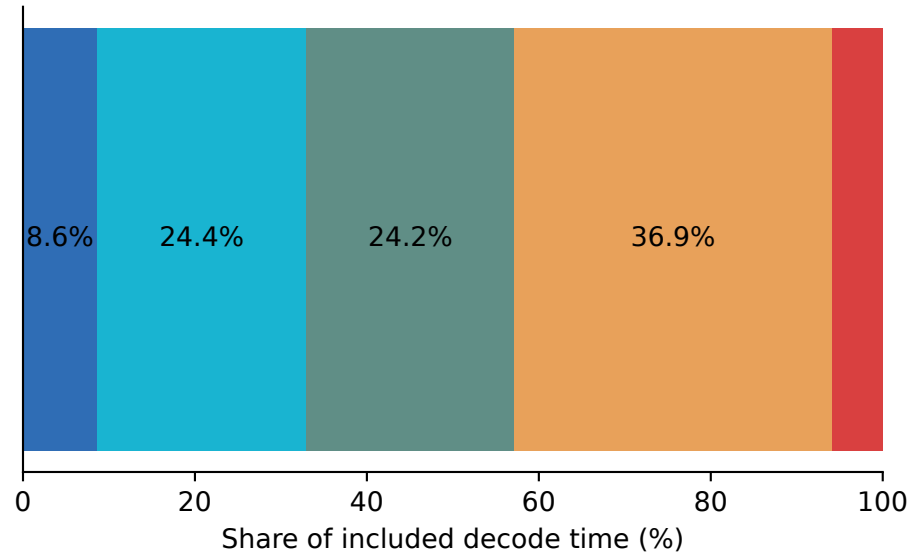
All saved passes; gray = warmup



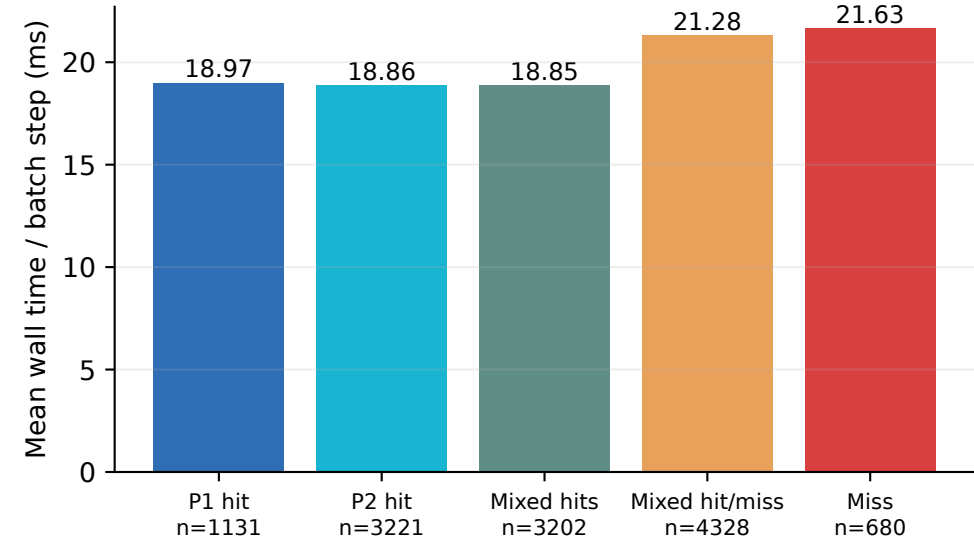
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b2_scale_duet
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
 Last pass: B=2, T=0.7, N=480 | AL*=2.149, hit=77.5%, TPS*=216.08

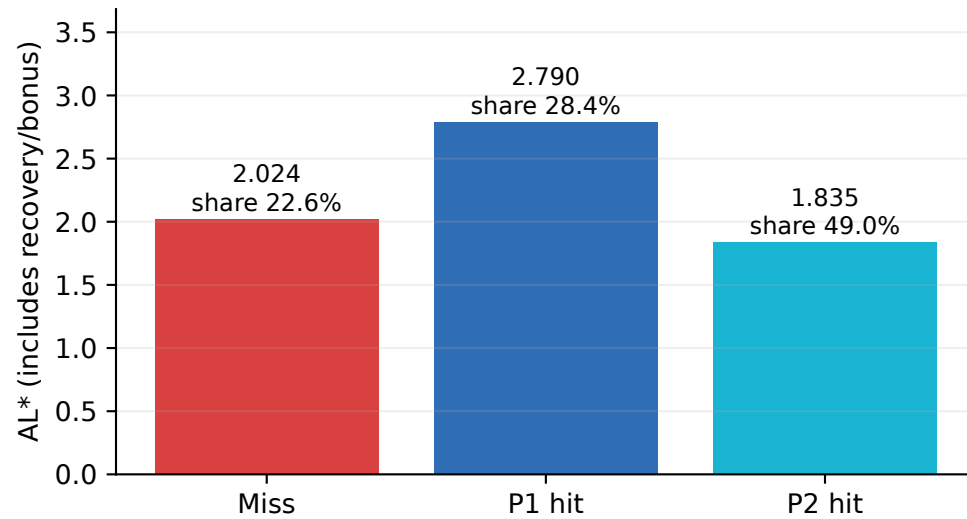
Measured wall time by batch cache status



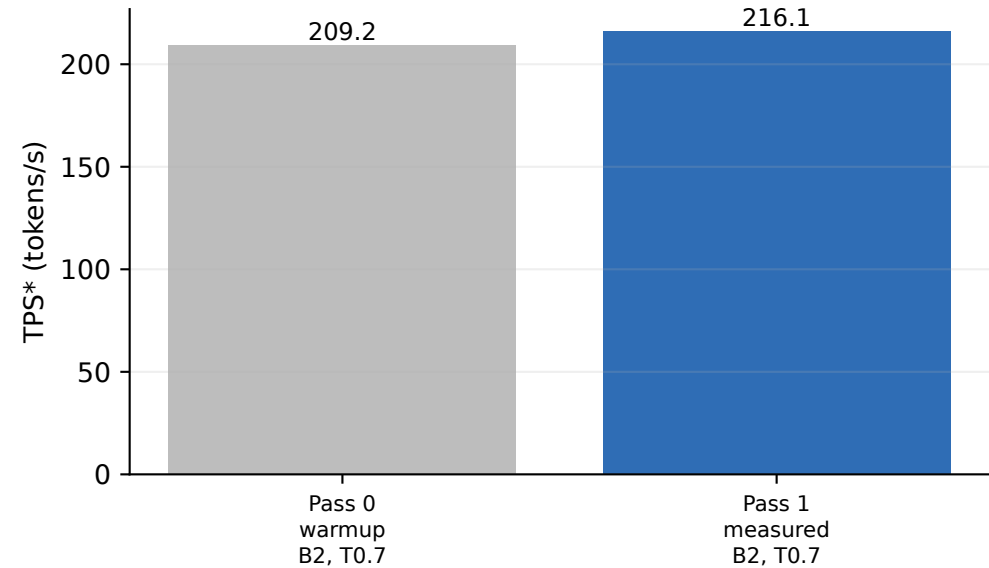
Boundary-free steps, including slow steps



Conditional AL* and source share

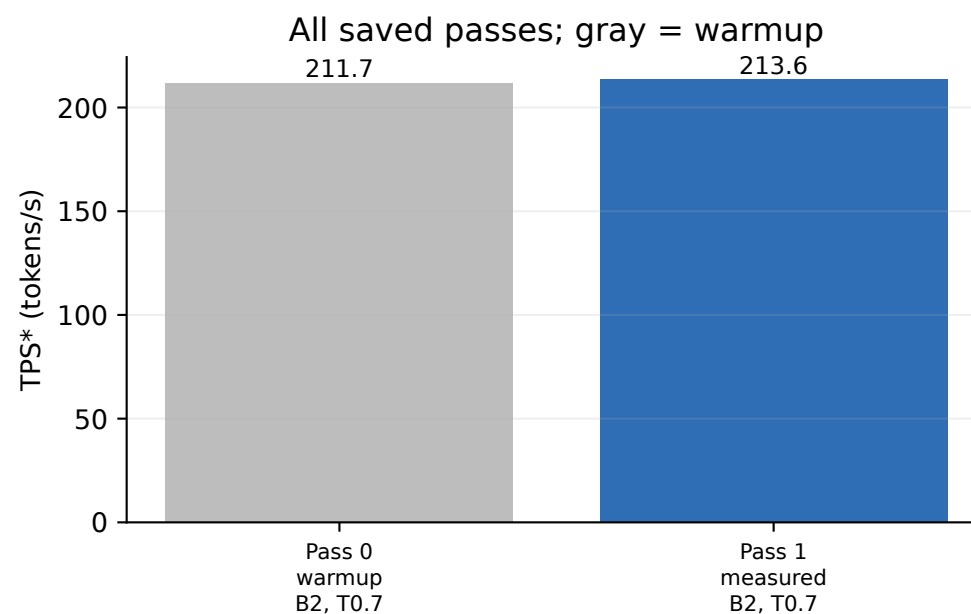
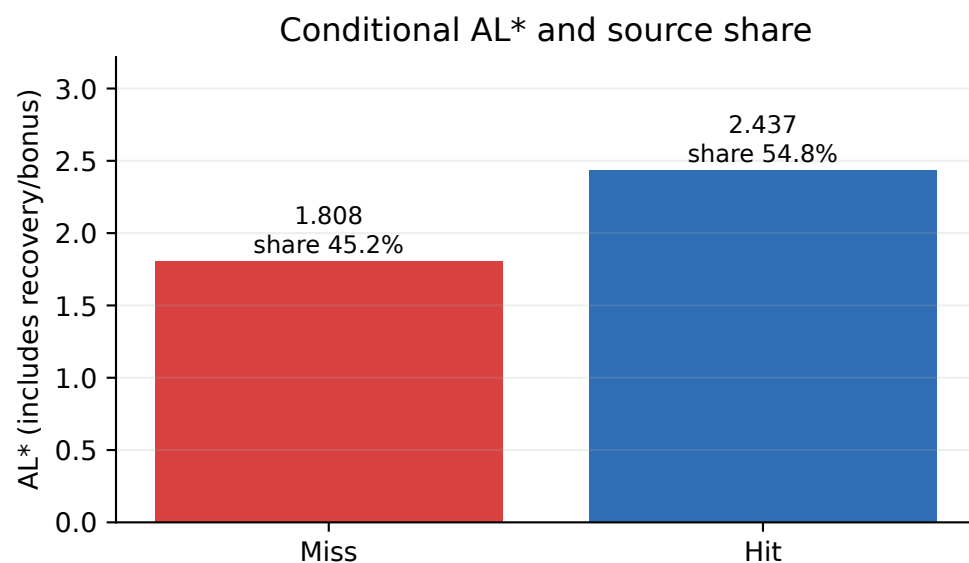
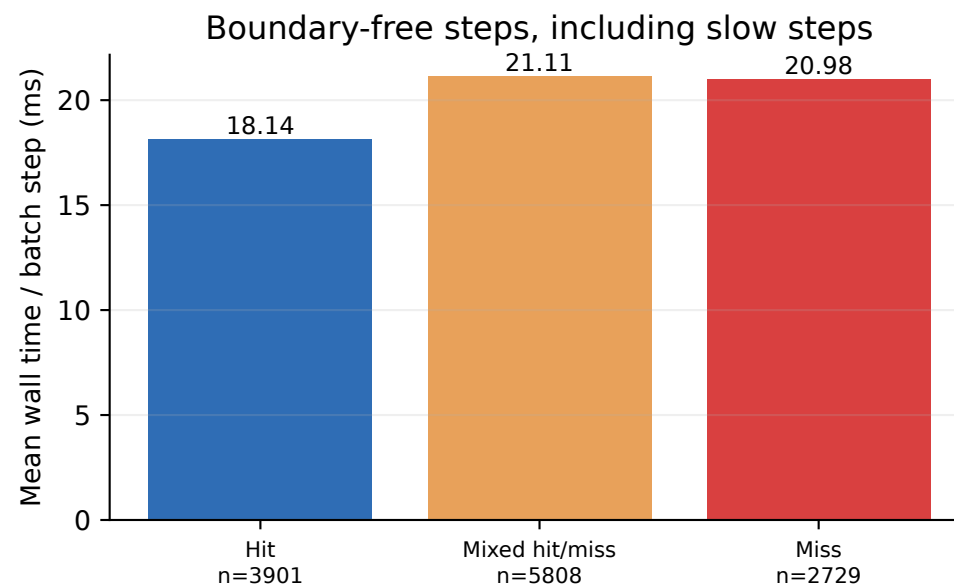
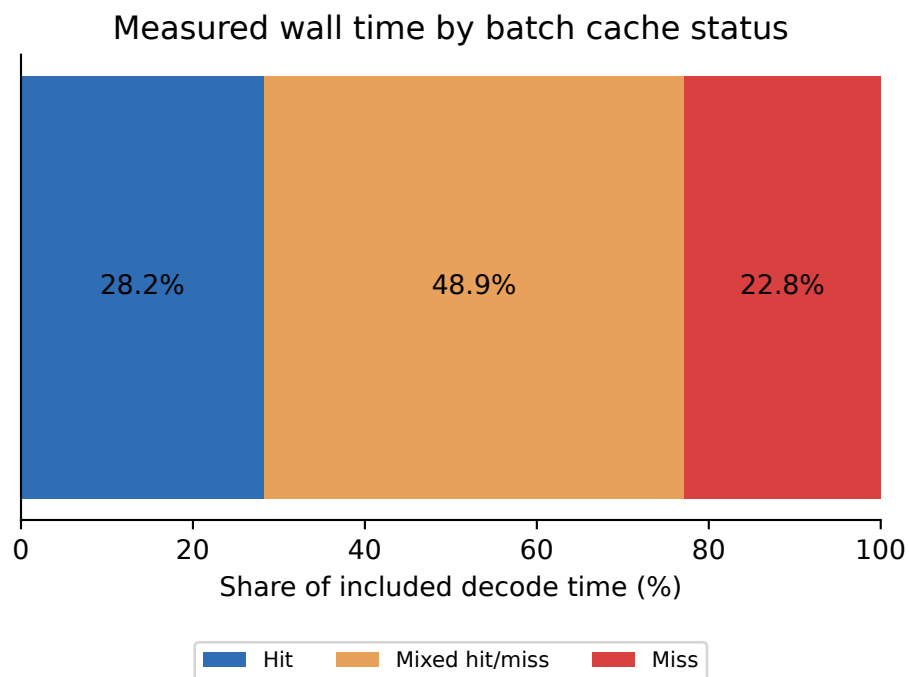


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

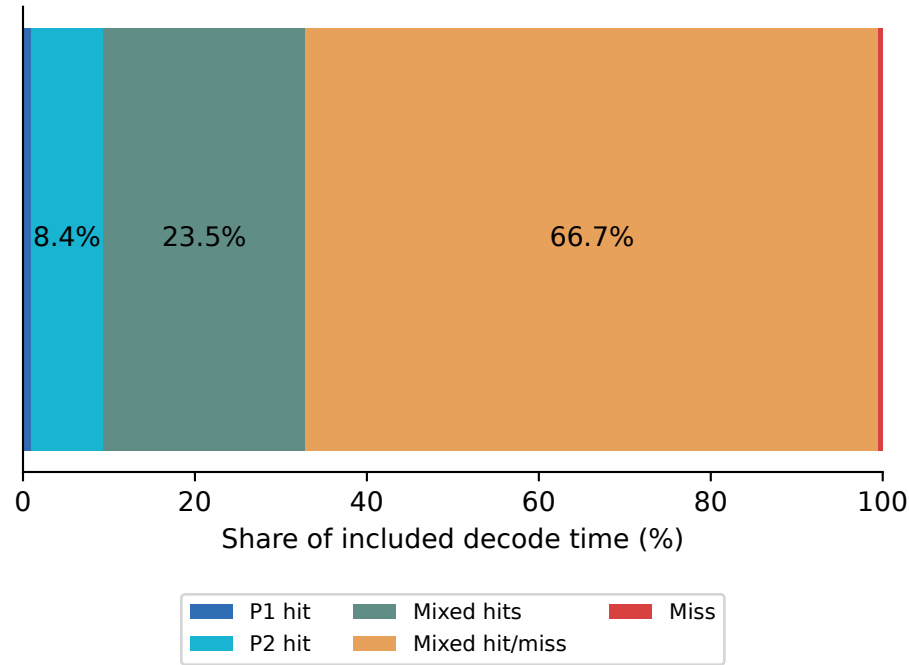
llama2_b2_scale_ssd
SSD: K=4, F=7
Last pass: B=2, T=0.7, N=480 | AL*=2.153, hit=55.0%, TPS*=213.61



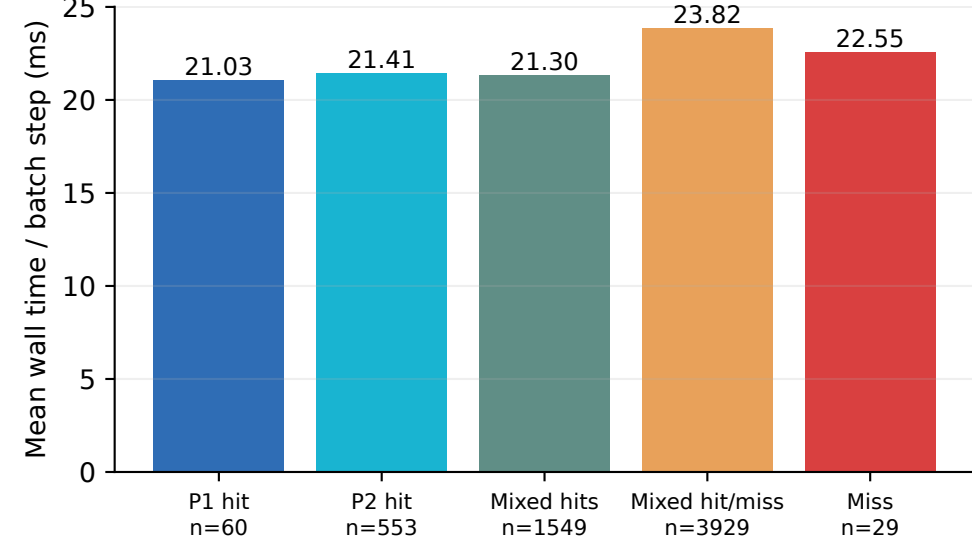
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b4_scale_duet
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
 Last pass: B=4, T=0.7, N=480 | AL*=2.108, hit=76.6%, TPS*=365.46

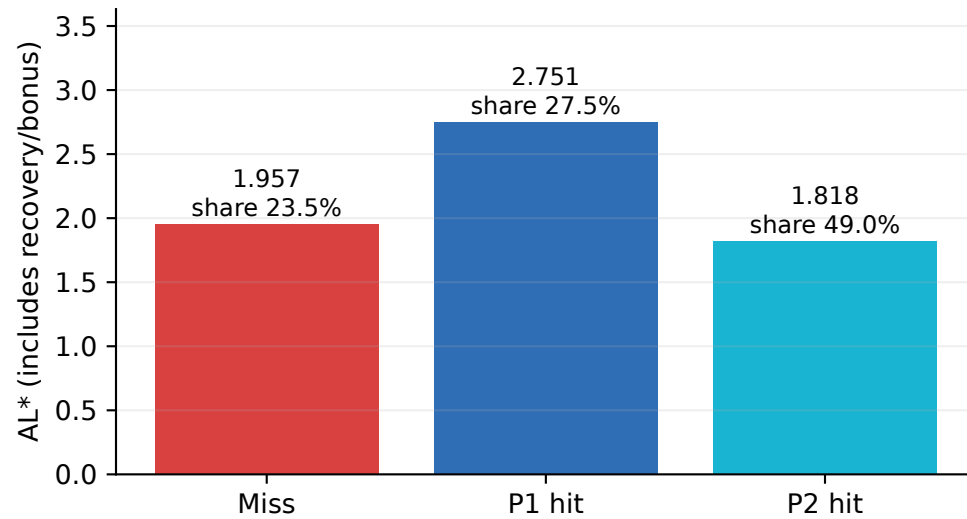
Measured wall time by batch cache status



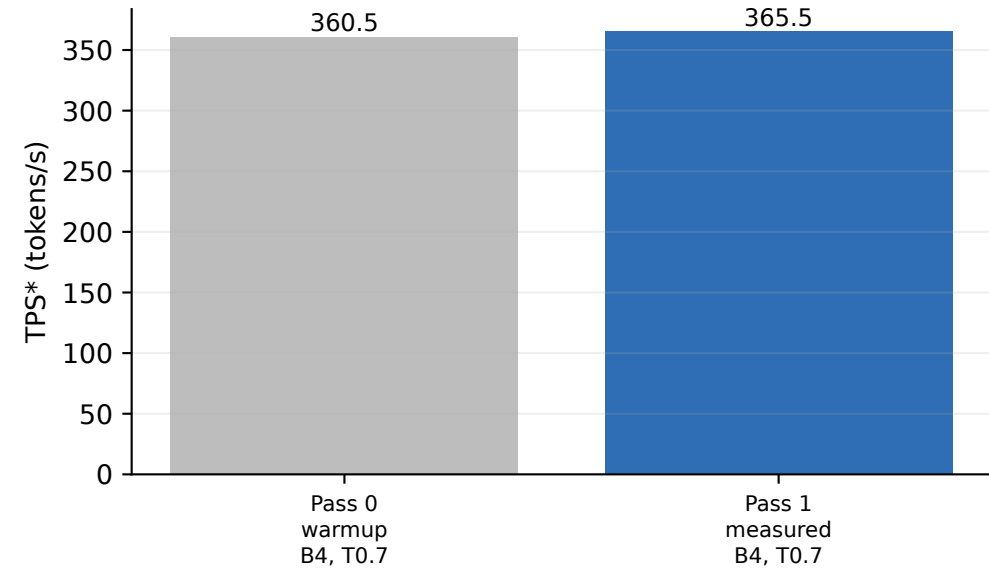
Boundary-free steps, including slow steps



Conditional AL* and source share



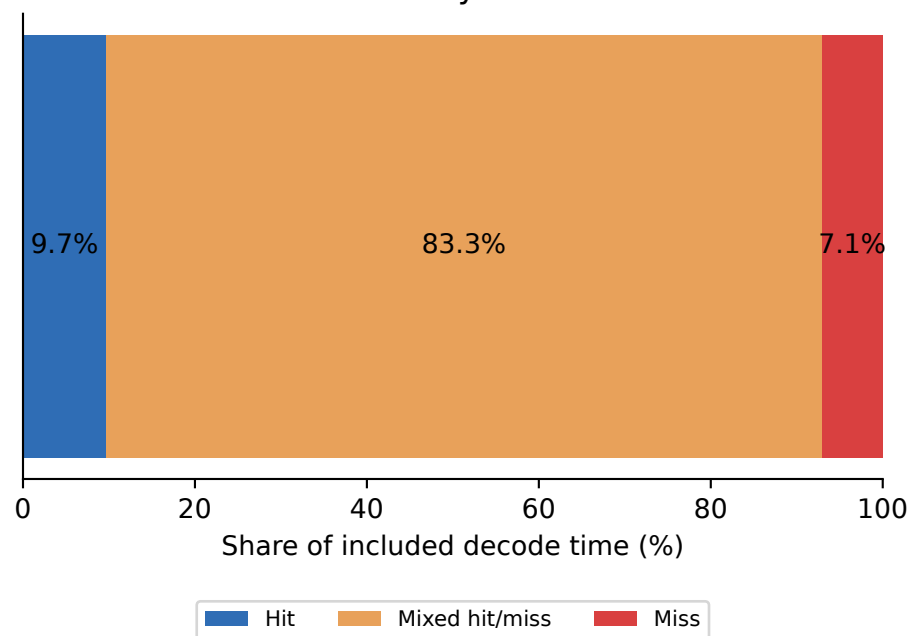
All saved passes; gray = warmup



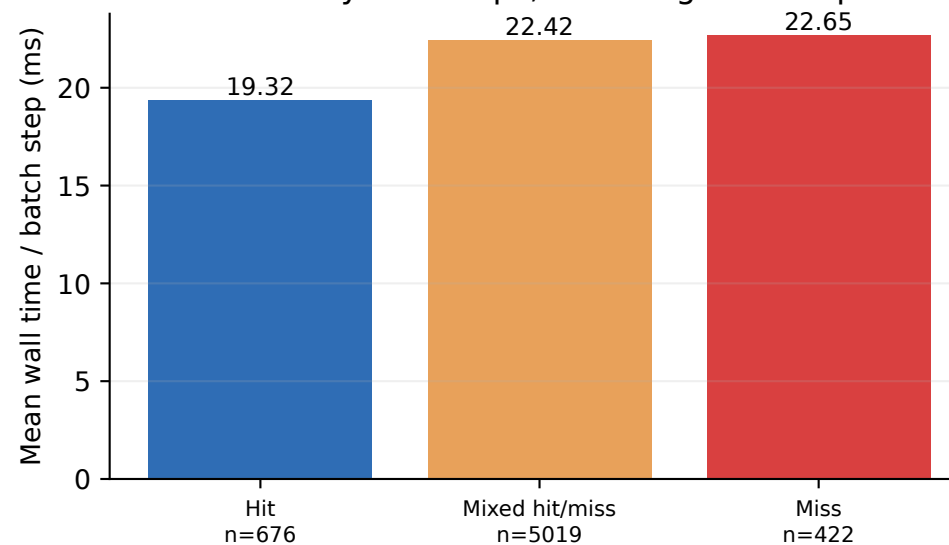
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b4_scale_ssd
SSD: K=4, F=7
Last pass: B=4, T=0.7, N=480 | AL*=2.123, hit=54.4%, TPS*=383.27

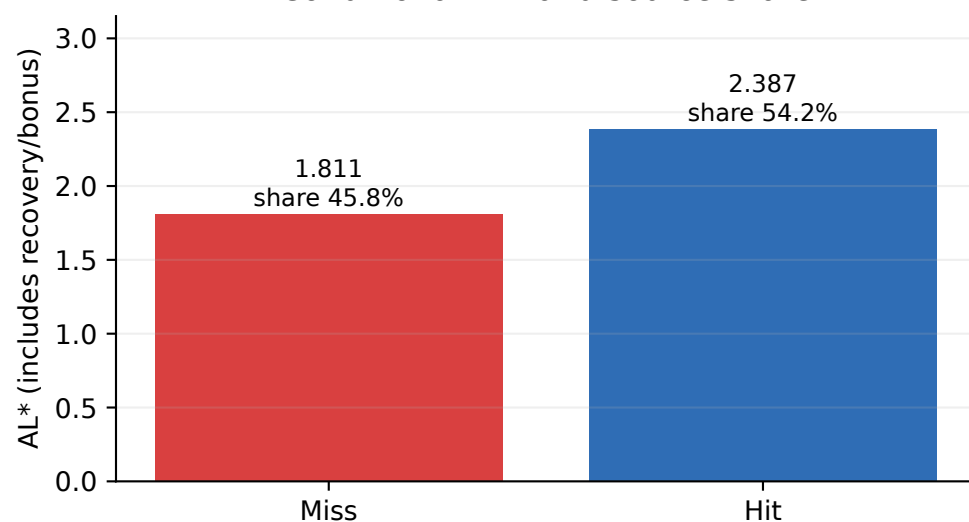
Measured wall time by batch cache status



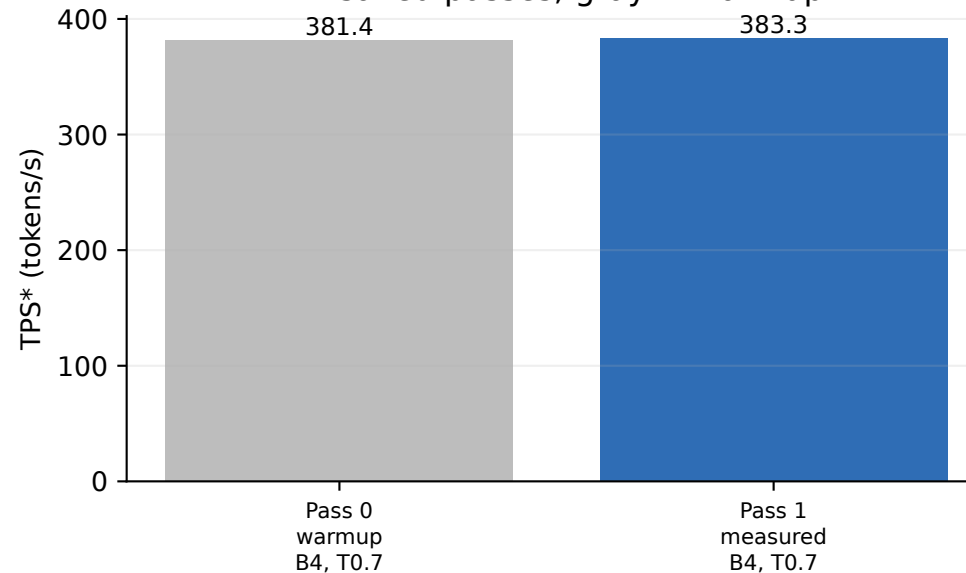
Boundary-free steps, including slow steps



Conditional AL* and source share

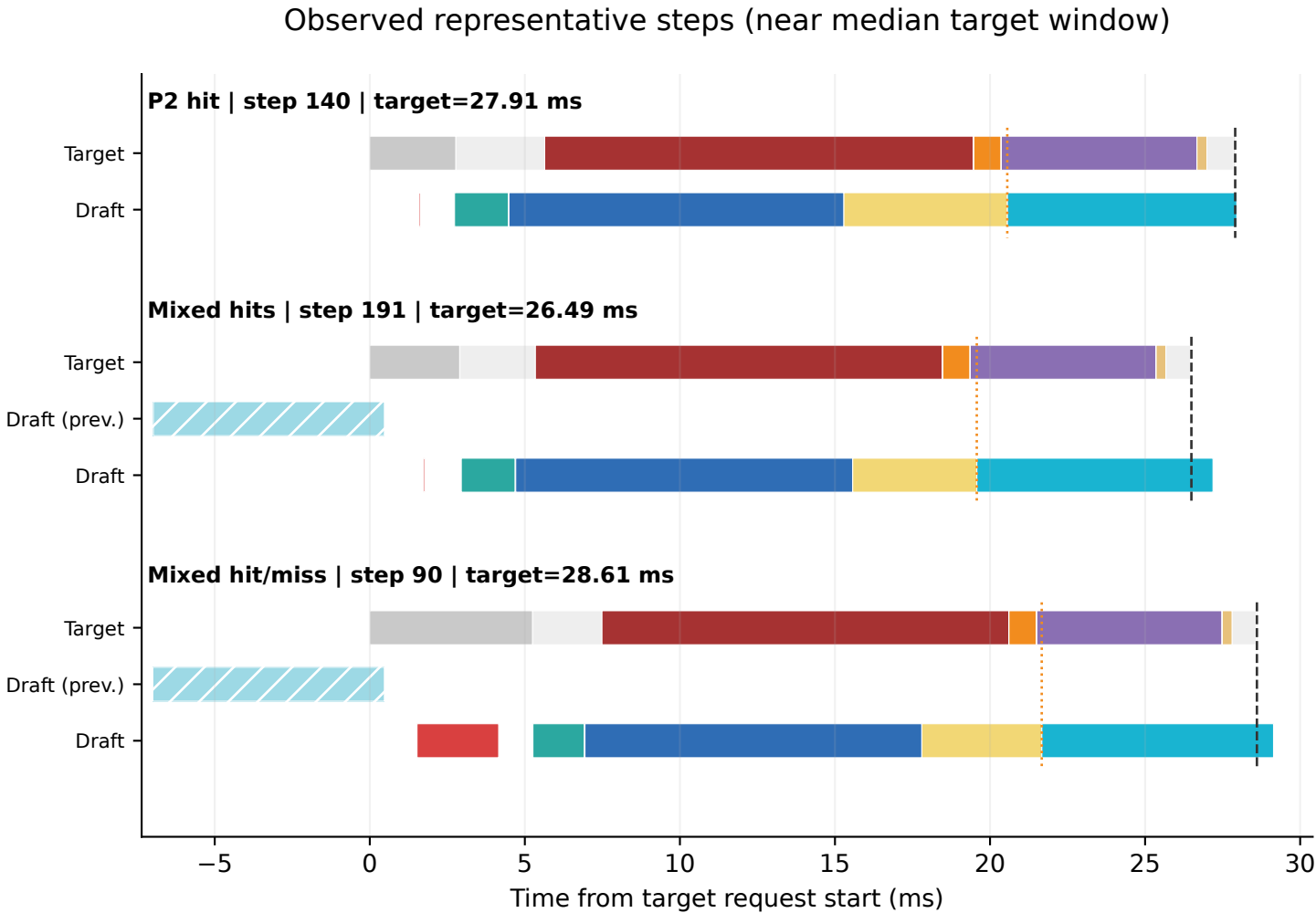
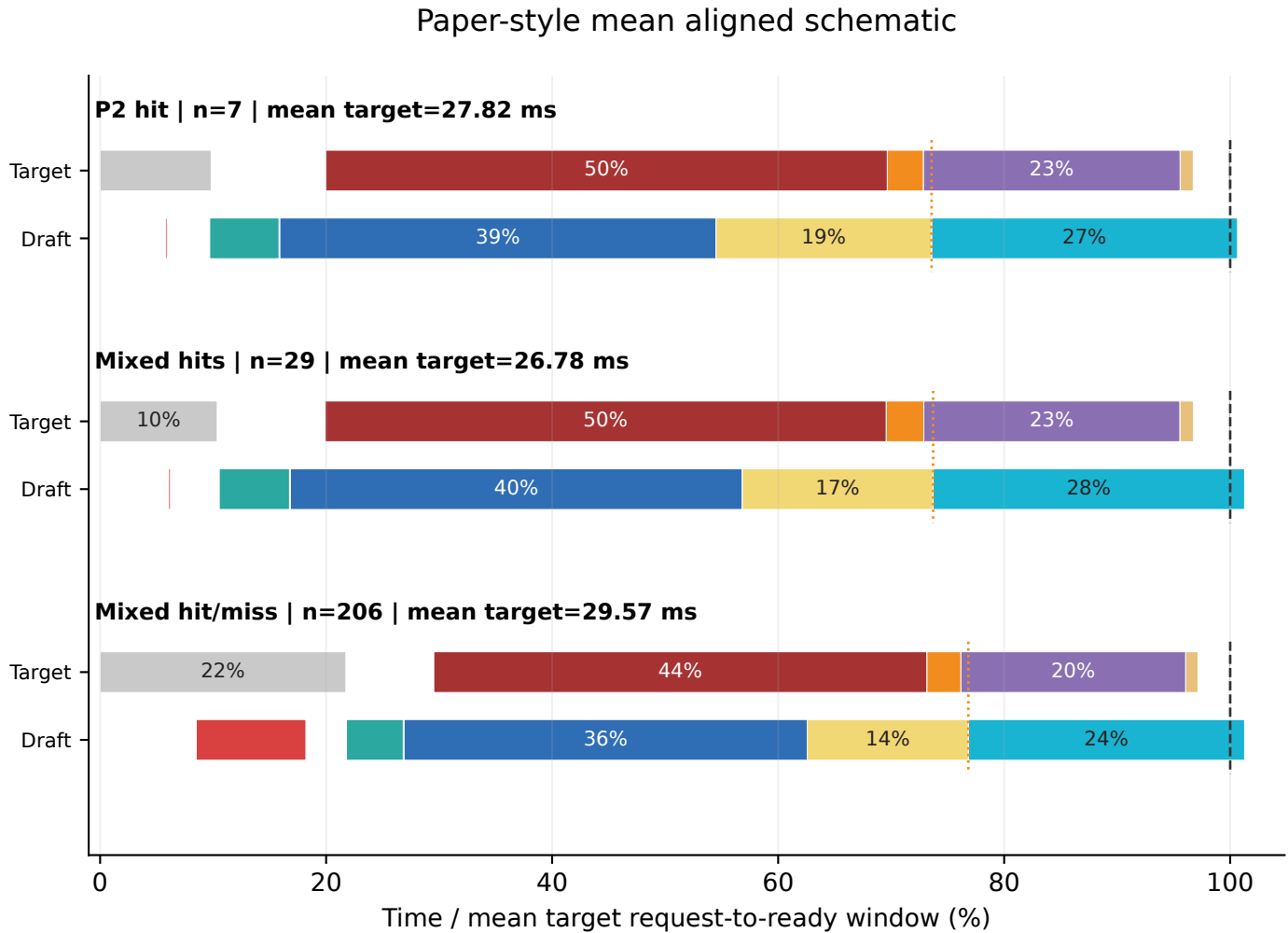
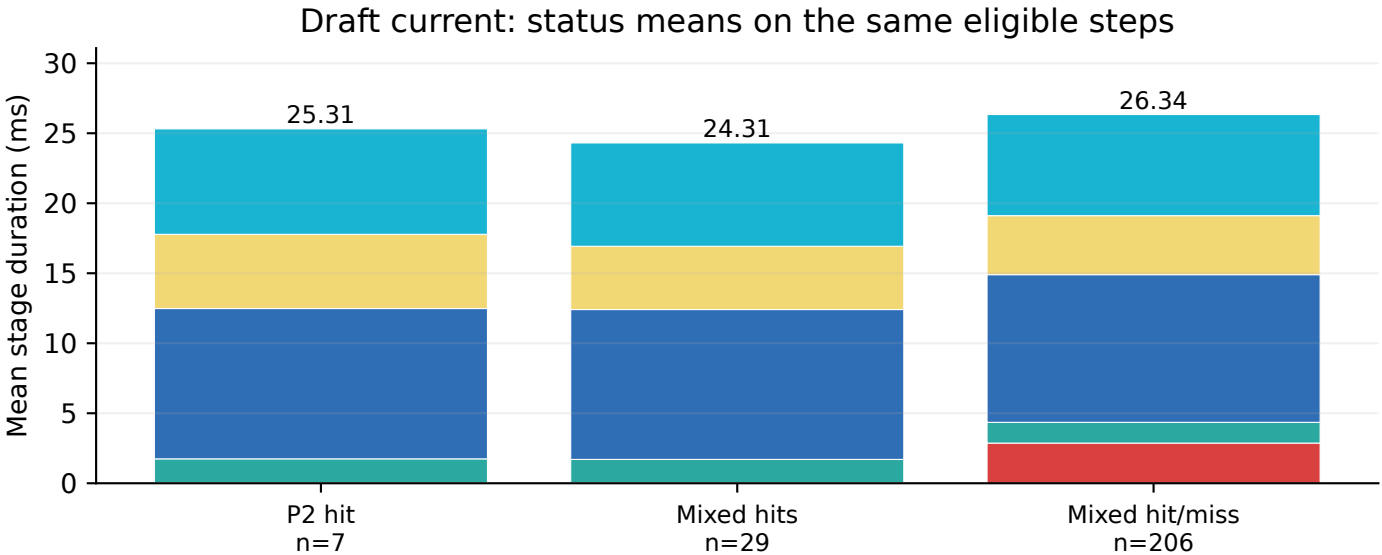
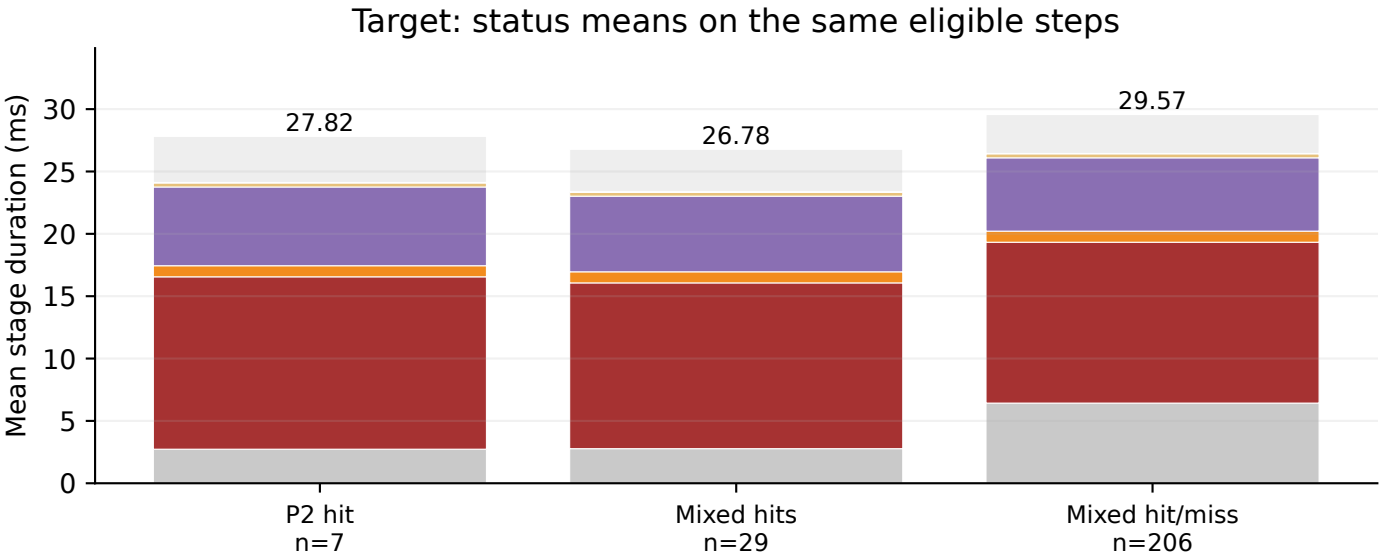


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_breakdown
DUET: K1/K2=3/2, exit=21 (0-based), $\bar{C}$ =3, $\bar{N1}/\bar{N2}$ =6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 242 eligible steps

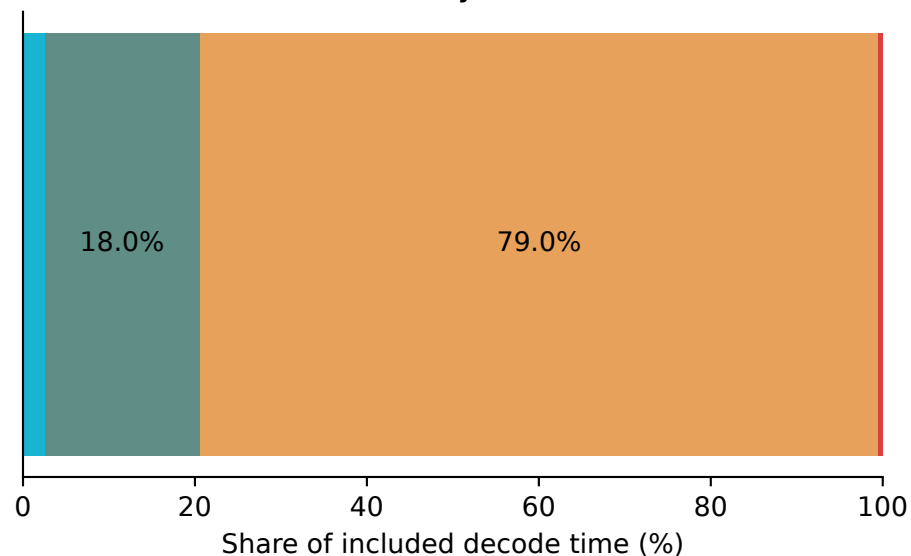


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

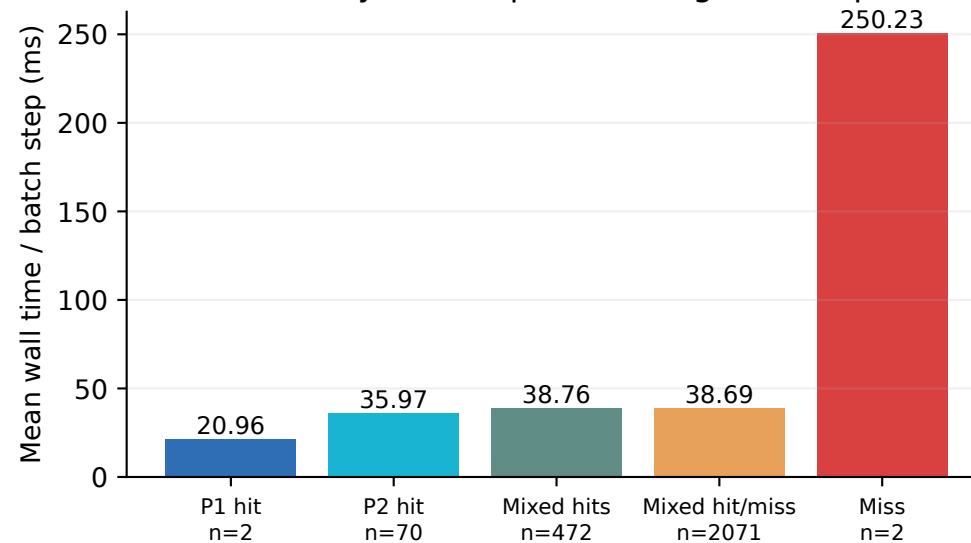
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_final_duet_al_r0
 DUET: K1/K2=6/2, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.299, hit=81.7%, TPS*=469.30

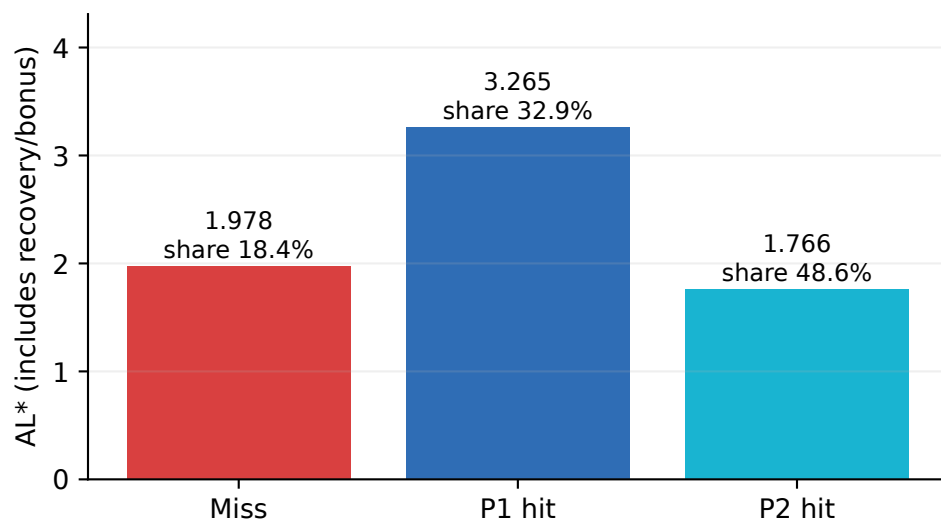
Measured wall time by batch cache status



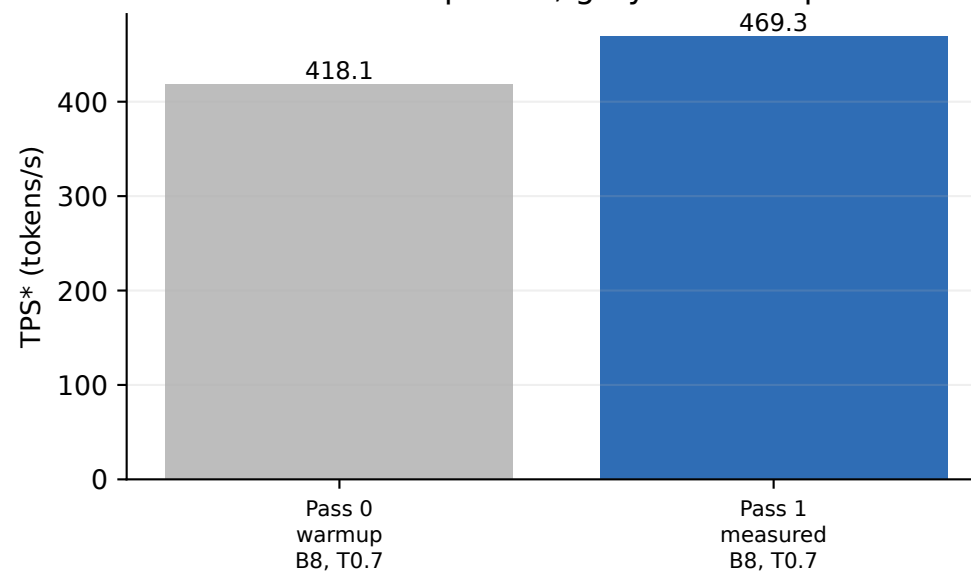
Boundary-free steps, including slow steps



Conditional AL* and source share



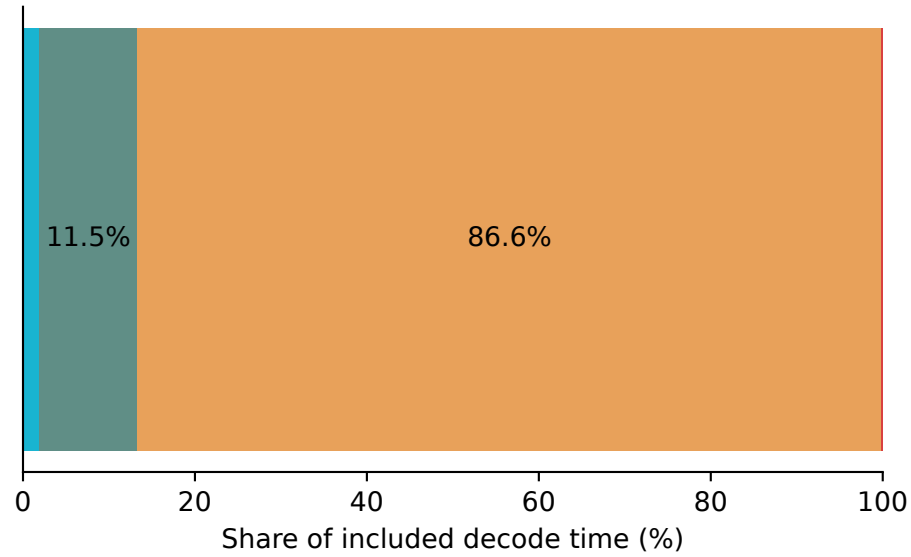
All saved passes; gray = warmup



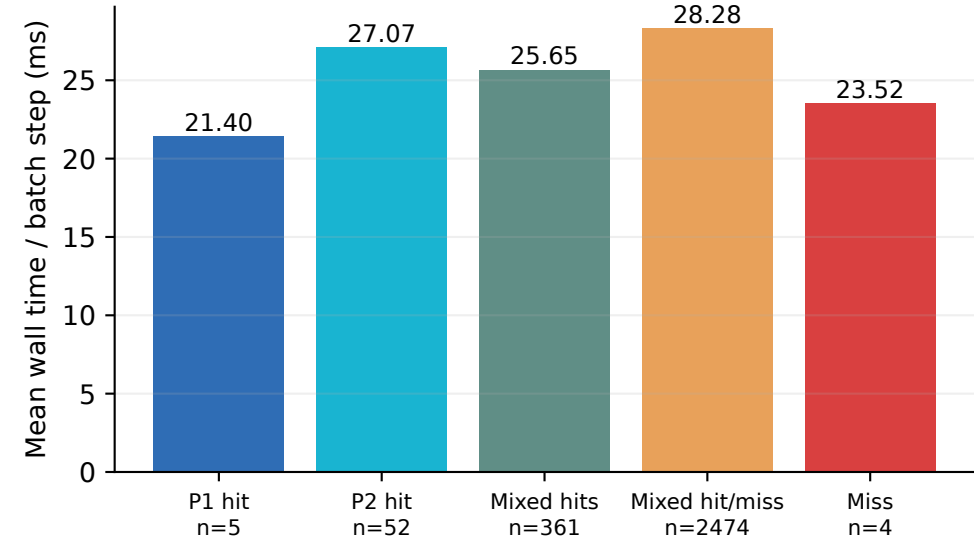
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_duet_fast_r0
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.124, hit=76.8%, TPS*=602.16

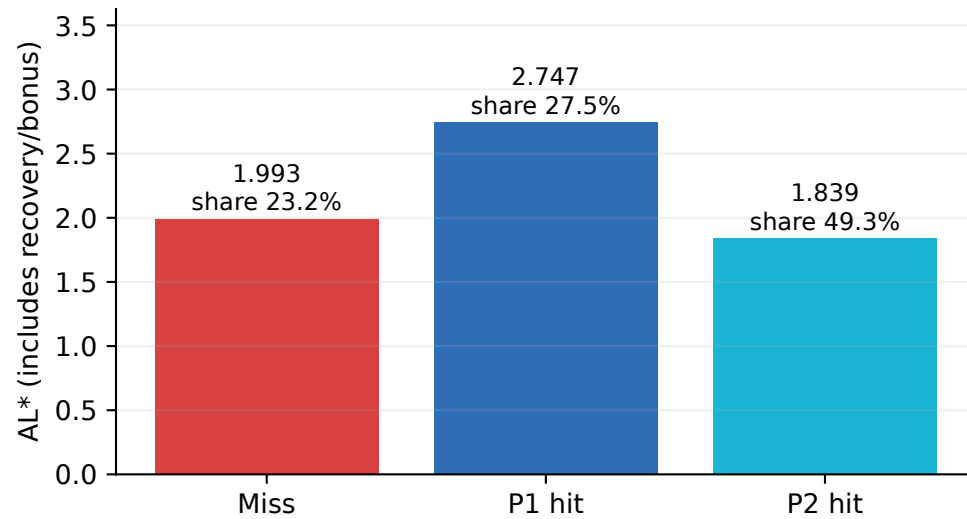
Measured wall time by batch cache status



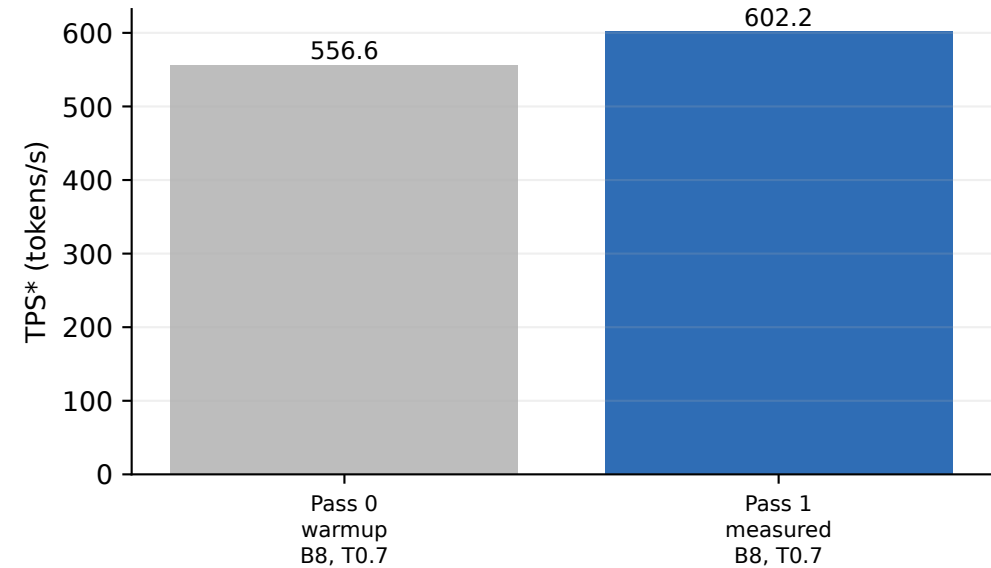
Boundary-free steps, including slow steps



Conditional AL* and source share



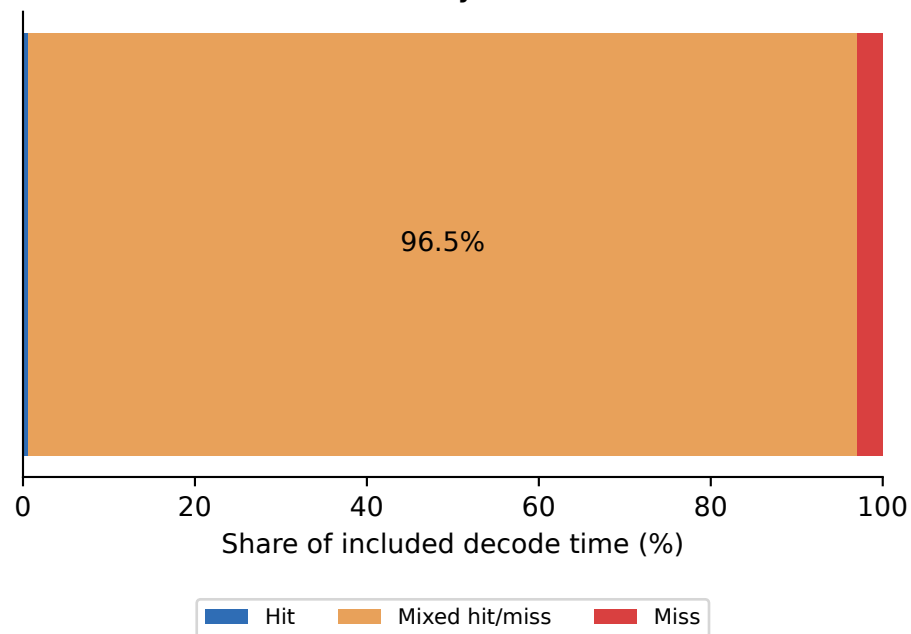
All saved passes; gray = warmup



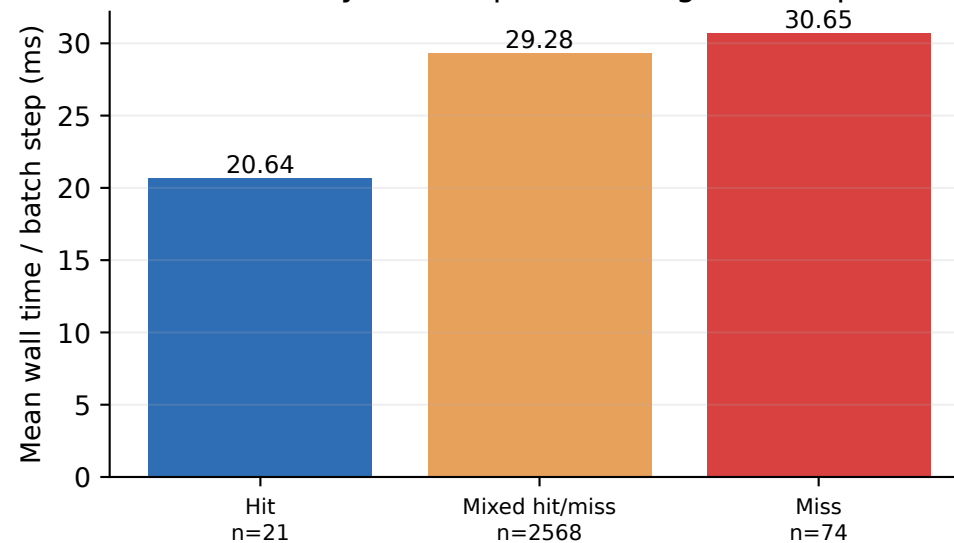
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_ssd_al_r0
SSD: K=8, F=5
Last pass: B=8, T=0.7, N=480 | AL*=2.304, hit=45.8%, TPS*=619.52

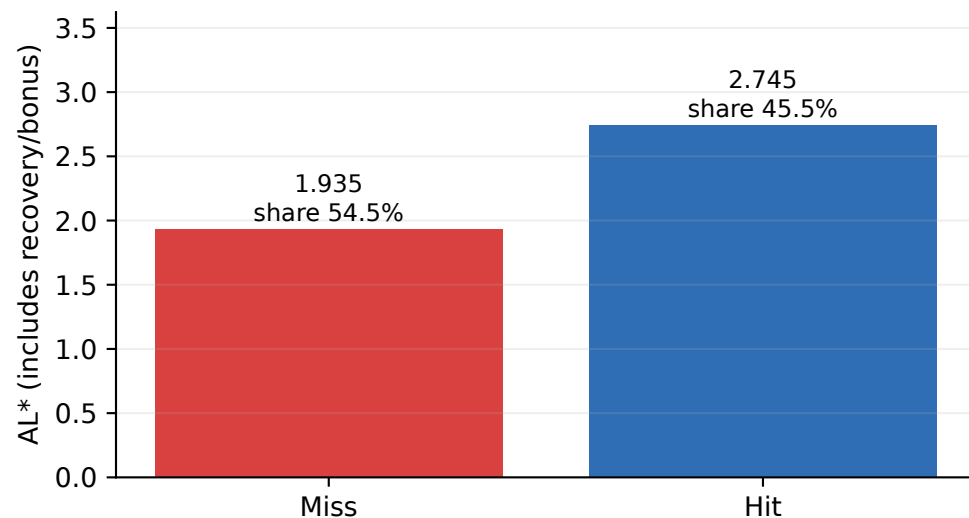
Measured wall time by batch cache status



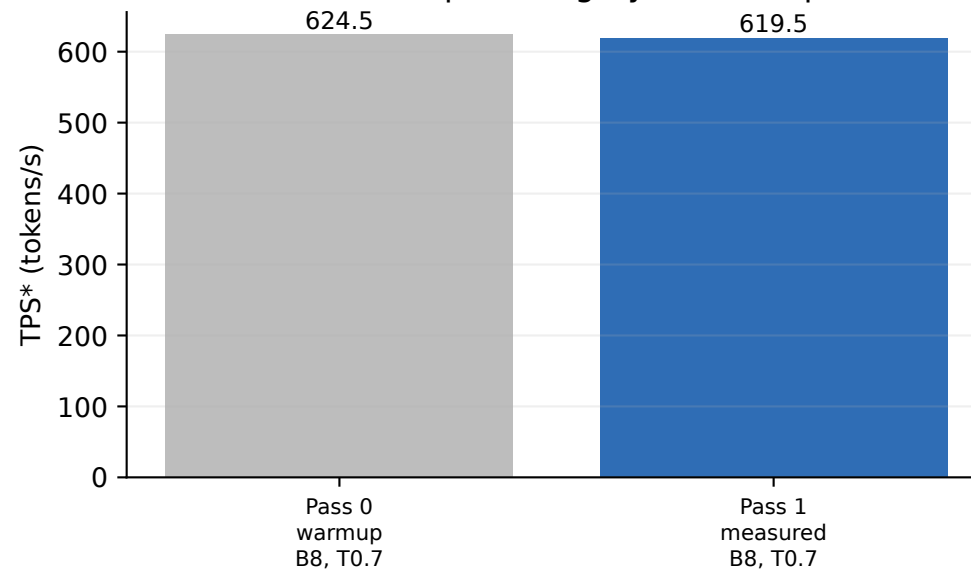
Boundary-free steps, including slow steps



Conditional AL* and source share



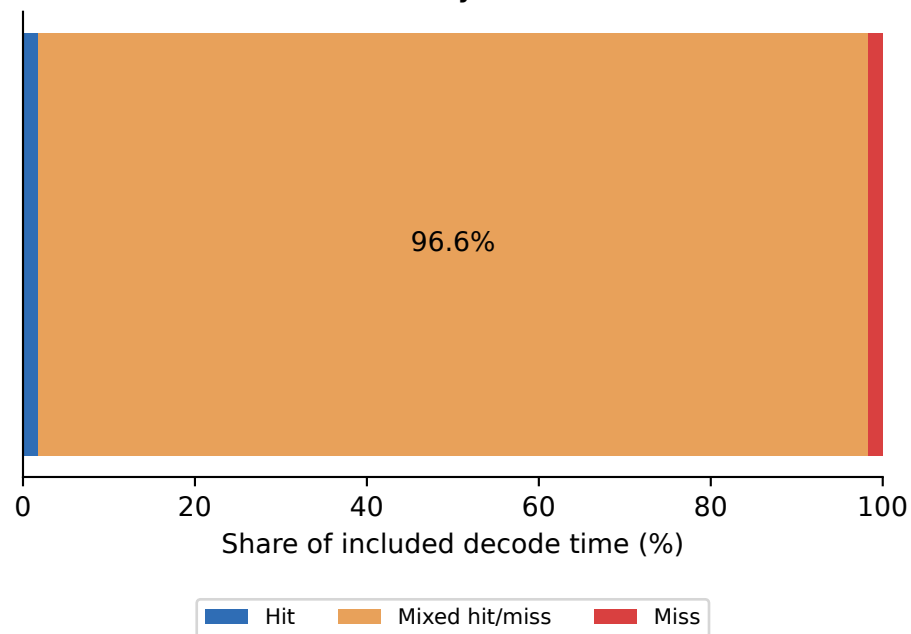
All saved passes; gray = warmup



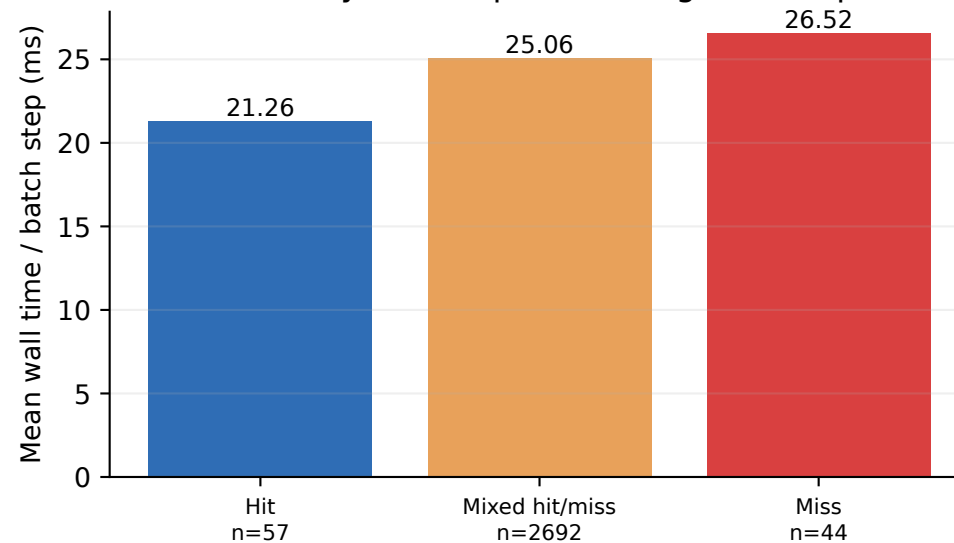
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_ssd_fast_r0
SSD: K=4, F=7
Last pass: B=8, T=0.7, N=480 | AL*=2.159, hit=54.5%, TPS*=683.09

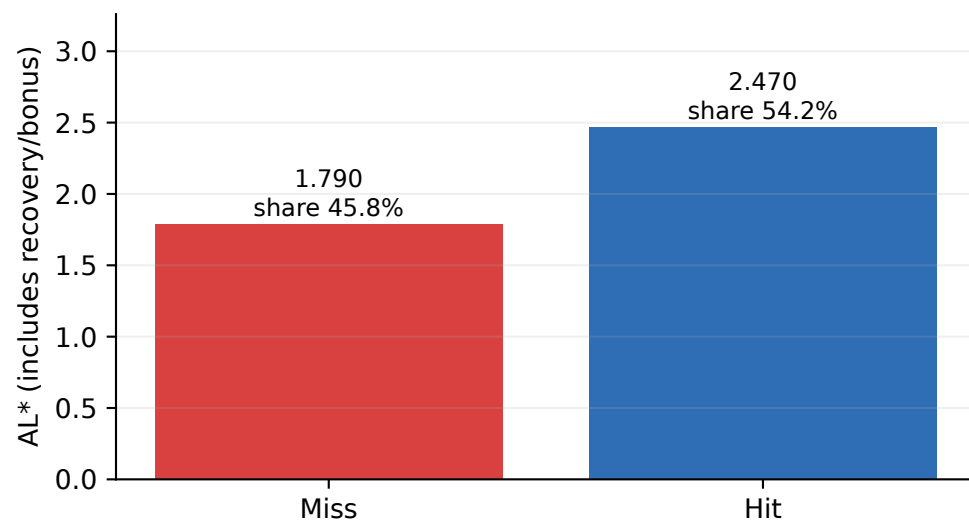
Measured wall time by batch cache status



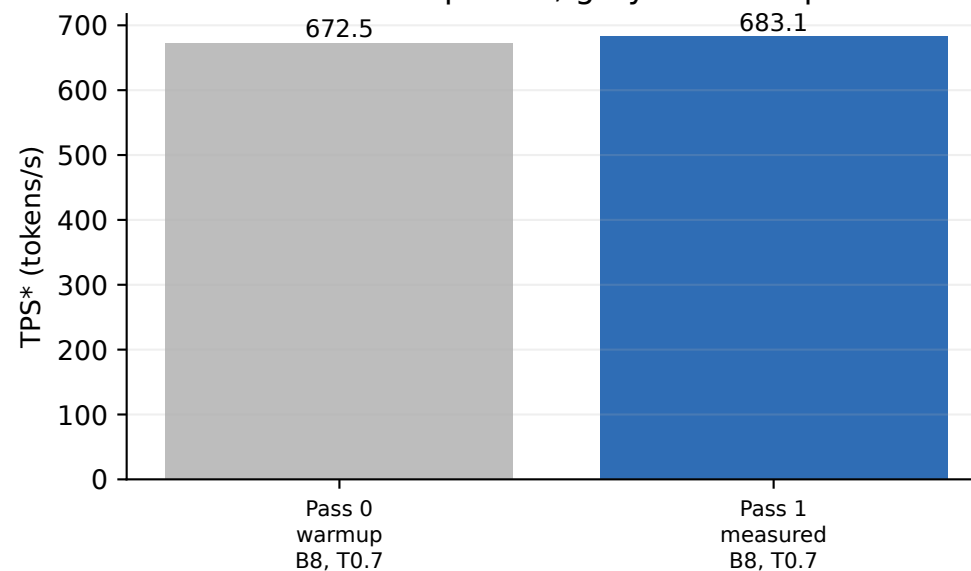
Boundary-free steps, including slow steps



Conditional AL* and source share



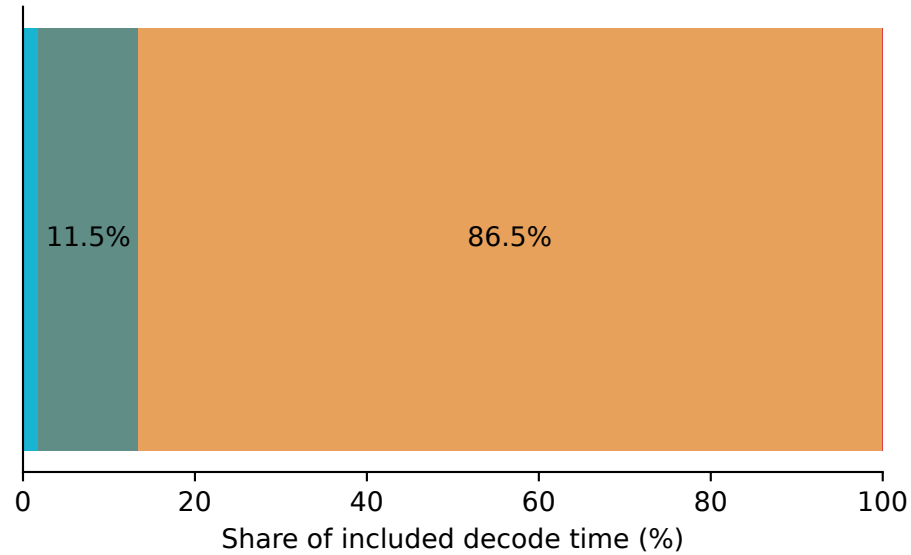
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

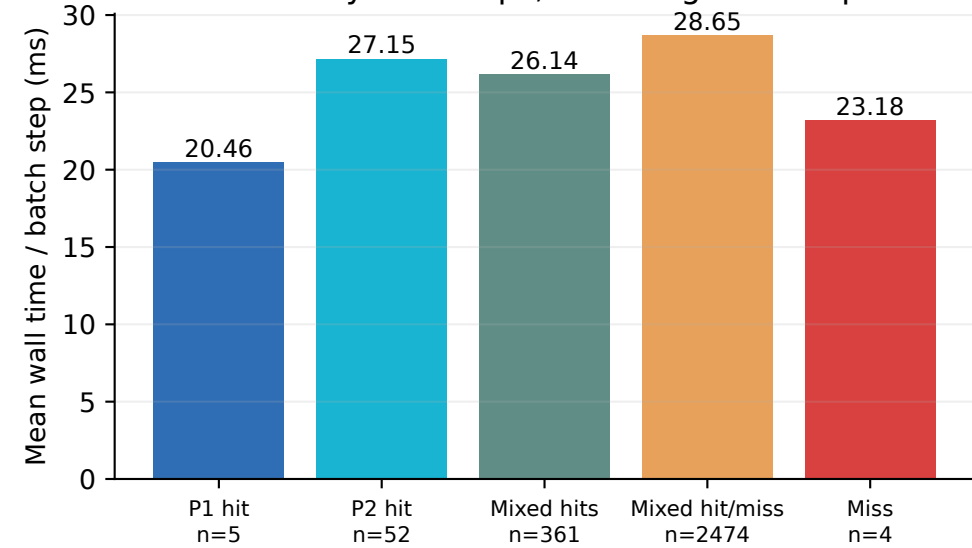
llama2_b8_implementation_off
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
Last pass: B=8, T=0.7, N=480 | AL*=2.124, hit=76.8%, TPS*=594.11

Measured wall time by batch cache status

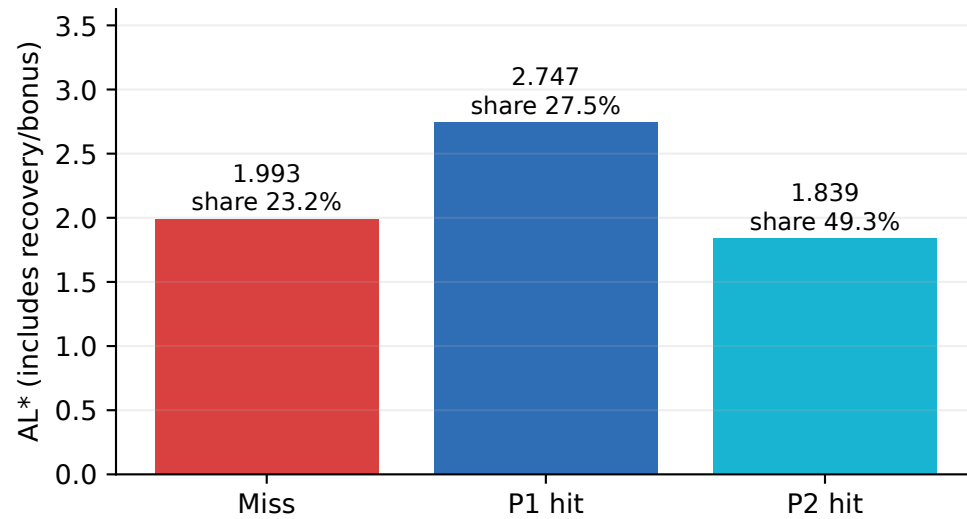


■ P1 hit ■ Mixed hits ■ Miss
■ P2 hit ■ Mixed hit/miss

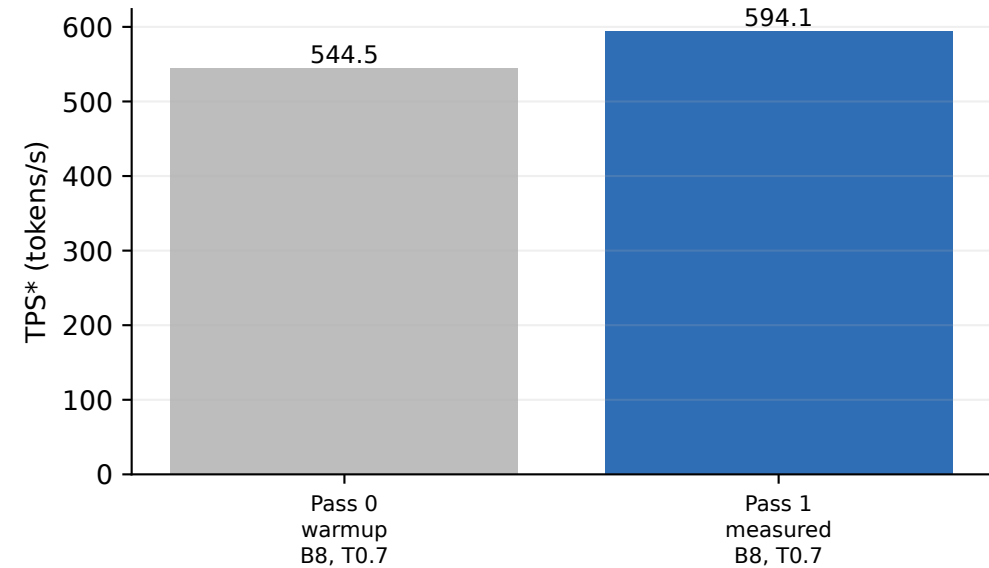
Boundary-free steps, including slow steps



Conditional AL* and source share



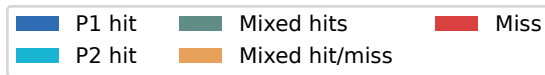
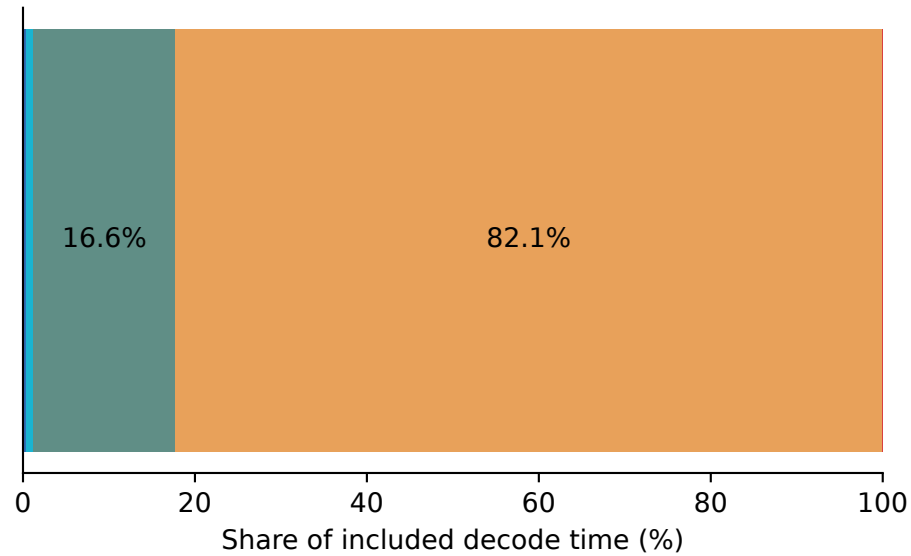
All saved passes; gray = warmup



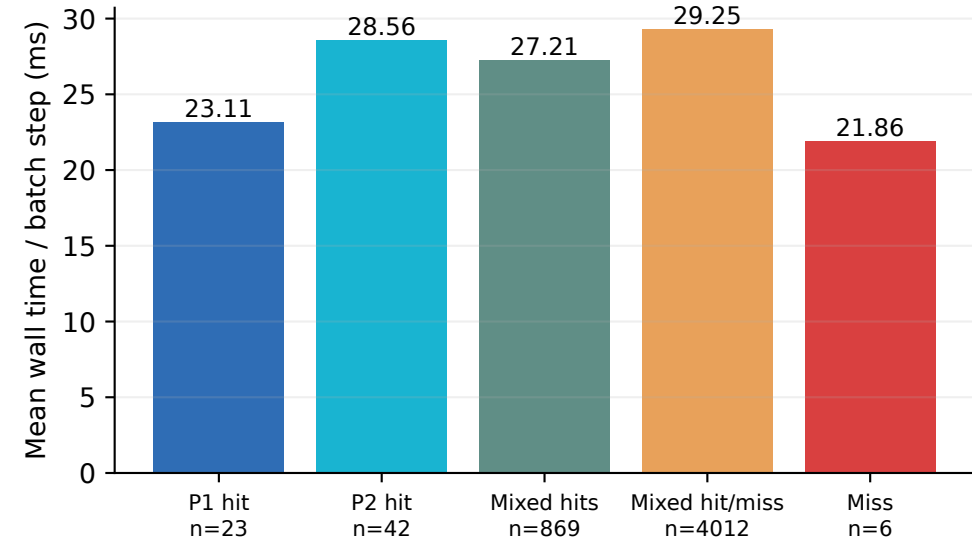
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_long_duet
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.359, hit=80.1%, TPS*=647.41

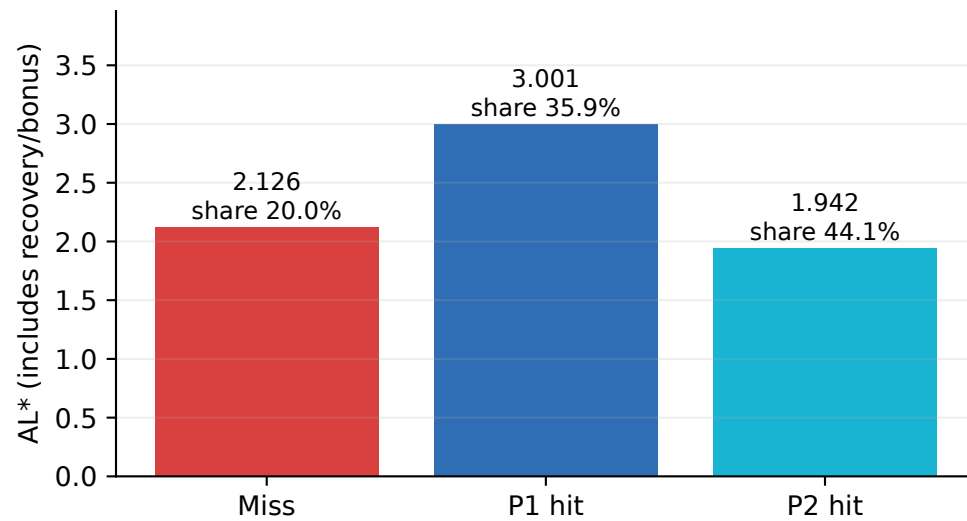
Measured wall time by batch cache status



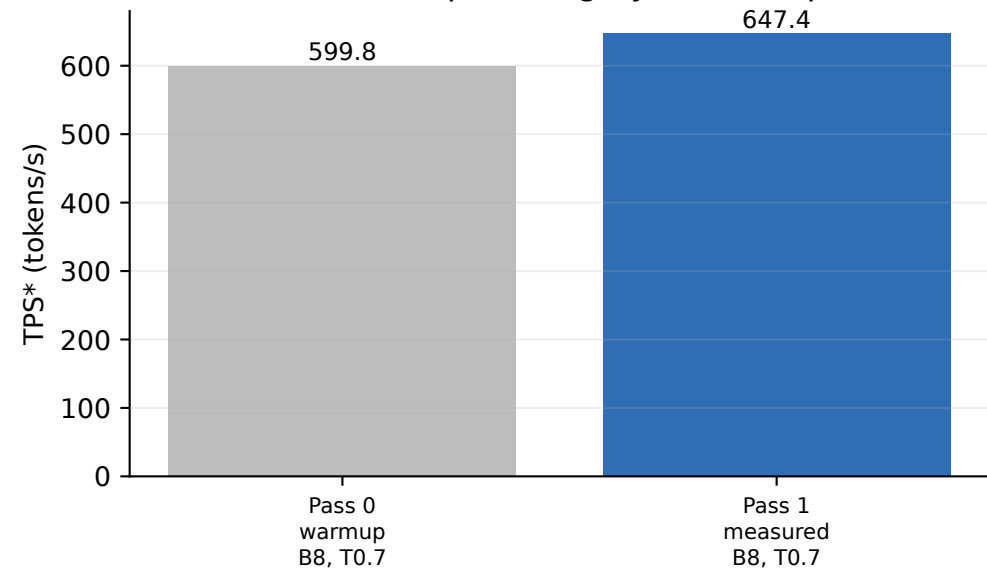
Boundary-free steps, including slow steps



Conditional AL* and source share



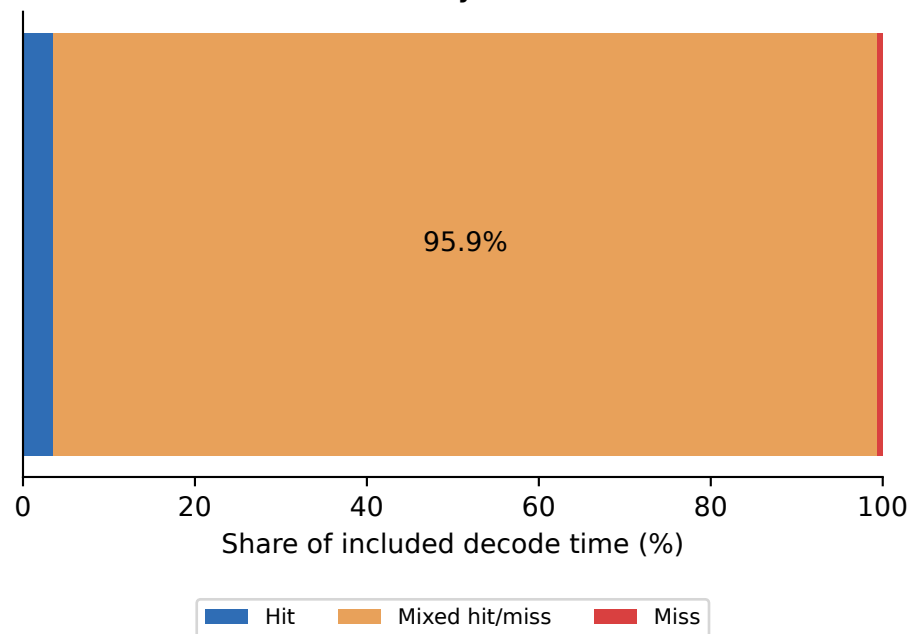
All saved passes; gray = warmup



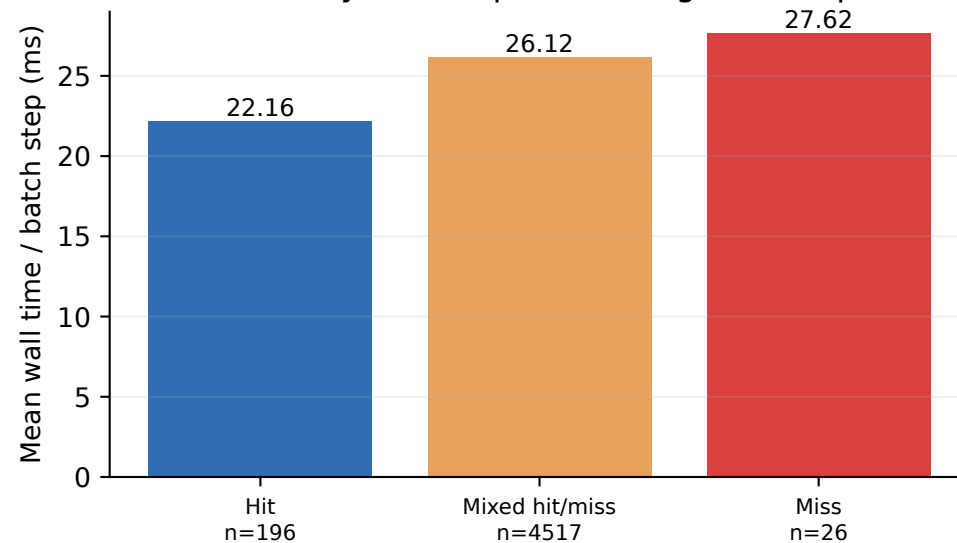
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_long_ssd
SSD: K=4, F=7
Last pass: B=8, T=0.7, N=480 | AL*=2.408, hit=60.8%, TPS*=735.69

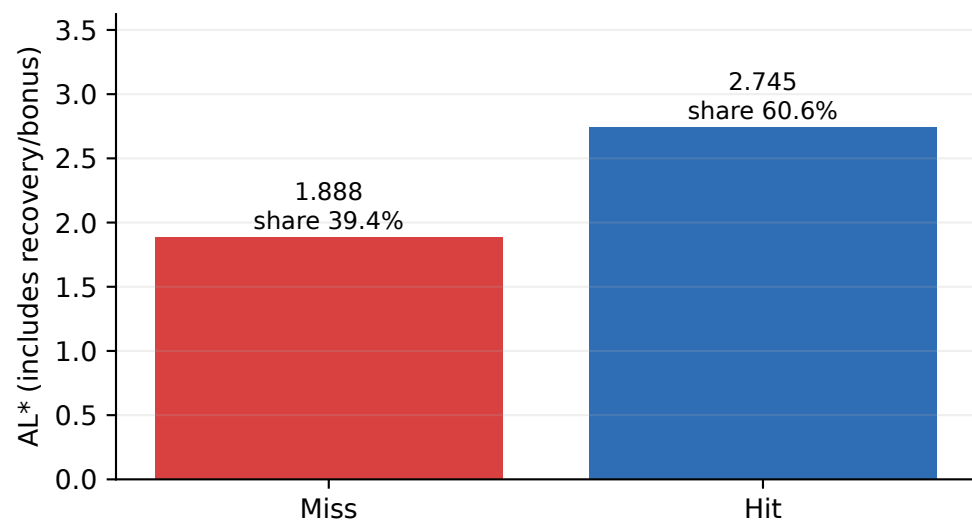
Measured wall time by batch cache status



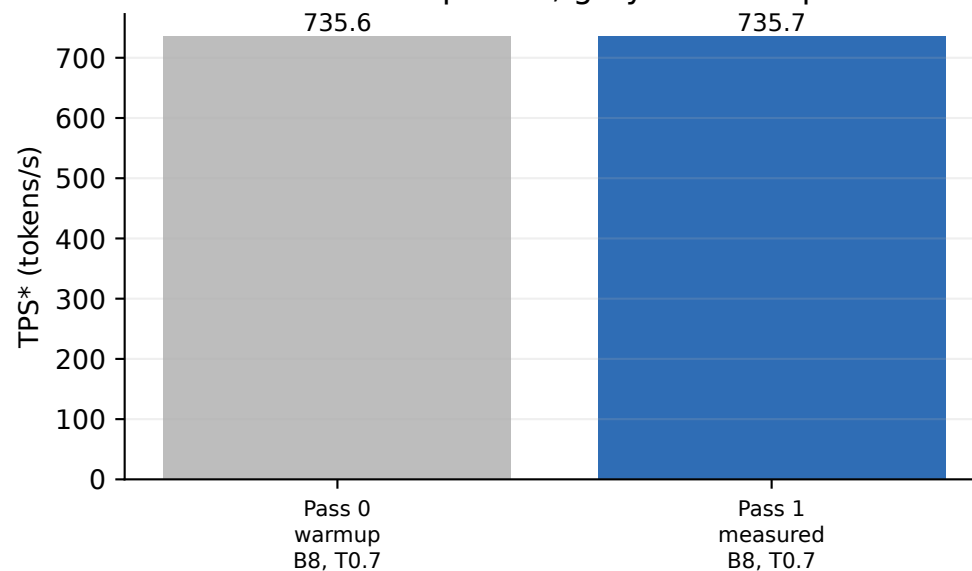
Boundary-free steps, including slow steps



Conditional AL* and source share



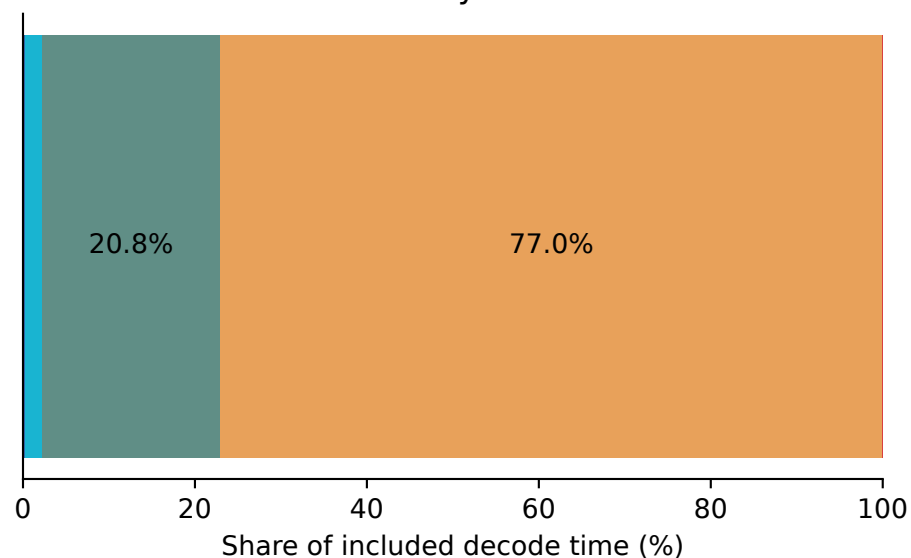
All saved passes; gray = warmup



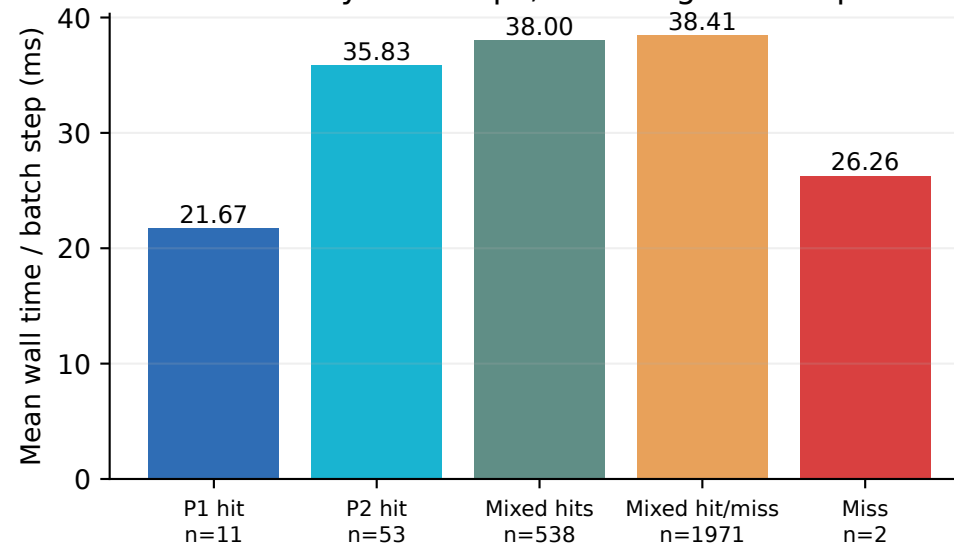
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_duet_al_r1
DUET: K1/K2=6/2, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
Last pass: B=8, T=0.7, N=480 | AL*=2.340, hit=82.3%, TPS*=483.18

Measured wall time by batch cache status

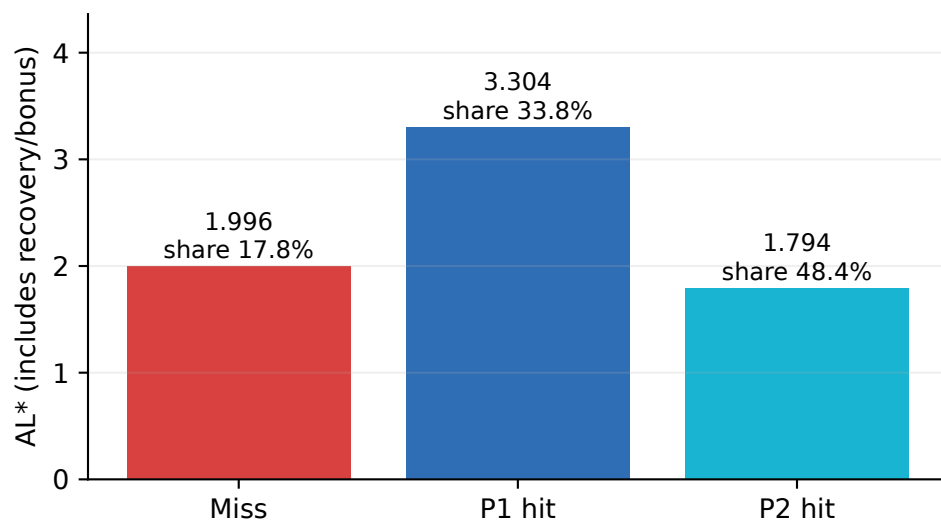


Boundary-free steps, including slow steps

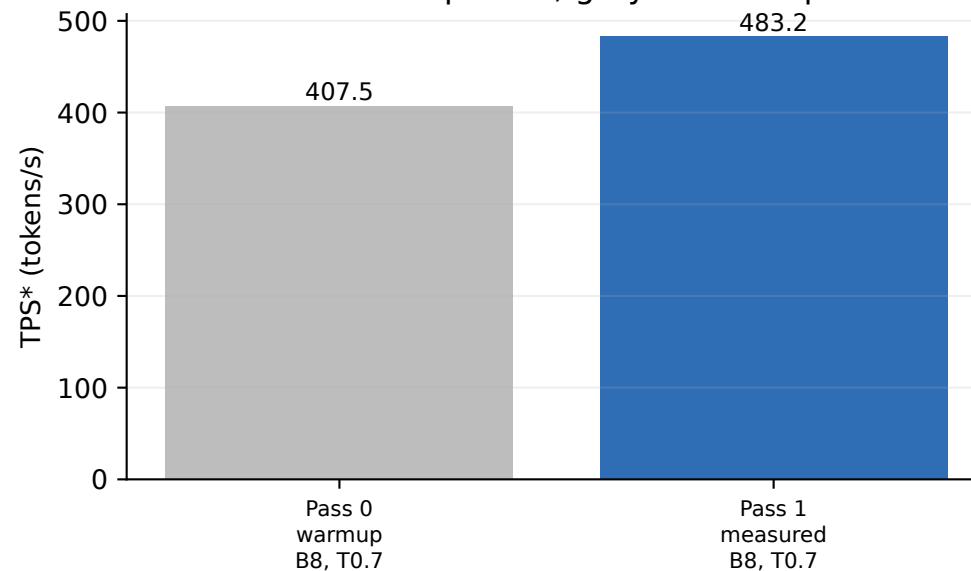


■ P1 hit ■ Mixed hits ■ Miss
■ P2 hit ■ Mixed hit/miss

Conditional AL* and source share



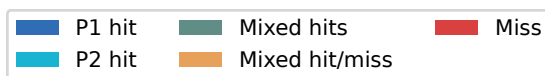
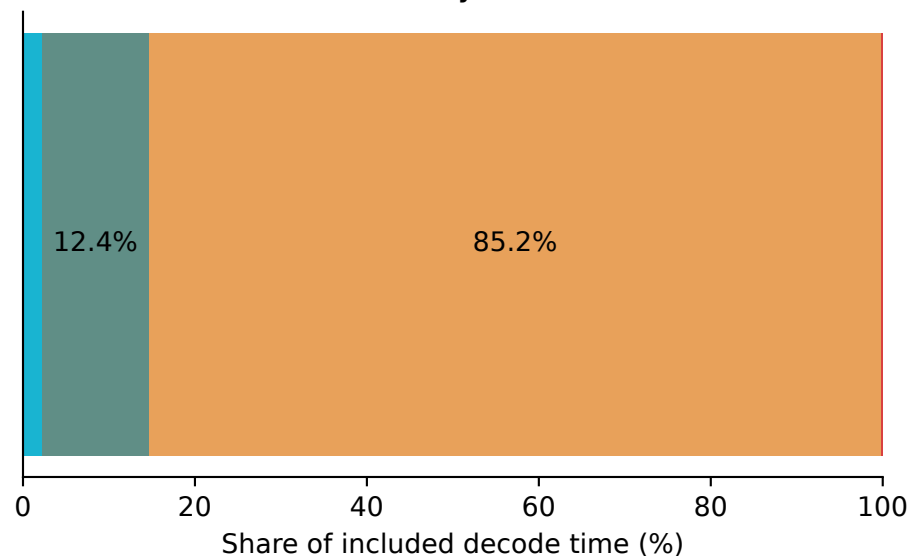
All saved passes; gray = warmup



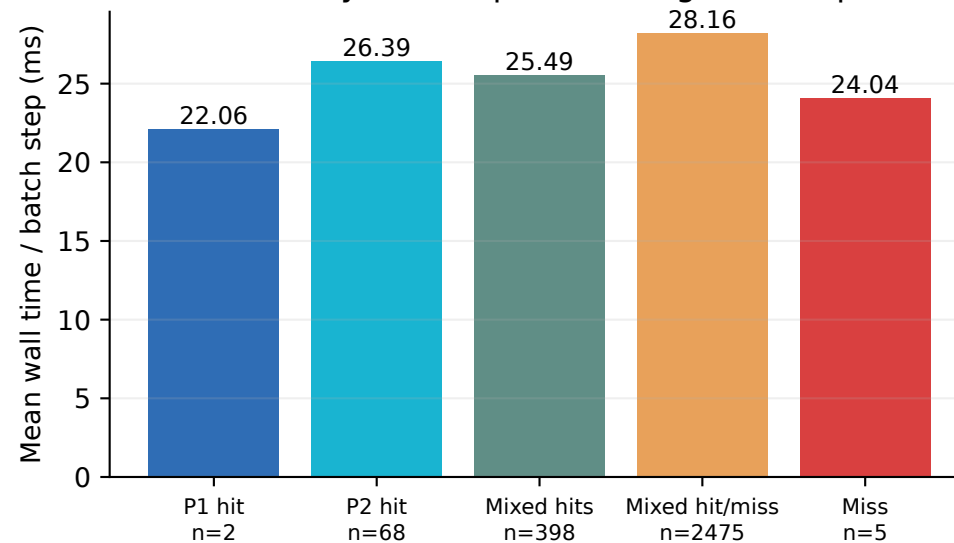
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_duet_fast_r1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
Last pass: B=8, T=0.7, N=480 | AL*=2.090, hit=77.1%, TPS*=594.65

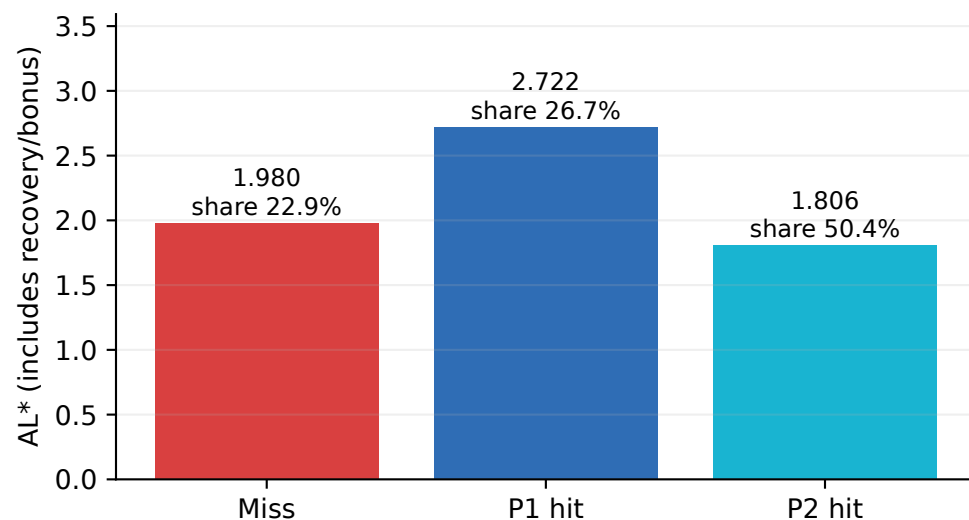
Measured wall time by batch cache status



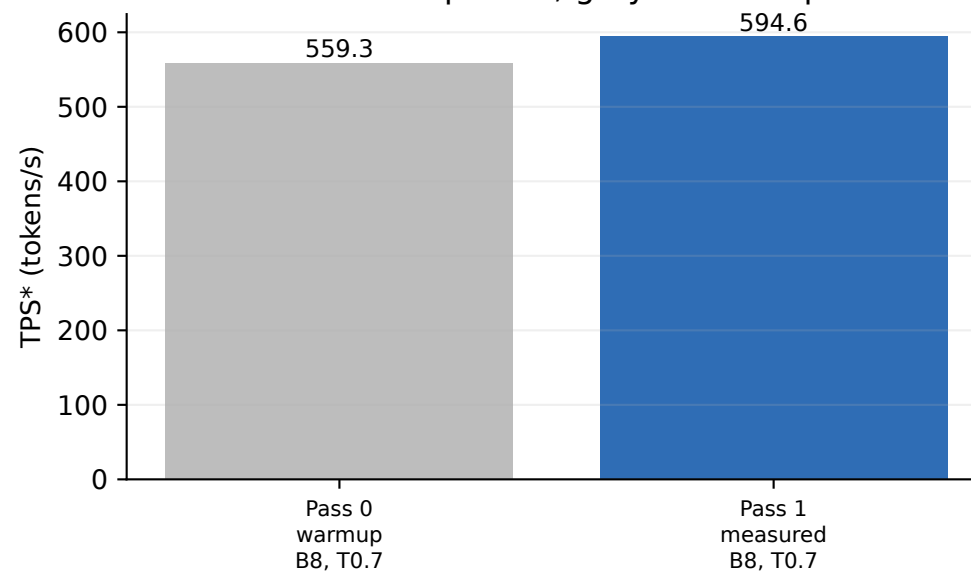
Boundary-free steps, including slow steps



Conditional AL* and source share



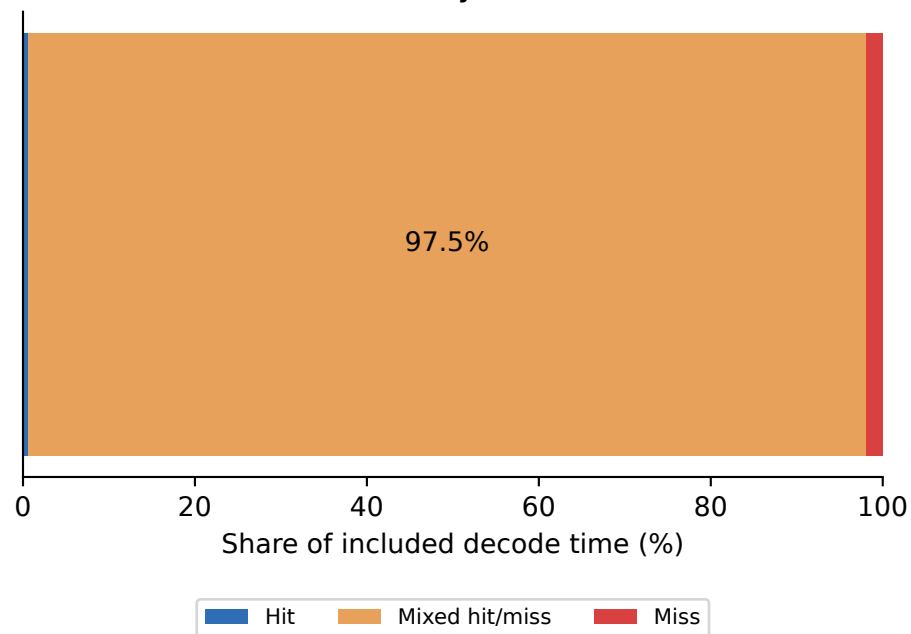
All saved passes; gray = warmup



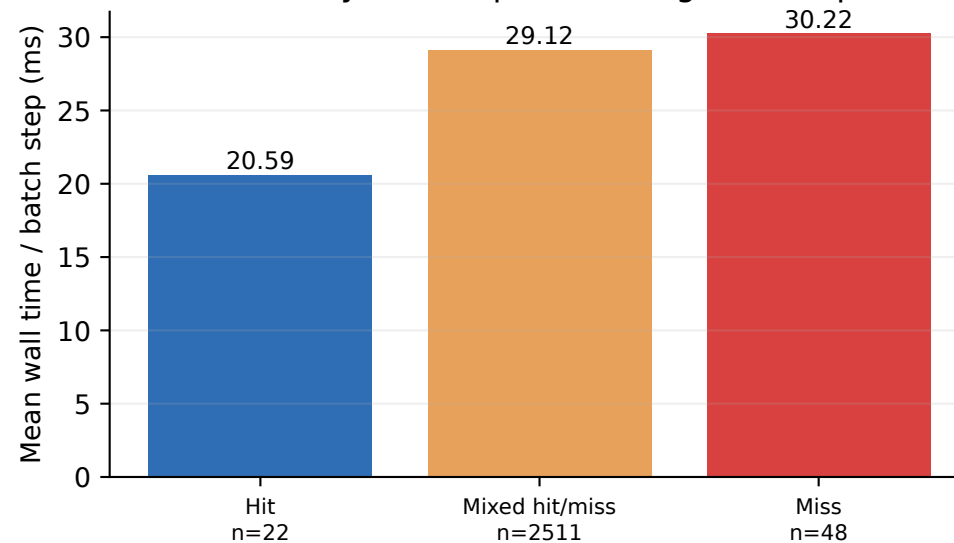
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_ssd_al_r1
SSD: K=8, F=5
Last pass: B=8, T=0.7, N=480 | AL*=2.317, hit=47.3%, TPS*=628.50

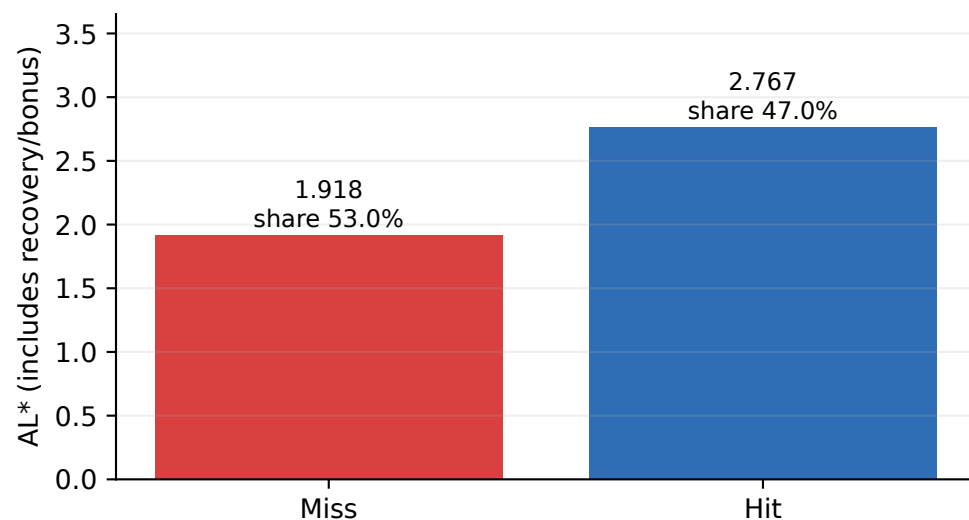
Measured wall time by batch cache status



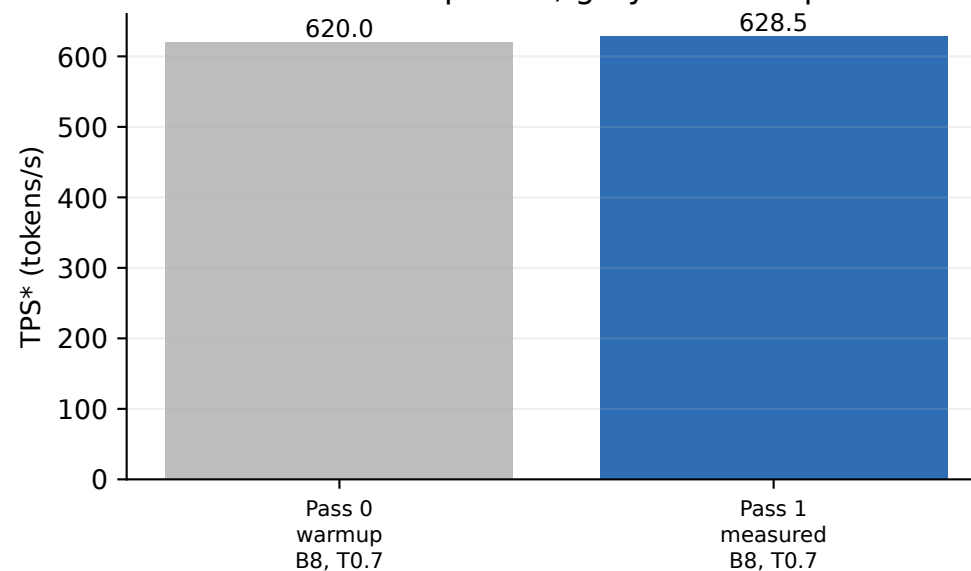
Boundary-free steps, including slow steps



Conditional AL* and source share



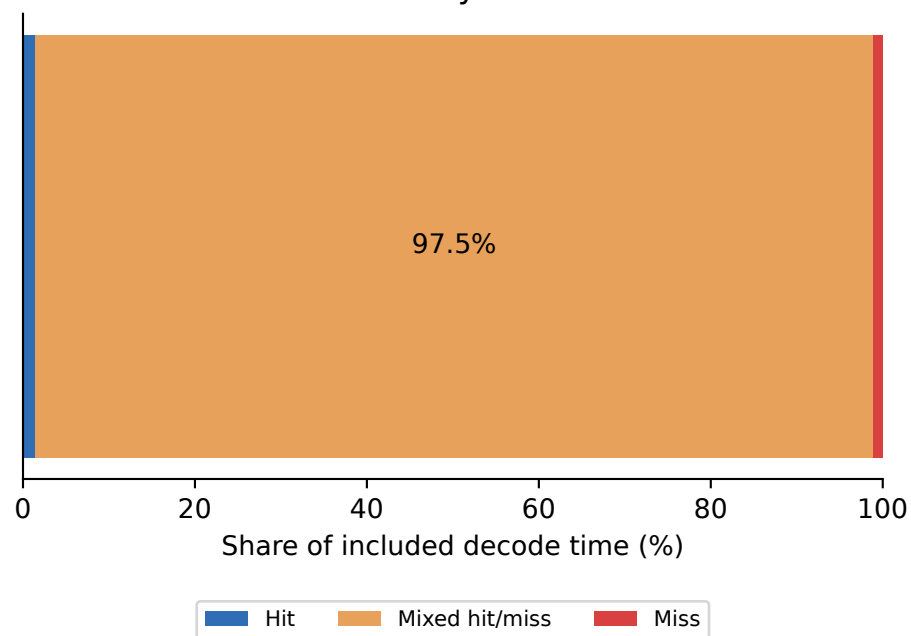
All saved passes; gray = warmup



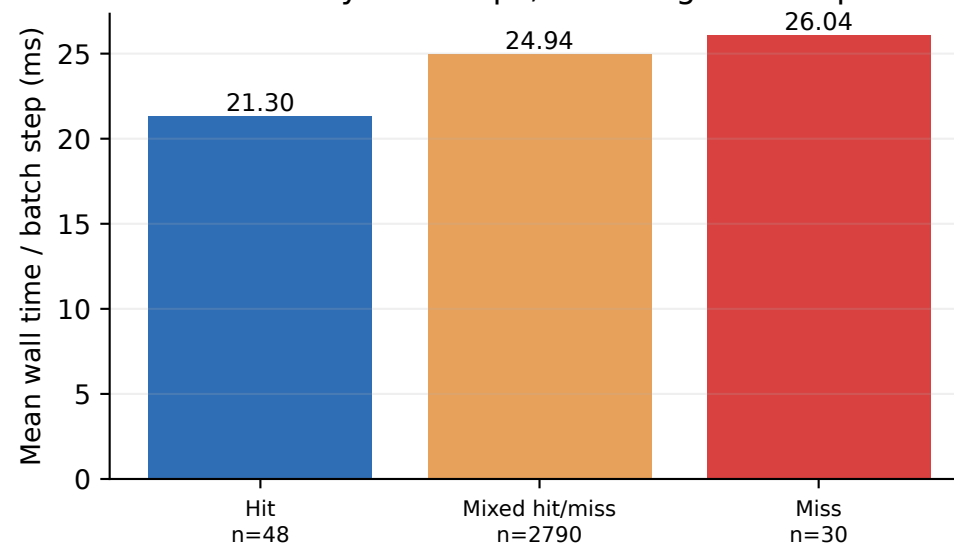
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_final_ssd_fast_r1
SSD: K=4, F=7
Last pass: B=8, T=0.7, N=480 | AL*=2.143, hit=54.3%, TPS*=681.79

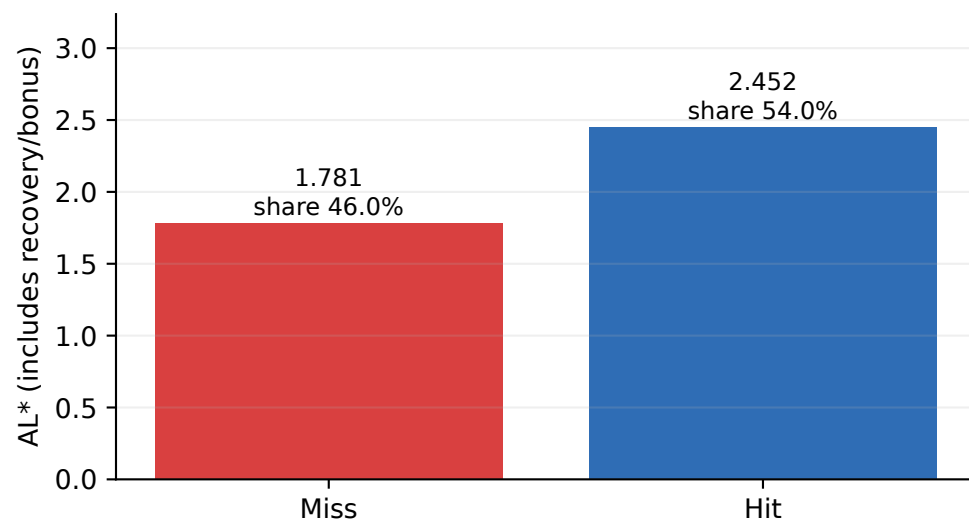
Measured wall time by batch cache status



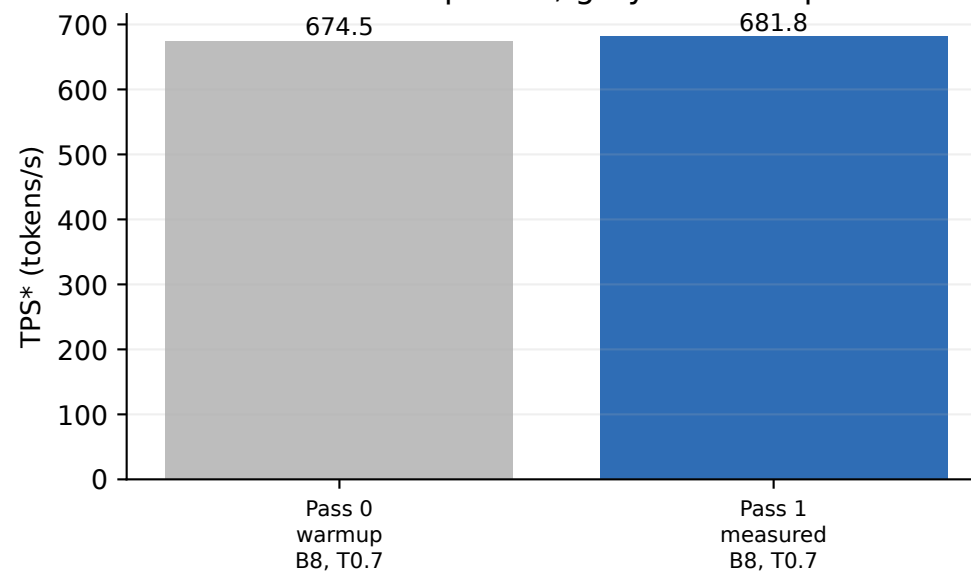
Boundary-free steps, including slow steps



Conditional AL* and source share



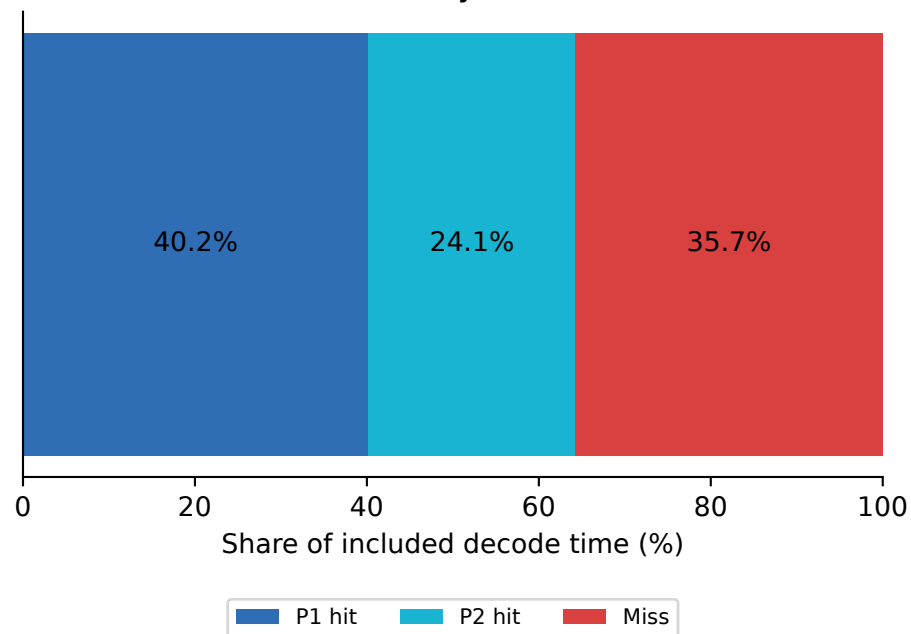
All saved passes; gray = warmup



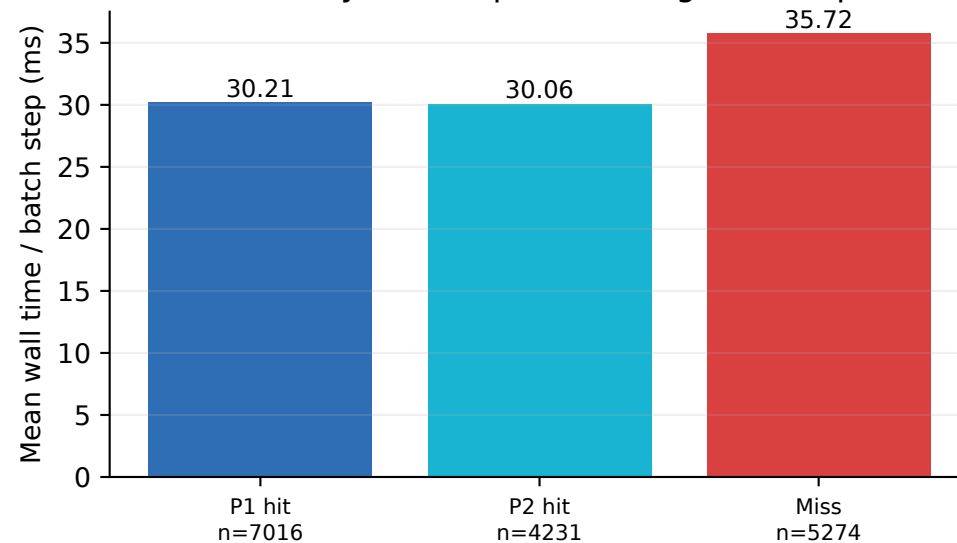
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_final_duet_al_r0
DUET: K1/K2=6/4, exit=16 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
Last pass: B=1, T=0.7, N=480 | AL*=3.000, hit=68.1%, TPS*=93.97

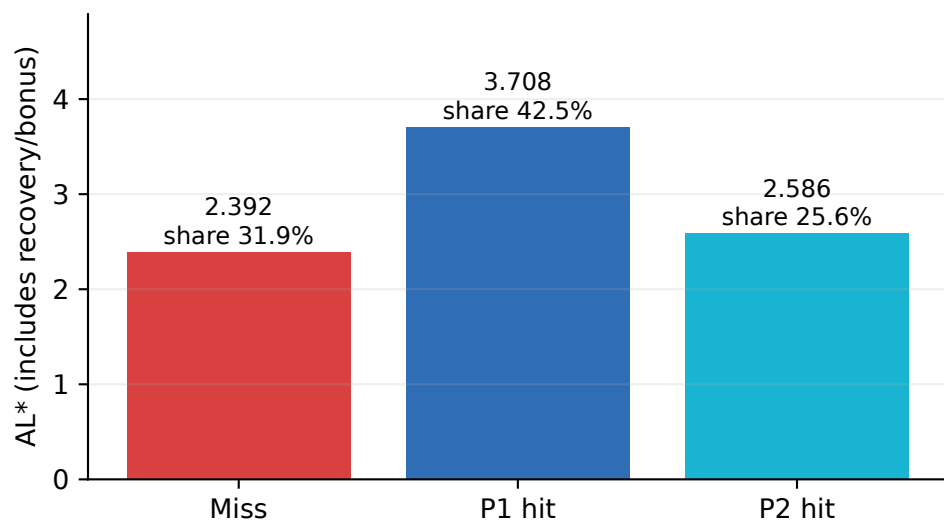
Measured wall time by batch cache status



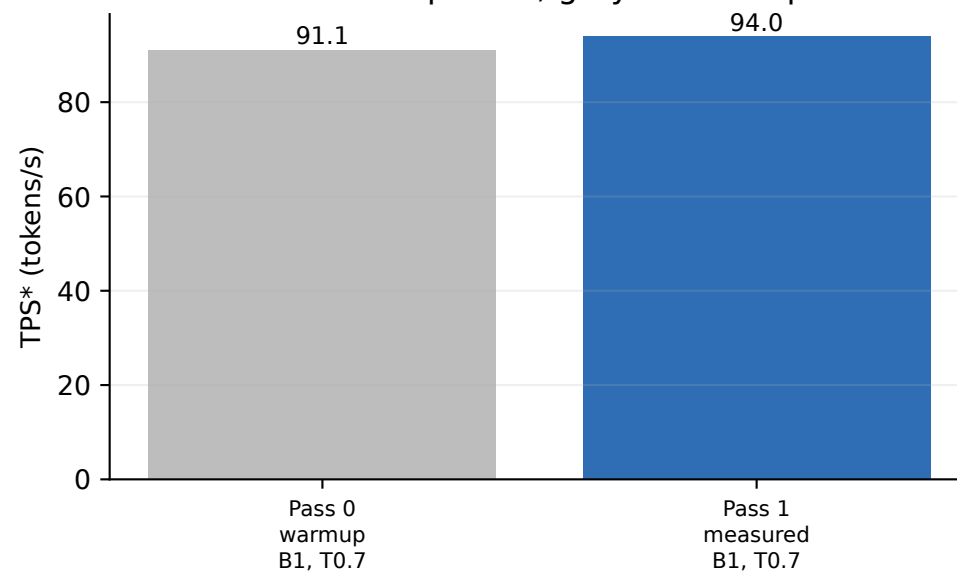
Boundary-free steps, including slow steps



Conditional AL* and source share



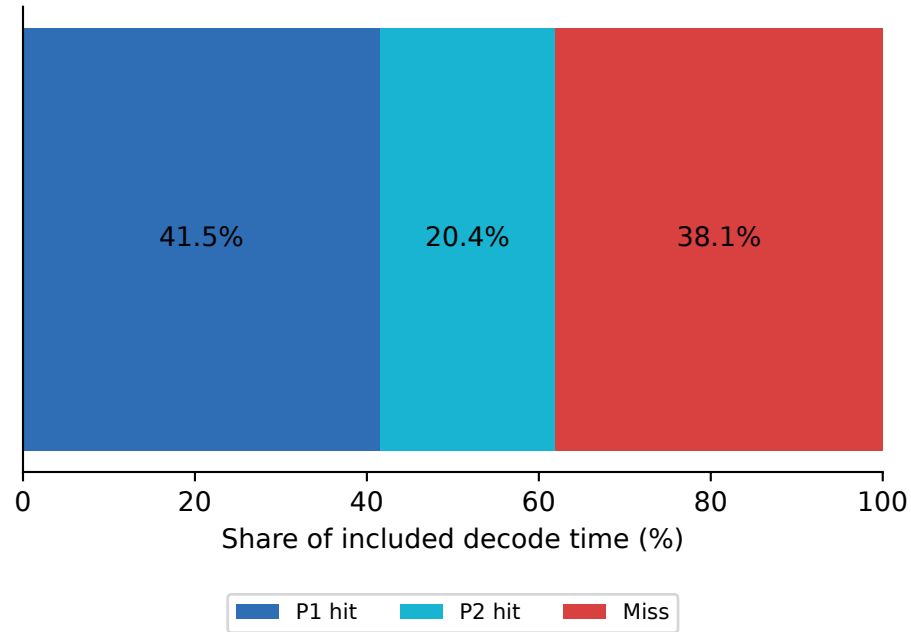
All saved passes; gray = warmup



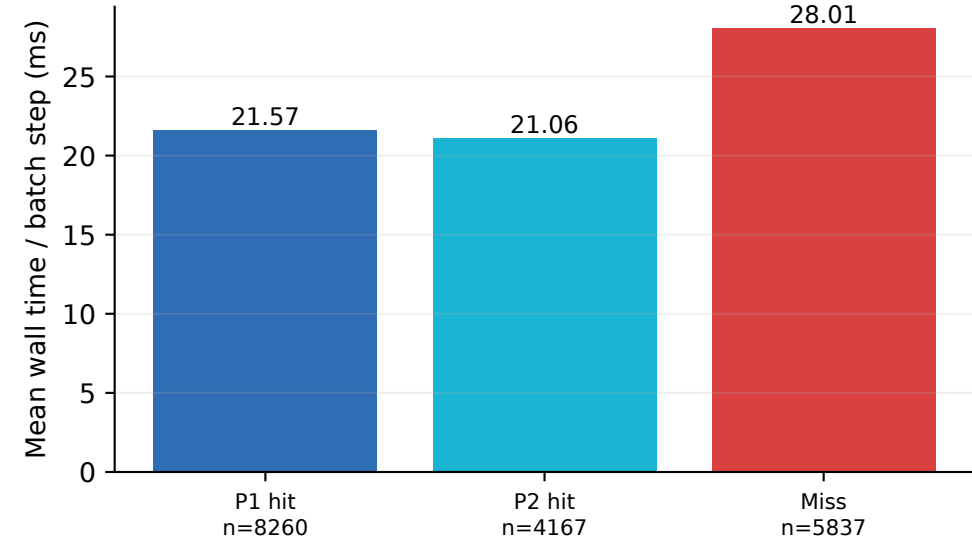
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_final_duet_fast_r0
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
Last pass: B=1, T=0.7, N=480 | AL*=2.679, hit=68.1%, TPS*=113.93

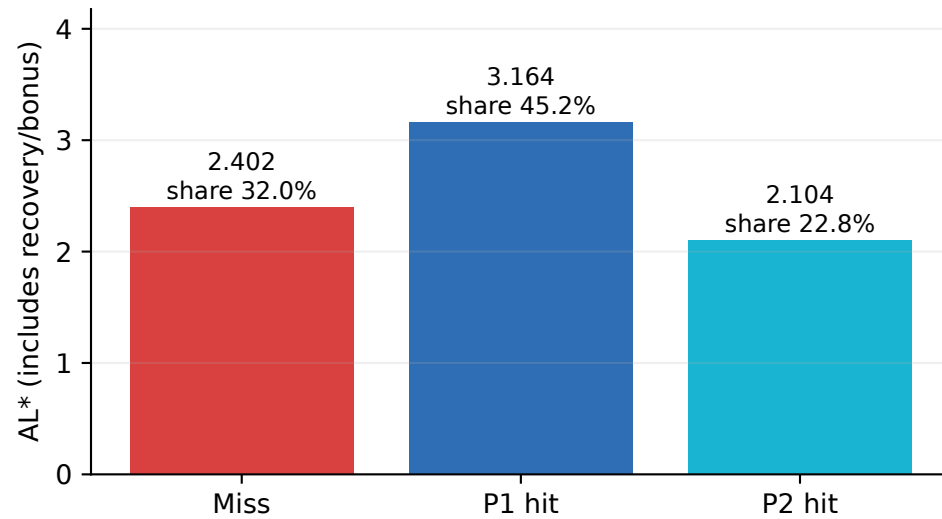
Measured wall time by batch cache status



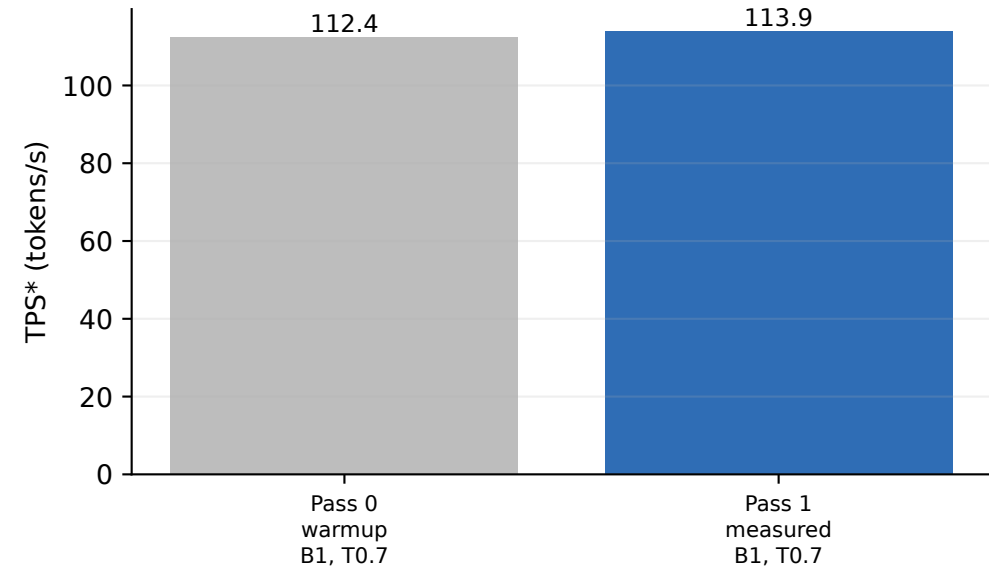
Boundary-free steps, including slow steps



Conditional AL* and source share

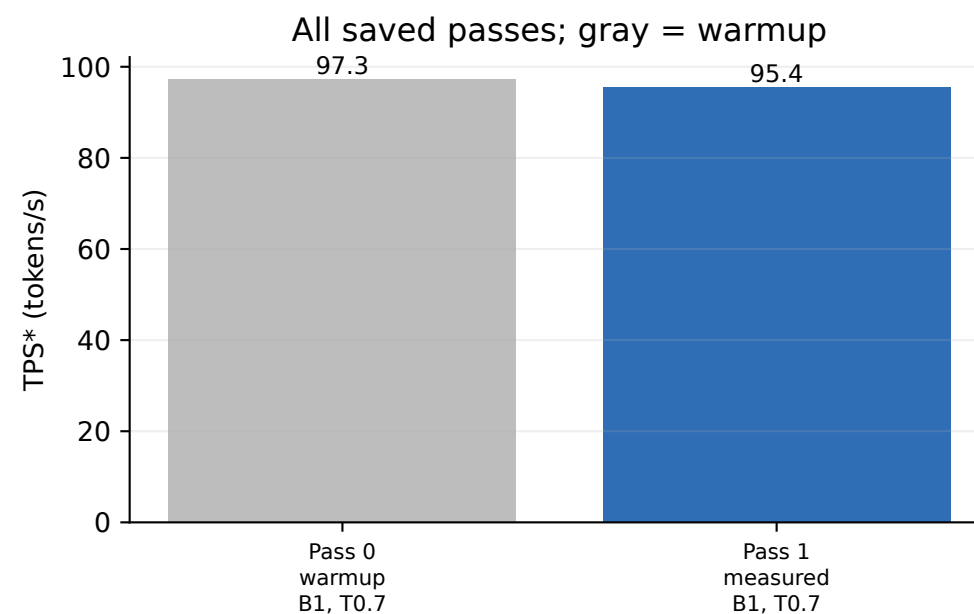
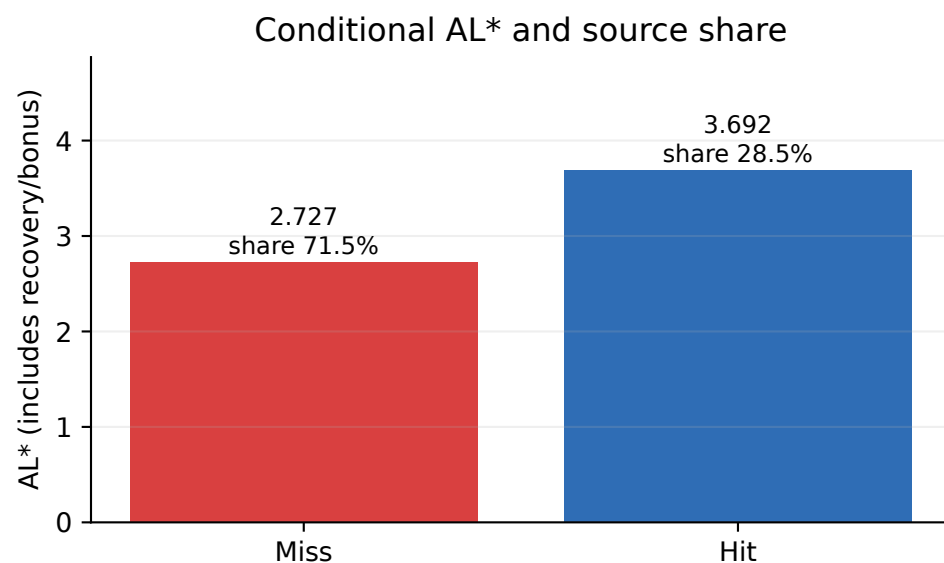
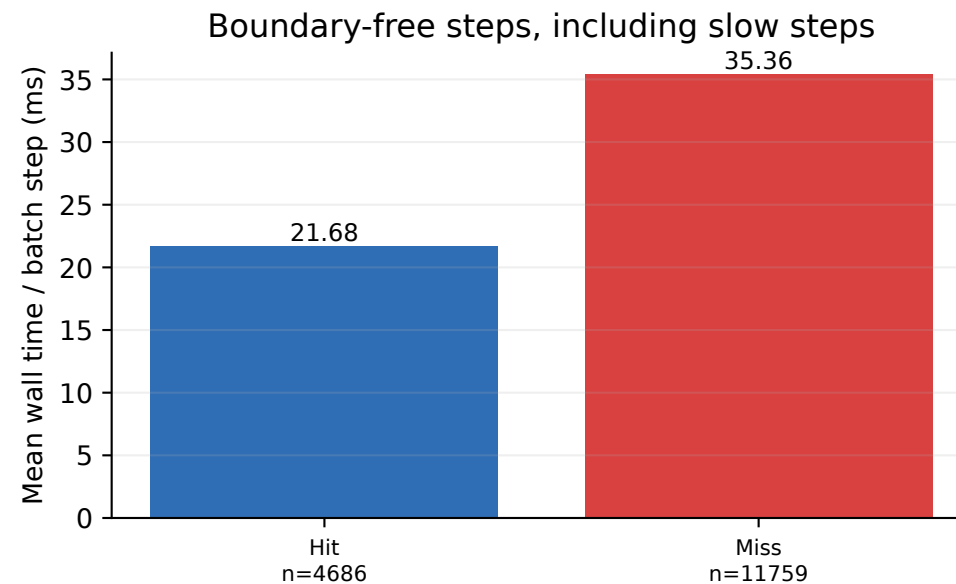
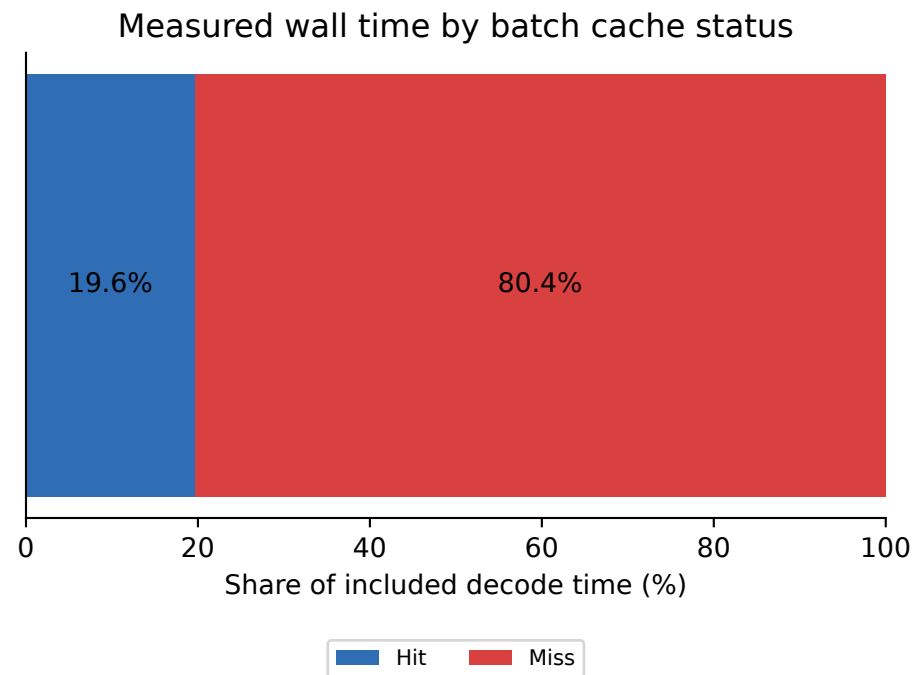


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

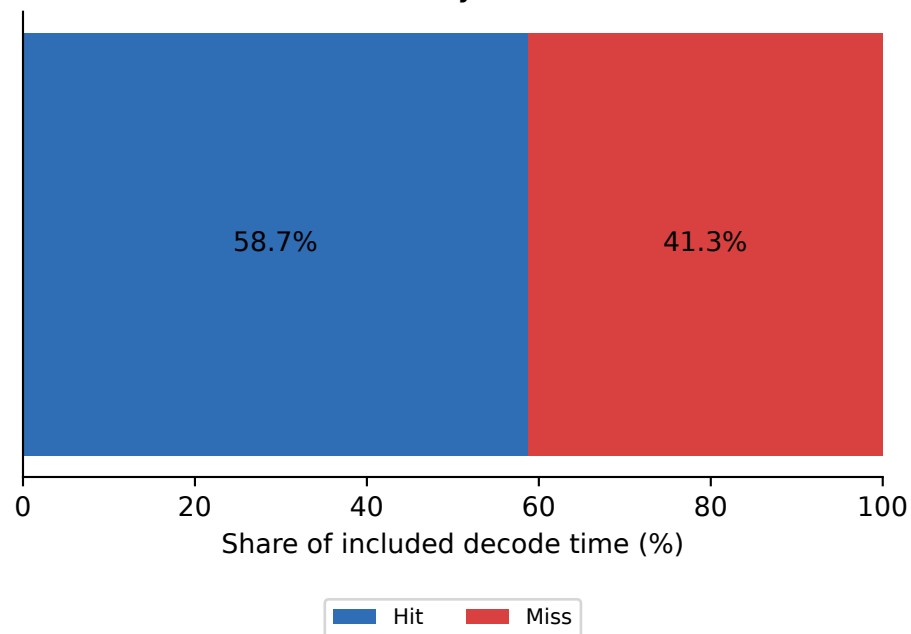
llama3_b1_final_ssd_al_r0
SSD: K=8, F=1
Last pass: B=1, T=0.7, N=480 | AL*=3.002, hit=28.7%, TPS*=95.42



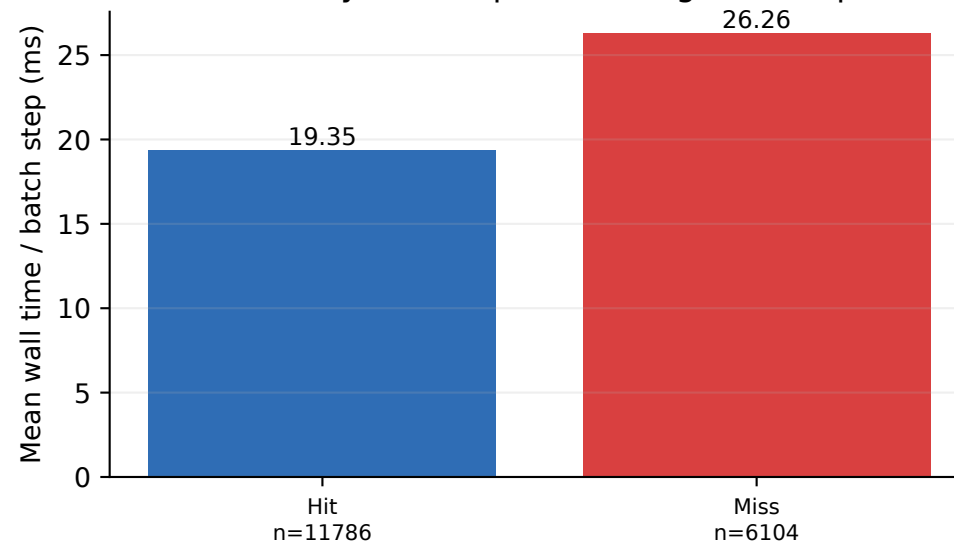
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_final_ssd_fast_r0
SSD: K=4, F=7
Last pass: B=1, T=0.7, N=480 | AL*=2.679, hit=65.8%, TPS*=123.40

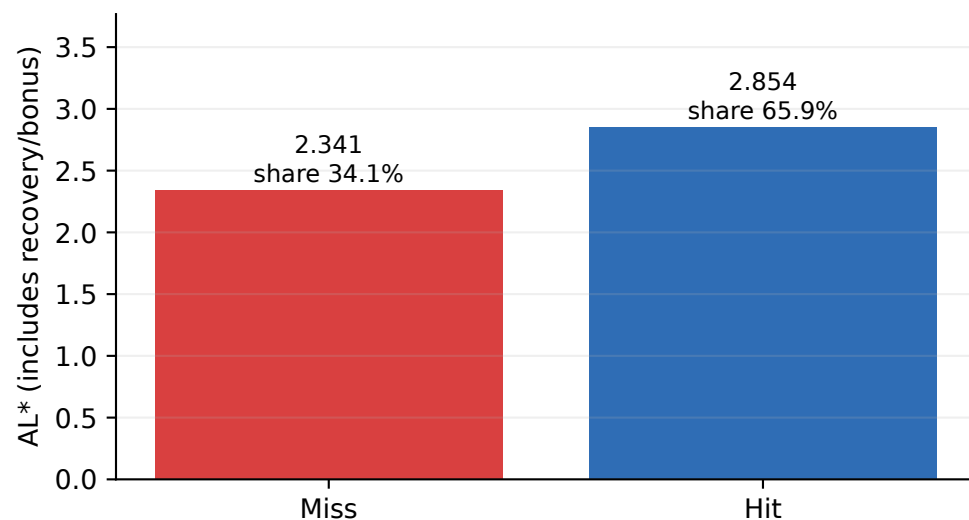
Measured wall time by batch cache status



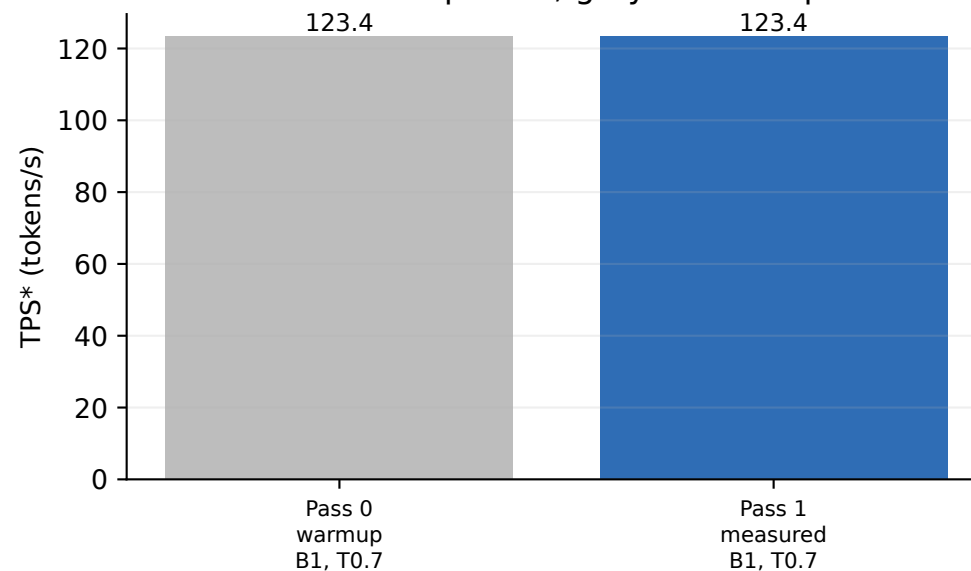
Boundary-free steps, including slow steps



Conditional AL* and source share



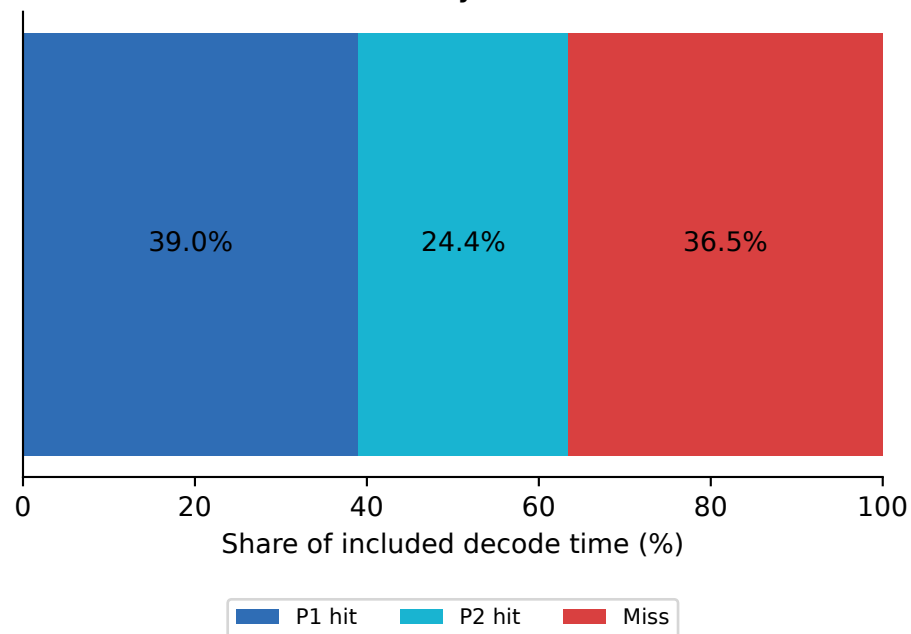
All saved passes; gray = warmup



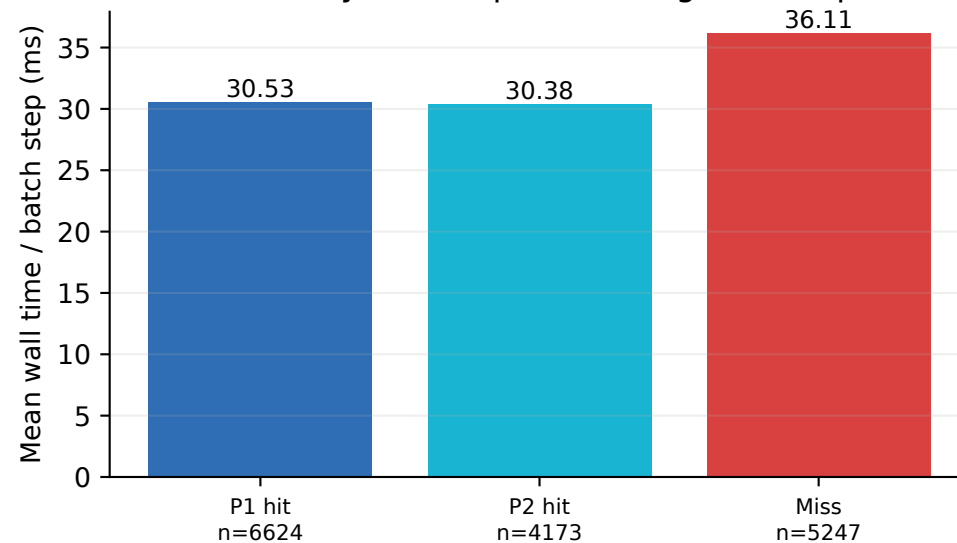
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_final_duet_al_r1
DUET: K1/K2=6/4, exit=16 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
Last pass: B=1, T=0.7, N=480 | AL*=2.952, hit=67.4%, TPS*=91.36

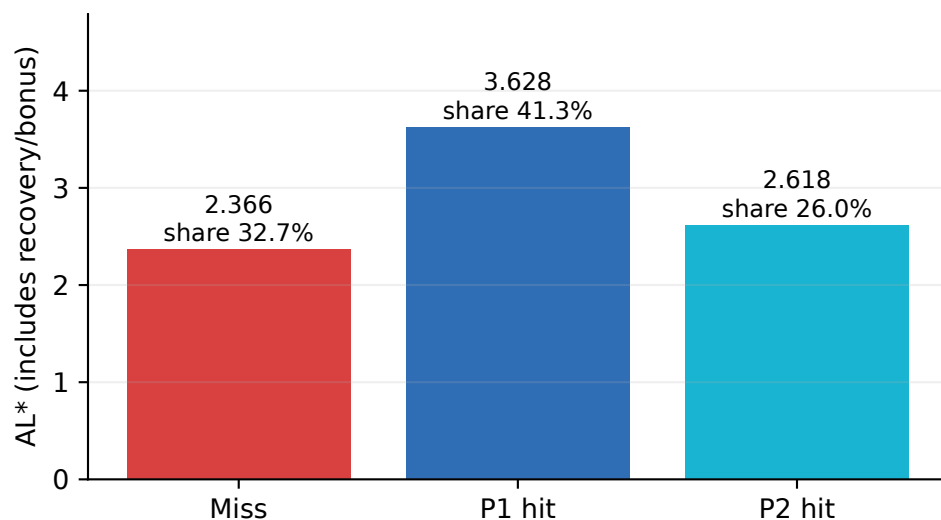
Measured wall time by batch cache status



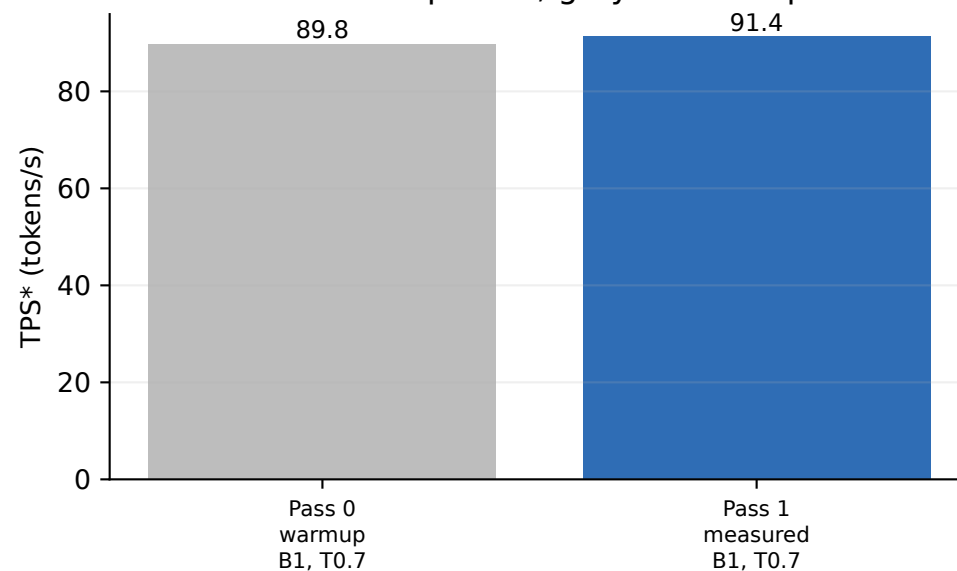
Boundary-free steps, including slow steps



Conditional AL* and source share



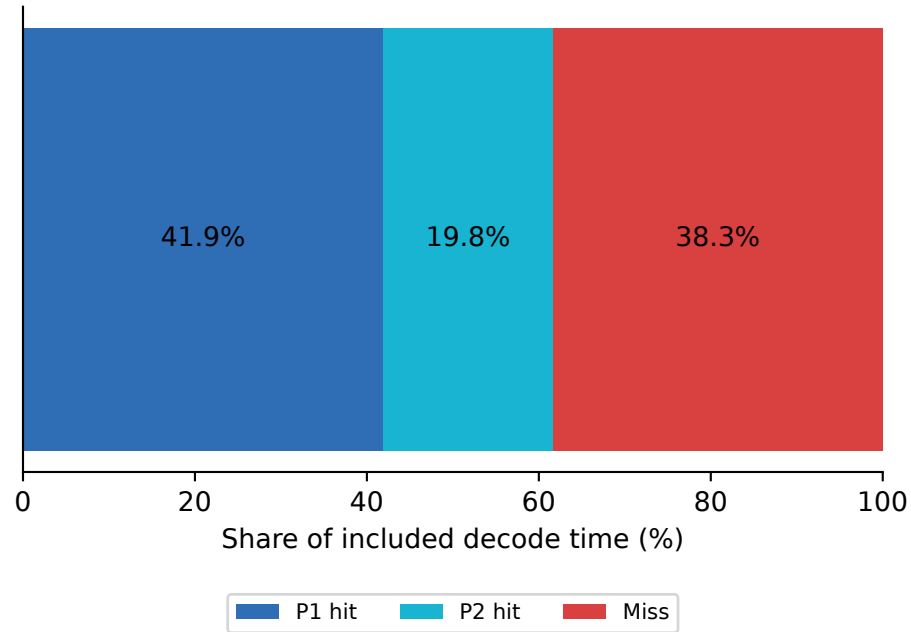
All saved passes; gray = warmup



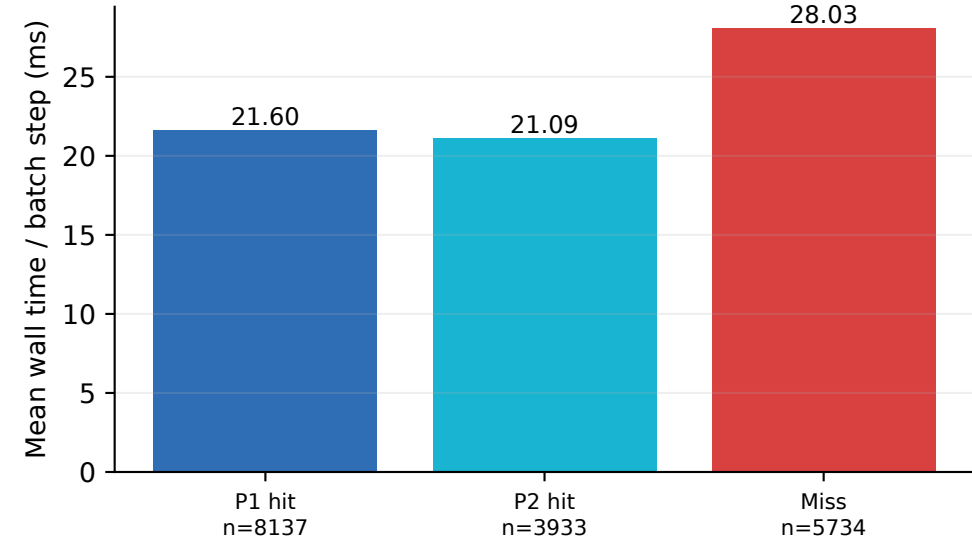
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_final_duet_fast_r1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
Last pass: B=1, T=0.7, N=480 | AL*=2.702, hit=67.7%, TPS*=114.69

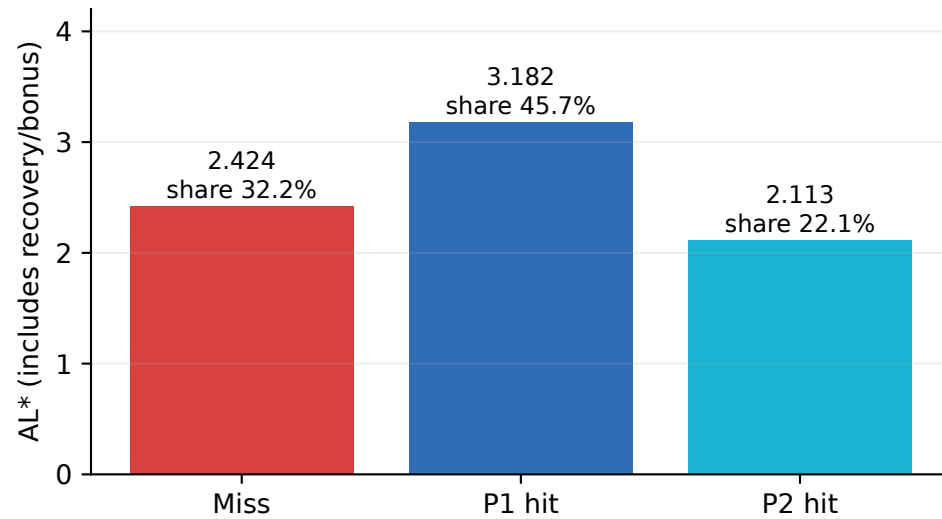
Measured wall time by batch cache status



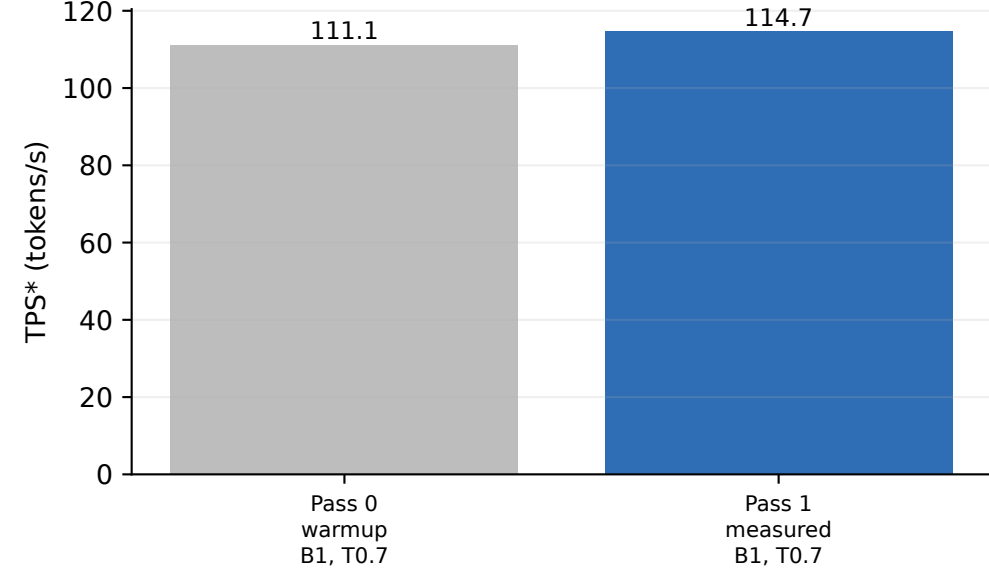
Boundary-free steps, including slow steps



Conditional AL* and source share

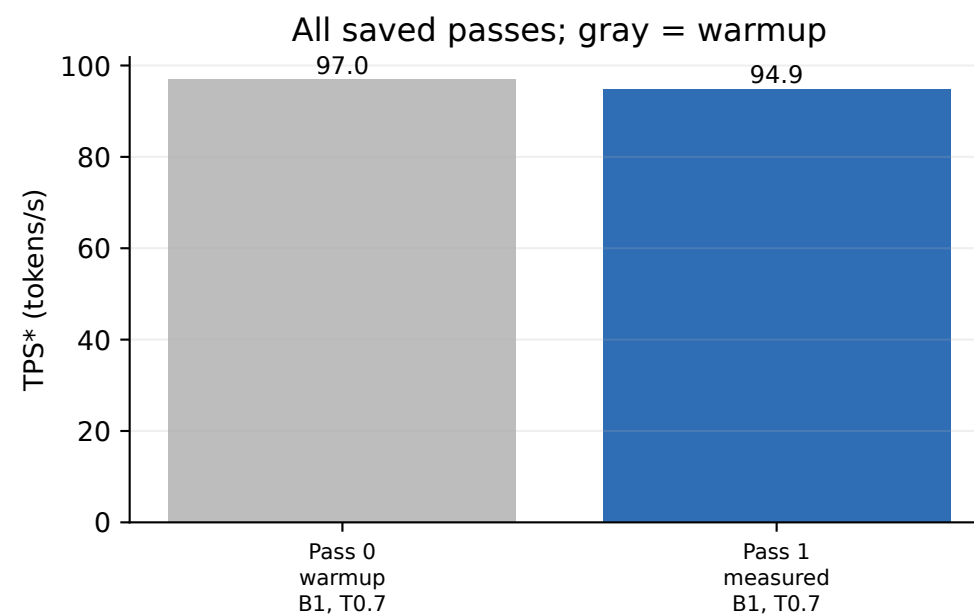
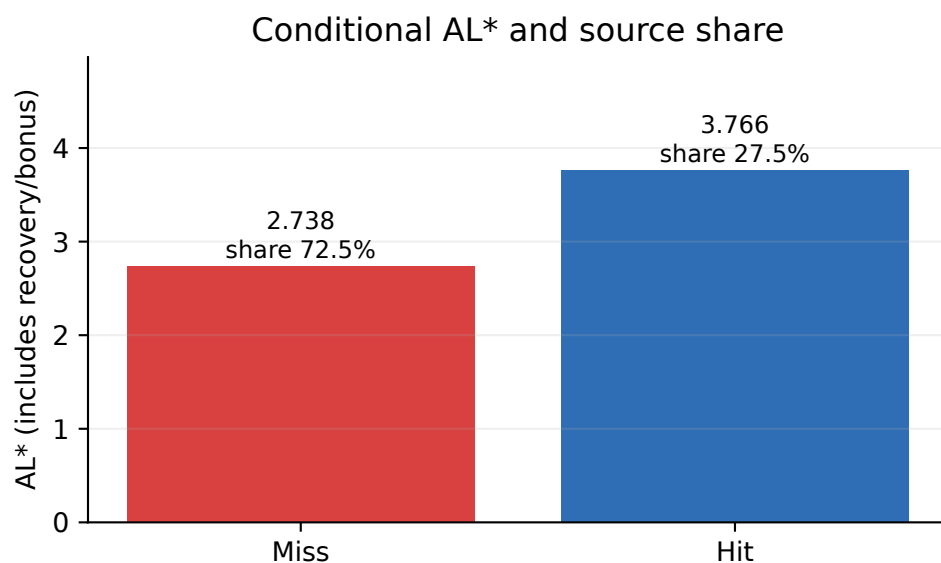
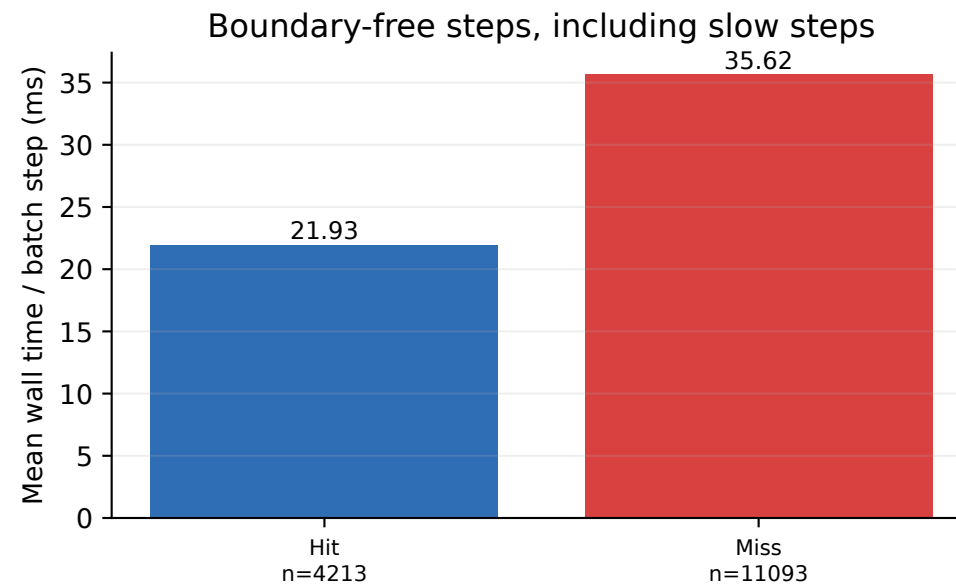
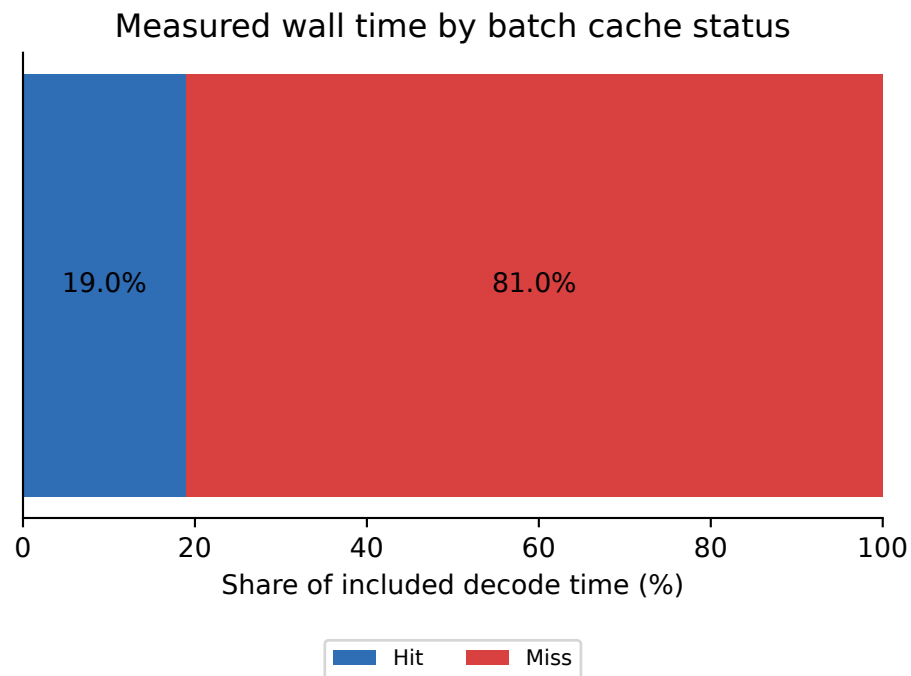


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

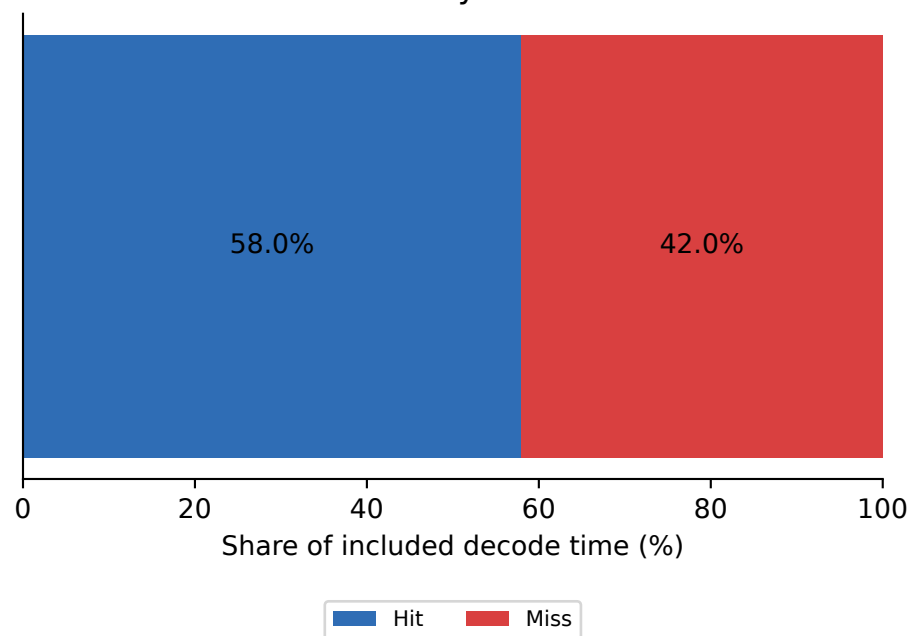
llama3_b1_final_ssd_al_r1
SSD: K=8, F=1
Last pass: B=1, T=0.7, N=480 | AL*=3.021, hit=27.8%, TPS*=94.86



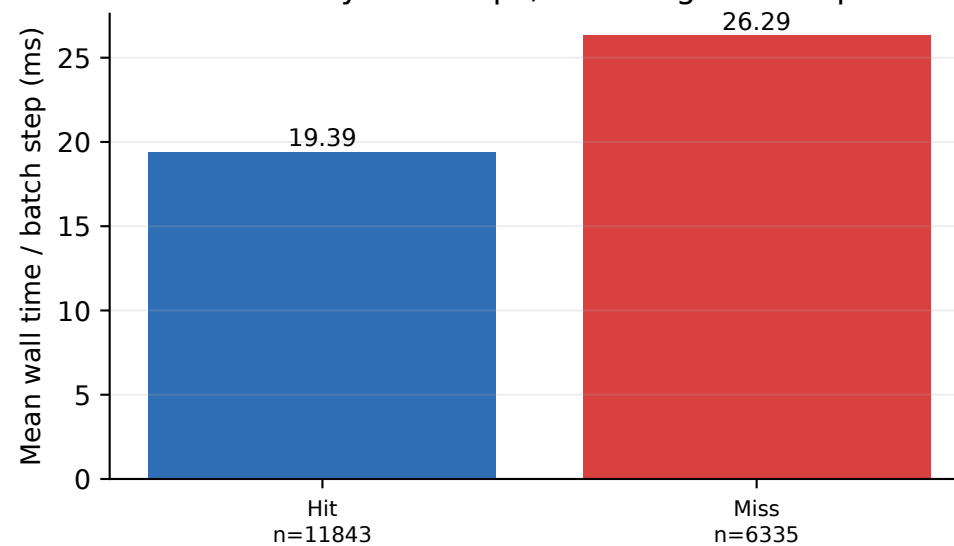
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_final_ssd_fast_r1
SSD: K=4, F=7
Last pass: B=1, T=0.7, N=480 | AL*=2.644, hit=65.2%, TPS*=121.30

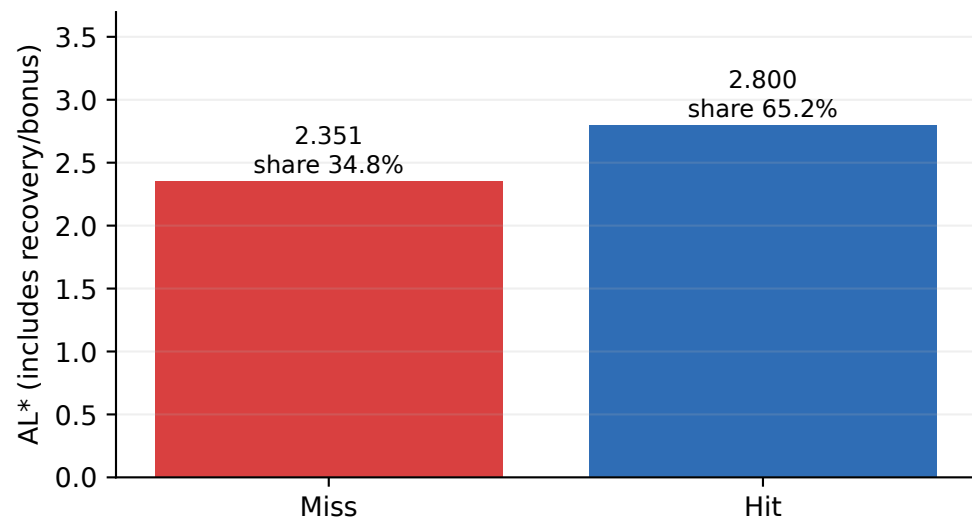
Measured wall time by batch cache status



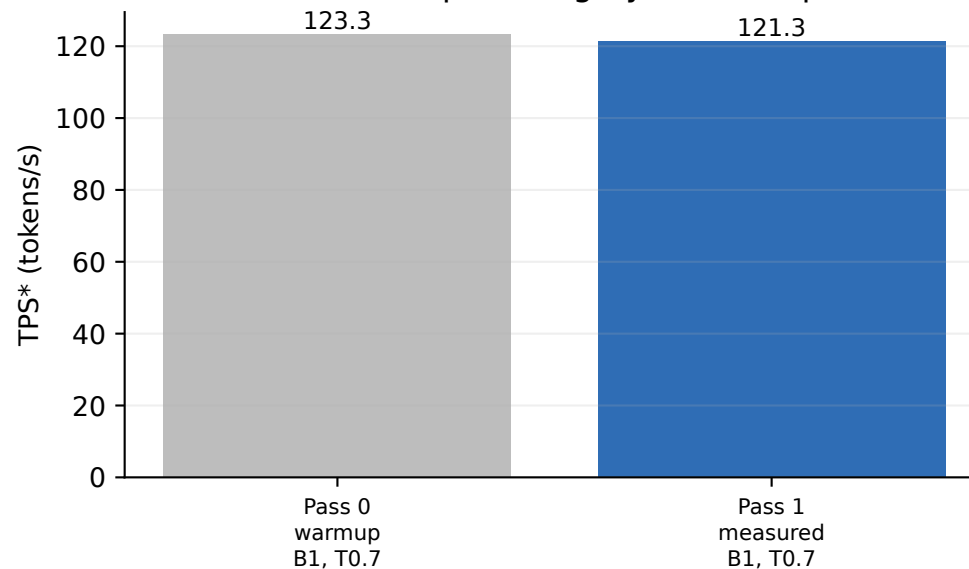
Boundary-free steps, including slow steps



Conditional AL* and source share



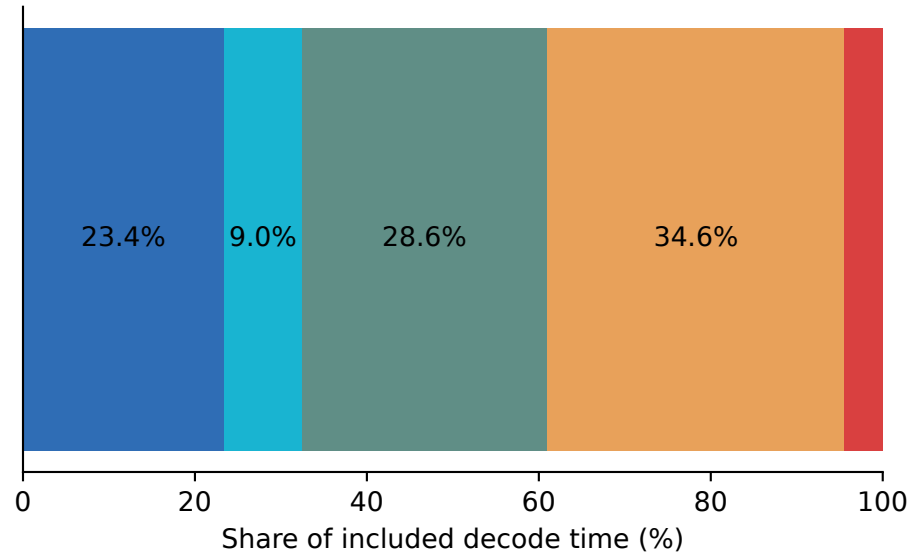
All saved passes; gray = warmup



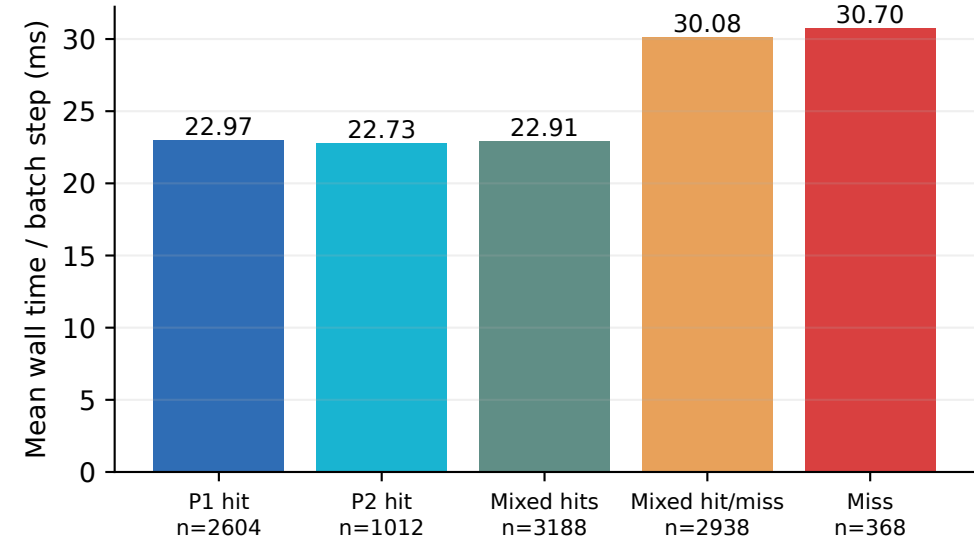
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b2_scale_duet
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
 Last pass: B=2, T=0.7, N=480 | AL*=2.389, hit=81.8%, TPS*=188.69

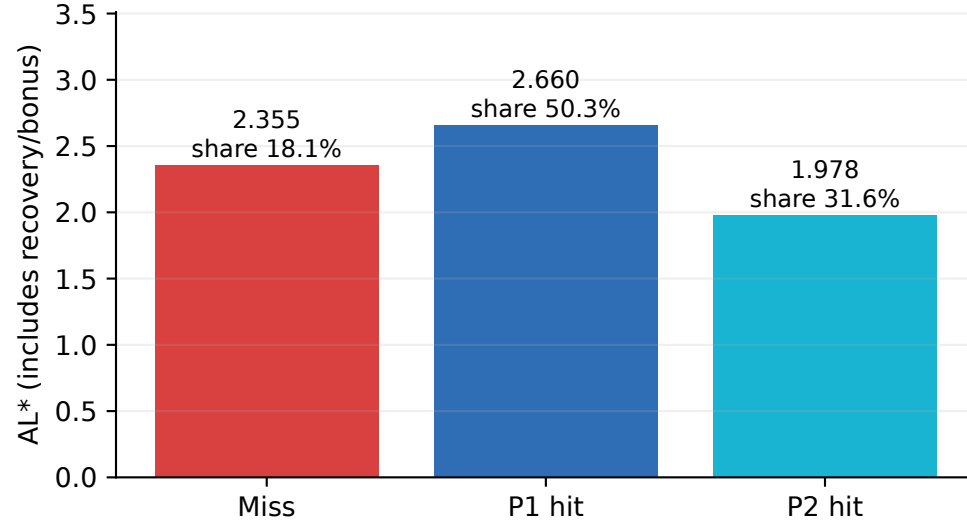
Measured wall time by batch cache status



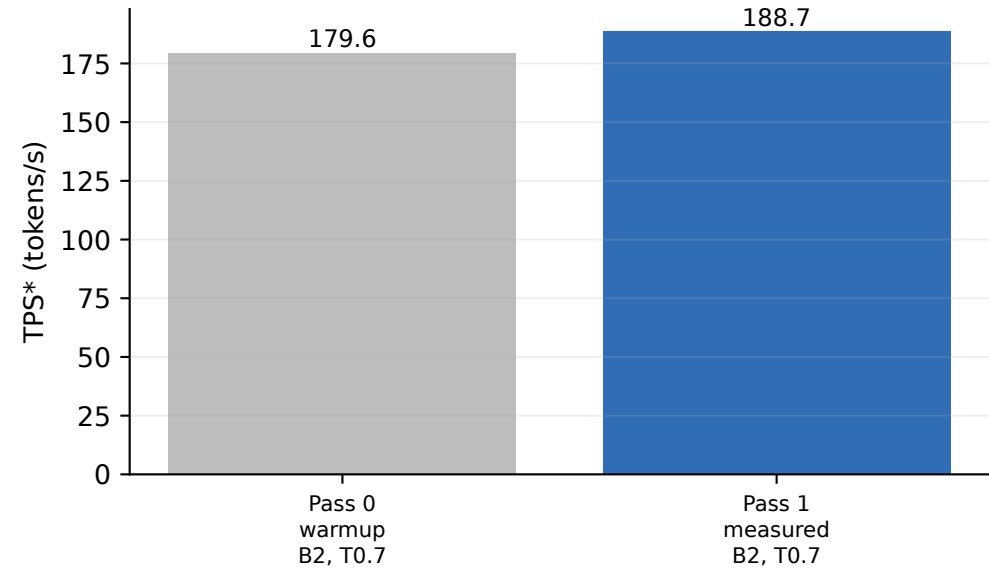
Boundary-free steps, including slow steps



Conditional AL* and source share



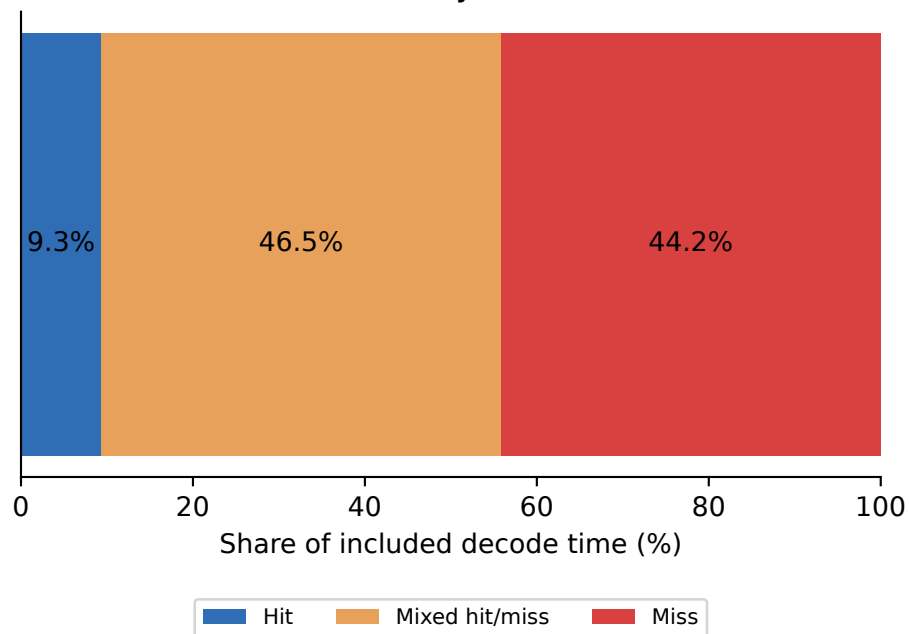
All saved passes; gray = warmup



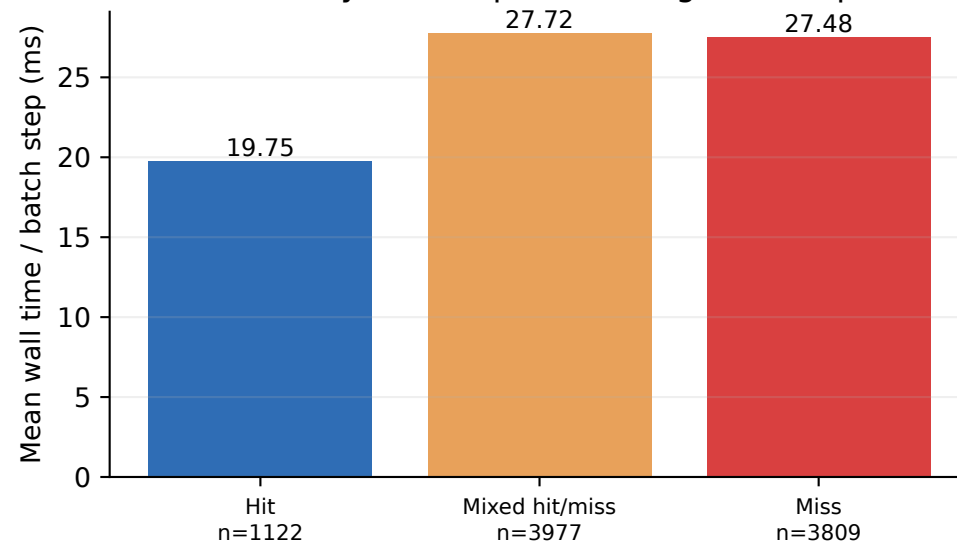
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b2_scale_ssd
SSD: K=4, F=1
Last pass: B=2, T=0.7, N=480 | AL*=2.675, hit=35.1%, TPS*=200.53

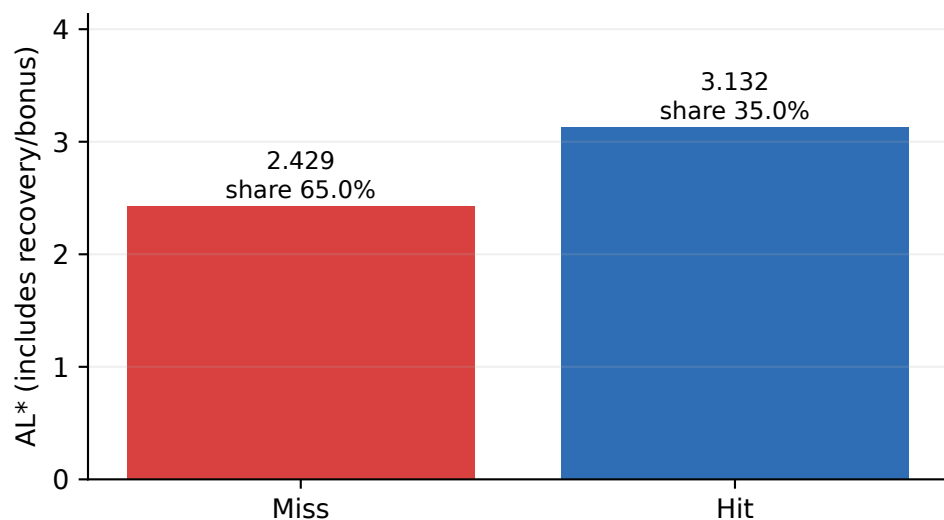
Measured wall time by batch cache status



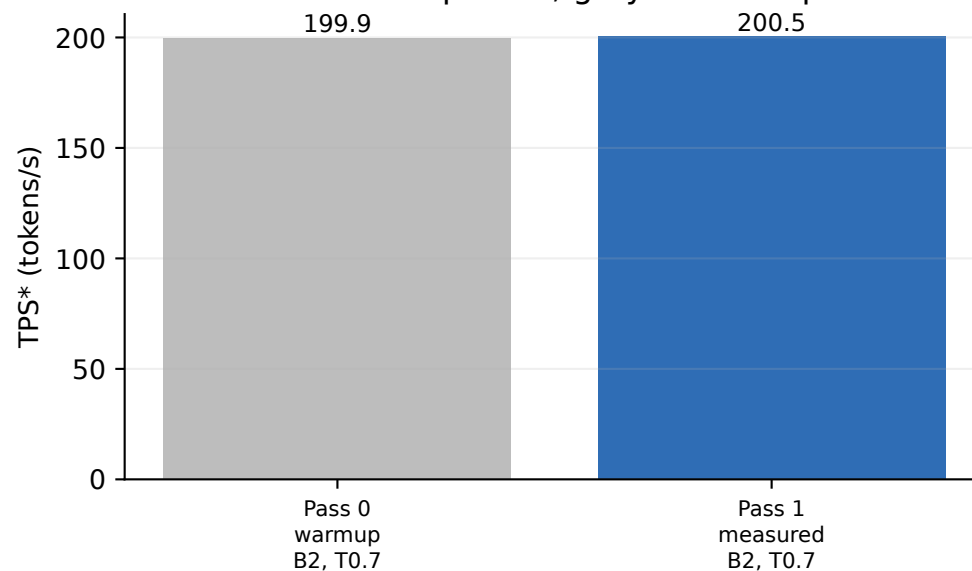
Boundary-free steps, including slow steps



Conditional AL* and source share



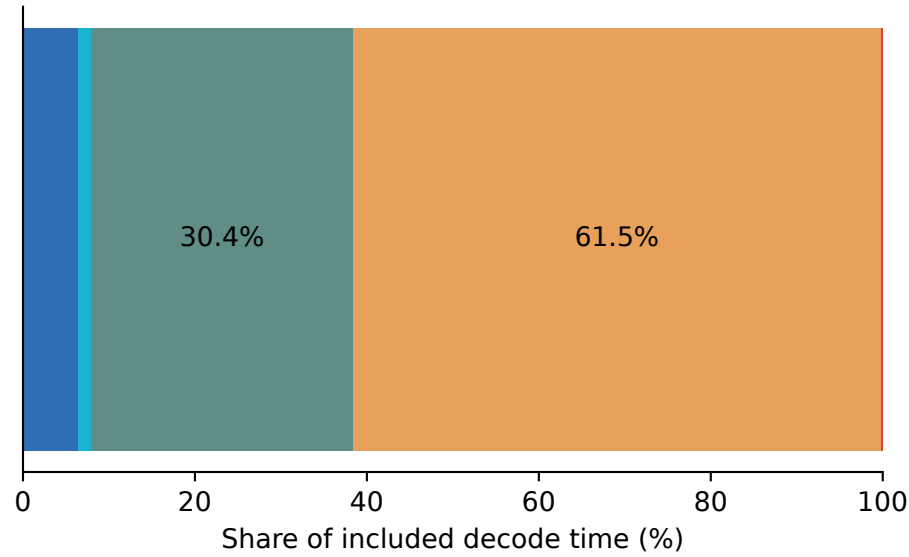
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

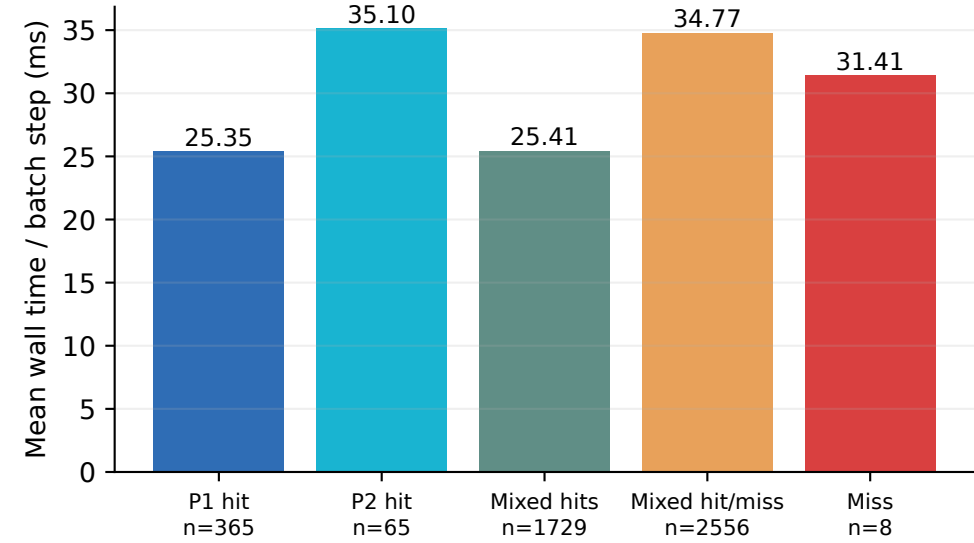
llama3_b4_scale_duet
 DUET: K1/K2=3/2, exit=26 (0-based), $\bar{C}=1$, N1/N2=3/2, U=2, W=4, trim/stream=0/0
 Last pass: B=4, T=0.7, N=480 | AL*=2.411, hit=81.8%, TPS*=312.43

Measured wall time by batch cache status

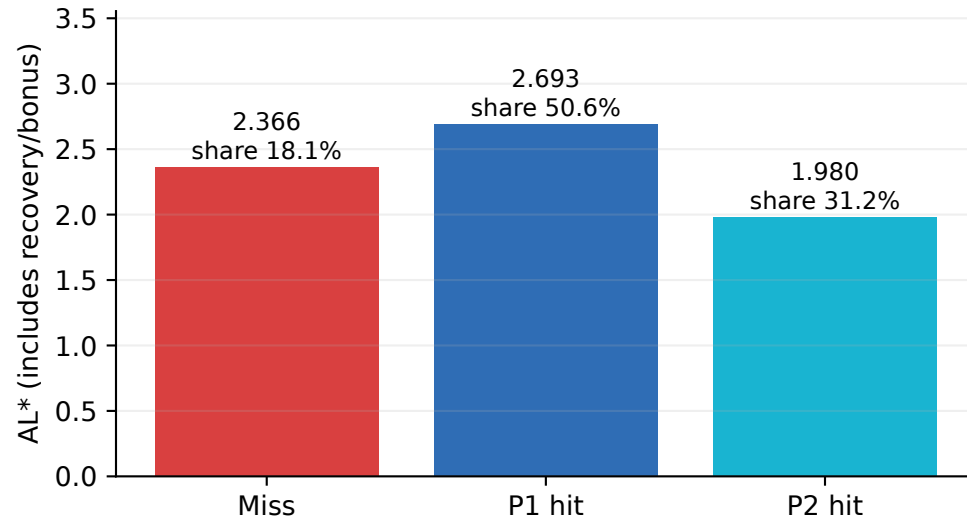


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

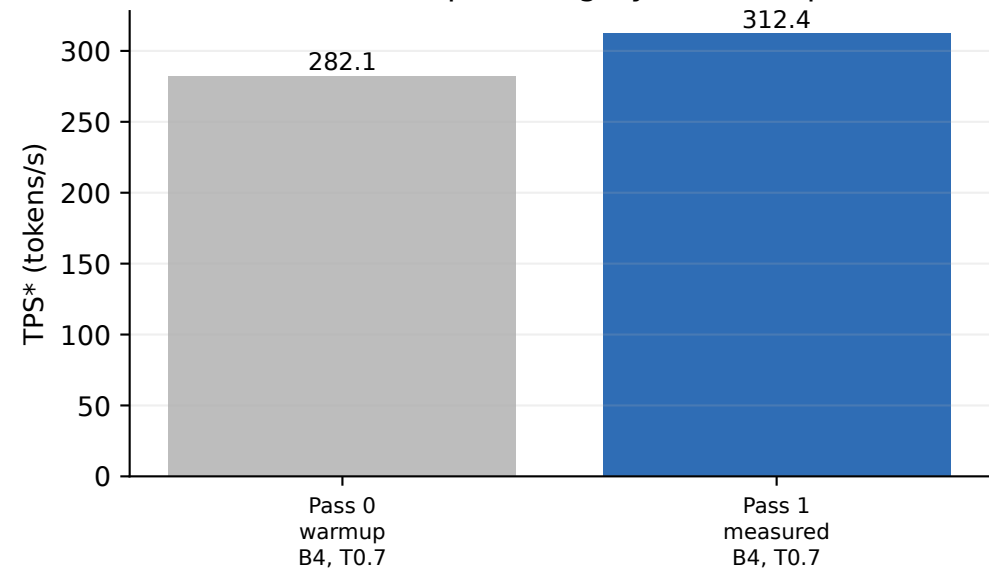
Boundary-free steps, including slow steps



Conditional AL* and source share



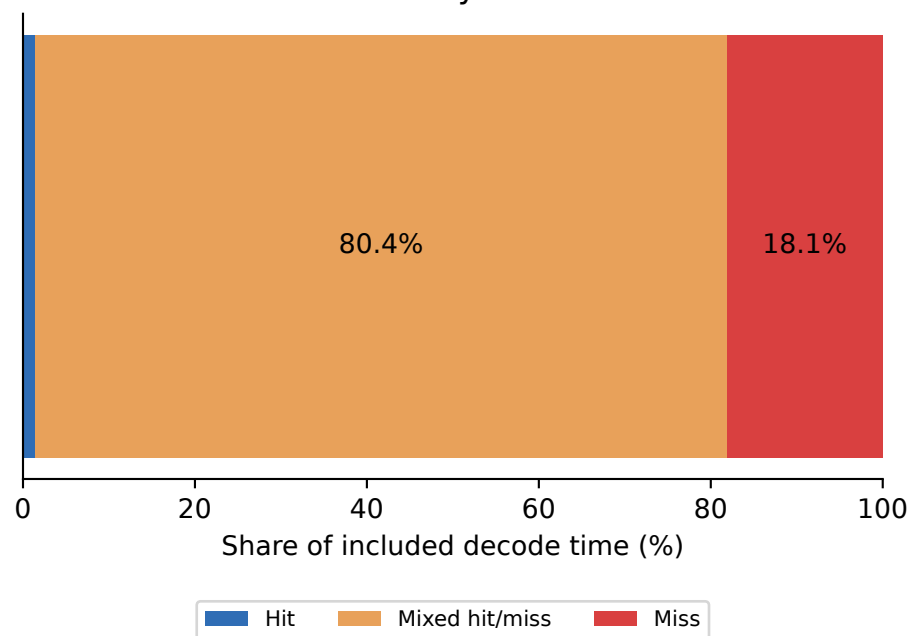
All saved passes; gray = warmup



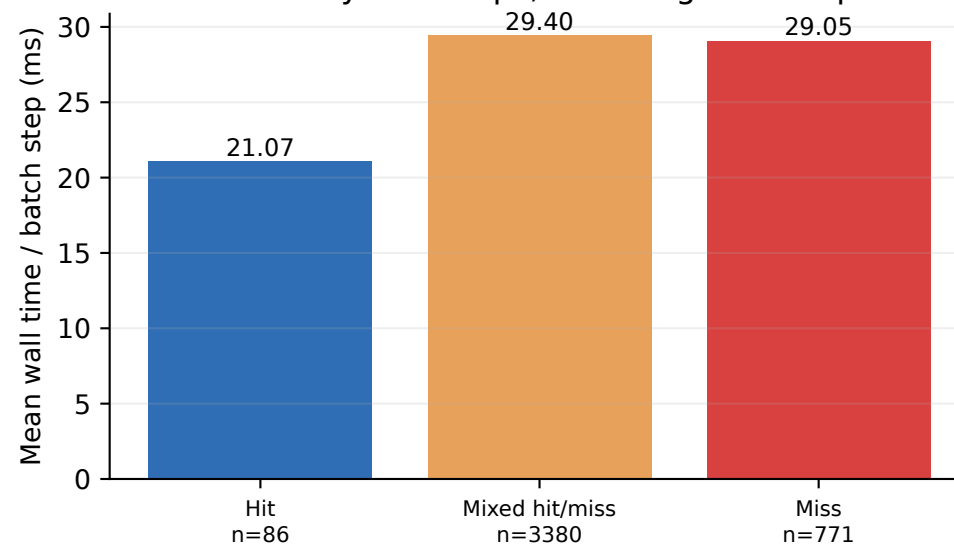
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b4_scale_ssd
SSD: K=4, F=1
Last pass: B=4, T=0.7, N=480 | AL*=2.696, hit=36.2%, TPS*=368.00

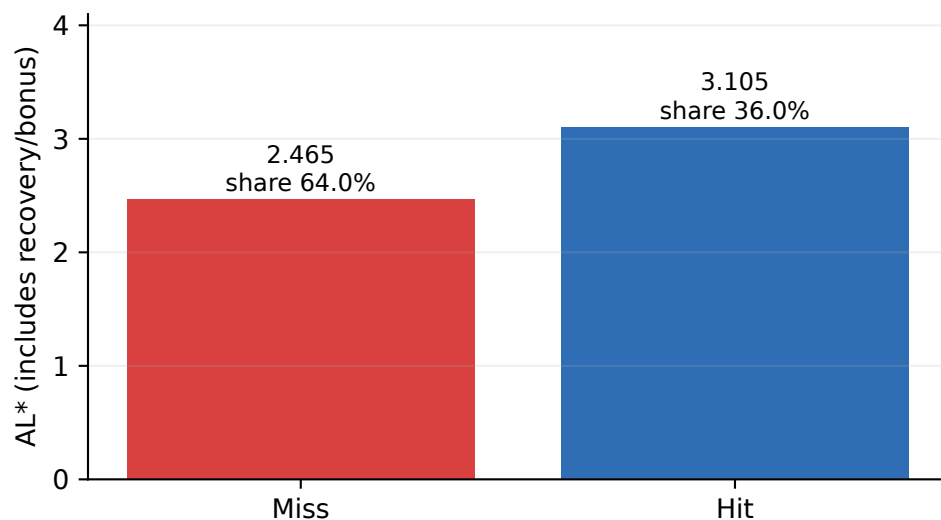
Measured wall time by batch cache status



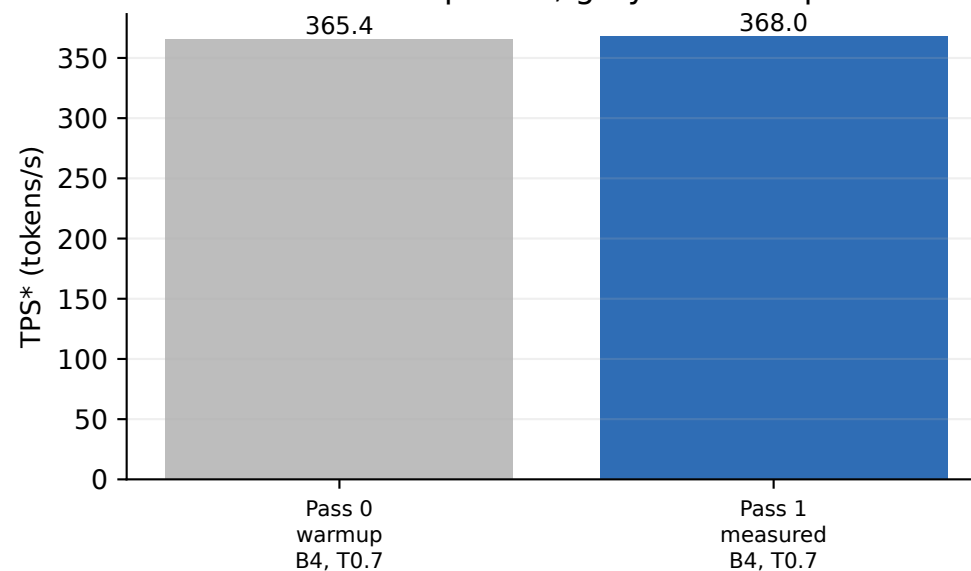
Boundary-free steps, including slow steps



Conditional AL* and source share

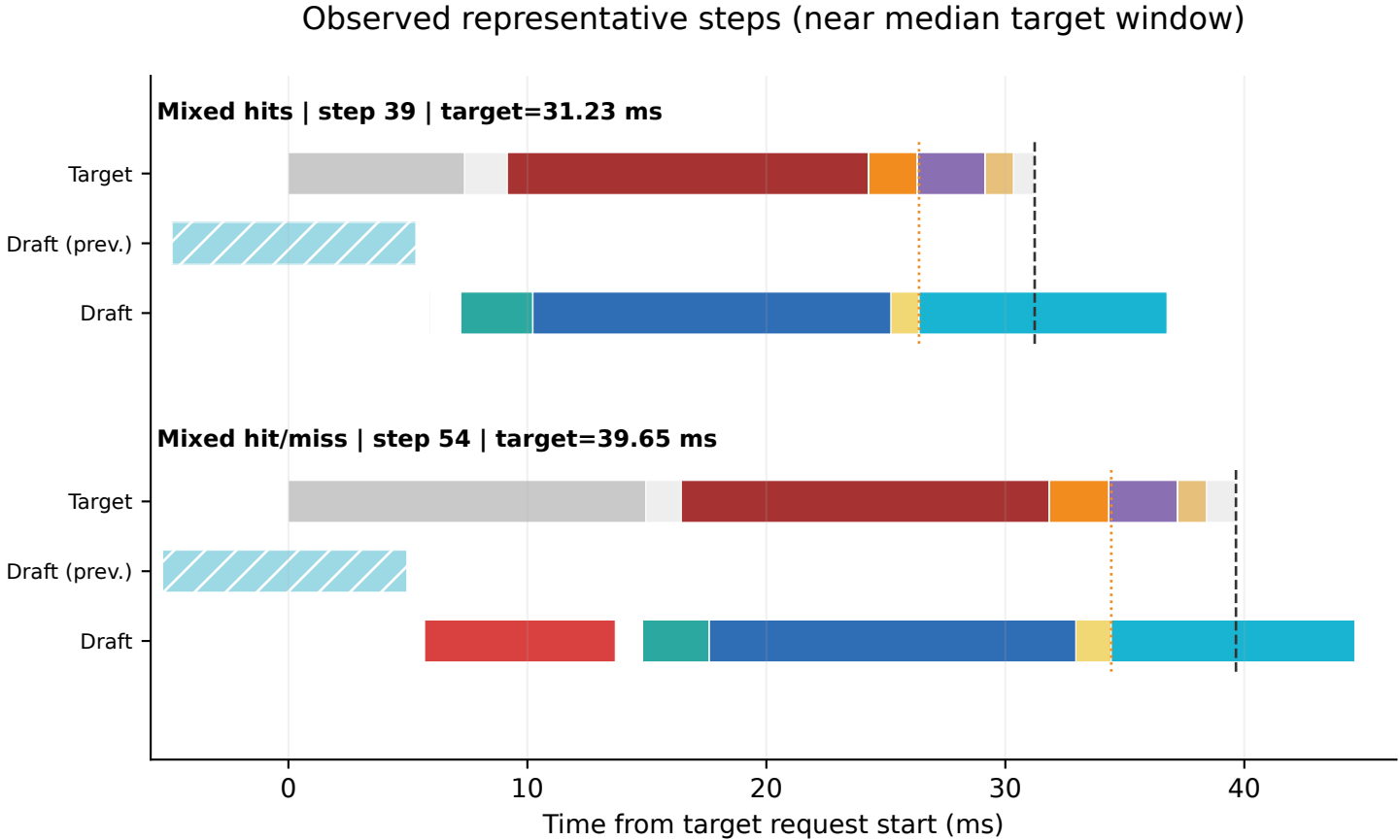
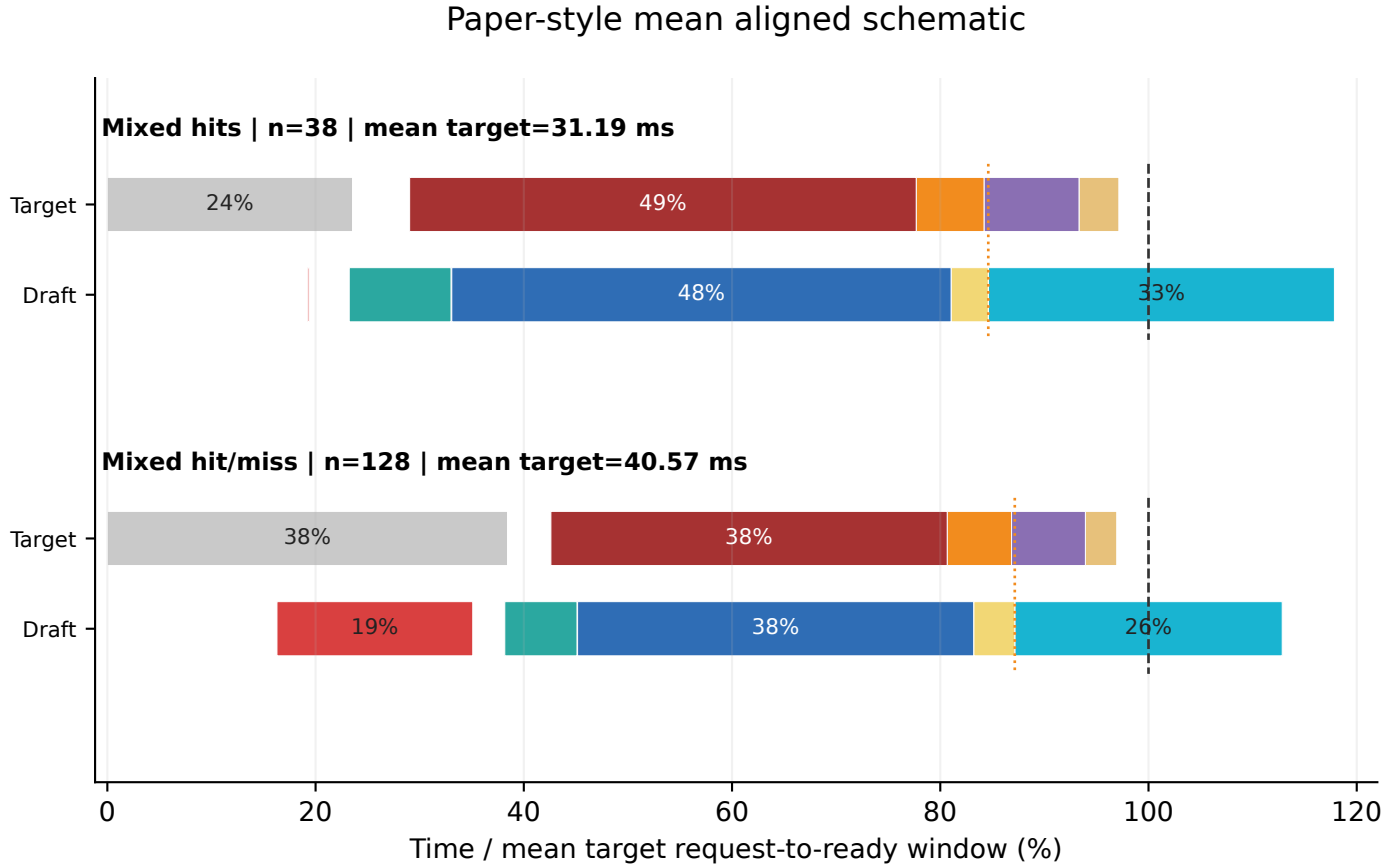
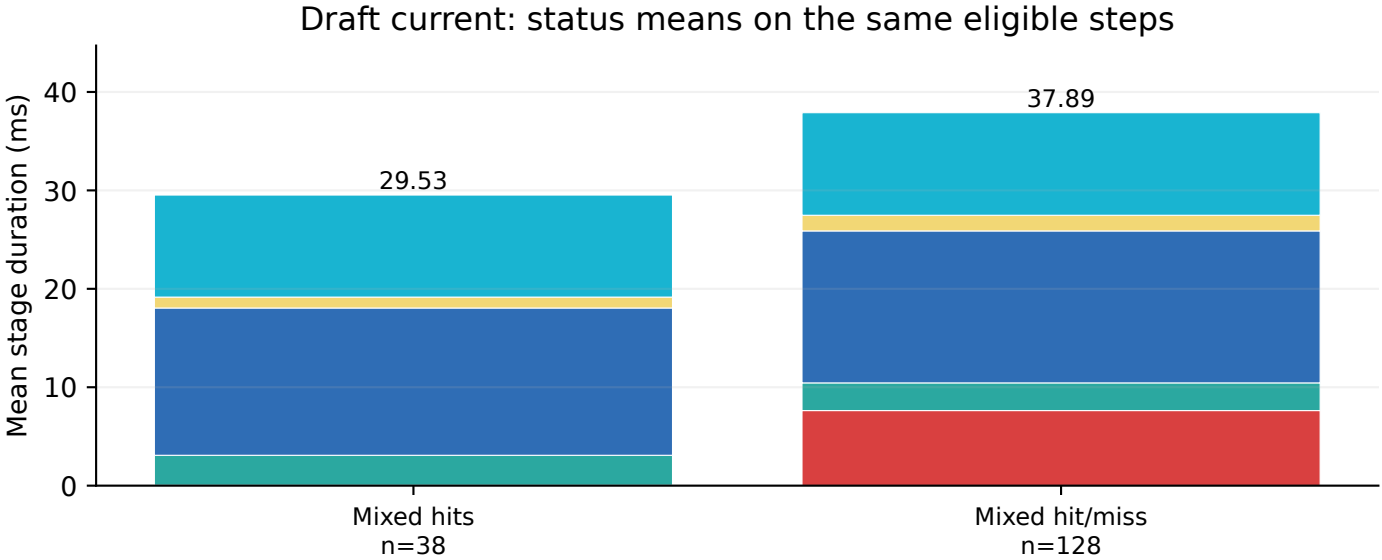
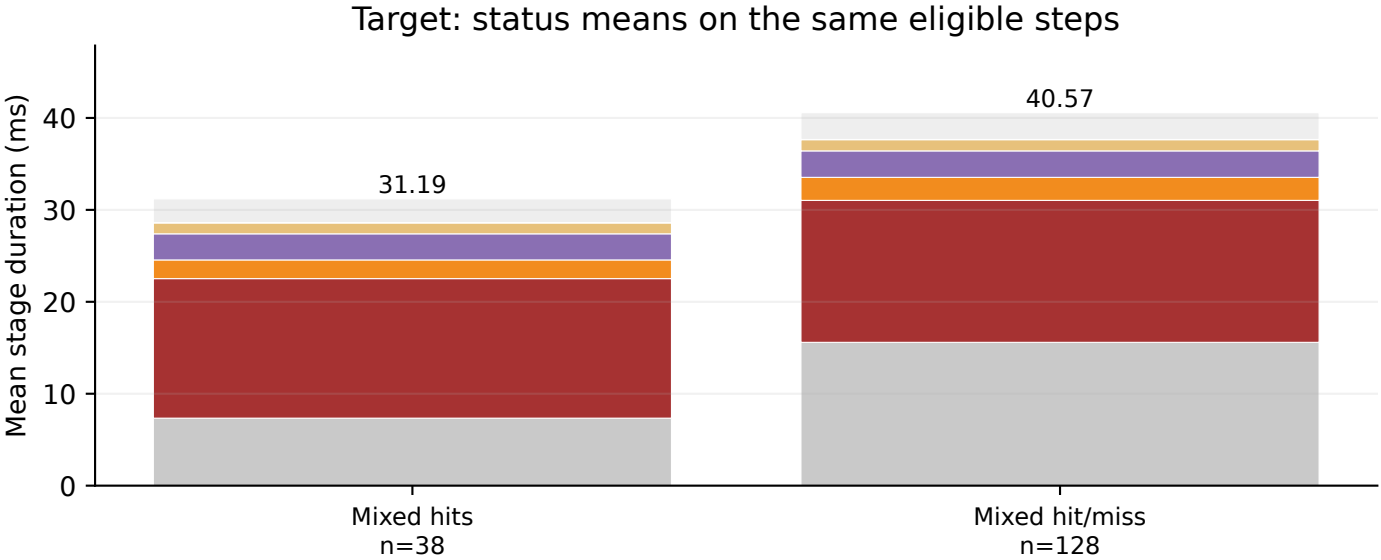


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_breakdown
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 166 eligible steps

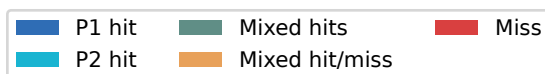
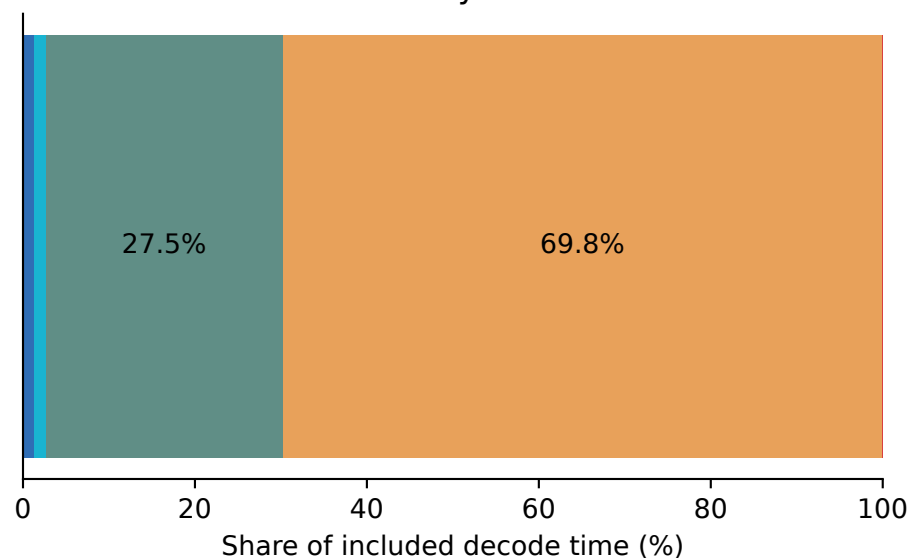


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

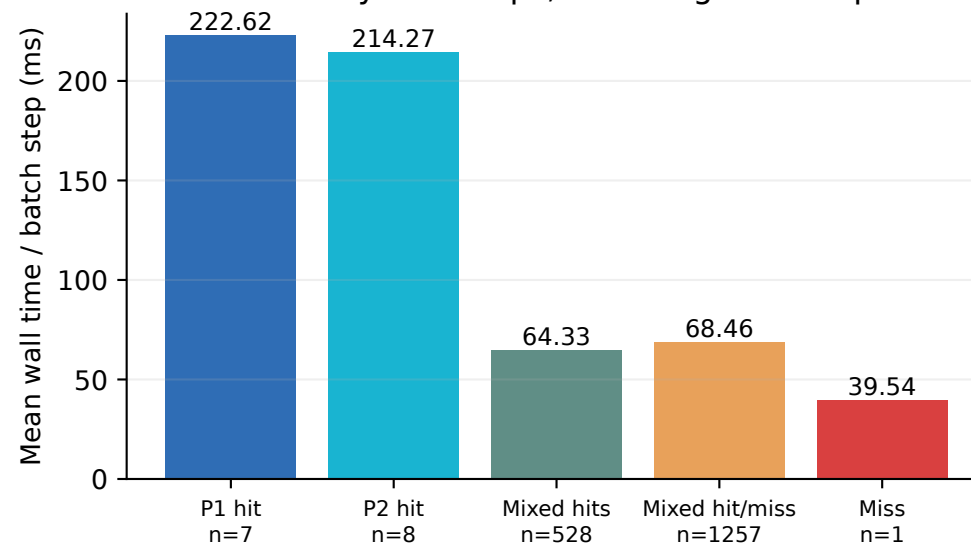
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_final_duet_al_r0
 DUET: K1/K2=6/3, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.903, hit=86.1%, TPS*=334.87

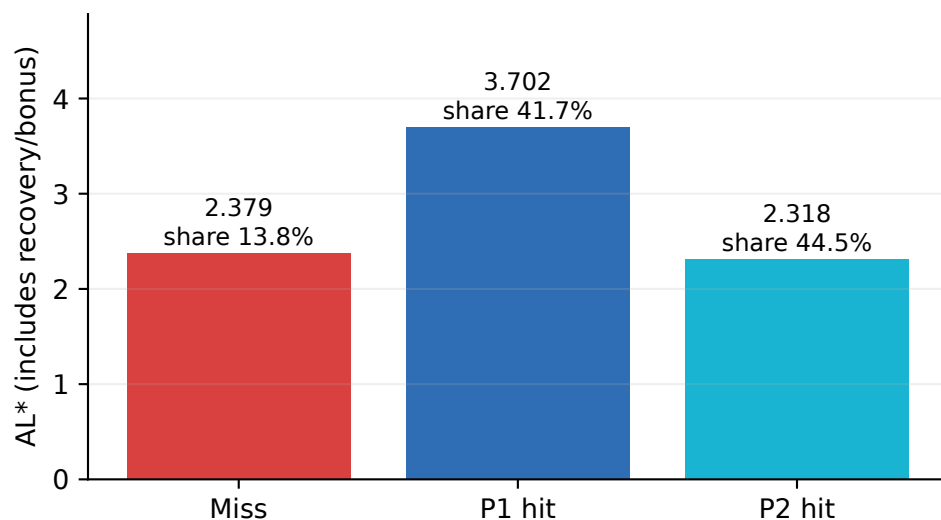
Measured wall time by batch cache status



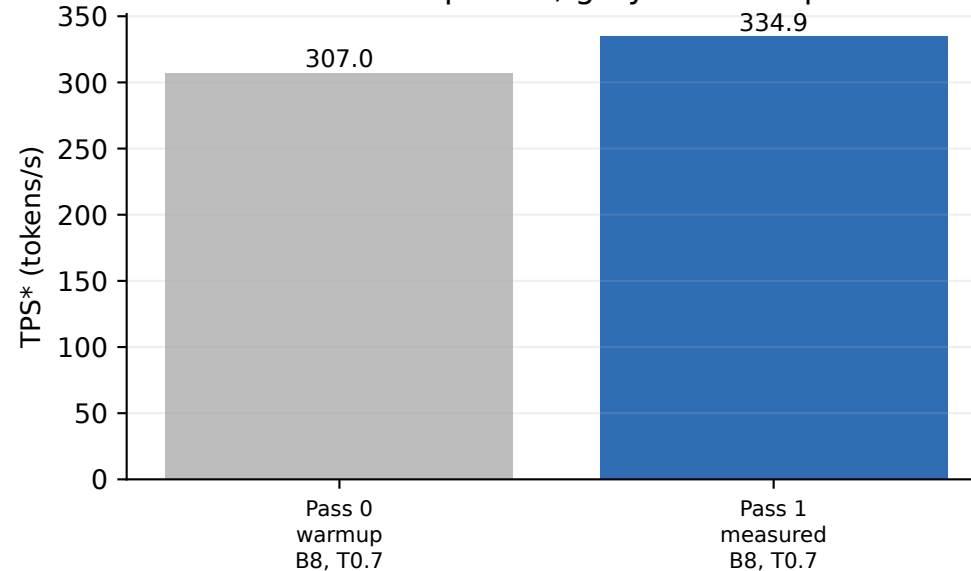
Boundary-free steps, including slow steps



Conditional AL* and source share



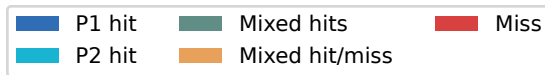
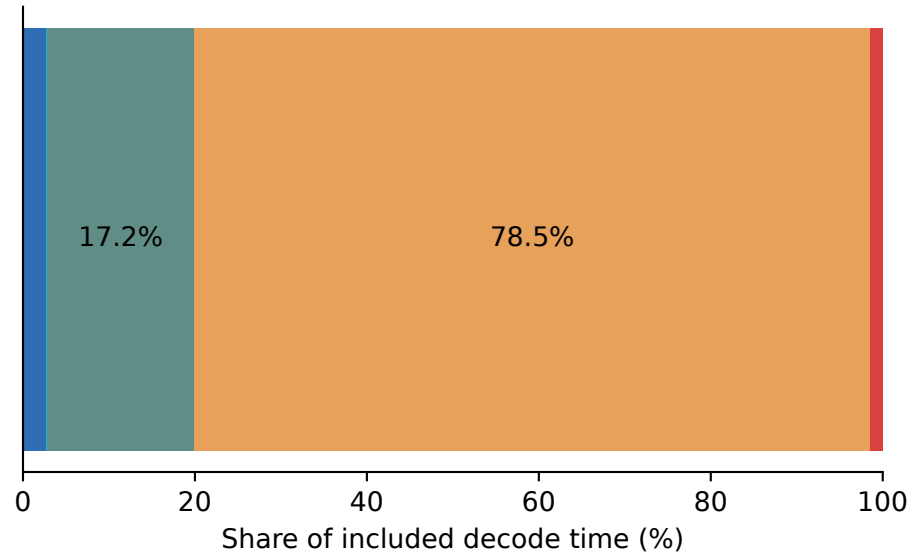
All saved passes; gray = warmup



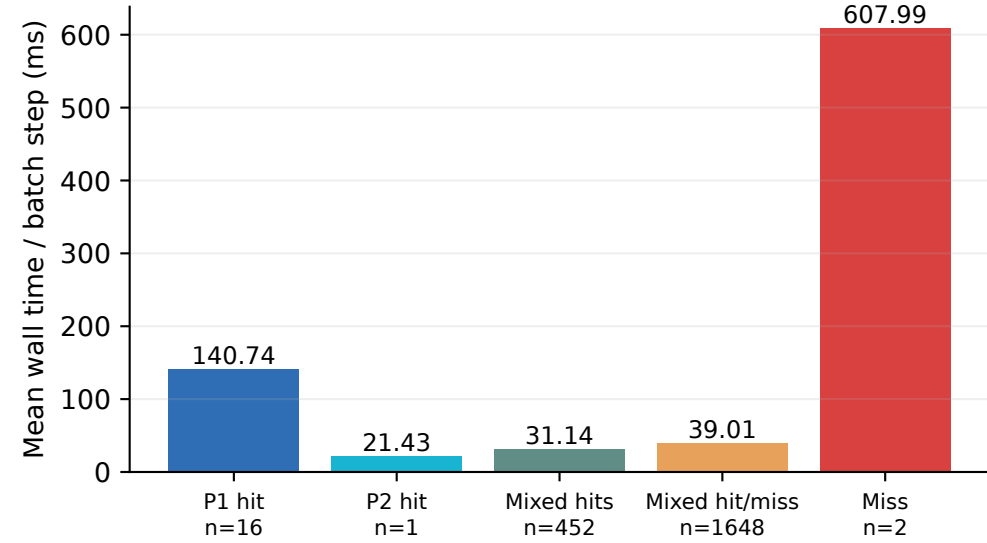
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_duet_fast_r0
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.429, hit=81.7%, TPS*=500.71

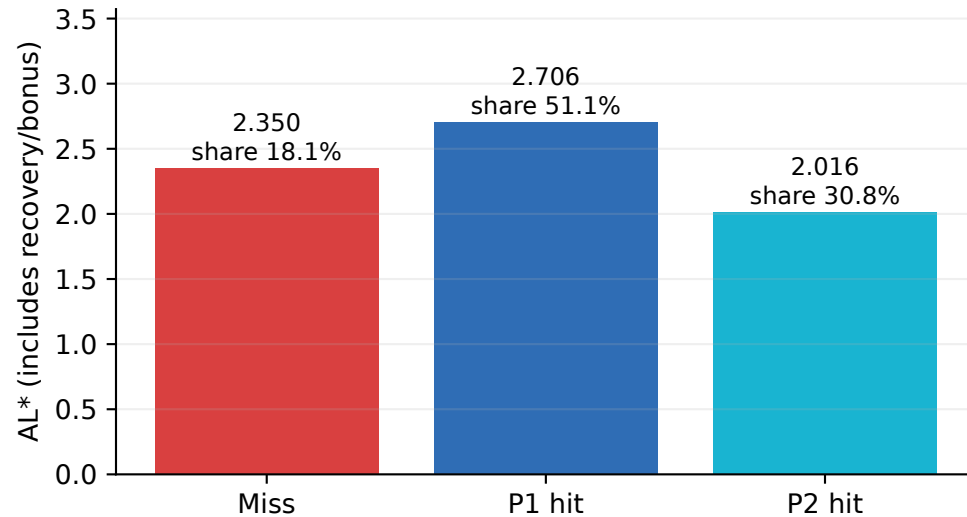
Measured wall time by batch cache status



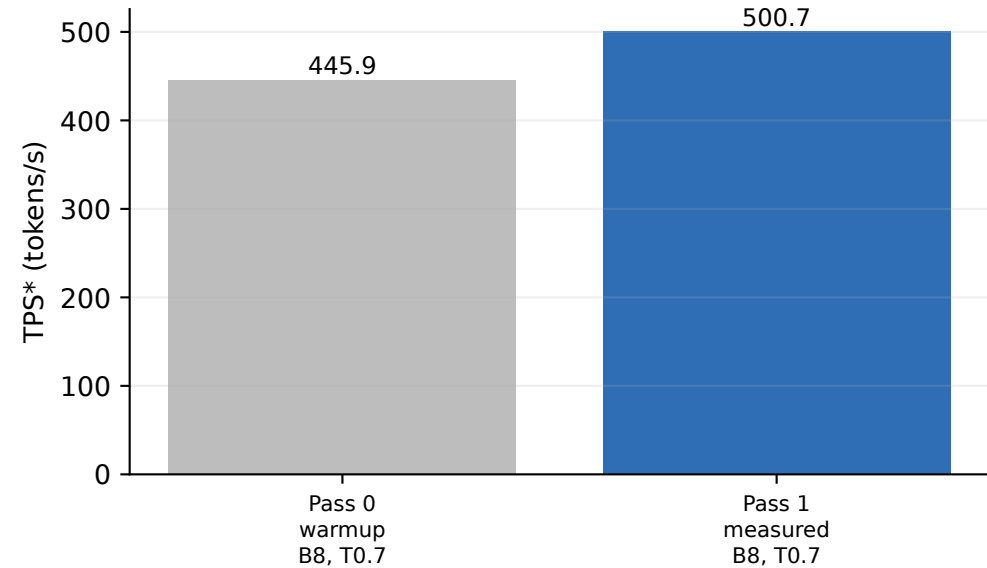
Boundary-free steps, including slow steps



Conditional AL* and source share



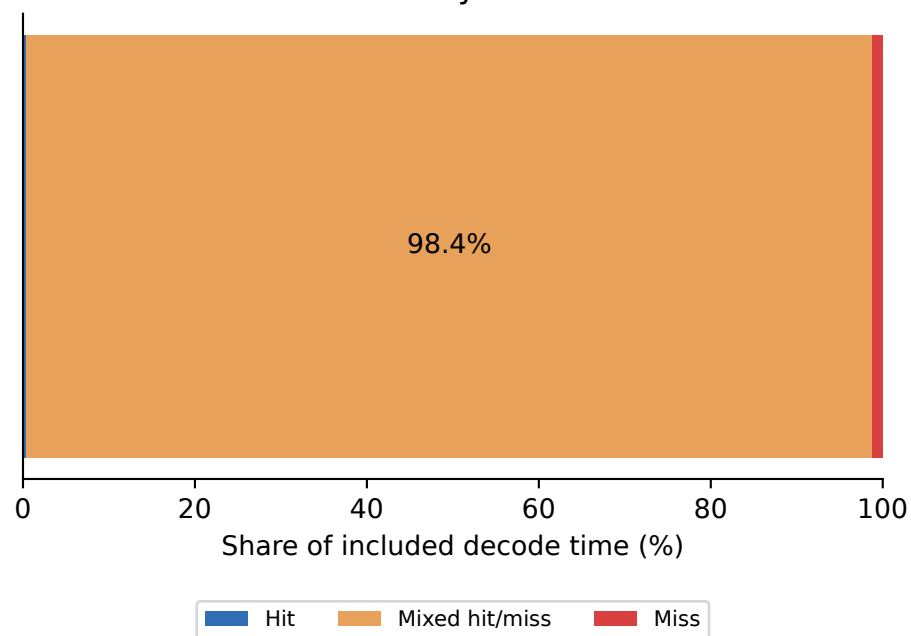
All saved passes; gray = warmup



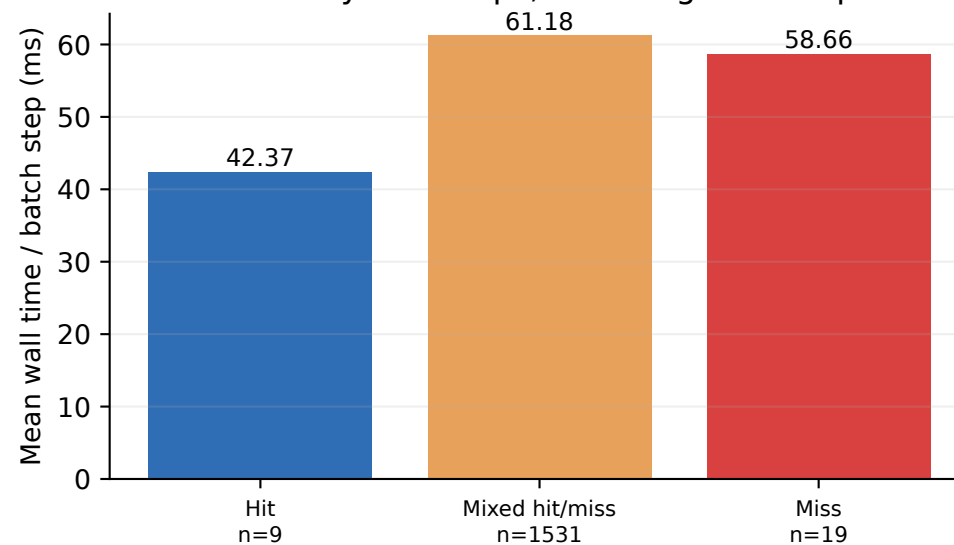
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_ssd_al_r0
SSD: K=8, F=3
Last pass: B=8, T=0.7, N=480 | AL*=3.101, hit=47.6%, TPS*=402.35

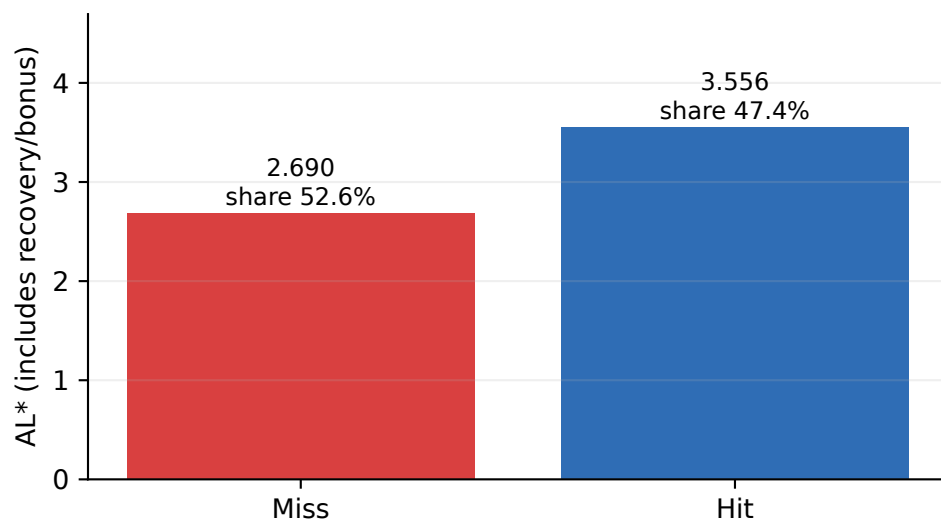
Measured wall time by batch cache status



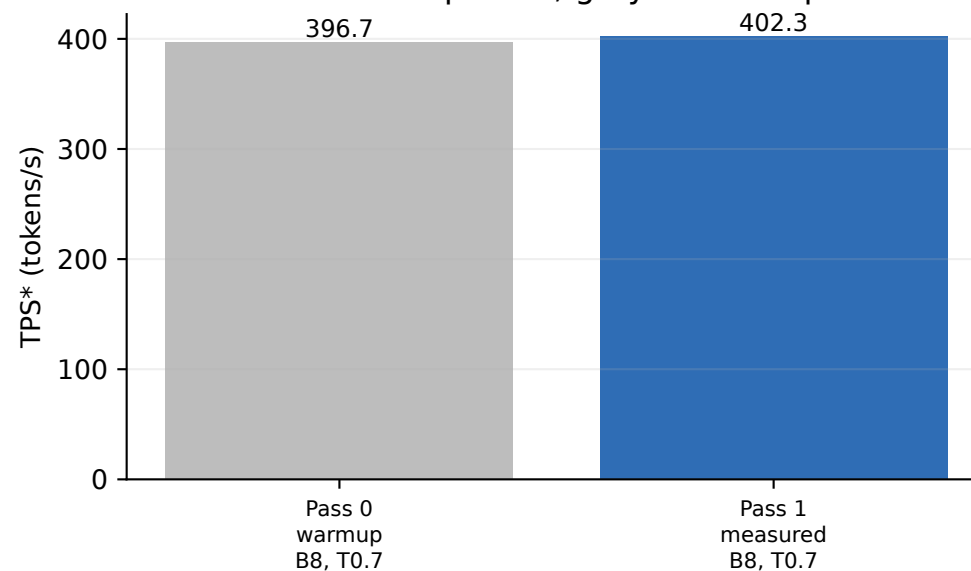
Boundary-free steps, including slow steps



Conditional AL* and source share



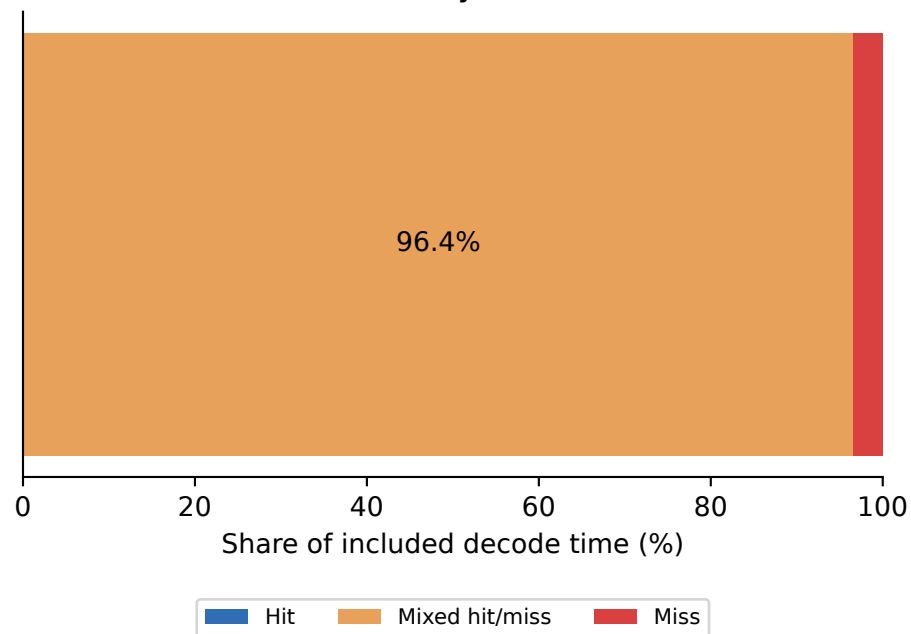
All saved passes; gray = warmup



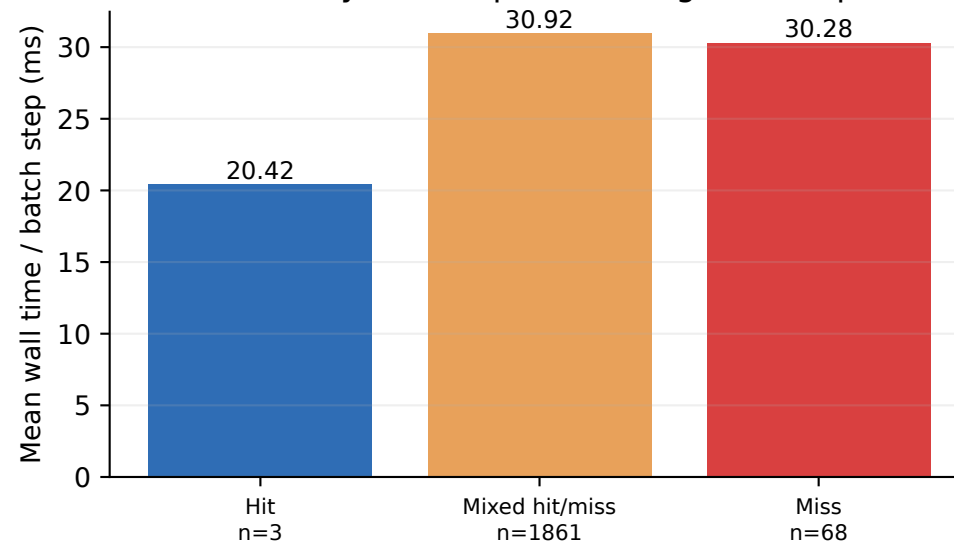
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_ssd_fast_r0
SSD: K=4, F=1
Last pass: B=8, T=0.7, N=480 | AL*=2.675, hit=35.5%, TPS*=684.36

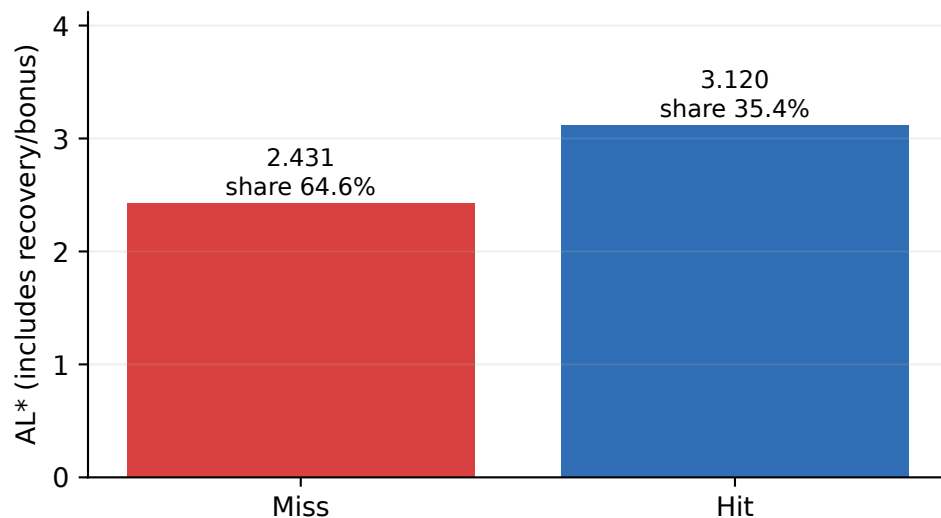
Measured wall time by batch cache status



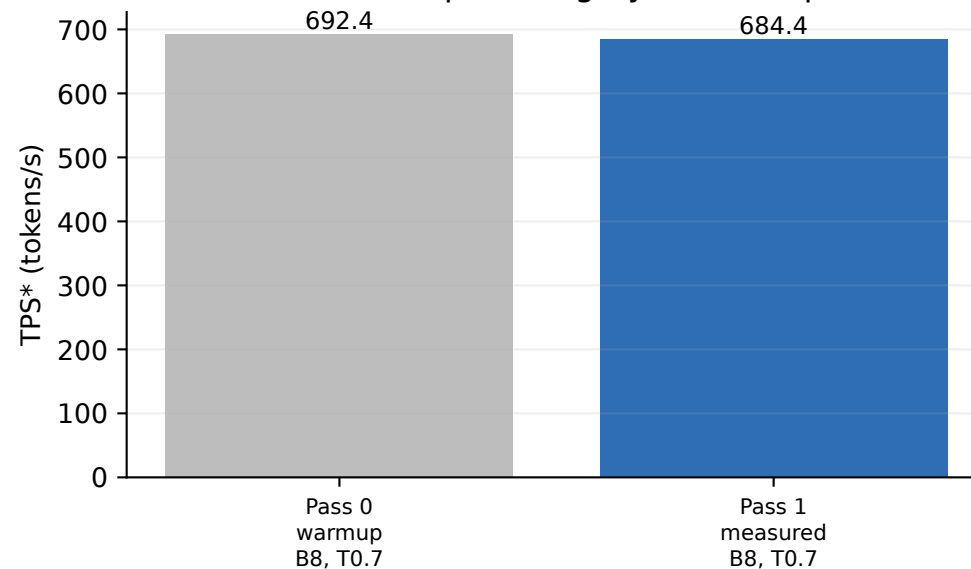
Boundary-free steps, including slow steps



Conditional AL* and source share



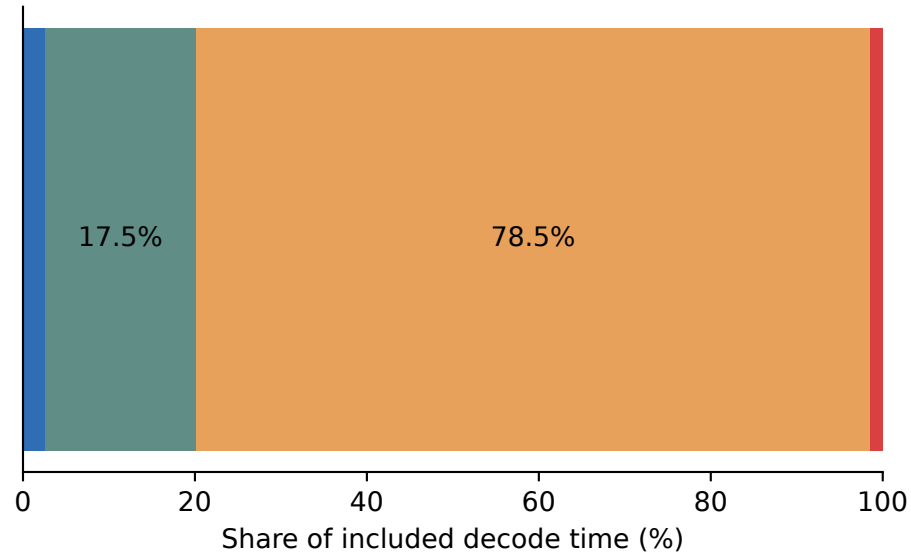
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

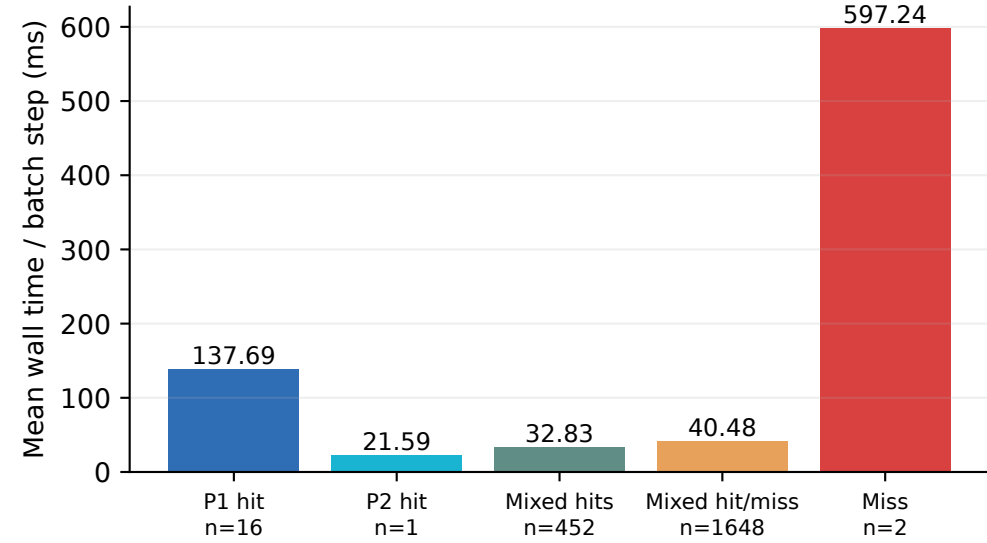
llama3_b8_implementation_off
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
Last pass: B=8, T=0.7, N=480 | AL*=2.429, hit=81.7%, TPS*=482.44

Measured wall time by batch cache status

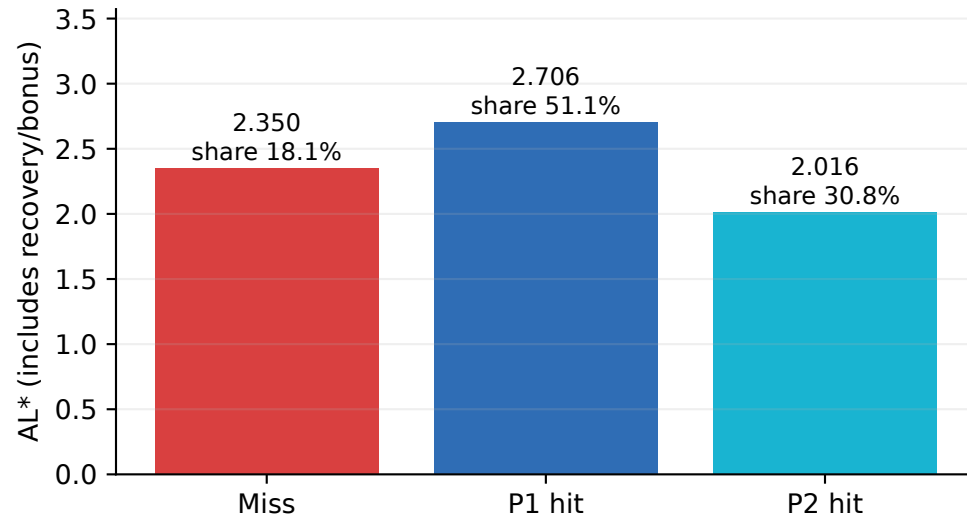


■ P1 hit ■ Mixed hits ■ Miss
■ P2 hit ■ Mixed hit/miss

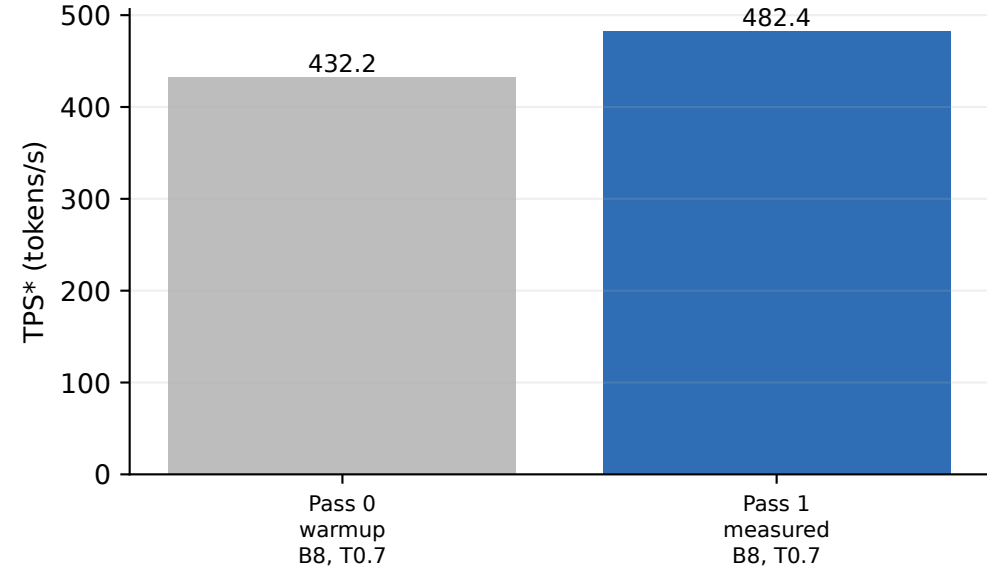
Boundary-free steps, including slow steps



Conditional AL* and source share



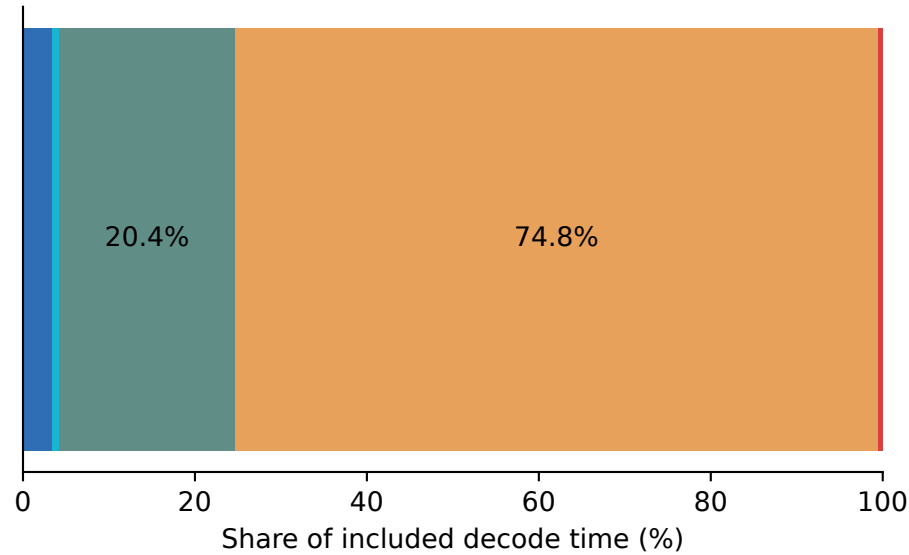
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

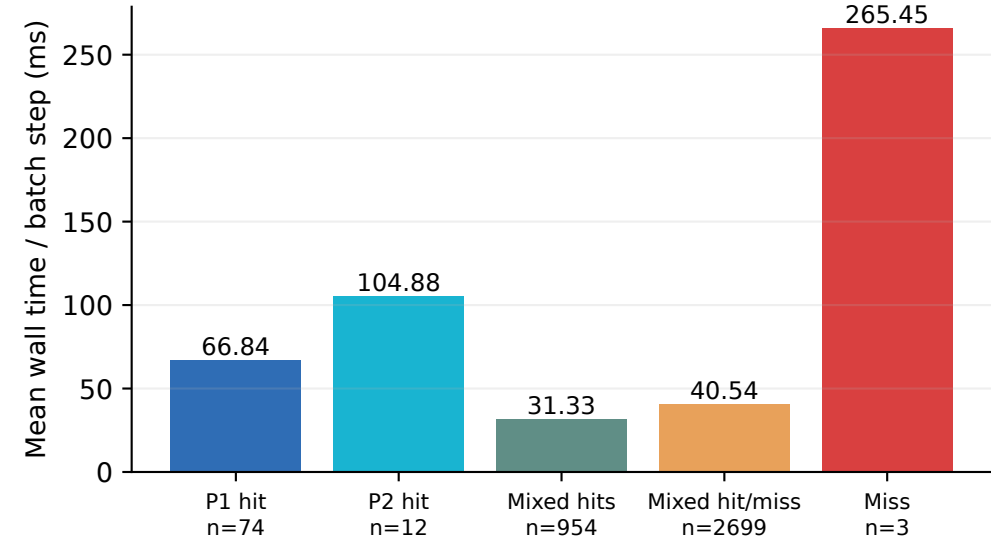
llama3_b8_long_duet
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.581, hit=84.8%, TPS*=518.81

Measured wall time by batch cache status

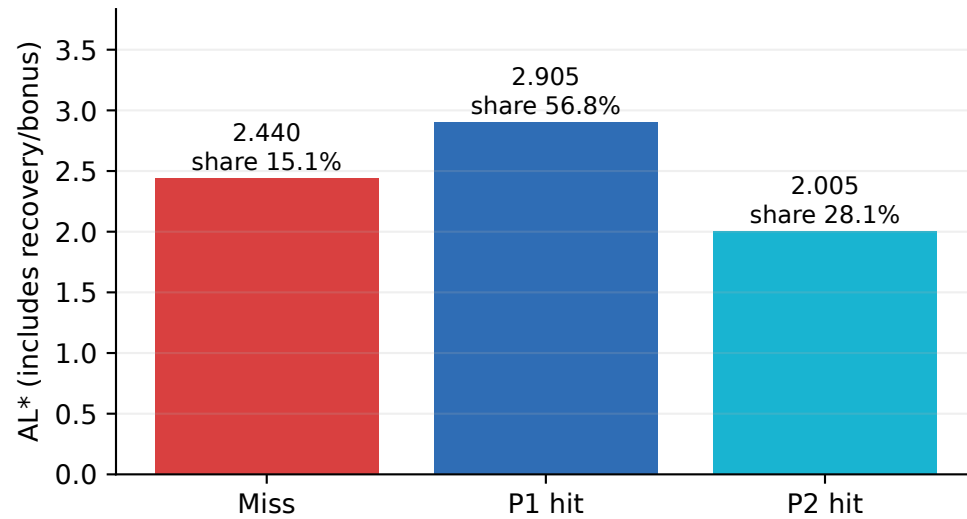


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

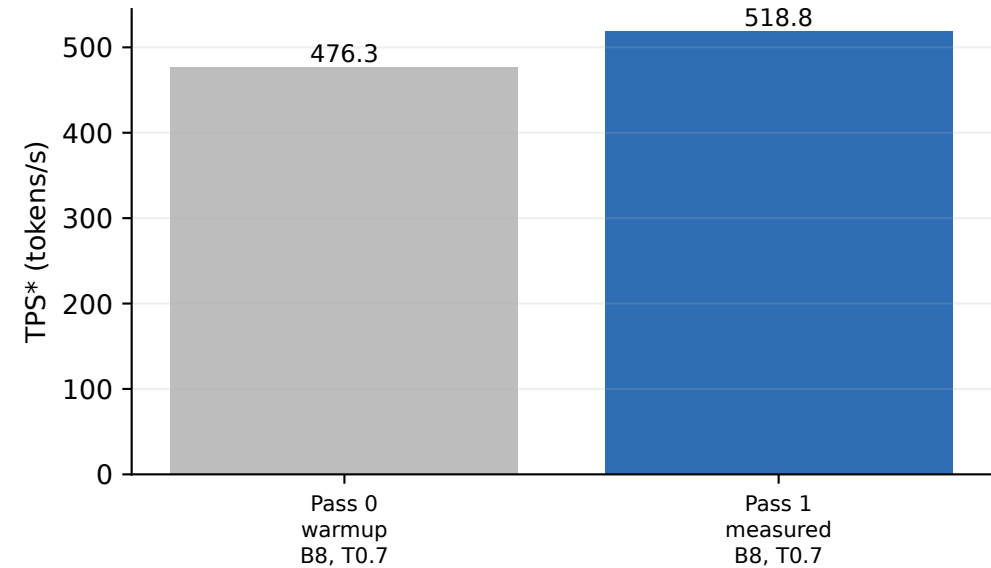
Boundary-free steps, including slow steps



Conditional AL* and source share



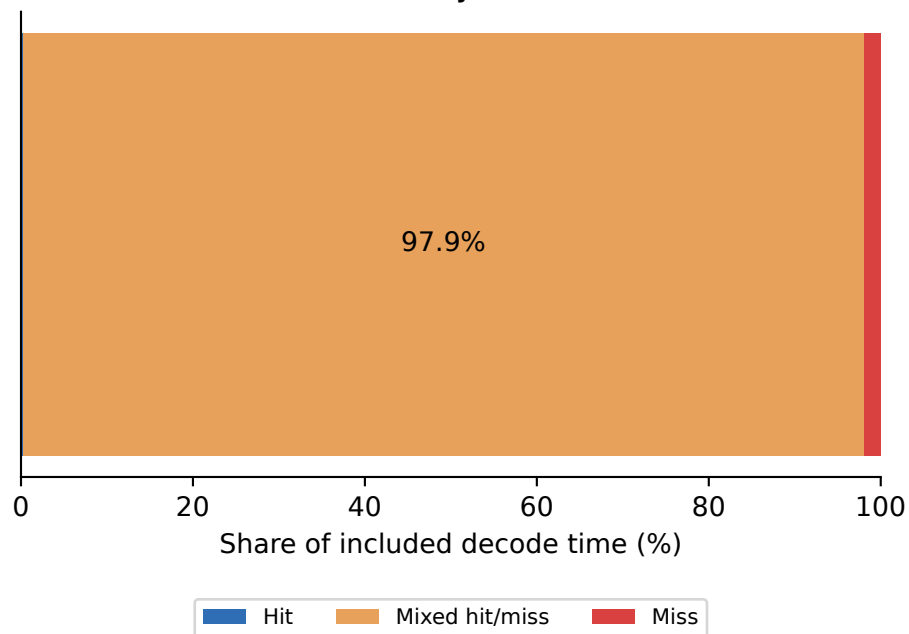
All saved passes; gray = warmup



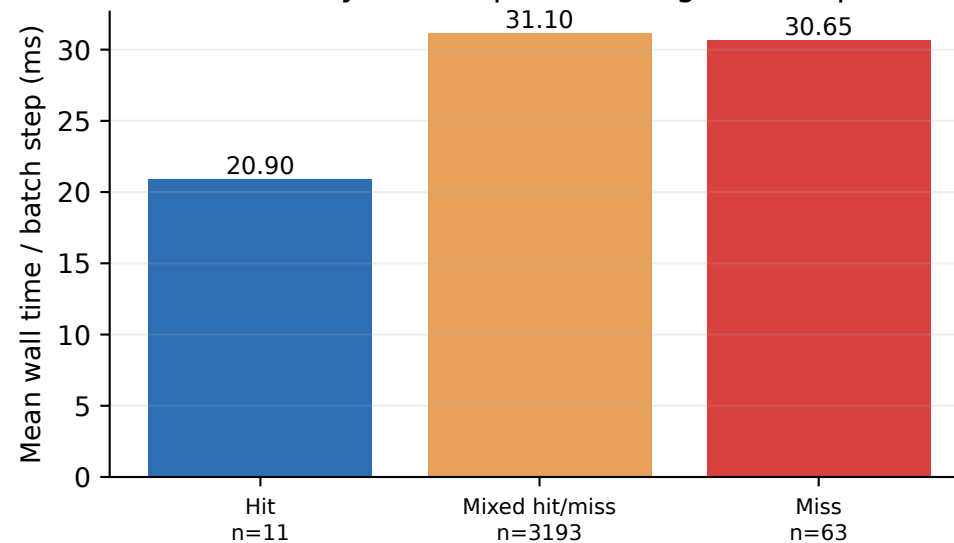
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_long_ssd
SSD: K=4, F=1
Last pass: B=8, T=0.7, N=480 | AL*=2.921, hit=42.6%, TPS*=745.26

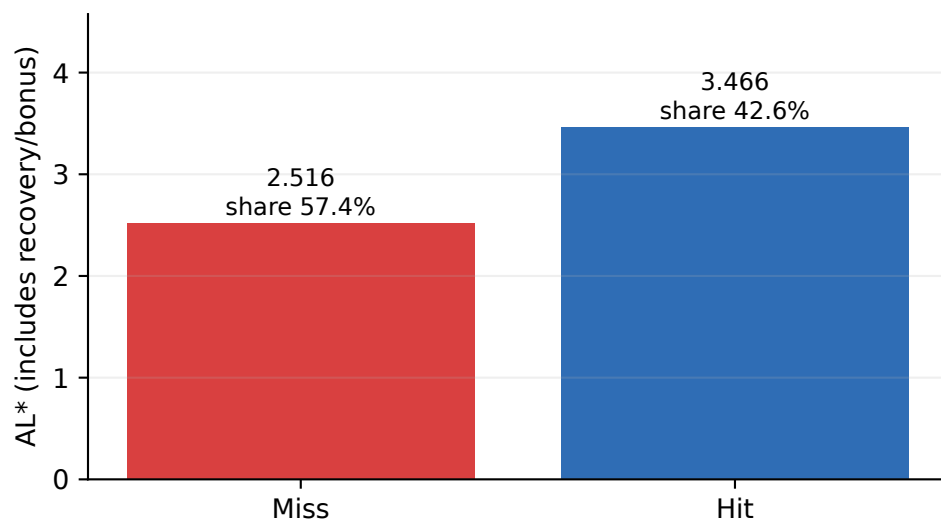
Measured wall time by batch cache status



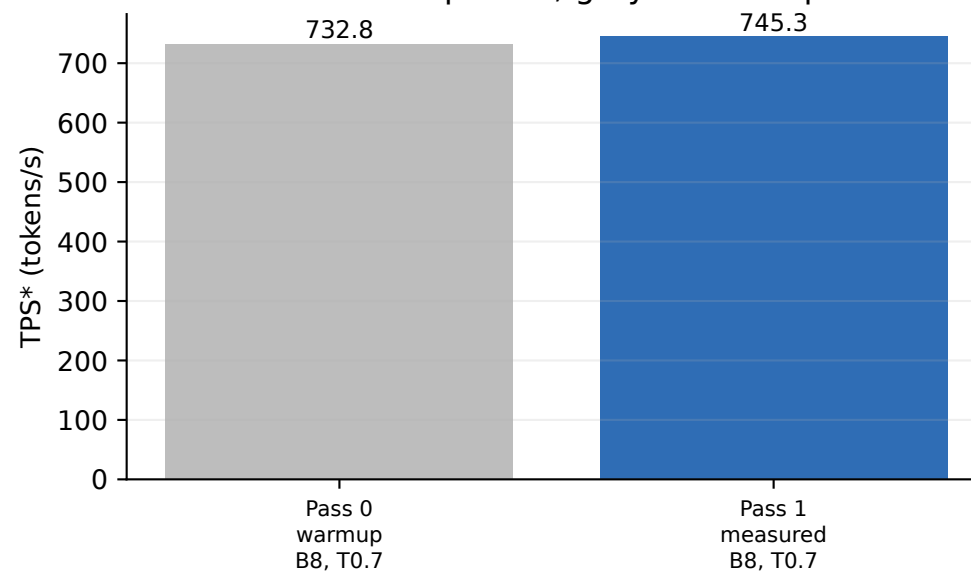
Boundary-free steps, including slow steps



Conditional AL* and source share



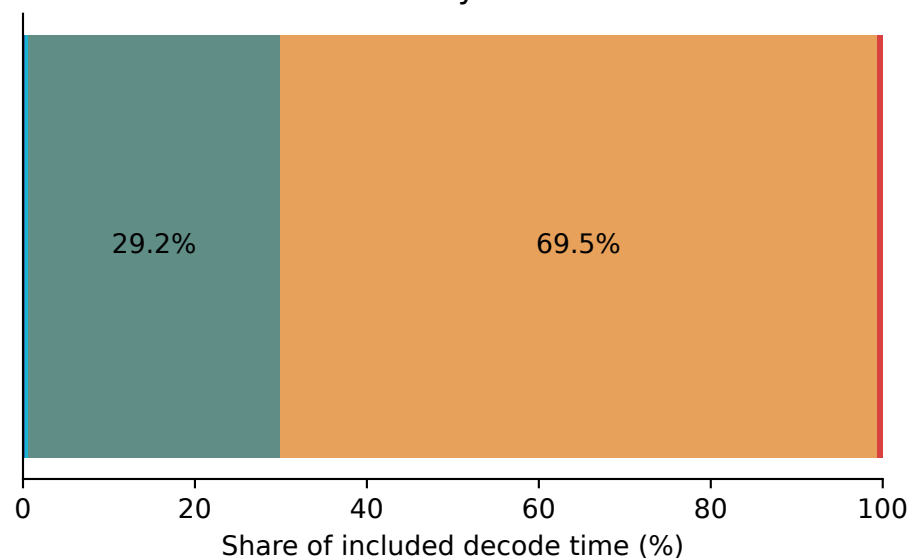
All saved passes; gray = warmup



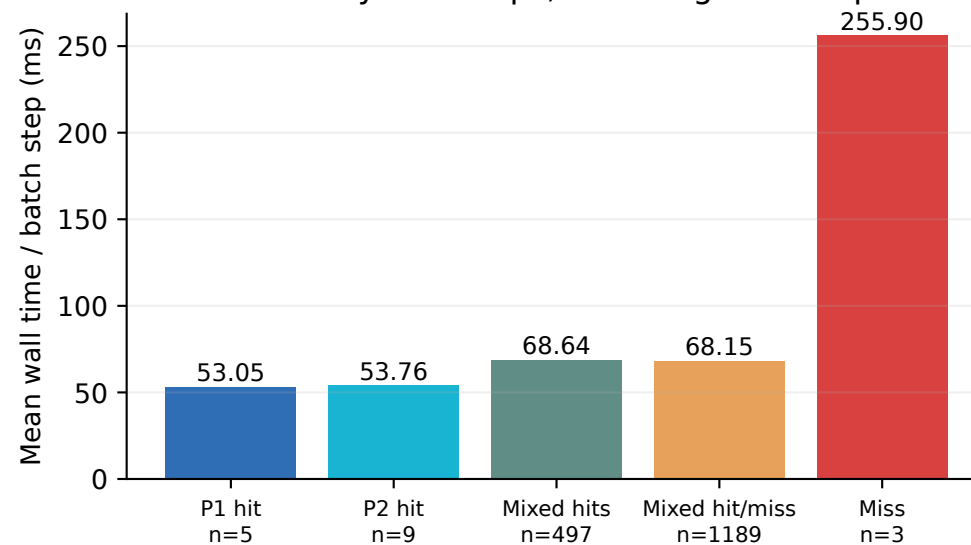
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_duet_al_r1
 DUET: K1/K2=6/3, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.897, hit=85.5%, TPS*=334.21

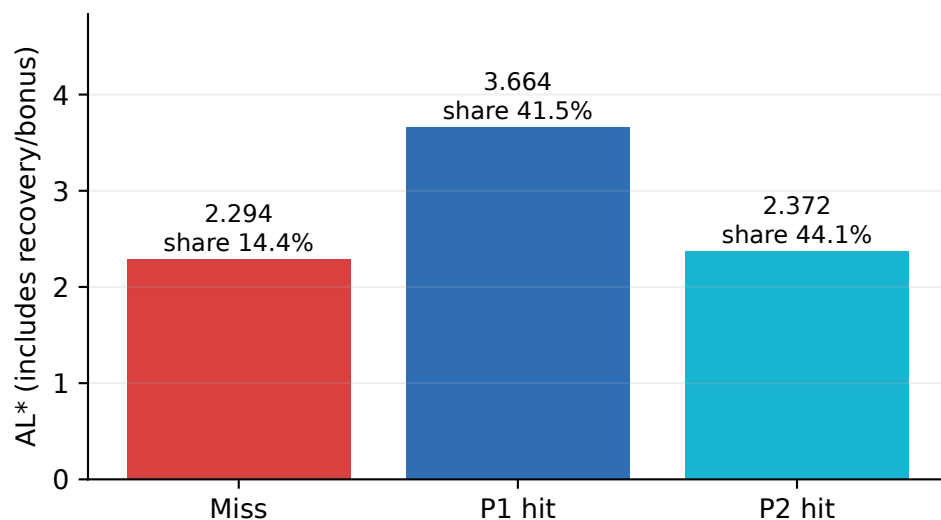
Measured wall time by batch cache status



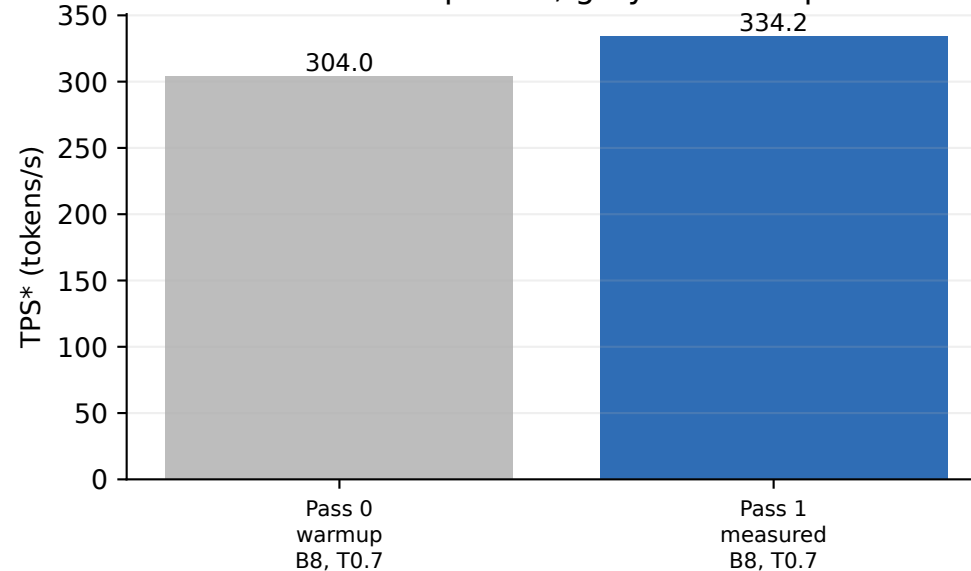
Boundary-free steps, including slow steps



Conditional AL* and source share



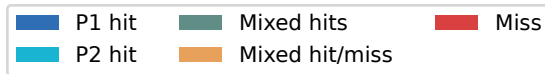
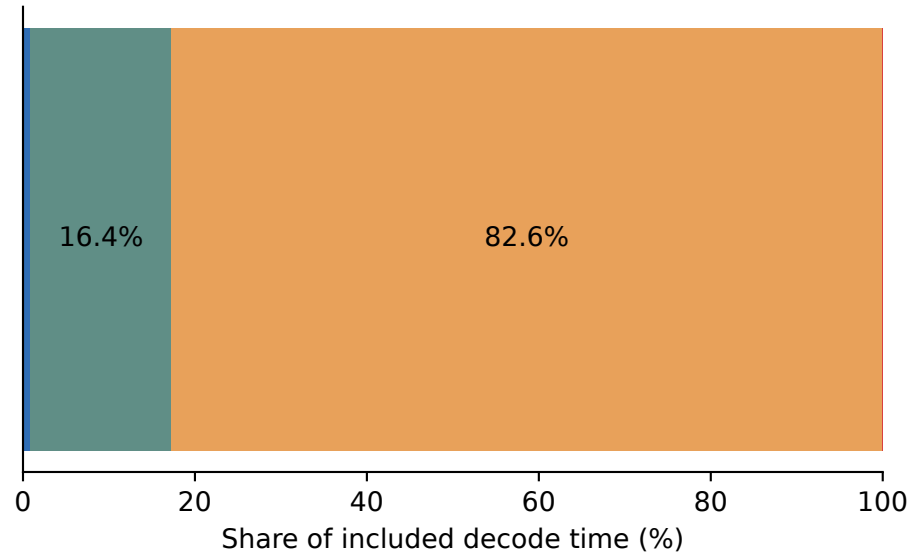
All saved passes; gray = warmup



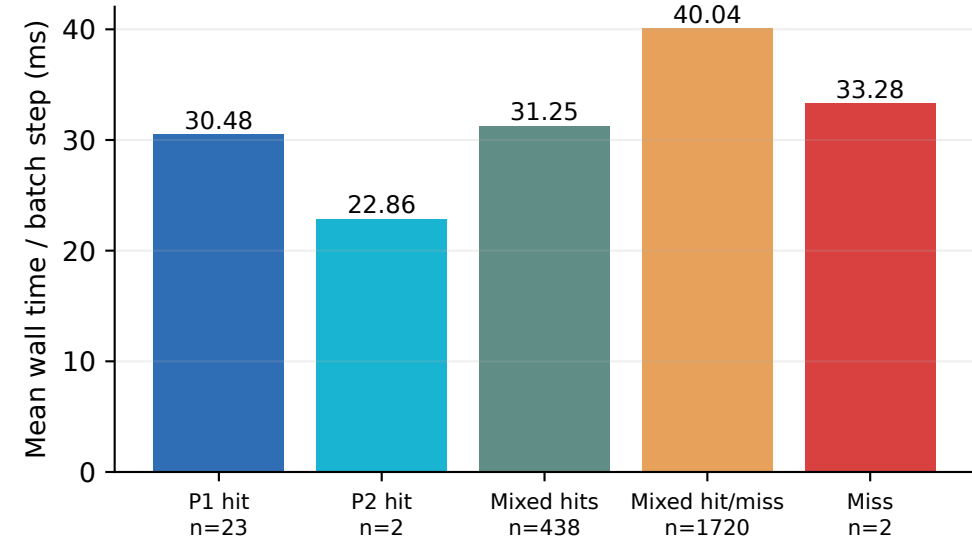
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_duet_fast_r1
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.386, hit=81.7%, TPS*=497.34

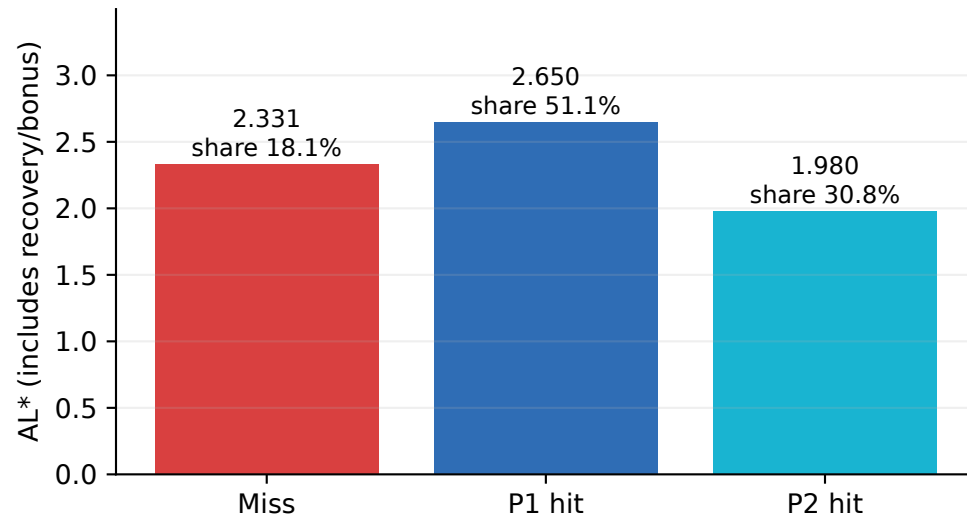
Measured wall time by batch cache status



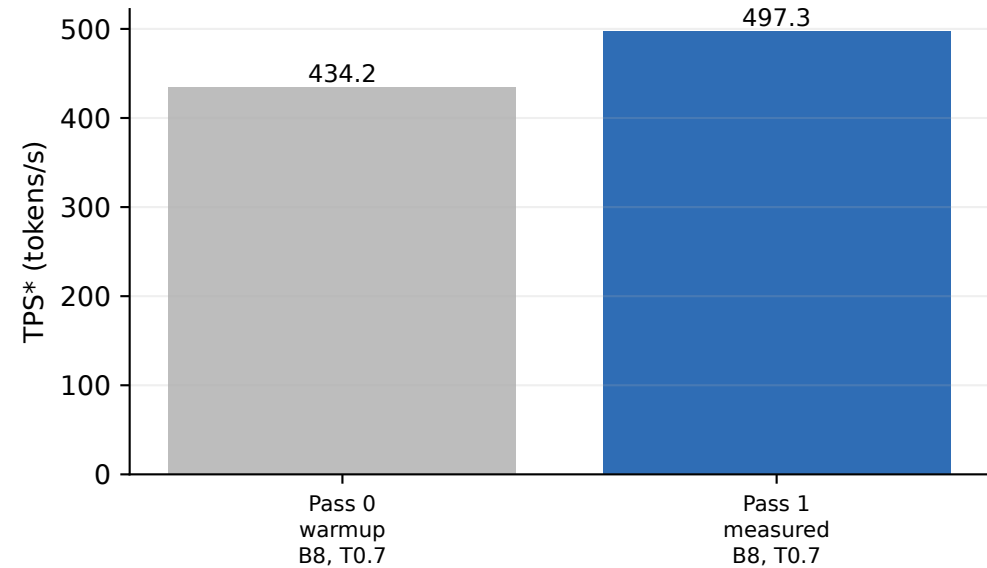
Boundary-free steps, including slow steps



Conditional AL* and source share



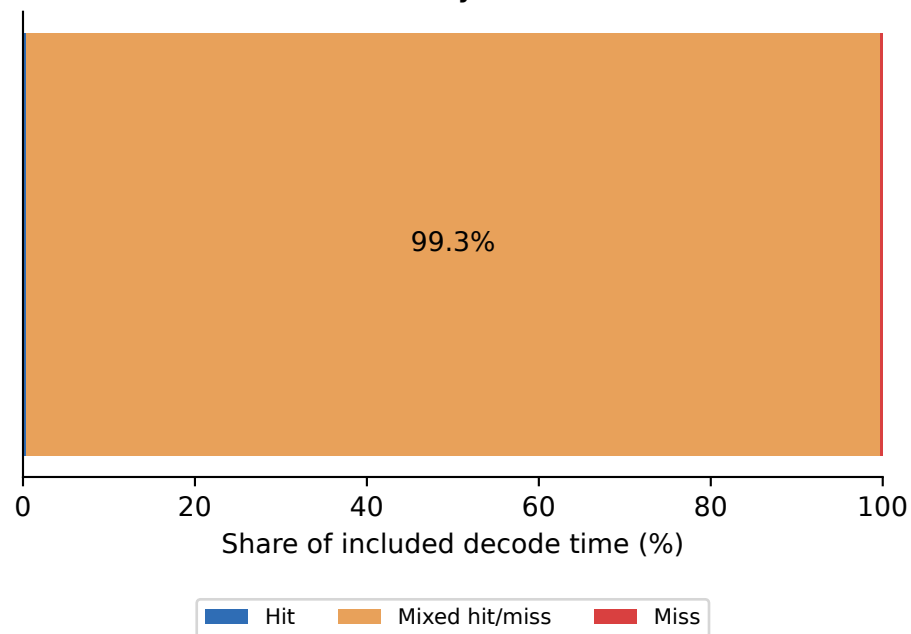
All saved passes; gray = warmup



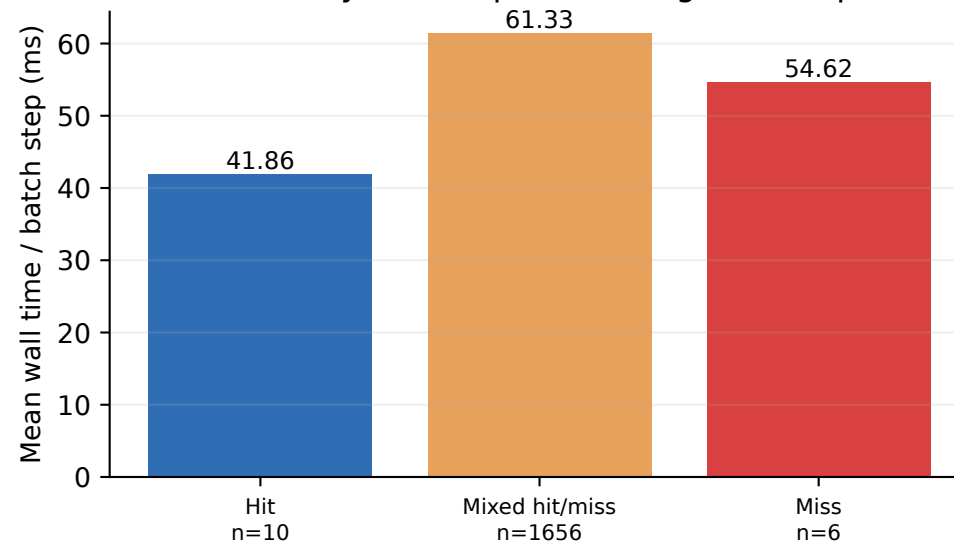
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_ssd_al_r1
SSD: K=8, F=3
Last pass: B=8, T=0.7, N=480 | AL*=3.030, hit=47.8%, TPS*=392.32

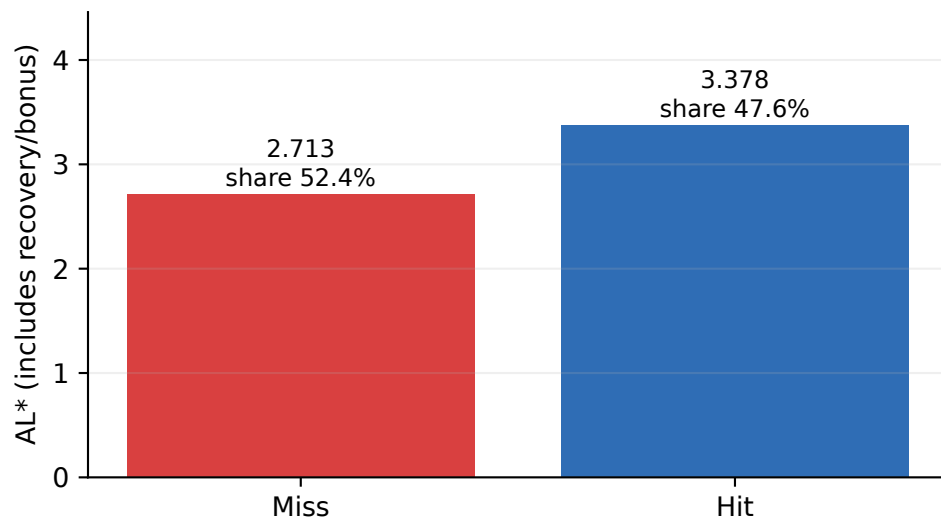
Measured wall time by batch cache status



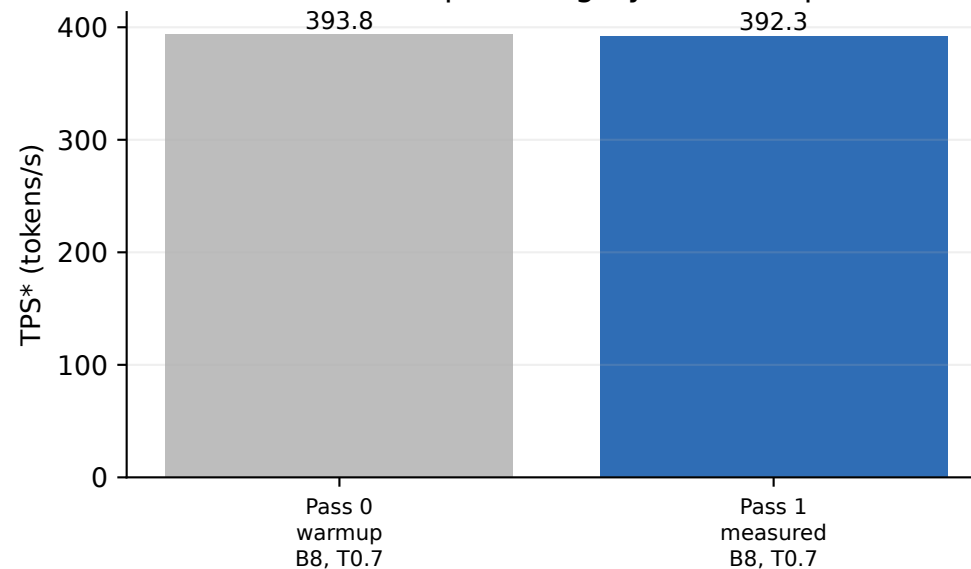
Boundary-free steps, including slow steps



Conditional AL* and source share



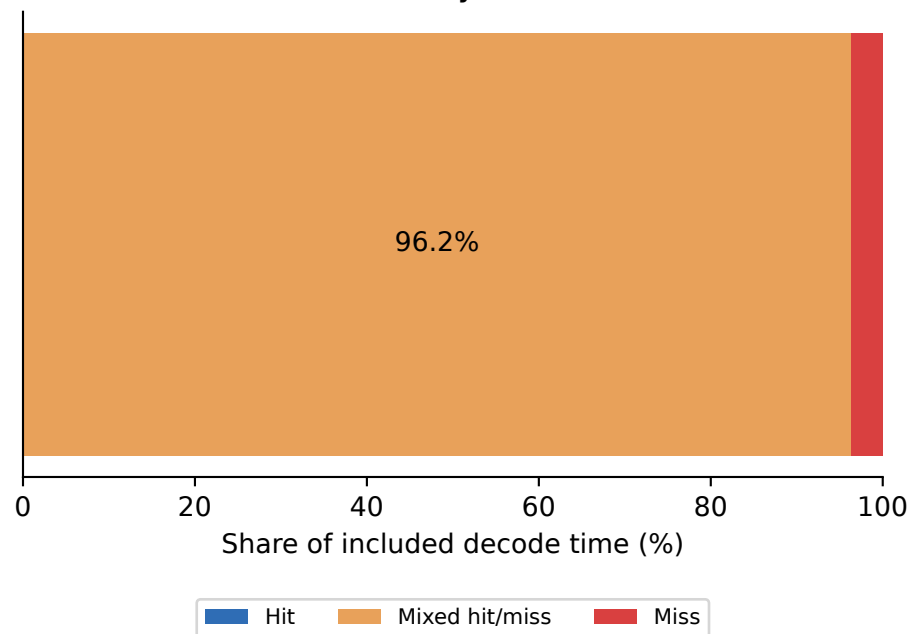
All saved passes; gray = warmup



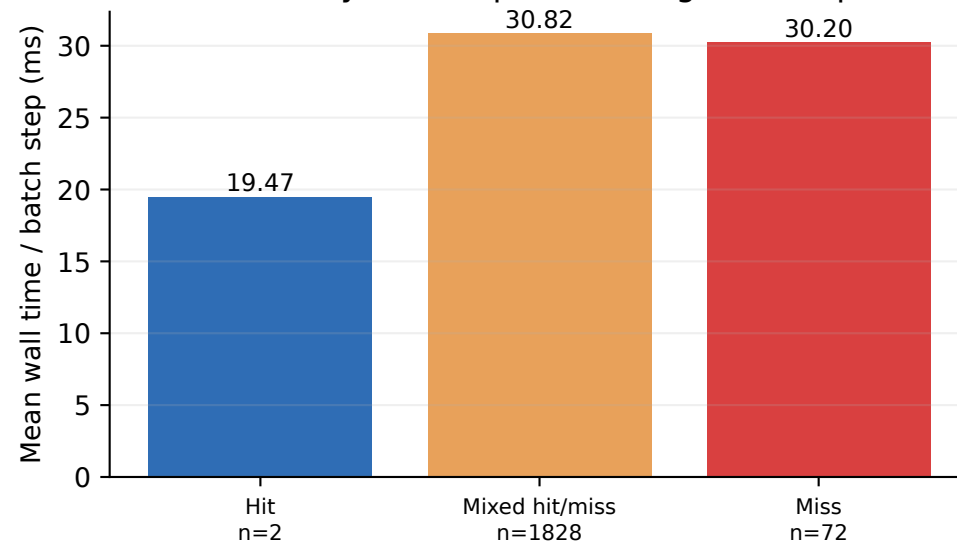
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_final_ssd_fast_r1
SSD: K=4, F=1
Last pass: B=8, T=0.7, N=480 | AL*=2.699, hit=35.5%, TPS*=694.42

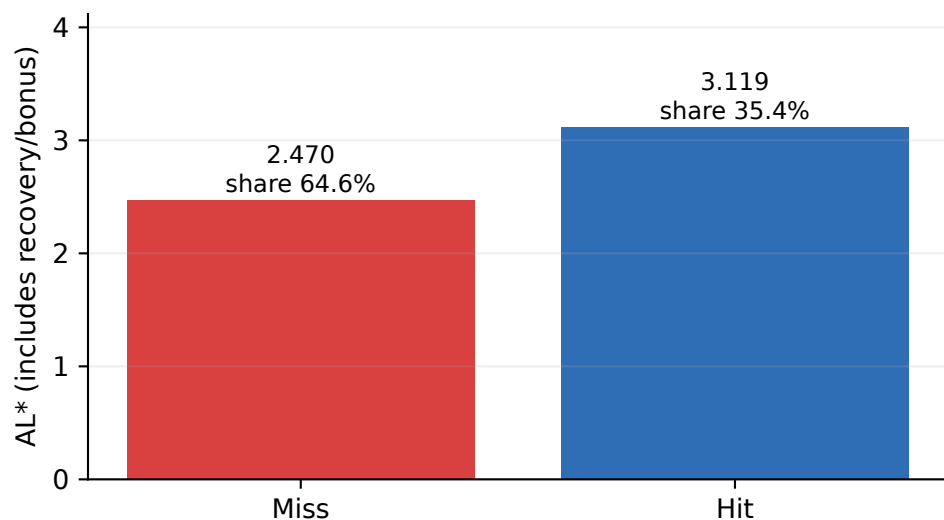
Measured wall time by batch cache status



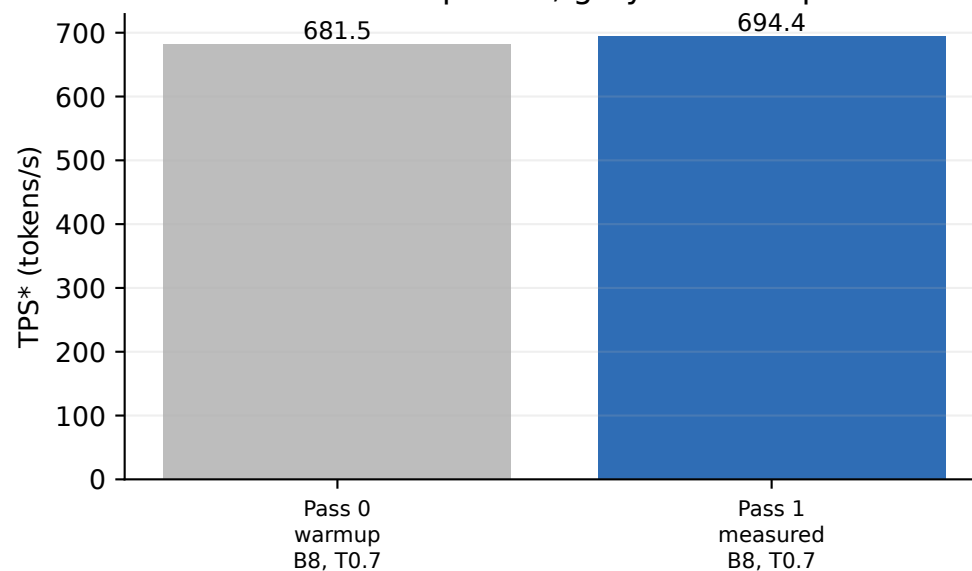
Boundary-free steps, including slow steps



Conditional AL* and source share

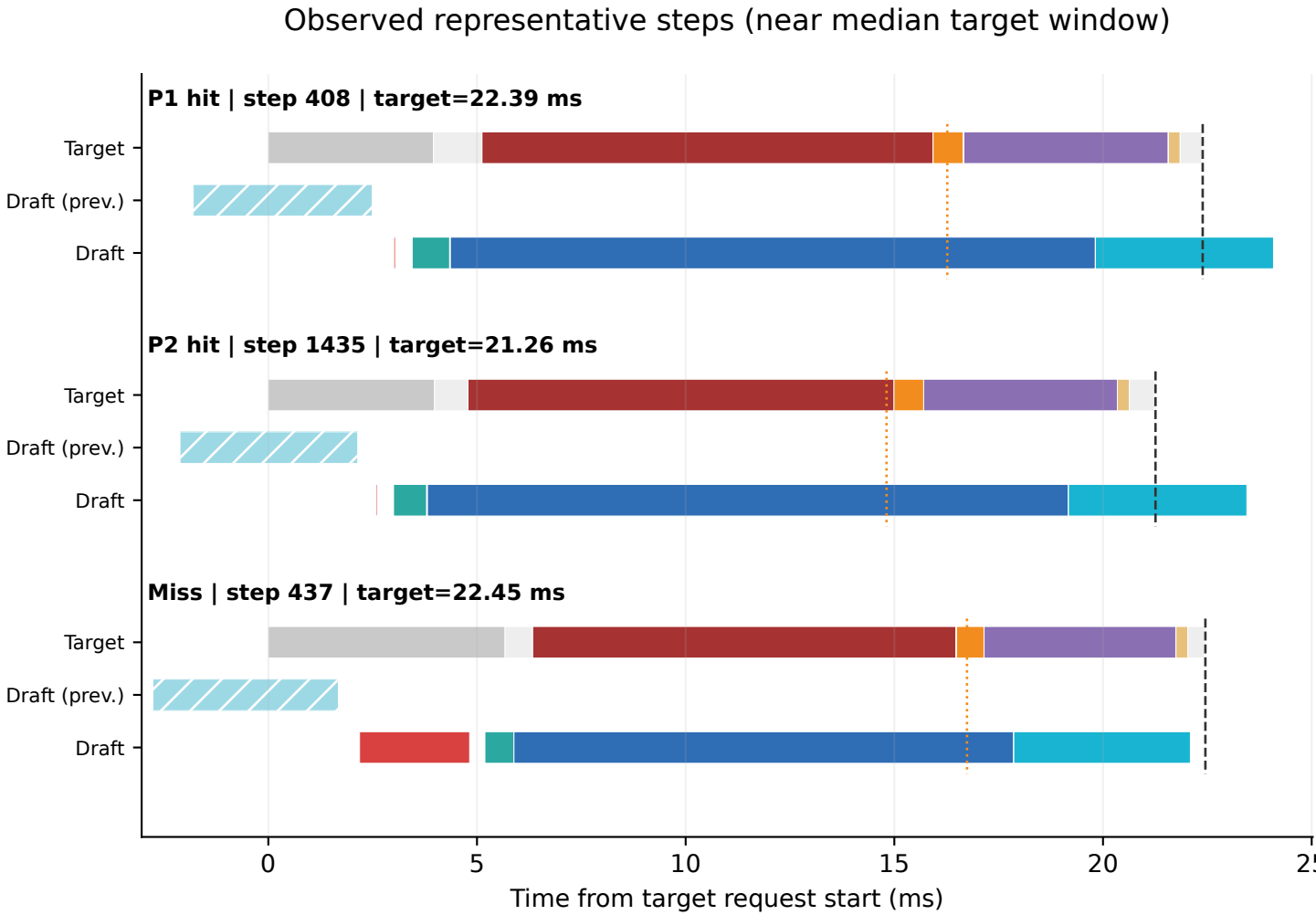
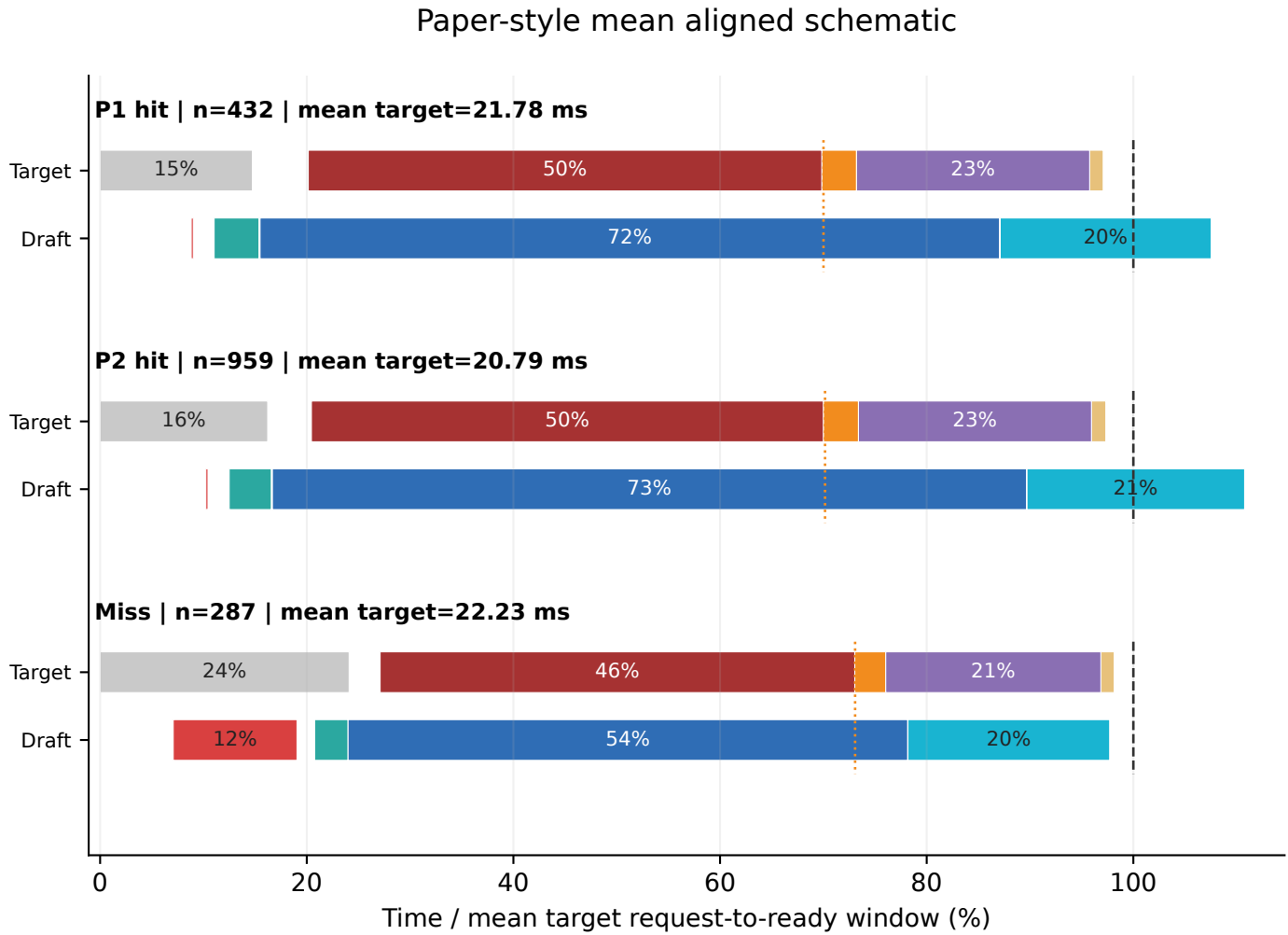
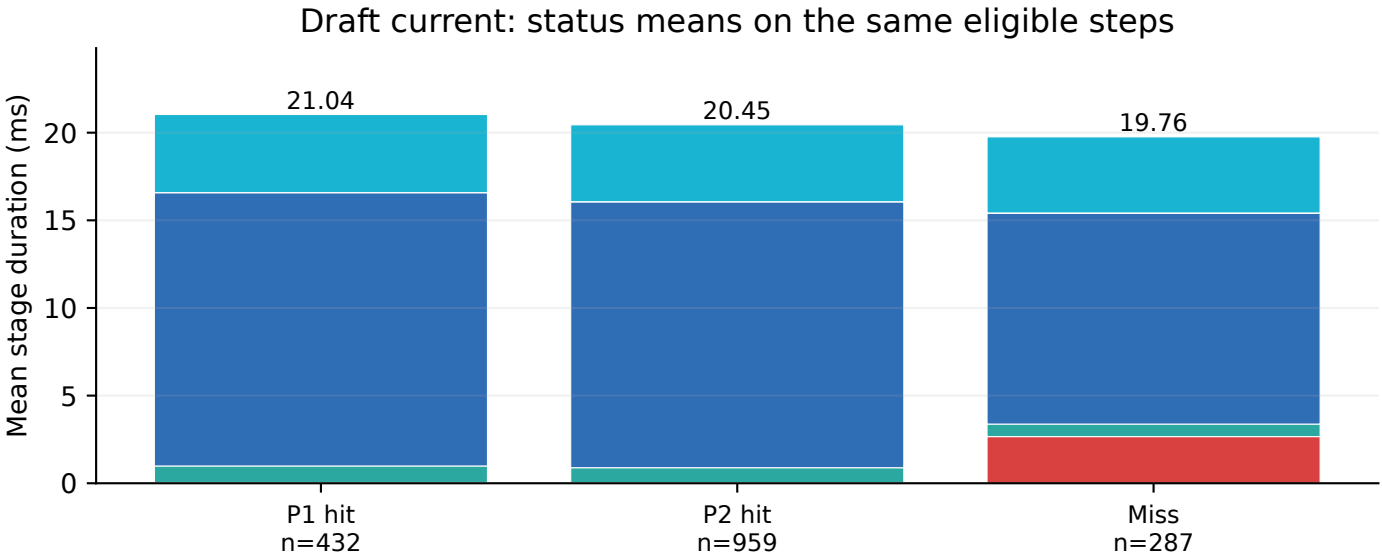
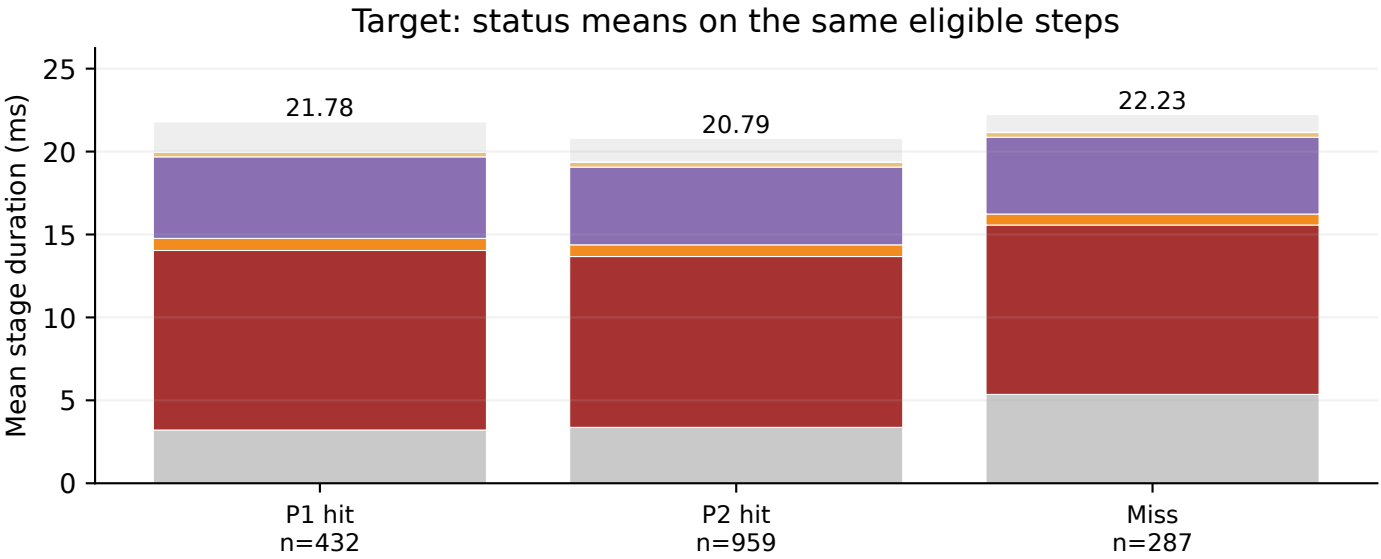


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

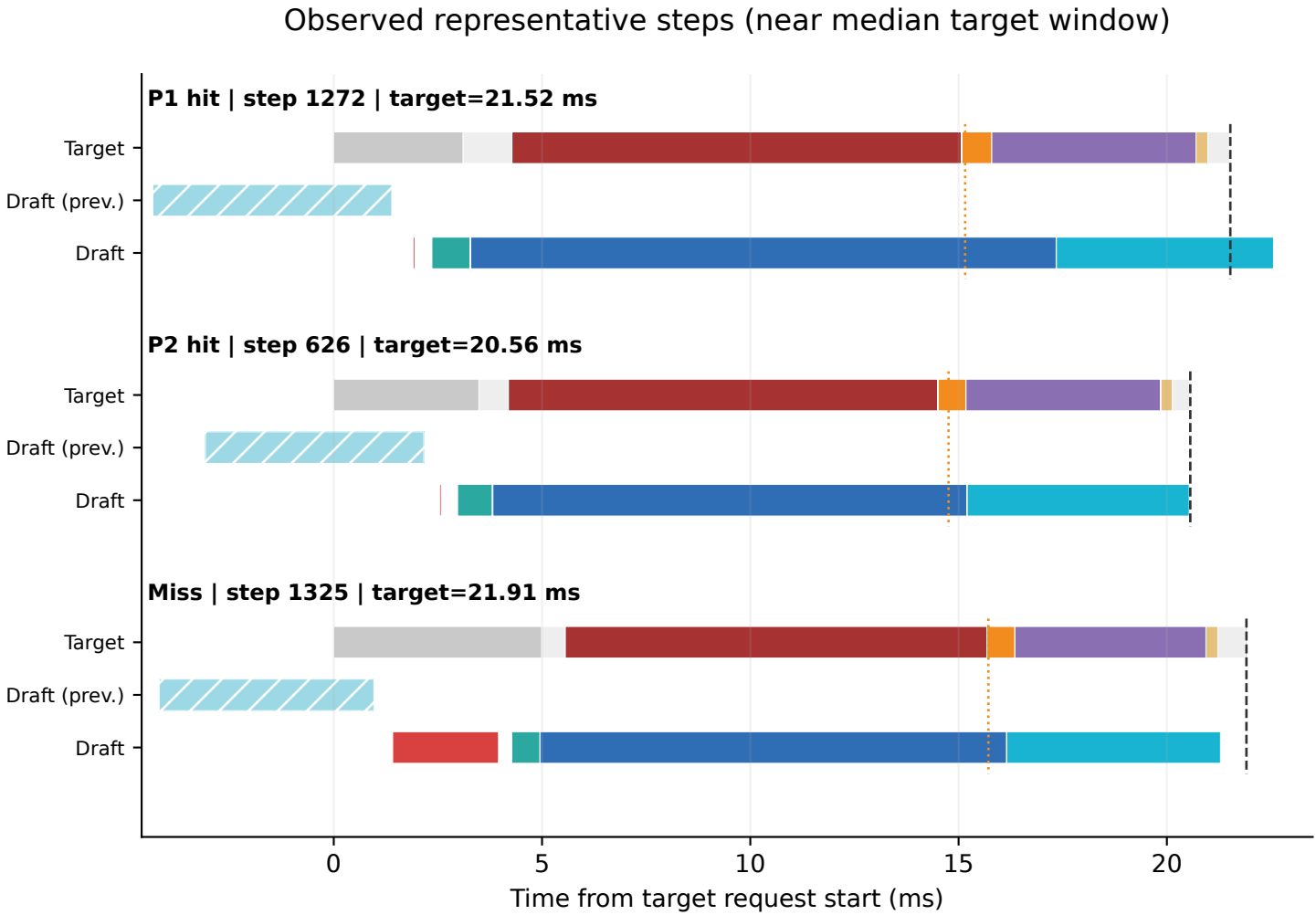
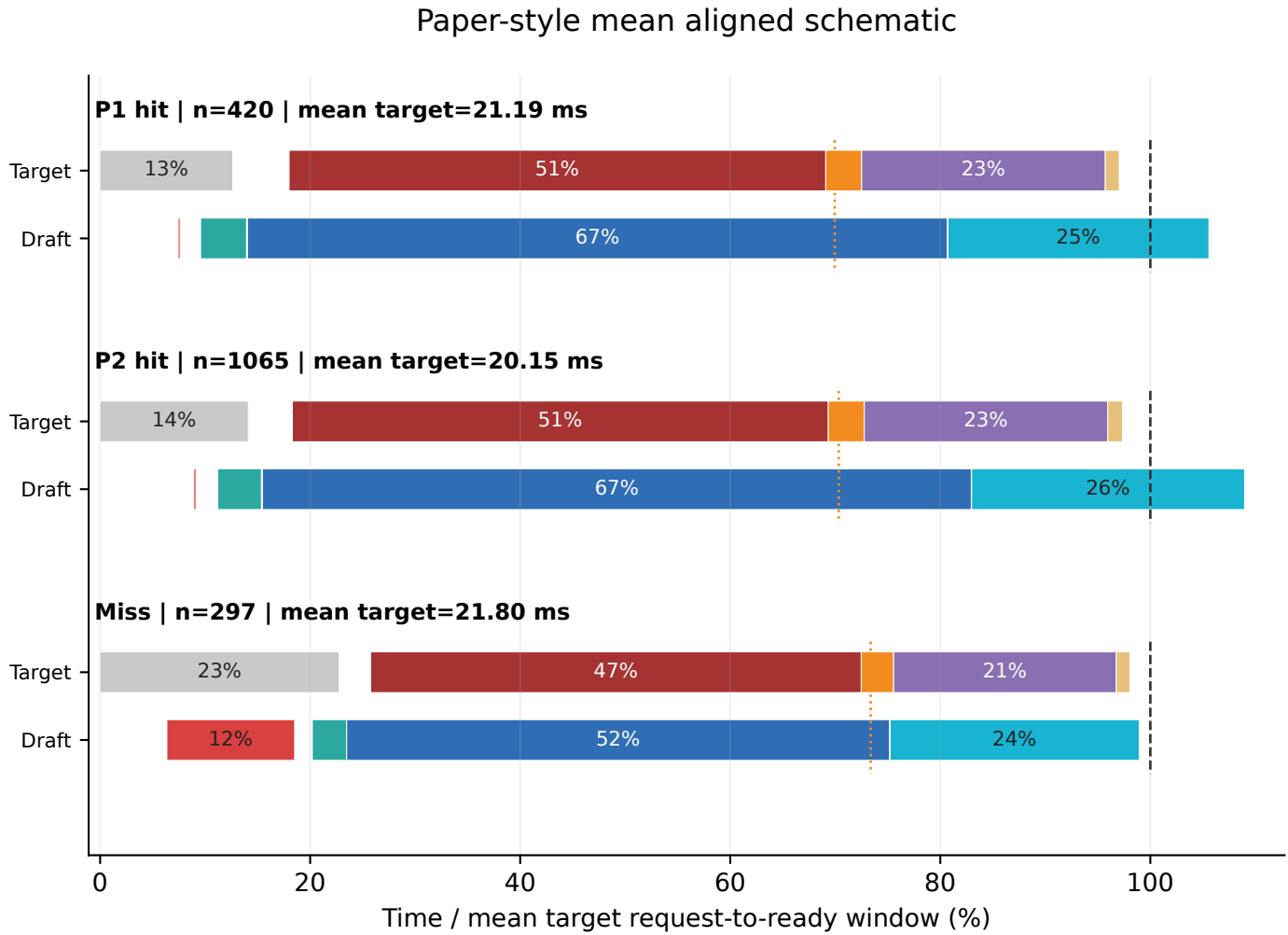
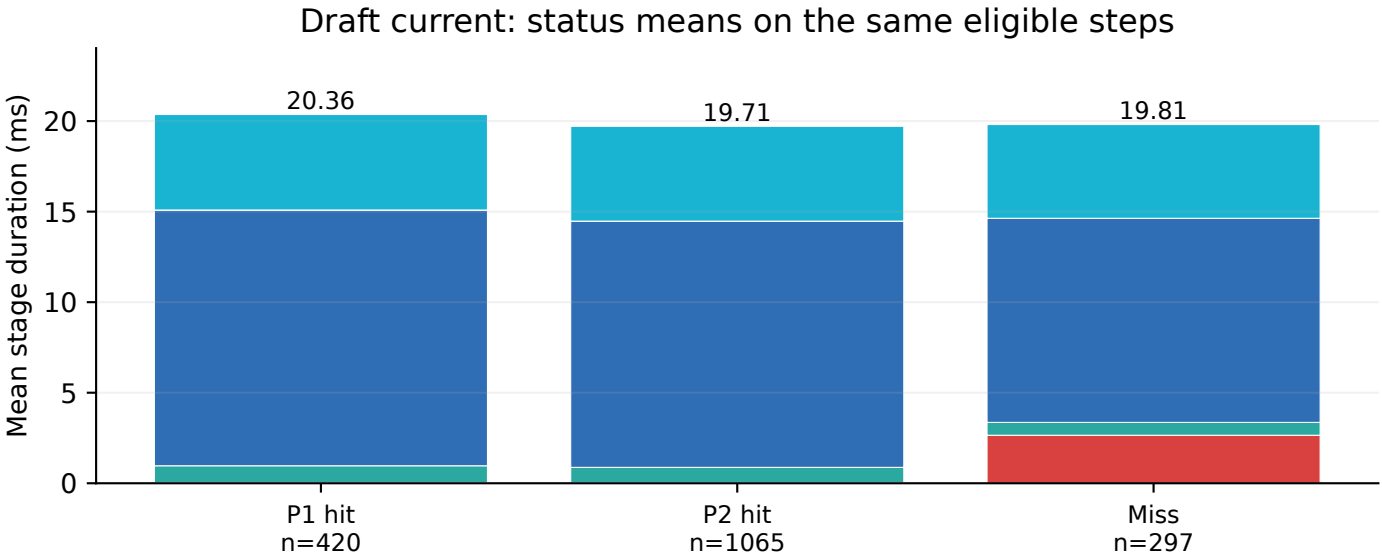
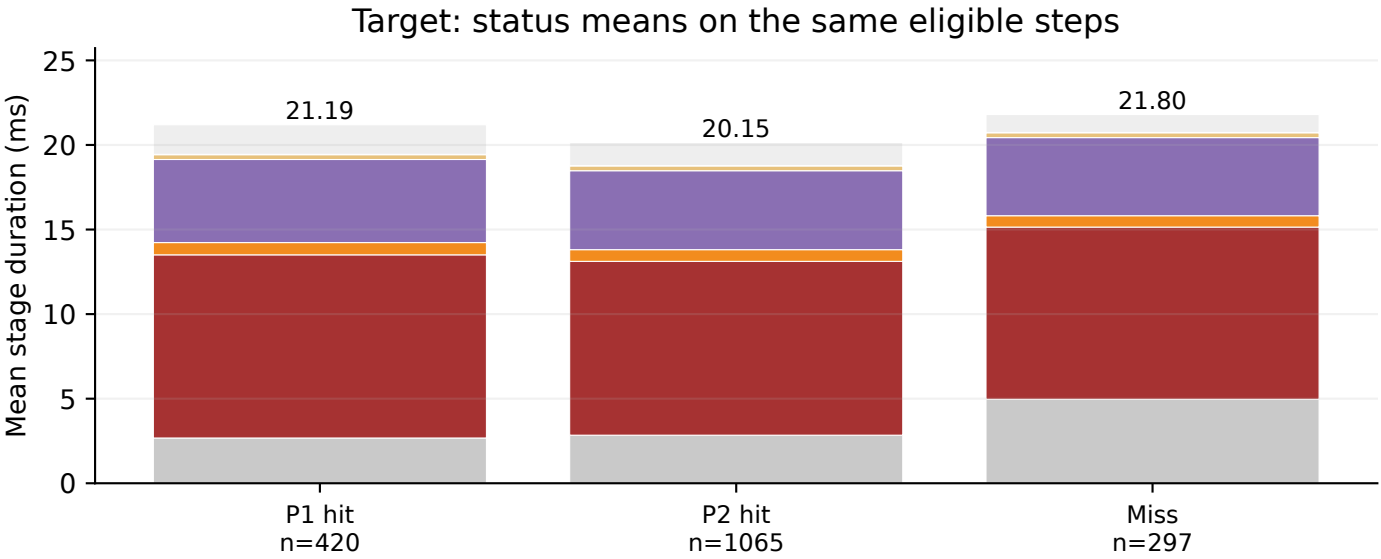
llama2_b1_combo_al_k1_13
DUET: K1/K2=13/4, exit=21 (0-based), C=3, N1/N2=16/8, U=2, W=14, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1678 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

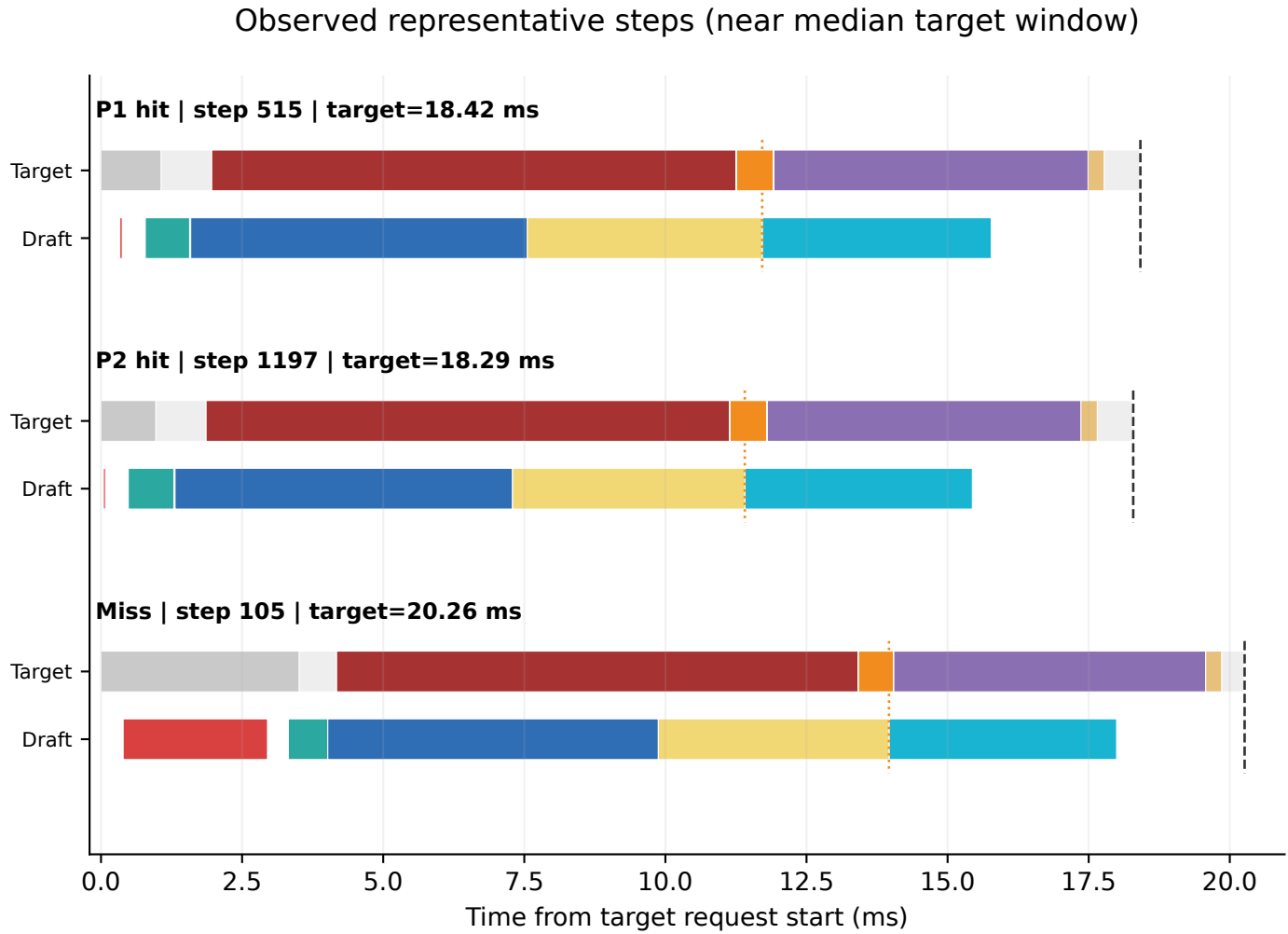
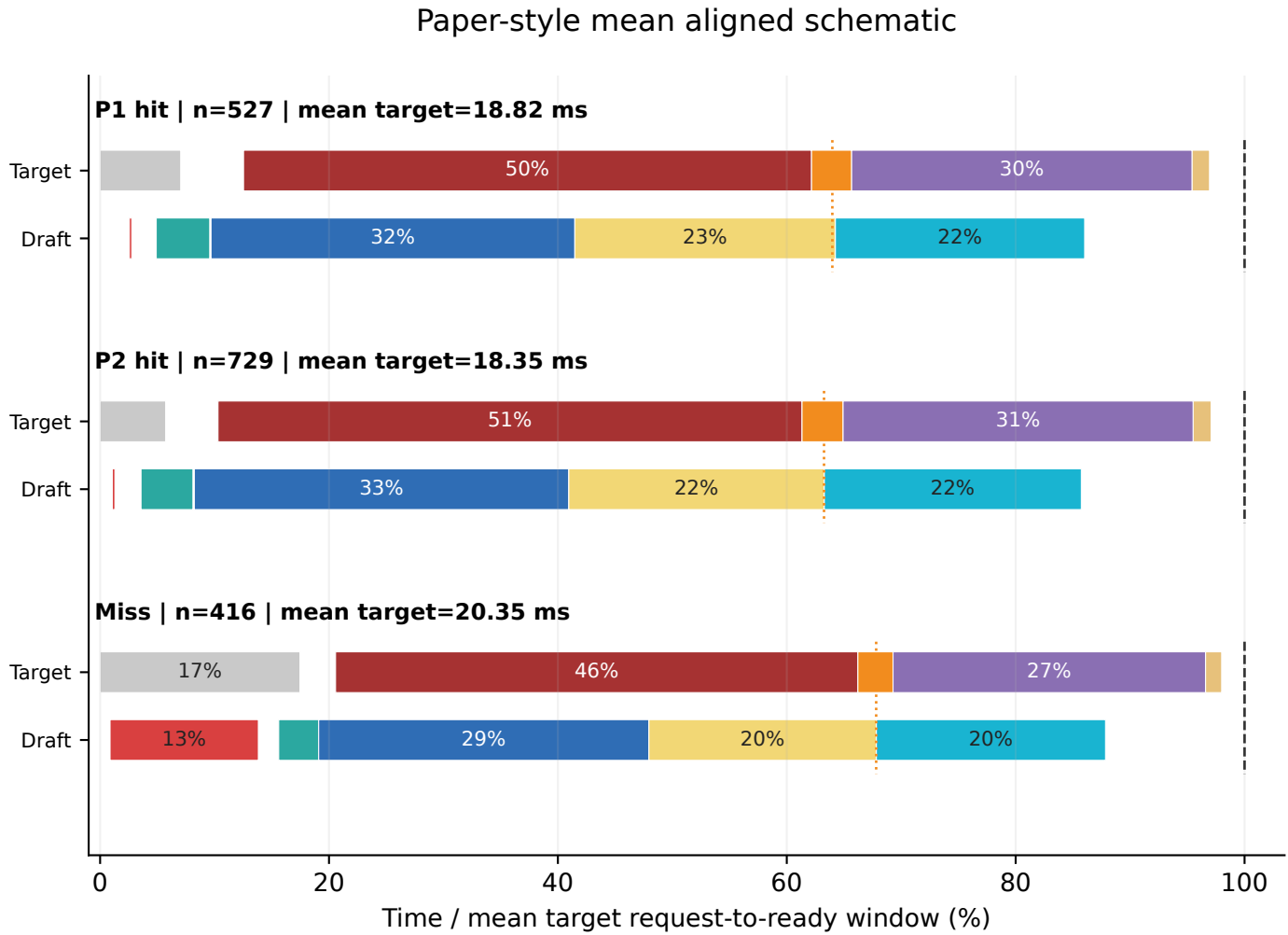
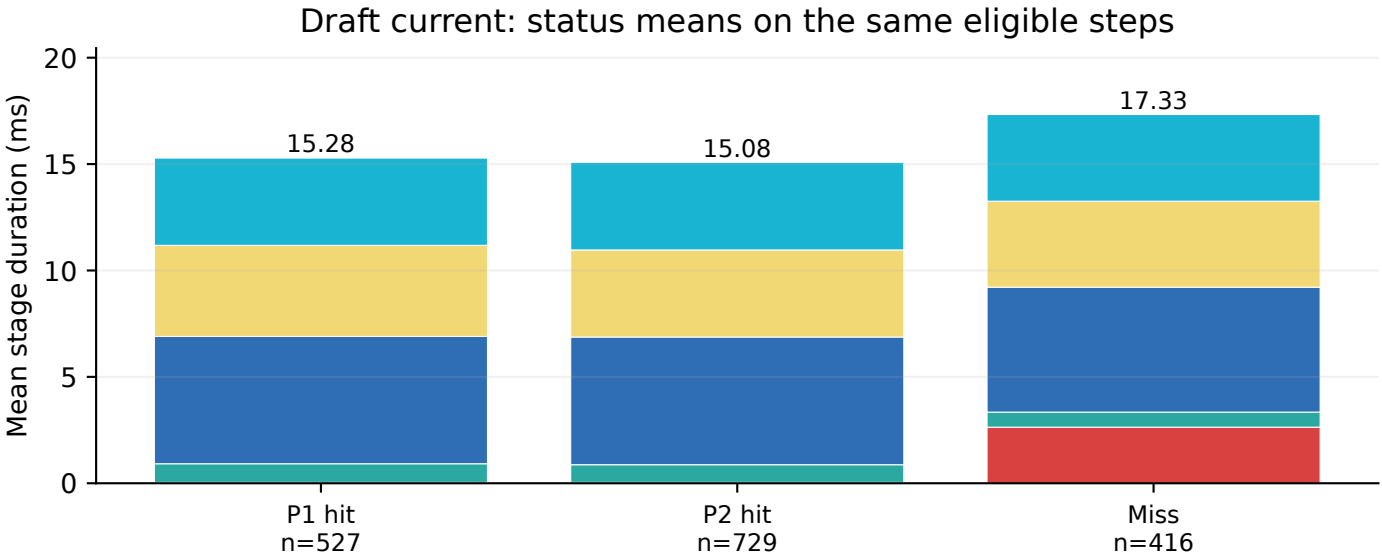
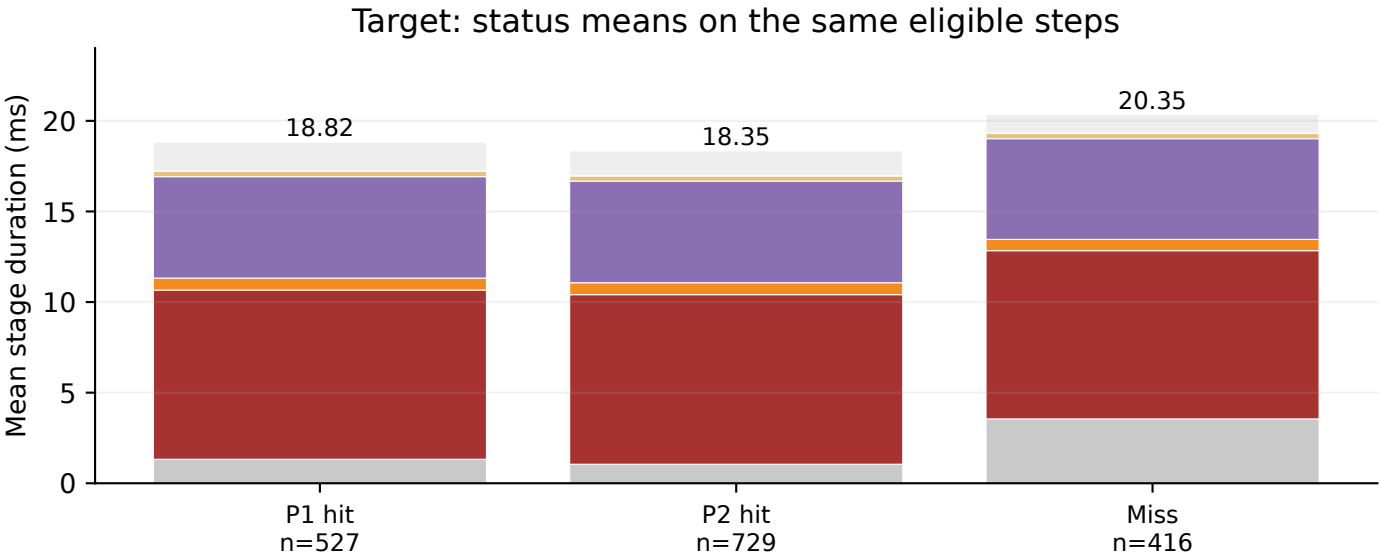
llama2_b1_combo_al_k2_5
DUET: K1/K2=12/5, exit=21 (0-based), C=3, N1/N2=16/8, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1782 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

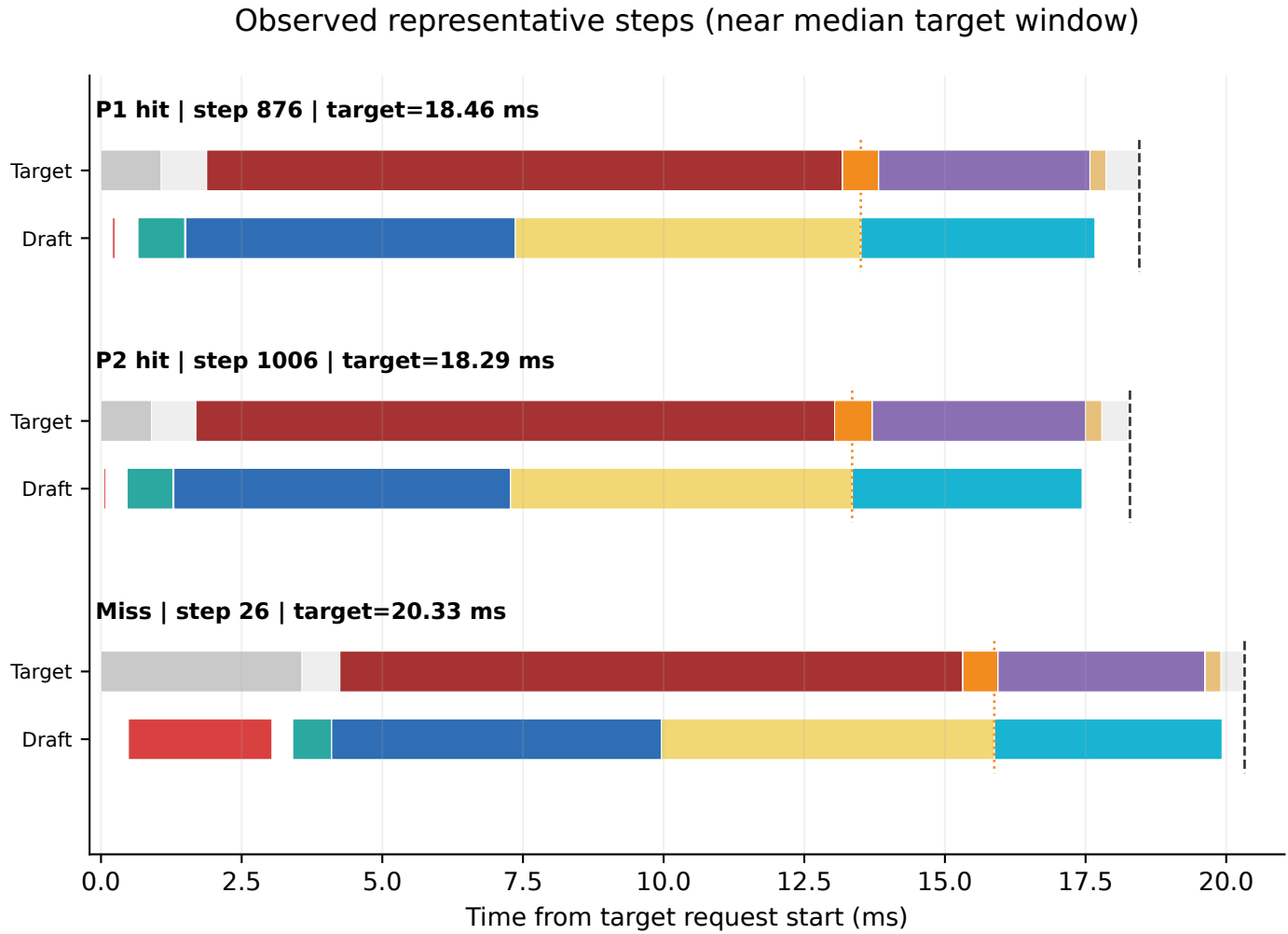
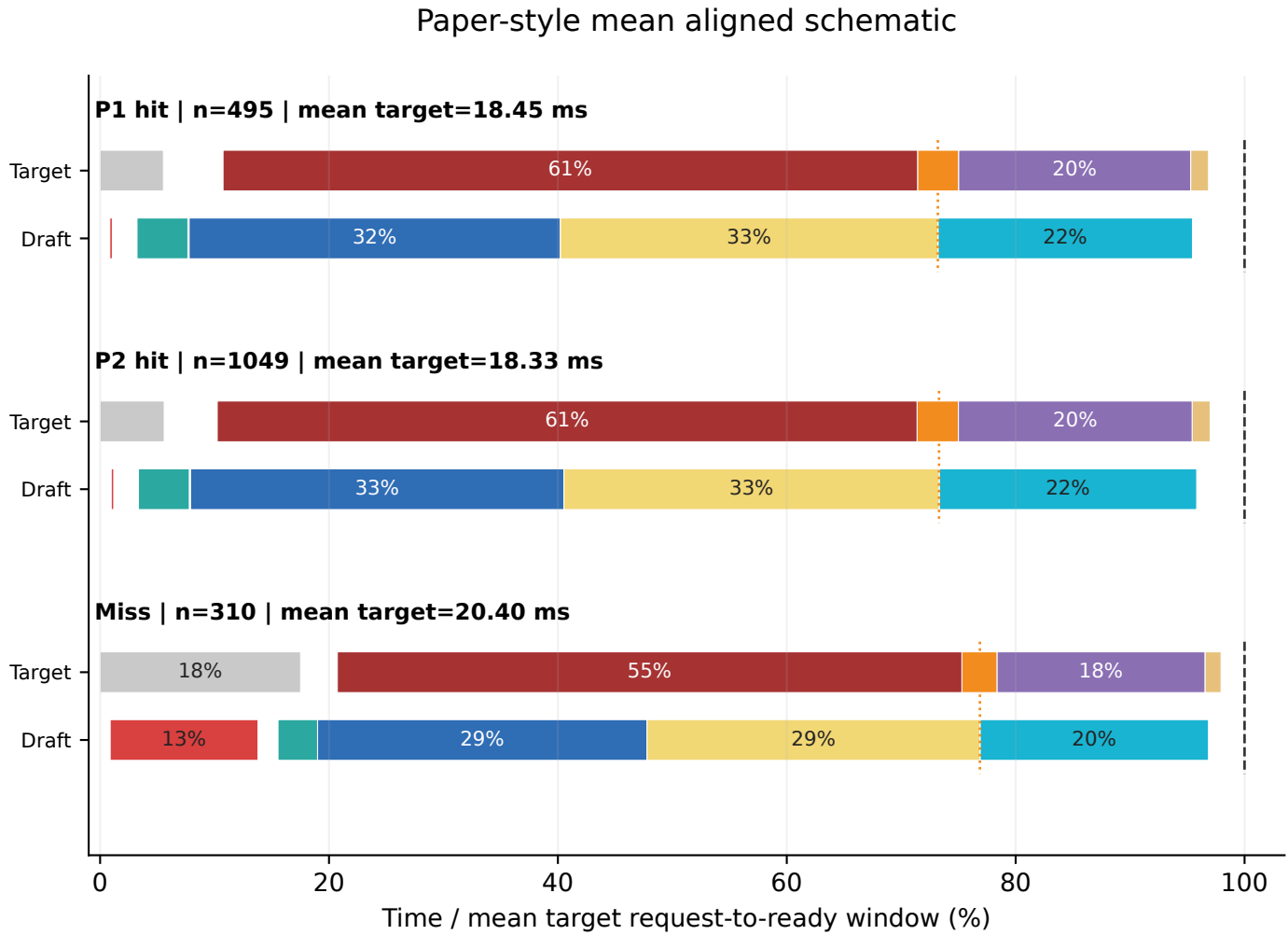
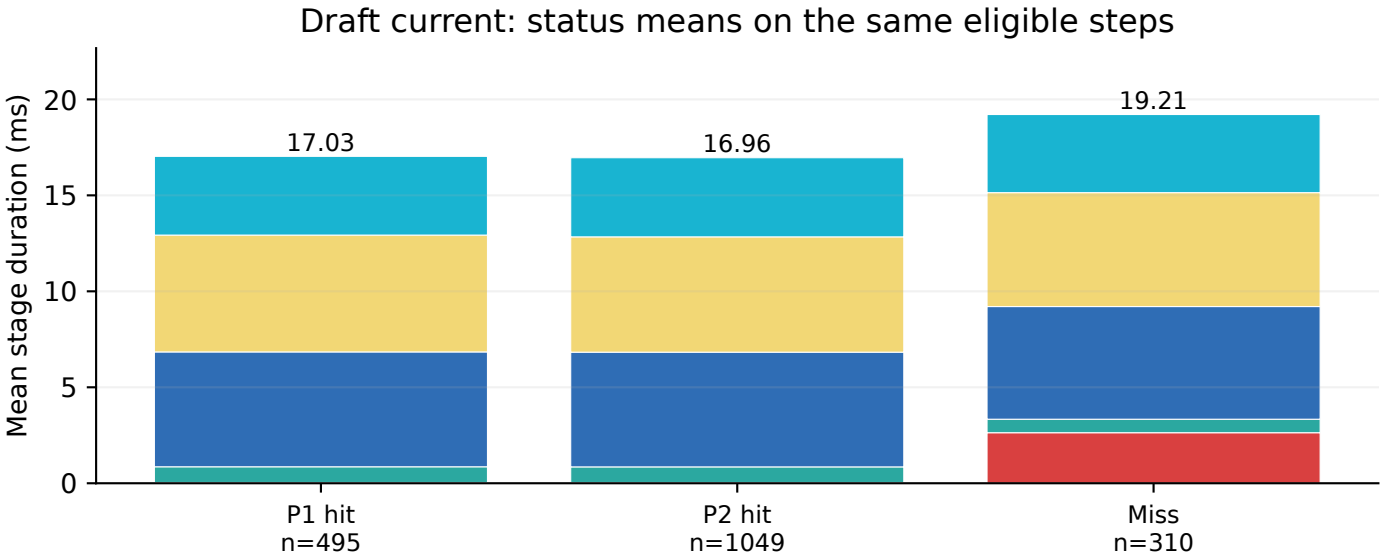
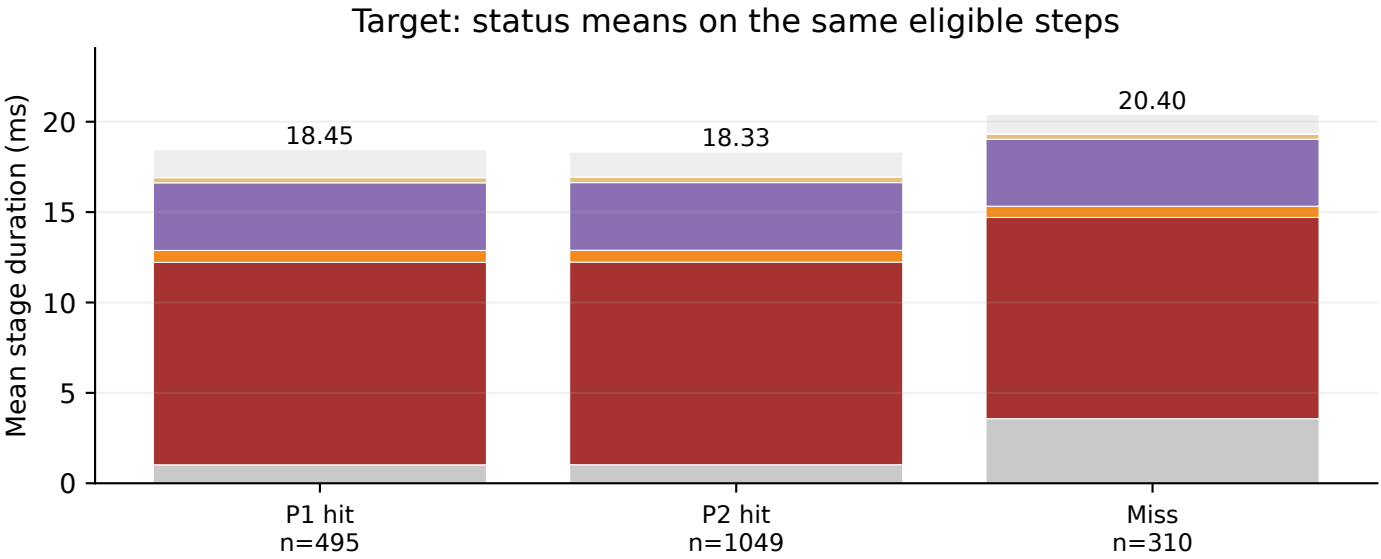
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_combo_e19
DUET: K1/K2=6/4, exit=19 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1672 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

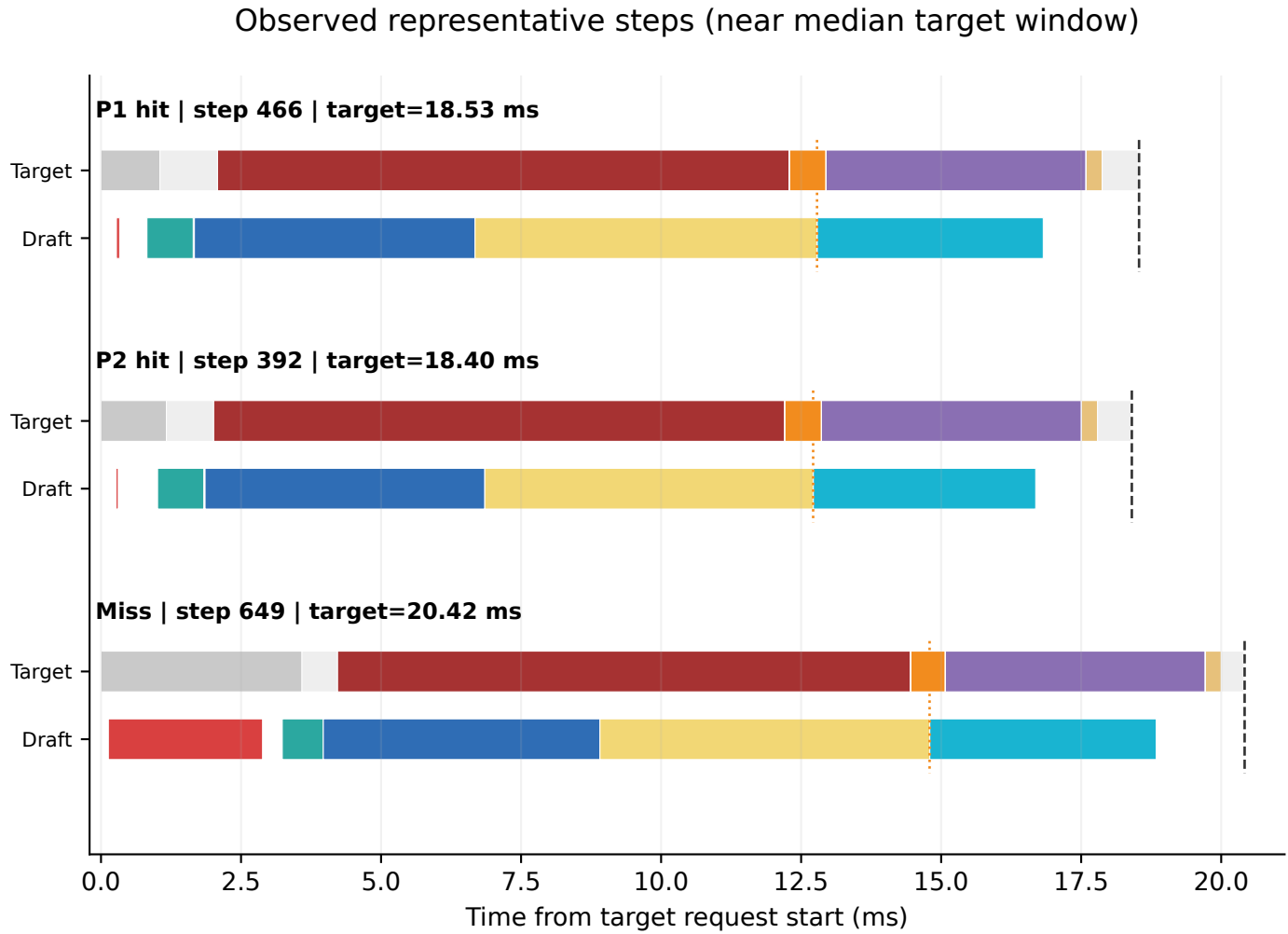
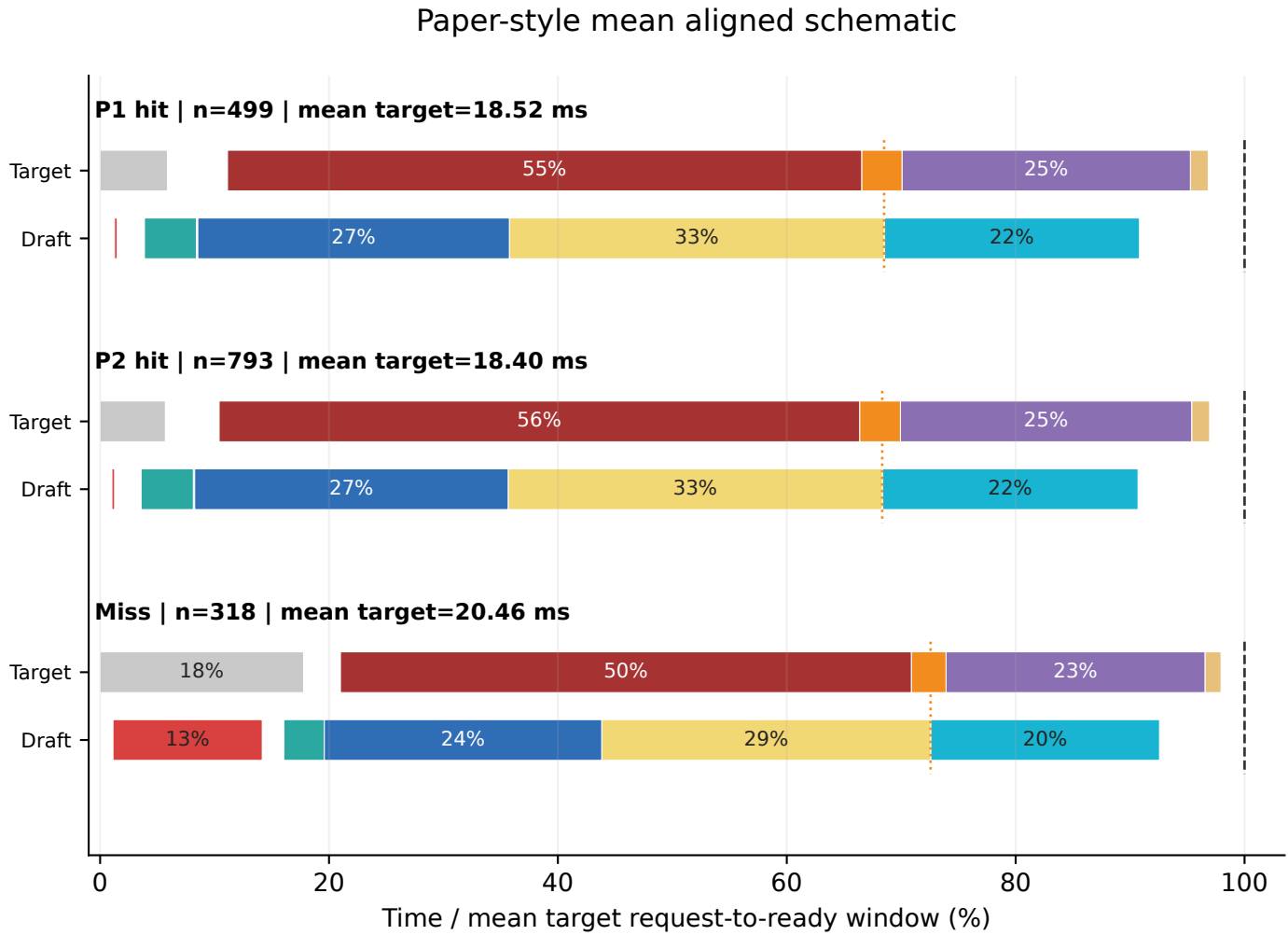
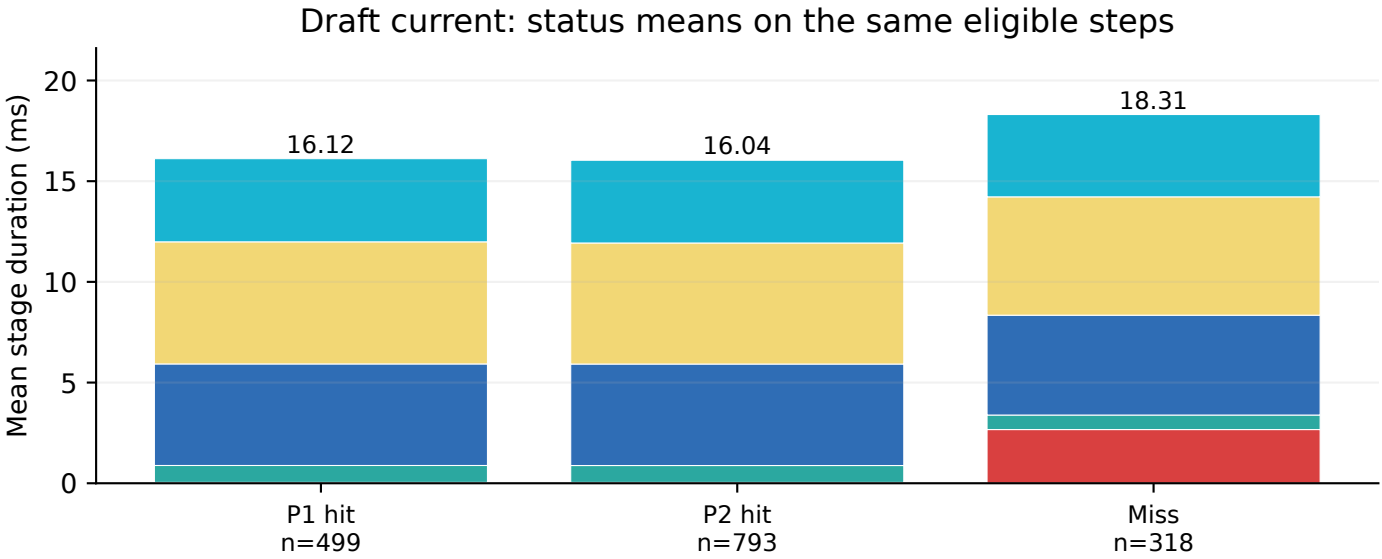
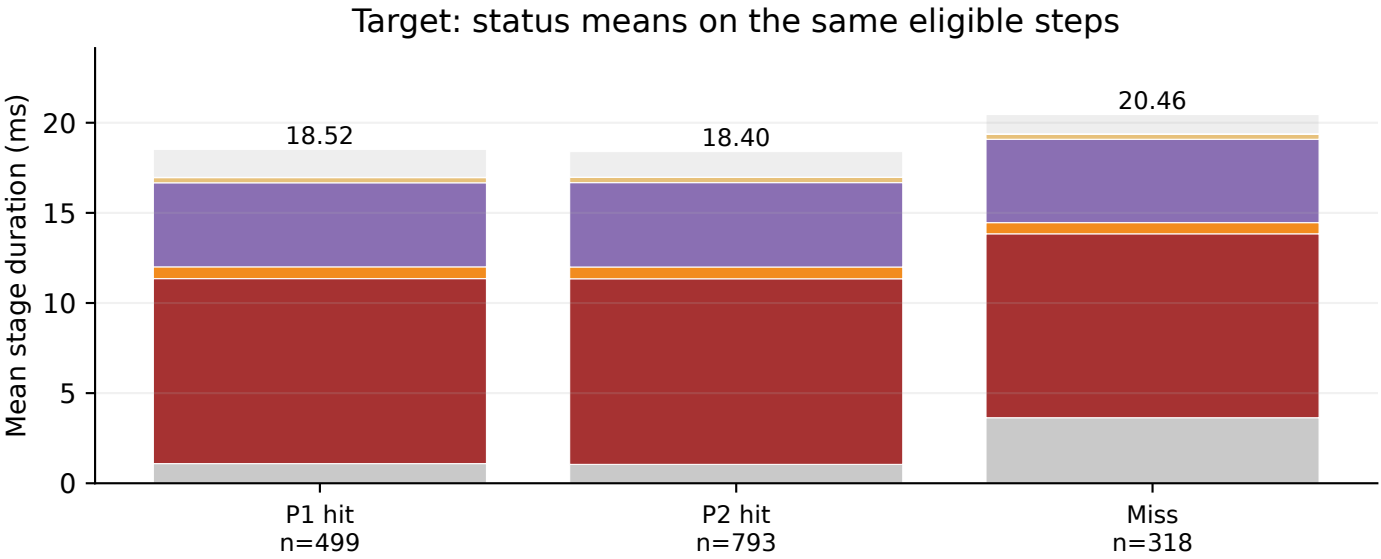
llama2_b1_combo_e23
DUET: K1/K2=6/4, exit=23 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1854 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

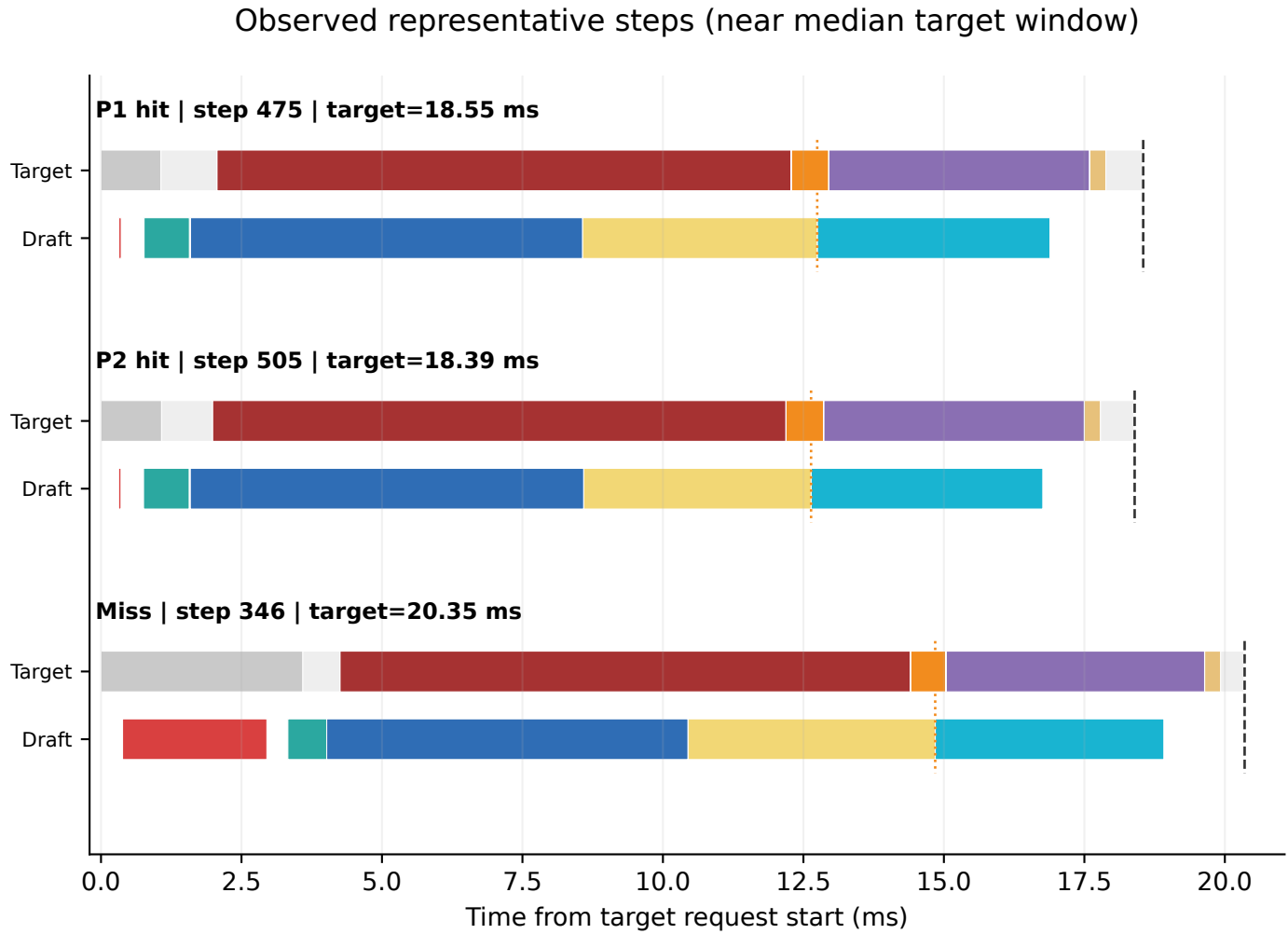
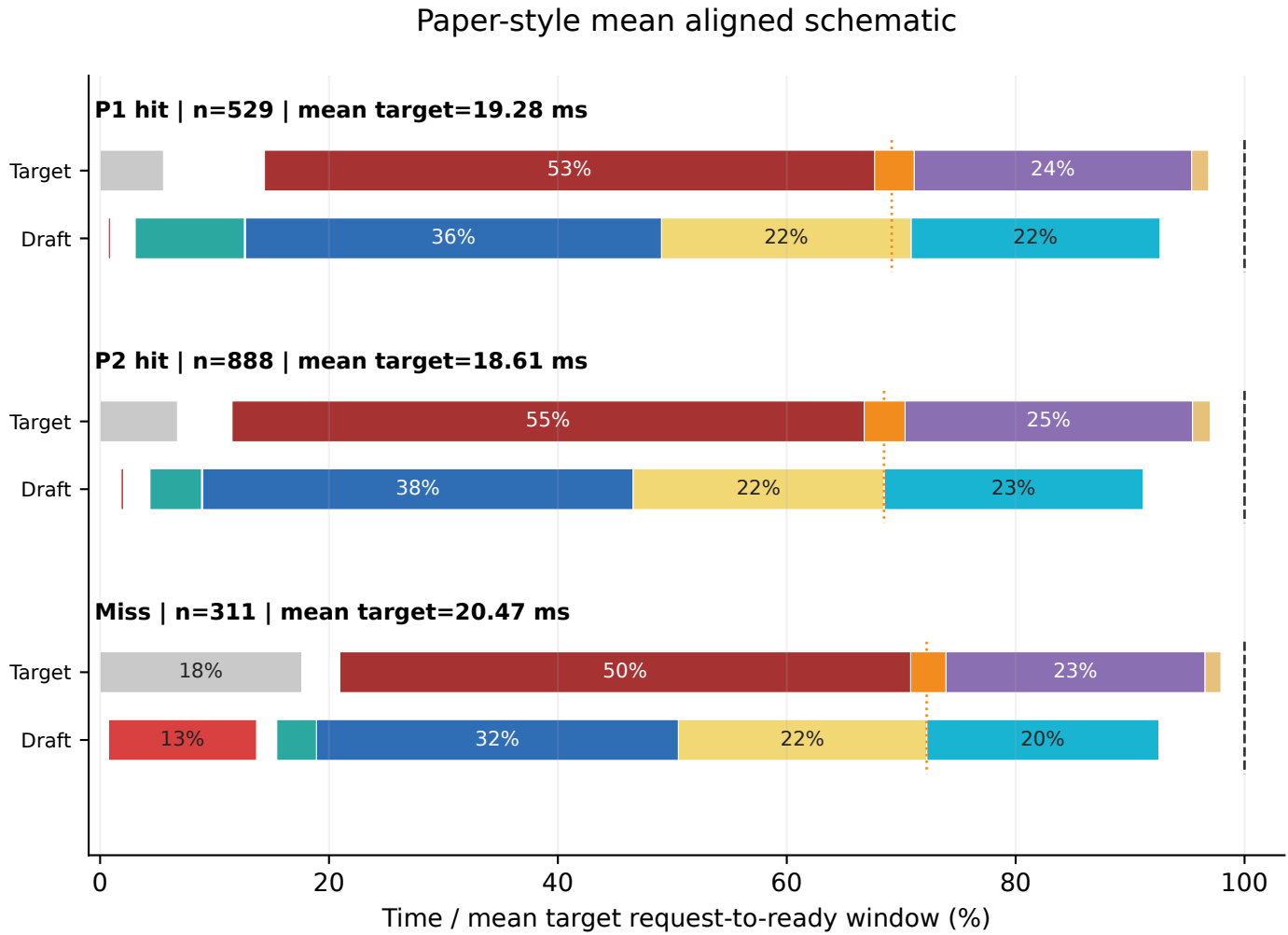
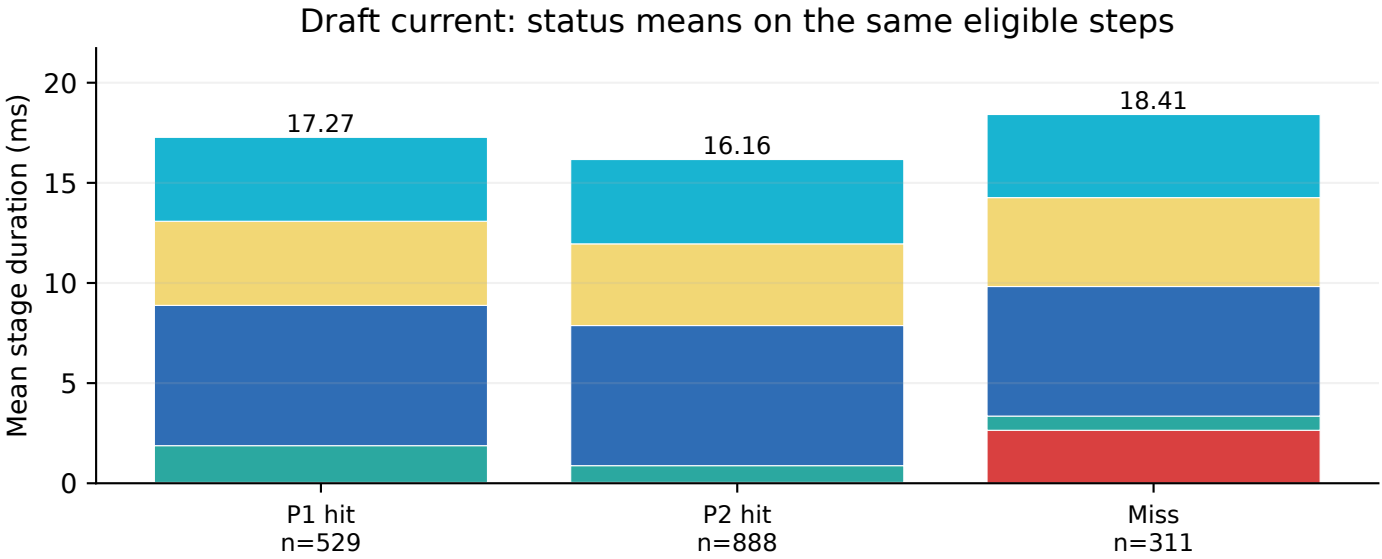
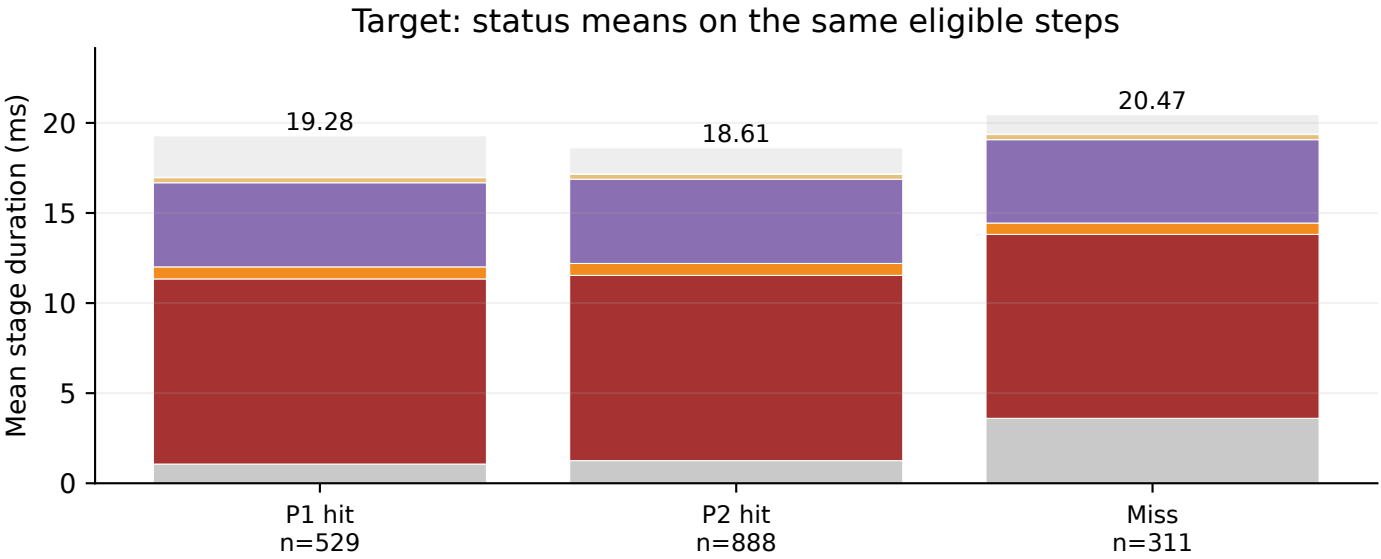
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_combo_k1_5
DUET: K1/K2=5/4, exit=21 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=6, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1610 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

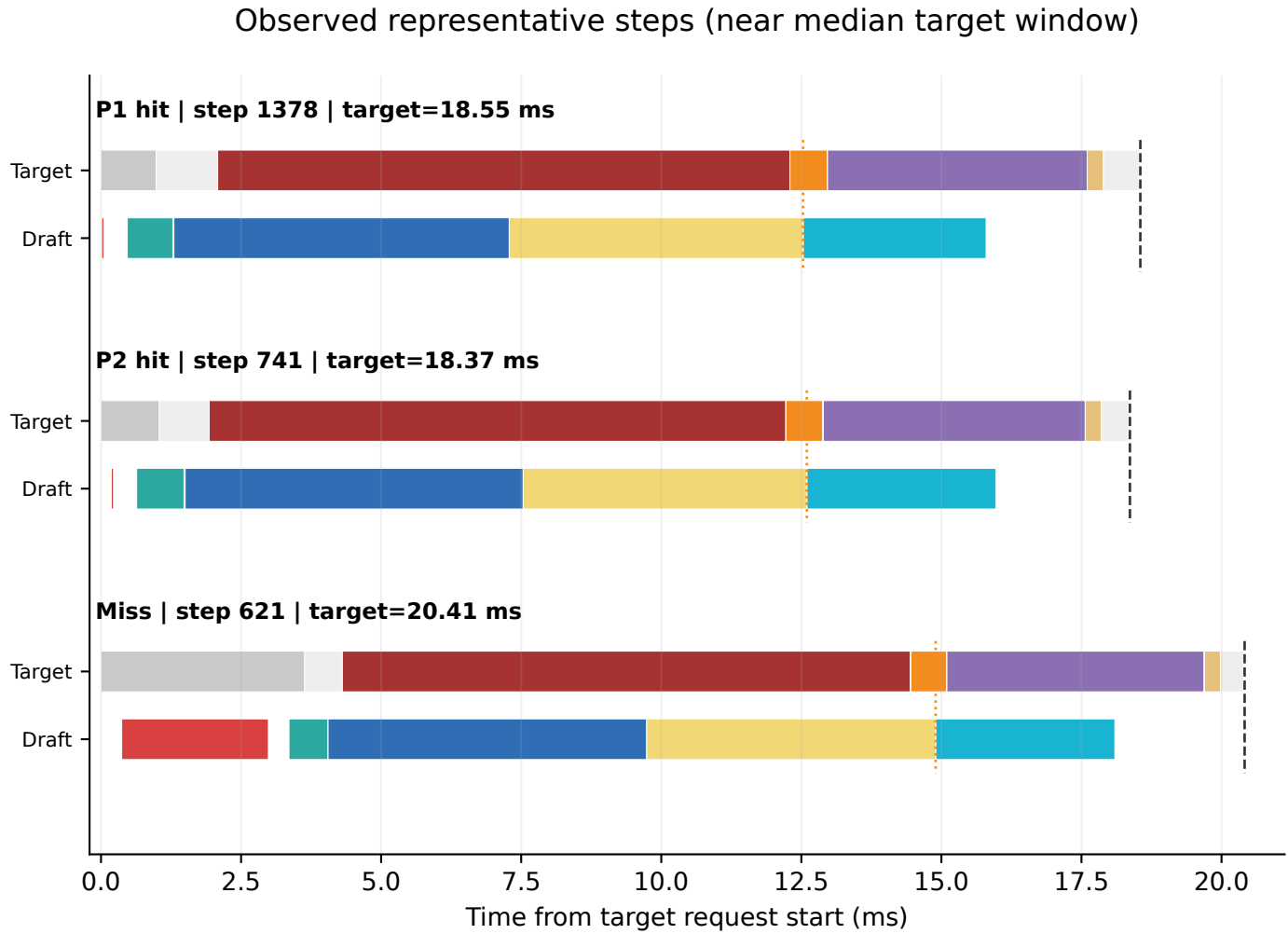
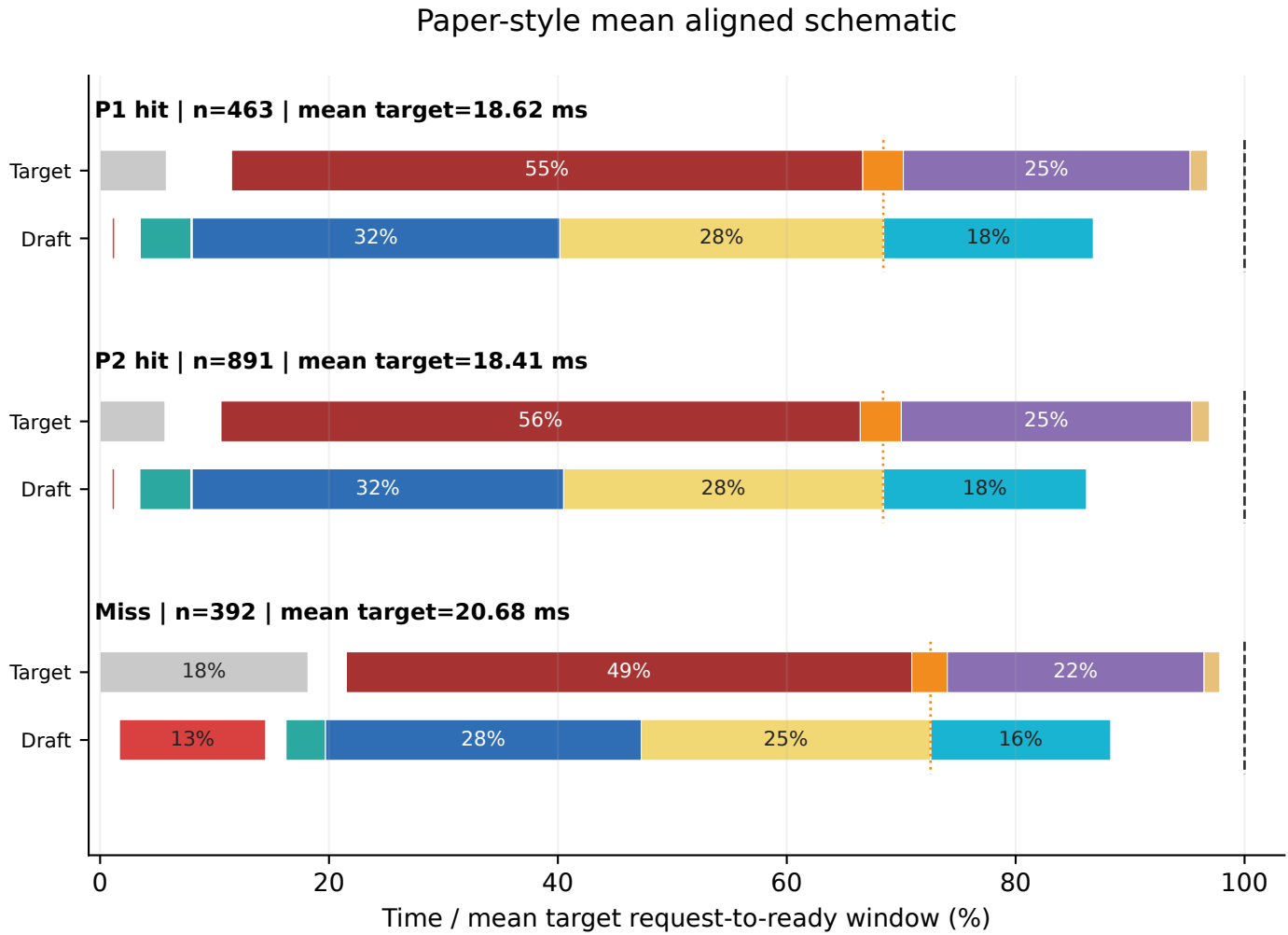
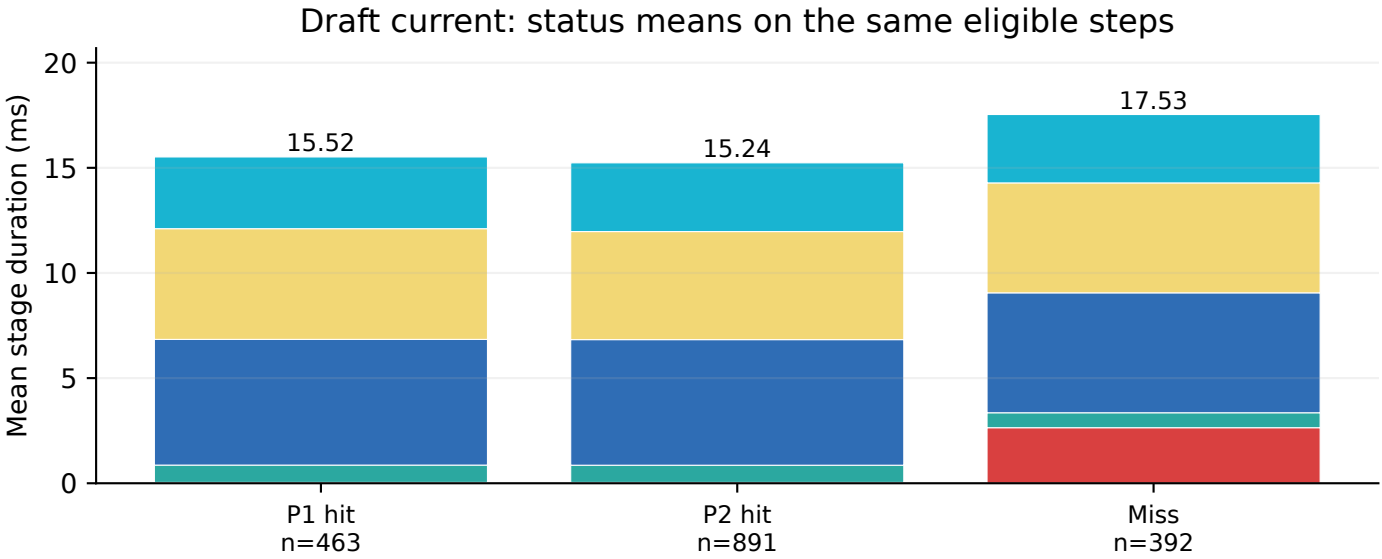
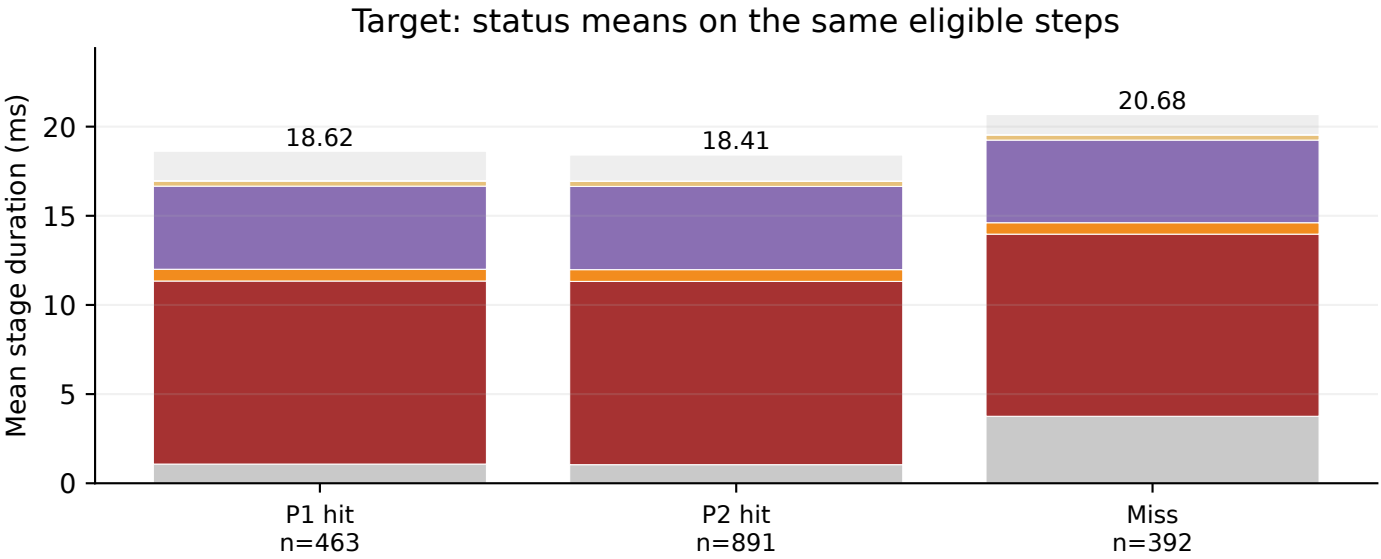
llama2_b1_combo_k1_7
DUET: K1/K2=7/4, exit=21 (0-based), $\bar{C}=3$, N1/N2=12/8, U=2, W=8, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1728 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

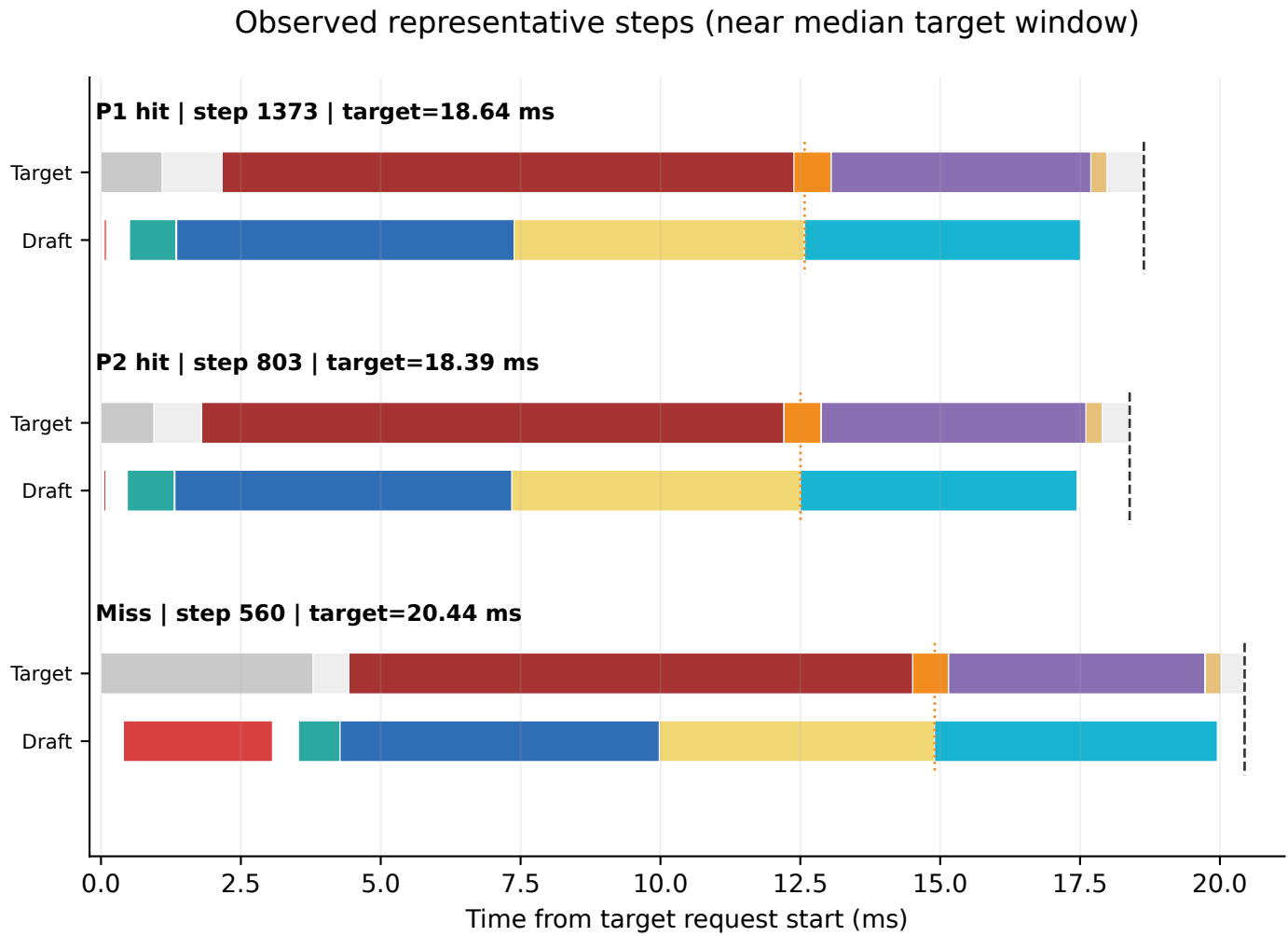
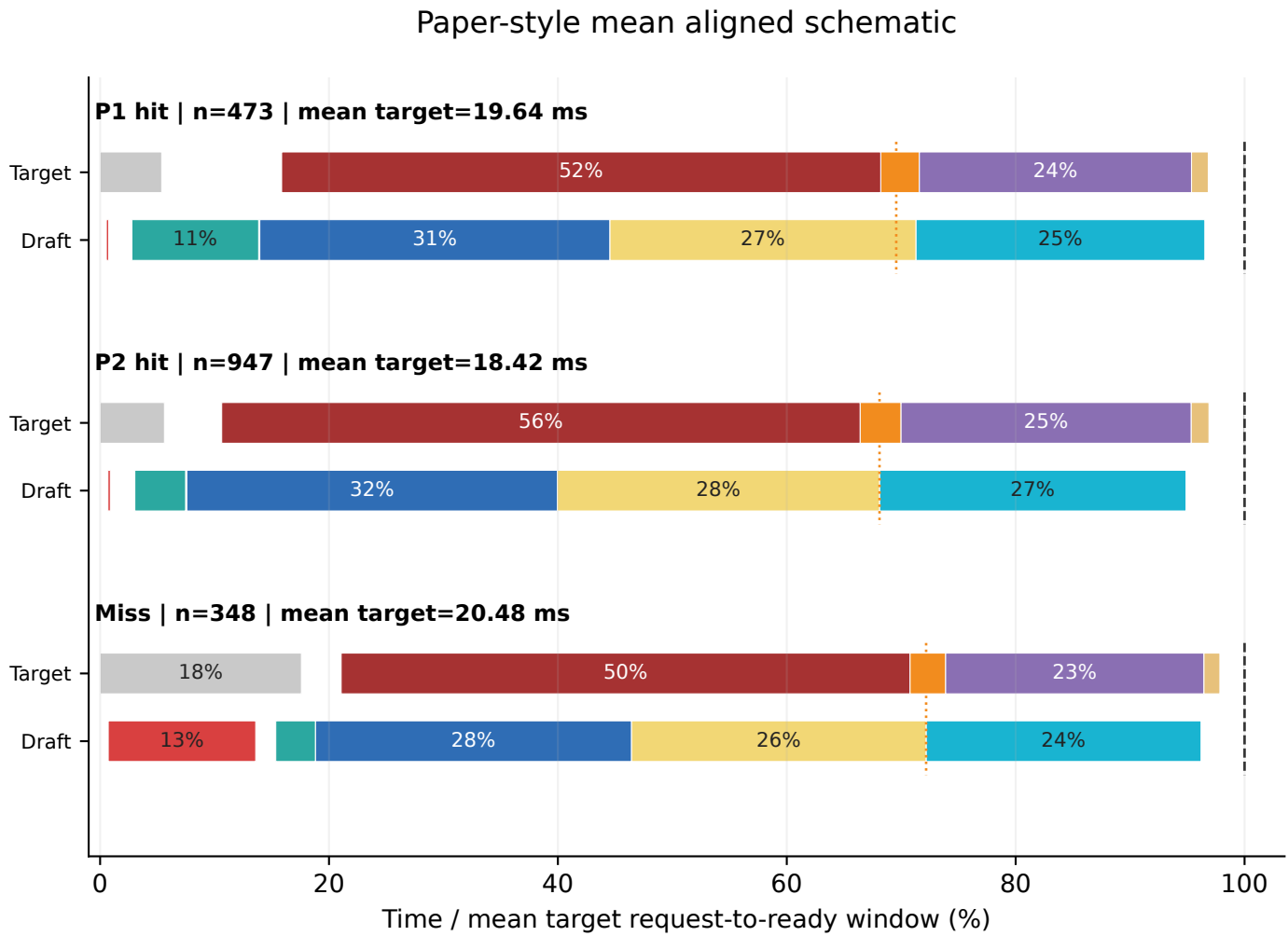
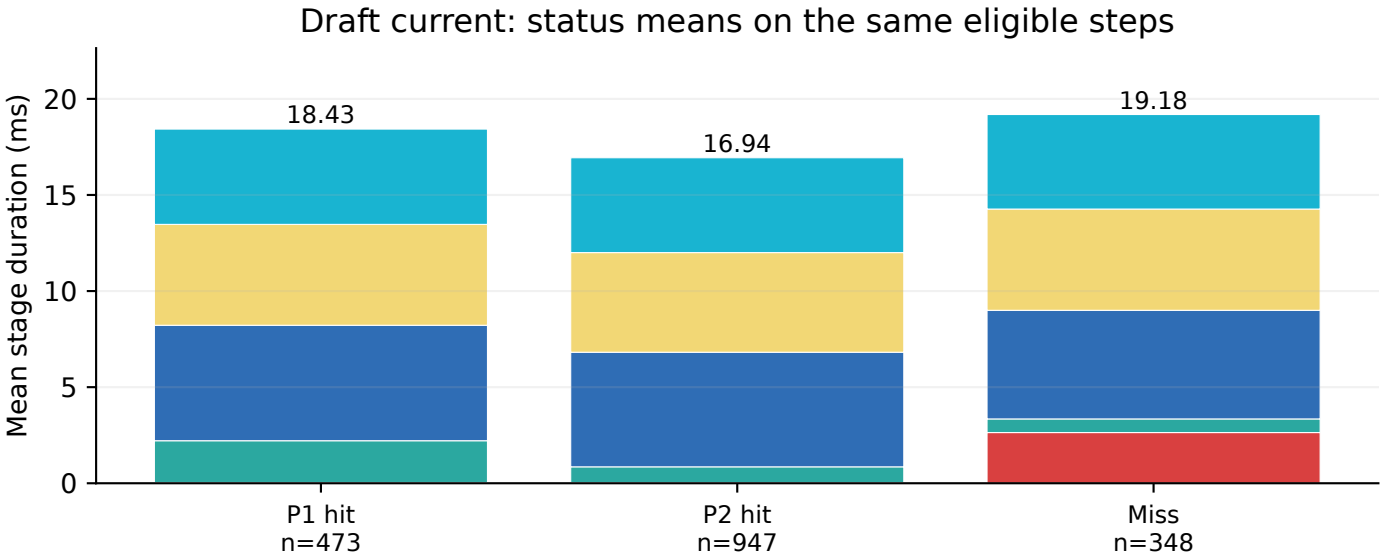
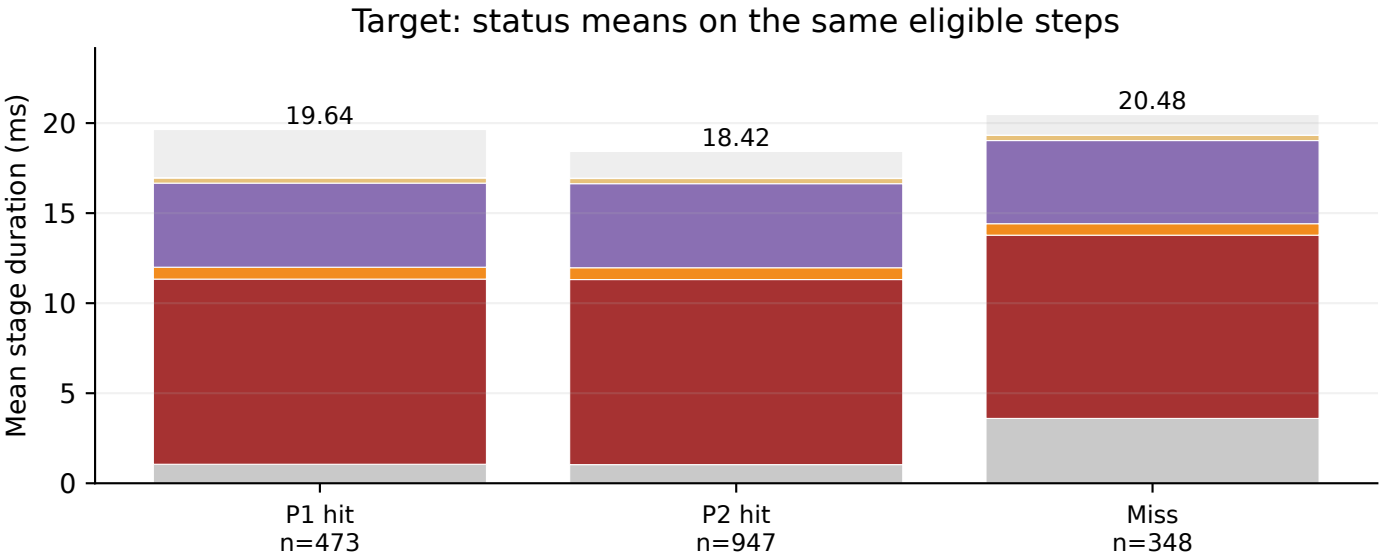
llama2_b1_combo_k2_3
DUET: K1/K2=6/3, exit=21 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1746 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

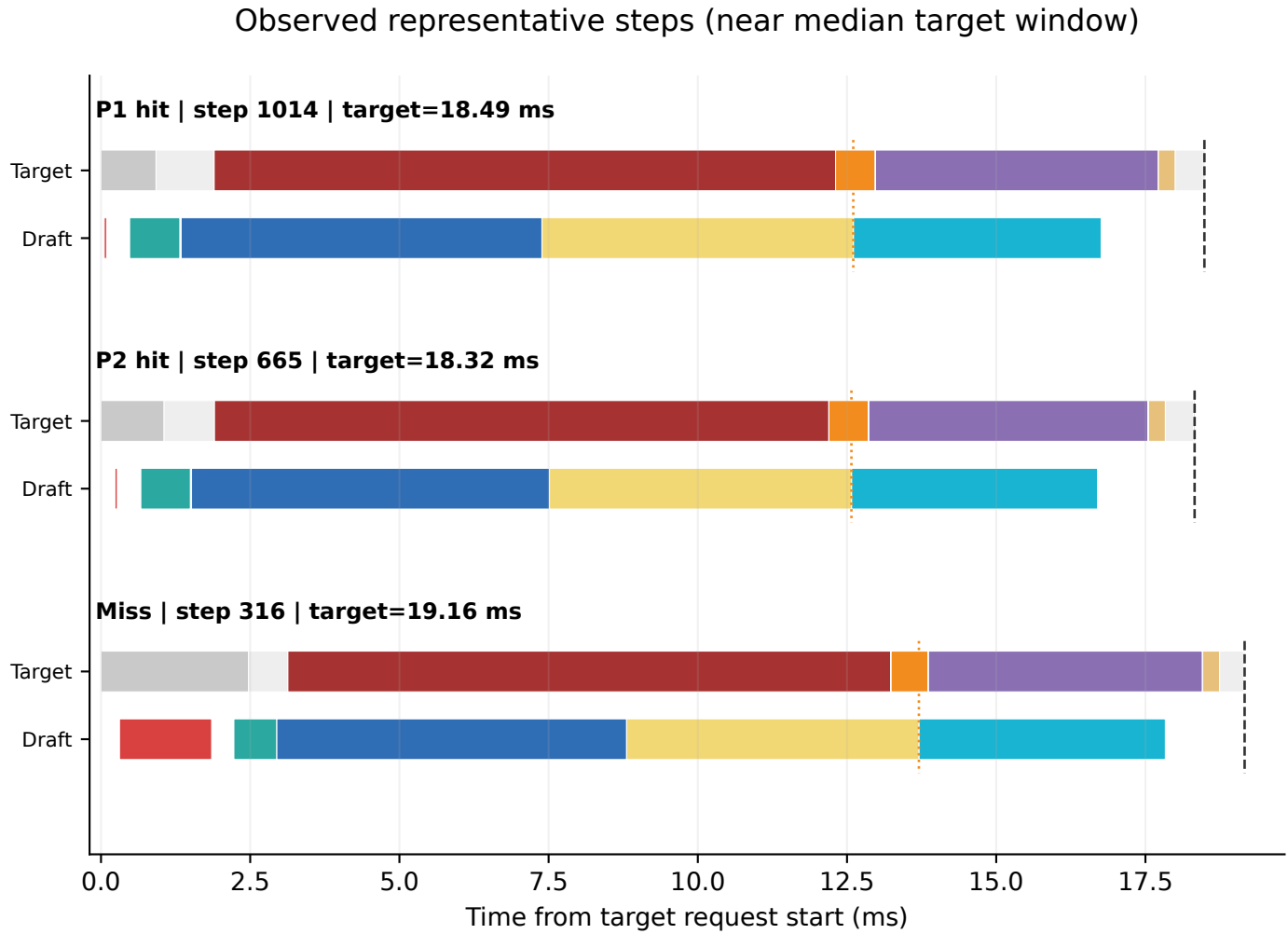
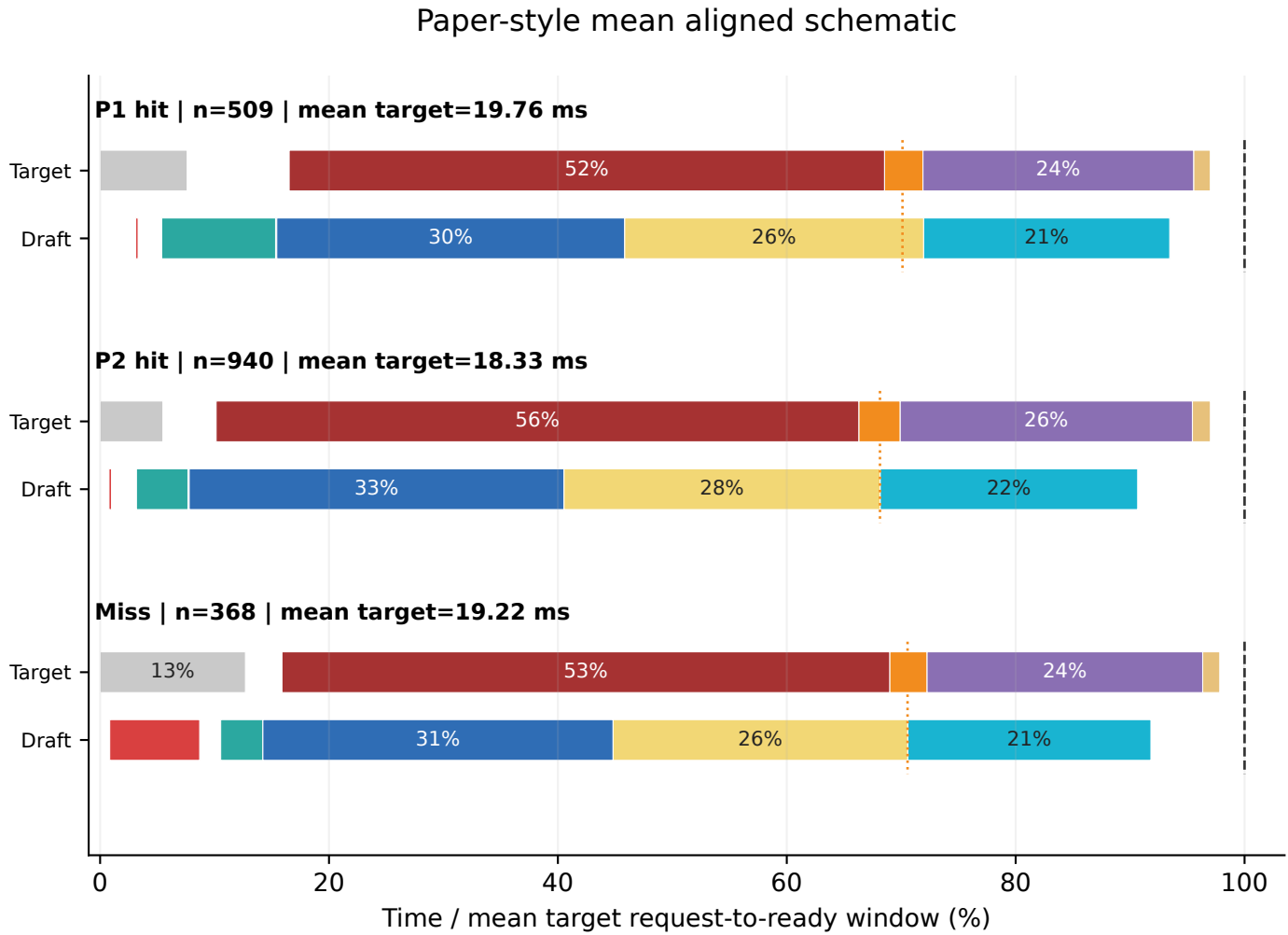
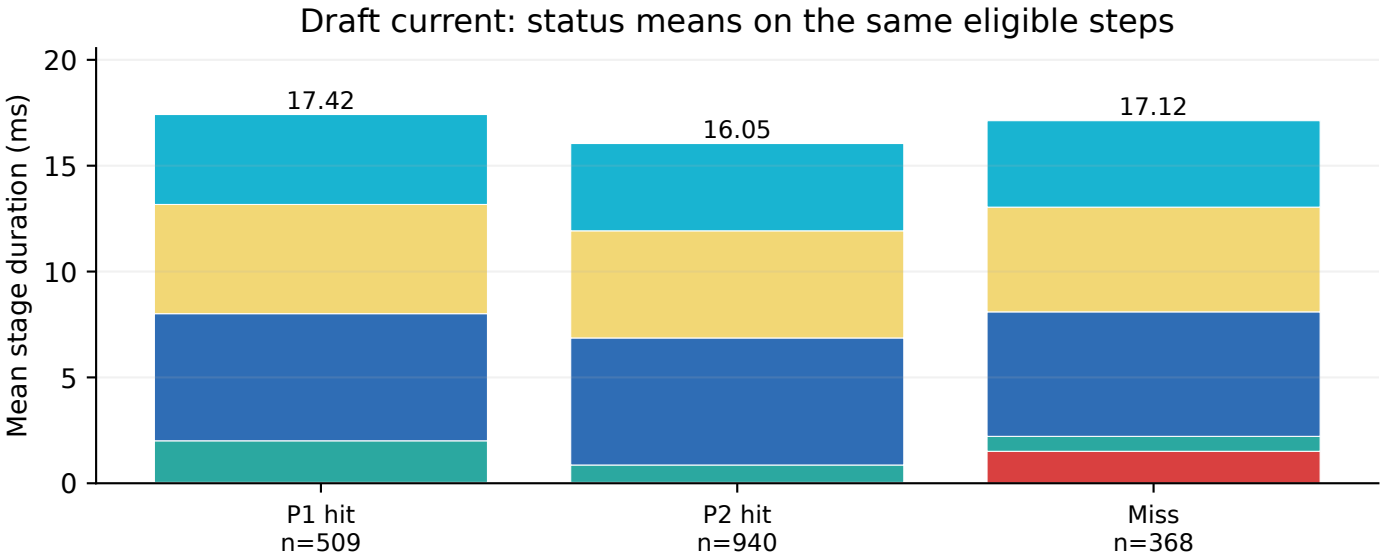
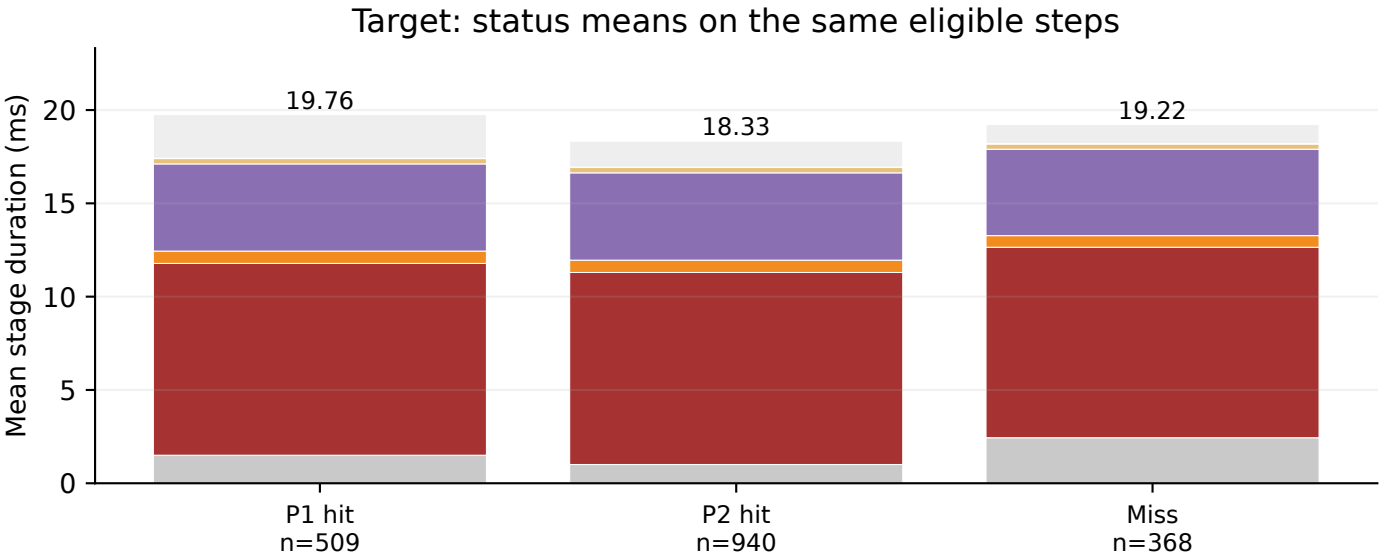
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_combo_k2_5
DUET: K1/K2=6/5, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1768 eligible steps



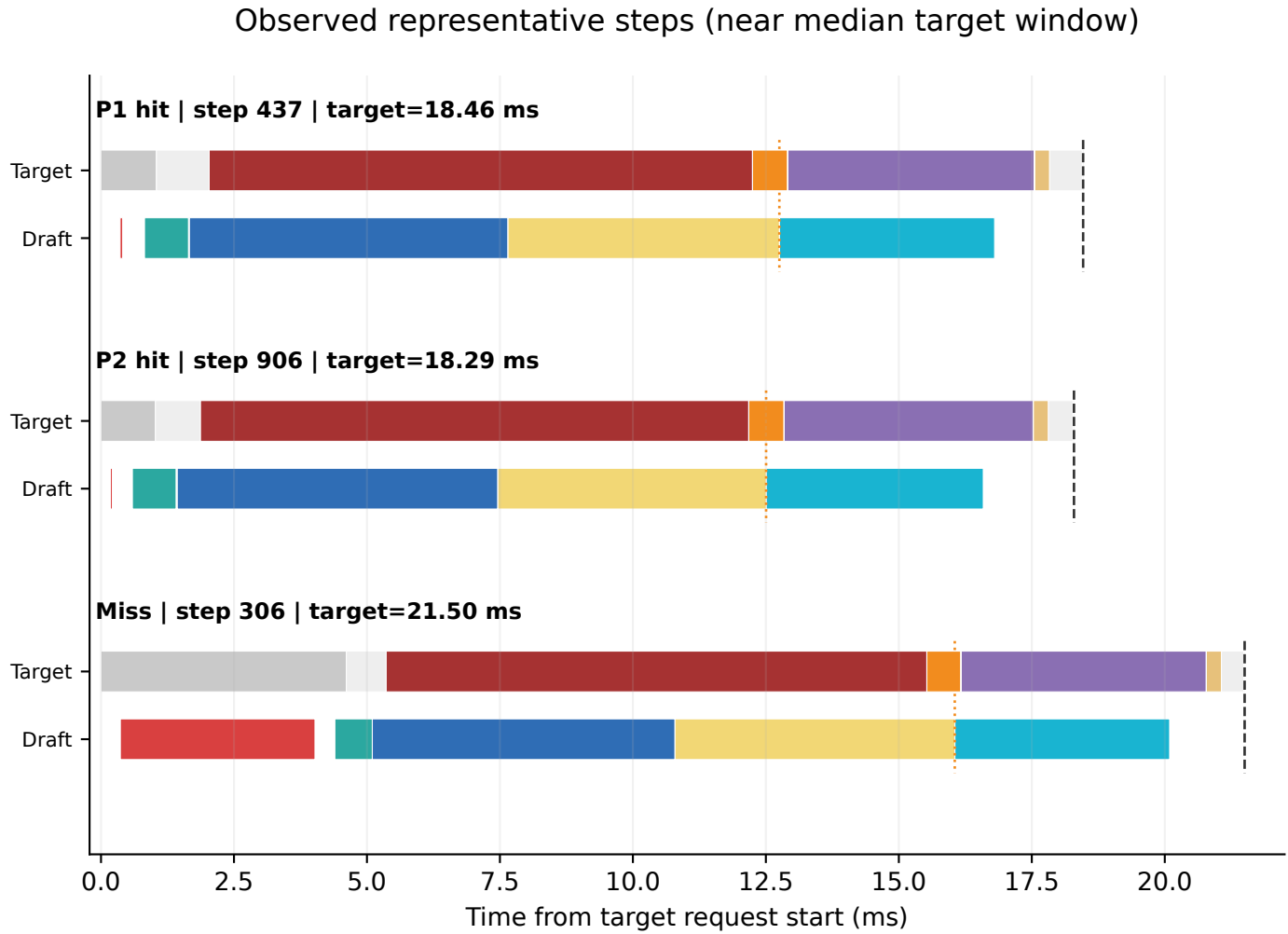
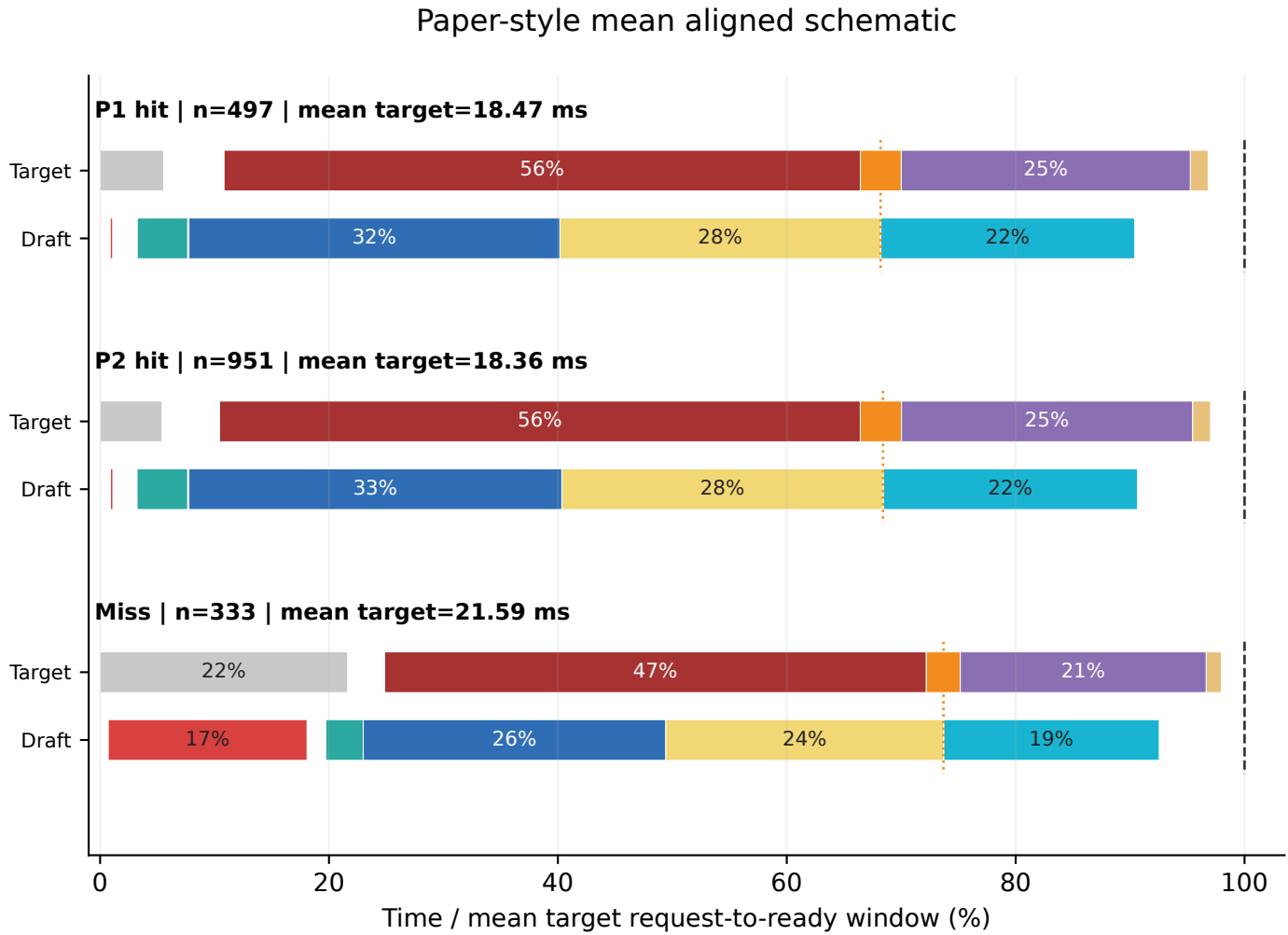
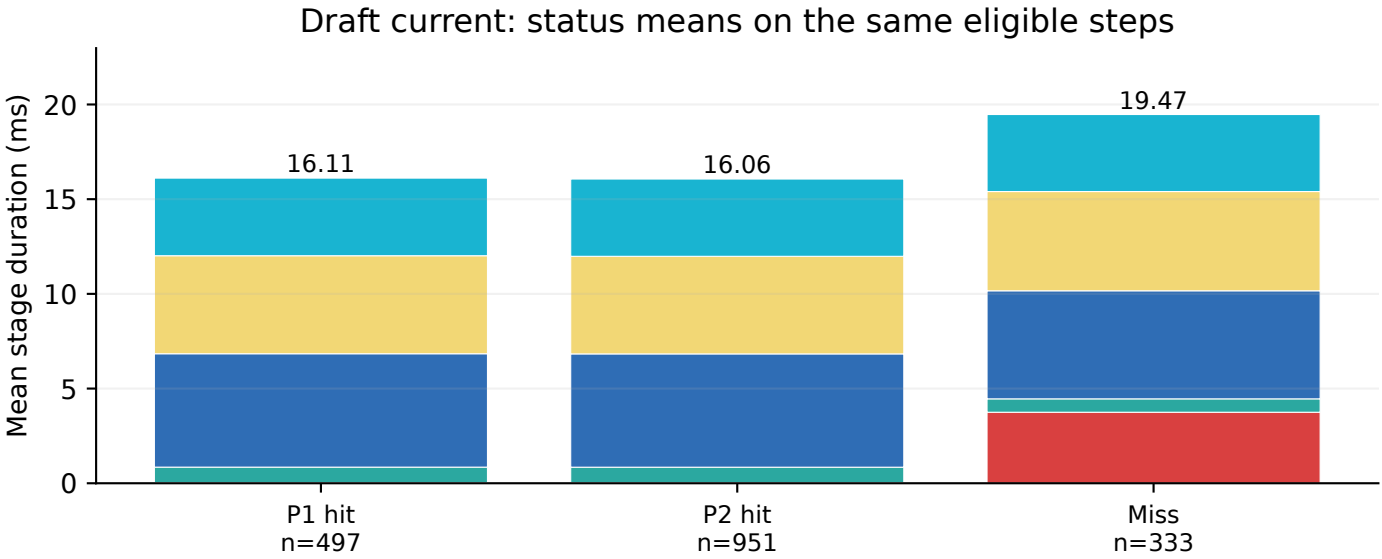
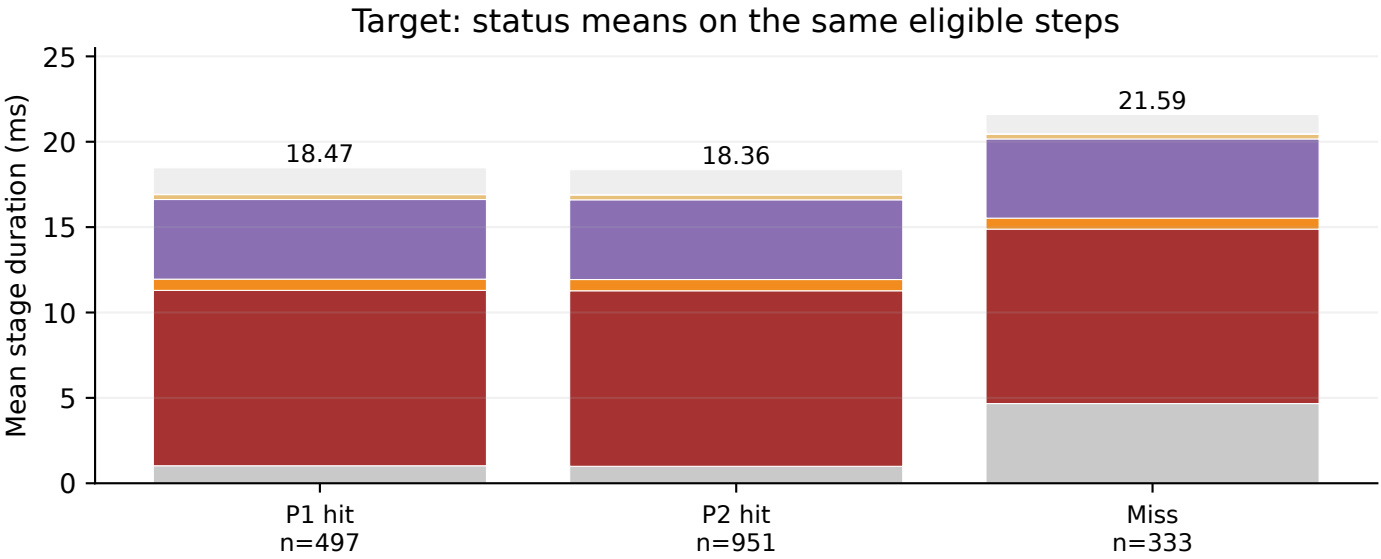
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_combo_miss2
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1817 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

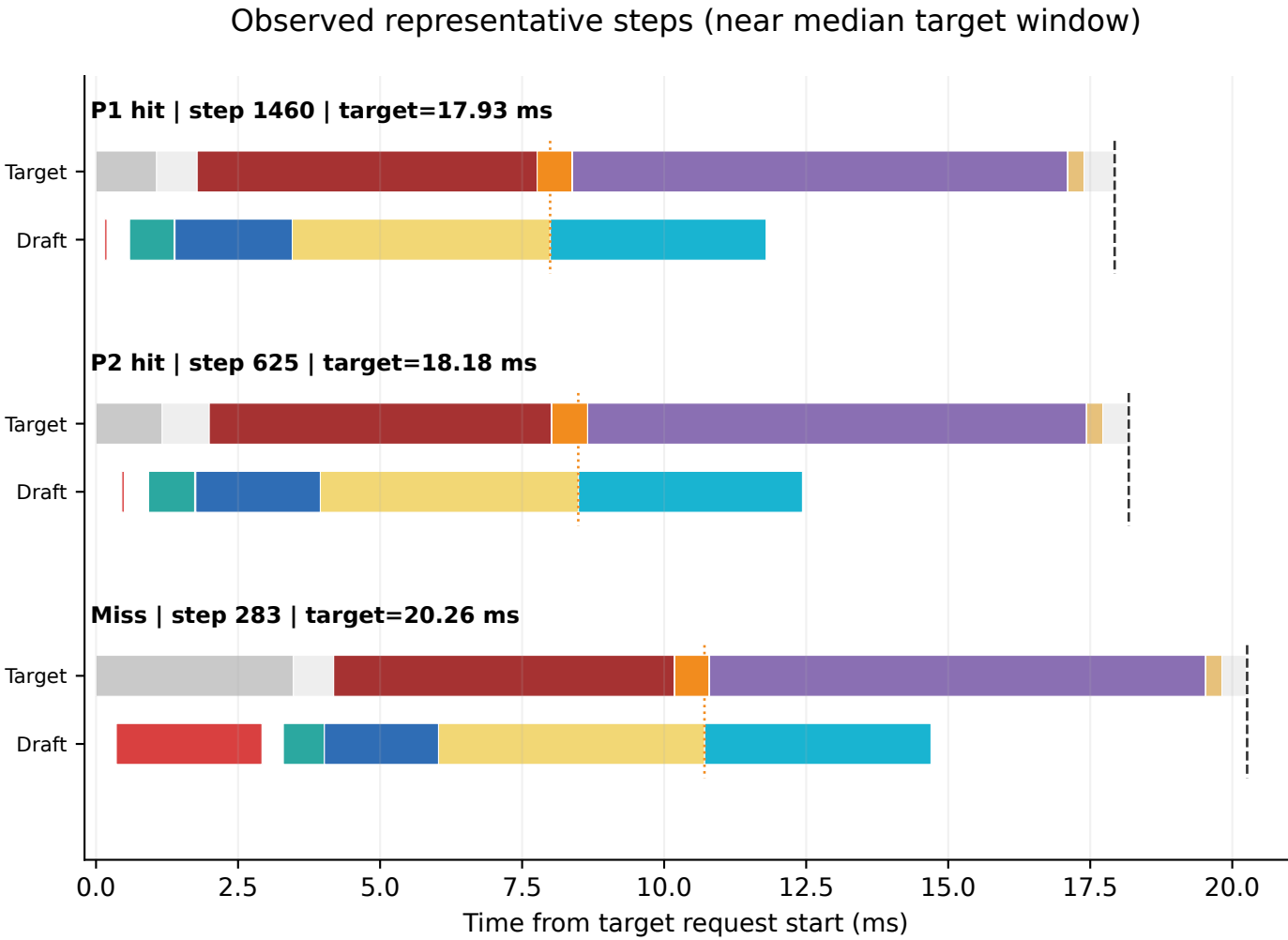
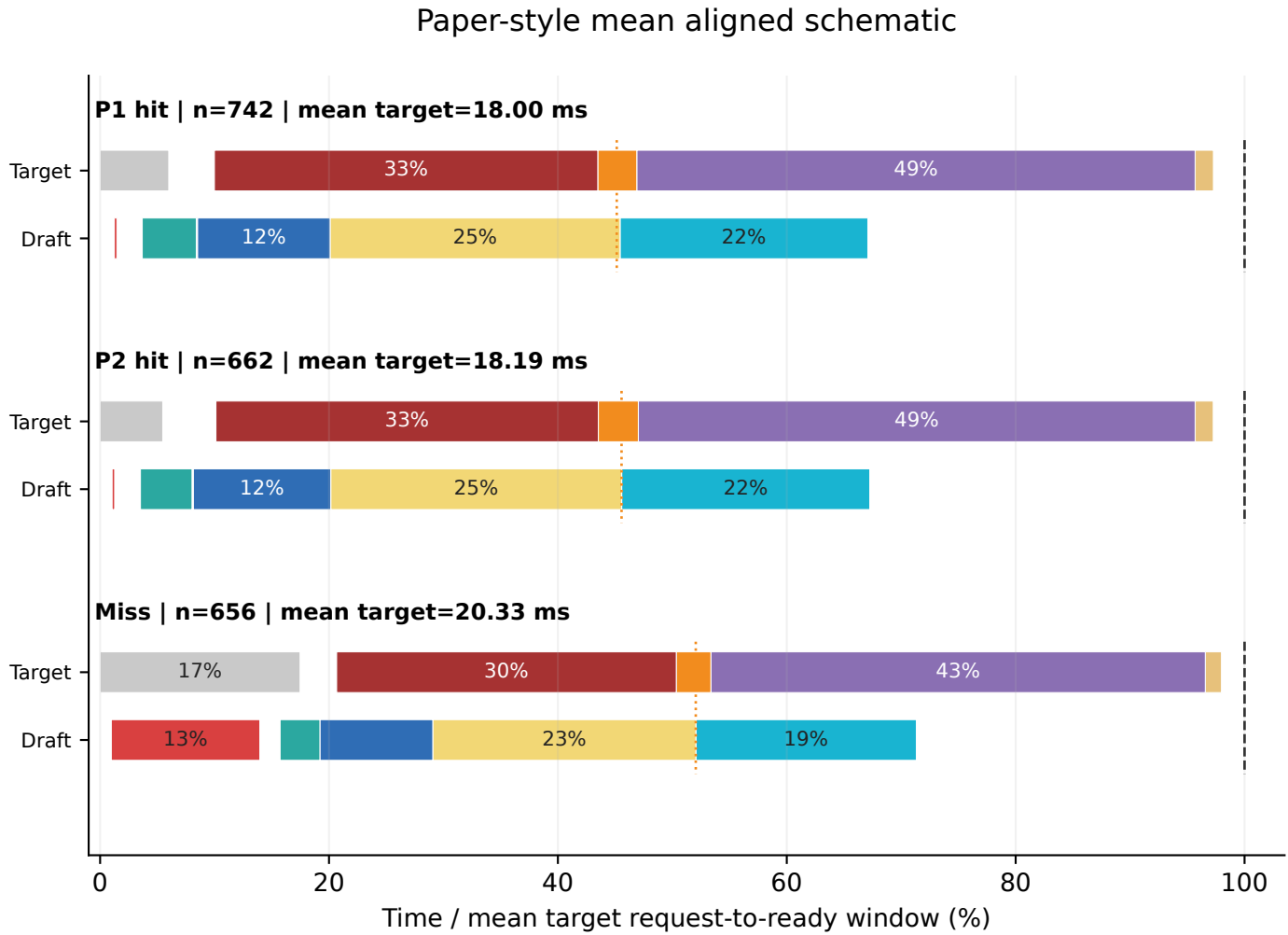
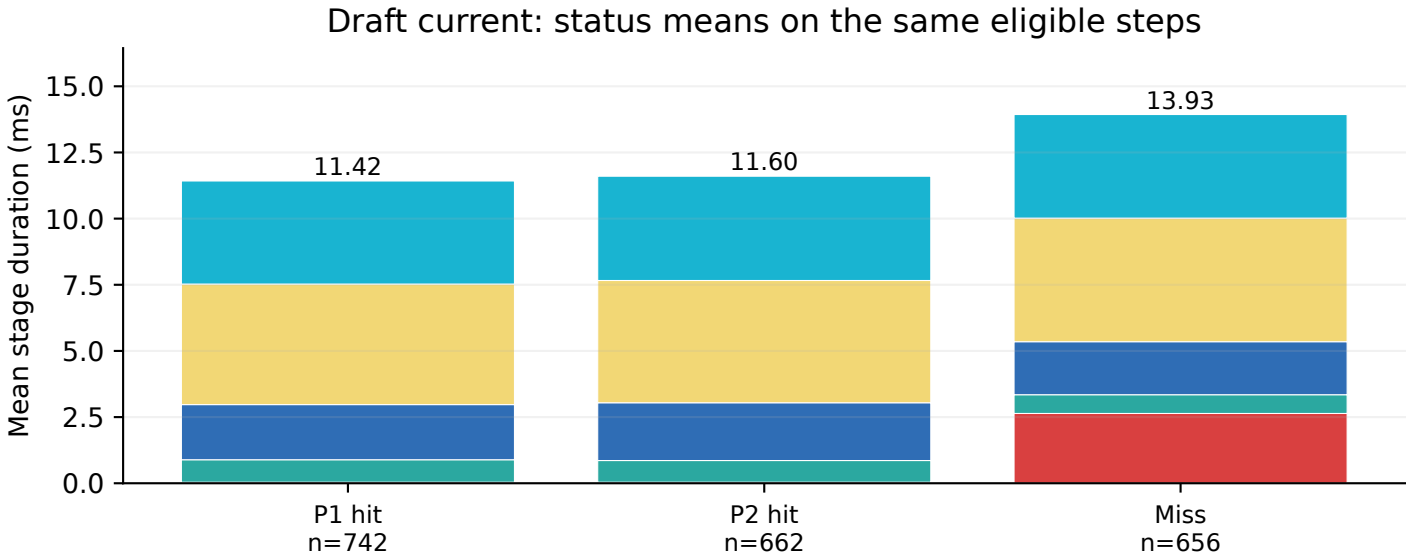
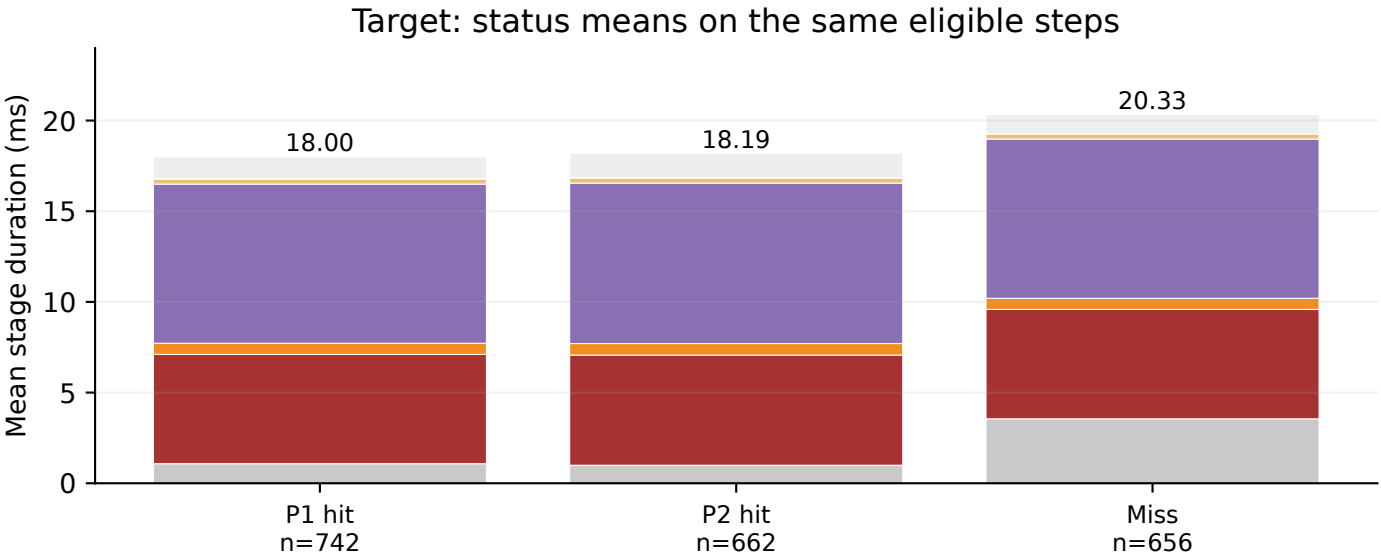
llama2_b1_combo_miss6
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1781 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

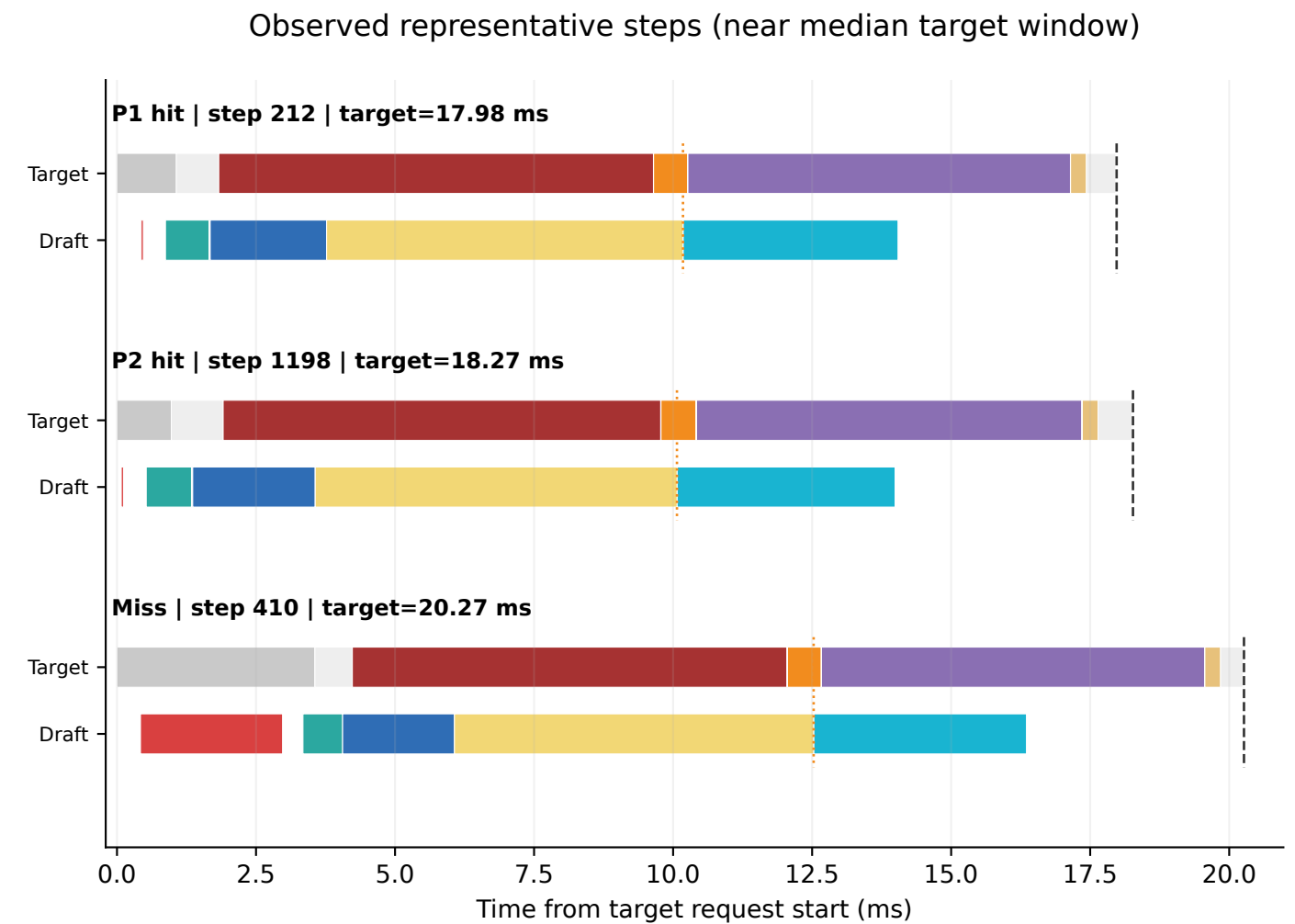
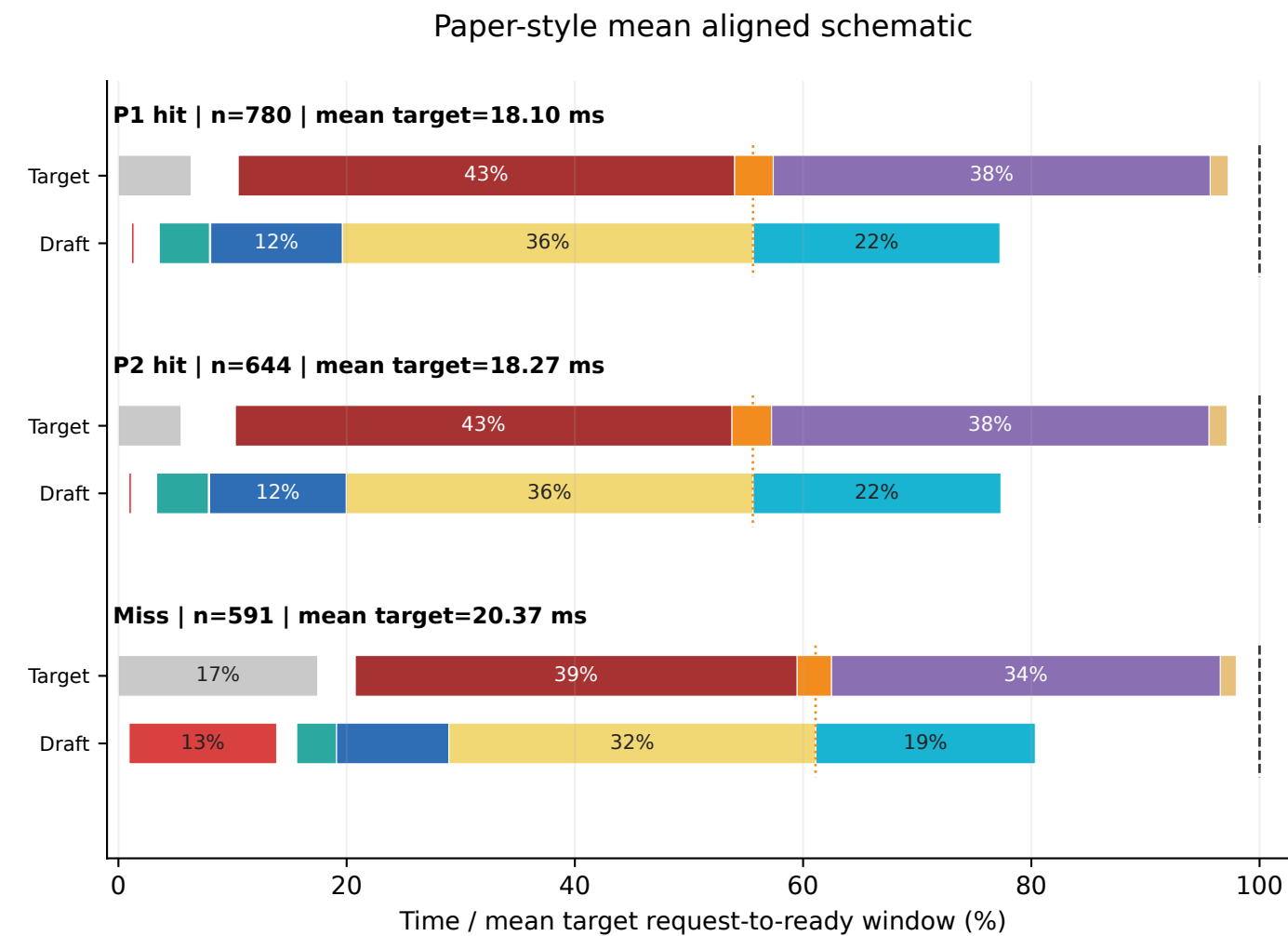
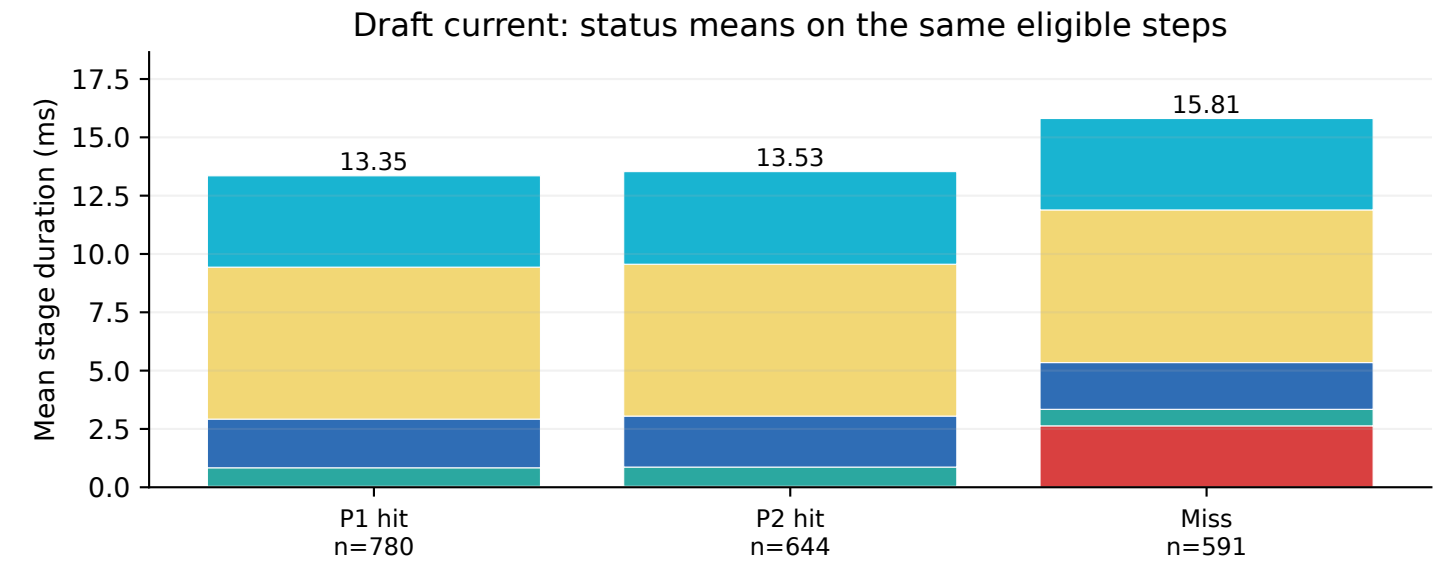
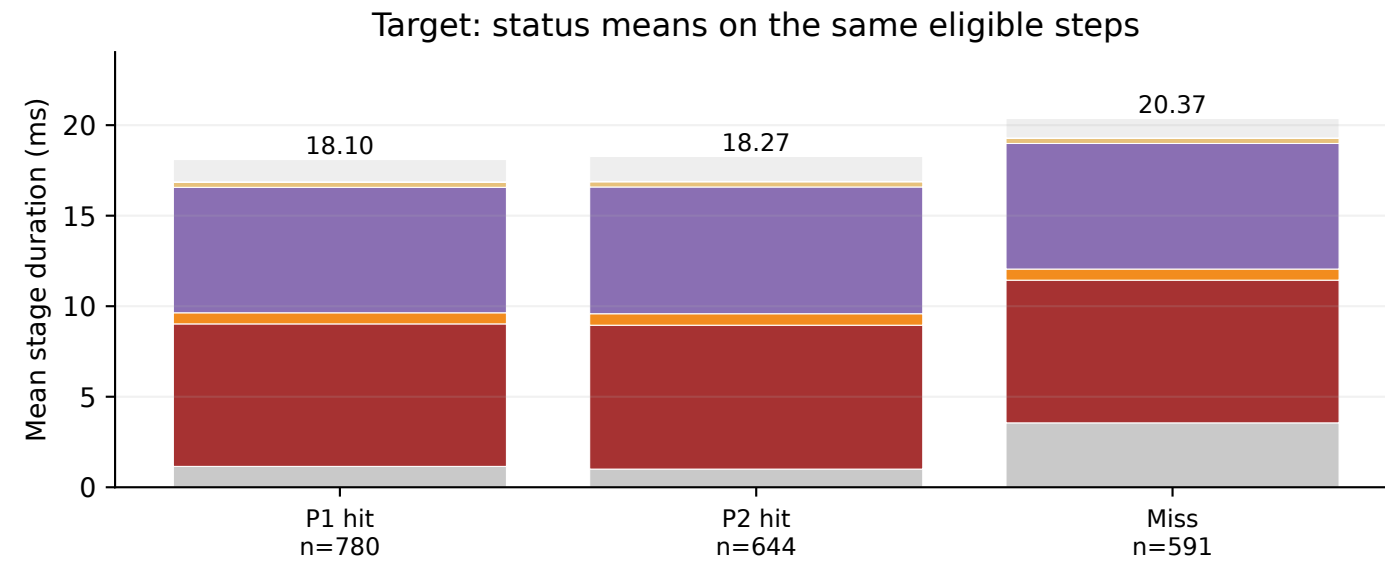
llama2_b1_combo_transfer_2_4_e12
DUET: K1/K2=2/4, exit=12 (0-based), C=3, N1/N2=4/8, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 2060 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

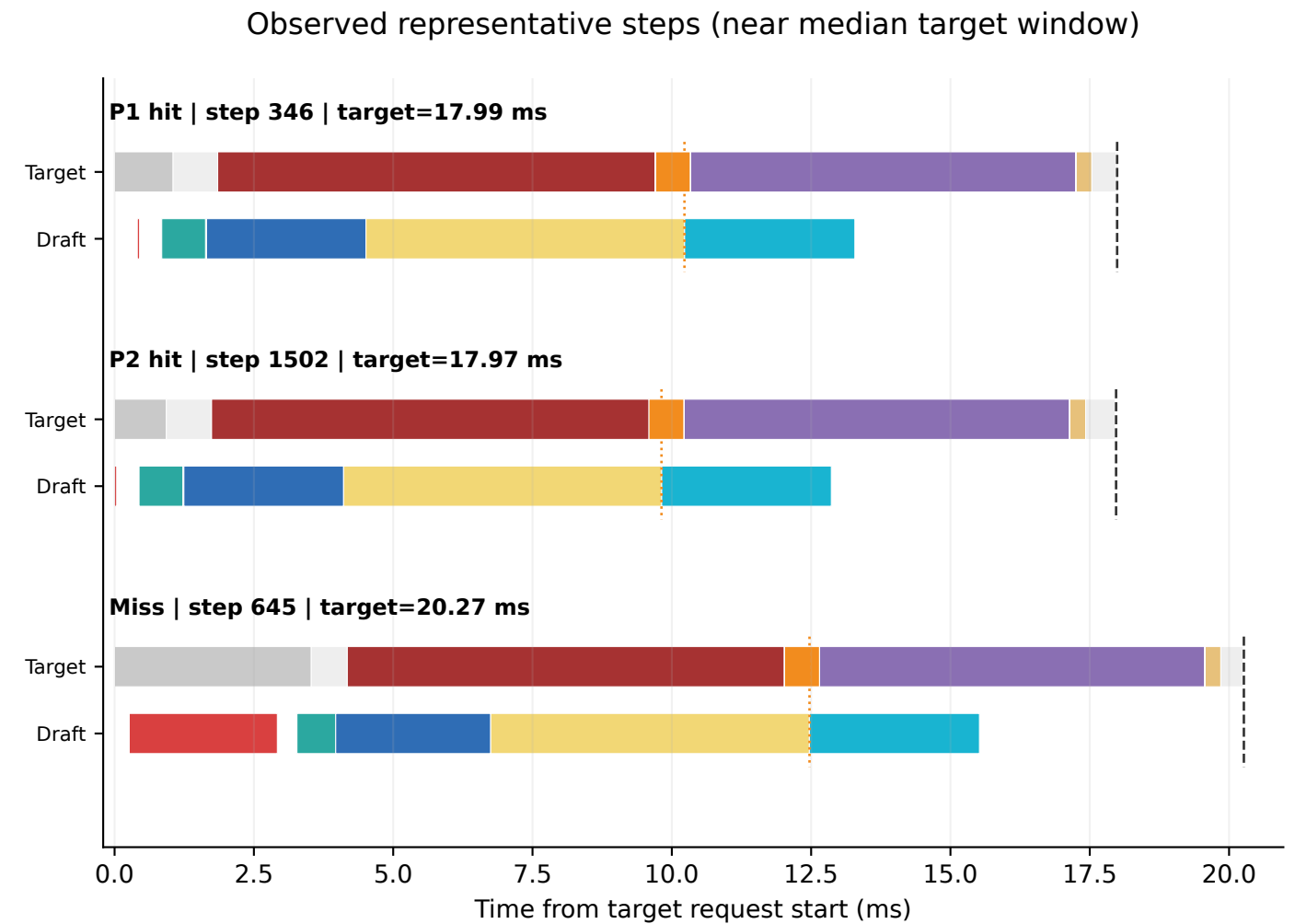
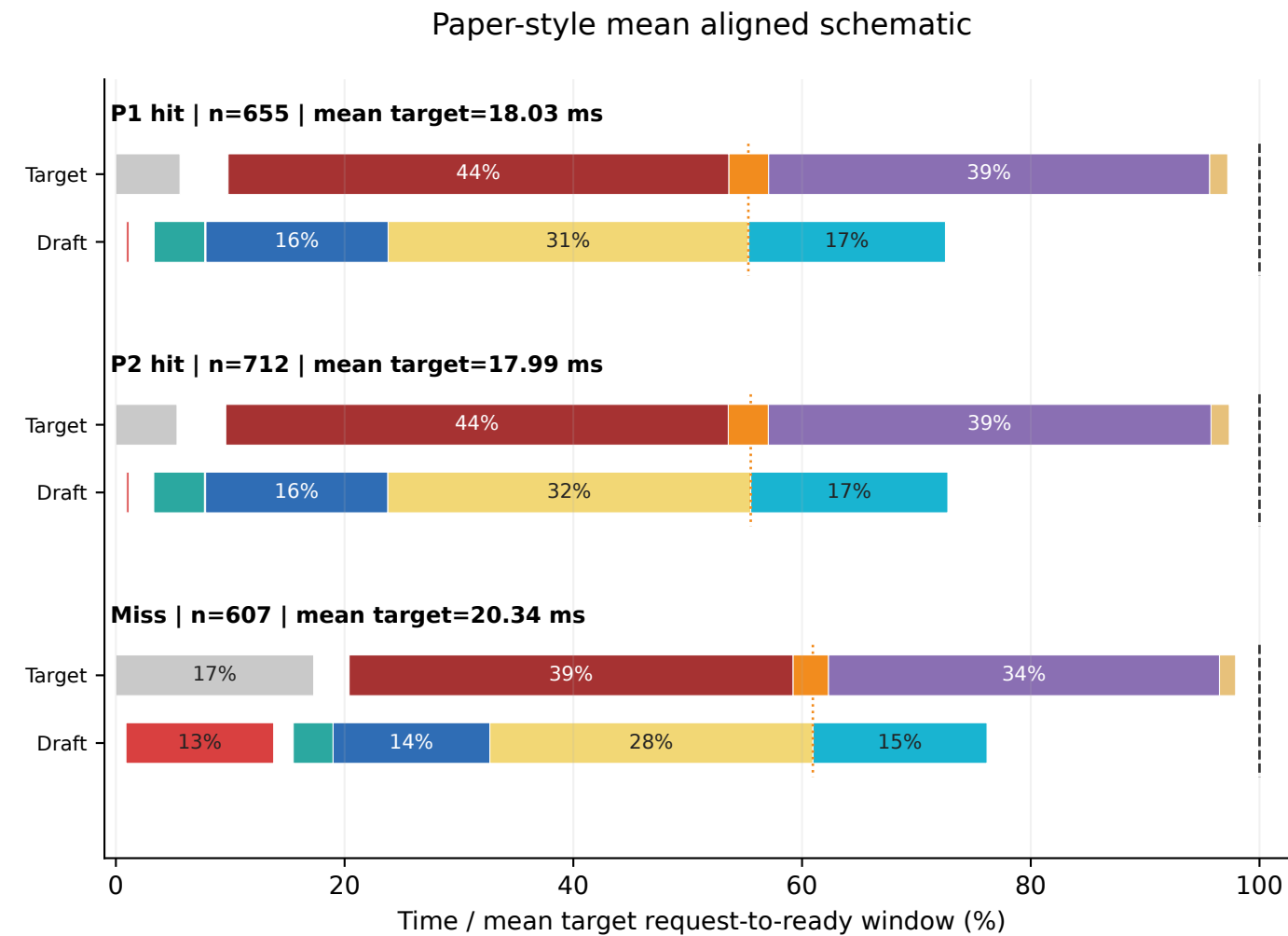
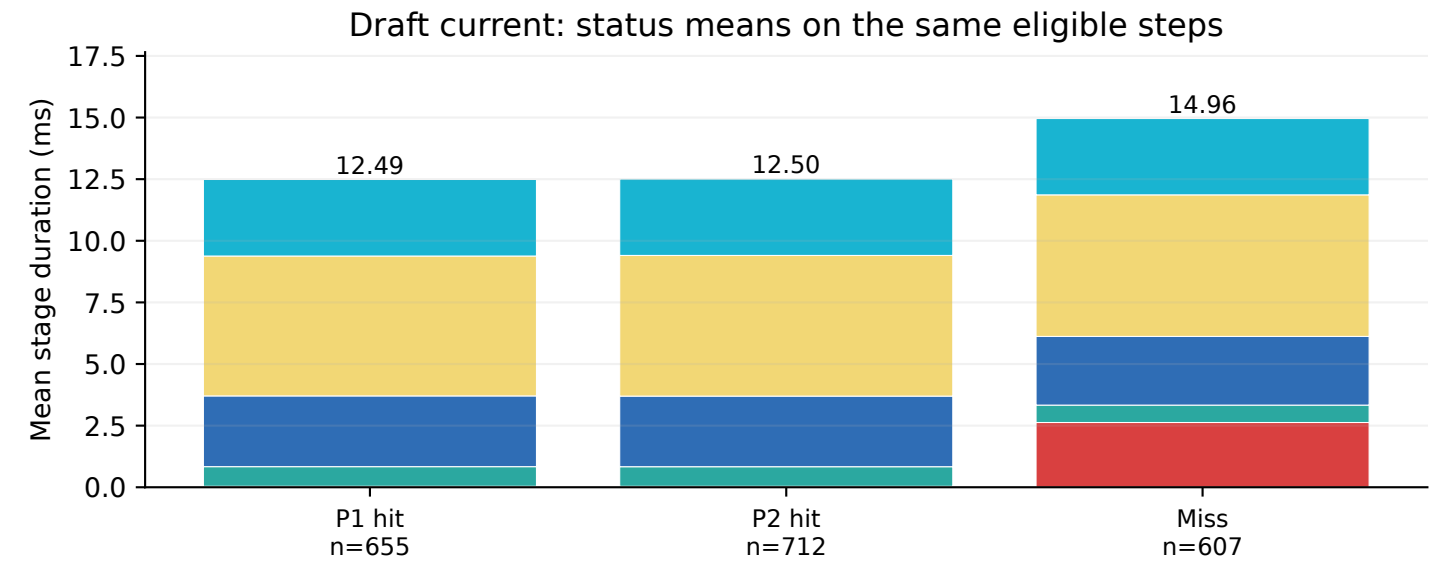
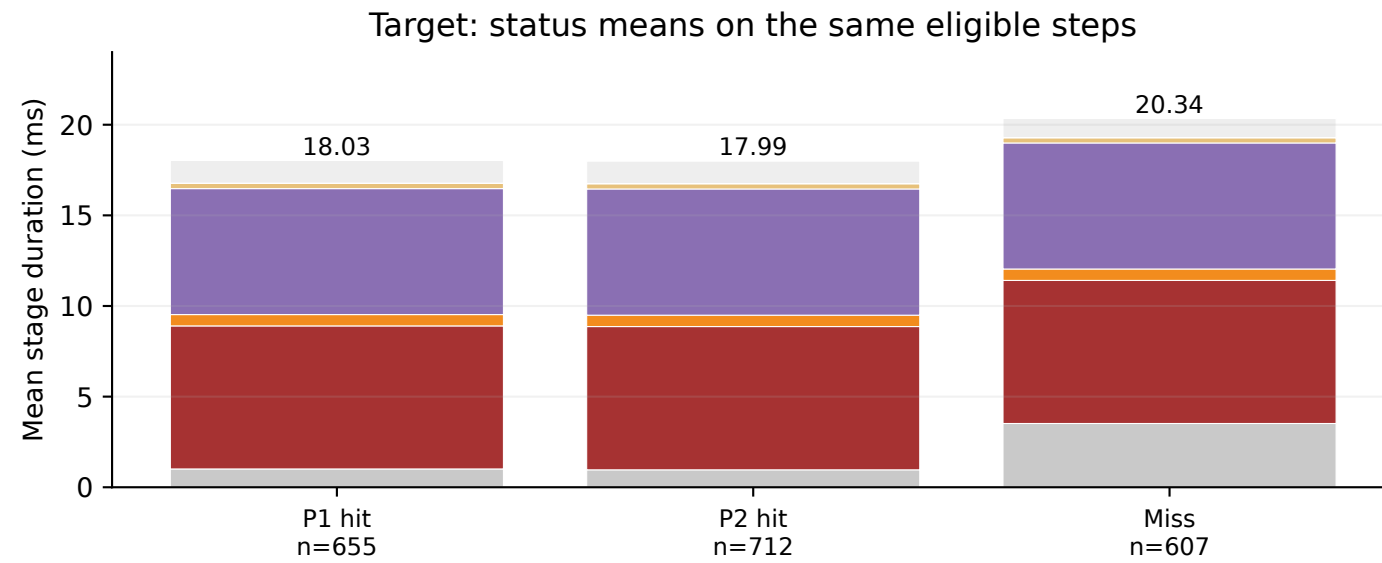
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_combo_transfer_2_4_e16
DUET: K1/K2=2/4, exit=16 (0-based), C=3, N1/N2=4/8, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 2015 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

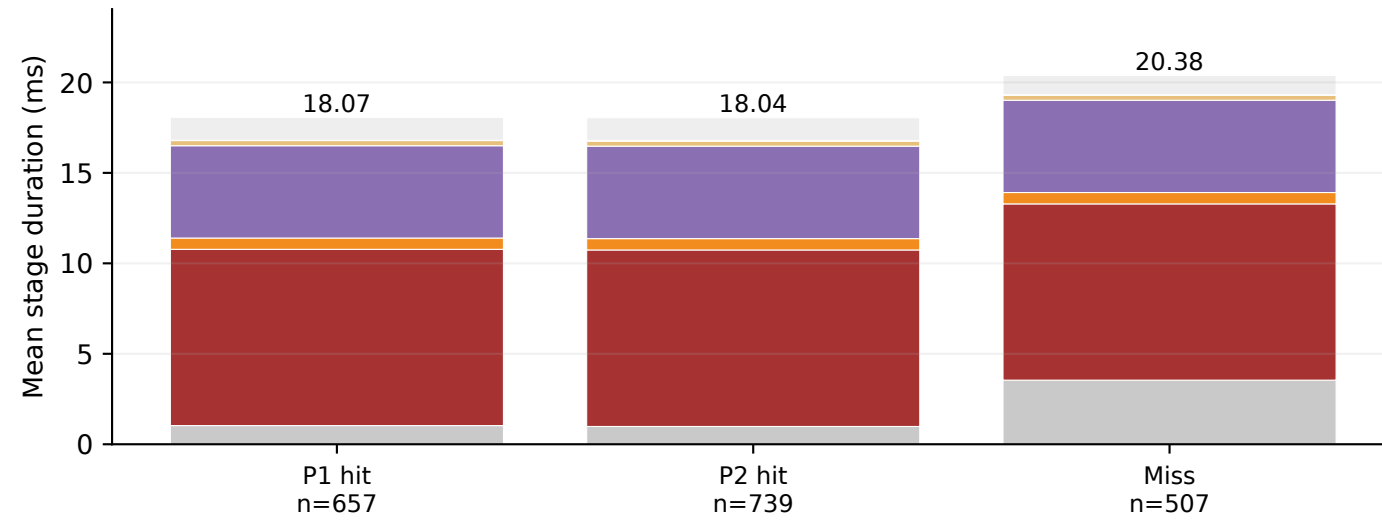
llama2_b1_combo_transfer_3_3_e16
DUET: K1/K2=3/3, exit=16 (0-based), C=3, N1/N2=6/6, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1974 eligible steps



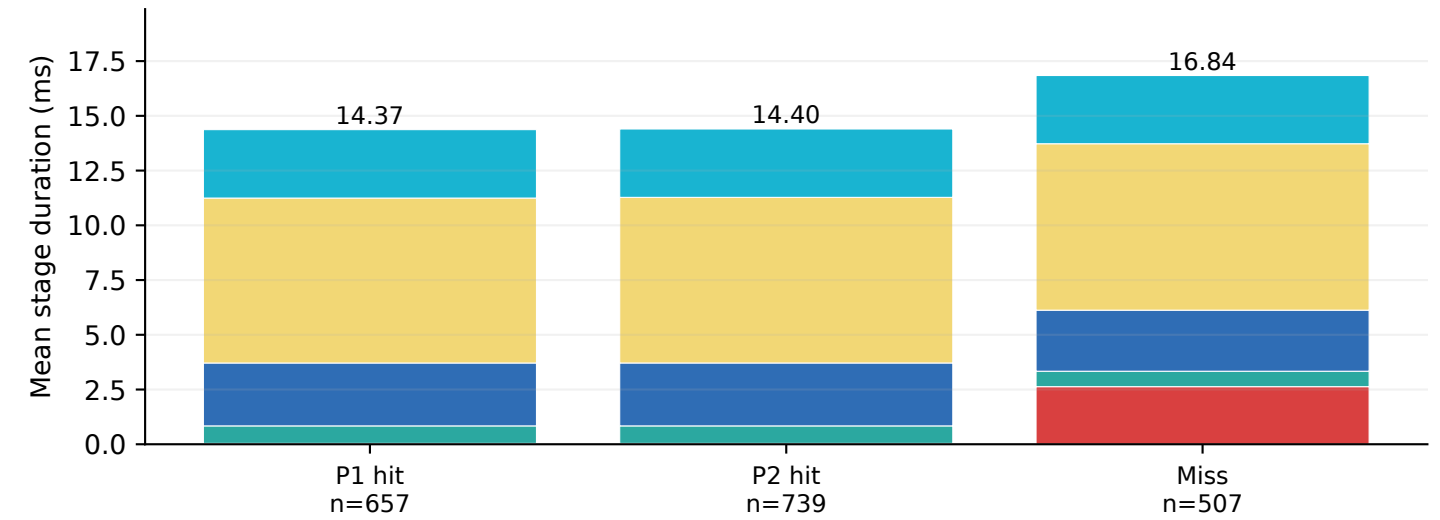
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_combo_transfer_3_3_e20
DUET: K1/K2=3/3, exit=20 (0-based), C=3, N1/N2=6/6, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1903 eligible steps

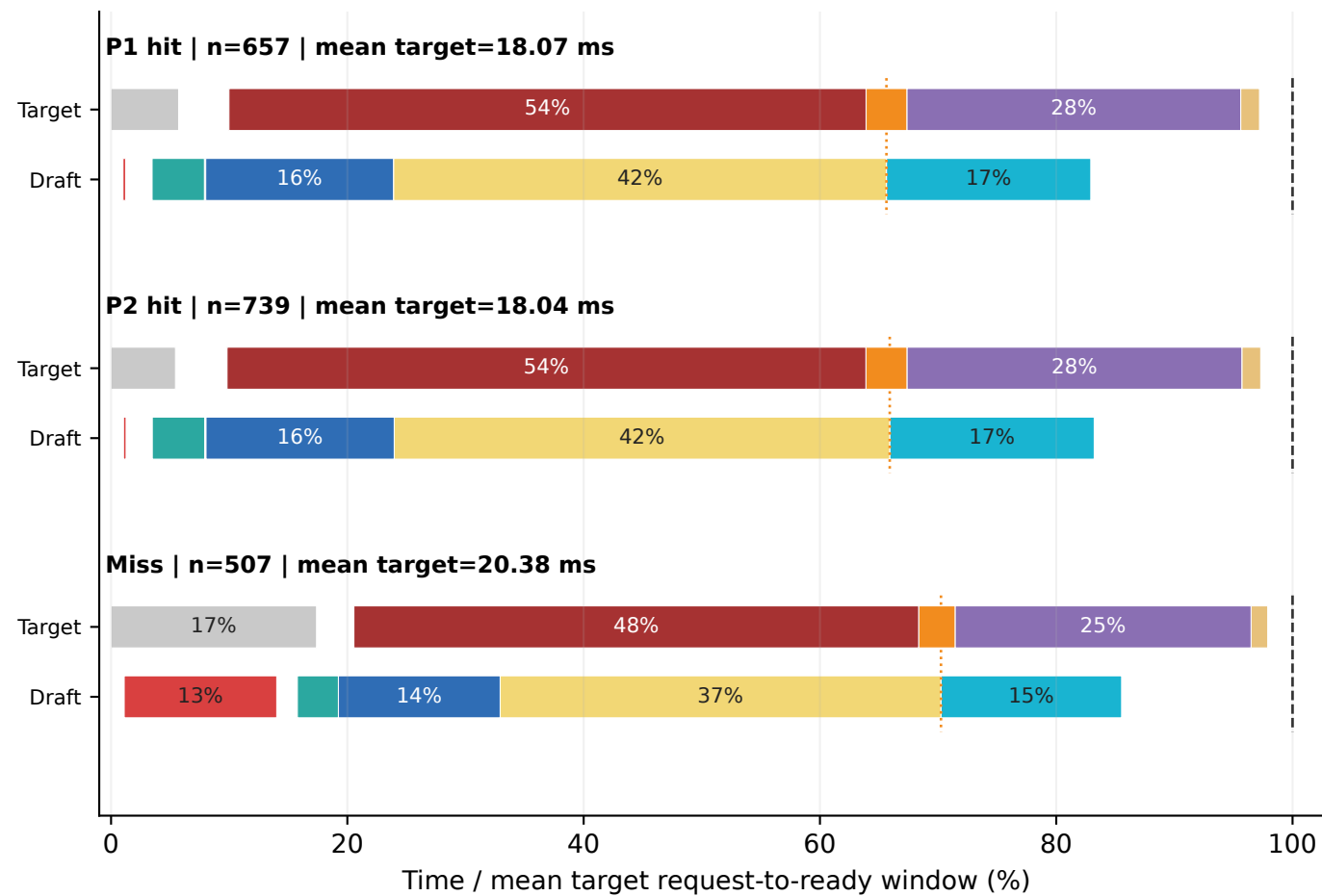
Target: status means on the same eligible steps



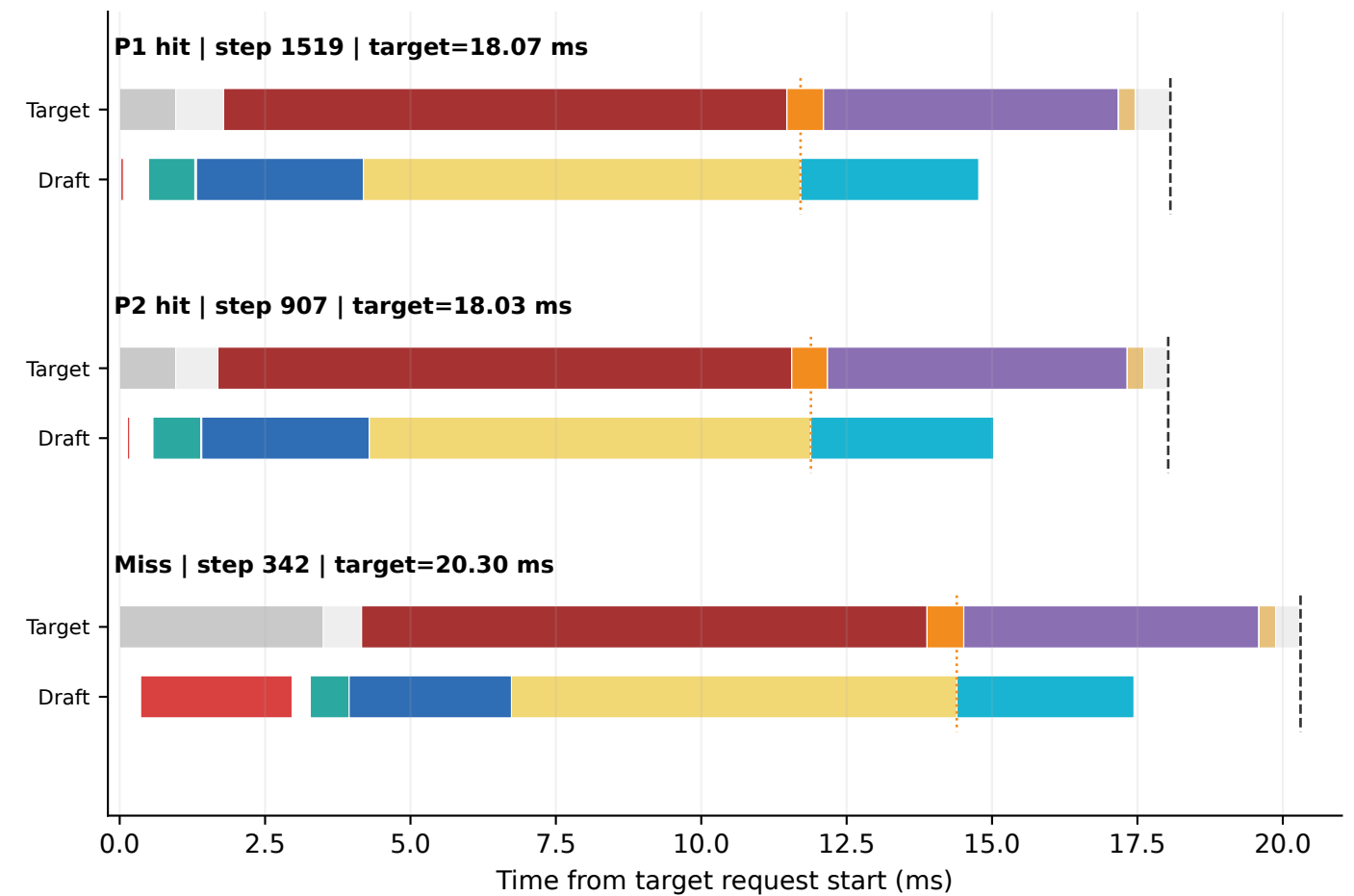
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



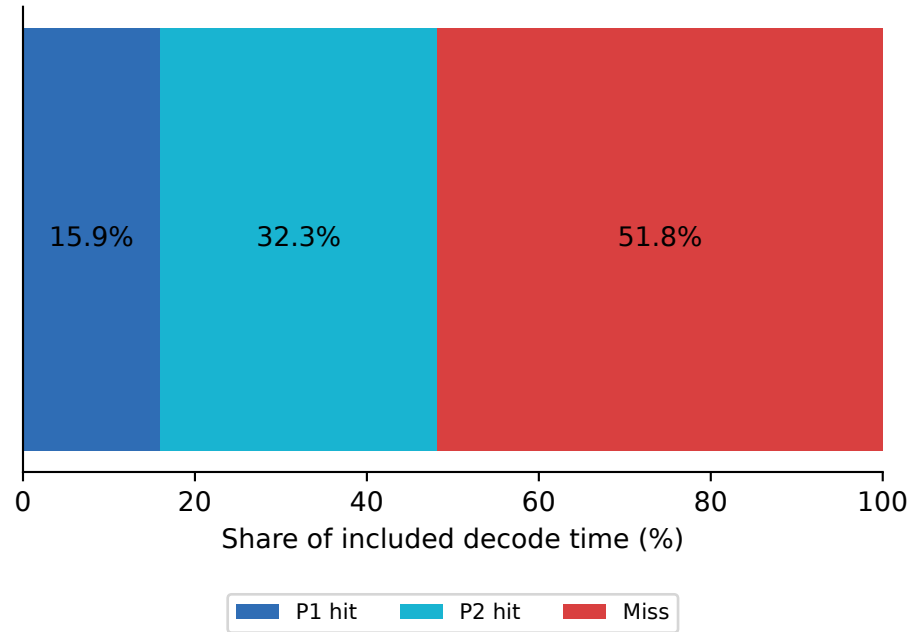
Observed representative steps (near median target window)



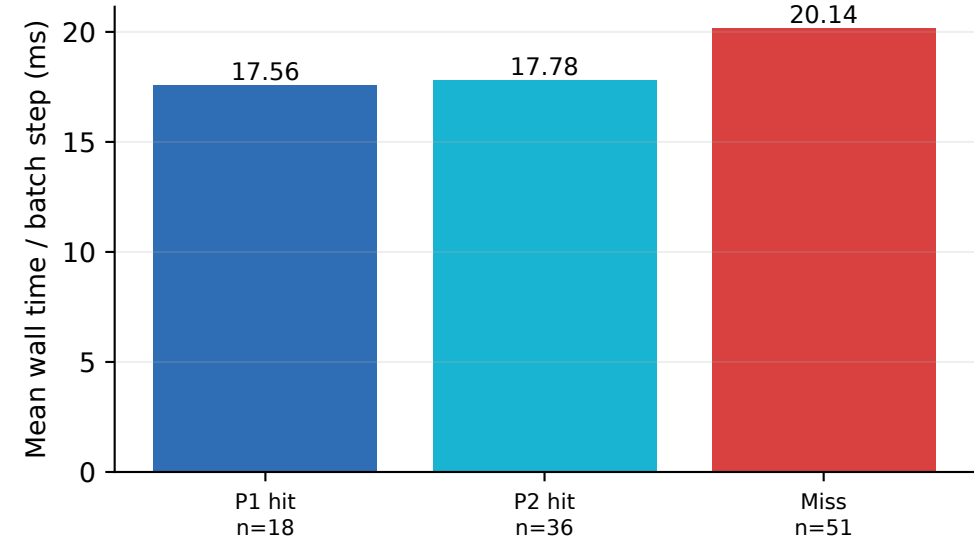
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_transfer_smoke
 DUET: K1/K2=1/3, exit=12 (0-based), C=3, N1/N2=2/6, U=2, W=3, trim/stream=0/0
 Last pass: B=1, T=0.7, N=8 | AL*=1.657, hit=51.3%, TPS*=87.74

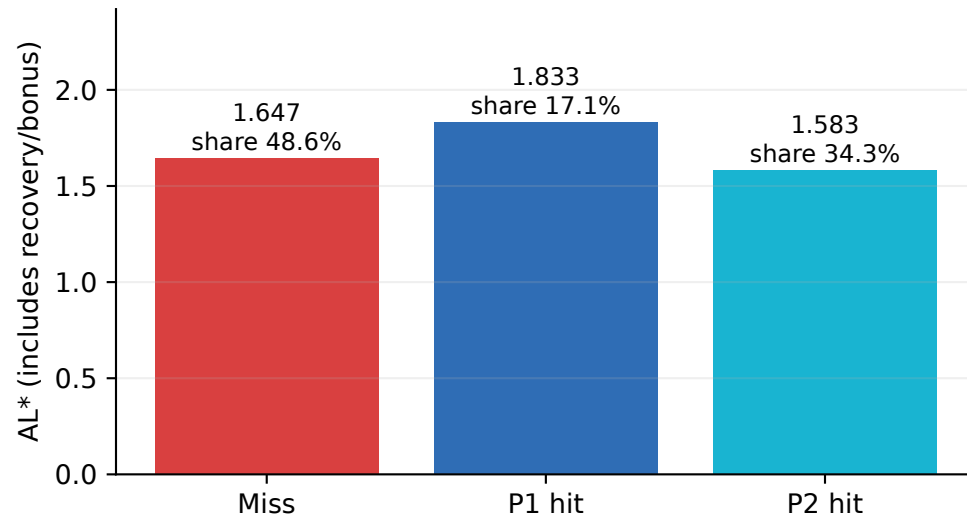
Measured wall time by batch cache status



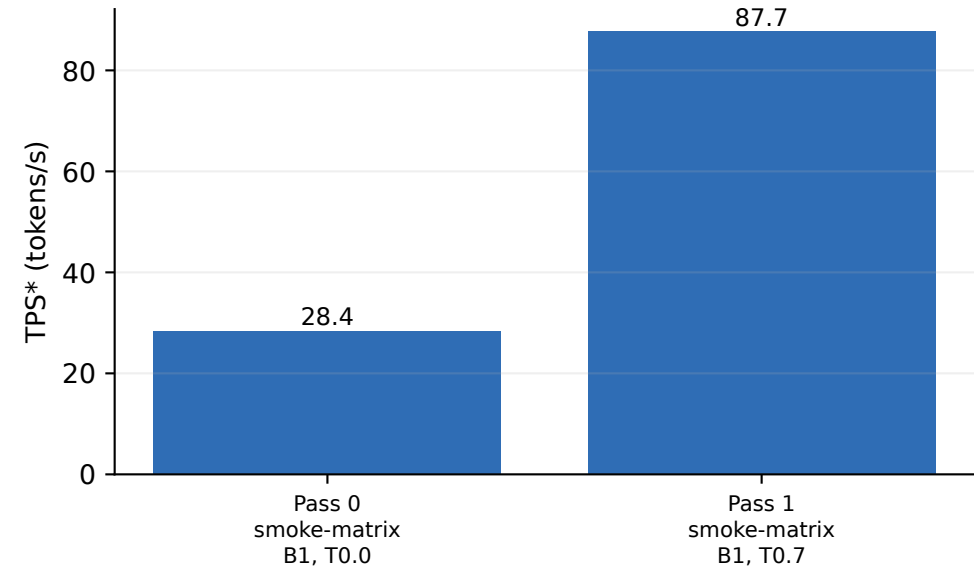
Boundary-free steps, including slow steps



Conditional AL* and source share



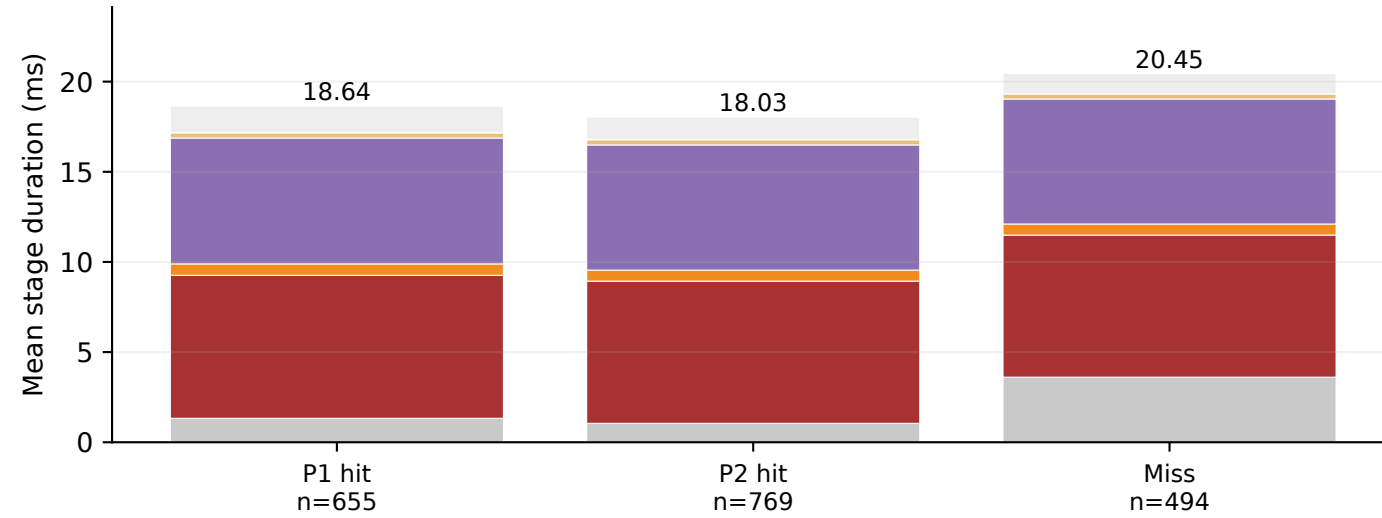
All saved passes; gray = warmup



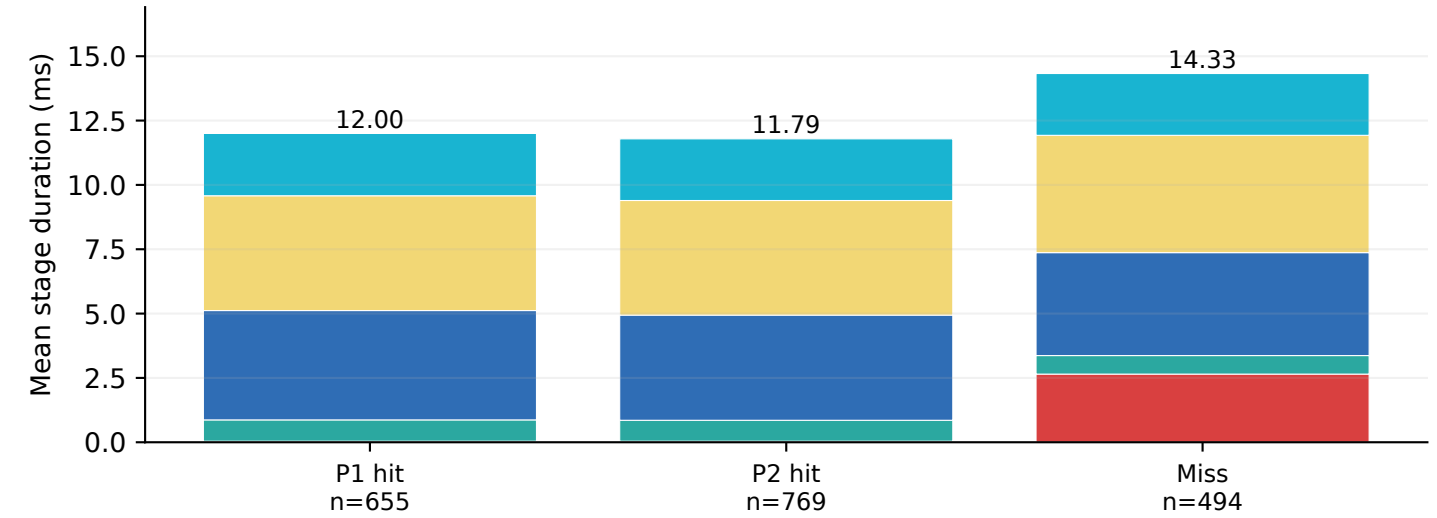
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_duet_e16_k4_2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1918 eligible steps

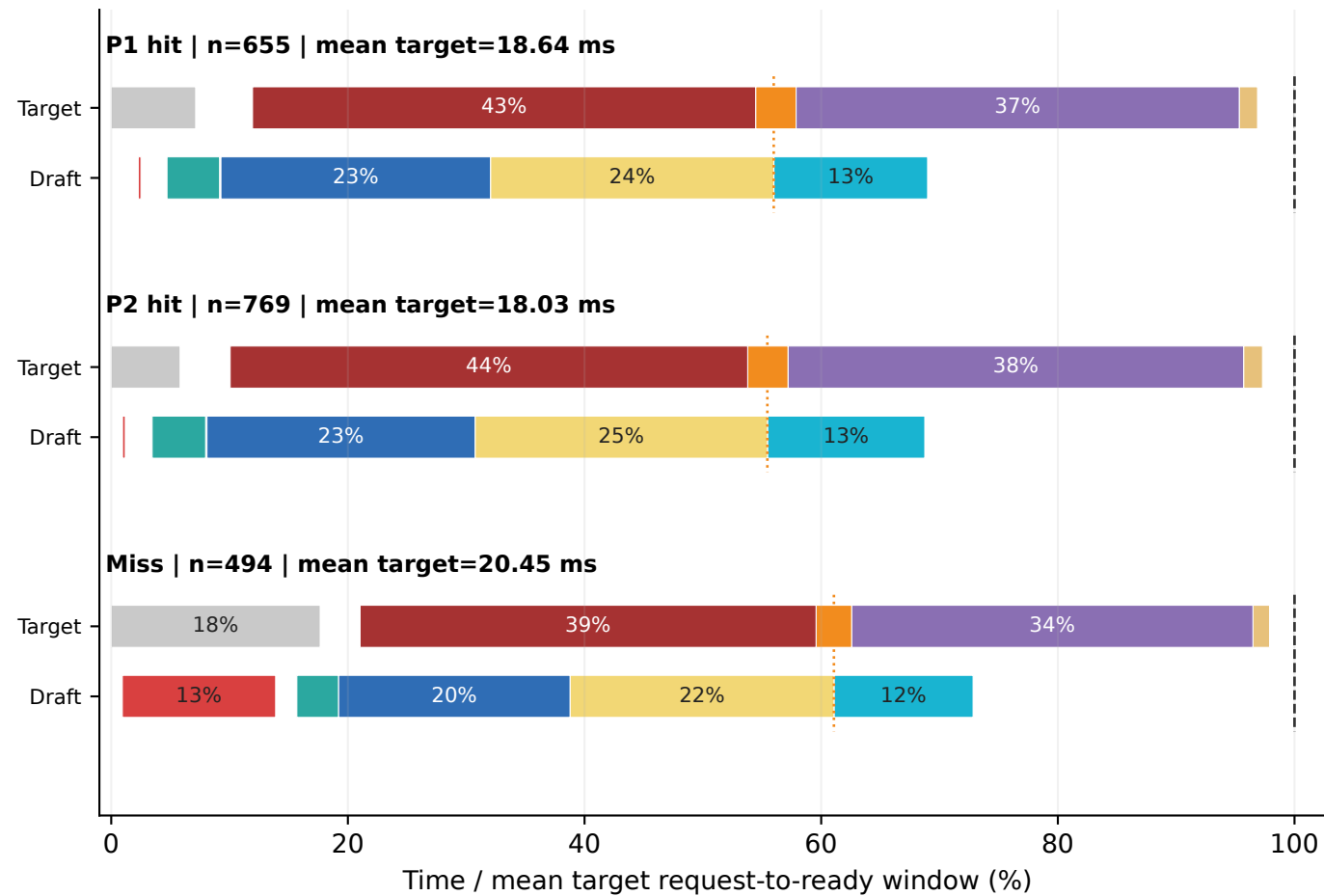
Target: status means on the same eligible steps



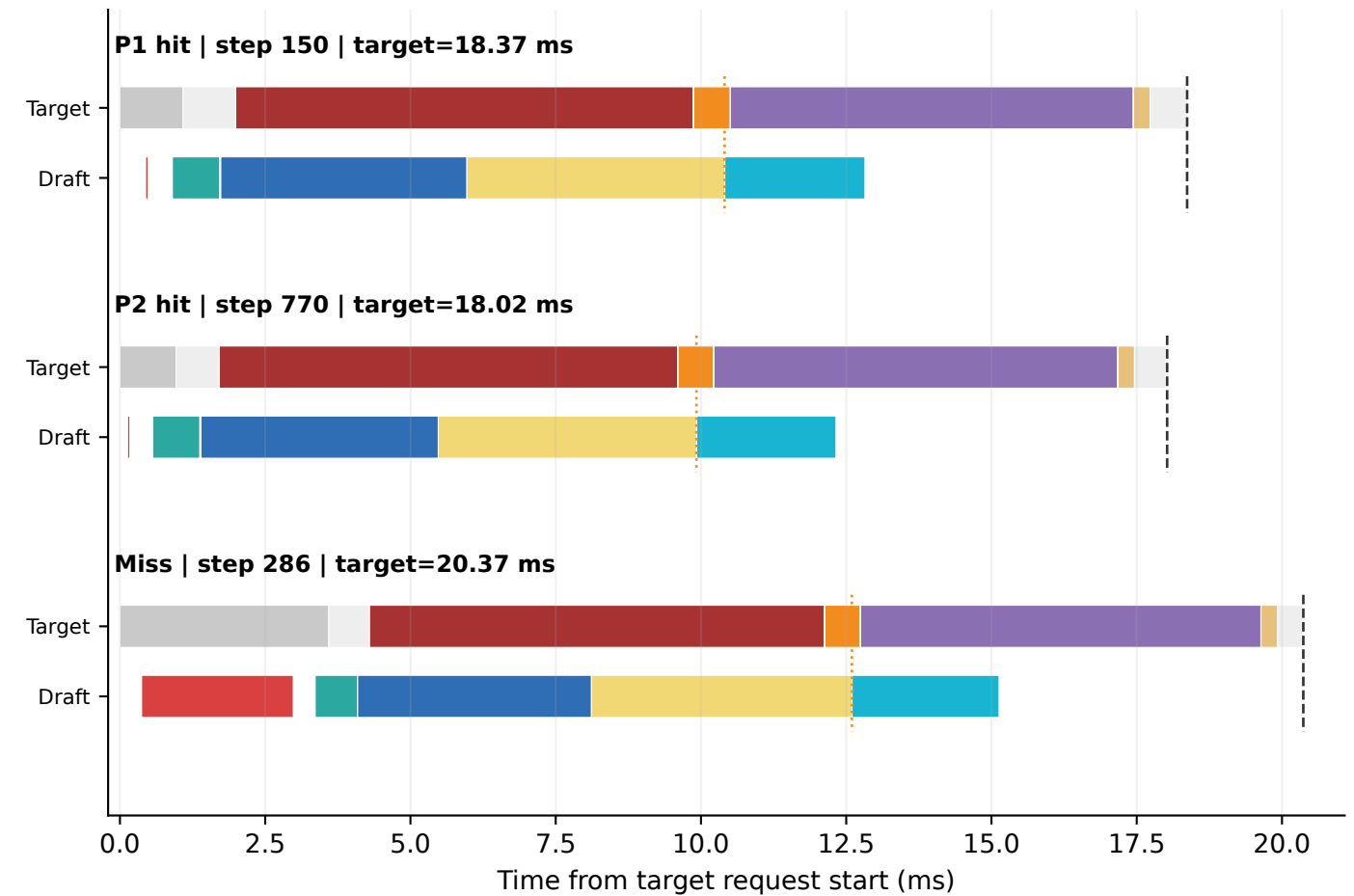
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic

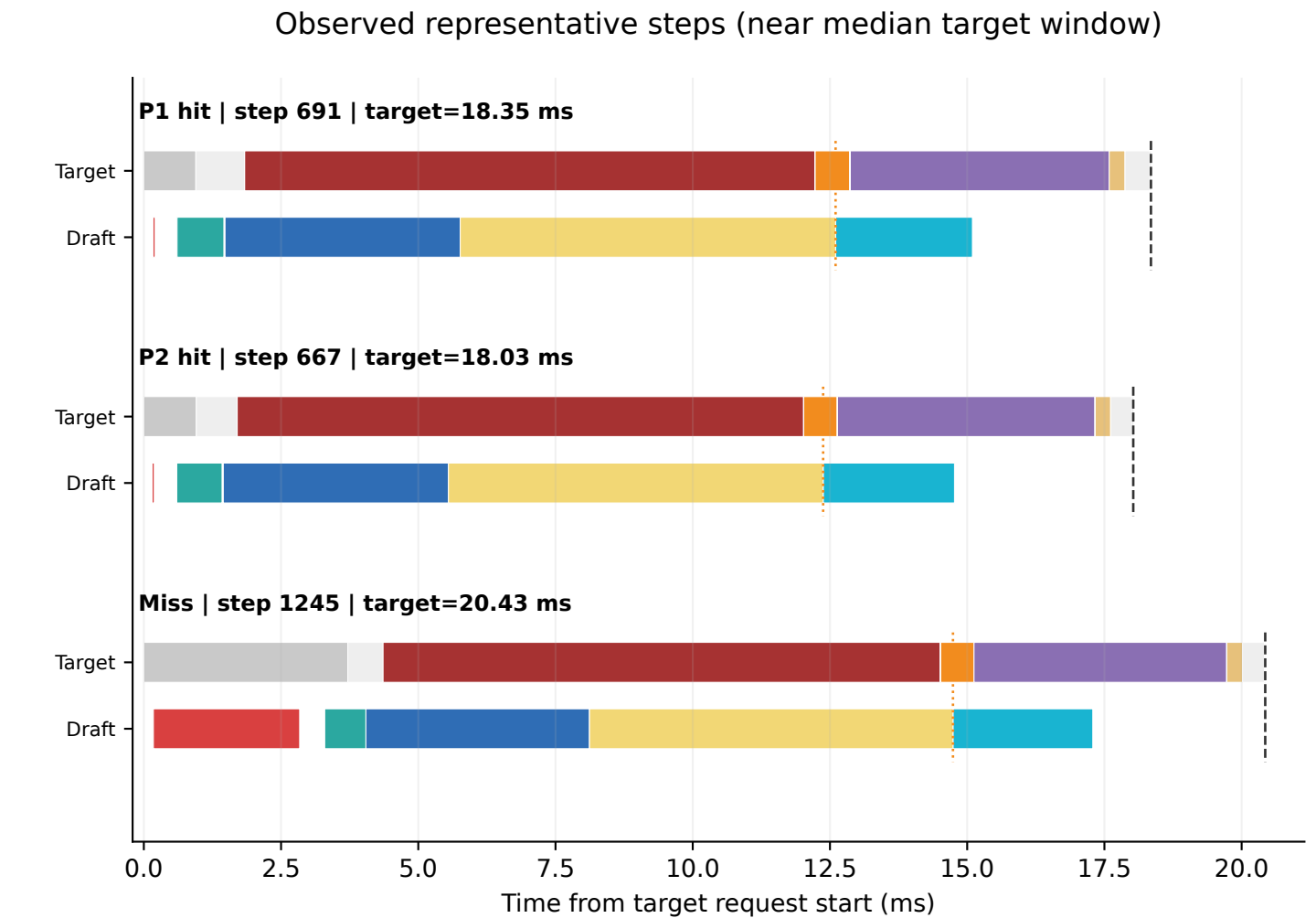
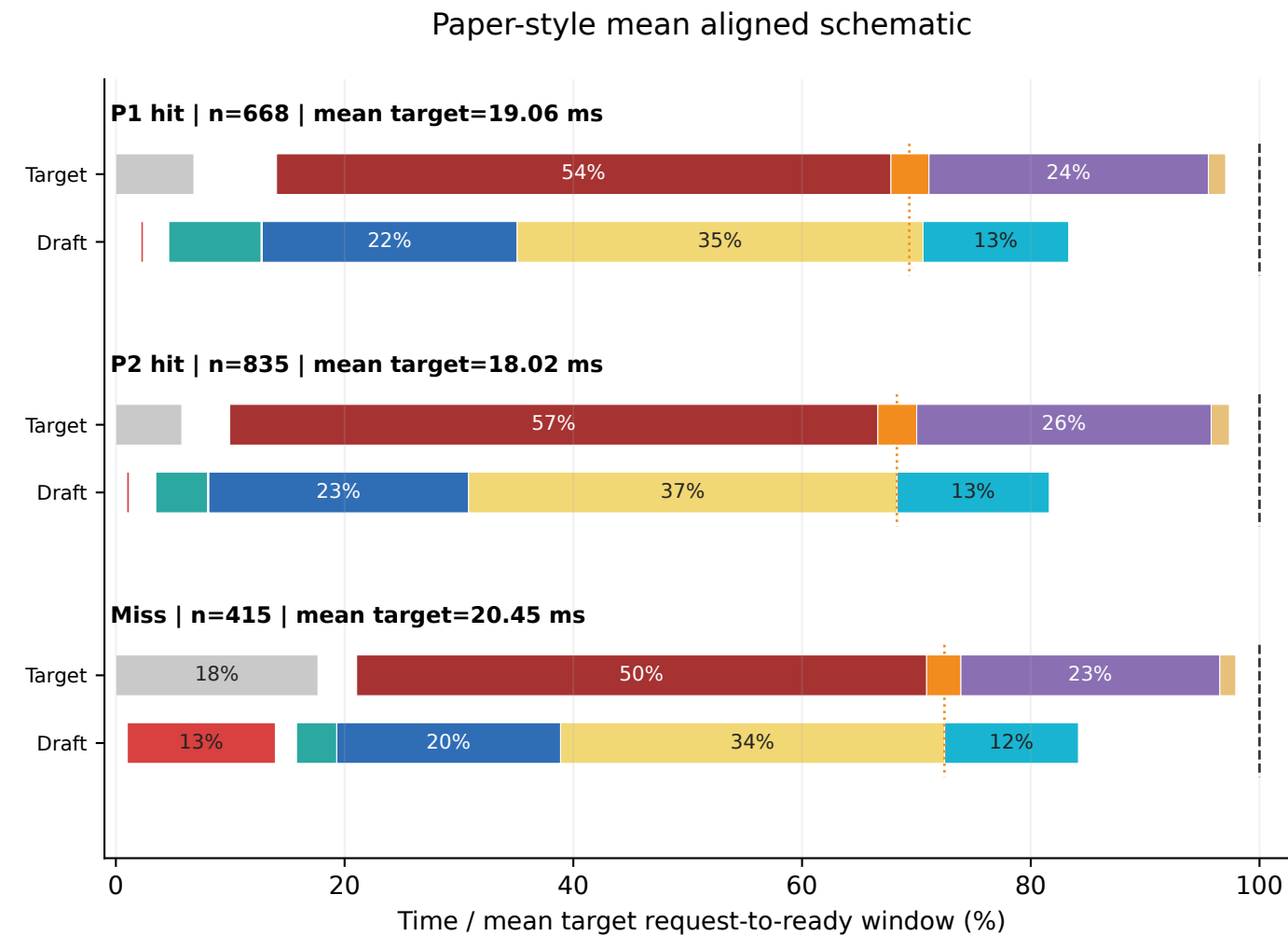
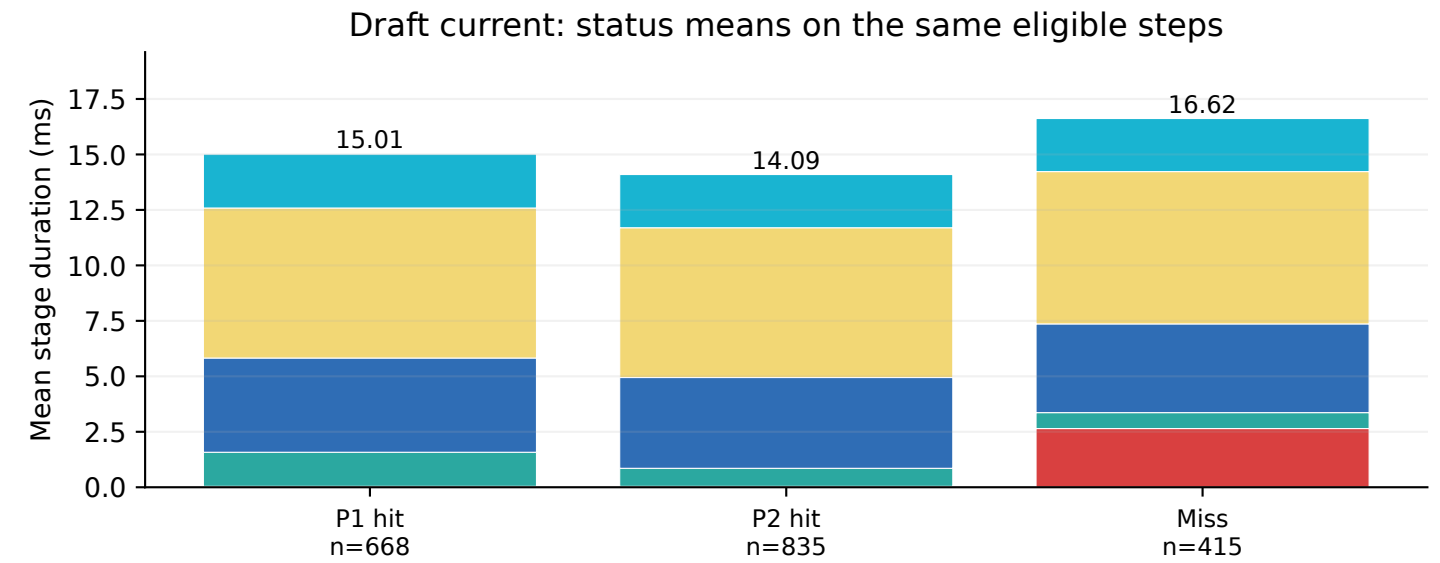
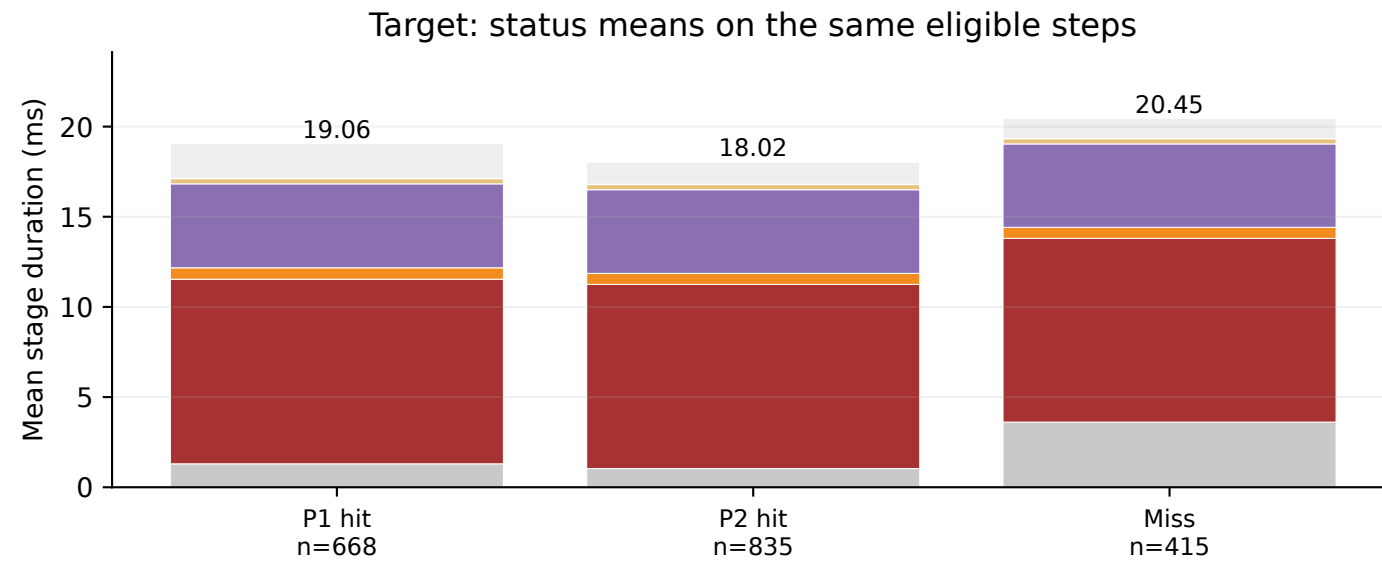


Observed representative steps (near median target window)



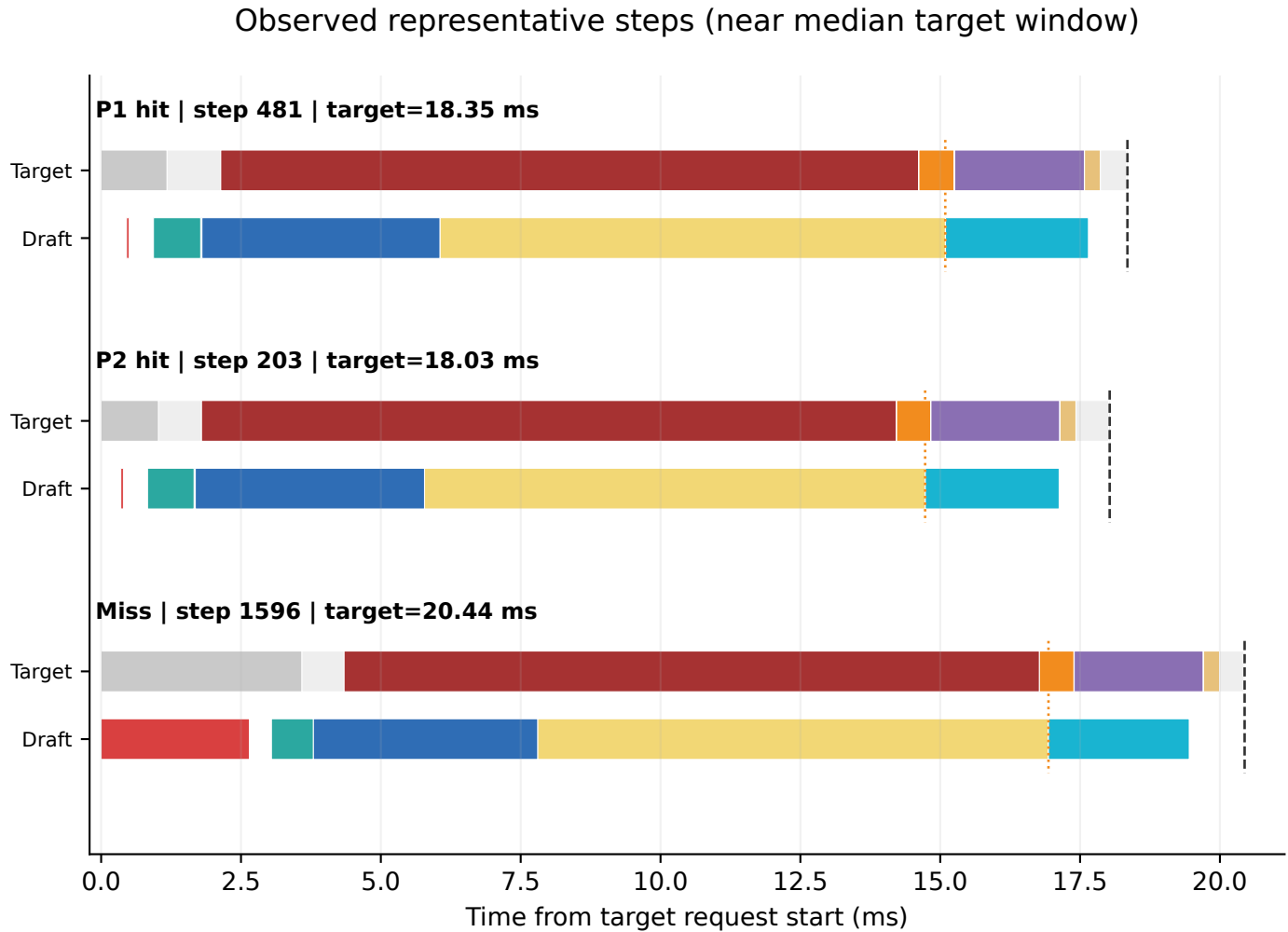
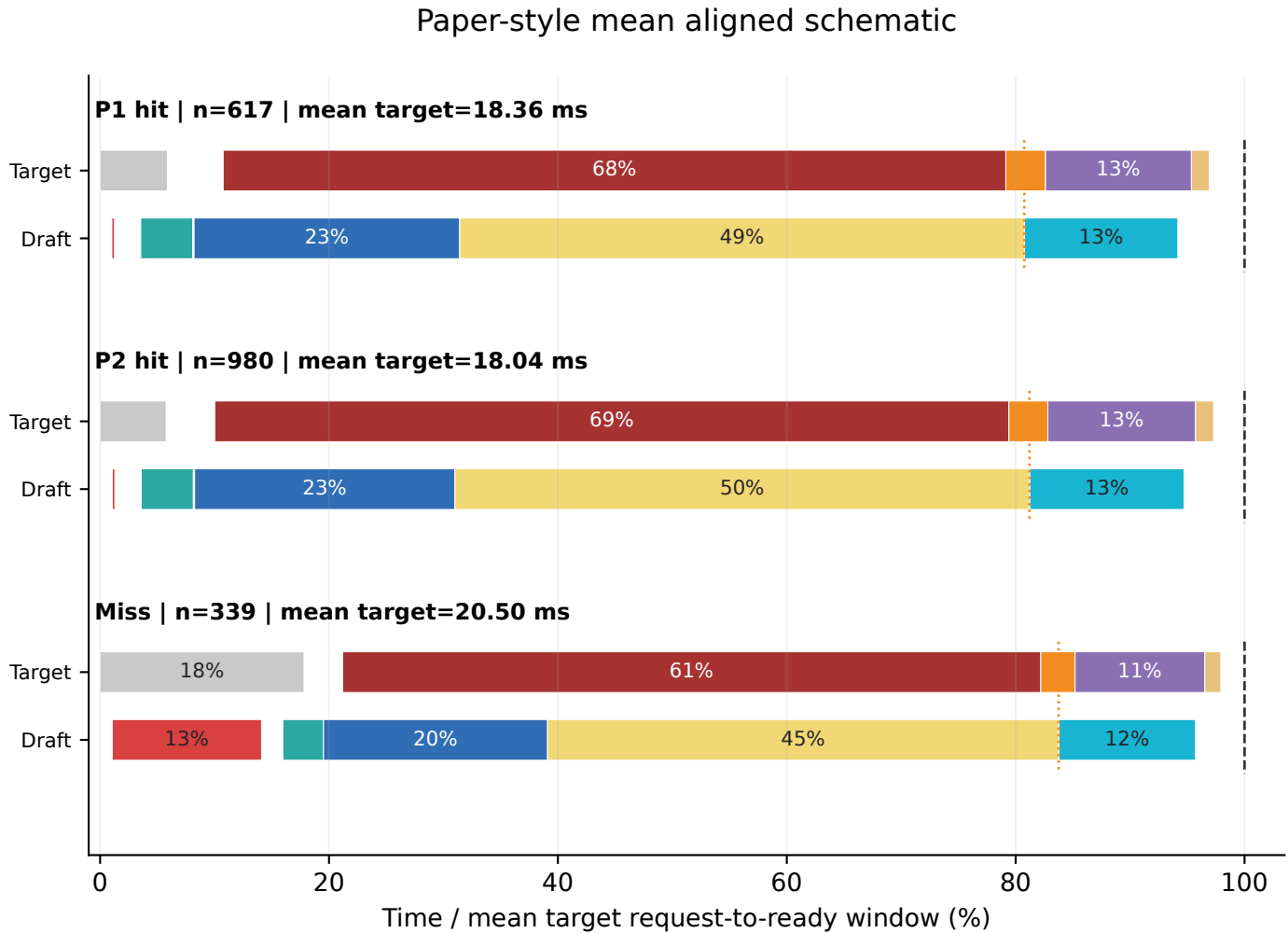
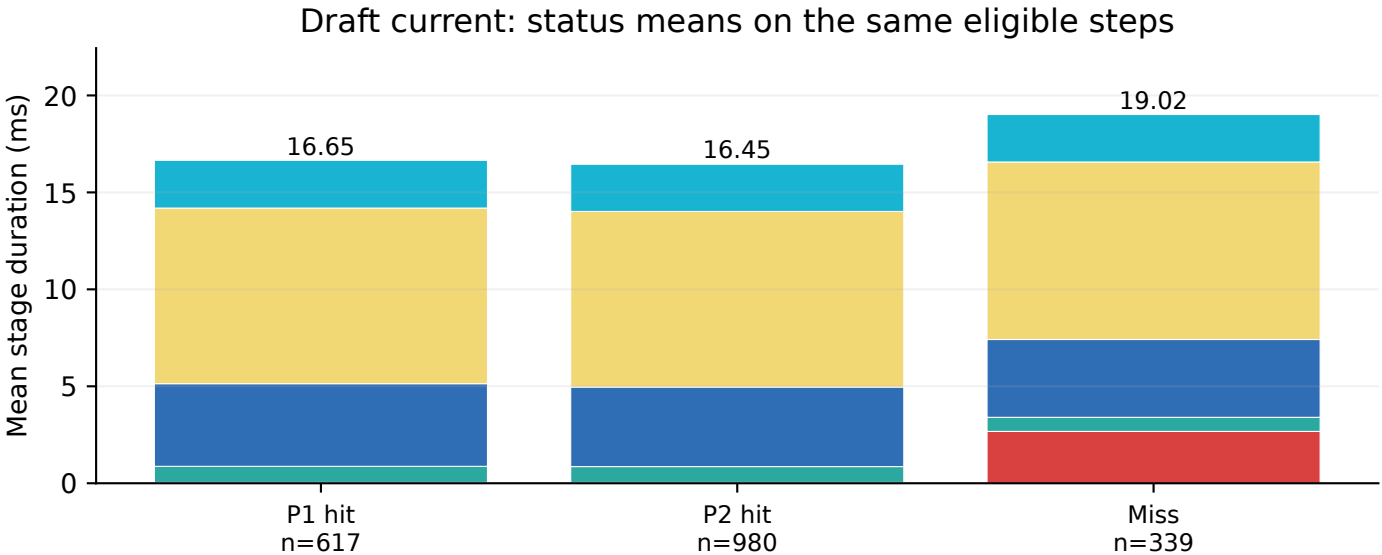
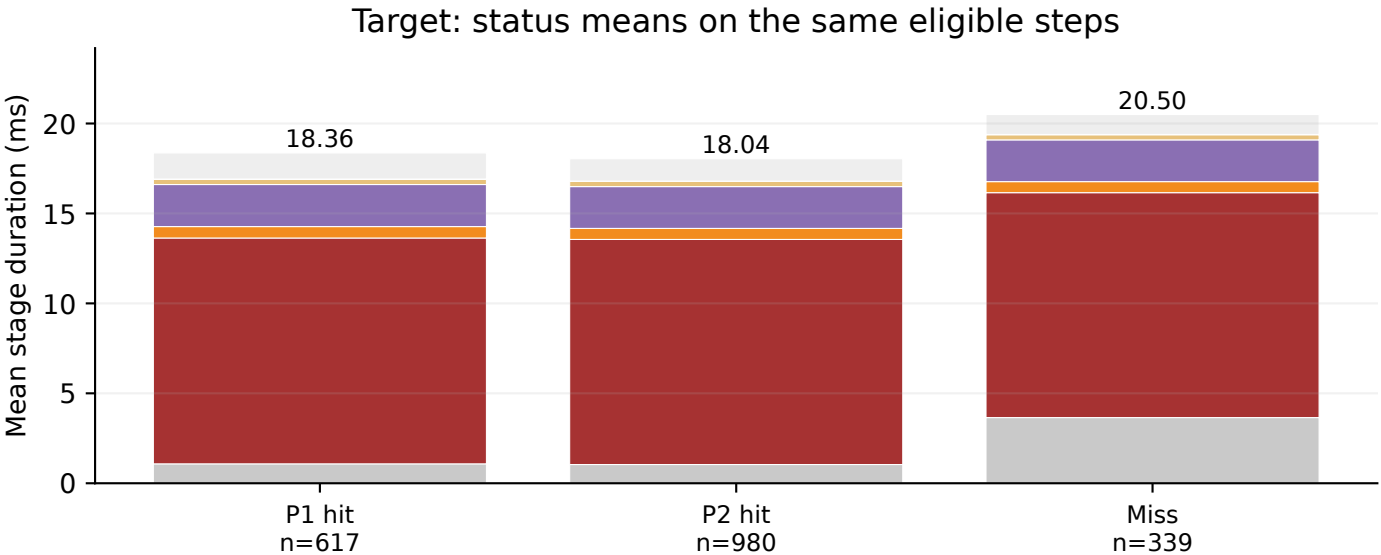
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_duet_e21_k4_2
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1918 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

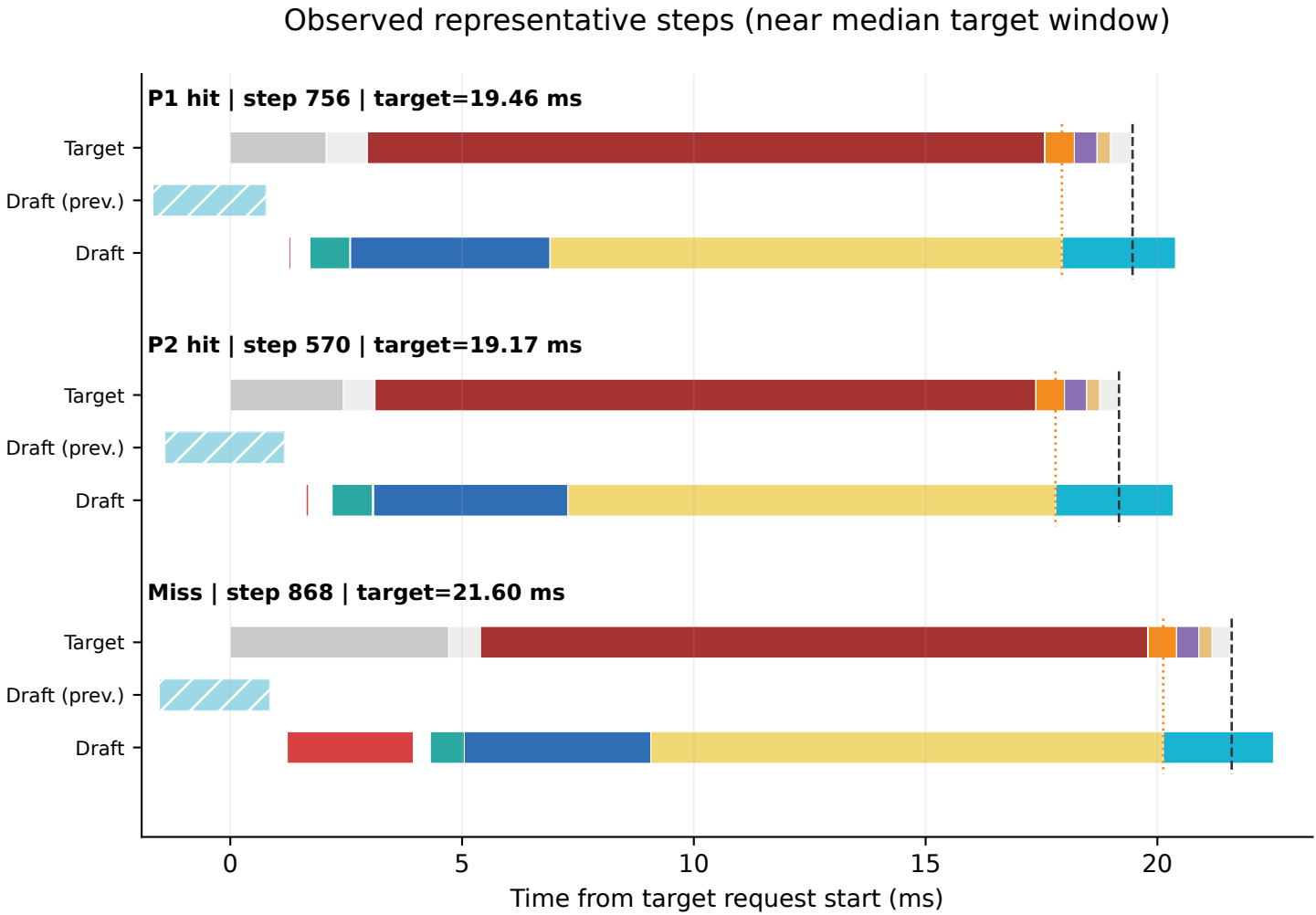
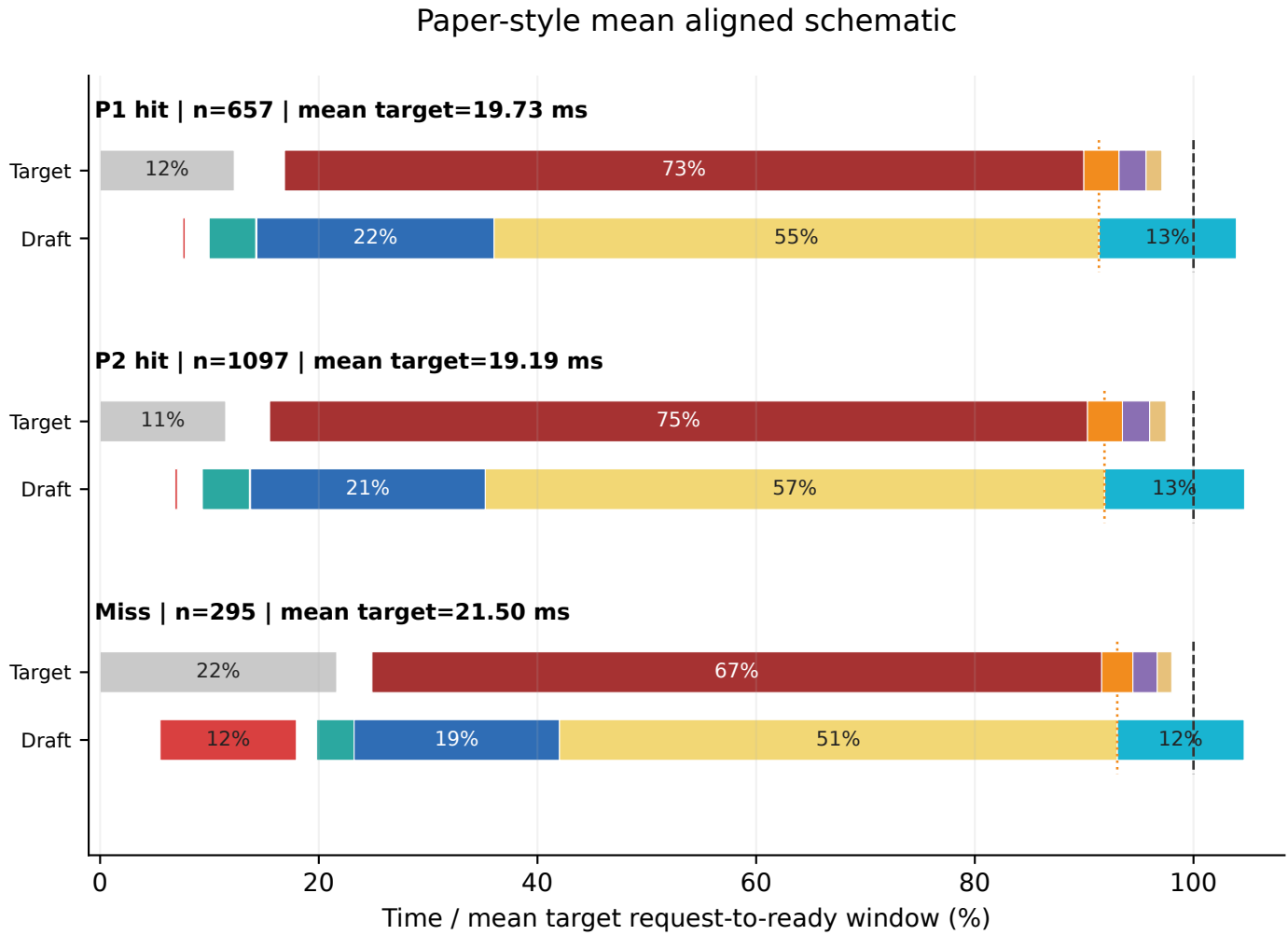
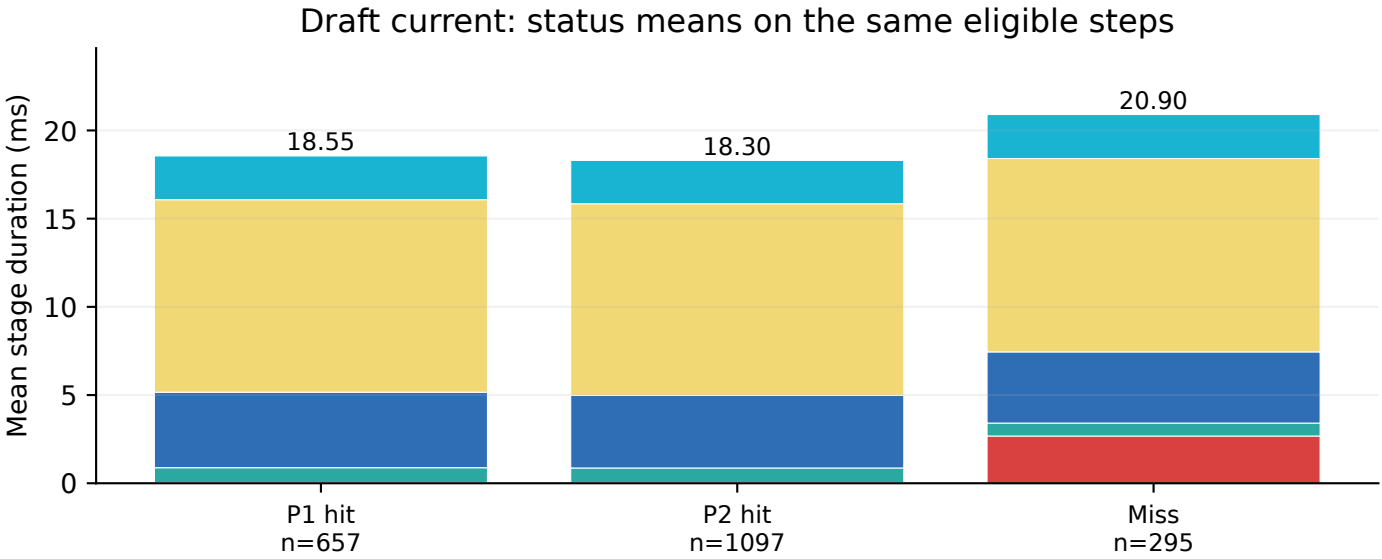
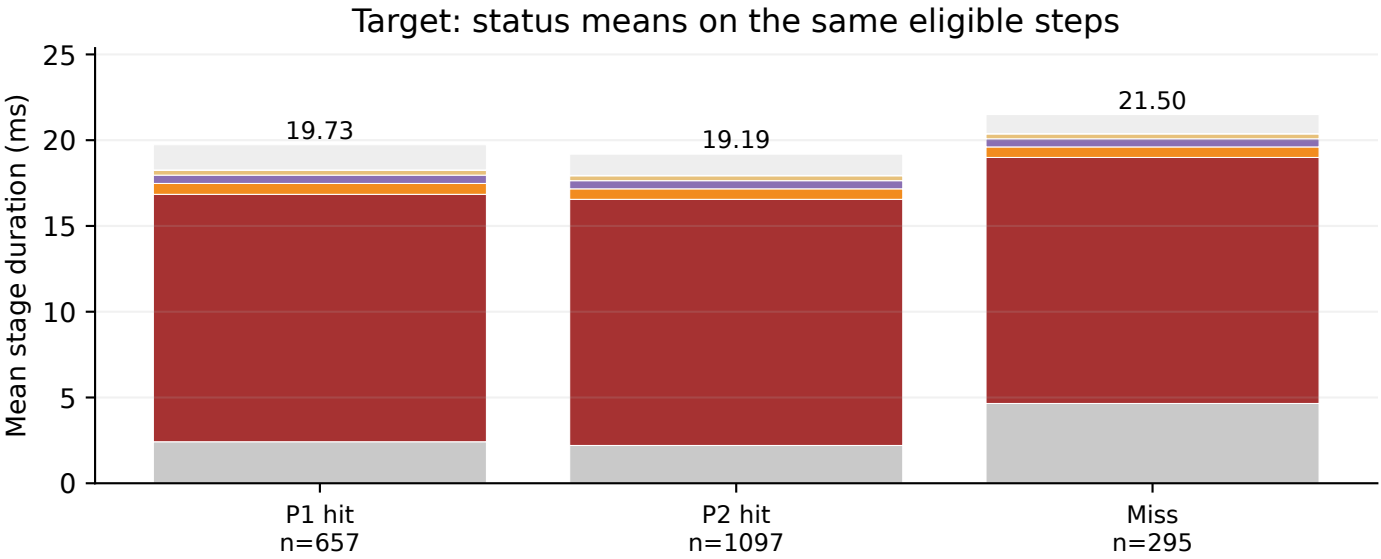
llama2_b1_duet_e26_k4_2
DUET: K1/K2=4/2, exit=26 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1936 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

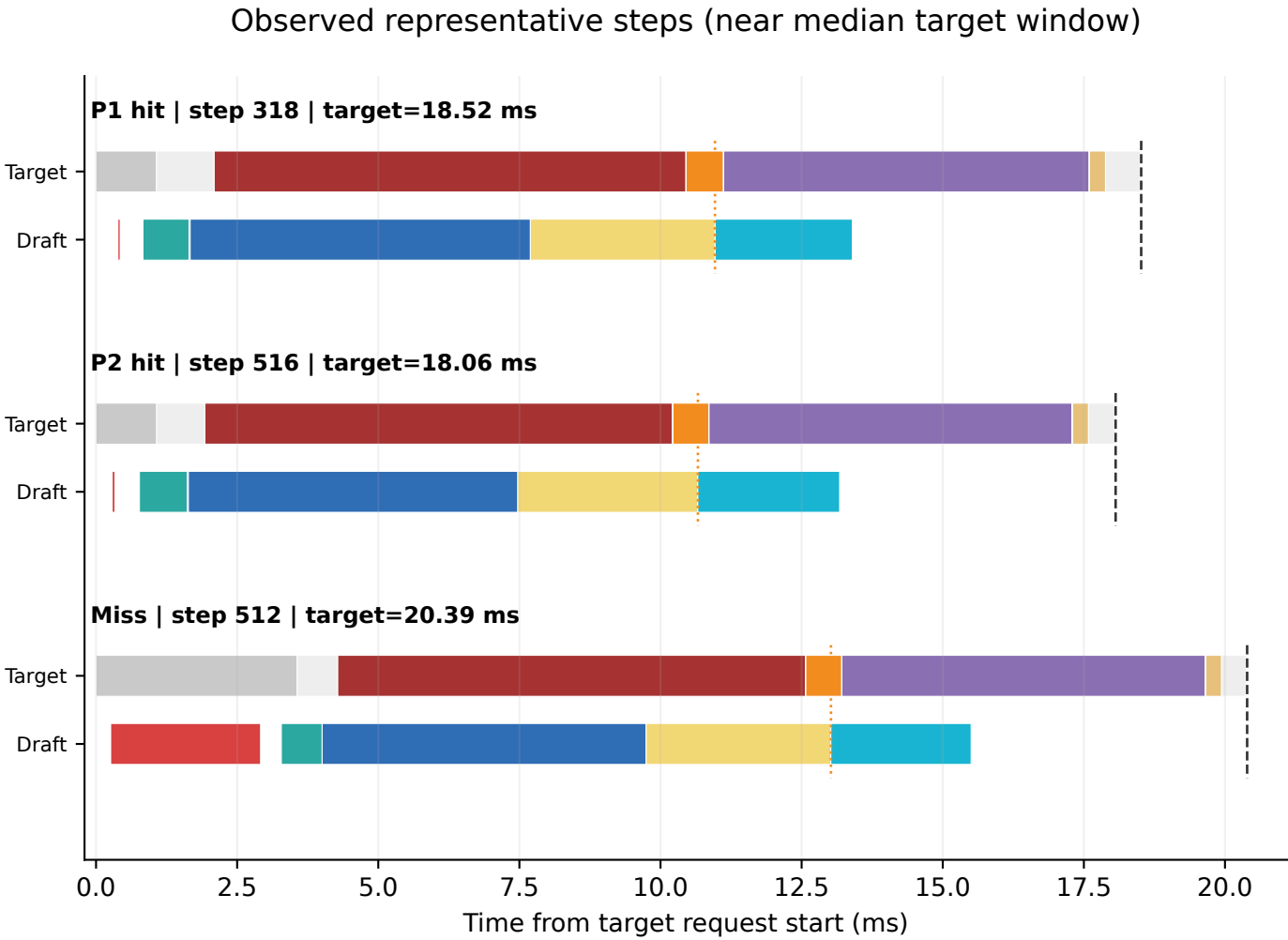
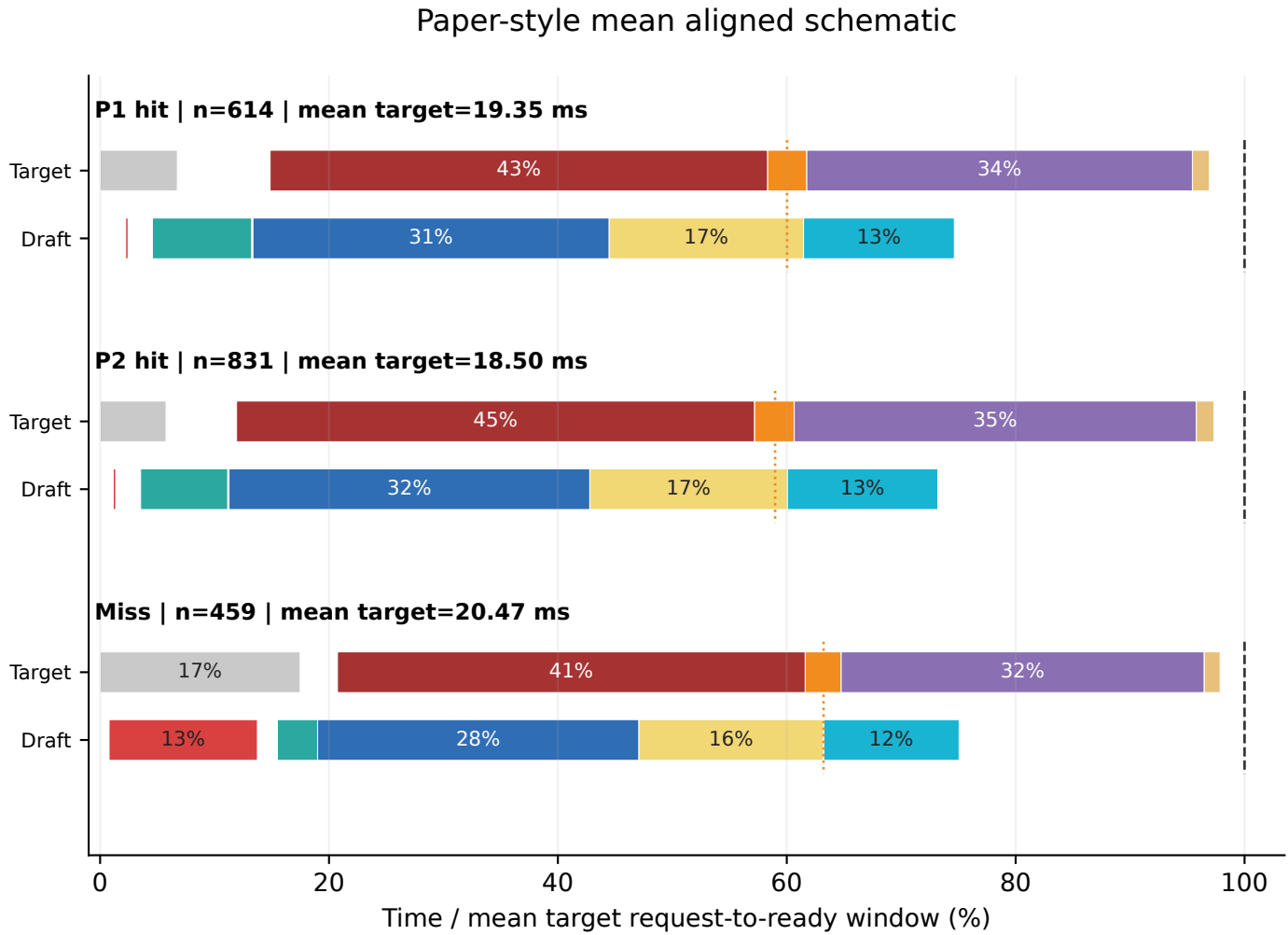
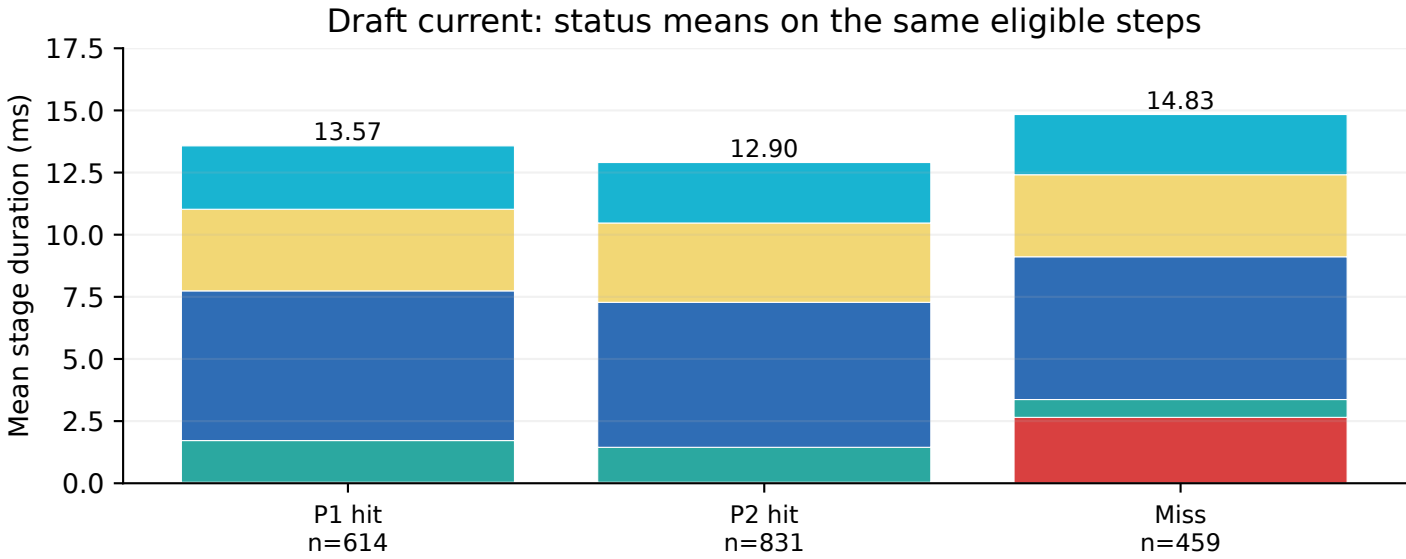
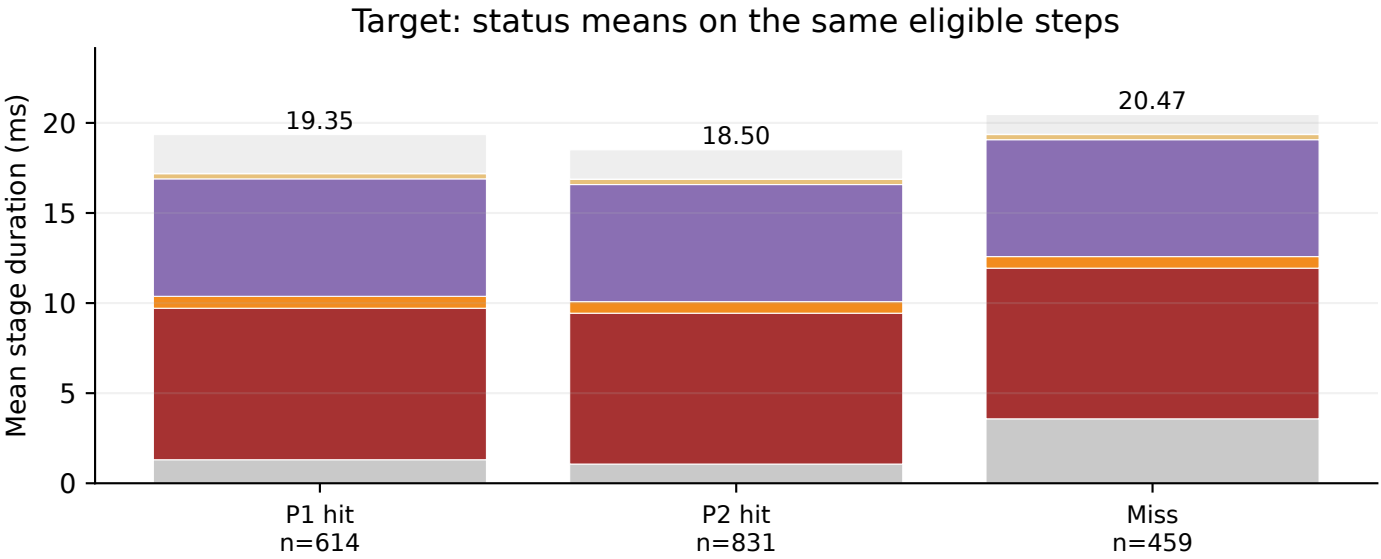
llama2_b1_duet_e30_k4_2
DUET: K1/K2=4/2, exit=30 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 2049 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

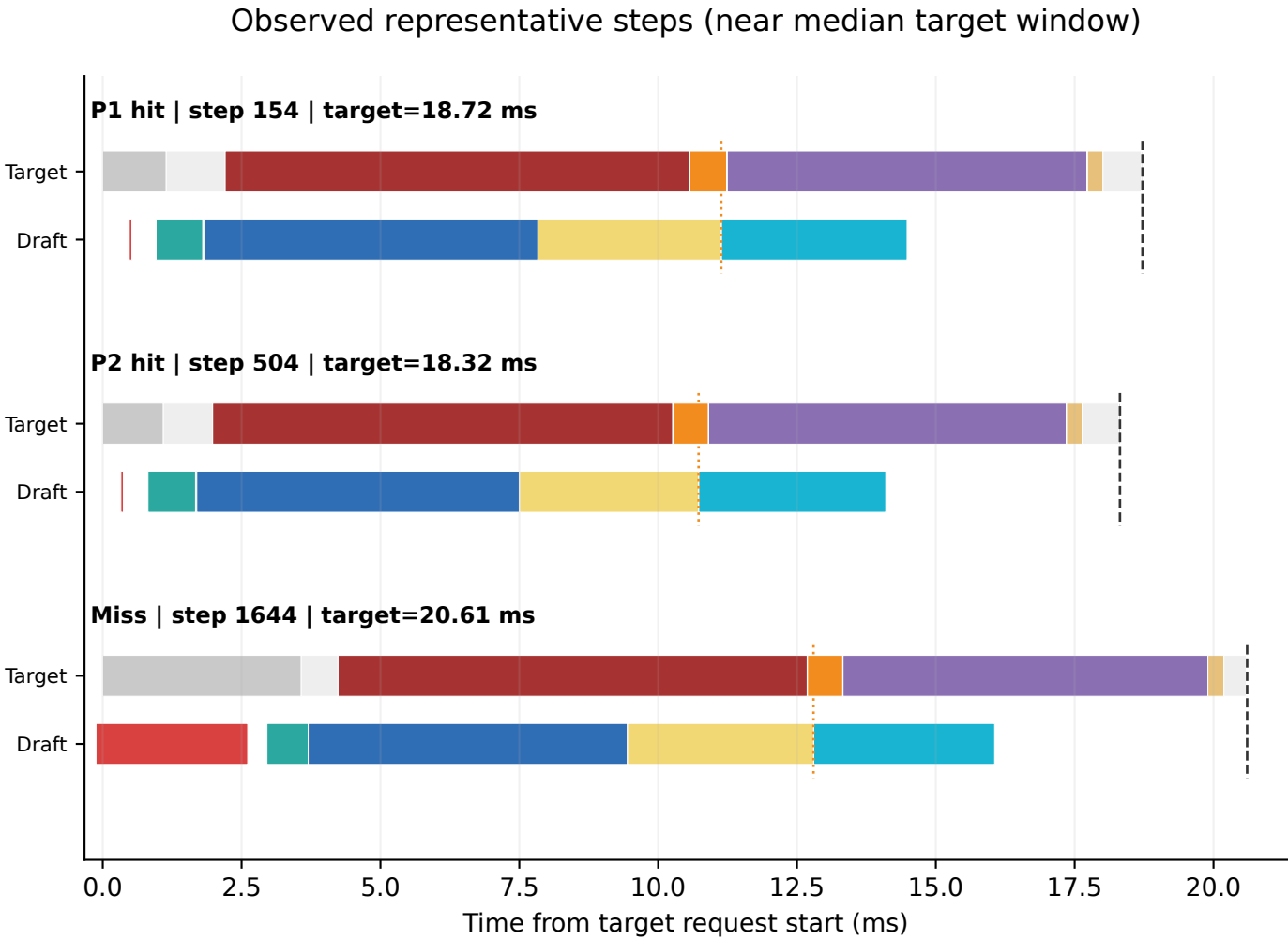
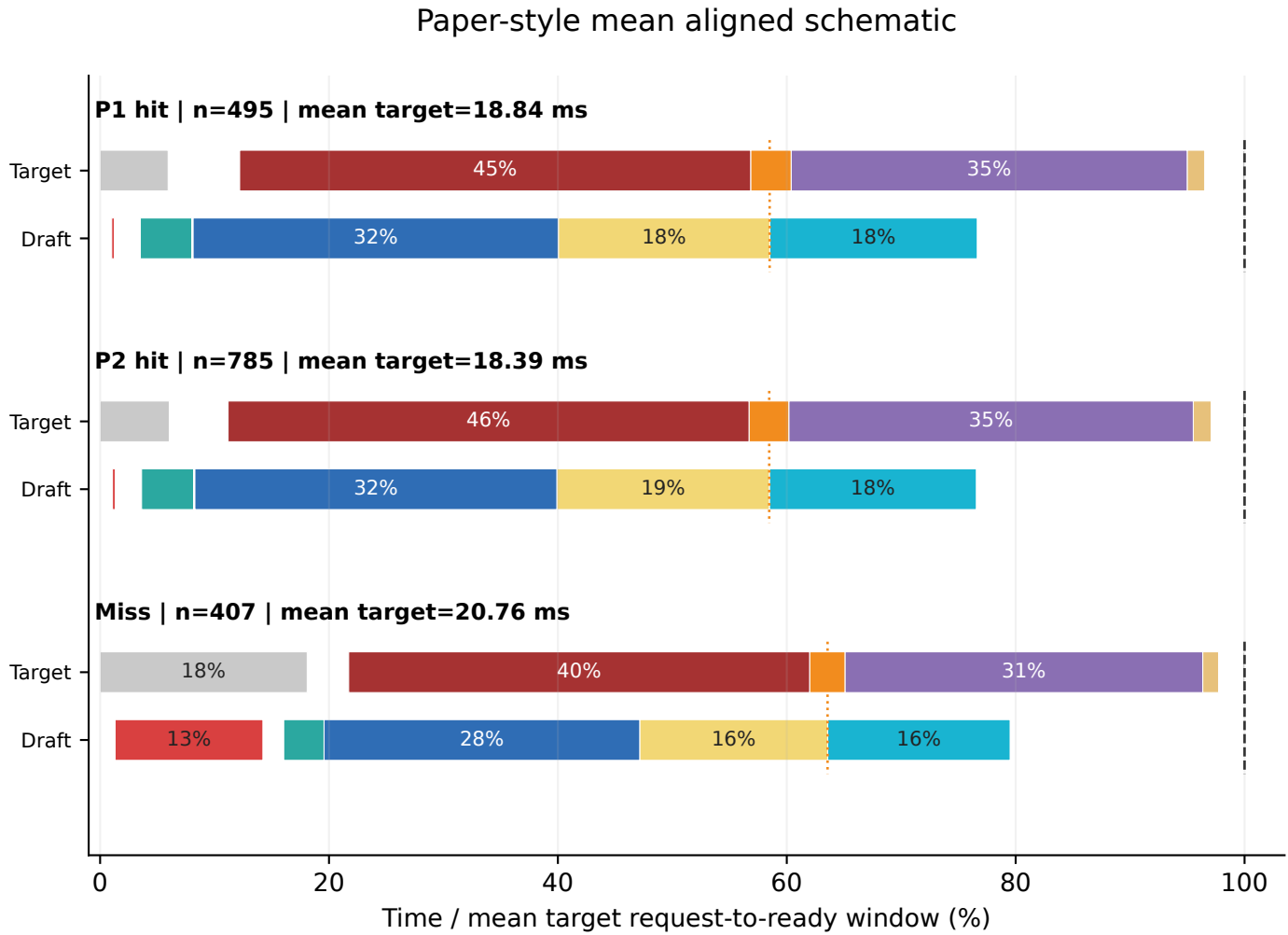
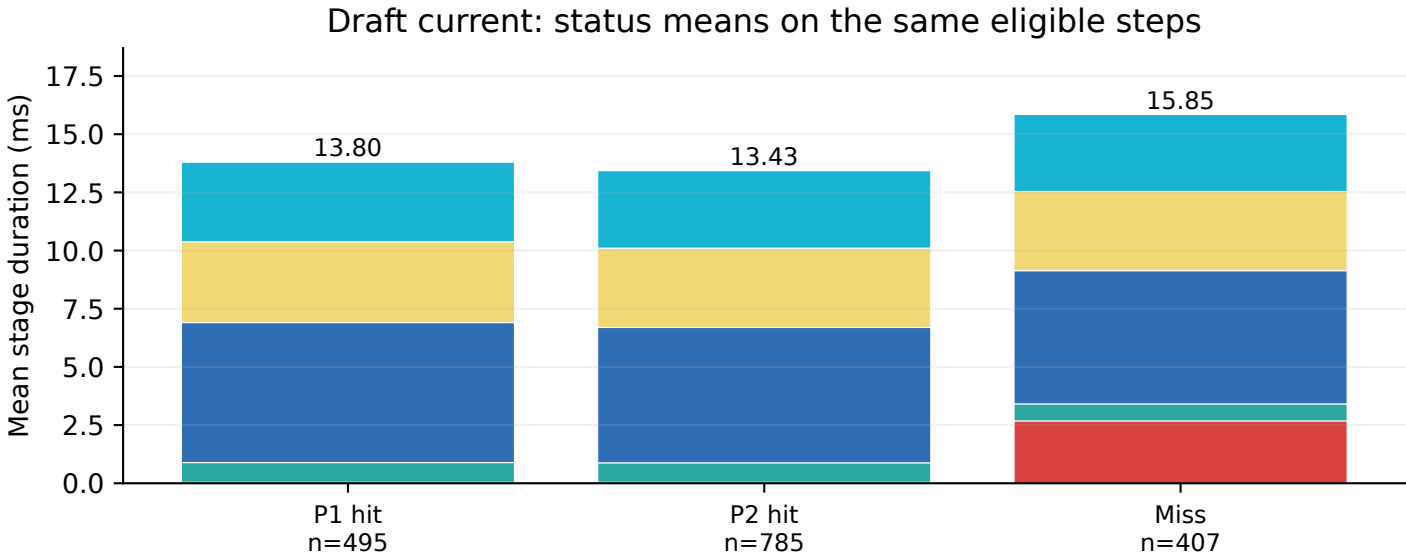
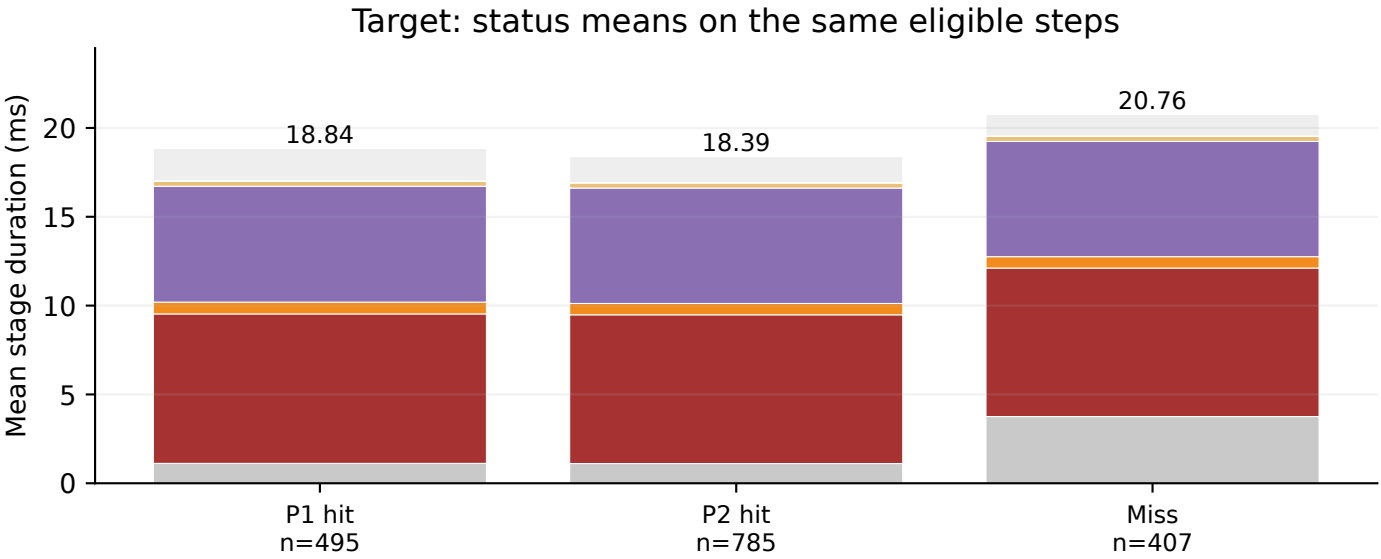
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e17_k6_2_r2
DUET: K1/K2=6/2, exit=17 (0-based), $\bar{C}$ =3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1904 eligible steps

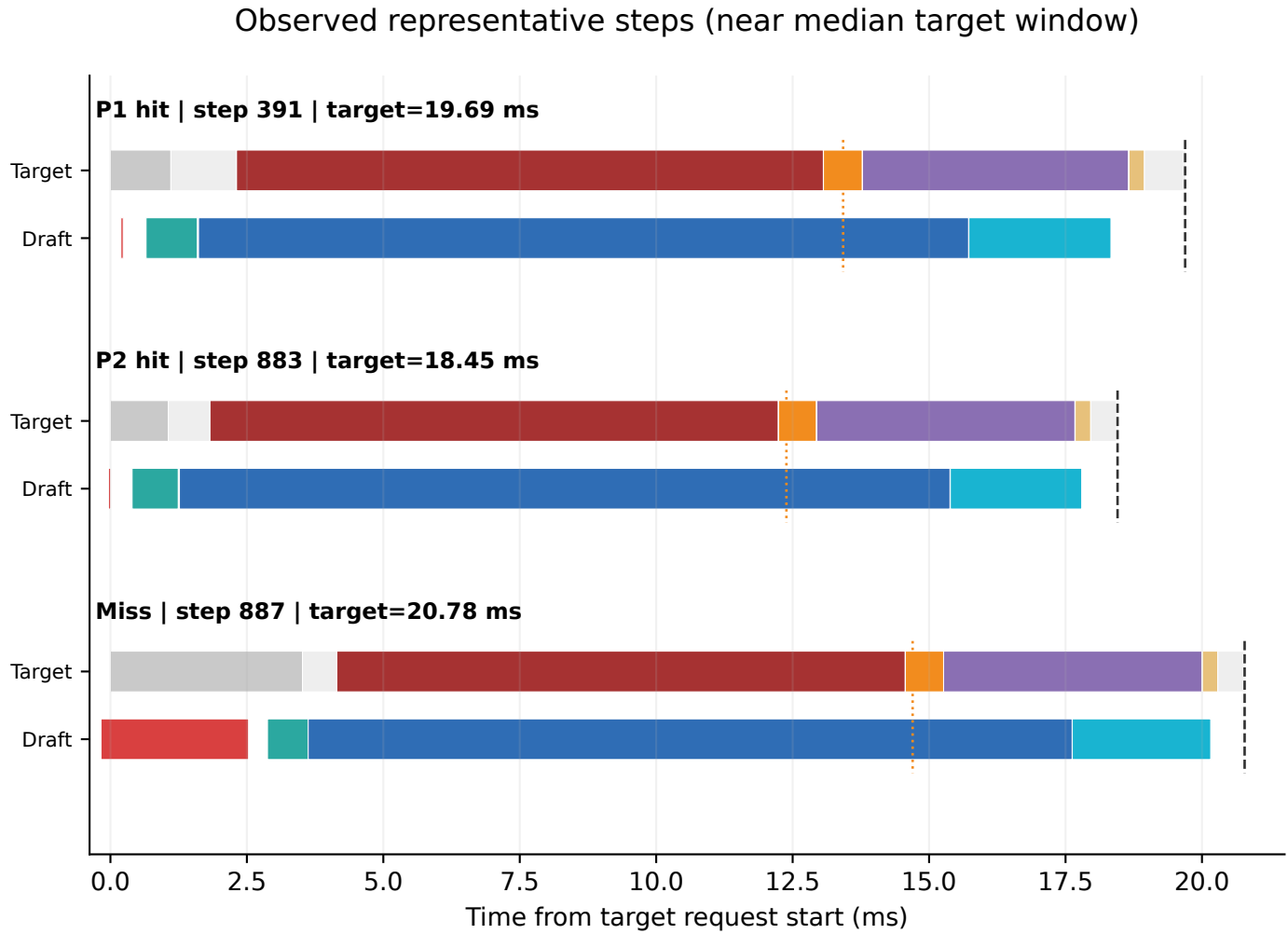
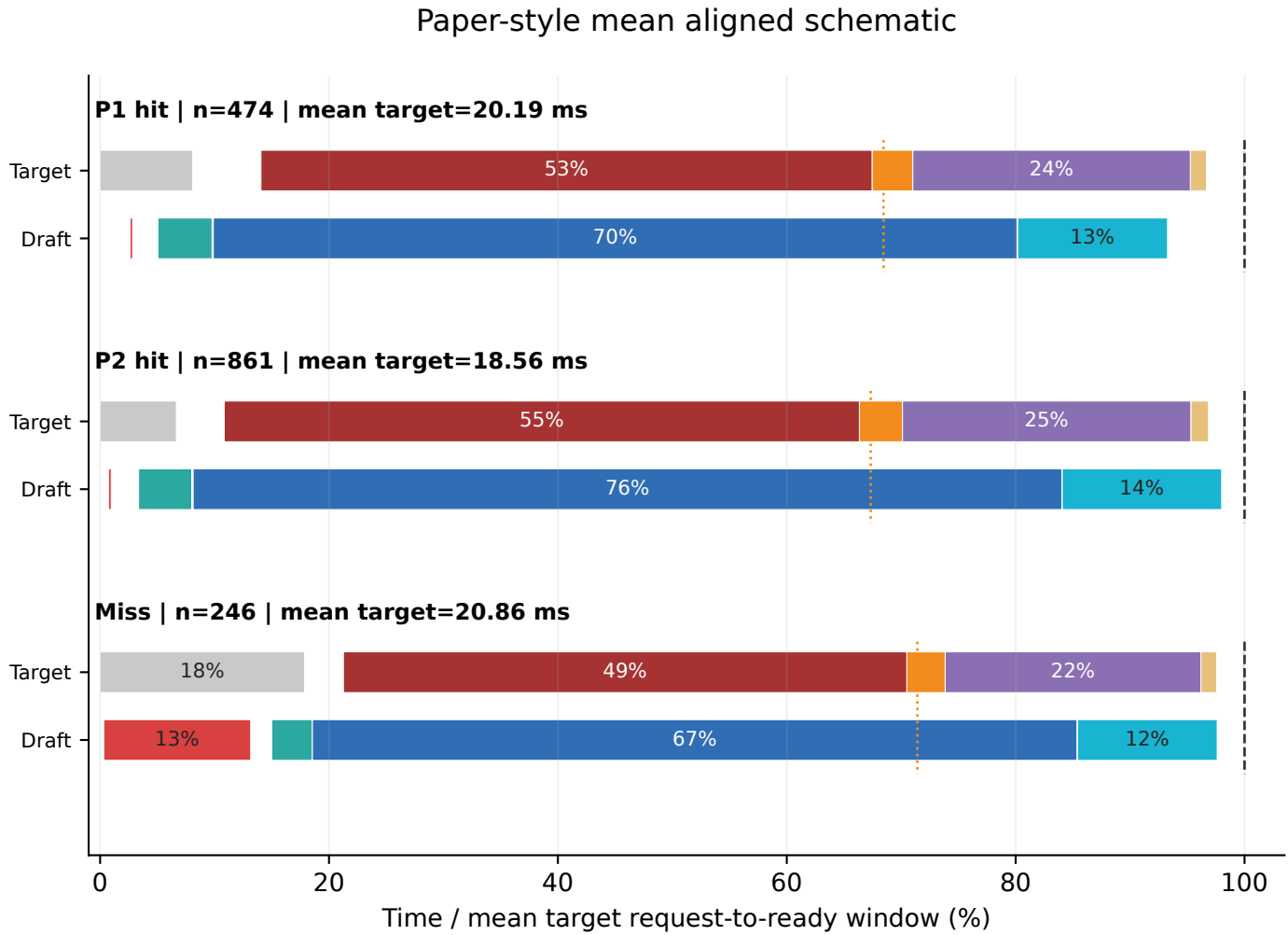
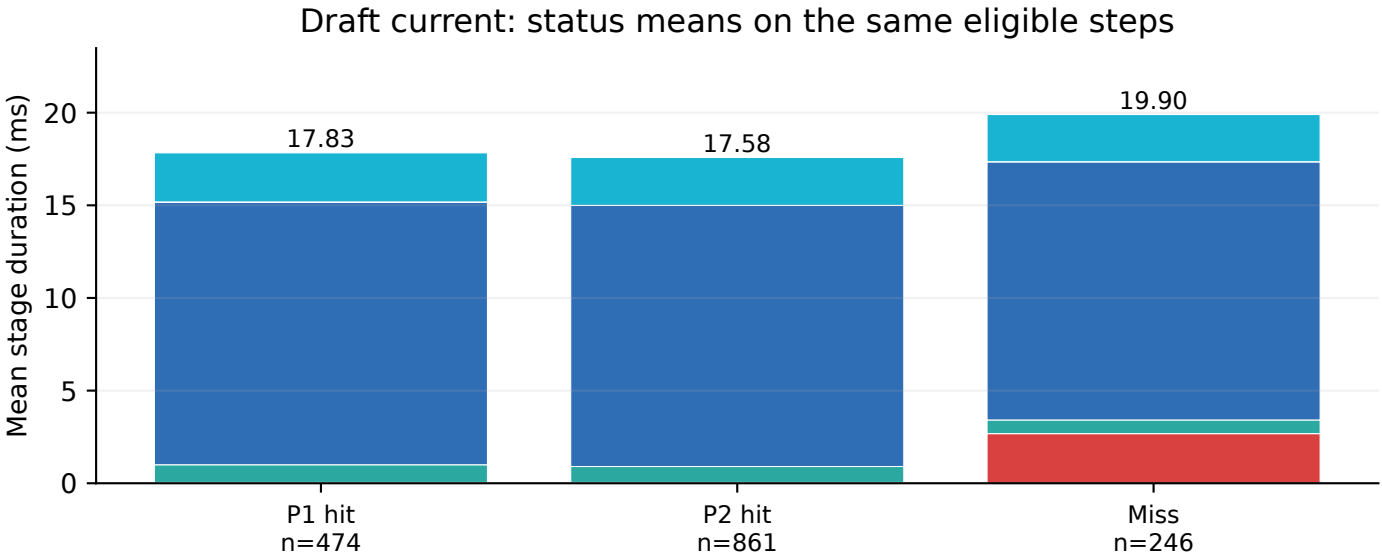
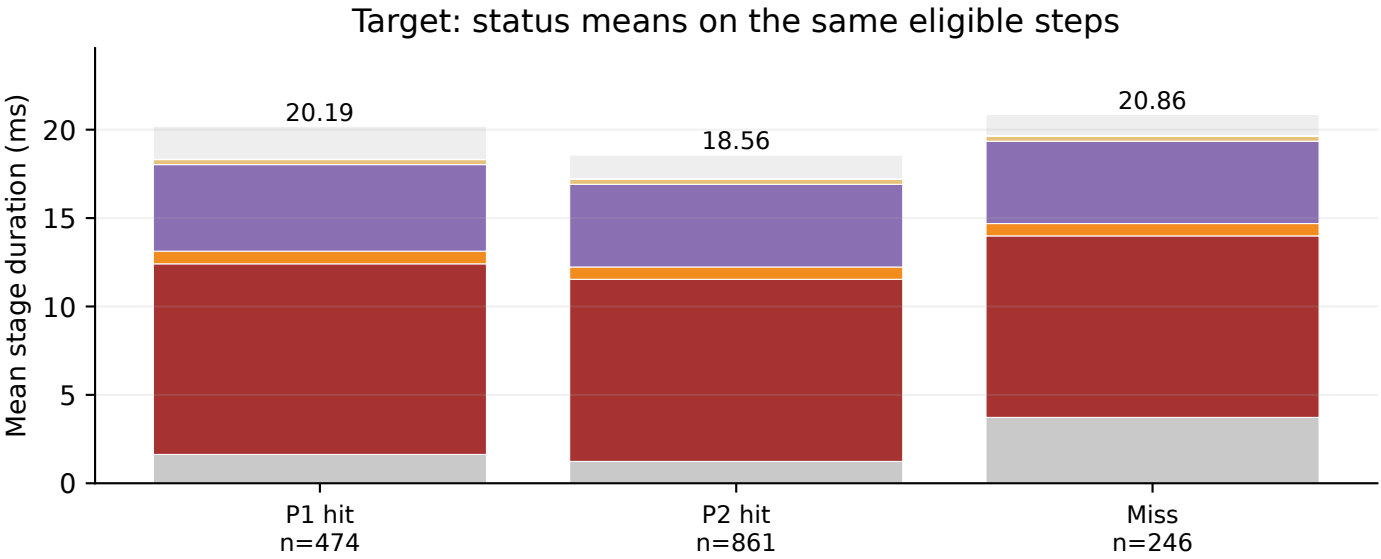


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

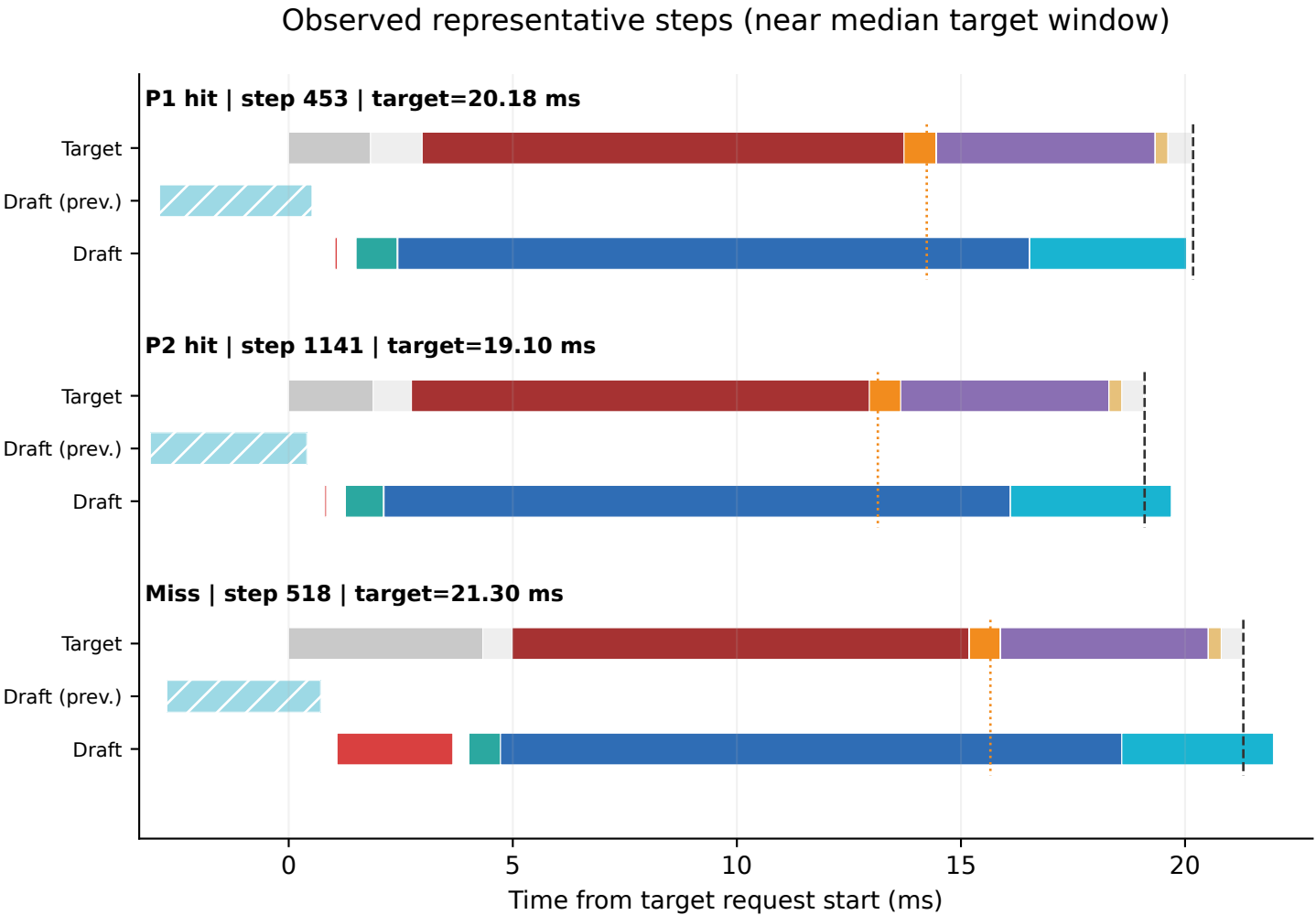
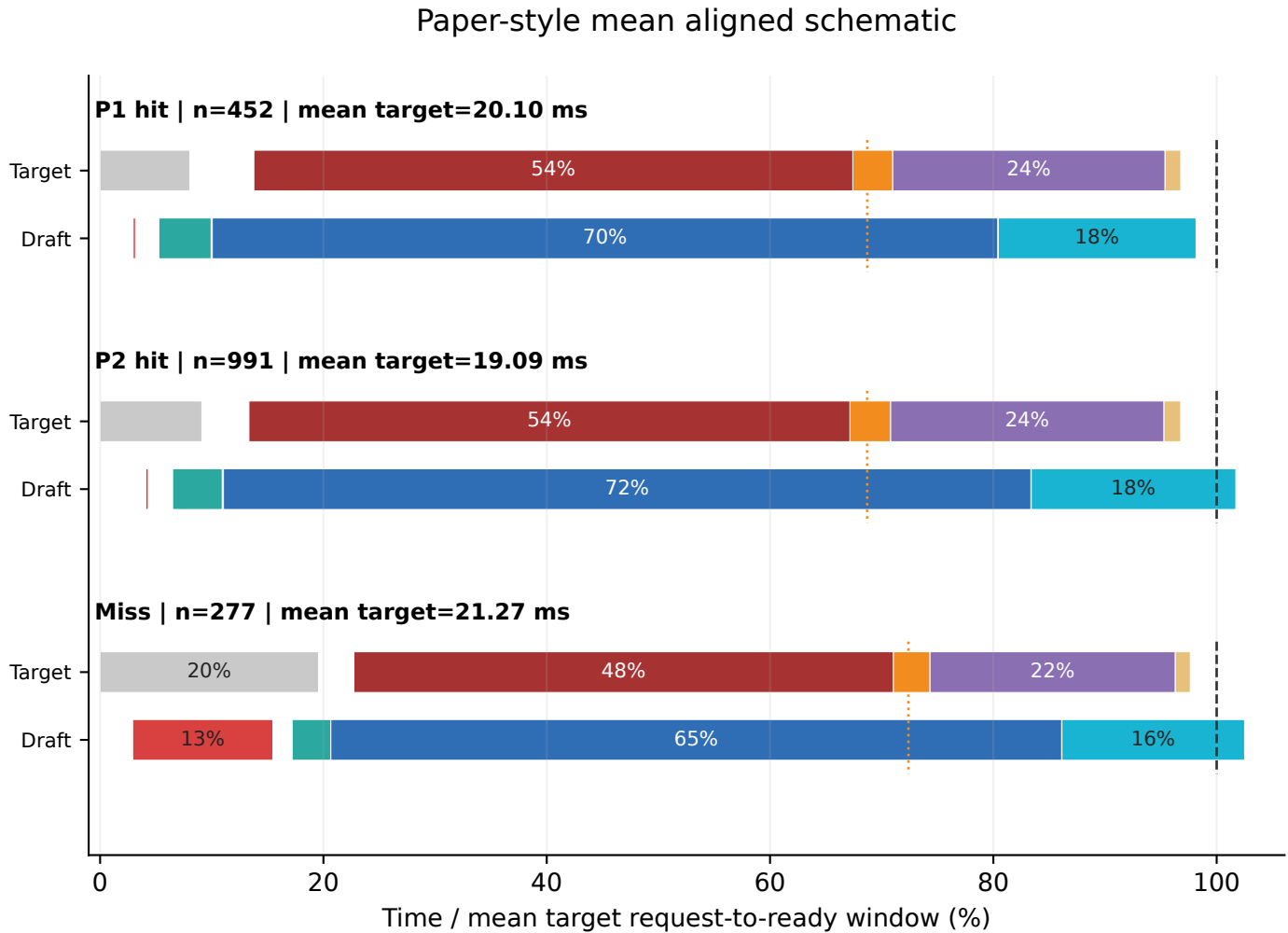
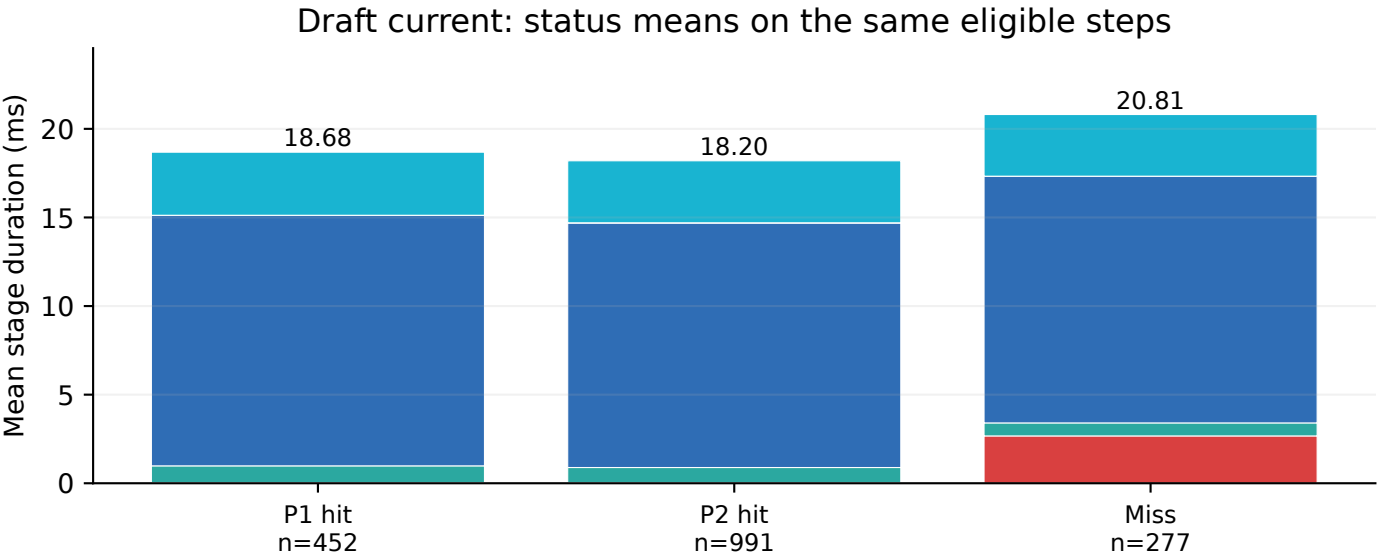
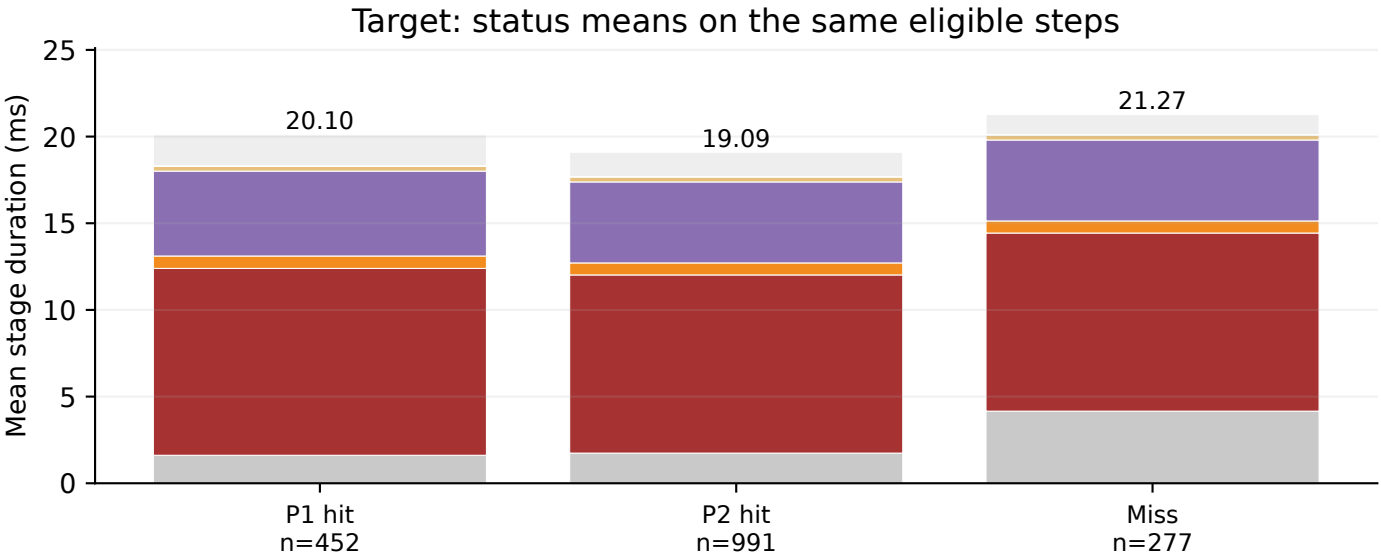


llama2_b1_refine_e21_k12_2_r2
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1581 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

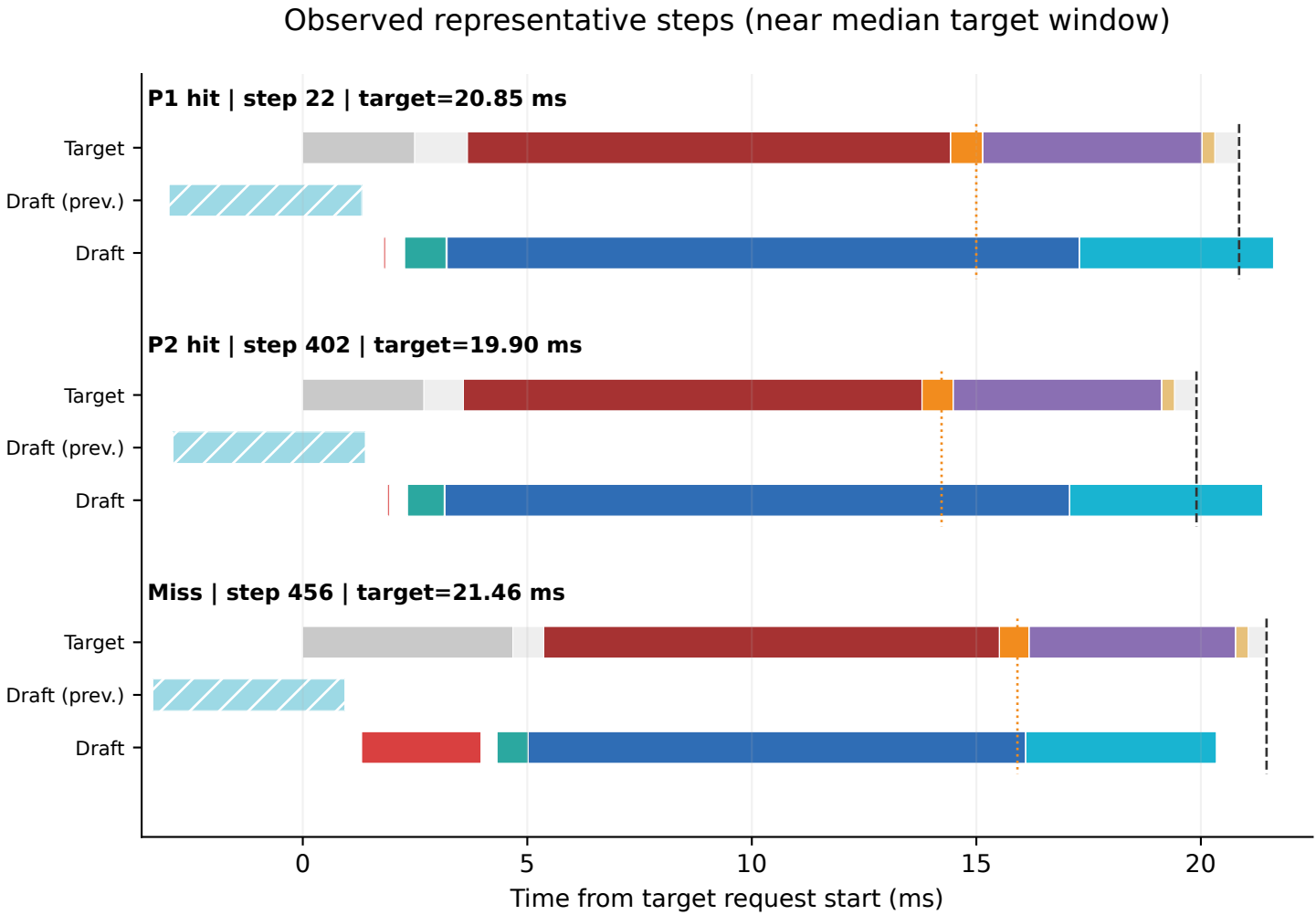
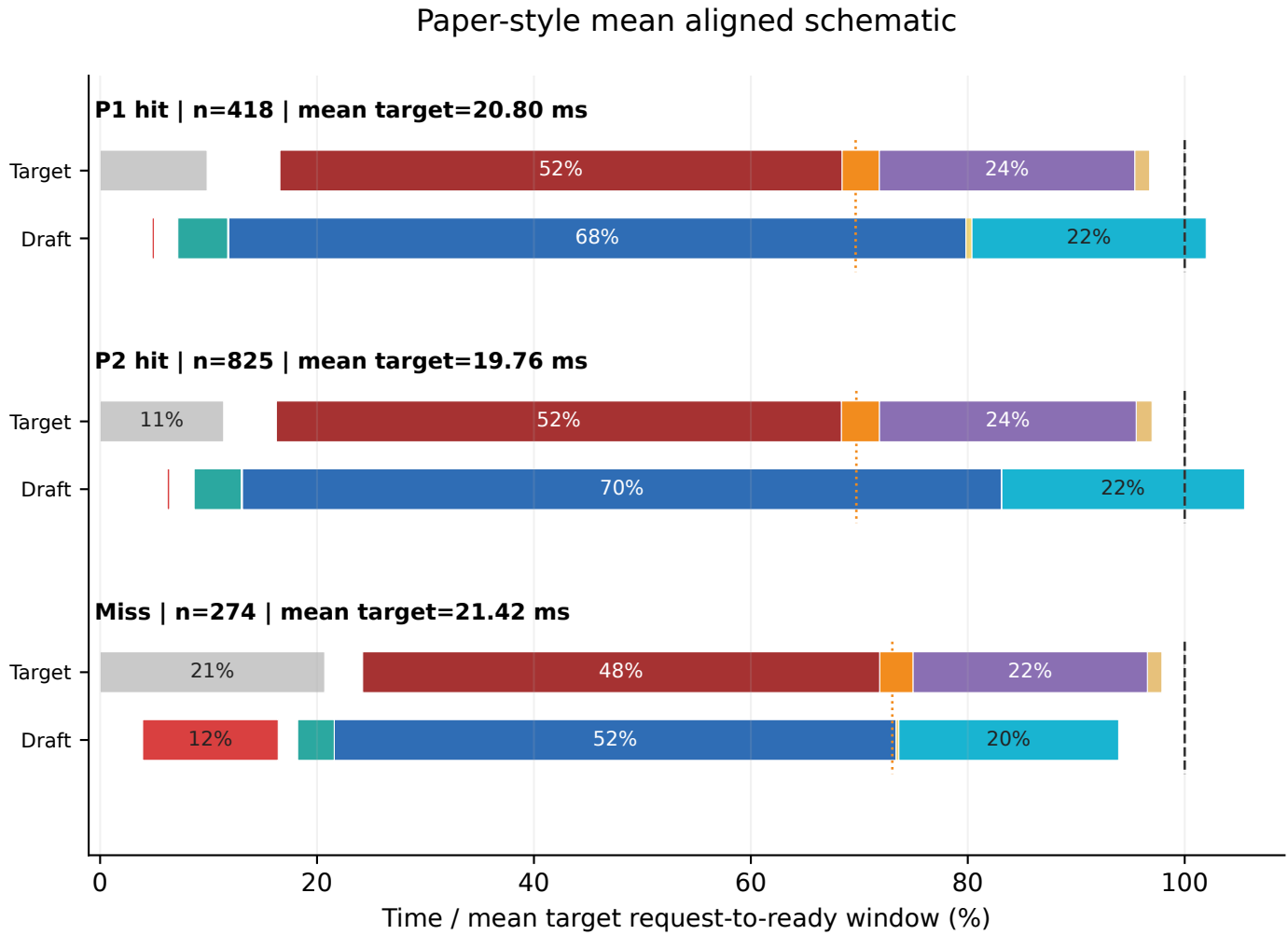
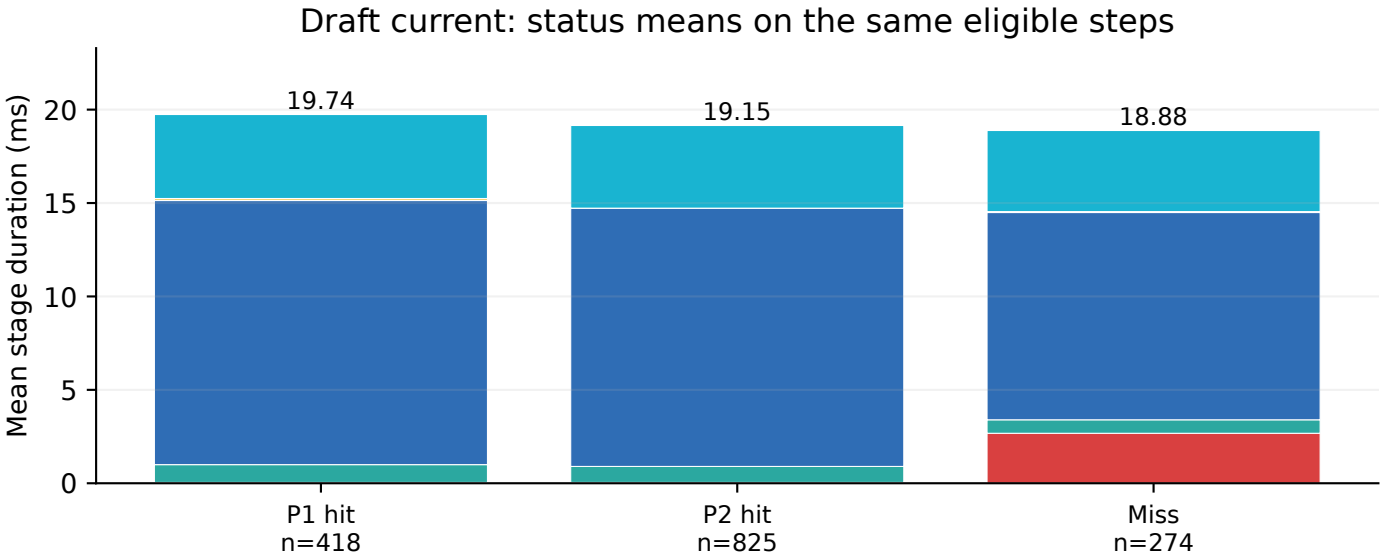
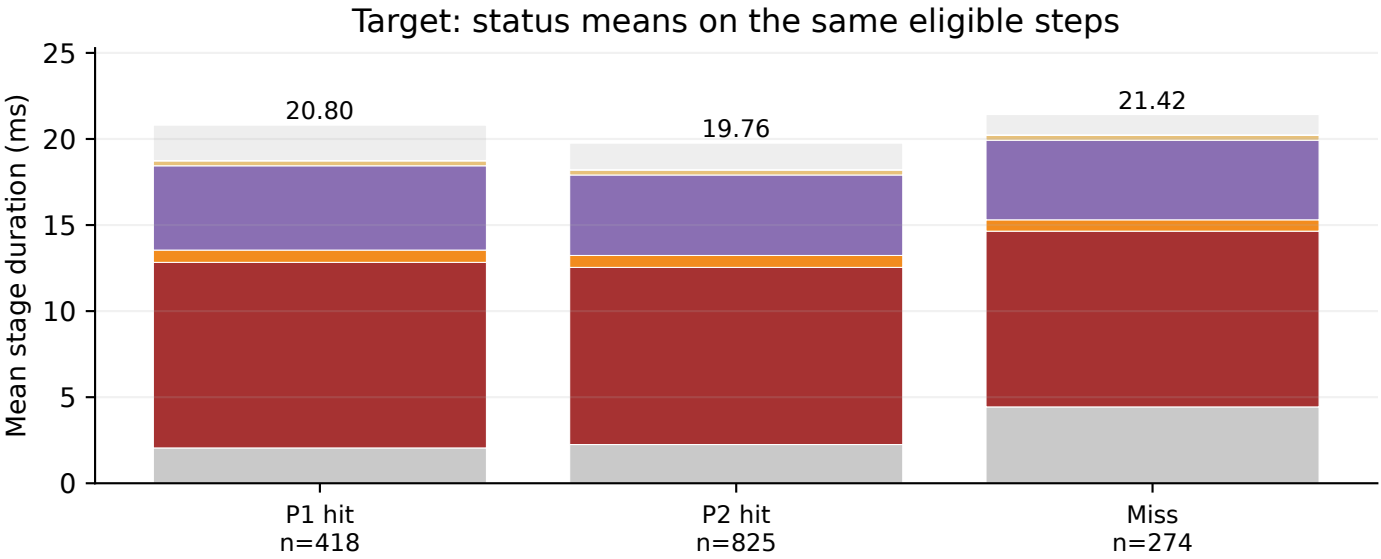
llama2_b1_refine_e21_k12_3_r2
DUET: K1/K2=12/3, exit=21 (0-based), C=3, N1/N2=16/6, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1720 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e21_k12_4_r2
DUET: K1/K2=12/4, exit=21 (0-based), C=3, N1/N2=16/8, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1517 eligible steps

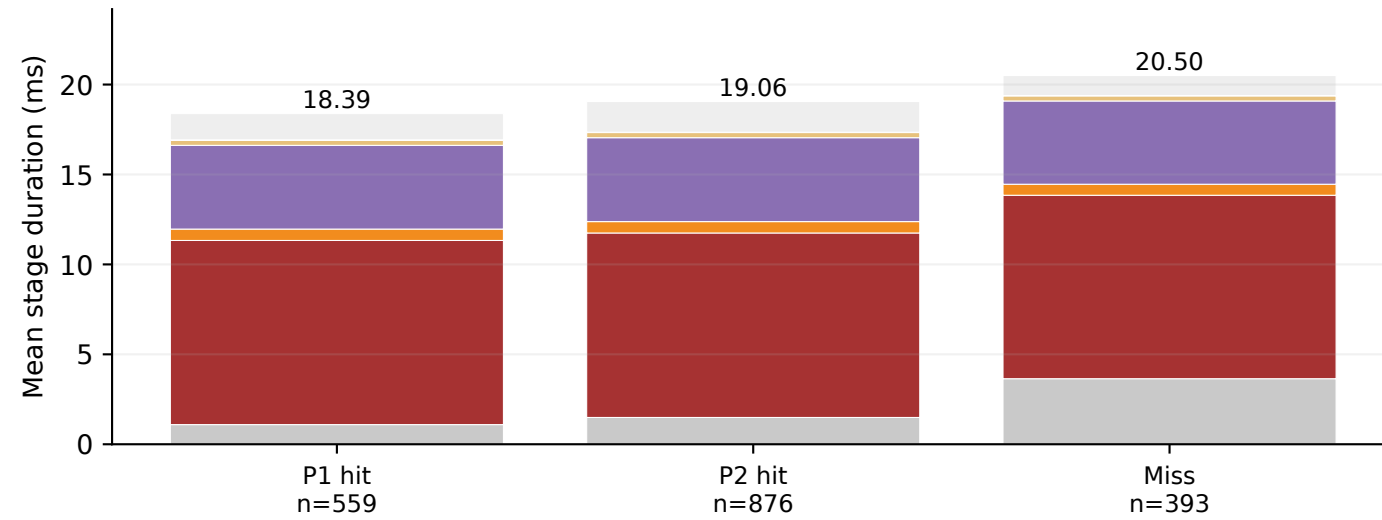


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

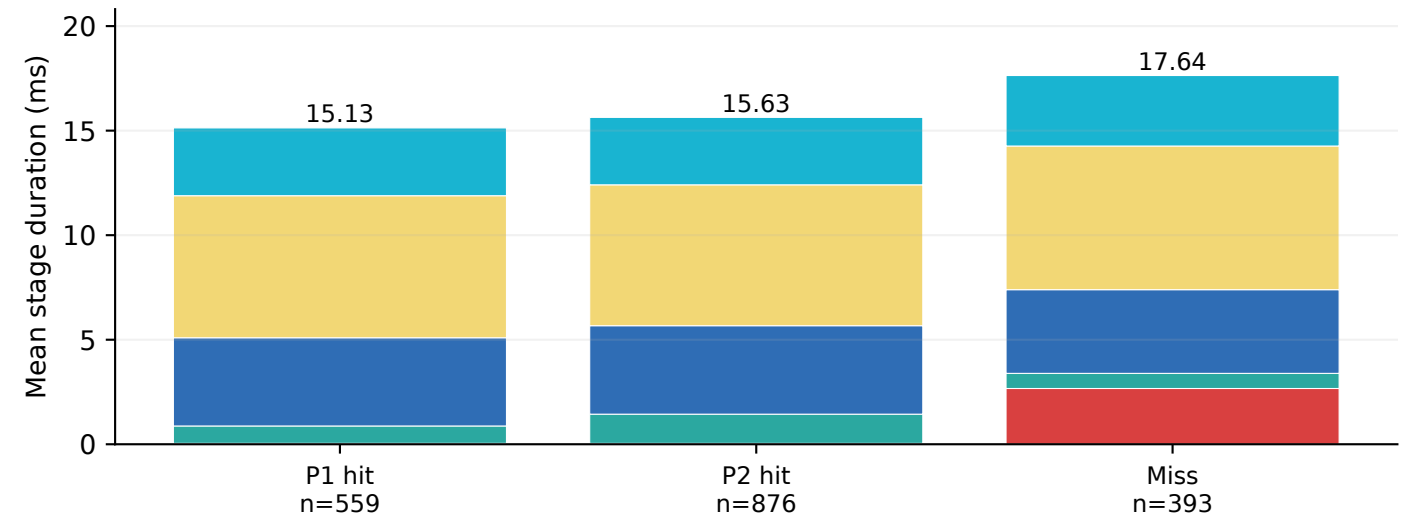
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e21_k4_3_r2
DUET: K1/K2=4/3, exit=21 (0-based), C=3, N1/N2=8/6, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1828 eligible steps

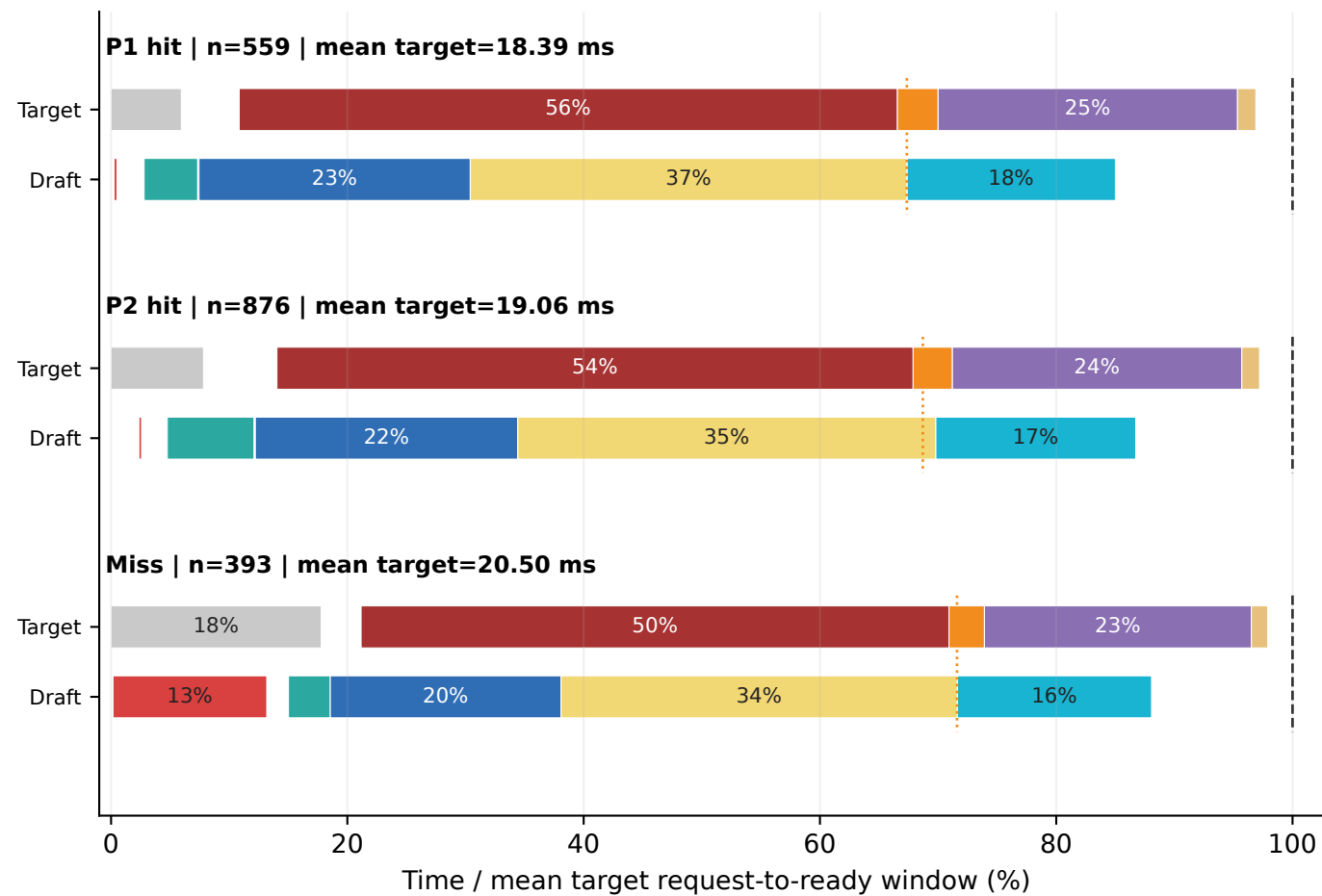
Target: status means on the same eligible steps



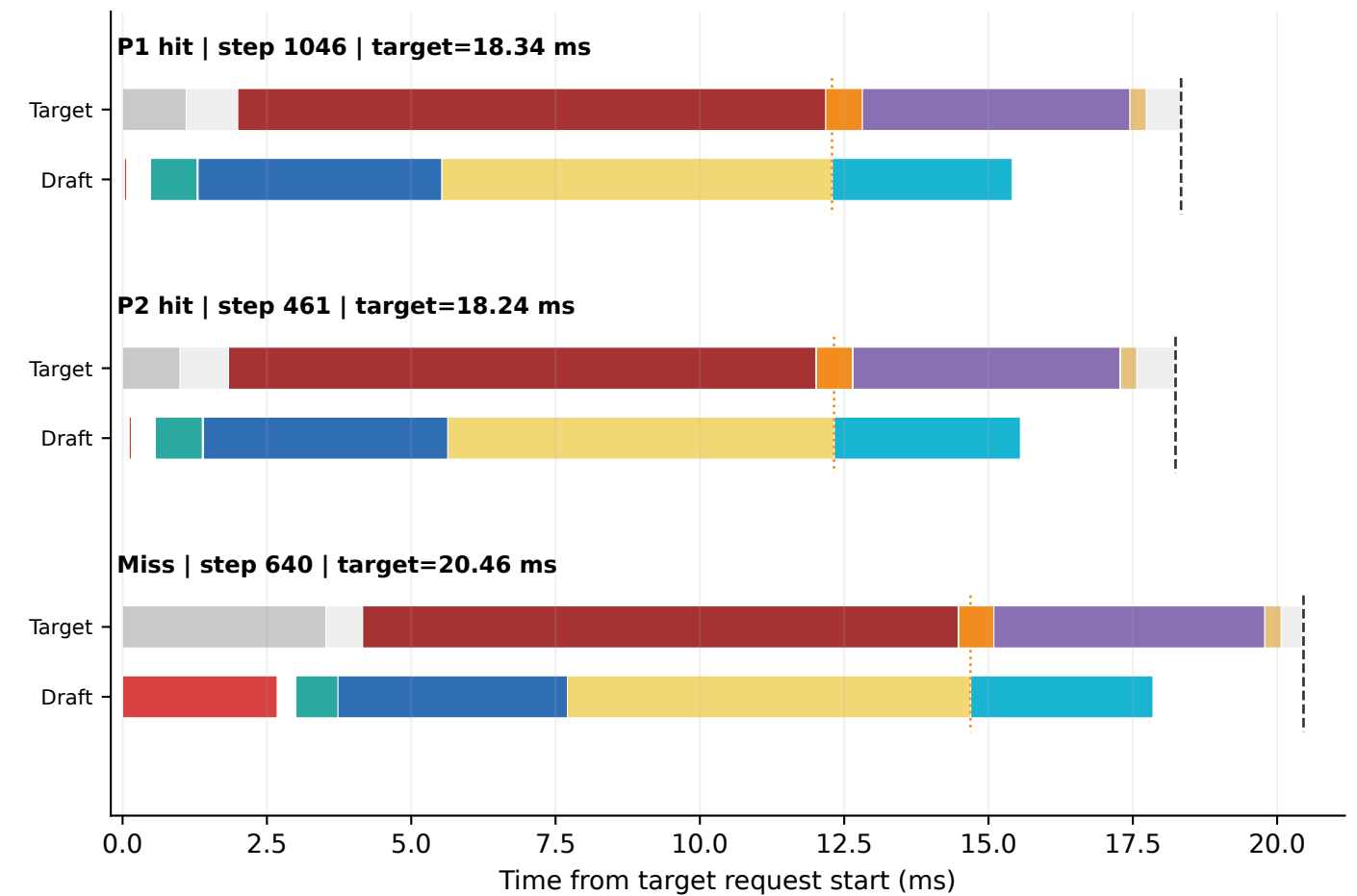
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



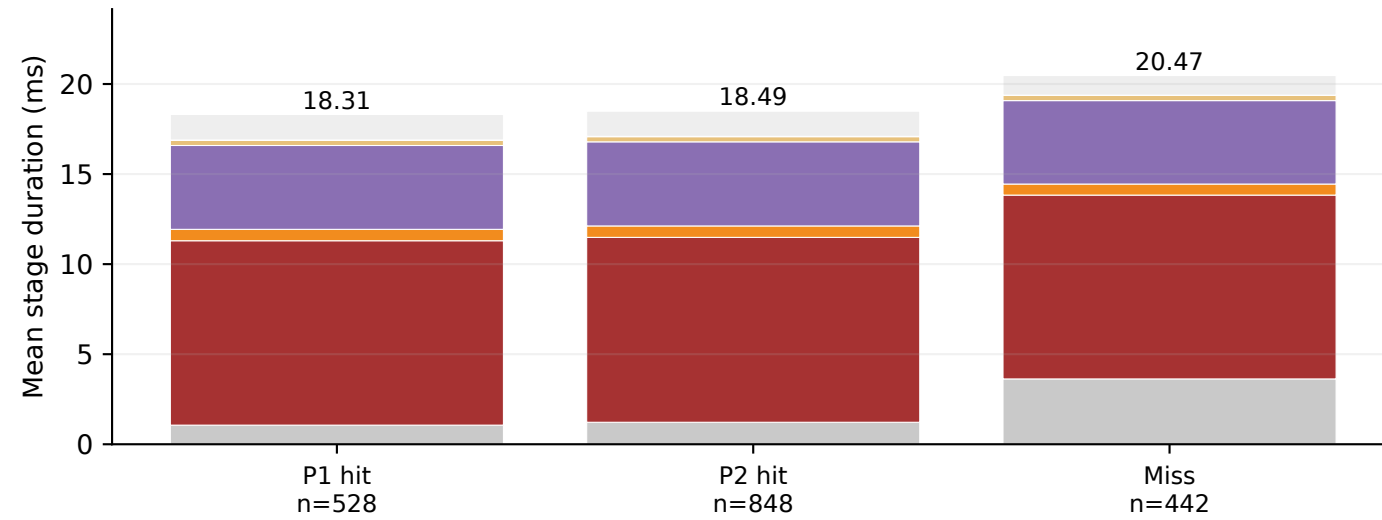
Observed representative steps (near median target window)



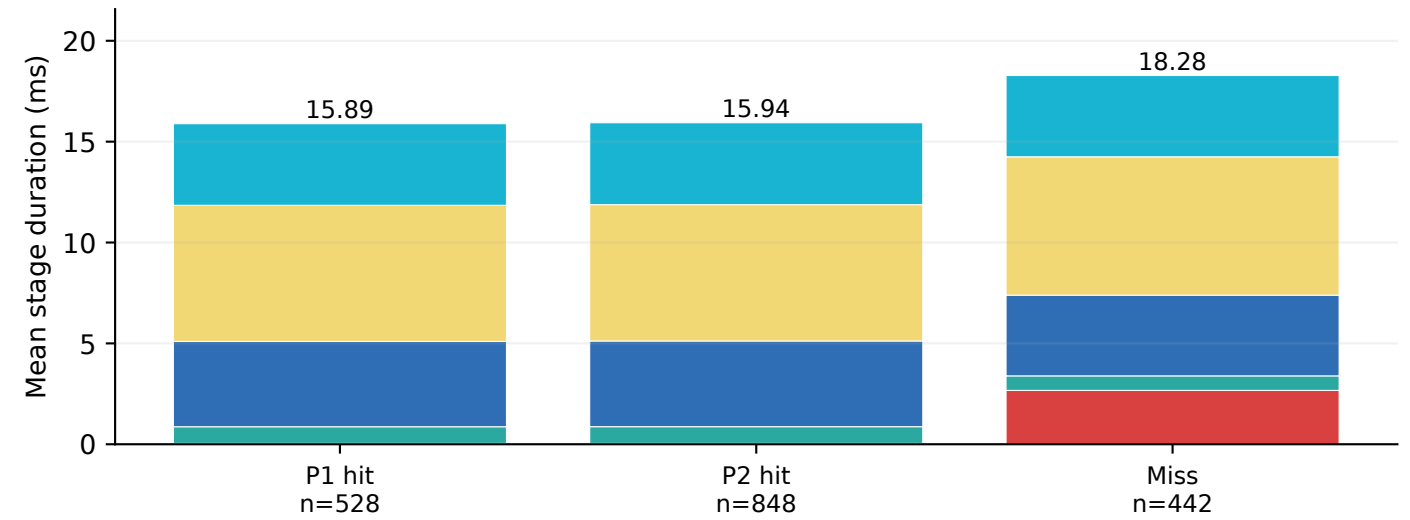
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e21_k4_4_r2
DUET: K1/K2=4/4, exit=21 (0-based), C=3, N1/N2=8/8, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1818 eligible steps

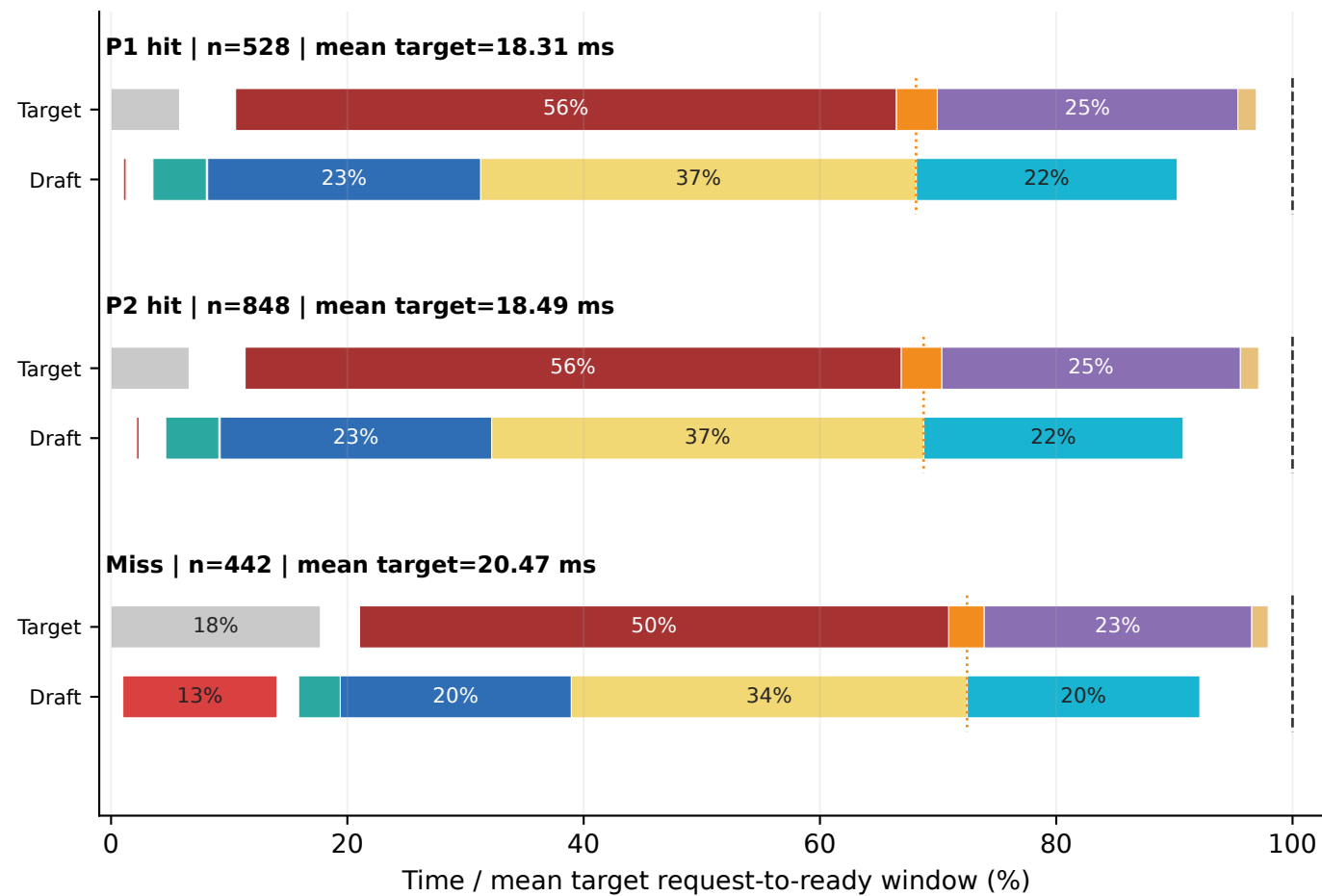
Target: status means on the same eligible steps



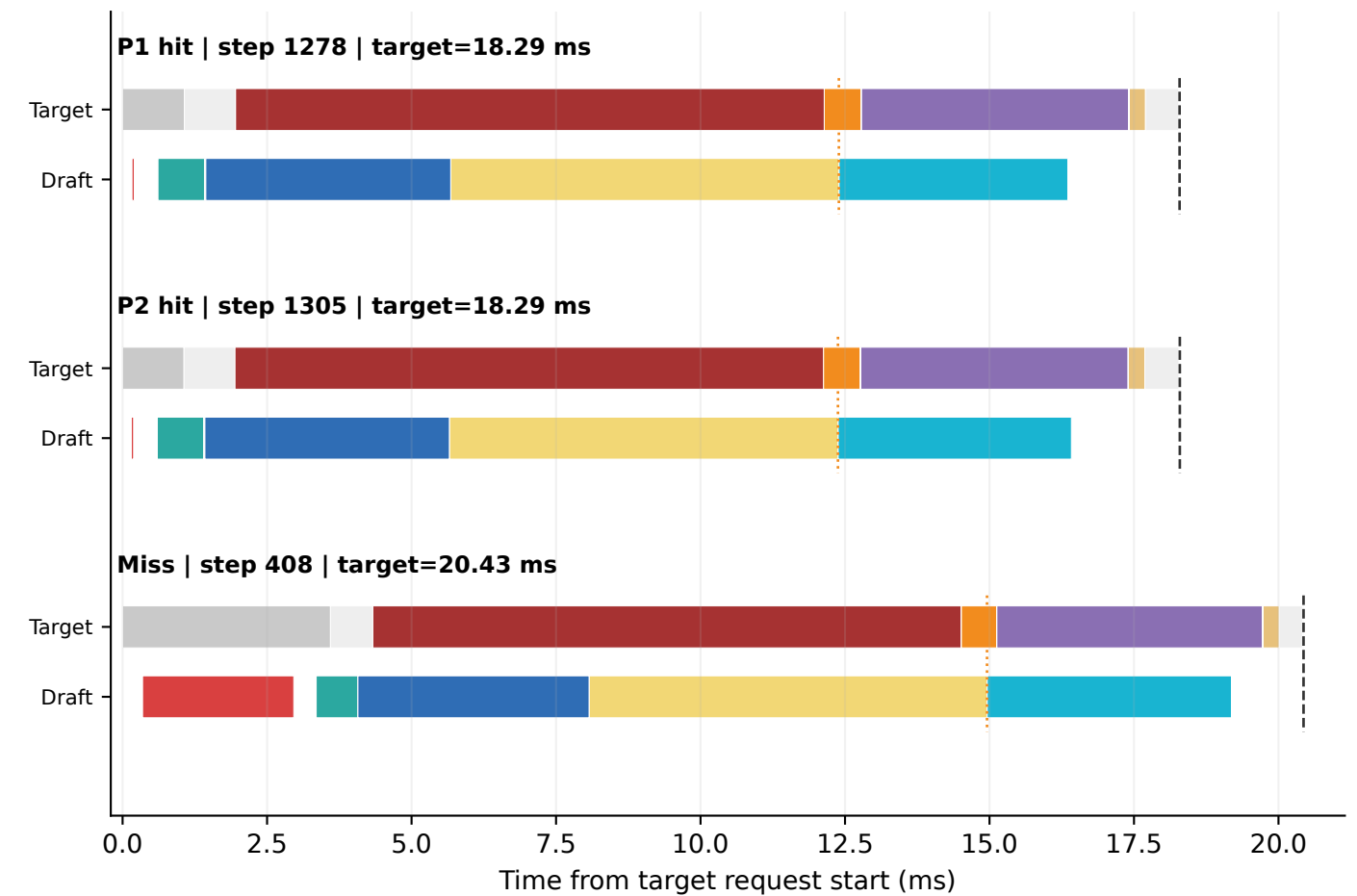
Draft current: status means on the same eligible steps



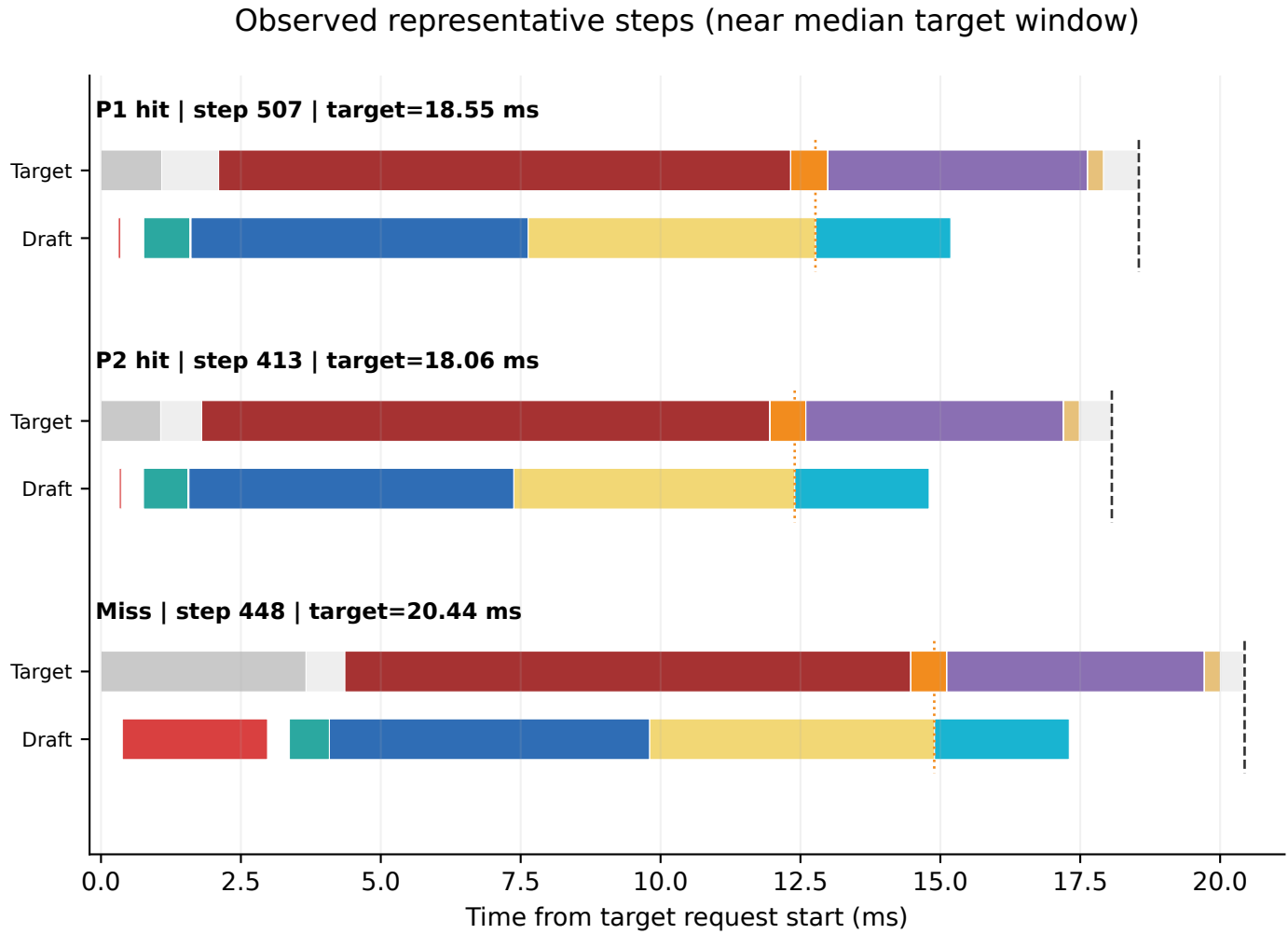
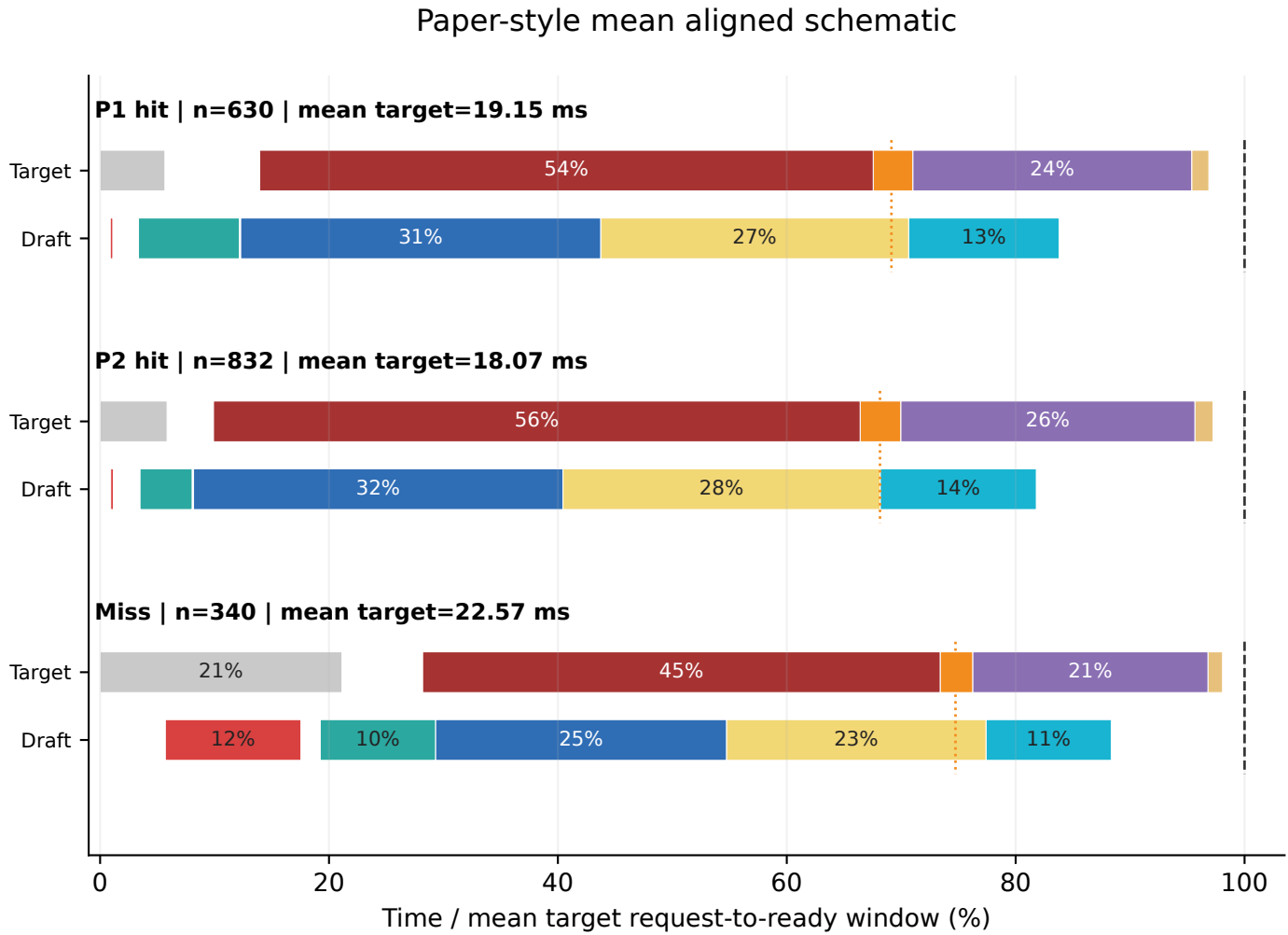
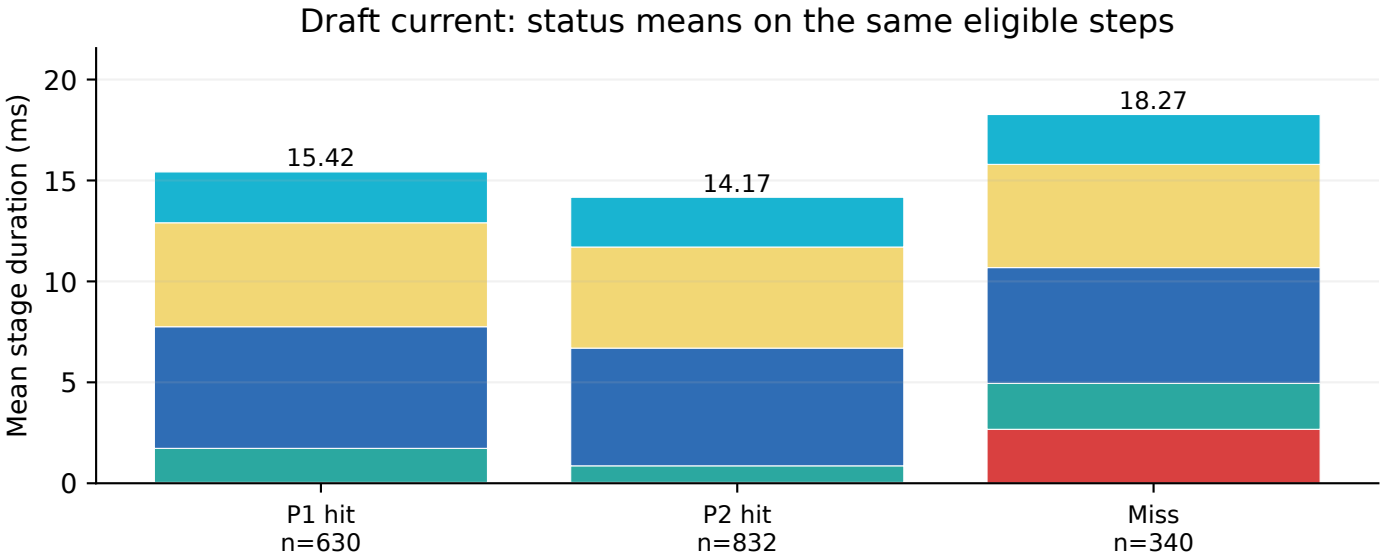
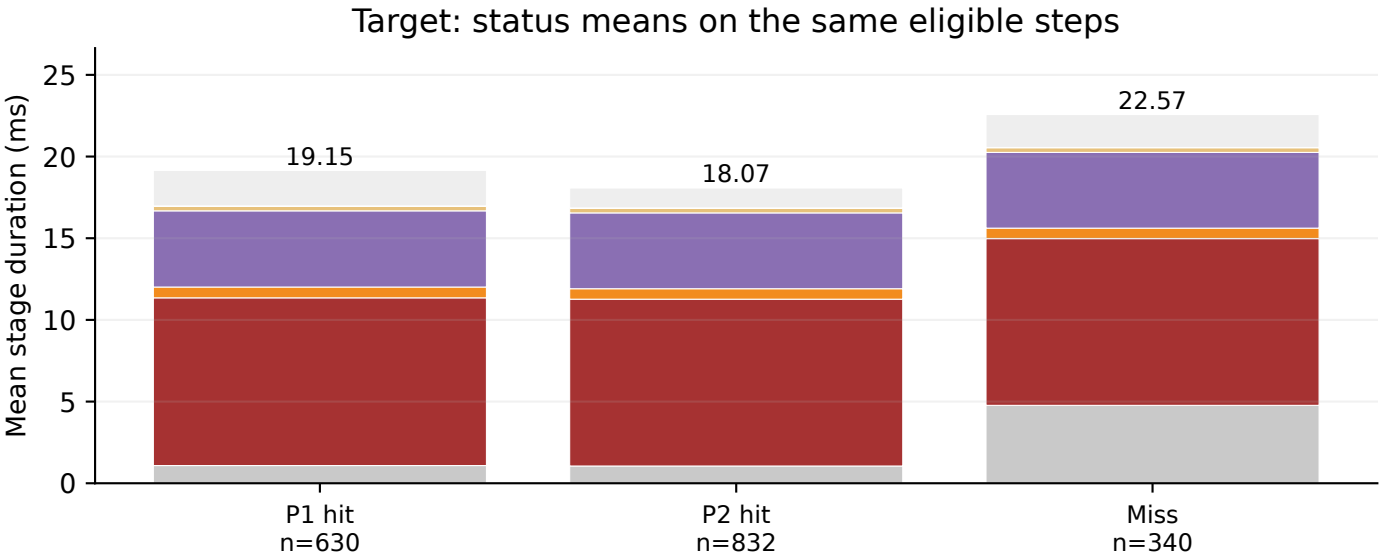
Paper-style mean aligned schematic



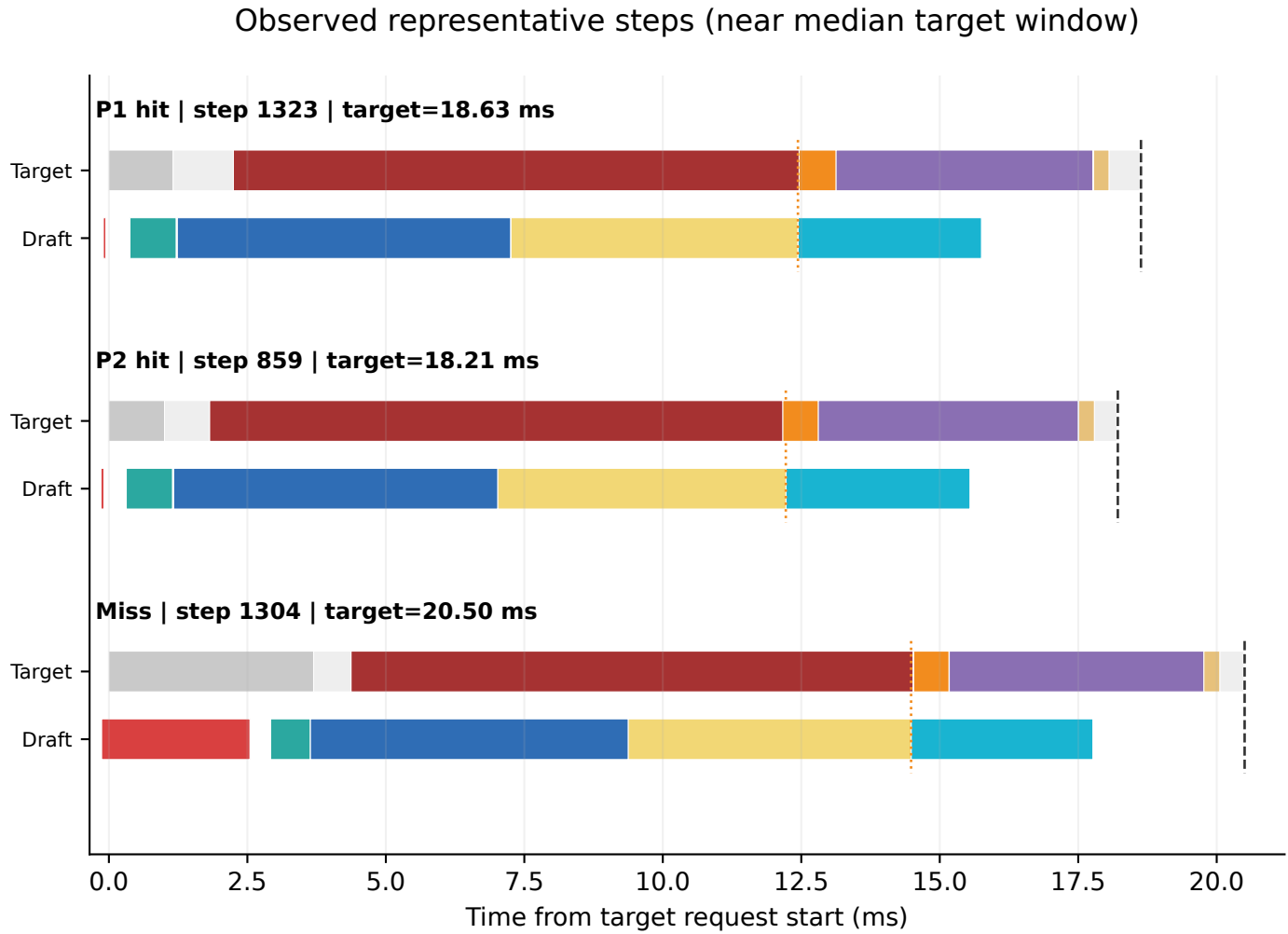
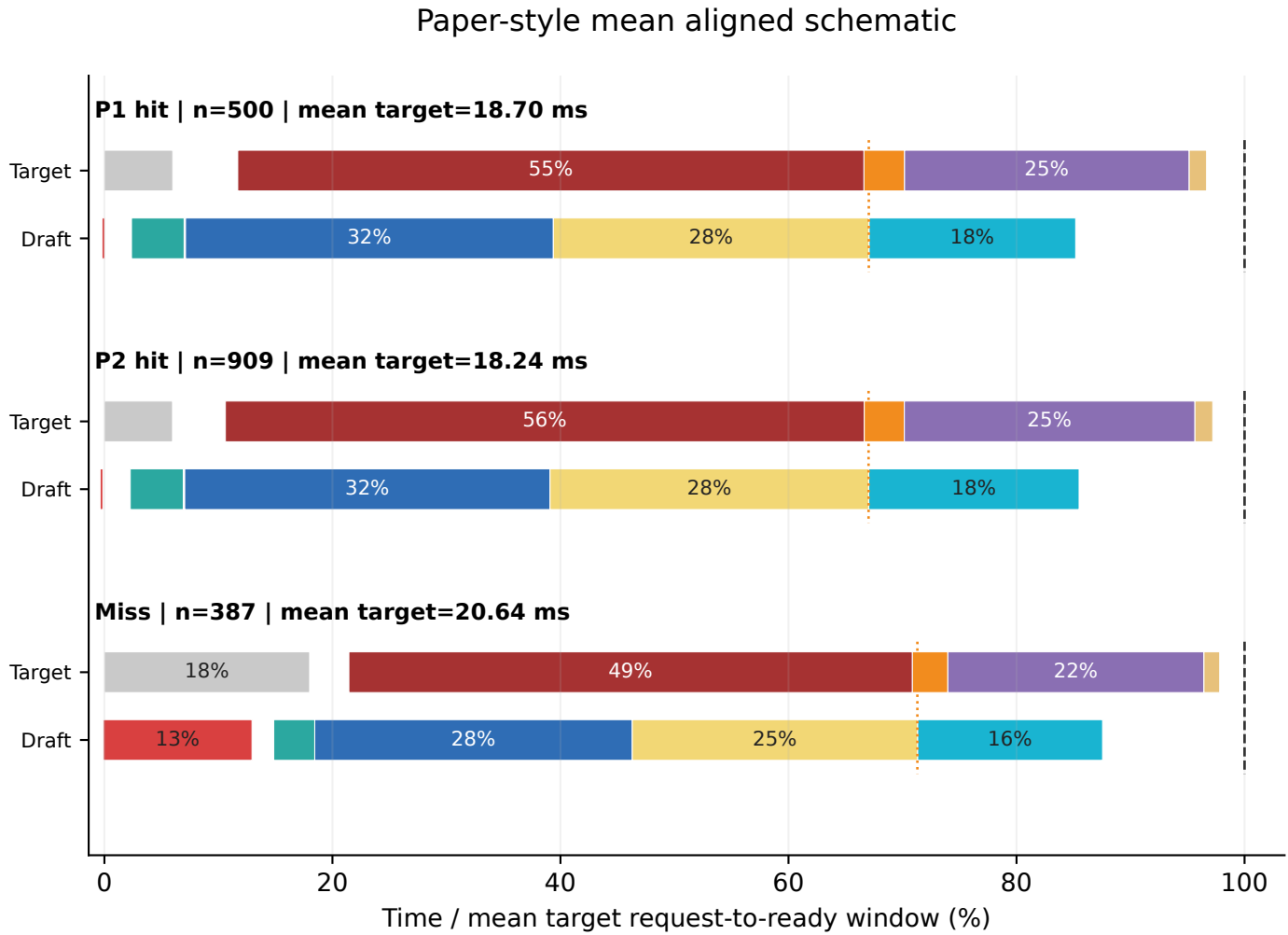
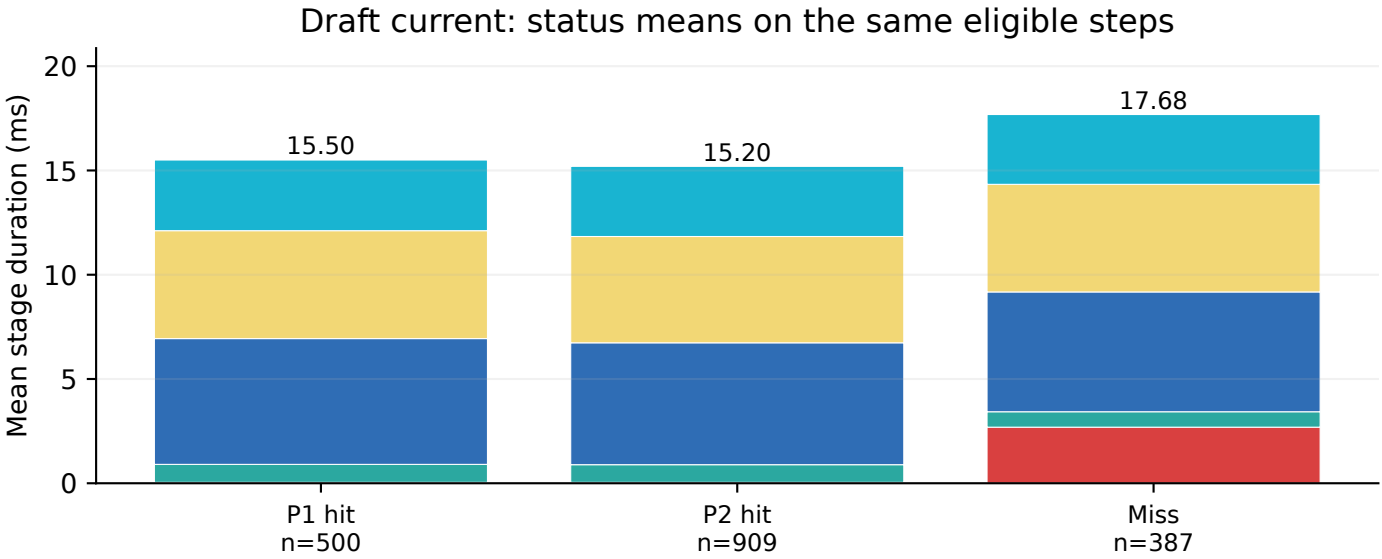
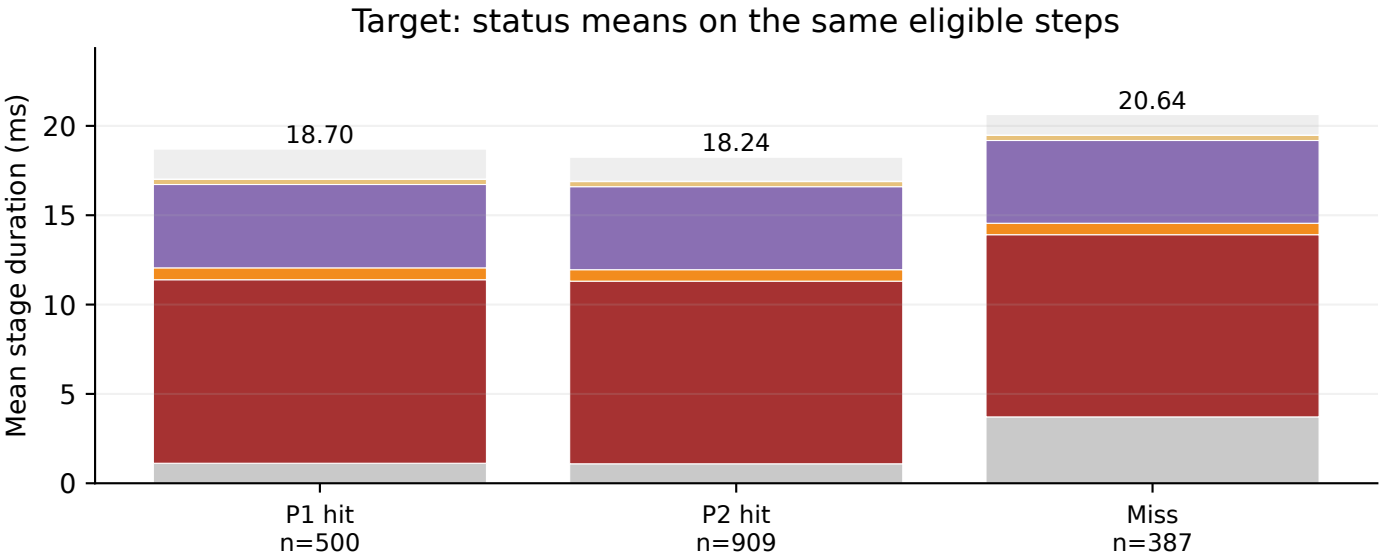
Observed representative steps (near median target window)



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



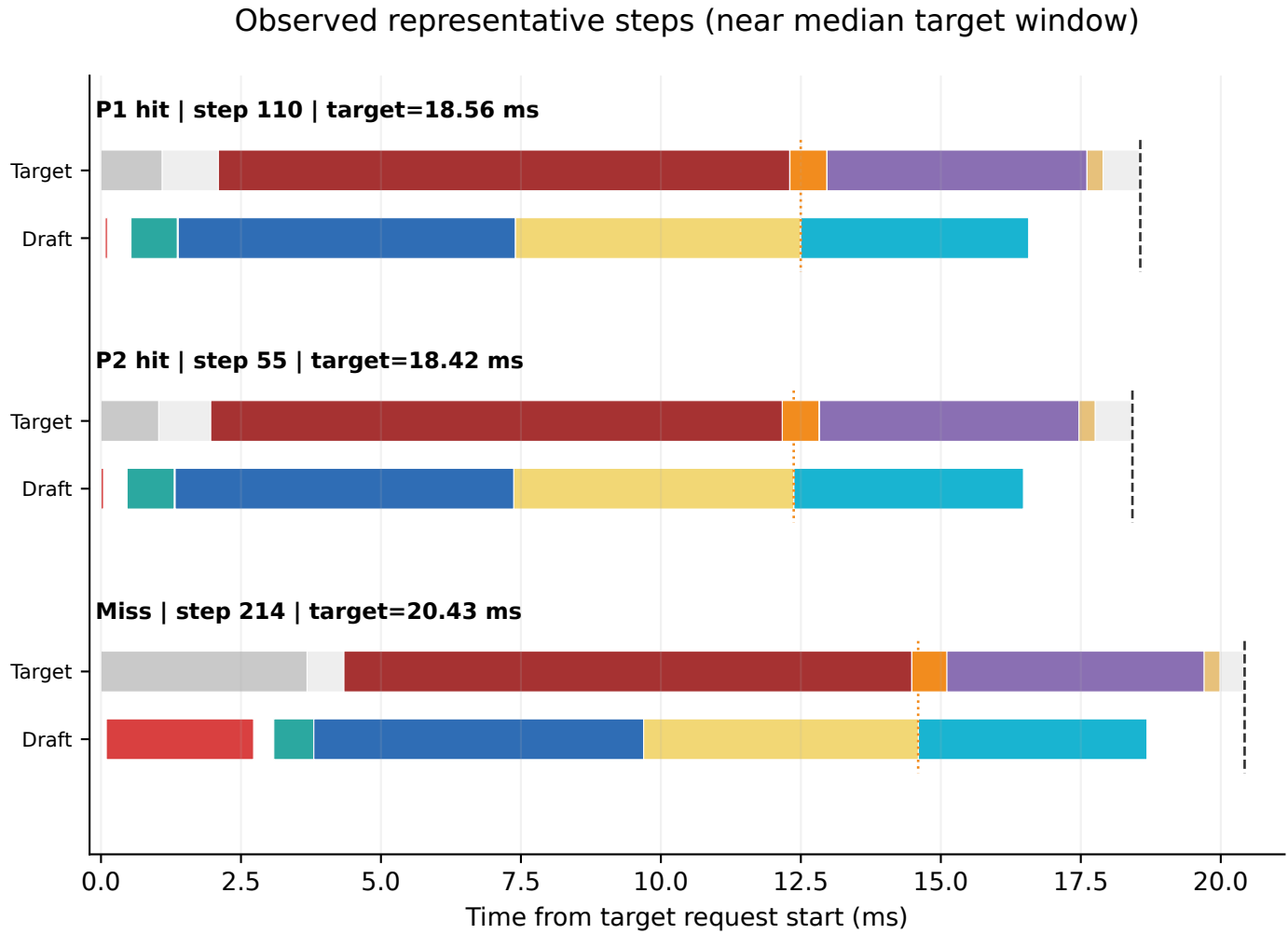
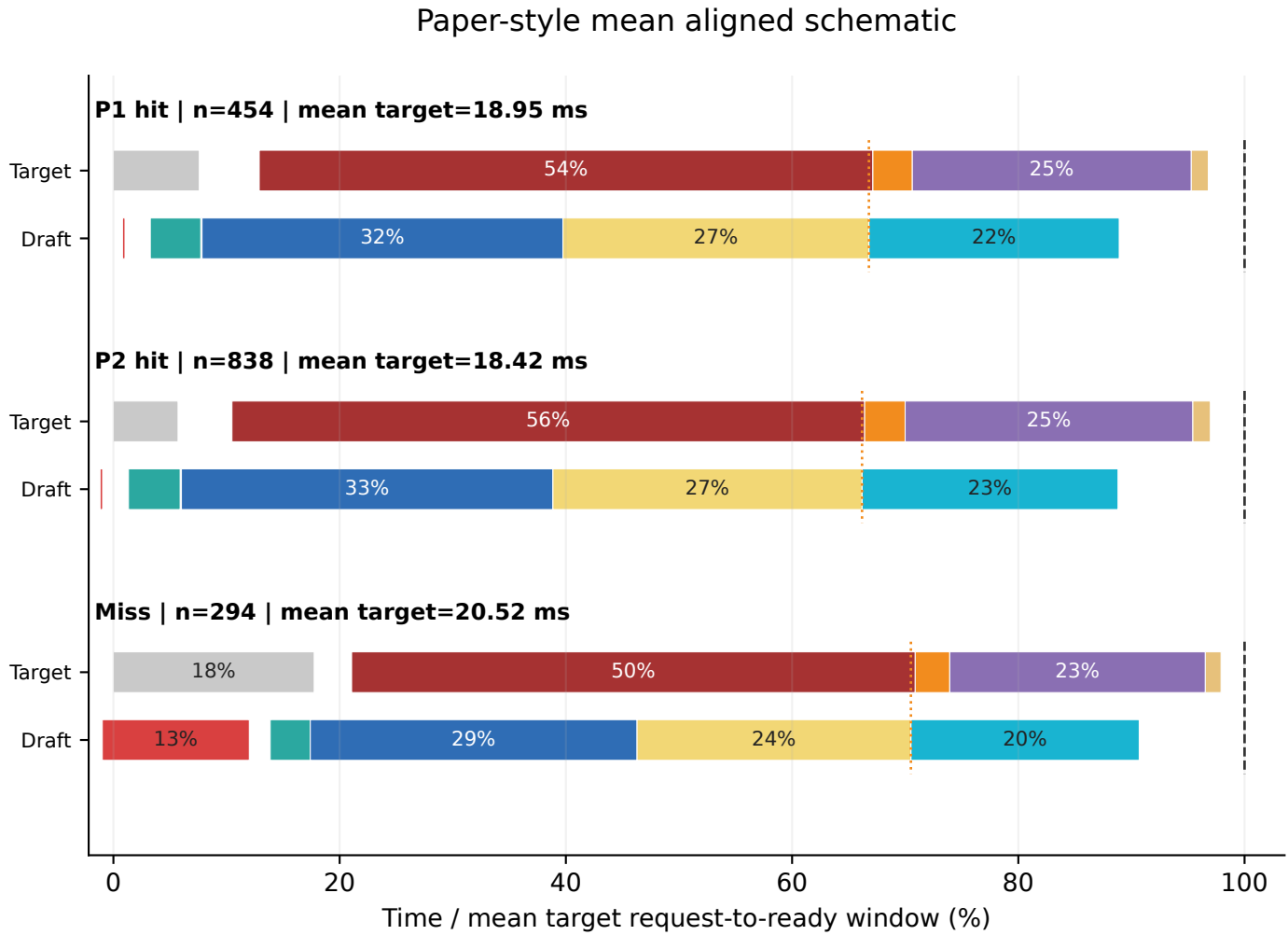
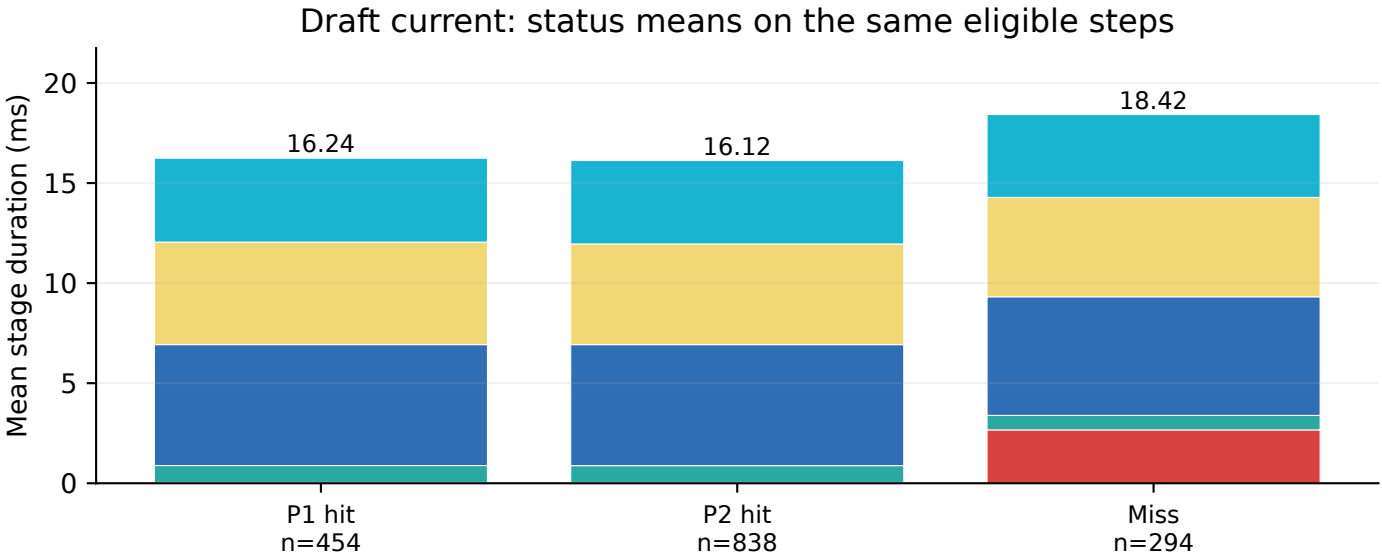
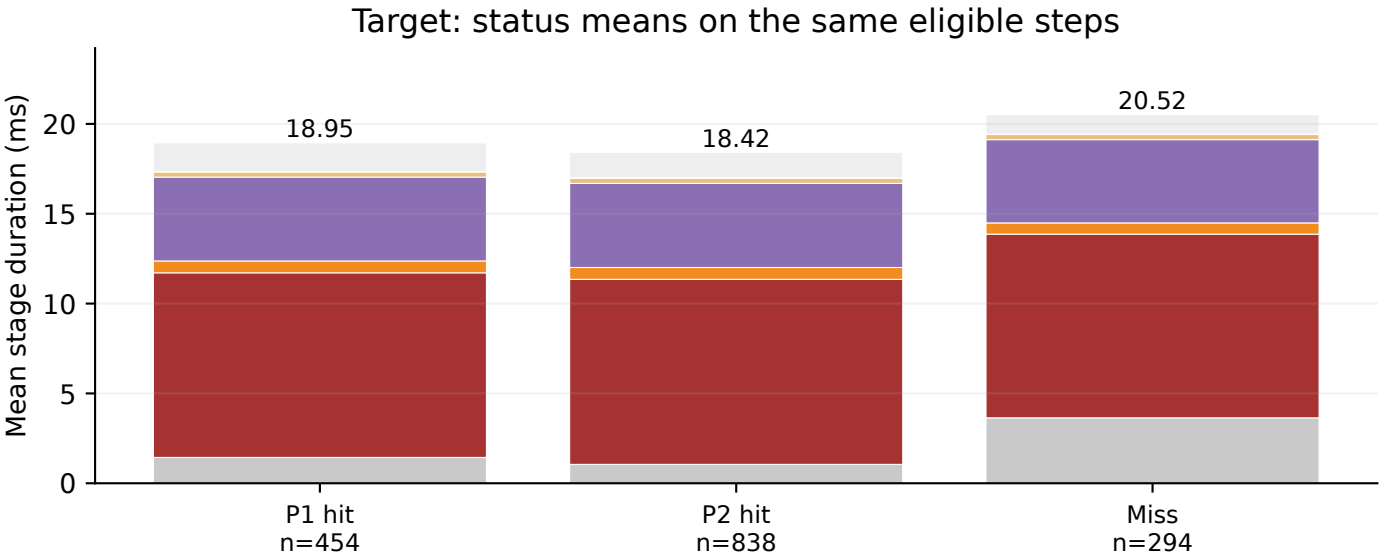
llama2_b1_refine_e21_k6_3_r2
DUET: K1/K2=6/3, exit=21 (0-based), C=3, N1/N2=12/6, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1796 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

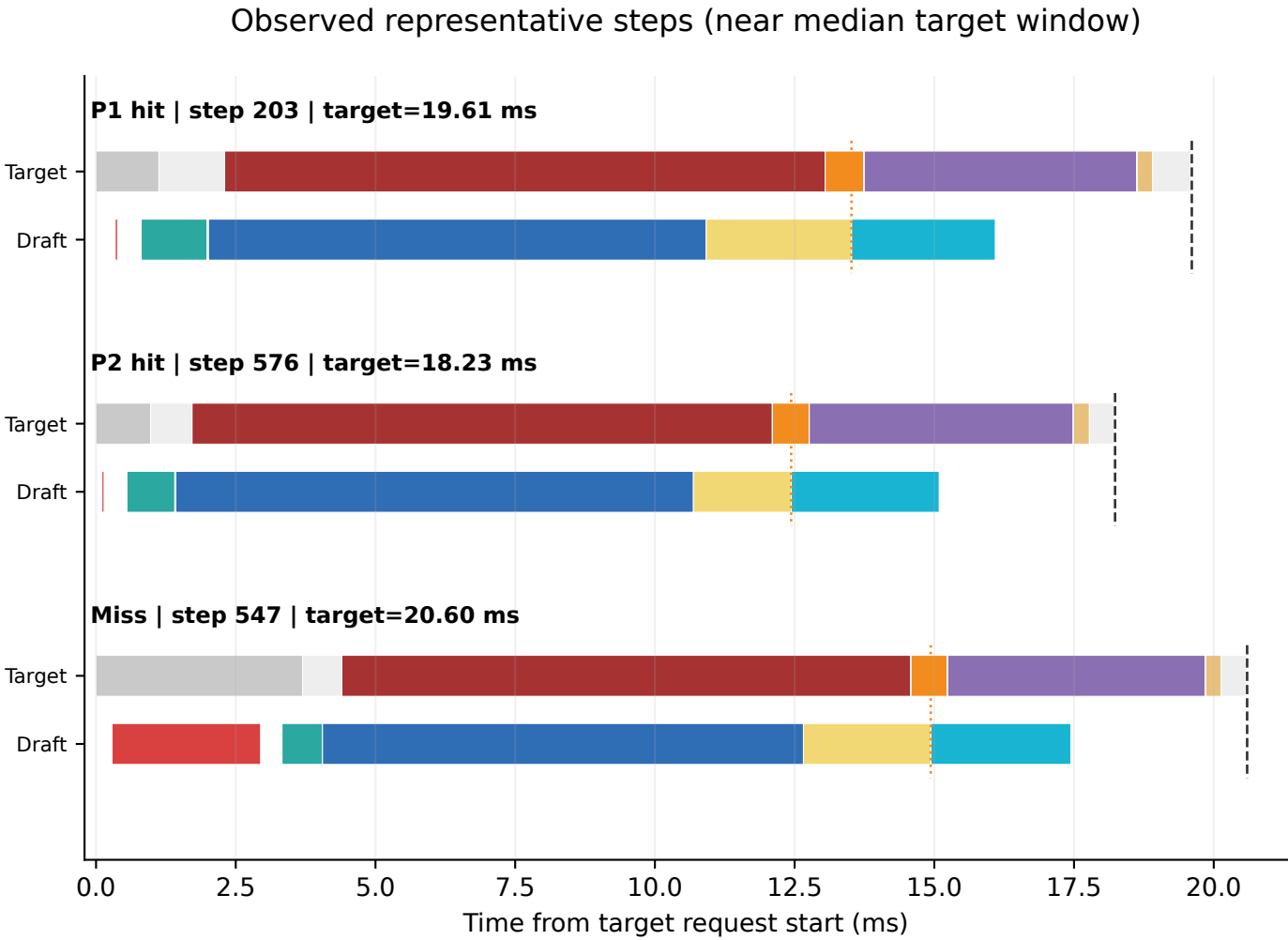
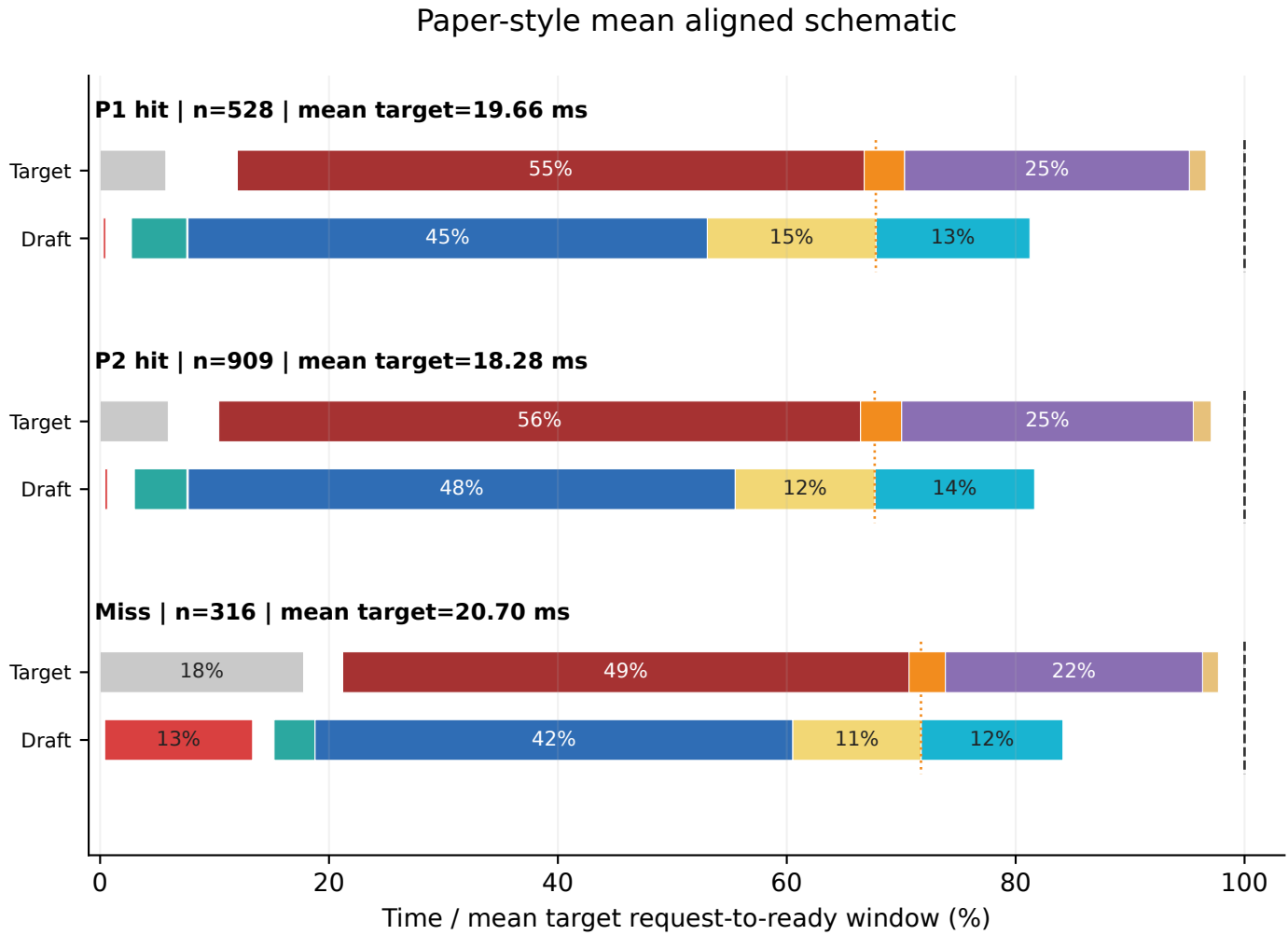
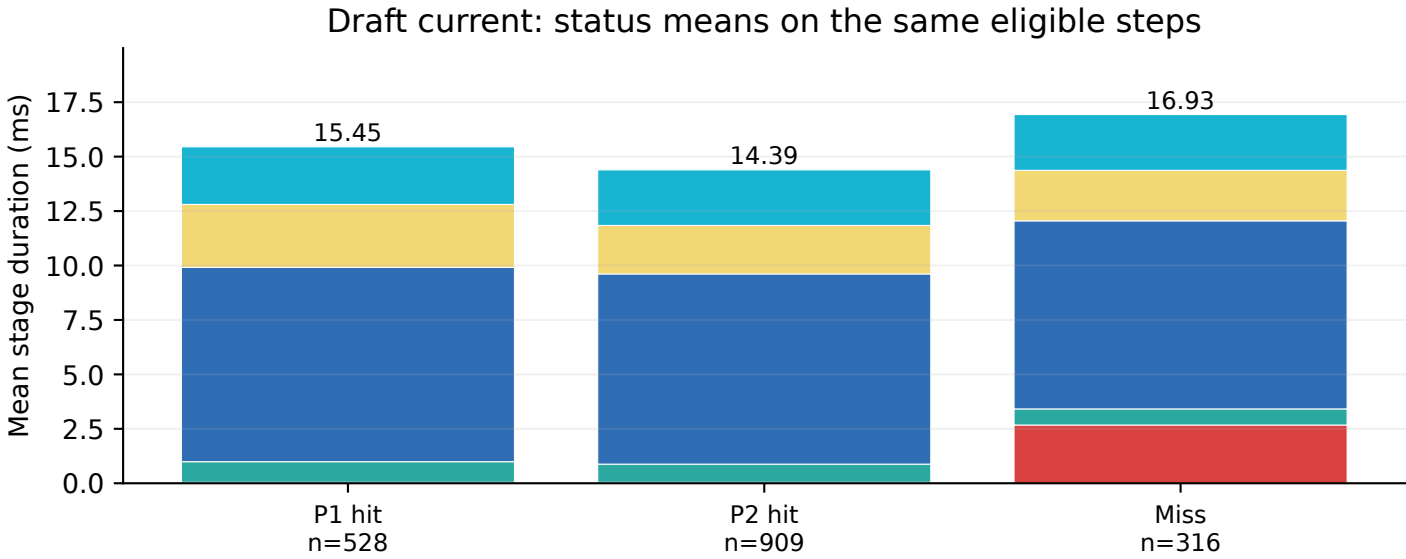
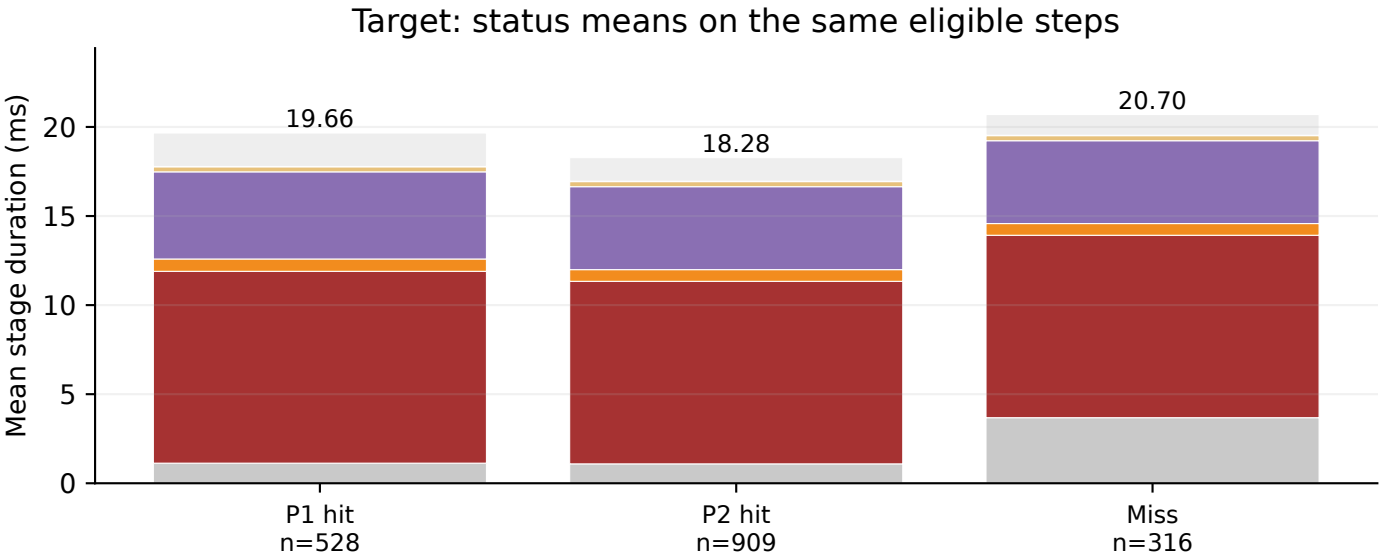
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e21_k6_4_r2
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1586 eligible steps



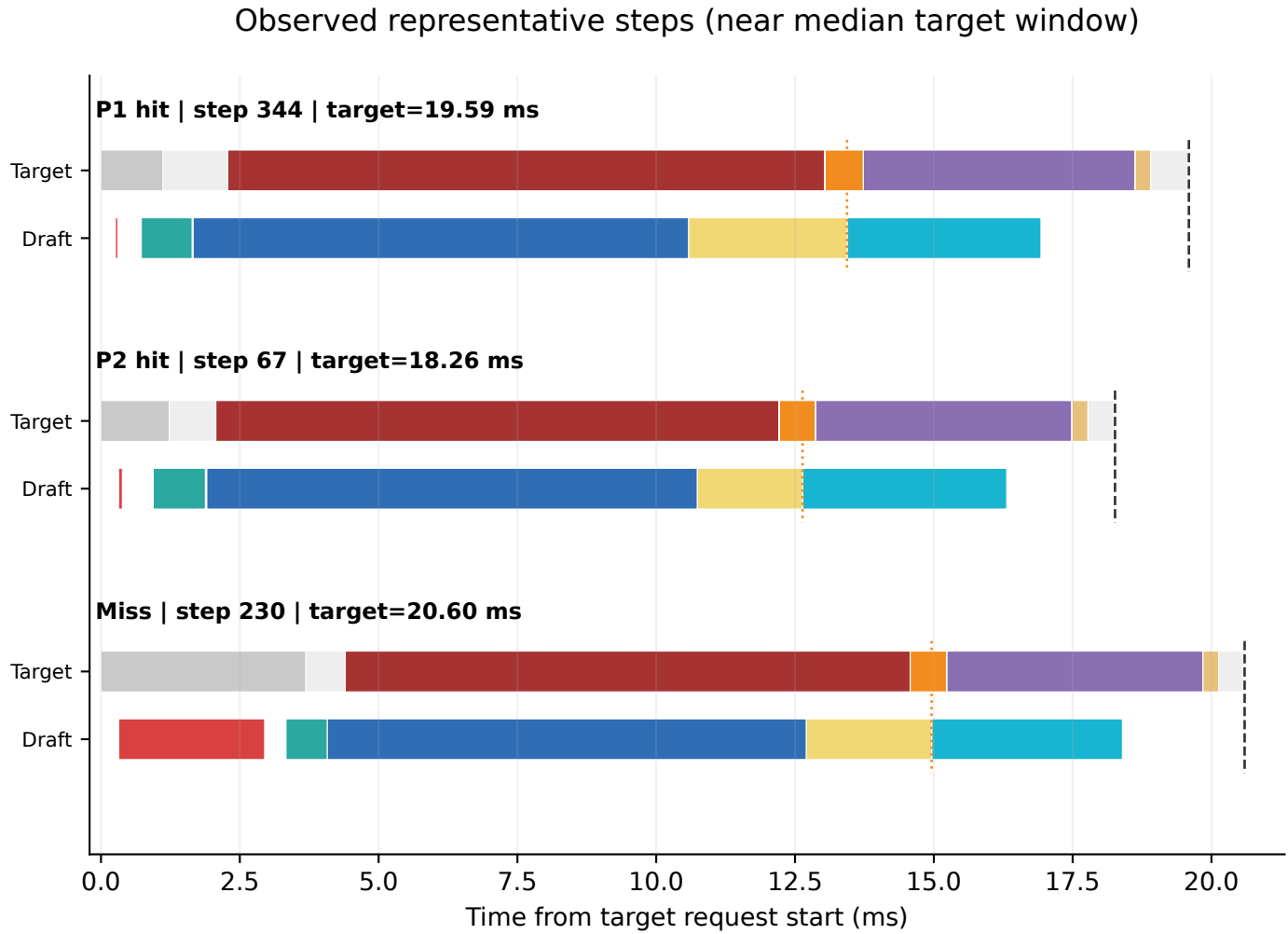
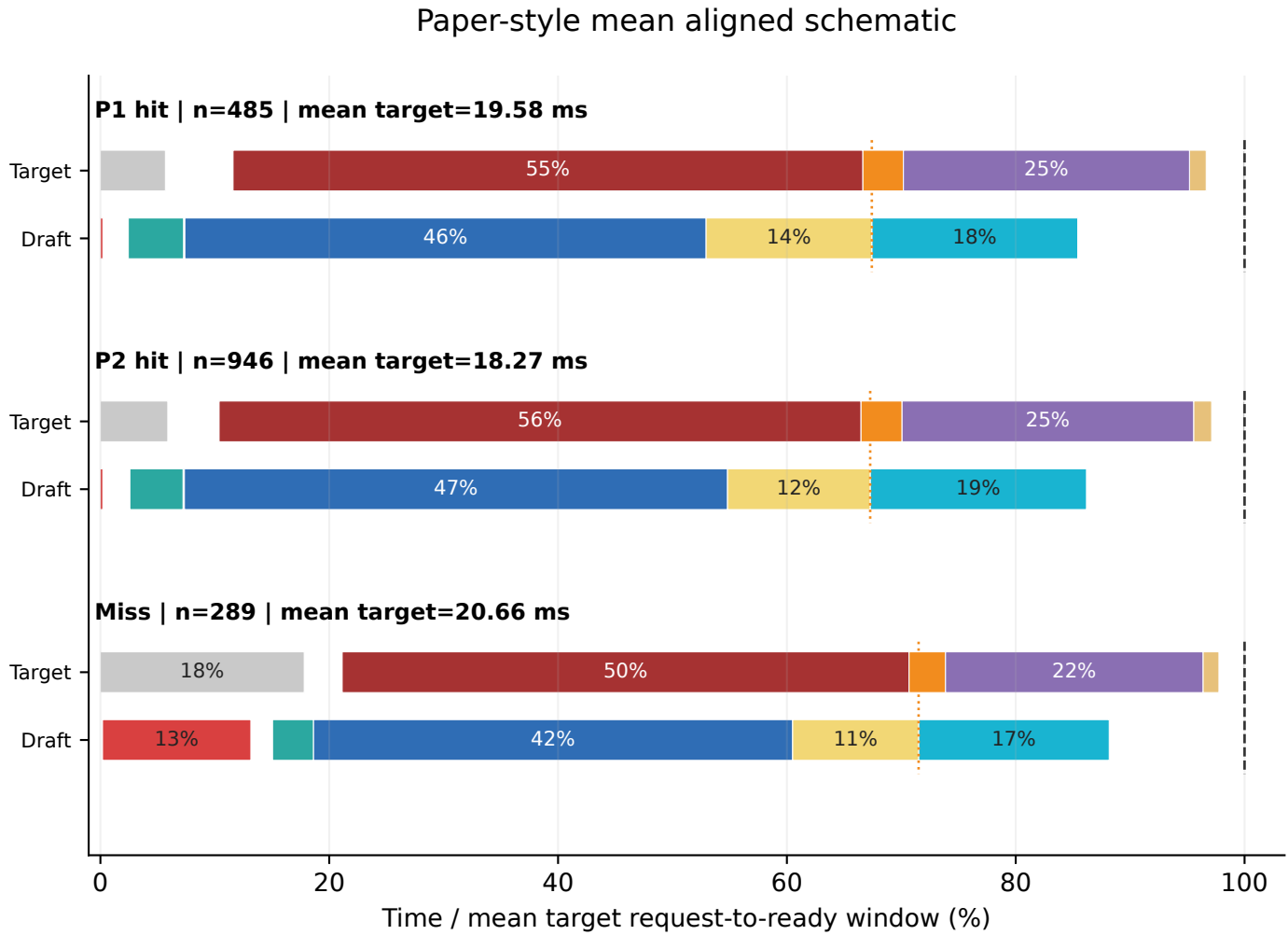
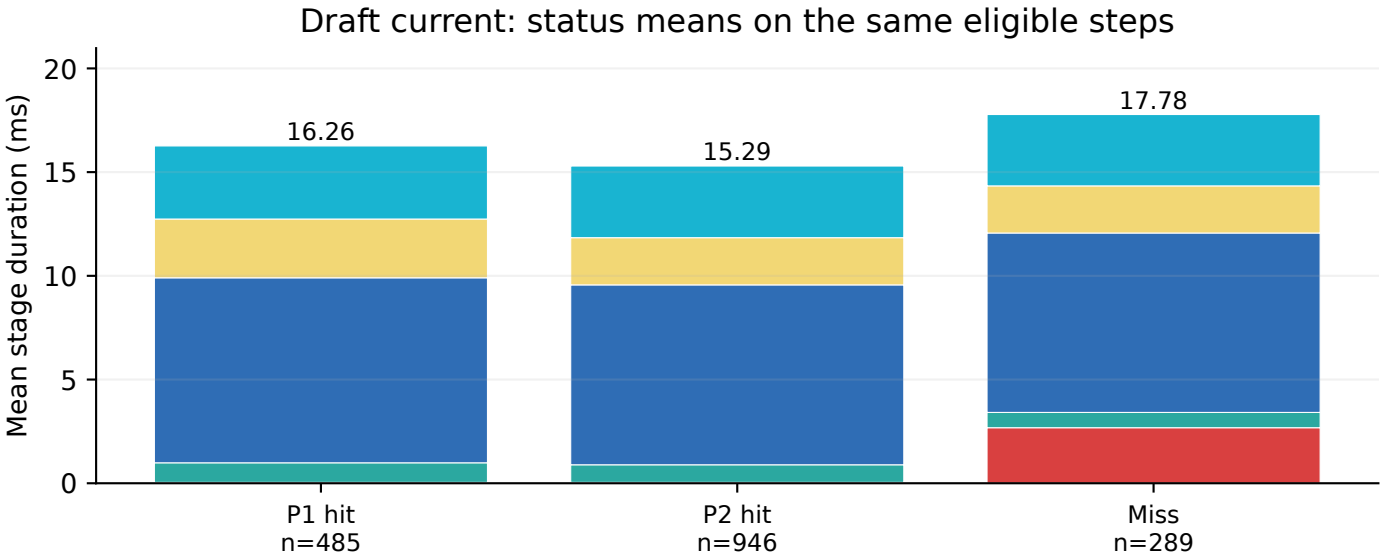
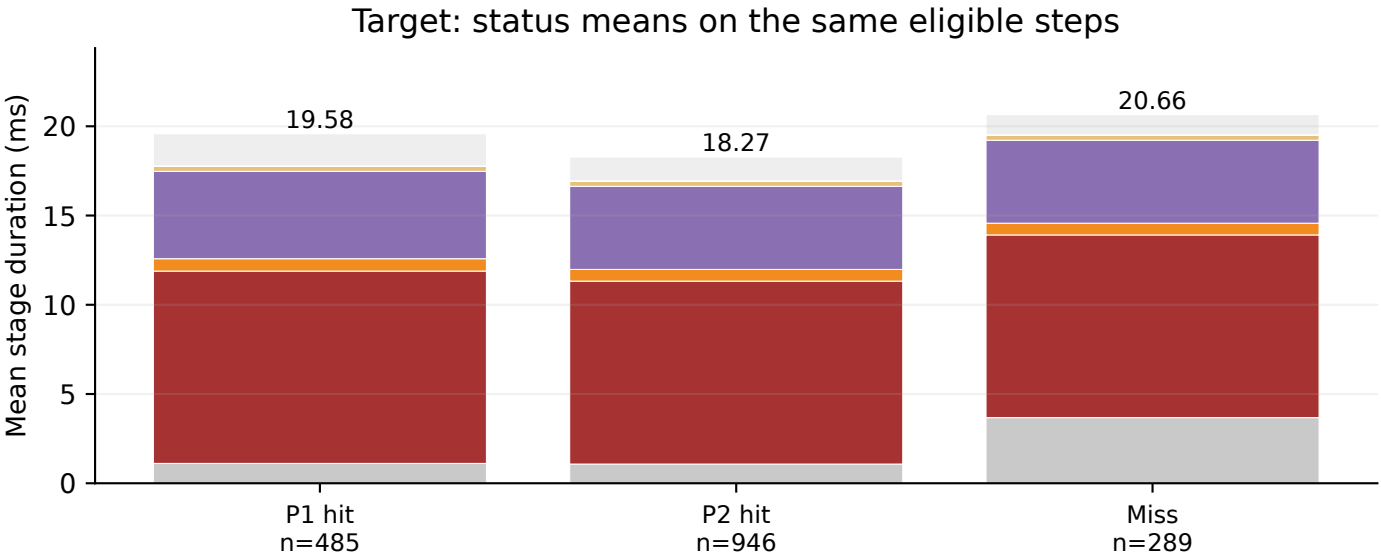
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e21_k8_2_r2
DUET: K1/K2=8/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=9, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1753 eligible steps

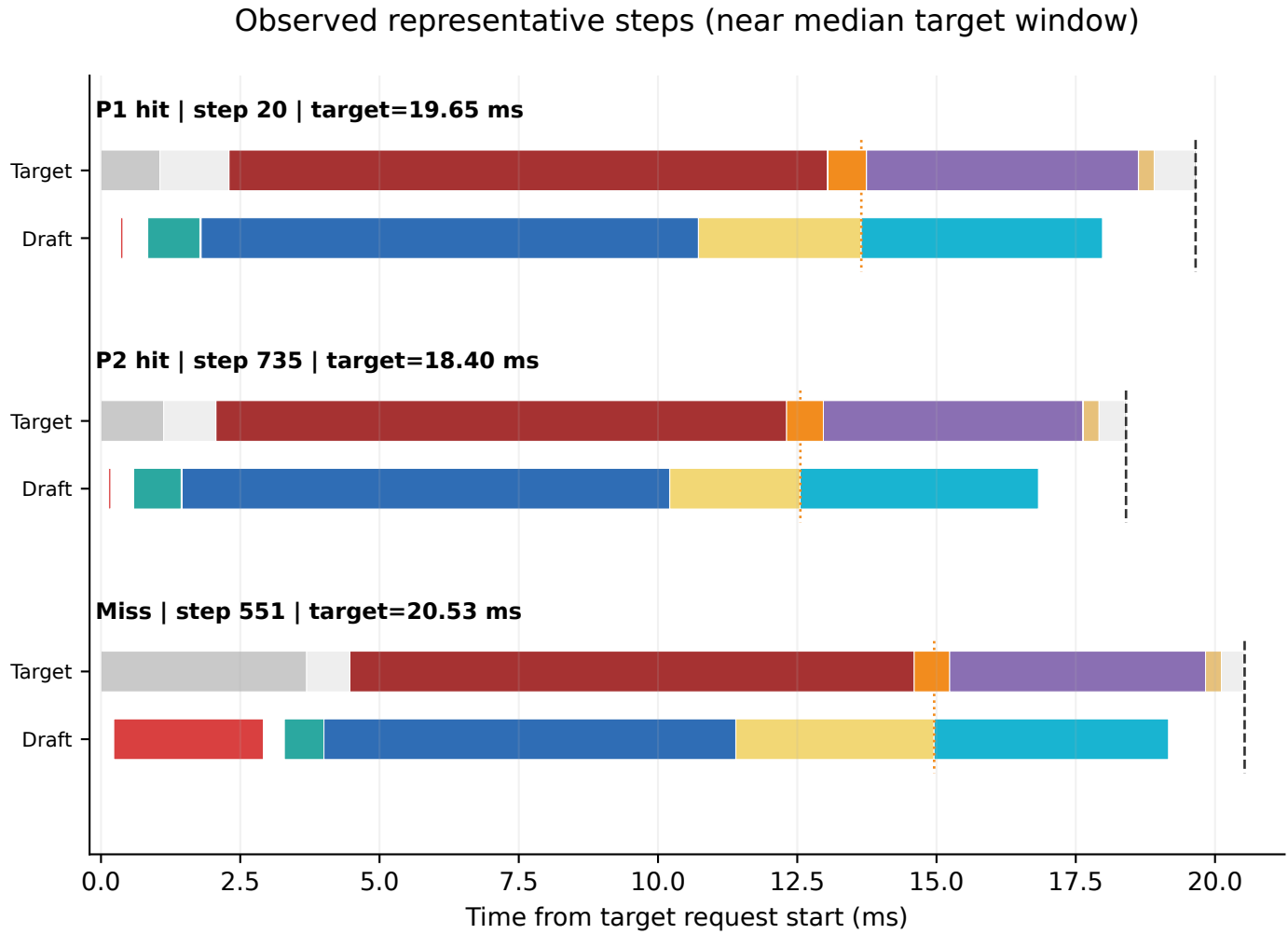
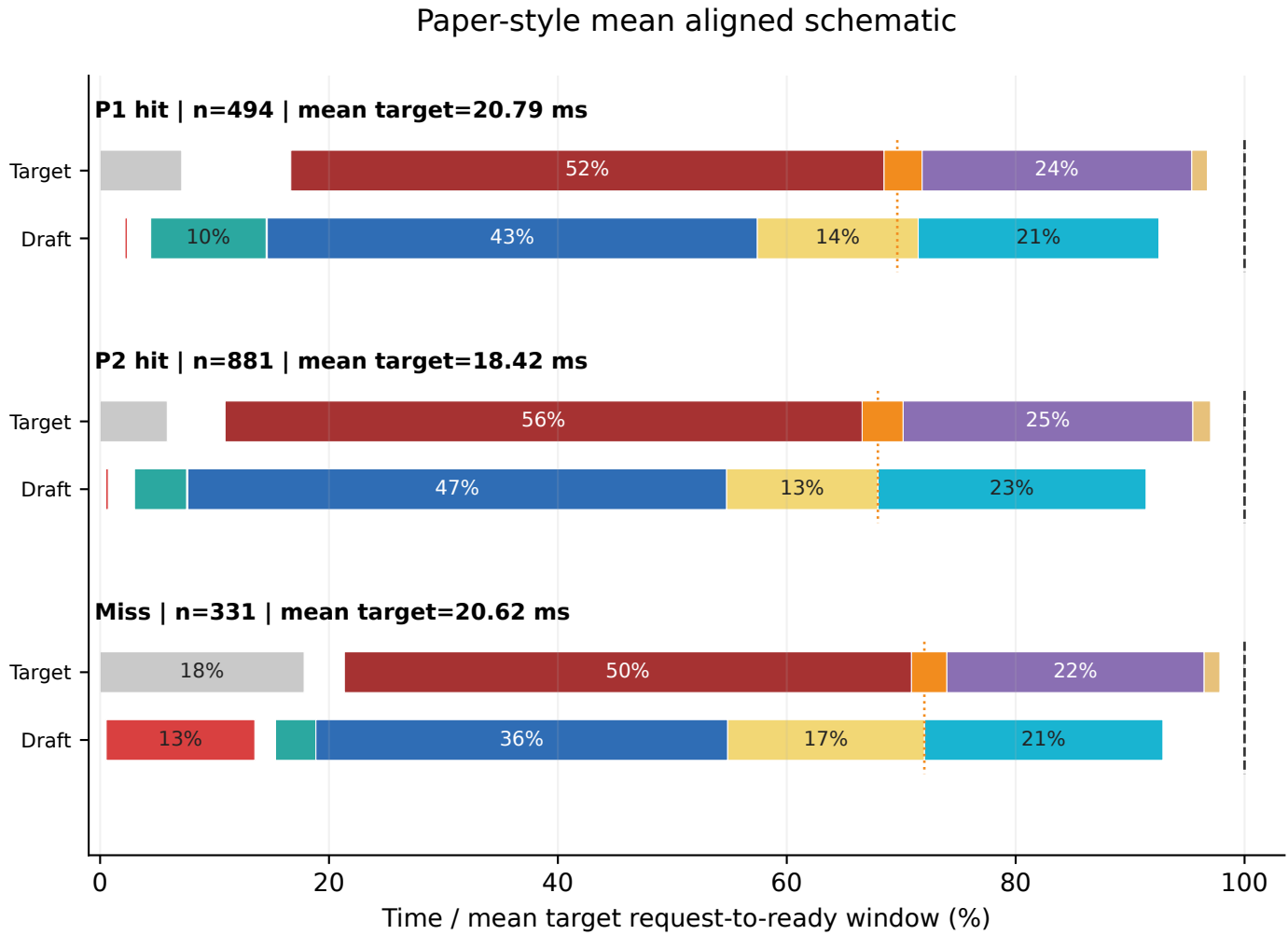
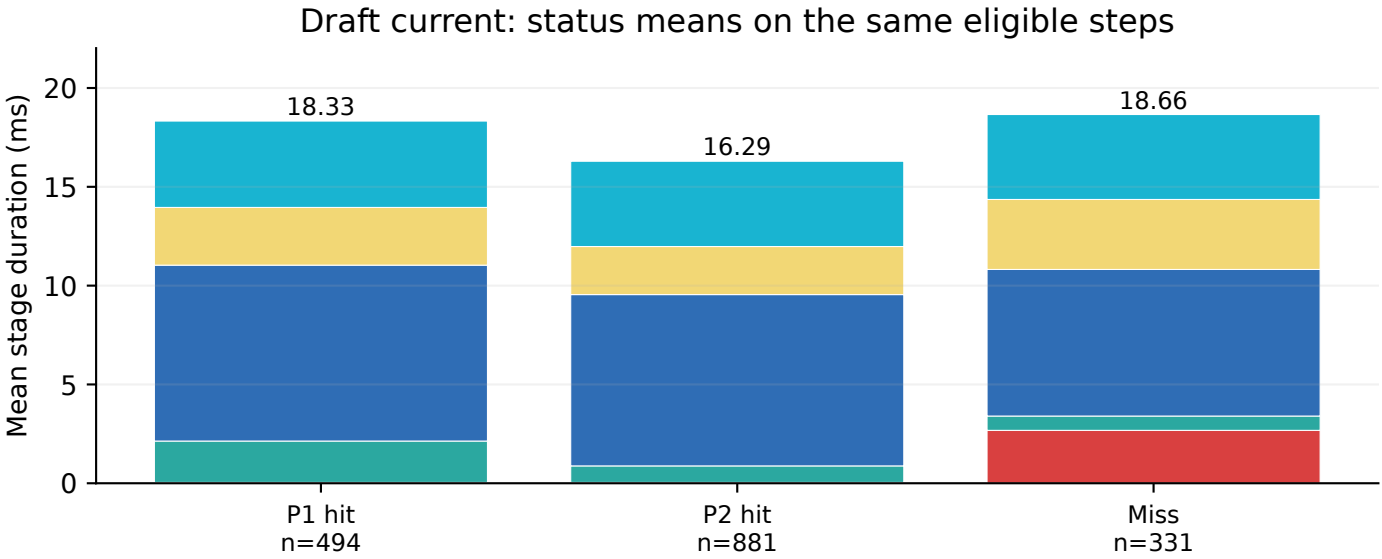
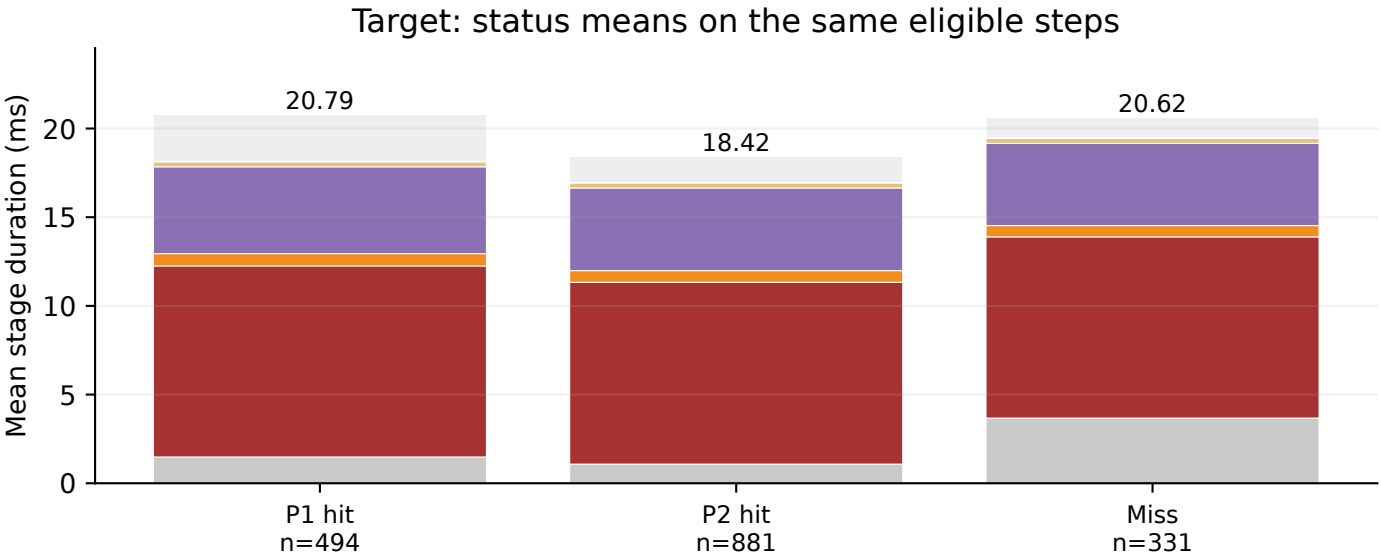


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

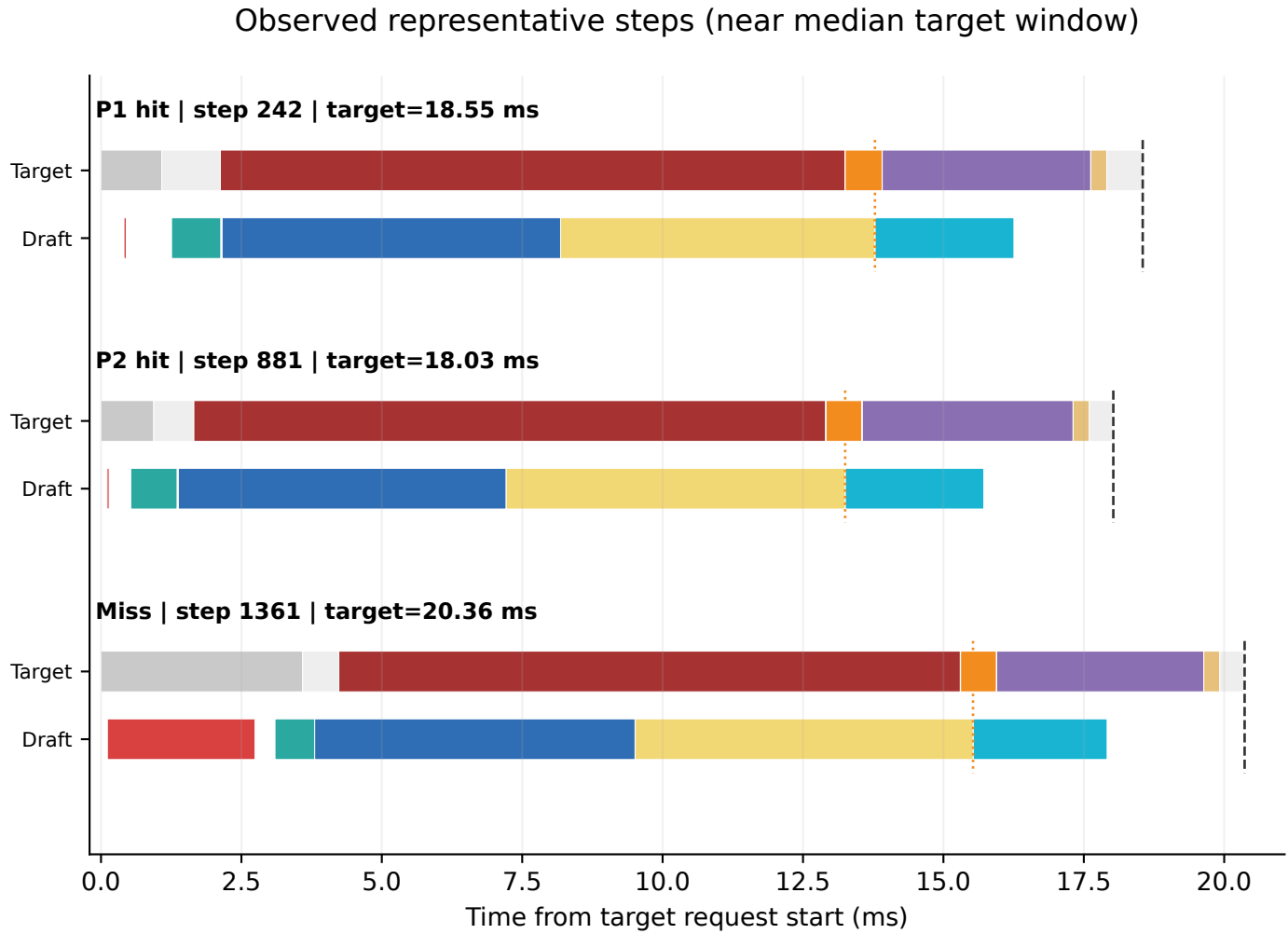
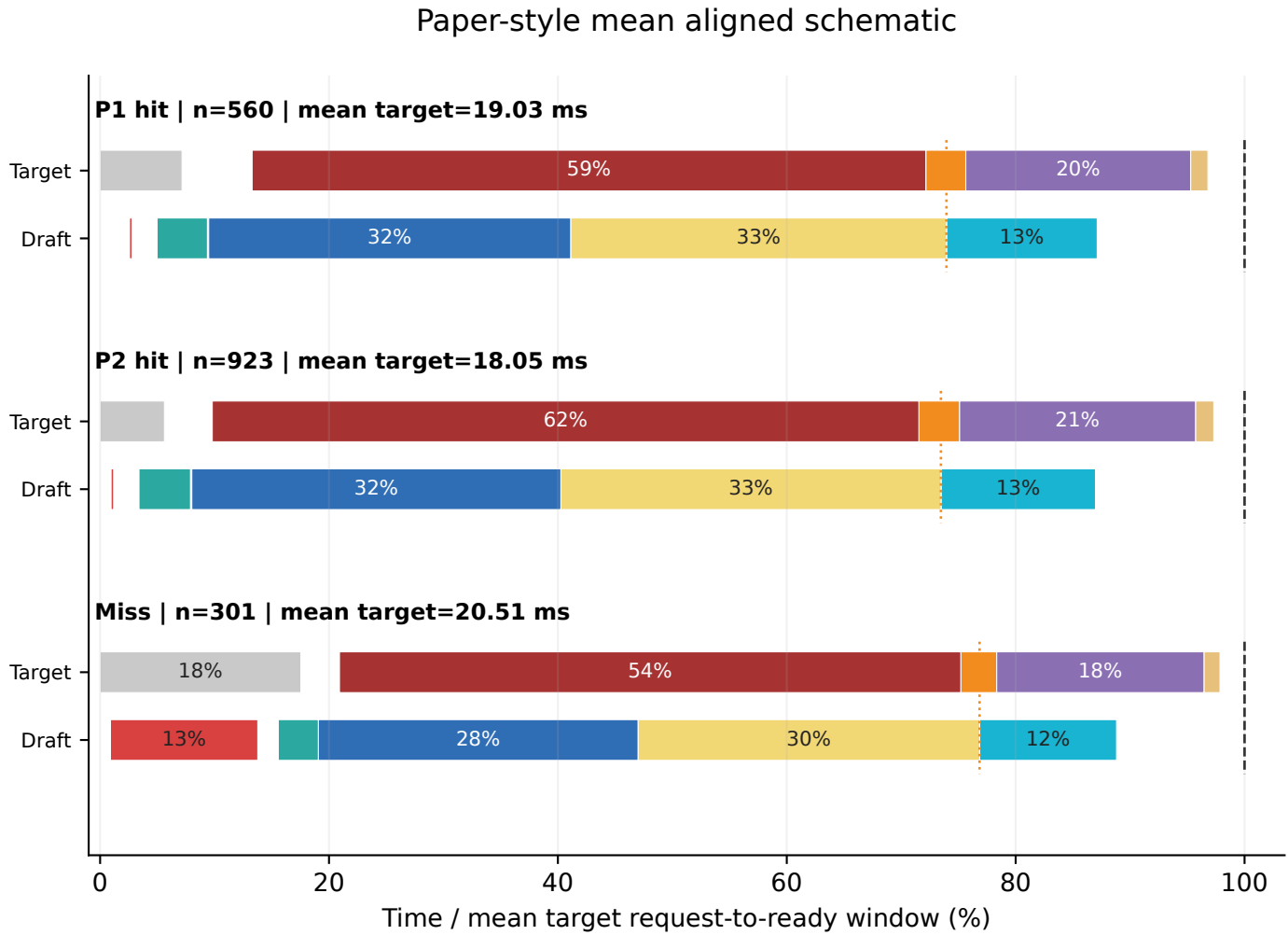
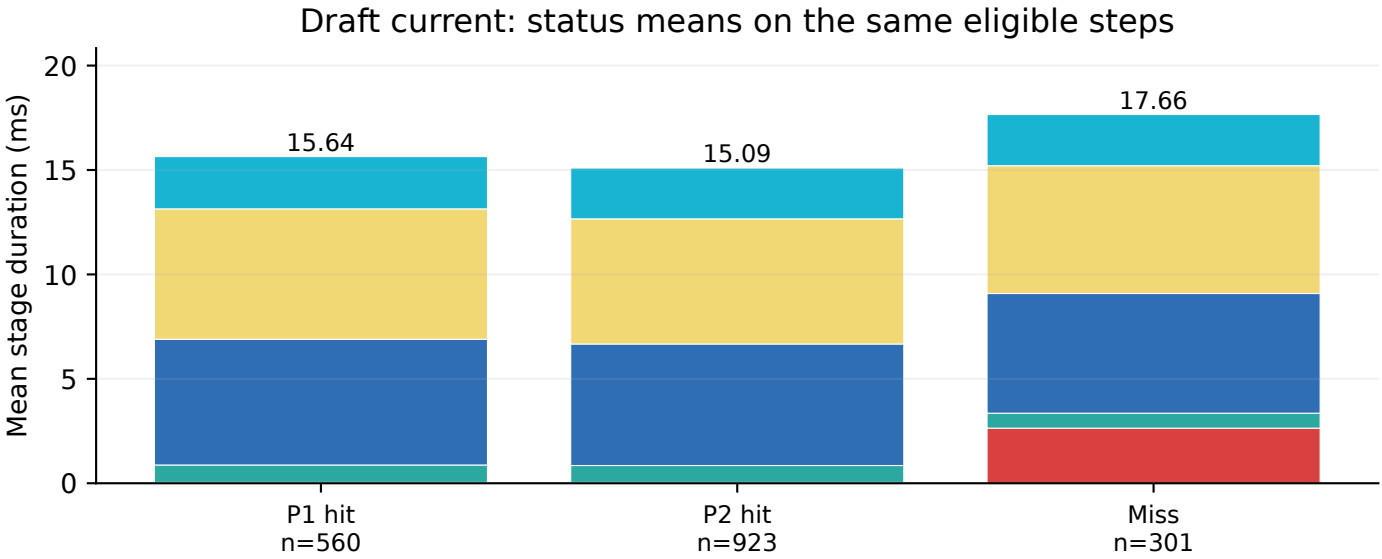
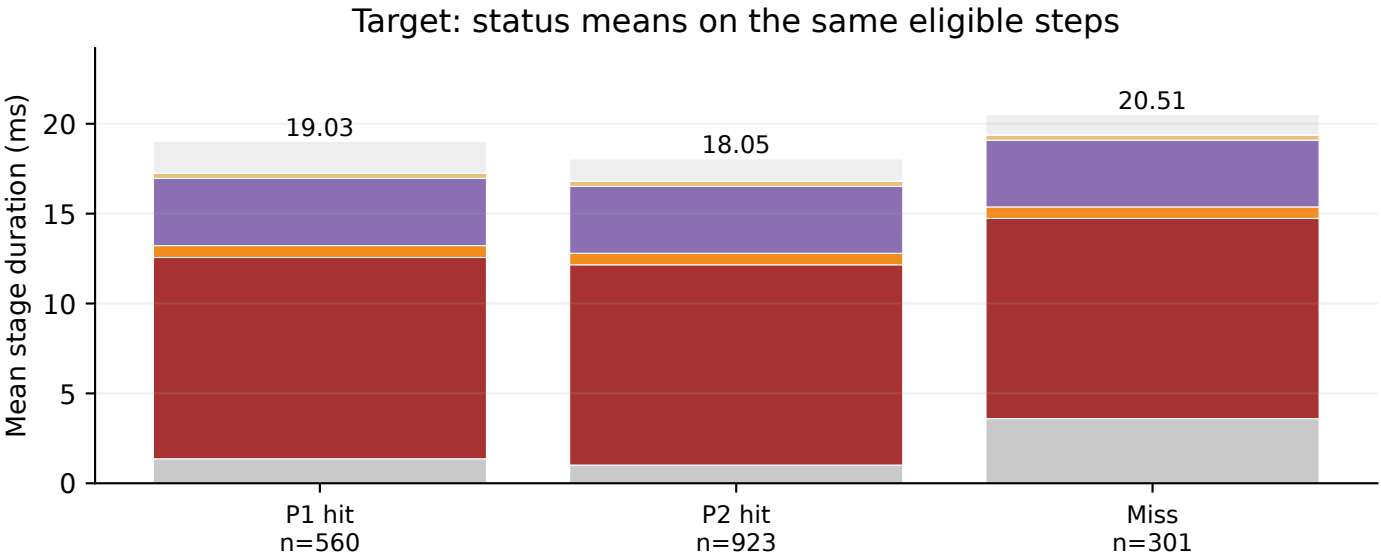
llama2_b1_refine_e21_k8_3_r2
DUET: K1/K2=8/3, exit=21 (0-based), C=3, N1/N2=16/6, U=2, W=9, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1720 eligible steps



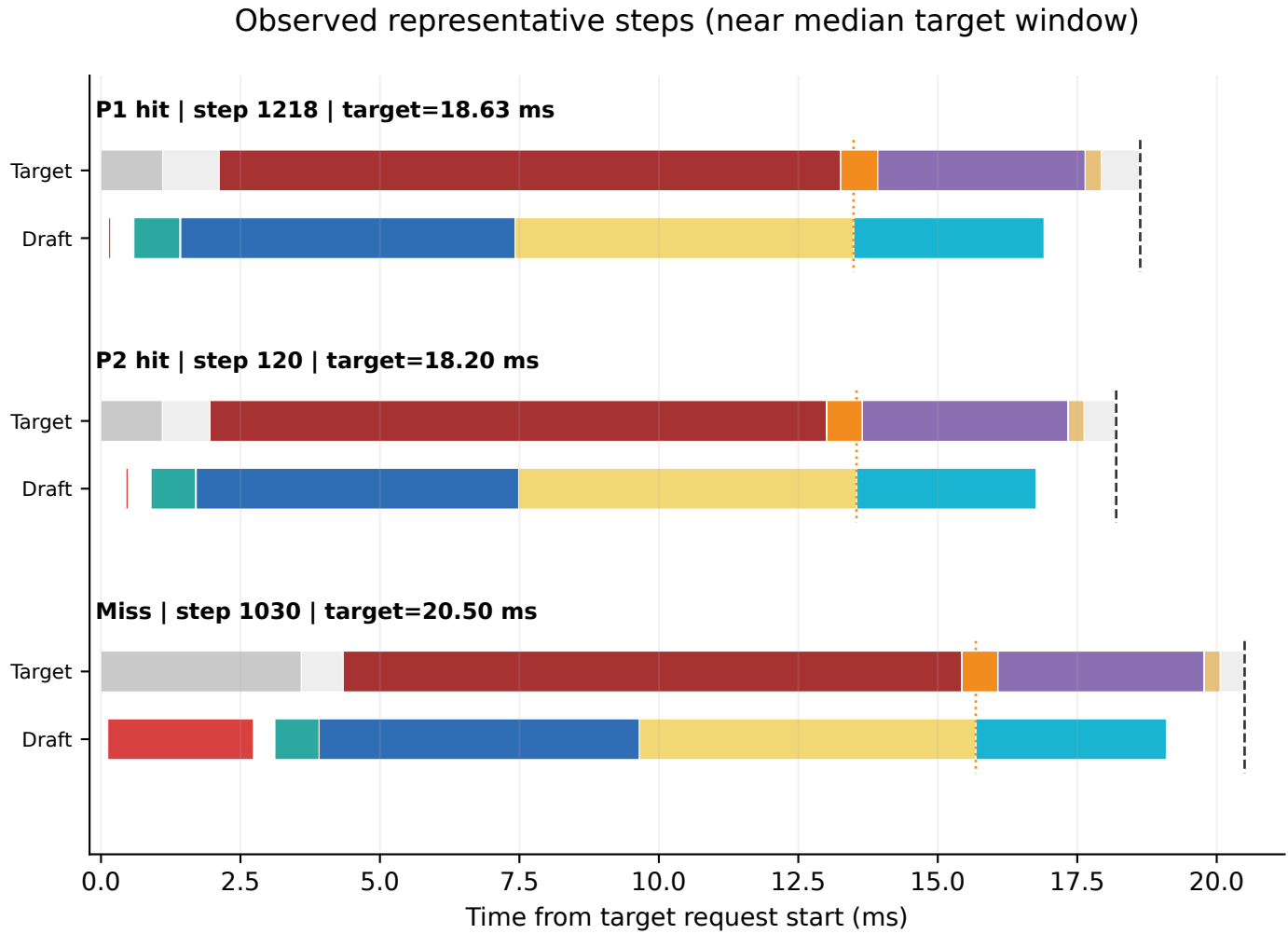
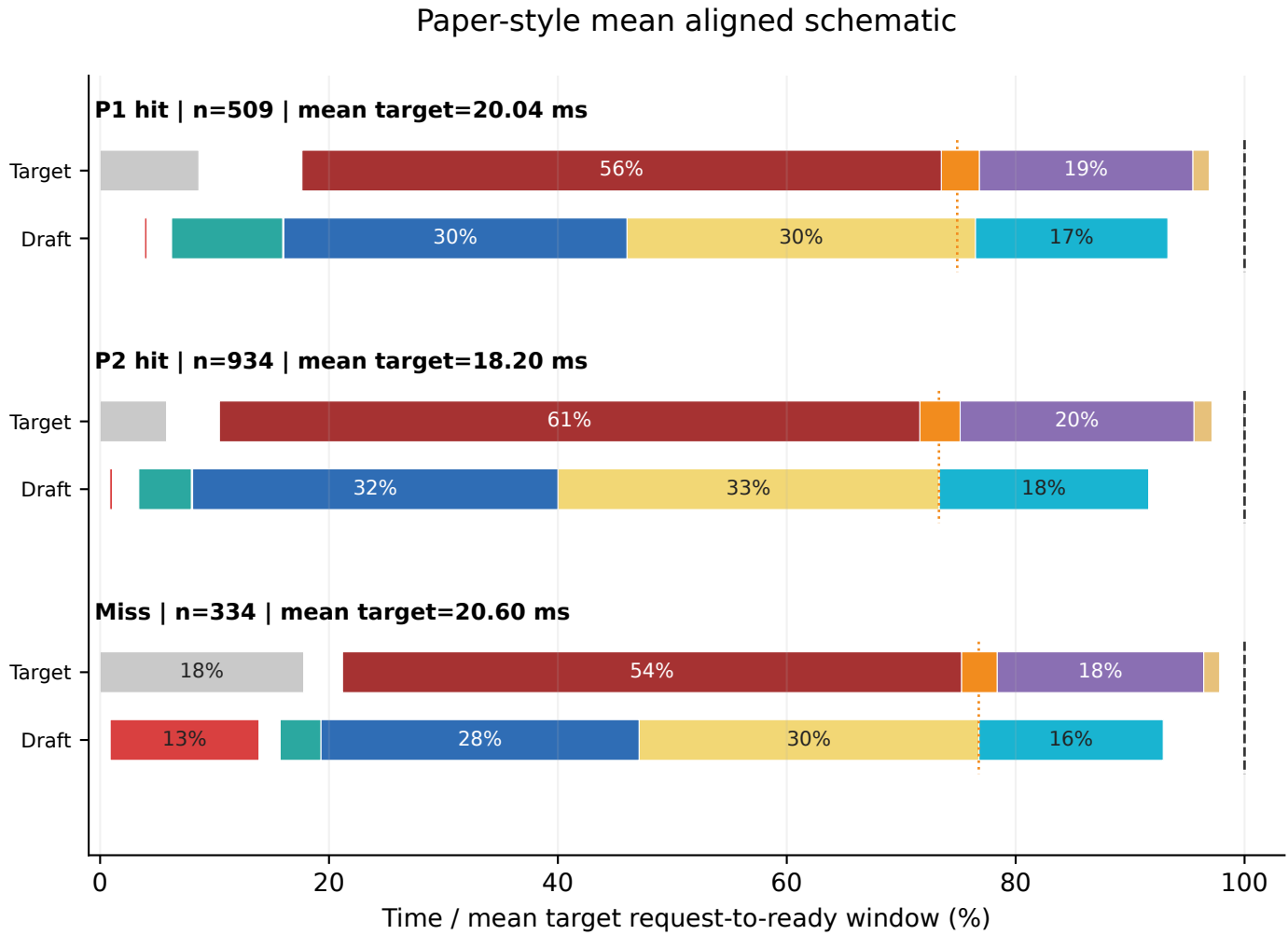
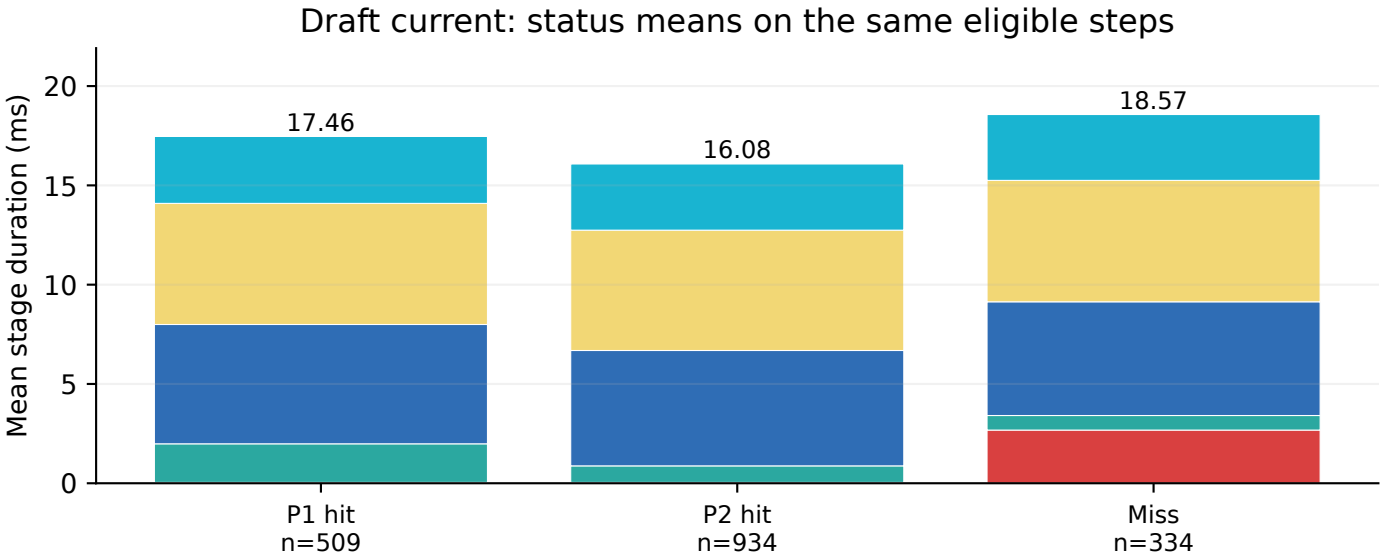
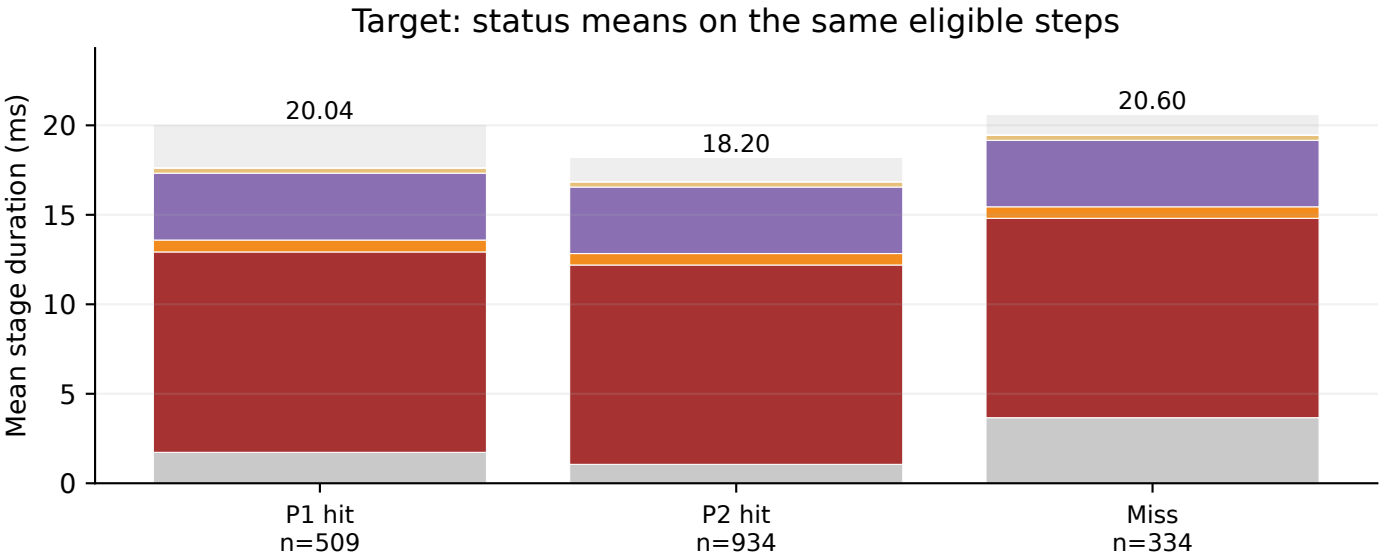
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

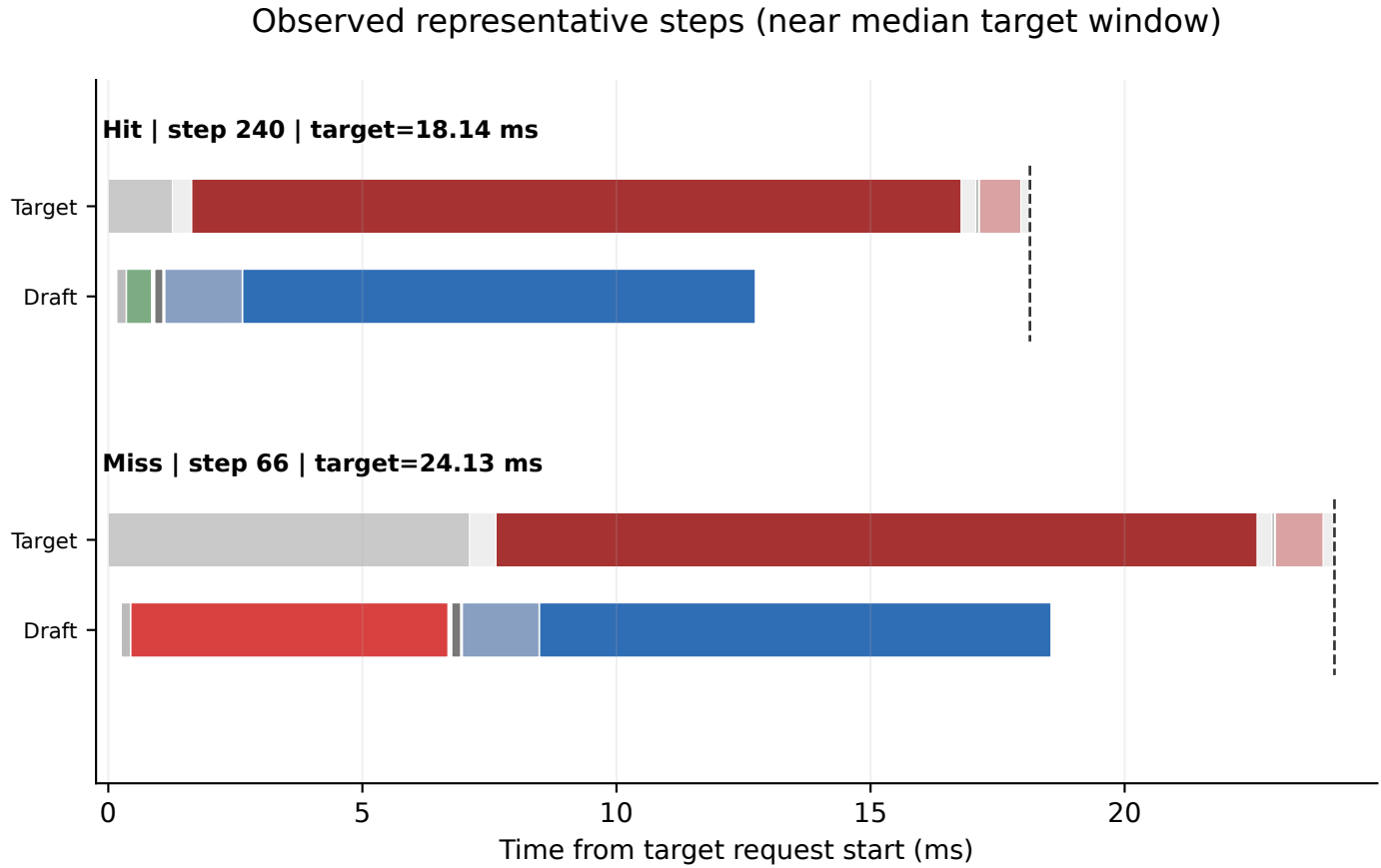
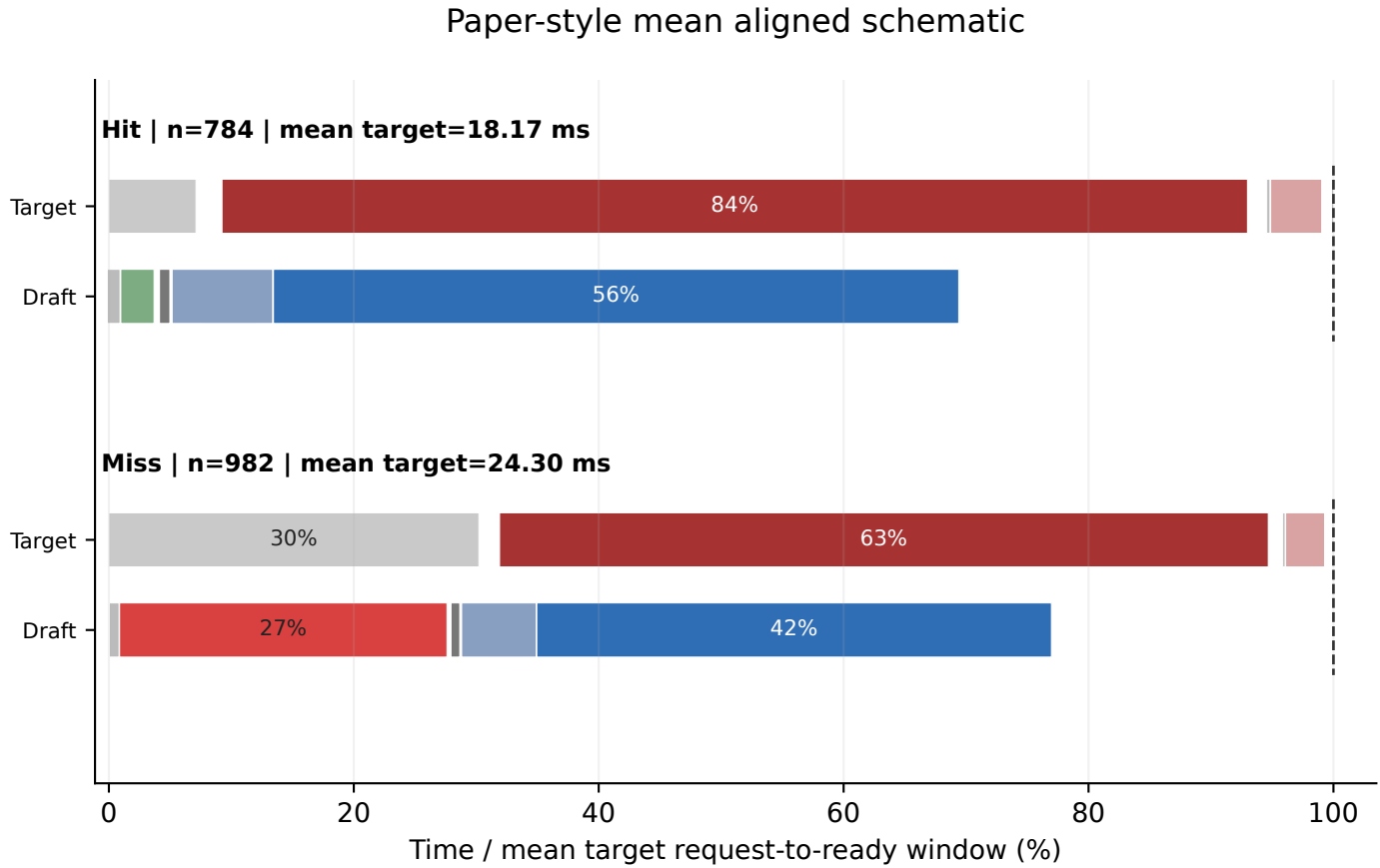
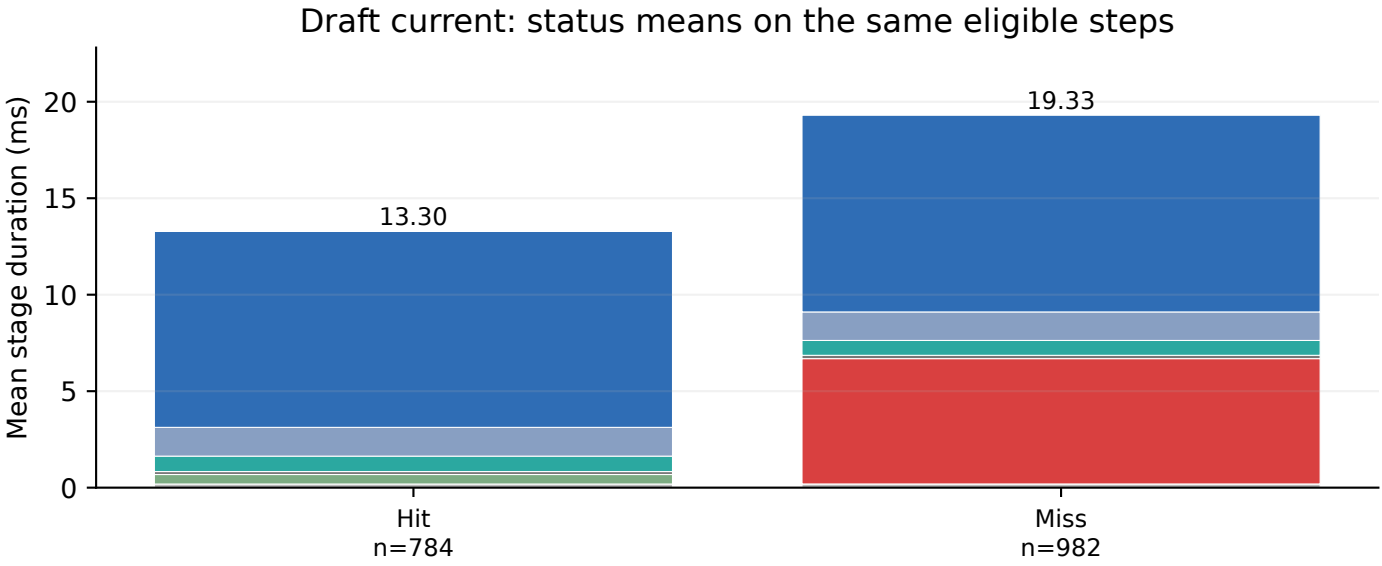
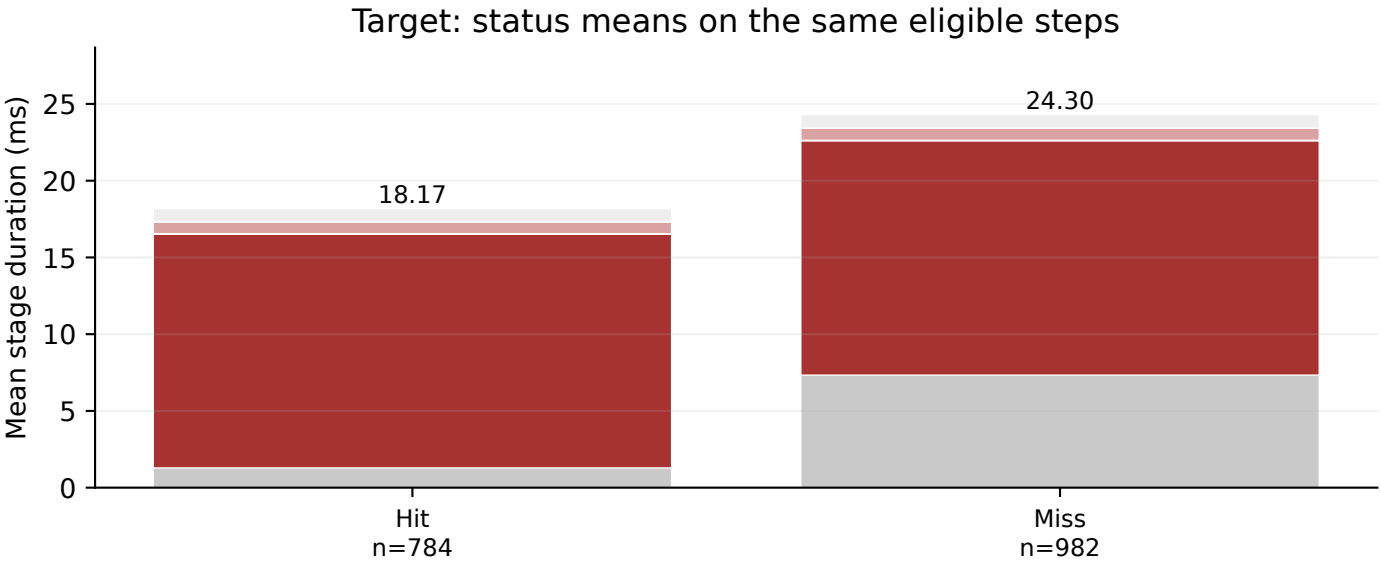


llama2_b1_refine_e23_k6_2_r2
DUET: K1/K2=6/2, exit=23 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1784 eligible steps



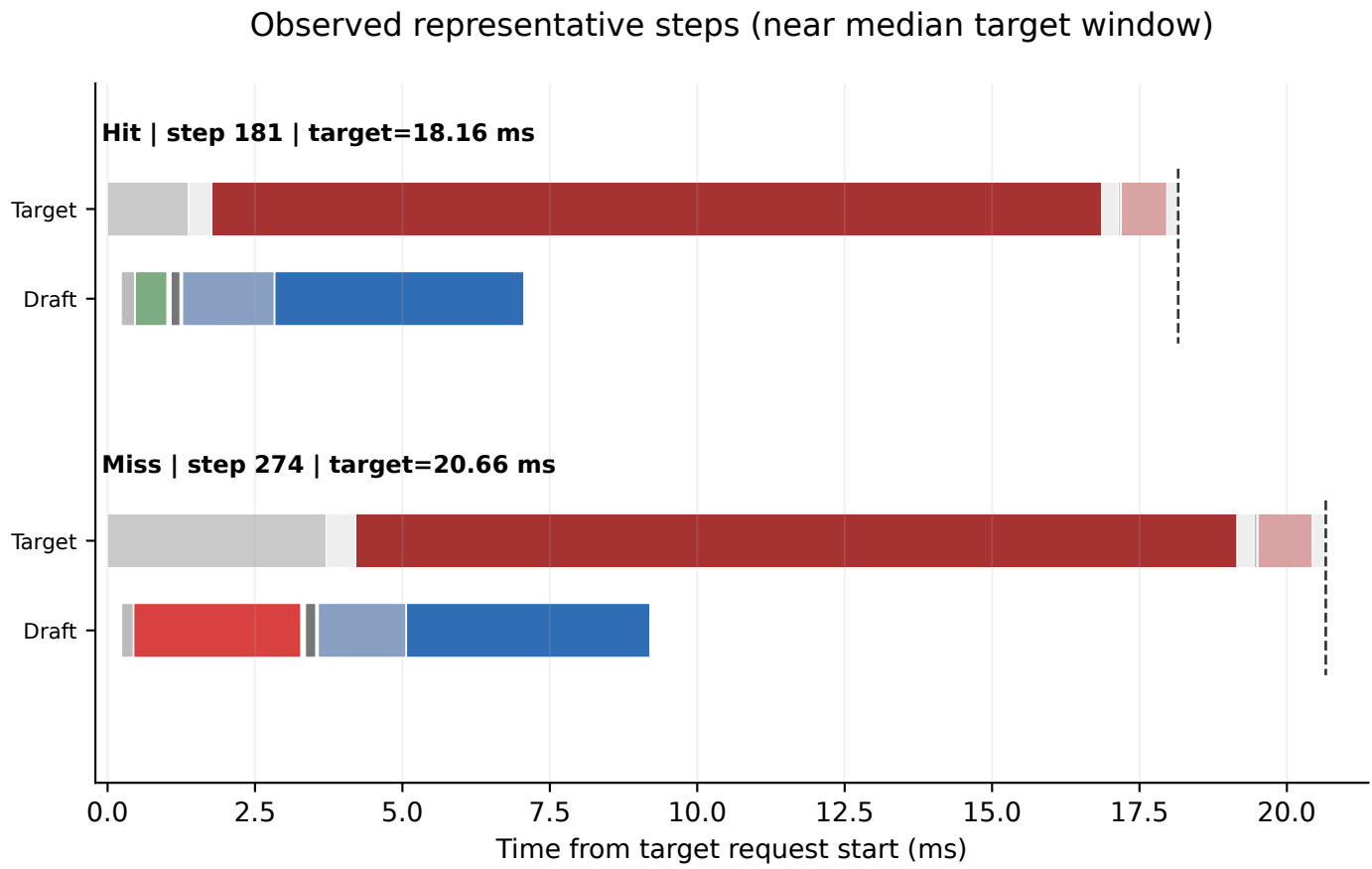
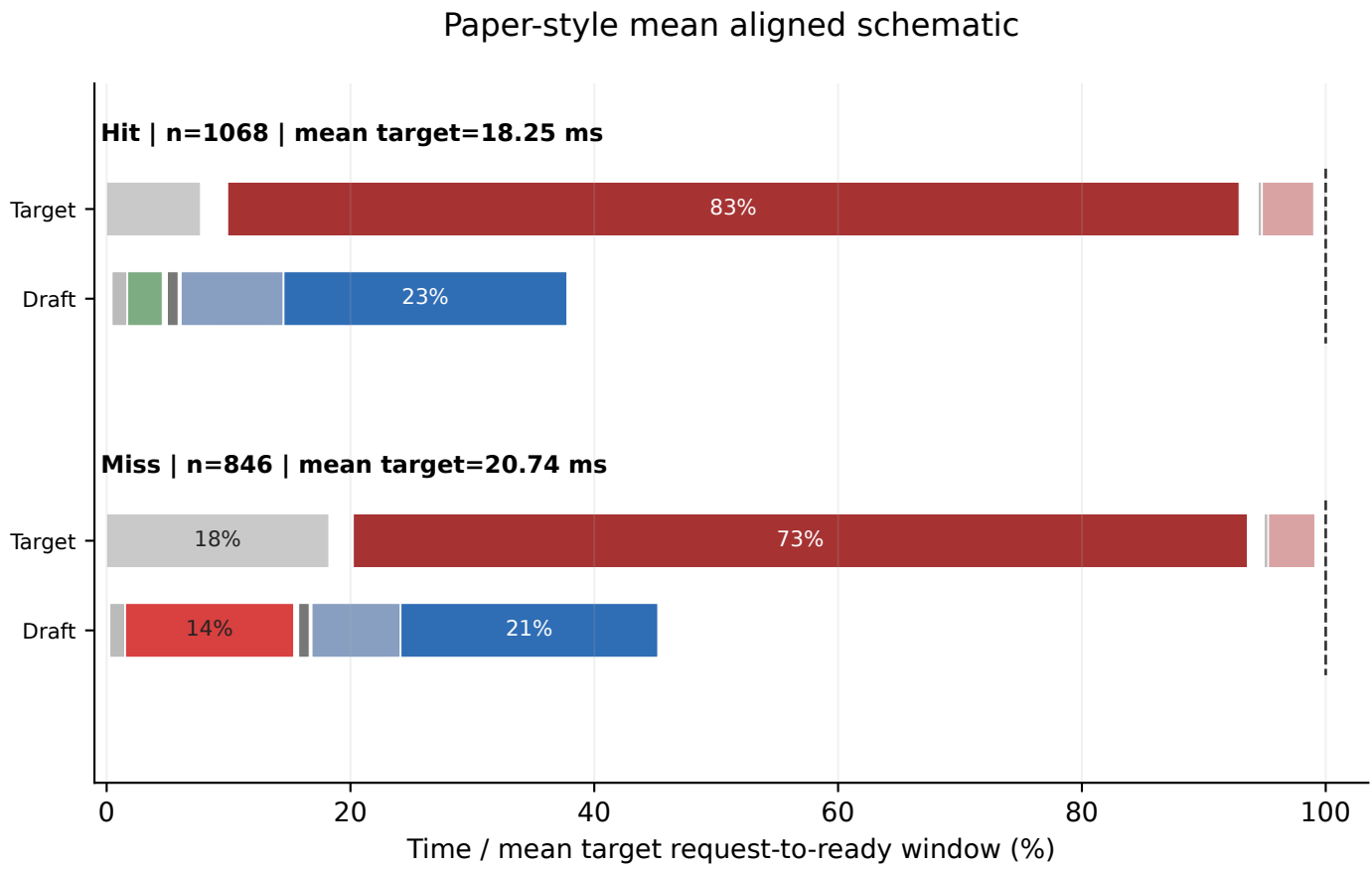
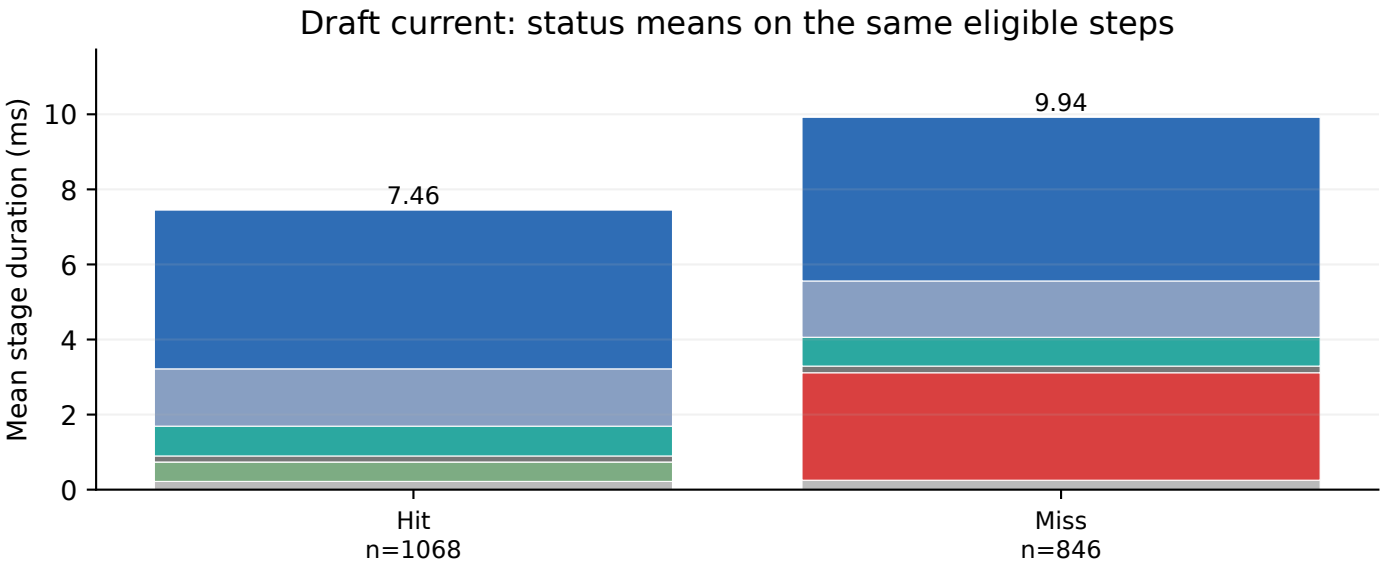
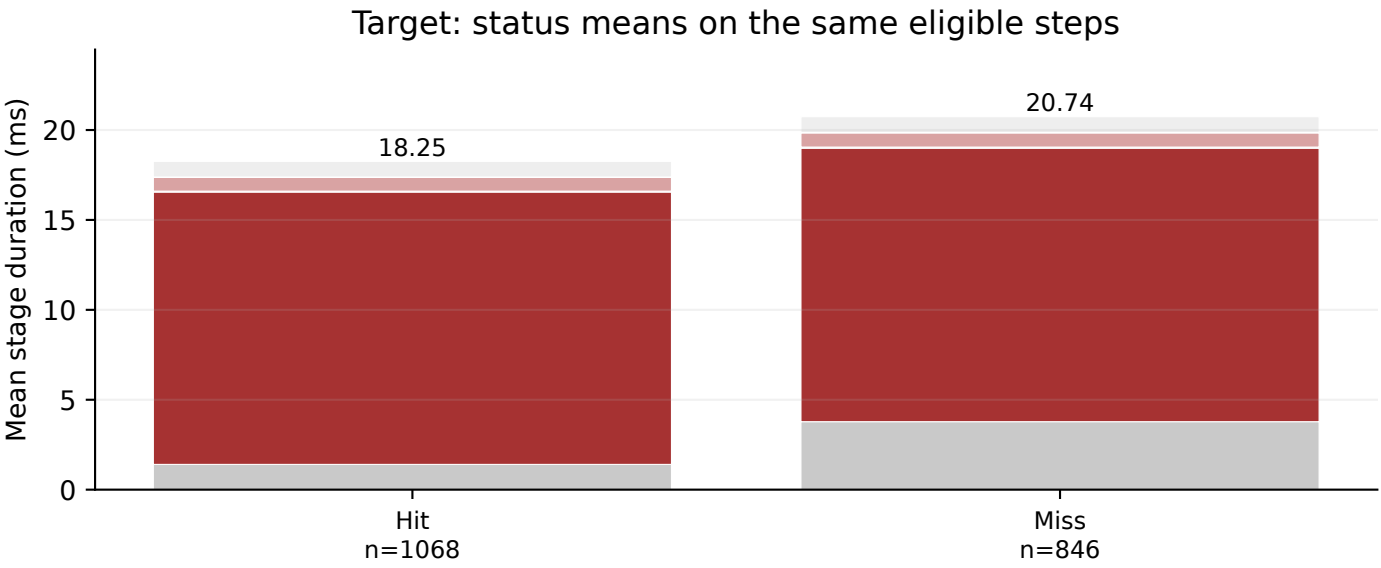
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
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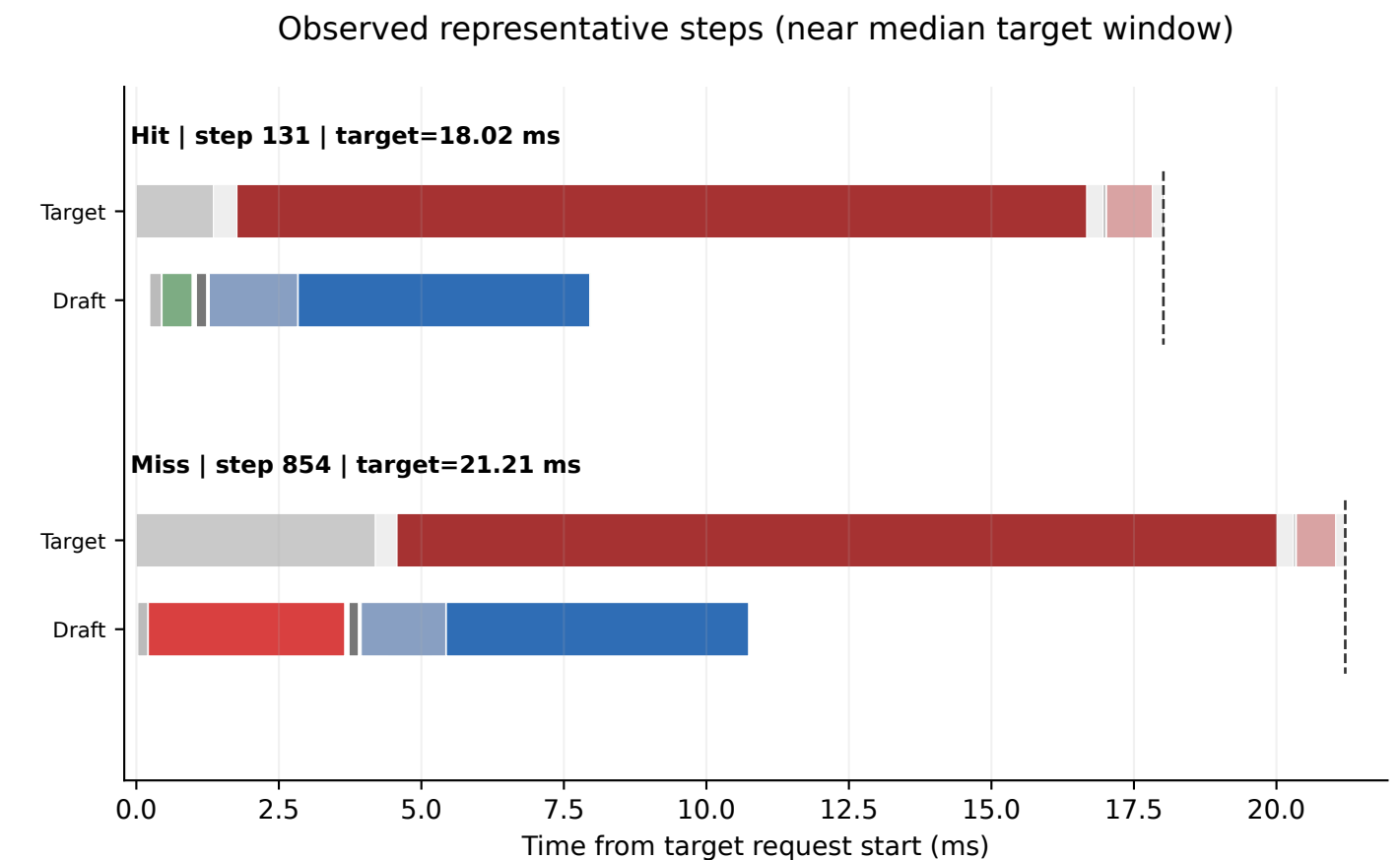
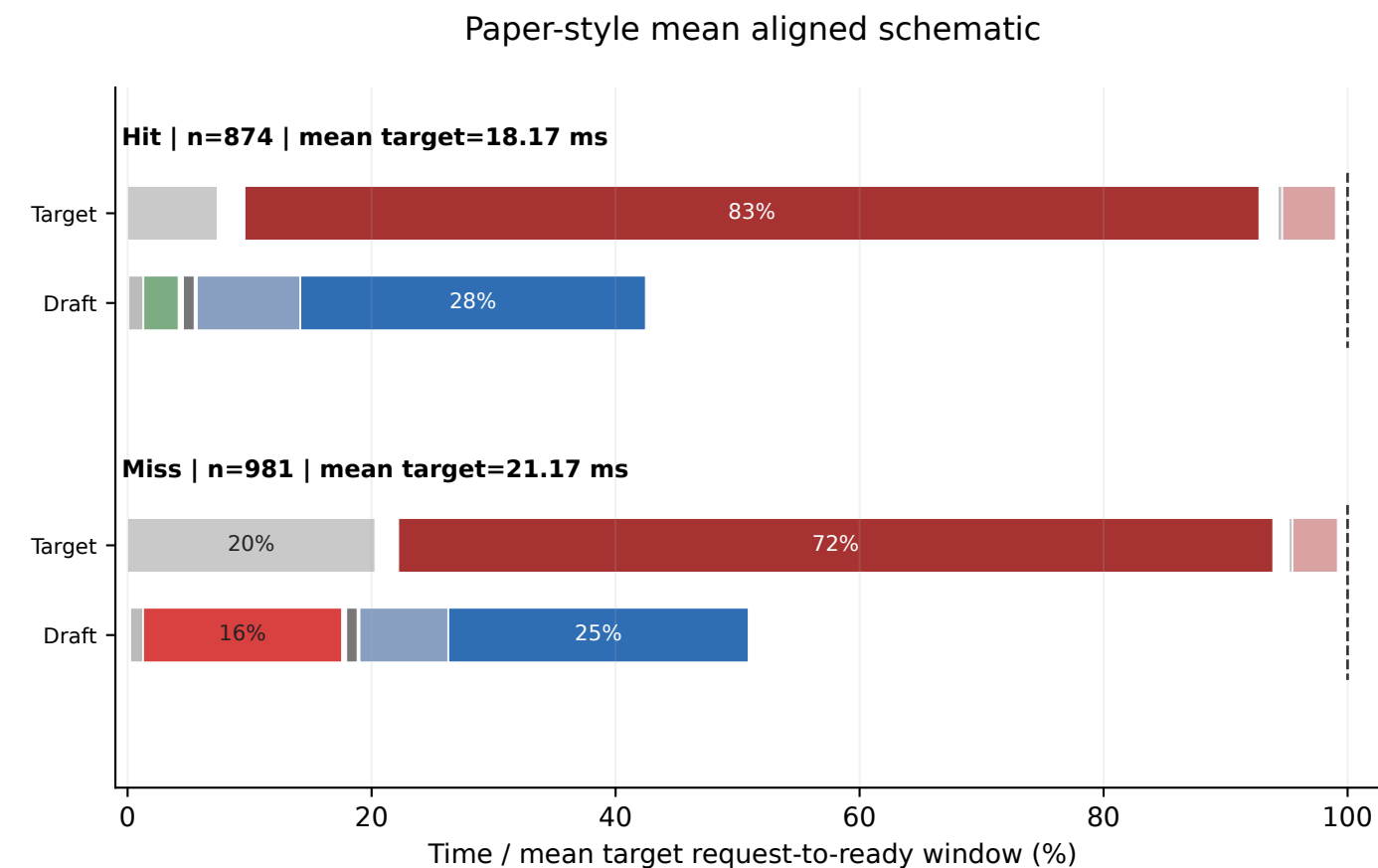
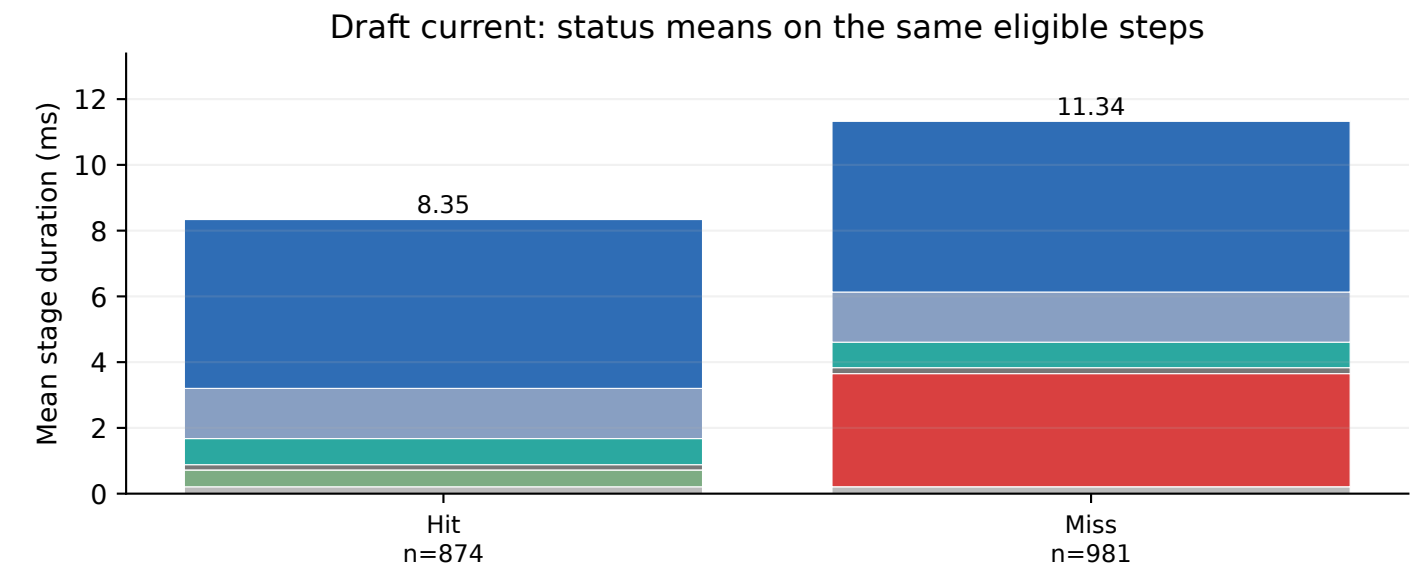
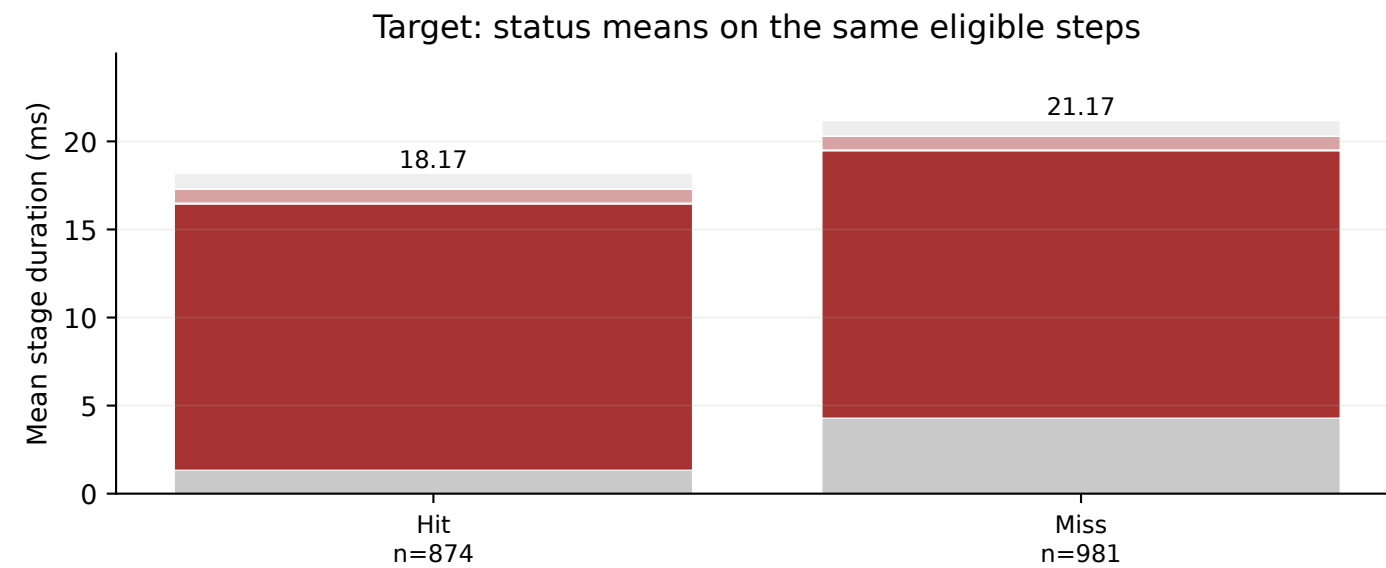
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

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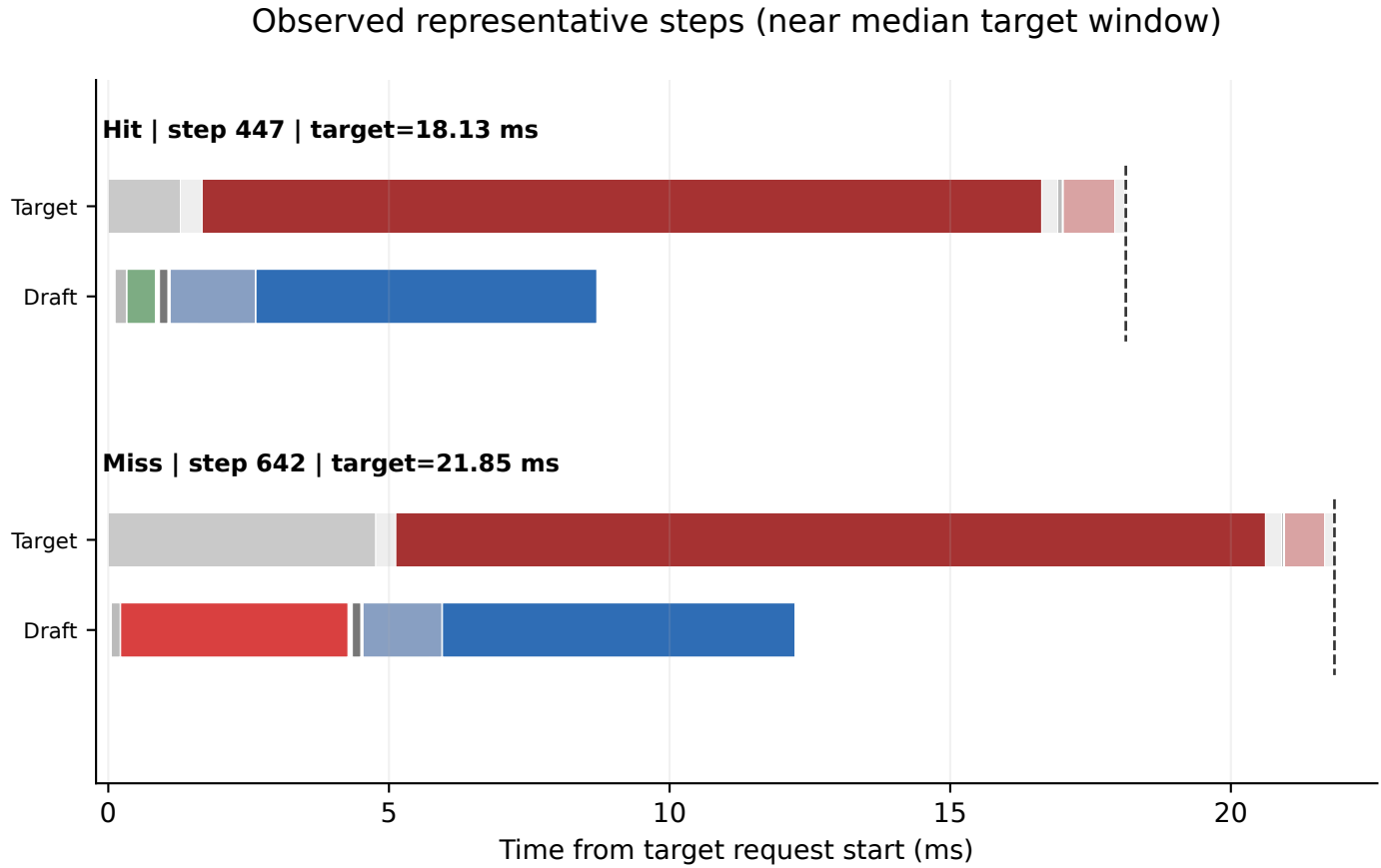
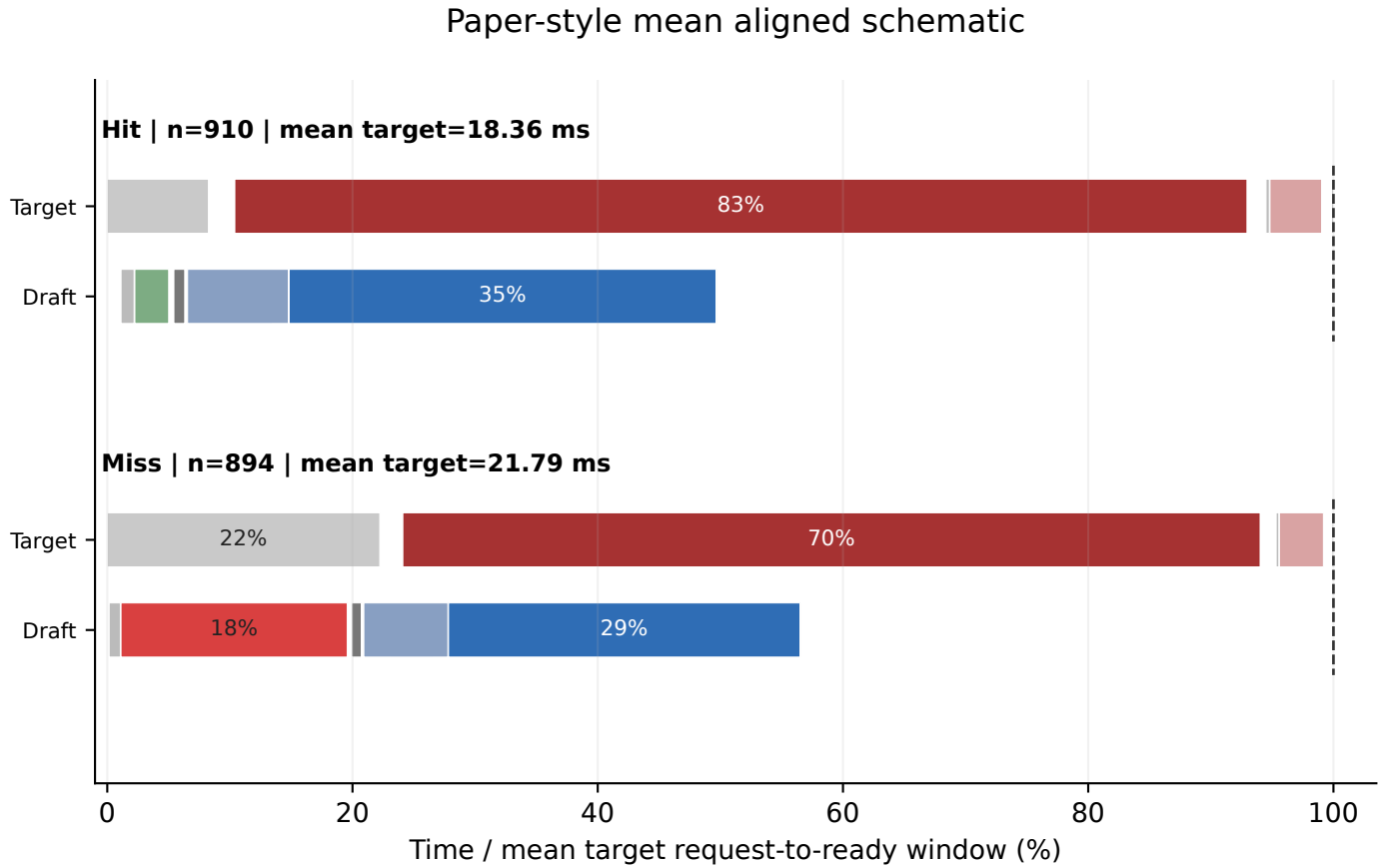
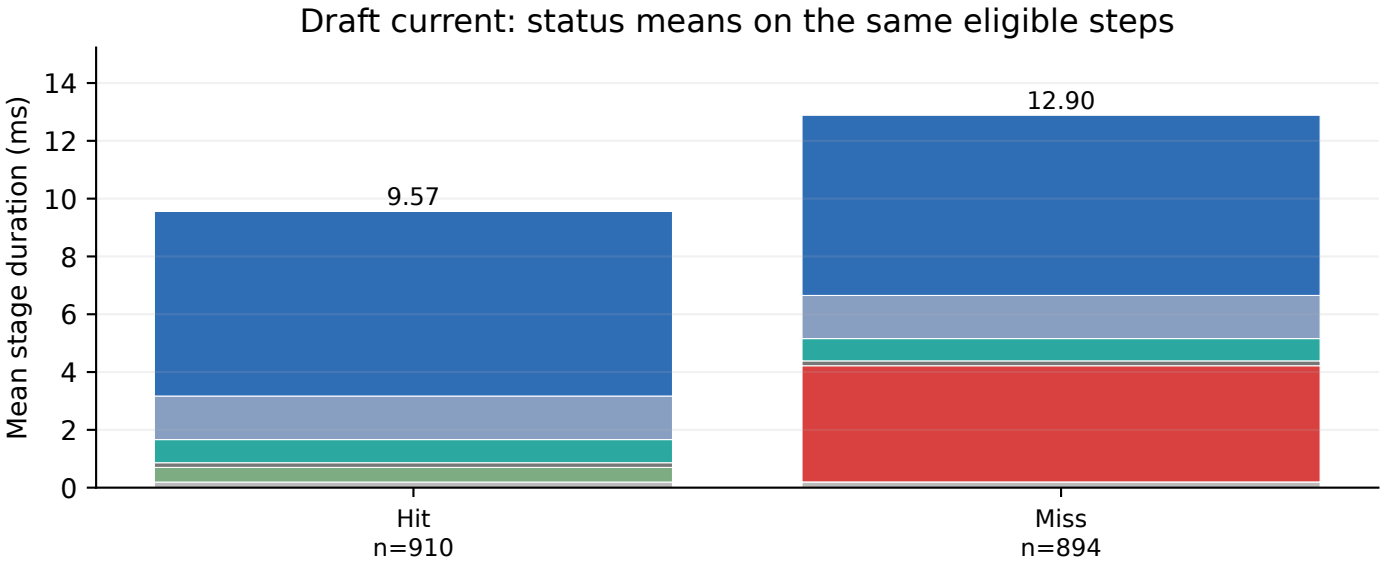
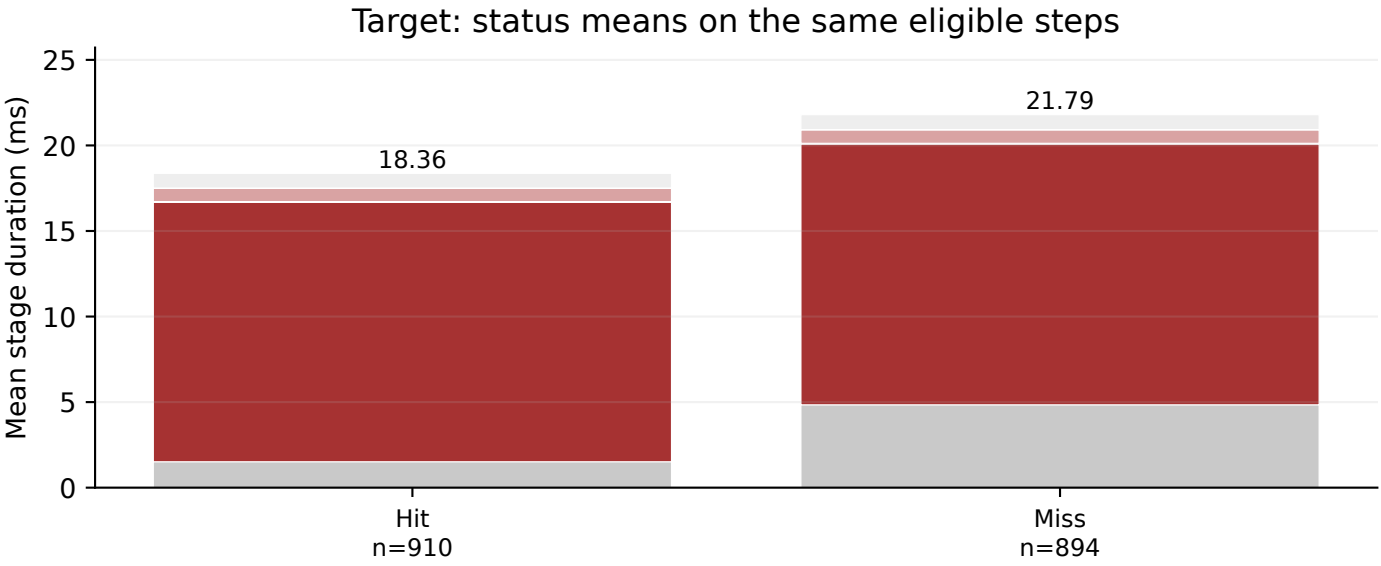
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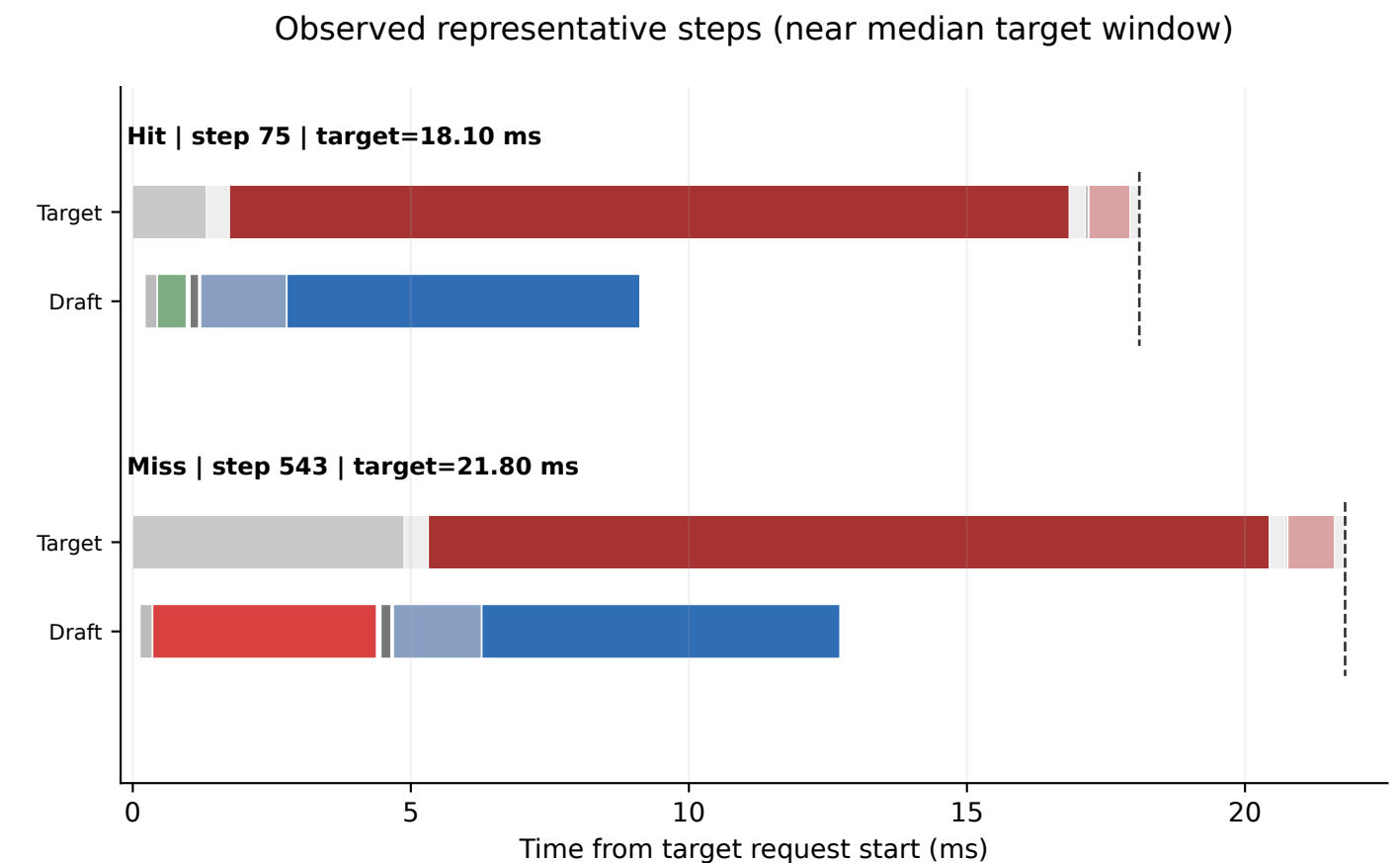
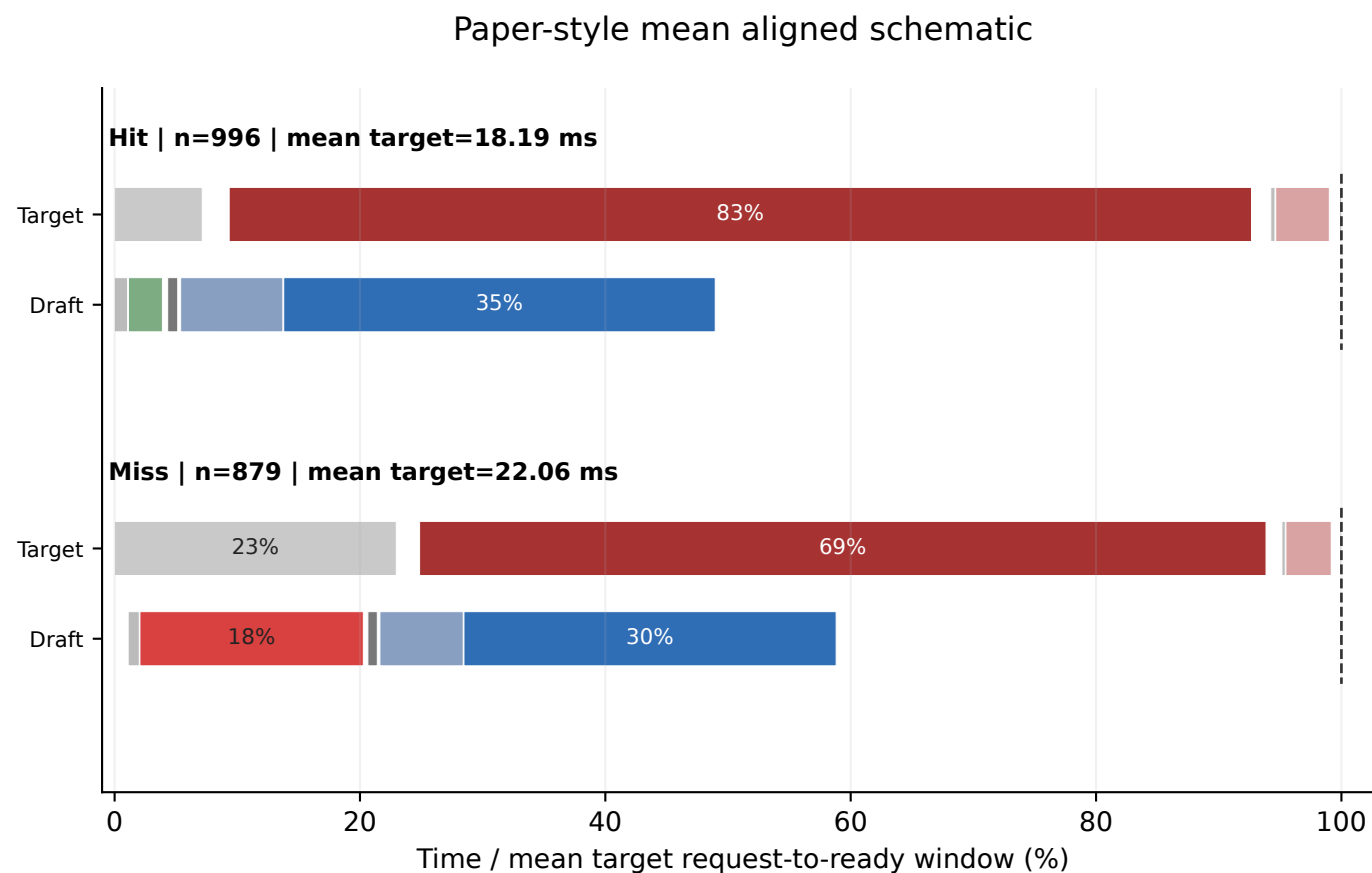
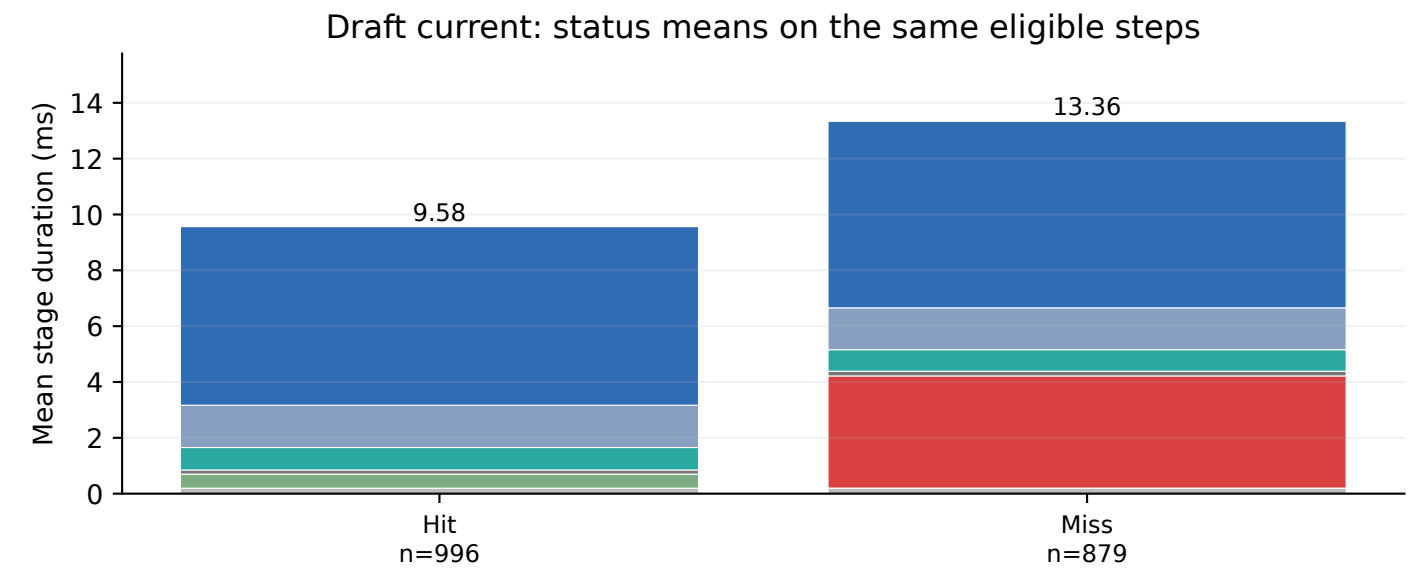
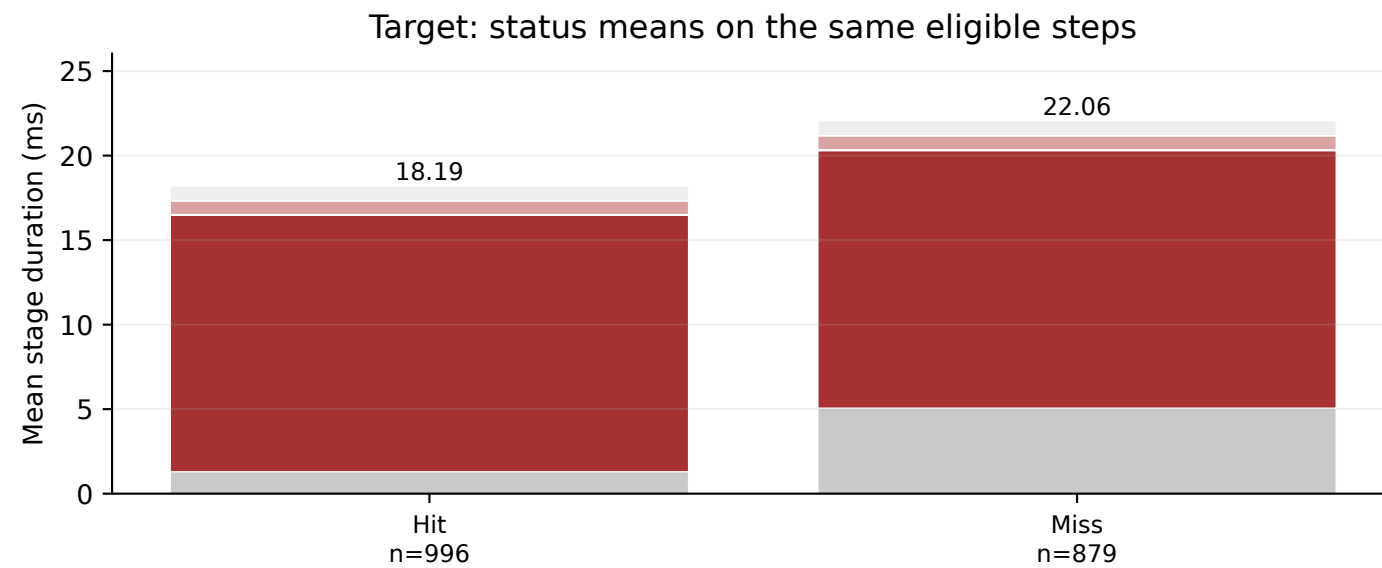
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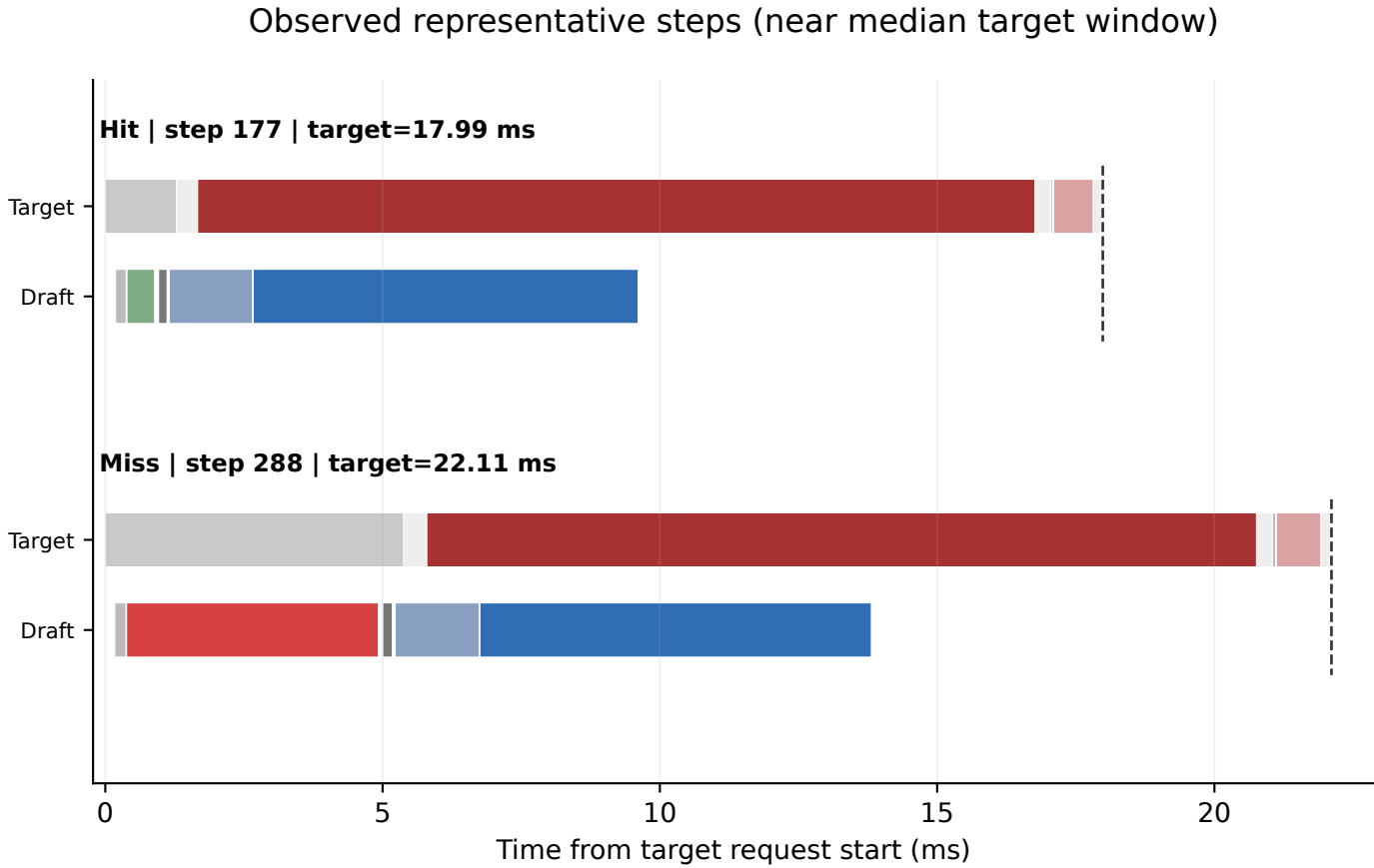
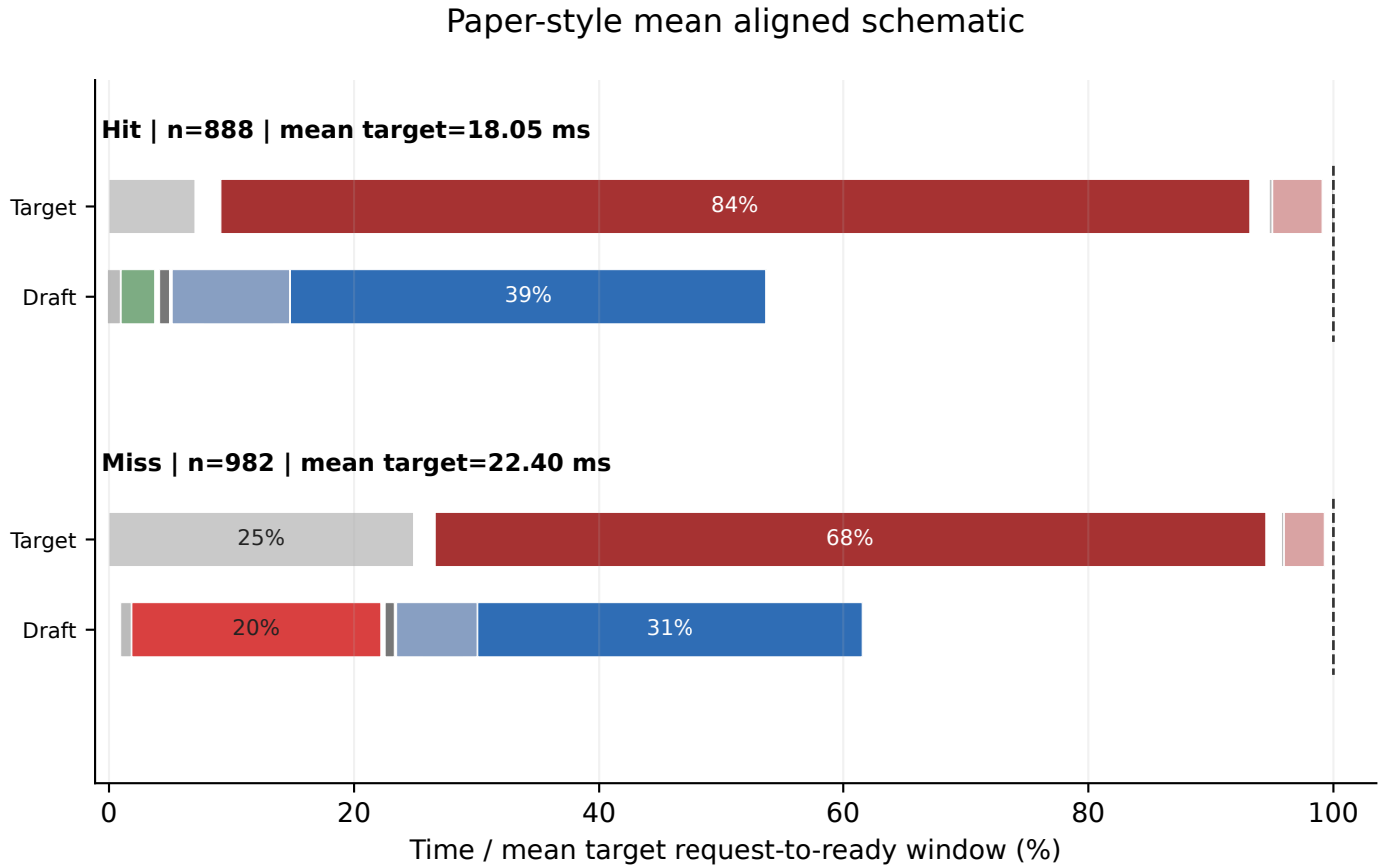
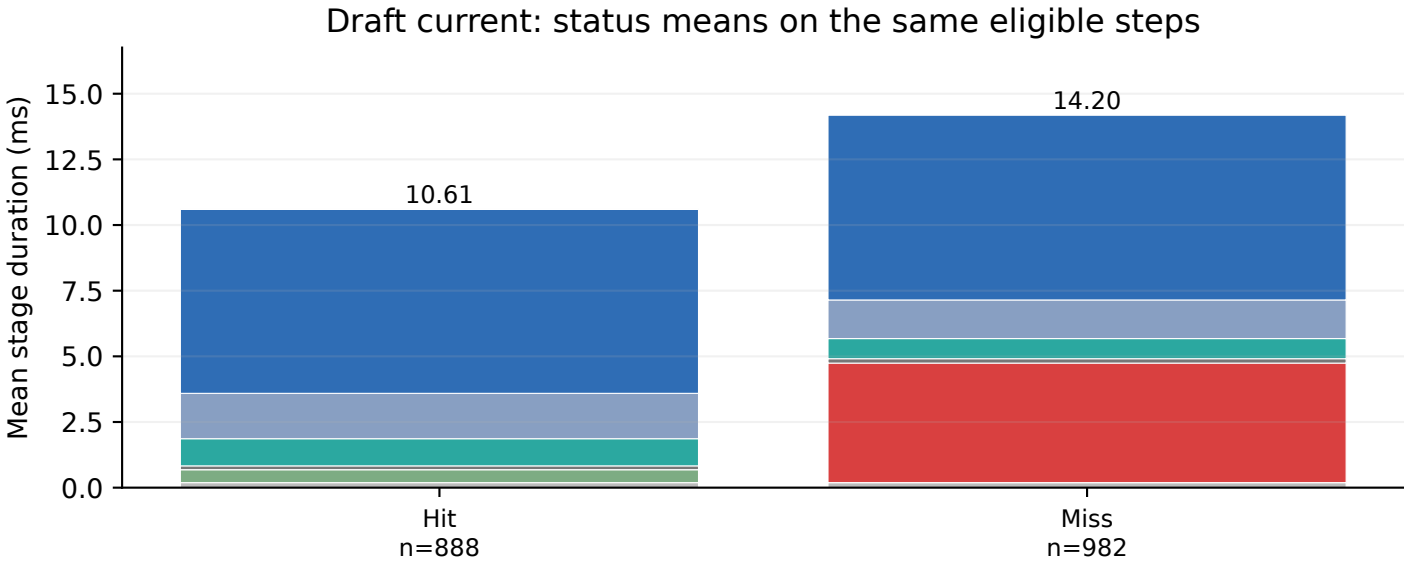
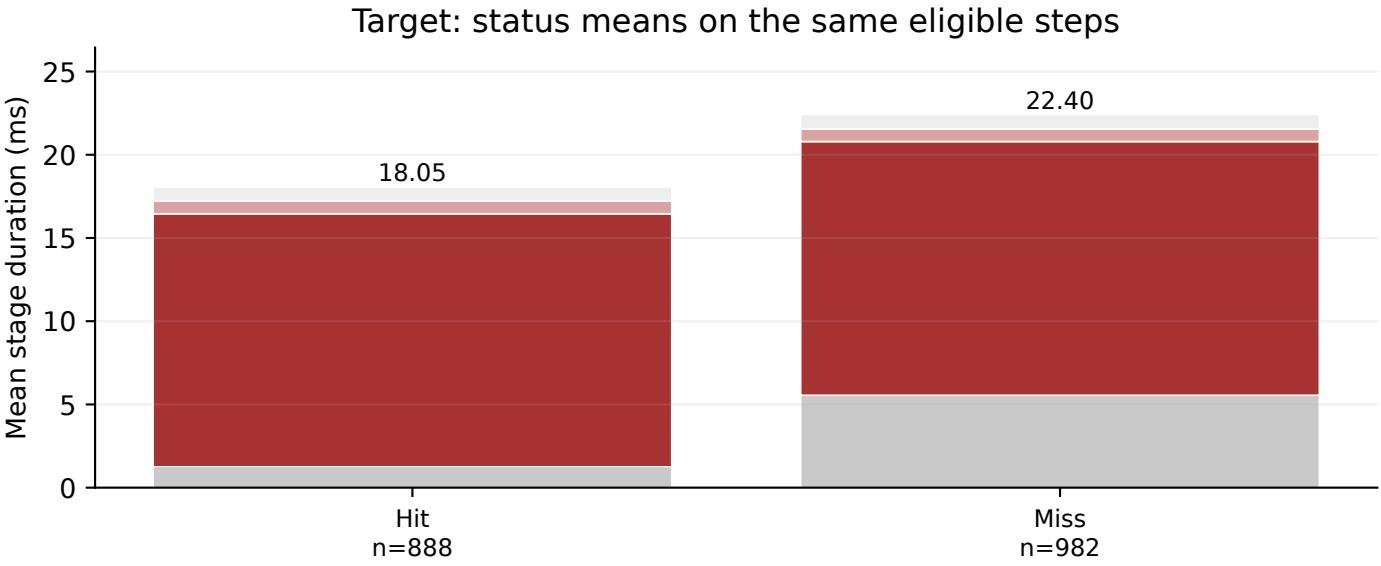
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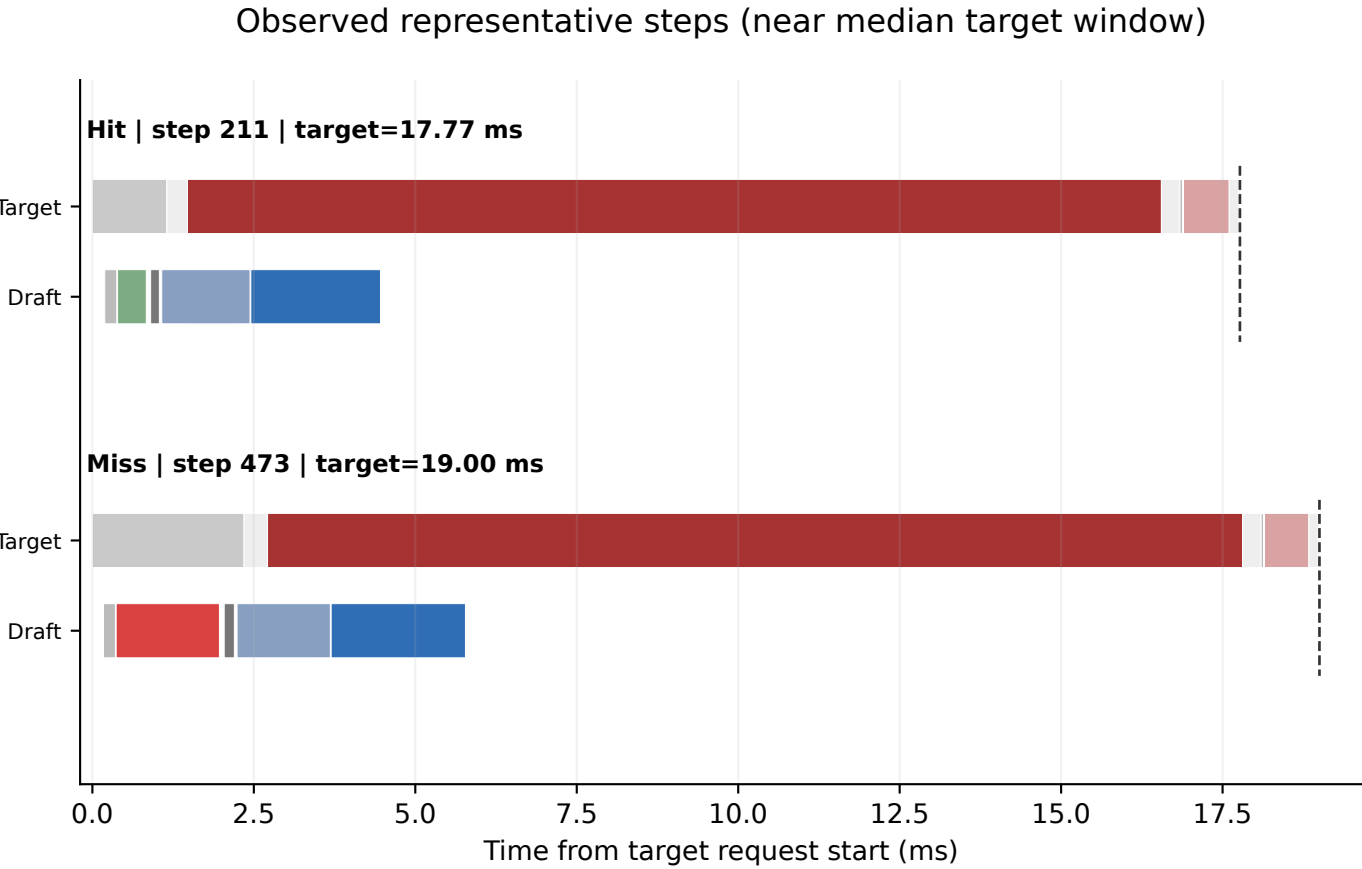
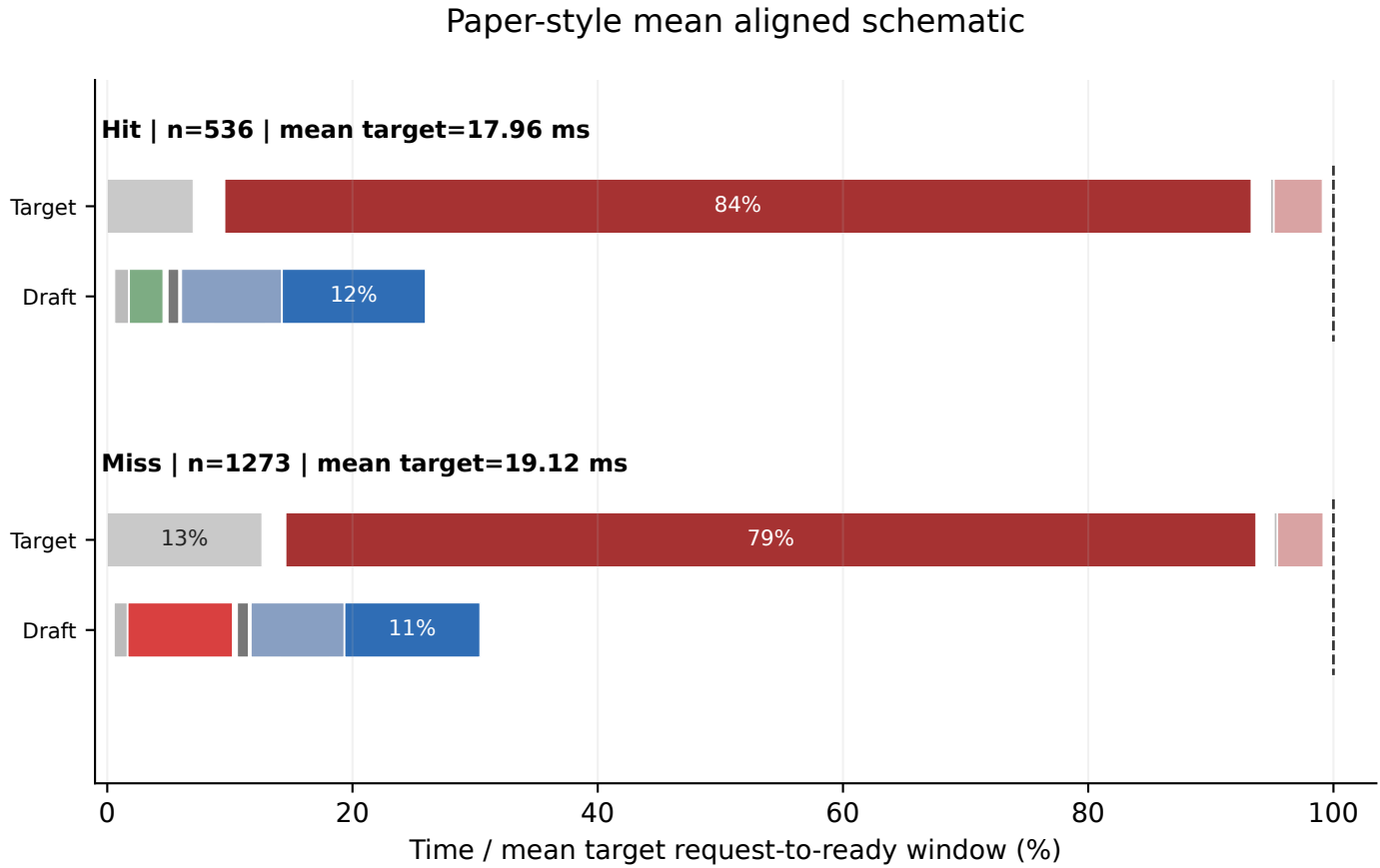
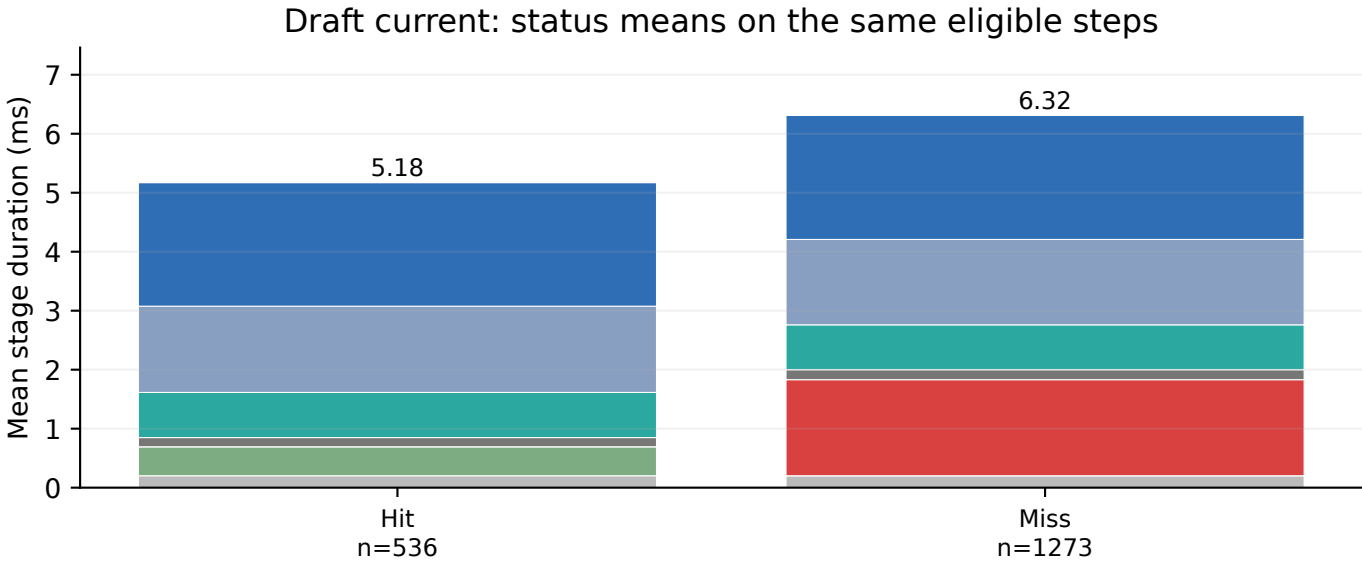
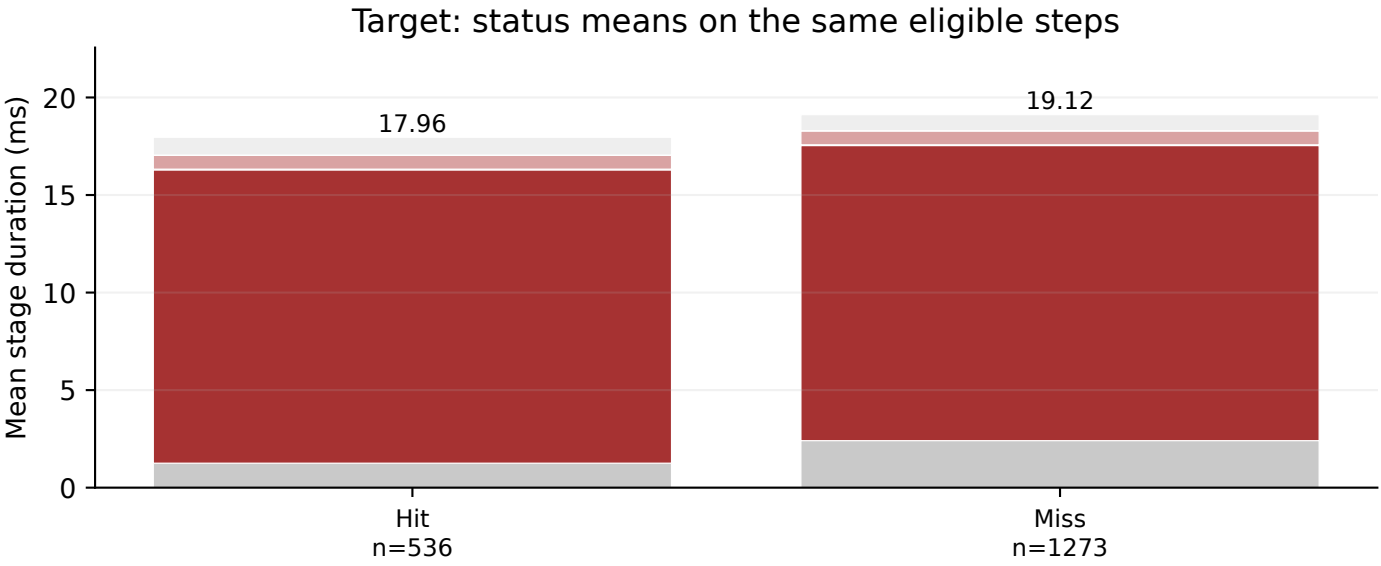
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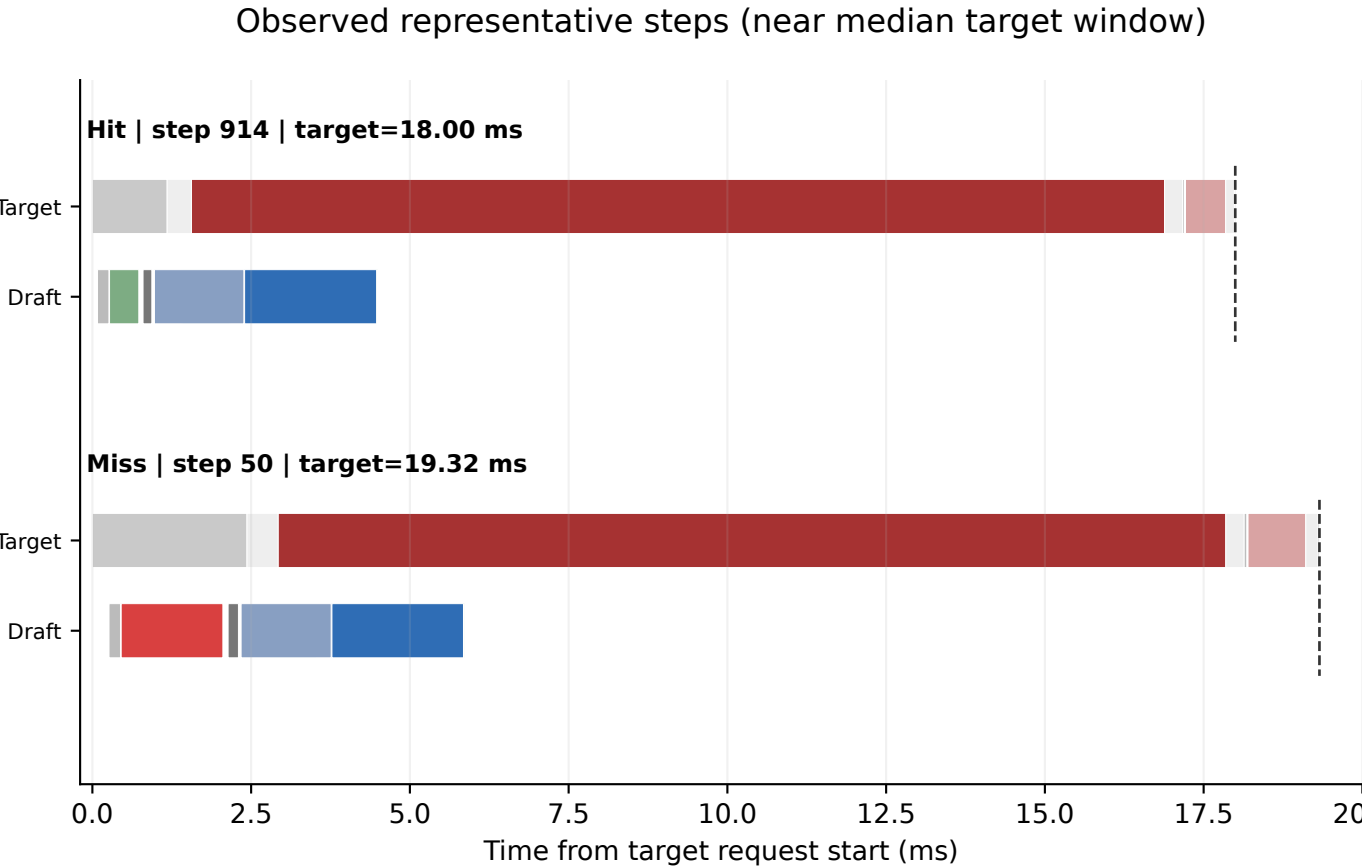
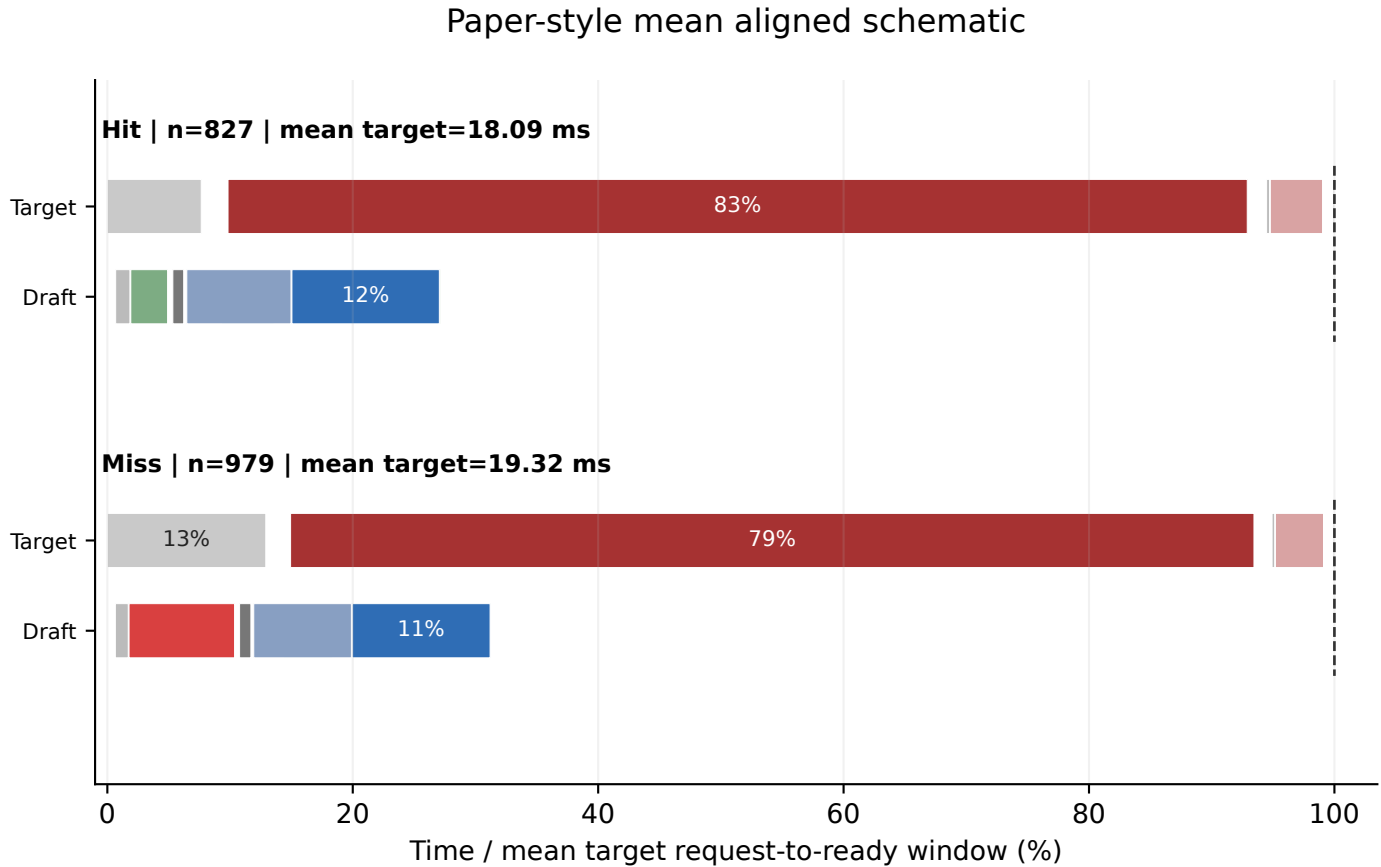
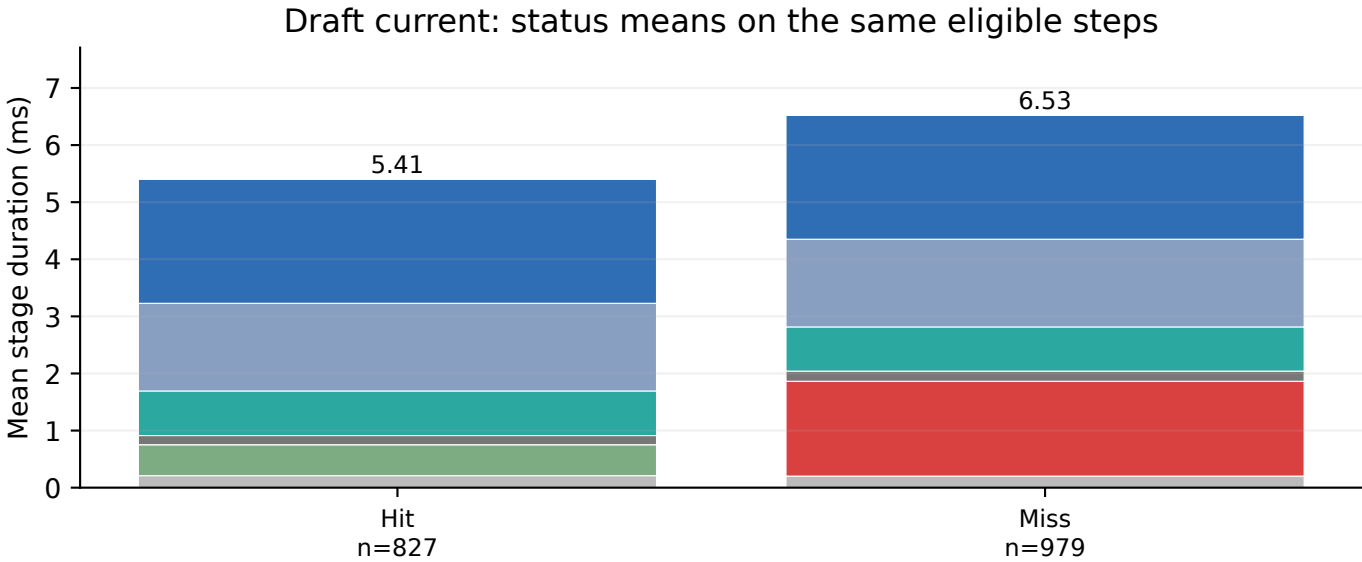
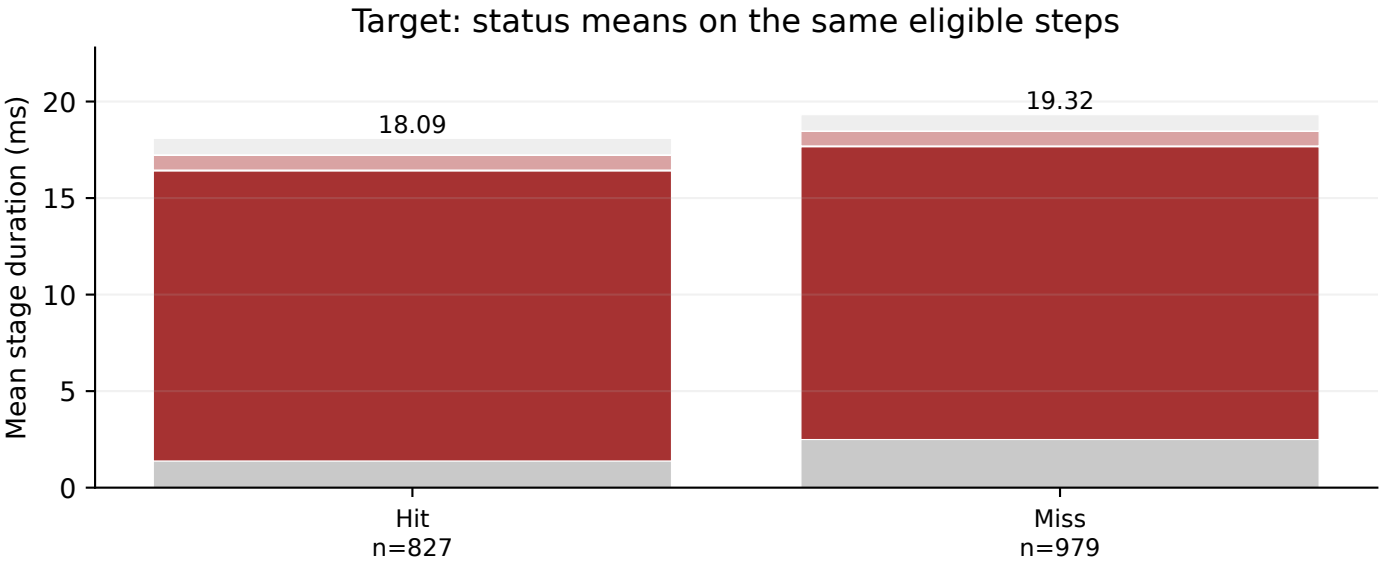
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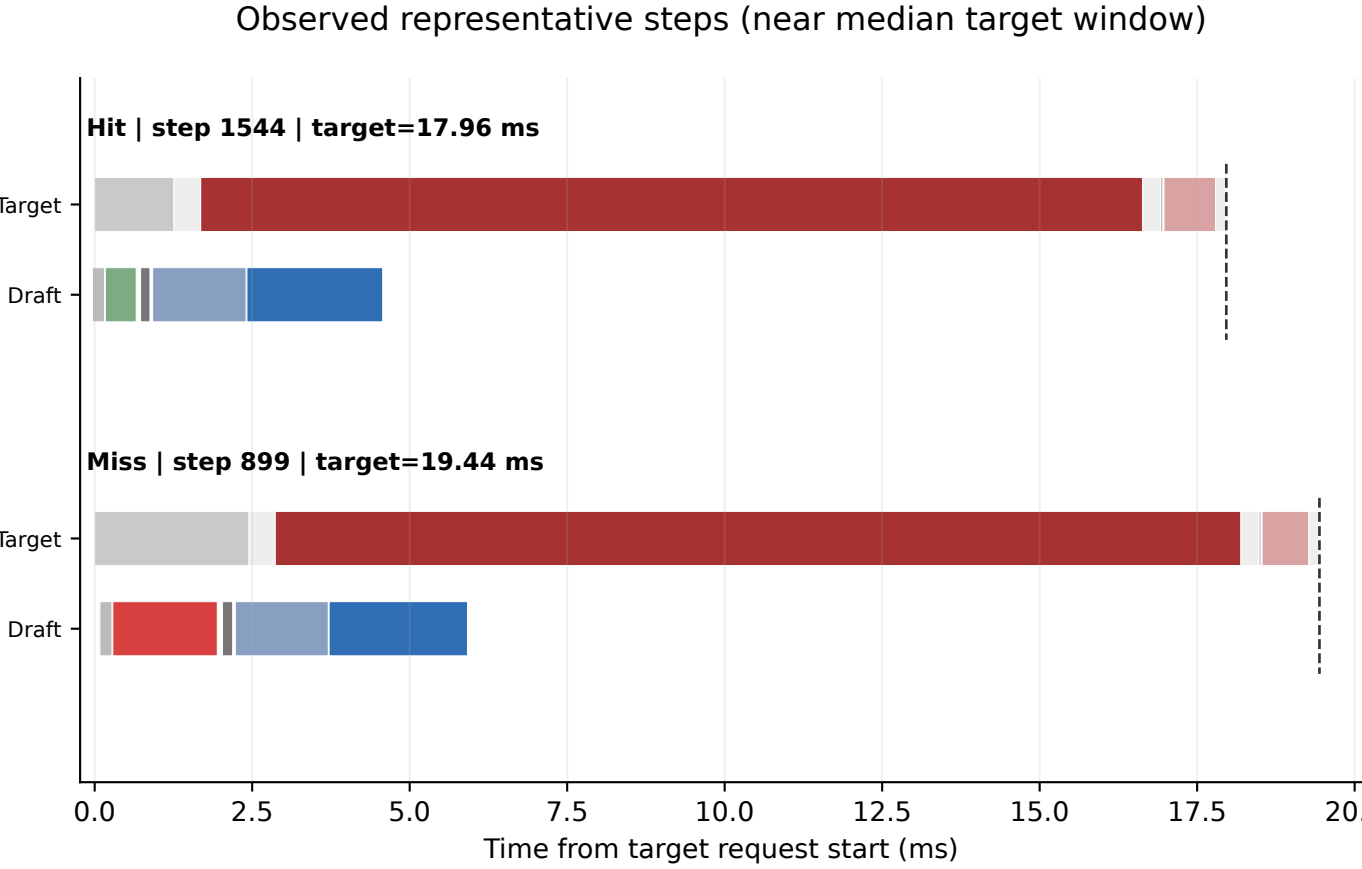
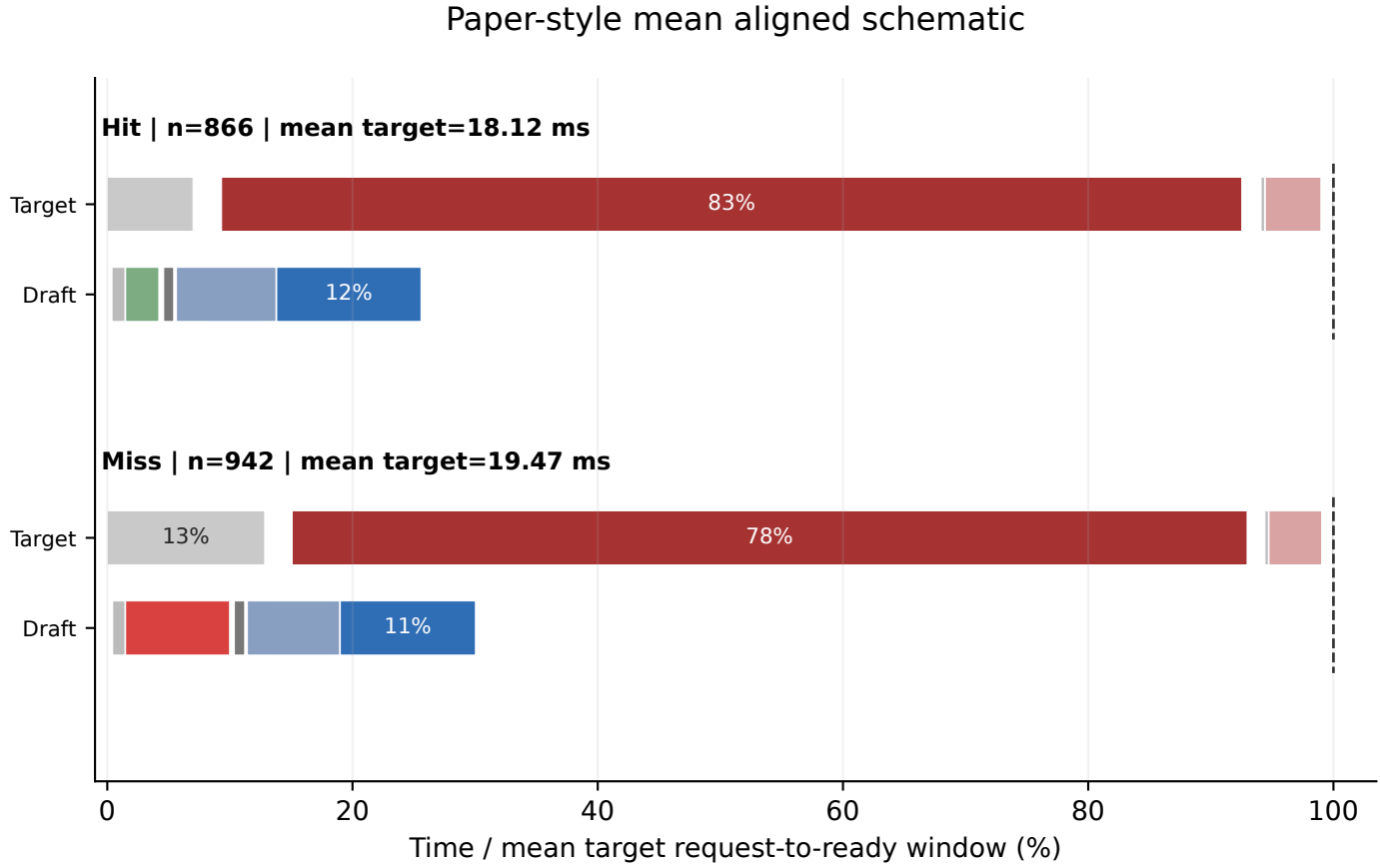
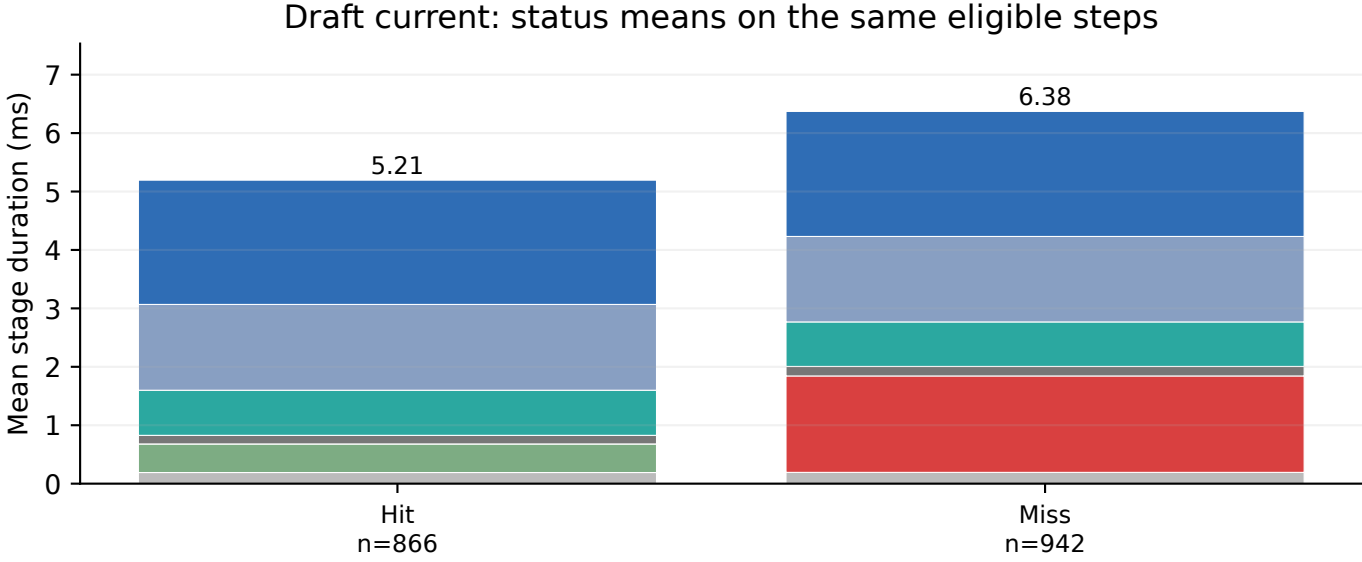
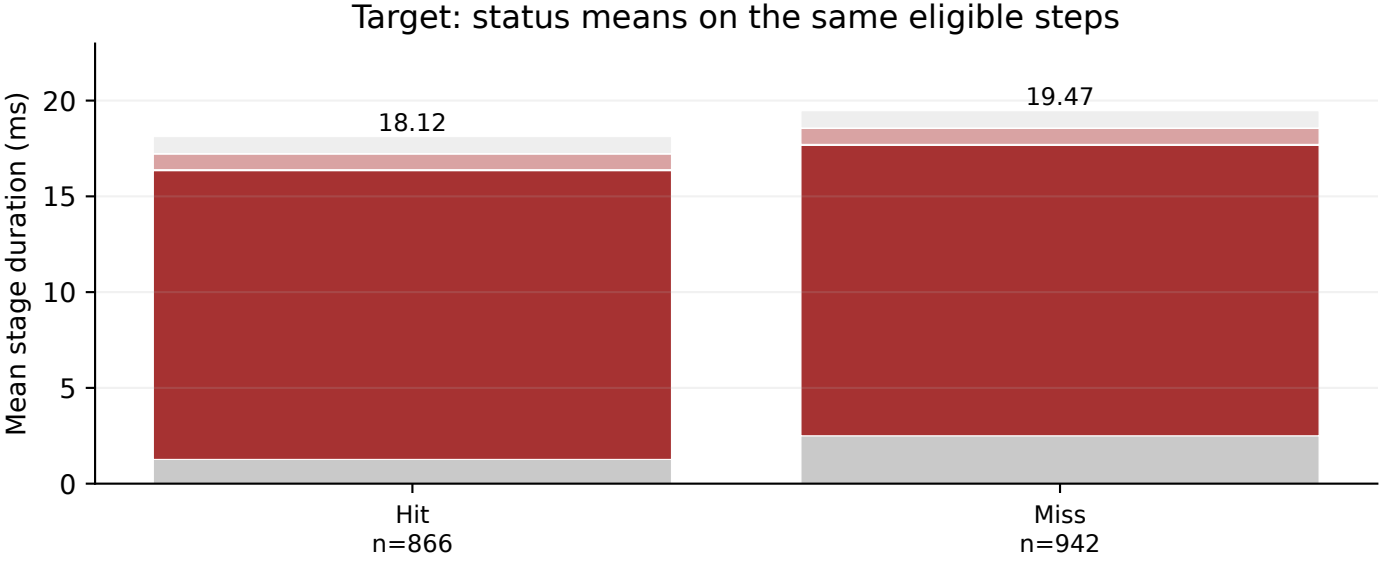
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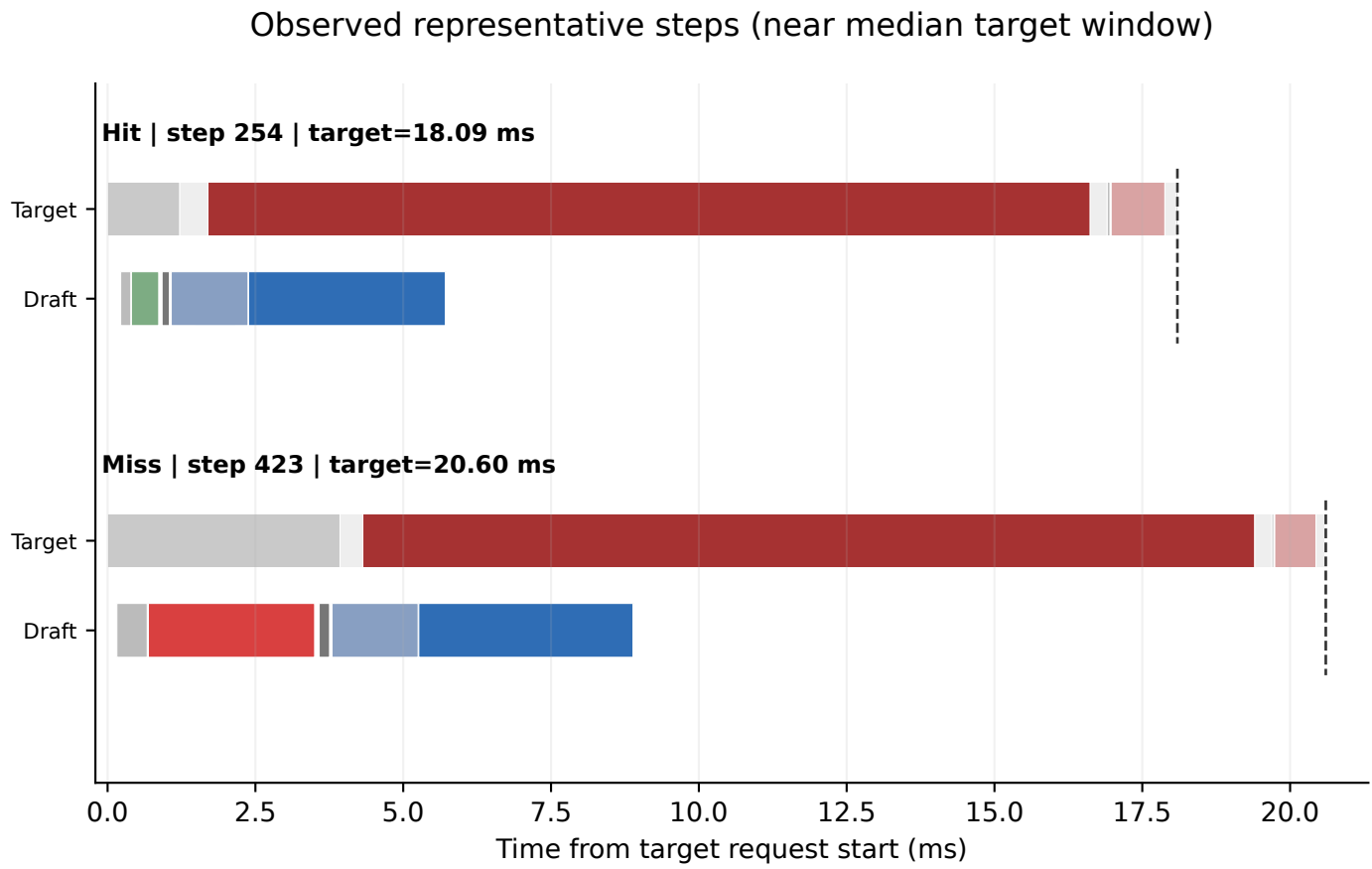
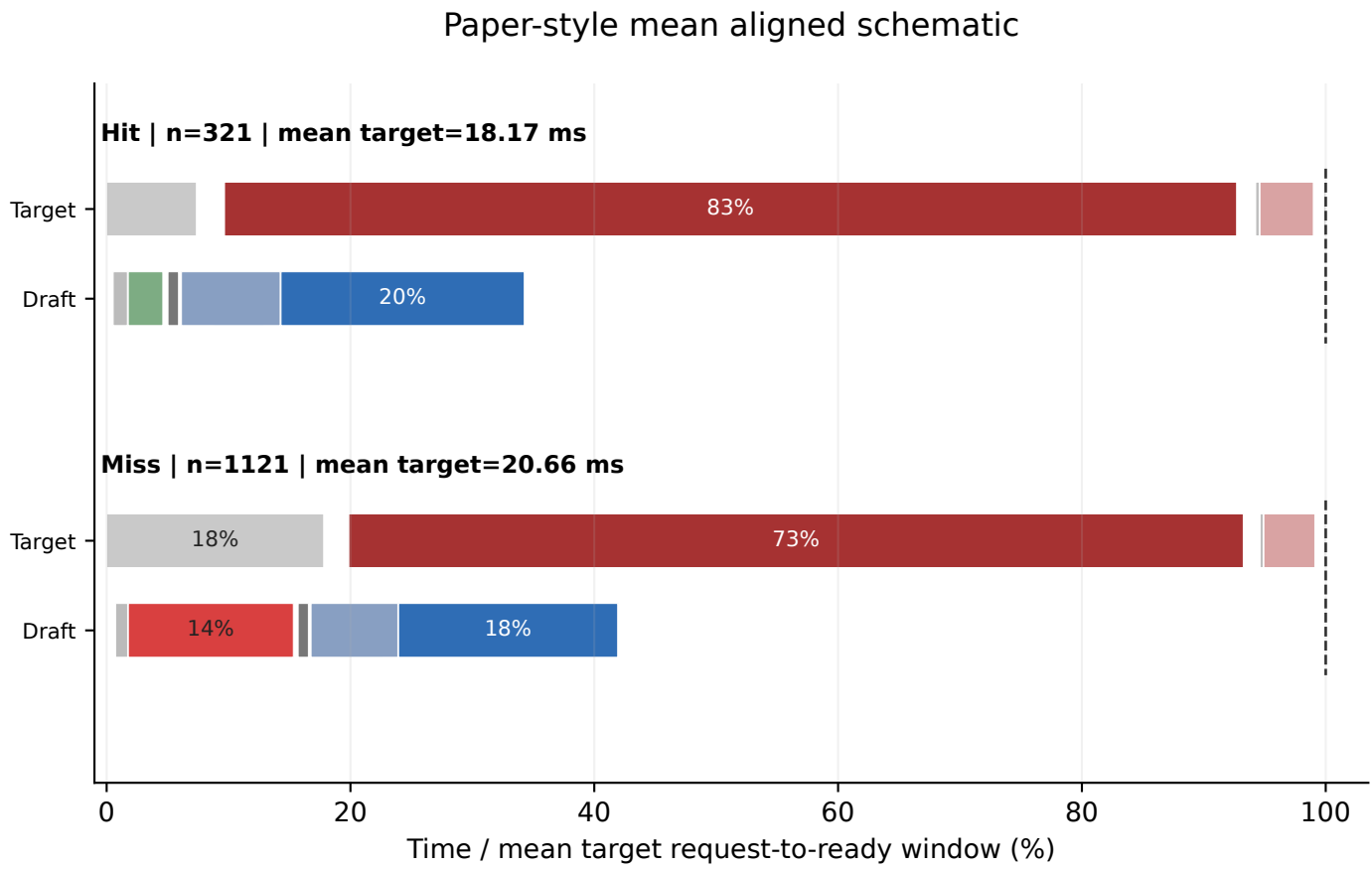
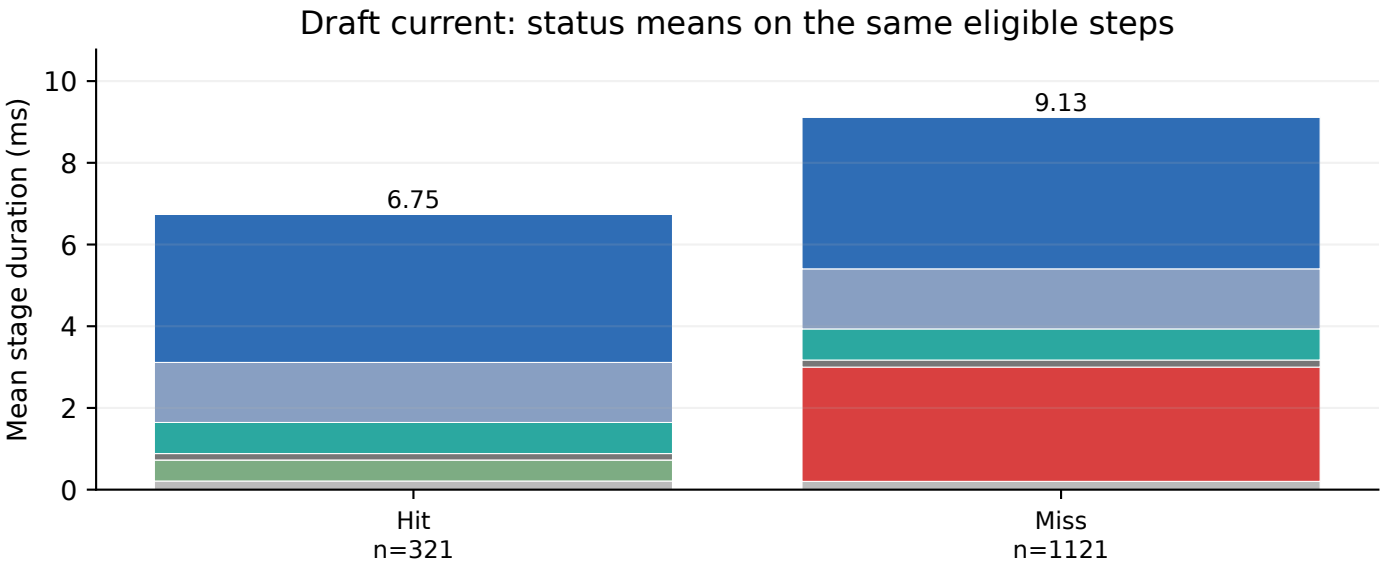
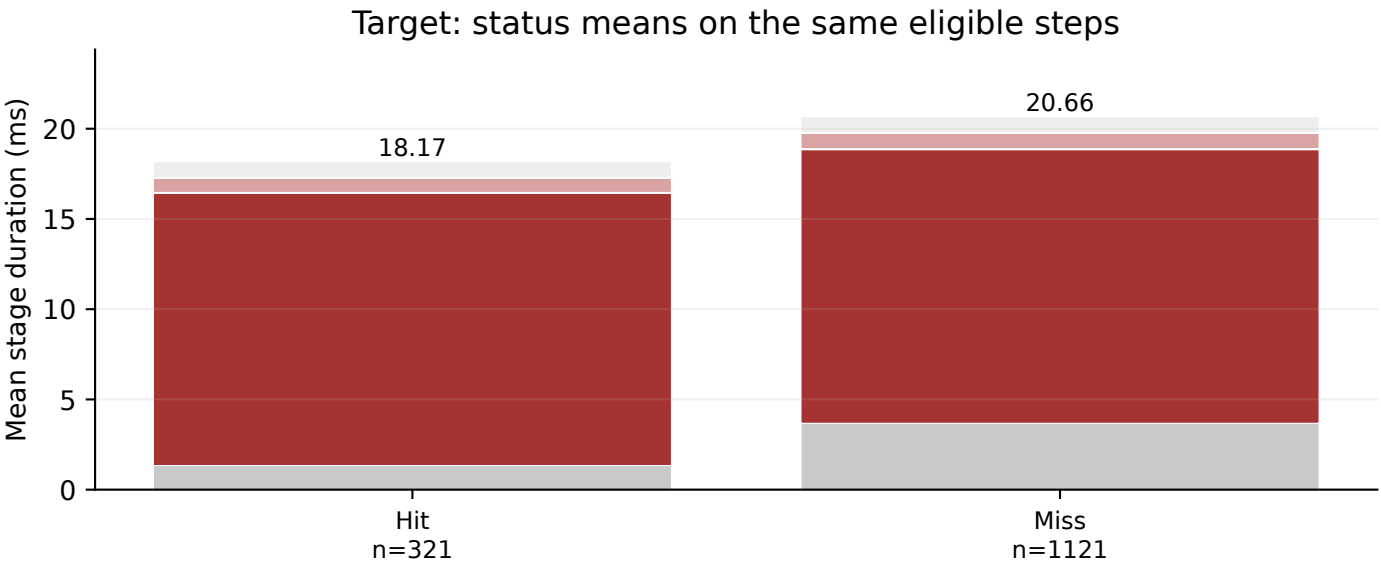
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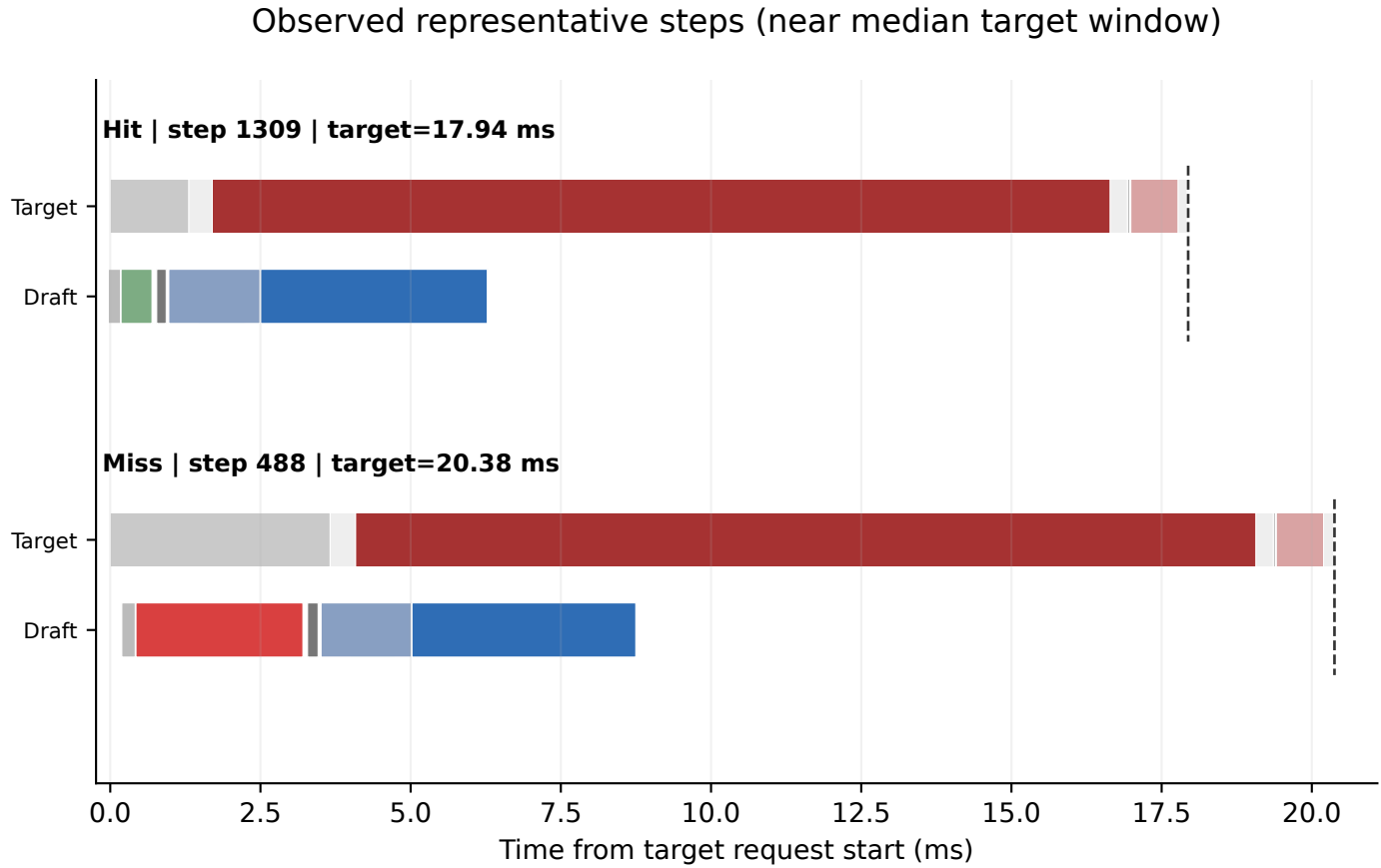
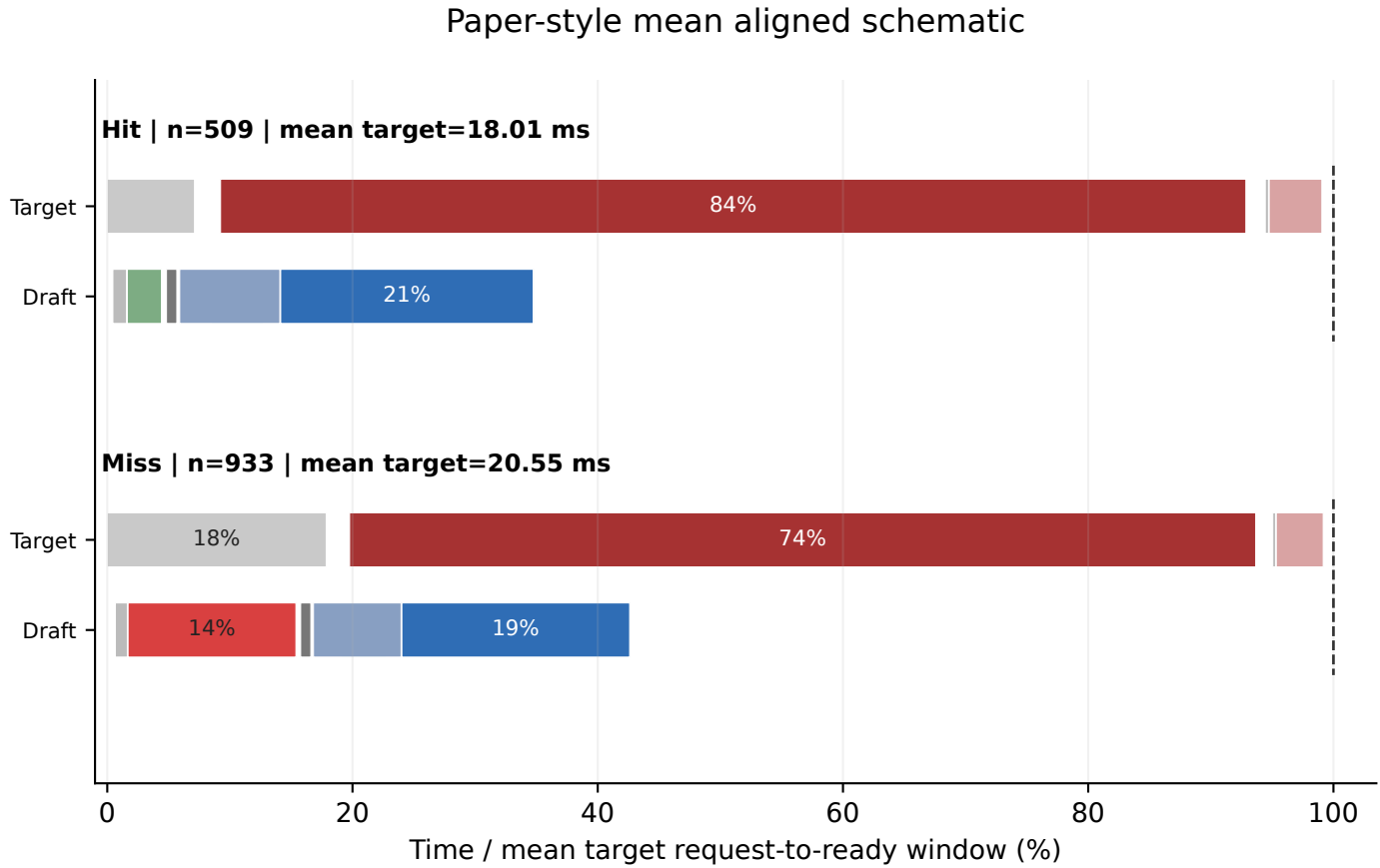
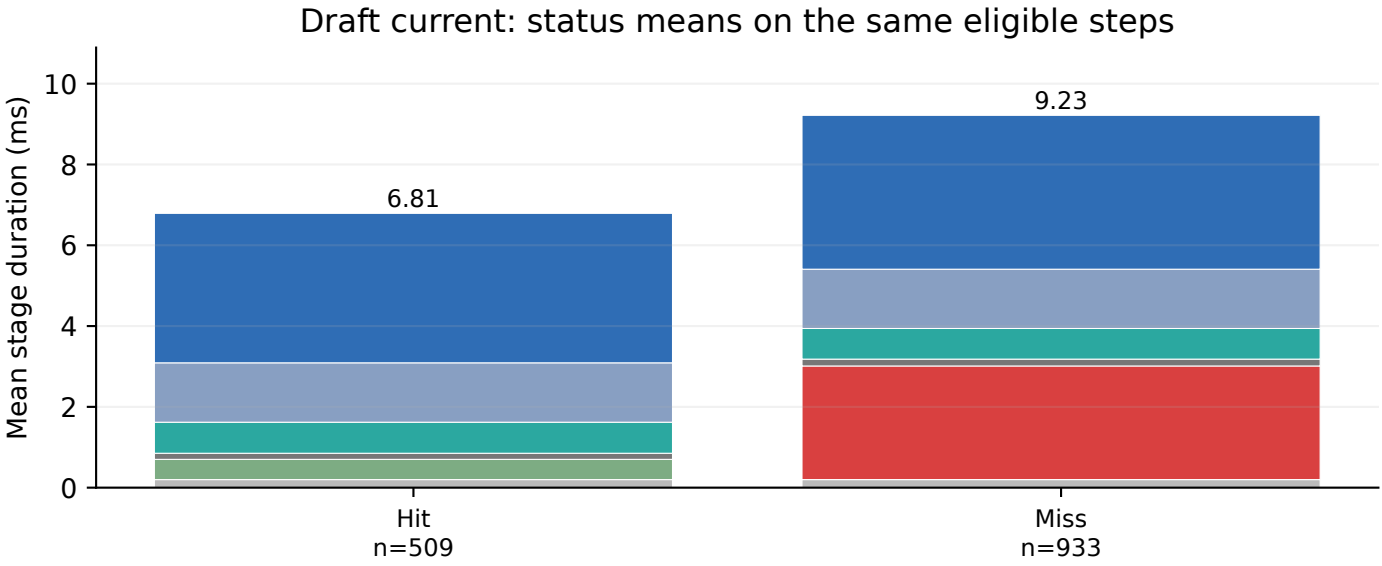
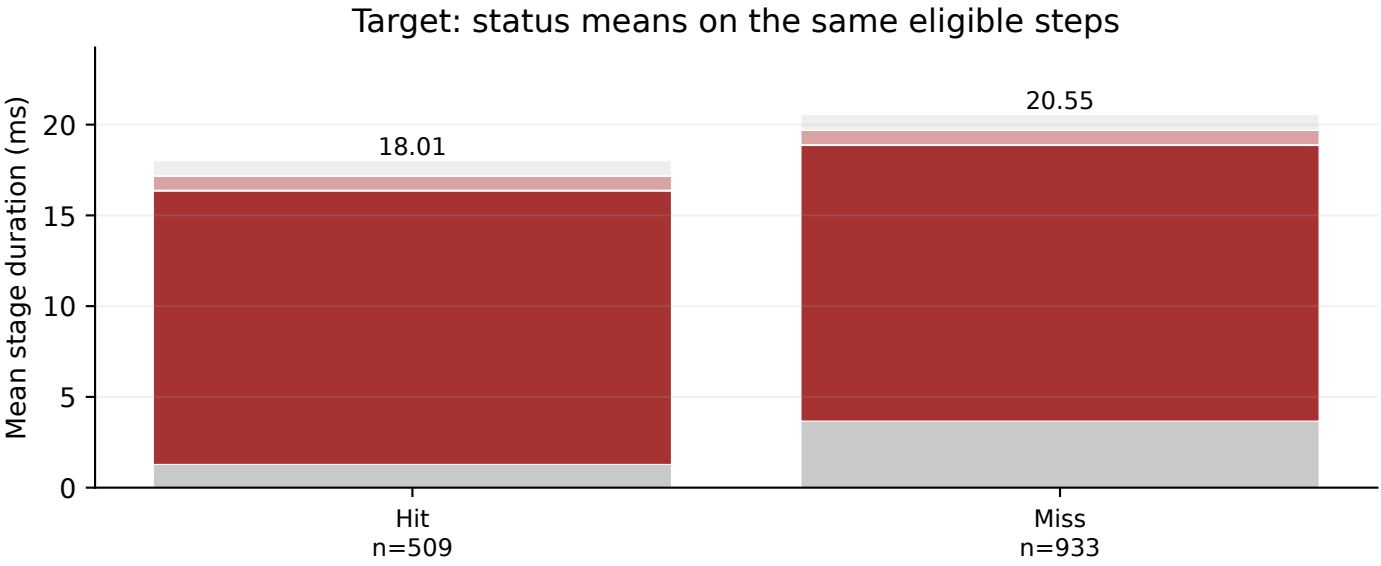
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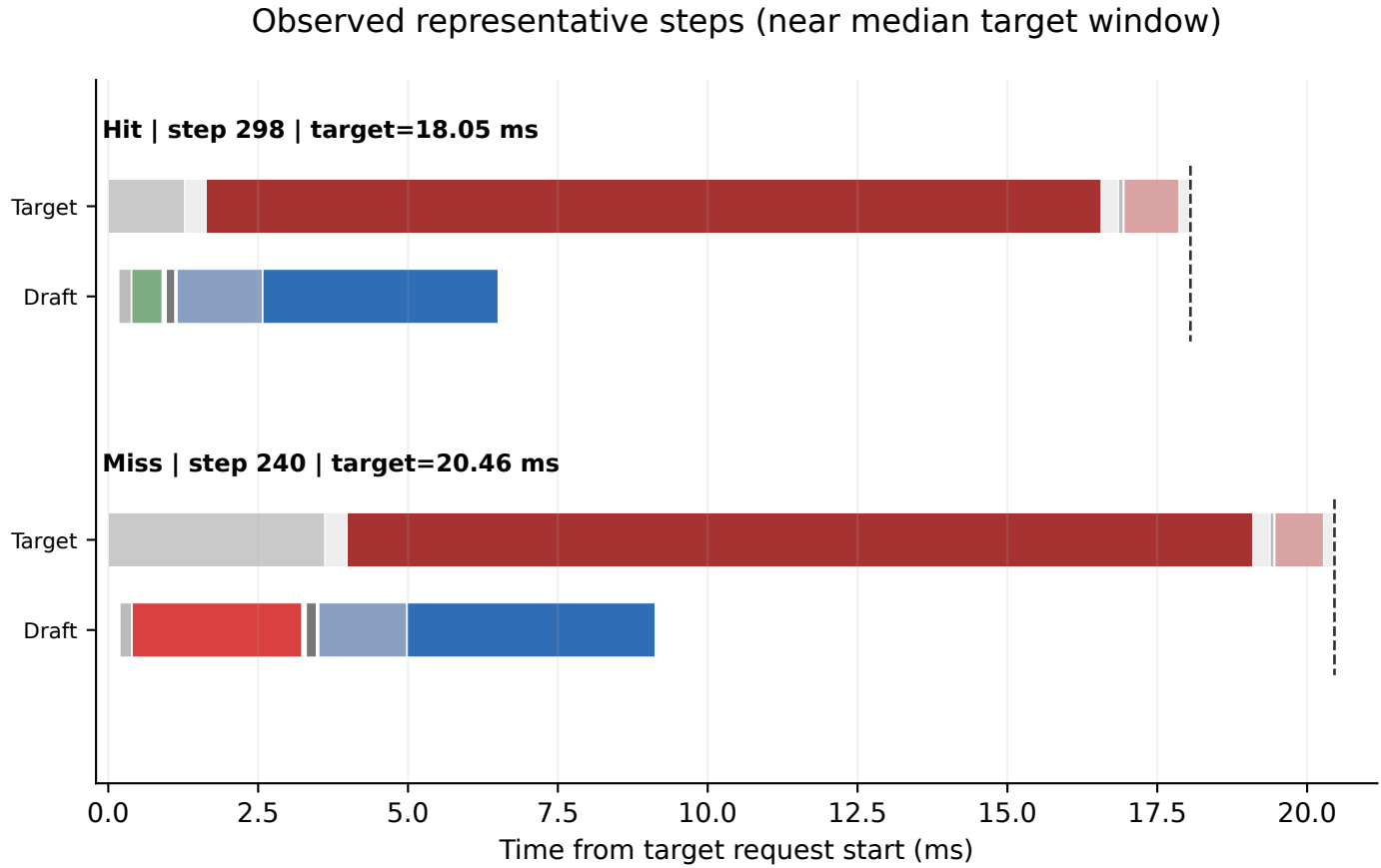
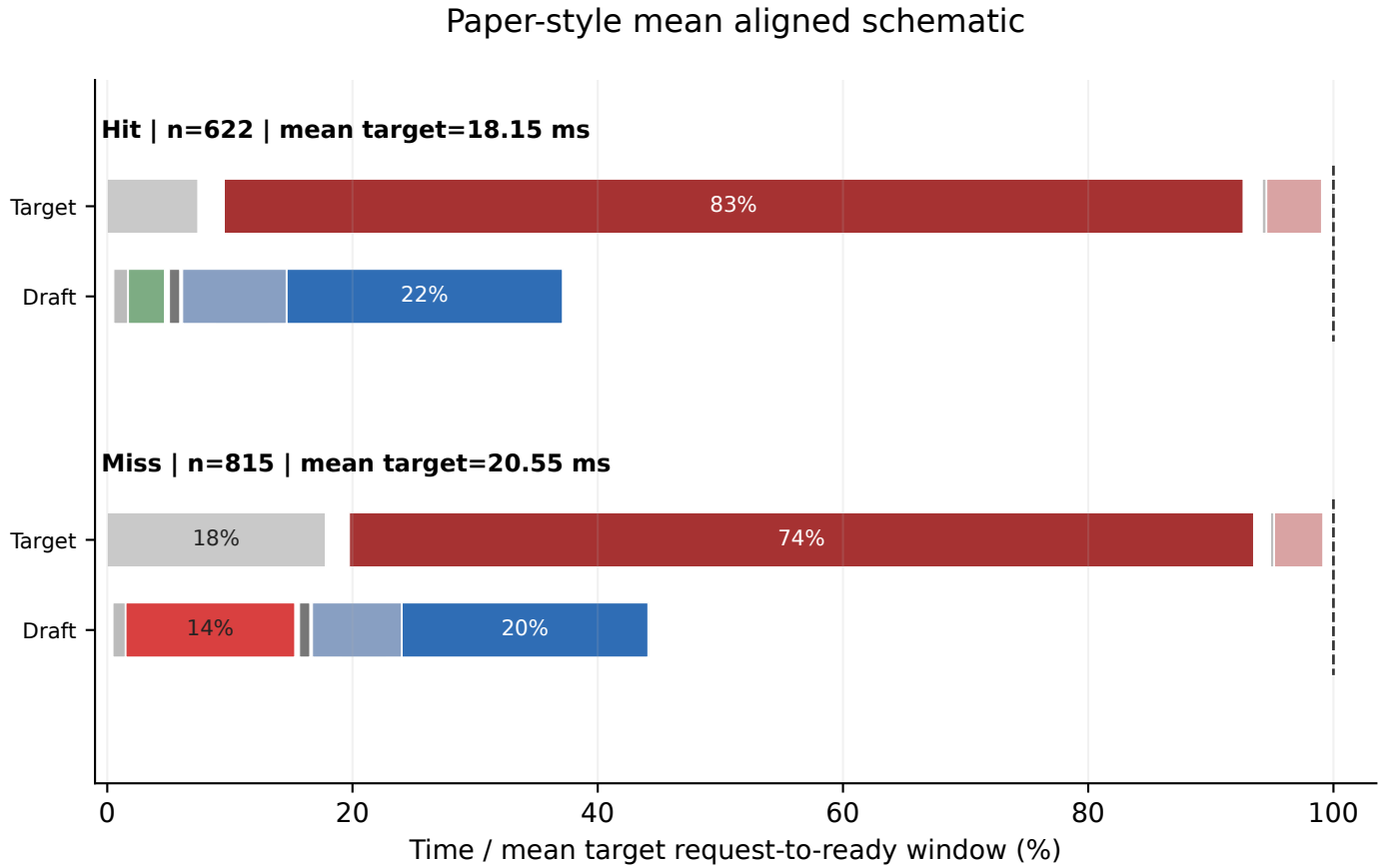
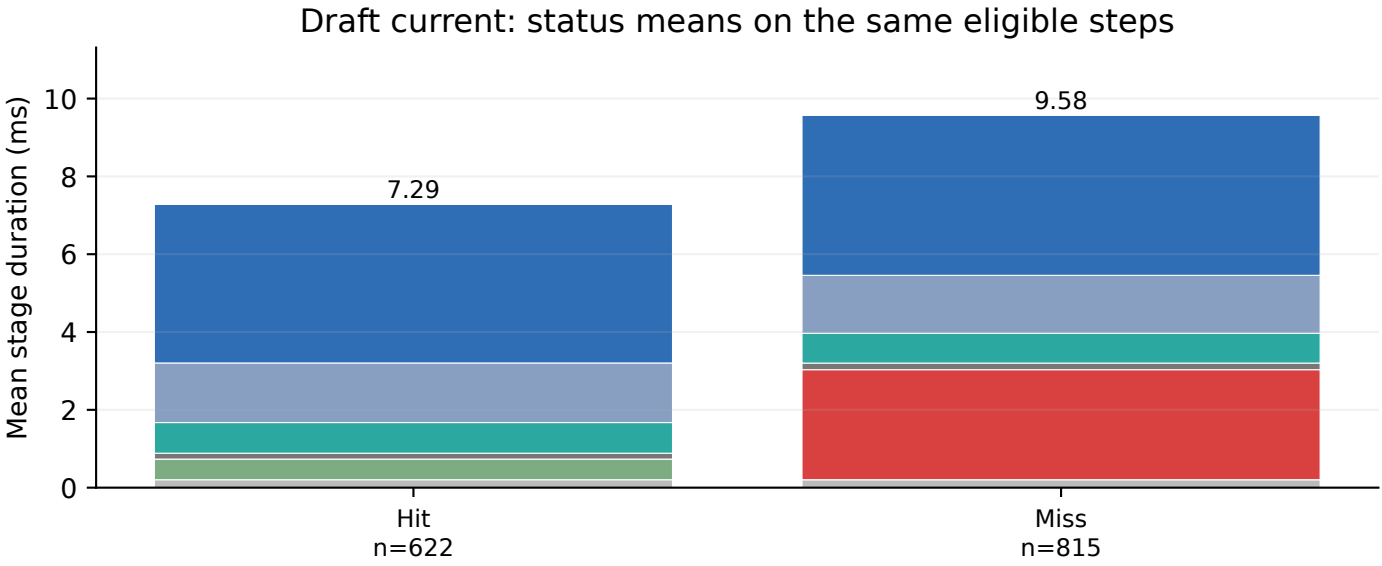
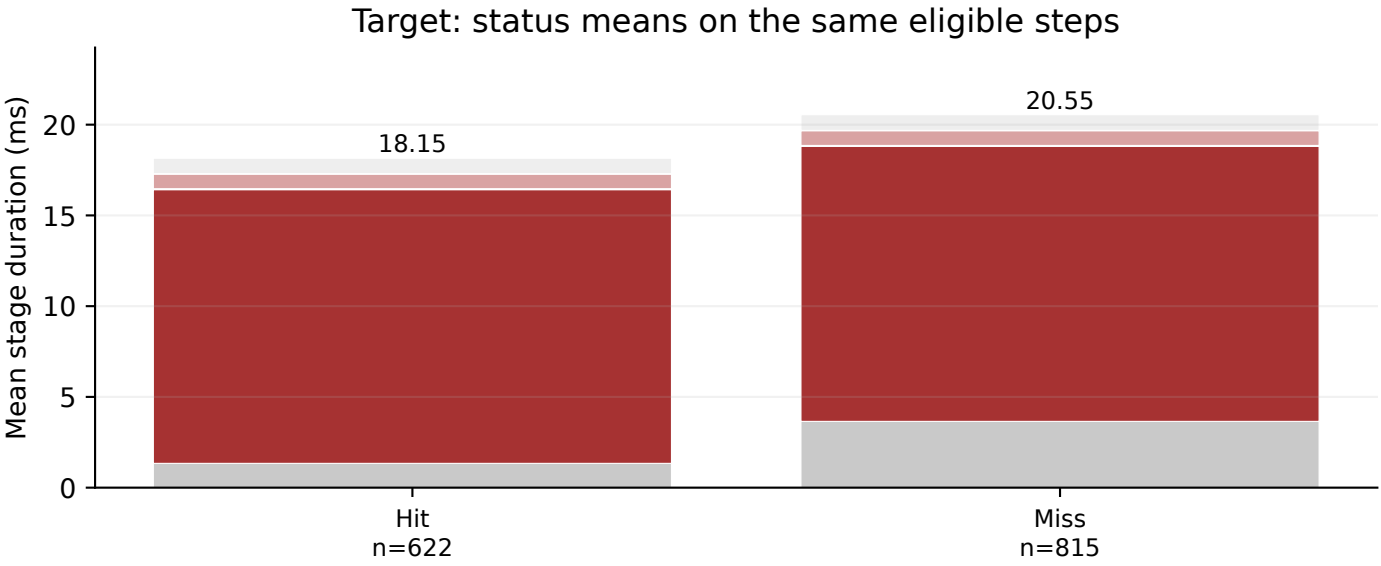
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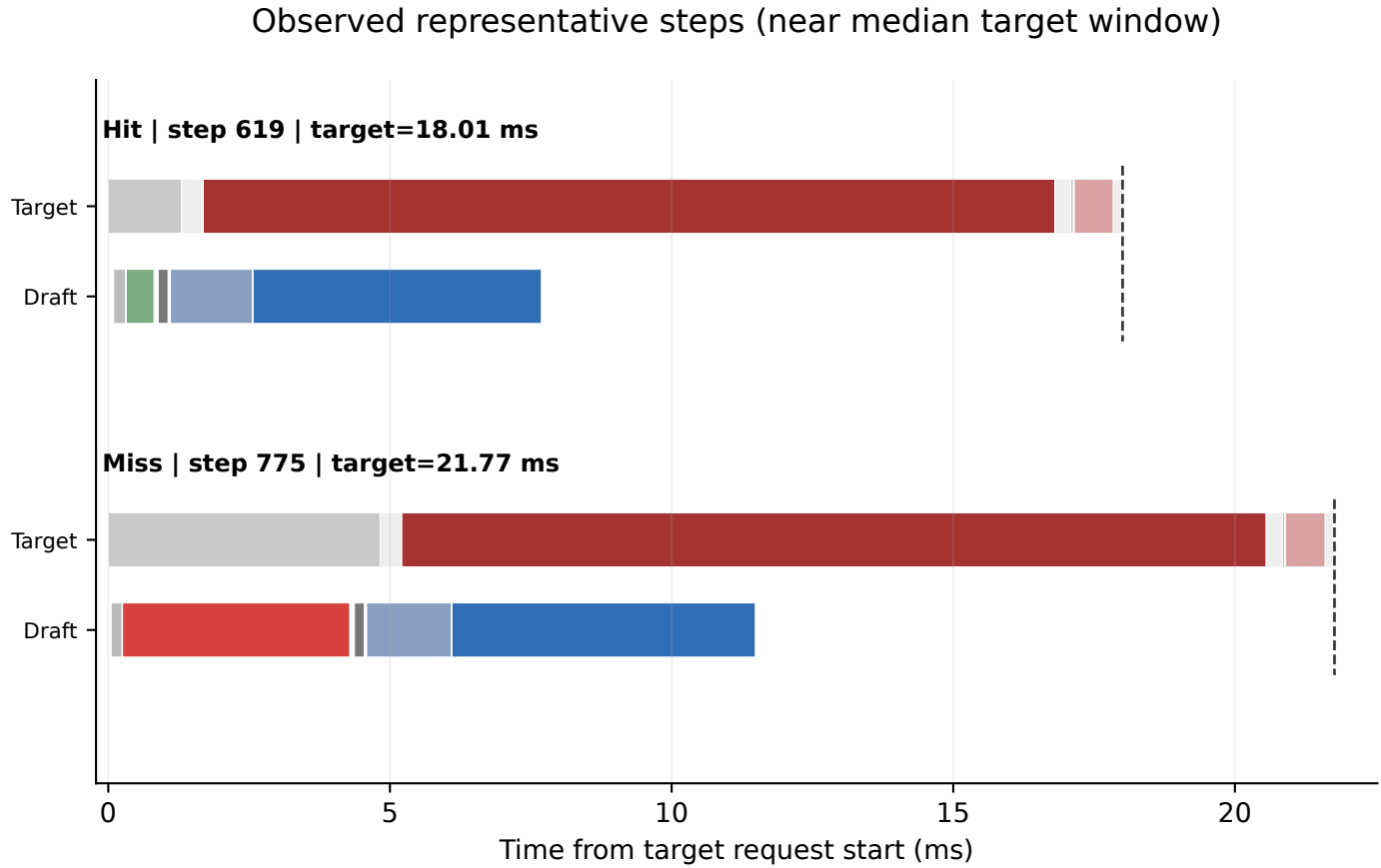
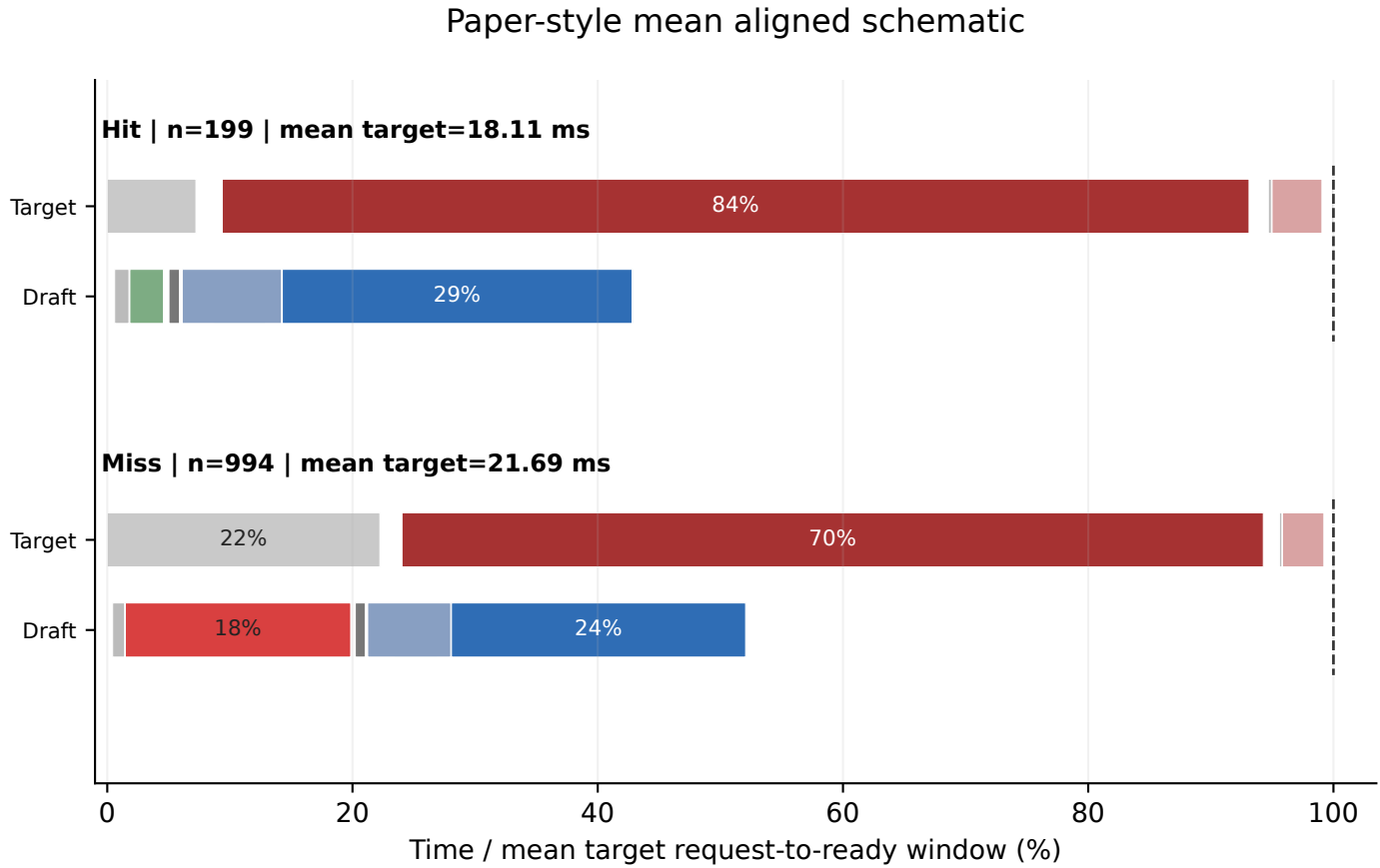
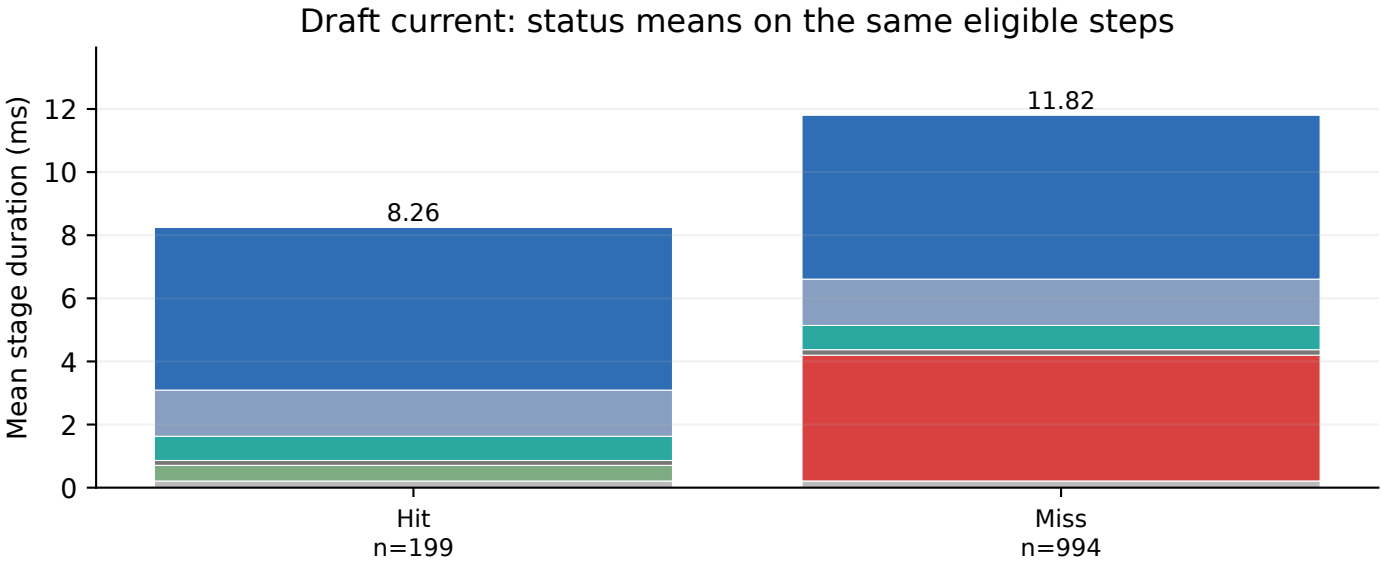
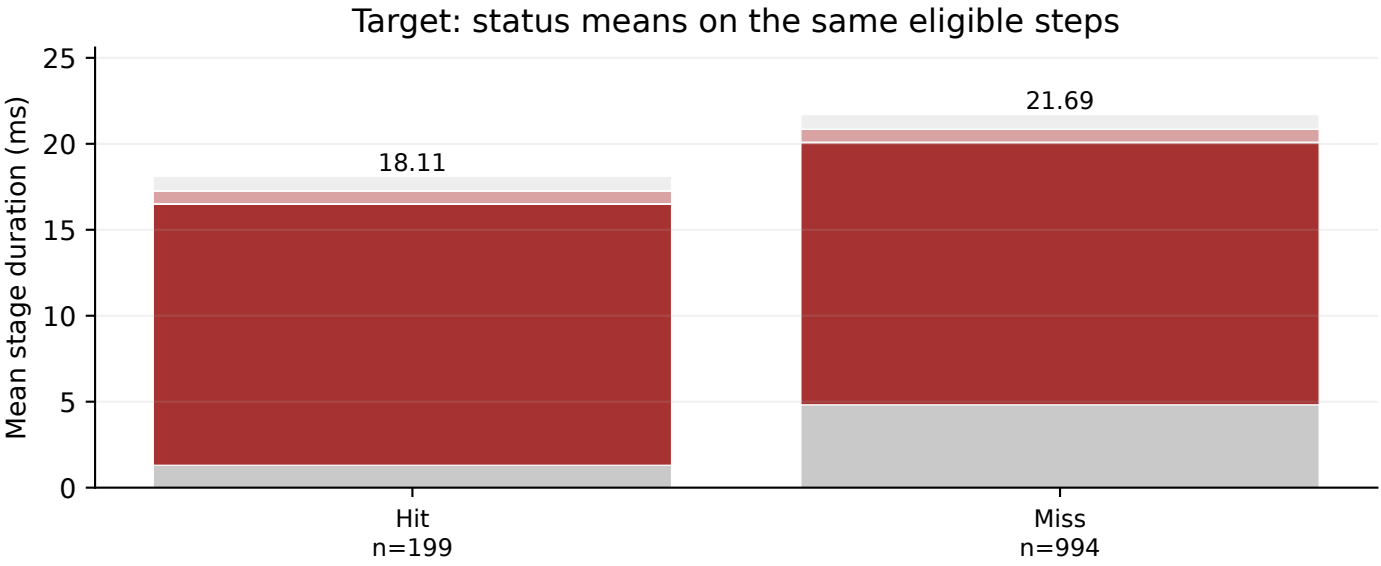
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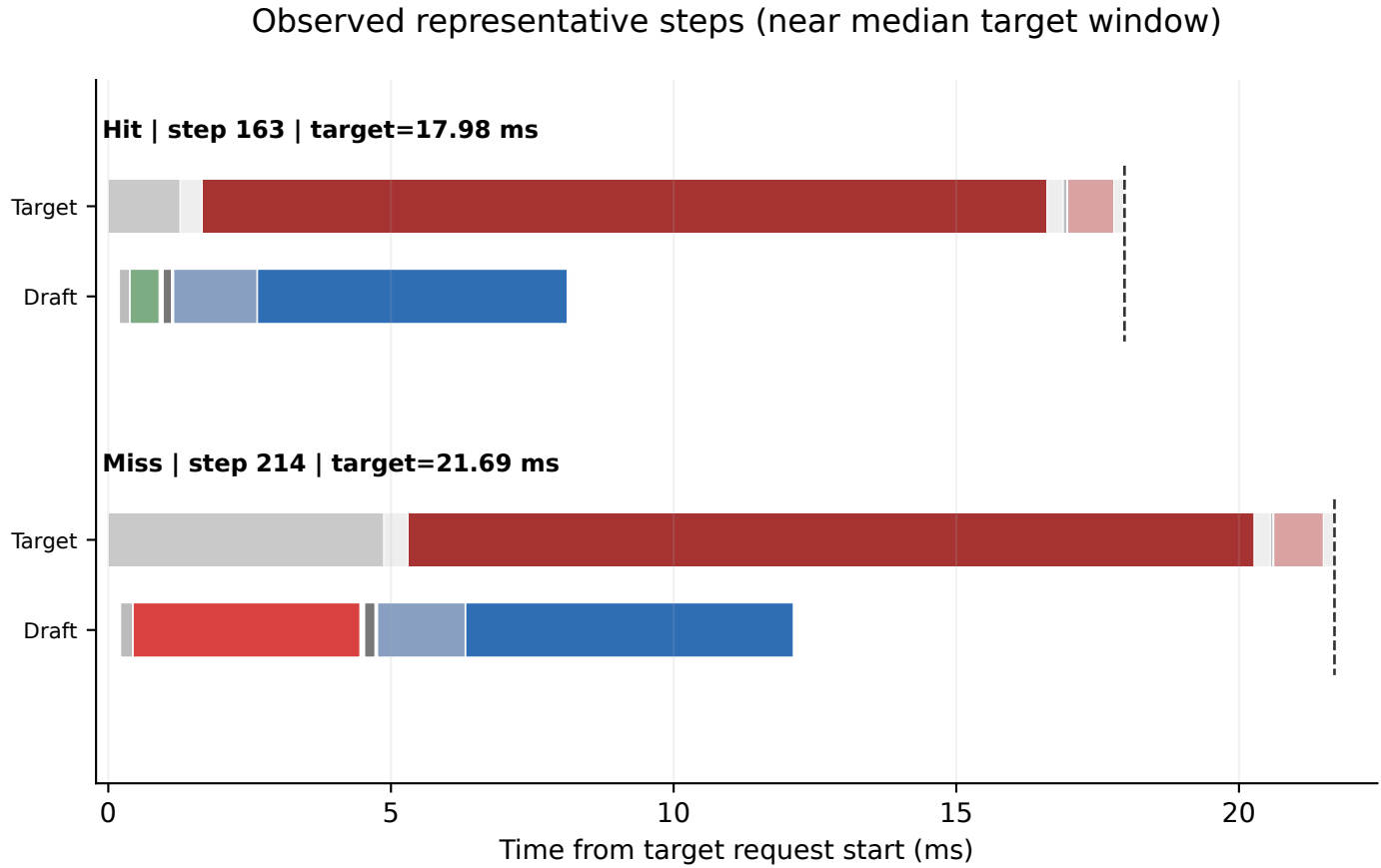
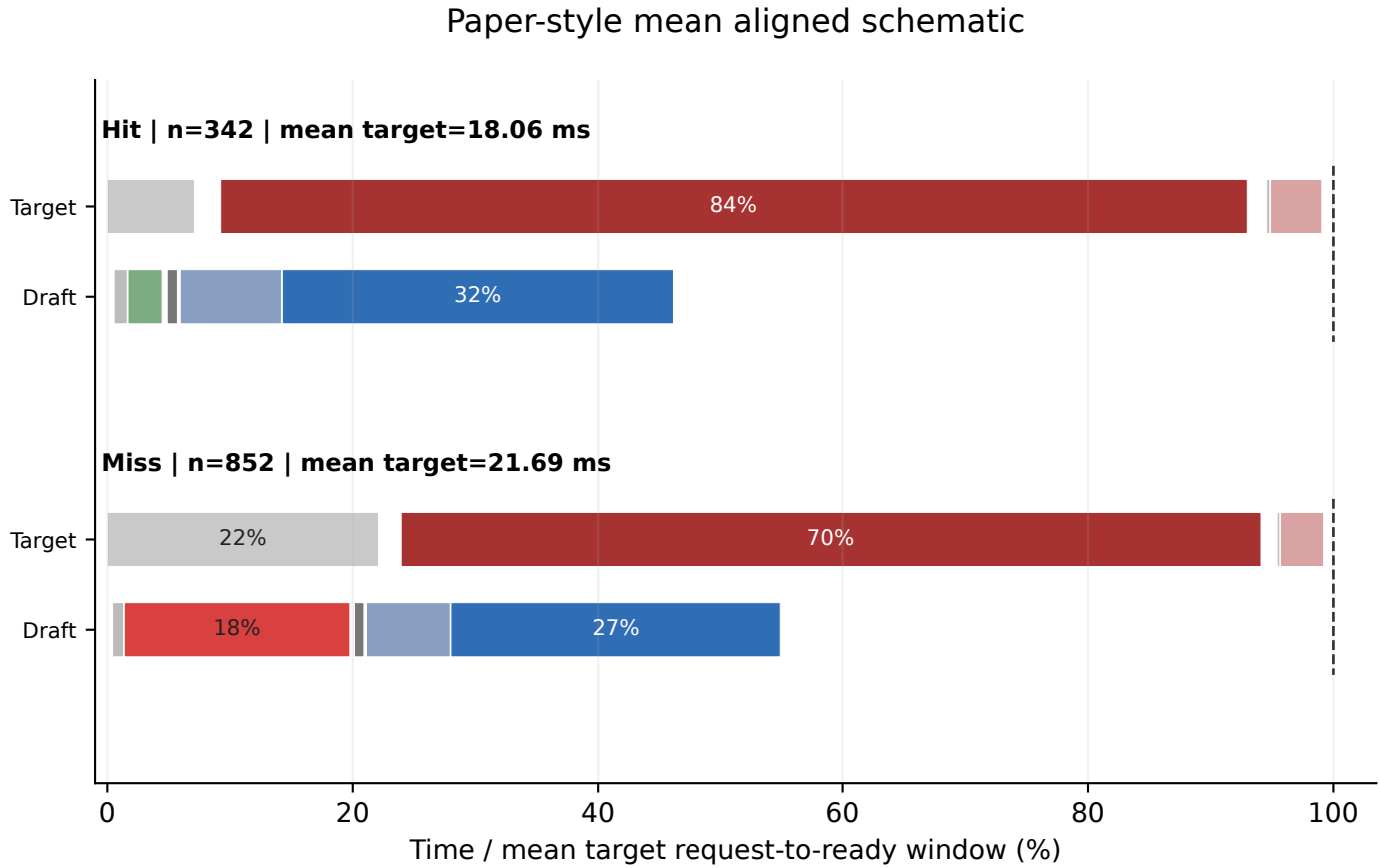
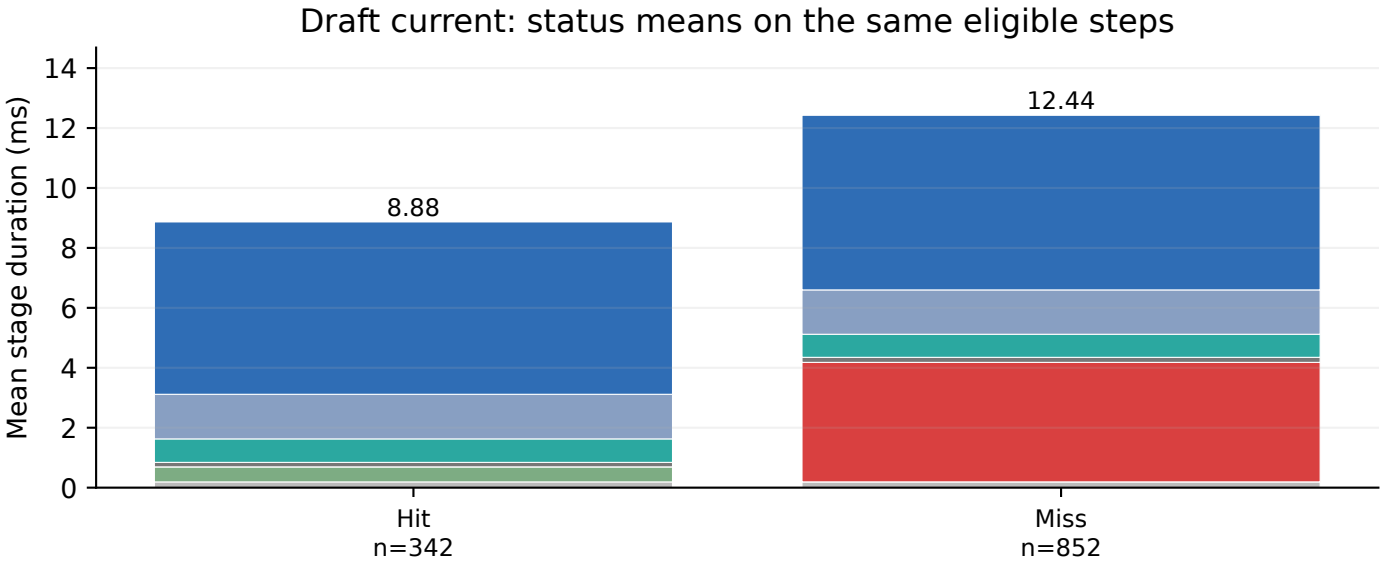
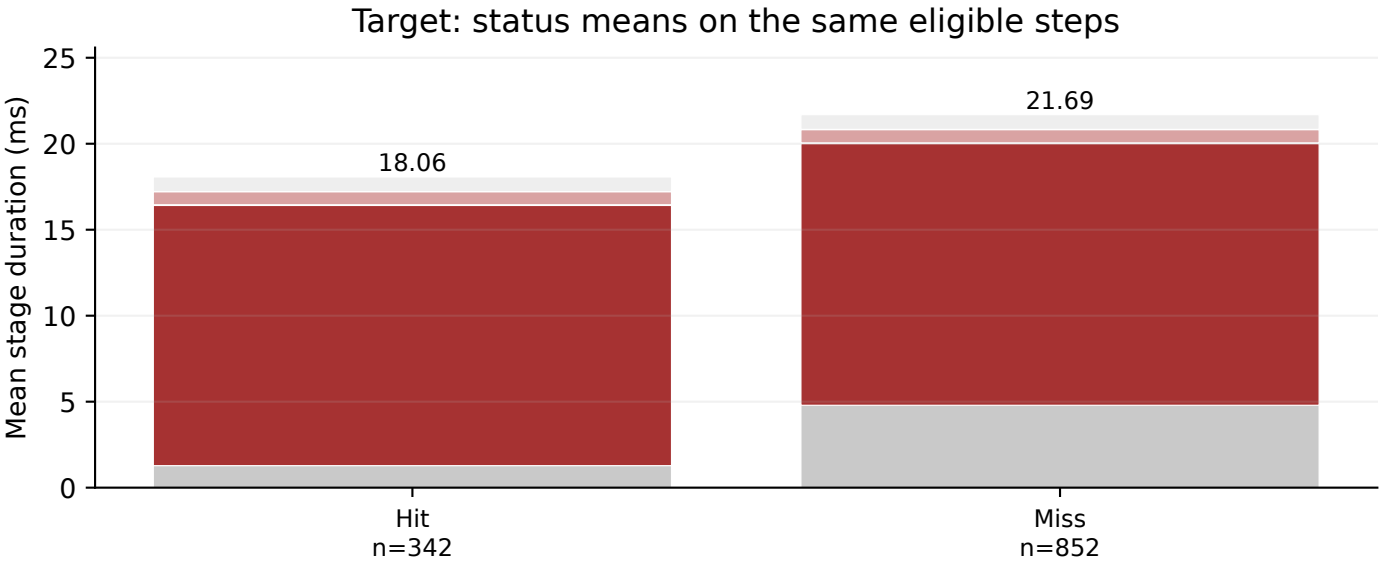
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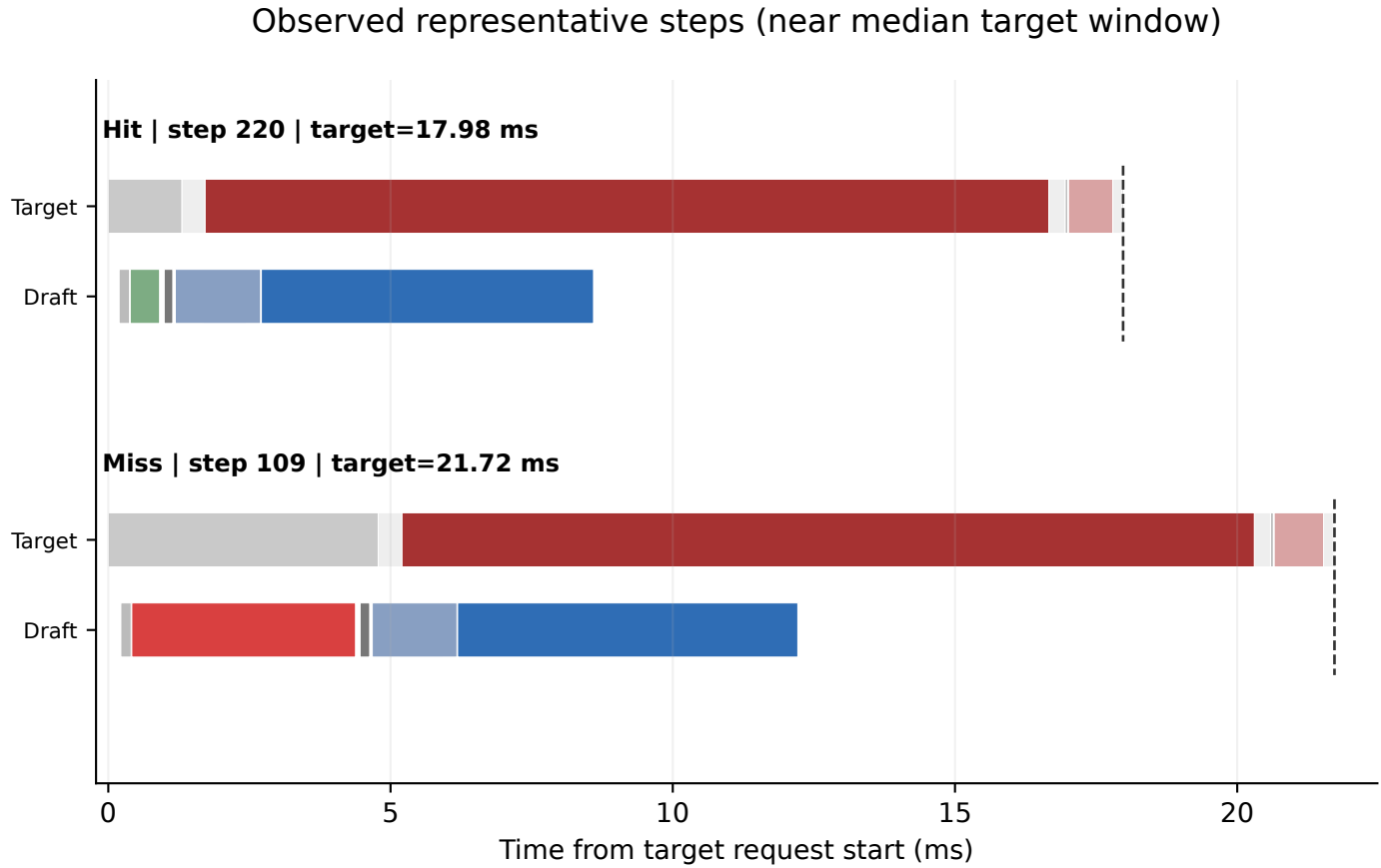
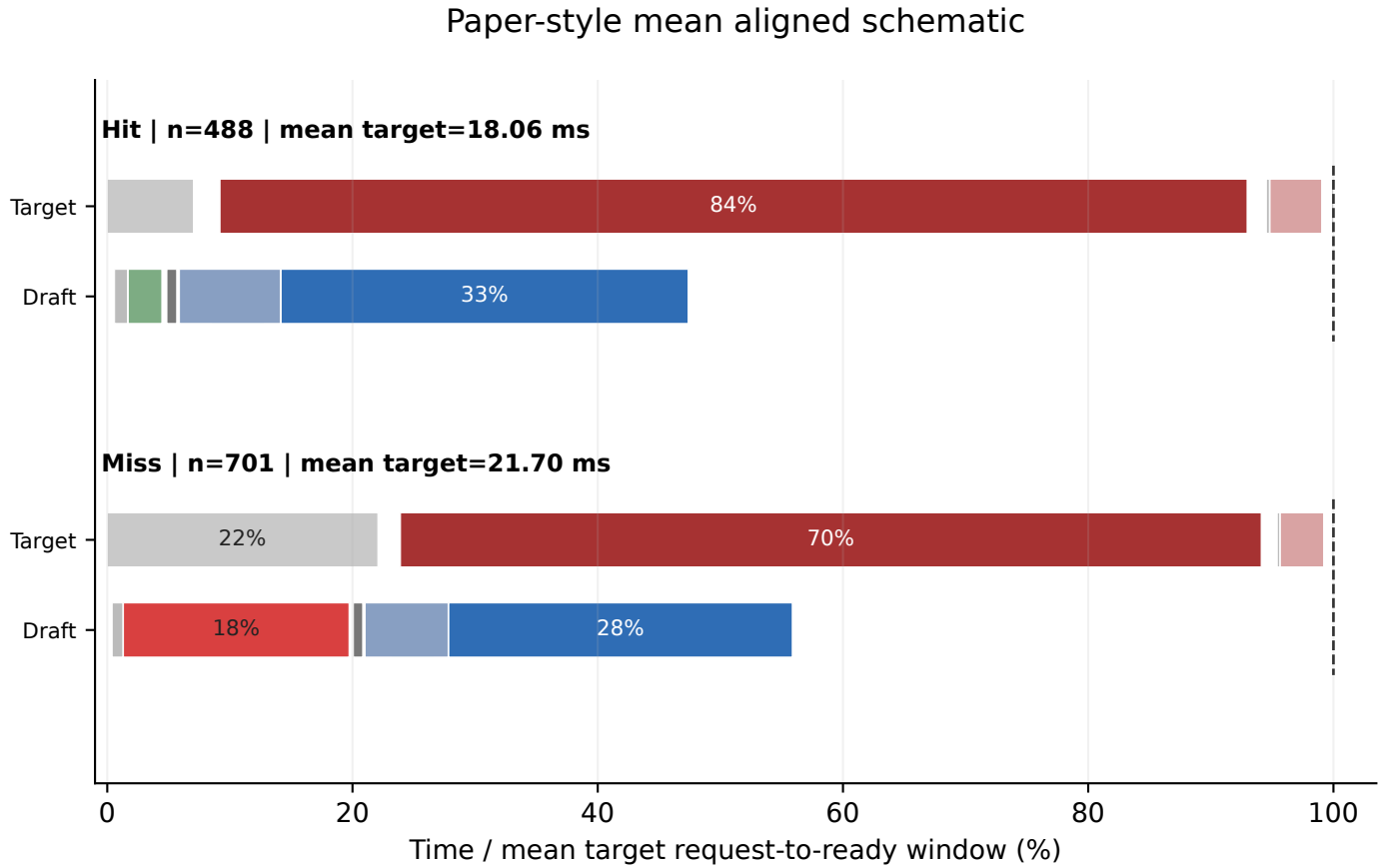
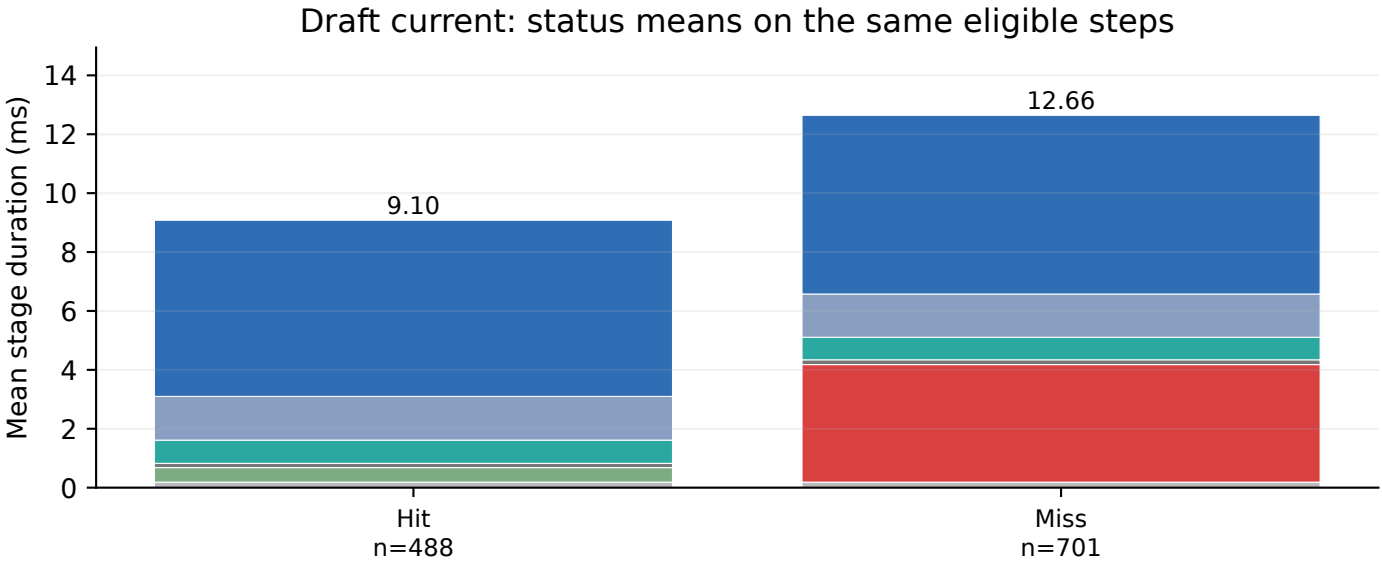
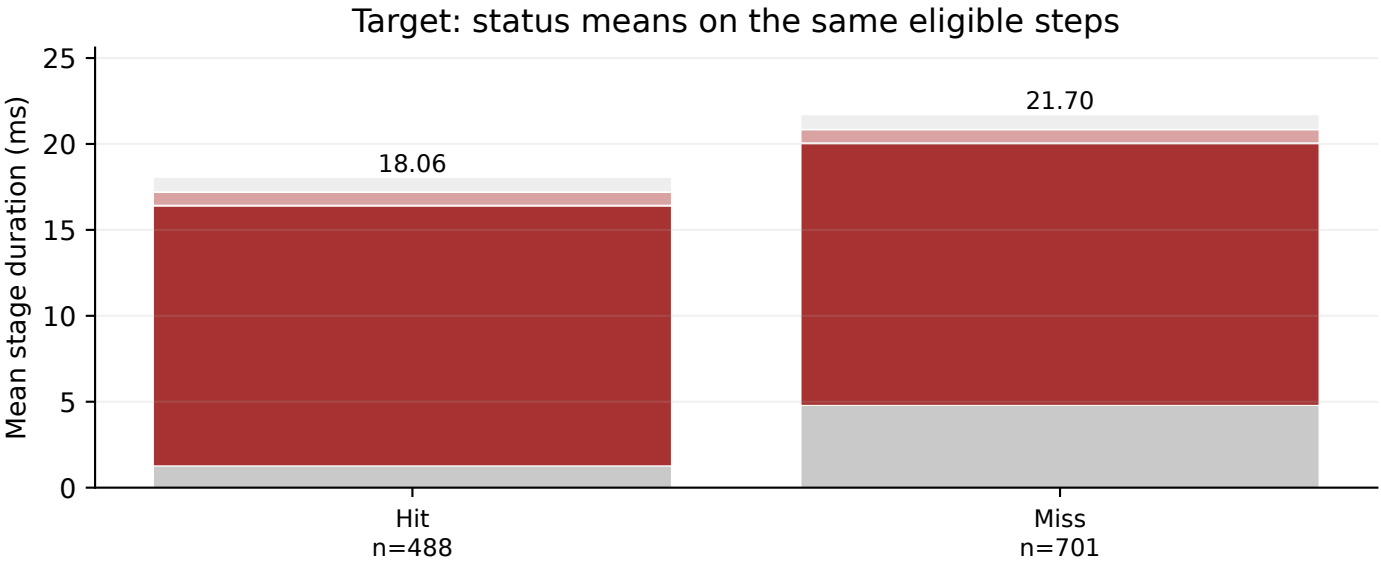
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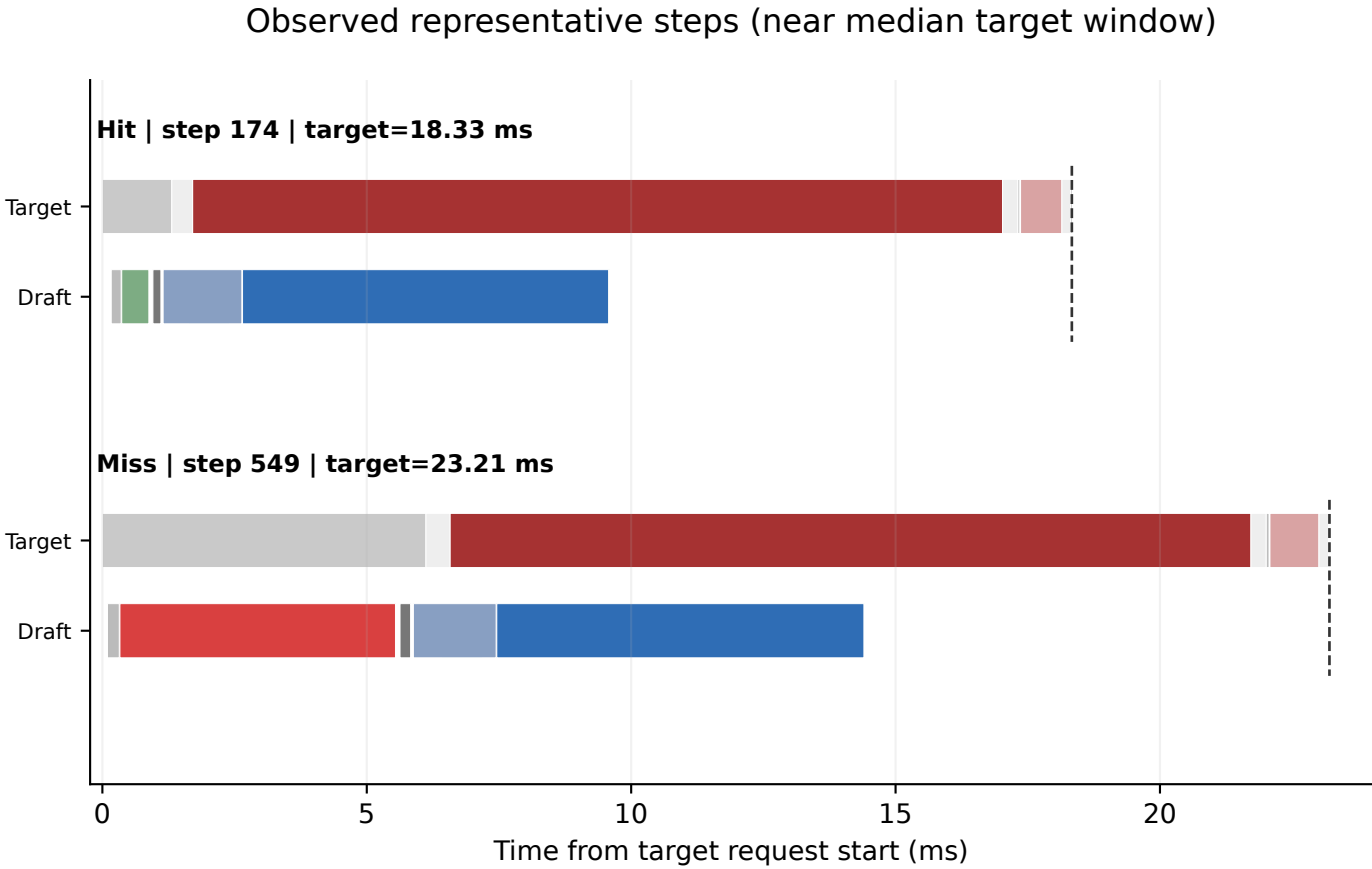
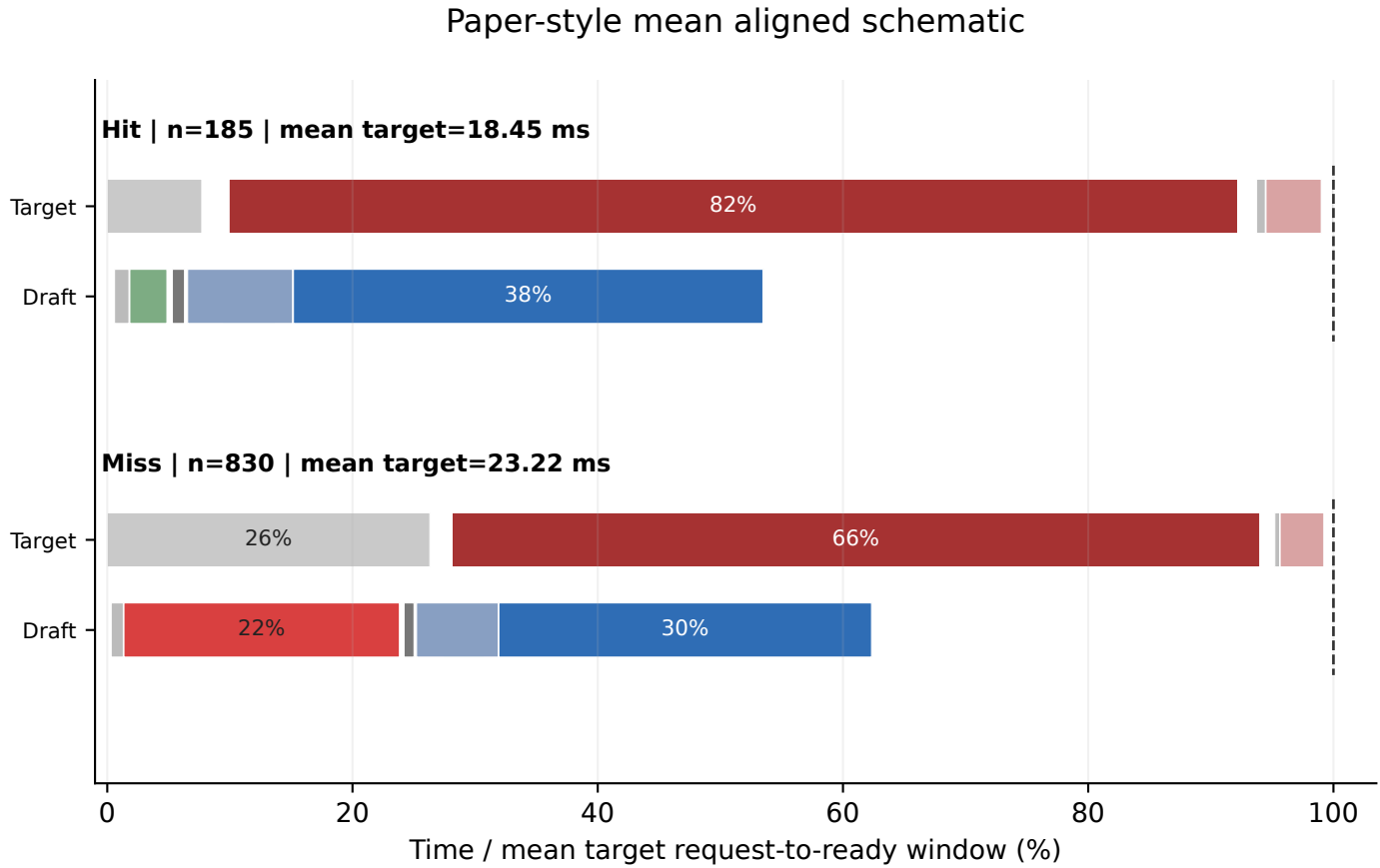
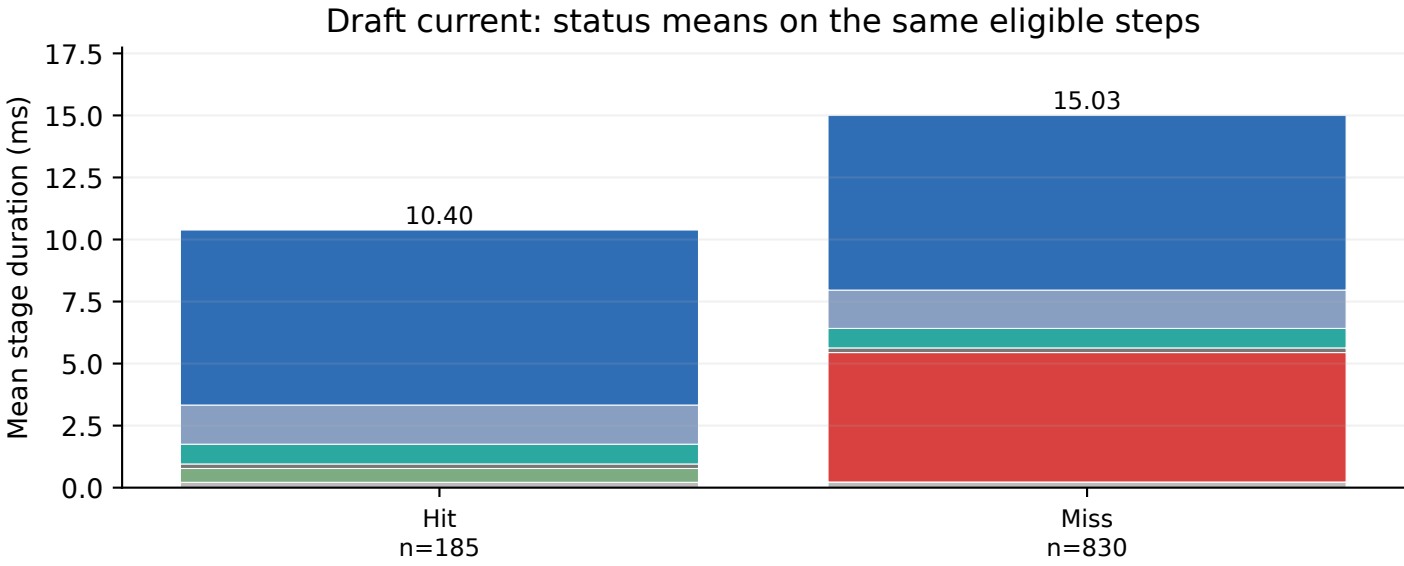
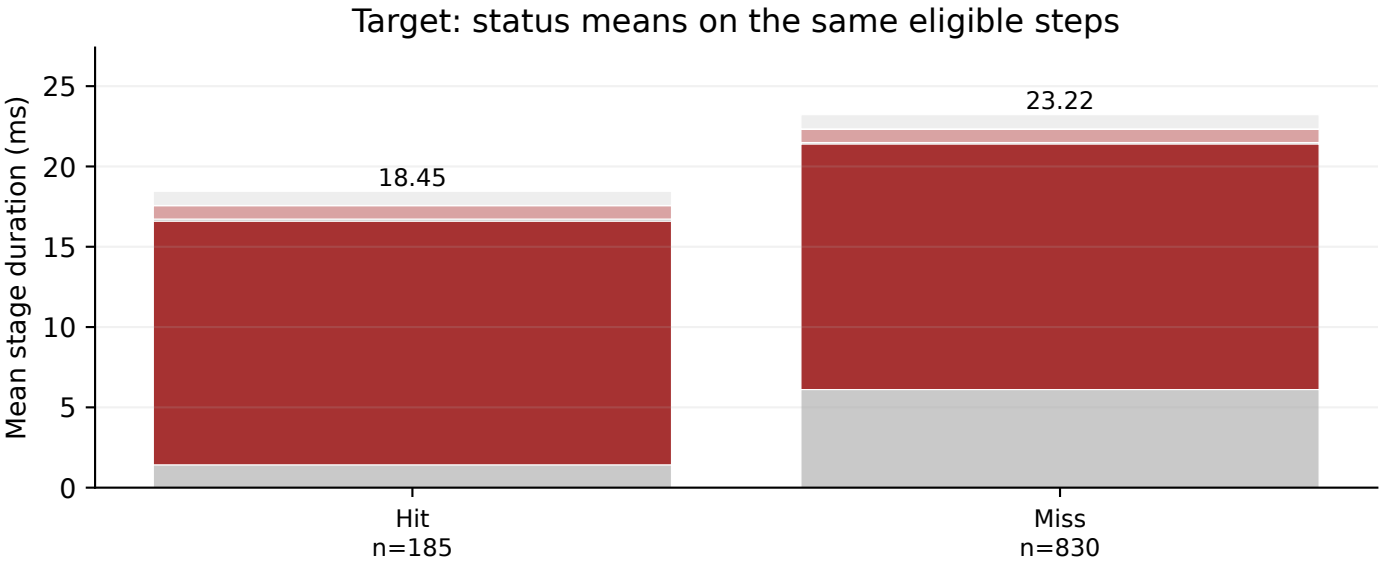
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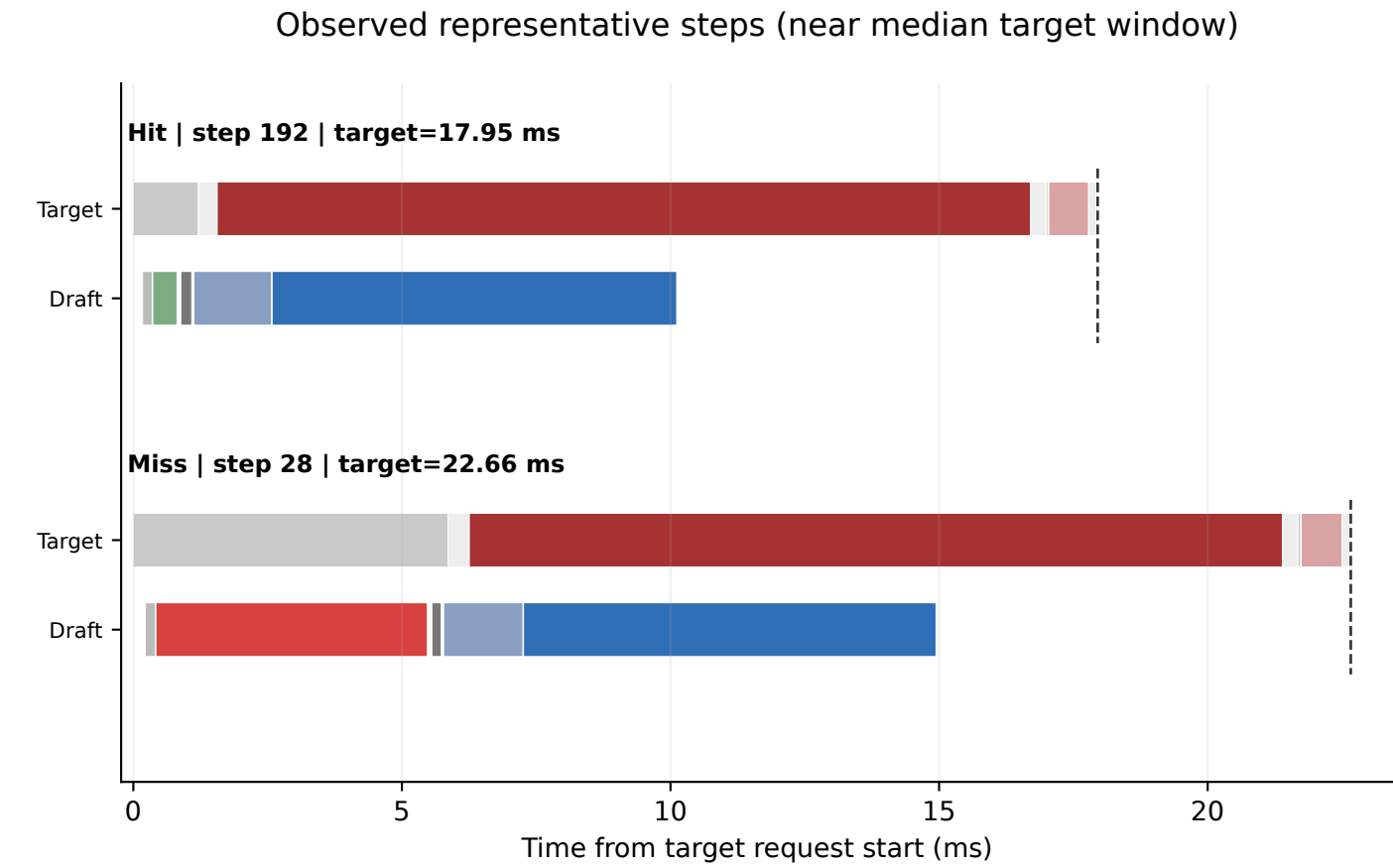
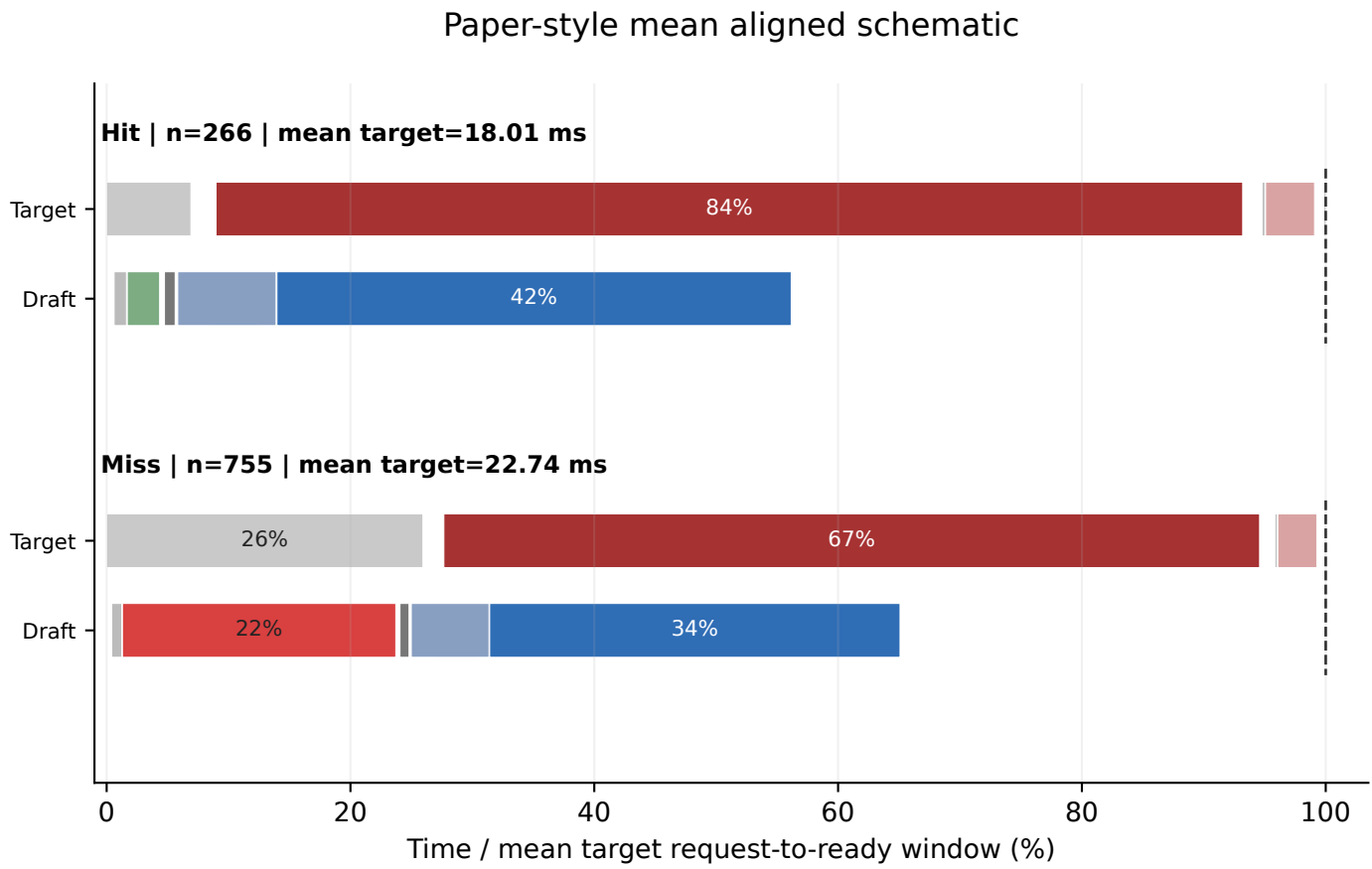
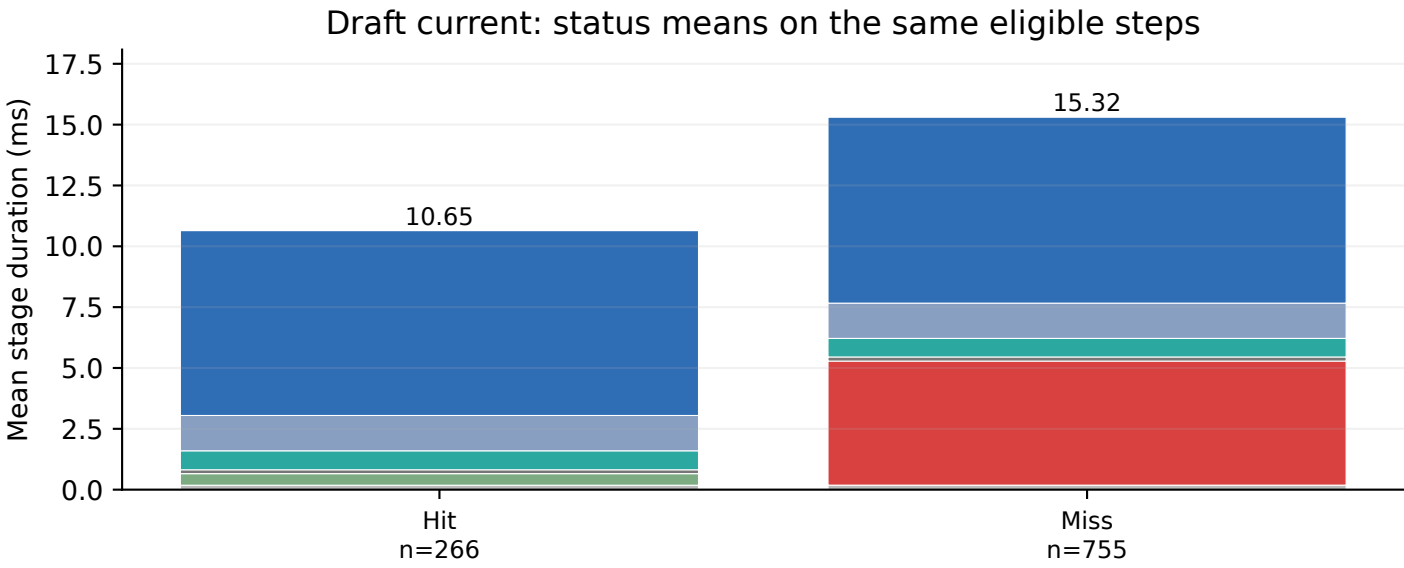
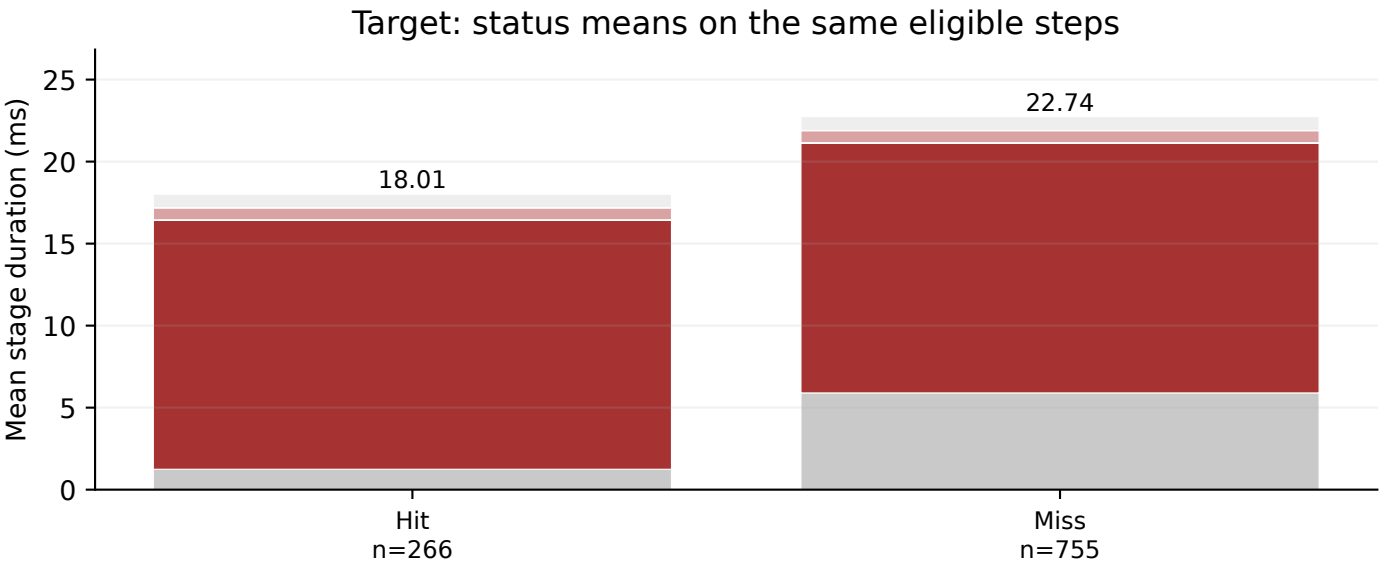
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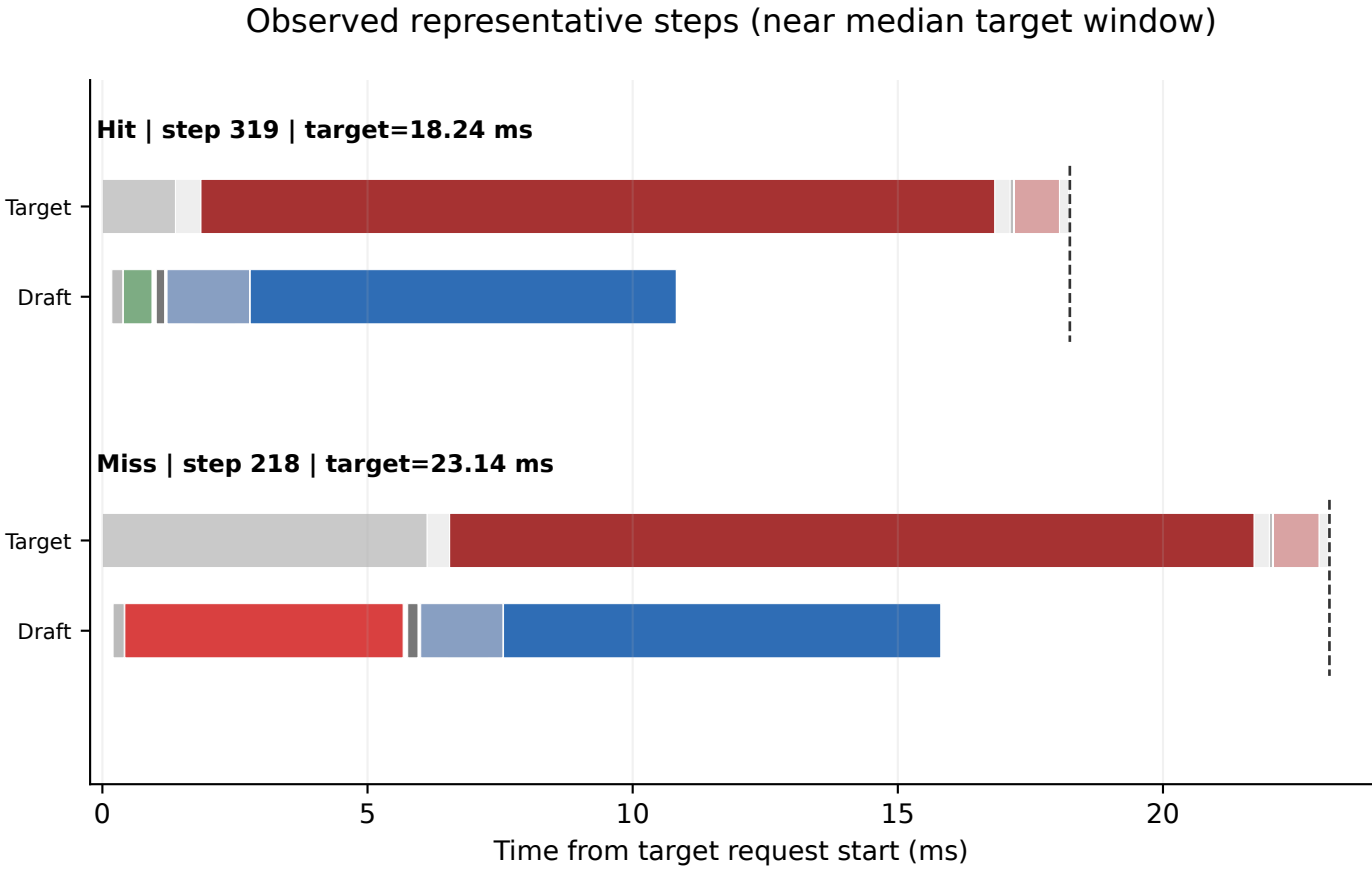
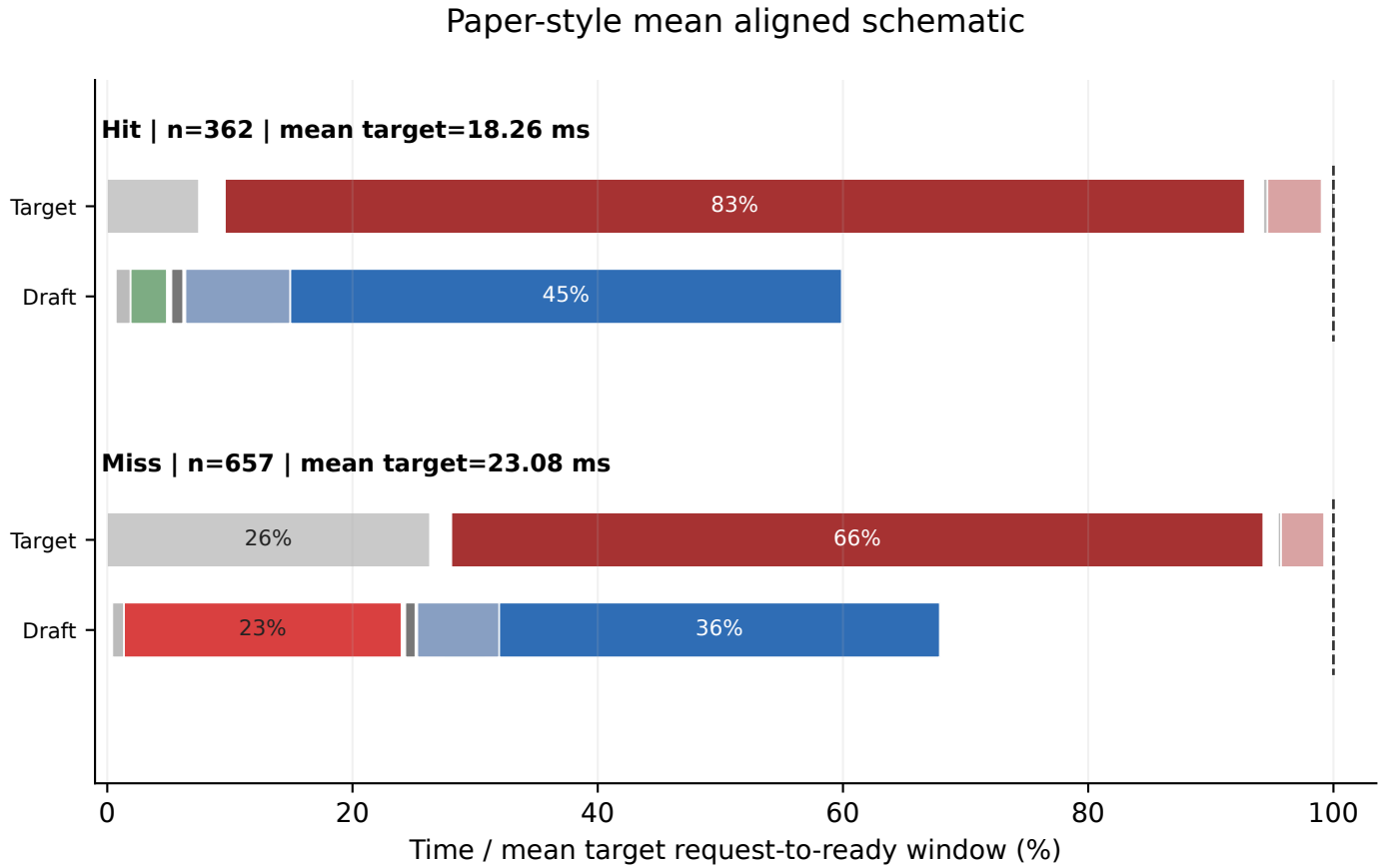
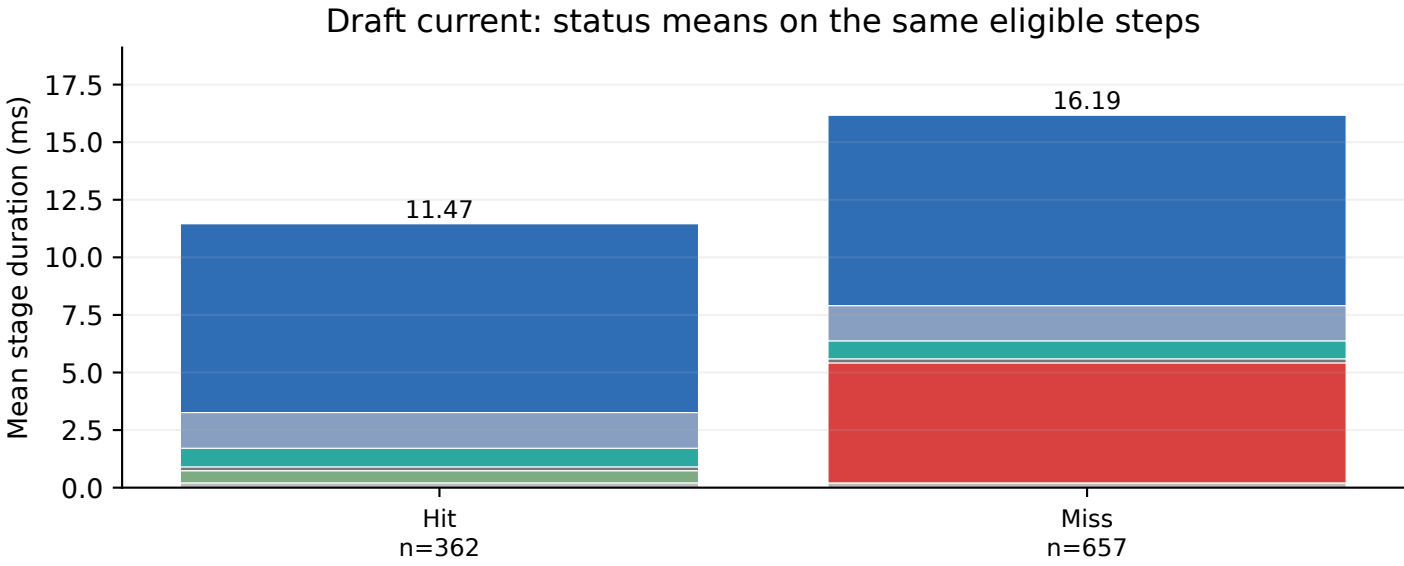
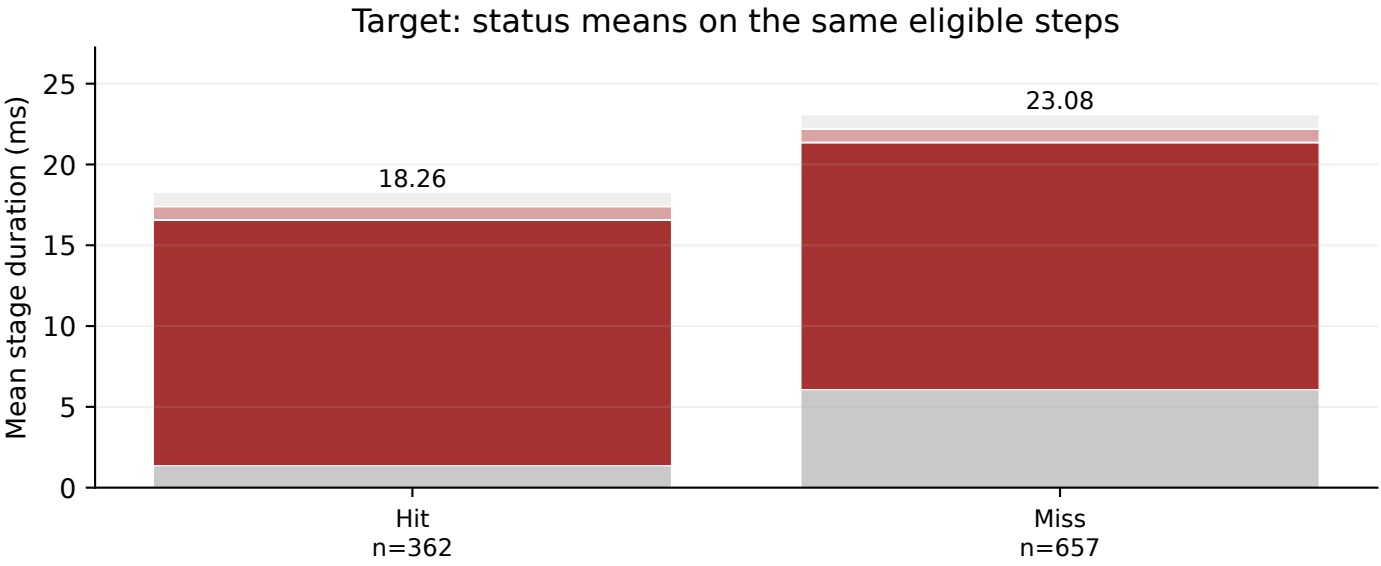
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

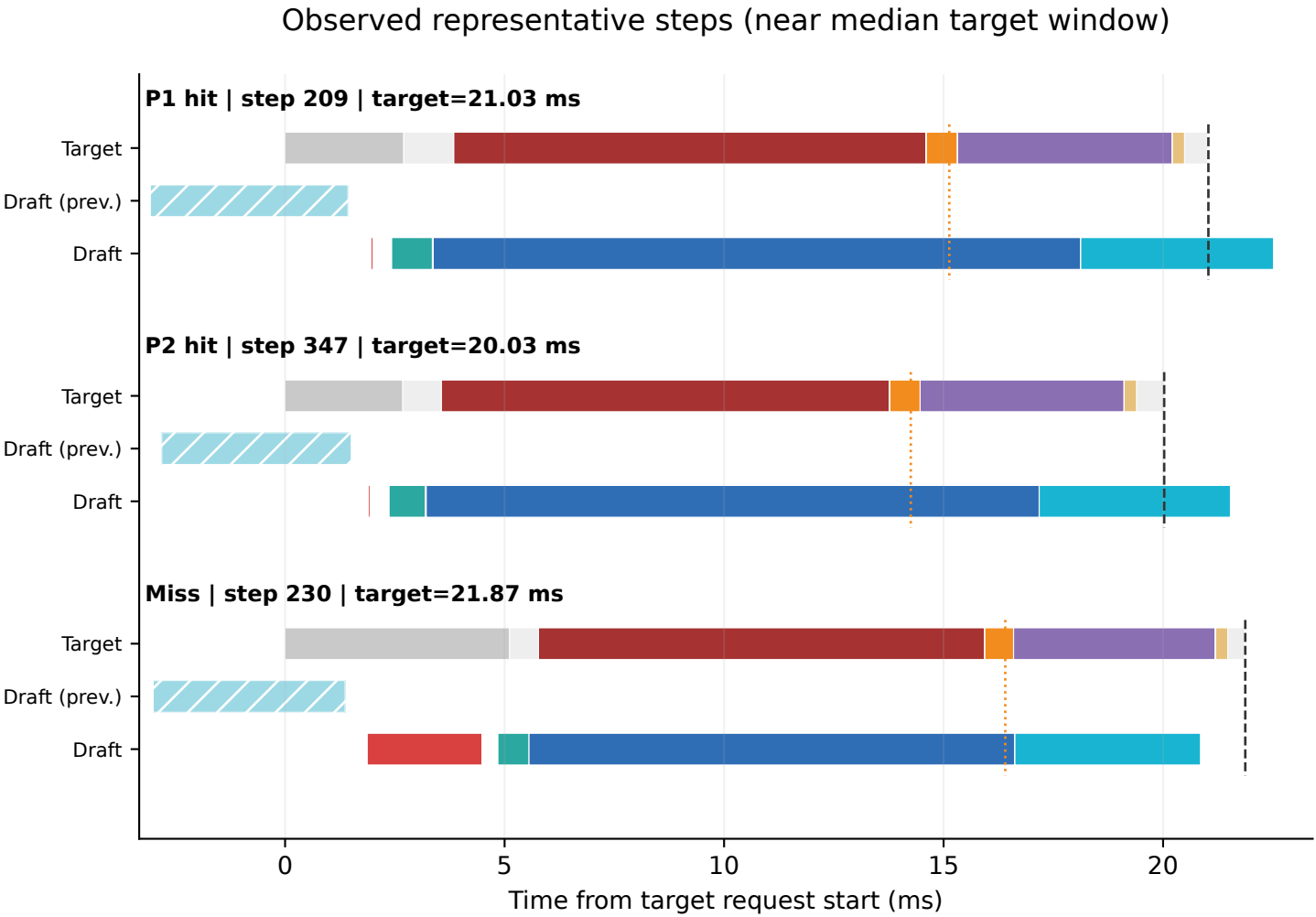
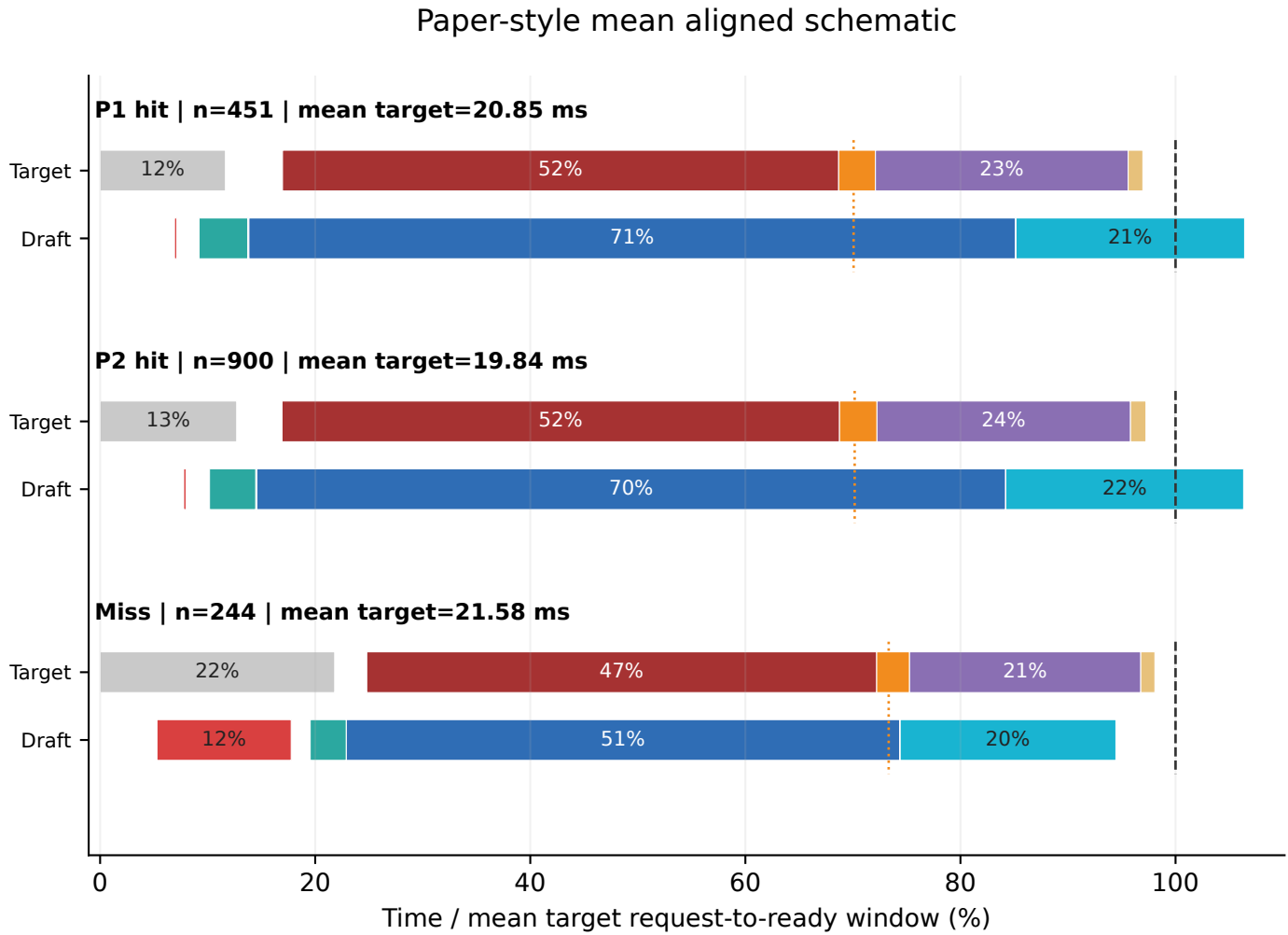
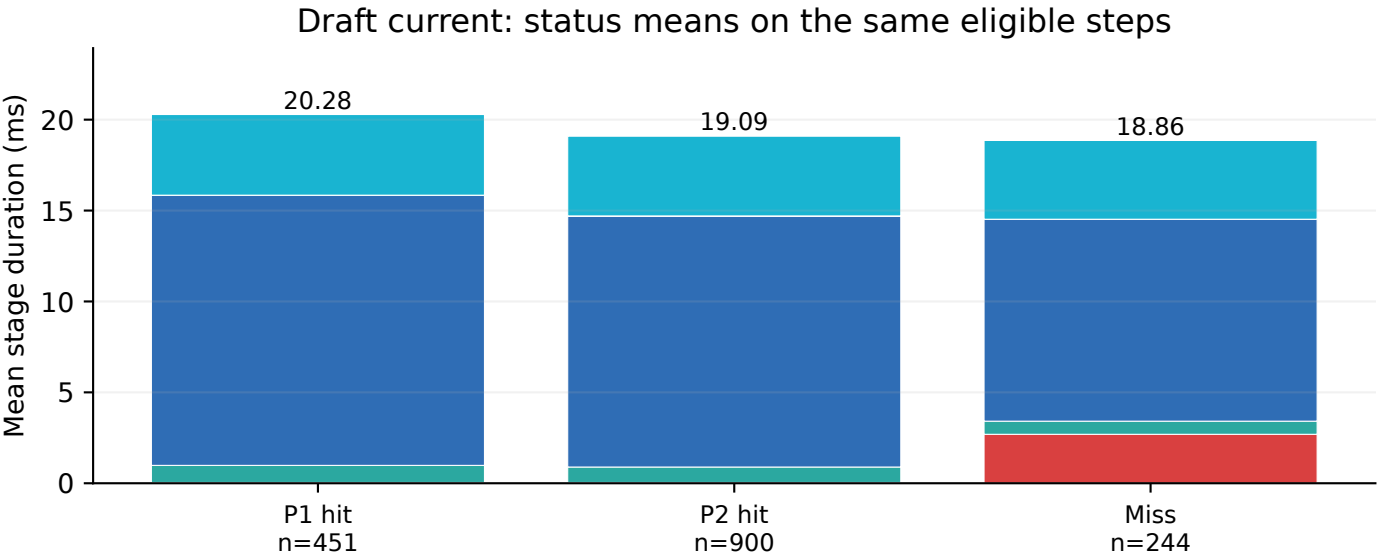
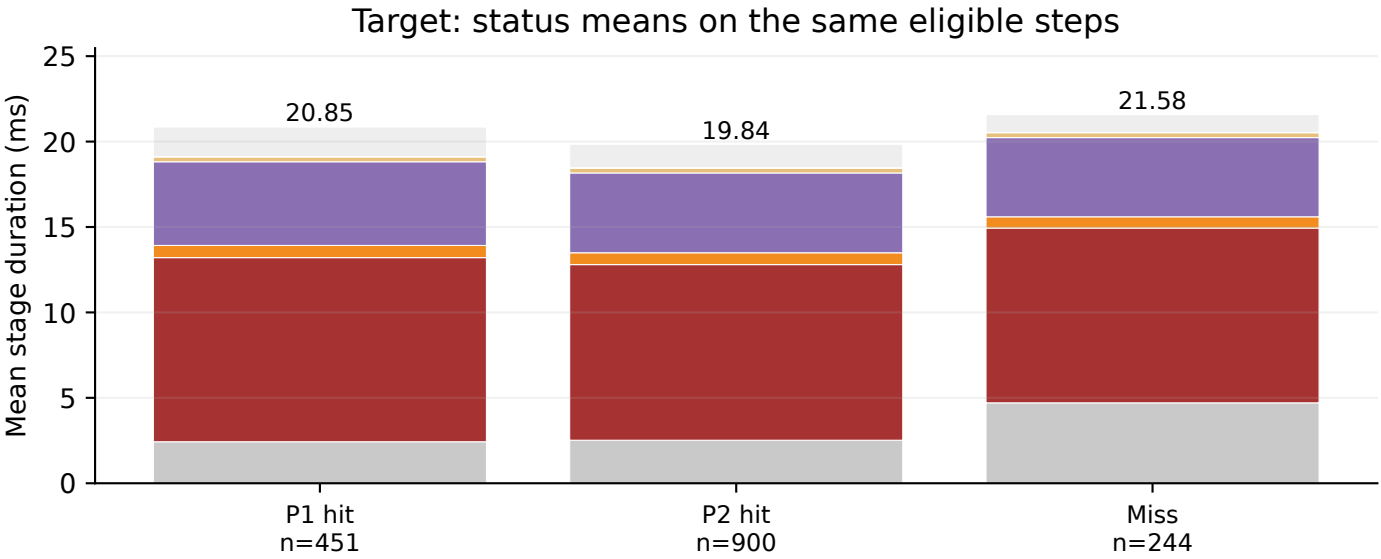




Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

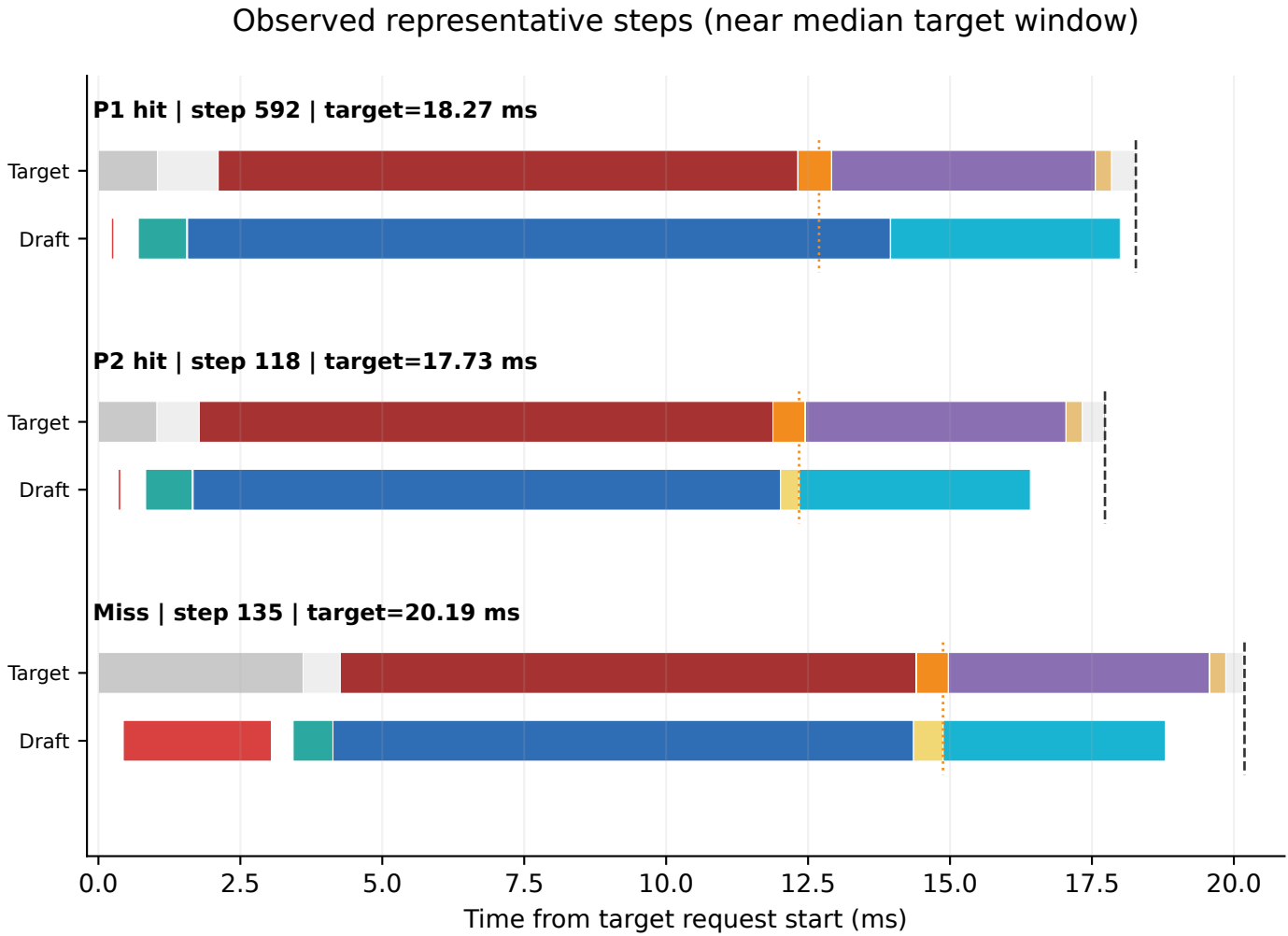
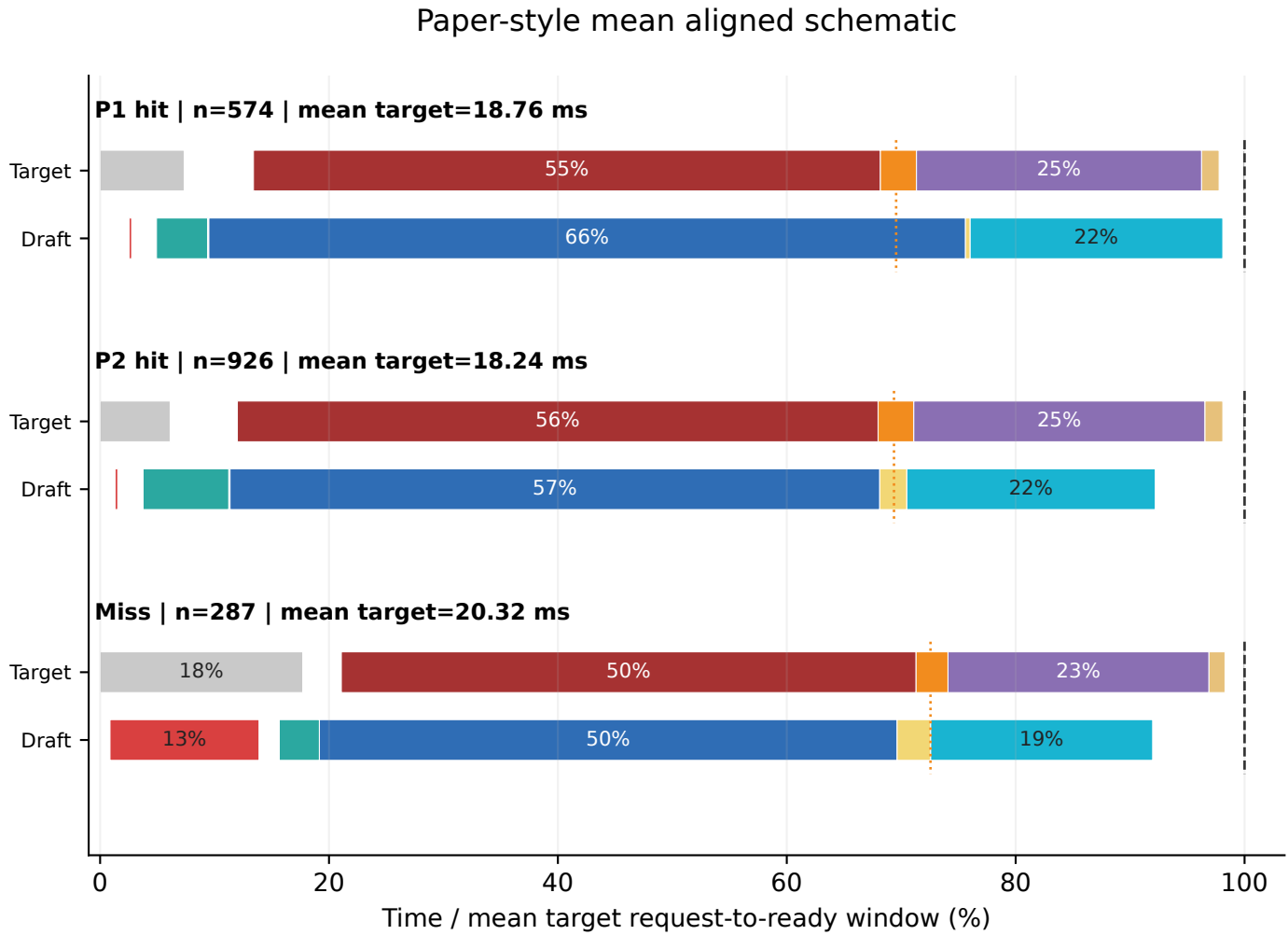
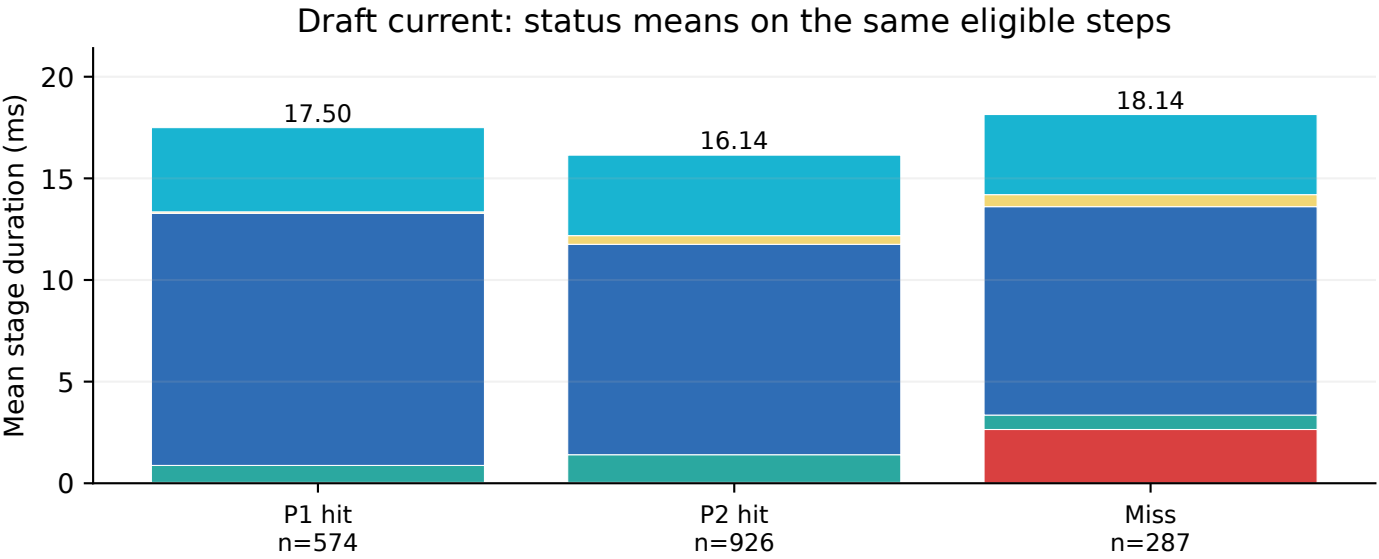
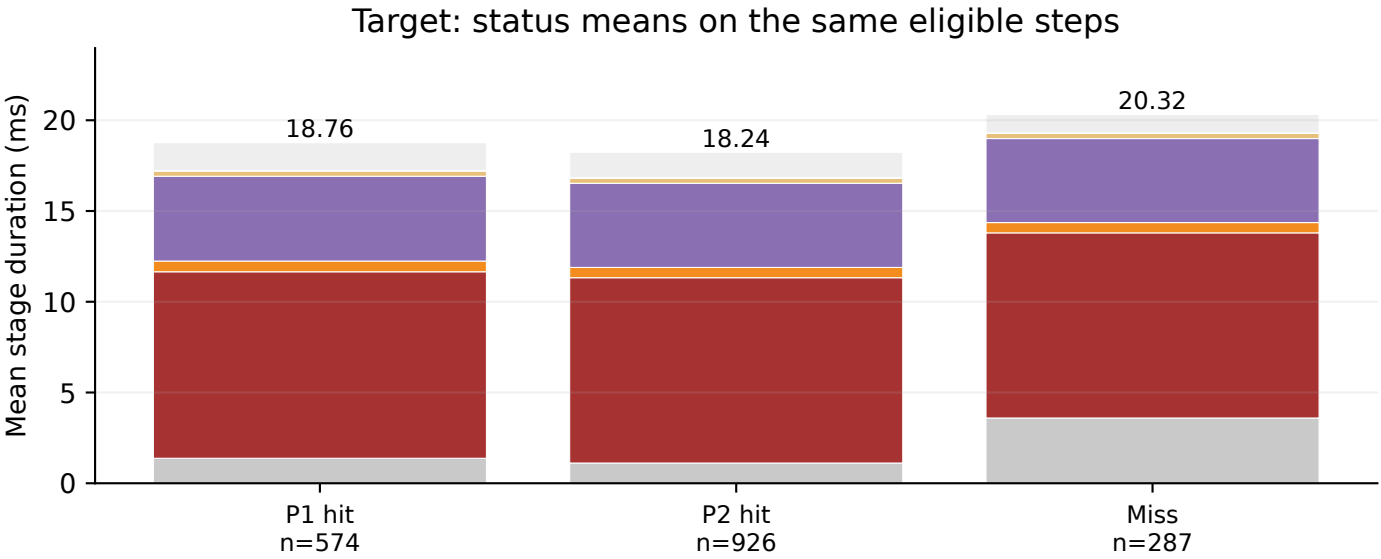
llama2_b1_tree_al_compute3
DUET: K1/K2=12/4, exit=21 (0-based), C=3, N1/N2=16/8, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1595 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

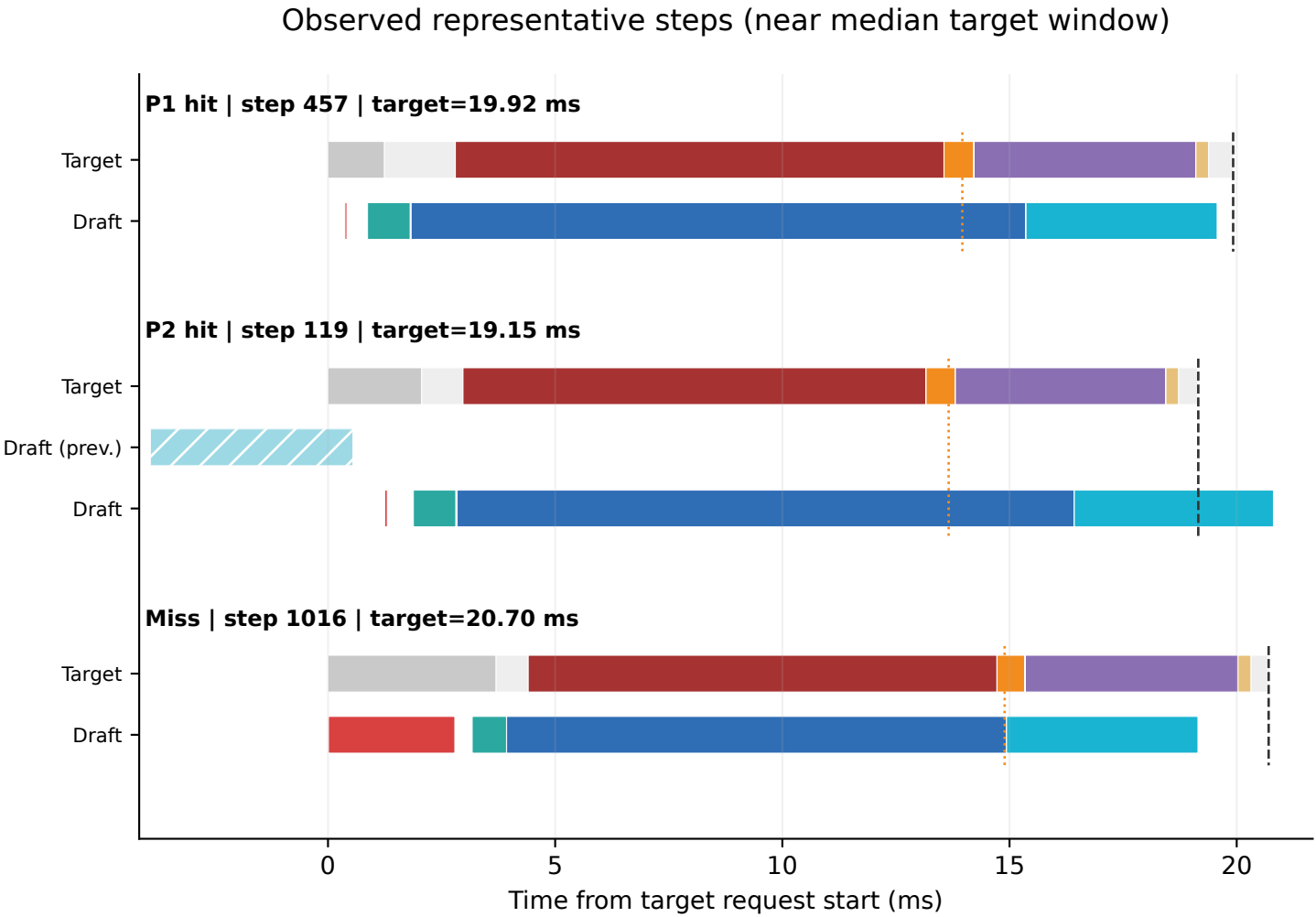
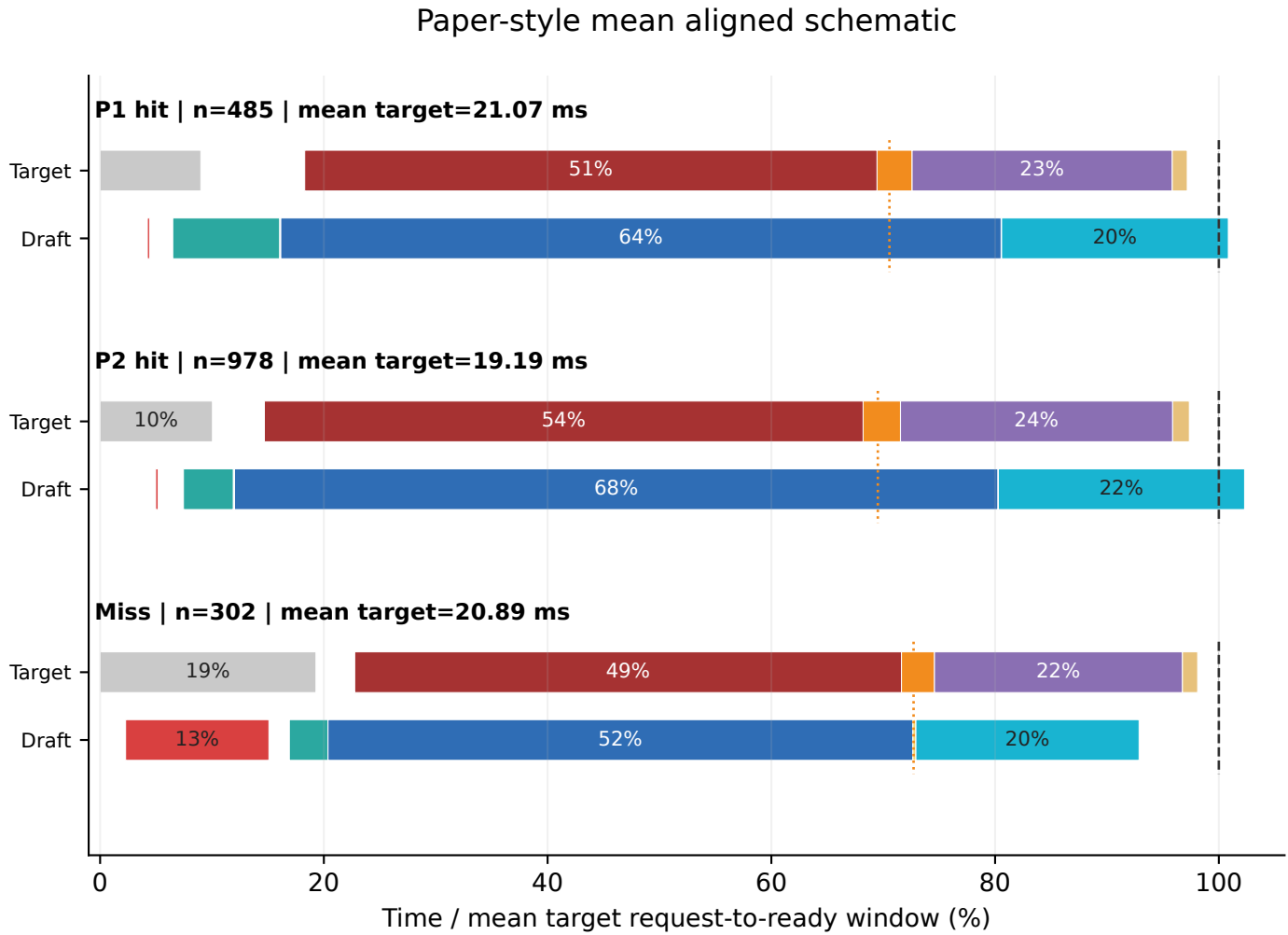
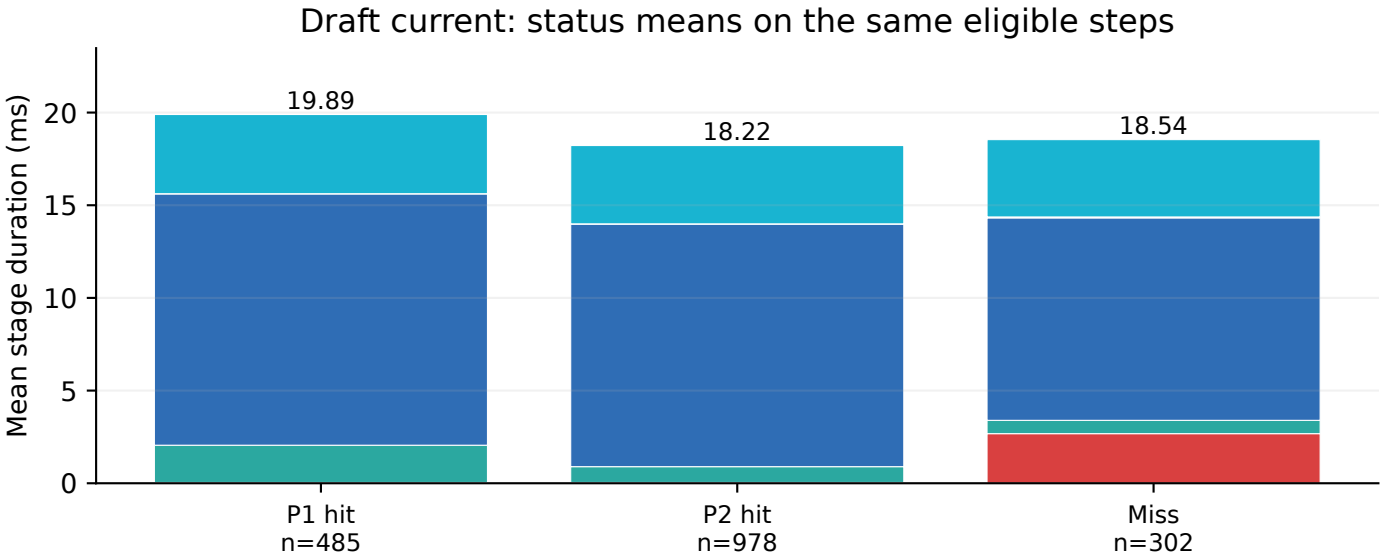
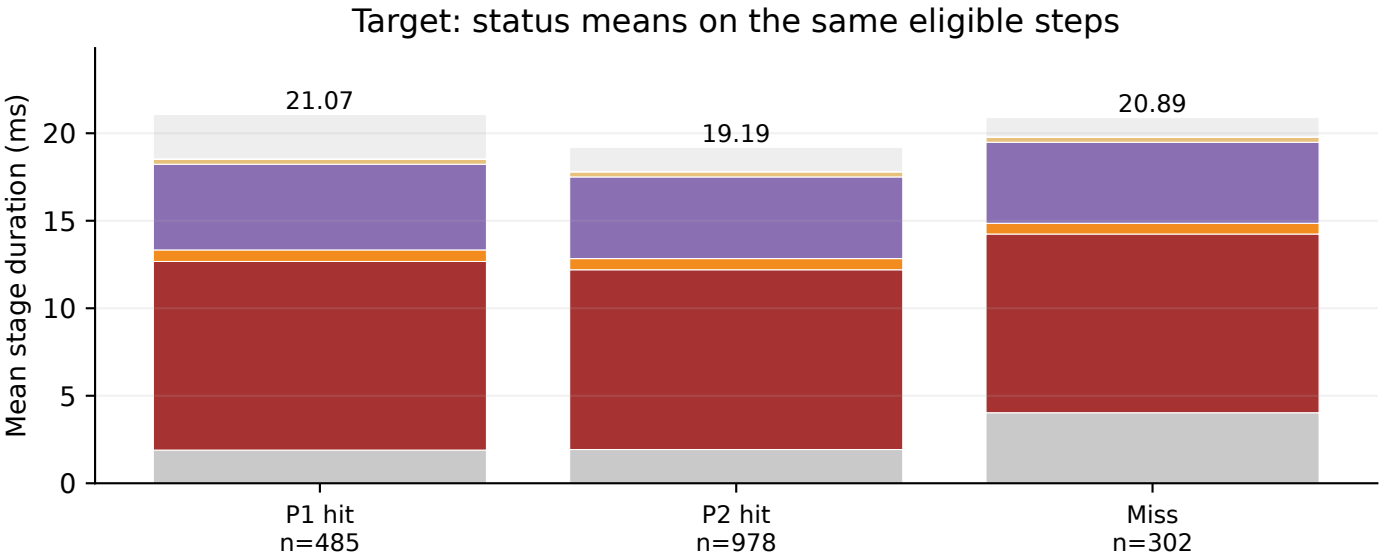
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_al_width1
DUET: K1/K2=12/4, exit=21 (0-based), C=1, N1/N2=12/4, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1787 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

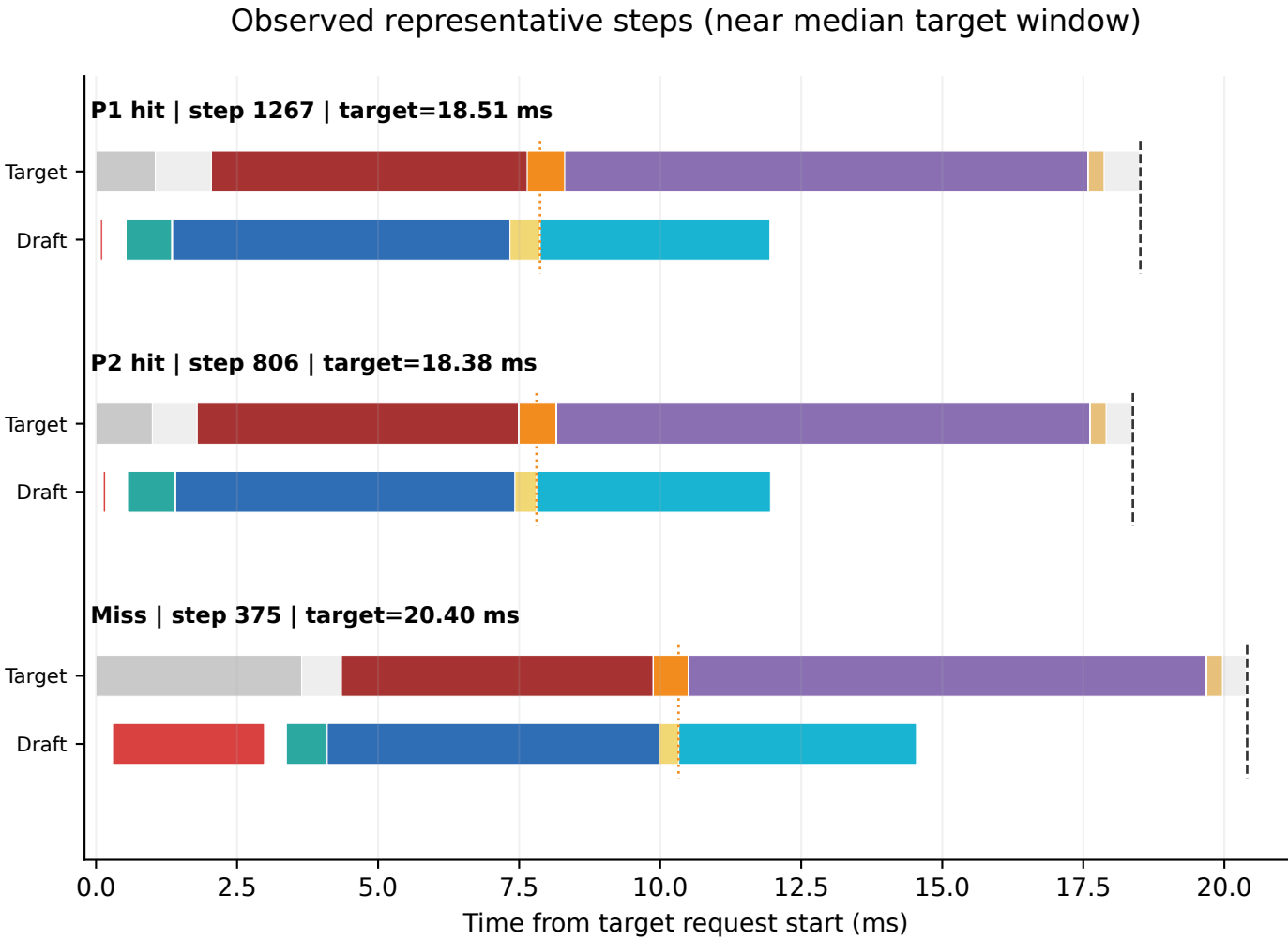
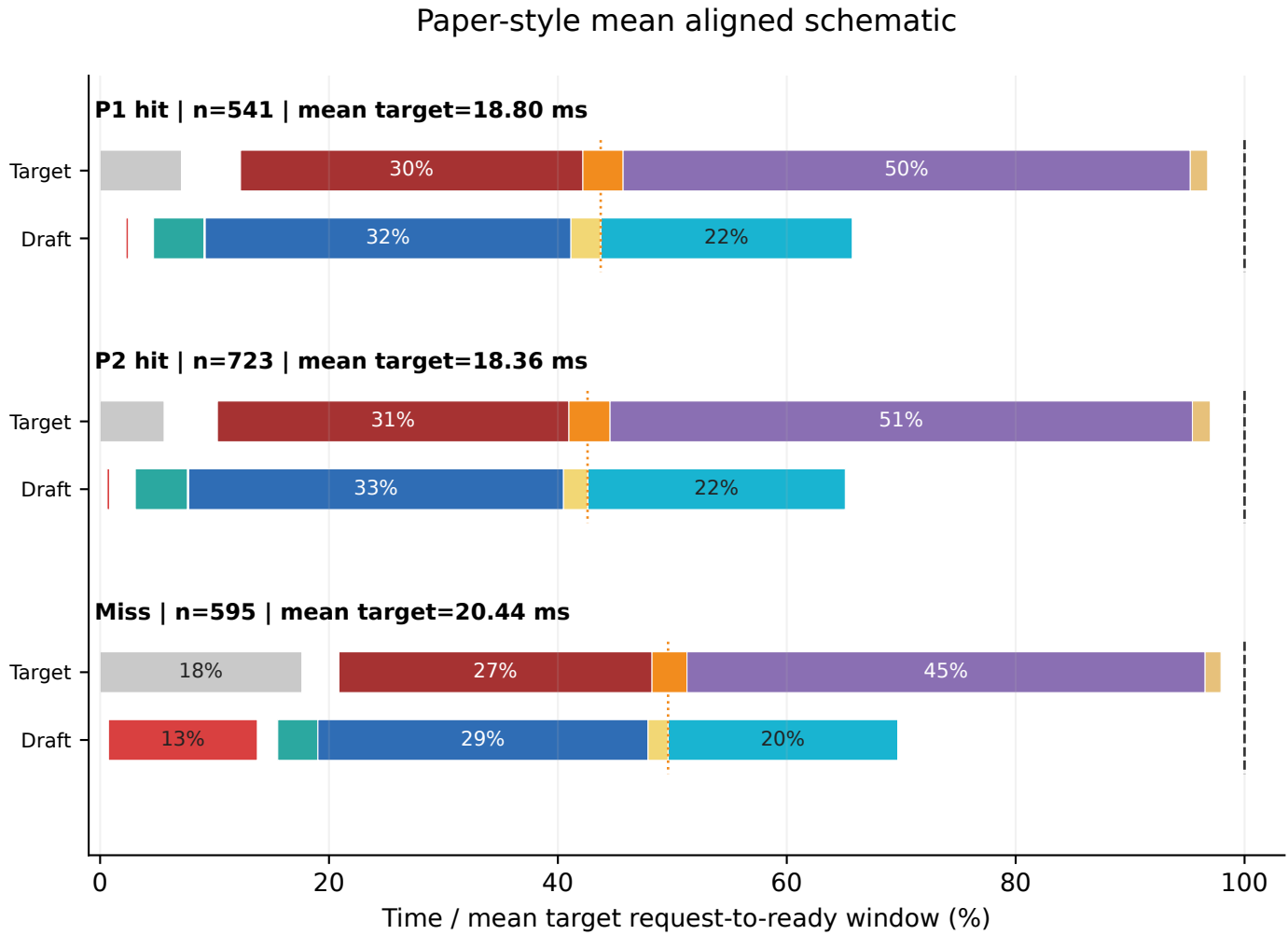
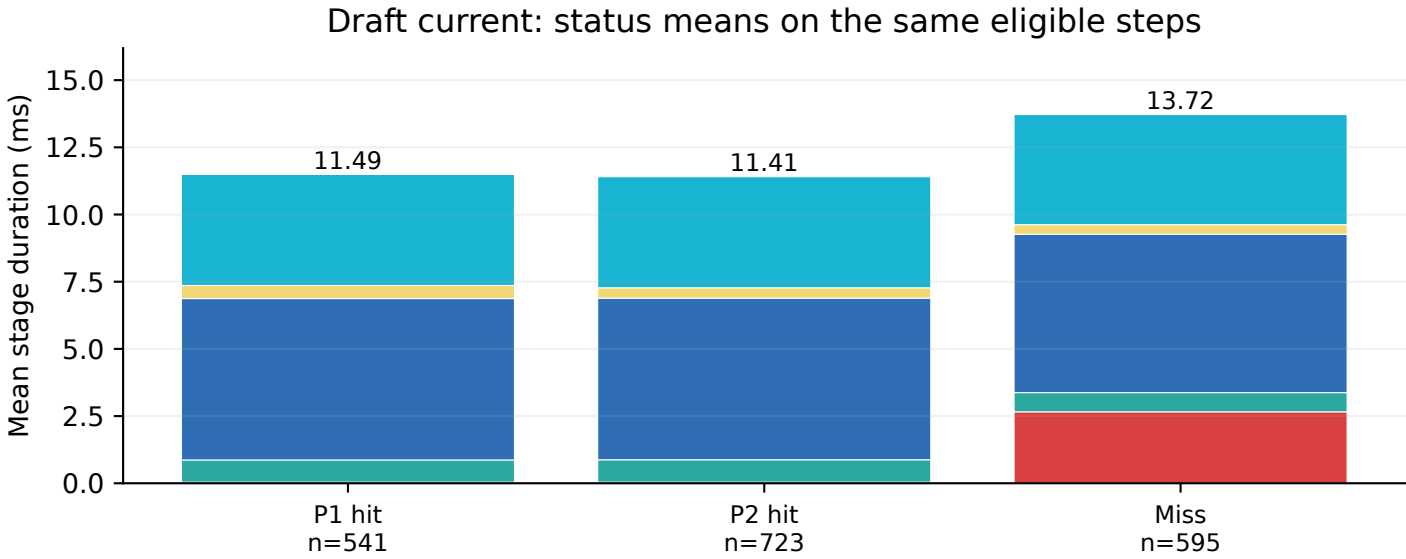
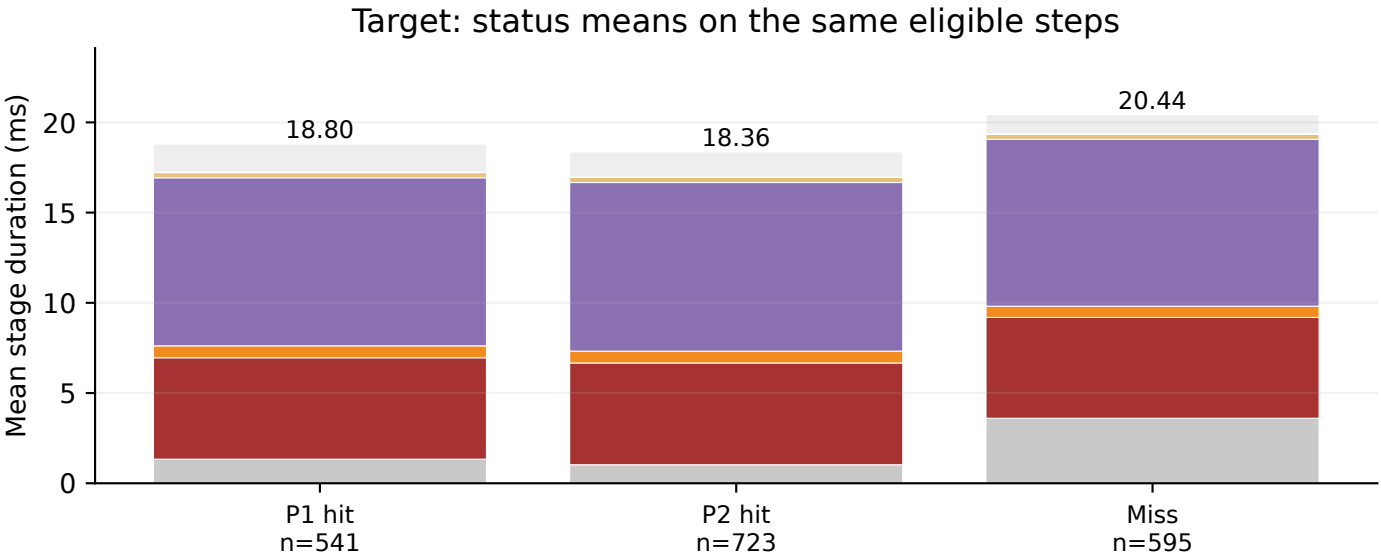
llama2_b1_tree_al_width2
DUET: K1/K2=12/4, exit=21 (0-based), C=2, N1/N2=16/8, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1765 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

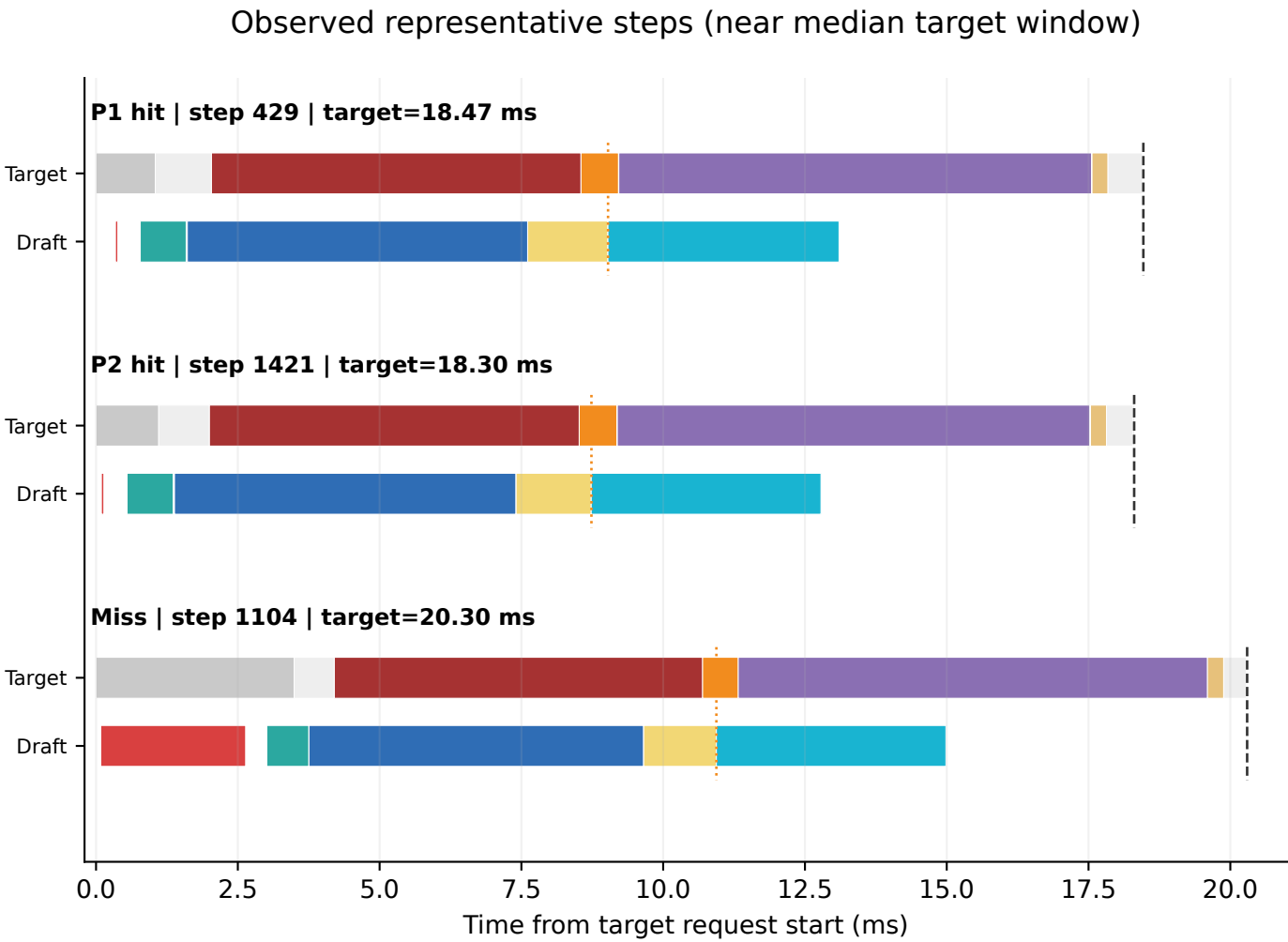
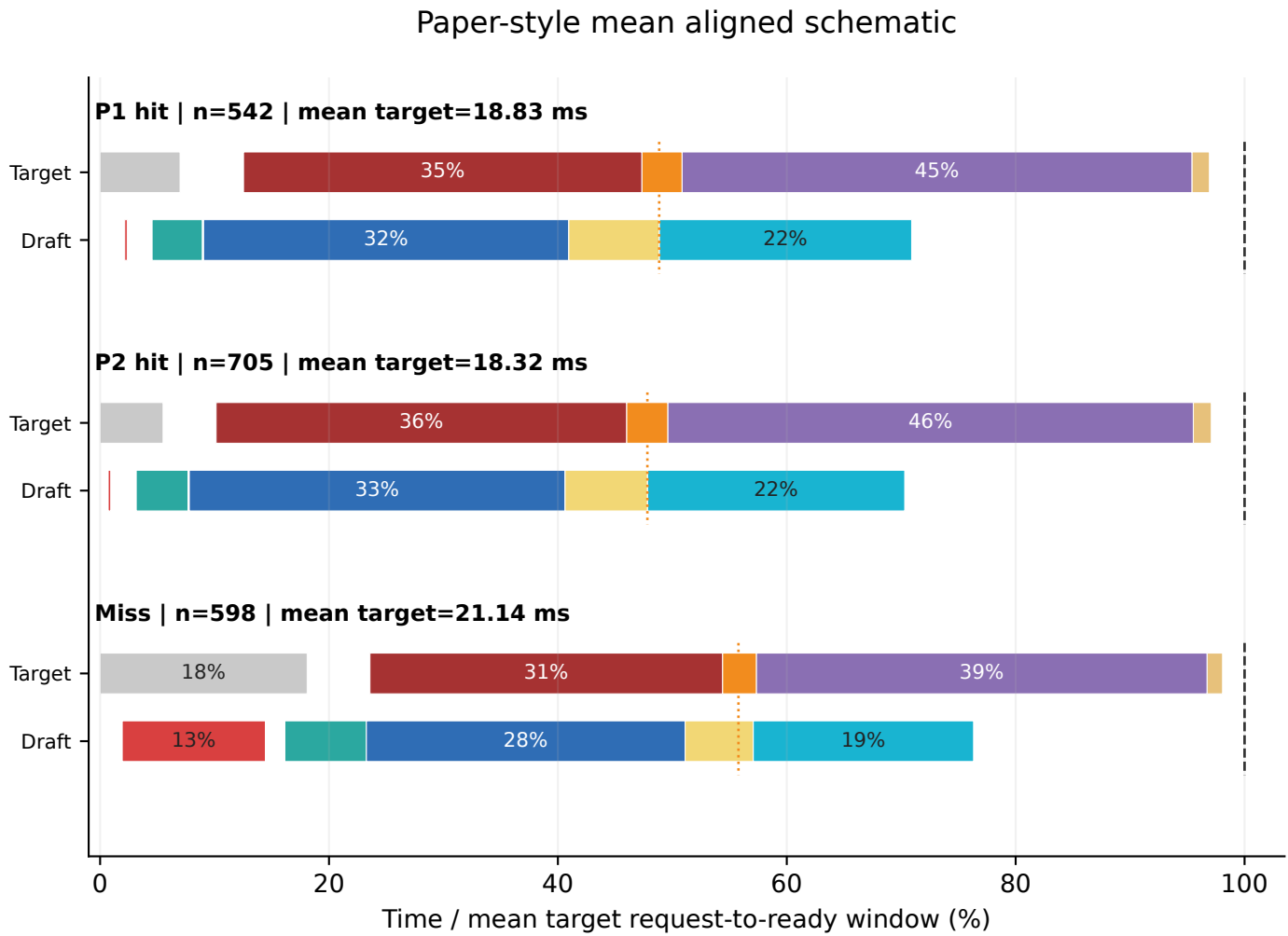
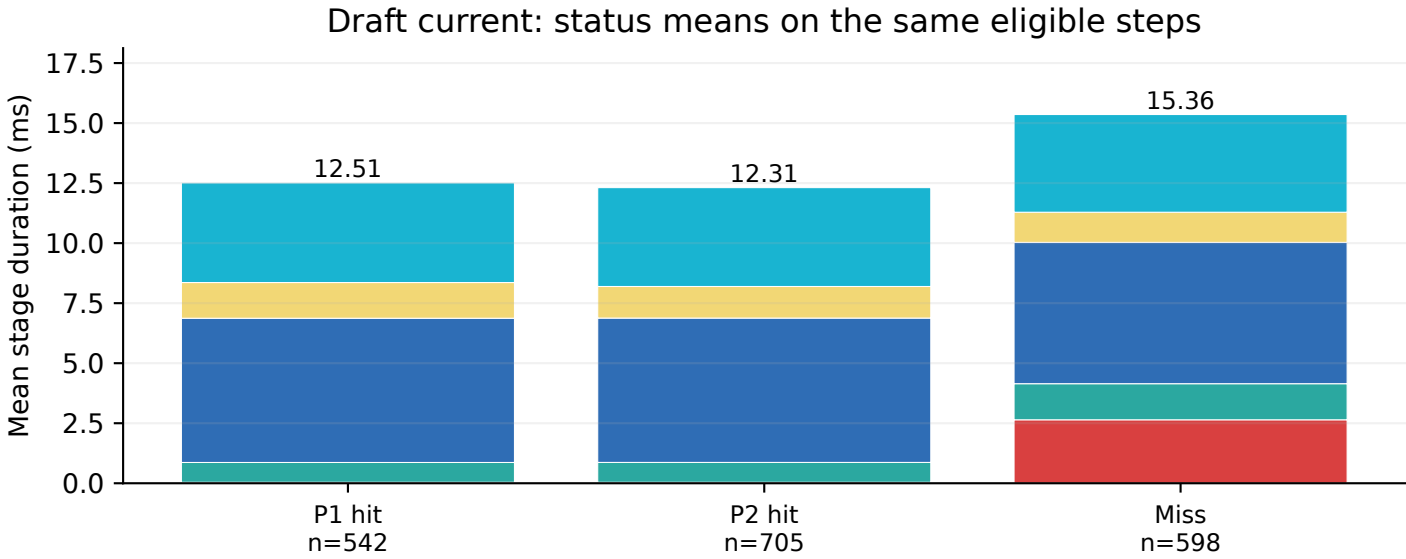
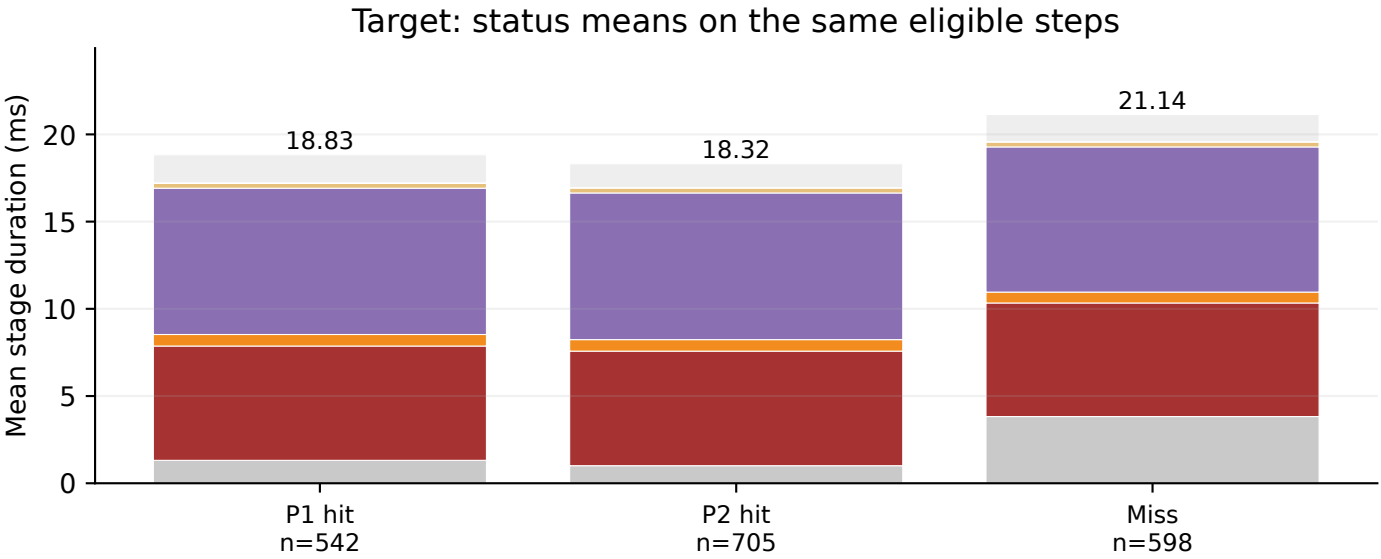
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_align_exit11
DUET: K1/K2=6/4, exit=11 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1859 eligible steps



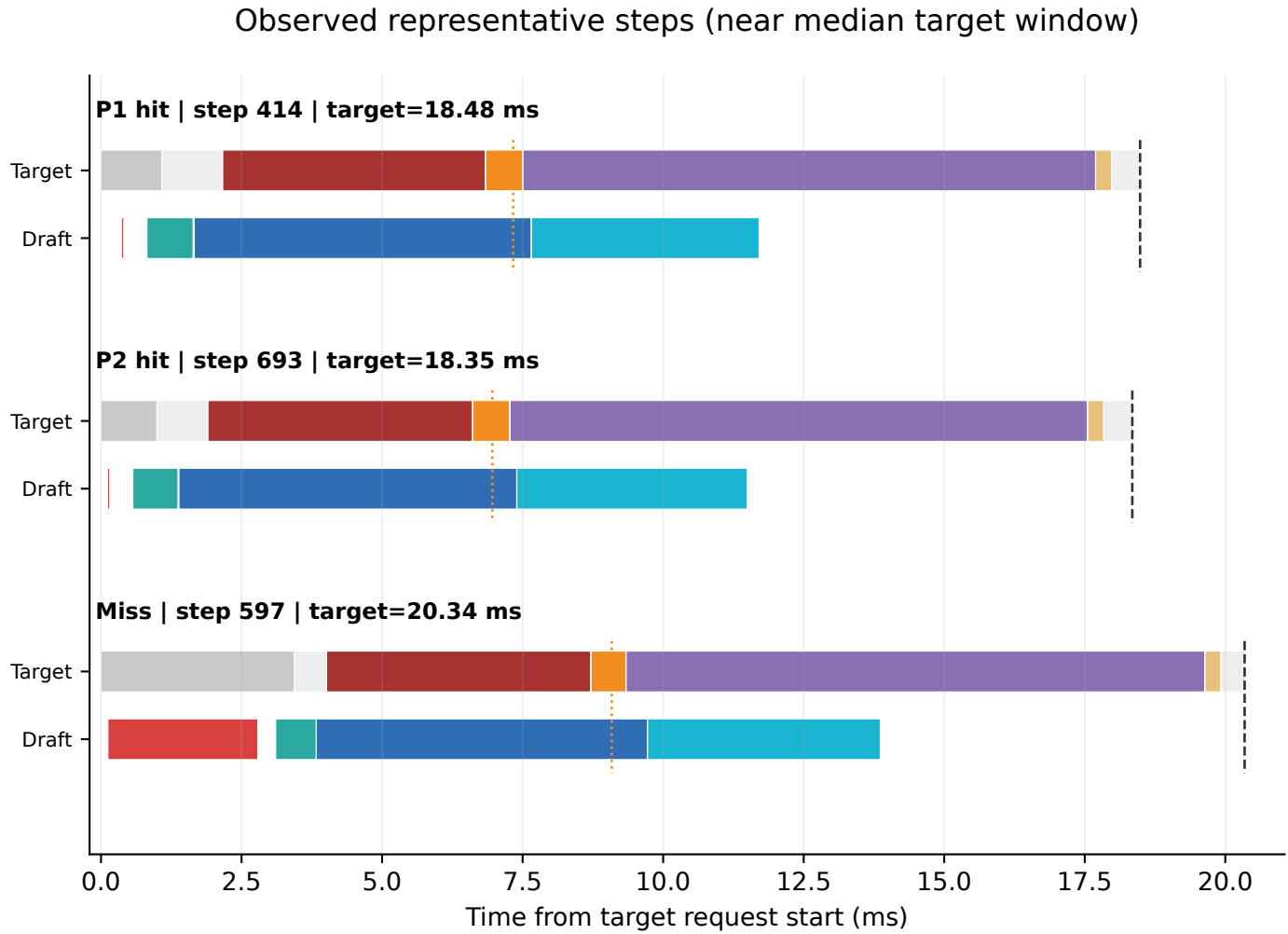
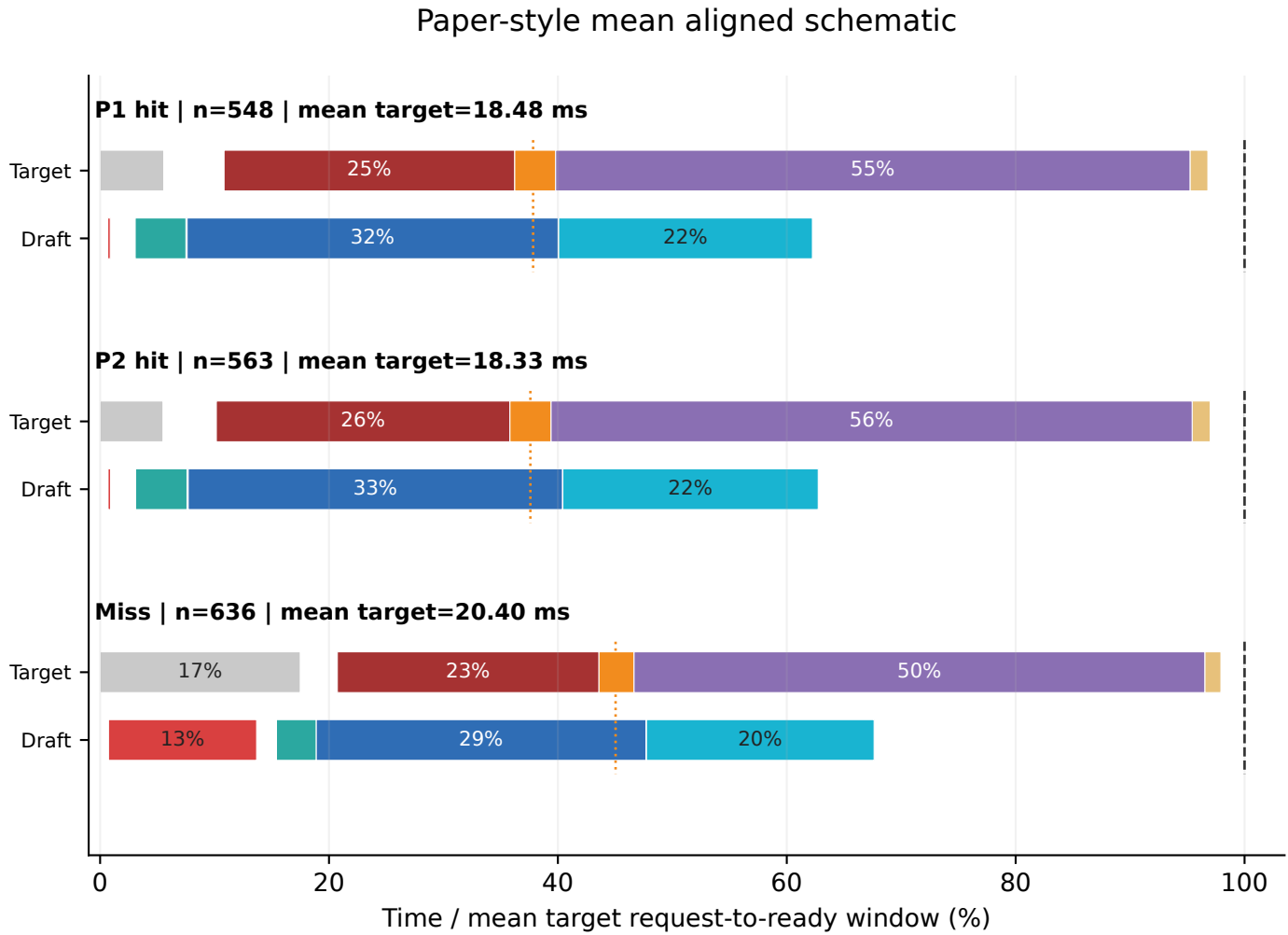
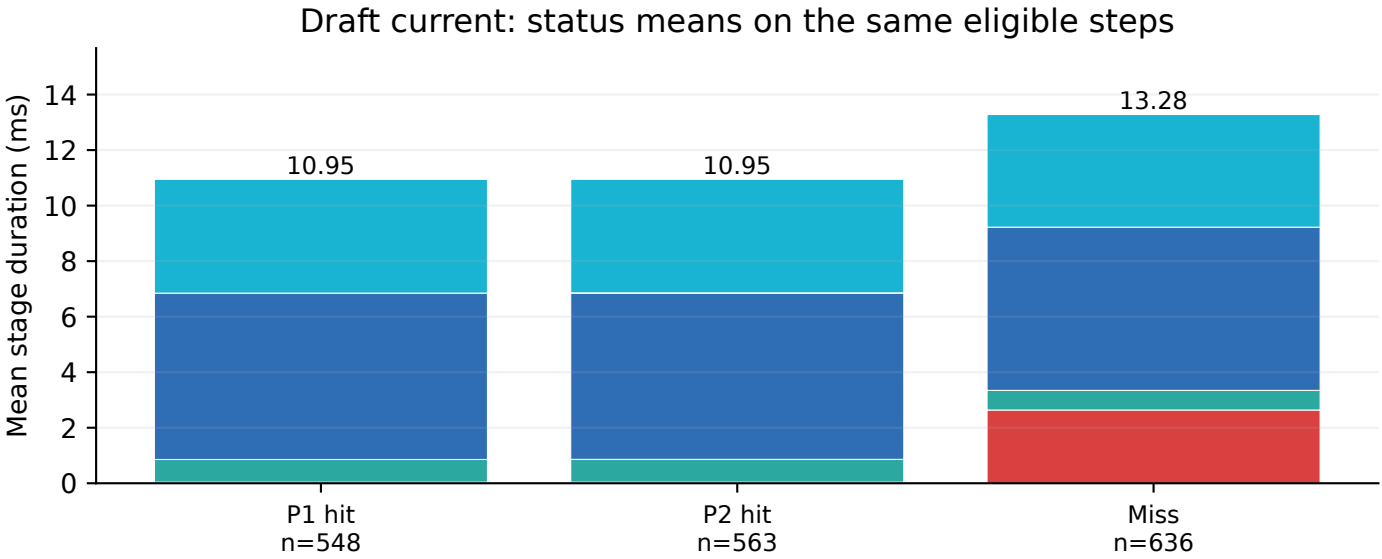
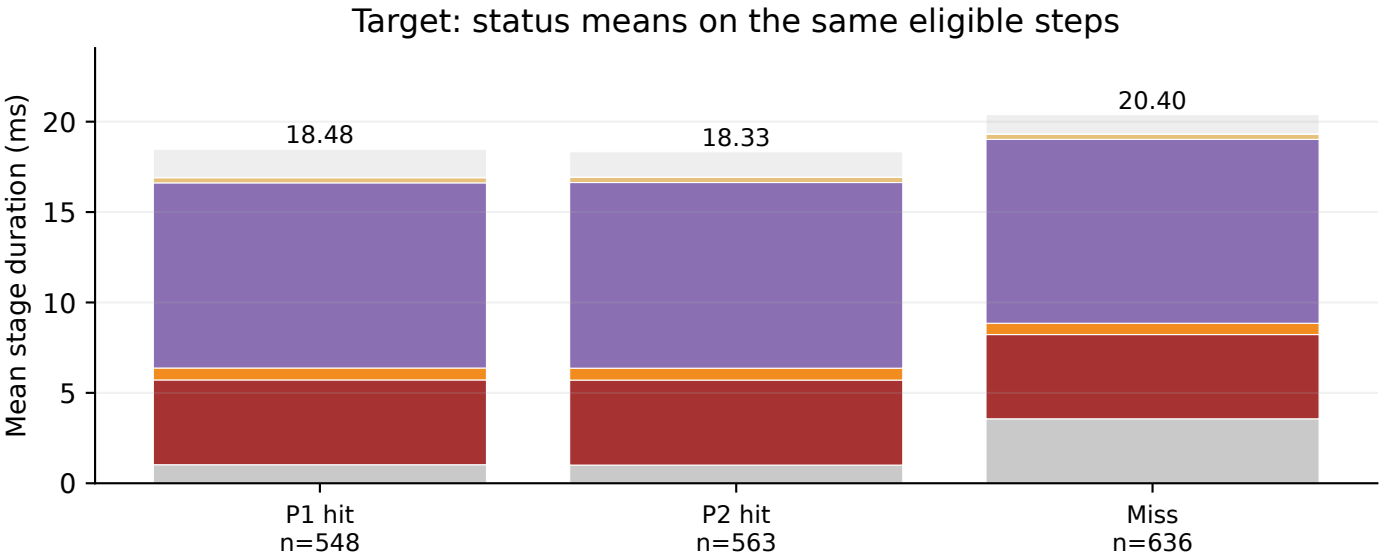
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_align_exit13
DUET: K1/K2=6/4, exit=13 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1845 eligible steps



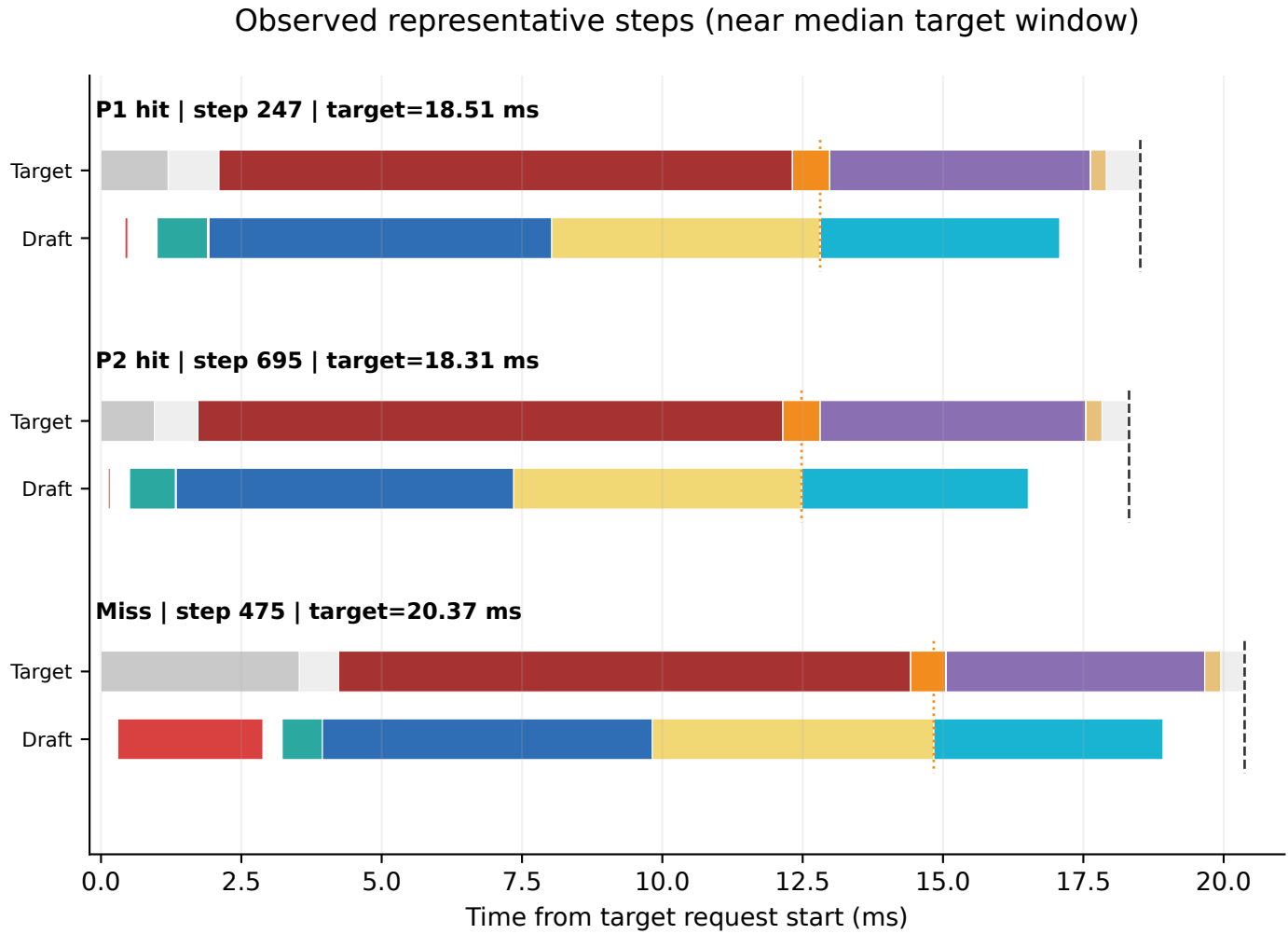
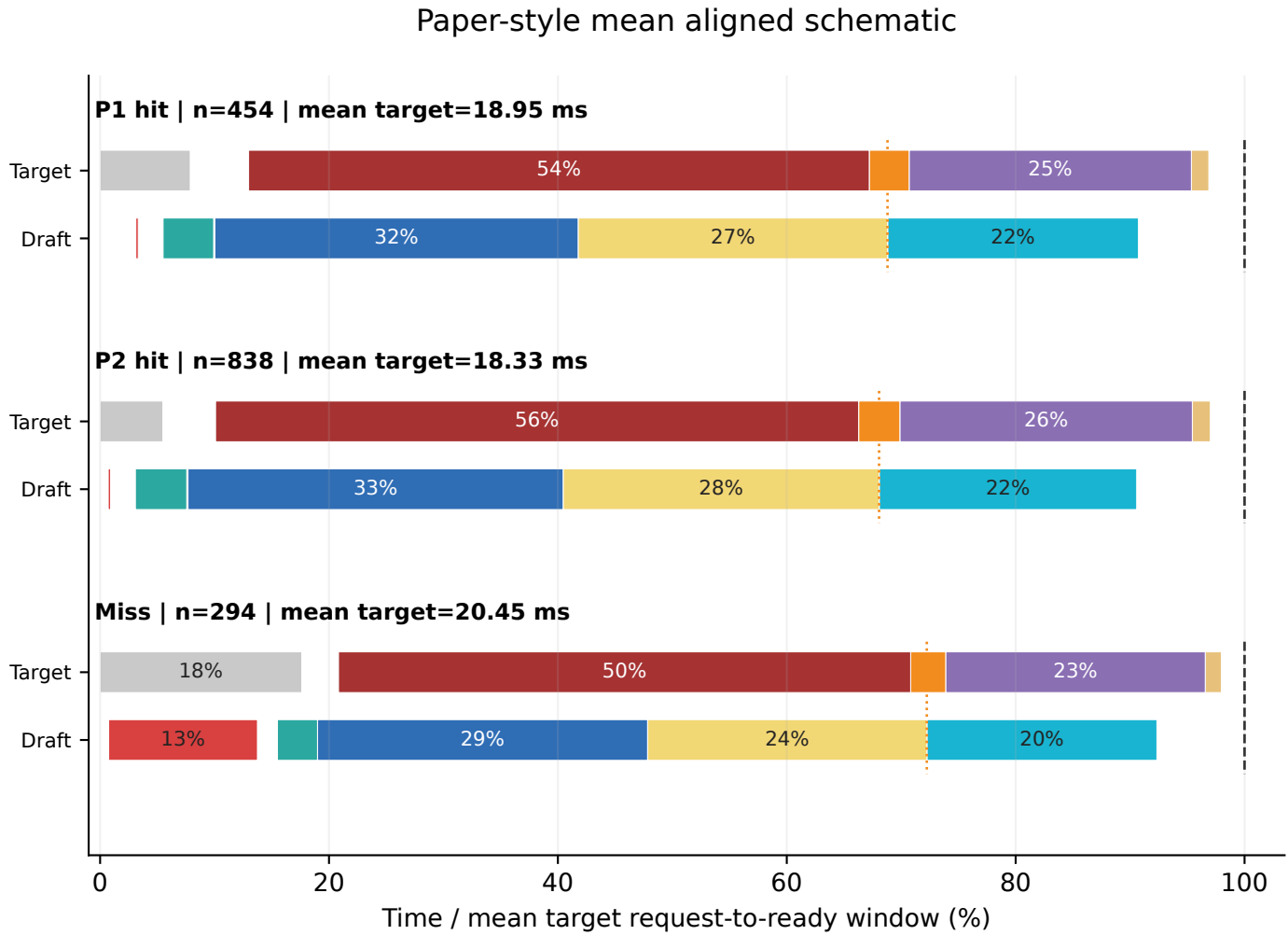
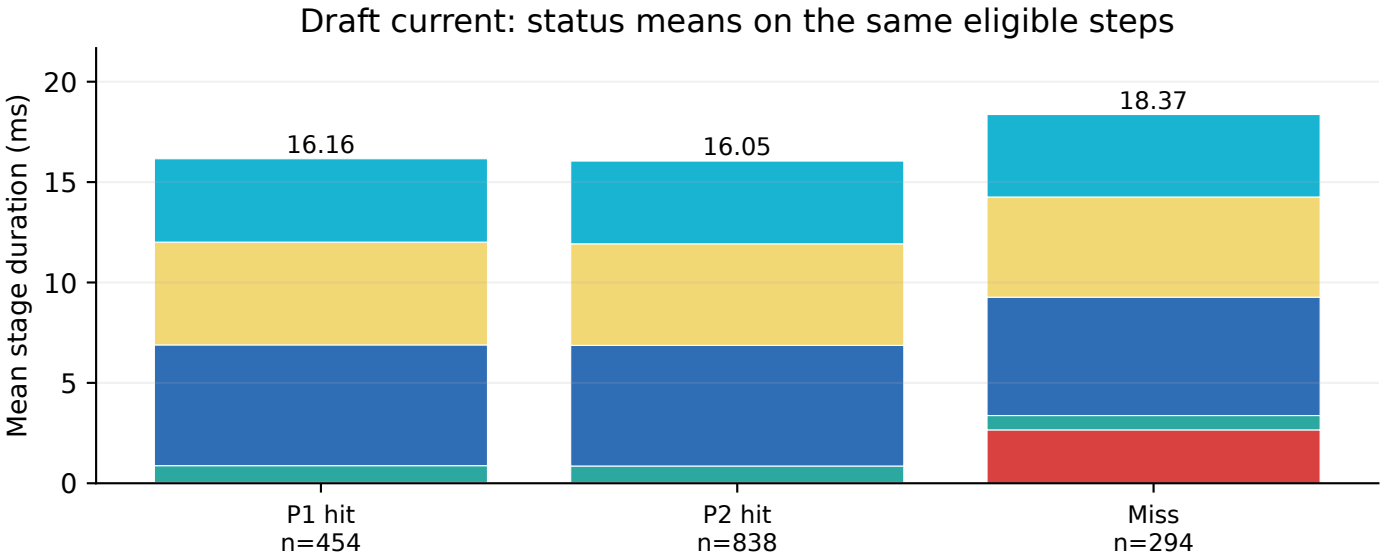
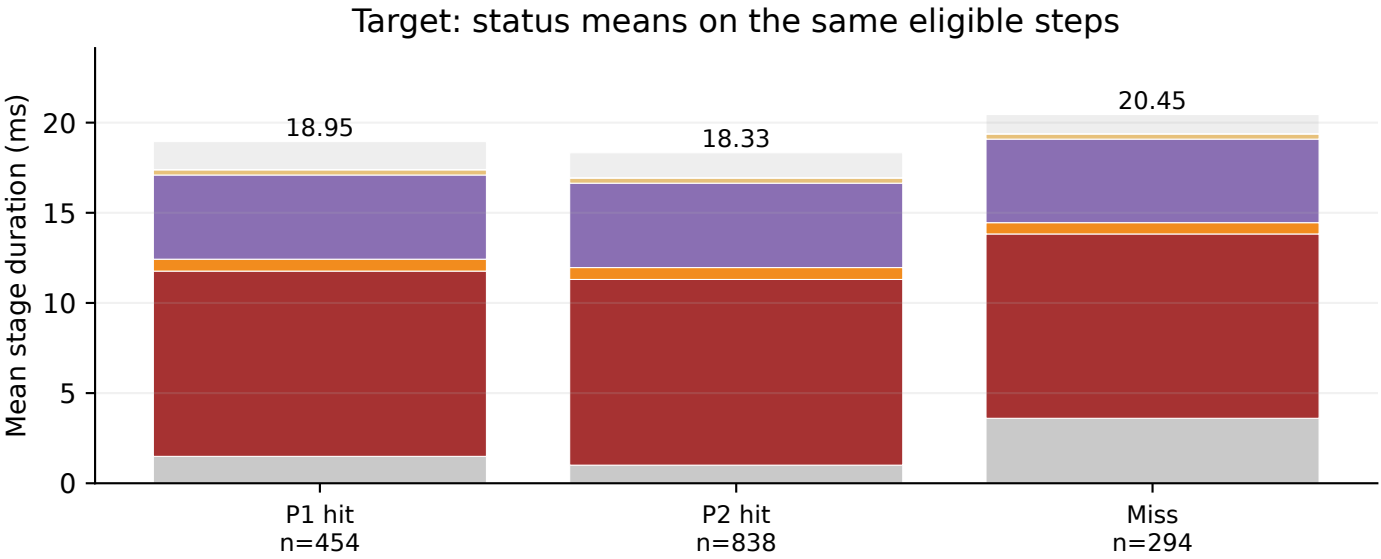
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_align_exit9
DUET: K1/K2=6/4, exit=9 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1747 eligible steps



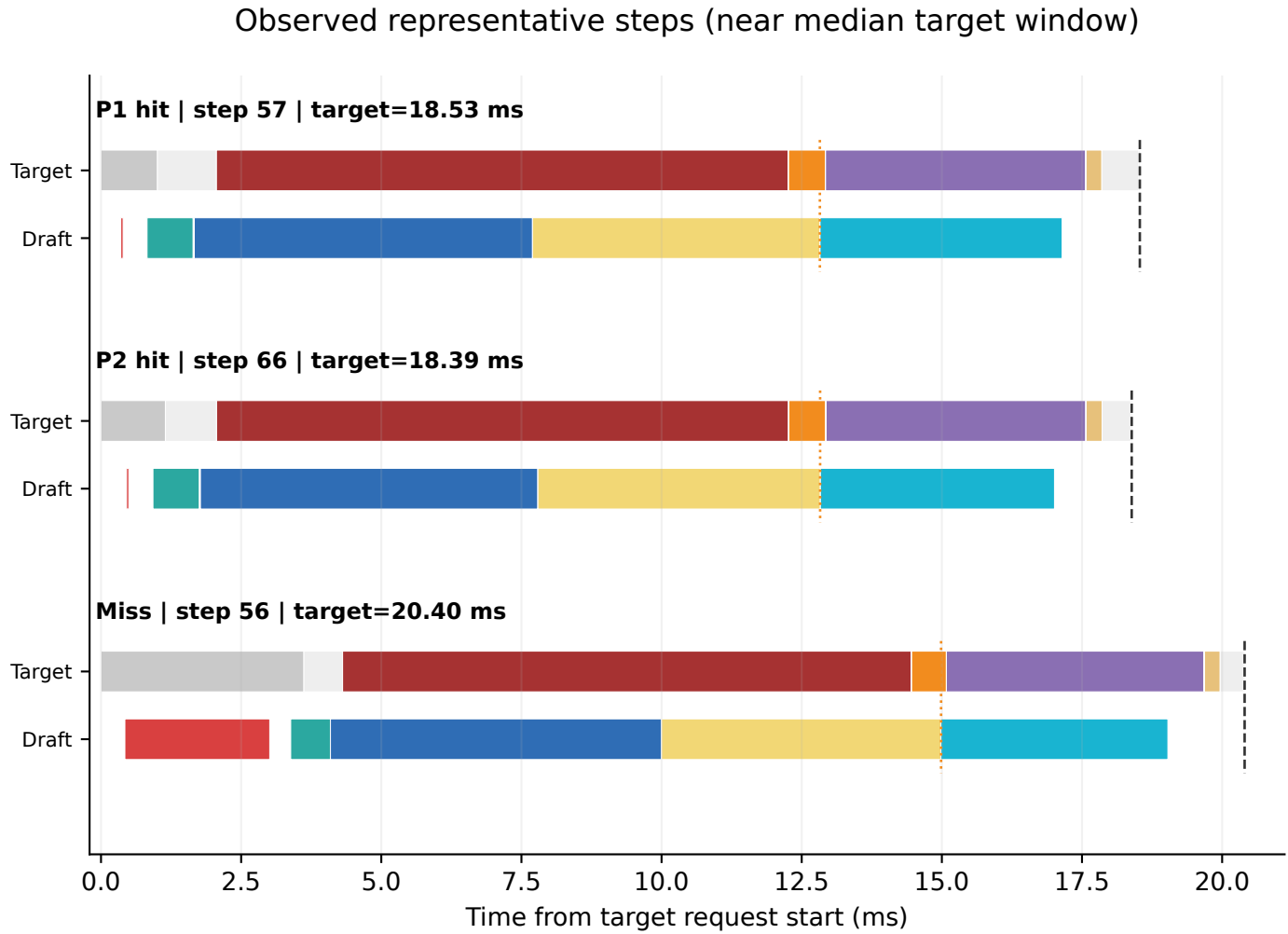
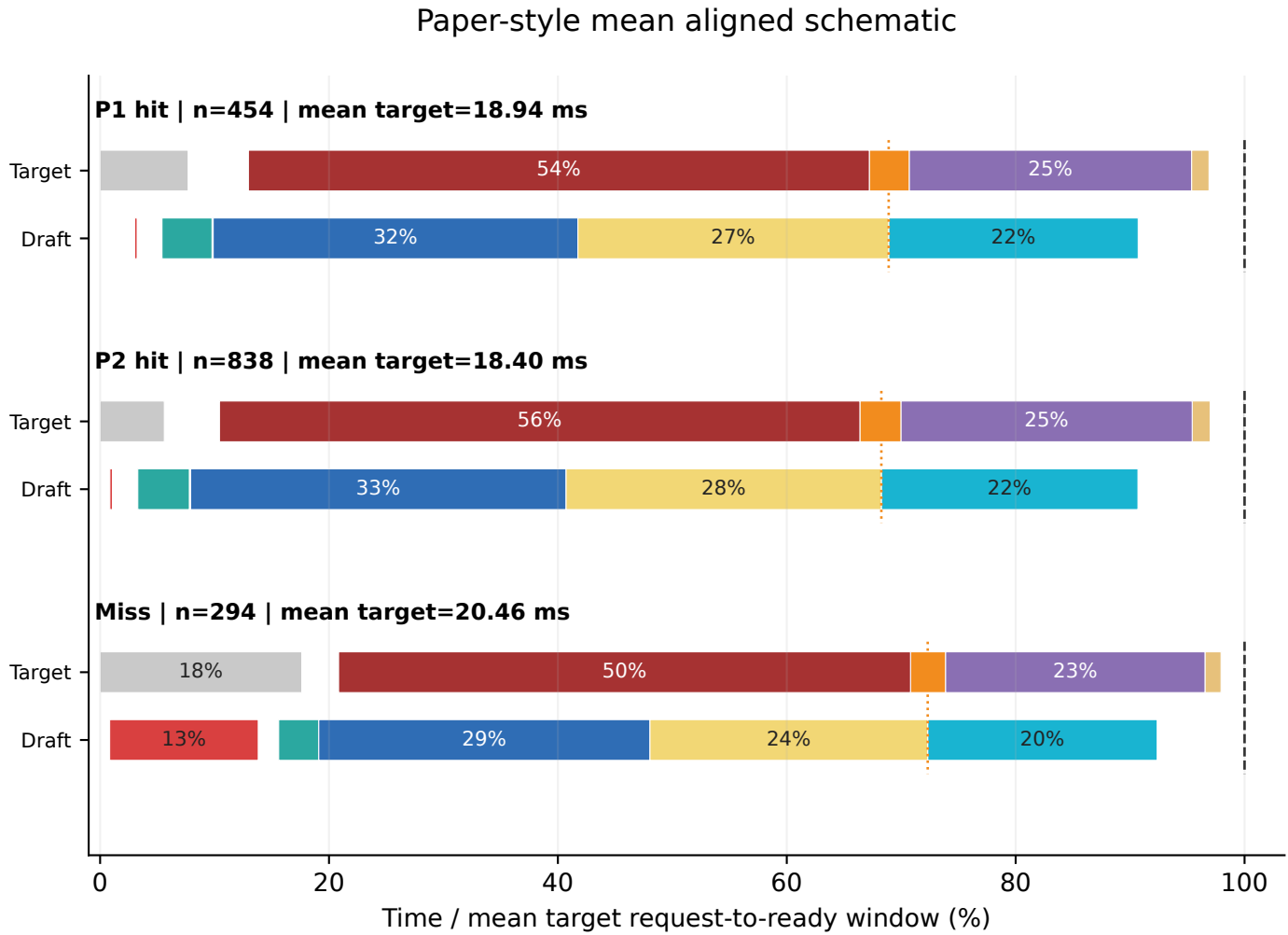
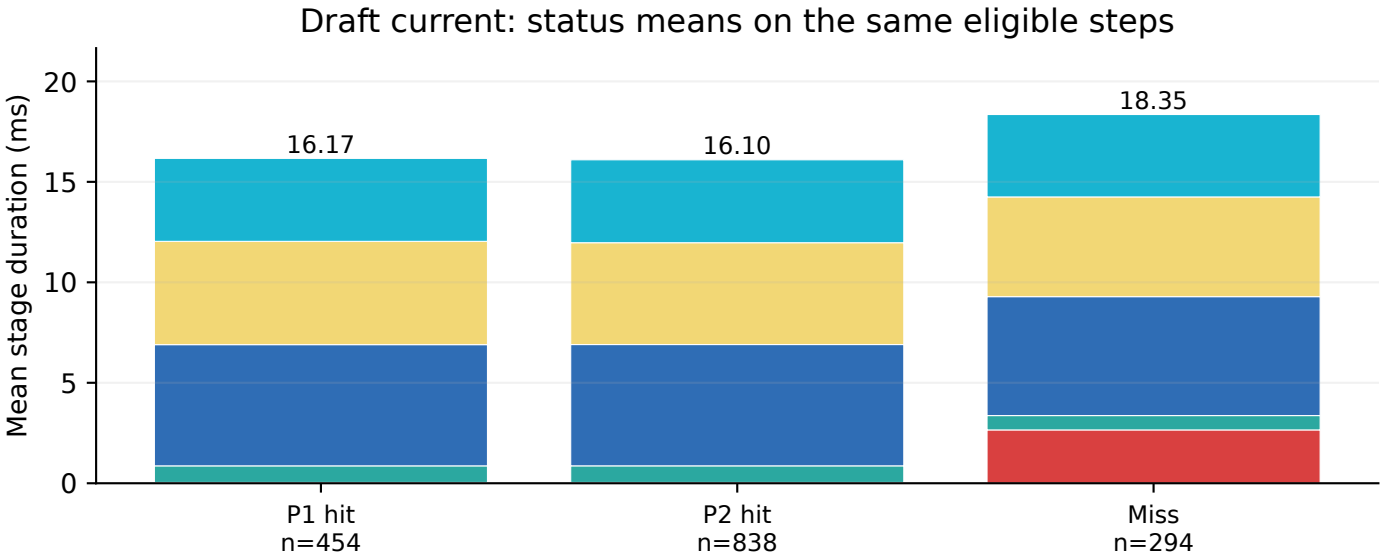
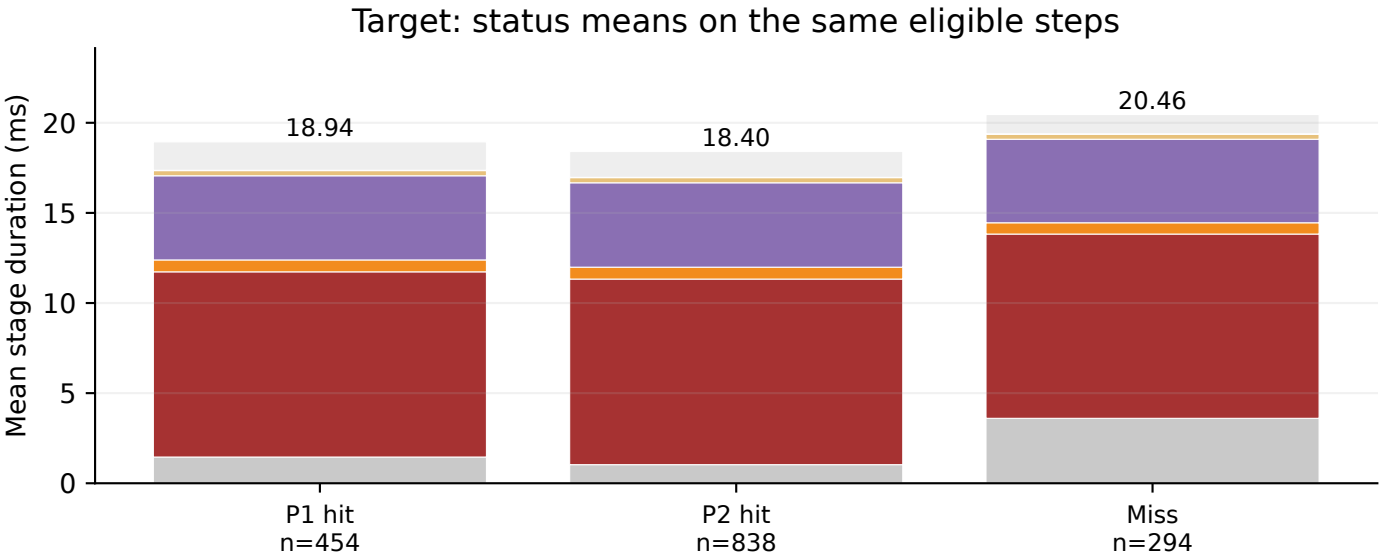
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_beta0
DUET: K1/K2=6/4, exit=21 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1586 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

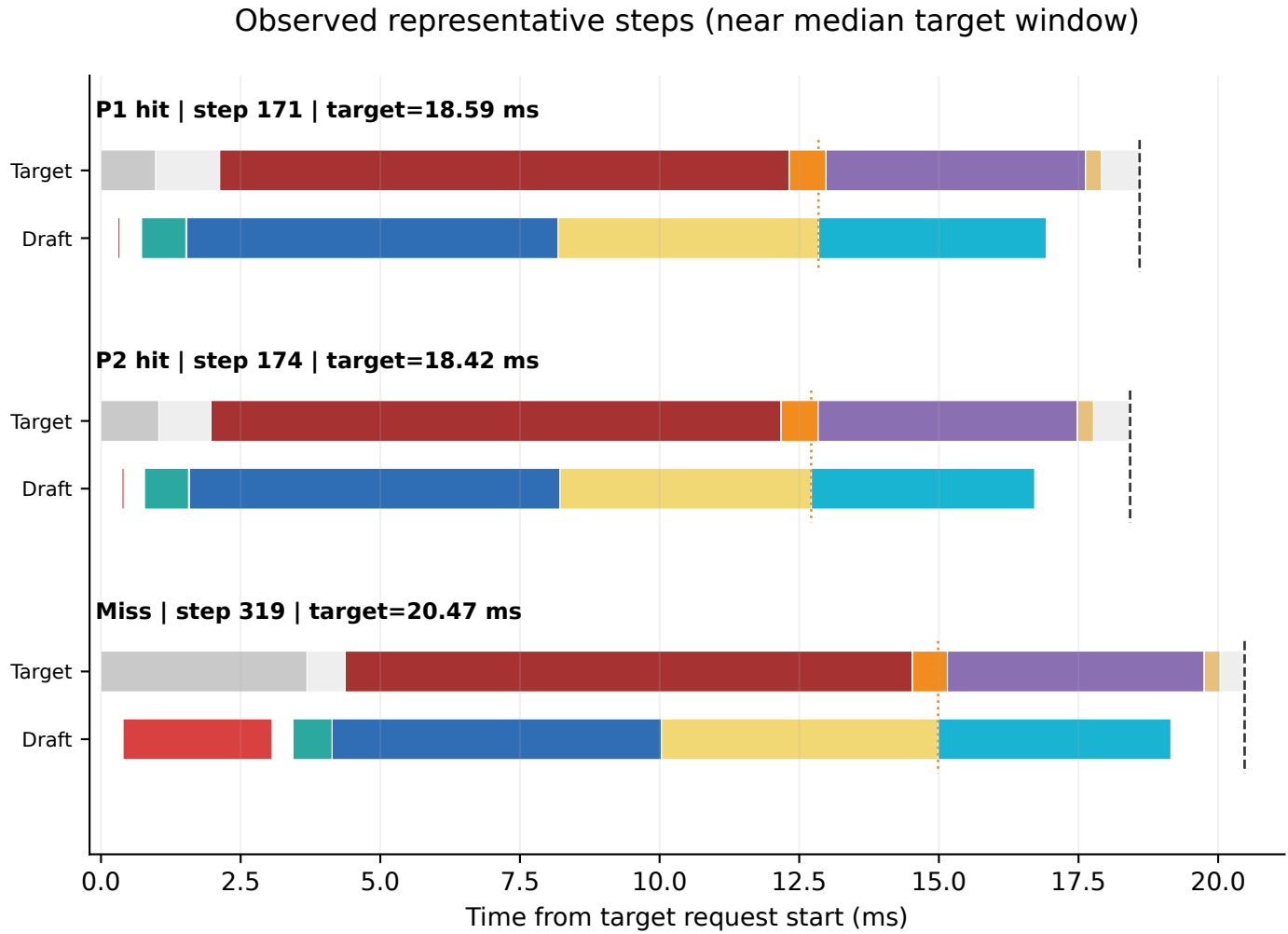
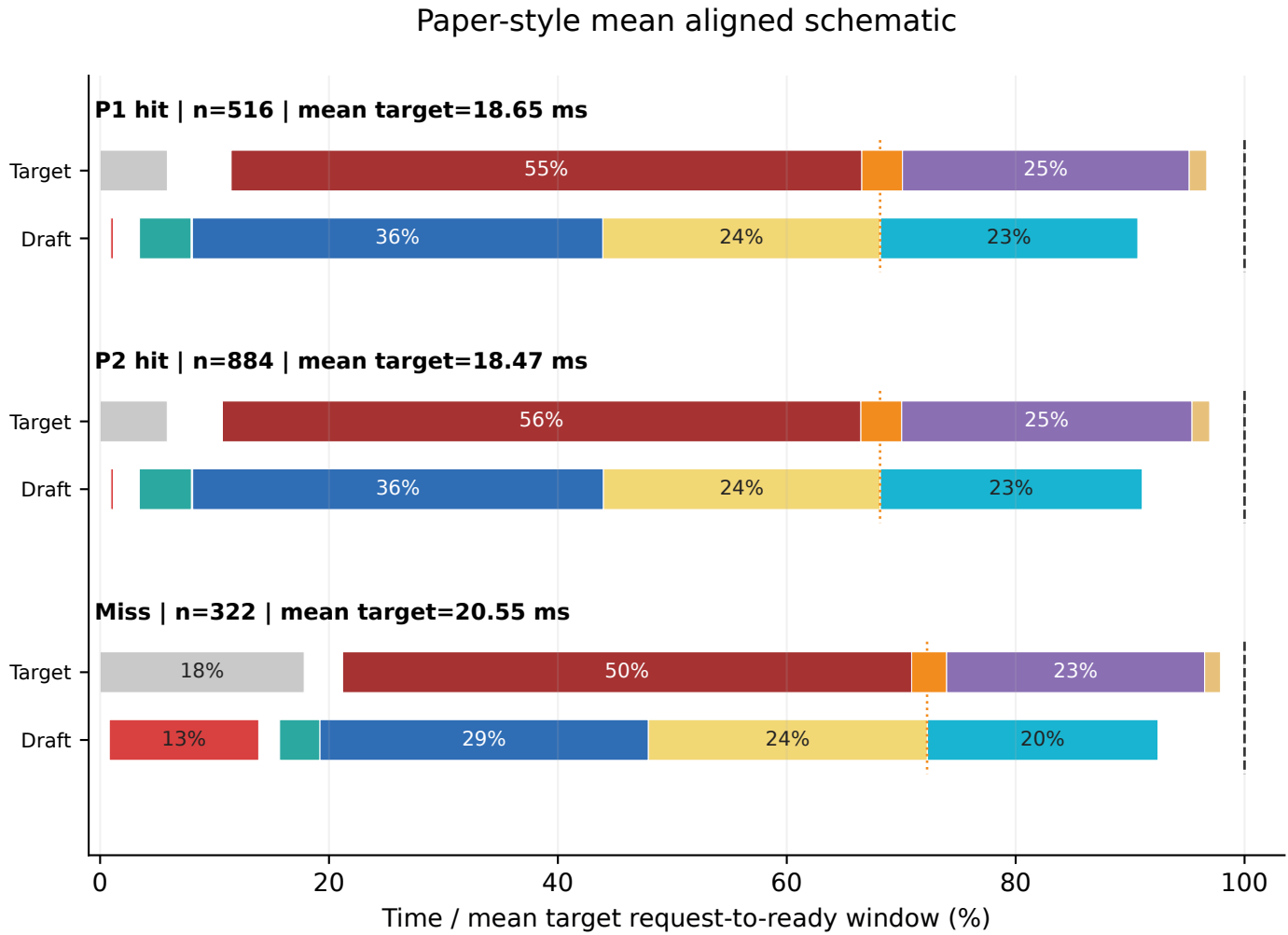
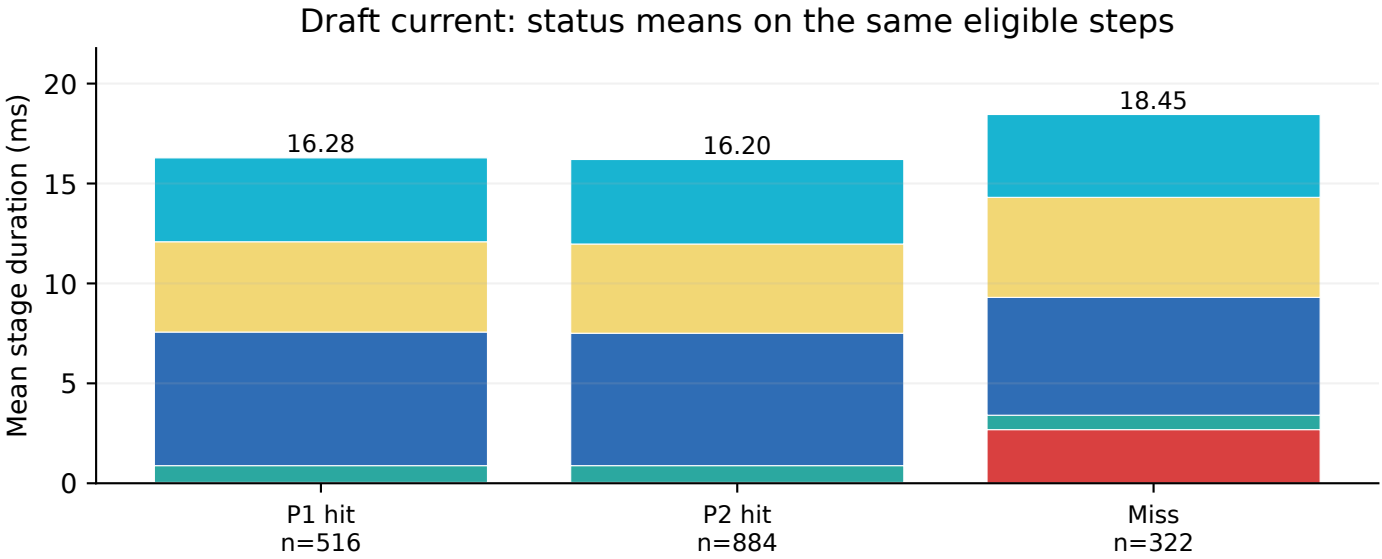
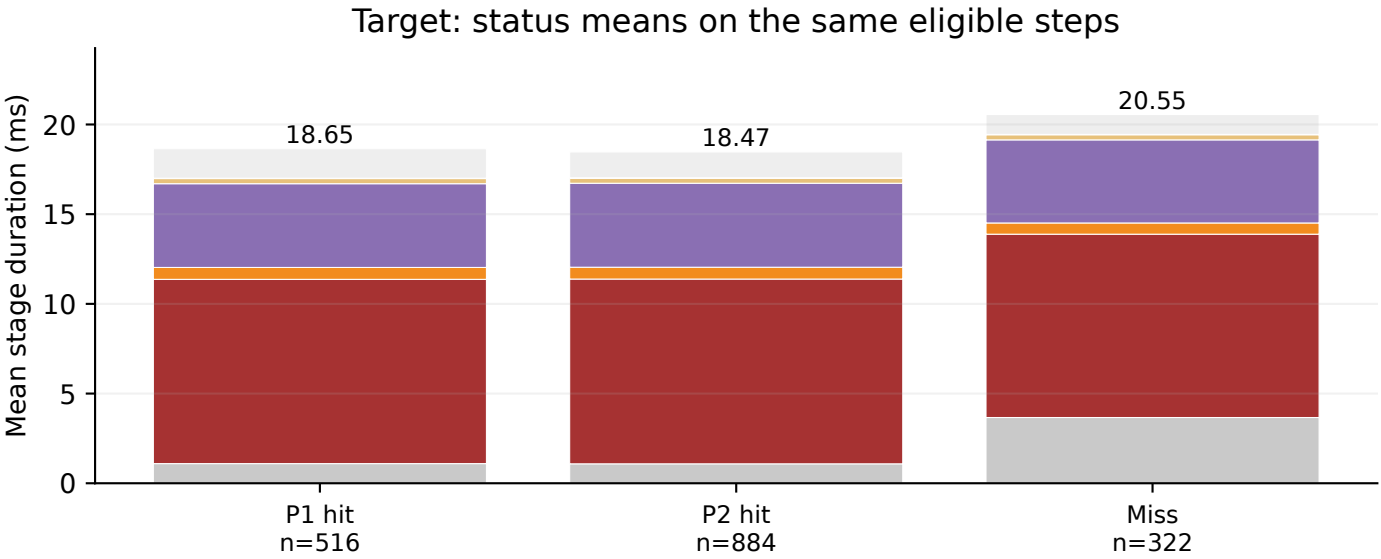
llama2_b1_tree_beta1
DUET: K1/K2=6/4, exit=21 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1586 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

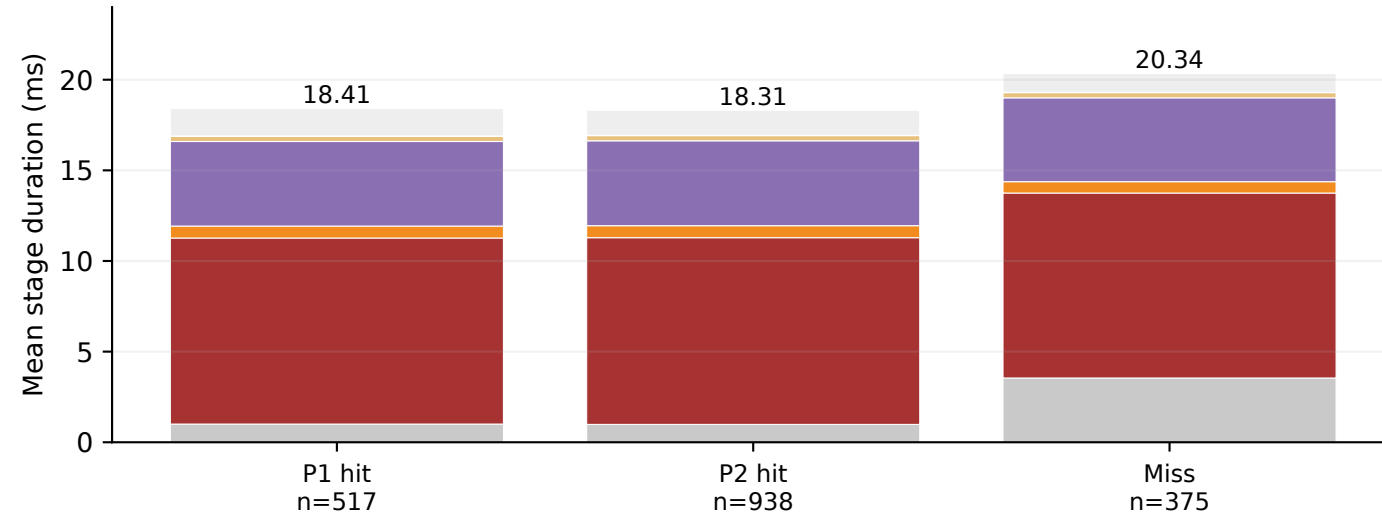
llama2_b1_tree_compute3
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1722 eligible steps



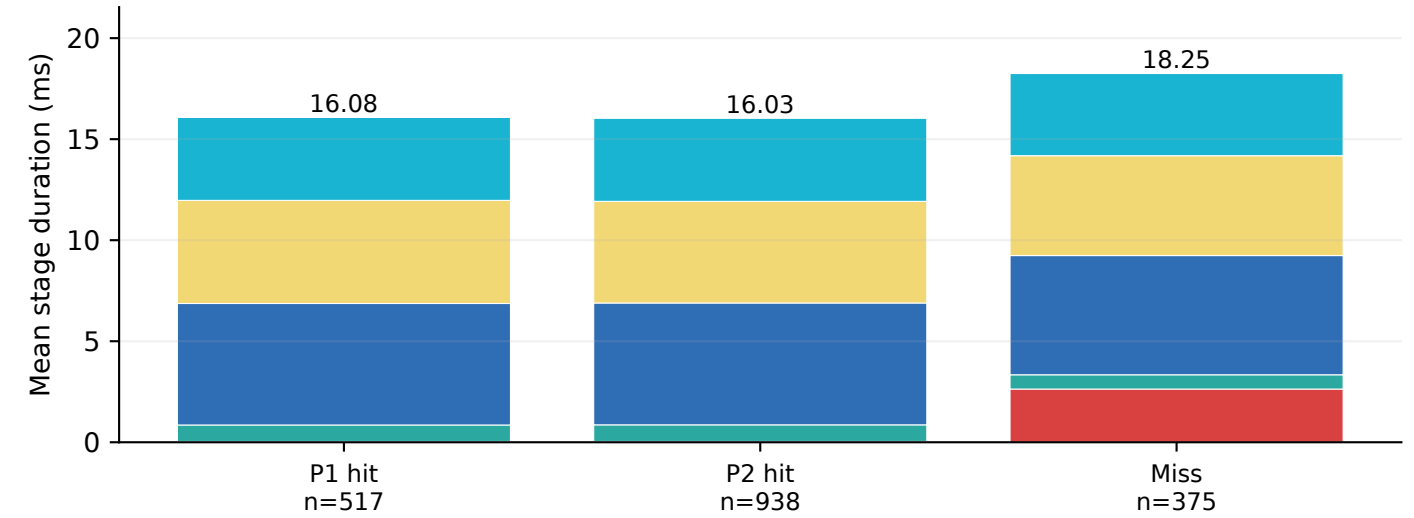
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_no_floors
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1830 eligible steps

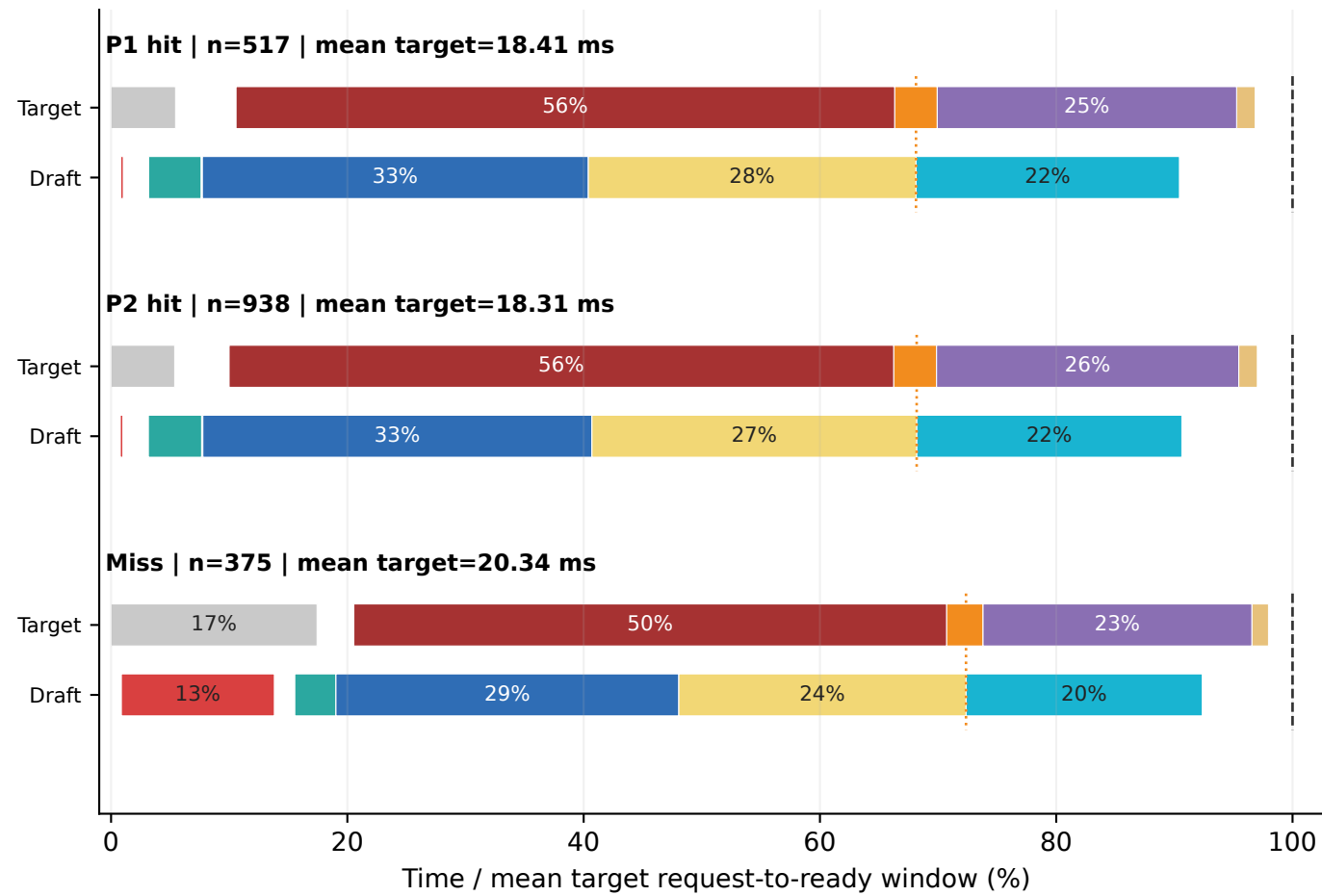
Target: status means on the same eligible steps



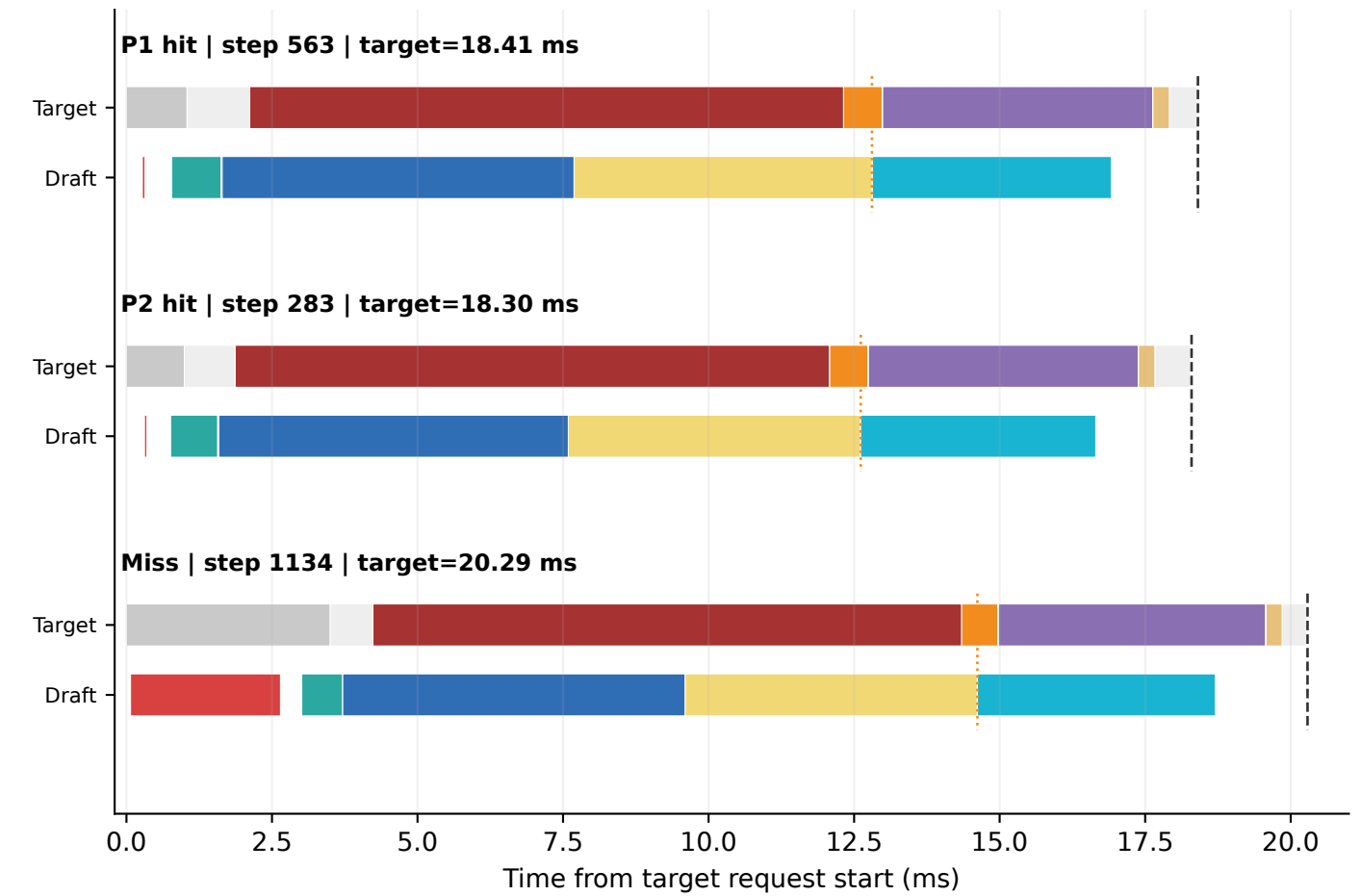
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic

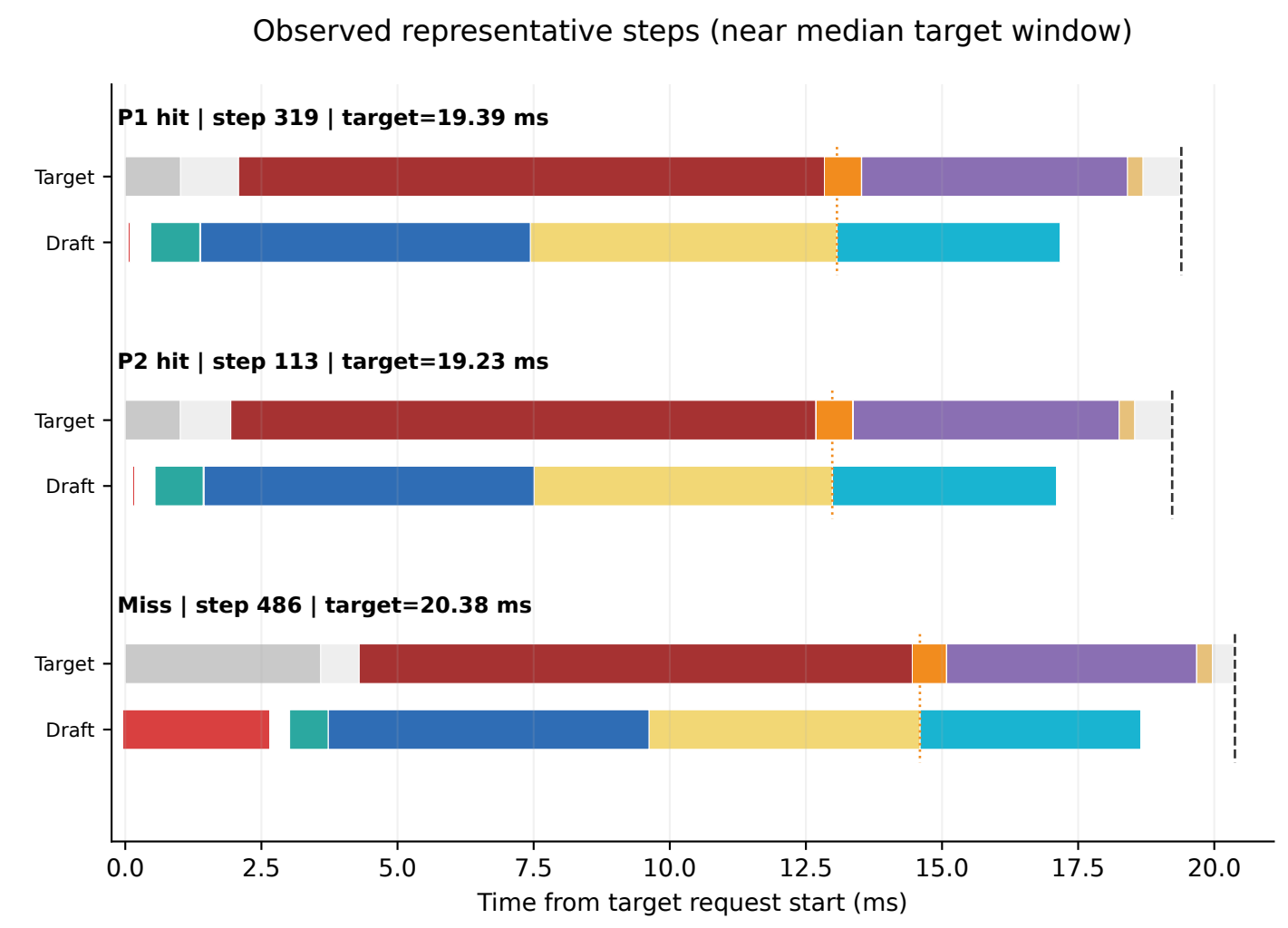
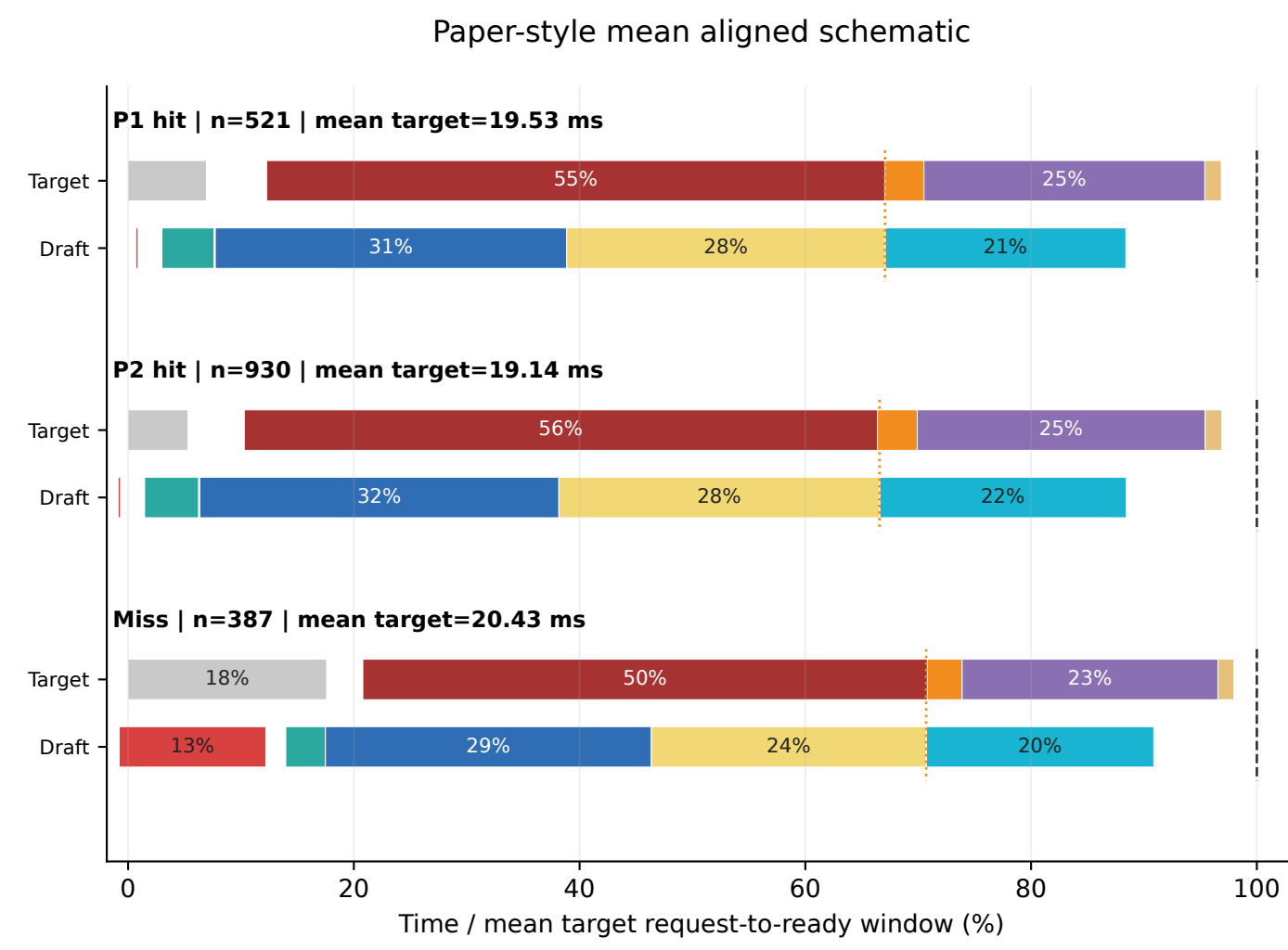
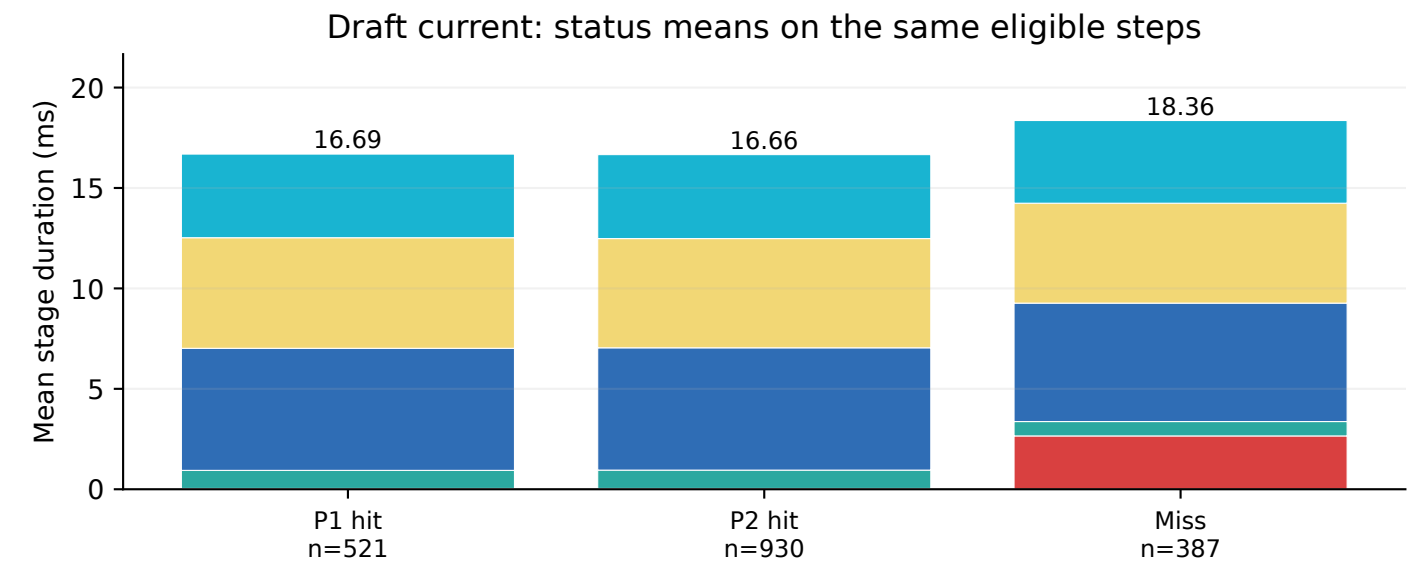
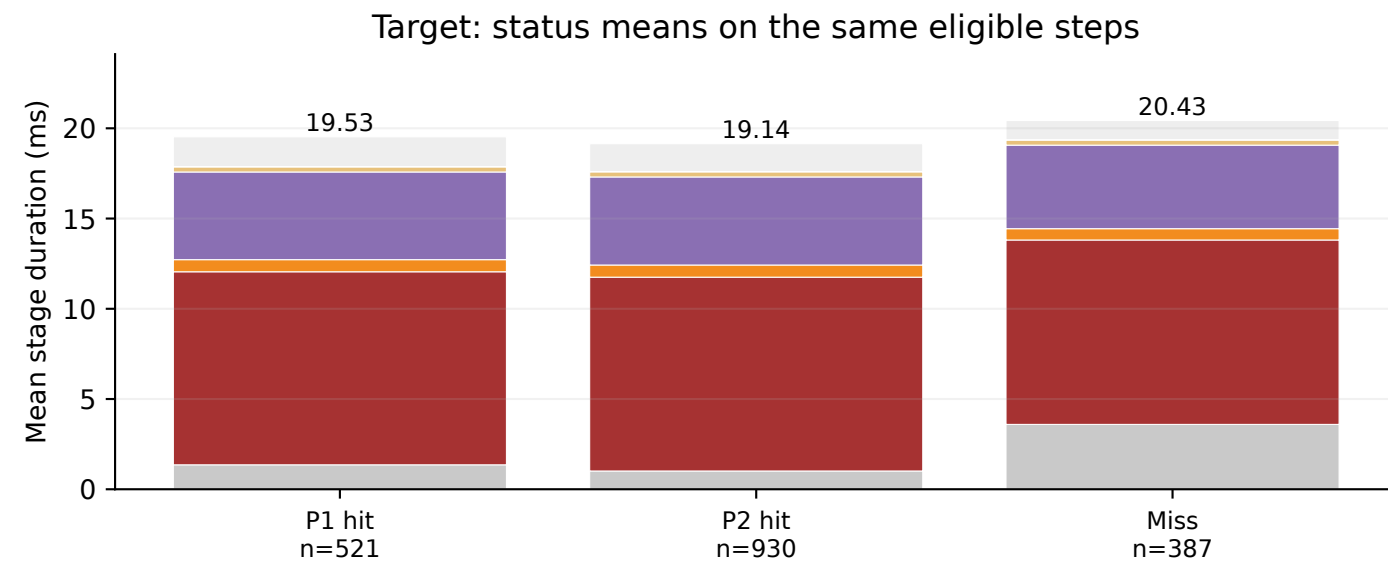


Observed representative steps (near median target window)



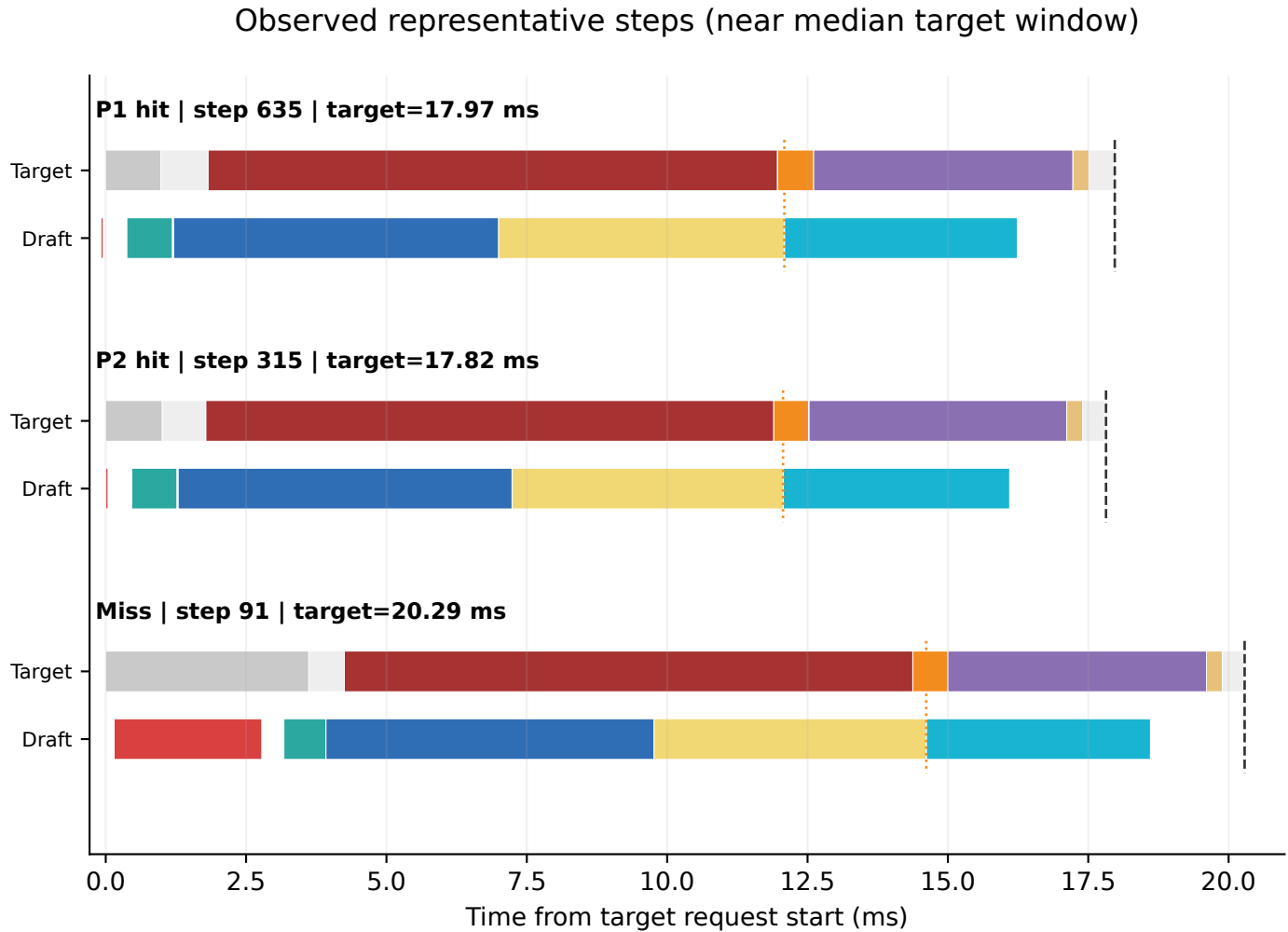
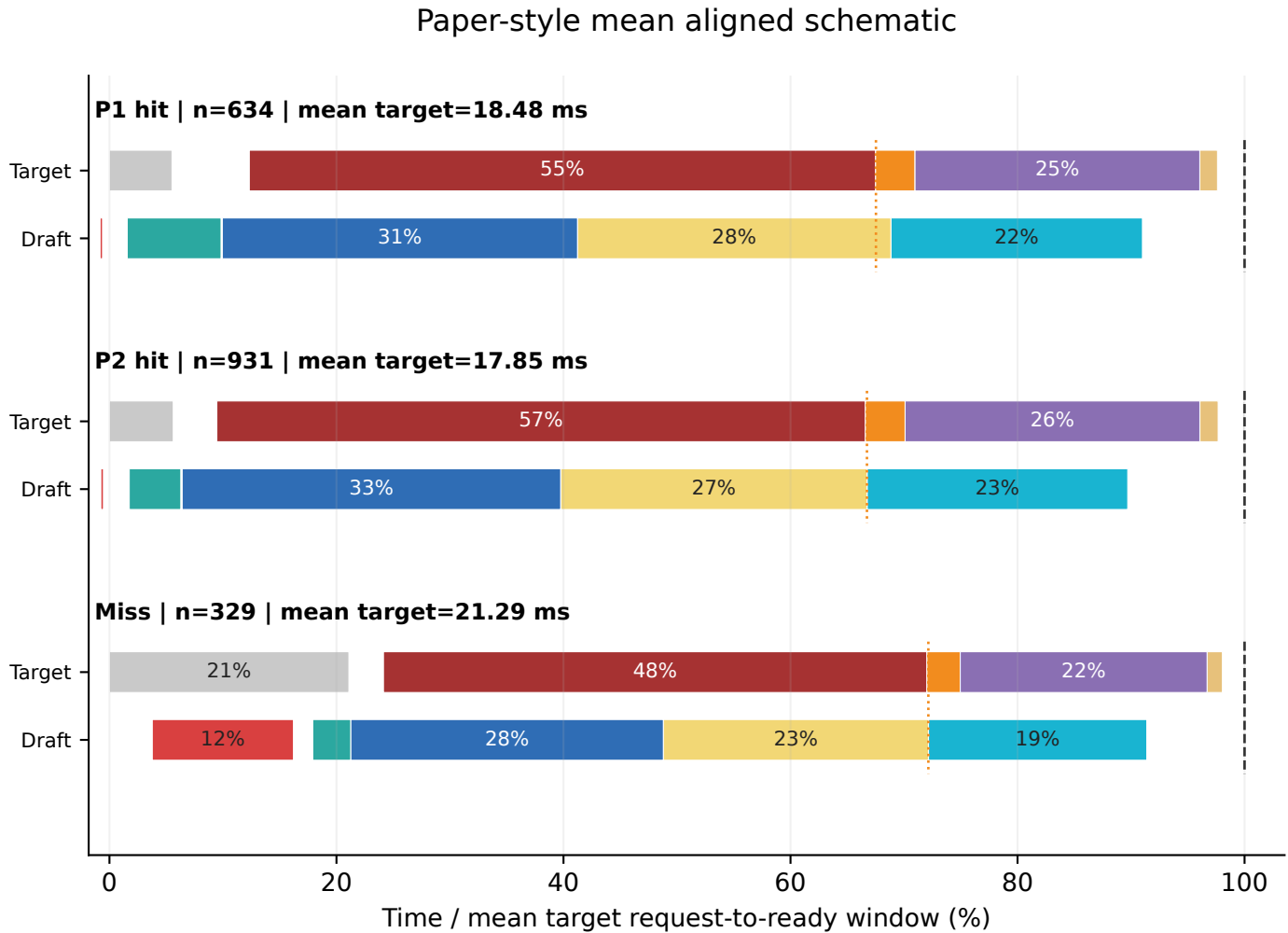
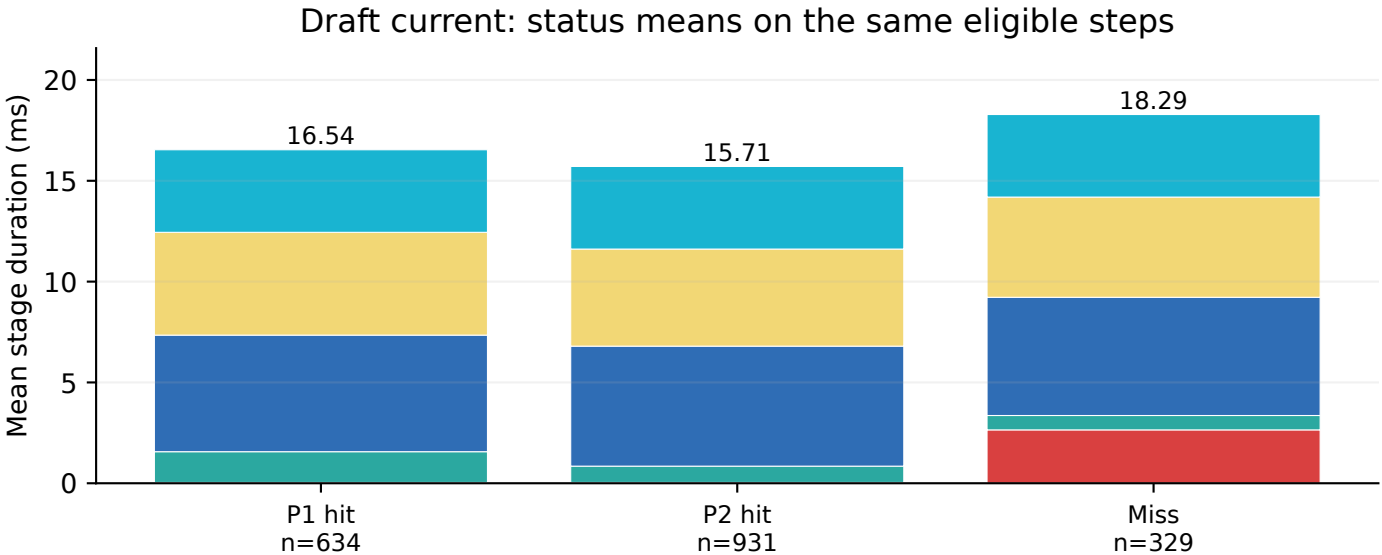
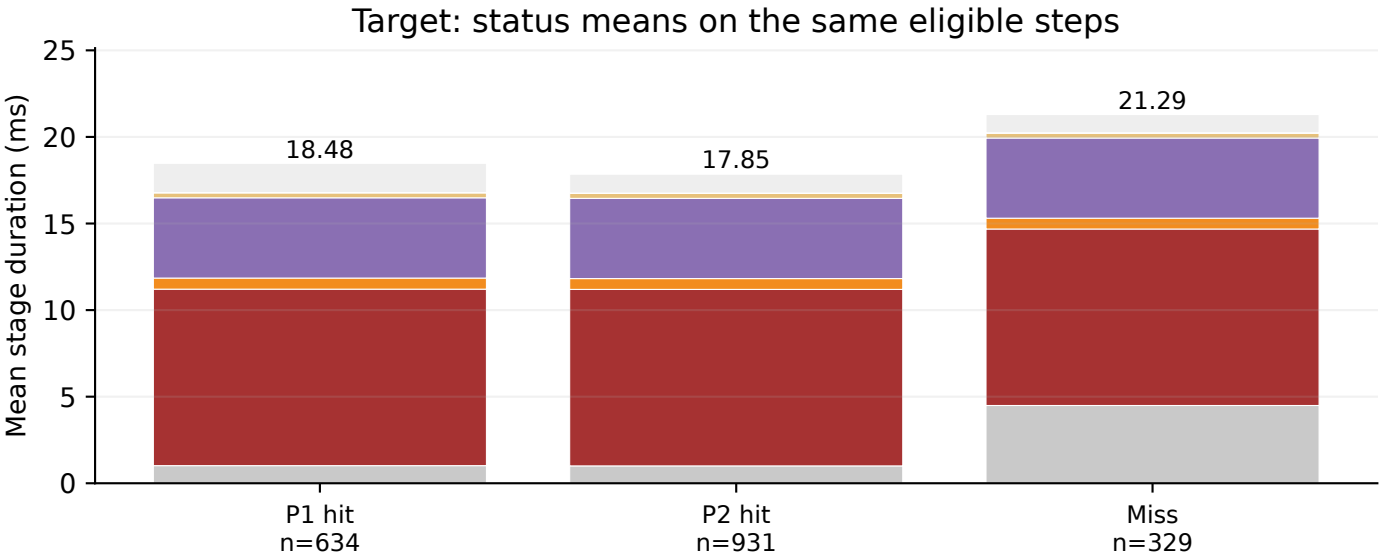
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_nodes_high
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=16/12, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1838 eligible steps



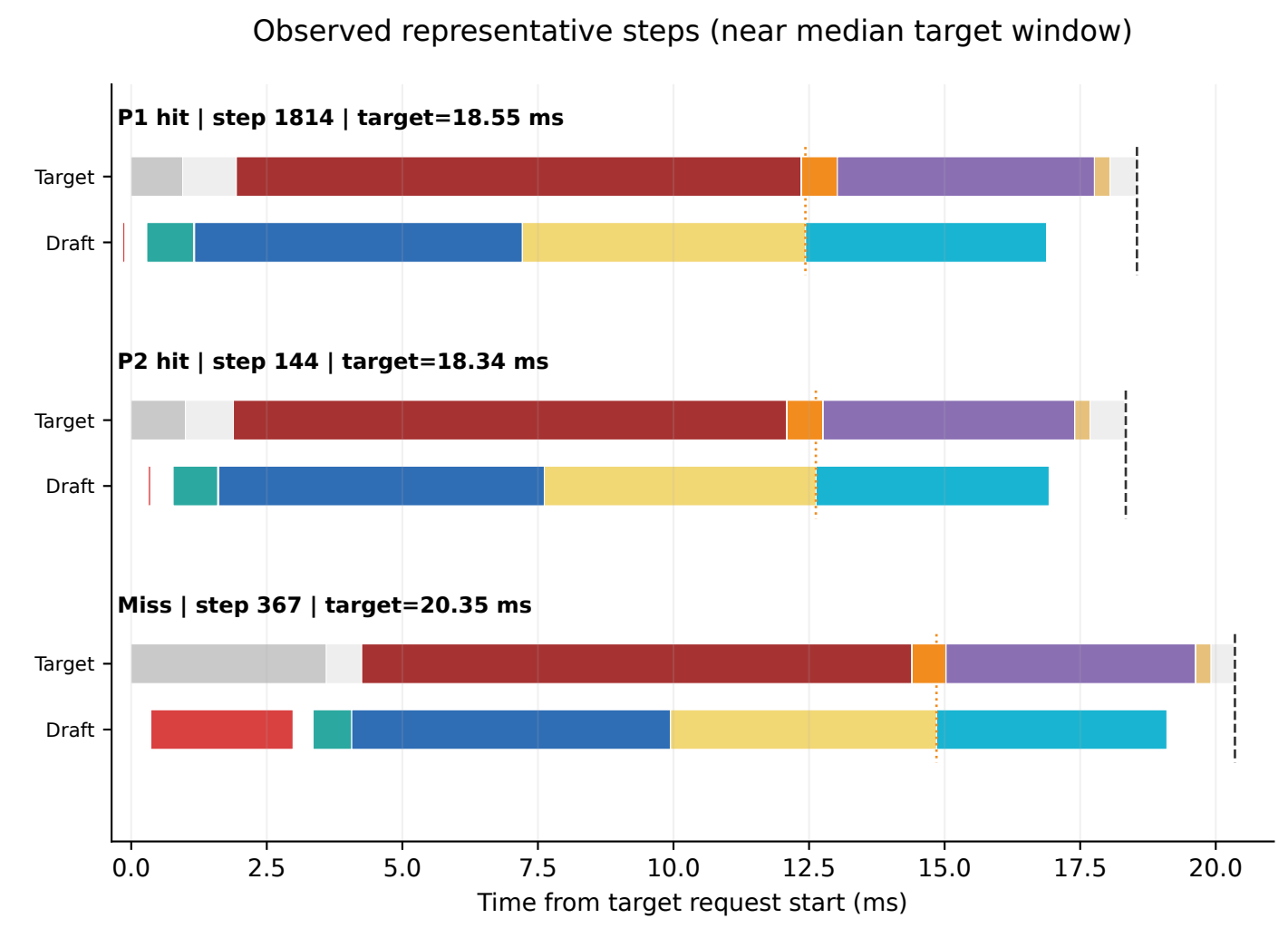
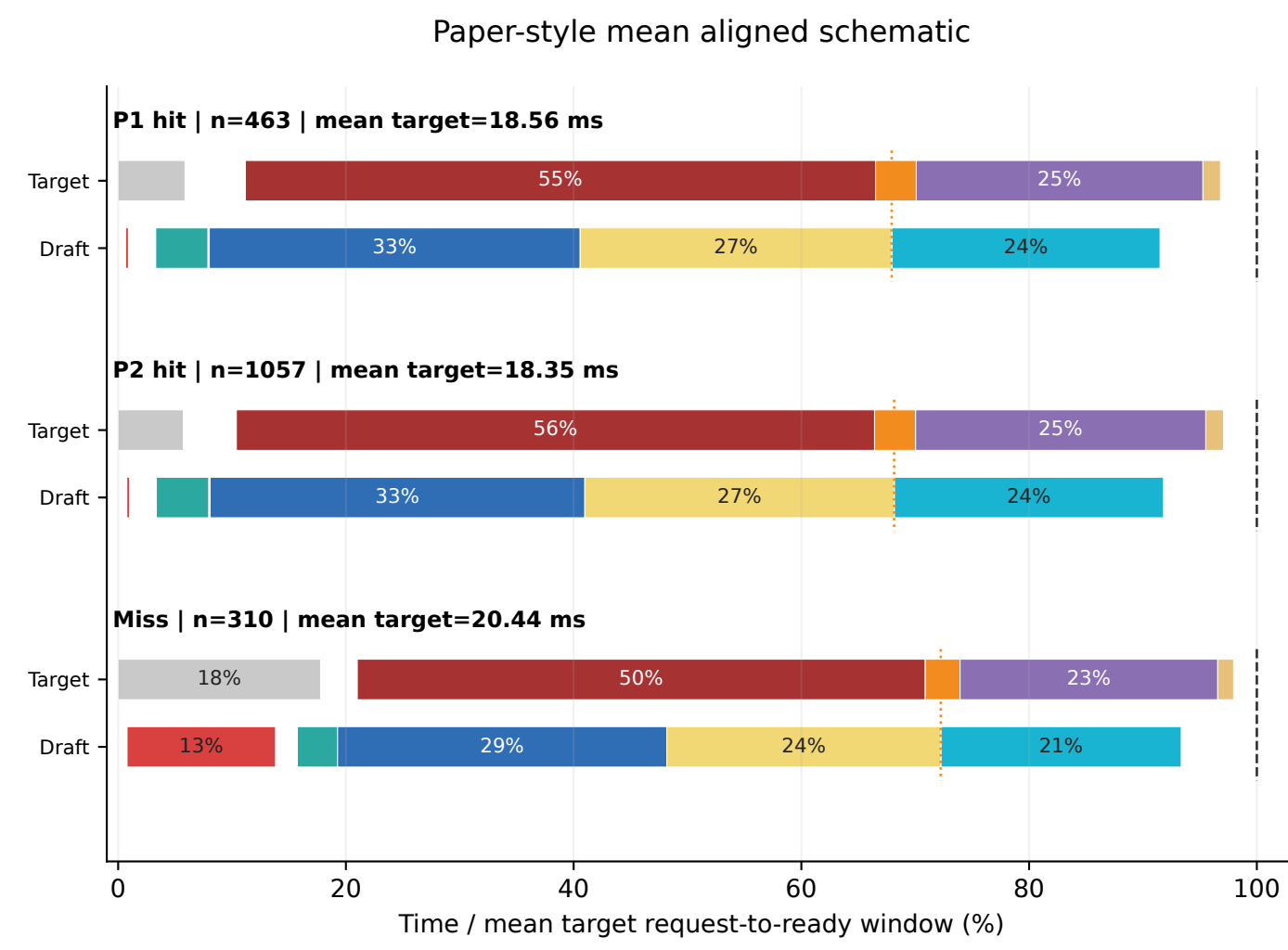
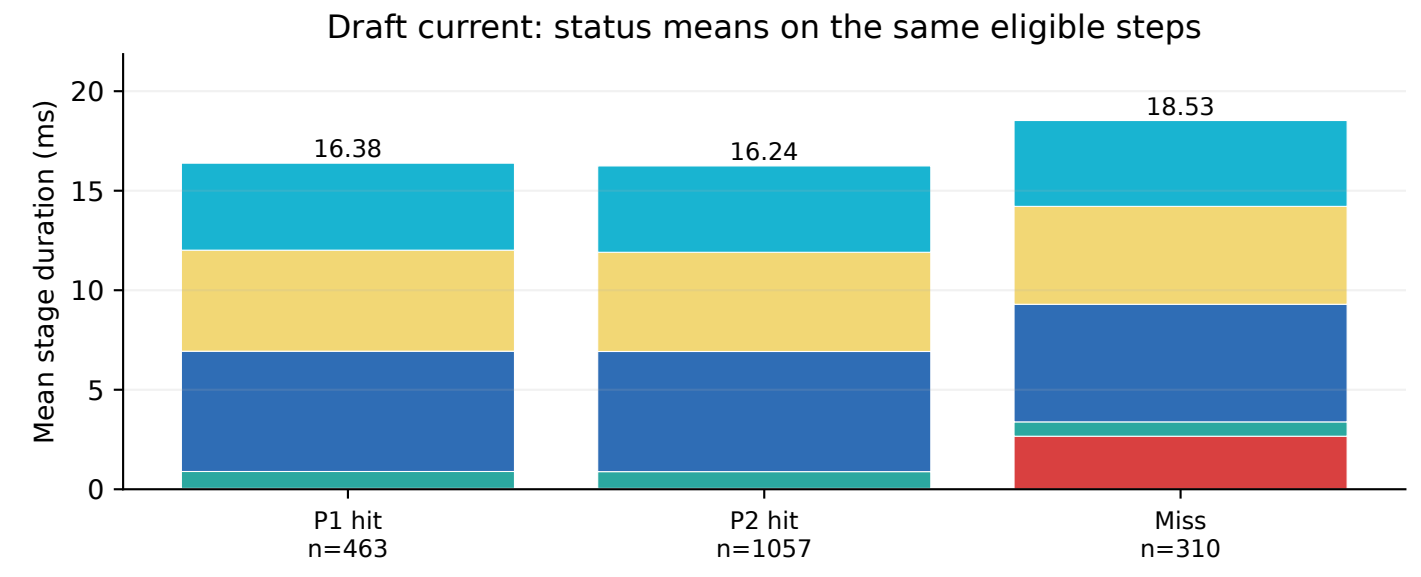
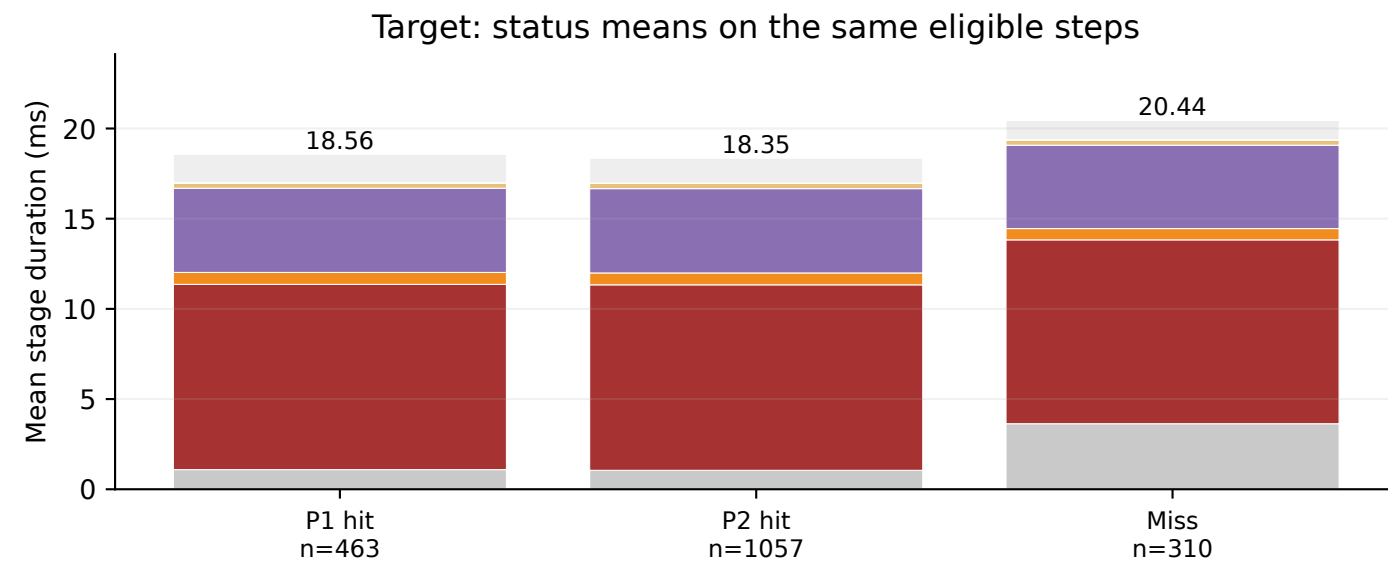
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_nodes_low
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=6/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1894 eligible steps



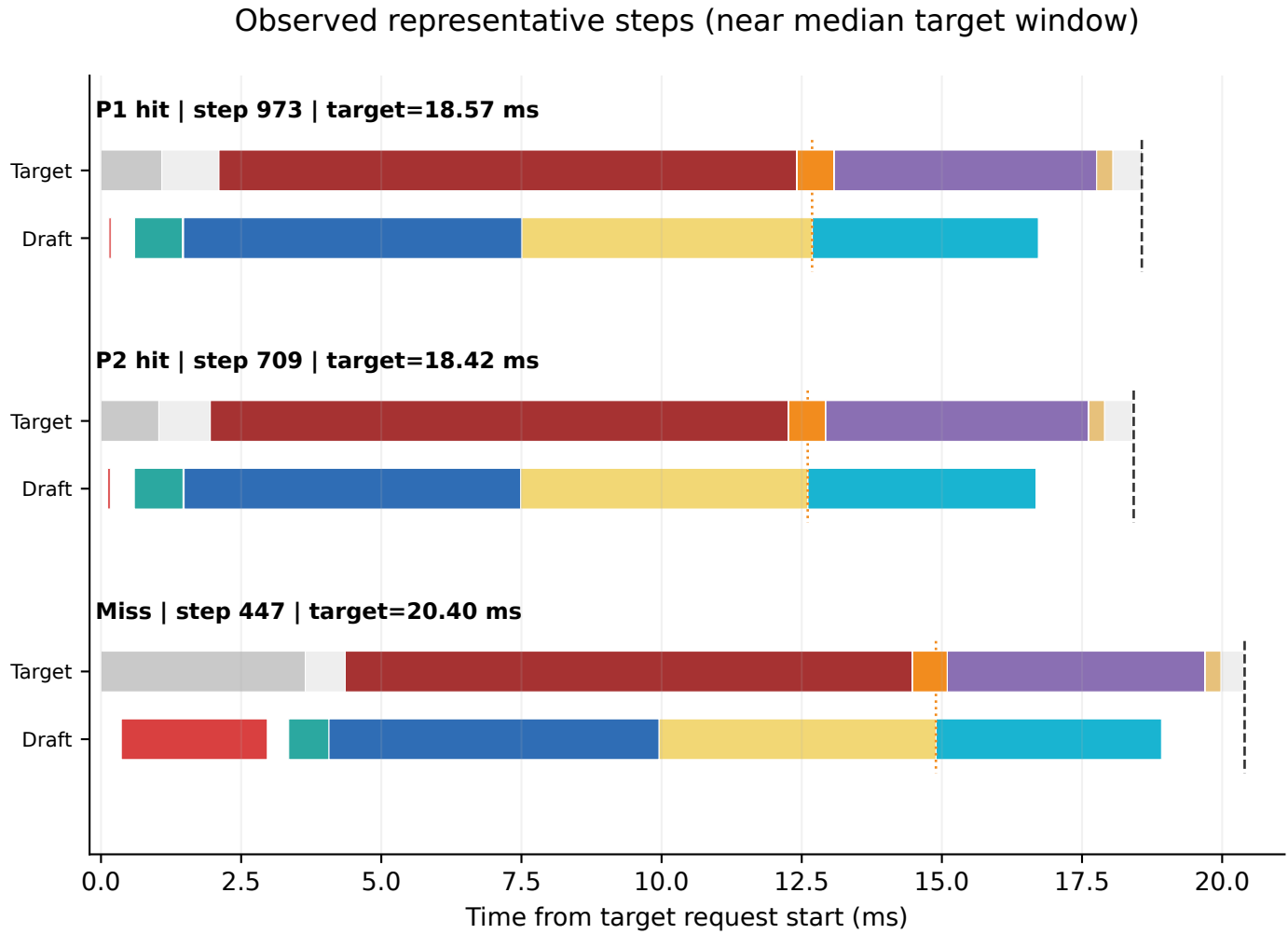
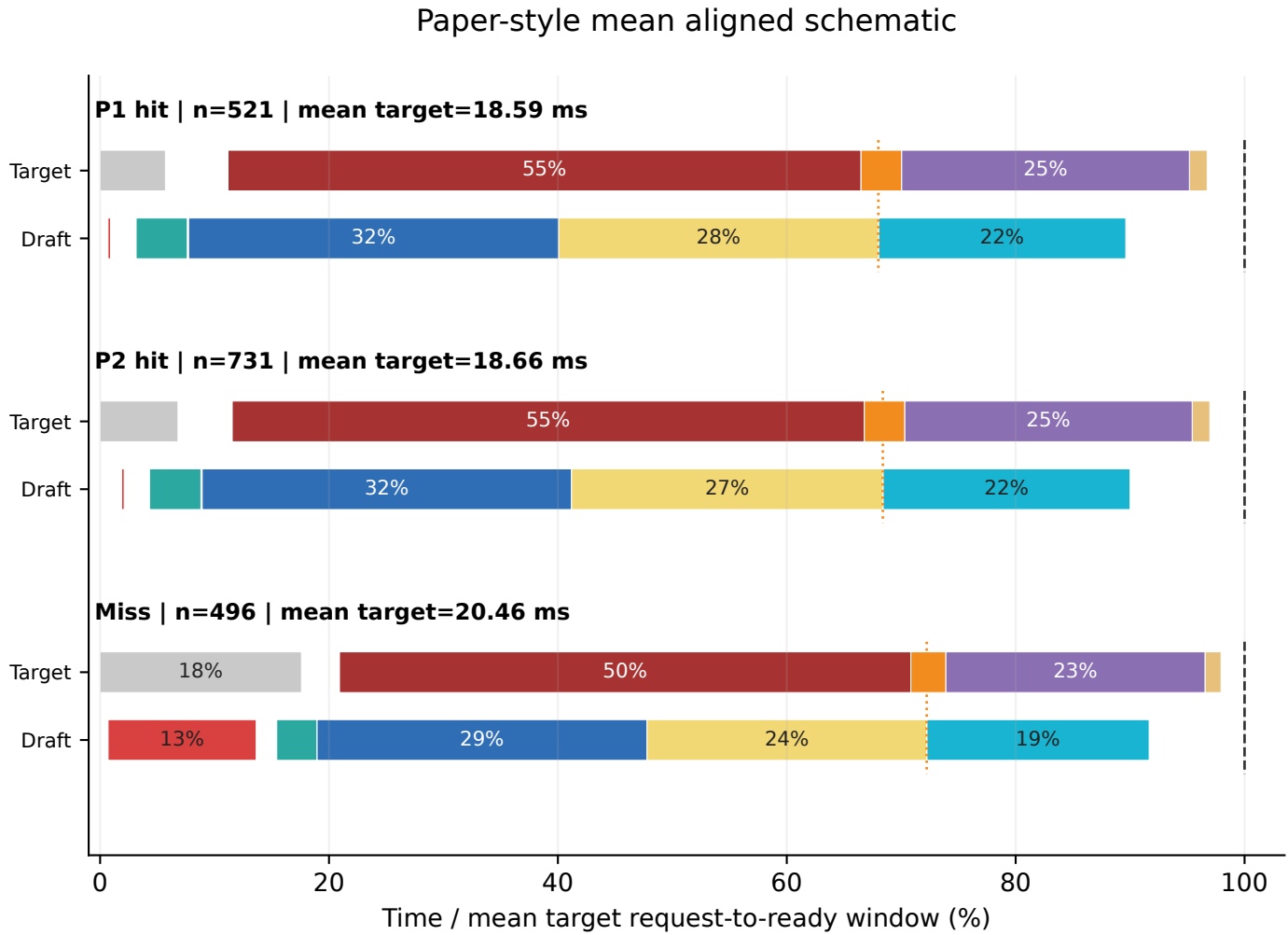
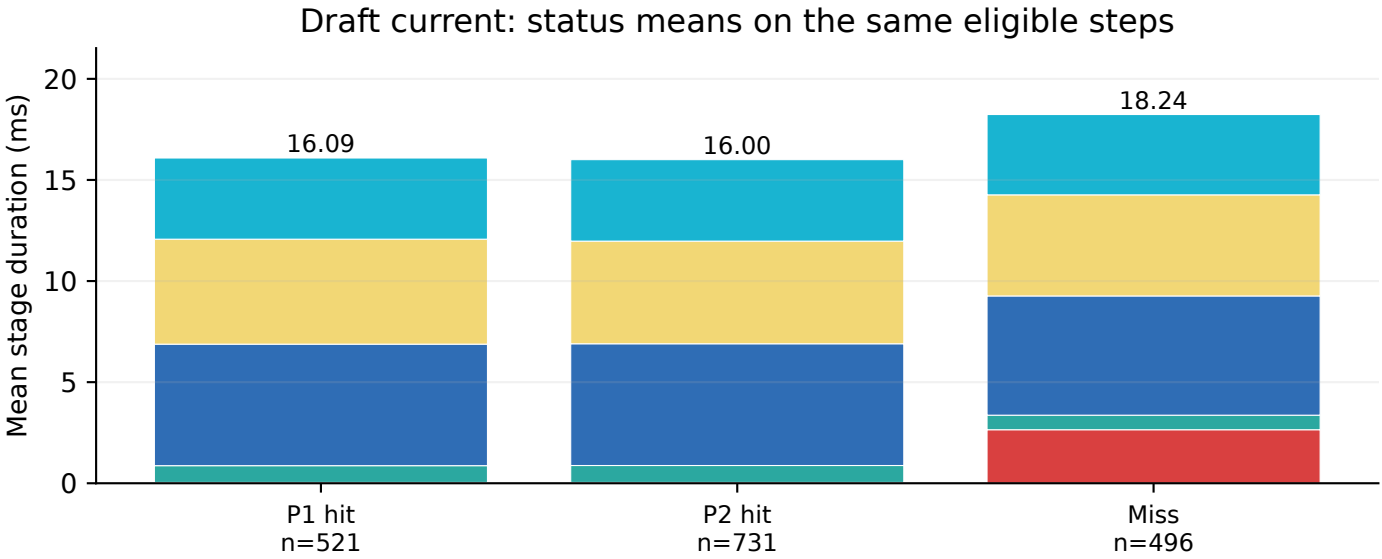
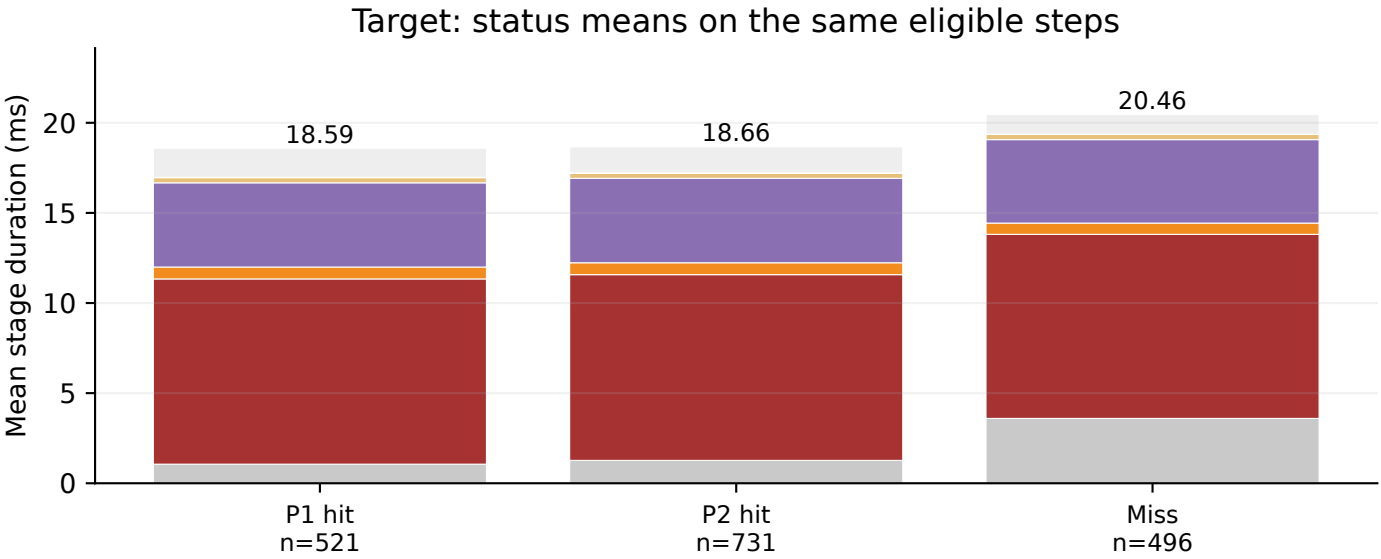
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_p2budget12
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=12, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1830 eligible steps



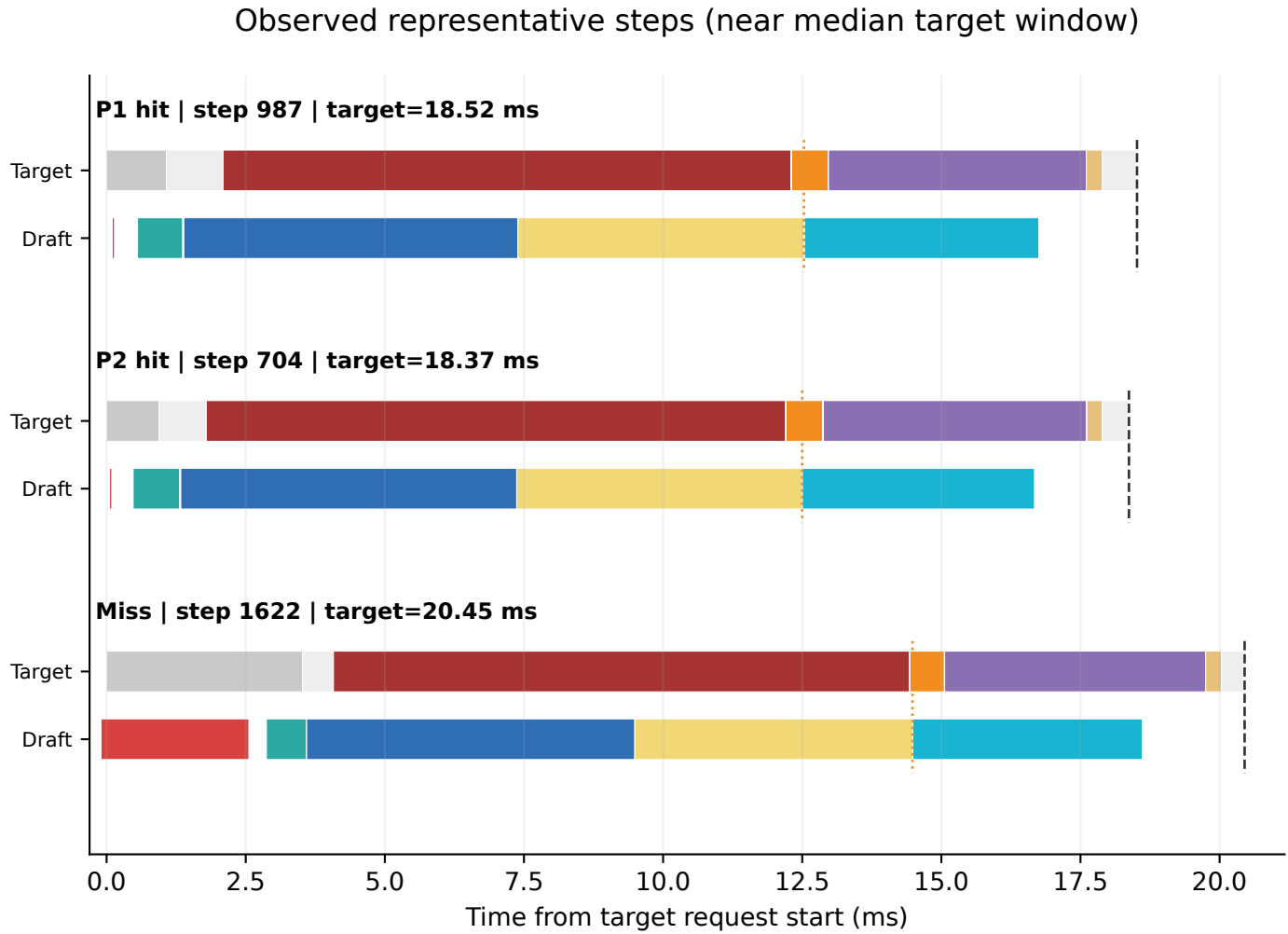
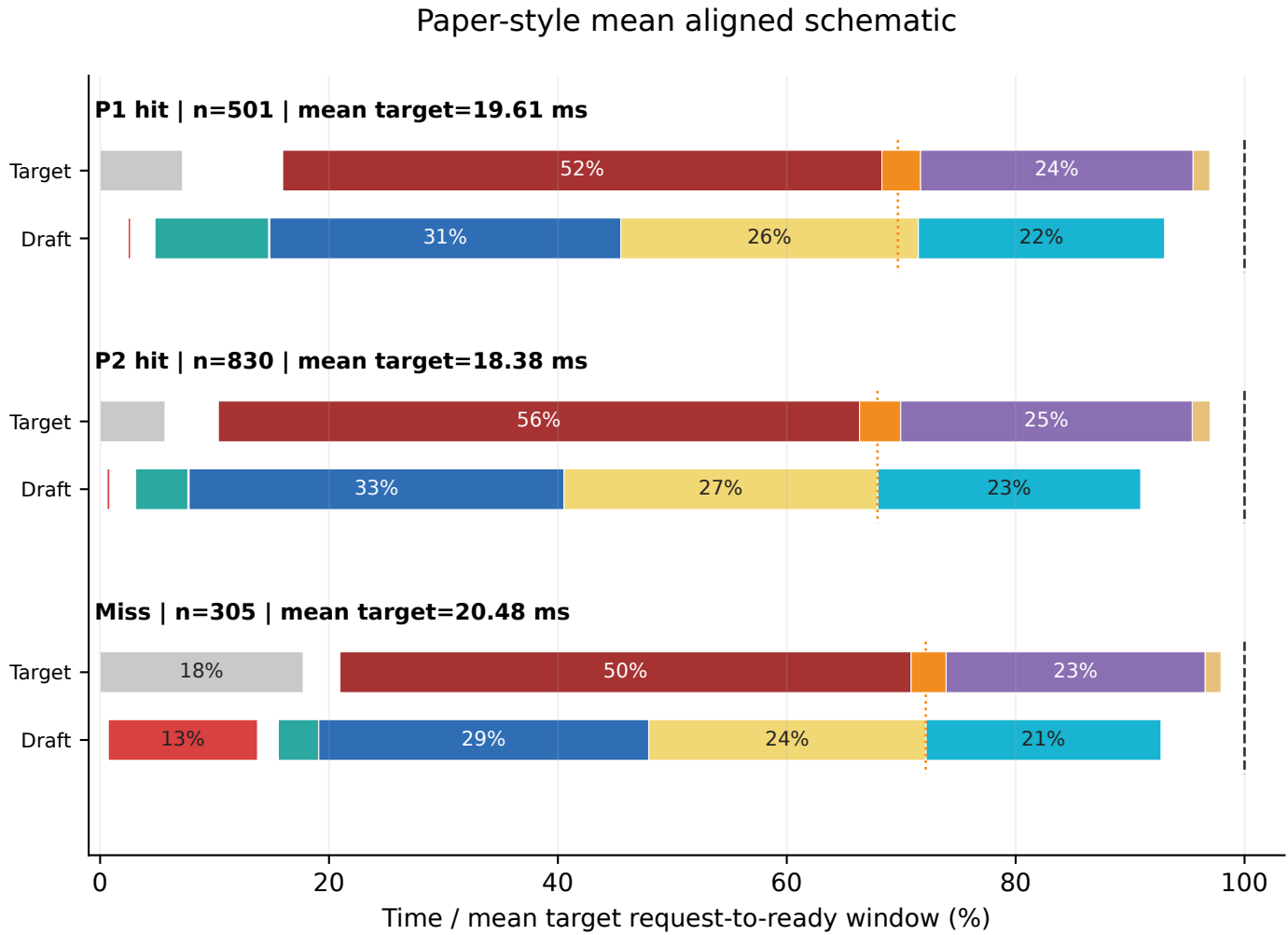
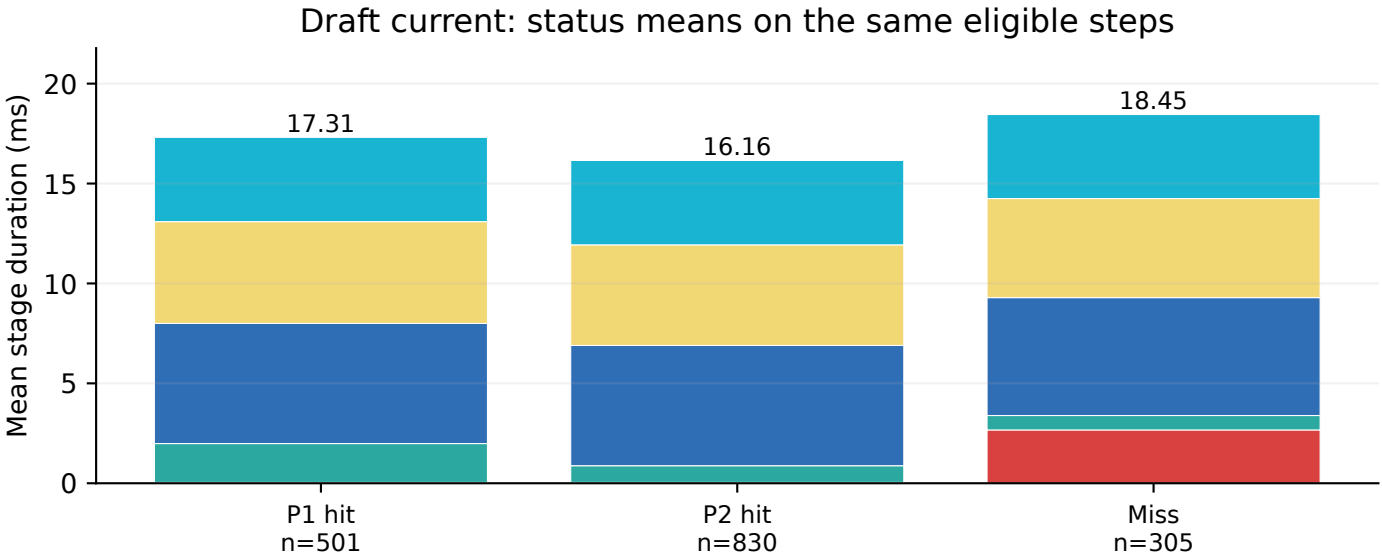
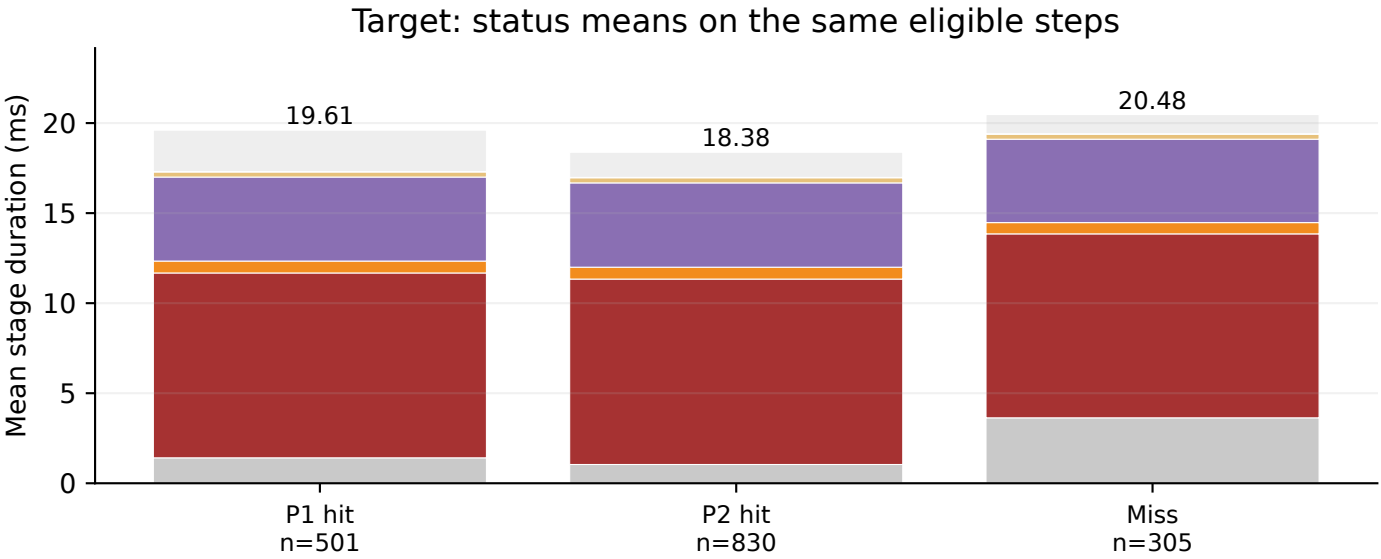
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_p2budget3
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1748 eligible steps



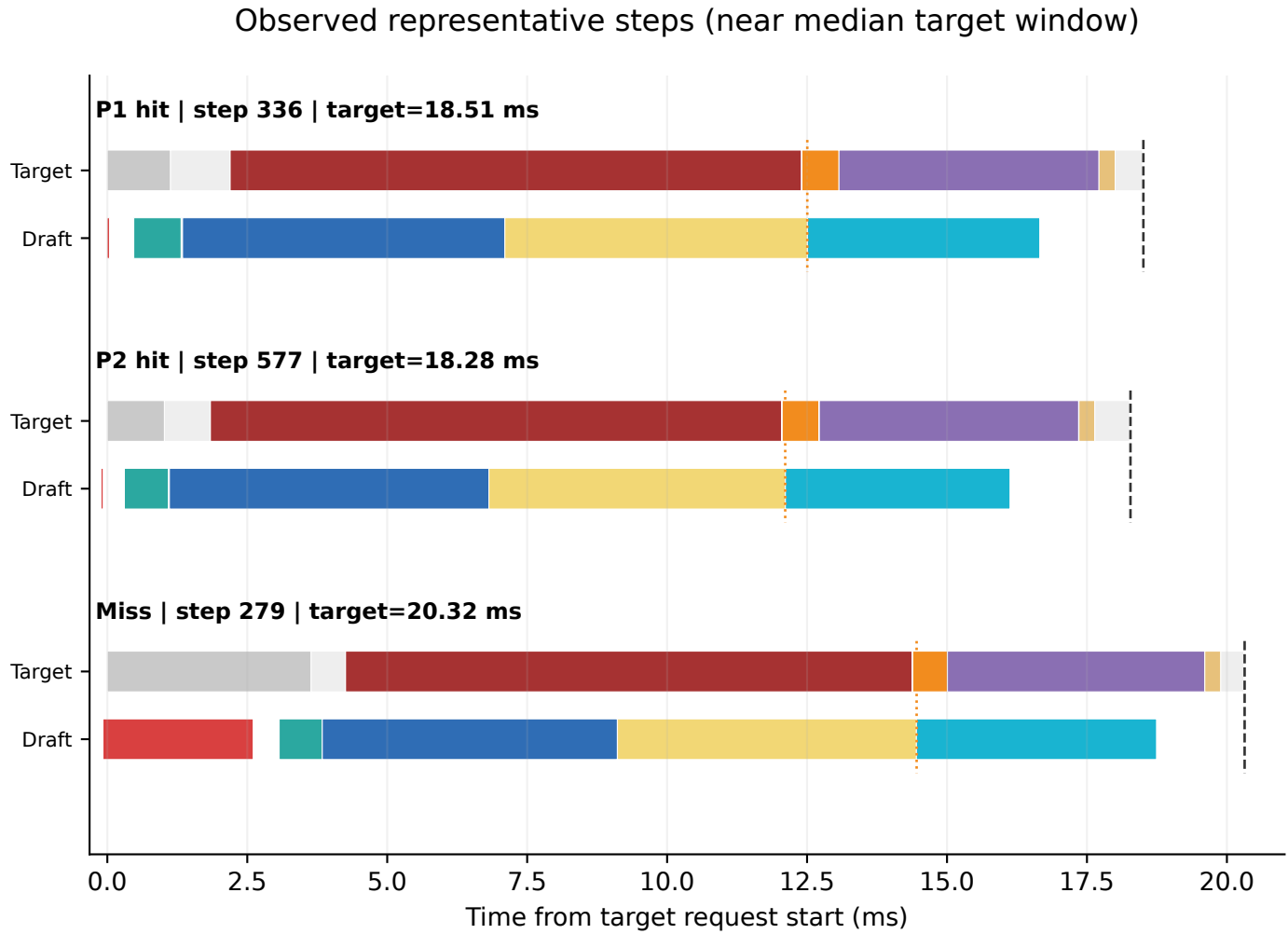
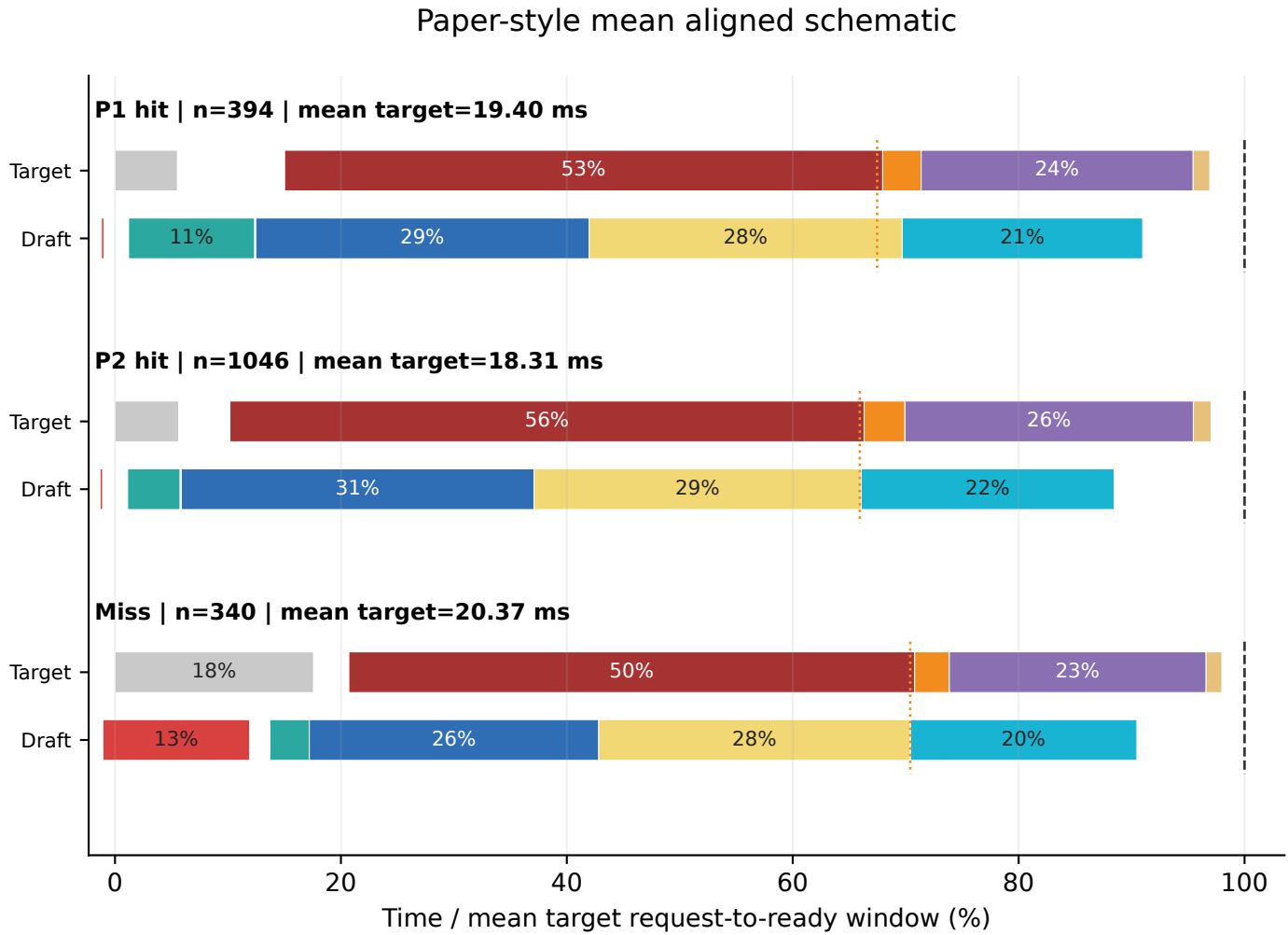
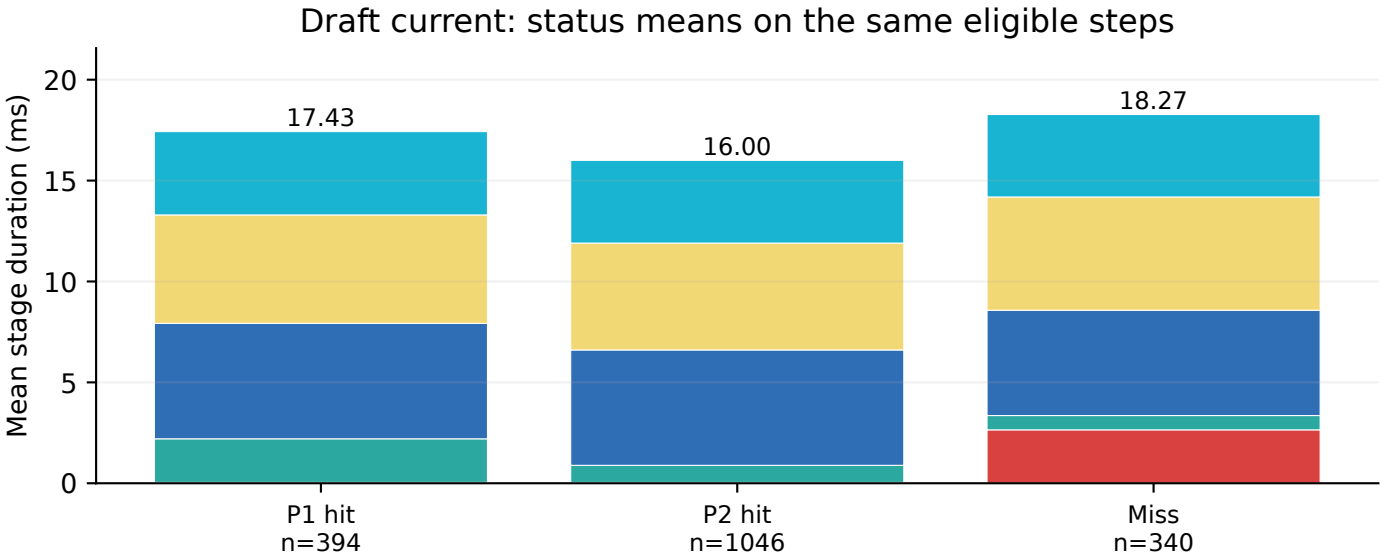
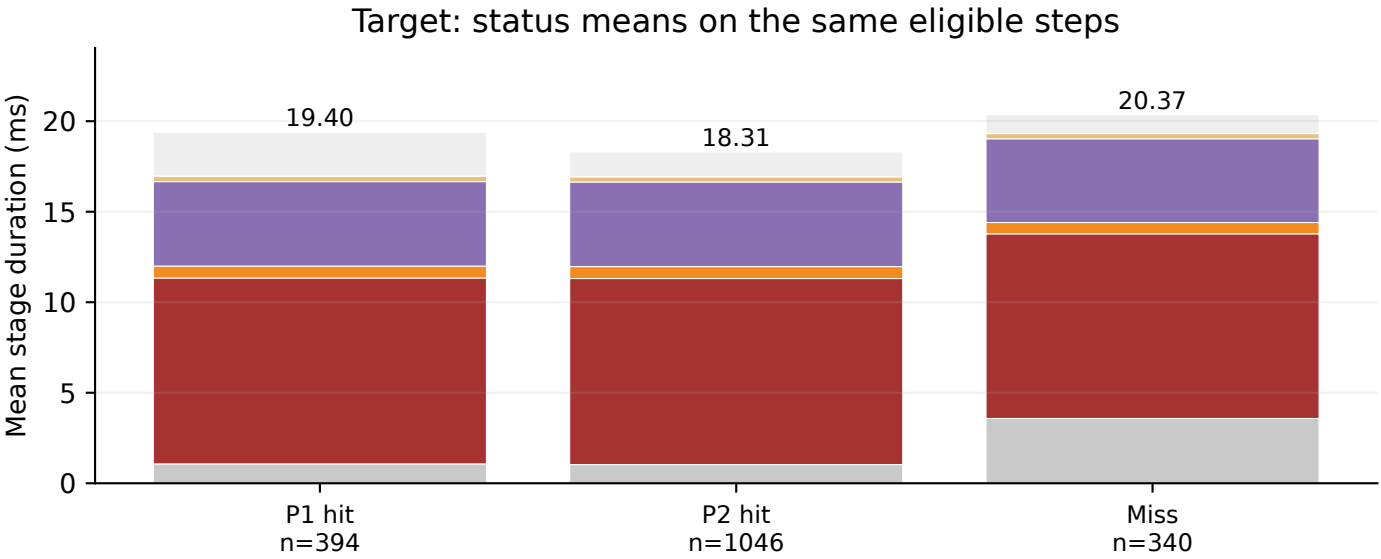
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_p2budget8
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=8, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1636 eligible steps

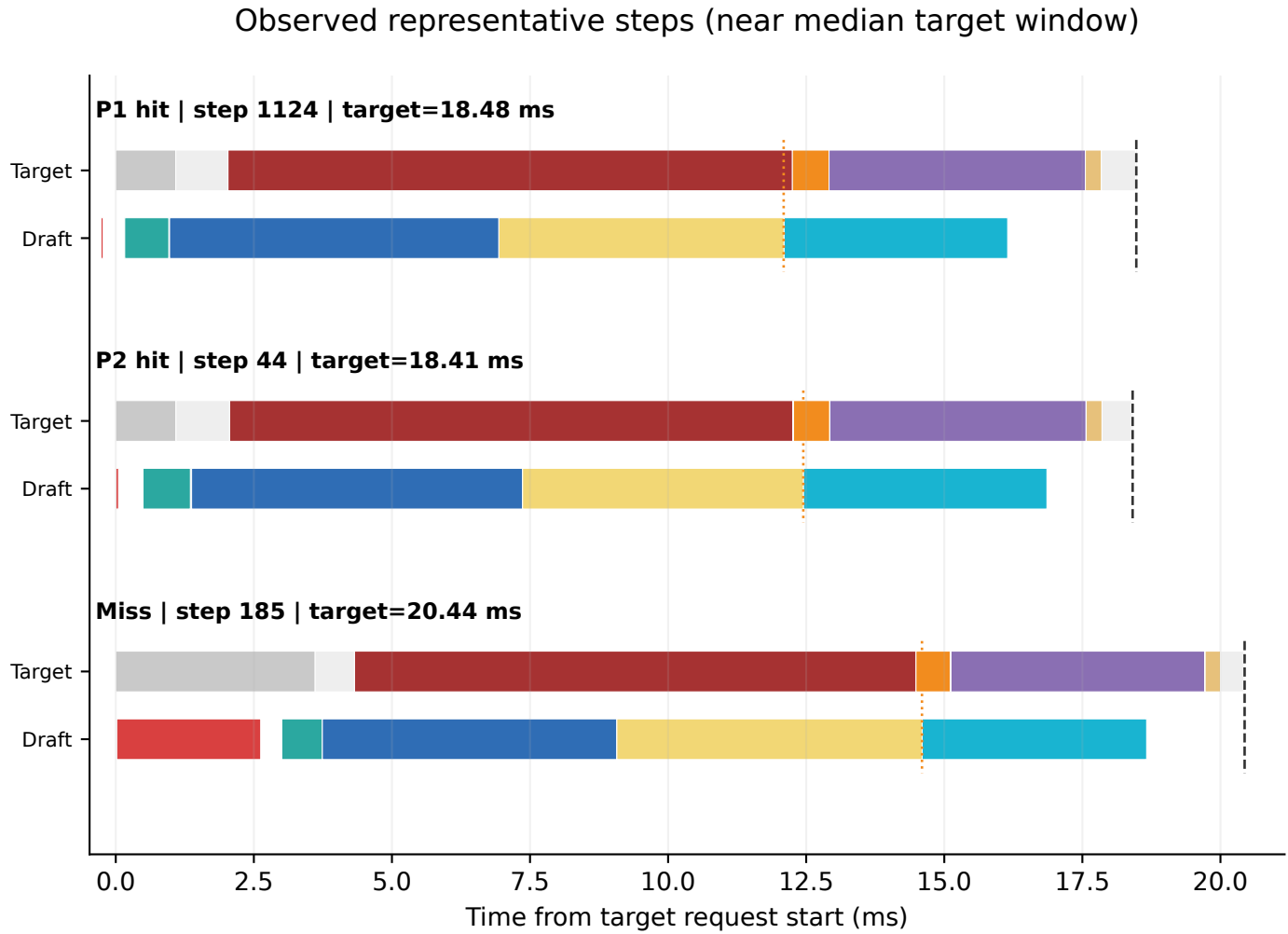
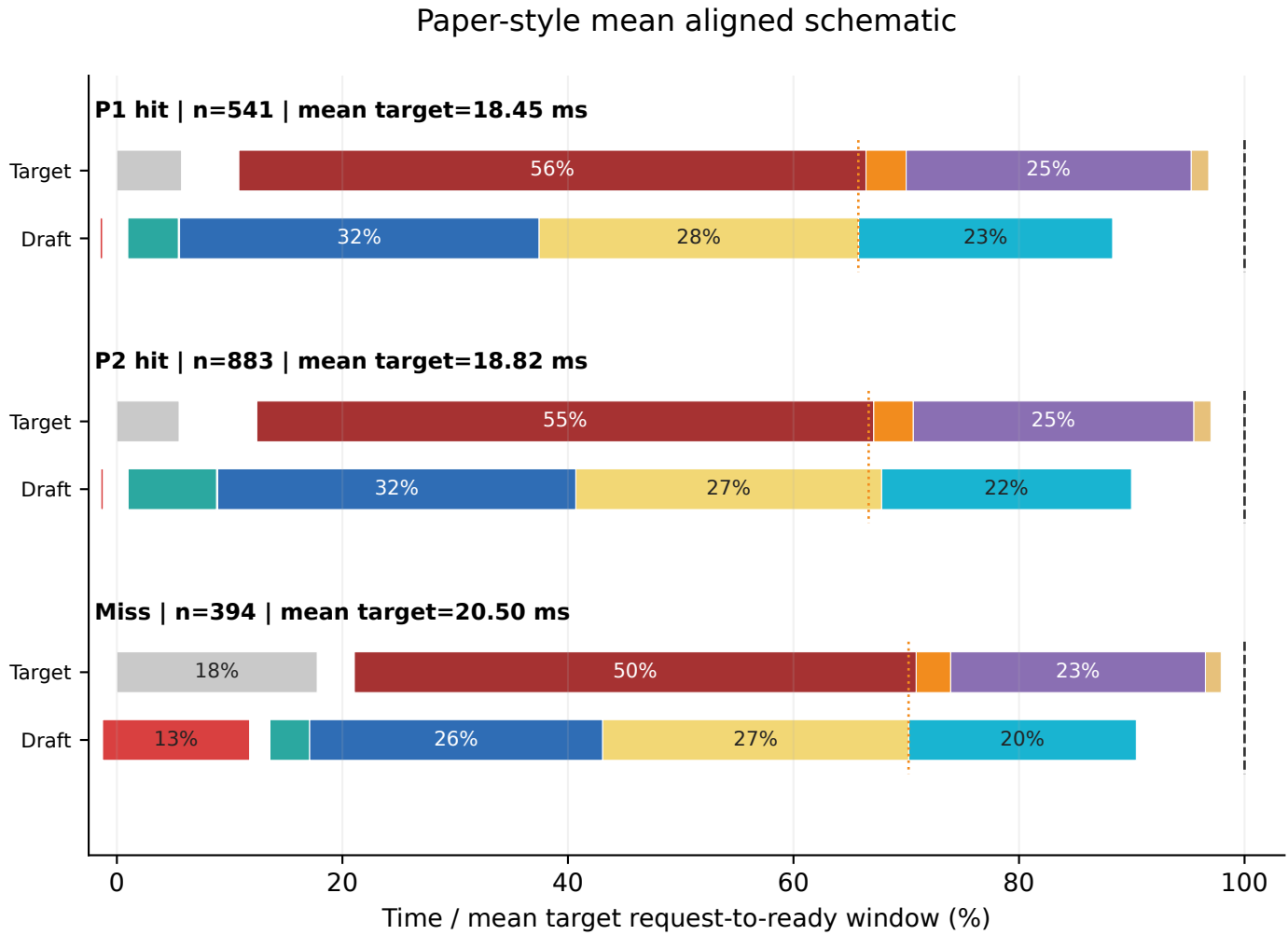
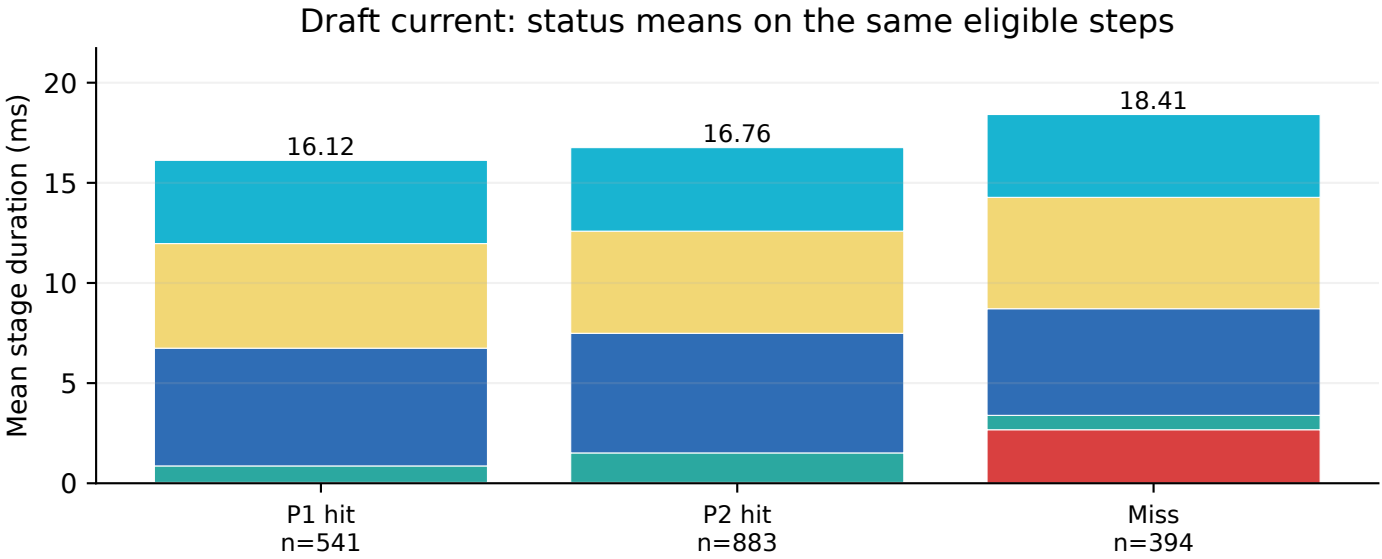
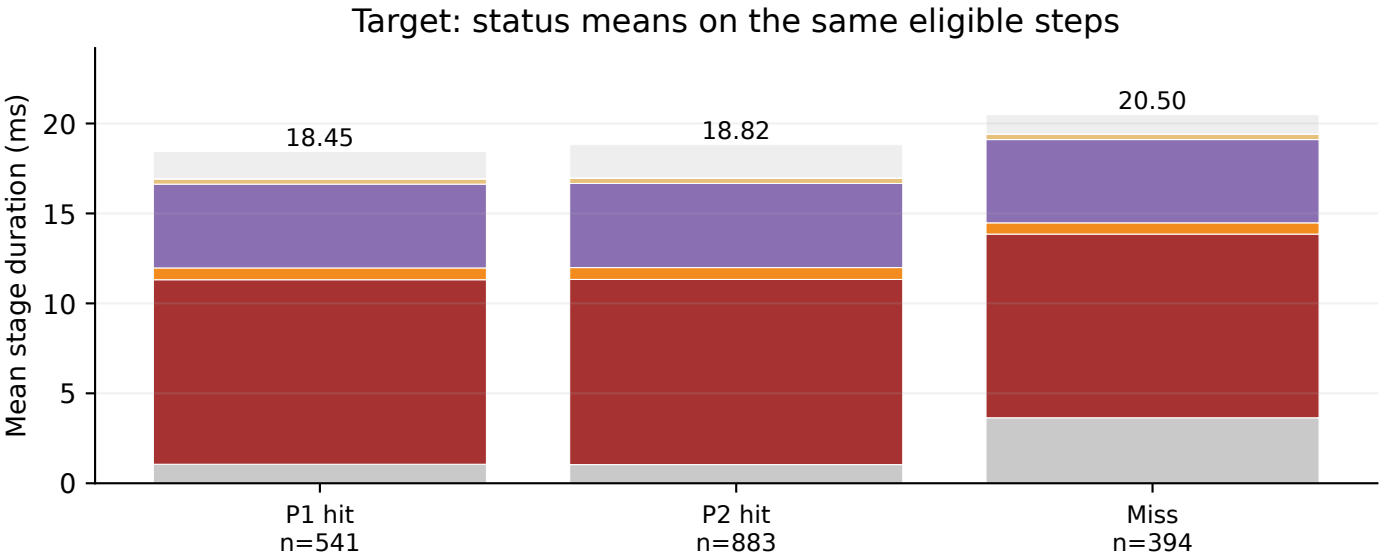


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

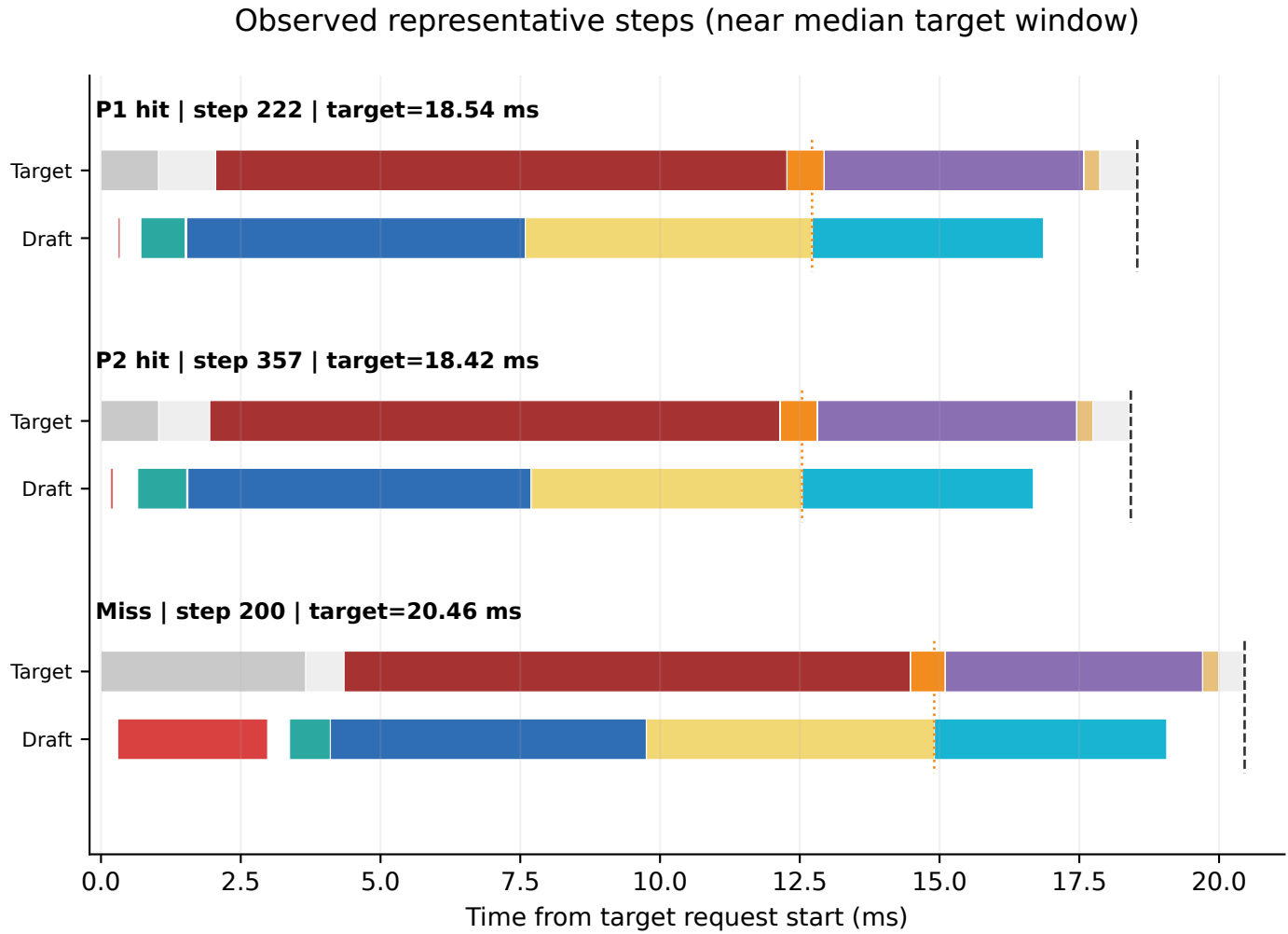
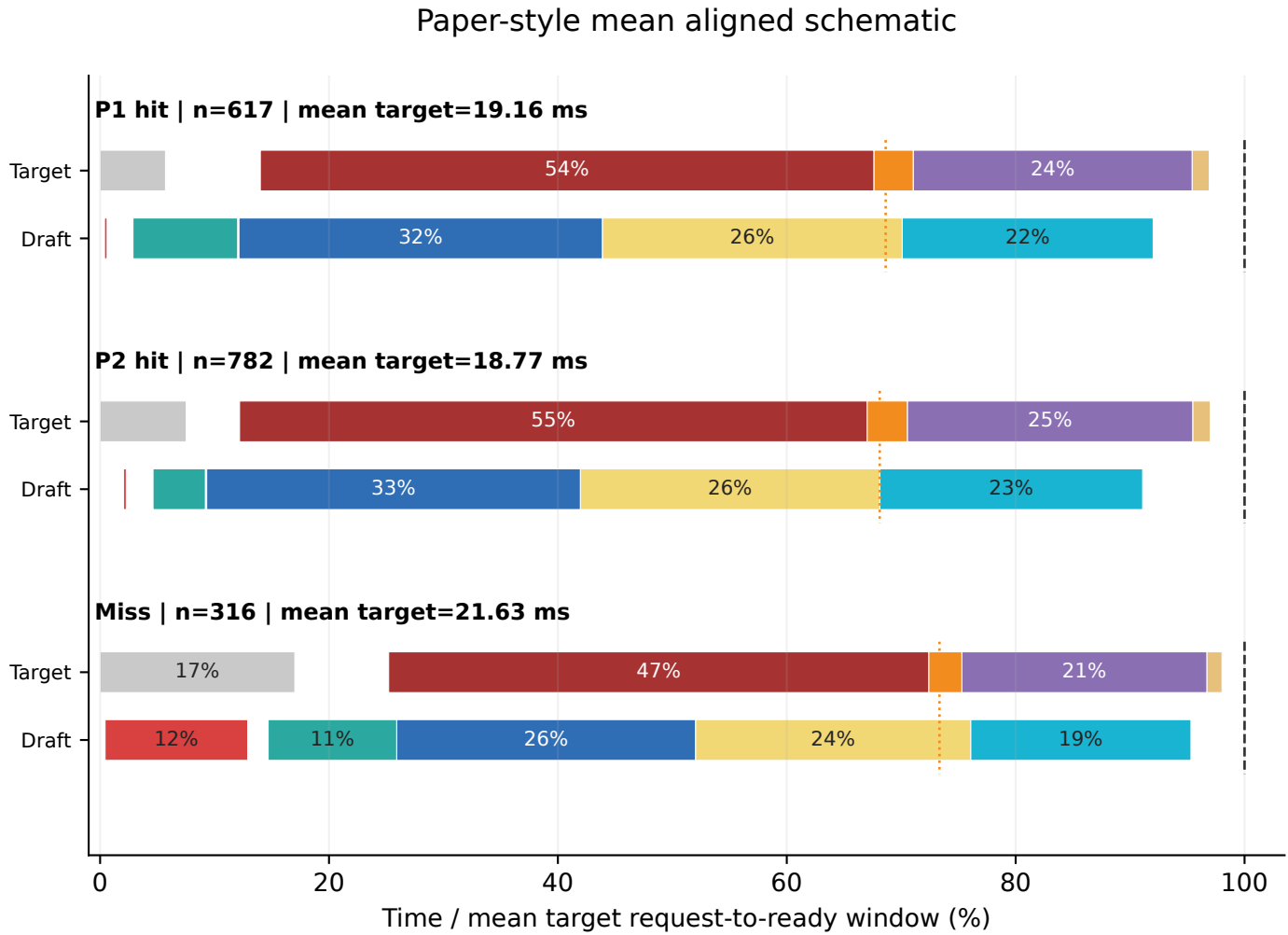
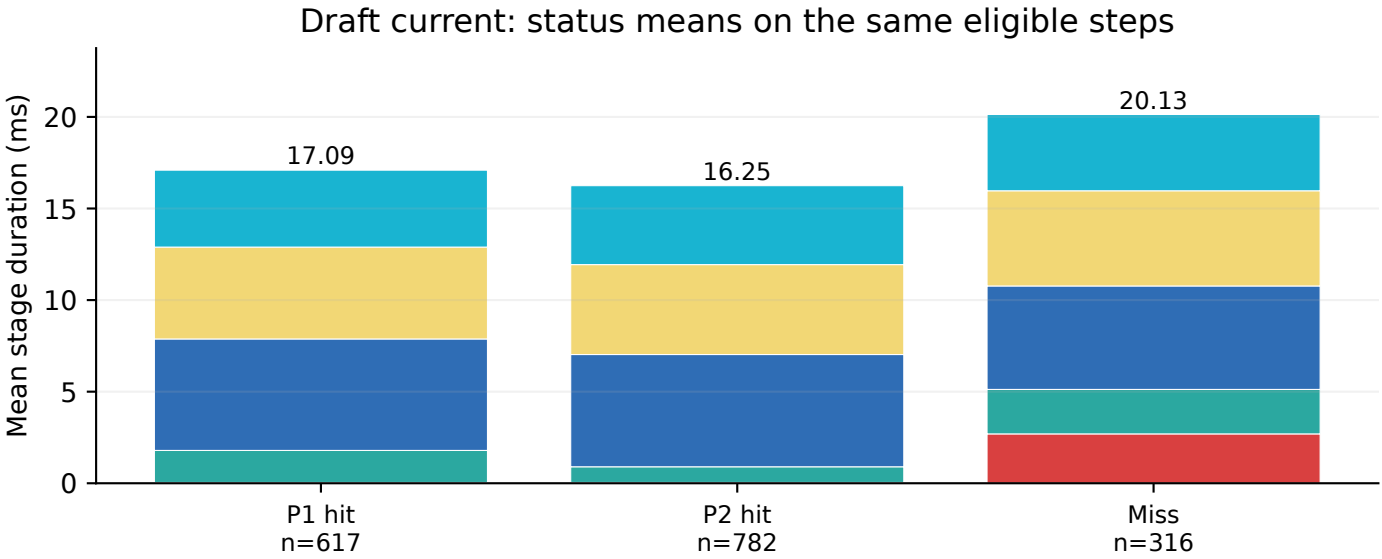
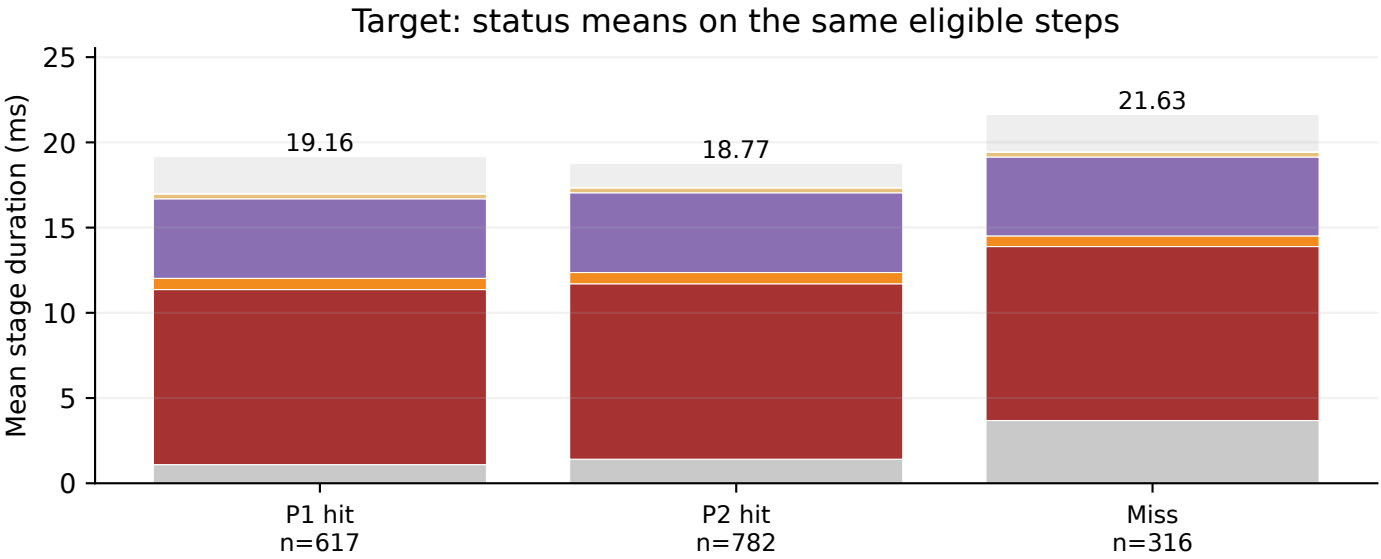
llama2_b1_tree_roots1_compute1
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=1, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1780 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



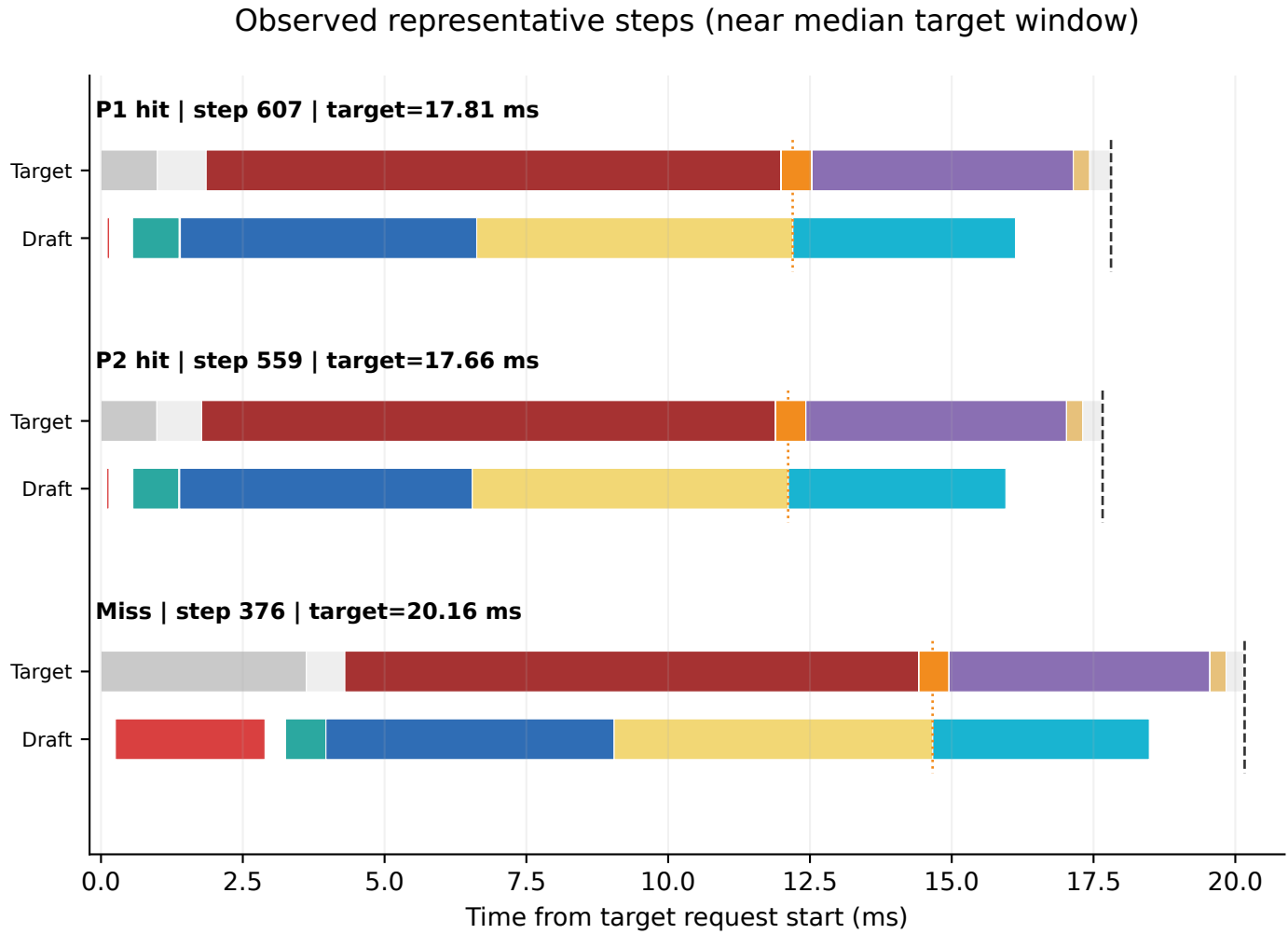
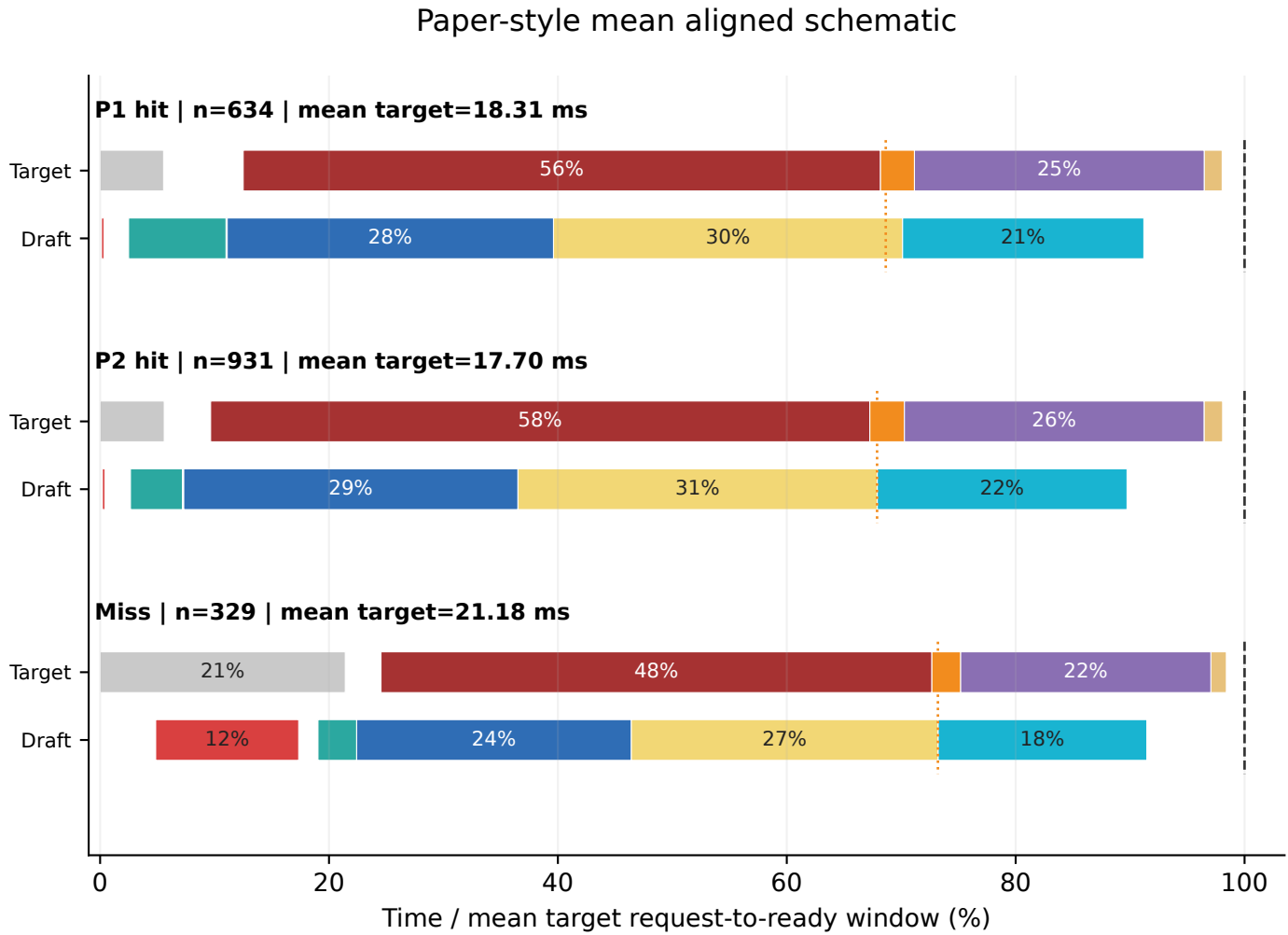
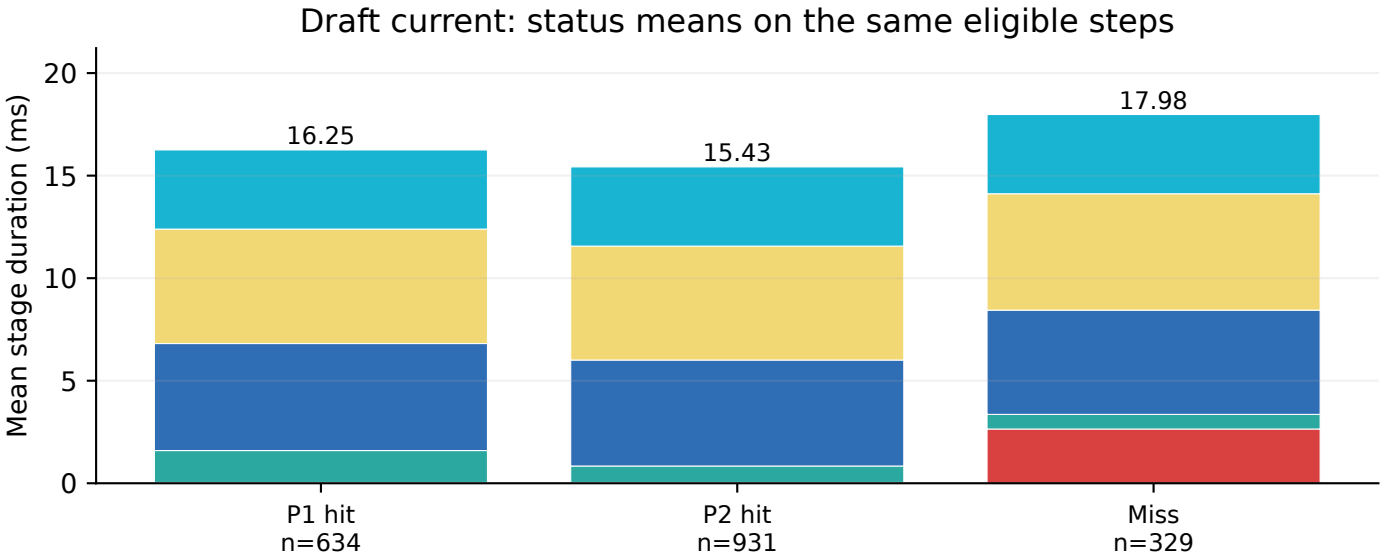
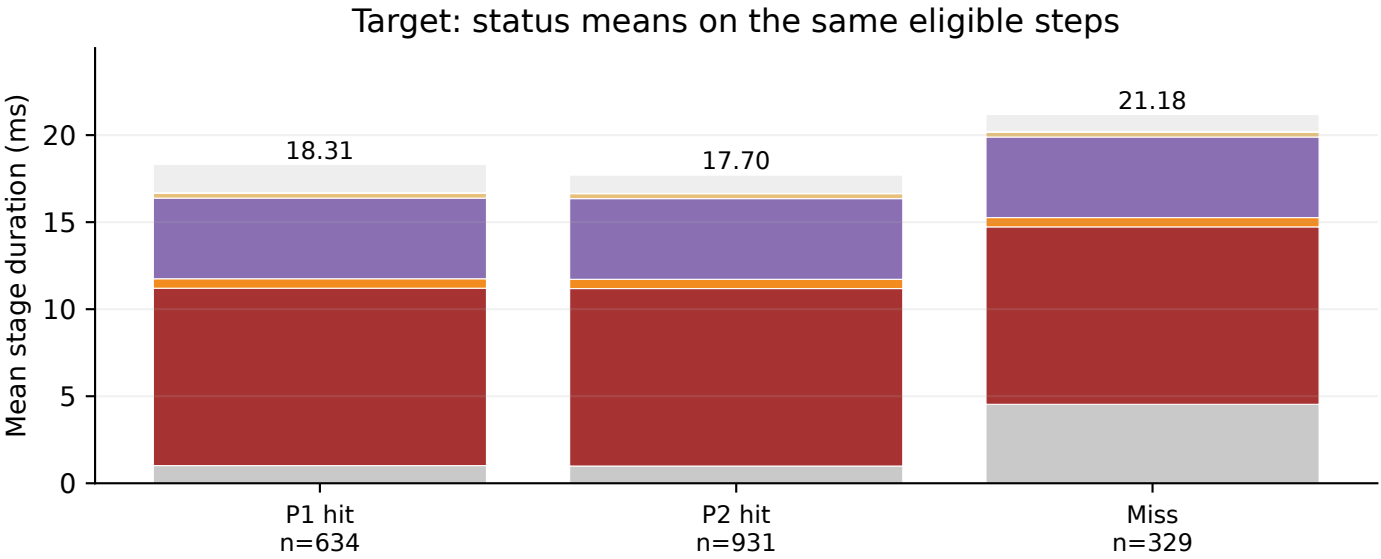
llama2_b1_tree_roots3
DUET: K1/K2=6/4, exit=21 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=3, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1715 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

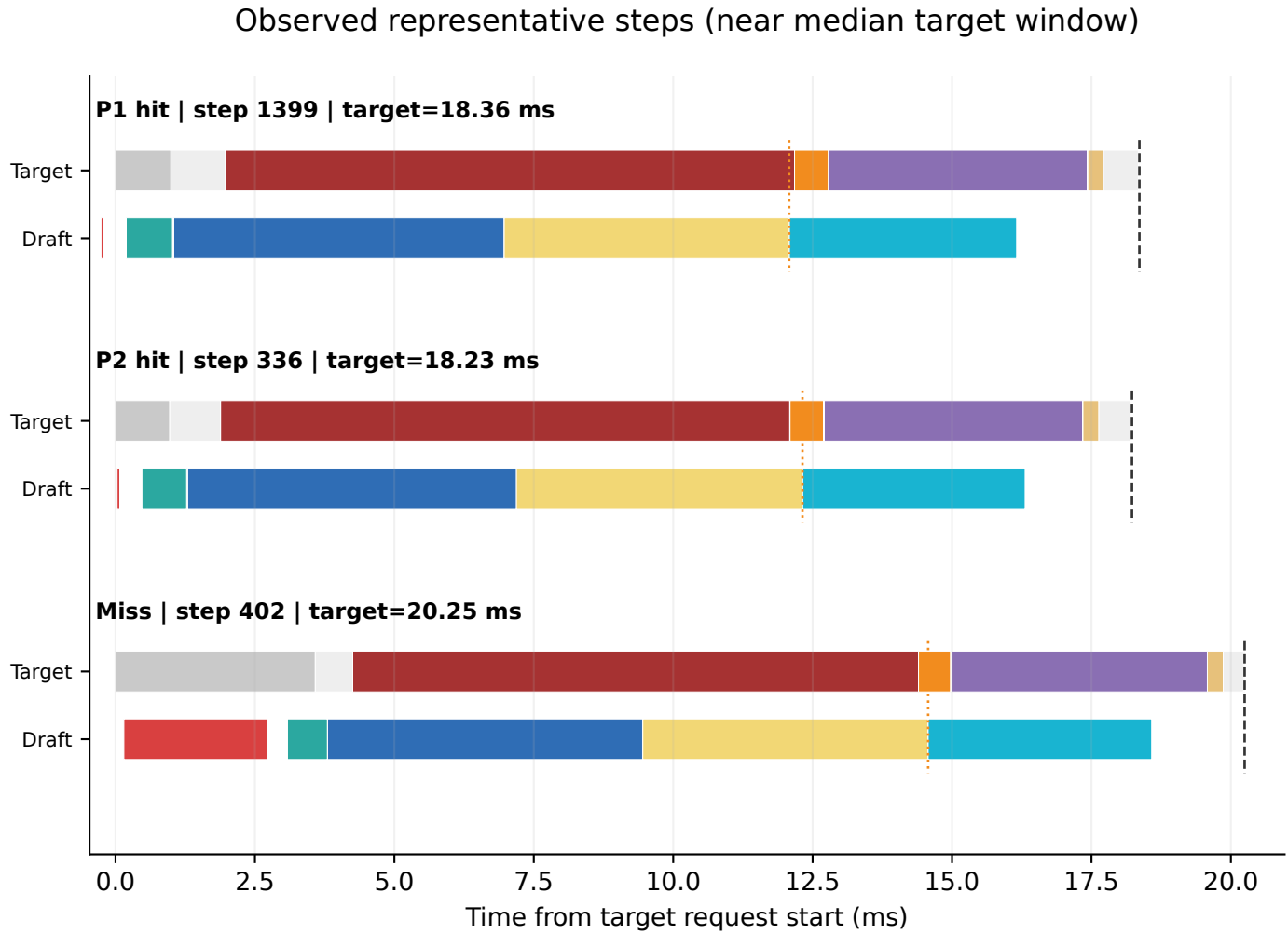
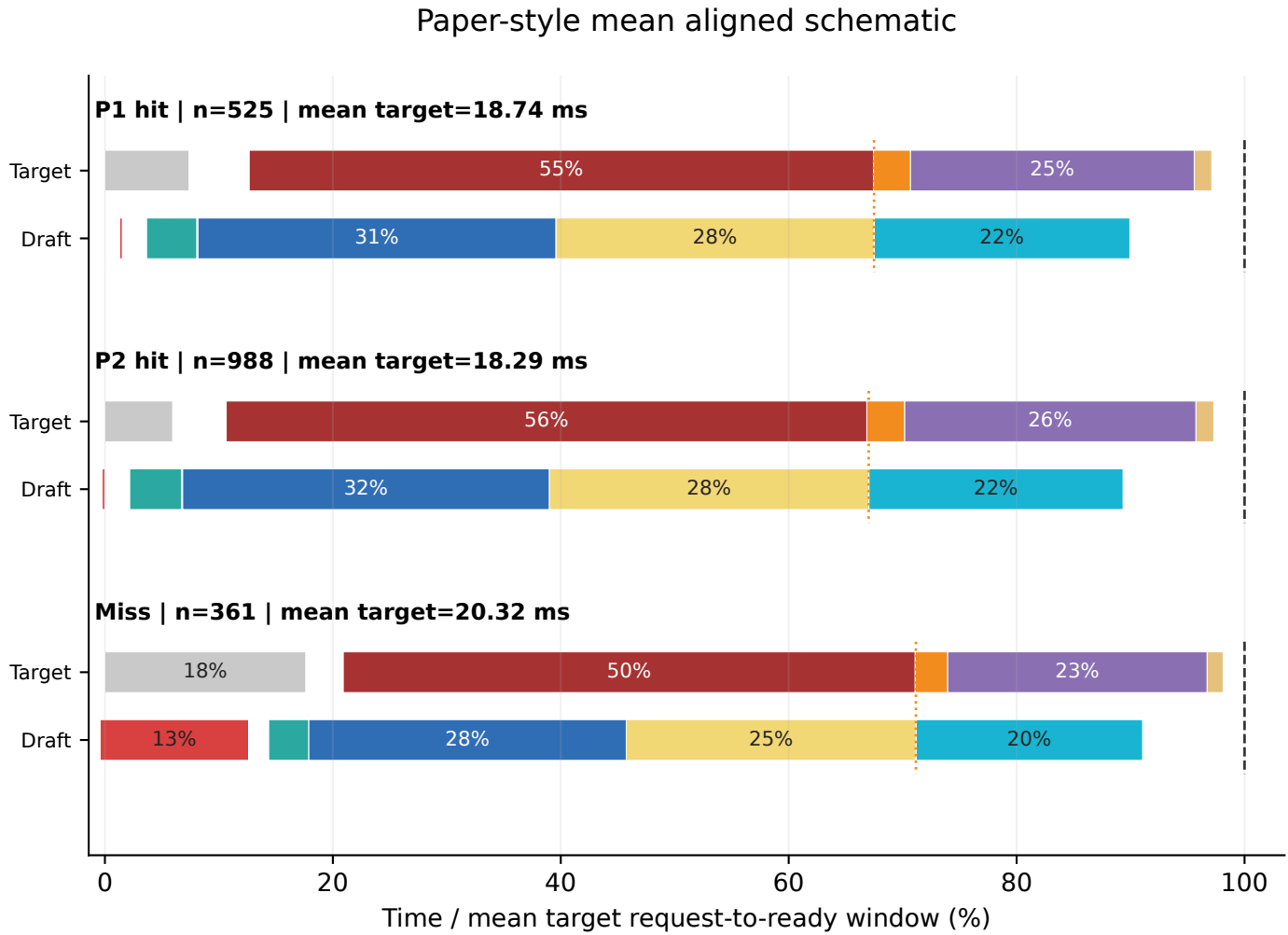
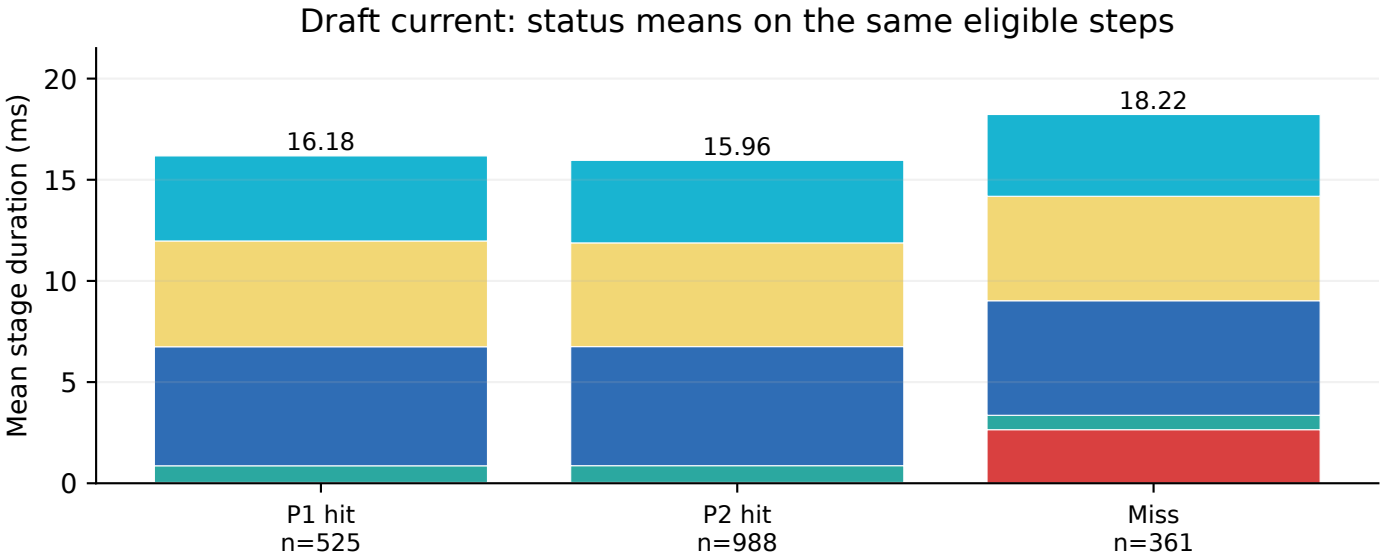
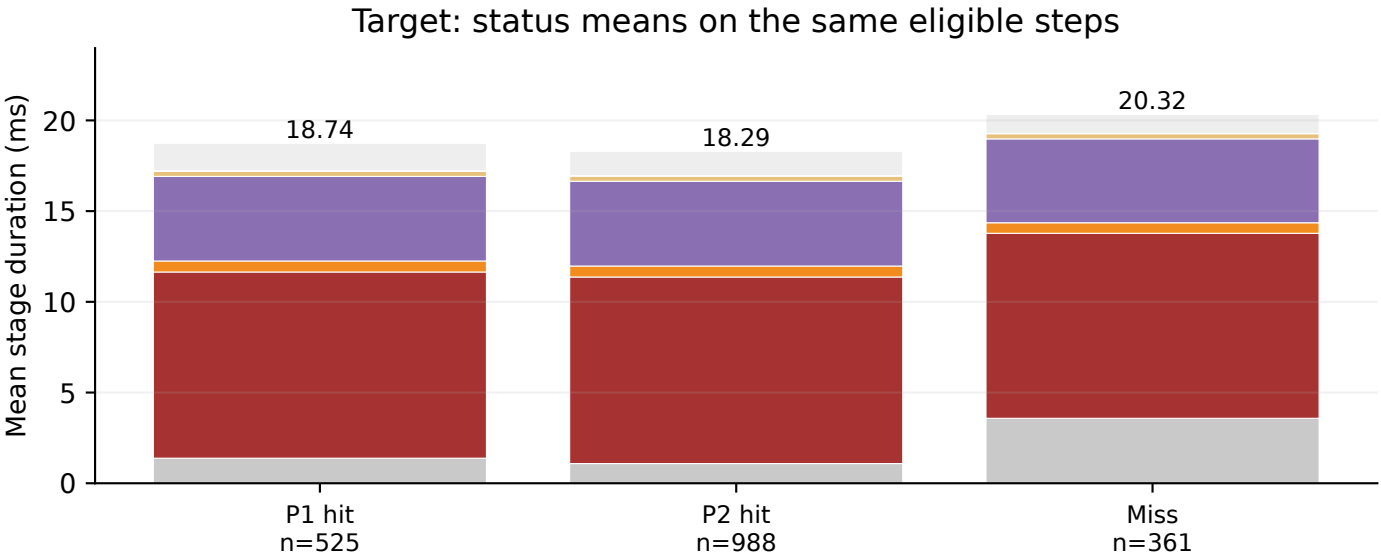
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_tree_width1
DUET: K1/K2=6/4, exit=21 (0-based), C=1, N1/N2=6/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1894 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

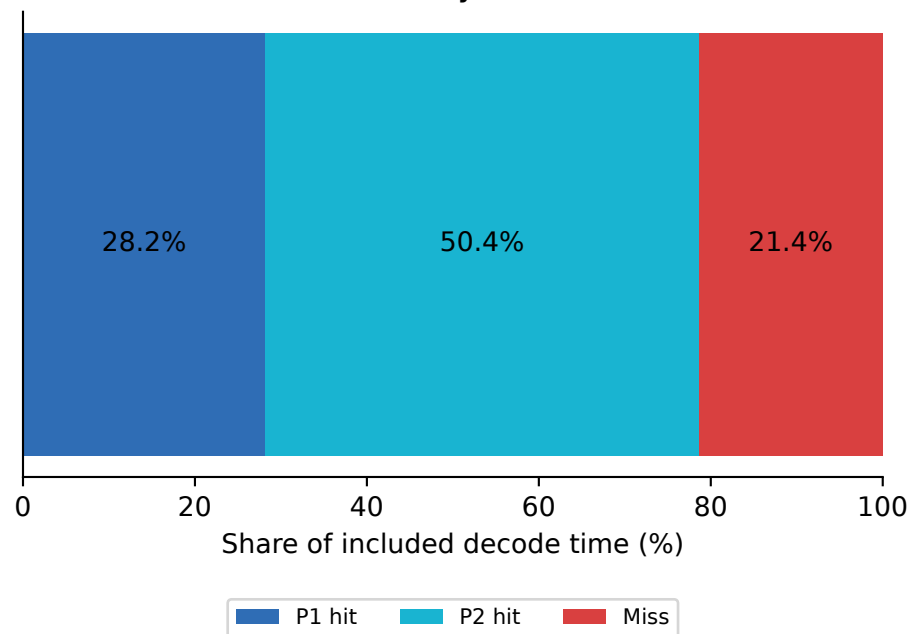
llama2_b1_tree_width2
DUET: K1/K2=6/4, exit=21 (0-based), $\bar{C}=2$, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1874 eligible steps



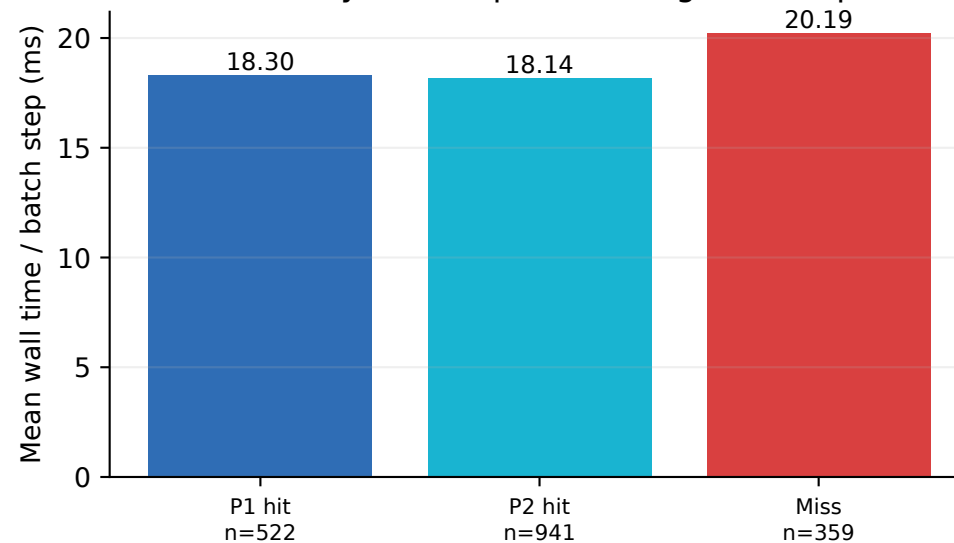
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_warm_duet_tree_1
DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=2.288, hit=80.4%, TPS*=123.07

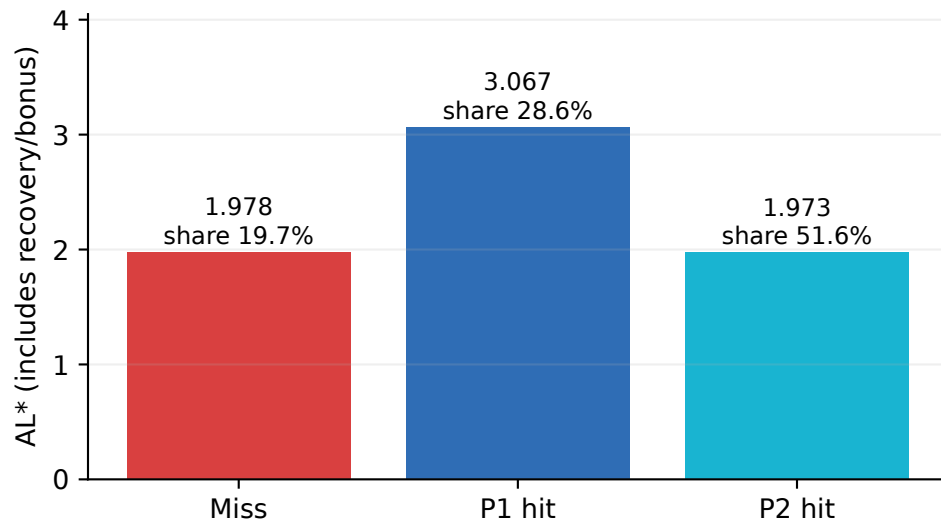
Measured wall time by batch cache status



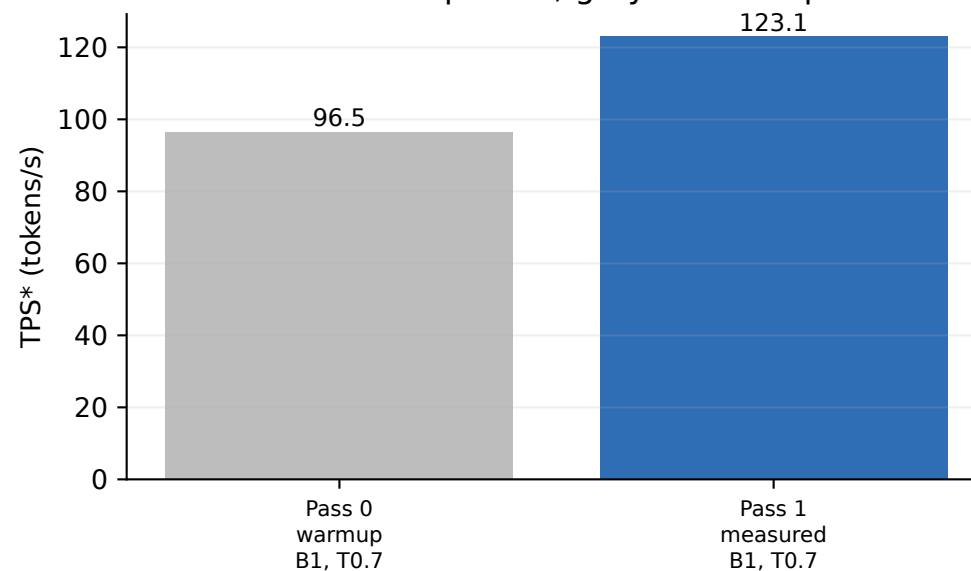
Boundary-free steps, including slow steps



Conditional AL* and source share



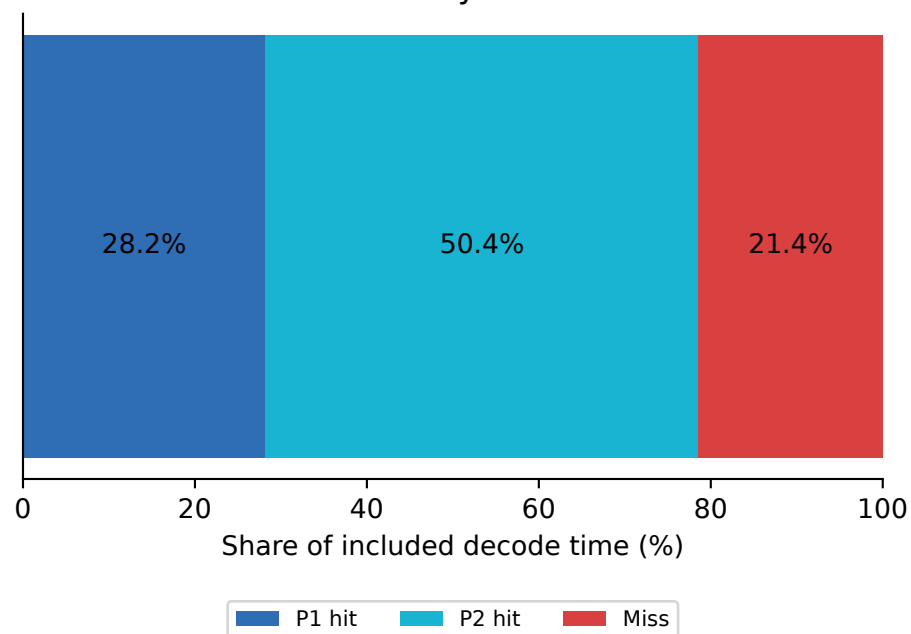
All saved passes; gray = warmup



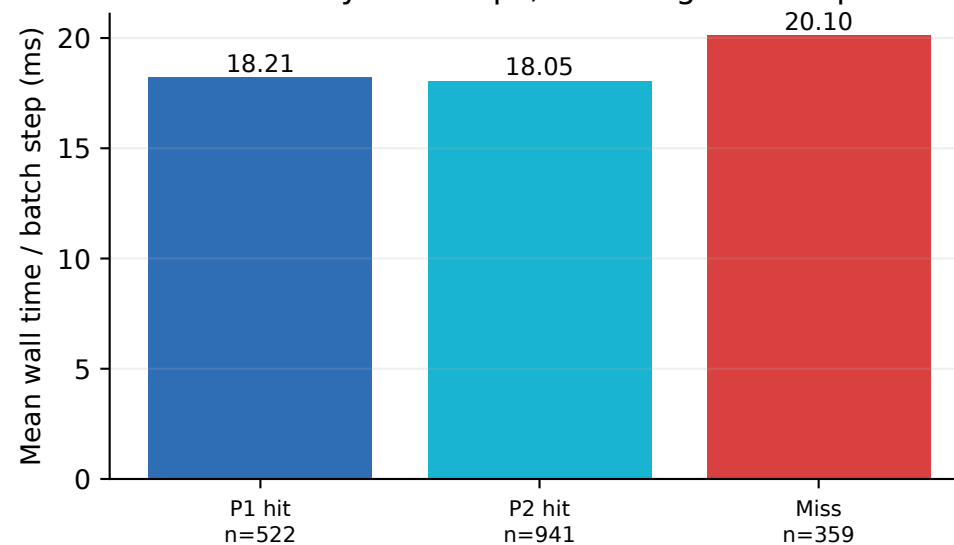
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_warm_duet_tree_2
 DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
 Last pass: B=1, T=0.7, N=48 | AL*=2.288, hit=80.4%, TPS*=123.66

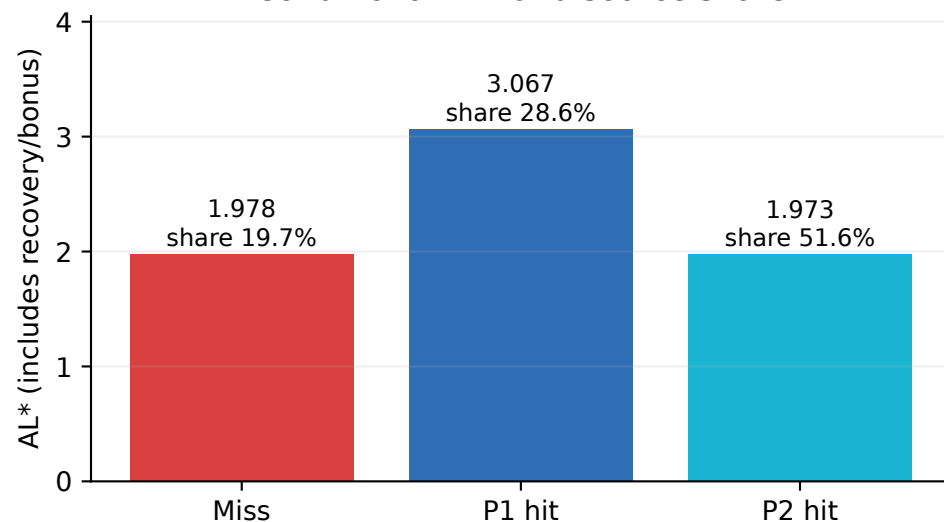
Measured wall time by batch cache status



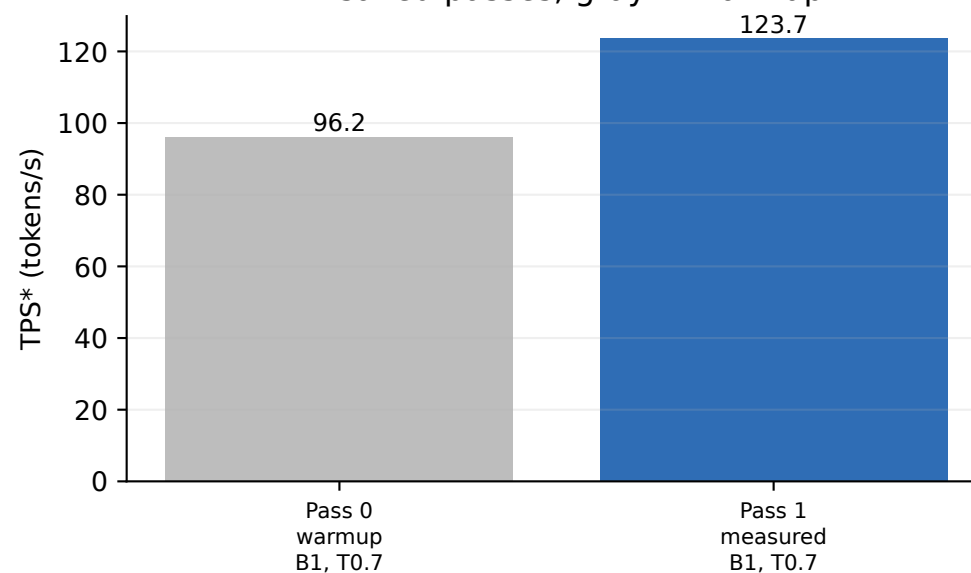
Boundary-free steps, including slow steps



Conditional AL* and source share

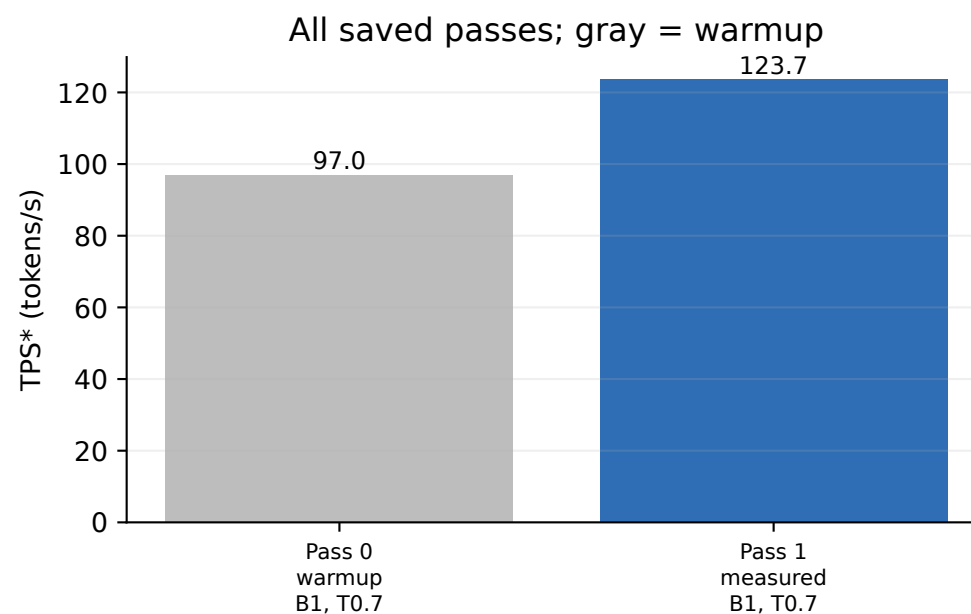
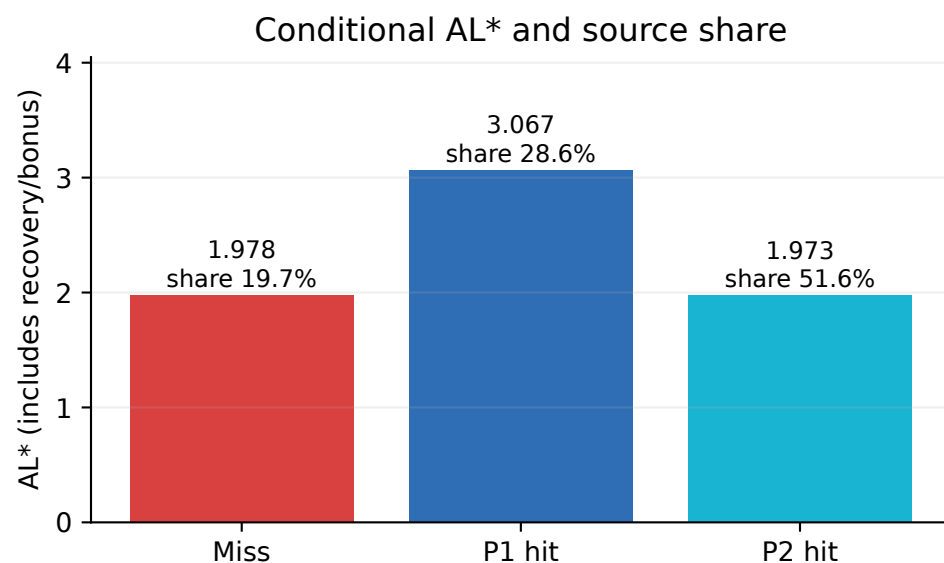
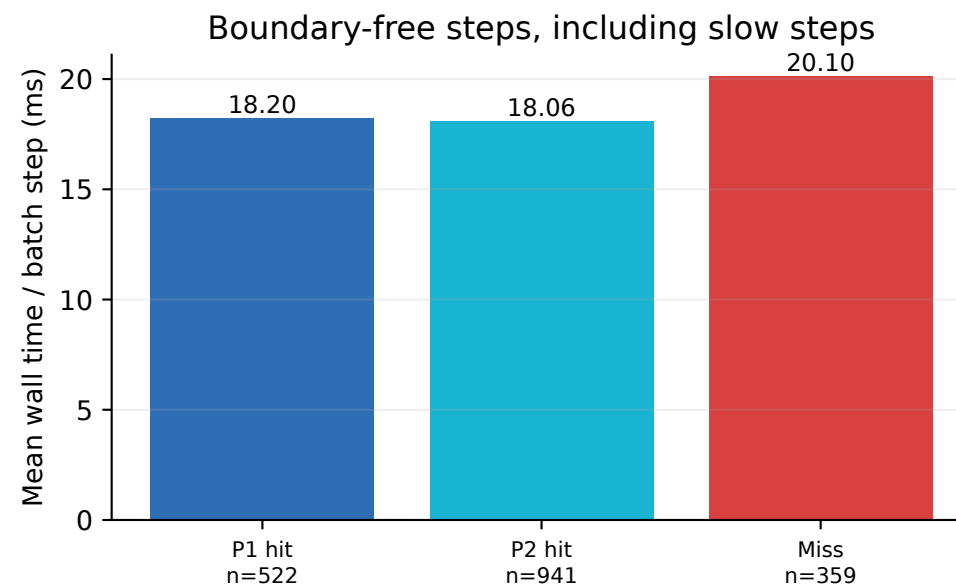
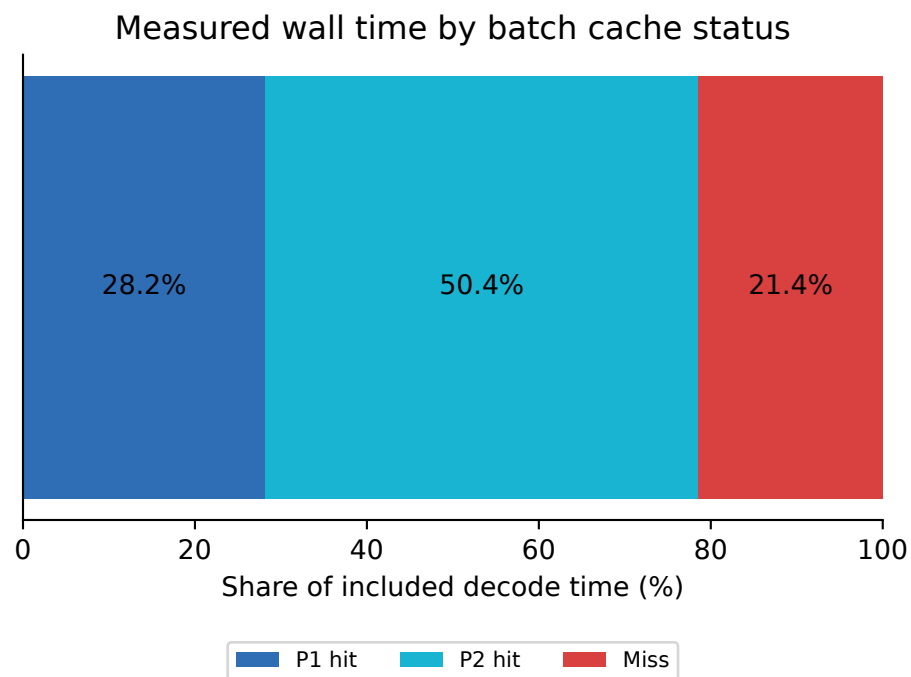


All saved passes; gray = warmup



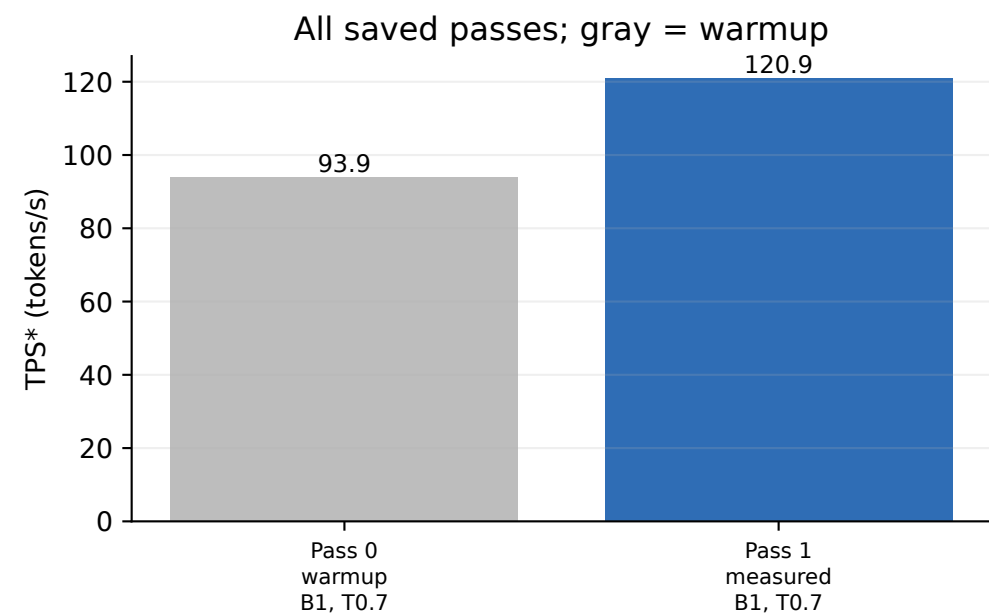
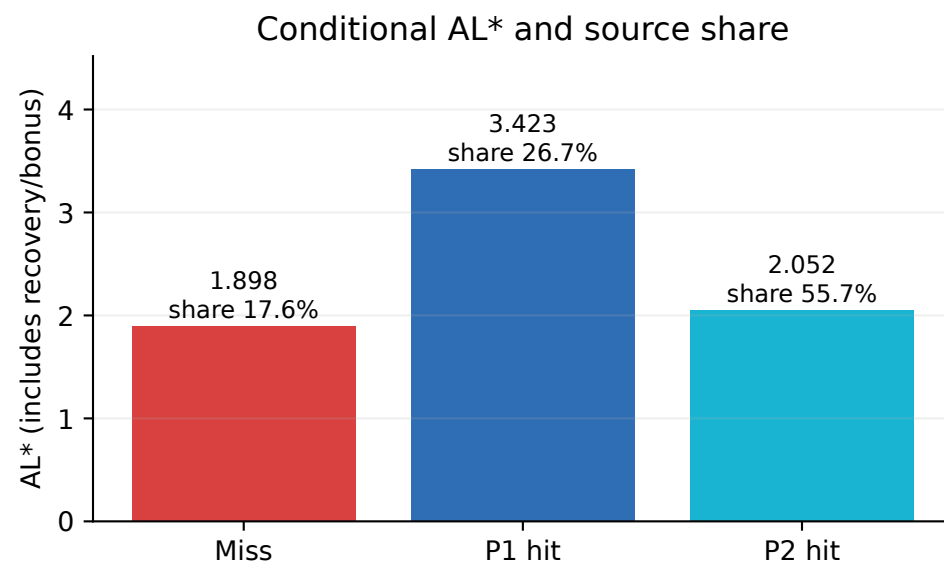
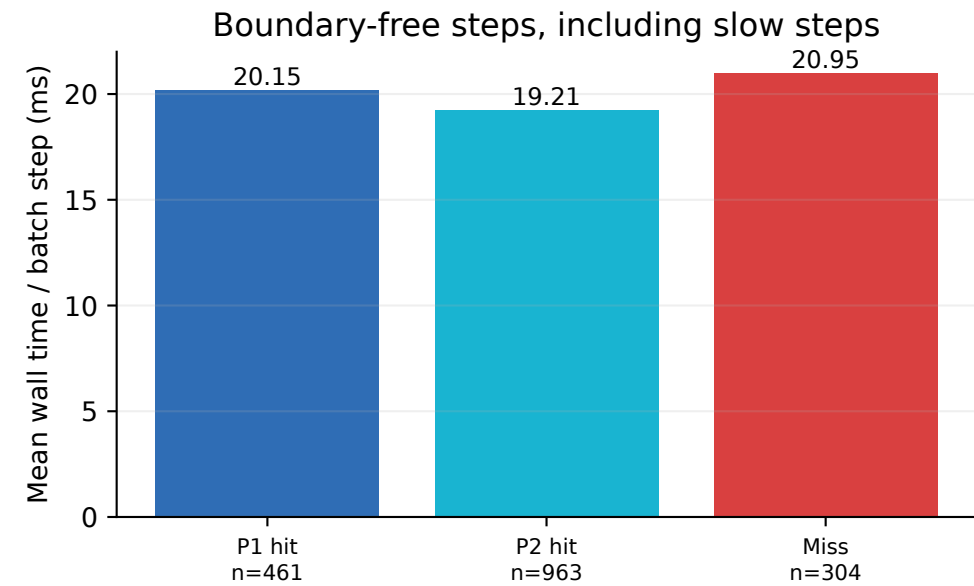
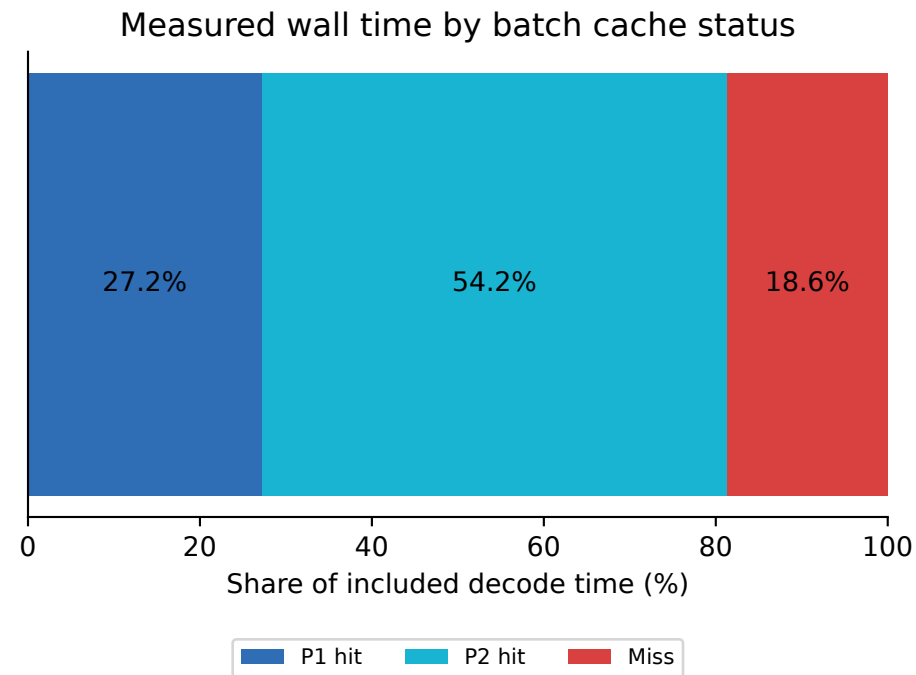
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_warm_duet_tree_3
 DUET: K1/K2=6/4, exit=21 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
 Last pass: B=1, T=0.7, N=48 | AL*=2.288, hit=80.4%, TPS*=123.67



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

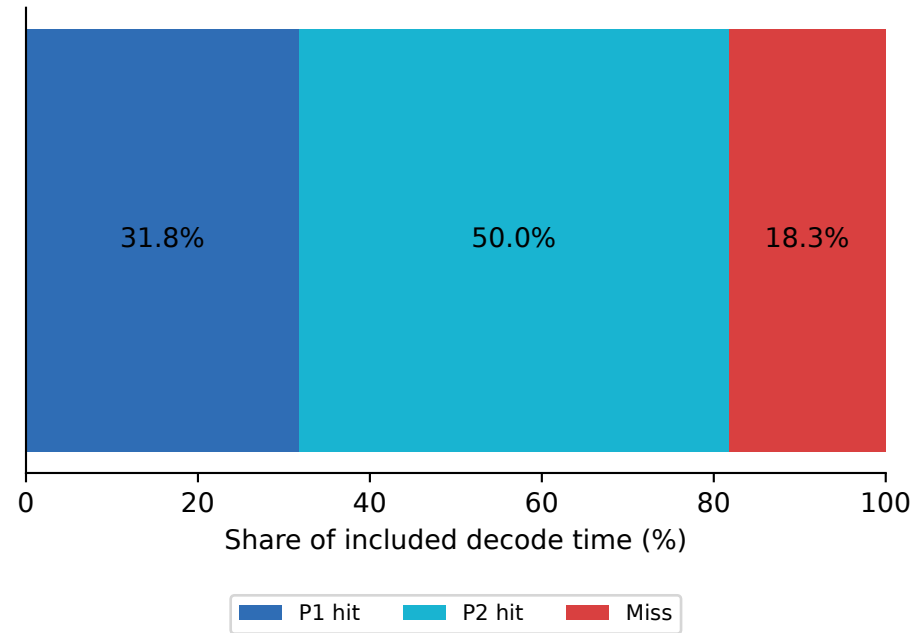
llama2_b1_warm_duet_tree_4
DUET: K1/K2=12/4, exit=21 (0-based), C=3, N1/N2=16/8, U=2, W=13, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=2.391, hit=82.7%, TPS*=120.95



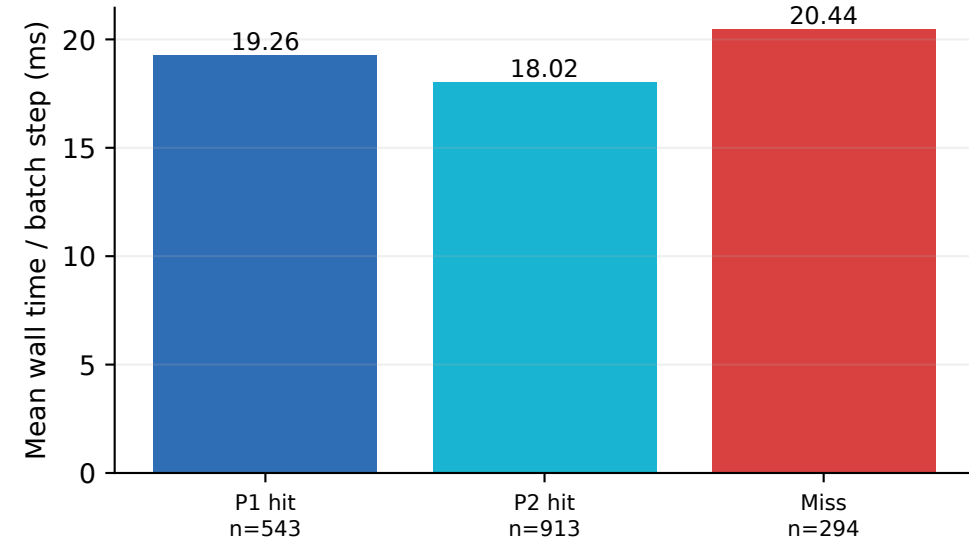
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_warm_duet_tree_5
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=2.397, hit=83.4%, TPS*=127.41

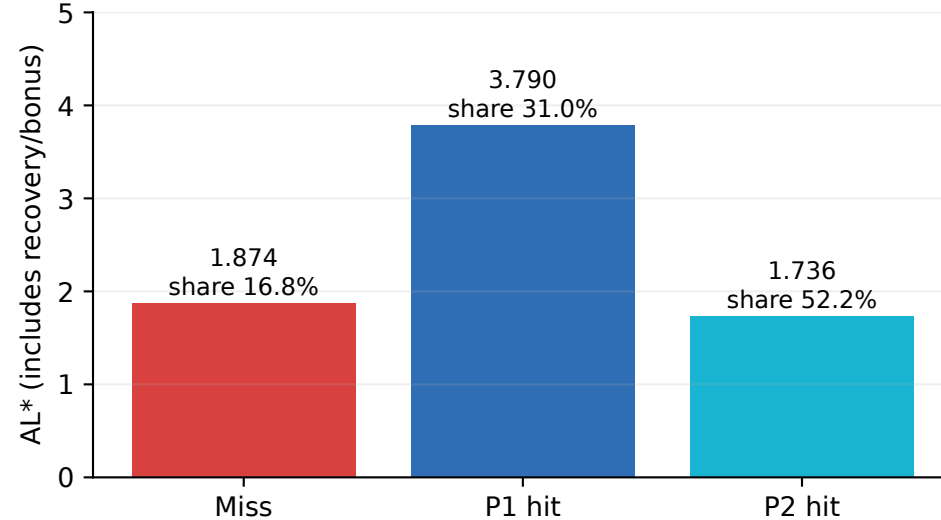
Measured wall time by batch cache status



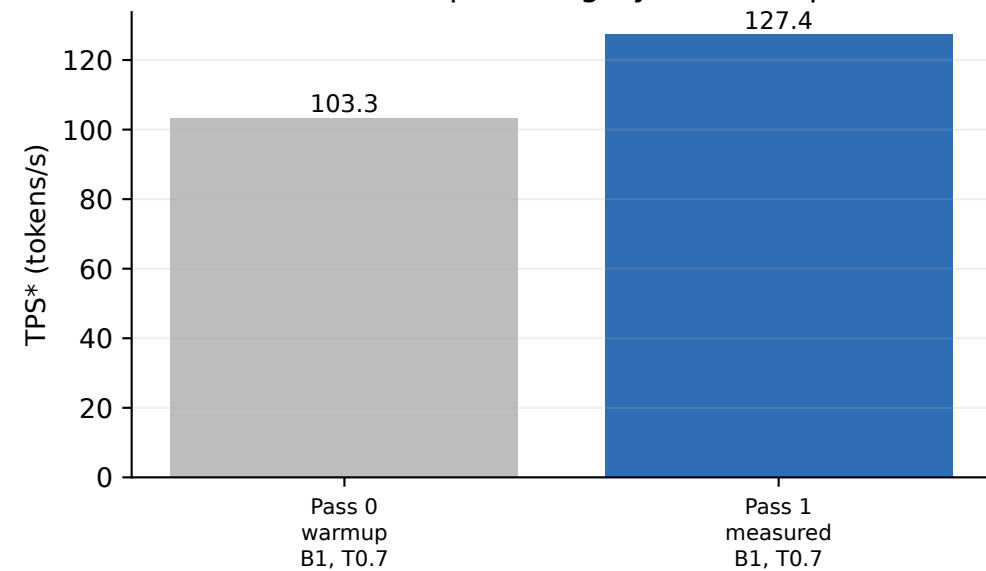
Boundary-free steps, including slow steps



Conditional AL* and source share



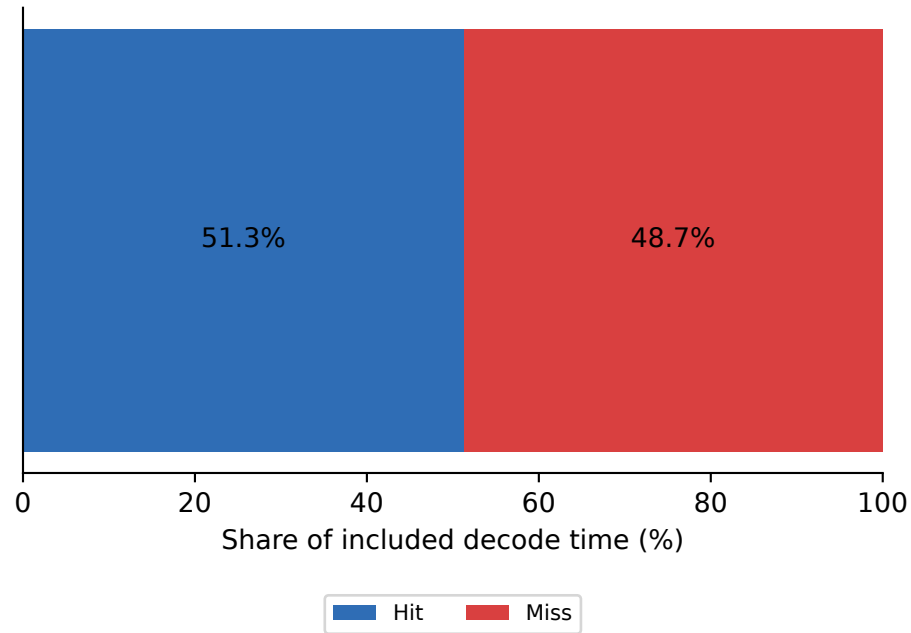
All saved passes; gray = warmup



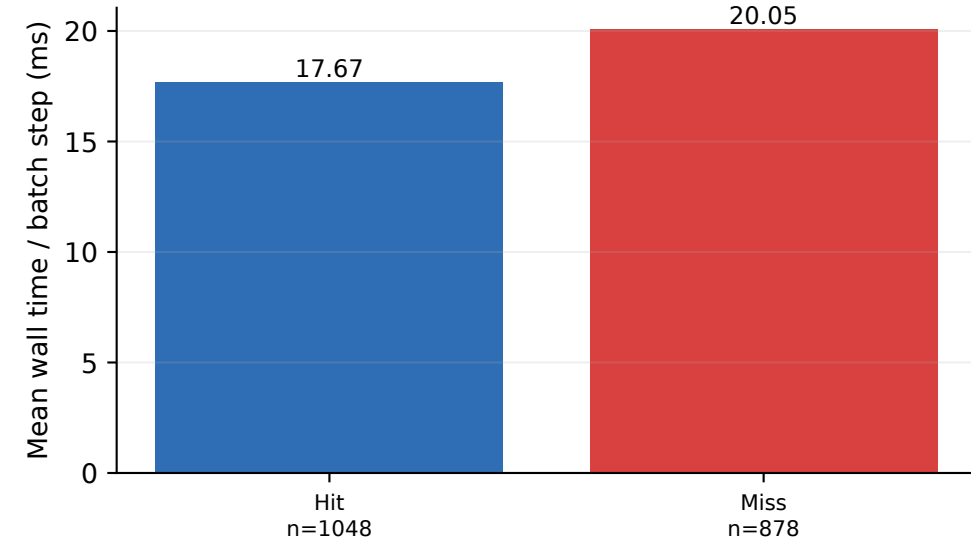
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_warm_ssd_1
SSD: K=4, F=7
Last pass: B=1, T=0.7, N=48 | AL*=2.184, hit=54.5%, TPS*=116.46

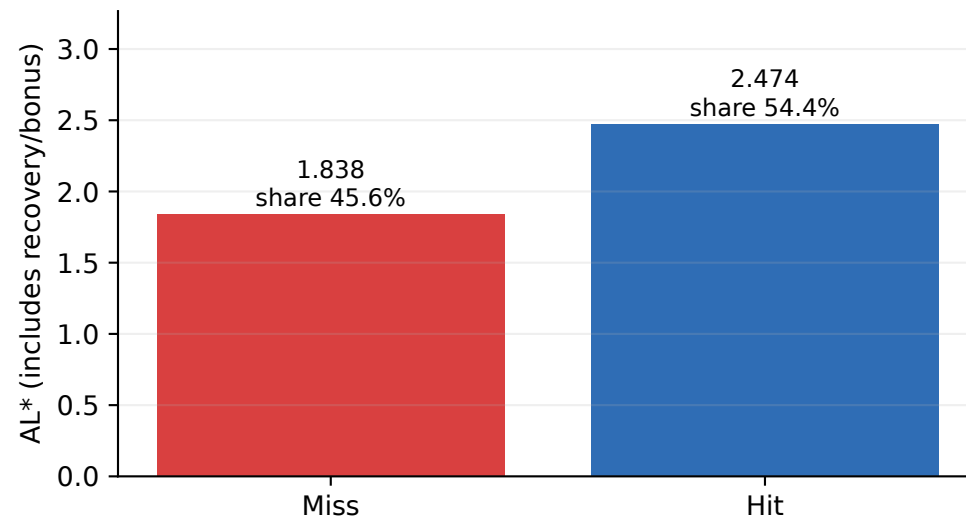
Measured wall time by batch cache status



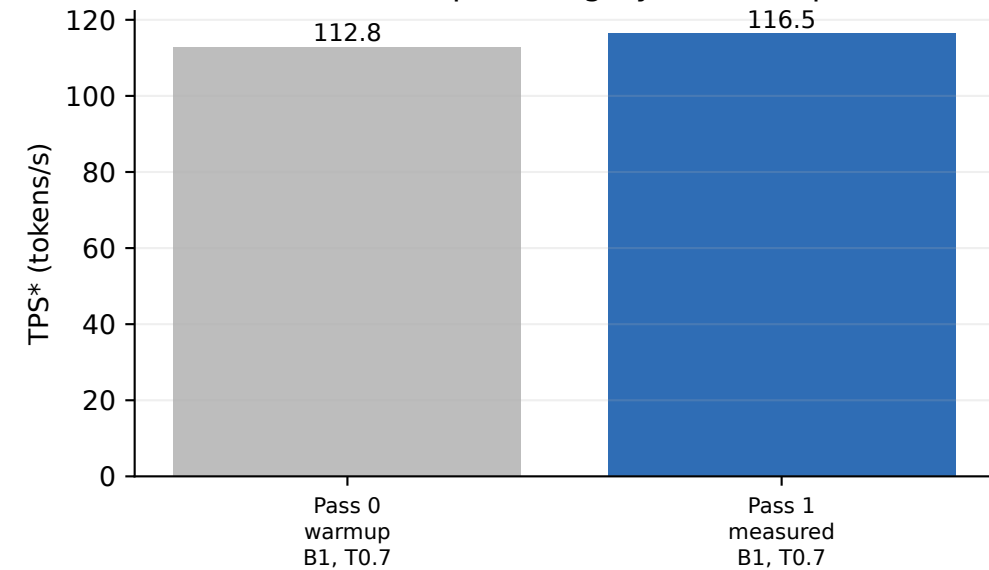
Boundary-free steps, including slow steps



Conditional AL* and source share

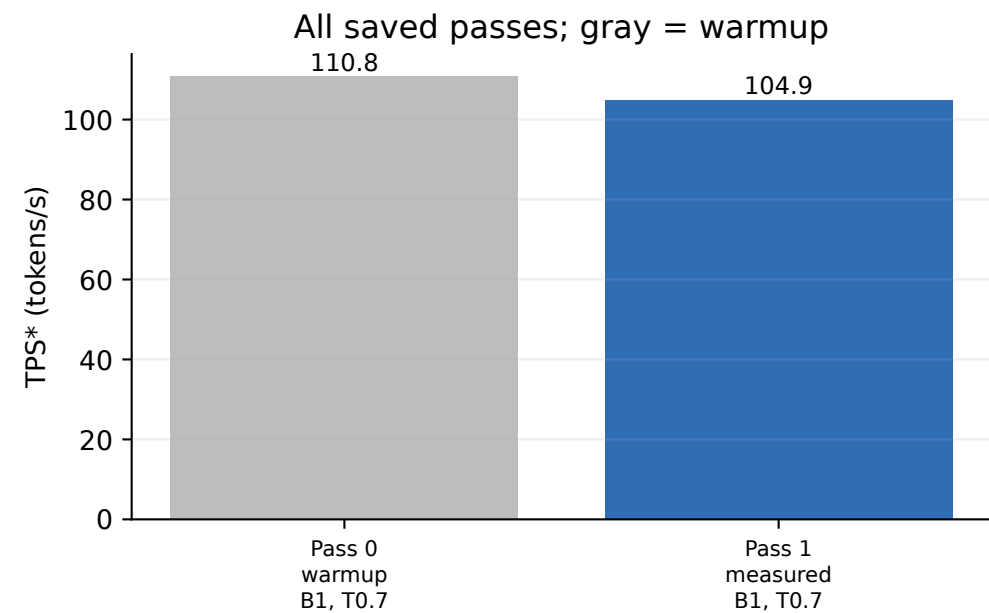
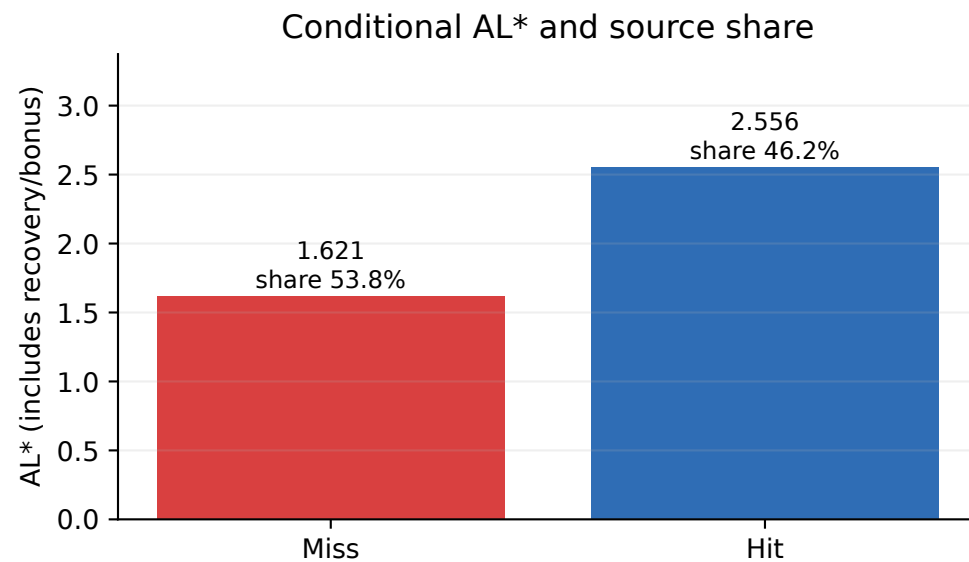
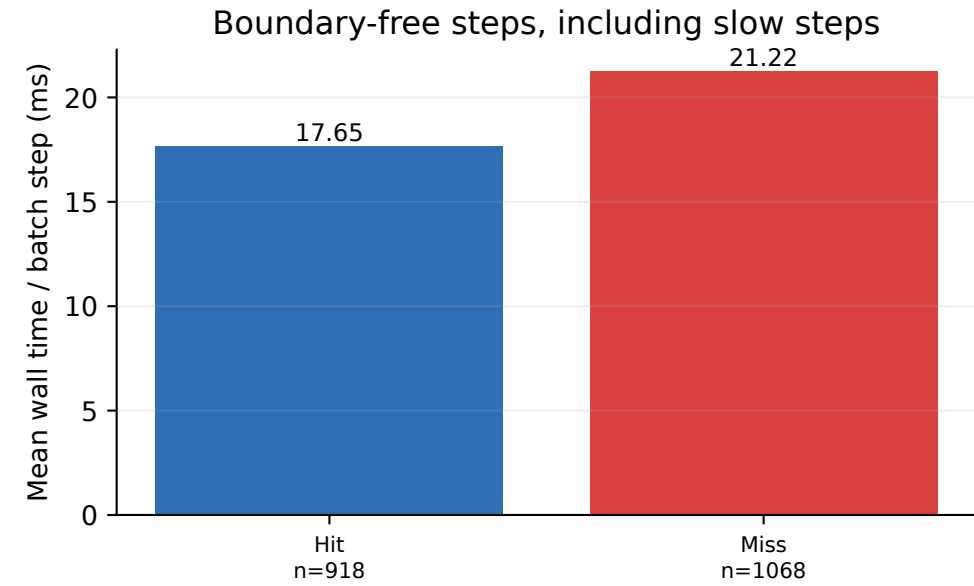
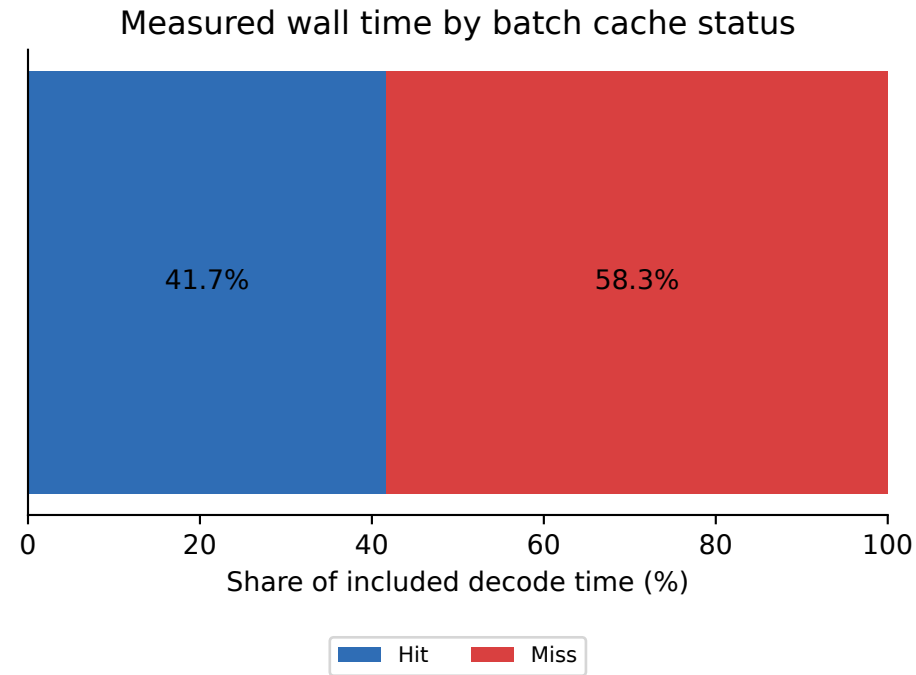


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

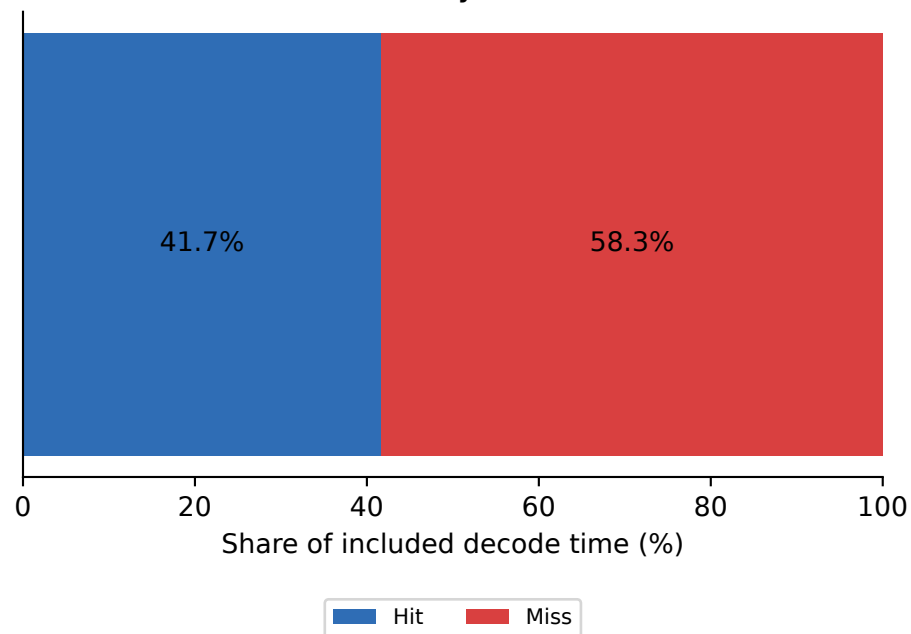
llama2_b1_warm_ssd_2
SSD: K=6, F=7
Last pass: B=1, T=0.7, N=48 | AL*=2.053, hit=46.6%, TPS*=104.89



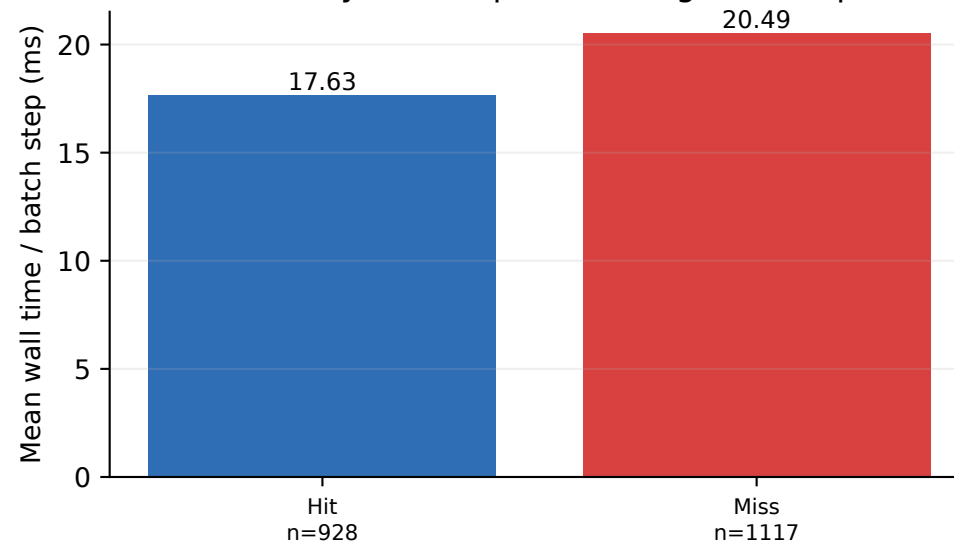
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_warm_ssd_3
SSD: K=5, F=5
Last pass: B=1, T=0.7, N=48 | AL*=2.113, hit=45.7%, TPS*=110.10

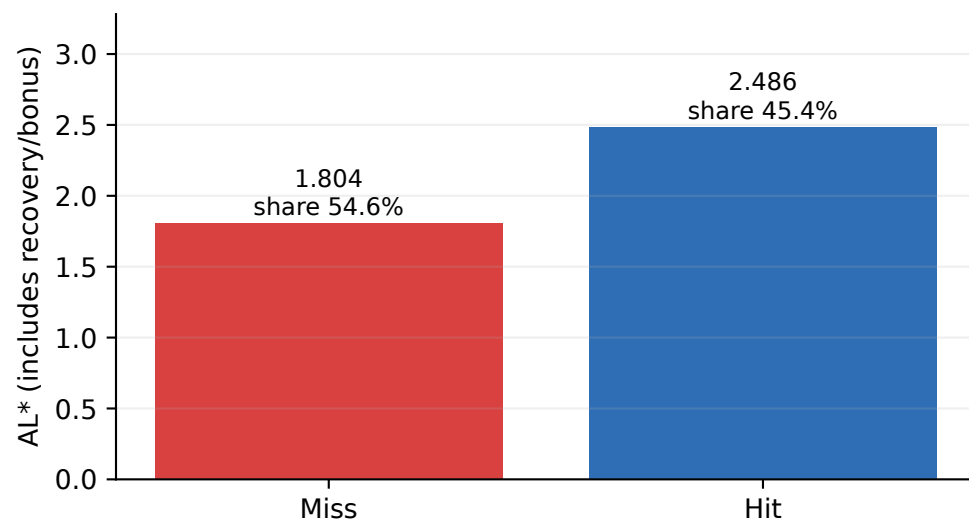
Measured wall time by batch cache status



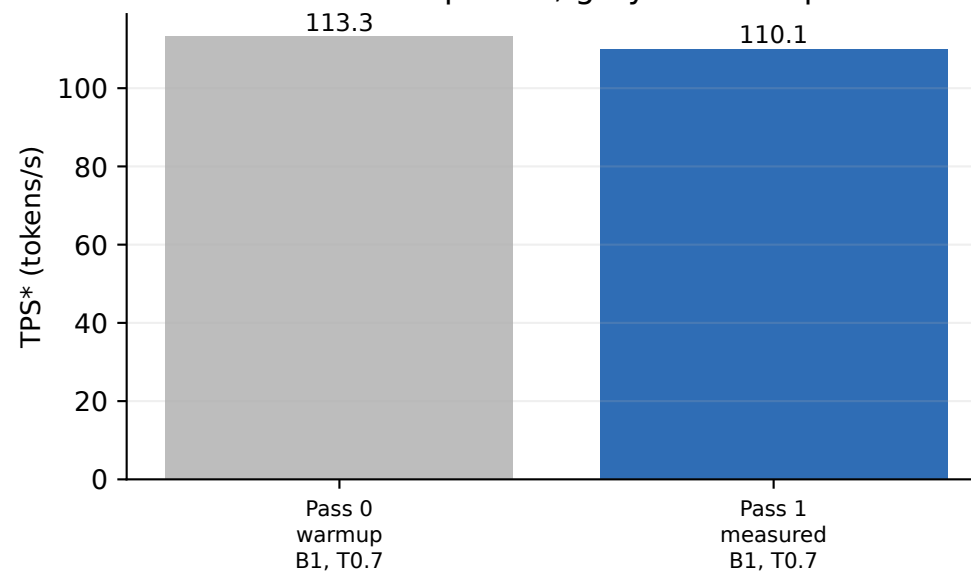
Boundary-free steps, including slow steps



Conditional AL* and source share

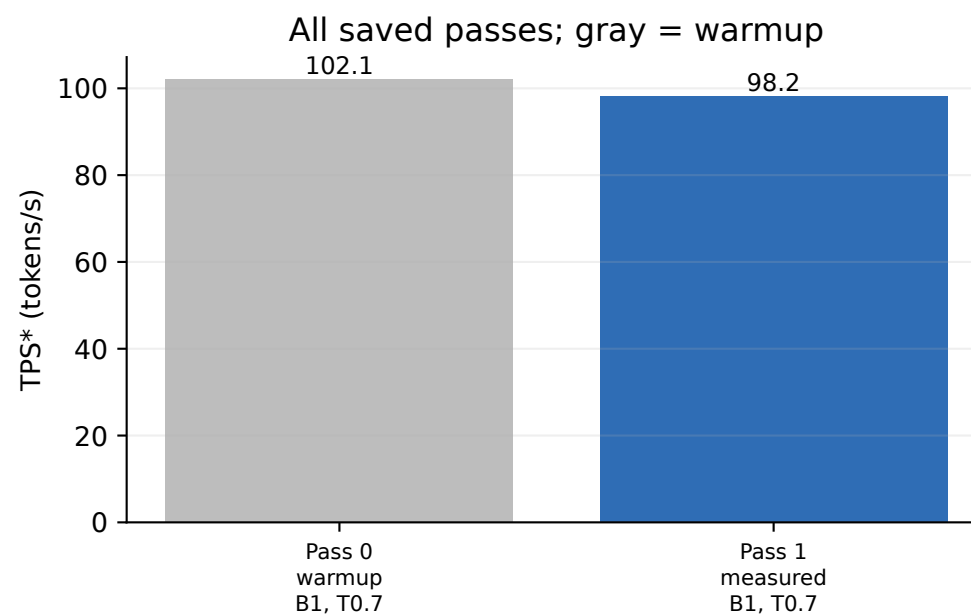
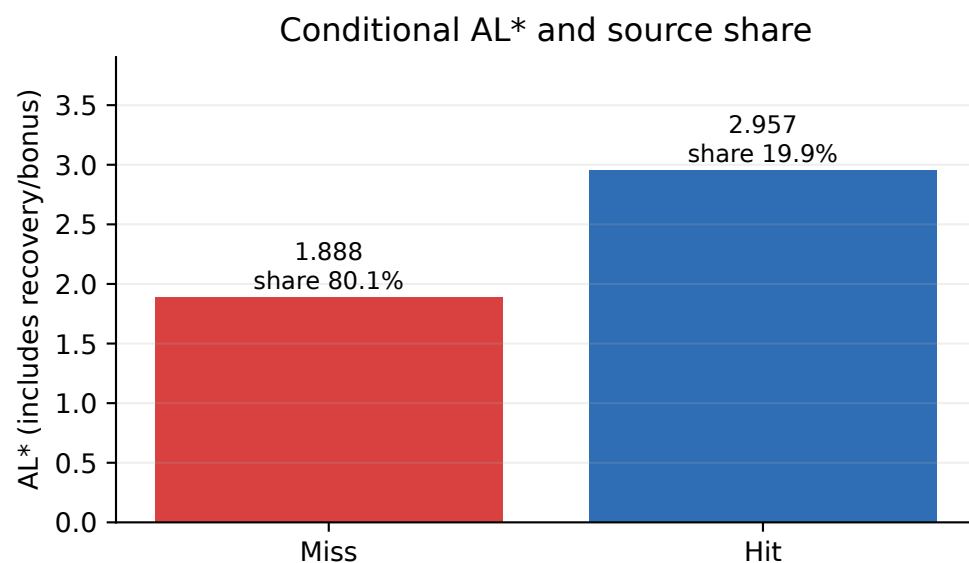
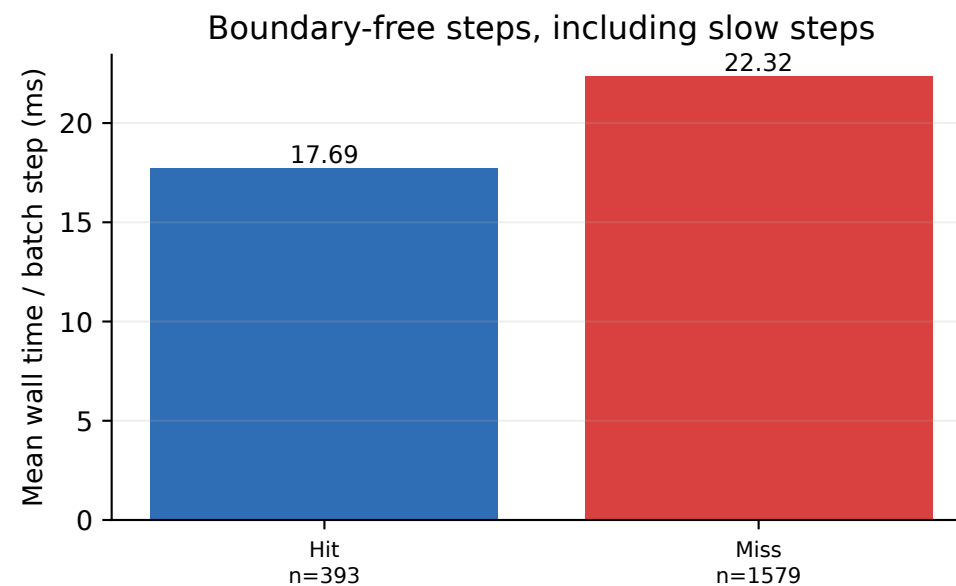
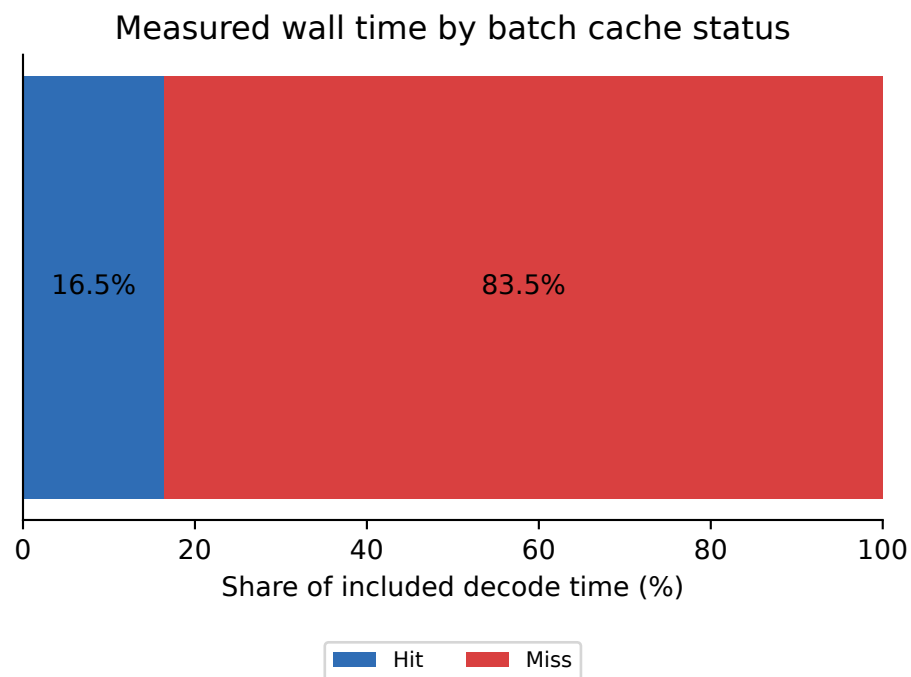


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

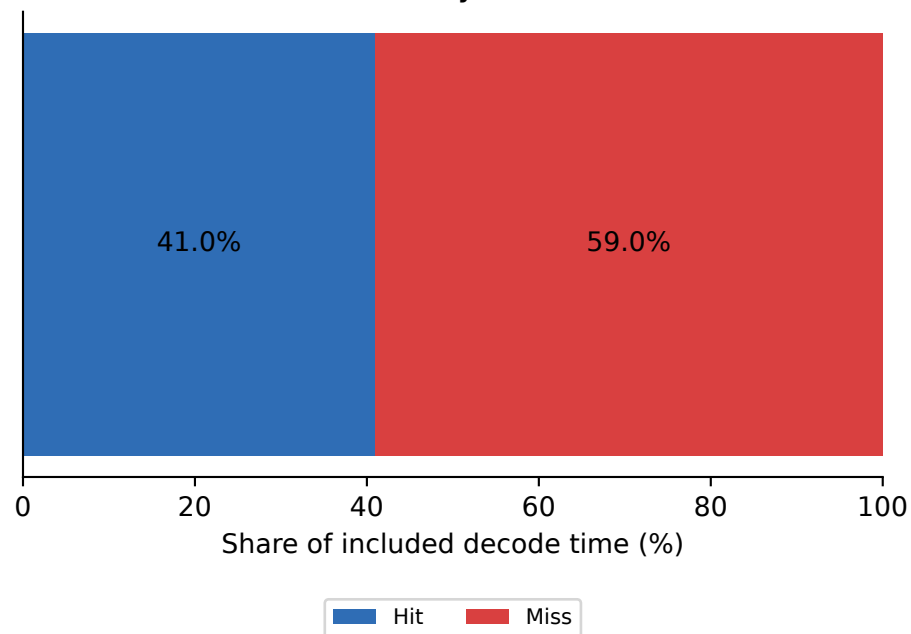
llama2_b1_warm_ssd_4
SSD: K=8, F=1
Last pass: B=1, T=0.7, N=48 | AL*=2.101, hit=20.1%, TPS*=98.17



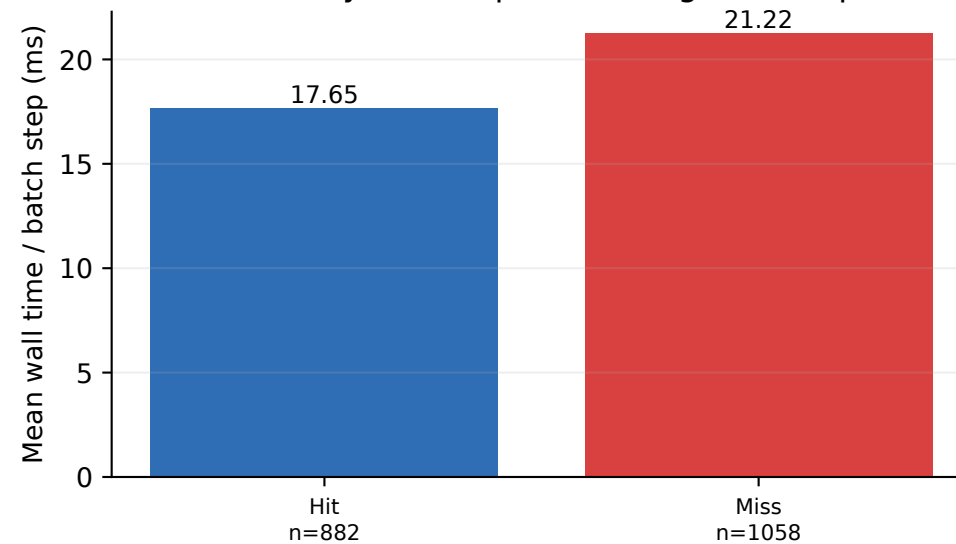
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_warm_ssd_5
SSD: K=6, F=5
Last pass: B=1, T=0.7, N=48 | AL*=2.223, hit=45.8%, TPS*=113.43

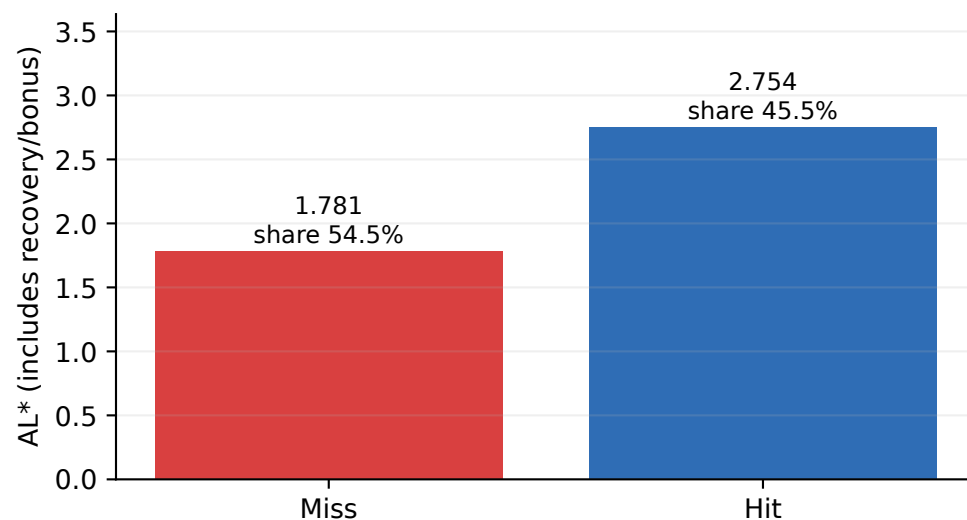
Measured wall time by batch cache status



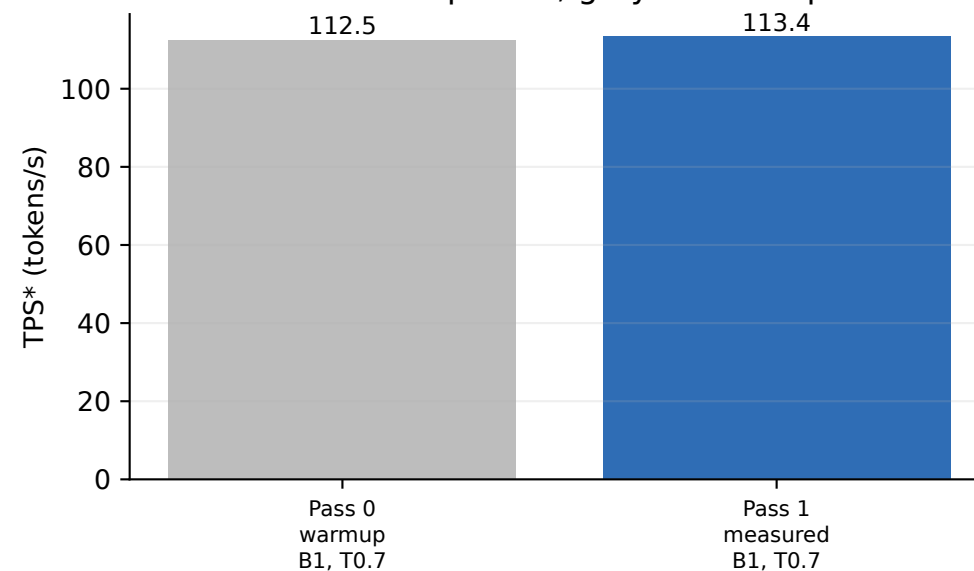
Boundary-free steps, including slow steps



Conditional AL* and source share

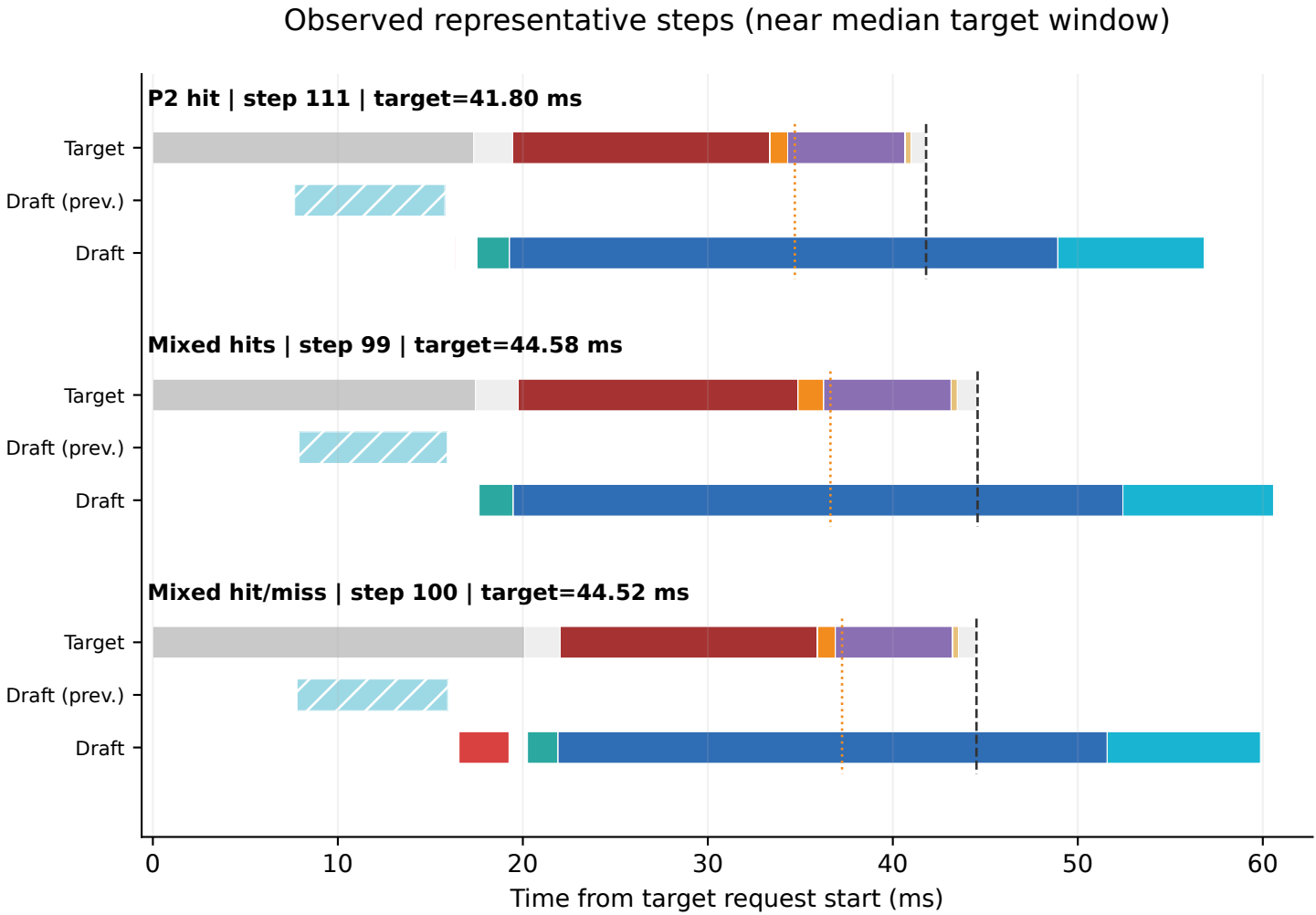
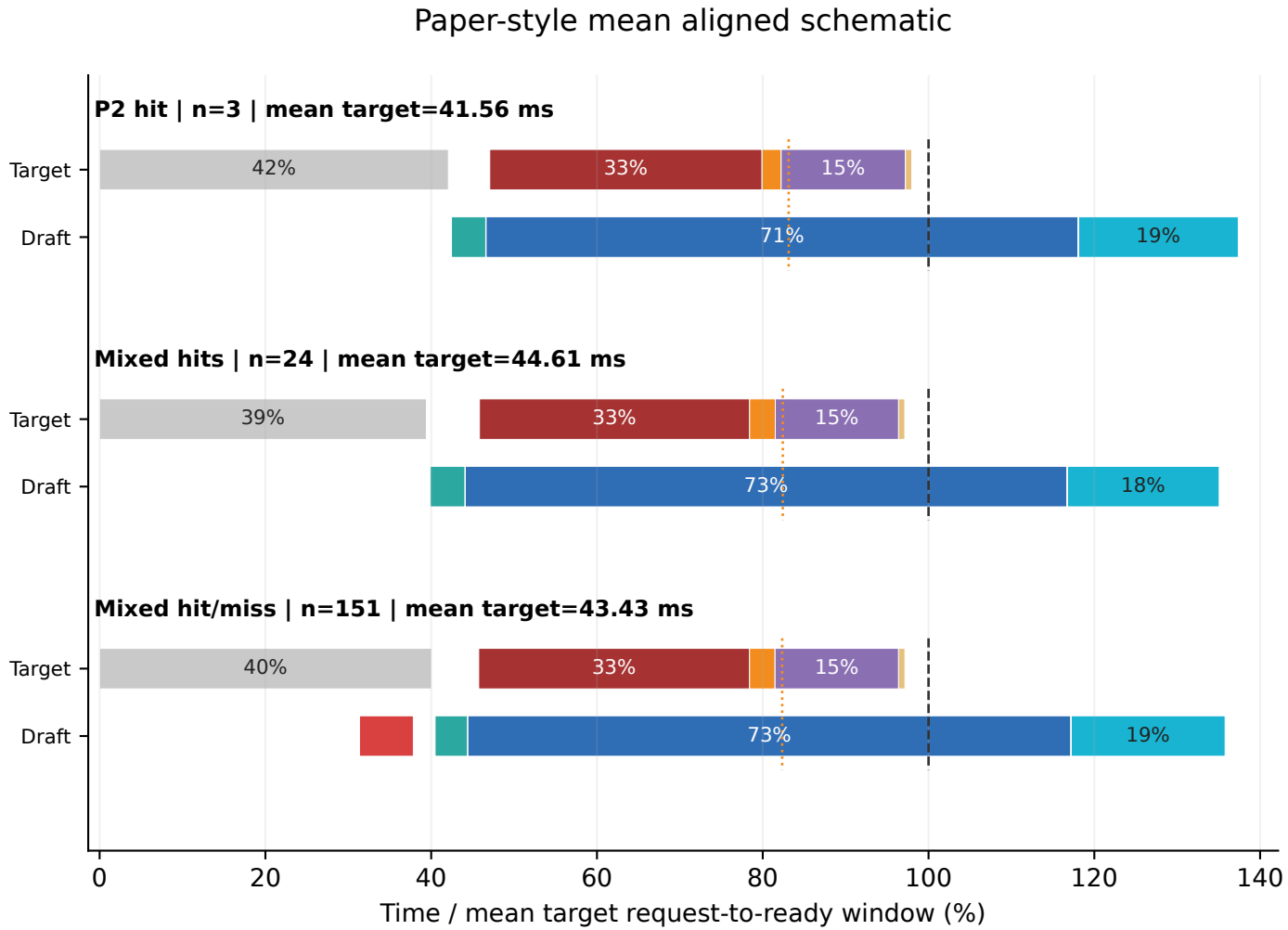
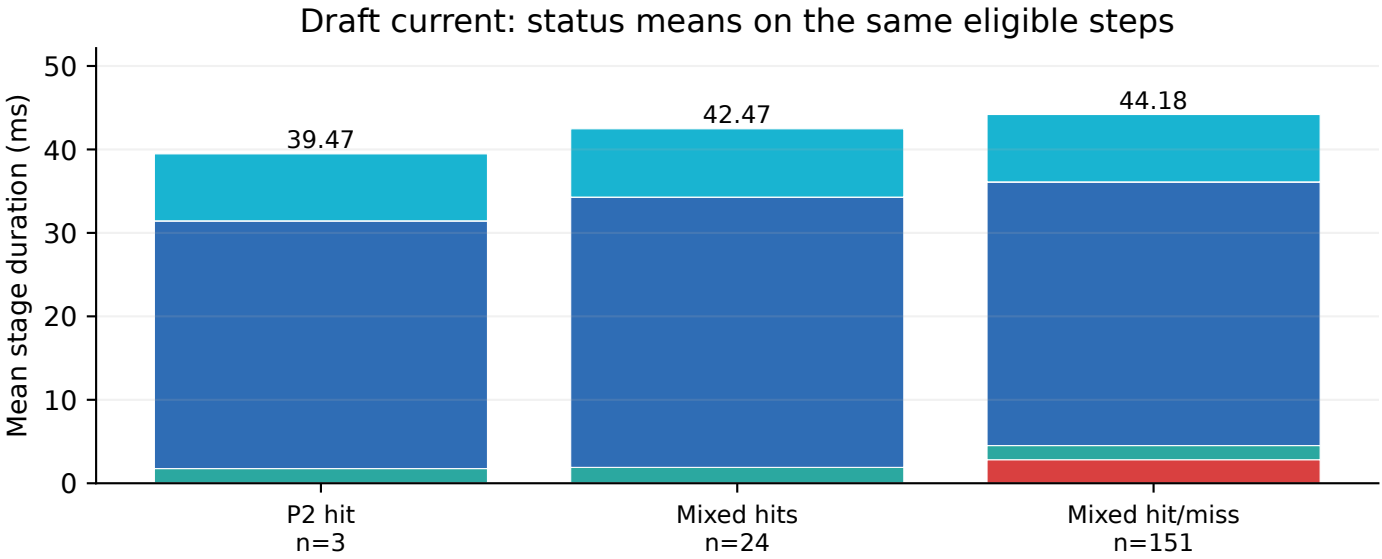
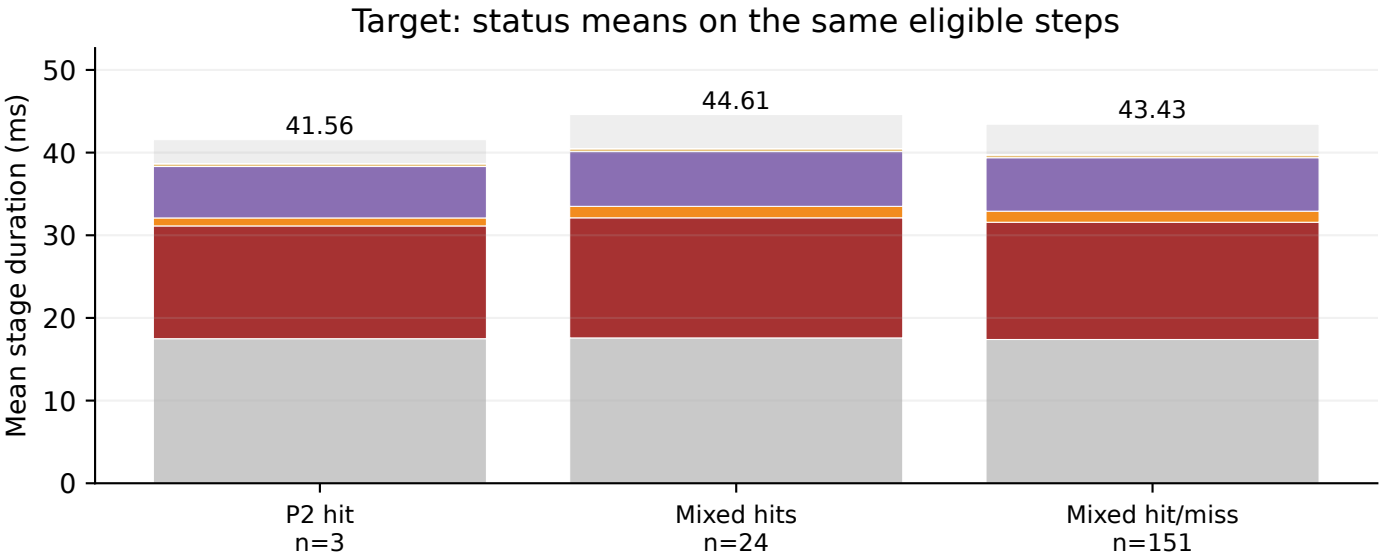


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

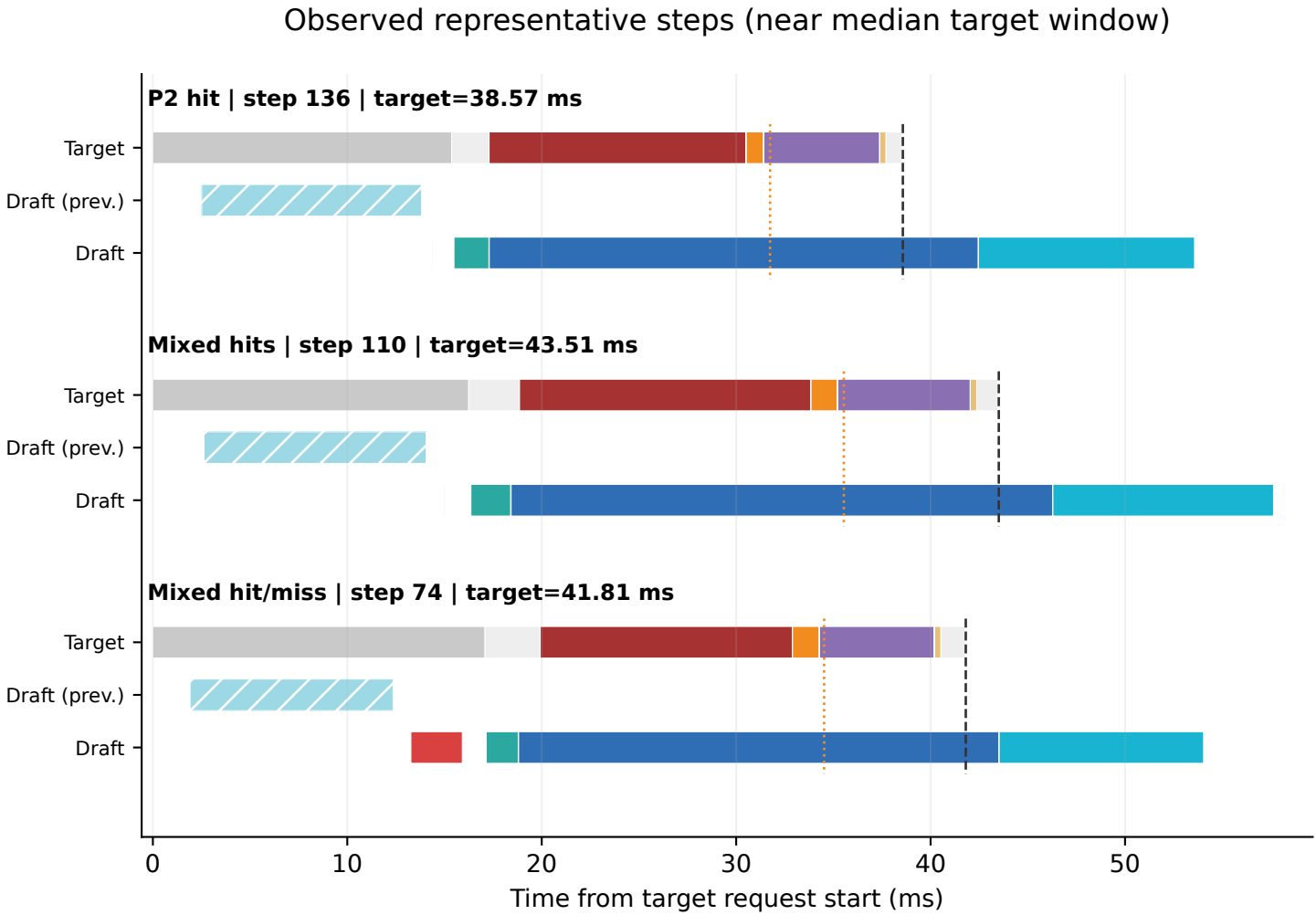
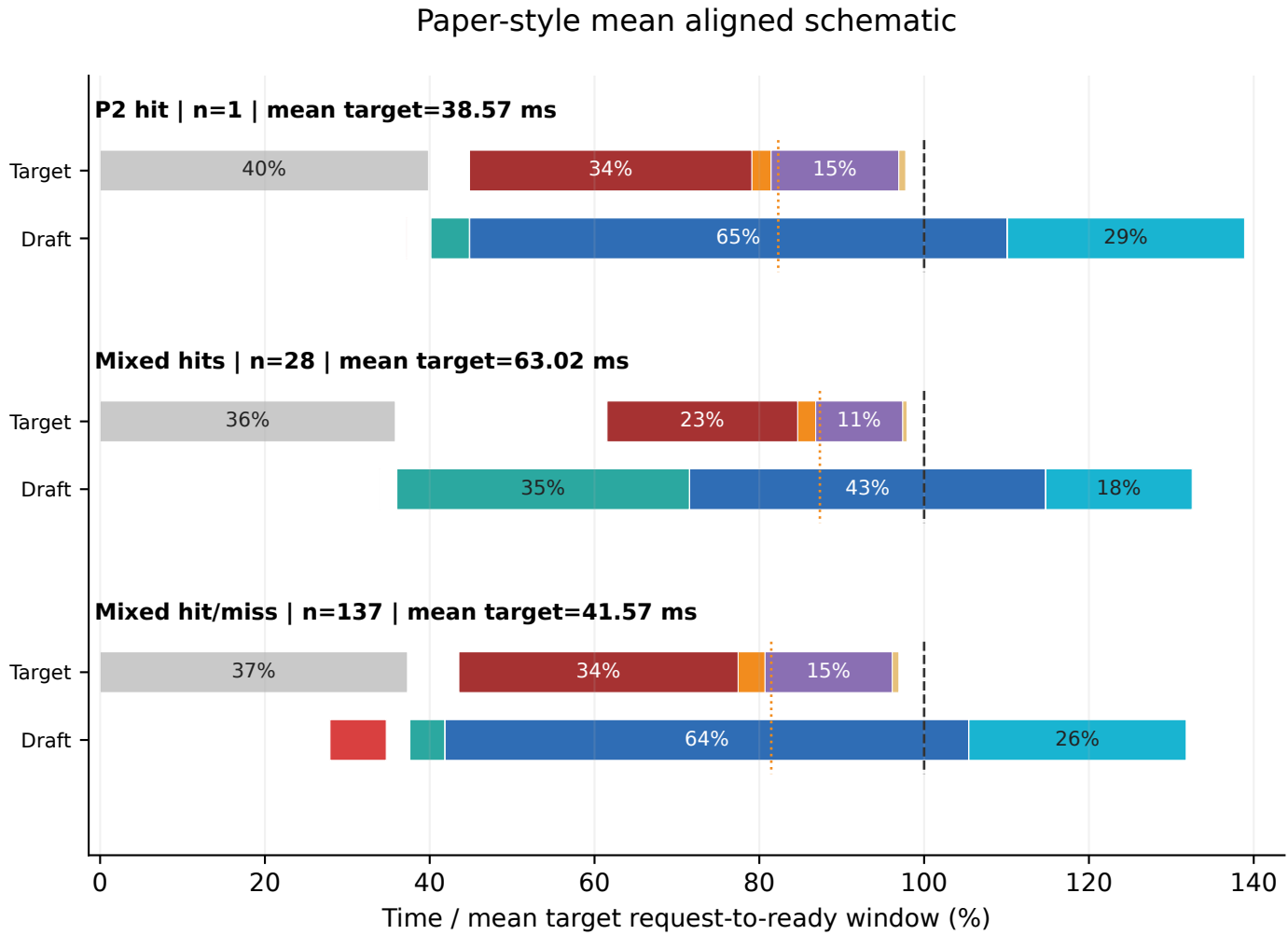
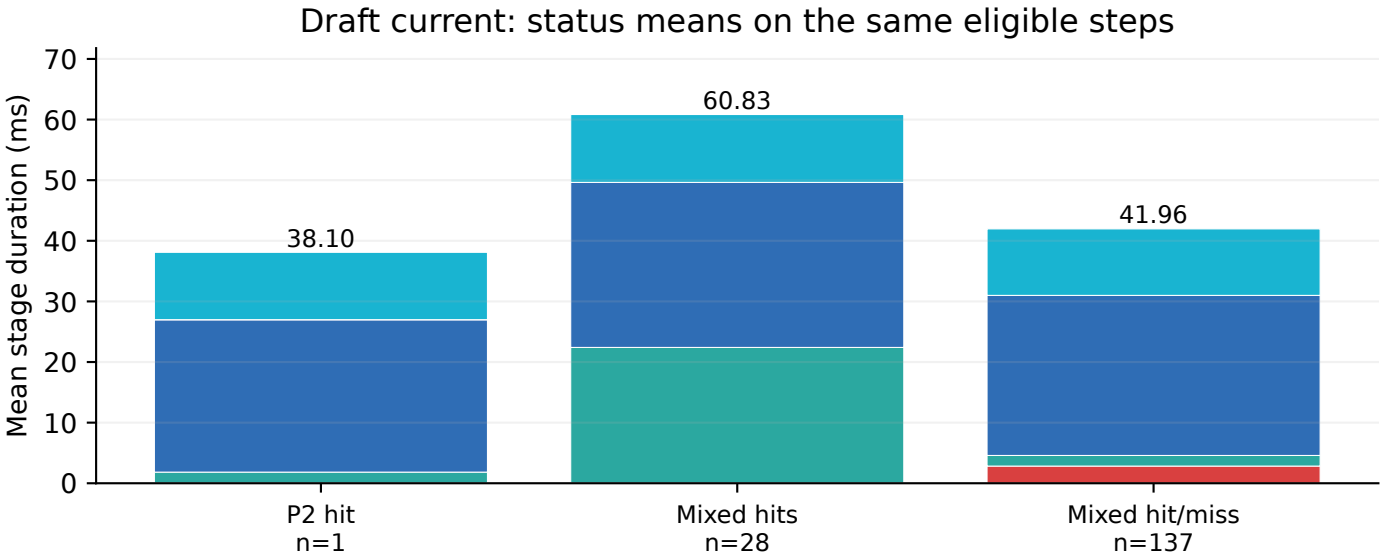
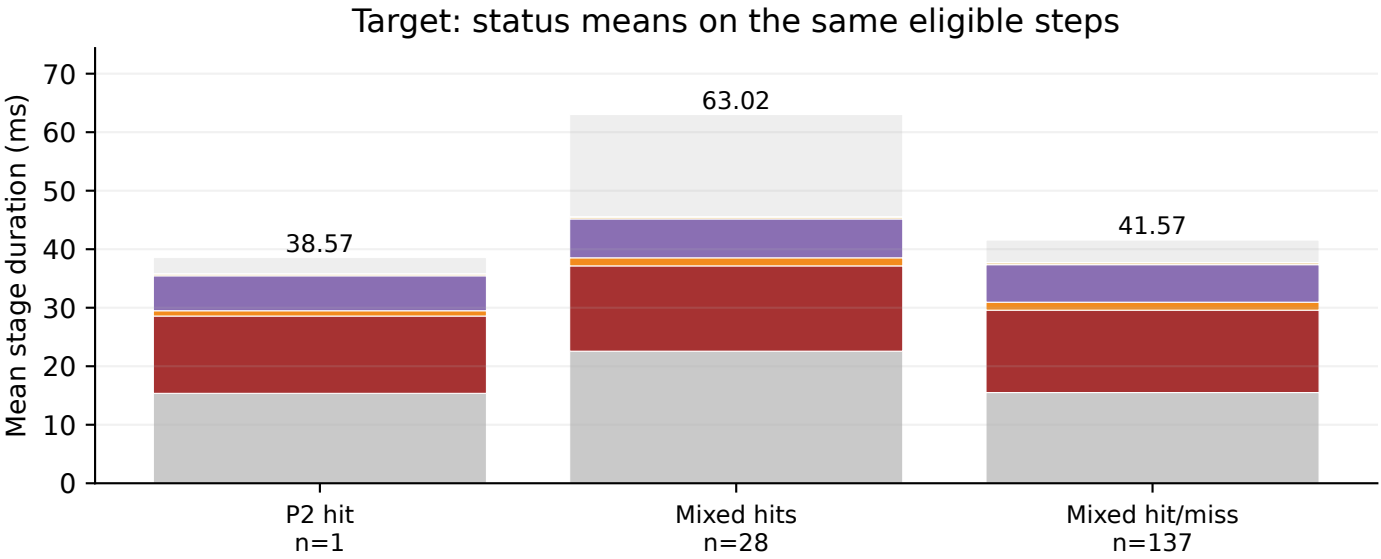
llama2_b8_combo_al_k1_7
DUET: K1/K2=7/2, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=8, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 178 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

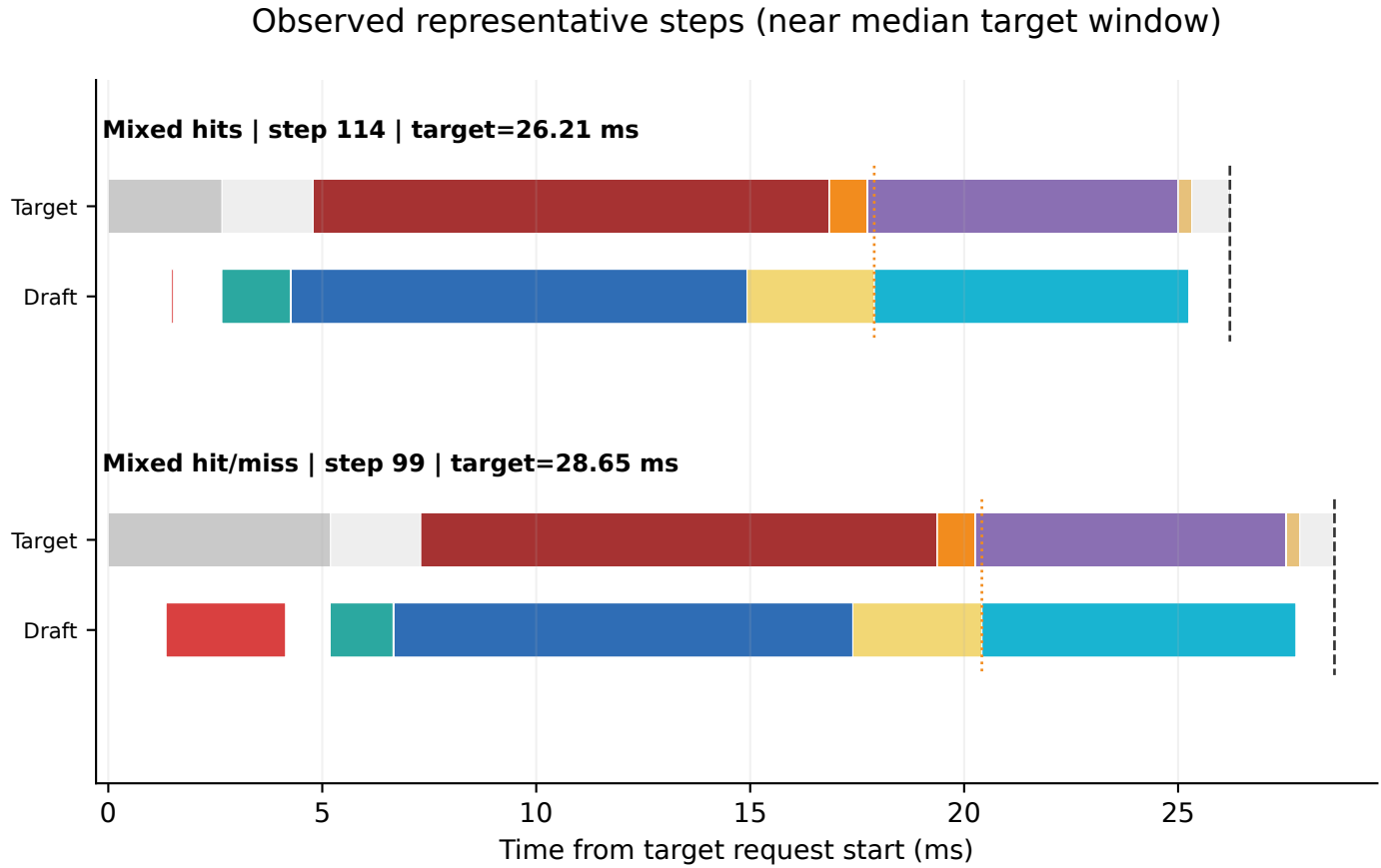
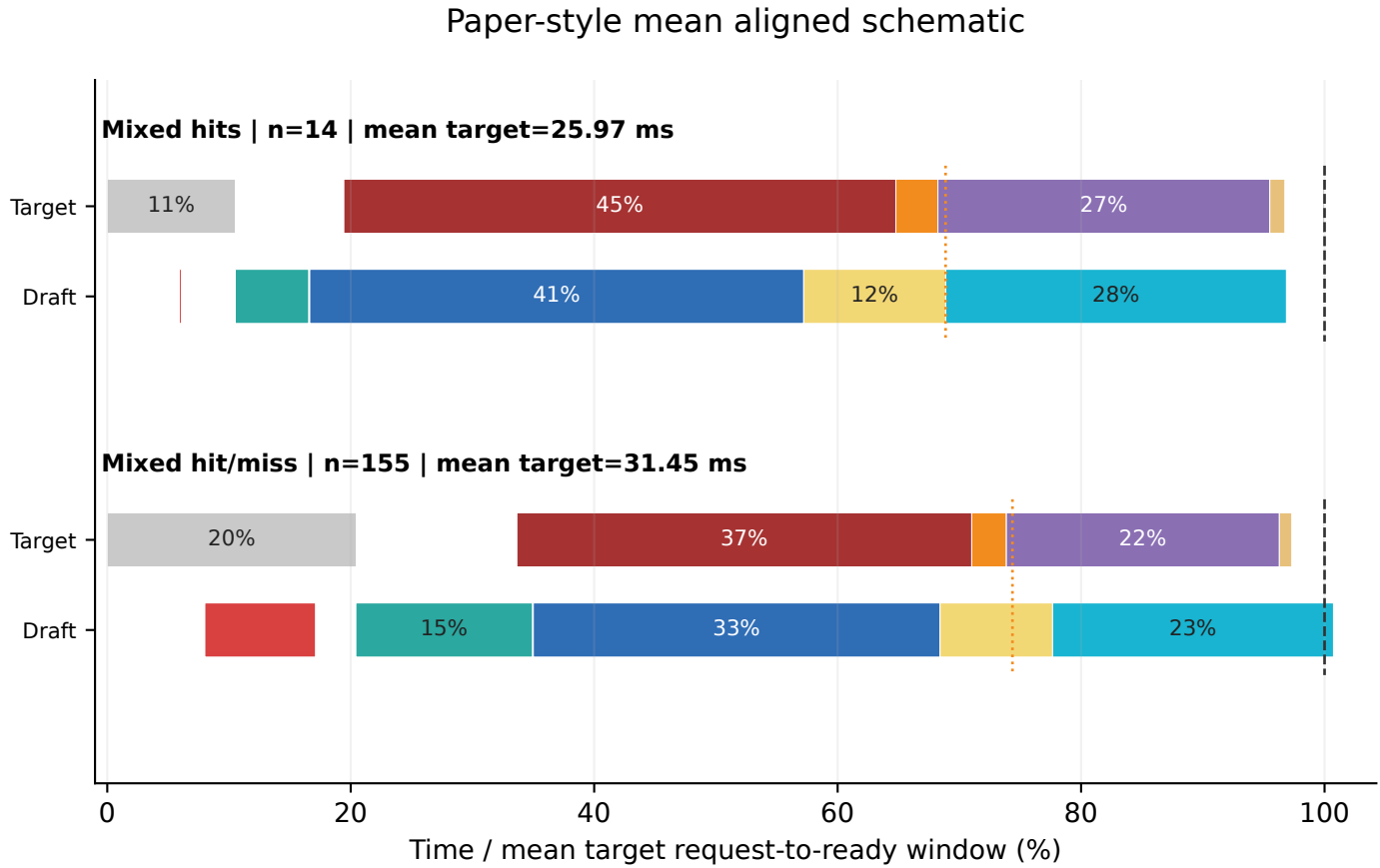
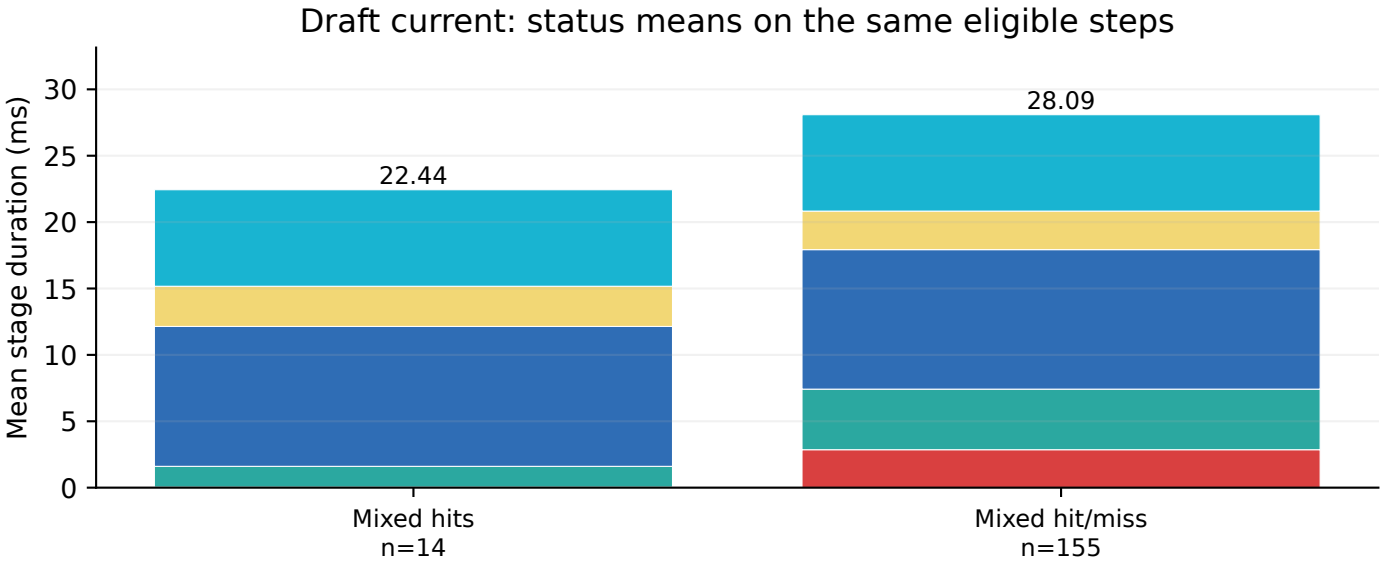
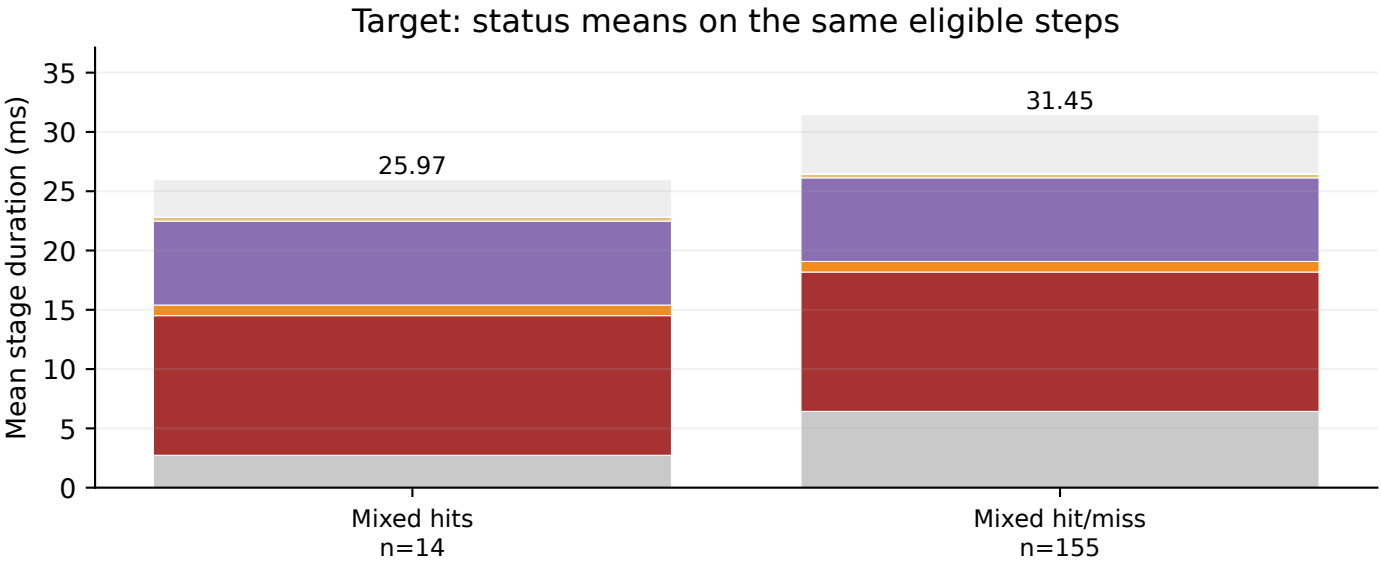
llama2_b8_combo_al_k2_3
DUET: K1/K2=6/3, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 166 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

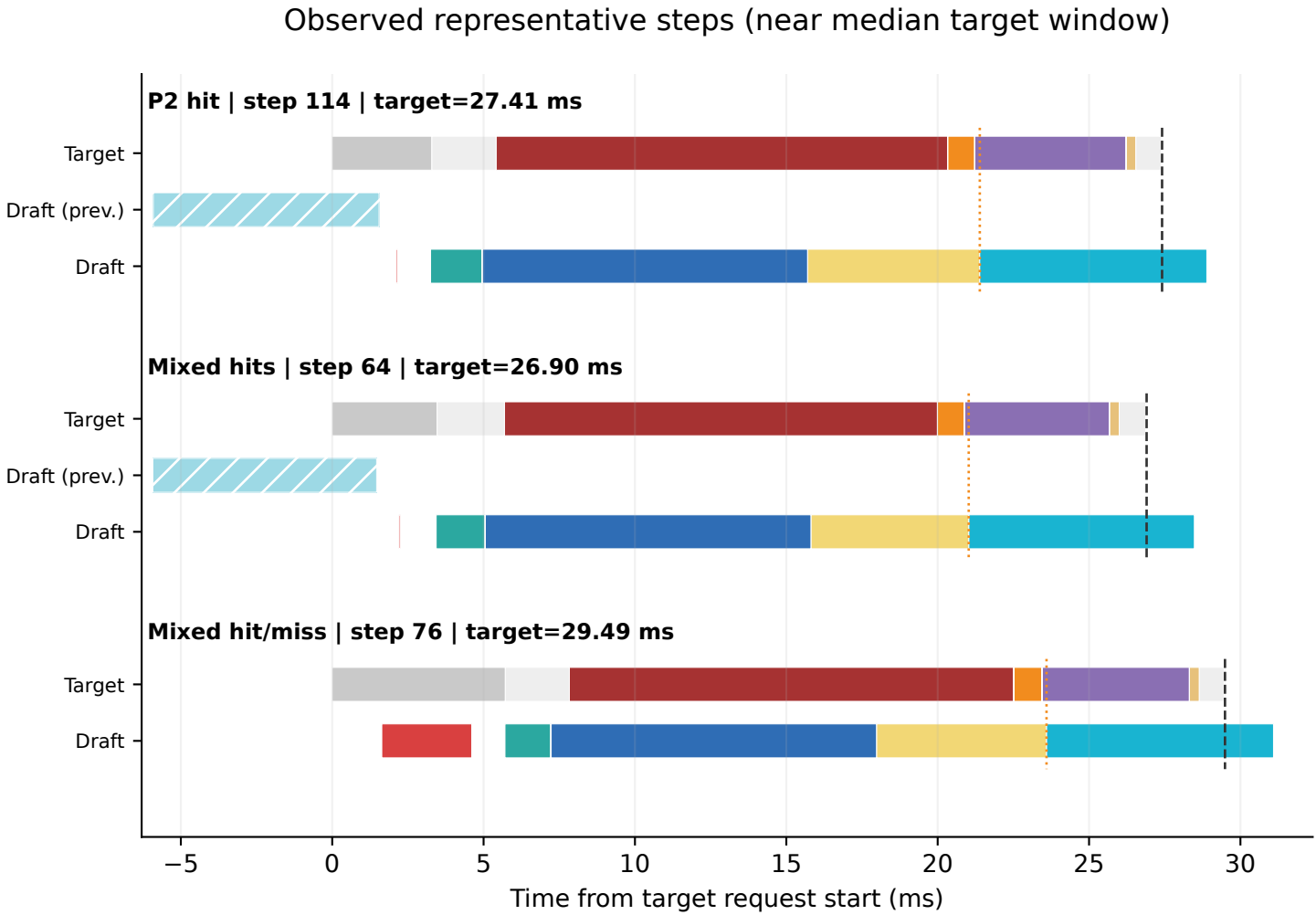
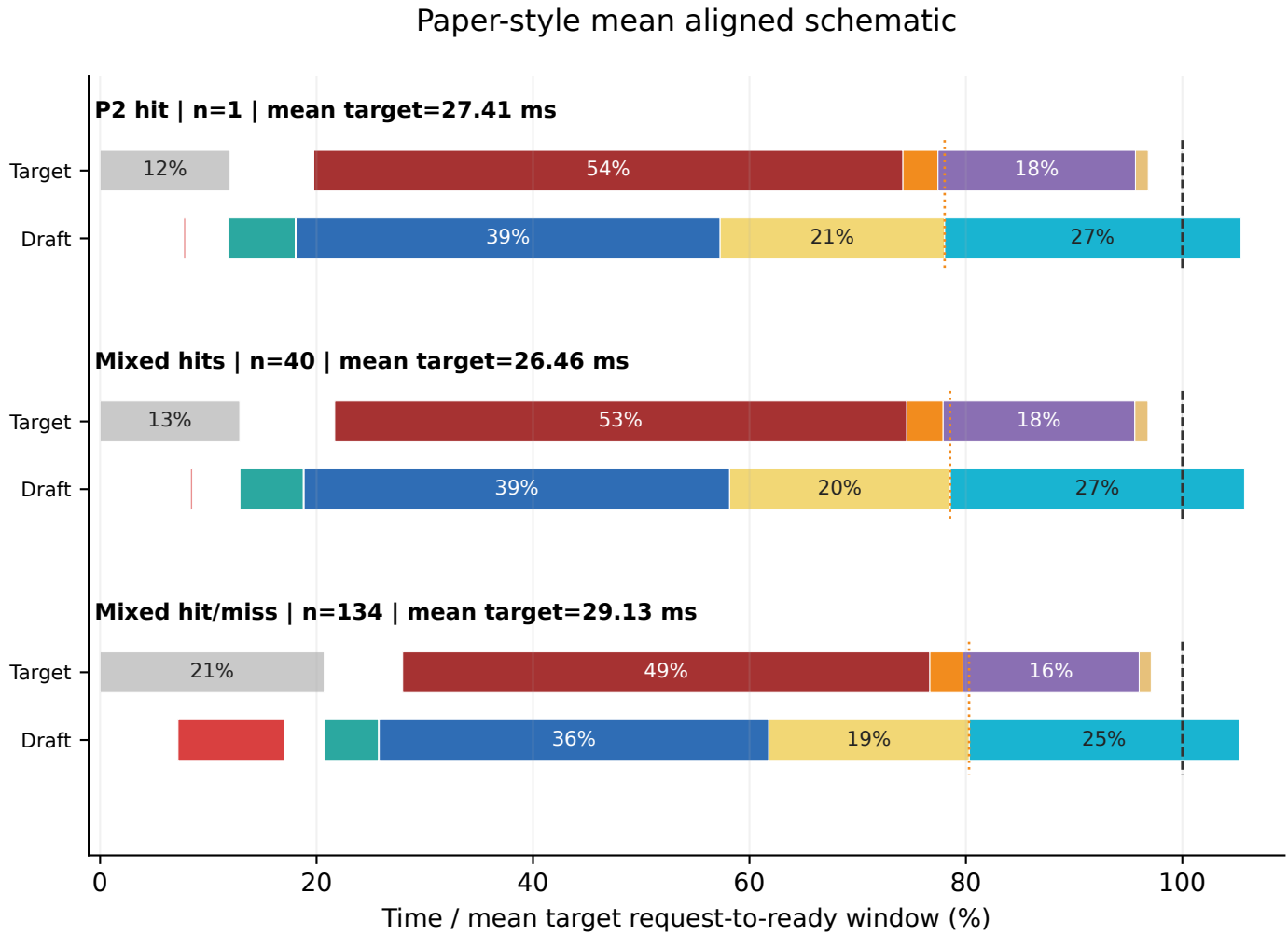
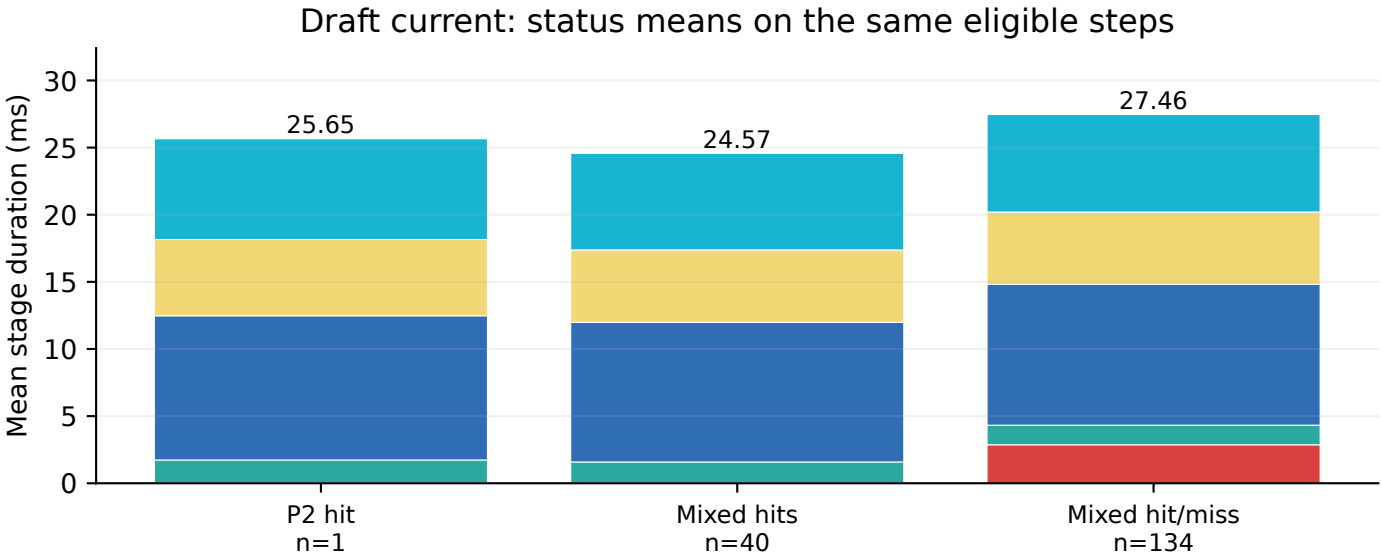
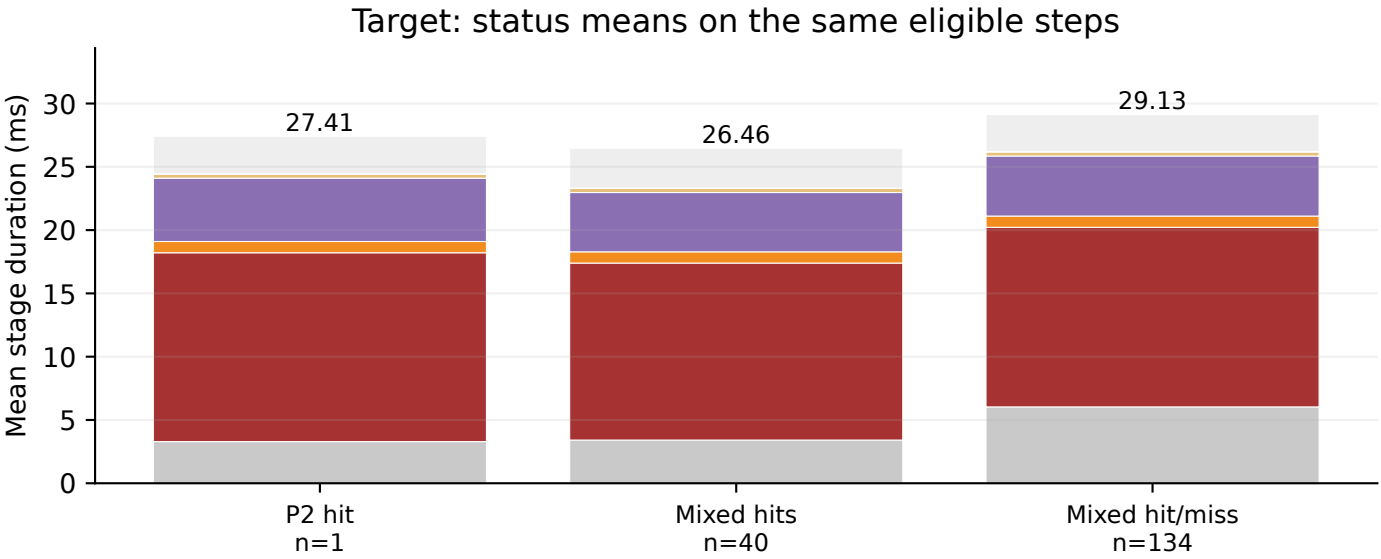
llama2_b8_combo_e19
DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 169 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

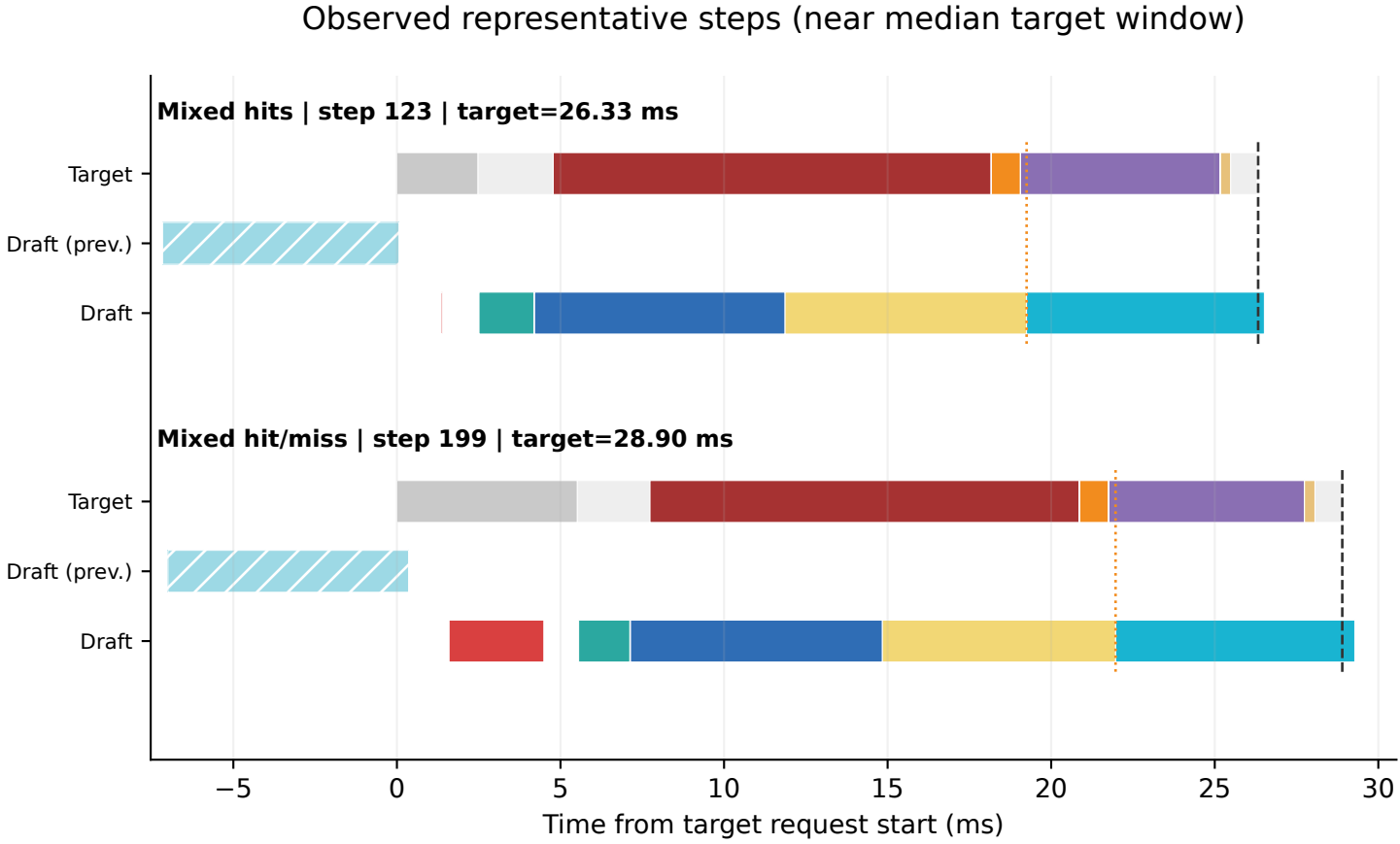
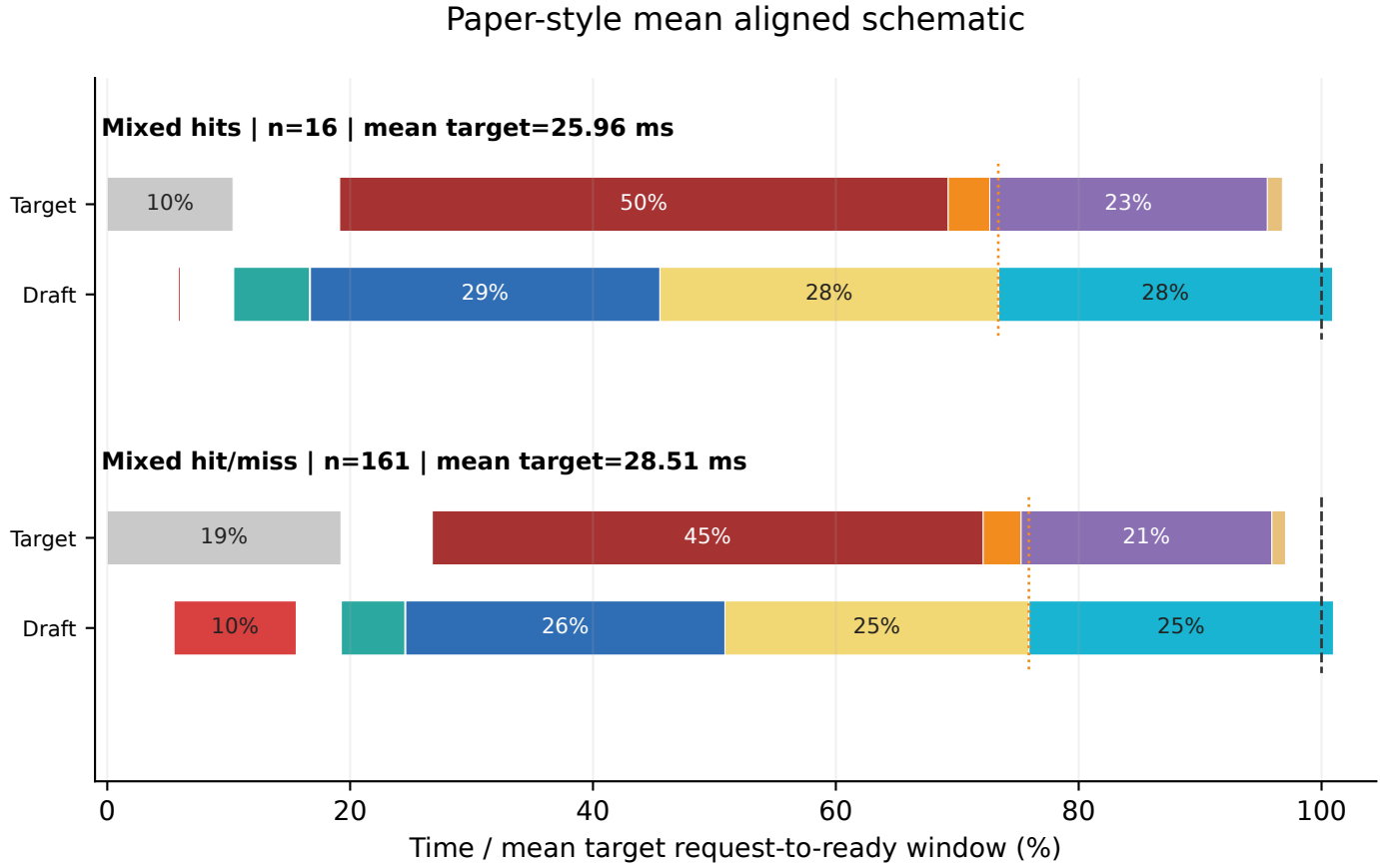
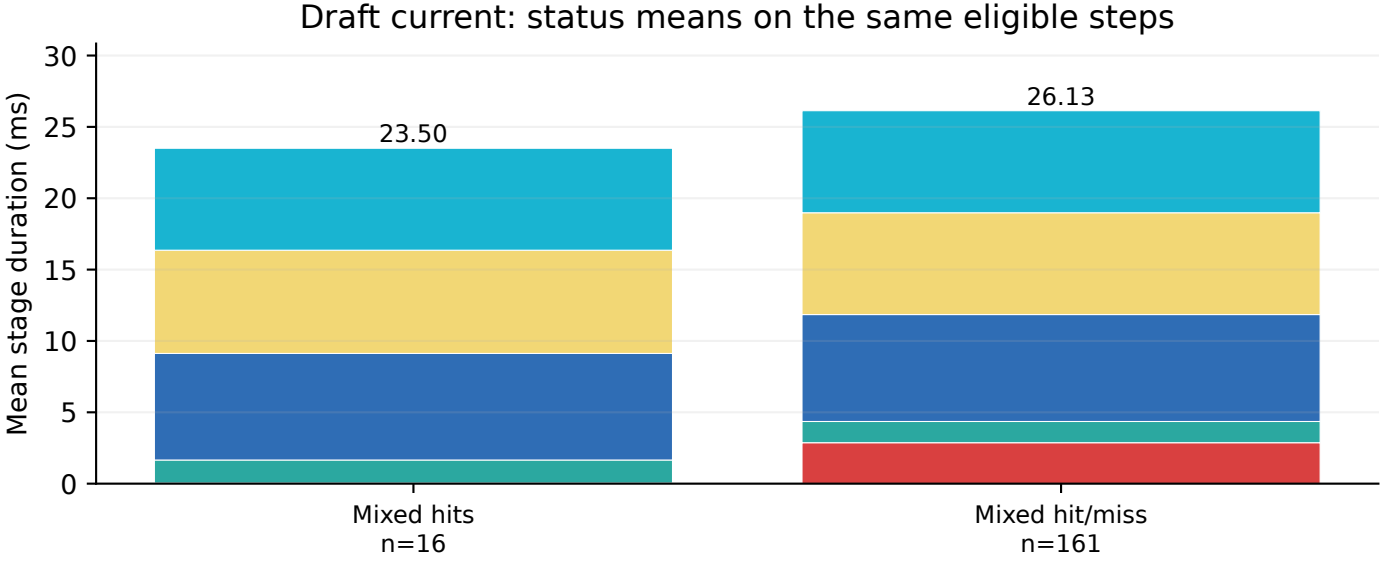
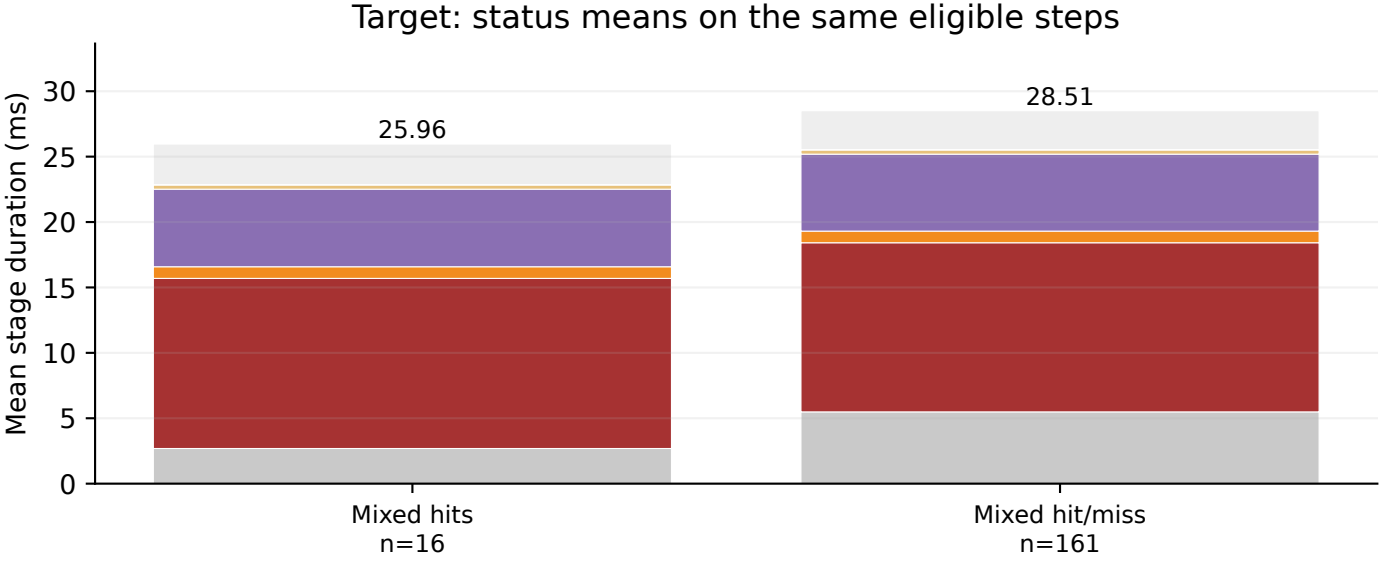
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_combo_e23
DUET: K1/K2=3/2, exit=23 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 175 eligible steps



Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

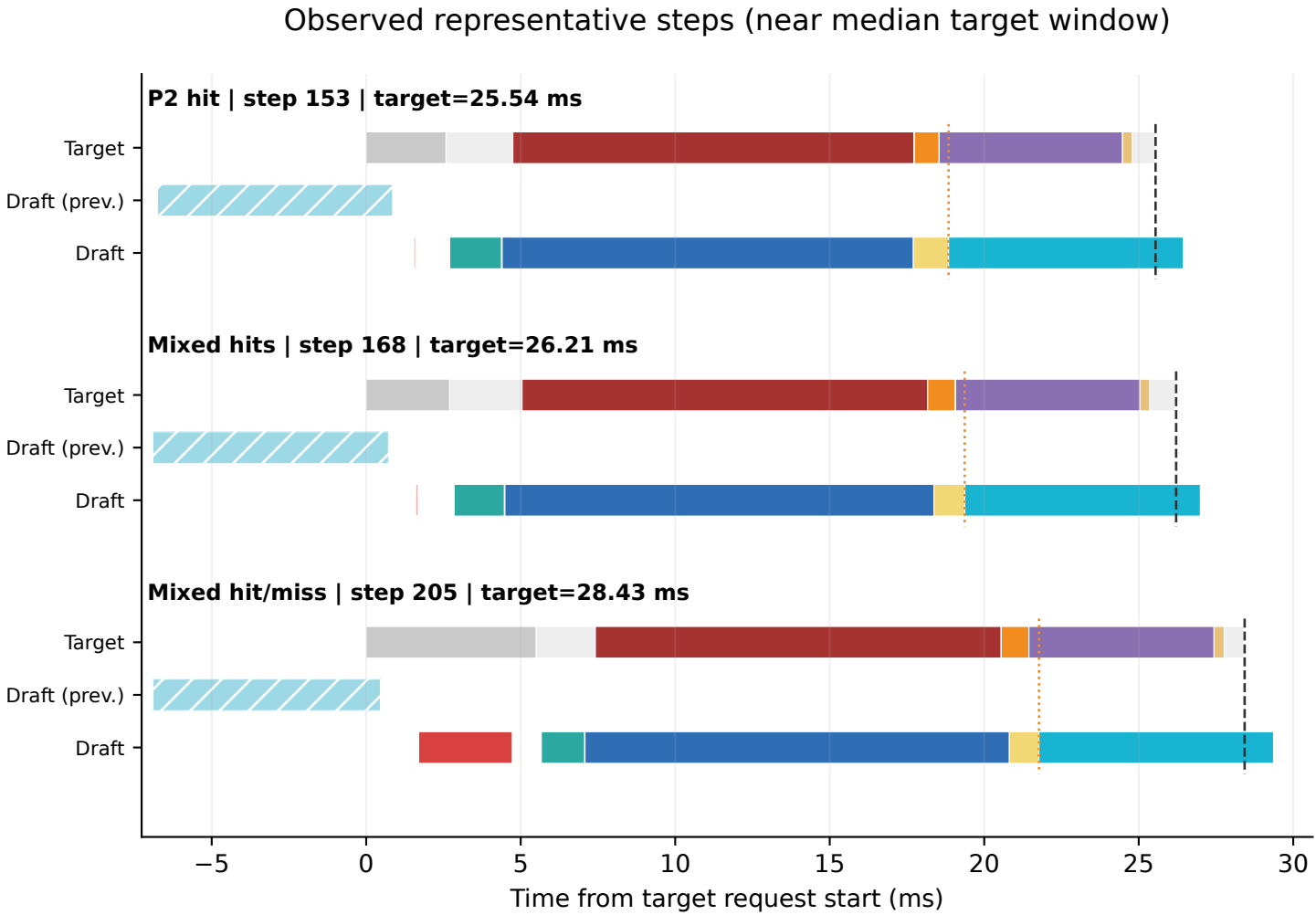
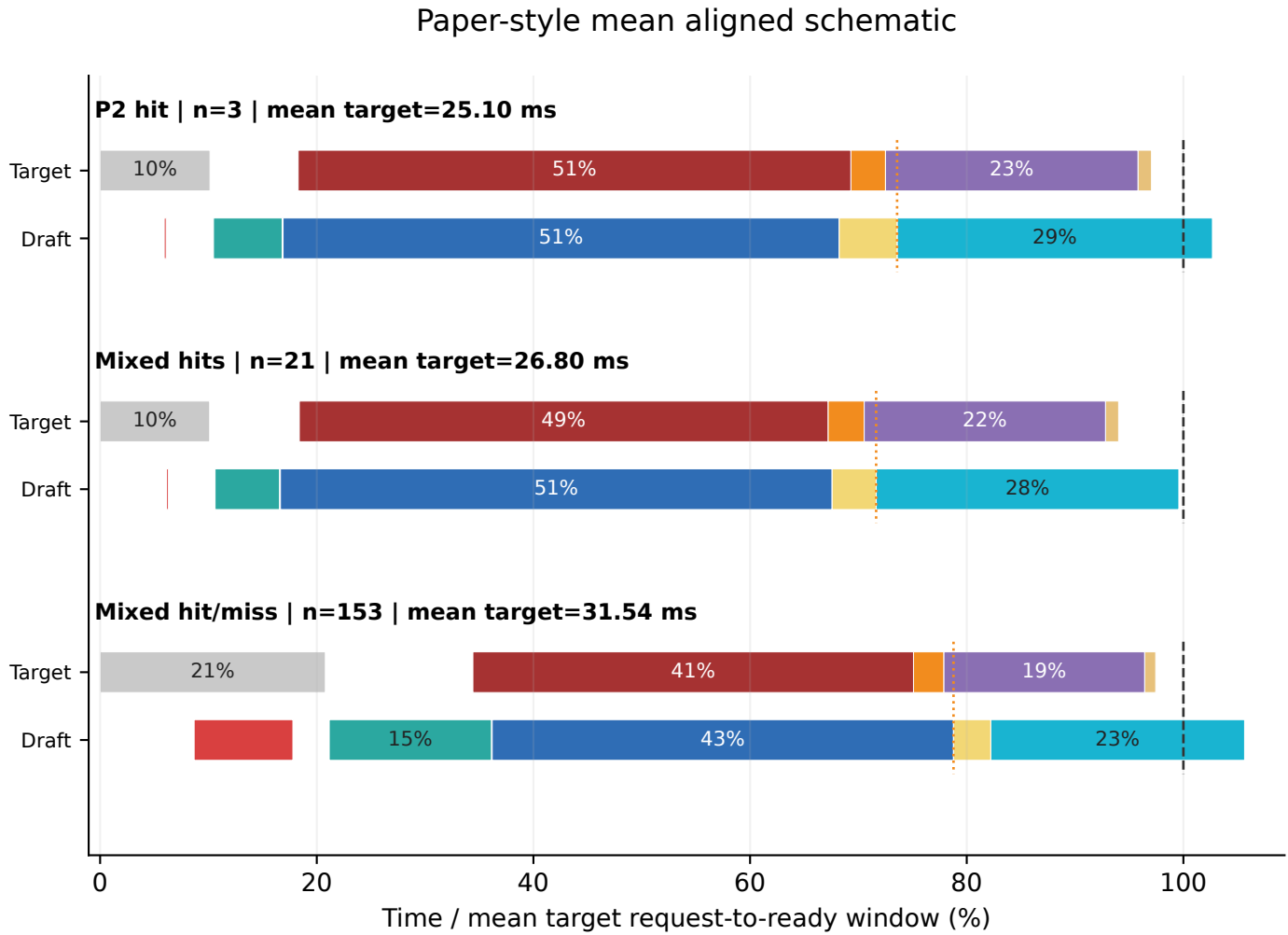
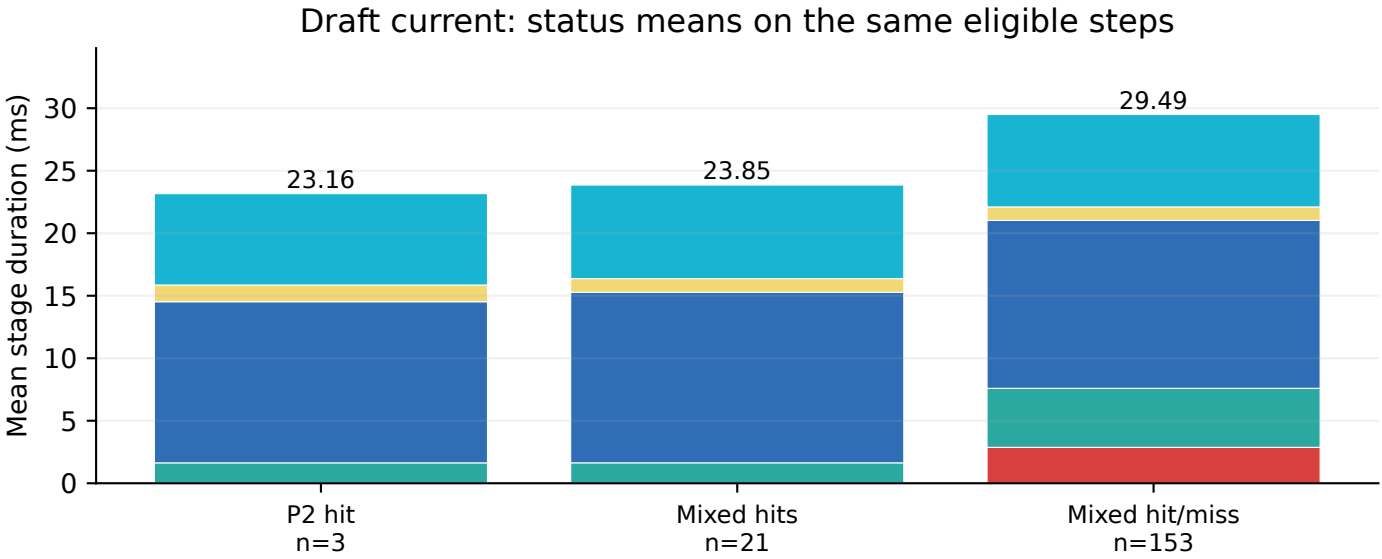
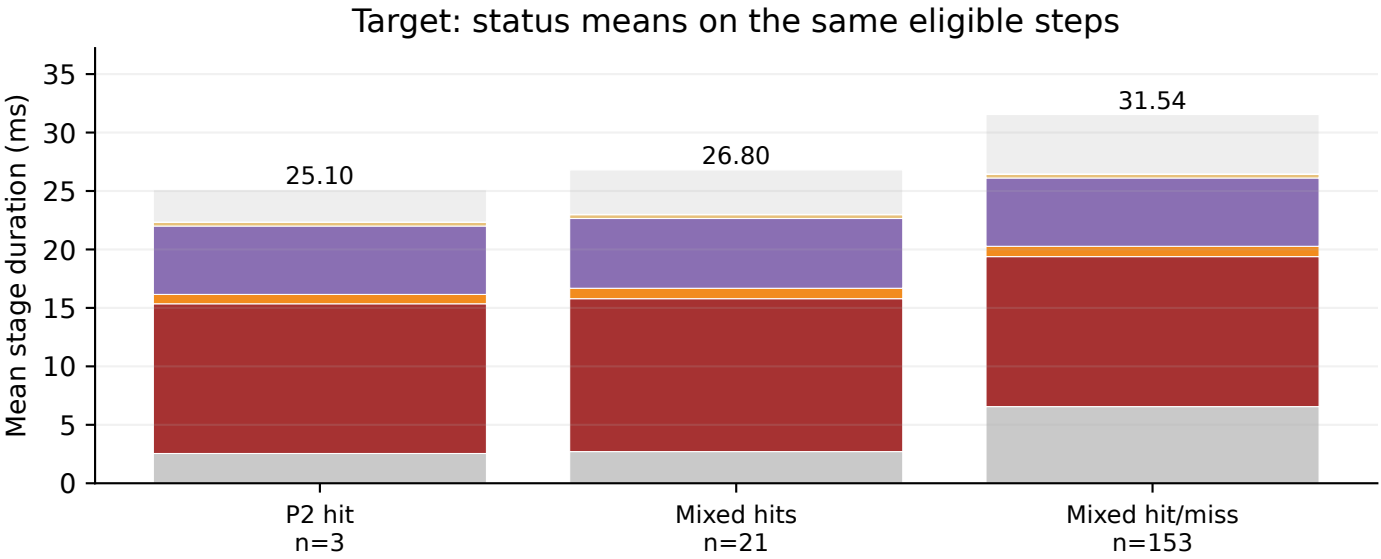
llama2_b8_combo_k1_2
DUET: K1/K2=2/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 177 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

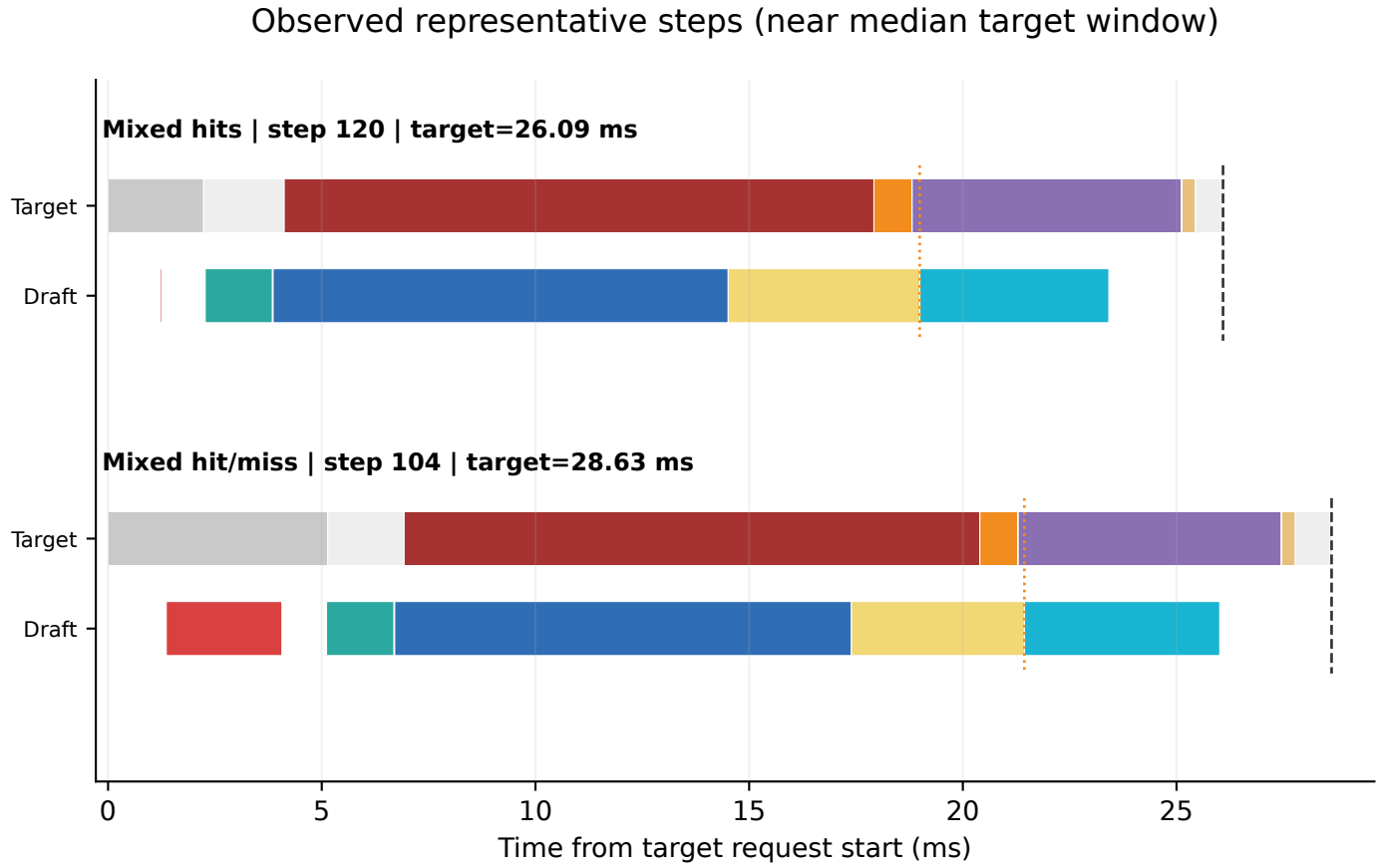
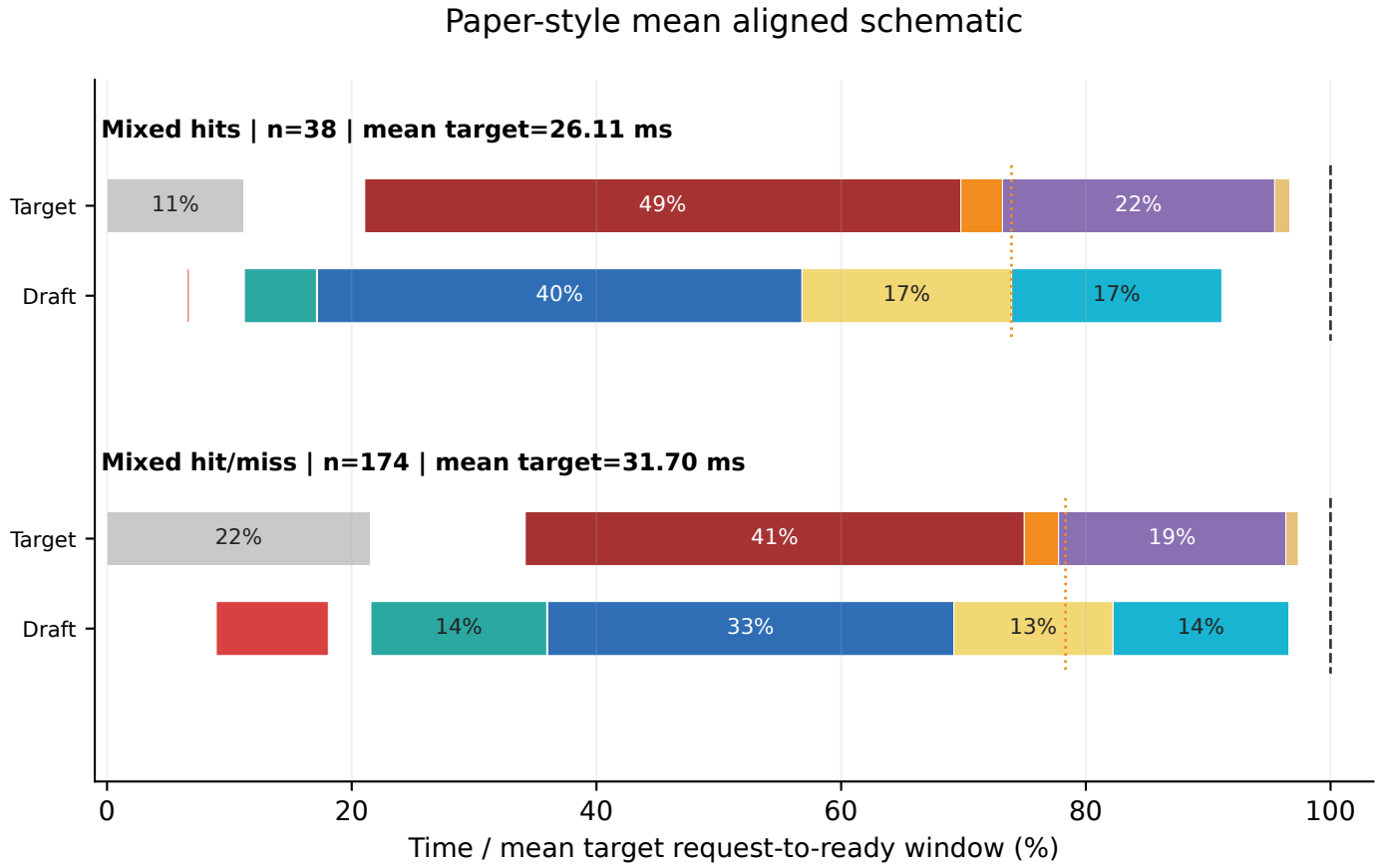
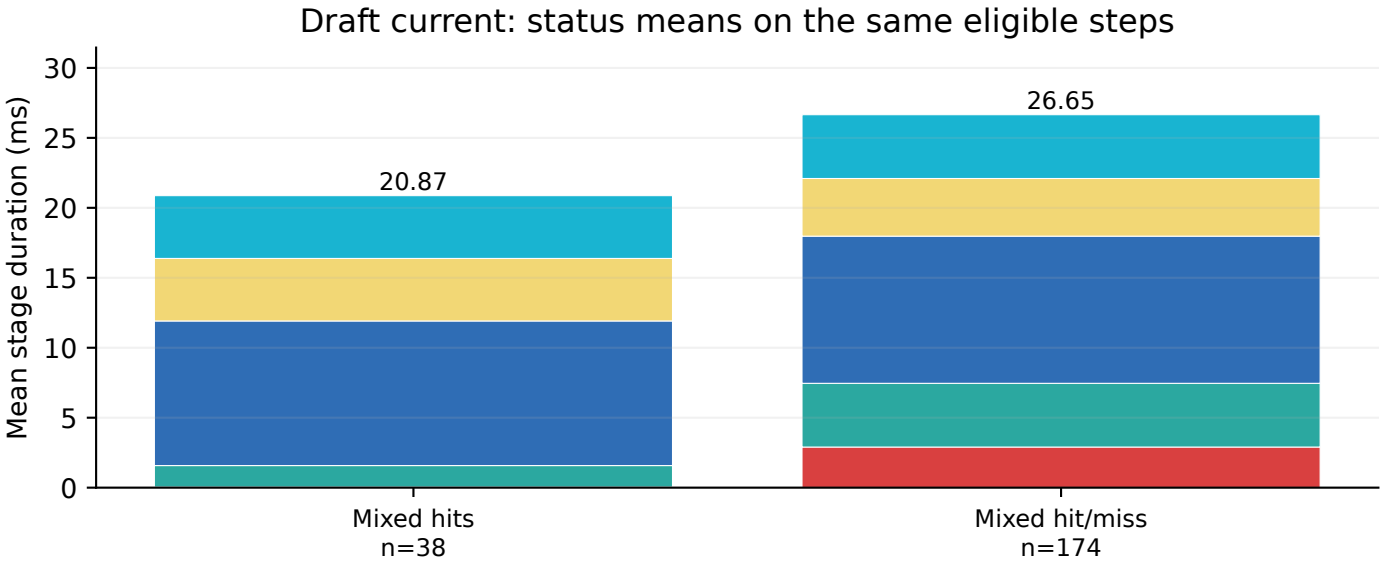
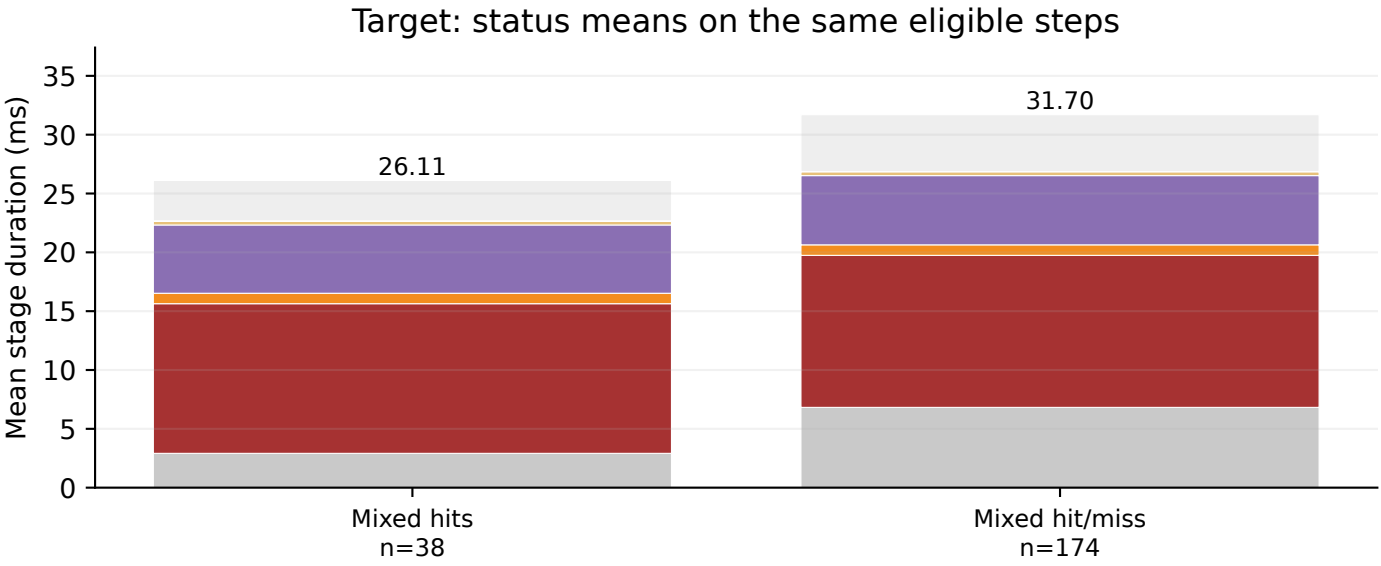
llama2_b8_combo_k1_4
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 177 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

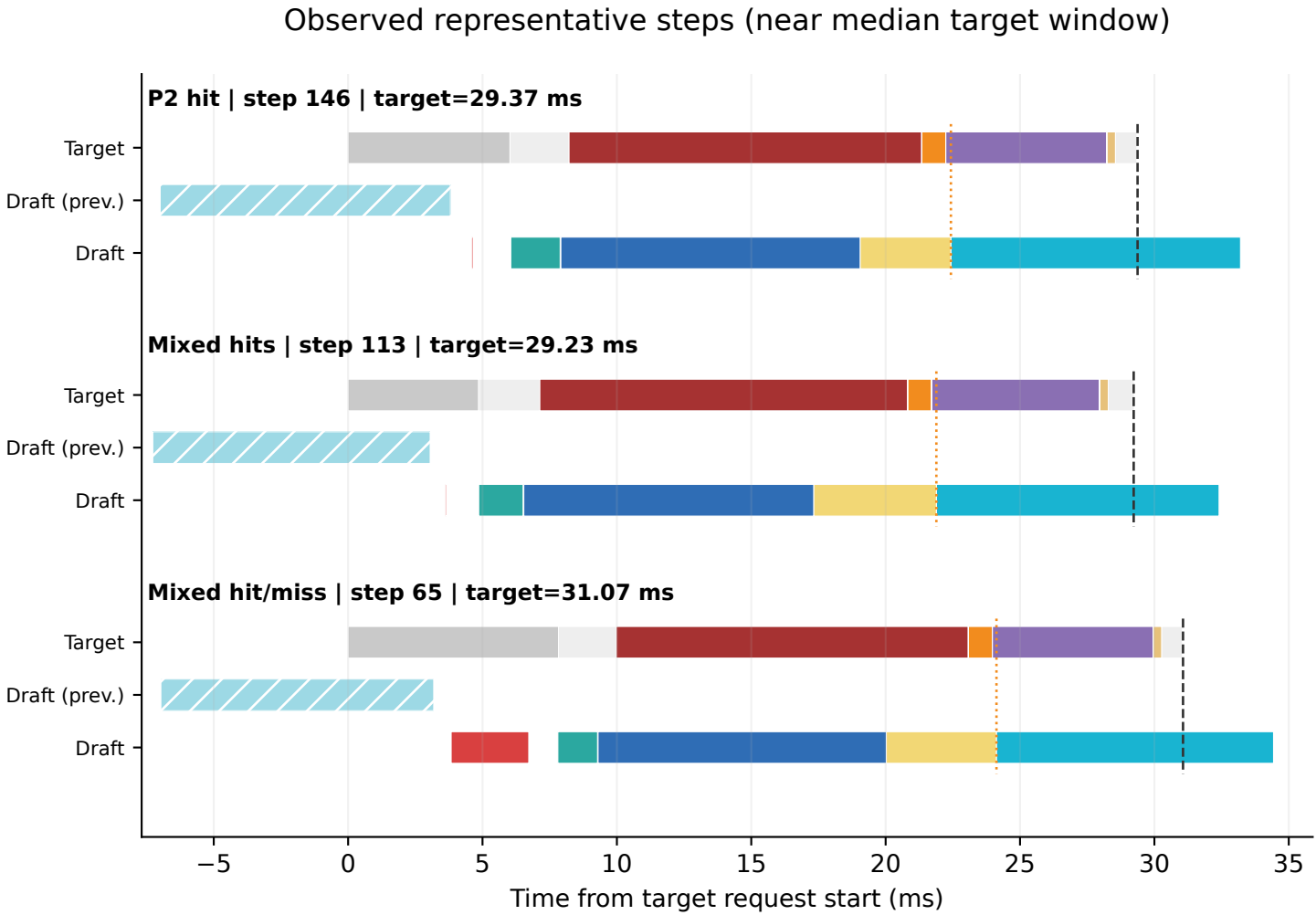
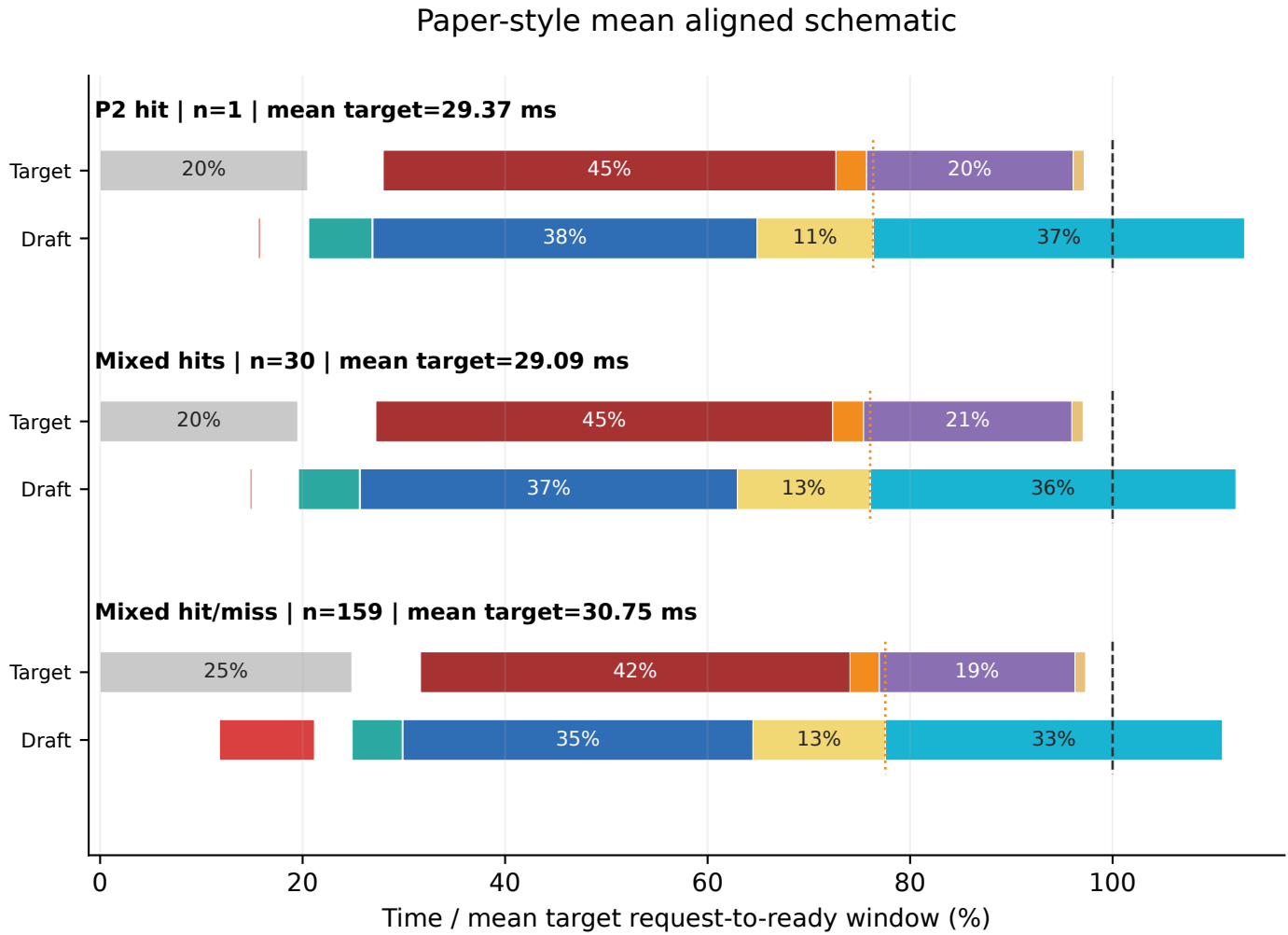
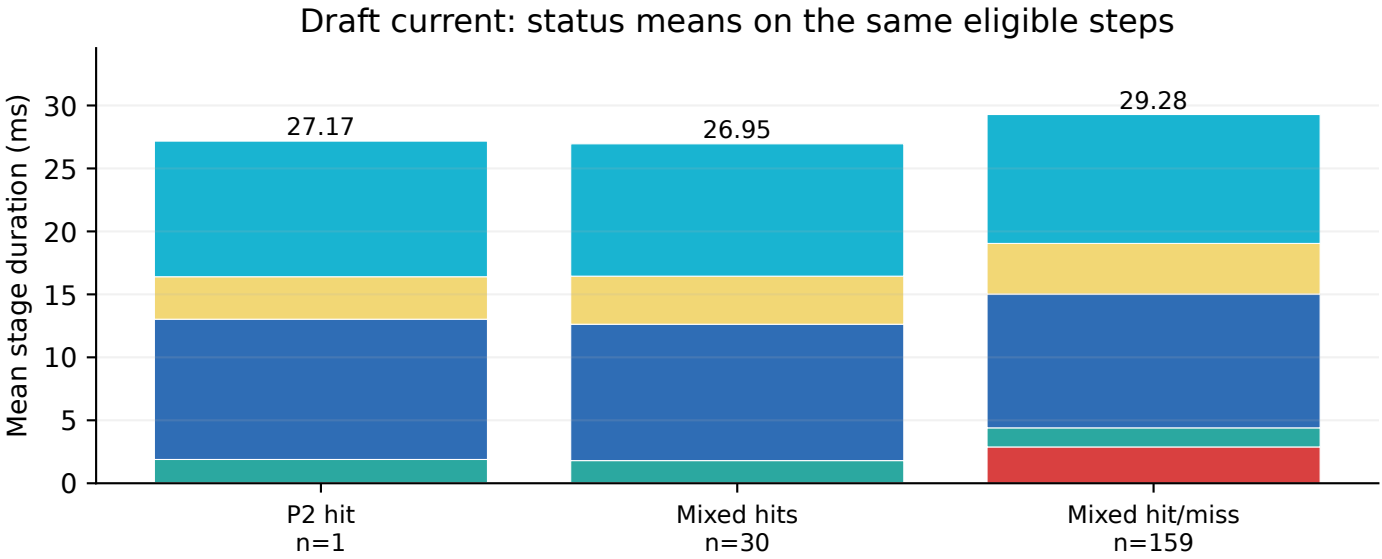
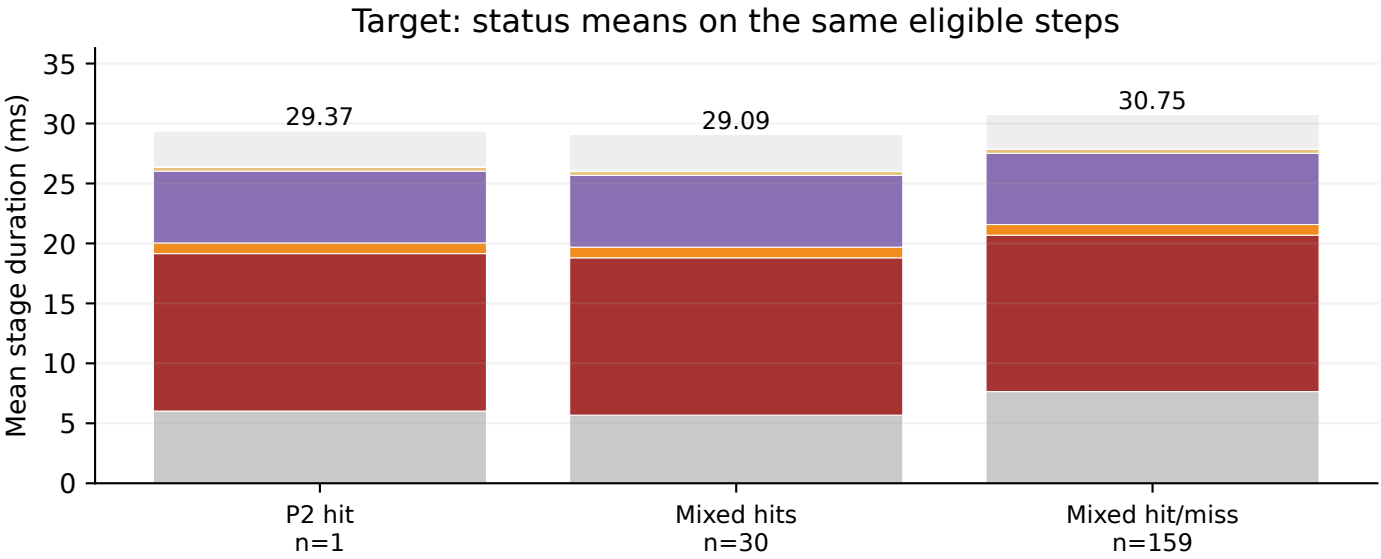
llama2_b8_combo_k2_1
DUET: K1/K2=3/1, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 212 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

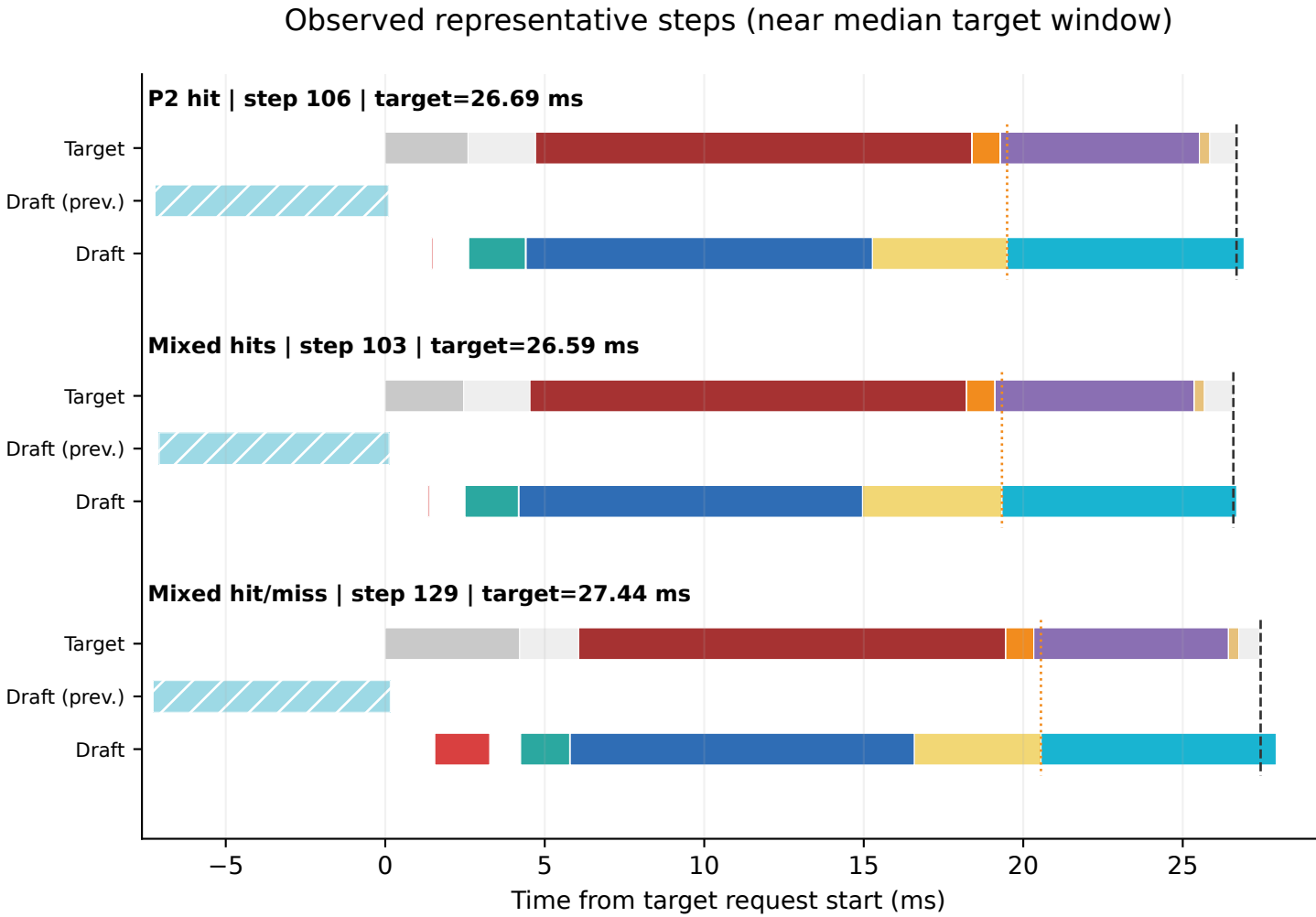
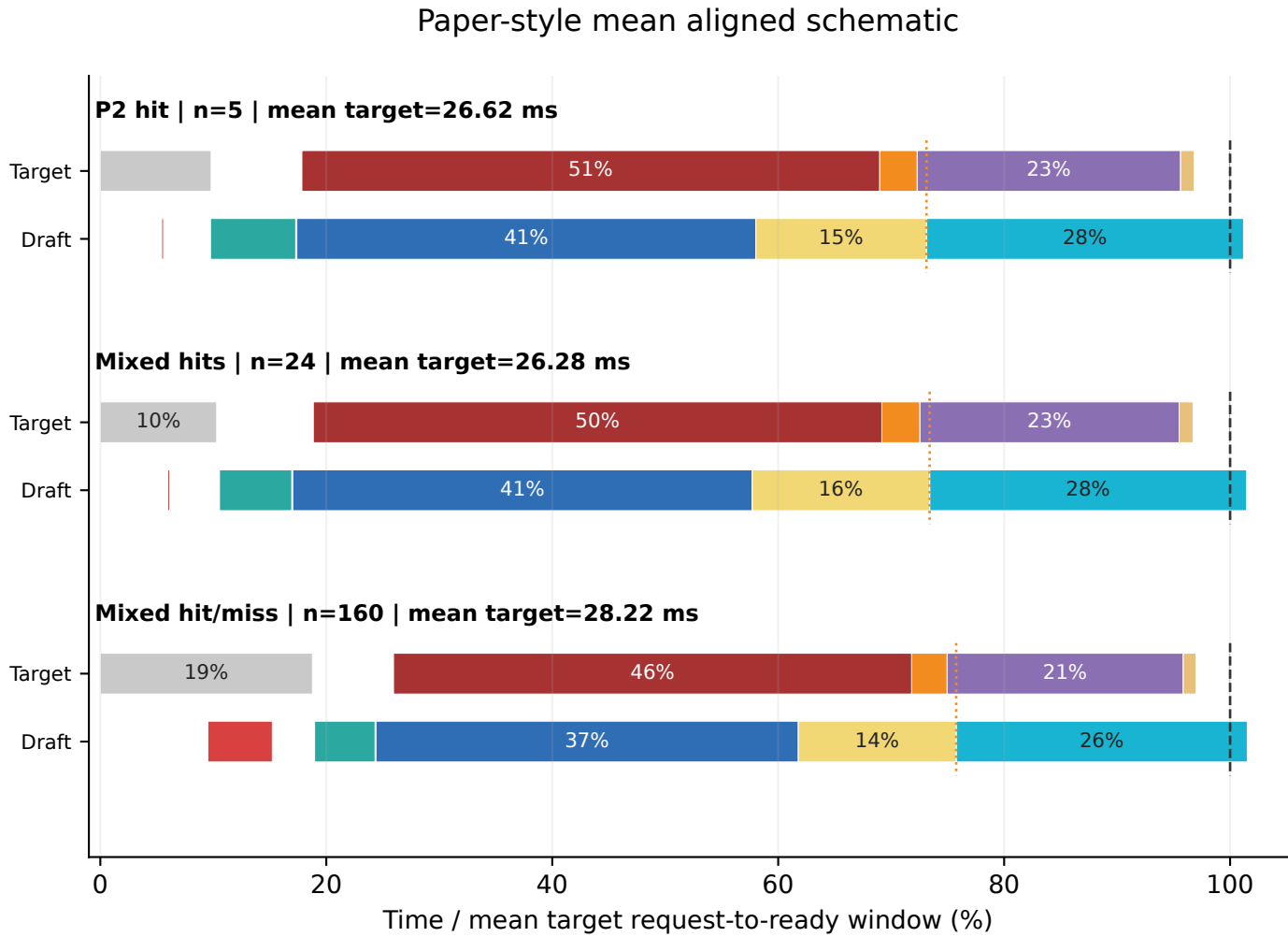
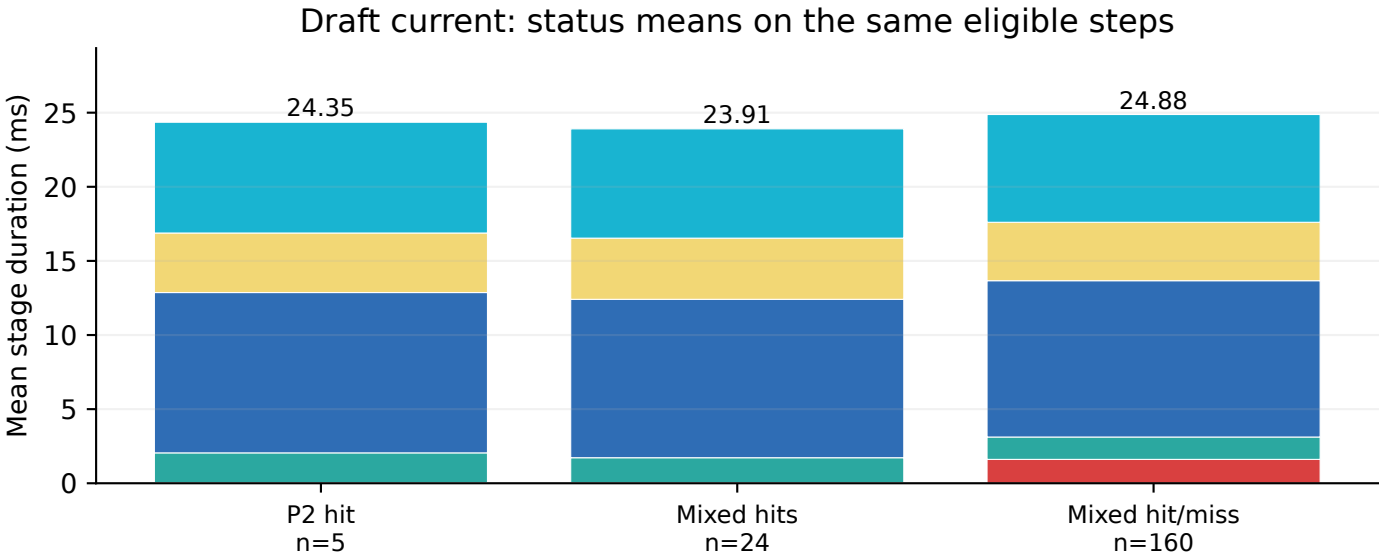
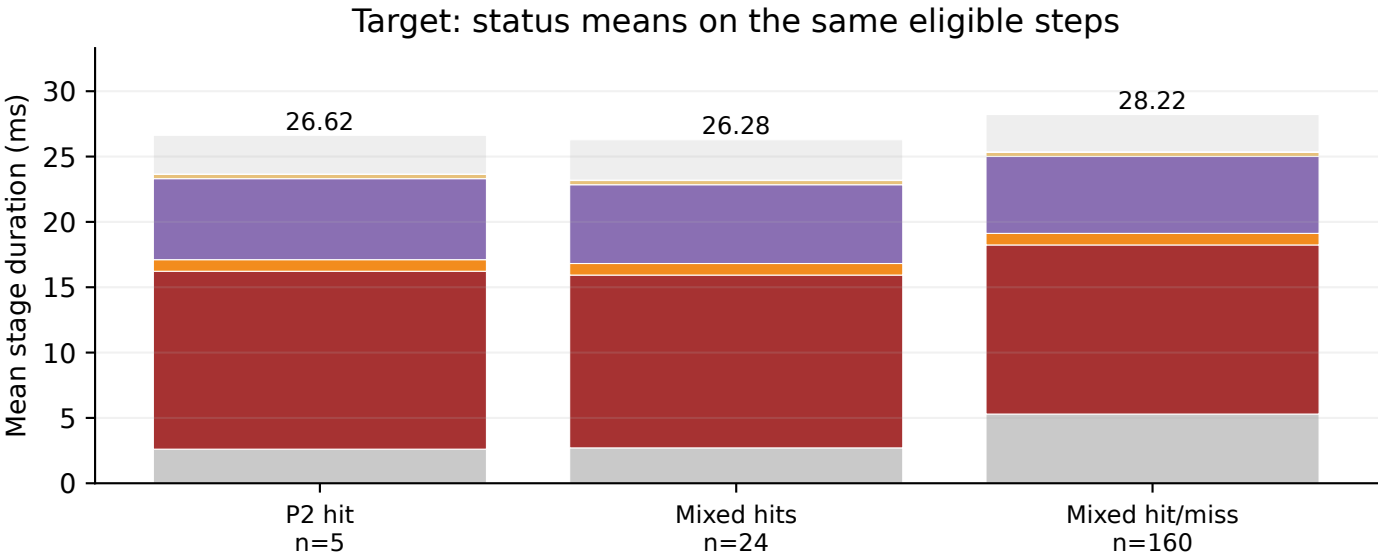
llama2_b8_combo_k2_3
DUET: K1/K2=3/3, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 190 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

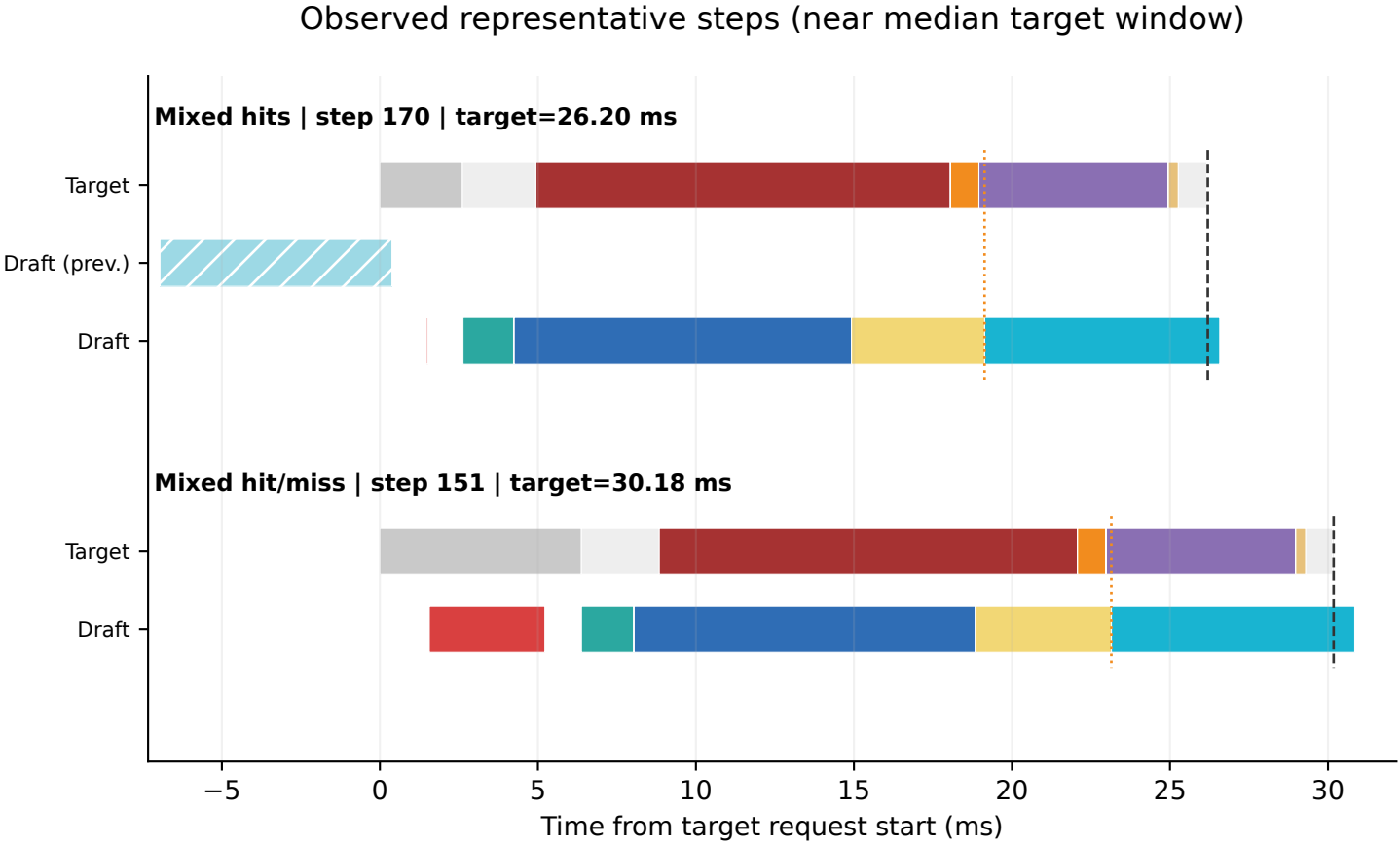
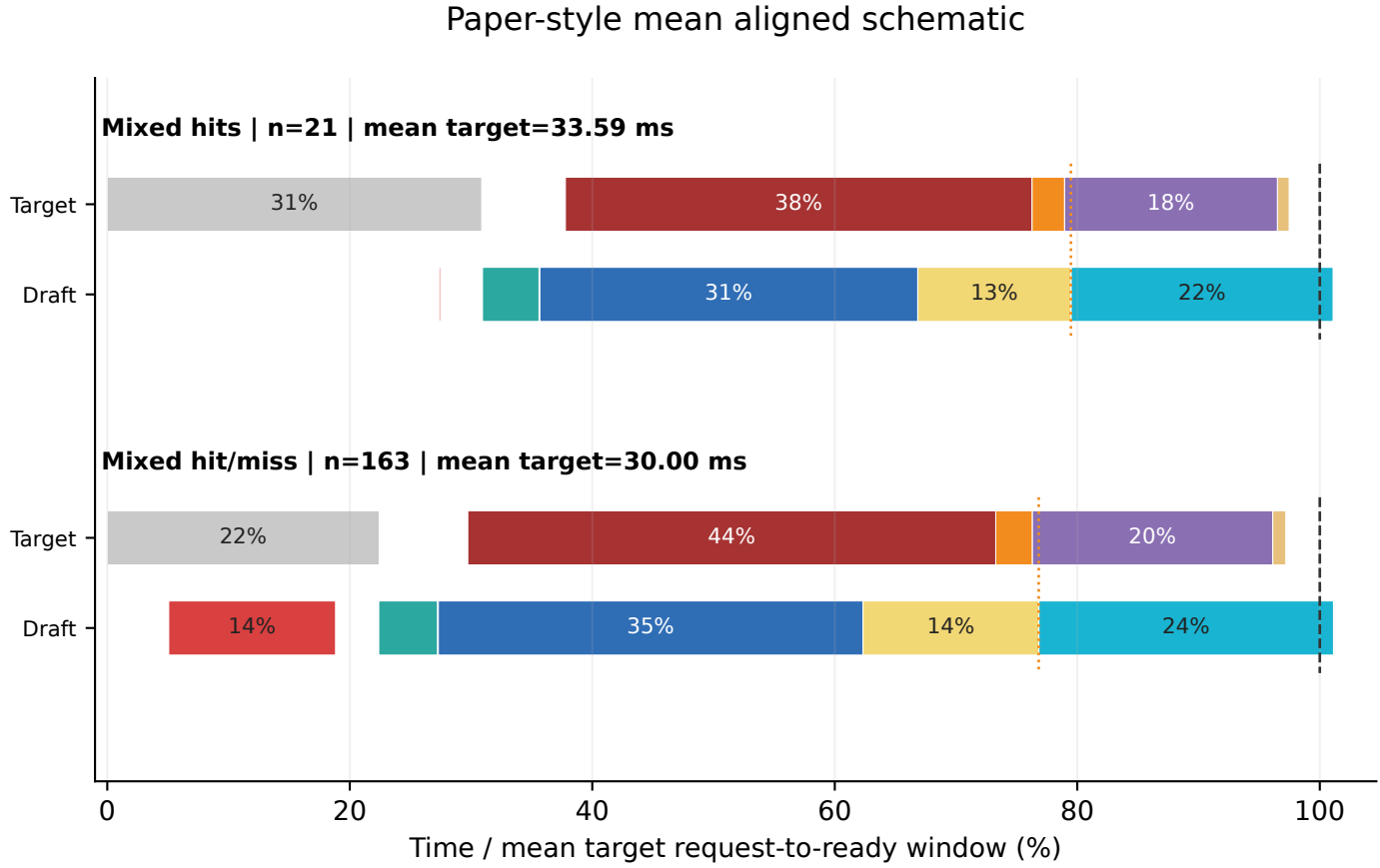
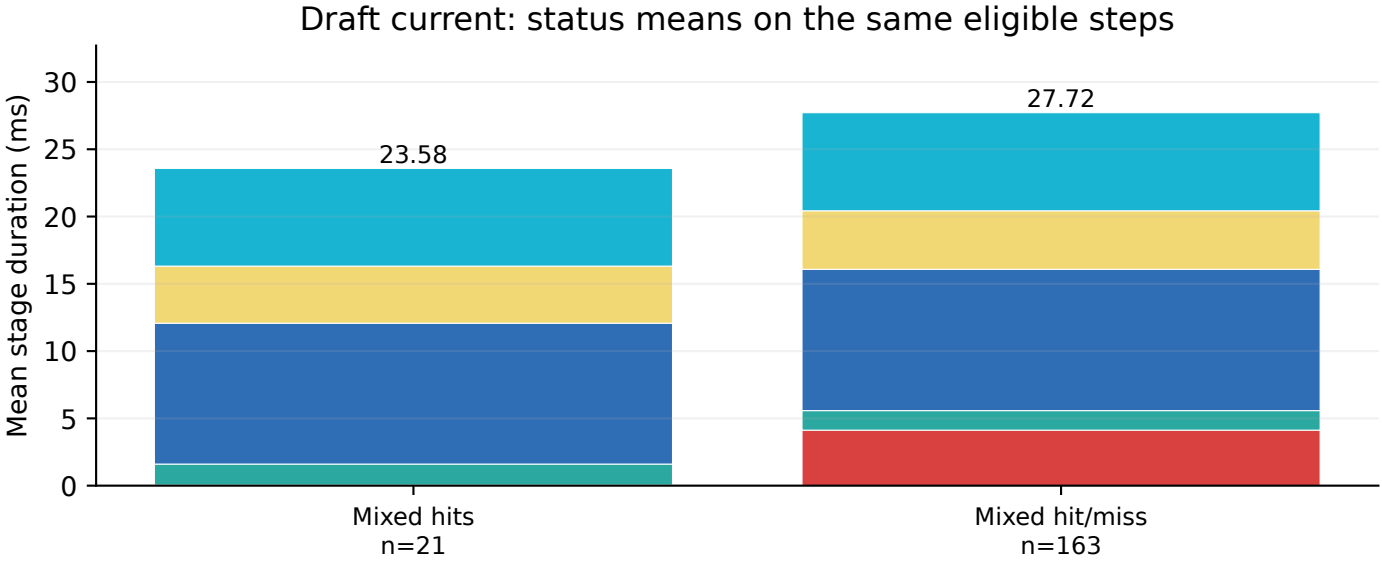
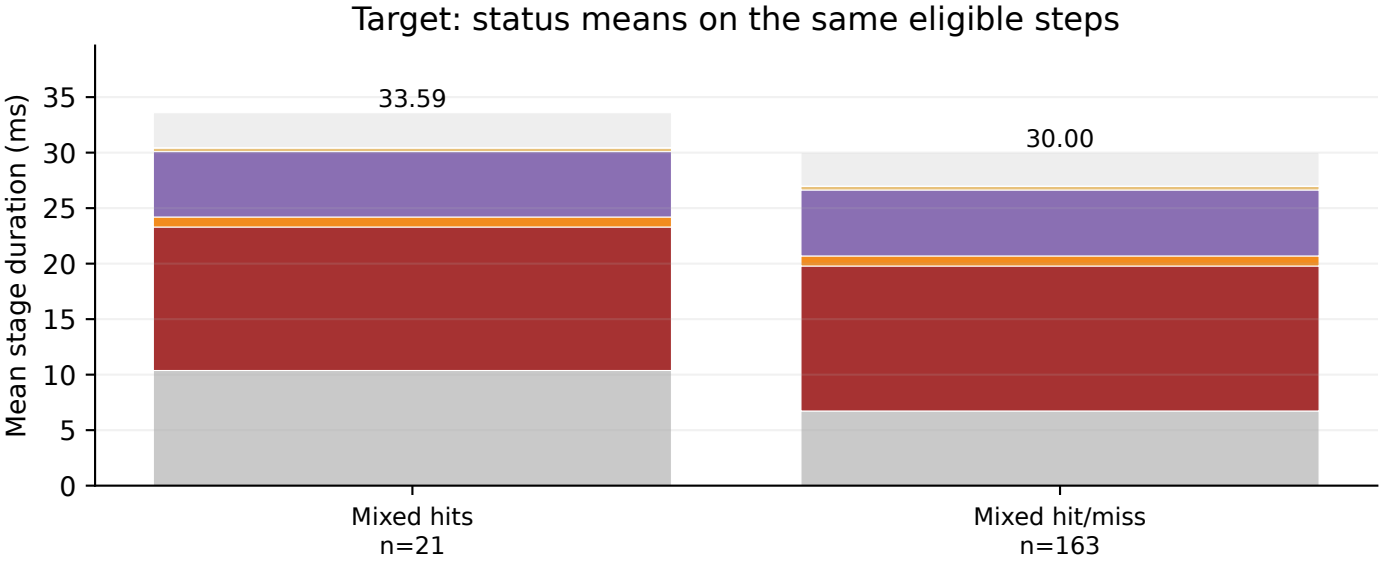
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_combo_miss2
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 189 eligible steps



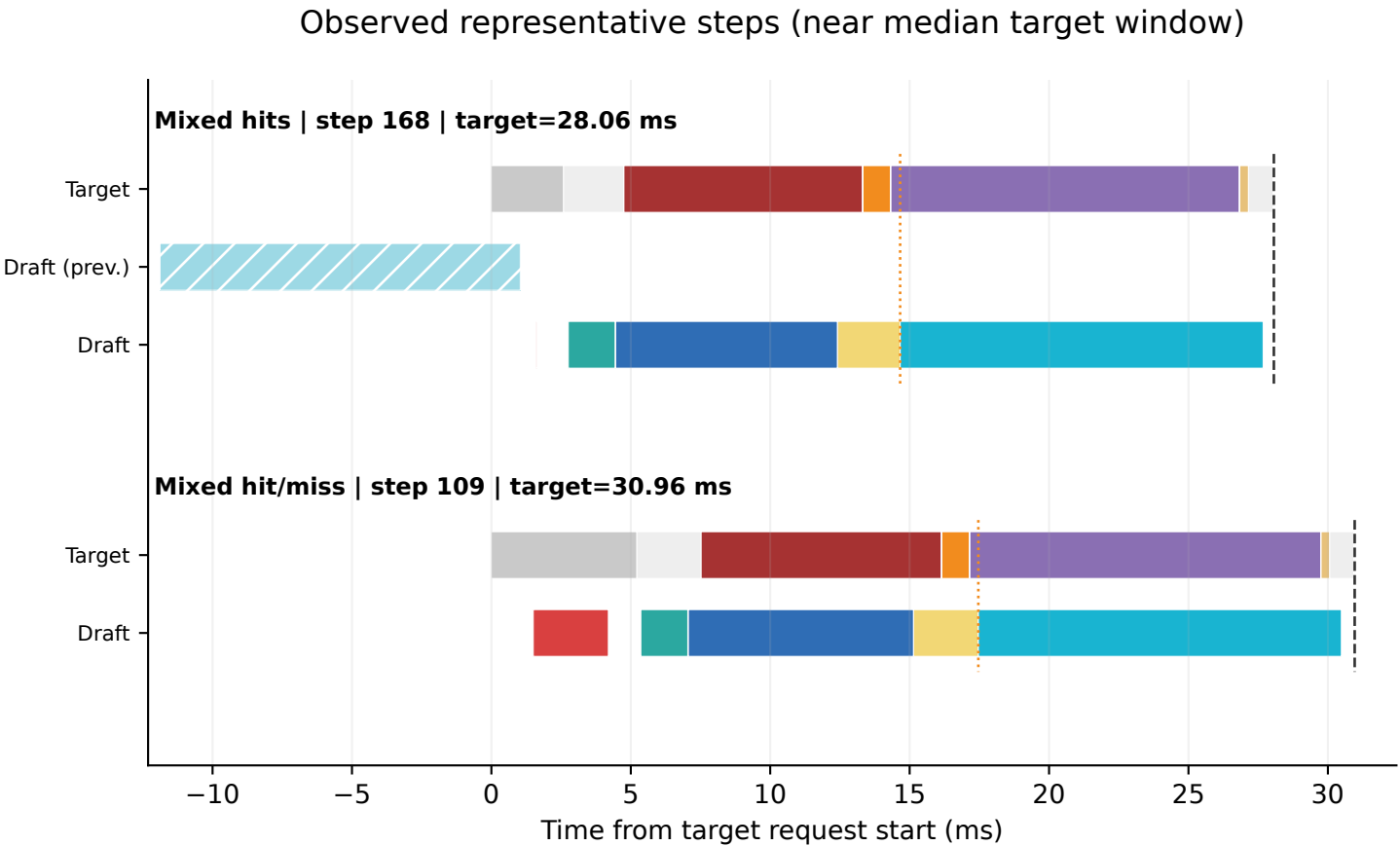
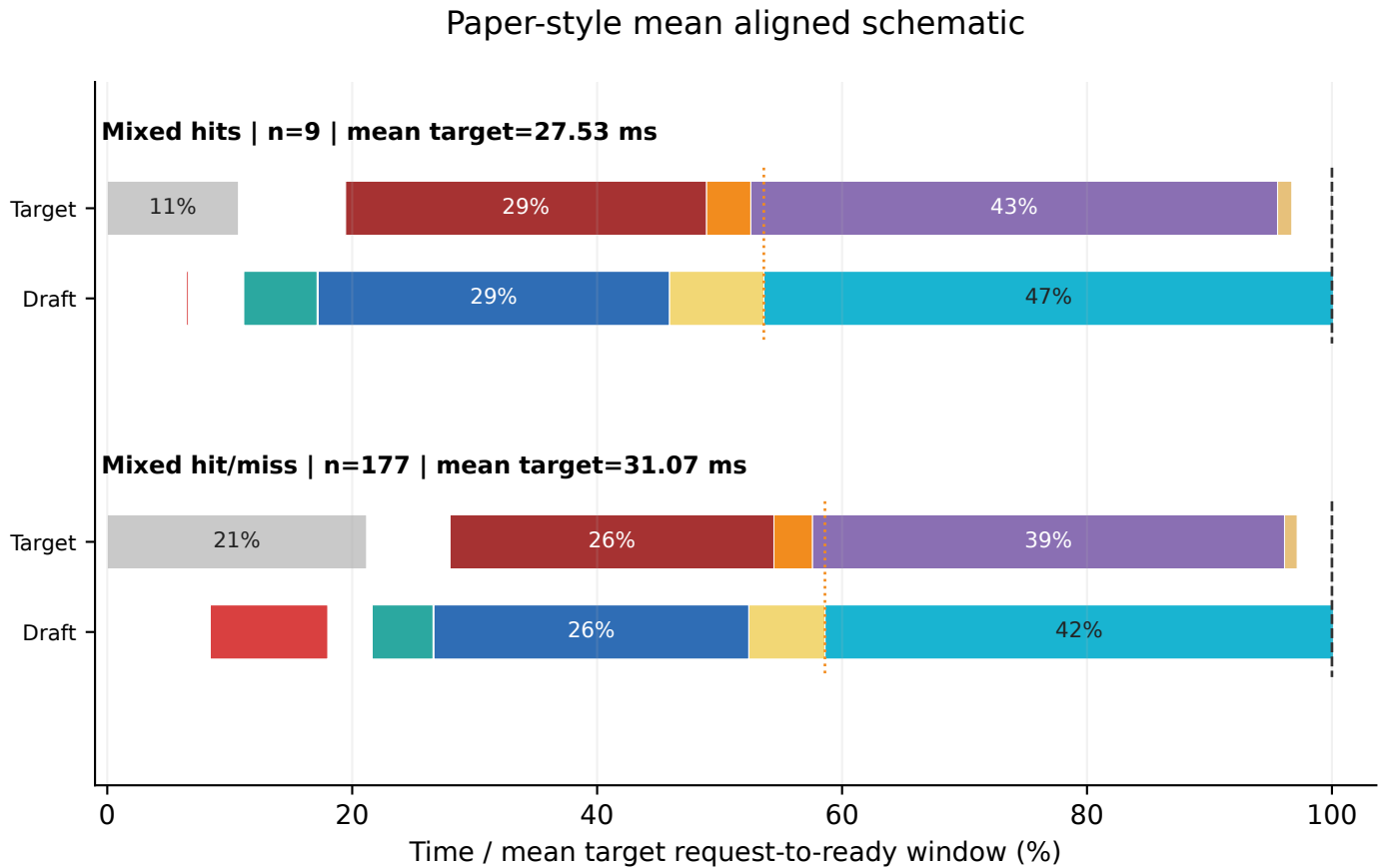
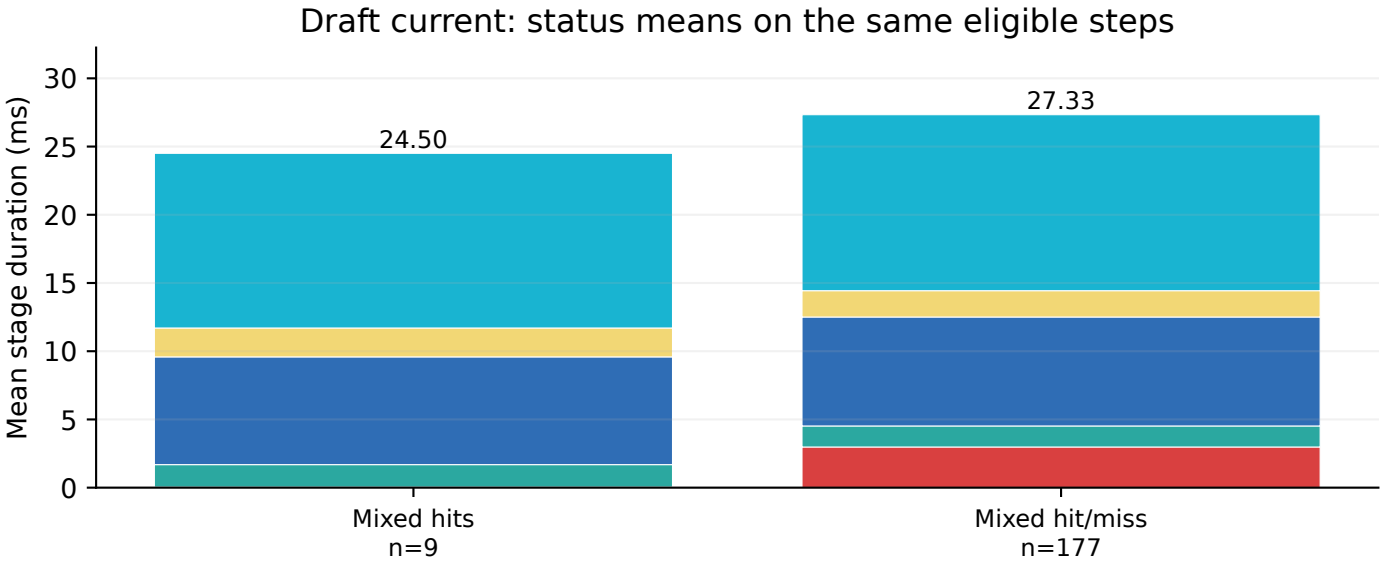
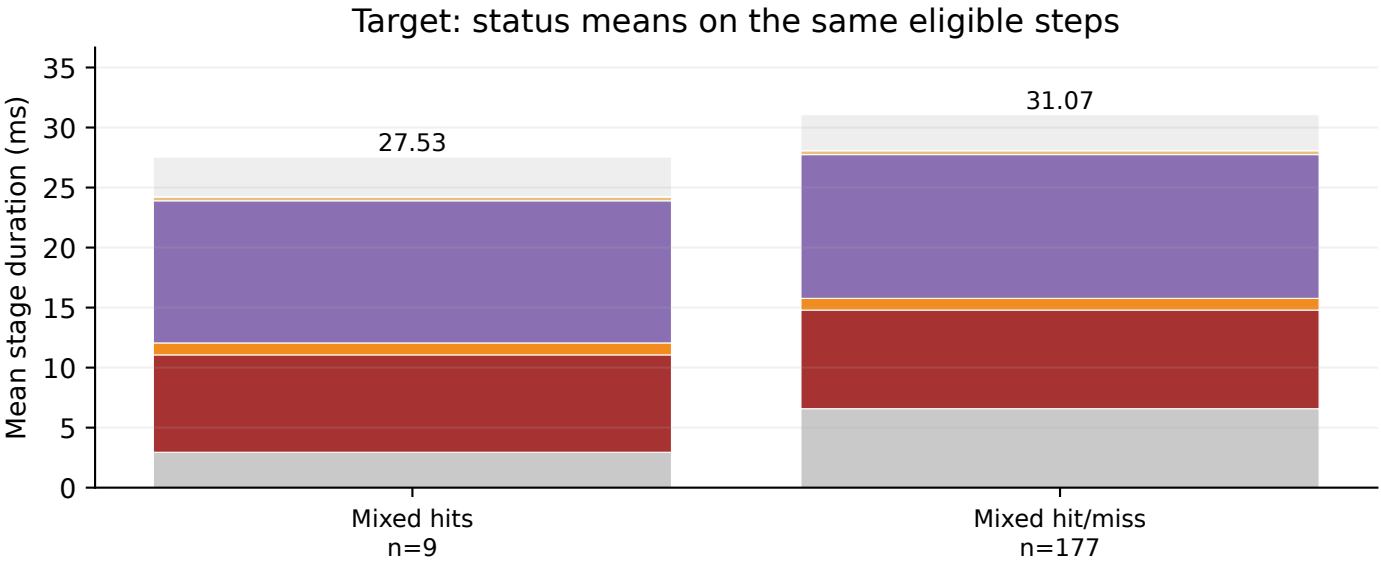
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_combo_miss6
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 184 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

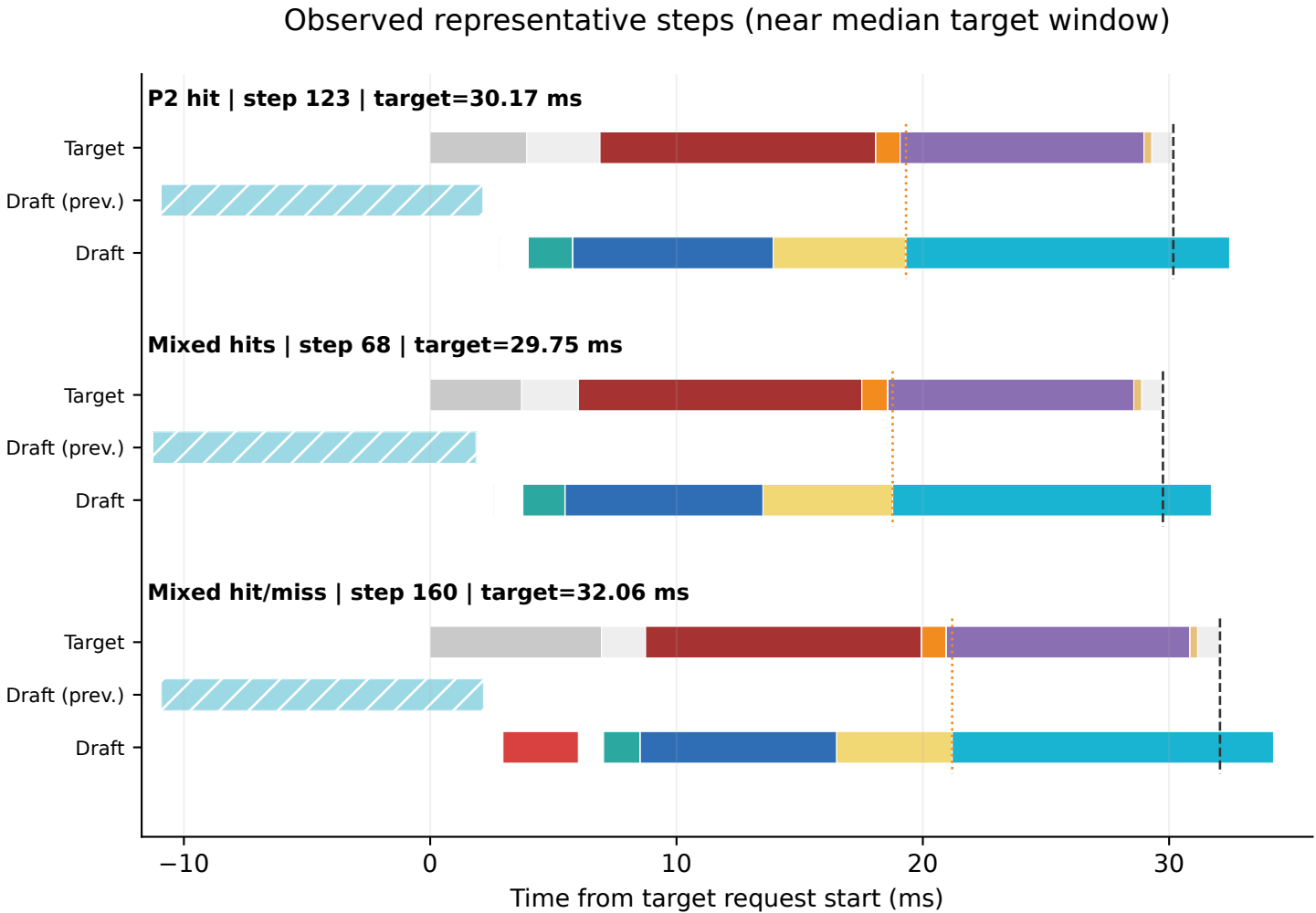
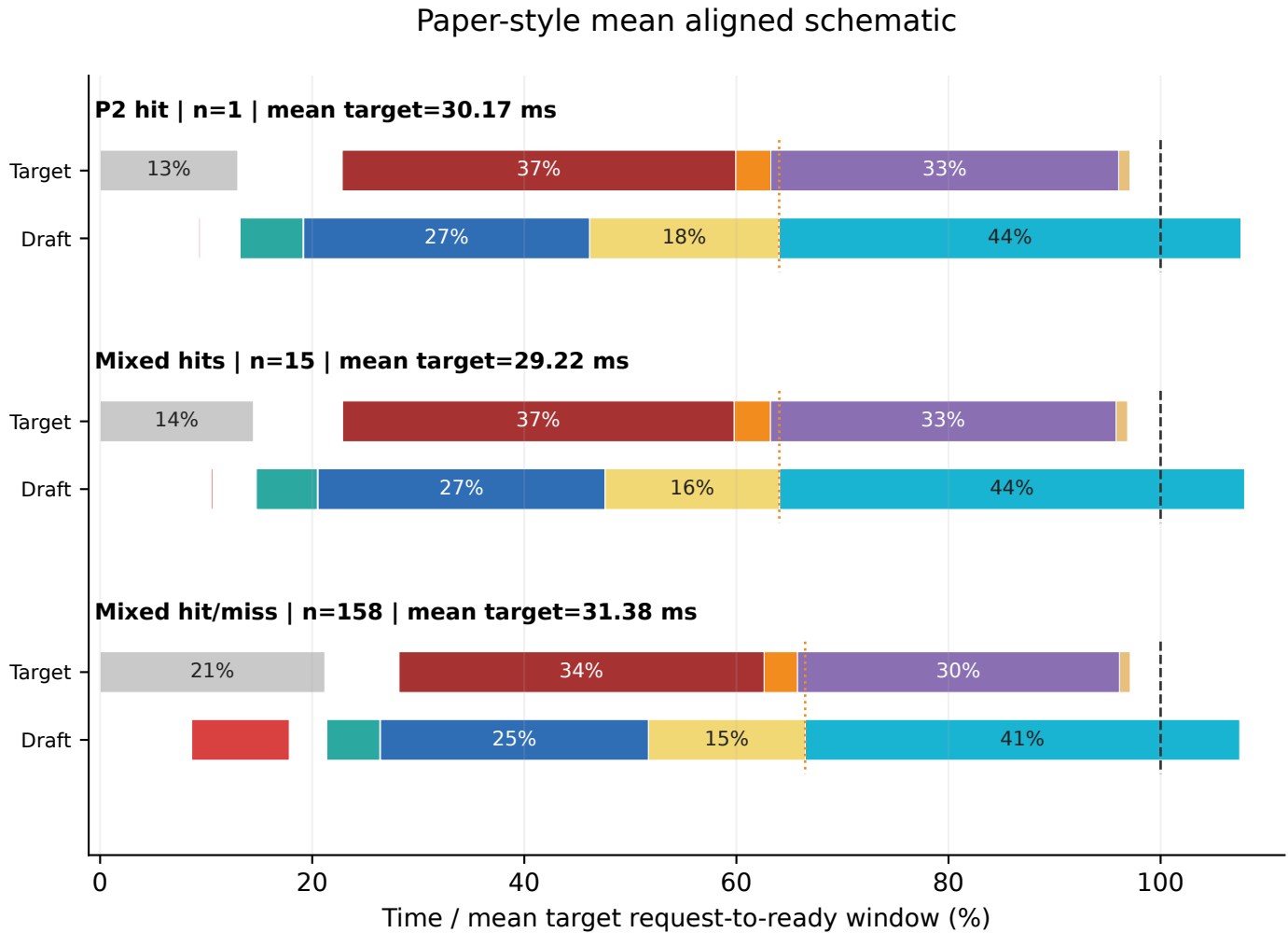
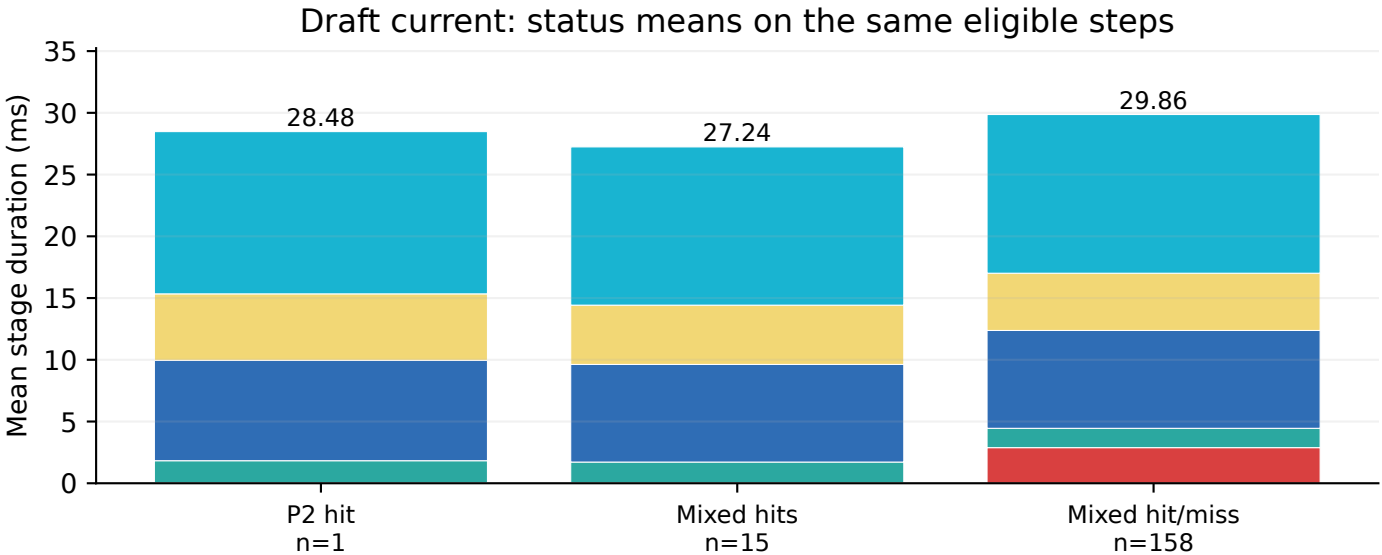
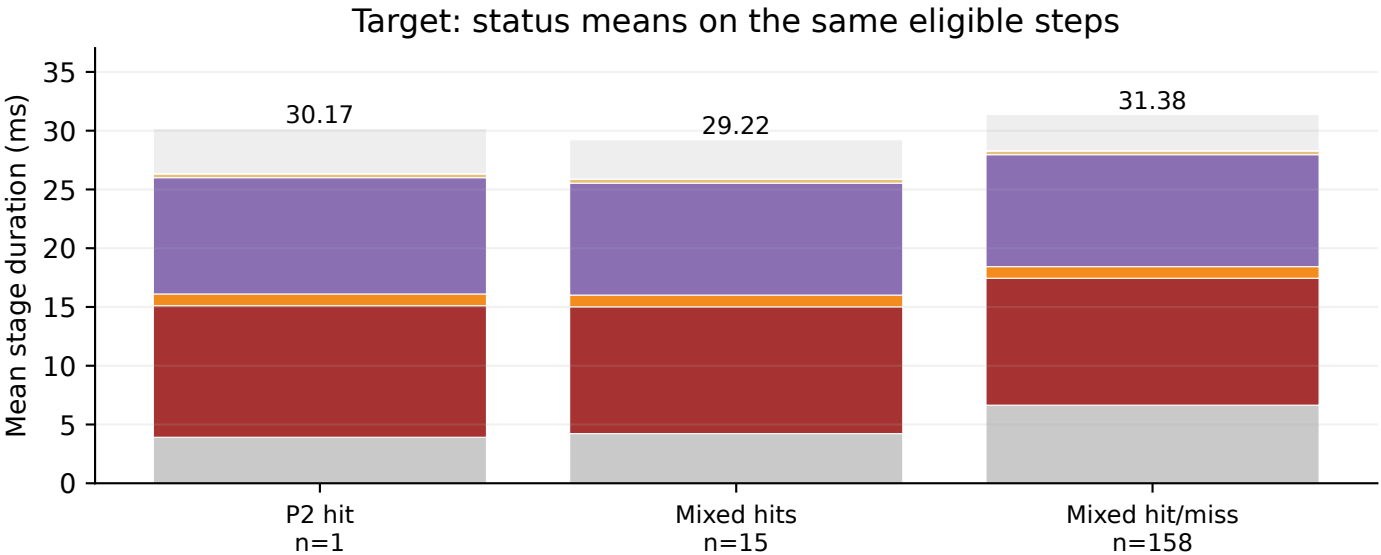
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

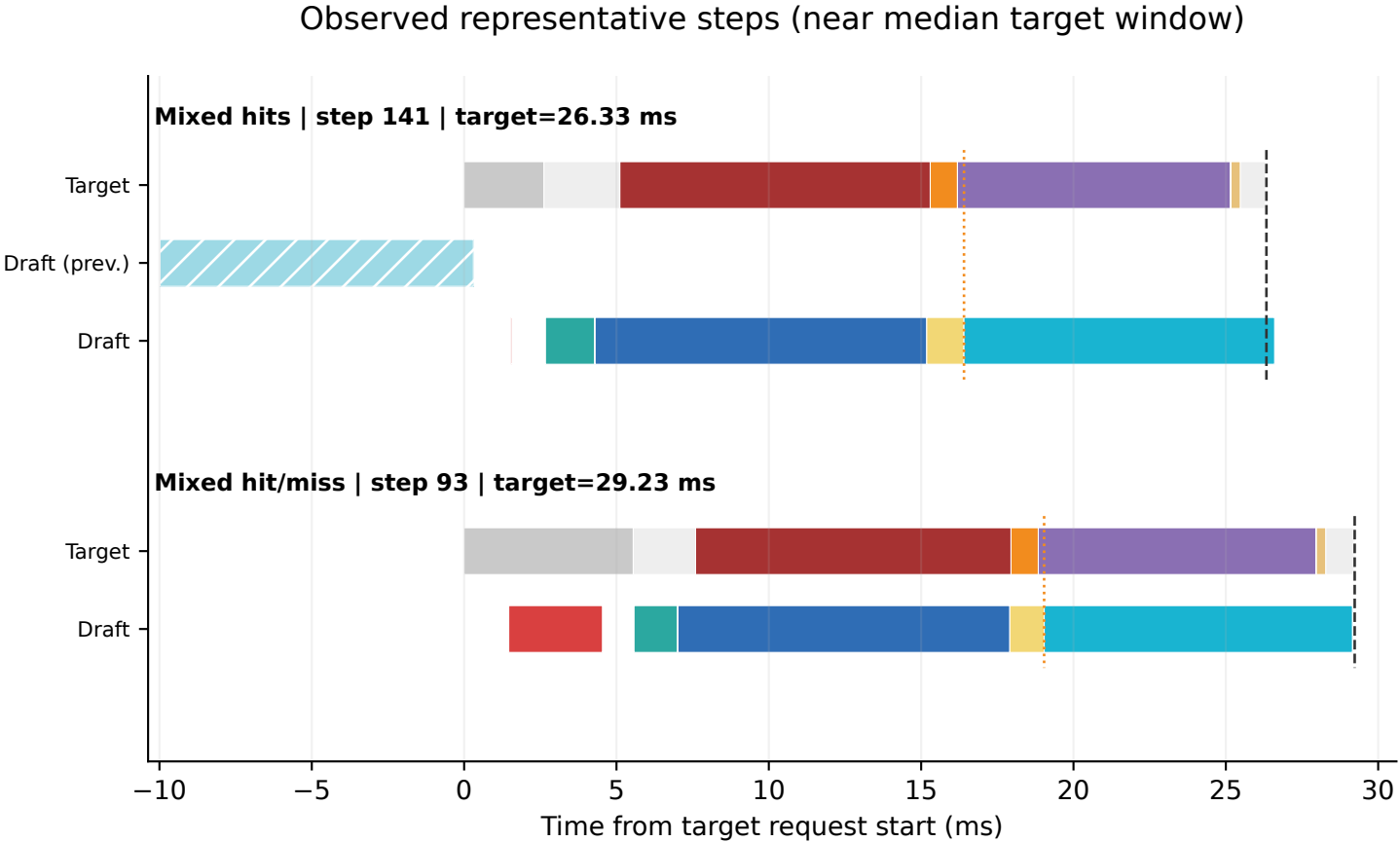
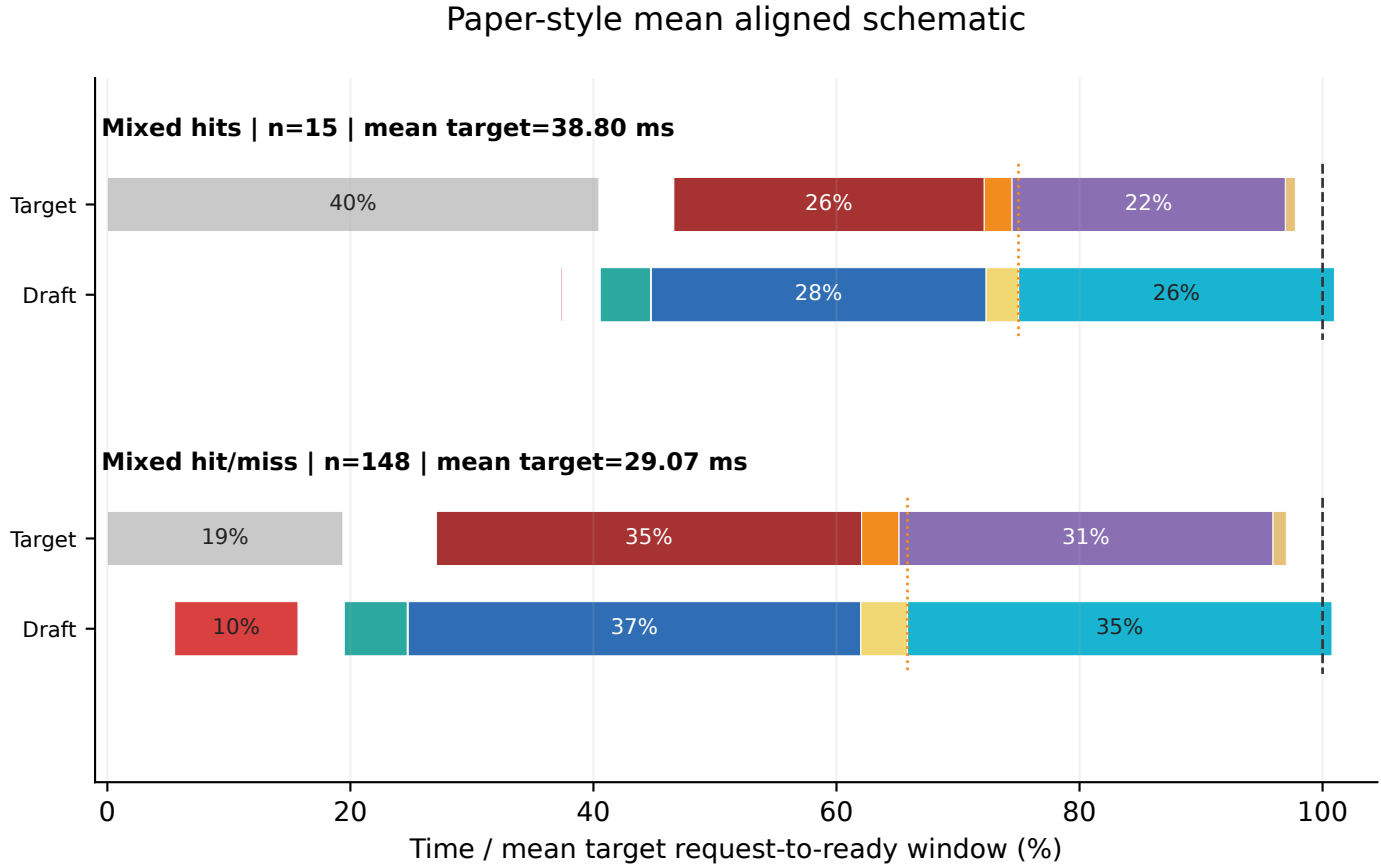
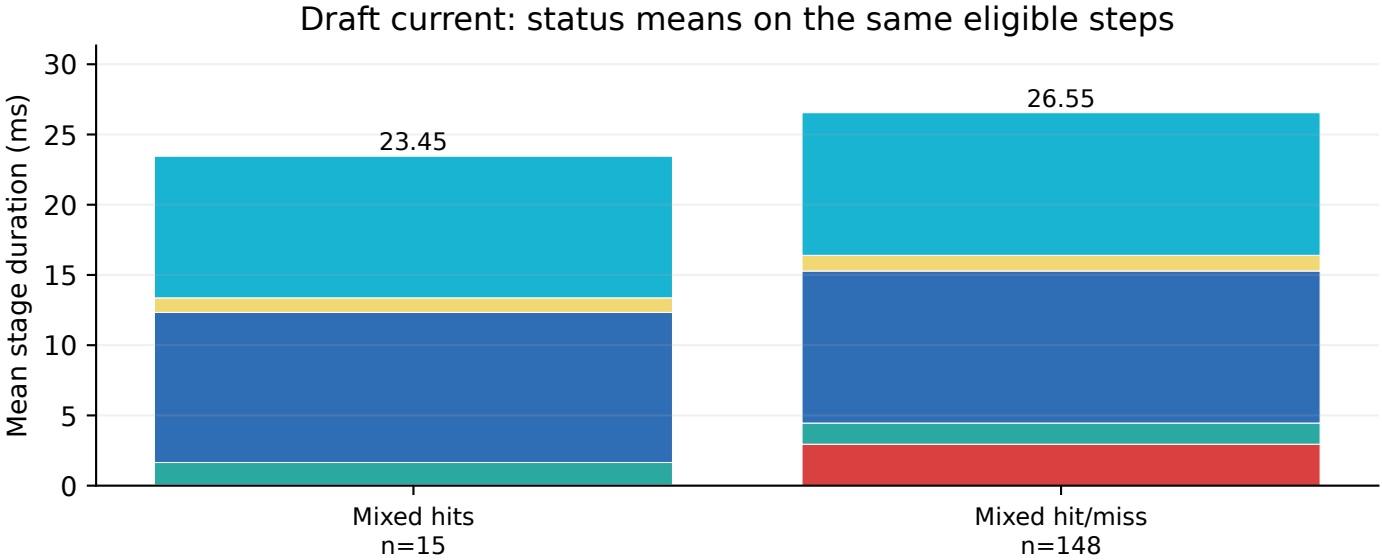
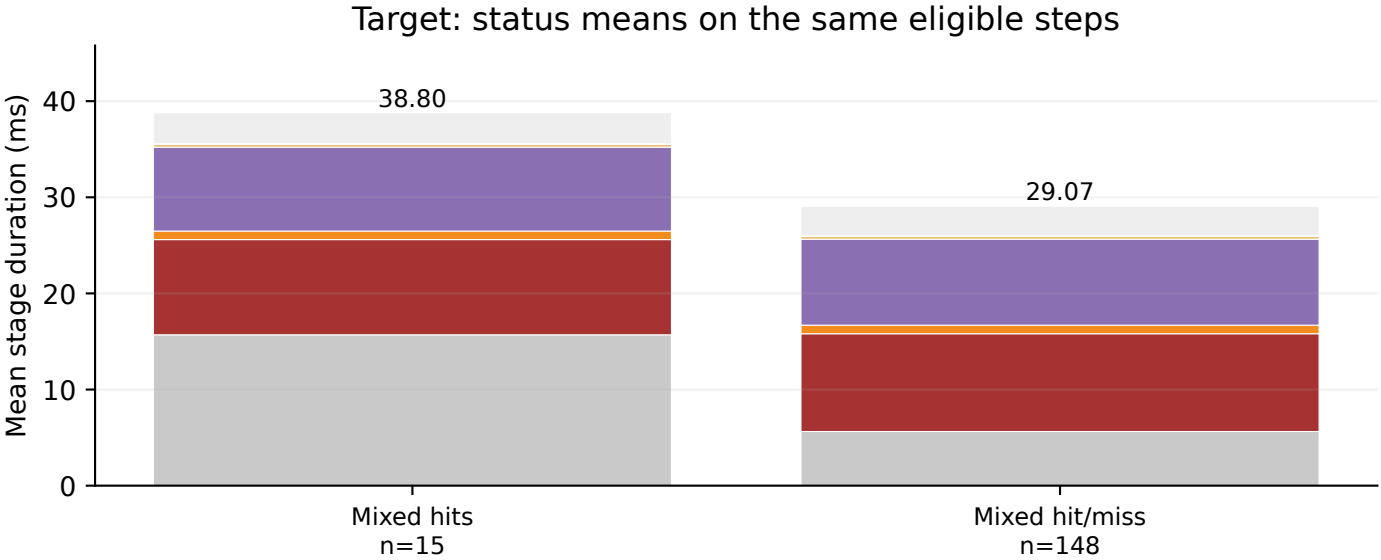
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_combo_transfer_2_4_e16
DUET: K1/K2=2/4, exit=16 (0-based), C=3, N1/N2=4/8, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 174 eligible steps



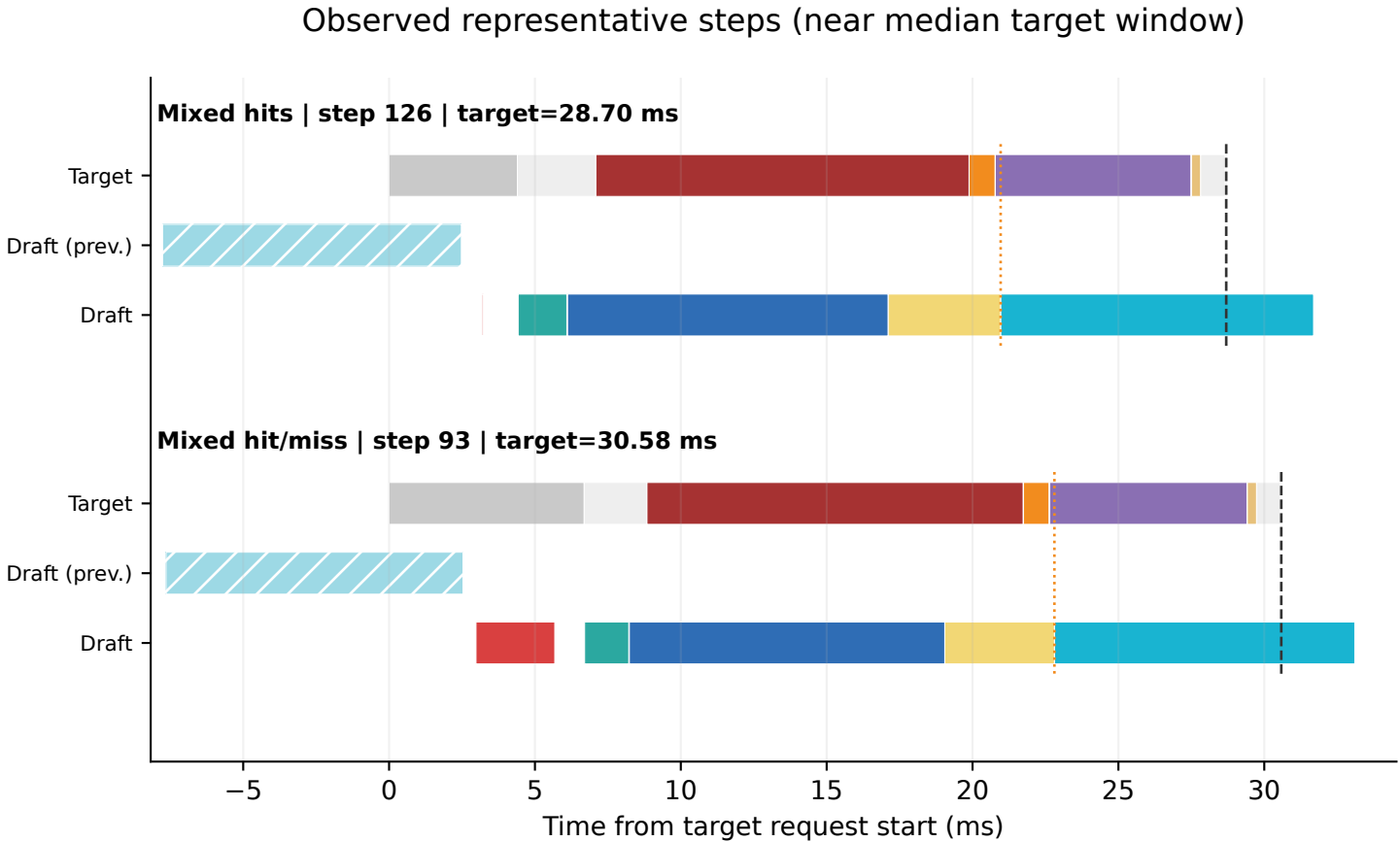
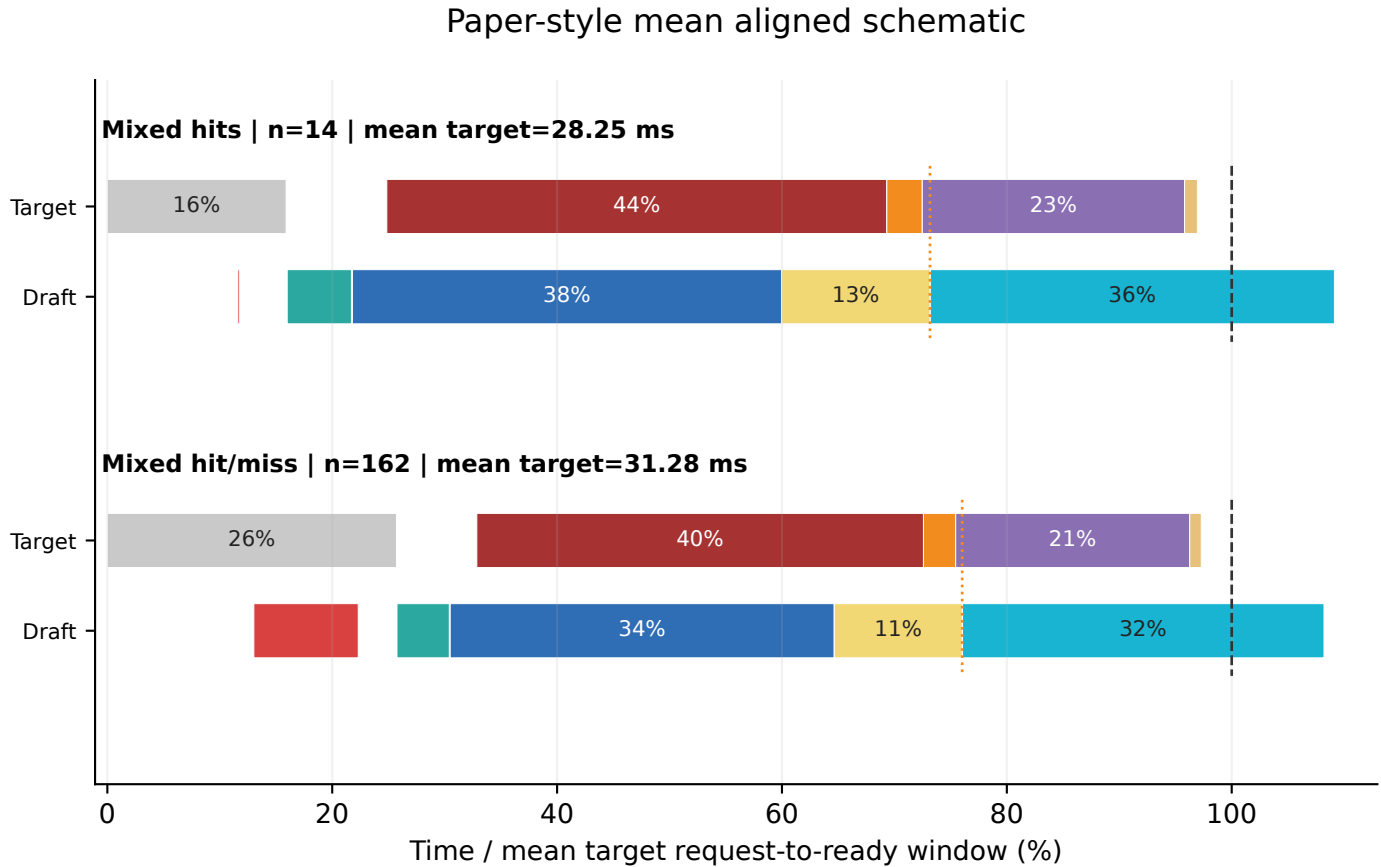
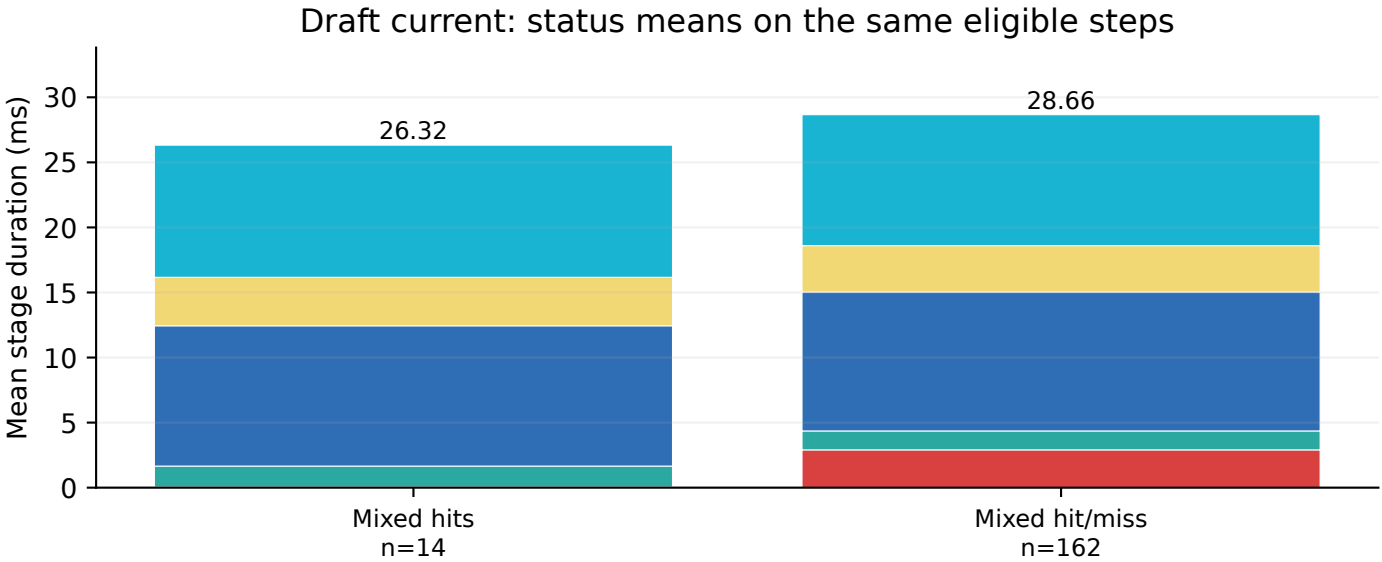
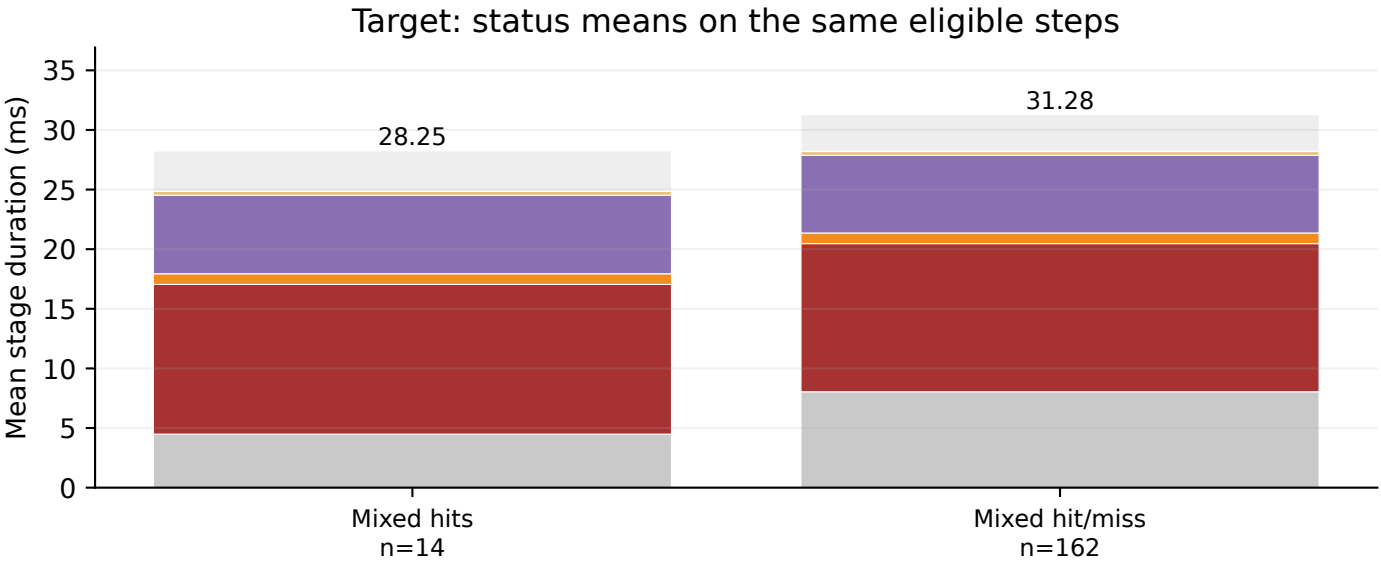
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

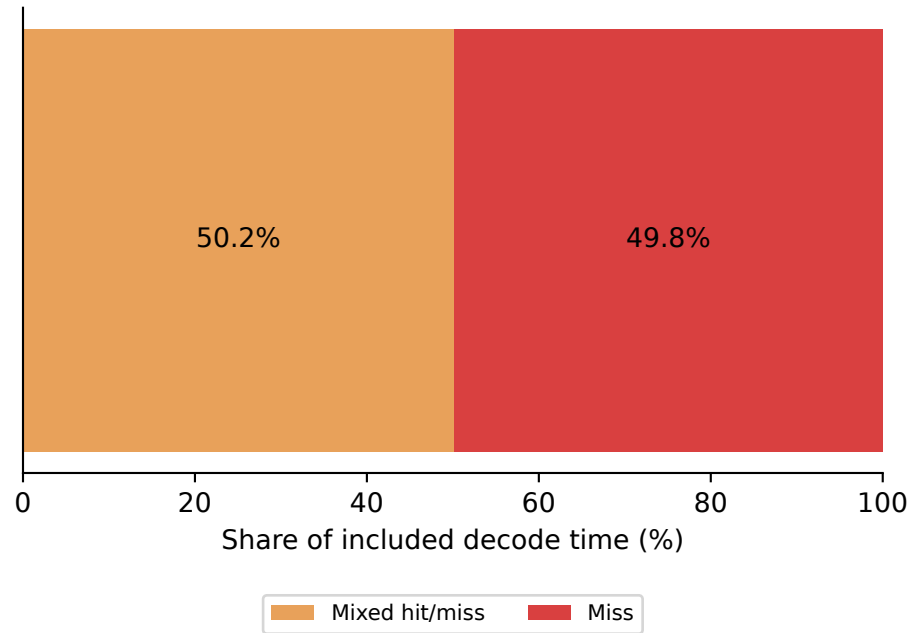


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

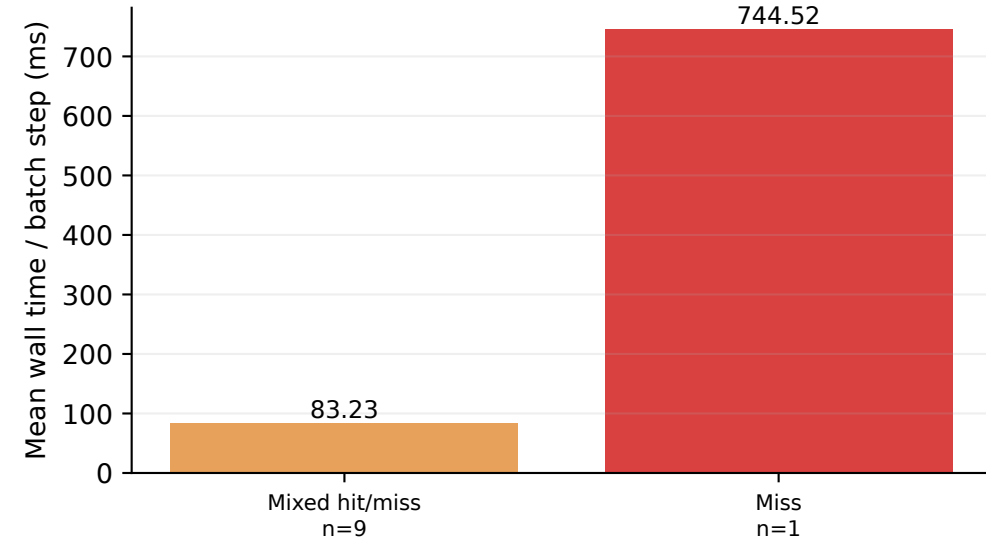
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_transfer_smoke
DUET: K1/K2=1/3, exit=12 (0-based), C=3, N1/N2=2/6, U=2, W=3, trim/stream=0/0
Last pass: B=8, T=0.7, N=8 | AL*=1.914, hit=59.1%, TPS*=85.70

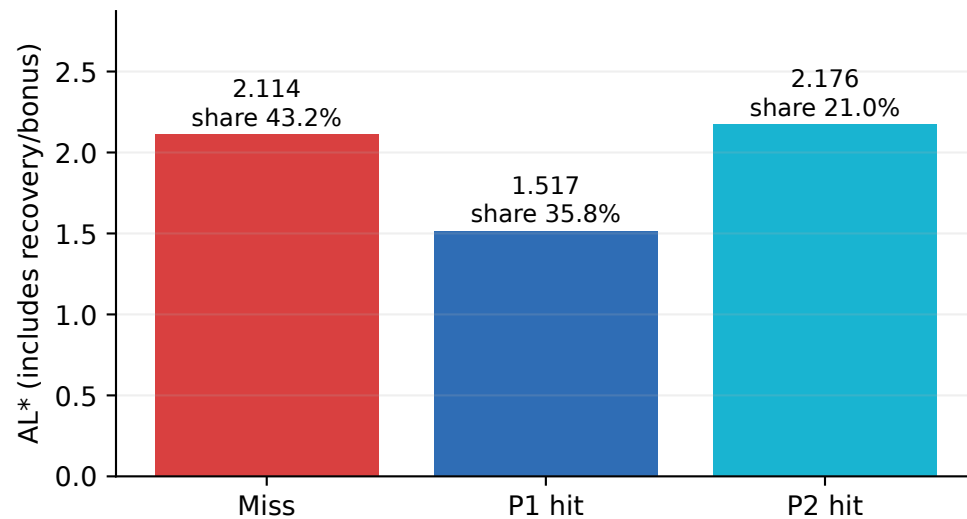
Measured wall time by batch cache status



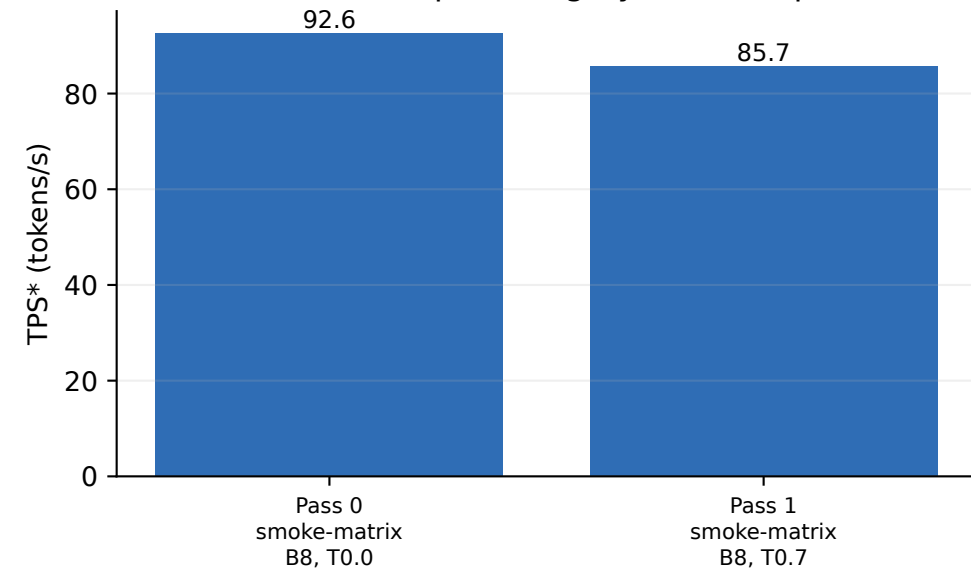
Boundary-free steps, including slow steps



Conditional AL* and source share

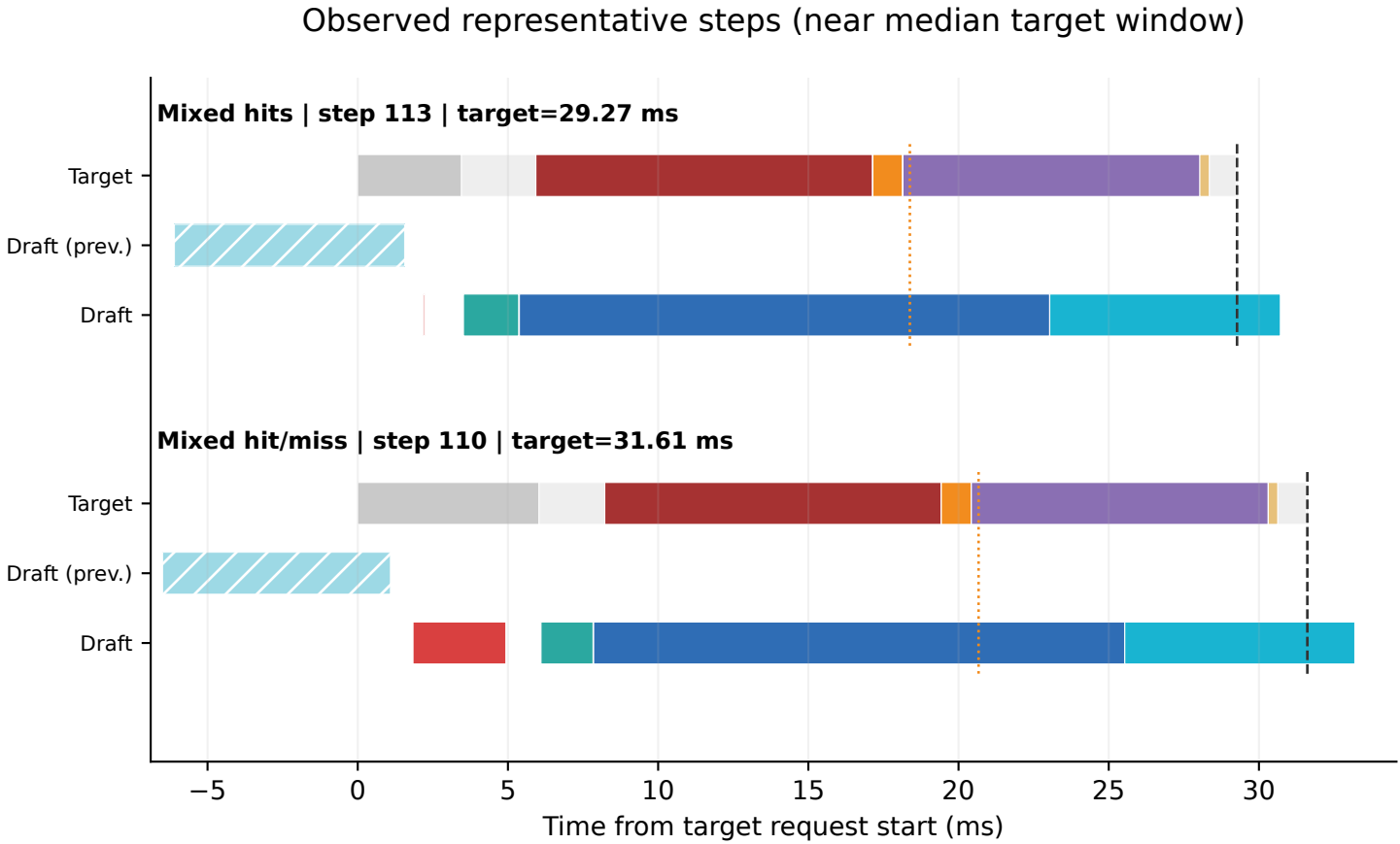
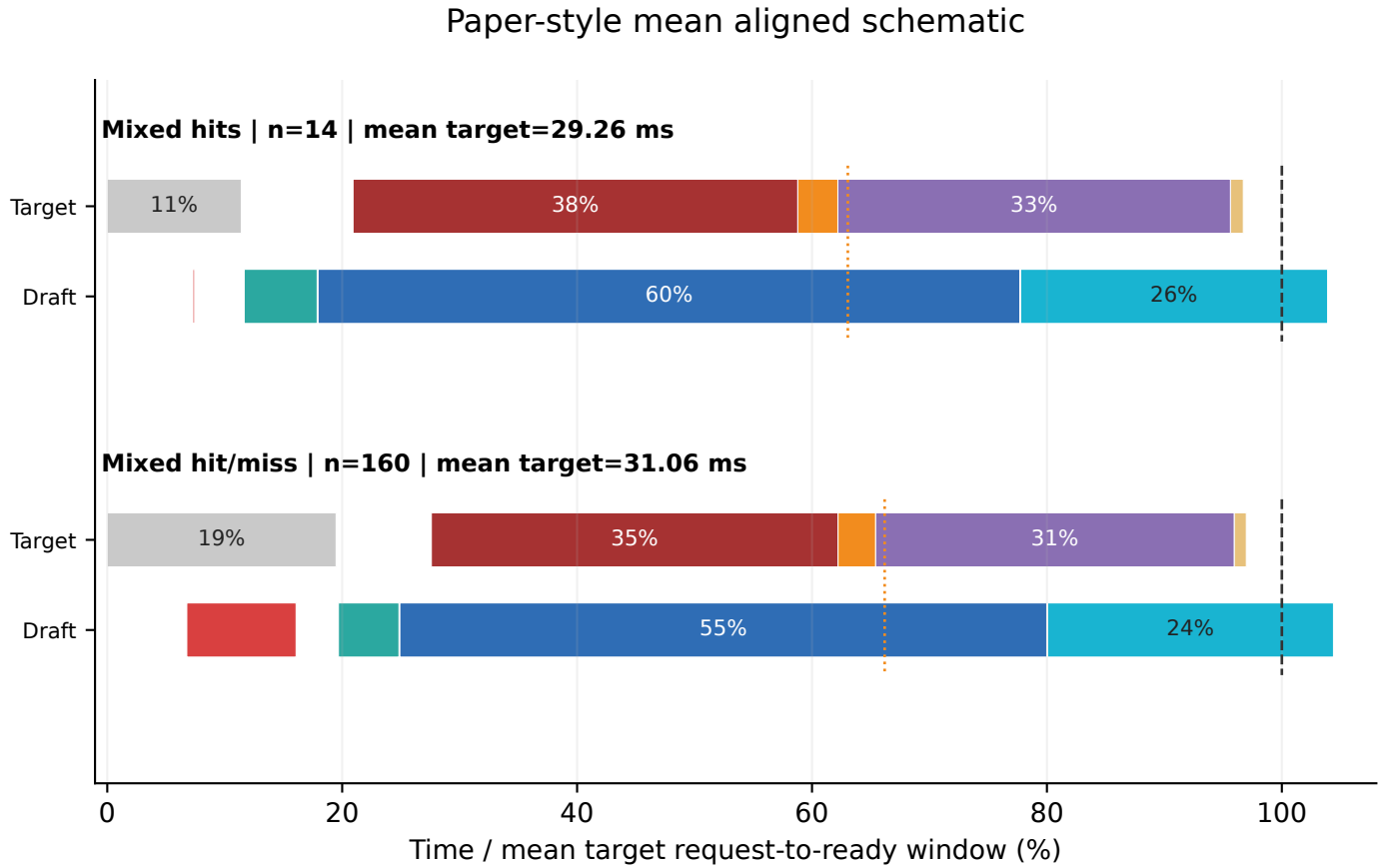
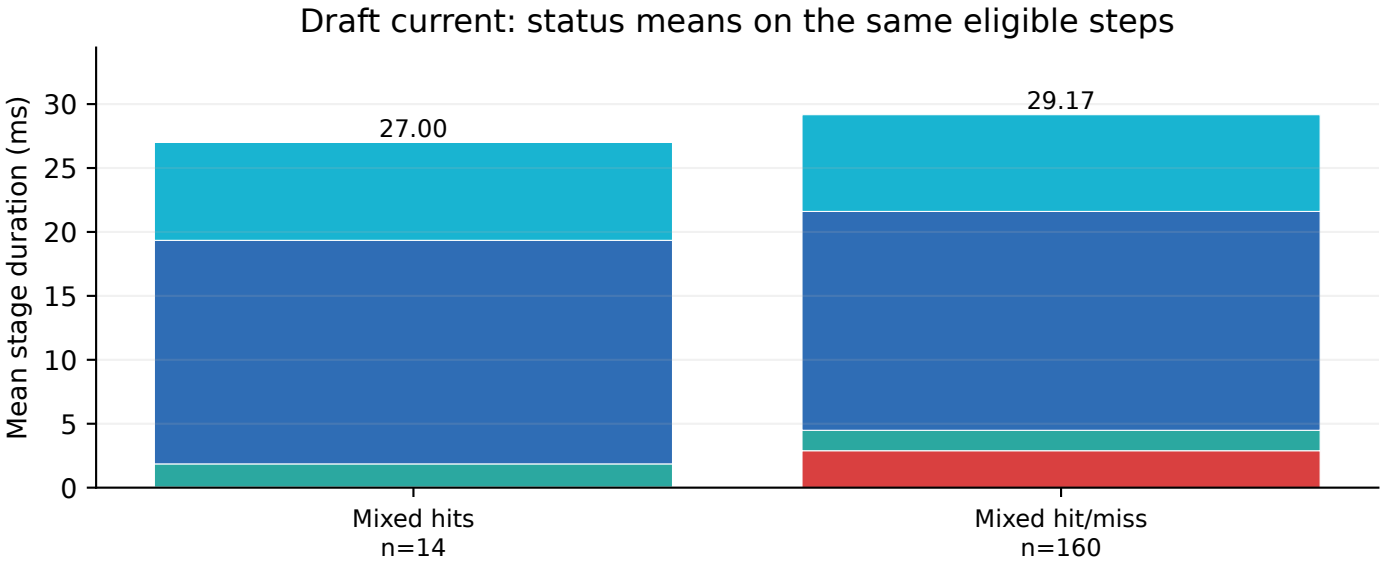
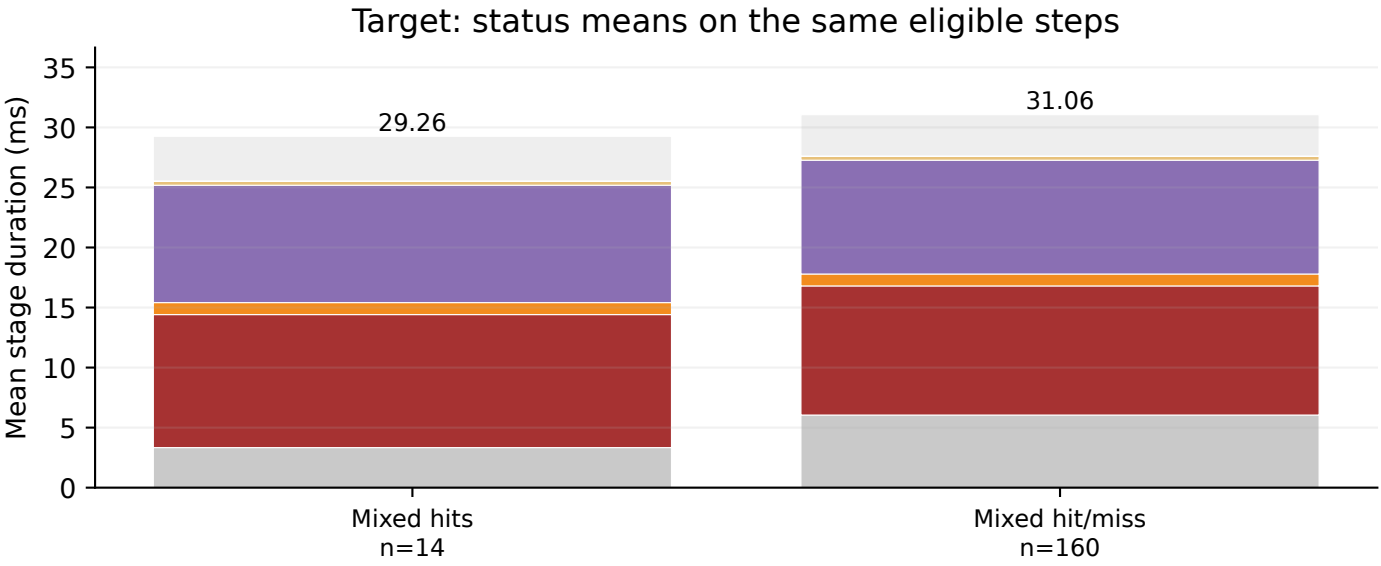


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

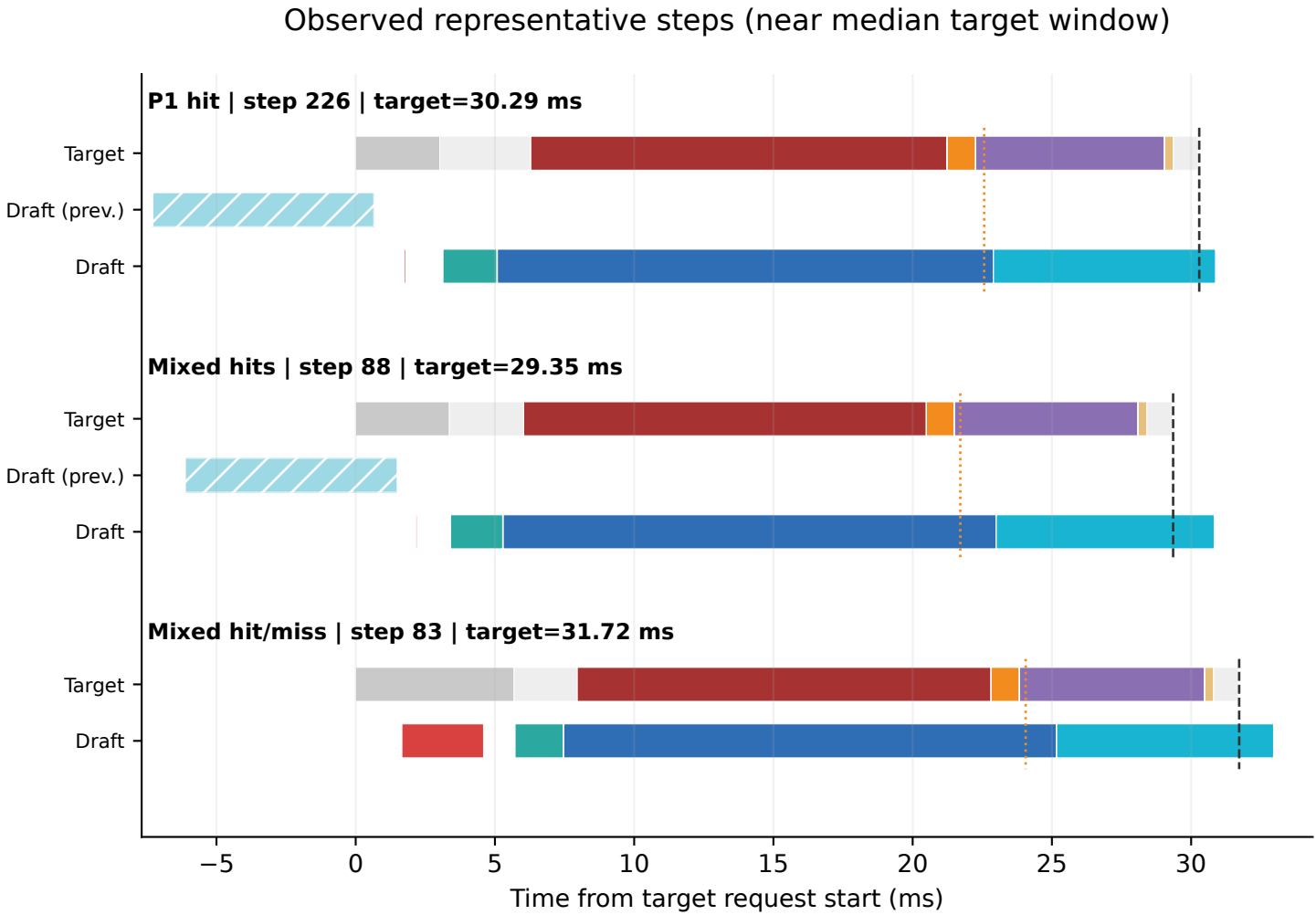
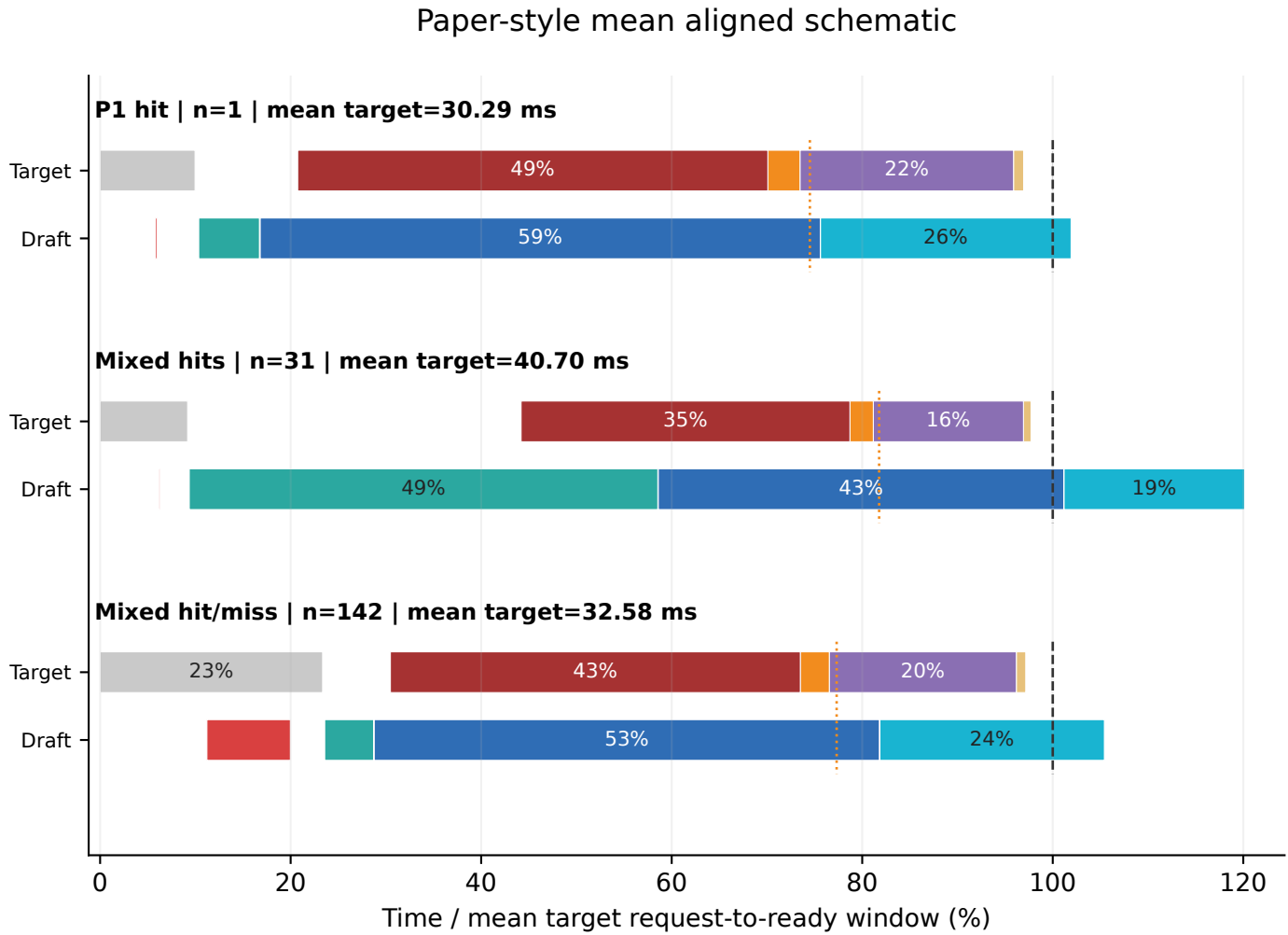
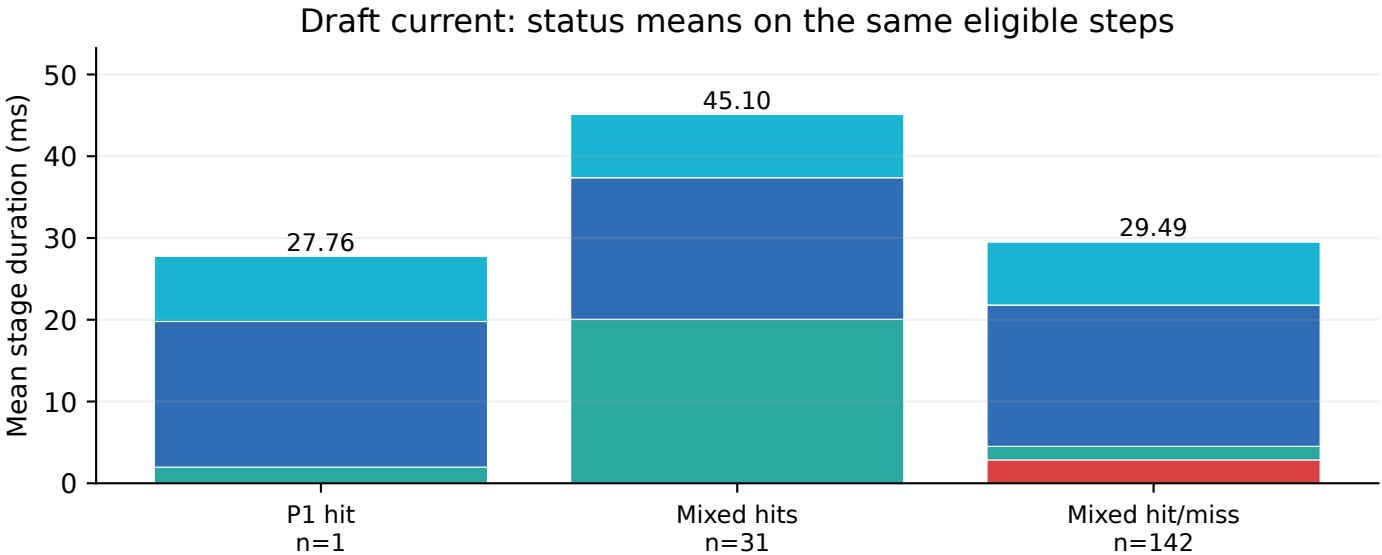
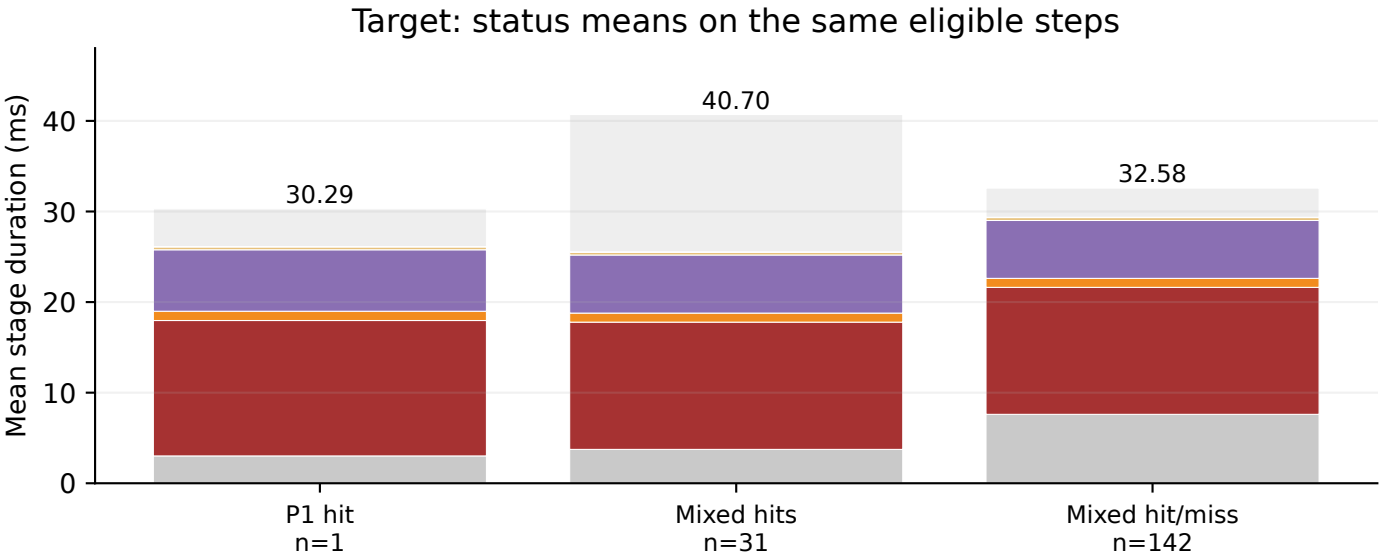
llama2_b8_duet_e16_k4_2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 174 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

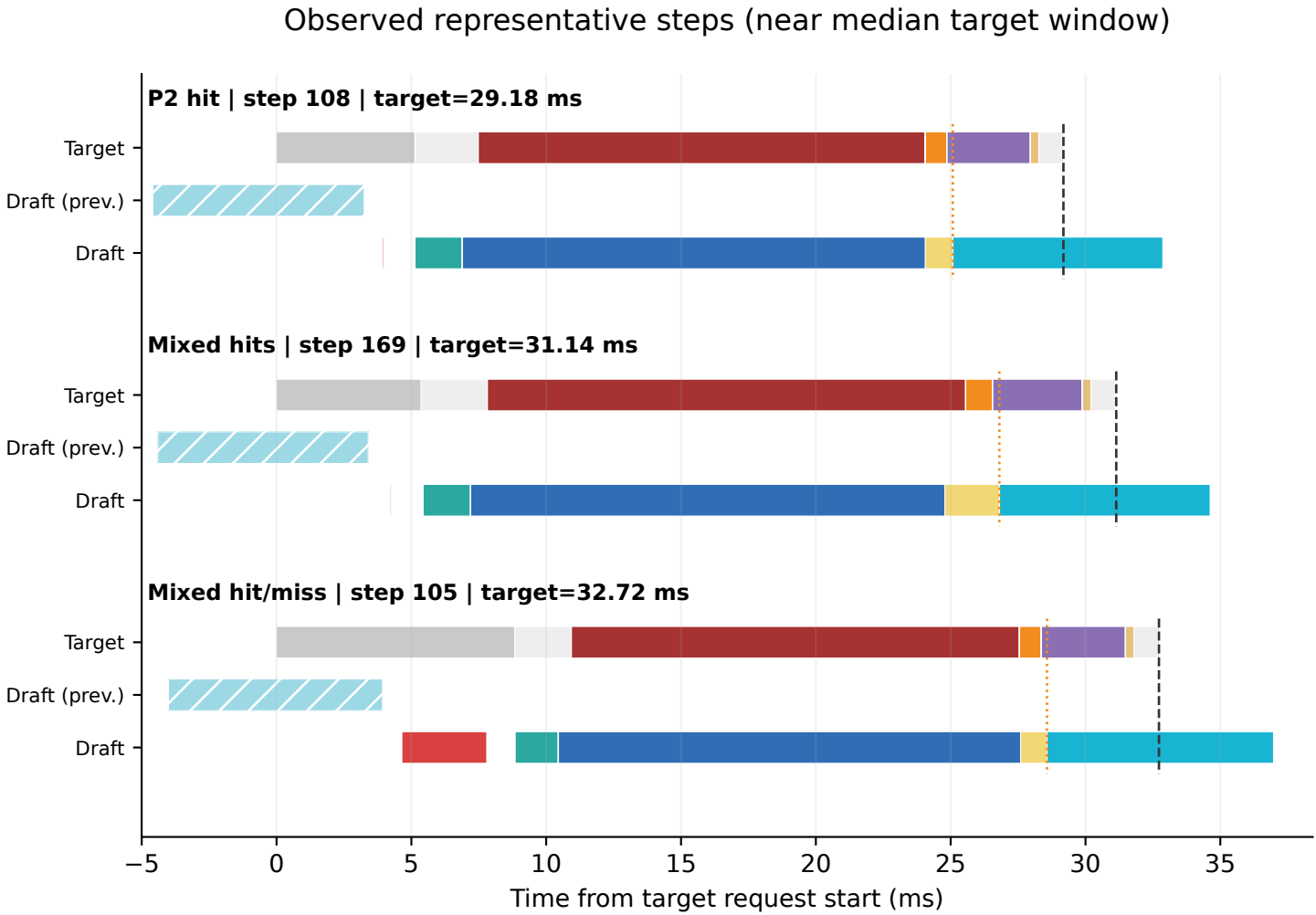
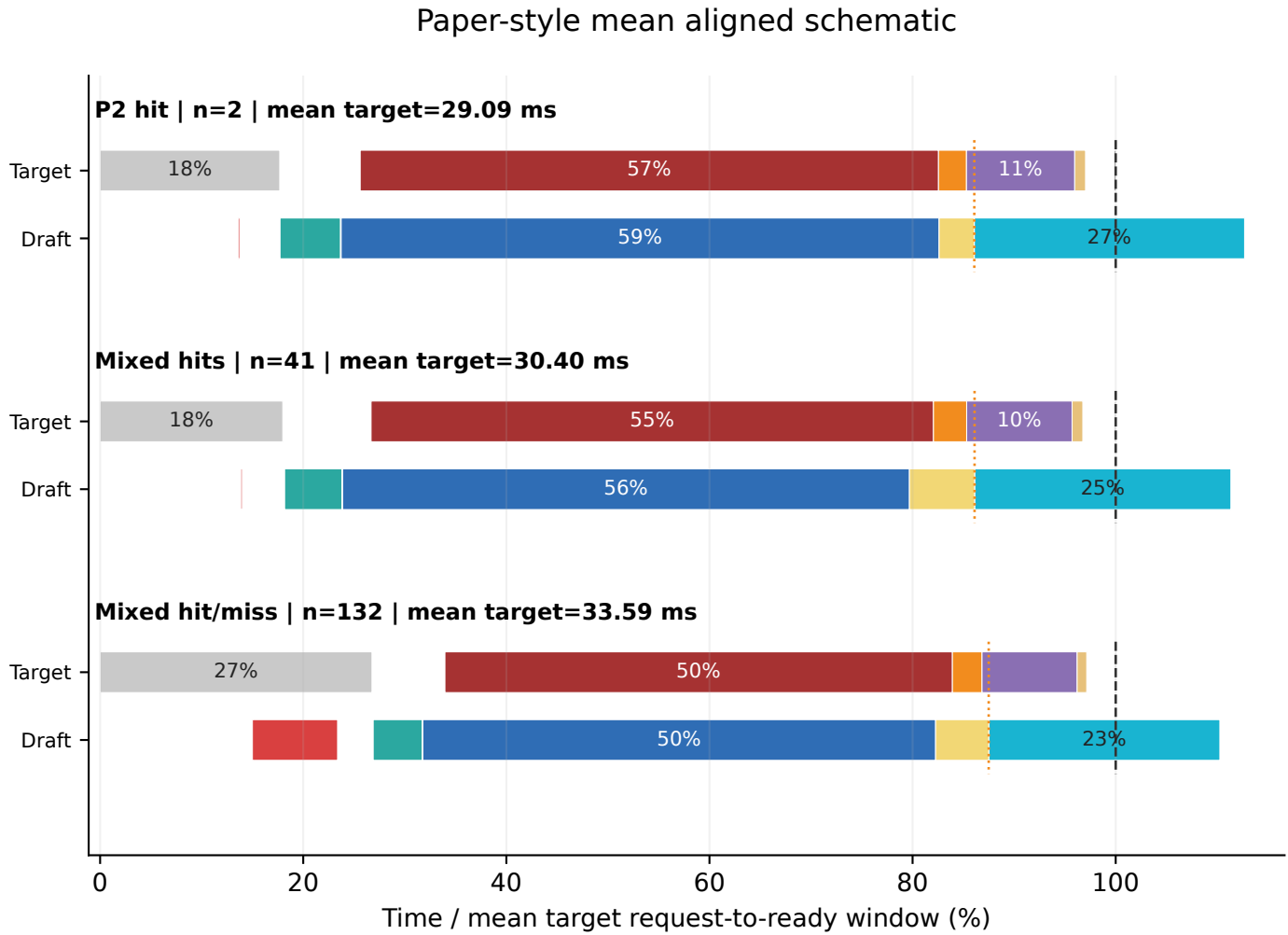
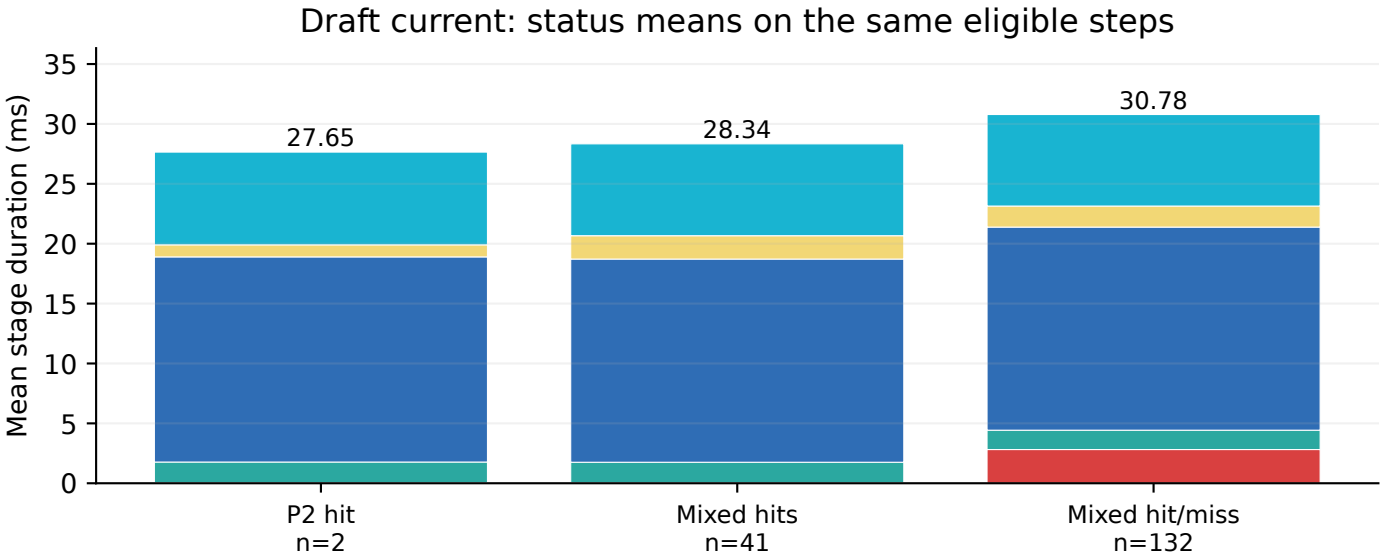
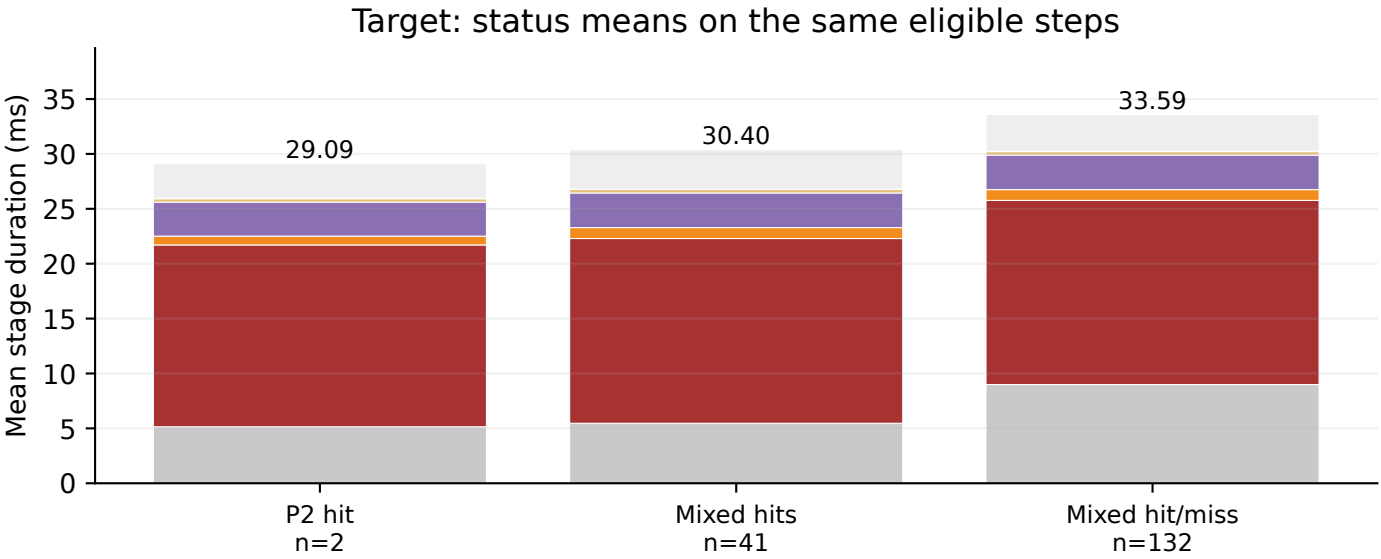
llama2_b8_duet_e21_k4_2
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 174 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

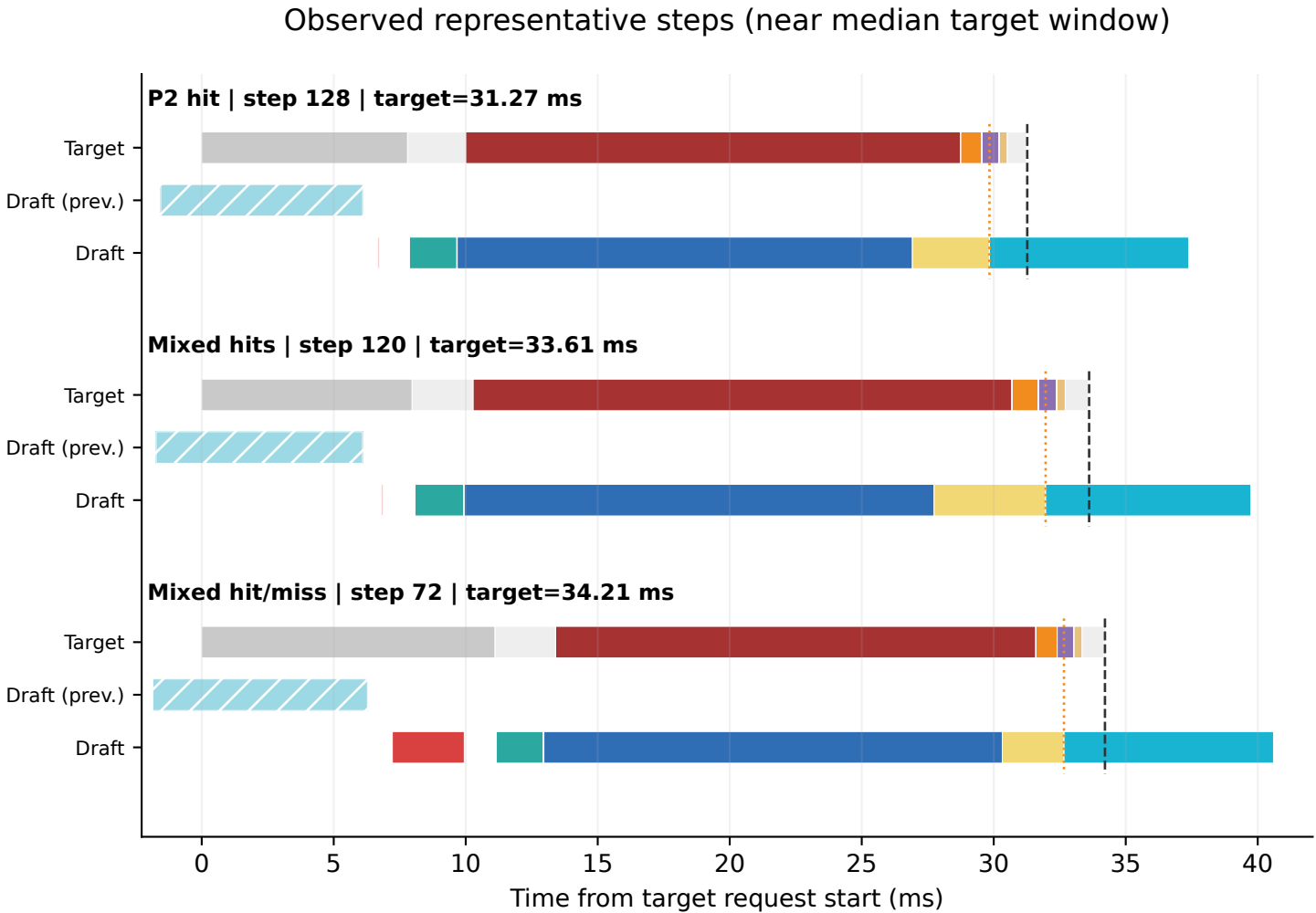
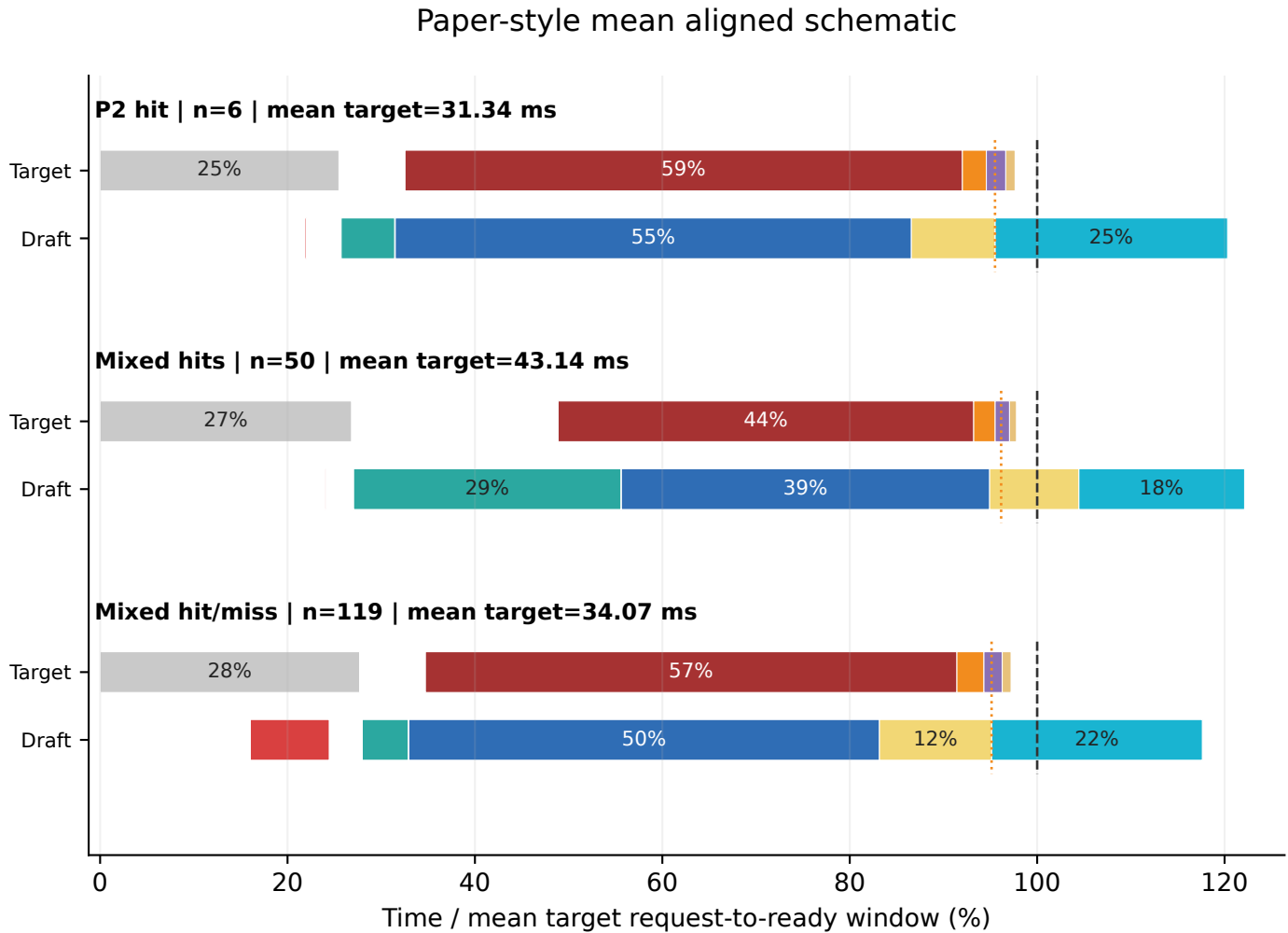
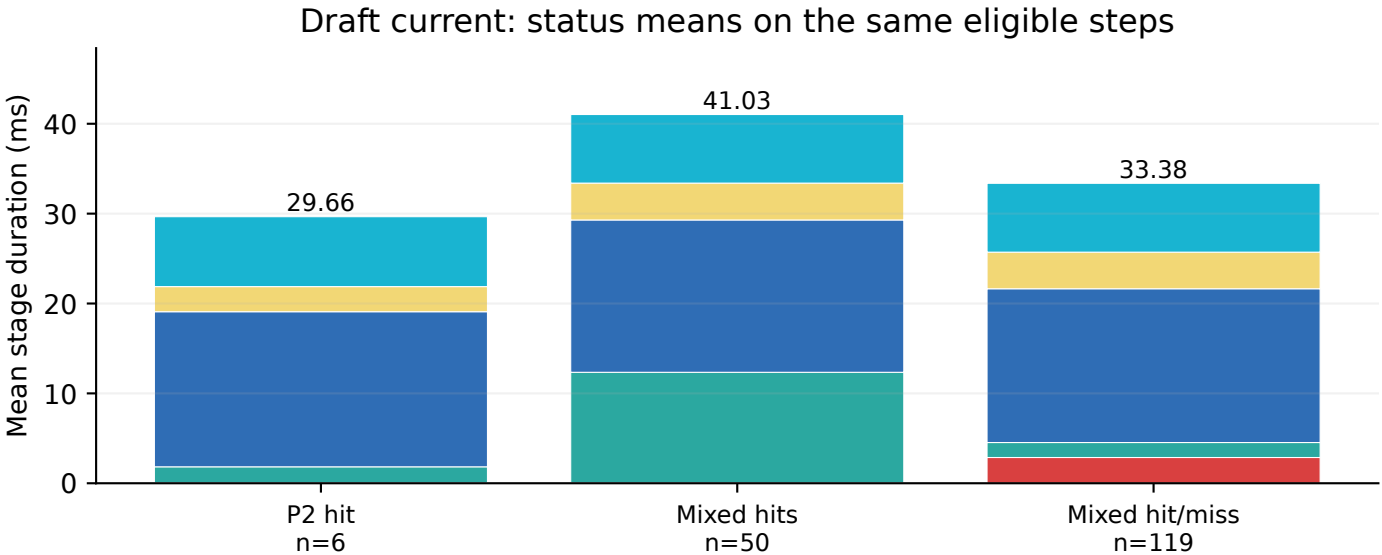
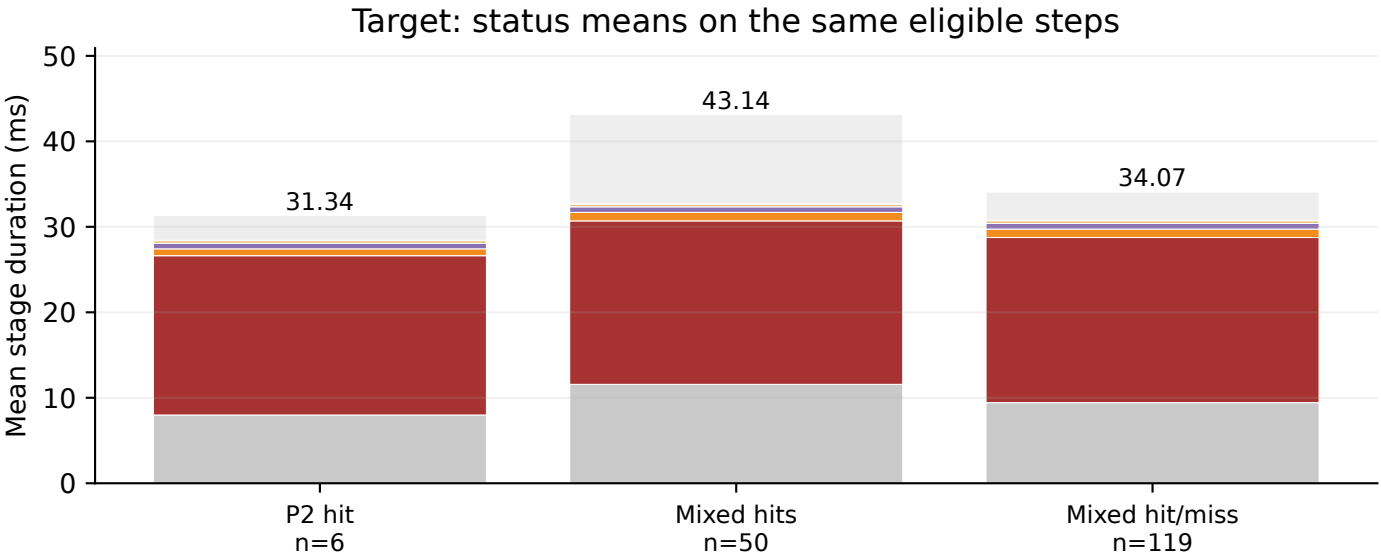
llama2_b8_duet_e26_k4_2
DUET: K1/K2=4/2, exit=26 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 175 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

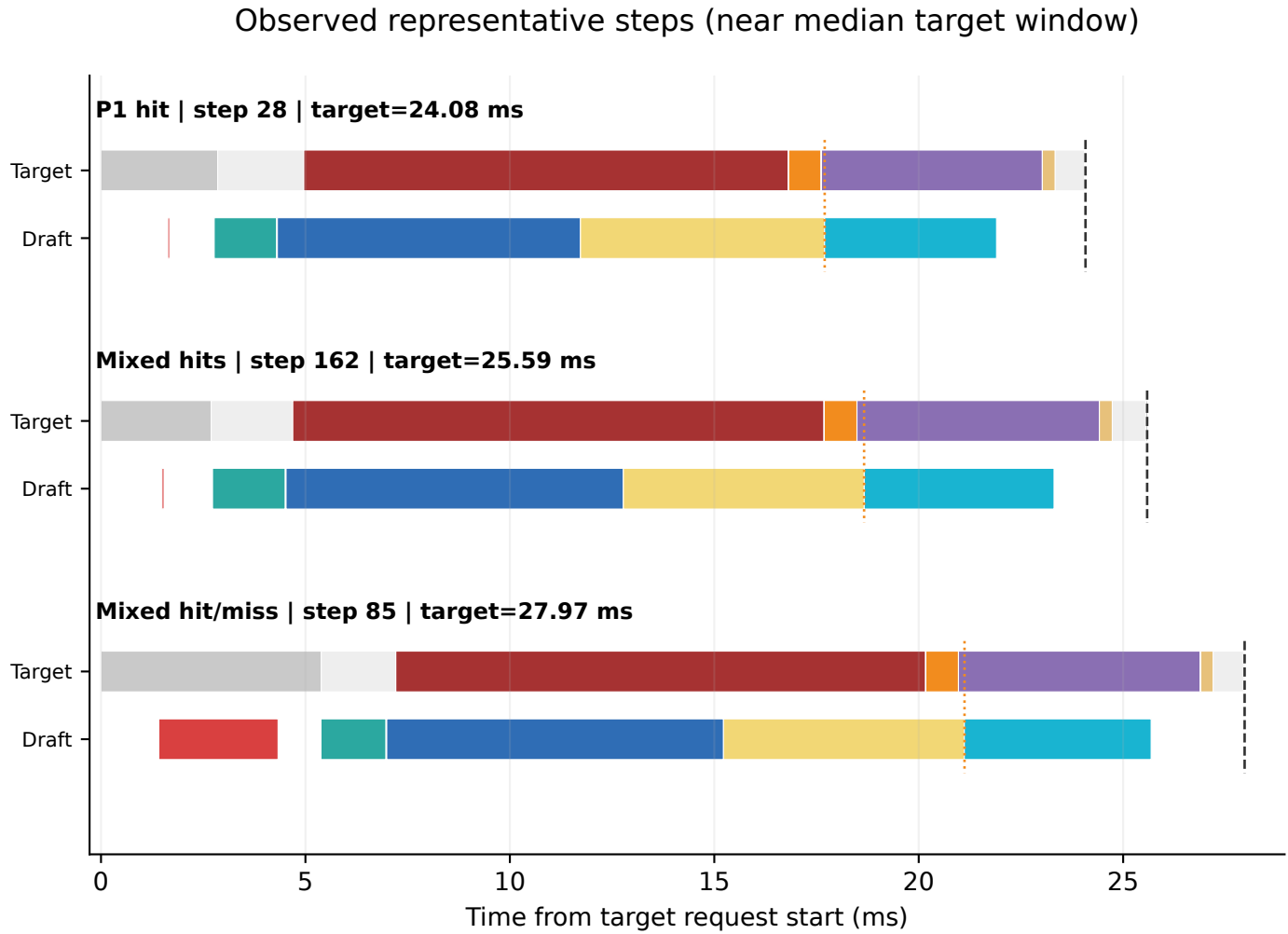
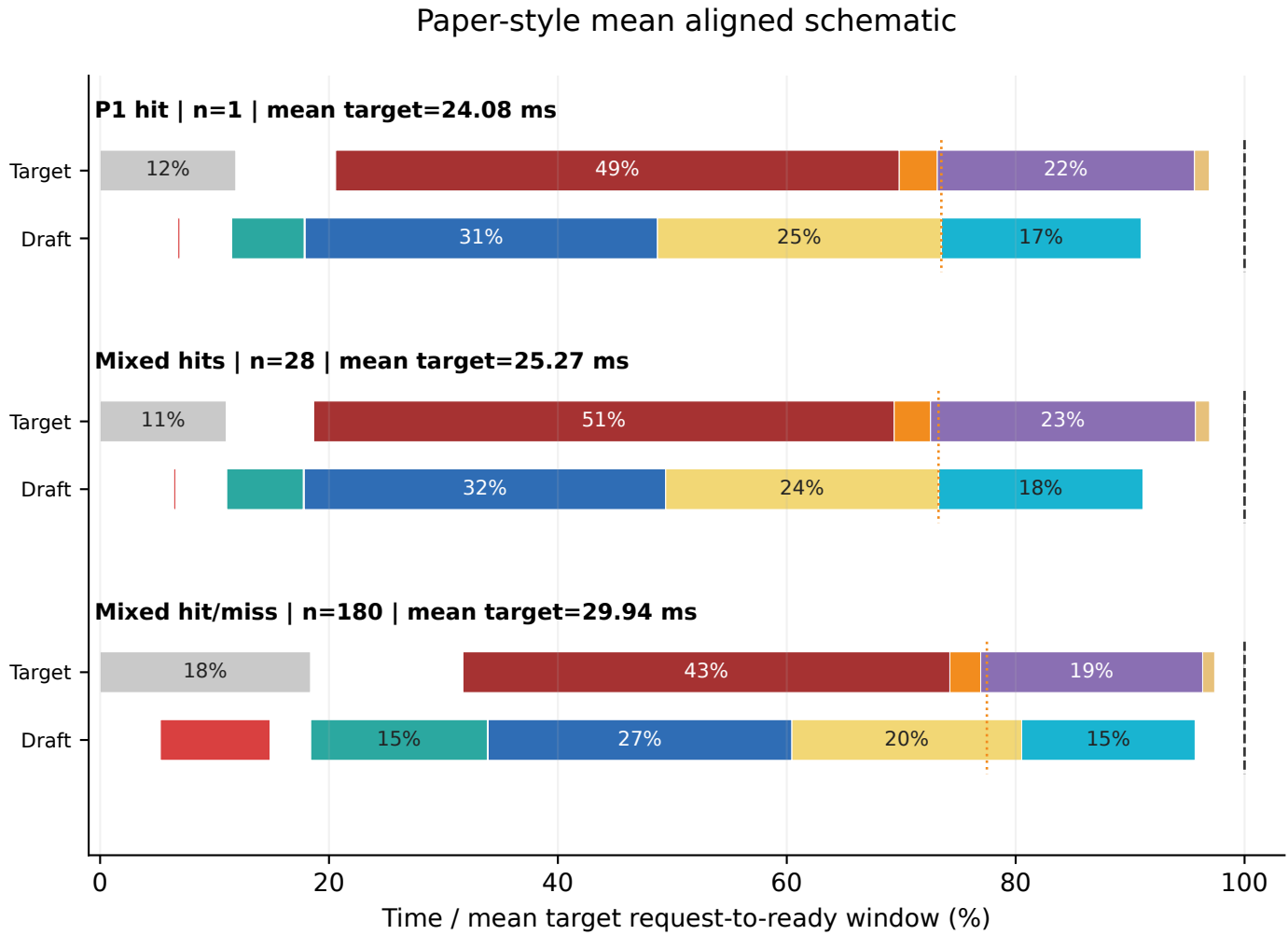
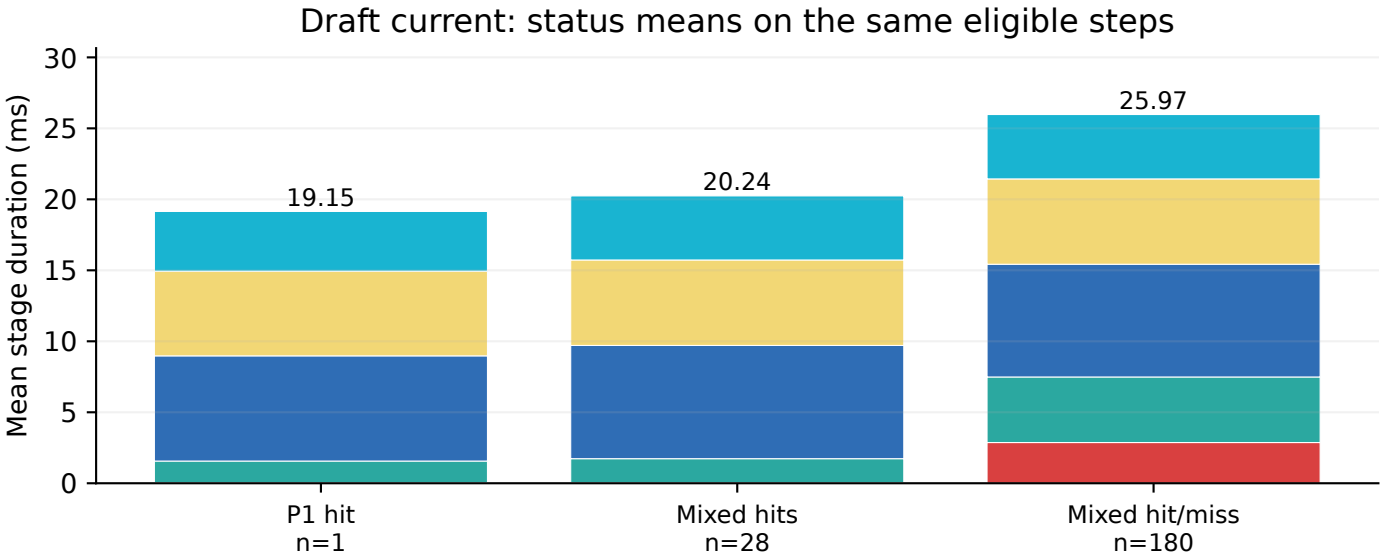
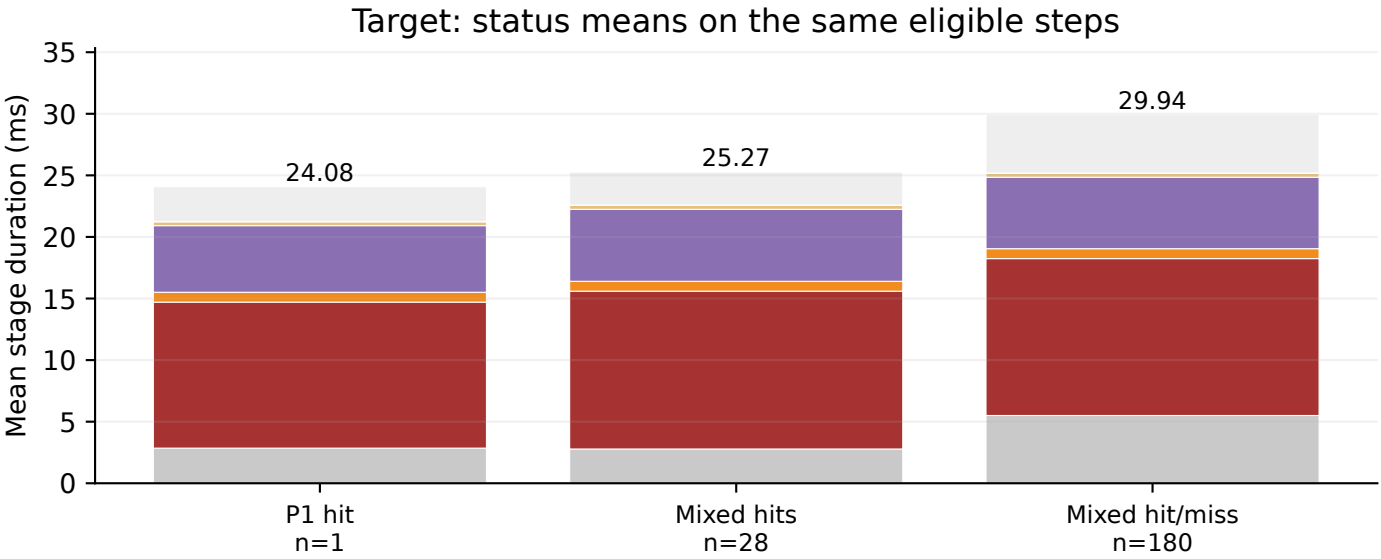
llama2_b8_duet_e30_k4_2
DUET: K1/K2=4/2, exit=30 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 175 eligible steps



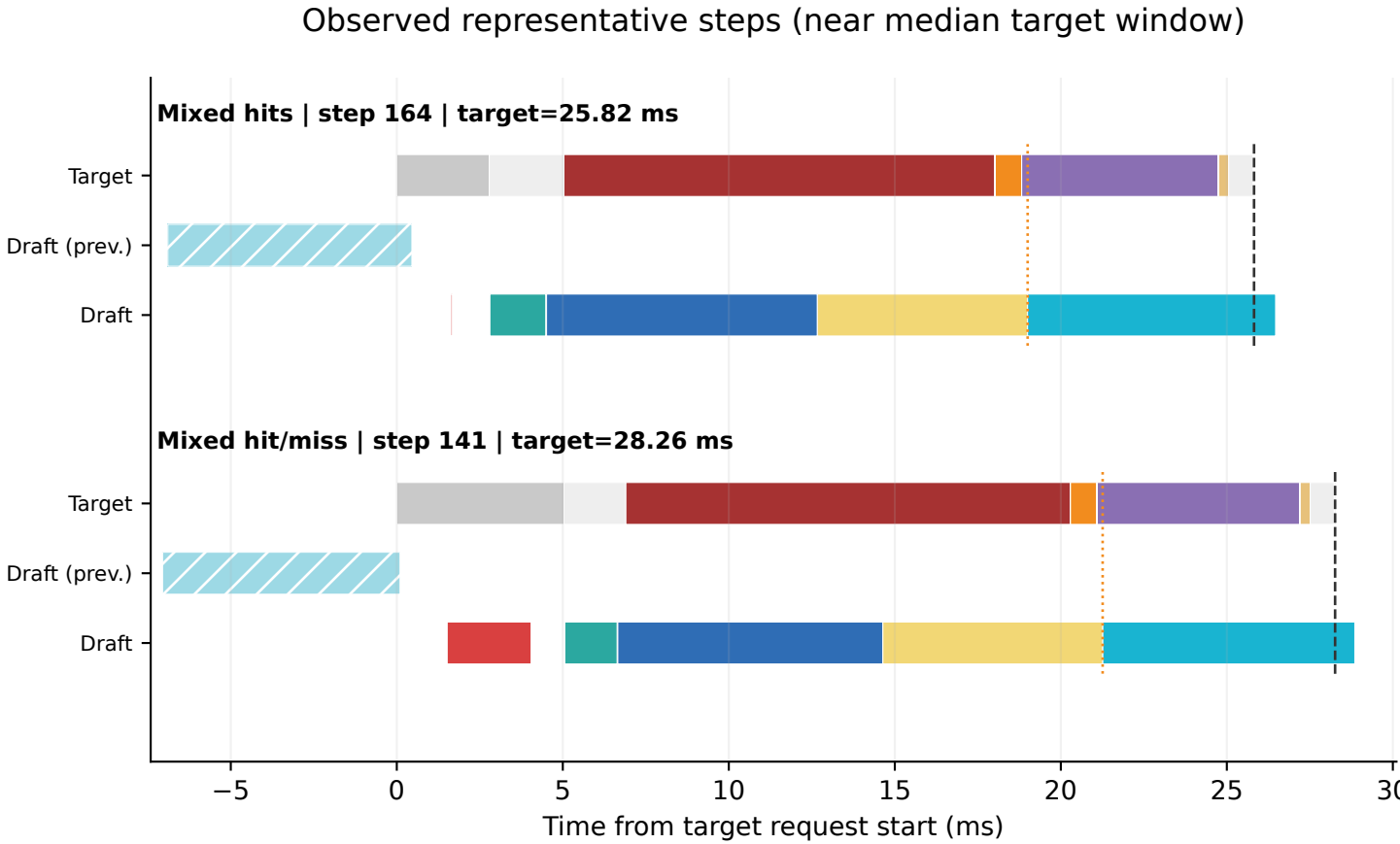
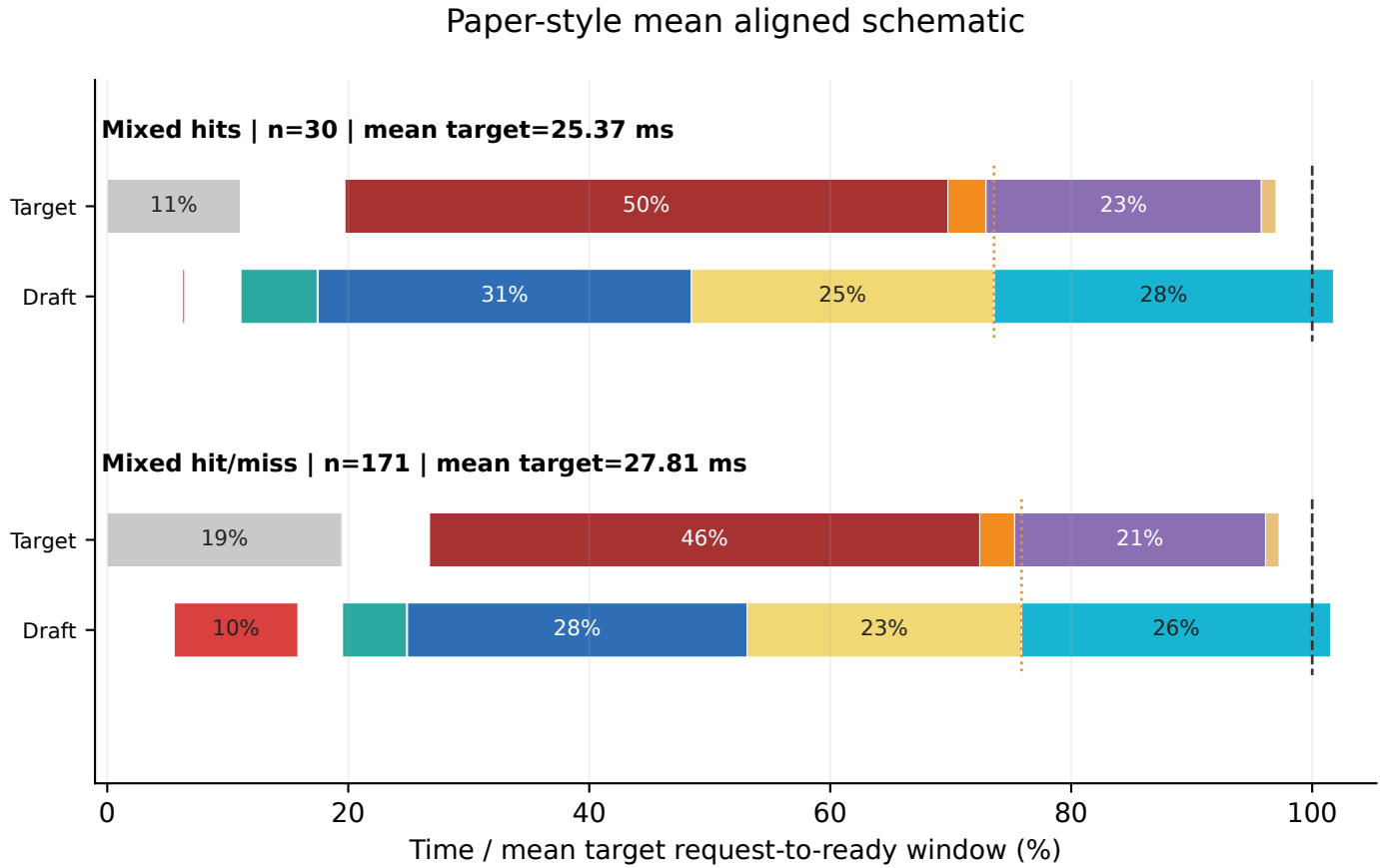
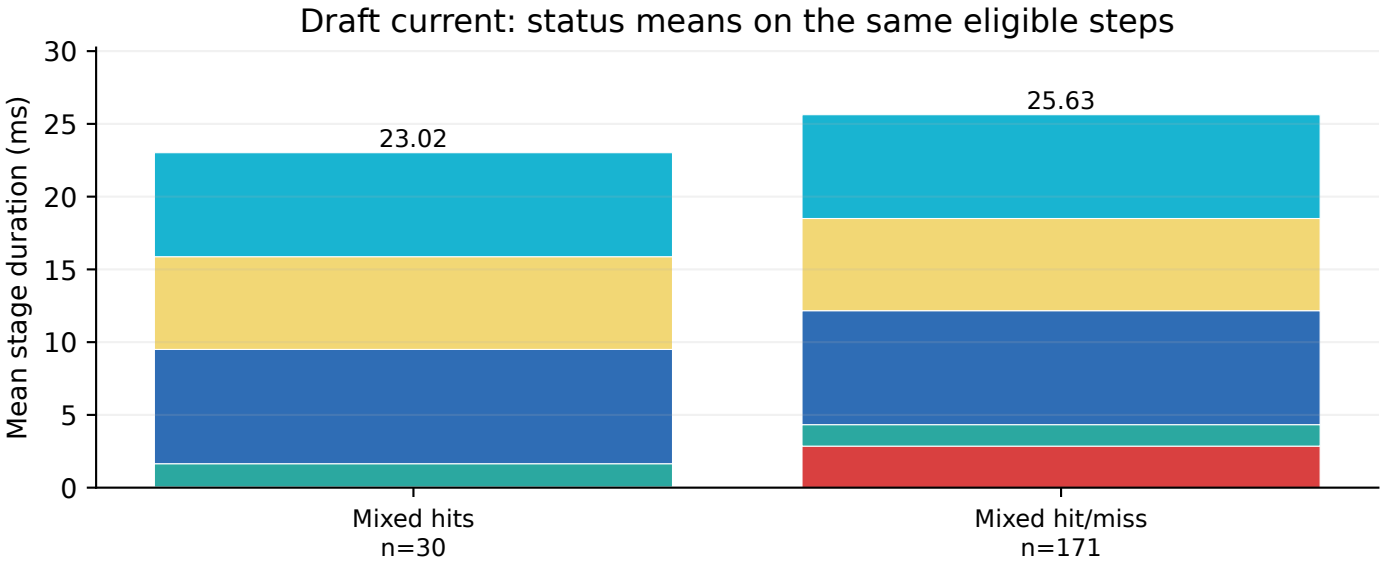
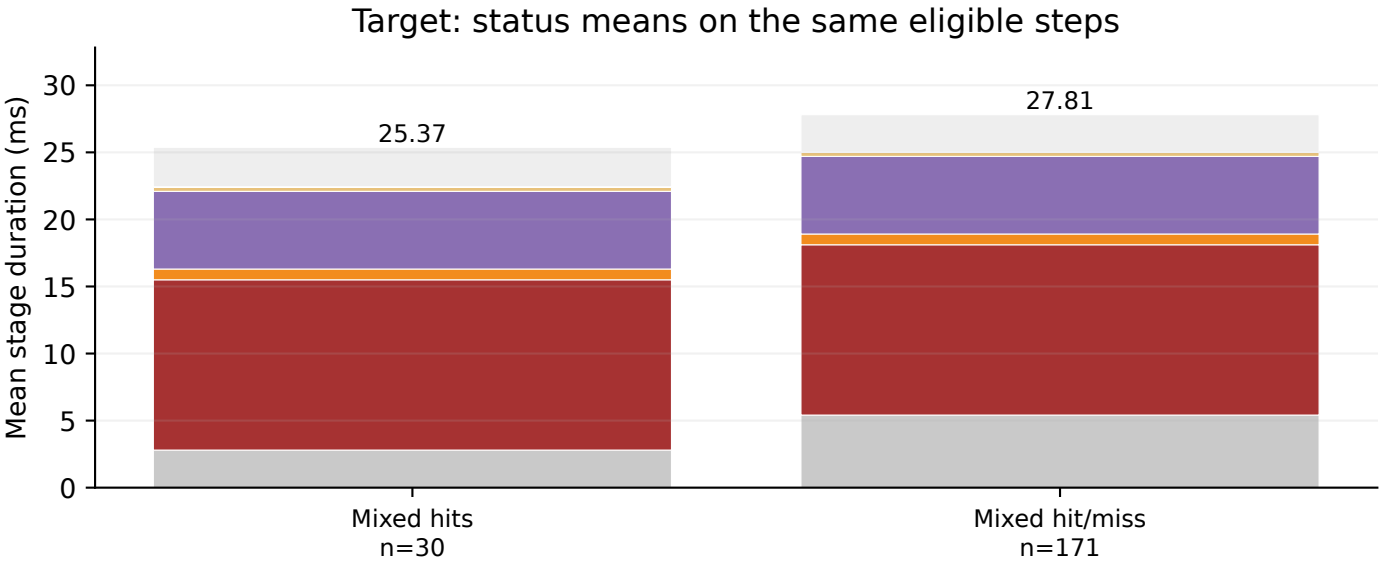
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e21_k2_1_r2
DUET: K1/K2=2/1, exit=21 (0-based), C=3, N1/N2=4/2, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 209 eligible steps



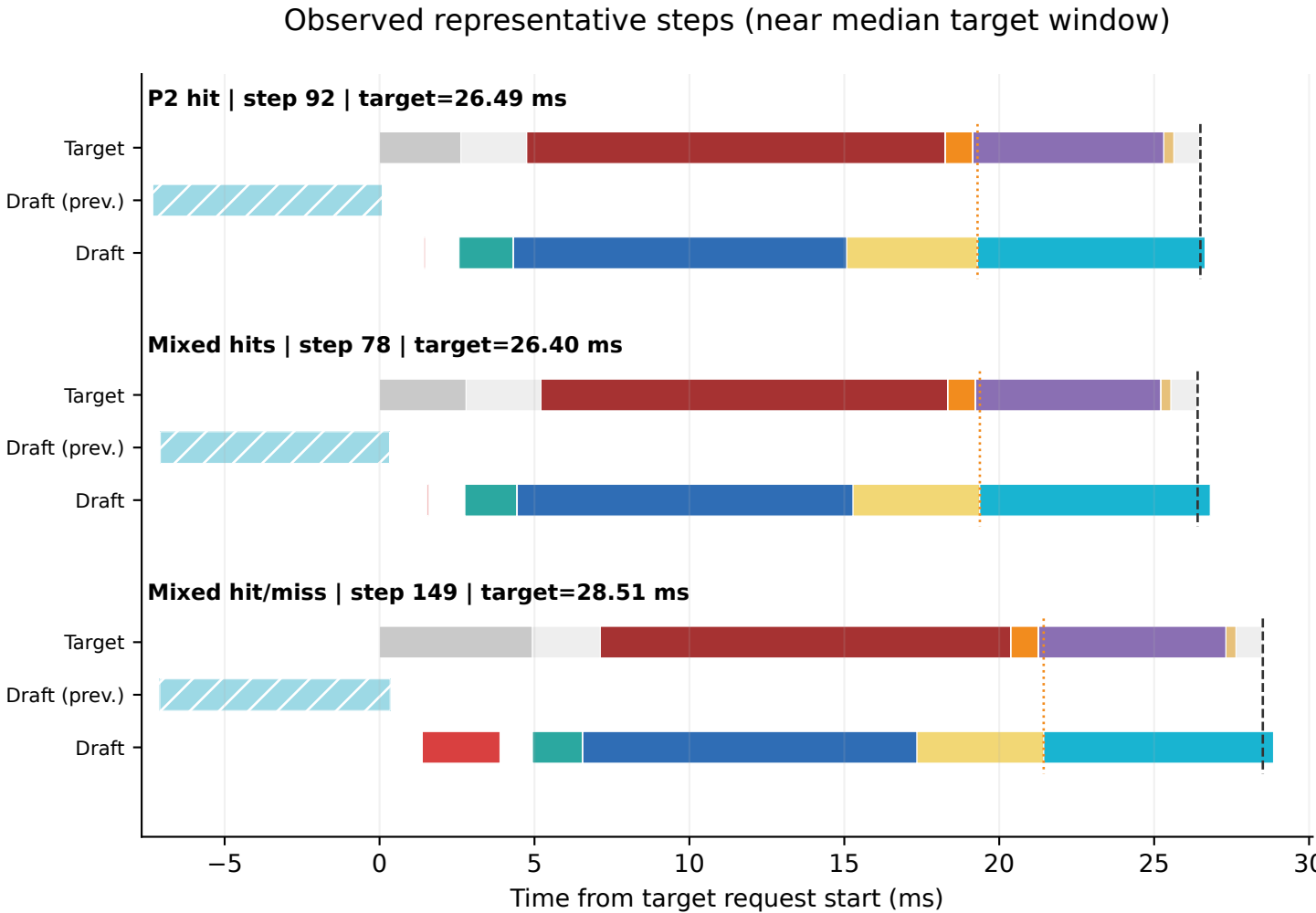
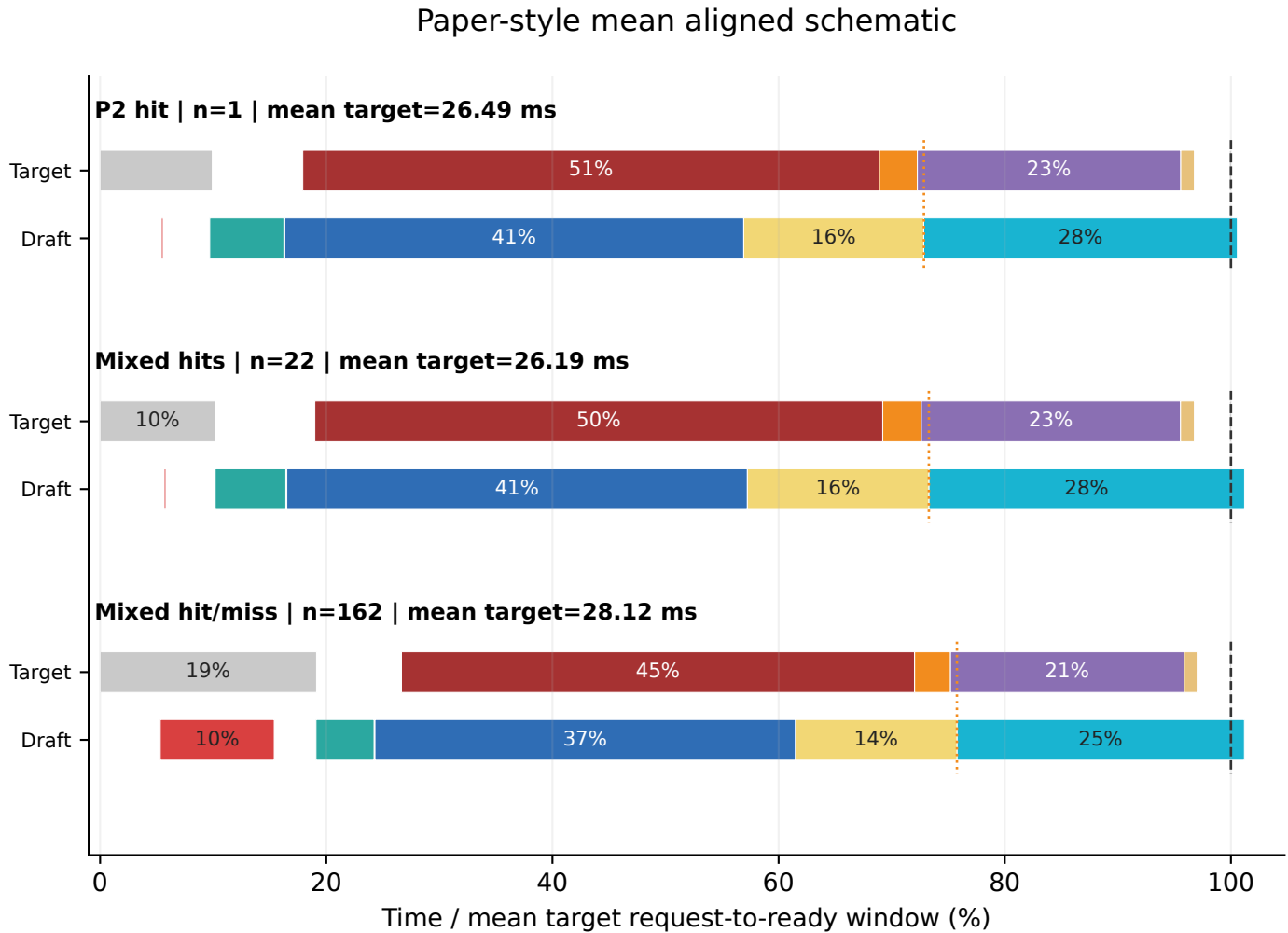
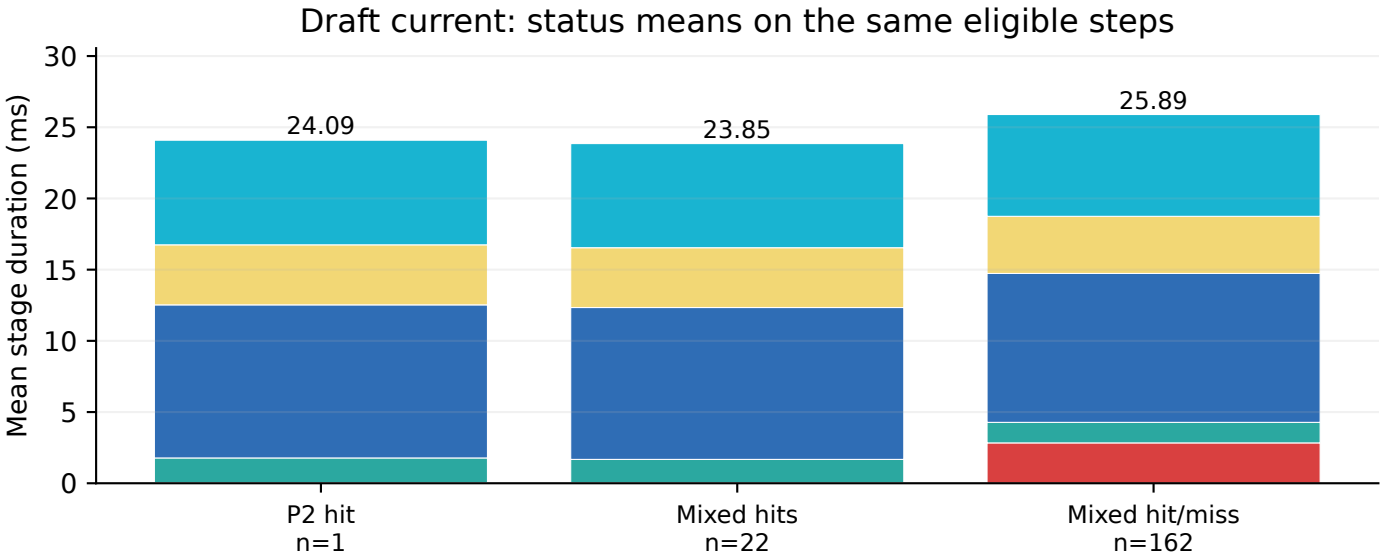
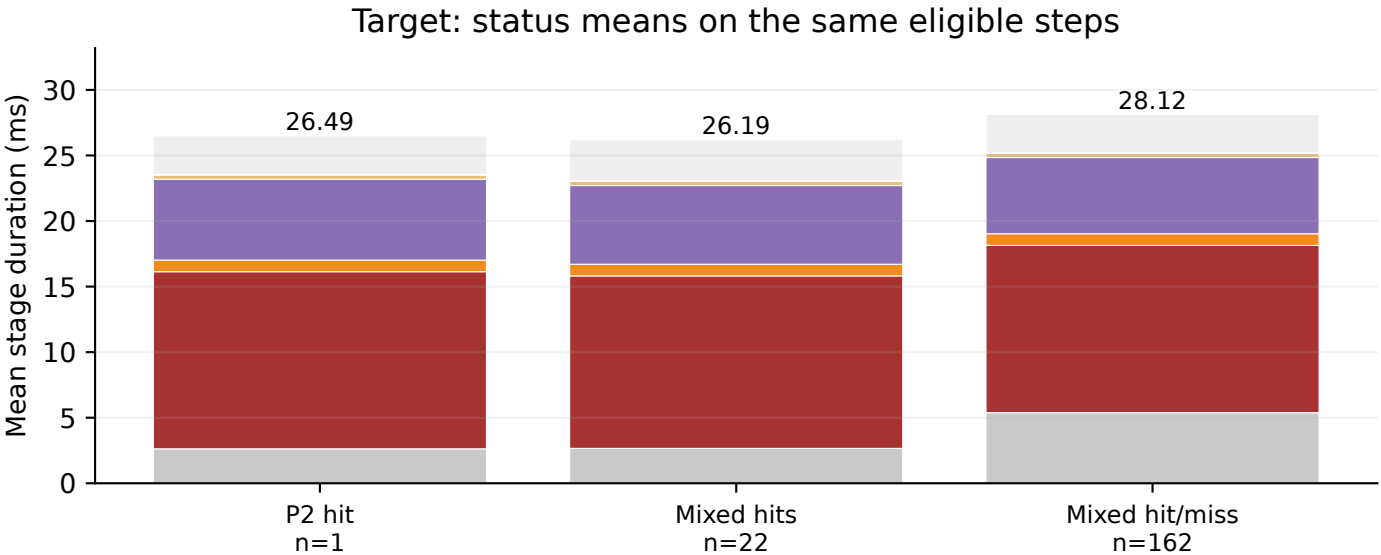
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

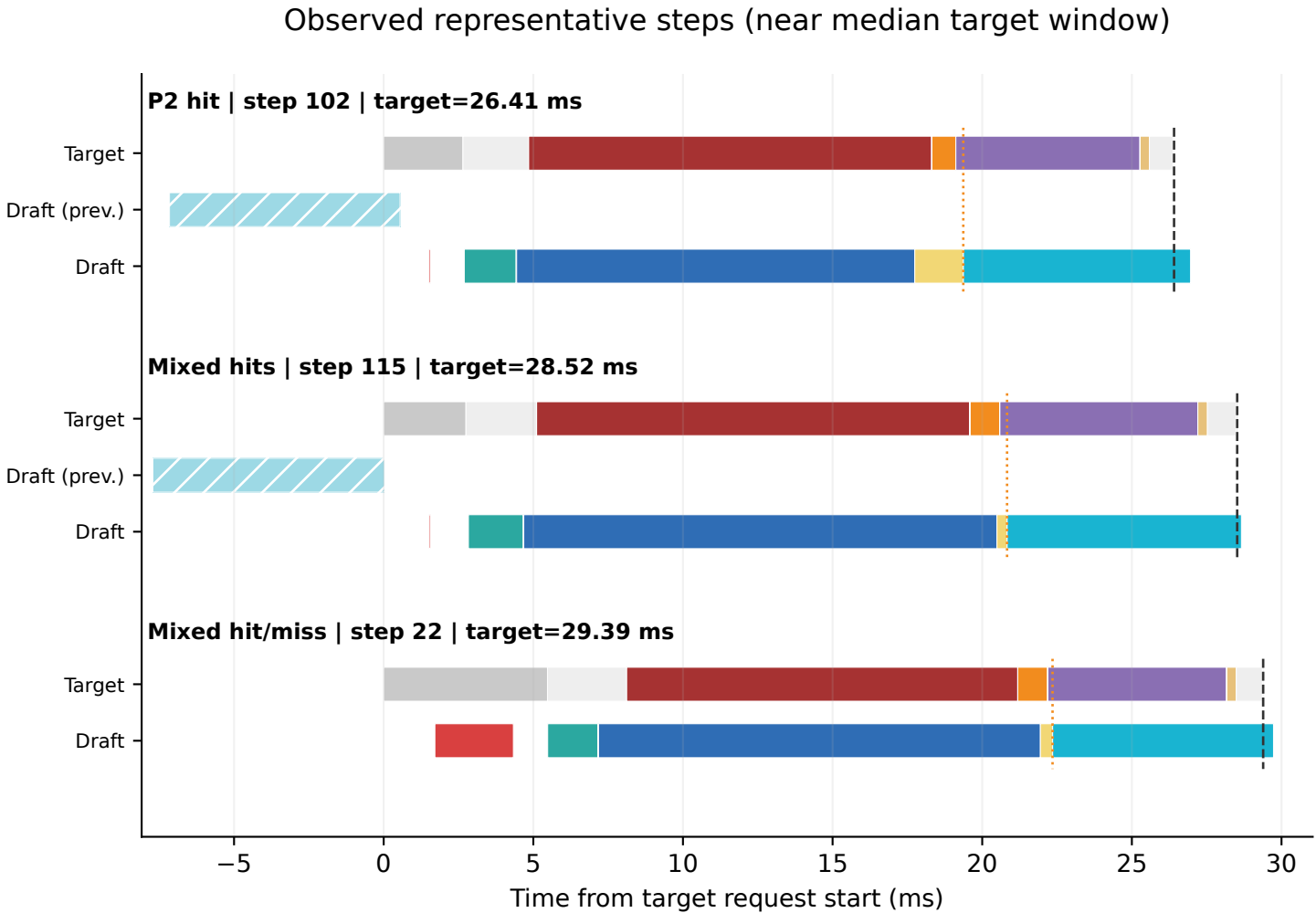
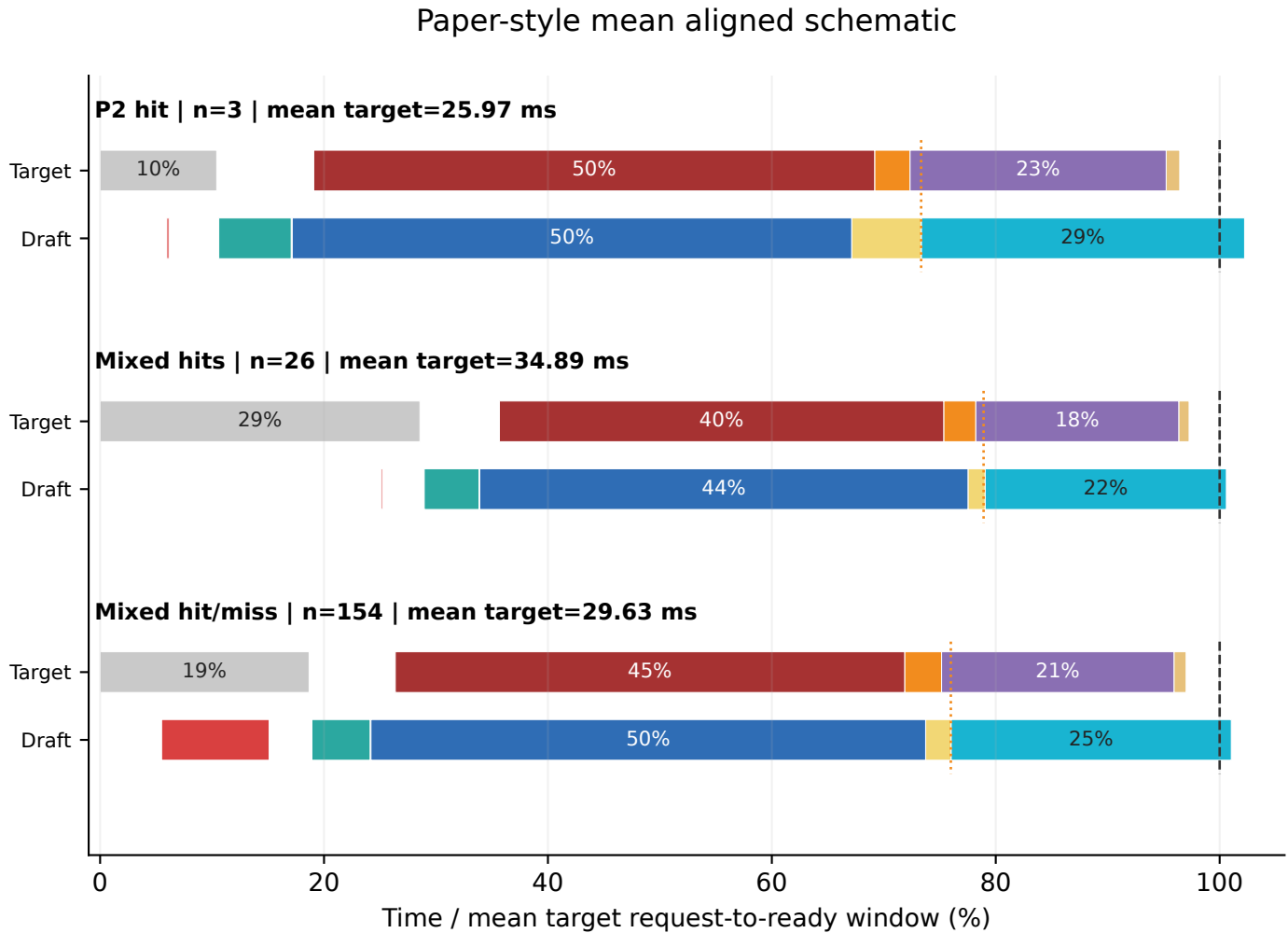
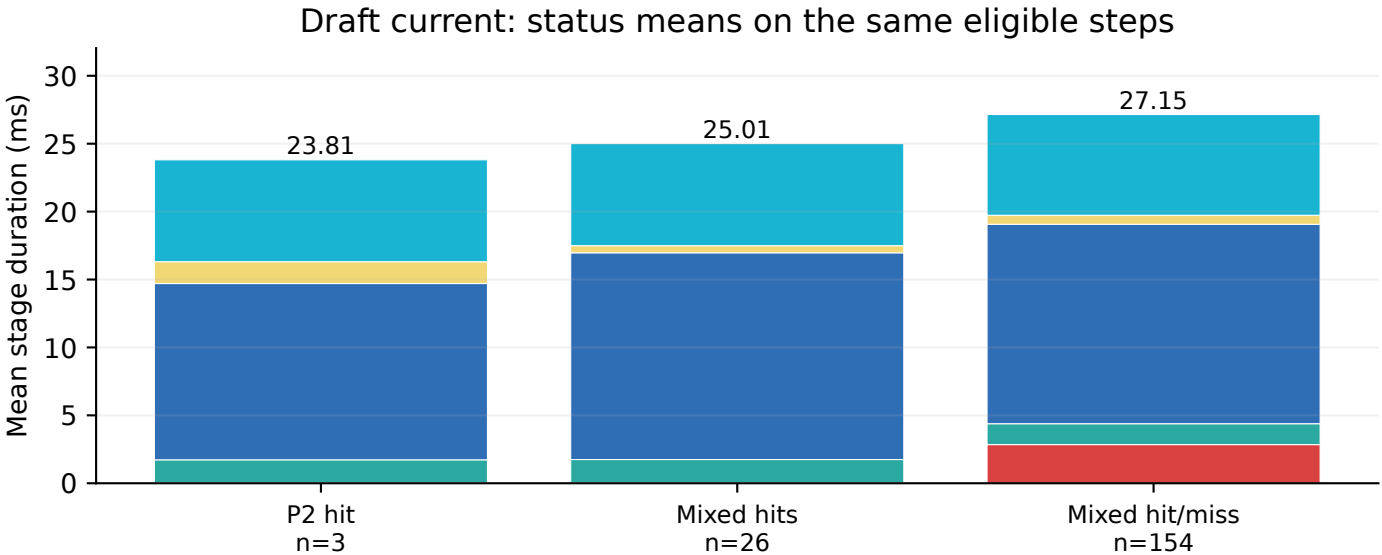
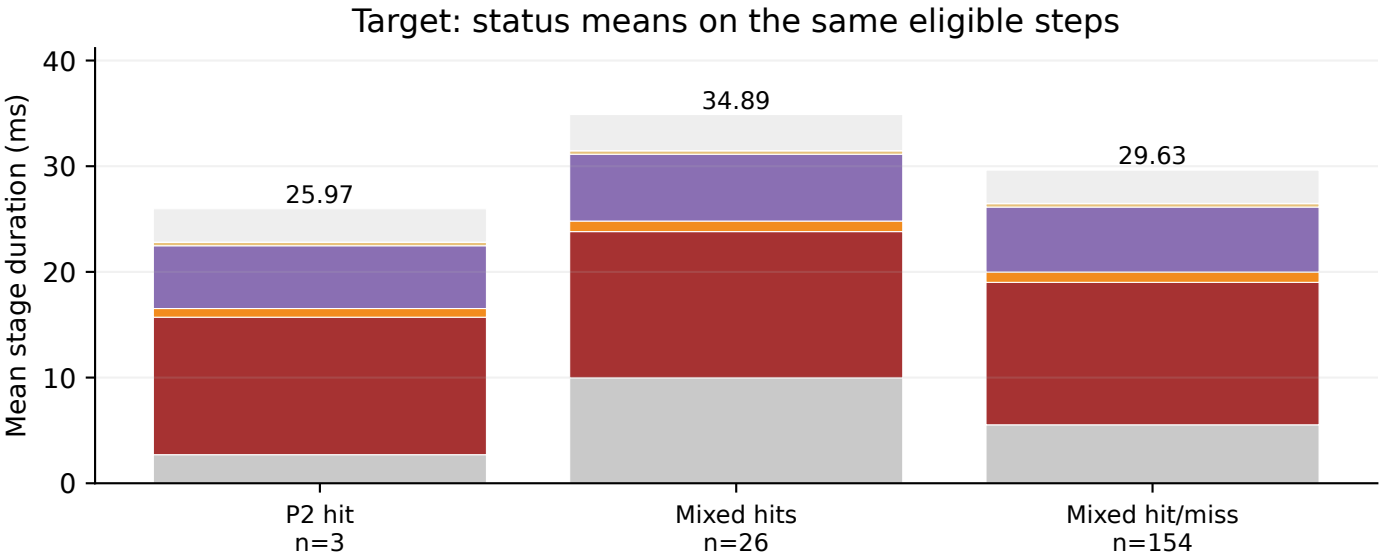
llama2_b8_refine_e21_k3_2_r1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 185 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

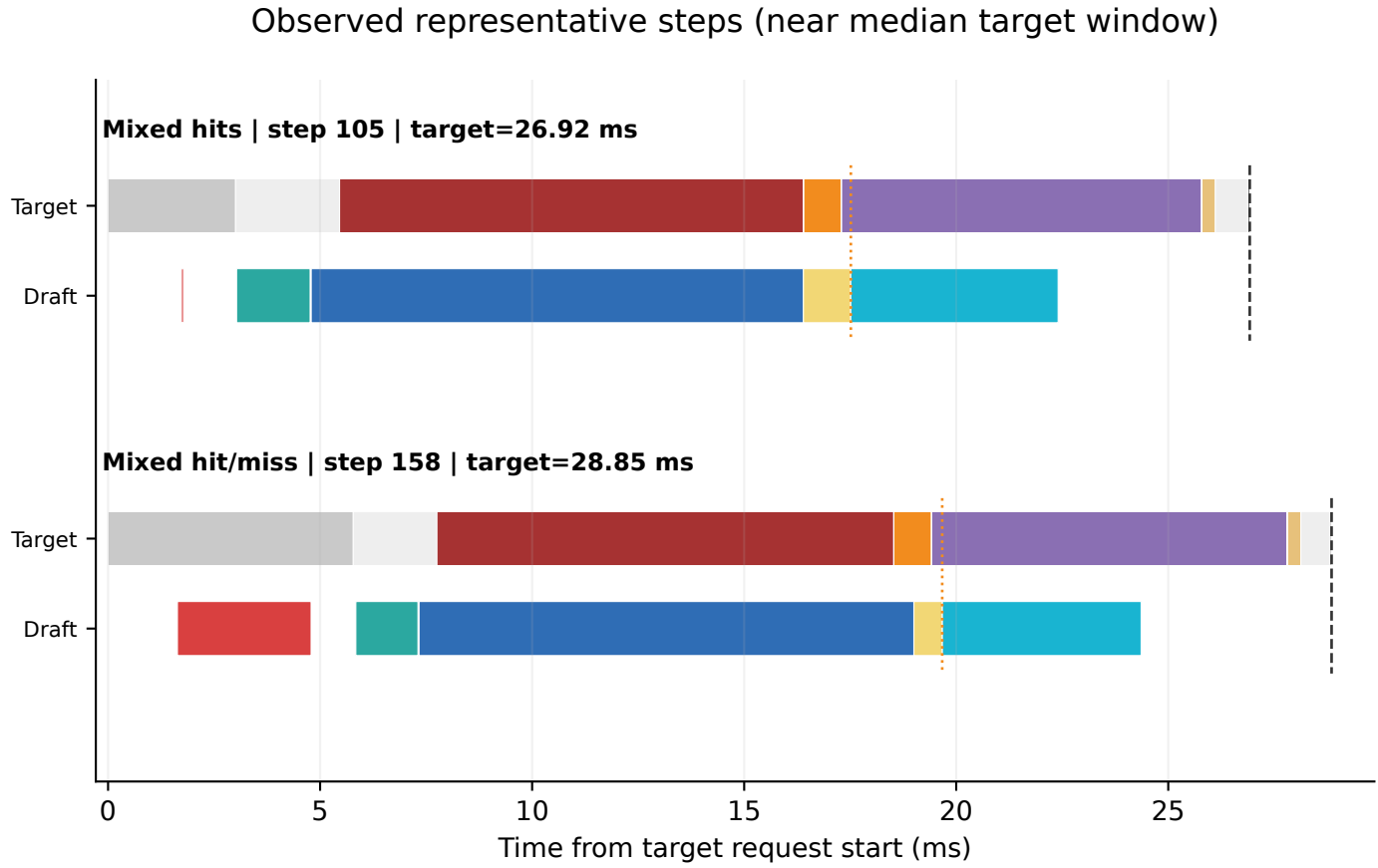
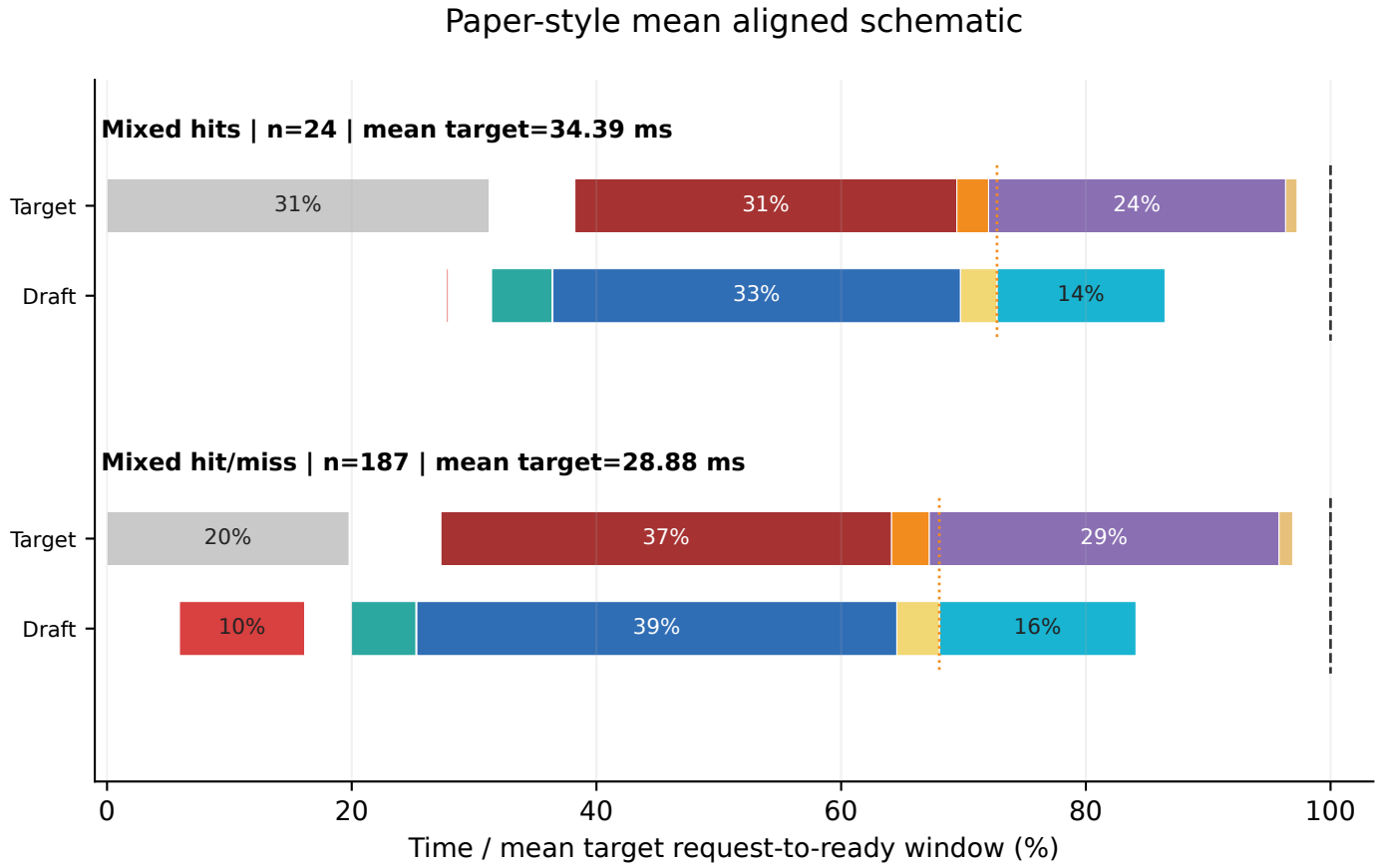
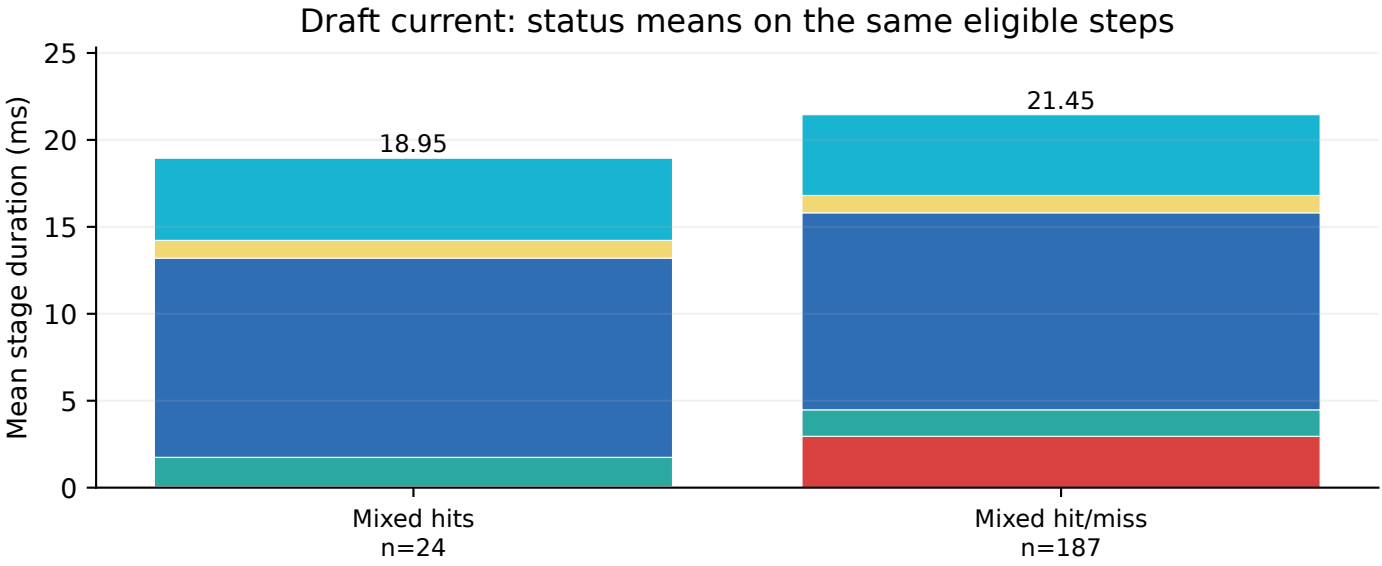
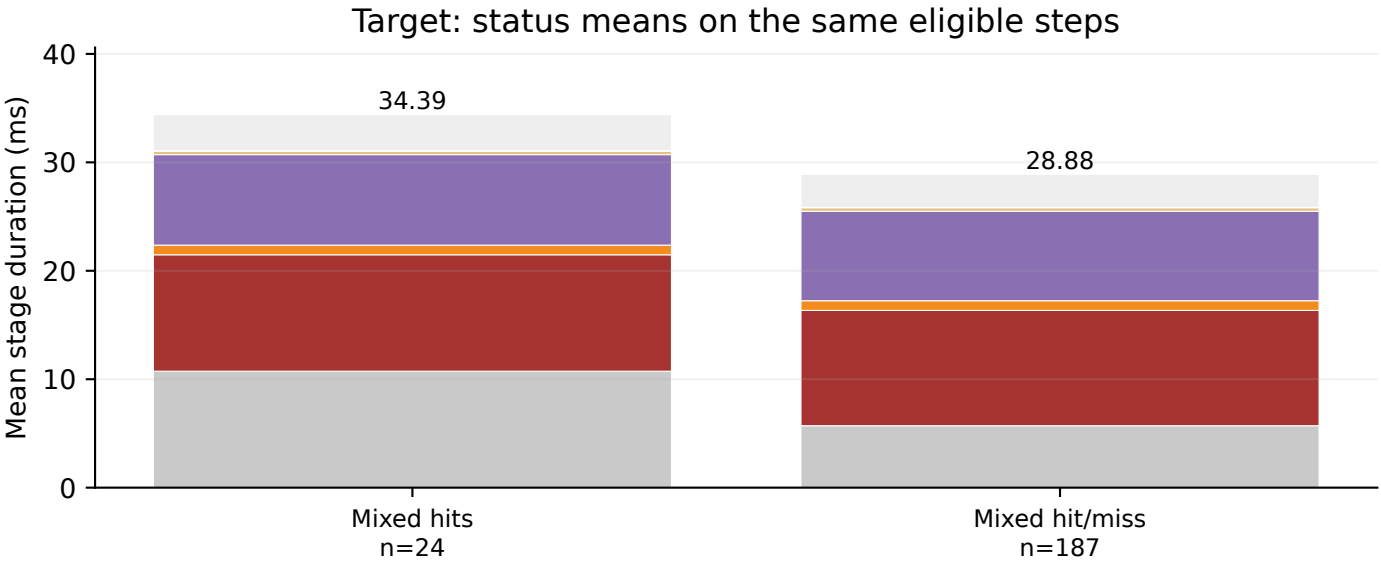
llama2_b8_refine_e21_k4_2_r1
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=1, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 183 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

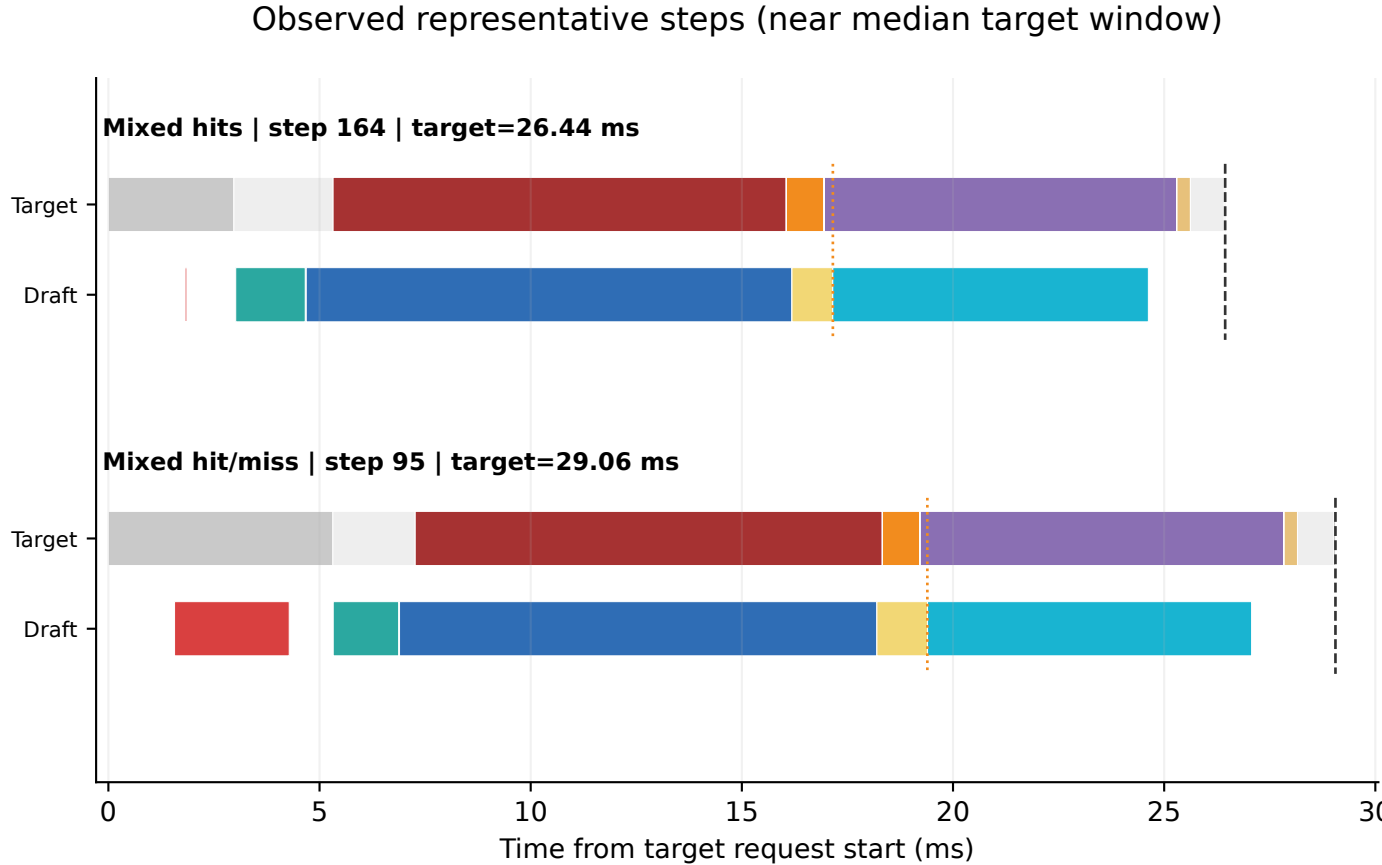
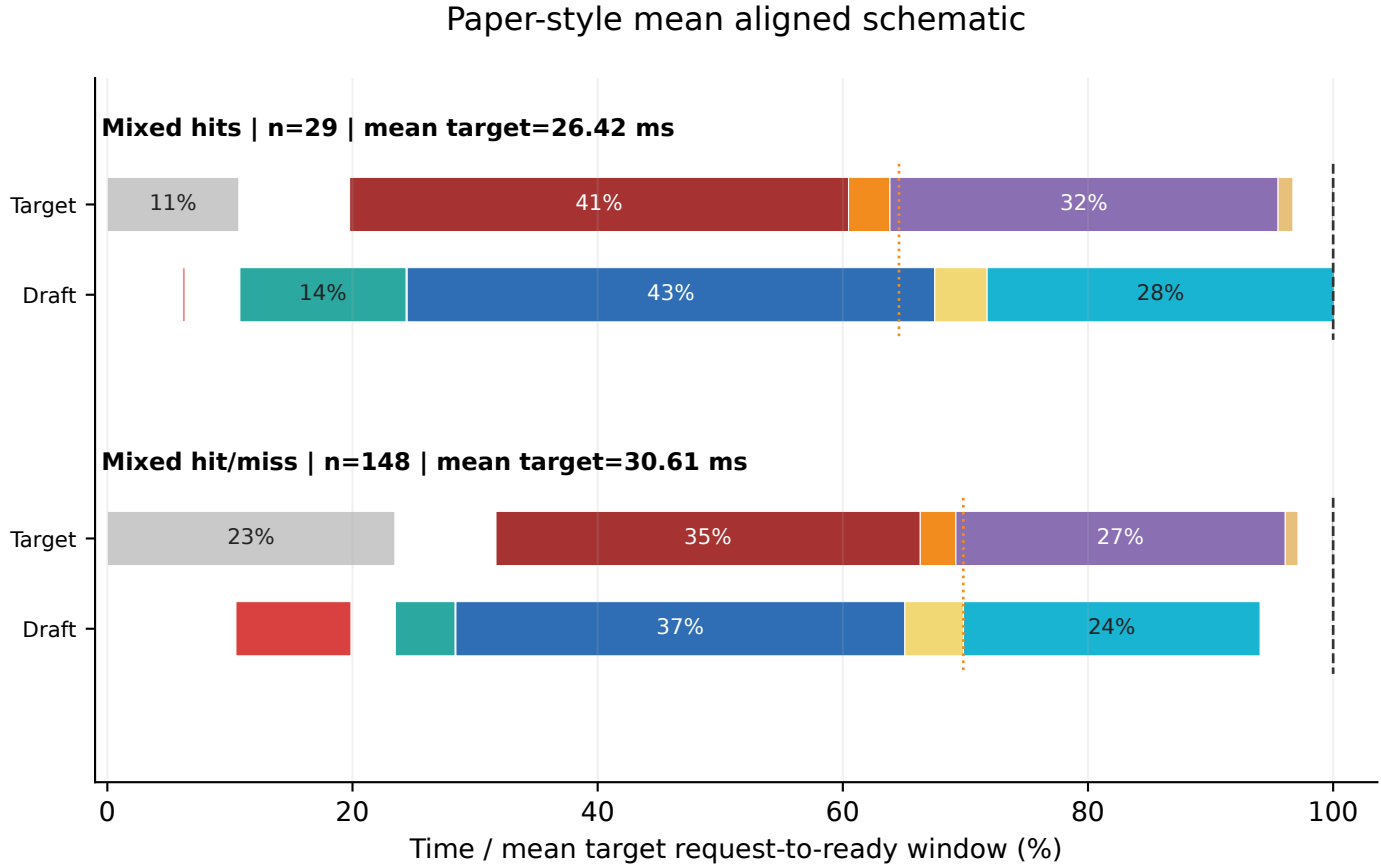
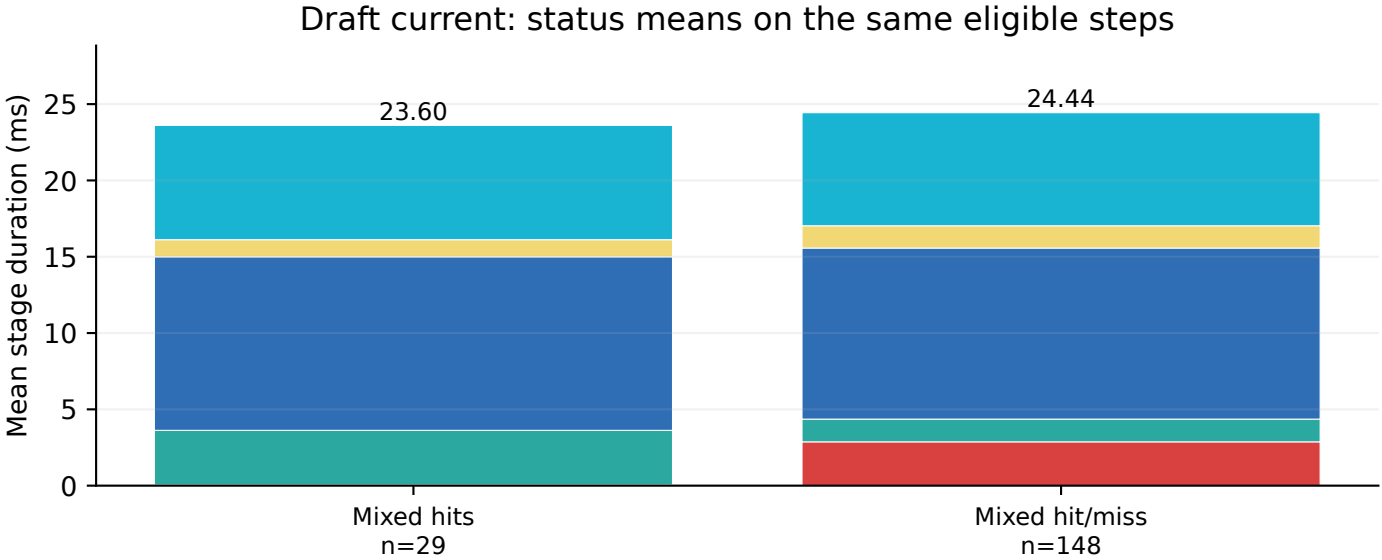
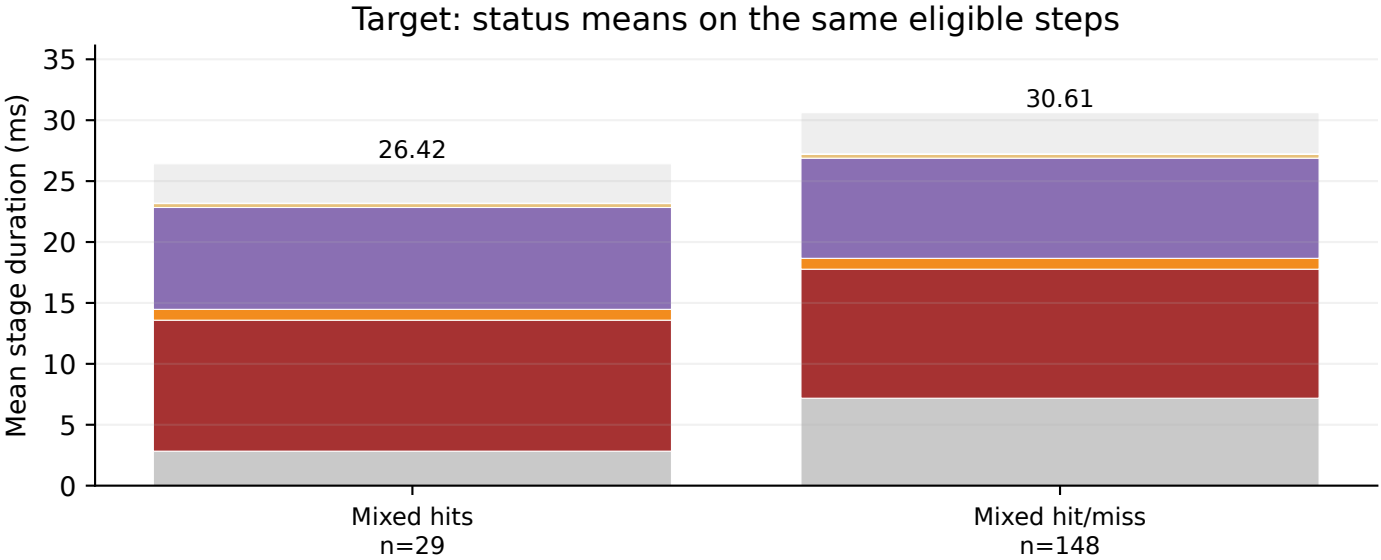
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e17_k3_1_r2
DUET: K1/K2=3/1, exit=17 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 211 eligible steps



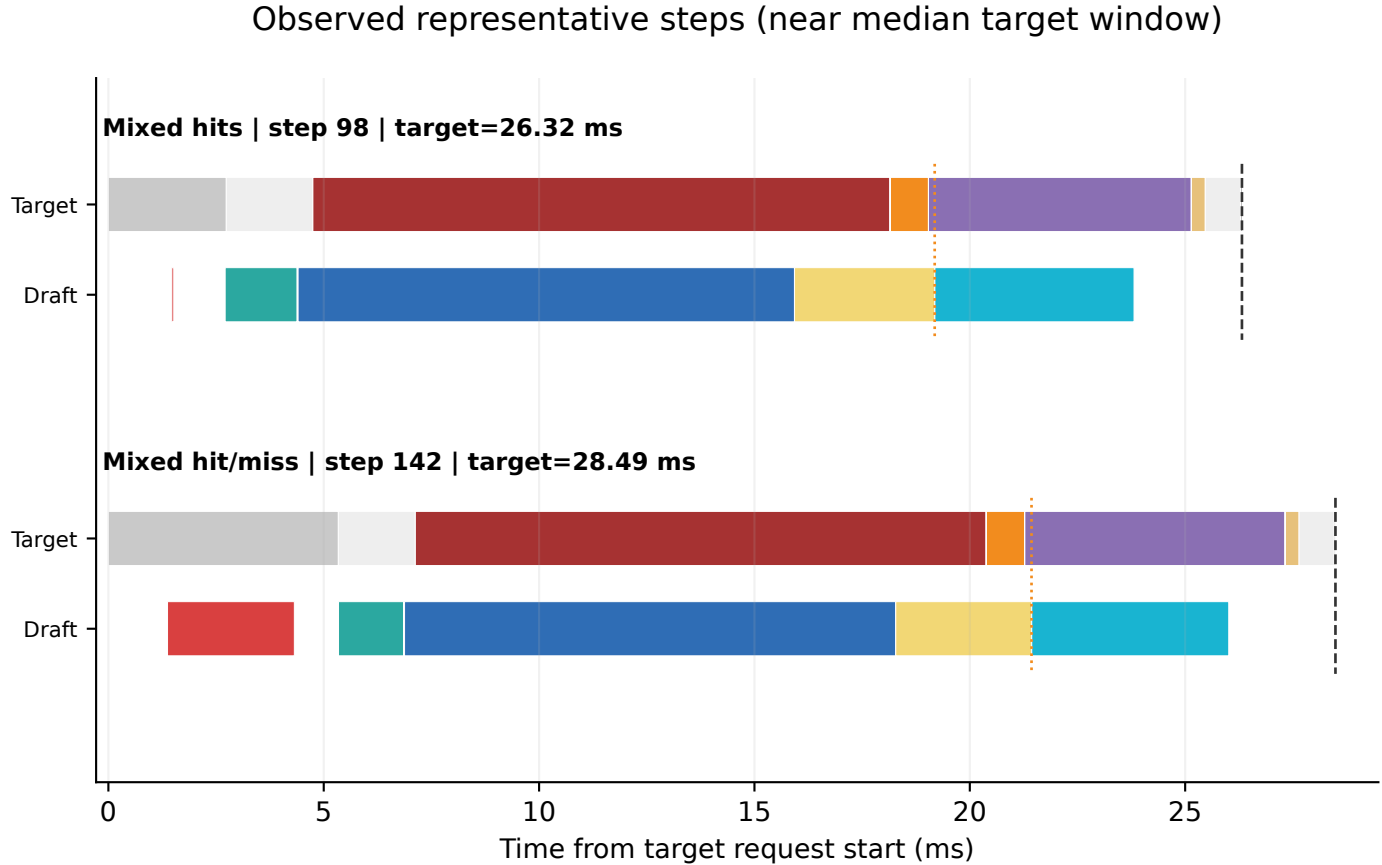
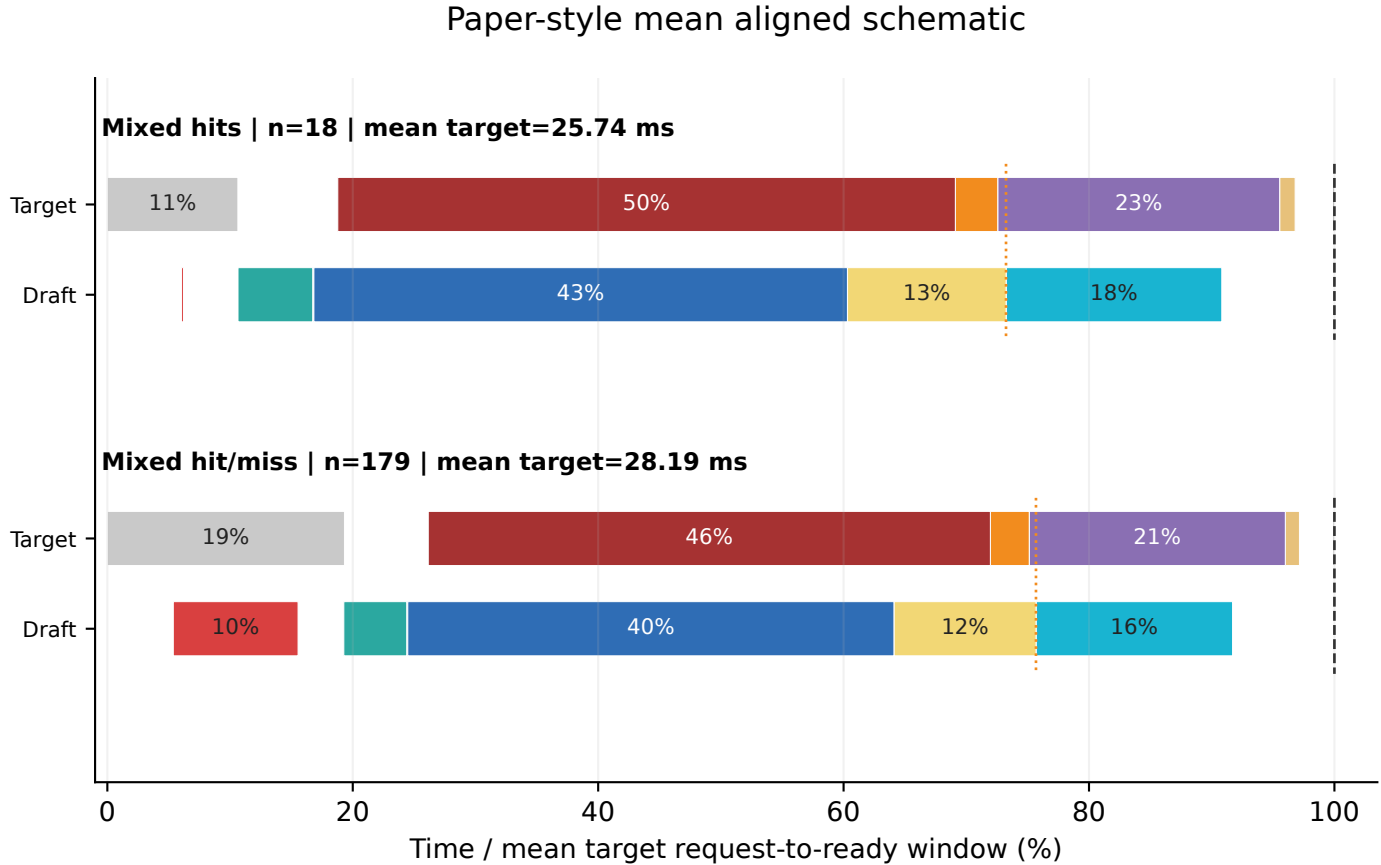
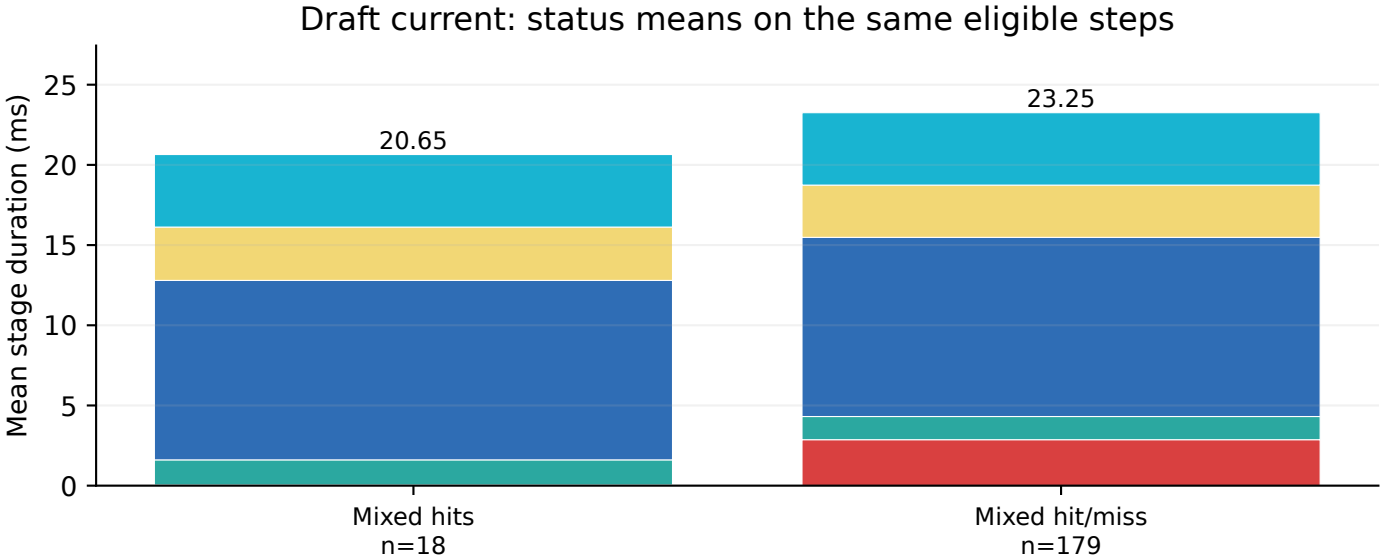
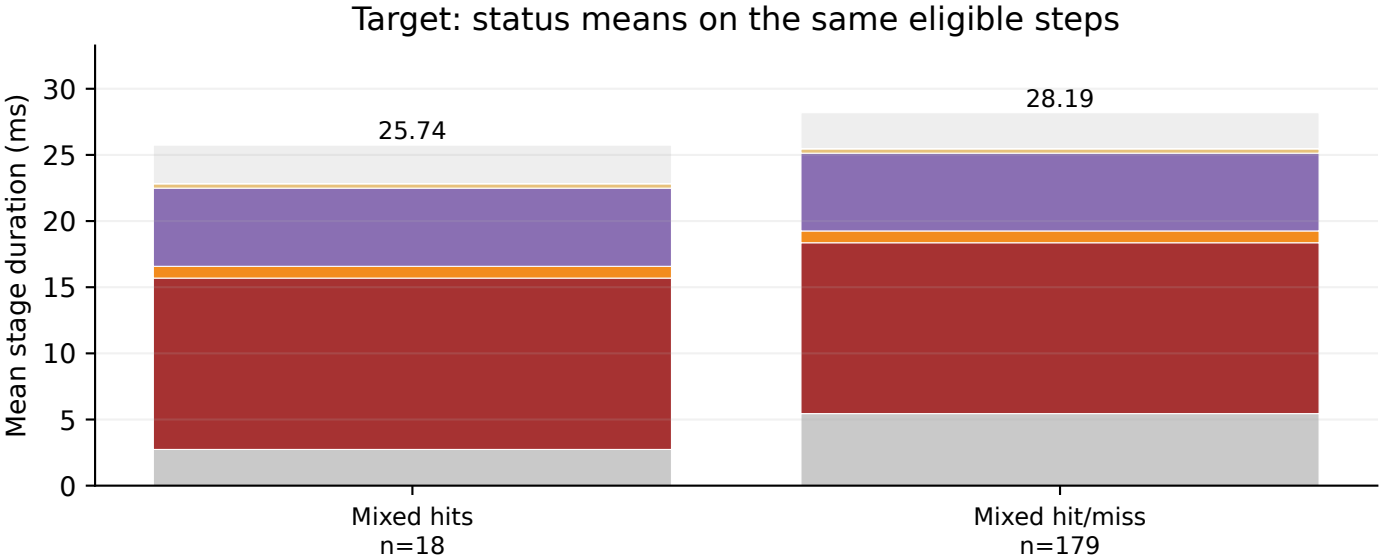
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e17_k3_2_r2
DUET: K1/K2=3/2, exit=17 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 177 eligible steps



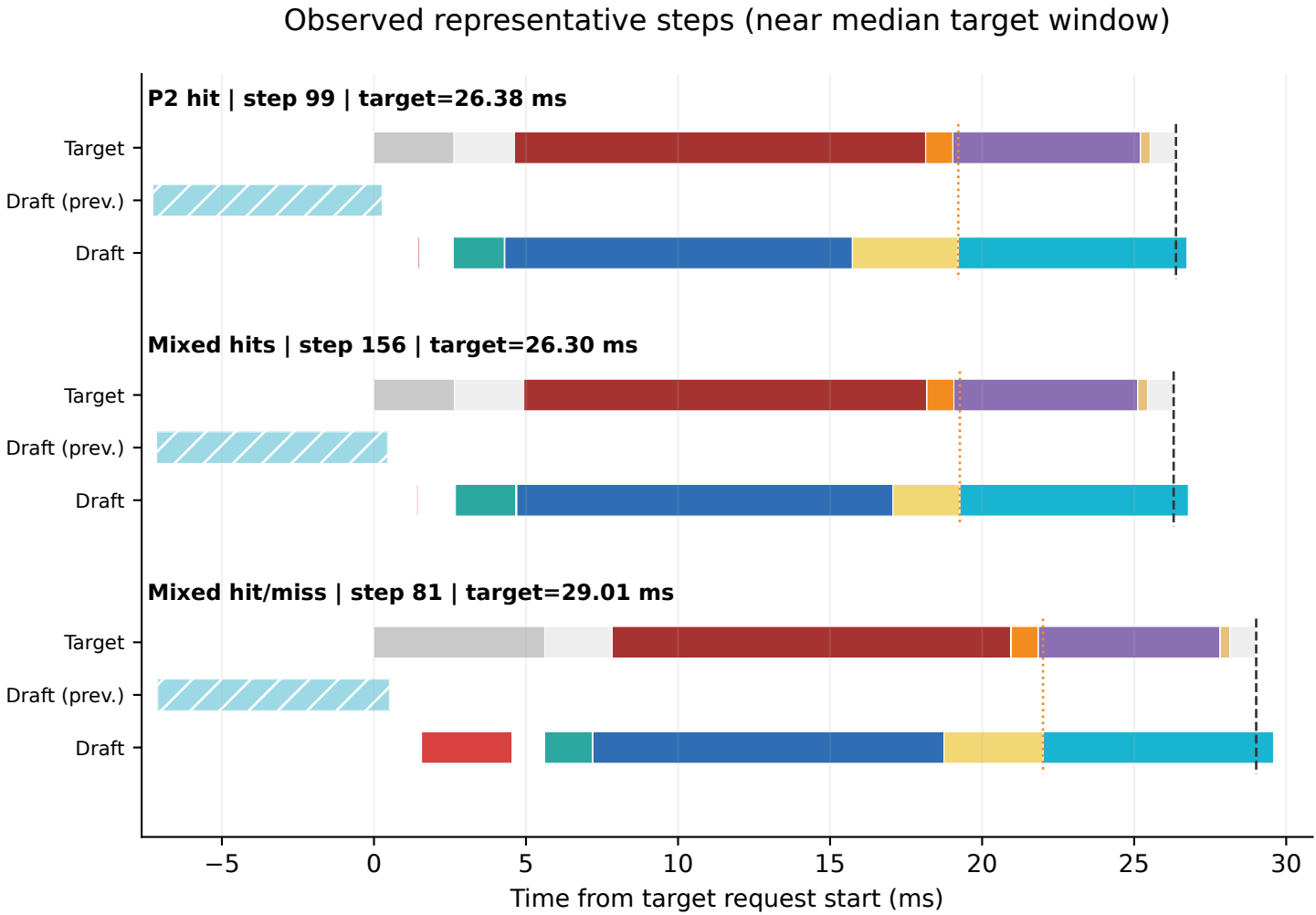
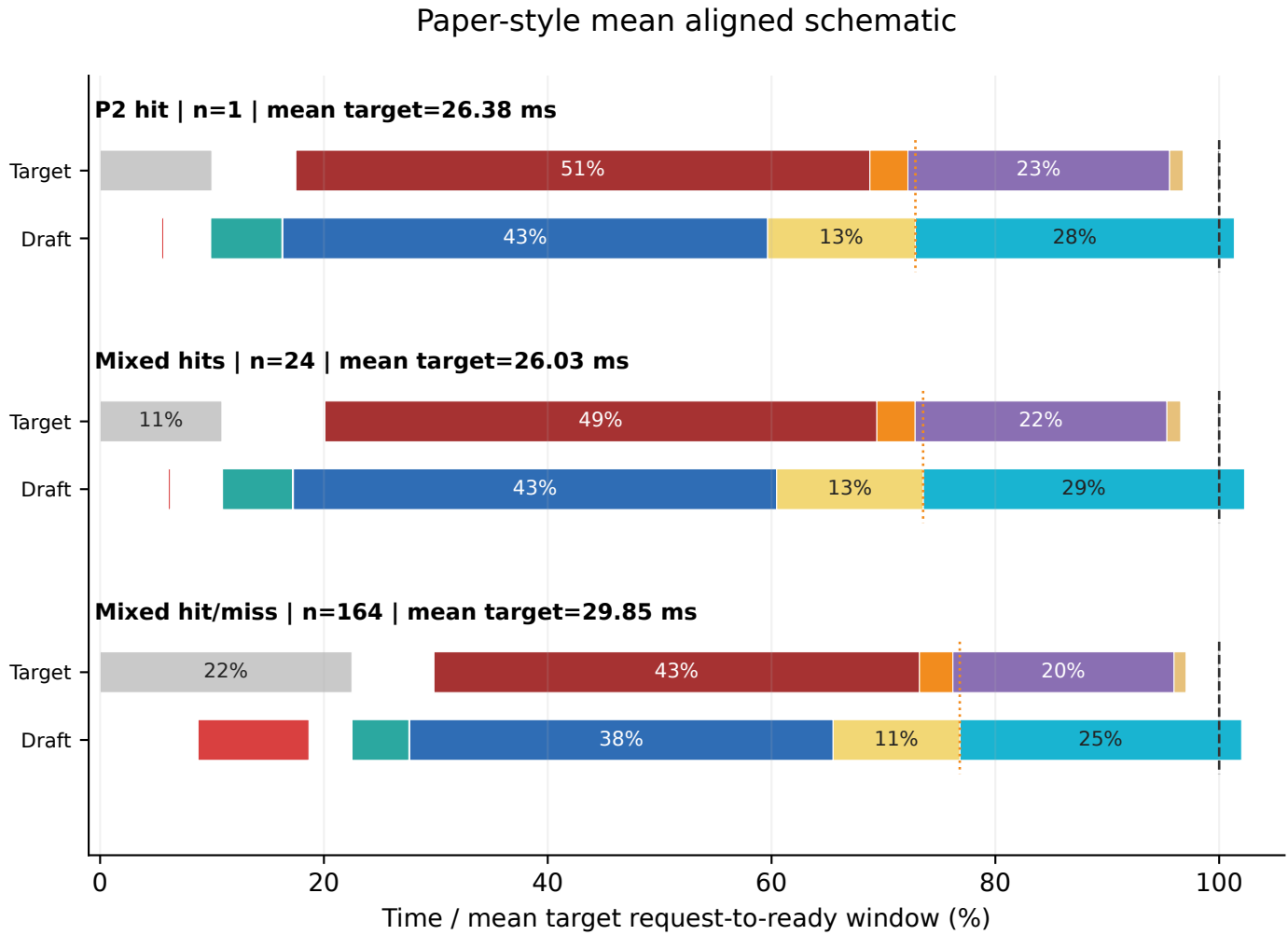
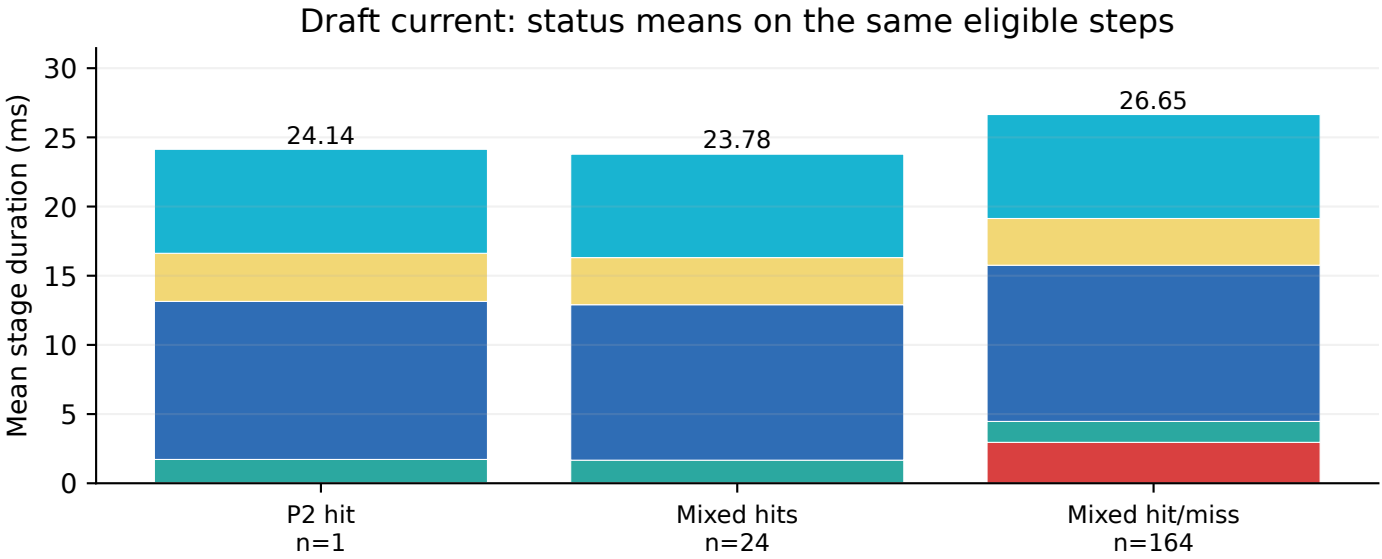
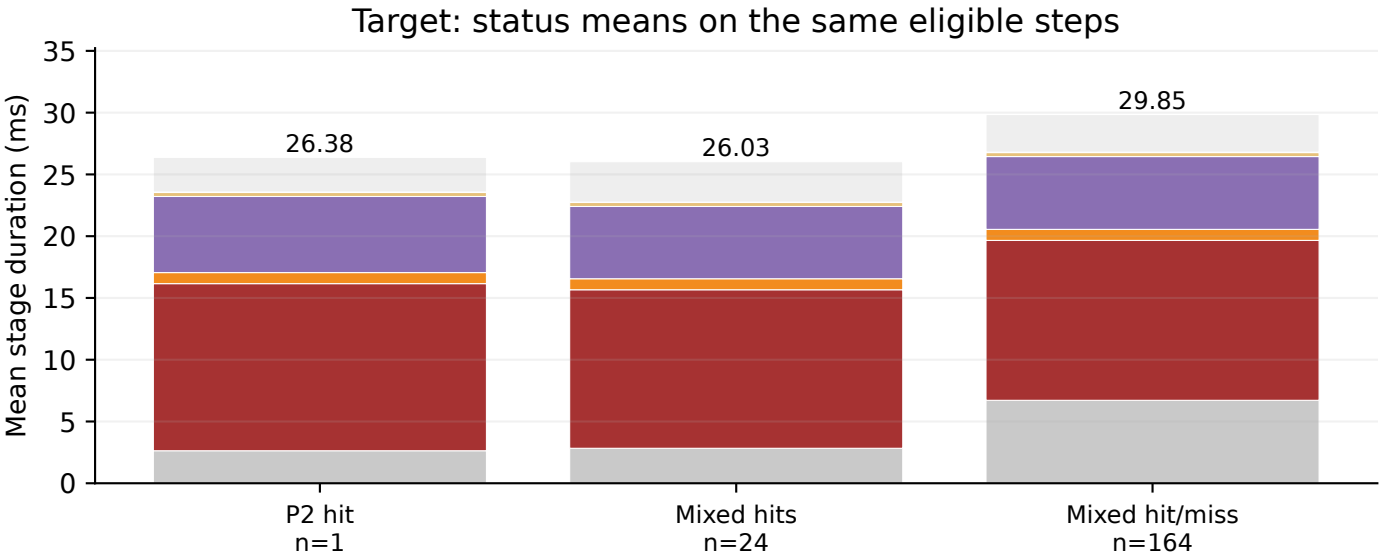
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e21_k3_1_r2
DUET: K1/K2=3/1, exit=21 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 197 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

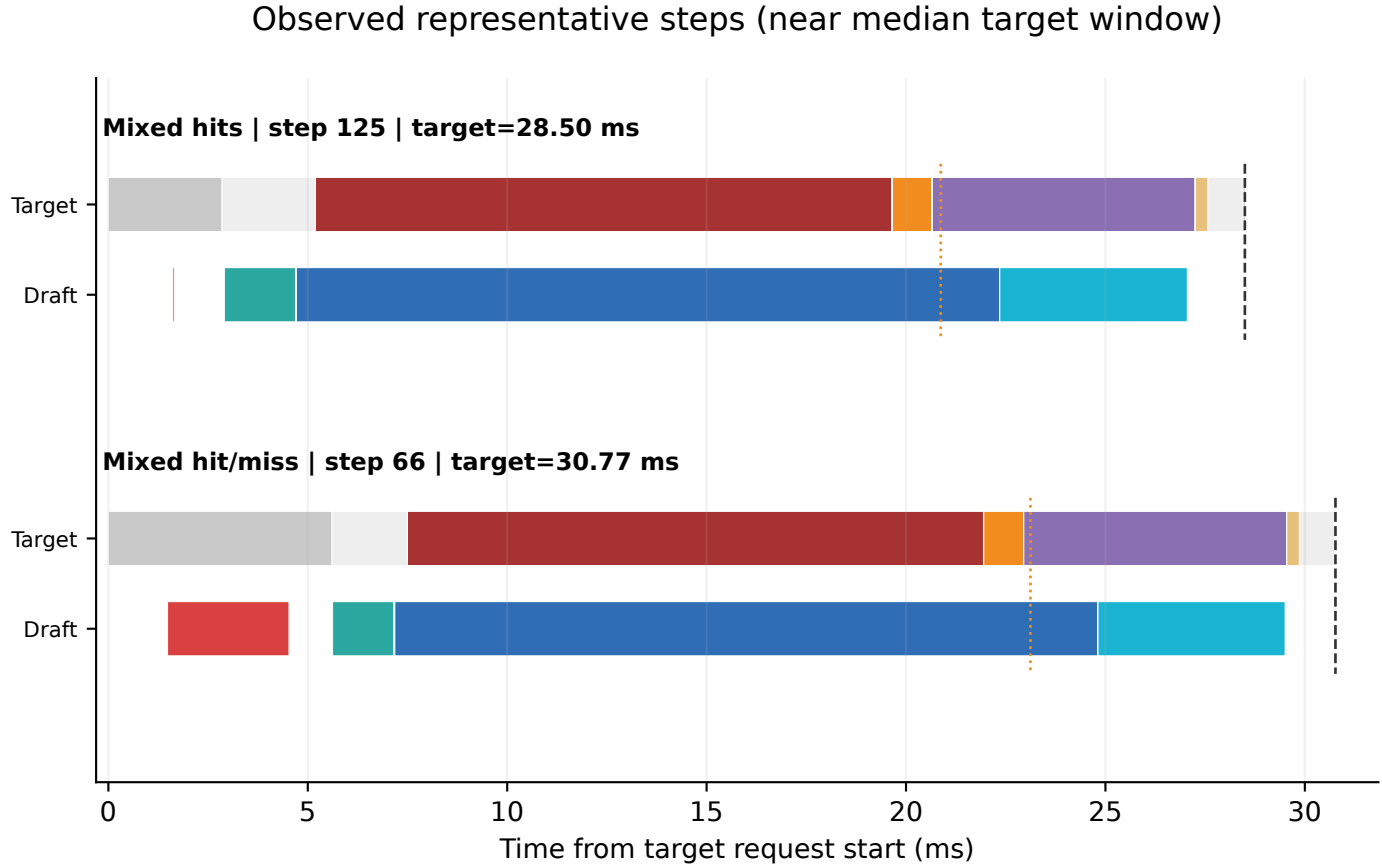
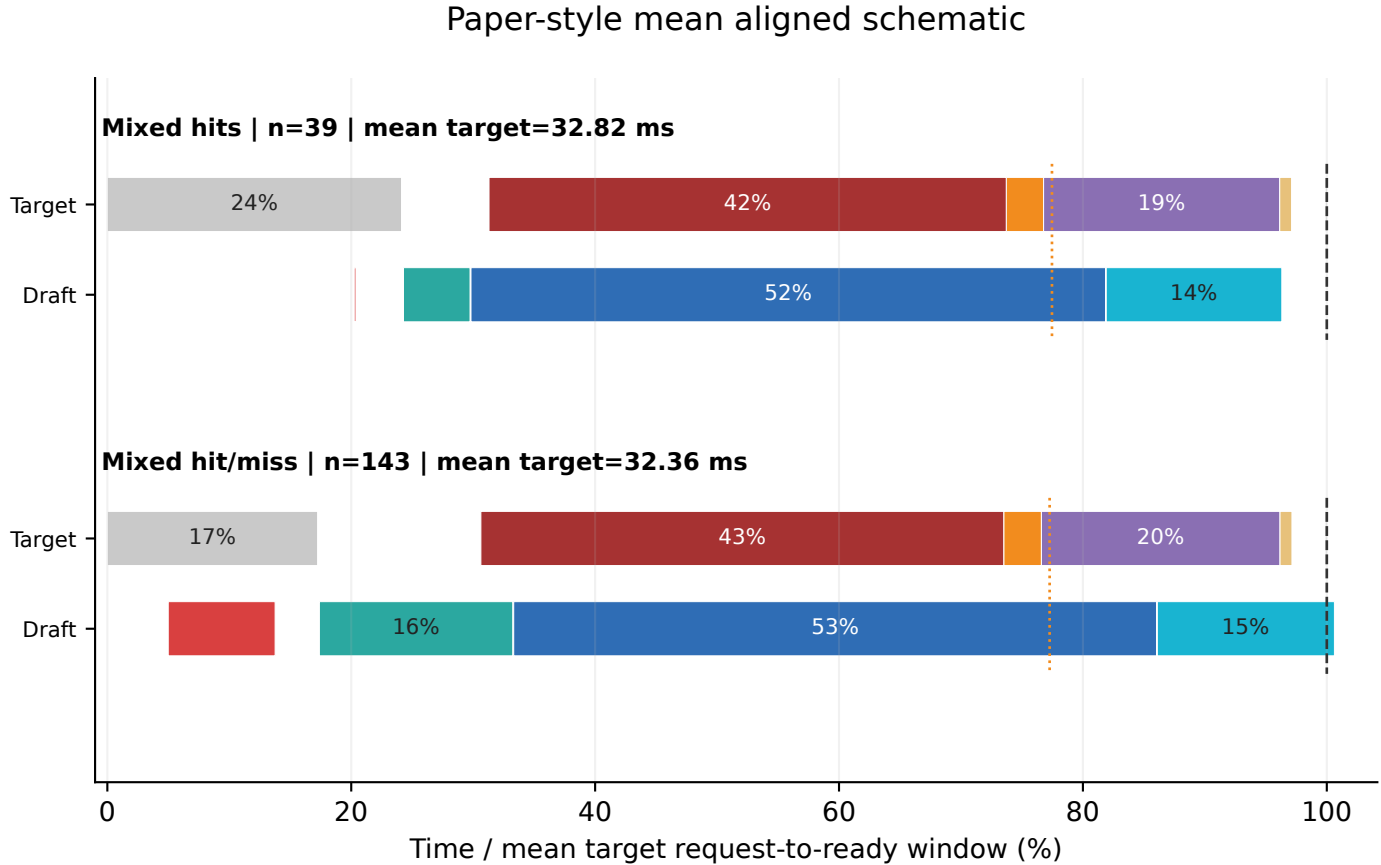
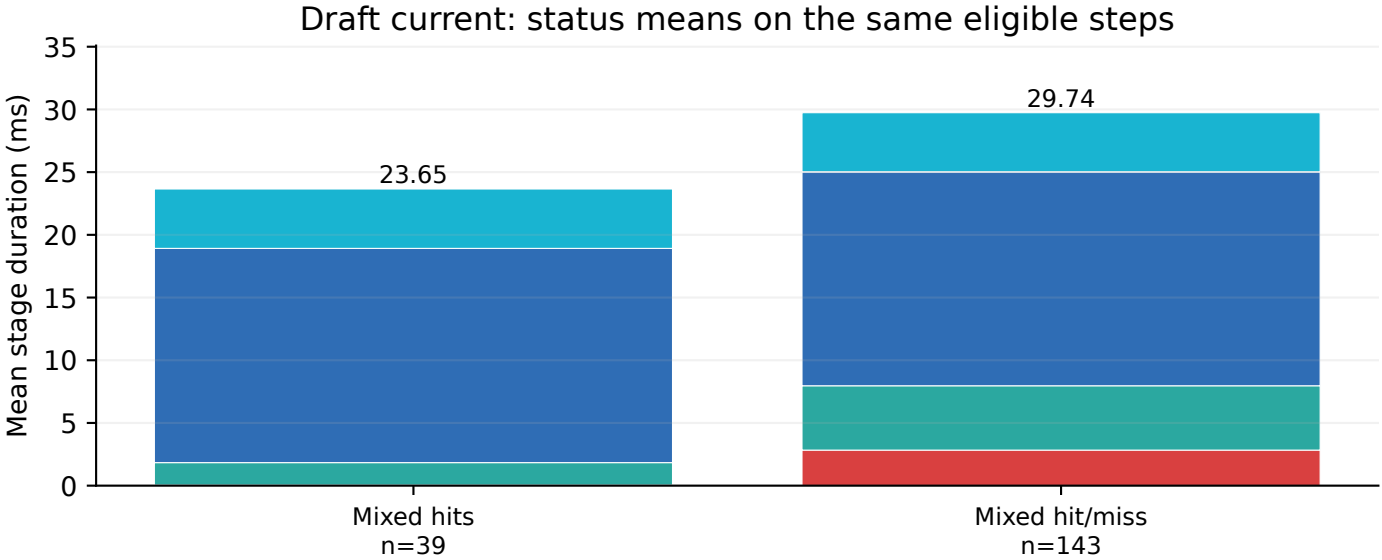
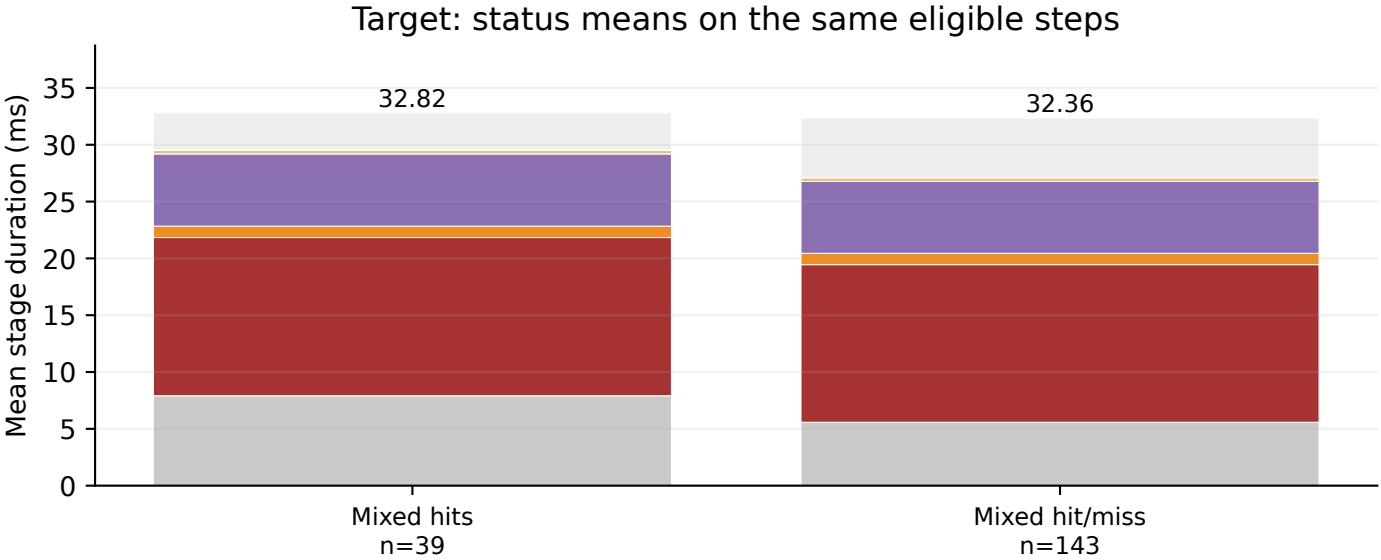
llama2_b8_refine_e21_k3_2_r2
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 189 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

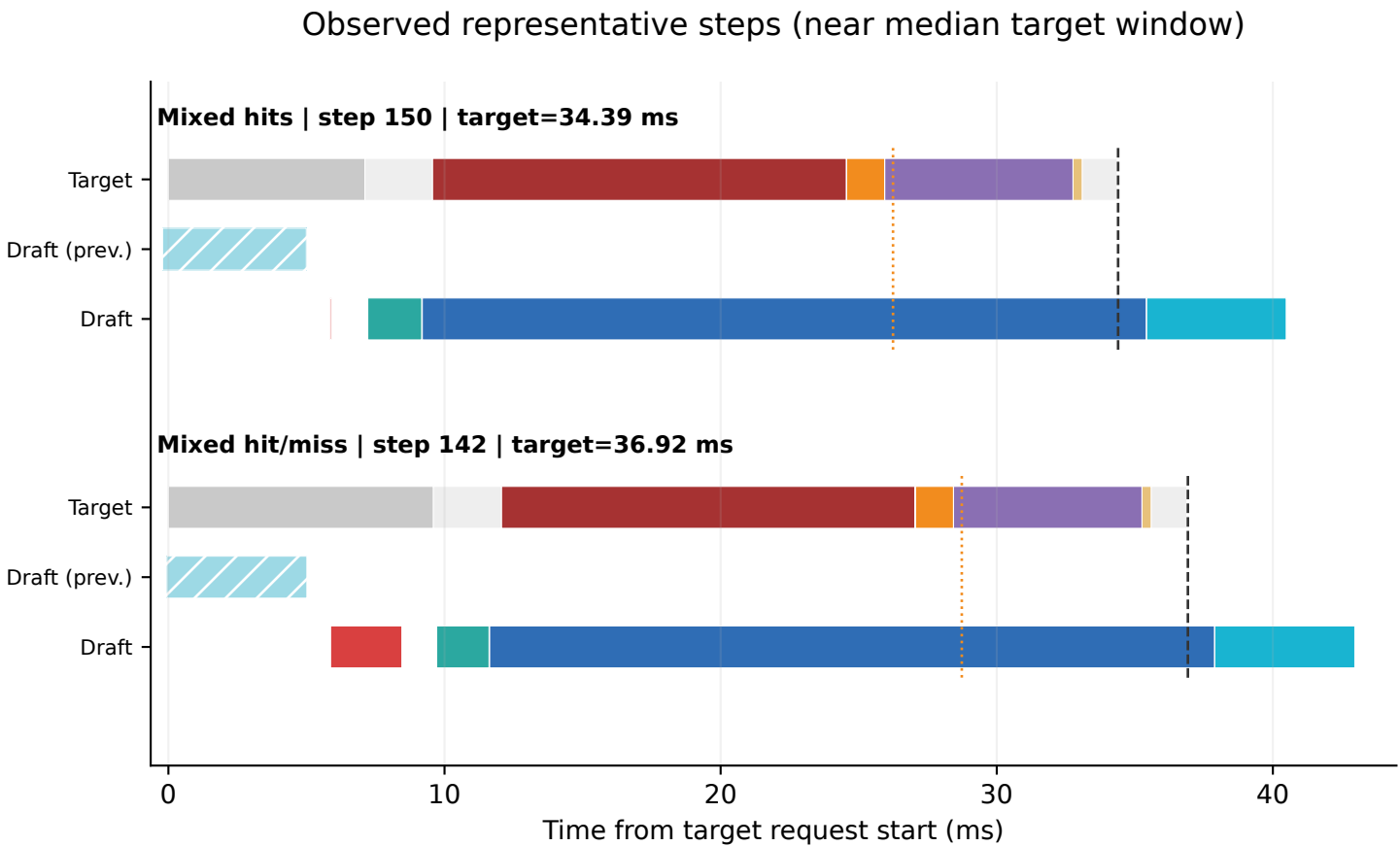
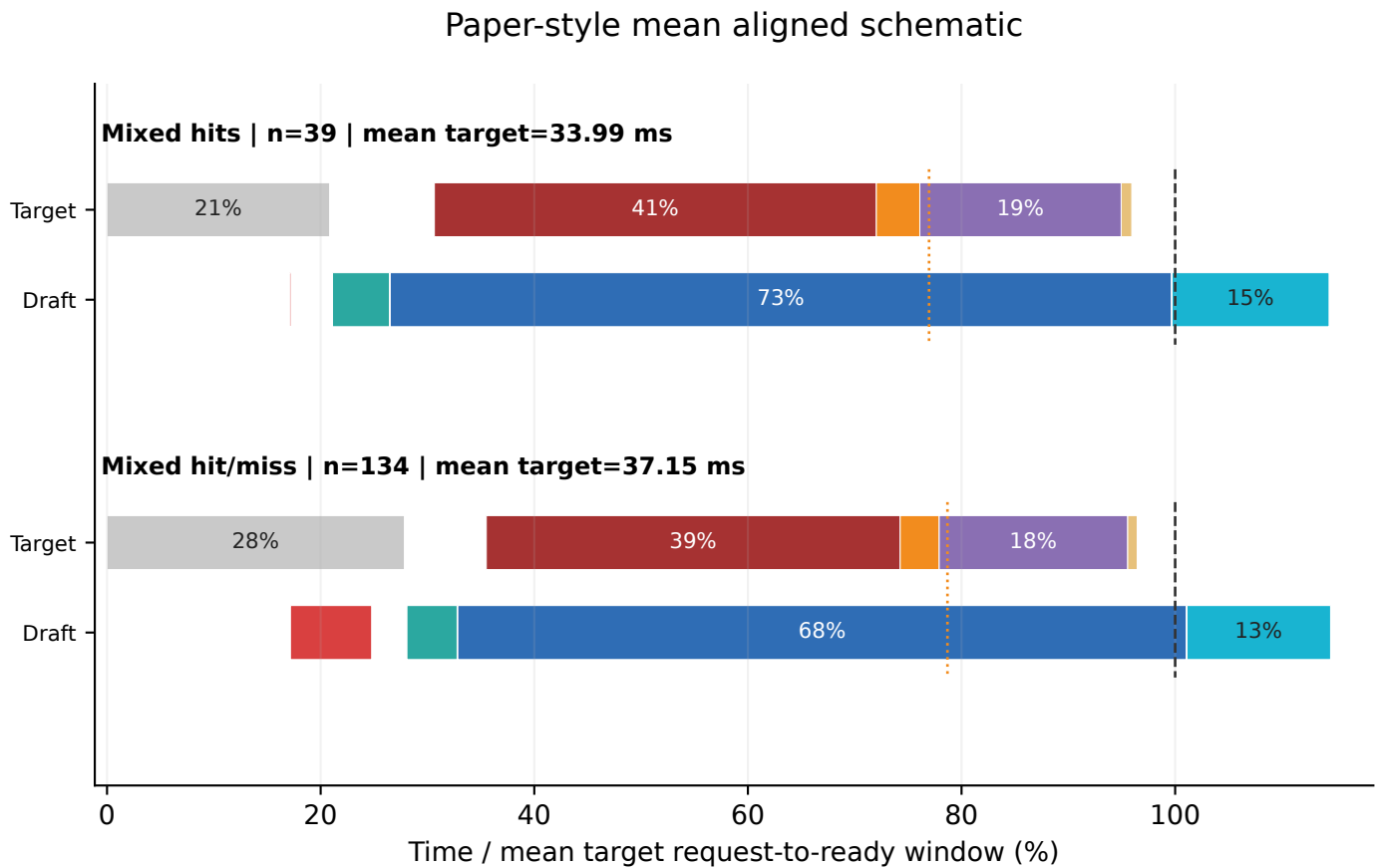
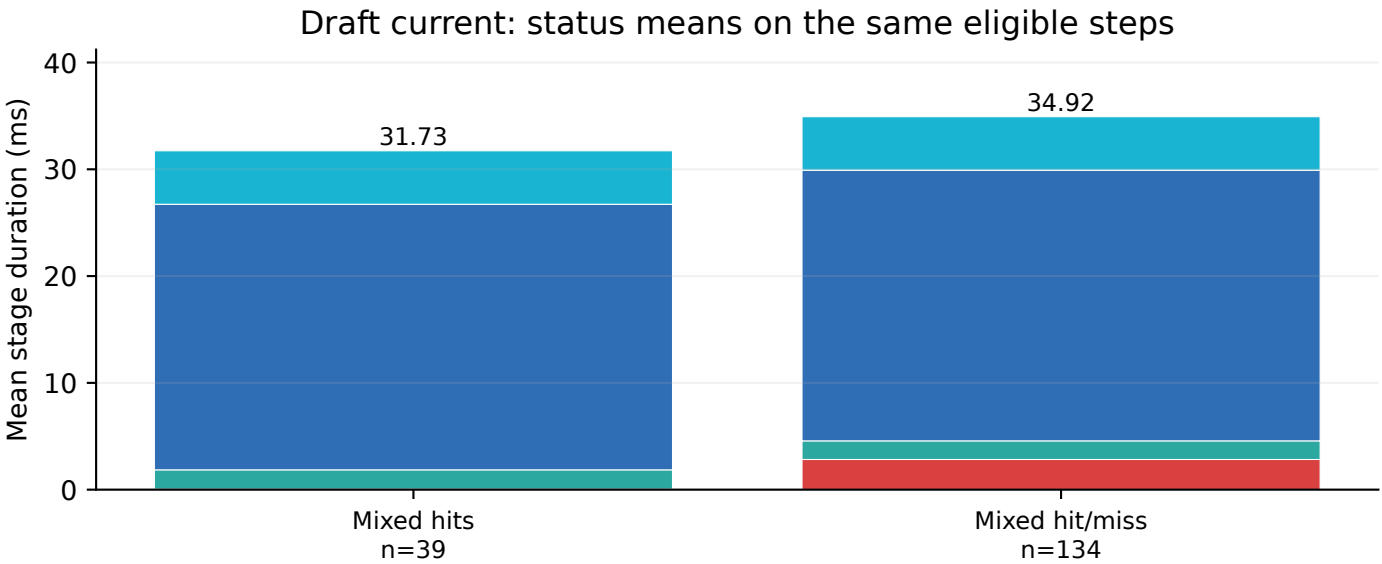
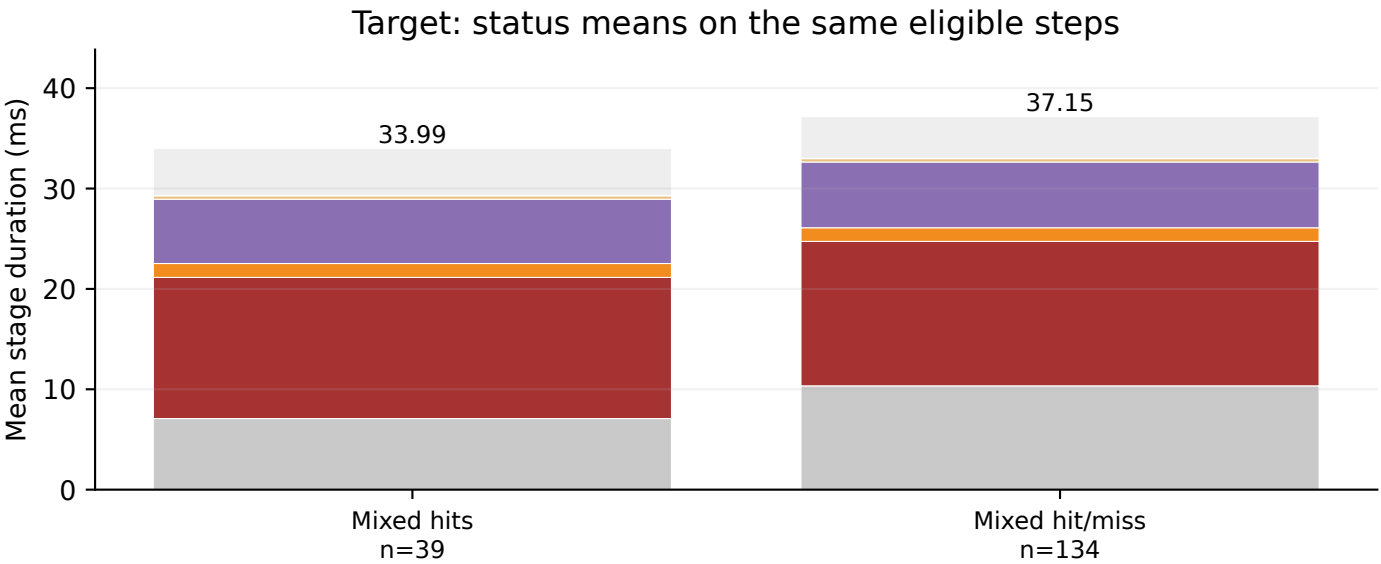
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e21_k4_1_r2
DUET: K1/K2=4/1, exit=21 (0-based), C=3, N1/N2=8/2, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 182 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e21_k6_1_r2
DUET: K1/K2=6/1, exit=21 (0-based), C=3, N1/N2=12/2, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 173 eligible steps

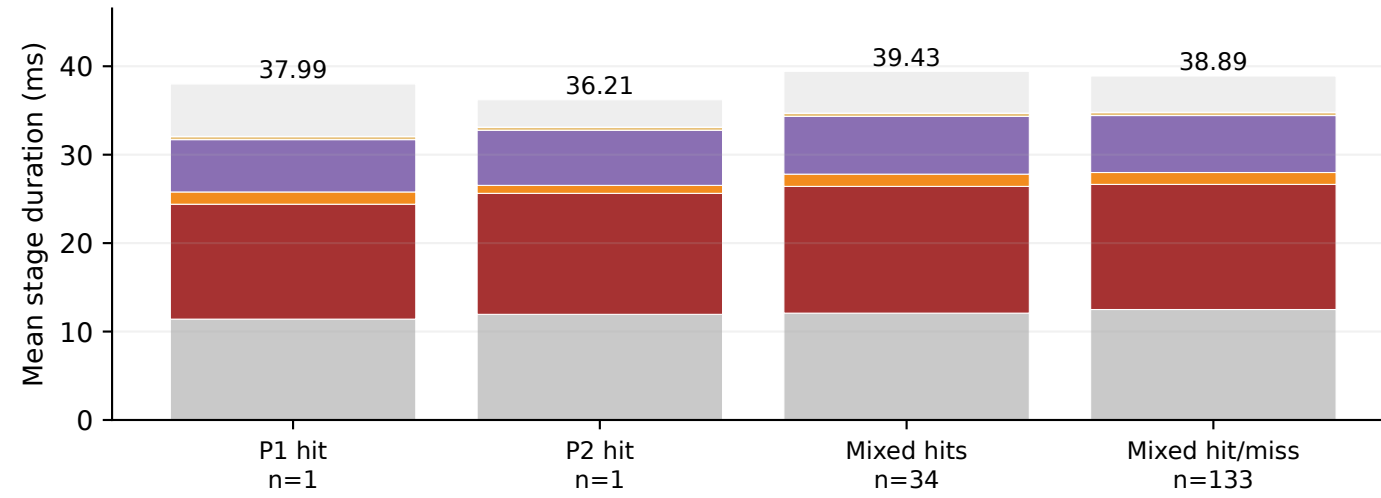


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

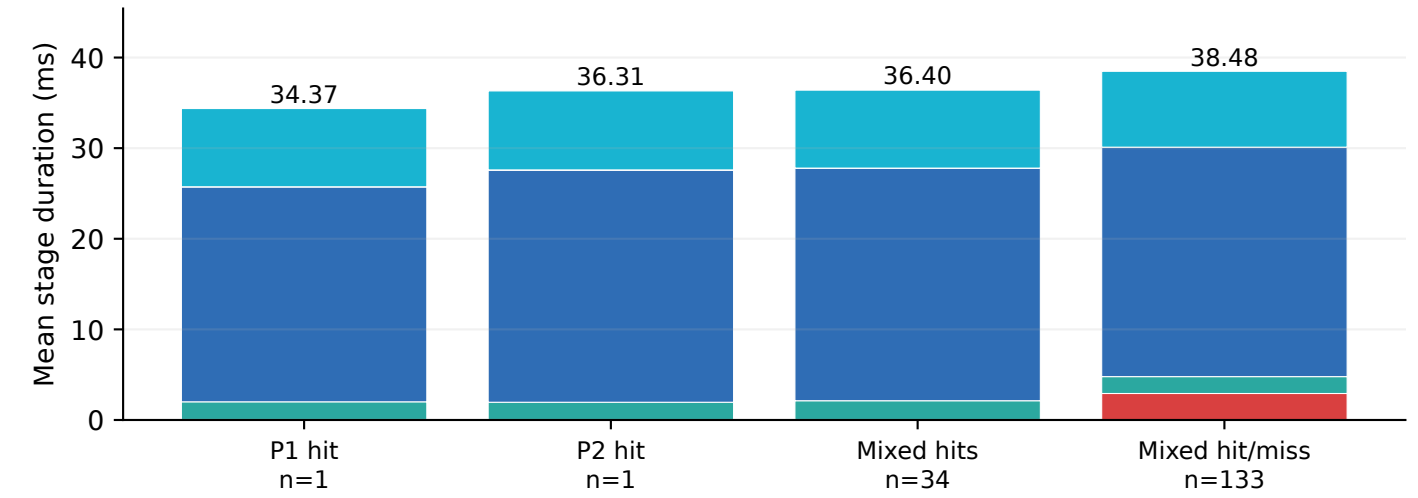
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_refine_e21_k6_2_r2
 DUET: K1/K2=6/2, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 169 eligible steps

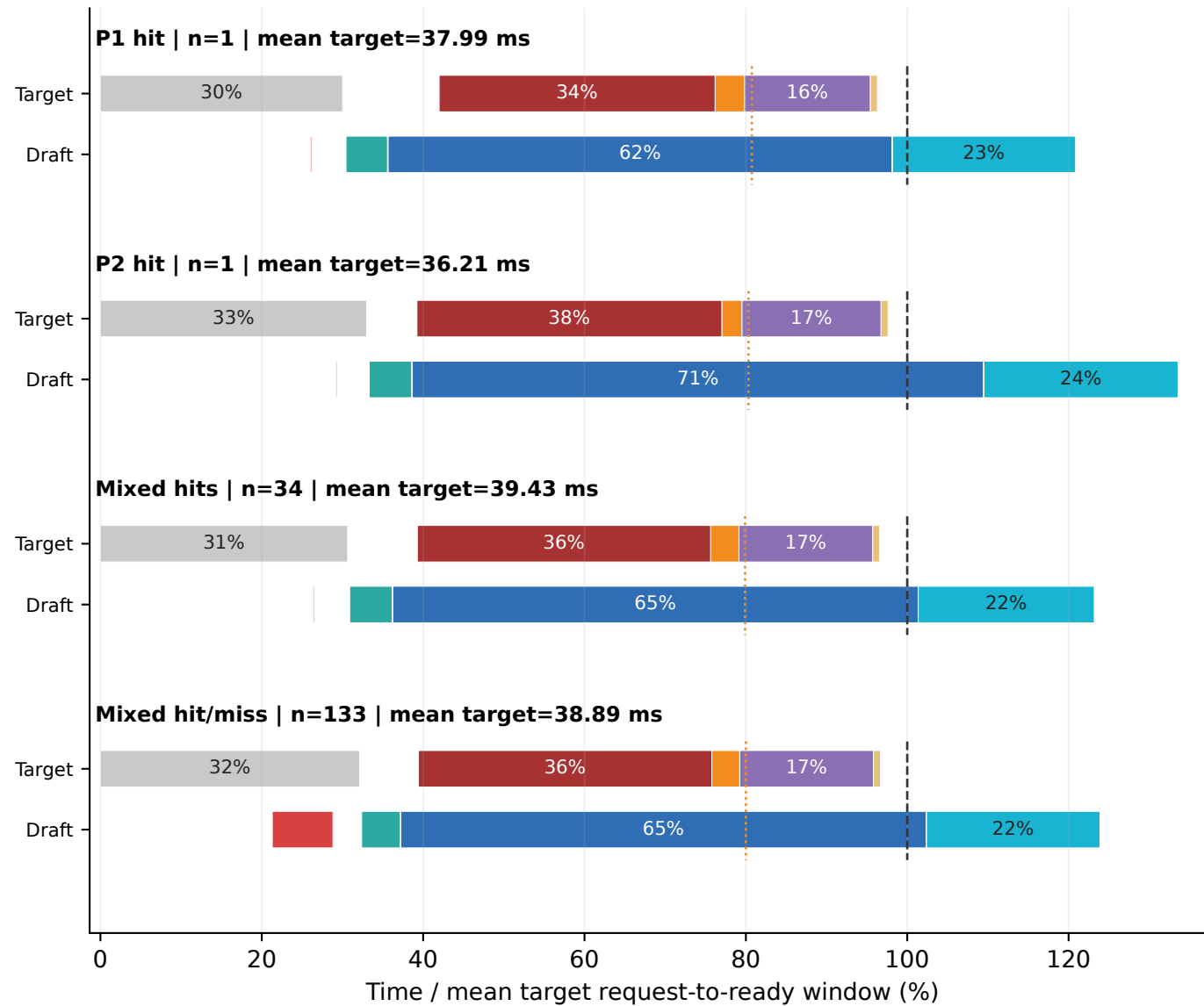
Target: status means on the same eligible steps



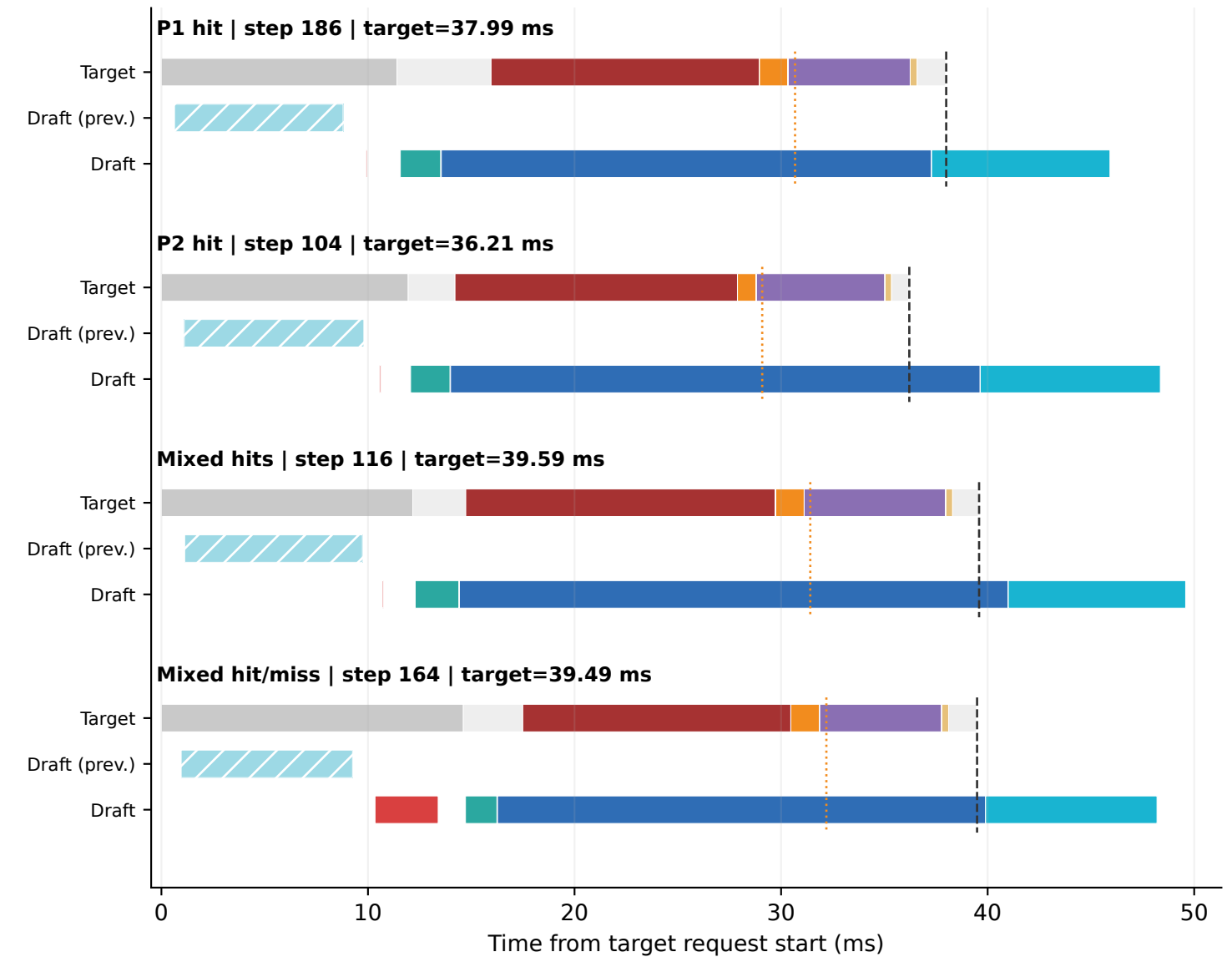
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic

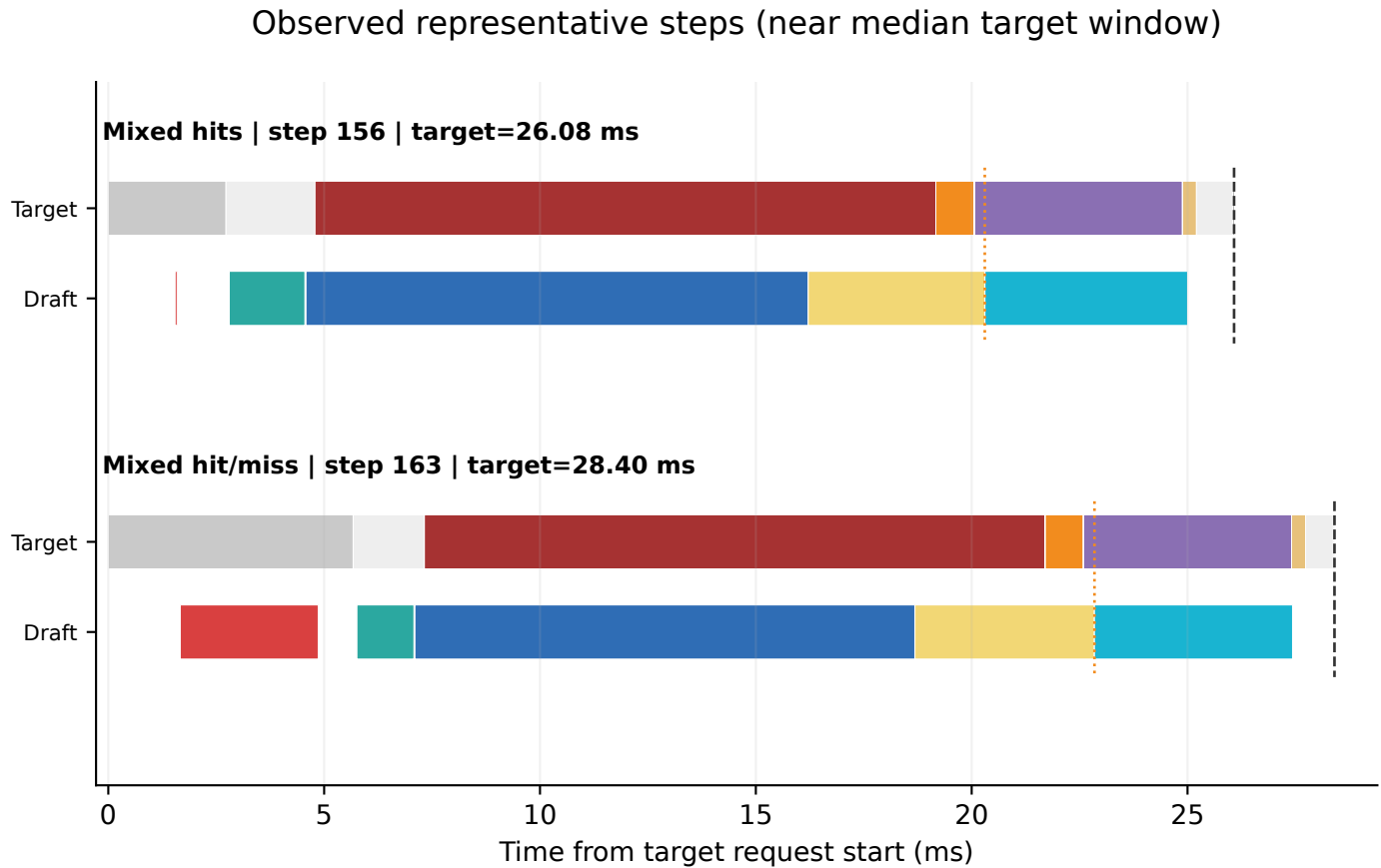
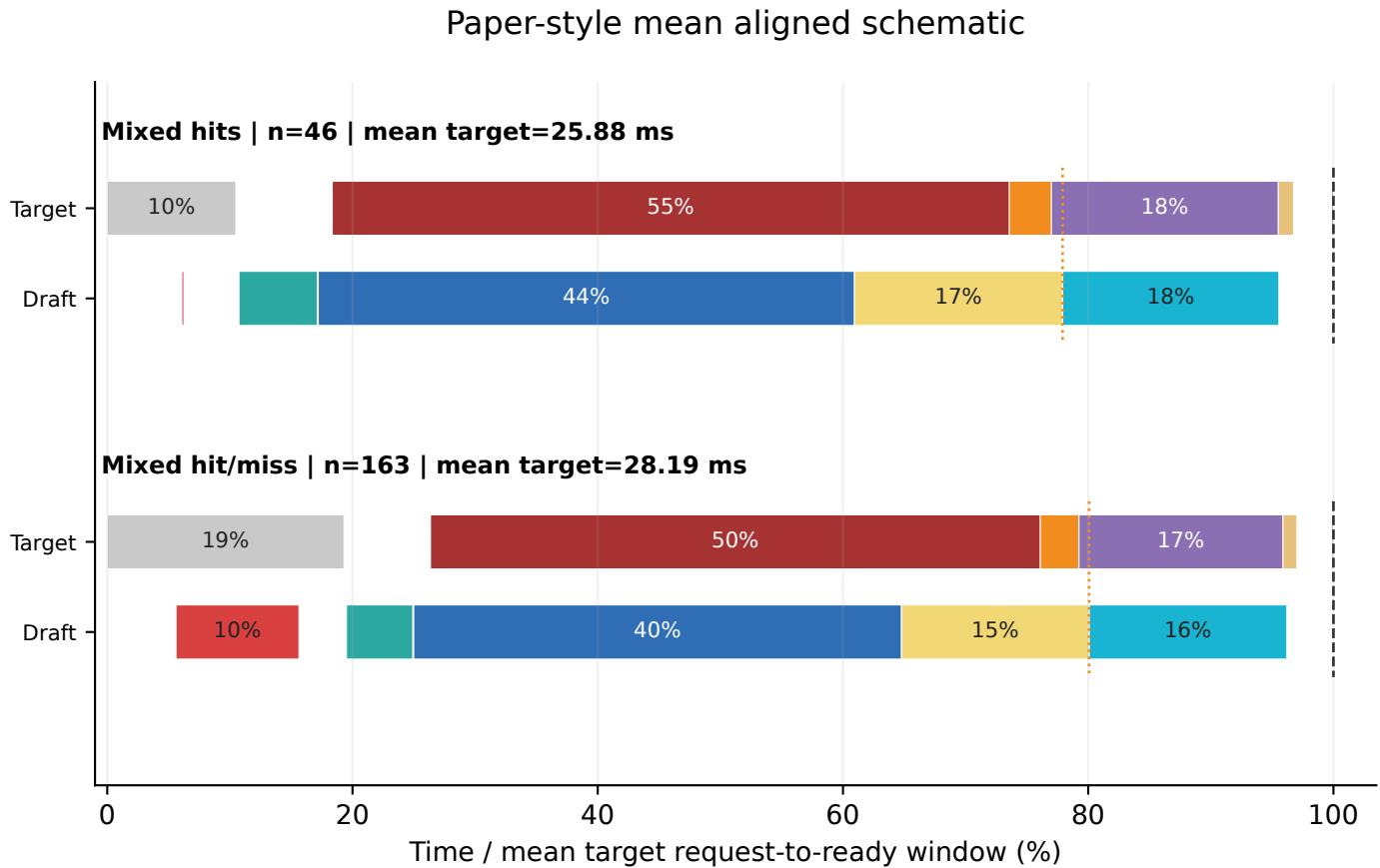
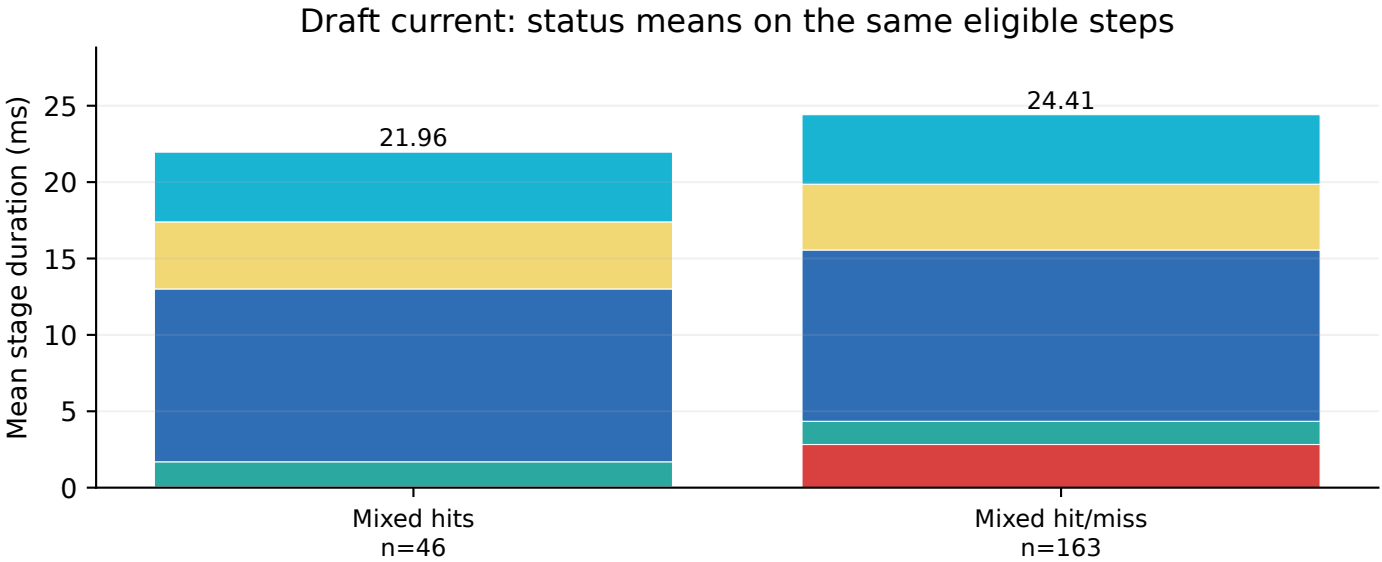
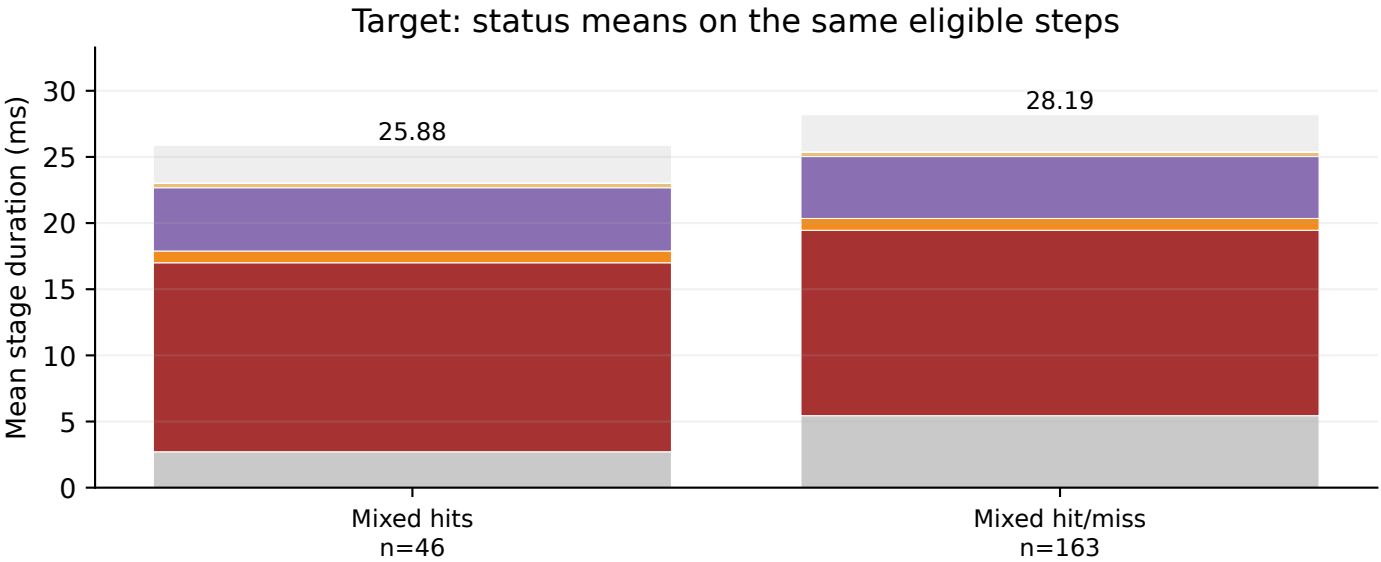


Observed representative steps (near median target window)

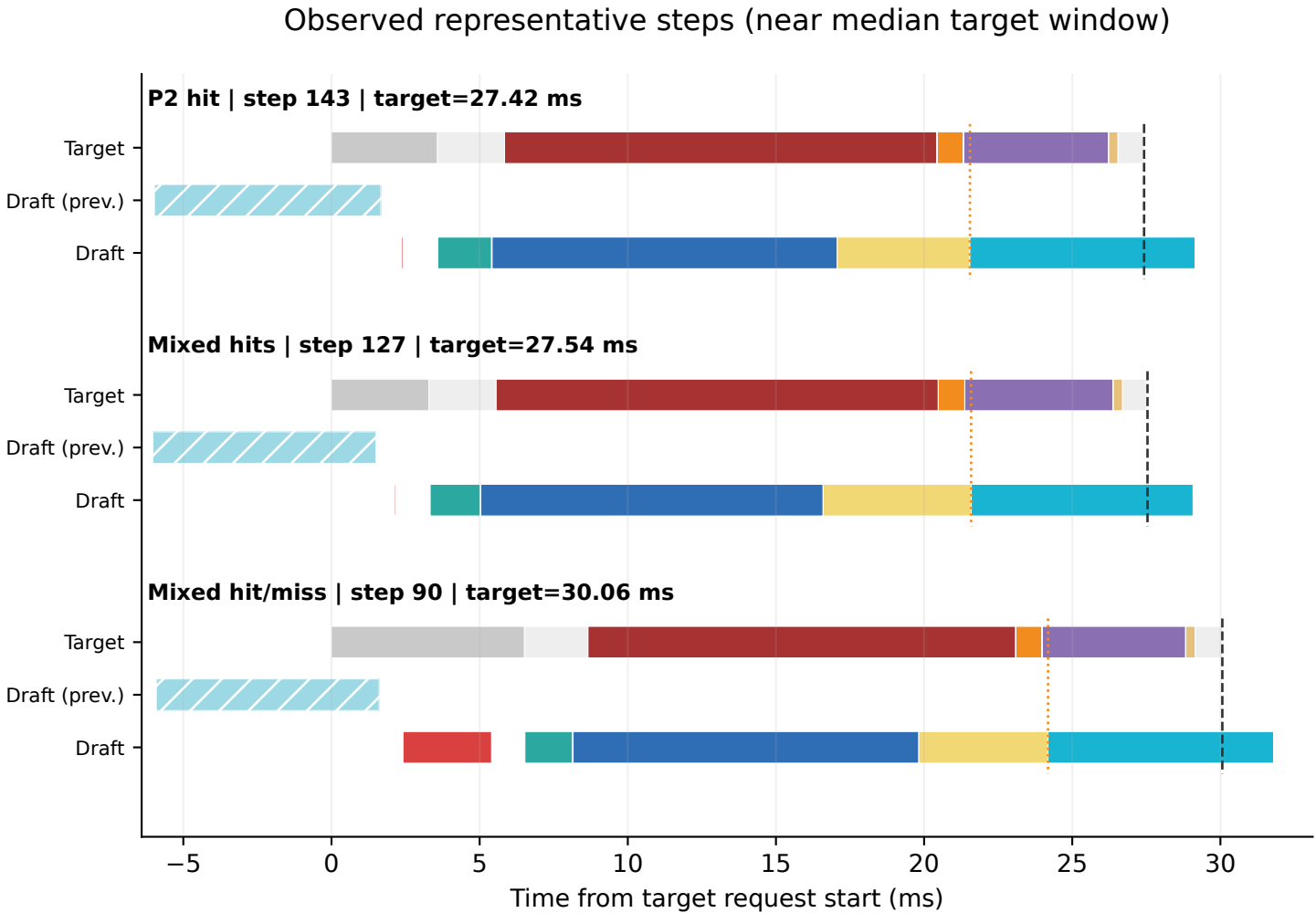
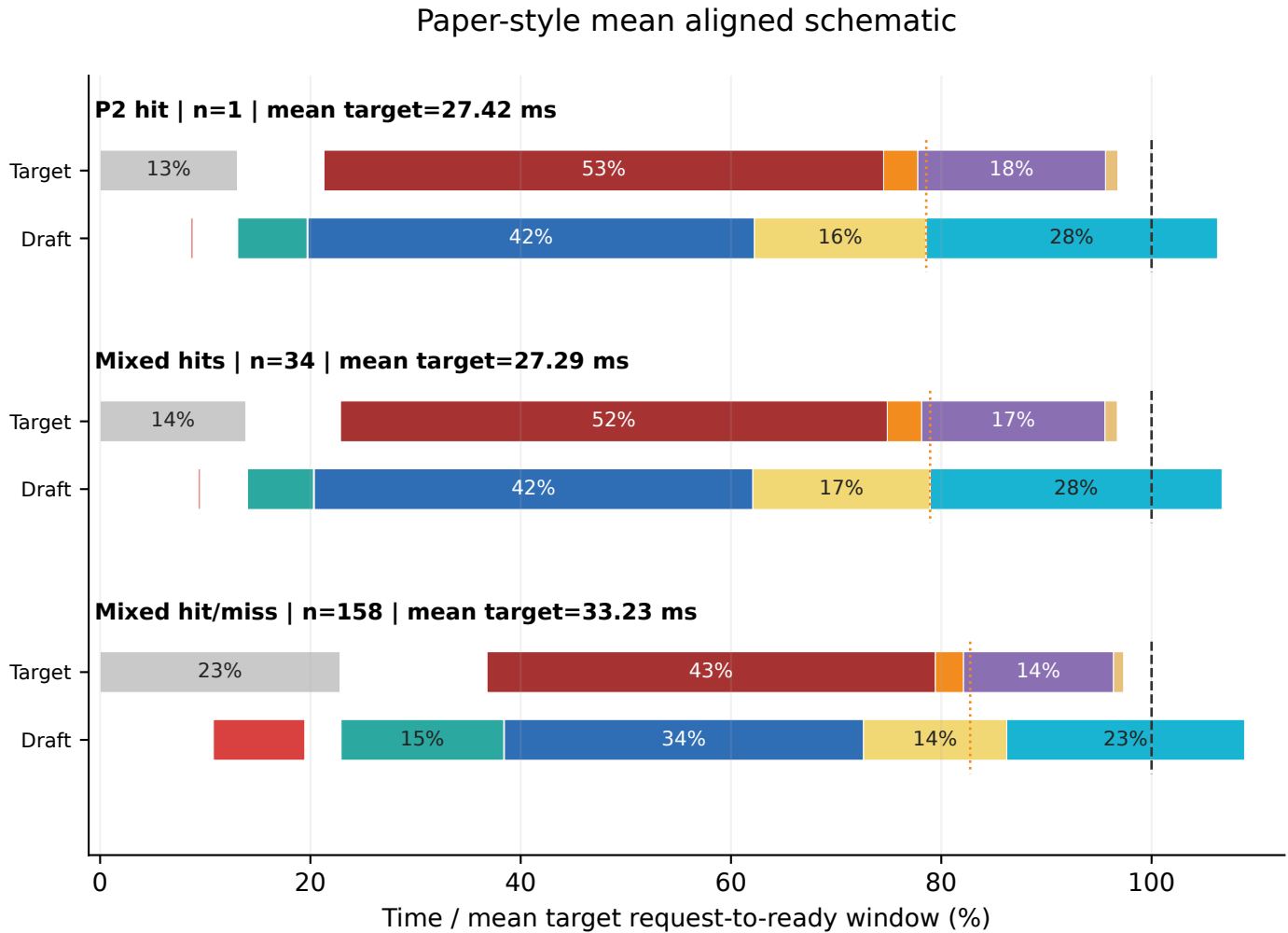
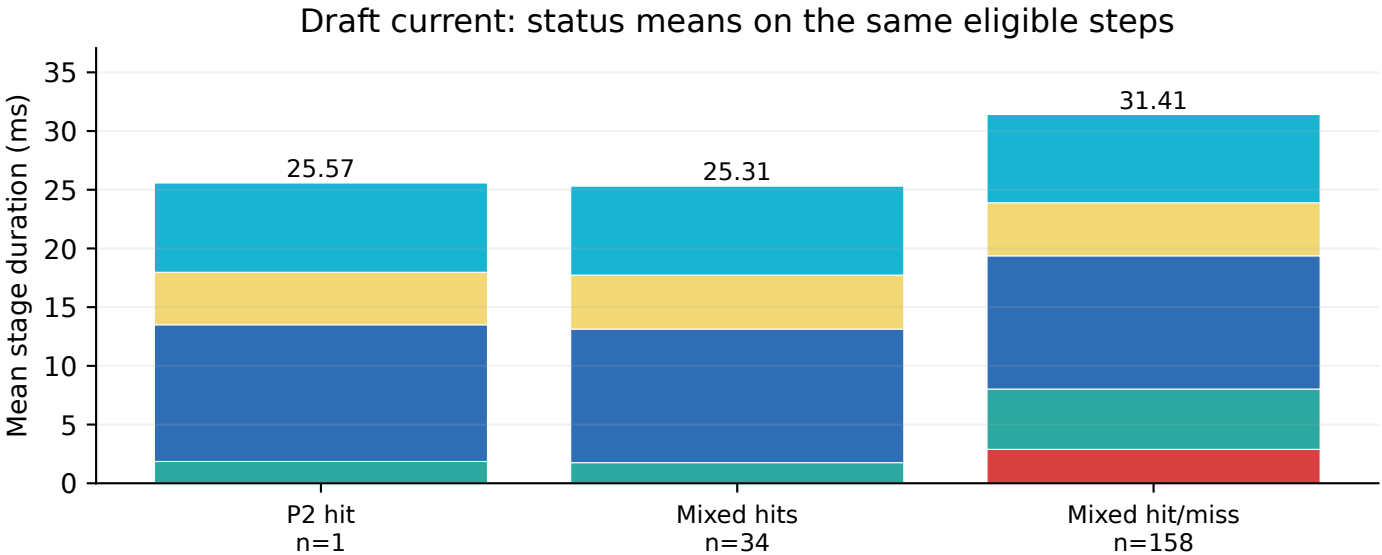
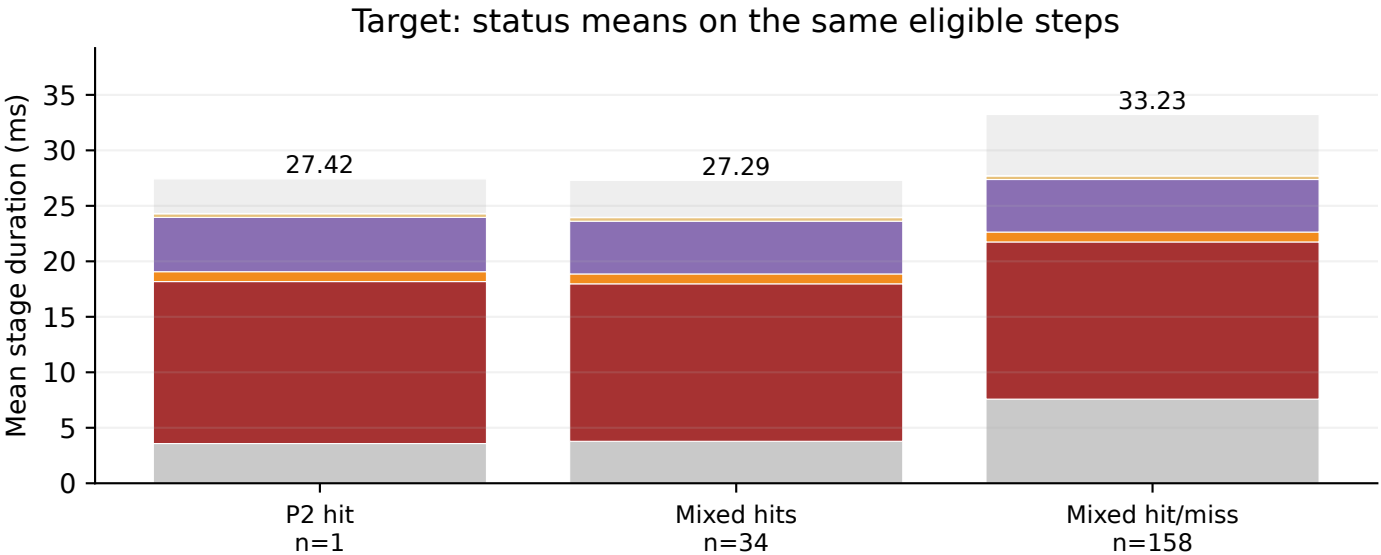


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
 Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
 Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
 Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

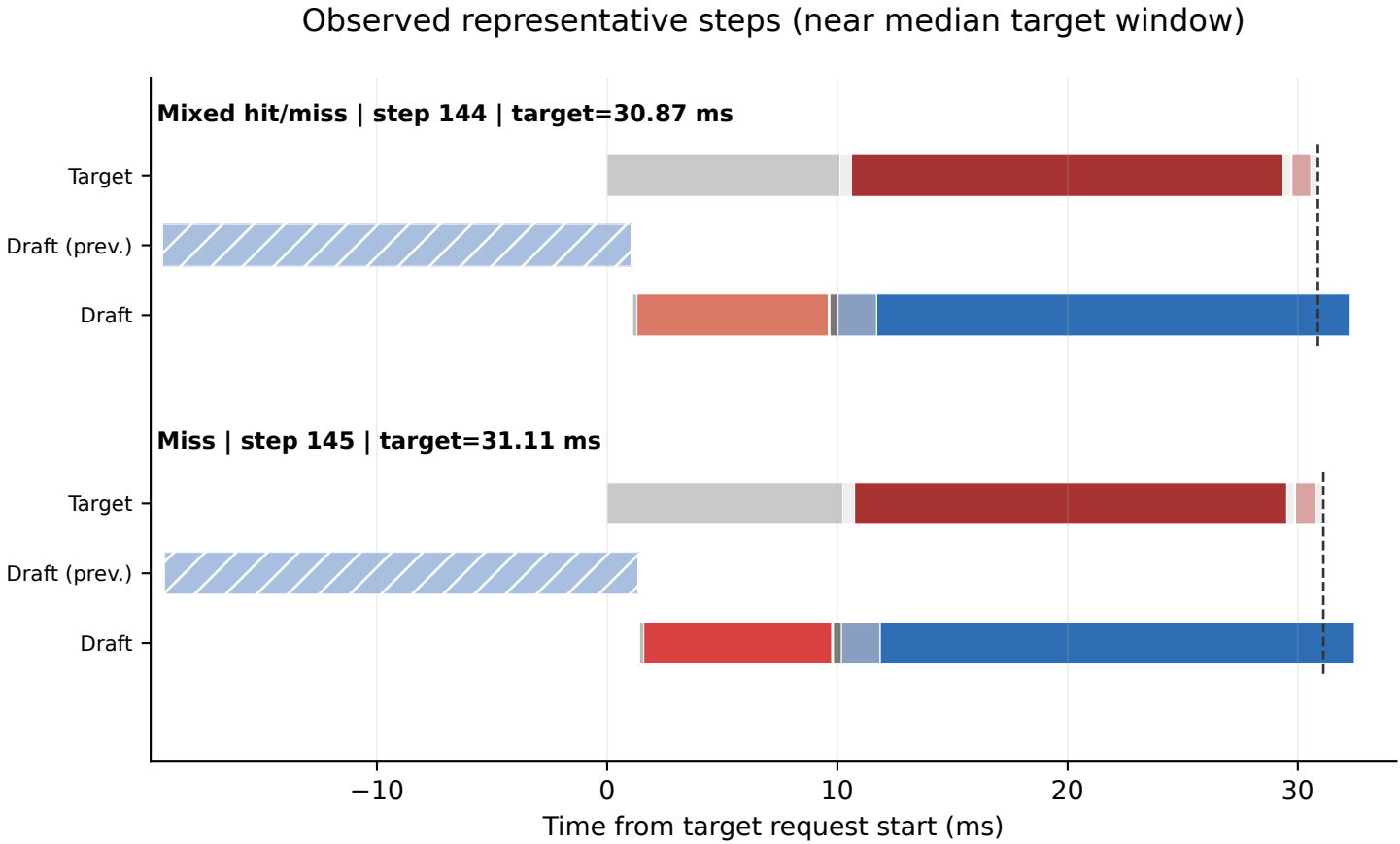
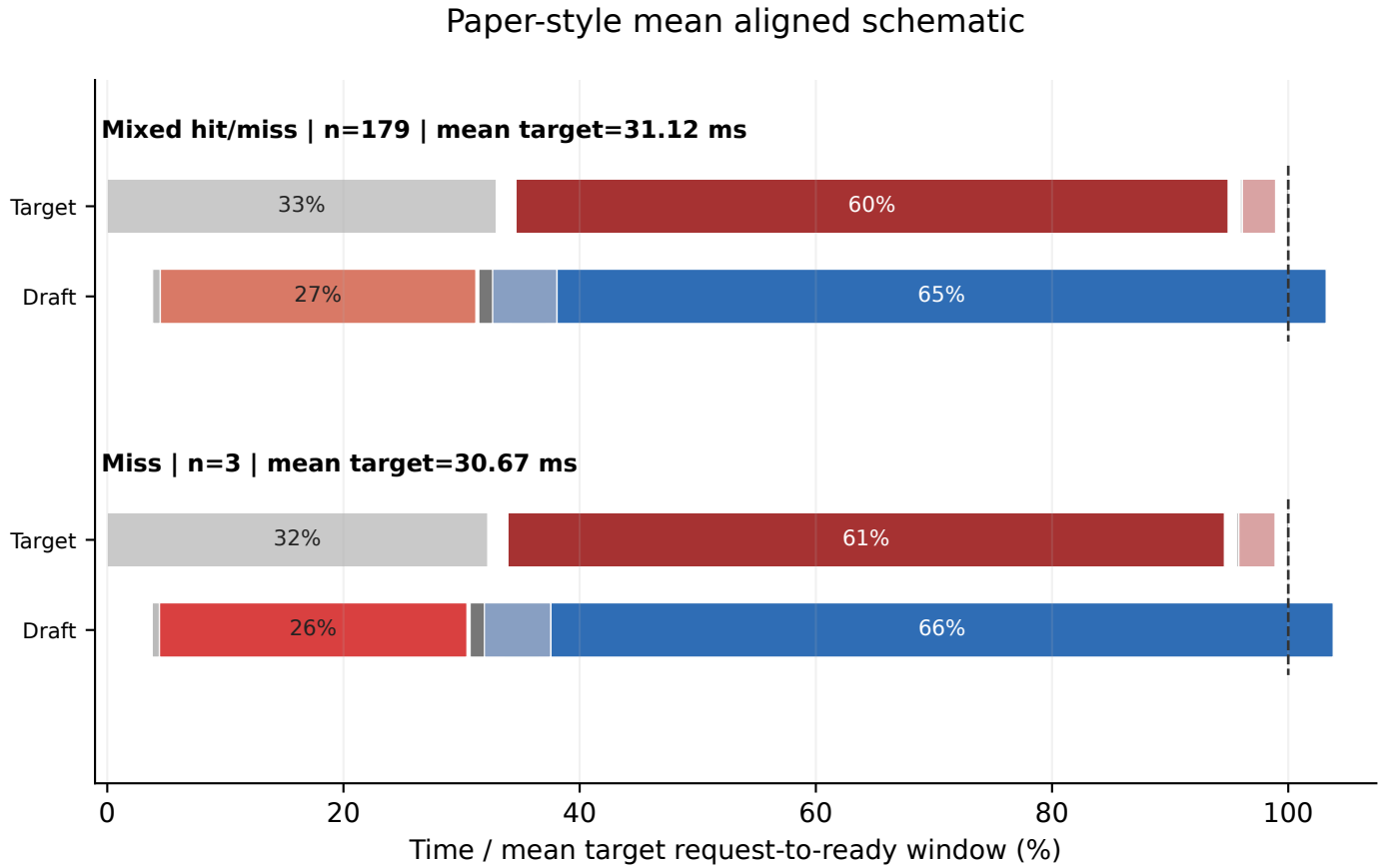
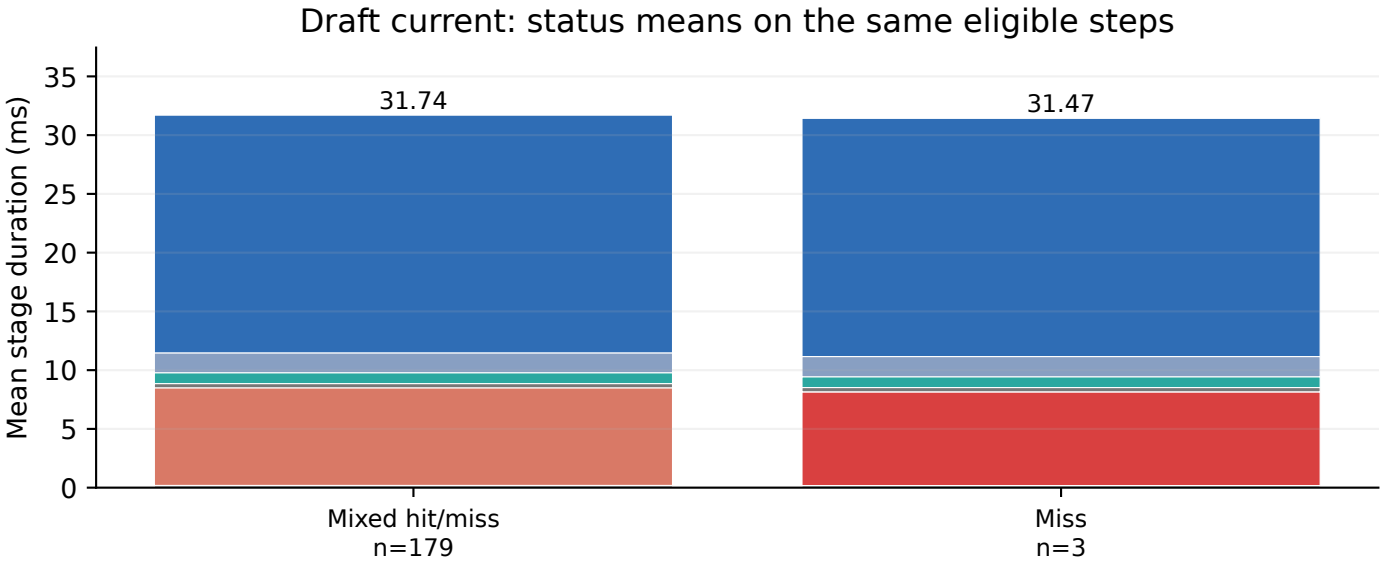
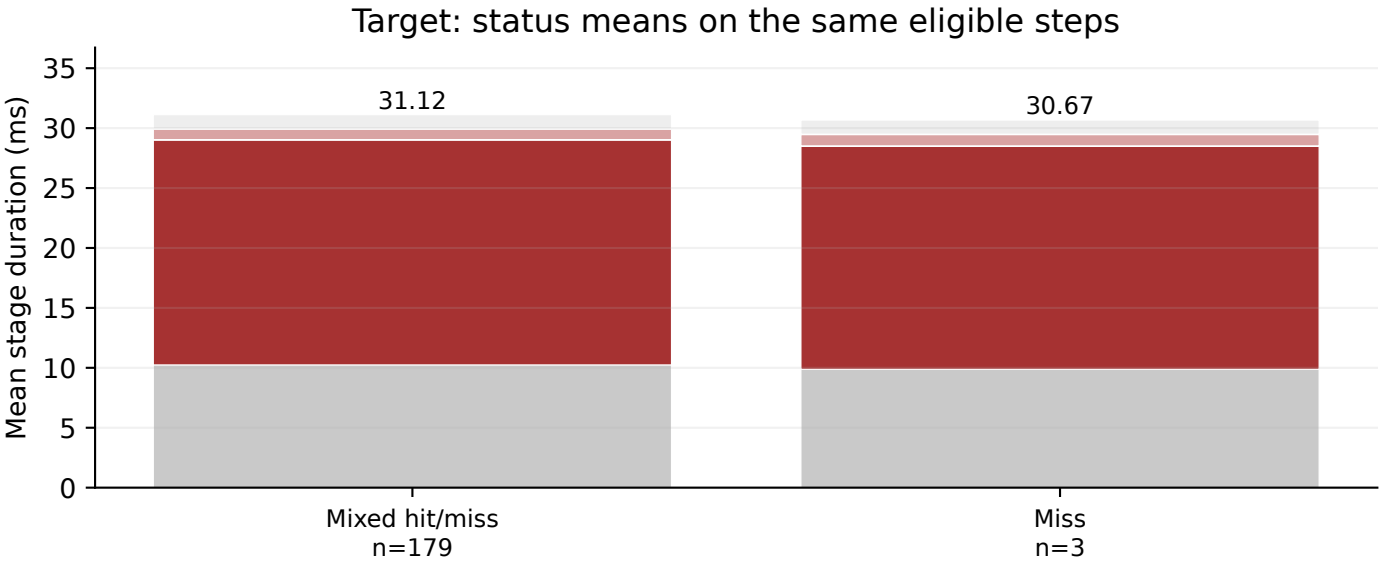


llama2_b8_refine_e23_k3_2_r2
DUET: K1/K2=3/2, exit=23 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 193 eligible steps



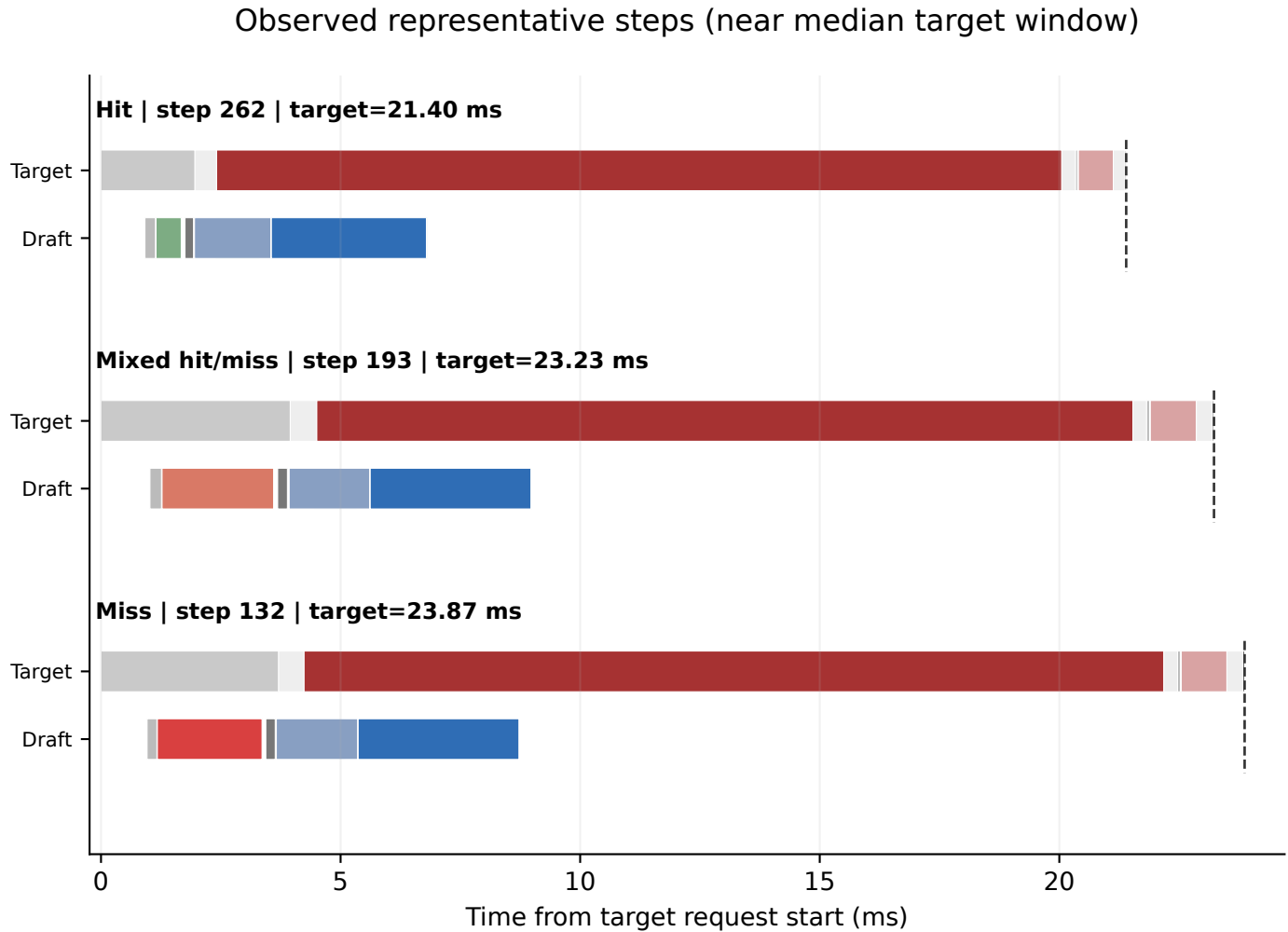
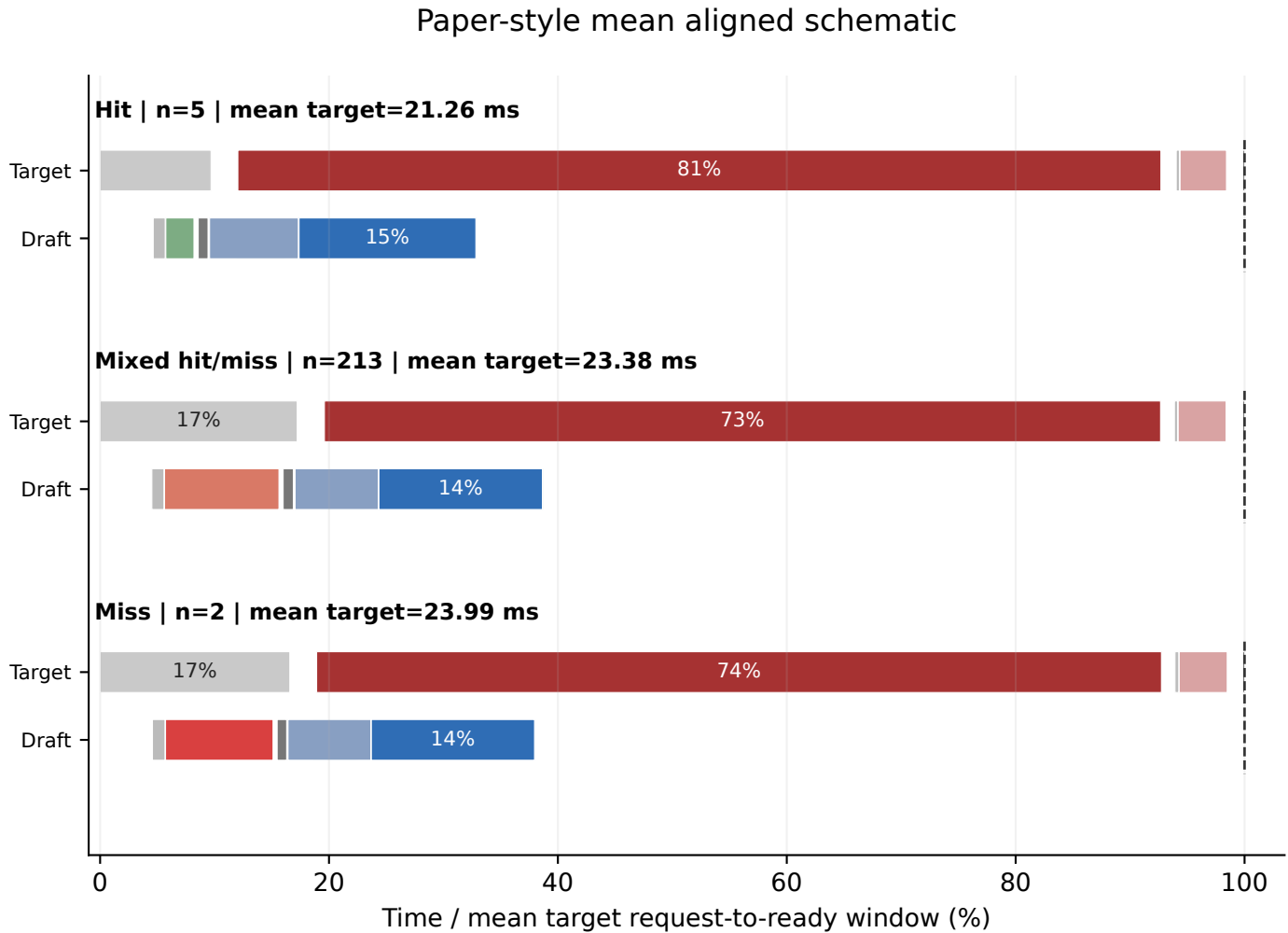
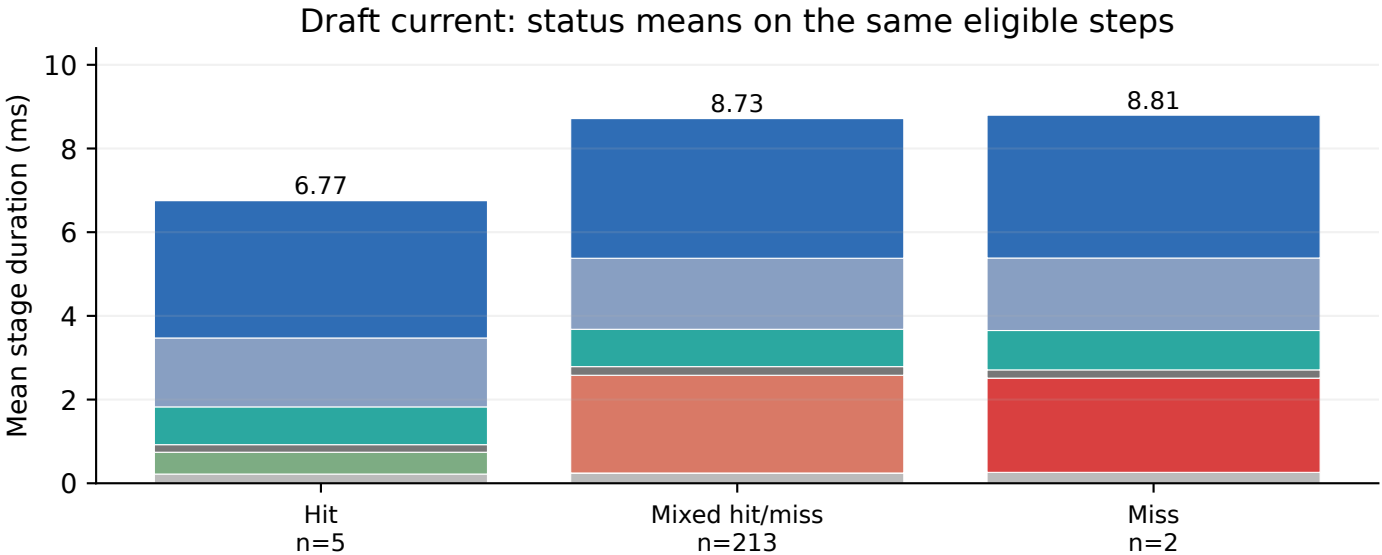
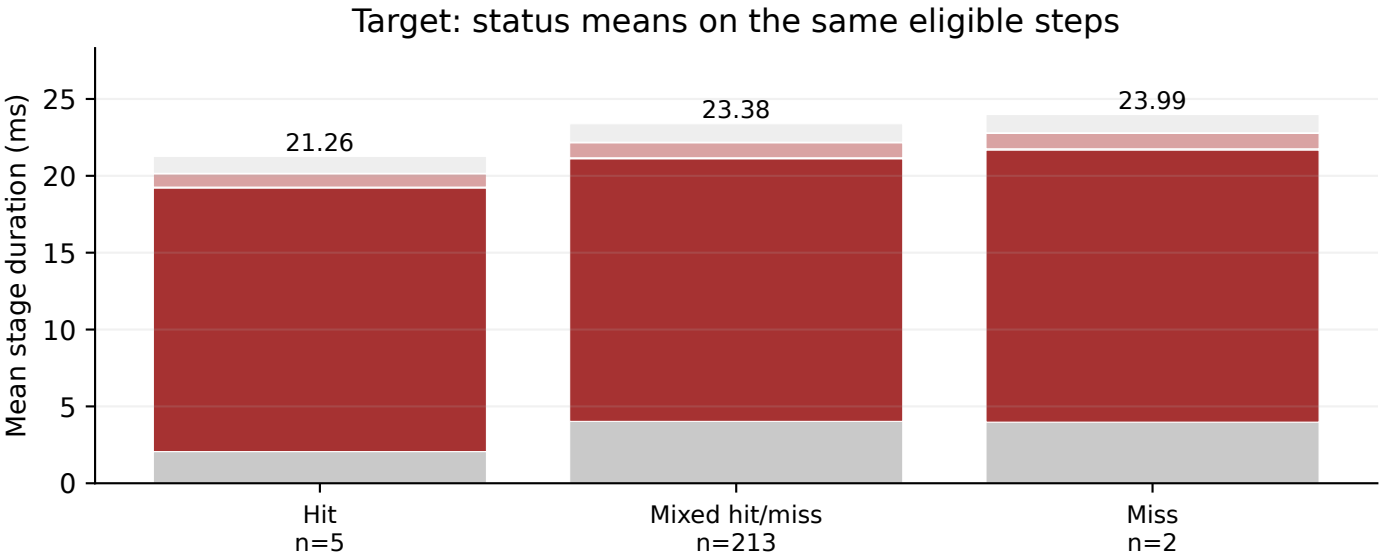
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



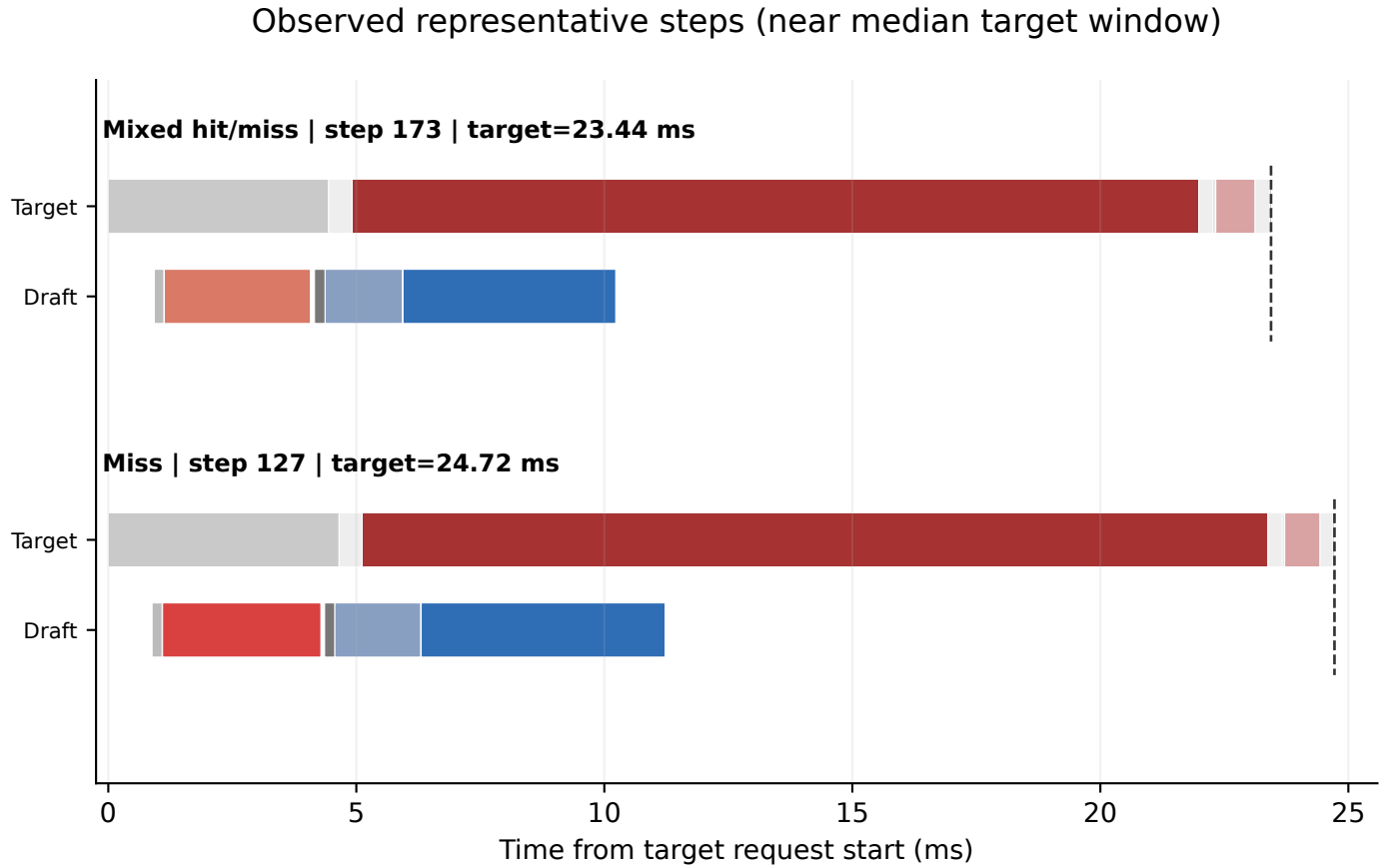
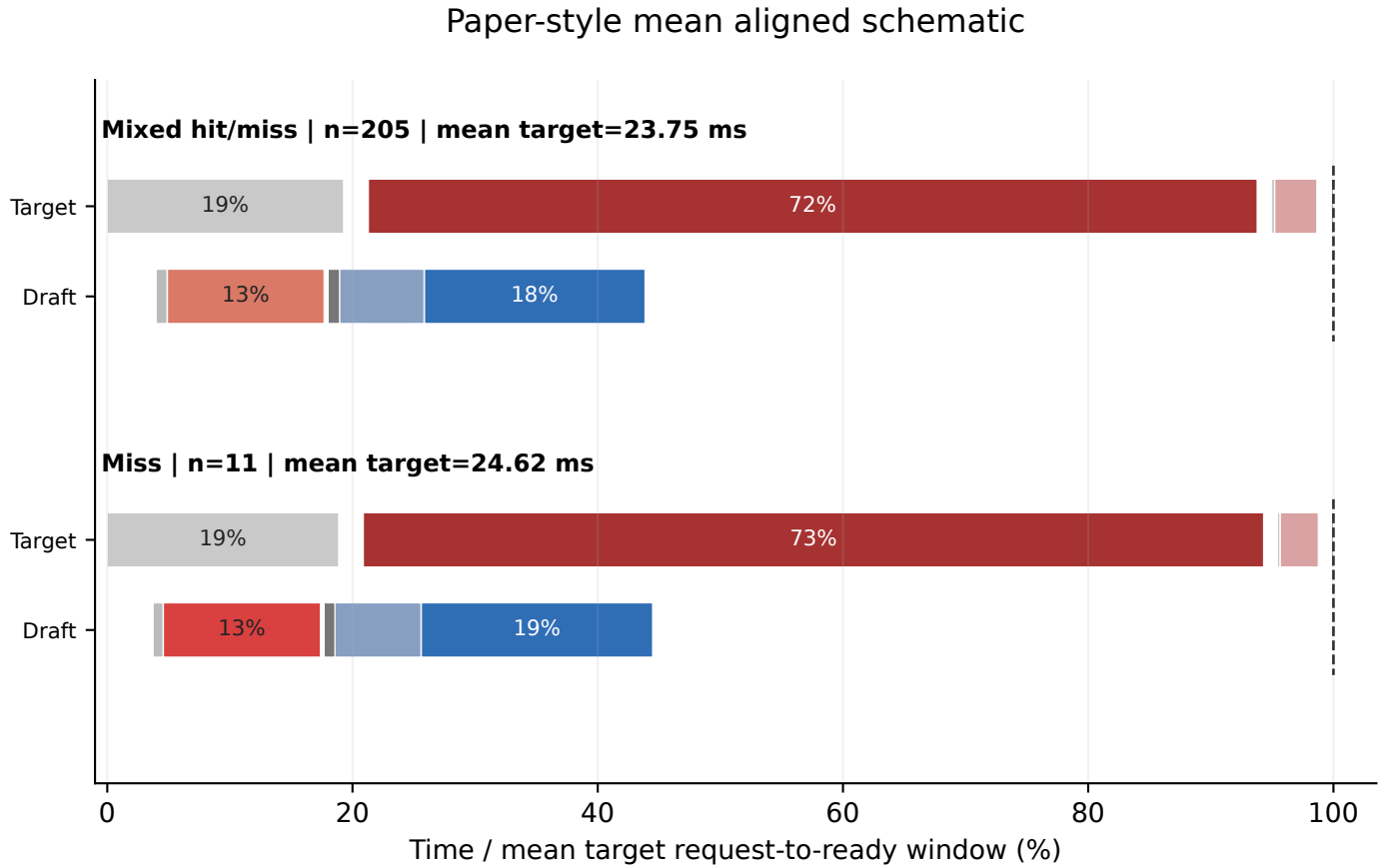
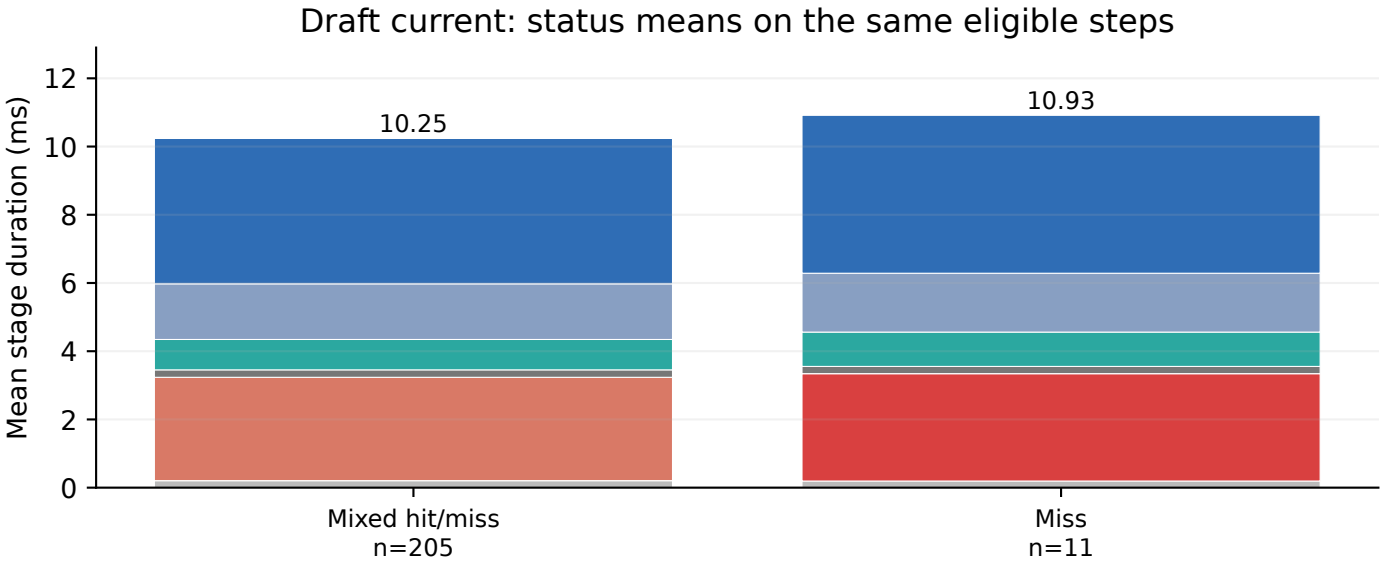
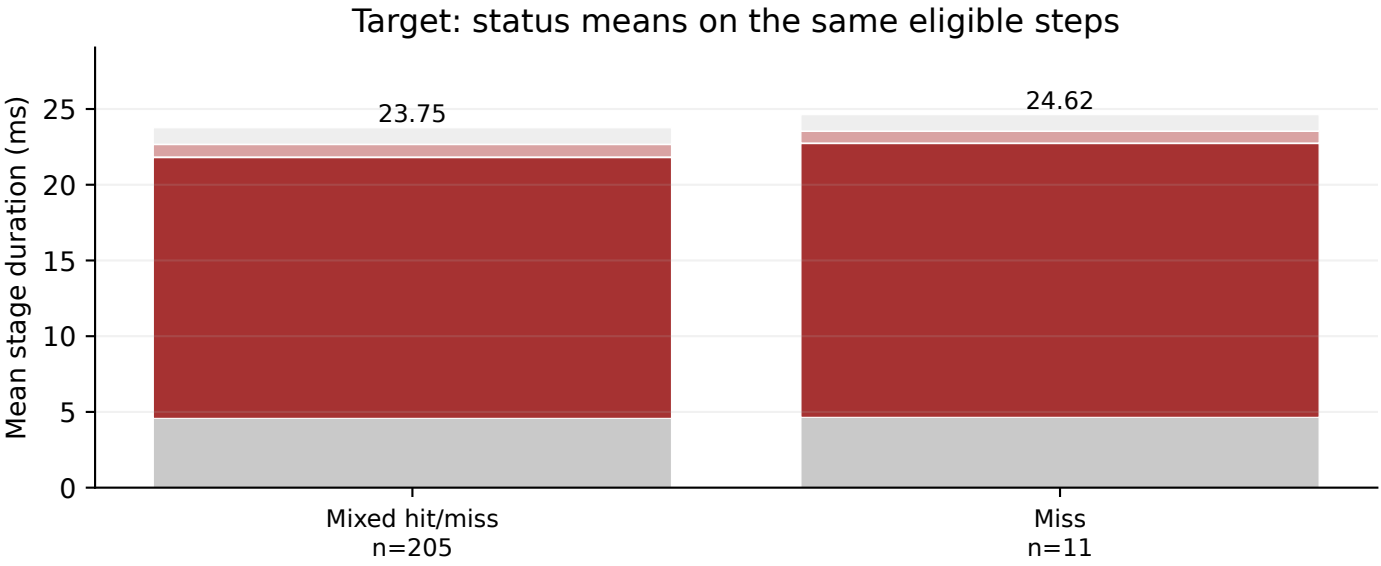
Wait / sync Accept prepare Miss response Request SSD decode Previous SSD decode Target ready
Target verify Postprocess Mixed response Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Context SSD build

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



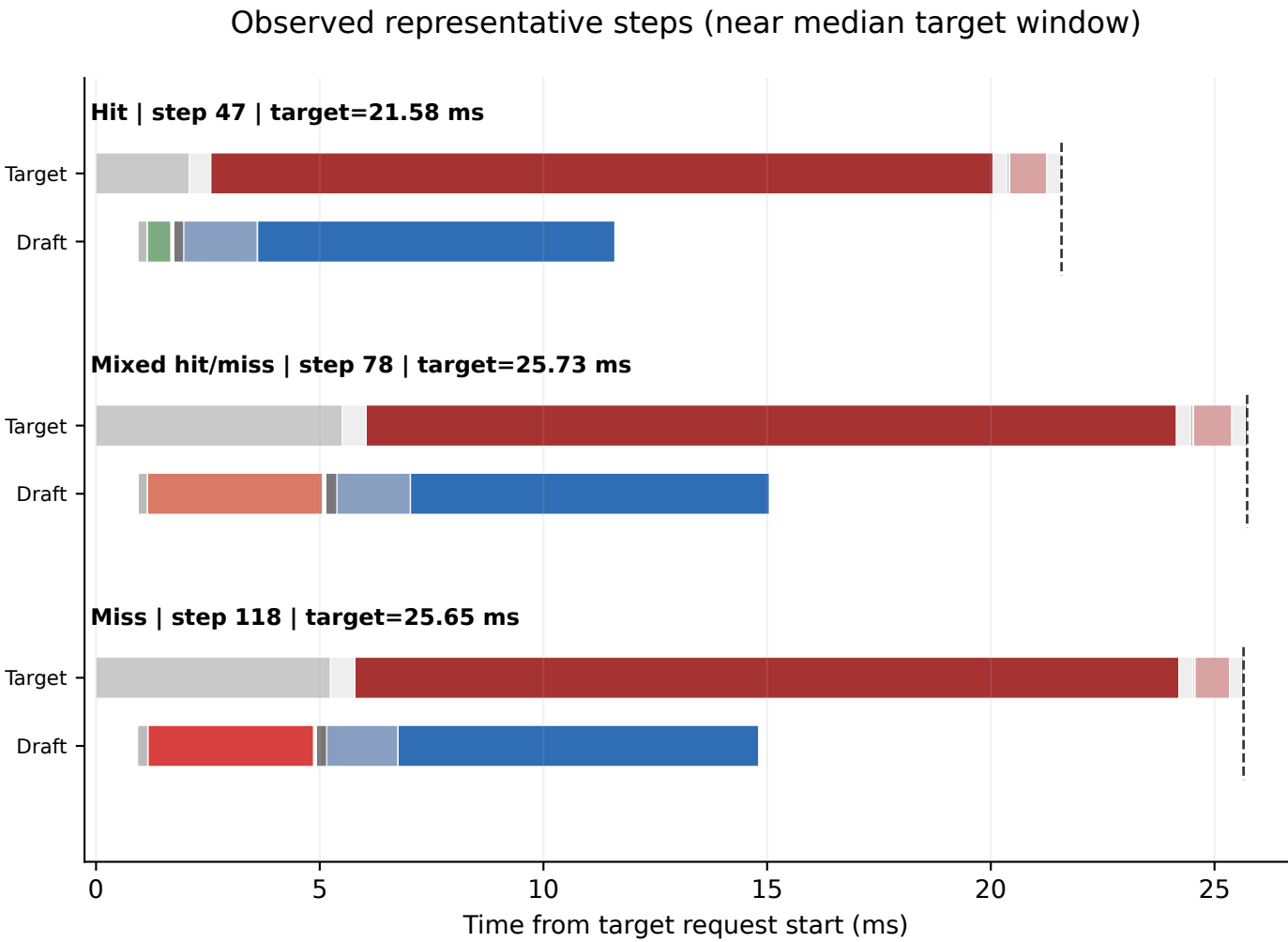
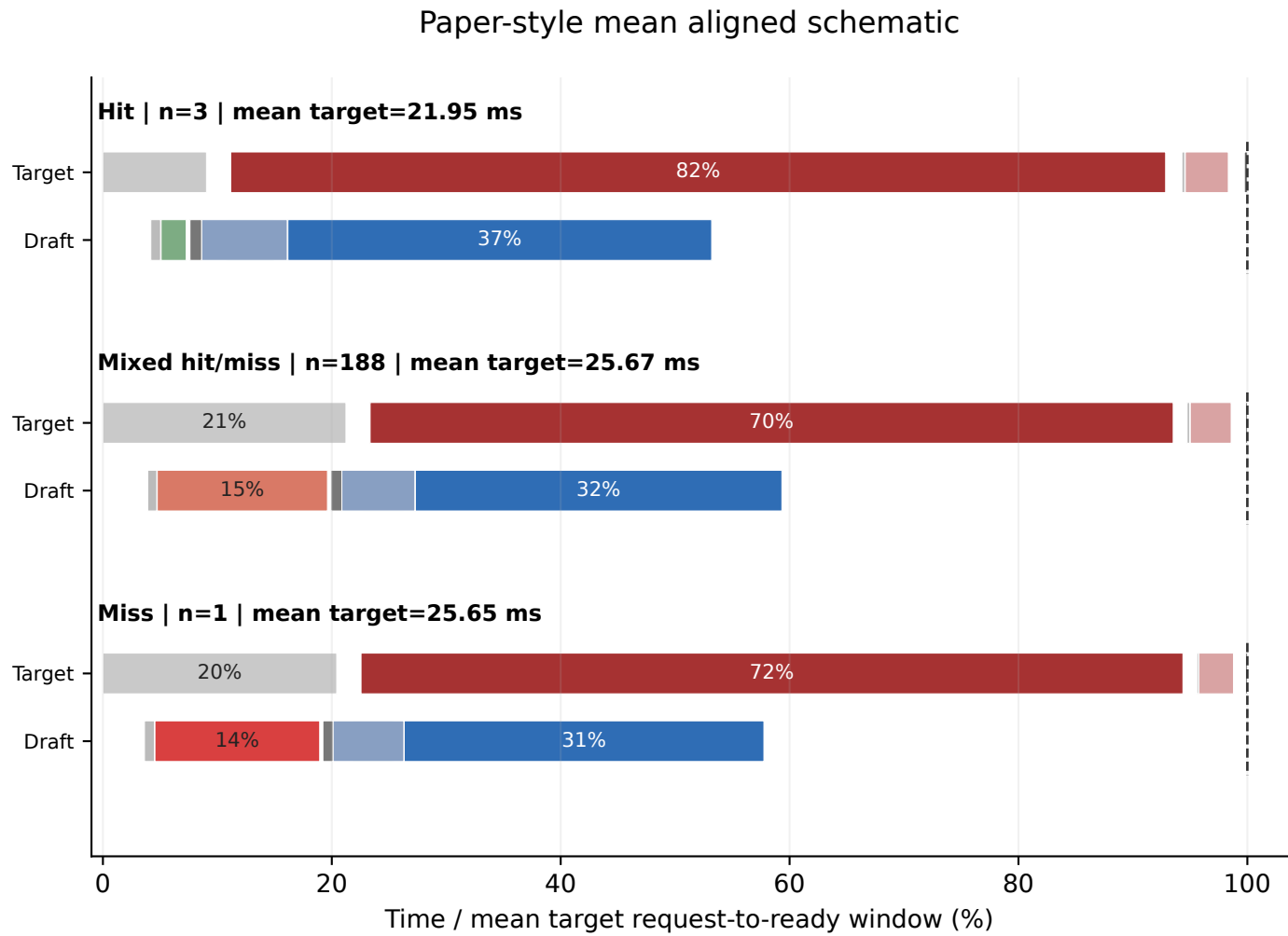
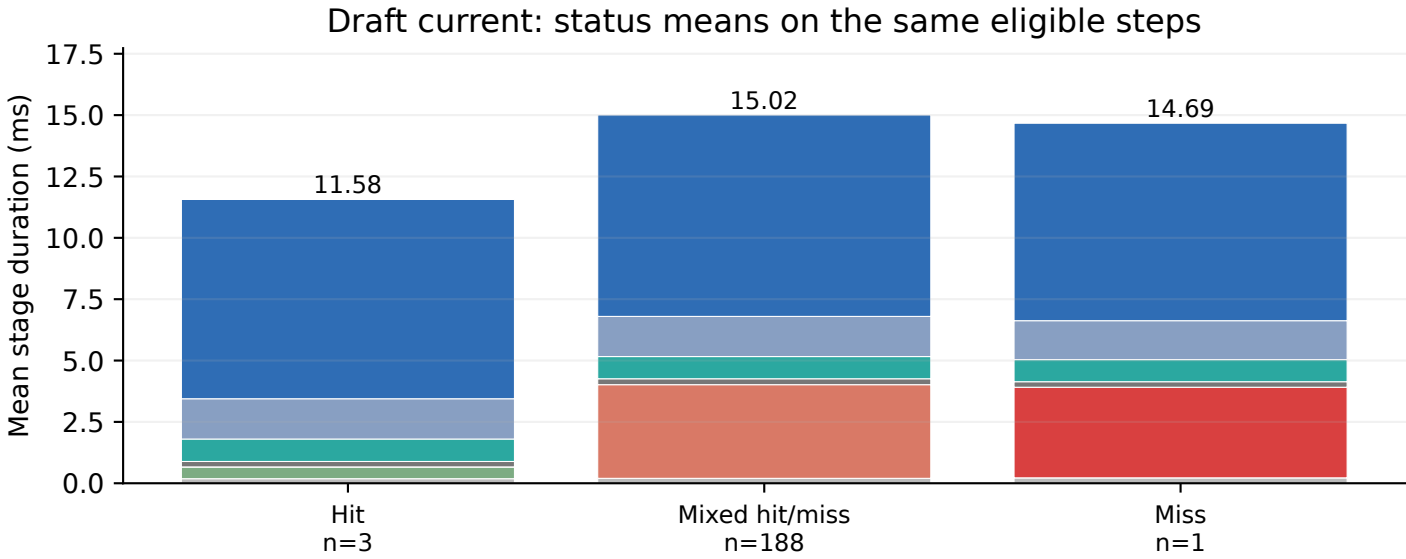
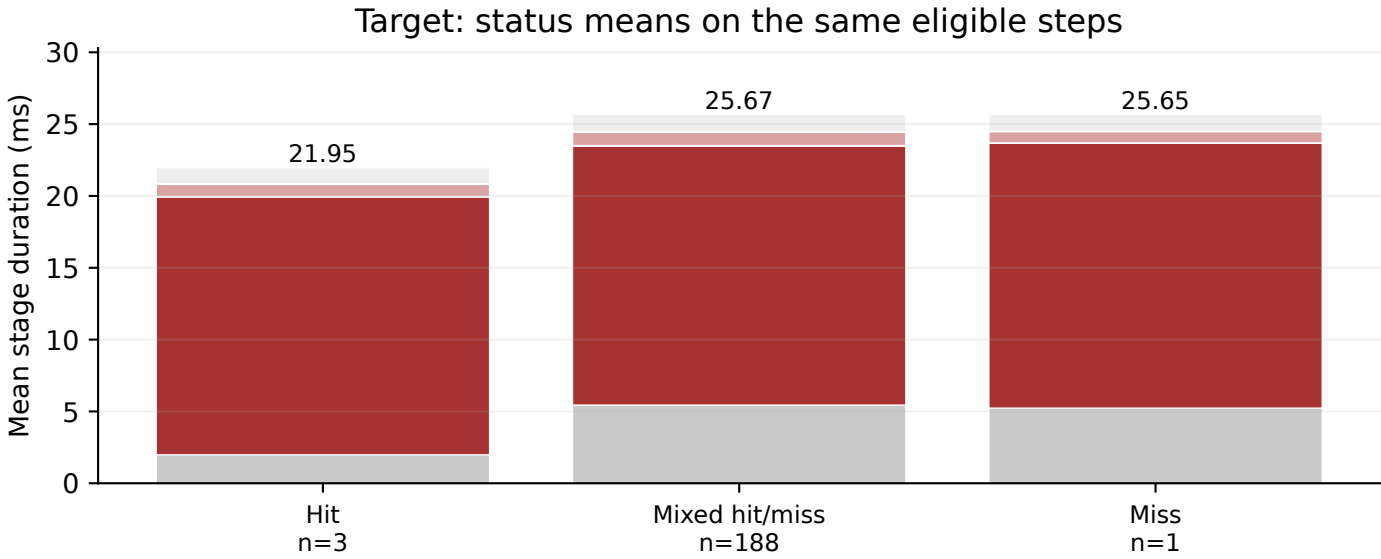
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

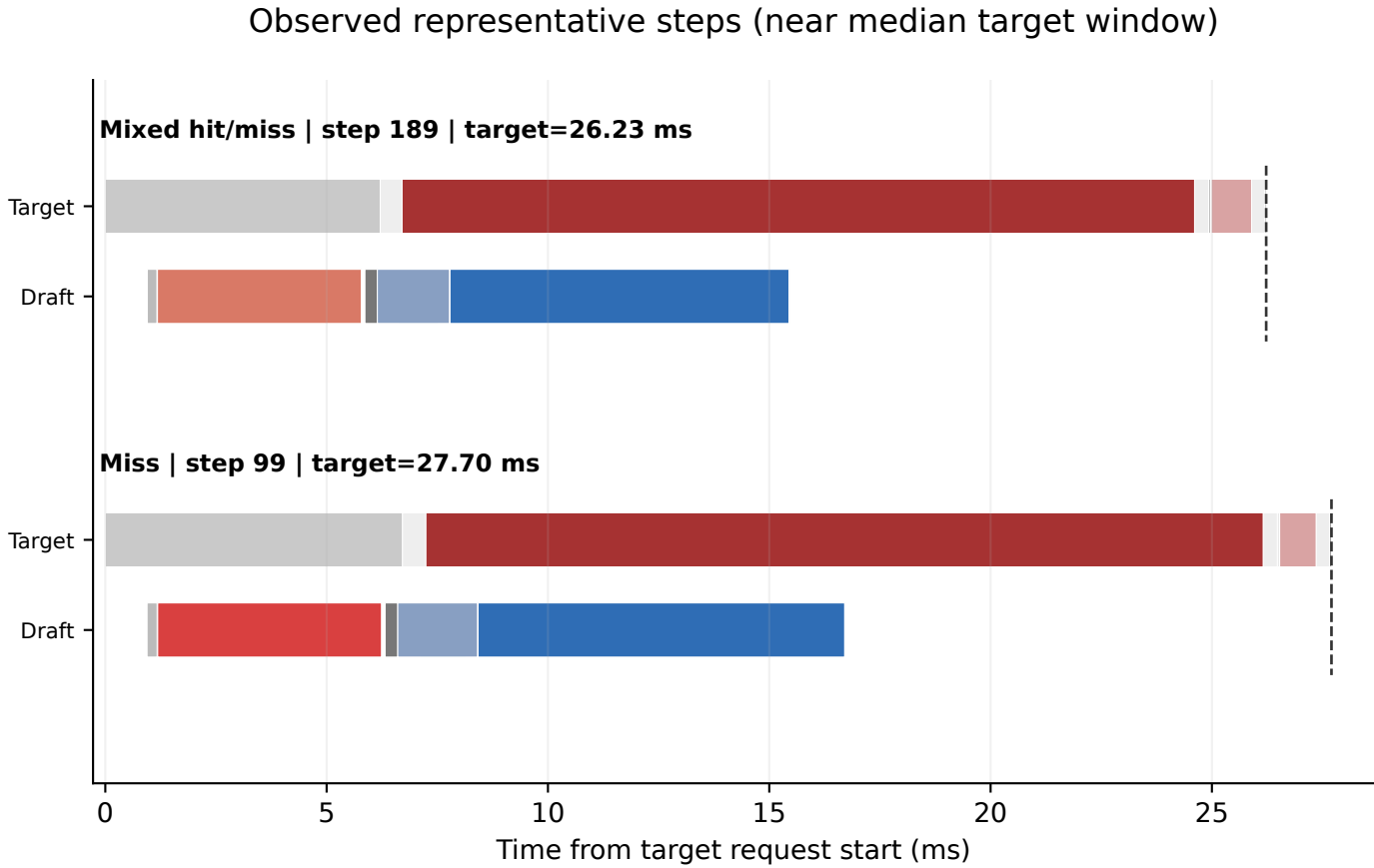
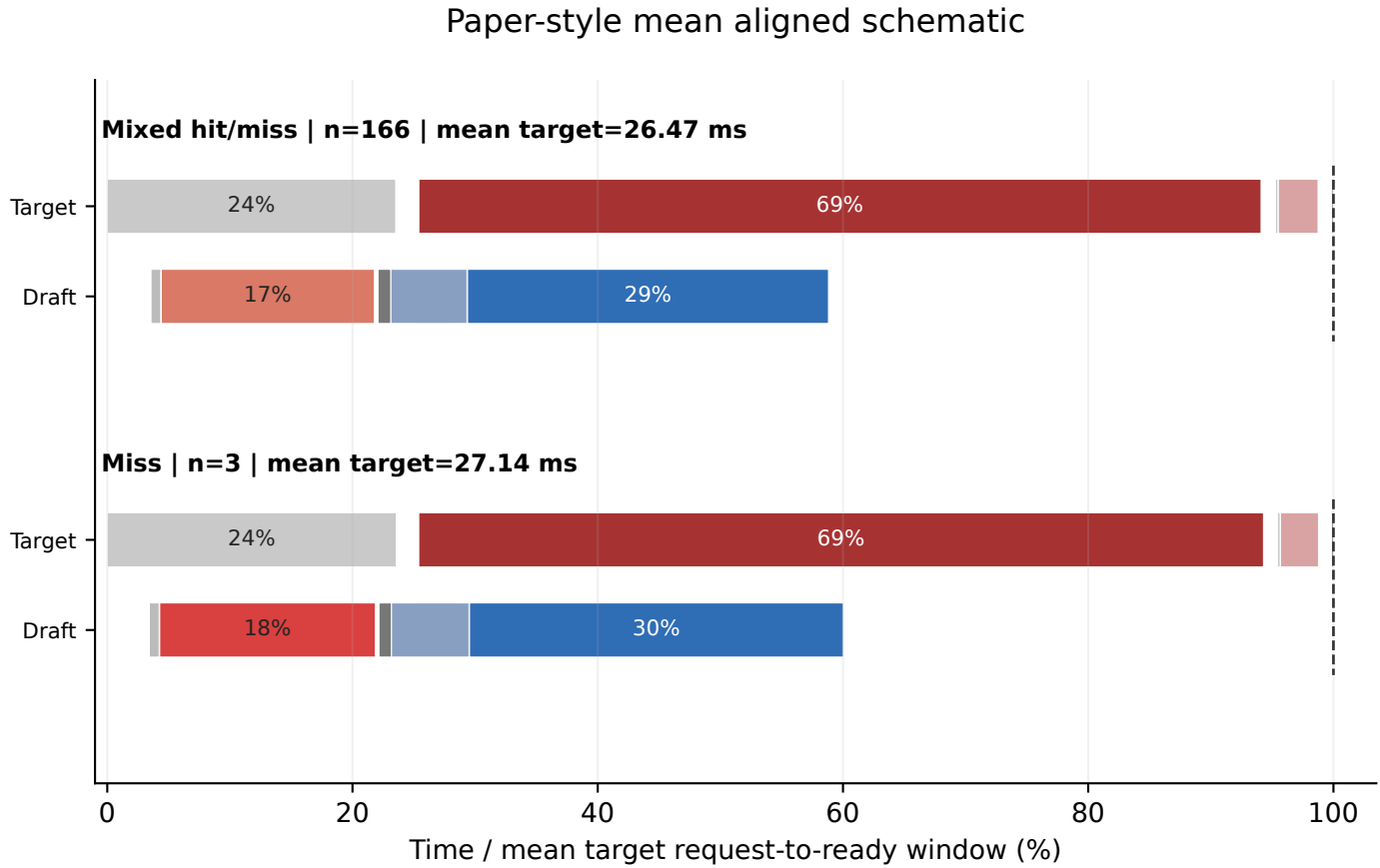
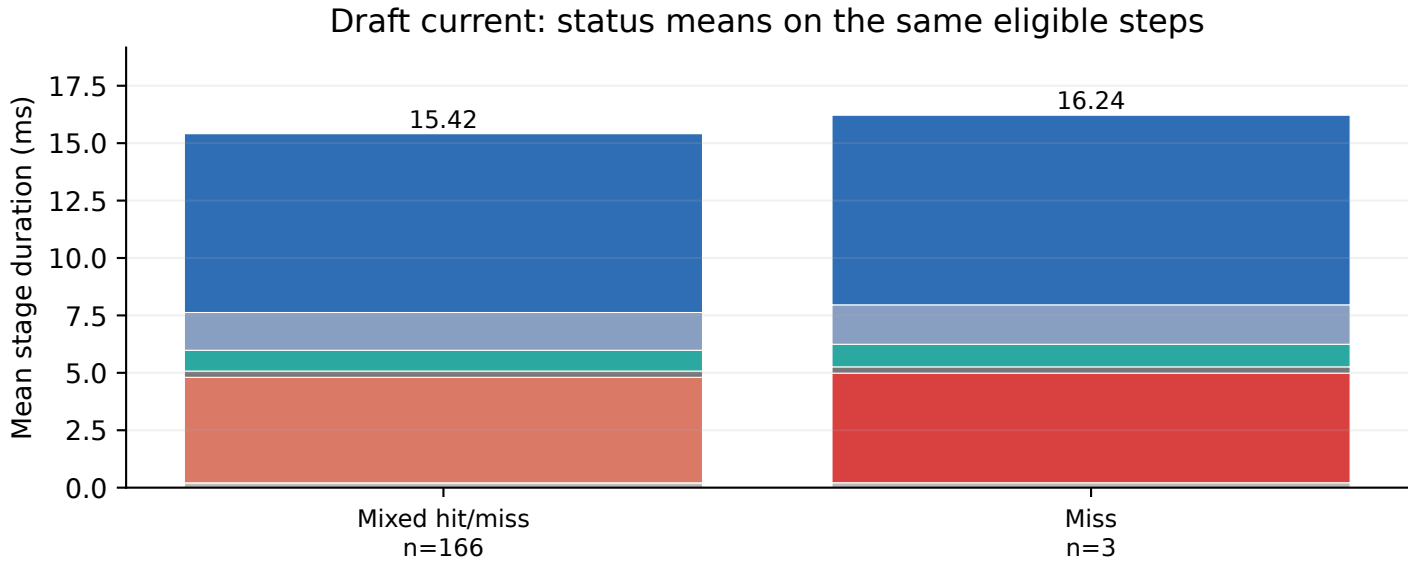
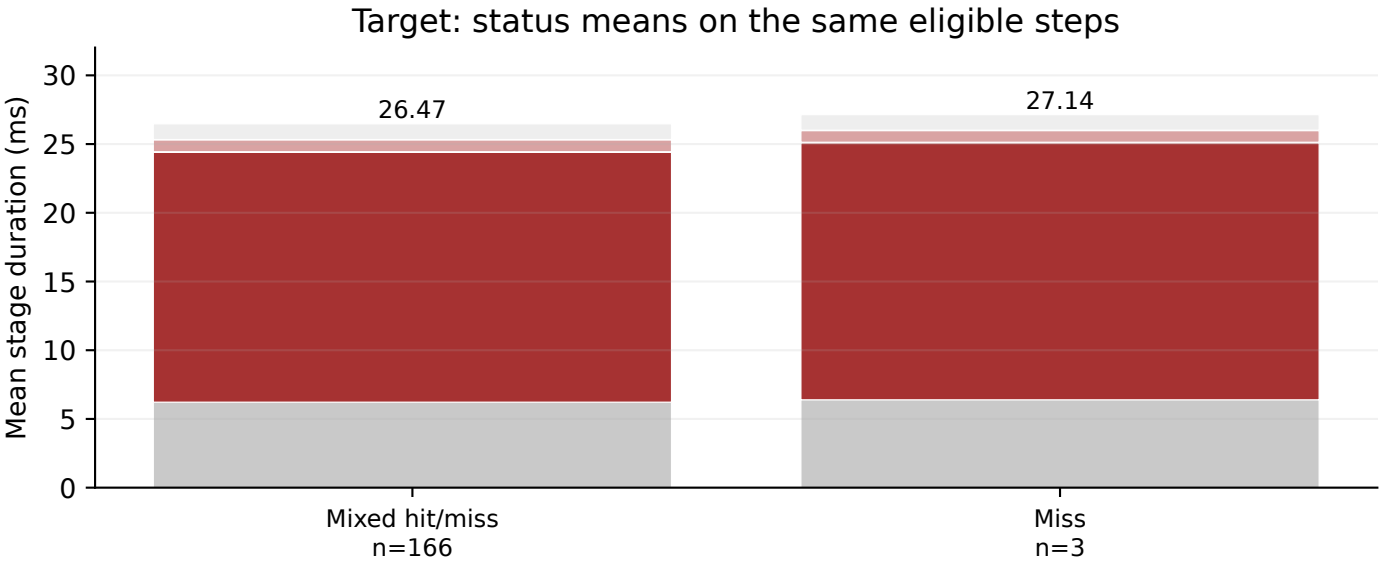
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

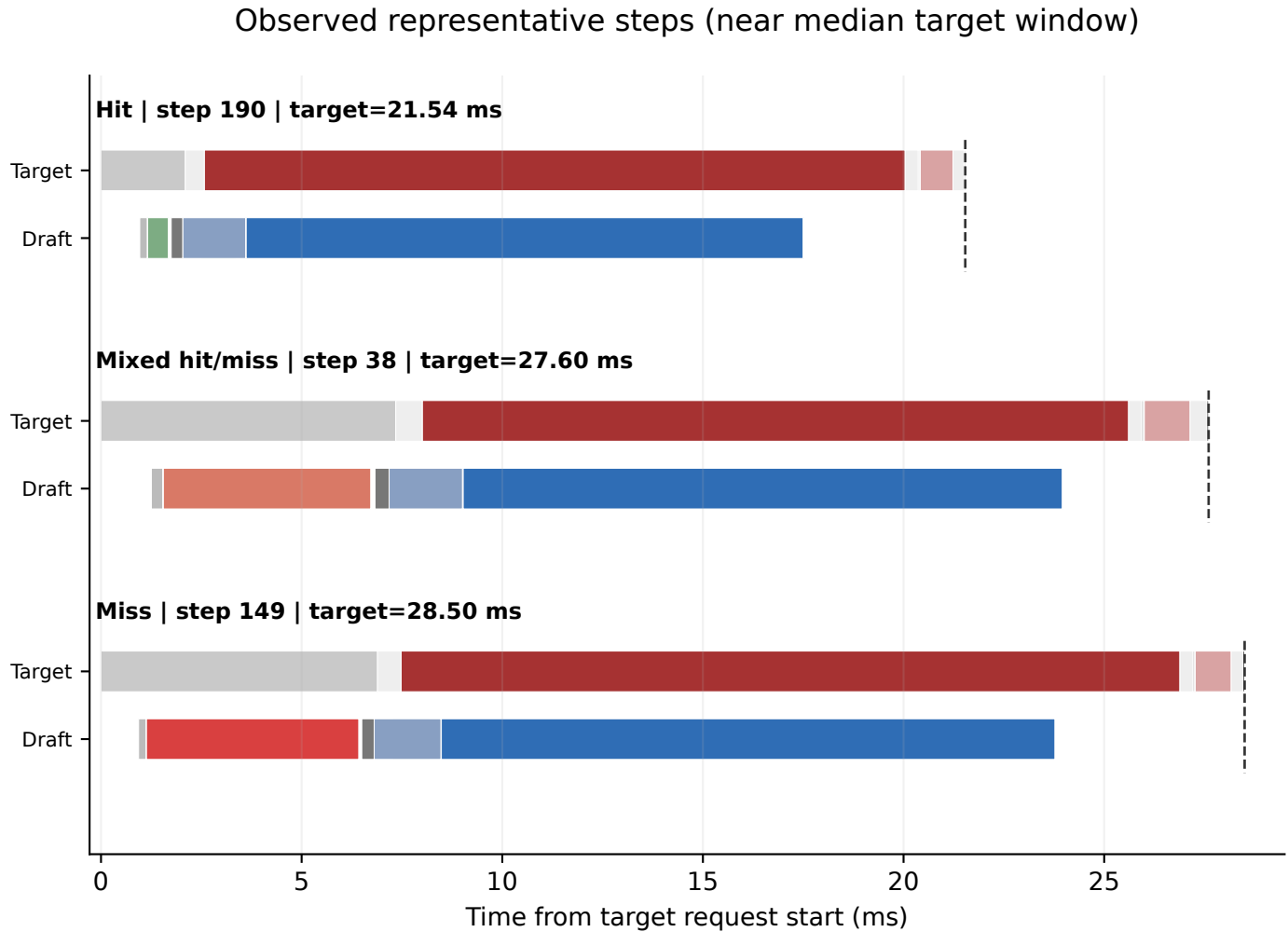
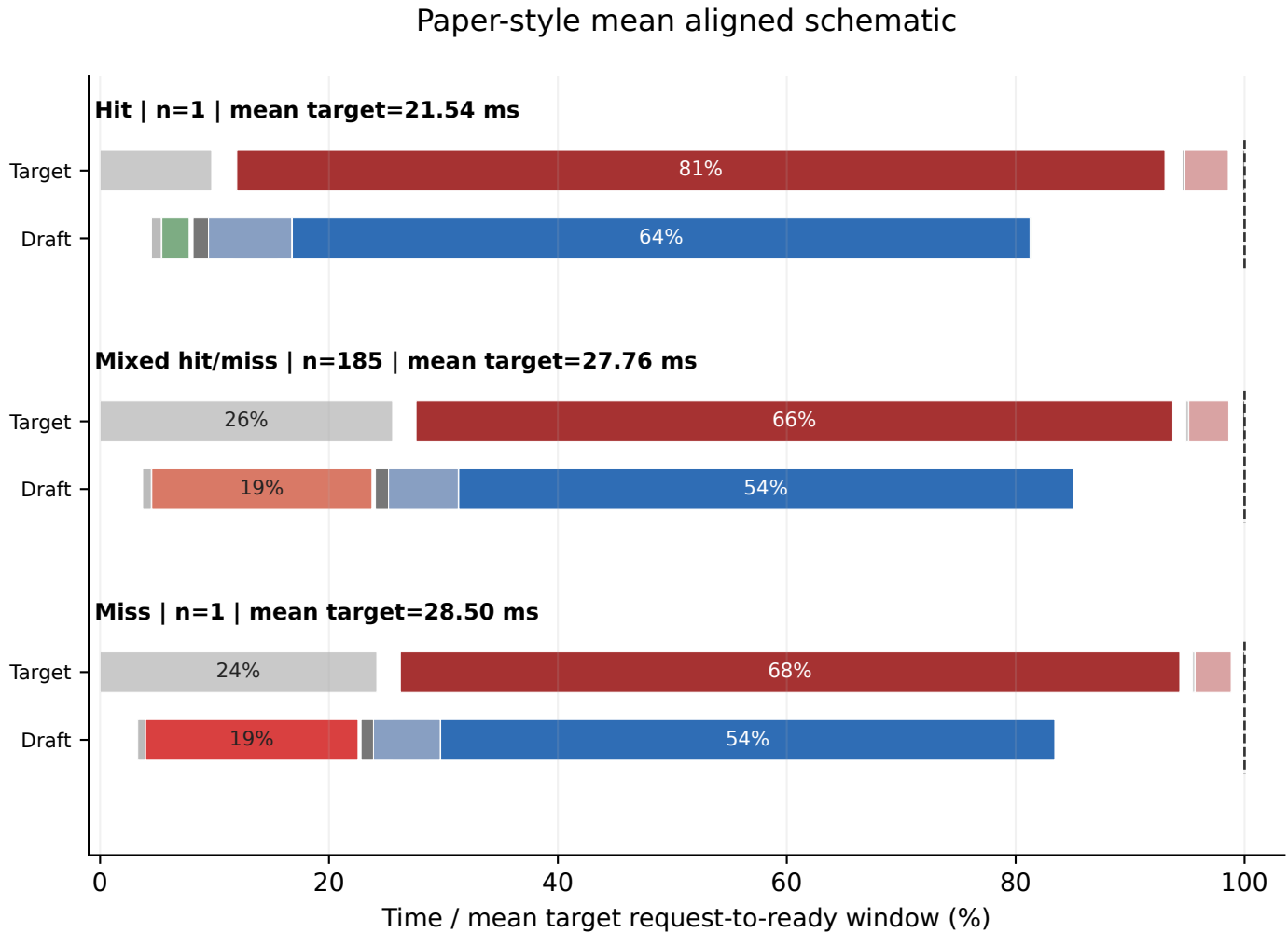
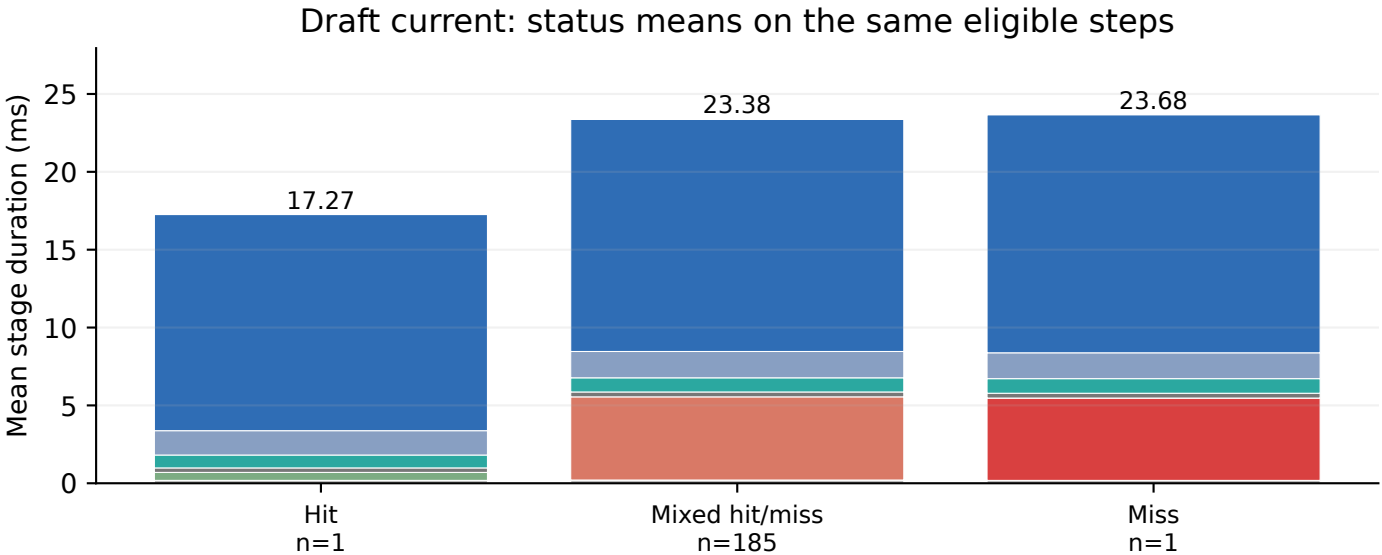
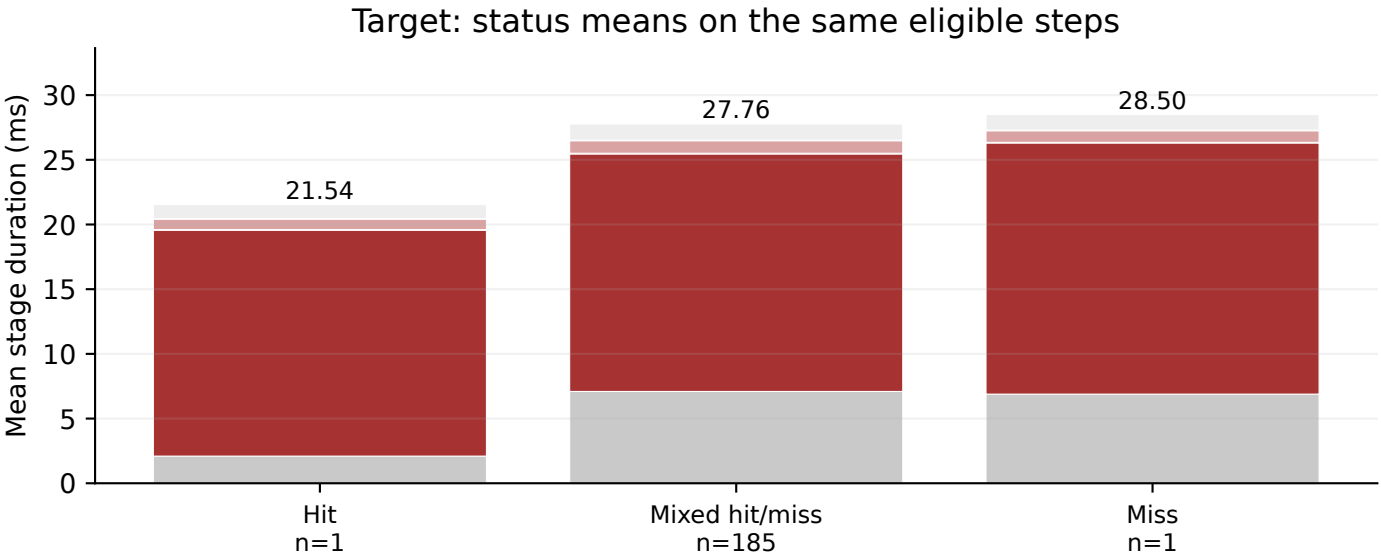
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.





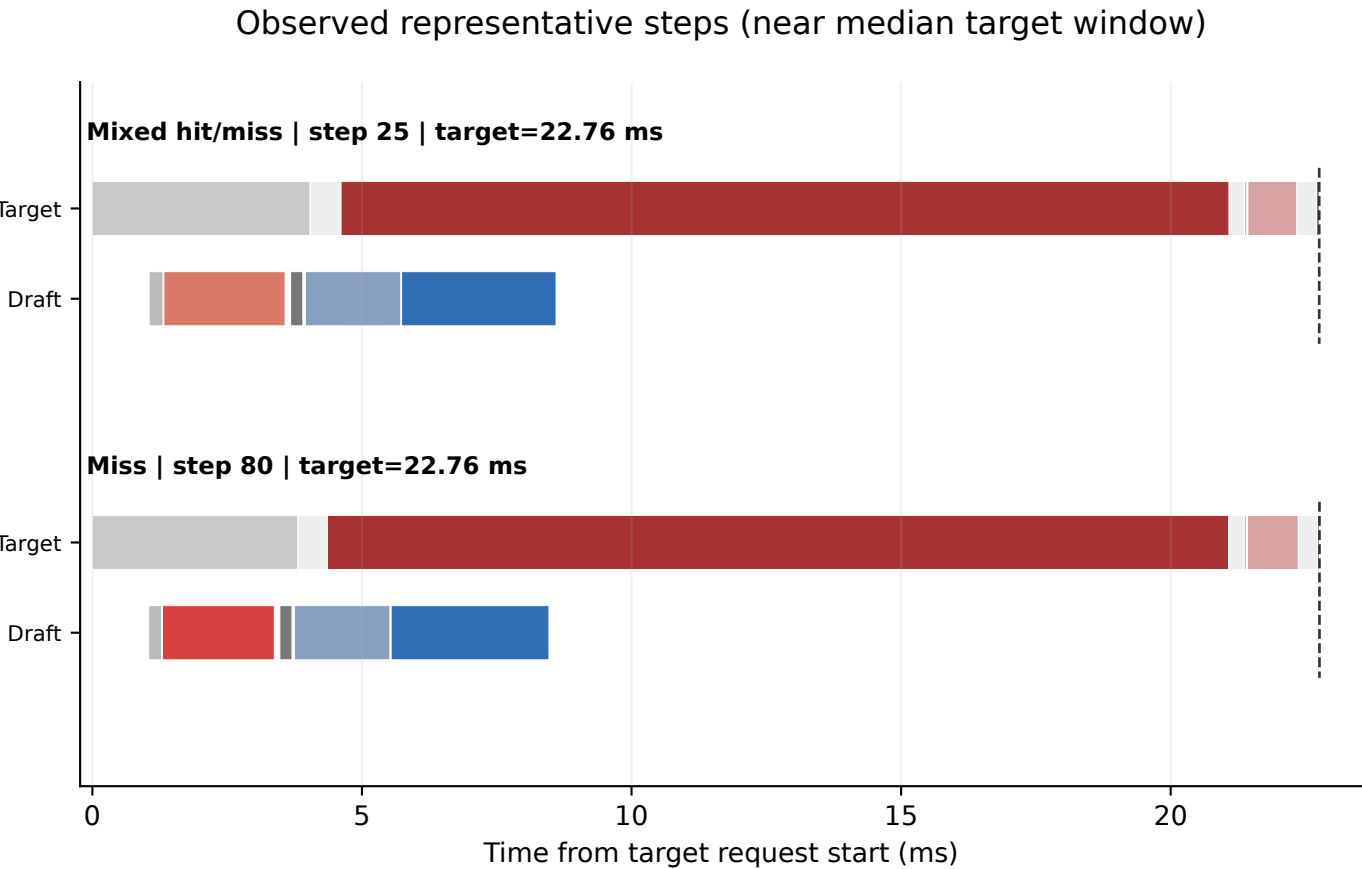
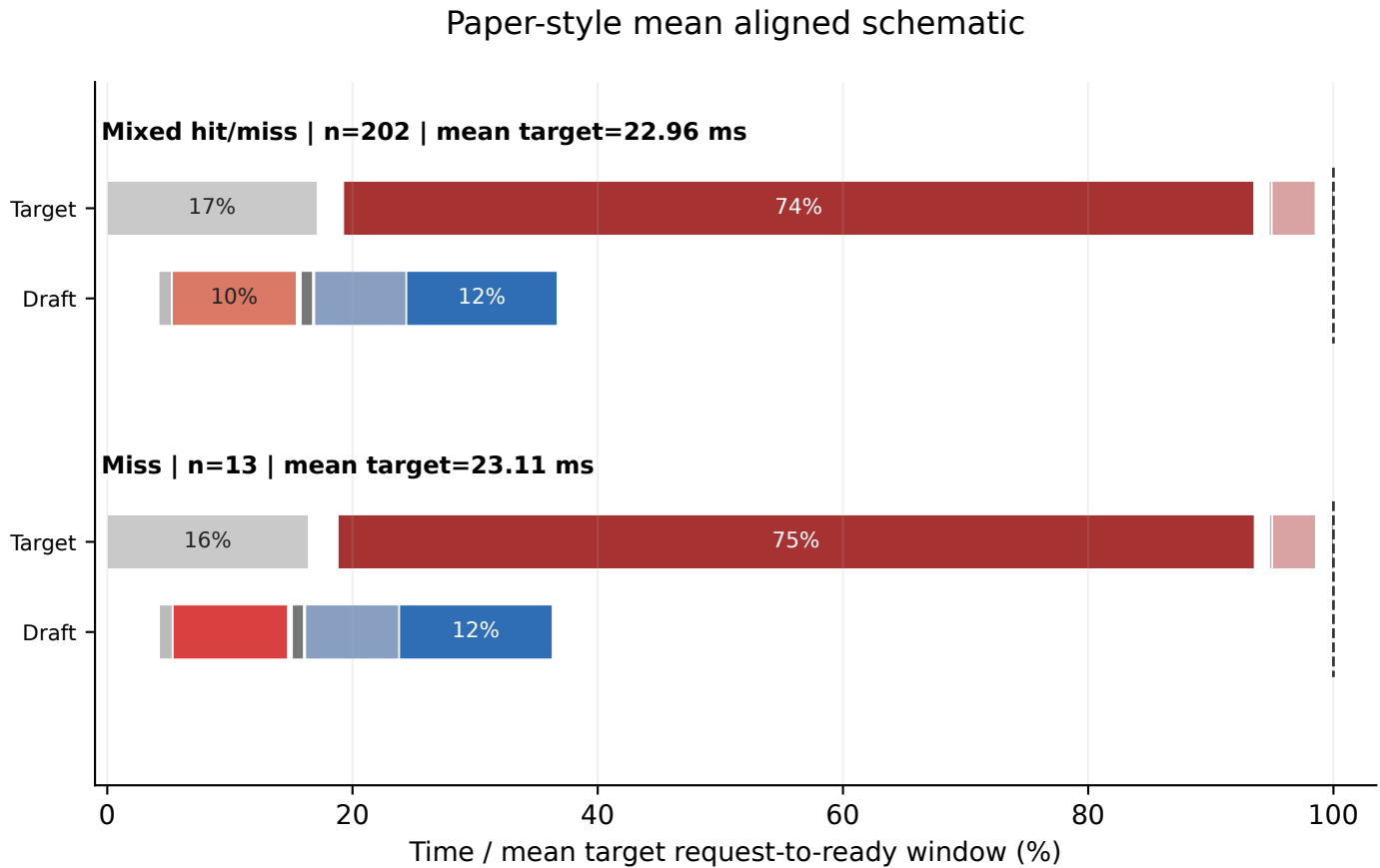
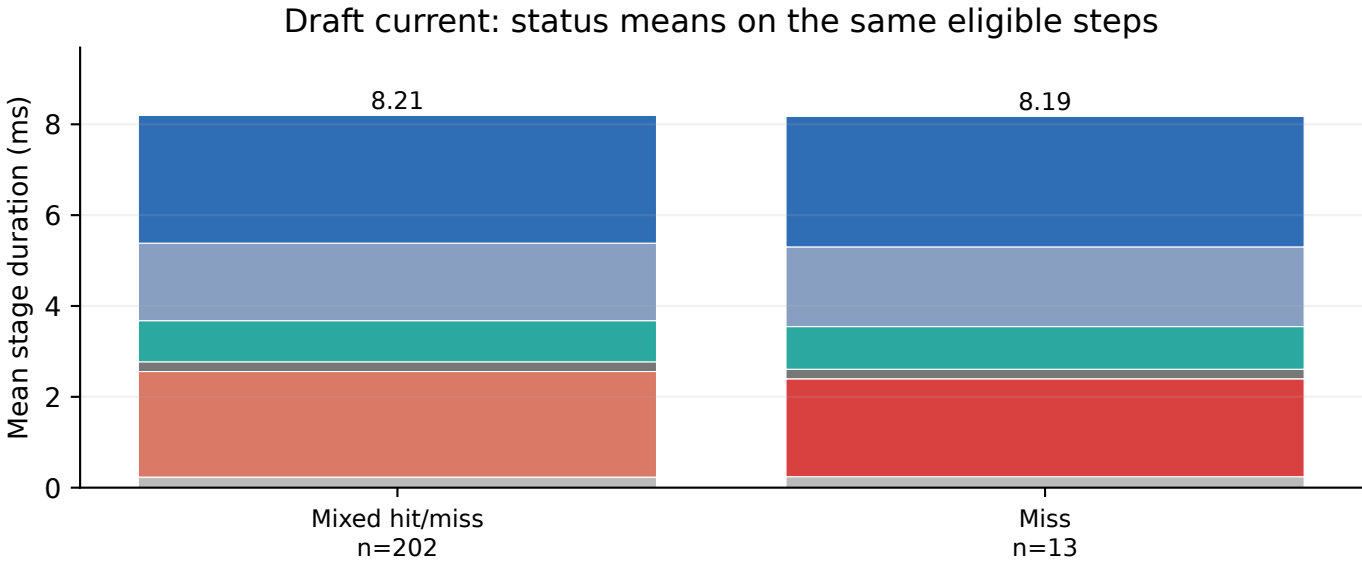
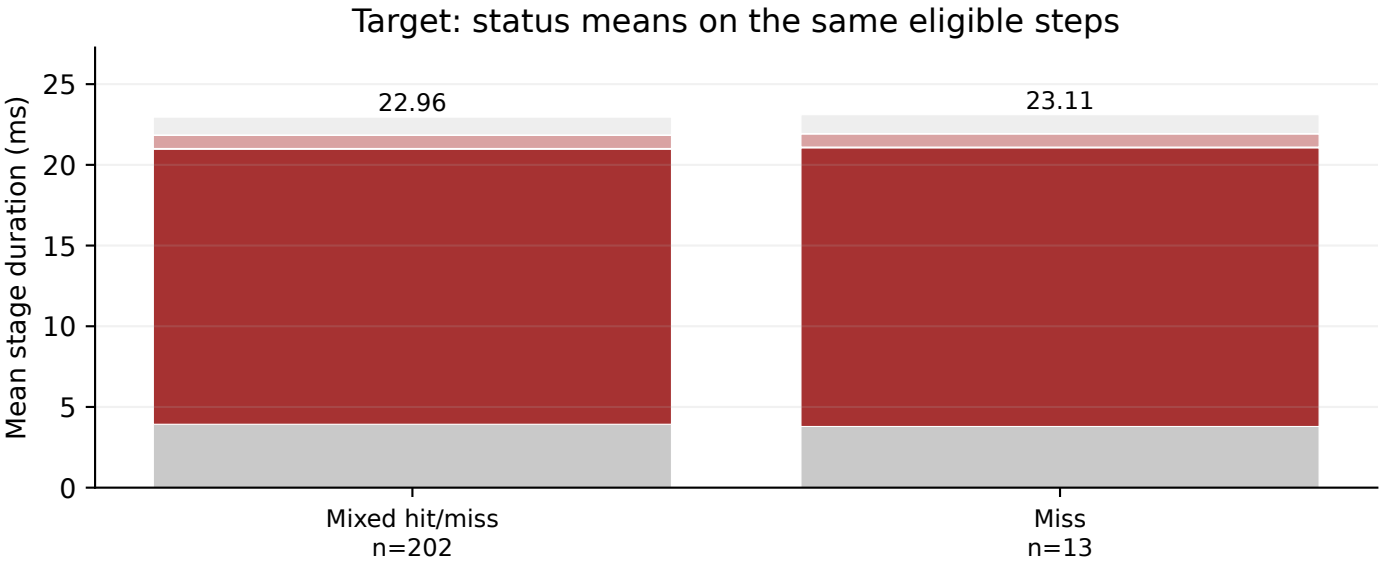
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



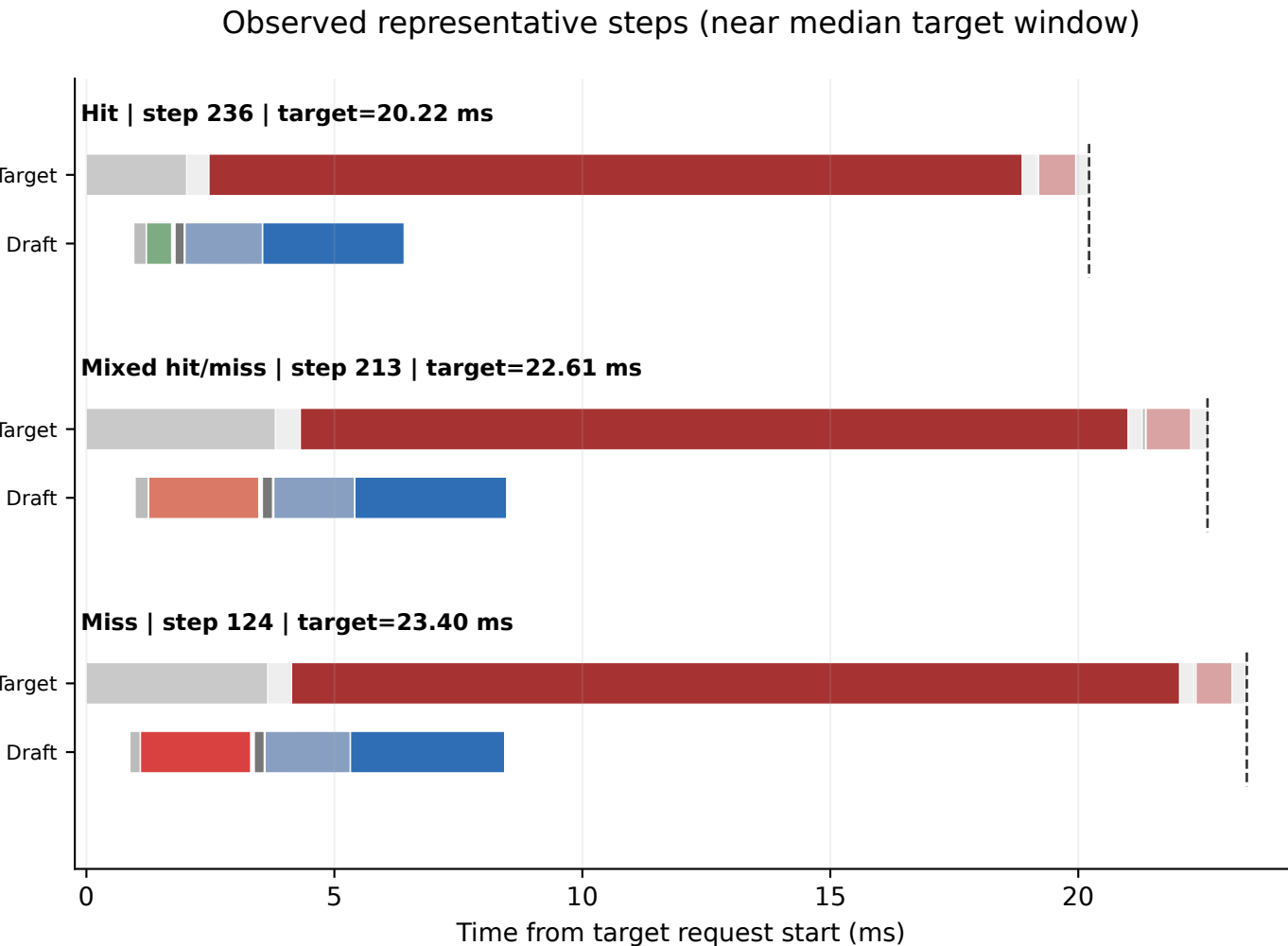
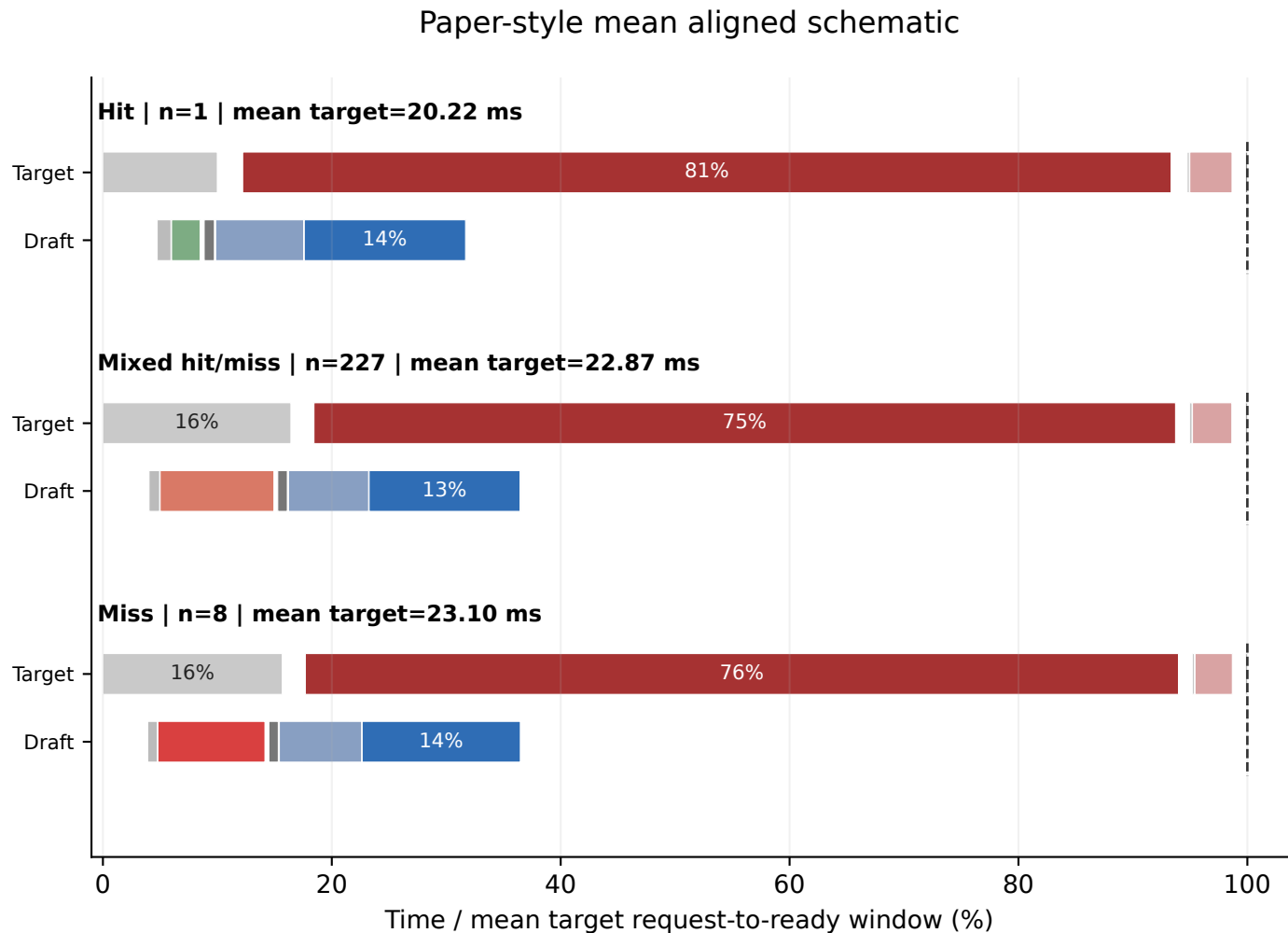
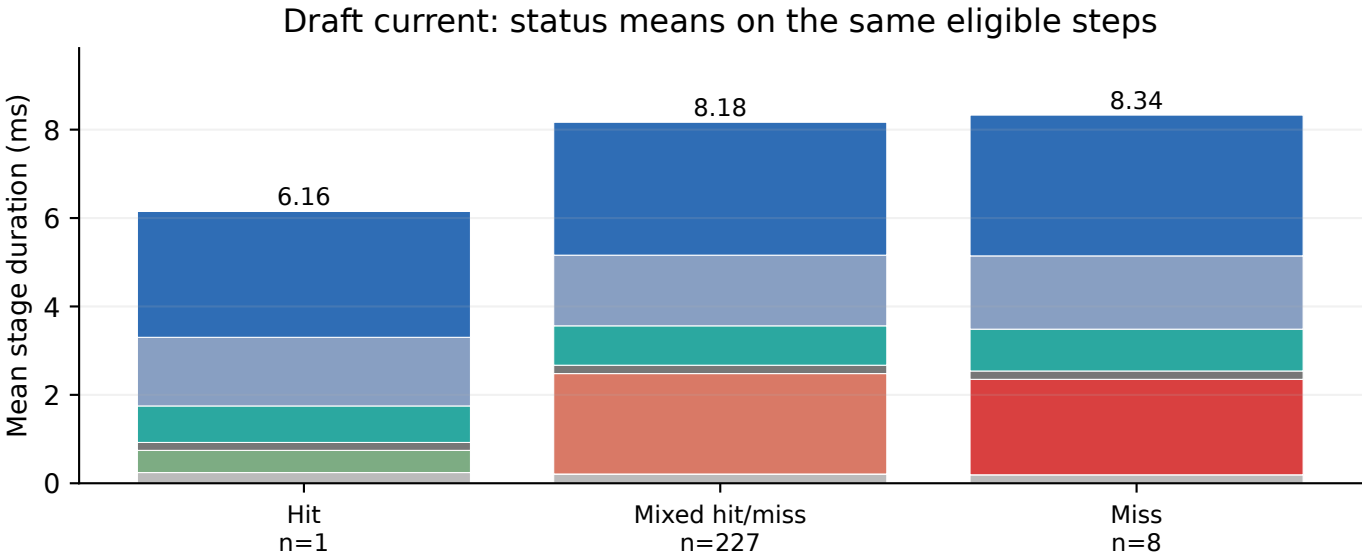
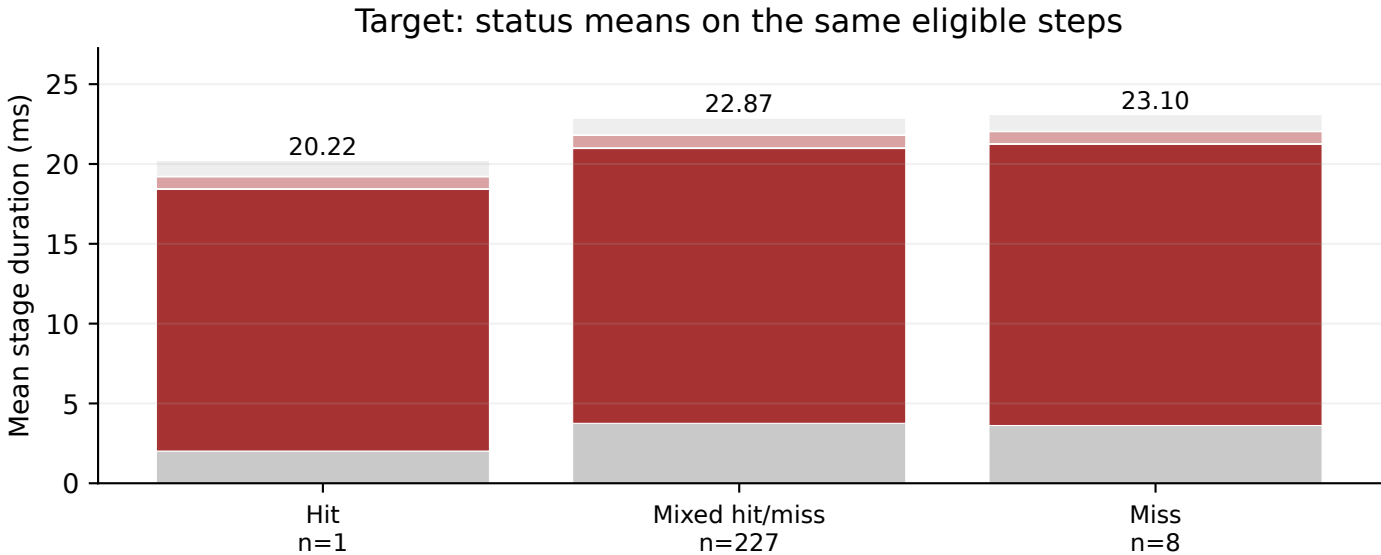
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

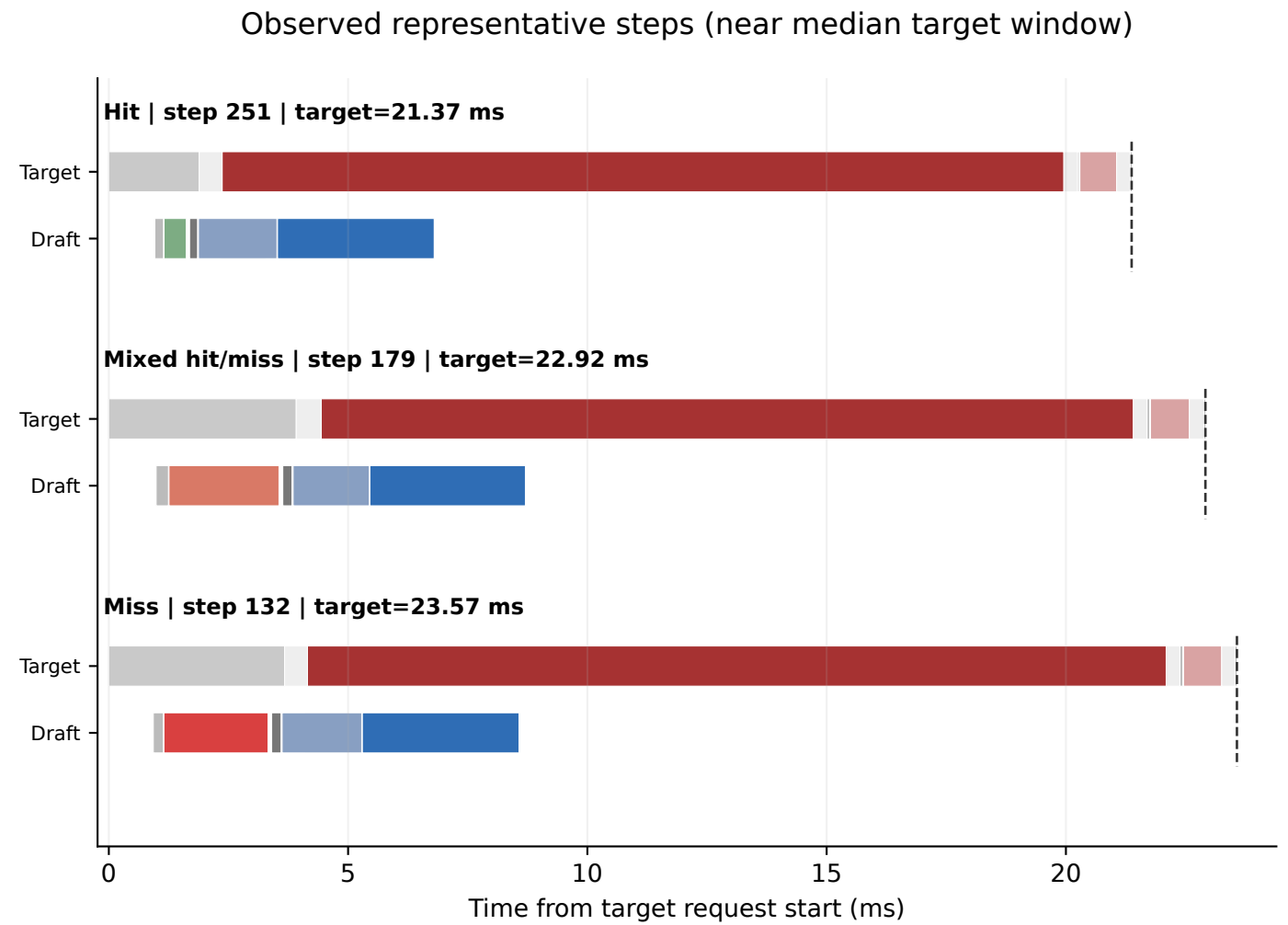
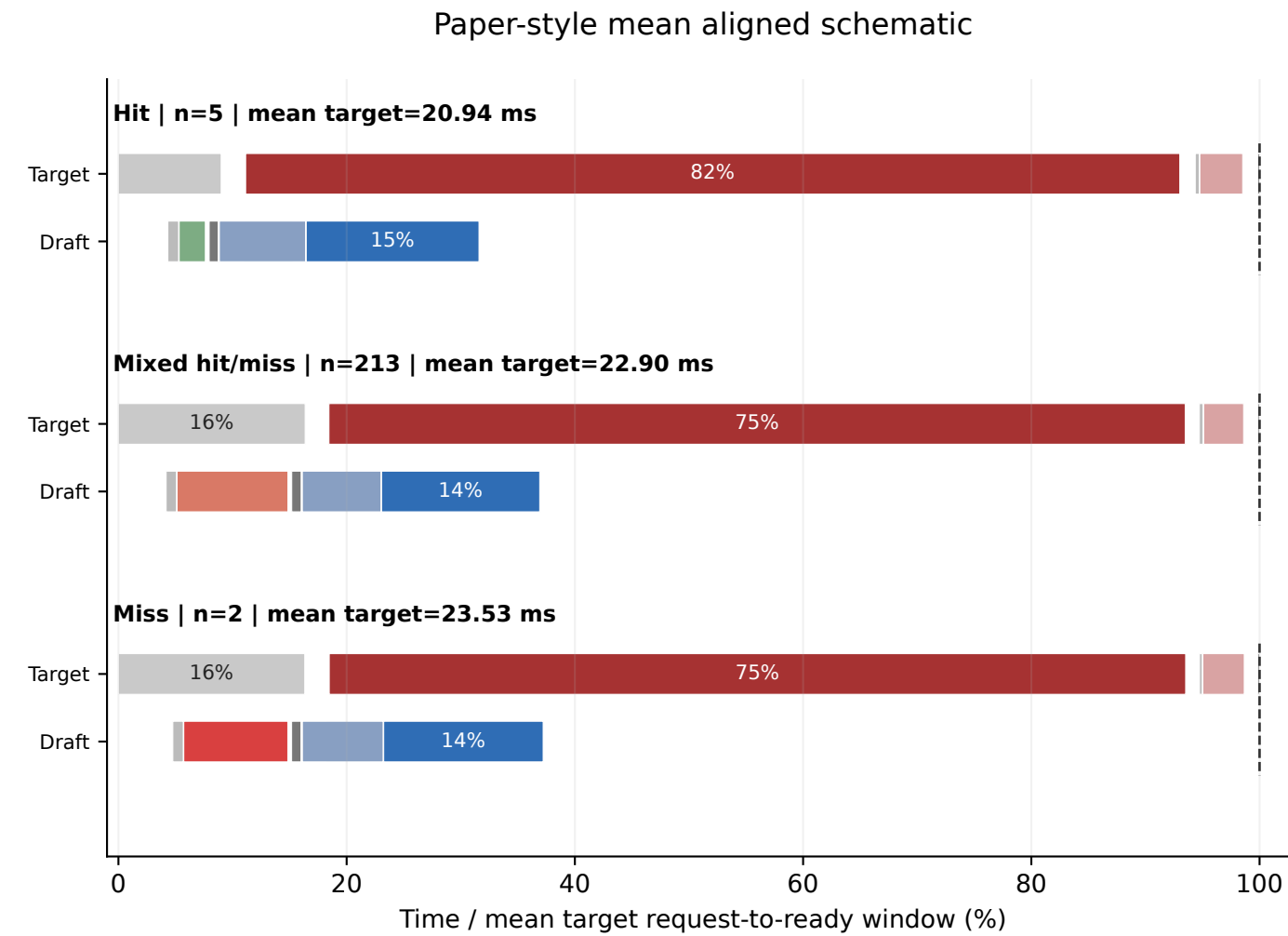
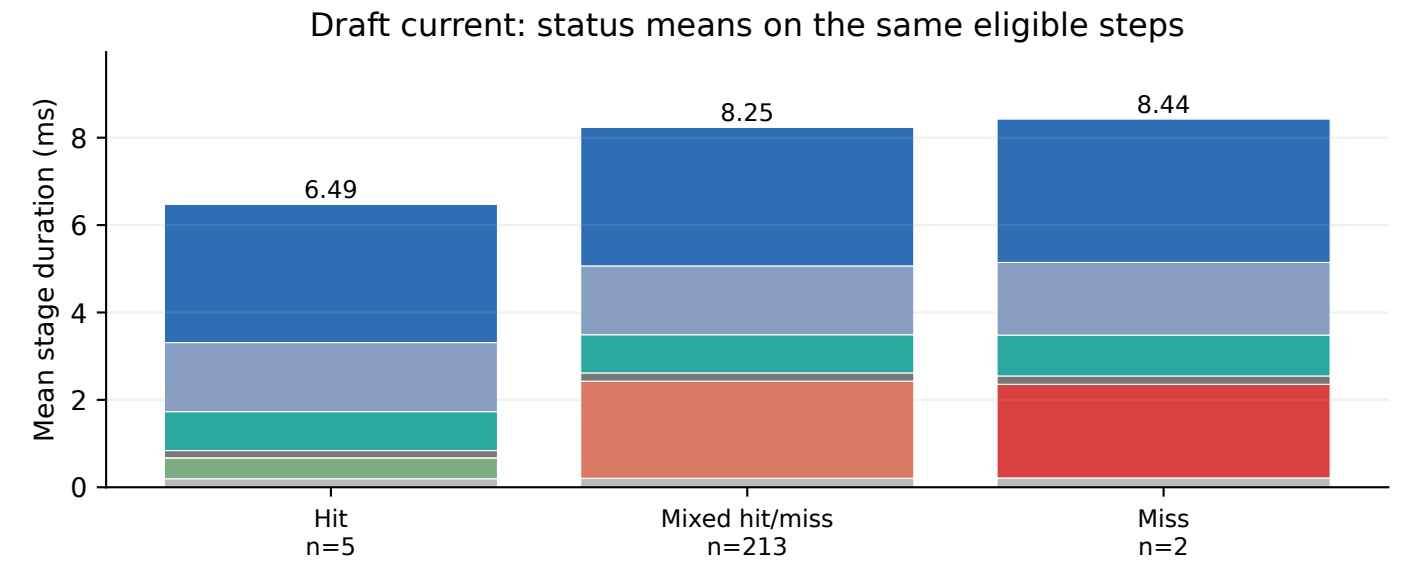
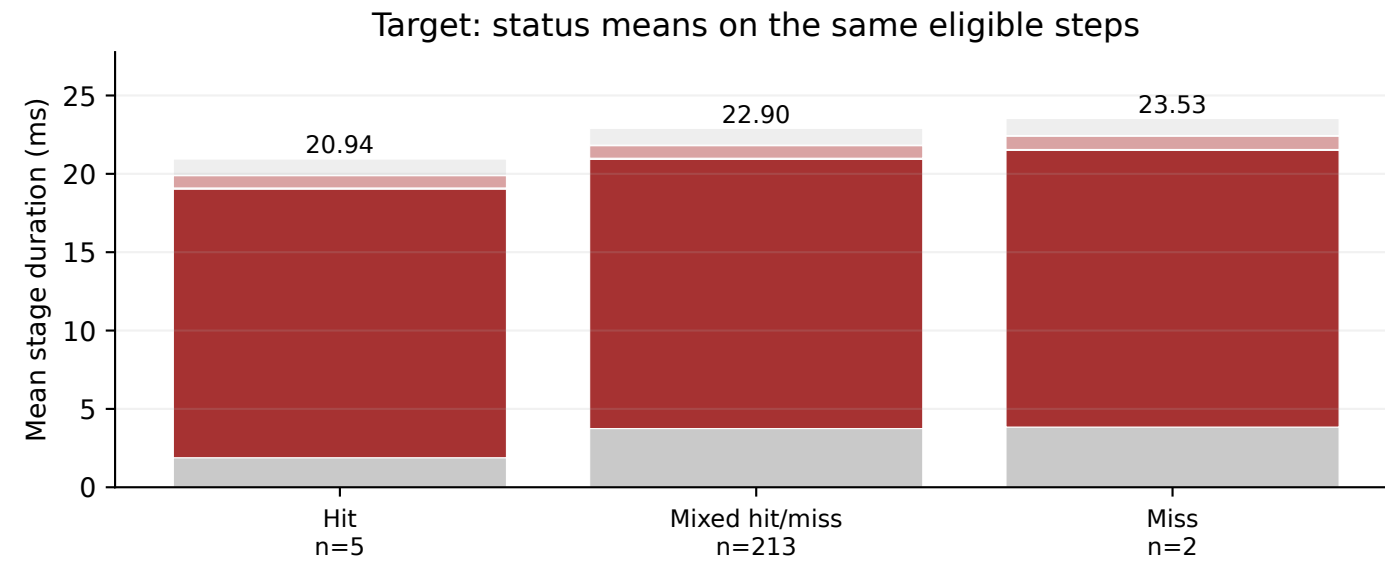
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

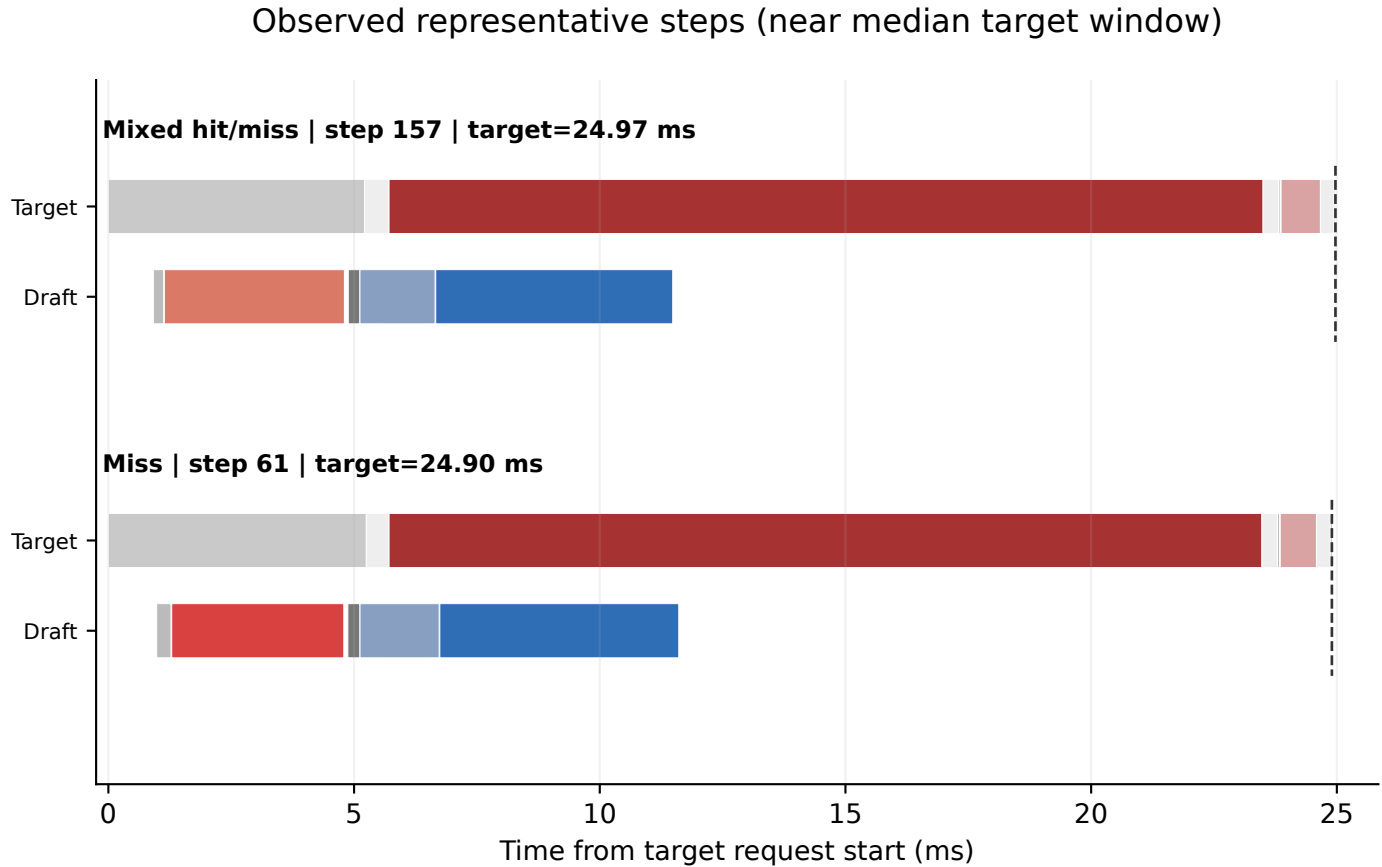
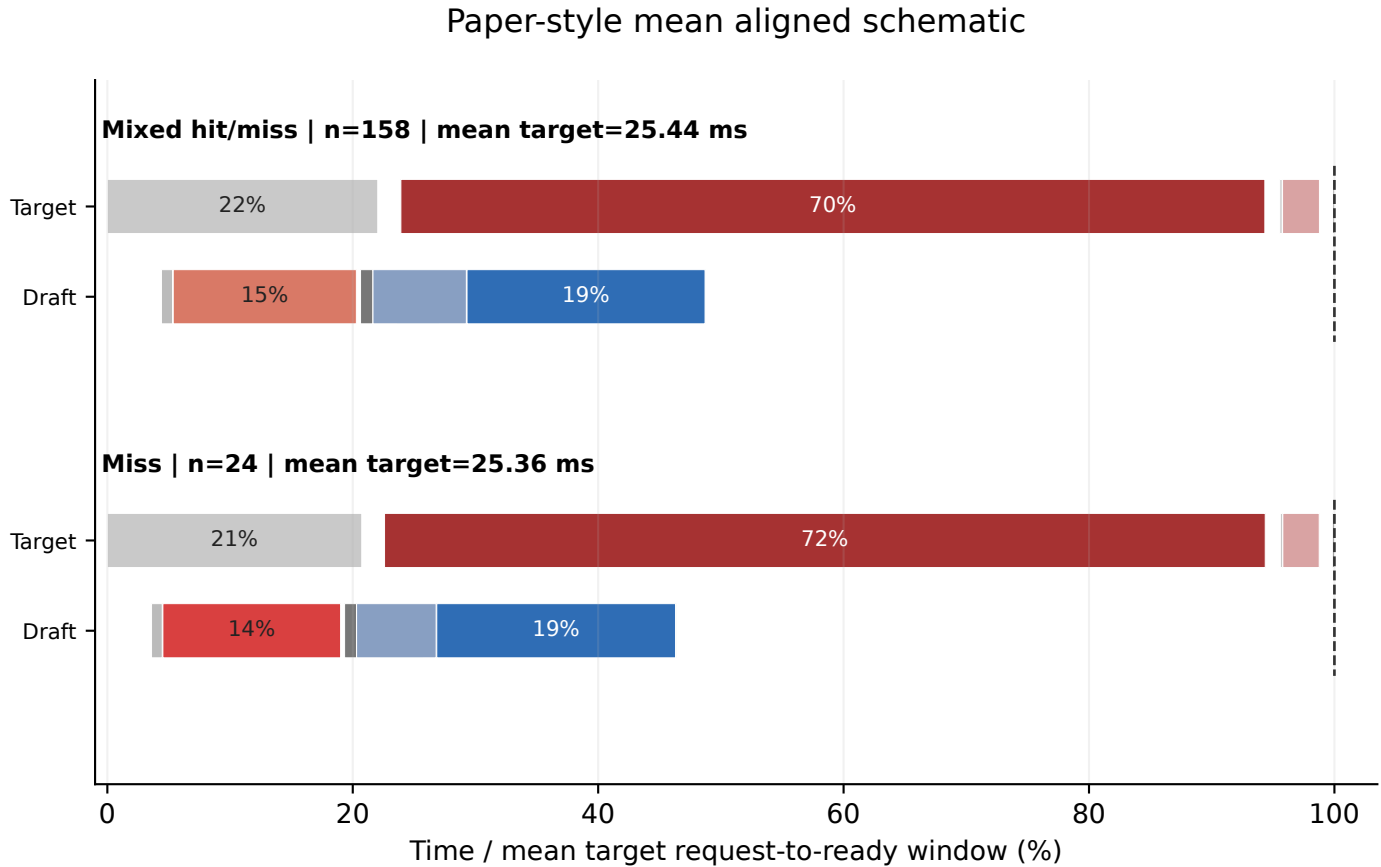
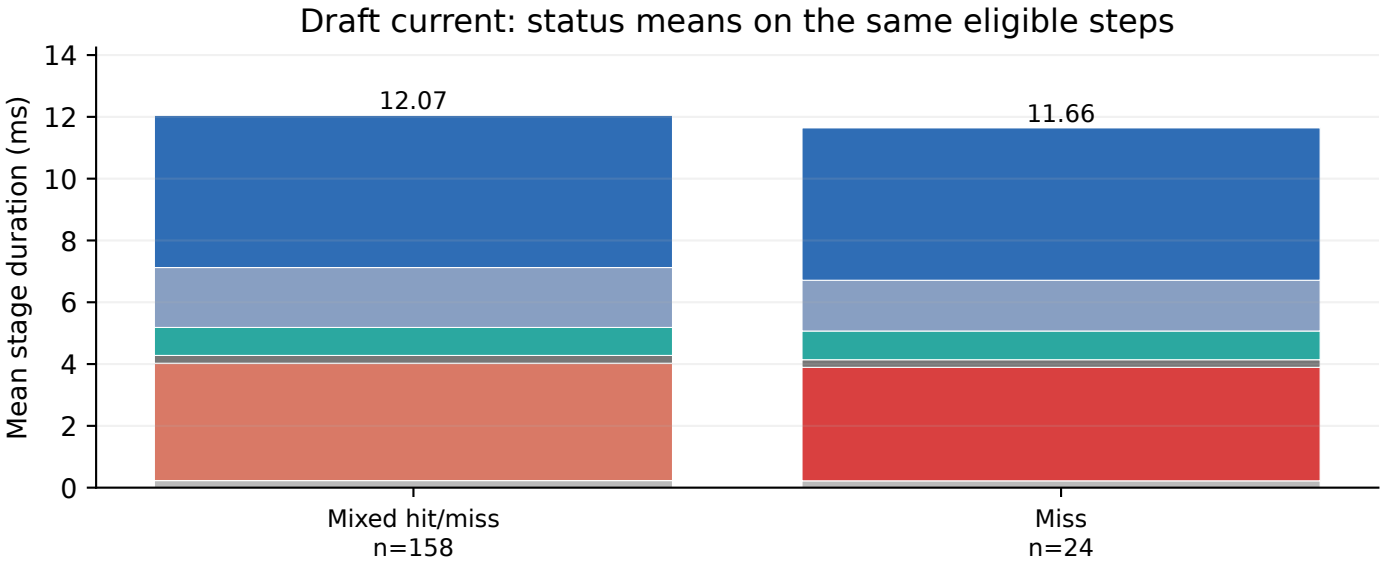
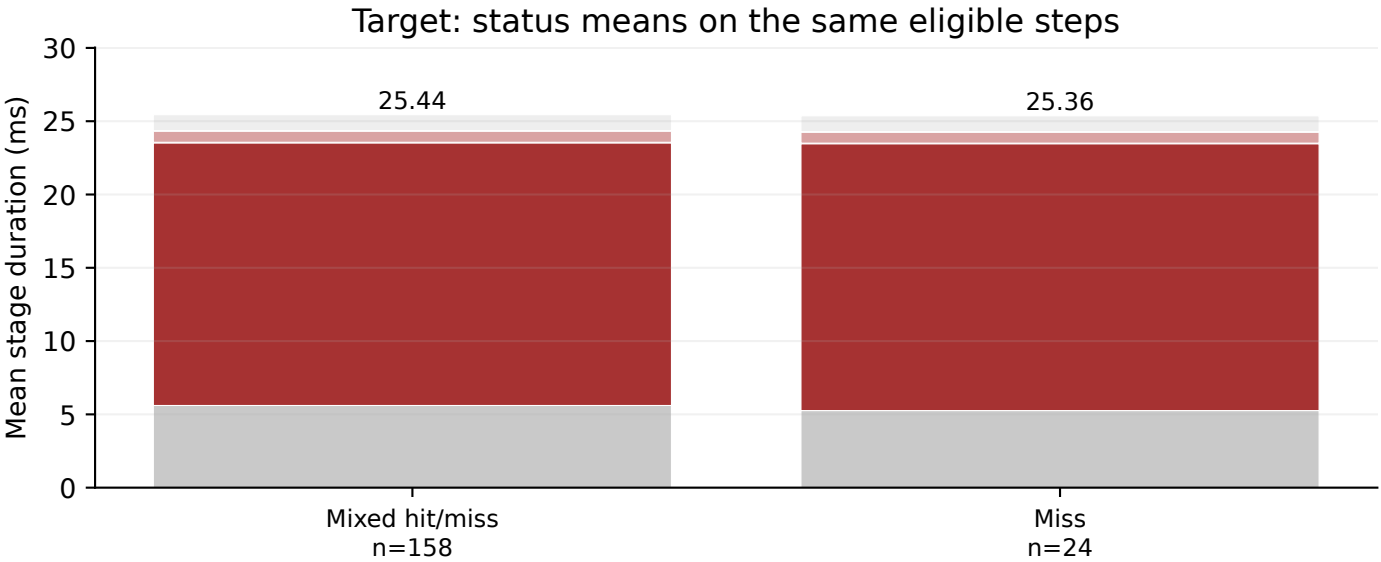
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

TIMING EXCLUDED (concurrent workload) | llama2_b8_ssd_k2_f5
SSD: K=2, F=5
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 220 eligible steps



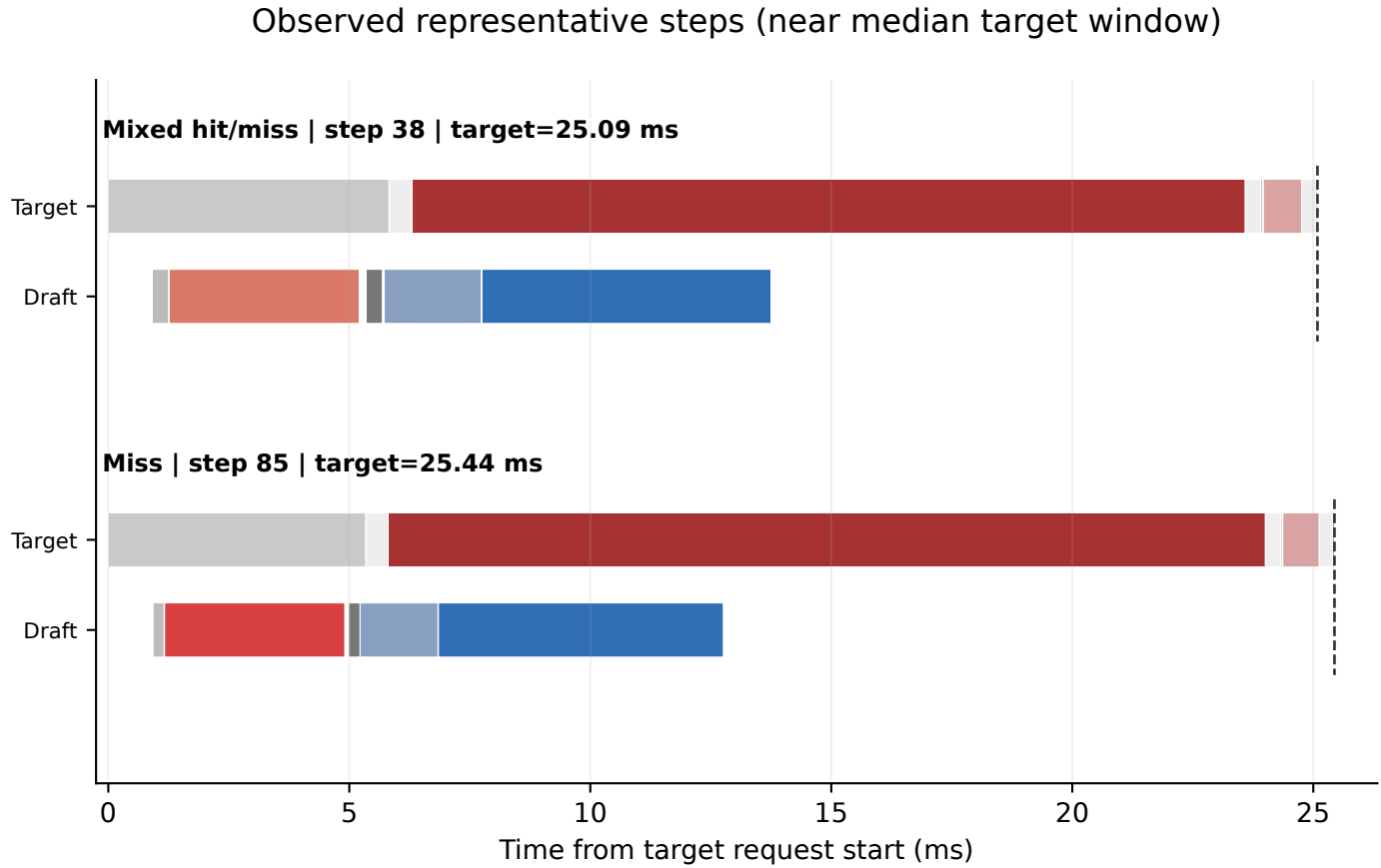
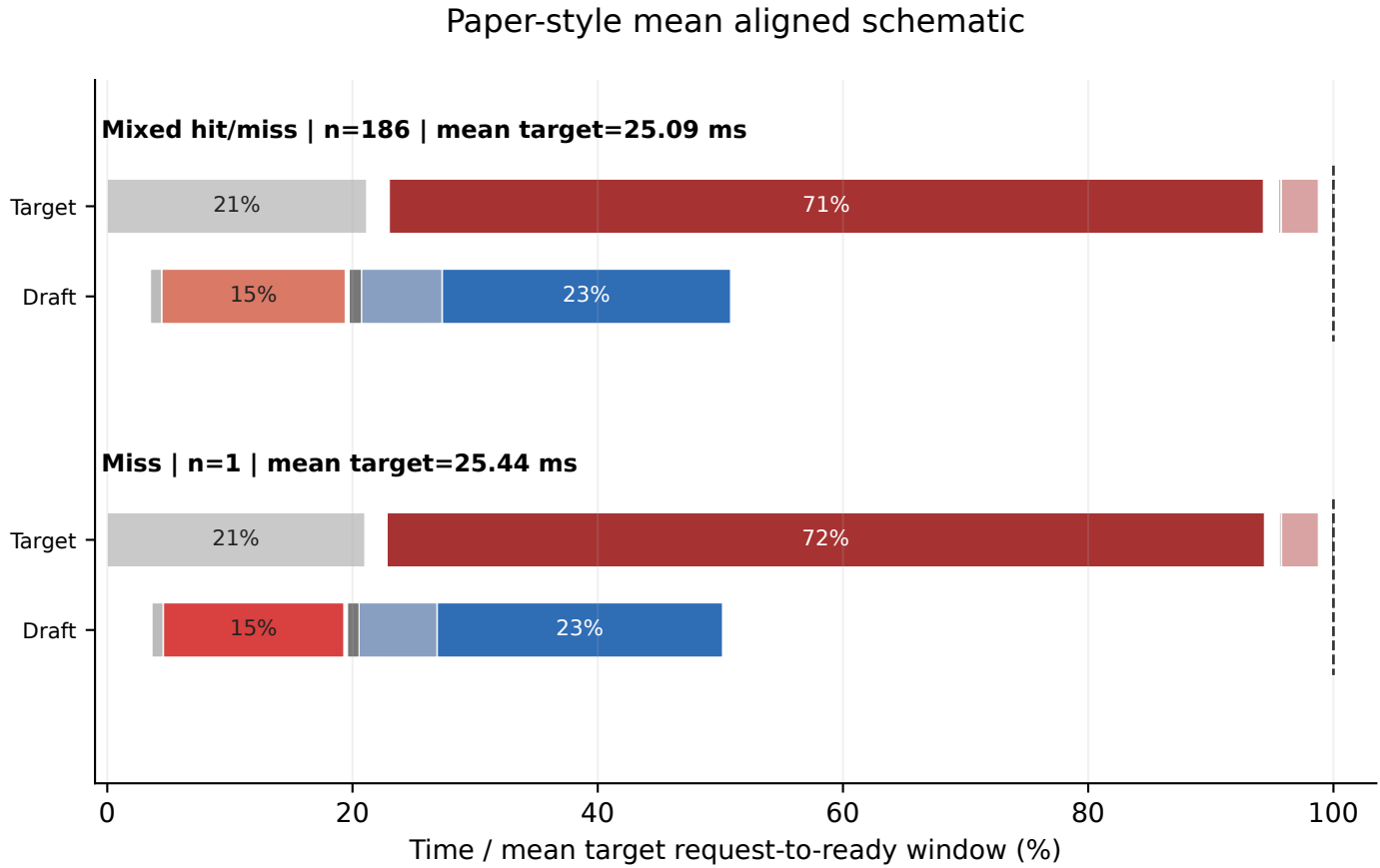
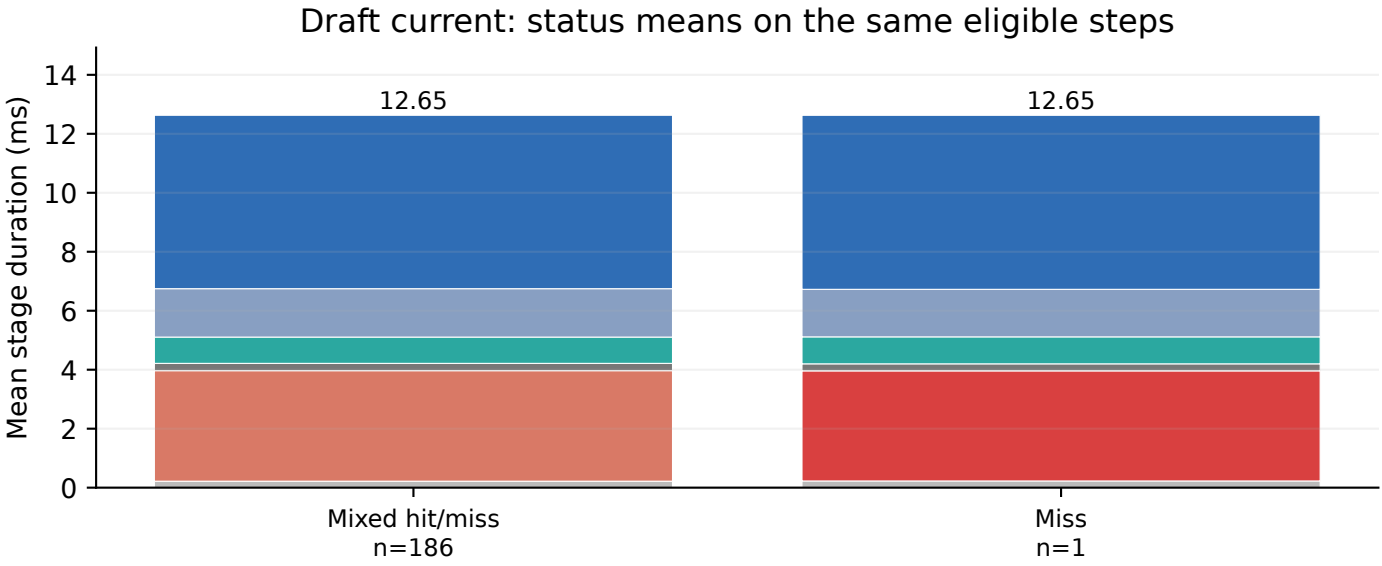
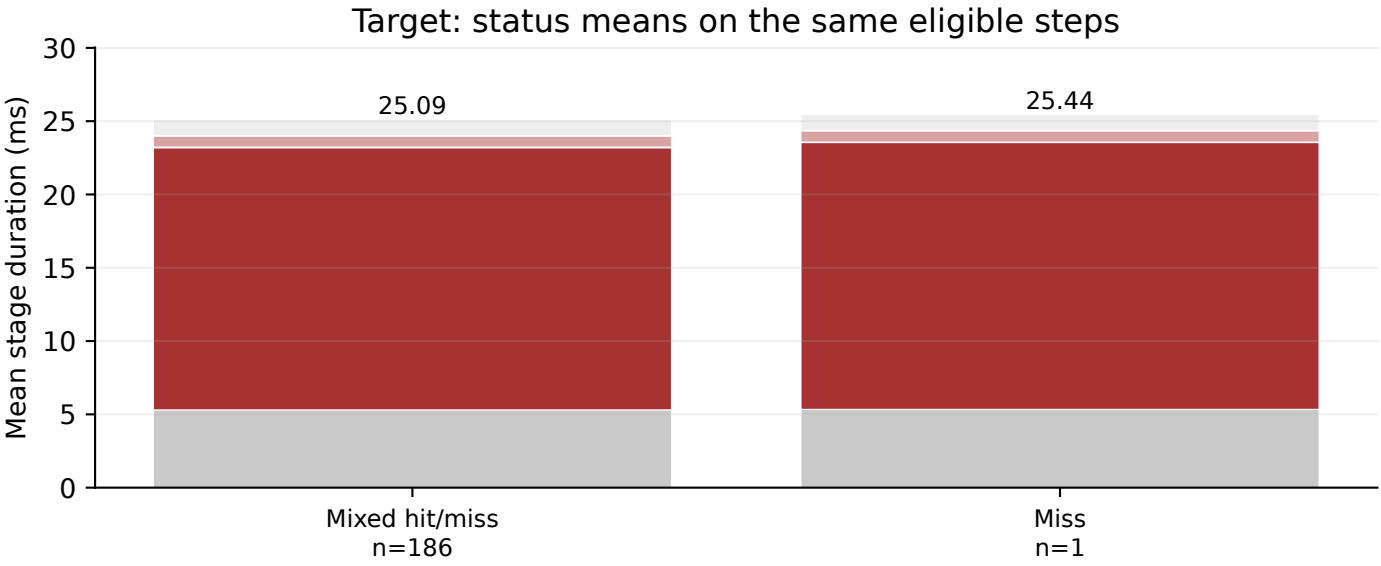
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



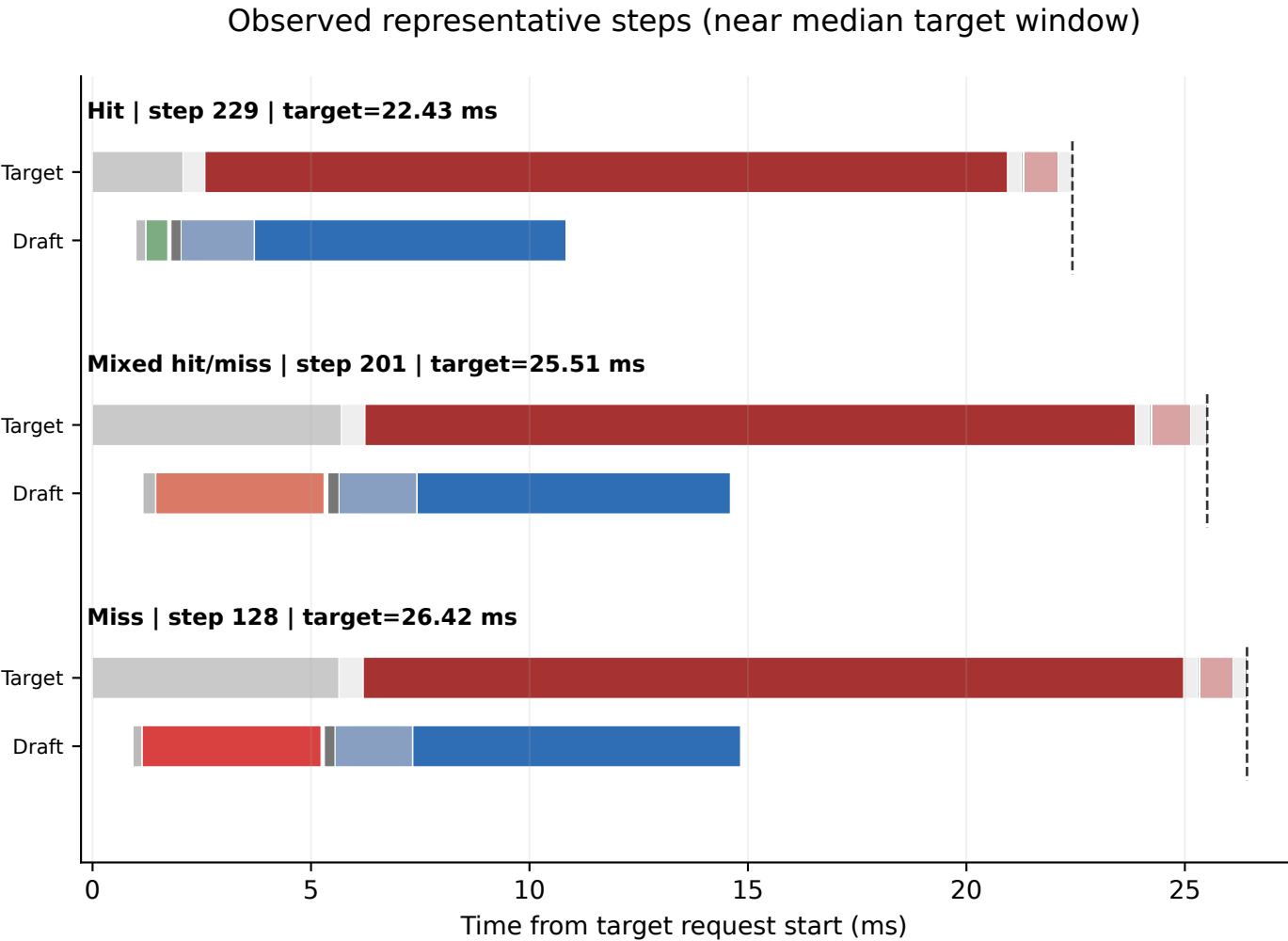
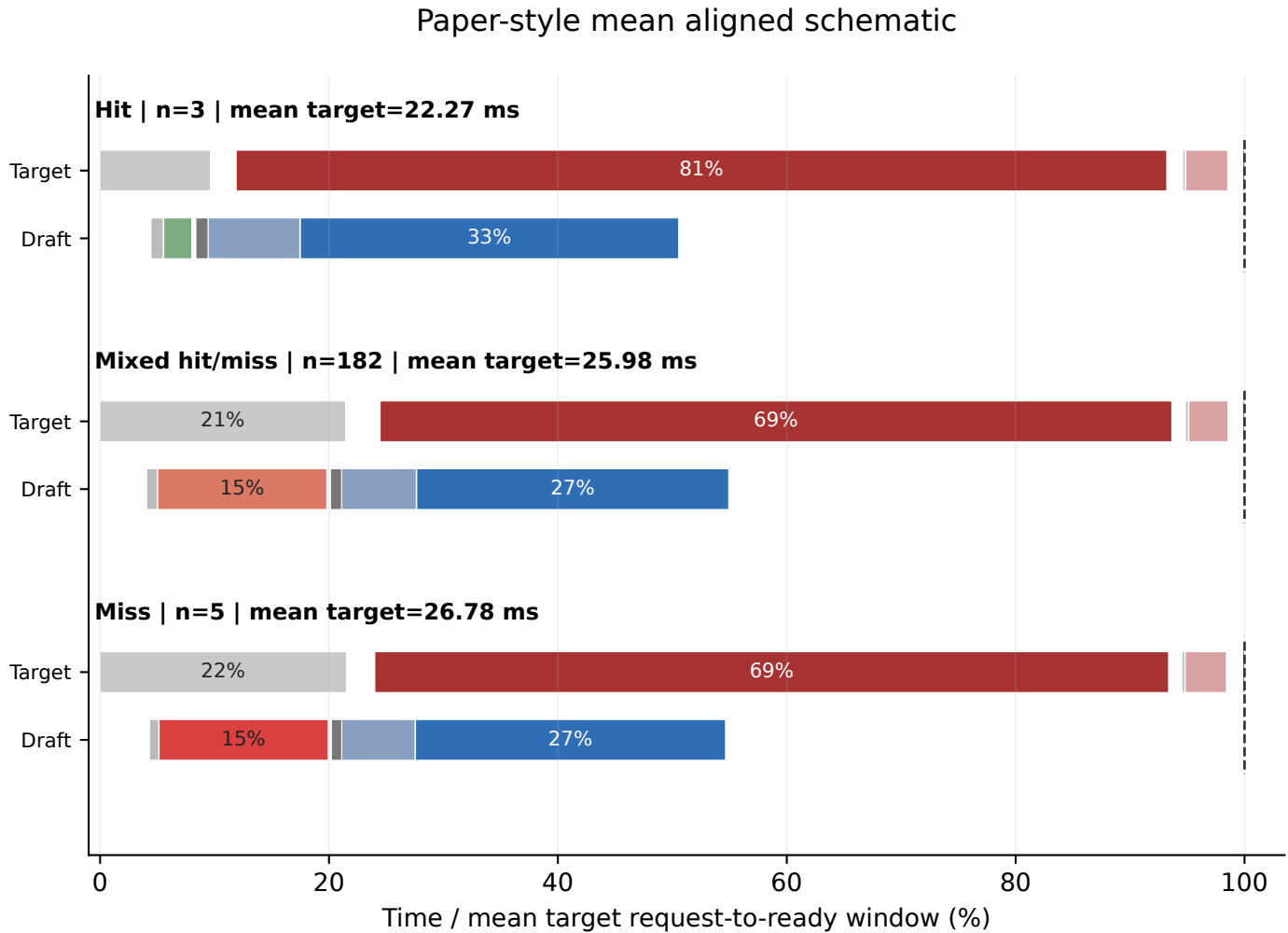
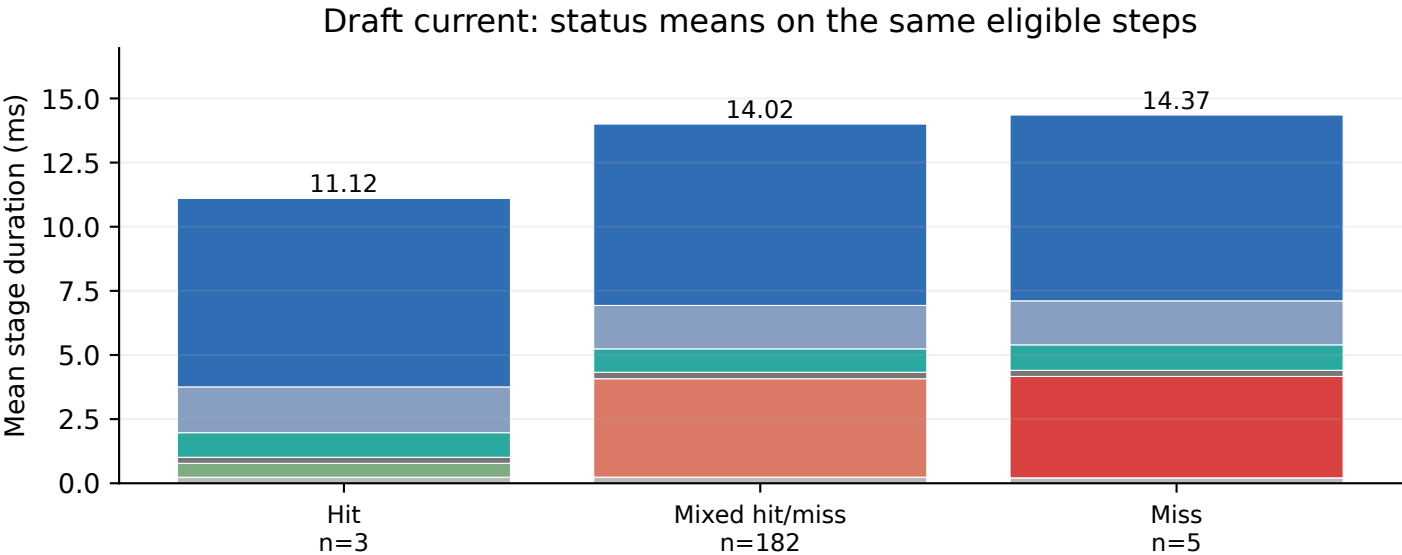
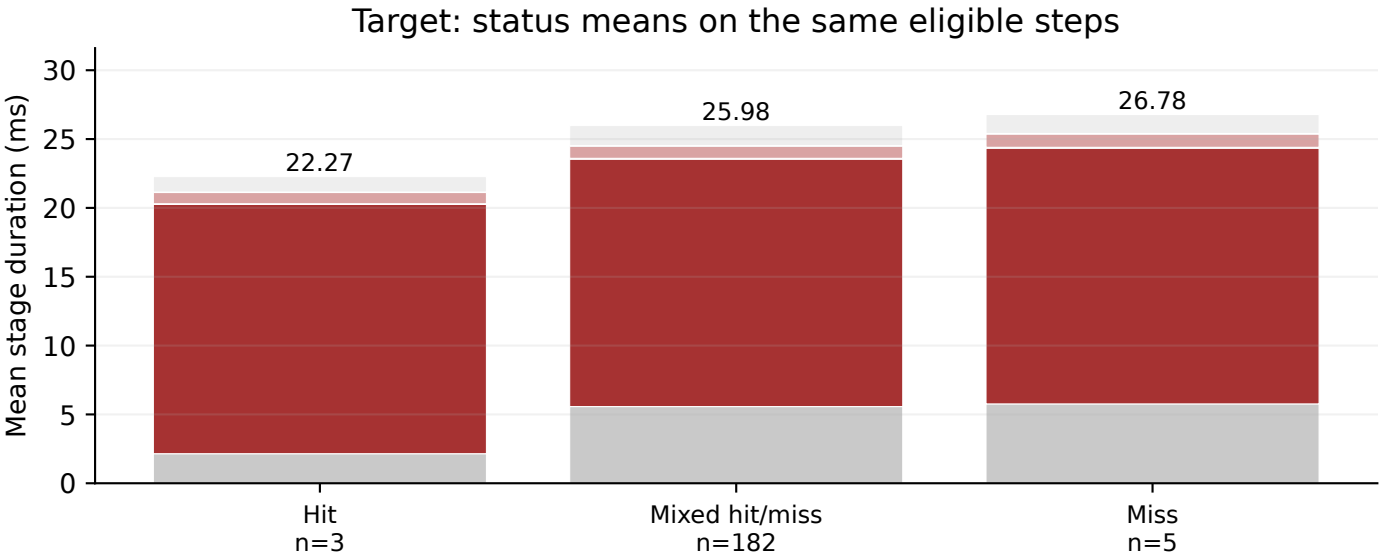
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



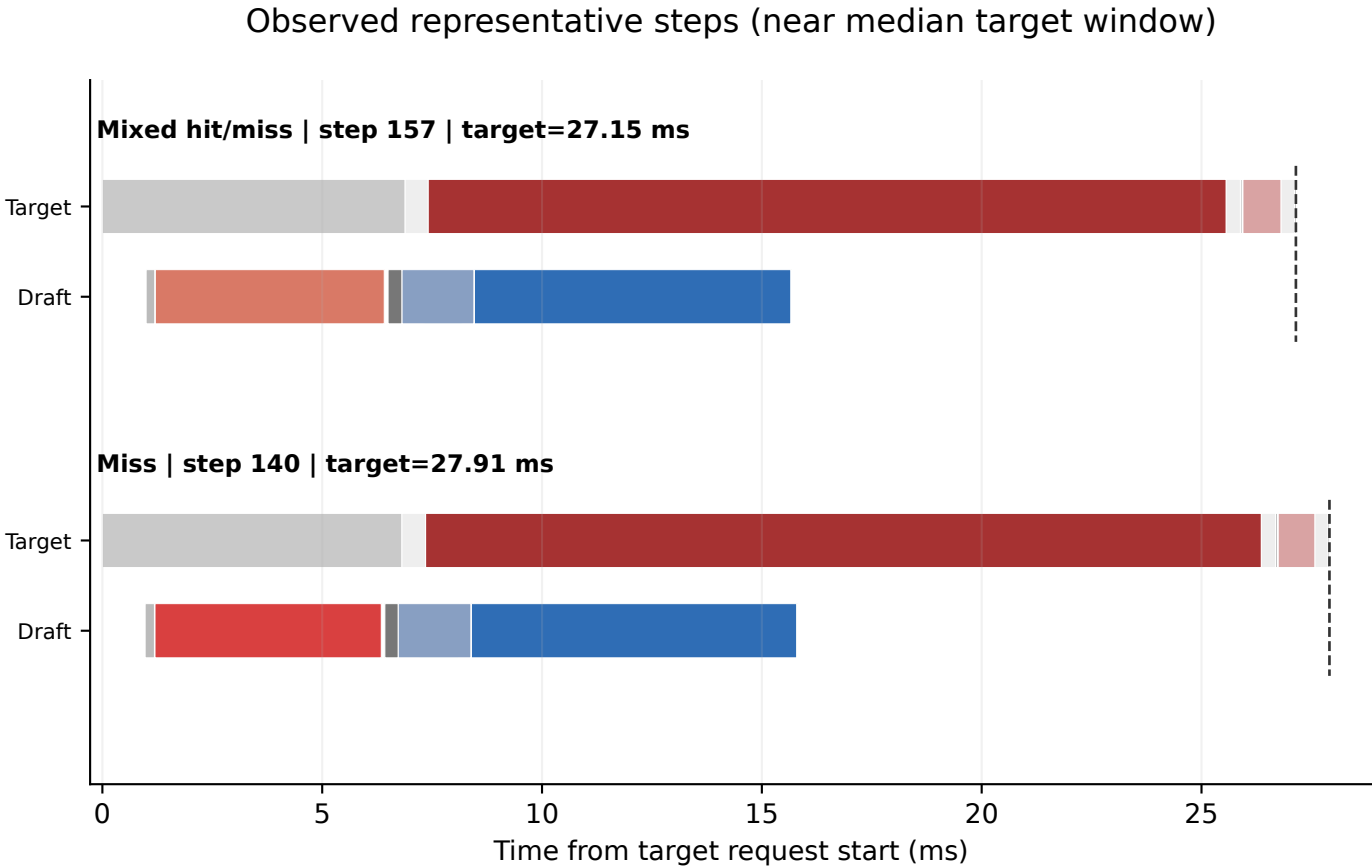
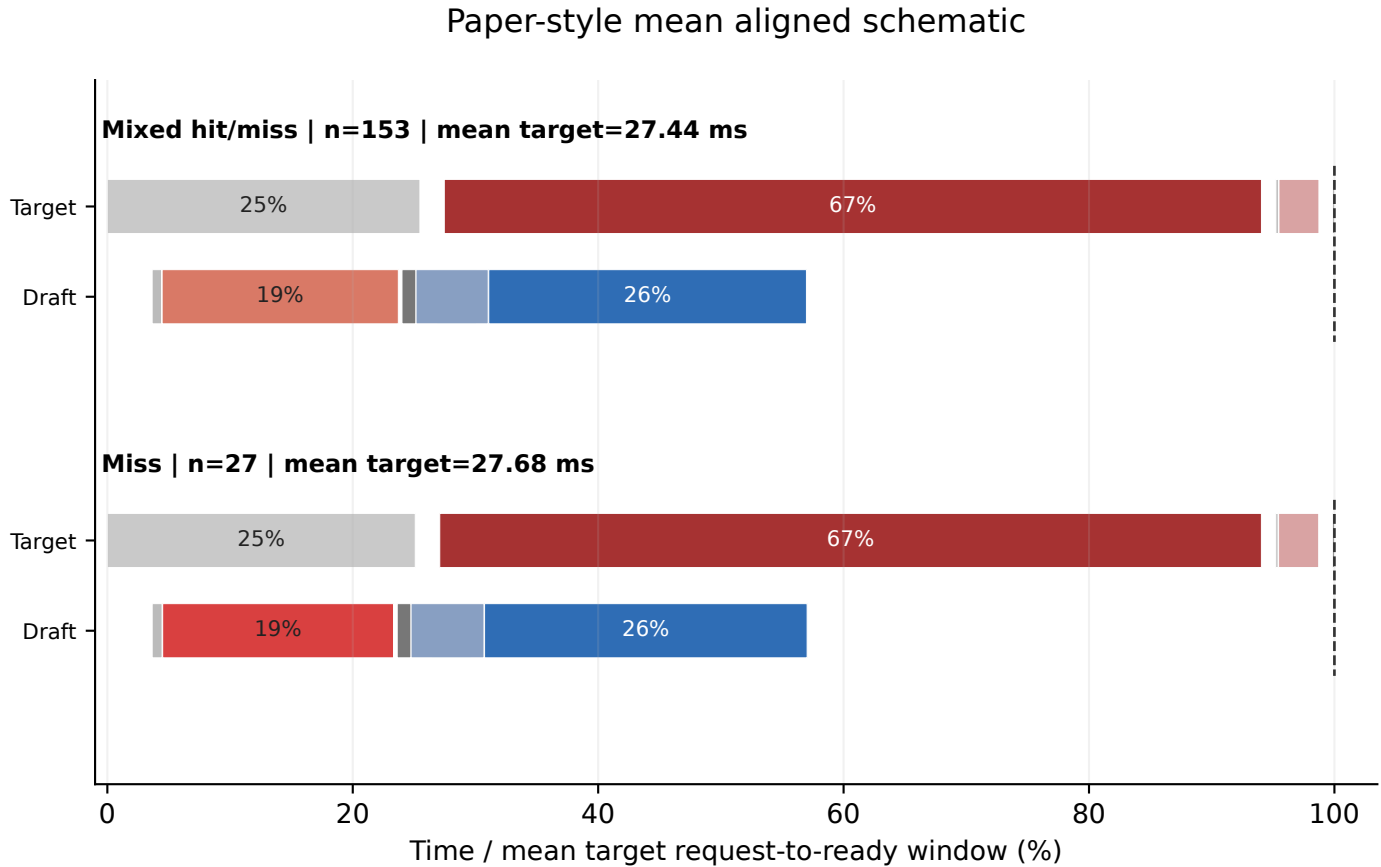
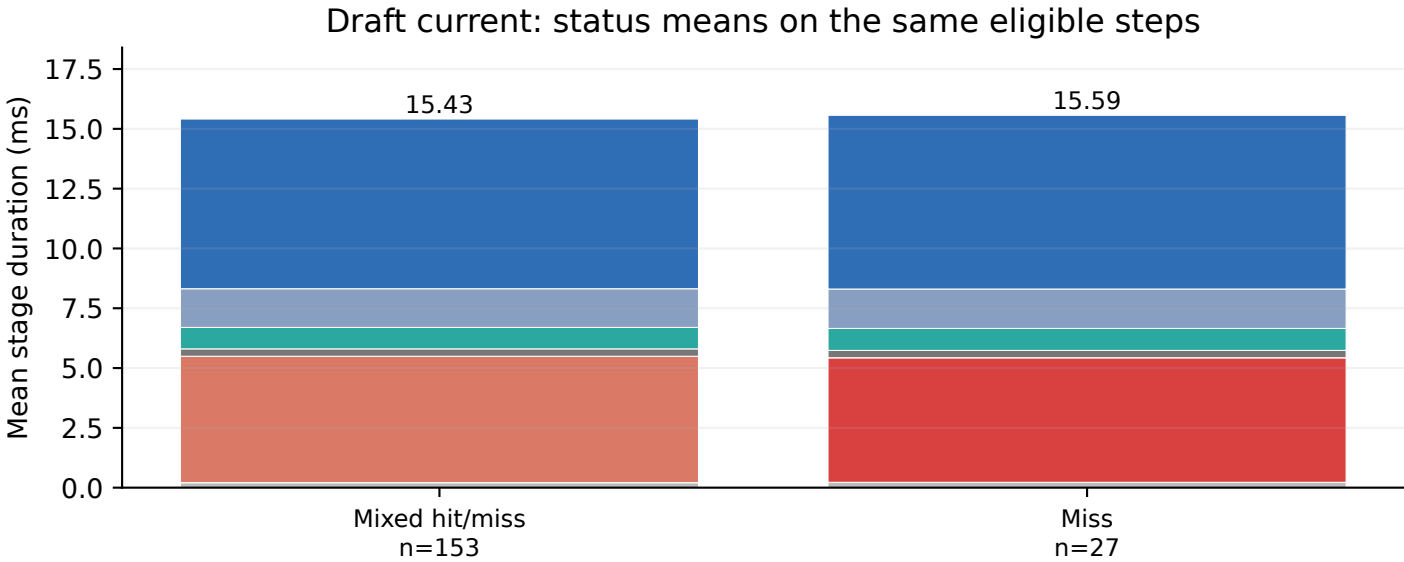
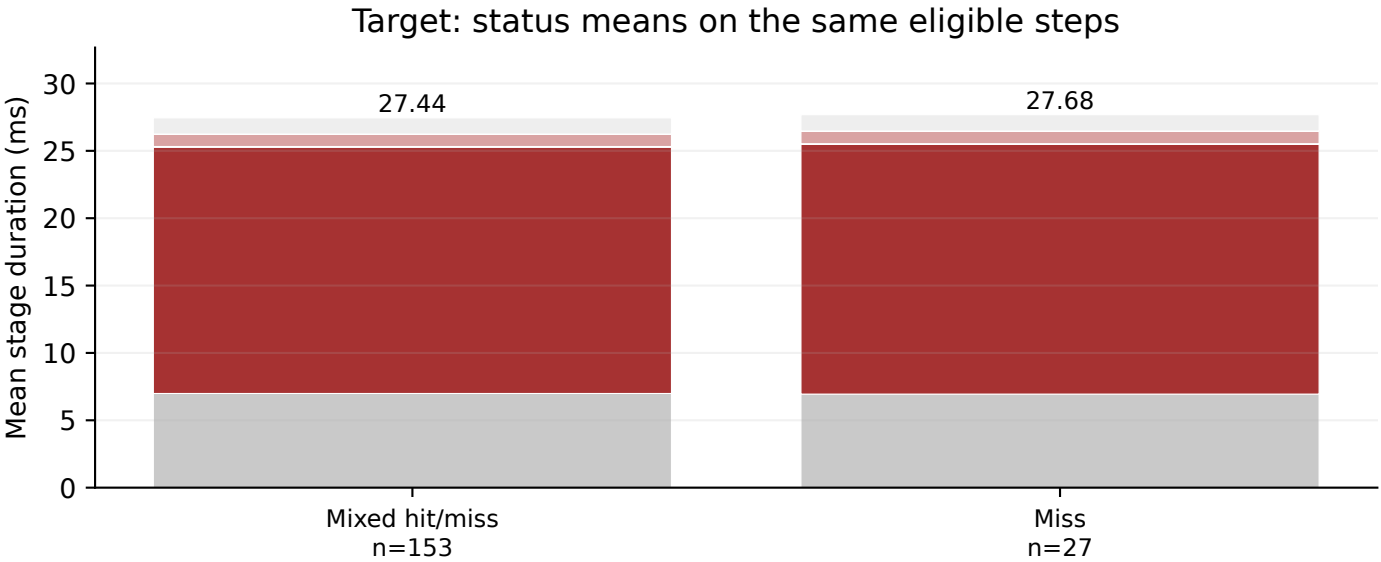
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



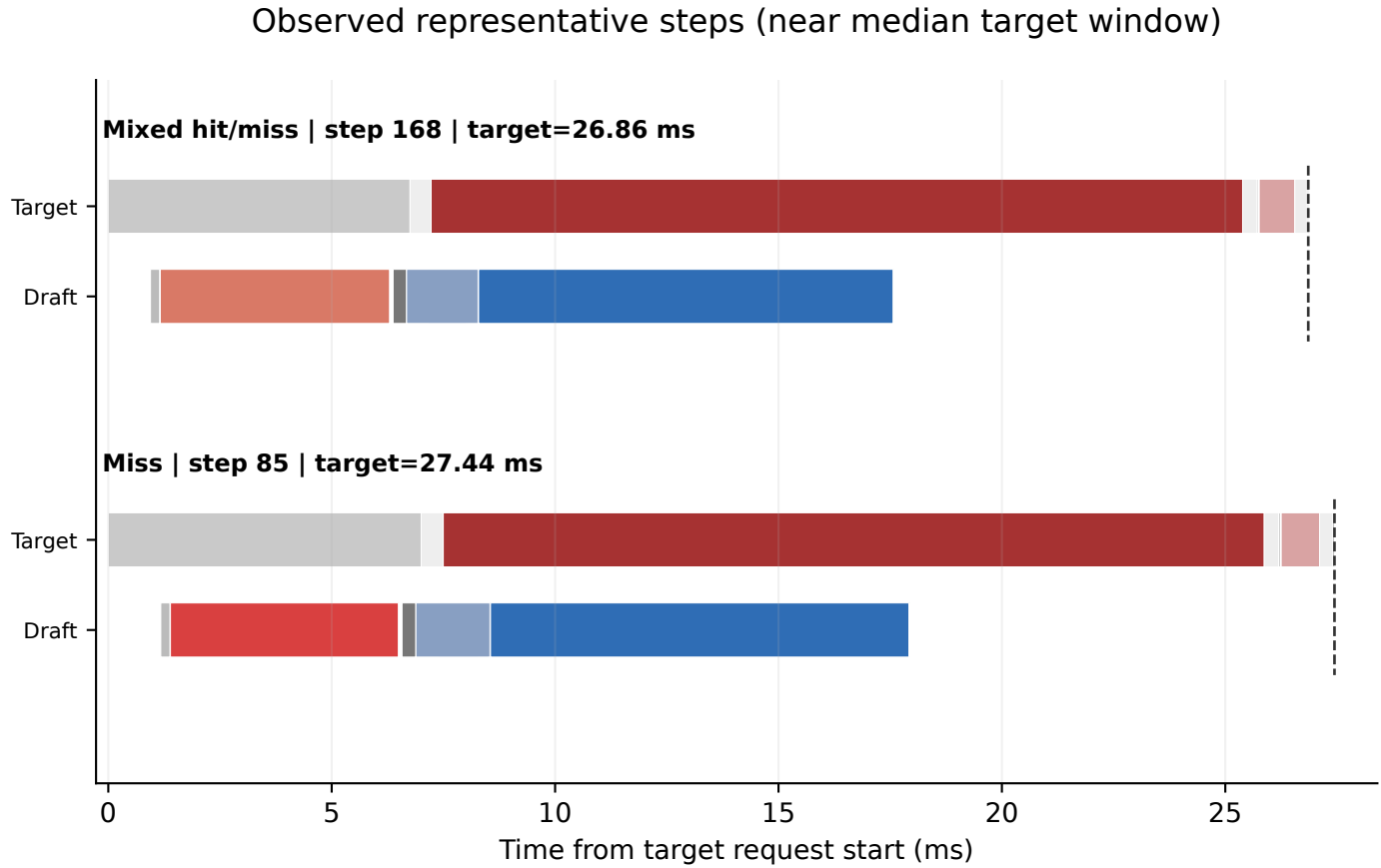
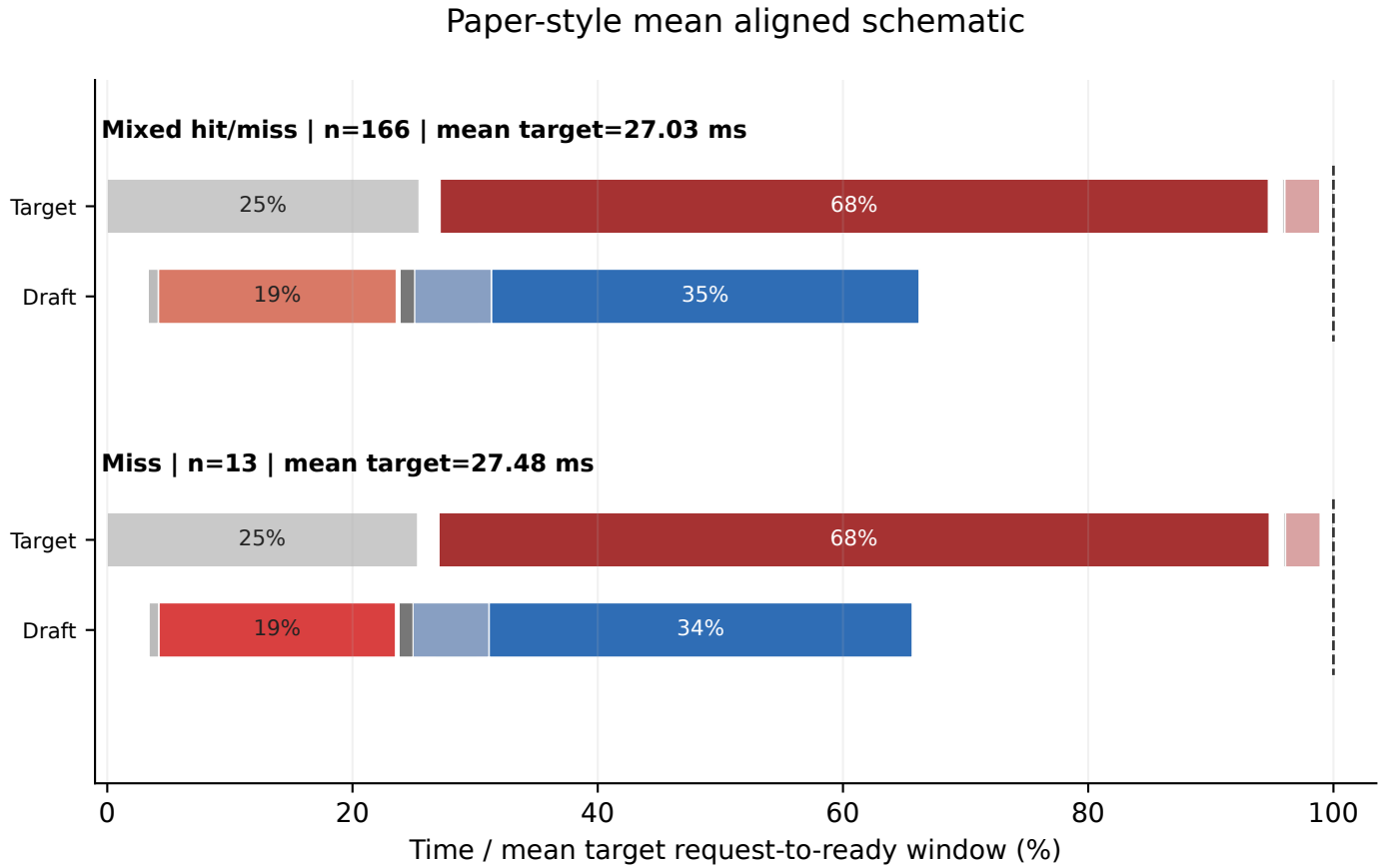
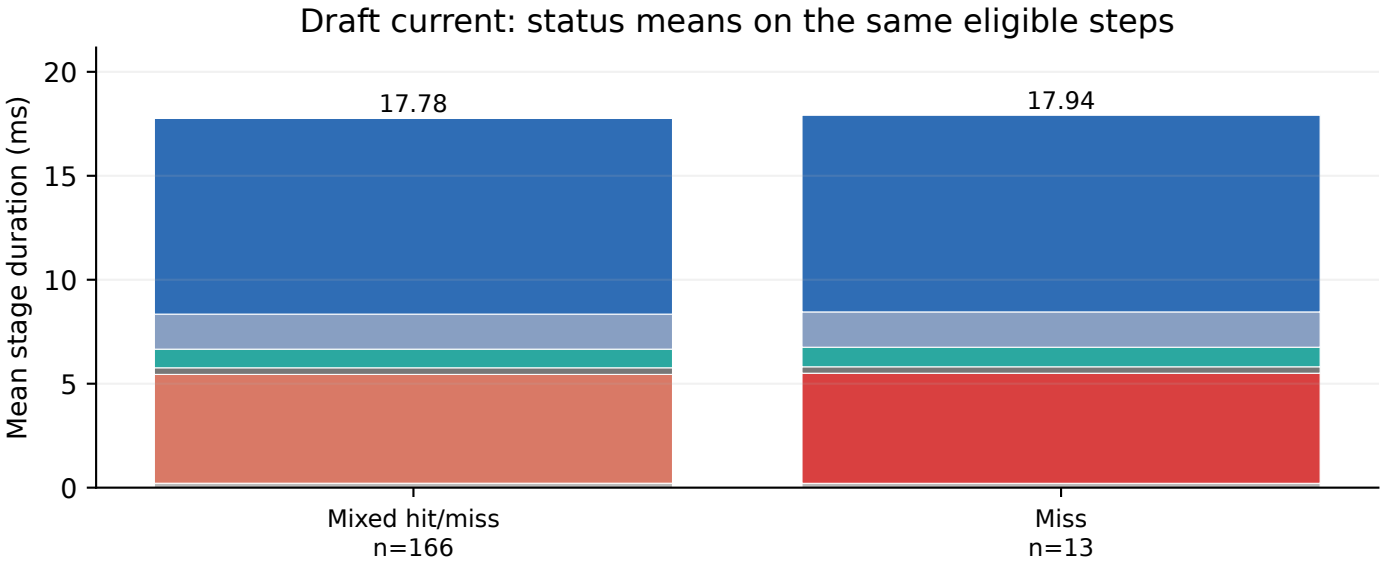
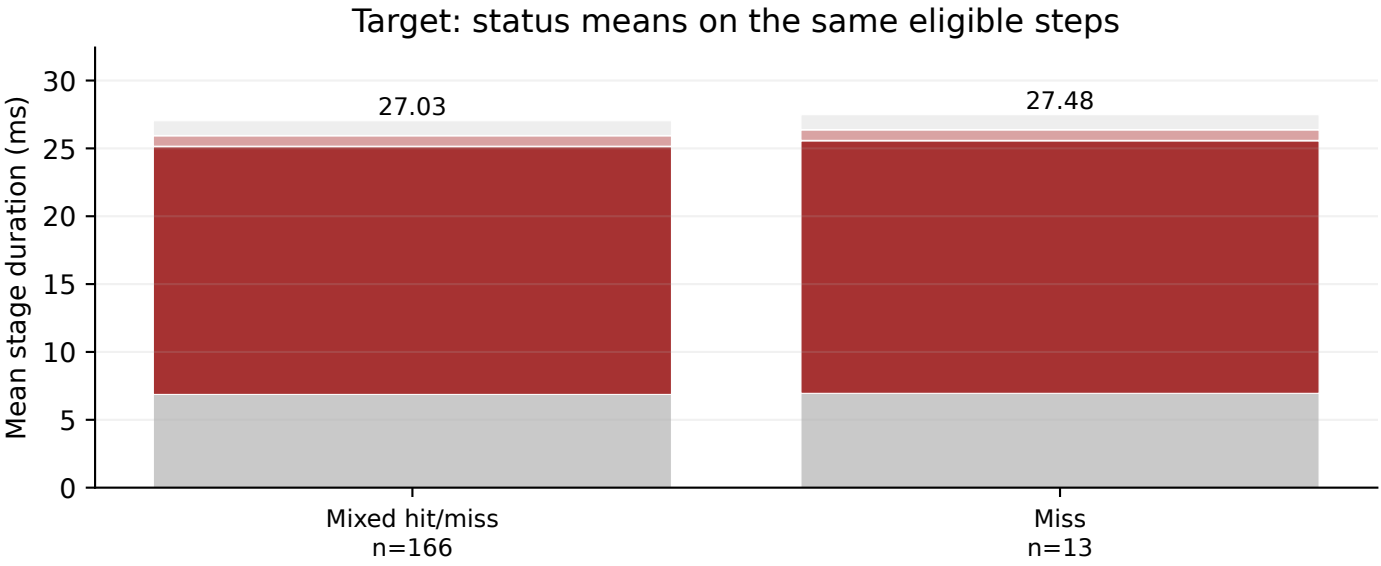
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



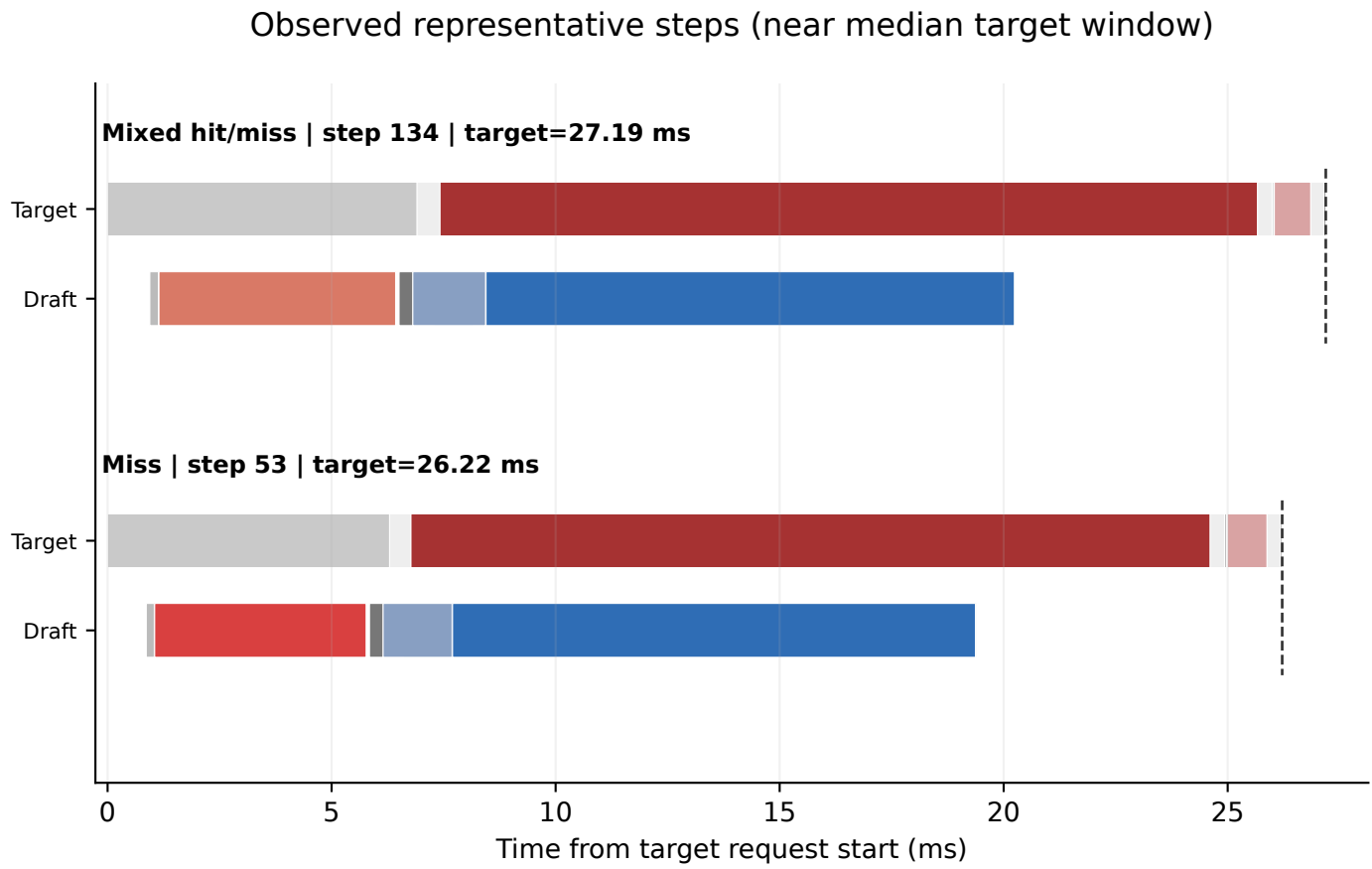
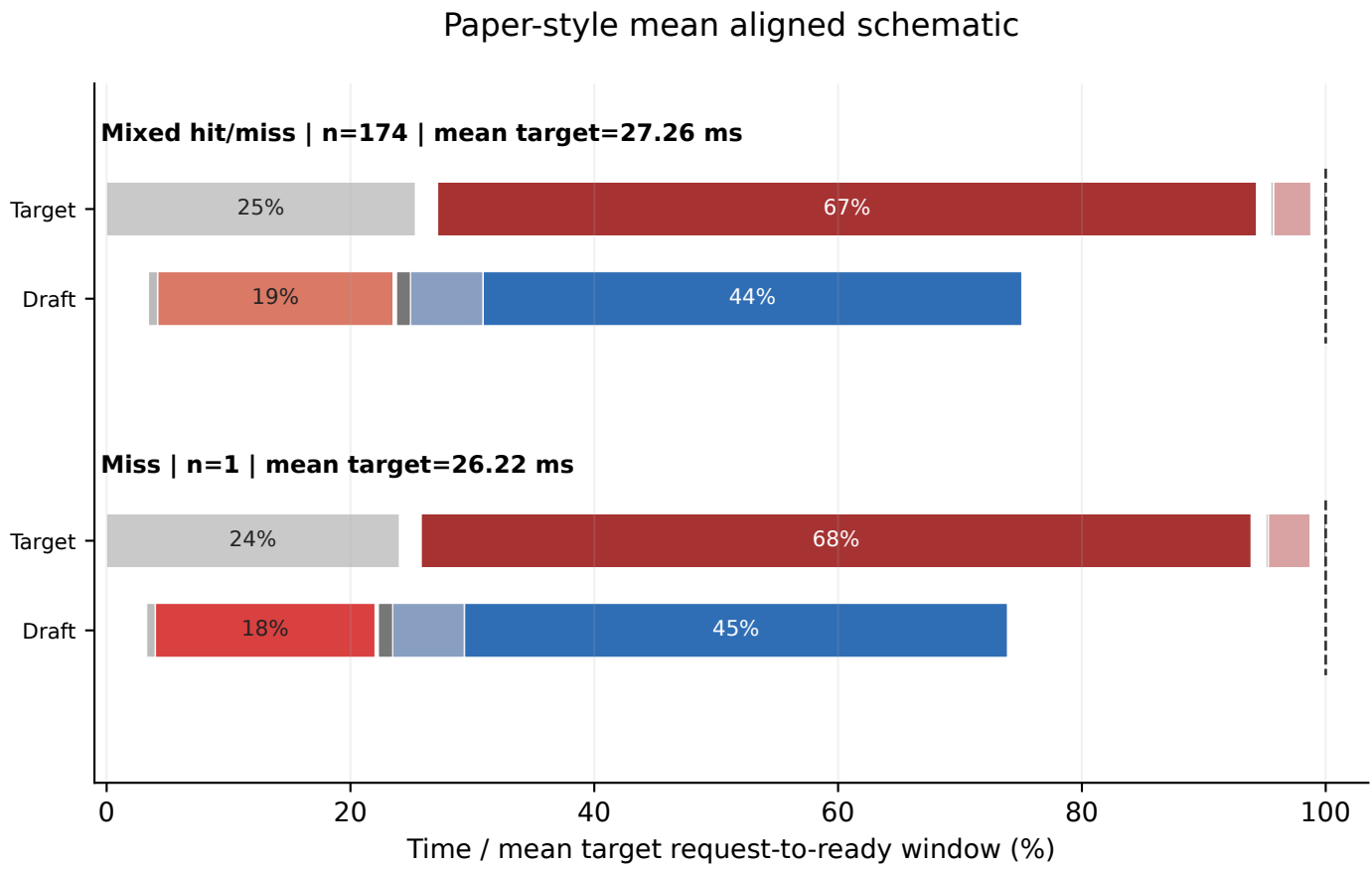
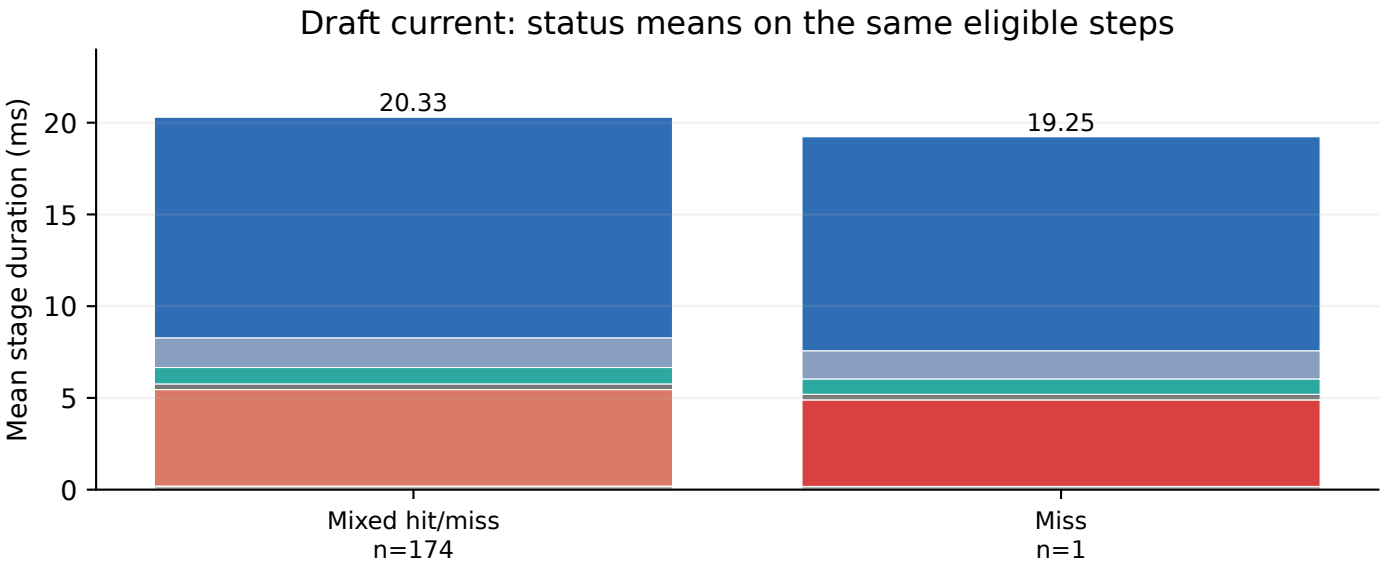
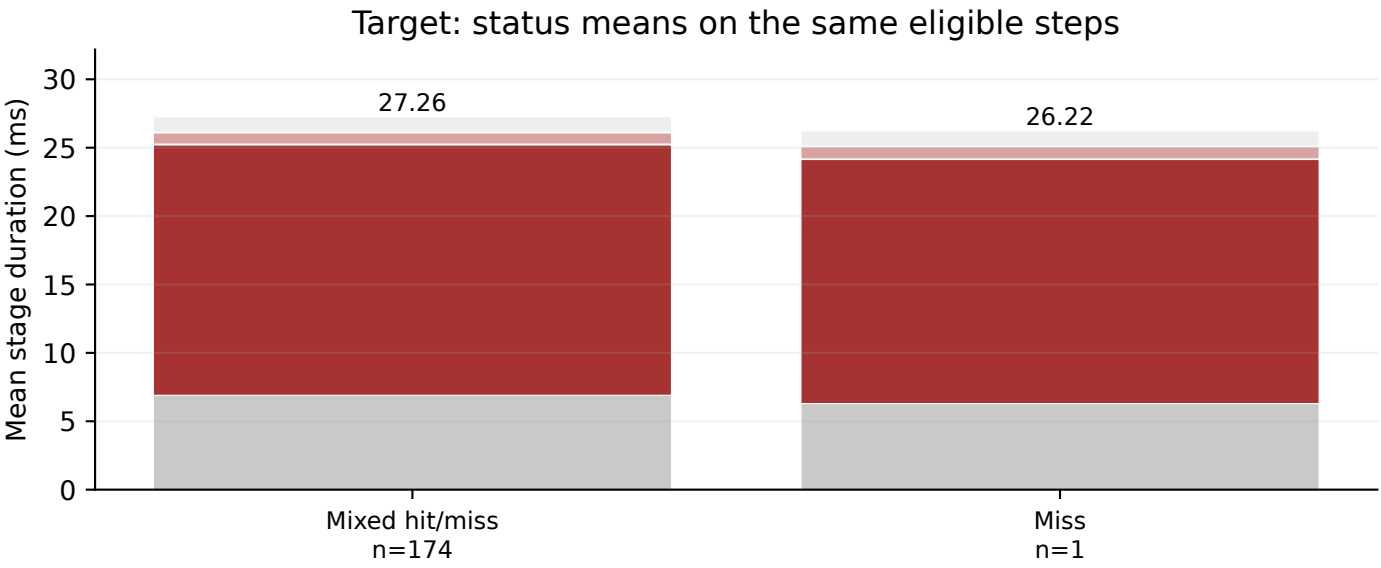
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



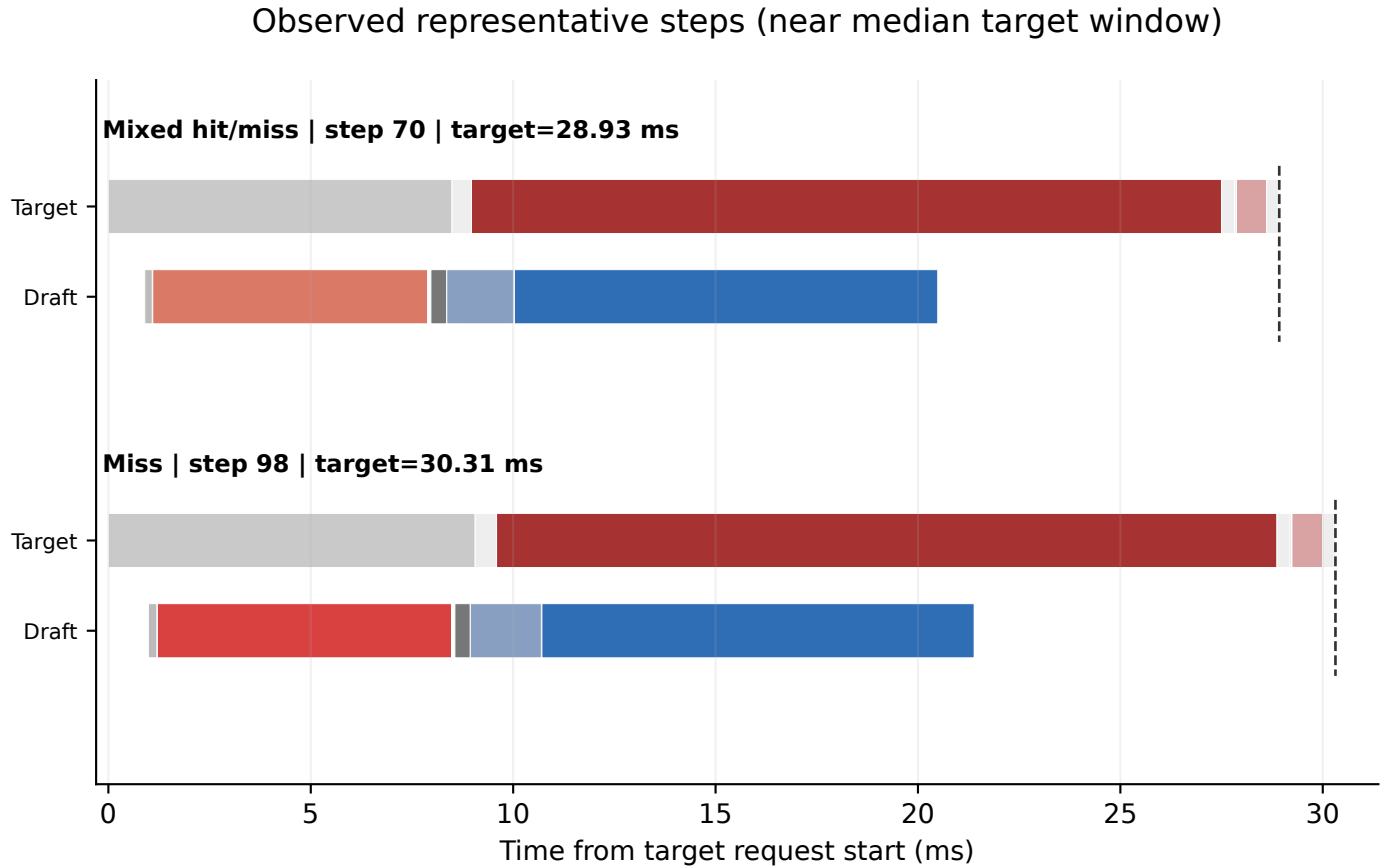
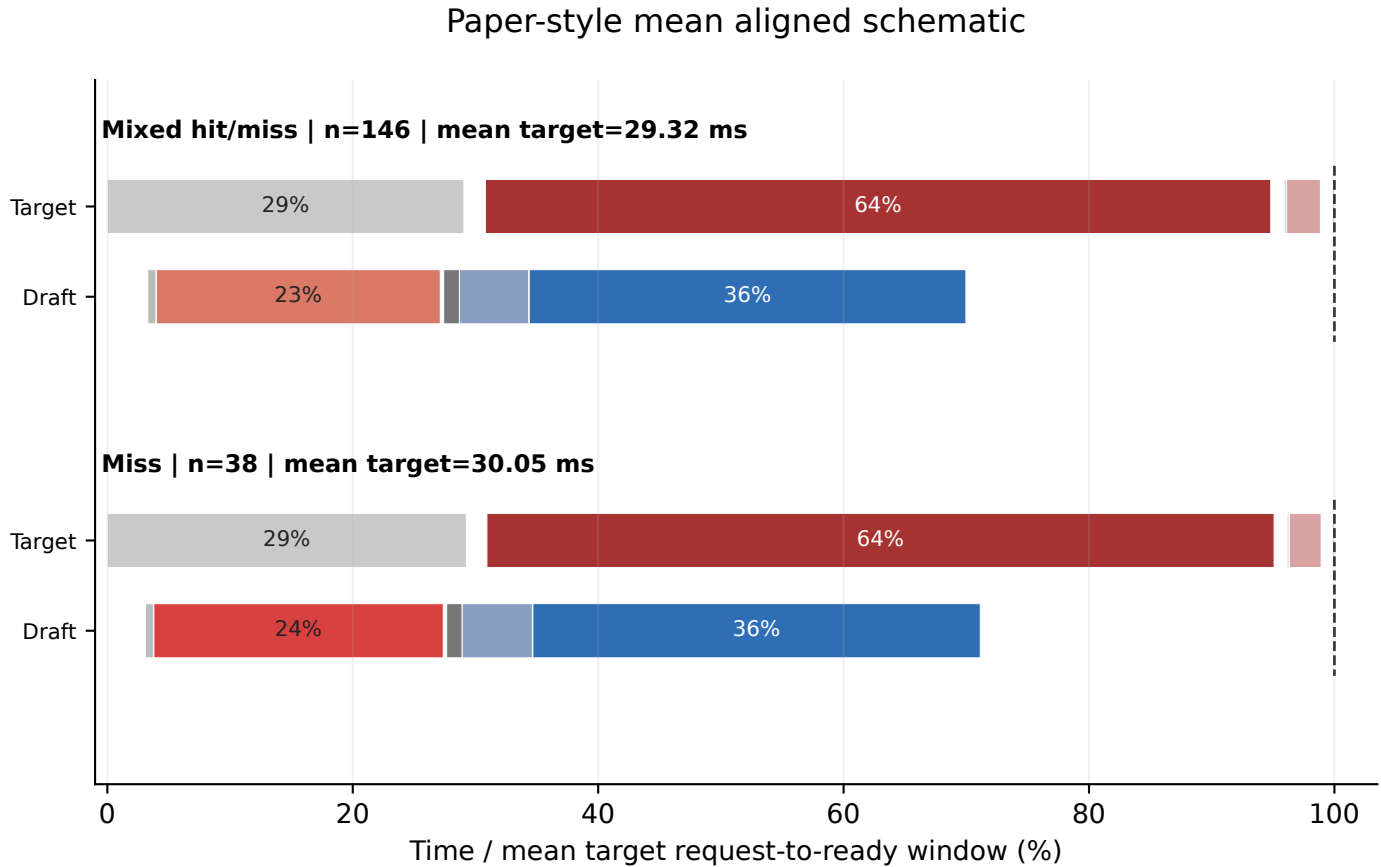
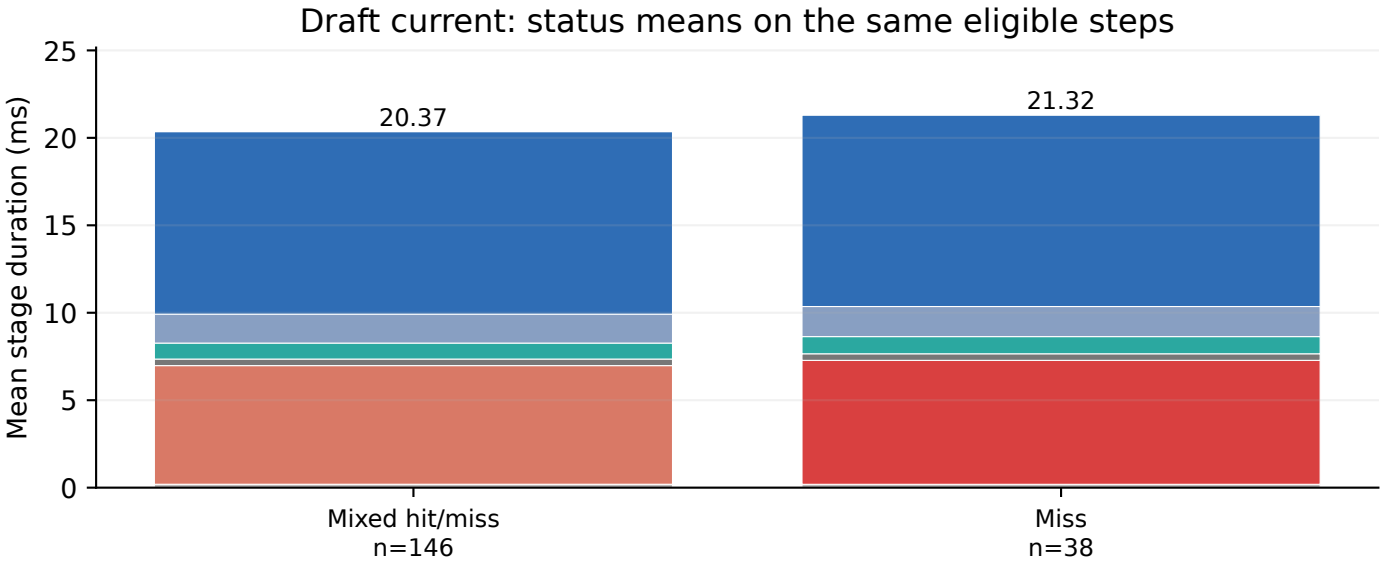
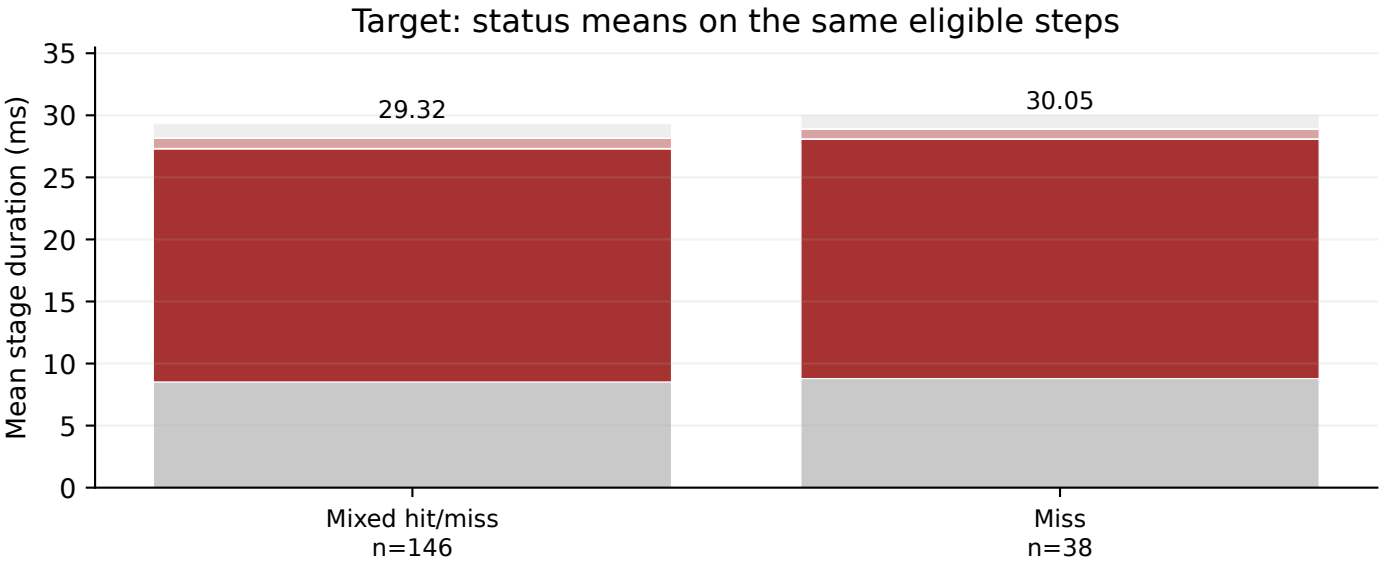
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



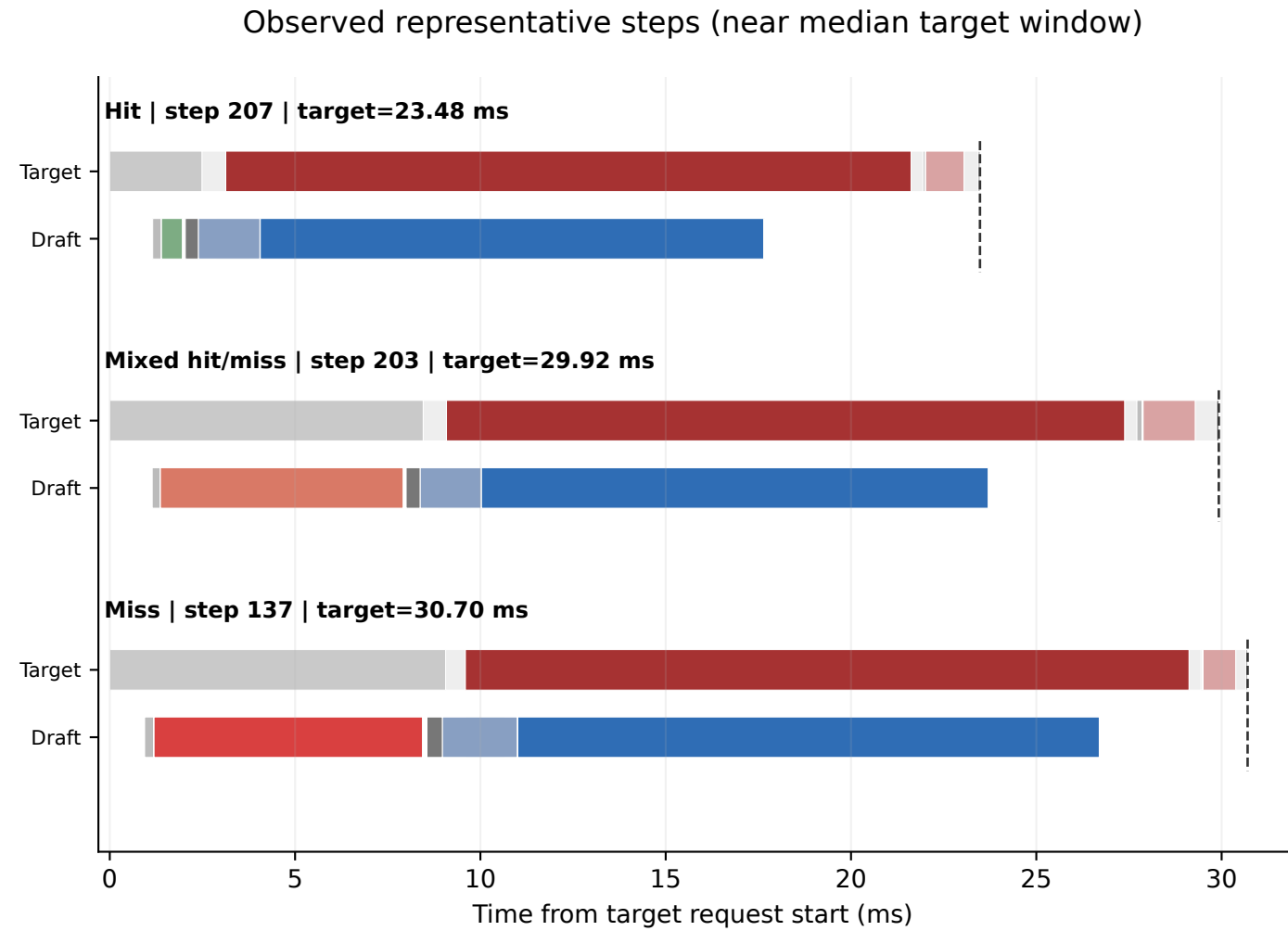
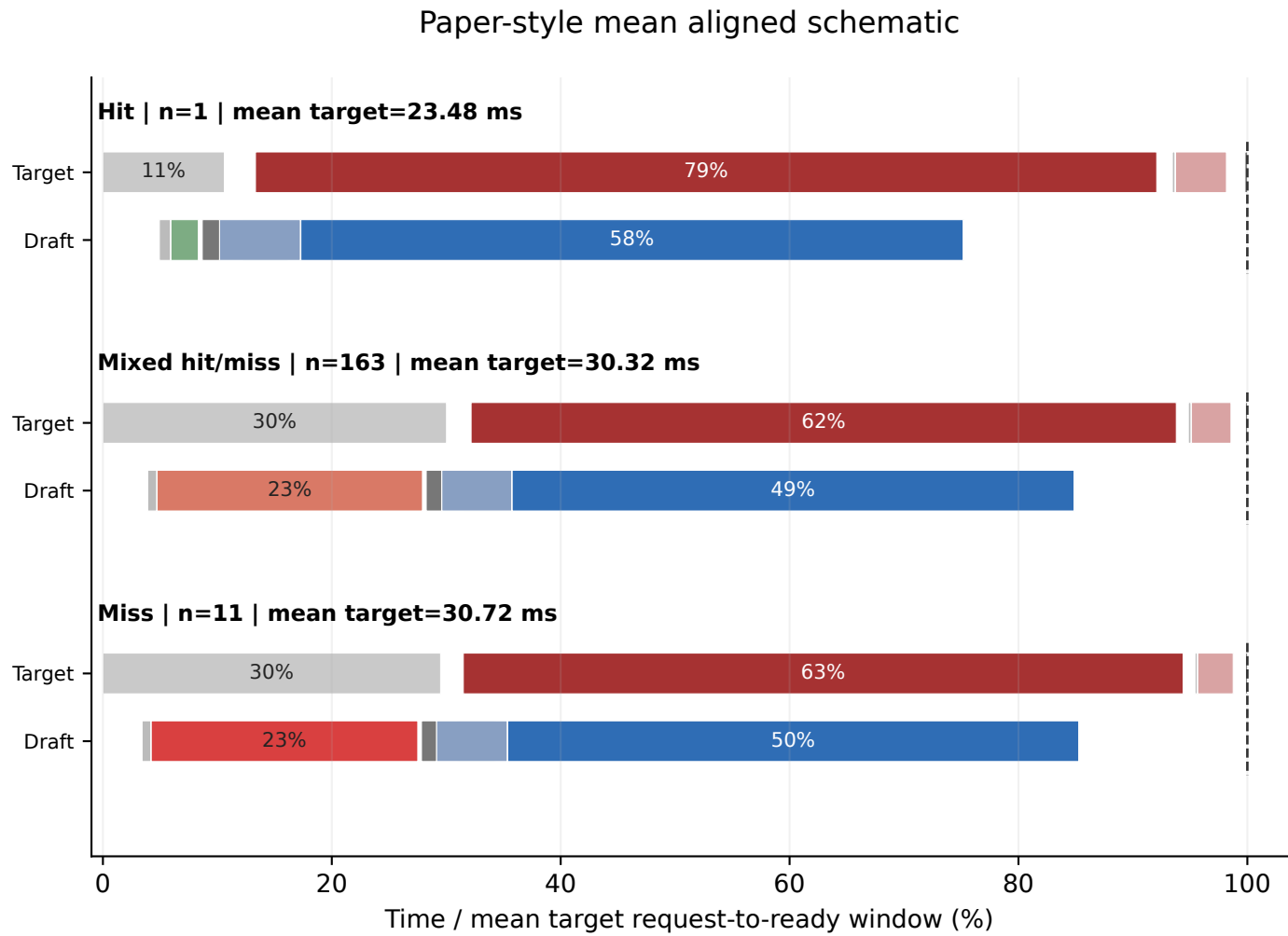
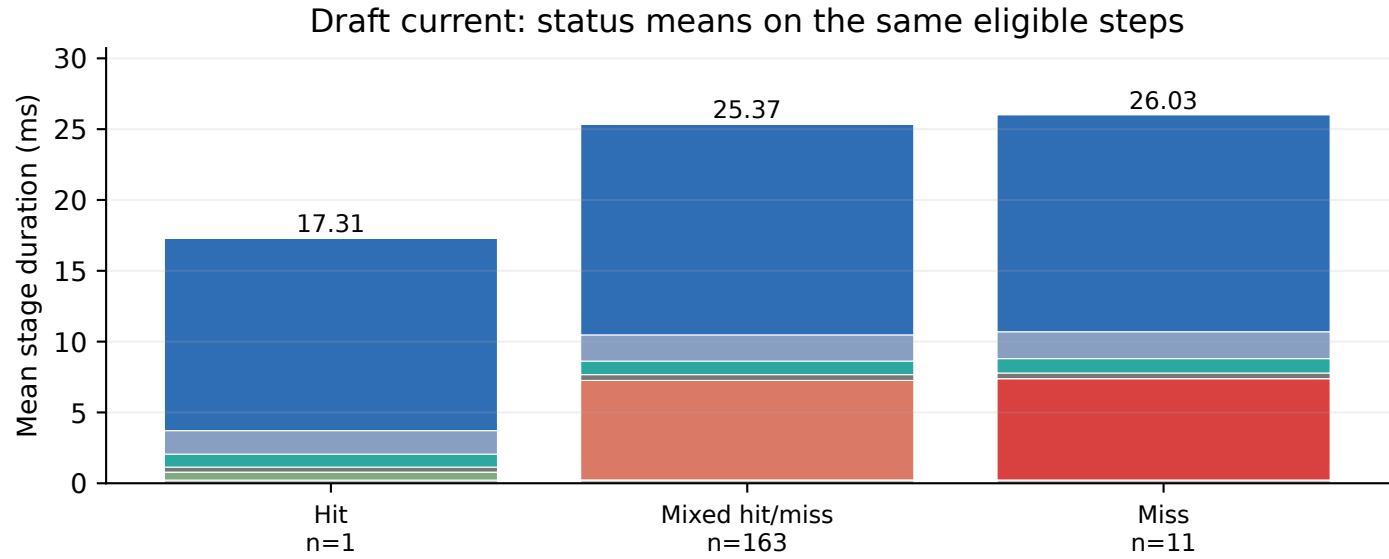
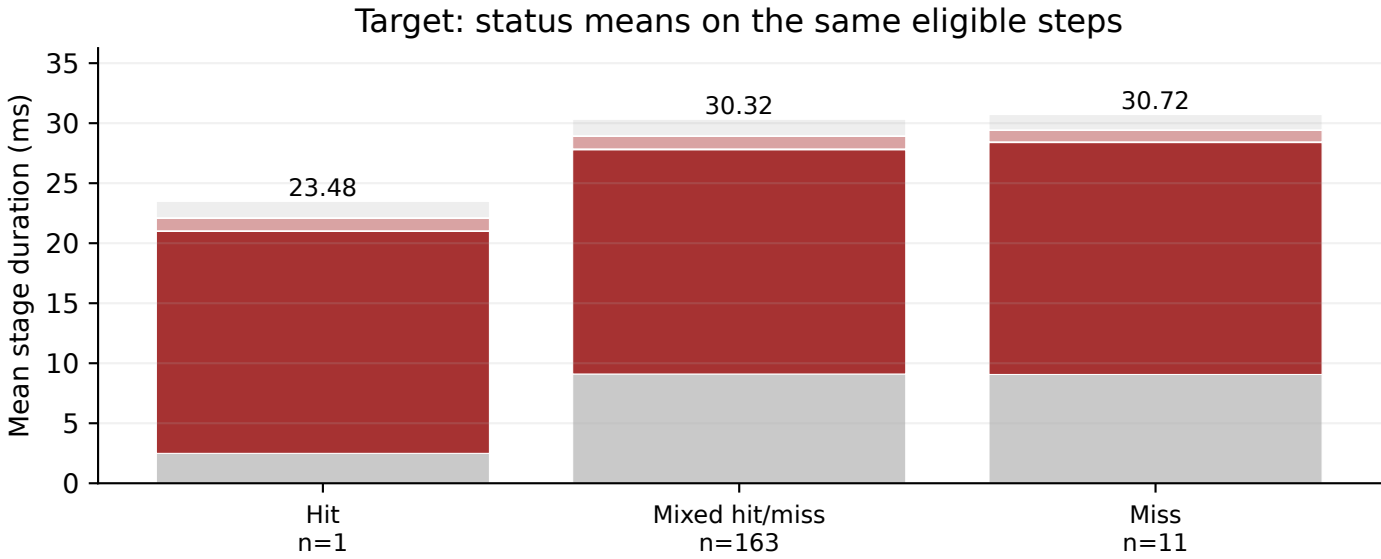
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

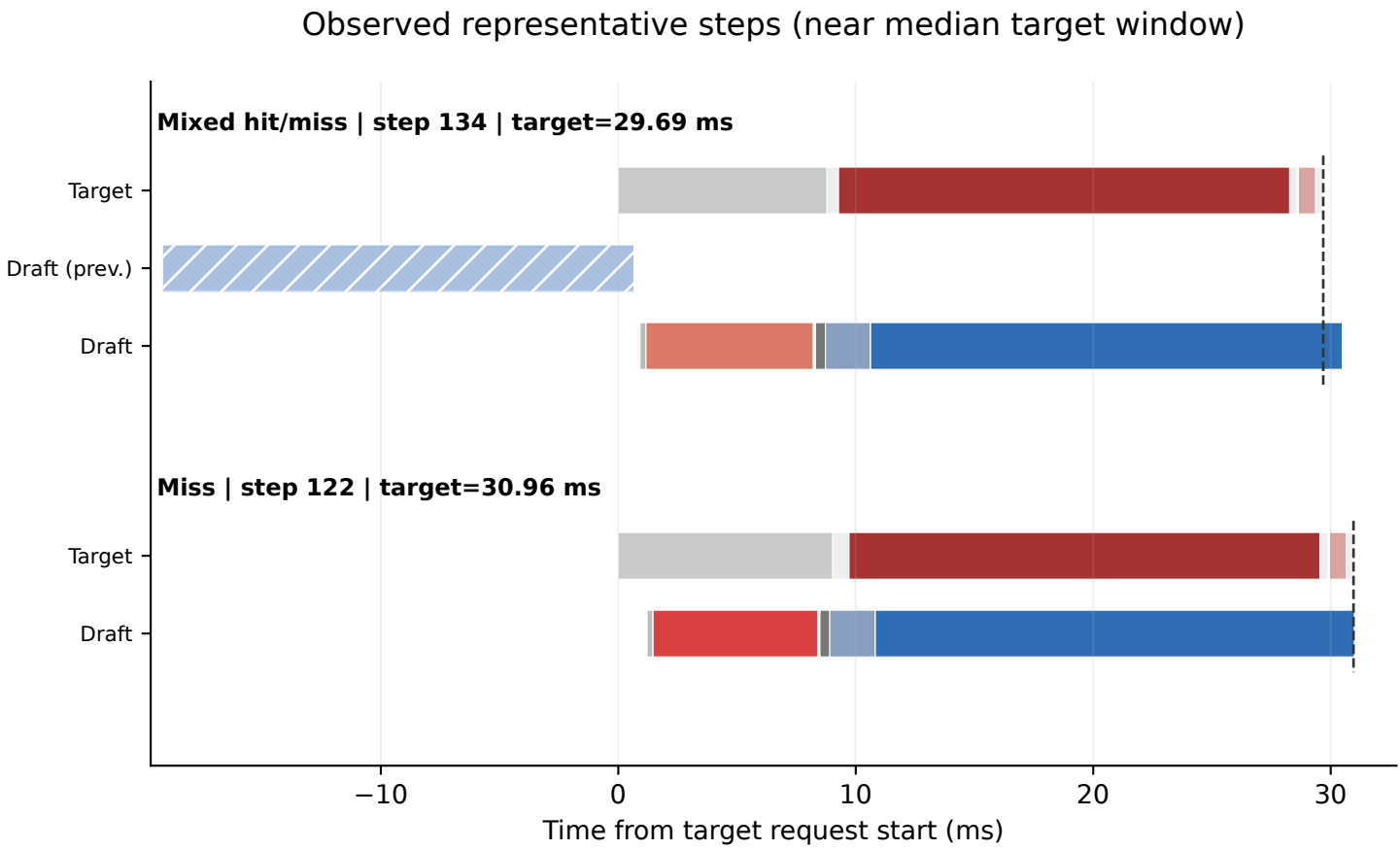
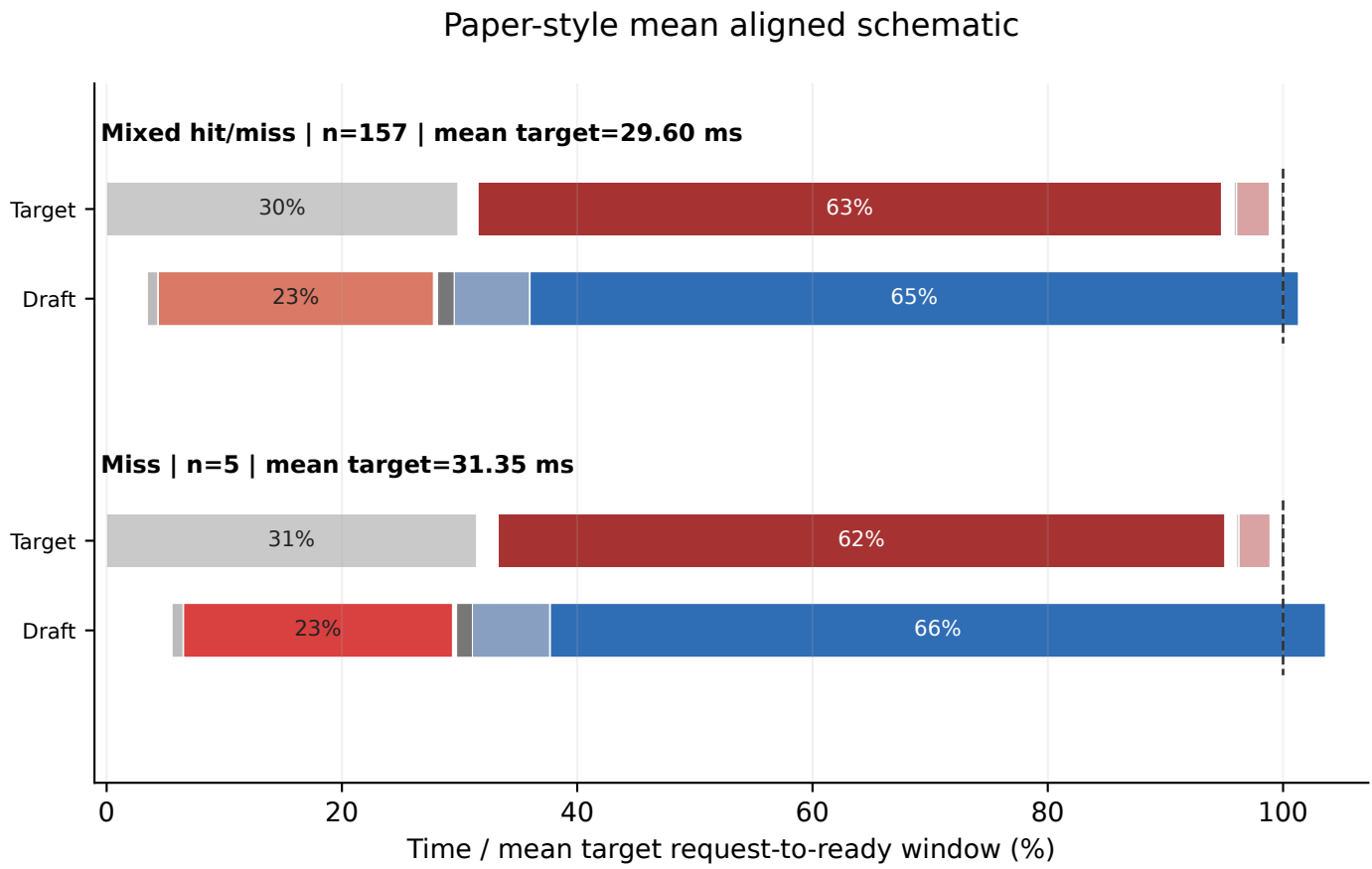
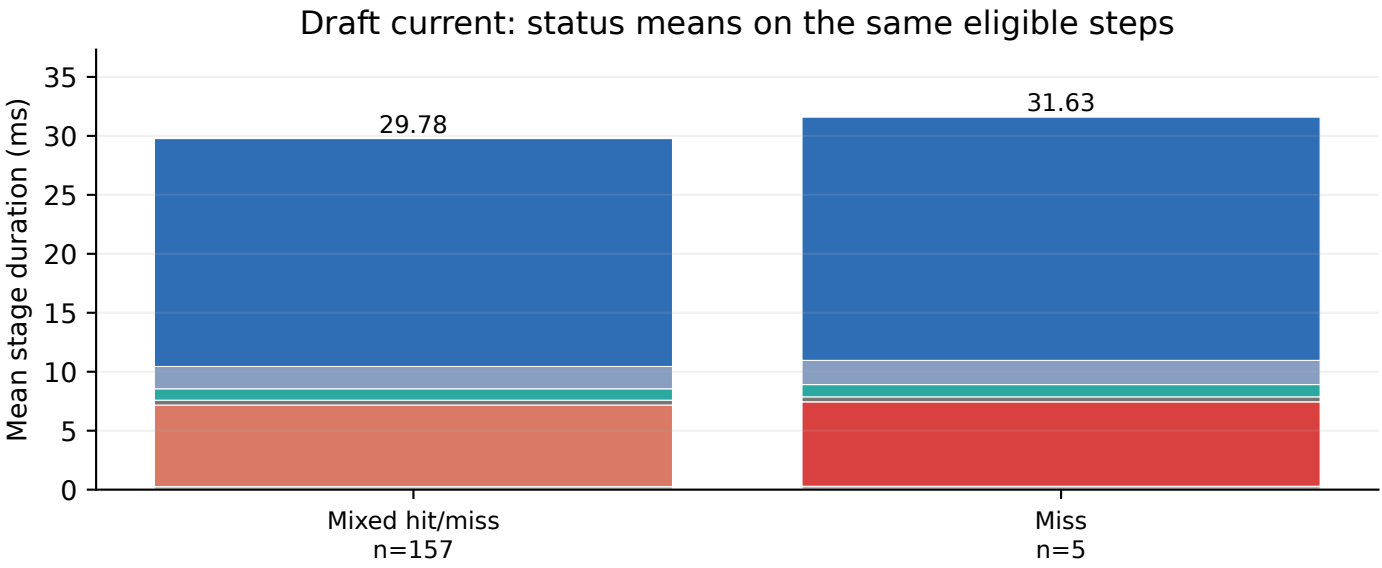
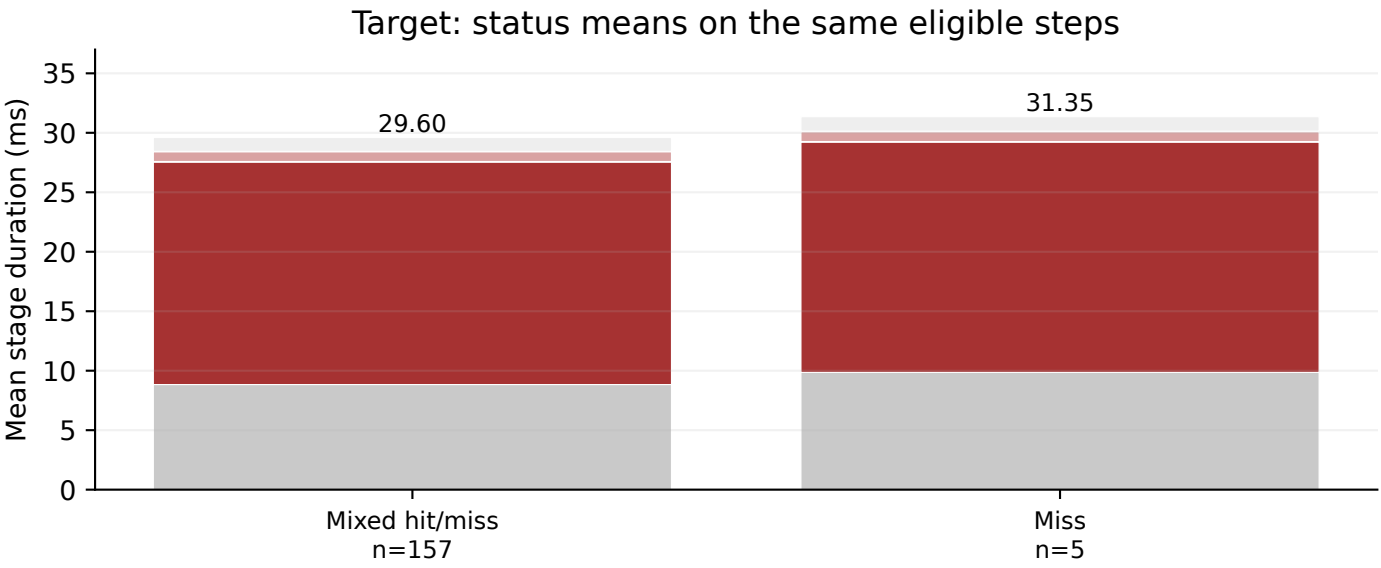
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



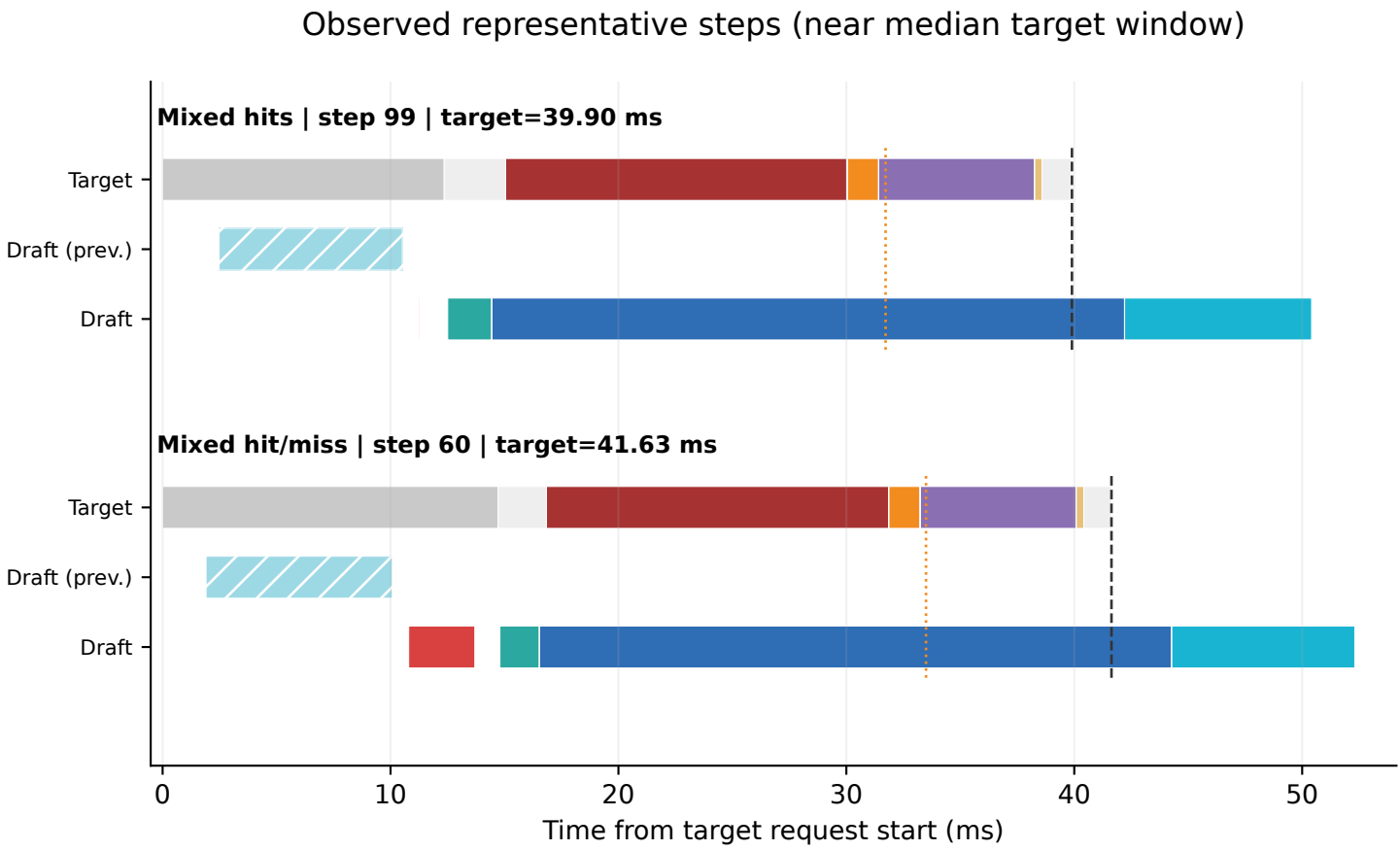
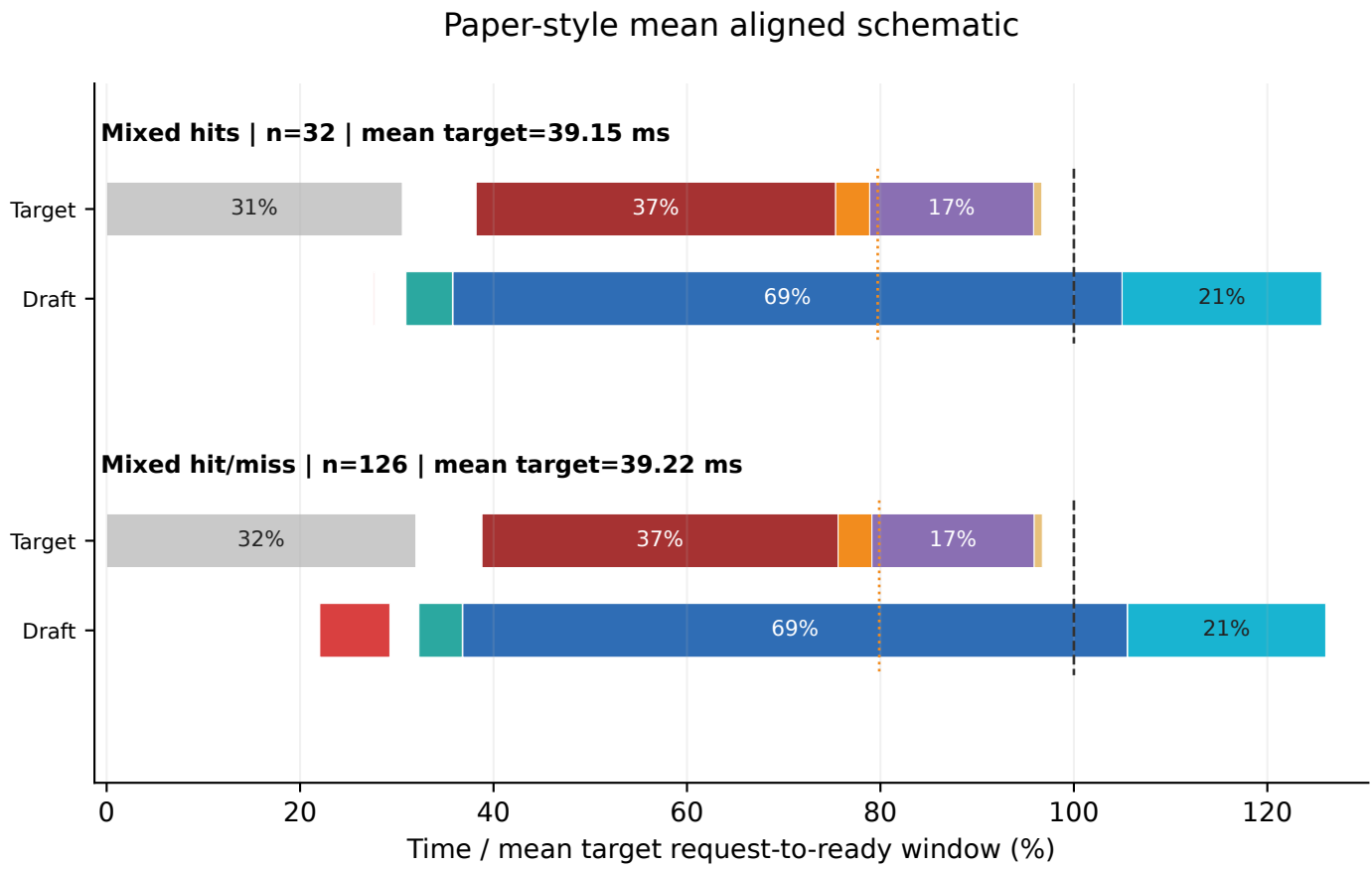
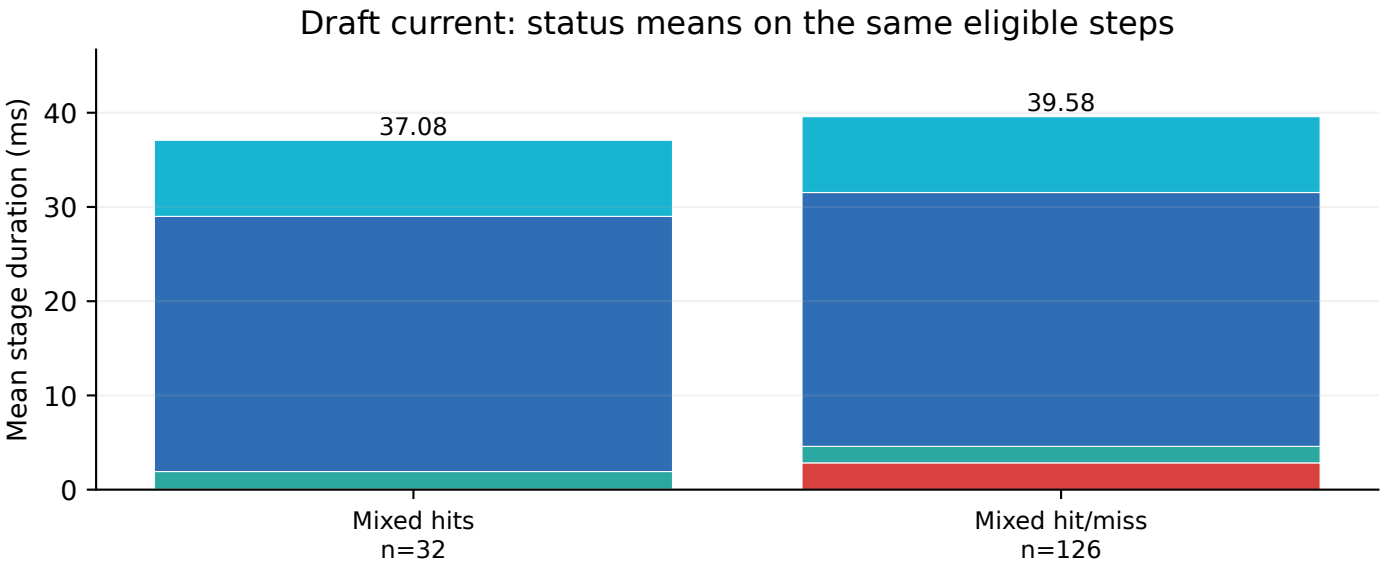
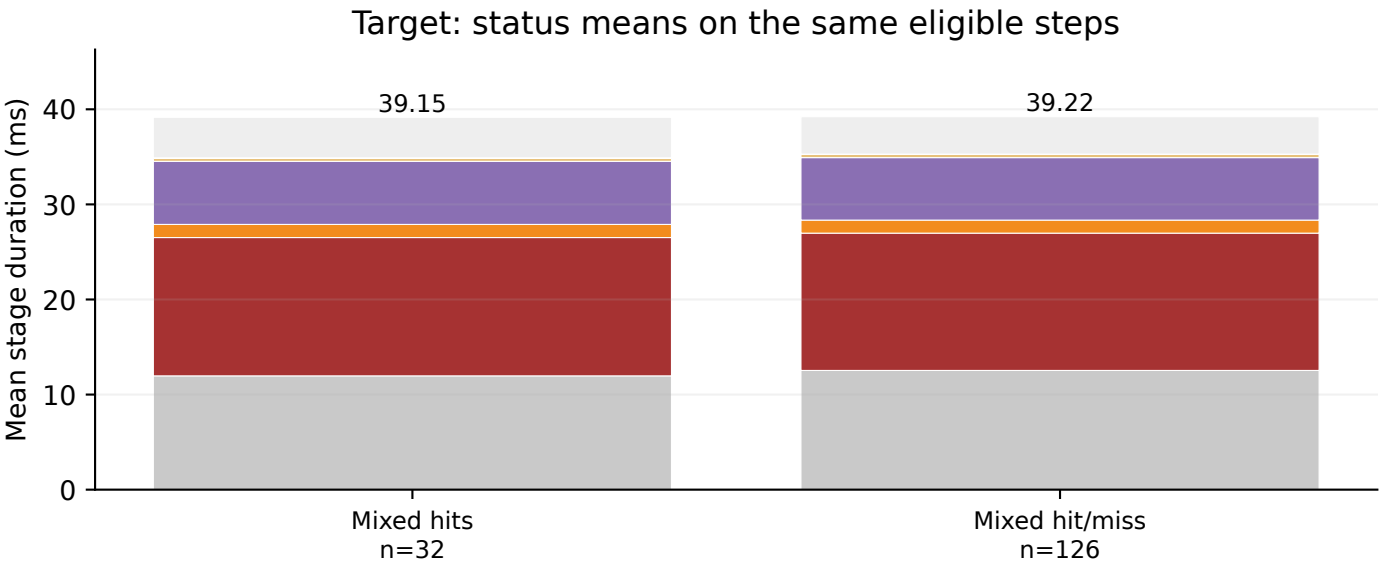
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.





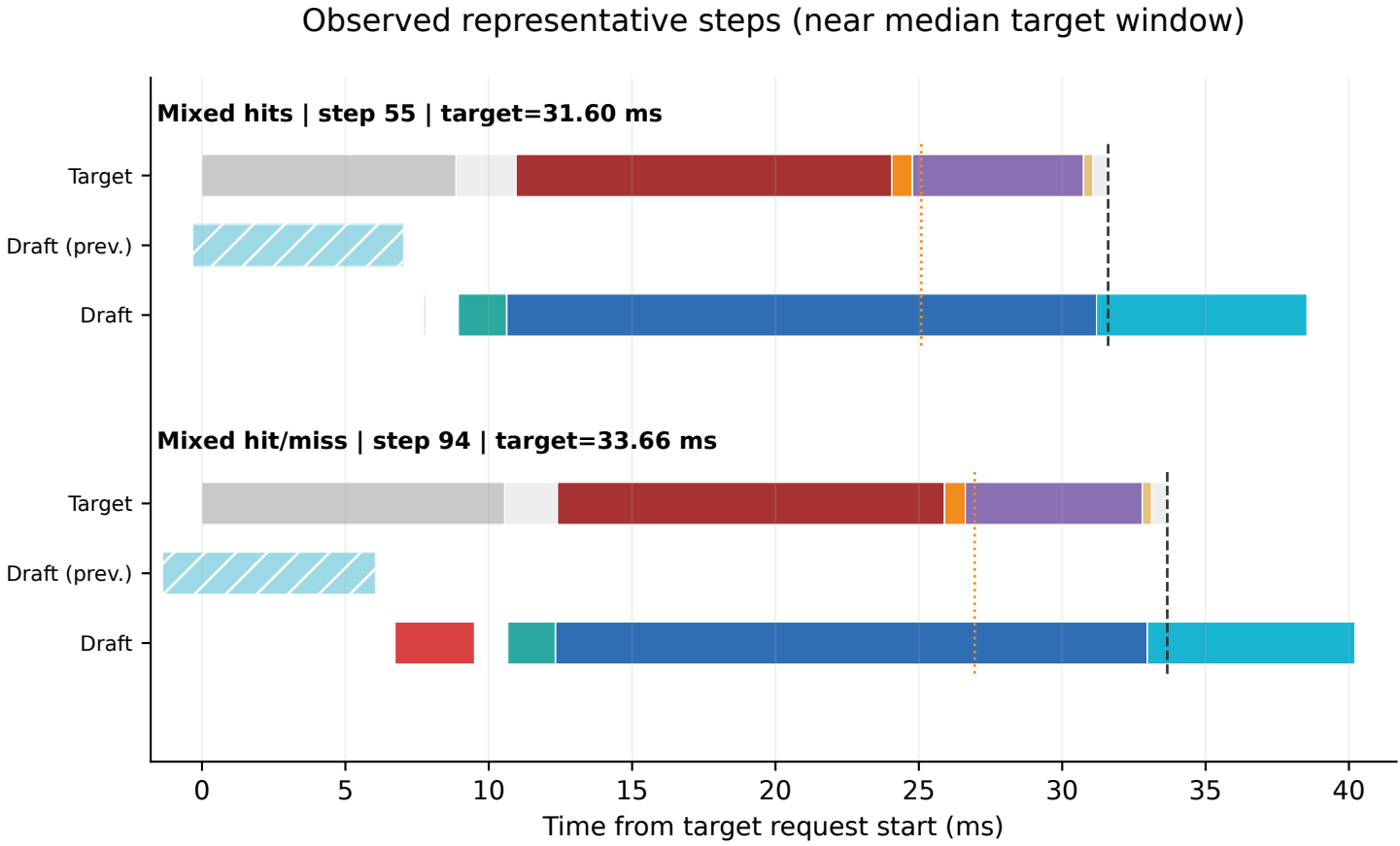
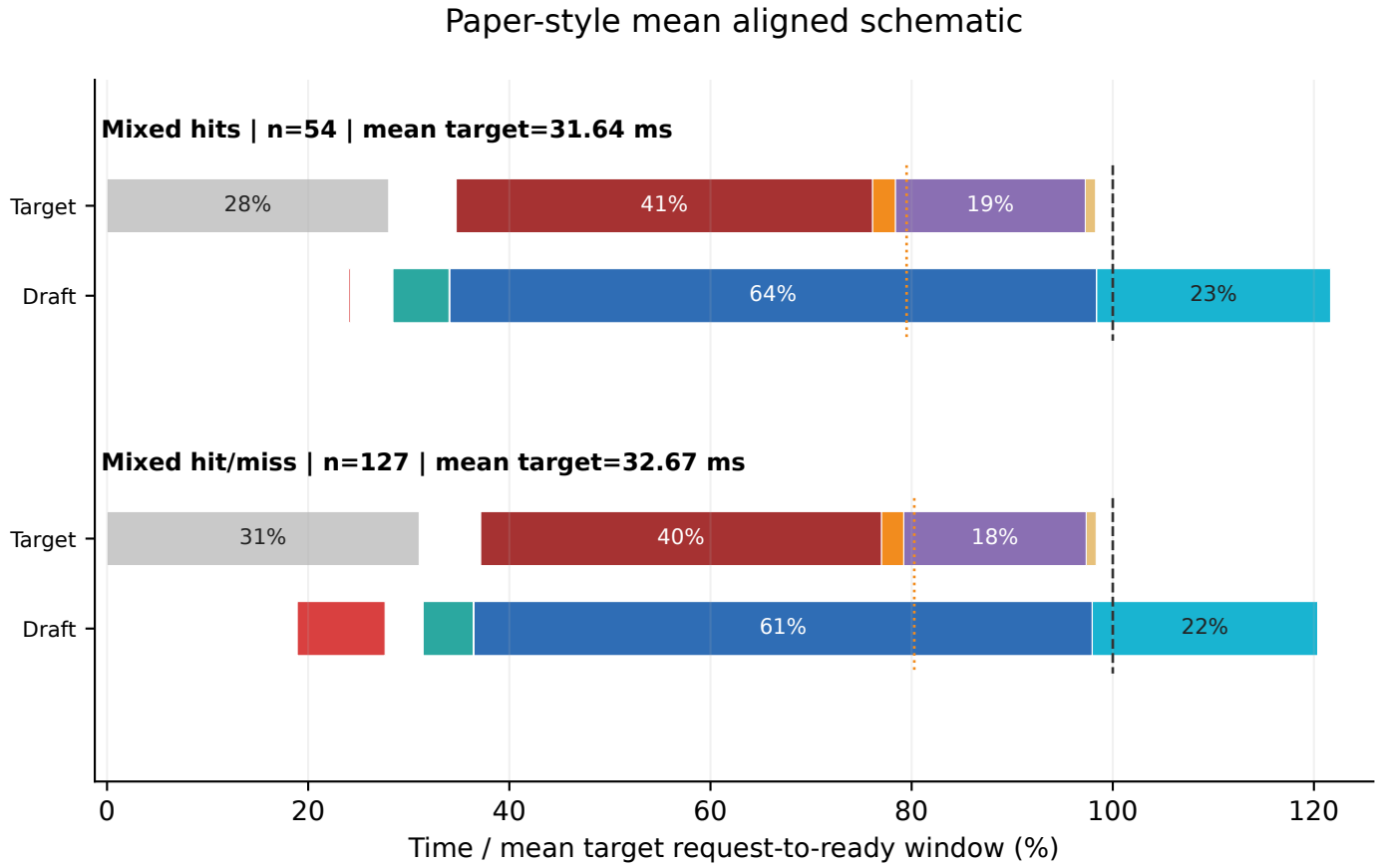
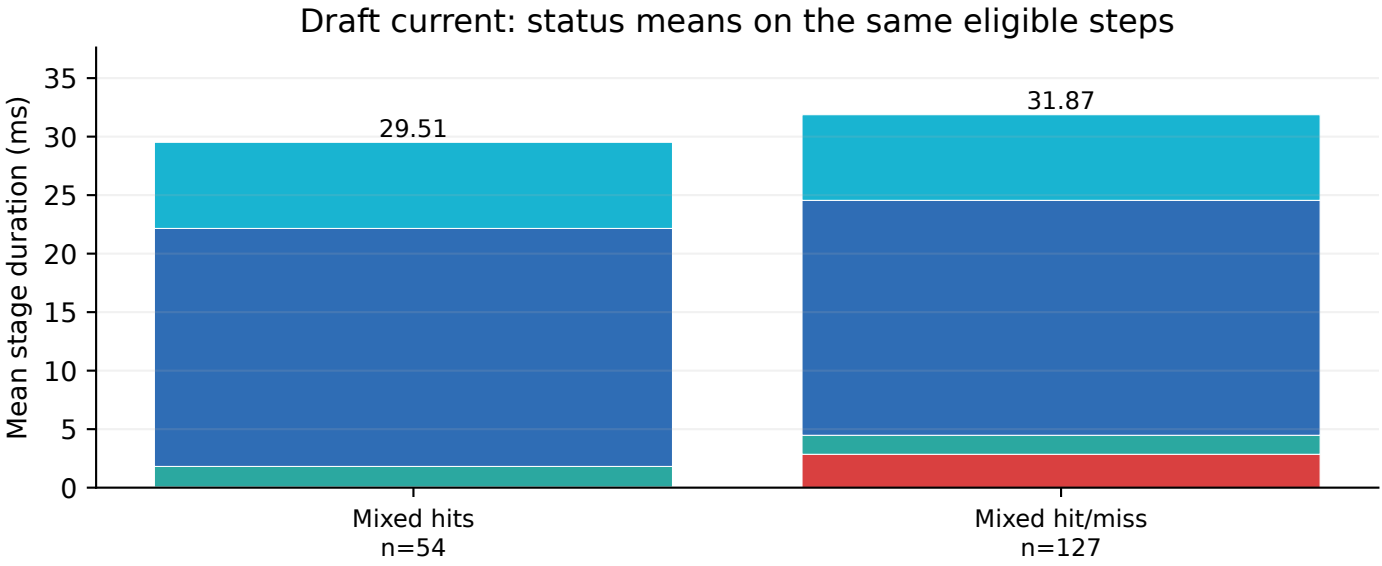
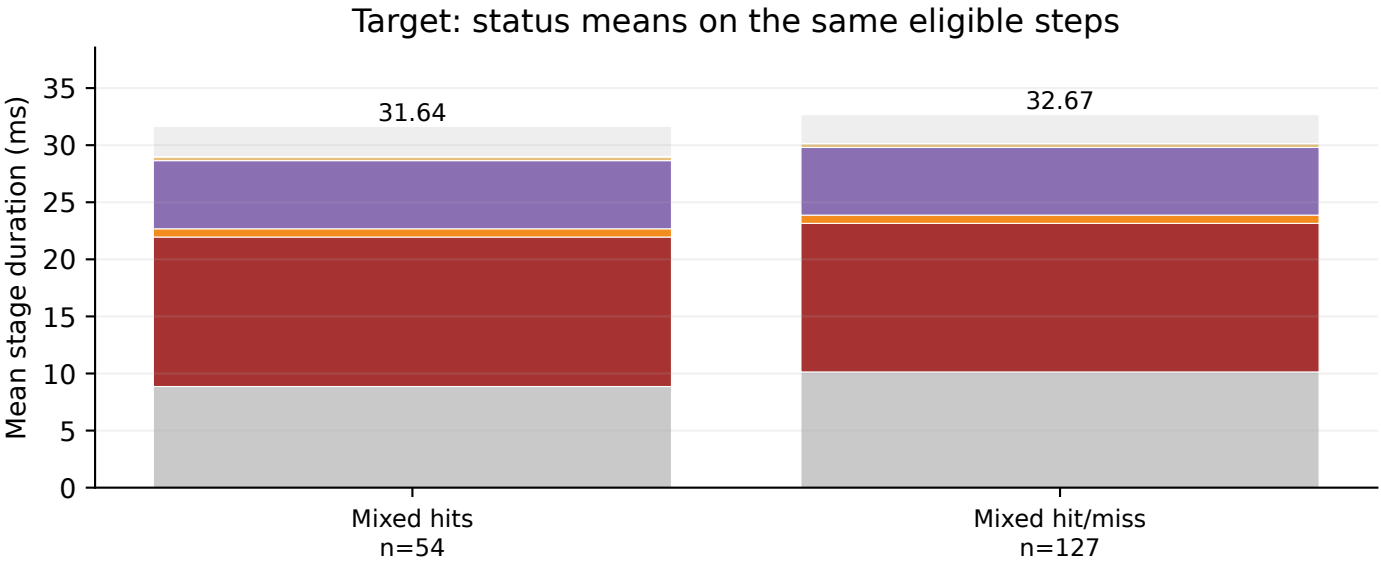
llama2_b8_tree_al_compute3
DUET: K1/K2=6/2, exit=21 (0-based), $\bar{C}$ =3, $\bar{N1}/\bar{N2}$ =12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 158 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

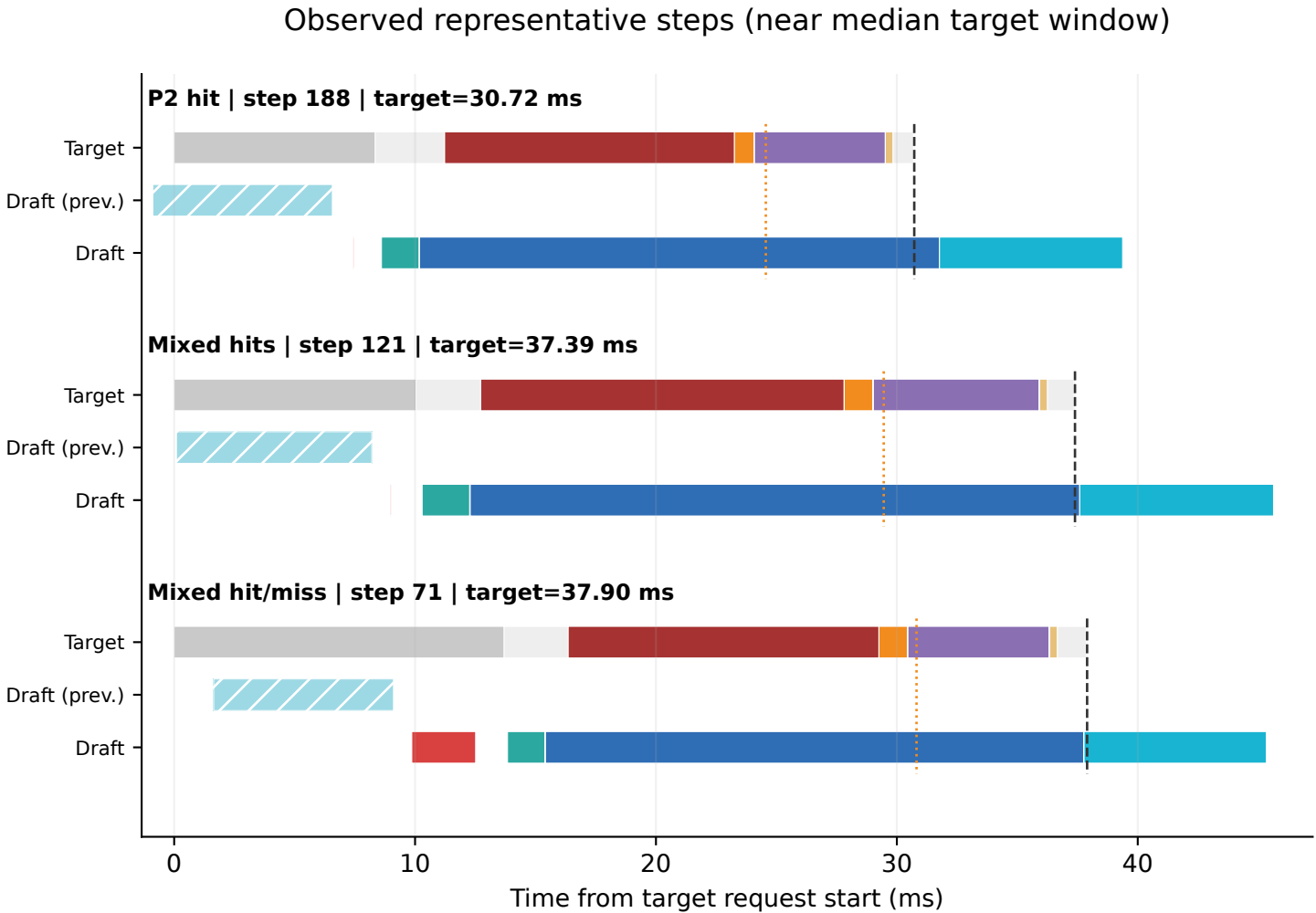
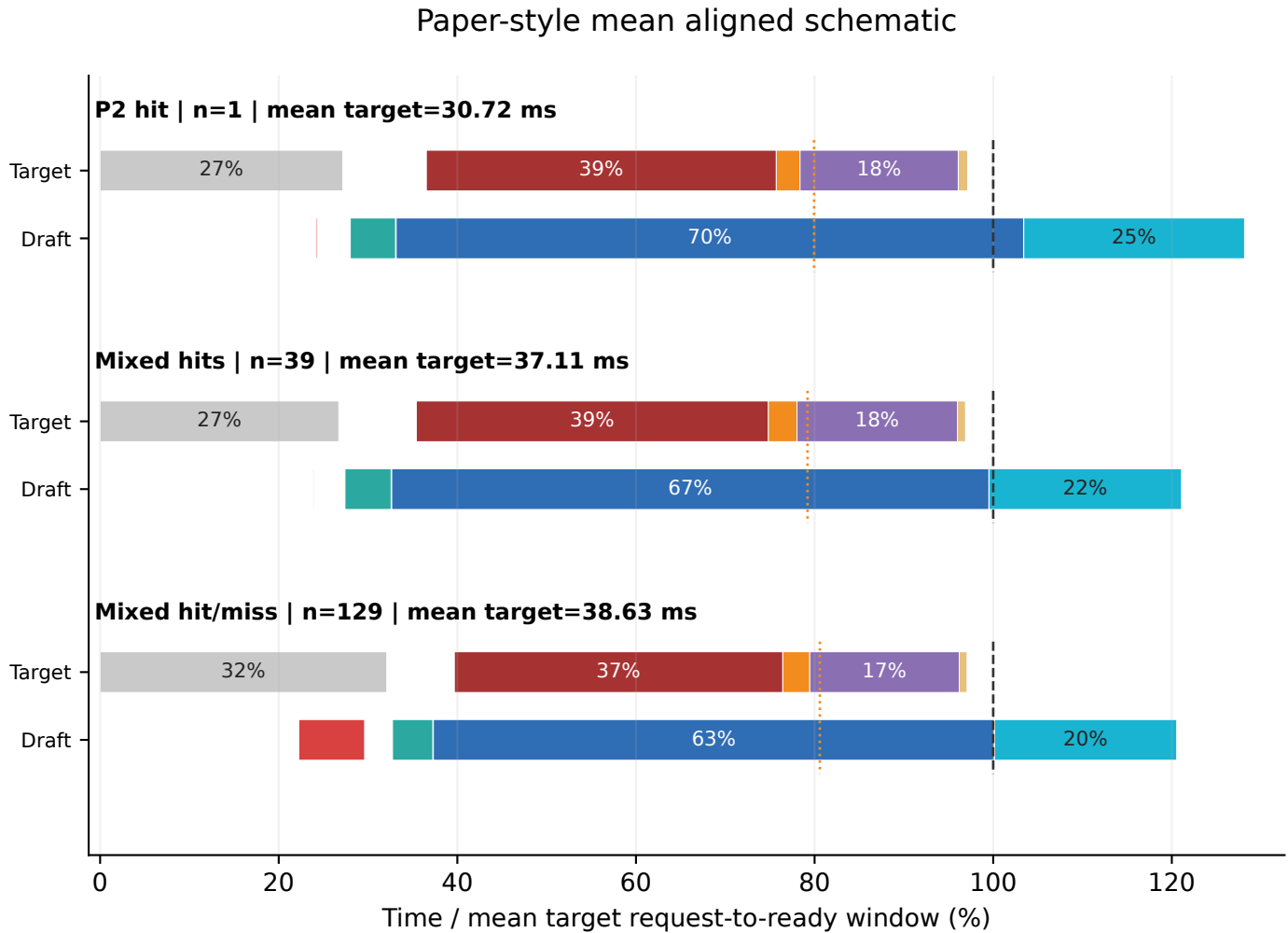
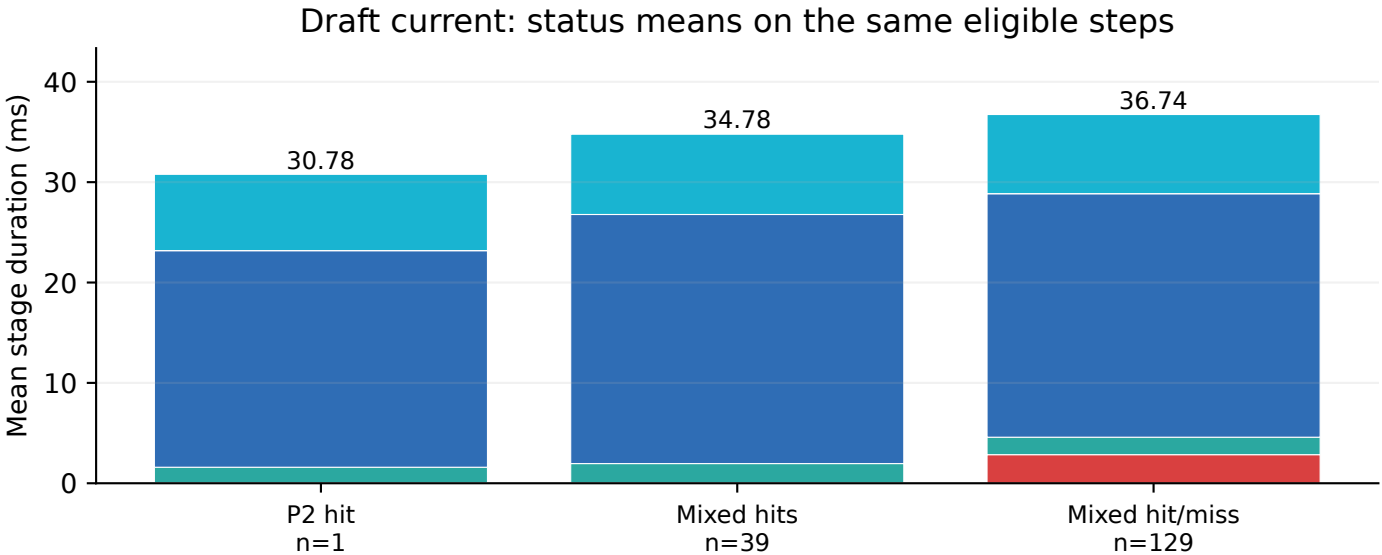
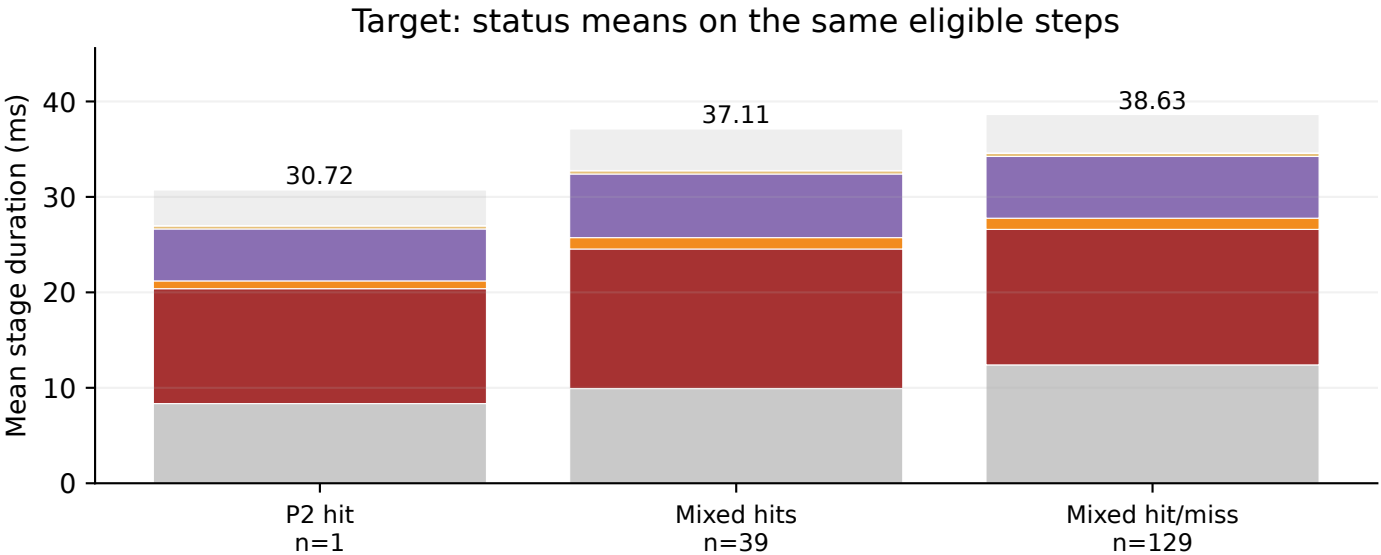
llama2_b8_tree_al_width1
DUET: K1/K2=6/2, exit=21 (0-based), C=1, N1/N2=6/2, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 181 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

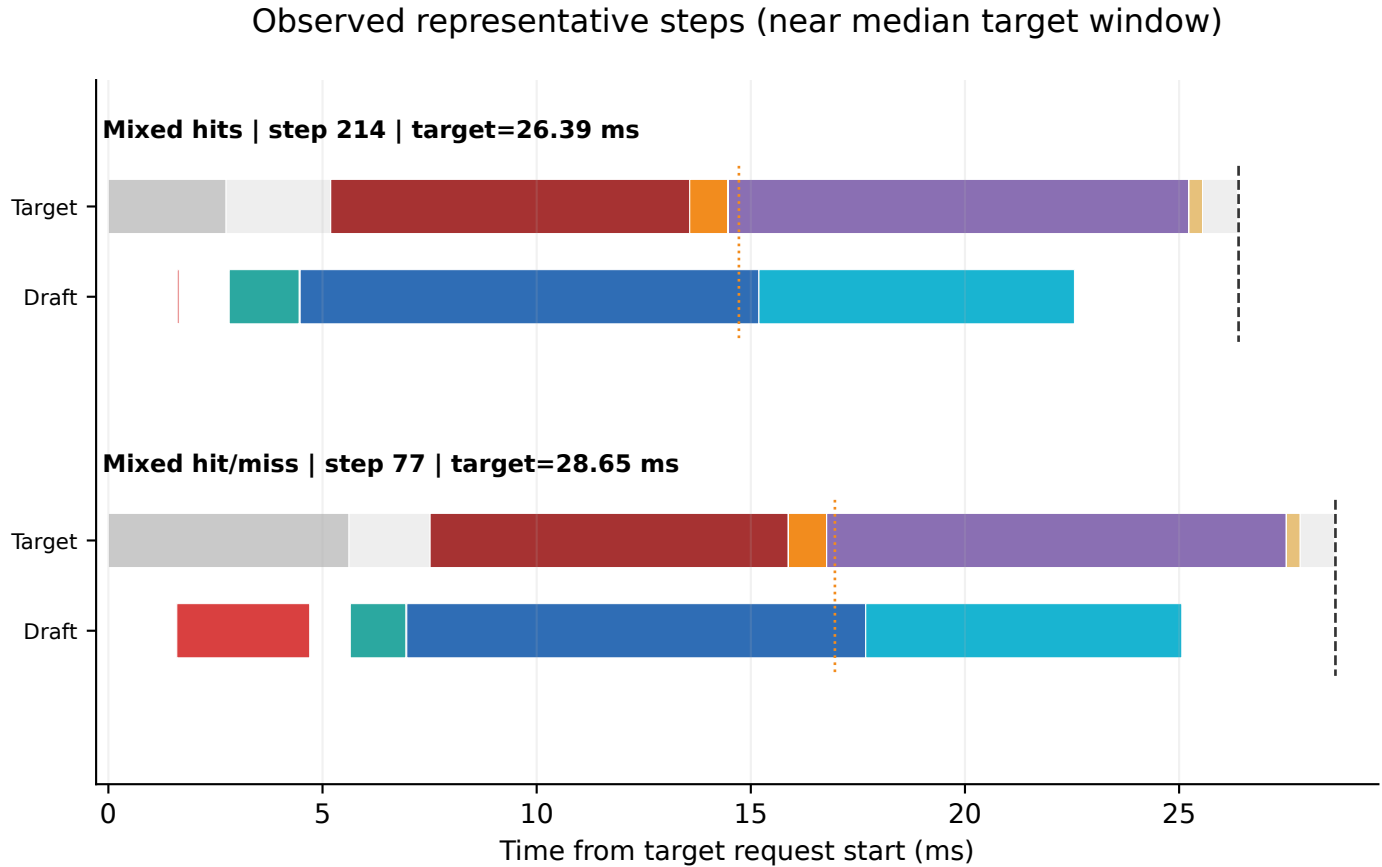
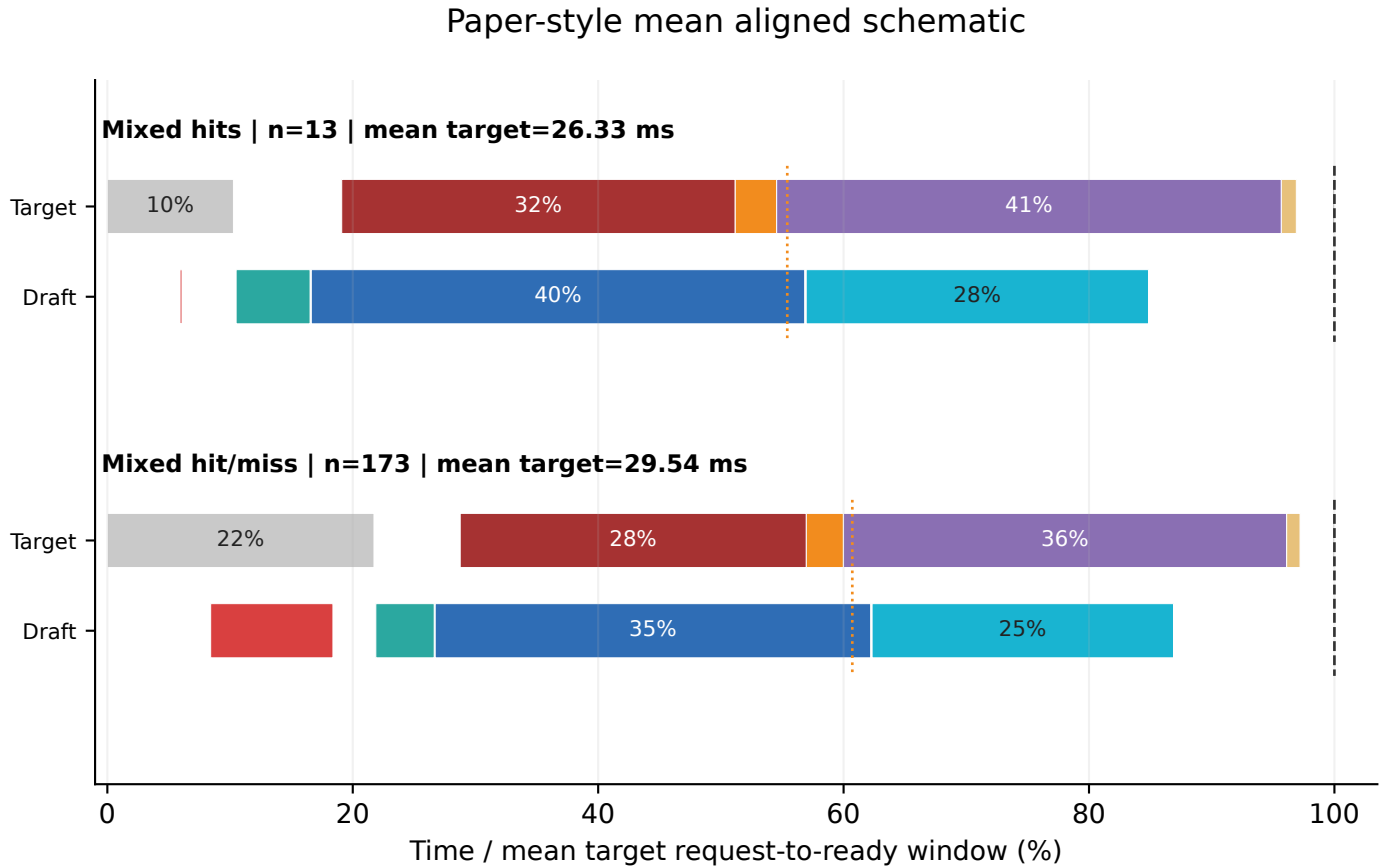
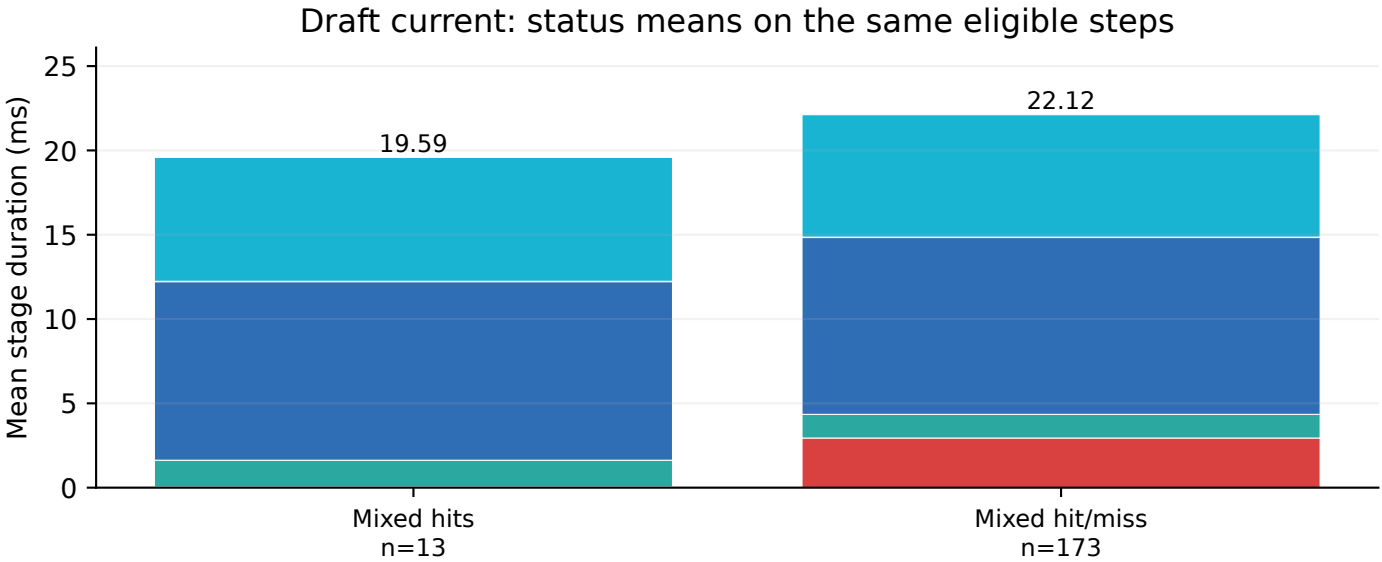
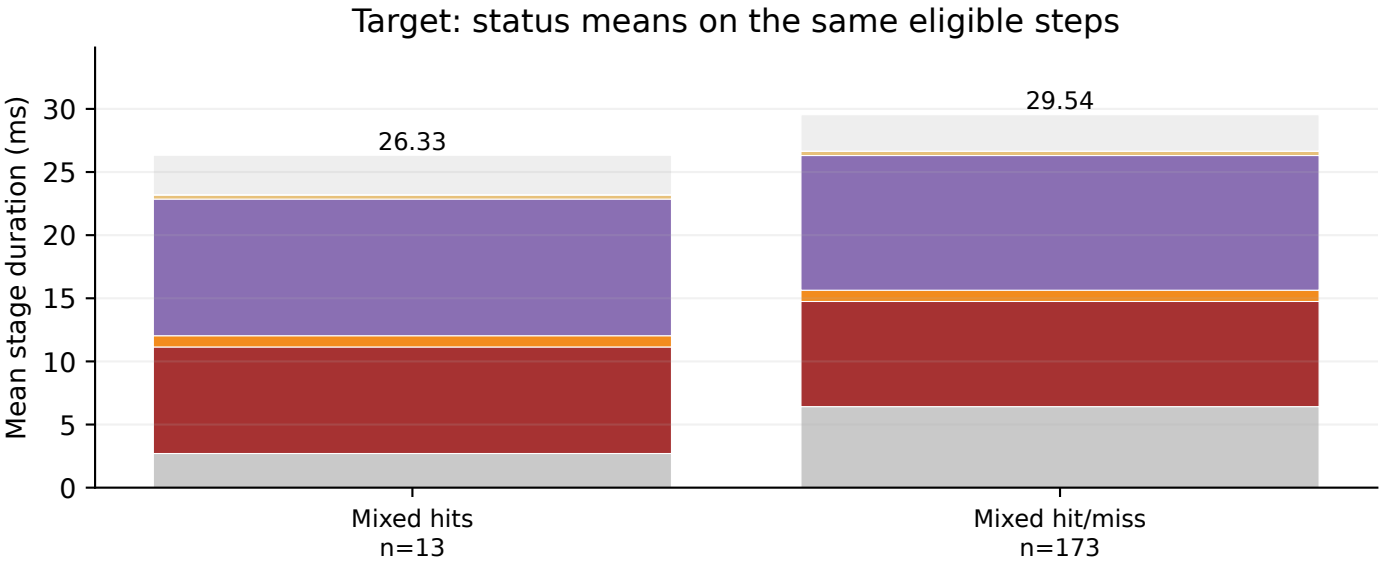
llama2_b8_tree_al_width2
DUET: K1/K2=6/2, exit=21 (0-based), C=2, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 169 eligible steps



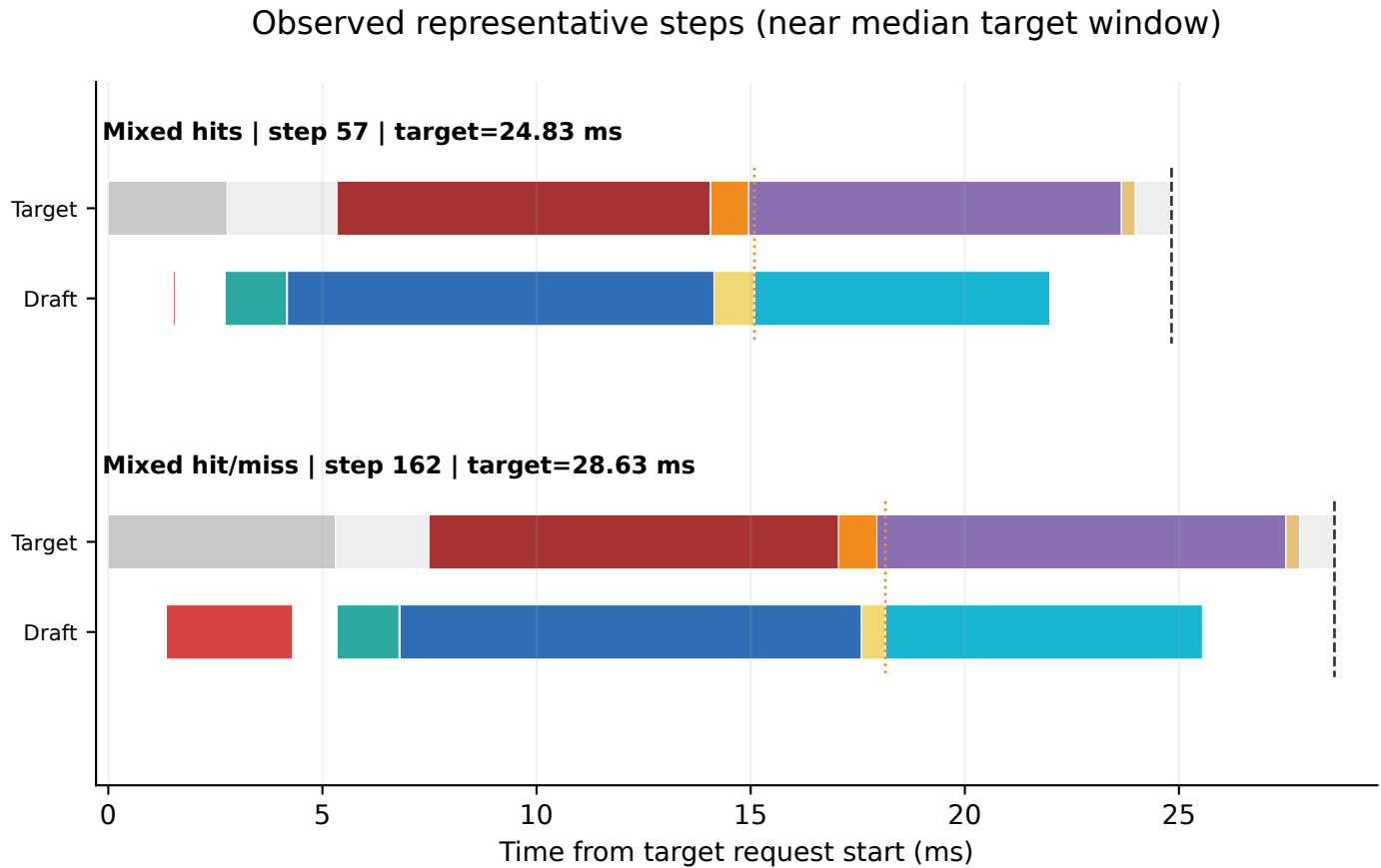
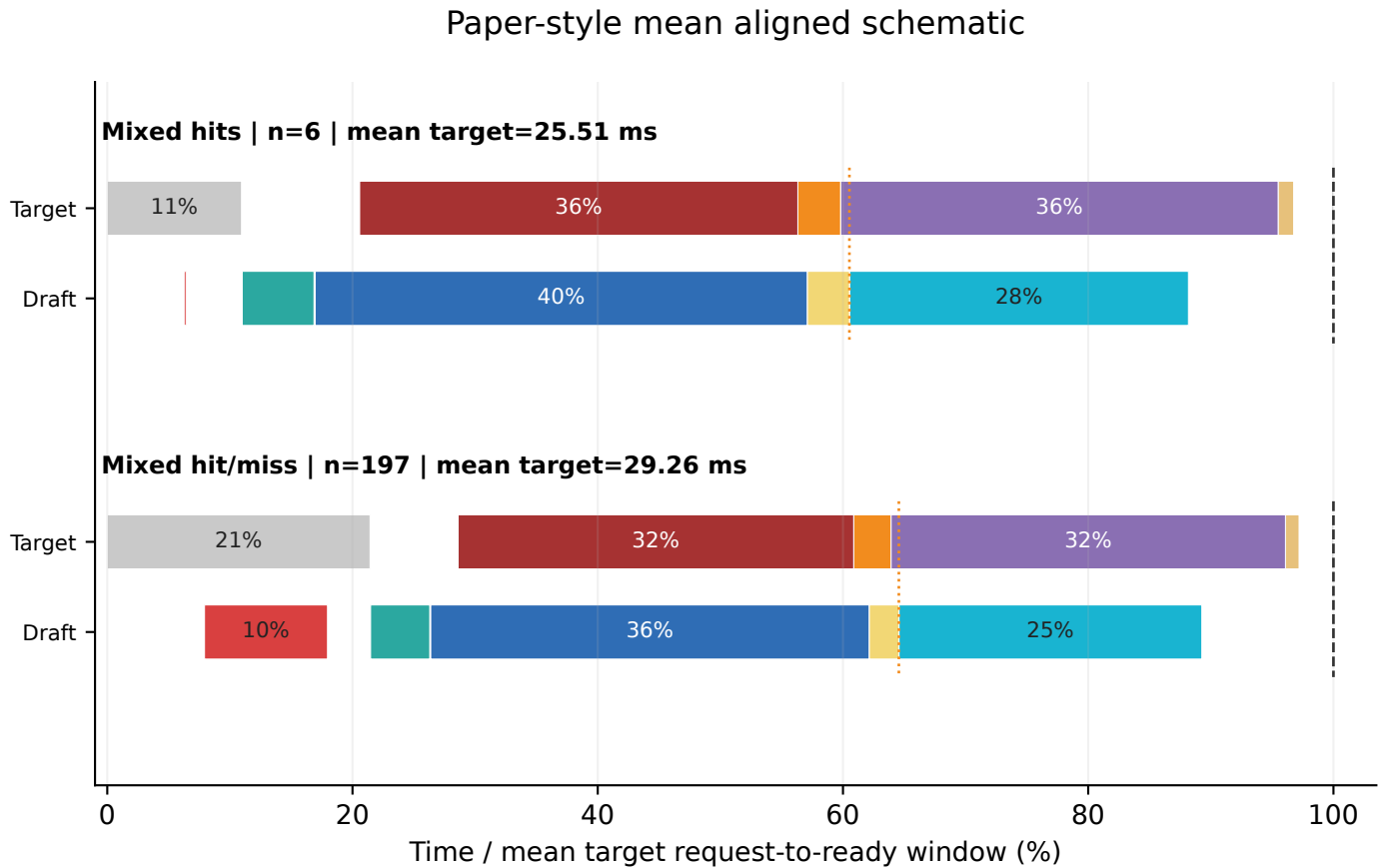
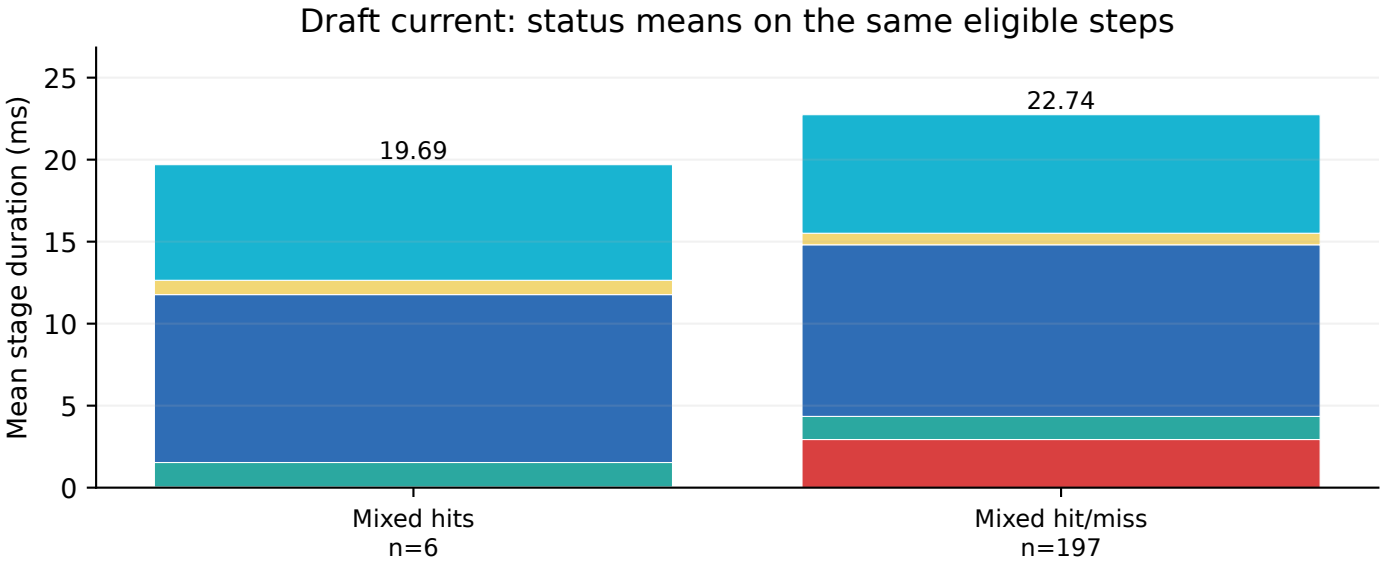
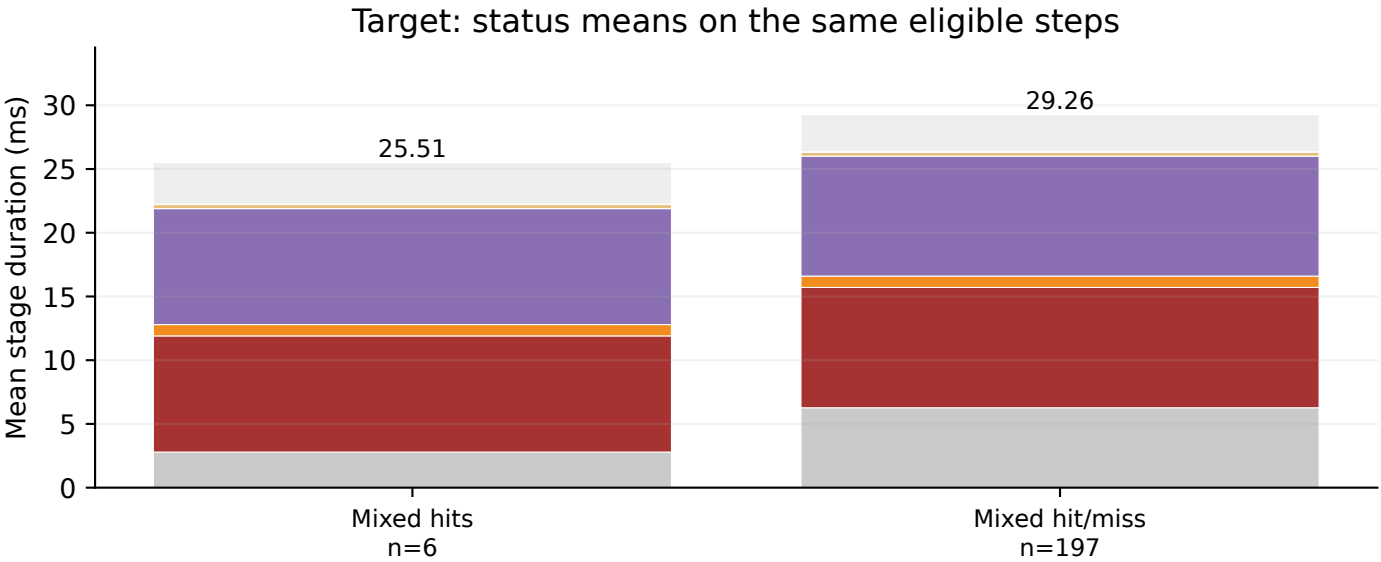
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

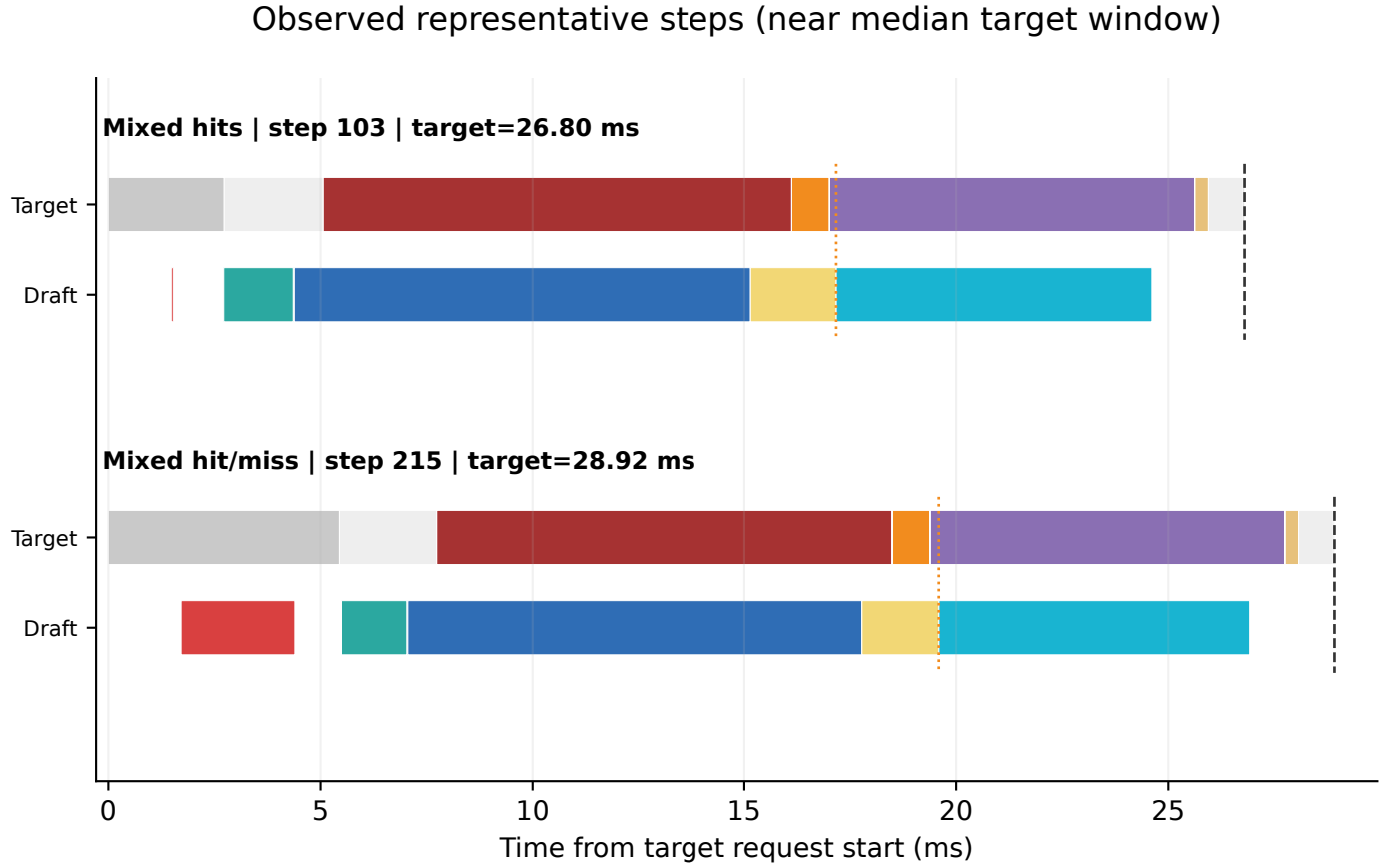
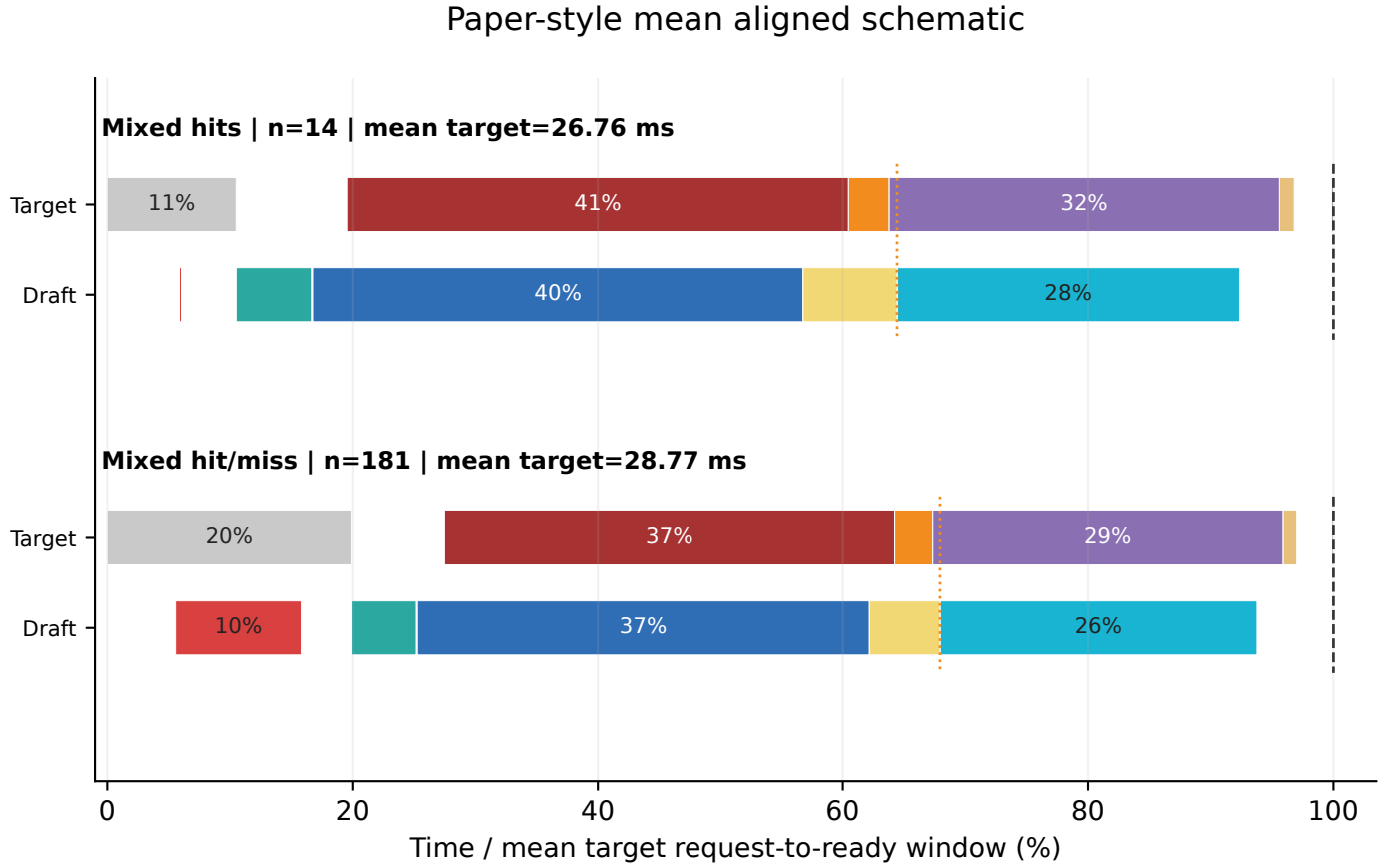
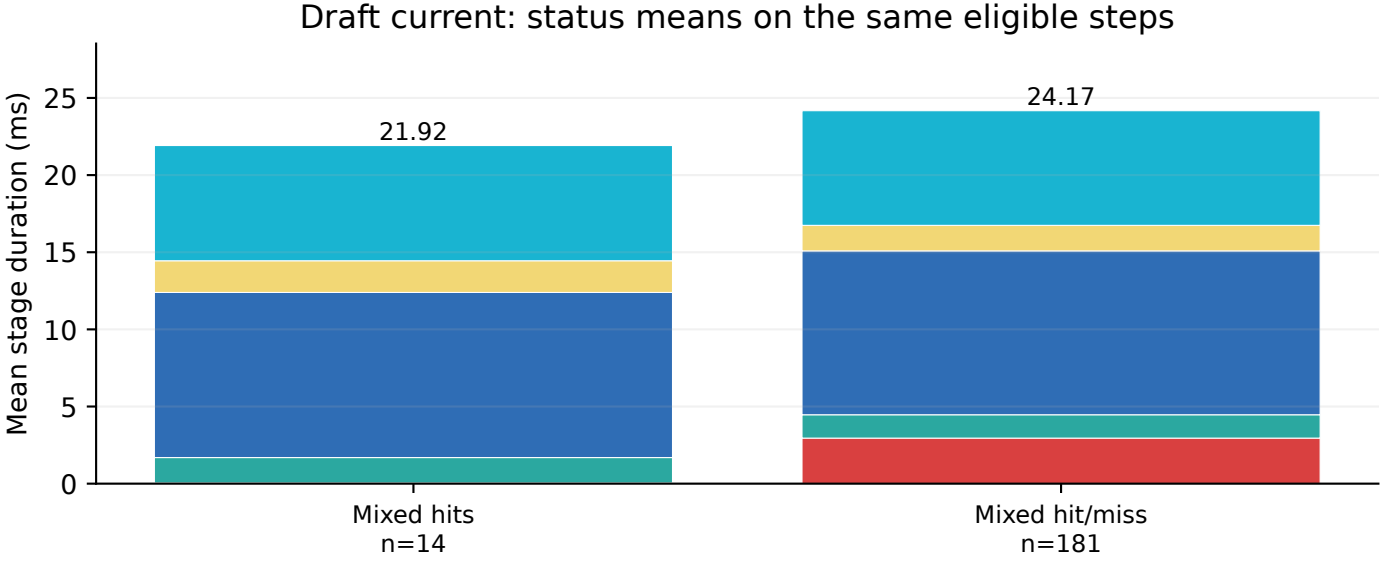
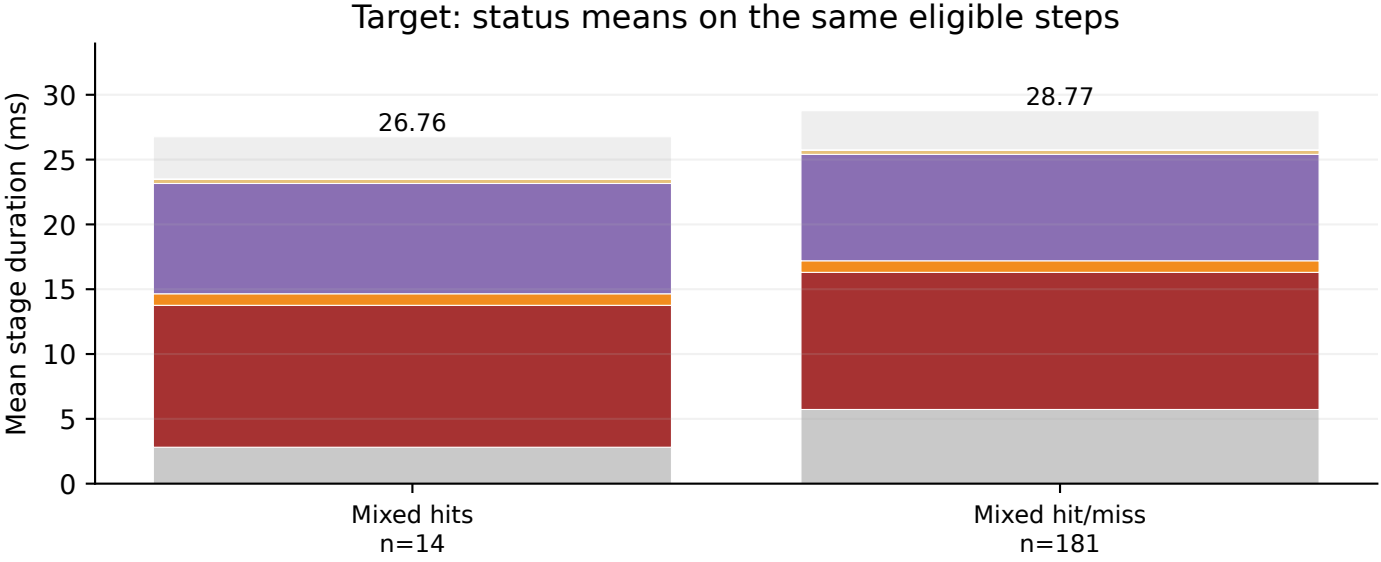
llama2_b8_tree_align_exit13
DUET: K1/K2=3/2, exit=13 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 186 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

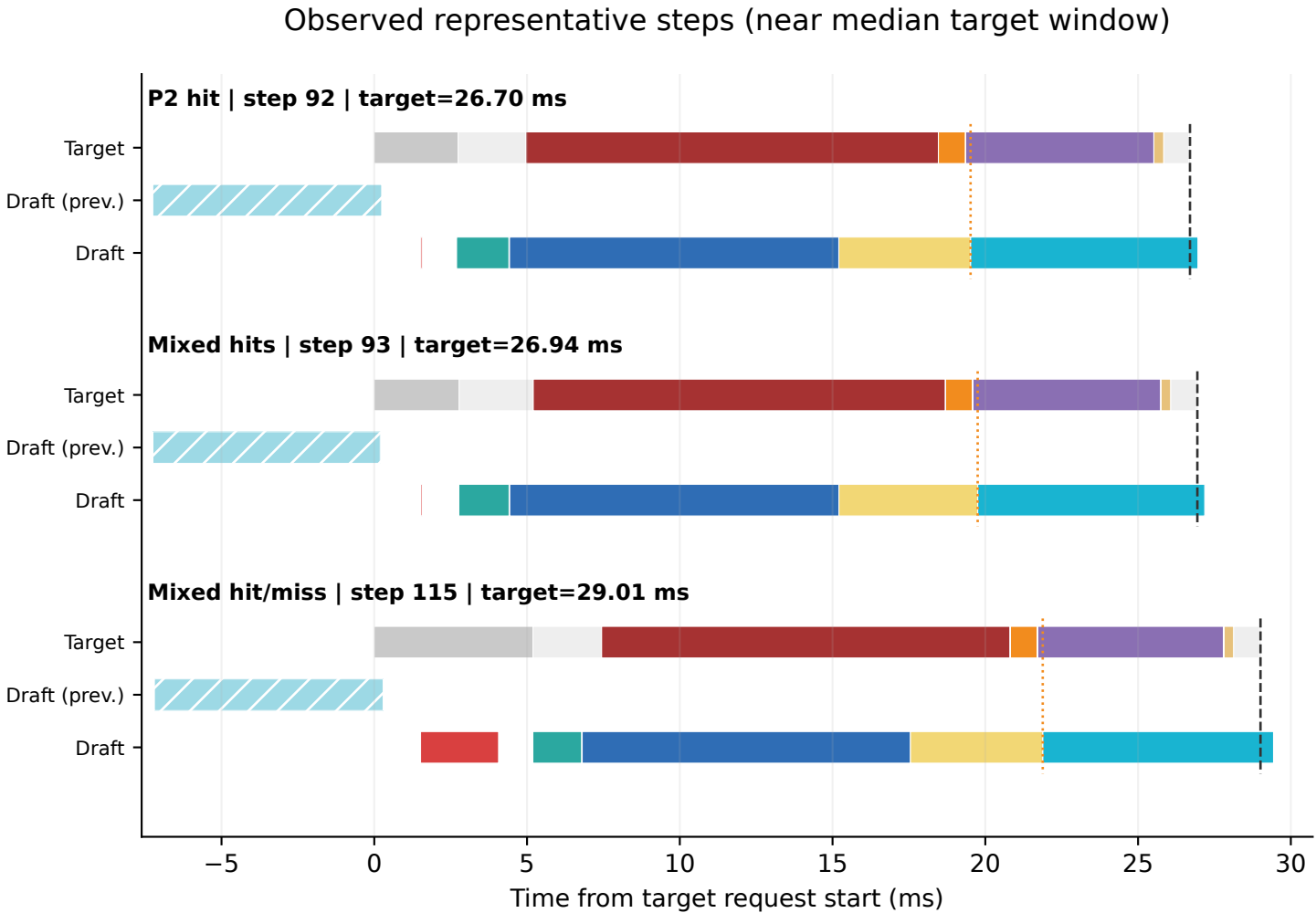
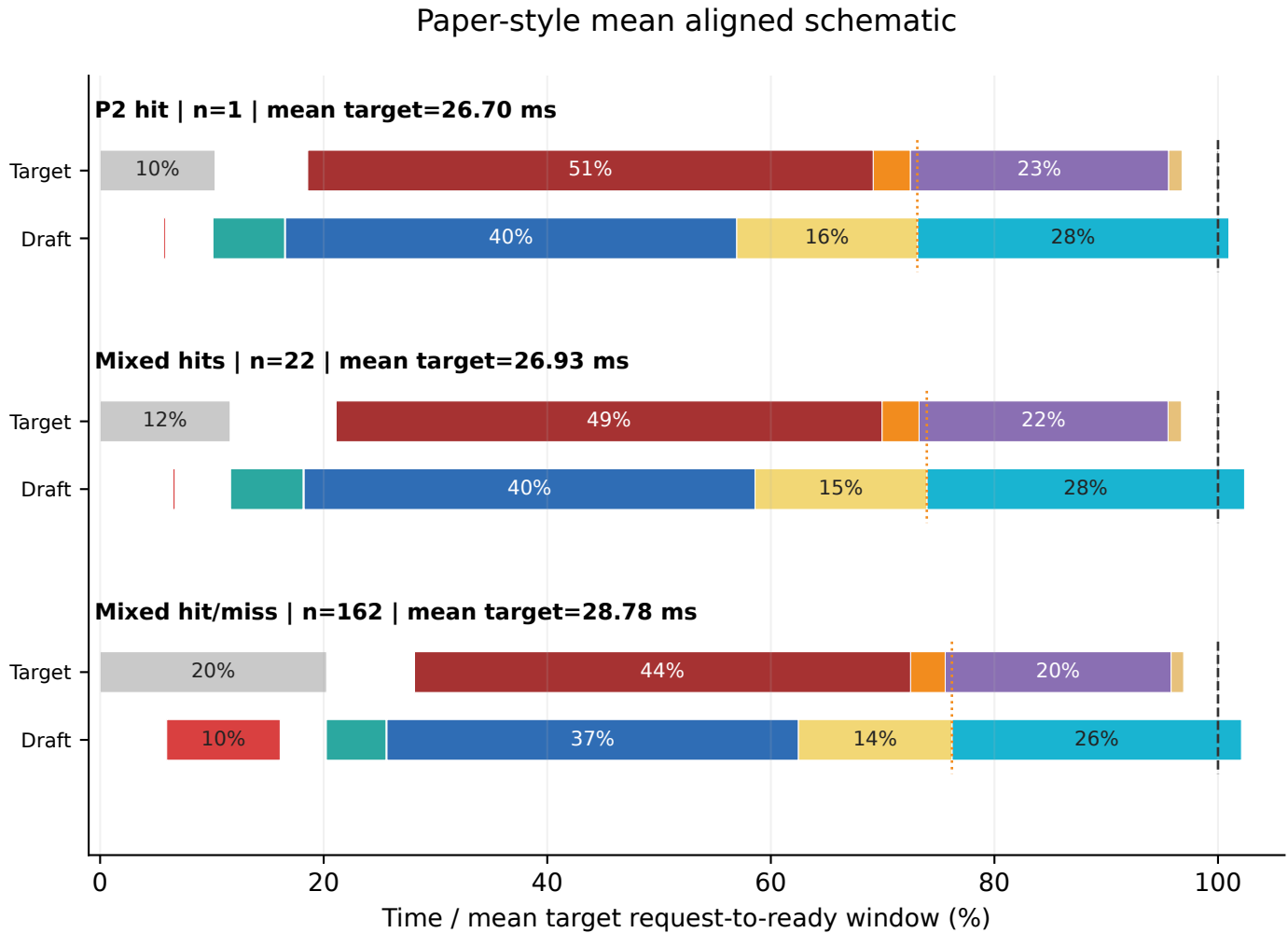
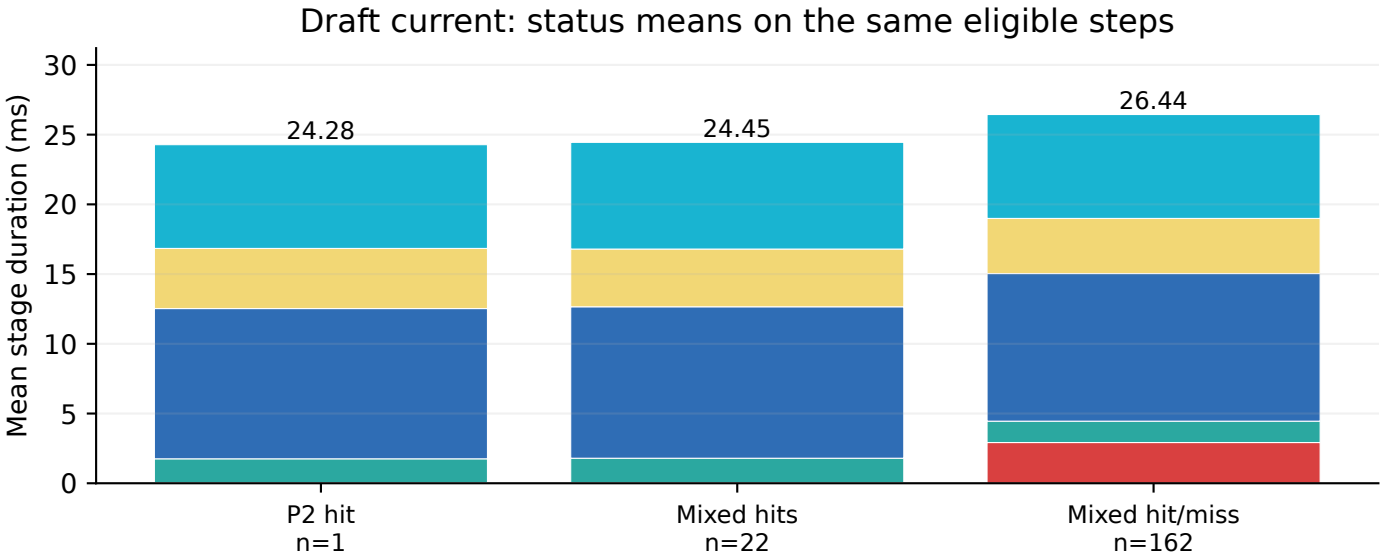
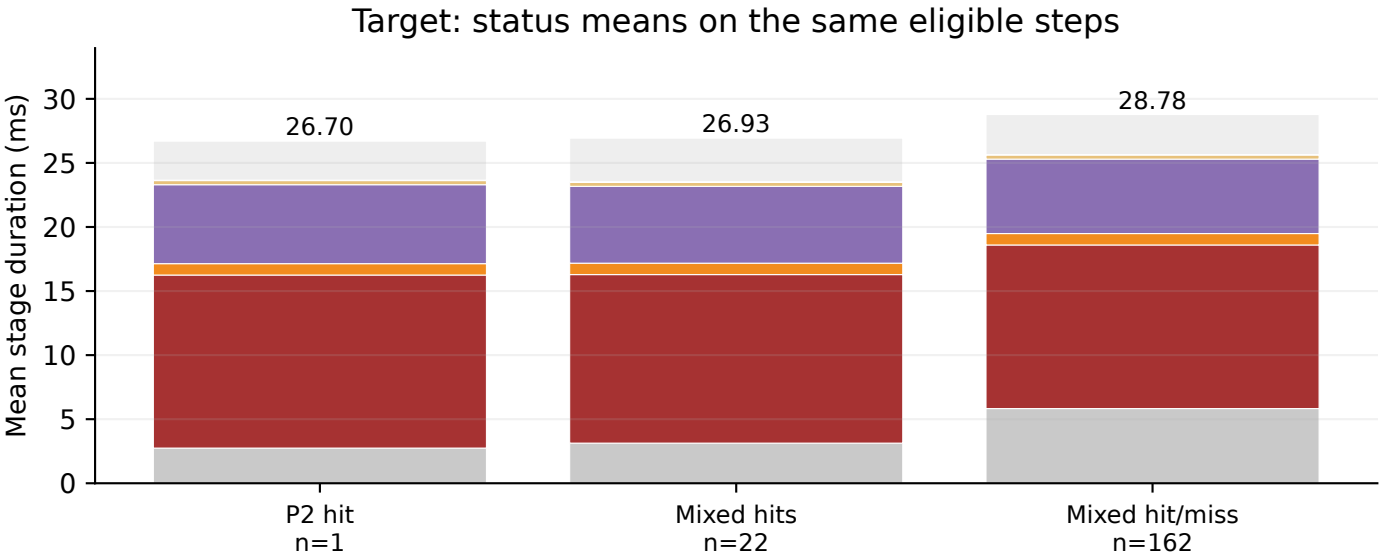


llama2_b8_tree_align_exit17
DUET: K1/K2=3/2, exit=17 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 195 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

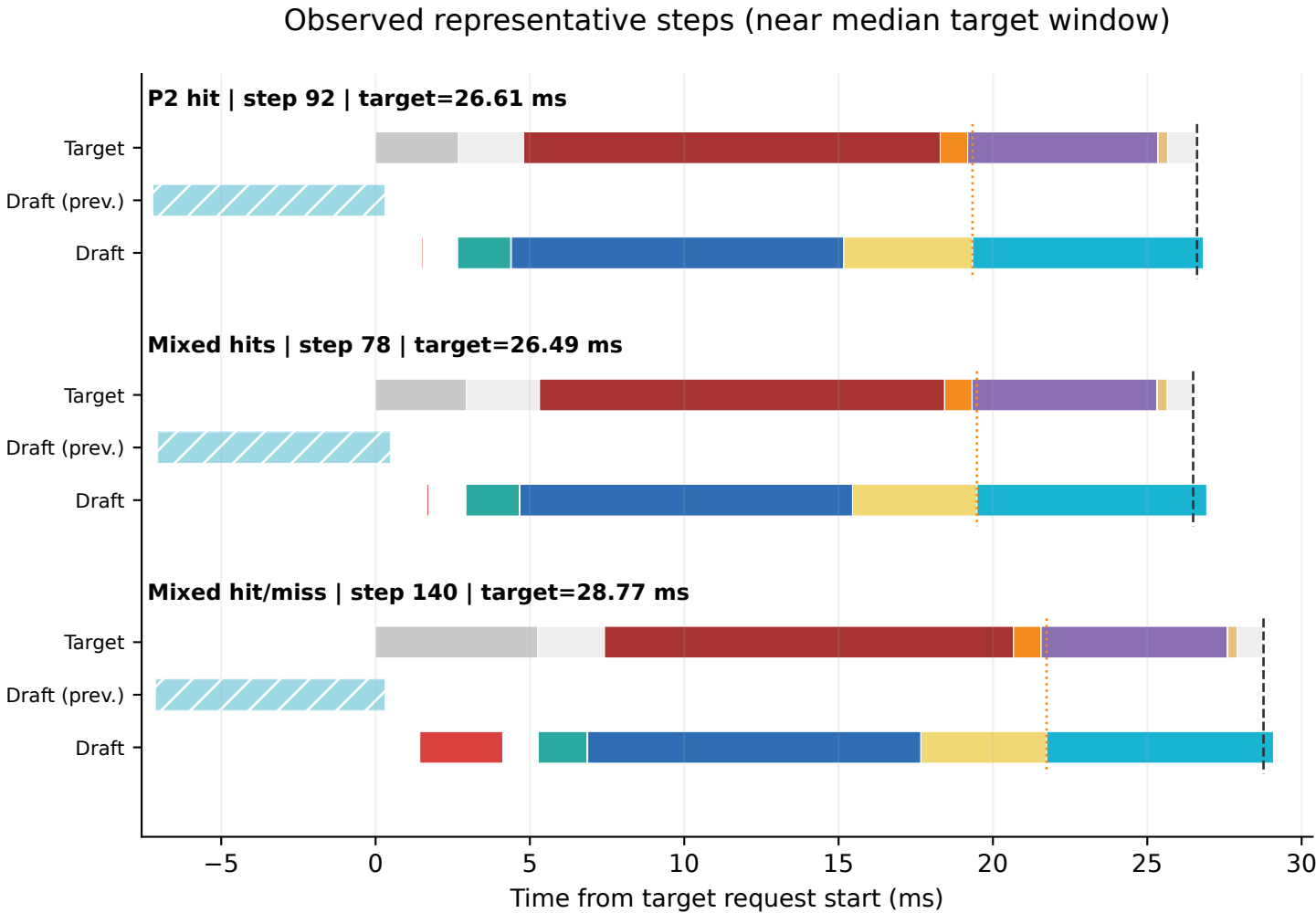
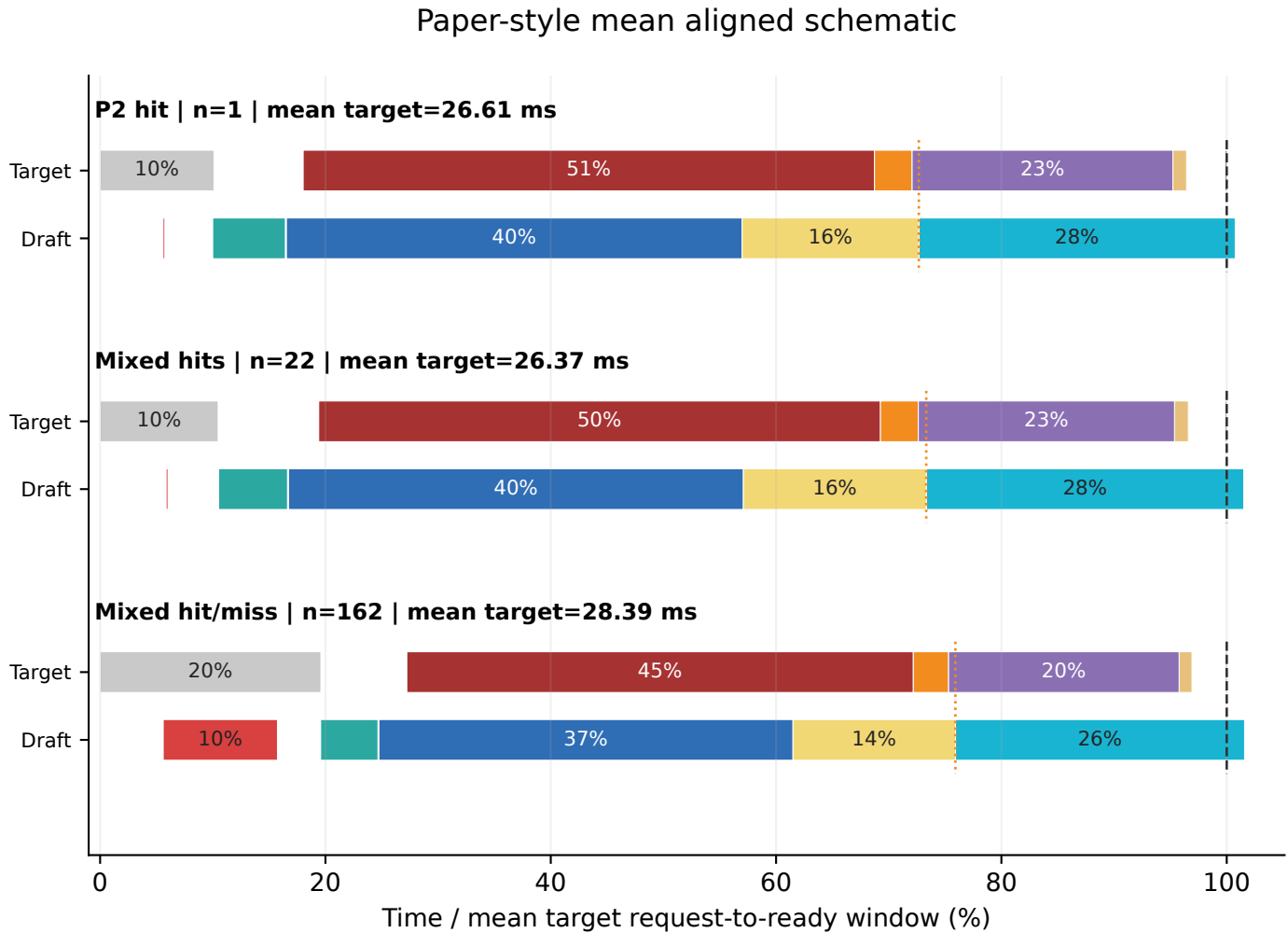
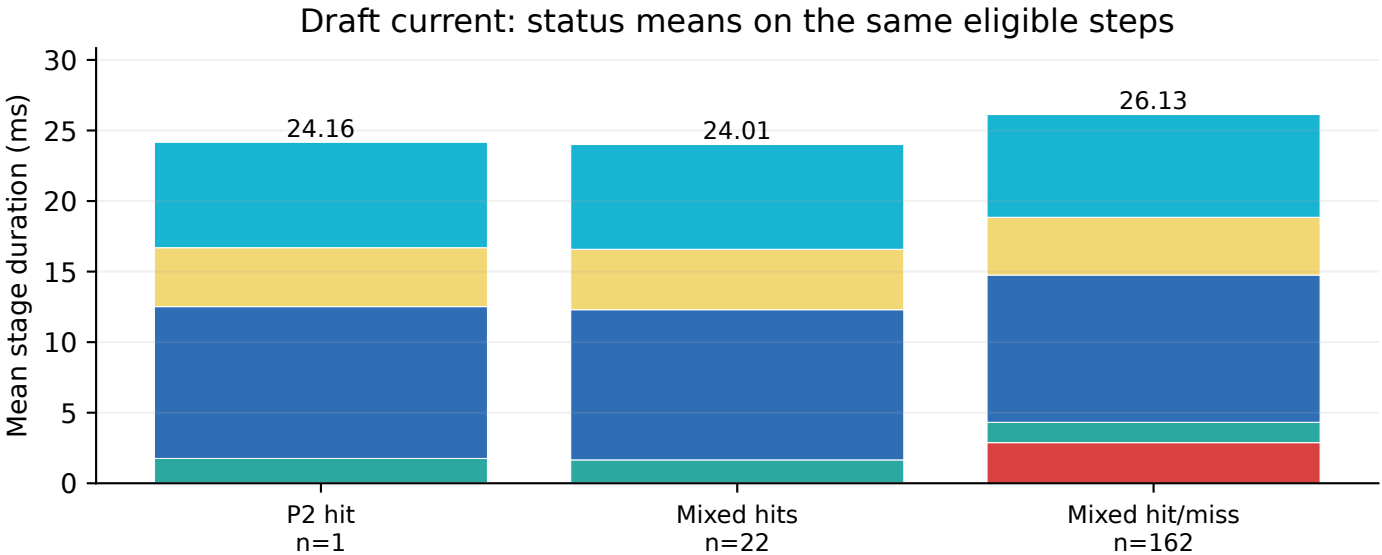
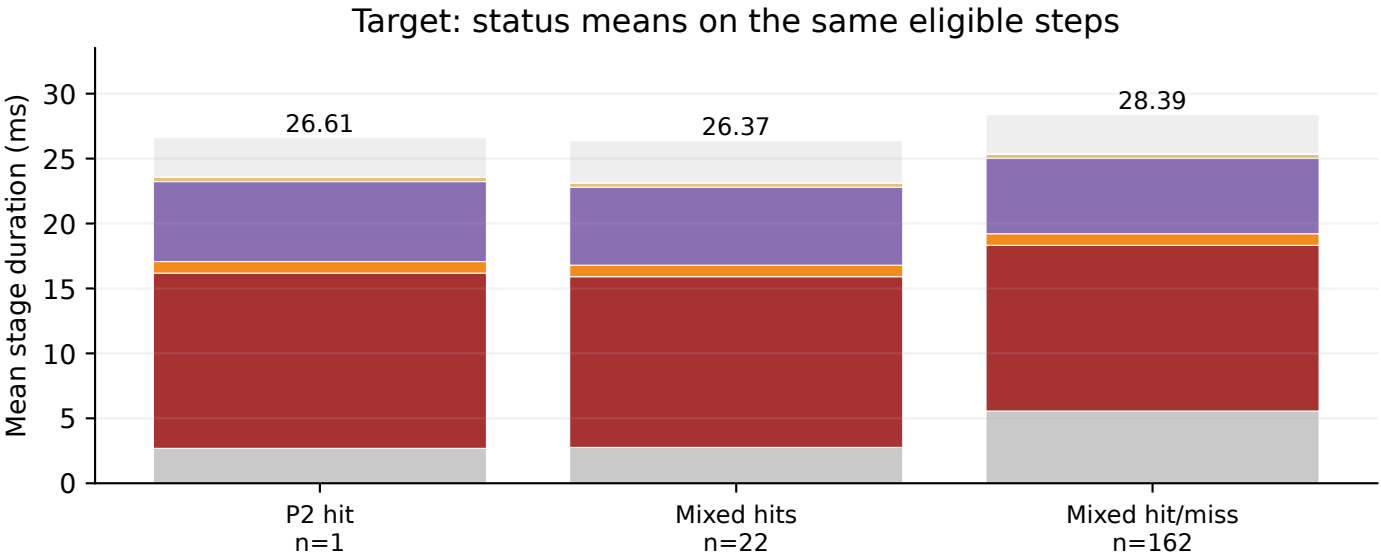
llama2_b8_tree_beta0
DUET: K1/K2=3/2, exit=21 (0-based), $\bar{C}=3$, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 185 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

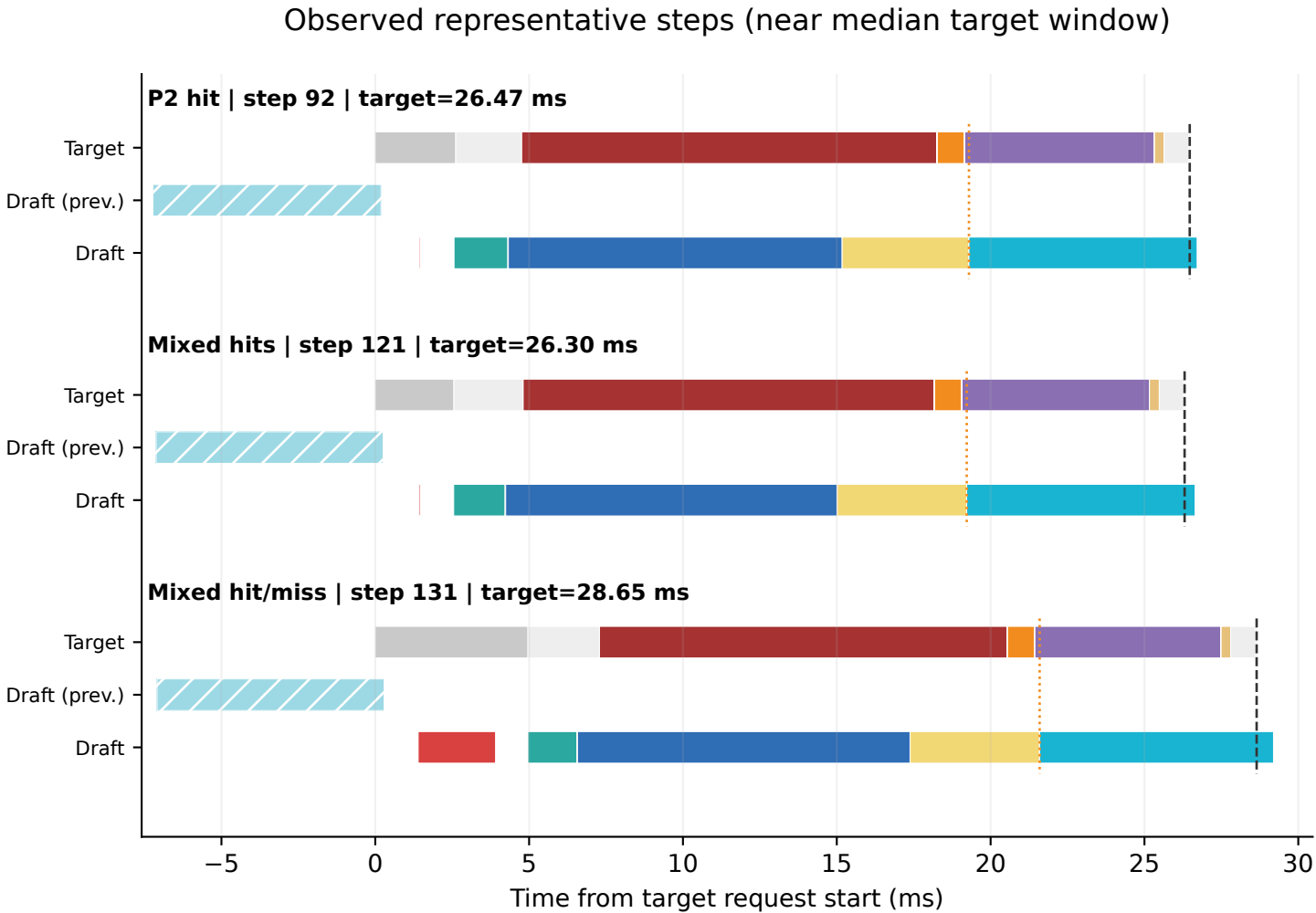
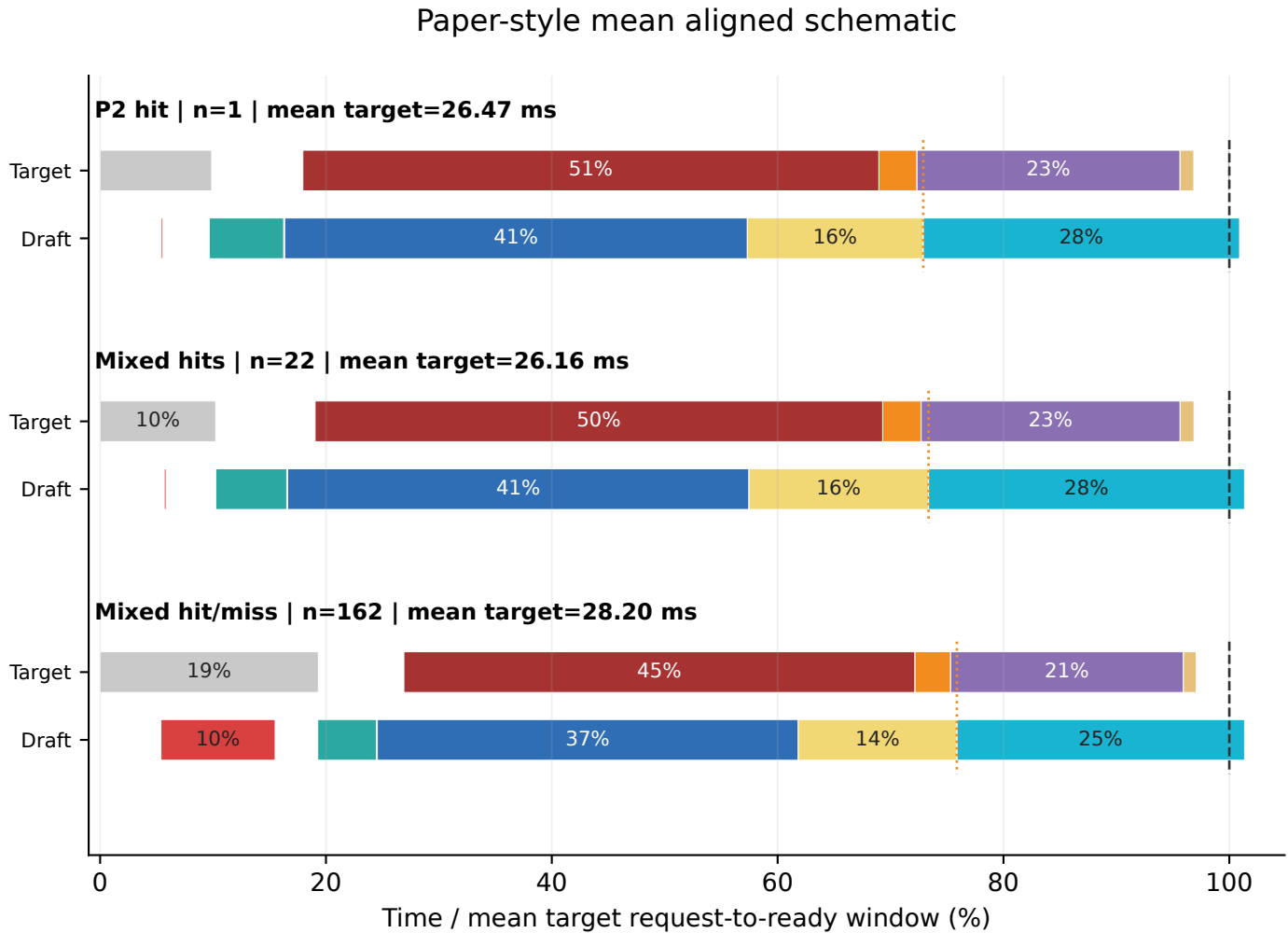
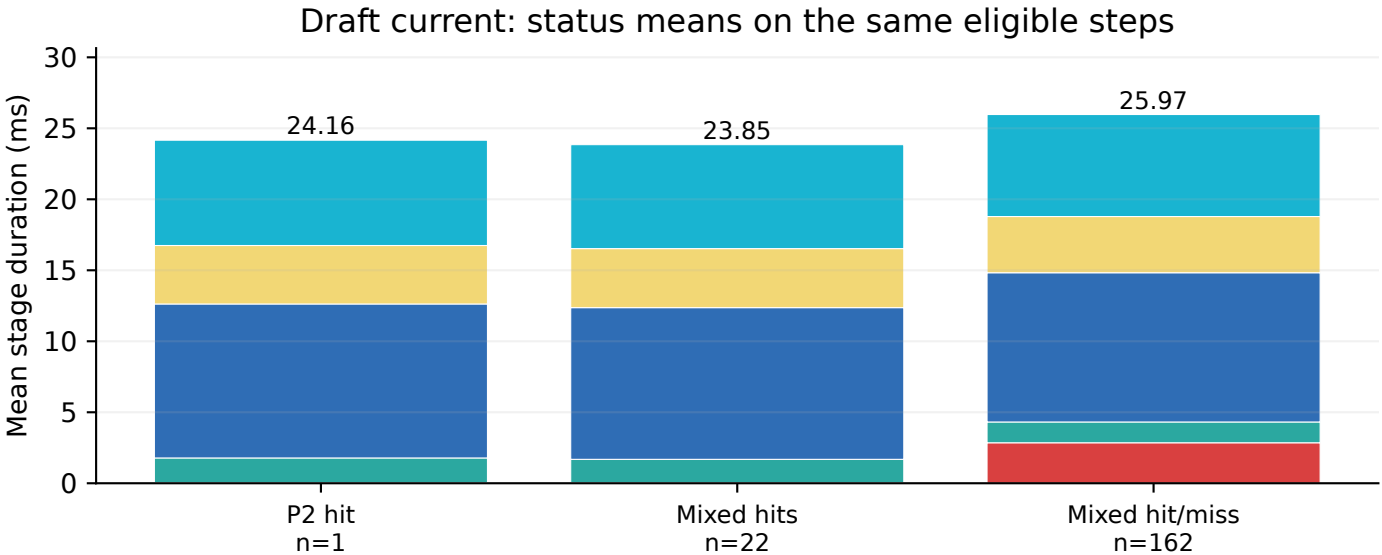
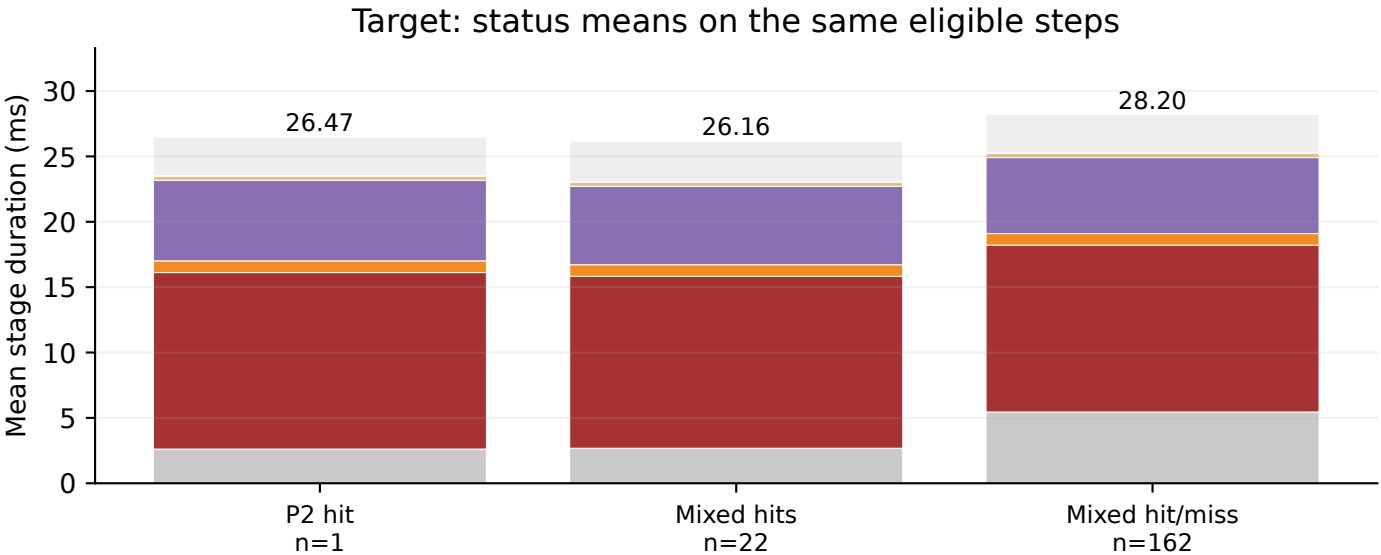
llama2_b8_tree_beta1
DUET: K1/K2=3/2, exit=21 (0-based), $\bar{C}=3$, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 185 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

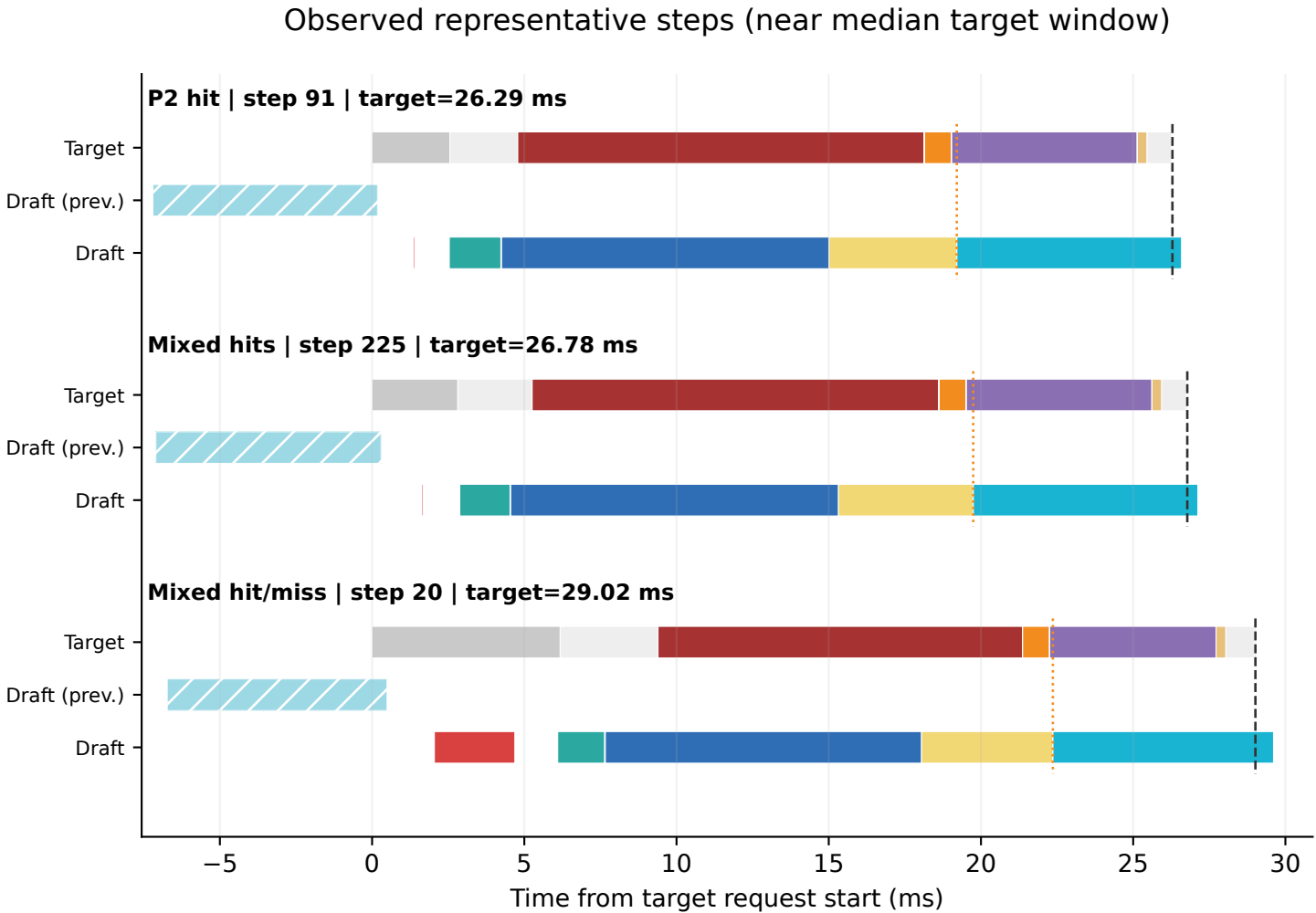
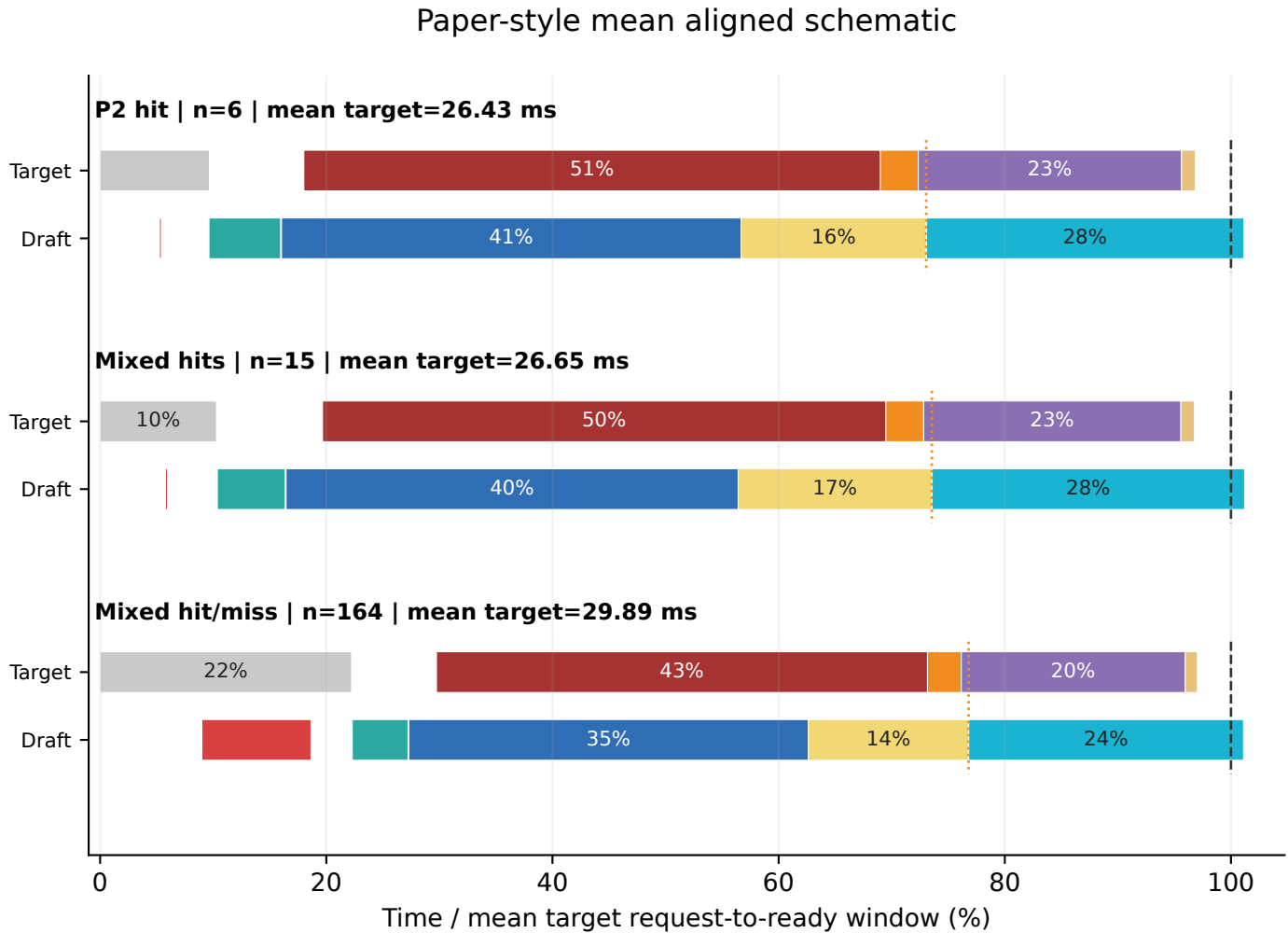
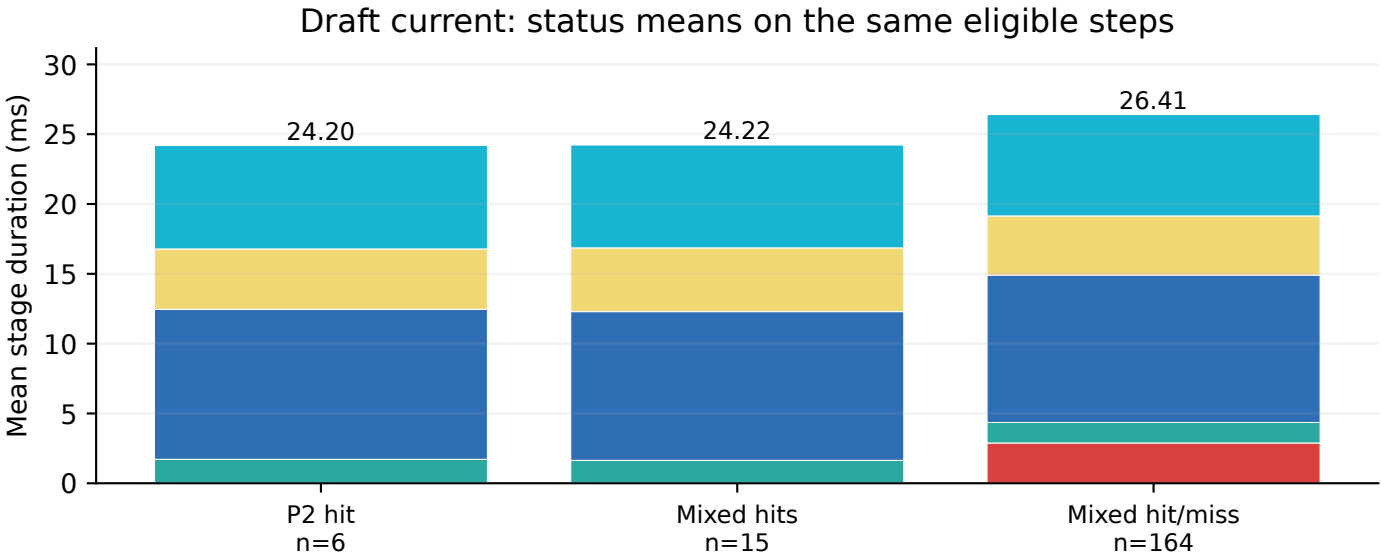
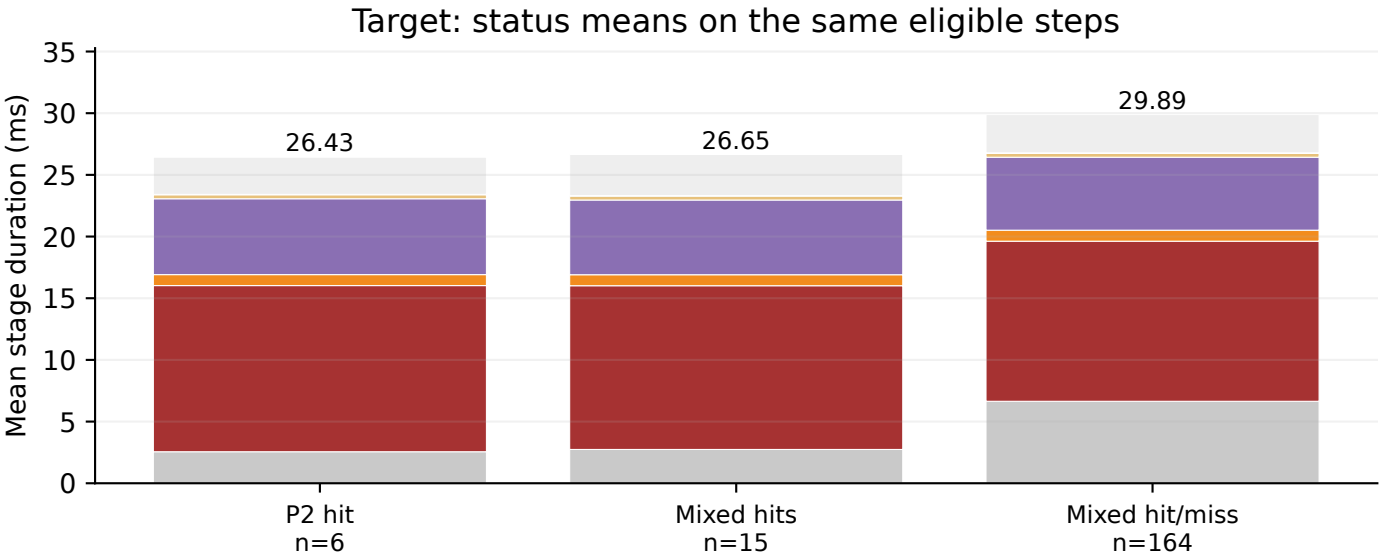
llama2_b8_tree_compute3
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 185 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

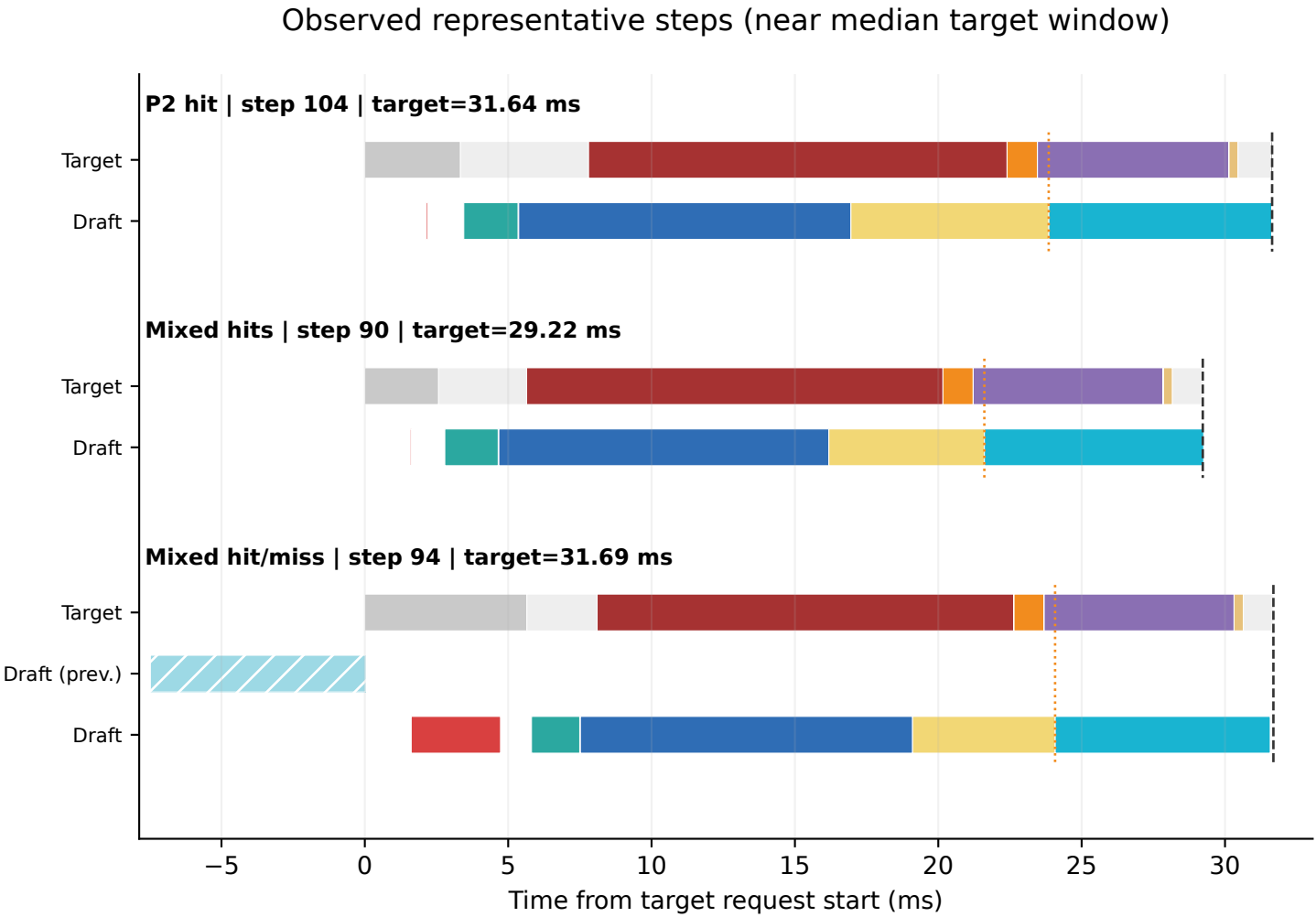
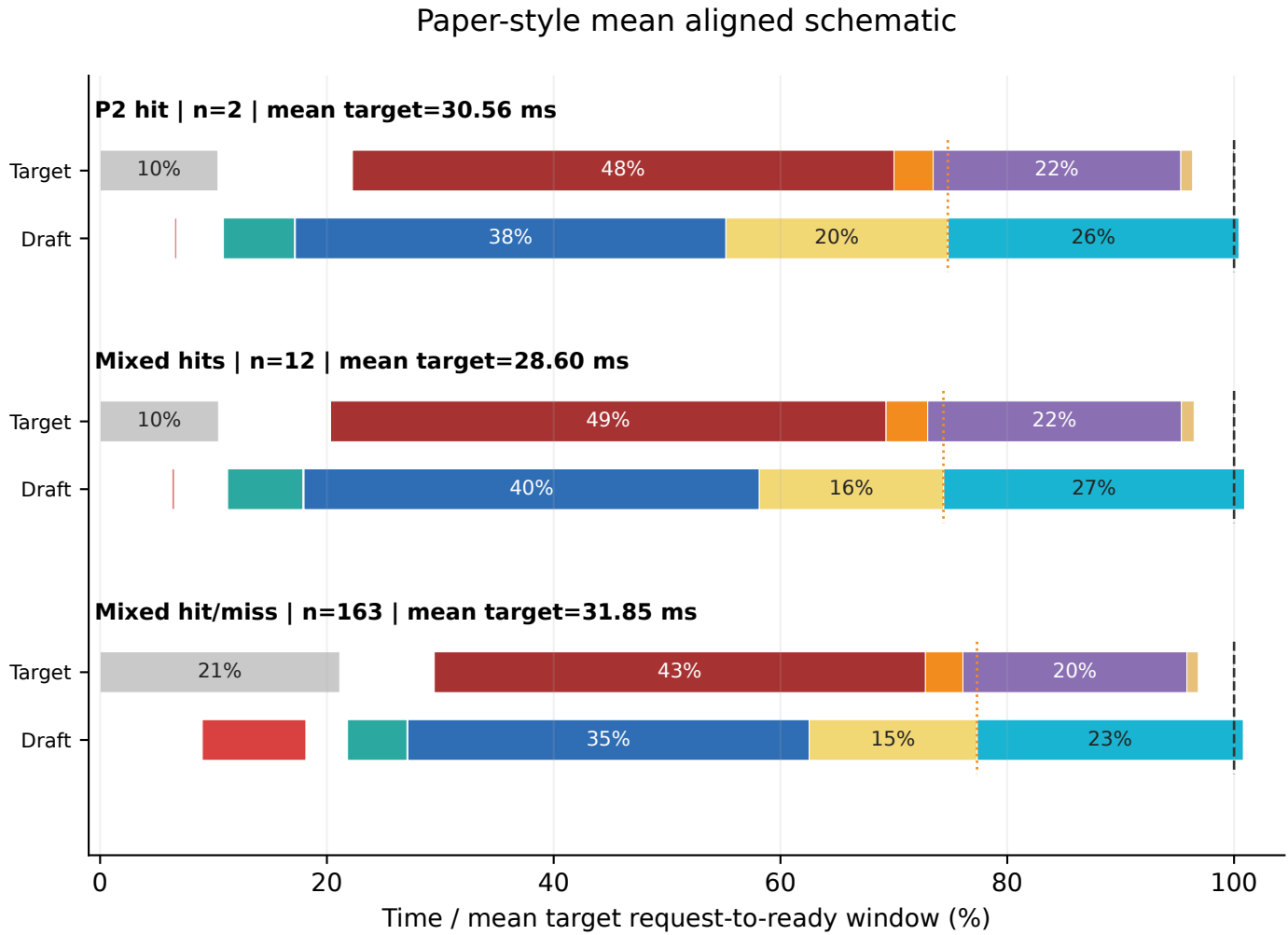
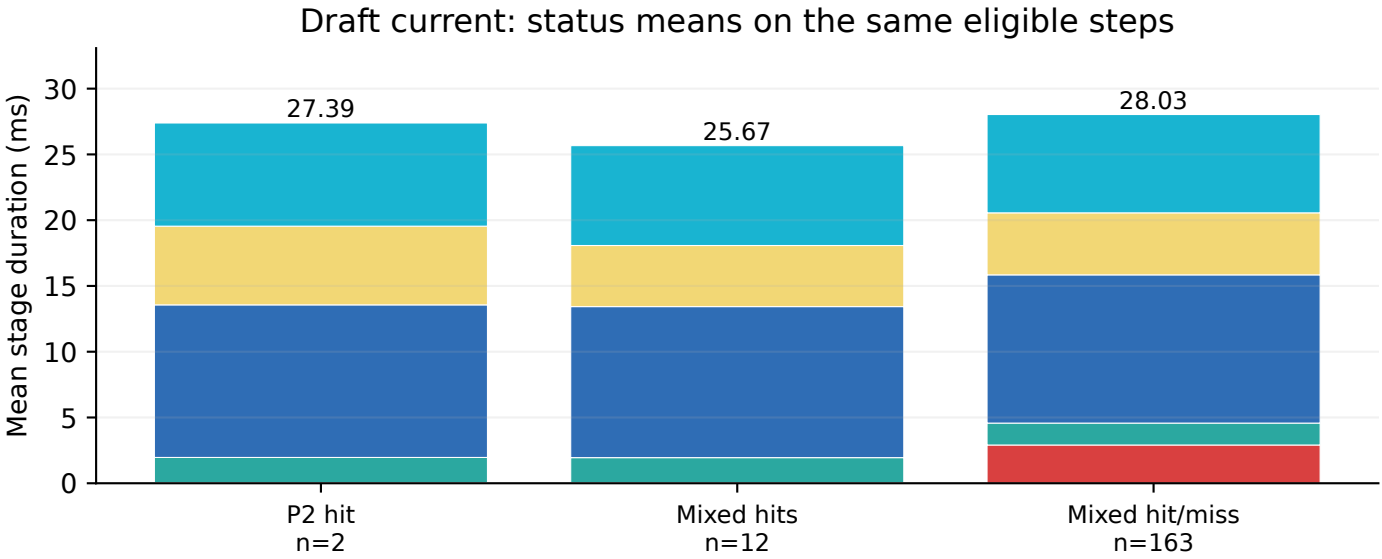
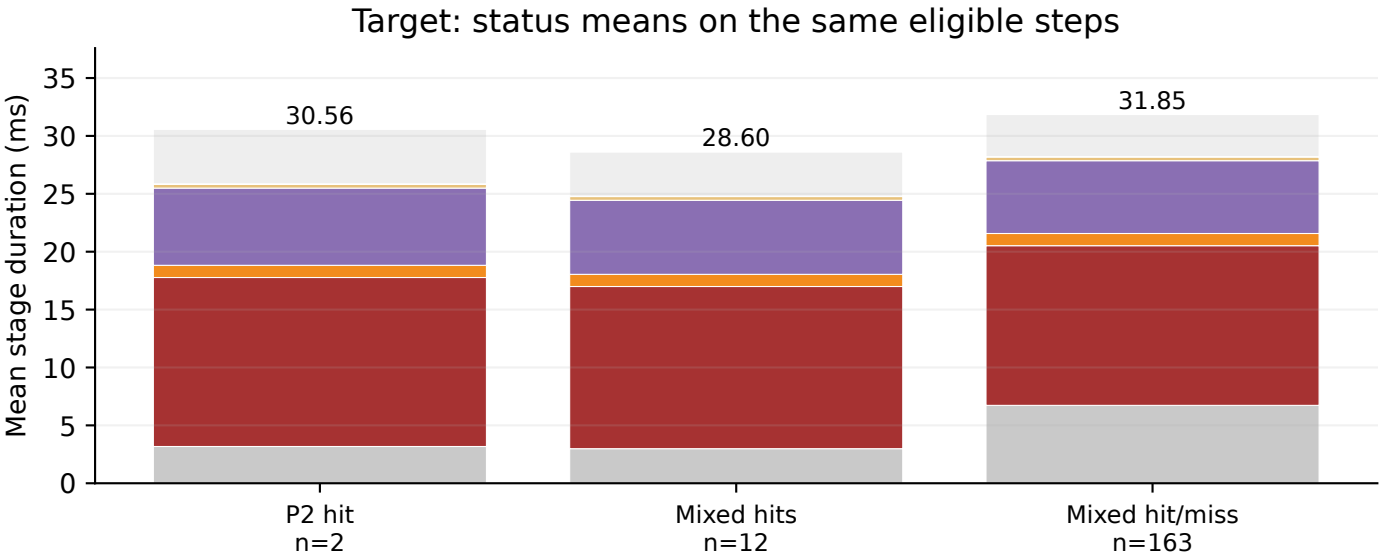
llama2_b8_tree_no_floors
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 185 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

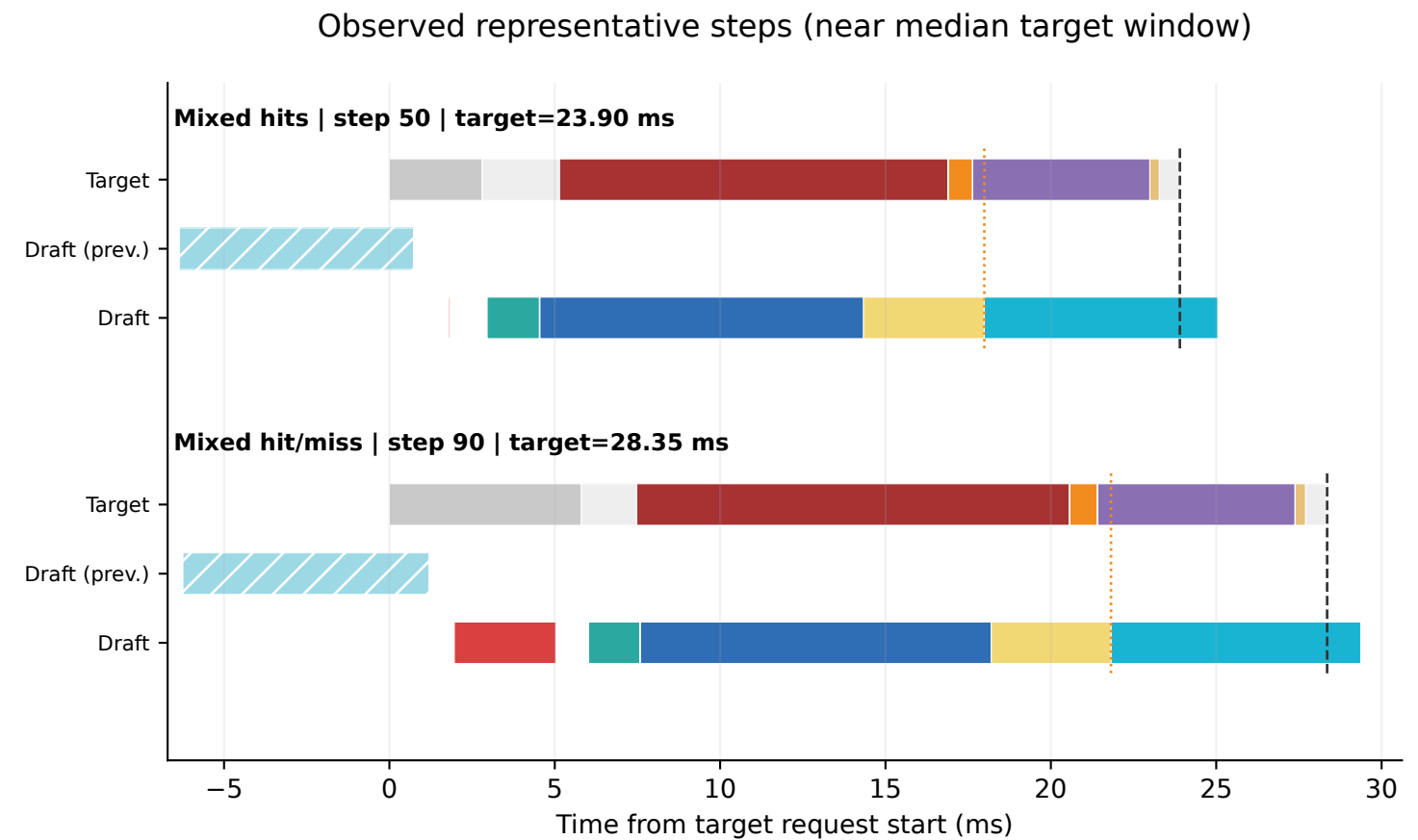
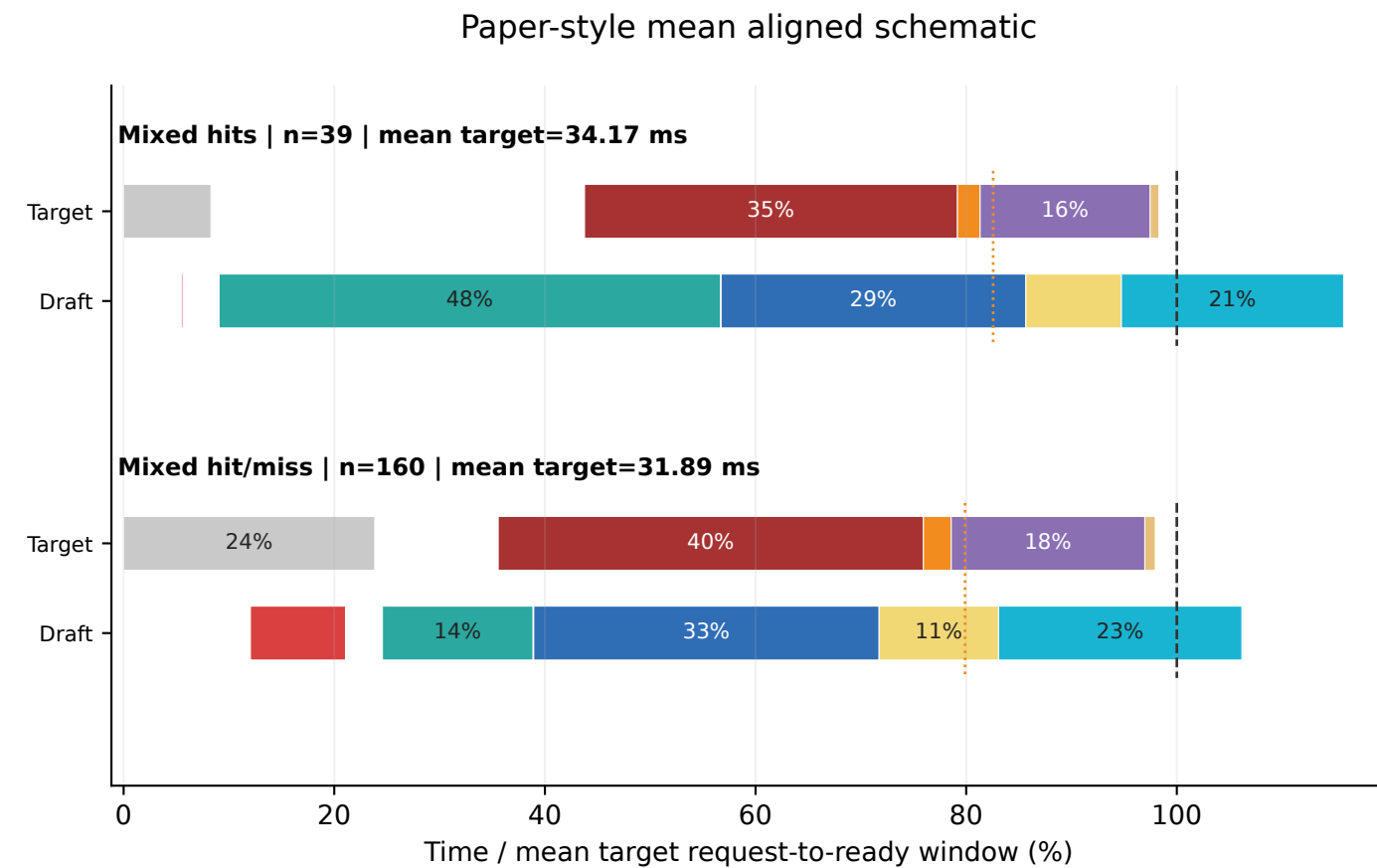
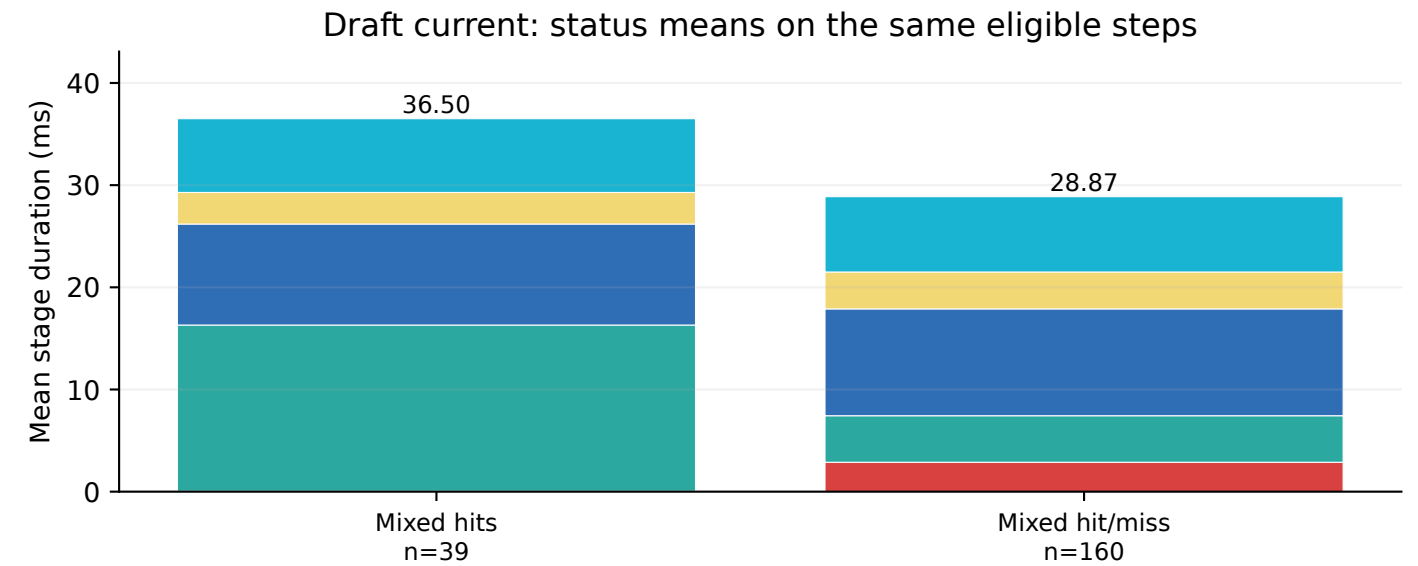
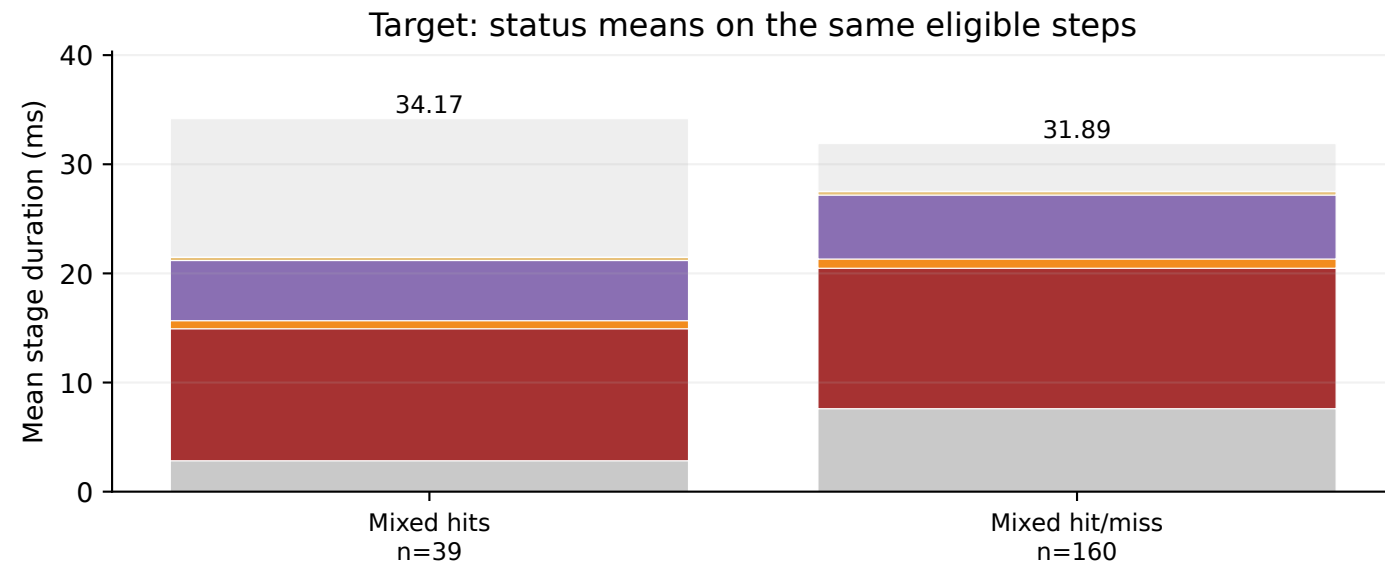
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_tree_nodes_high
DUET: K1/K2=3/2, exit=21 (0-based), $\bar{C}$ =3, $\bar{N1}/\bar{N2}$ =9/6, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 177 eligible steps



Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_tree_nodes_low
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=3/2, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 199 eligible steps

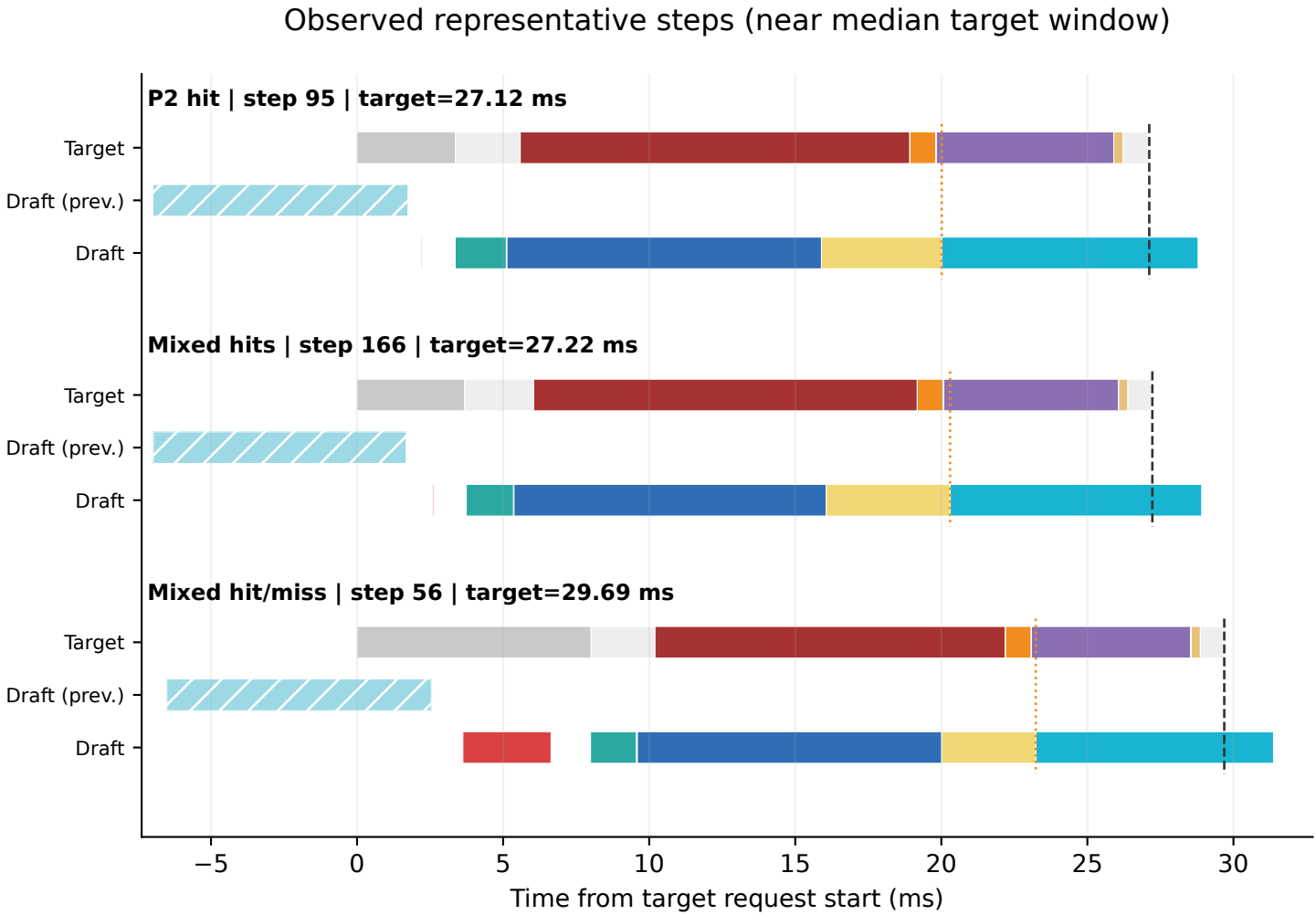
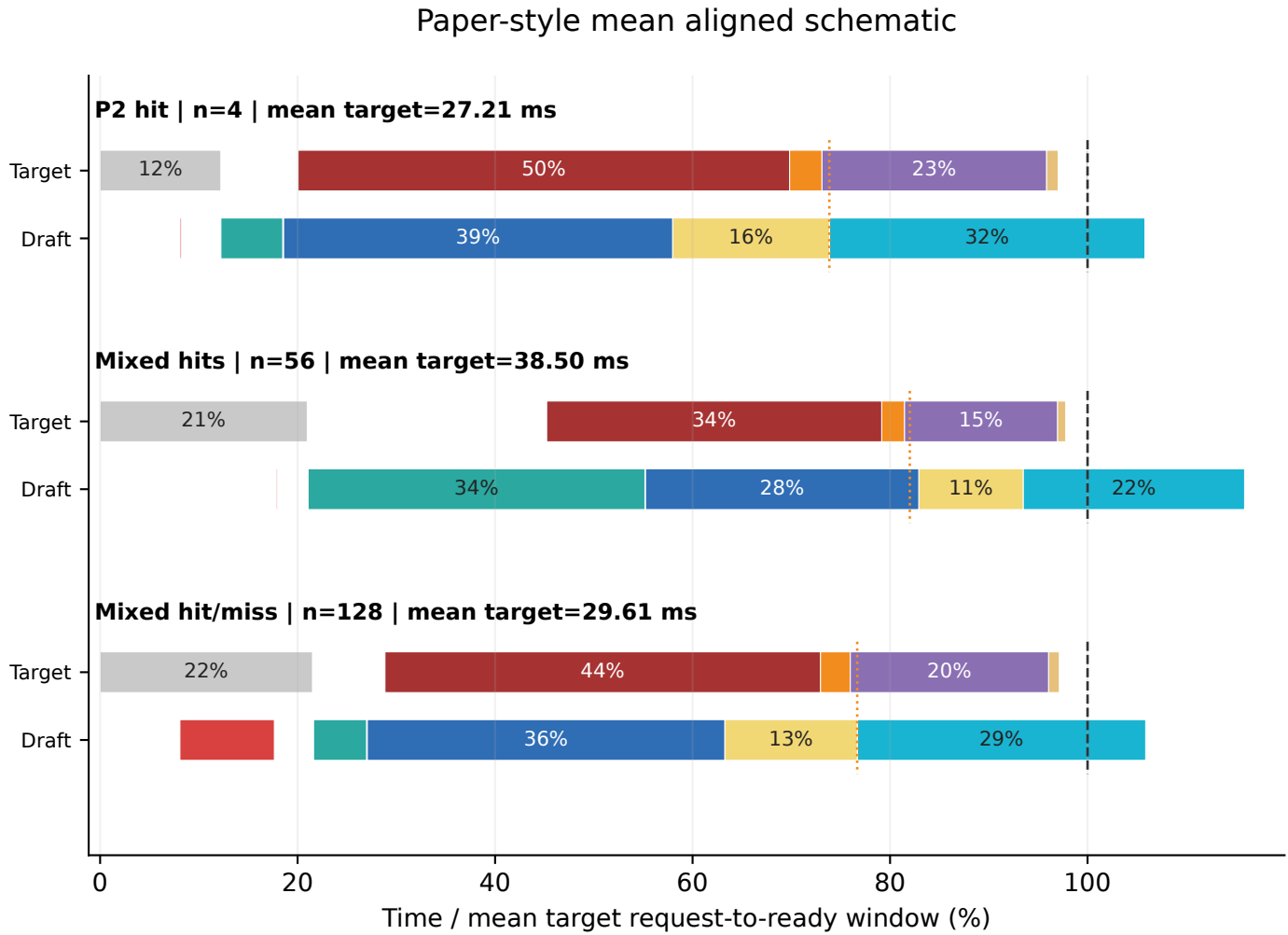
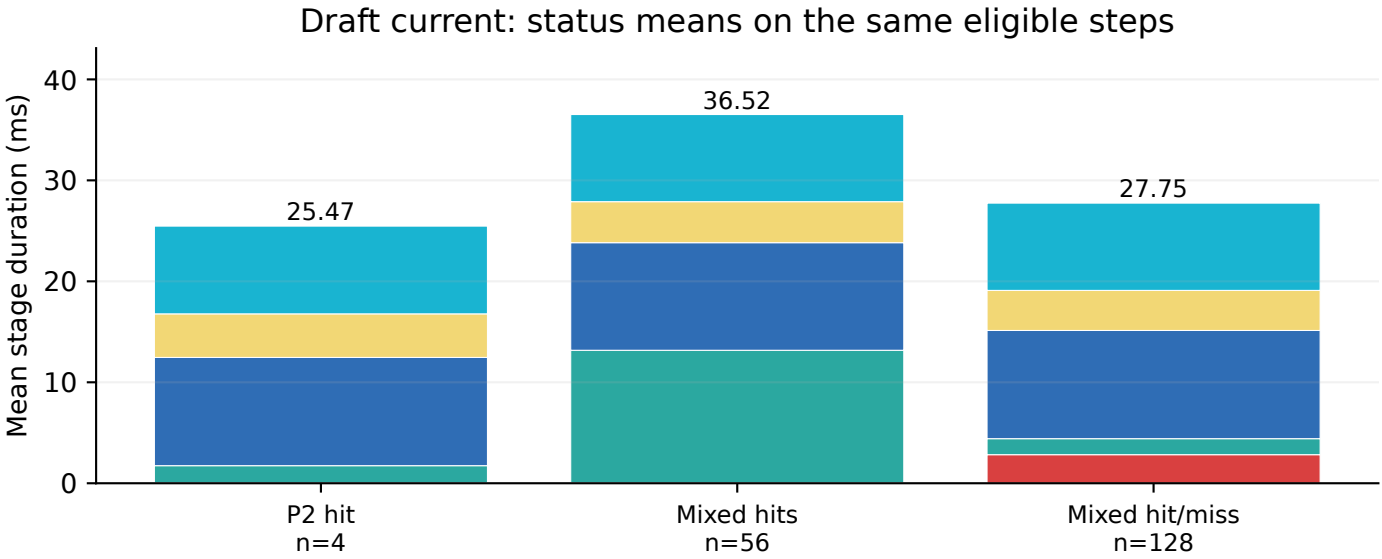
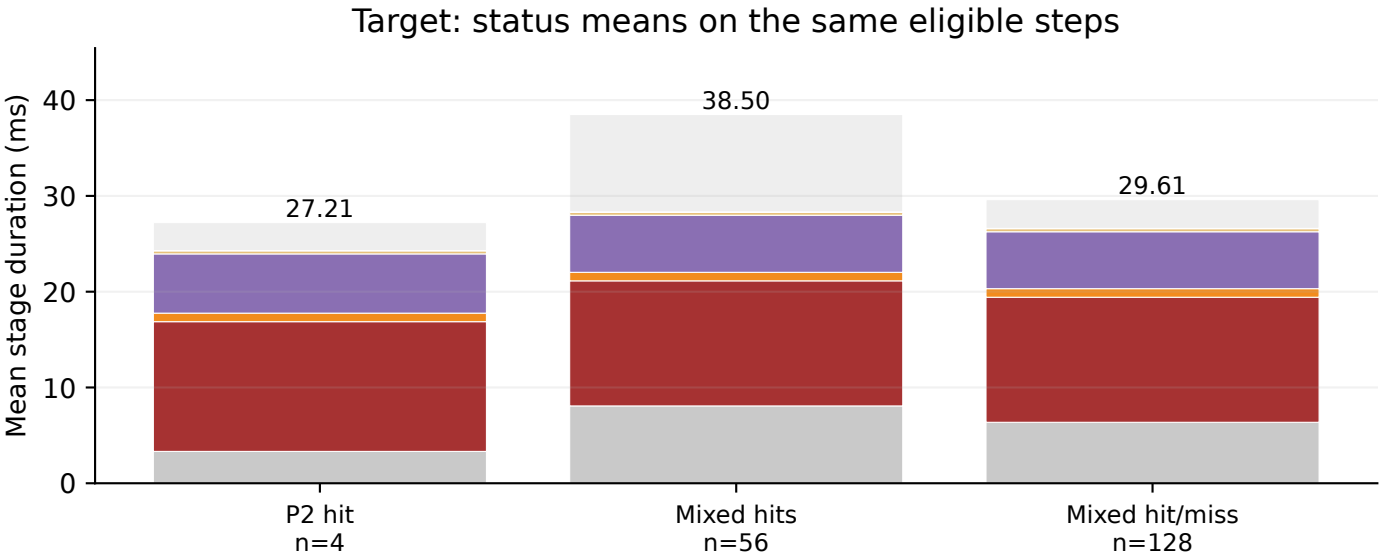


Legend:

Wait / sync	Proxy / launch	Postprocess	Miss draft	P1	Proxy wait	Target ready
Target pre	Final logits	Unlabelled gap	Context	P2	Previous P2	Proxy received
Target post						

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack. Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

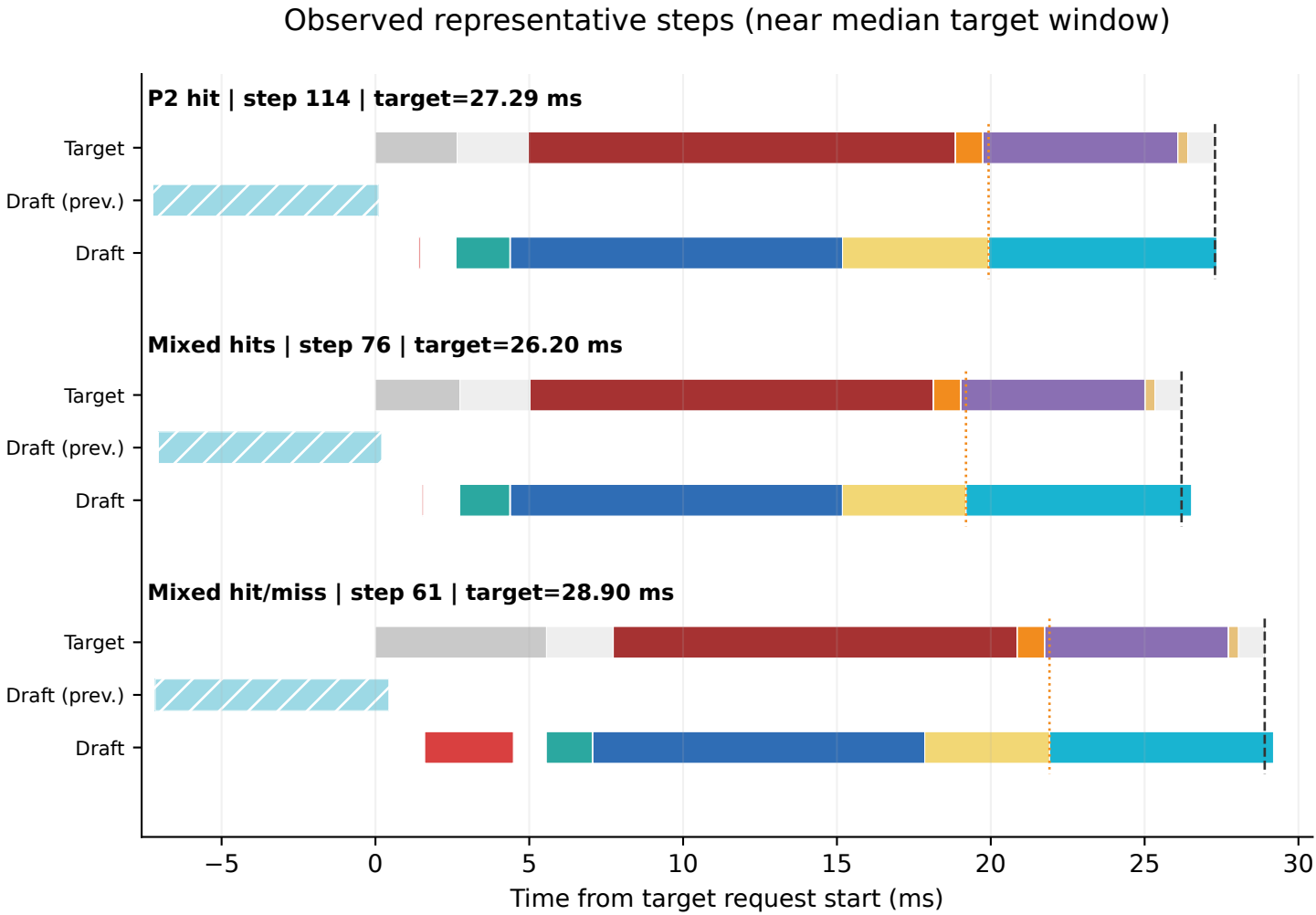
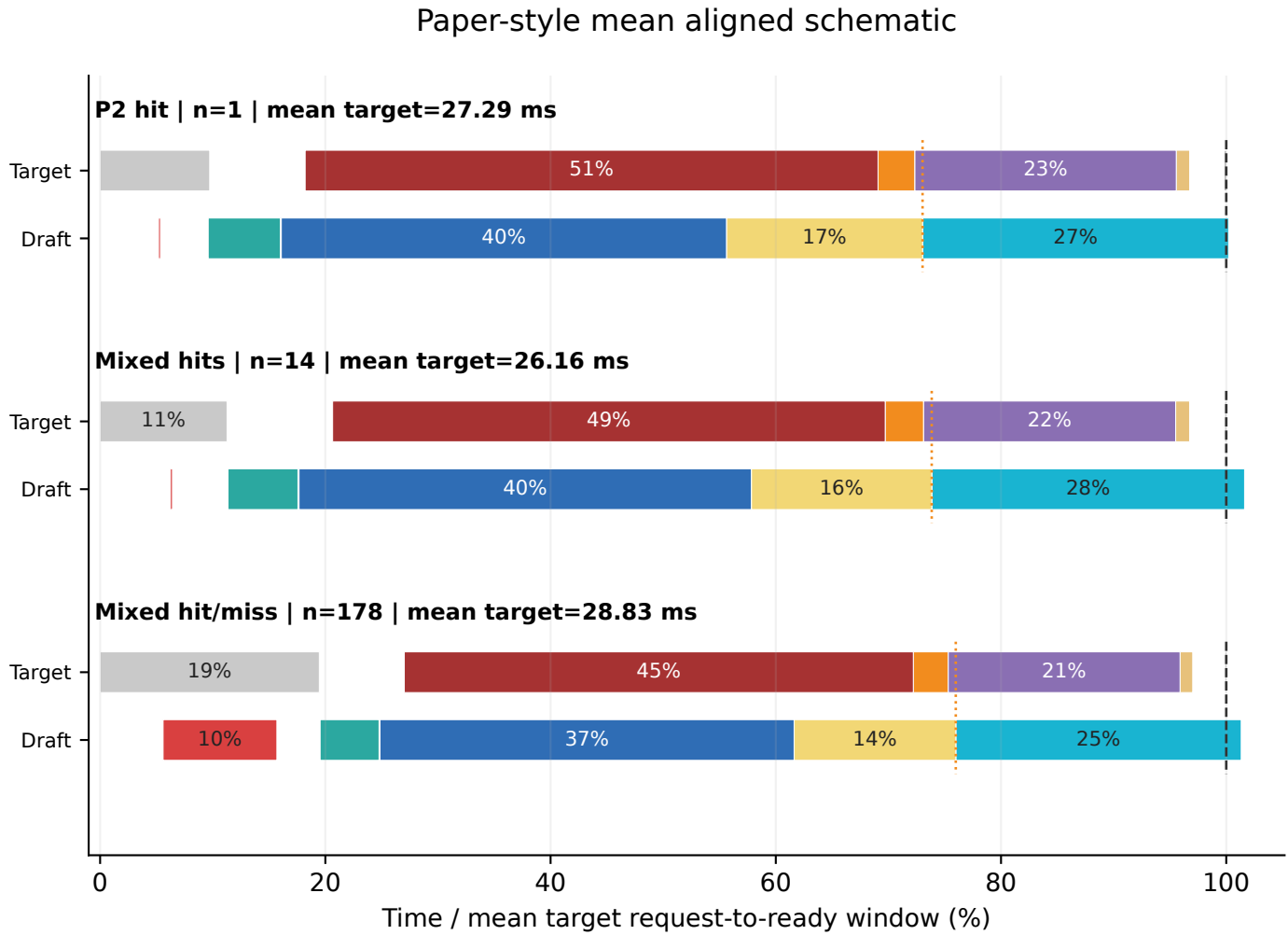
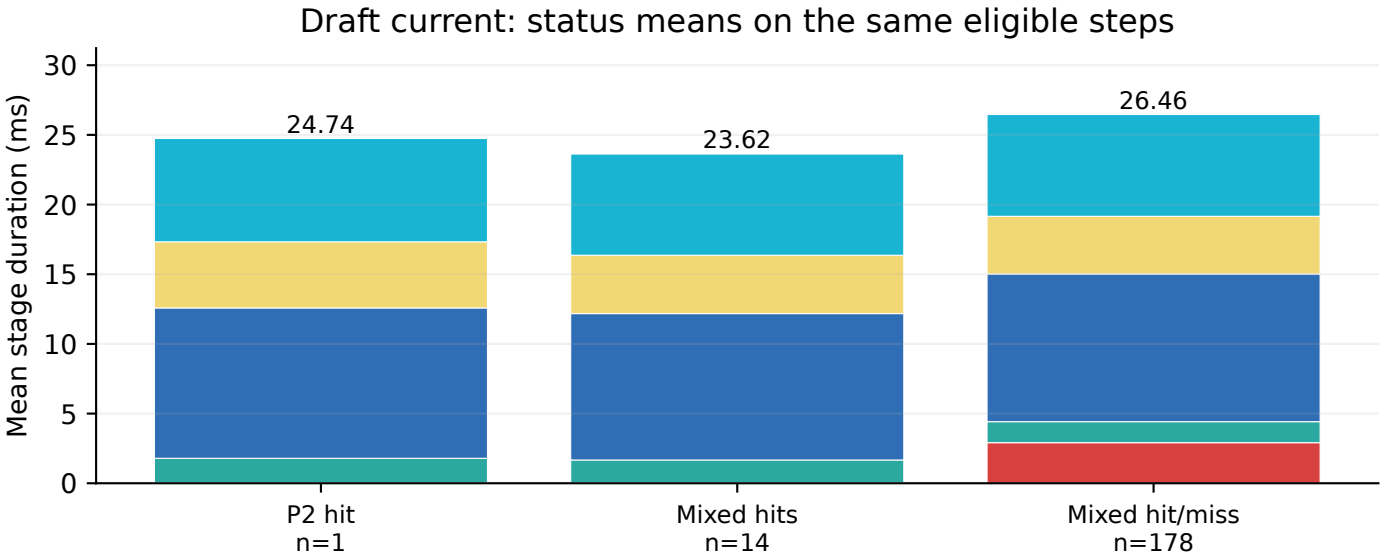
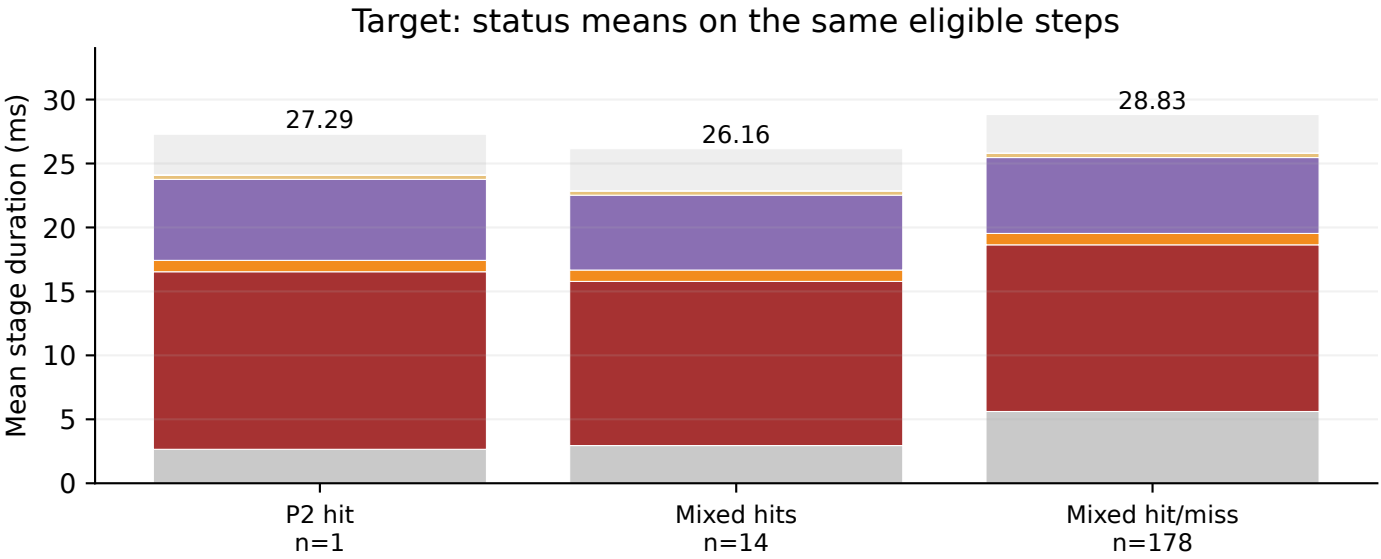
llama2_b8_tree_p2budget12
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=12, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 188 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_tree_p2budget3
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 193 eligible steps

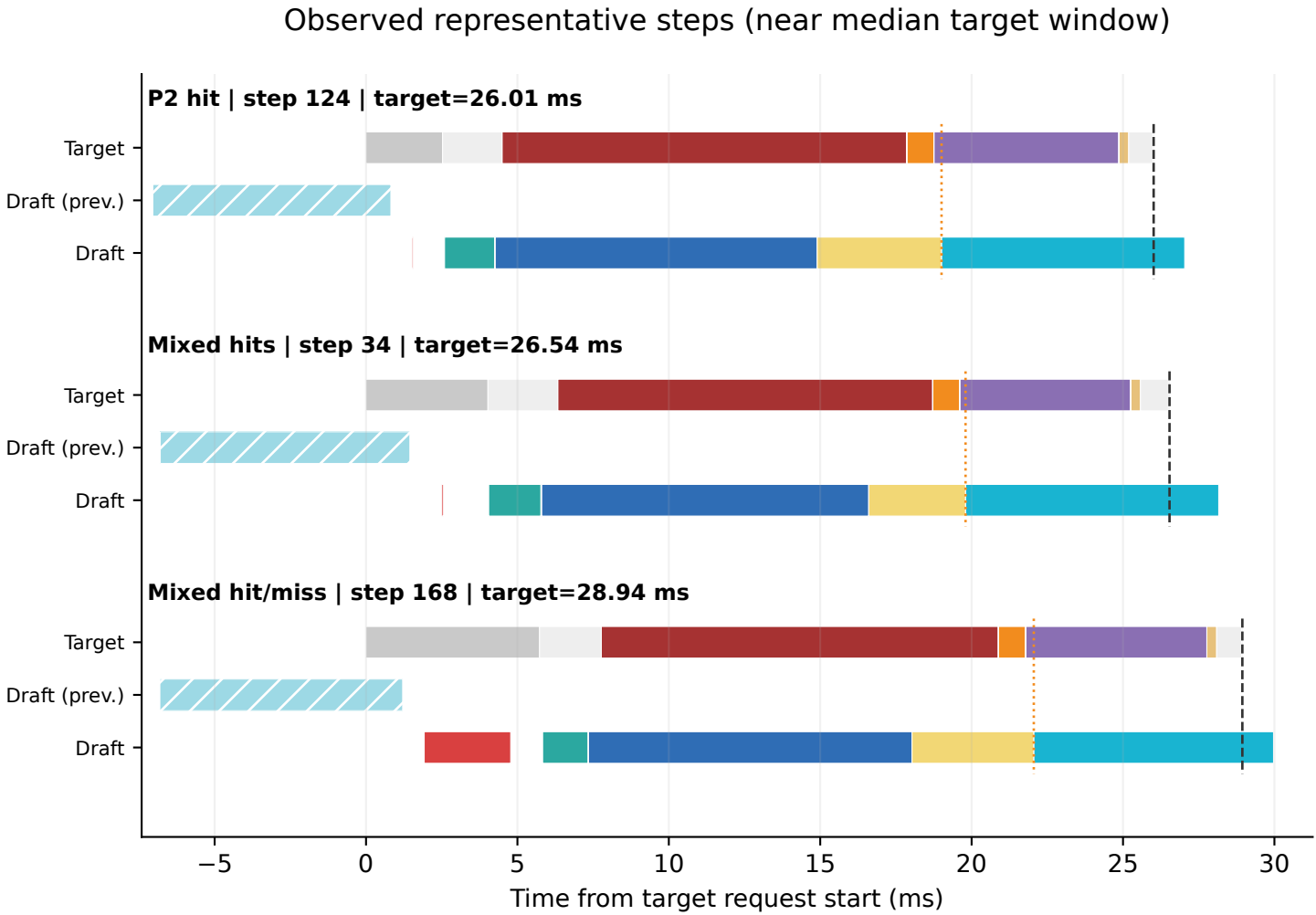
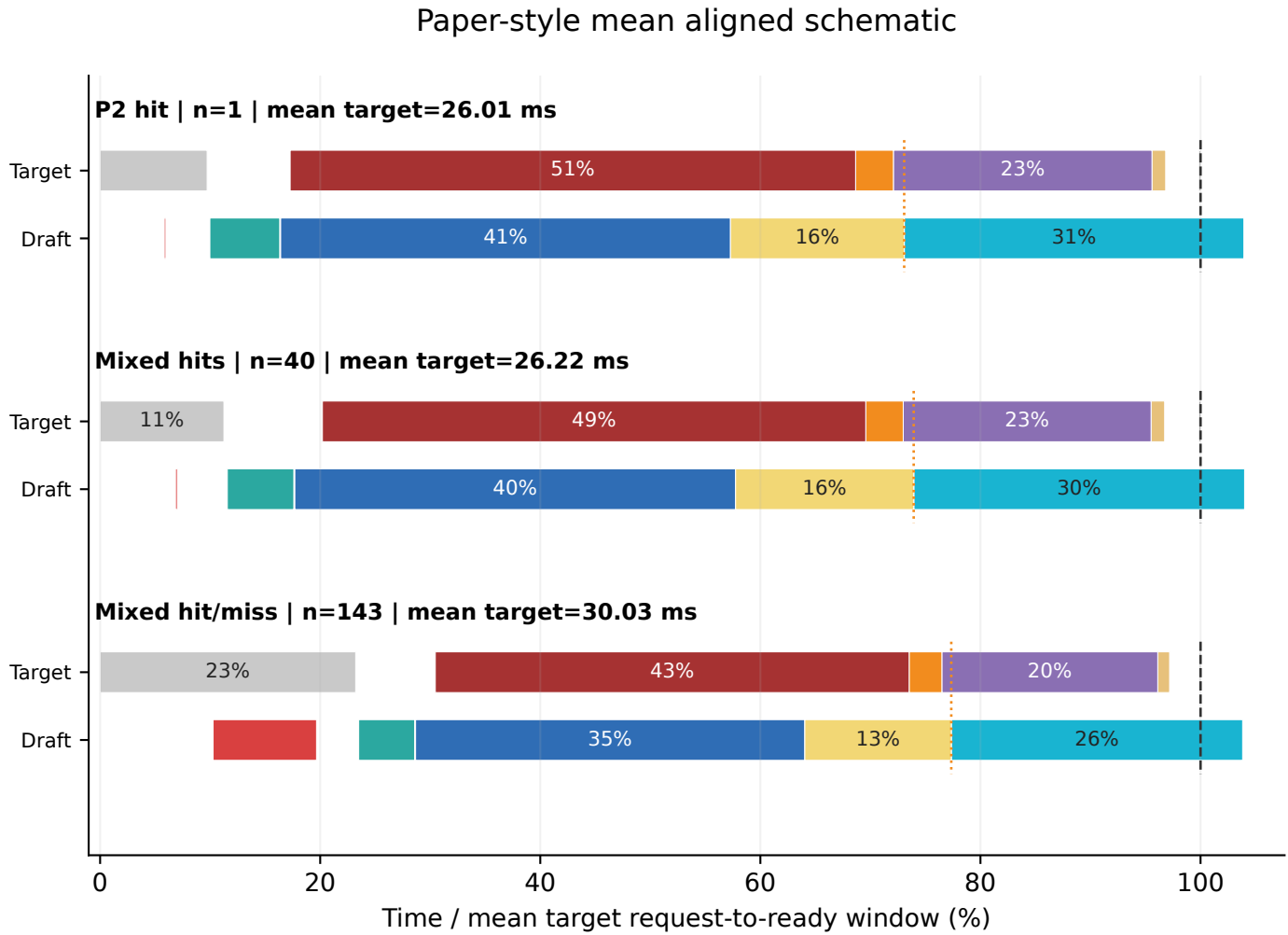
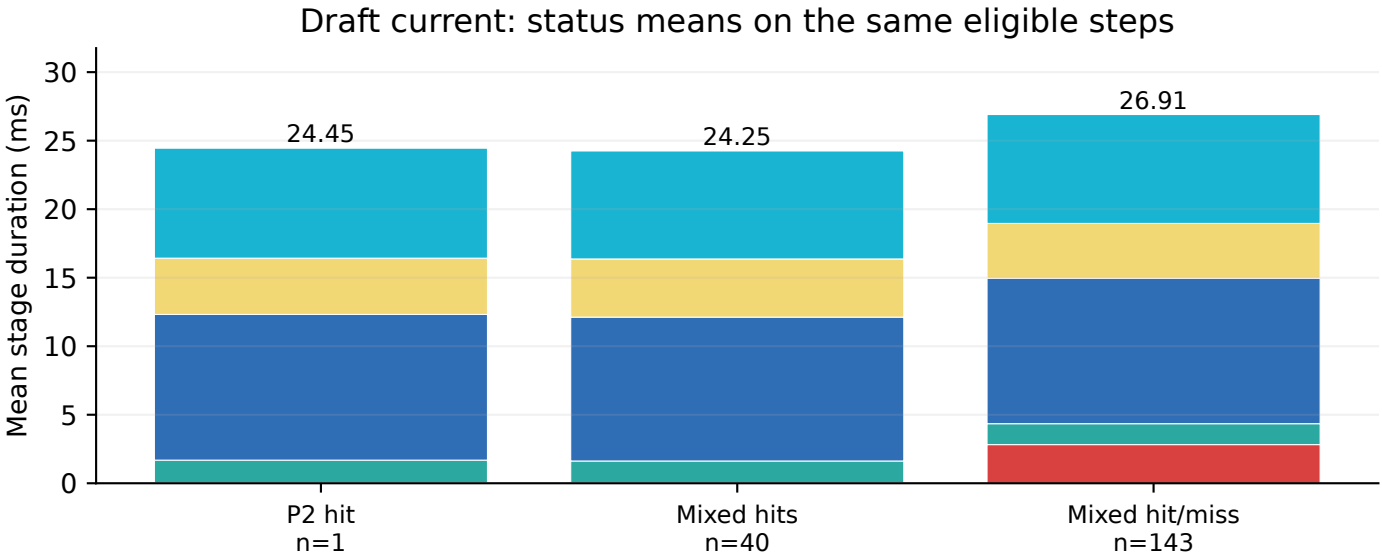
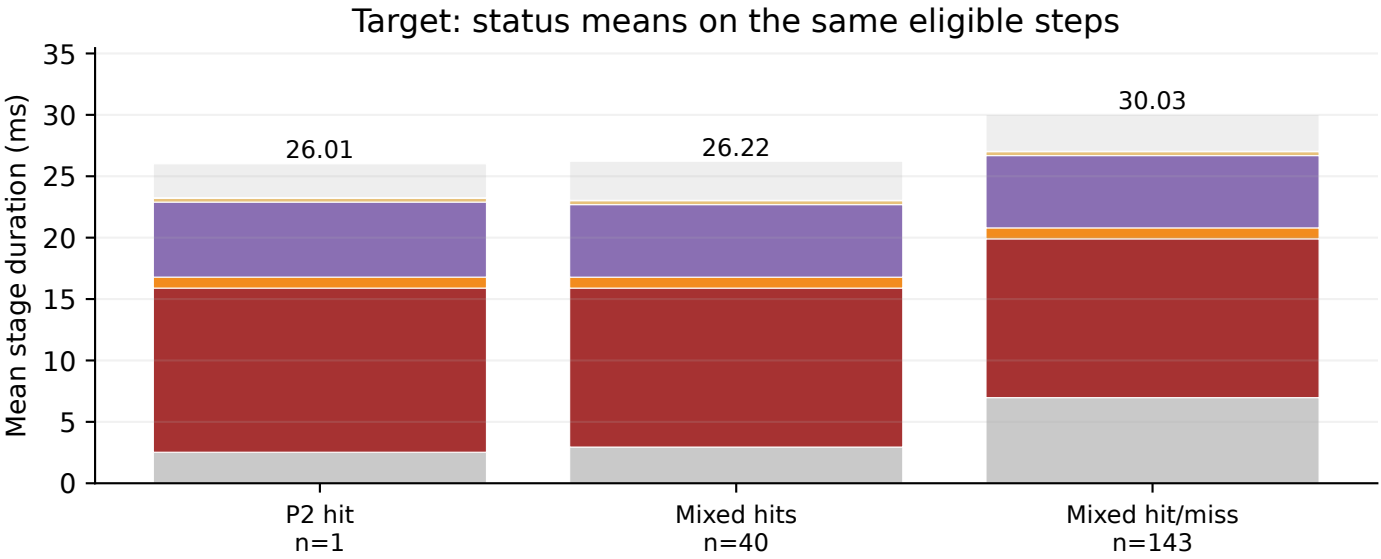


Legend:

- Wait / sync
- Proxy / launch
- Postprocess
- Miss draft
- P1
- Proxy wait
- Target ready
- Target pre
- Final logits
- Unlabelled gap
- Context
- P2
- Previous P2
- Proxy received
- Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

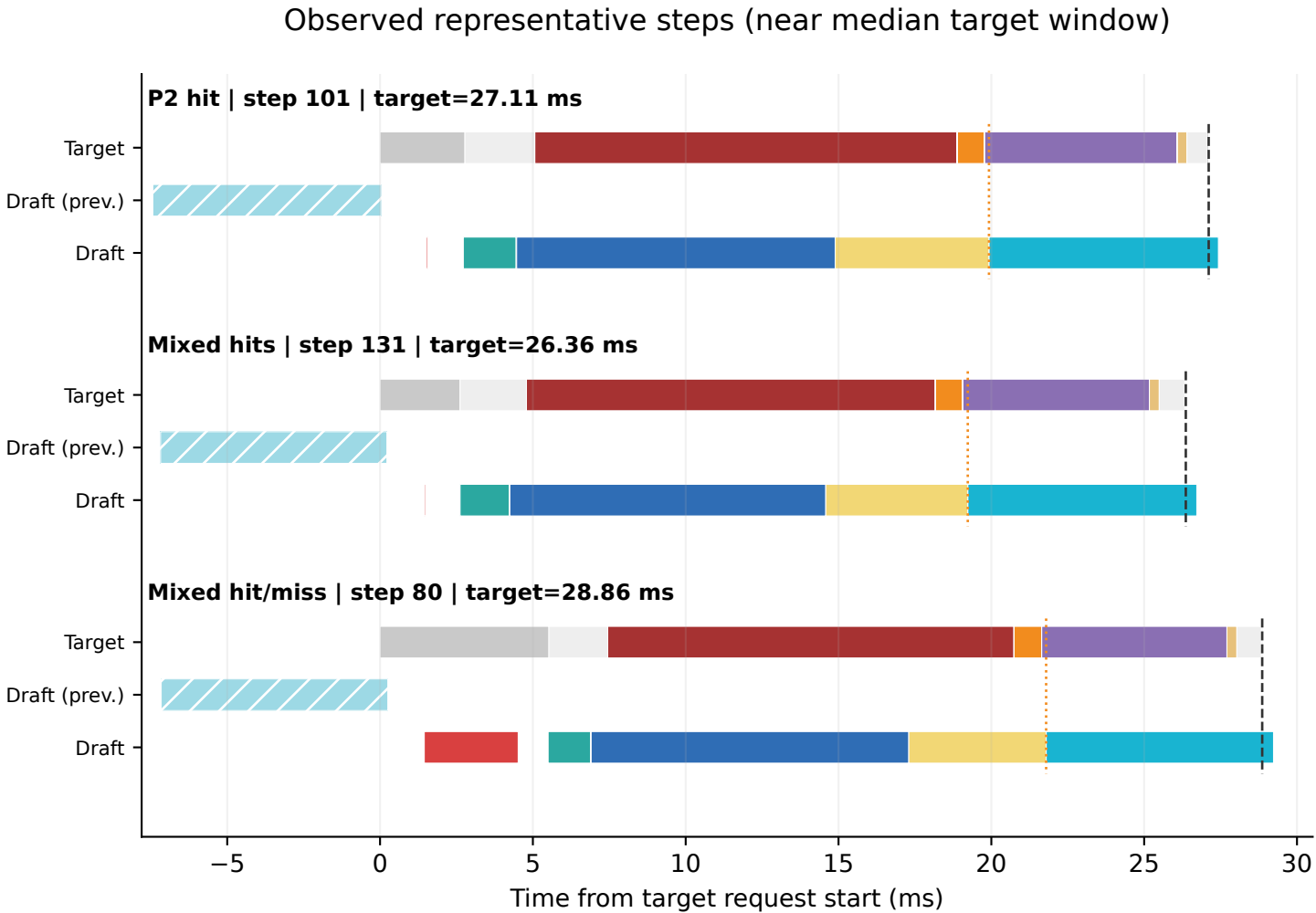
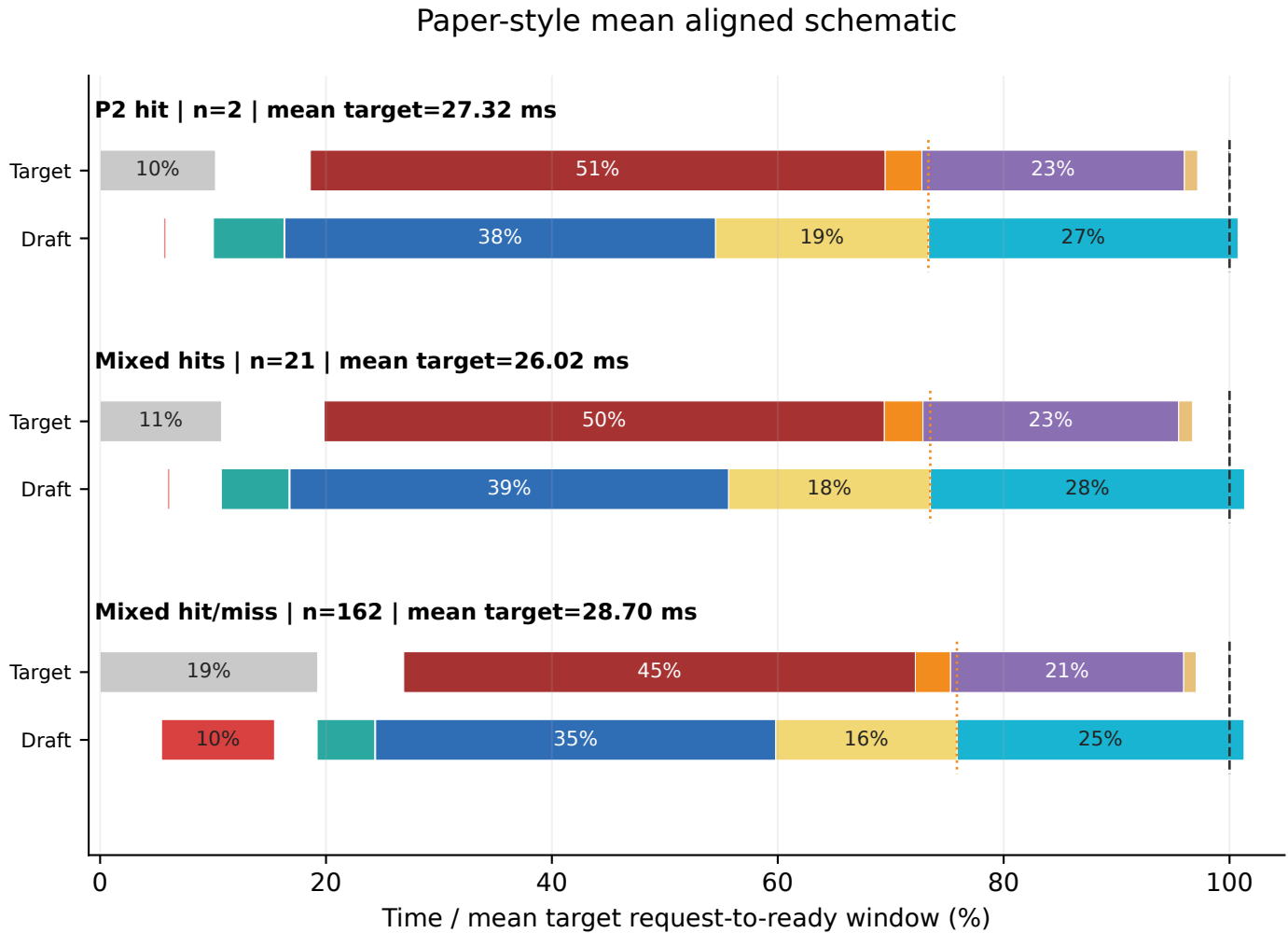
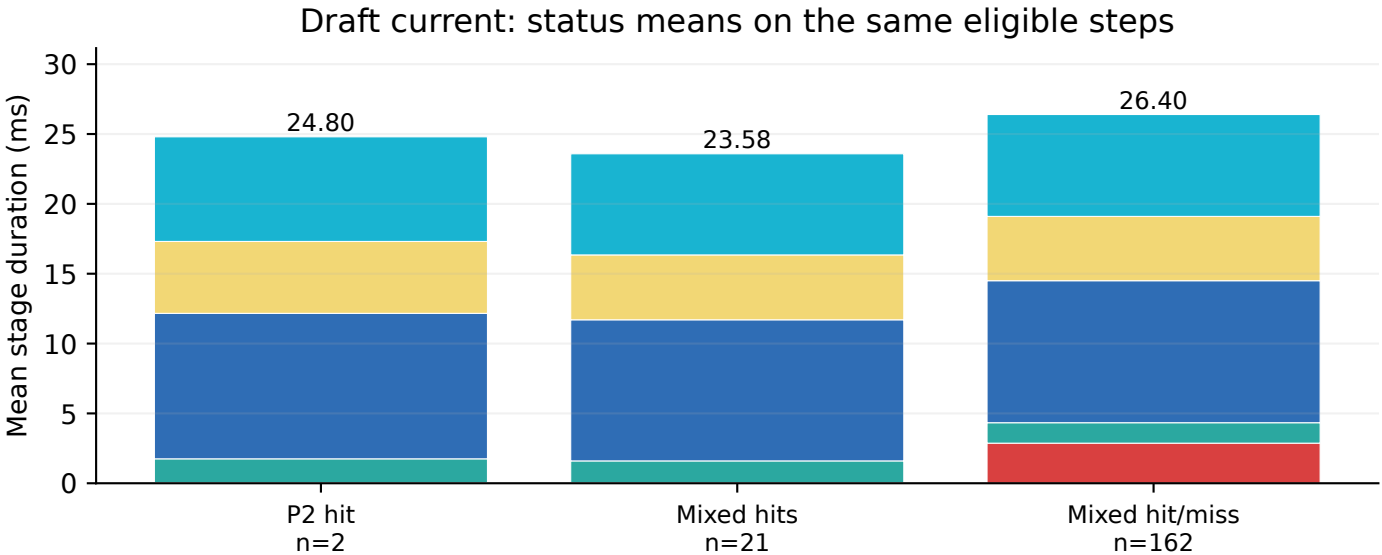
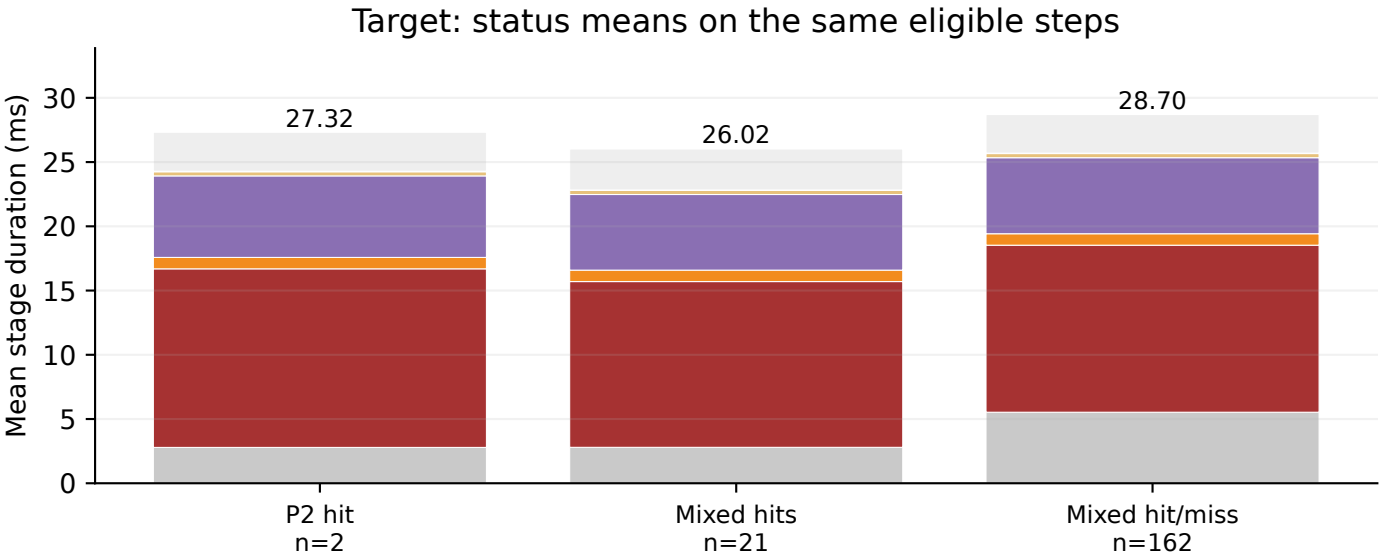
llama2_b8_tree_p2budget8
DUET: K1/K2=3/2, exit=21 (0-based), $\bar{C}$ =3, N1/N2=6/4, U=1, W=8, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 184 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

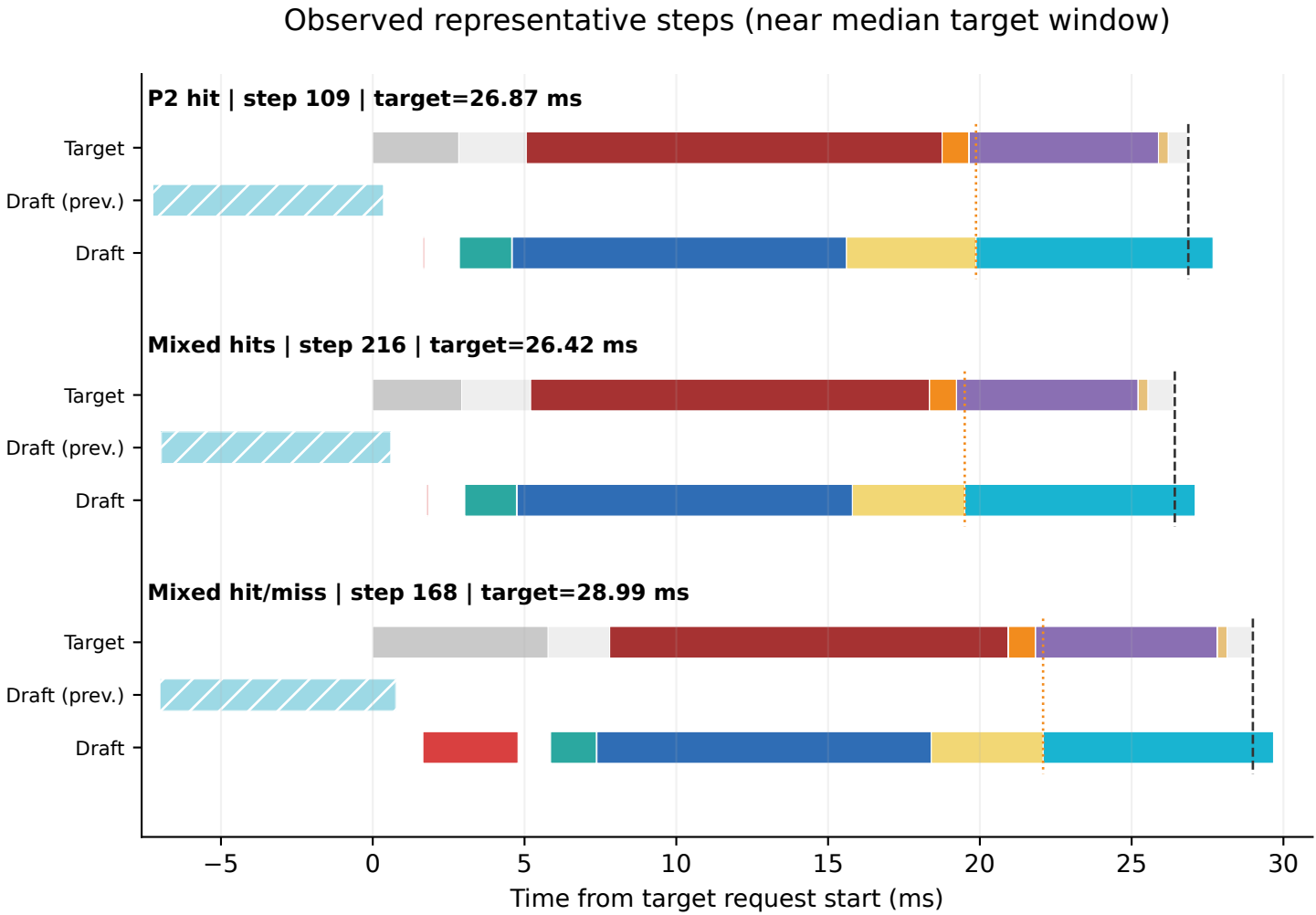
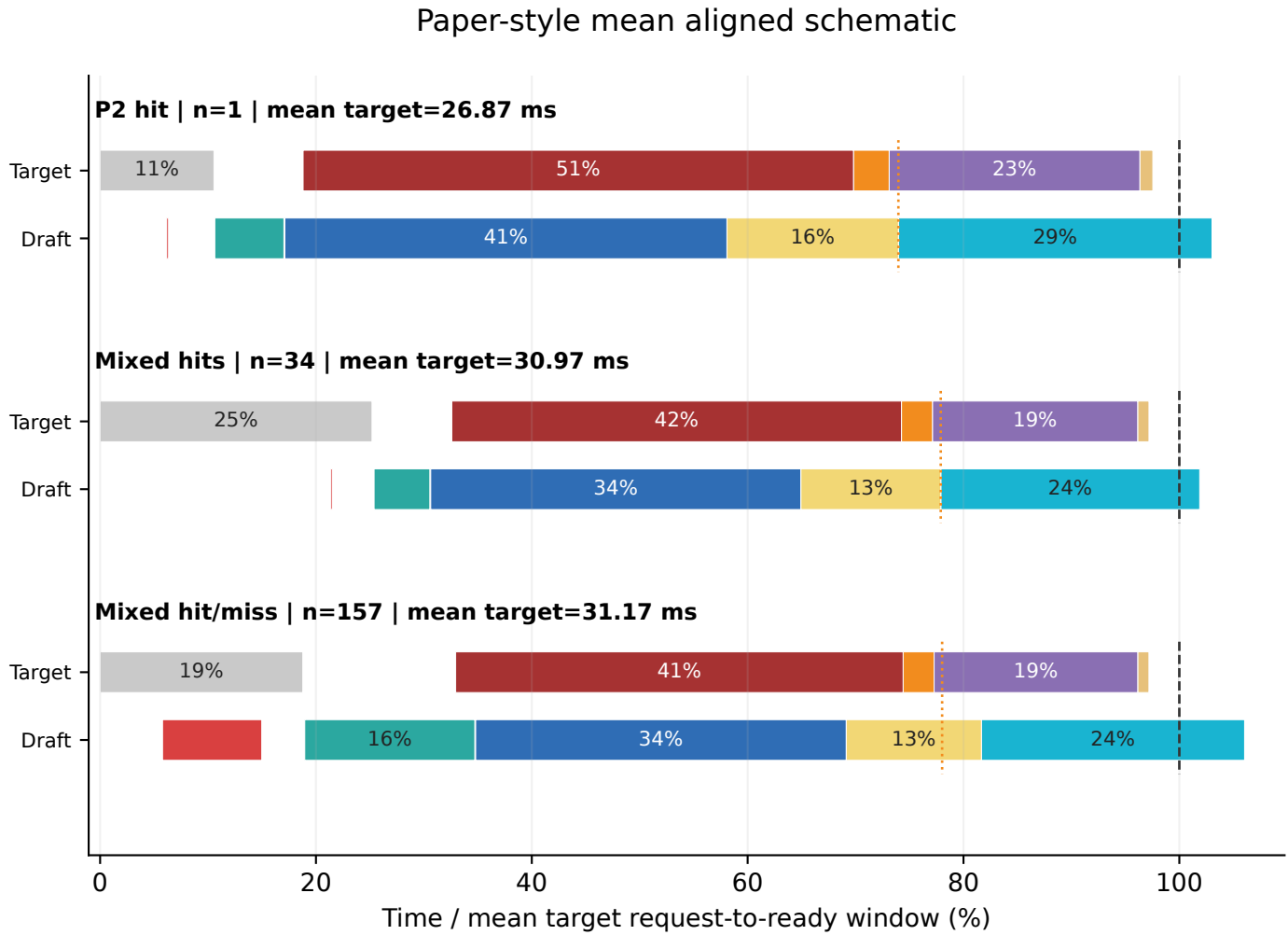
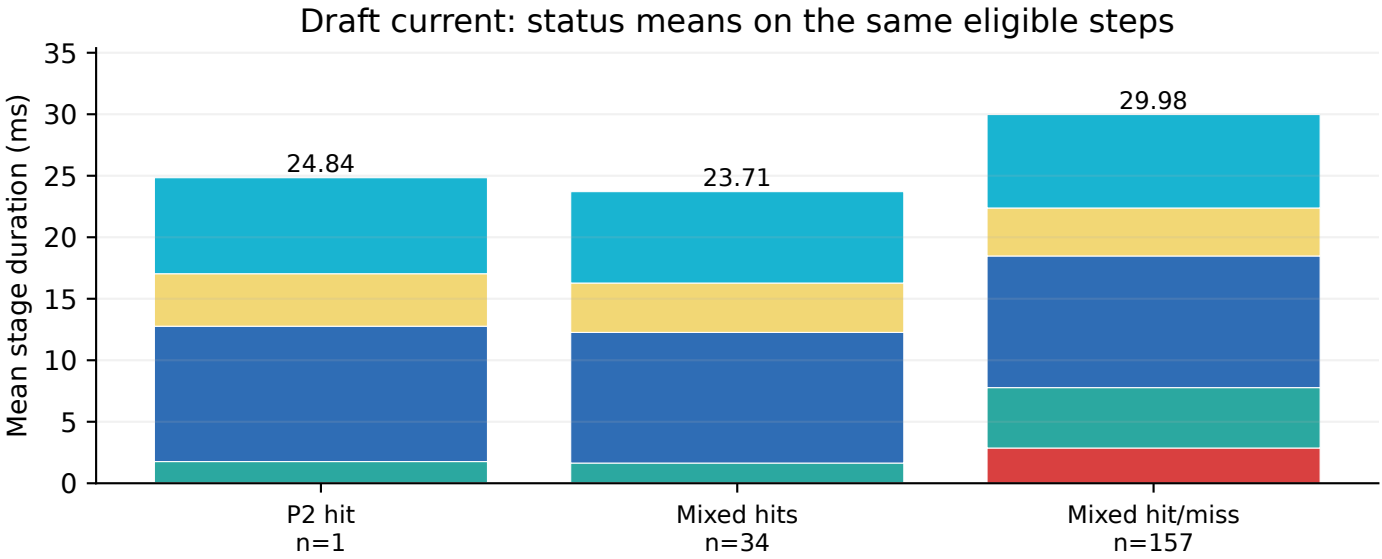
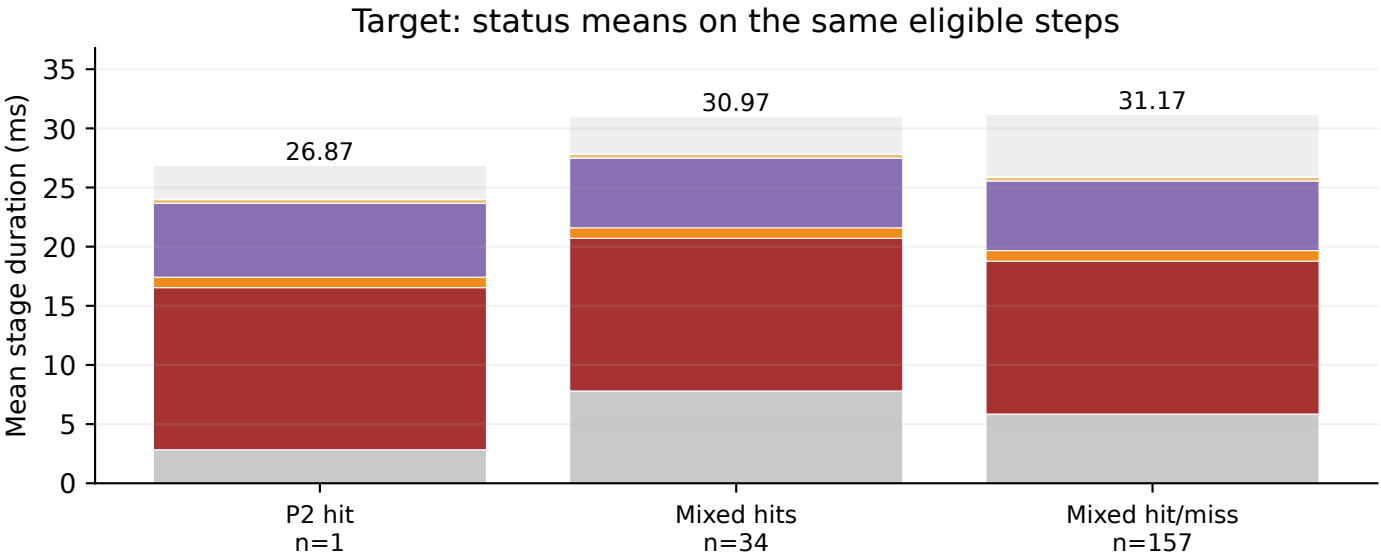
llama2_b8_tree_roots1_compute1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 185 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

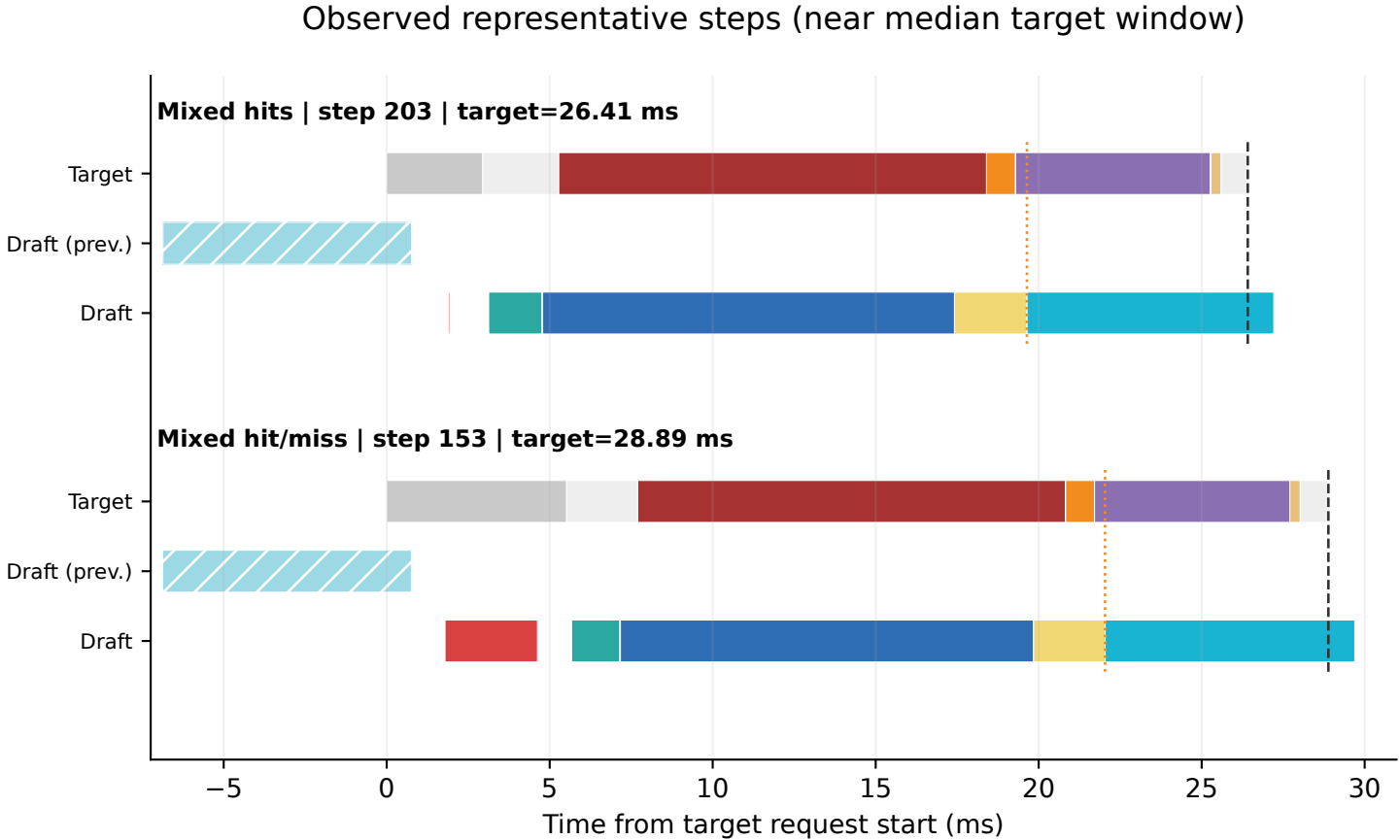
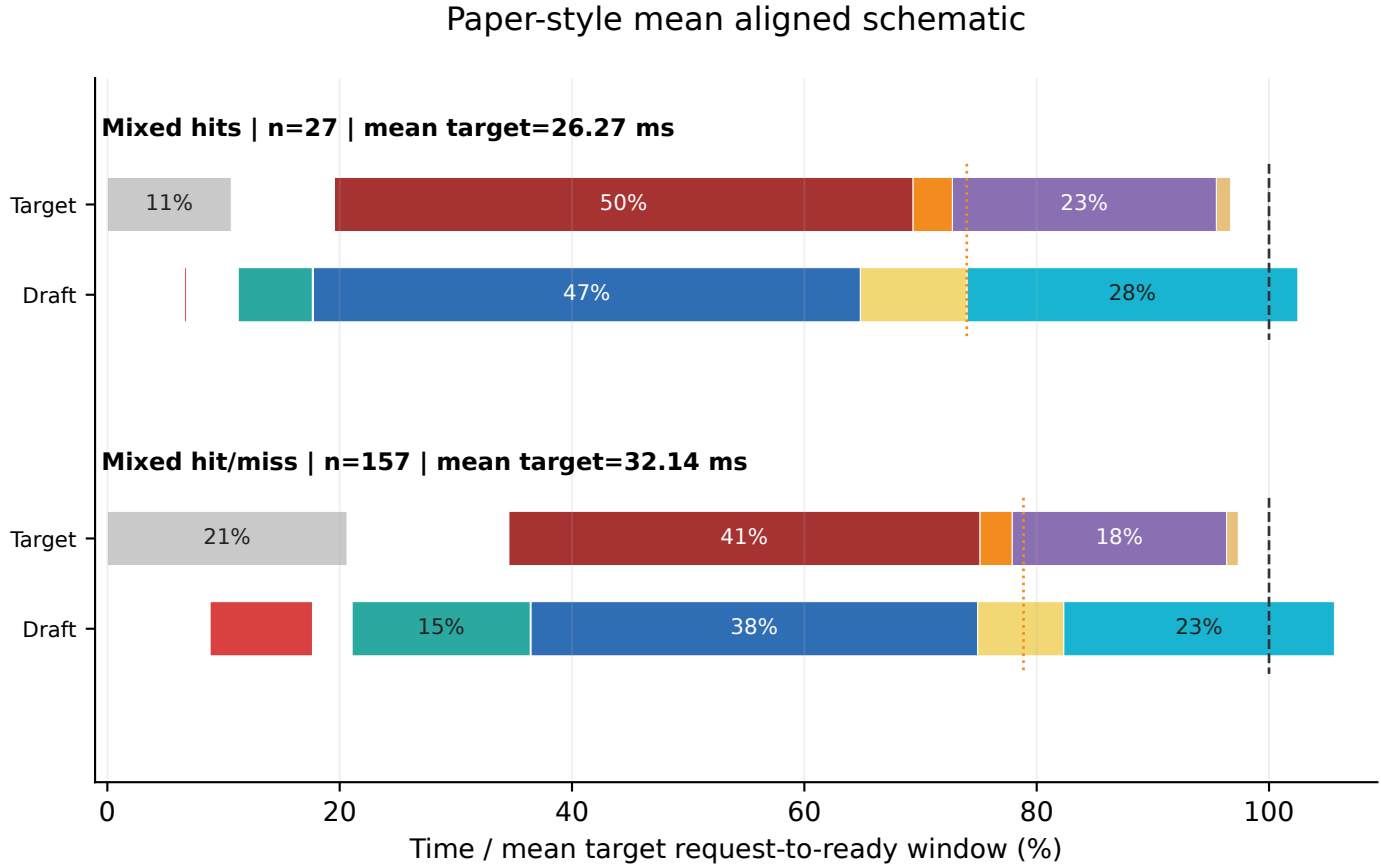
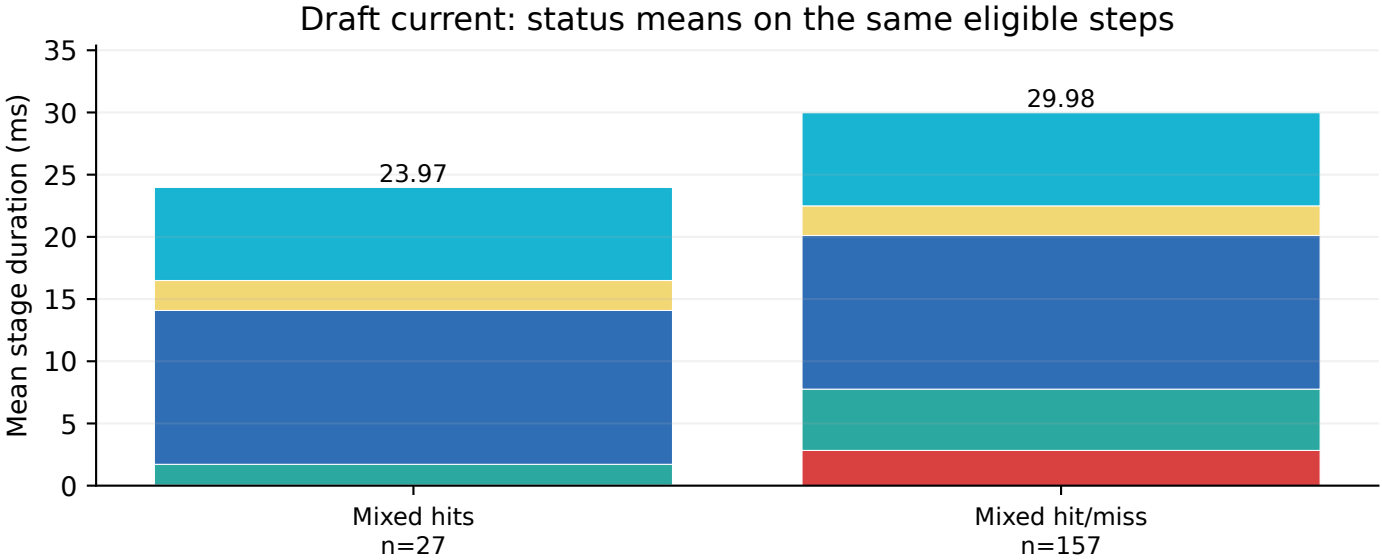
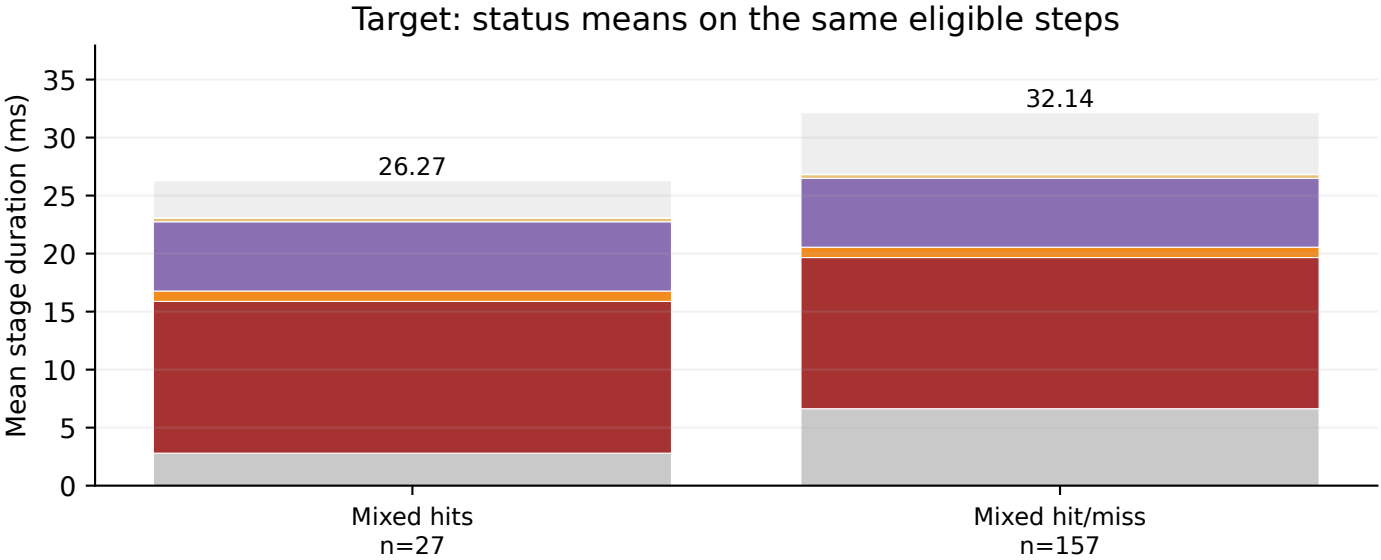
llama2_b8_tree_roots2_compute1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 192 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

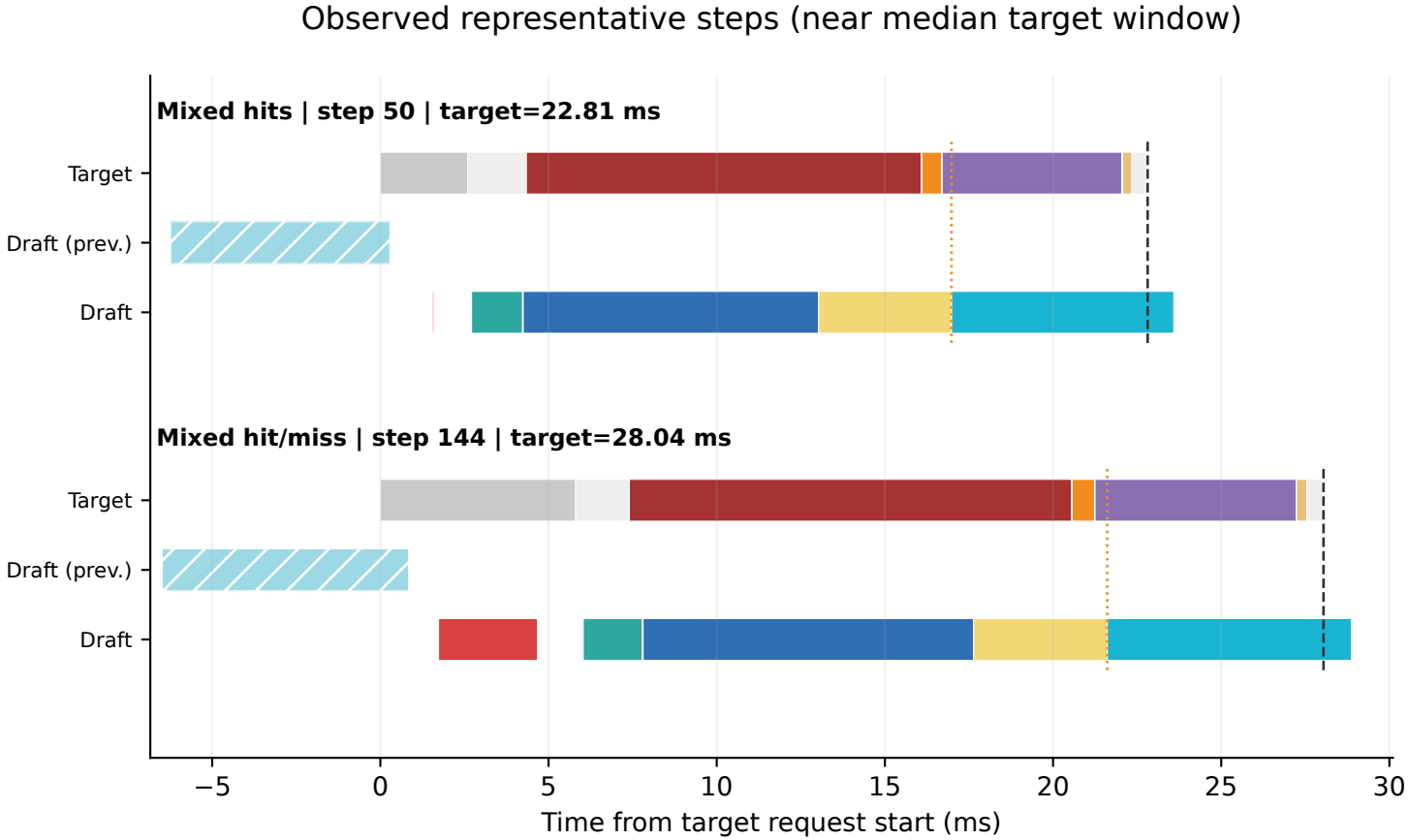
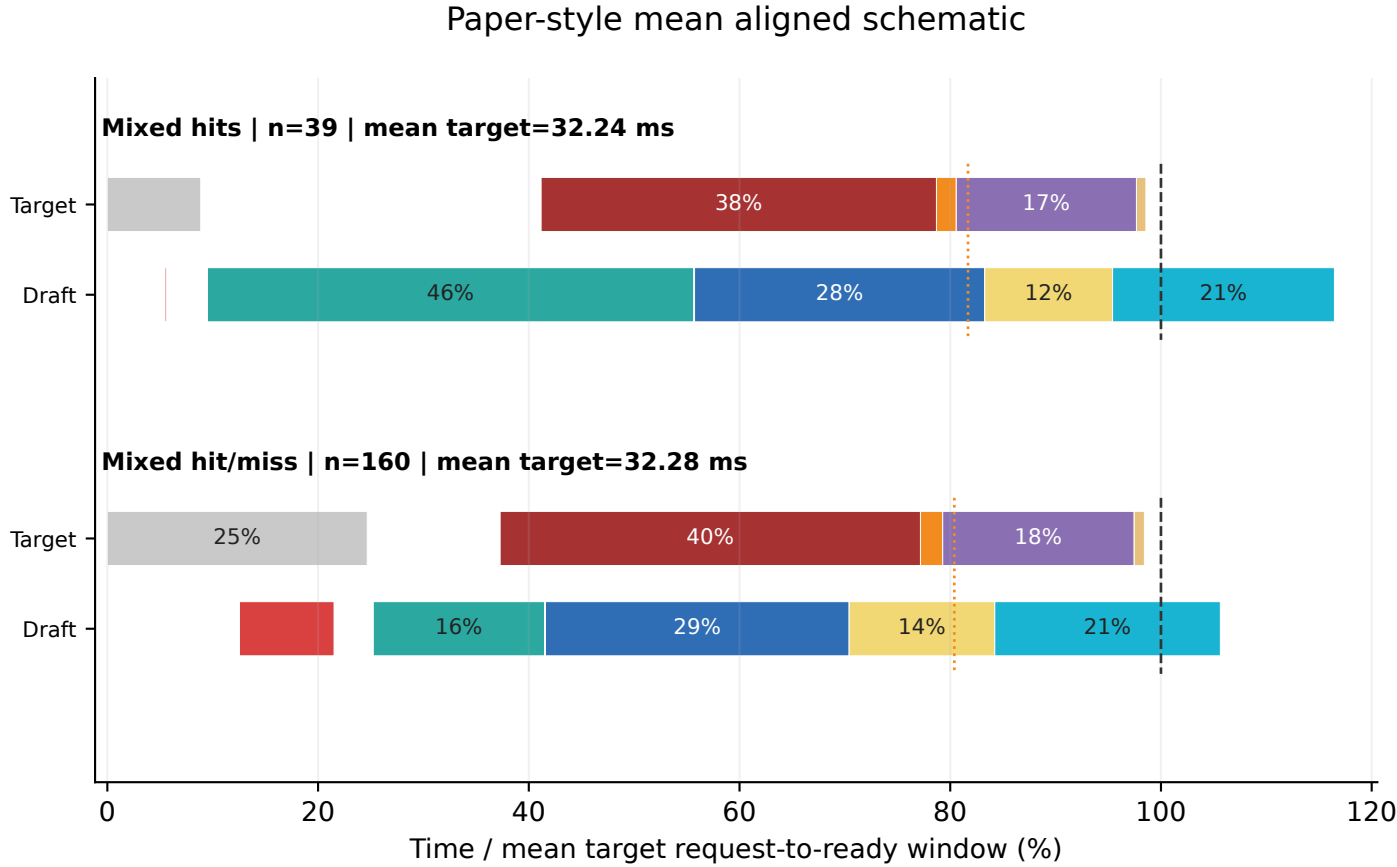
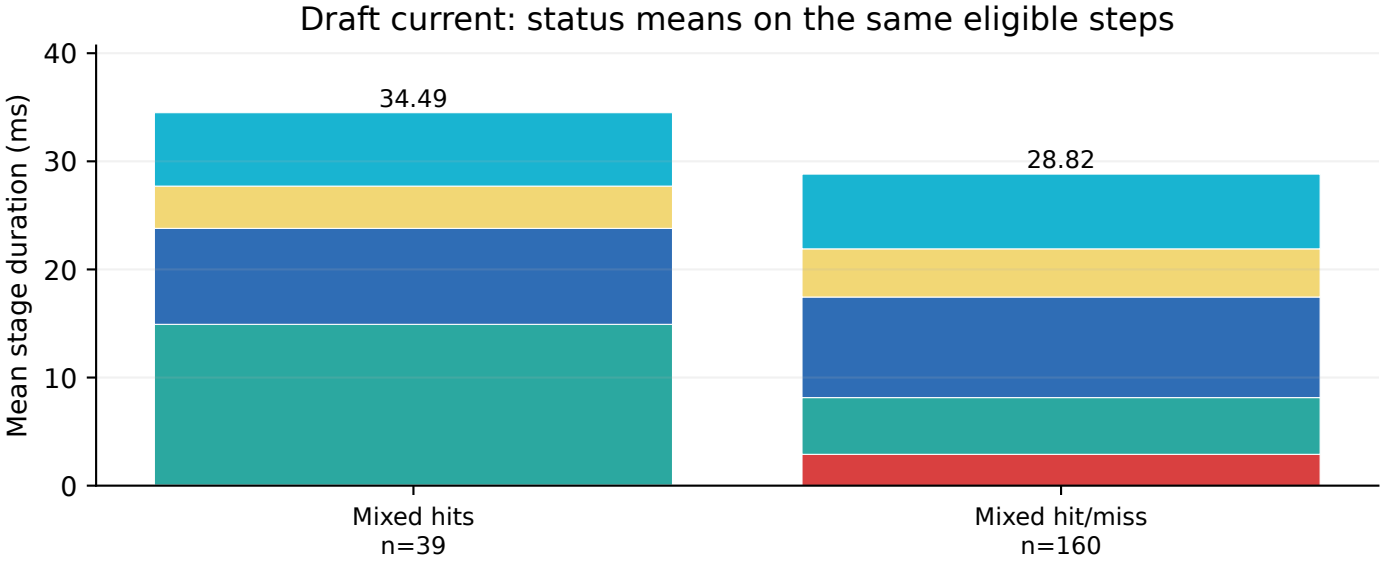
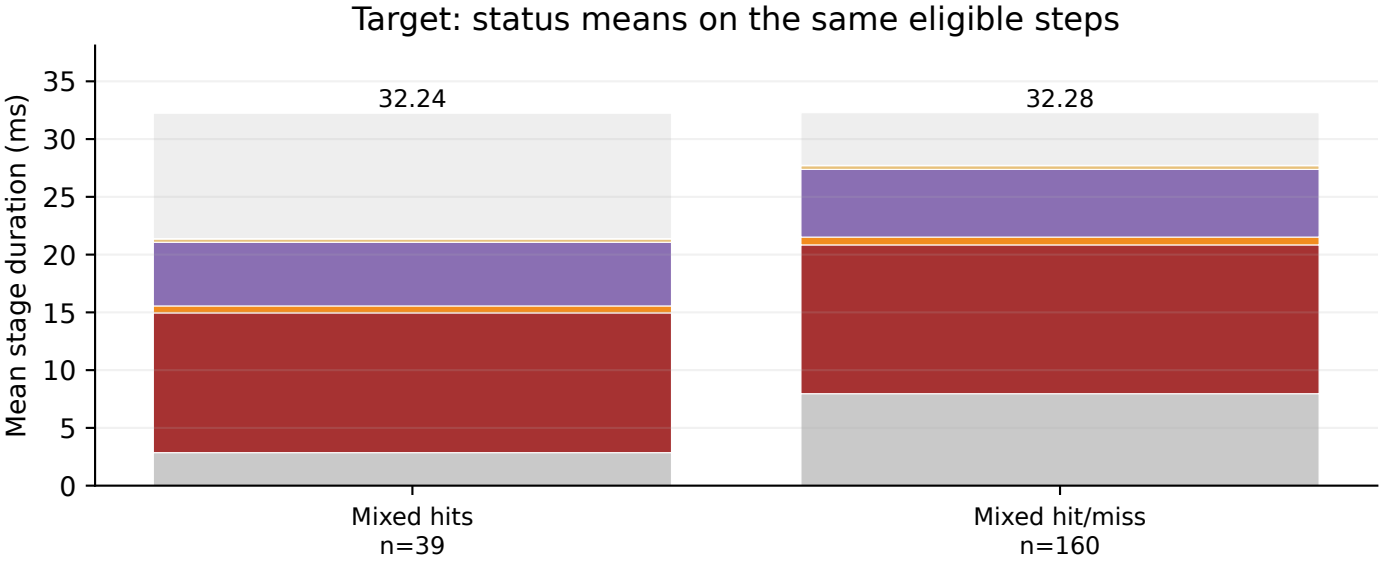
llama2_b8_tree_roots3
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=3, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 184 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

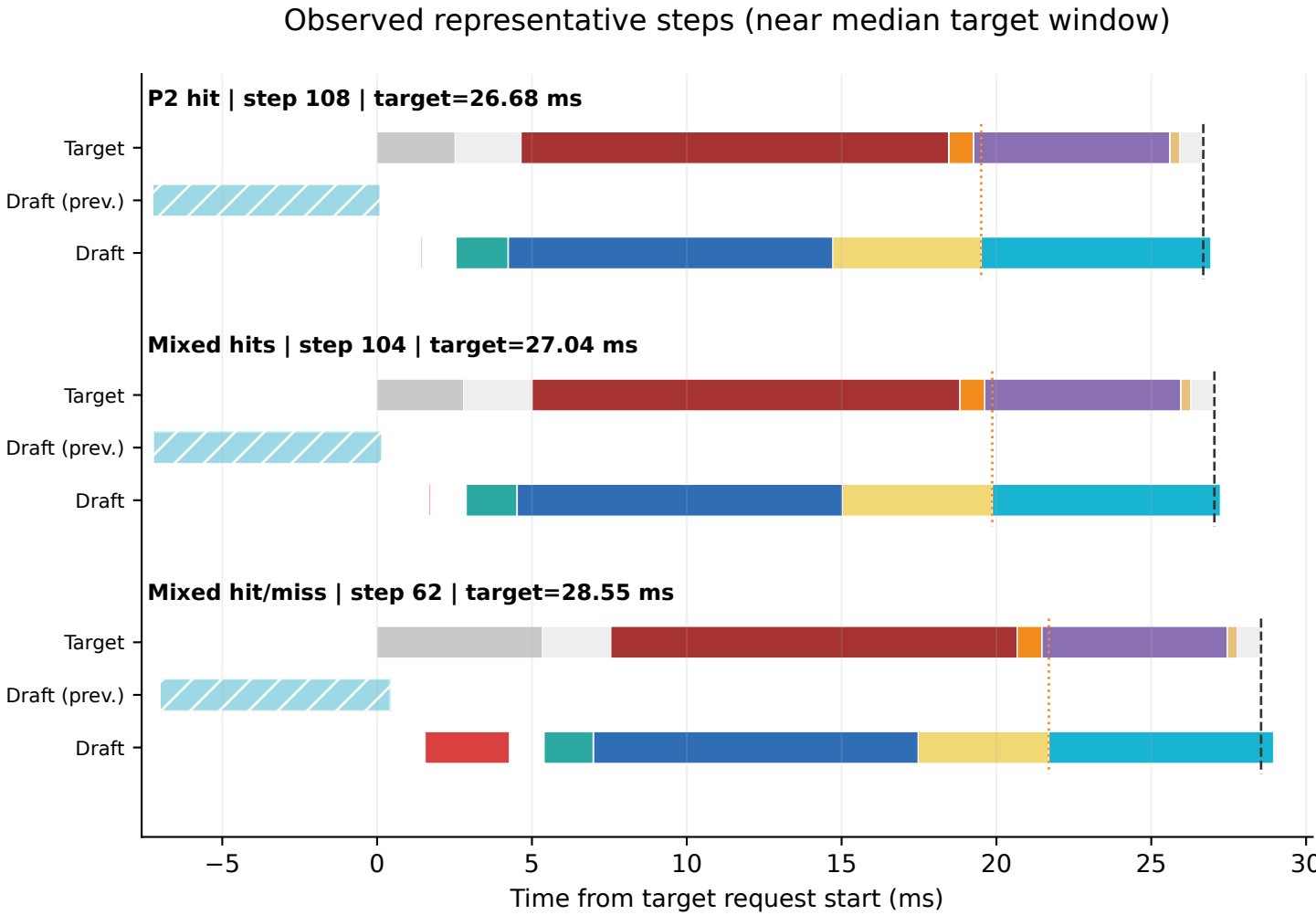
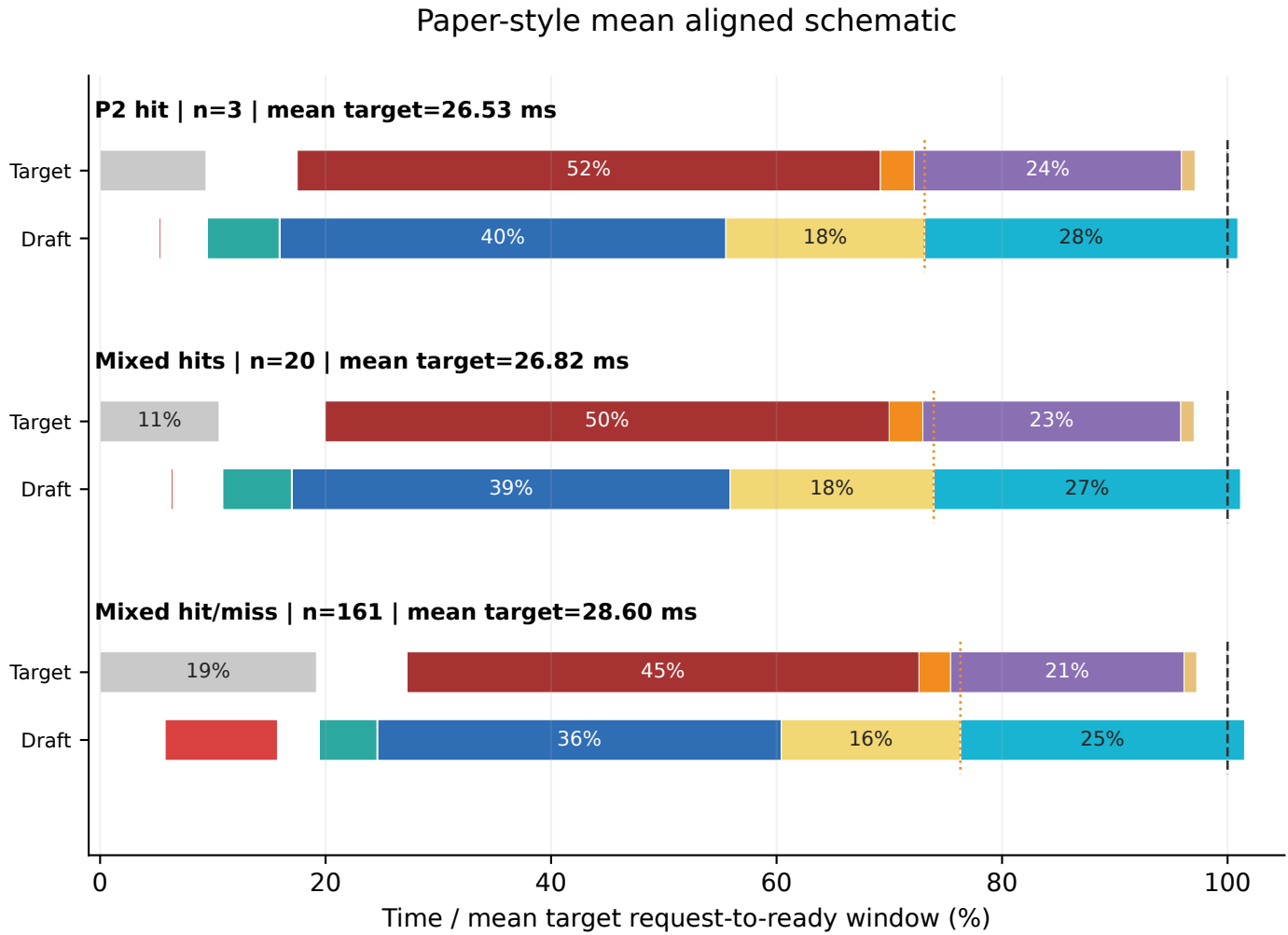
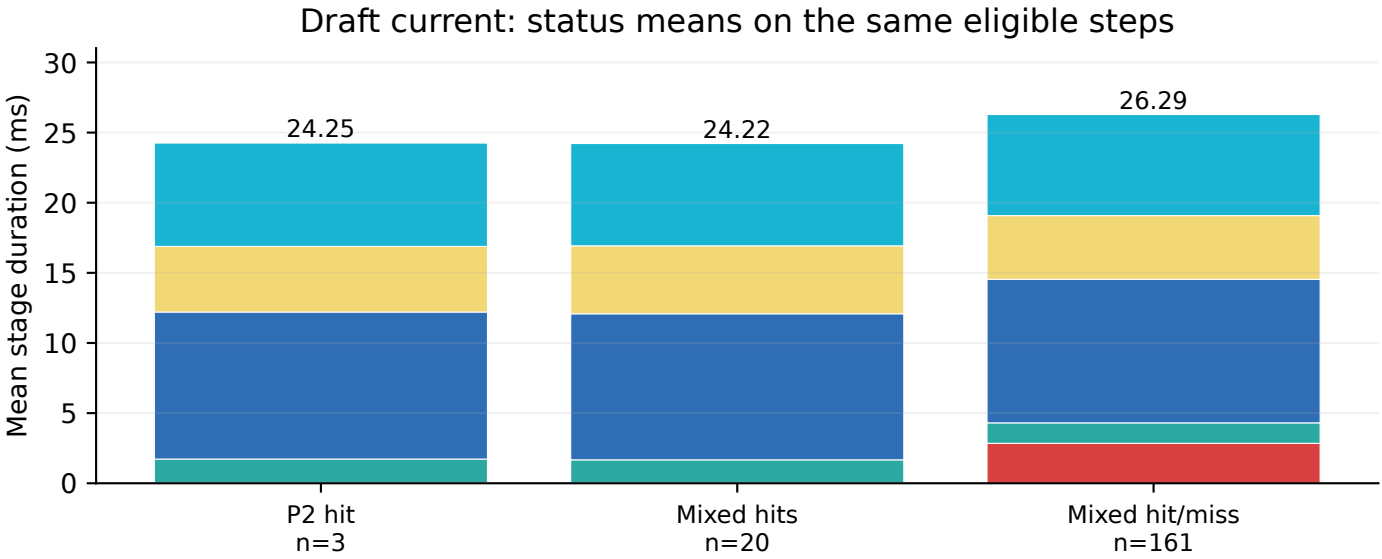
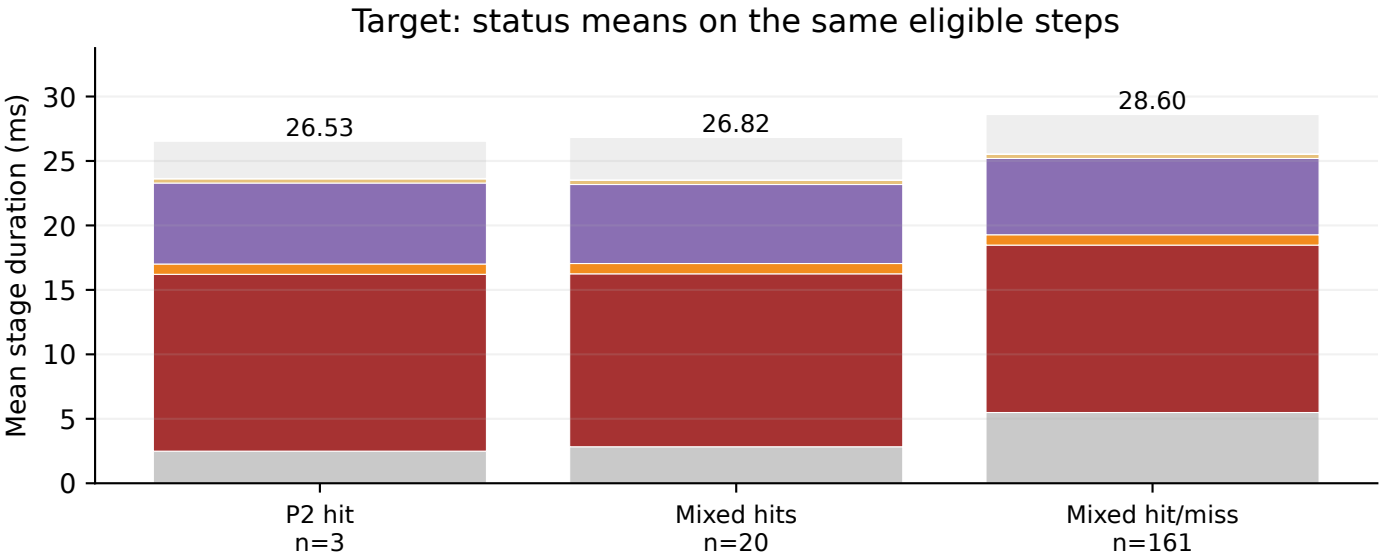
llama2_b8_tree_width1
DUET: K1/K2=3/2, exit=21 (0-based), C=1, N1/N2=3/2, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 199 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_tree_width2
DUET: K1/K2=3/2, exit=21 (0-based), C=2, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 184 eligible steps

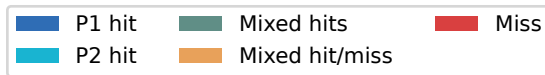
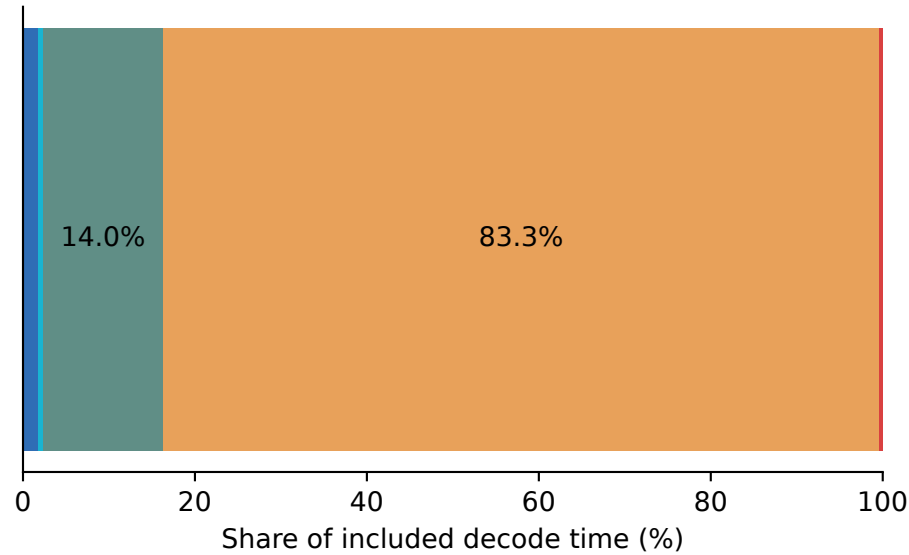


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

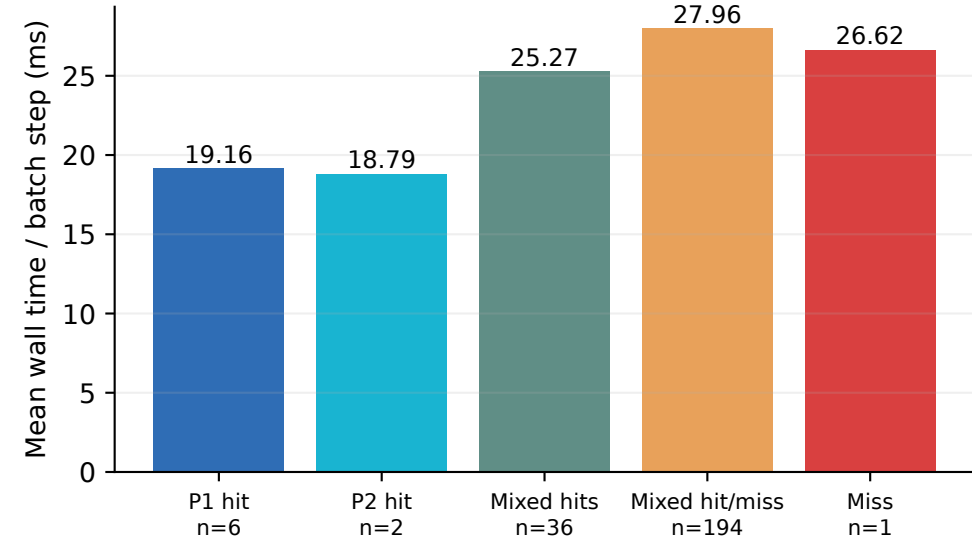
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_warm_duet_tree_1
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.123, hit=76.8%, TPS*=575.19

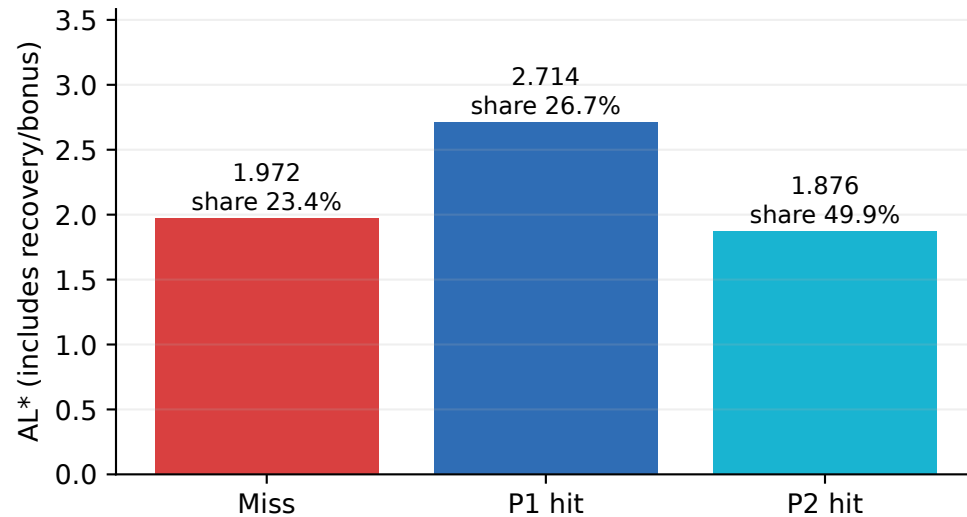
Measured wall time by batch cache status



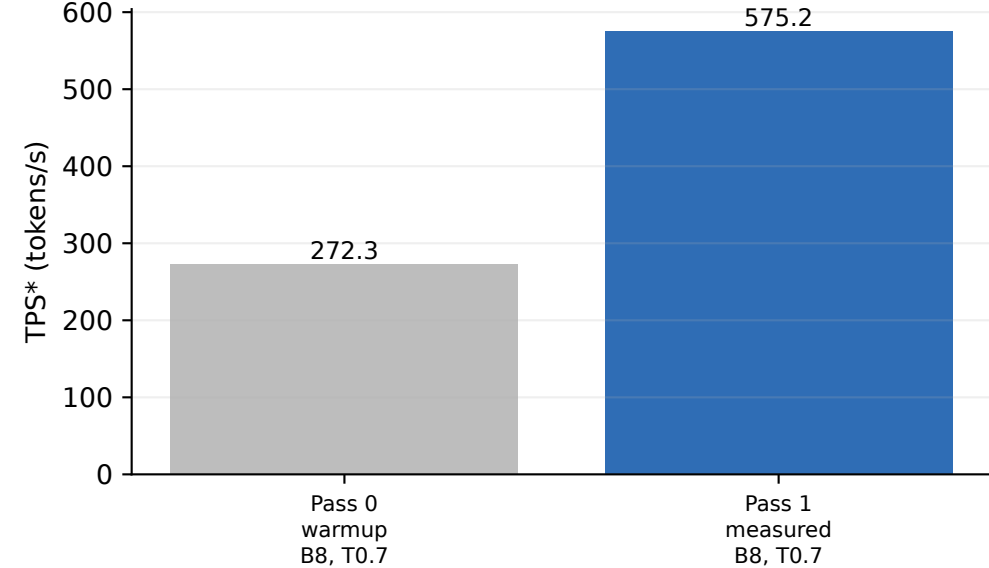
Boundary-free steps, including slow steps



Conditional AL* and source share



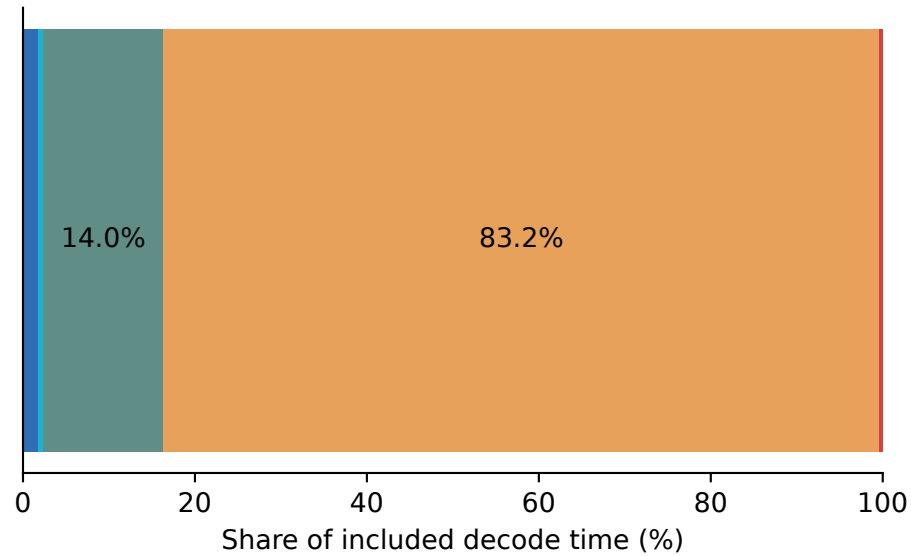
All saved passes; gray = warmup



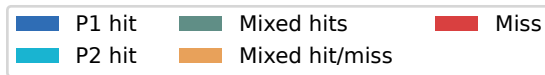
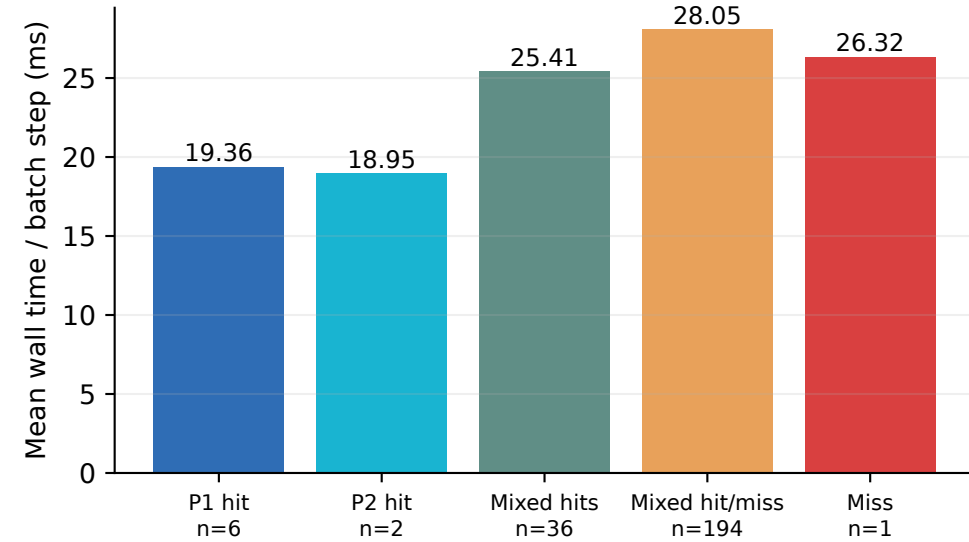
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_duet_tree_2
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.123, hit=76.8%, TPS*=573.14

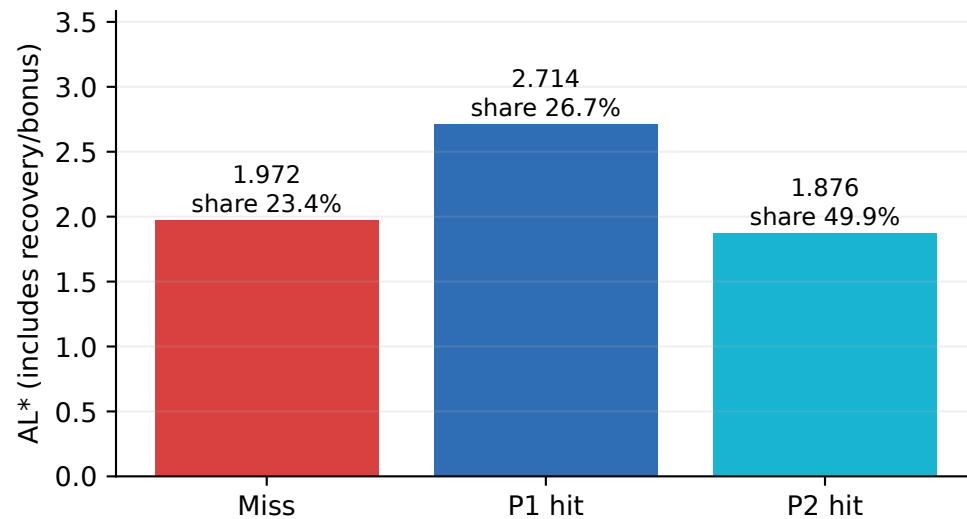
Measured wall time by batch cache status



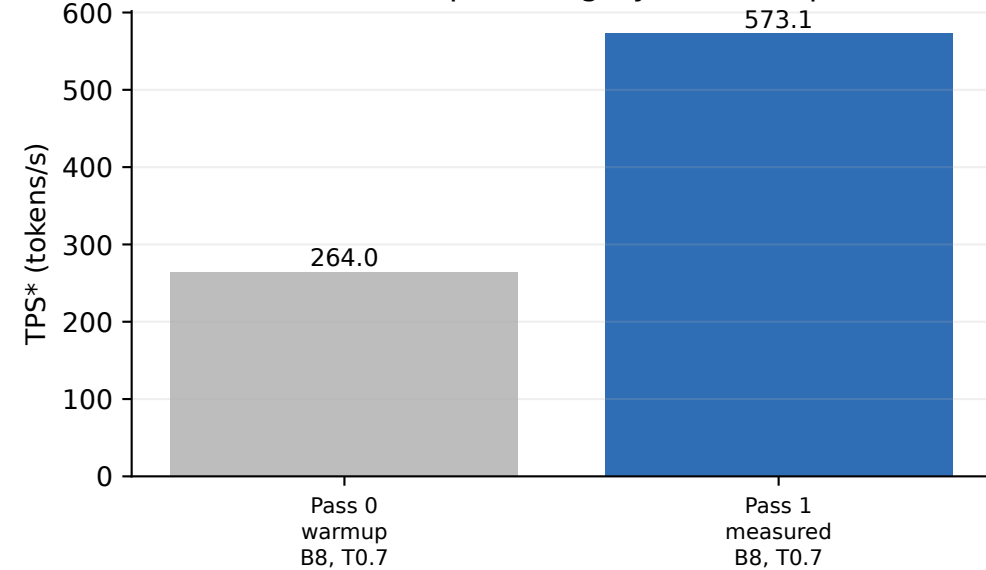
Boundary-free steps, including slow steps



Conditional AL* and source share



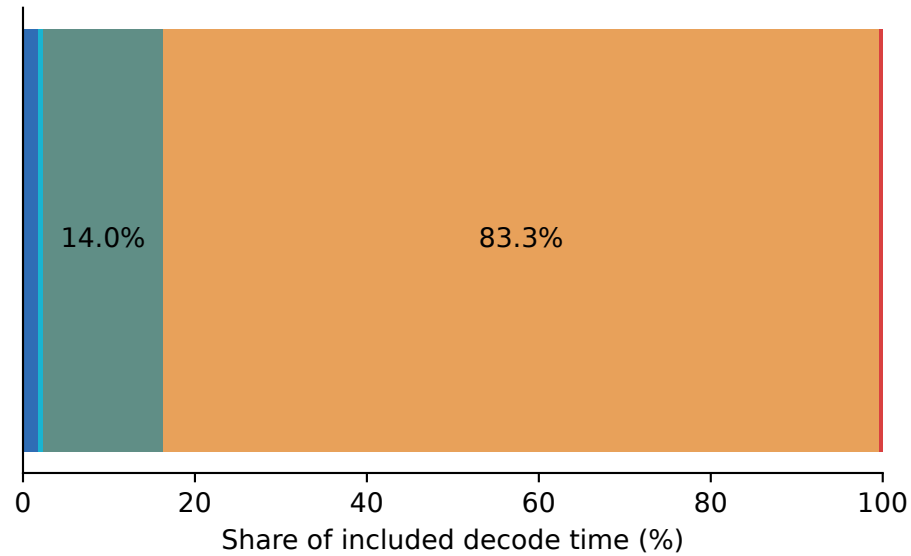
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

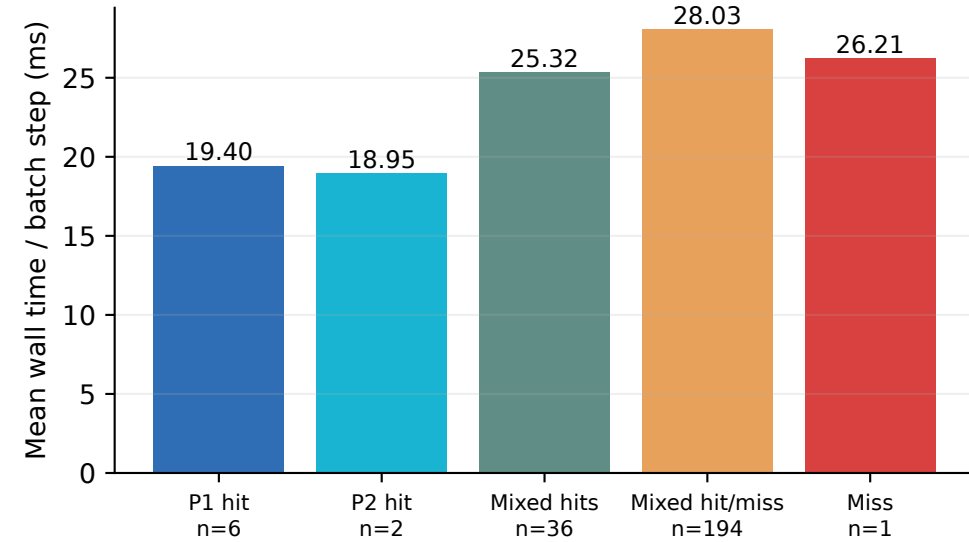
llama2_b8_warm_duet_tree_3
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
Last pass: B=8, T=0.7, N=48 | AL*=2.123, hit=76.8%, TPS*=573.72

Measured wall time by batch cache status

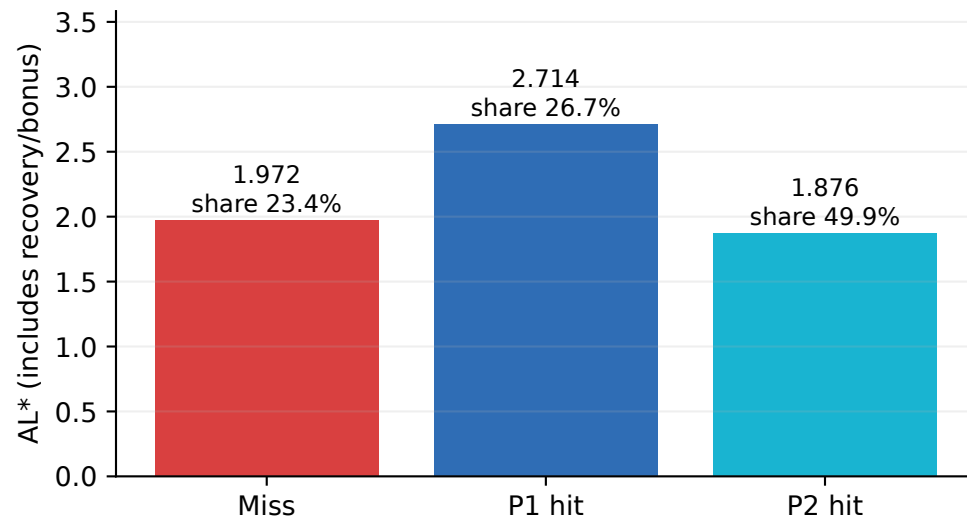


■ P1 hit ■ Mixed hits ■ Miss
■ P2 hit ■ Mixed hit/miss

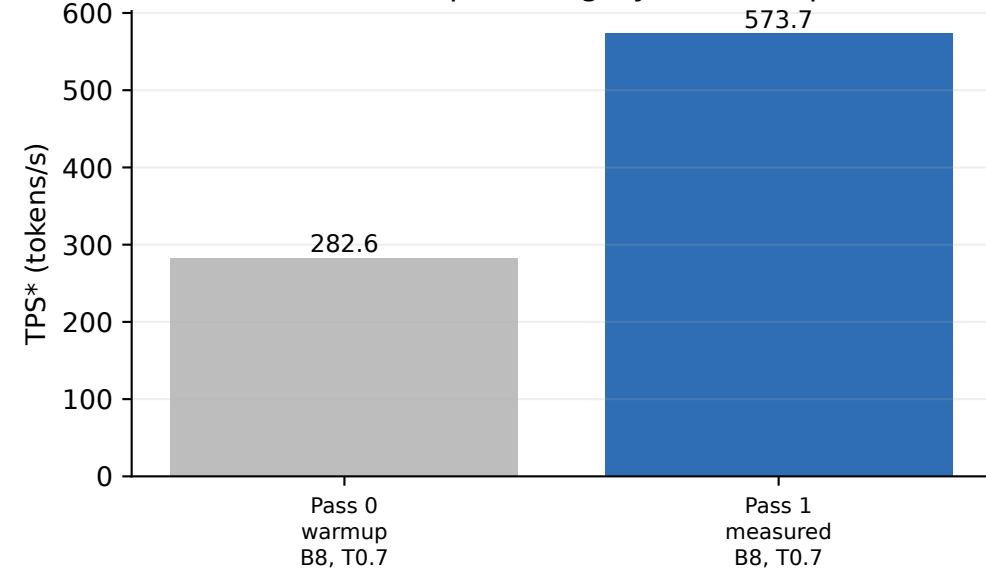
Boundary-free steps, including slow steps



Conditional AL* and source share



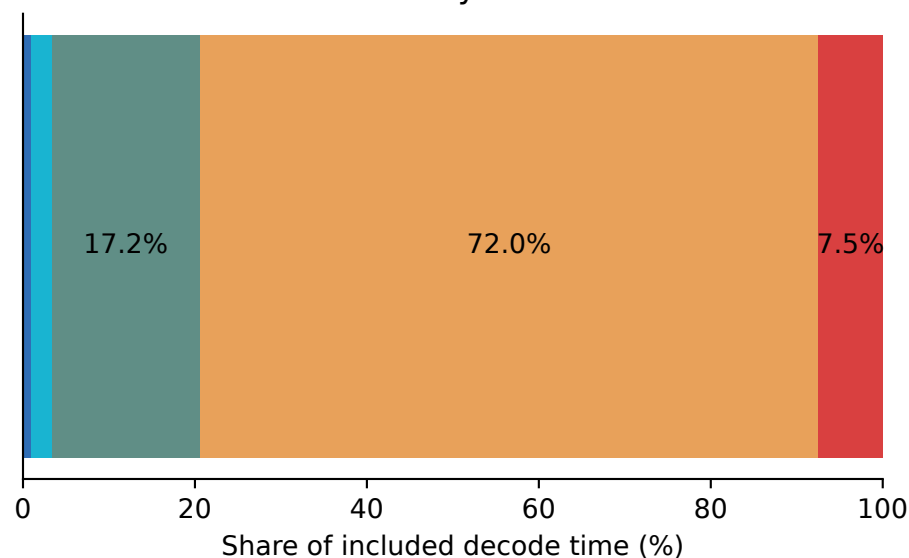
All saved passes; gray = warmup



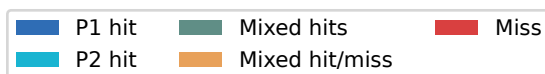
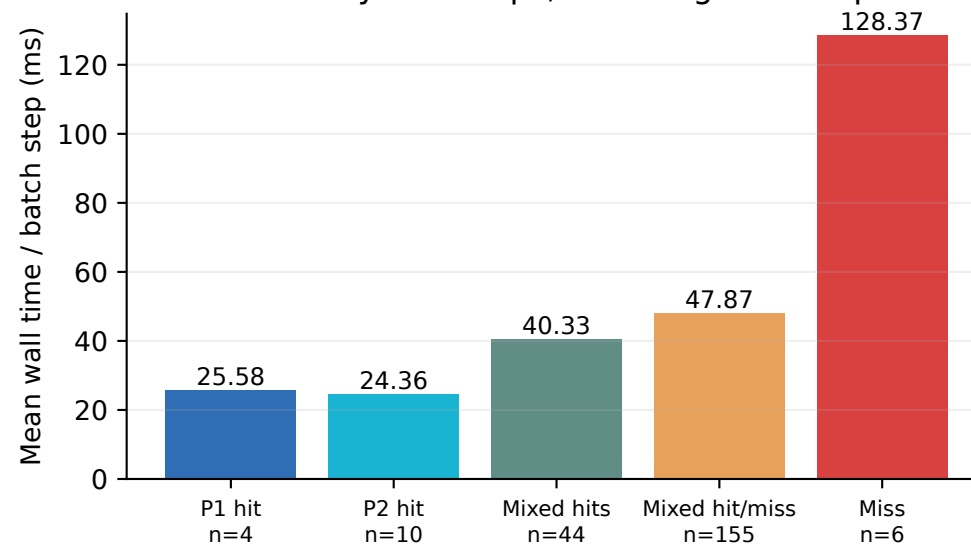
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_duet_tree_4
 DUET: K1/K2=6/3, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.281, hit=80.9%, TPS*=350.51

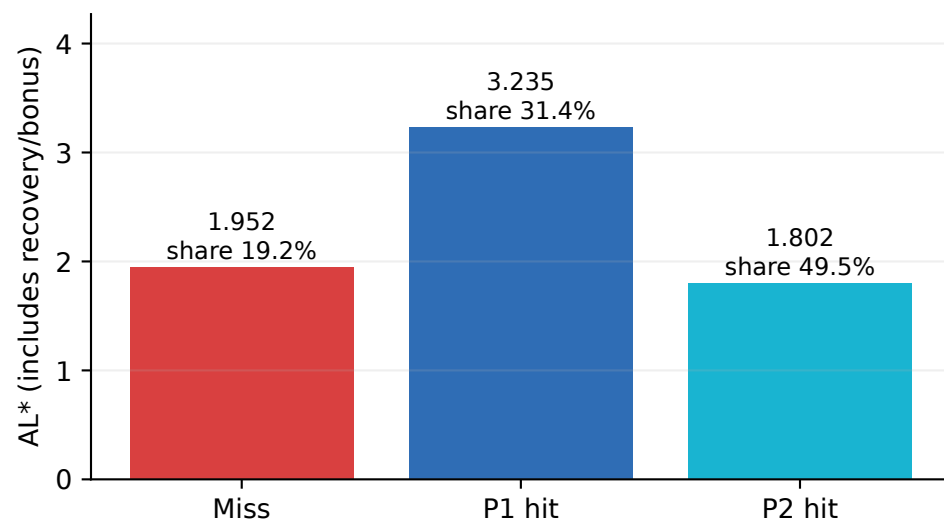
Measured wall time by batch cache status



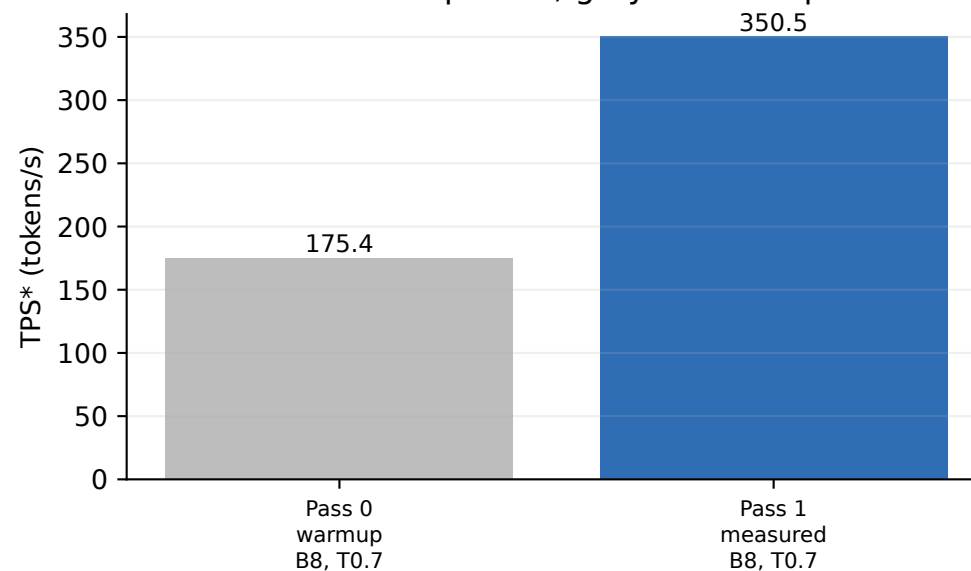
Boundary-free steps, including slow steps



Conditional AL* and source share



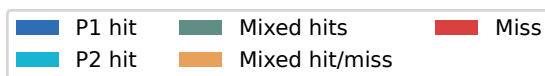
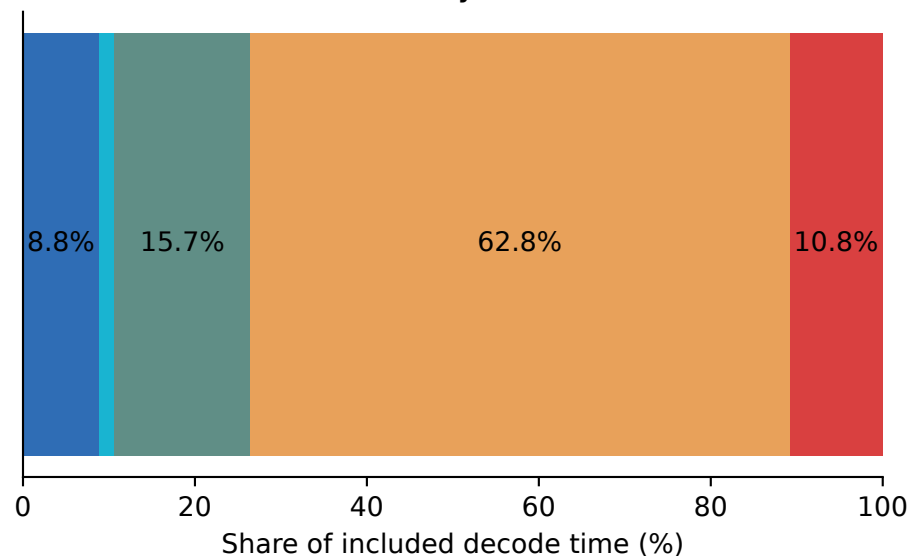
All saved passes; gray = warmup



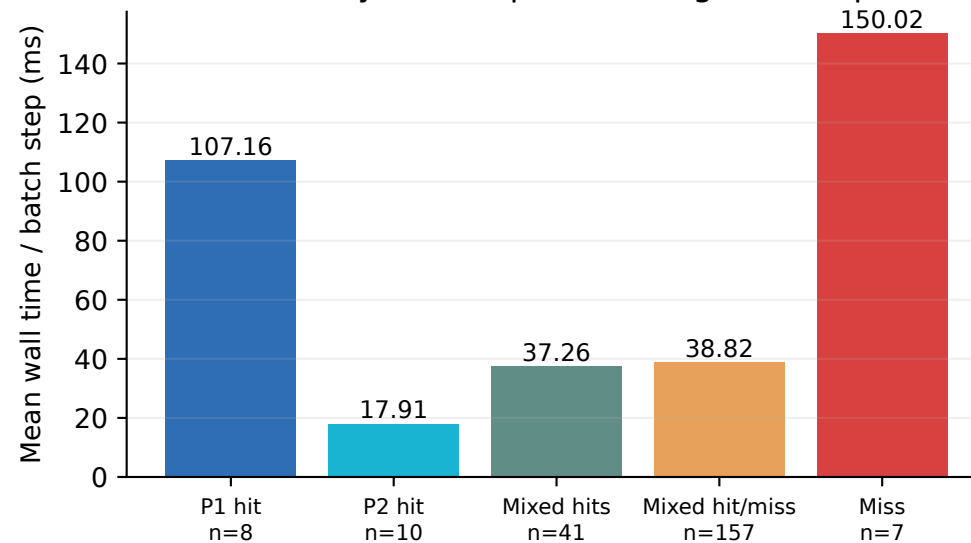
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_duet_tree_5
 DUET: K1/K2=6/2, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.313, hit=81.4%, TPS*=372.76

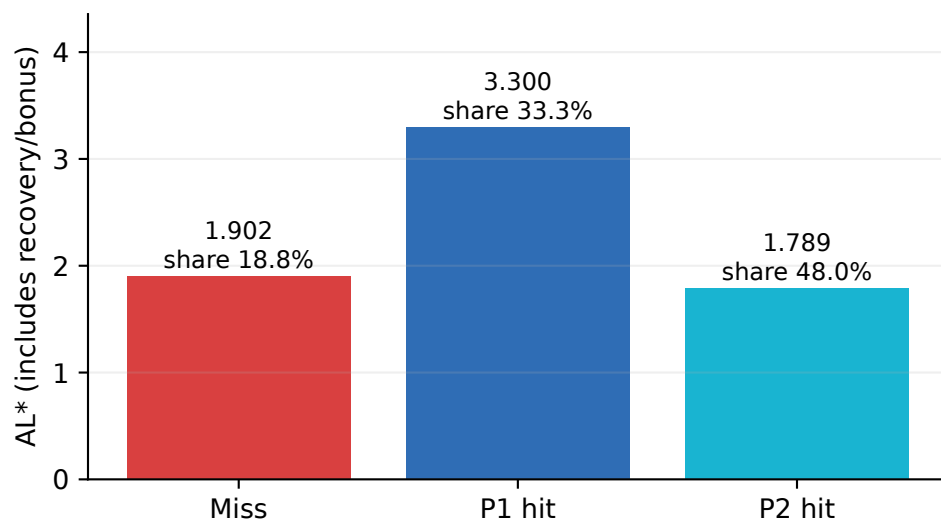
Measured wall time by batch cache status



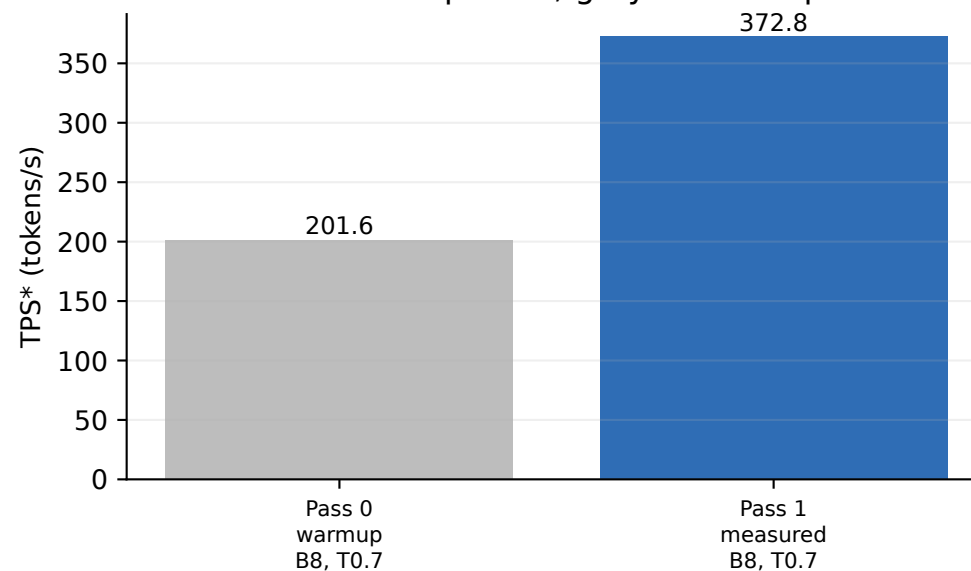
Boundary-free steps, including slow steps



Conditional AL* and source share



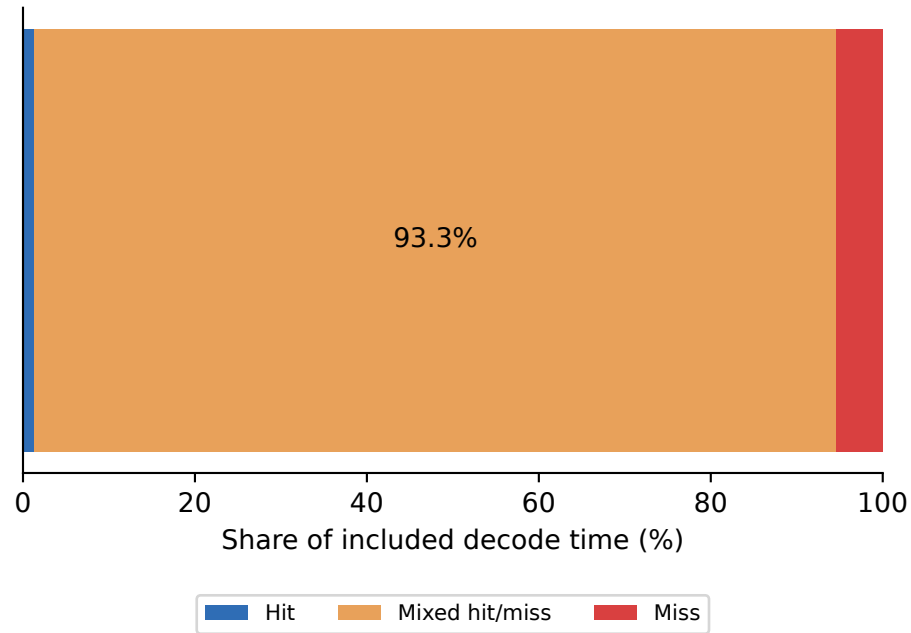
All saved passes; gray = warmup



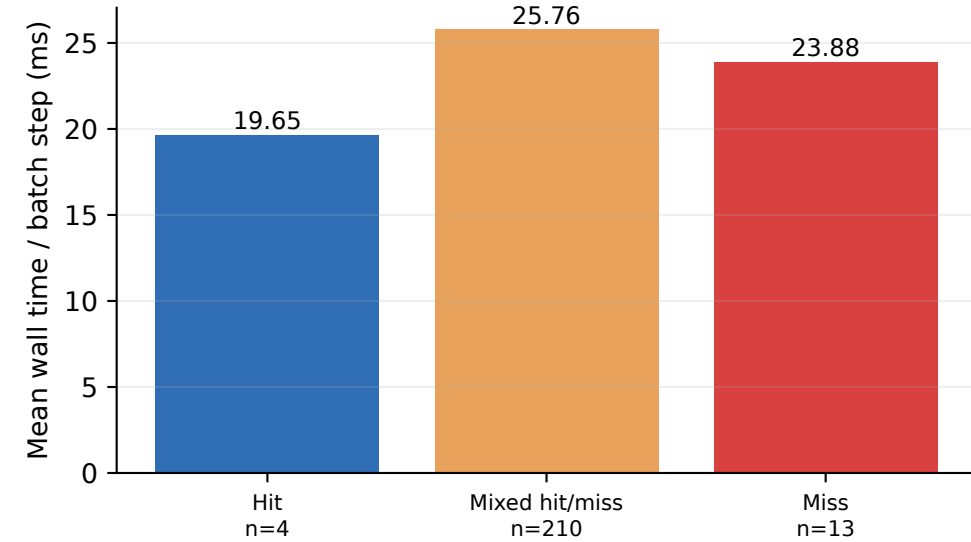
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_ssd_1
SSD: K=5, F=3
Last pass: B=8, T=0.7, N=48 | AL*=2.057, hit=37.5%, TPS*=587.76

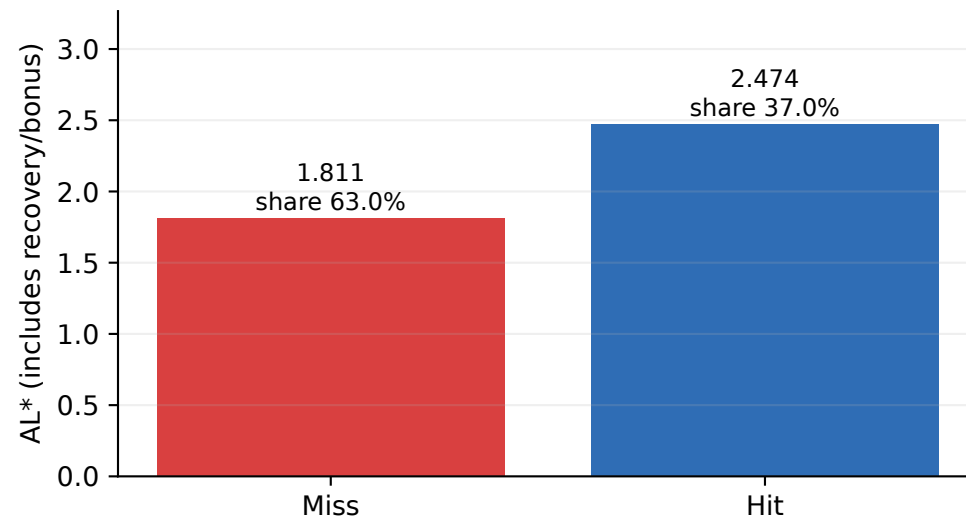
Measured wall time by batch cache status



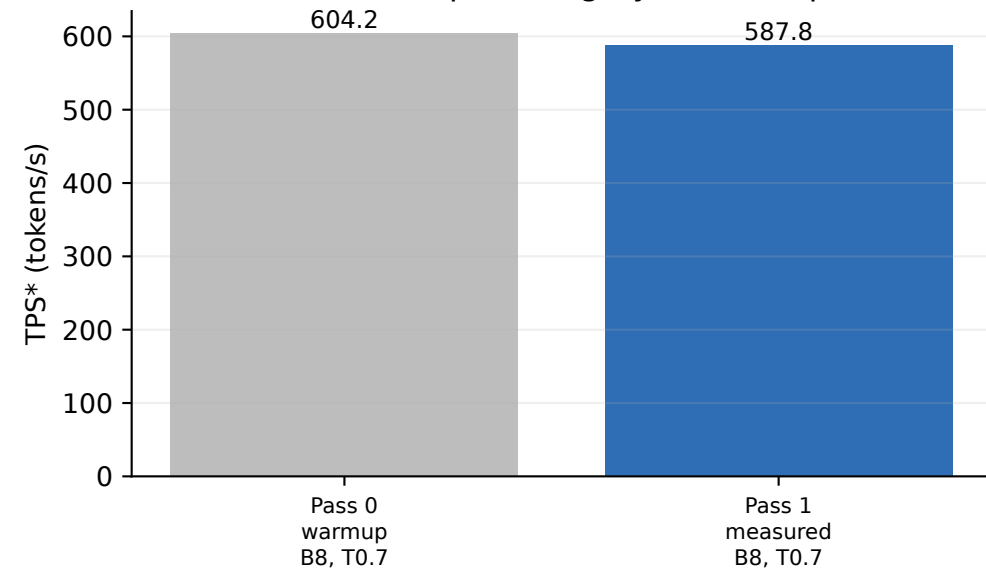
Boundary-free steps, including slow steps



Conditional AL* and source share



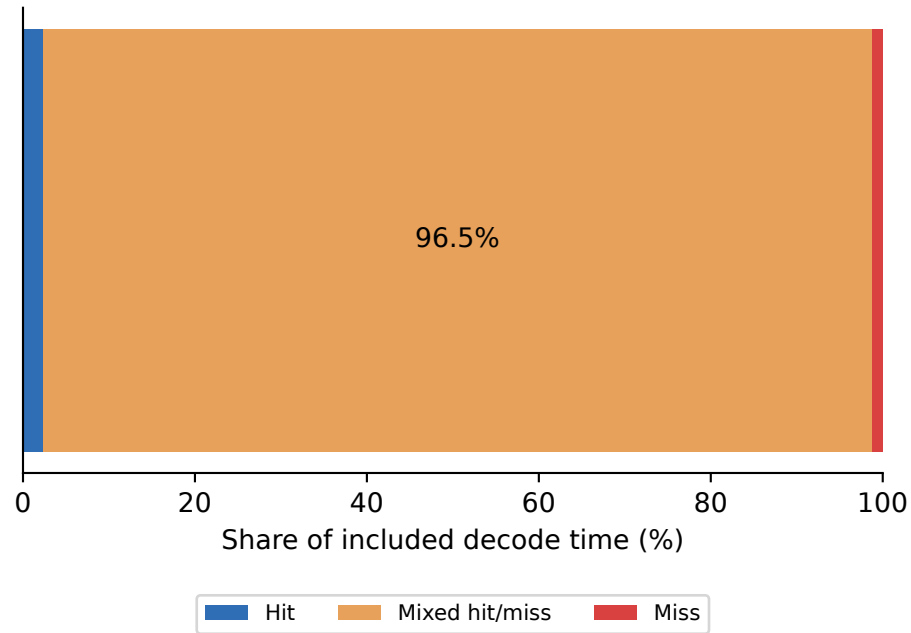
All saved passes; gray = warmup



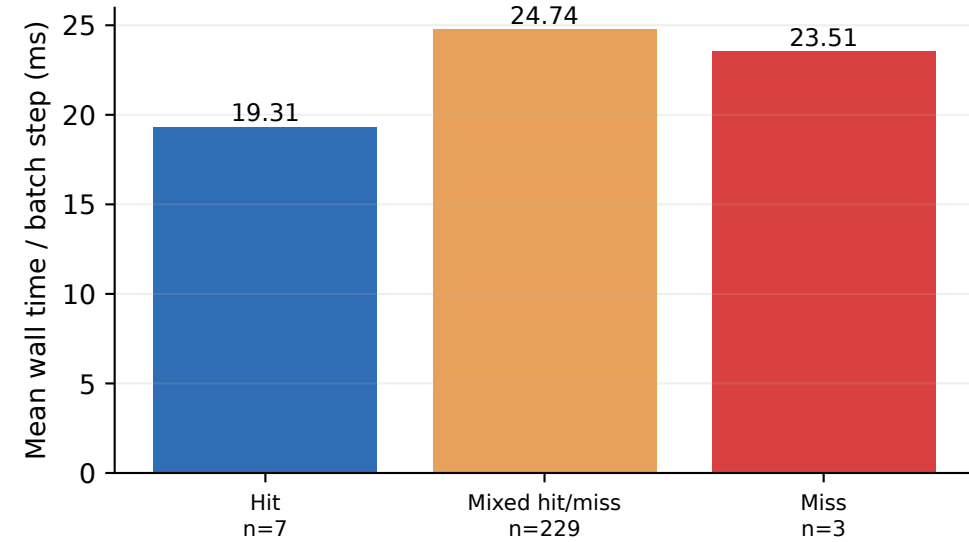
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_ssd_2
SSD: K=4, F=7
Last pass: B=8, T=0.7, N=48 | AL*=2.126, hit=52.5%, TPS*=650.61

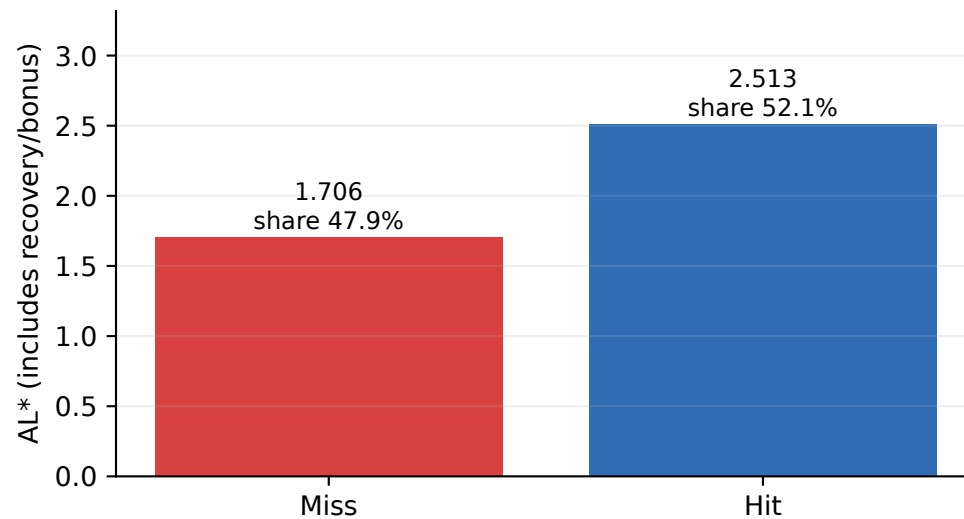
Measured wall time by batch cache status



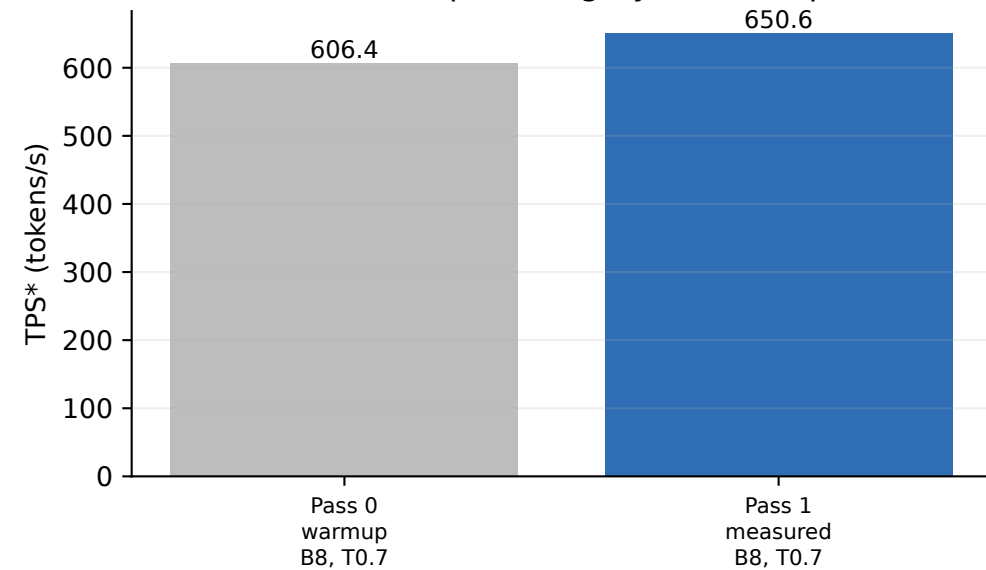
Boundary-free steps, including slow steps



Conditional AL* and source share



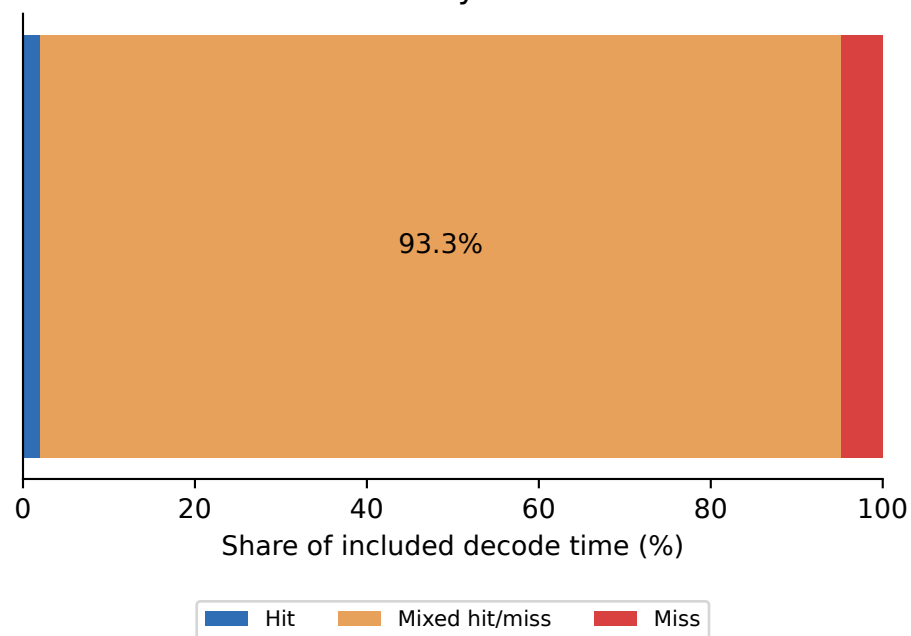
All saved passes; gray = warmup



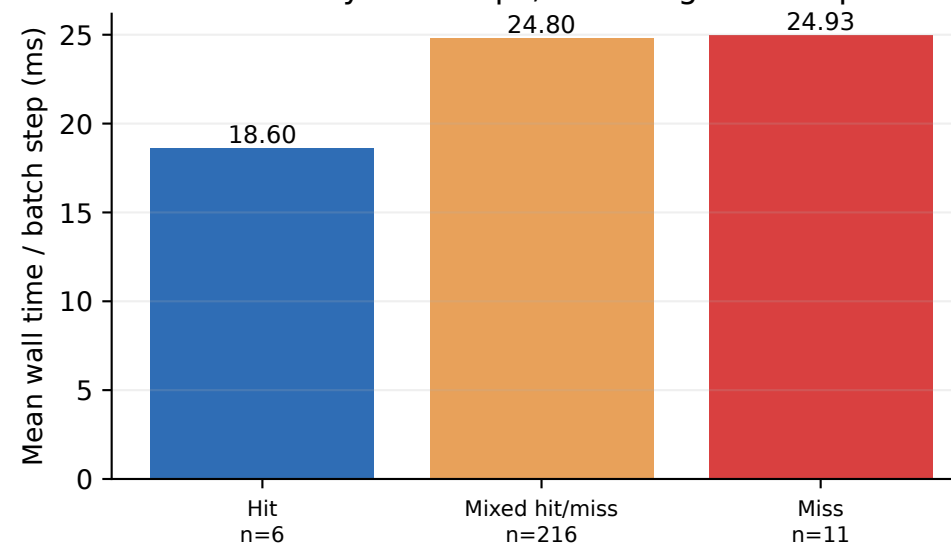
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_ssd_3
SSD: K=4, F=3
Last pass: B=8, T=0.7, N=48 | AL*=2.025, hit=39.0%, TPS*=621.26

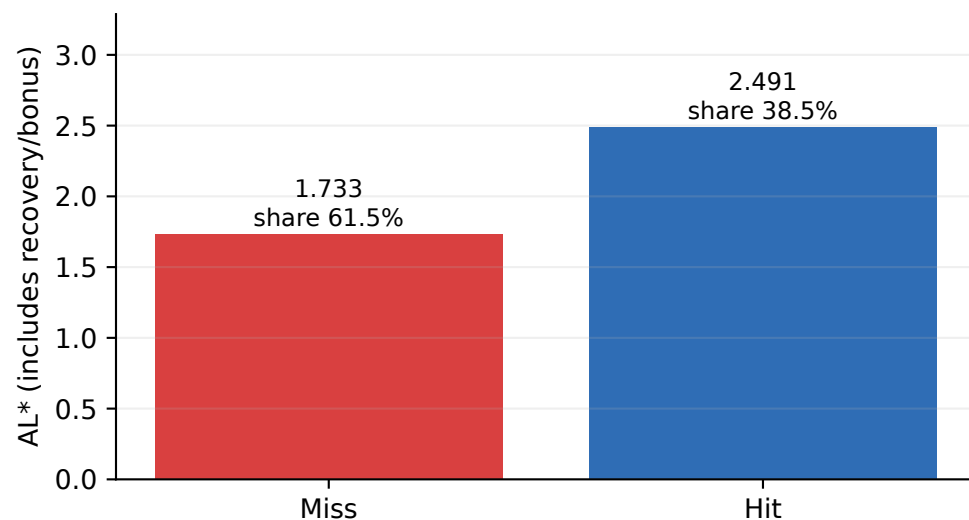
Measured wall time by batch cache status



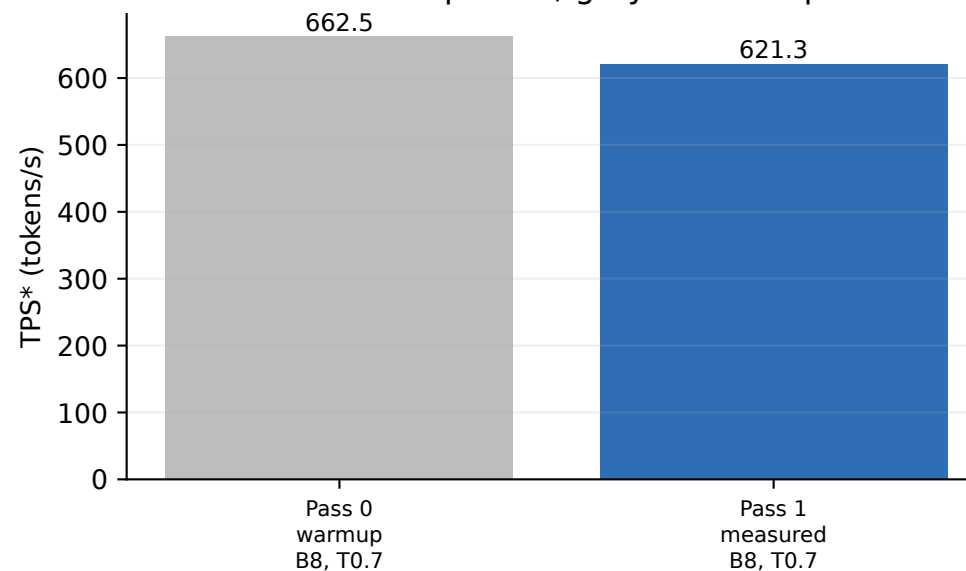
Boundary-free steps, including slow steps



Conditional AL* and source share



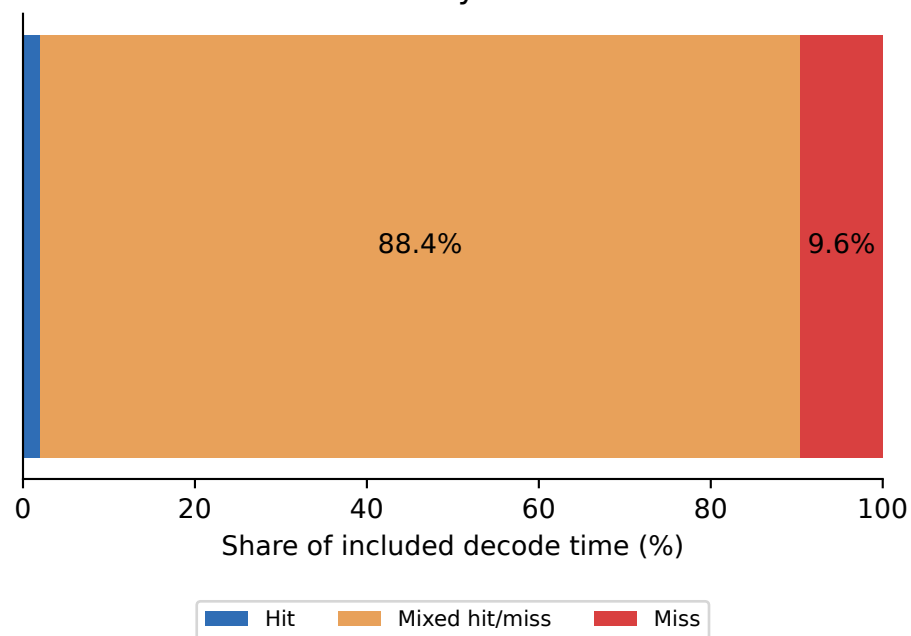
All saved passes; gray = warmup



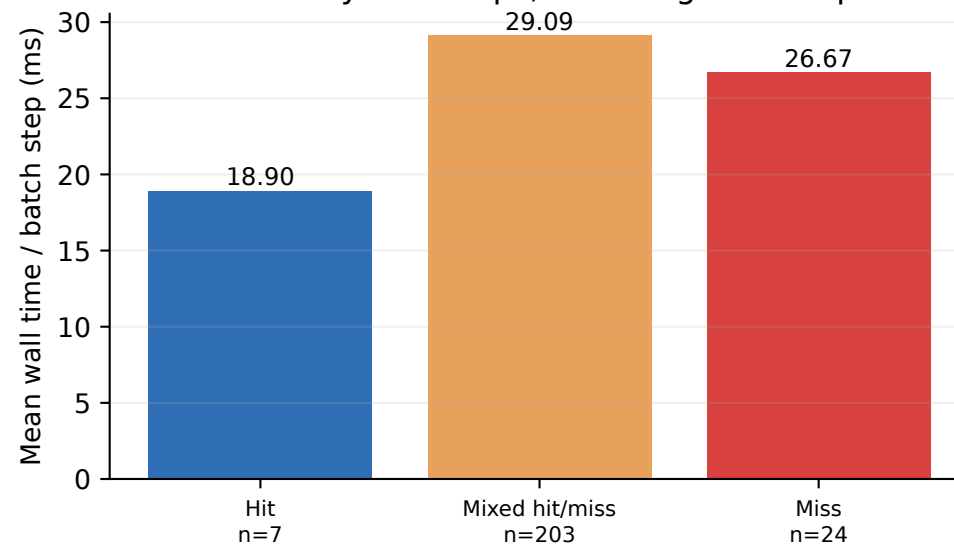
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_ssd_4
SSD: K=8, F=3
Last pass: B=8, T=0.7, N=48 | AL*=2.184, hit=36.3%, TPS*=552.96

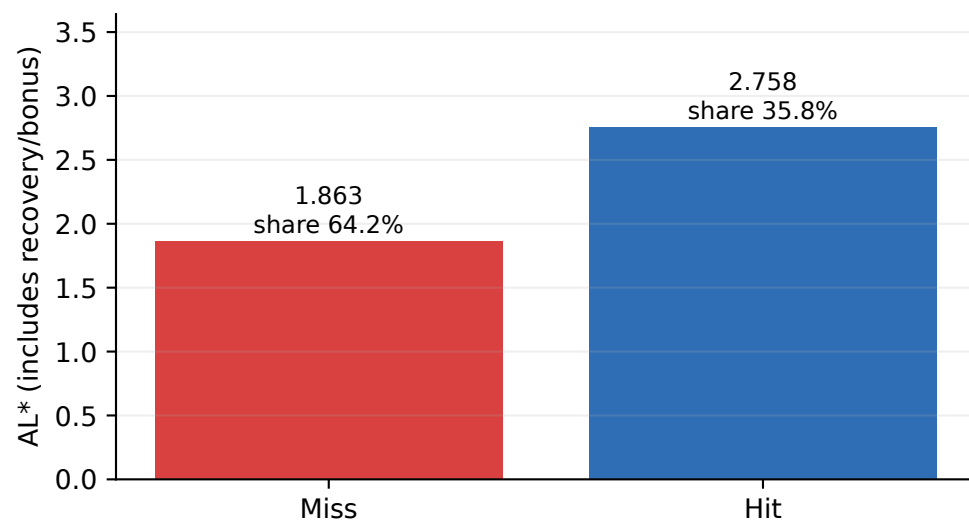
Measured wall time by batch cache status



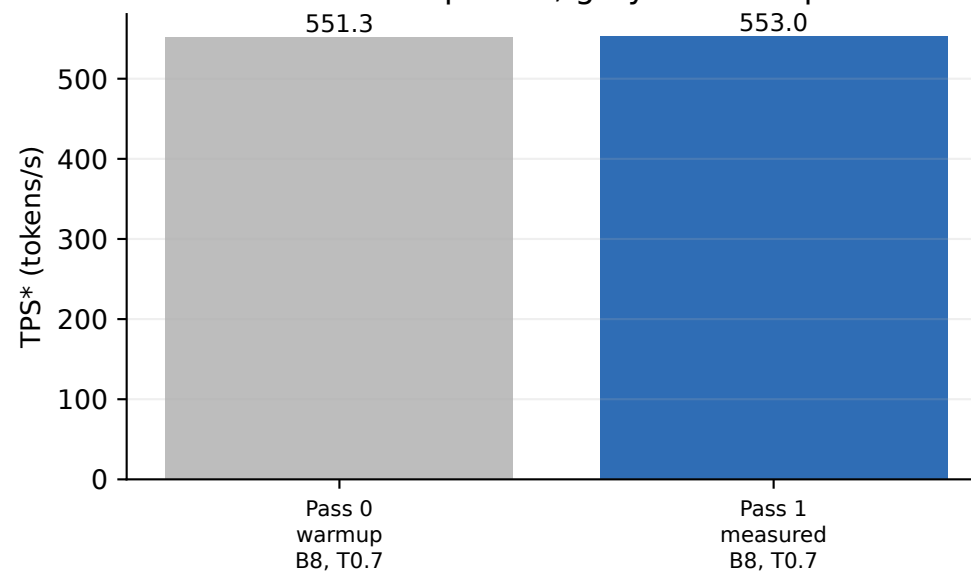
Boundary-free steps, including slow steps



Conditional AL* and source share



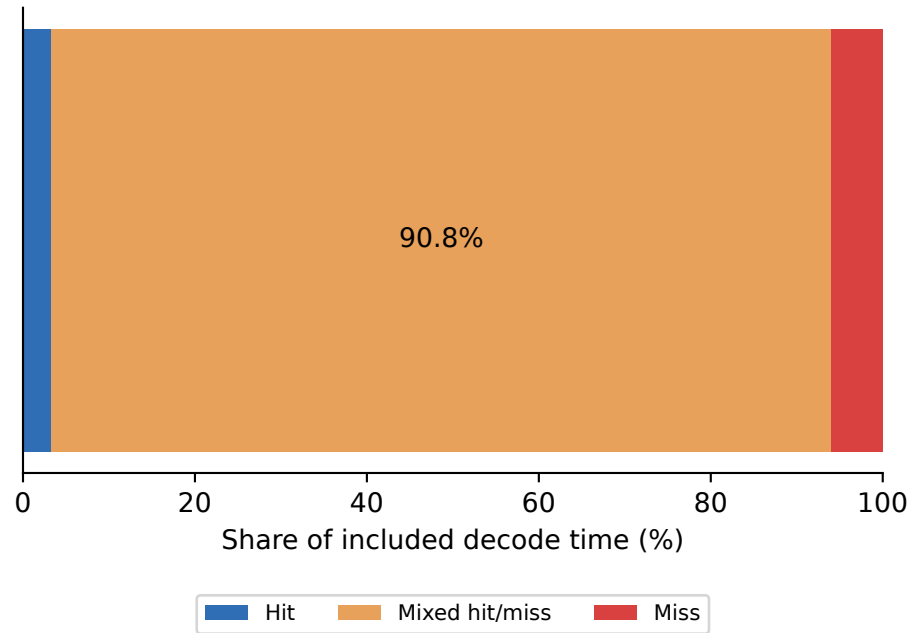
All saved passes; gray = warmup



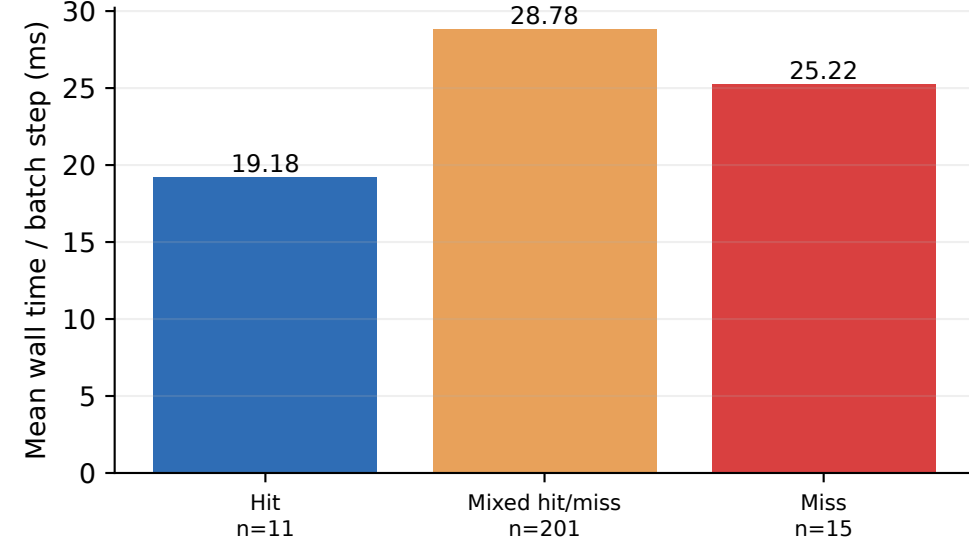
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_warm_ssd_5
SSD: K=8, F=5
Last pass: B=8, T=0.7, N=48 | AL*=2.275, hit=43.9%, TPS*=584.92

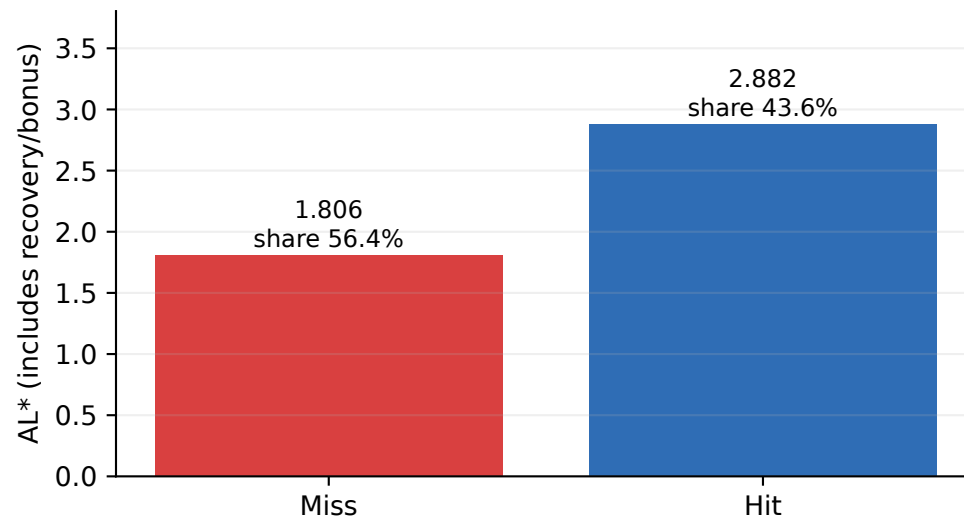
Measured wall time by batch cache status



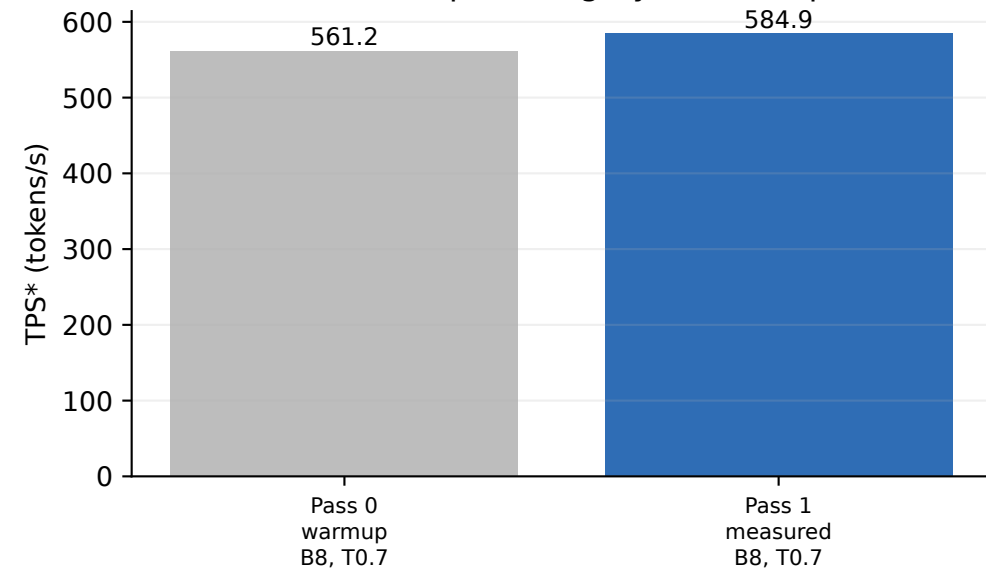
Boundary-free steps, including slow steps



Conditional AL* and source share

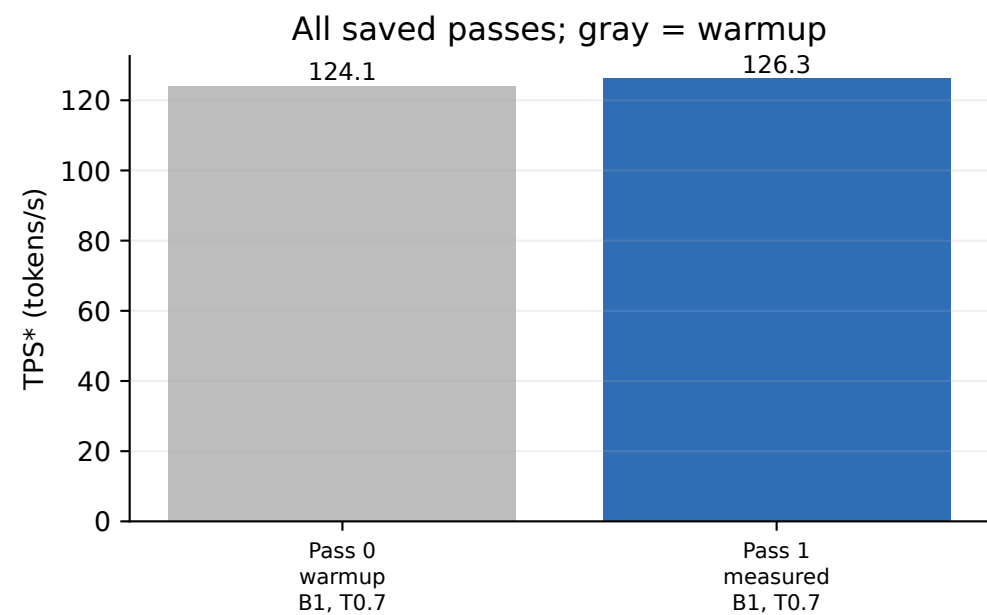
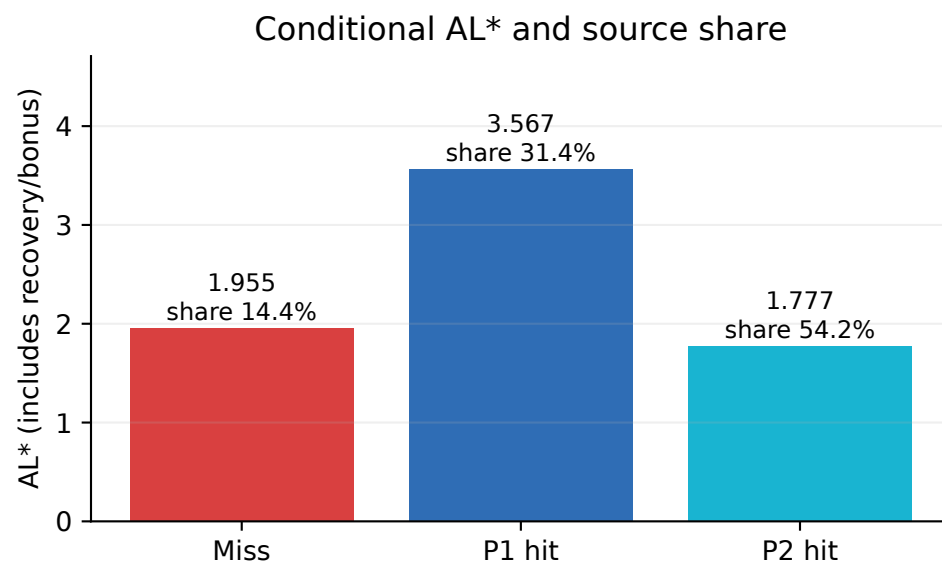
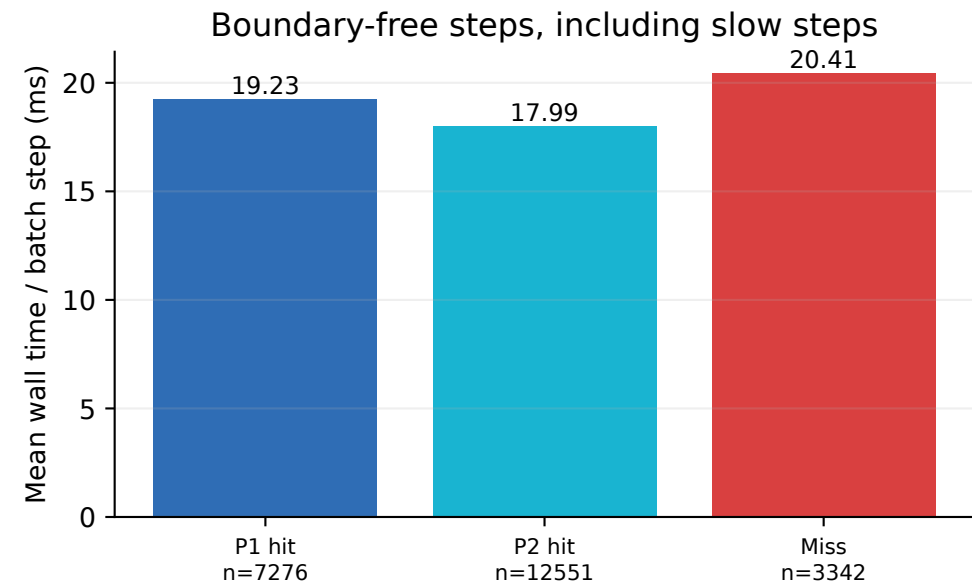
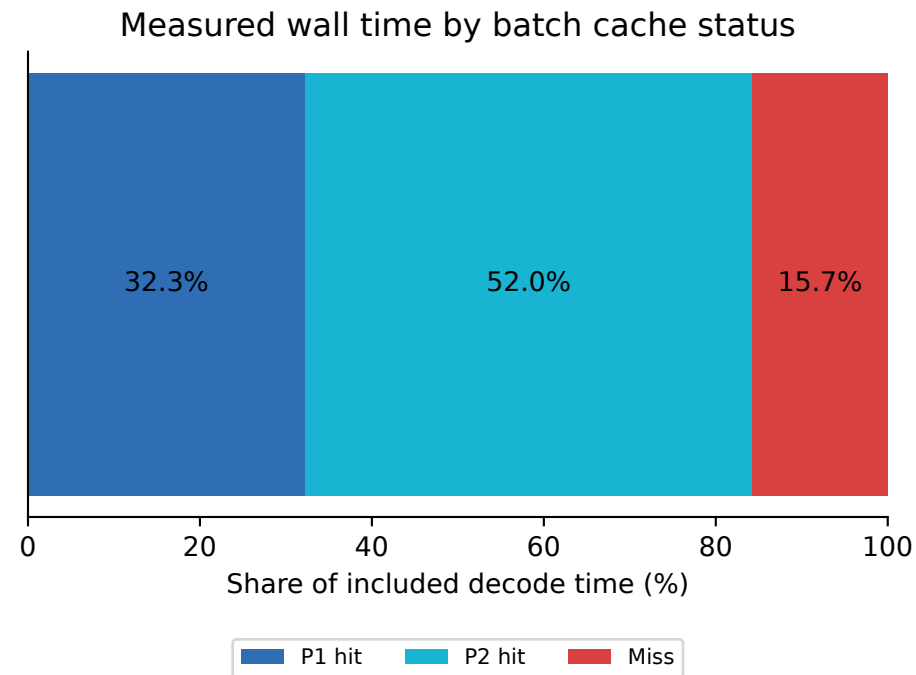


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

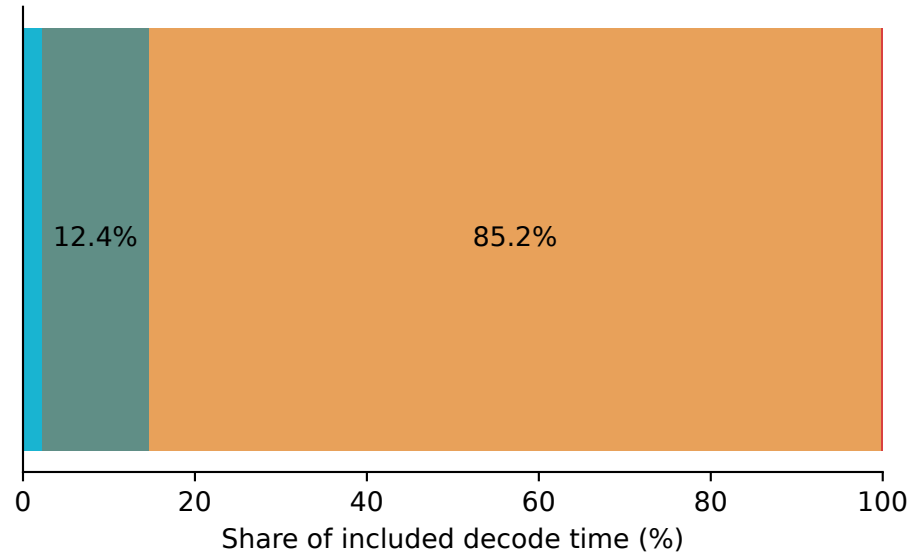
llama2_b1_opt_fast_r0
 DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/0
 Last pass: B=1, T=0.7, N=480 | AL*=2.365, hit=85.7%, TPS*=126.27



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

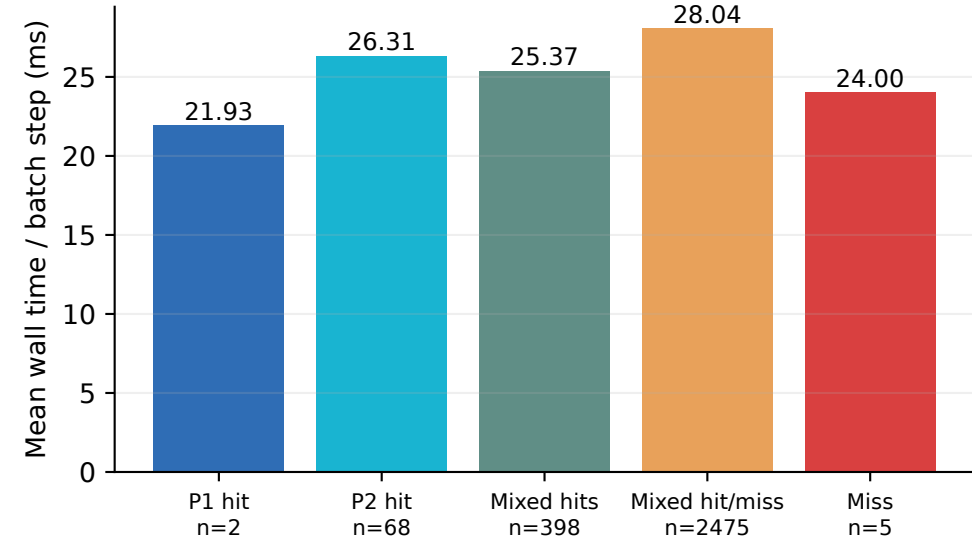
llama2_b8_opt_fast_r1
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.090, hit=77.1%, TPS*=597.28

Measured wall time by batch cache status

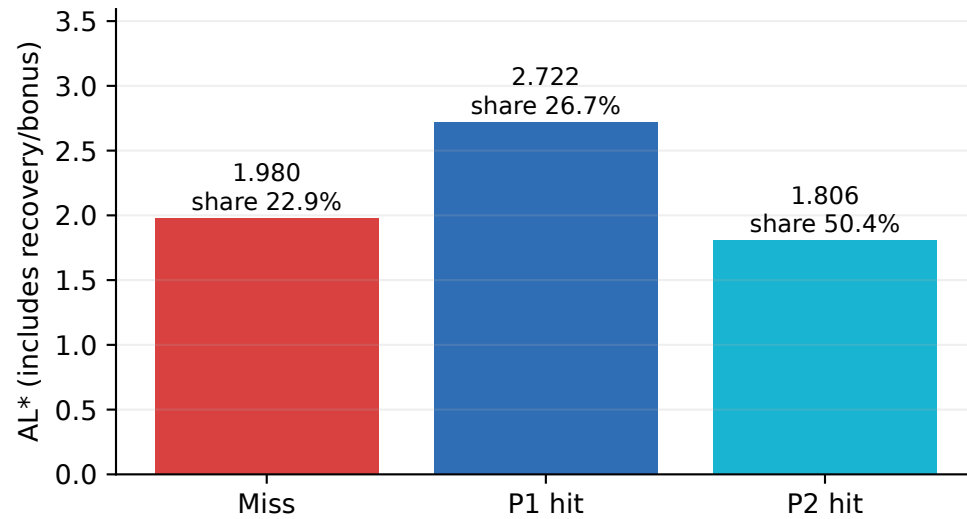


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

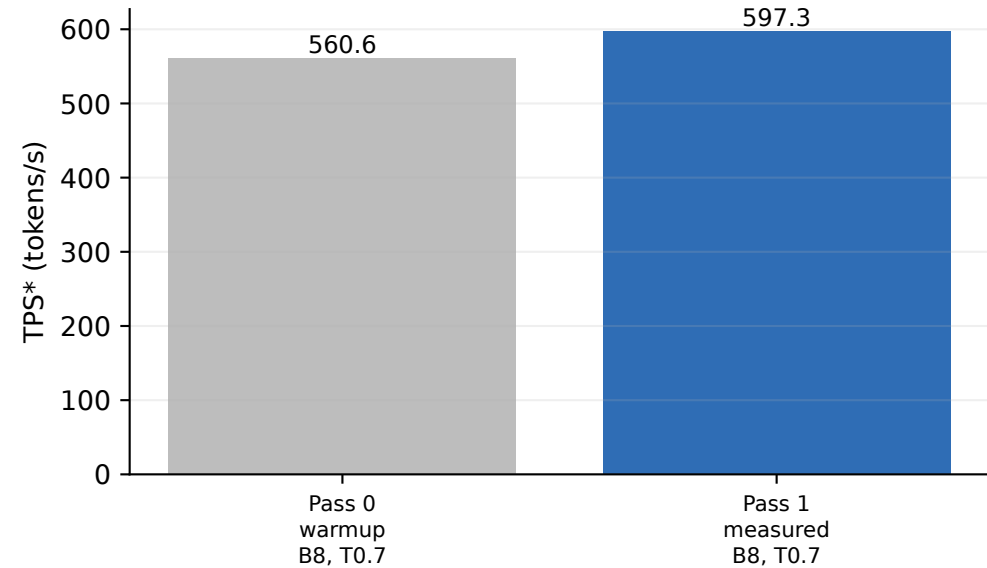
Boundary-free steps, including slow steps



Conditional AL* and source share

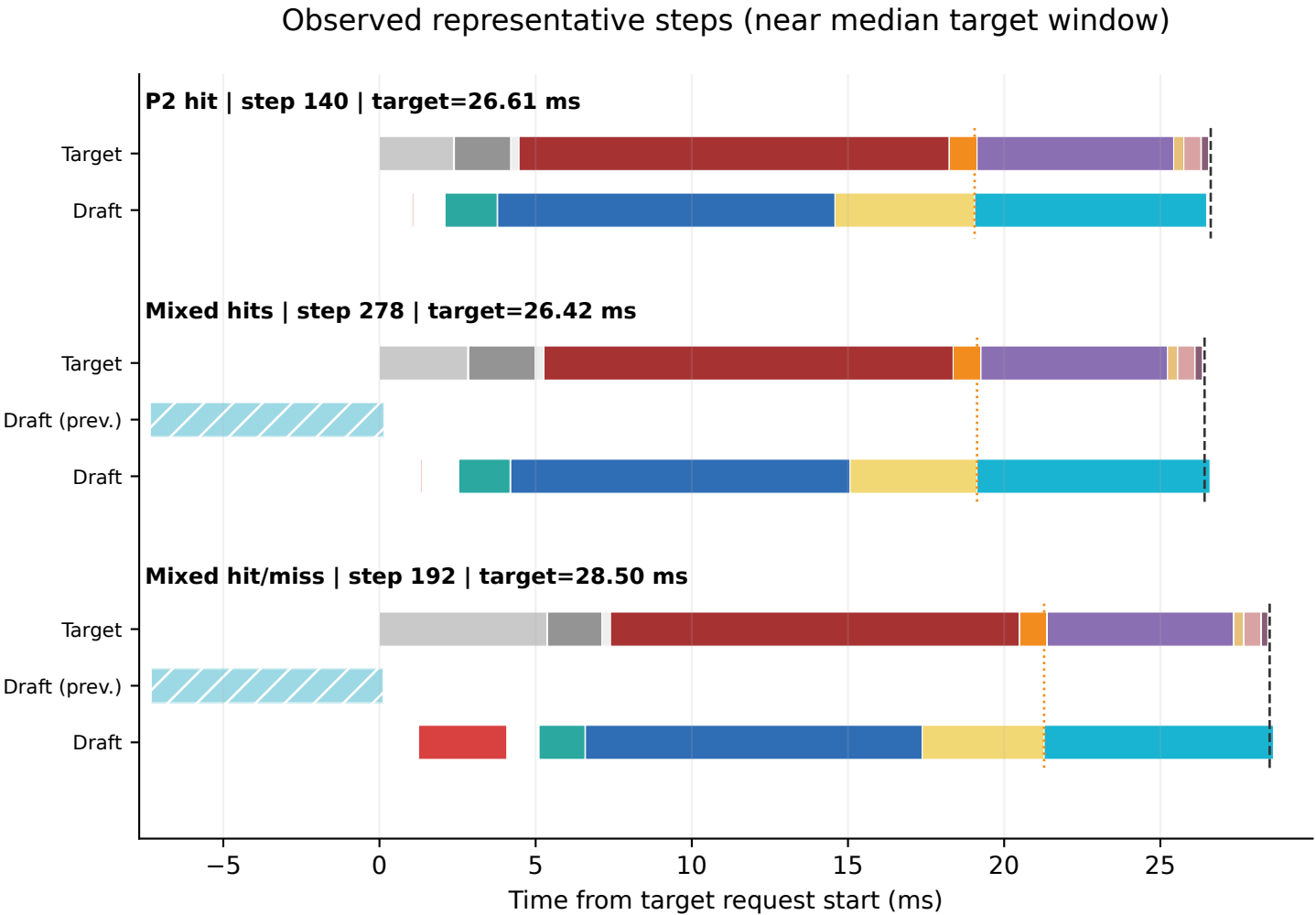
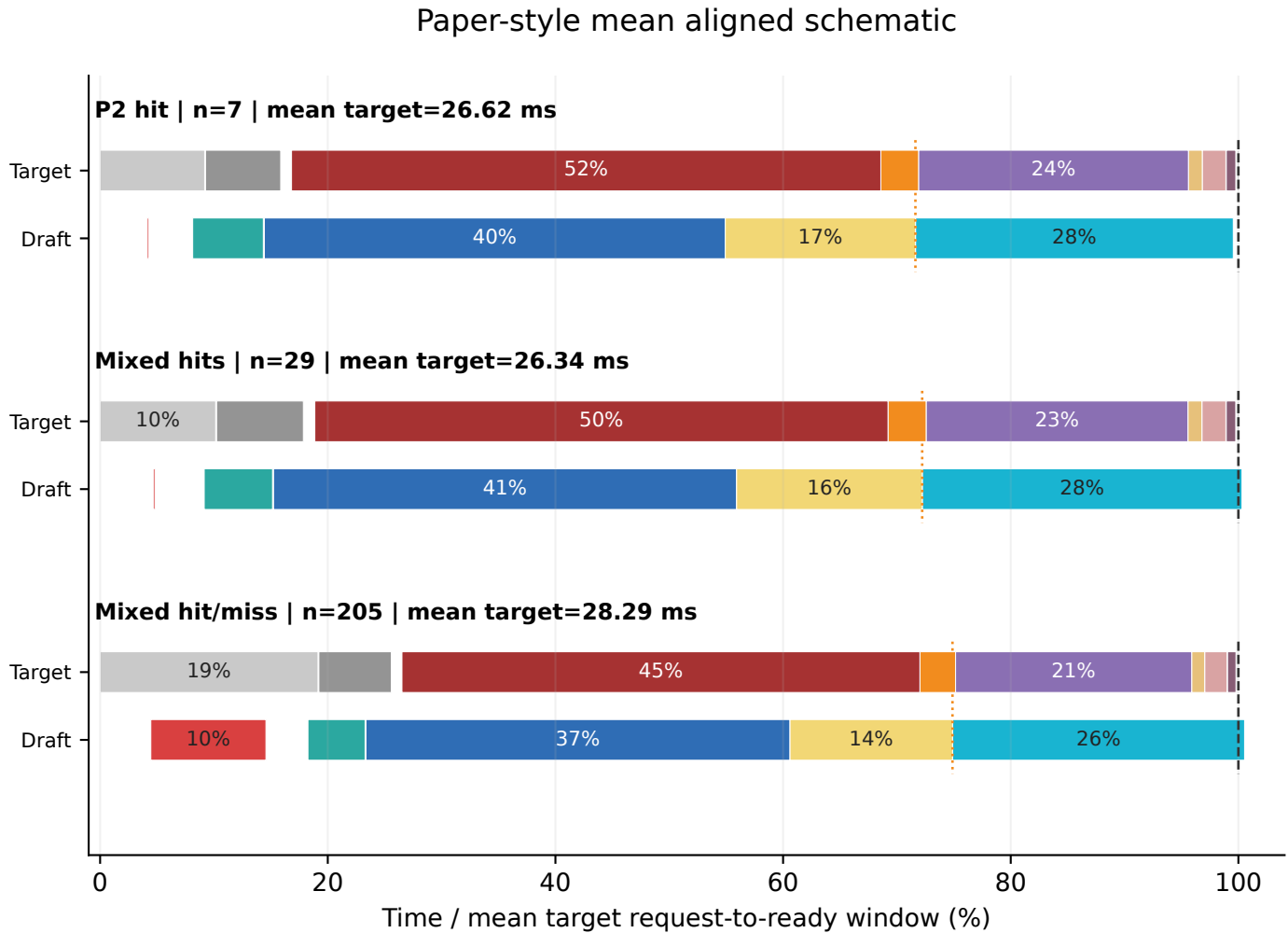
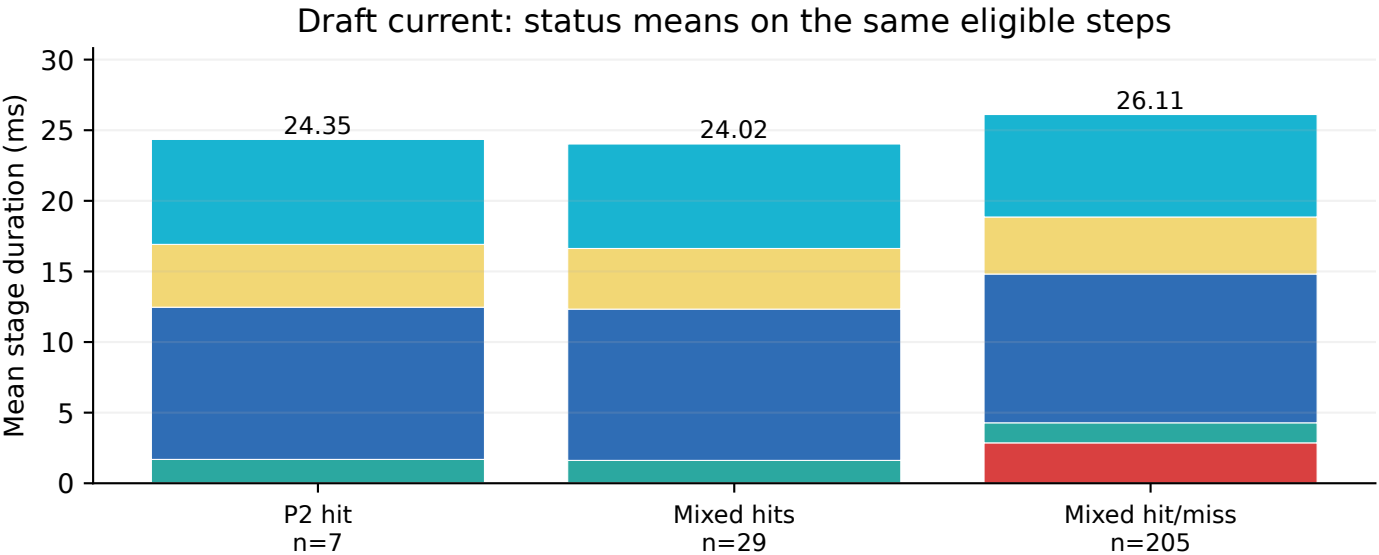
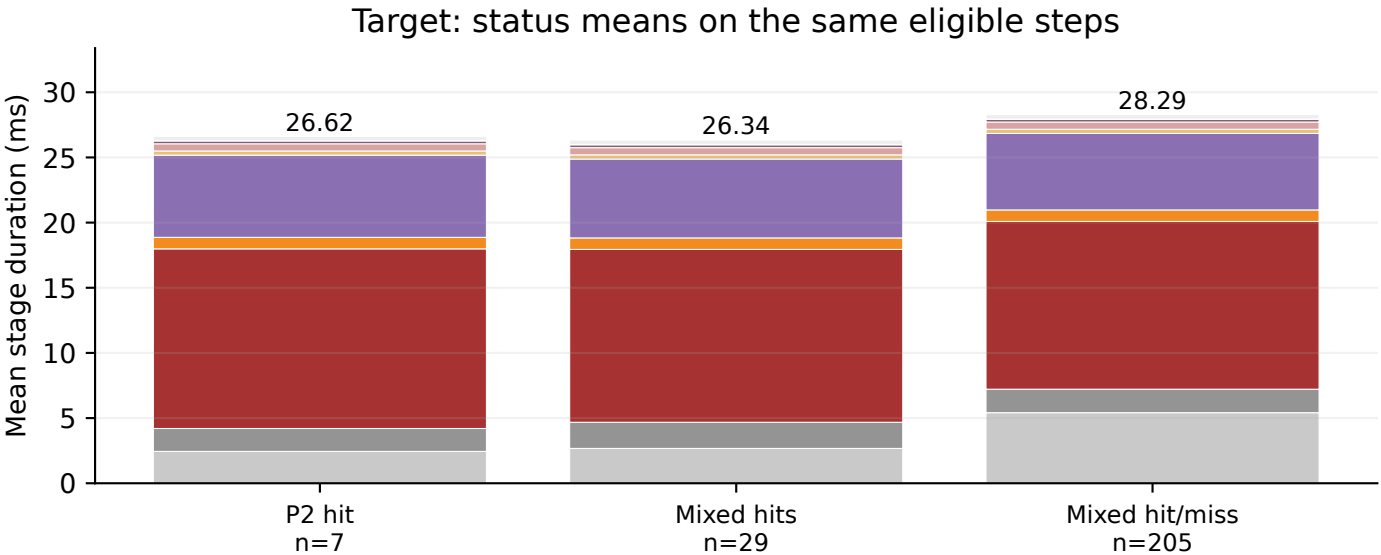


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

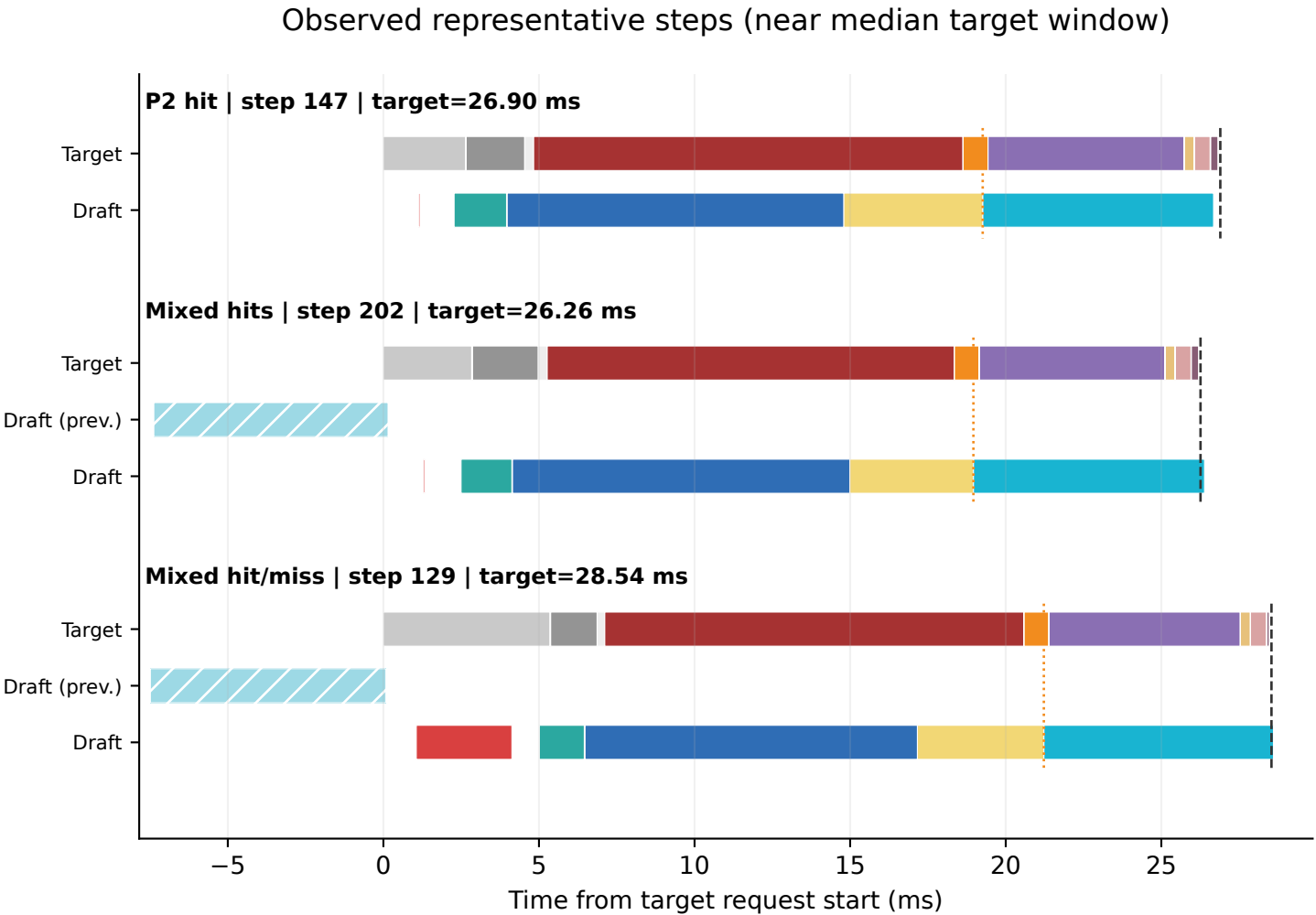
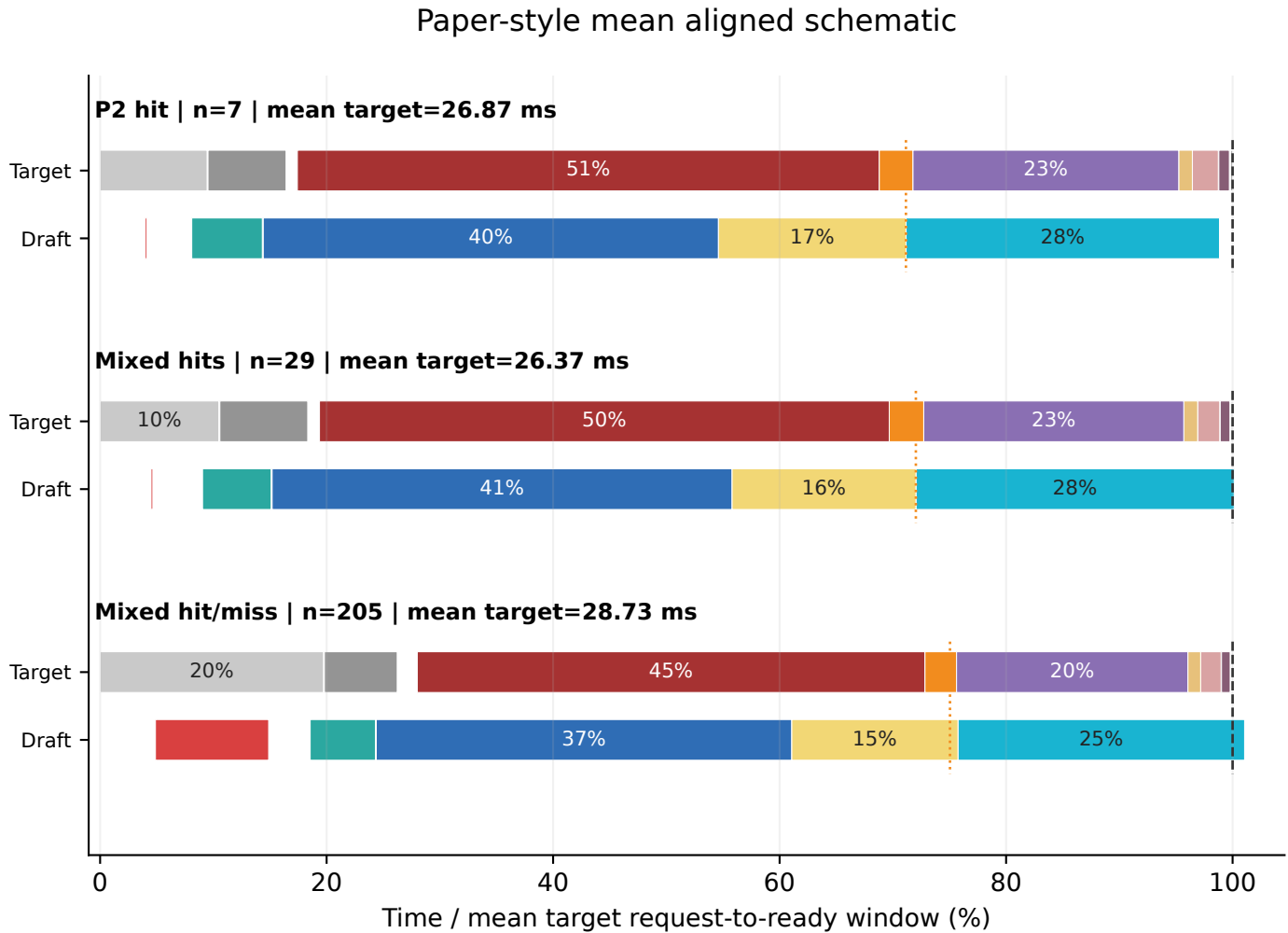
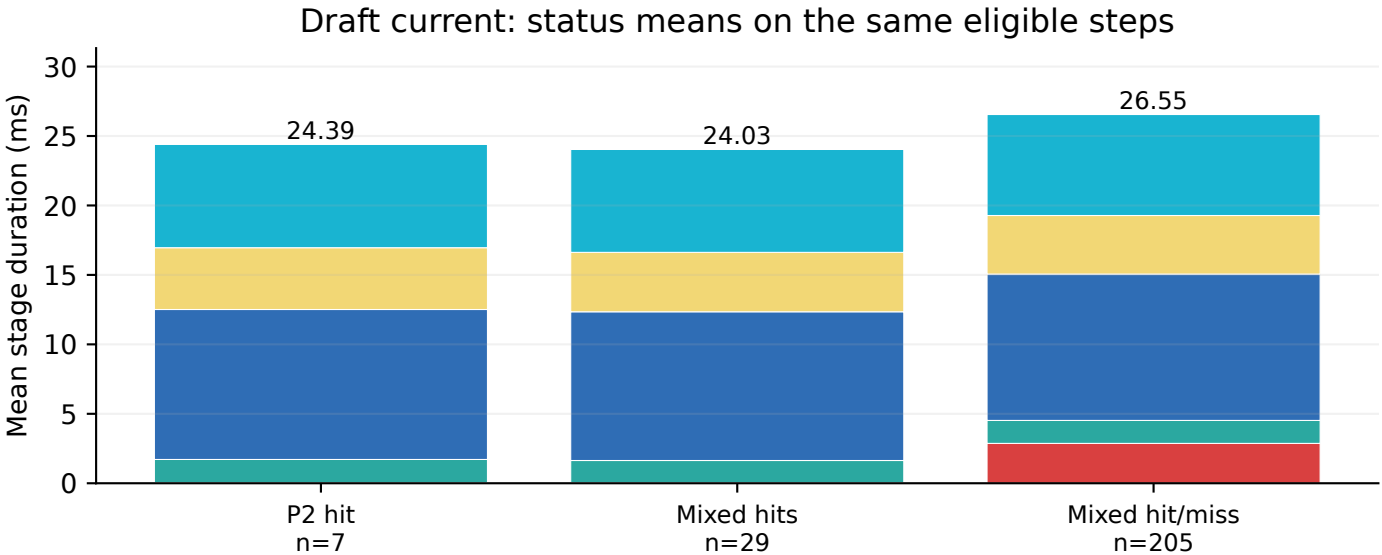
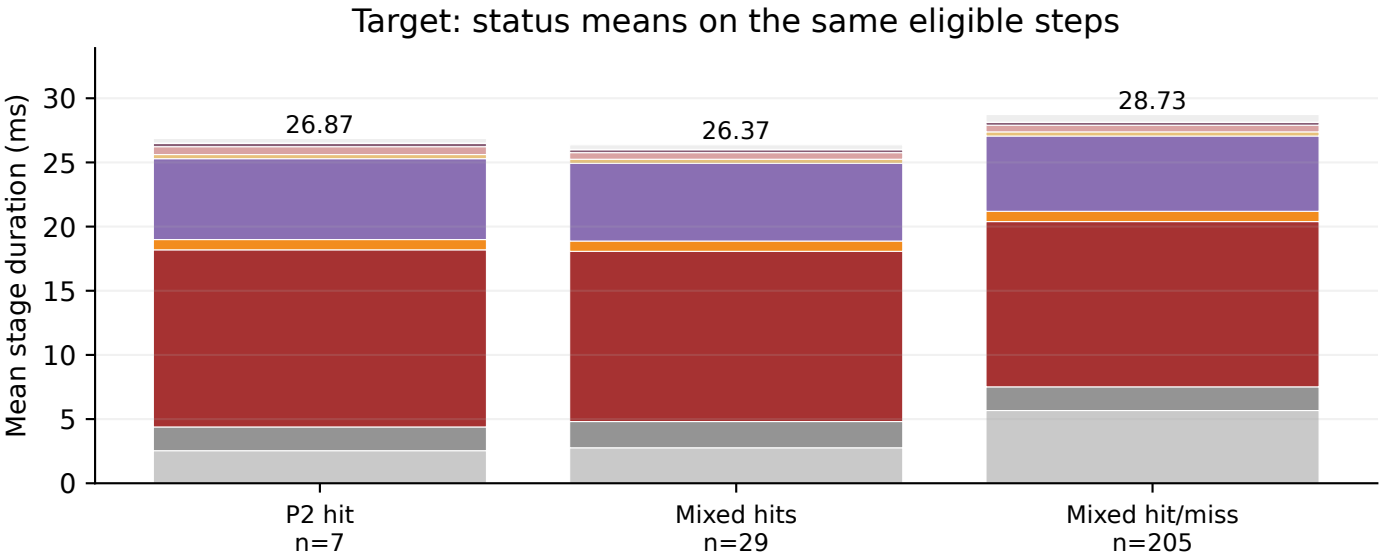
llama2_b8_opt_l0_profile_trim0
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 241 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_opt_l0_profile_trim1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 241 eligible steps



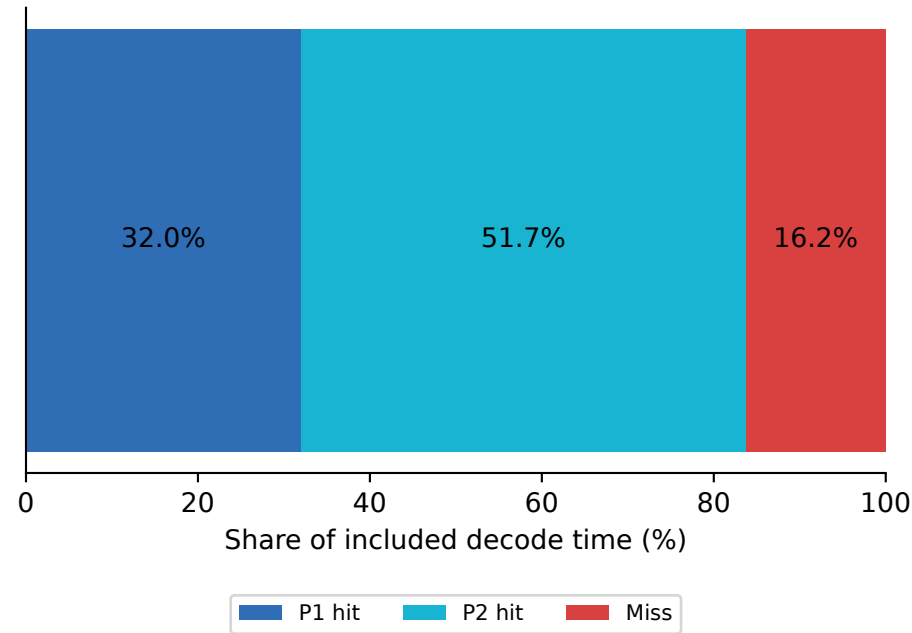
Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received

Target pre Final logits Postprocess Context

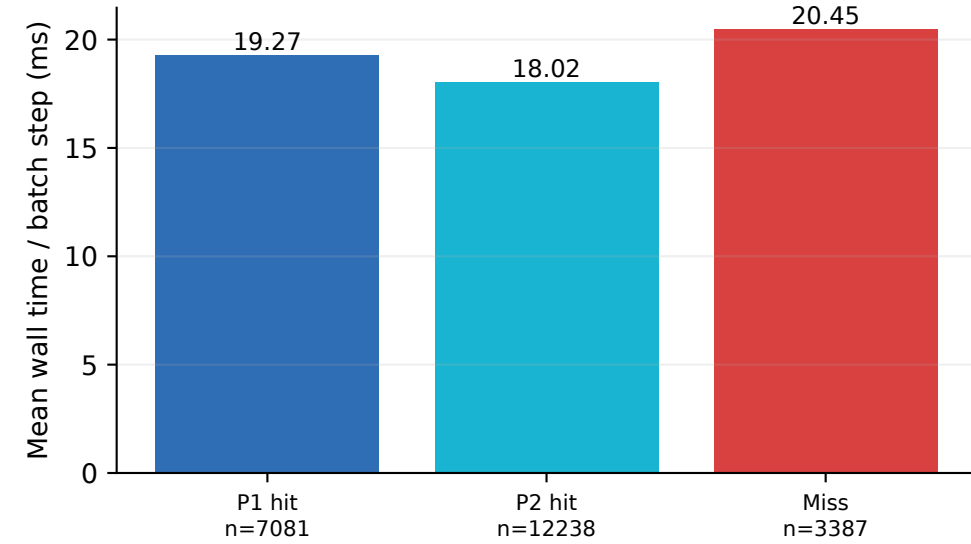
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_opt_fast_r1
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/0
Last pass: B=1, T=0.7, N=480 | AL*=2.366, hit=85.1%, TPS*=126.03

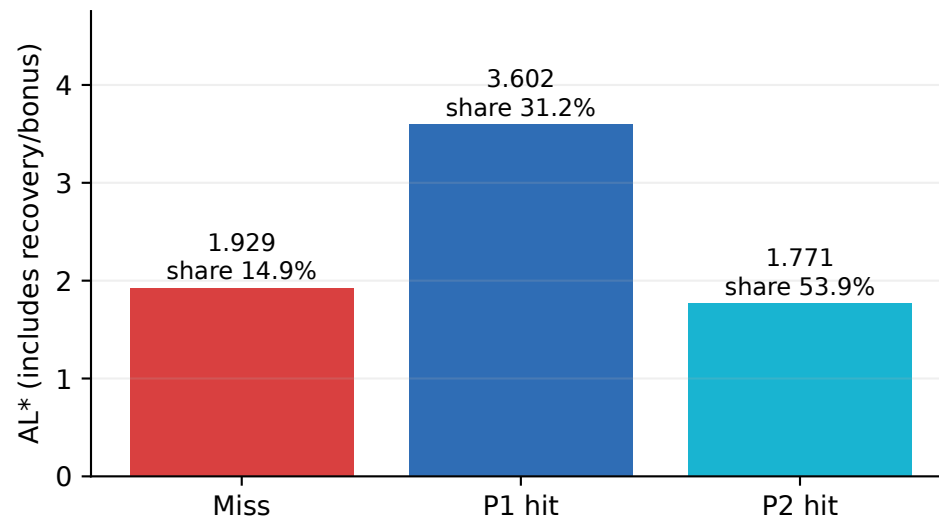
Measured wall time by batch cache status



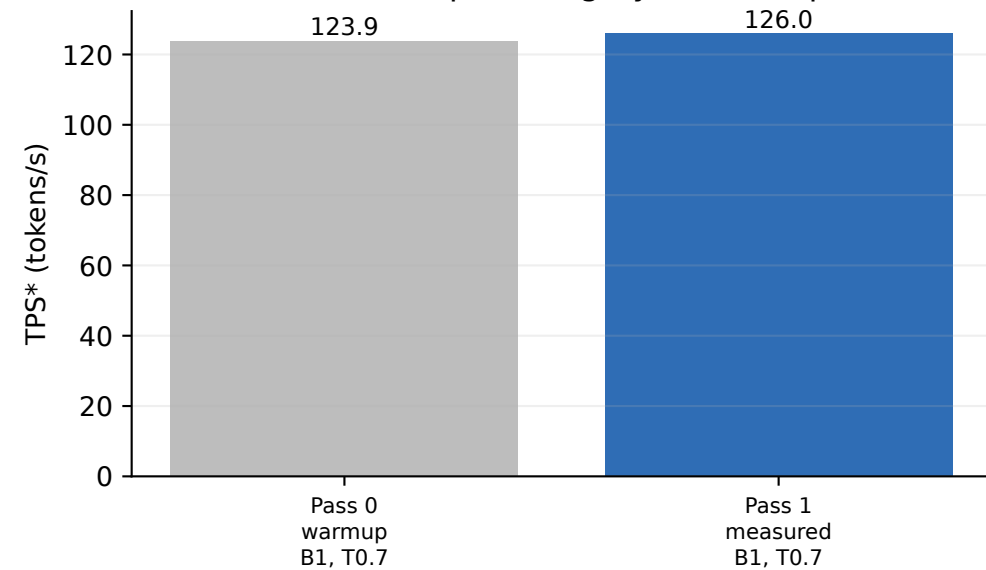
Boundary-free steps, including slow steps



Conditional AL* and source share



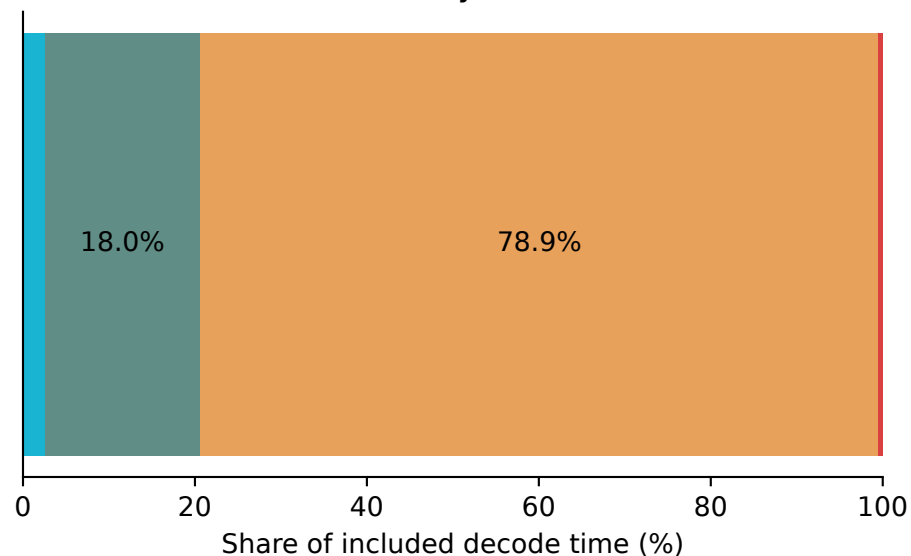
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

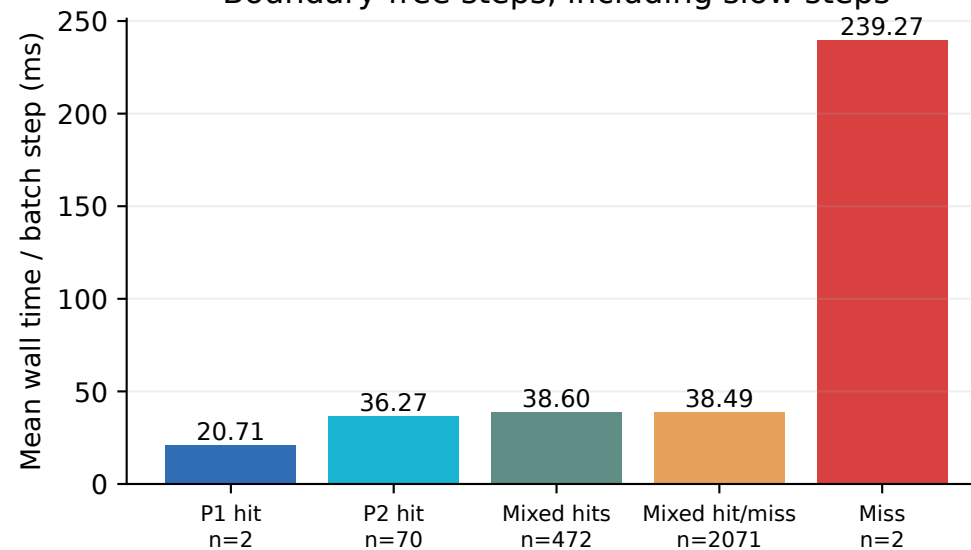
llama2_b8_opt_al_r0
 DUET: K1/K2=6/2, exit=21 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.299, hit=81.7%, TPS*=471.57

Measured wall time by batch cache status

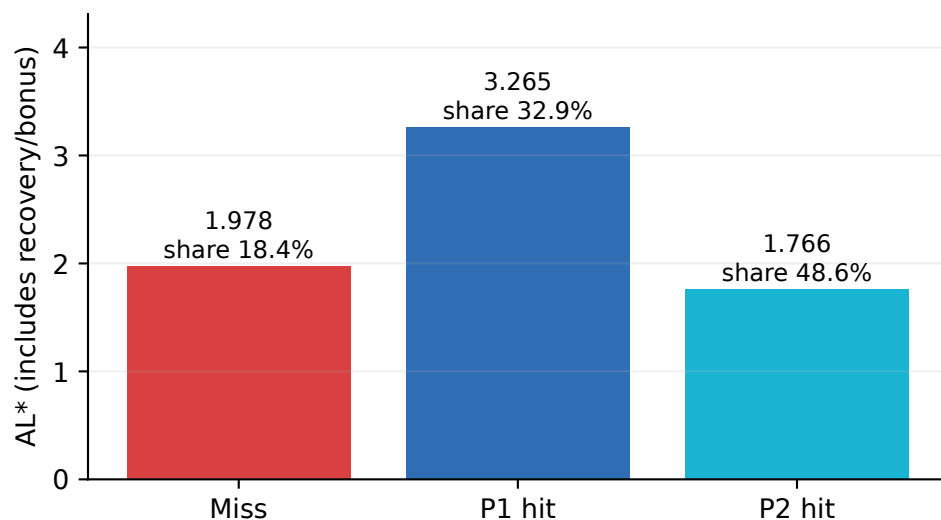


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

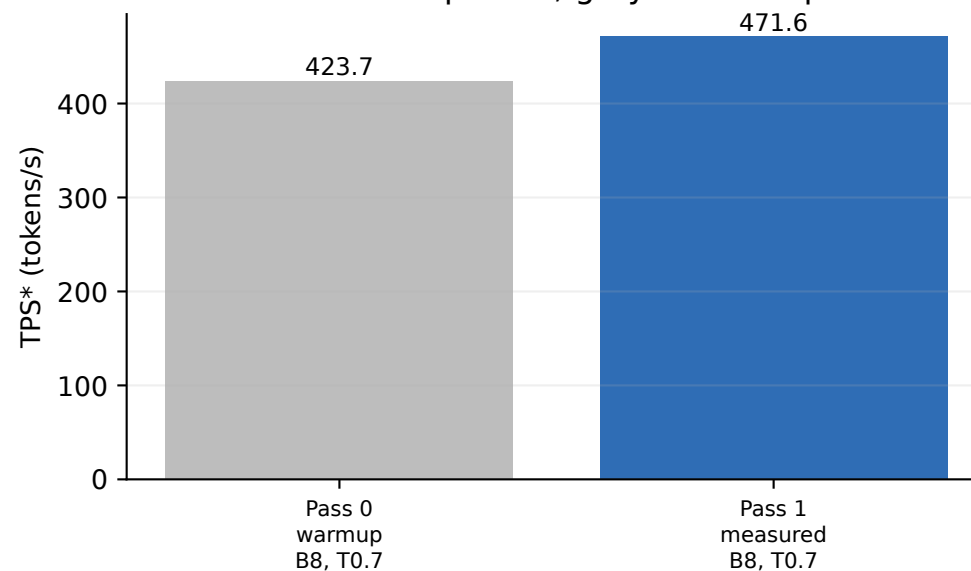
Boundary-free steps, including slow steps



Conditional AL* and source share



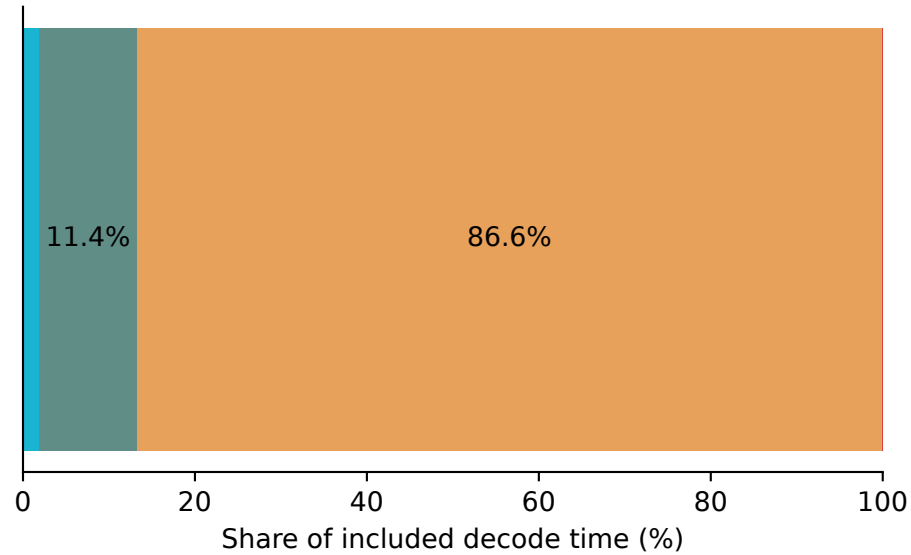
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

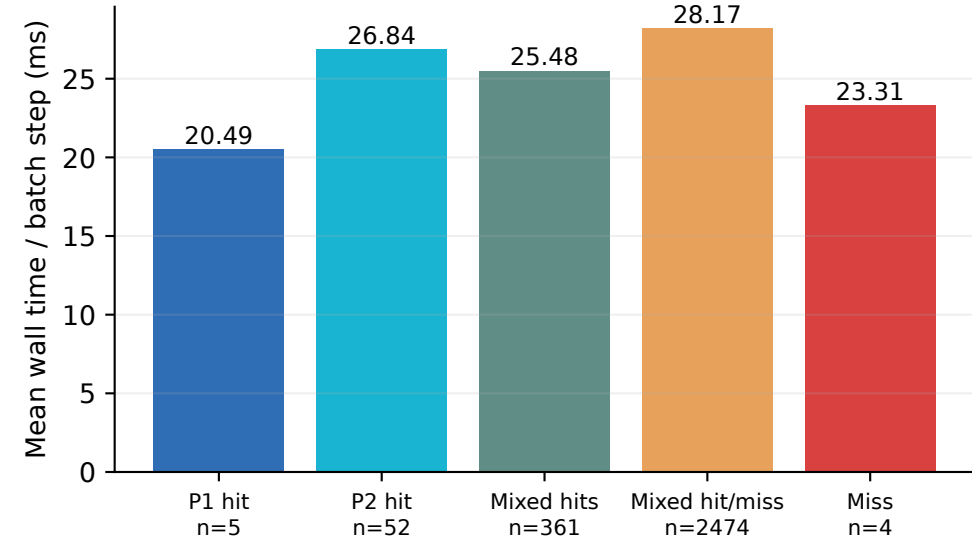
llama2_b8_opt_fast_r0
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.124, hit=76.8%, TPS*=604.86

Measured wall time by batch cache status

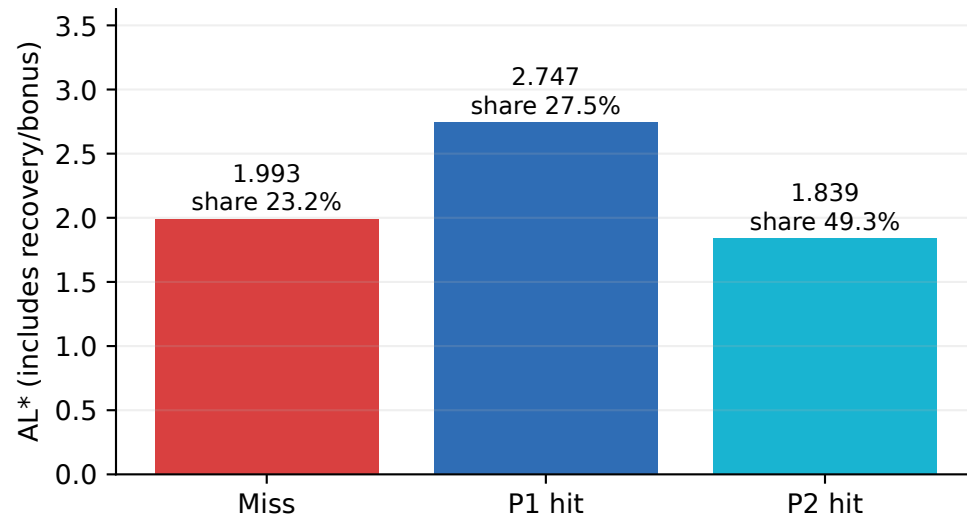


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

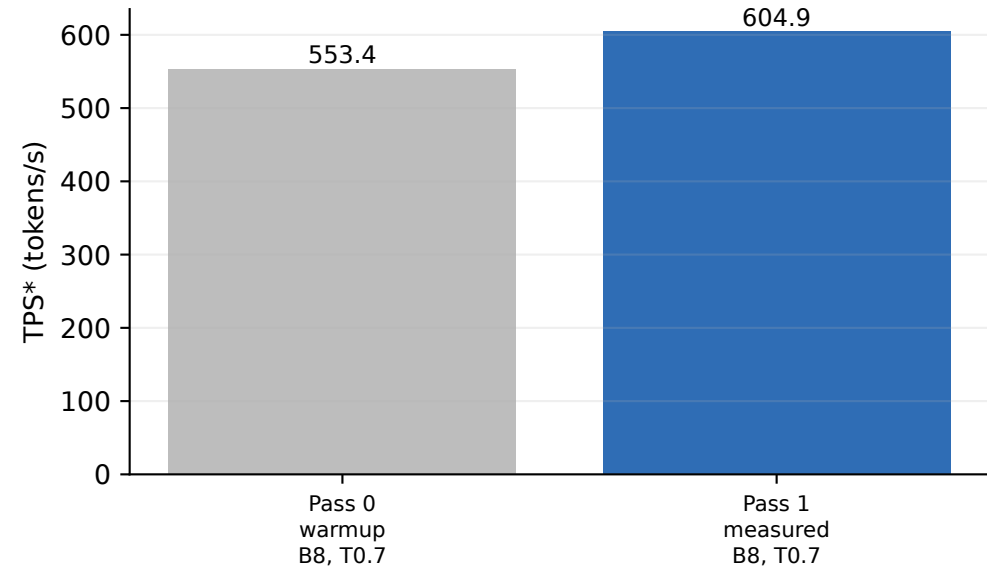
Boundary-free steps, including slow steps



Conditional AL* and source share

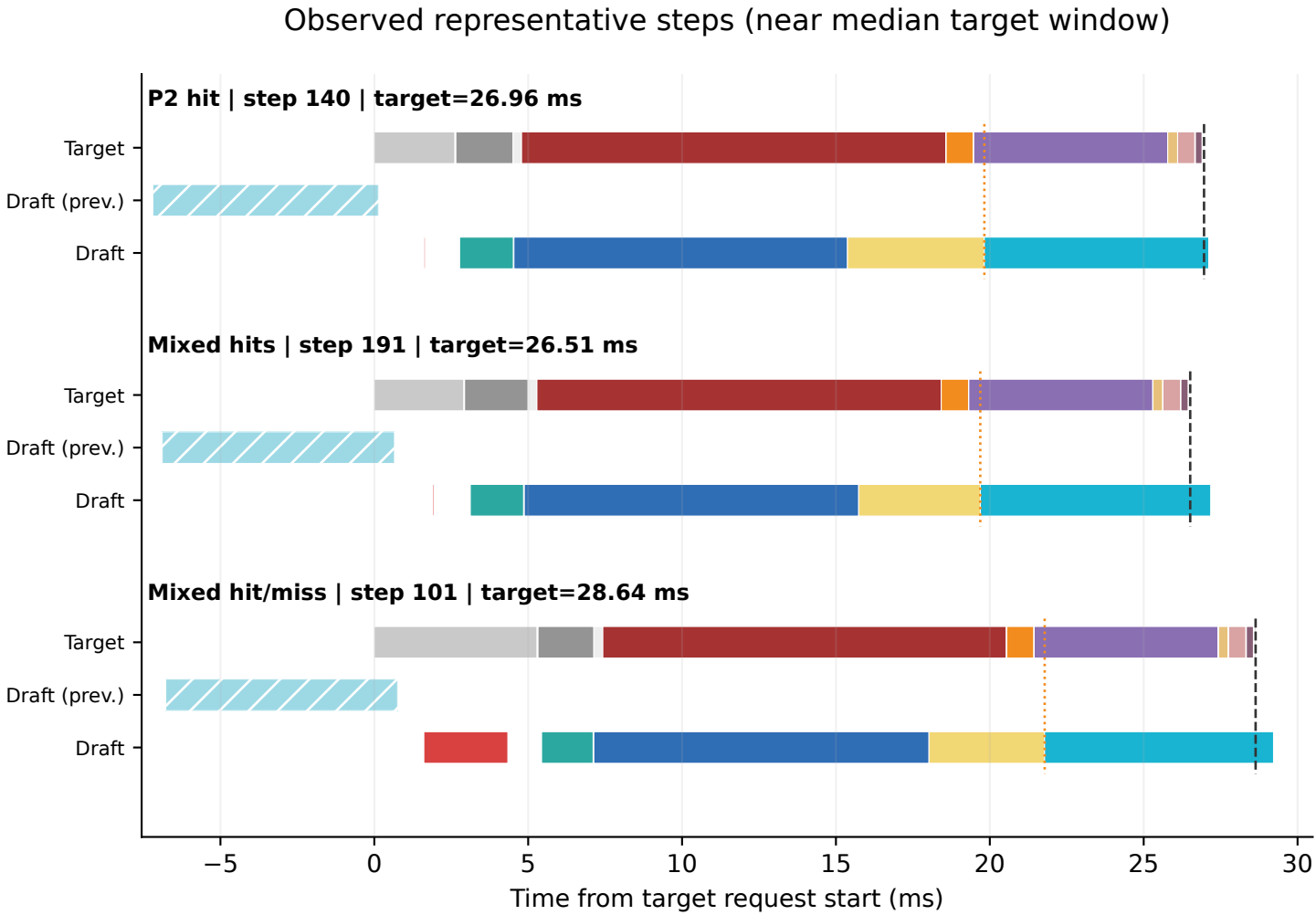
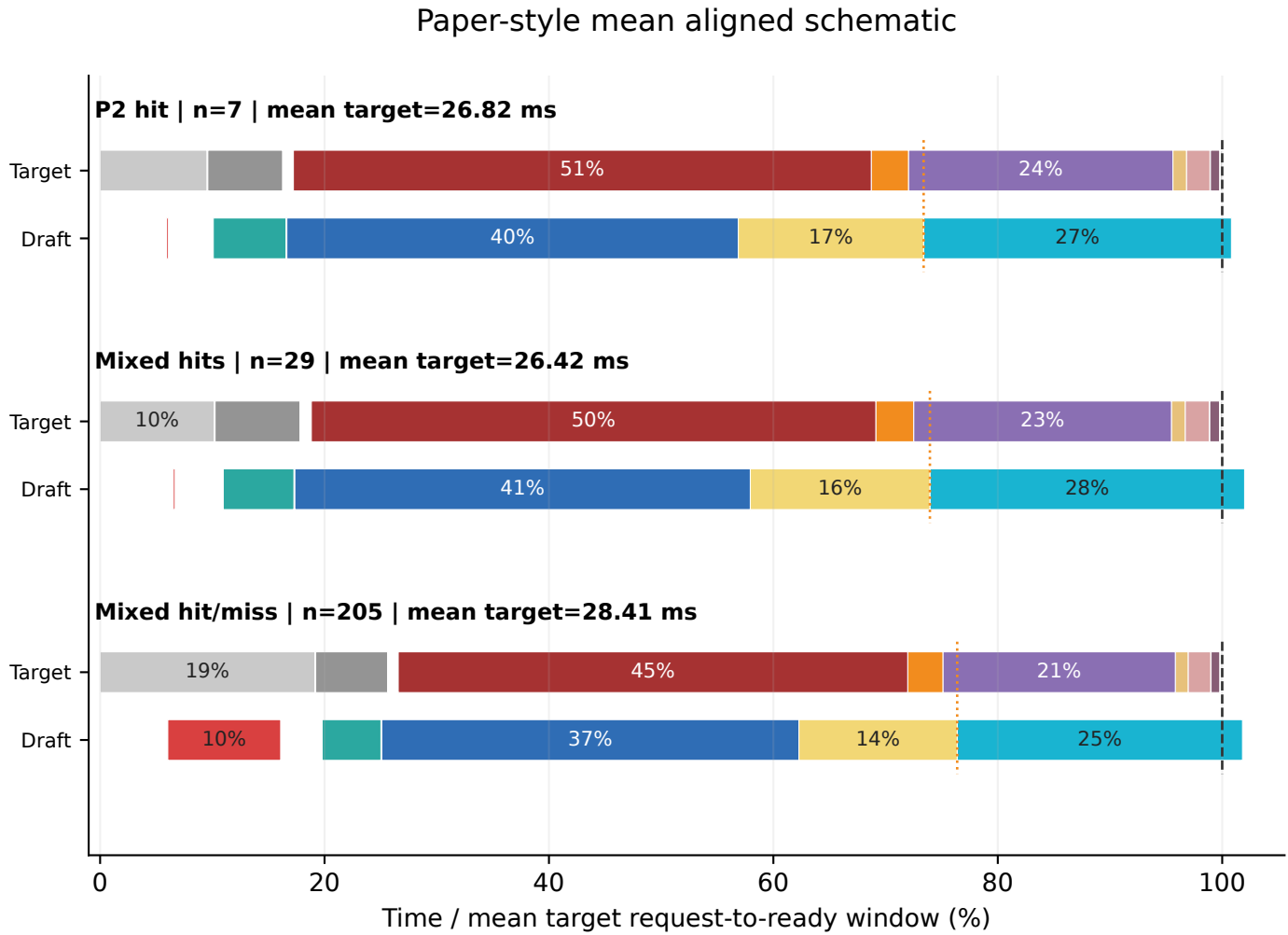
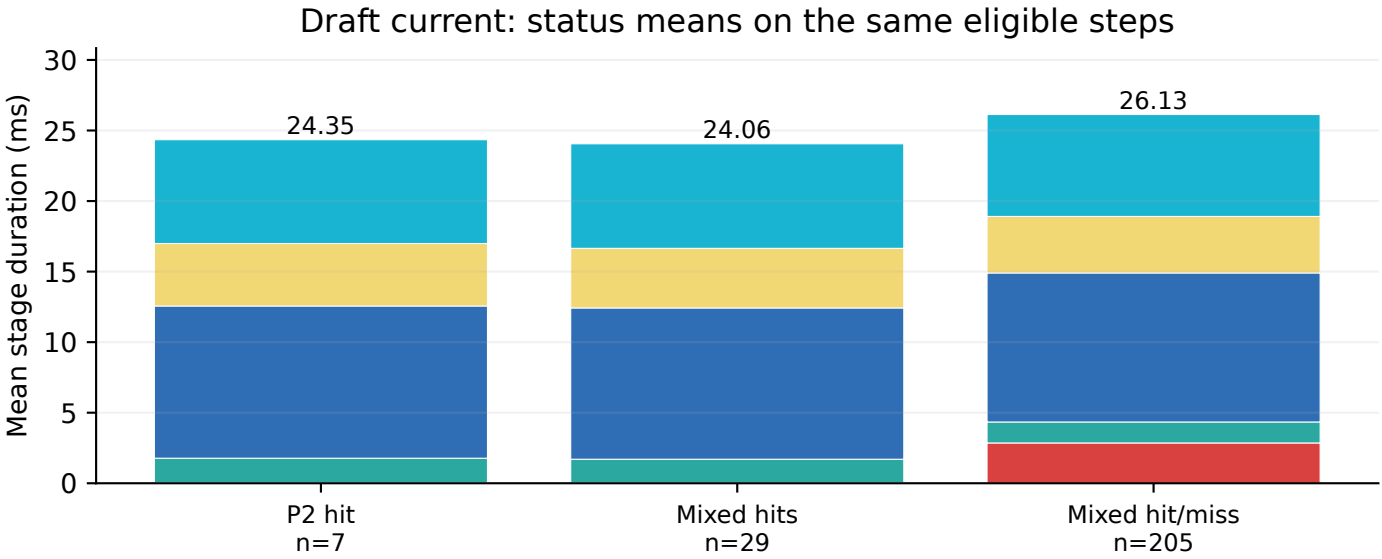
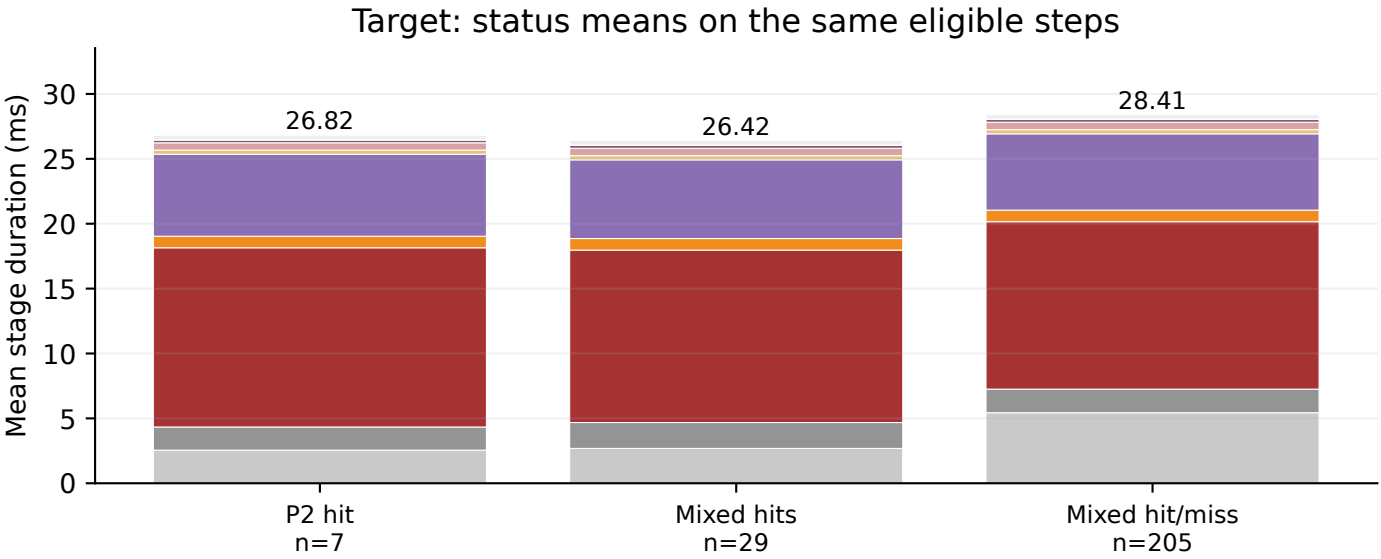


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_opt_l1_profile_trim0
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 241 eligible steps

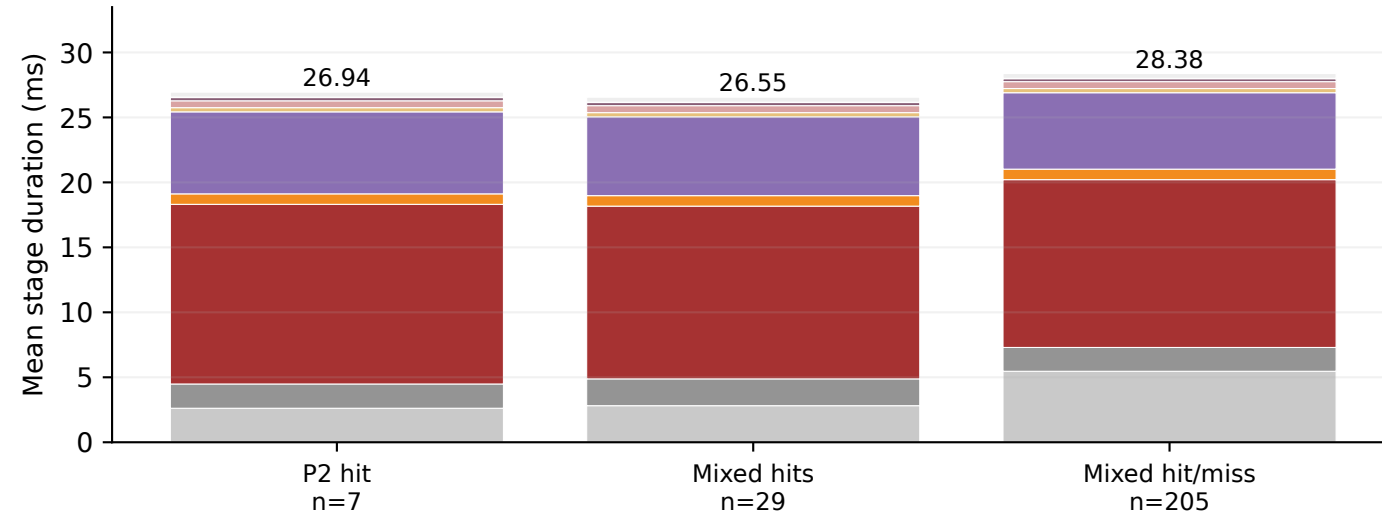


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

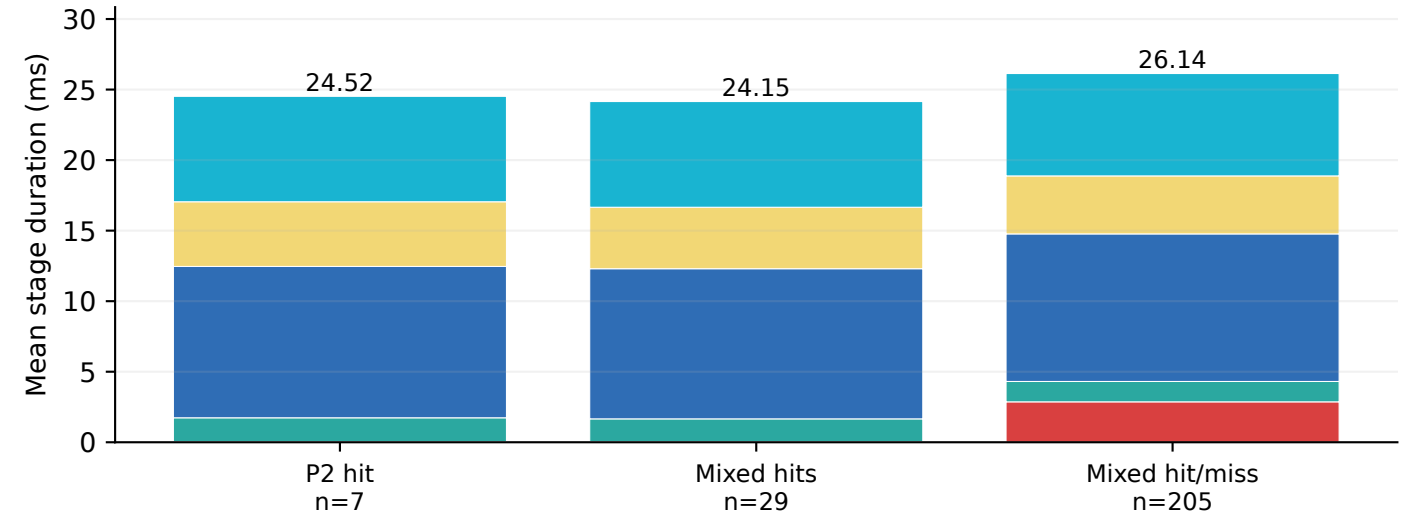
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_opt_l1_profile_trim1
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
 B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 241 eligible steps

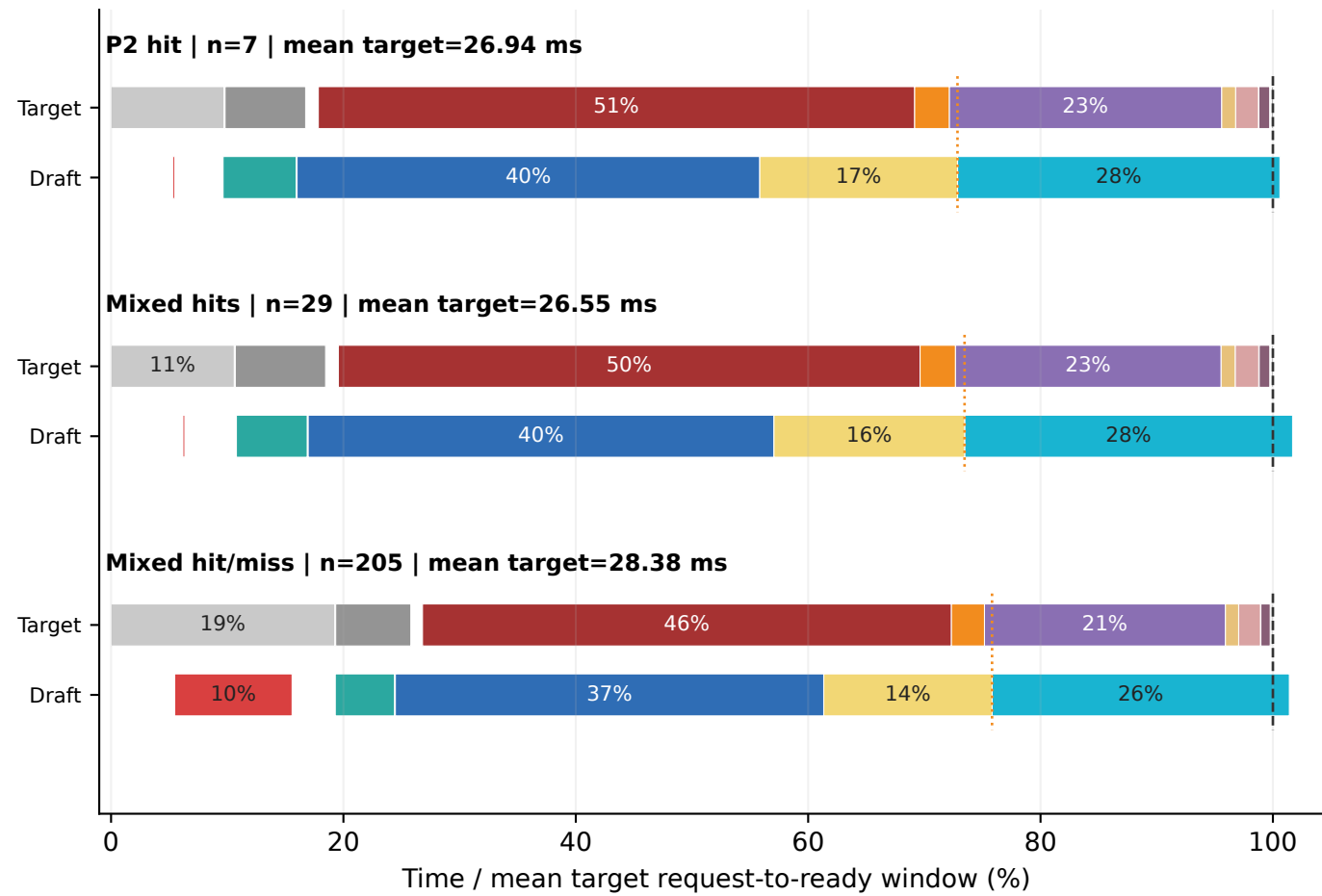
Target: status means on the same eligible steps



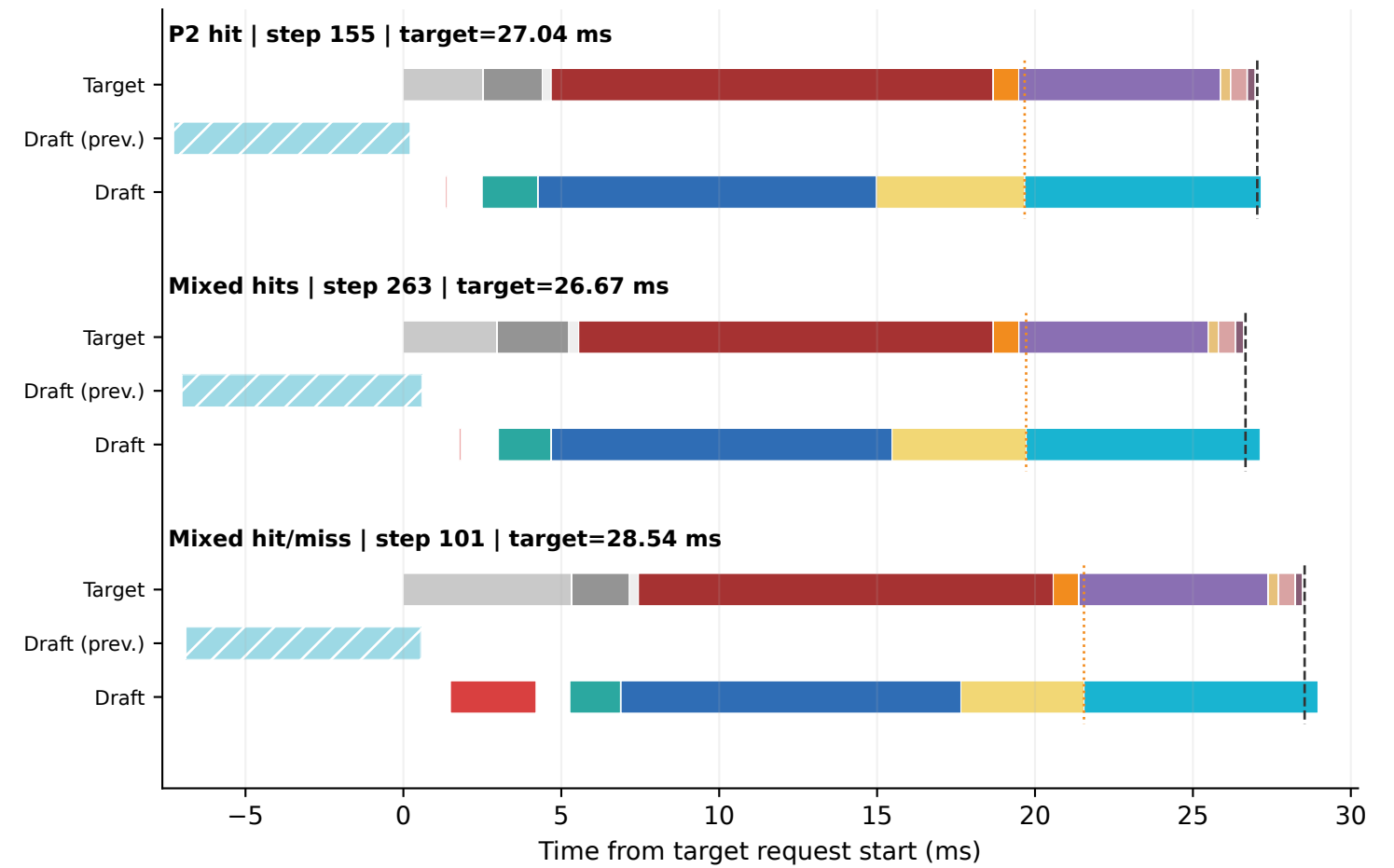
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

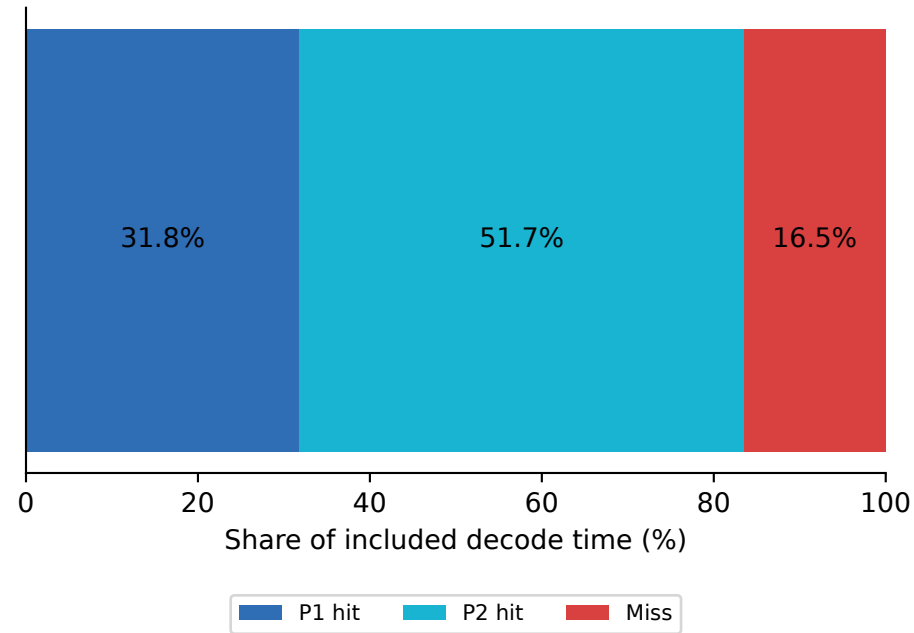


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
 Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
 Target pre Final logits Postprocess Context

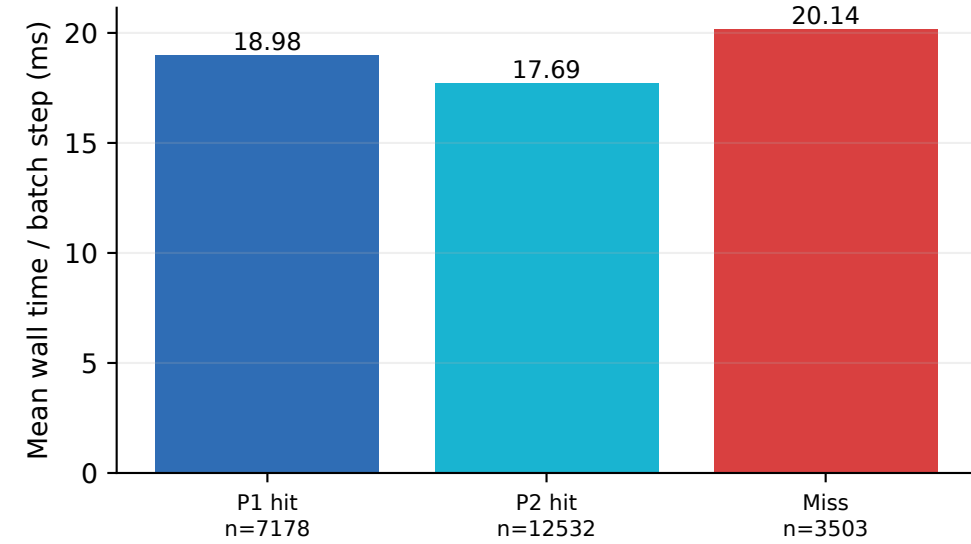
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
 Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_postopt_final_r0
DUET: K1/K2=11/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=12, trim/stream=1/1
Last pass: B=1, T=0.7, N=480 | AL*=2.344, hit=85.0%, TPS*=127.03

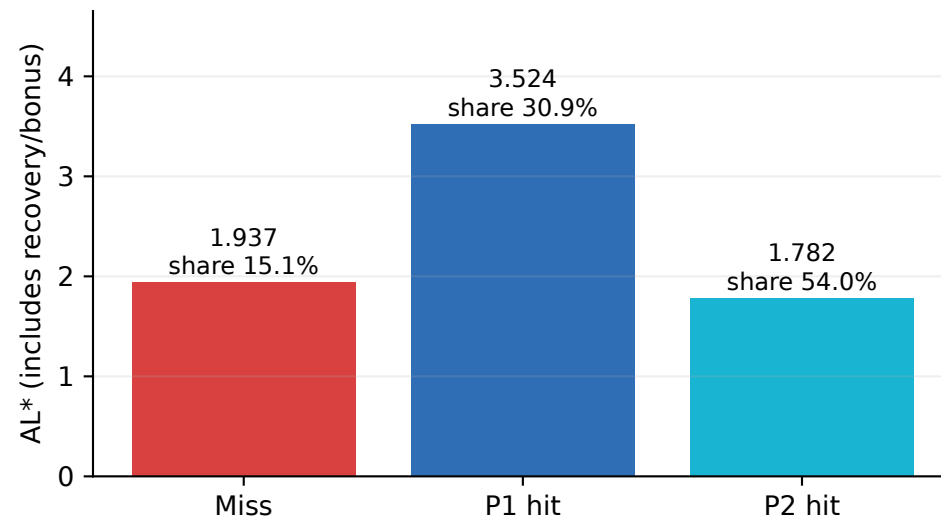
Measured wall time by batch cache status



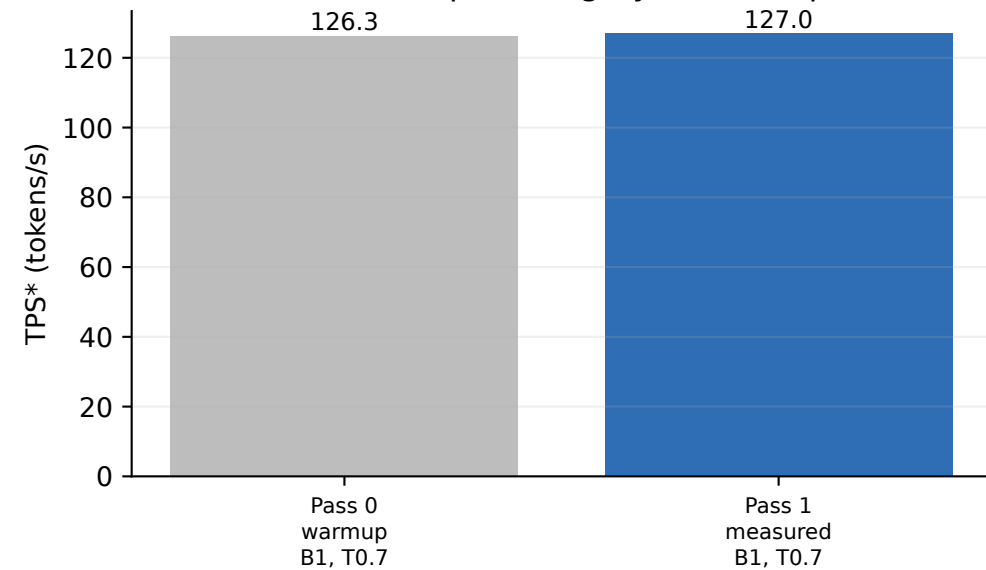
Boundary-free steps, including slow steps



Conditional AL* and source share



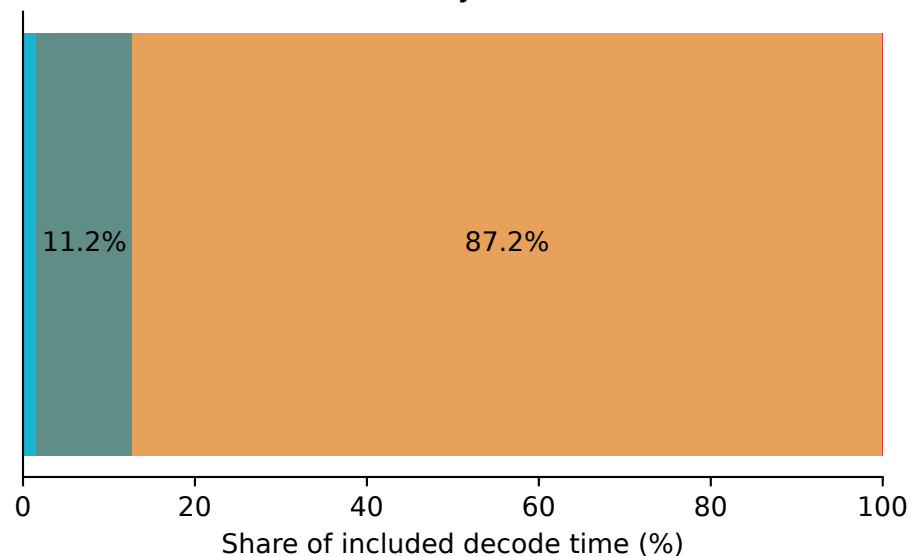
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

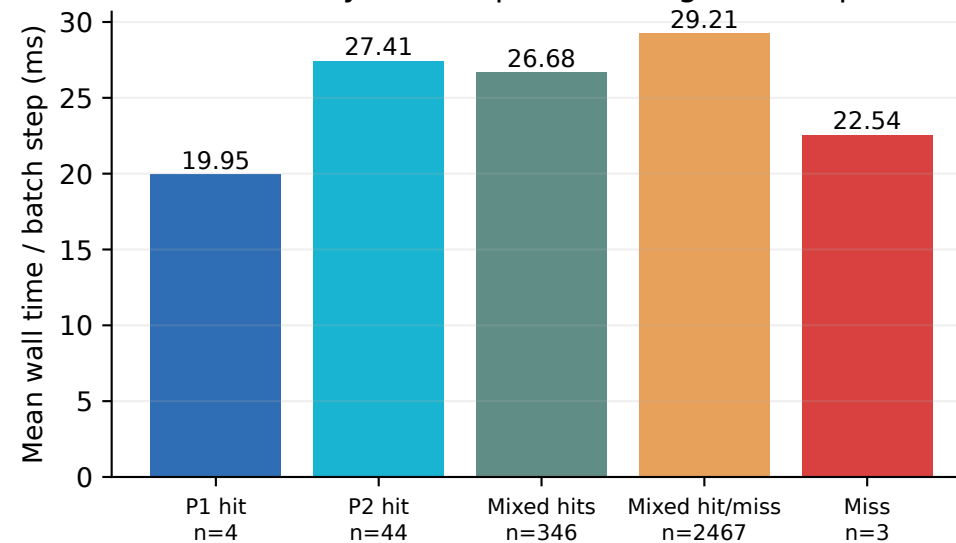
llama2_b8_postopt_final_r1
 DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.135, hit=75.8%, TPS*=585.30

Measured wall time by batch cache status

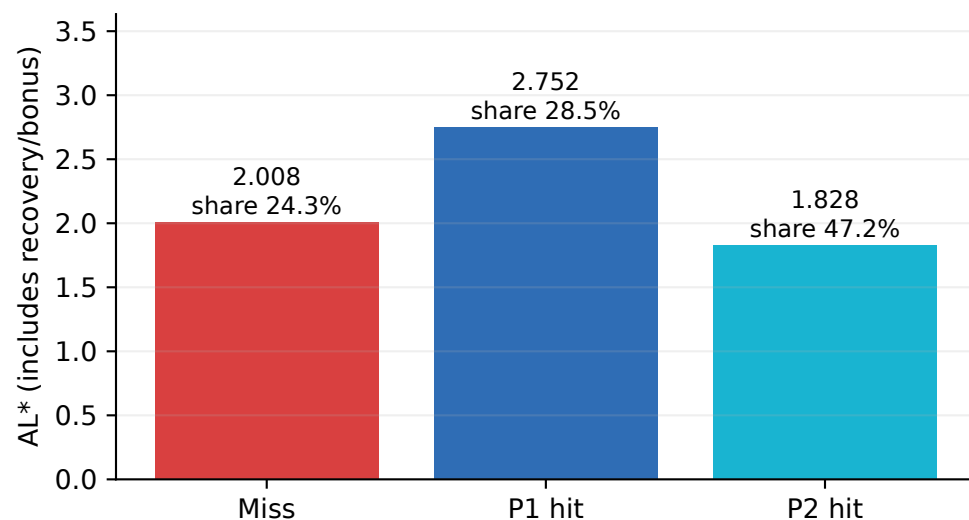


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

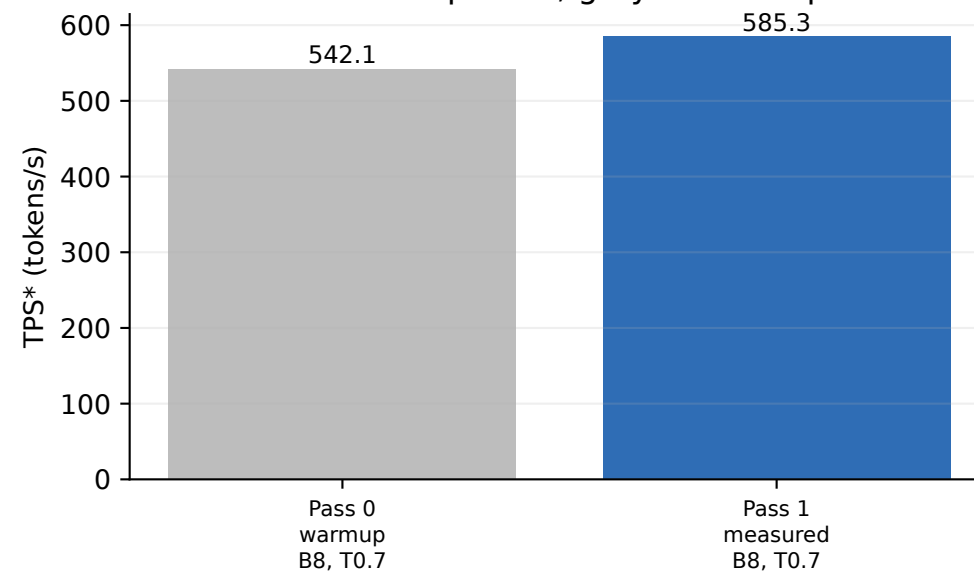
Boundary-free steps, including slow steps



Conditional AL* and source share



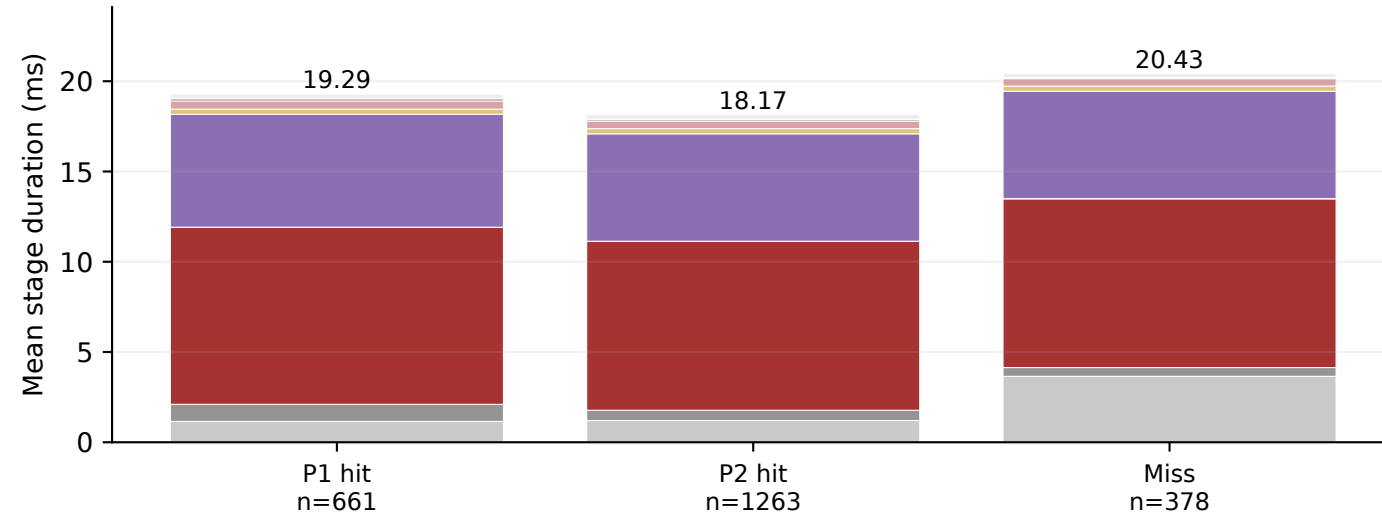
All saved passes; gray = warmup



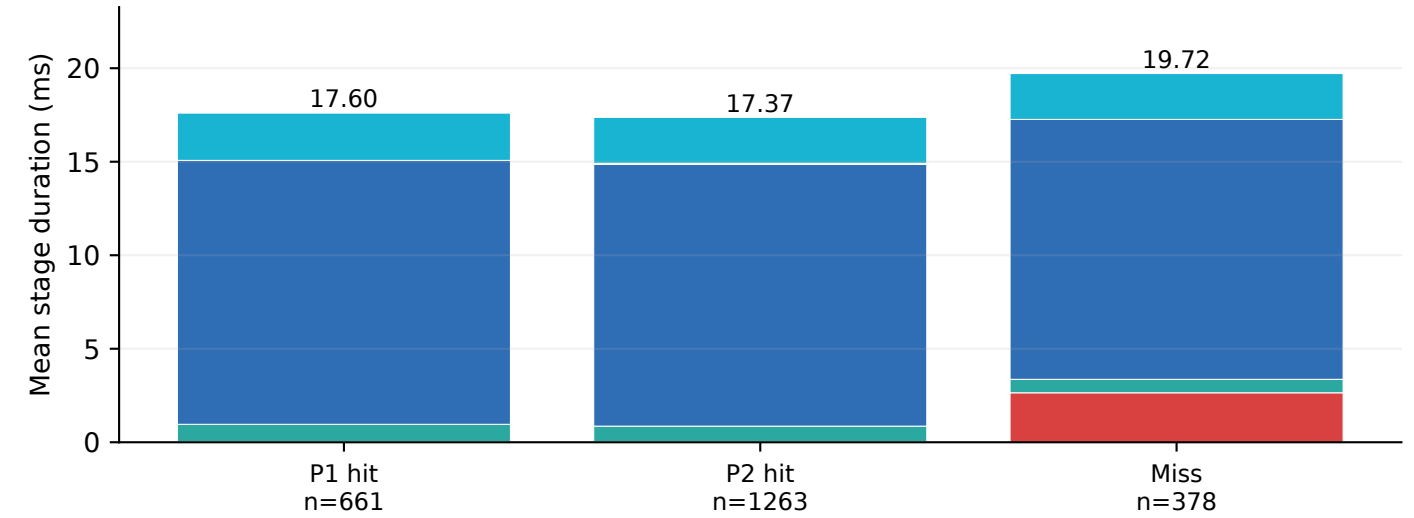
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_postopt_profile_exit_earlier
DUET: K1/K2=12/2, exit=19 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2302 eligible steps

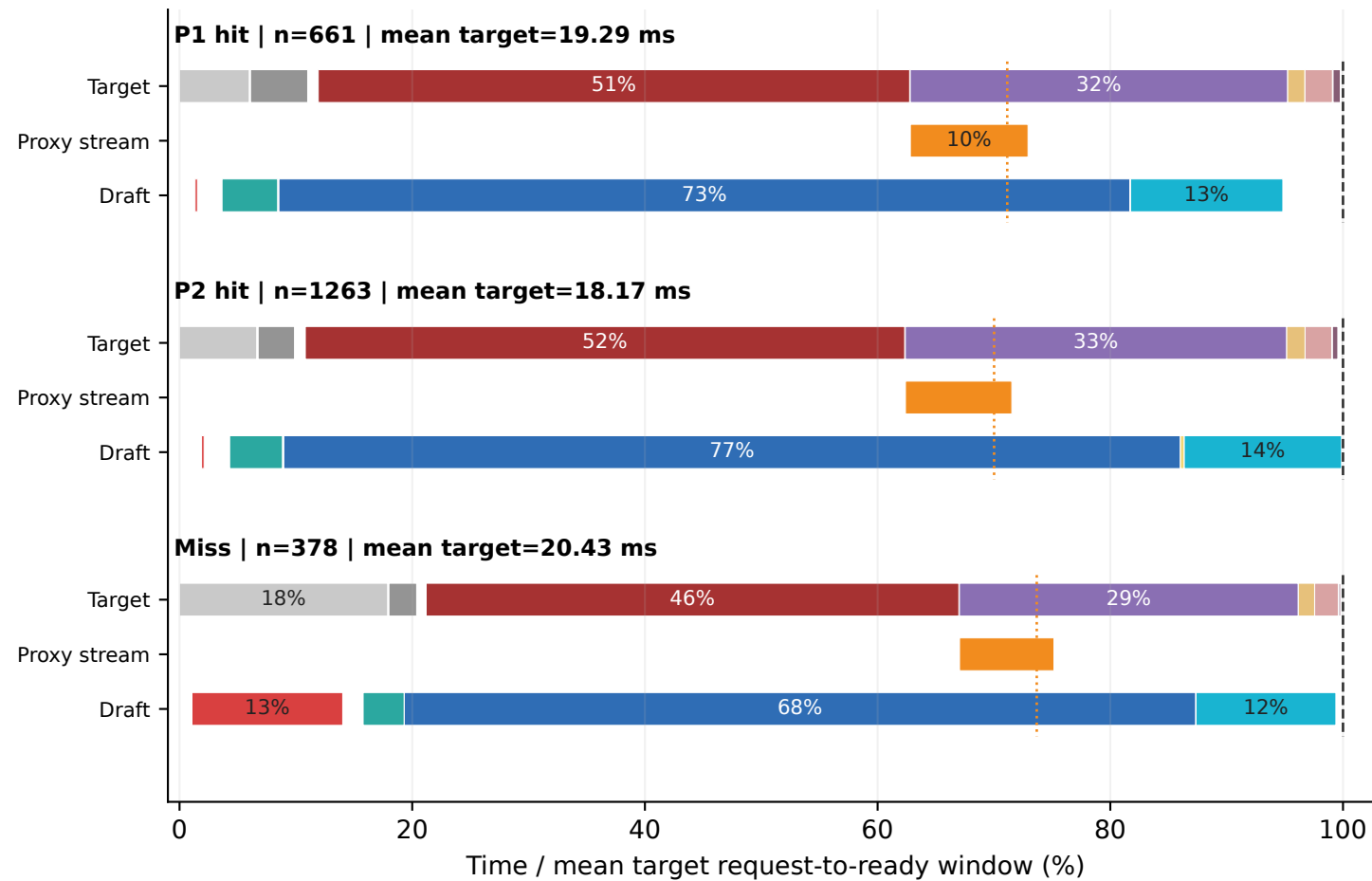
Target: status means on the same eligible steps



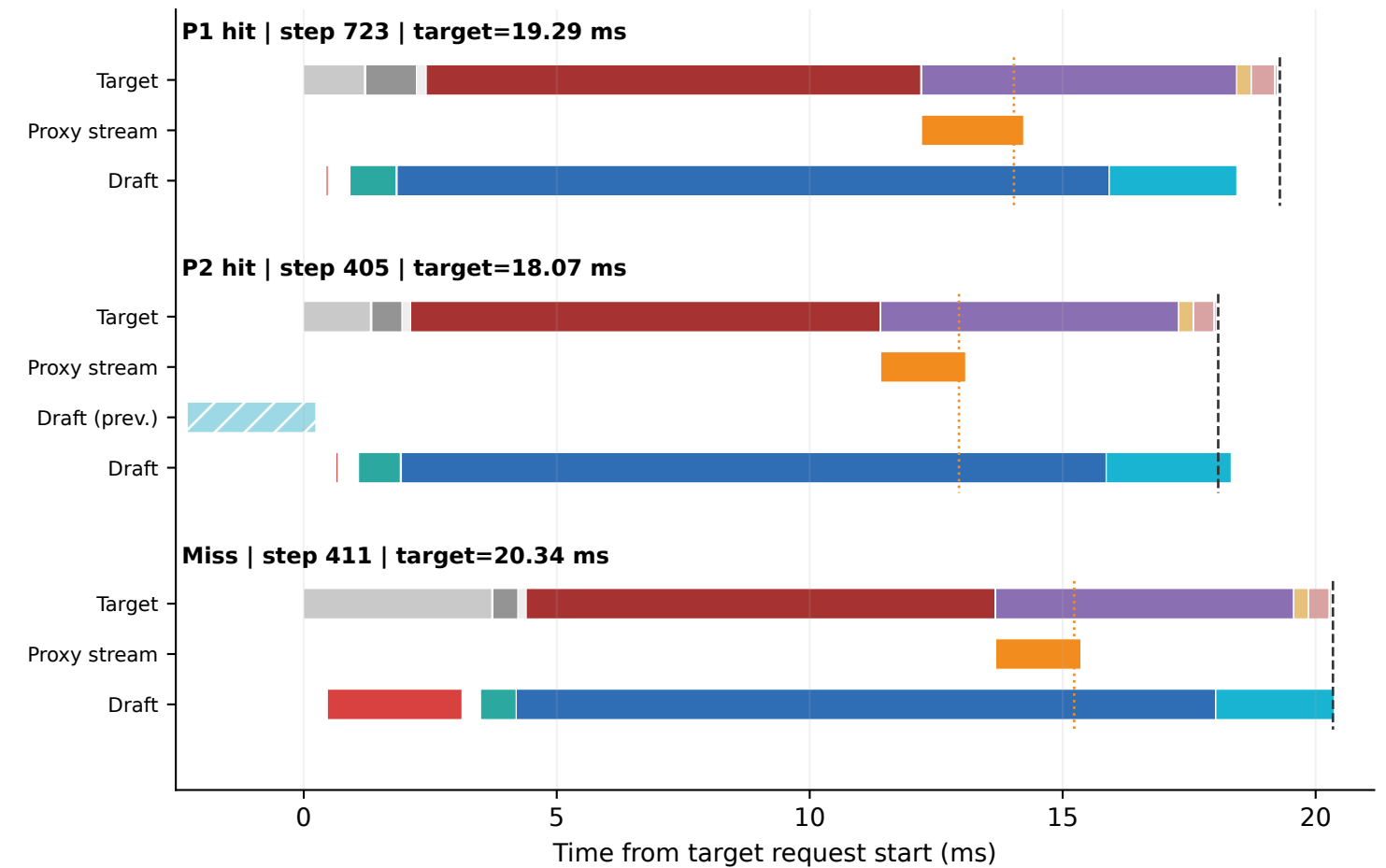
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



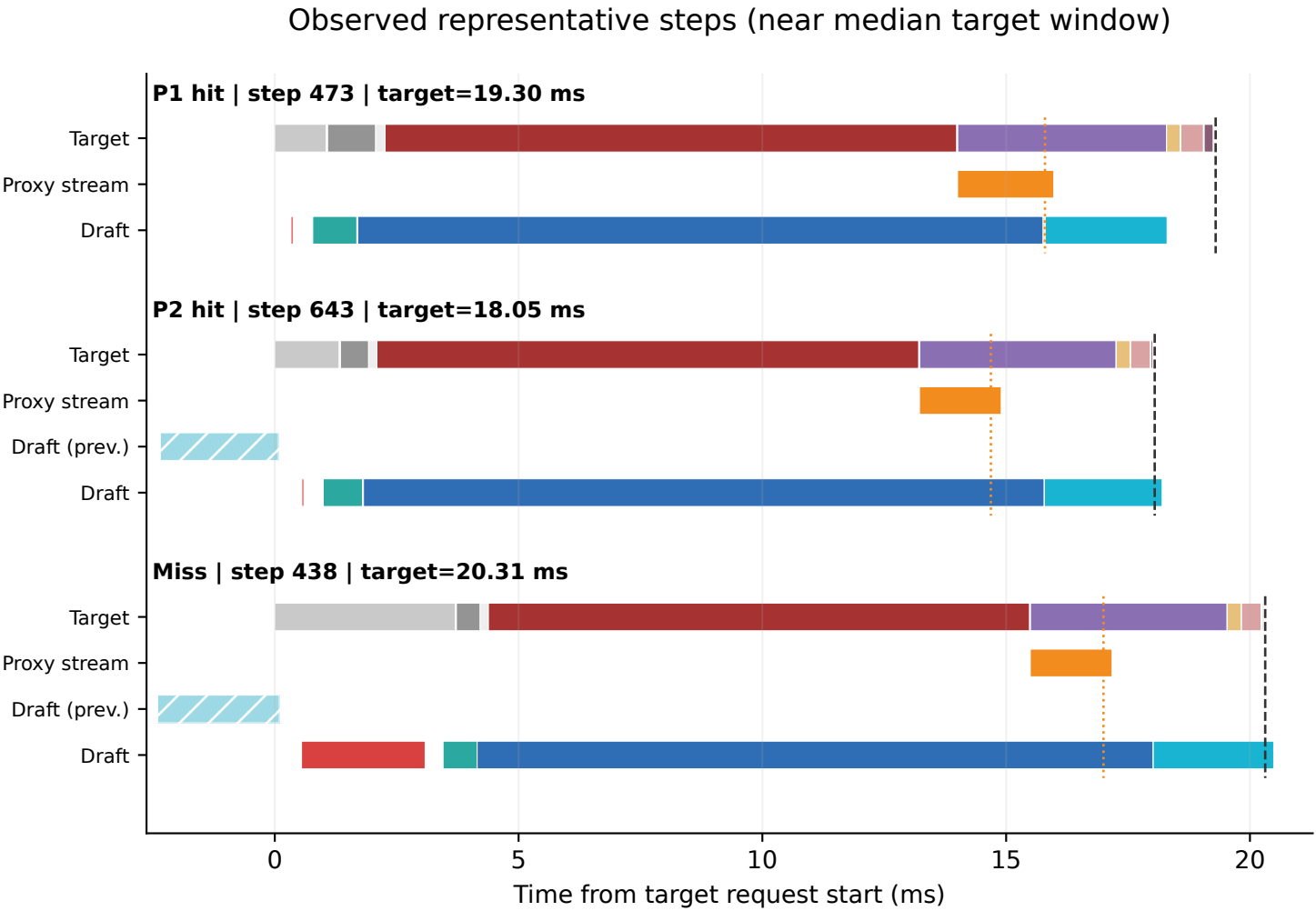
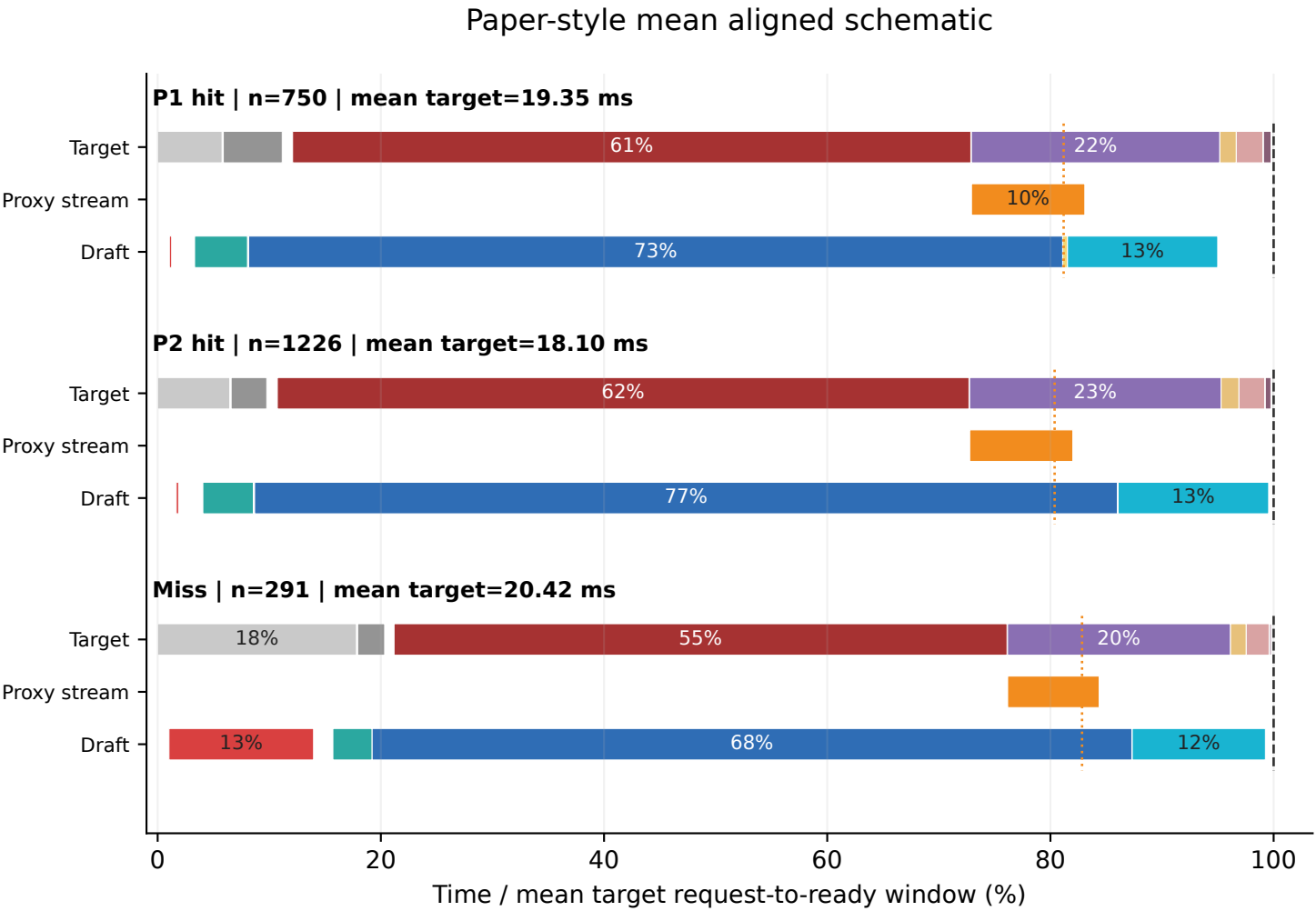
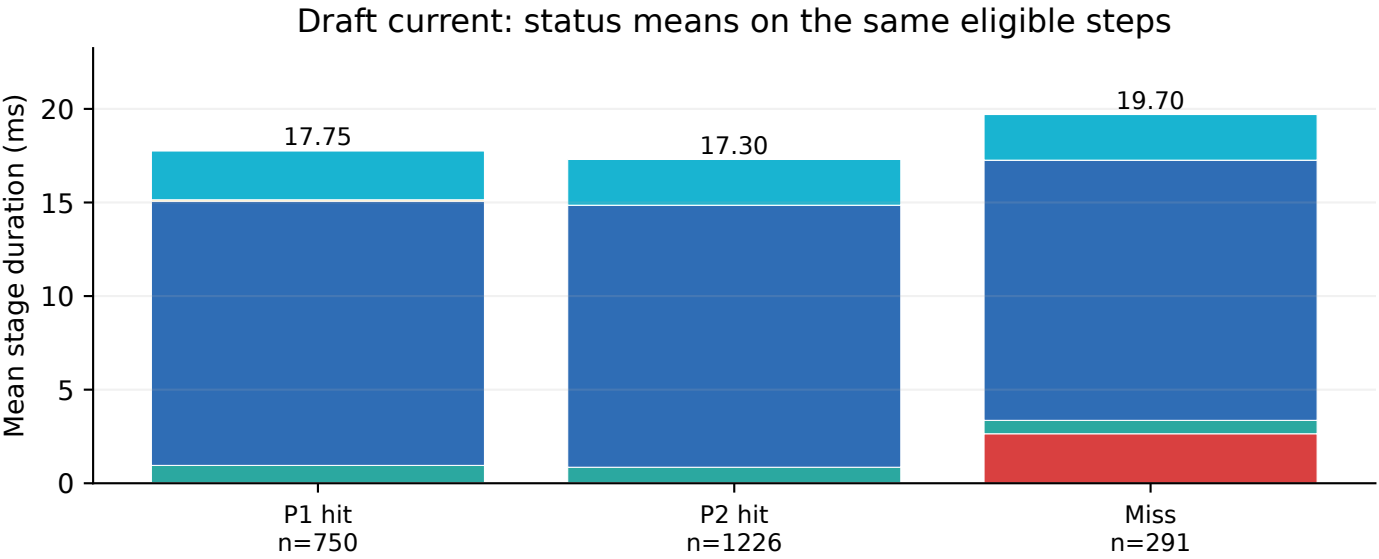
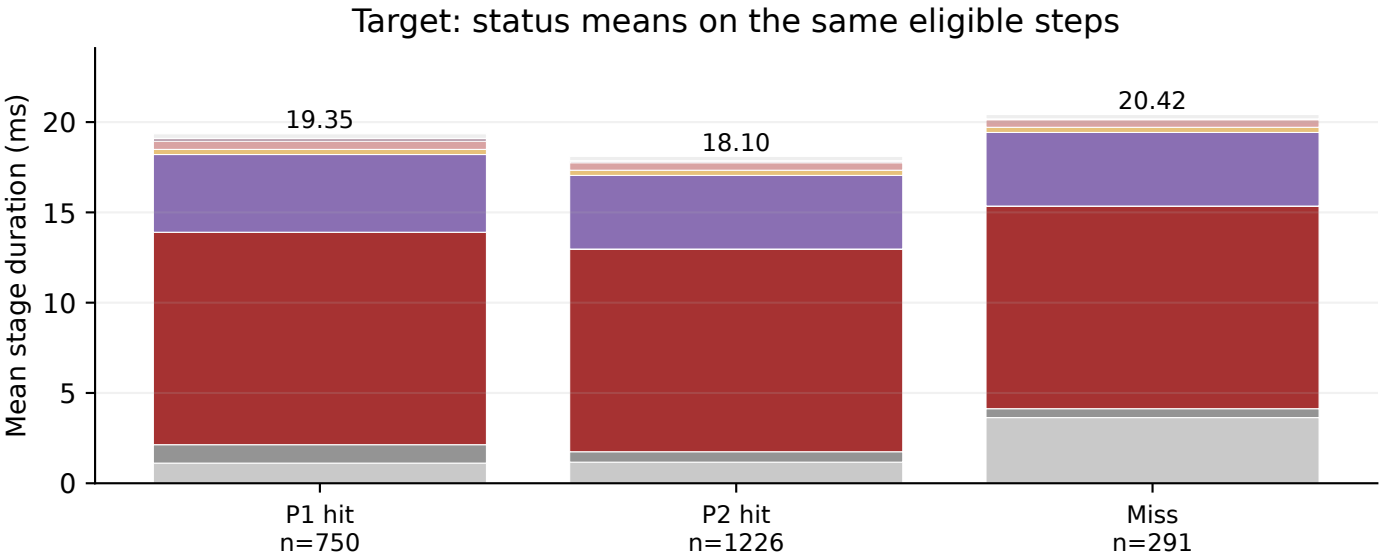
Observed representative steps (near median target window)



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

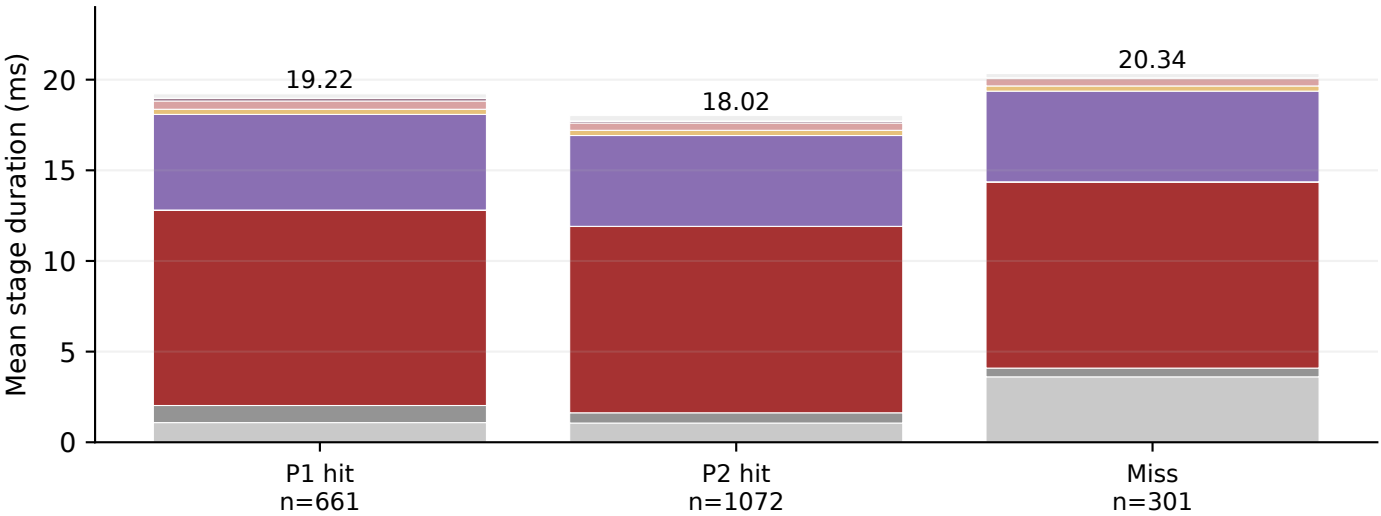
llama2_b1_postopt_profile_exit_later
DUET: K1/K2=12/2, exit=23 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2267 eligible steps



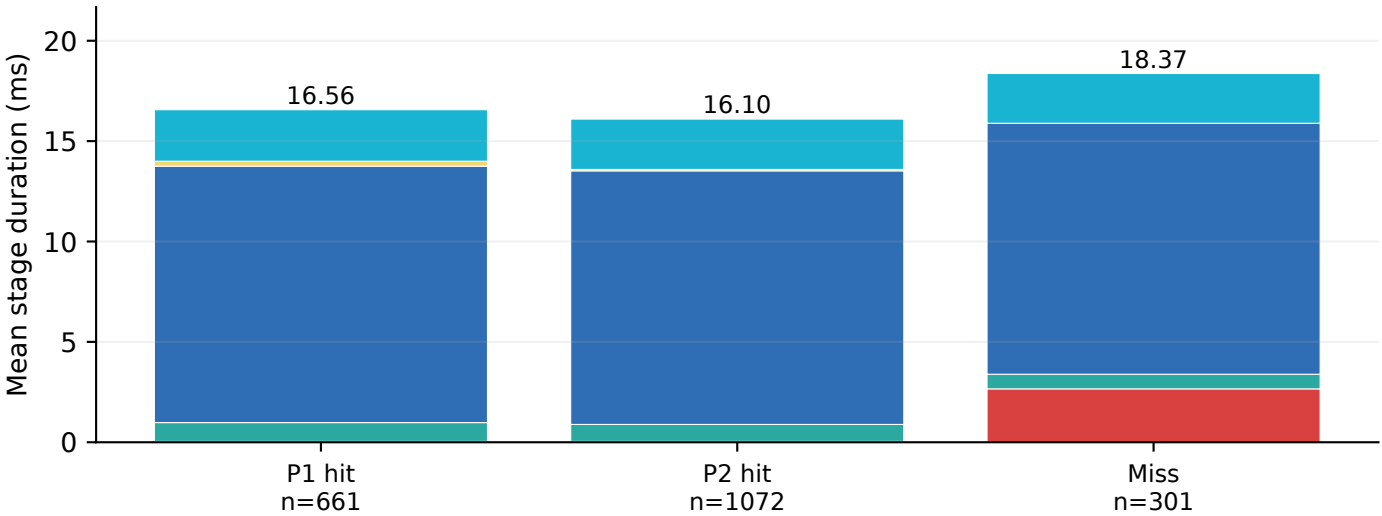
Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

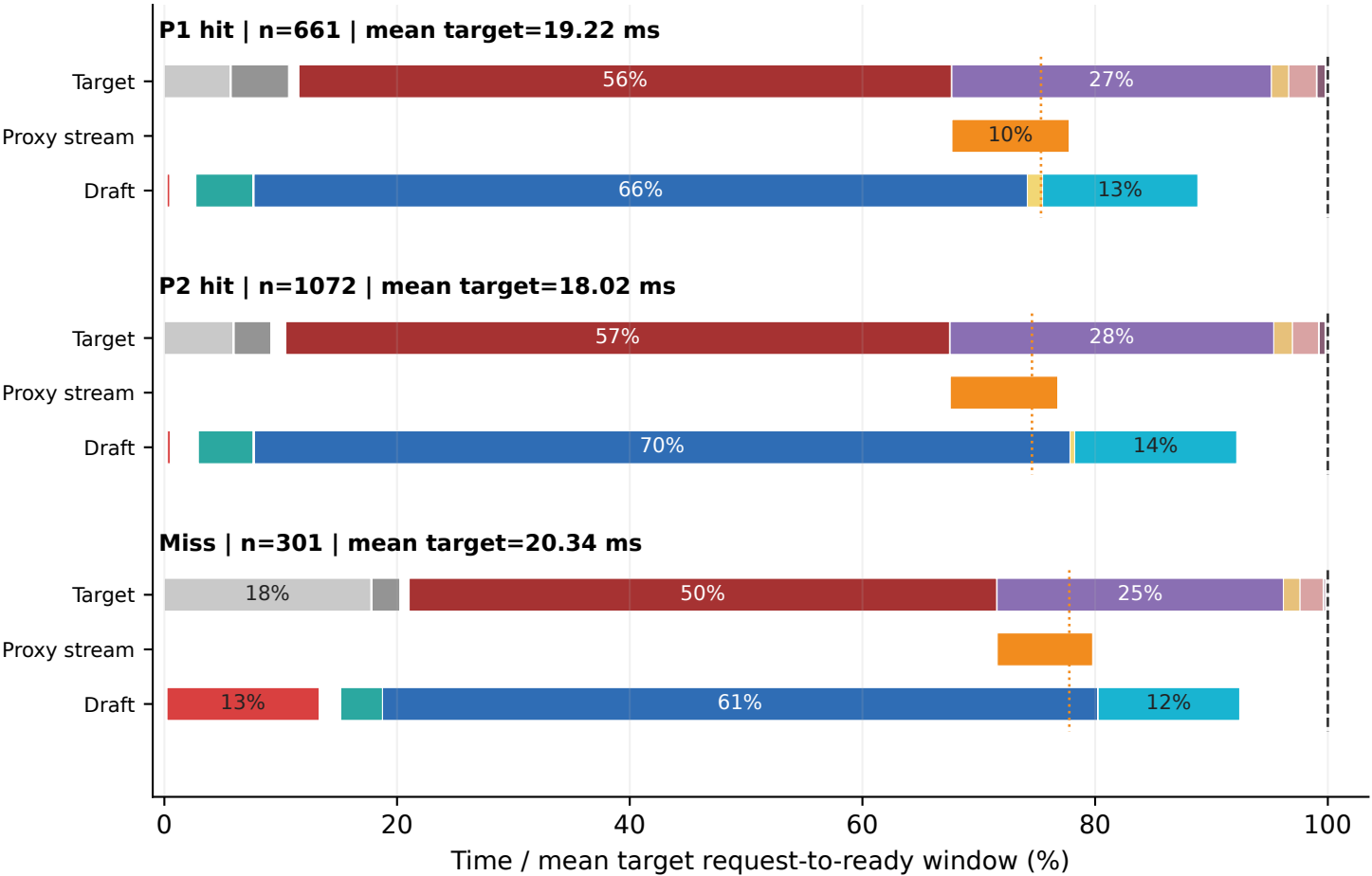
Target: status means on the same eligible steps



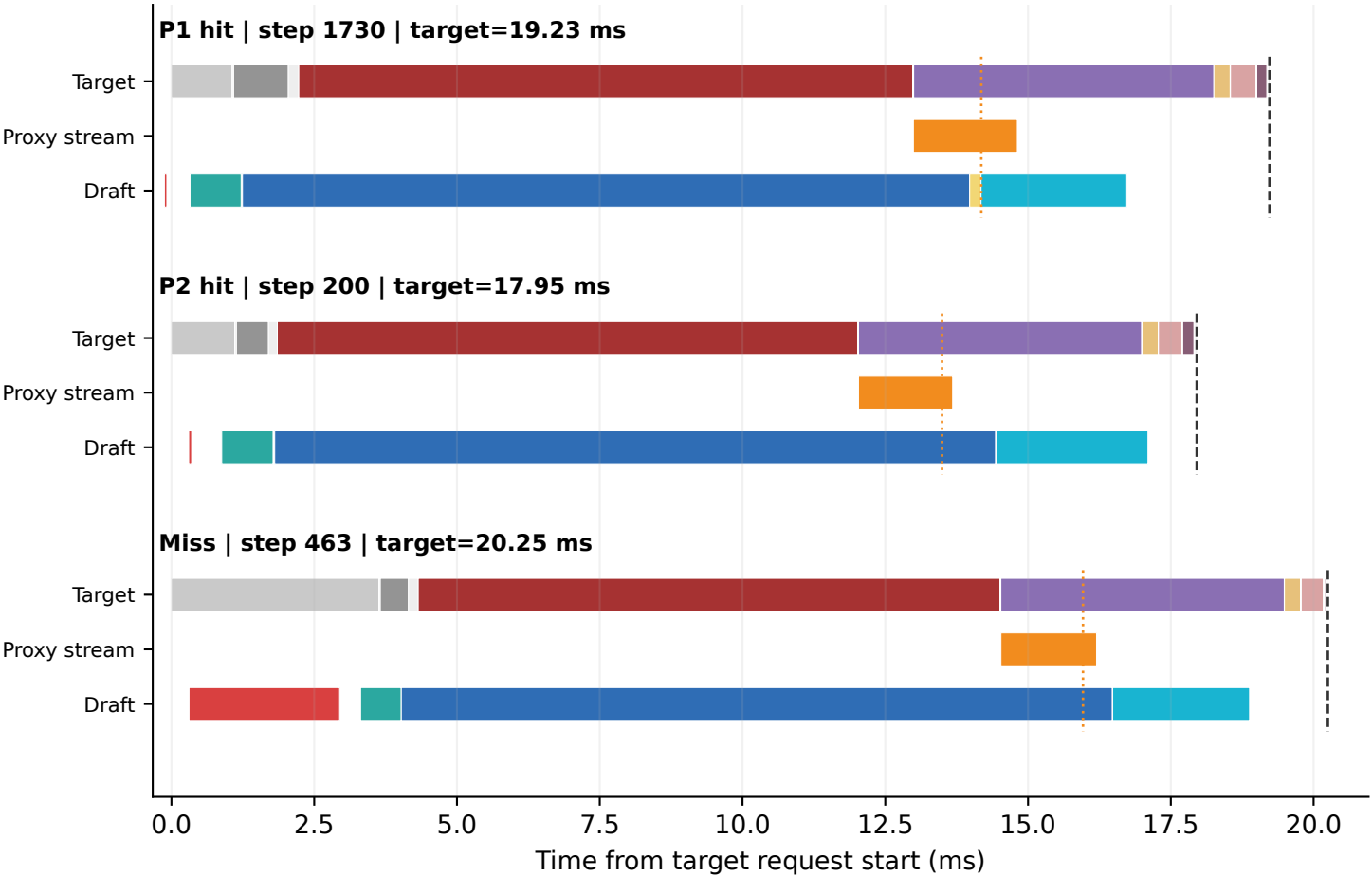
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



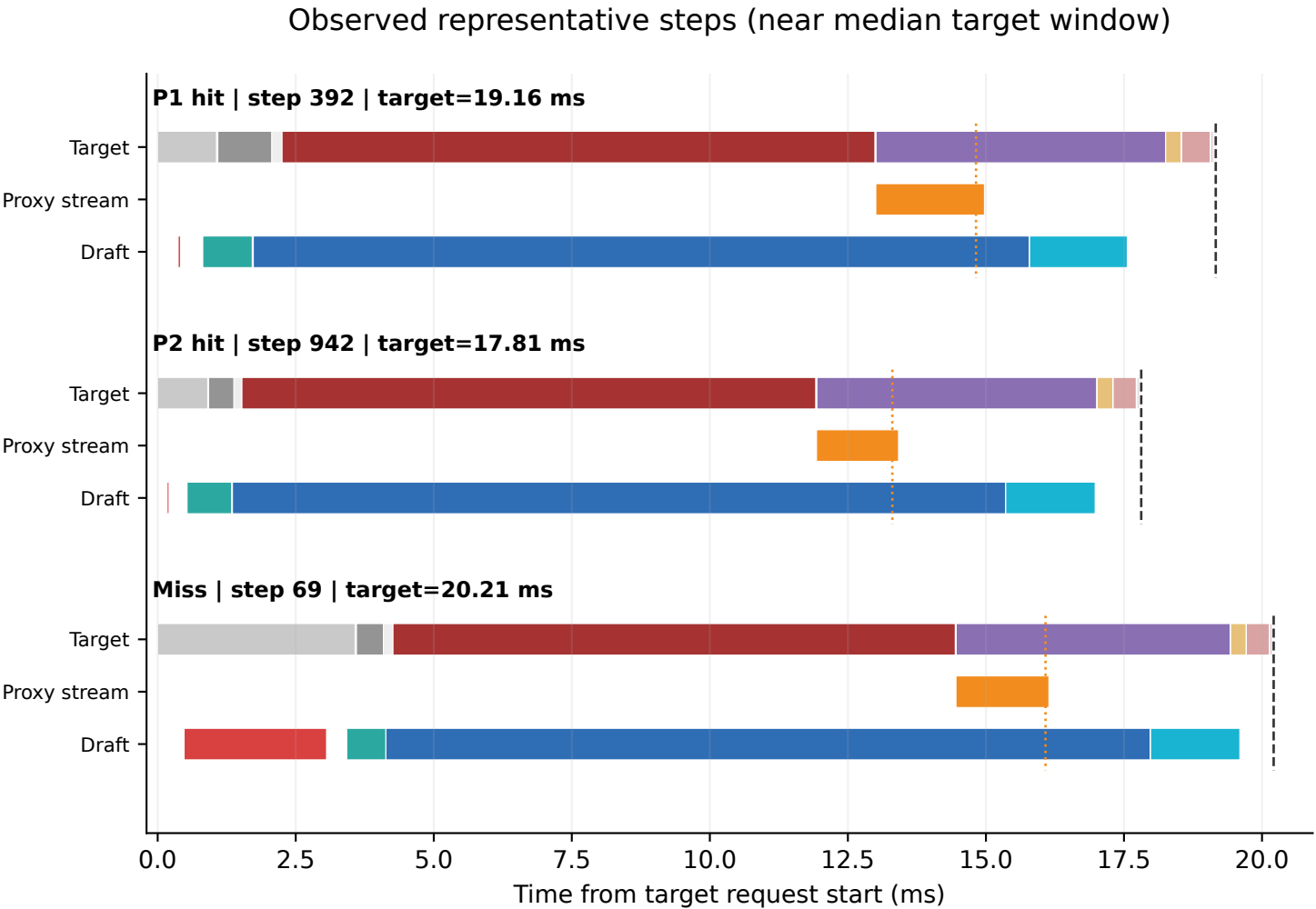
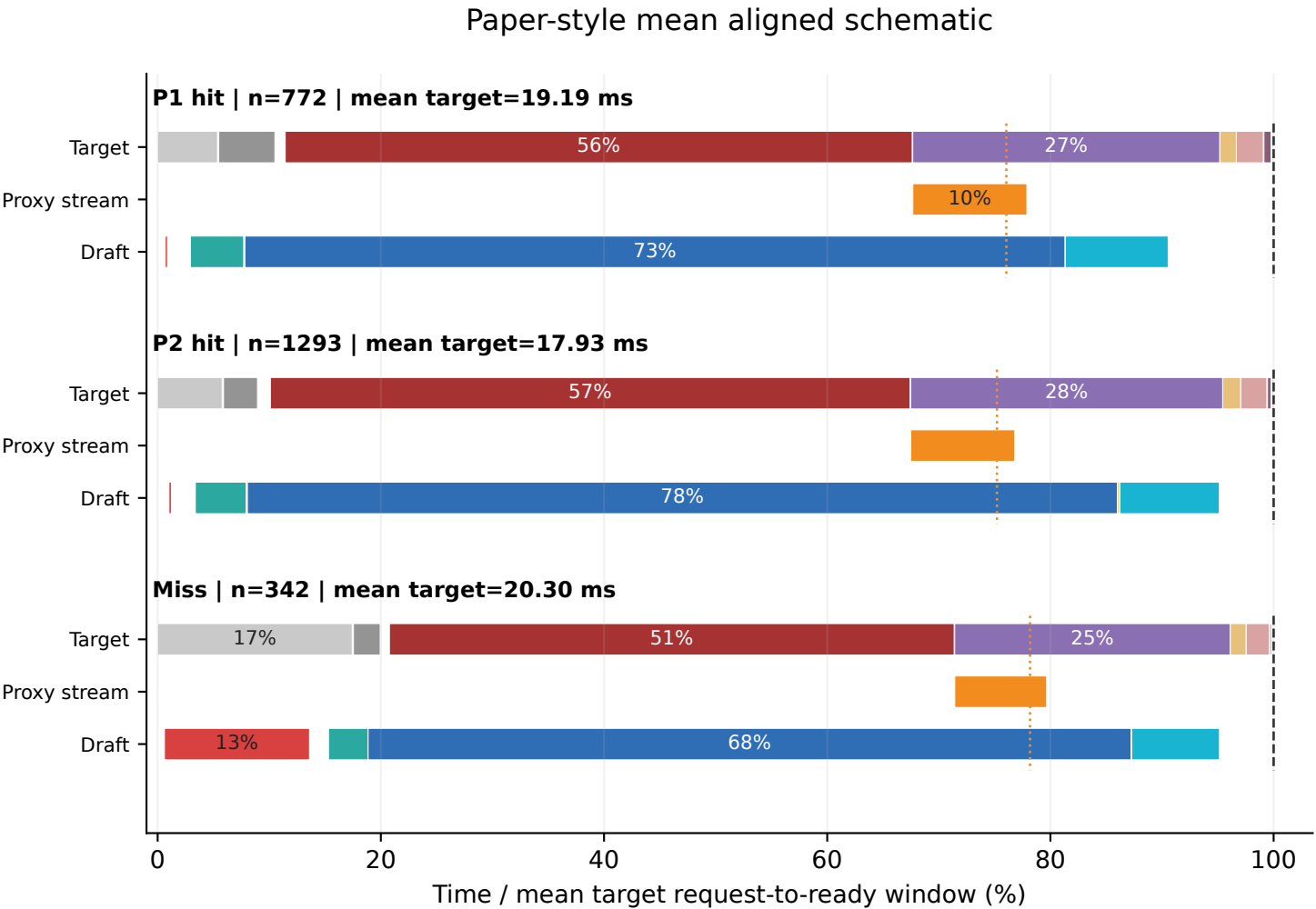
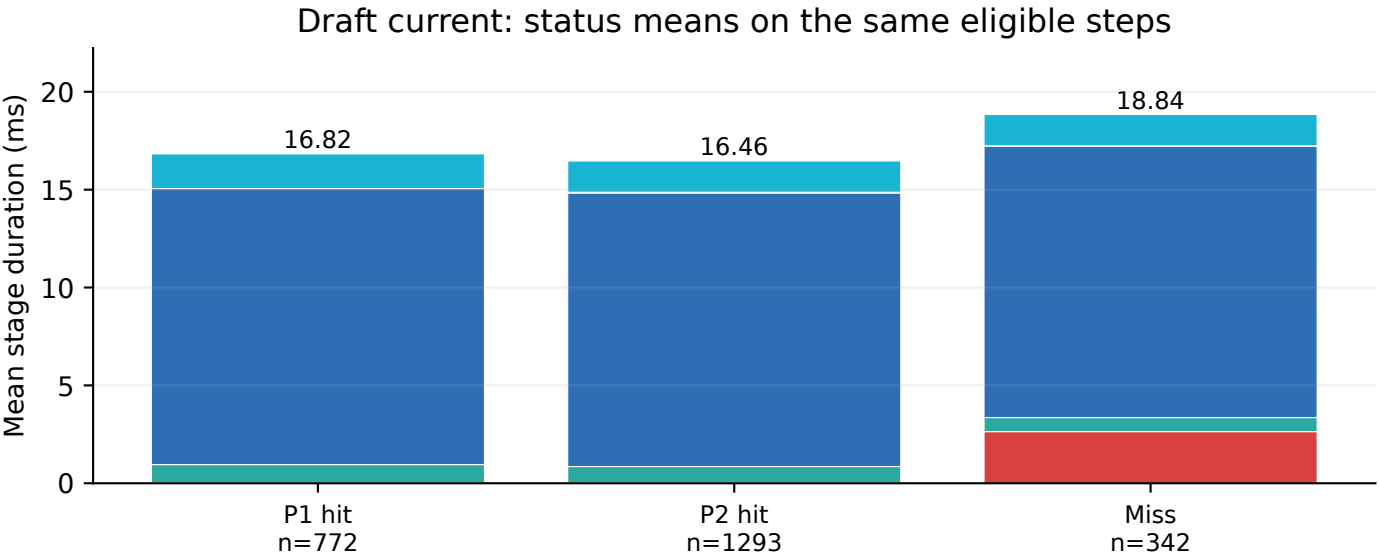
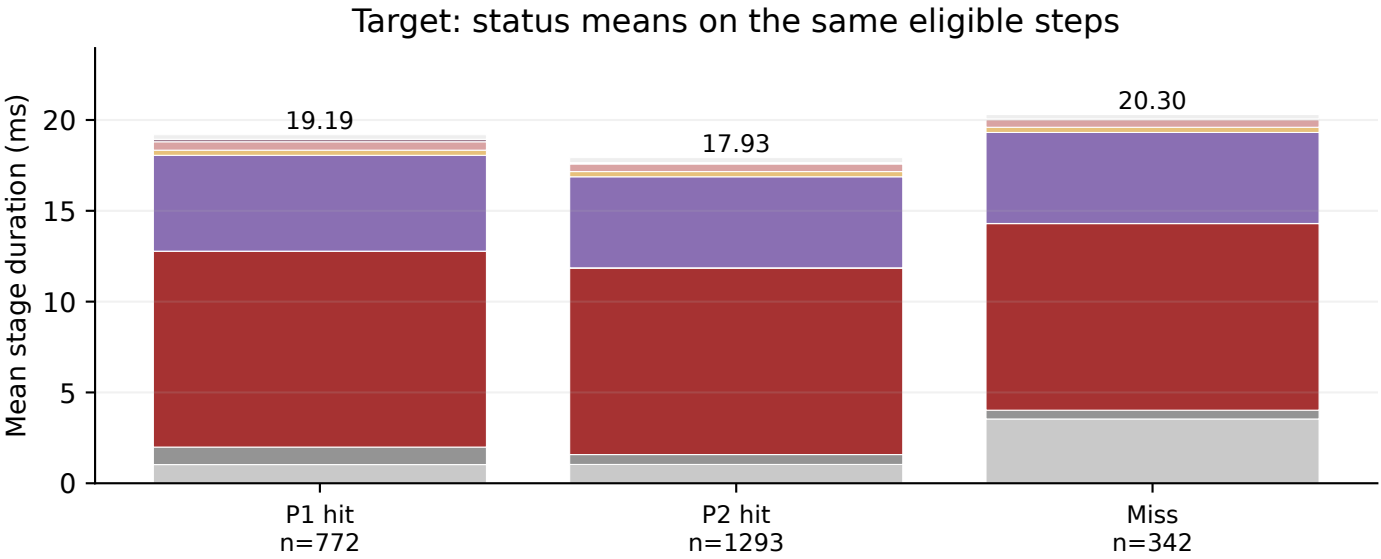
Observed representative steps (near median target window)



Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_postopt_profile_k2_down
DUET: K1/K2=12/1, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2407 eligible steps

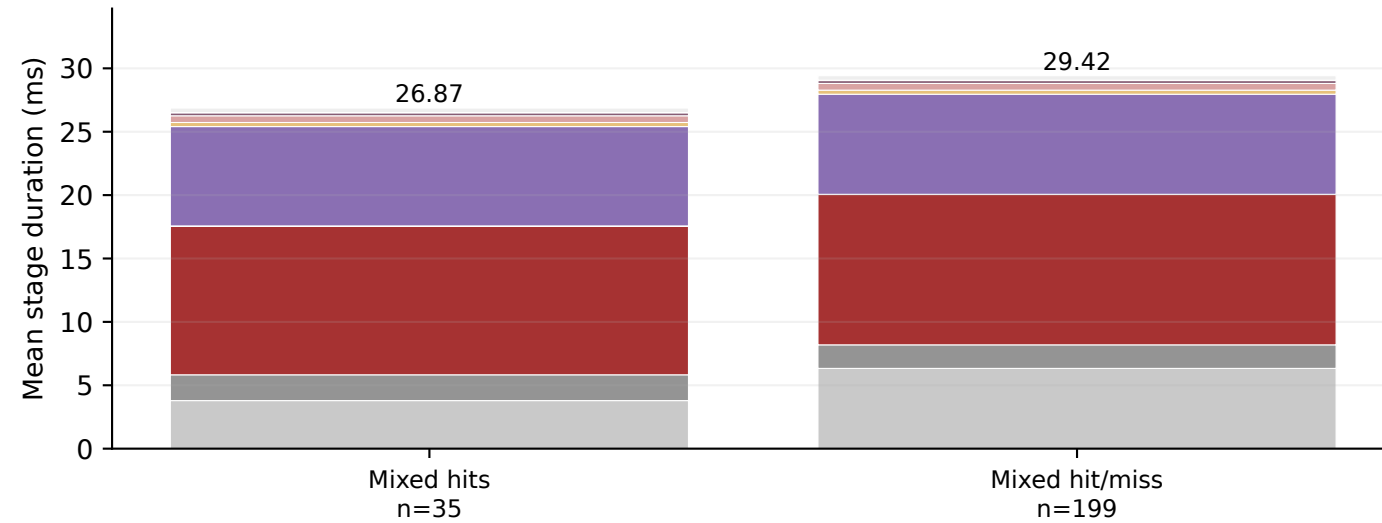


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

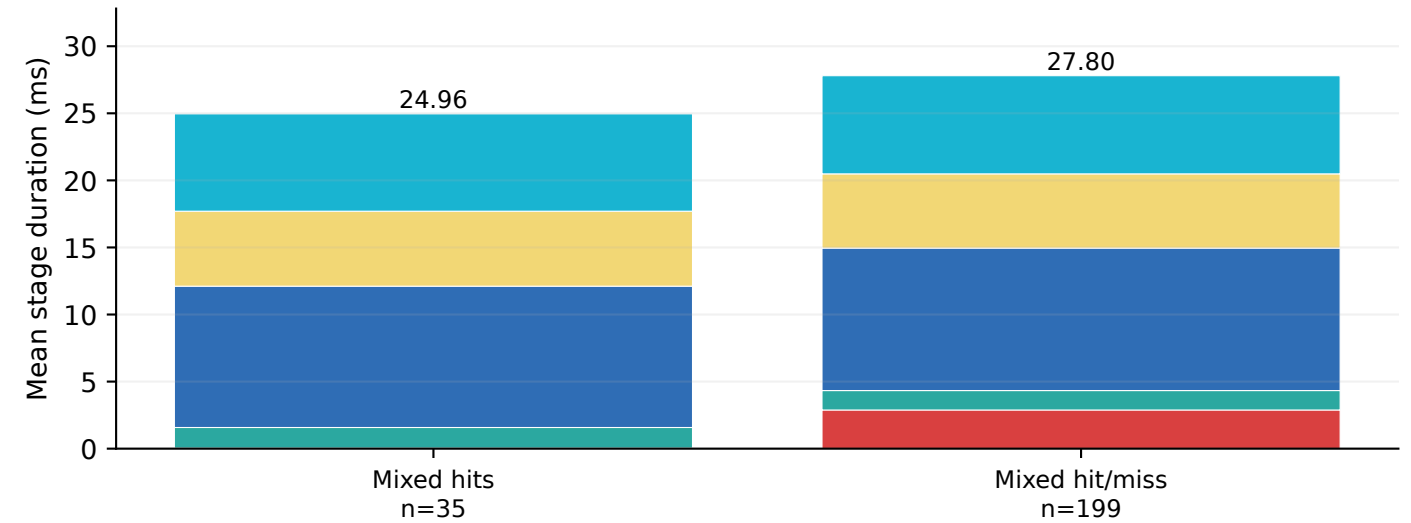
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_postopt_profile_exit_earlier
DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 234 eligible steps

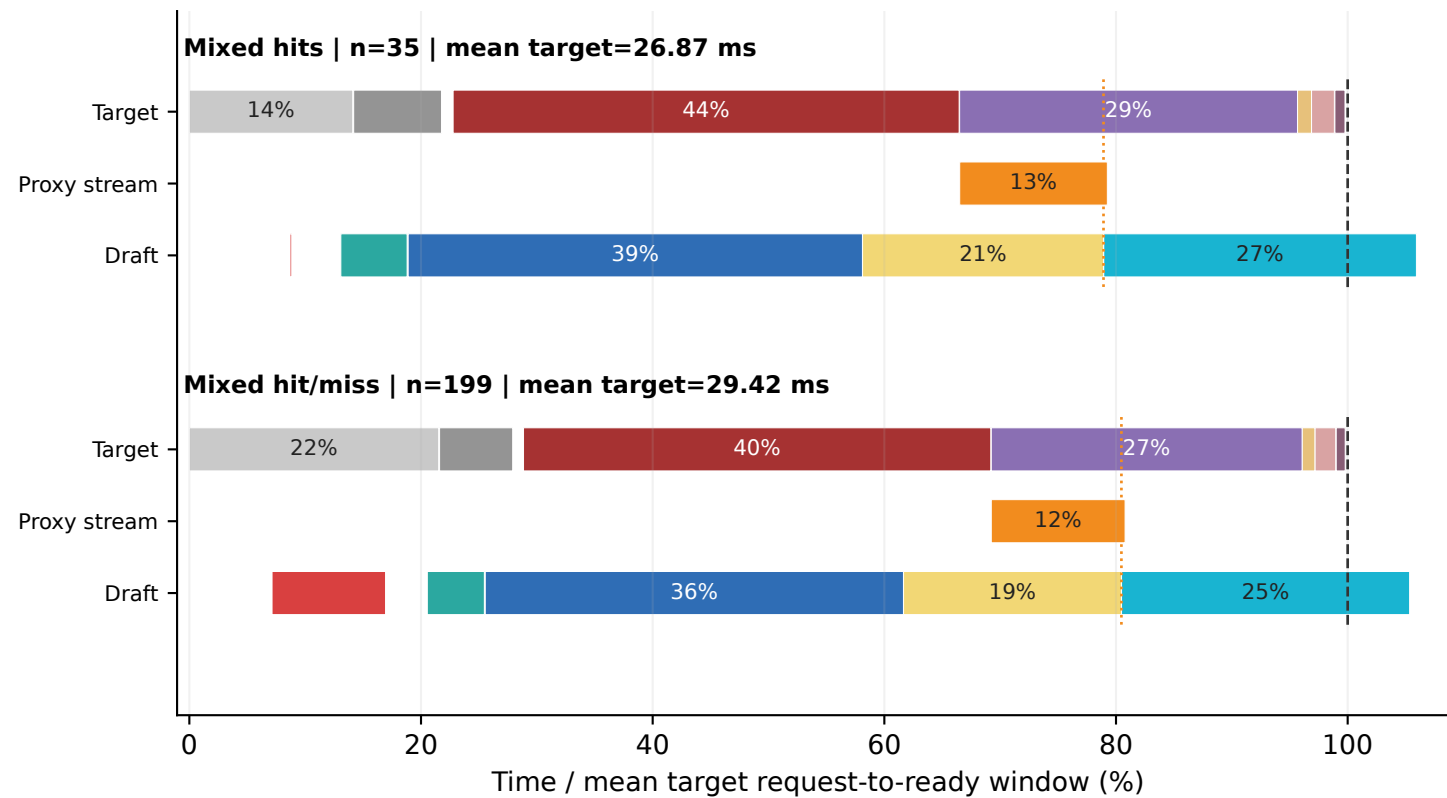
Target: status means on the same eligible steps



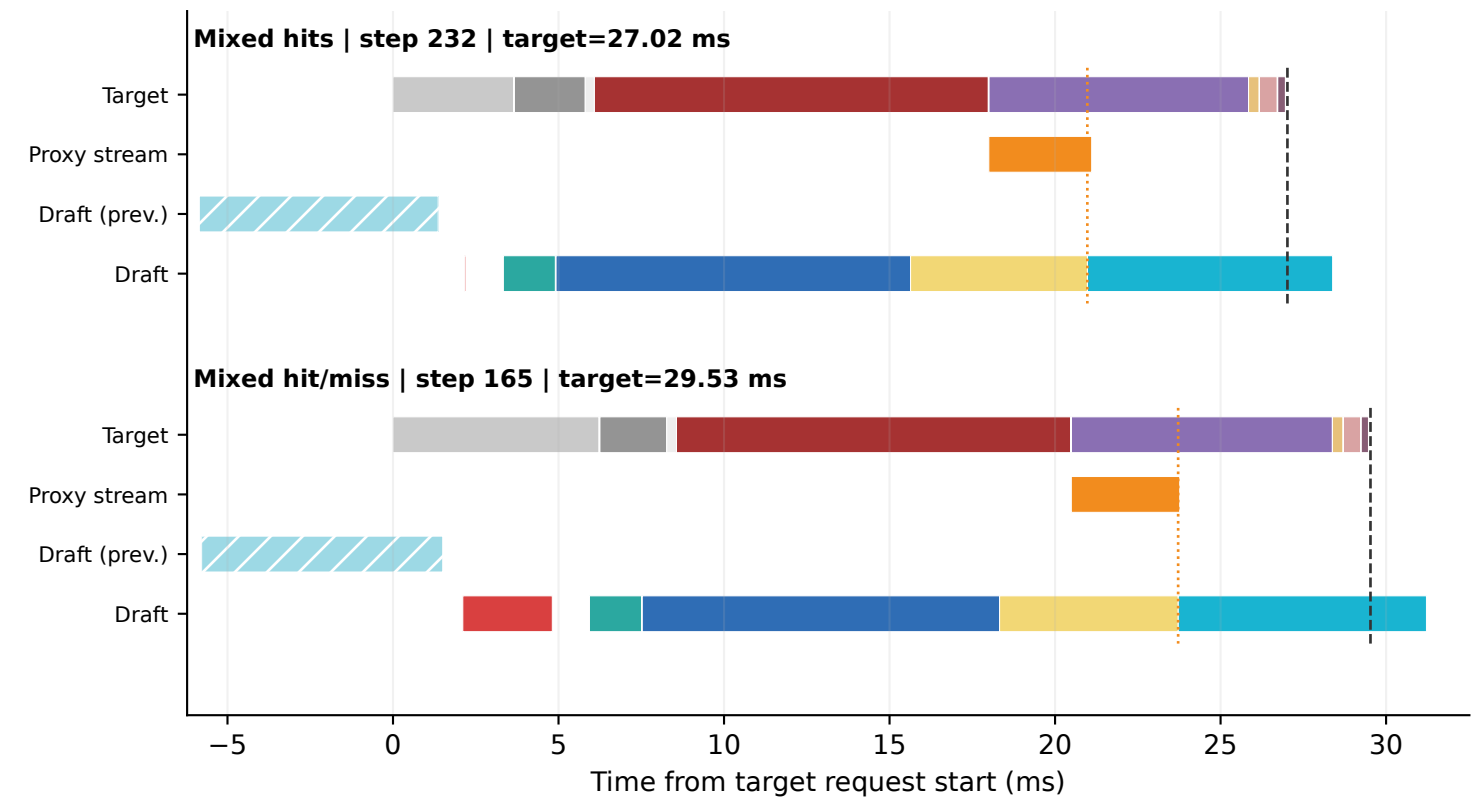
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

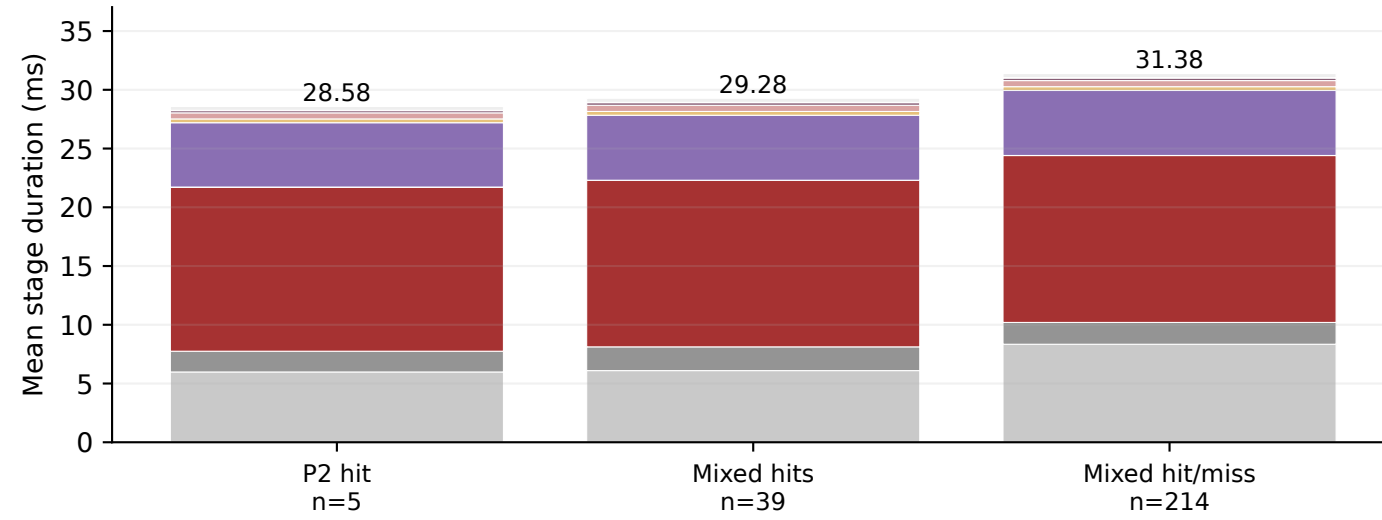


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

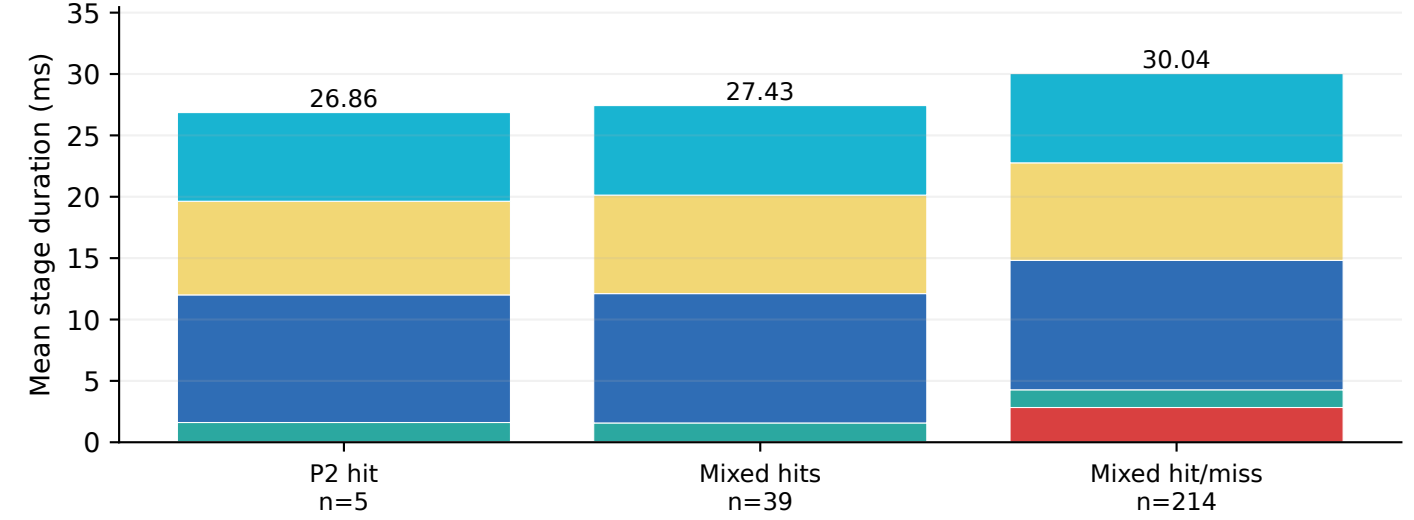
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_postopt_profile_exit_later
 DUET: K1/K2=3/2, exit=23 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 258 eligible steps

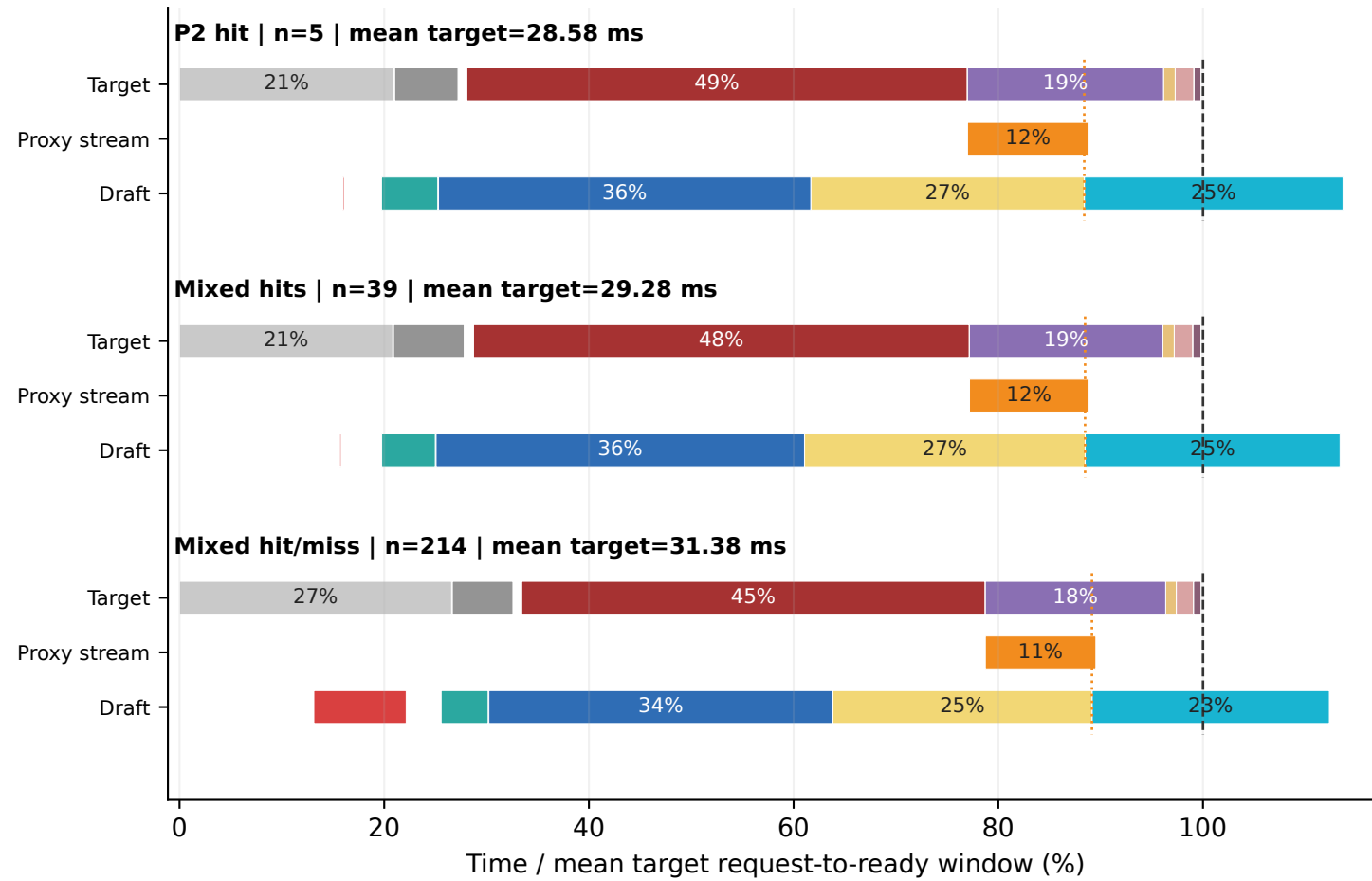
Target: status means on the same eligible steps



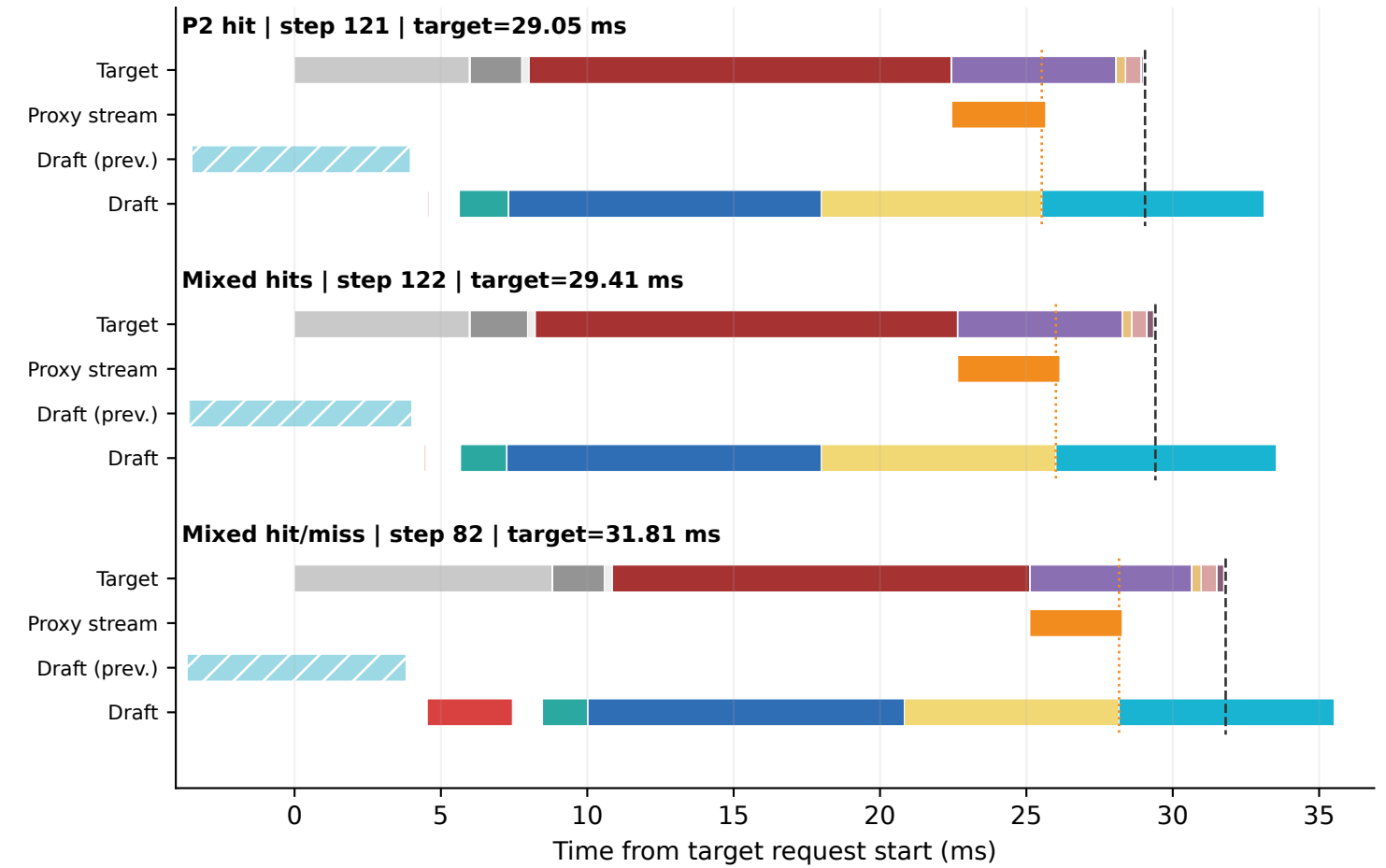
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



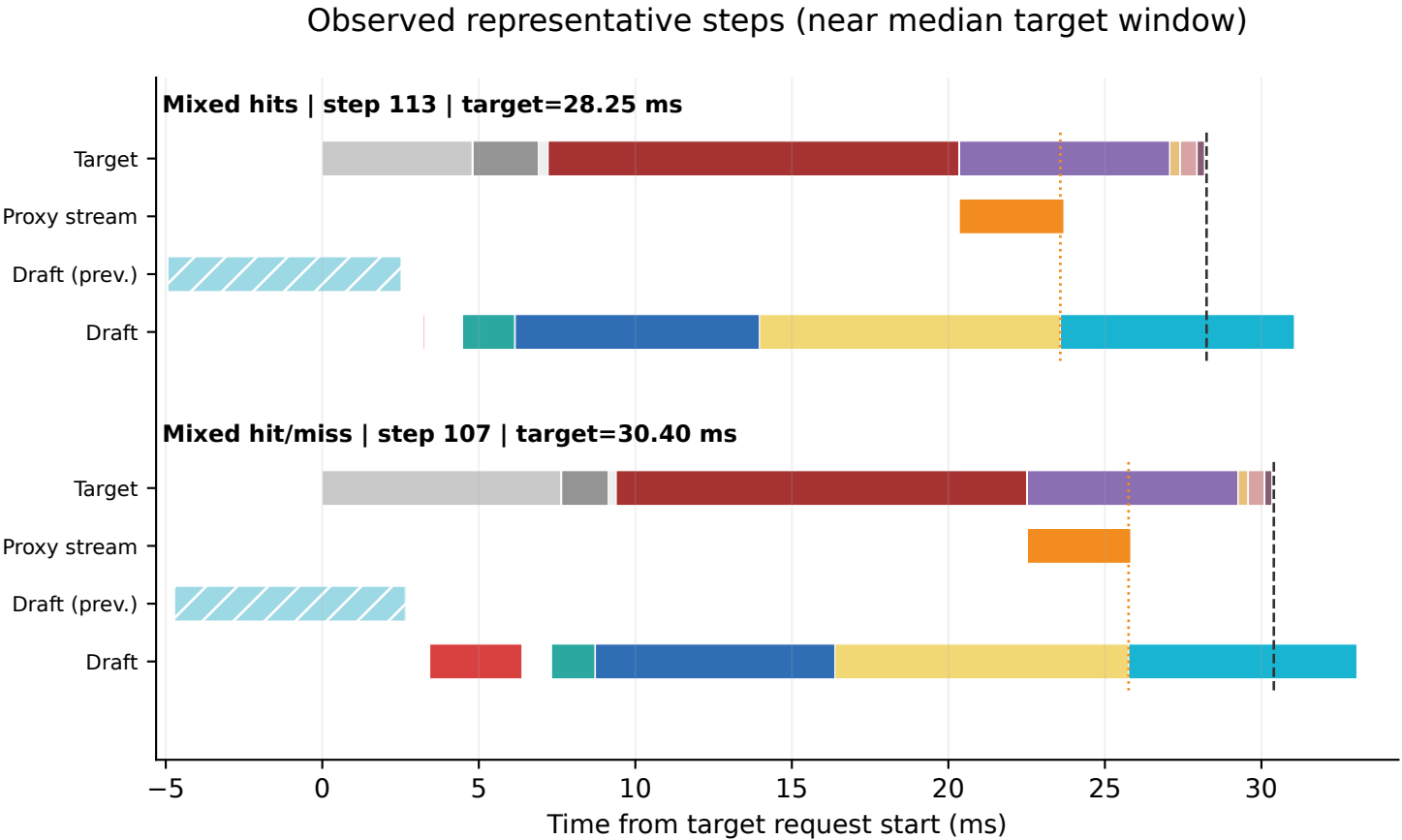
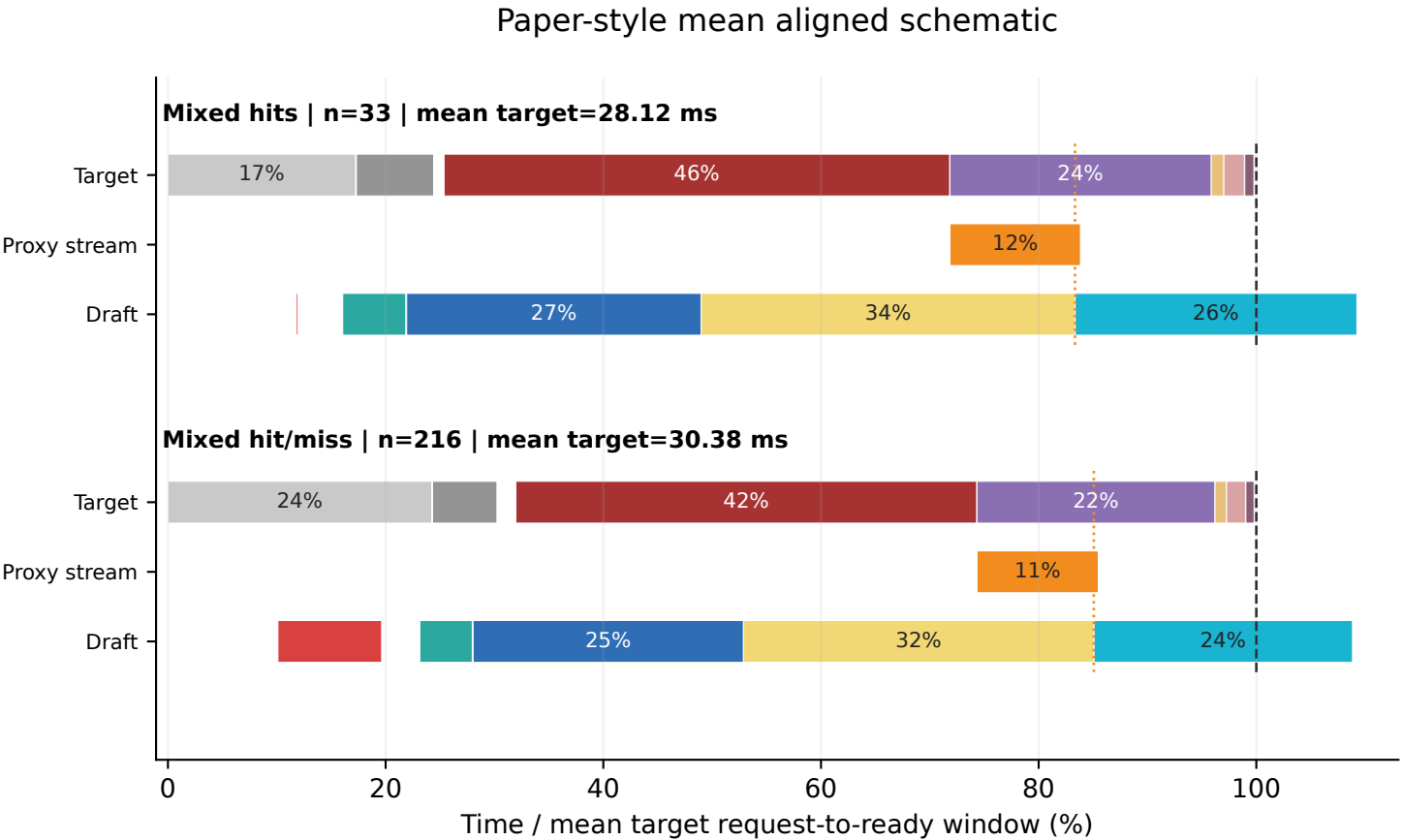
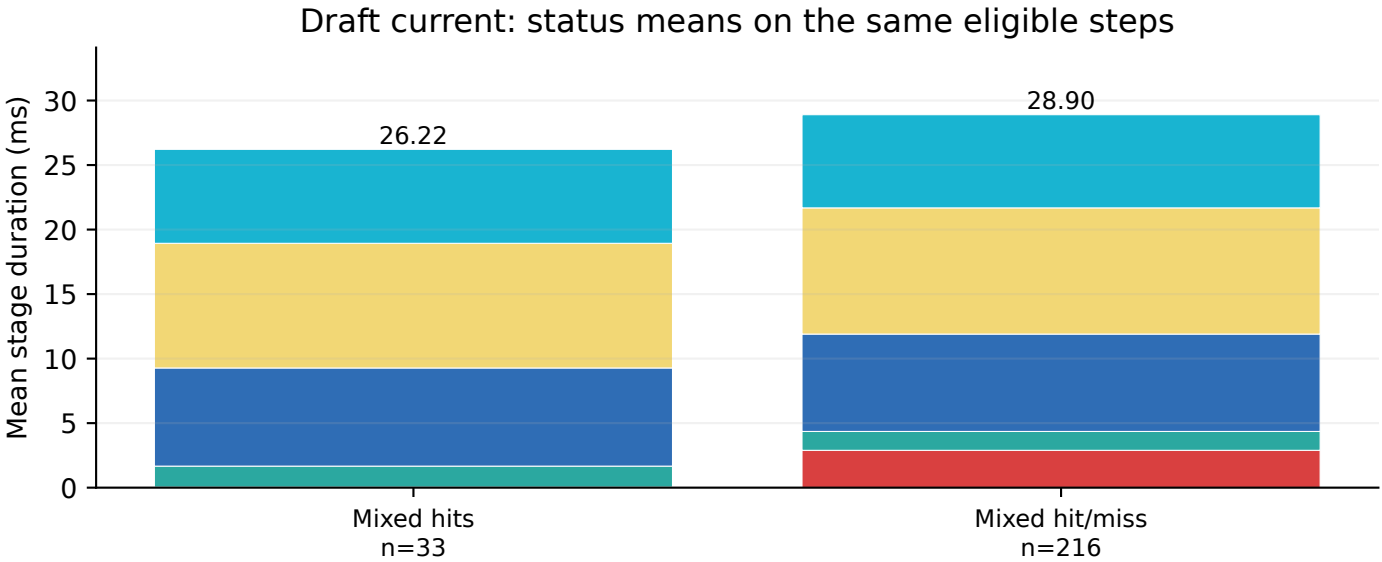
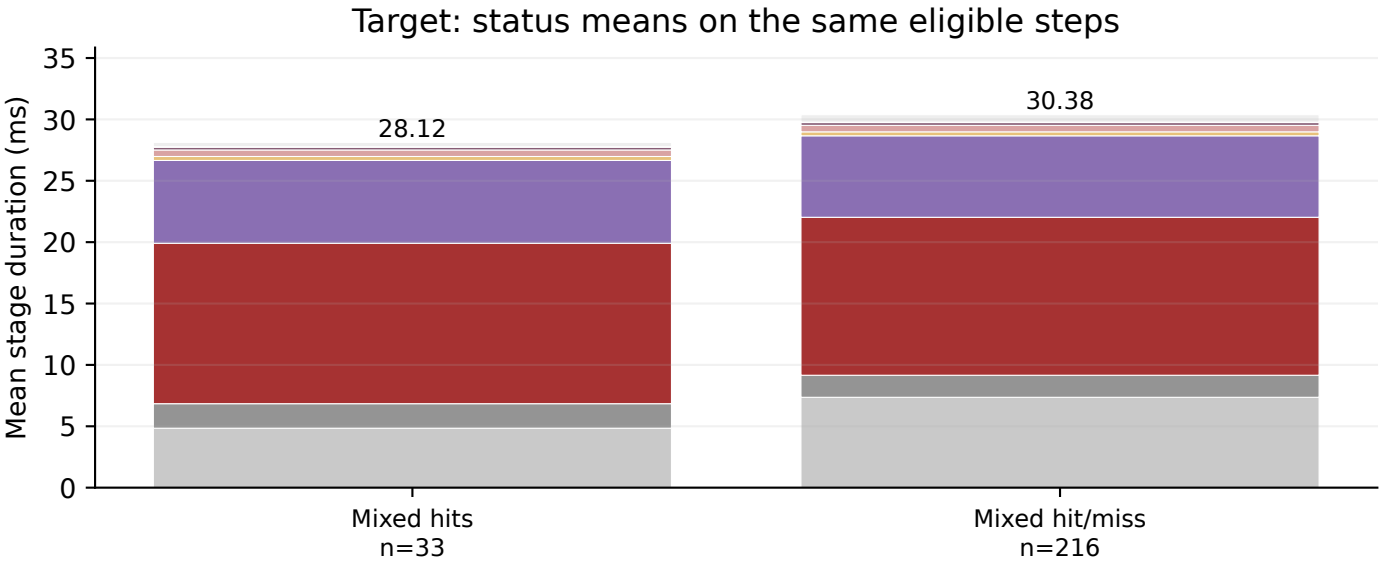
Observed representative steps (near median target window)



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
 Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
 Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
 Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

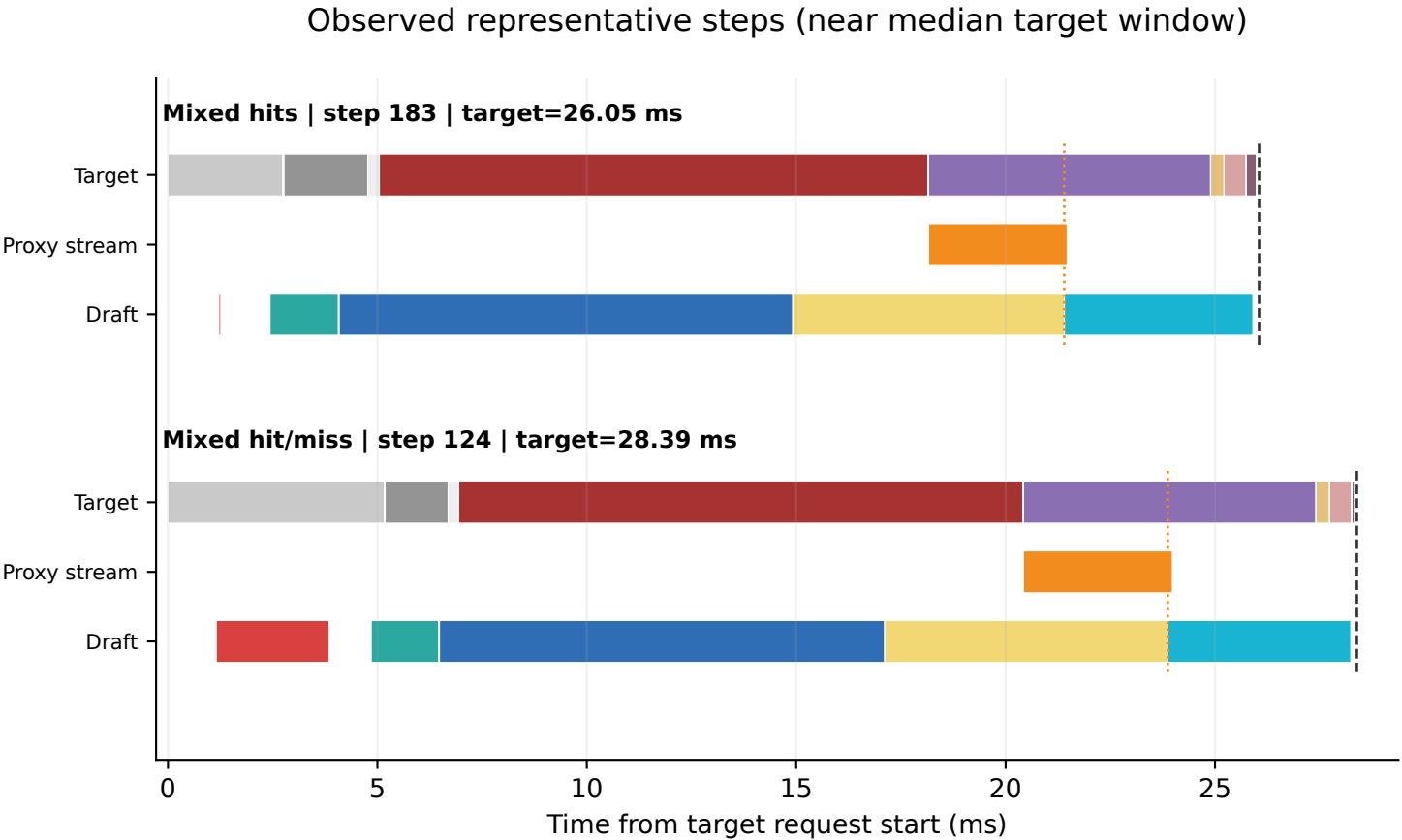
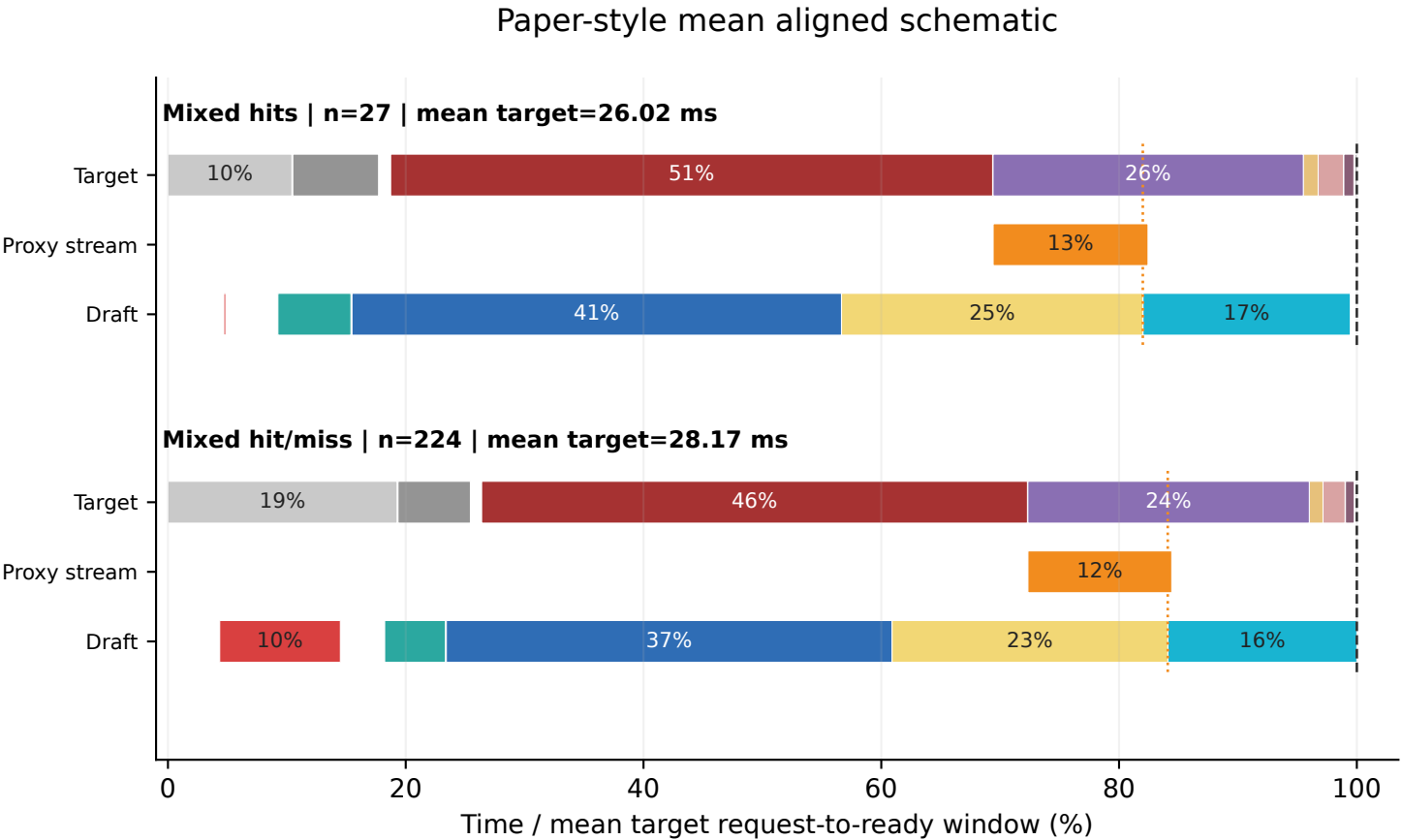
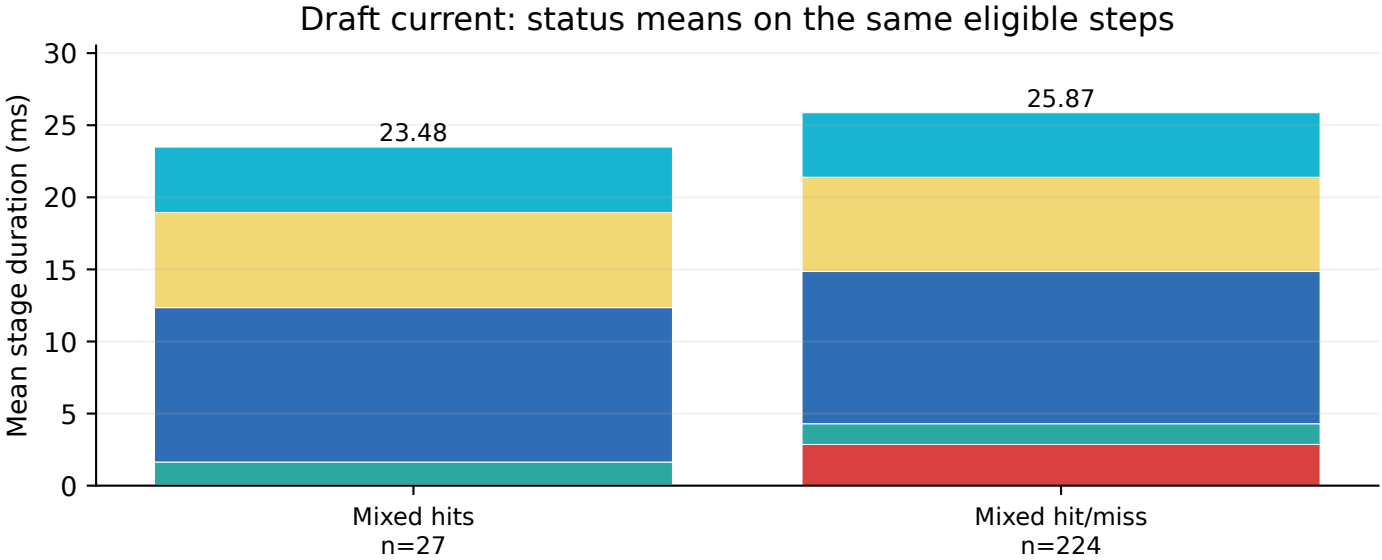
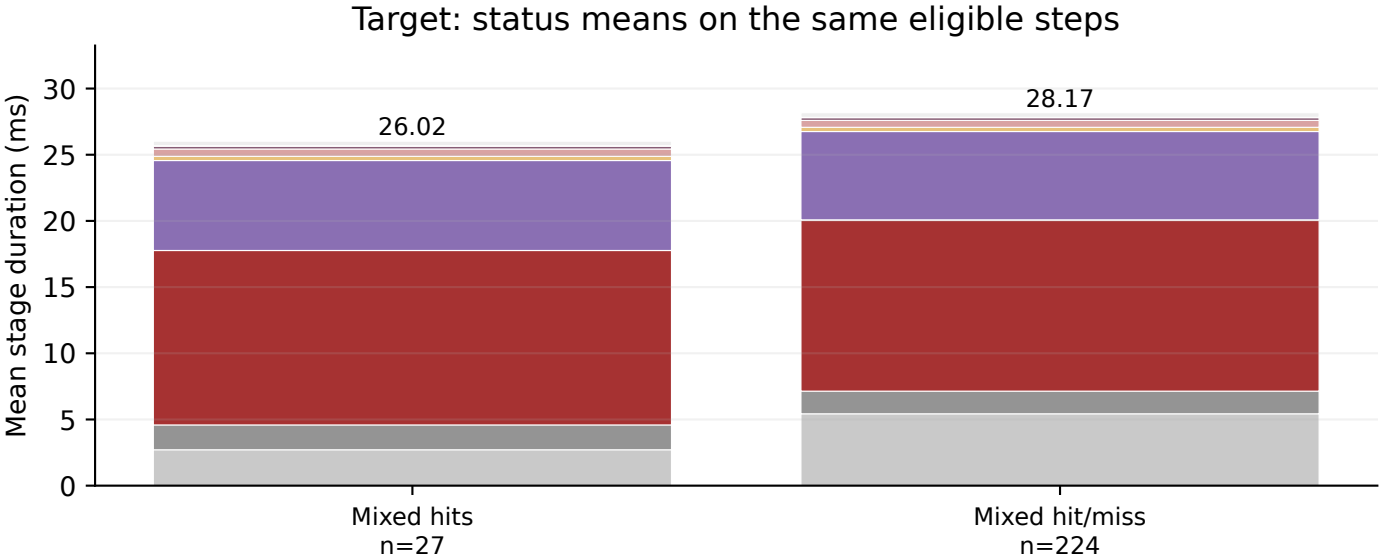
llama2_b8_postopt_profile_k1_down
DUET: K1/K2=2/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=3, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 249 eligible steps



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_postopt_profile_k2_down
DUET: K1/K2=3/1, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 251 eligible steps

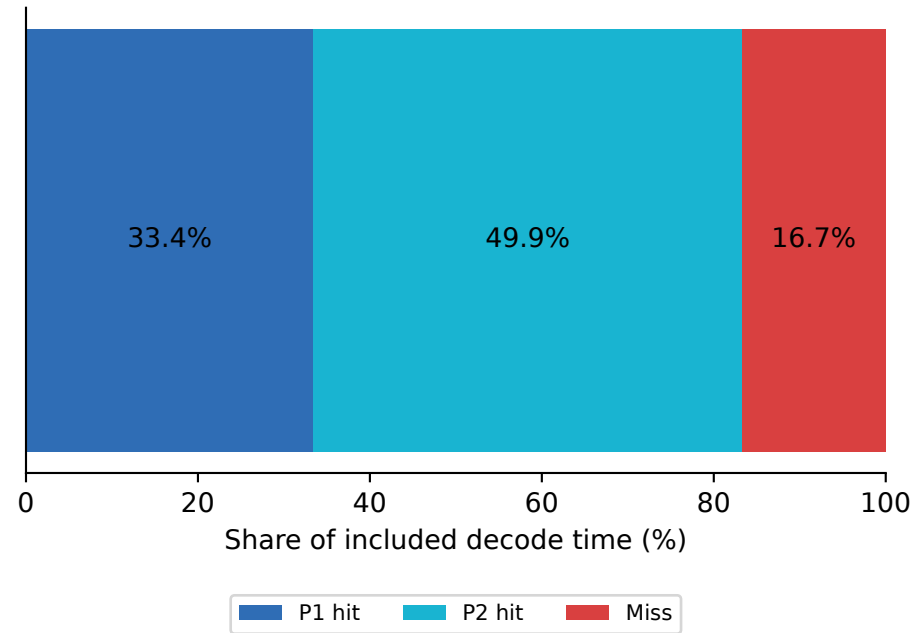


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

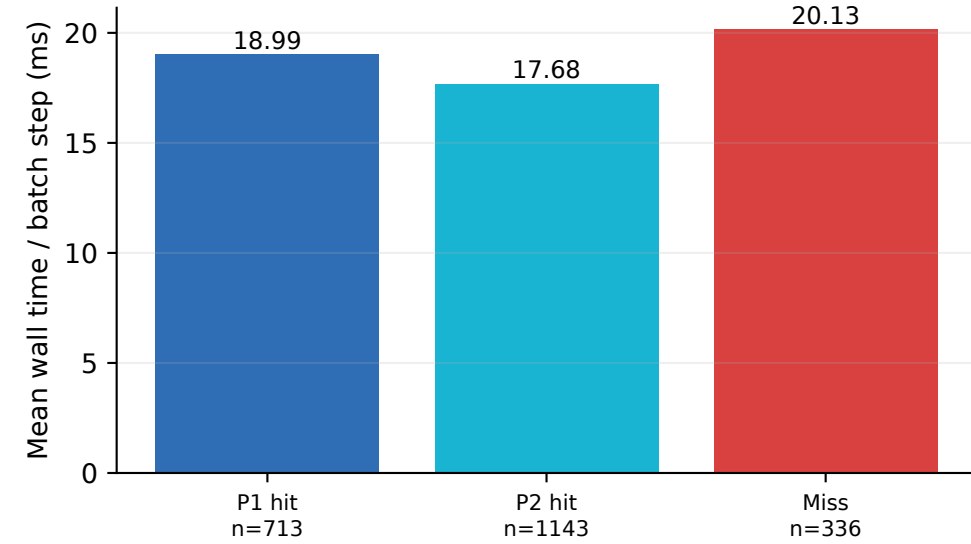
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_postopt_warm_0
DUET: K1/K2=11/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=12, trim/stream=1/1
Last pass: B=1, T=0.7, N=48 | AL*=2.465, hit=84.8%, TPS*=133.37

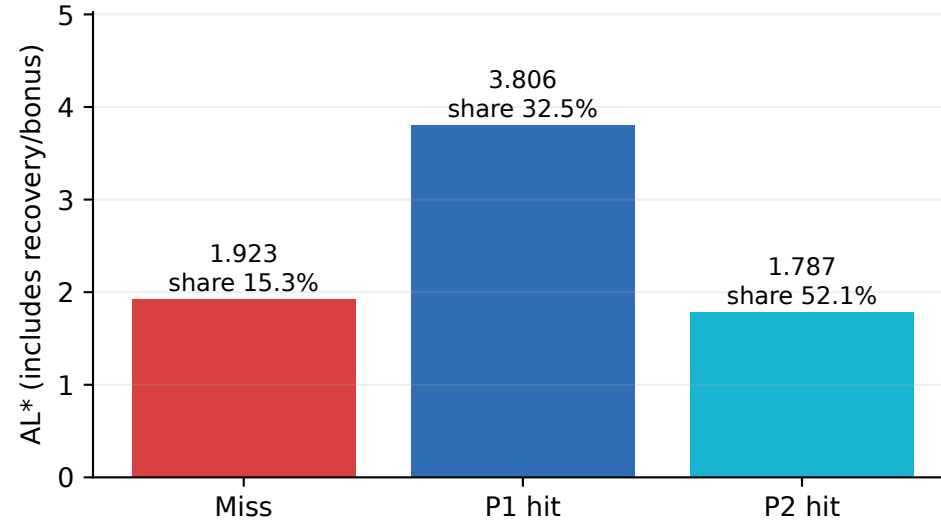
Measured wall time by batch cache status



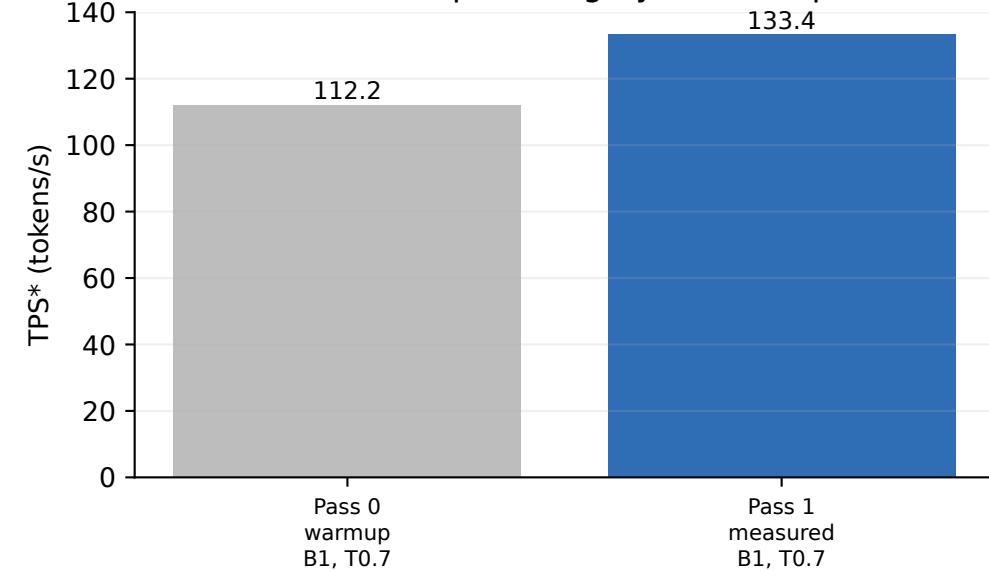
Boundary-free steps, including slow steps



Conditional AL* and source share



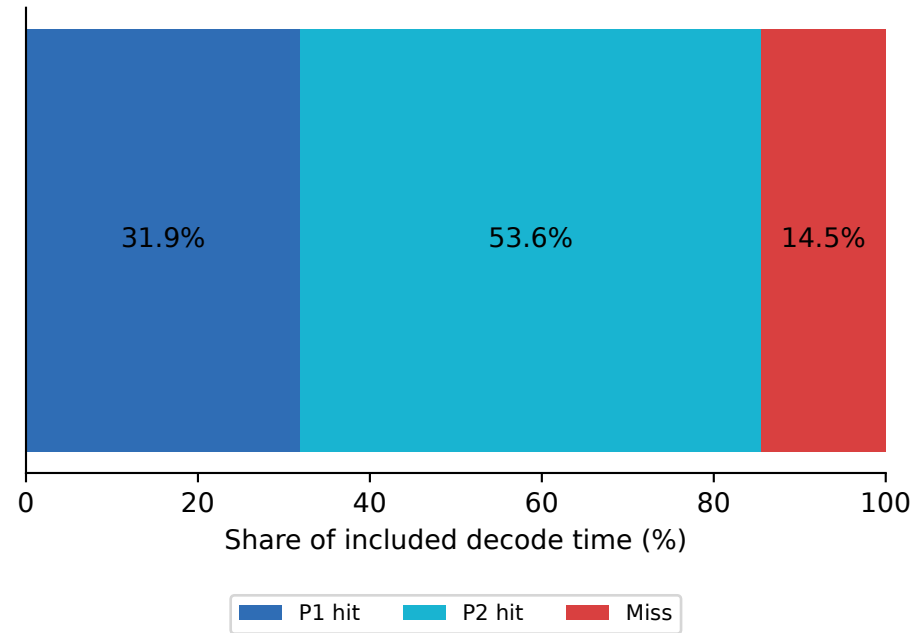
All saved passes; gray = warmup



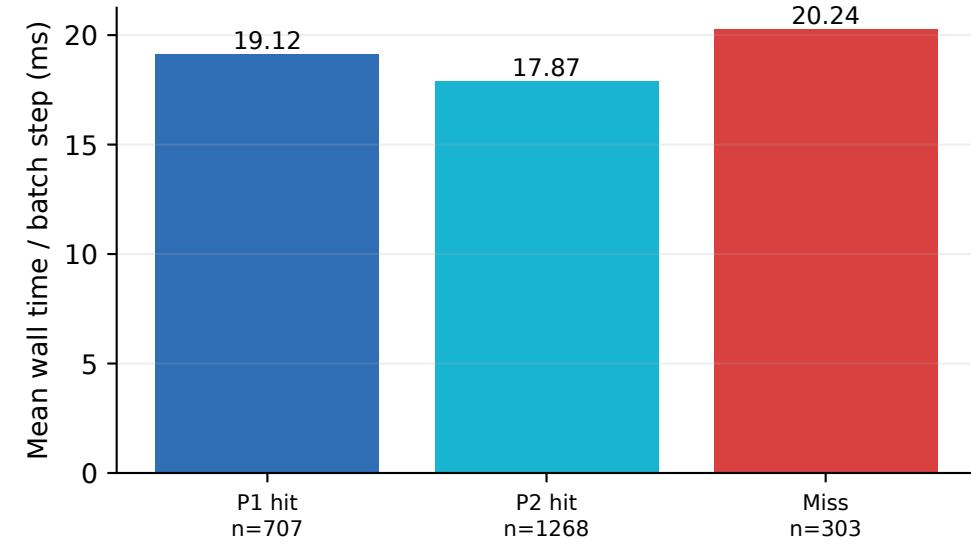
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_postopt_warm_1
DUET: K1/K2=12/2, exit=23 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
Last pass: B=1, T=0.7, N=48 | AL*=2.400, hit=86.9%, TPS*=129.25

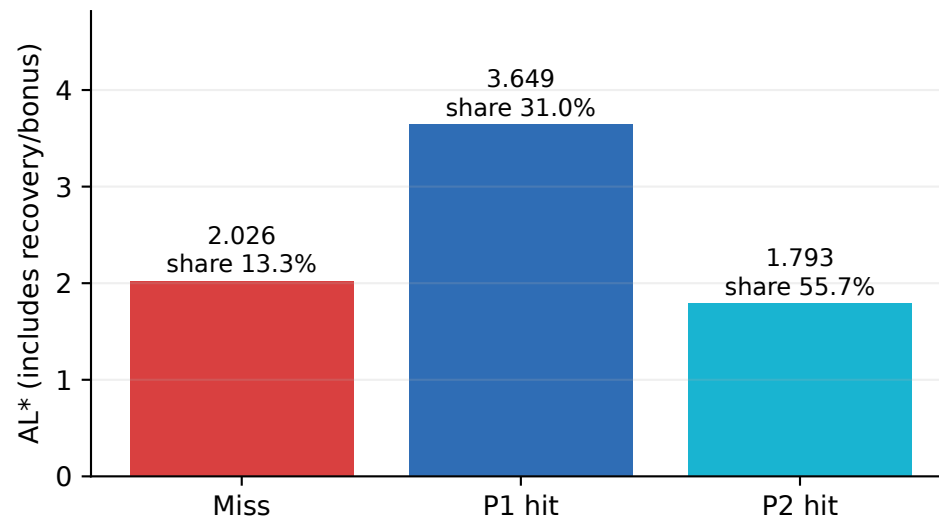
Measured wall time by batch cache status



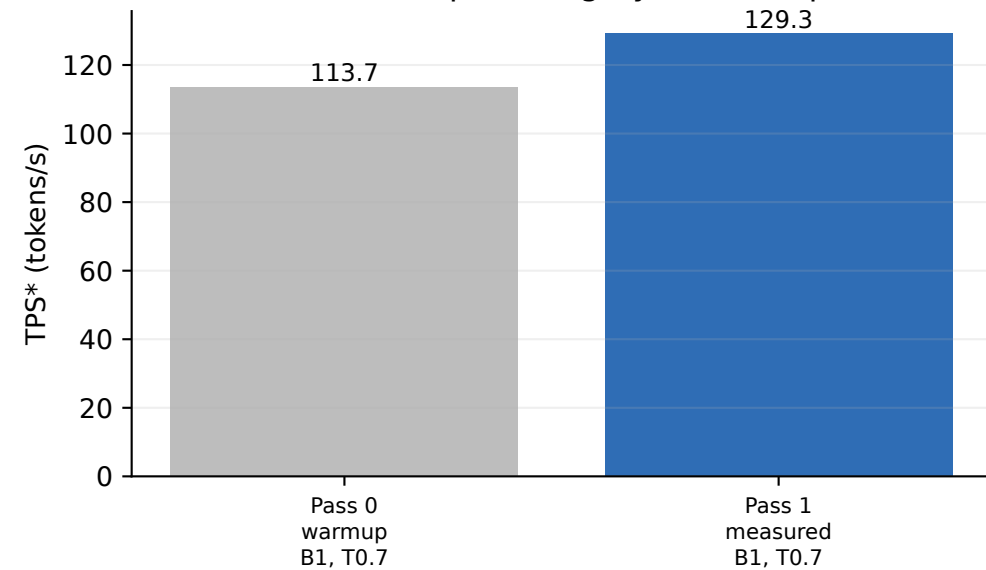
Boundary-free steps, including slow steps



Conditional AL* and source share



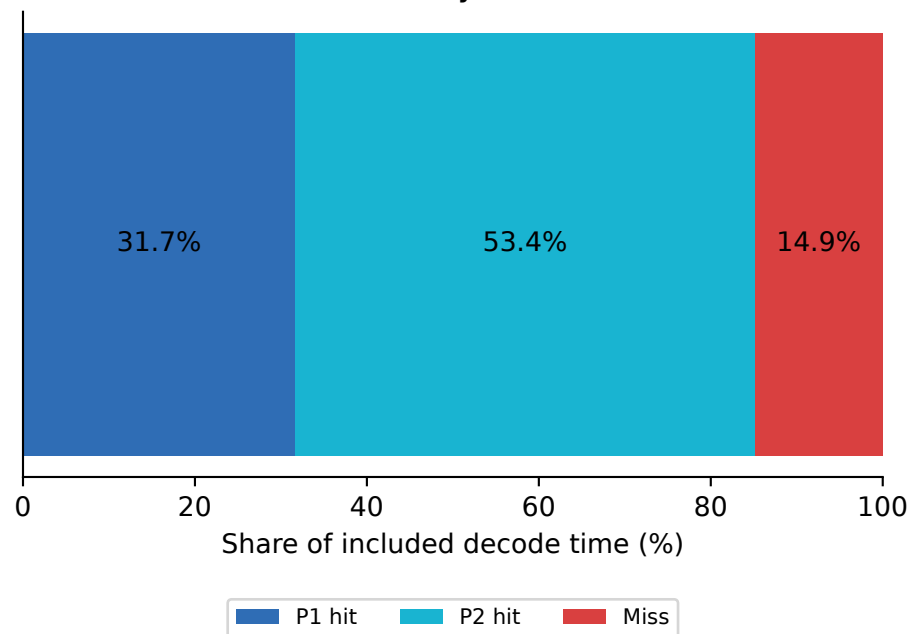
All saved passes; gray = warmup



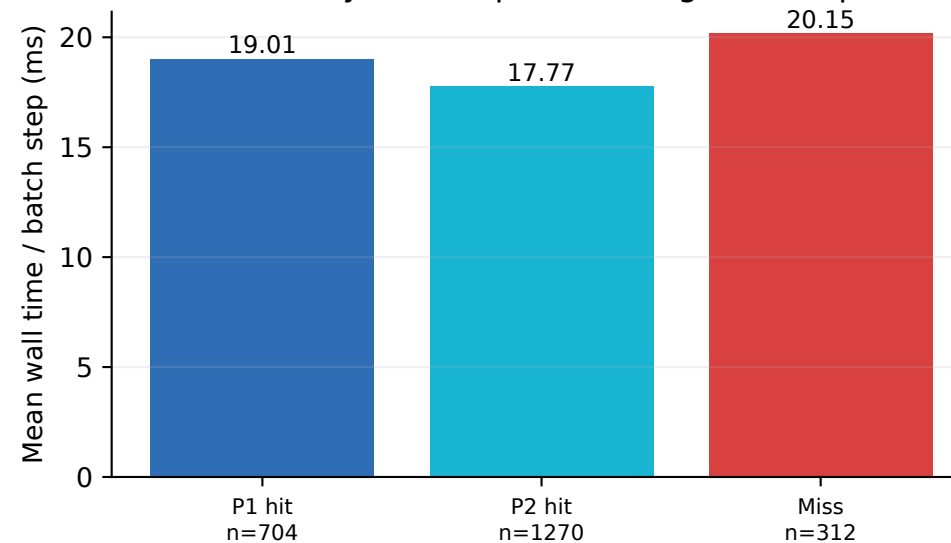
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_postopt_warm_2
 DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
 Last pass: B=1, T=0.7, N=48 | AL*=2.350, hit=86.4%, TPS*=127.21

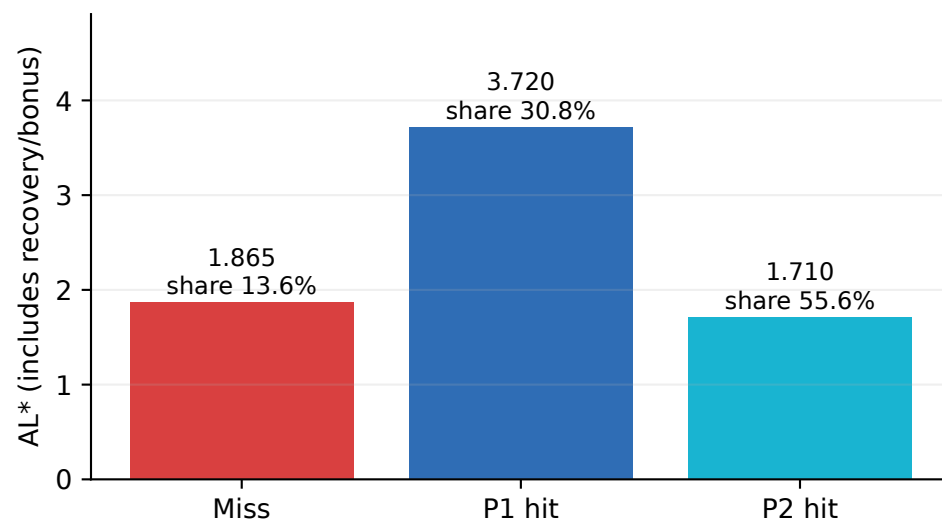
Measured wall time by batch cache status



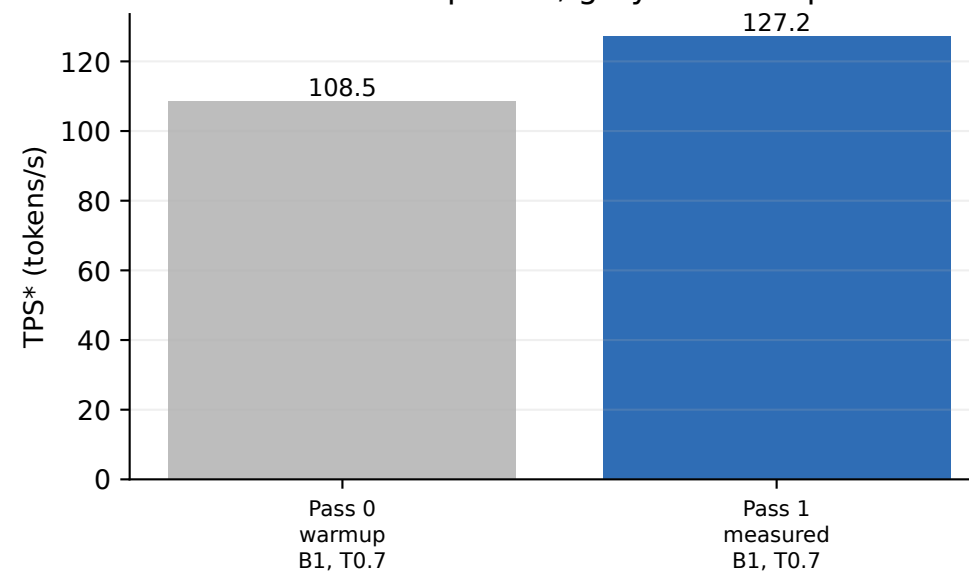
Boundary-free steps, including slow steps



Conditional AL* and source share



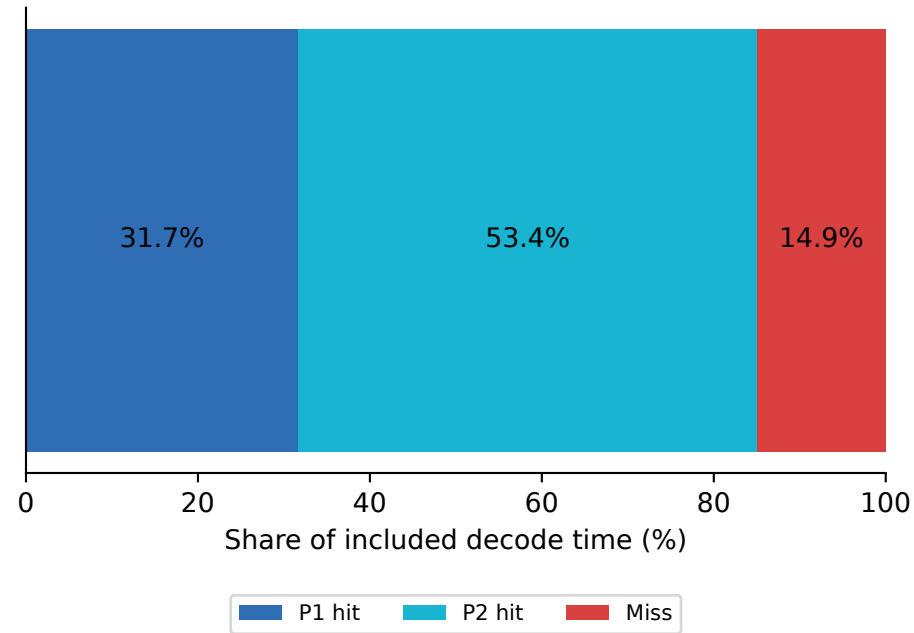
All saved passes; gray = warmup



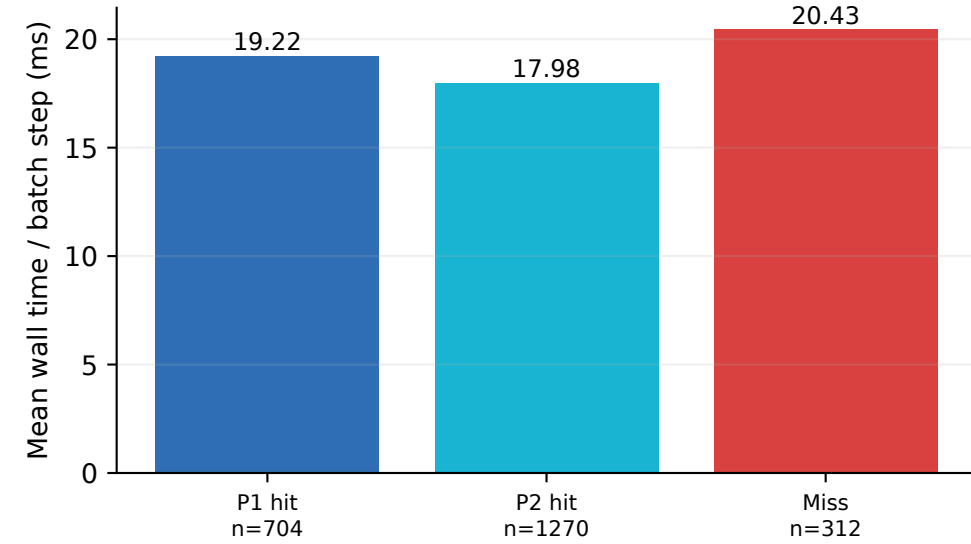
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_postopt_warm_3
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/0
Last pass: B=1, T=0.7, N=48 | AL*=2.350, hit=86.4%, TPS*=125.72

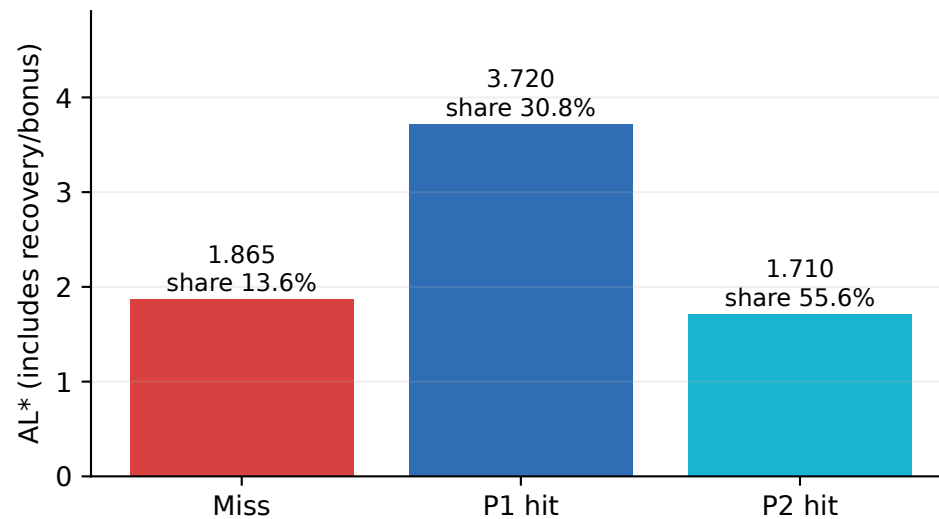
Measured wall time by batch cache status



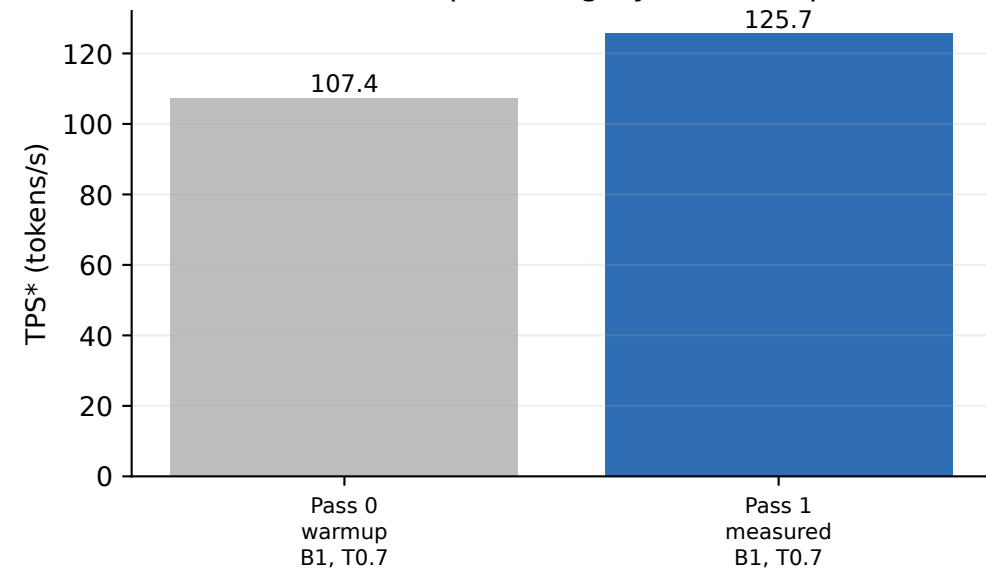
Boundary-free steps, including slow steps



Conditional AL* and source share



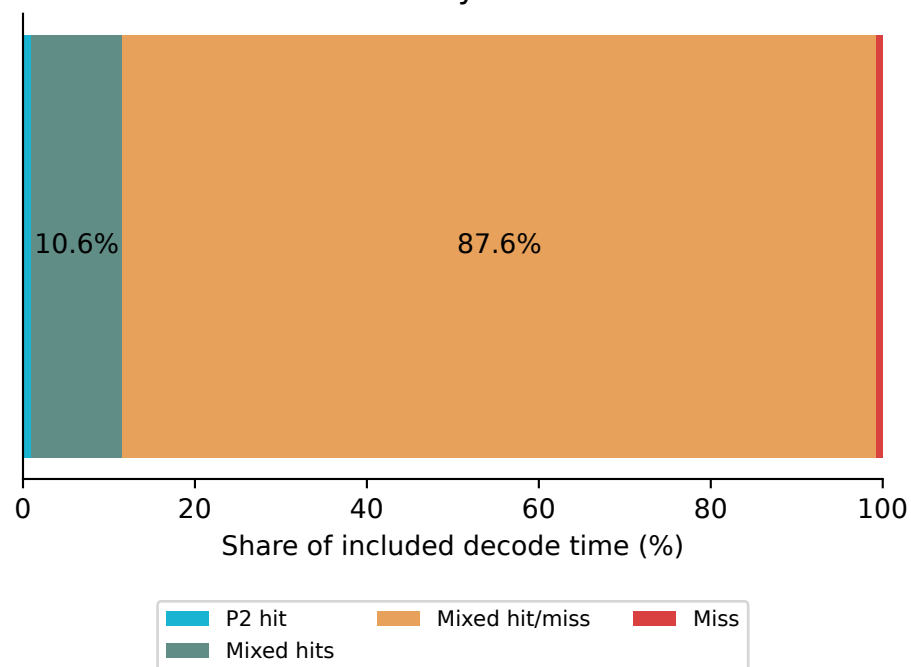
All saved passes; gray = warmup



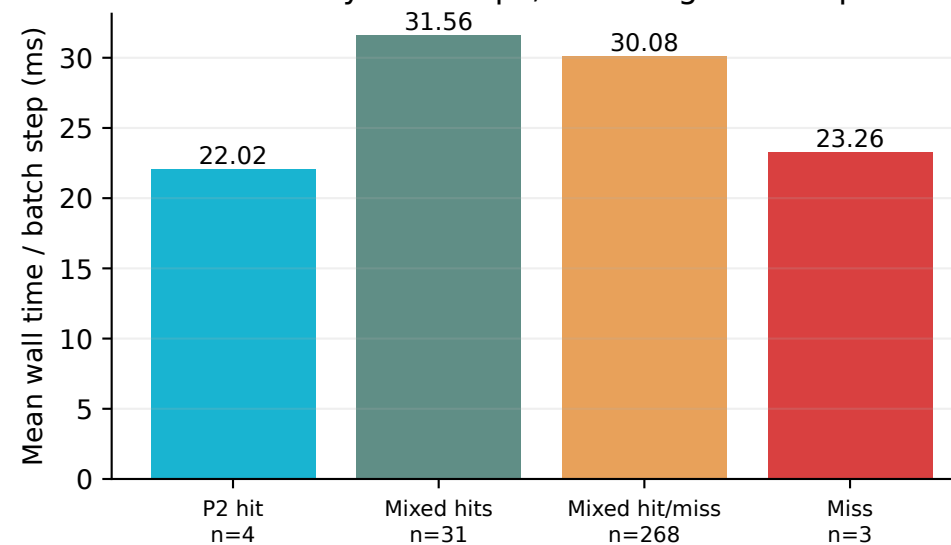
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_postopt_warm_0
 DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.149, hit=75.6%, TPS*=539.96

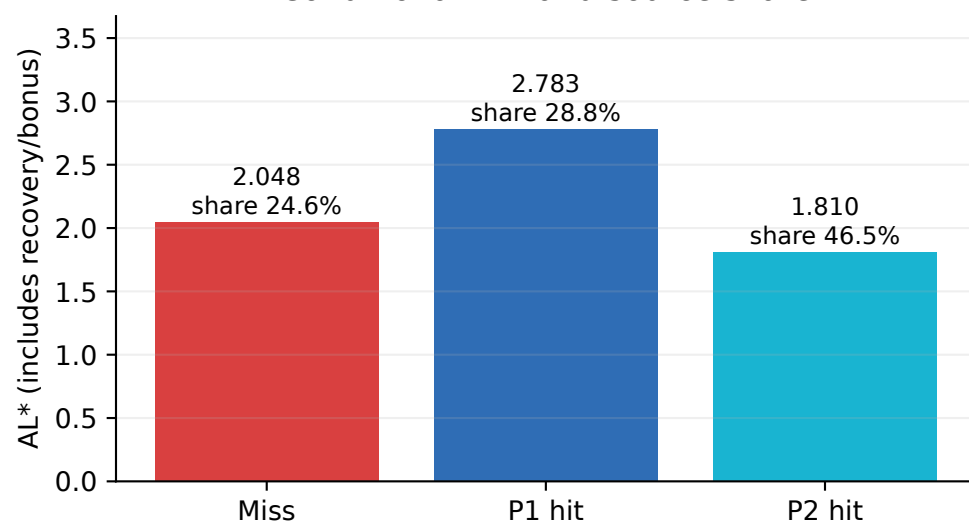
Measured wall time by batch cache status



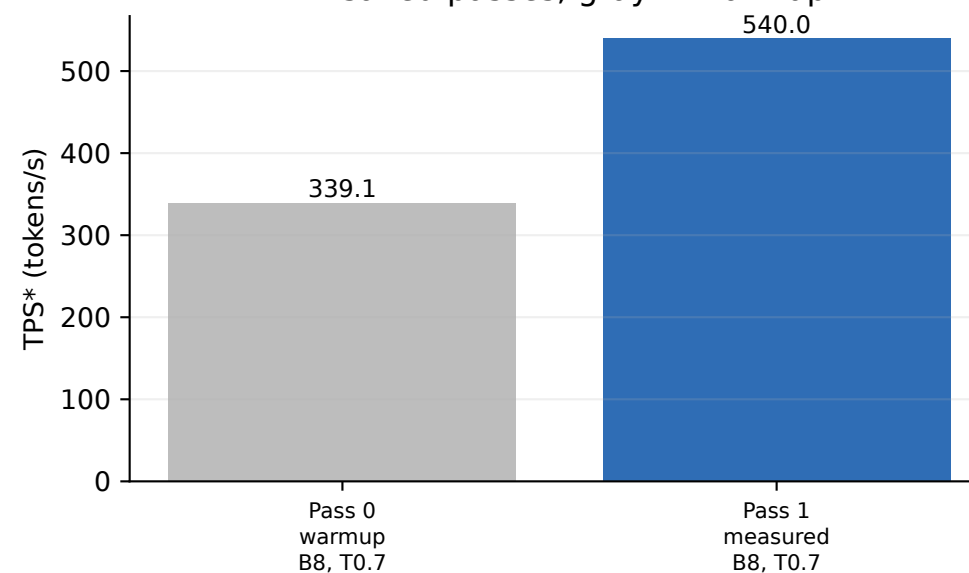
Boundary-free steps, including slow steps



Conditional AL* and source share



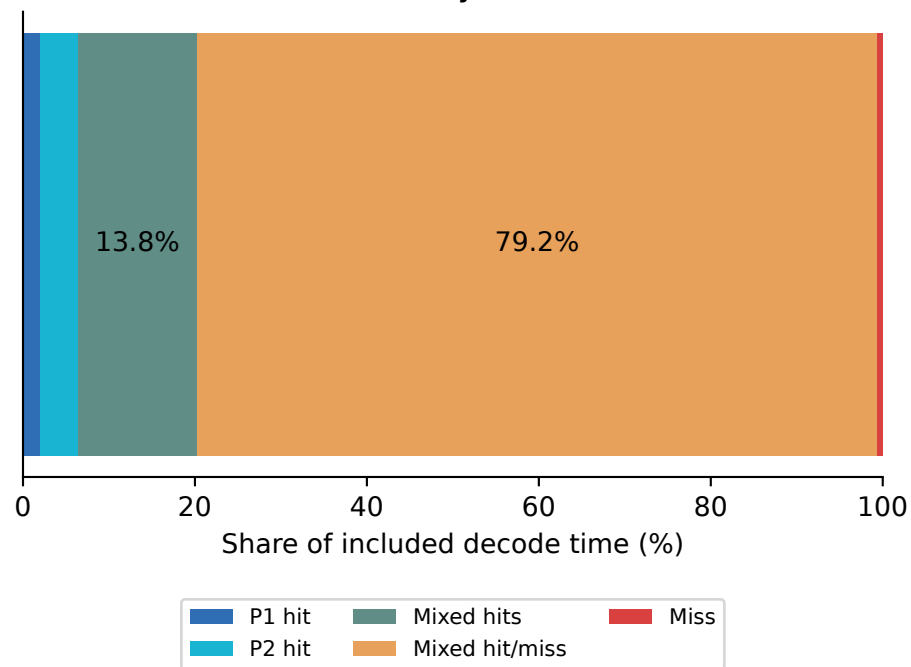
All saved passes; gray = warmup



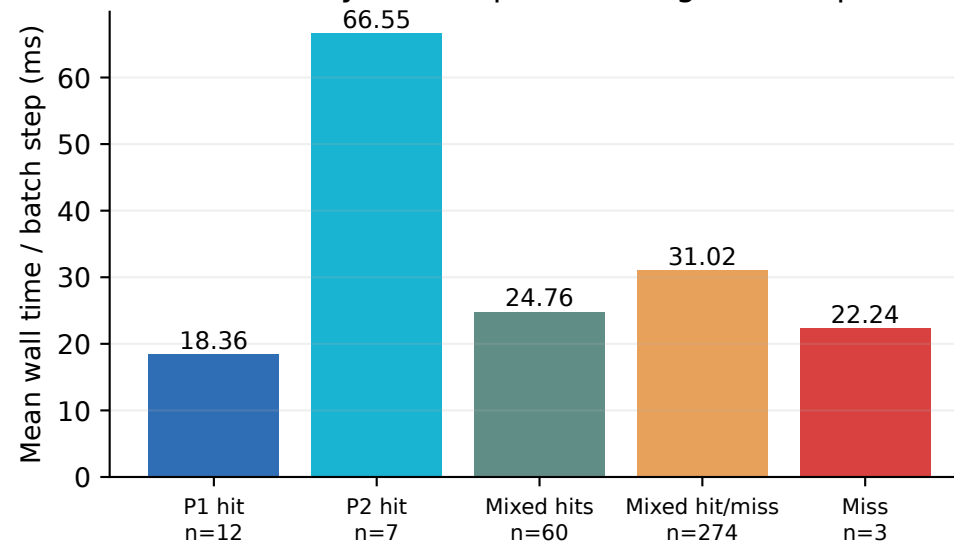
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_postopt_warm_1
 DUET: K1/K2=3/1, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=1.996, hit=79.8%, TPS*=489.14

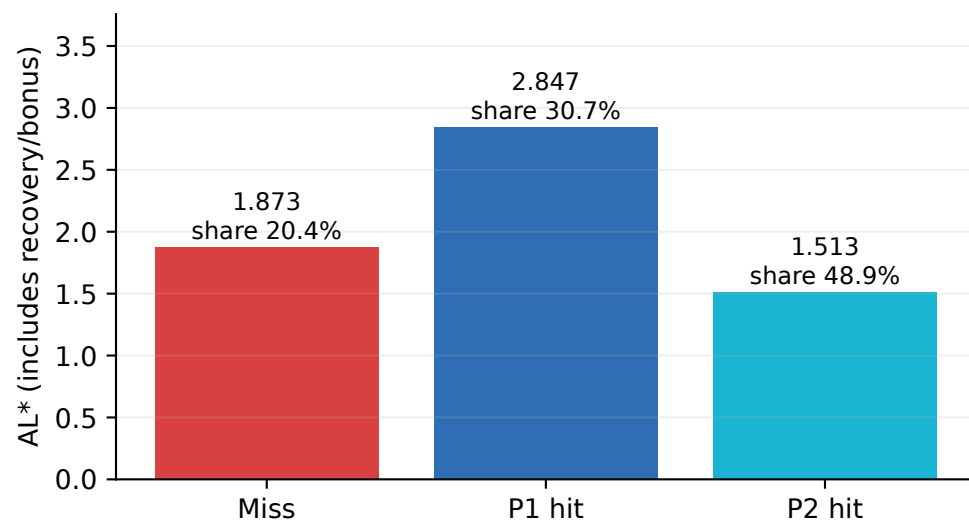
Measured wall time by batch cache status



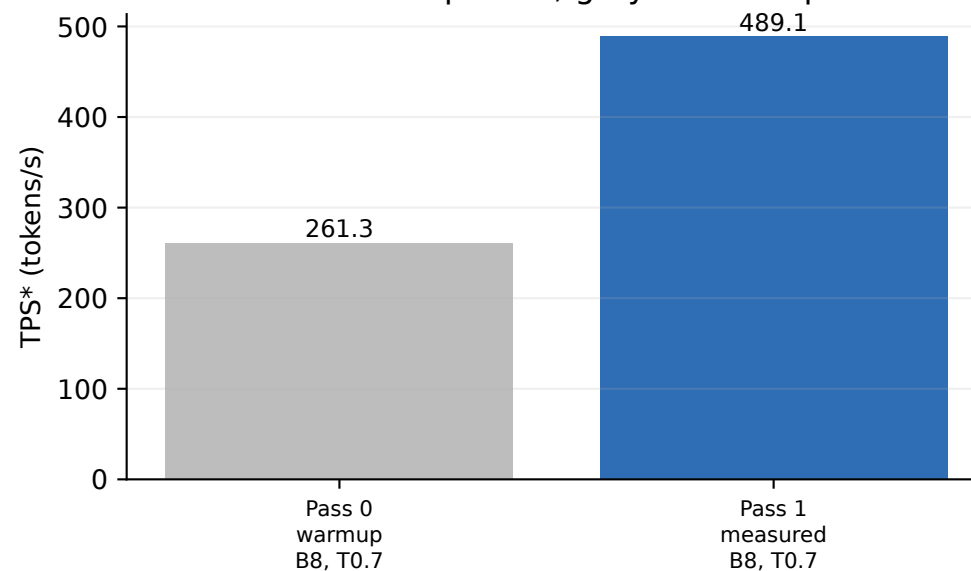
Boundary-free steps, including slow steps



Conditional AL* and source share



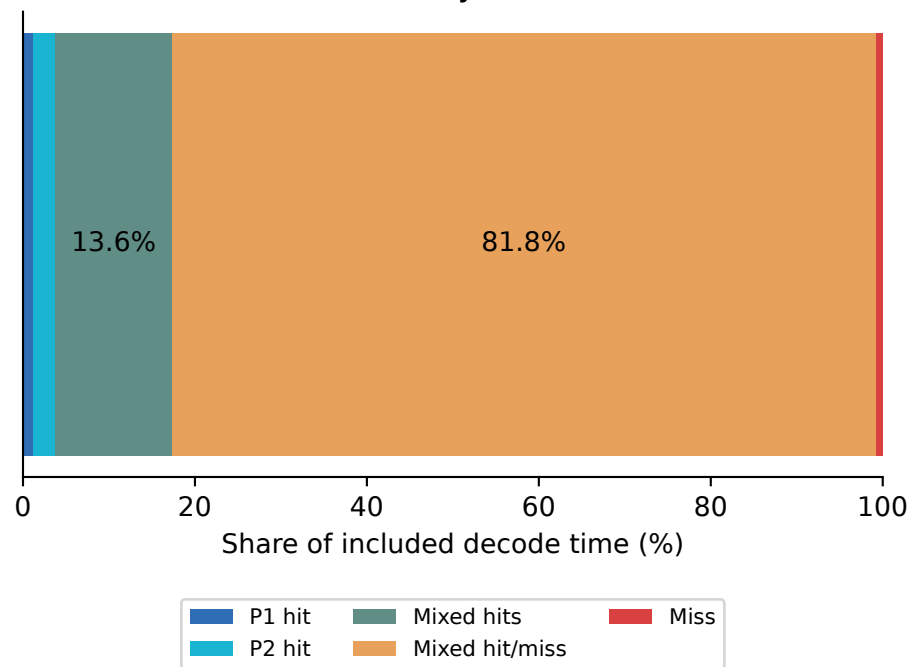
All saved passes; gray = warmup



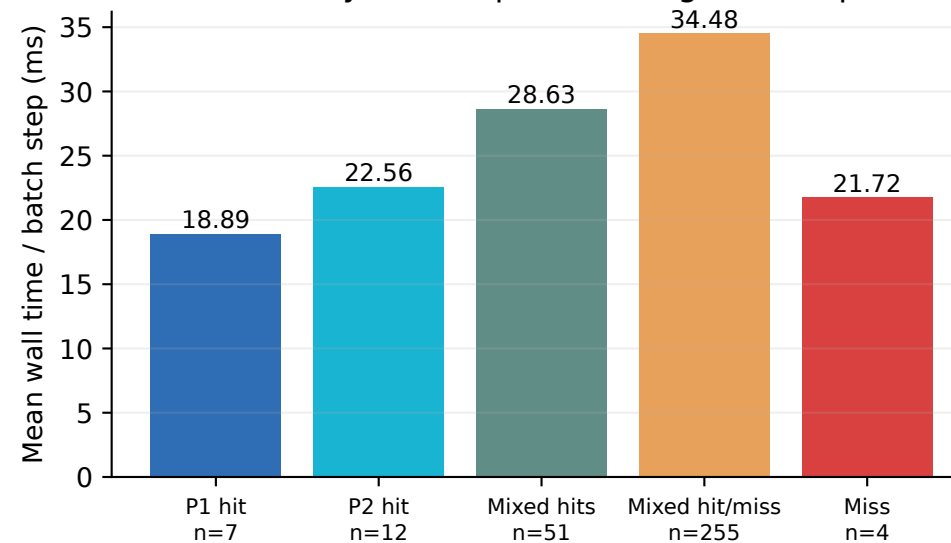
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_postopt_warm_2
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.125, hit=78.2%, TPS*=477.78

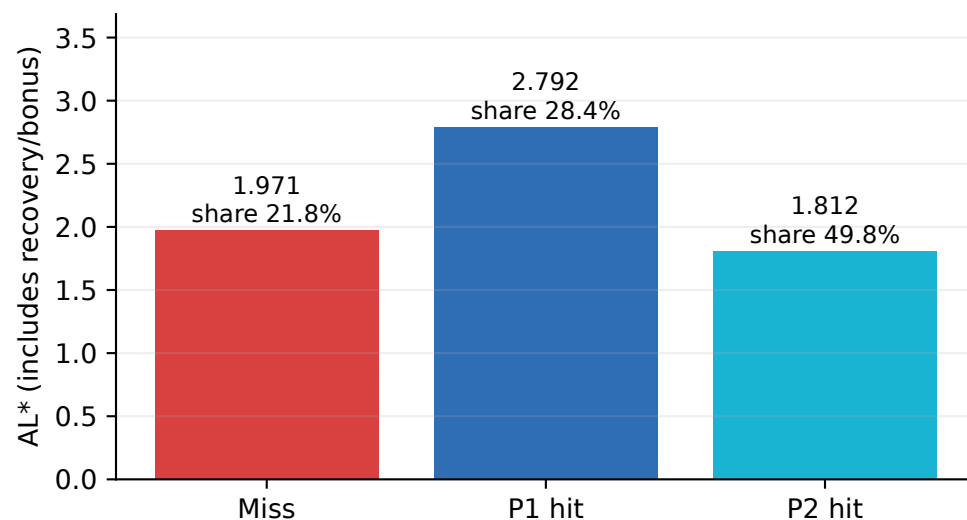
Measured wall time by batch cache status



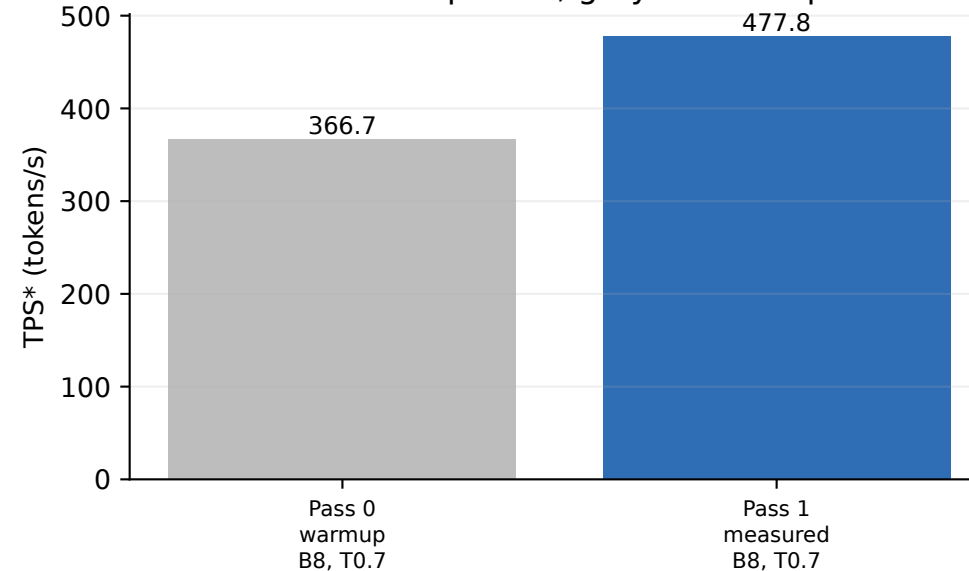
Boundary-free steps, including slow steps



Conditional AL* and source share



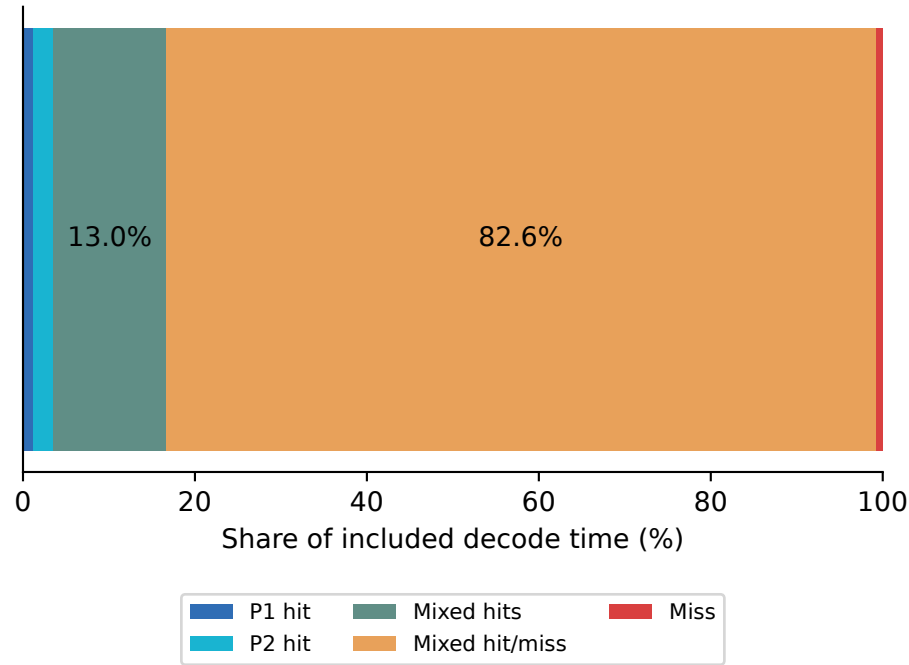
All saved passes; gray = warmup



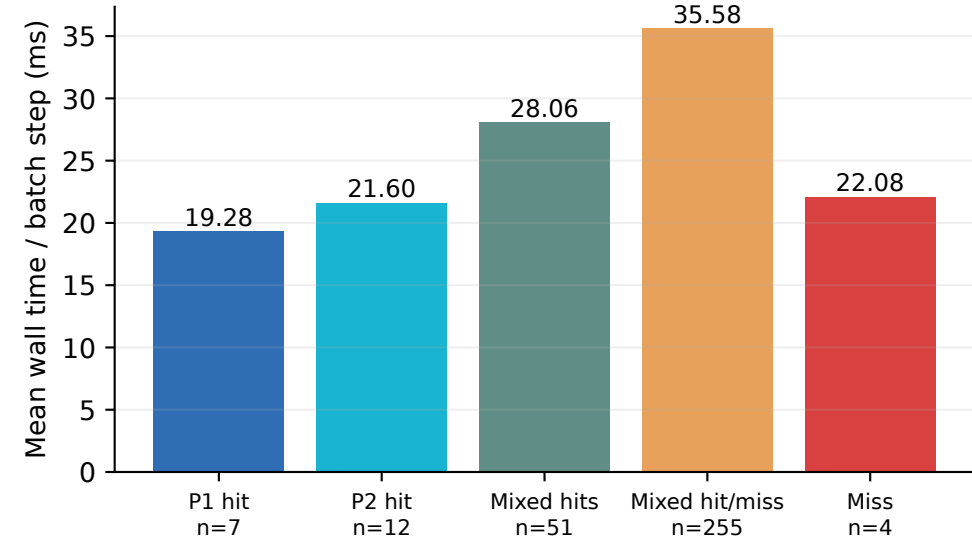
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_postopt_warm_3
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.125, hit=78.2%, TPS*=467.13

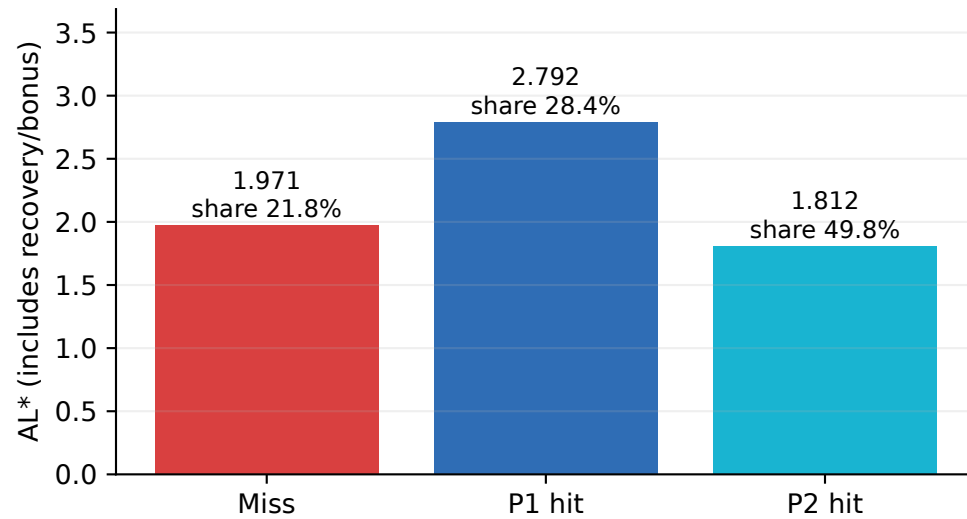
Measured wall time by batch cache status



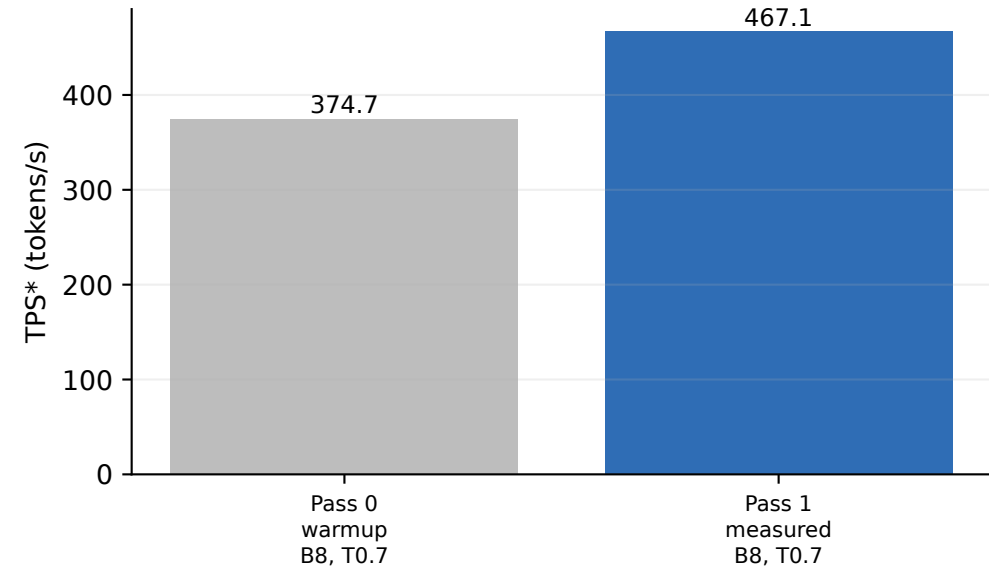
Boundary-free steps, including slow steps



Conditional AL* and source share



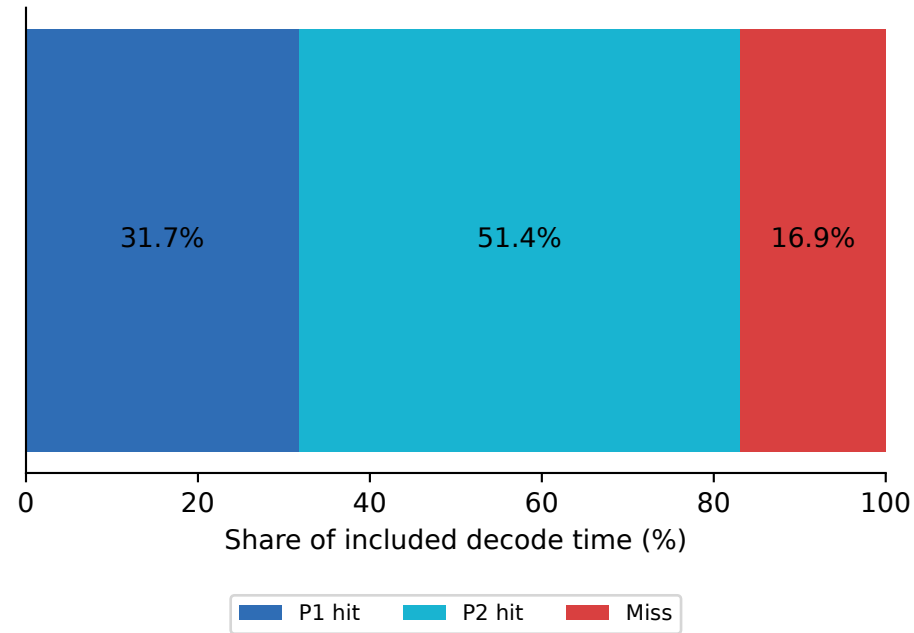
All saved passes; gray = warmup



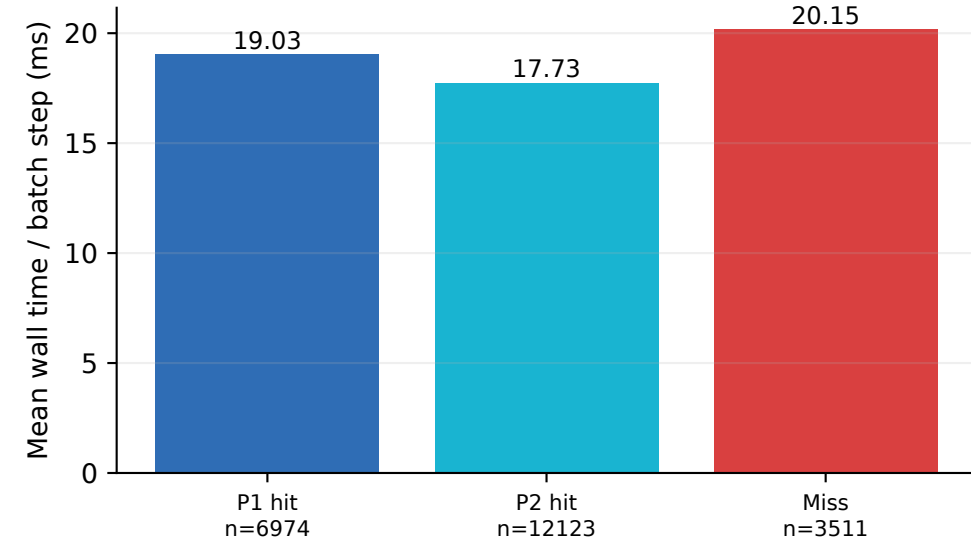
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_postopt_final_r1
DUET: K1/K2=11/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=12, trim/stream=1/1
Last pass: B=1, T=0.7, N=480 | AL*=2.347, hit=84.5%, TPS*=126.84

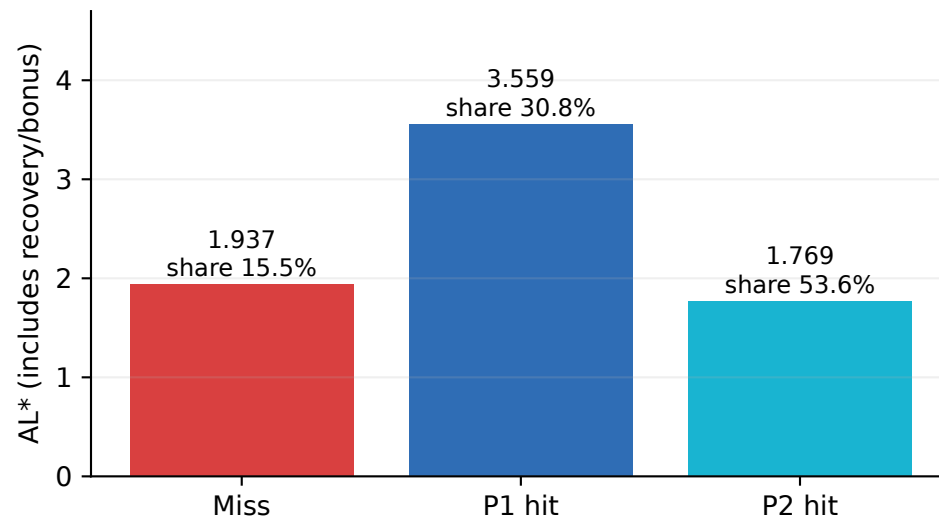
Measured wall time by batch cache status



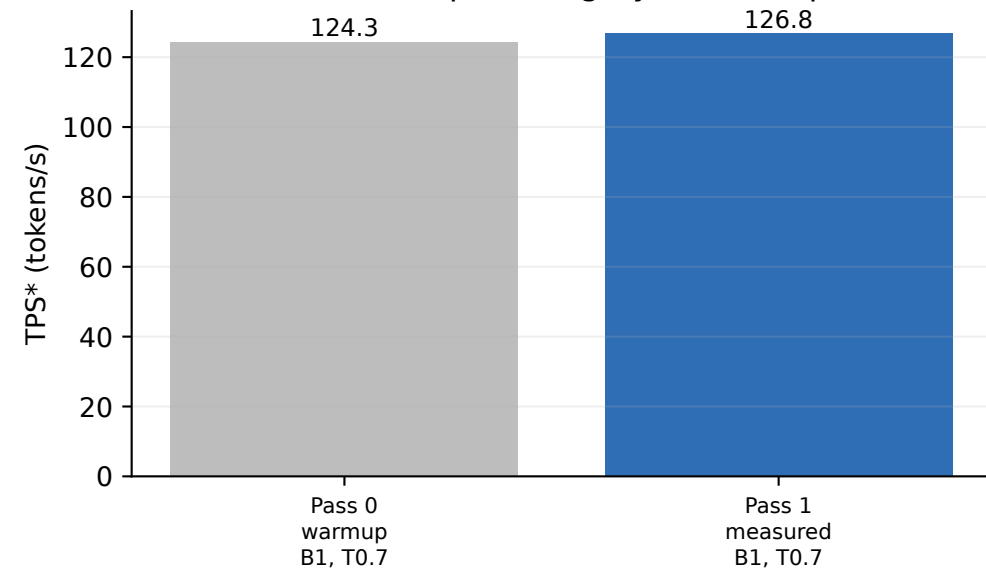
Boundary-free steps, including slow steps



Conditional AL* and source share



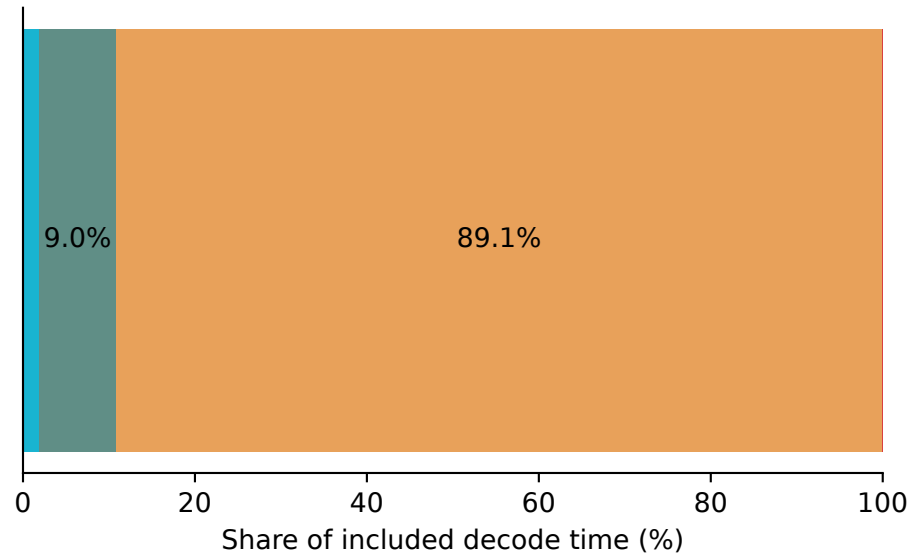
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

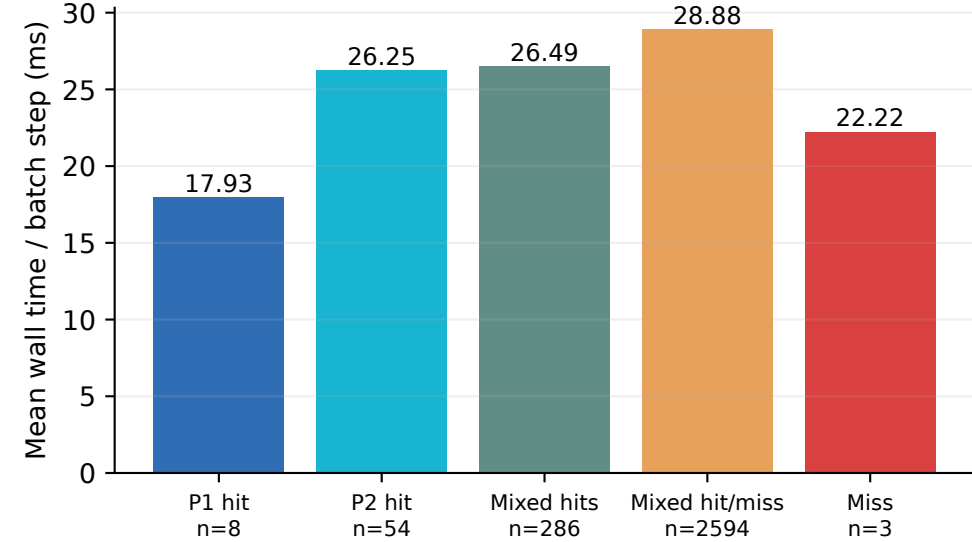
llama2_b8_postopt_final_r0
 DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.107, hit=74.2%, TPS*=583.12

Measured wall time by batch cache status

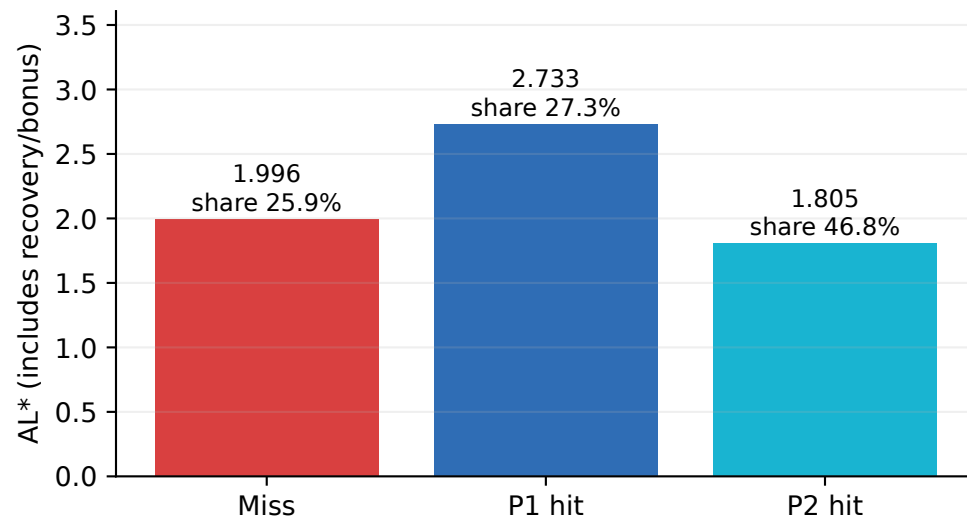


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

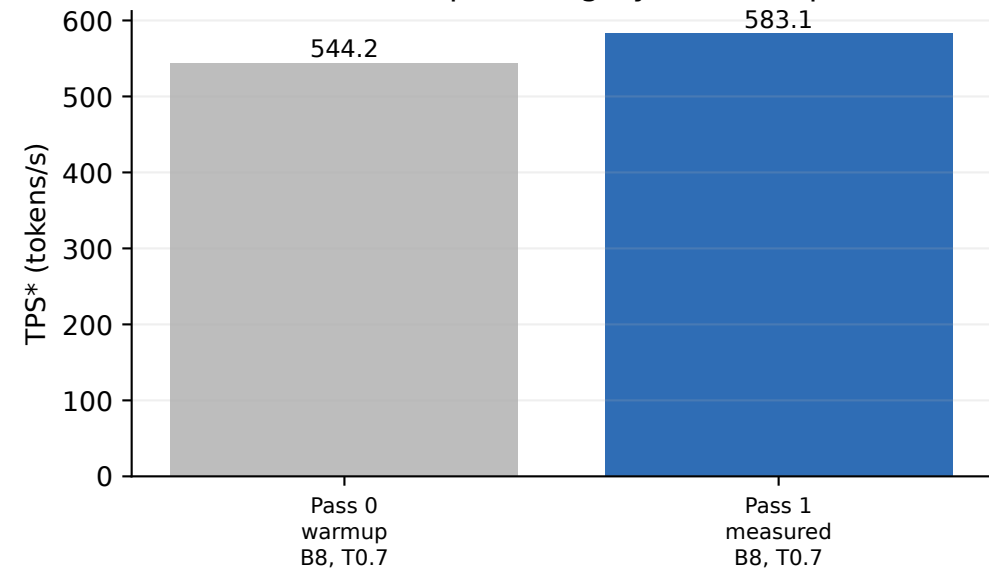
Boundary-free steps, including slow steps



Conditional AL* and source share



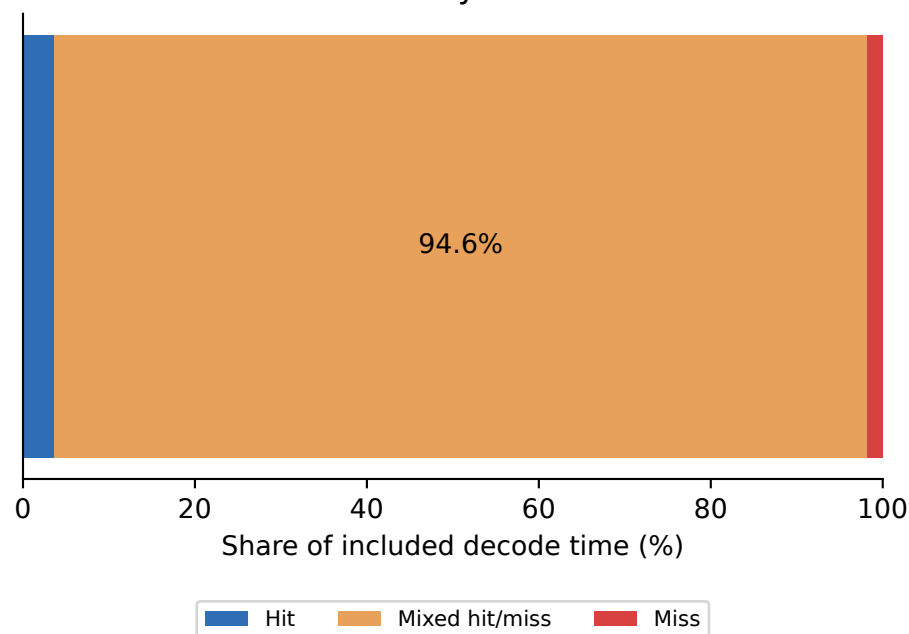
All saved passes; gray = warmup



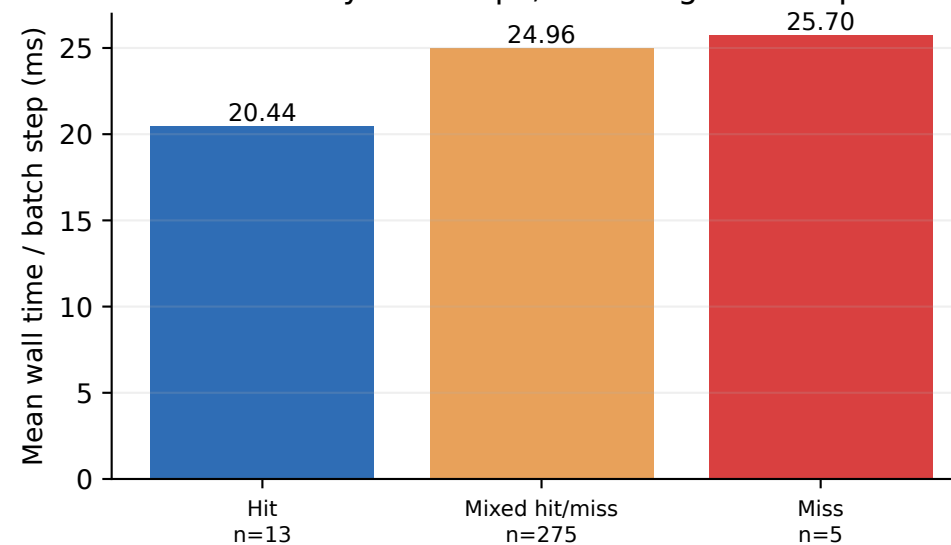
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stability_ssd_control
SSD: K=4, F=7
Last pass: B=8, T=0.7, N=48 | AL*=2.167, hit=52.9%, TPS*=668.37

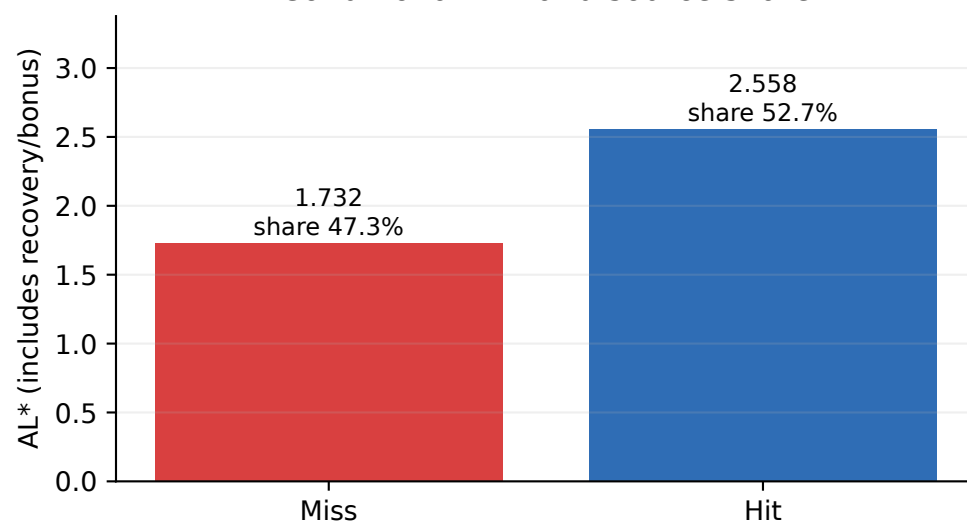
Measured wall time by batch cache status



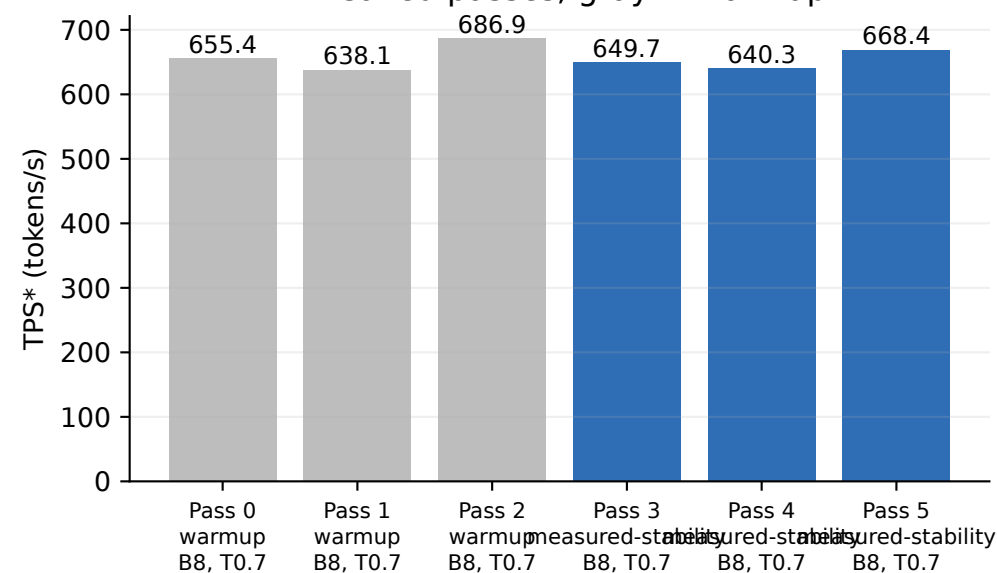
Boundary-free steps, including slow steps



Conditional AL* and source share



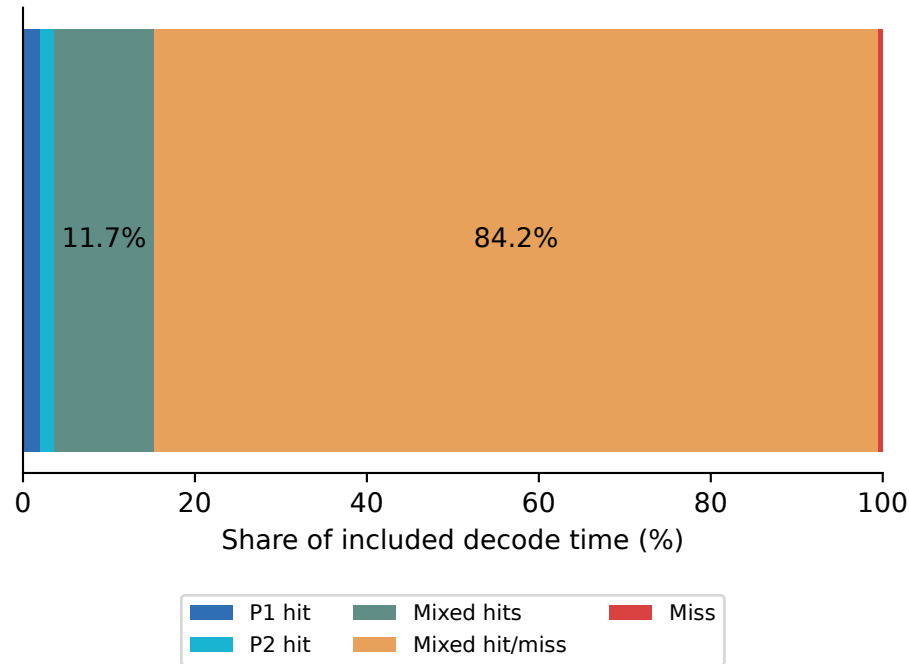
All saved passes; gray = warmup



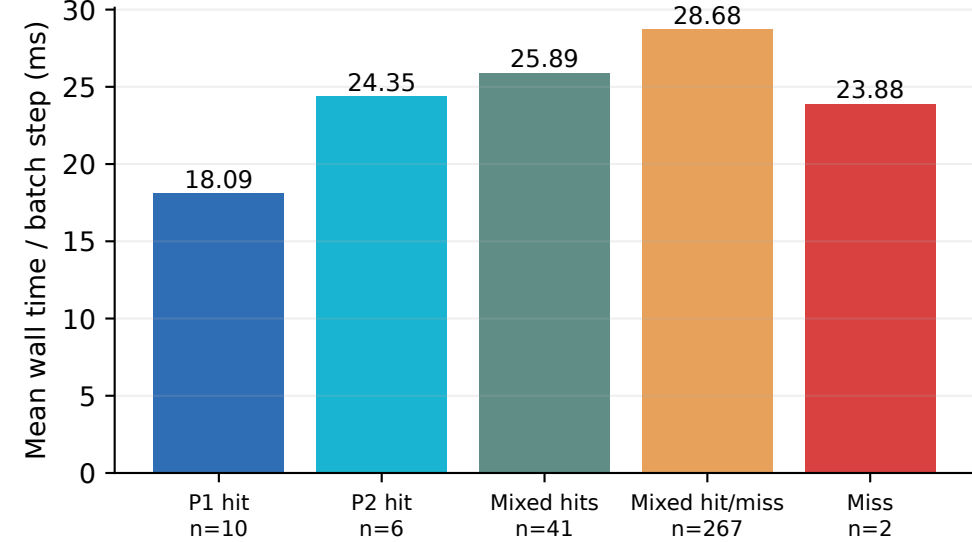
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stability_warm_0
 DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.130, hit=77.2%, TPS*=552.93

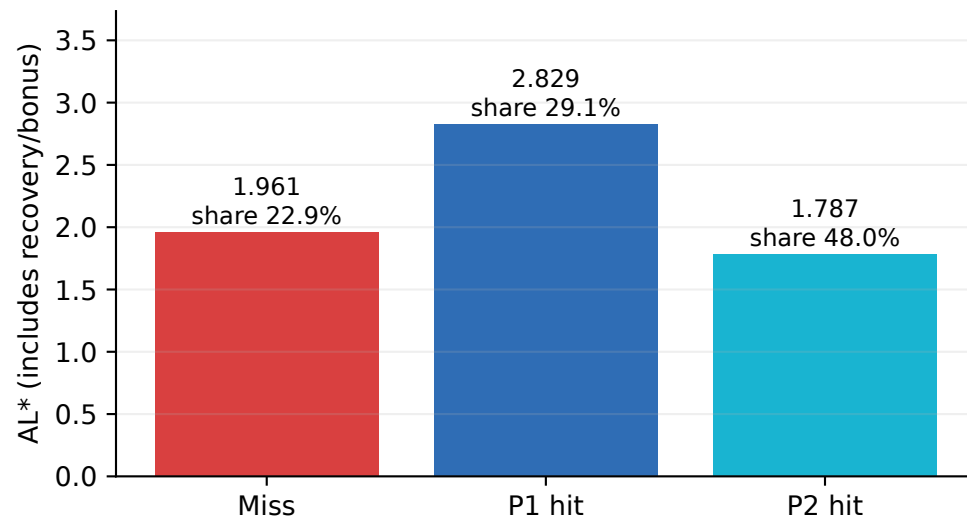
Measured wall time by batch cache status



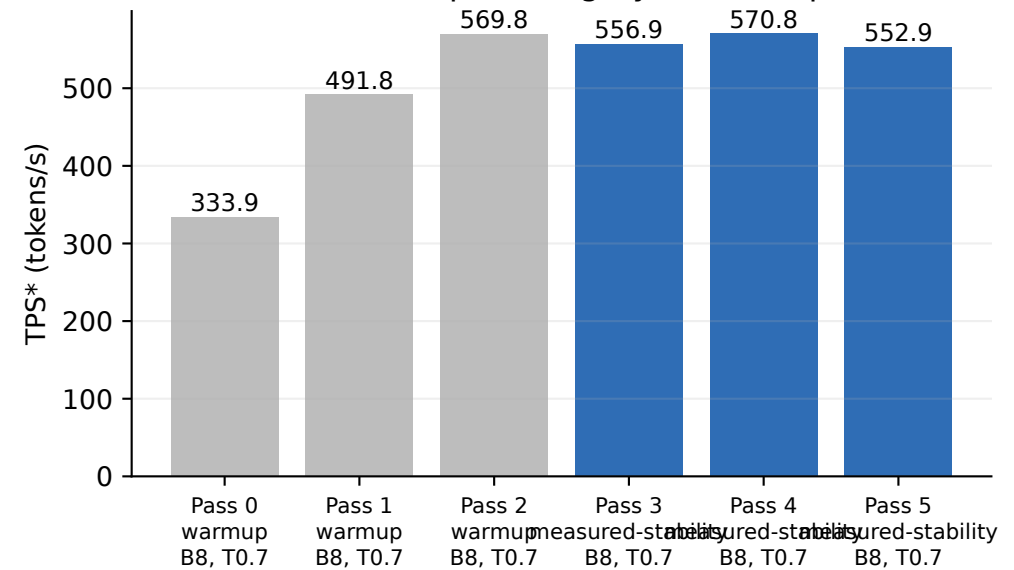
Boundary-free steps, including slow steps



Conditional AL* and source share



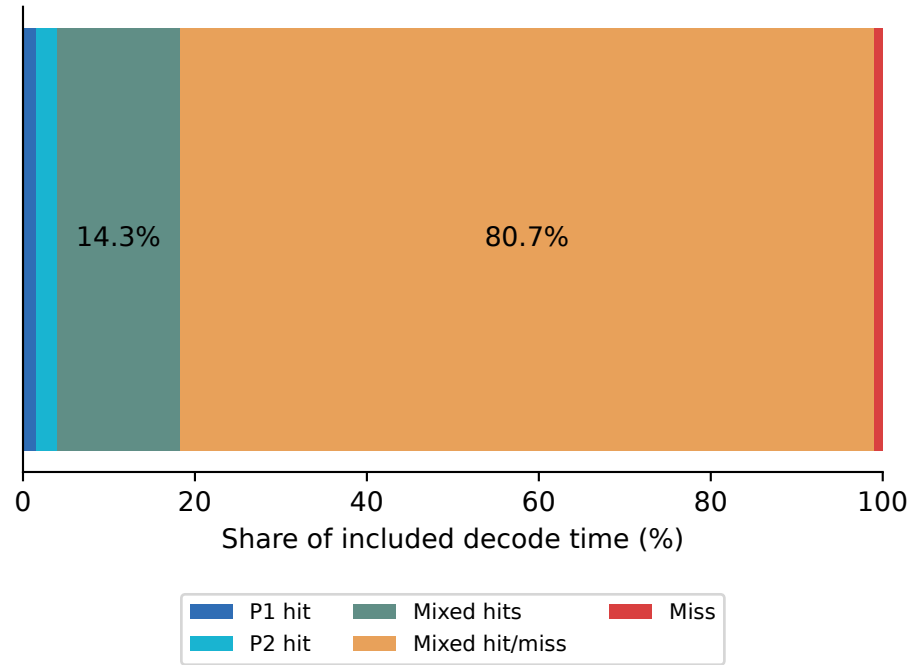
All saved passes; gray = warmup



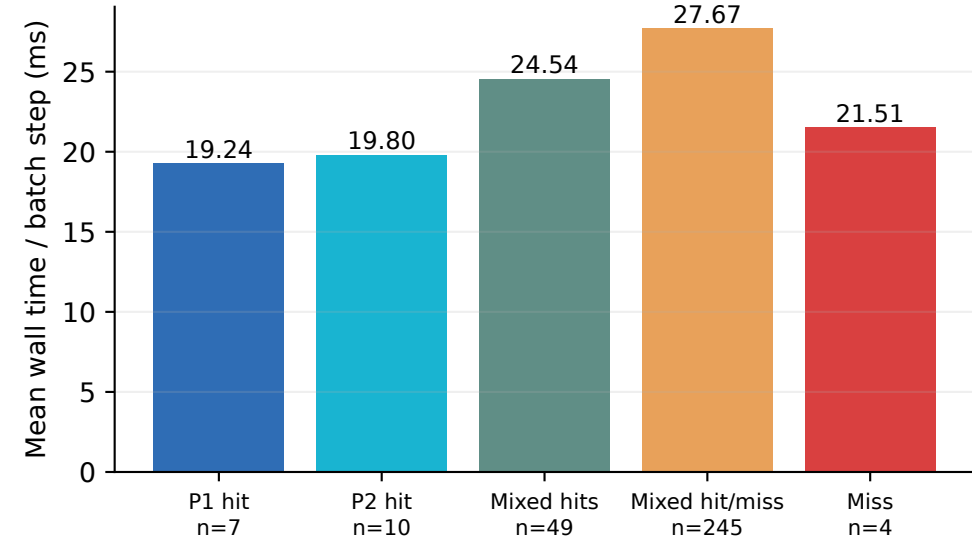
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stability_warm_1
 DUET: K1/K2=3/1, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.019, hit=77.9%, TPS*=546.63

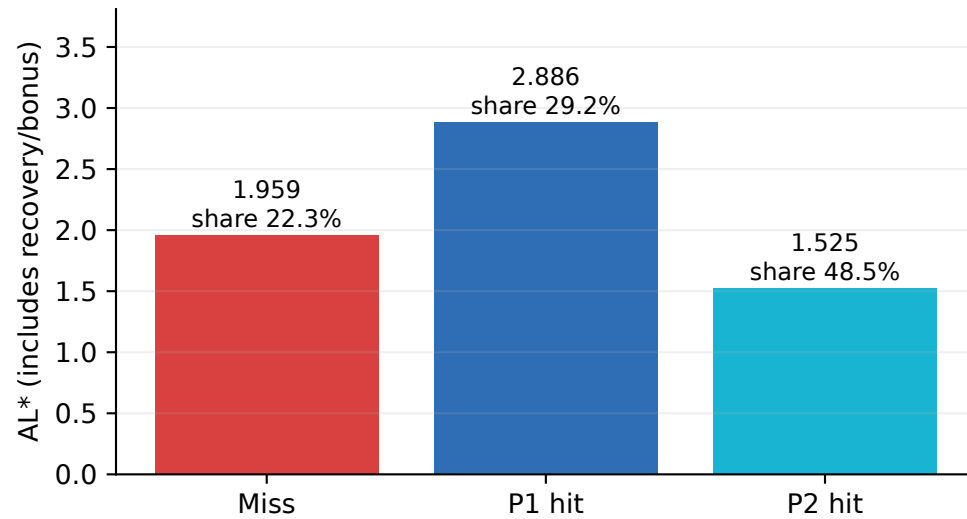
Measured wall time by batch cache status



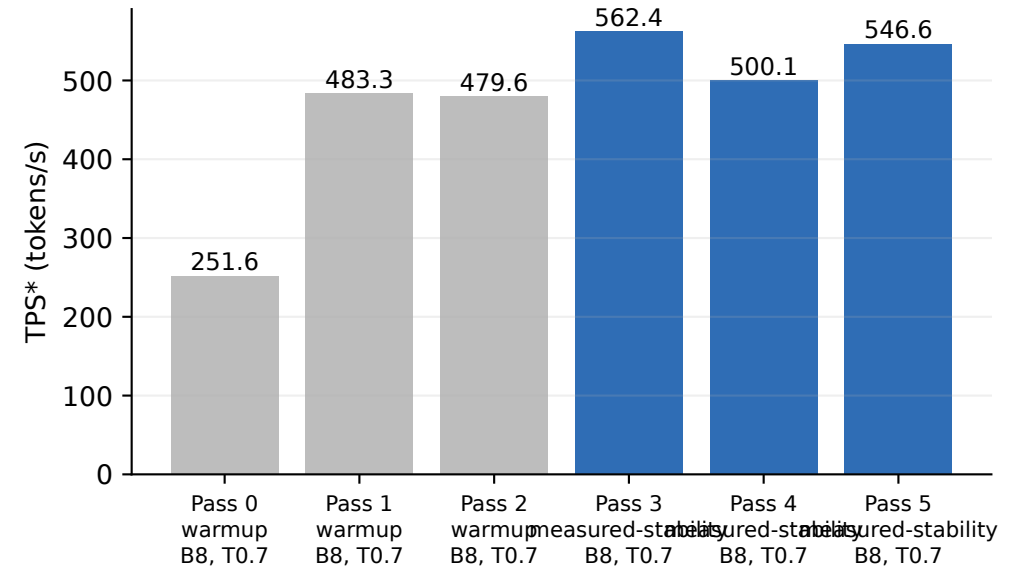
Boundary-free steps, including slow steps



Conditional AL* and source share



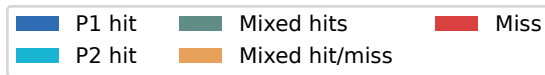
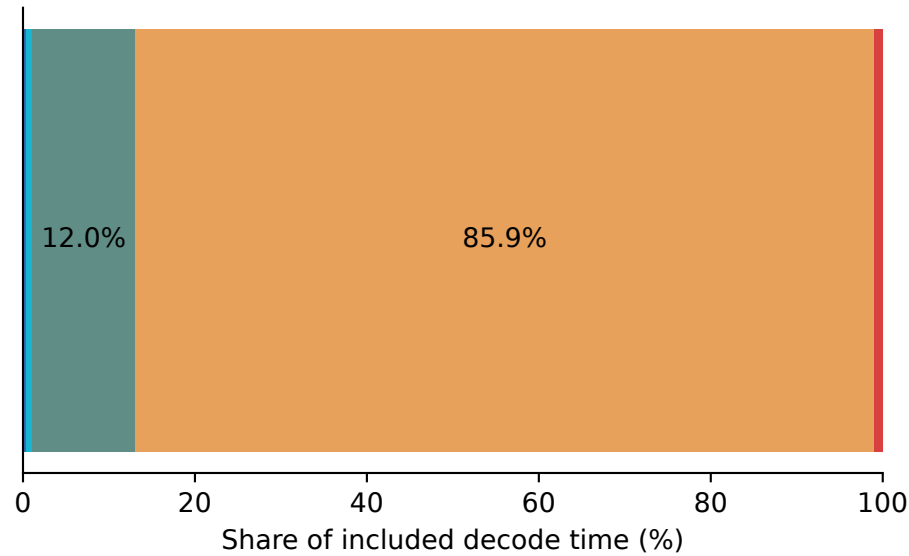
All saved passes; gray = warmup



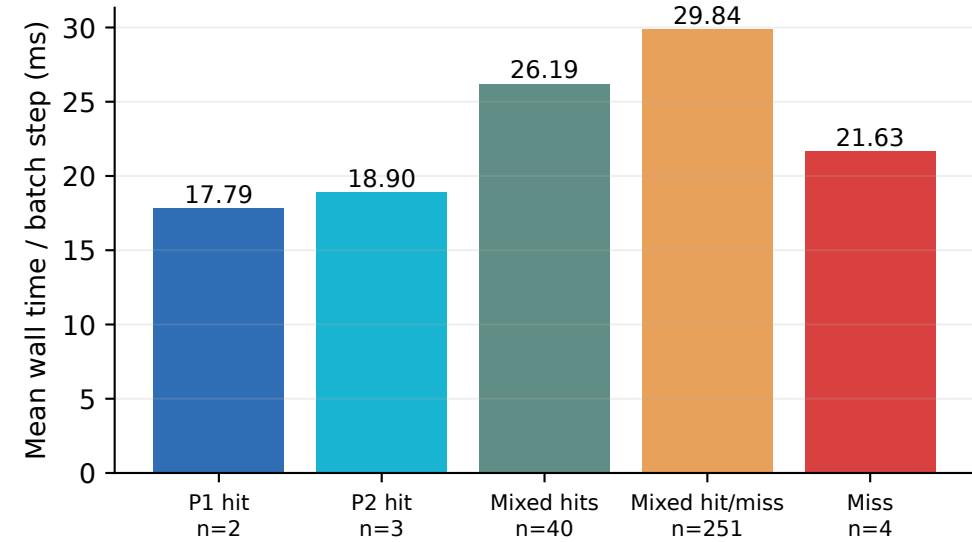
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stability_warm_2
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.110, hit=76.2%, TPS*=545.58

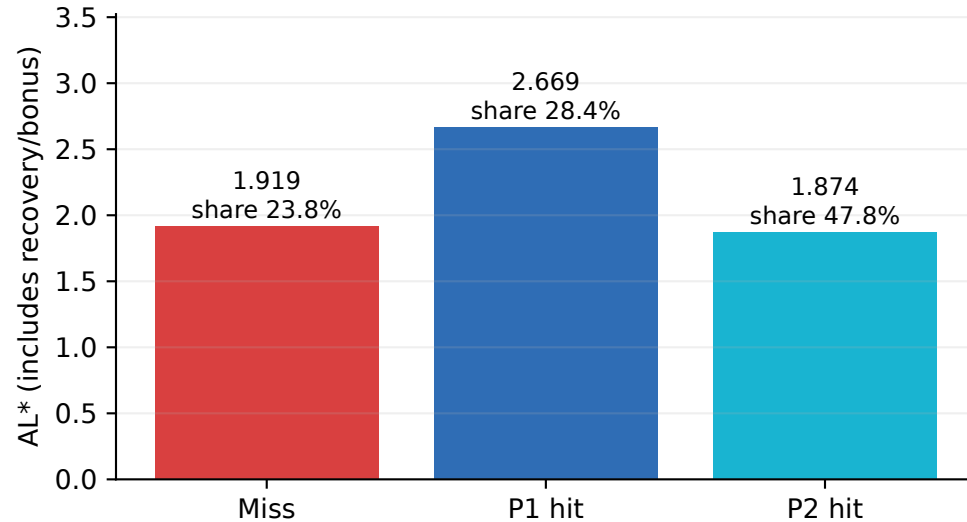
Measured wall time by batch cache status



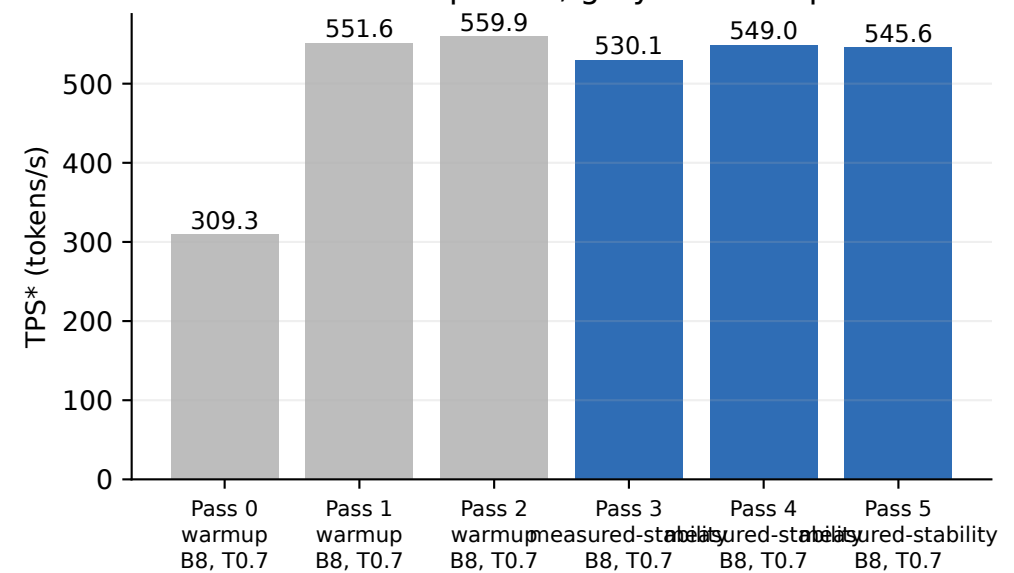
Boundary-free steps, including slow steps



Conditional AL* and source share



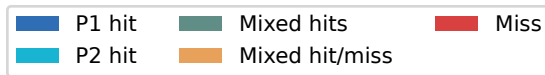
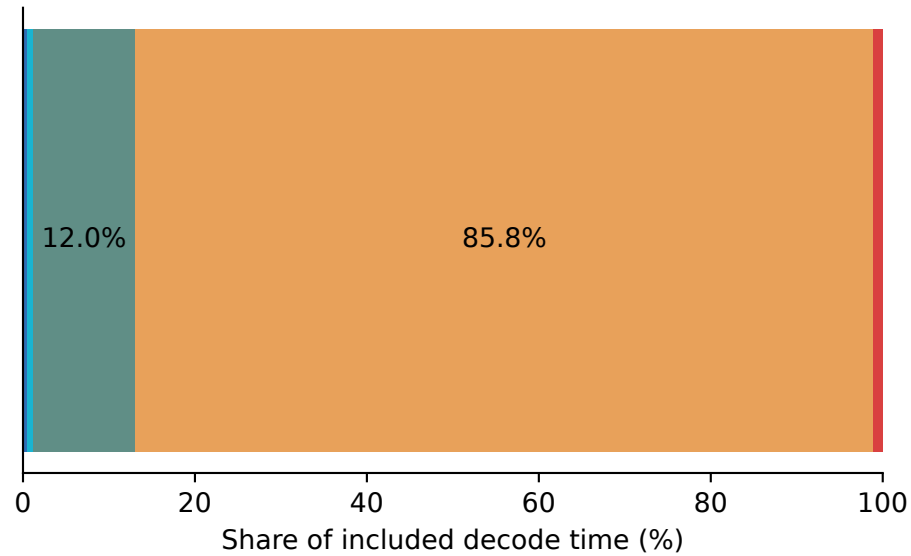
All saved passes; gray = warmup



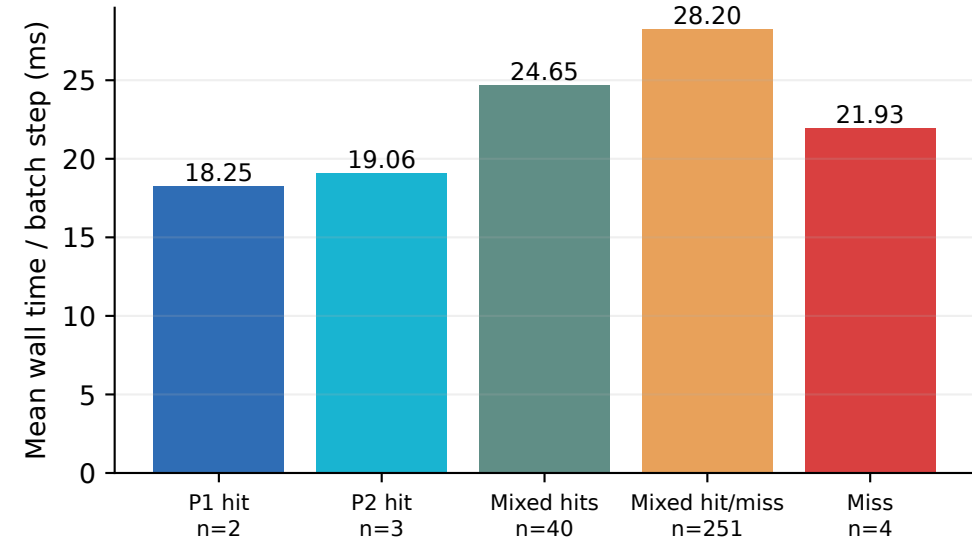
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stability_warm_3
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.110, hit=76.2%, TPS*=576.61

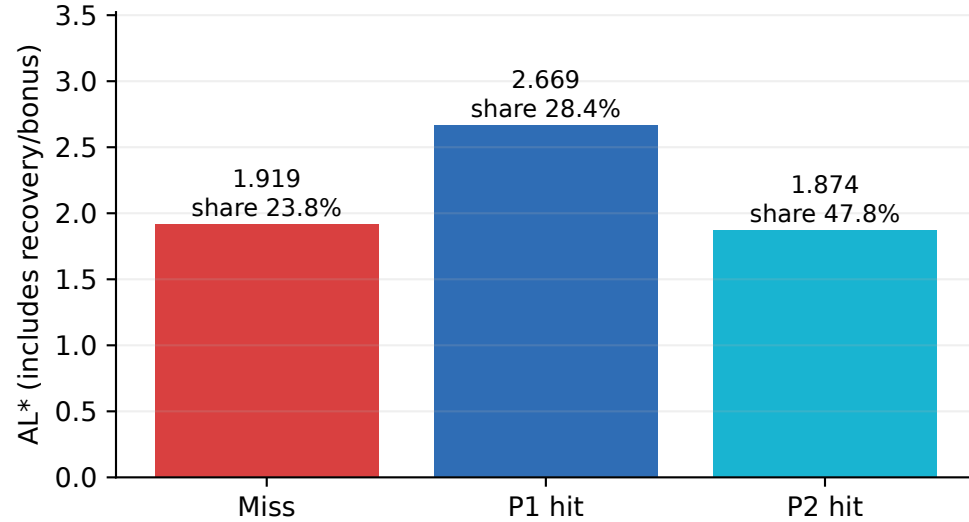
Measured wall time by batch cache status



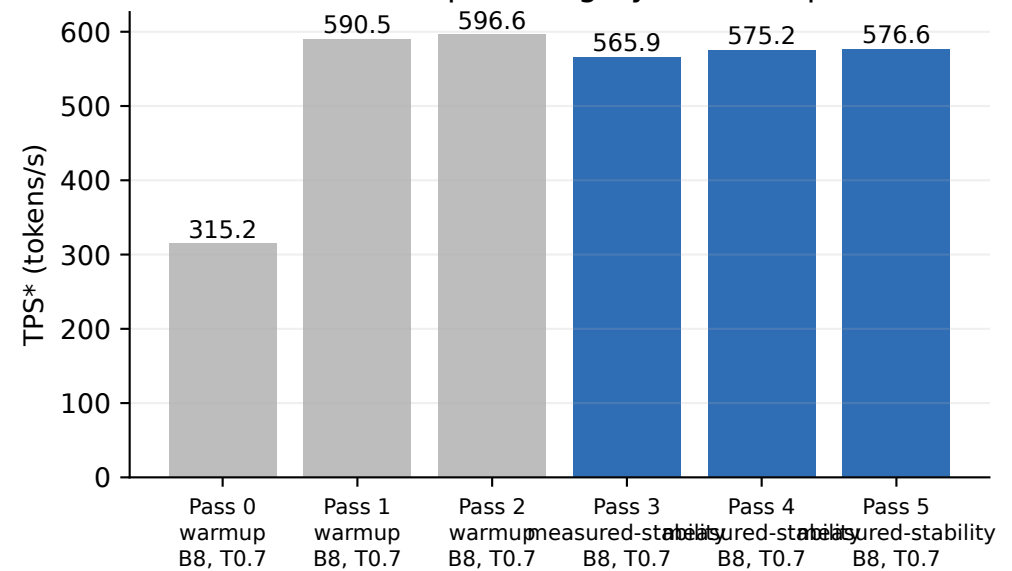
Boundary-free steps, including slow steps



Conditional AL* and source share

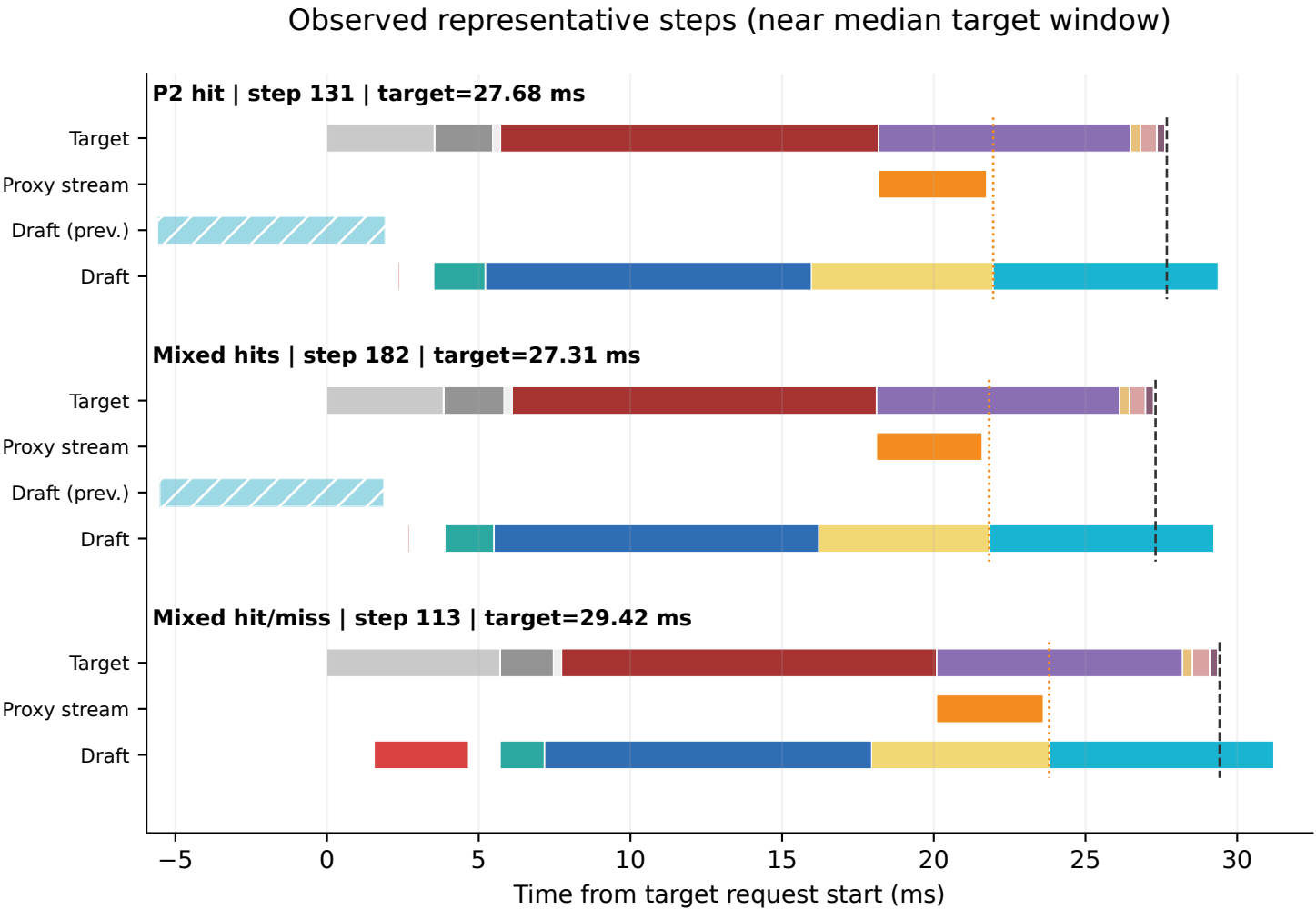
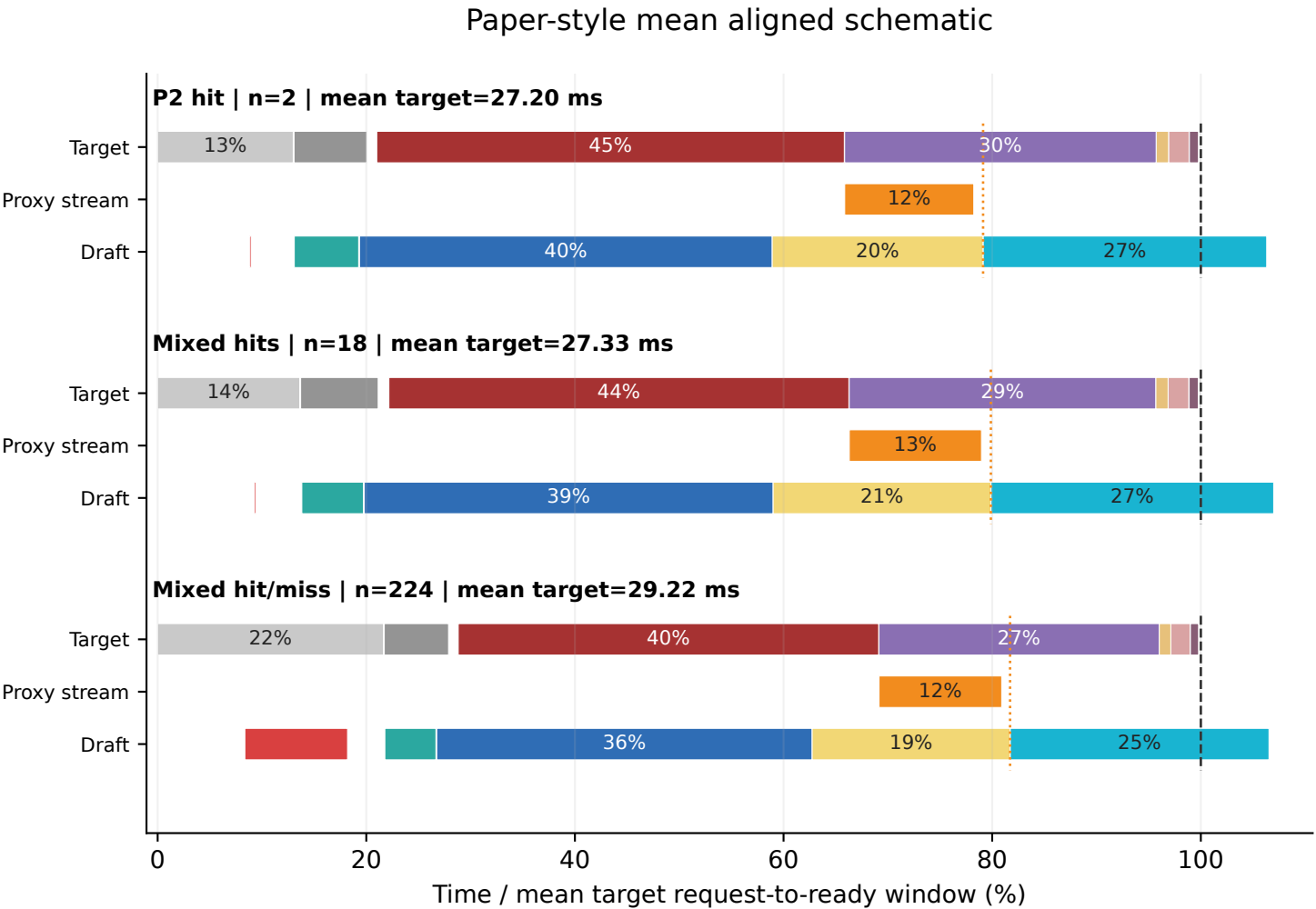
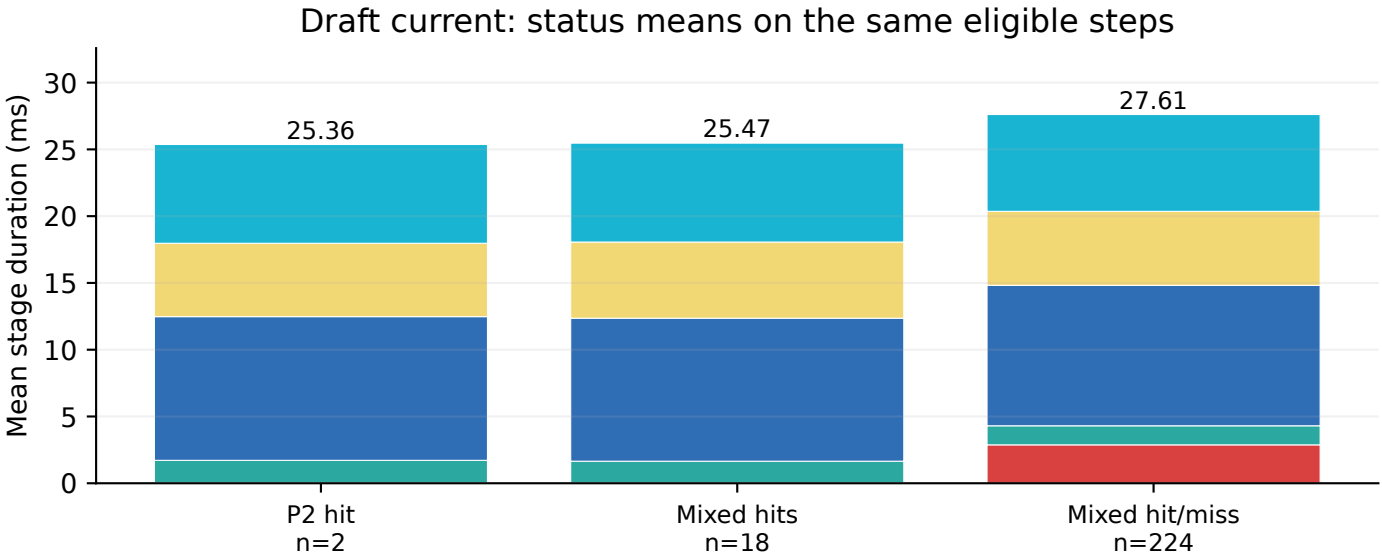
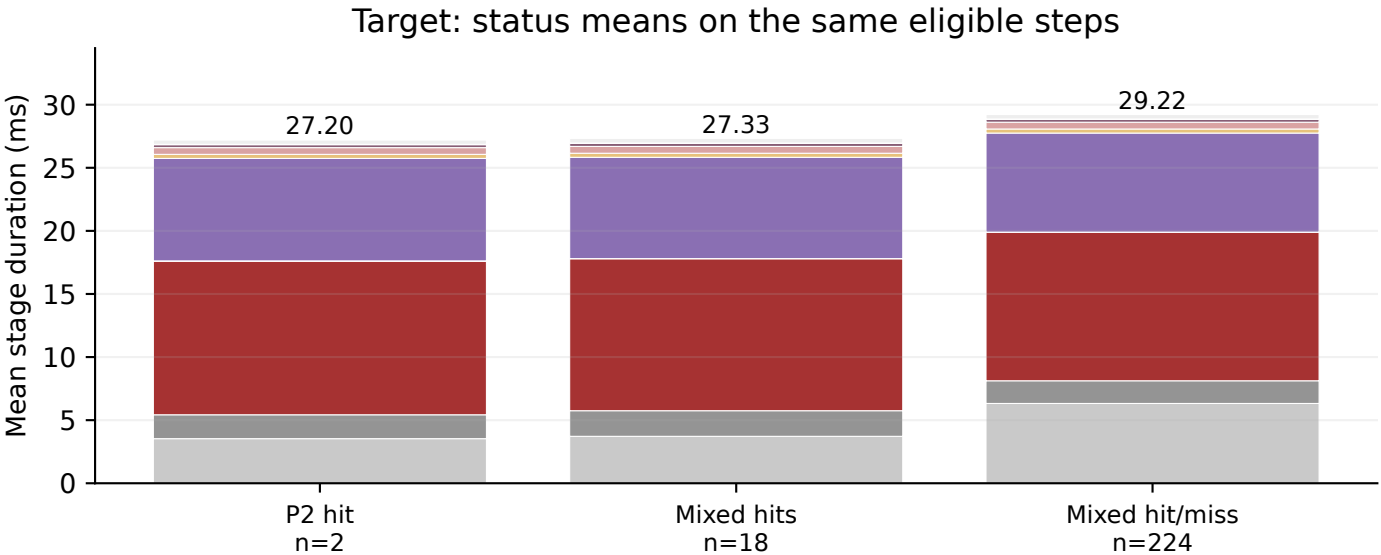


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

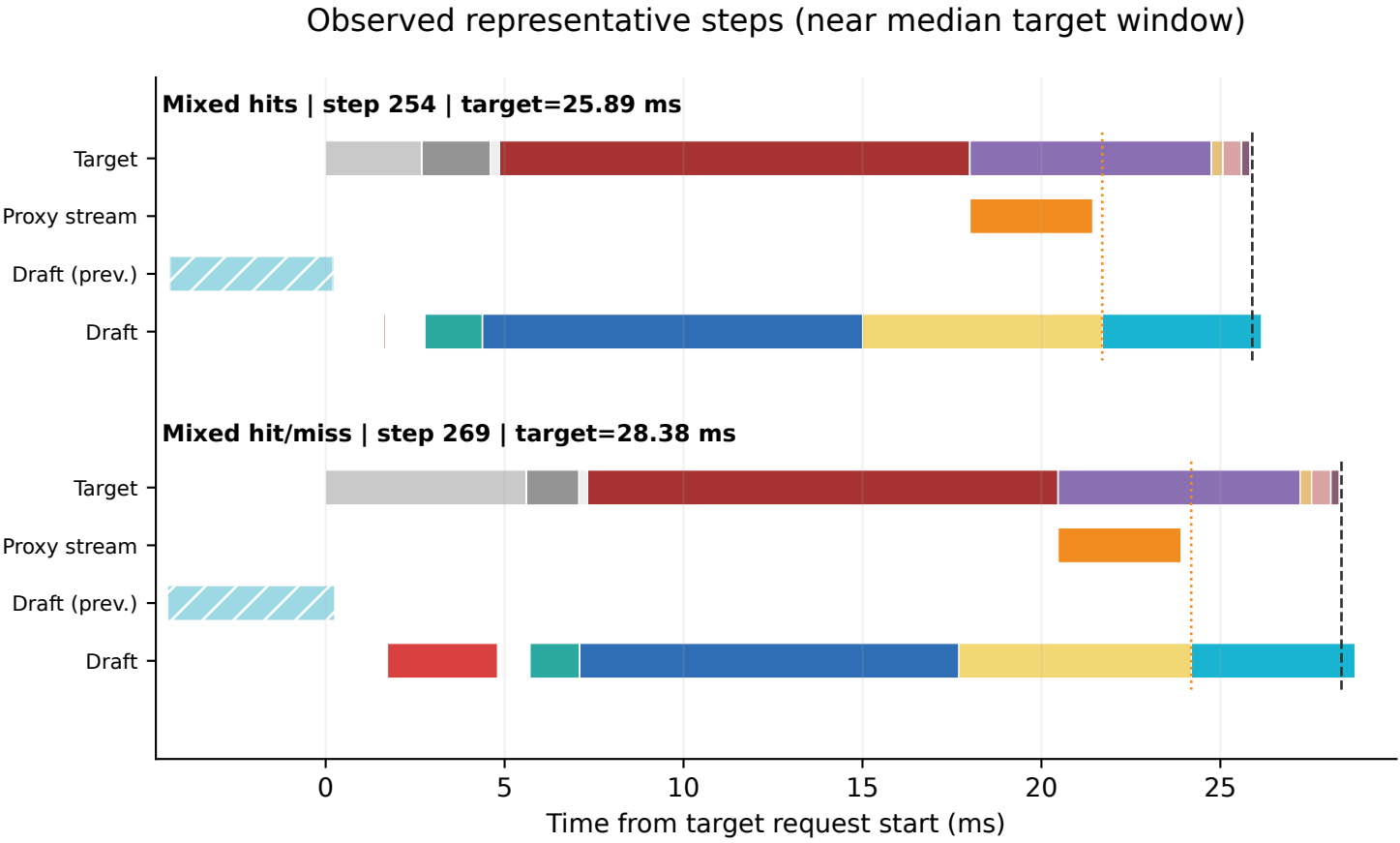
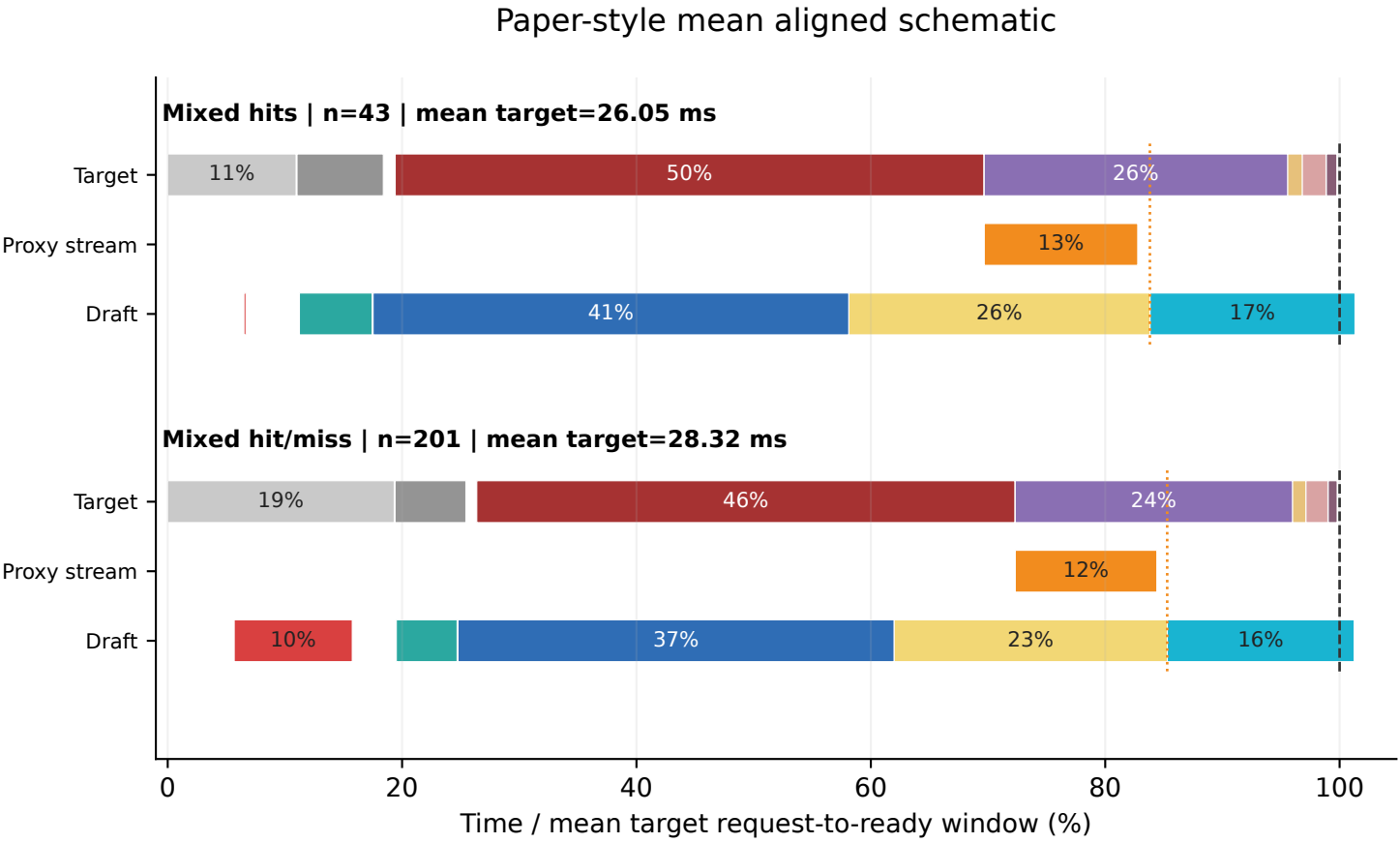
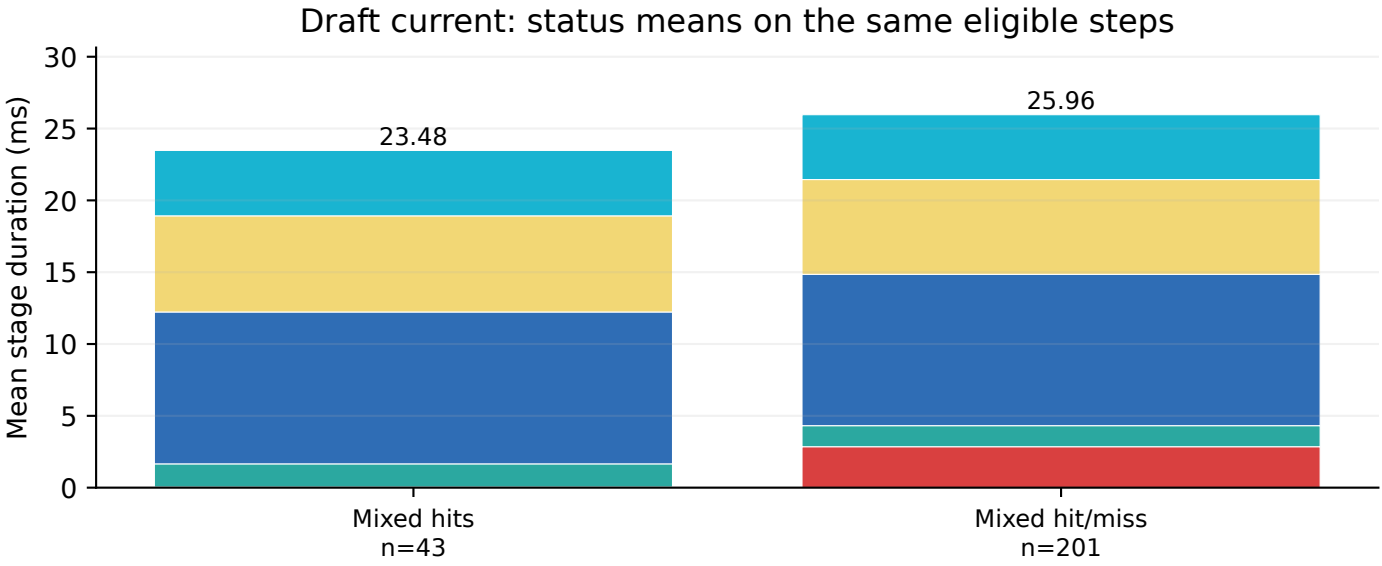
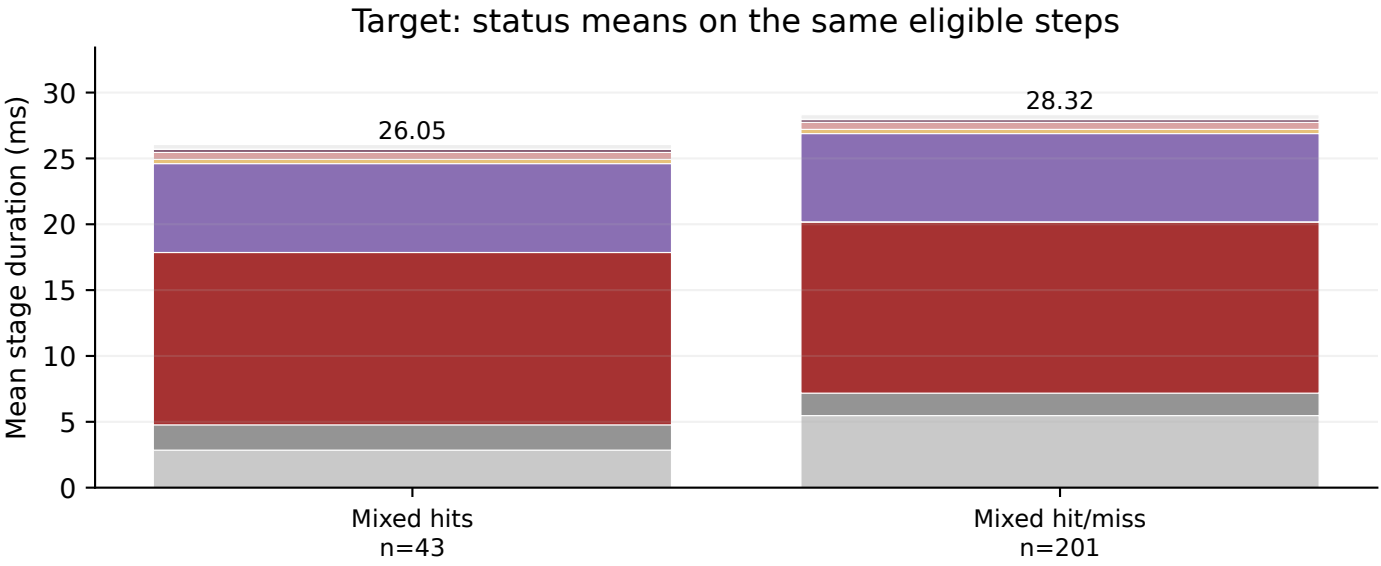
llama2_b8_stability_profile_0
DUET: K1/K2=3/2, exit=19 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 244 eligible steps



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_stability_profile_1
DUET: K1/K2=3/1, exit=21 (0-based), $\bar{C}$ =3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 244 eligible steps

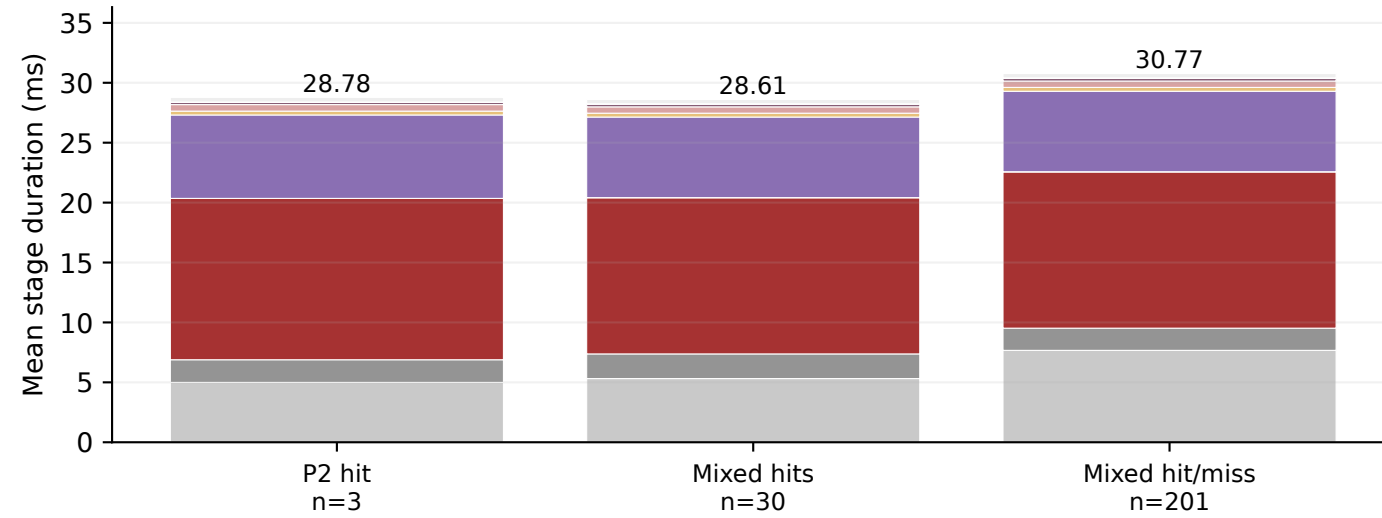


Wait / sync Prepare Target pre Target post Proxy / launch Proxy compute Final logits Accept Commit Postprocess Unlabelled gap Context P1 P2 Previous P2 Proxy wait Target ready Proxy received

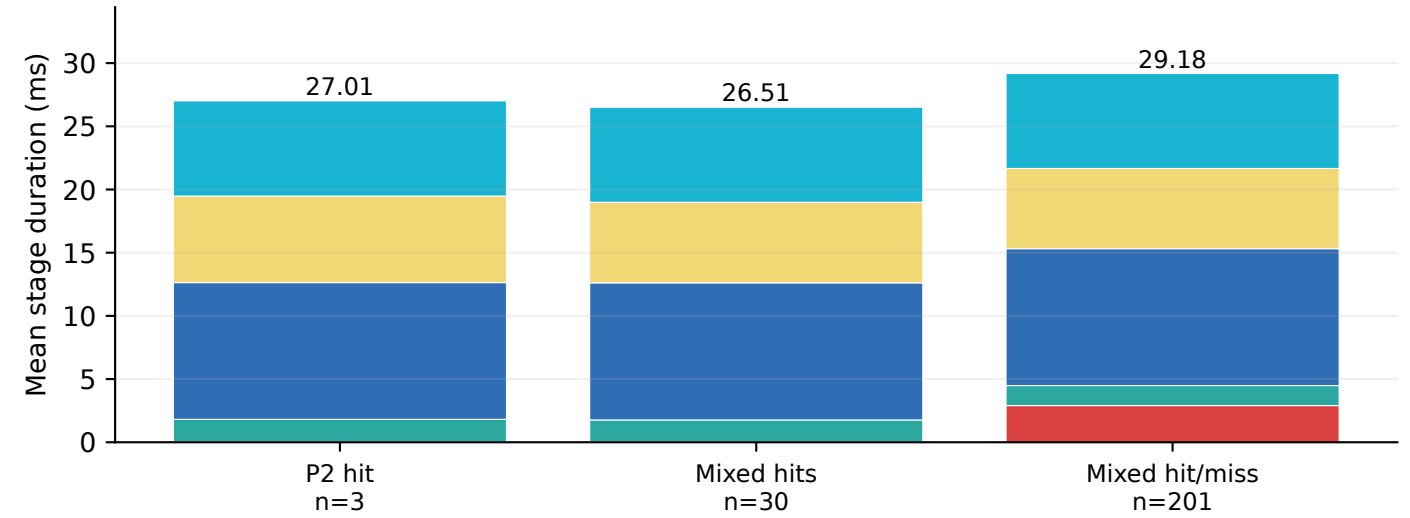
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_stability_profile_2
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 234 eligible steps

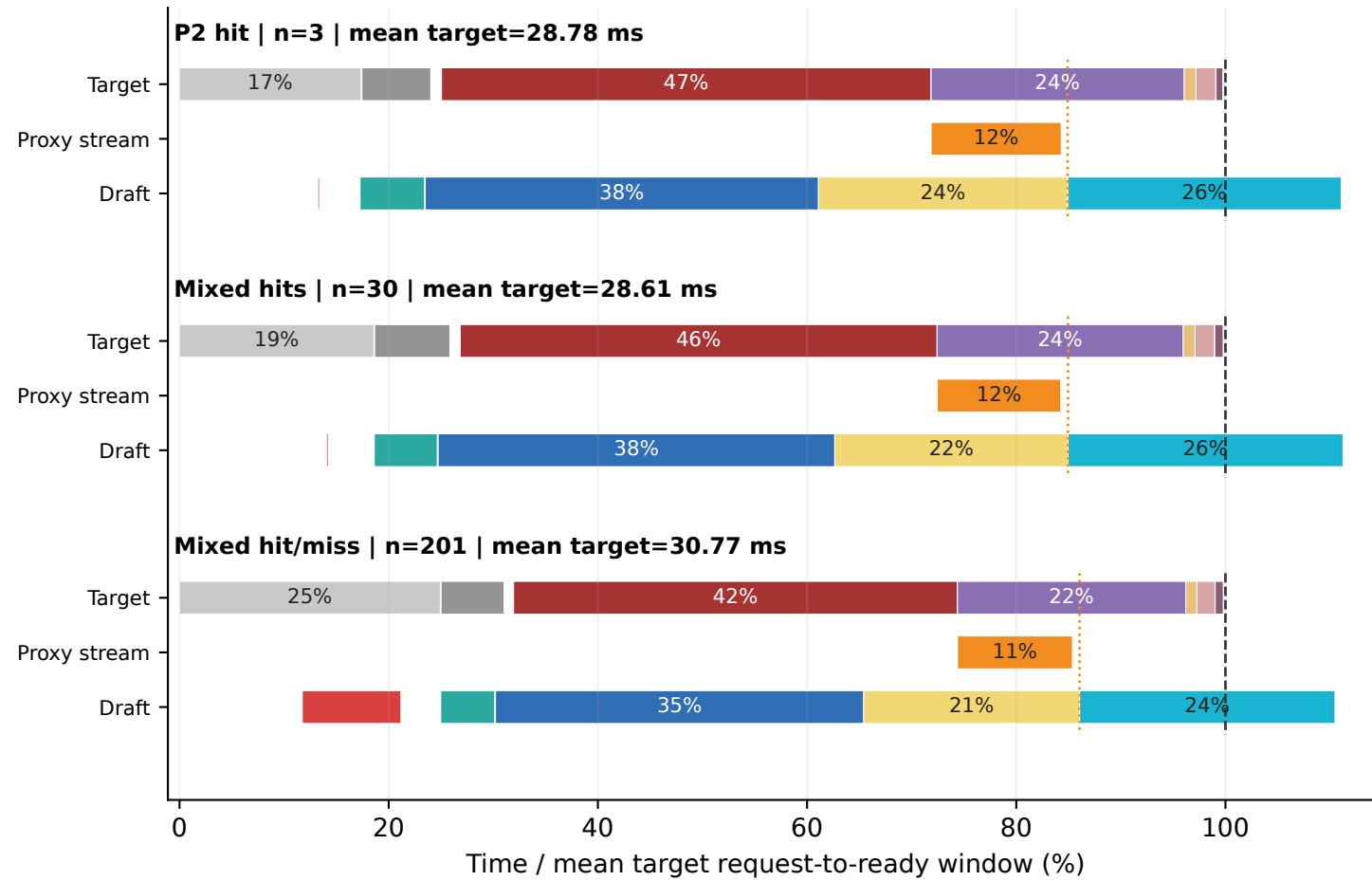
Target: status means on the same eligible steps



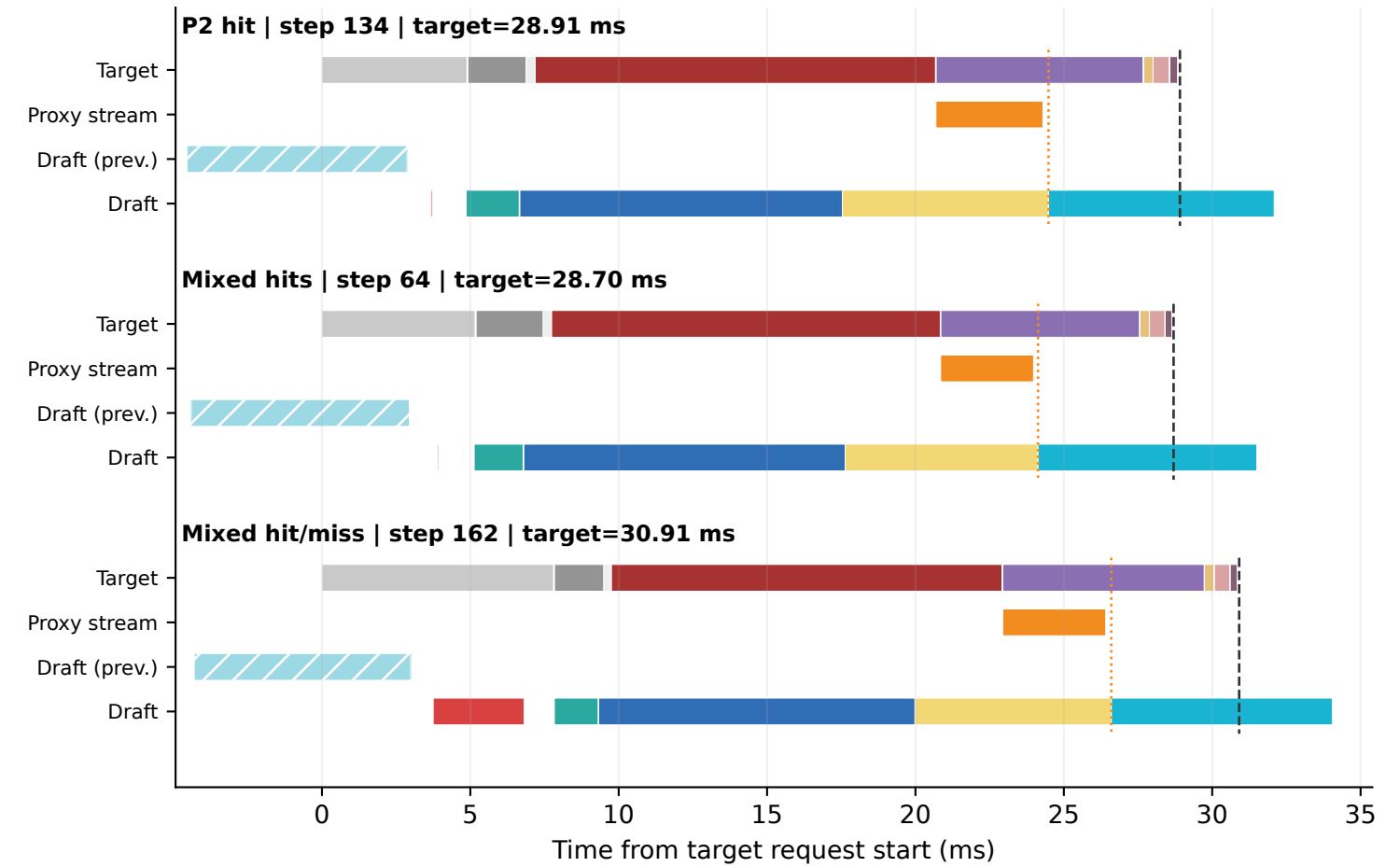
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



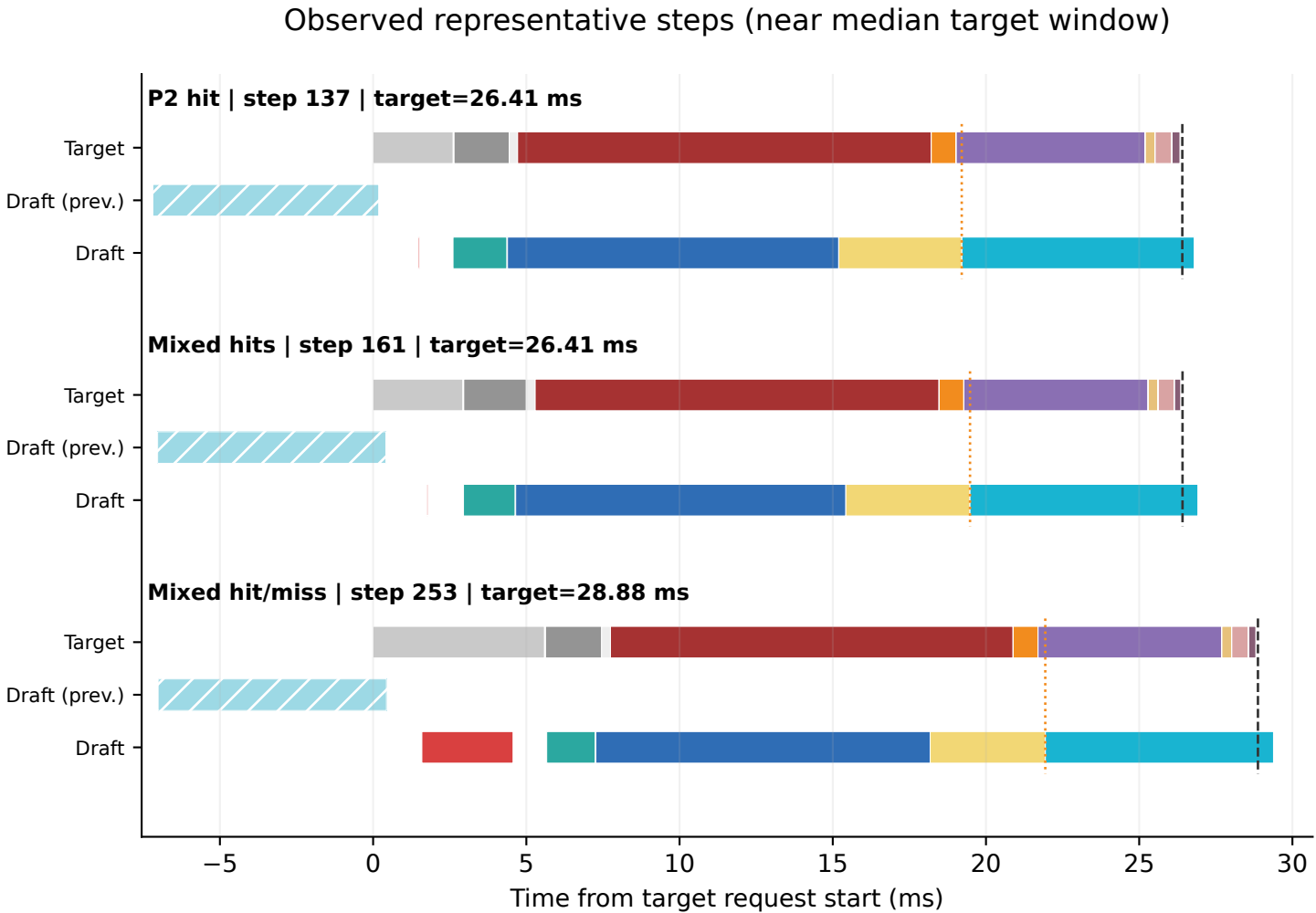
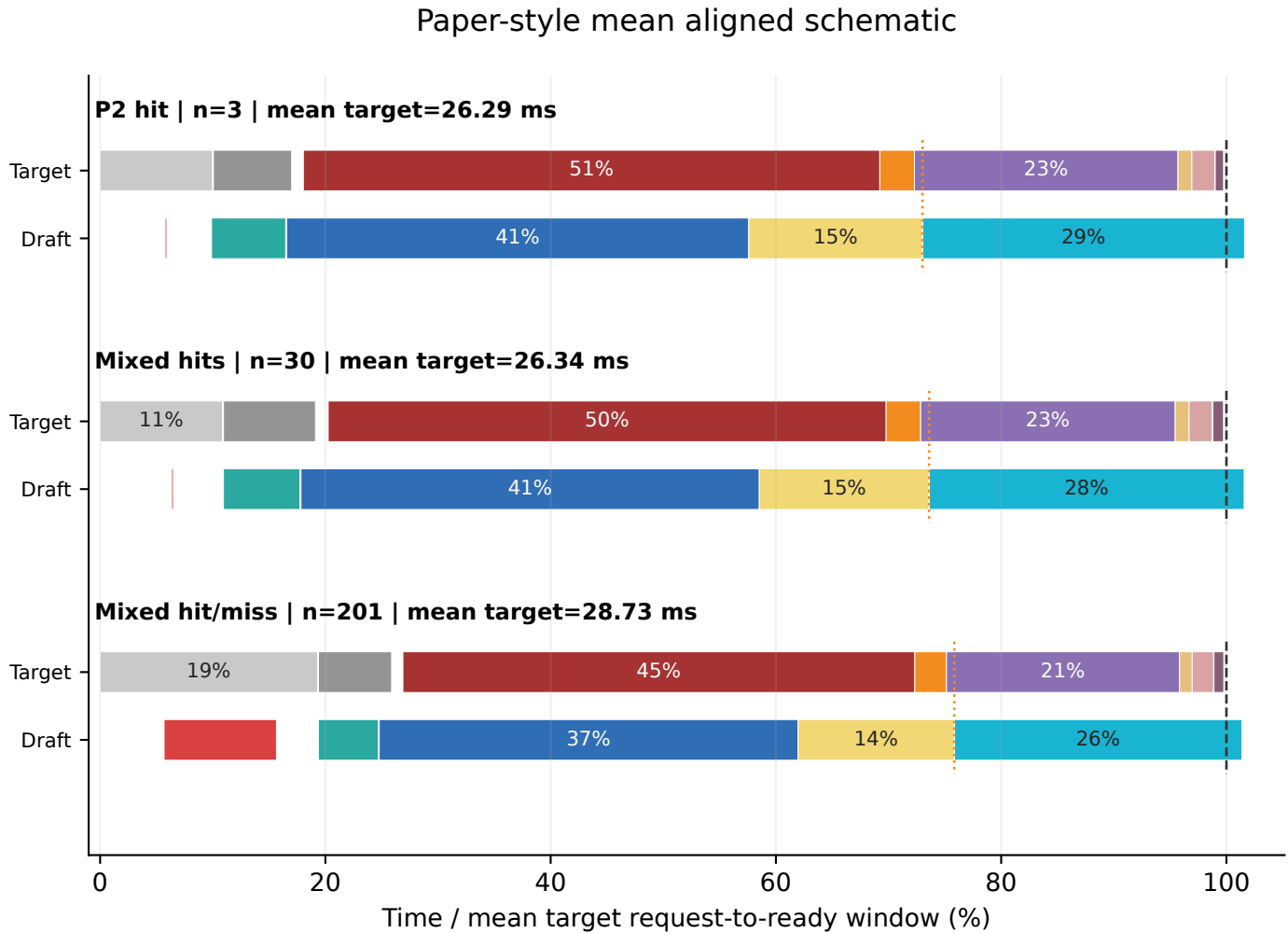
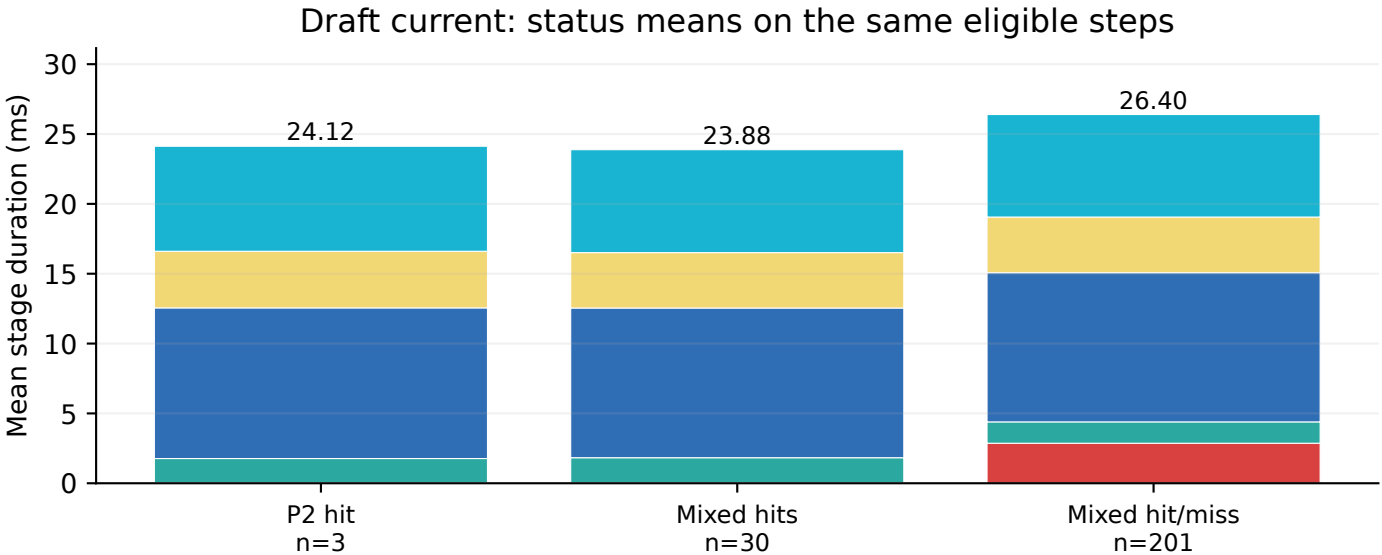
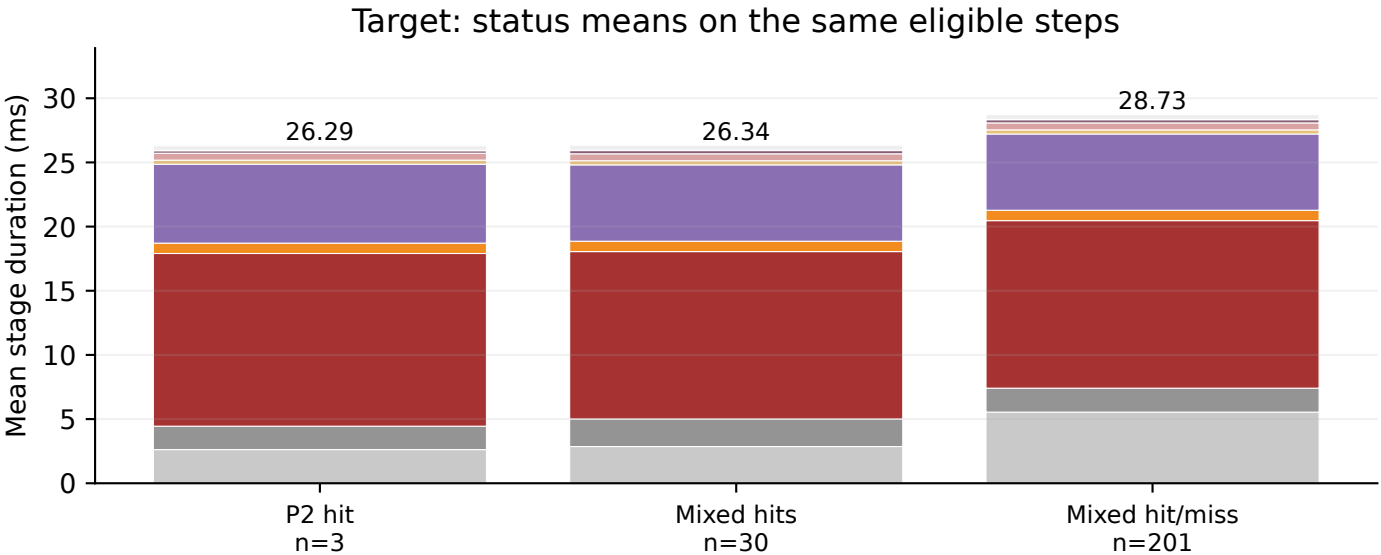
Observed representative steps (near median target window)



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_stability_profile_3
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 234 eligible steps

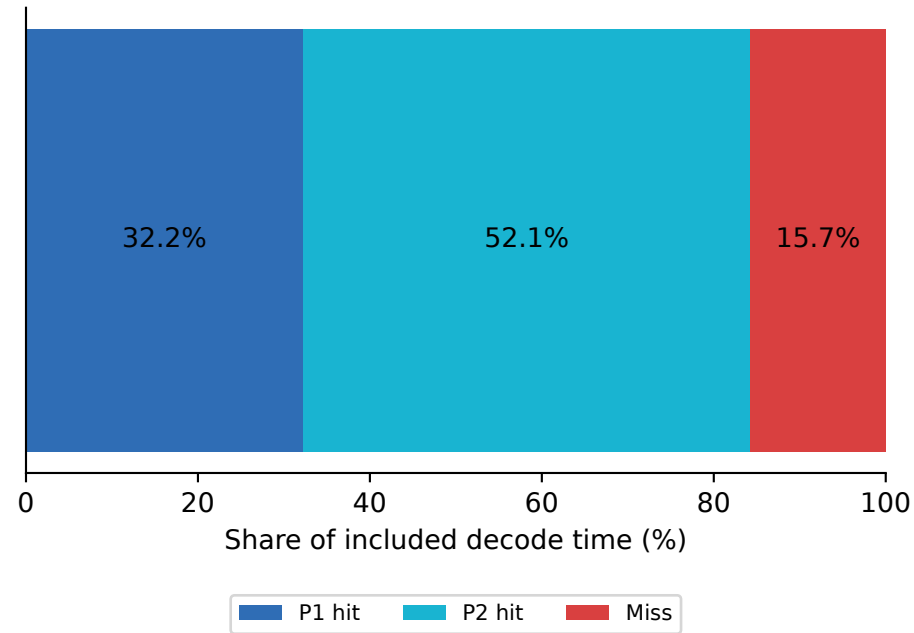


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

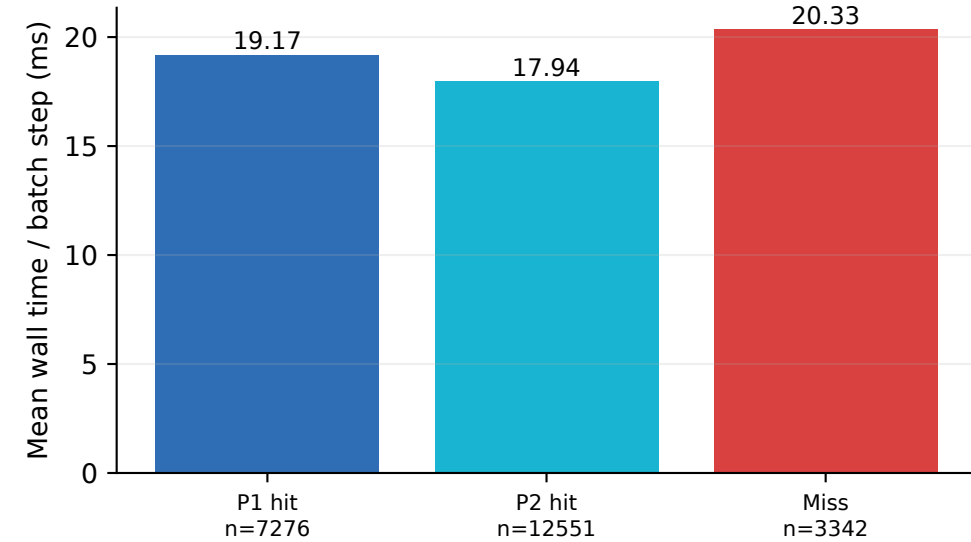
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_stream_fast_r0
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
Last pass: B=1, T=0.7, N=480 | AL*=2.365, hit=85.7%, TPS*=126.65

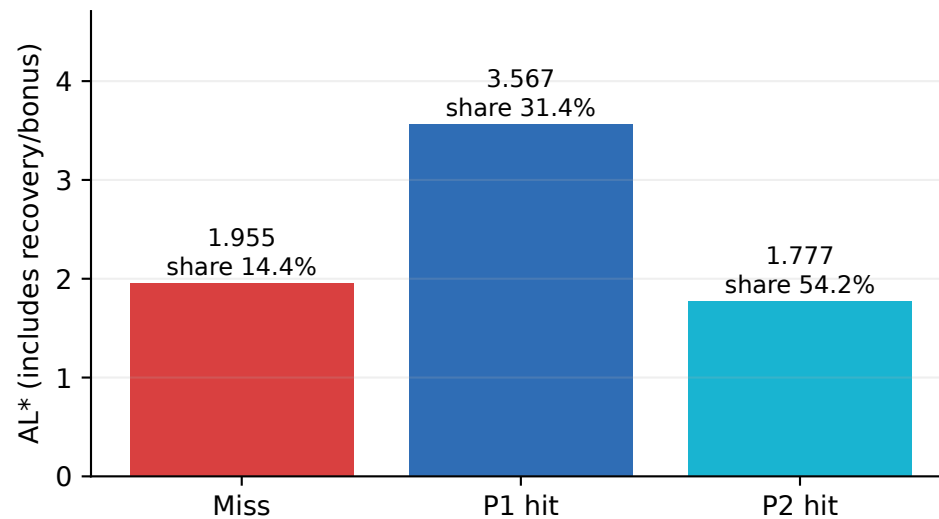
Measured wall time by batch cache status



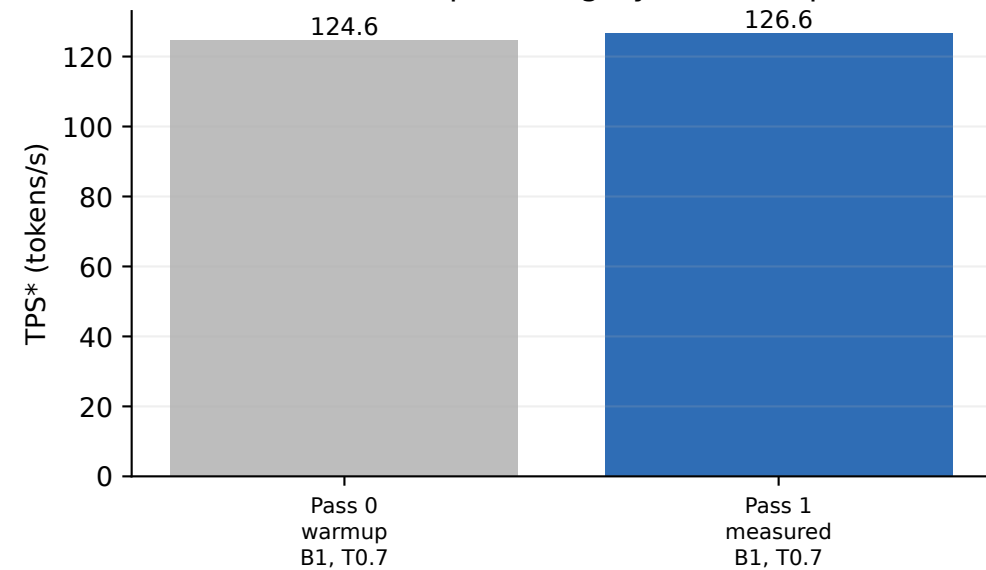
Boundary-free steps, including slow steps



Conditional AL* and source share



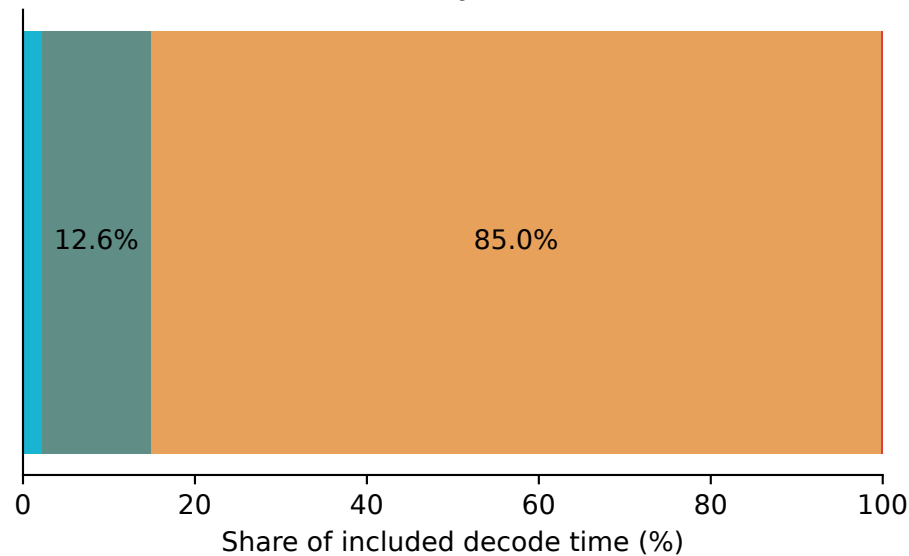
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

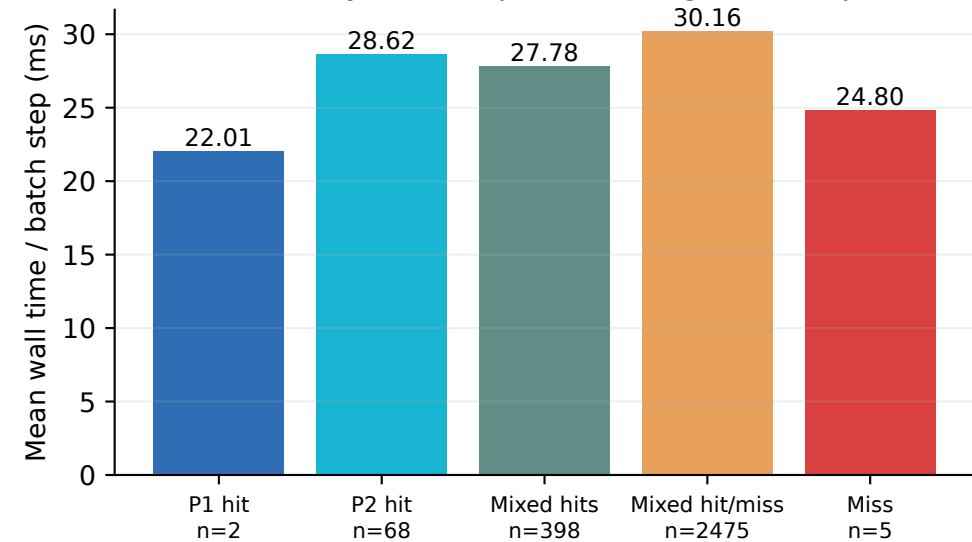
llama2_b8_stream_fast_r1
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.090, hit=77.1%, TPS*=553.94

Measured wall time by batch cache status

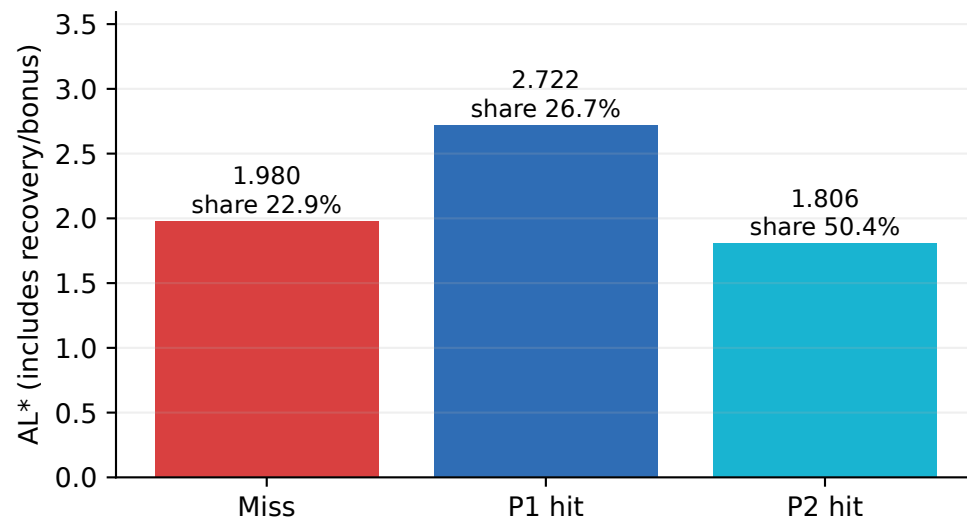


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

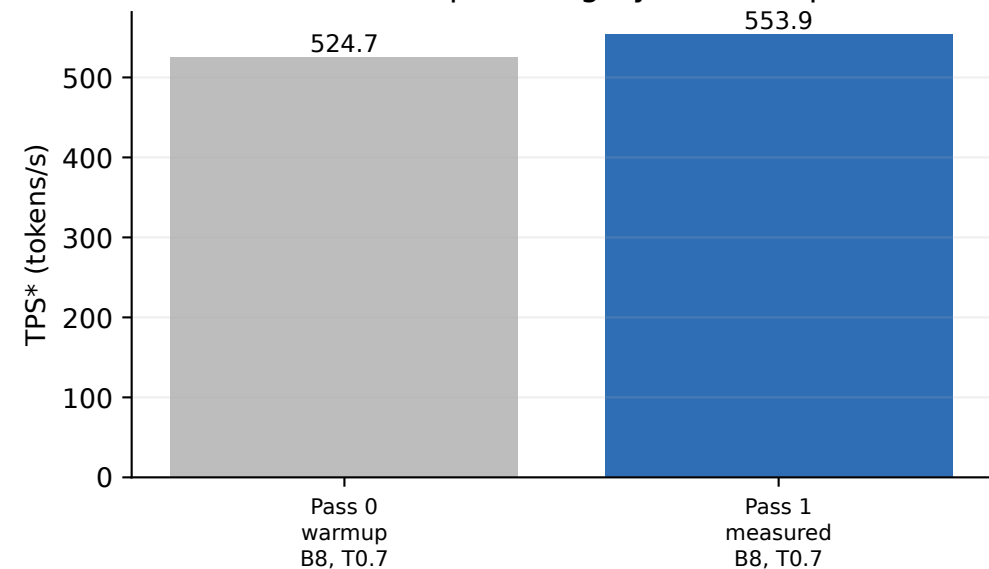
Boundary-free steps, including slow steps



Conditional AL* and source share

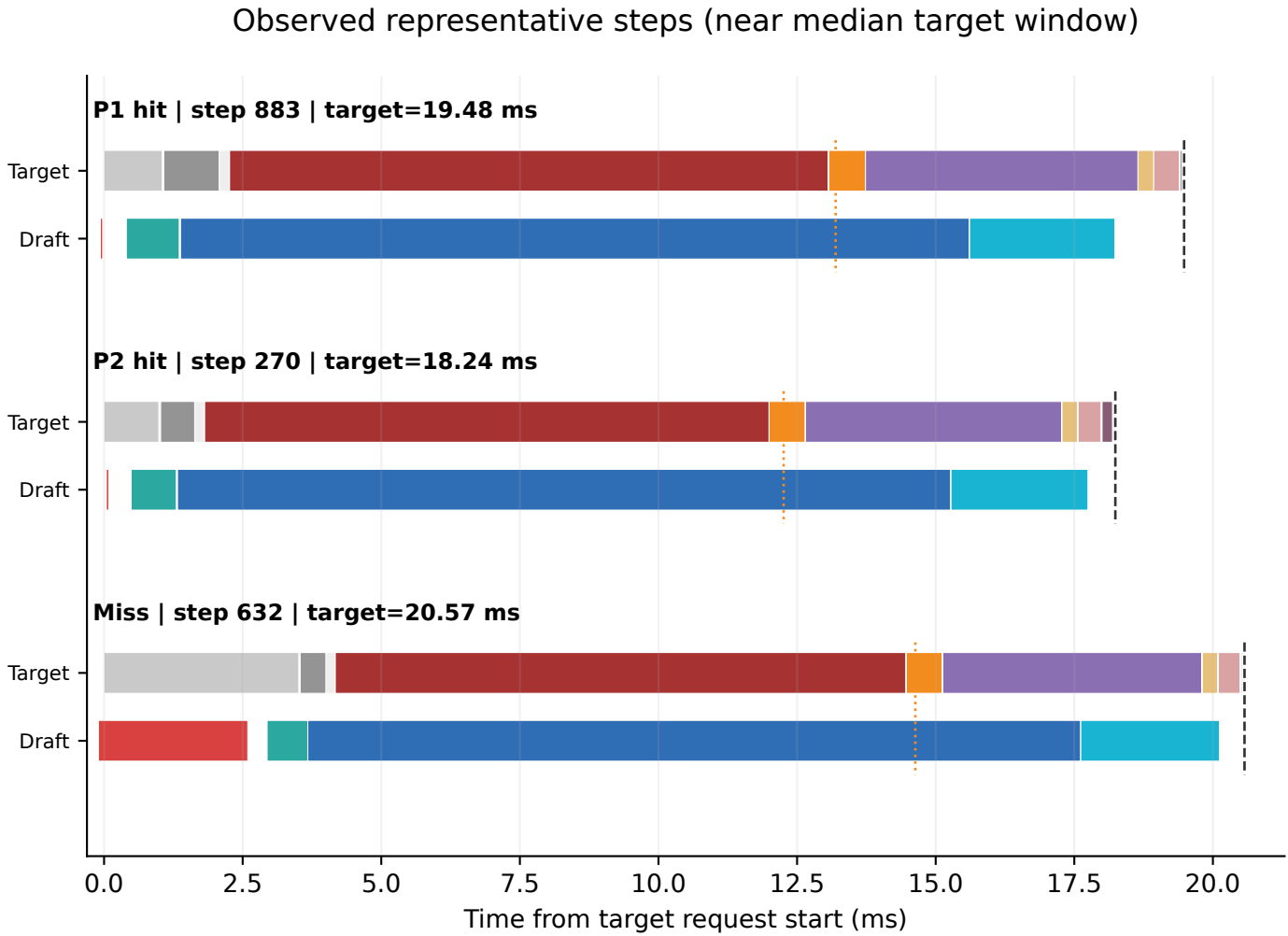
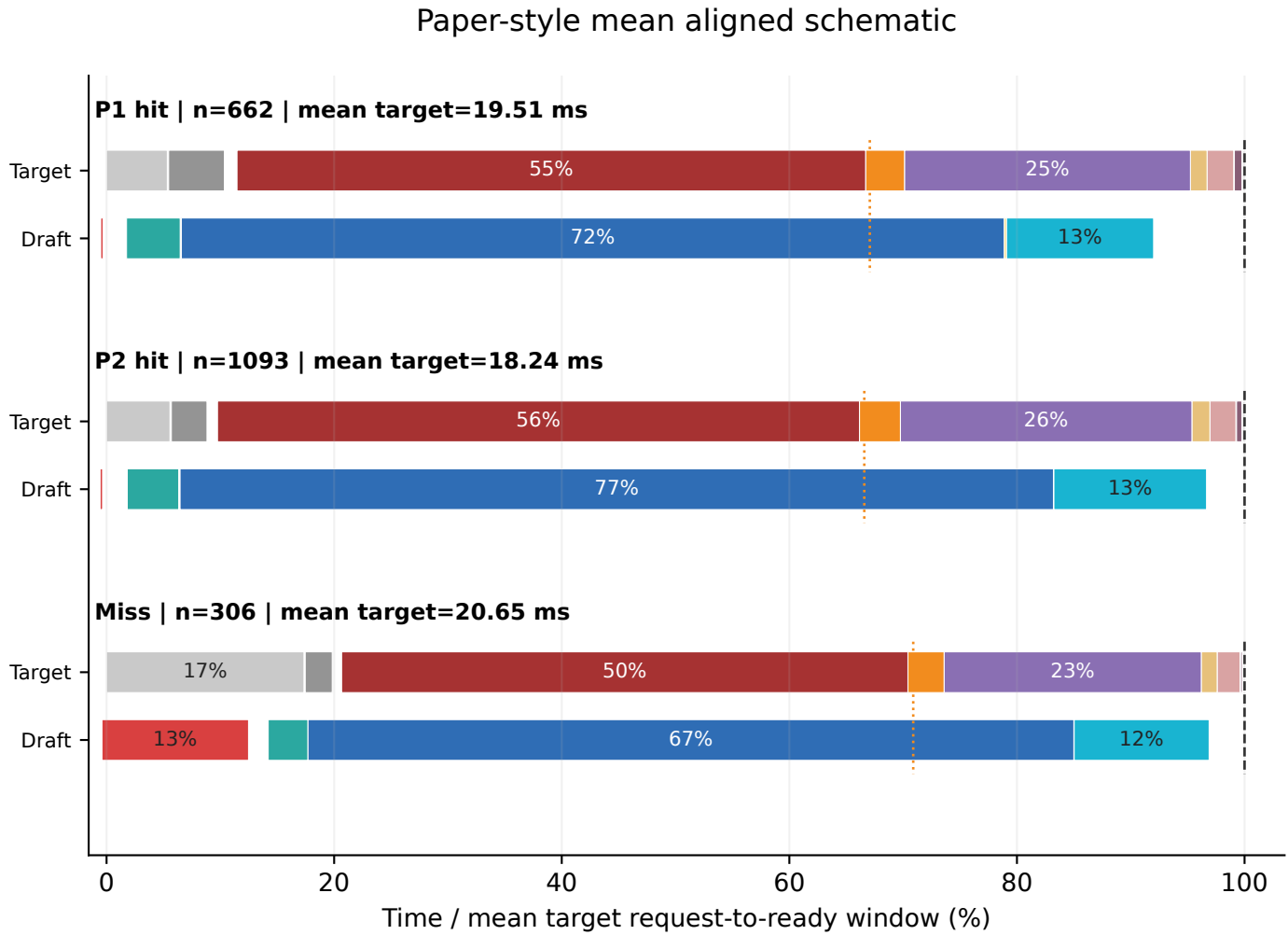
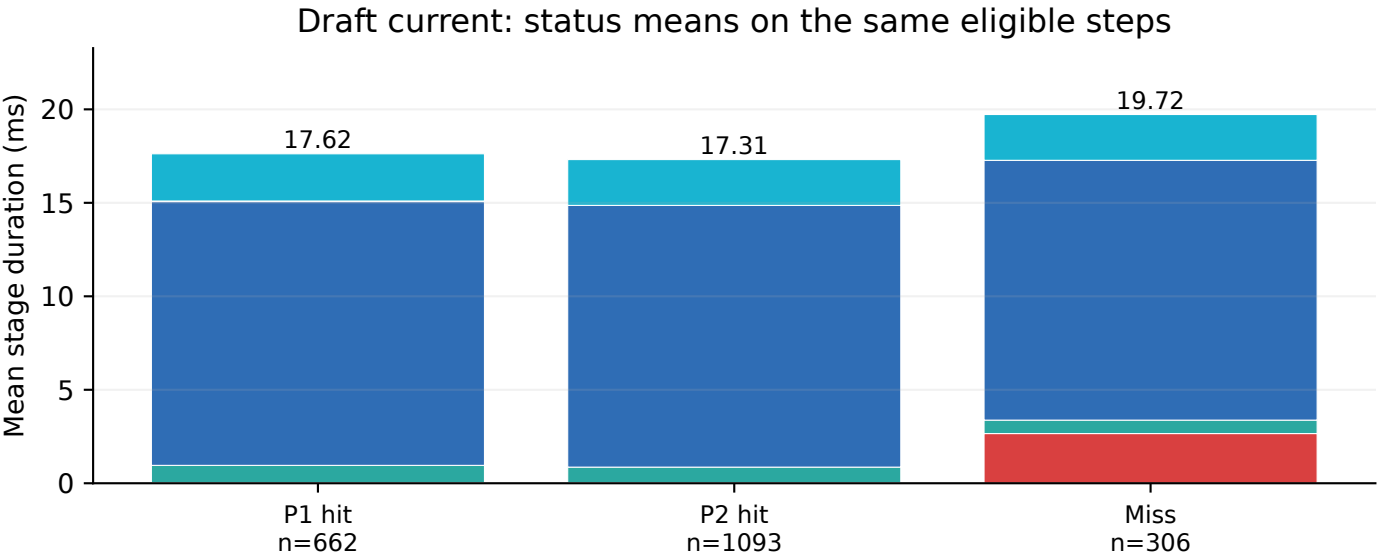
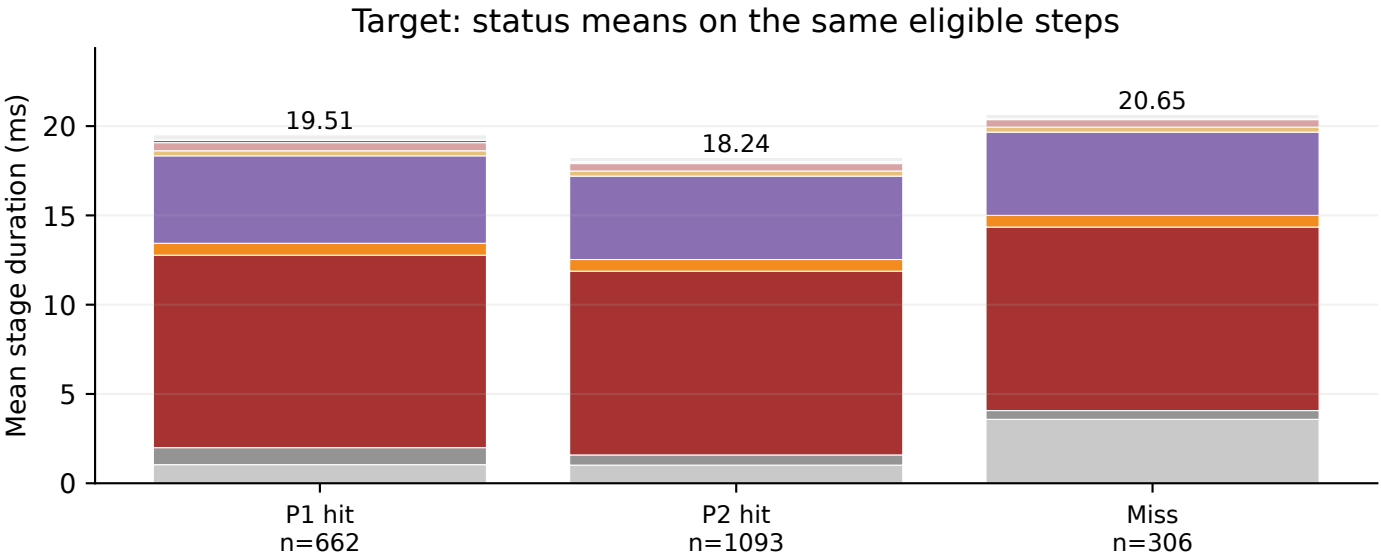


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

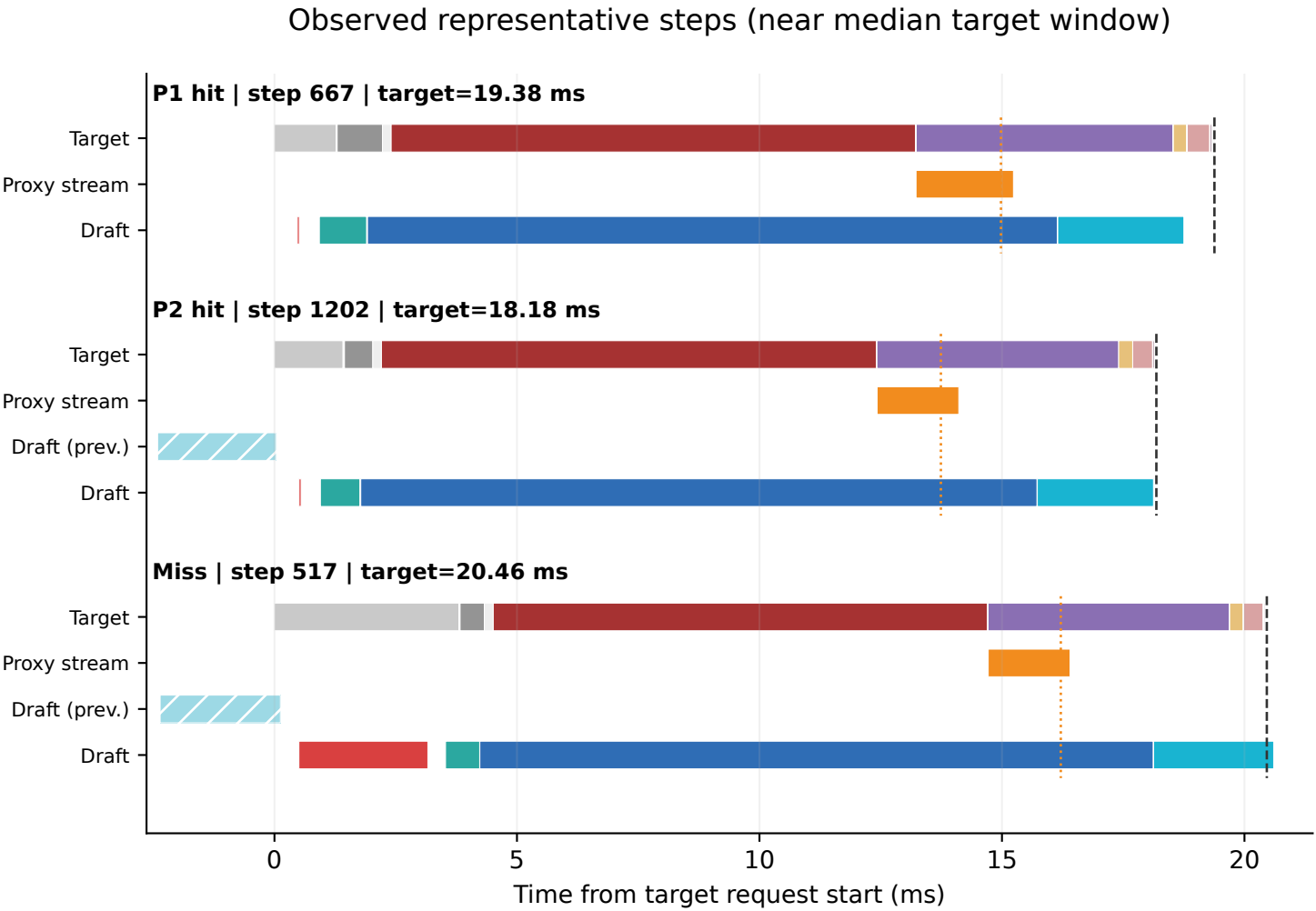
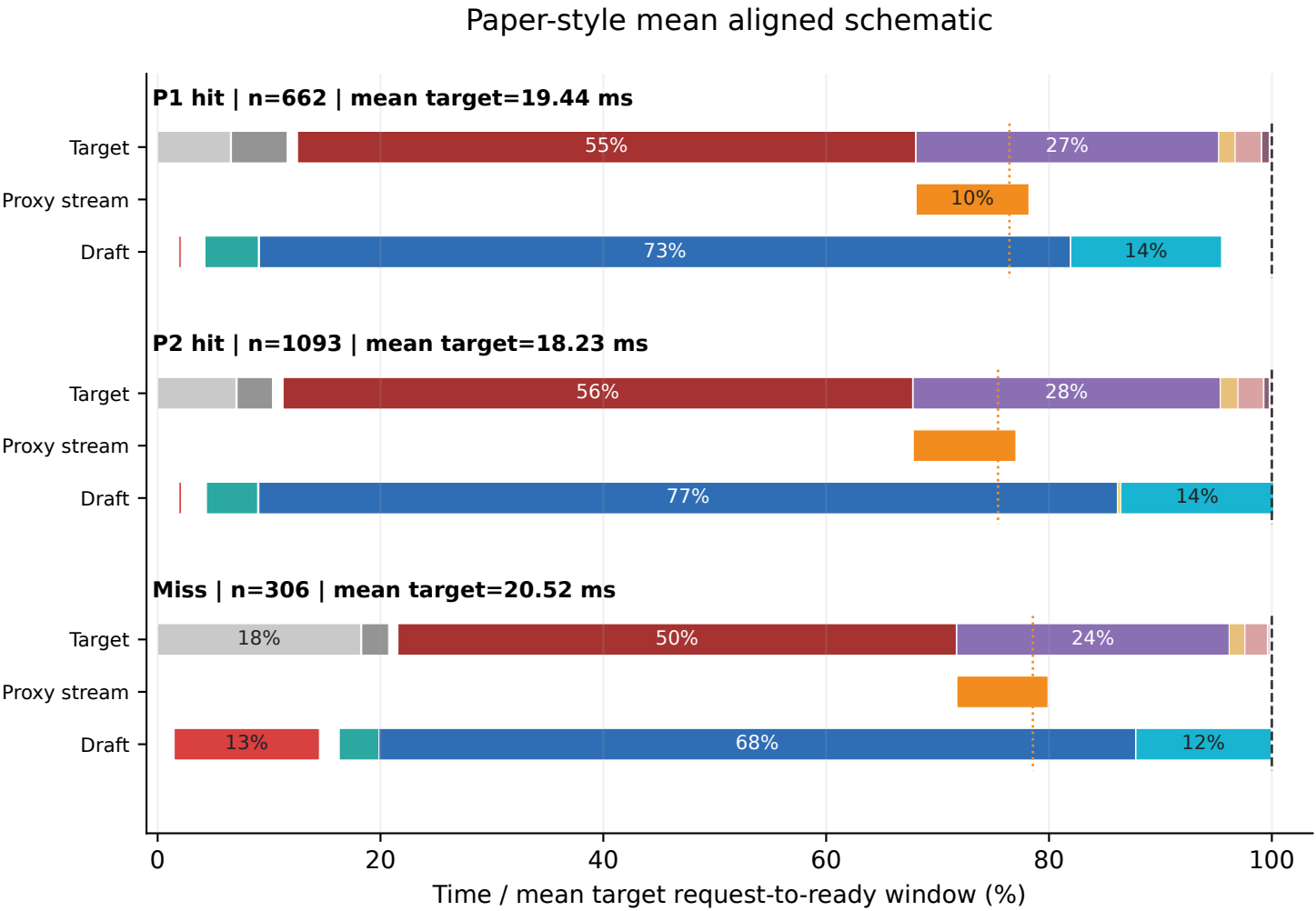
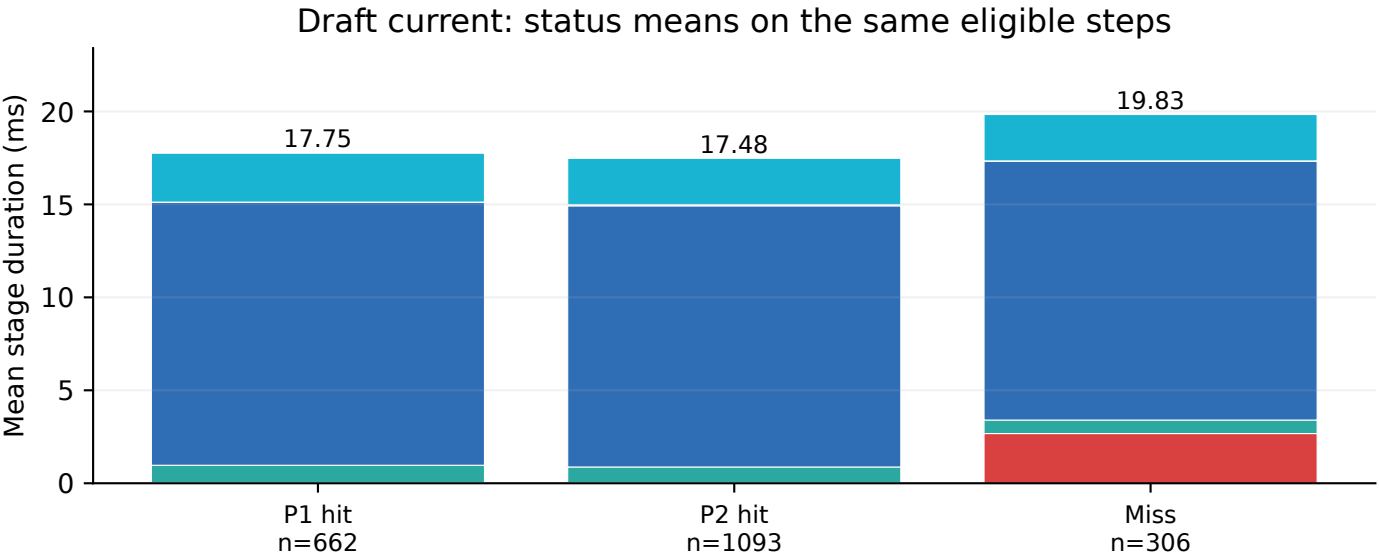
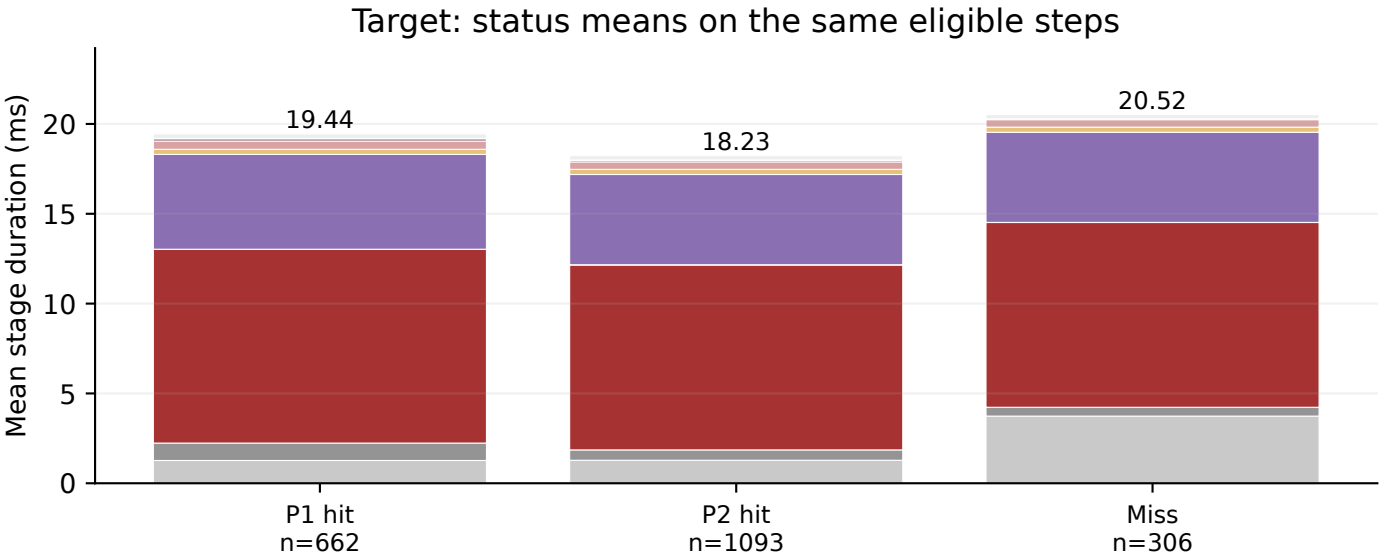
llama2_b1_stream_l0_profile_s0
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/0
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2061 eligible steps



Wait / sync Prepare Target pre Target post Proxy / launch Final logits Postprocess Unlabelled gap Miss draft Context P1 P2 Proxy wait Target ready Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_stream_l0_profile_s1
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2061 eligible steps

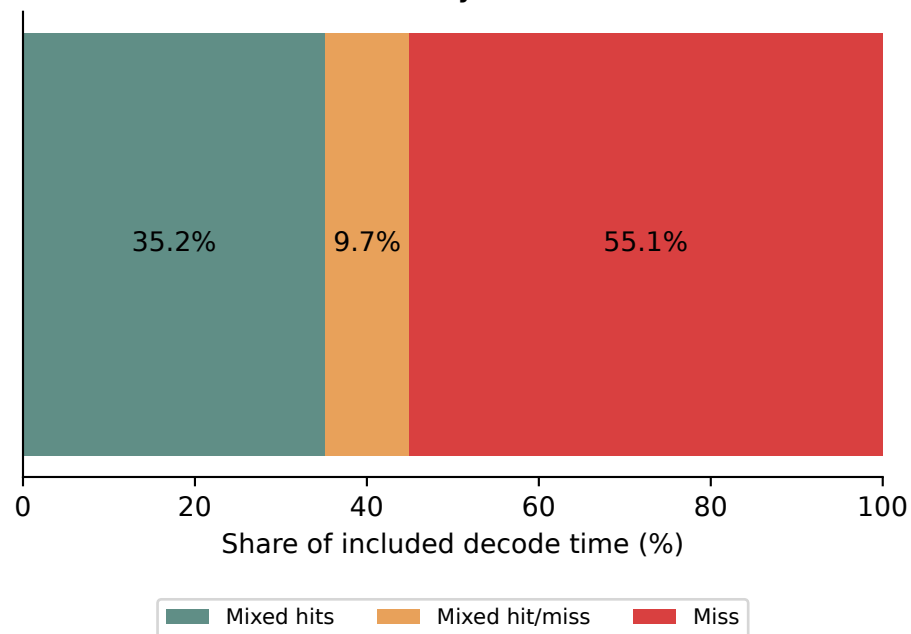


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

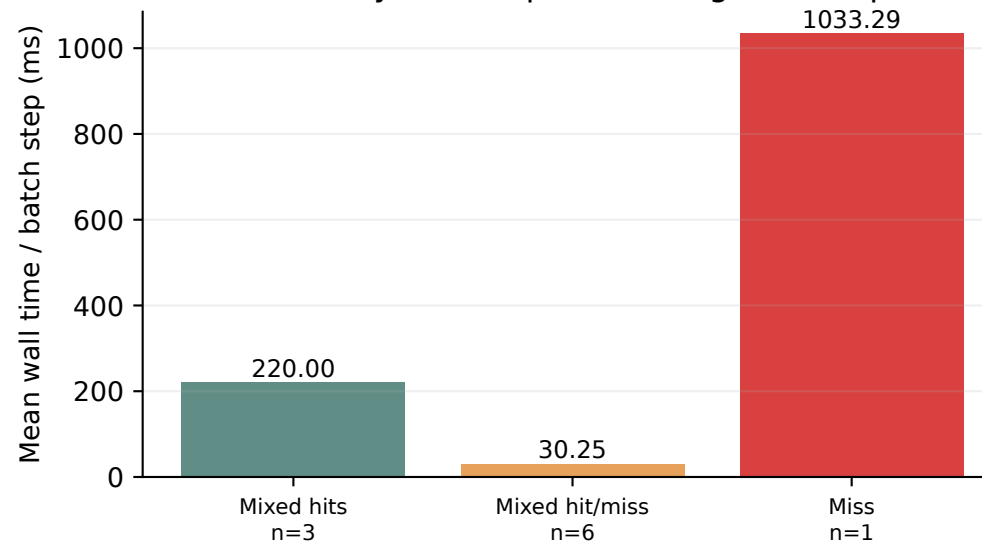
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_stream_l0_greedy_smoke
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.0, N=8 | AL*=1.870, hit=75.0%, TPS*=68.27

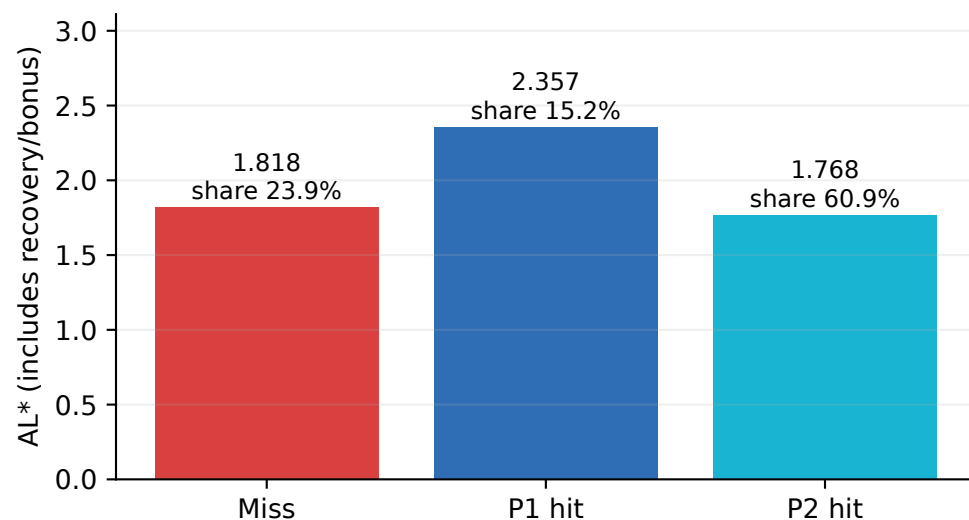
Measured wall time by batch cache status



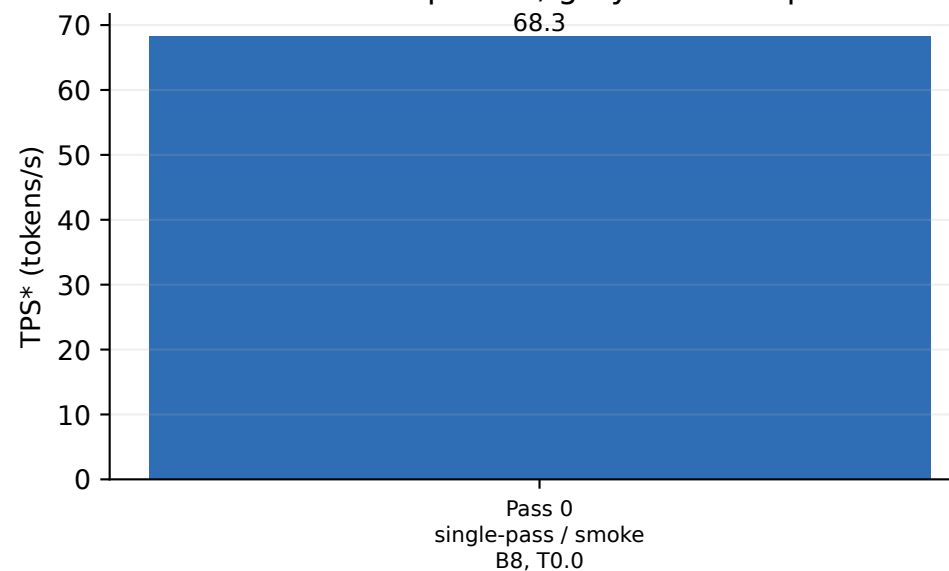
Boundary-free steps, including slow steps



Conditional AL* and source share

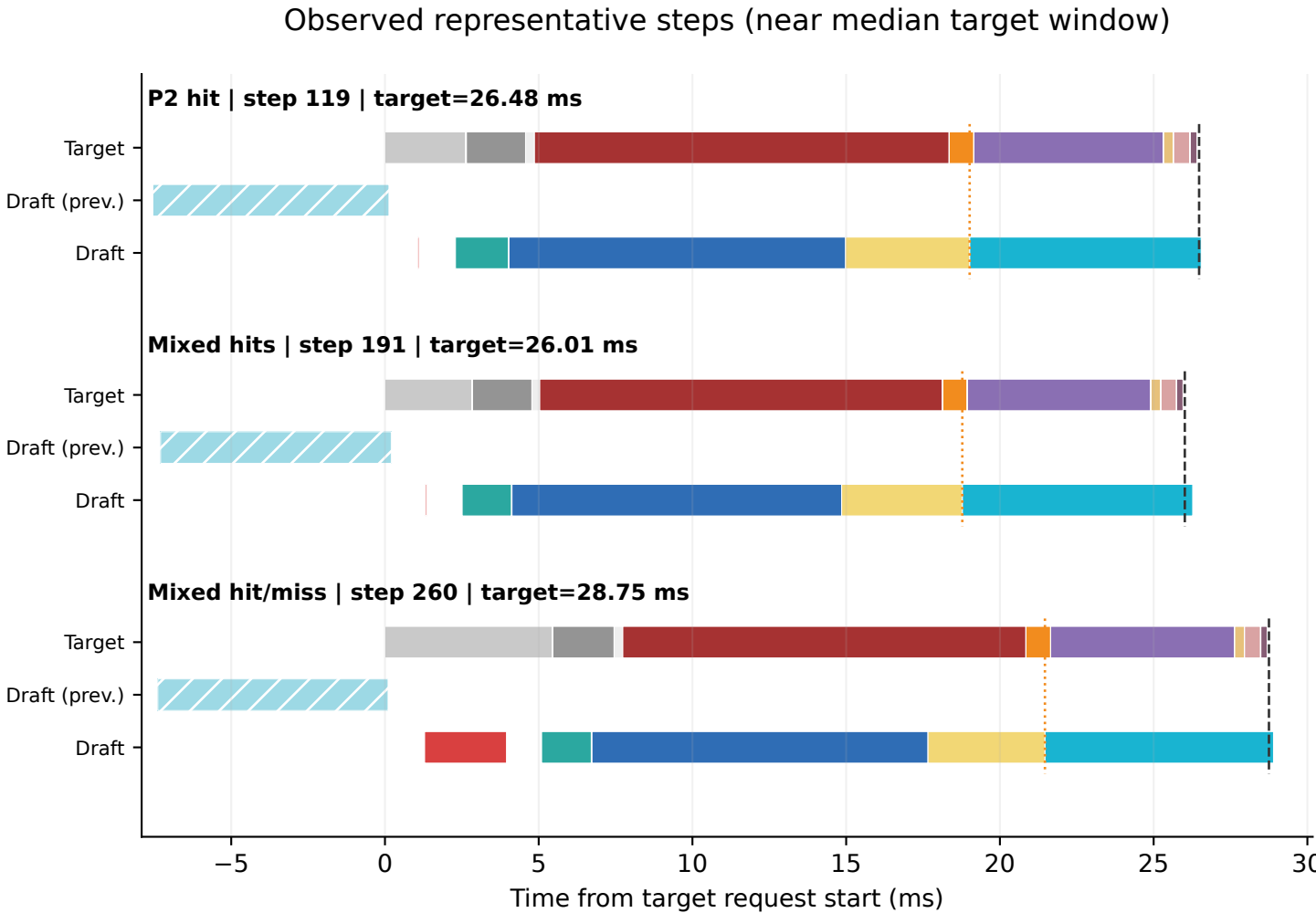
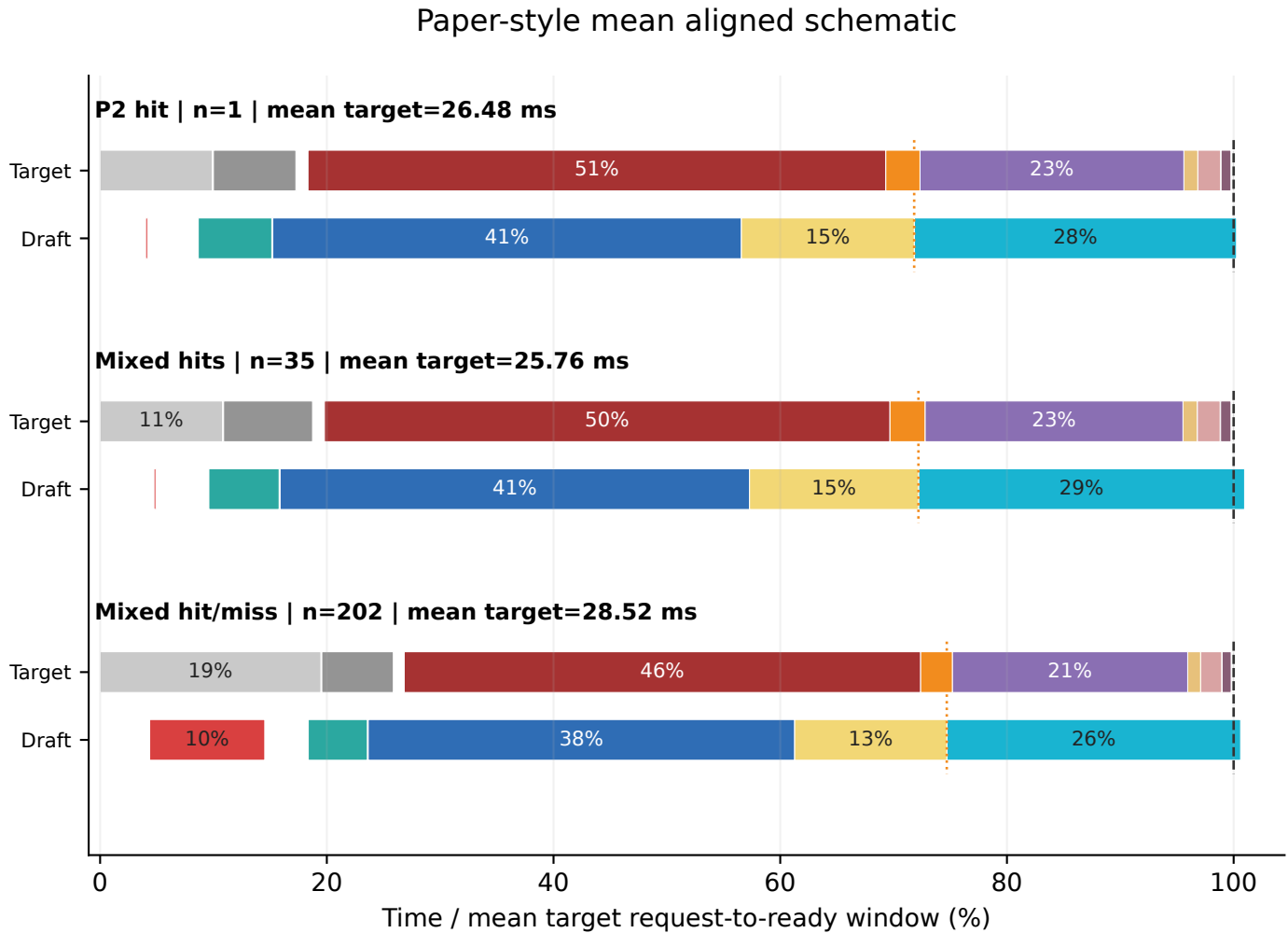
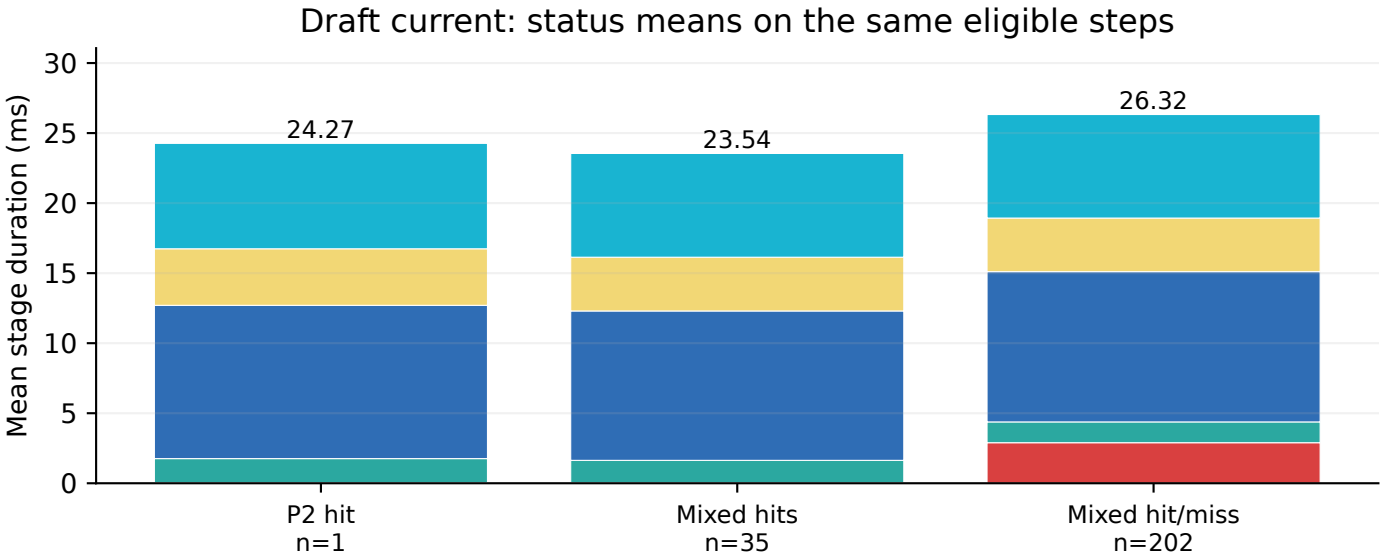
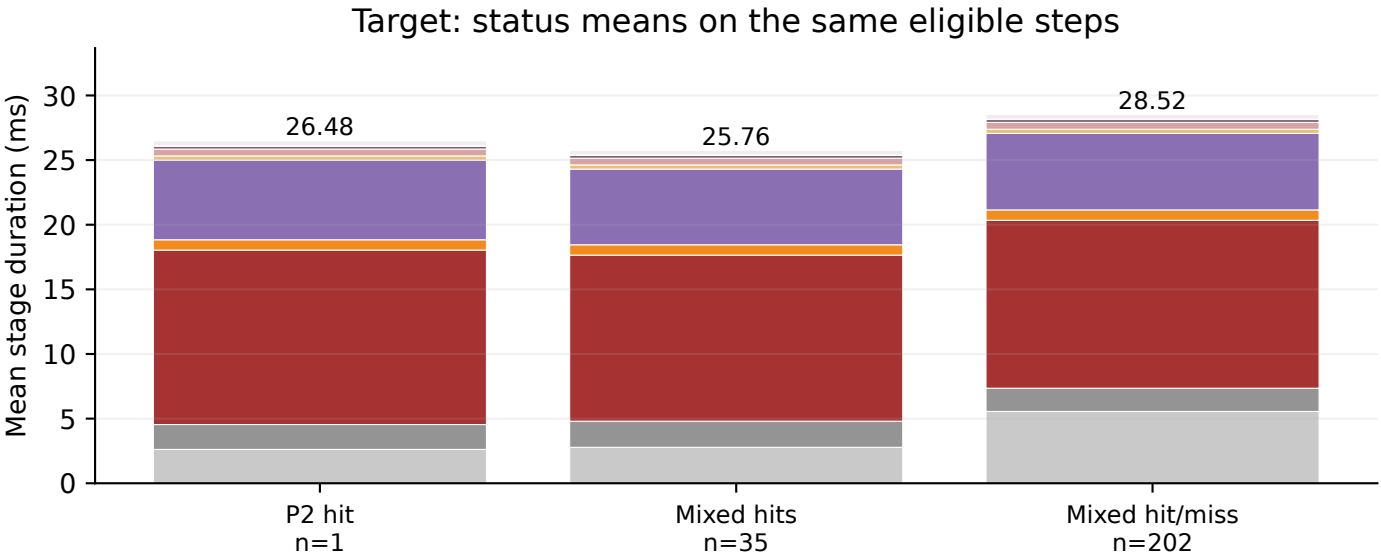


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stream_l0_profile_s0
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 238 eligible steps

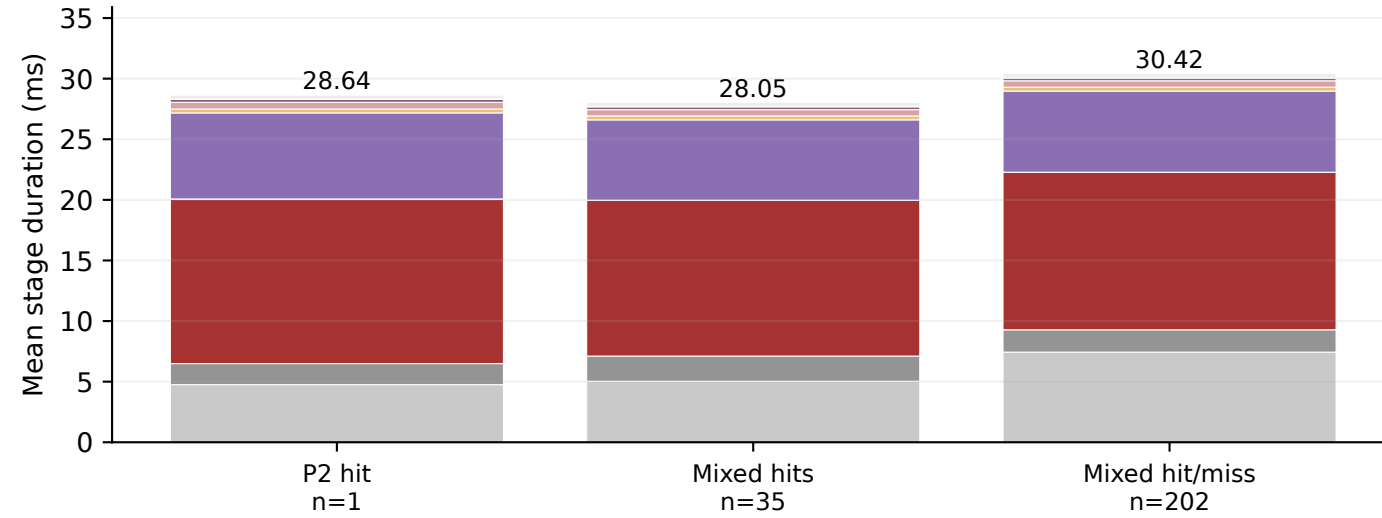


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

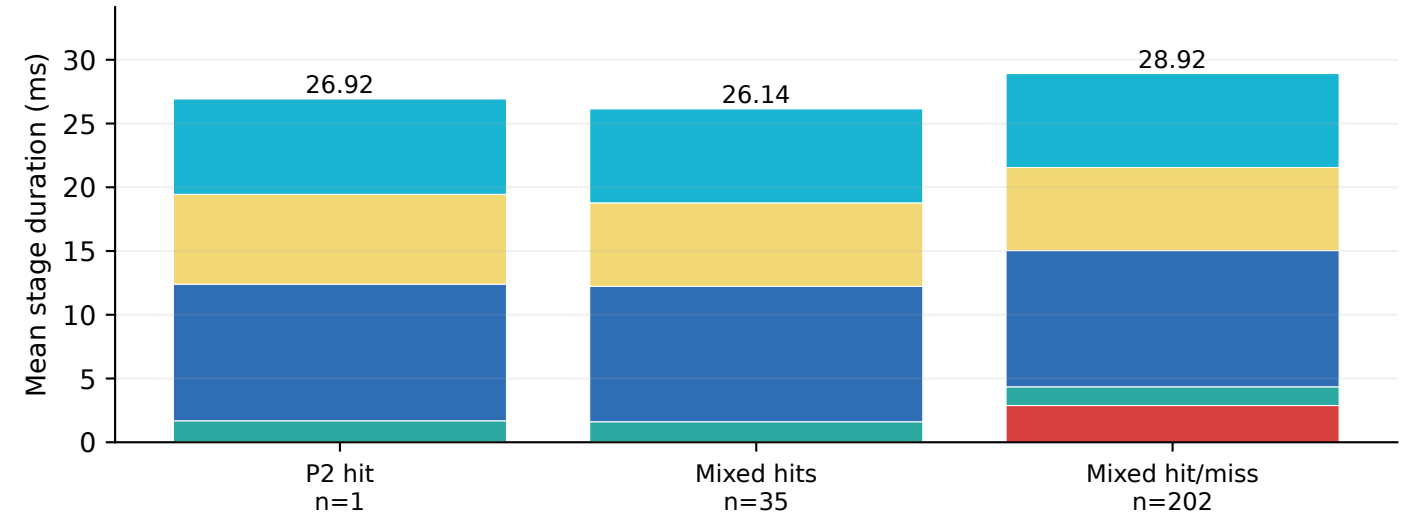
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_stream_l0_profile_s1
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 238 eligible steps

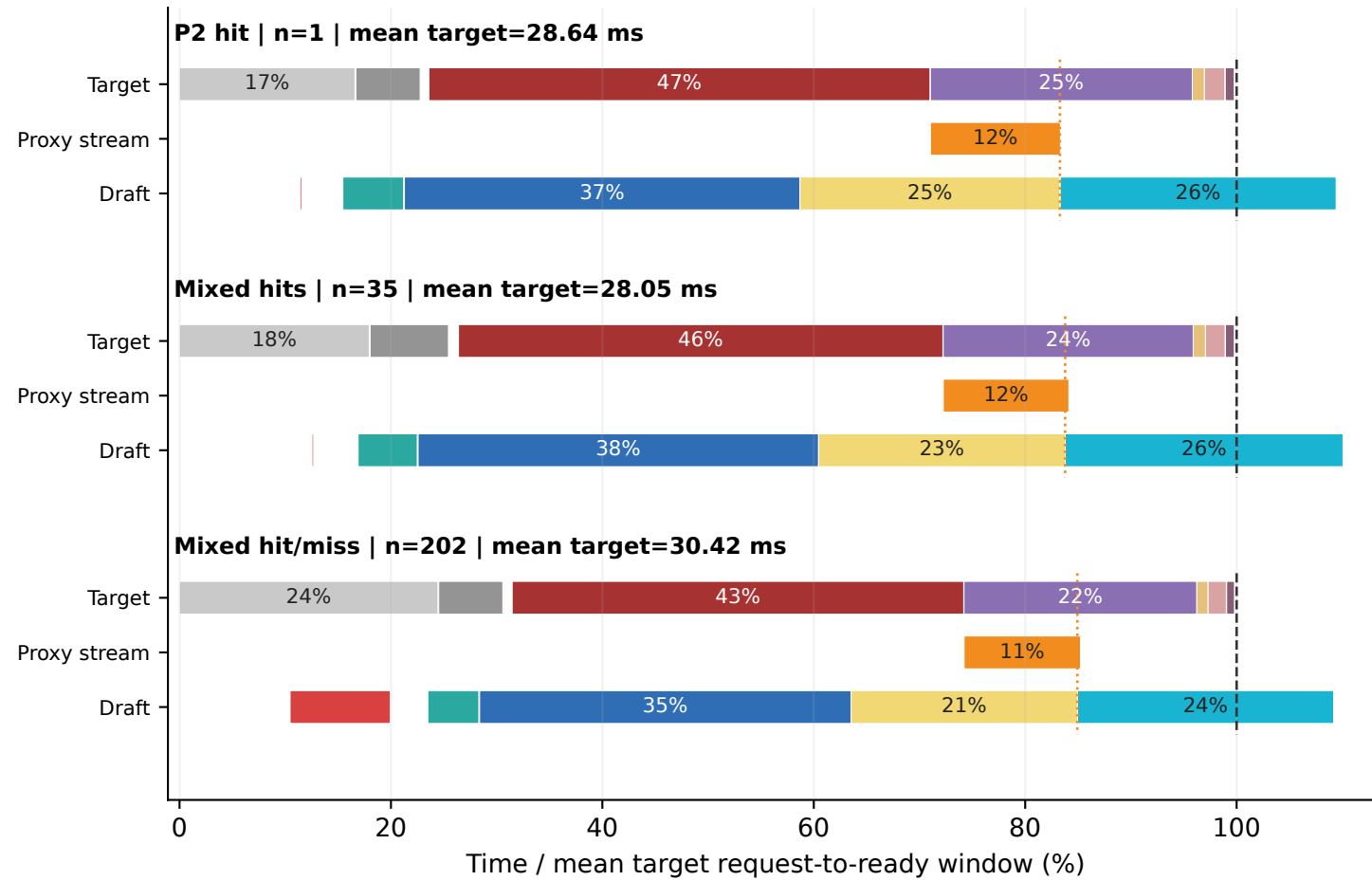
Target: status means on the same eligible steps



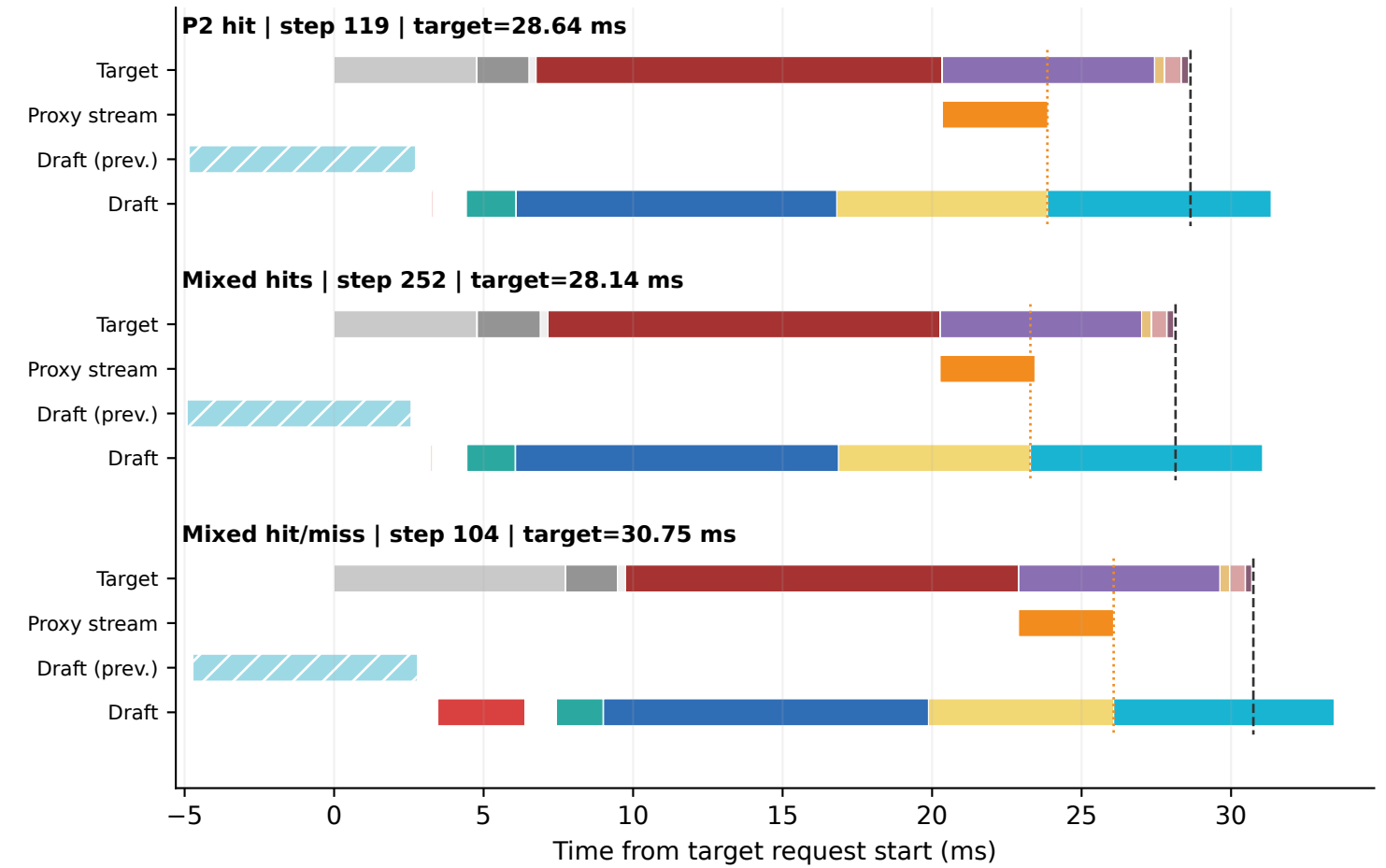
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

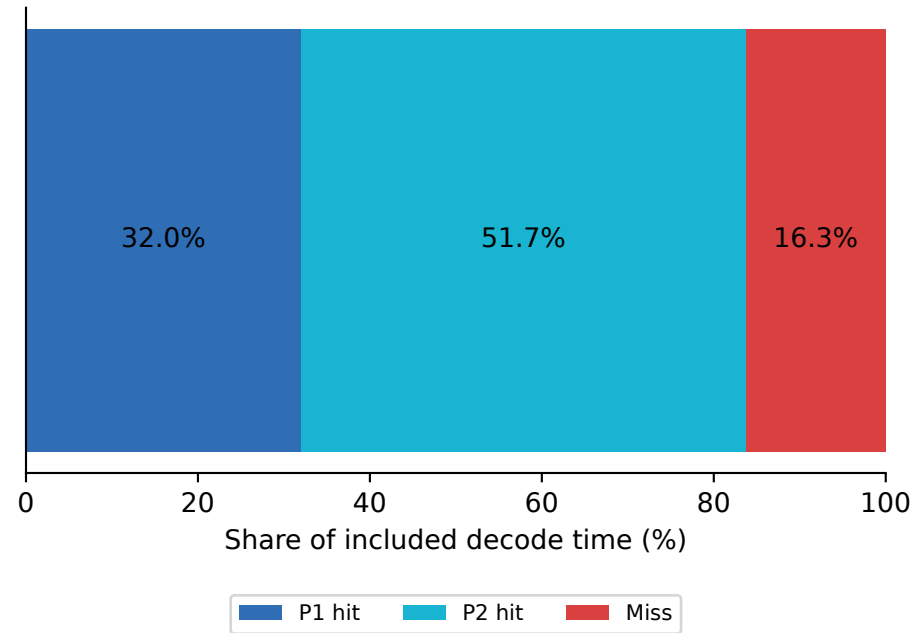


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
 Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
 Target pre Proxy compute Commit Miss draft P2

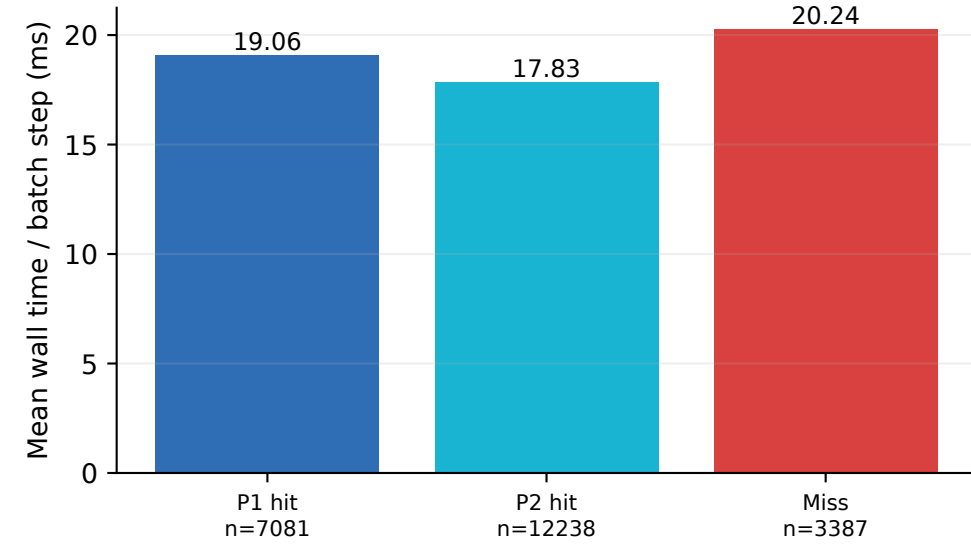
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
 Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_stream_fast_r1
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
Last pass: B=1, T=0.7, N=480 | AL*=2.366, hit=85.1%, TPS*=127.38

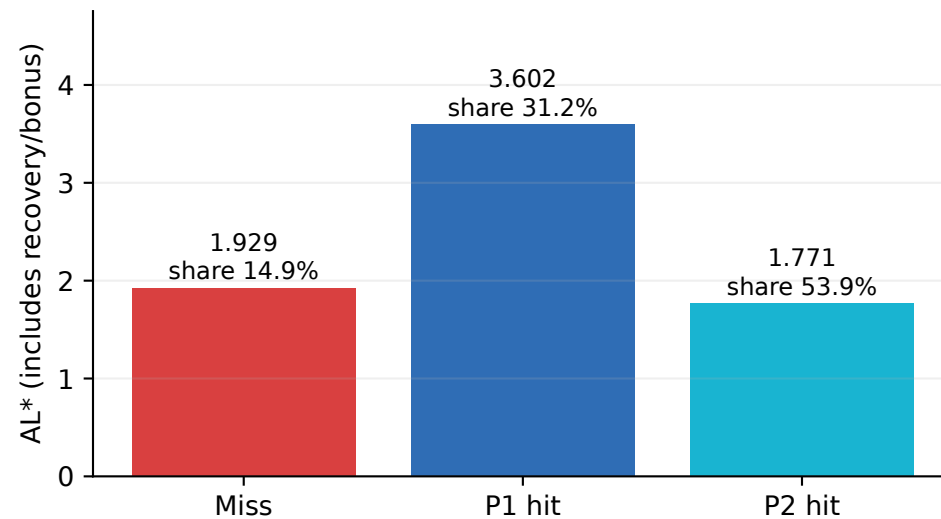
Measured wall time by batch cache status



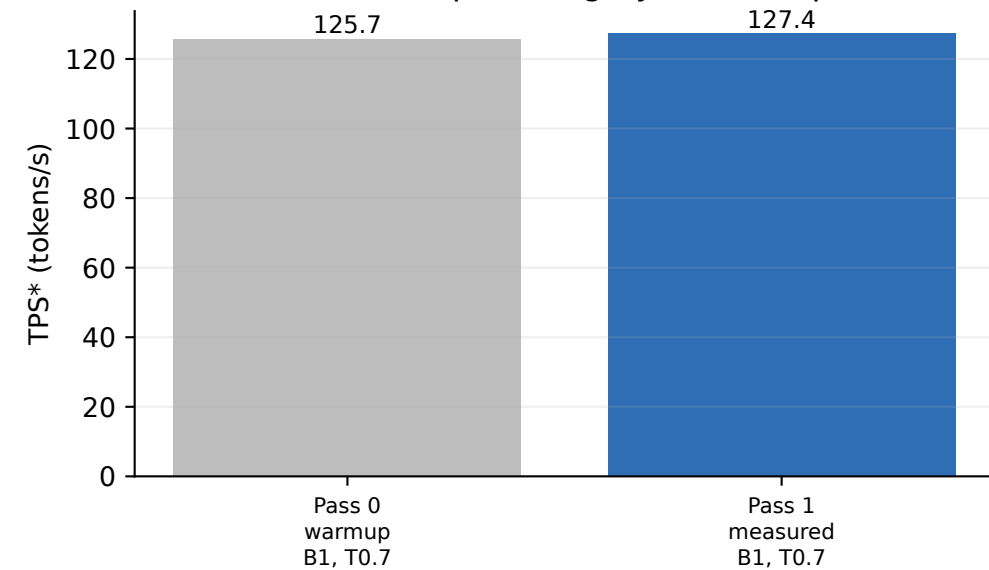
Boundary-free steps, including slow steps



Conditional AL* and source share



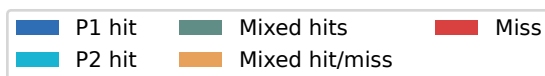
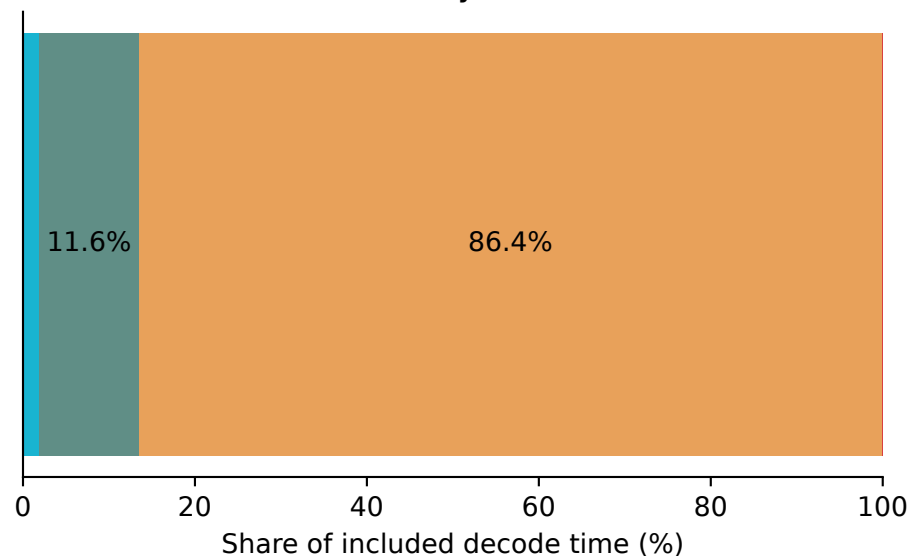
All saved passes; gray = warmup



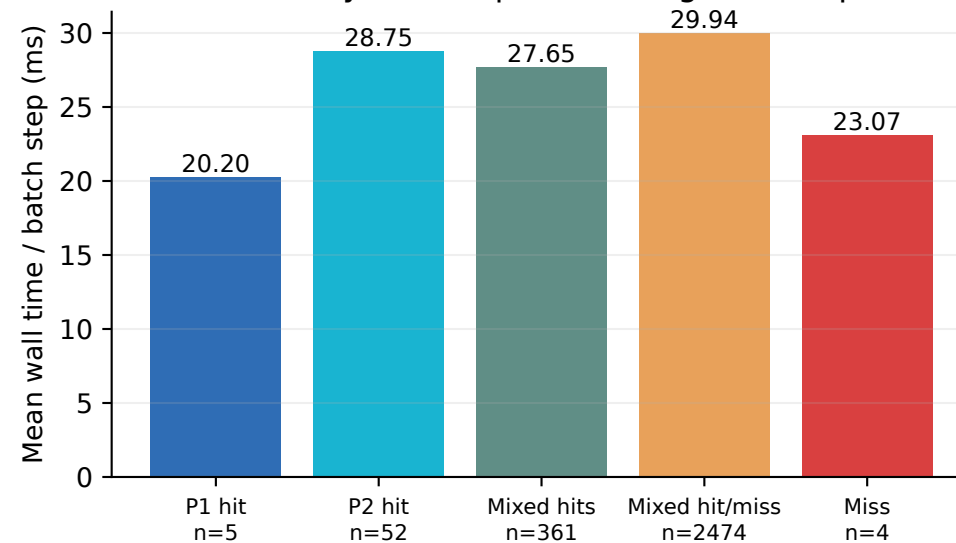
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b8_stream_fast_r0
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.124, hit=76.8%, TPS*=567.70

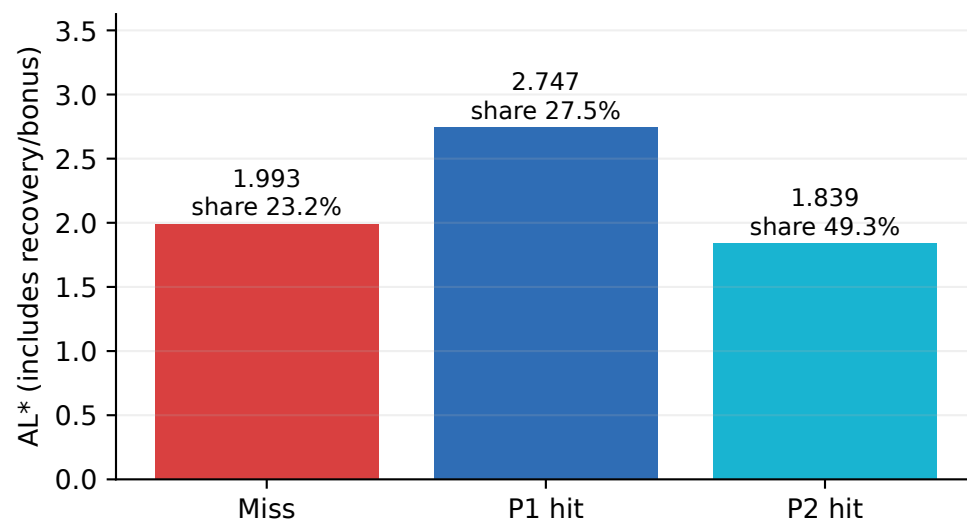
Measured wall time by batch cache status



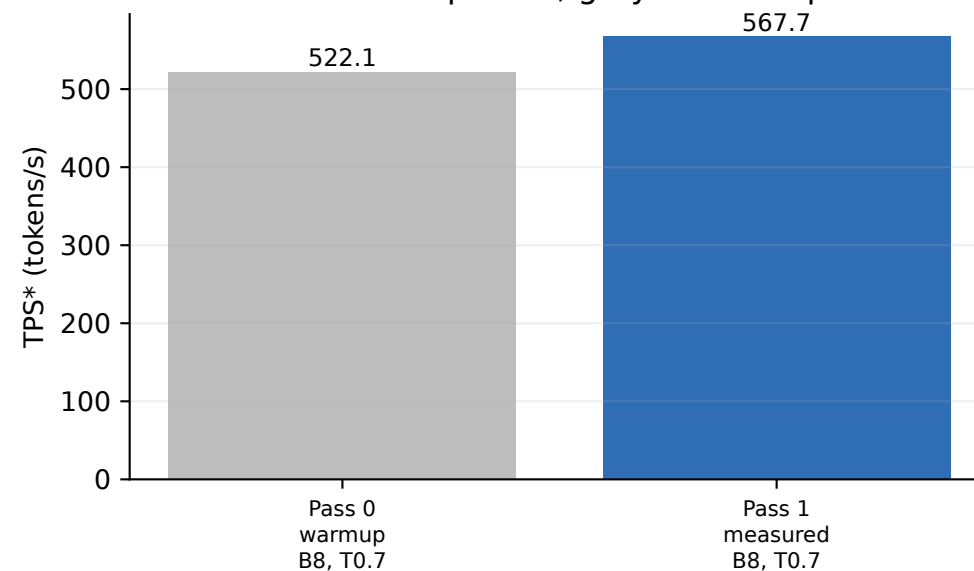
Boundary-free steps, including slow steps



Conditional AL* and source share

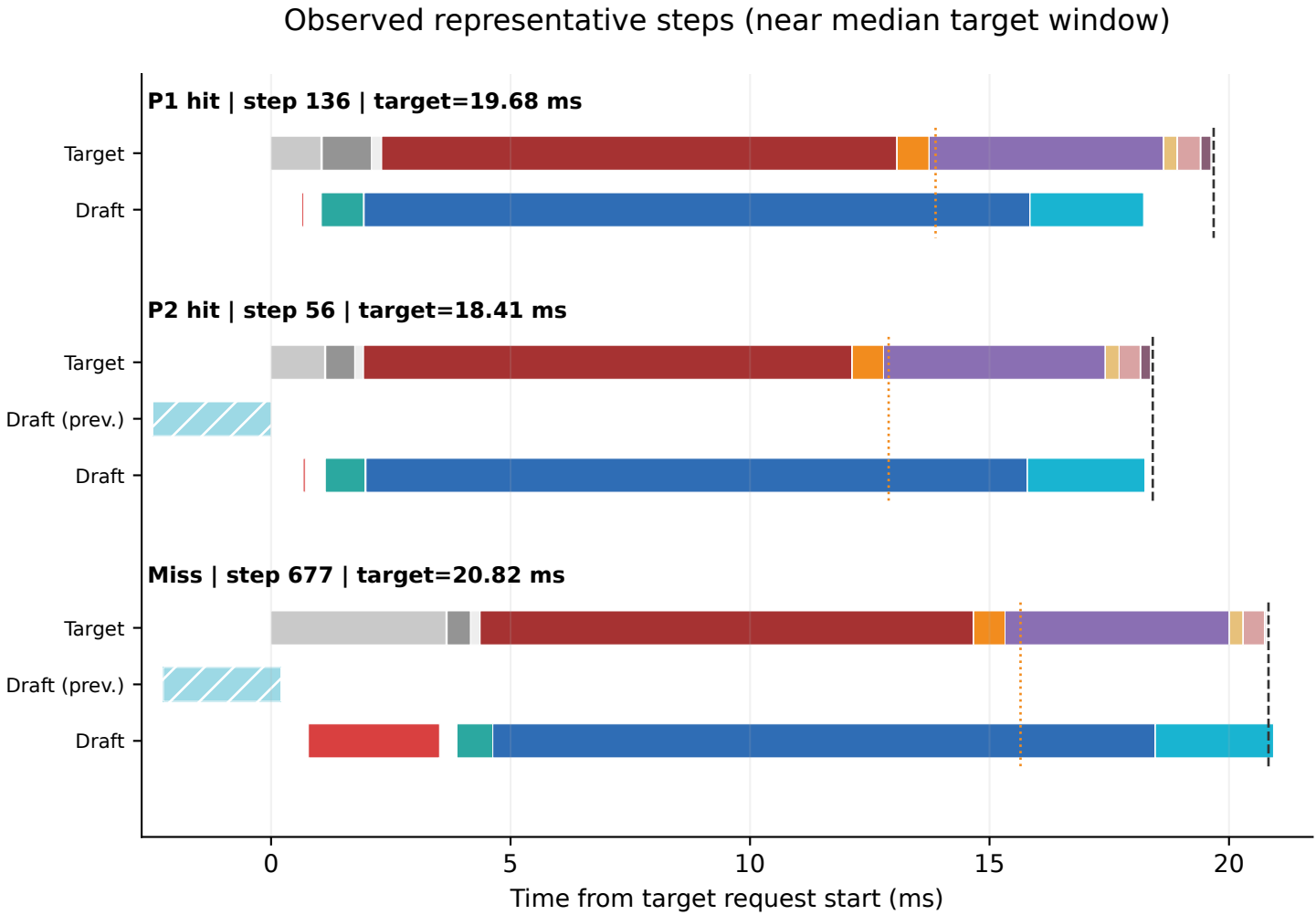
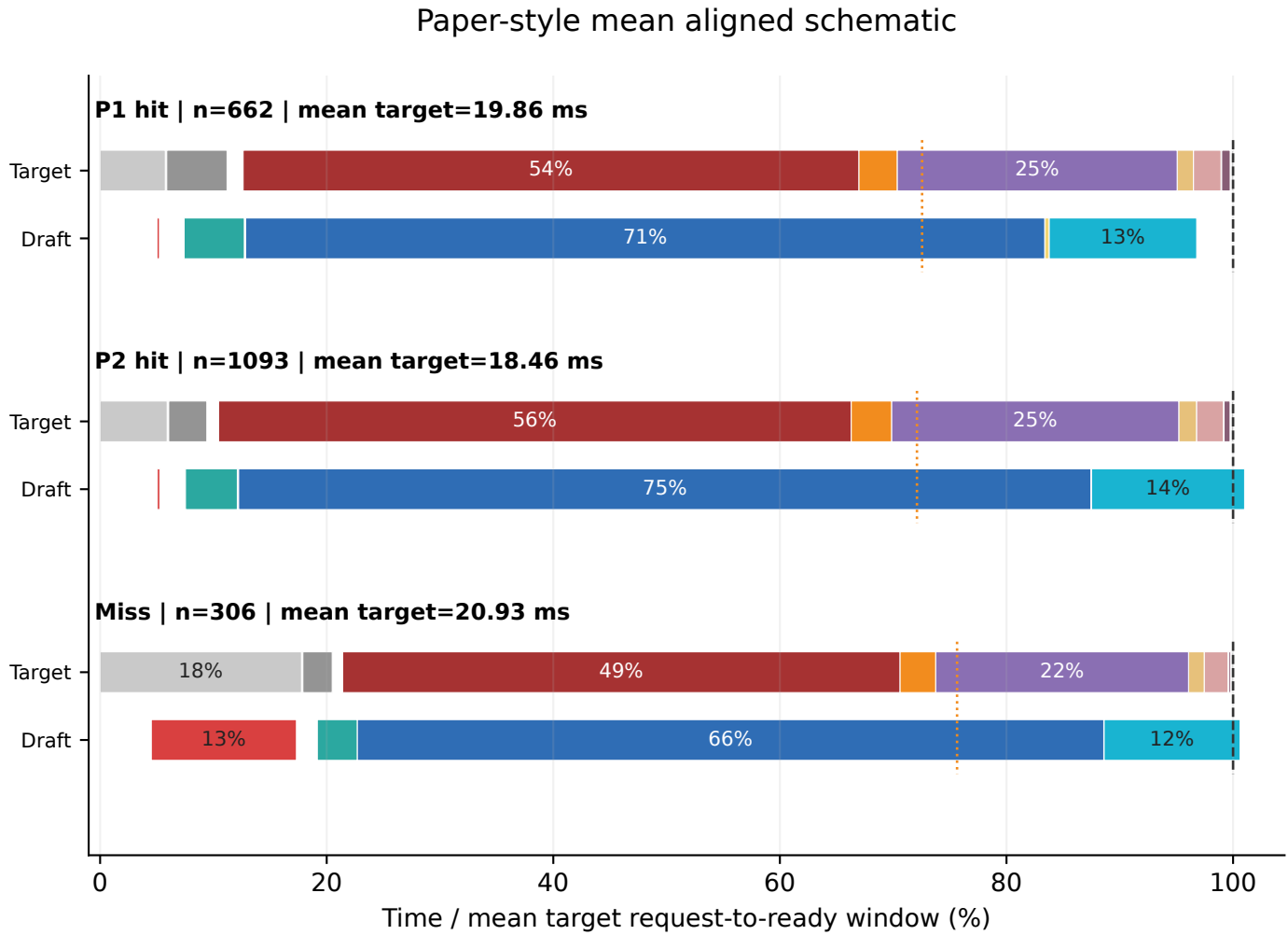
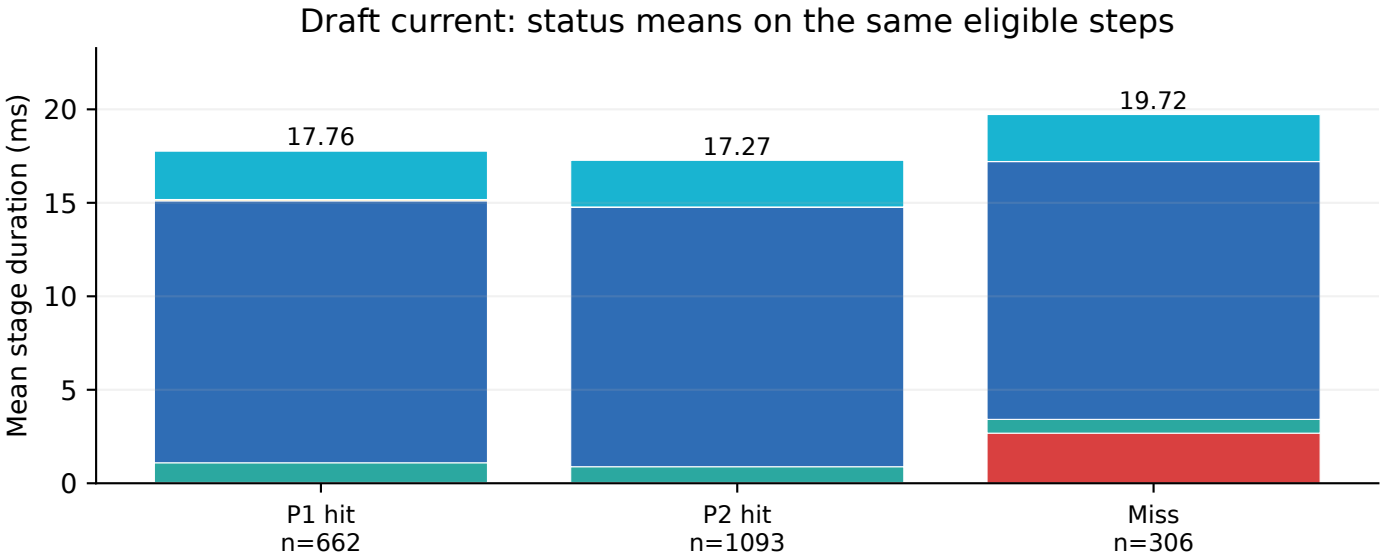
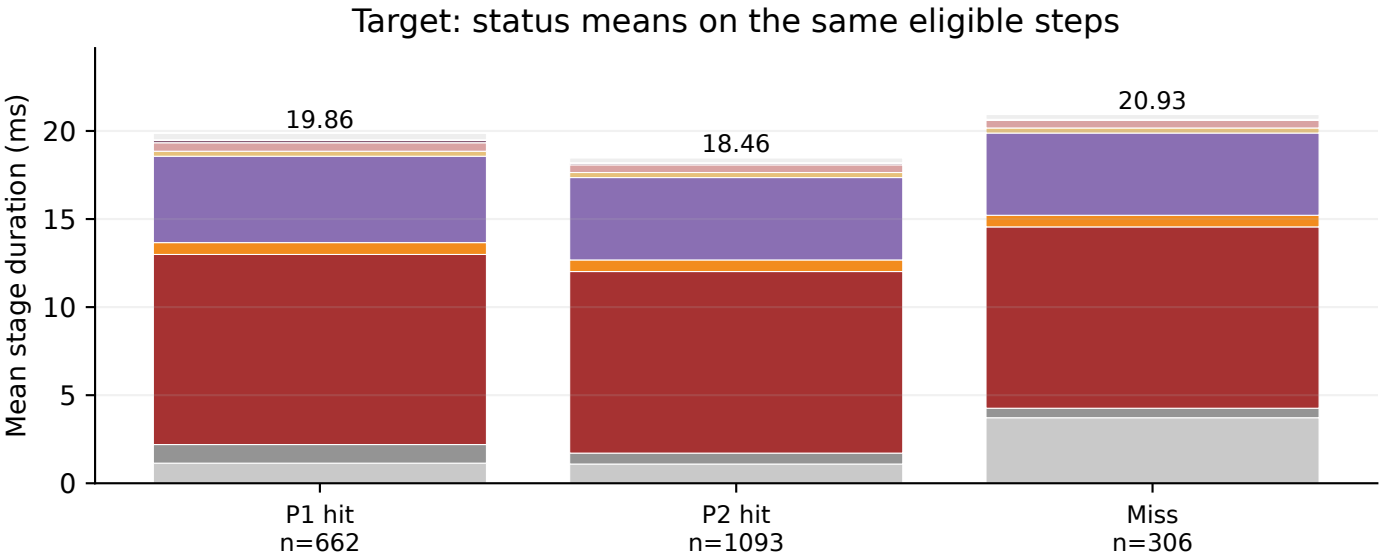


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama2_b1_stream_l1_profile_s0
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/0
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2061 eligible steps



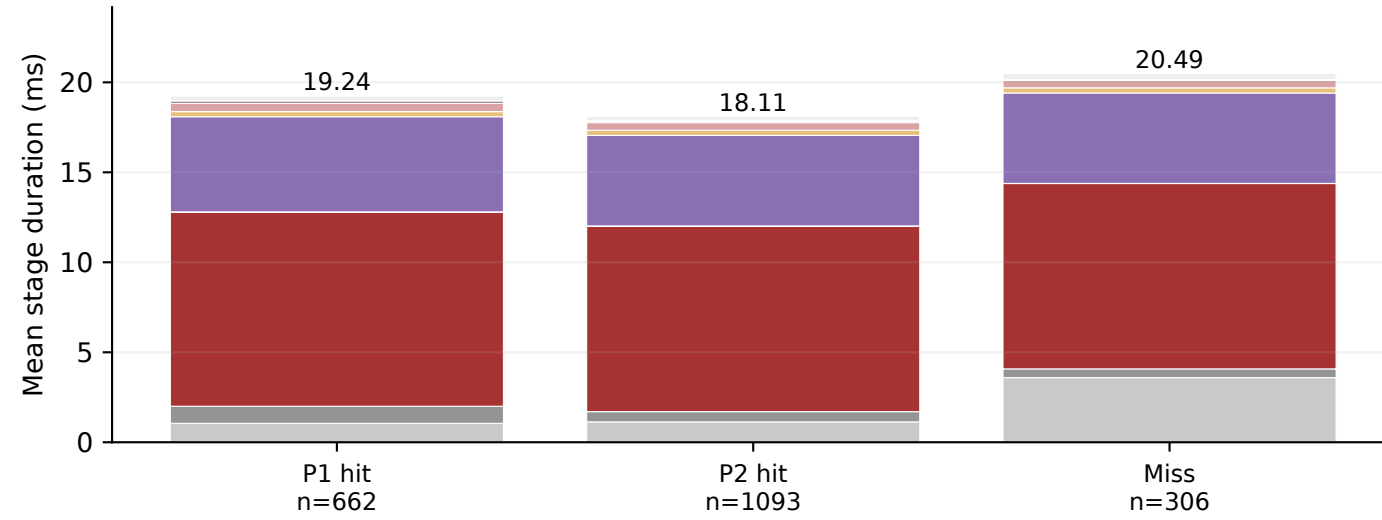
Legend:

- Wait / sync (grey)
- Prepare (dark grey)
- Target pre (red)
- Target post (purple)
- Proxy / launch (orange)
- Final logits (yellow)
- Accept (pink)
- Commit (dark purple)
- Postprocess (dark grey)
- Unlabelled gap (light grey)
- Miss draft (red)
- Previous P2 (light blue)
- P1 (blue)
- P2 (cyan)
- Proxy wait (yellow)
- Context (green)
- Target ready (dashed line)
- Proxy received (dotted line)

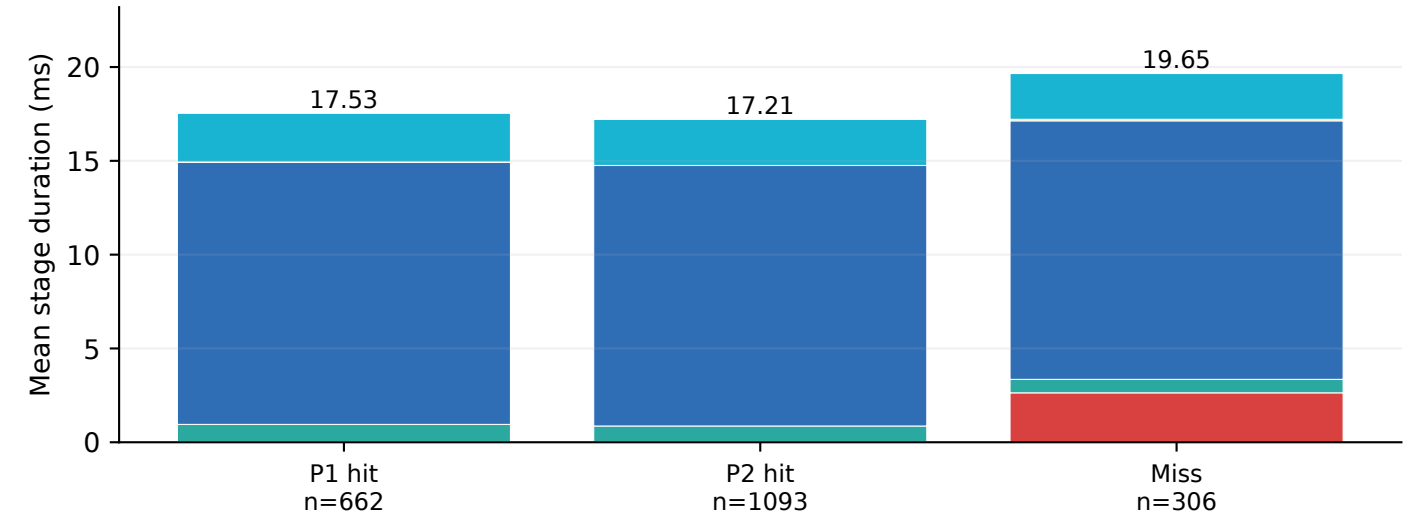
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_stream_l1_profile_s1
 DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=1/1
 B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2061 eligible steps

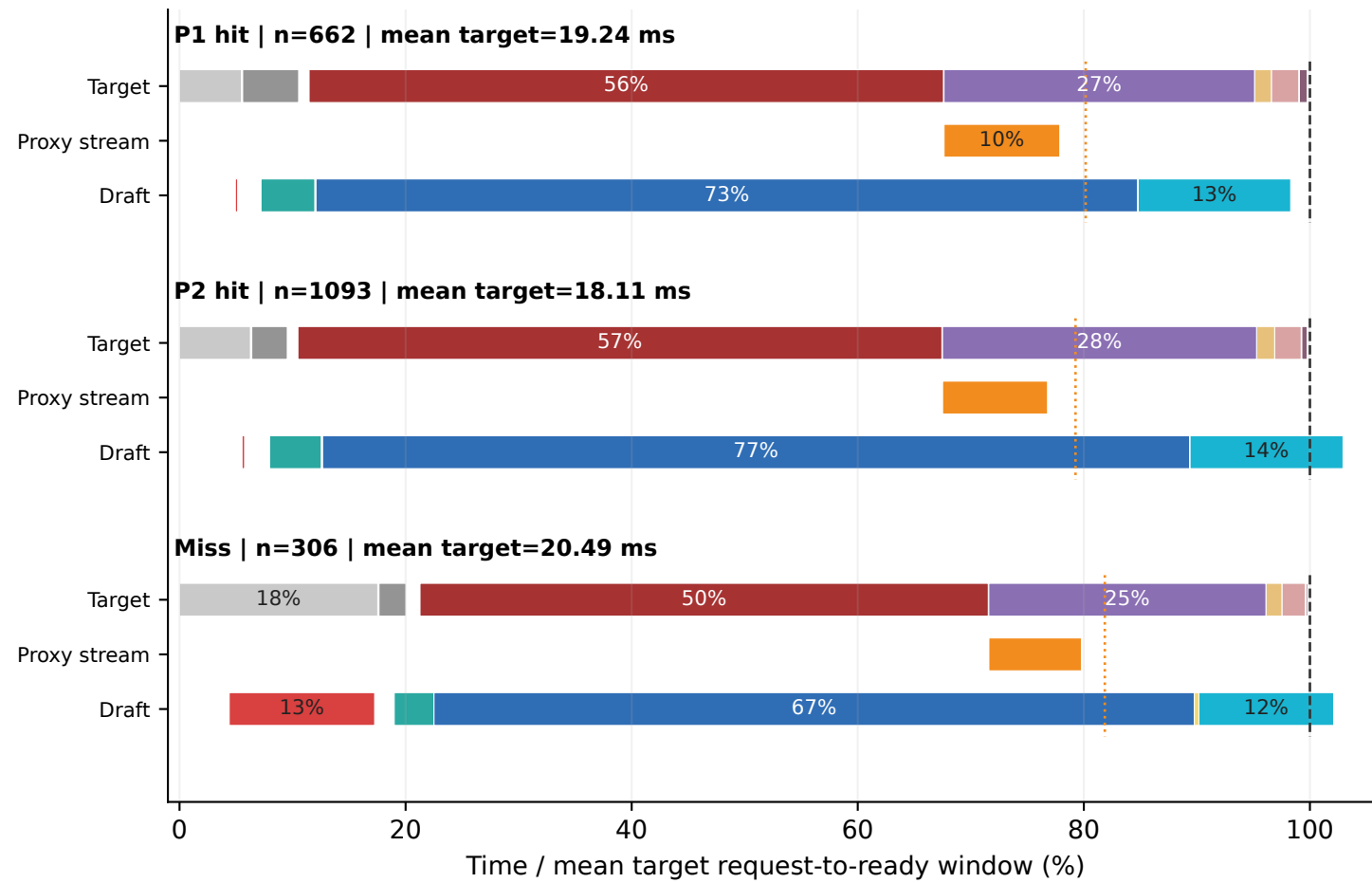
Target: status means on the same eligible steps



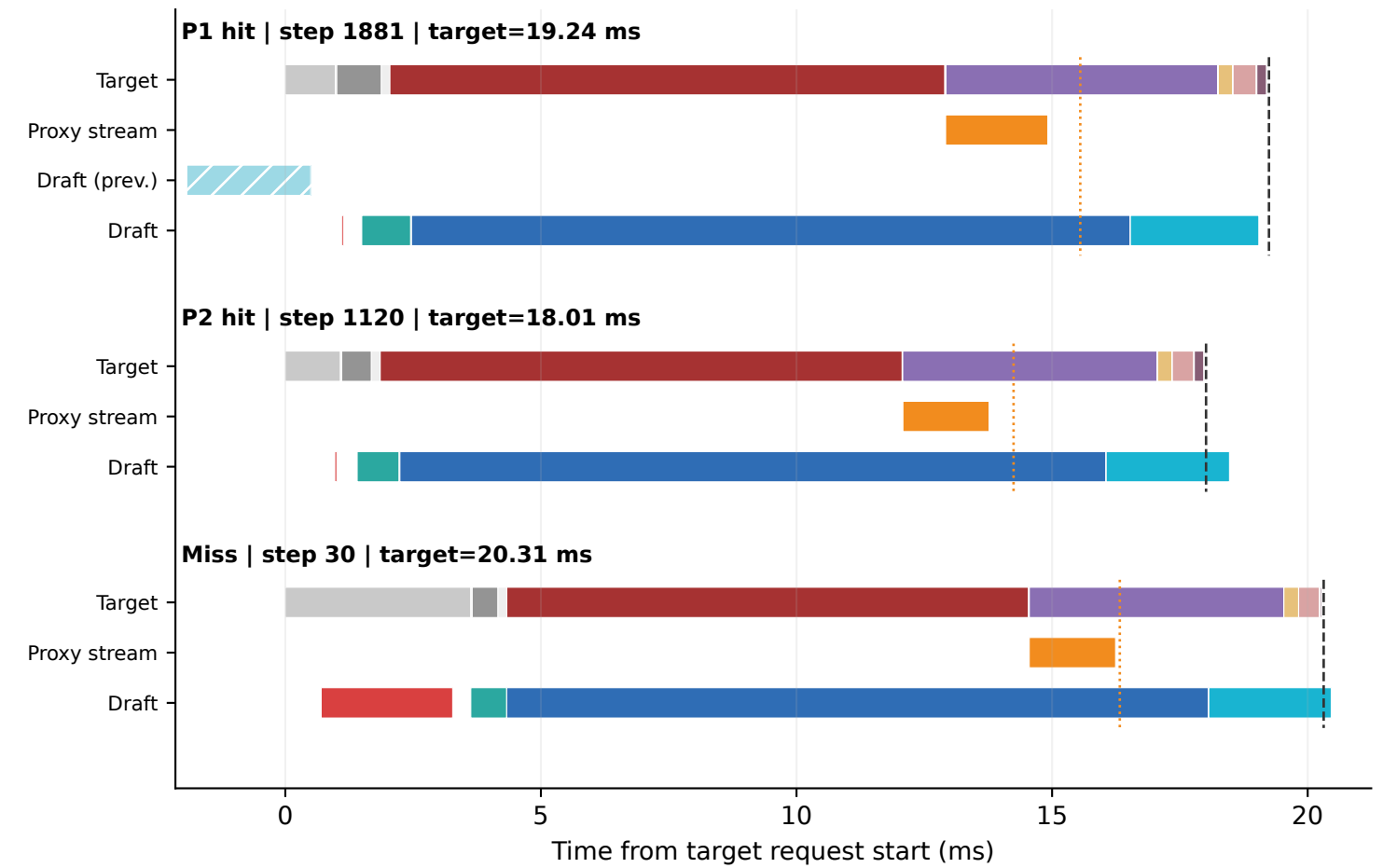
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

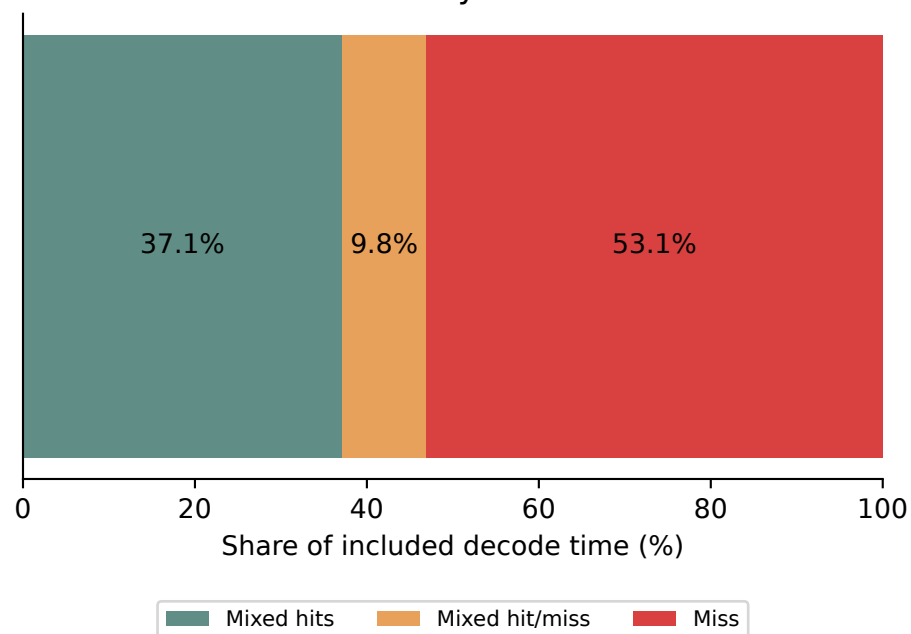


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
 Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
 Target pre Proxy compute Commit Miss draft P2

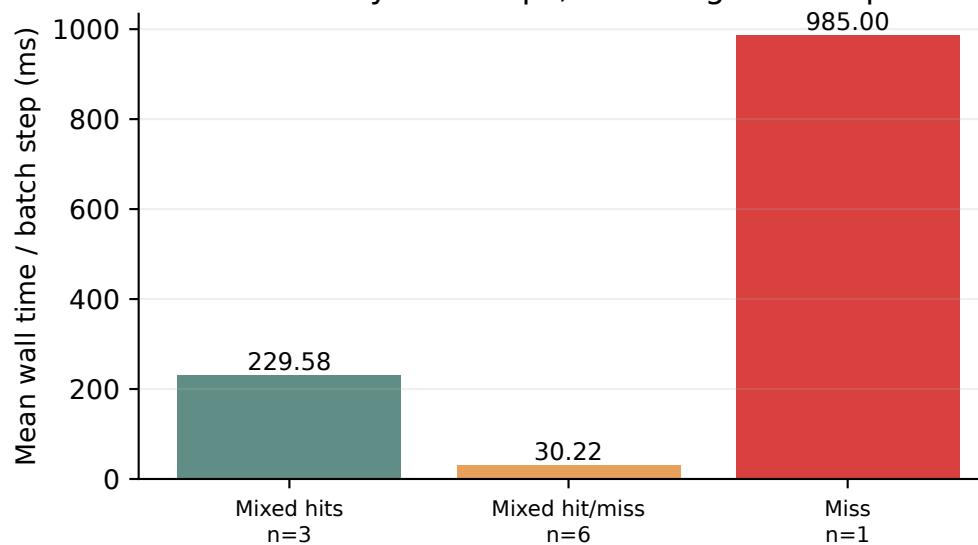
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
 Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_stream_l1_greedy_smoke
 DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
 Last pass: B=8, T=0.0, N=8 | AL*=1.870, hit=75.0%, TPS*=69.00

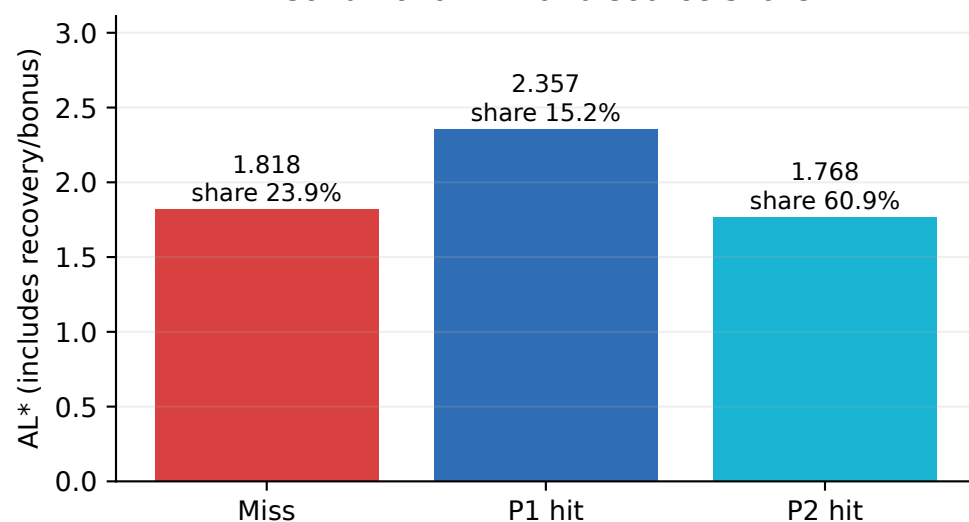
Measured wall time by batch cache status



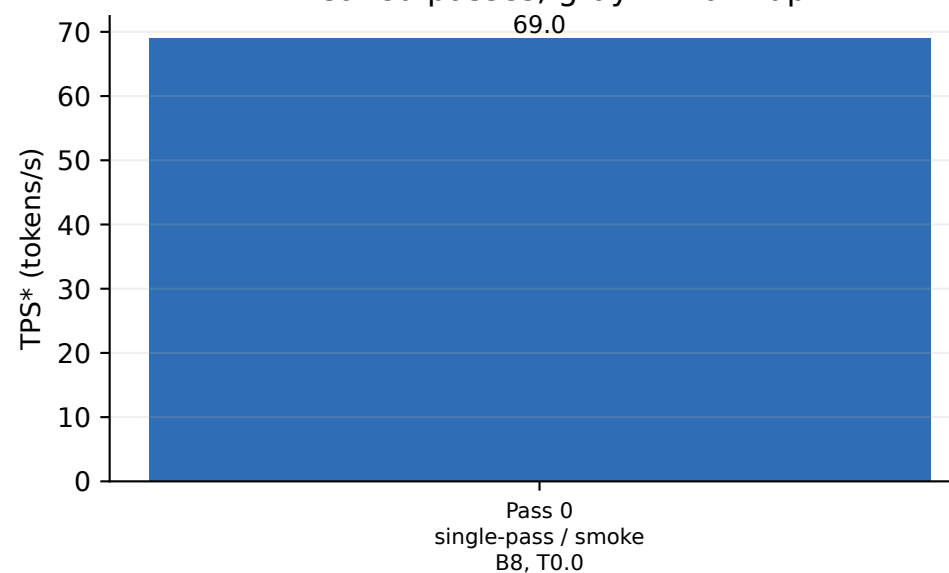
Boundary-free steps, including slow steps



Conditional AL* and source share

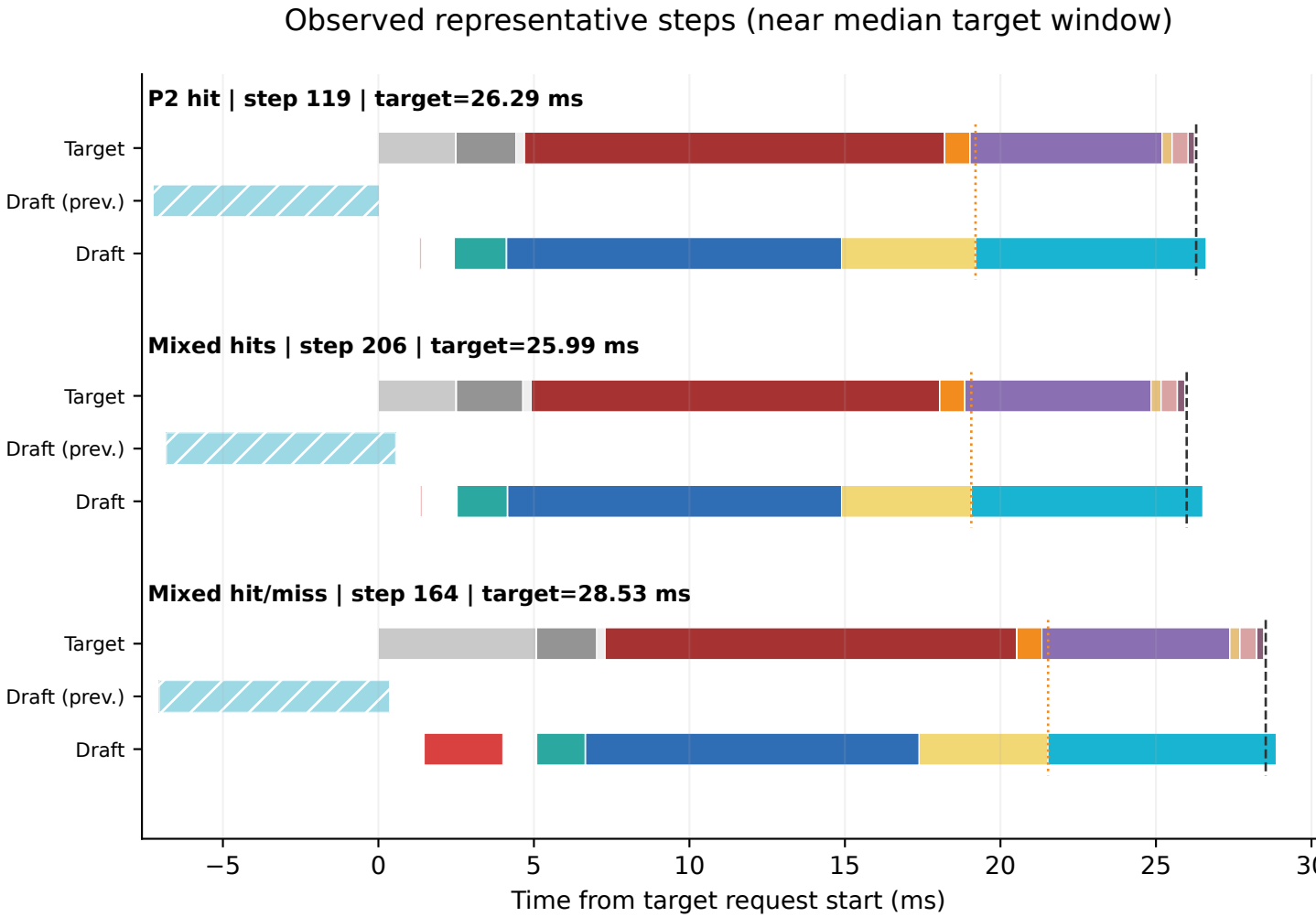
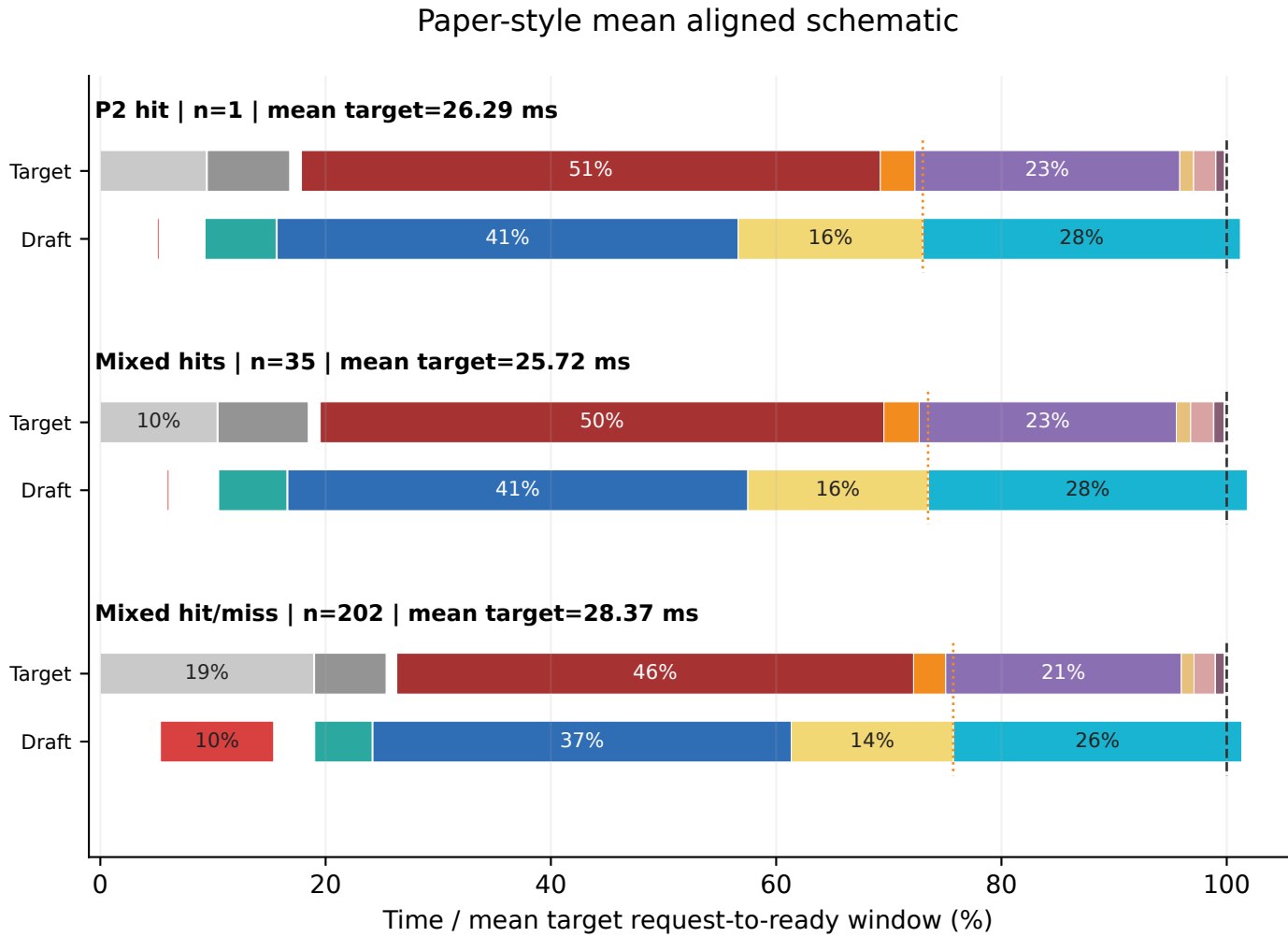
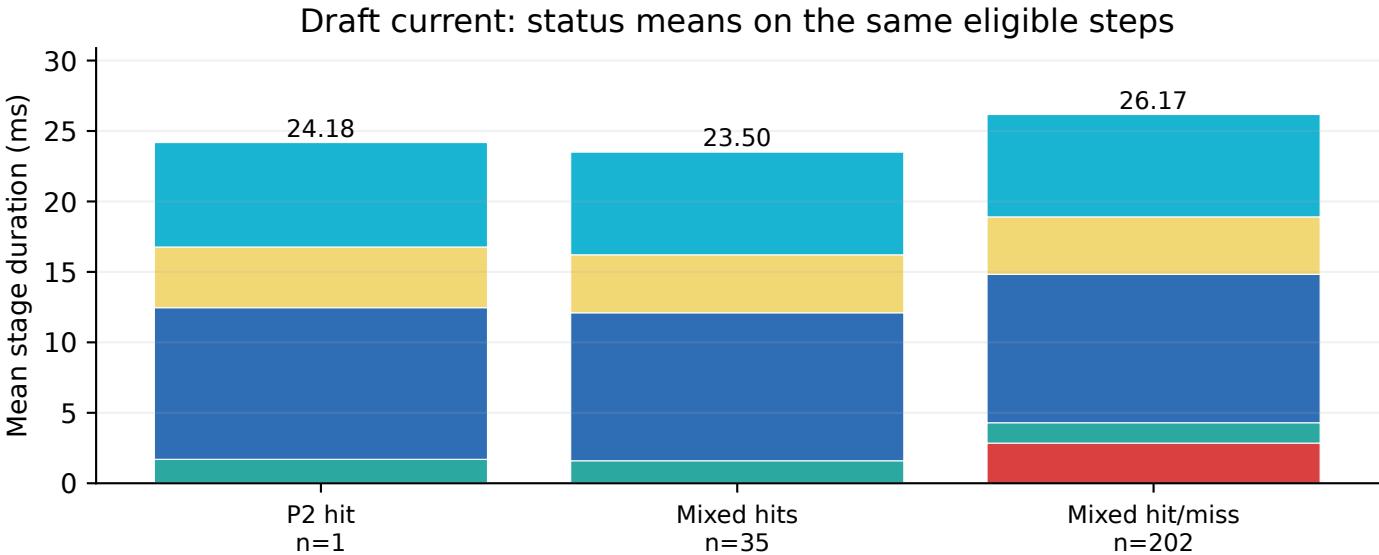
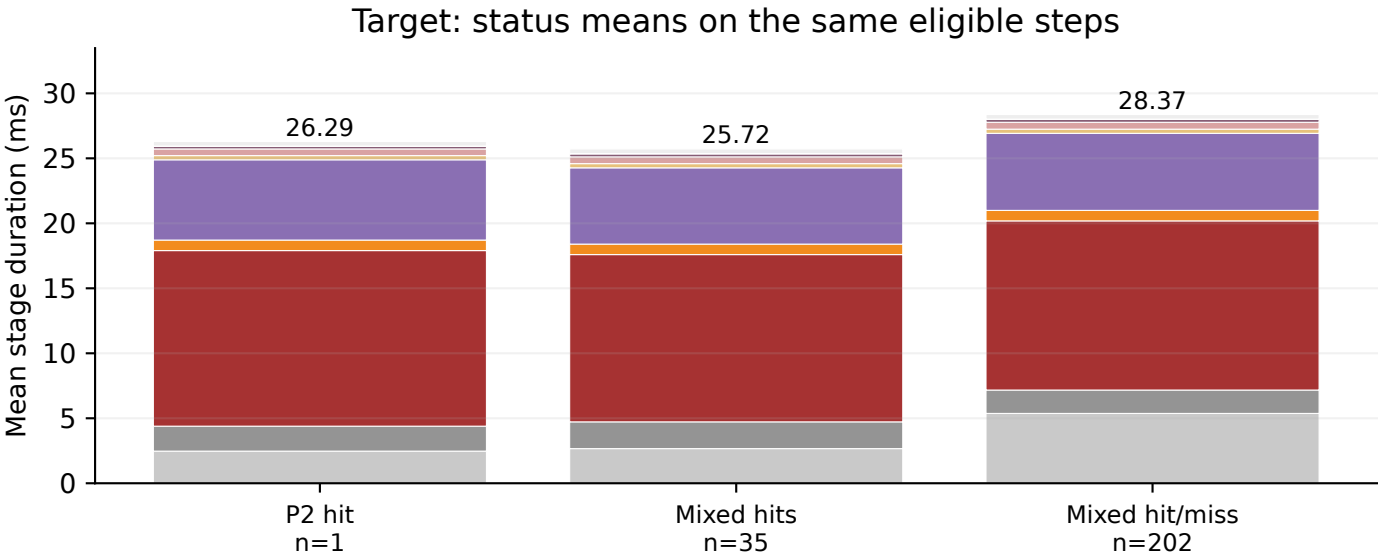


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

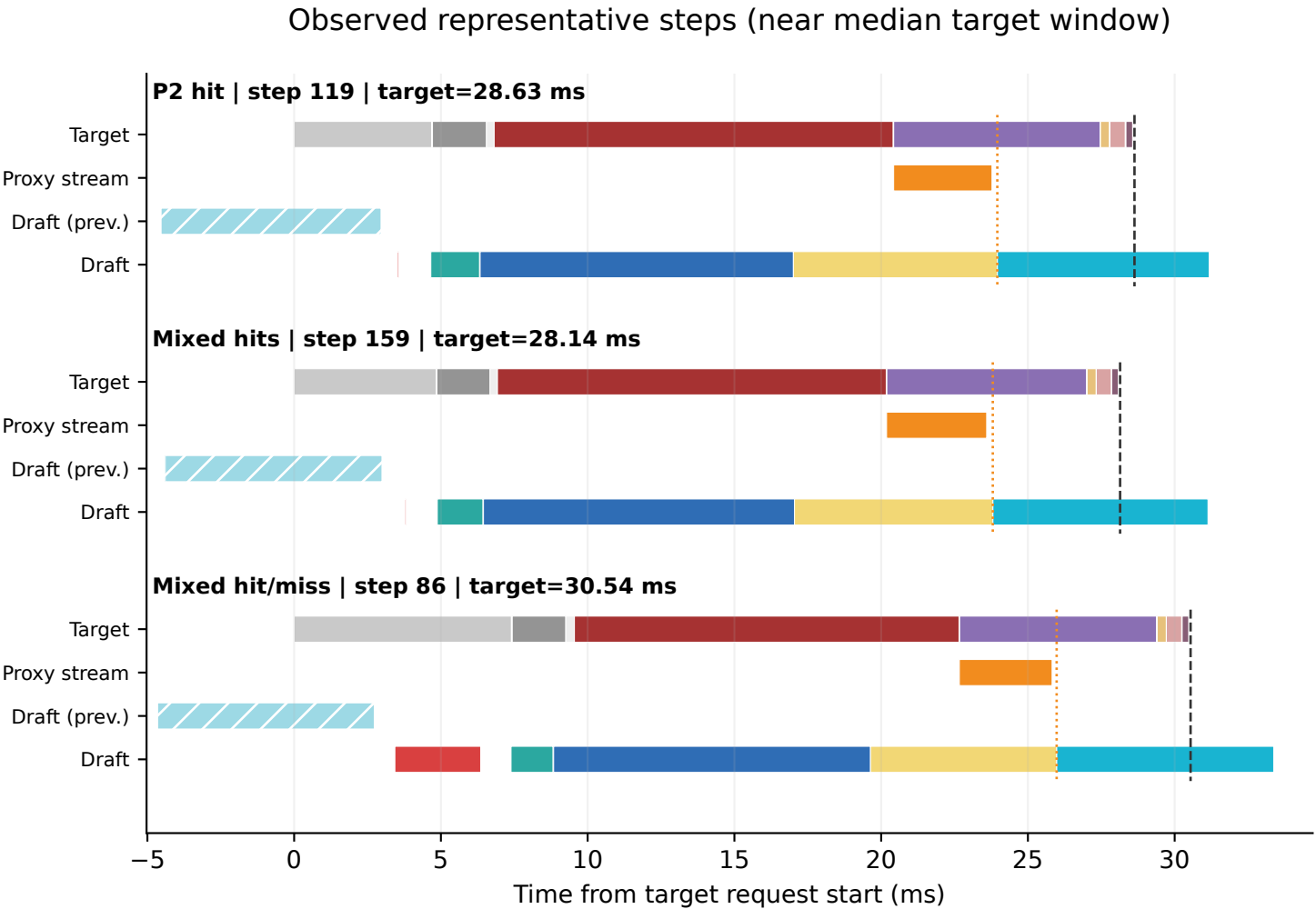
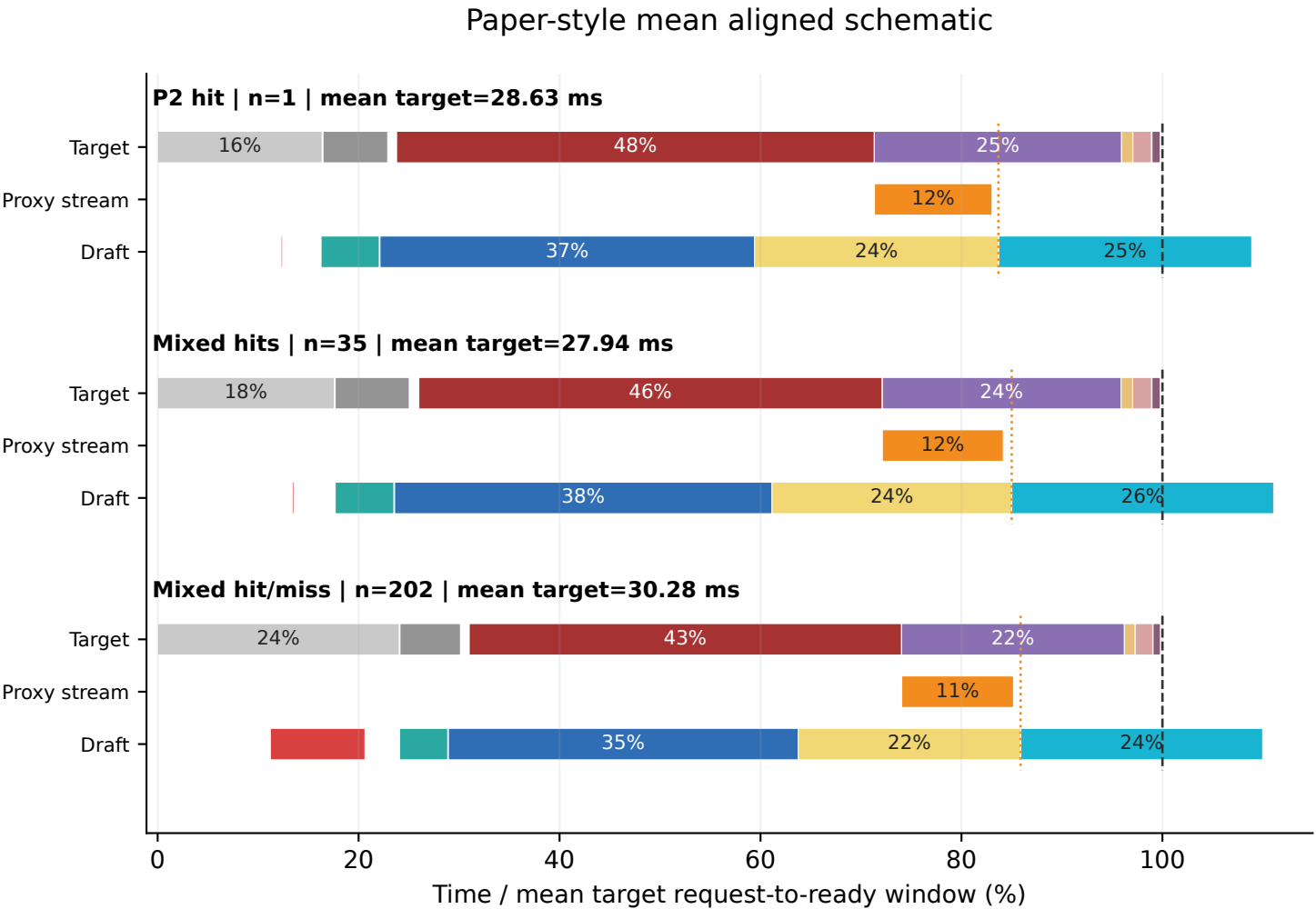
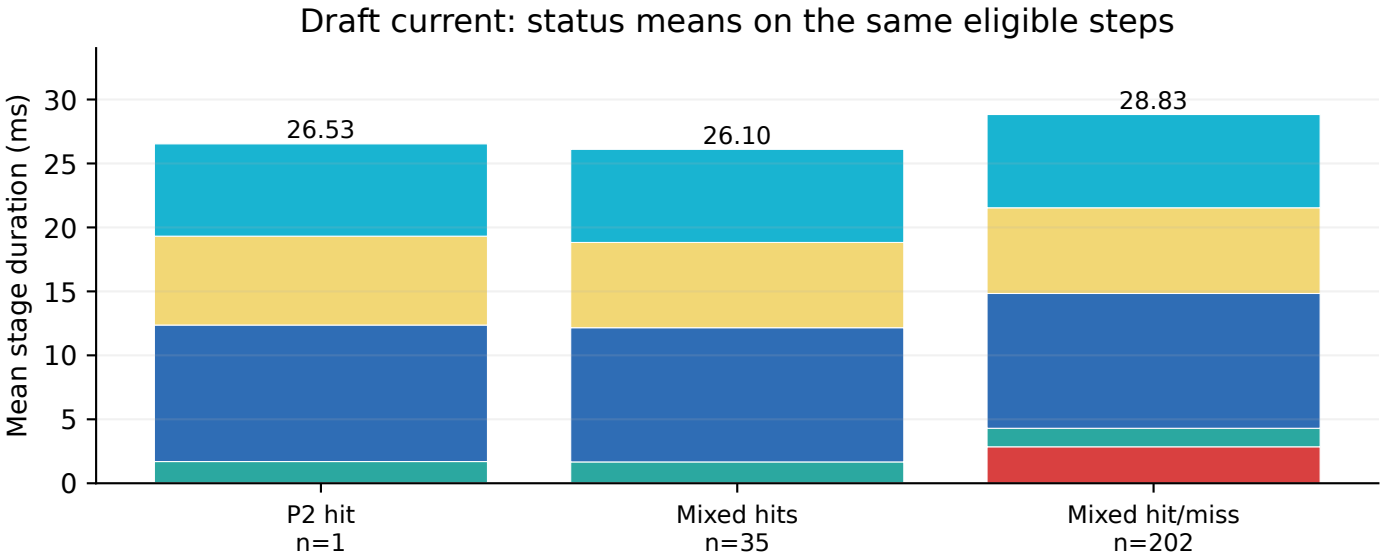
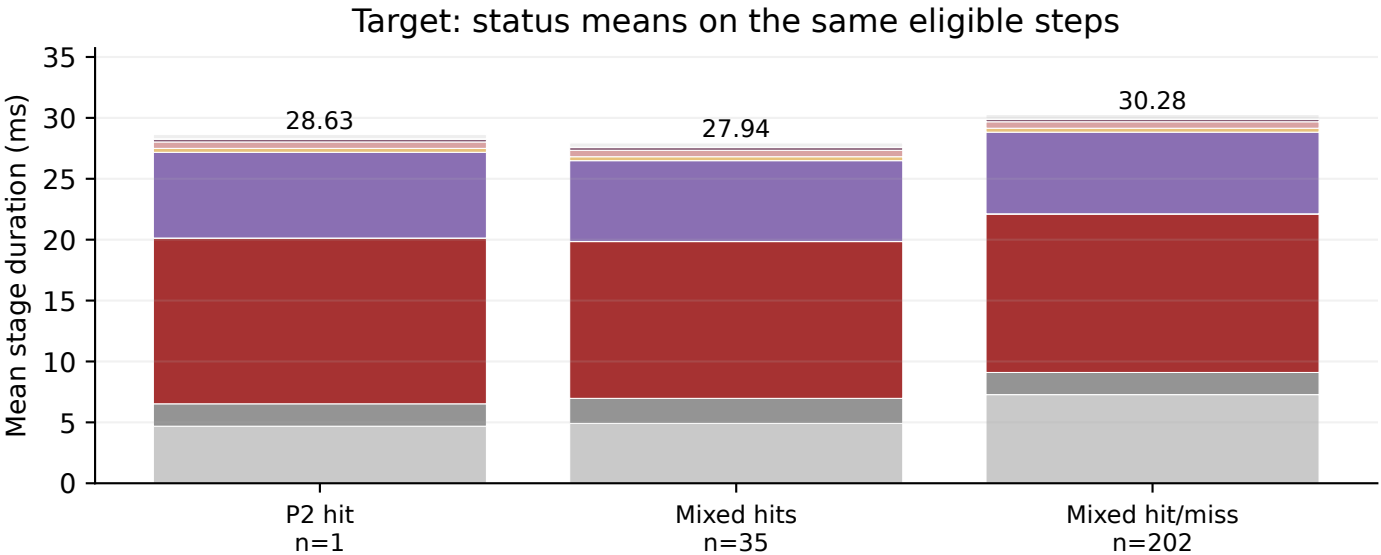
llama2_b8_stream_l1_profile_s0
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 238 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

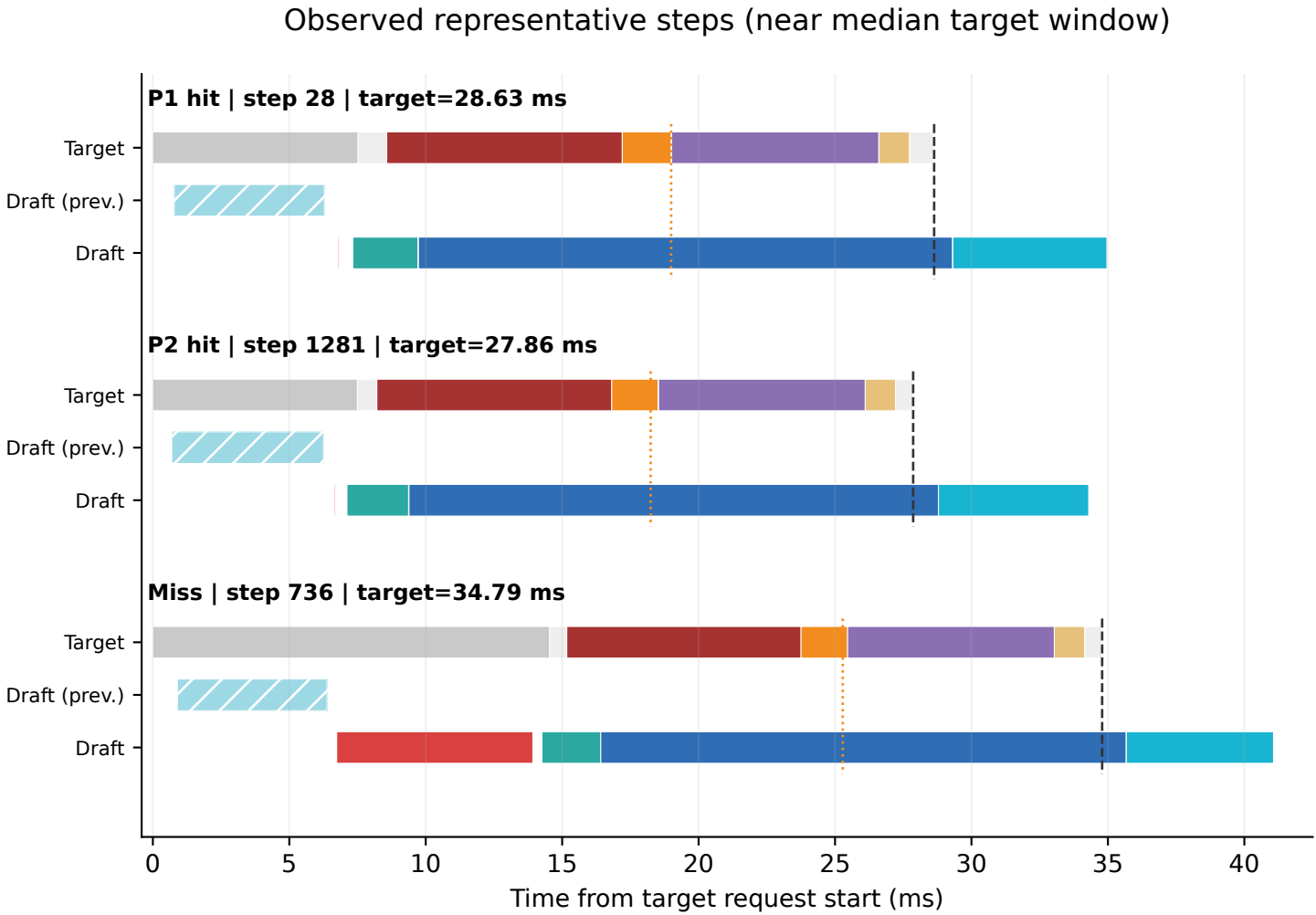
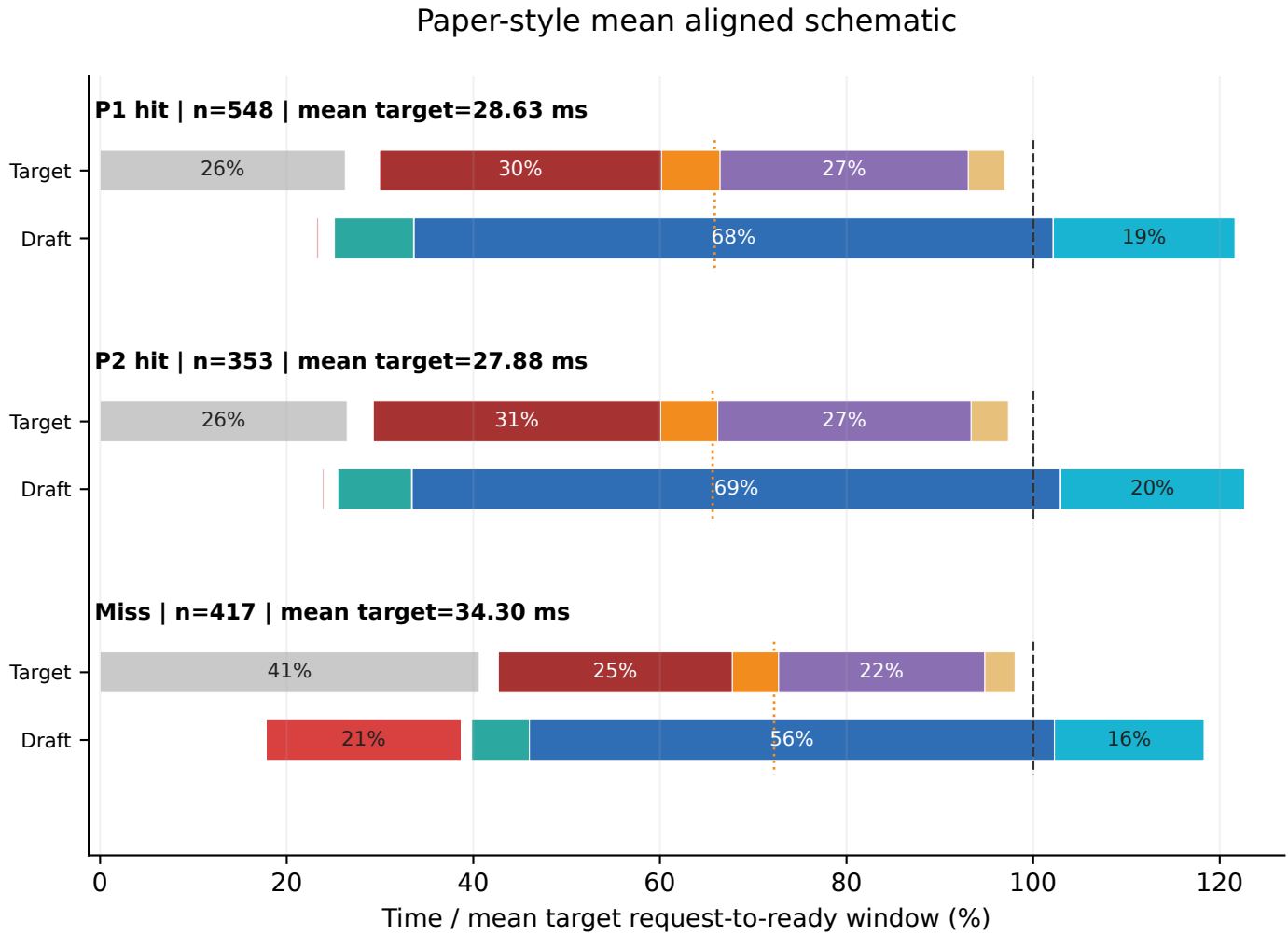
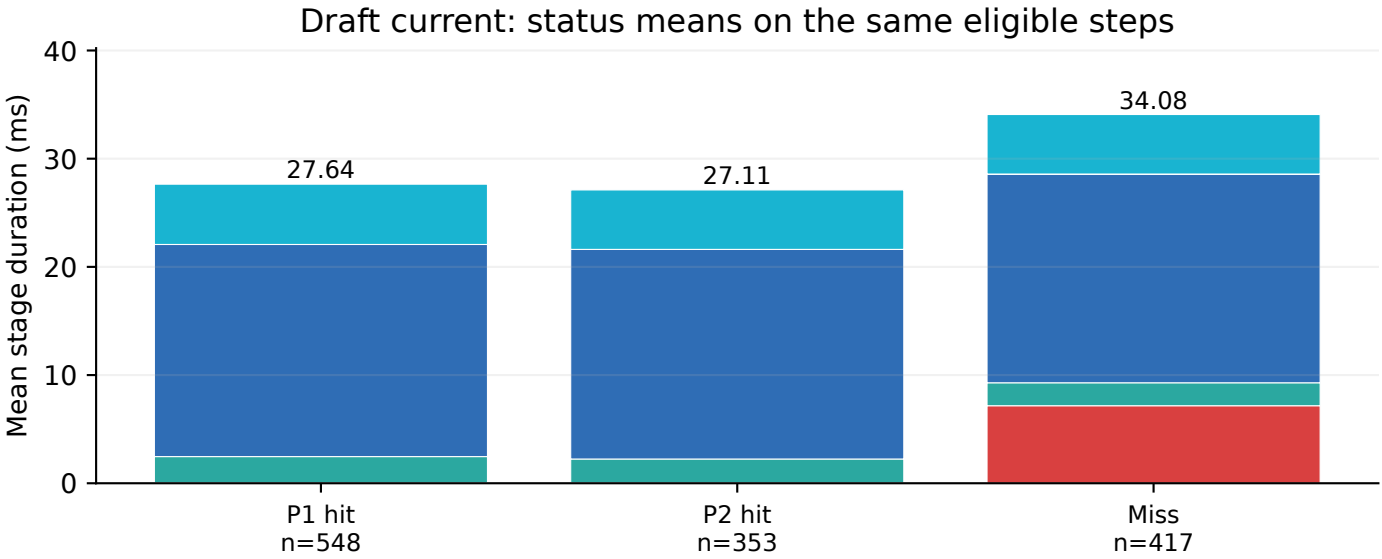
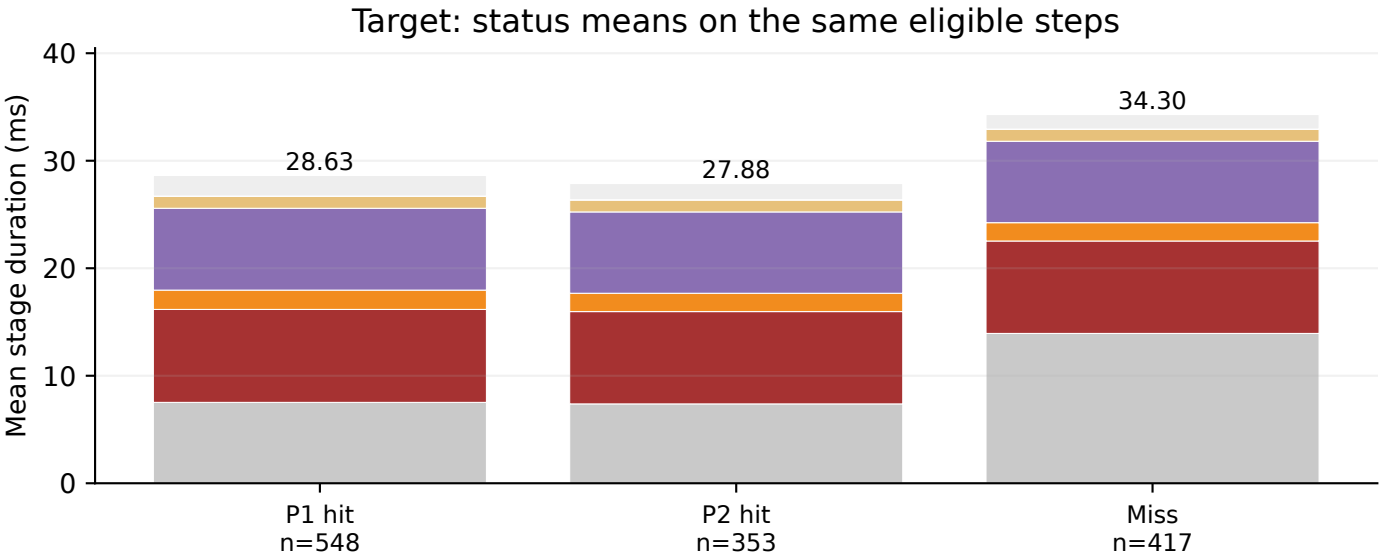
llama2_b8_stream_l1_profile_s1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 238 eligible steps



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

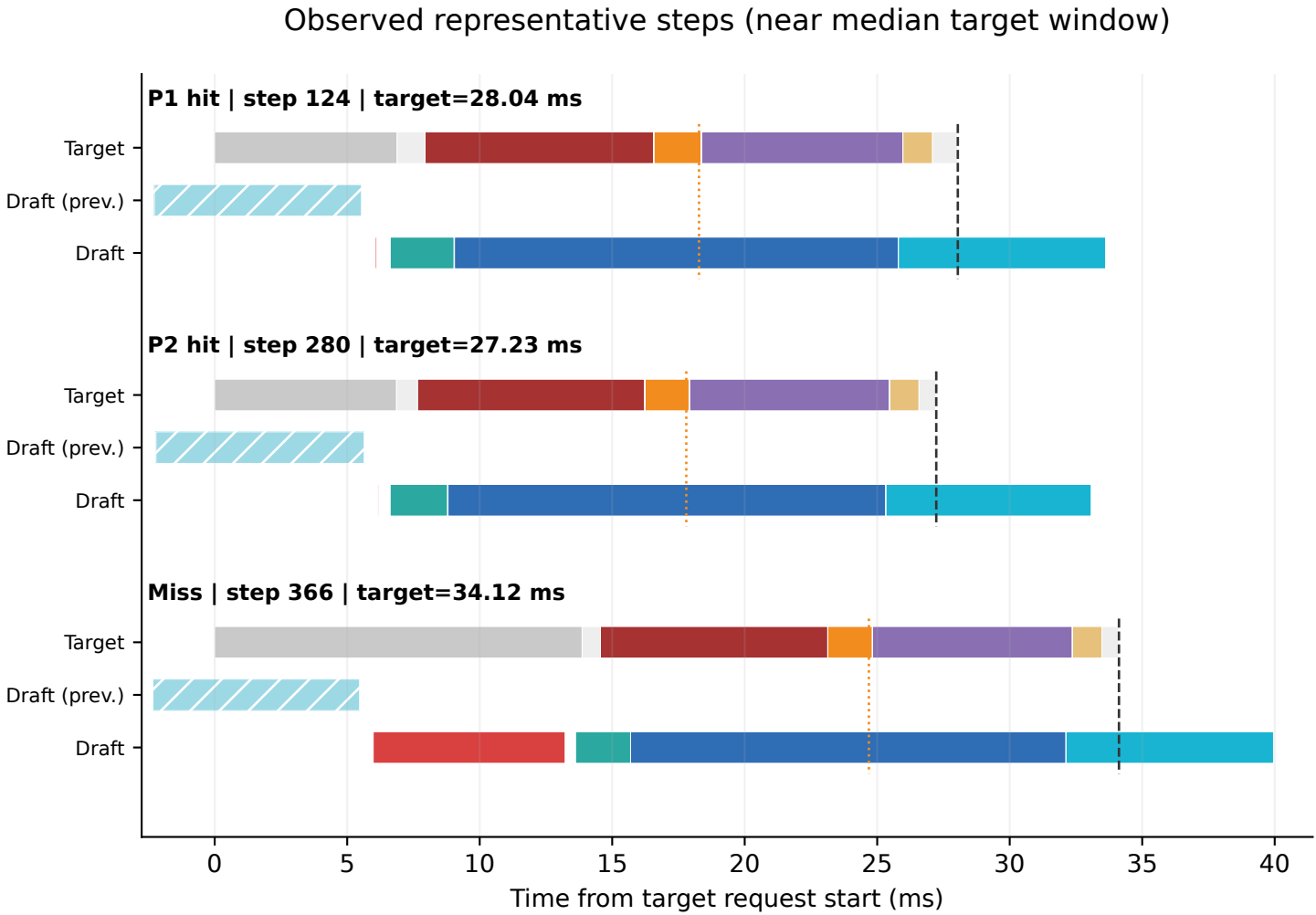
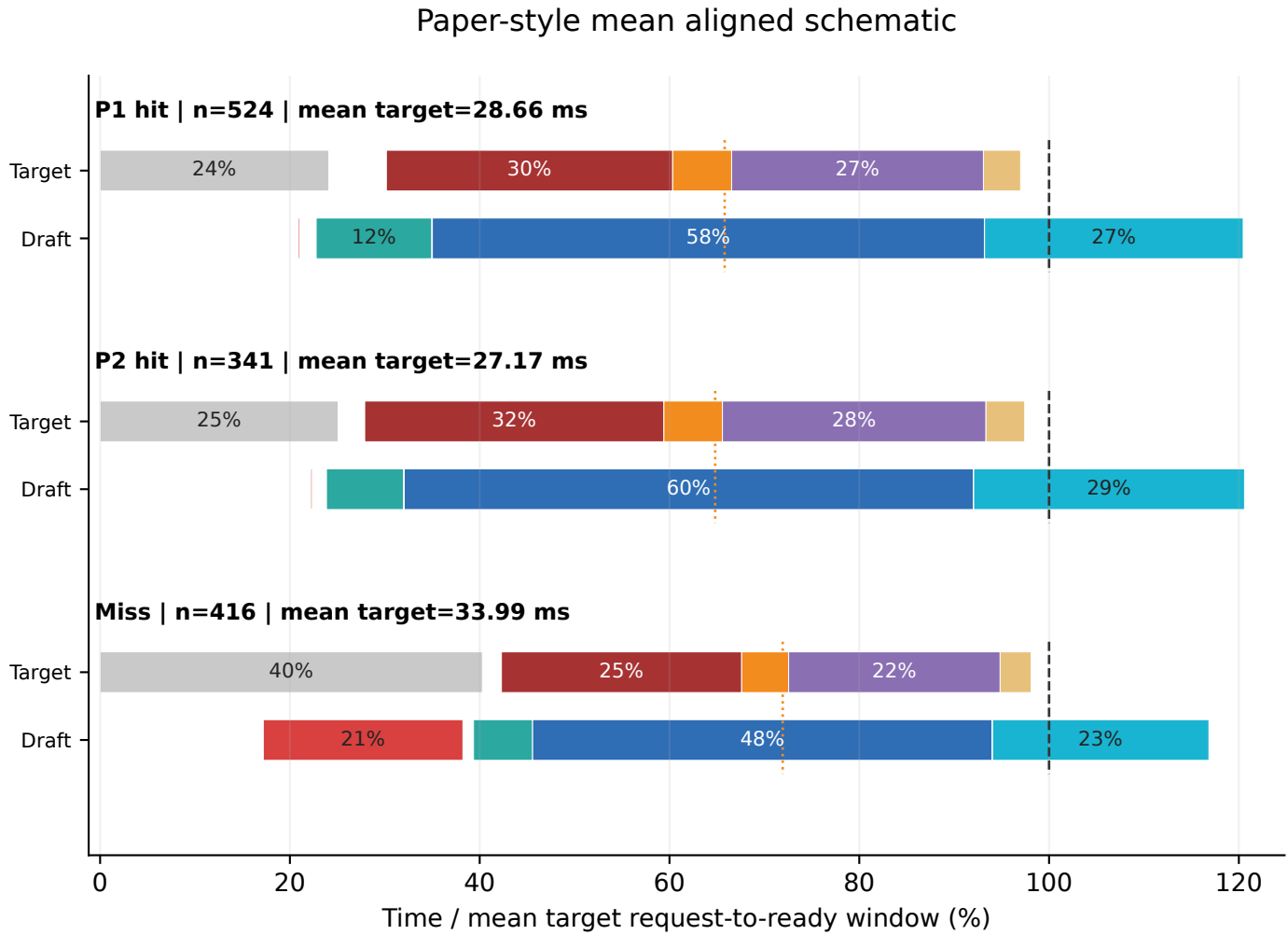
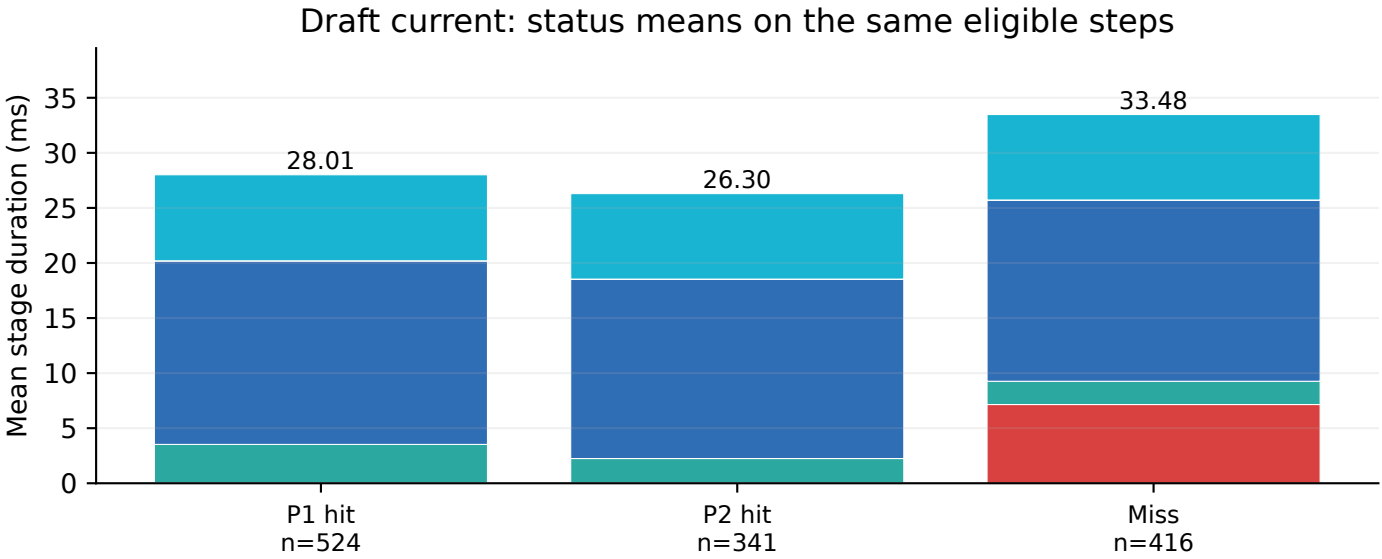
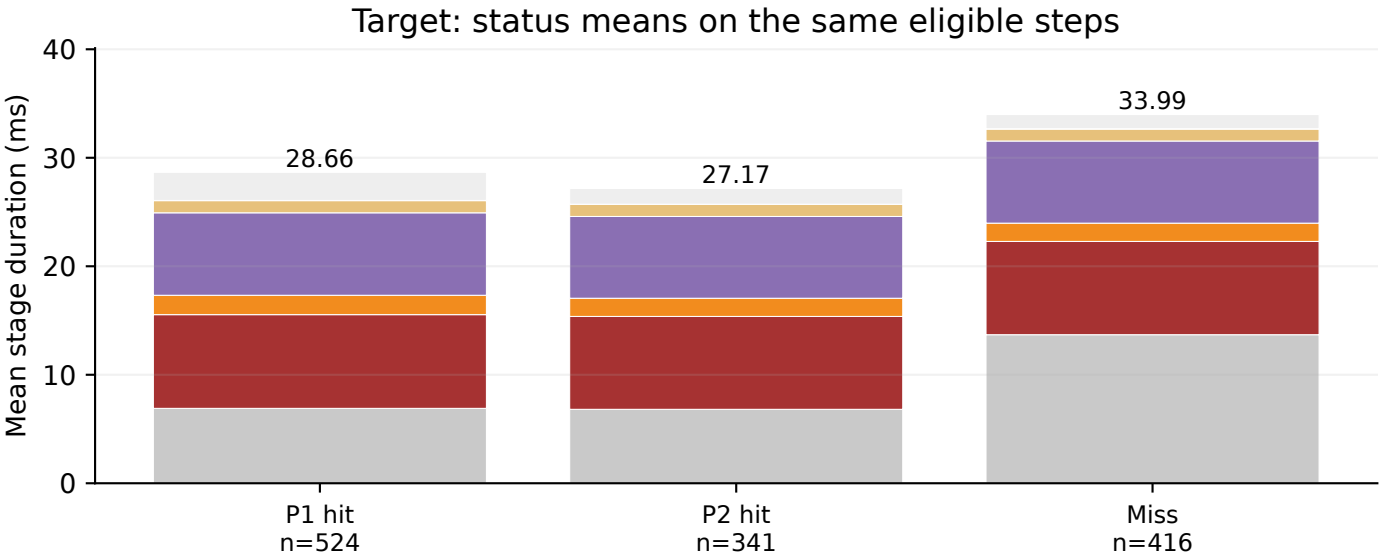
llama3_b1_combo_al_k1_7
DUET: K1/K2=7/2, exit=16 (0-based), C=3, N1/N2=12/4, U=2, W=8, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1318 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

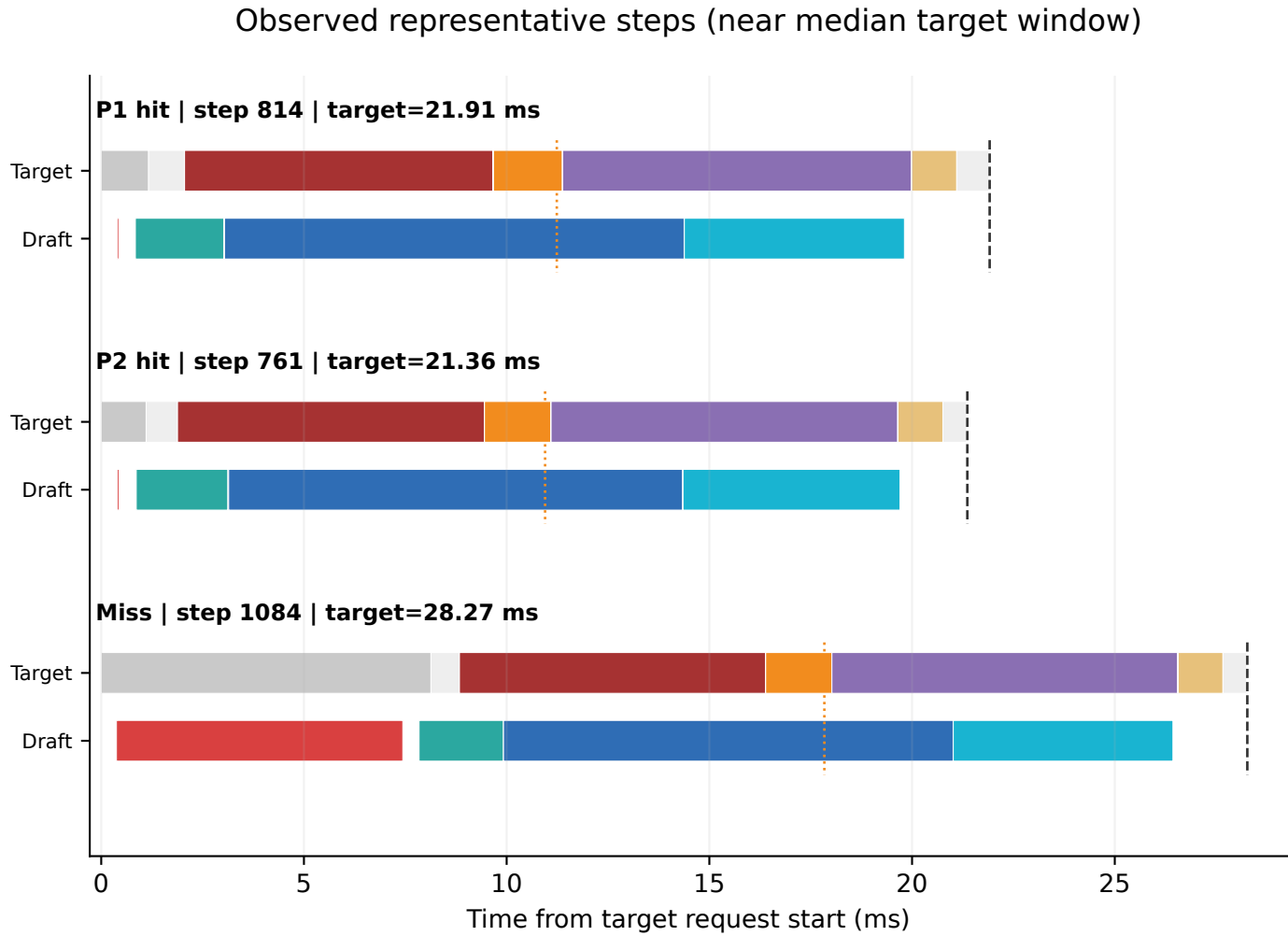
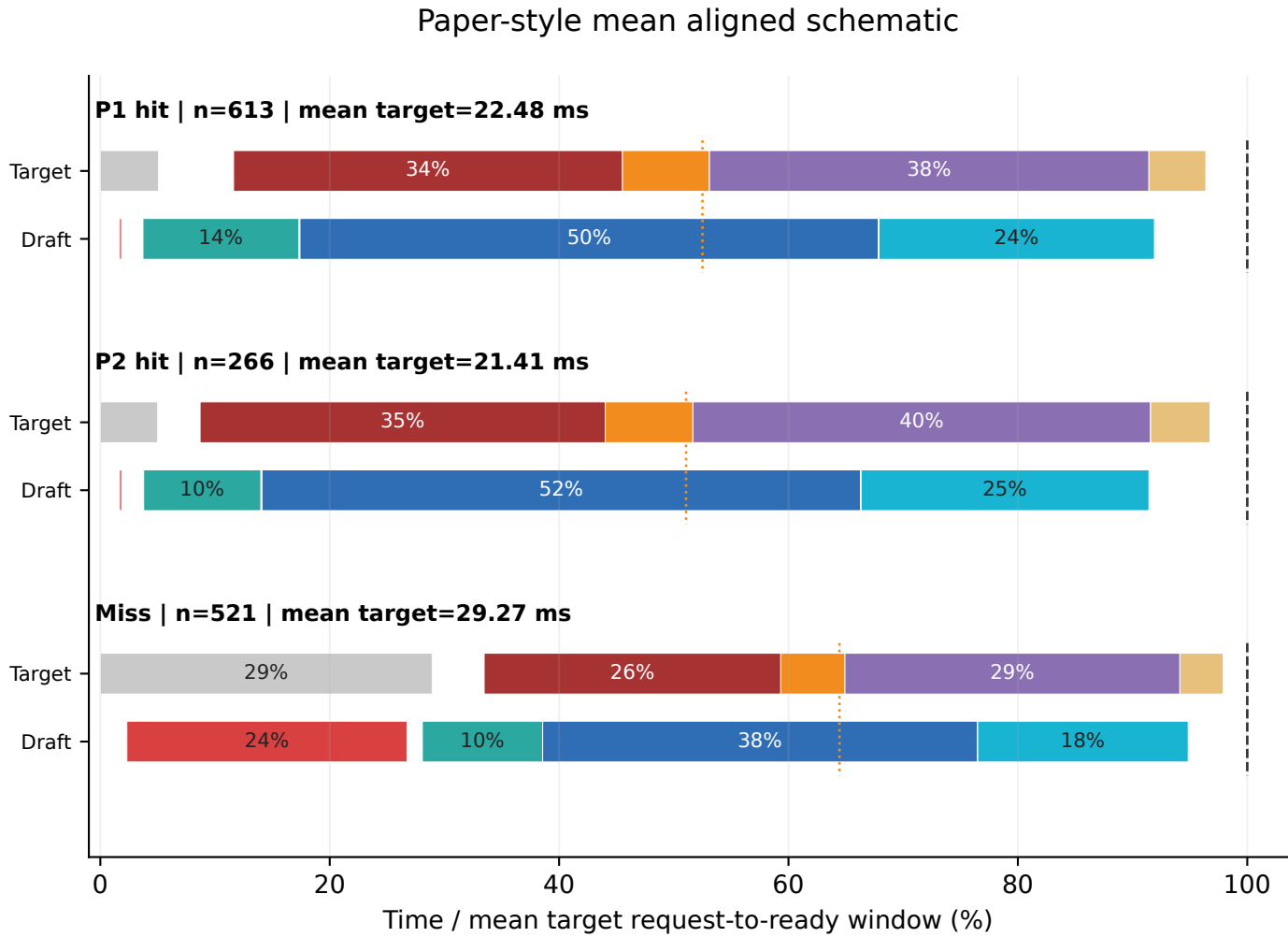
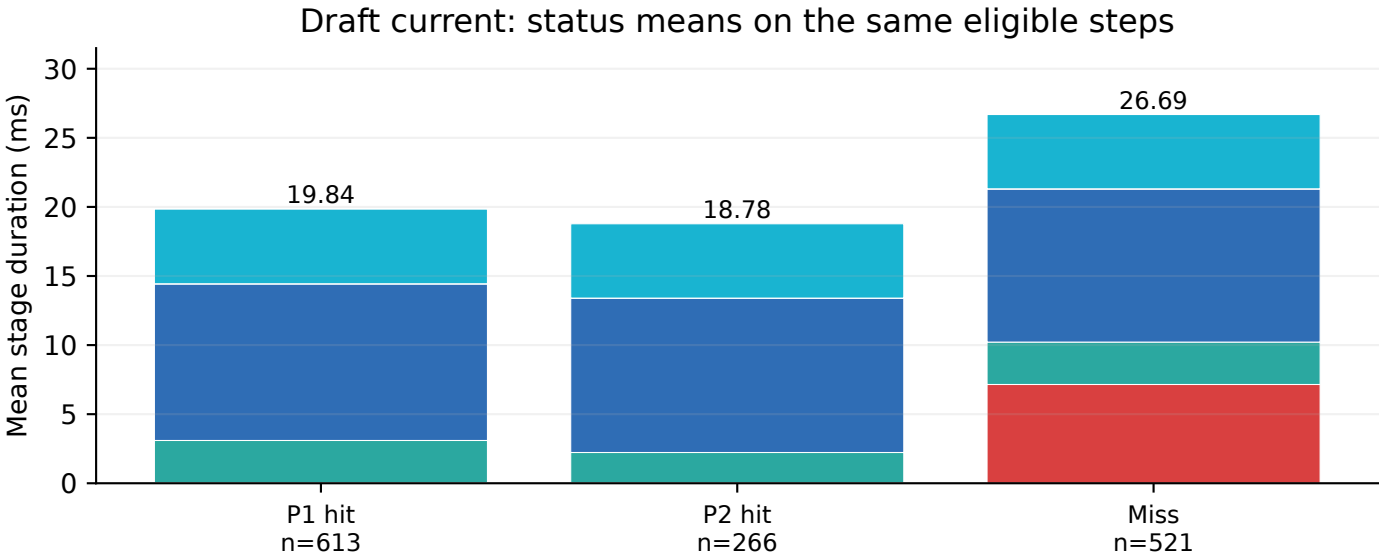
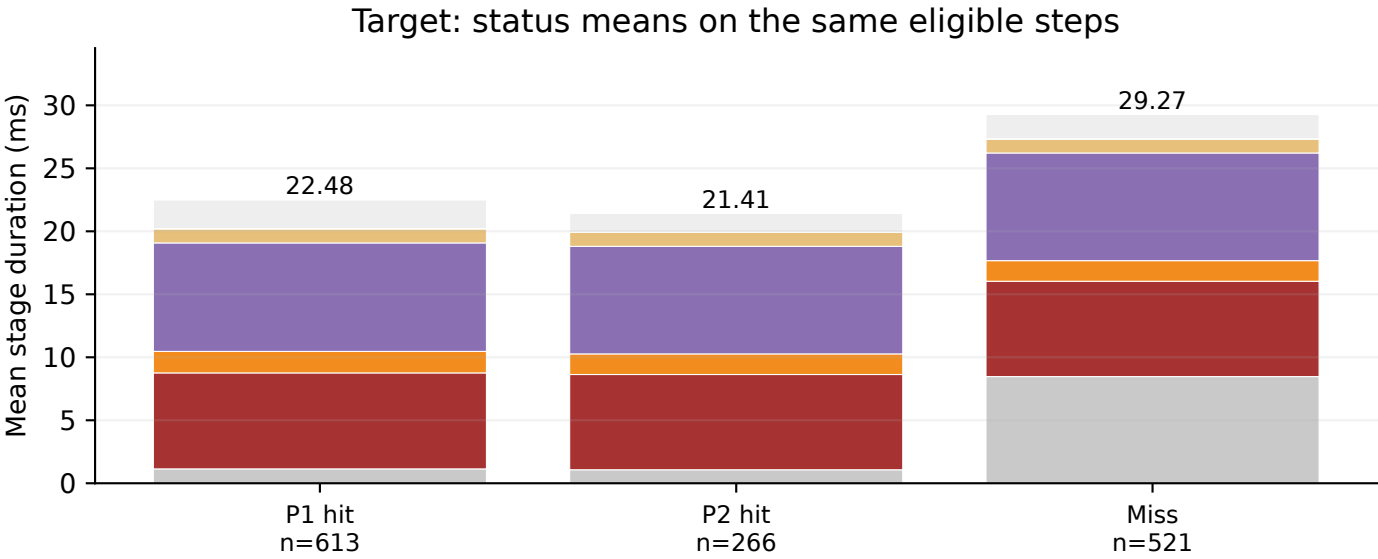
llama3_b1_combo_al_k2_3
DUET: K1/K2=6/3, exit=16 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1281 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

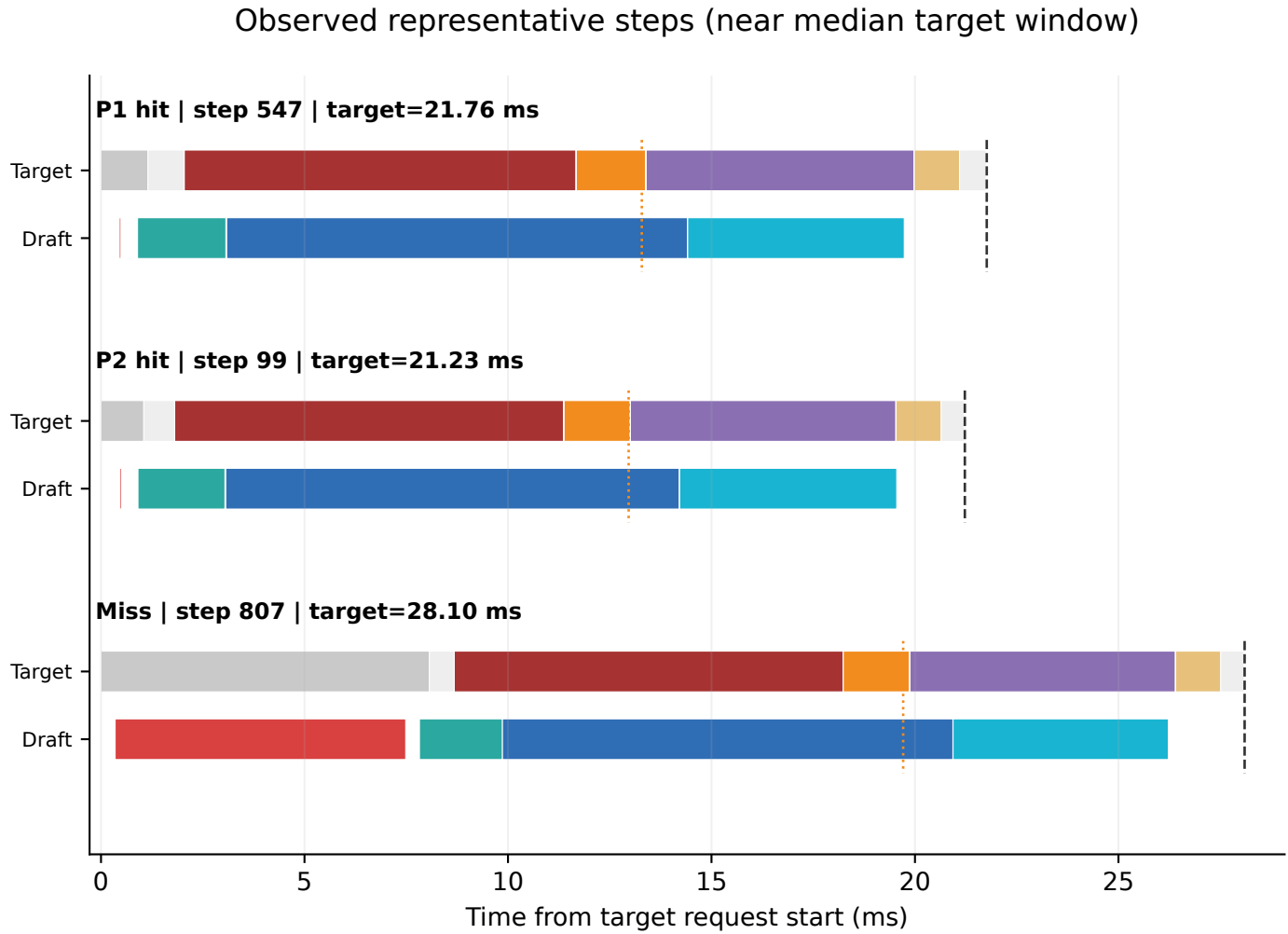
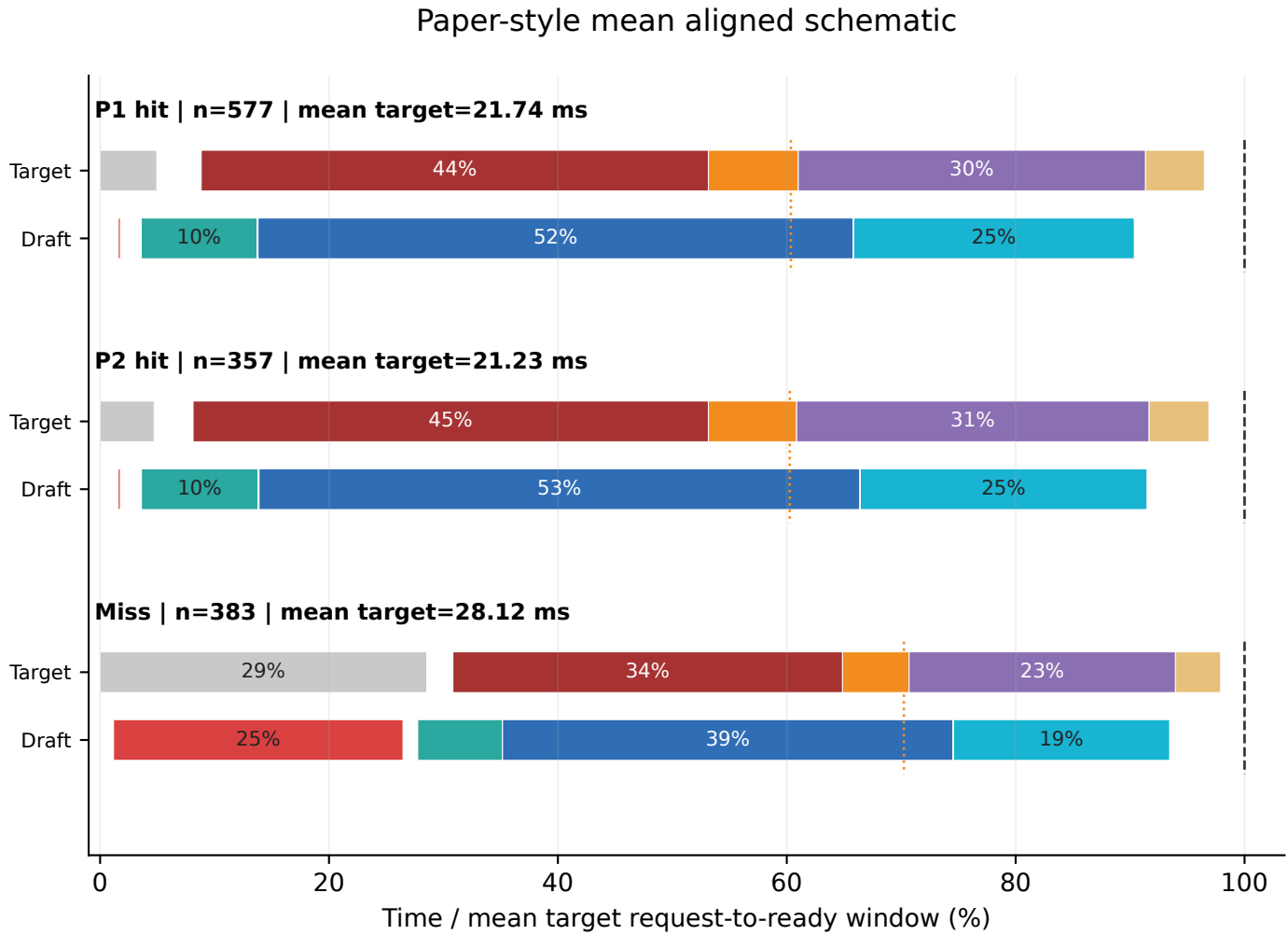
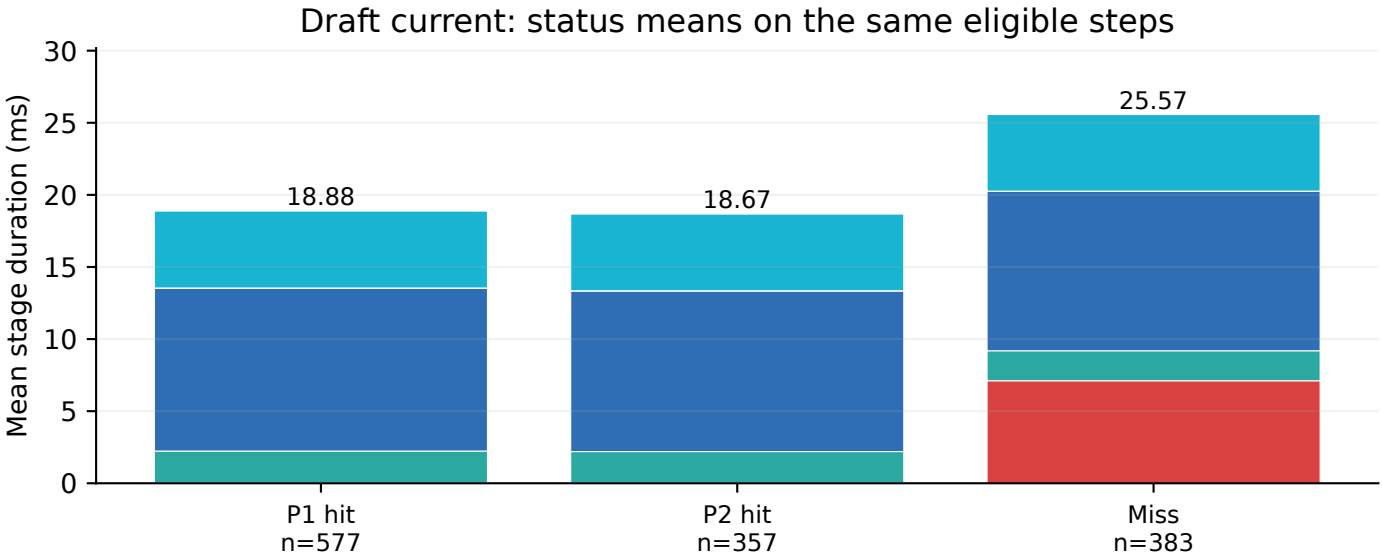
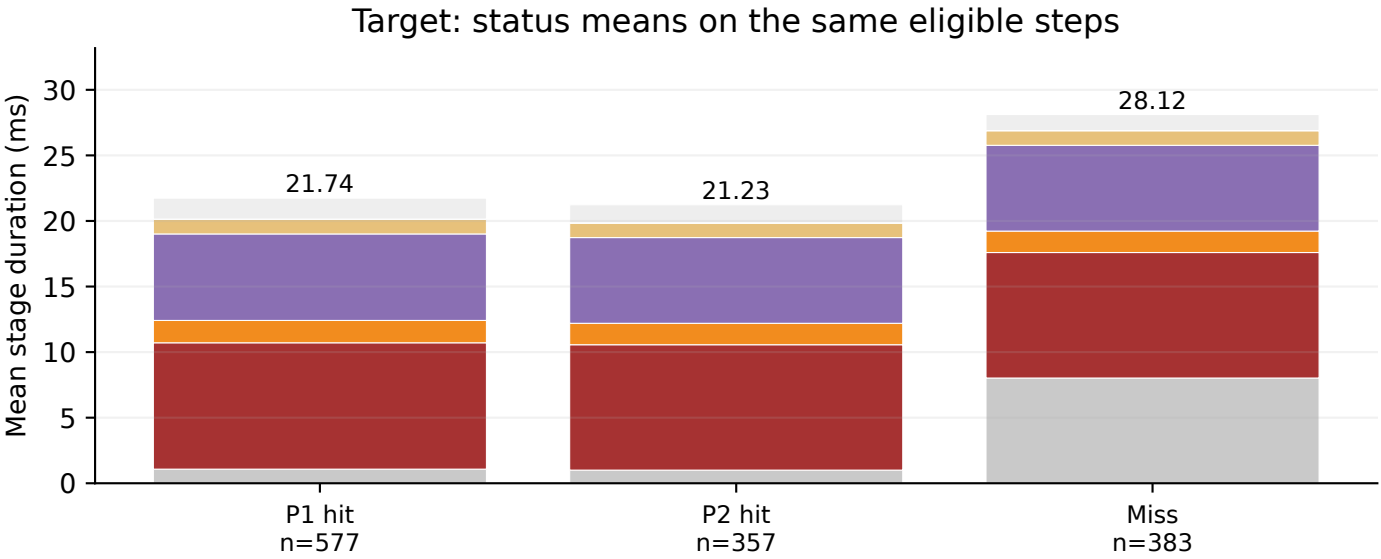
llama3_b1_combo_e14
DUET: K1/K2=4/2, exit=14 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1400 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

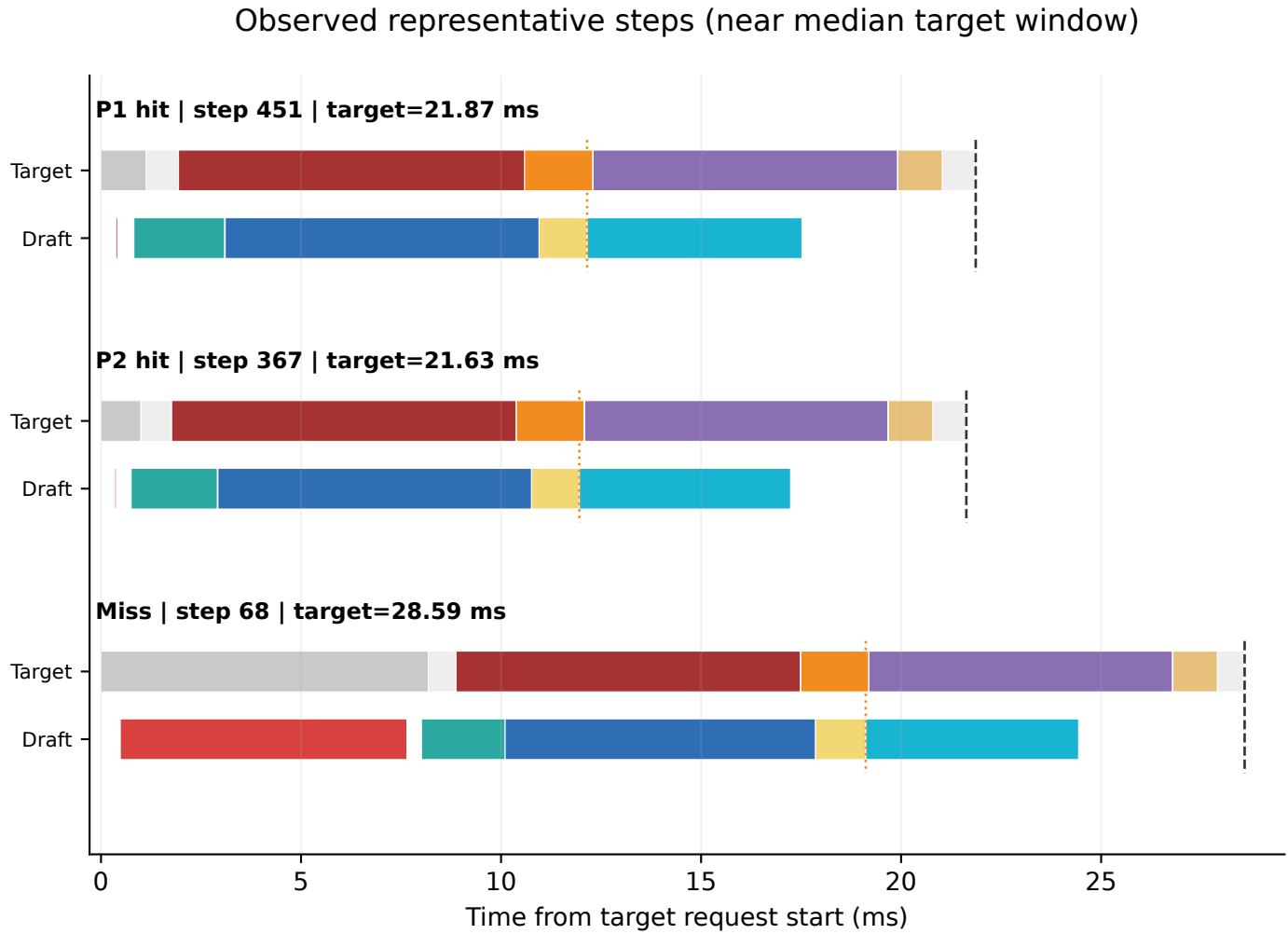
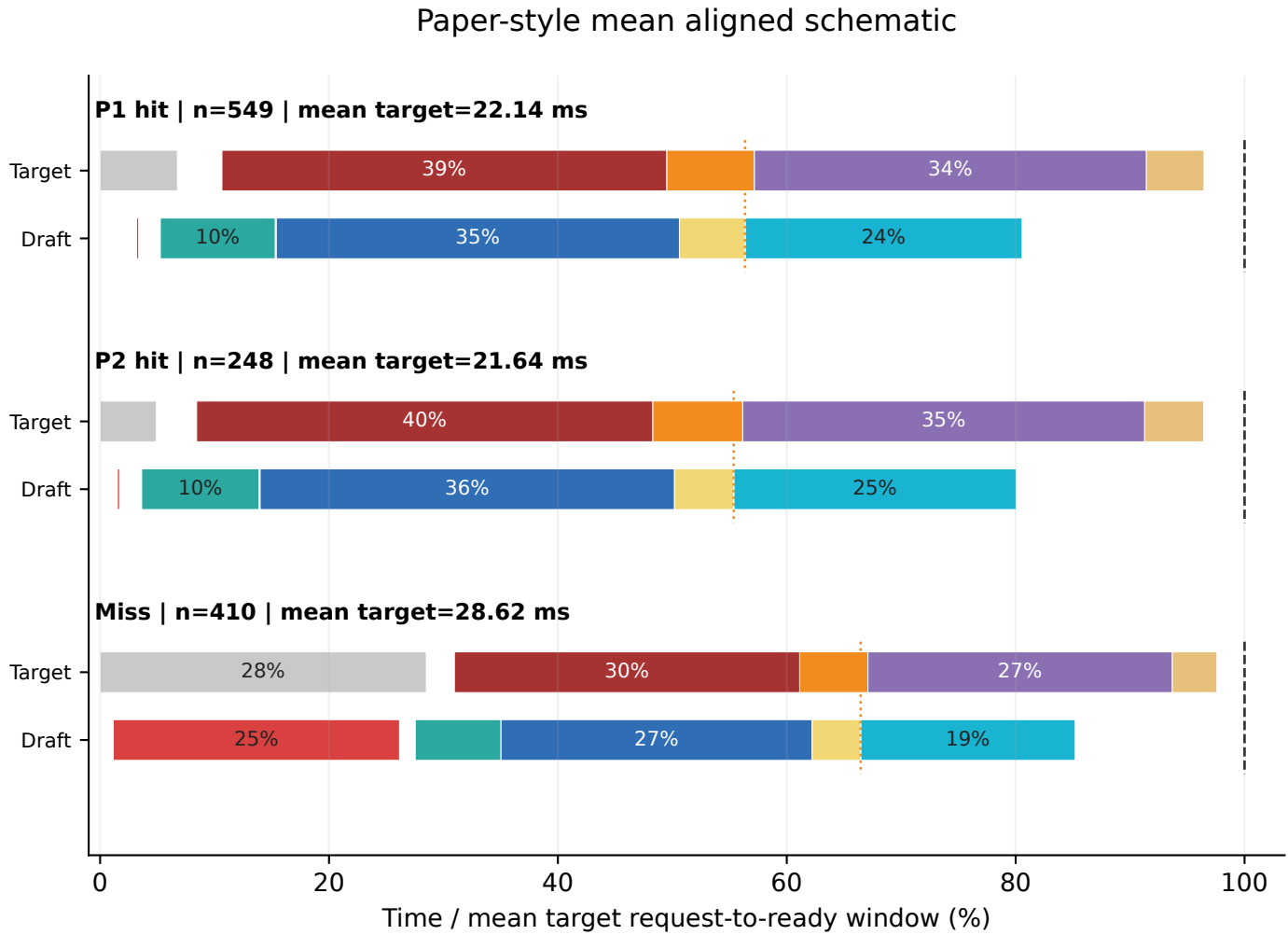
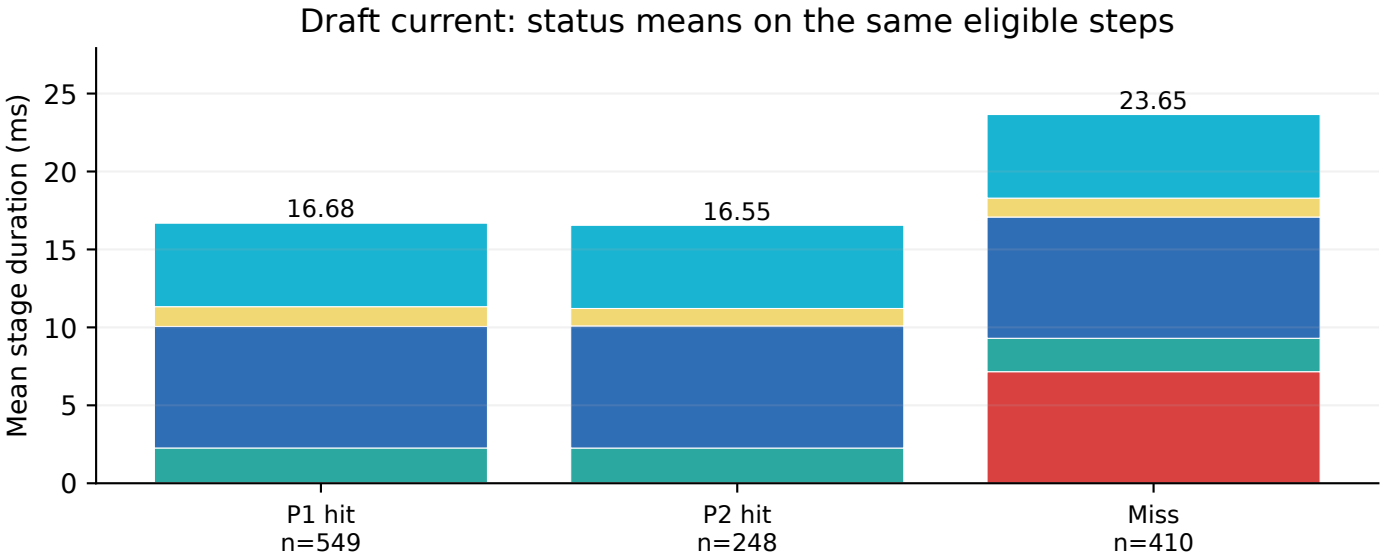
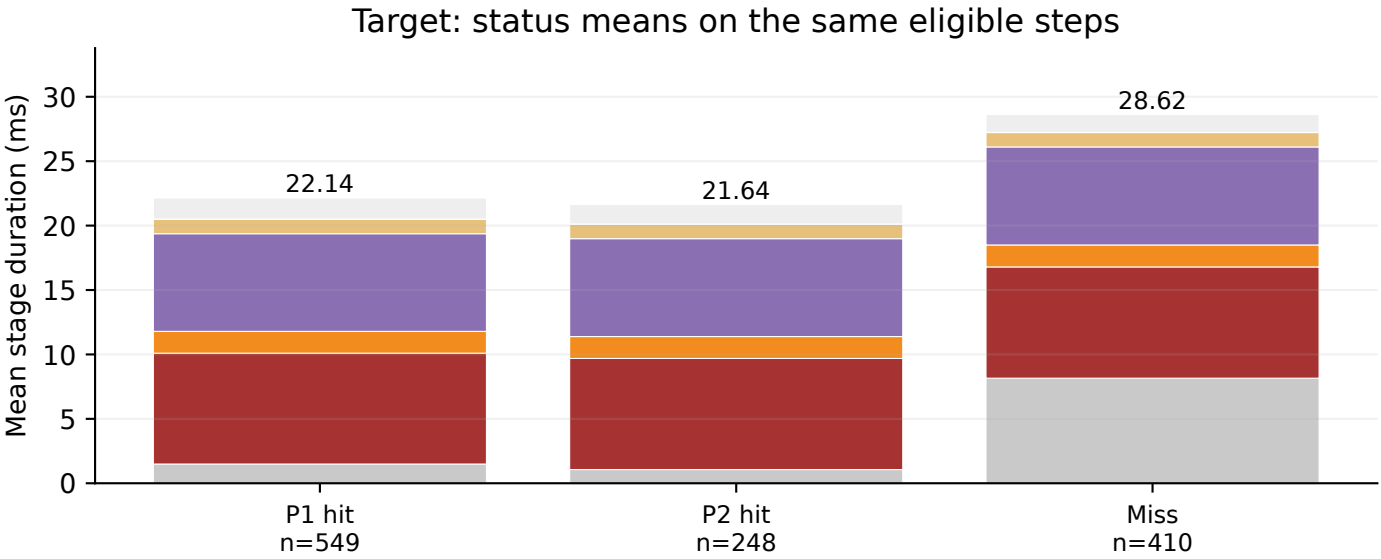
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_e18
DUET: K1/K2=4/2, exit=18 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1317 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

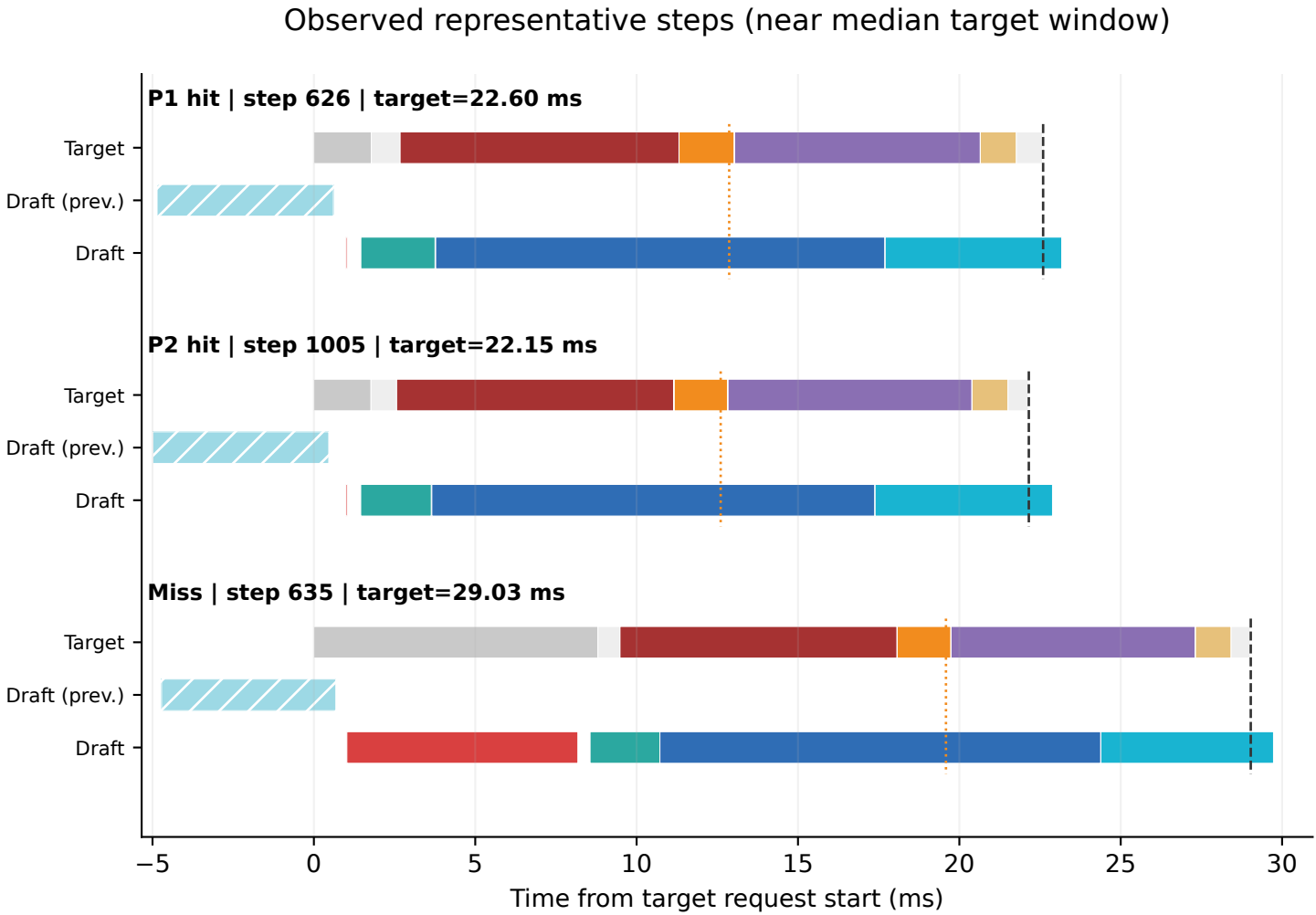
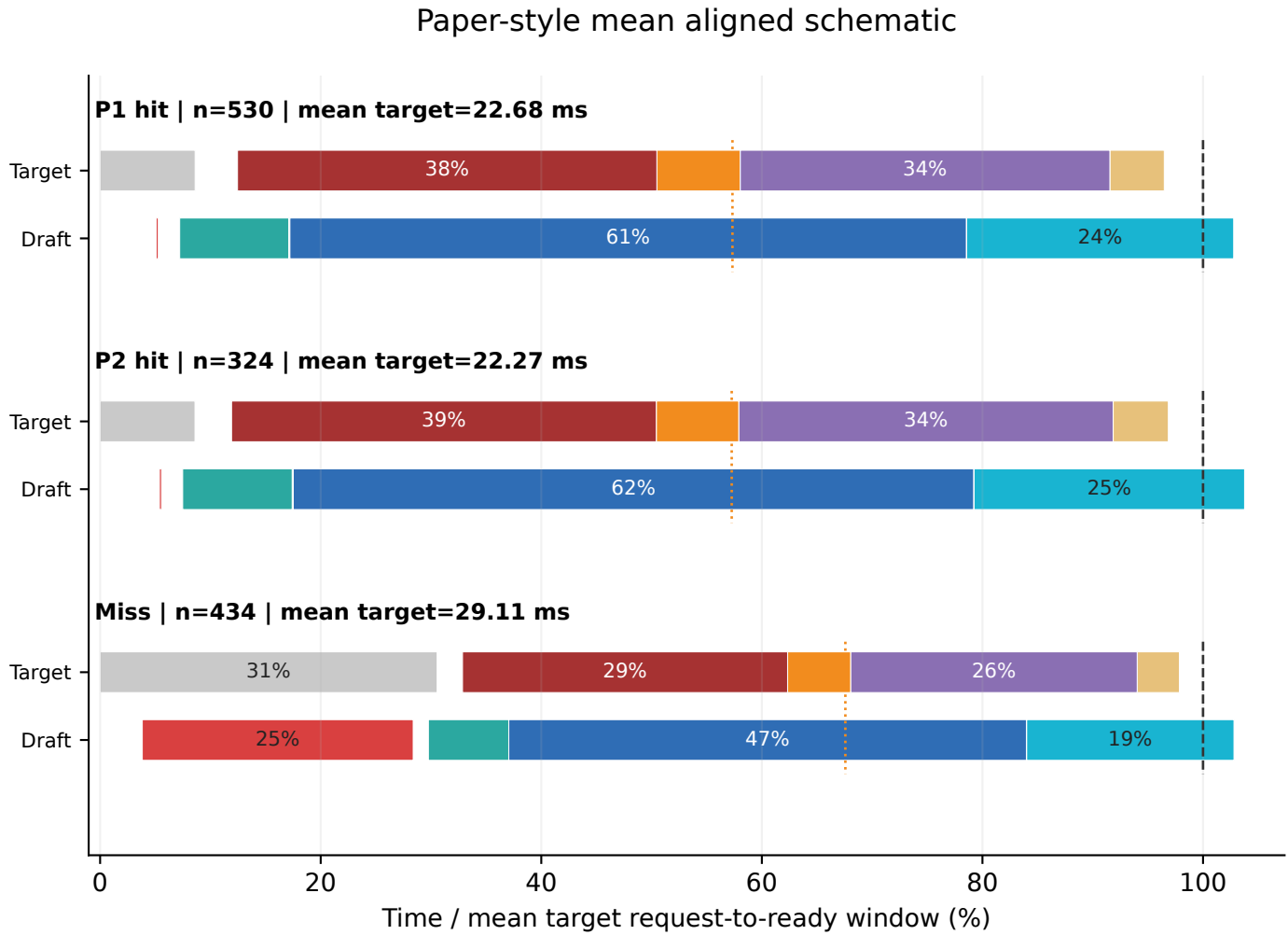
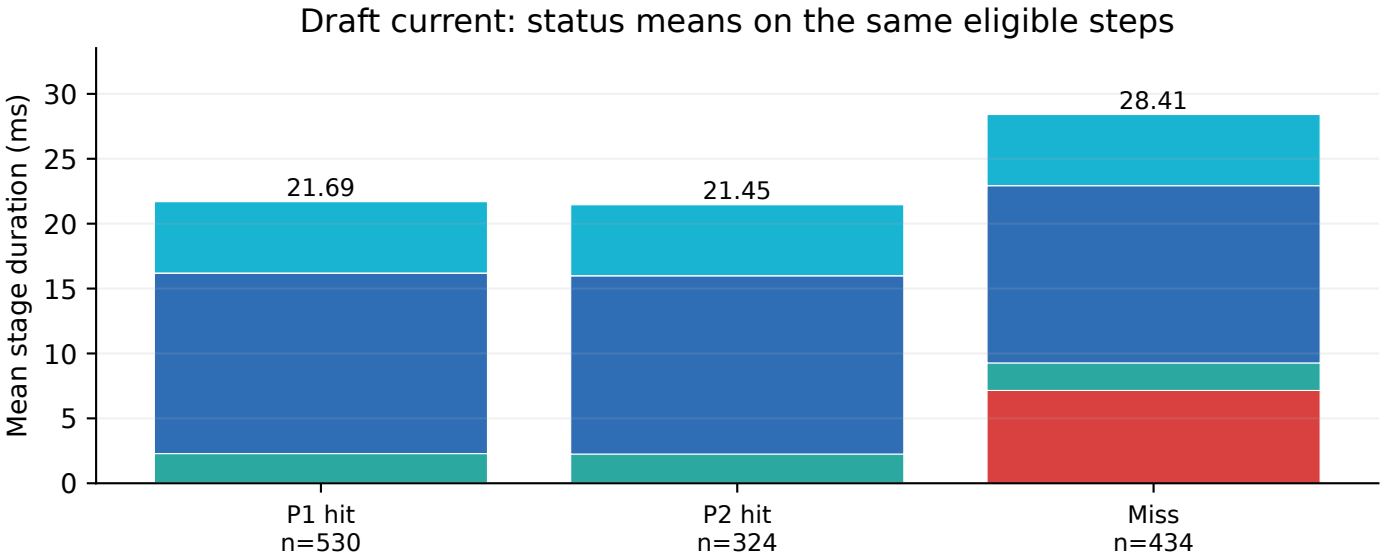
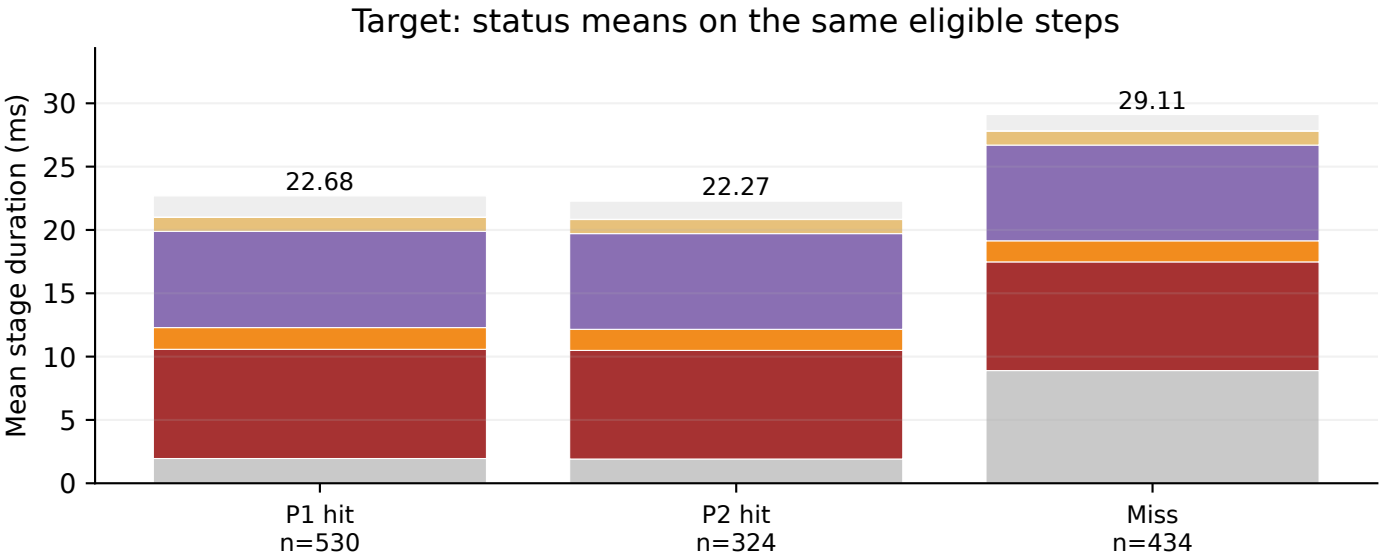
llama3_b1_combo_k1_3
DUET: K1/K2=3/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1207 eligible steps



Legend: Wait / sync (grey), Target pre (red), Proxy / launch (orange), Target post (purple), Final logits (yellow), Unlabelled gap (light grey), Context (teal), P1 (blue), Proxy wait (yellow), P2 (cyan), Miss draft (red), Target ready (dashed line), Proxy received (dotted line).

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

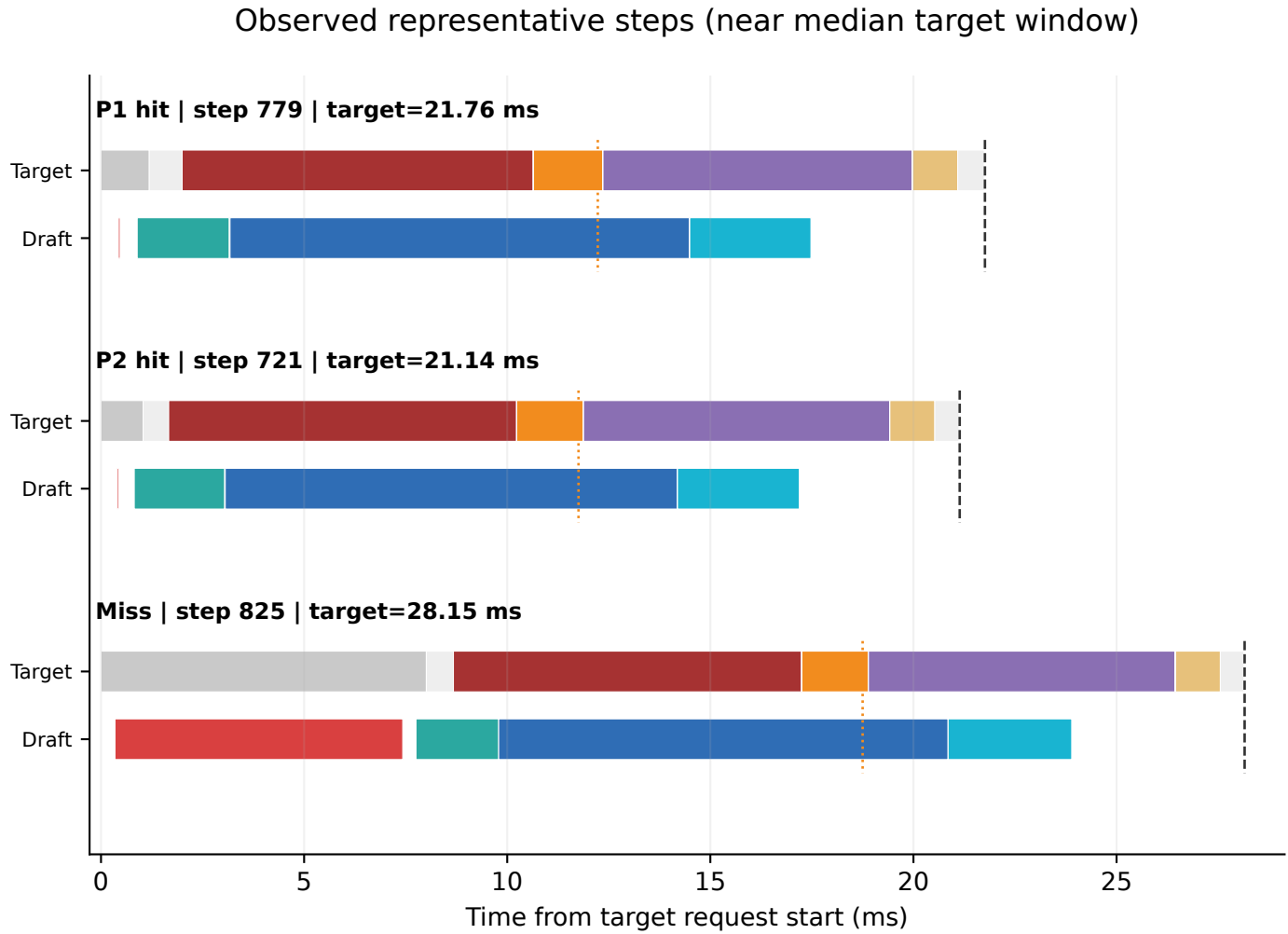
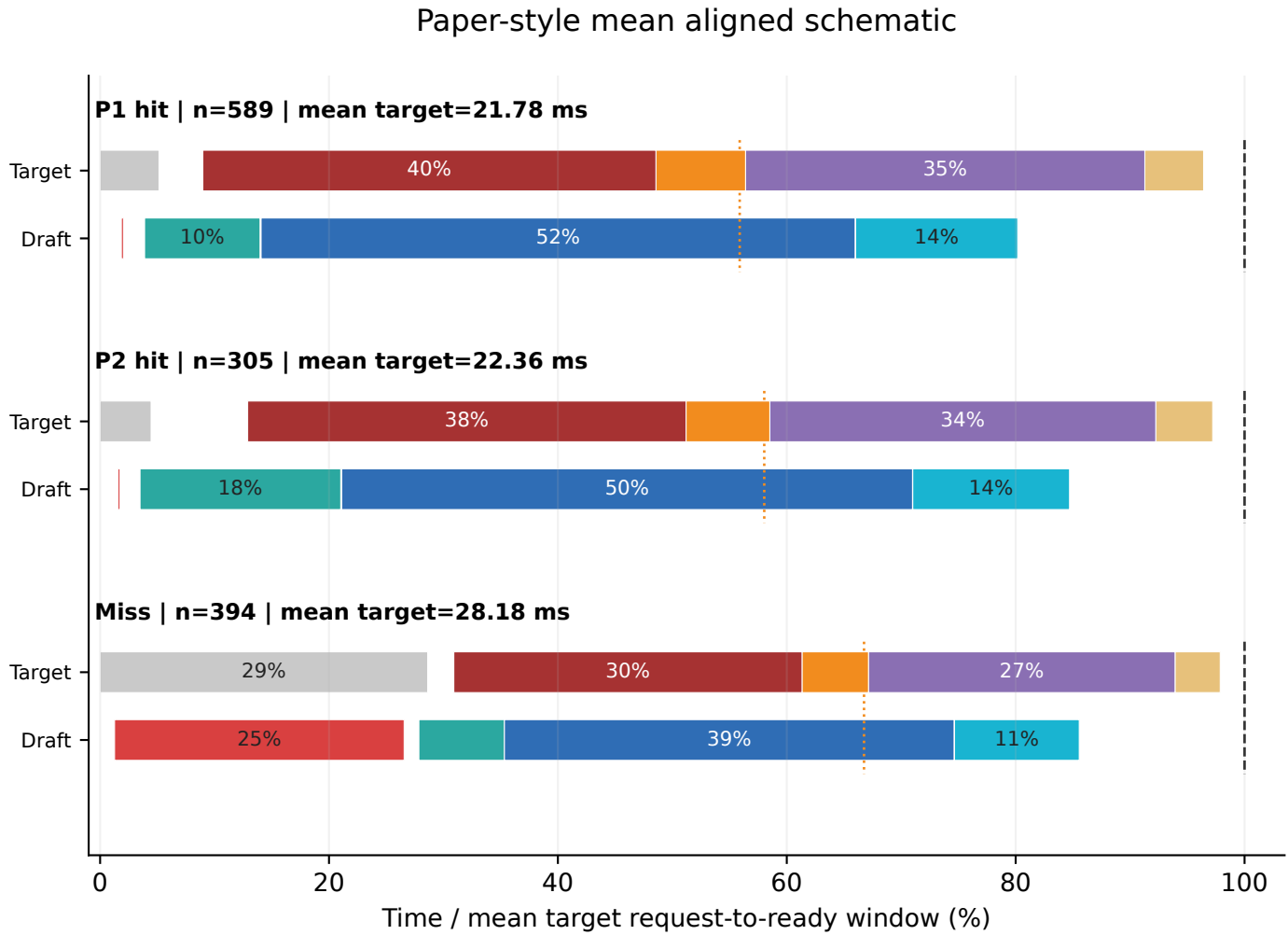
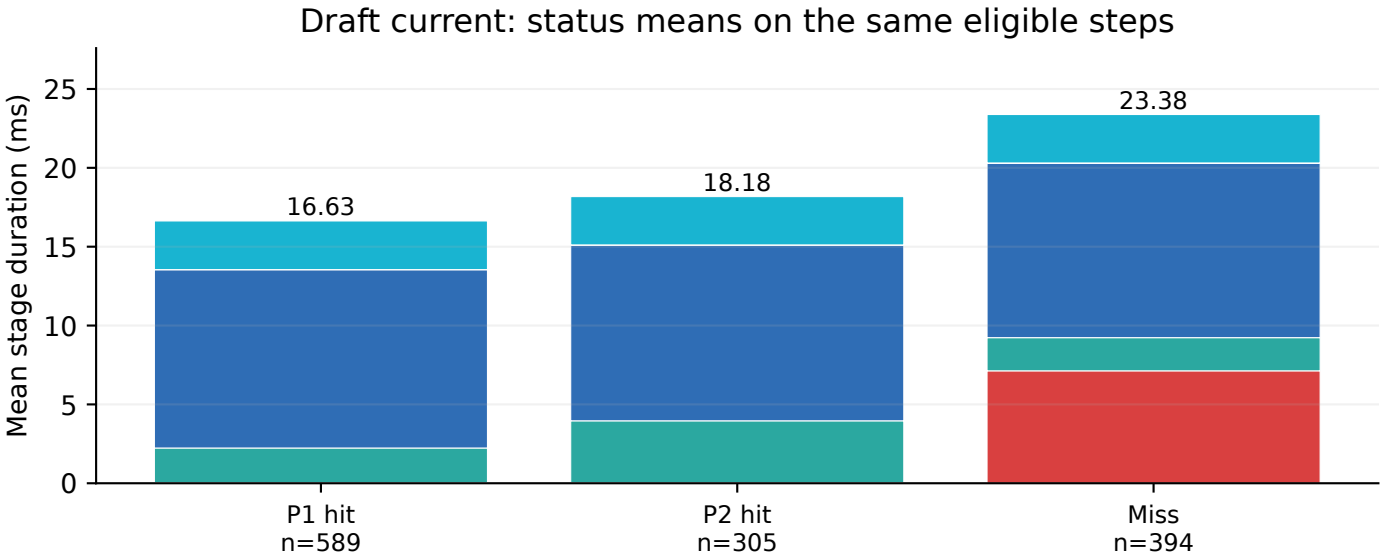
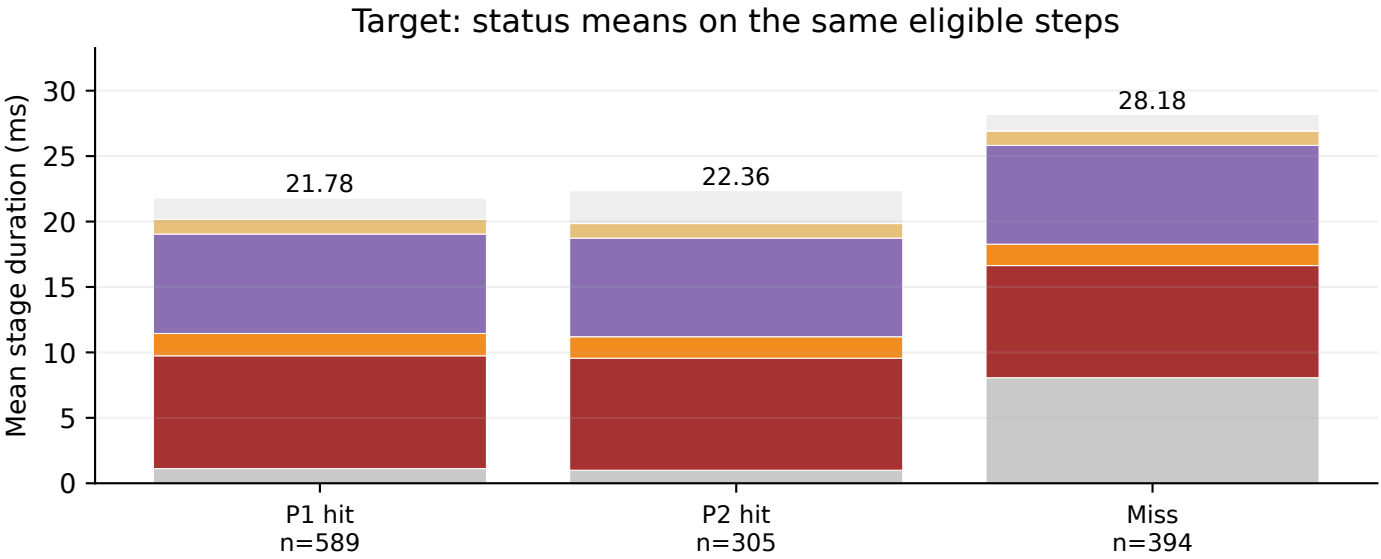
llama3_b1_combo_k1_5
DUET: K1/K2=5/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=6, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1288 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

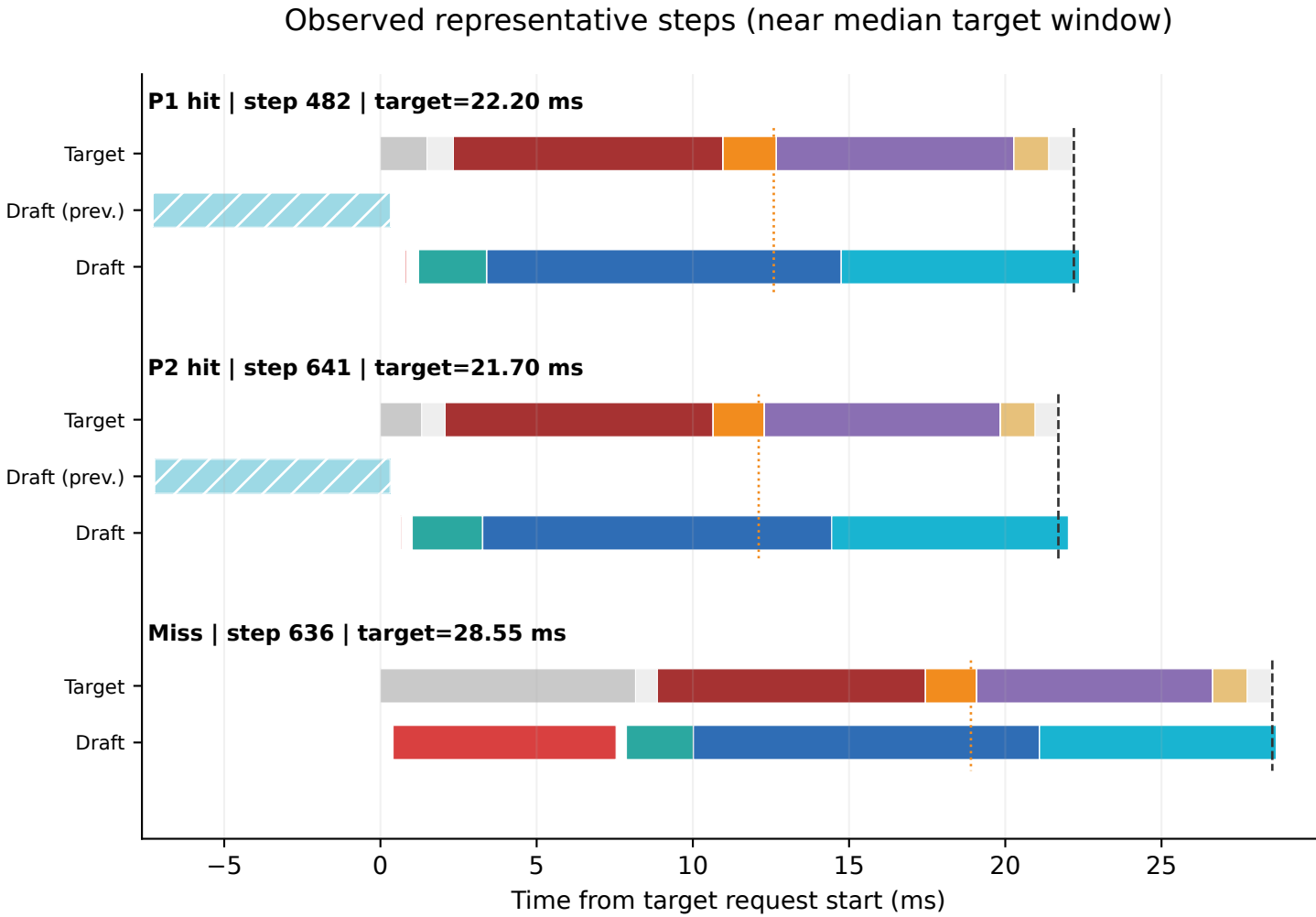
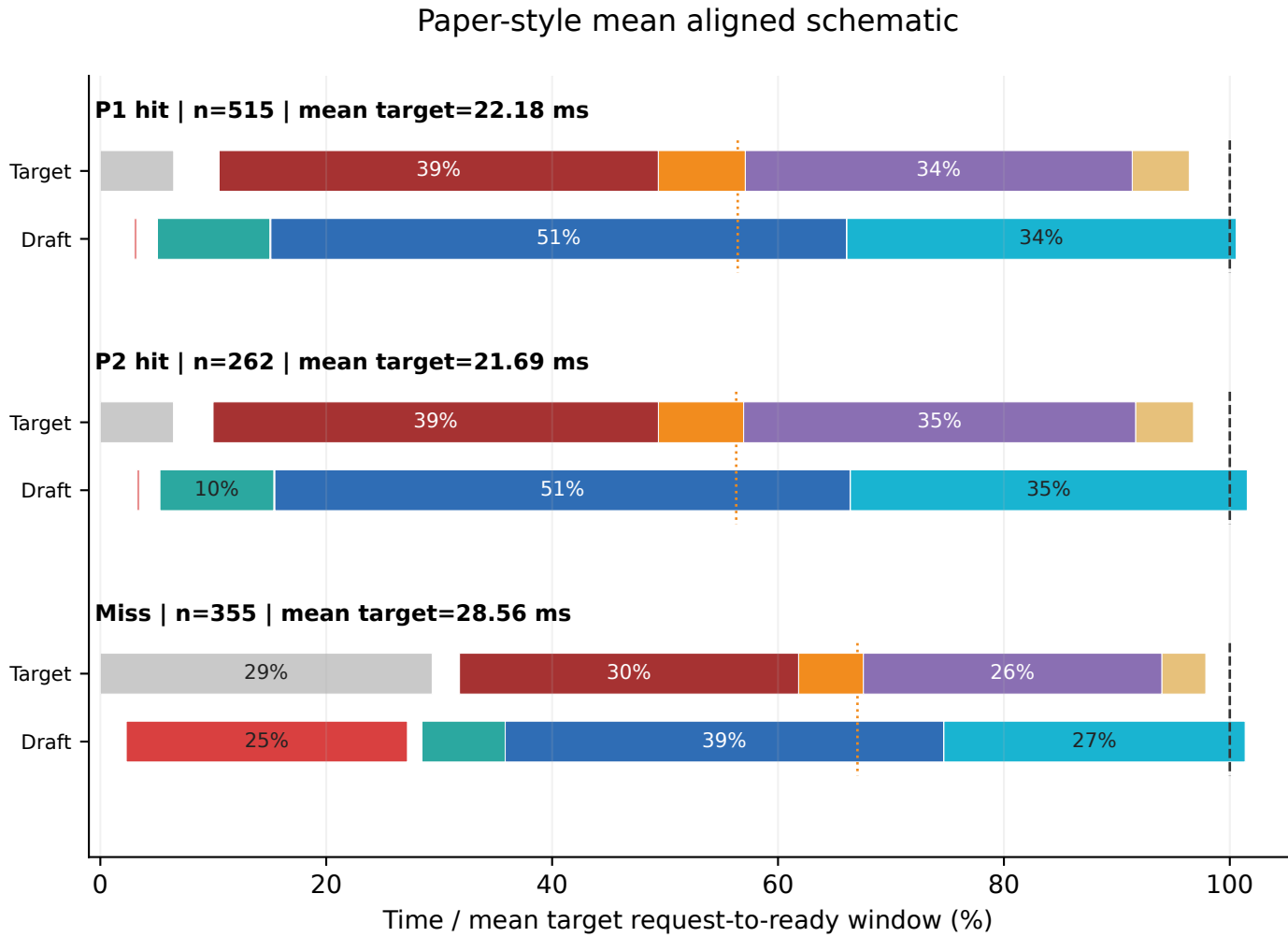
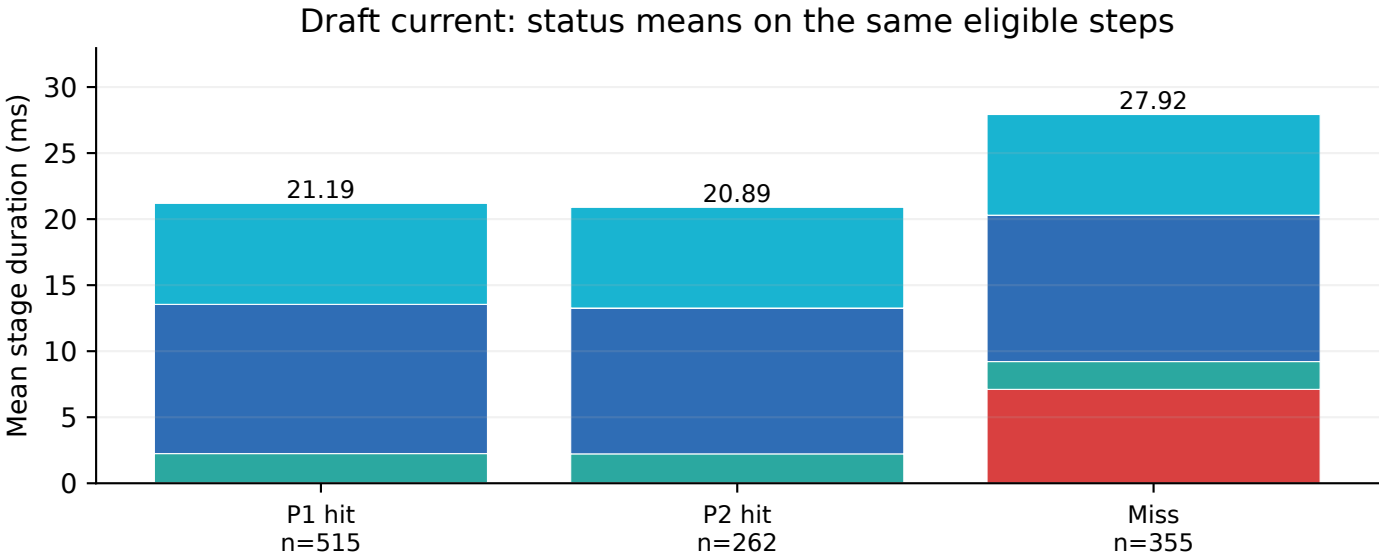
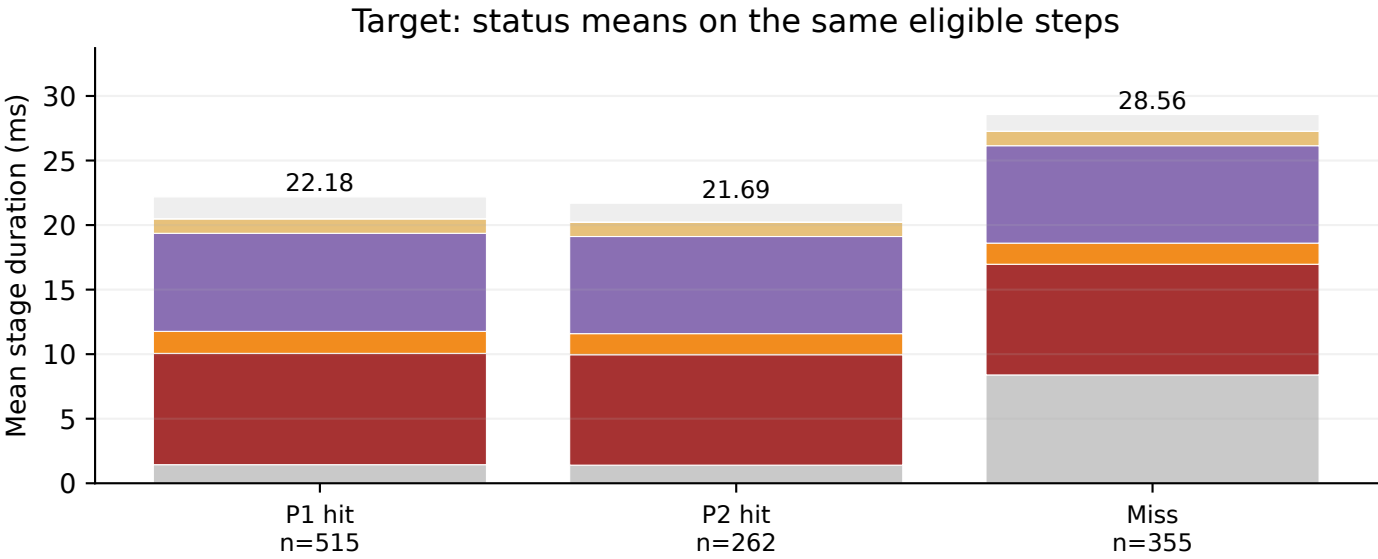
llama3_b1_combo_k2_1
DUET: K1/K2=4/1, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1288 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

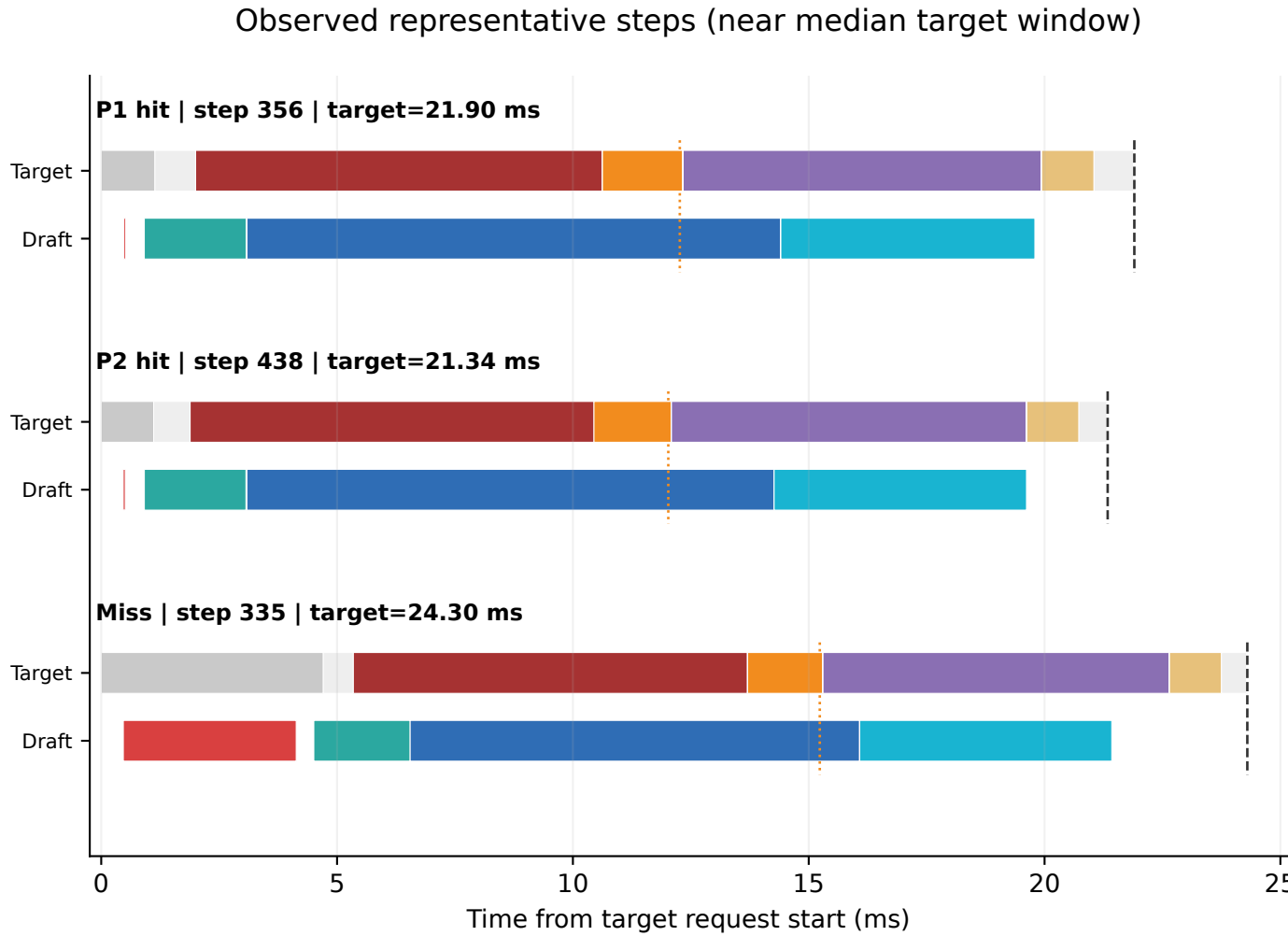
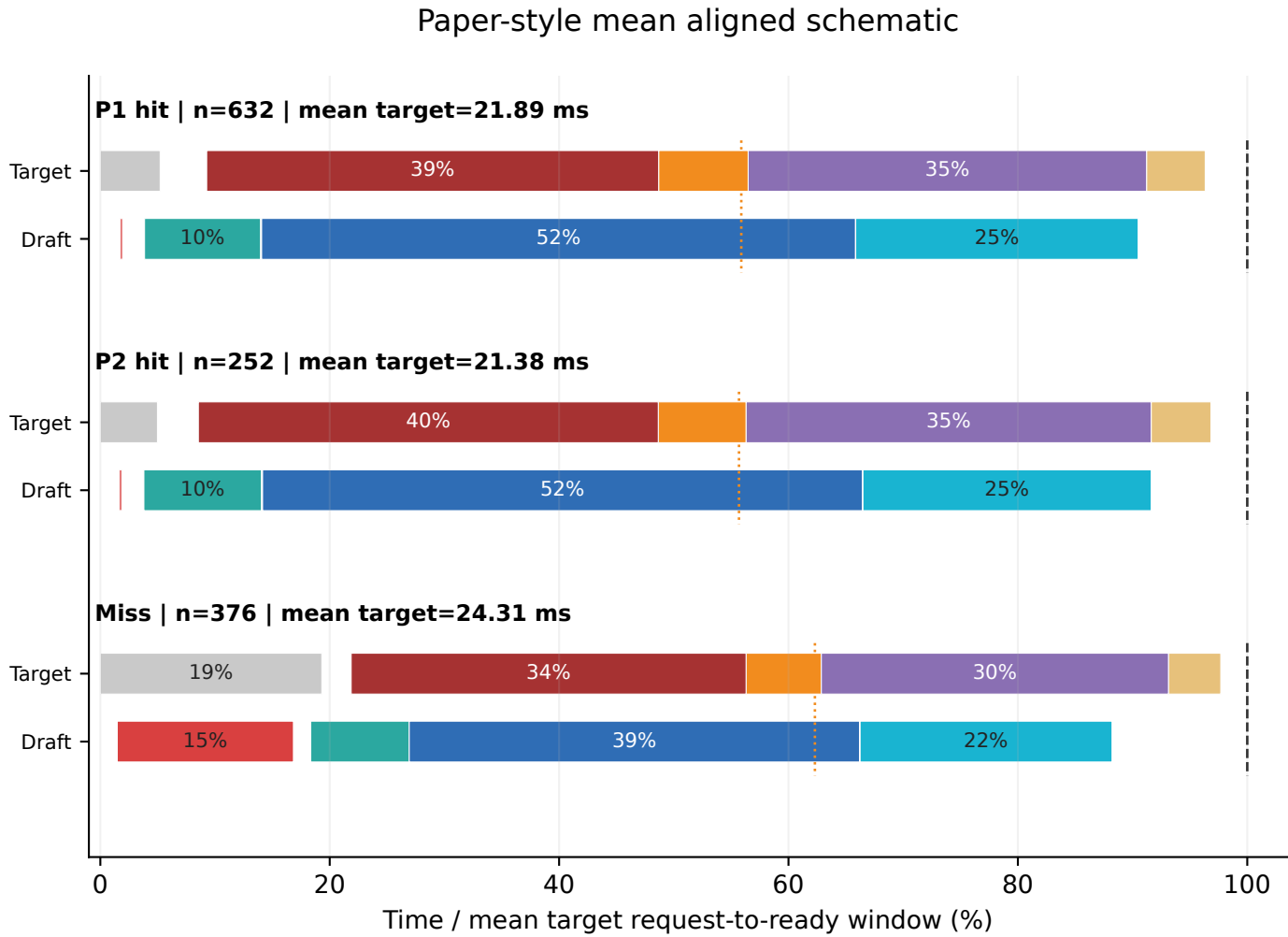
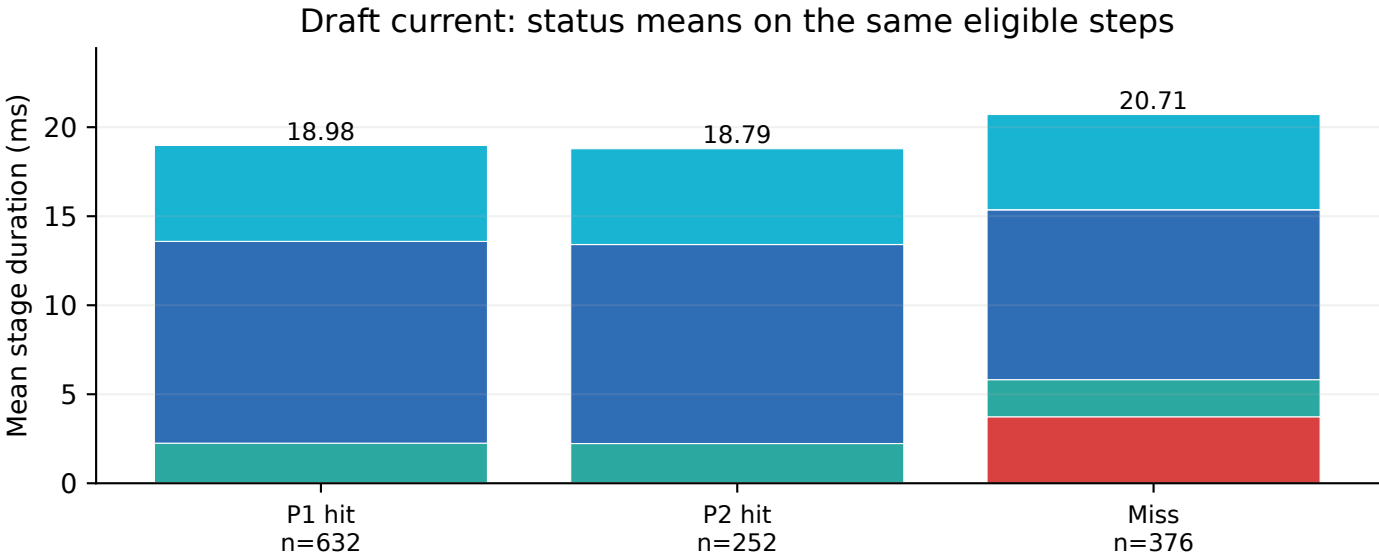
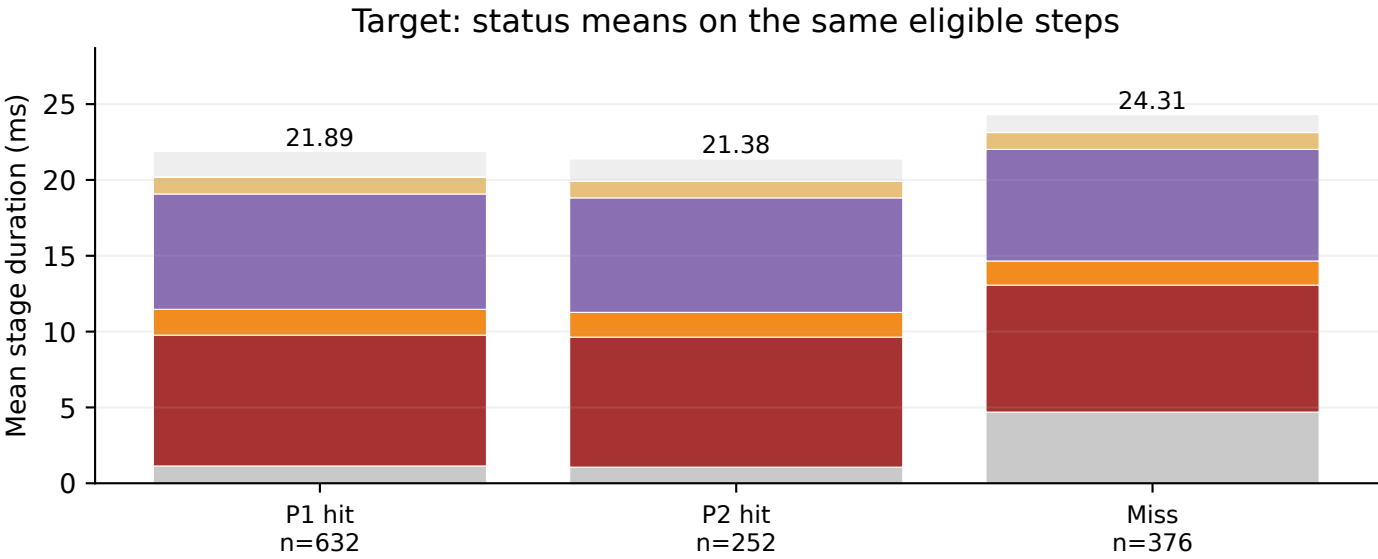
llama3_b1_combo_k2_3
DUET: K1/K2=4/3, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1132 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

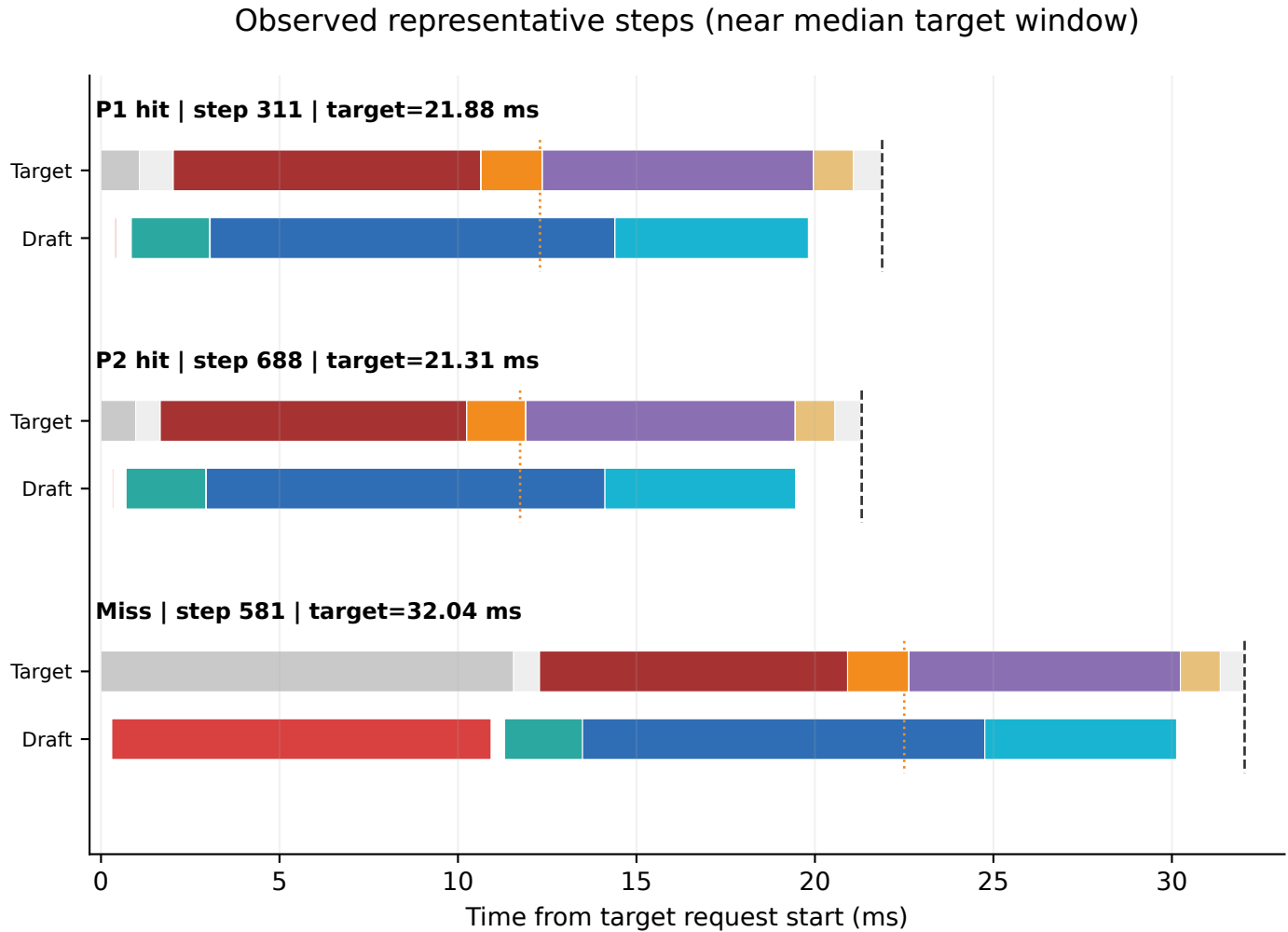
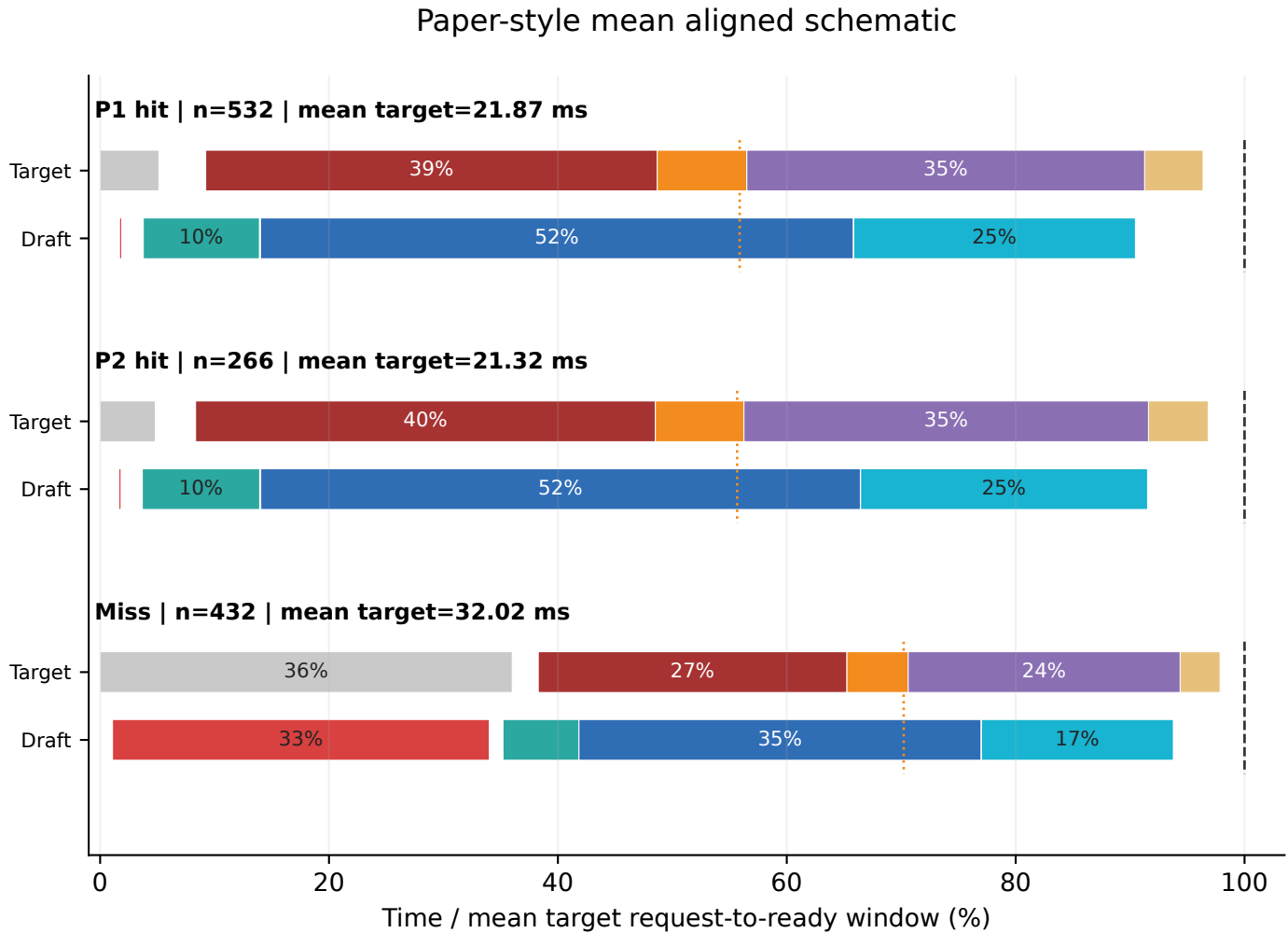
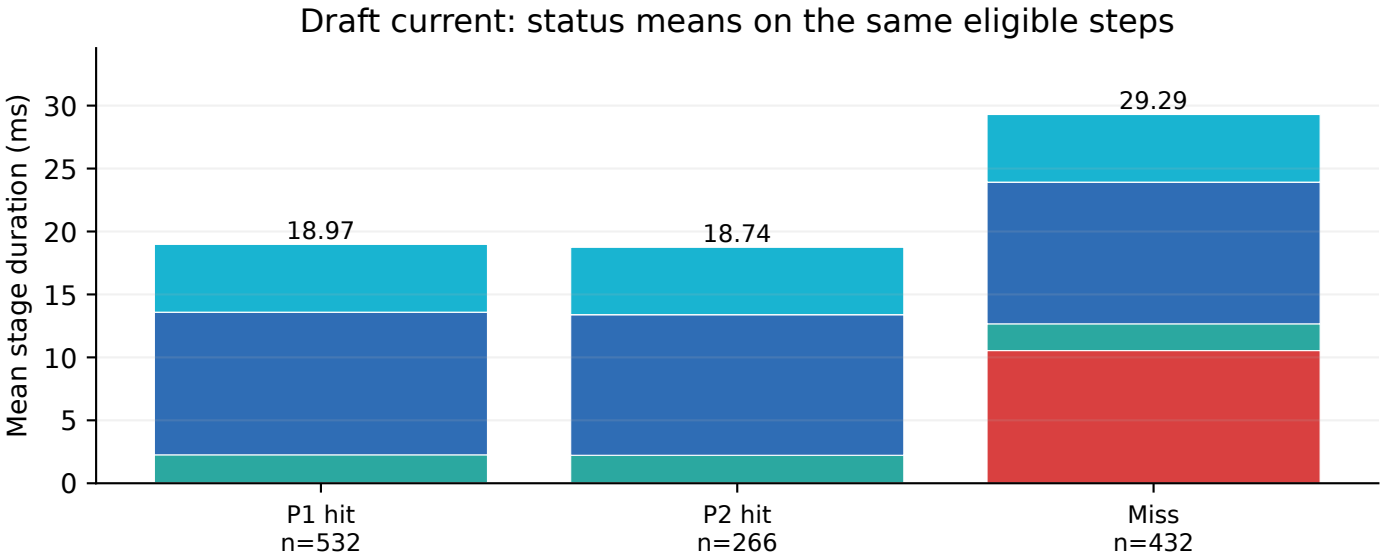
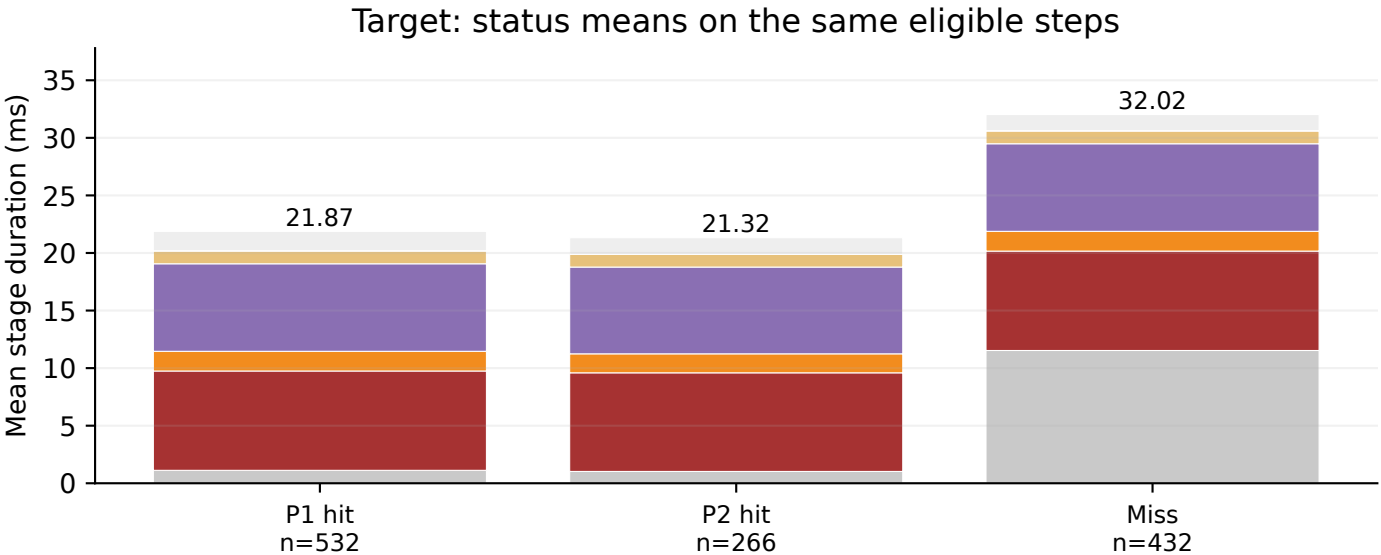
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_miss2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1260 eligible steps



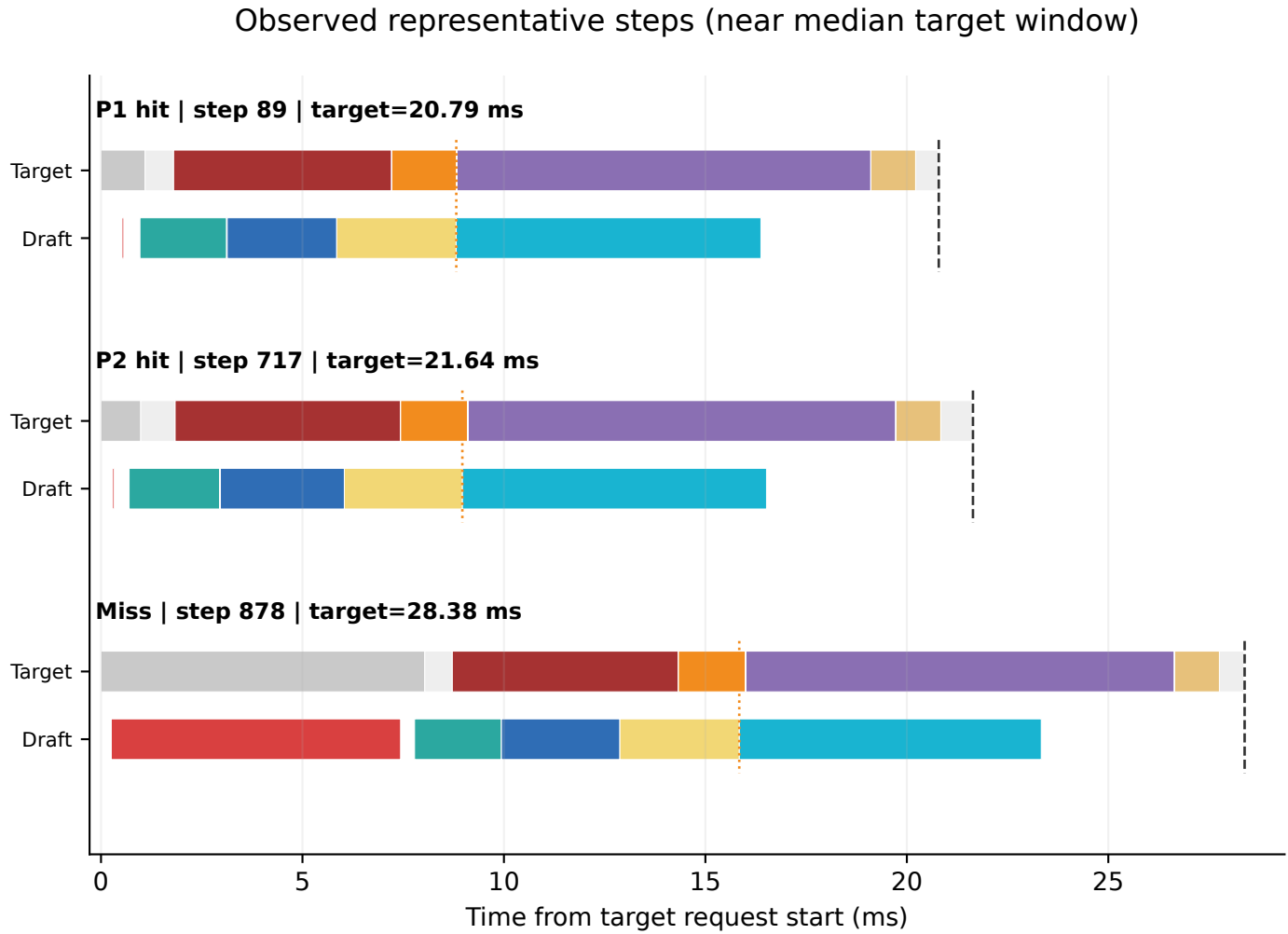
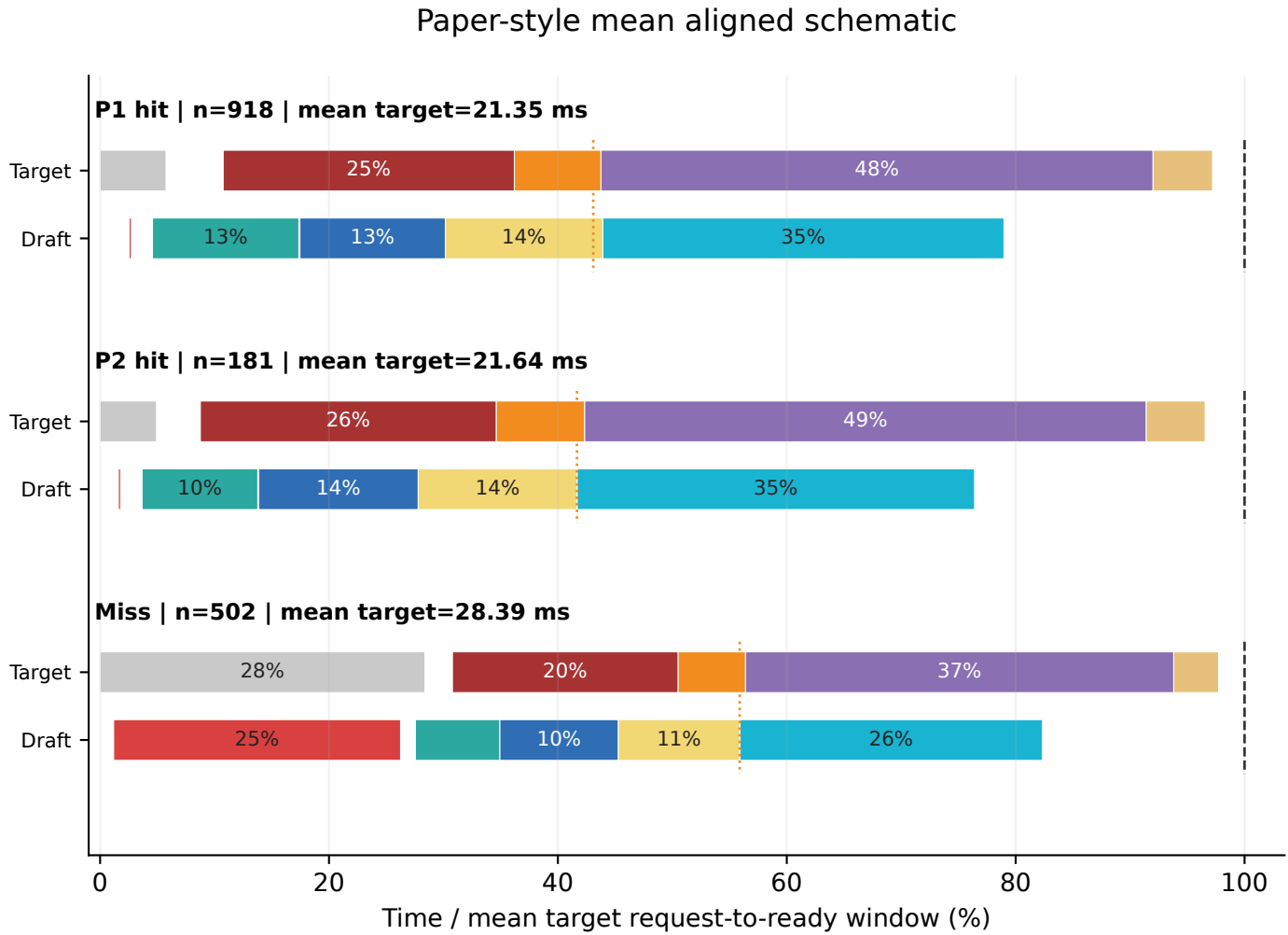
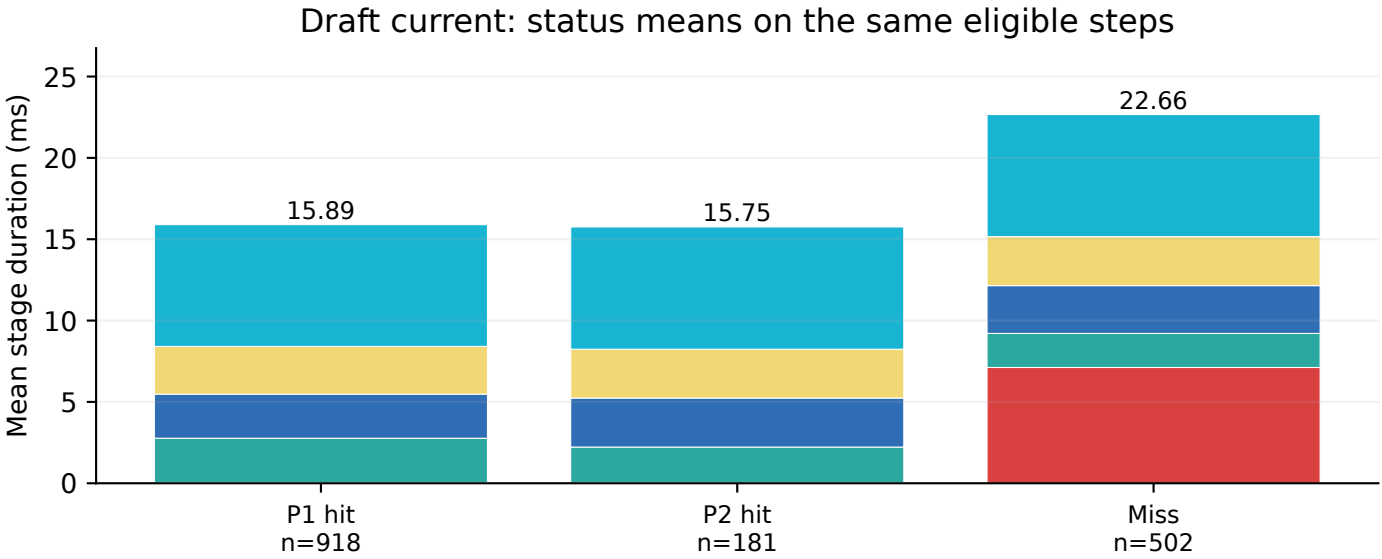
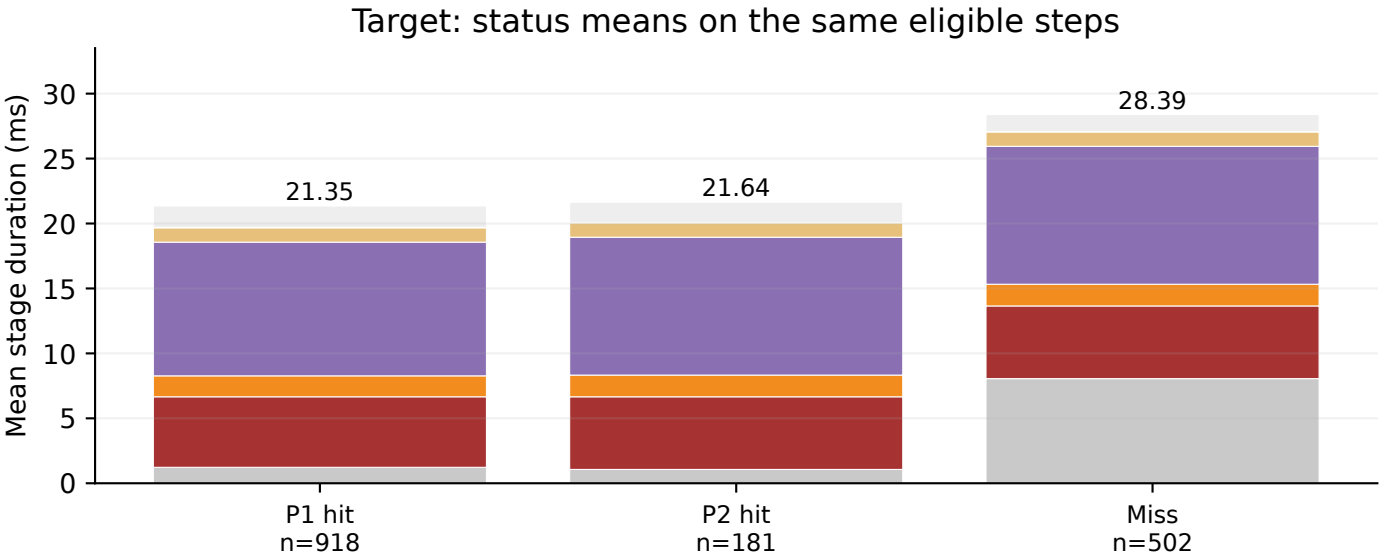
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_miss6
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1230 eligible steps



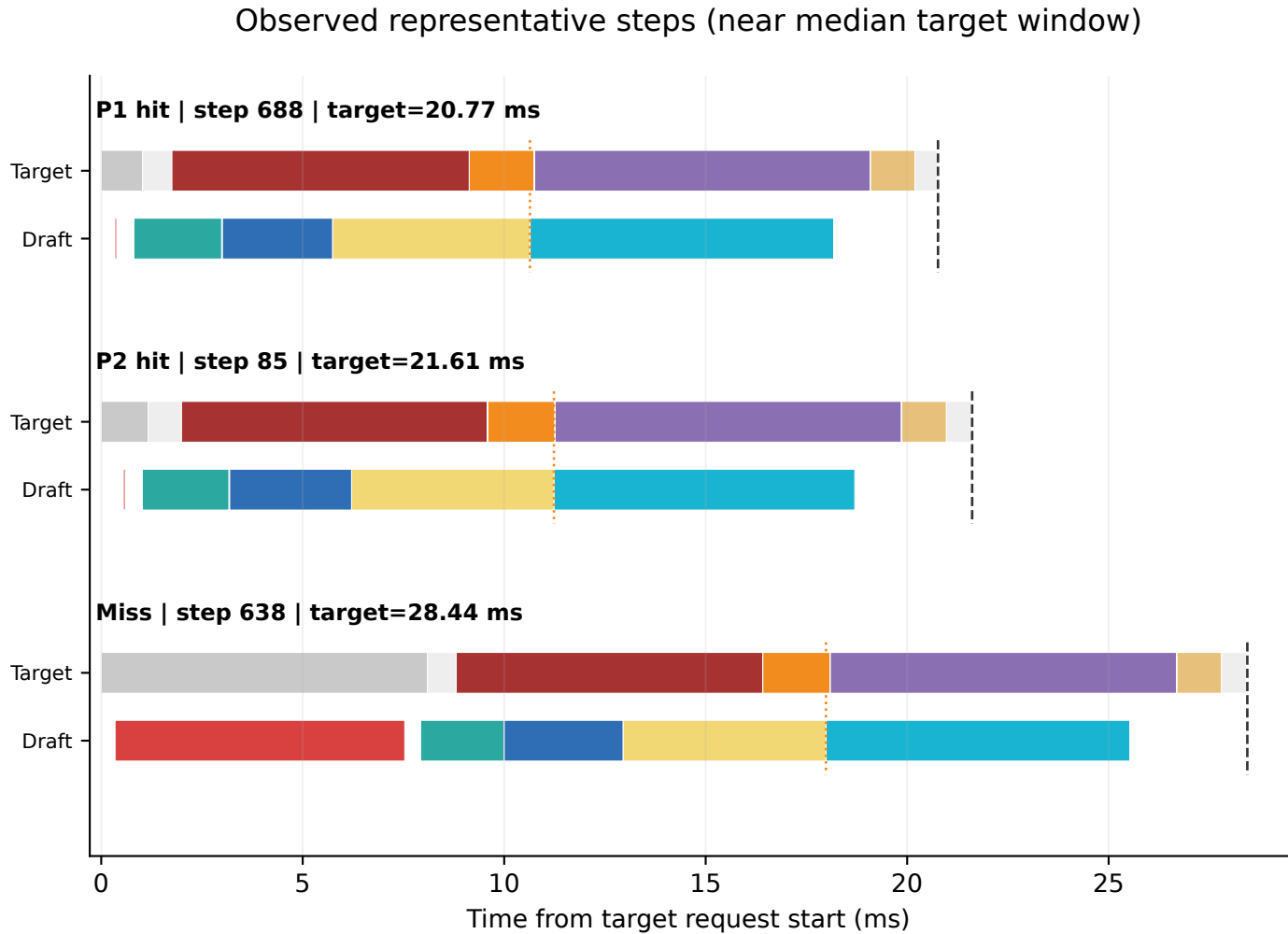
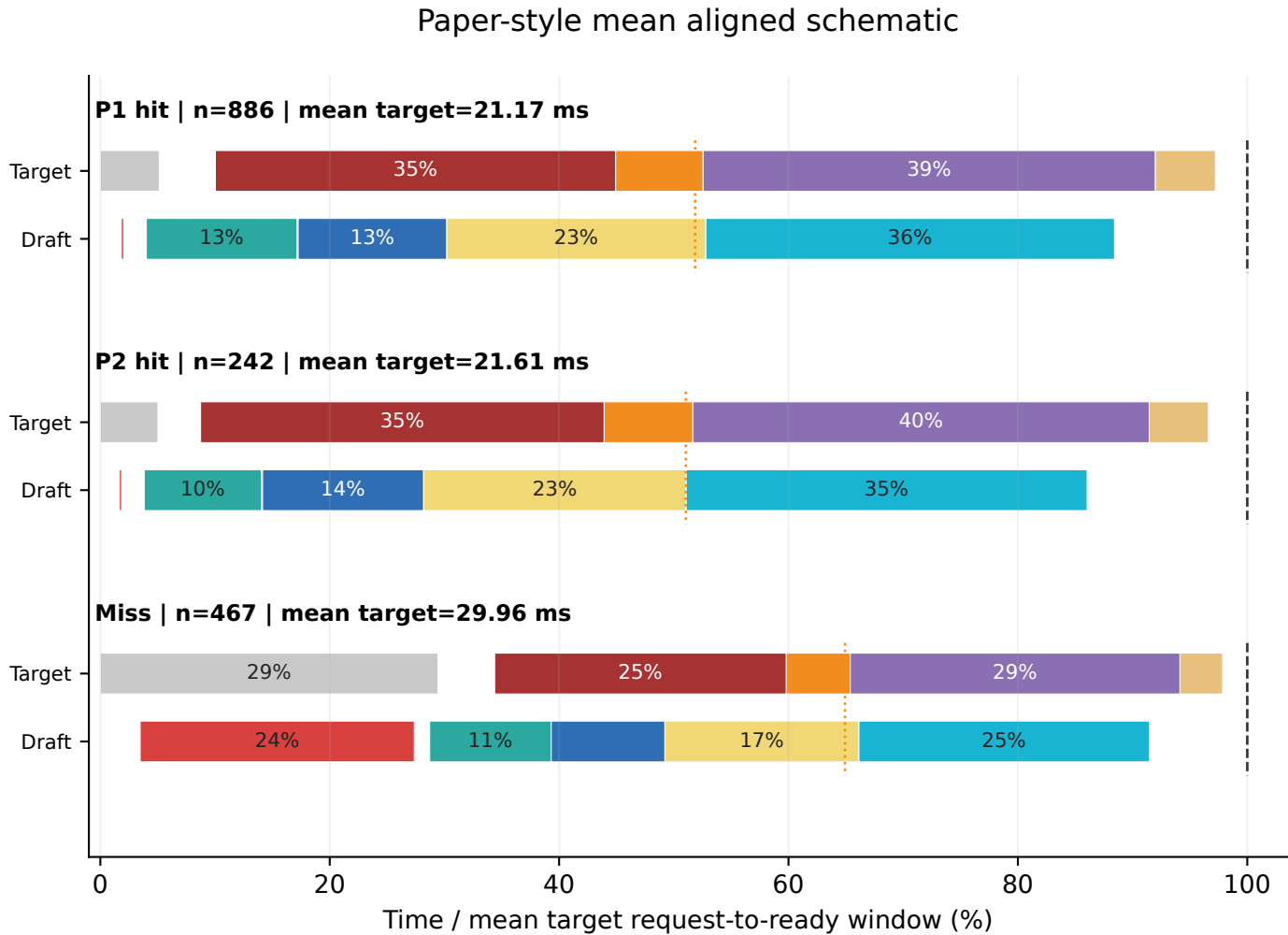
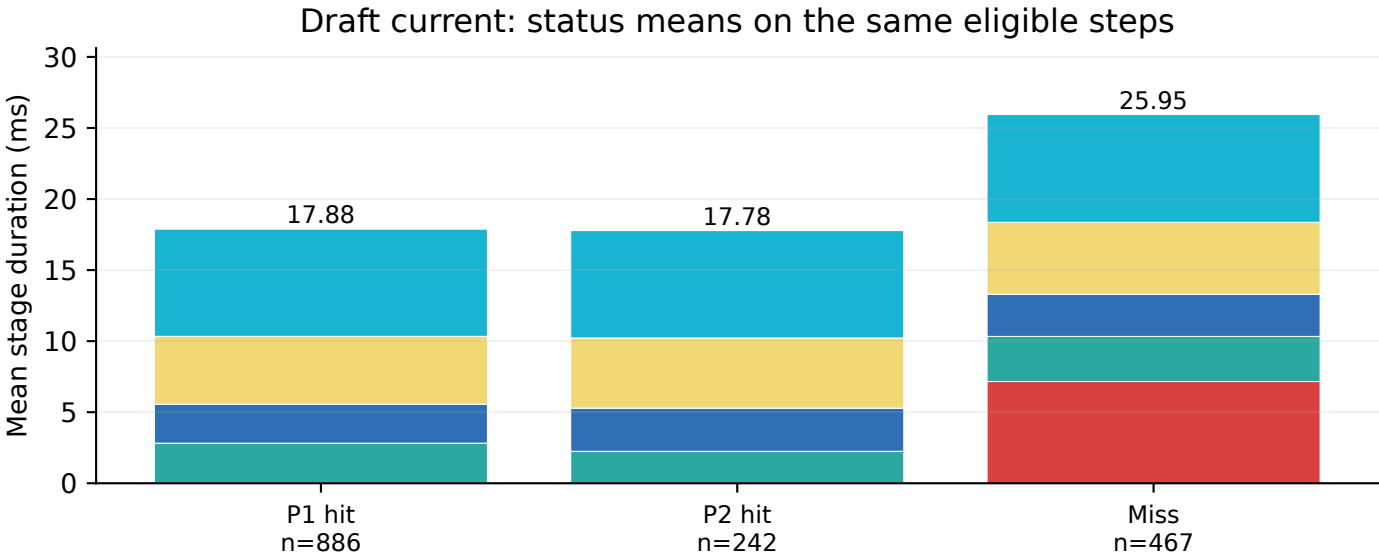
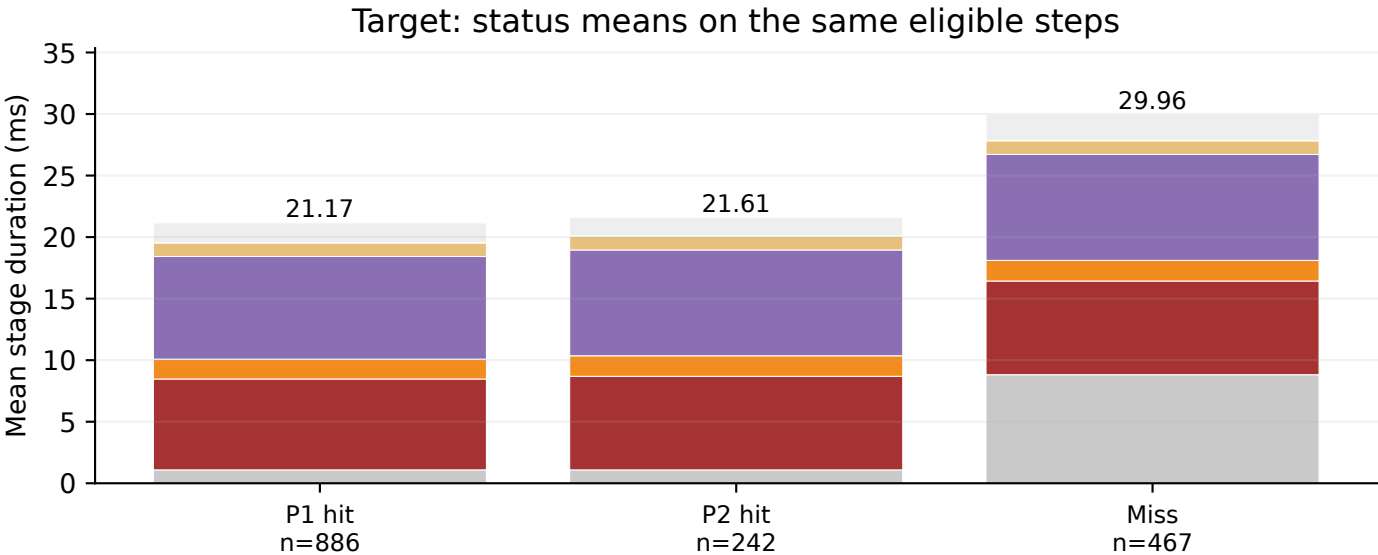
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_transfer_1_3_e10
DUET: K1/K2=1/3, exit=10 (0-based), C=3, N1/N2=2/6, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1601 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

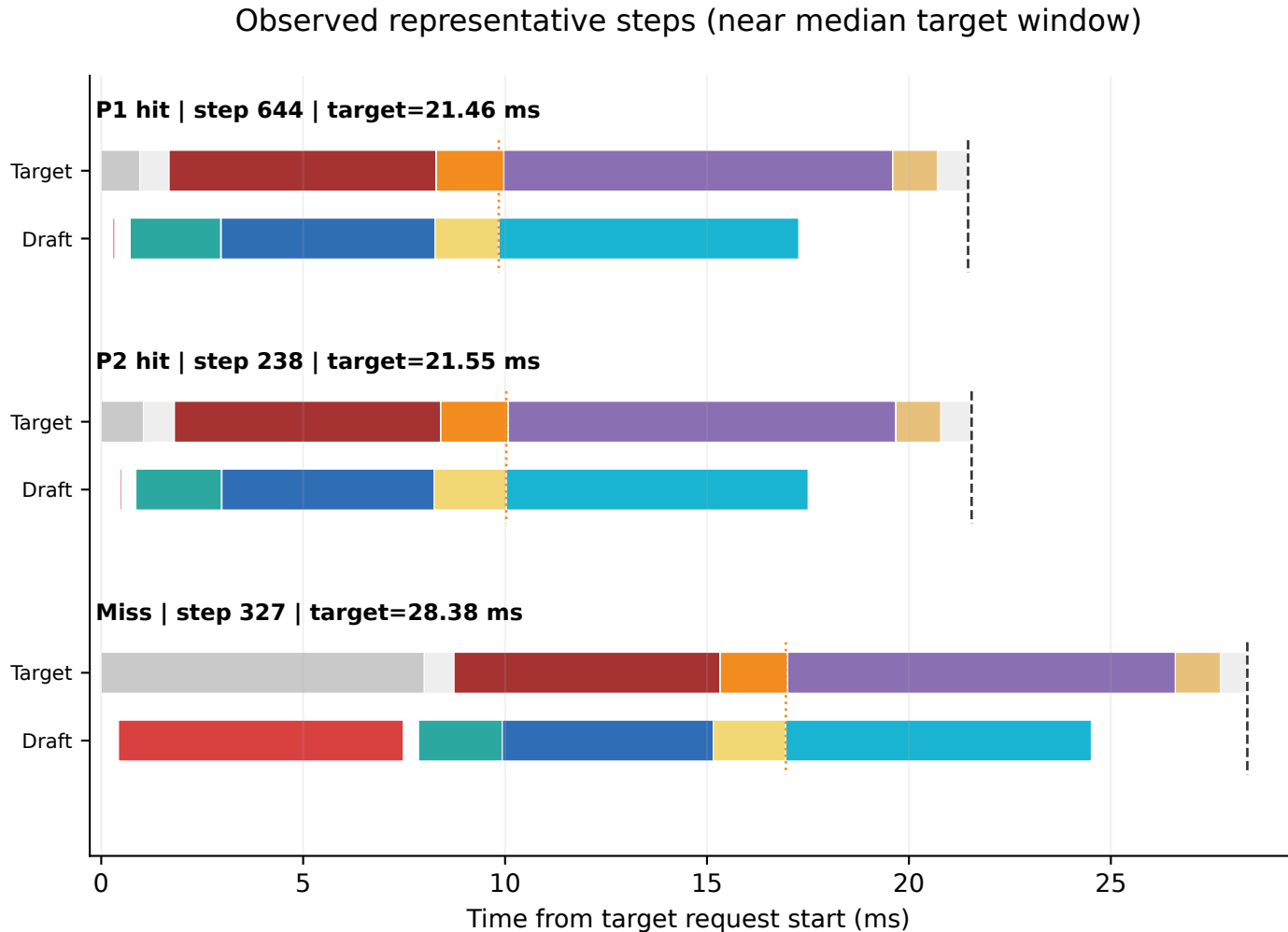
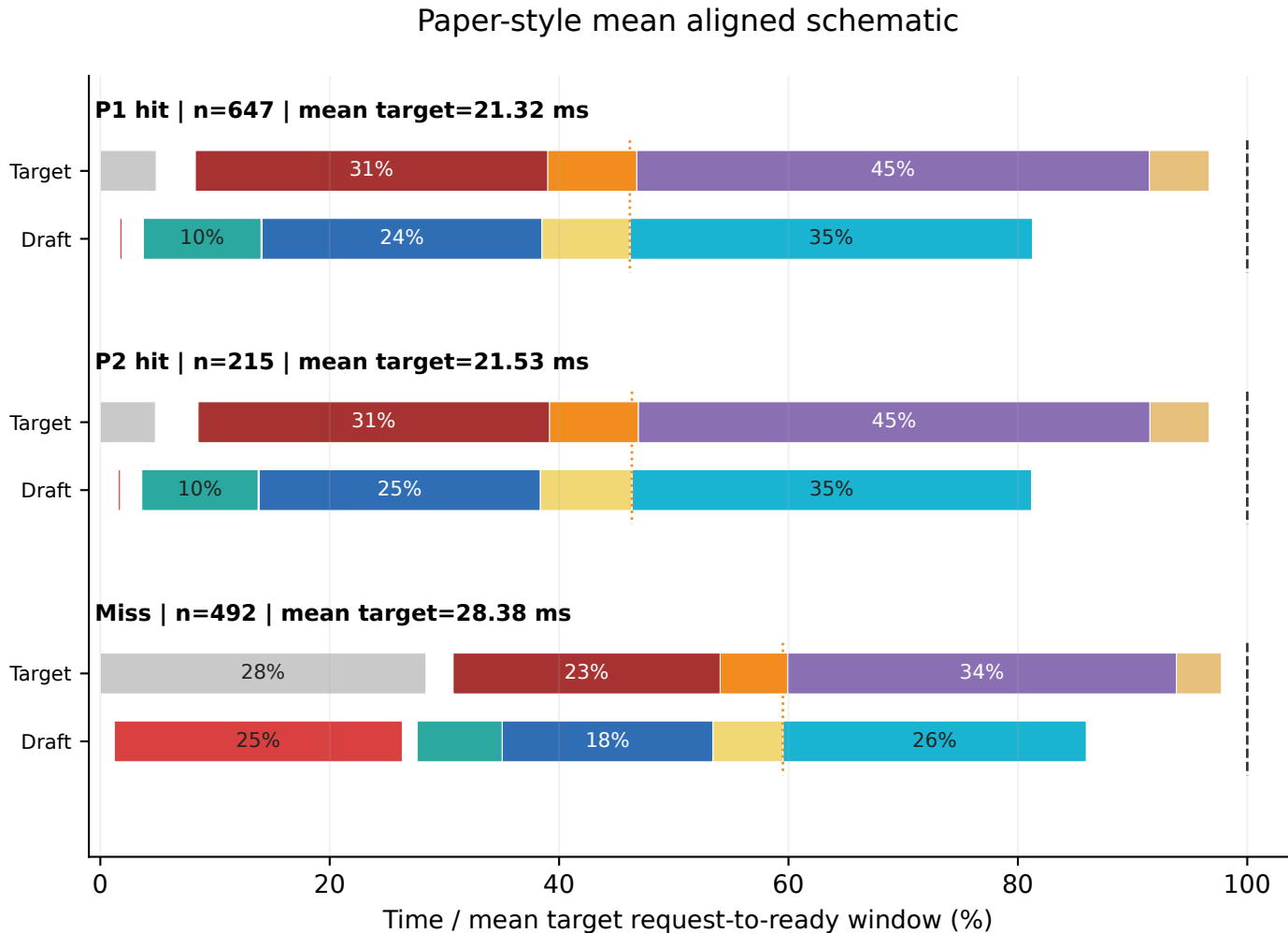
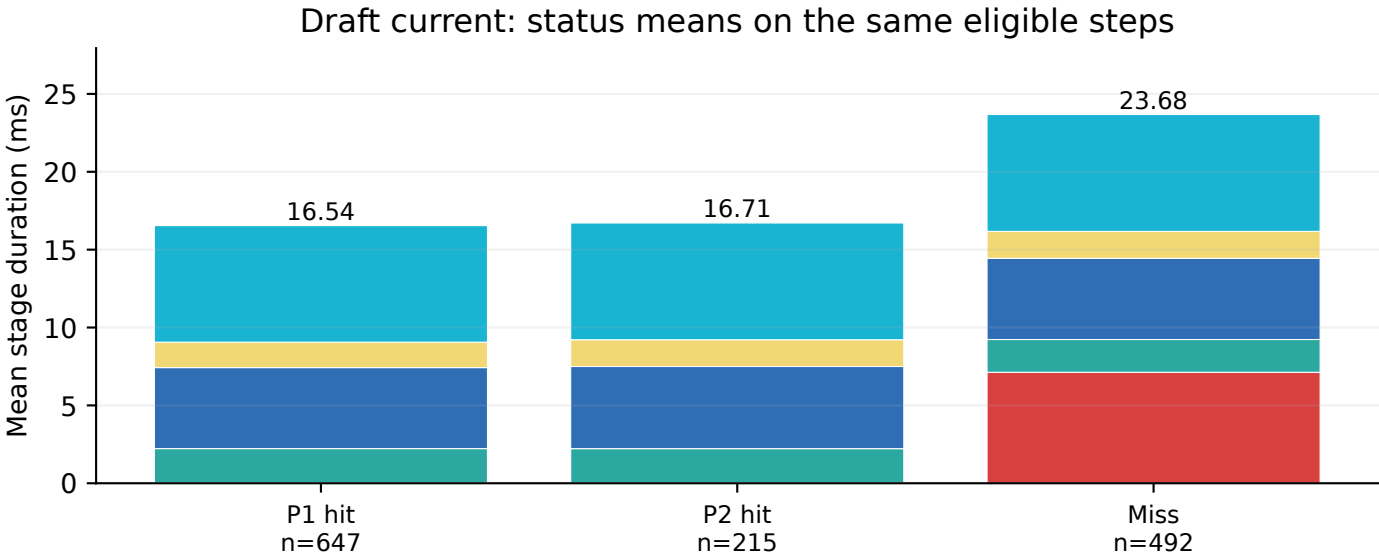
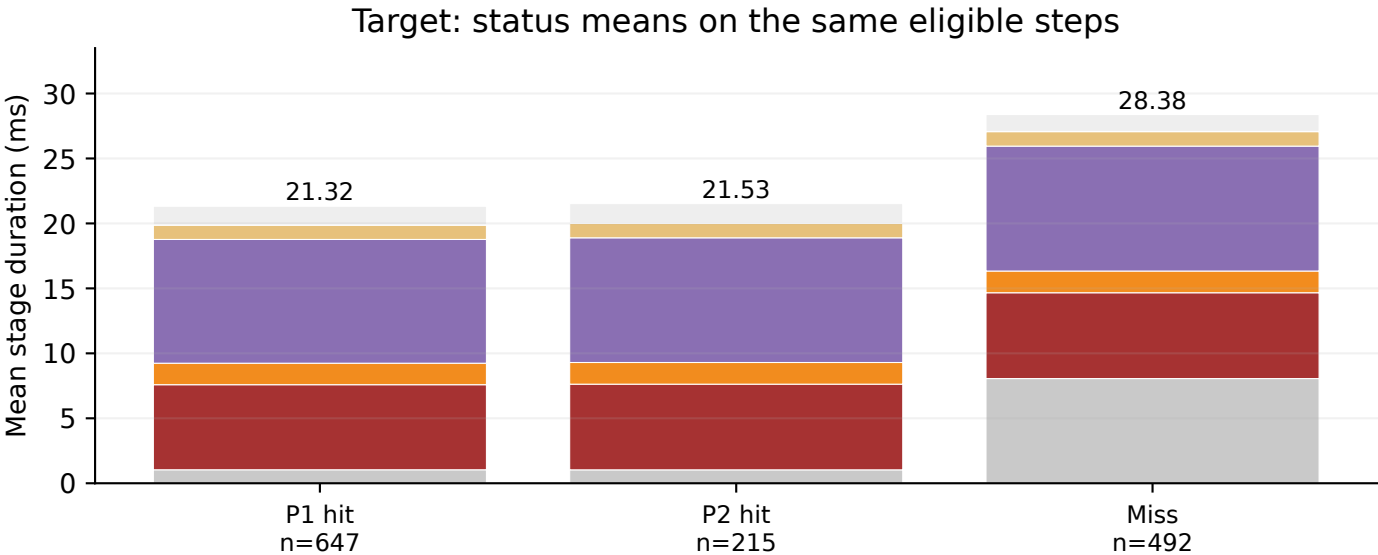
llama3_b1_combo_transfer_1_3_e14
DUET: K1/K2=1/3, exit=14 (0-based), C=3, N1/N2=2/6, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1595 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

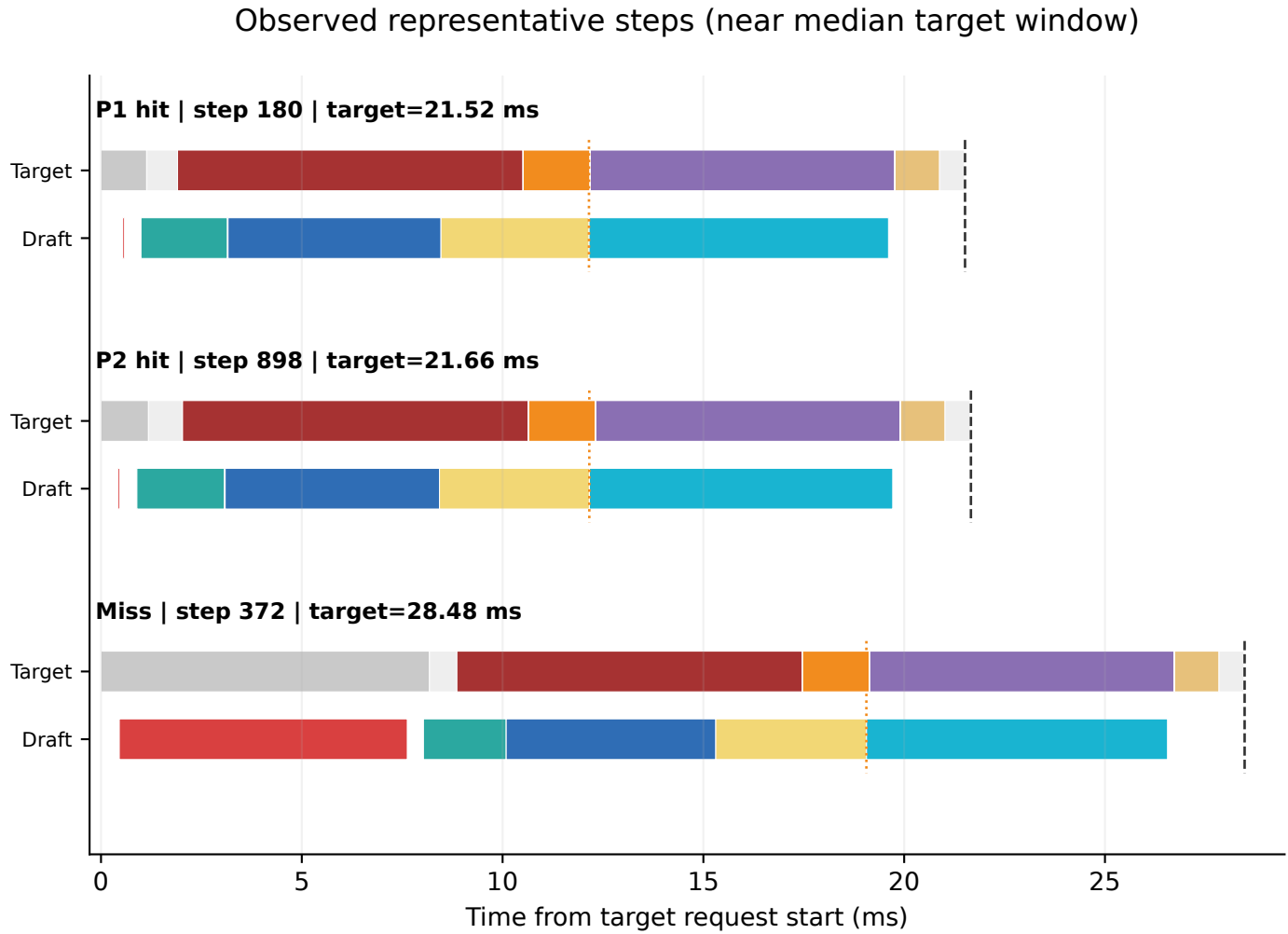
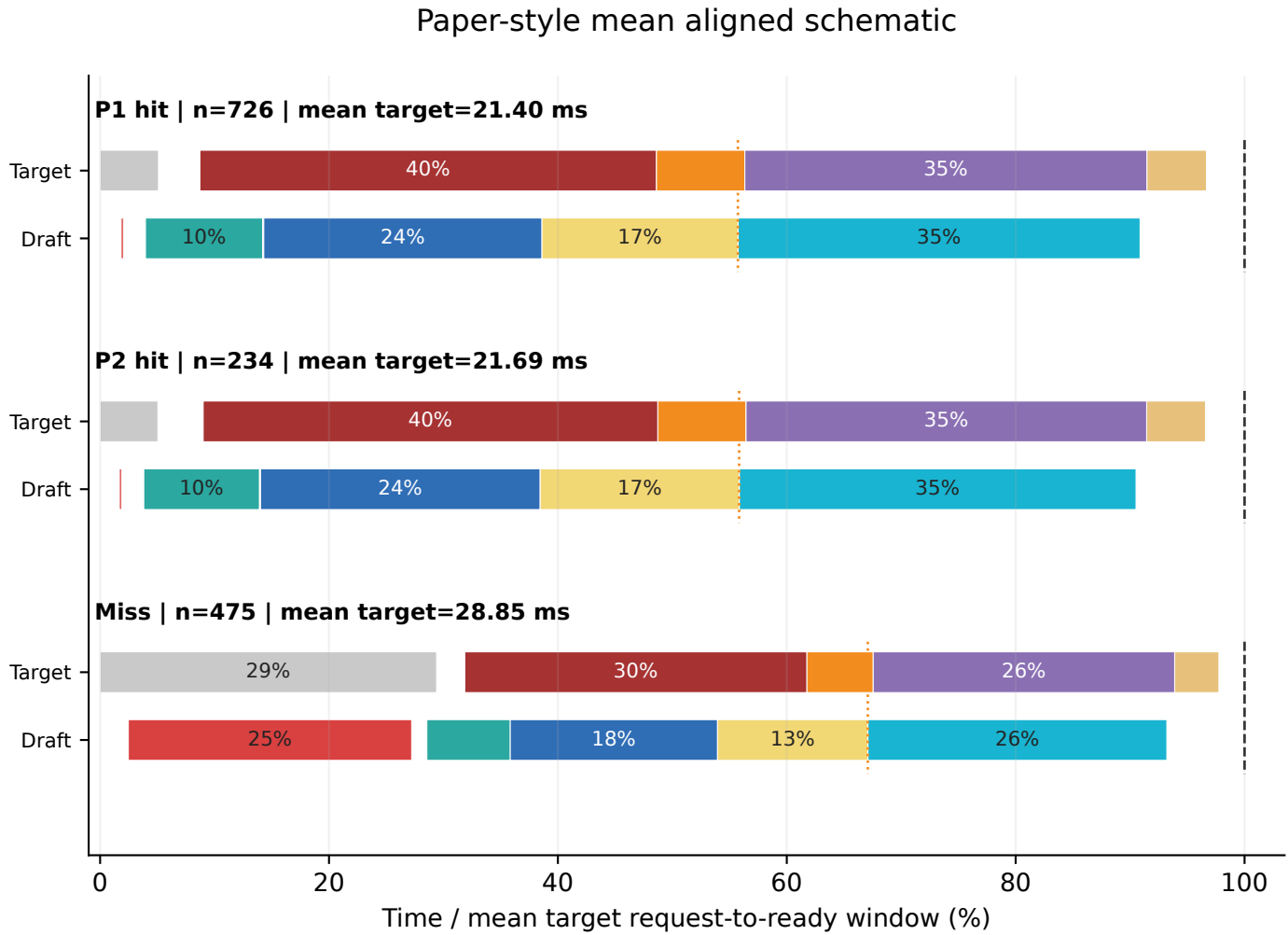
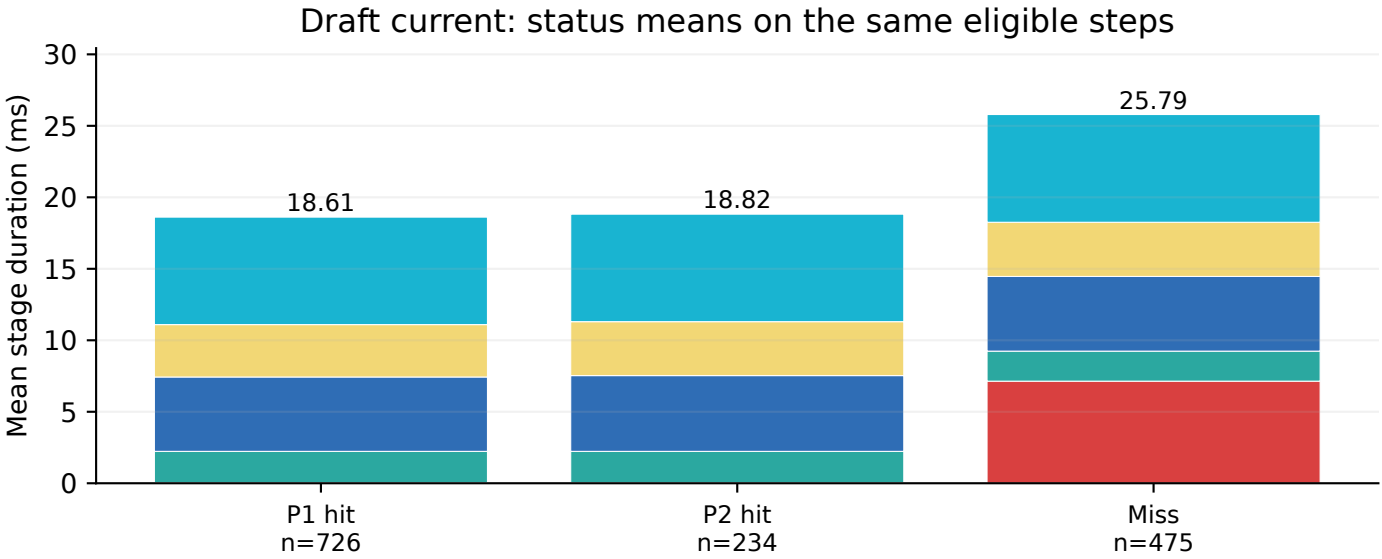
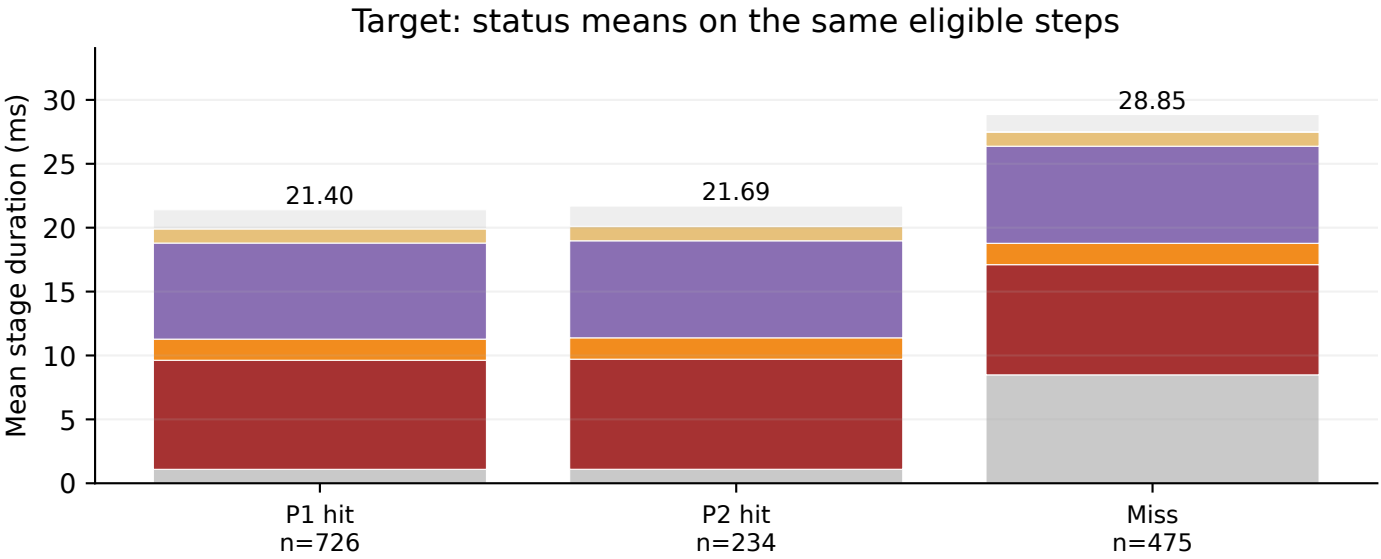
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_transfer_2_3_e12
DUET: K1/K2=2/3, exit=12 (0-based), C=3, N1/N2=4/6, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1354 eligible steps



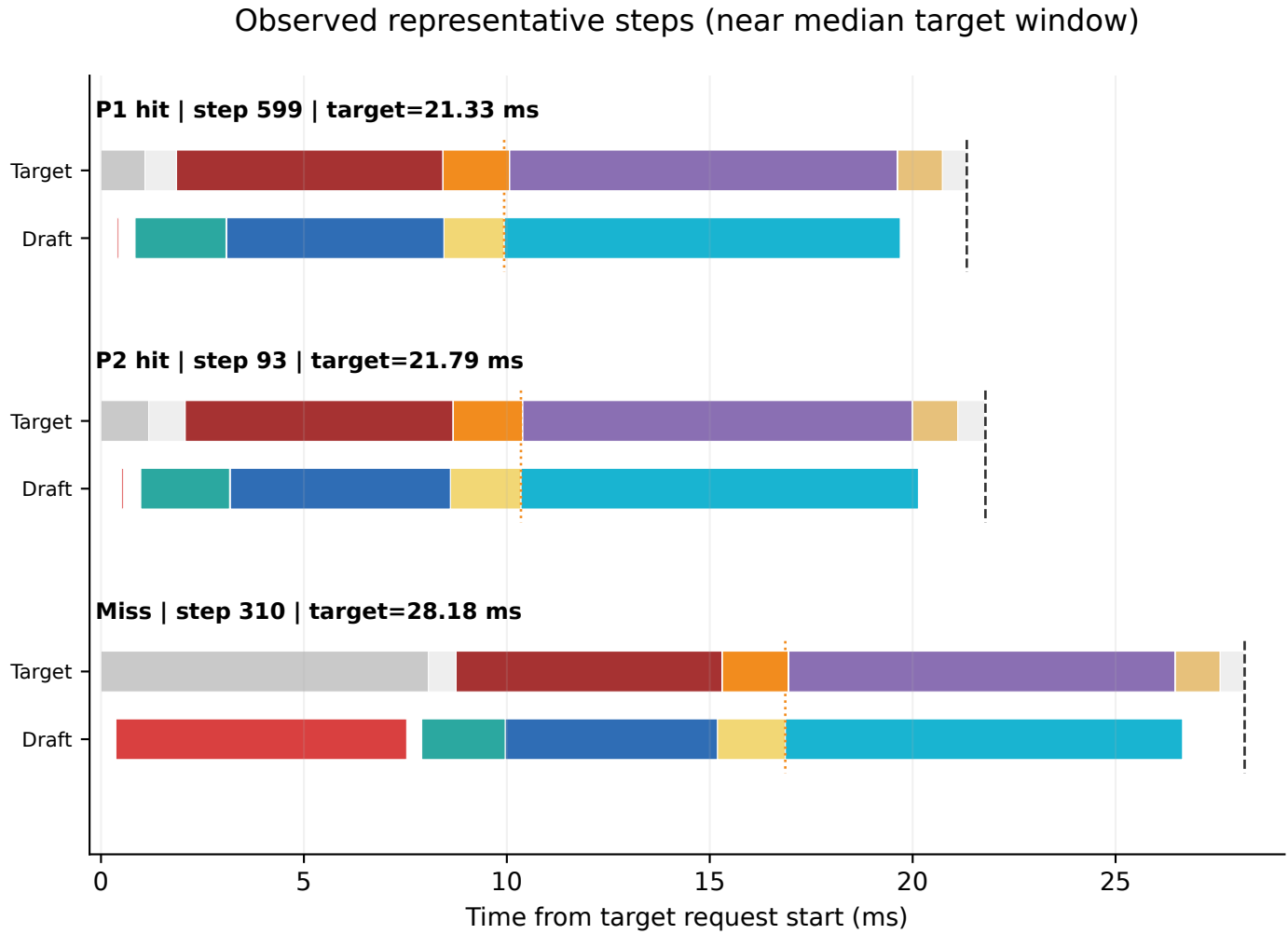
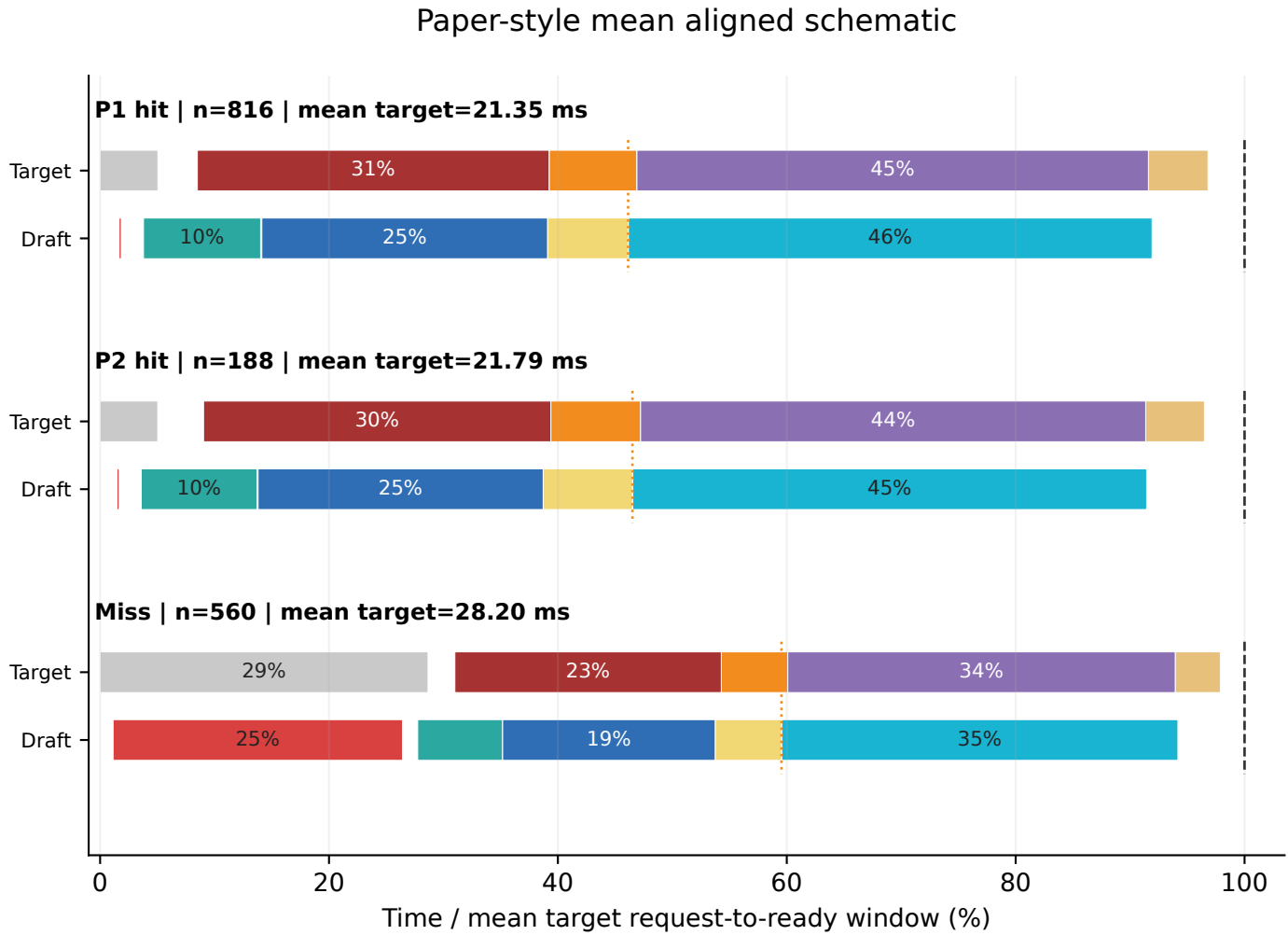
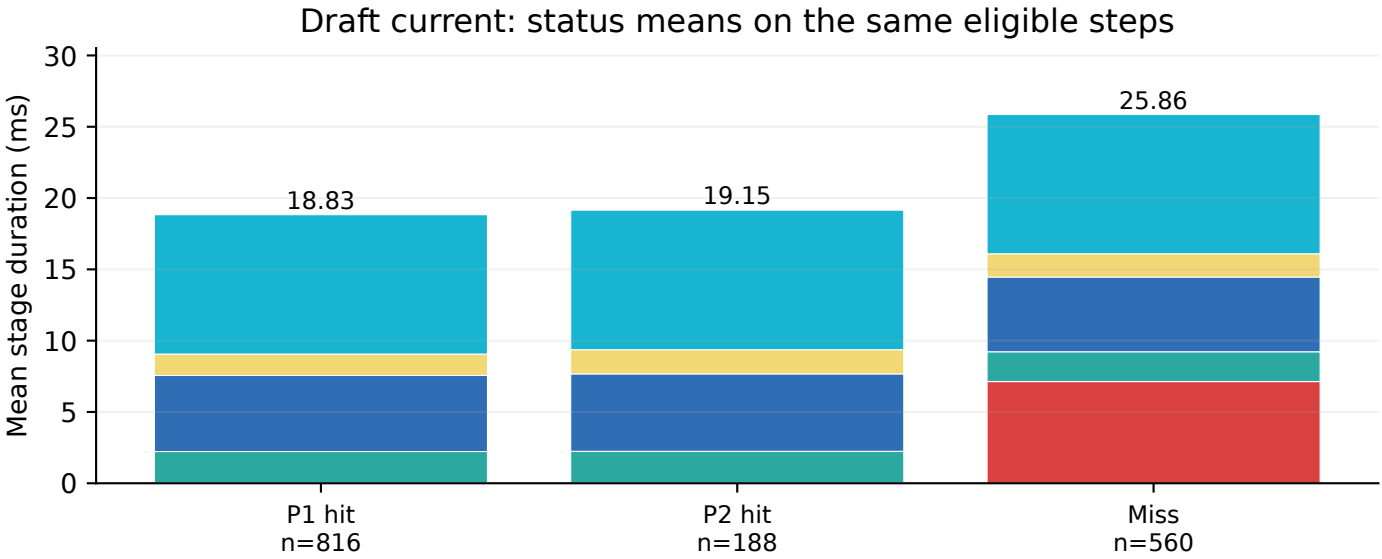
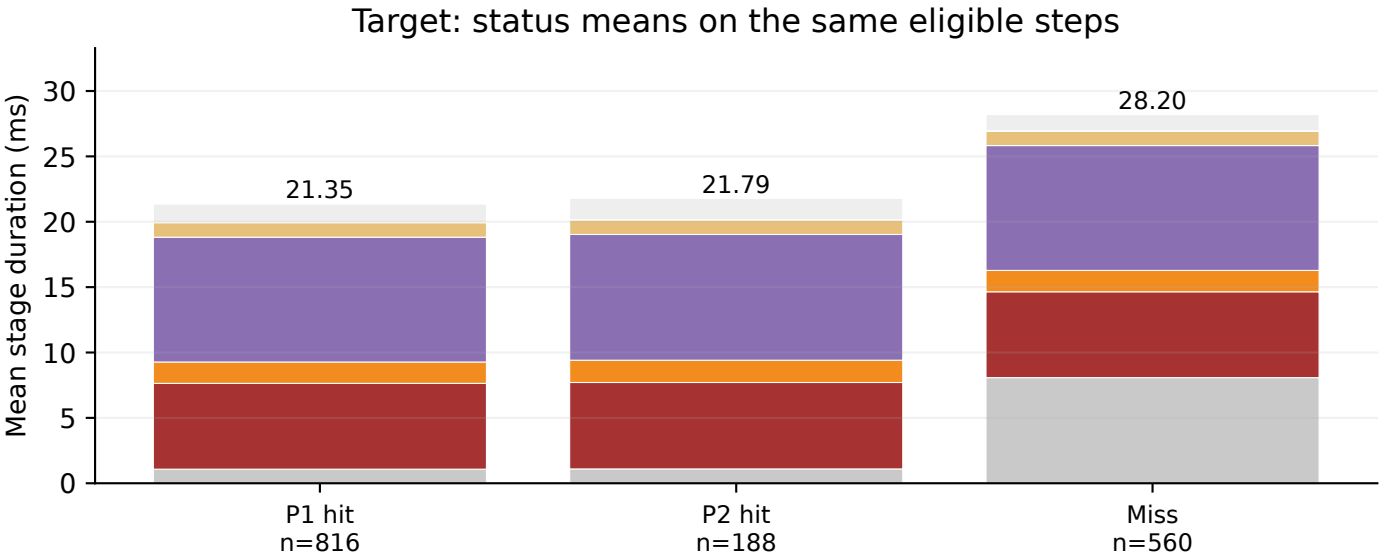
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_transfer_2_3_e16
DUET: K1/K2=2/3, exit=16 (0-based), C=3, N1/N2=4/6, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1435 eligible steps



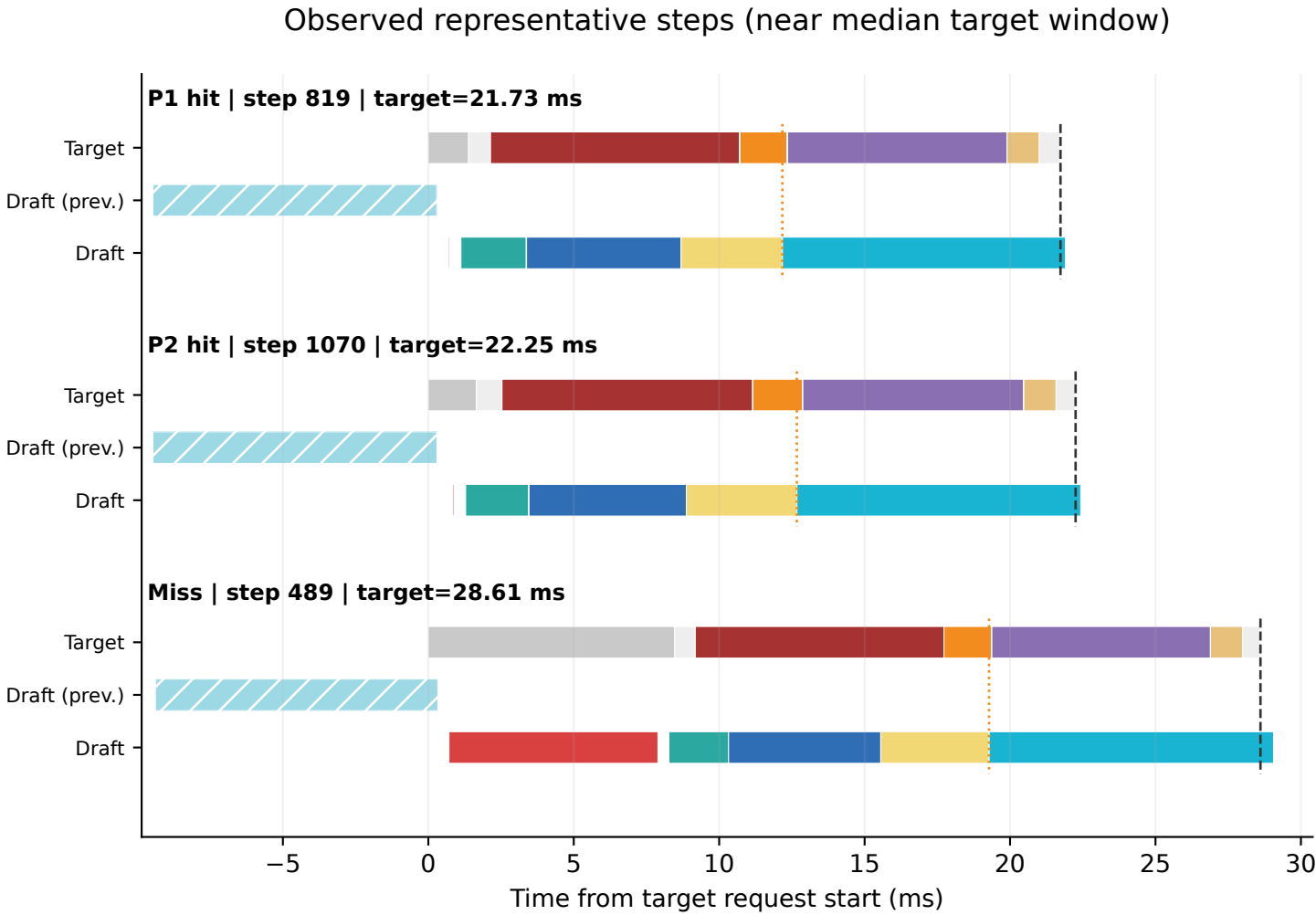
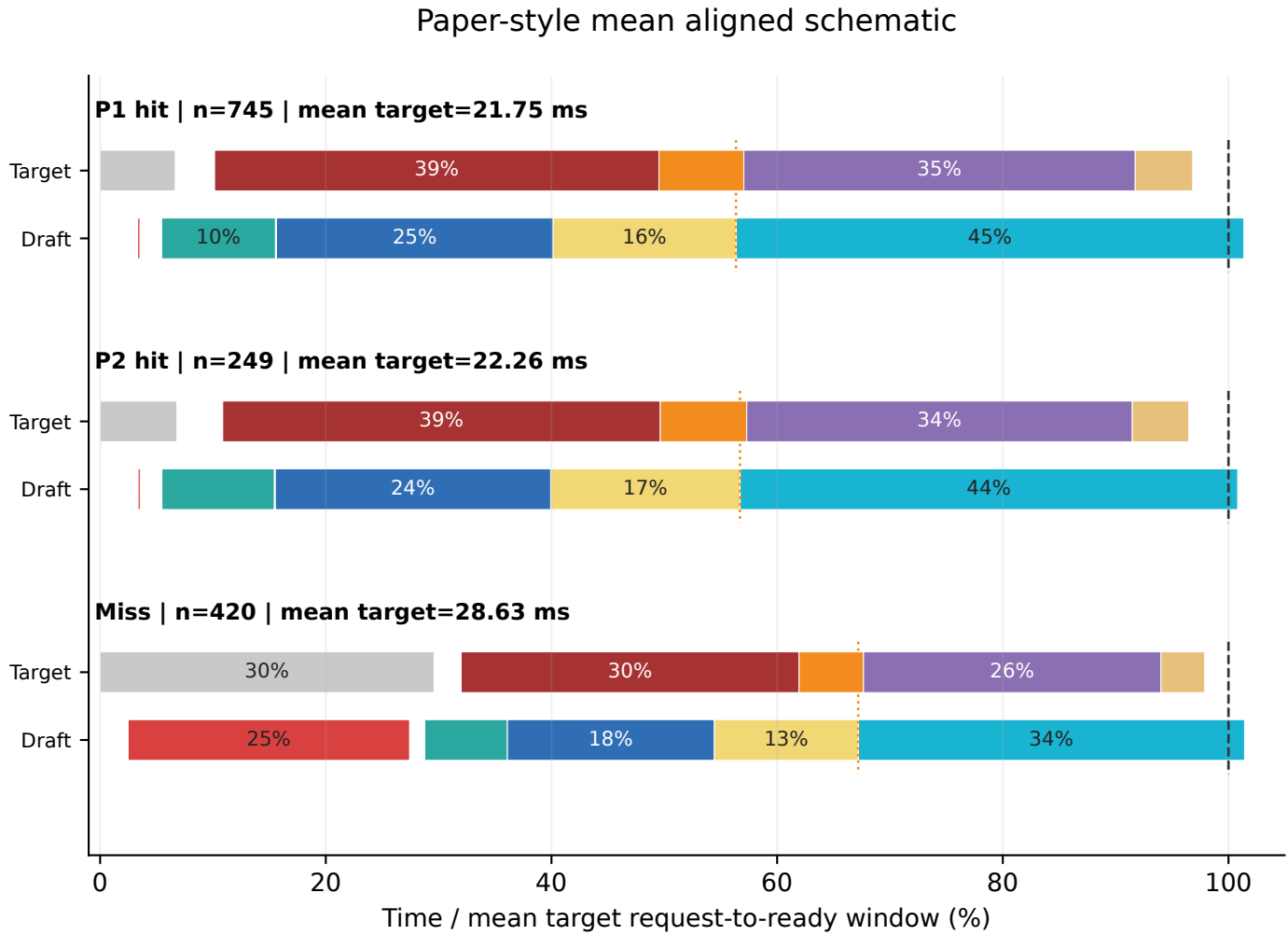
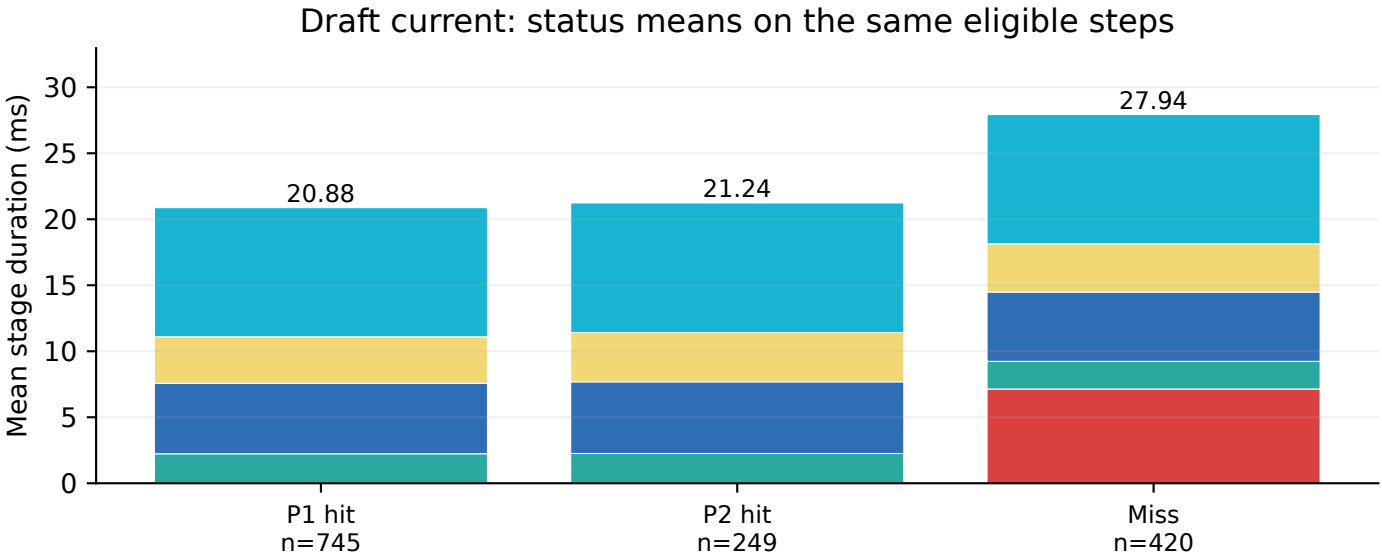
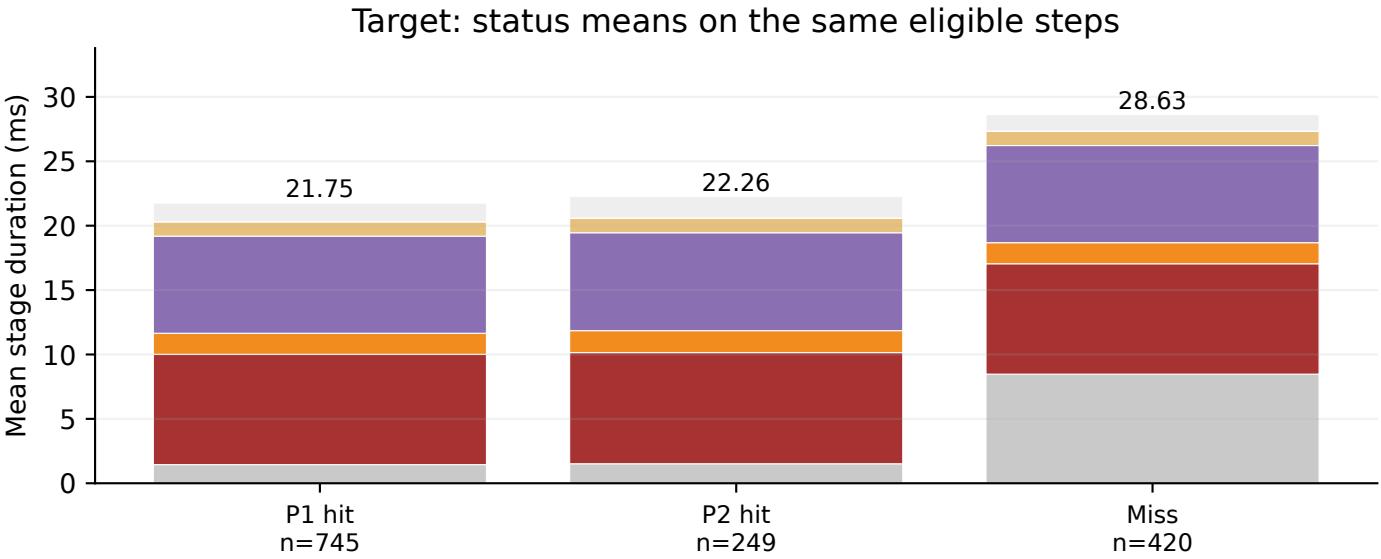
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_transfer_2_4_e12
DUET: K1/K2=2/4, exit=12 (0-based), C=3, N1/N2=4/8, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1564 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_combo_transfer_2_4_e16
DUET: K1/K2=2/4, exit=16 (0-based), C=3, N1/N2=4/8, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1414 eligible steps

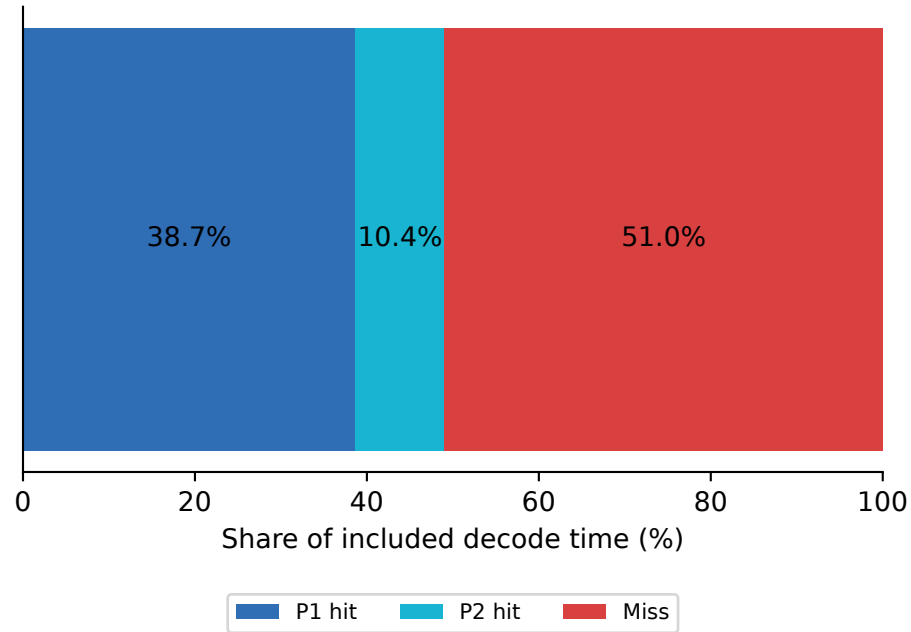


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

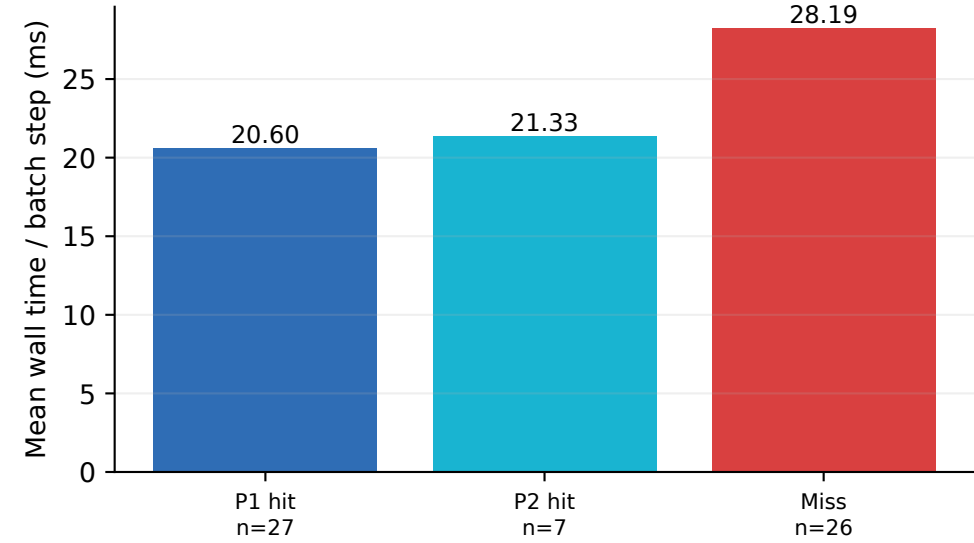
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_transfer_smoke
DUET: K1/K2=1/3, exit=12 (0-based), C=3, N1/N2=2/6, U=2, W=3, trim/stream=0/0
Last pass: B=1, T=0.7, N=8 | AL*=2.267, hit=55.2%, TPS*=94.55

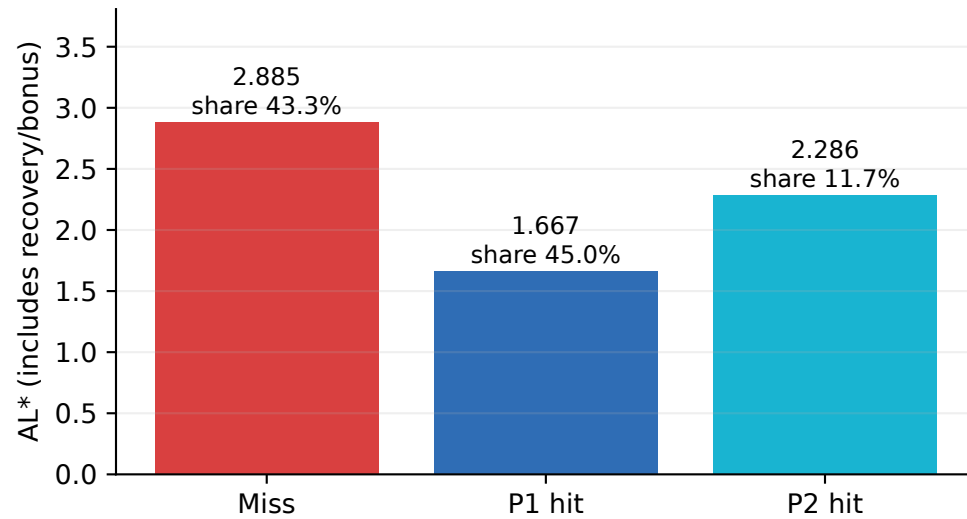
Measured wall time by batch cache status



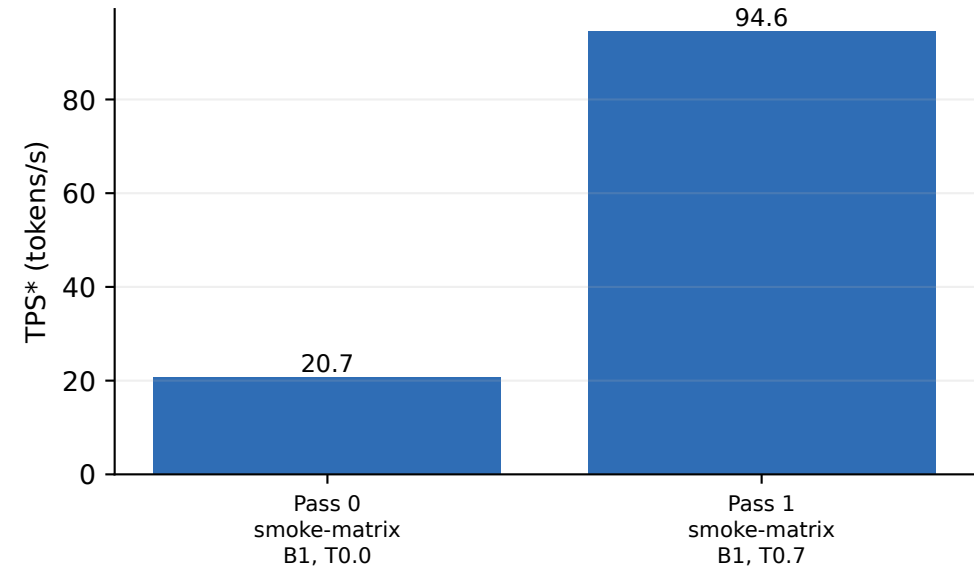
Boundary-free steps, including slow steps



Conditional AL* and source share

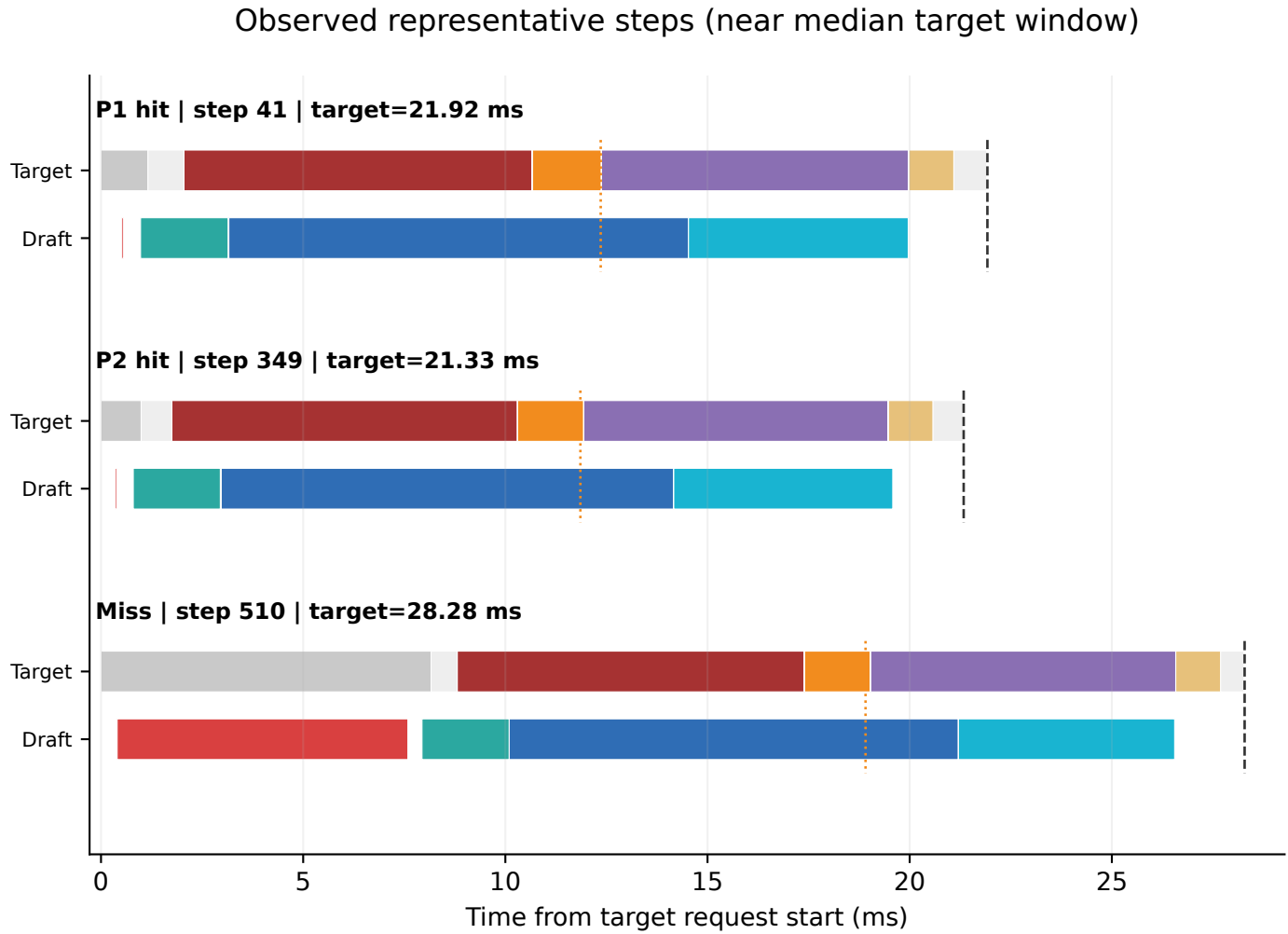
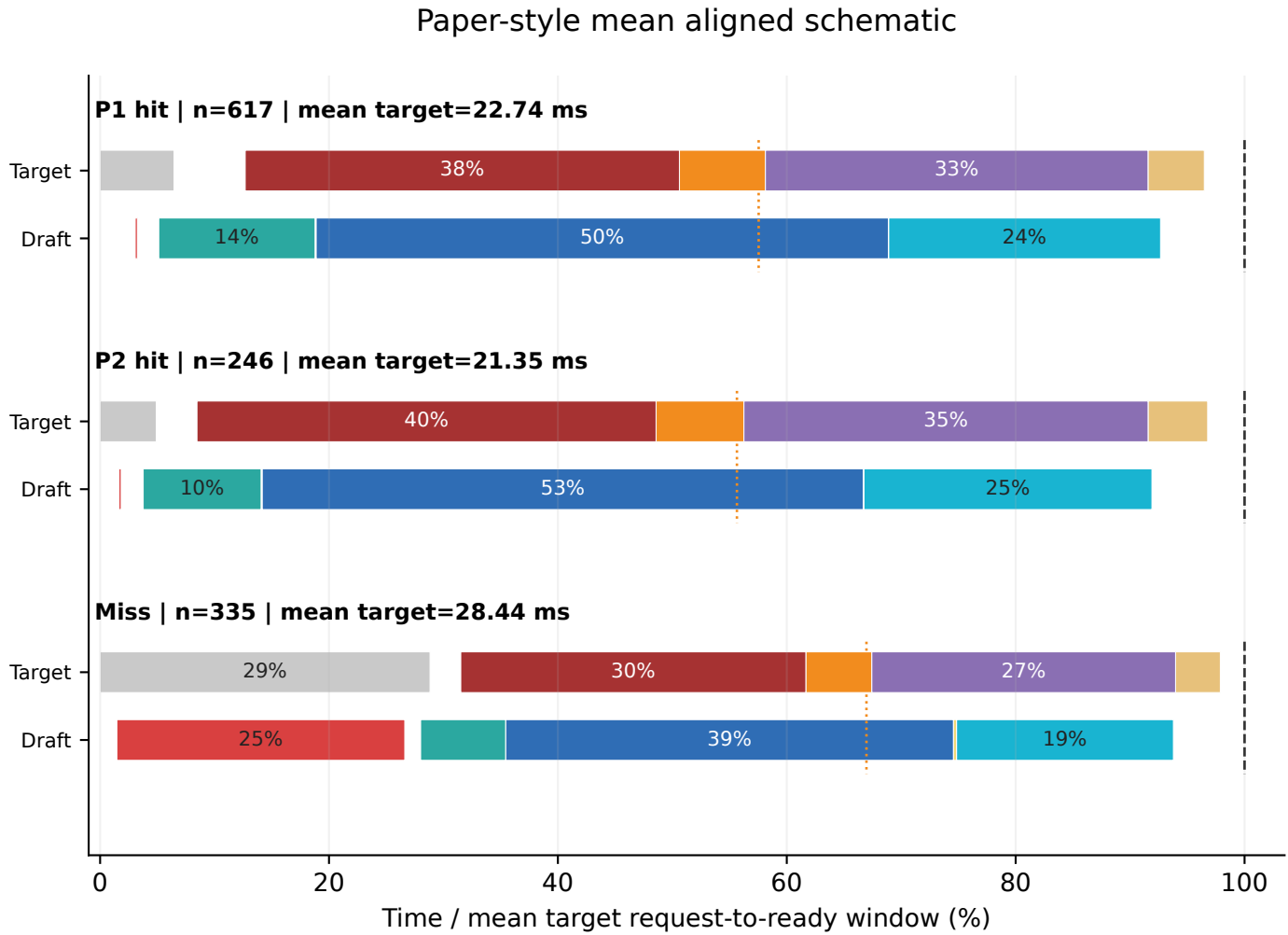
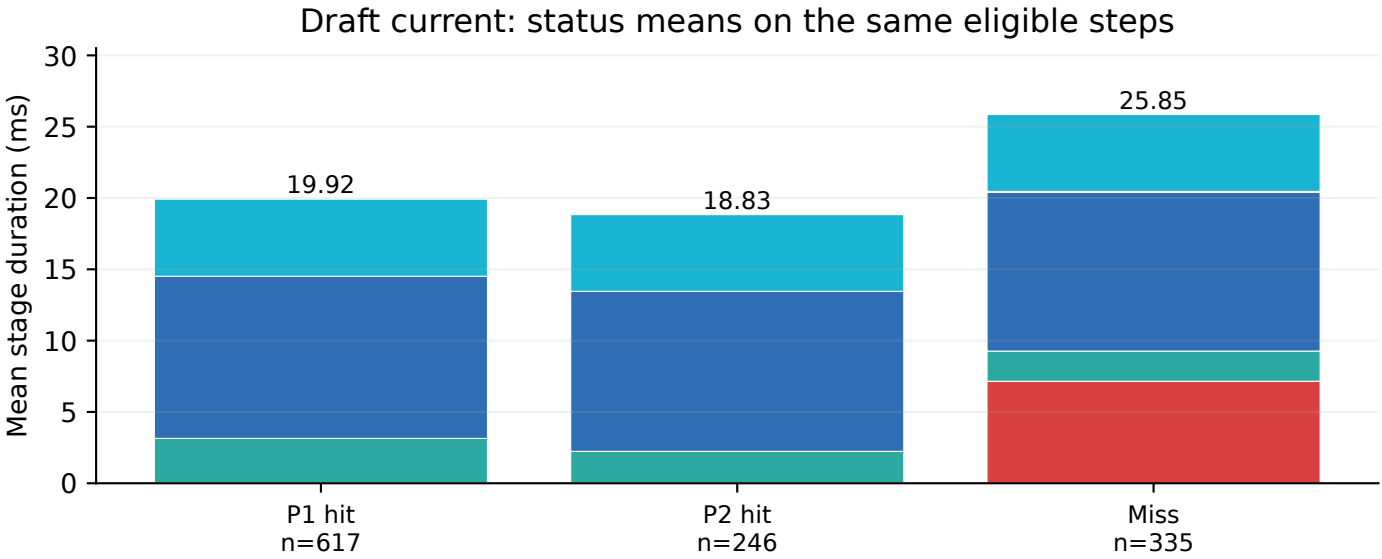
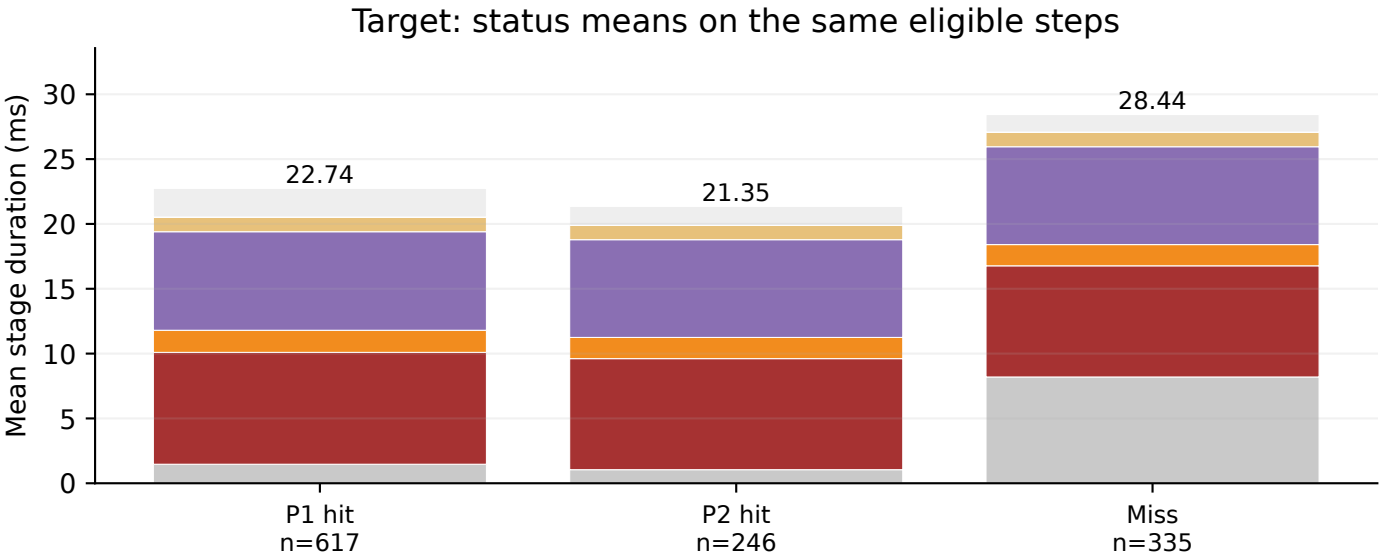


All saved passes; gray = warmup



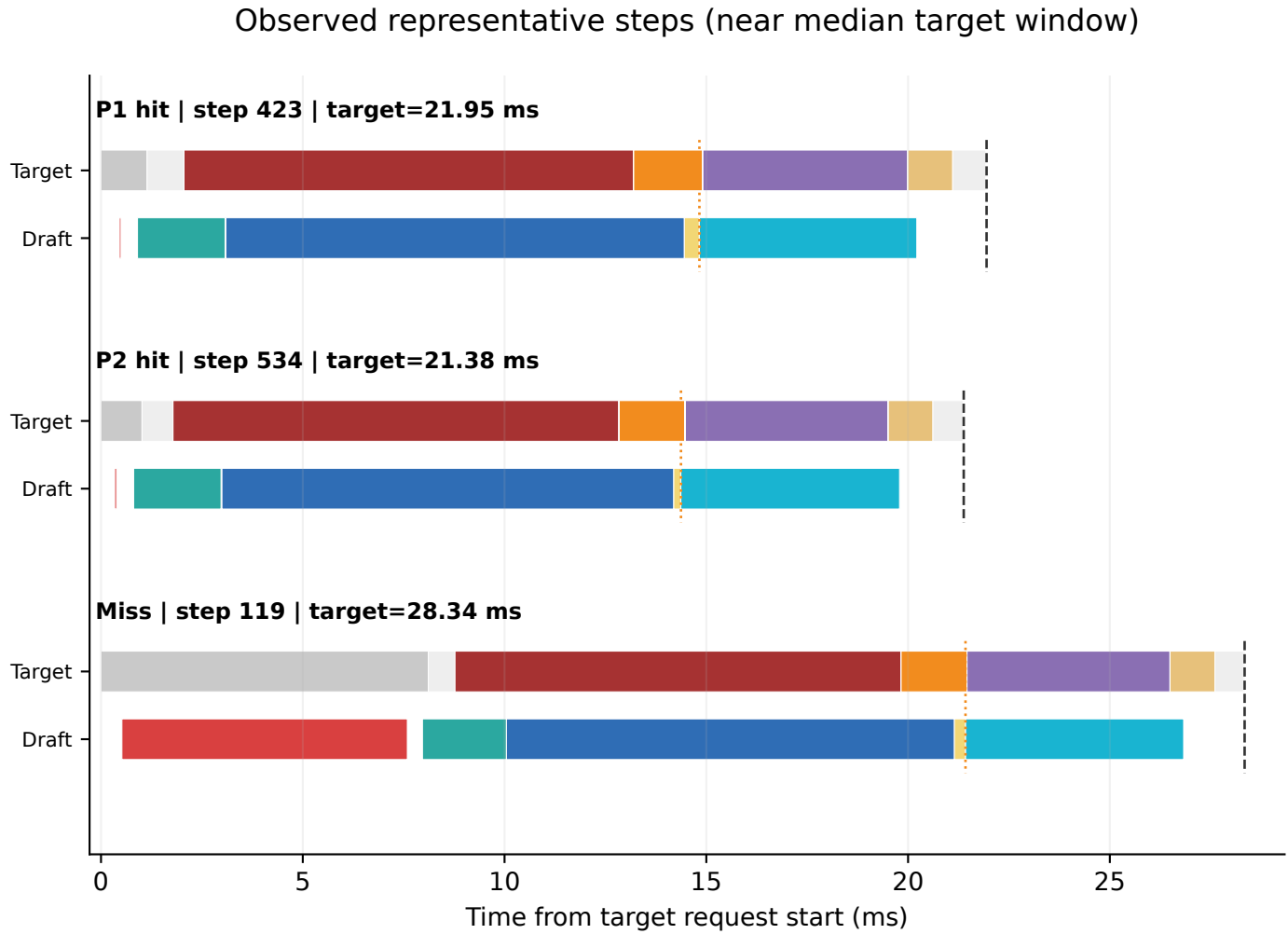
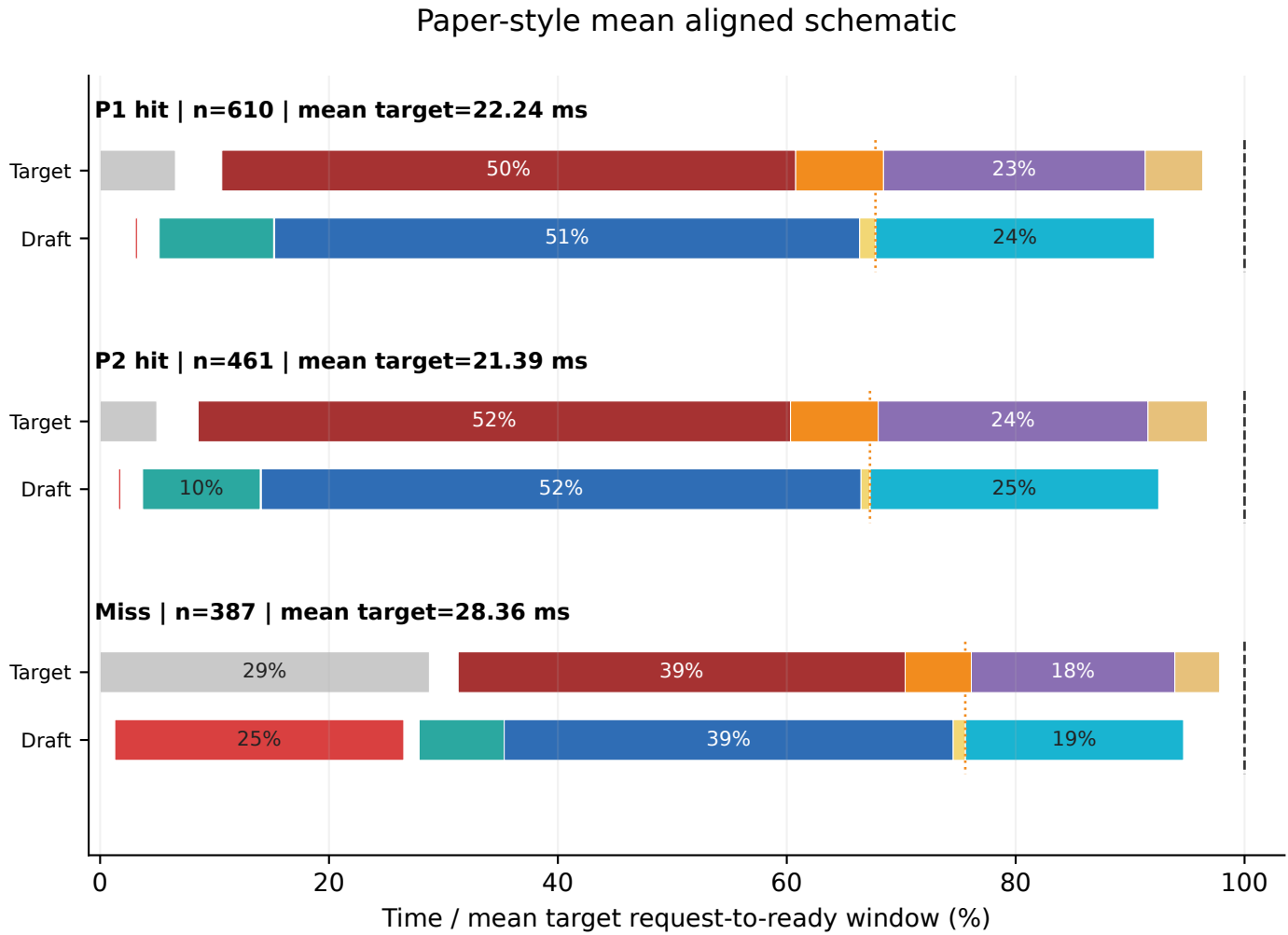
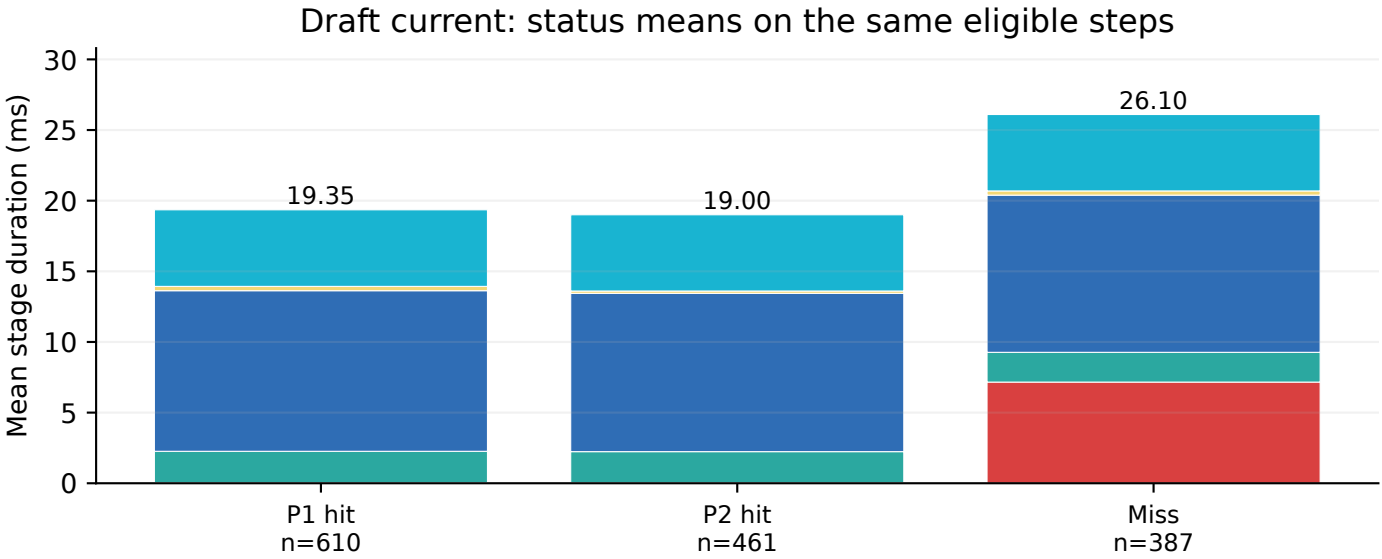
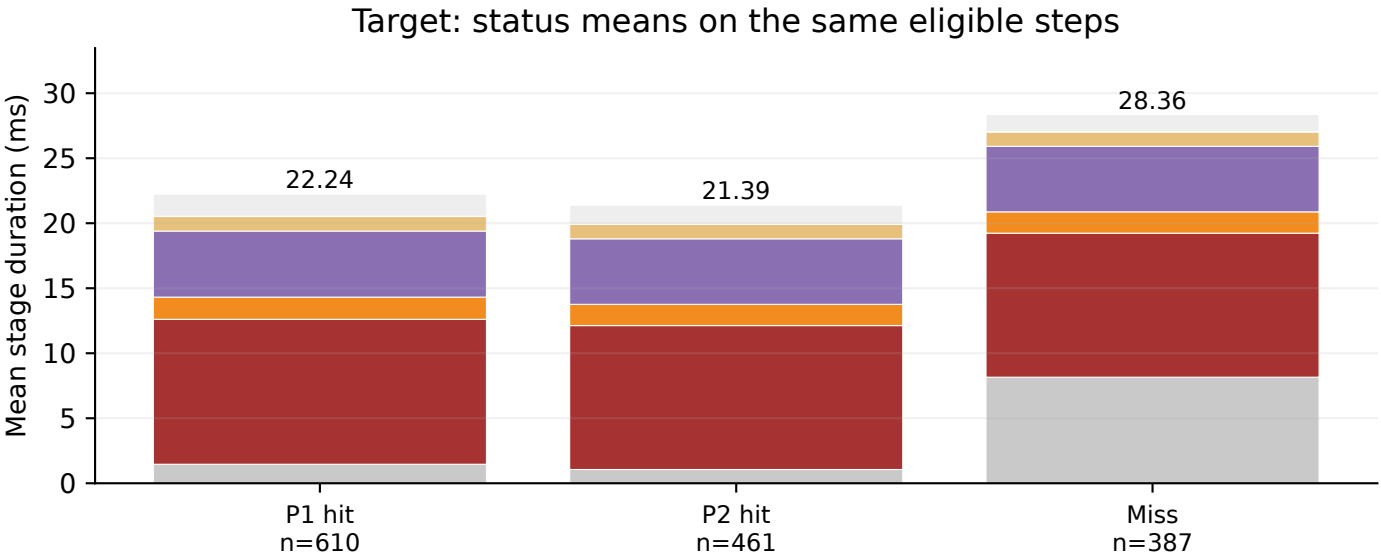
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_duet_e16_k4_2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1198 eligible steps



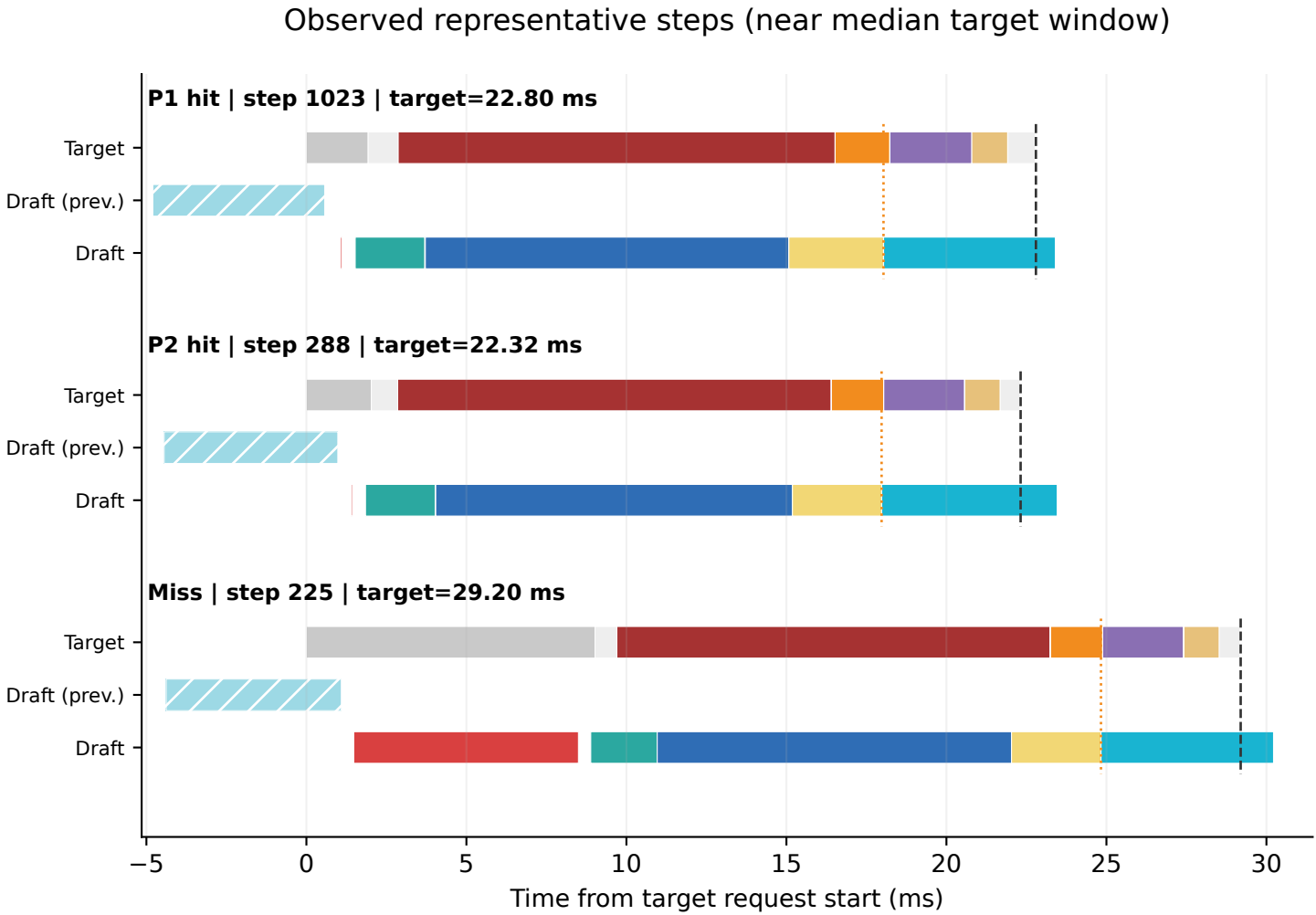
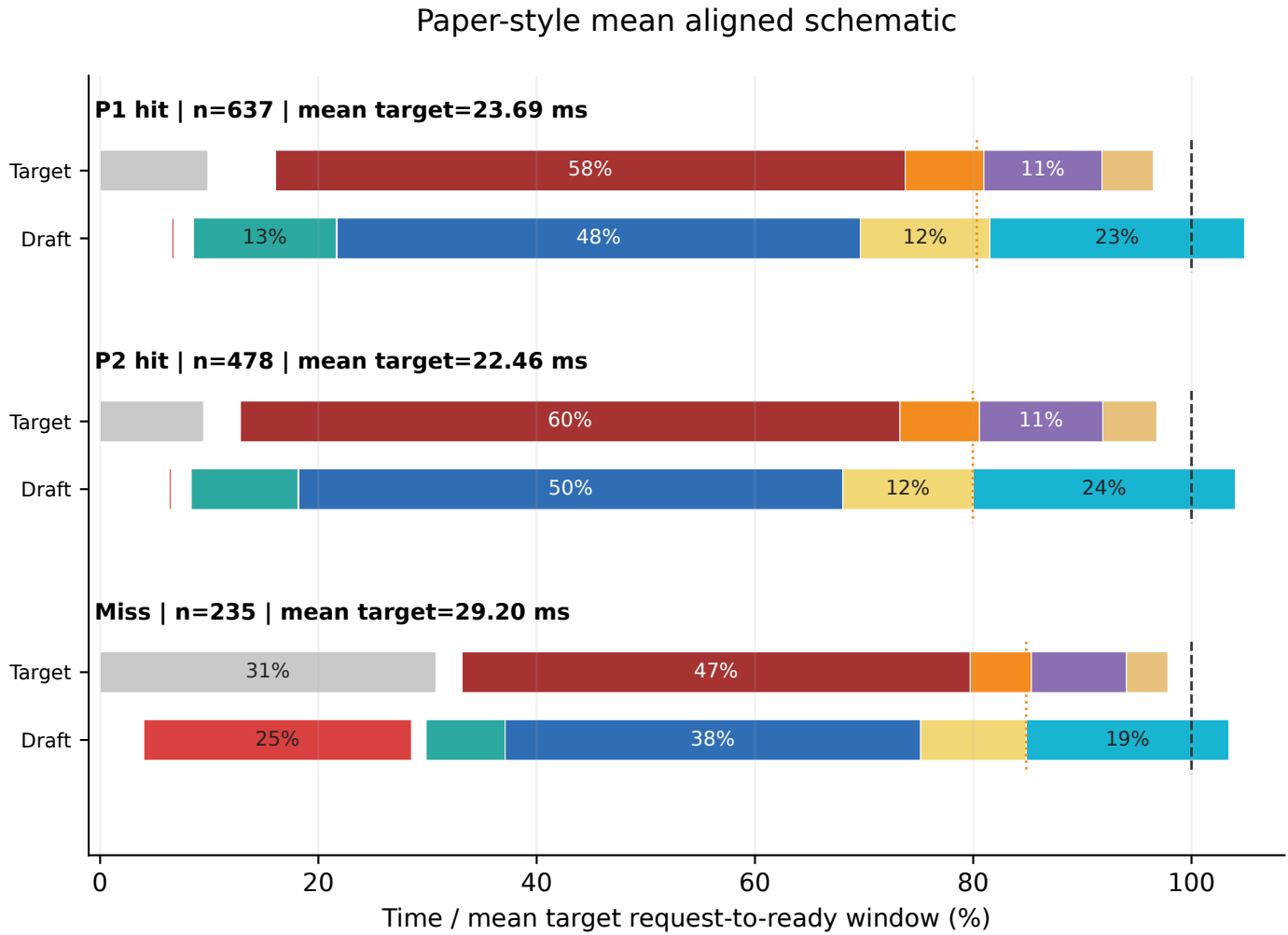
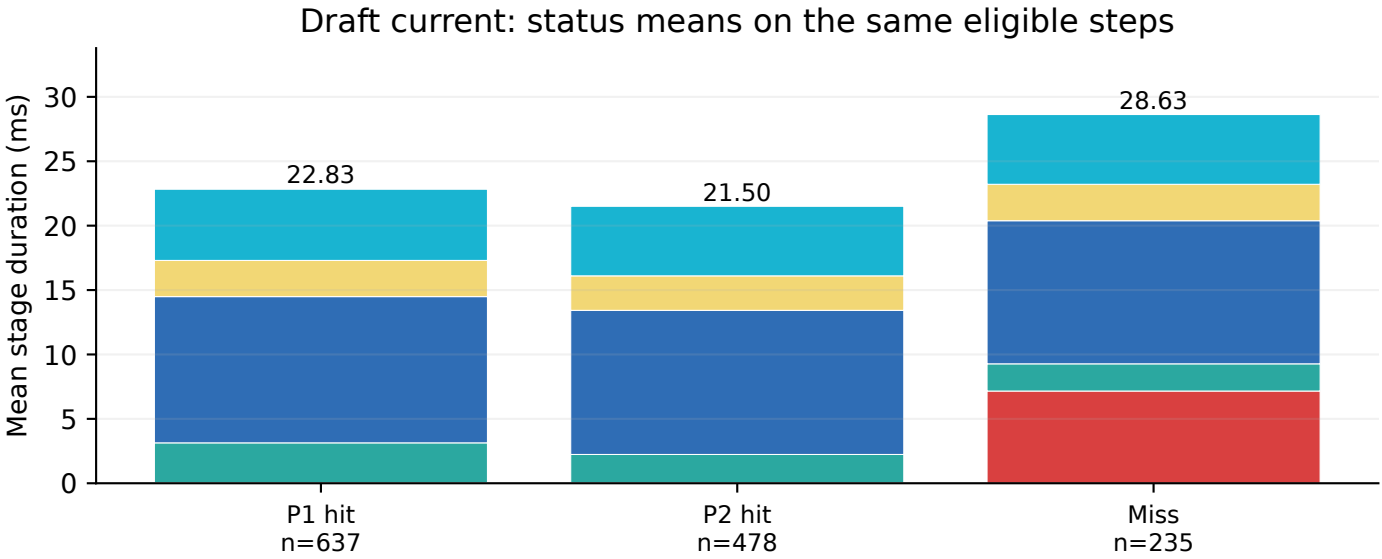
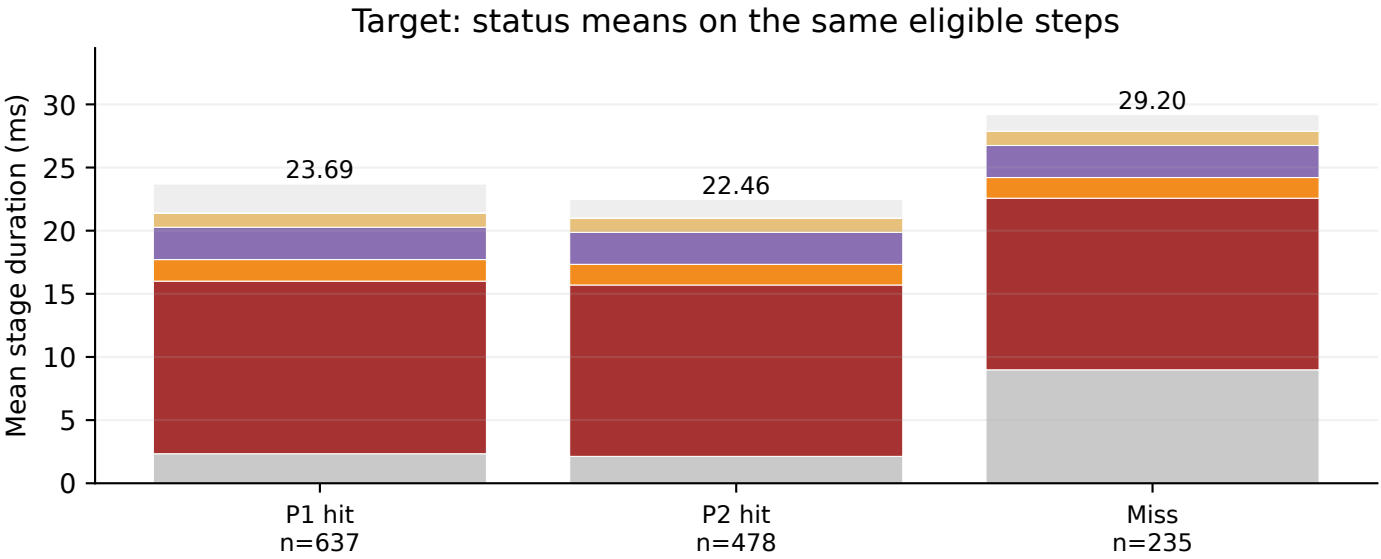
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_duet_e21_k4_2
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1458 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

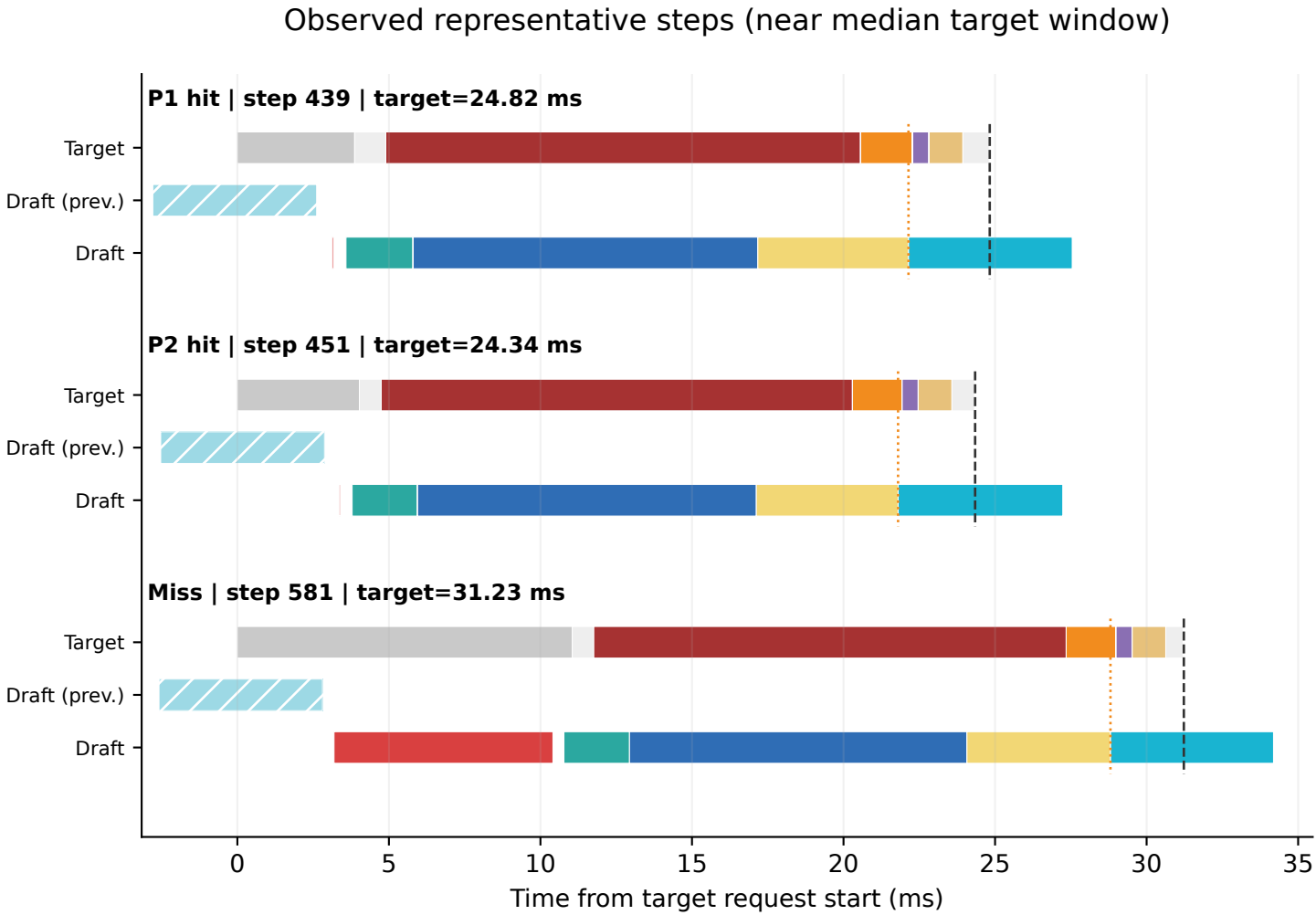
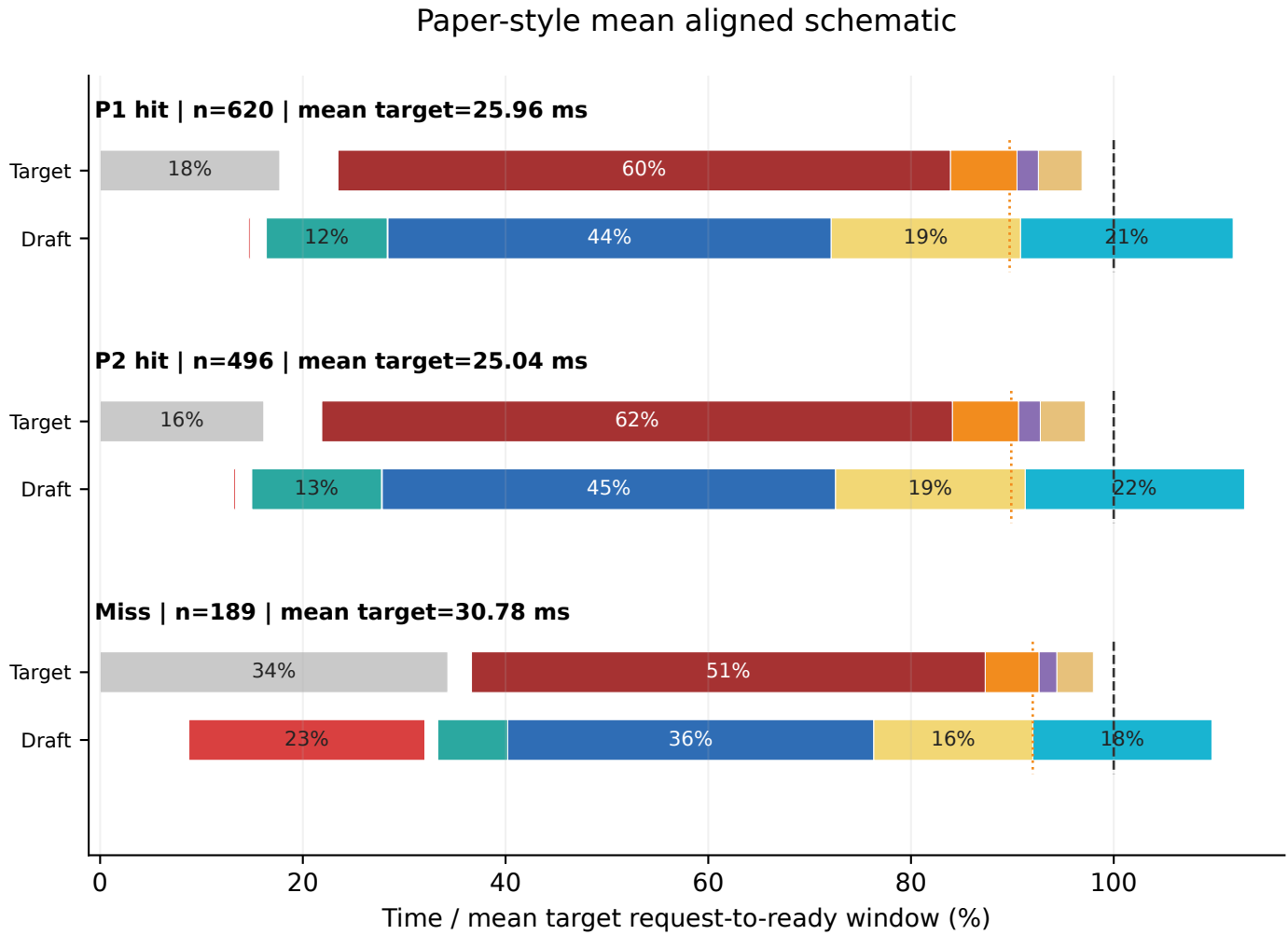
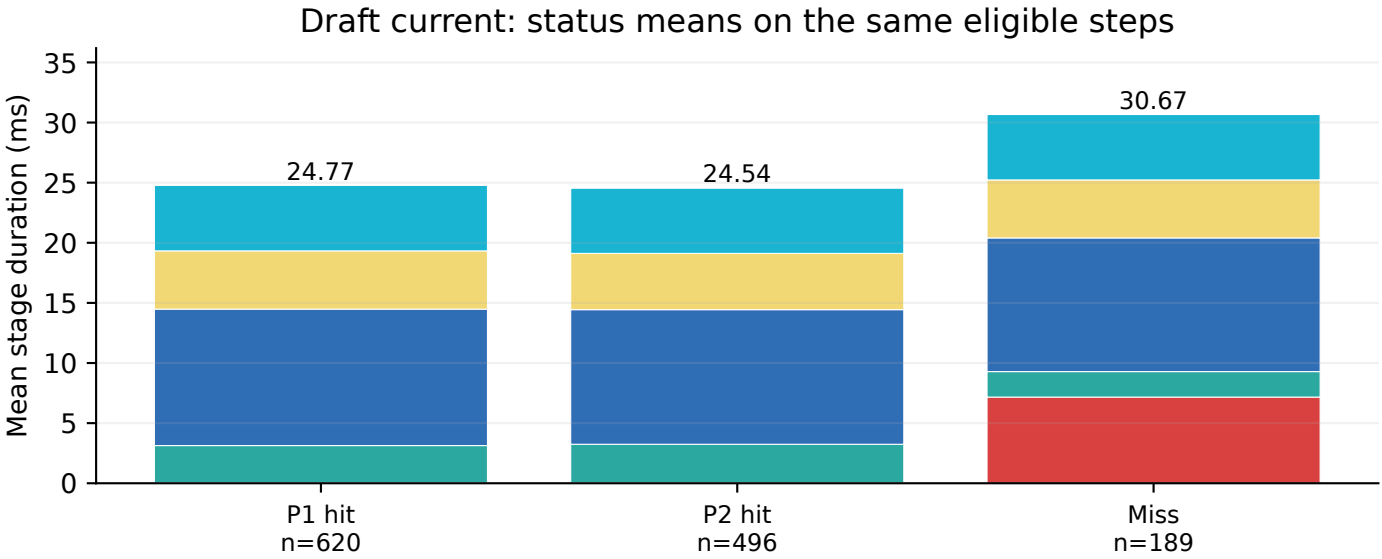
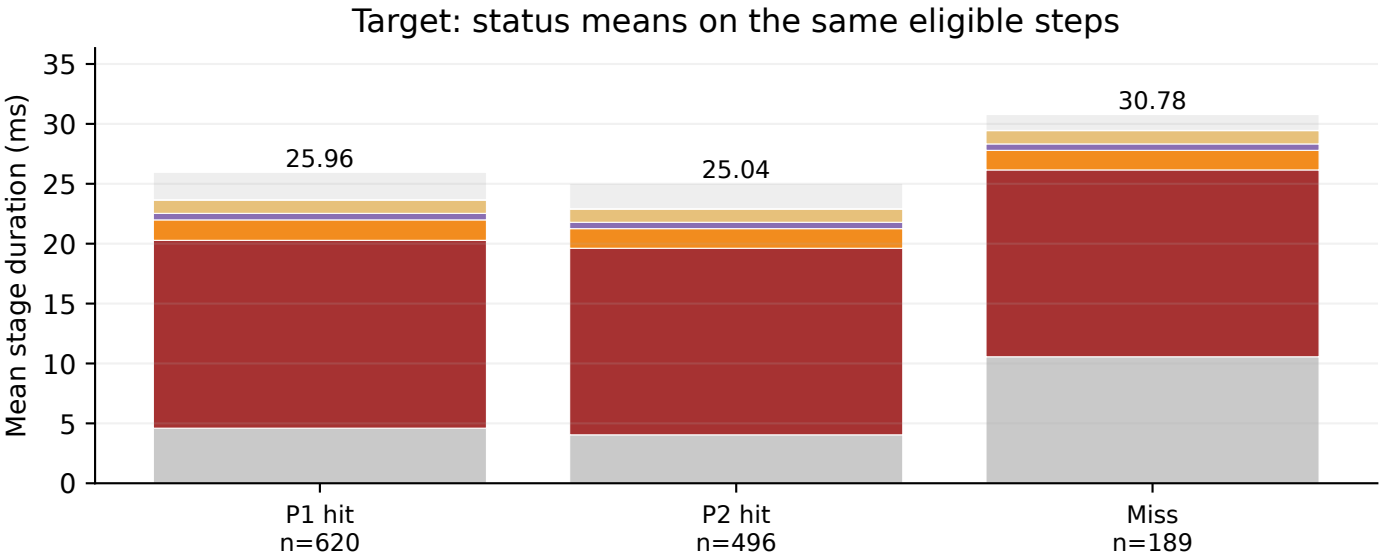
llama3_b1_duet_e26_k4_2
DUET: K1/K2=4/2, exit=26 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1350 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

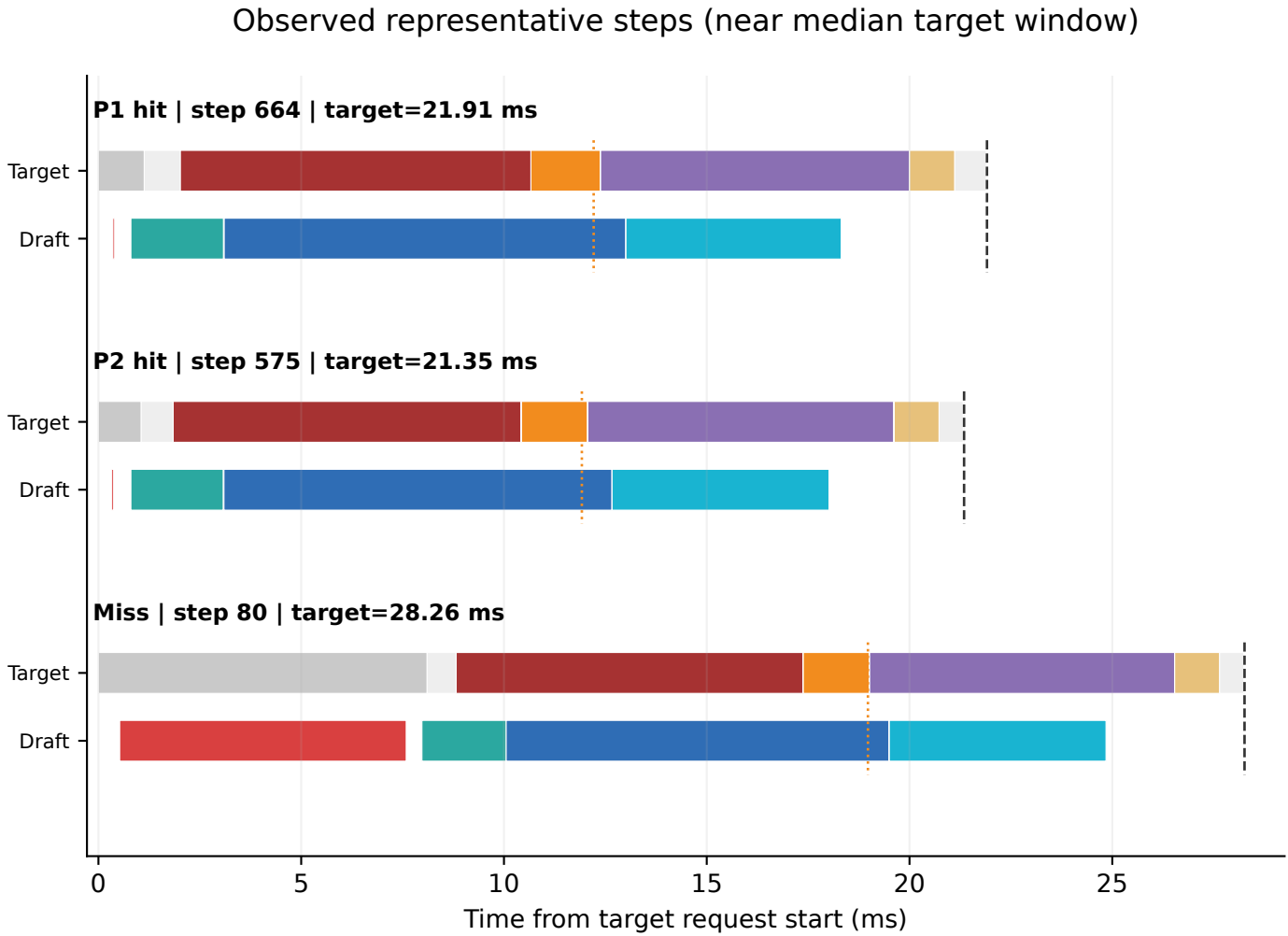
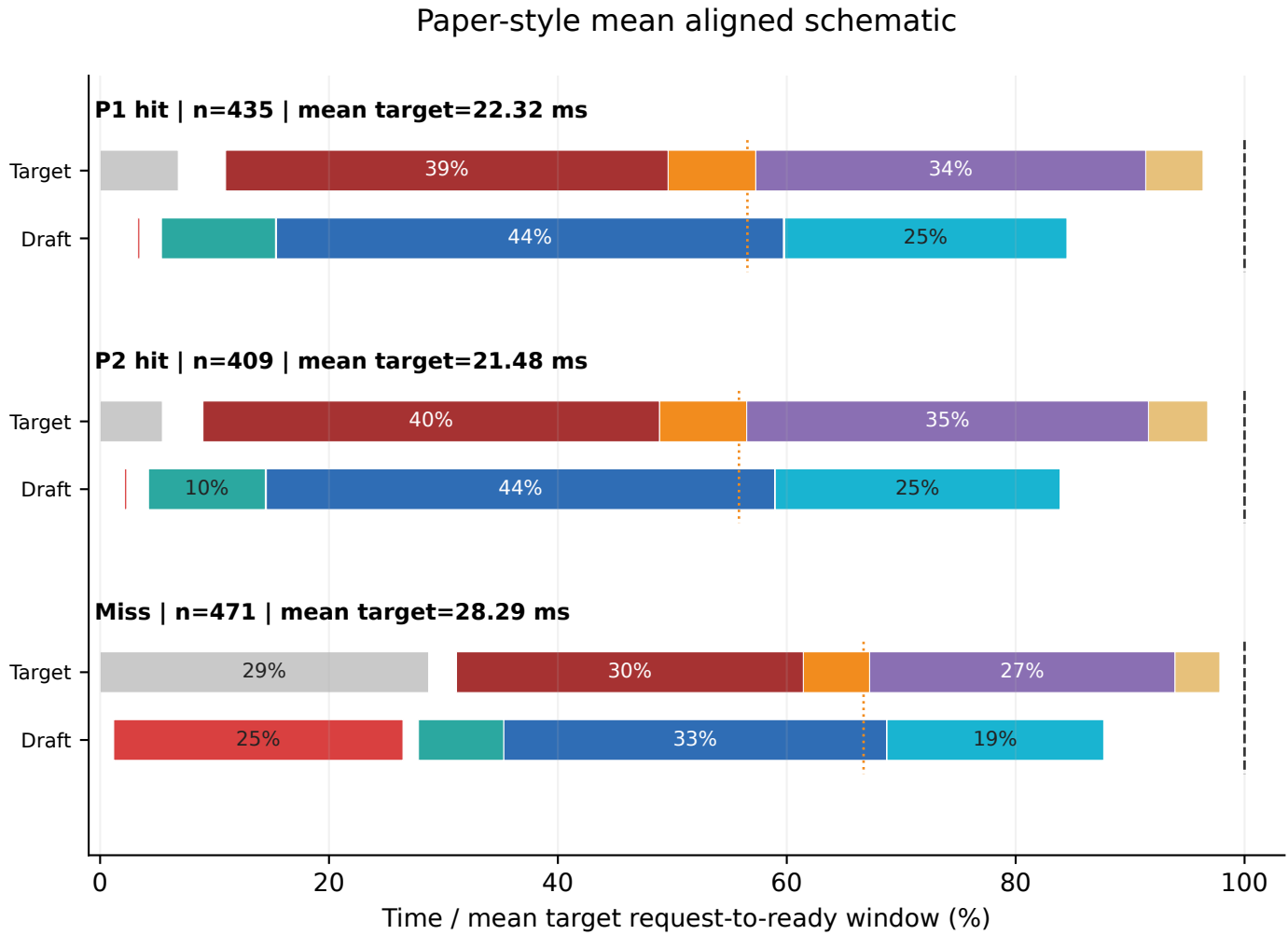
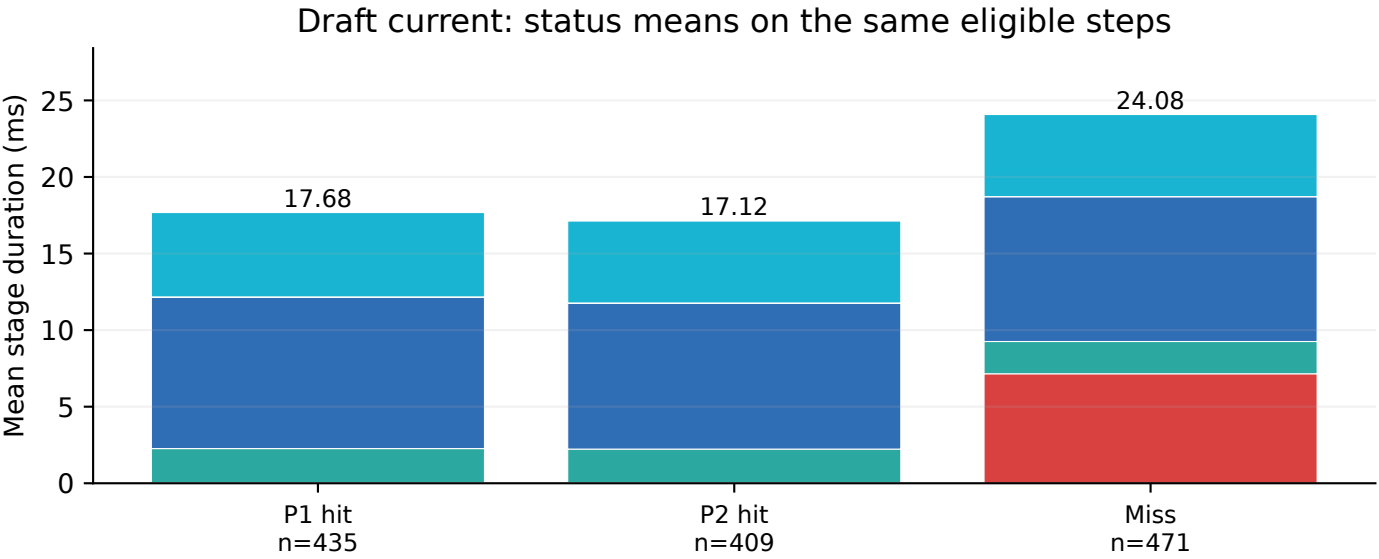
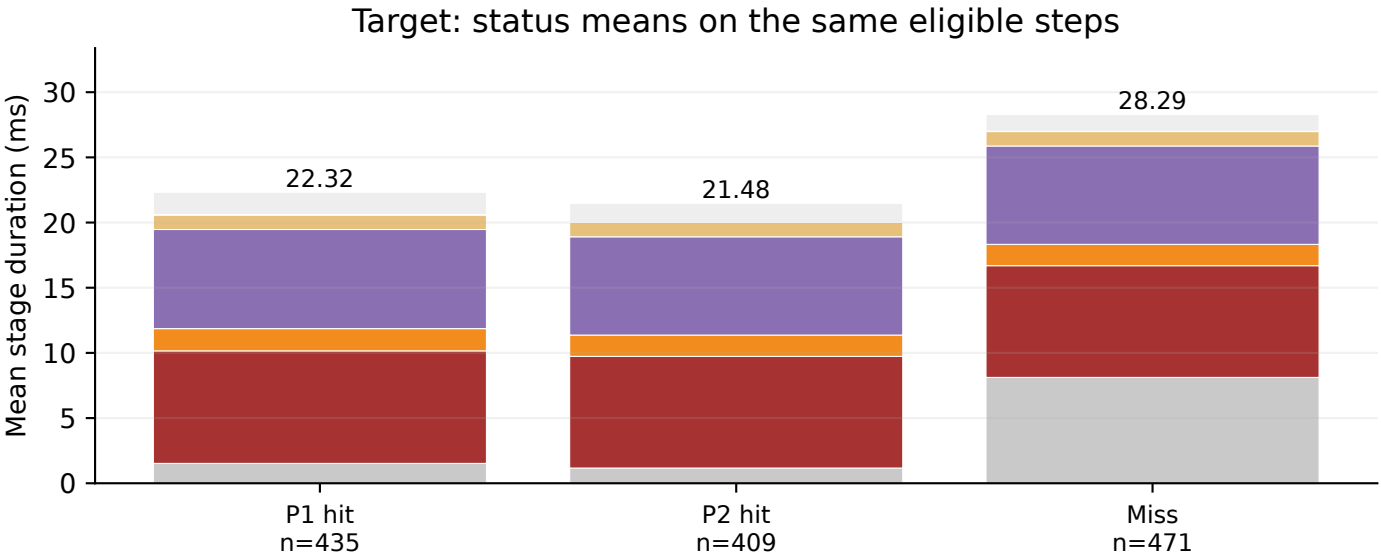
llama3_b1_duet_e30_k4_2
DUET: K1/K2=4/2, exit=30 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1305 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

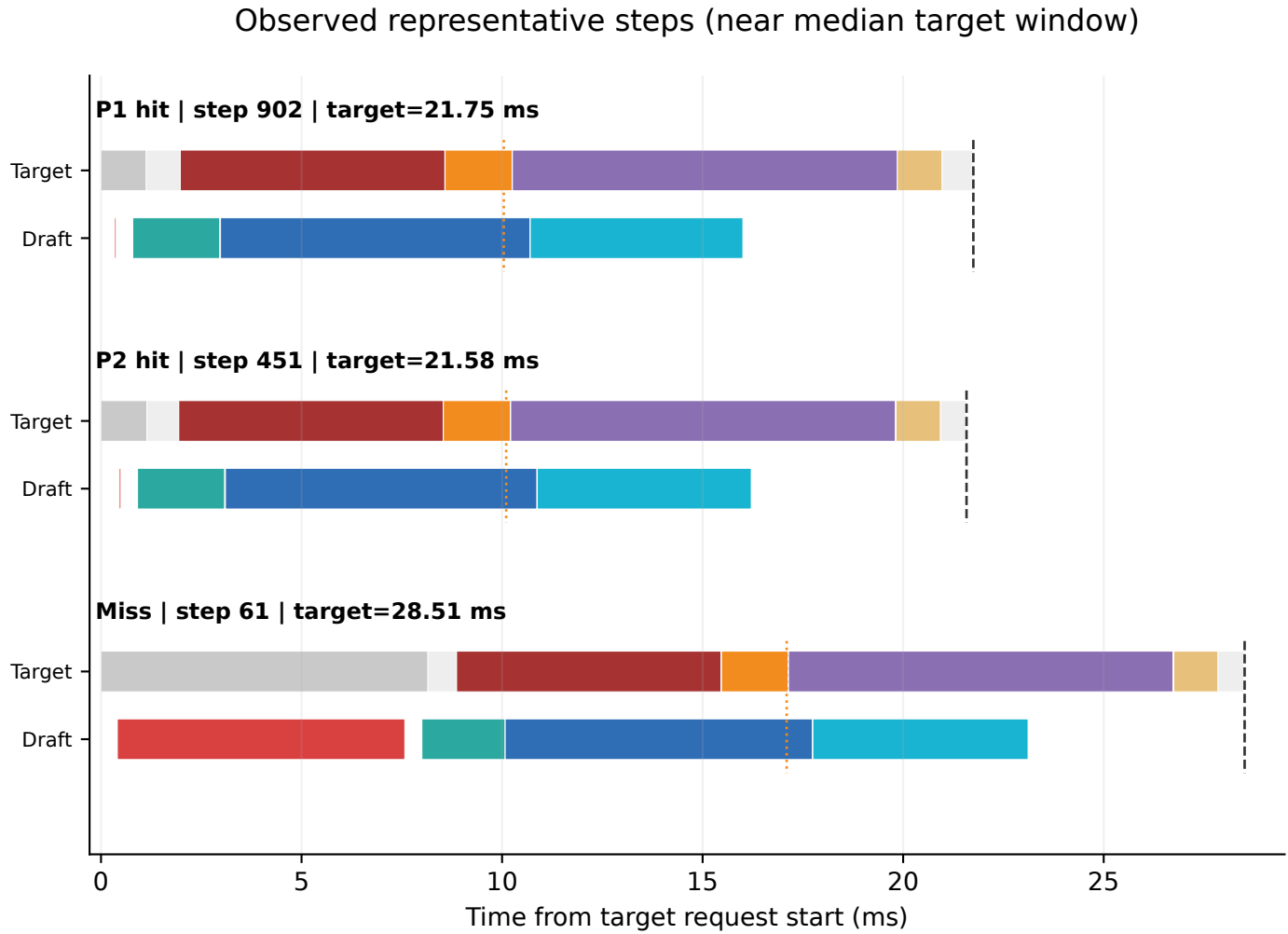
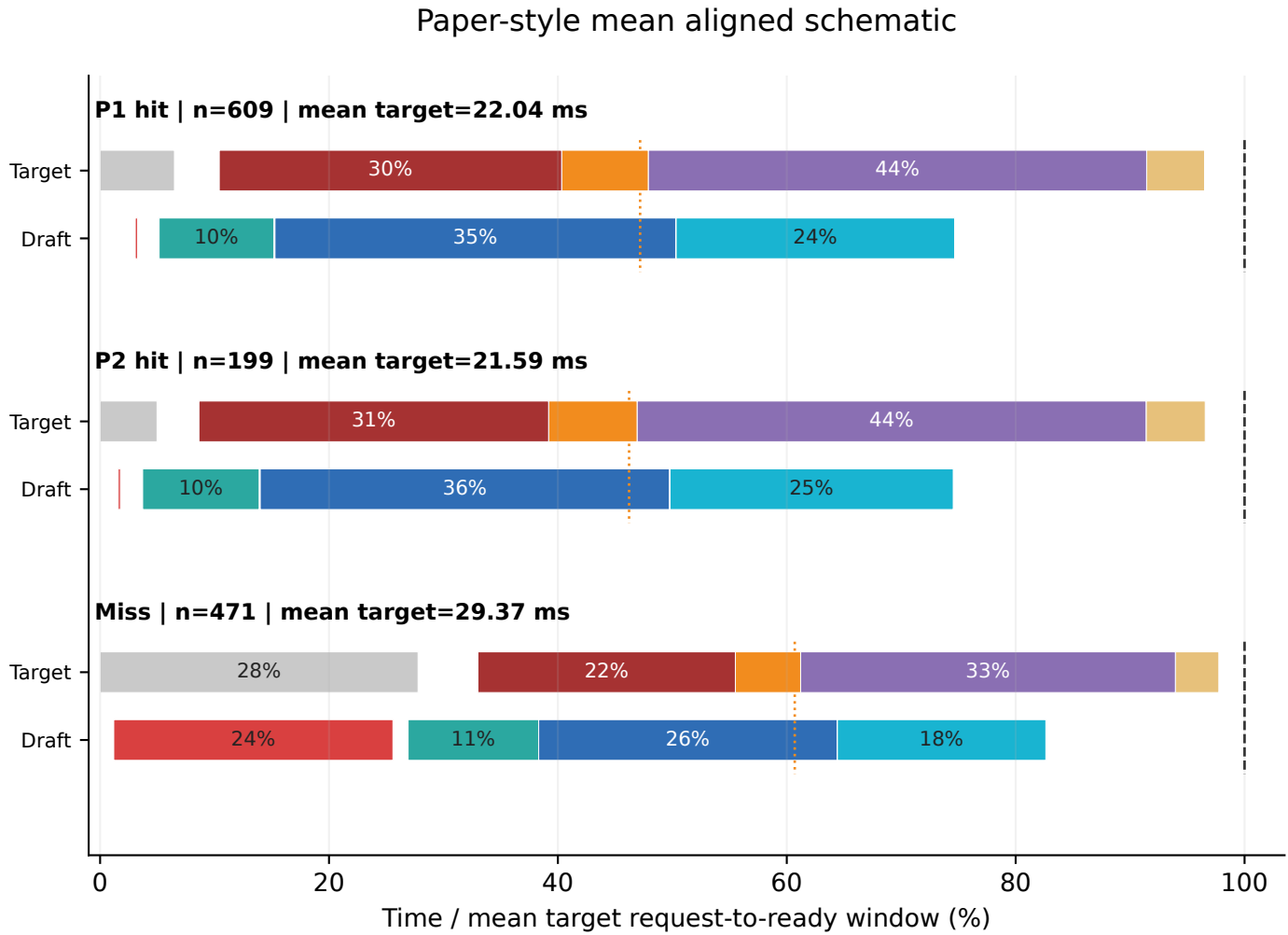
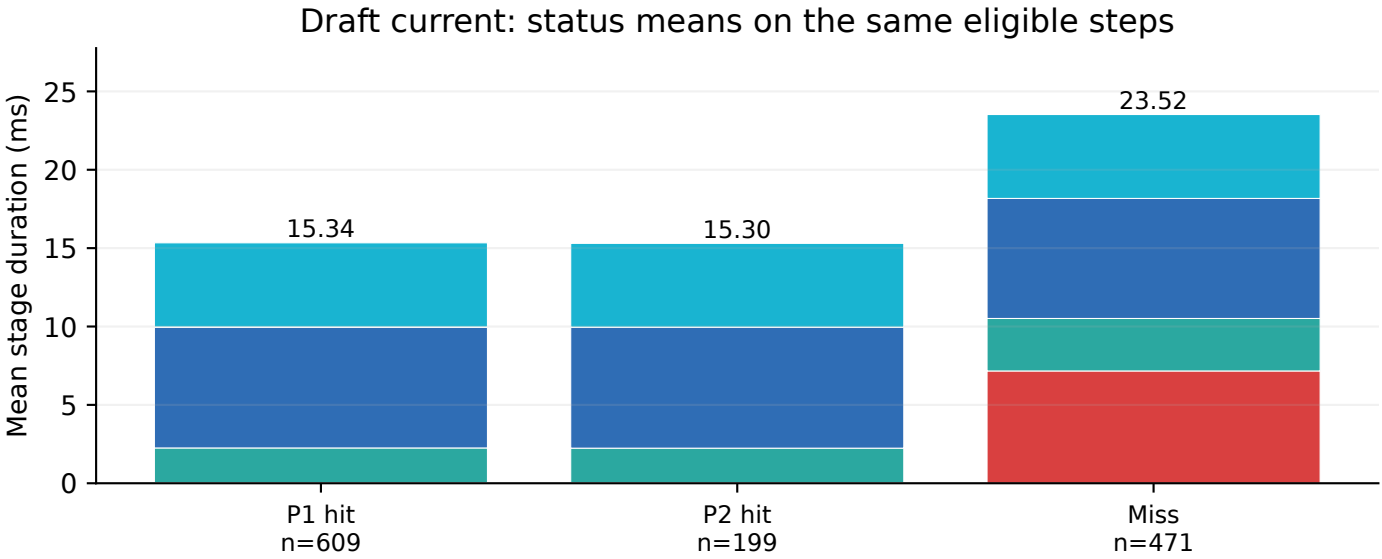
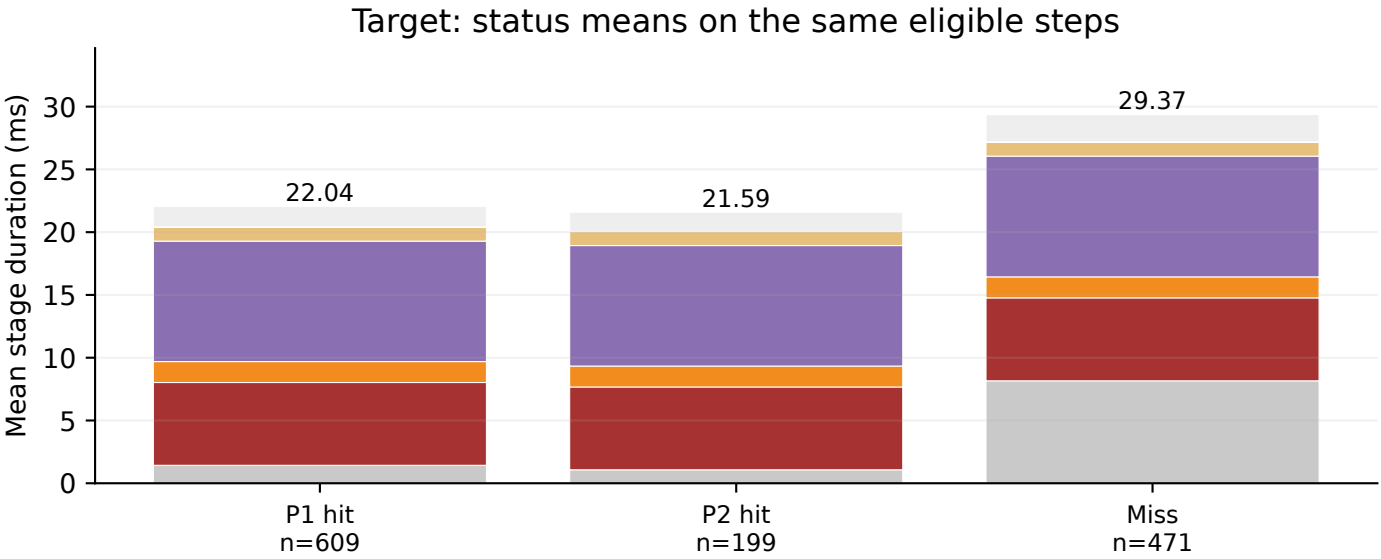
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k4_2_r1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=1, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1315 eligible steps



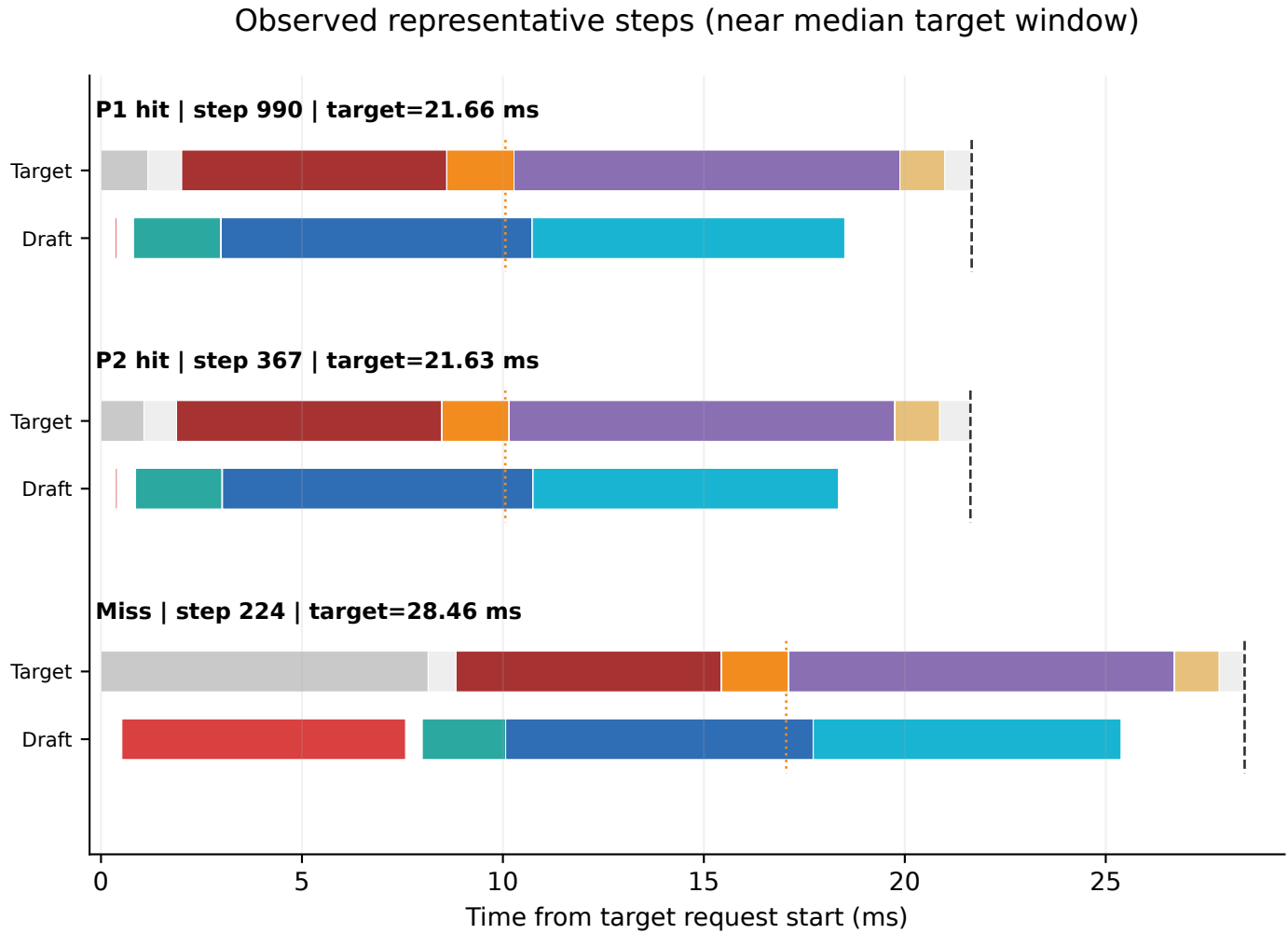
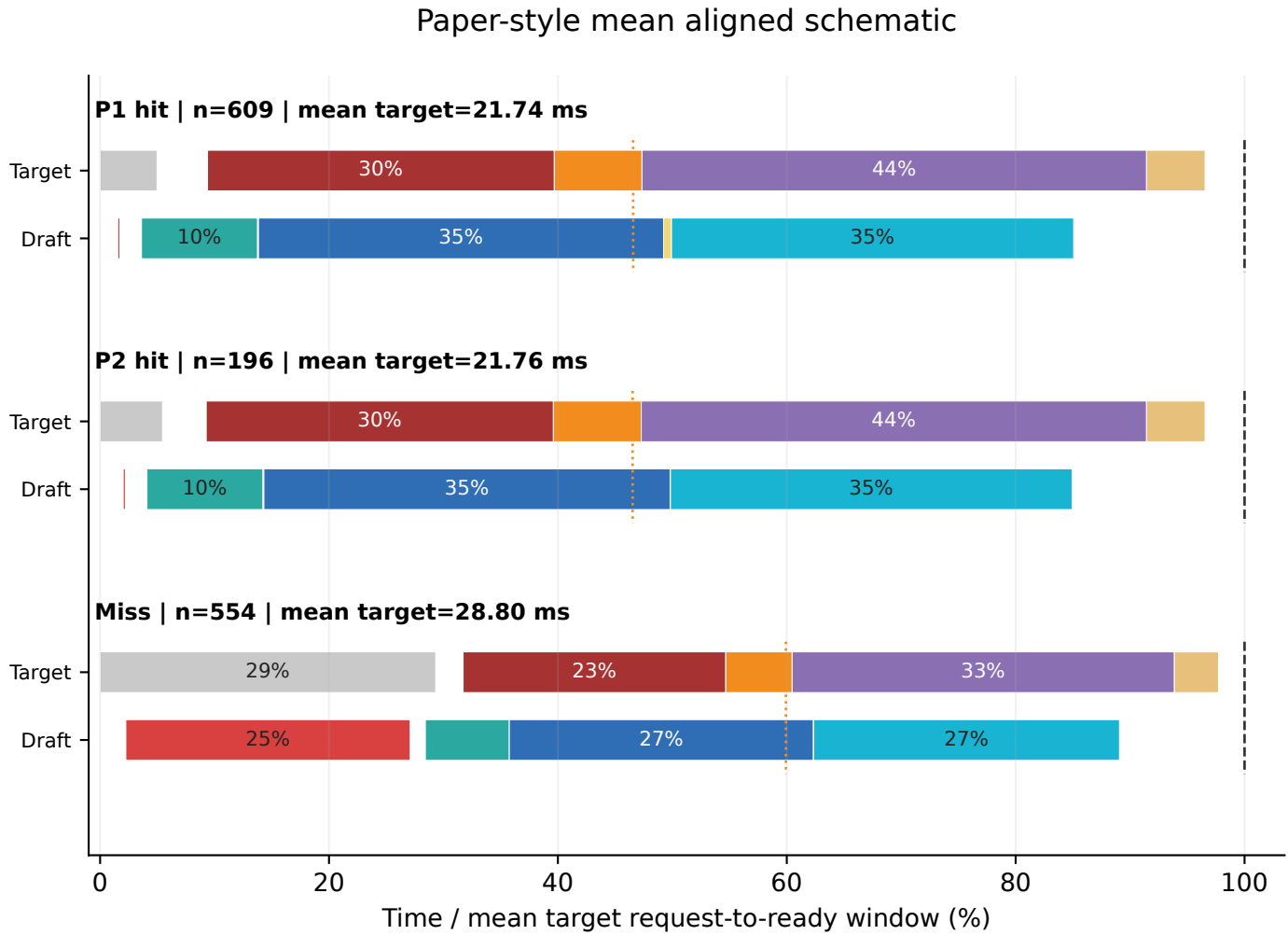
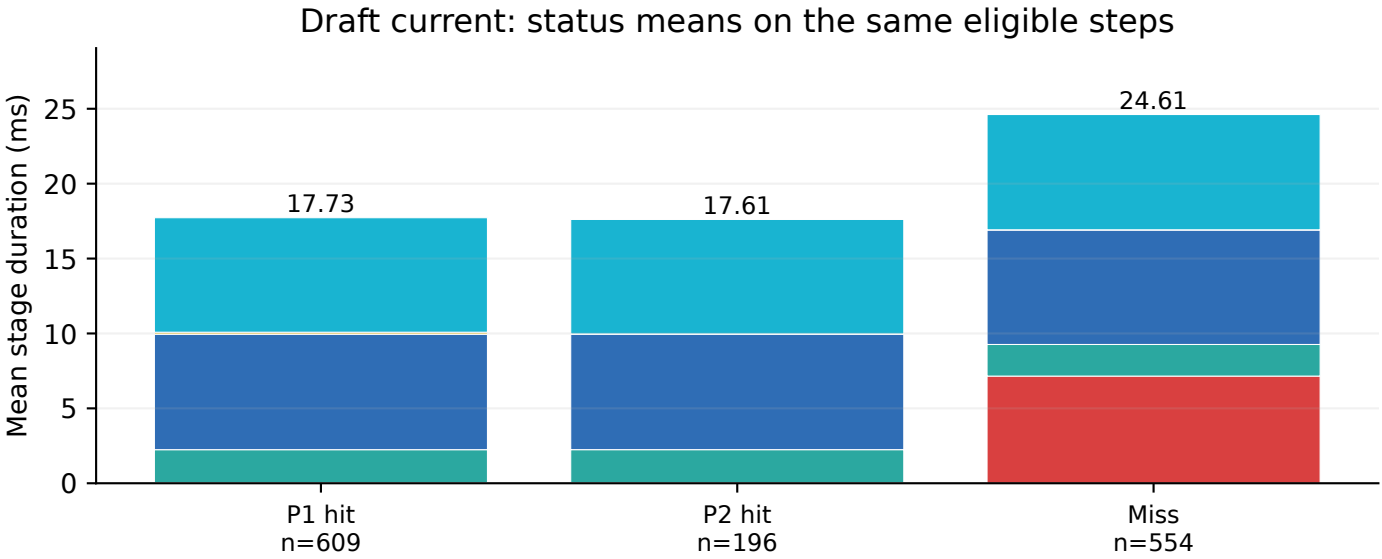
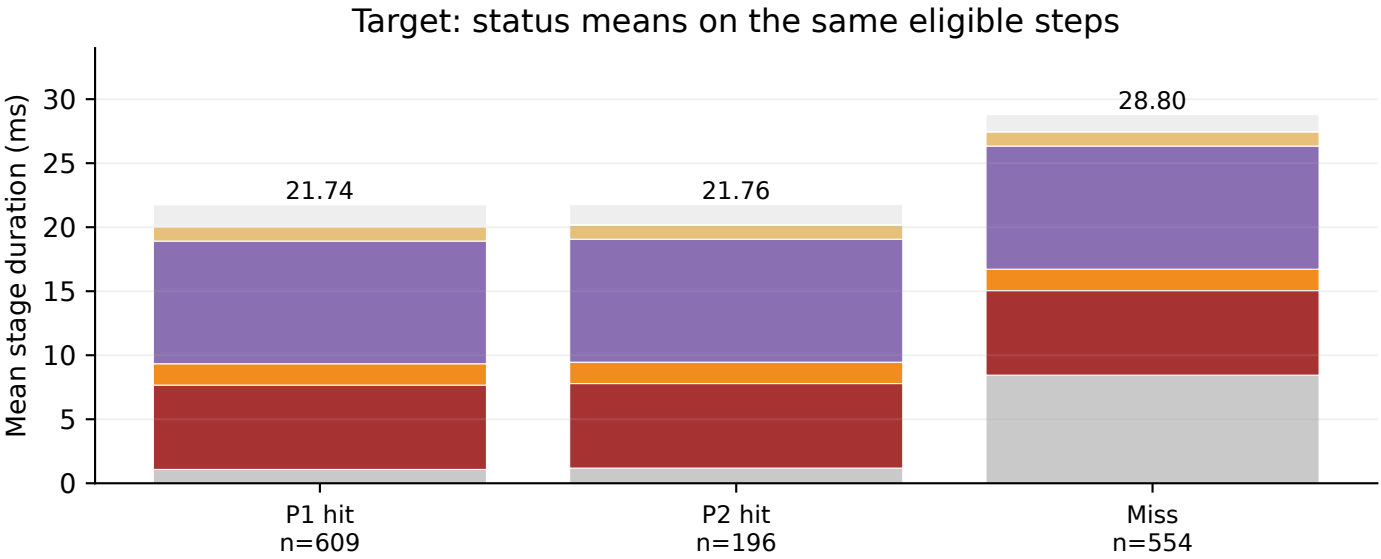
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e12_k3_2_r2
DUET: K1/K2=3/2, exit=12 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1279 eligible steps



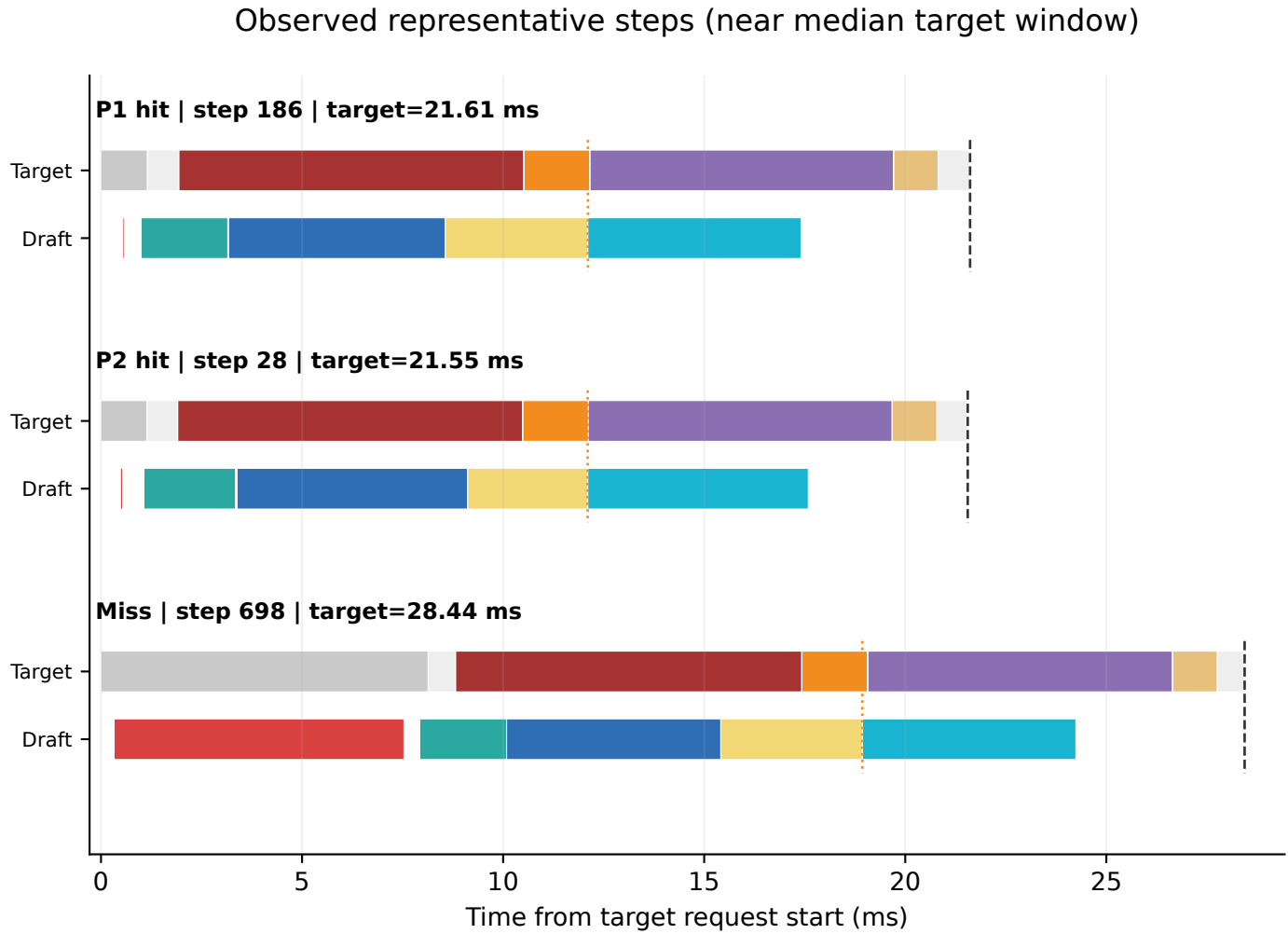
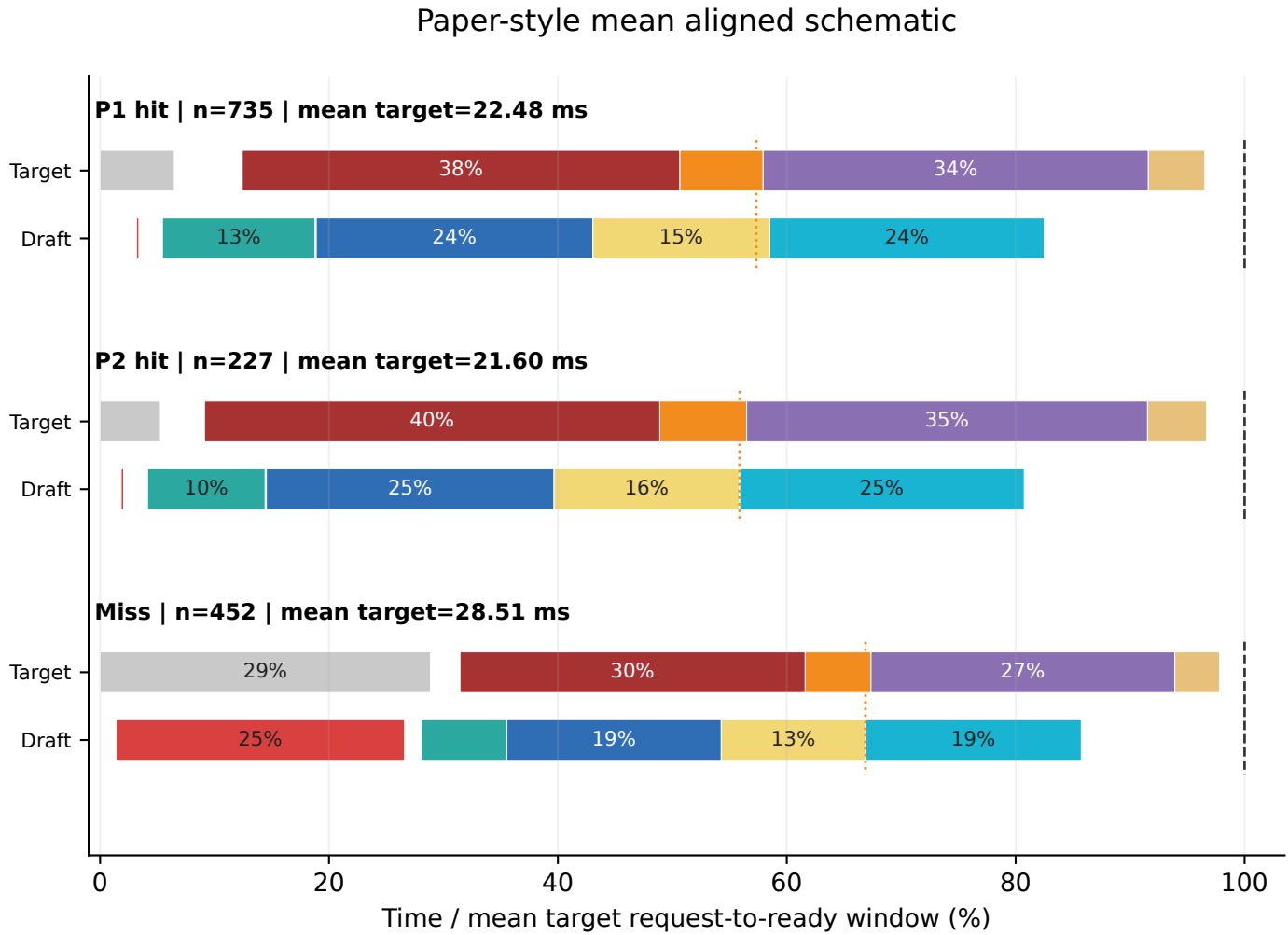
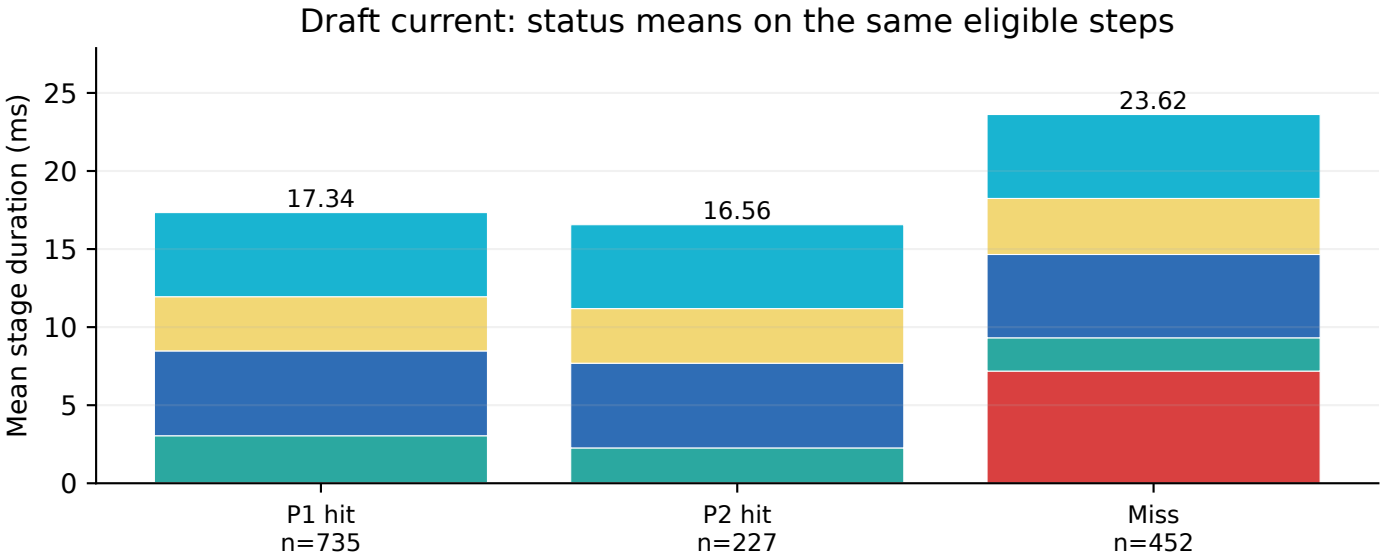
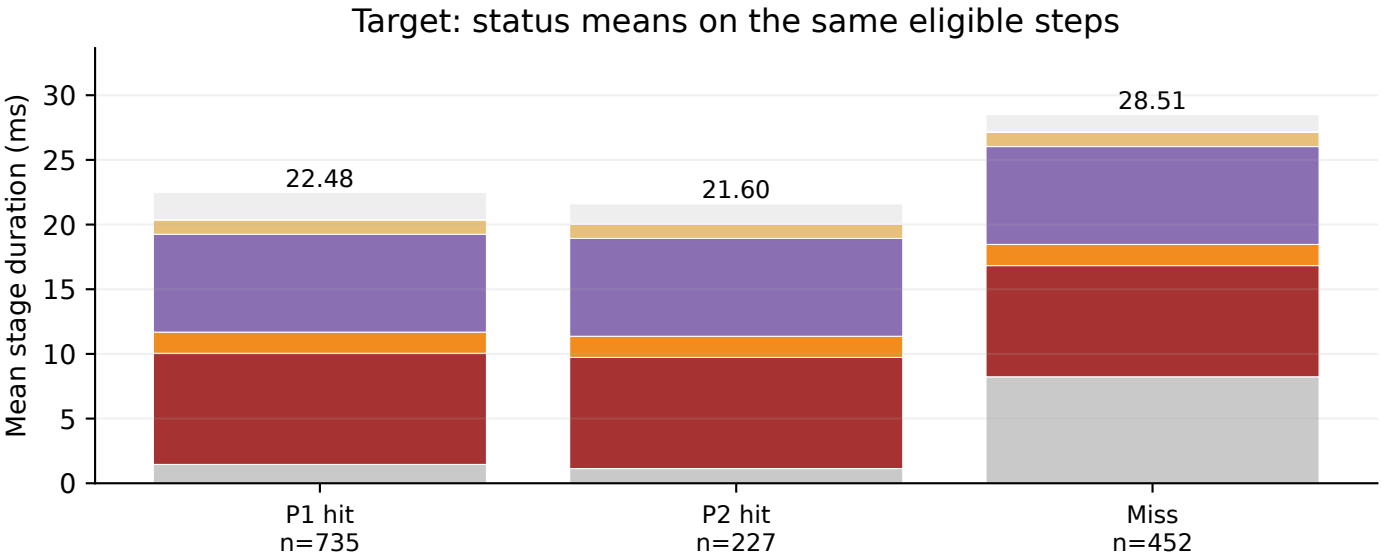
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e12_k3_3_r2
DUET: K1/K2=3/3, exit=12 (0-based), C=3, N1/N2=6/6, U=2, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1359 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k2_2_r2
DUET: K1/K2=2/2, exit=16 (0-based), C=3, N1/N2=4/4, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1414 eligible steps

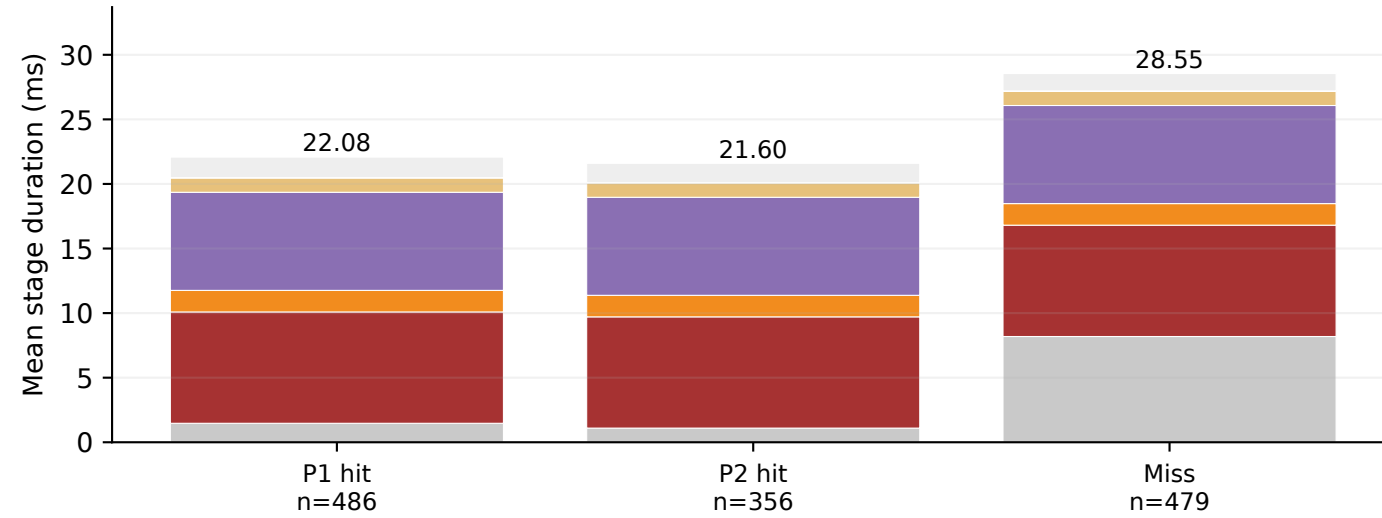


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

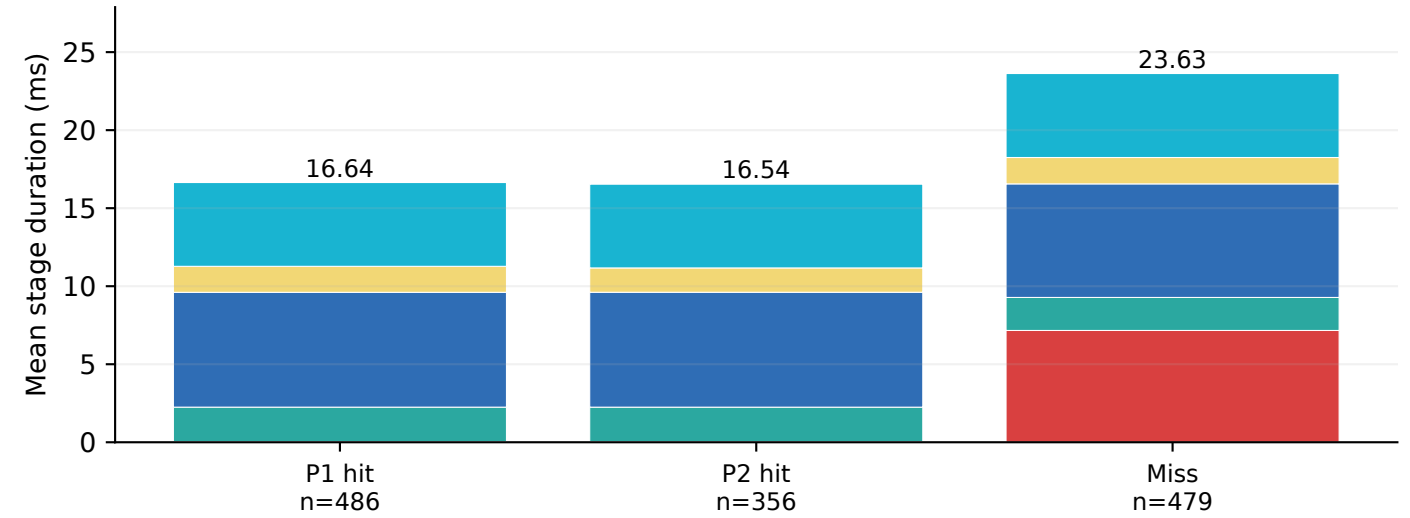
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k3_2_r1
DUET: K1/K2=3/2, exit=16 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1321 eligible steps

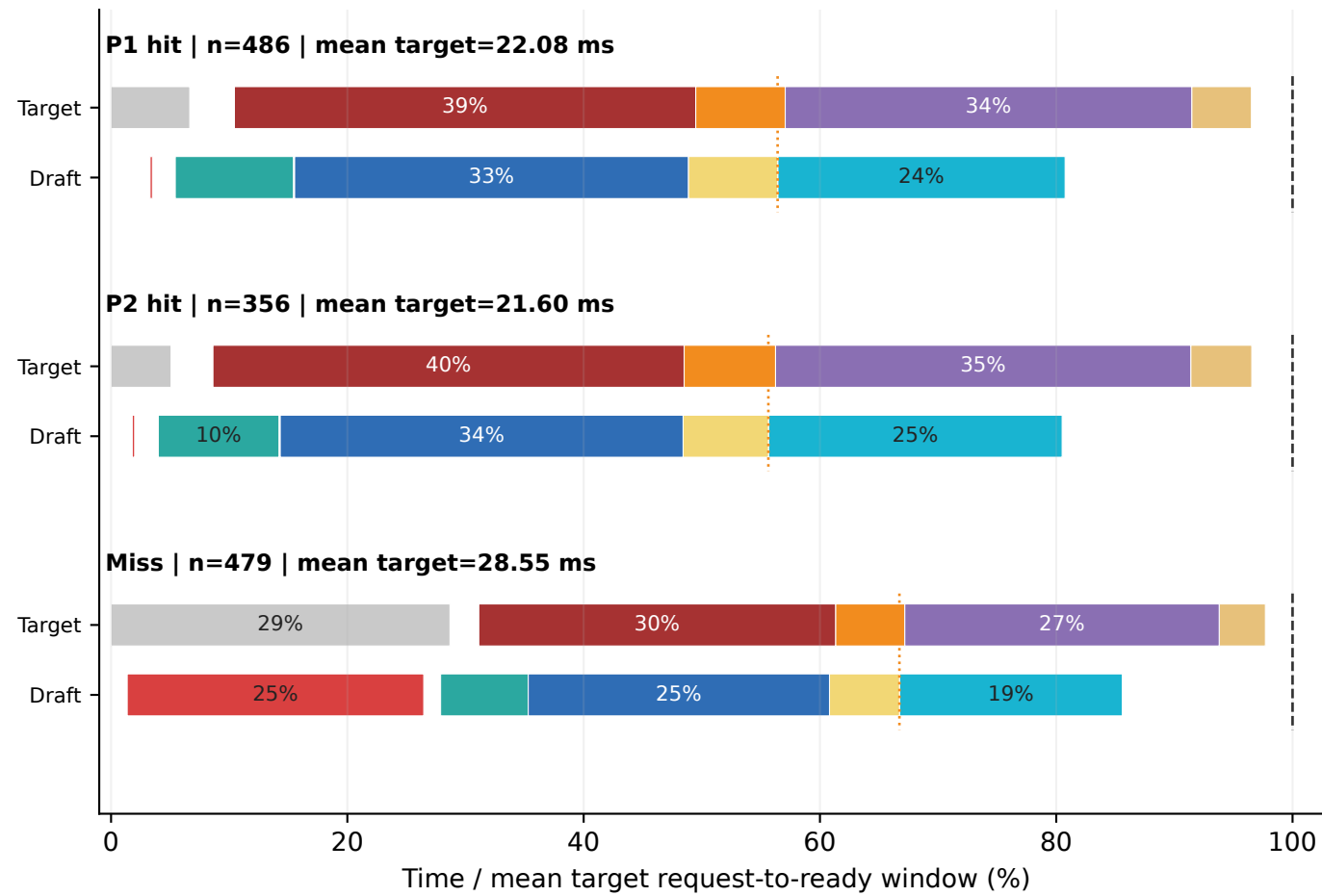
Target: status means on the same eligible steps



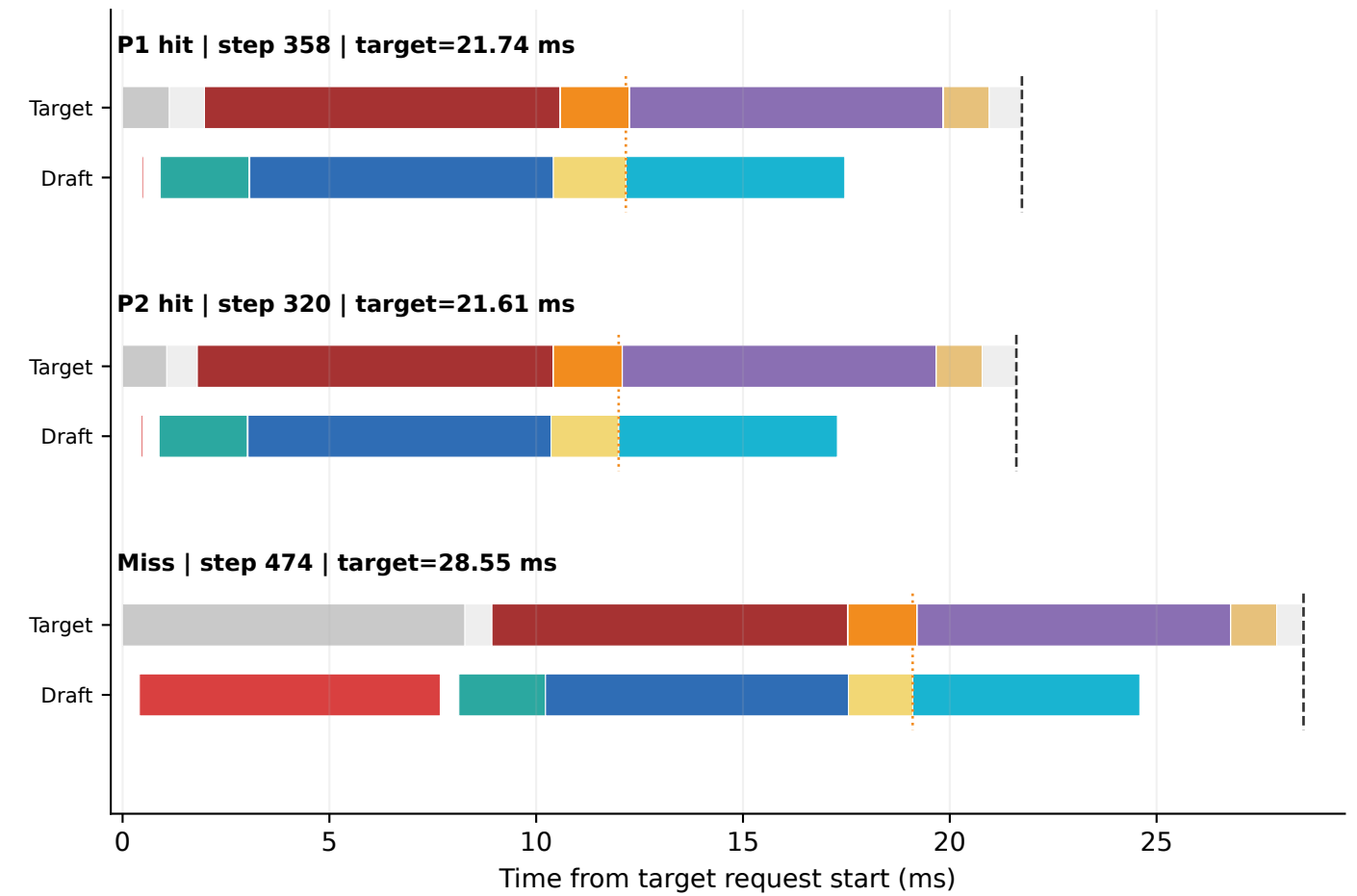
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



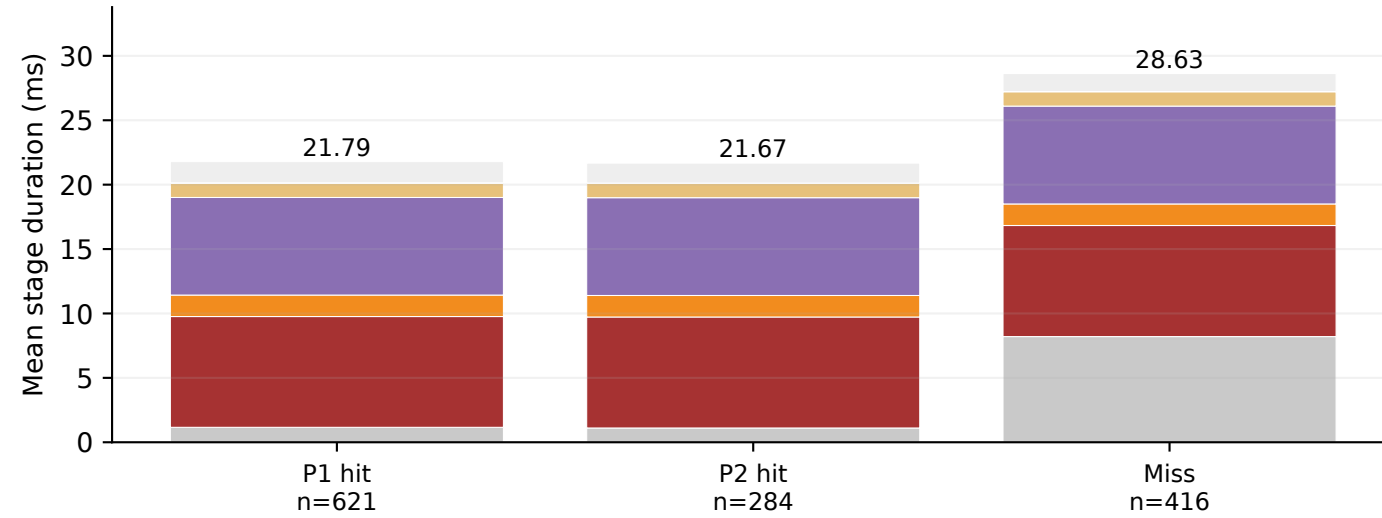
Observed representative steps (near median target window)



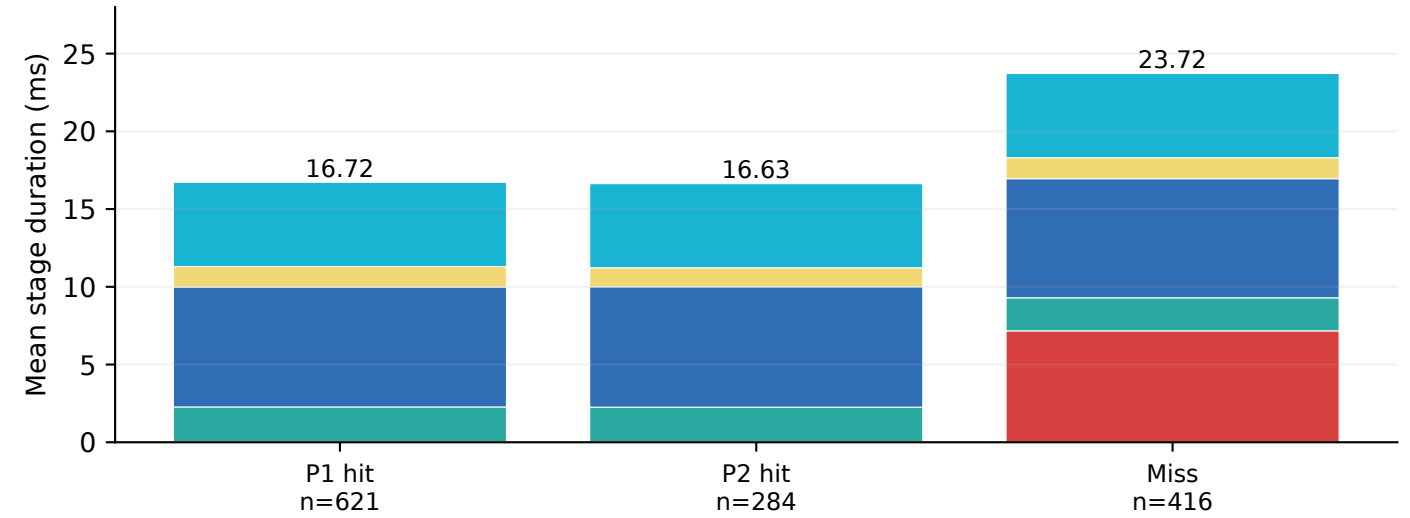
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k3_2_r2
DUET: K1/K2=3/2, exit=16 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1321 eligible steps

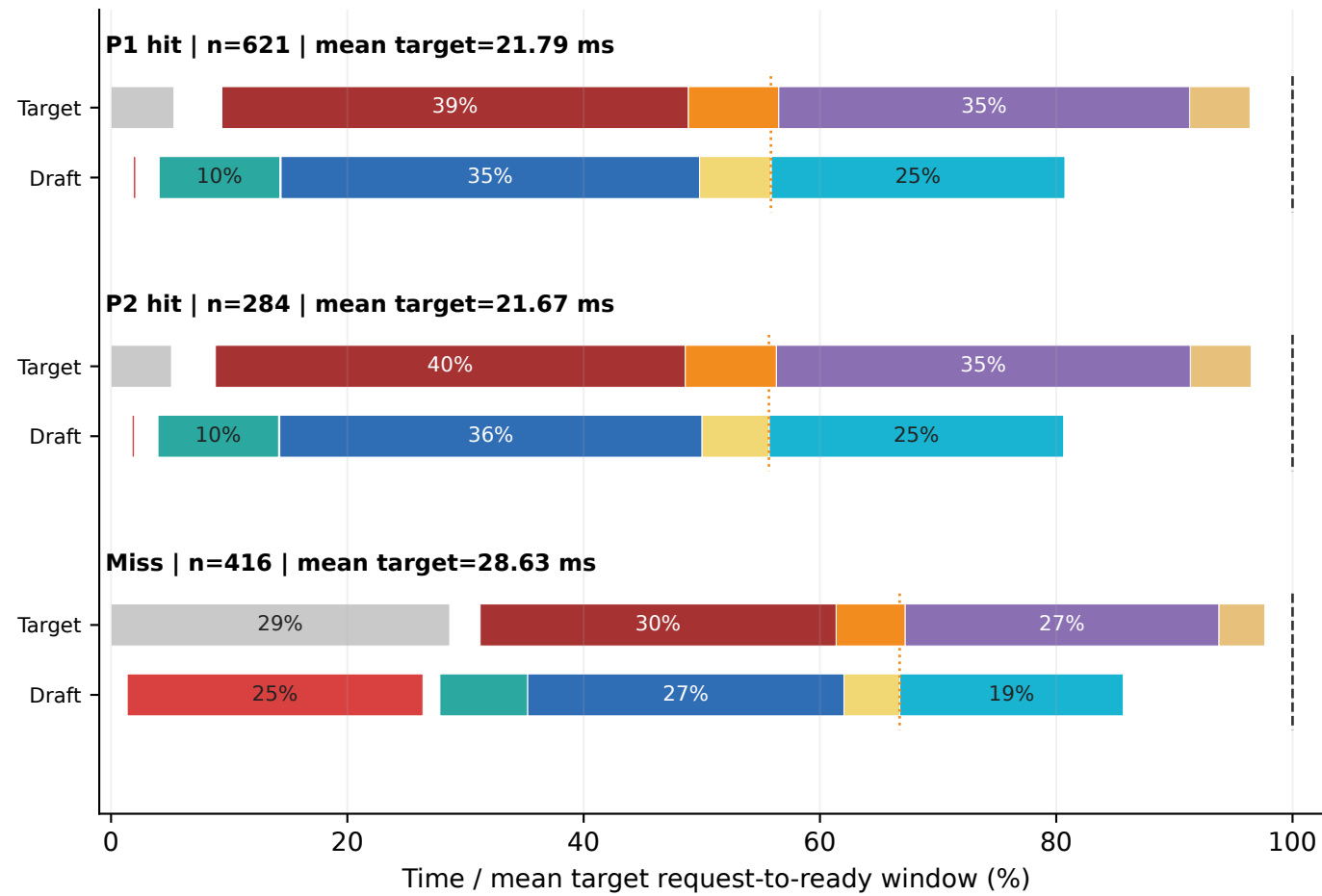
Target: status means on the same eligible steps



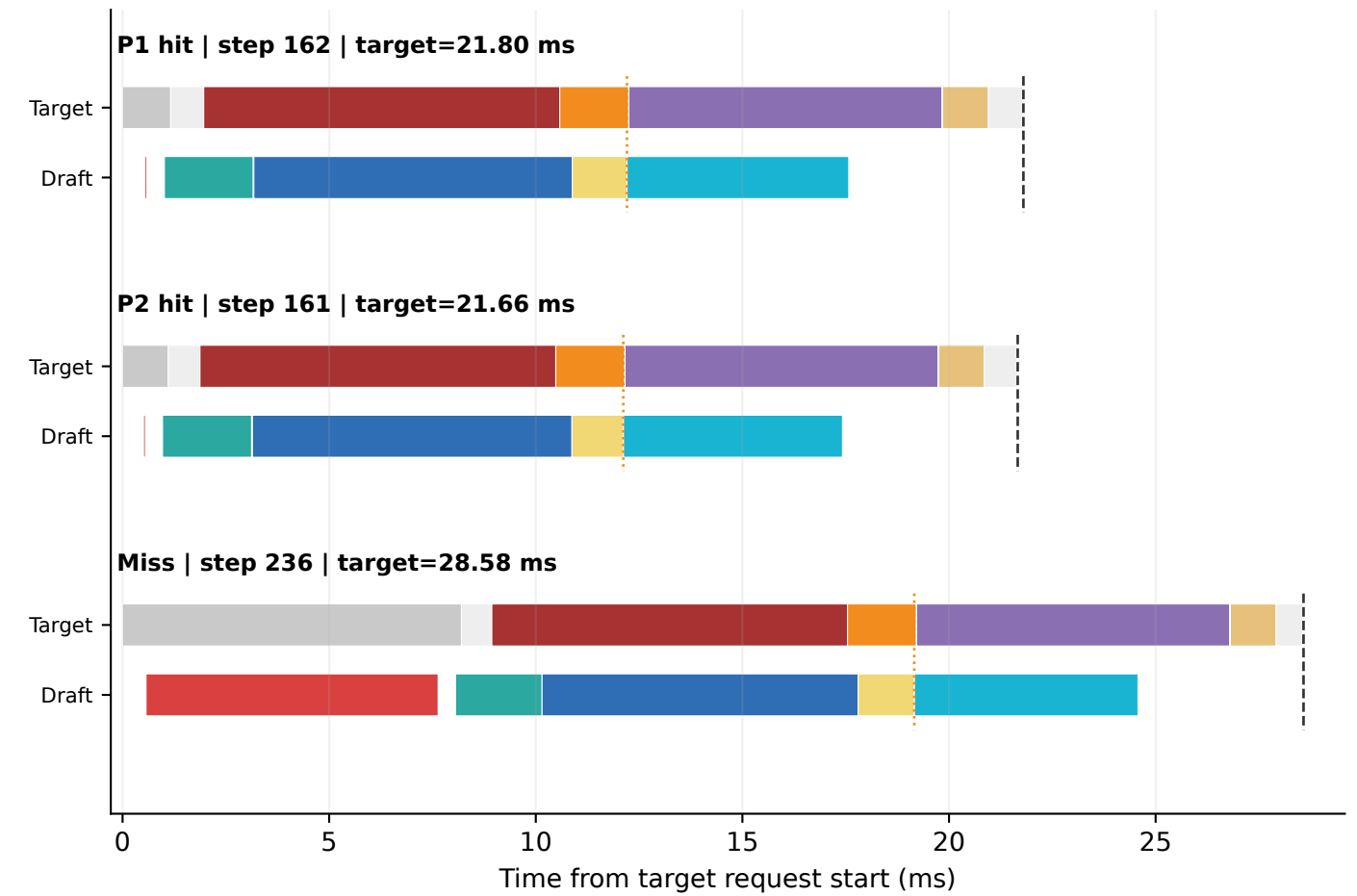
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic

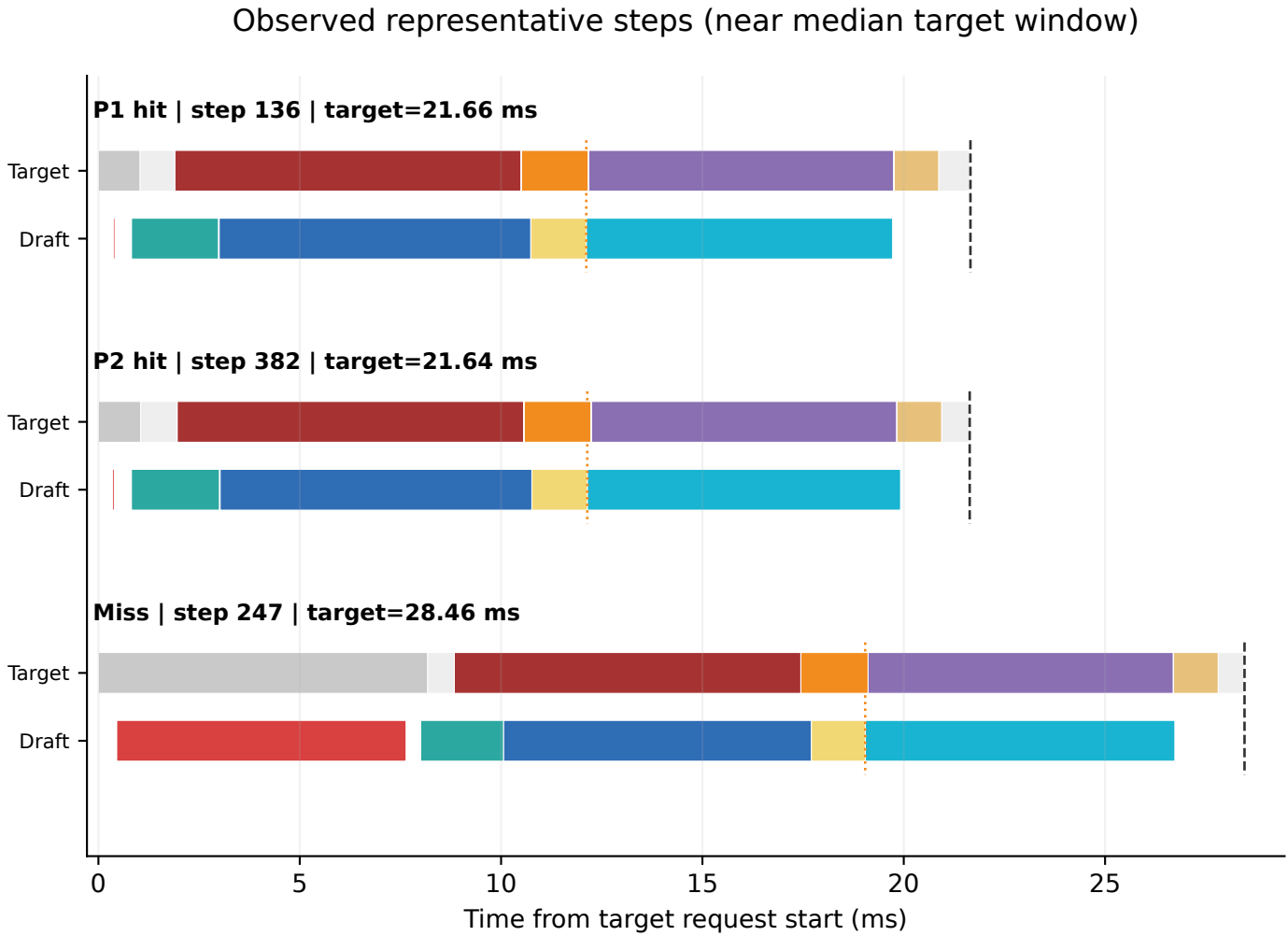
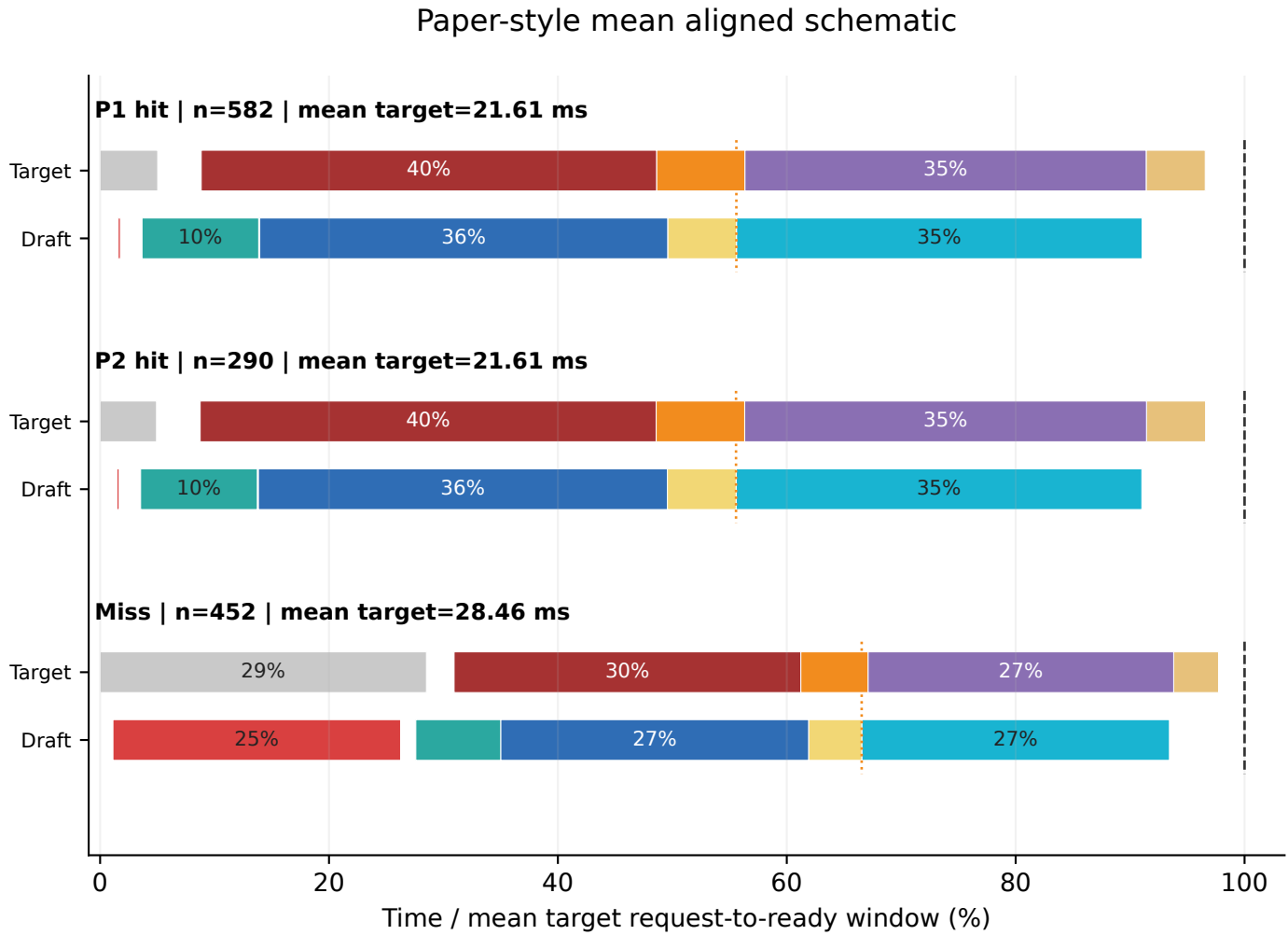
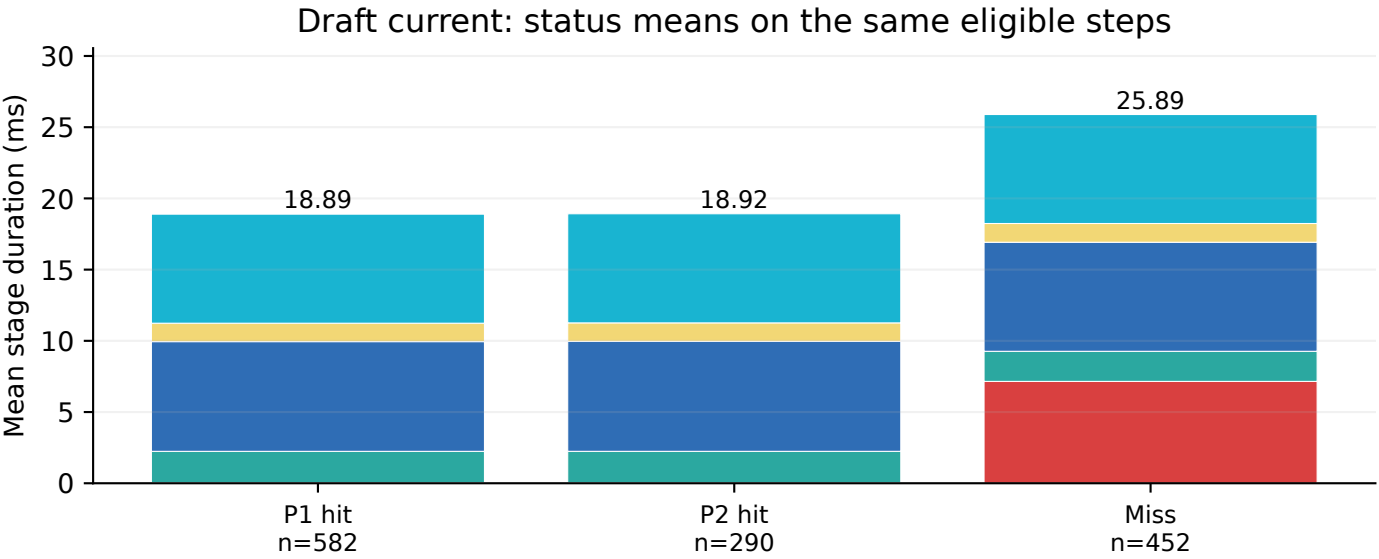
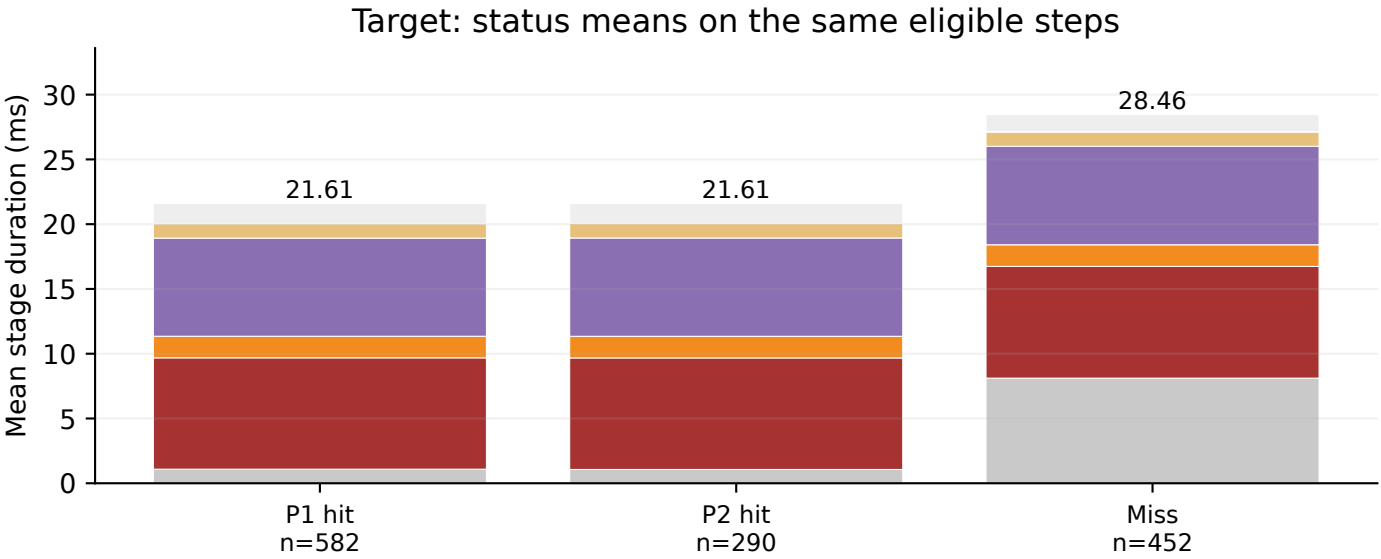


Observed representative steps (near median target window)



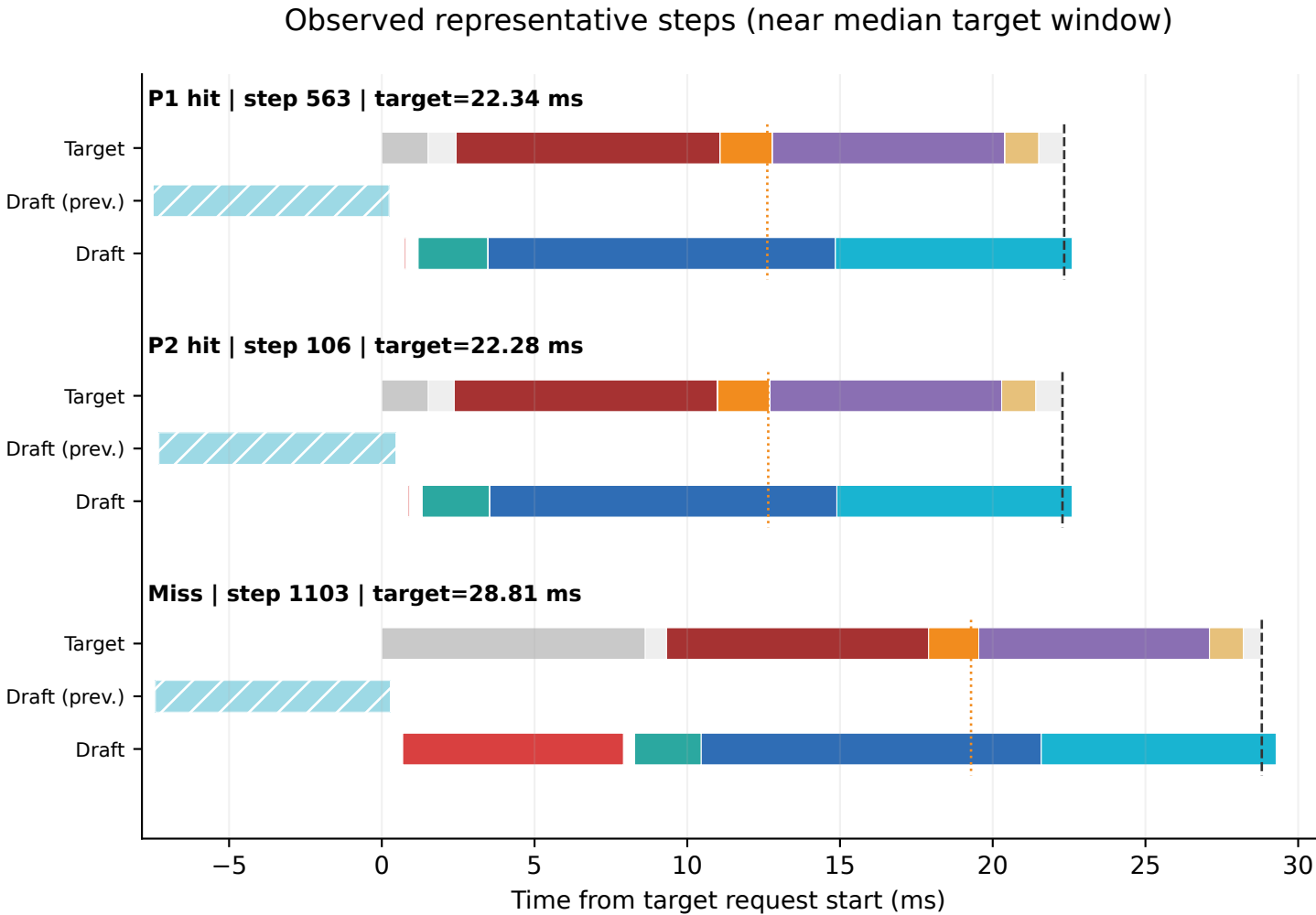
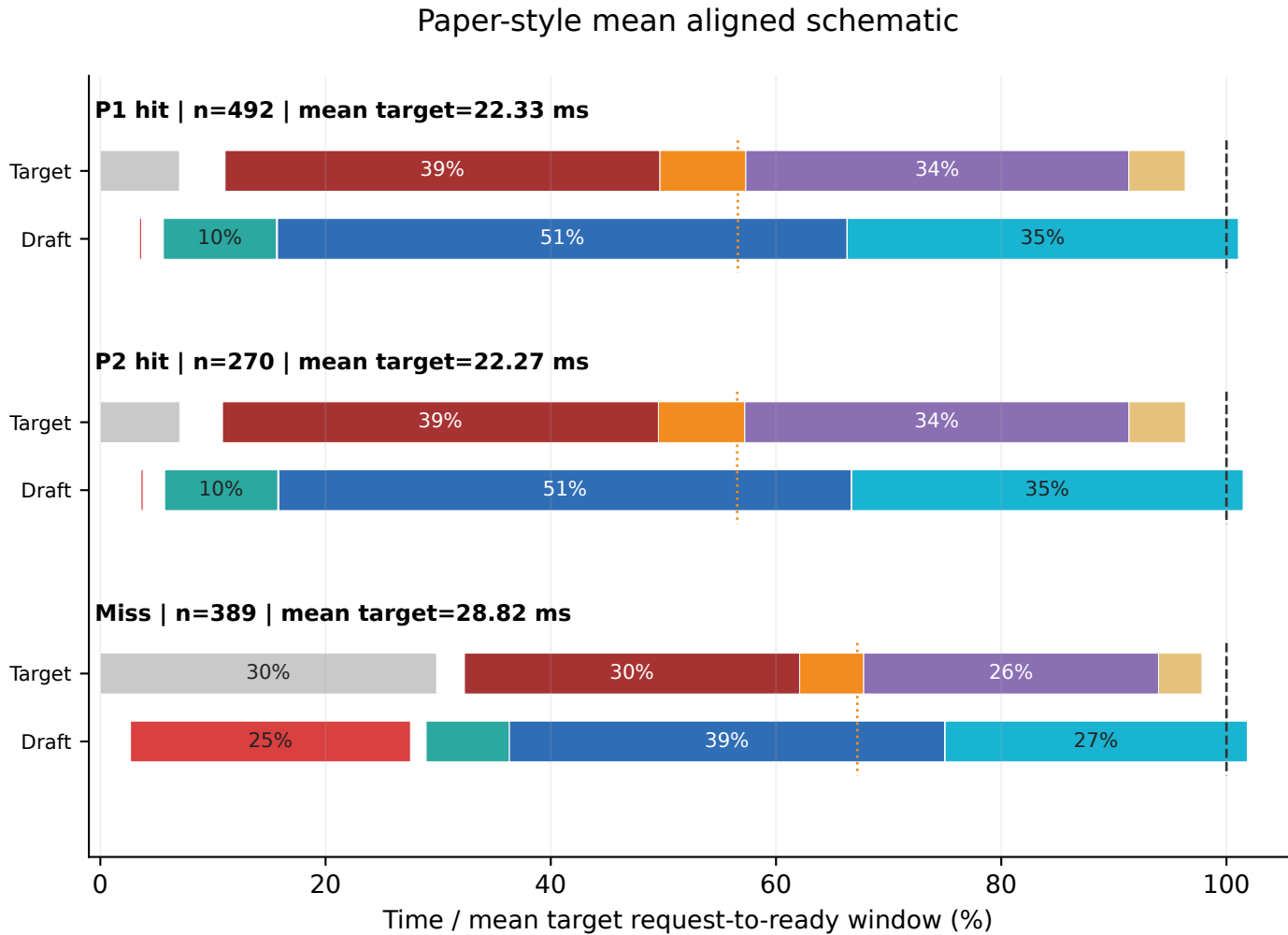
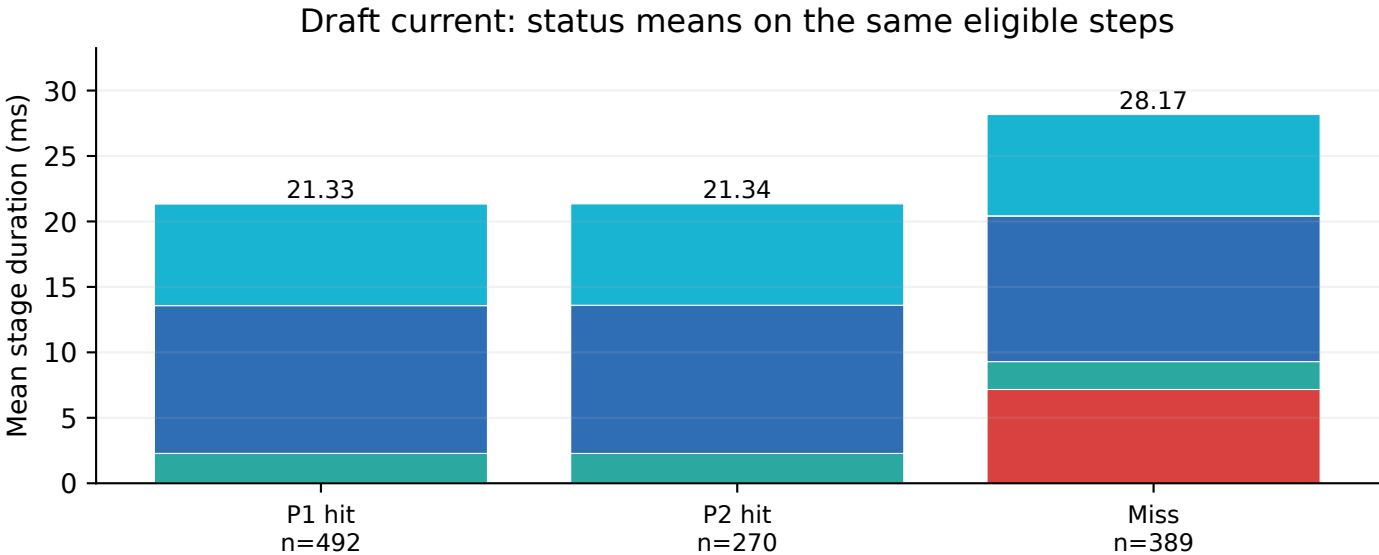
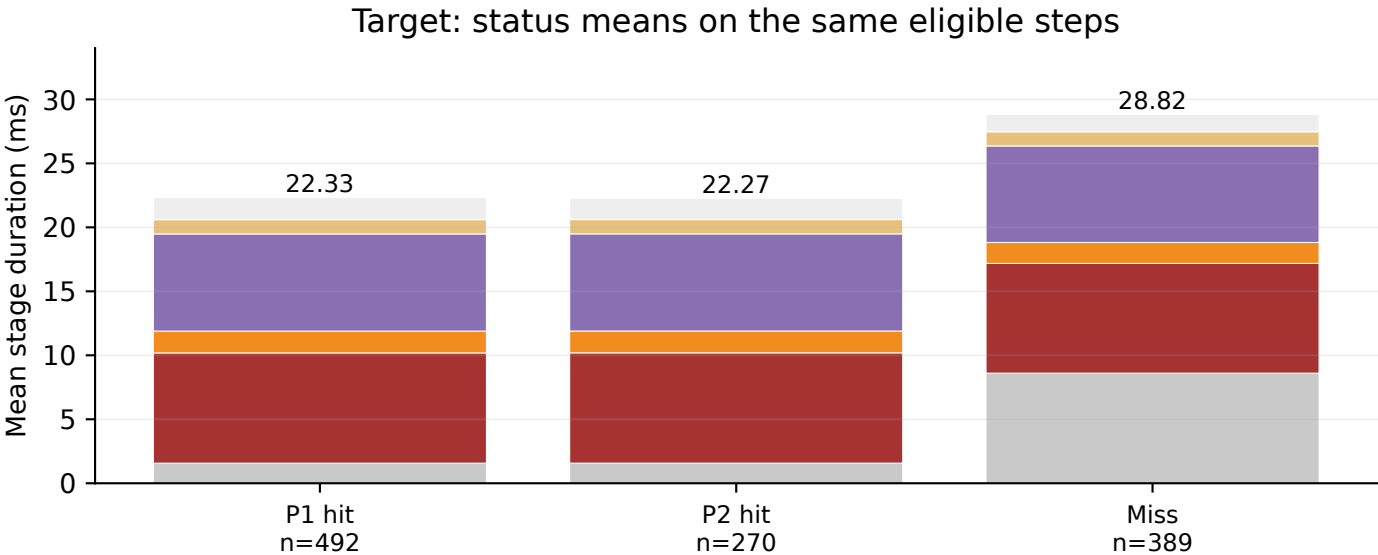
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k3_3_r2
DUET: K1/K2=3/3, exit=16 (0-based), C=3, N1/N2=6/6, U=2, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1324 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

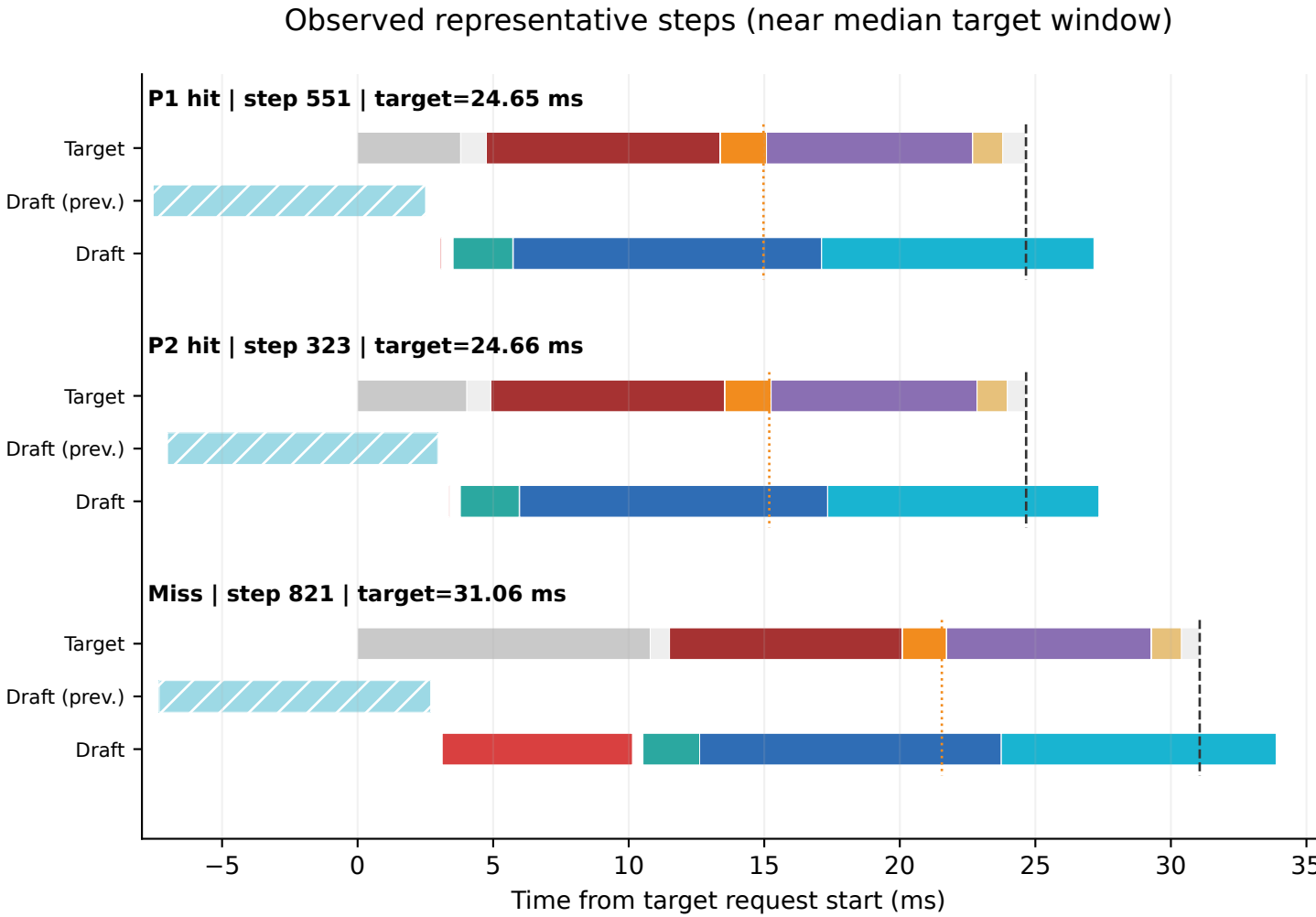
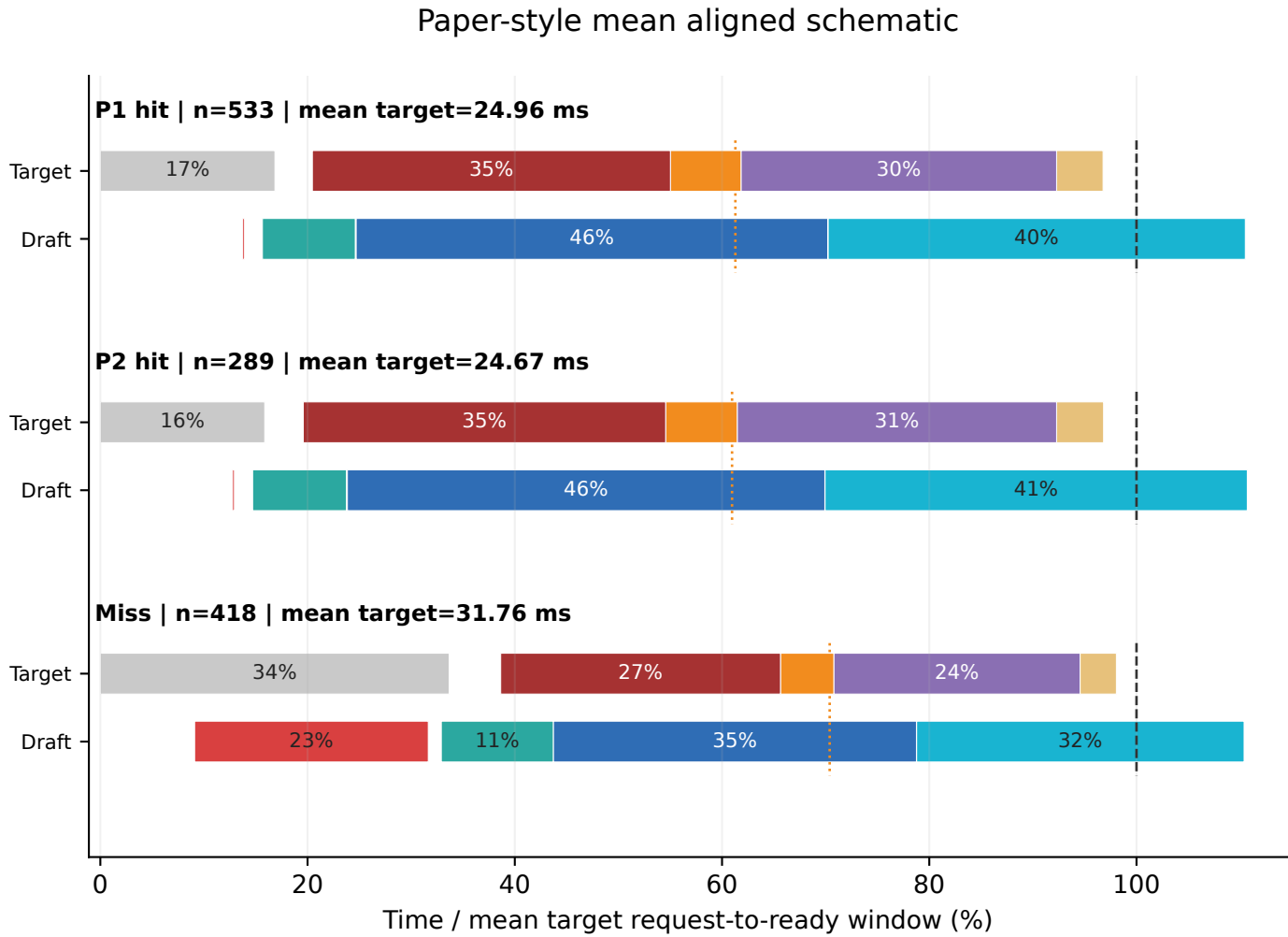
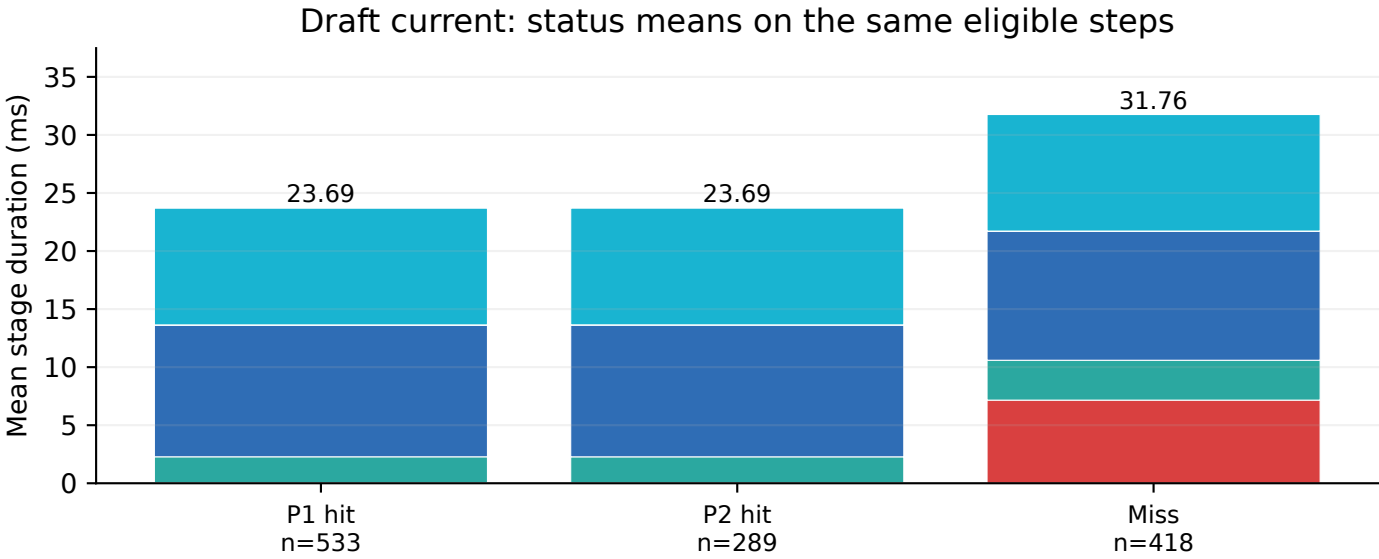
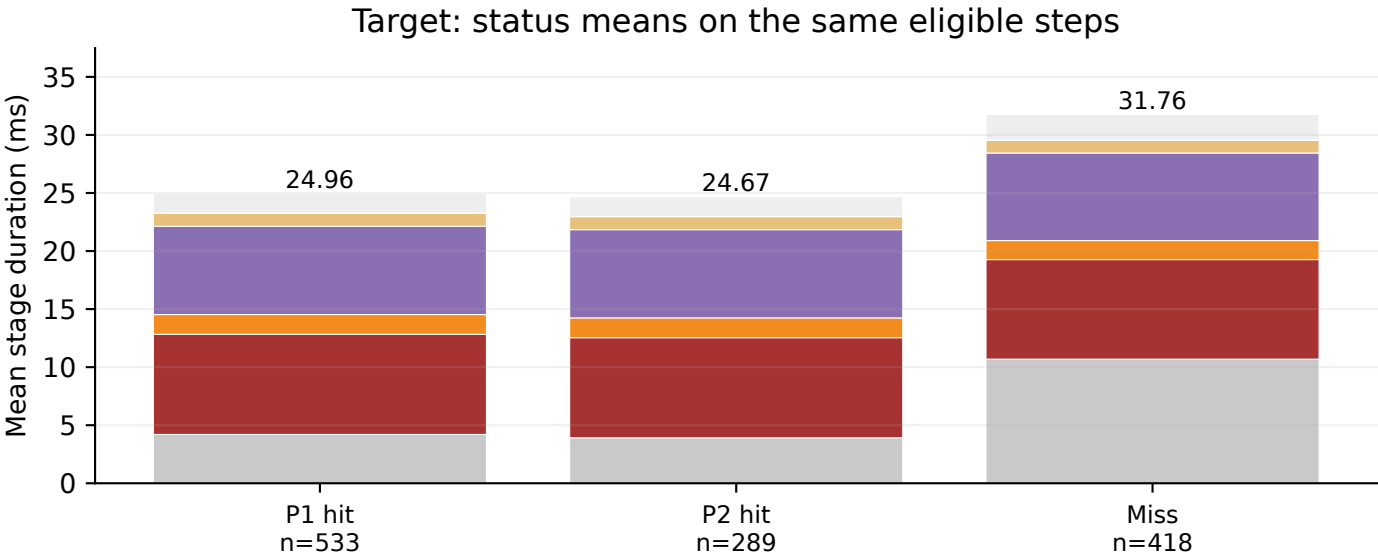
llama3_b1_refine_e16_k4_3_r2
DUET: K1/K2=4/3, exit=16 (0-based), C=3, N1/N2=8/6, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1151 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

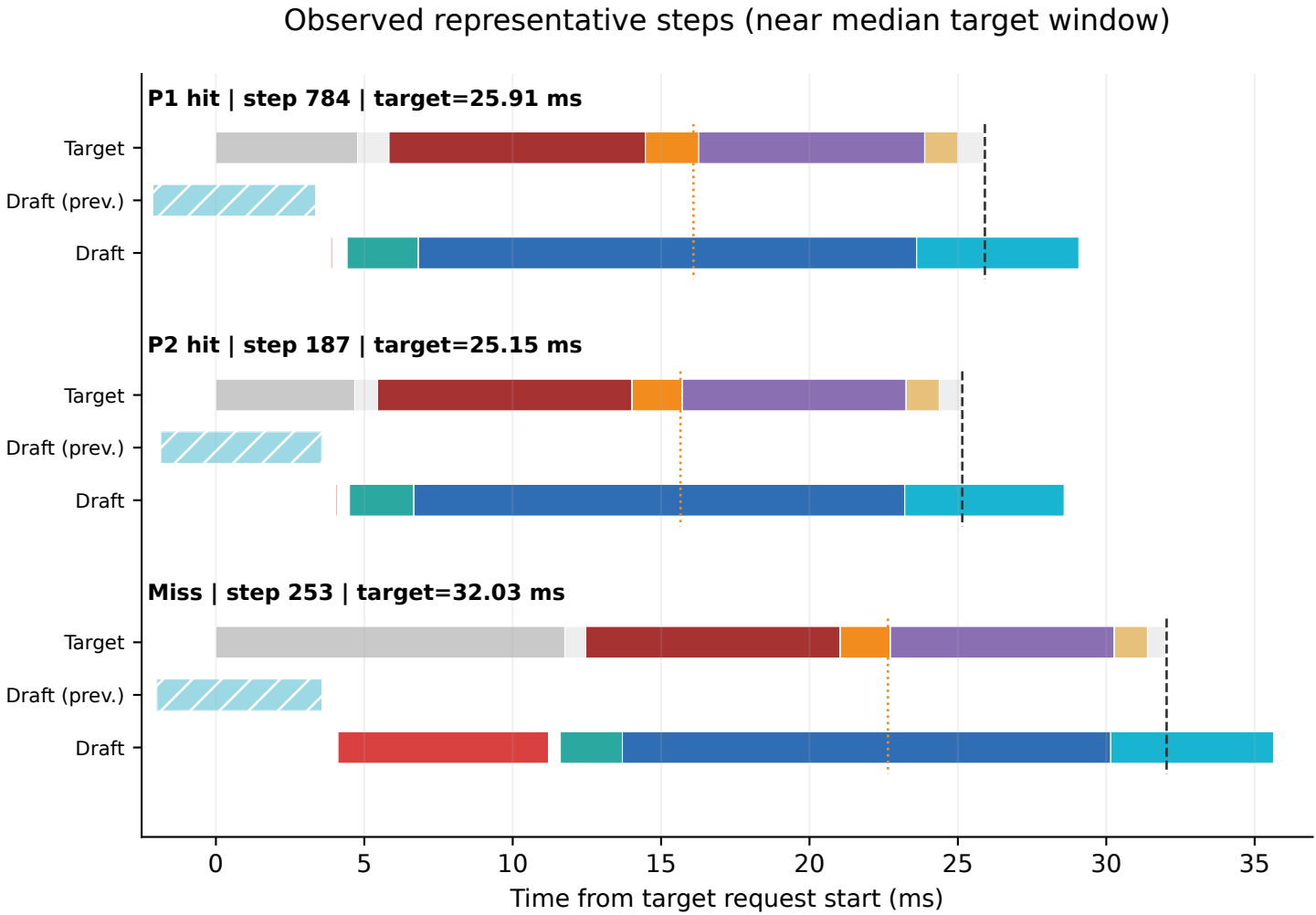
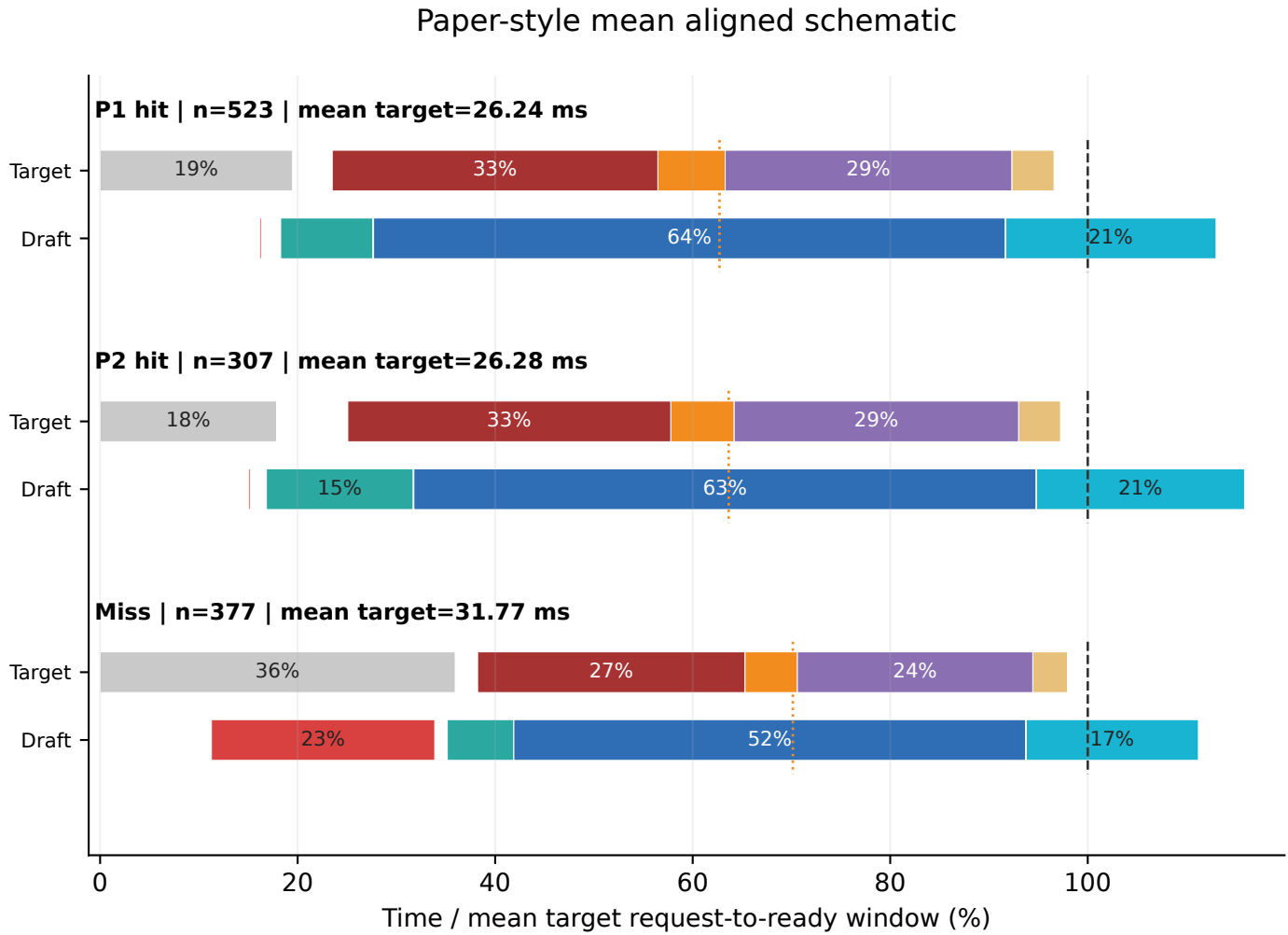
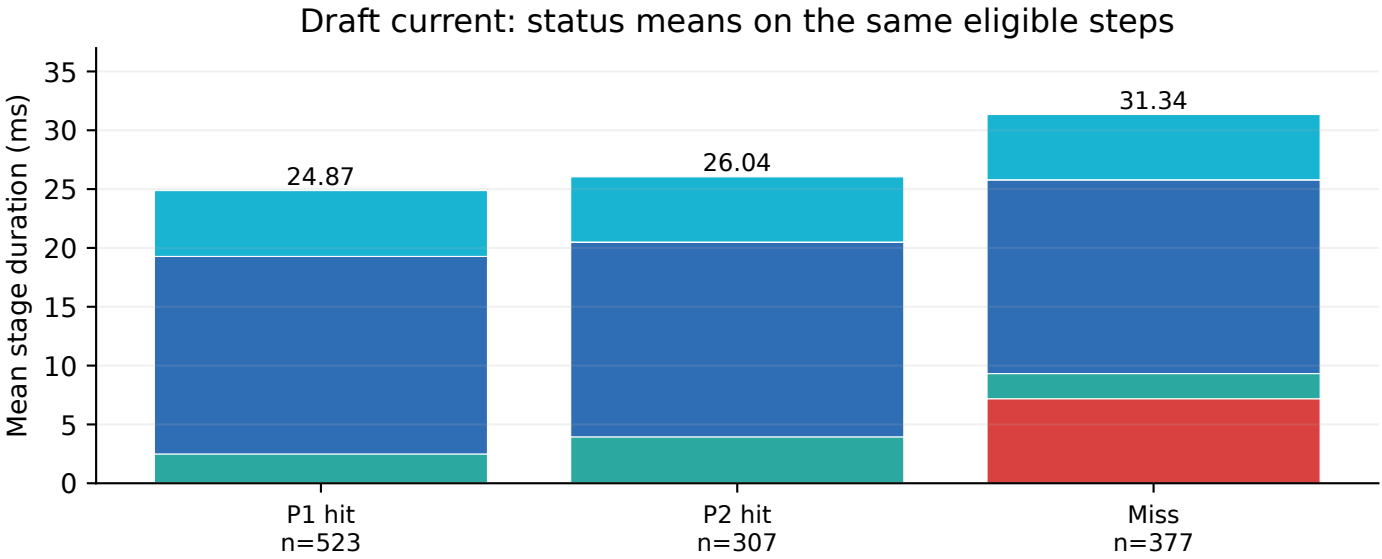
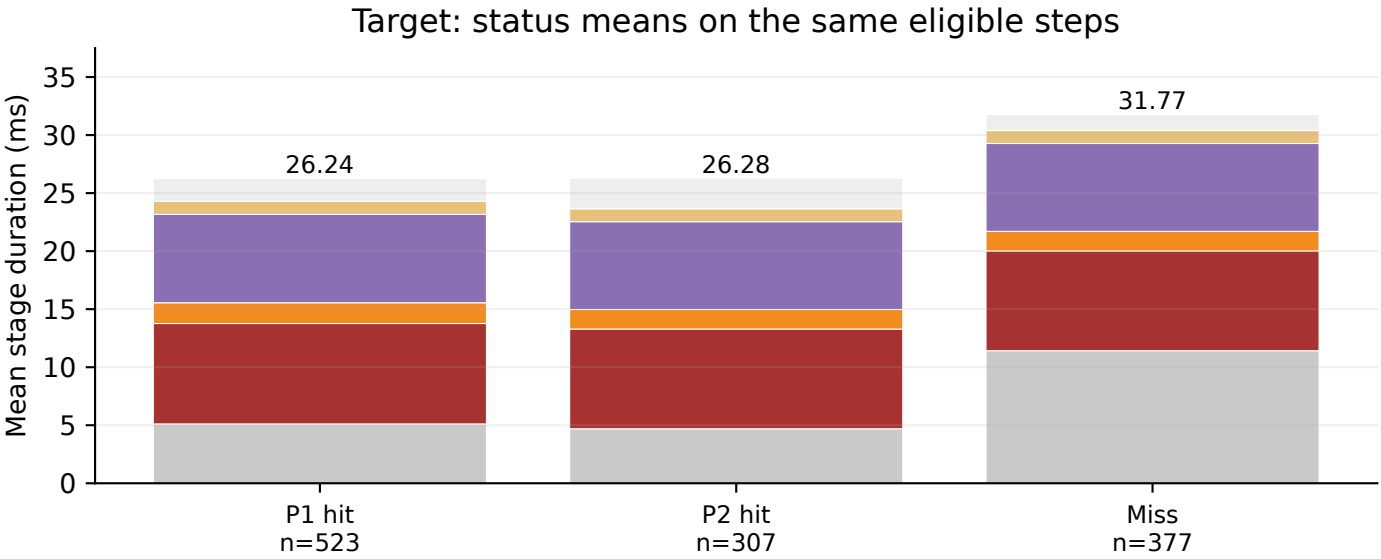
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k4_4_r2
DUET: K1/K2=4/4, exit=16 (0-based), C=3, N1/N2=8/8, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1240 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

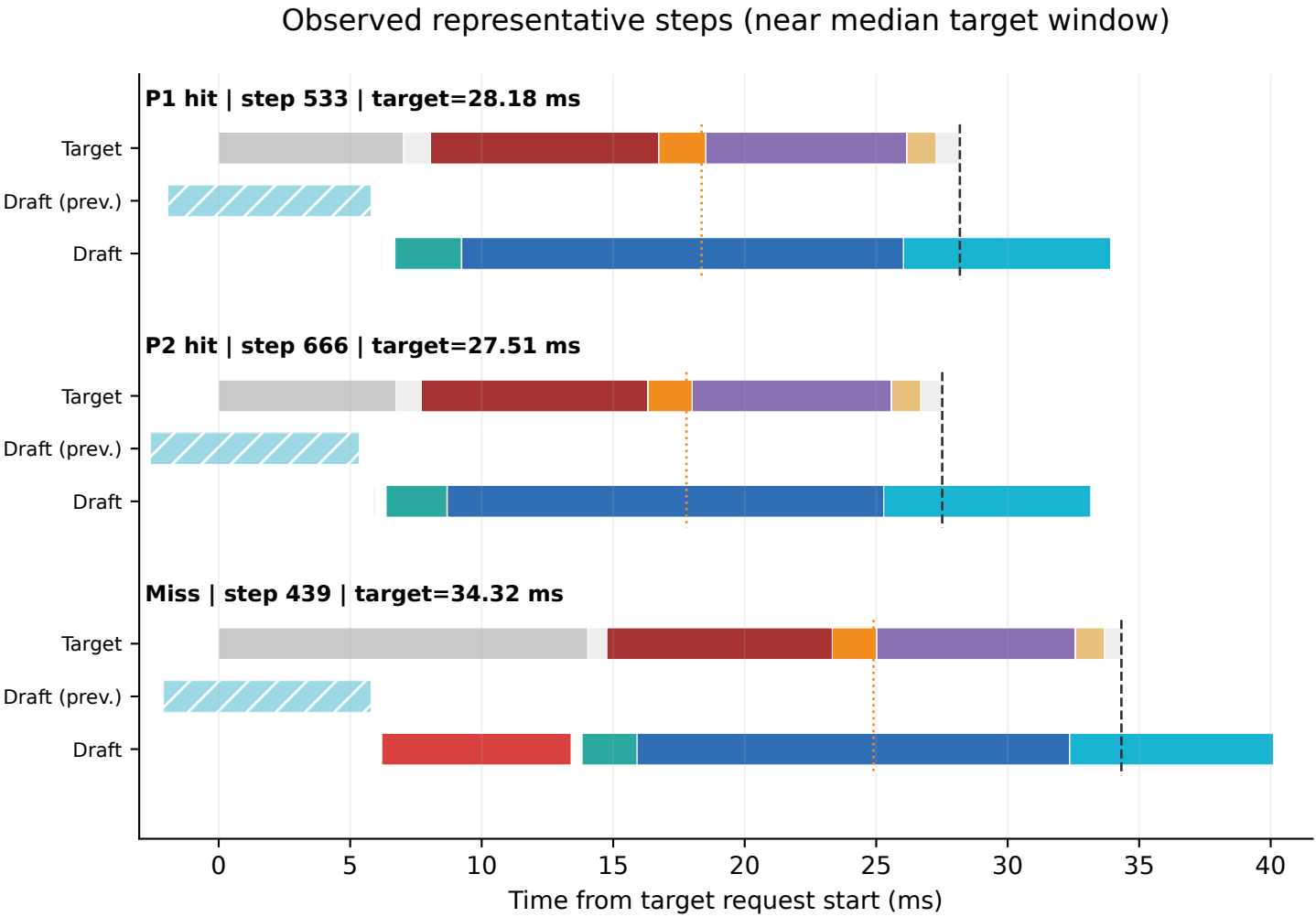
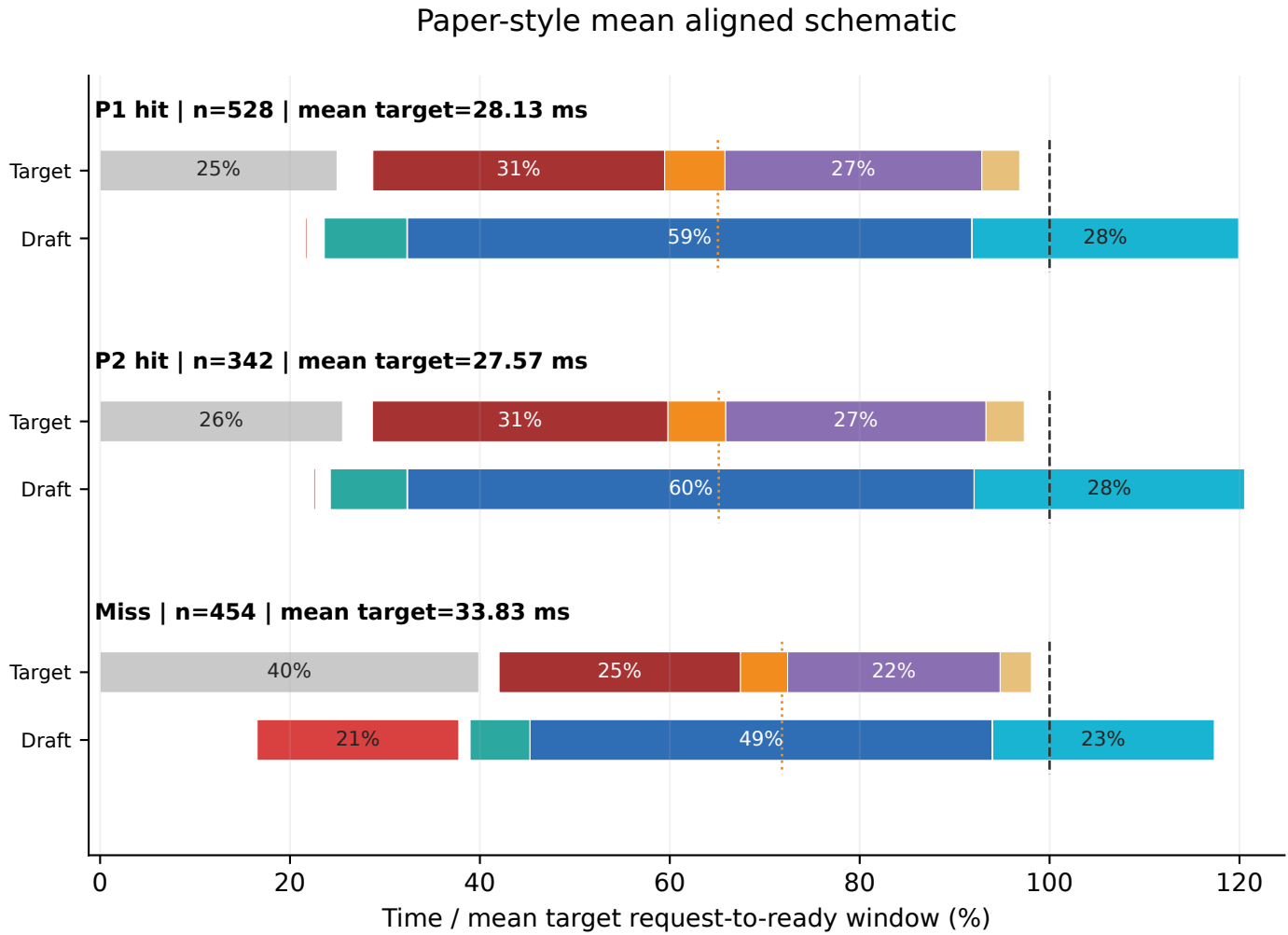
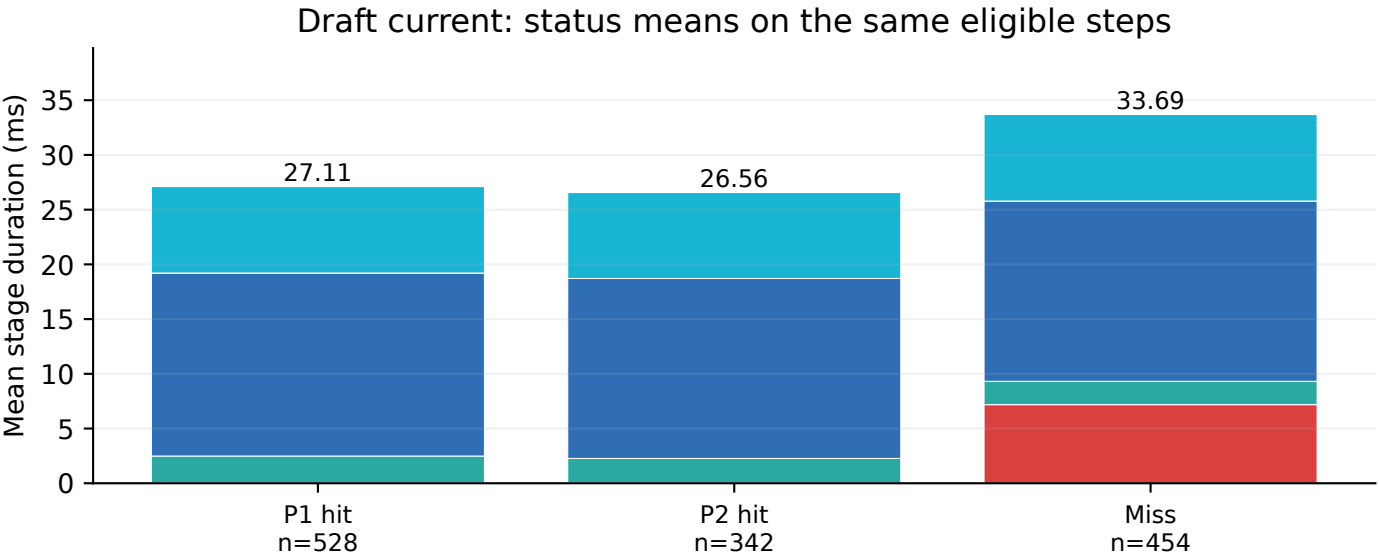
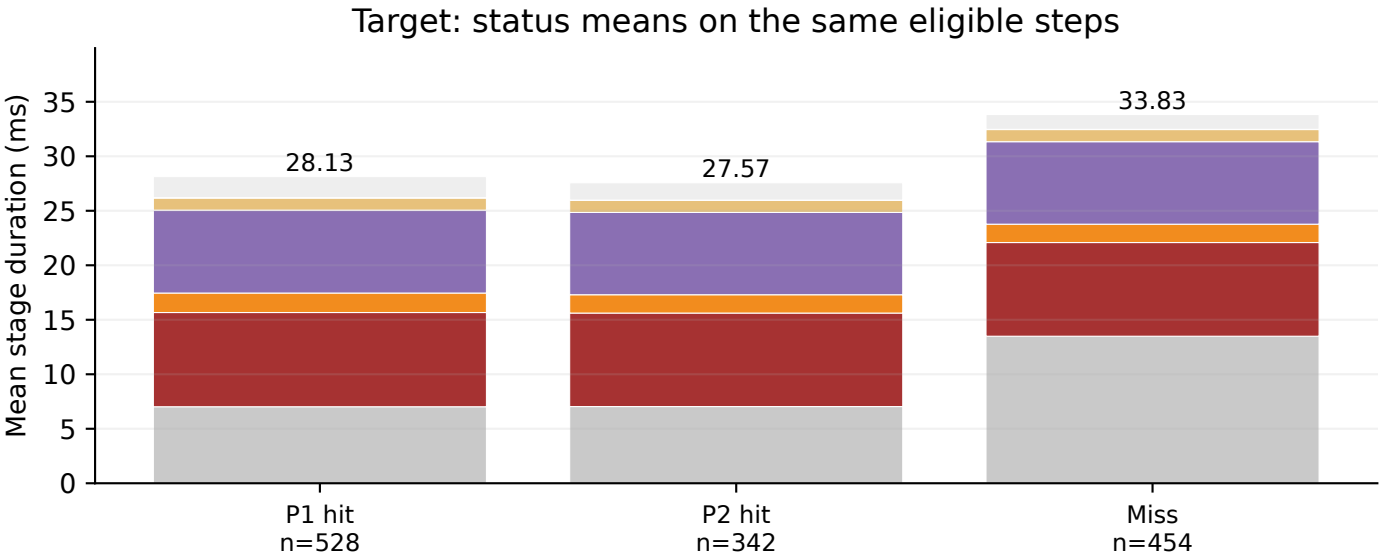
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

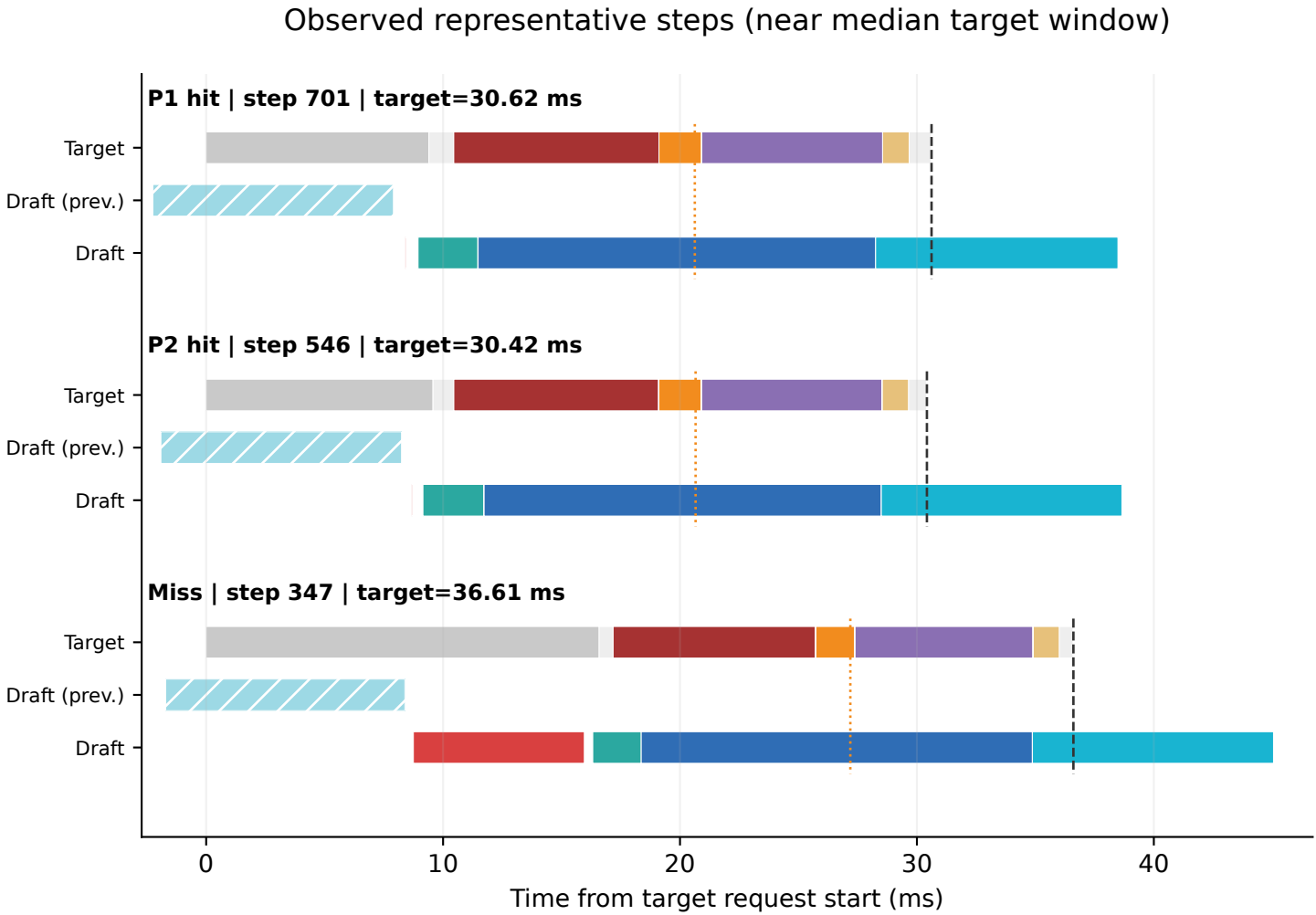
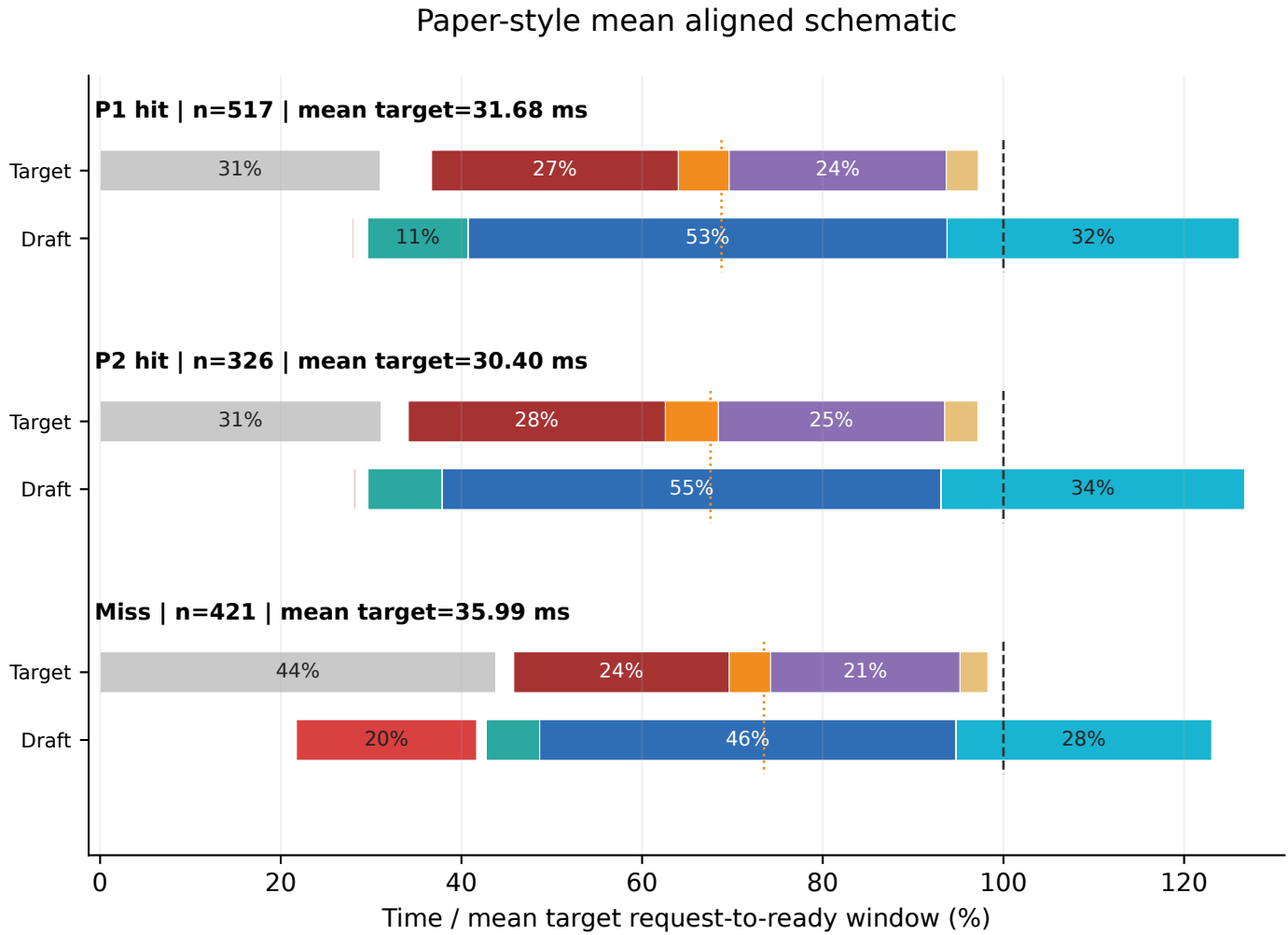
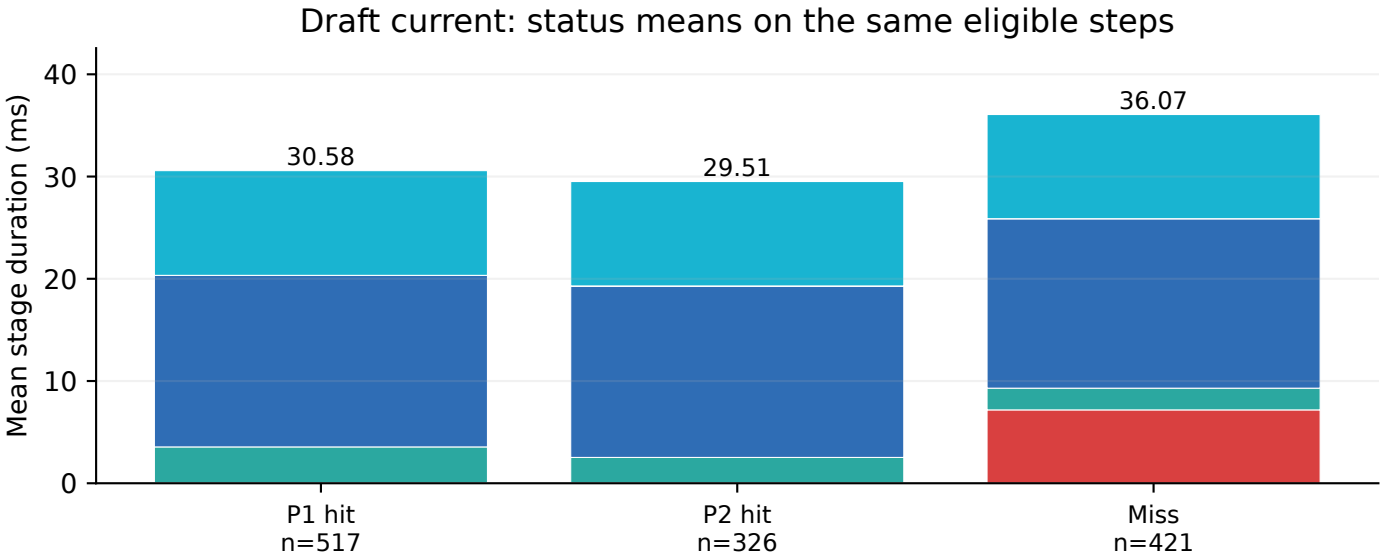
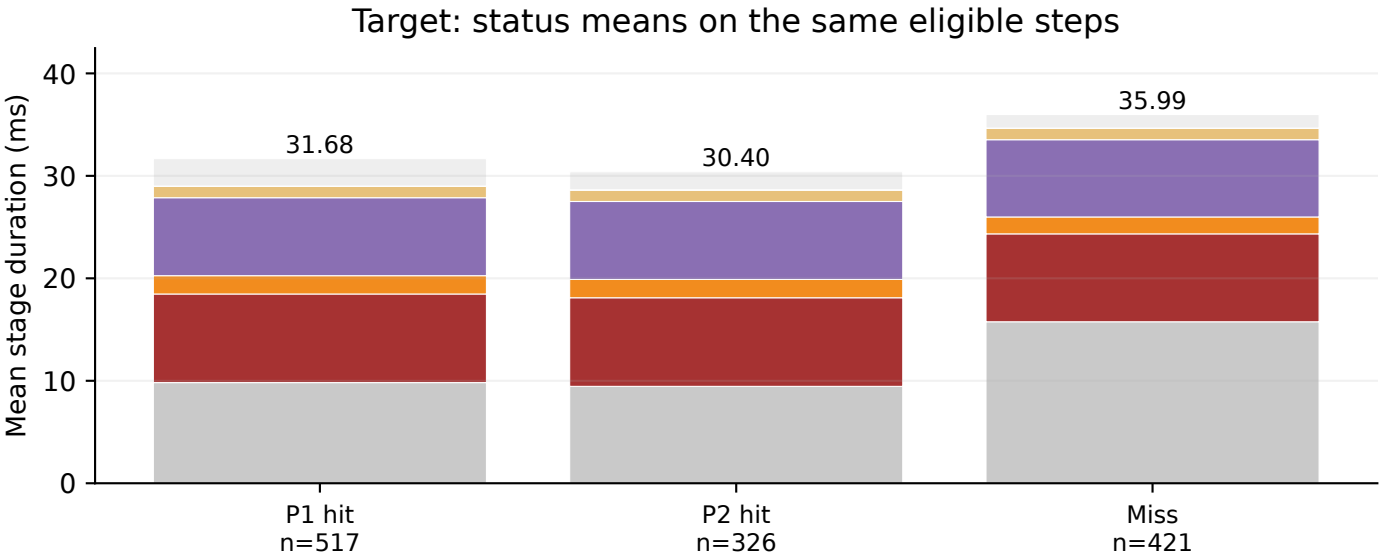
llama3_b1_refine_e16_k6_3_r2
DUET: K1/K2=6/3, exit=16 (0-based), $\bar{C}$ =3, N1/N2=12/6, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1324 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

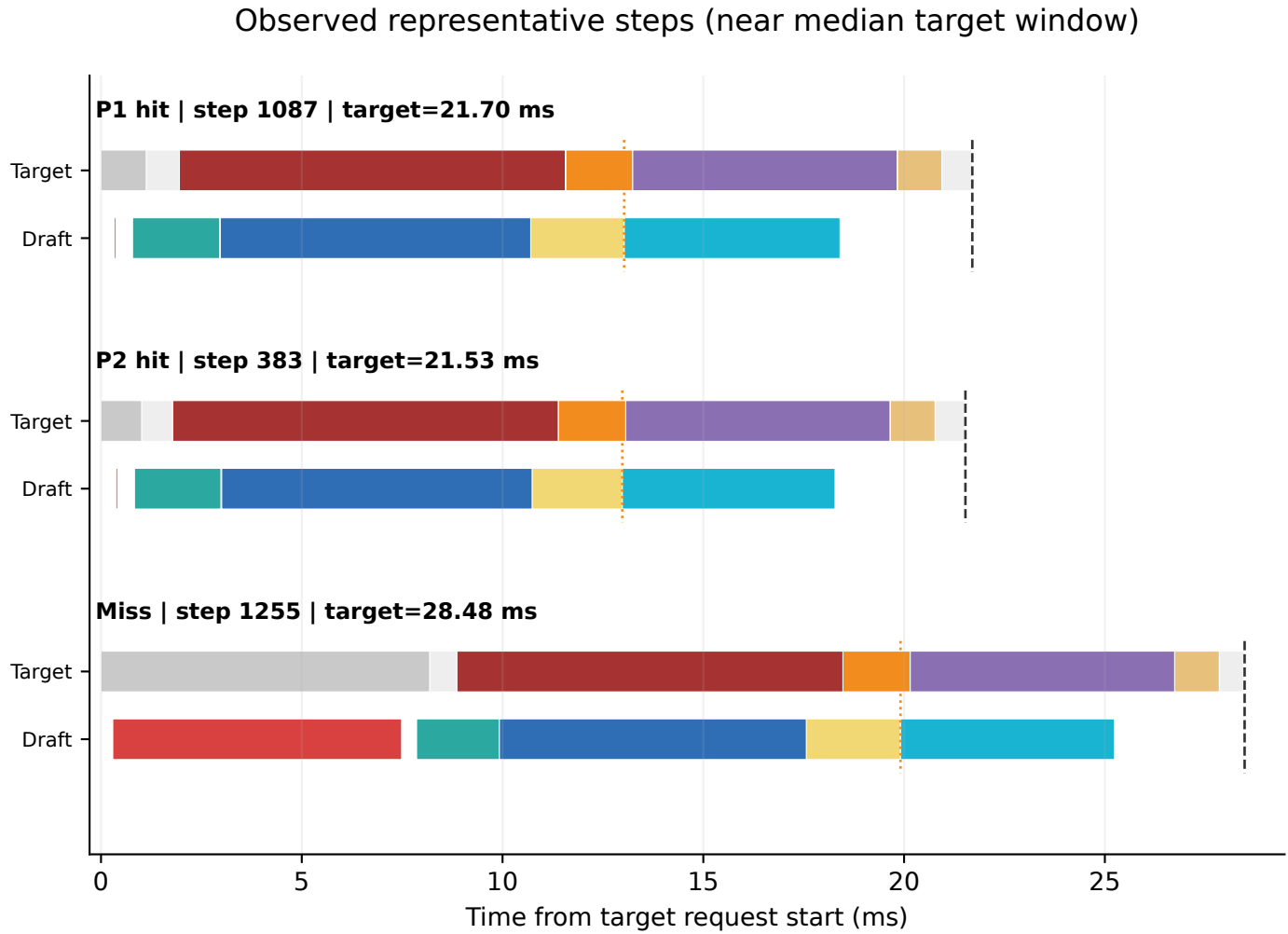
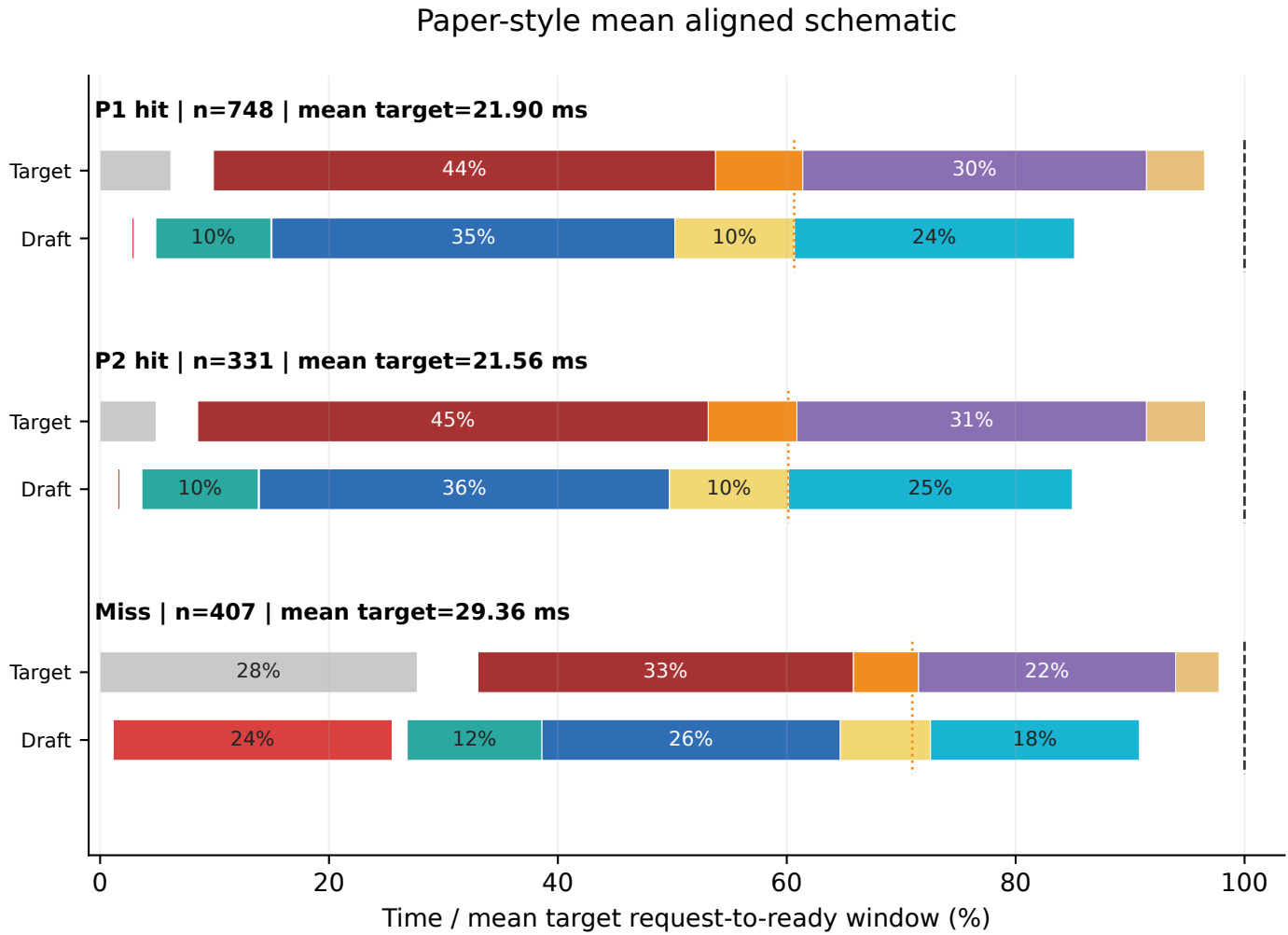
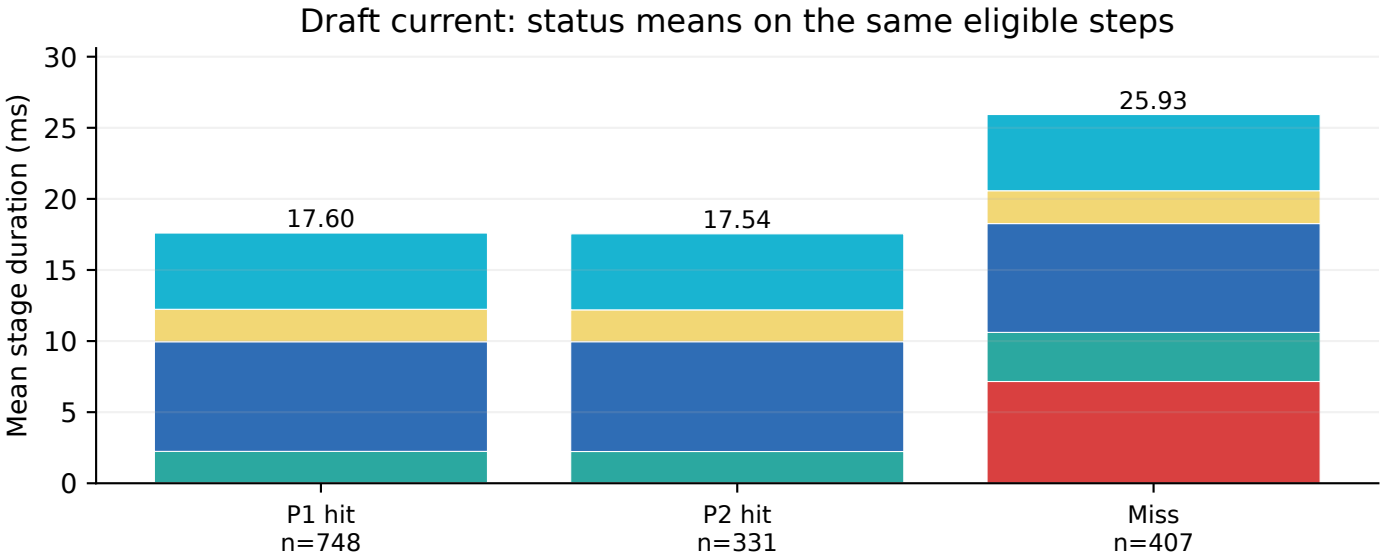
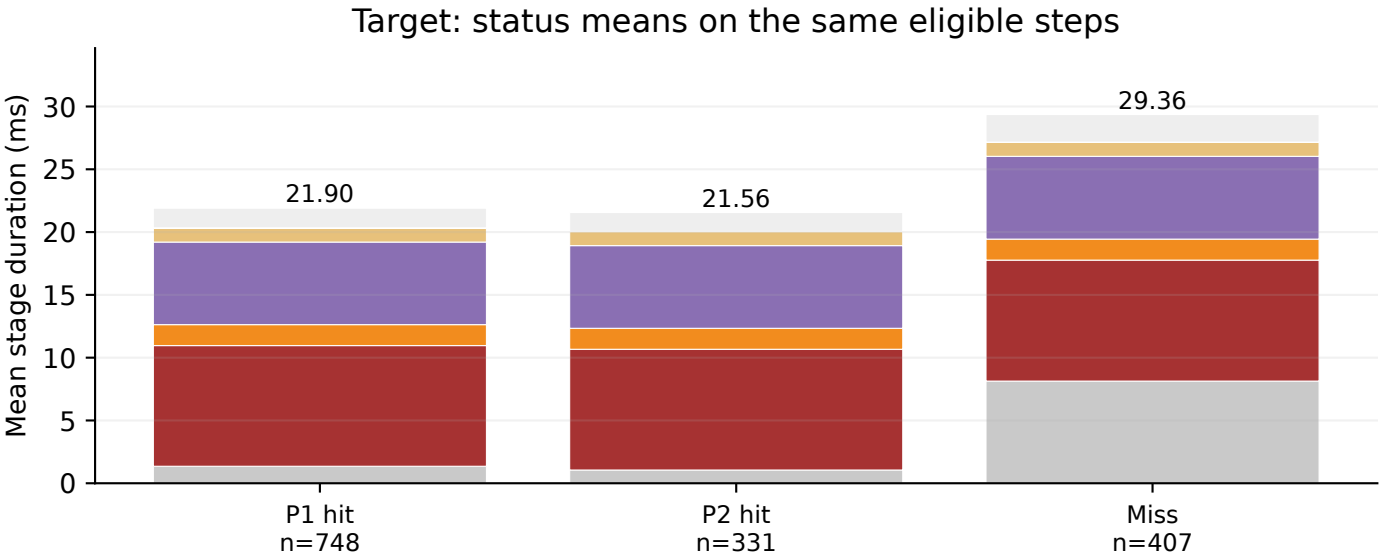
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k6_4_r2
DUET: K1/K2=6/4, exit=16 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1264 eligible steps

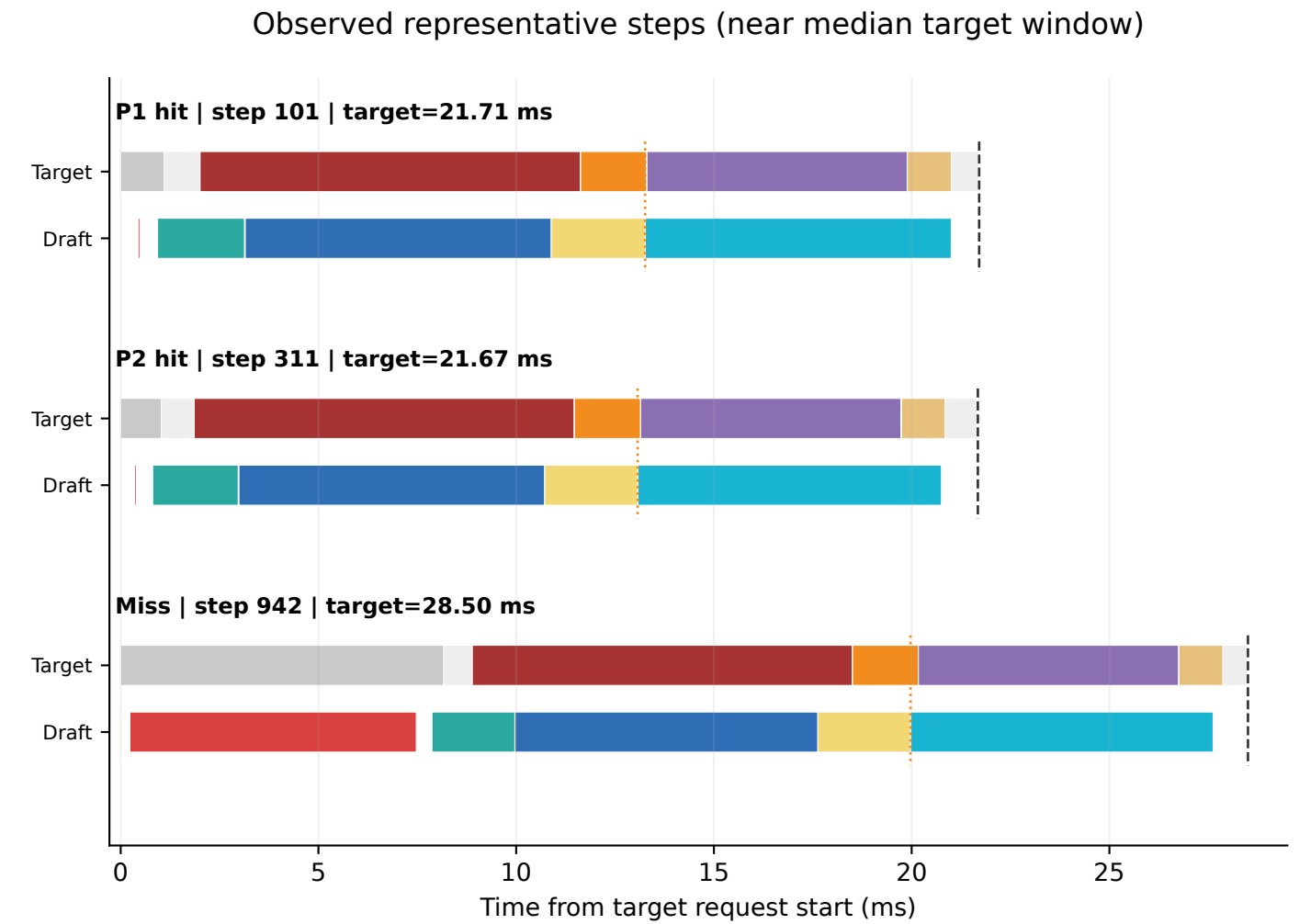
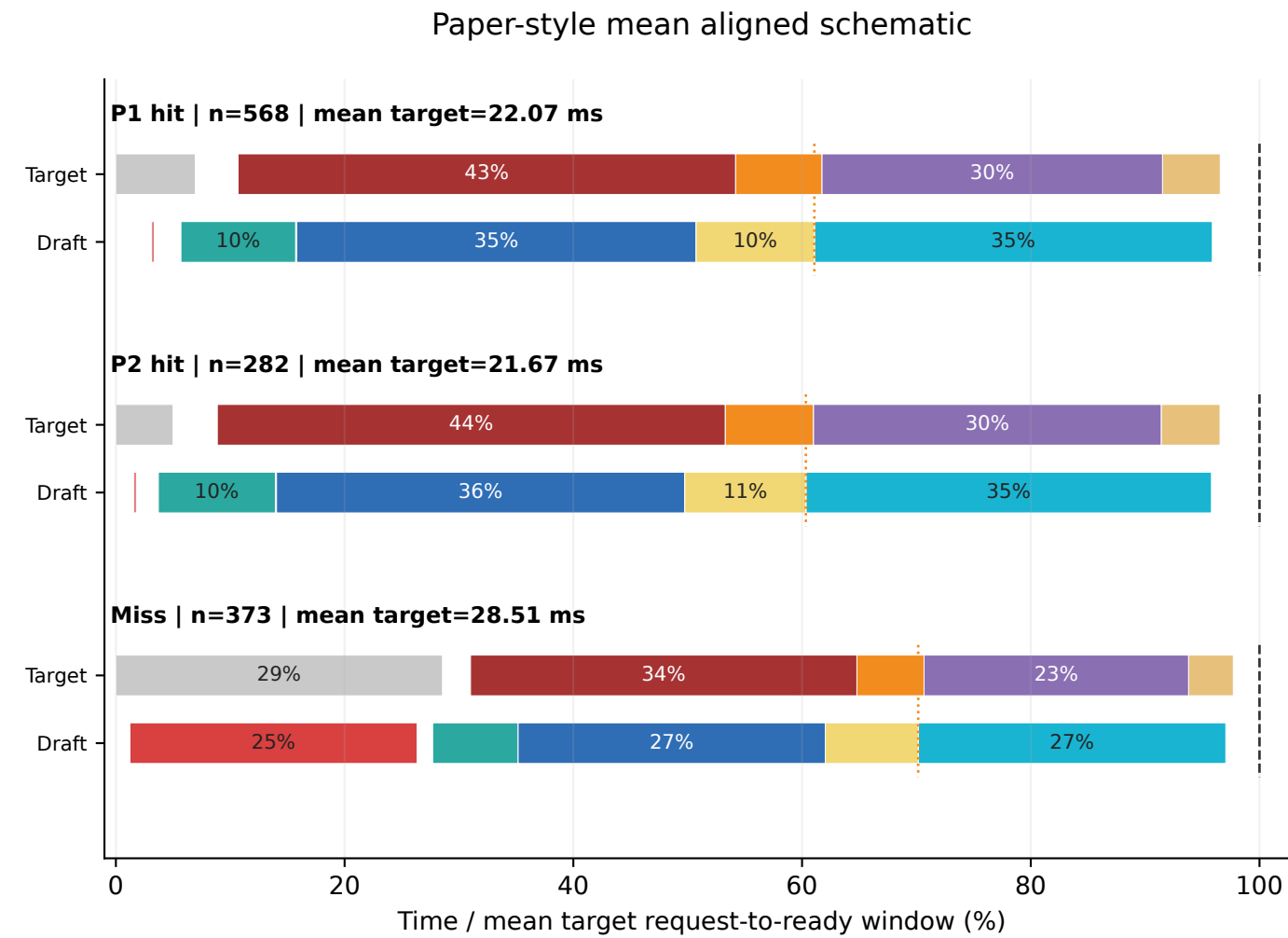
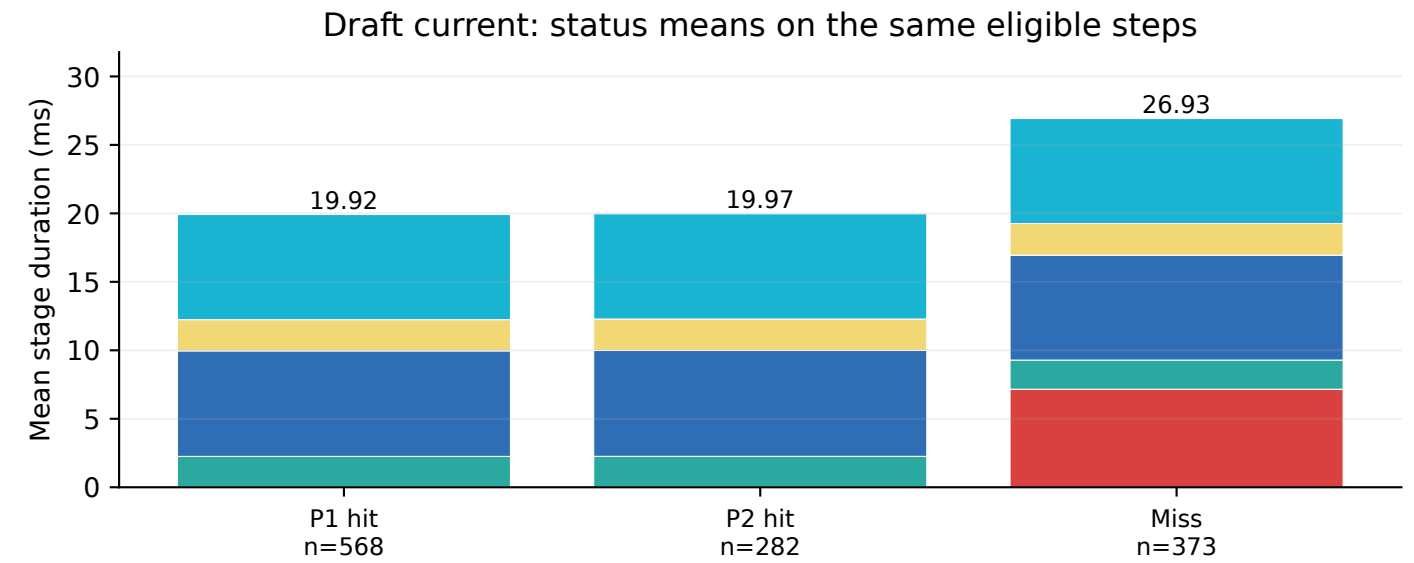
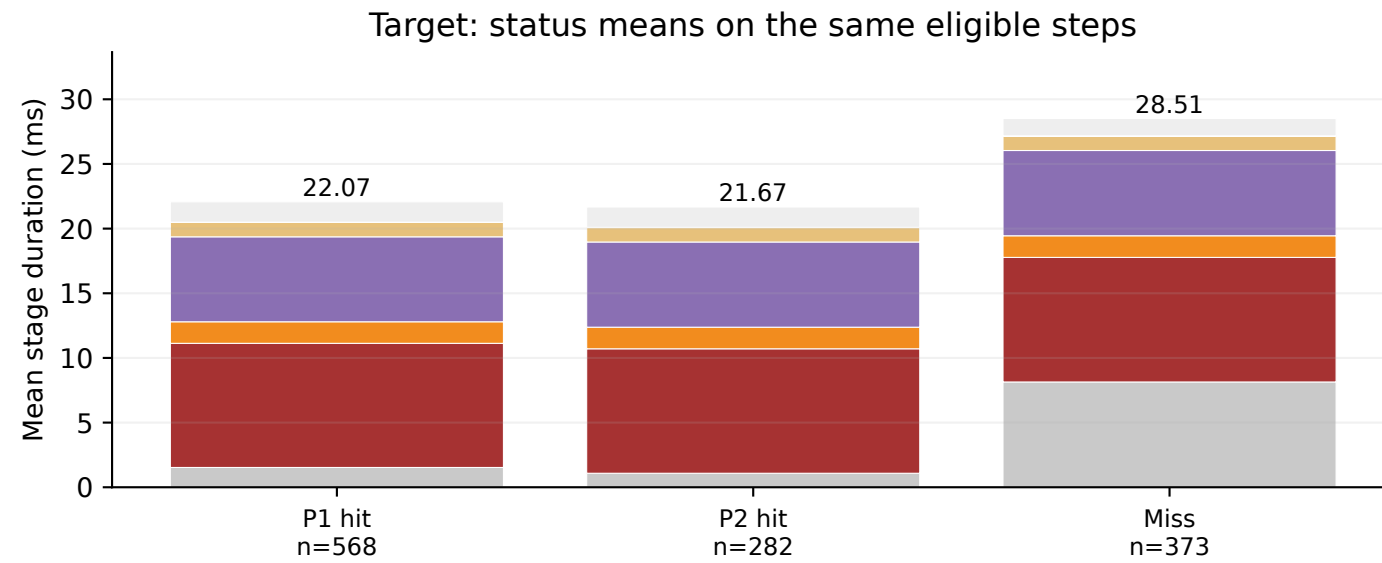


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

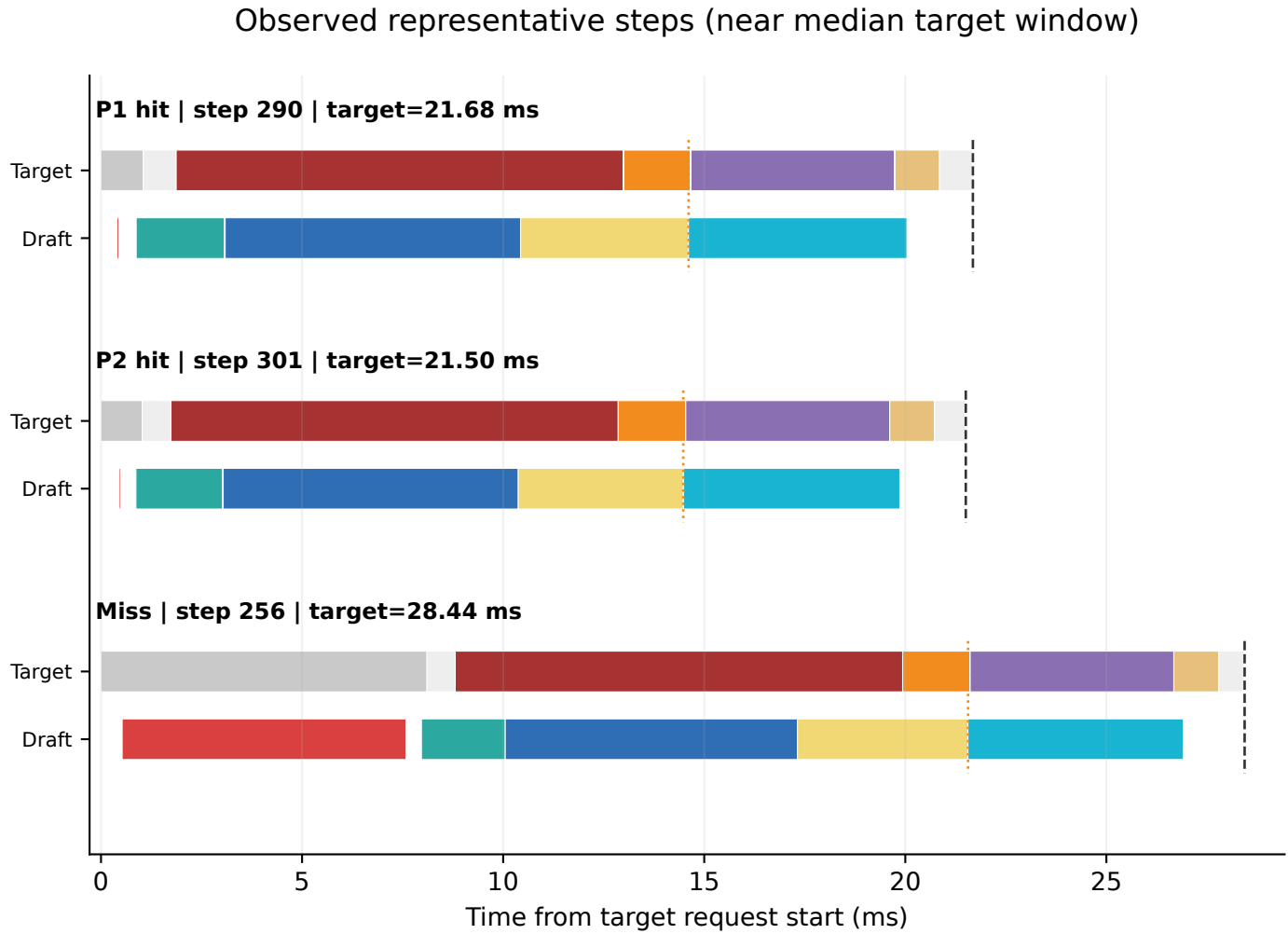
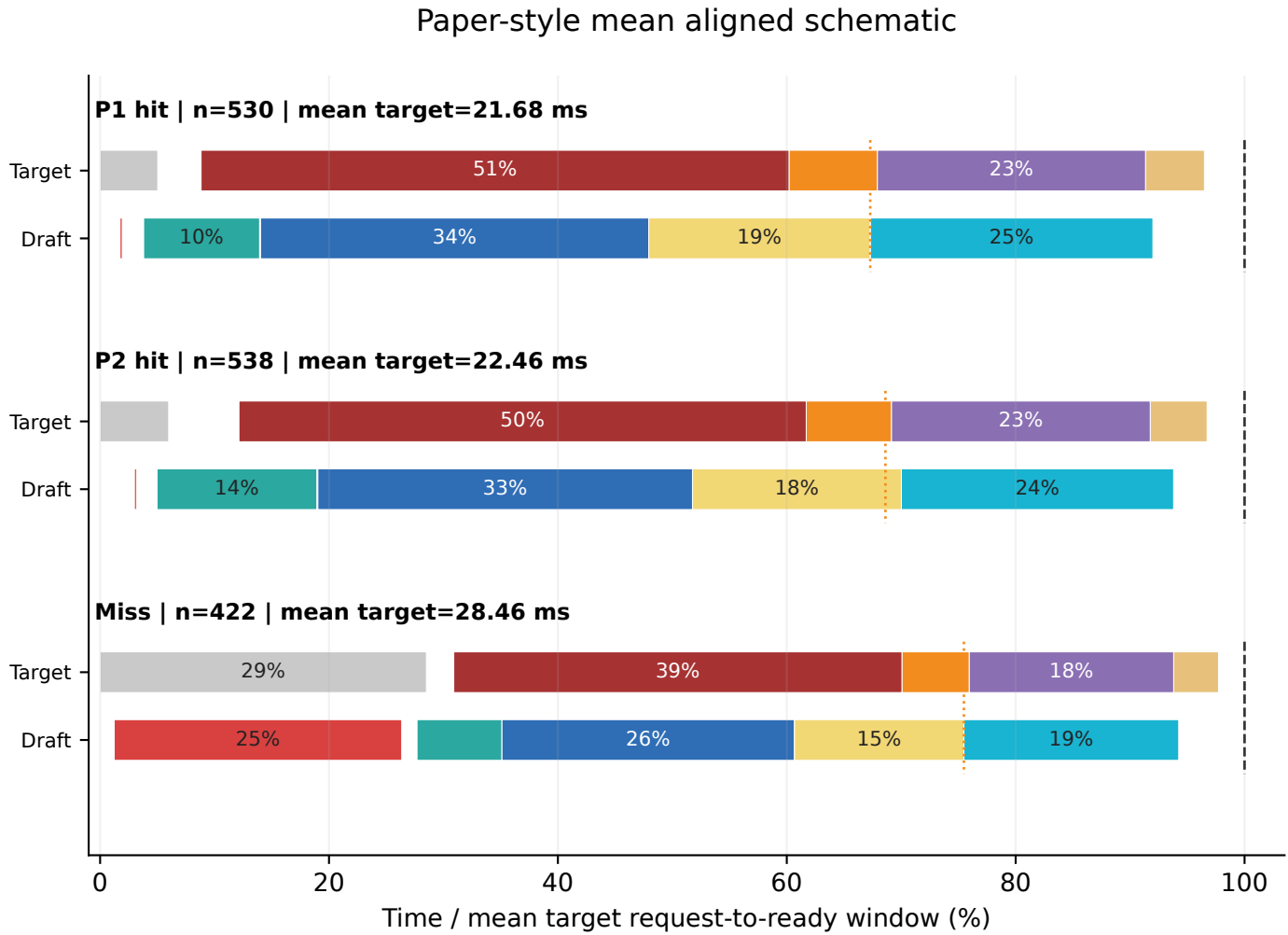
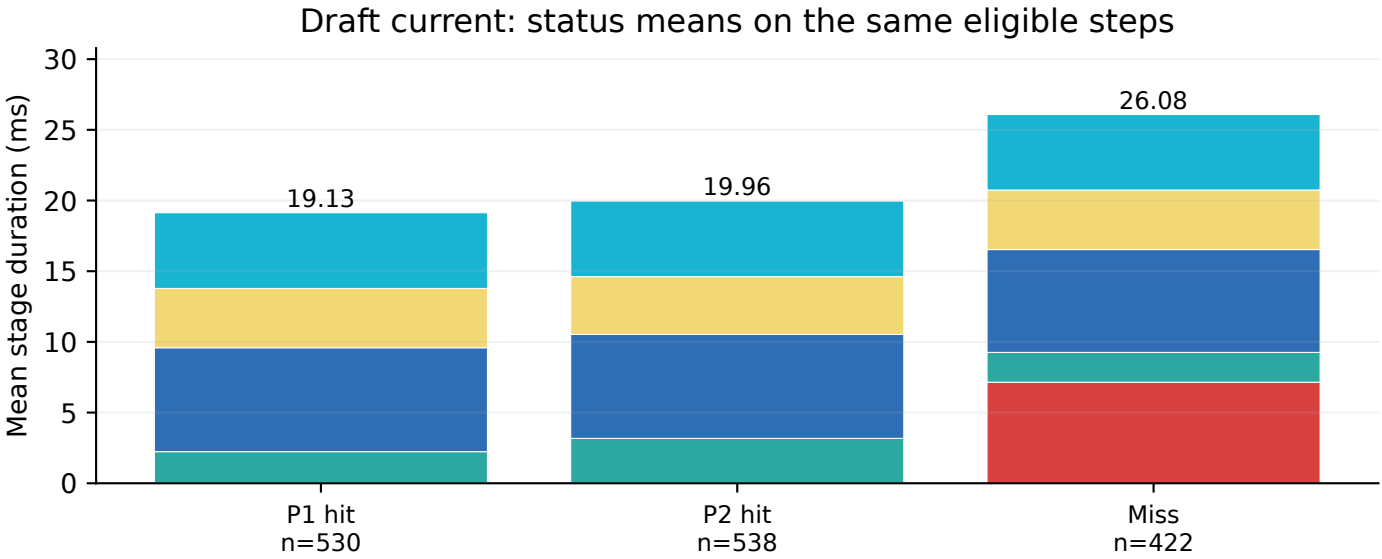
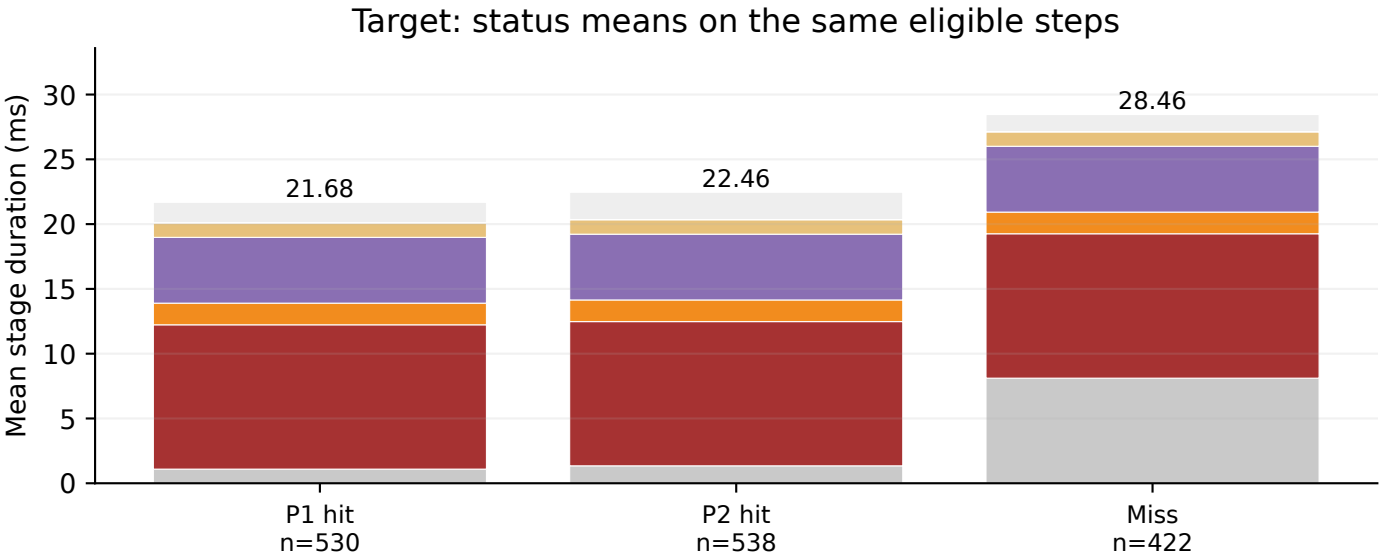


llama3_b1_refine_e18_k3_3_r2
DUET: K1/K2=3/3, exit=18 (0-based), C=3, N1/N2=6/6, U=2, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1223 eligible steps



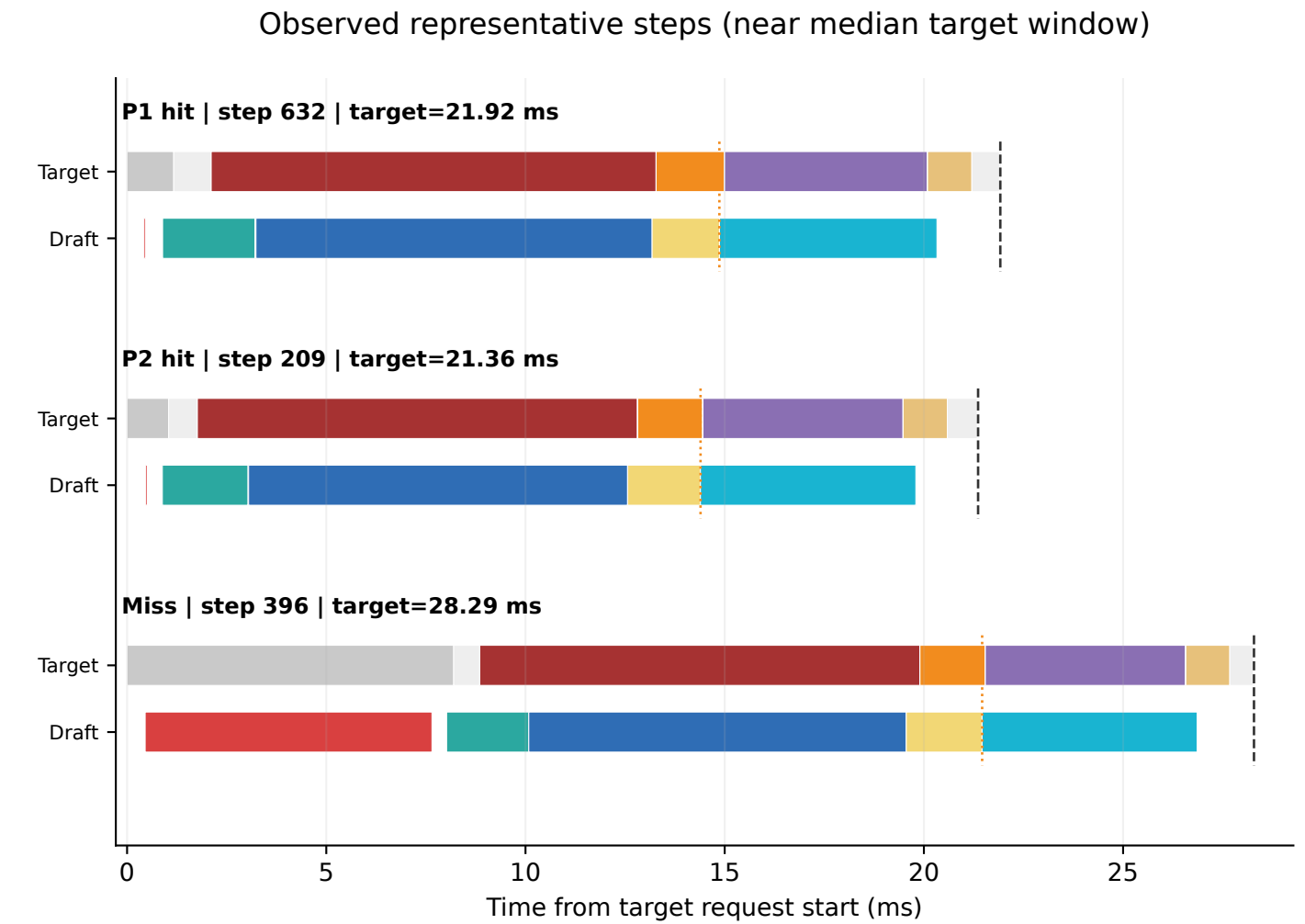
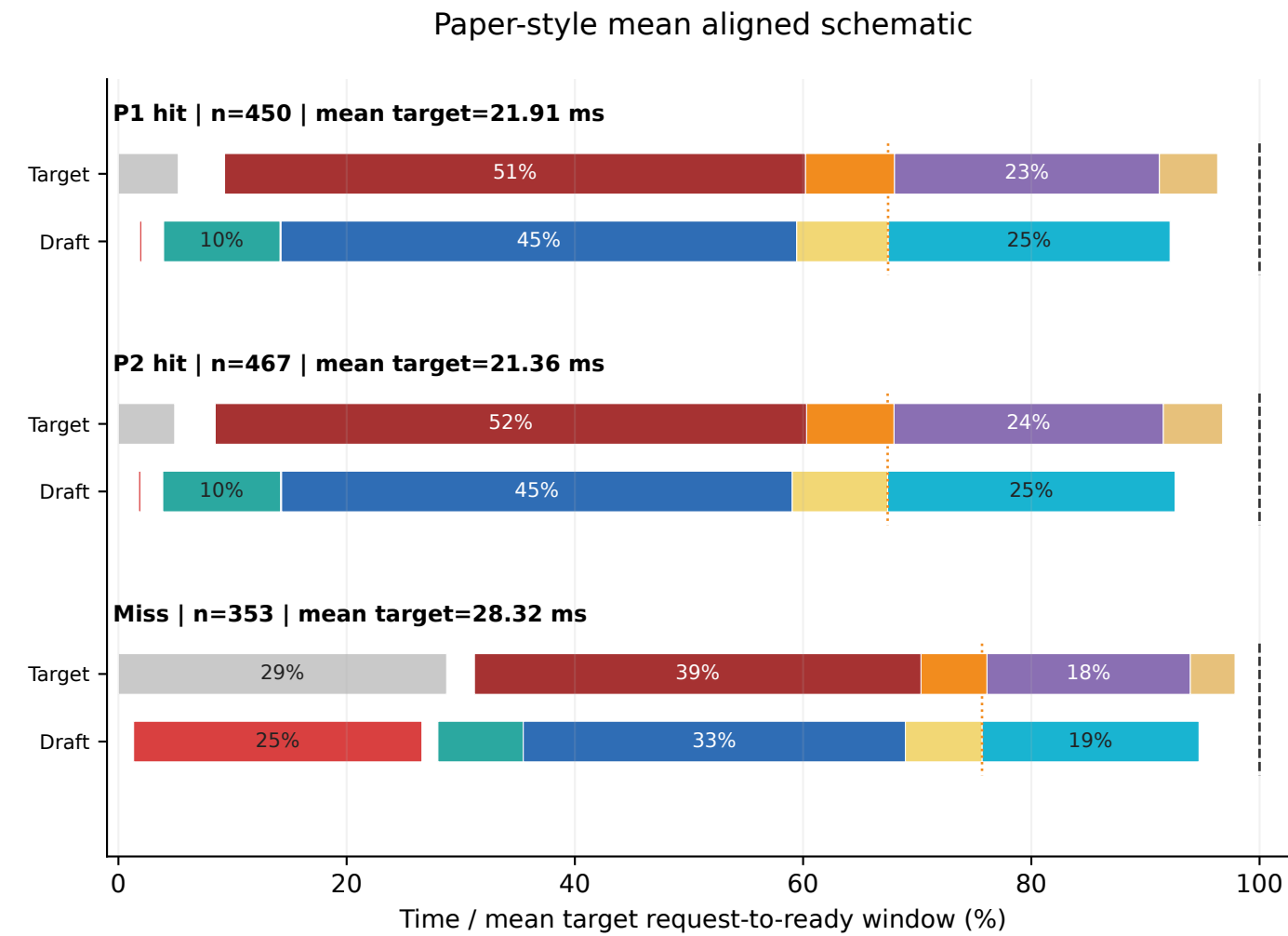
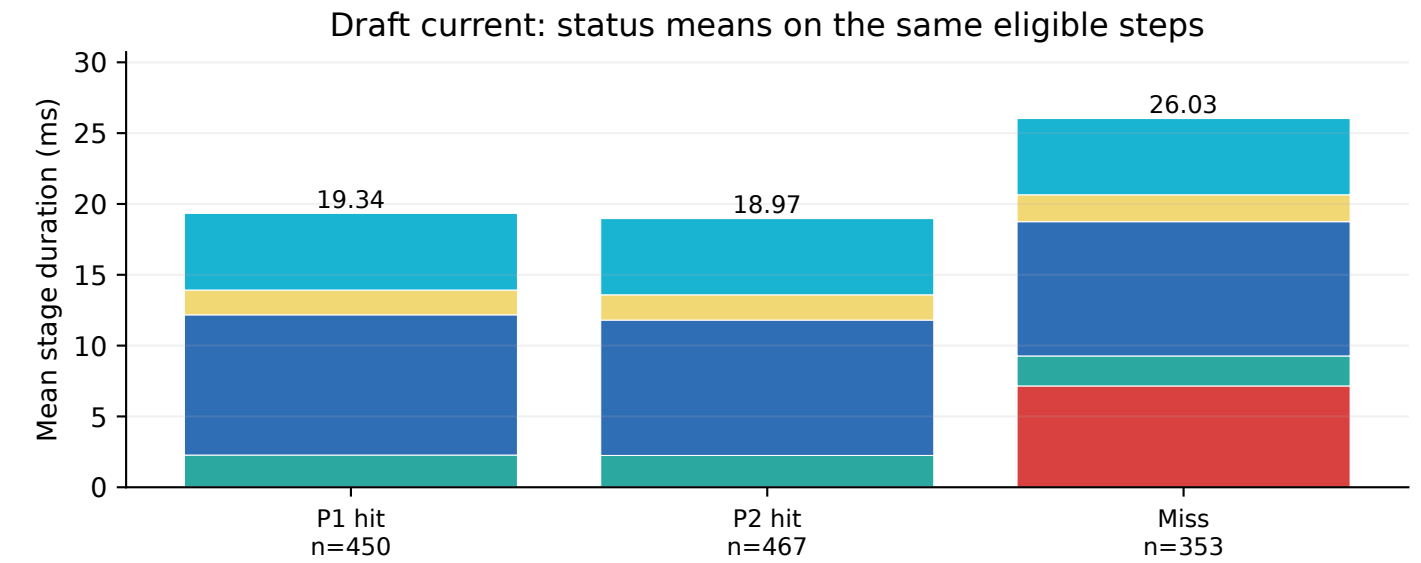
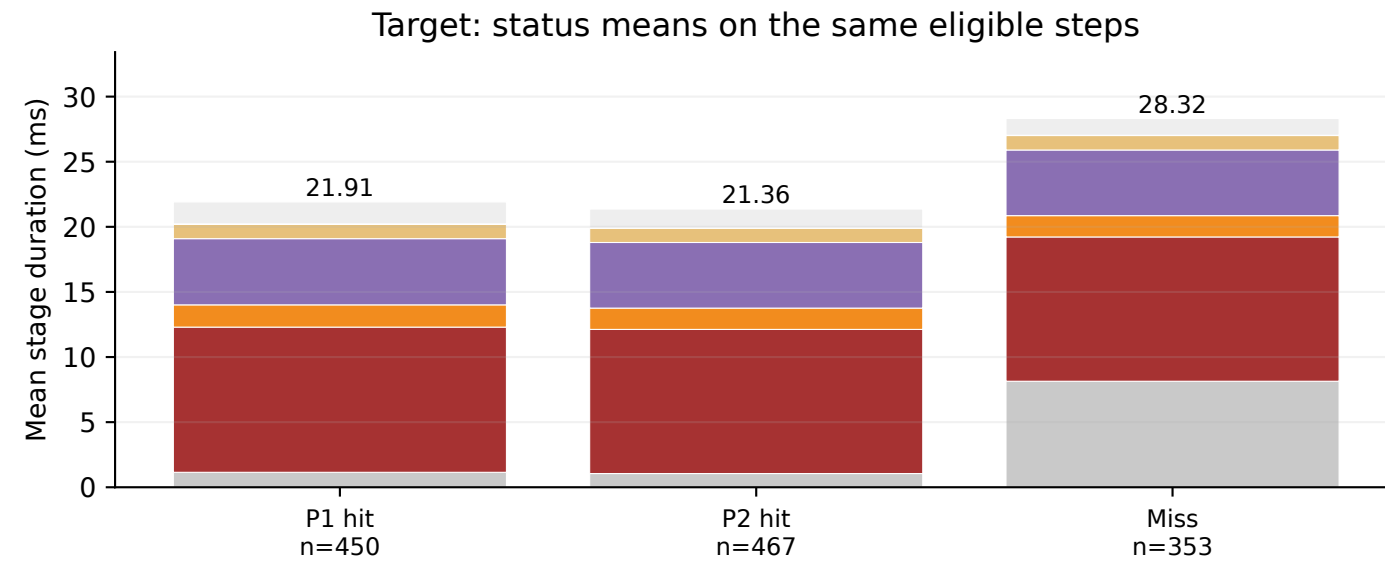
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e21_k3_2_r1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1490 eligible steps

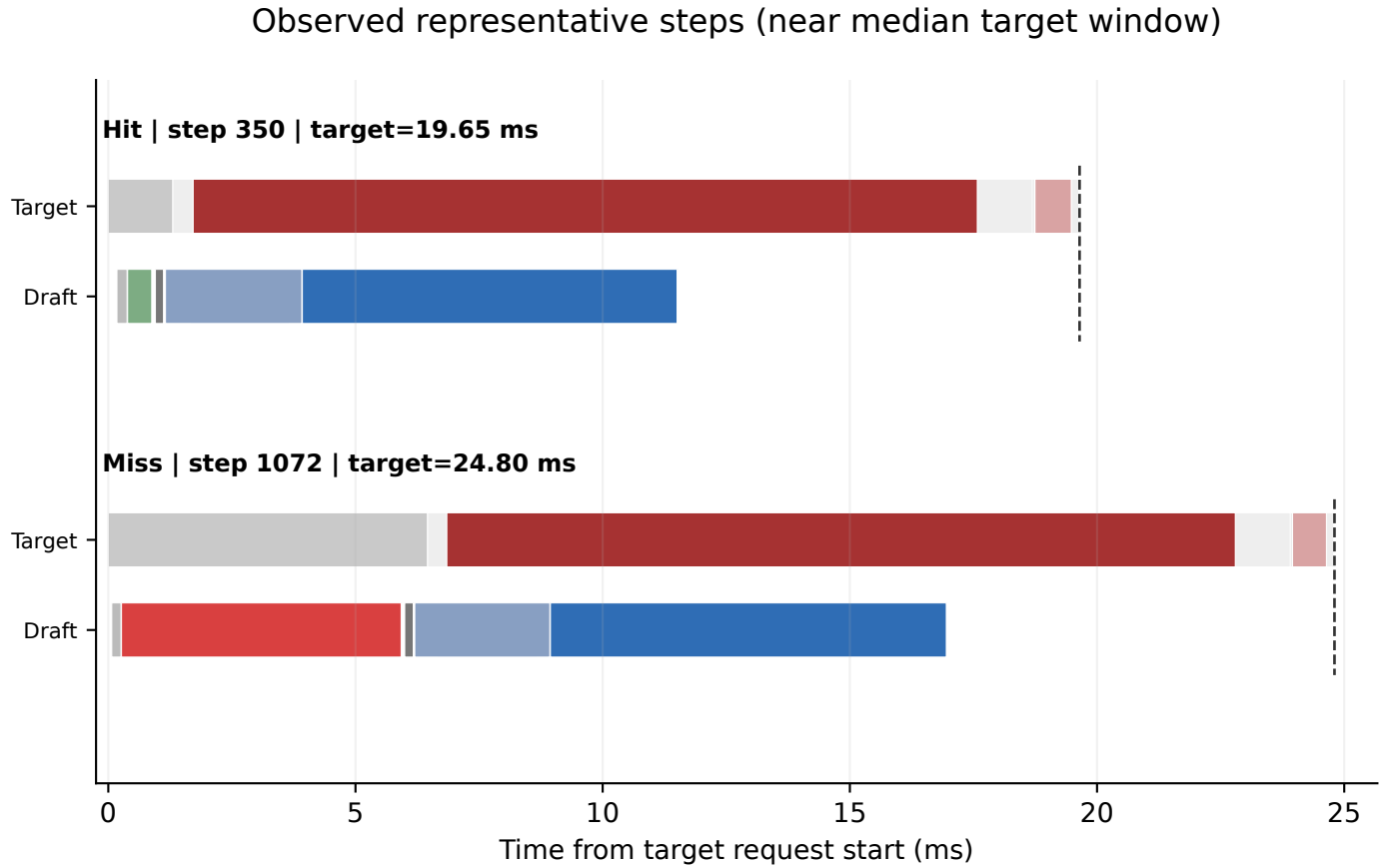
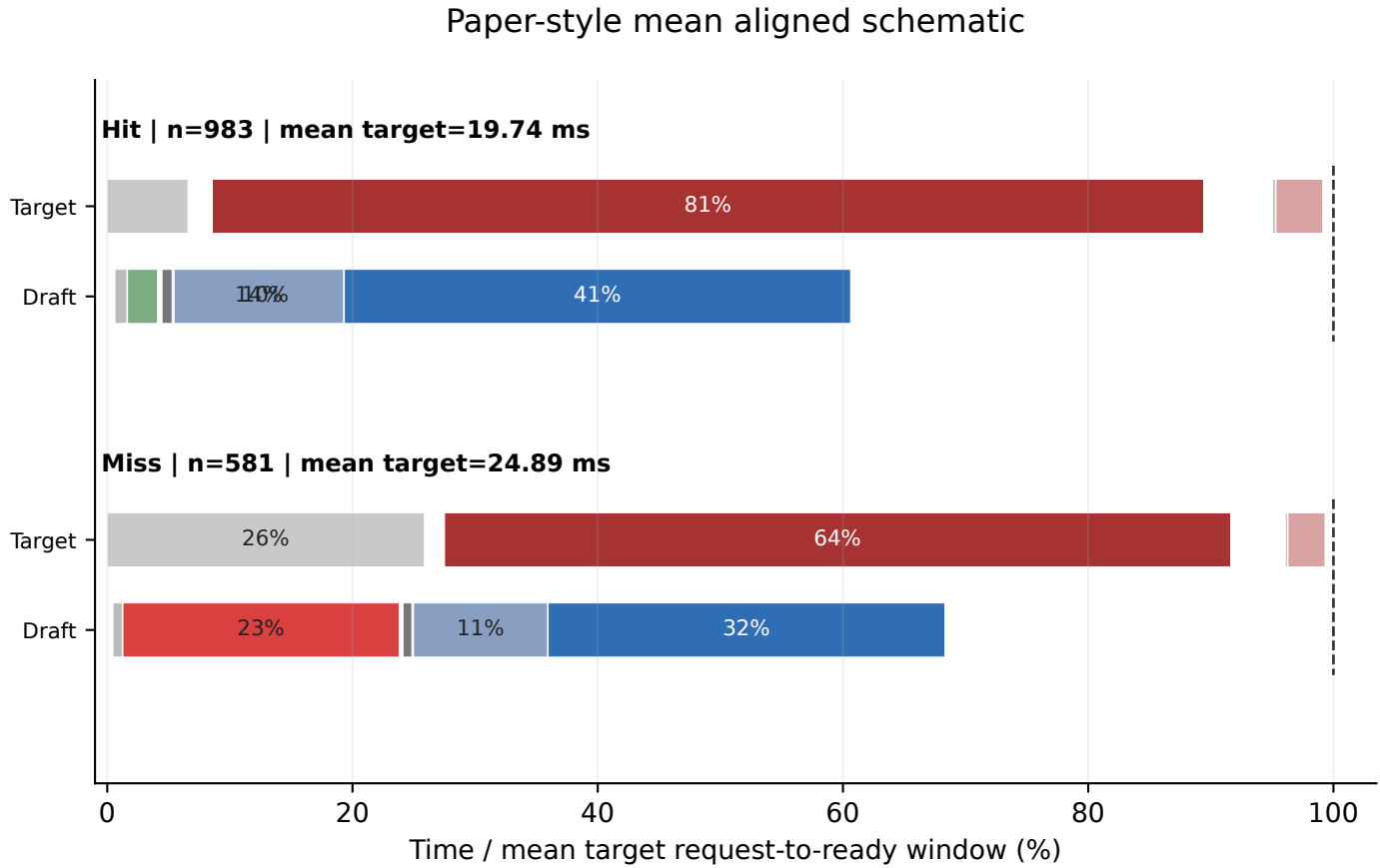
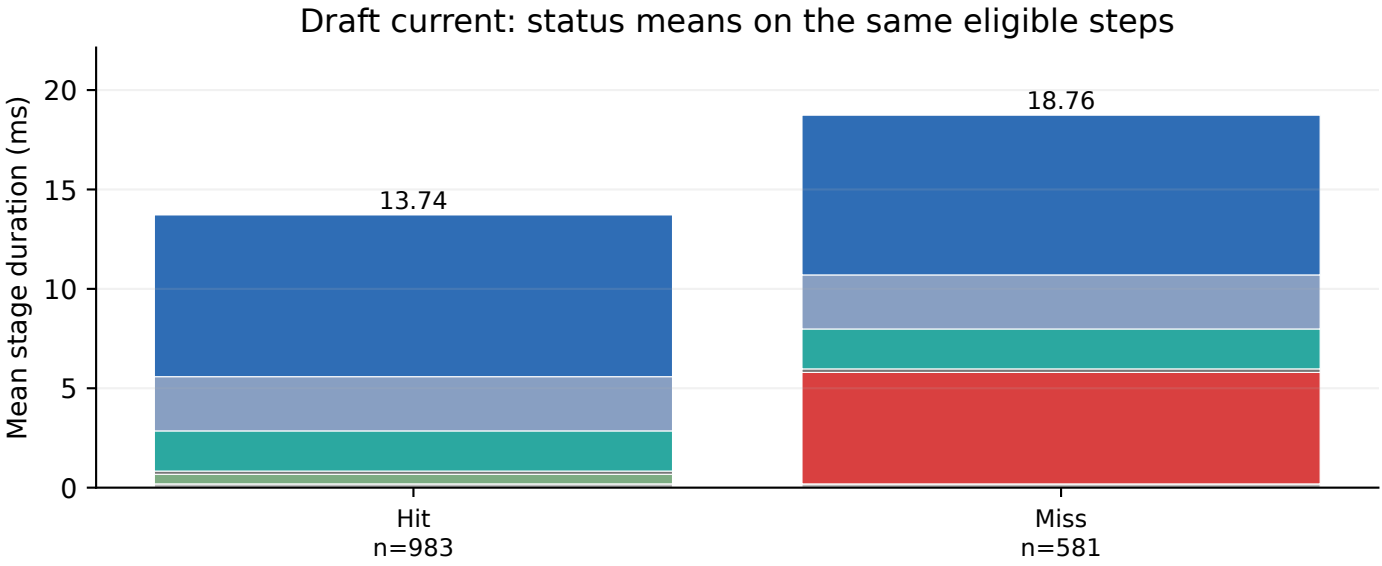
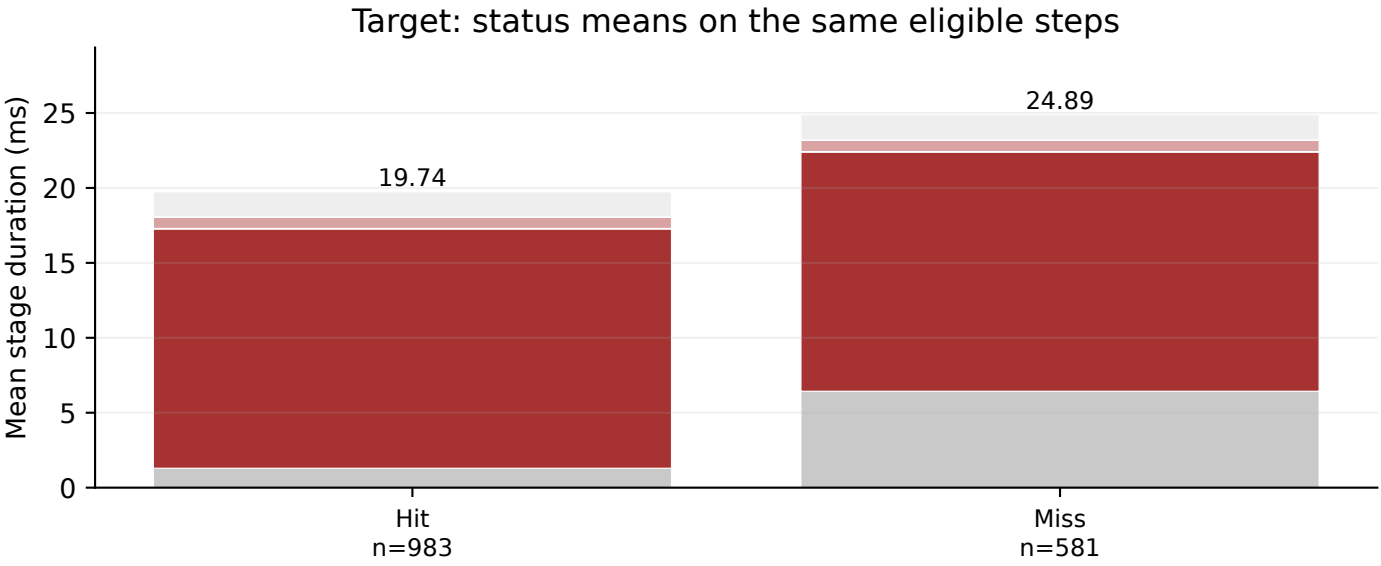


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e21_k4_2_r1
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=1, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1270 eligible steps

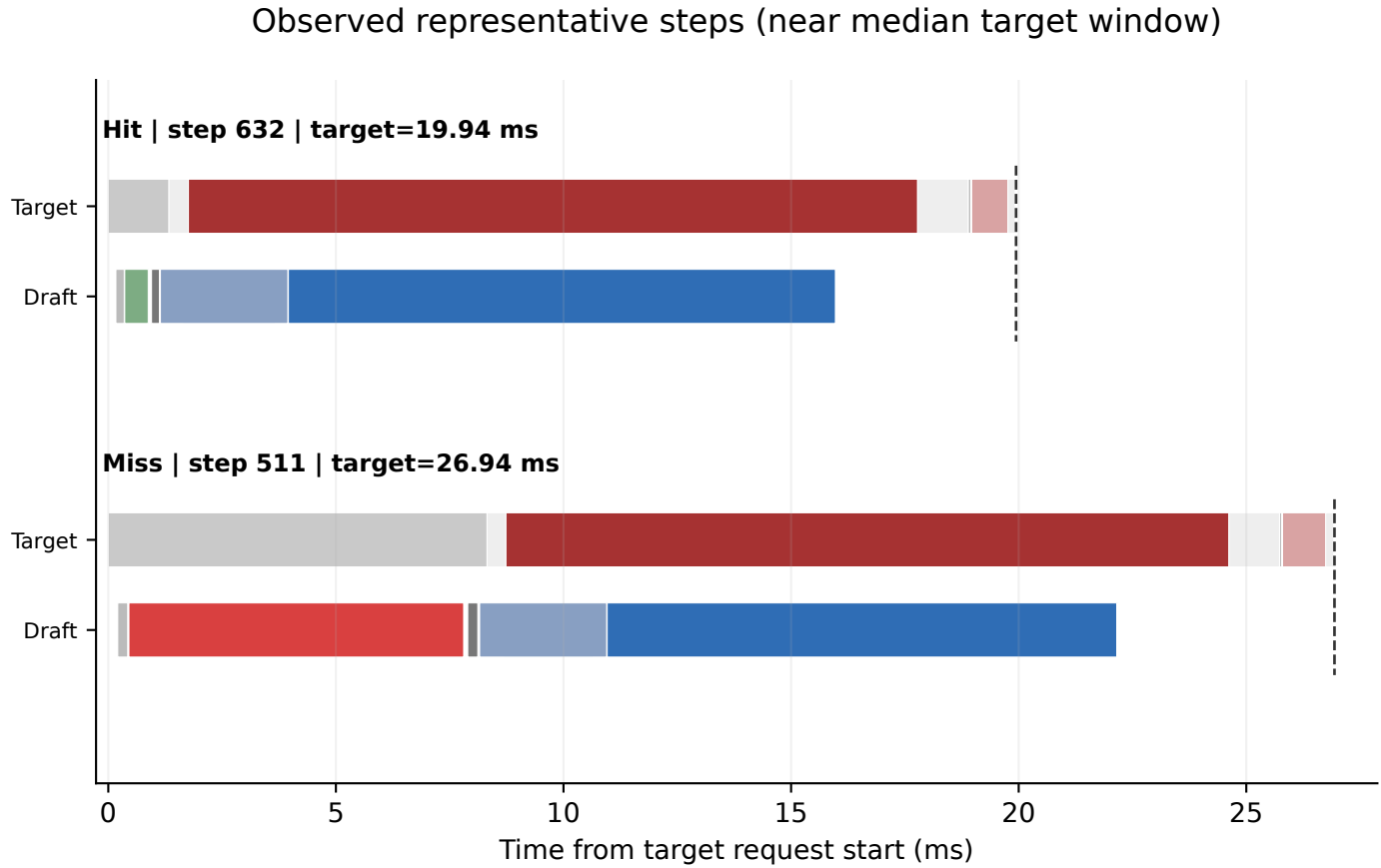
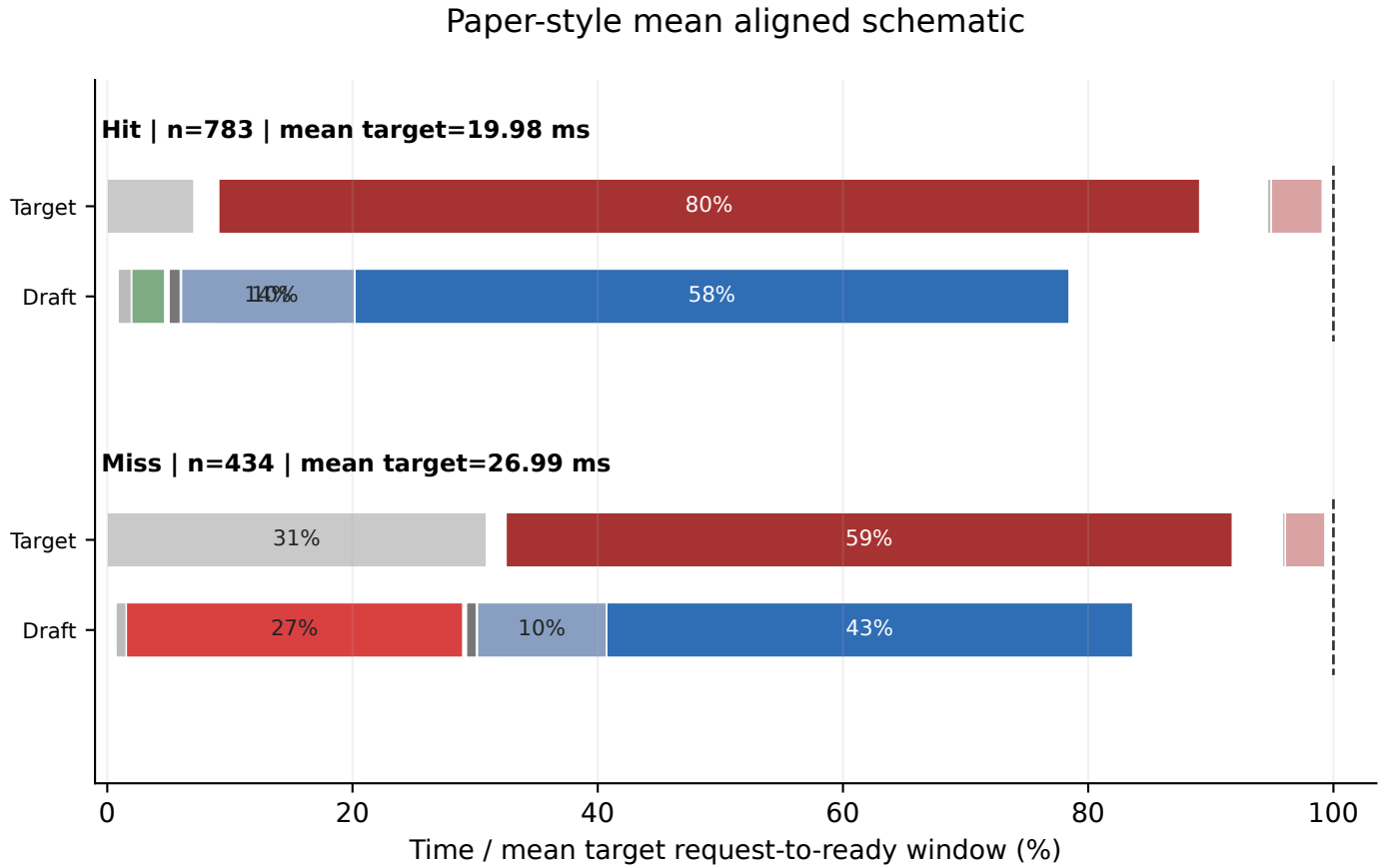
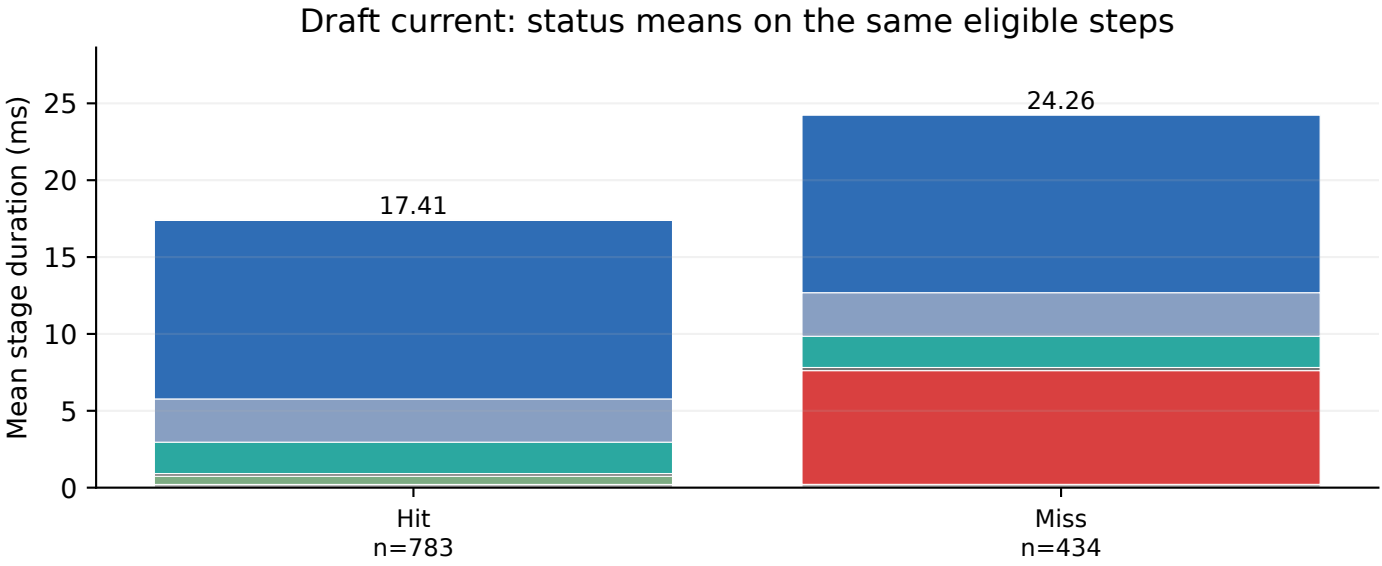
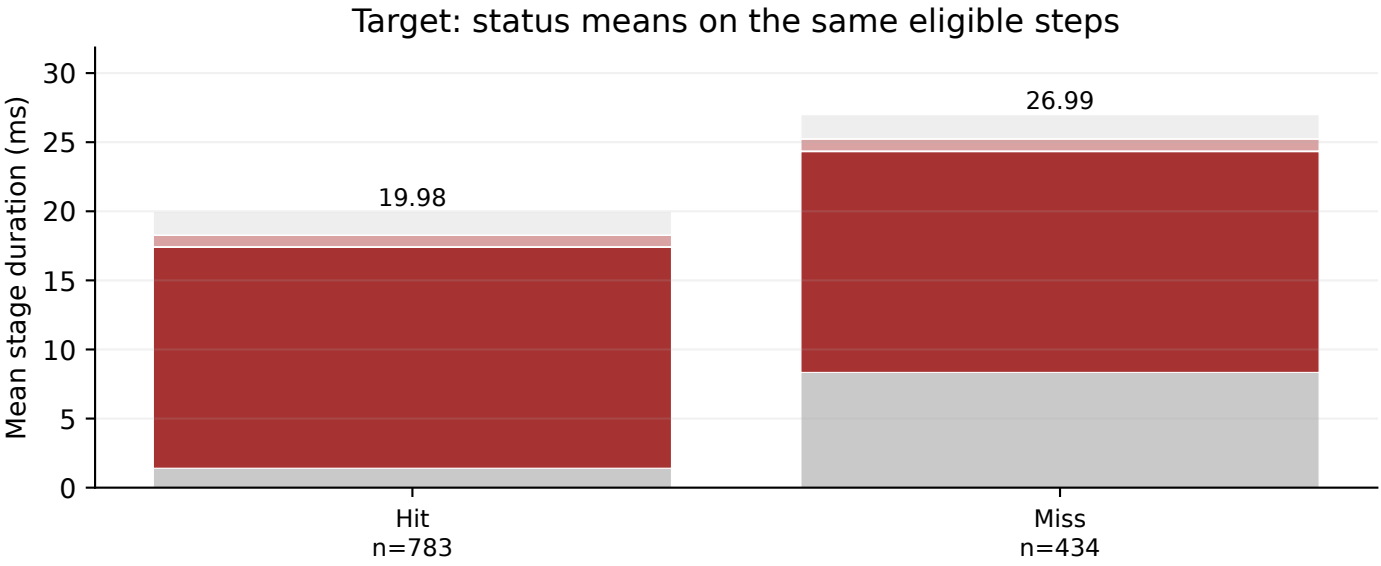


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



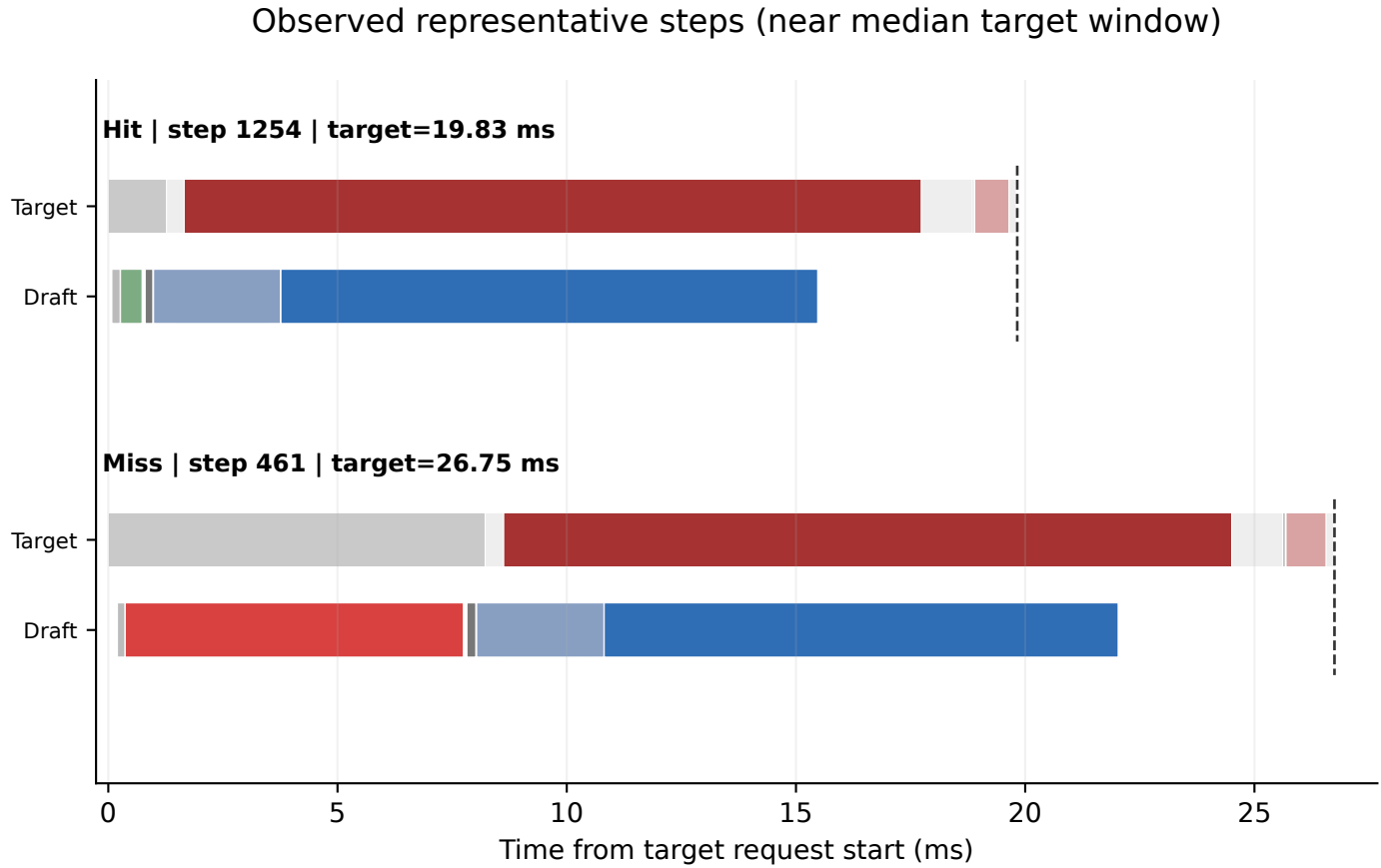
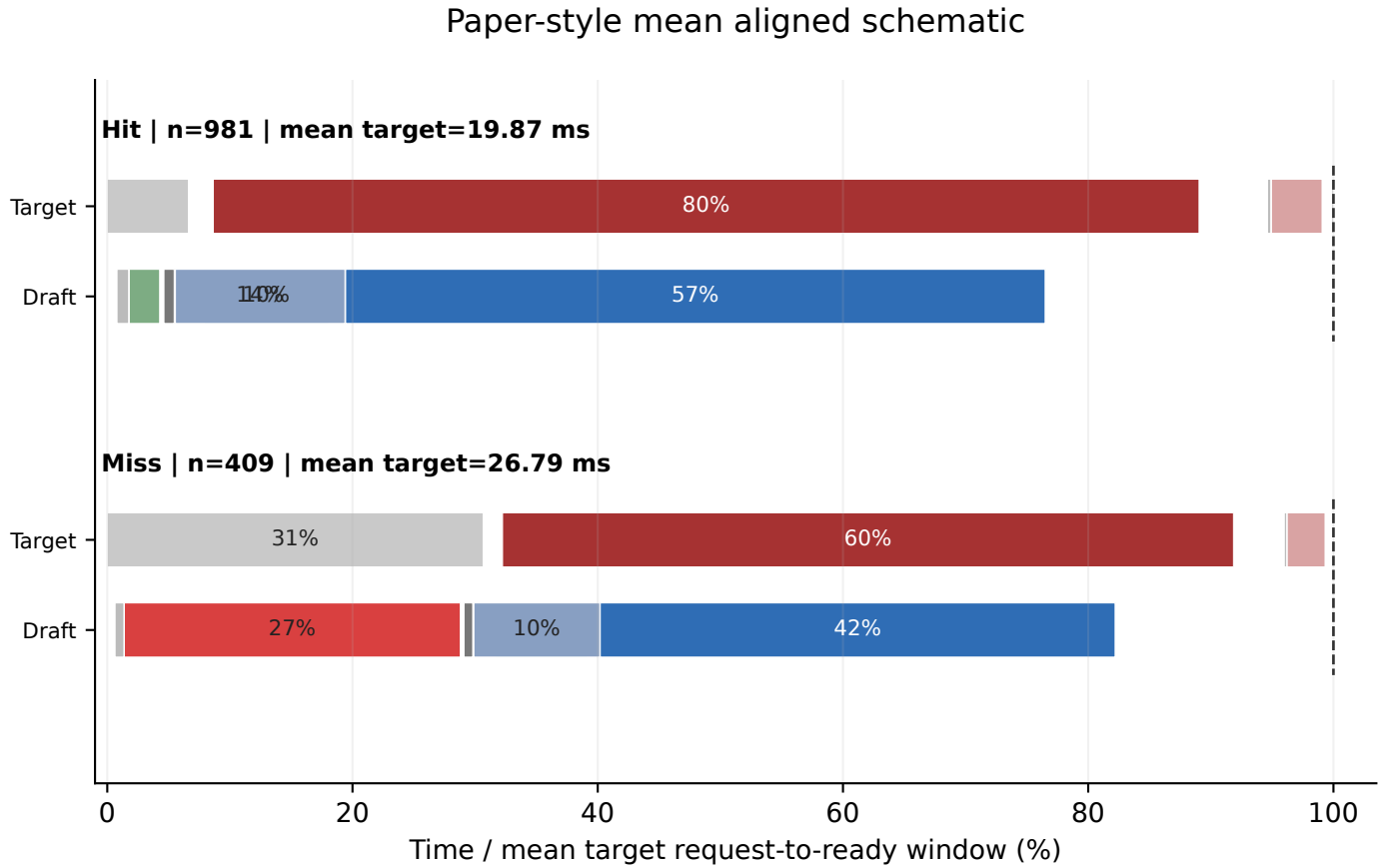
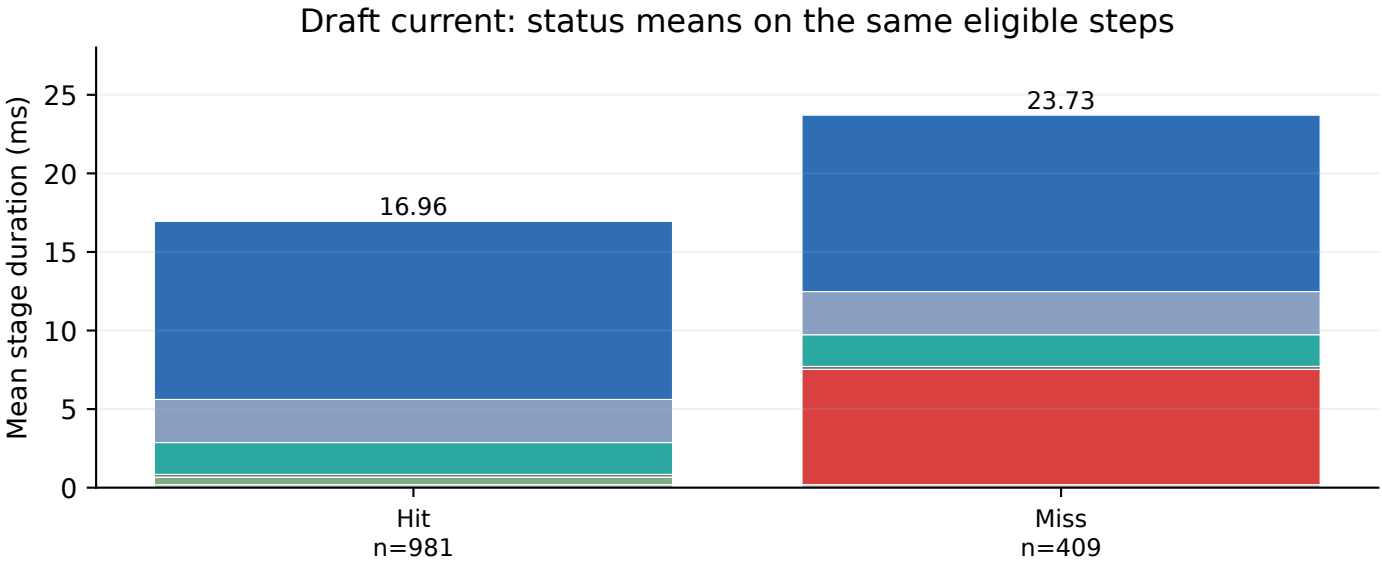
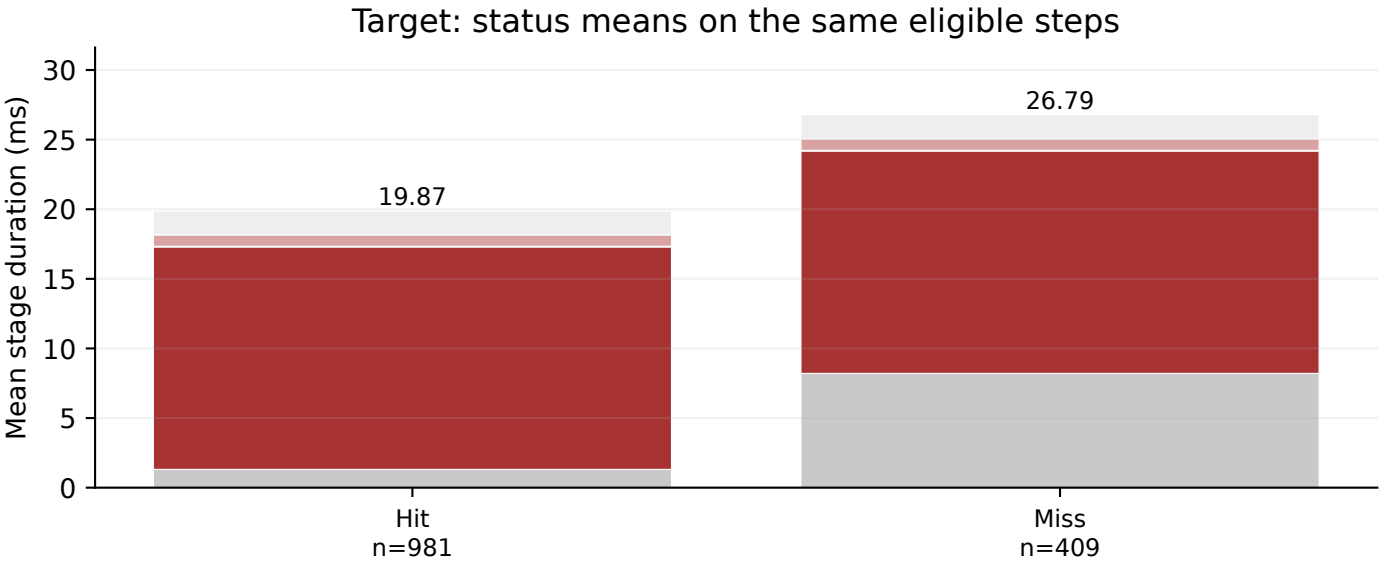
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



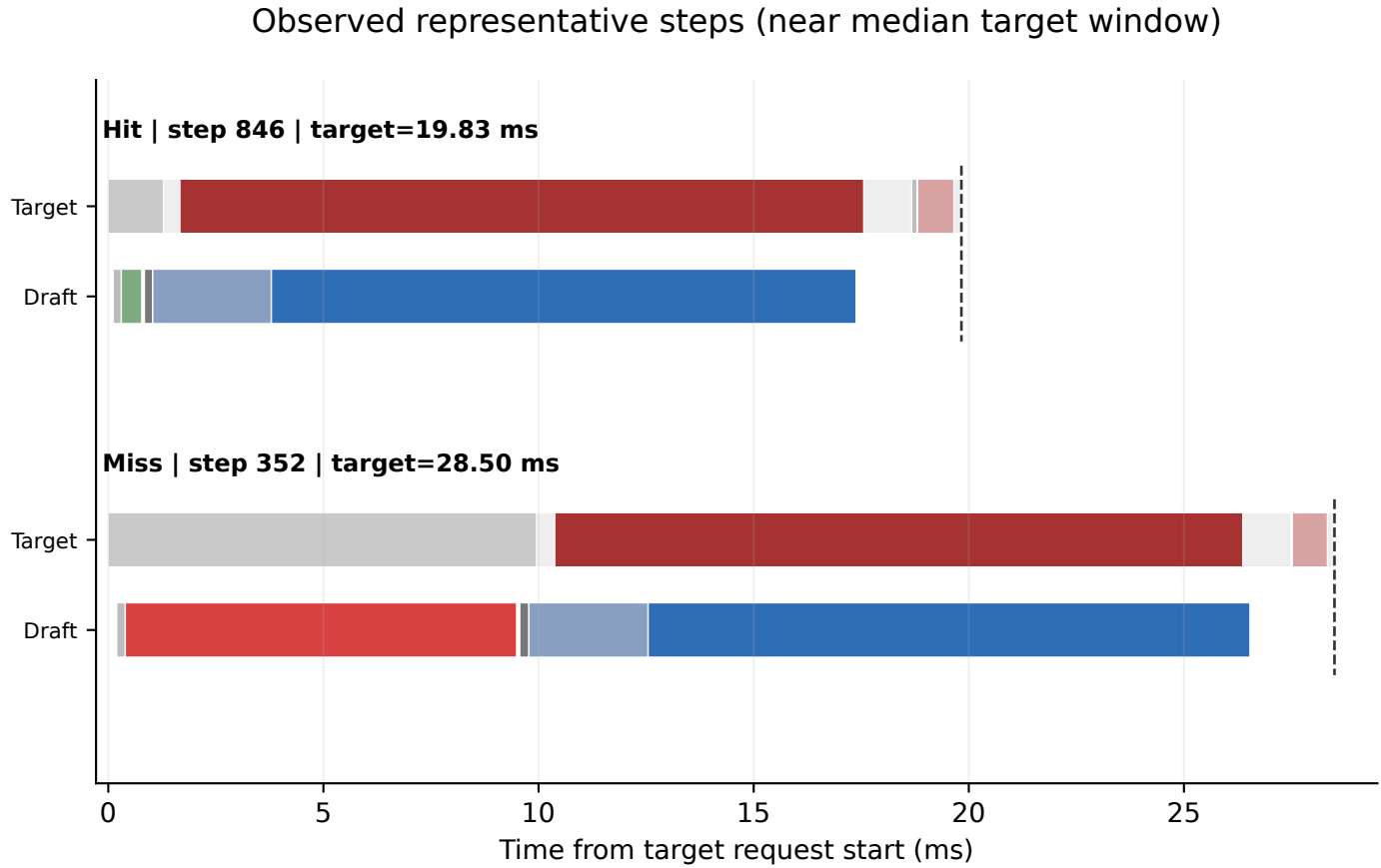
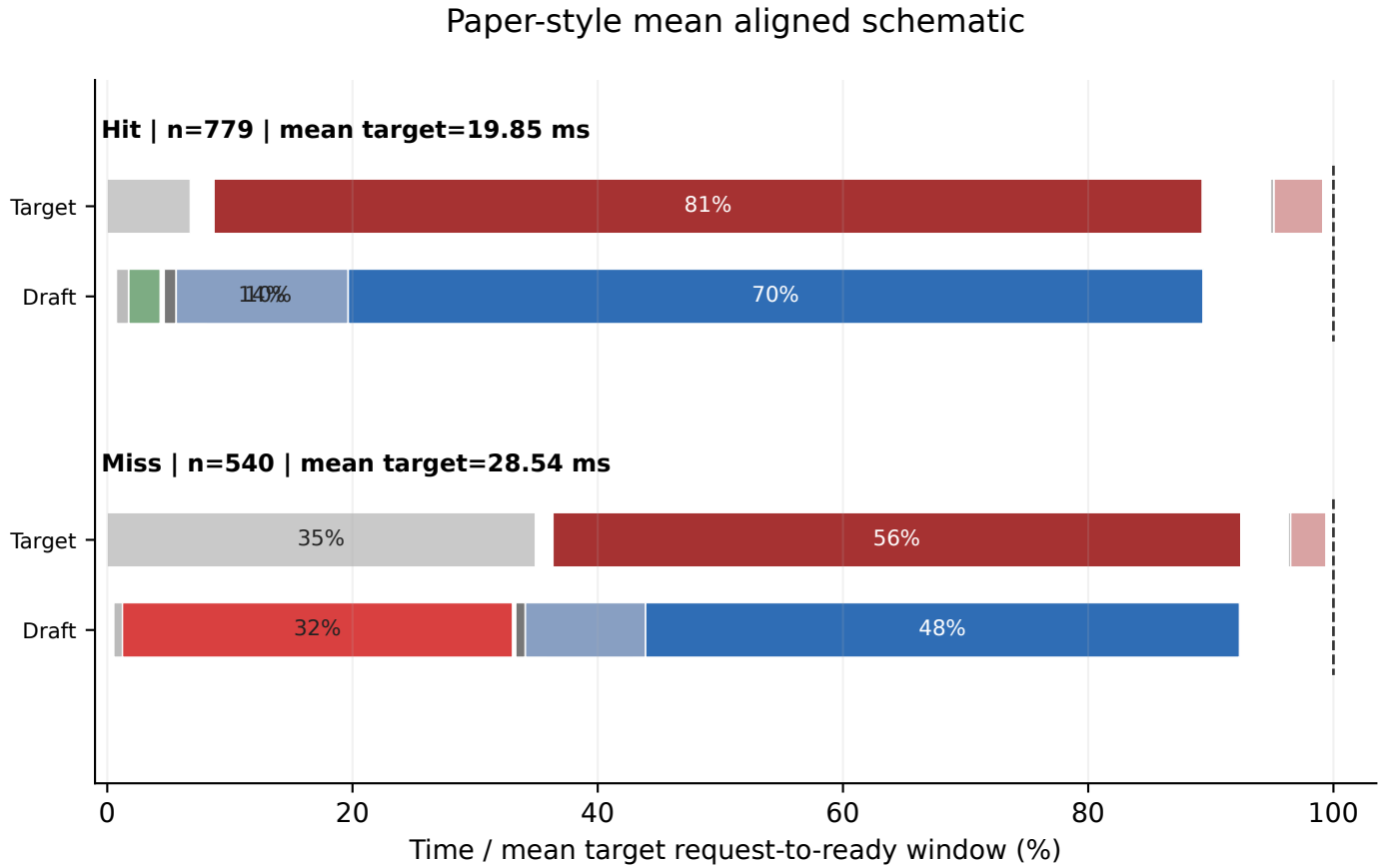
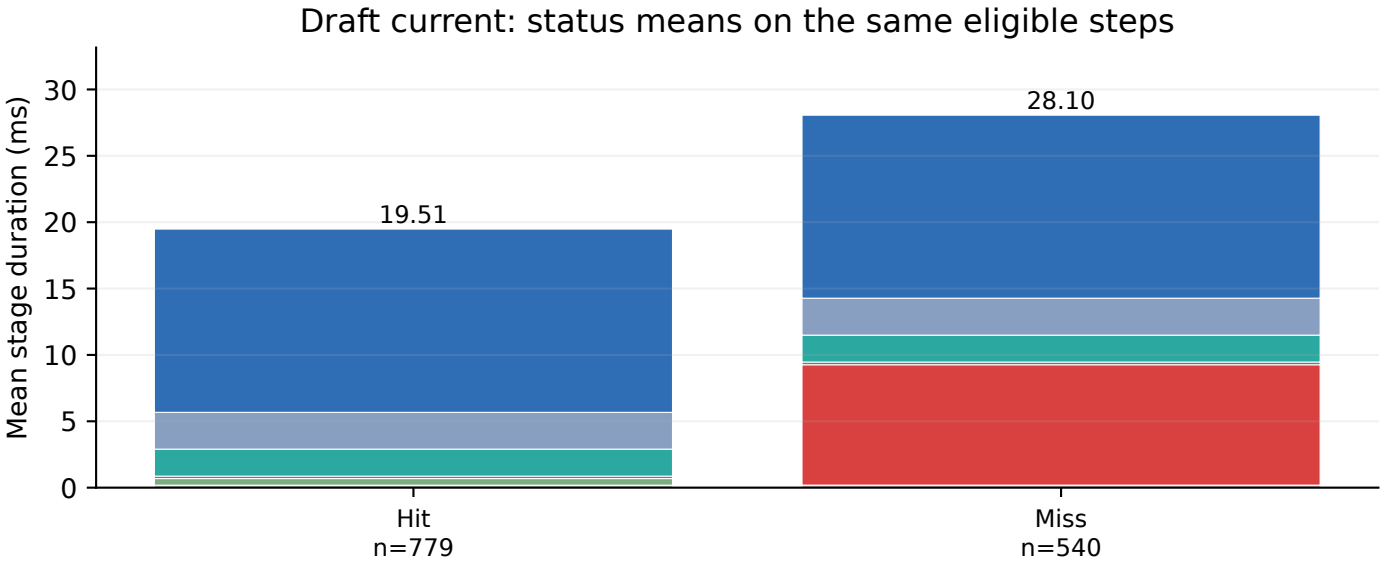
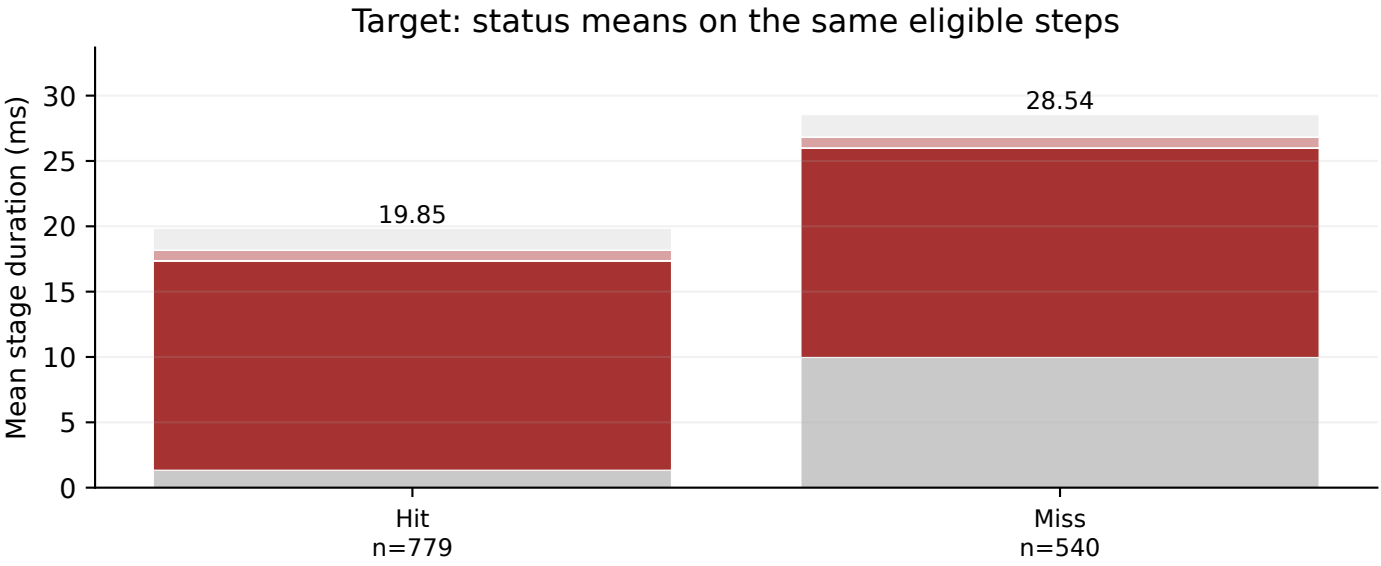
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



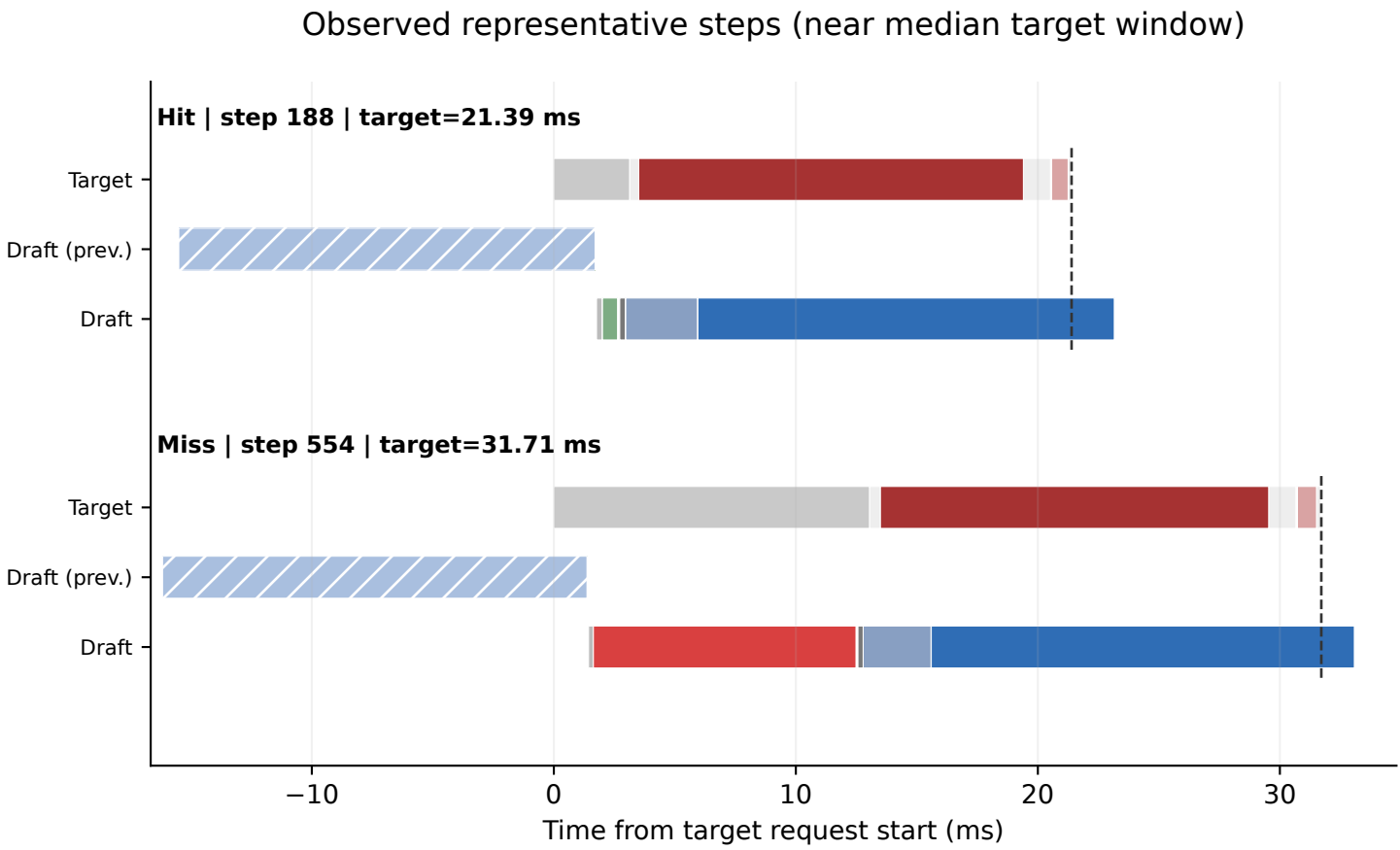
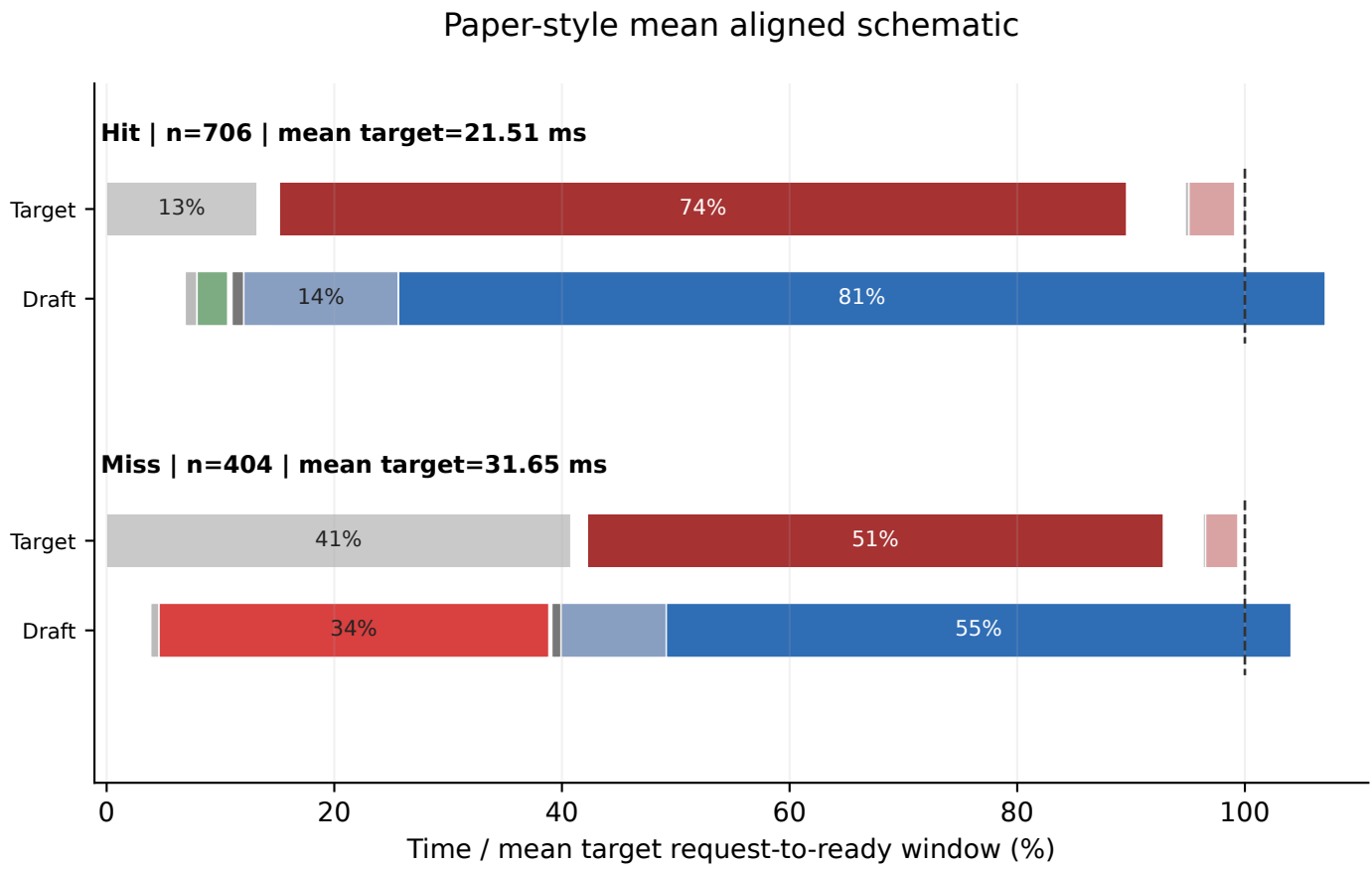
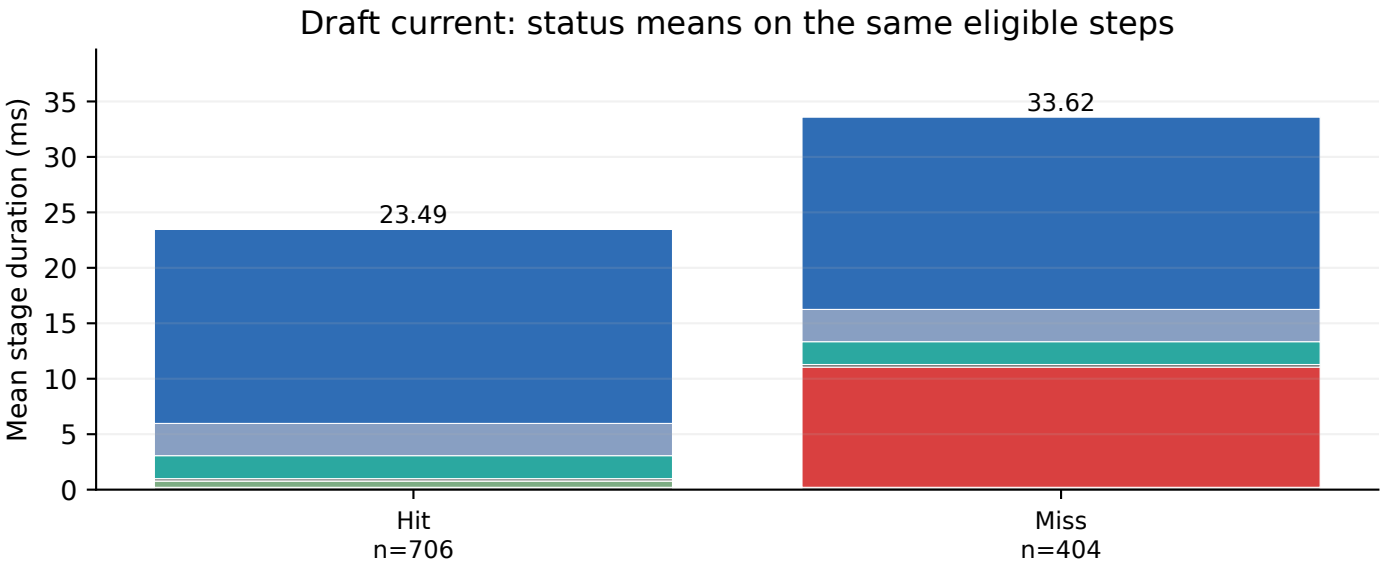
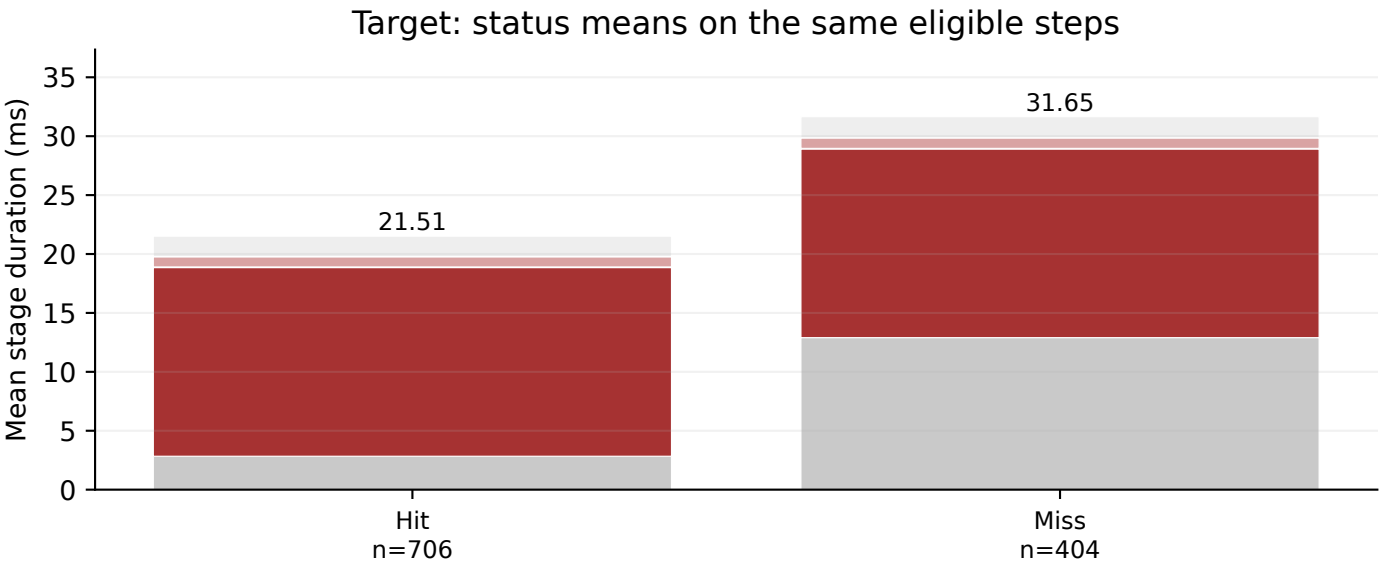
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

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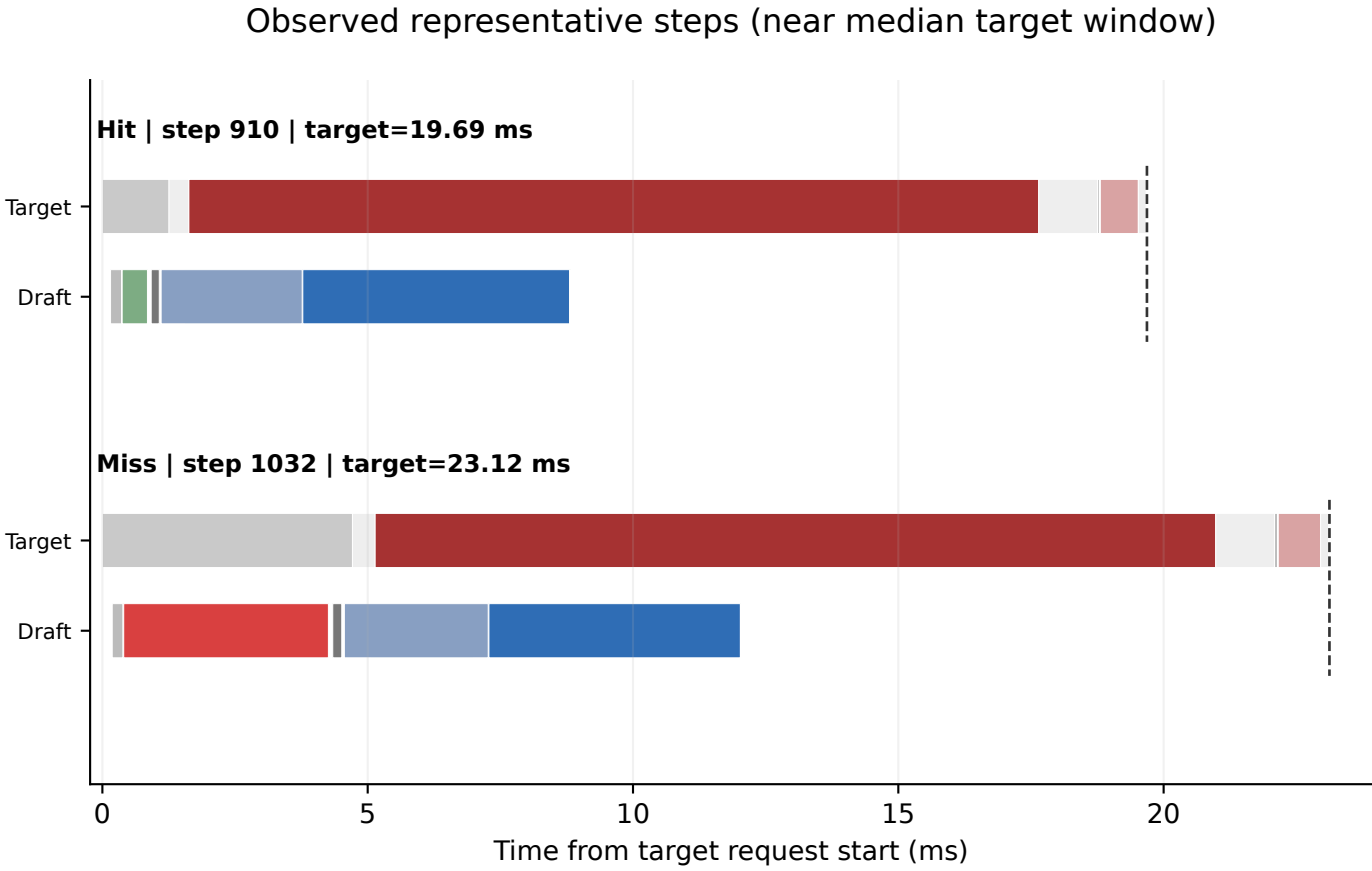
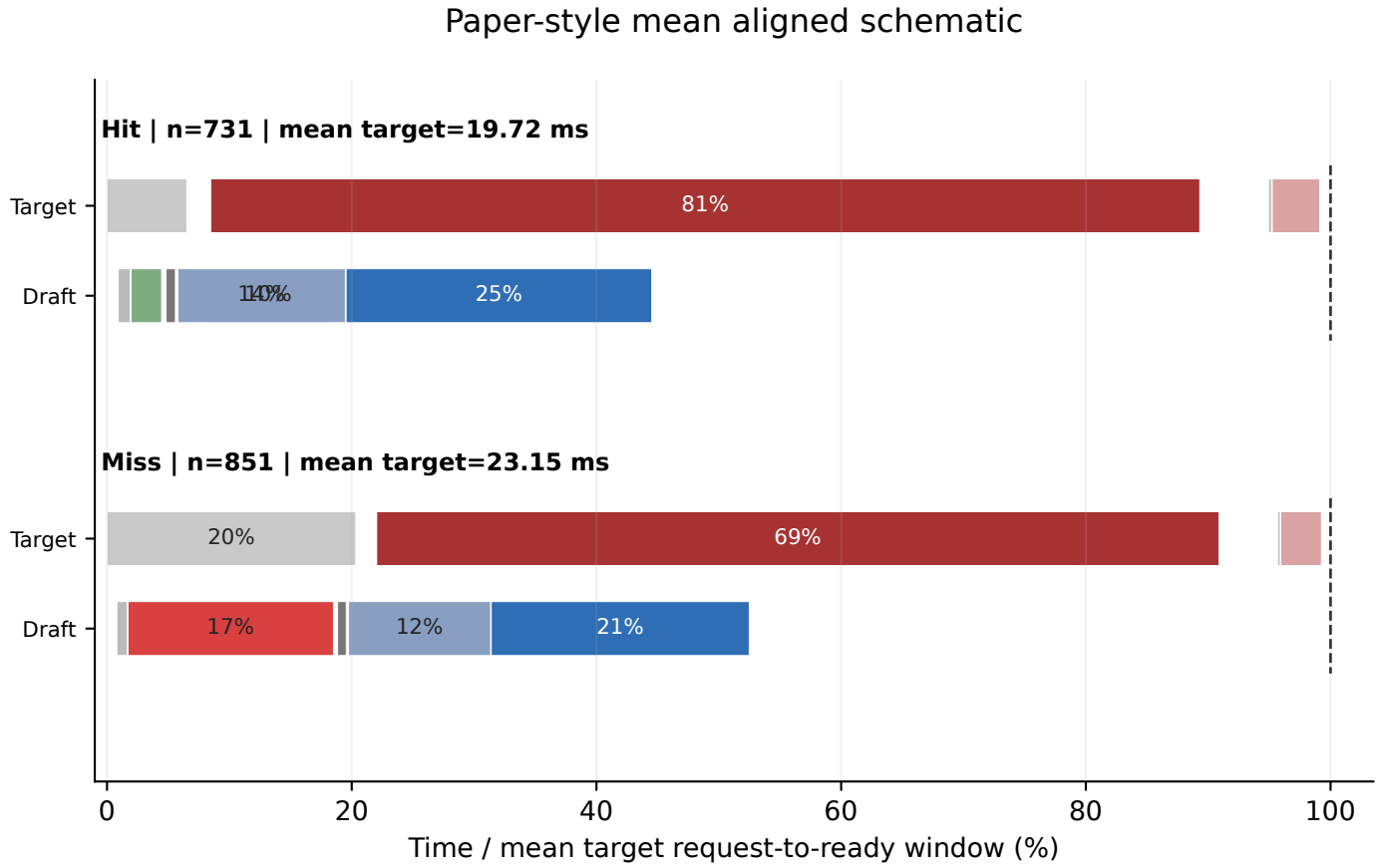
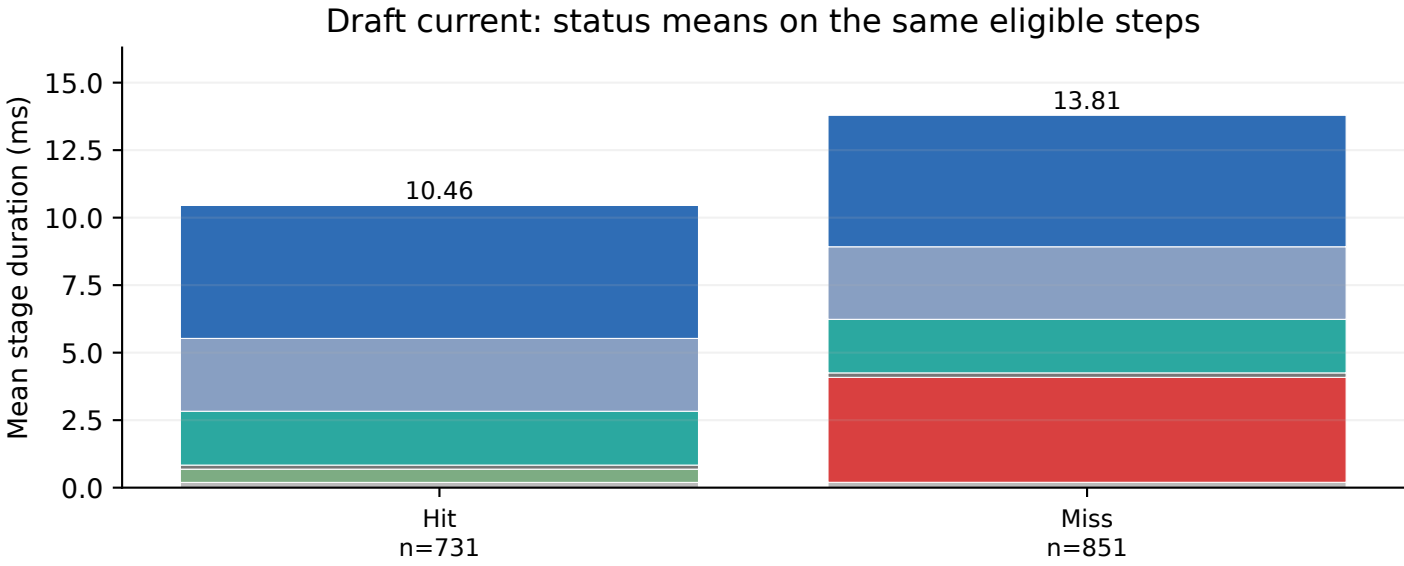
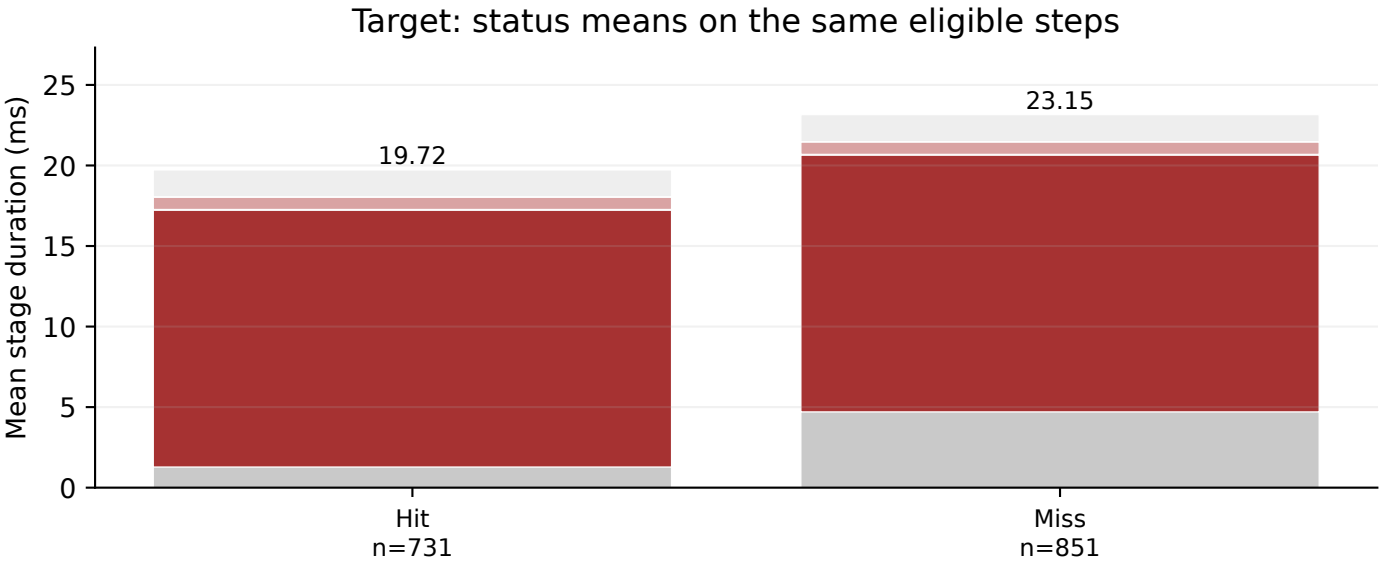
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
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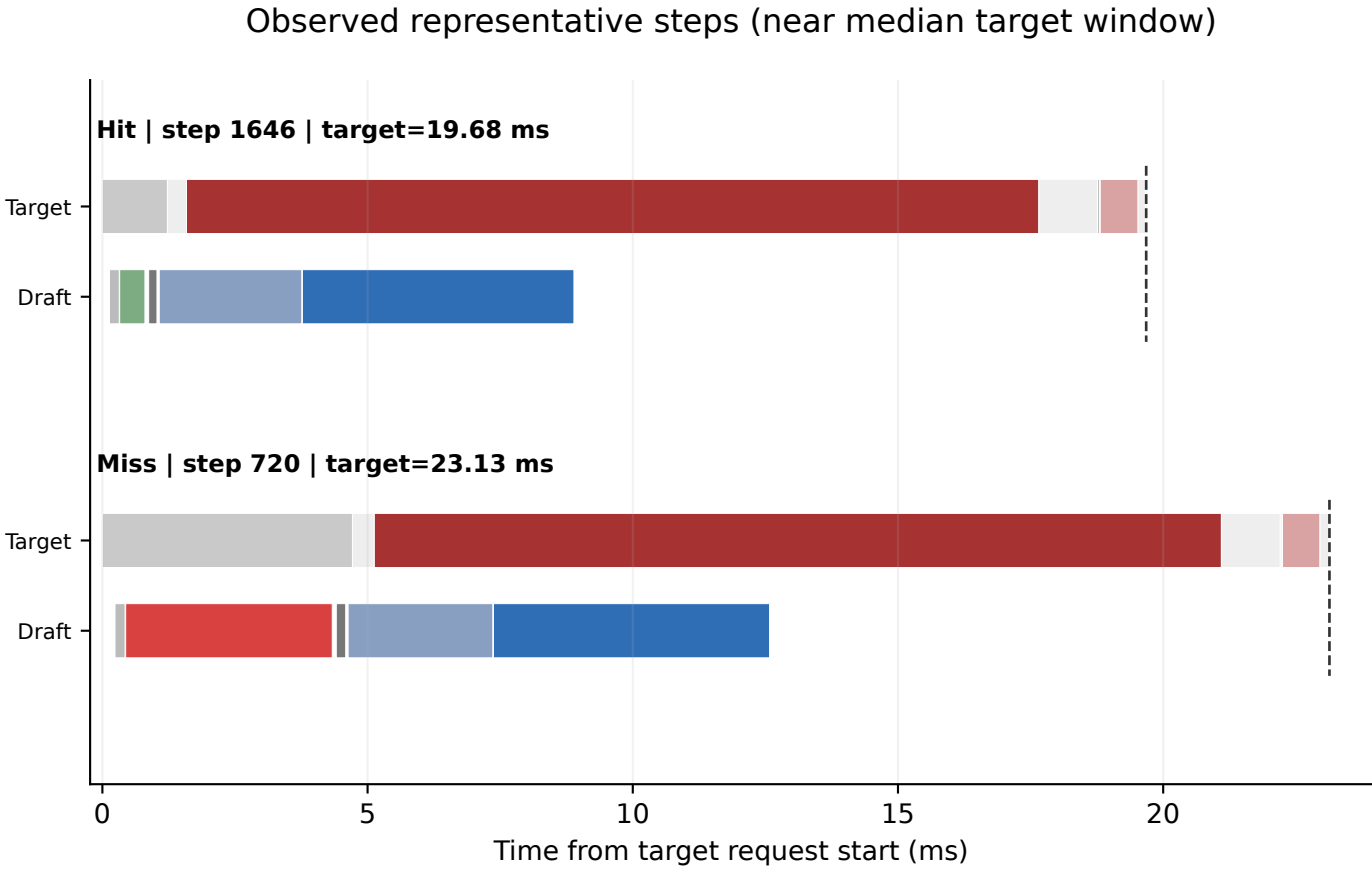
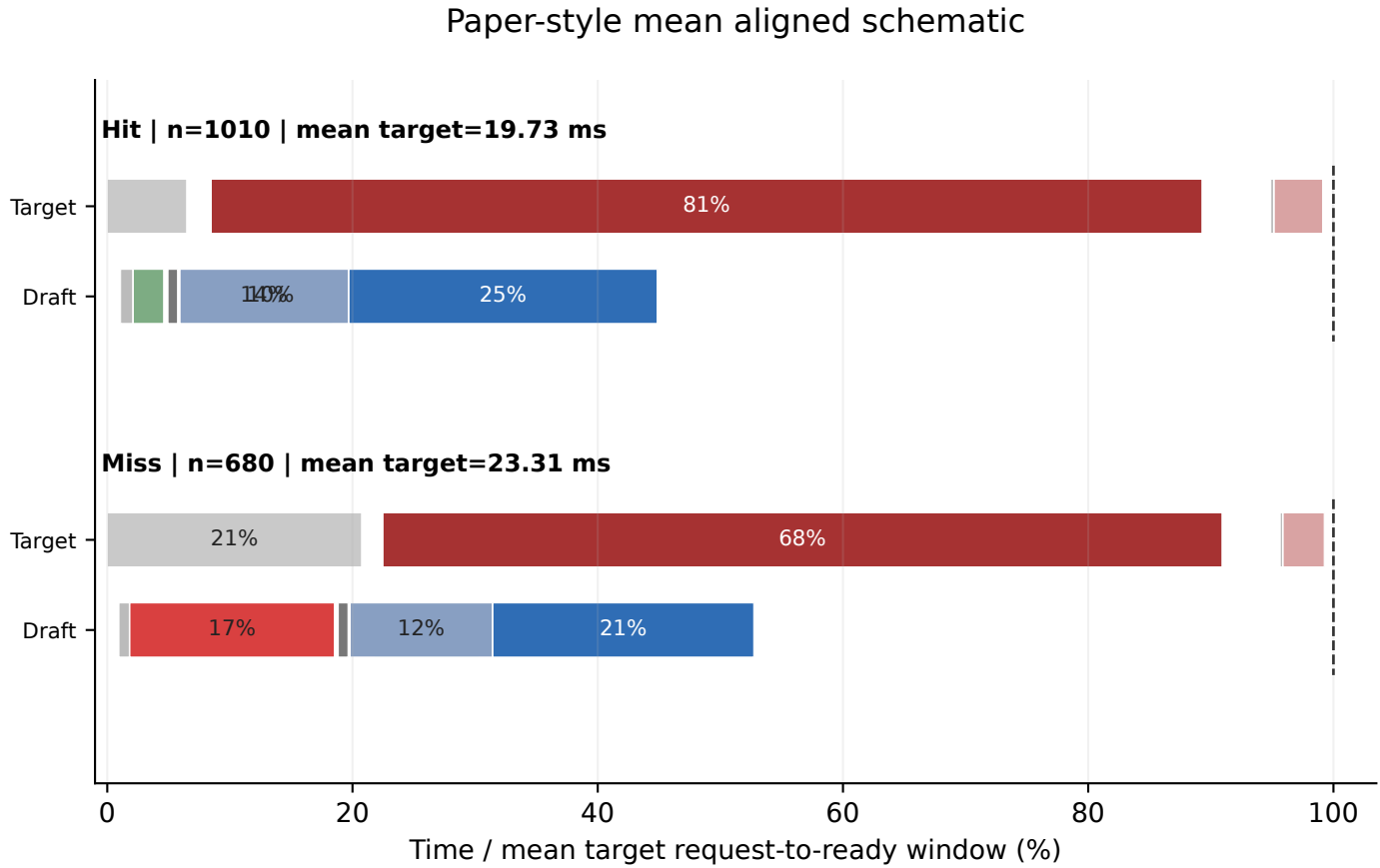
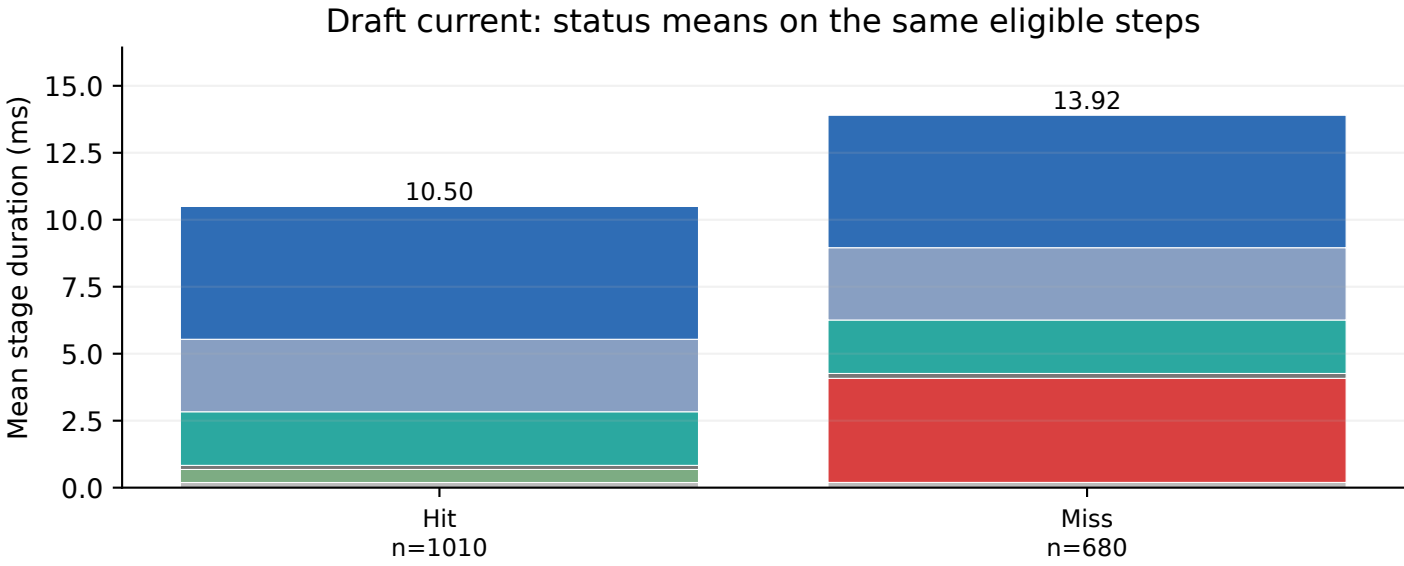
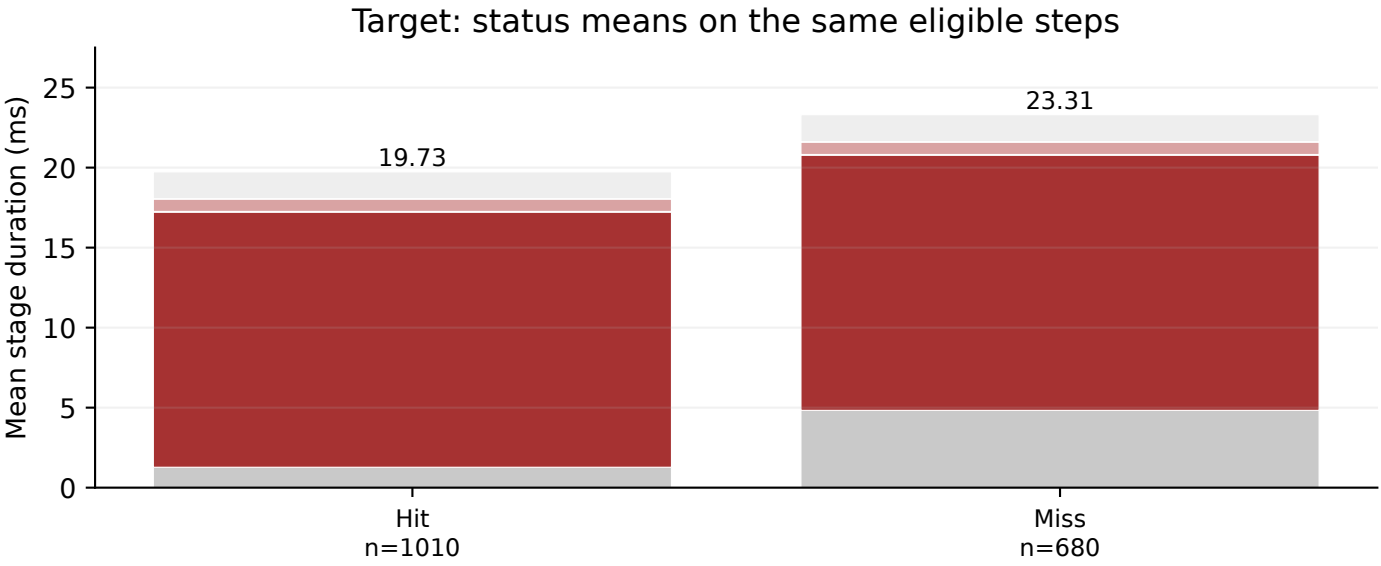
Wait / sync Accept prepare Miss response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

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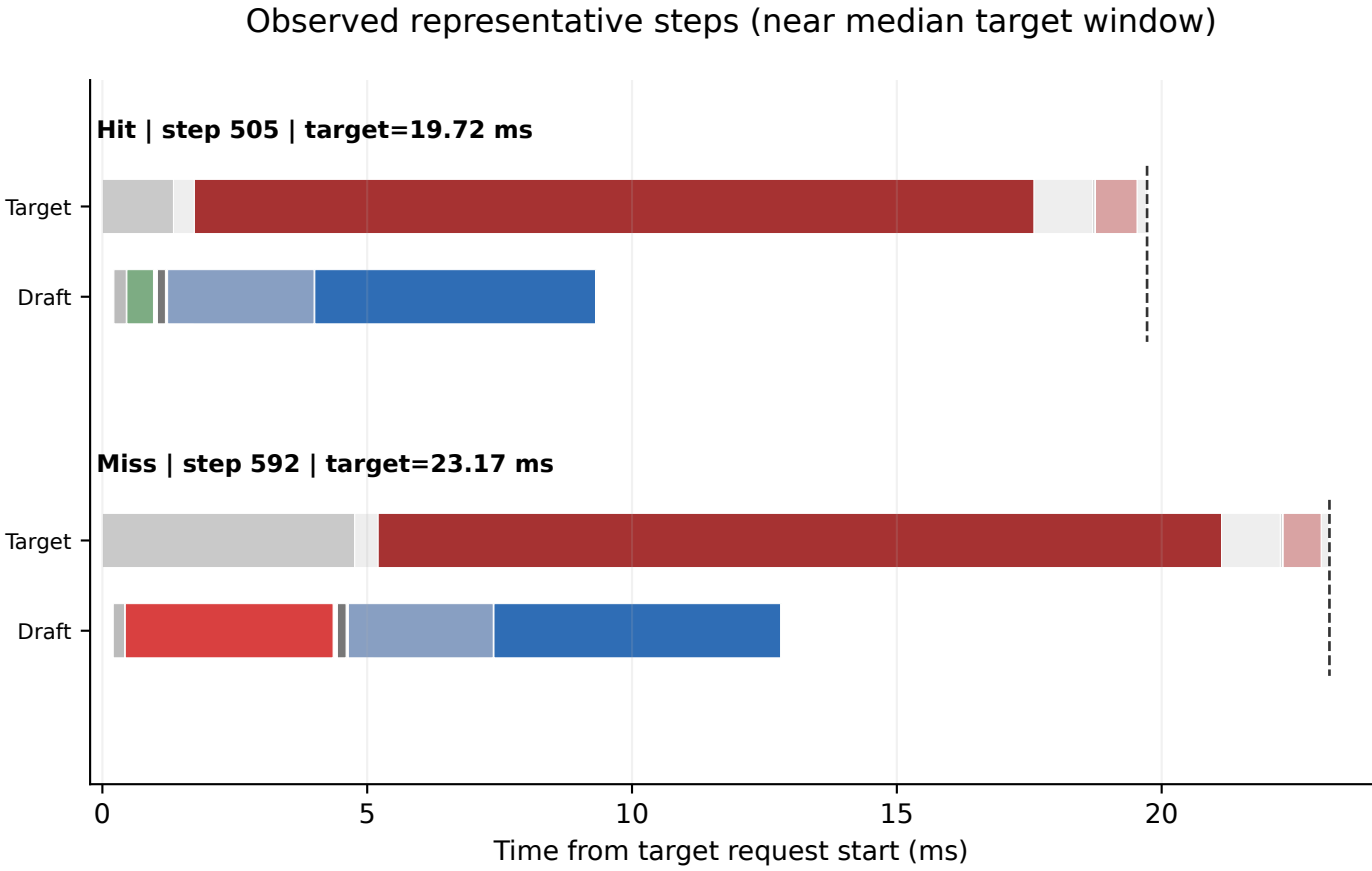
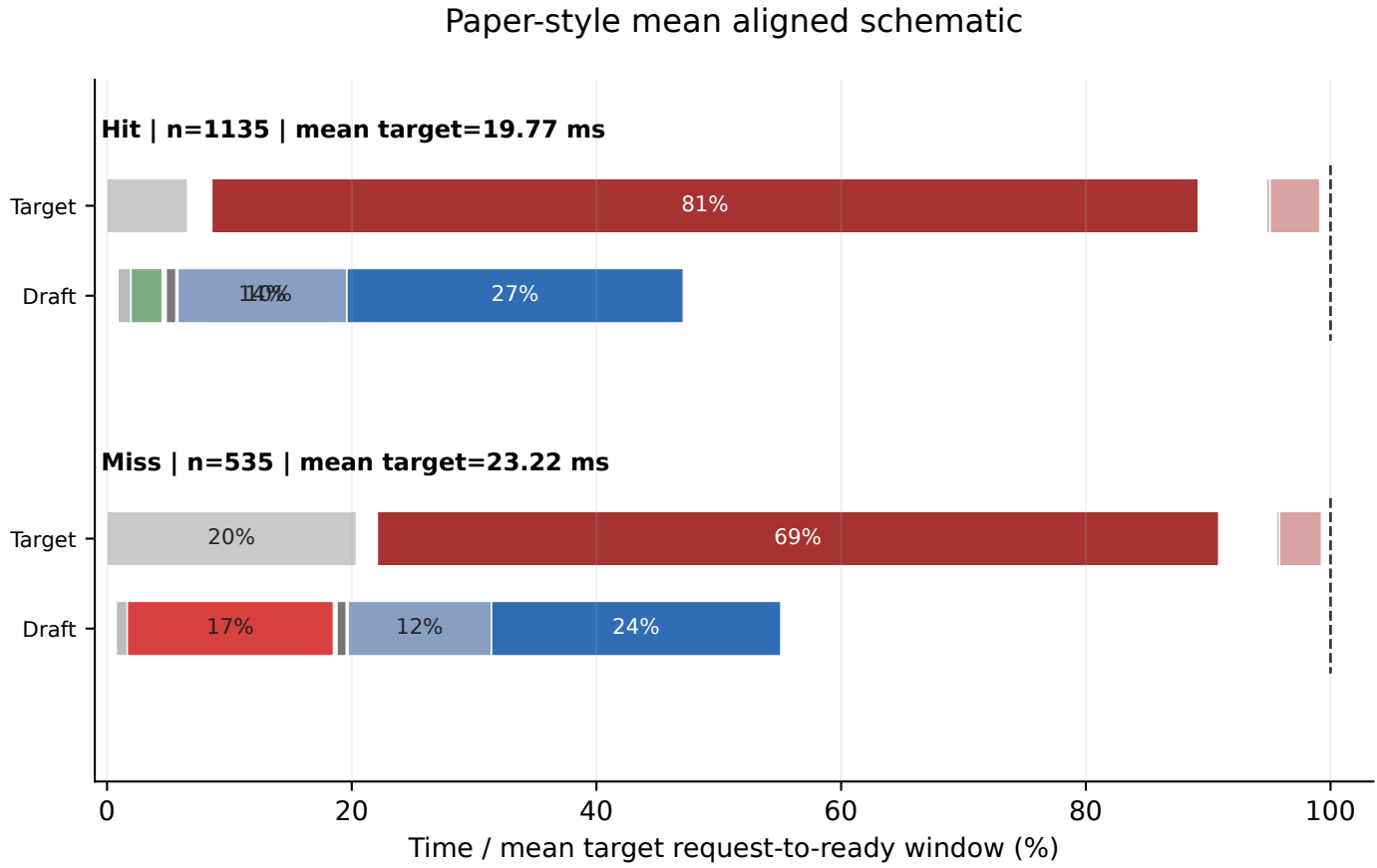
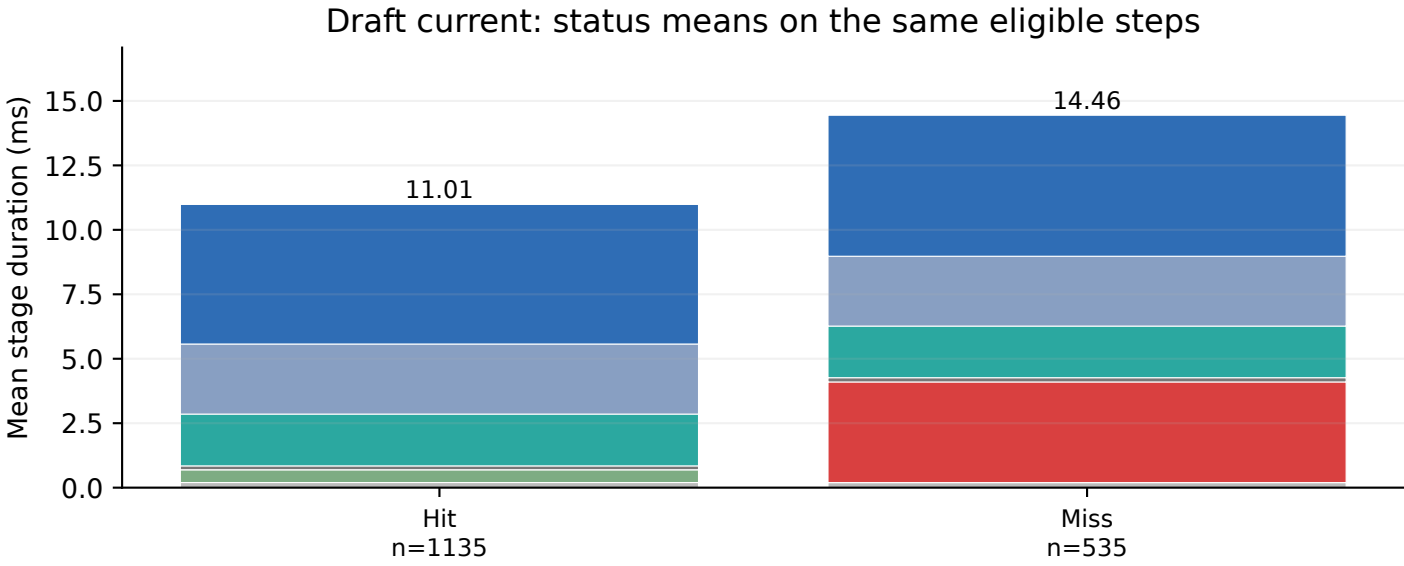
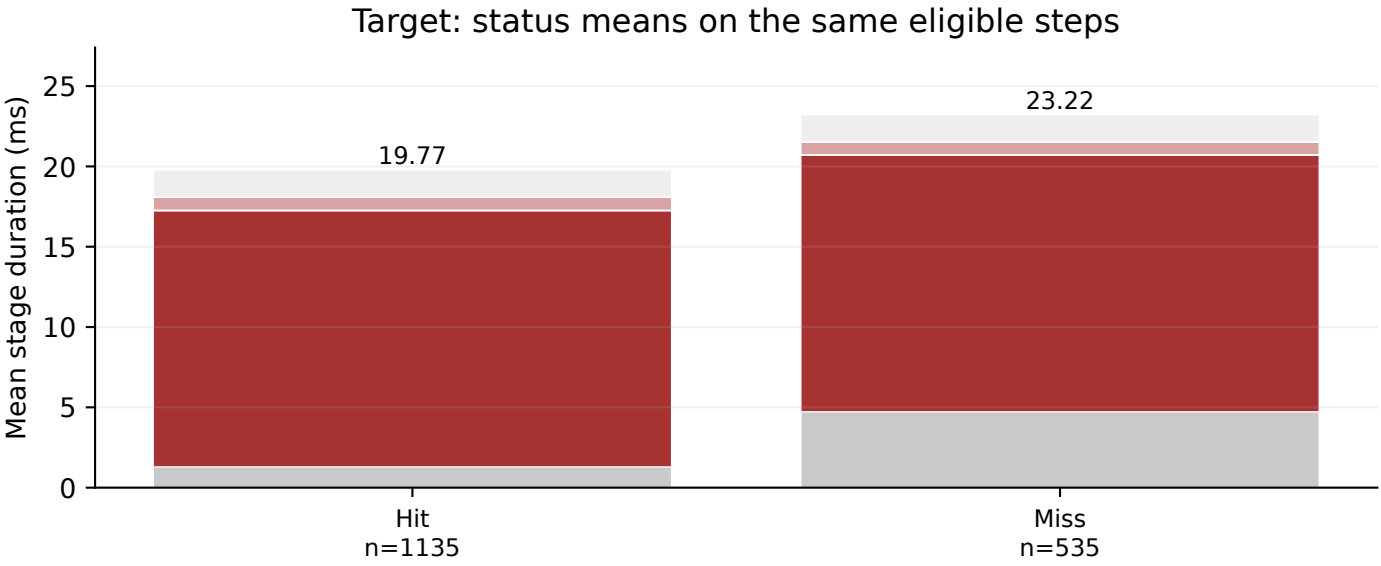
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

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Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



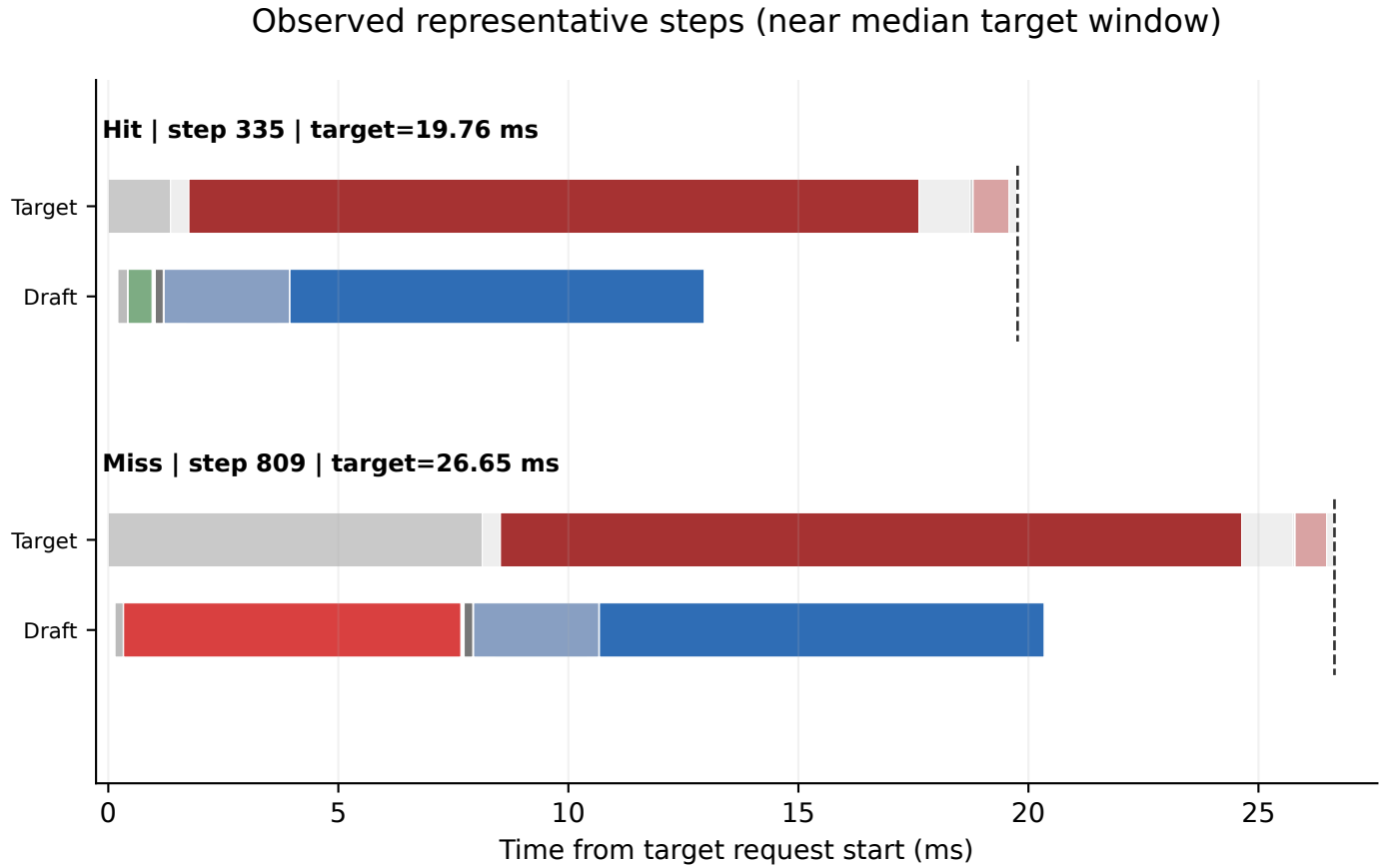
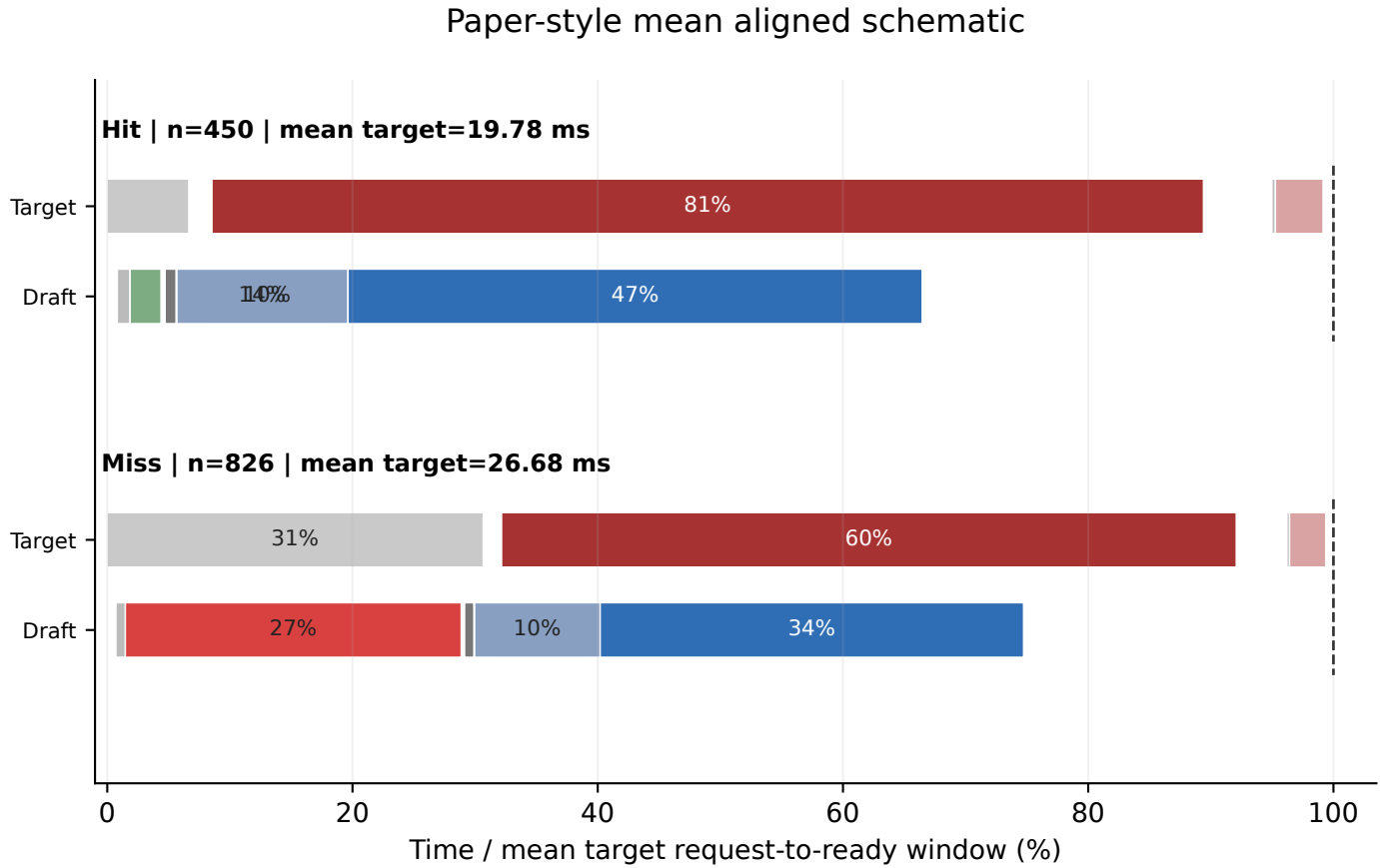
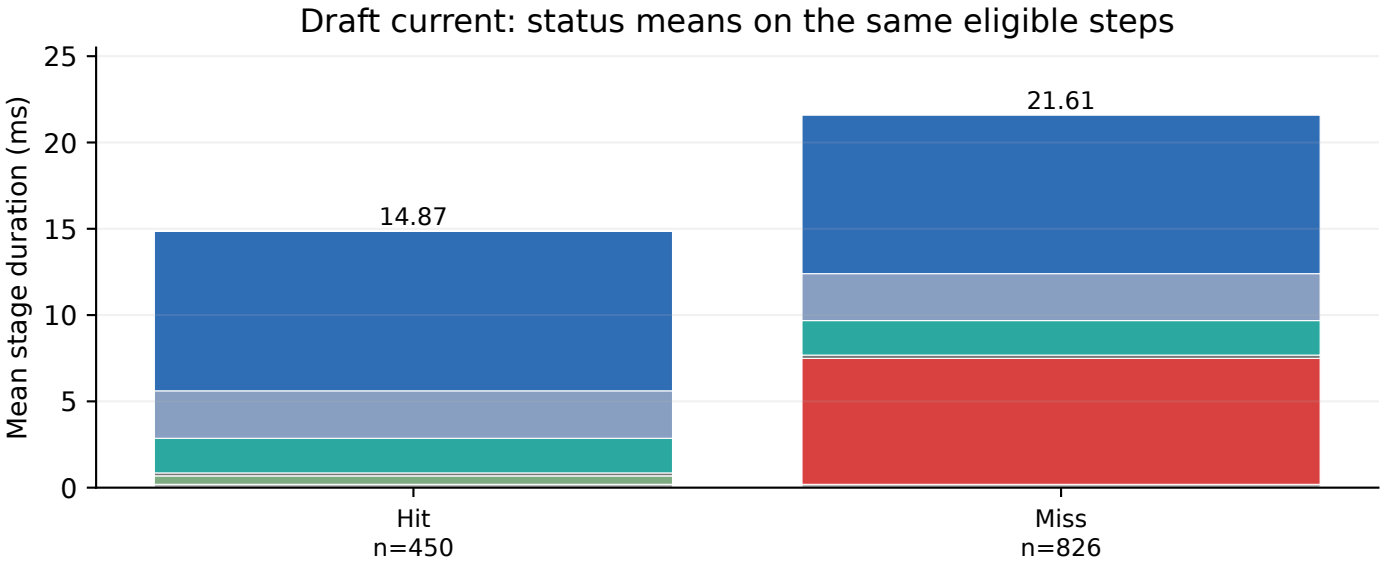
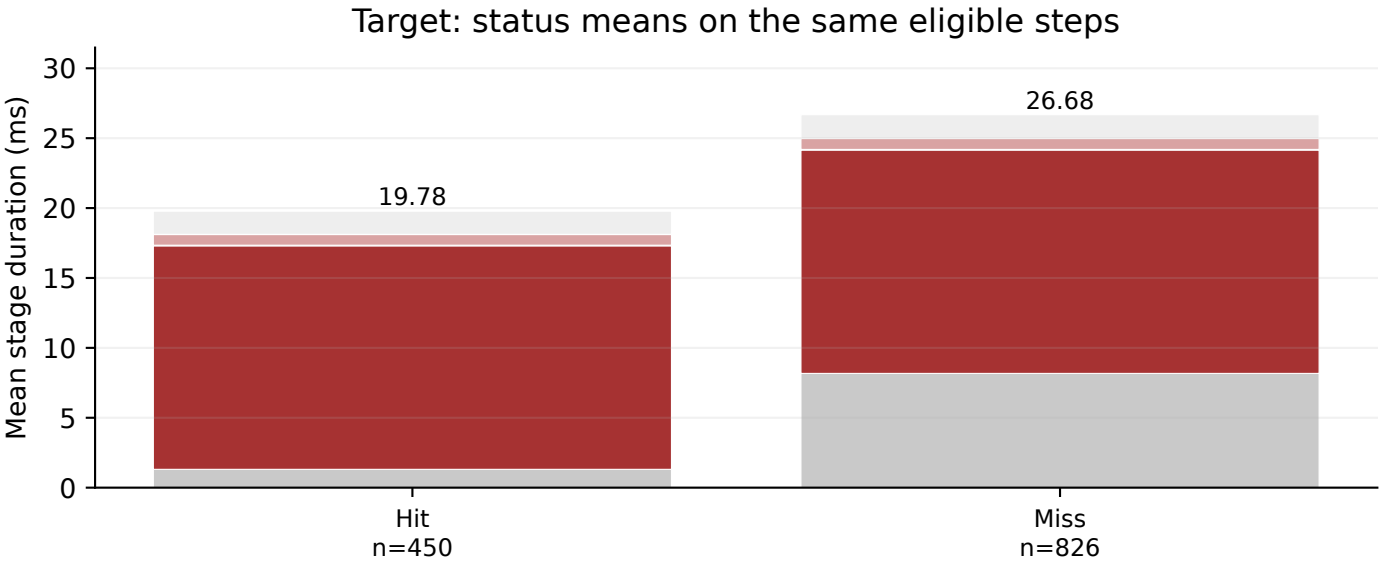
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

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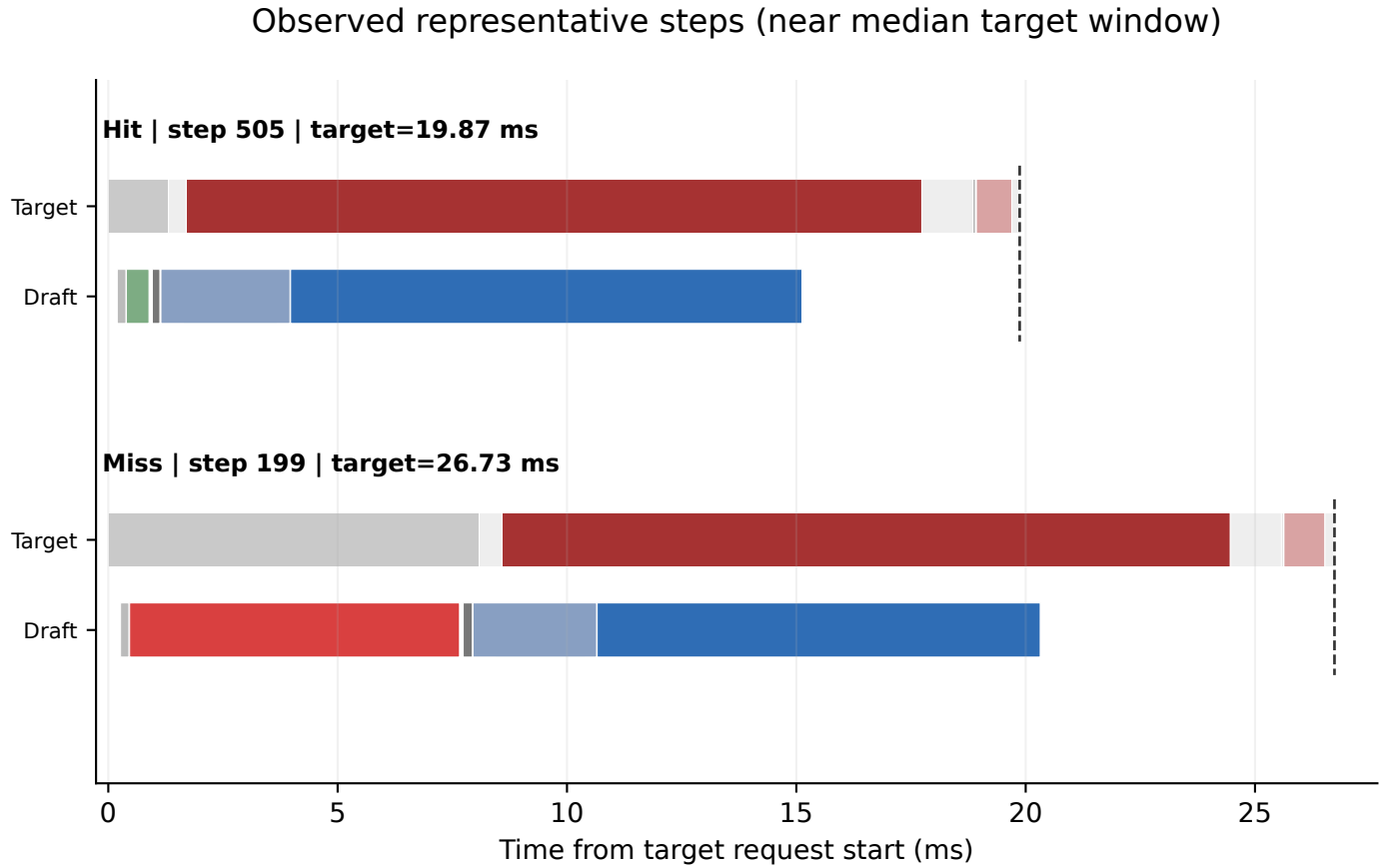
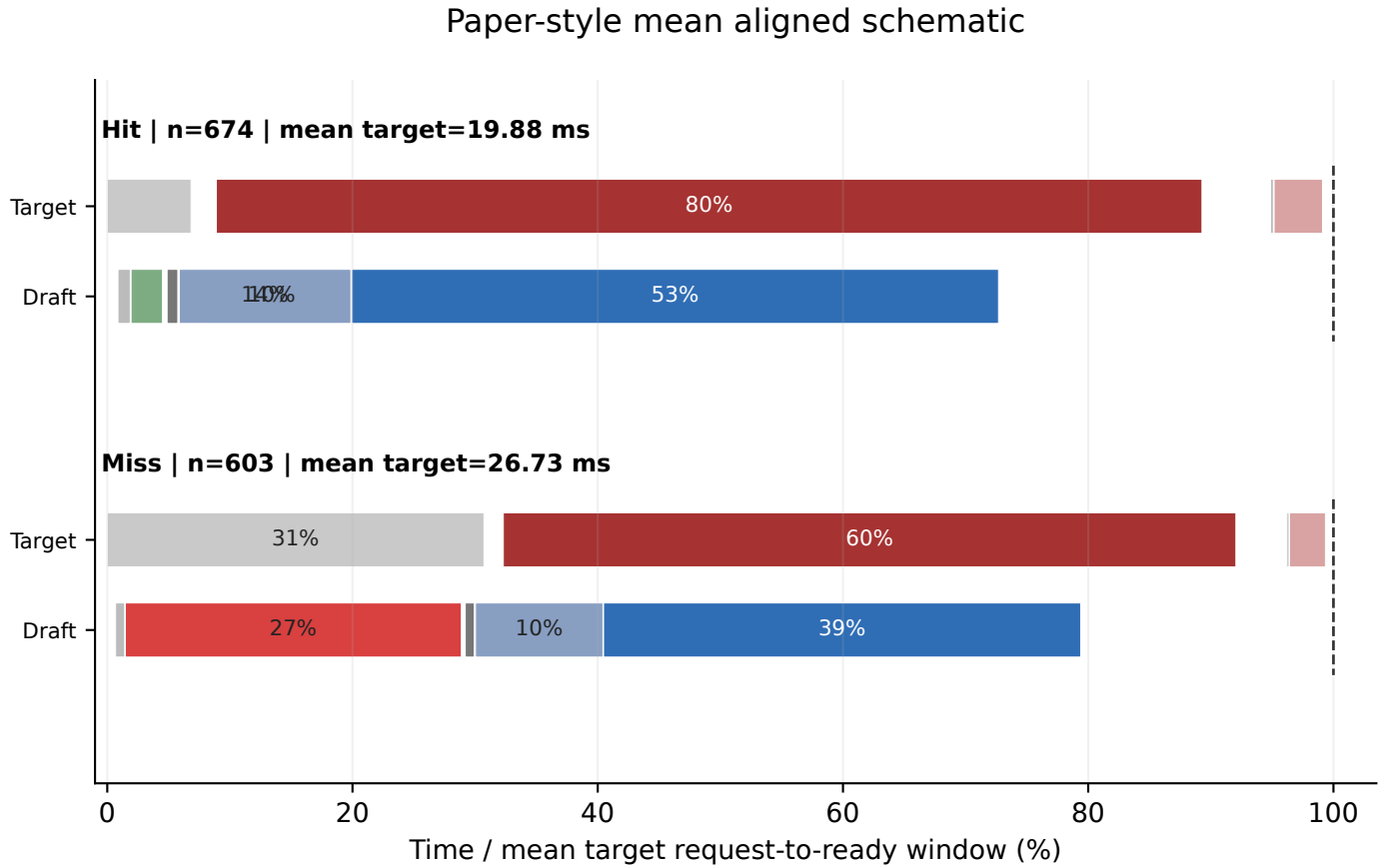
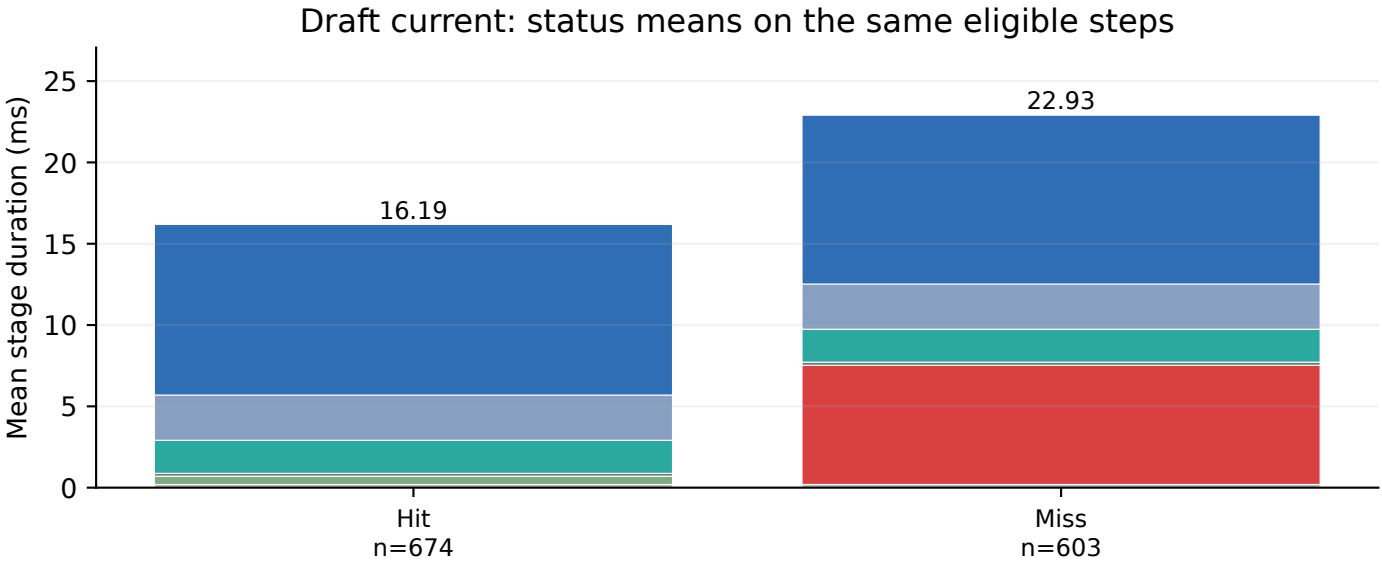
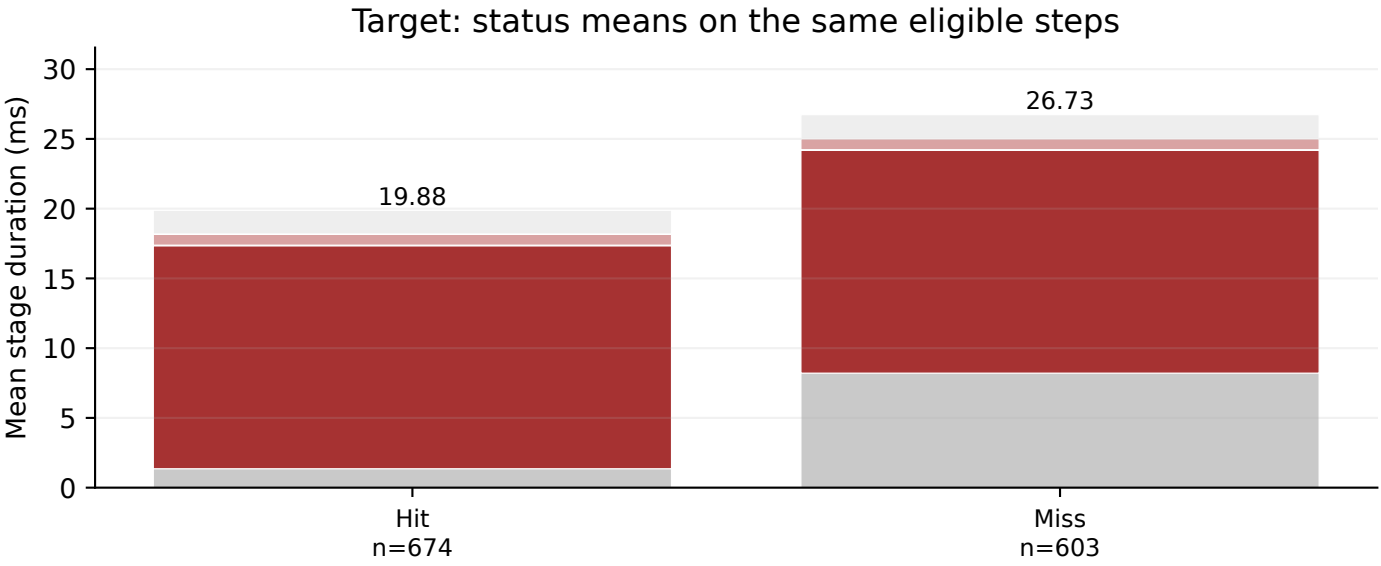
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

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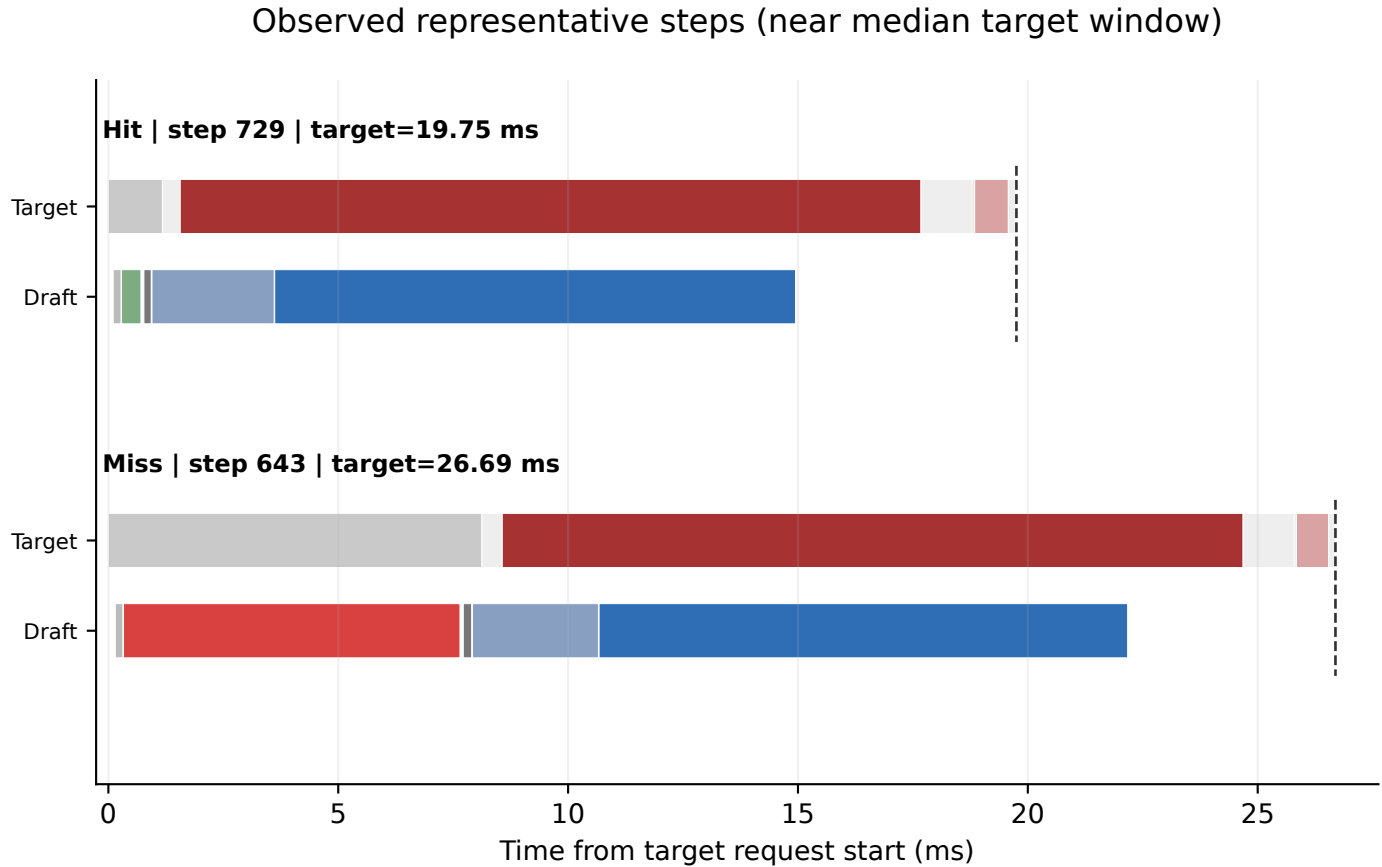
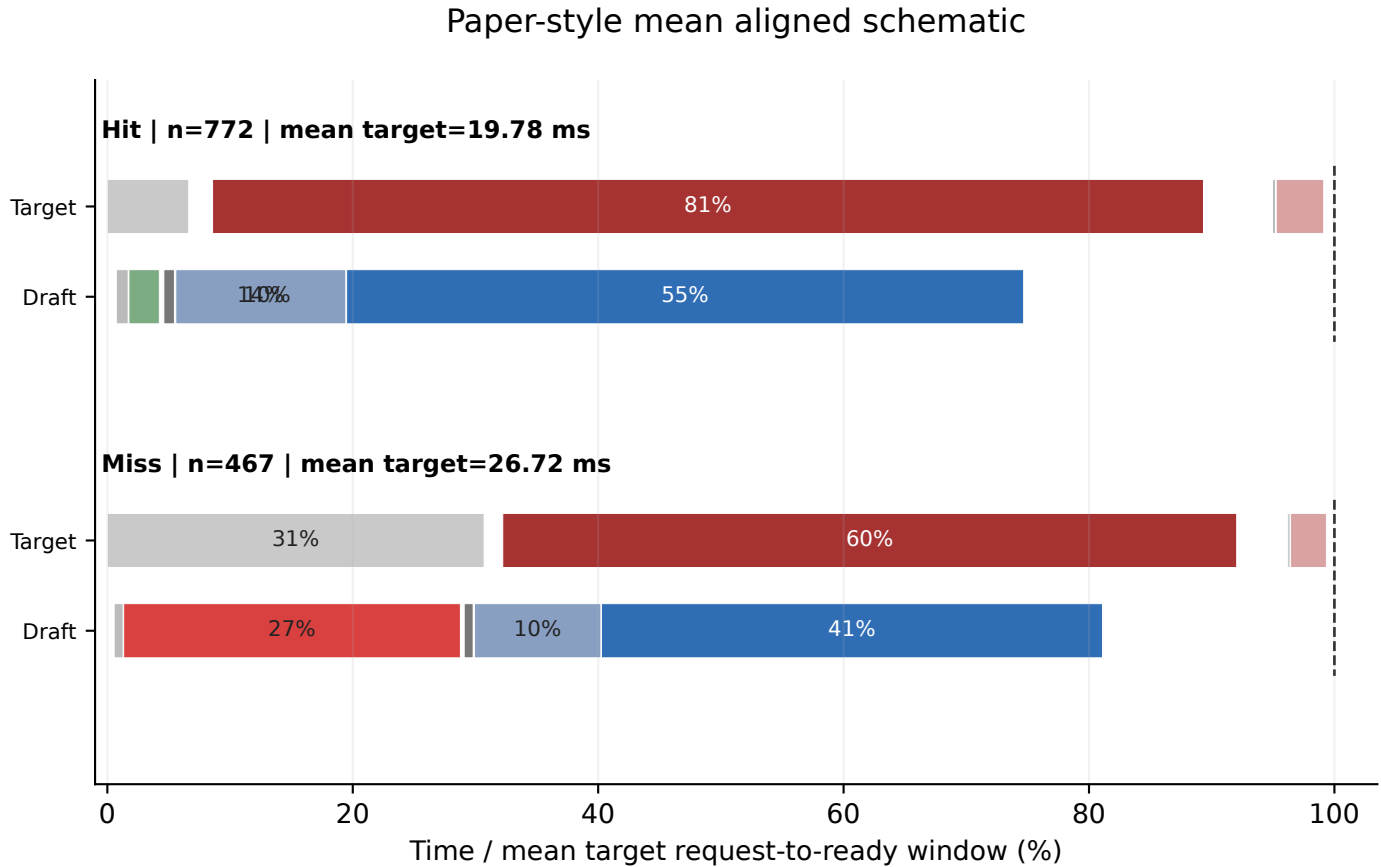
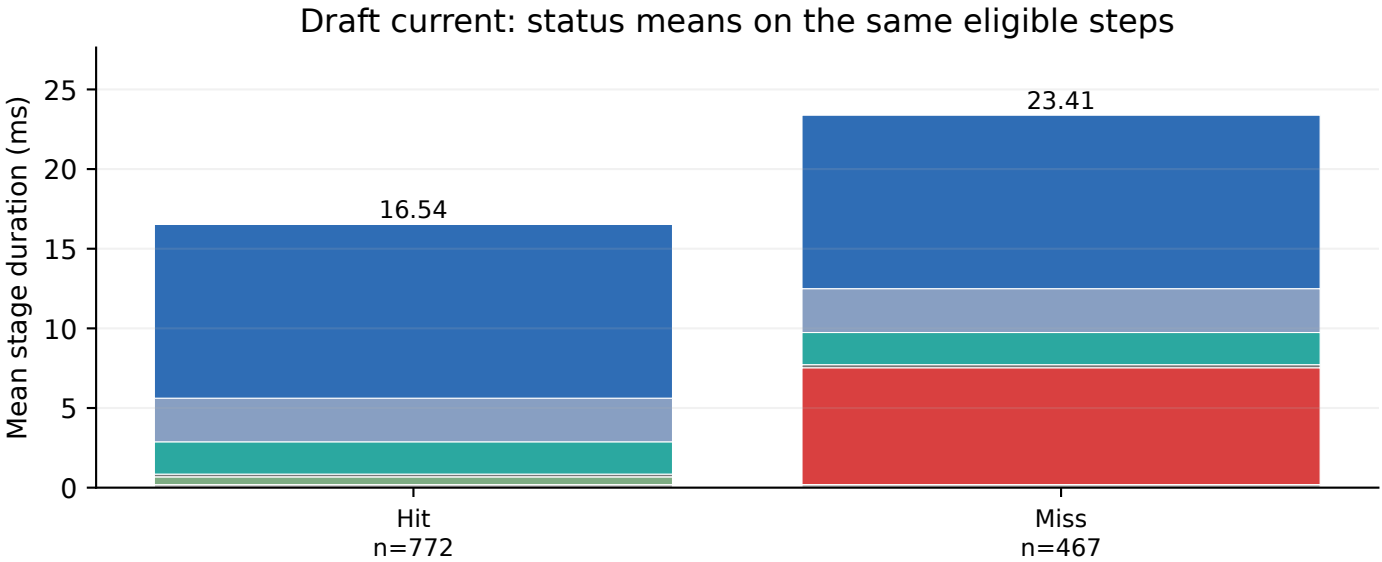
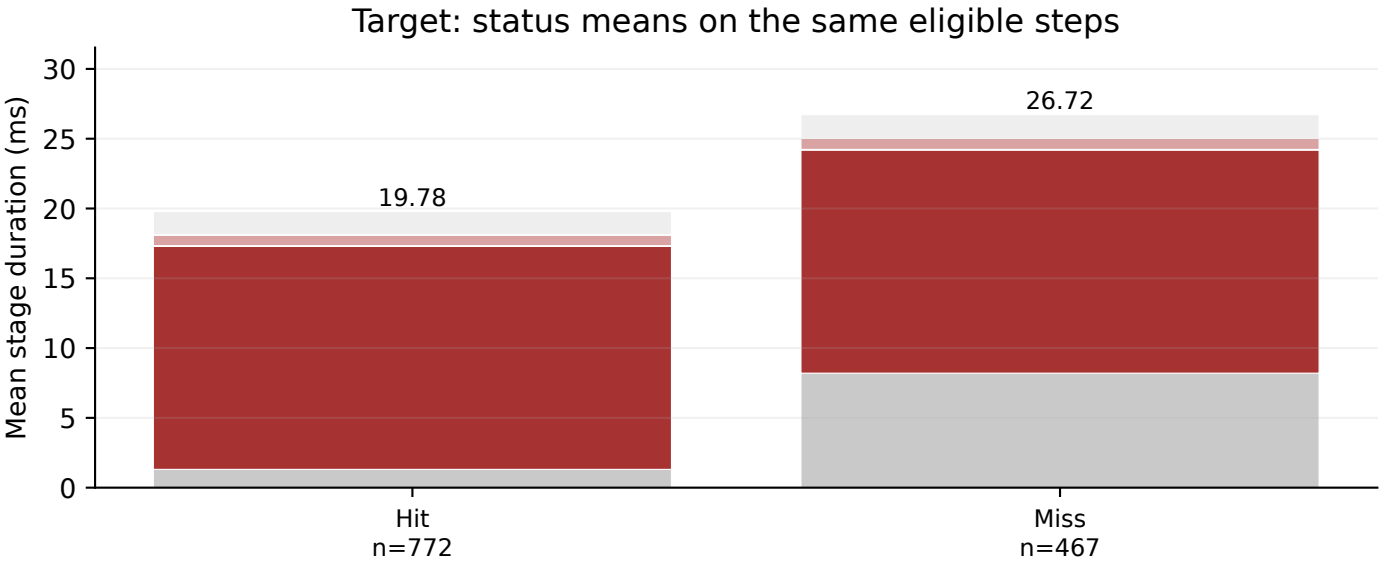
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

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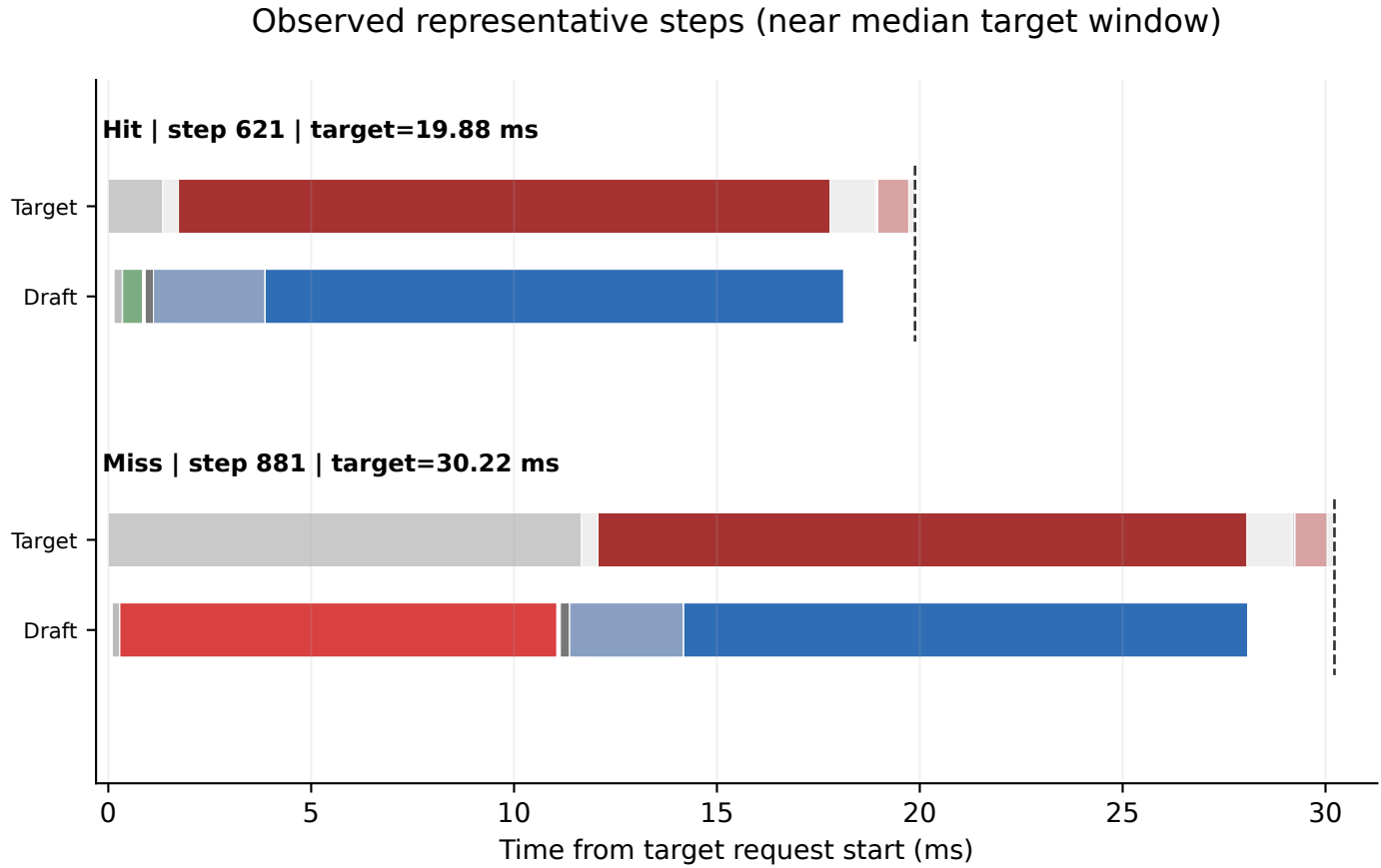
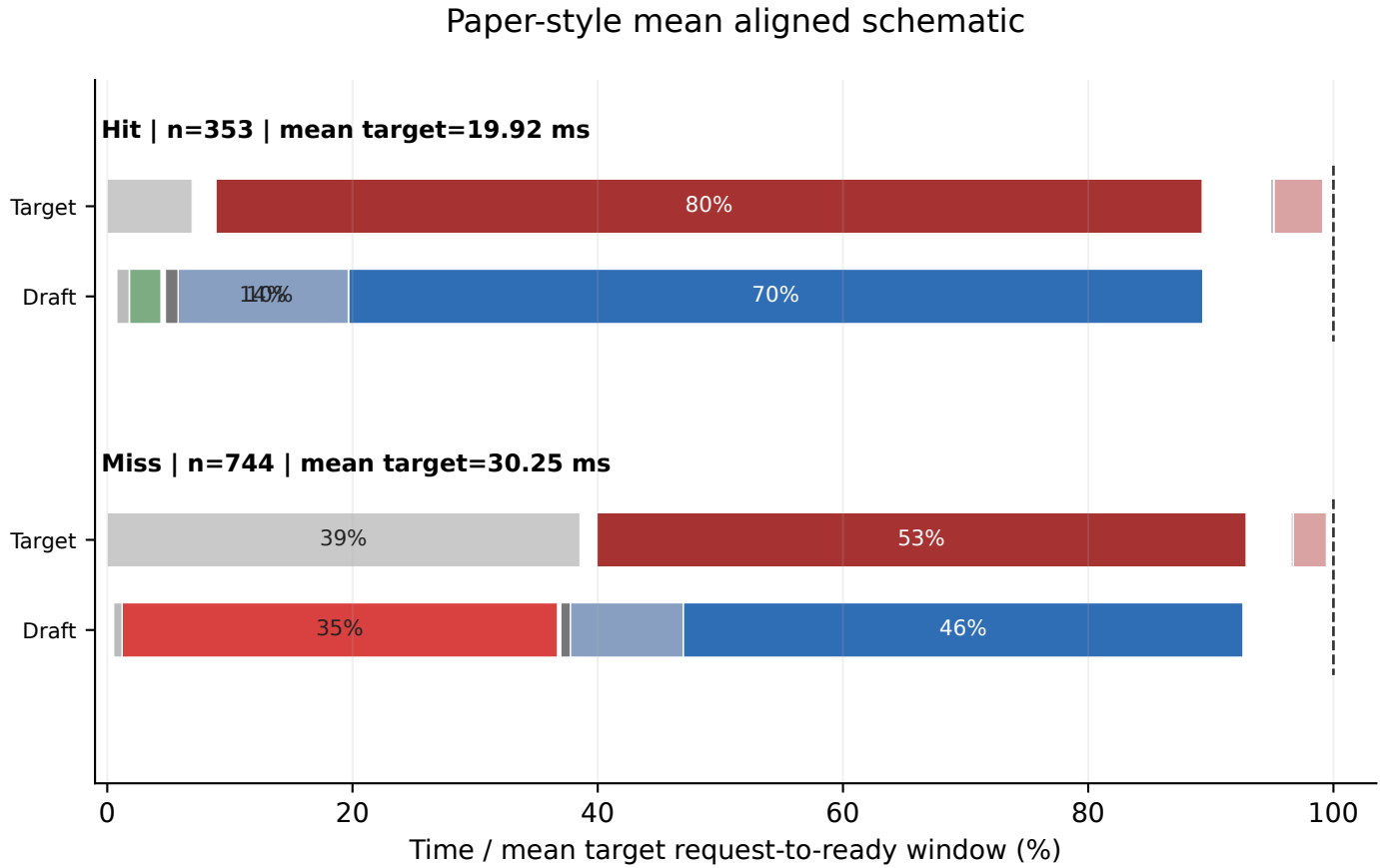
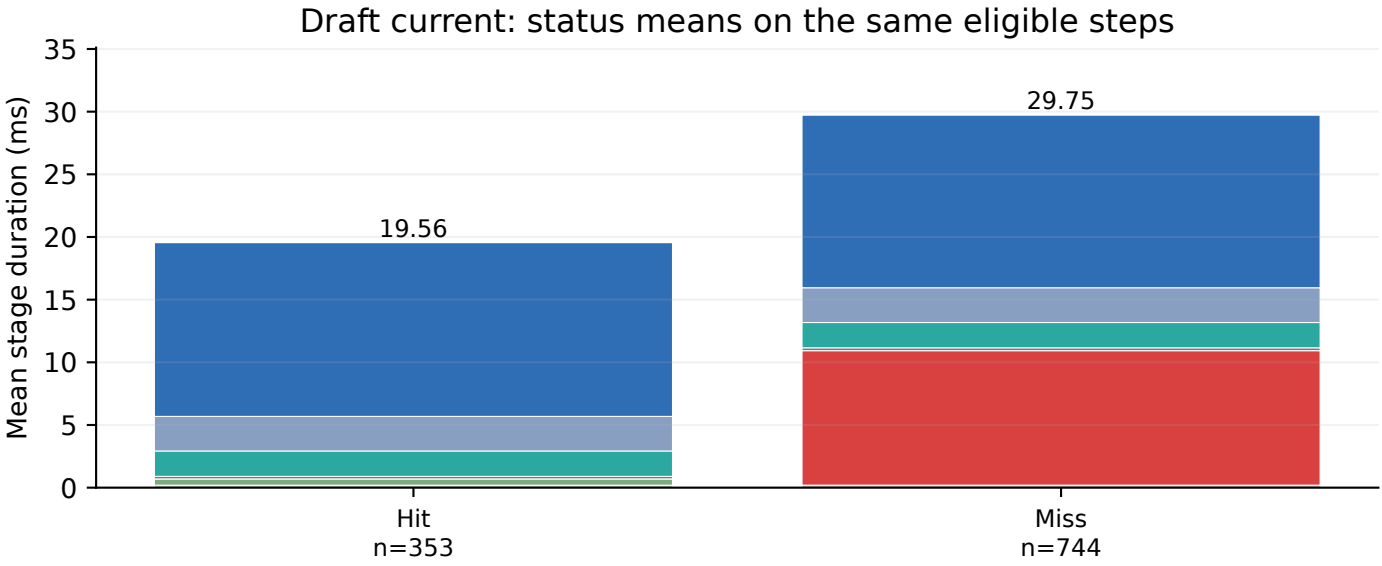
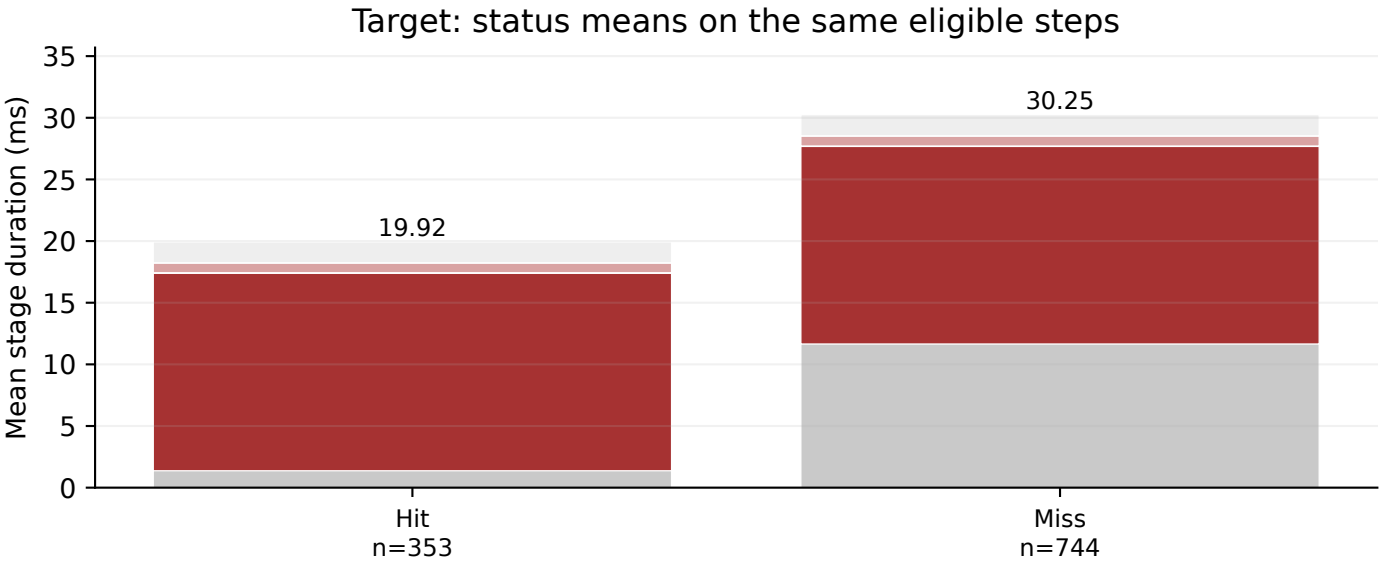
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
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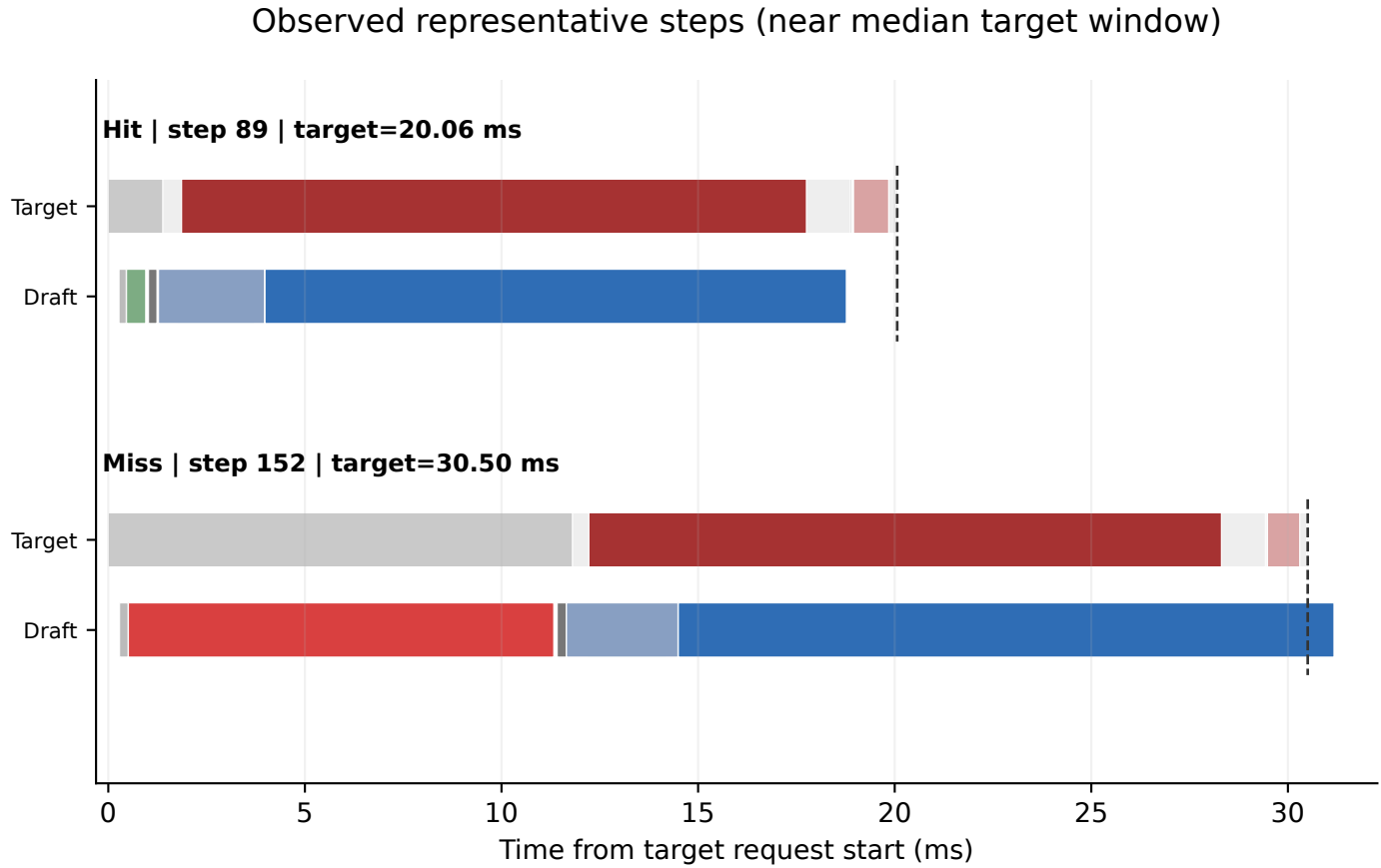
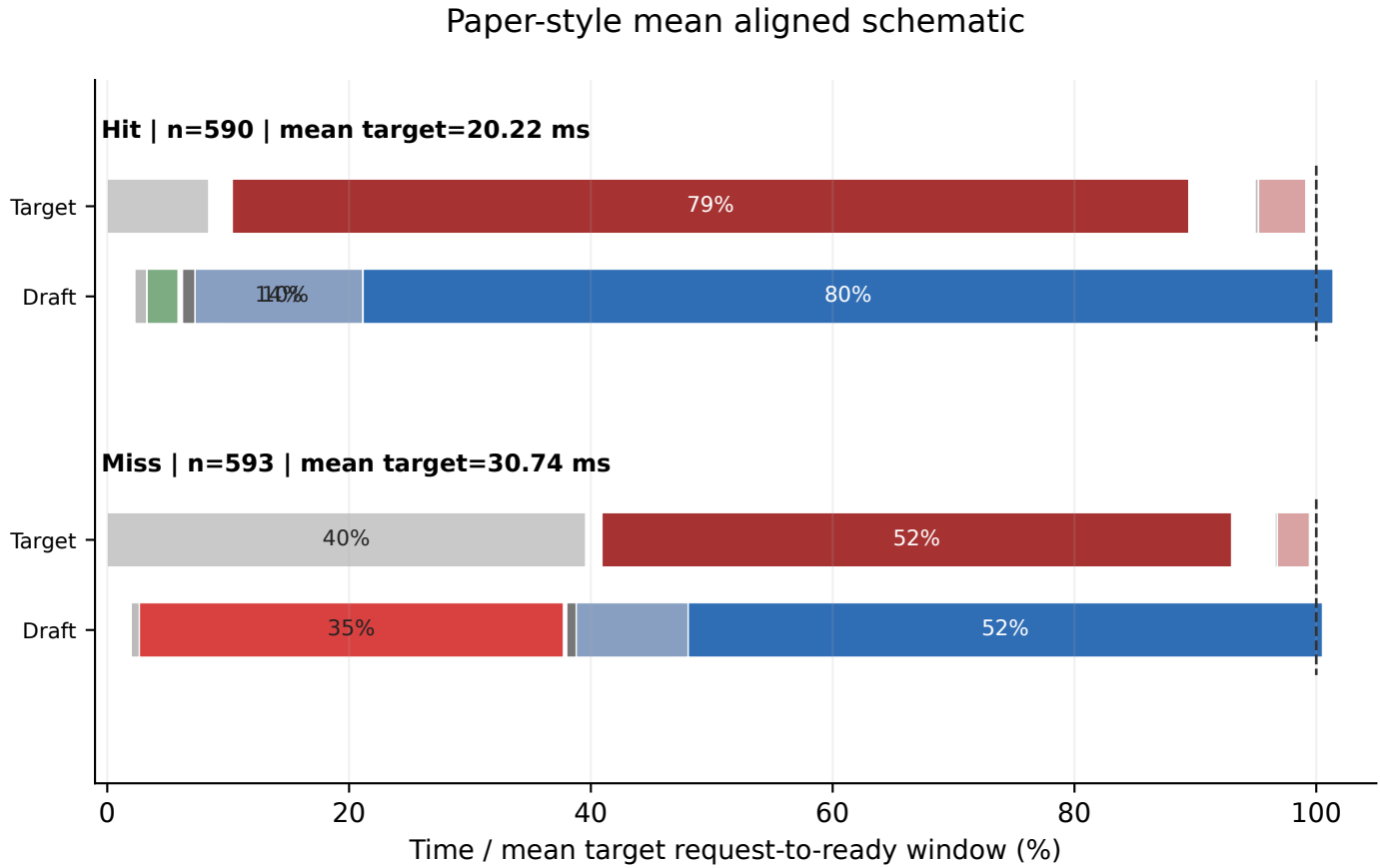
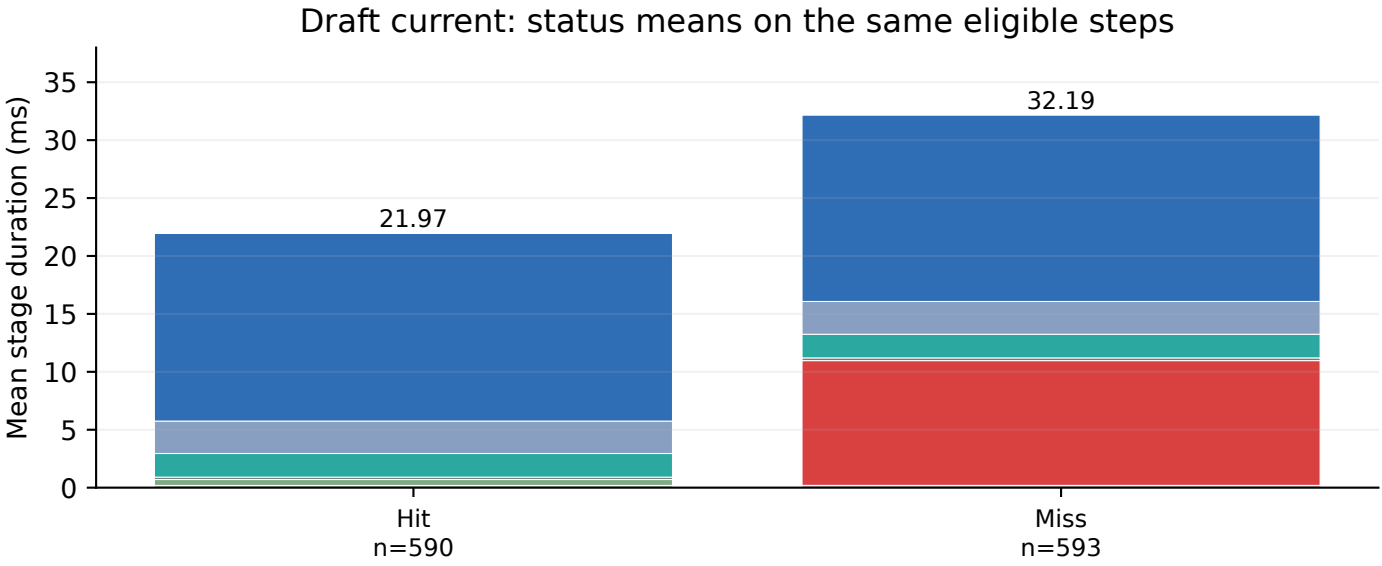
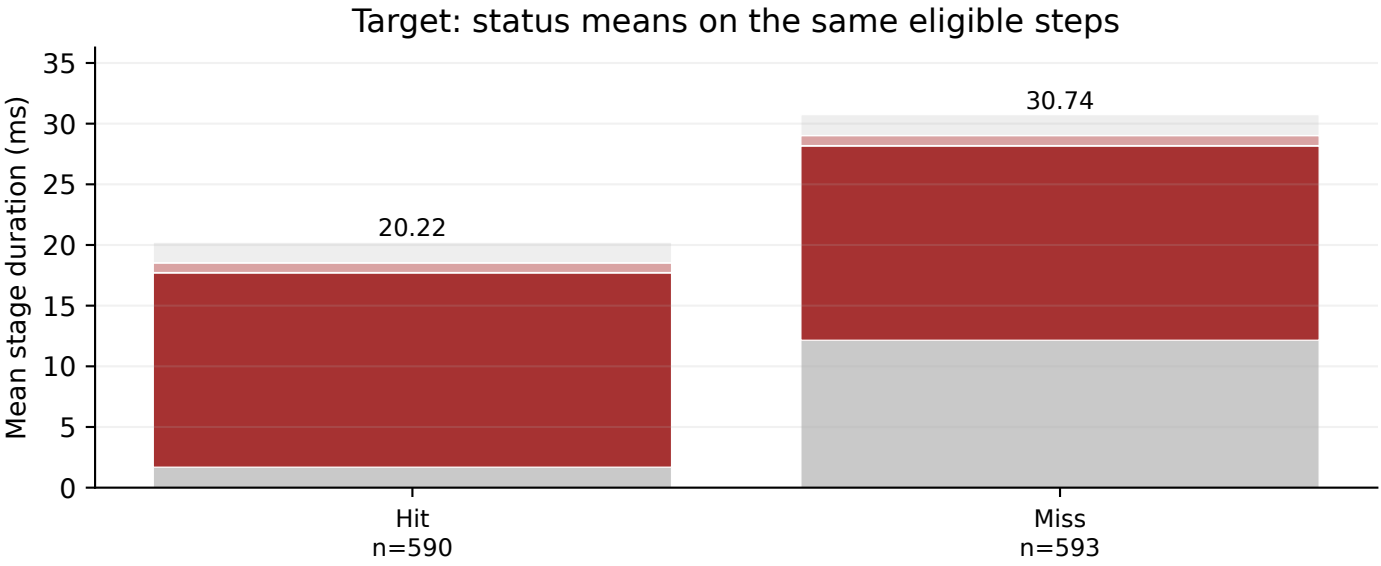
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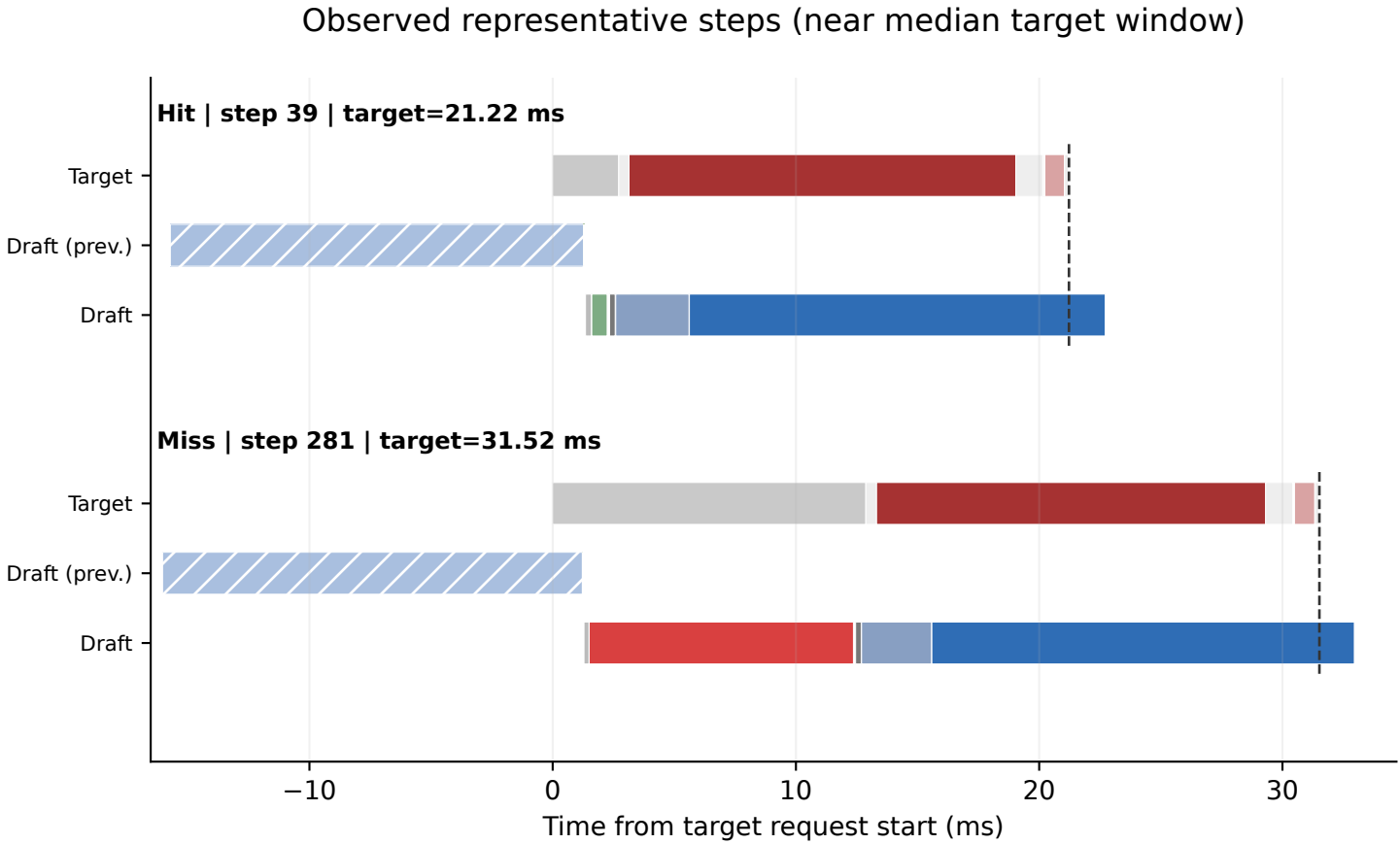
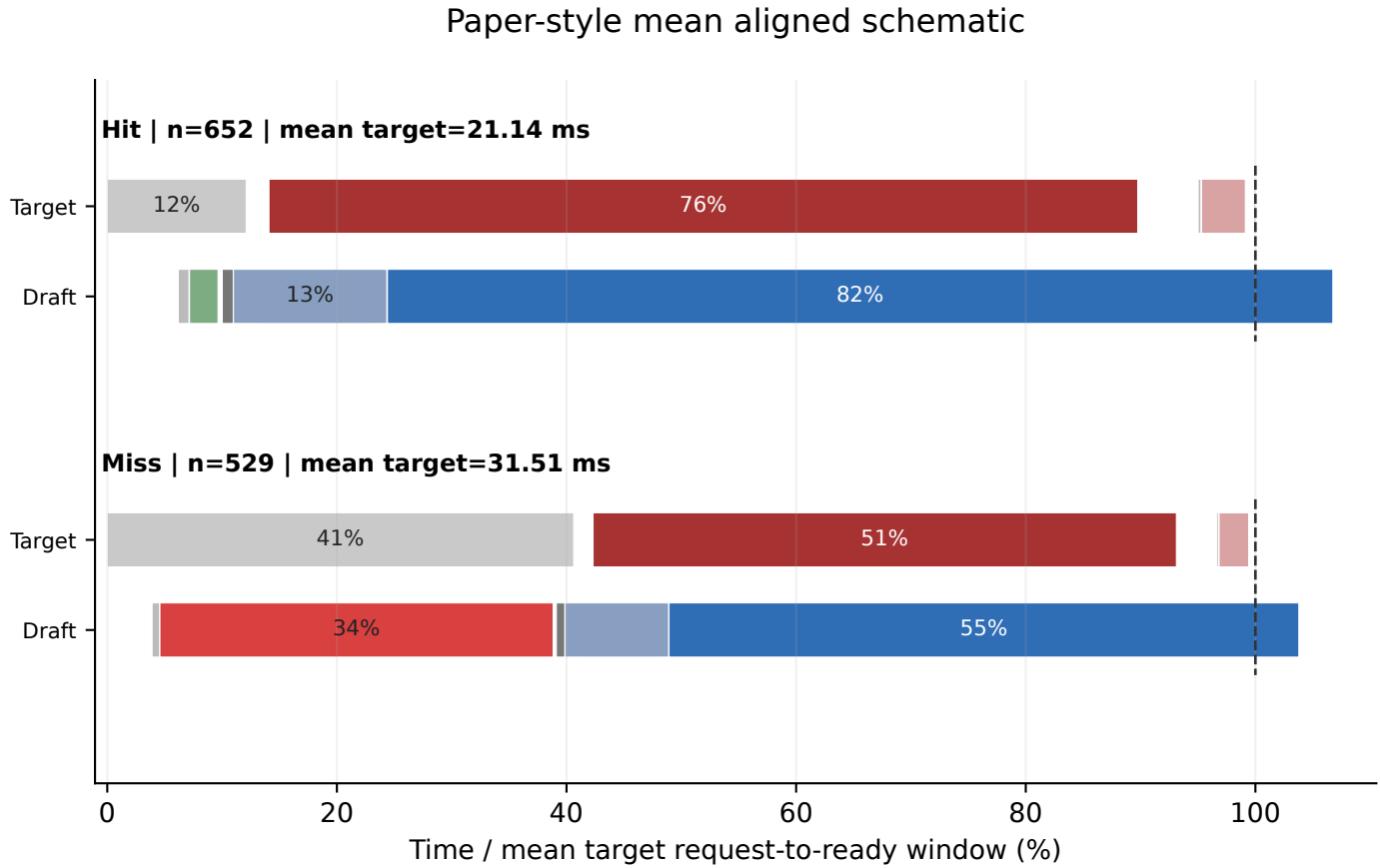
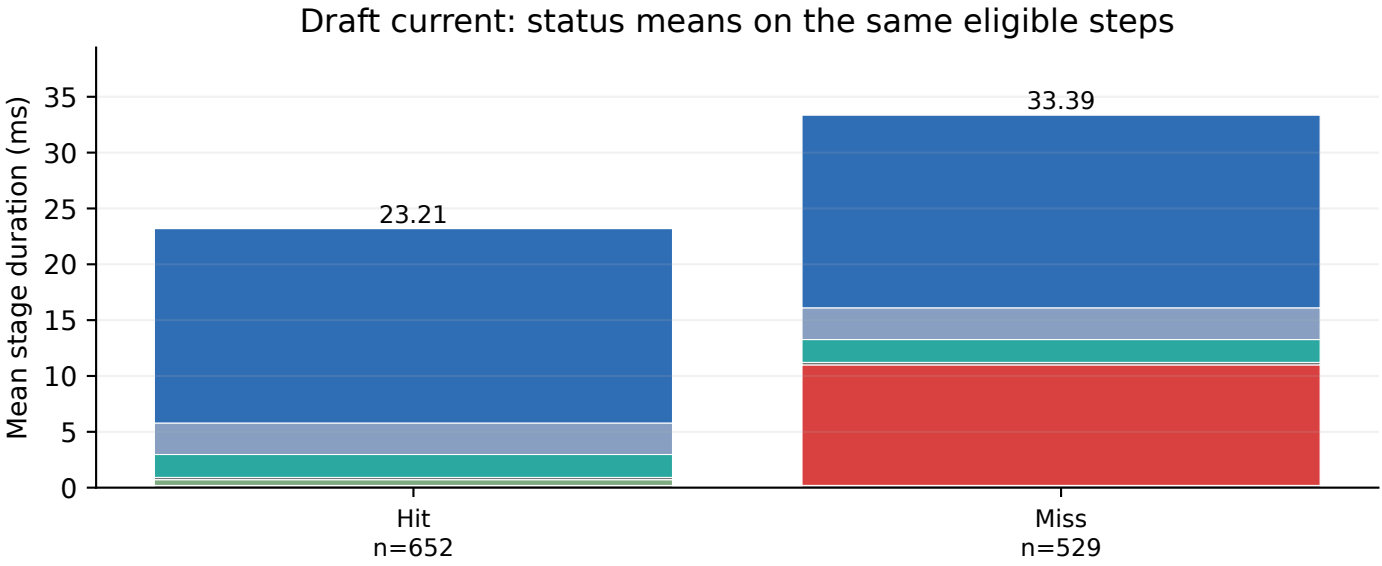
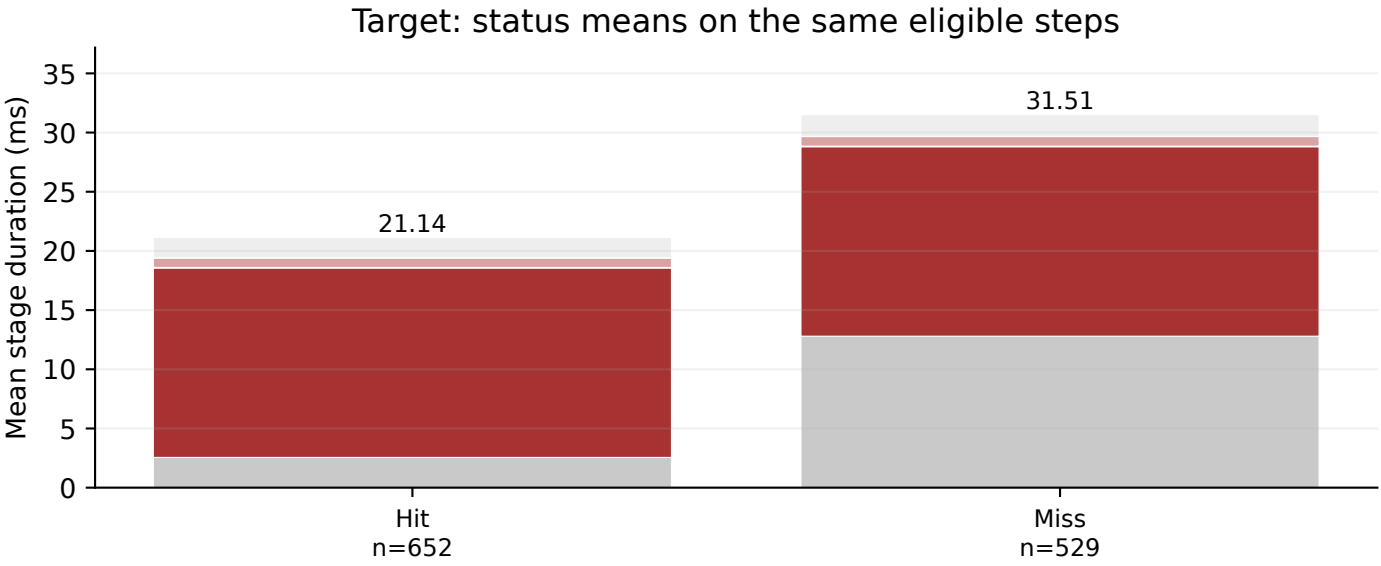
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
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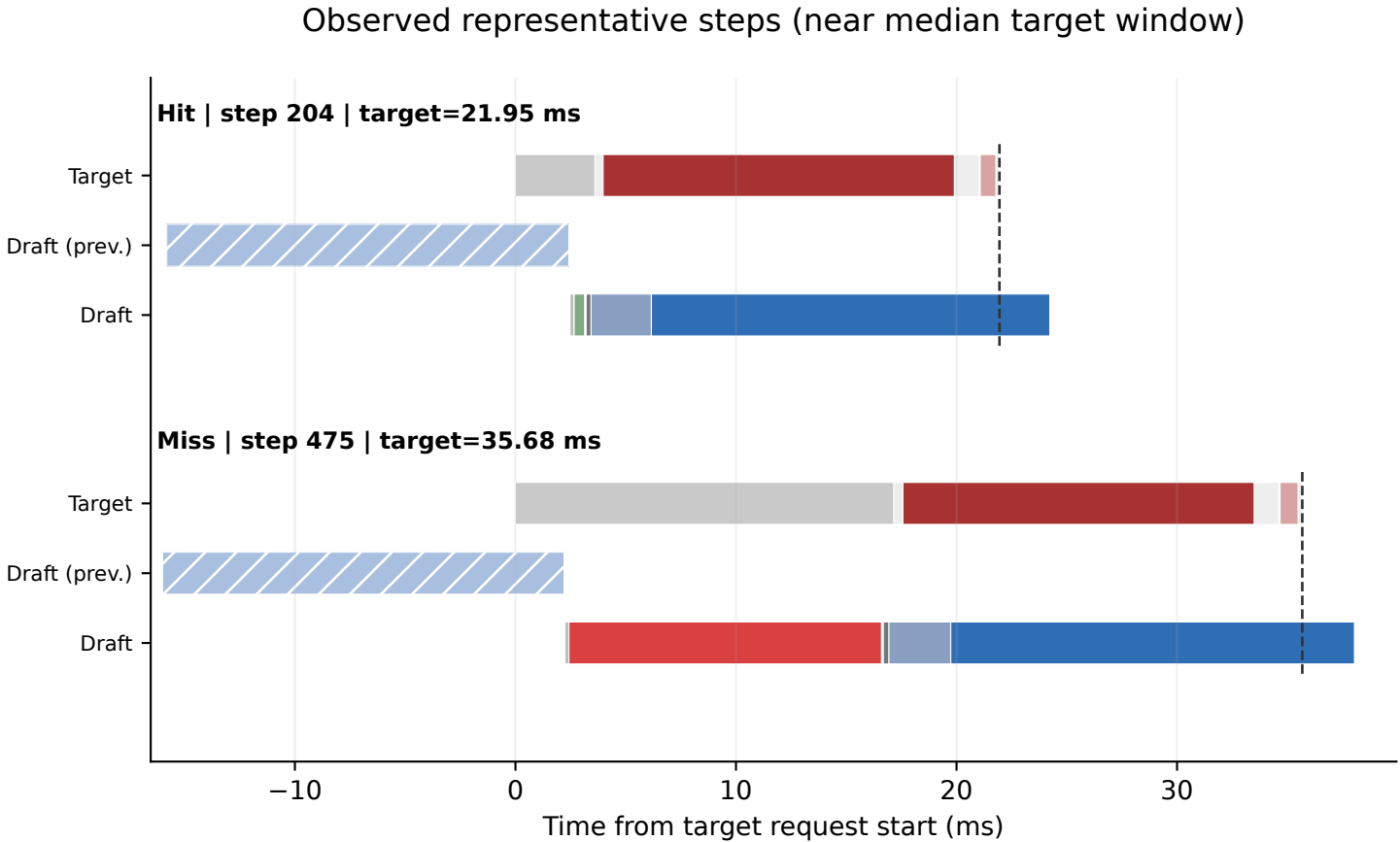
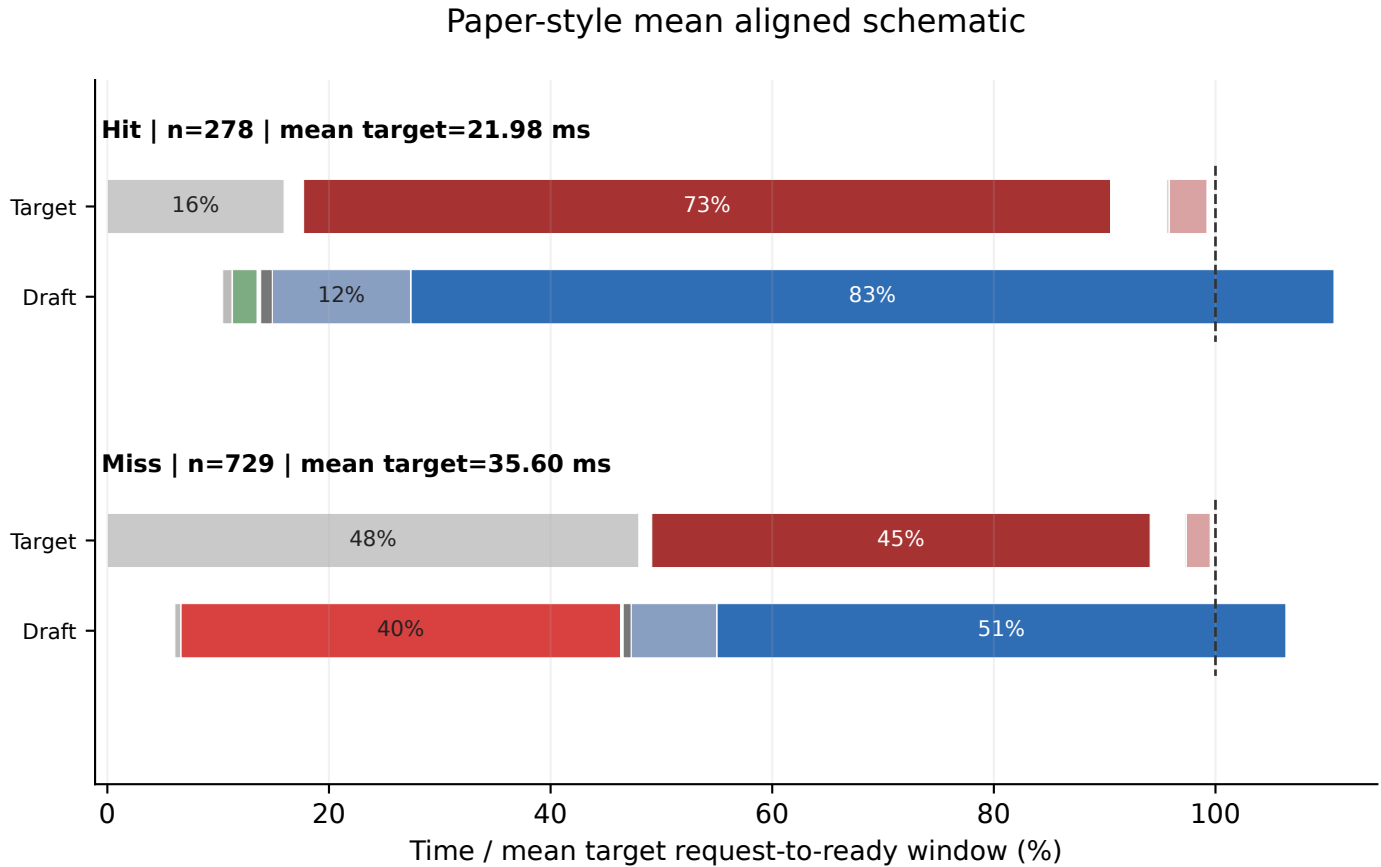
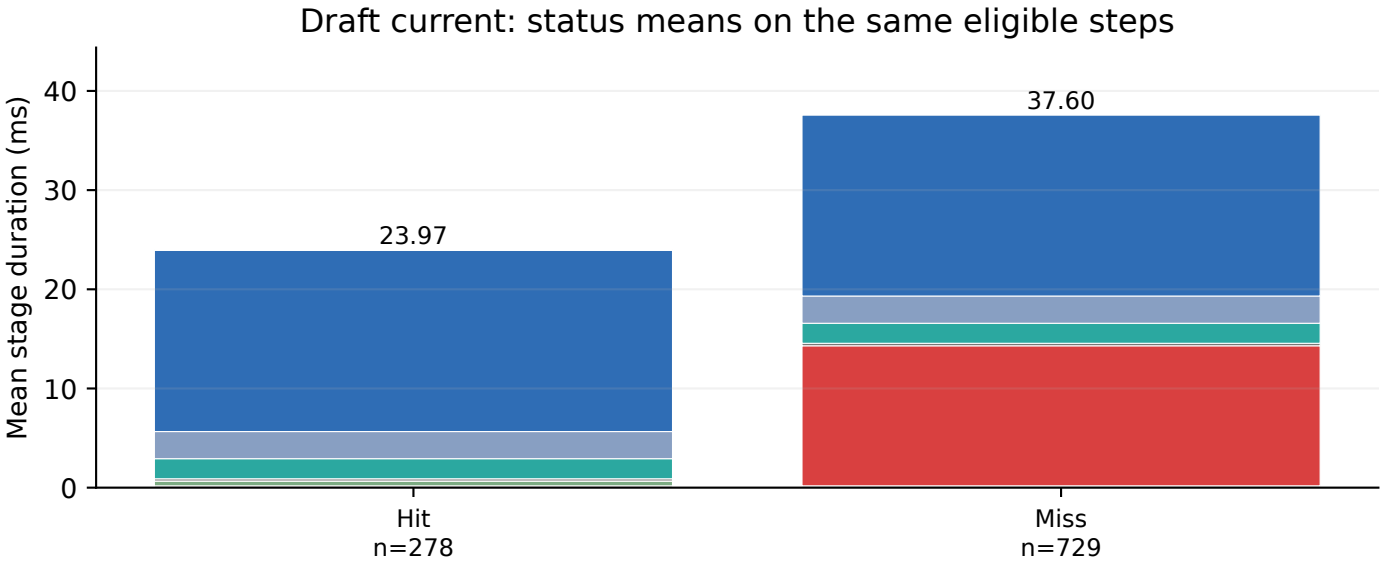
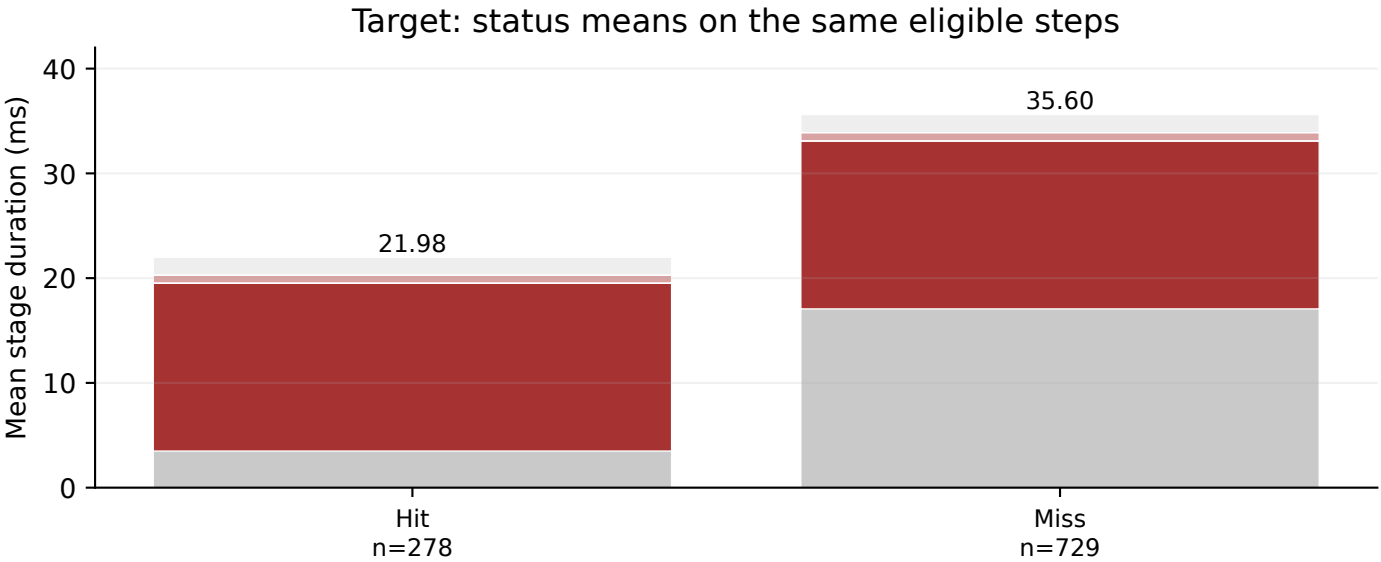
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
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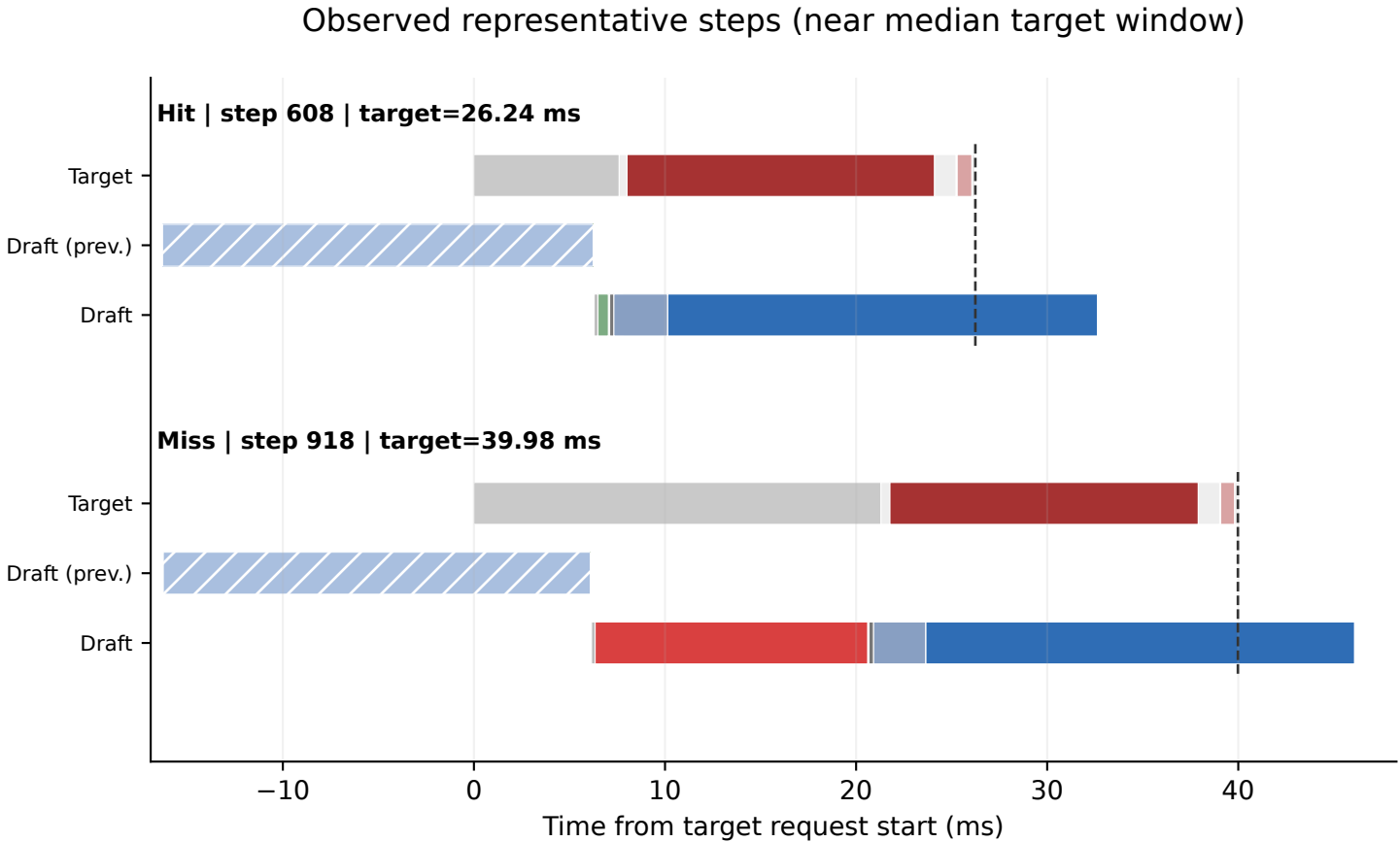
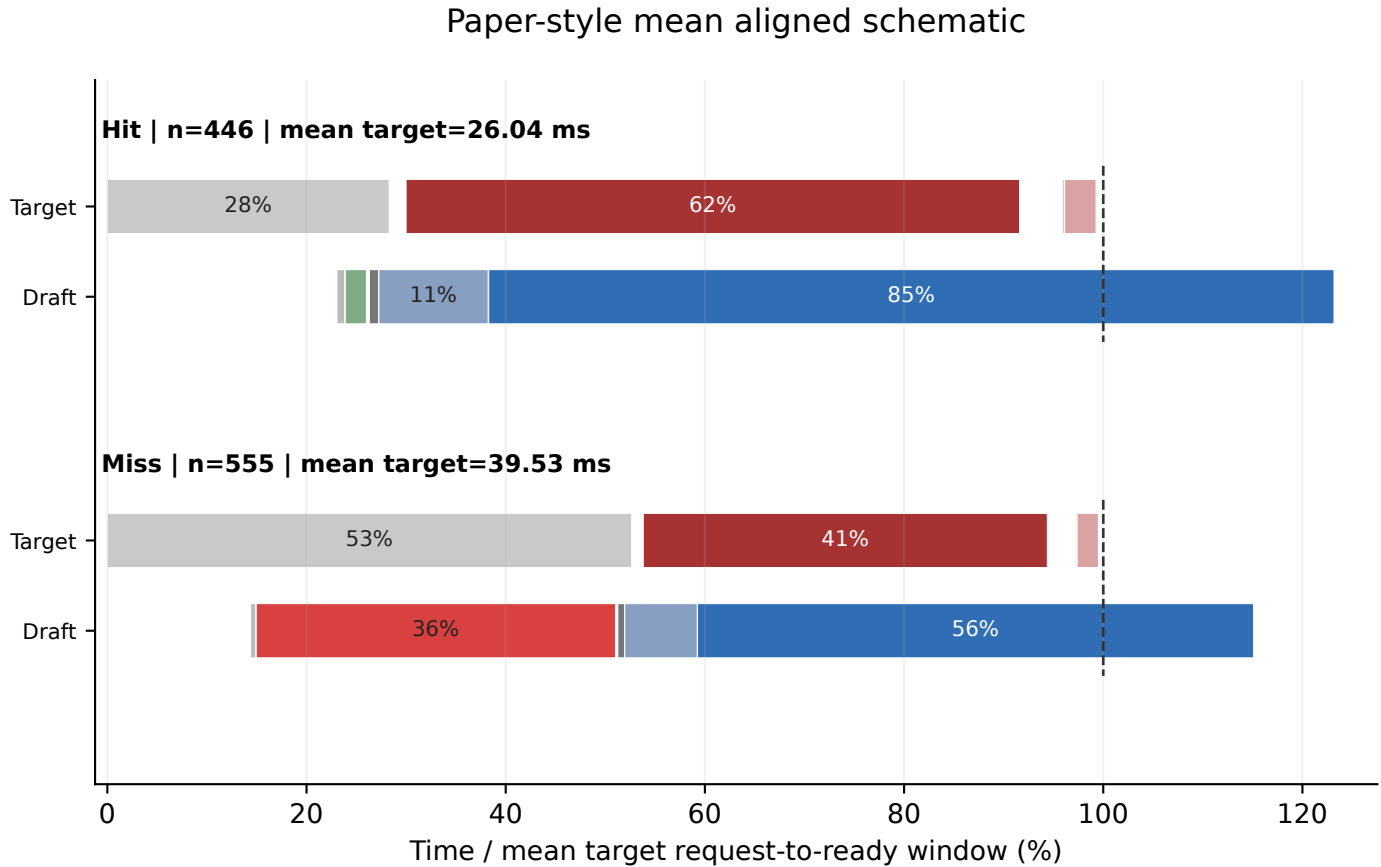
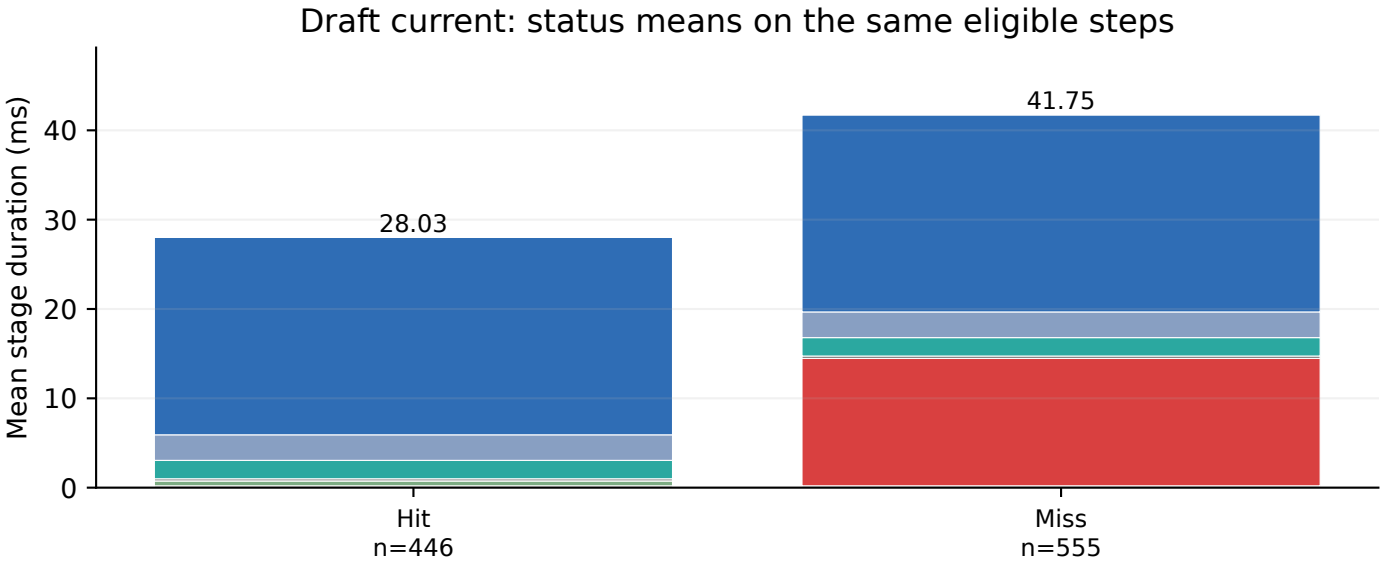
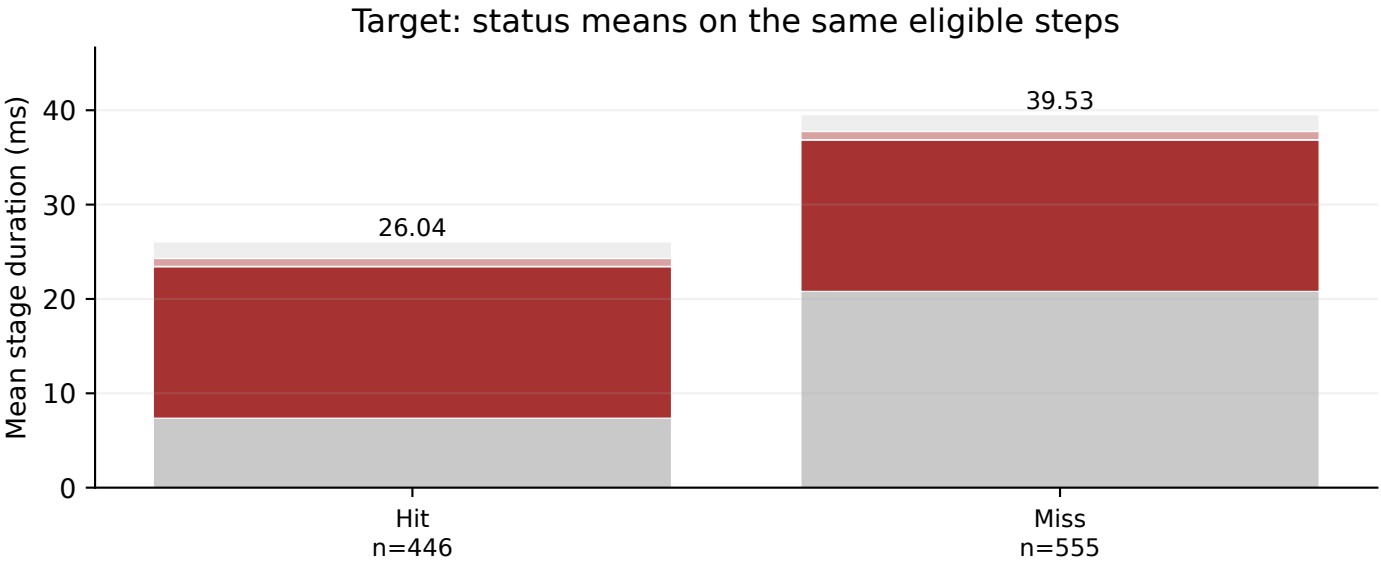
Wait / sync Accept prepare Miss response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

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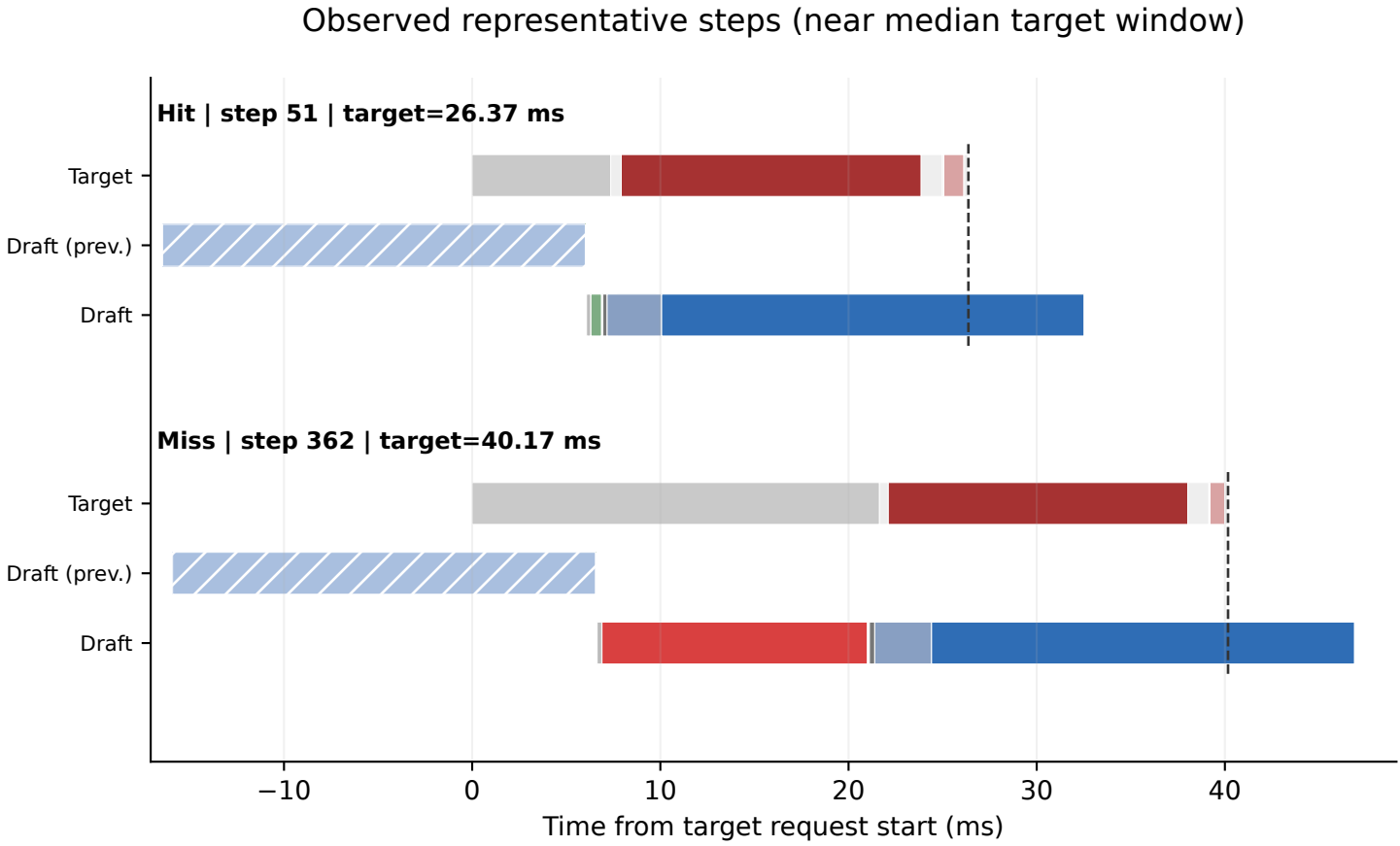
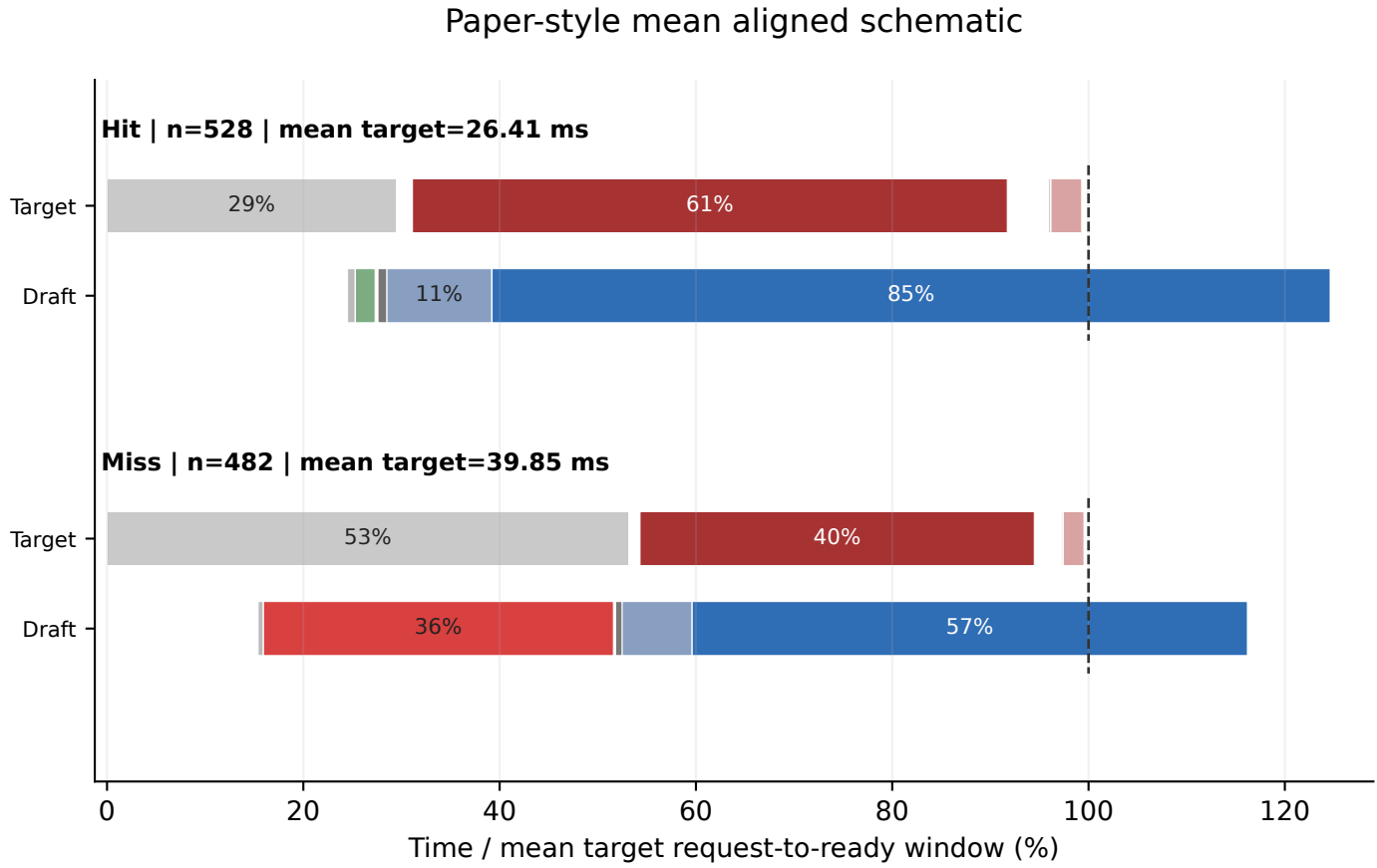
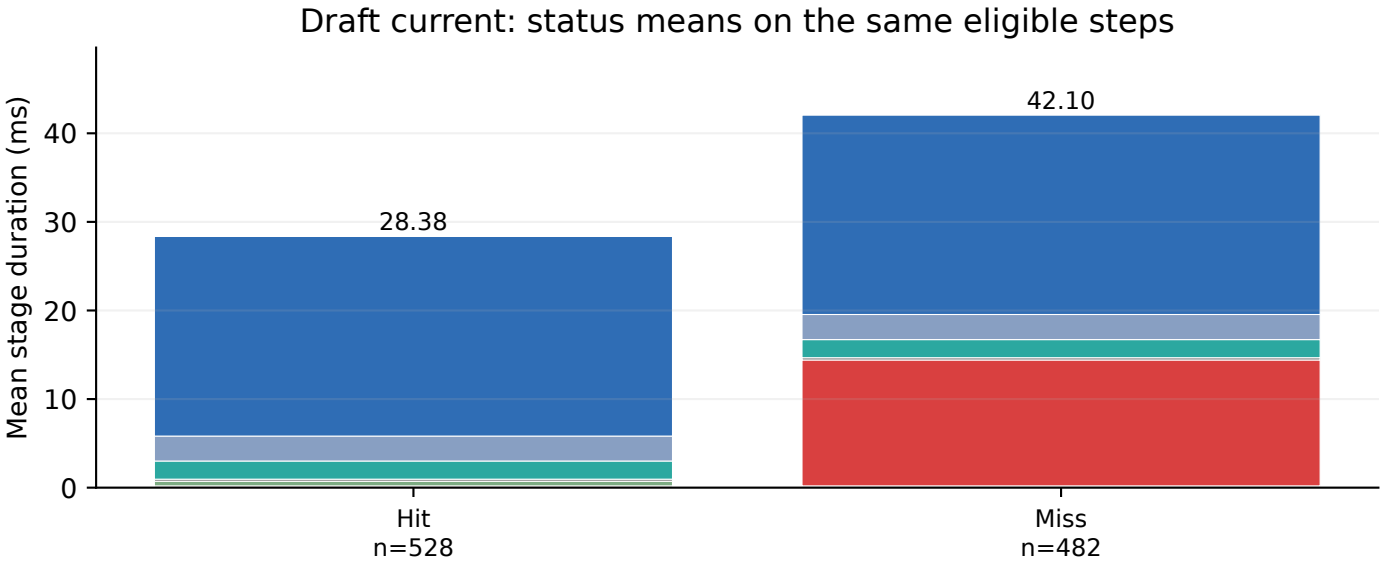
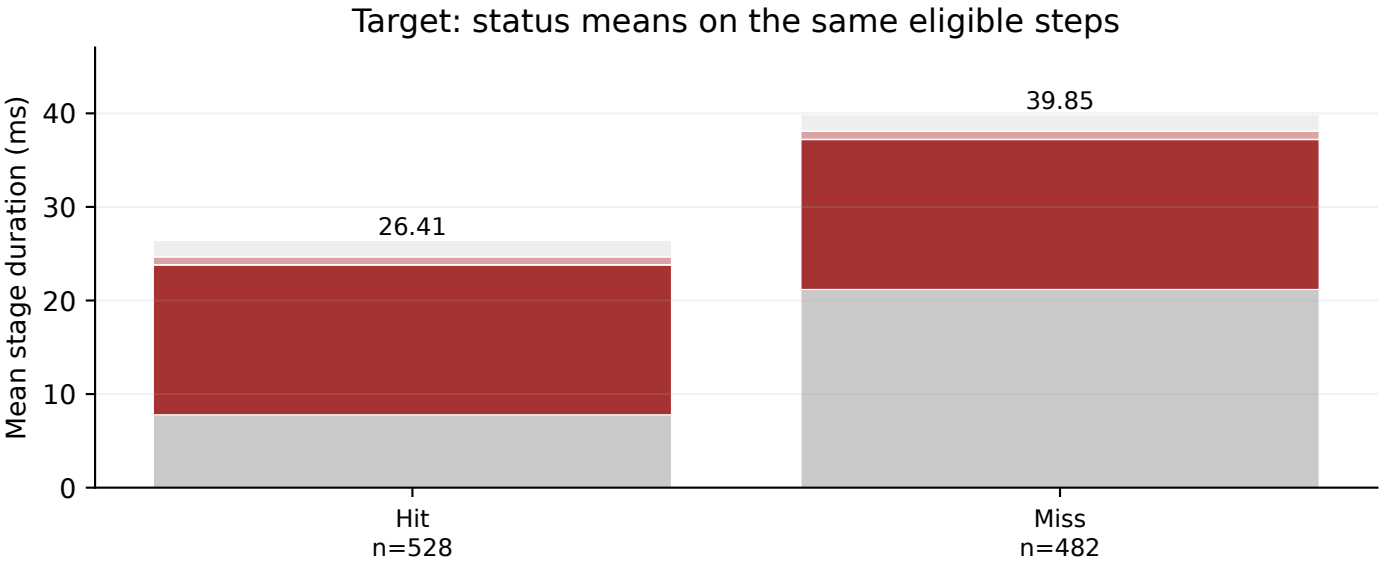
Wait / sync Accept prepare Miss response Cache response SSD decode Previous SSD decode Target ready
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Wait / sync Accept prepare Miss response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

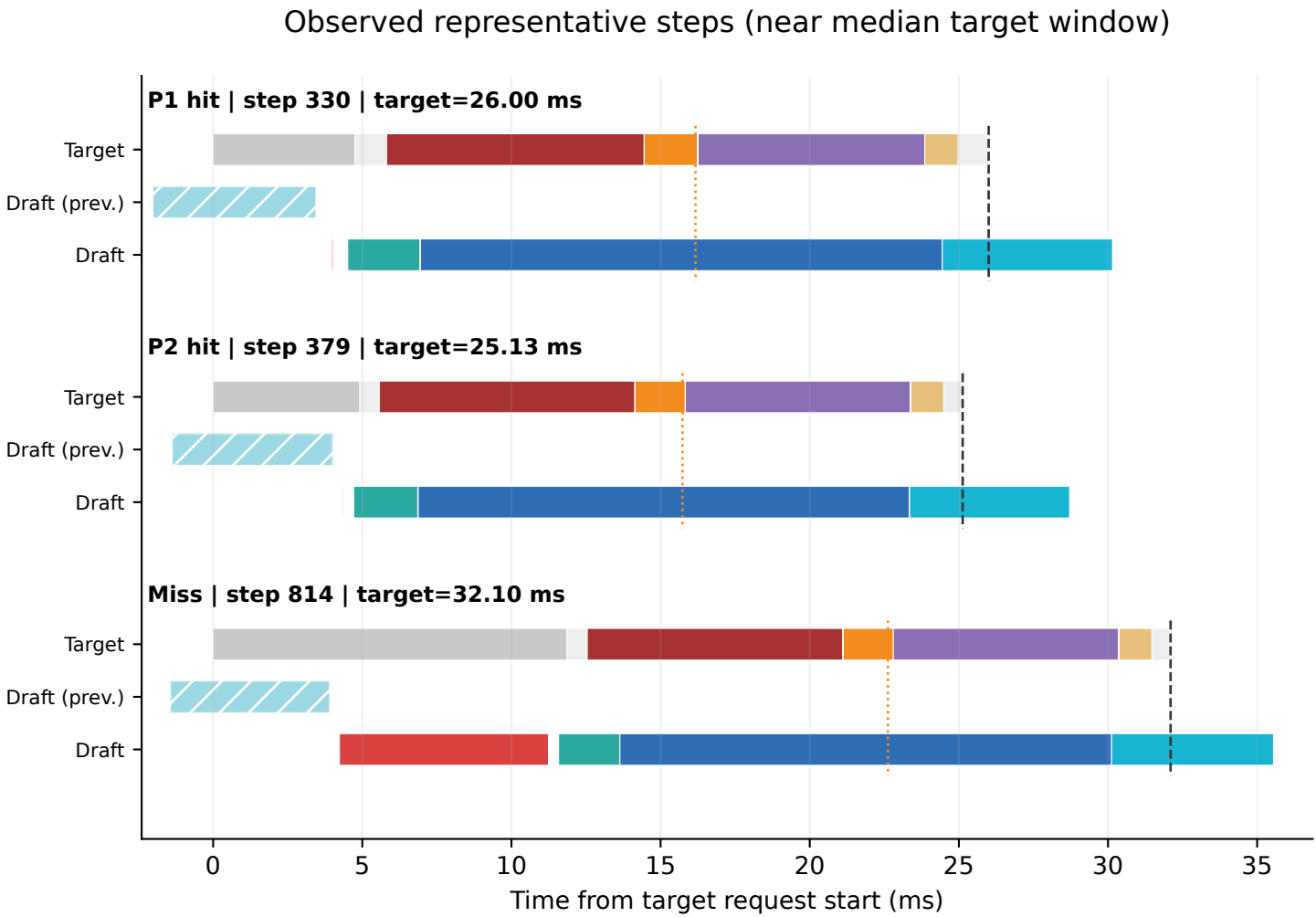
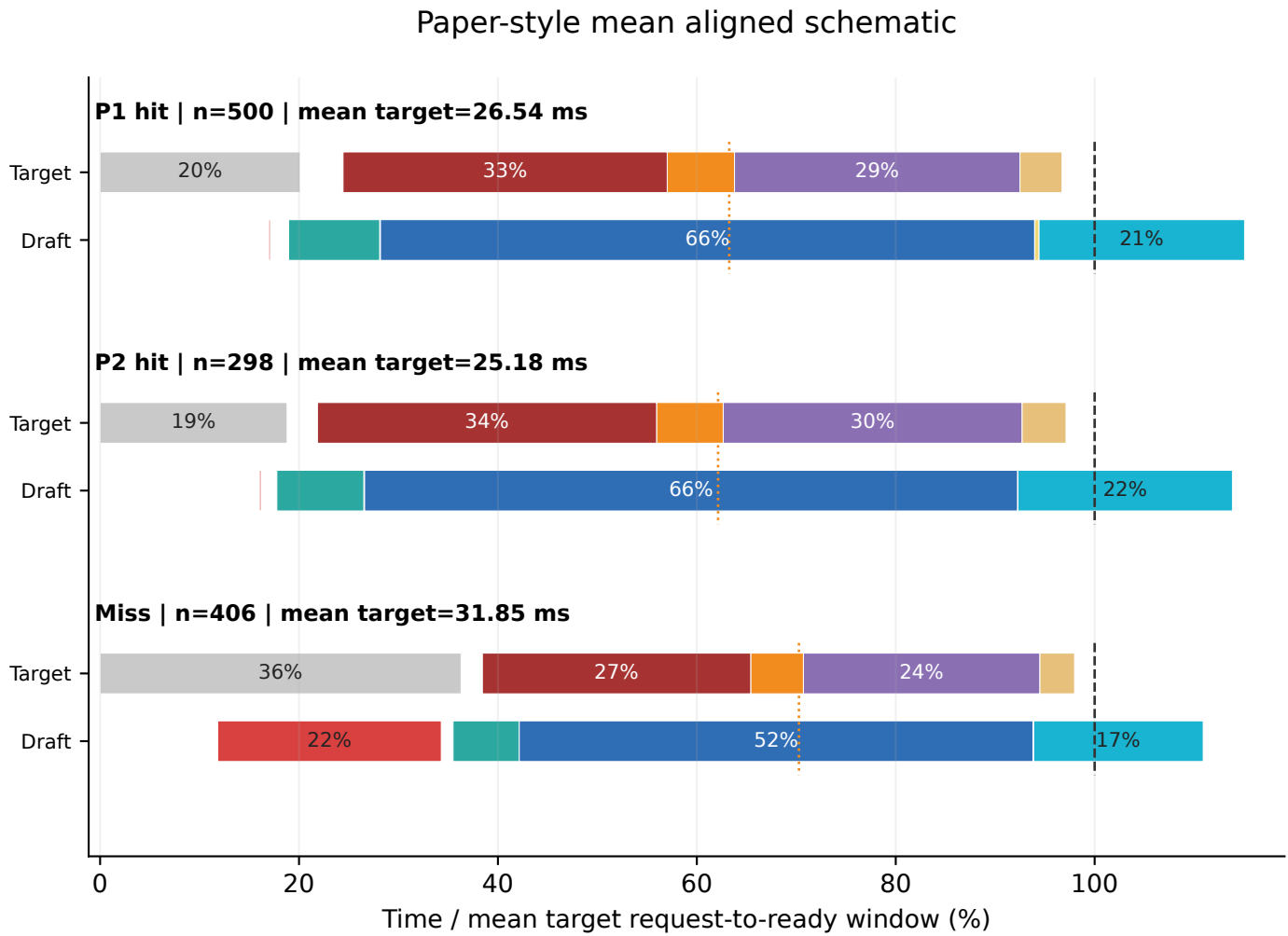
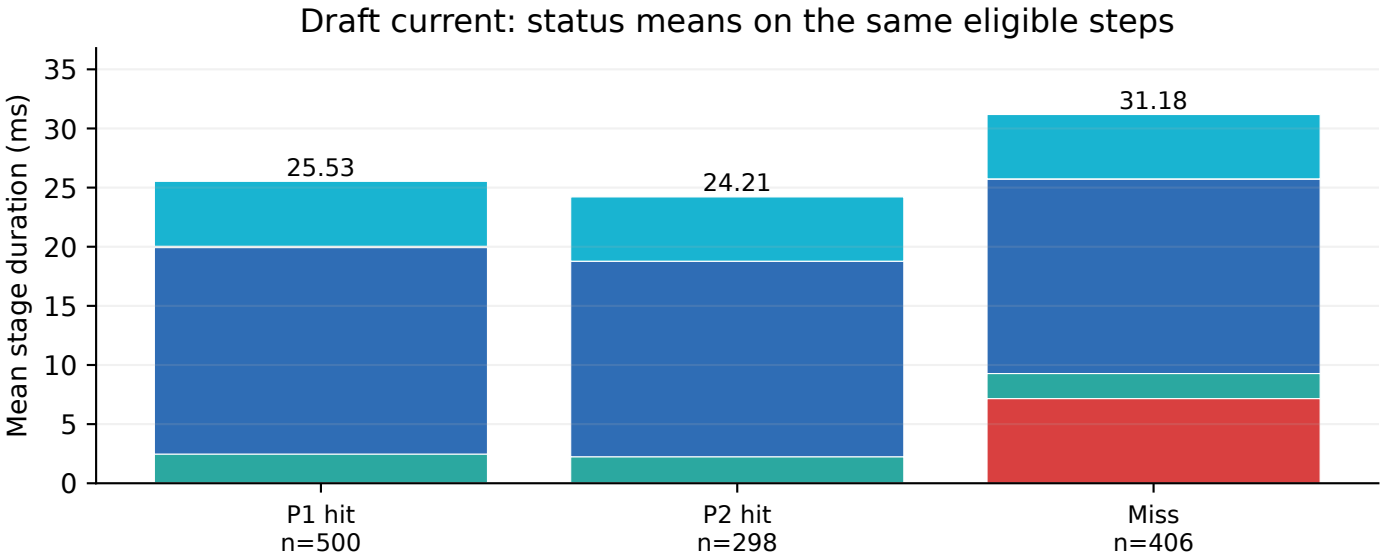
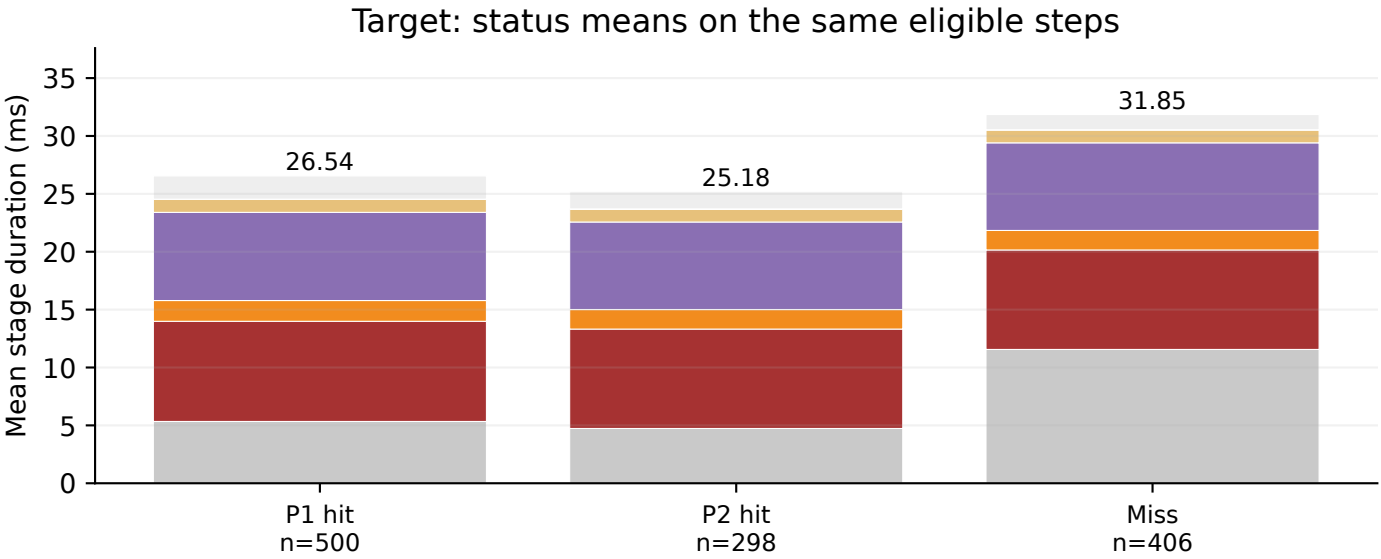
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

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Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

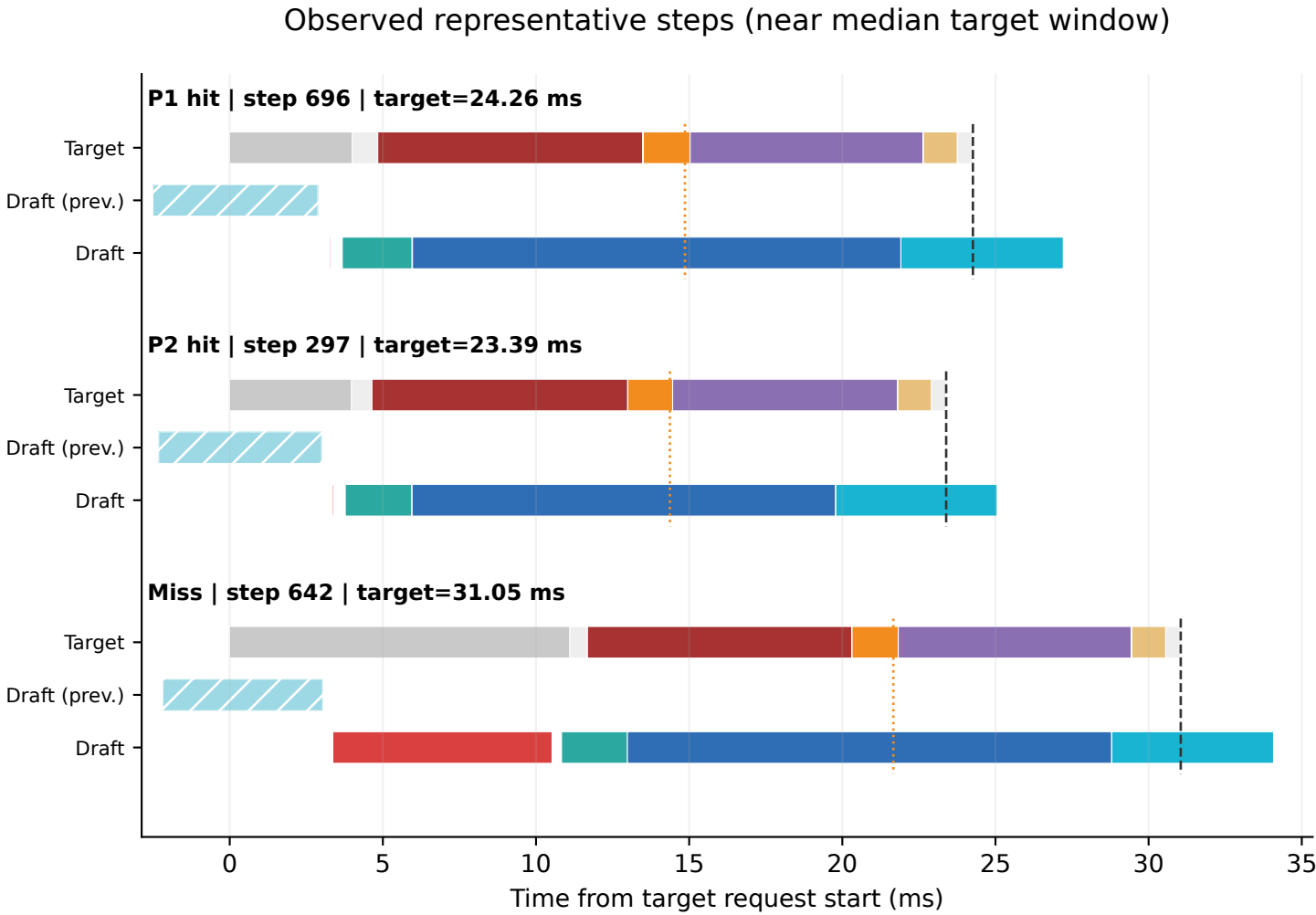
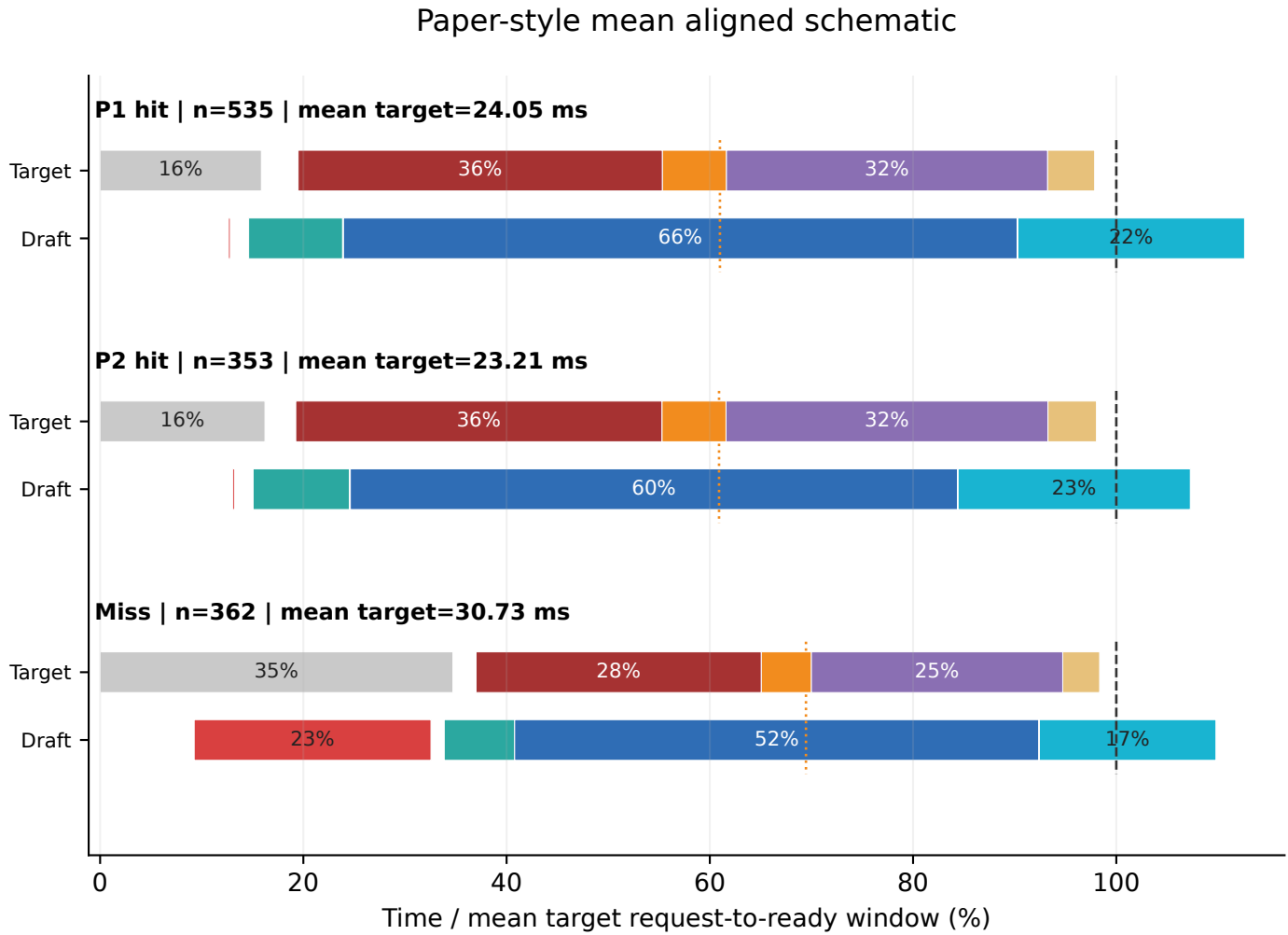
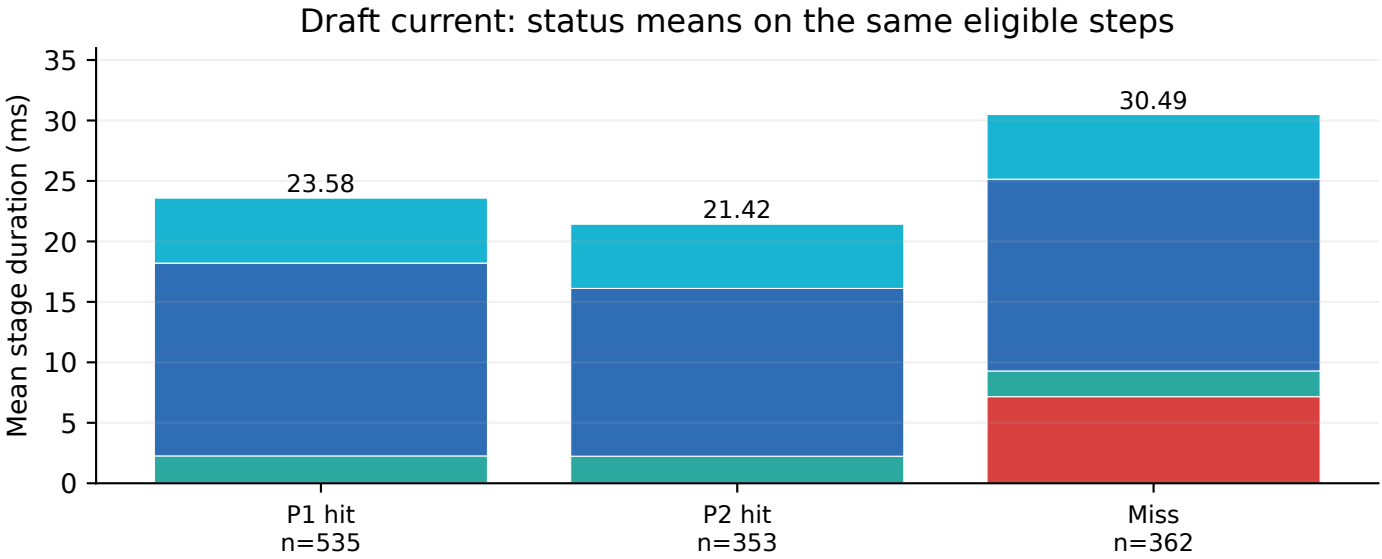
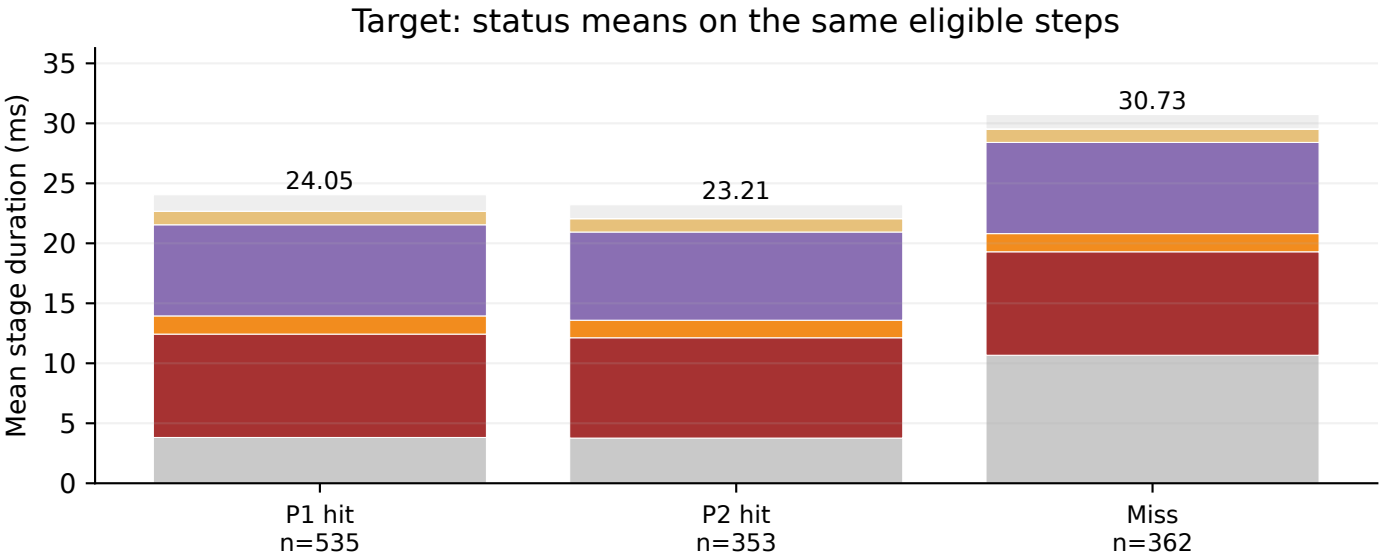
llama3_b1_tree_al_compute3
DUET: K1/K2=6/2, exit=16 (0-based), $\bar{C}$ =3, $\bar{N1}/\bar{N2}$ =12/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1204 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

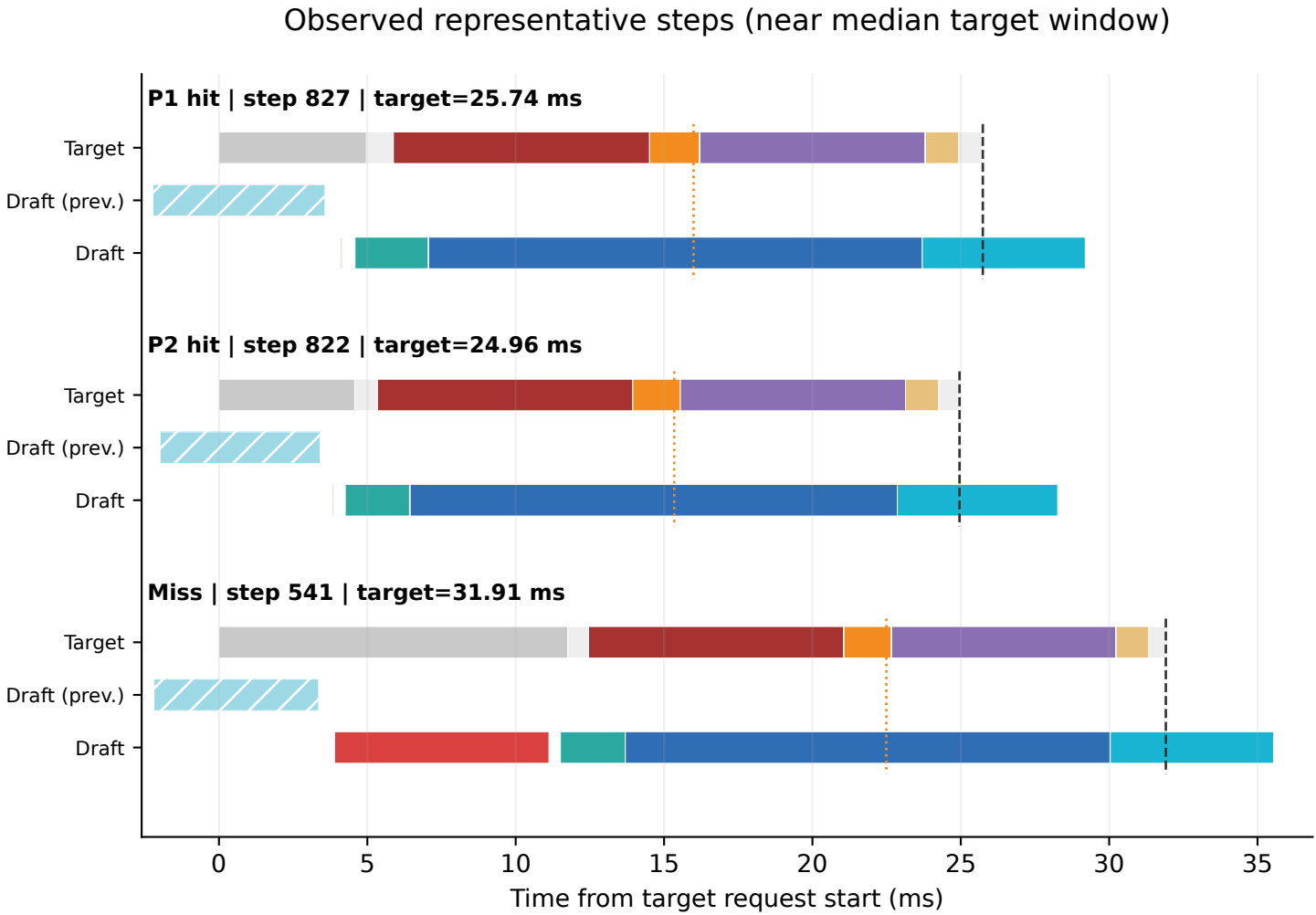
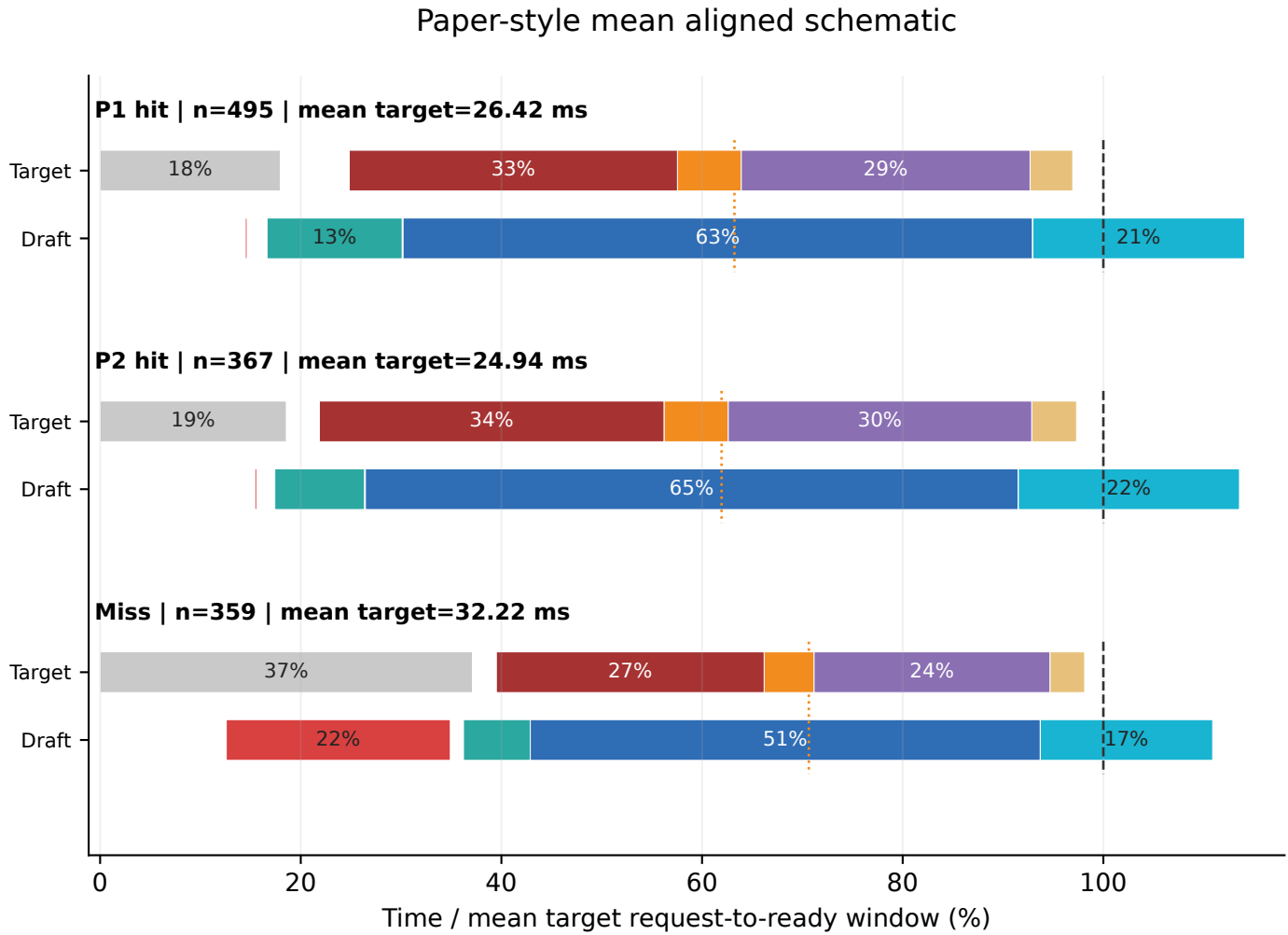
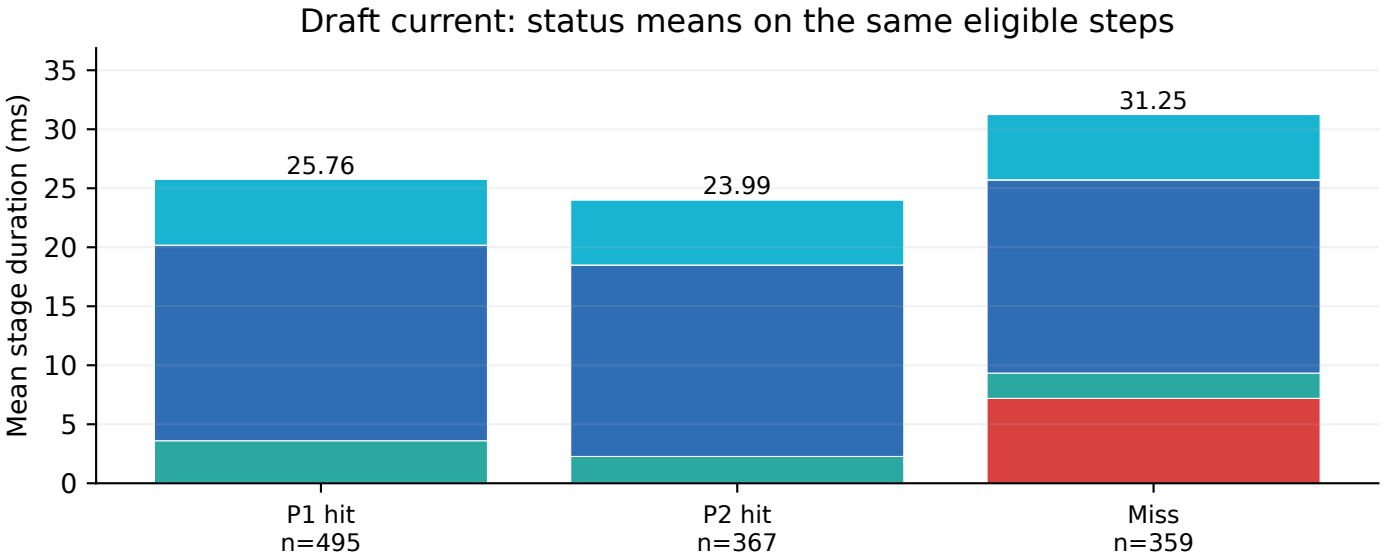
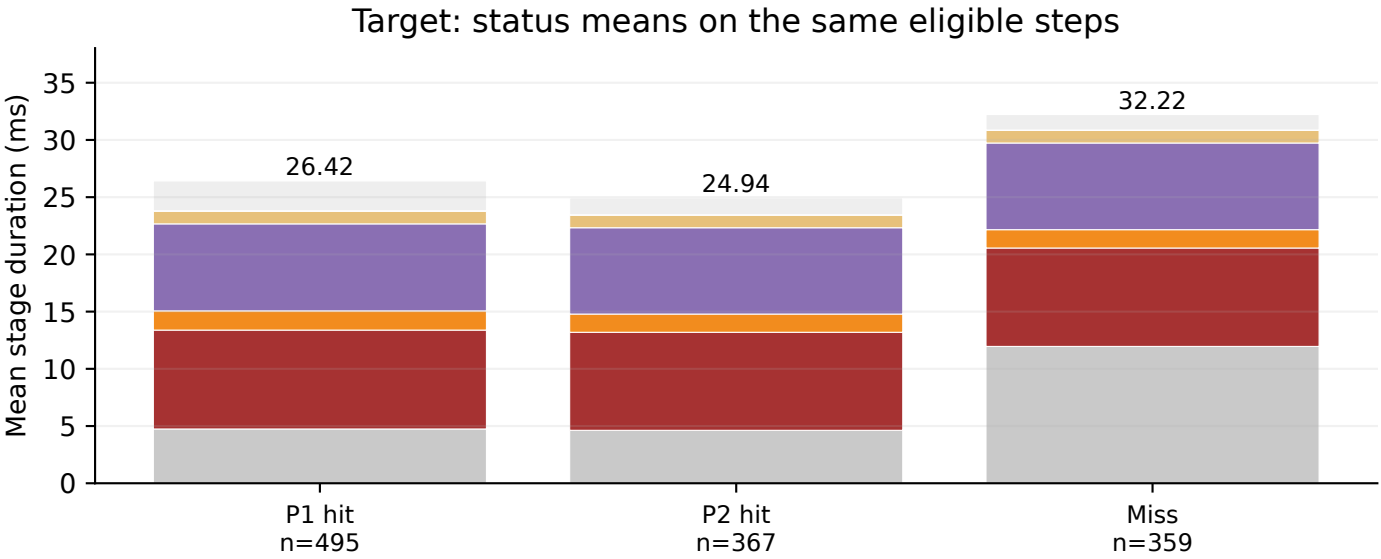
llama3_b1_tree_al_width1
DUET: K1/K2=6/2, exit=16 (0-based), C=1, N1/N2=6/2, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1250 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

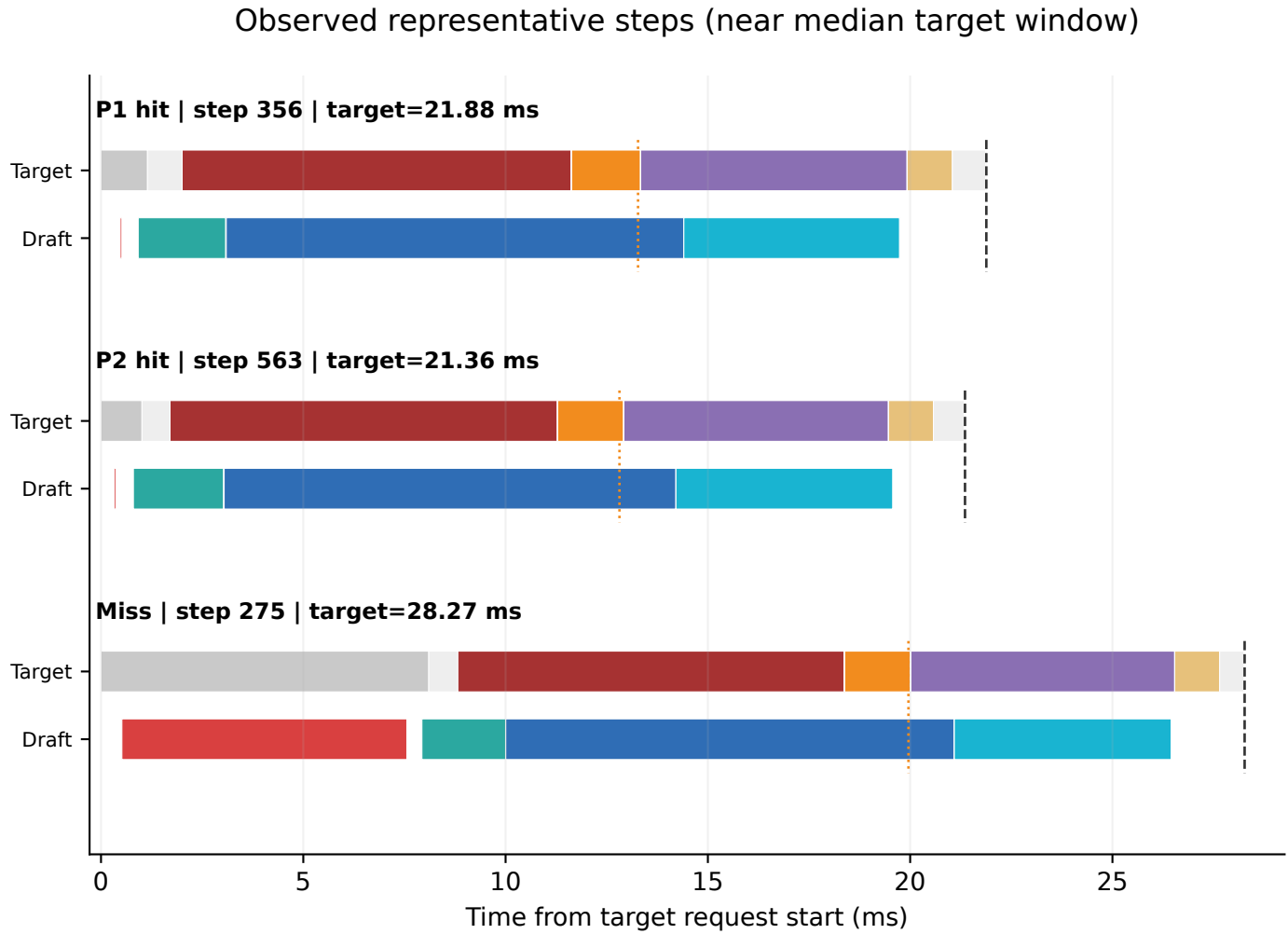
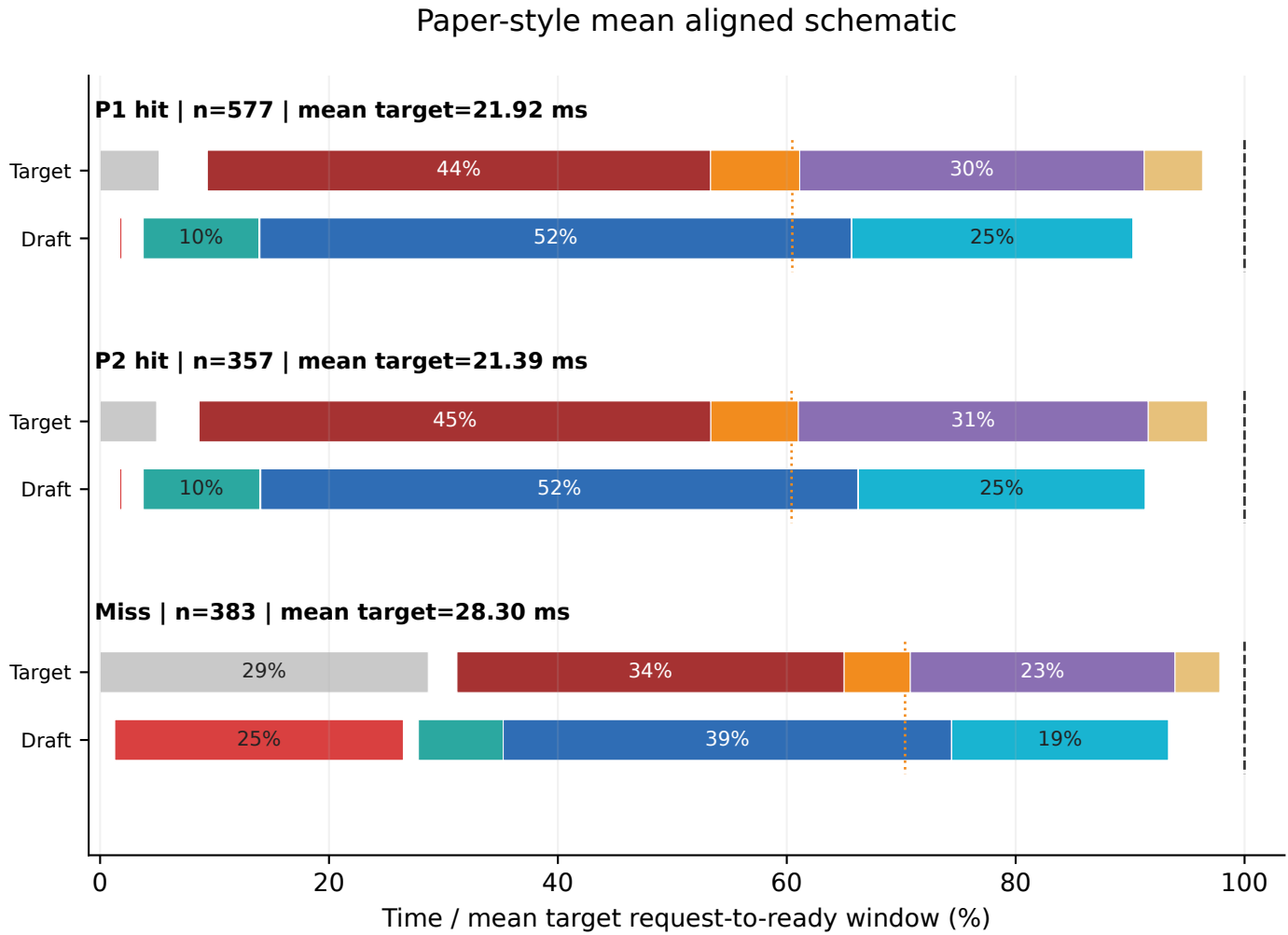
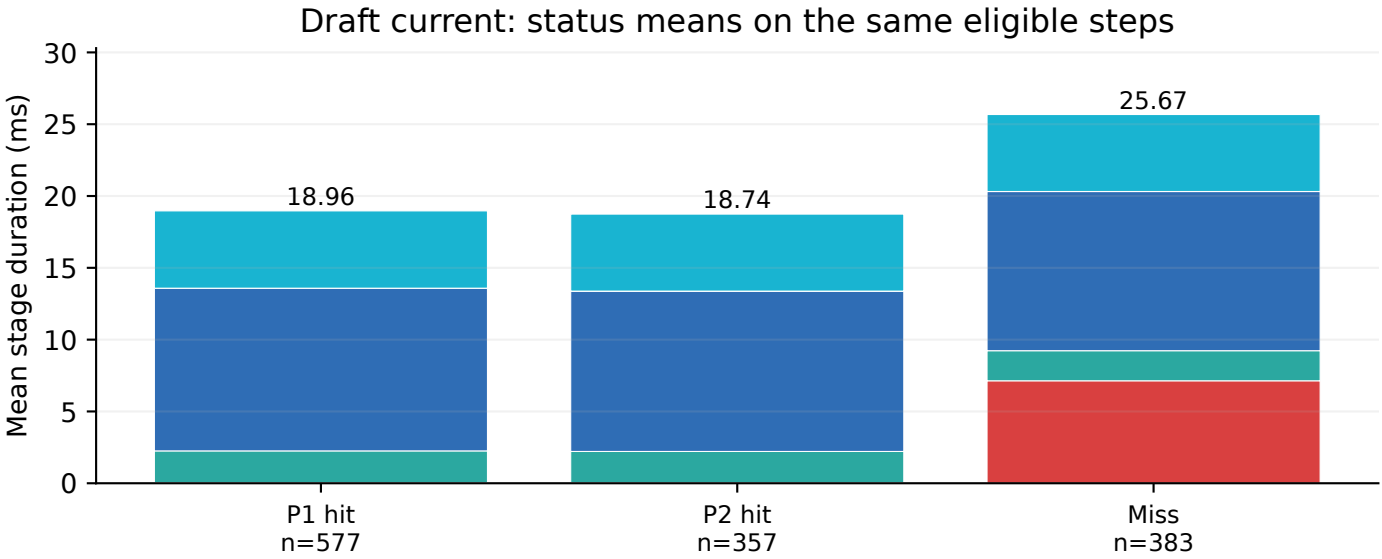
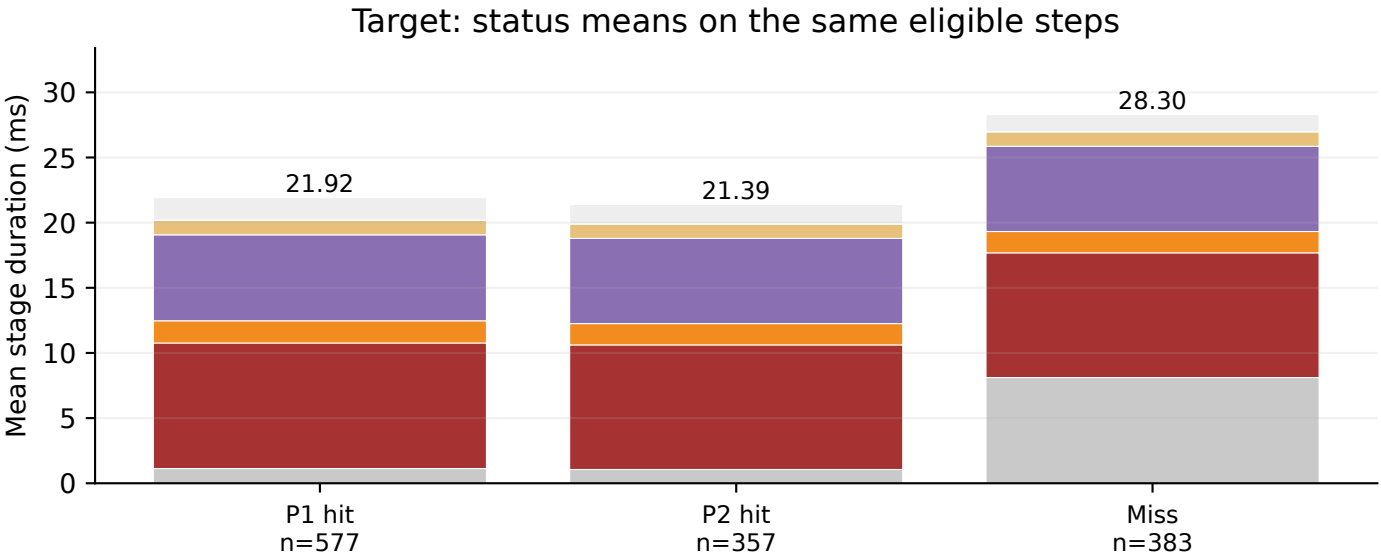
llama3_b1_tree_al_width2
DUET: K1/K2=6/2, exit=16 (0-based), C=2, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1221 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

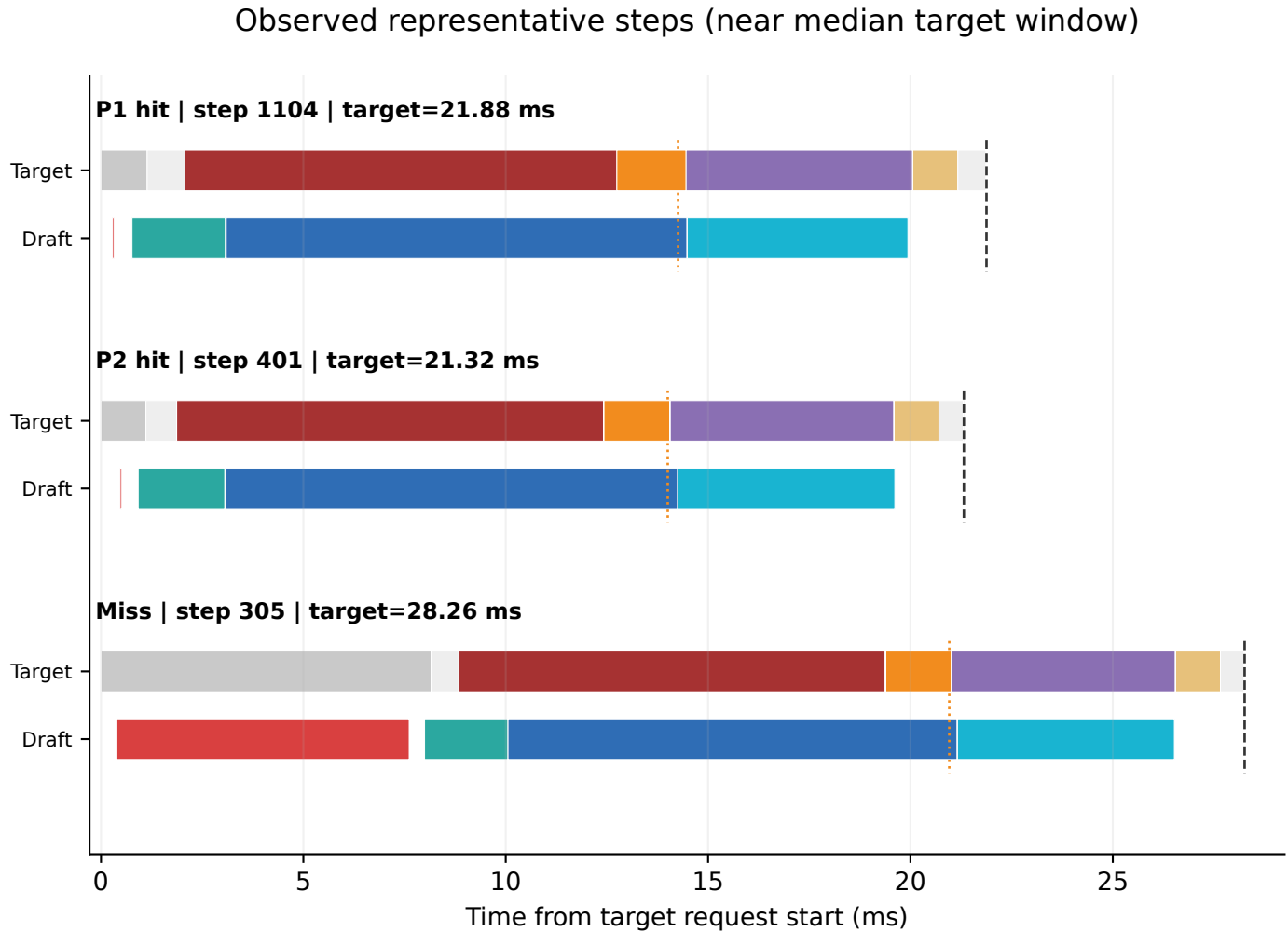
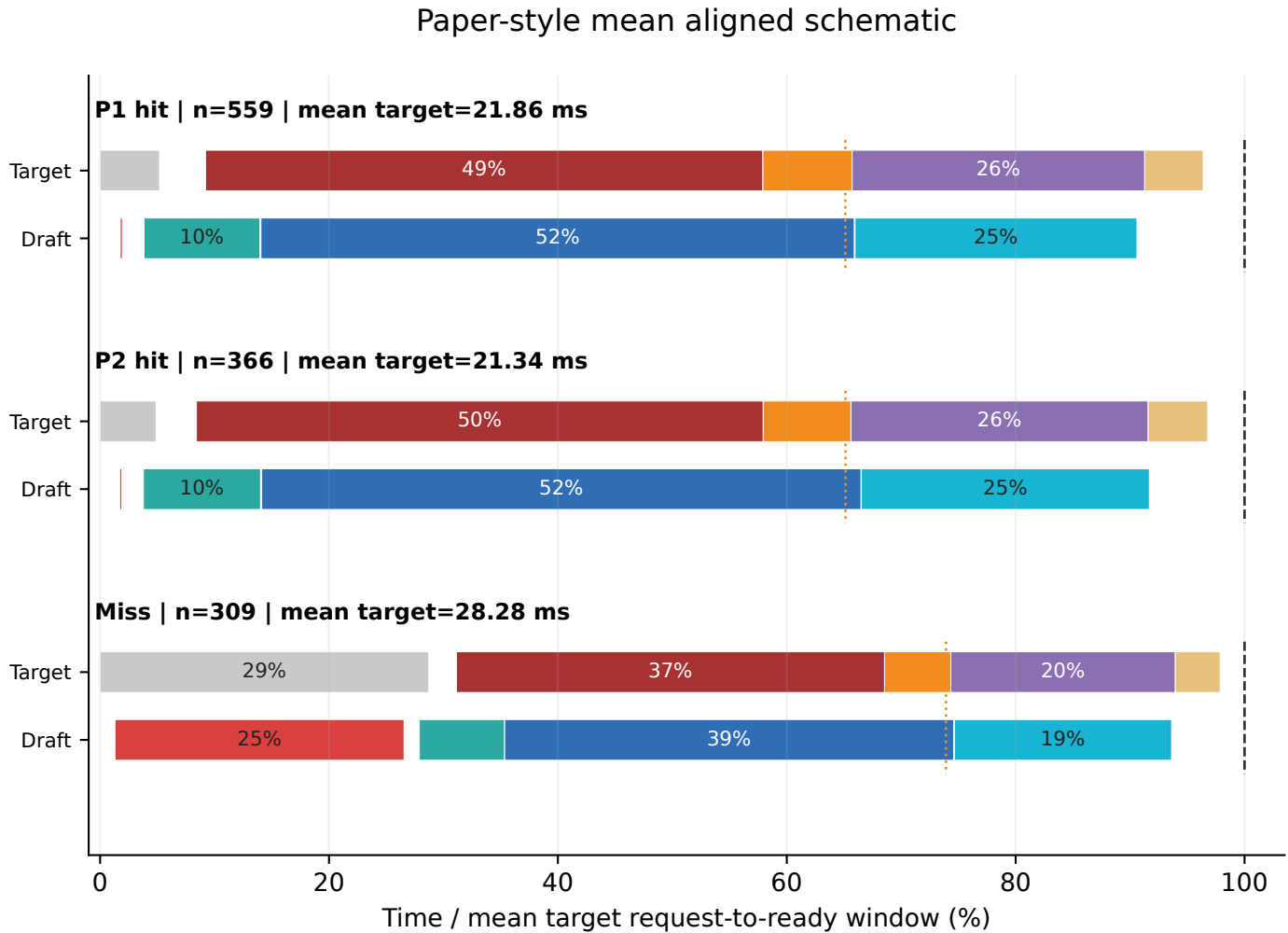
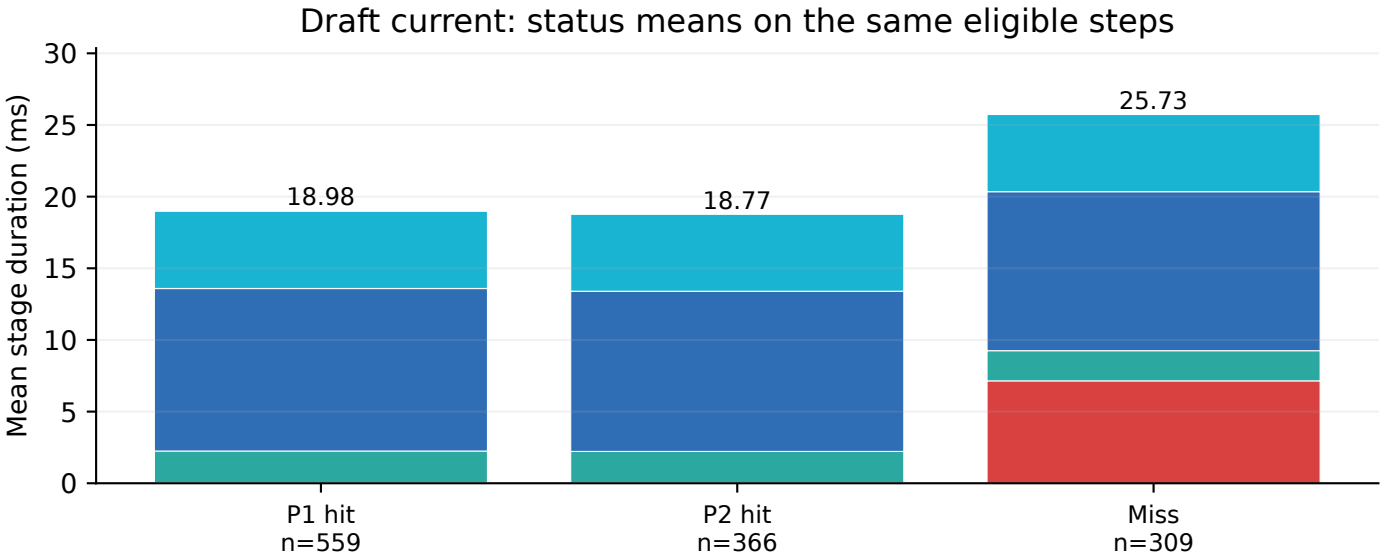
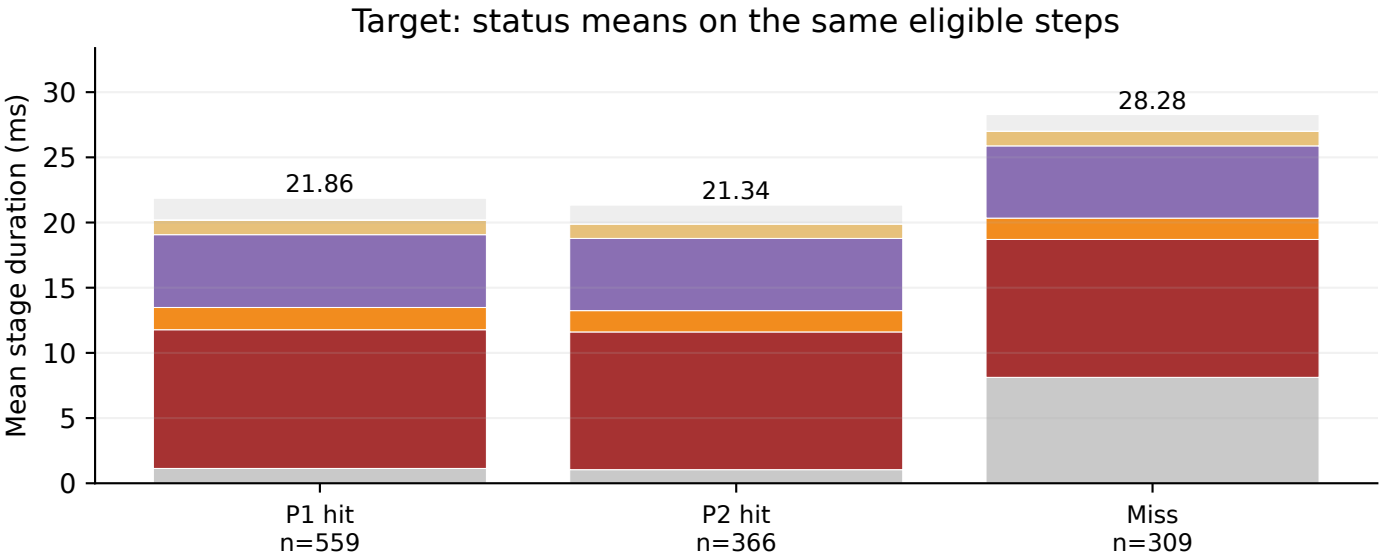
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_align_exit18
DUET: K1/K2=4/2, exit=18 (0-based), $\bar{C}$ =3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1317 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

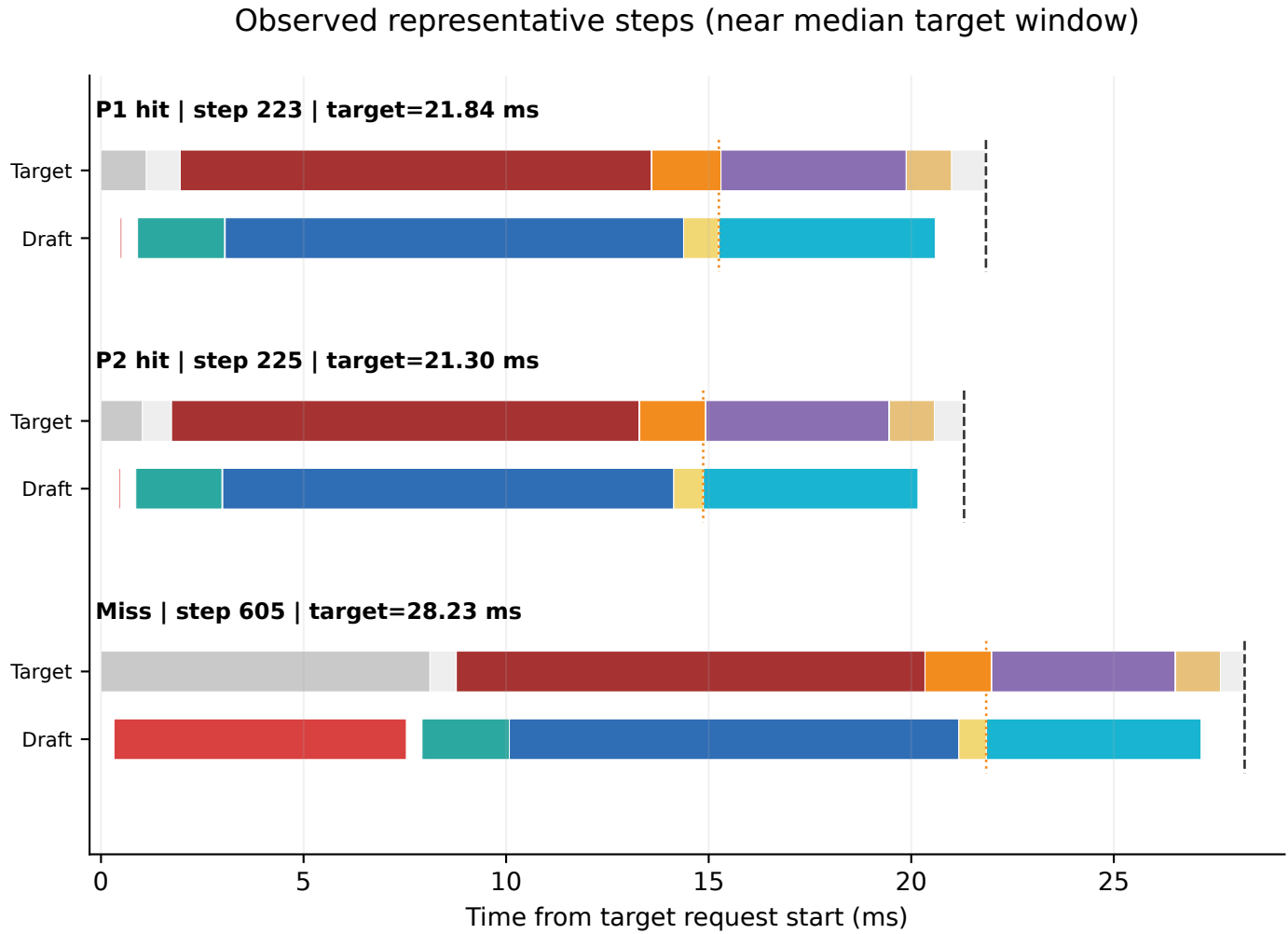
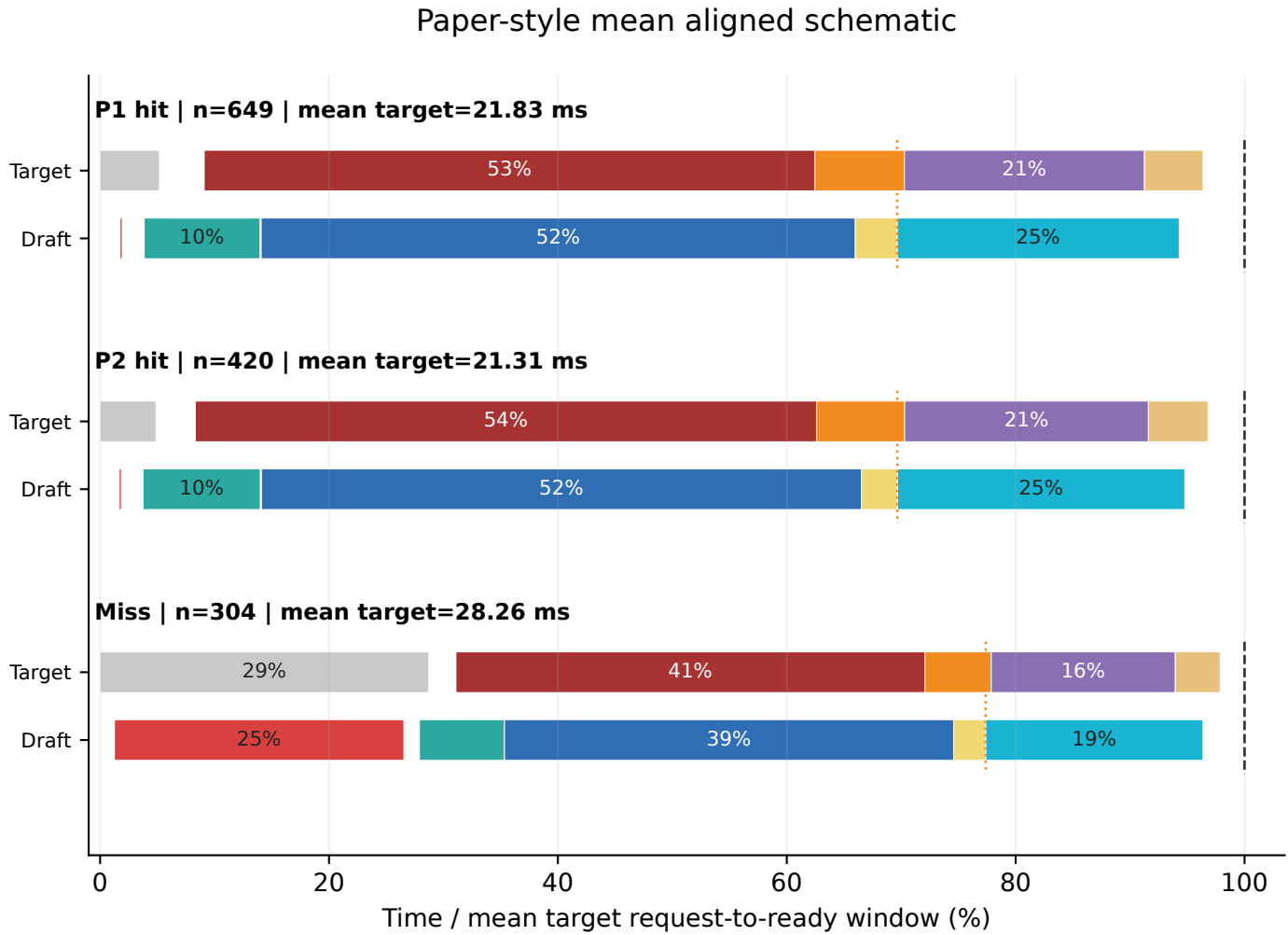
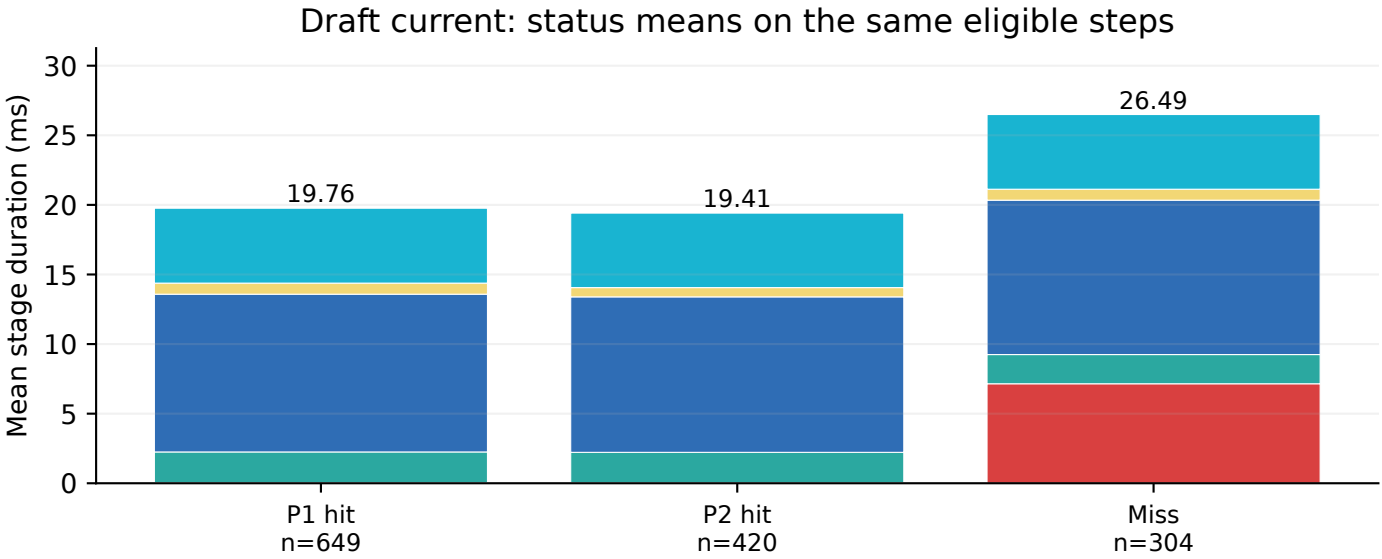
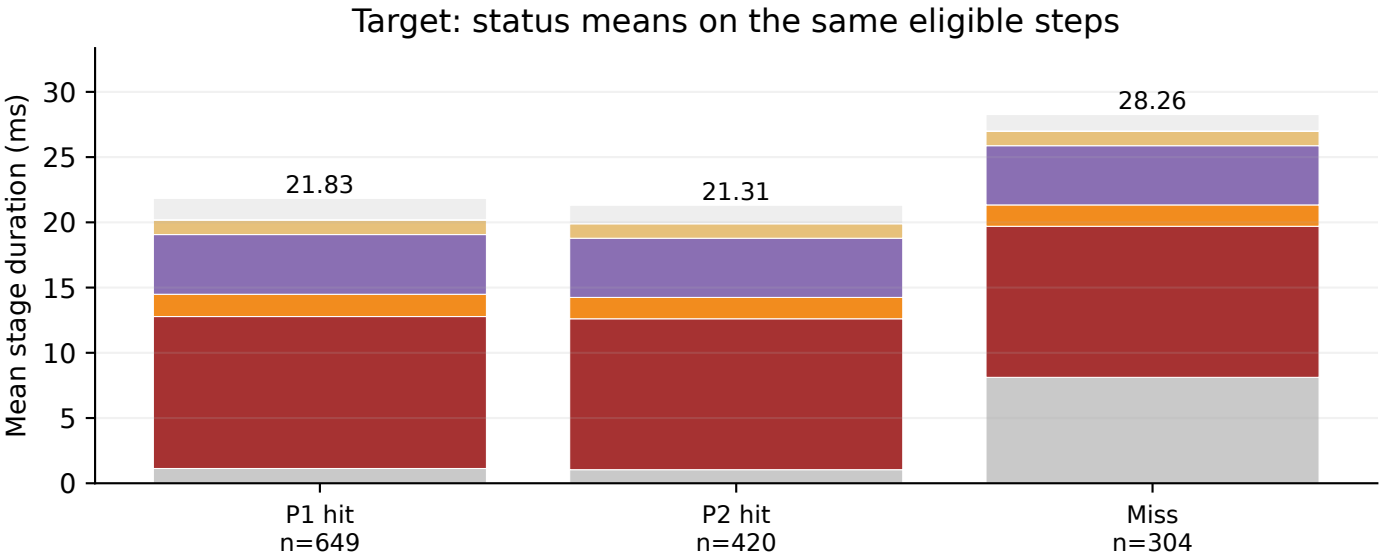
llama3_b1_tree_align_exit20
DUET: K1/K2=4/2, exit=20 (0-based), $\bar{C}$ =3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1234 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

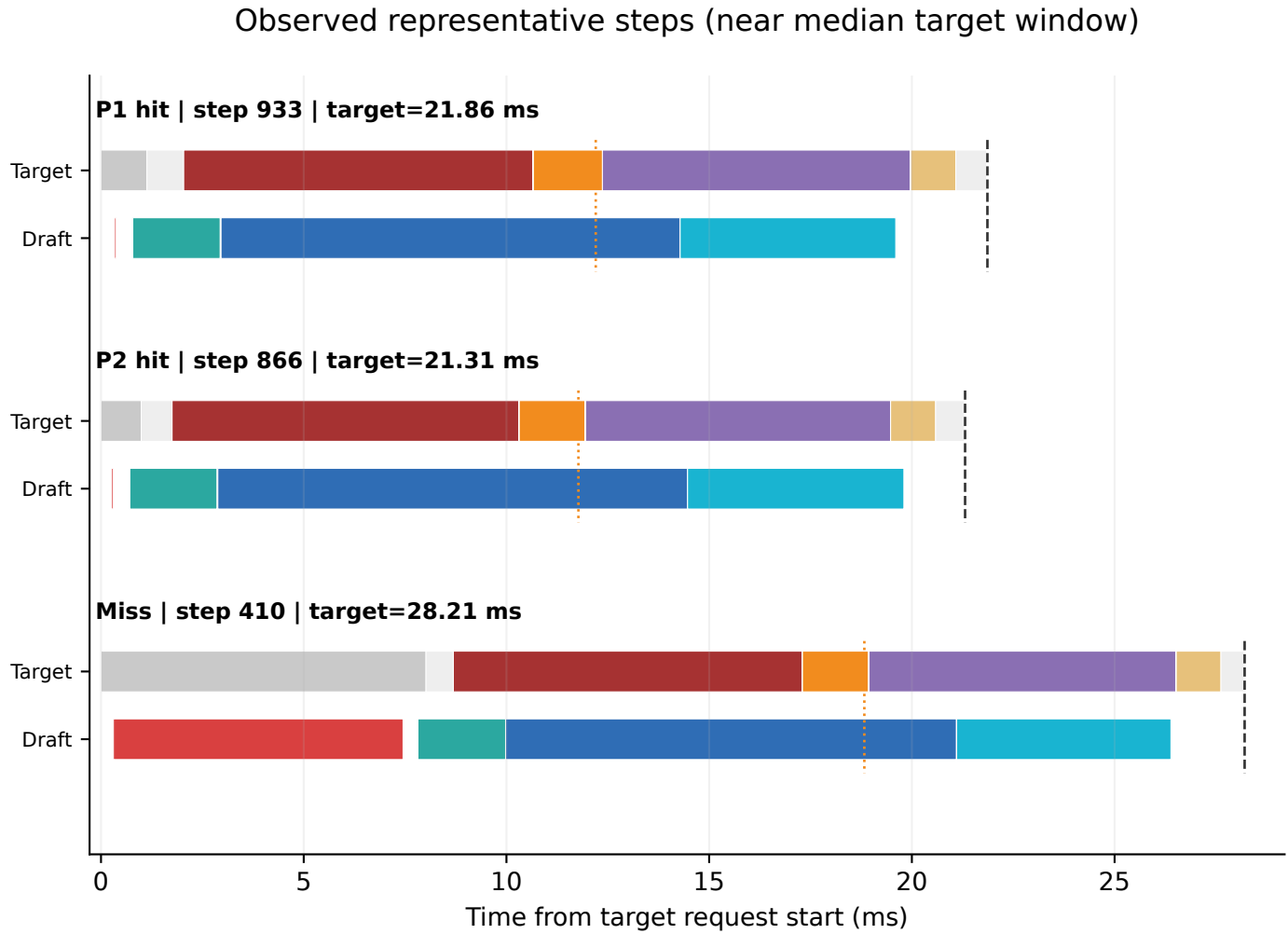
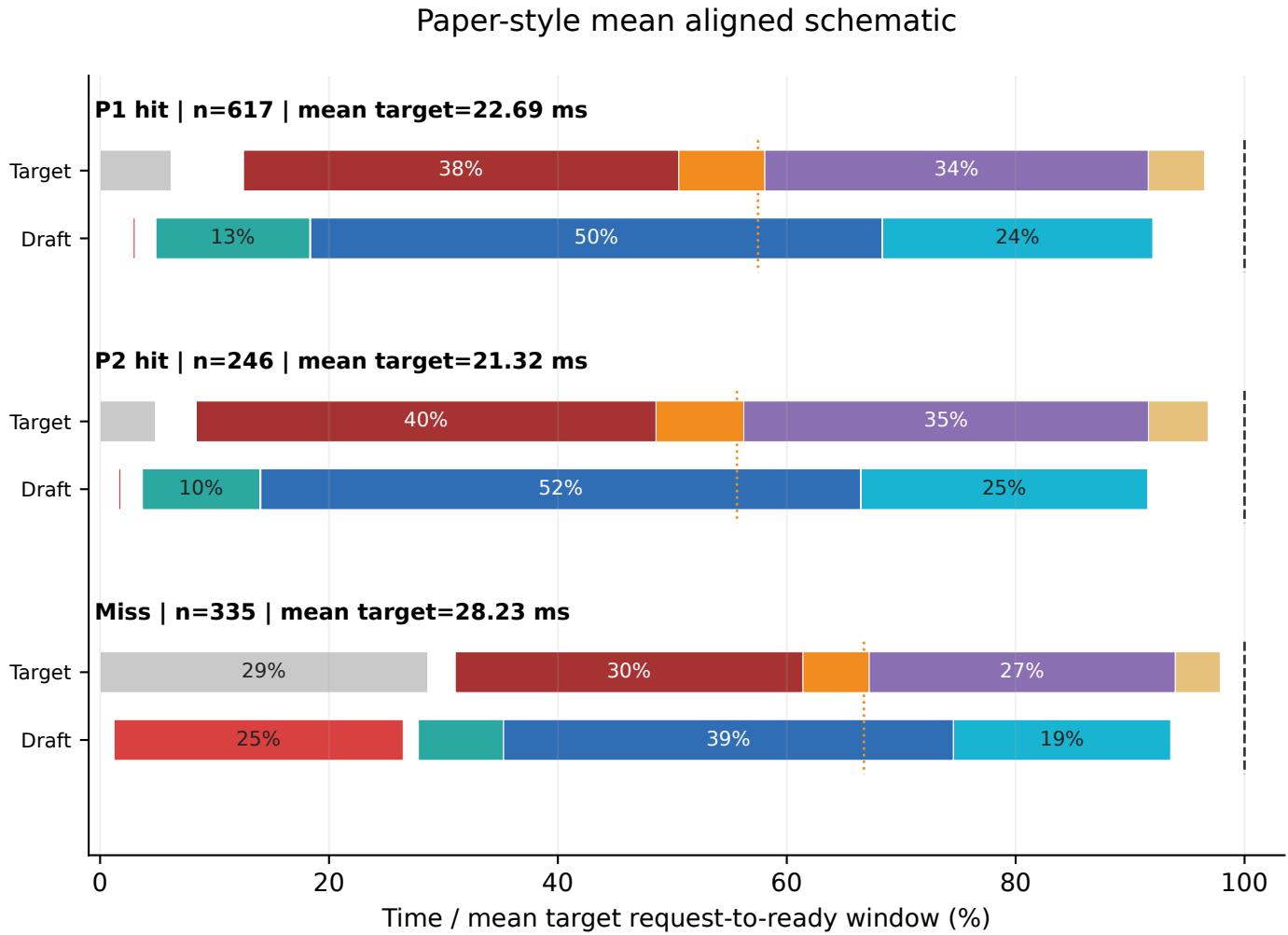
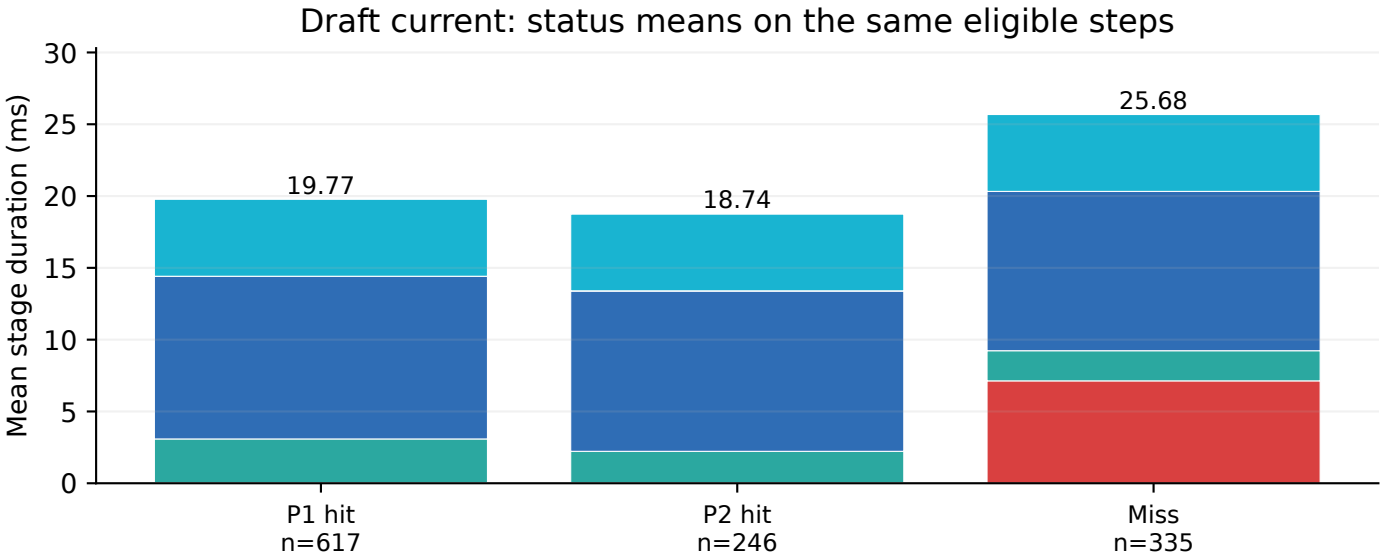
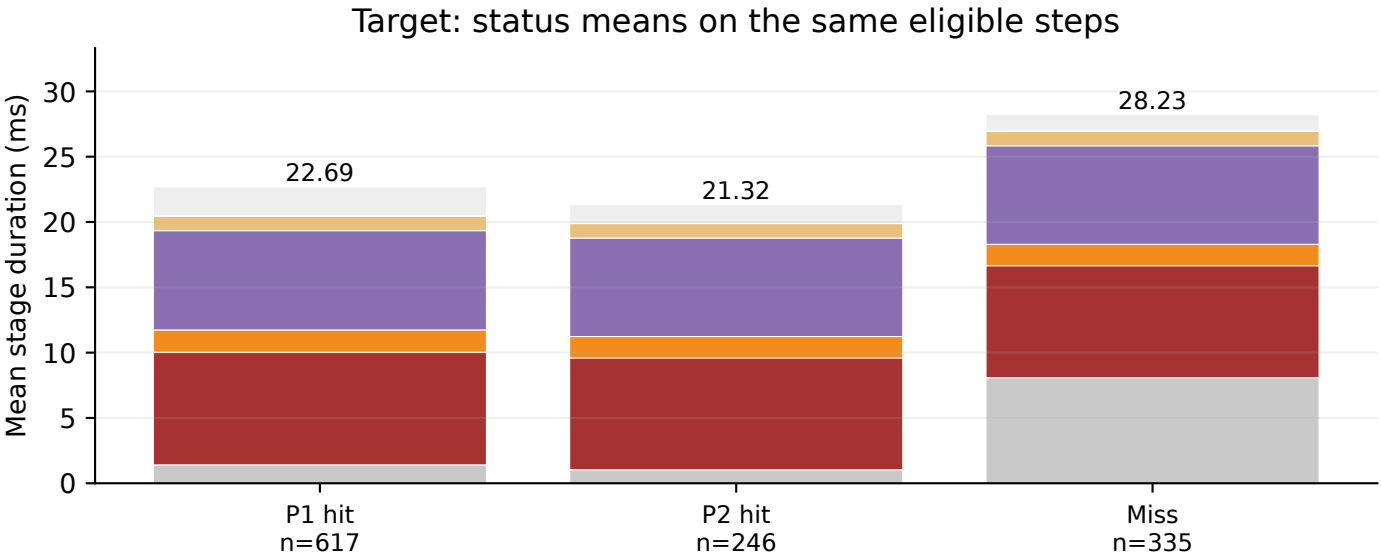
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_align_exit22
DUET: K1/K2=4/2, exit=22 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1373 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

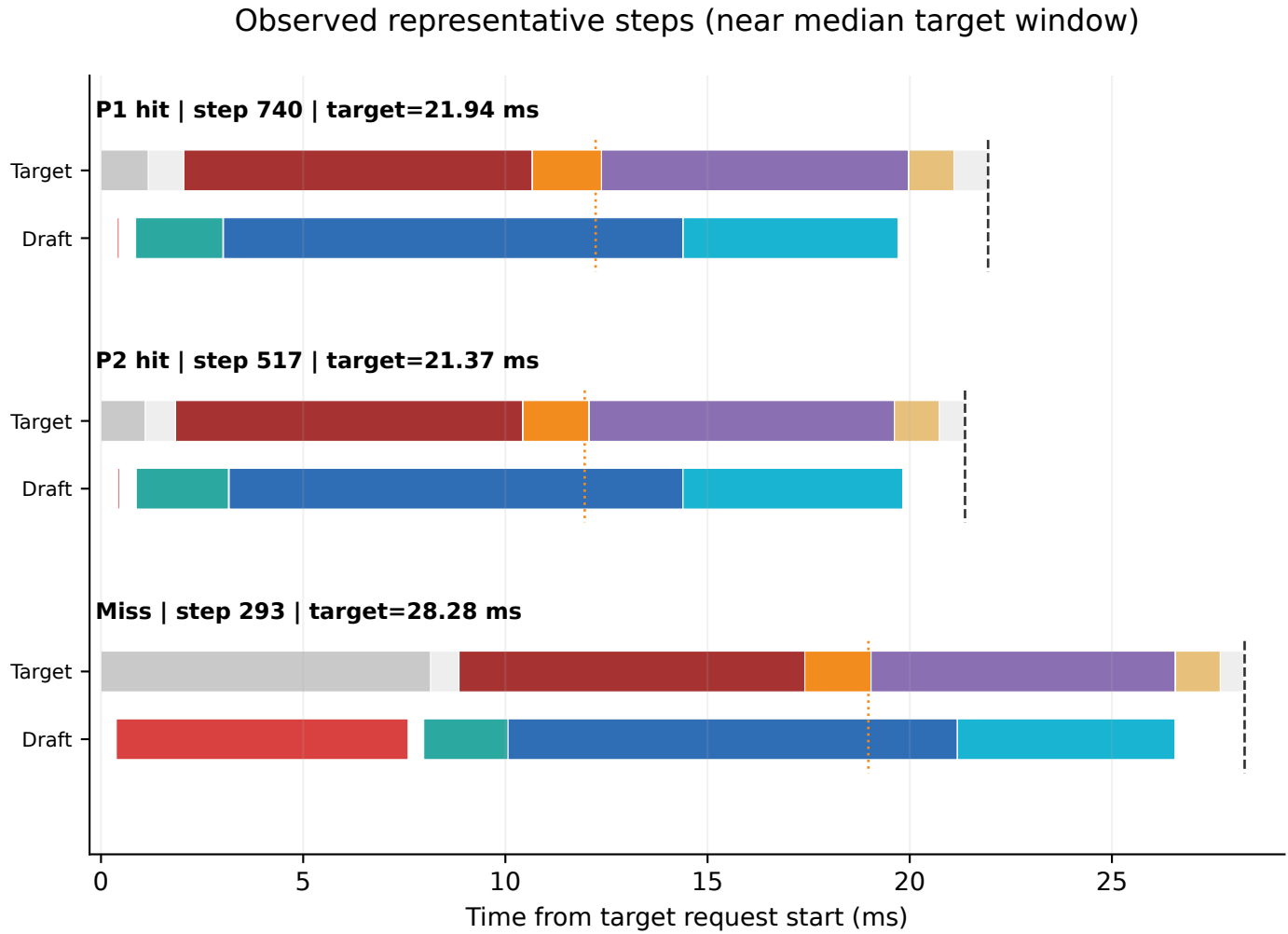
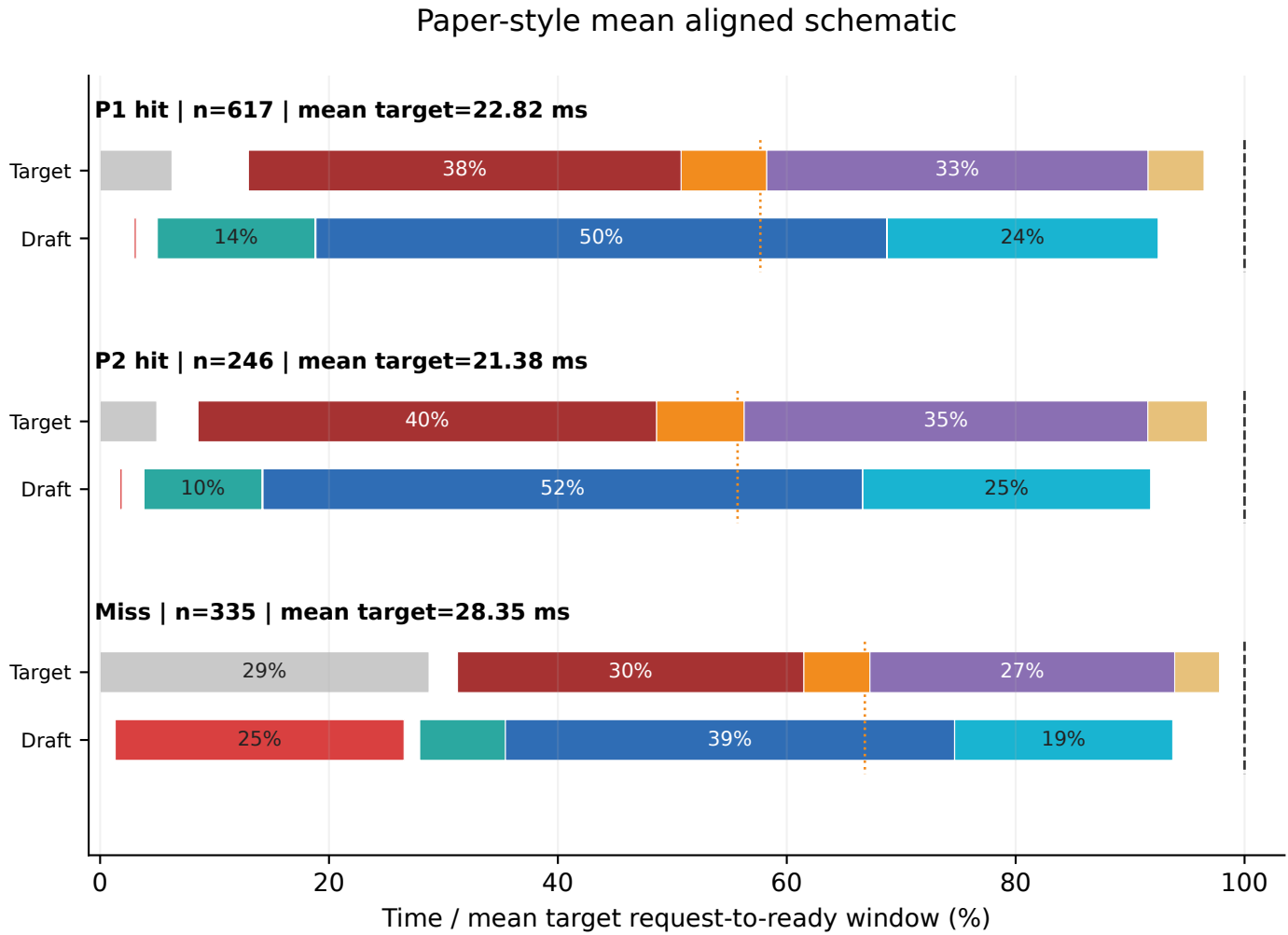
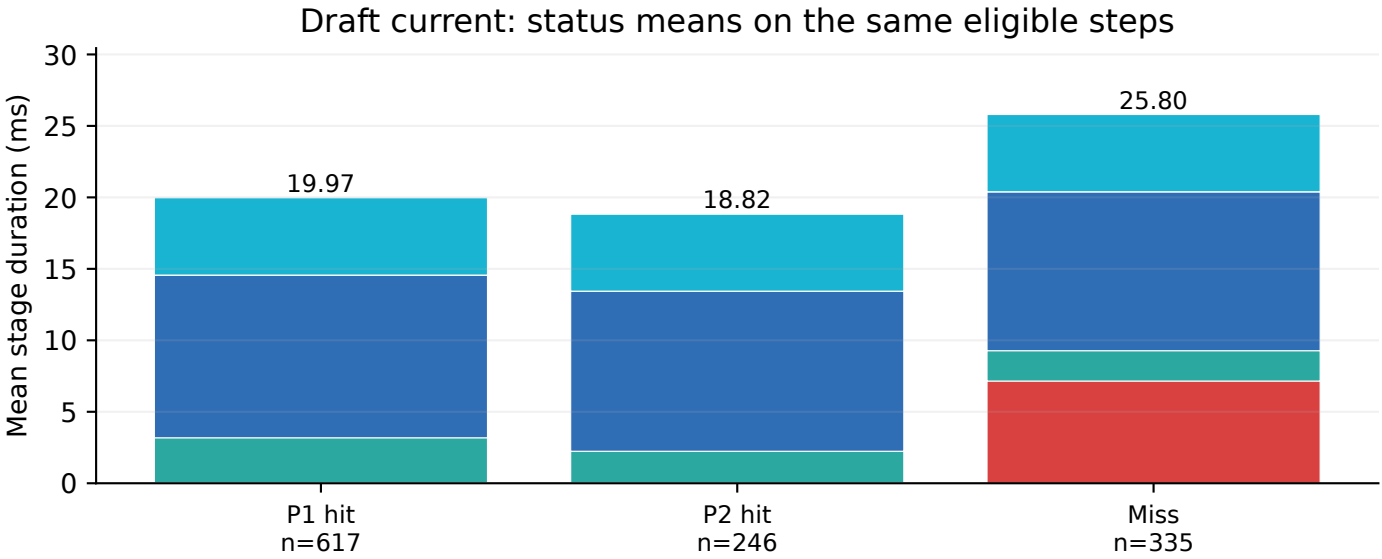
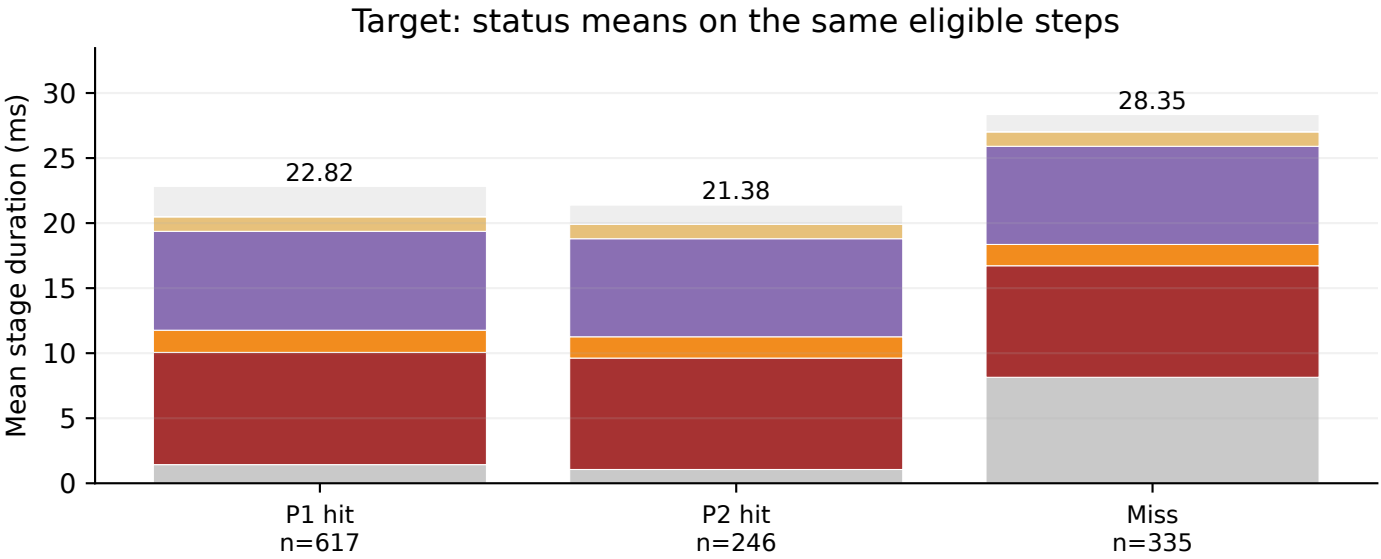
llama3_b1_tree_beta0
DUET: K1/K2=4/2, exit=16 (0-based), $\bar{C}=3$, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1198 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

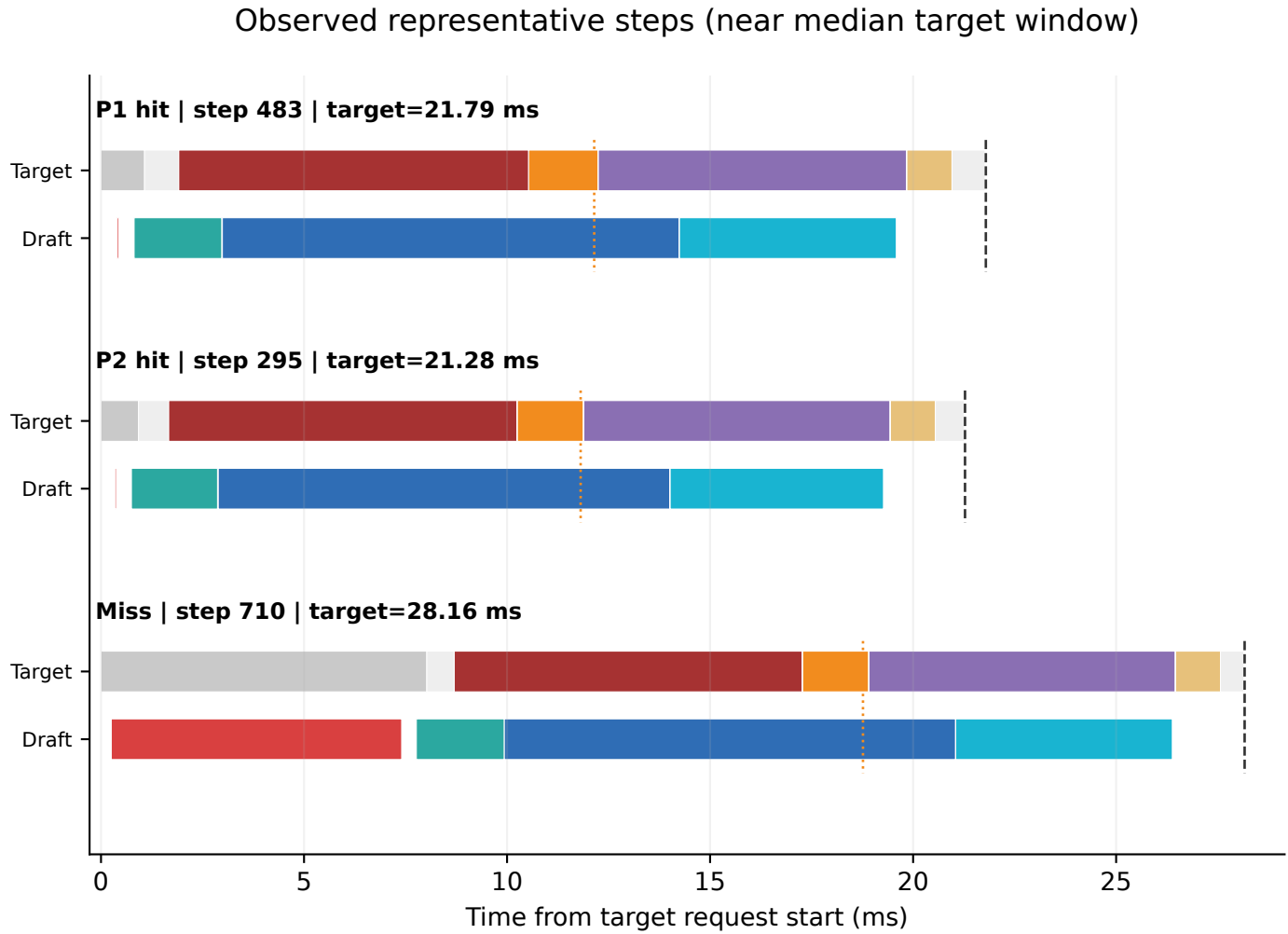
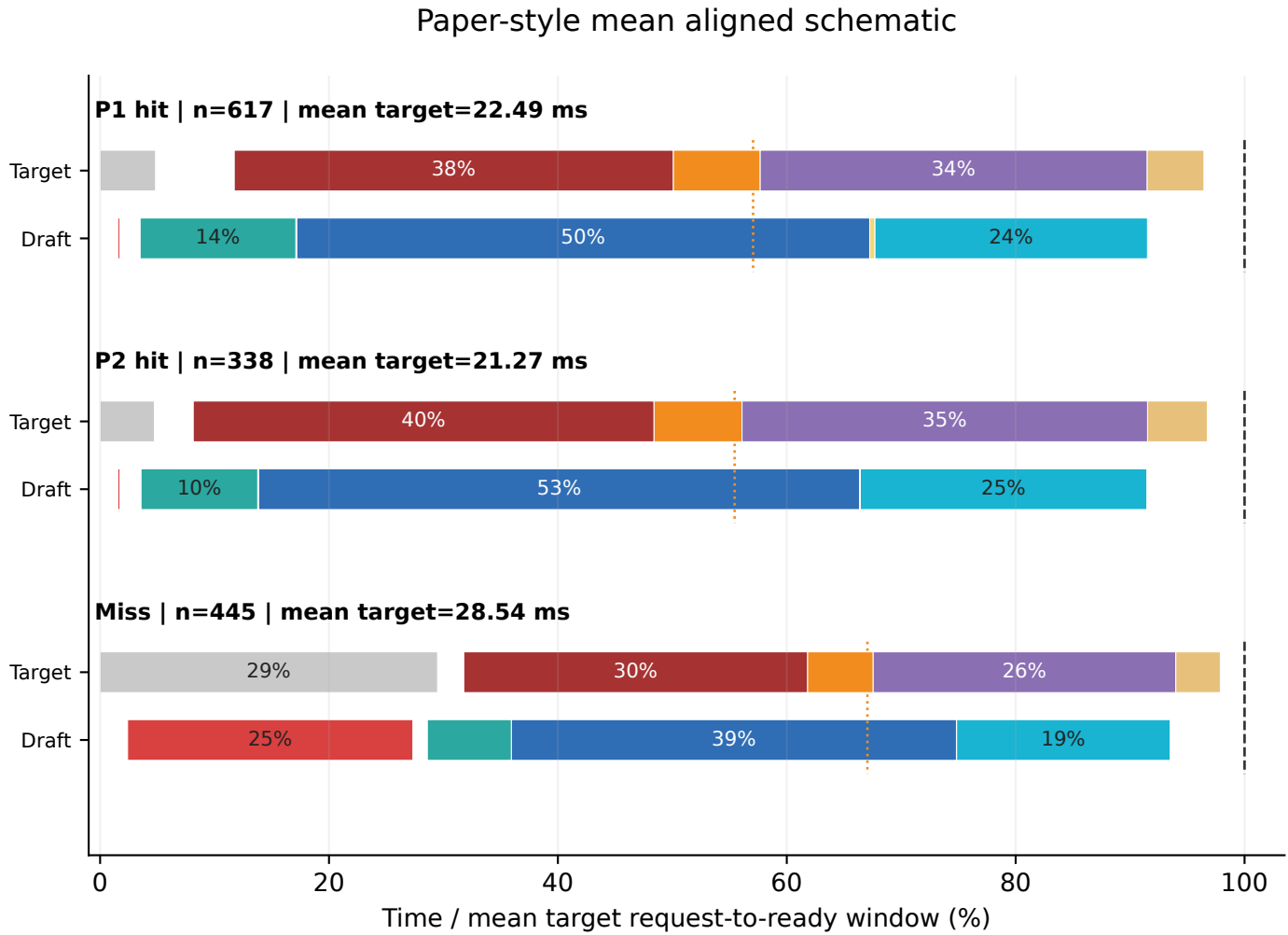
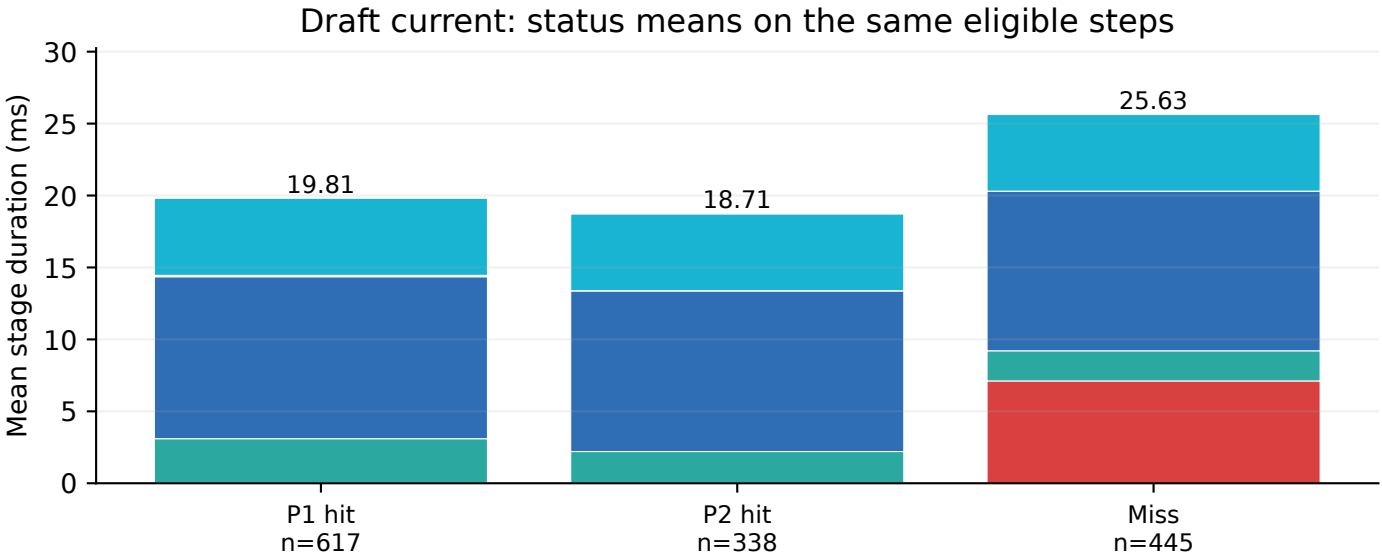
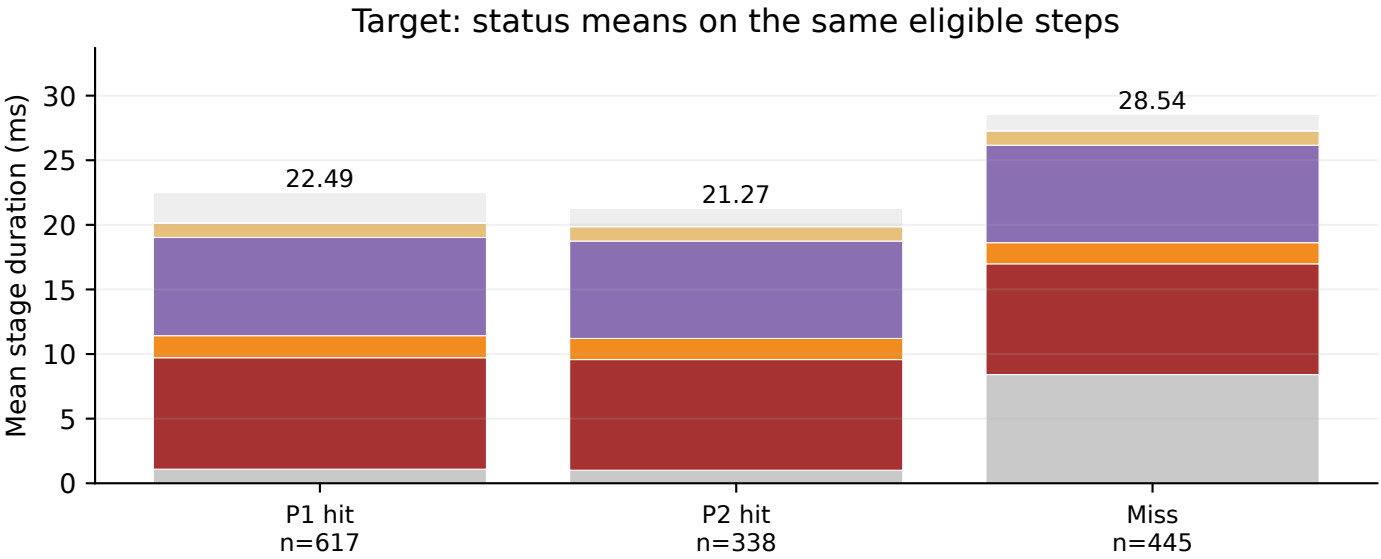
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_beta1
DUET: K1/K2=4/2, exit=16 (0-based), $\bar{C}=3$, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1198 eligible steps



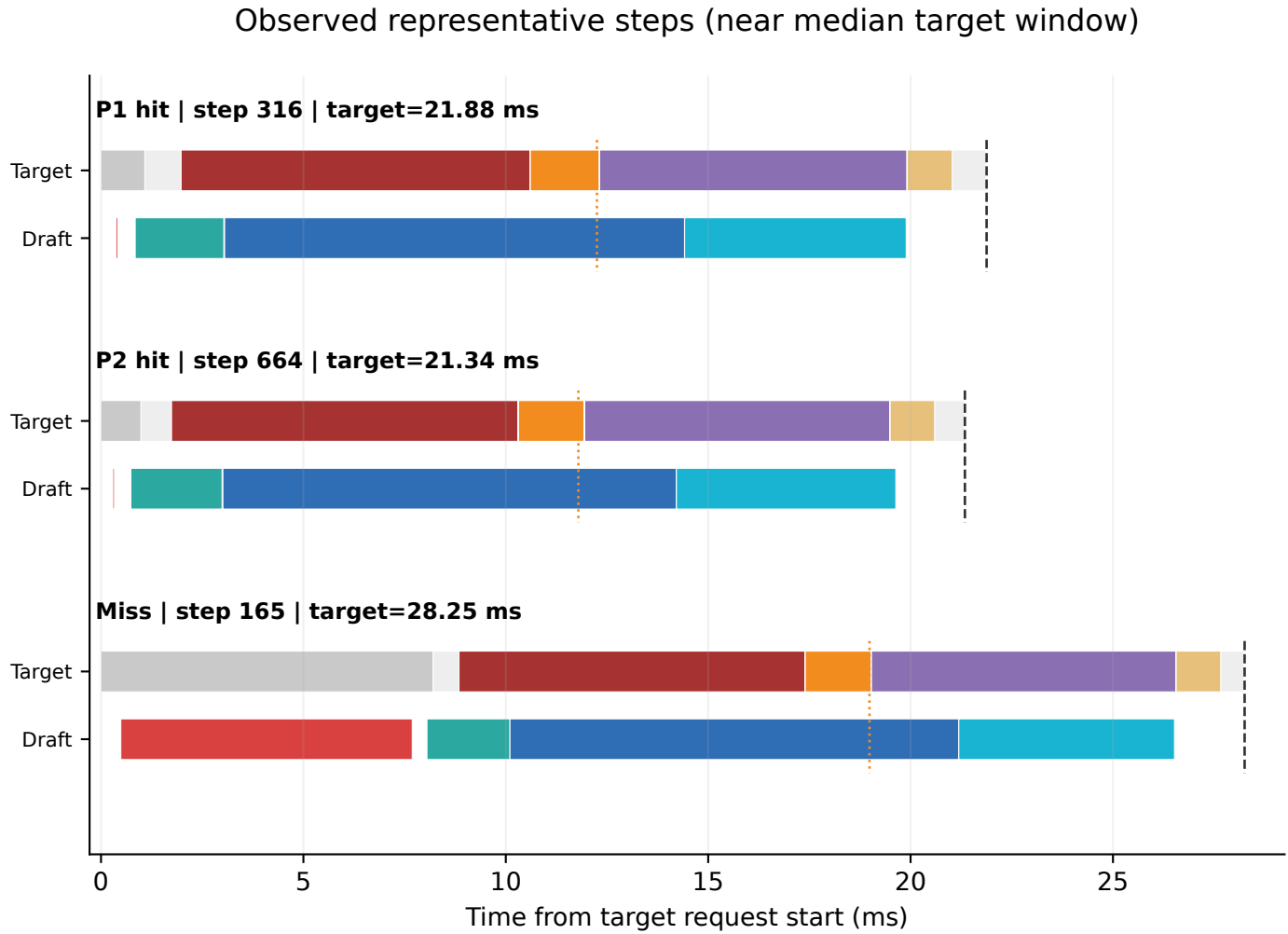
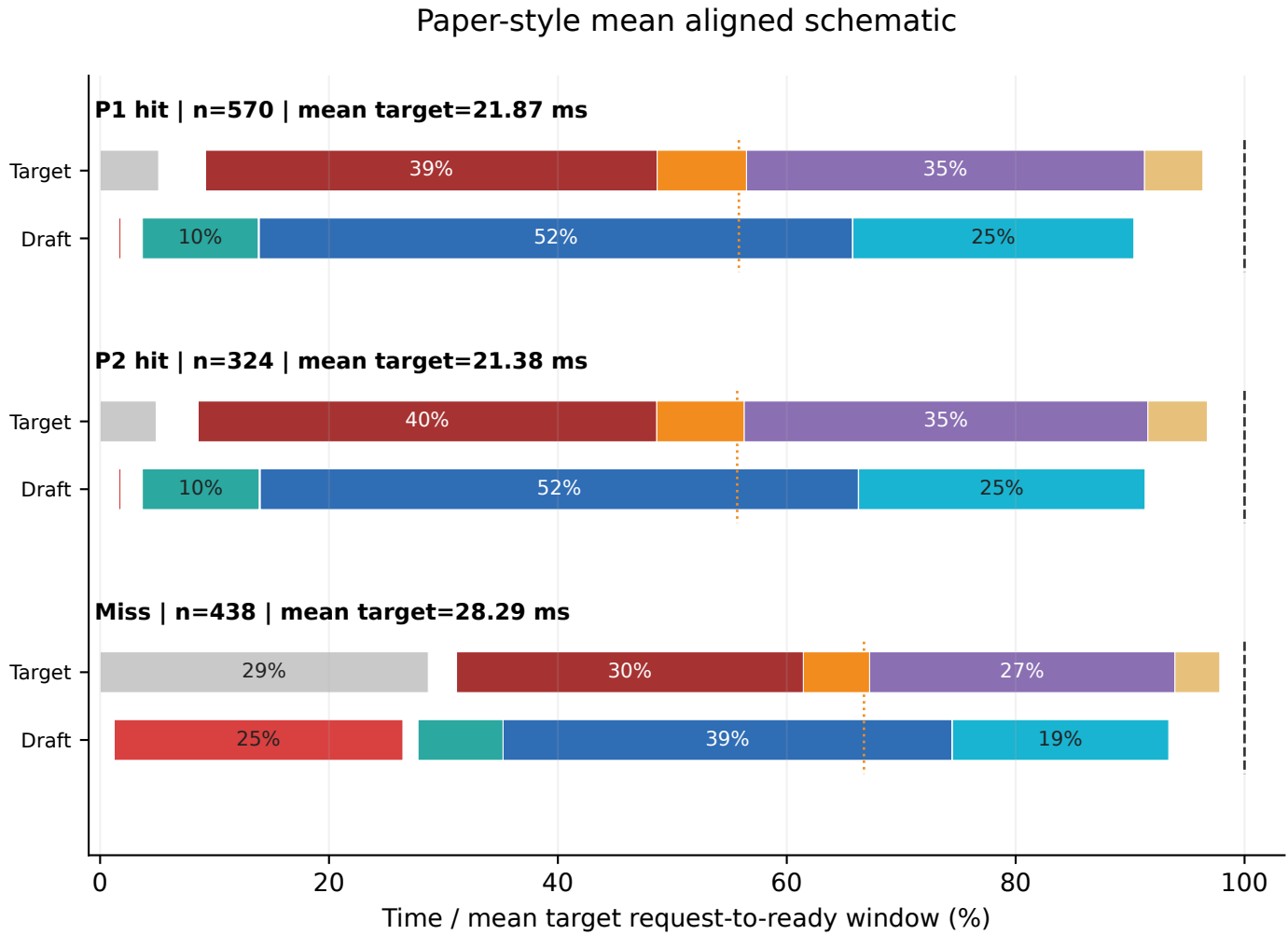
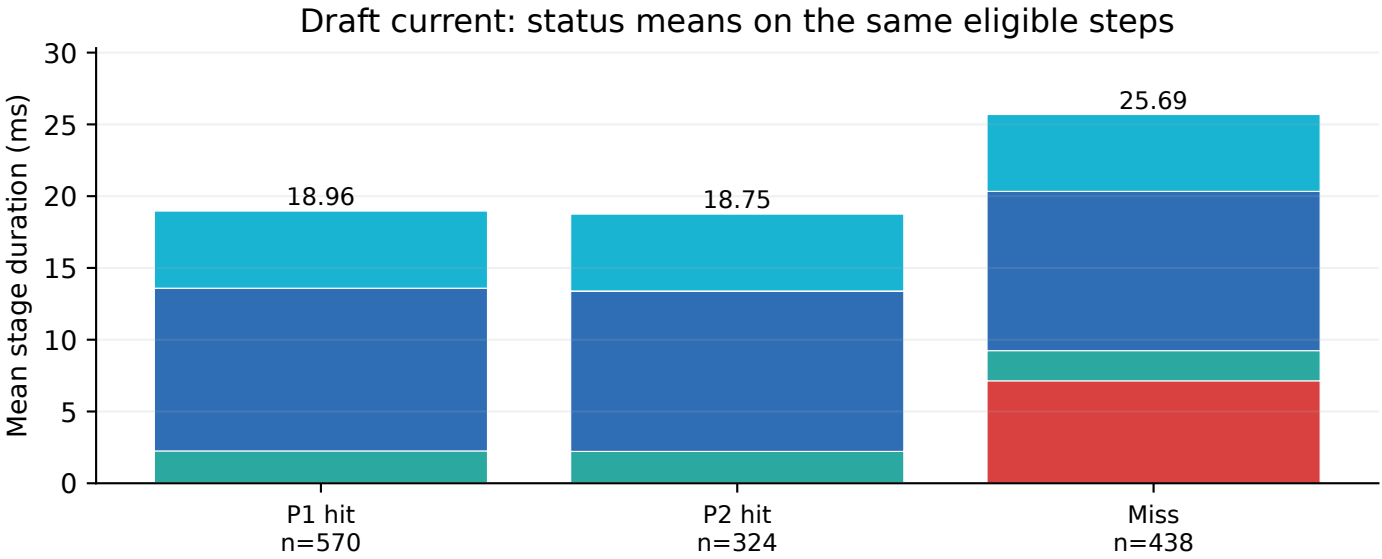
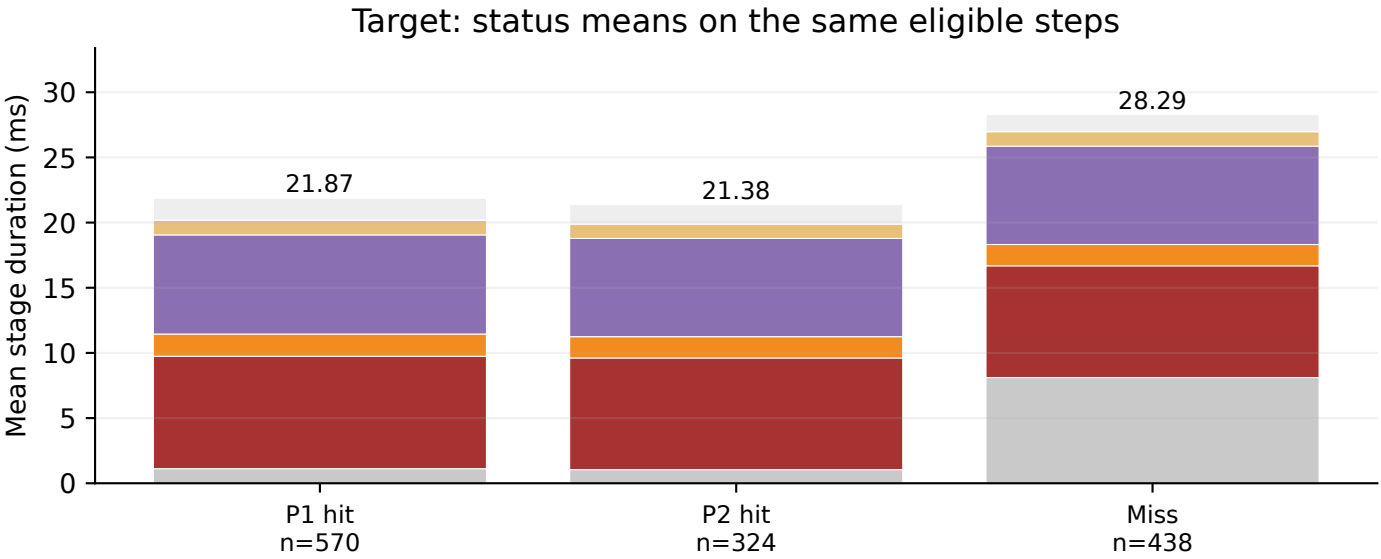
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_compute3
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1400 eligible steps



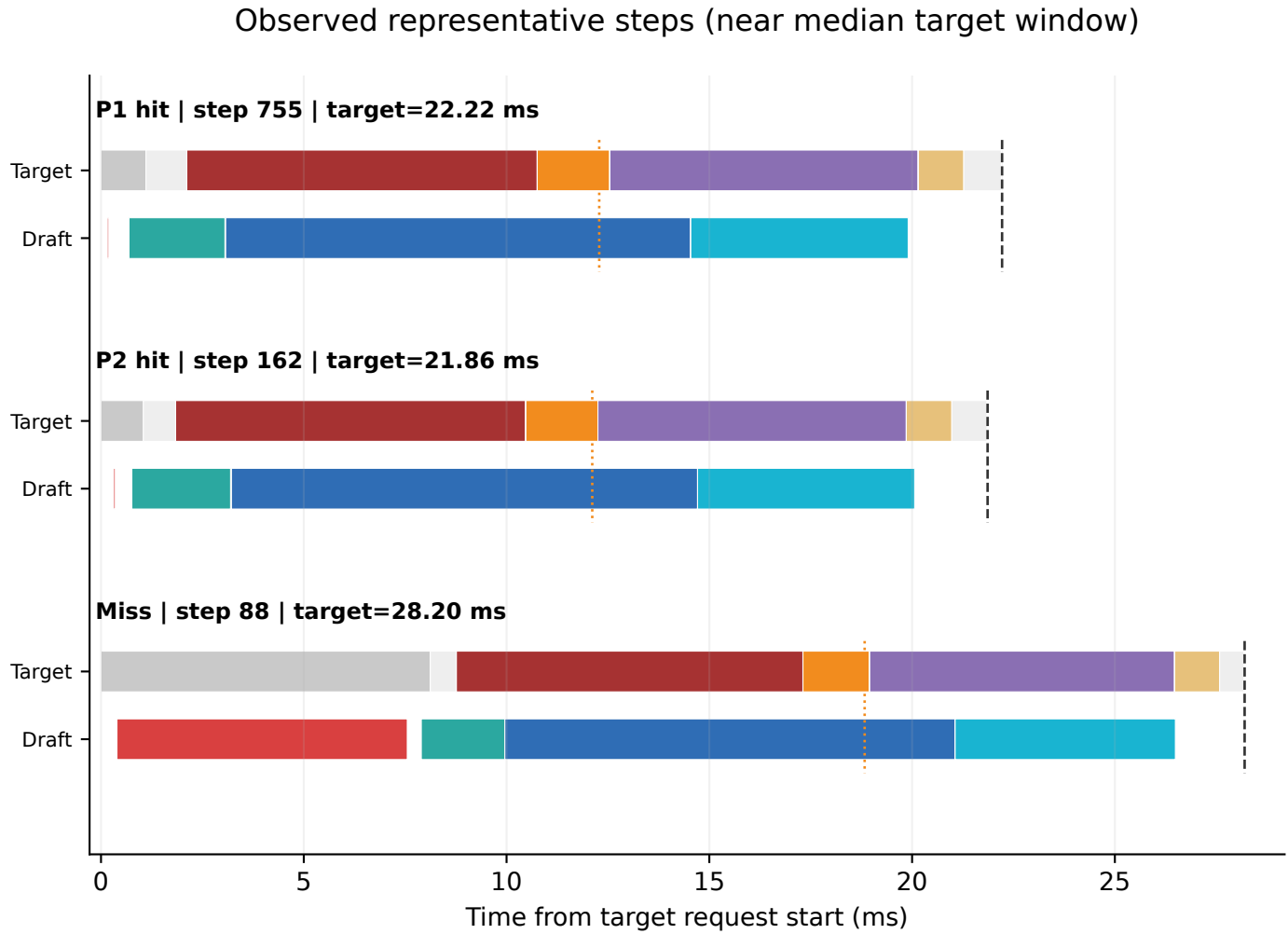
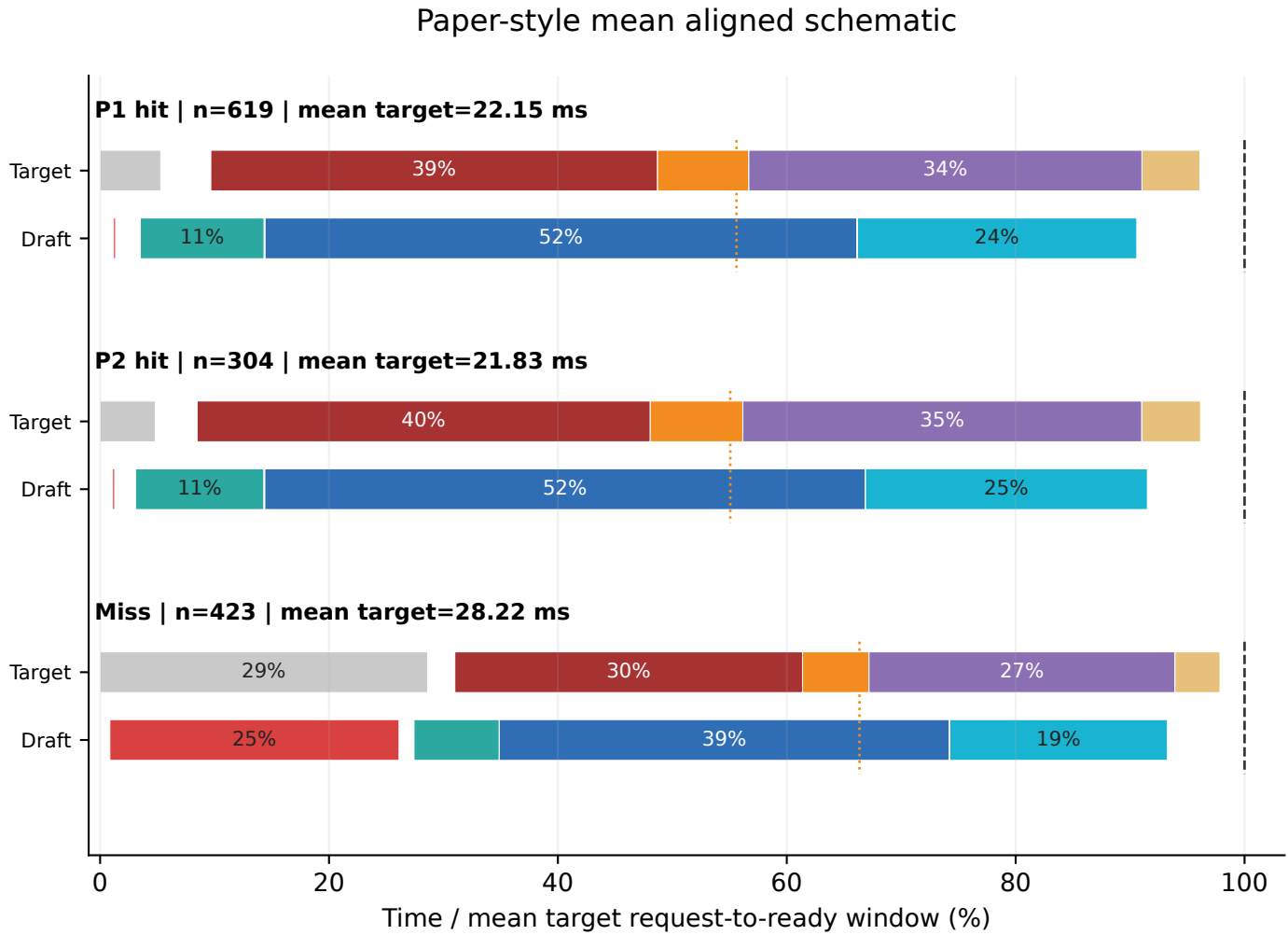
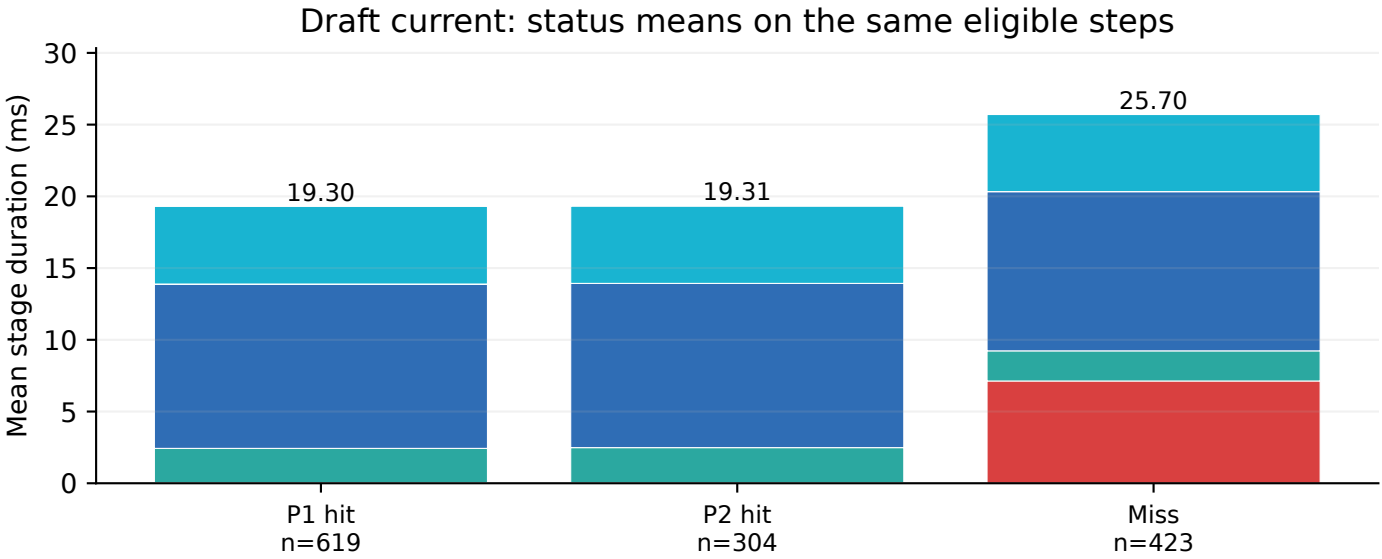
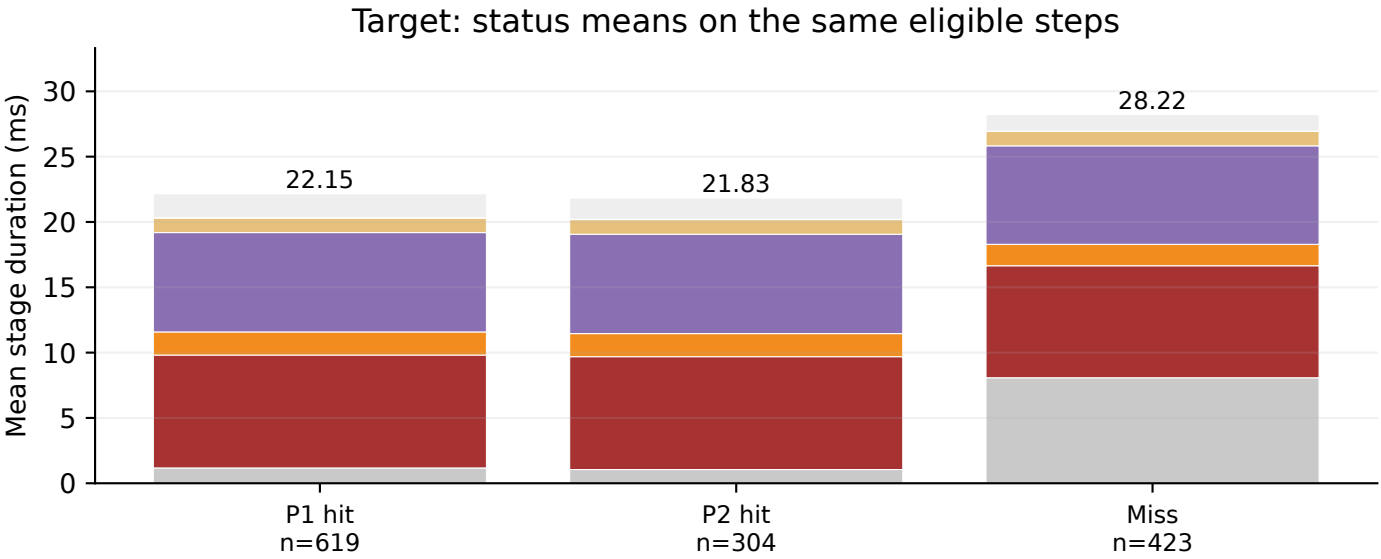
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_no_floors
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1332 eligible steps



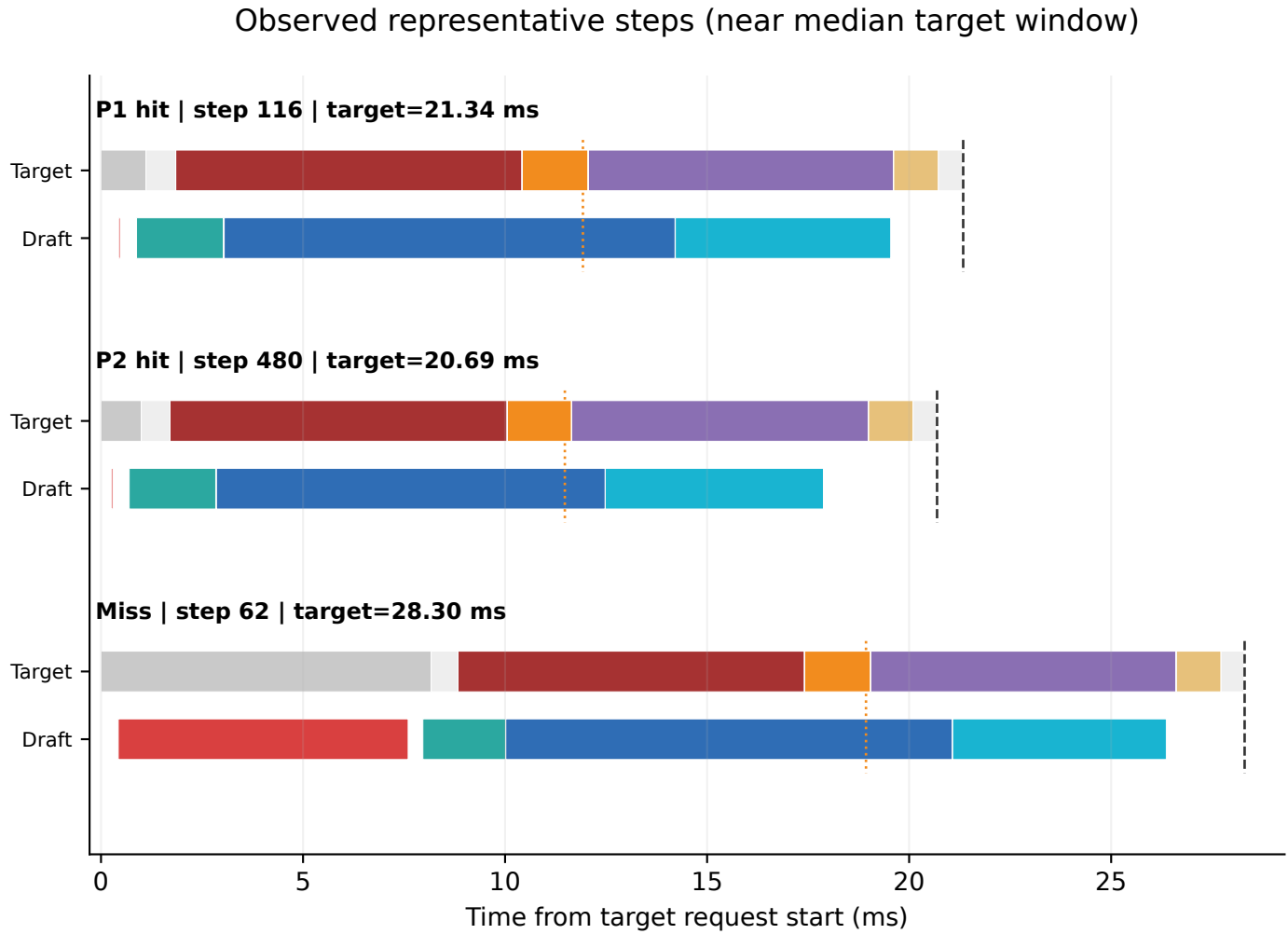
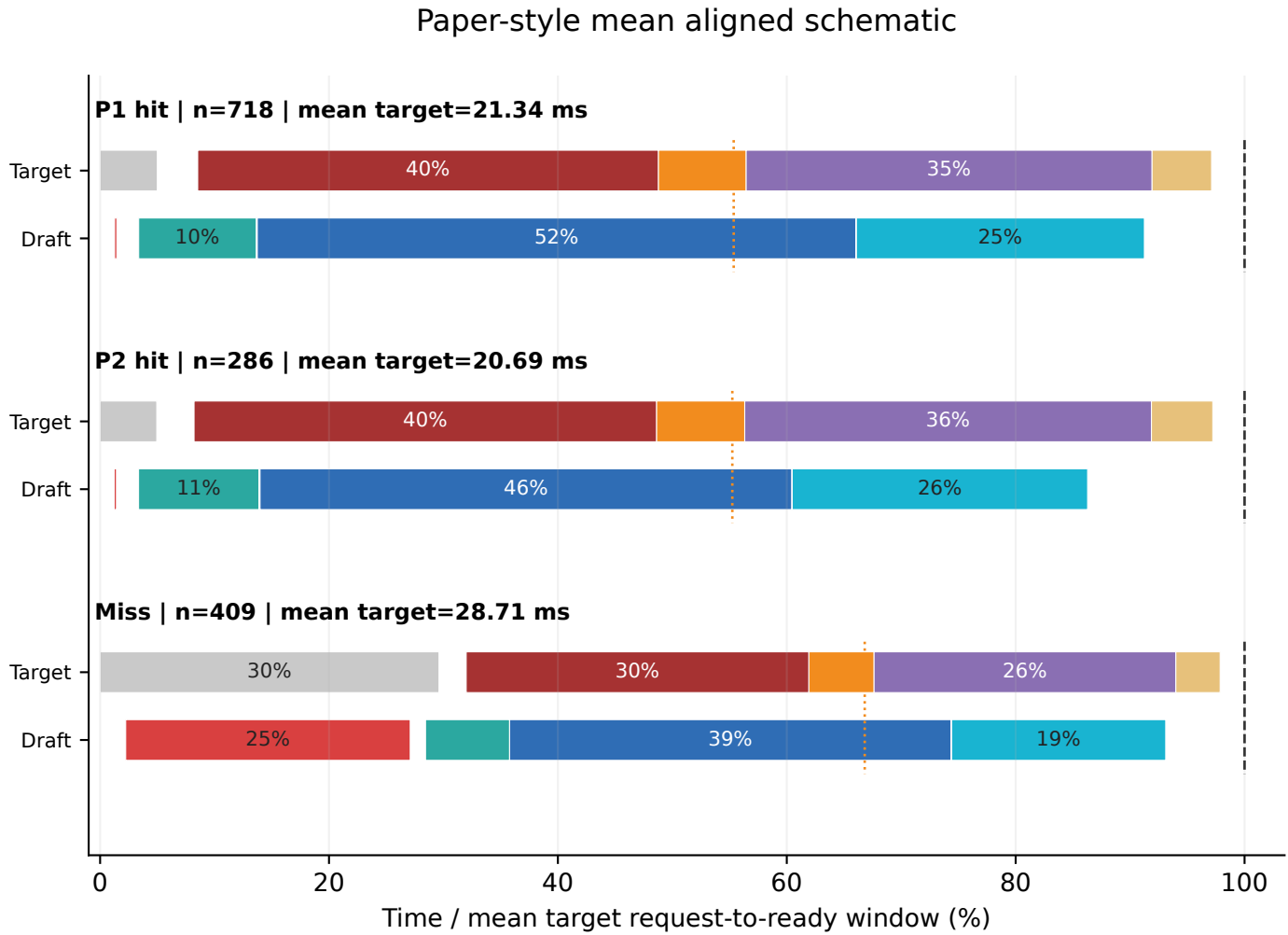
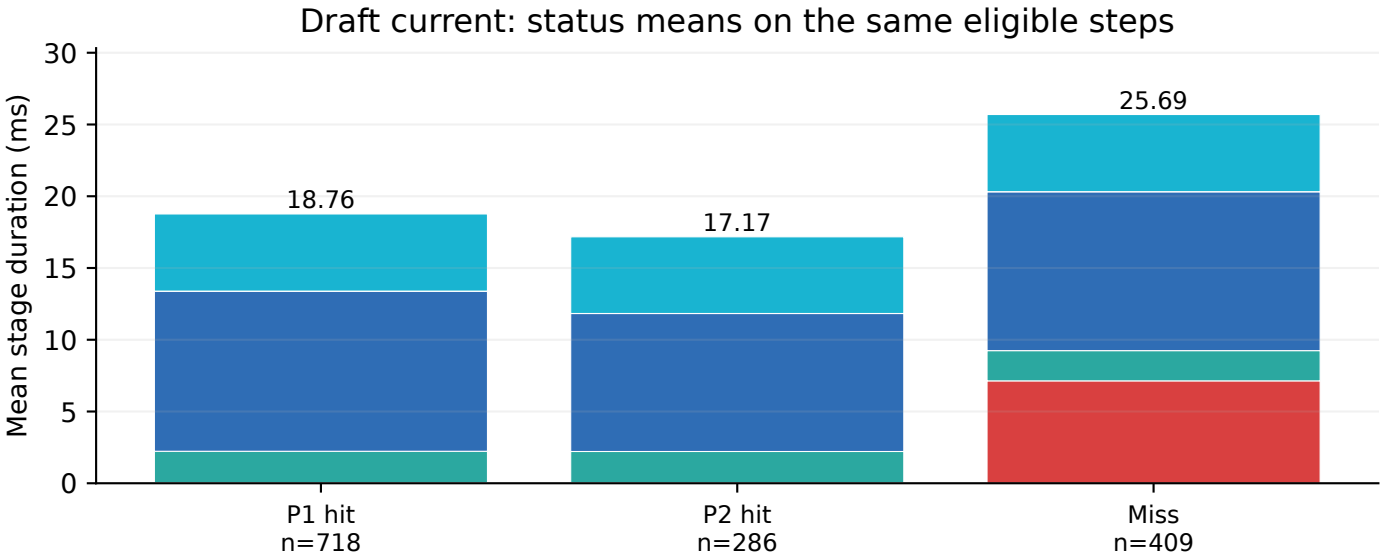
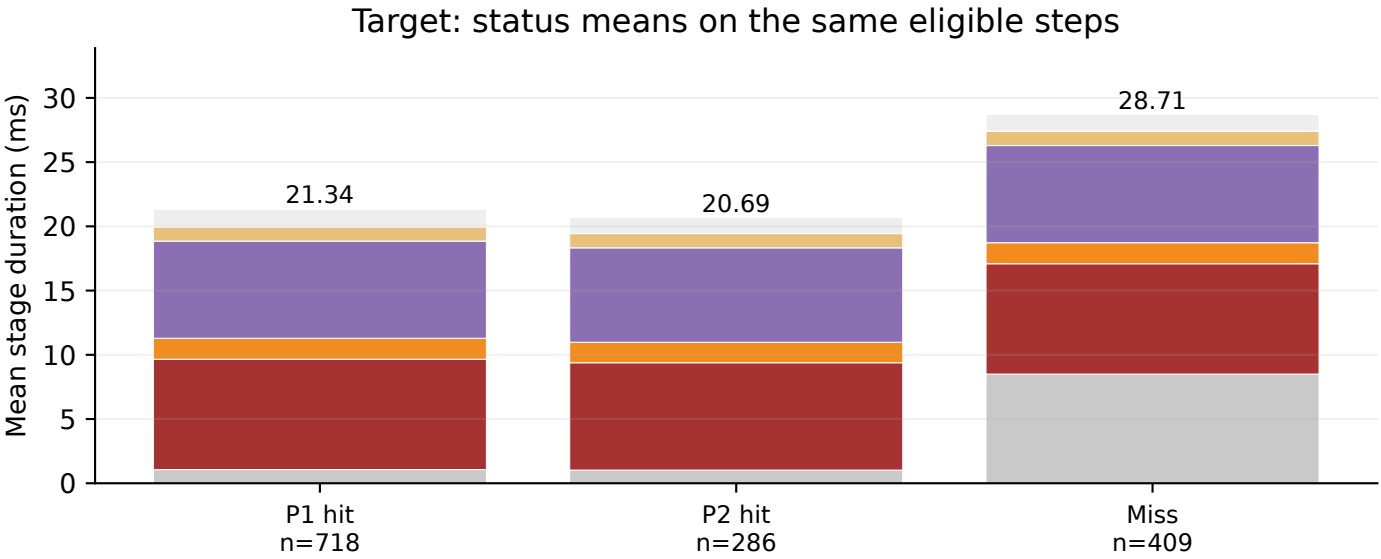
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_nodes_high
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=12/6, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1346 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

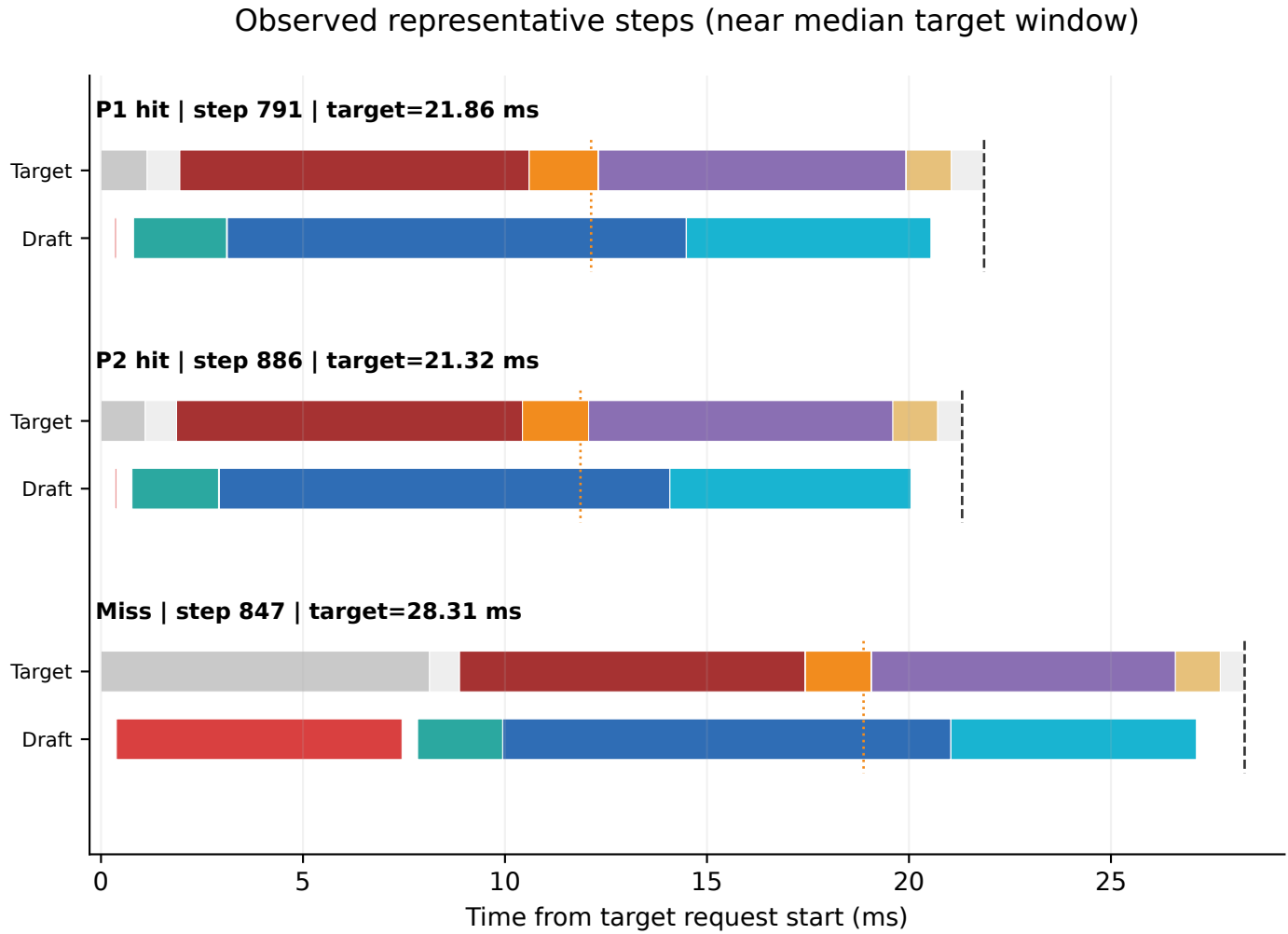
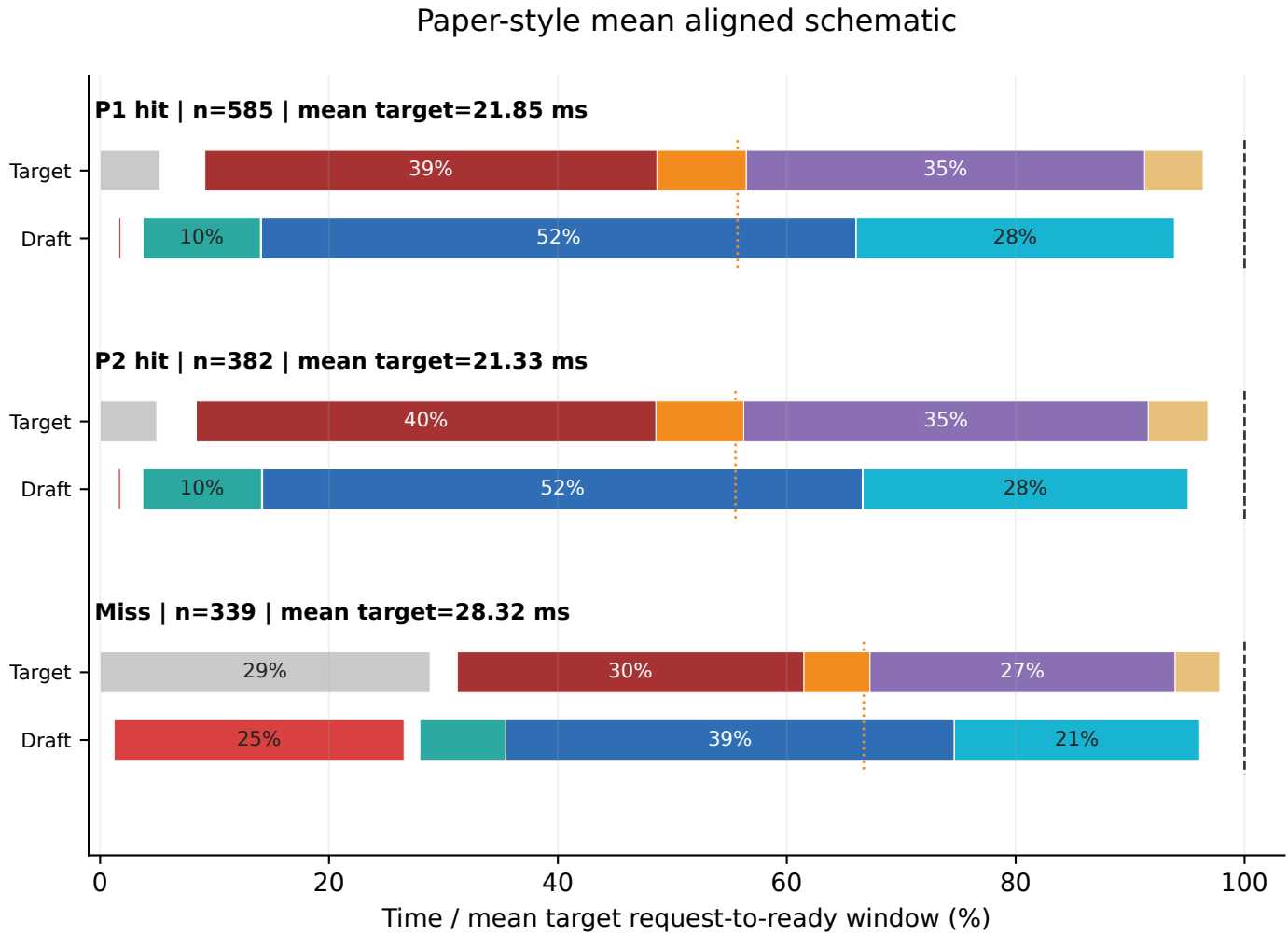
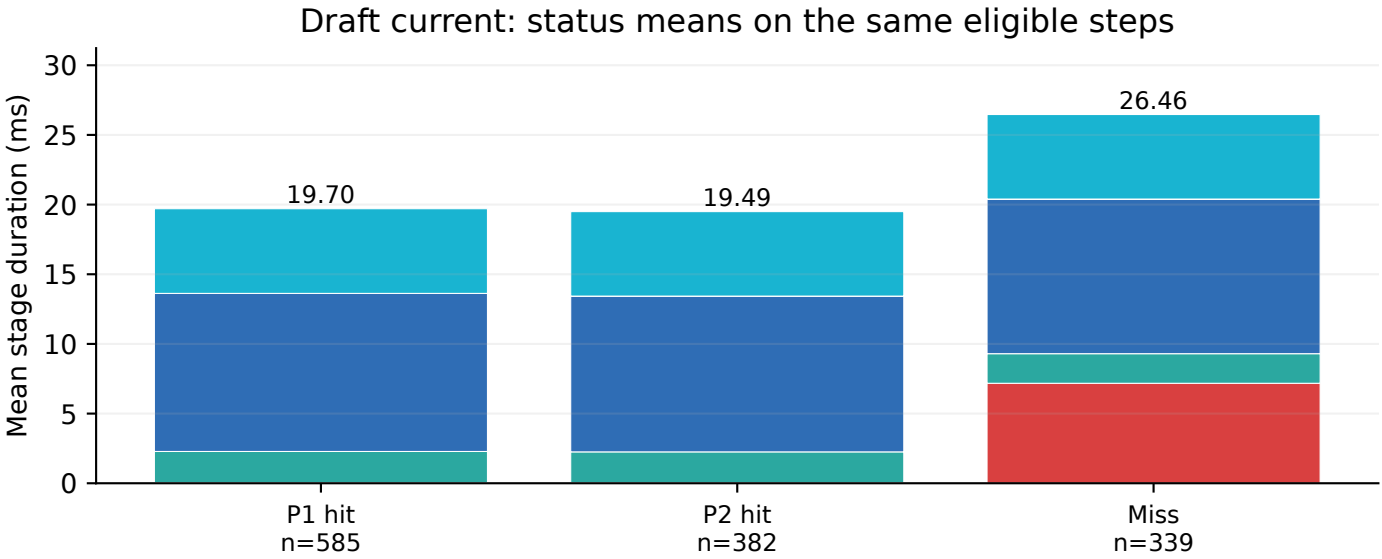
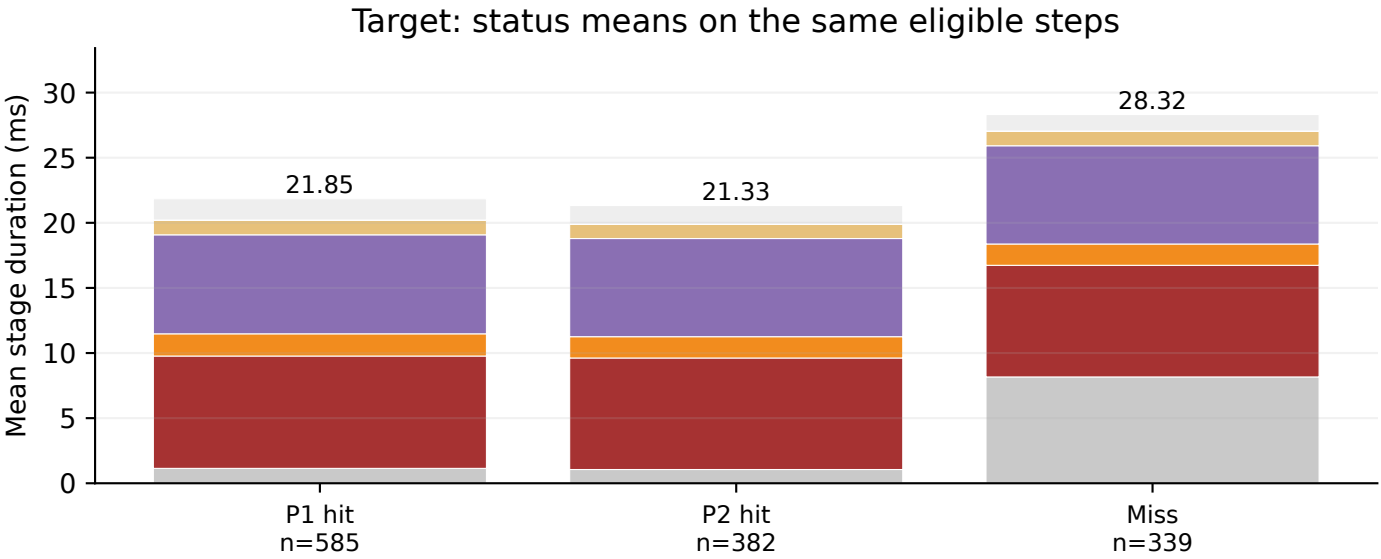
llama3_b1_tree_nodes_low
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=4/2, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1413 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

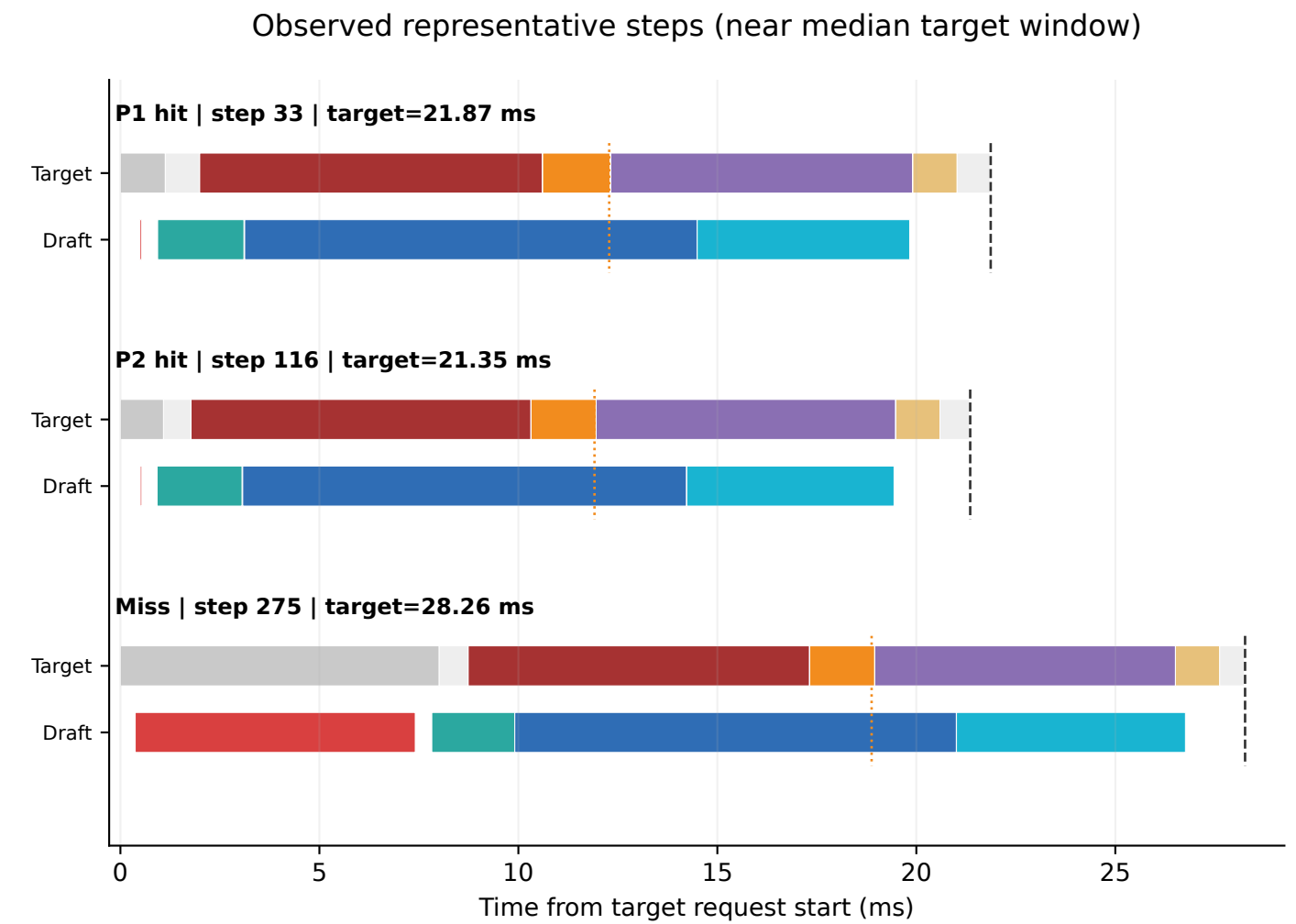
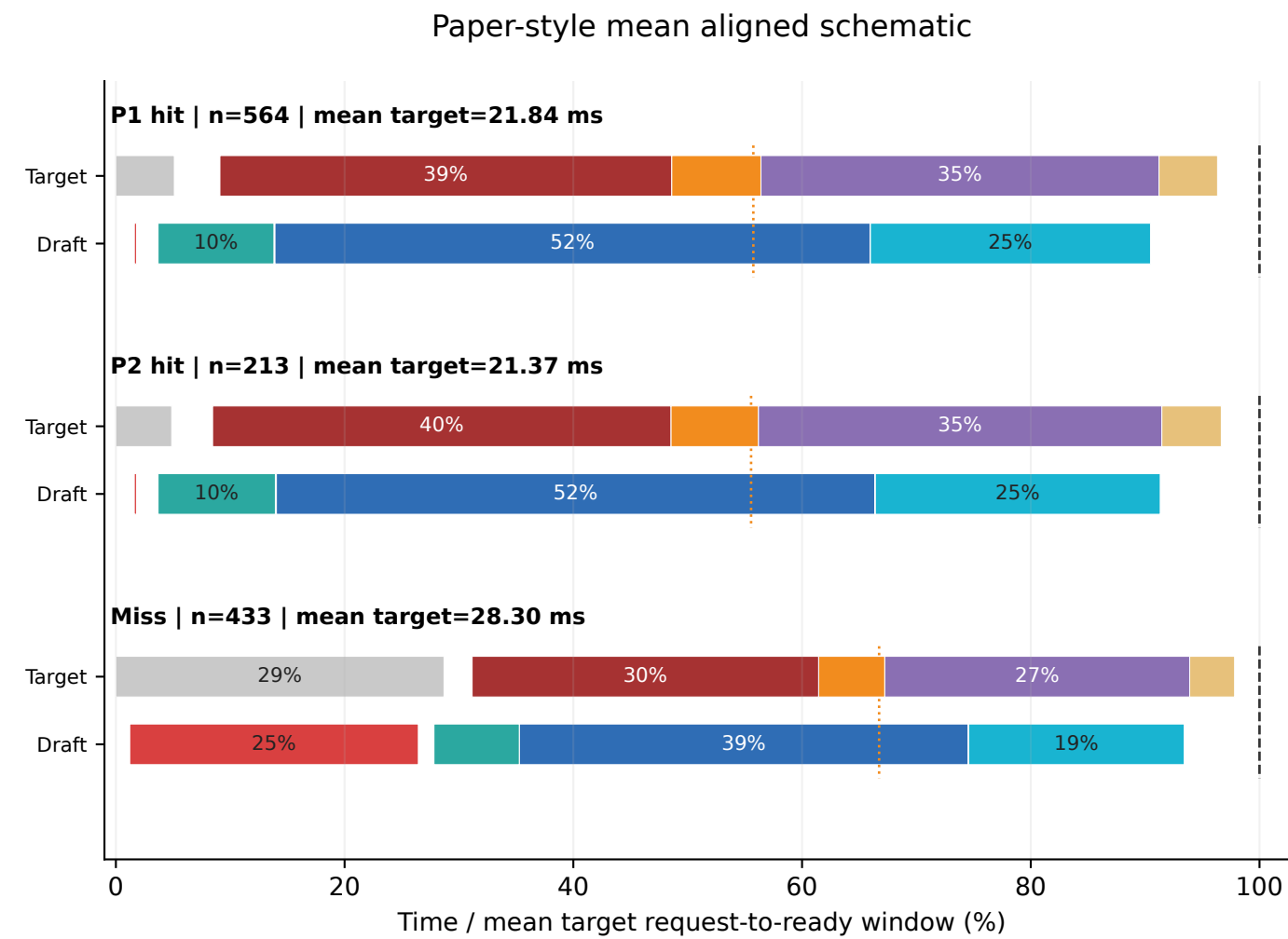
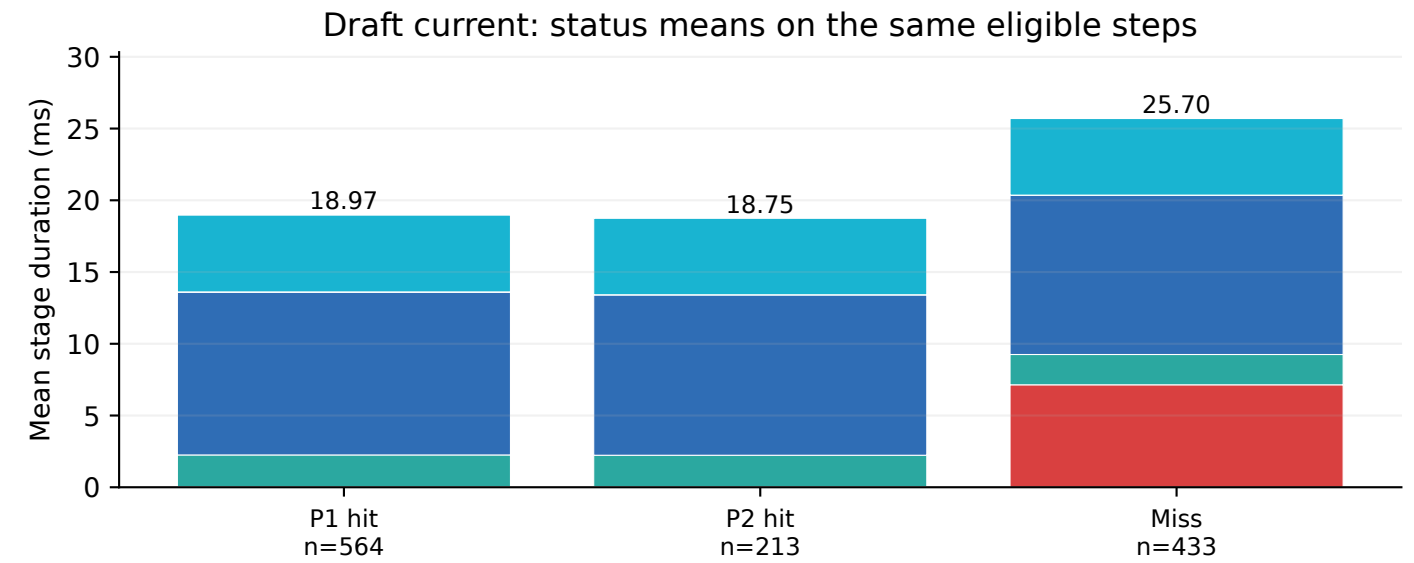
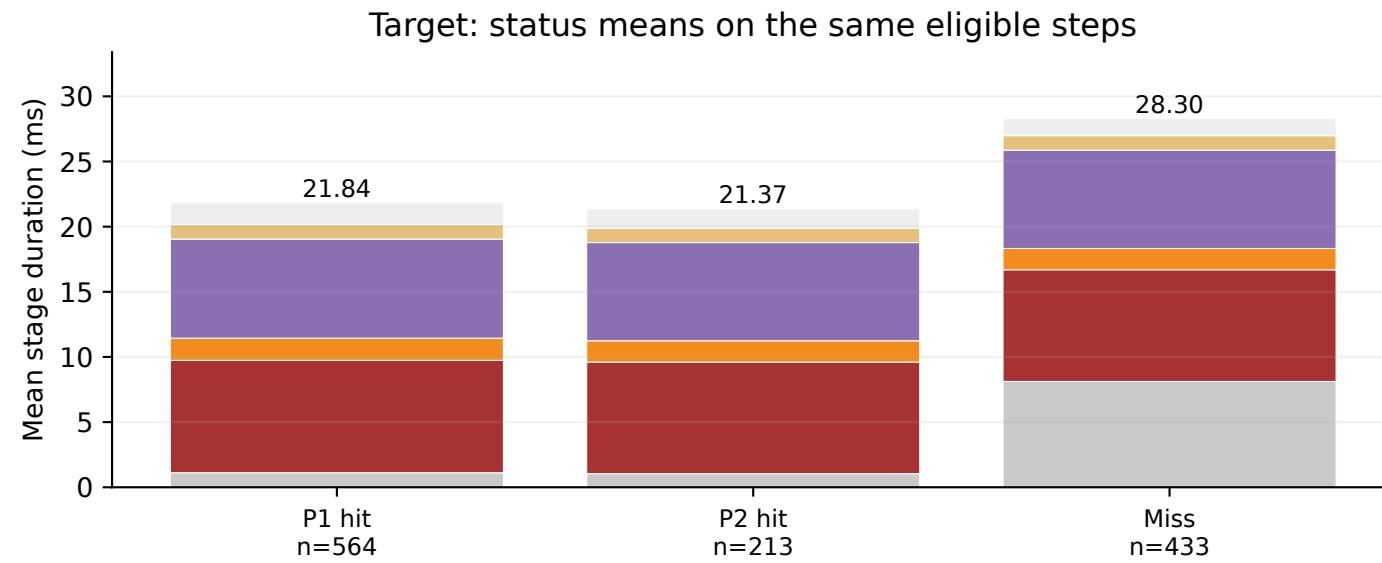
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_p2budget12
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=12, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1306 eligible steps



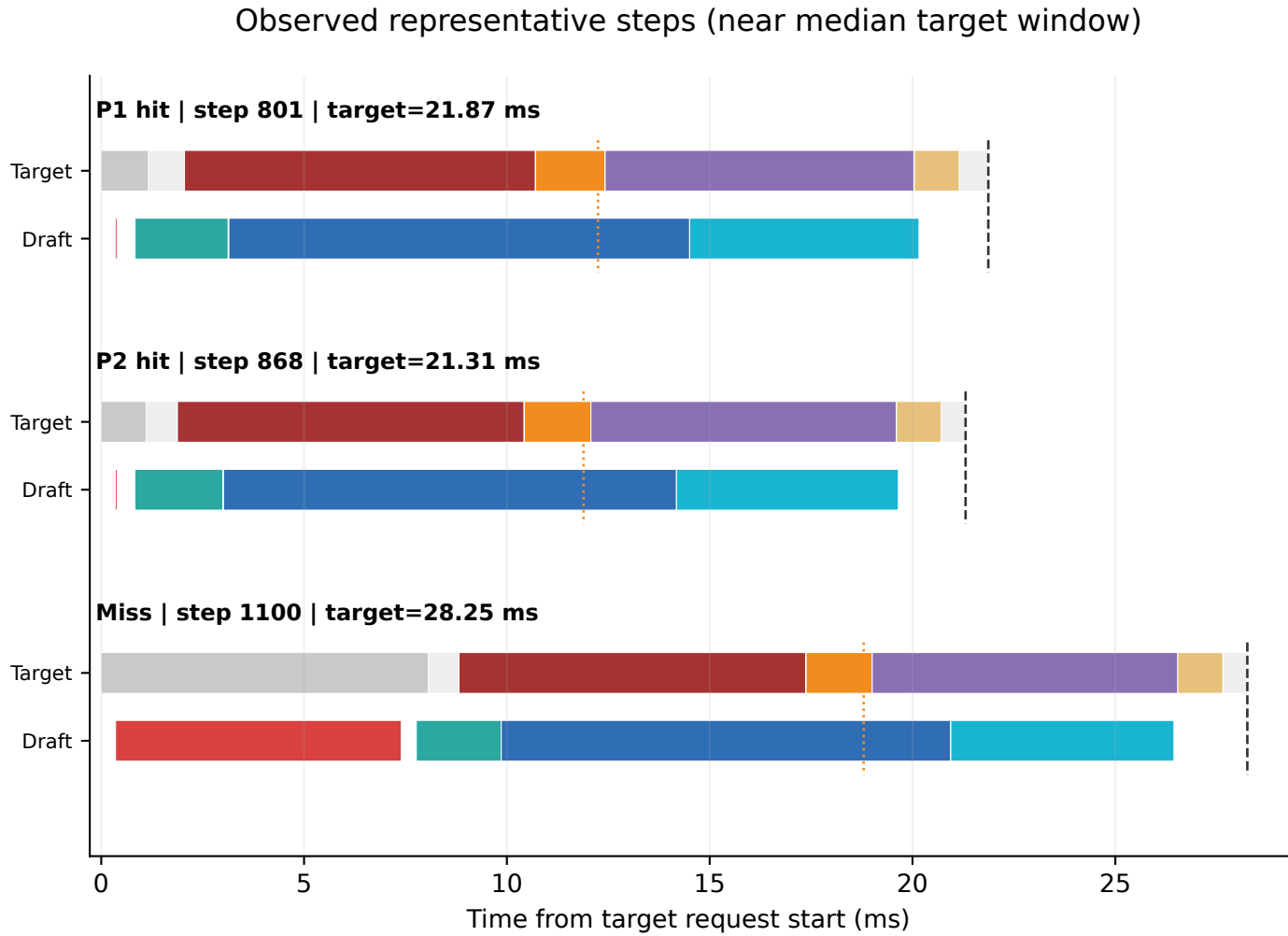
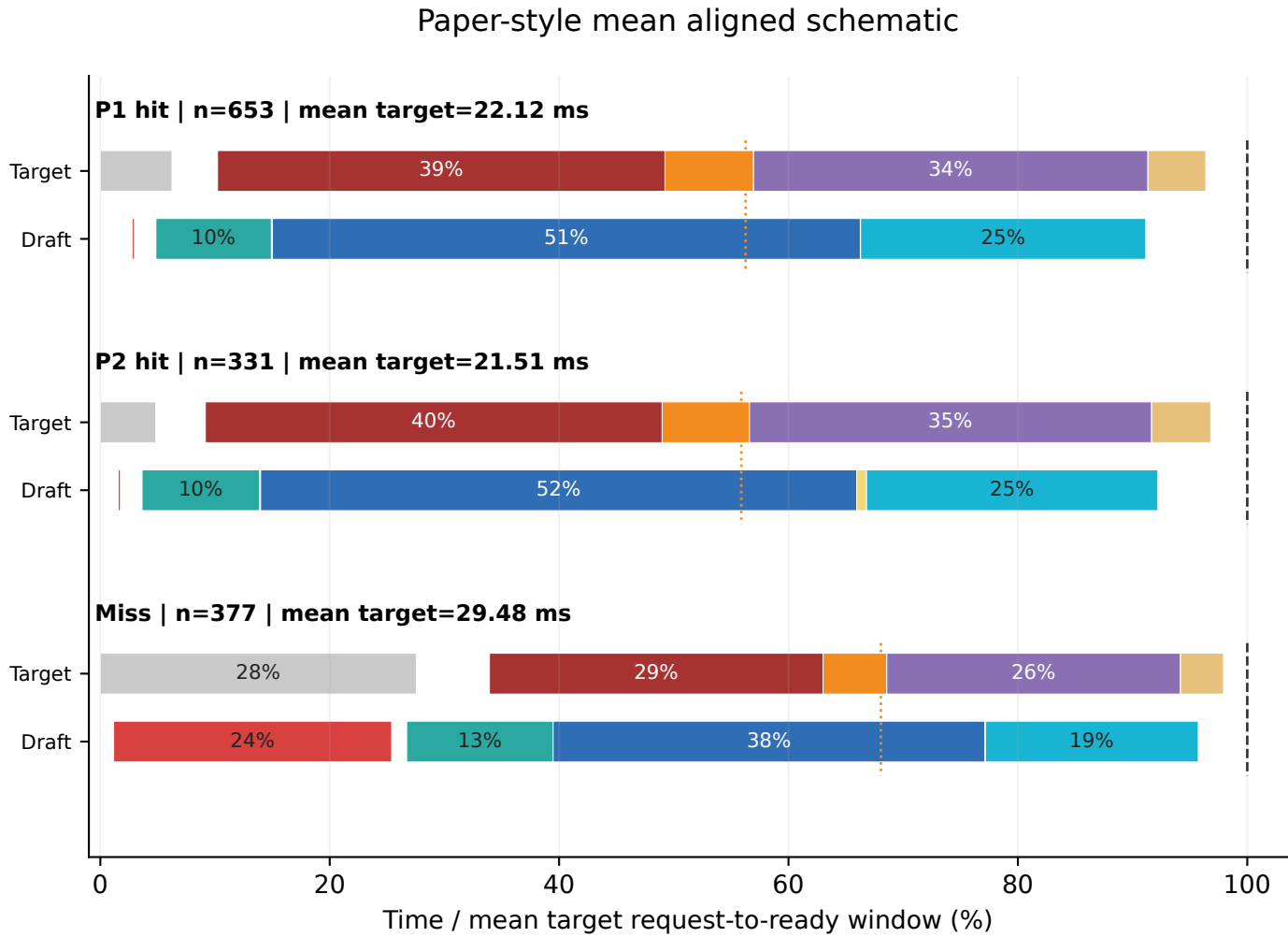
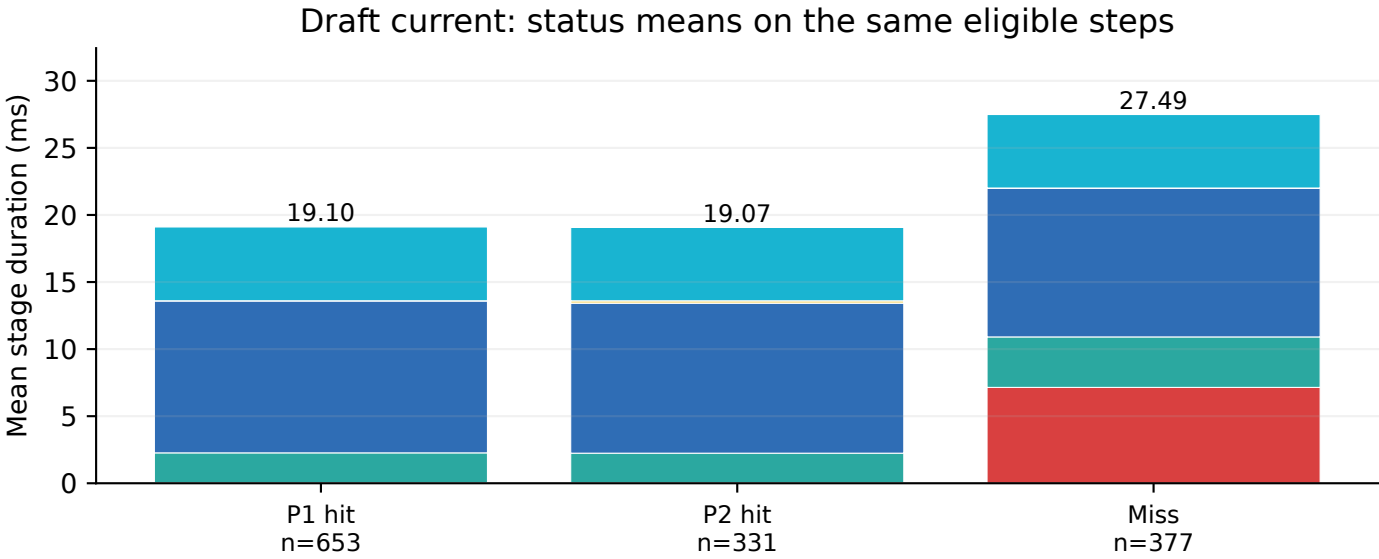
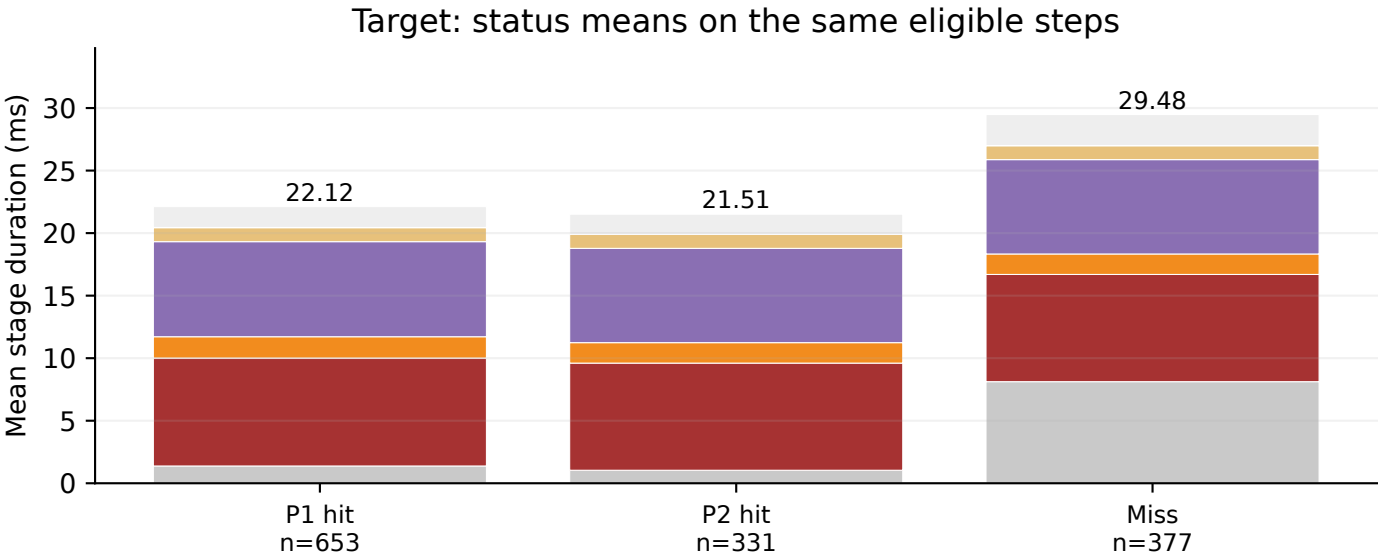
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_p2budget3
DUET: K1/K2=4/2, exit=16 (0-based), $\bar{C}$ =3, N1/N2=8/4, U=2, W=3, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1210 eligible steps

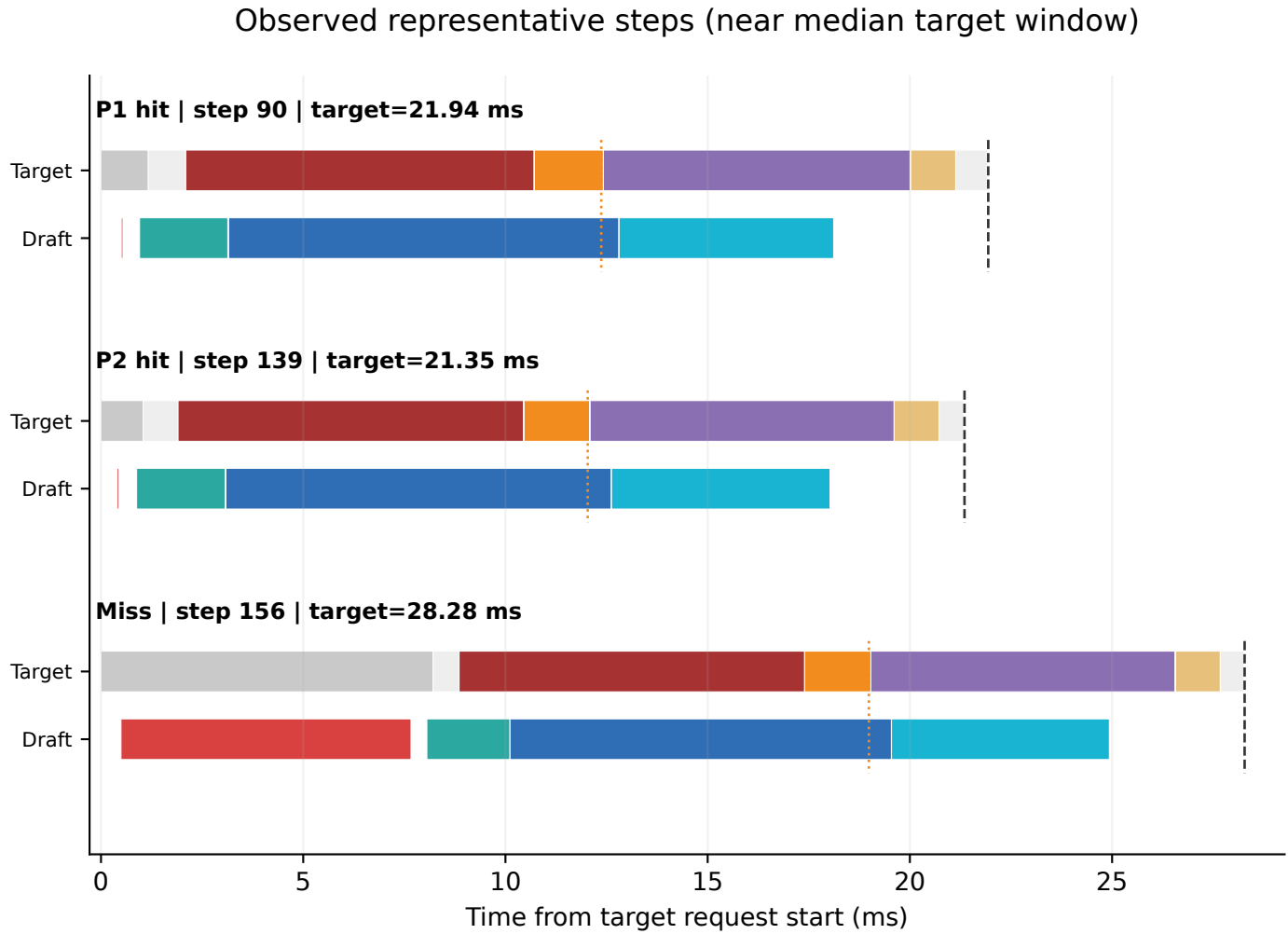
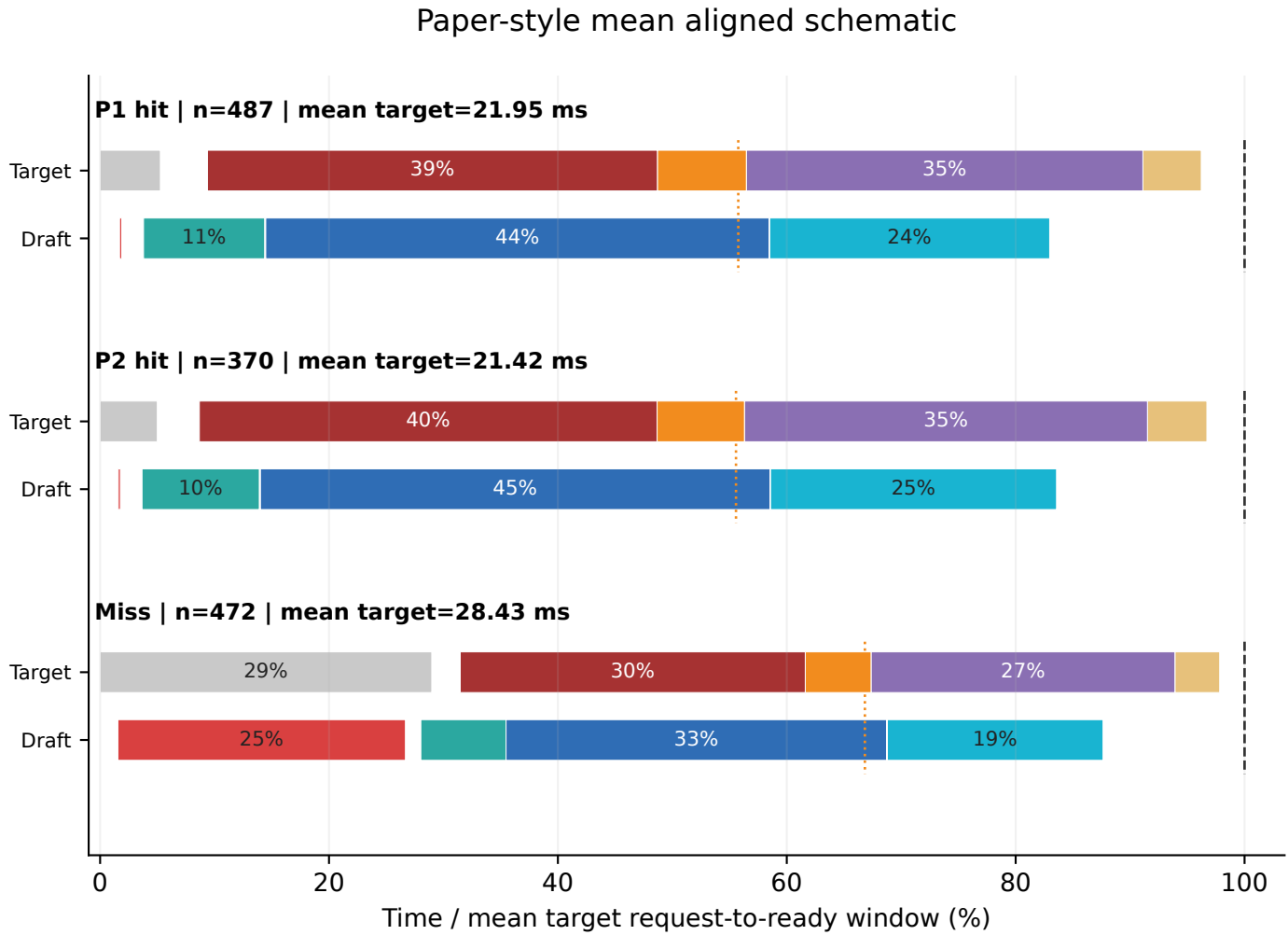
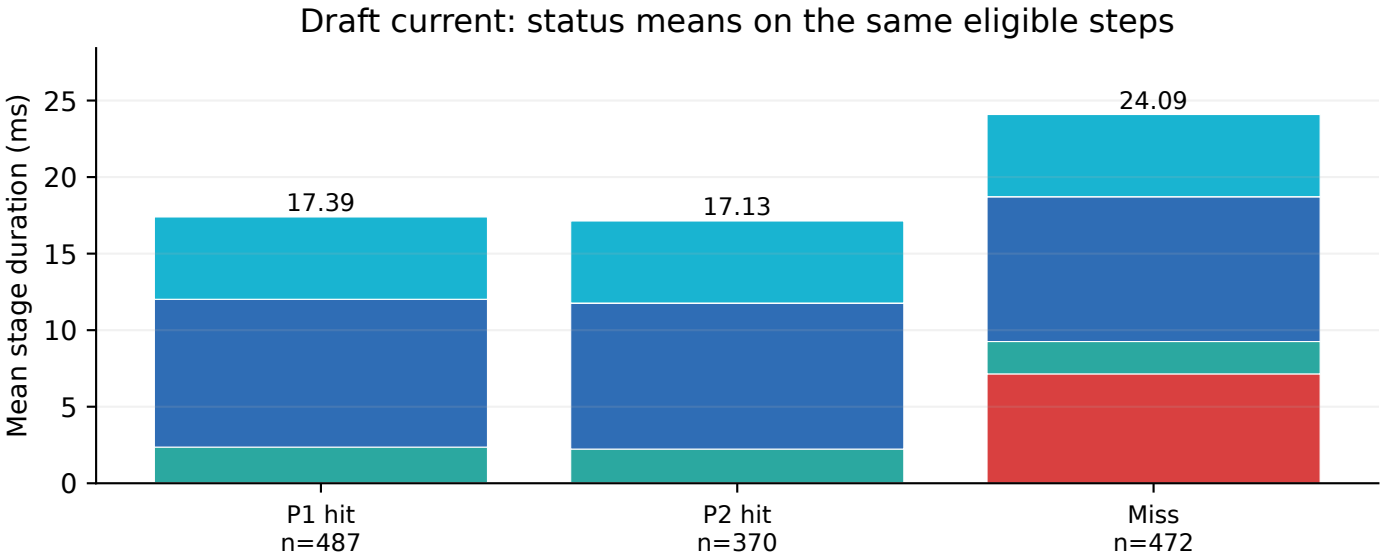
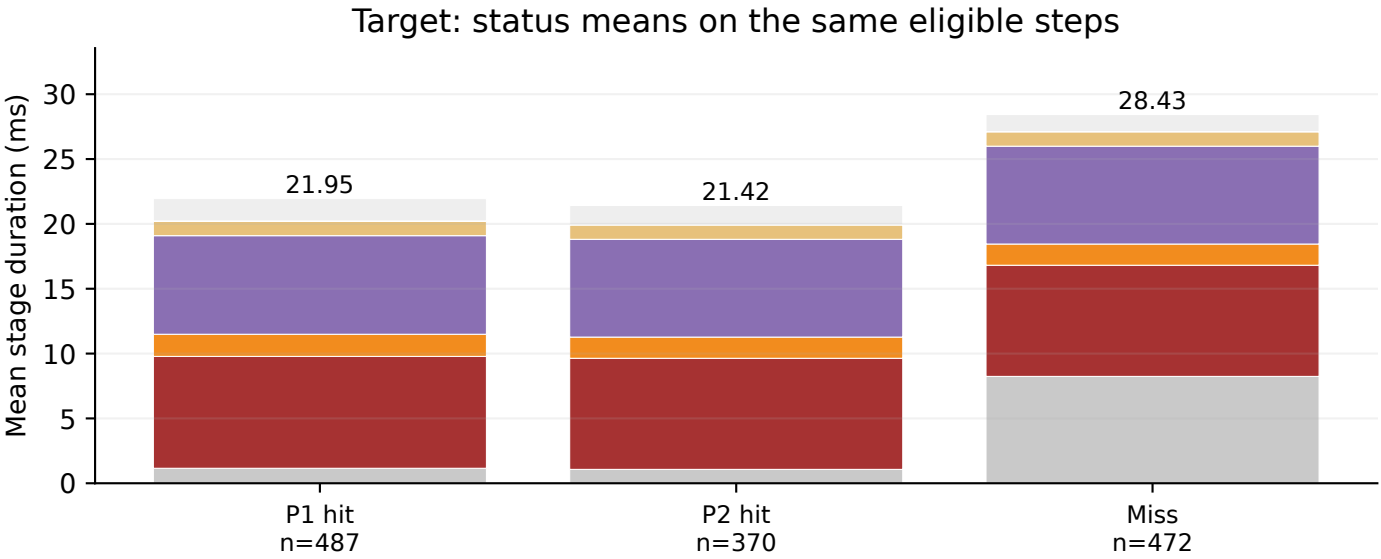


Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

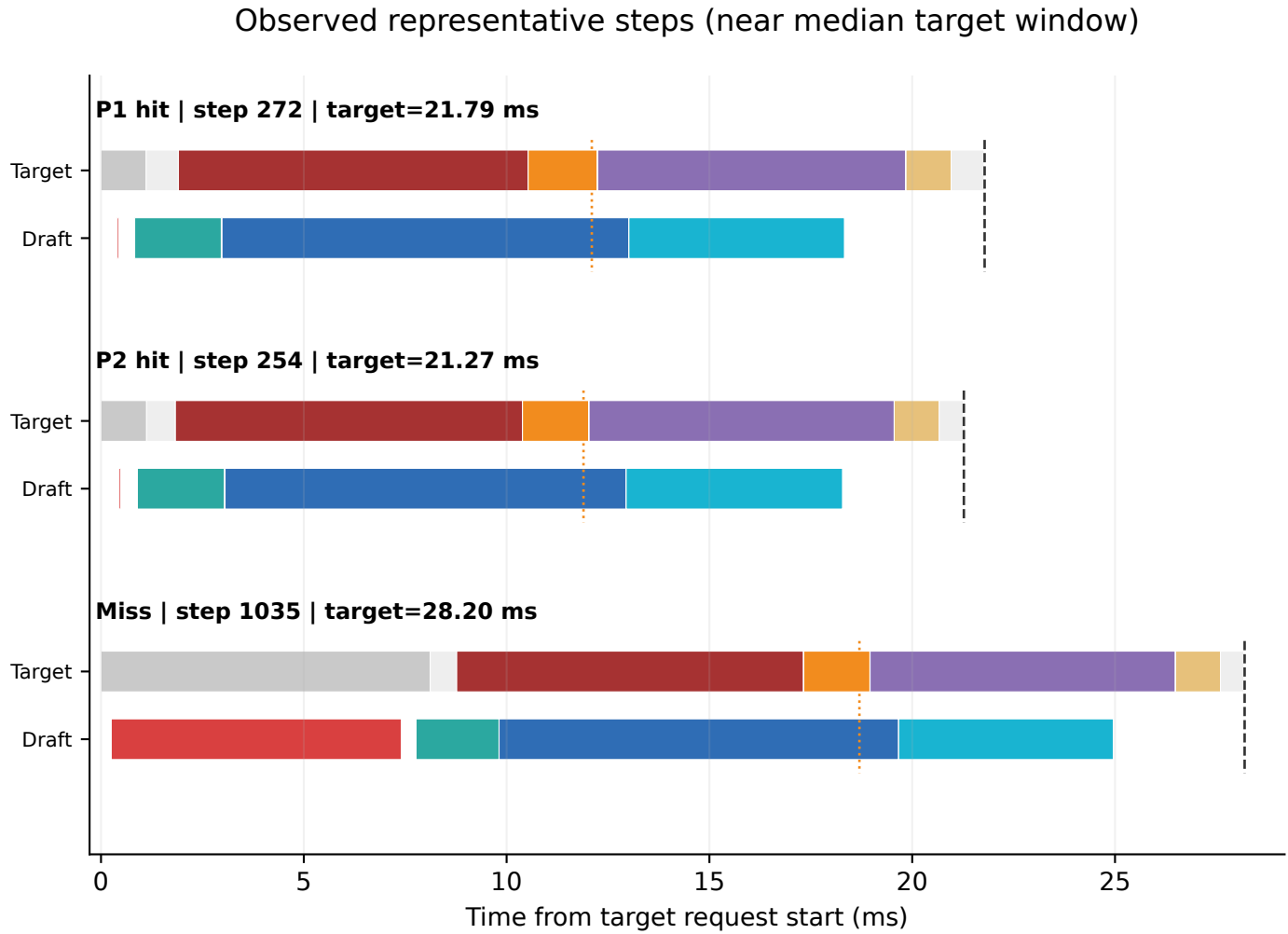
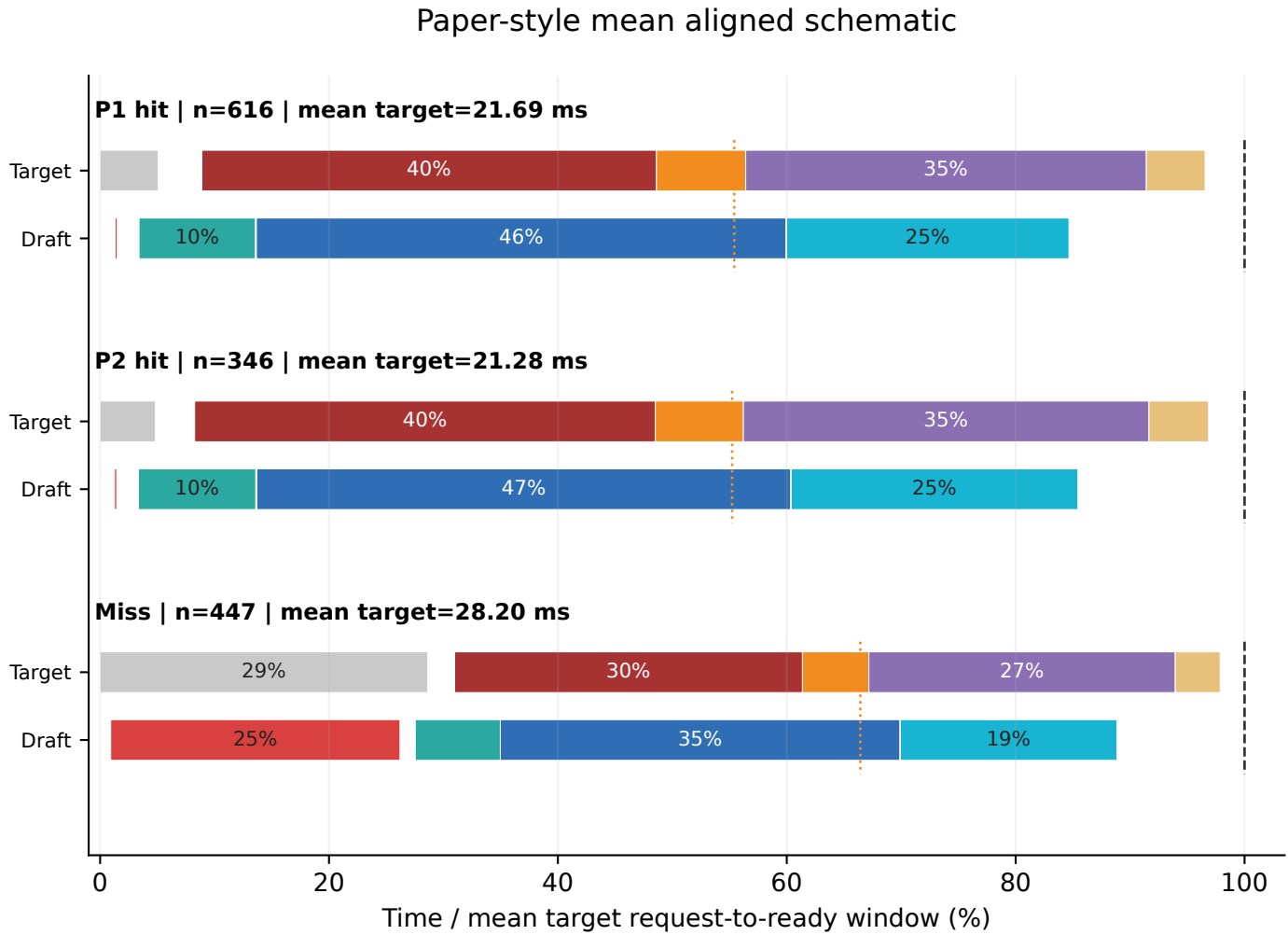
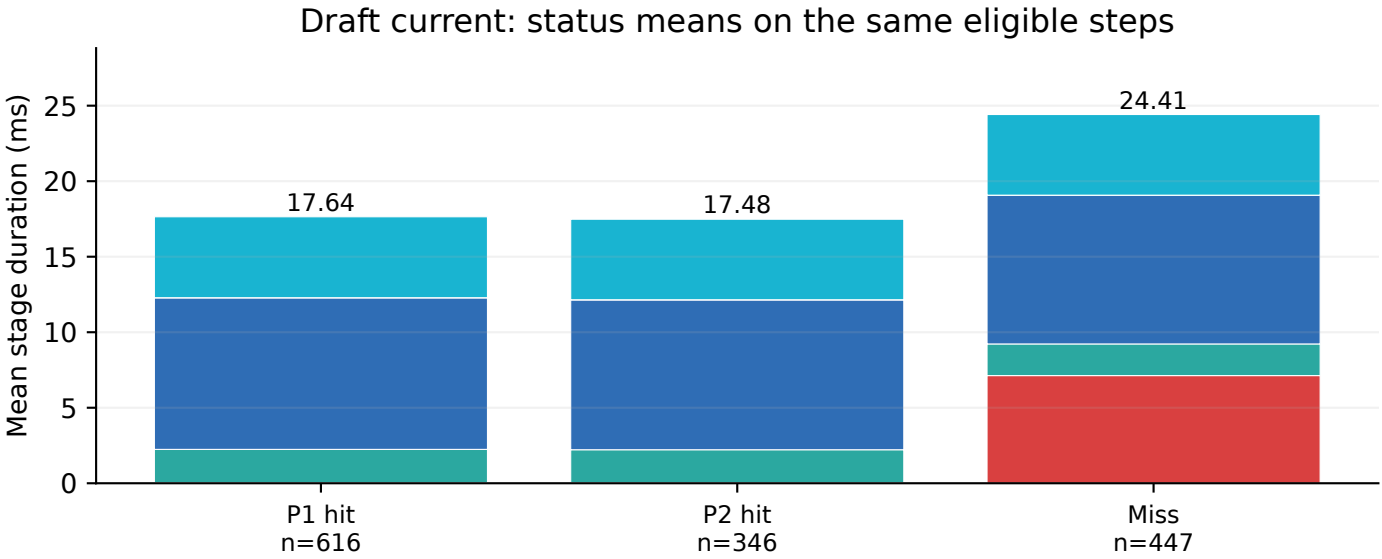
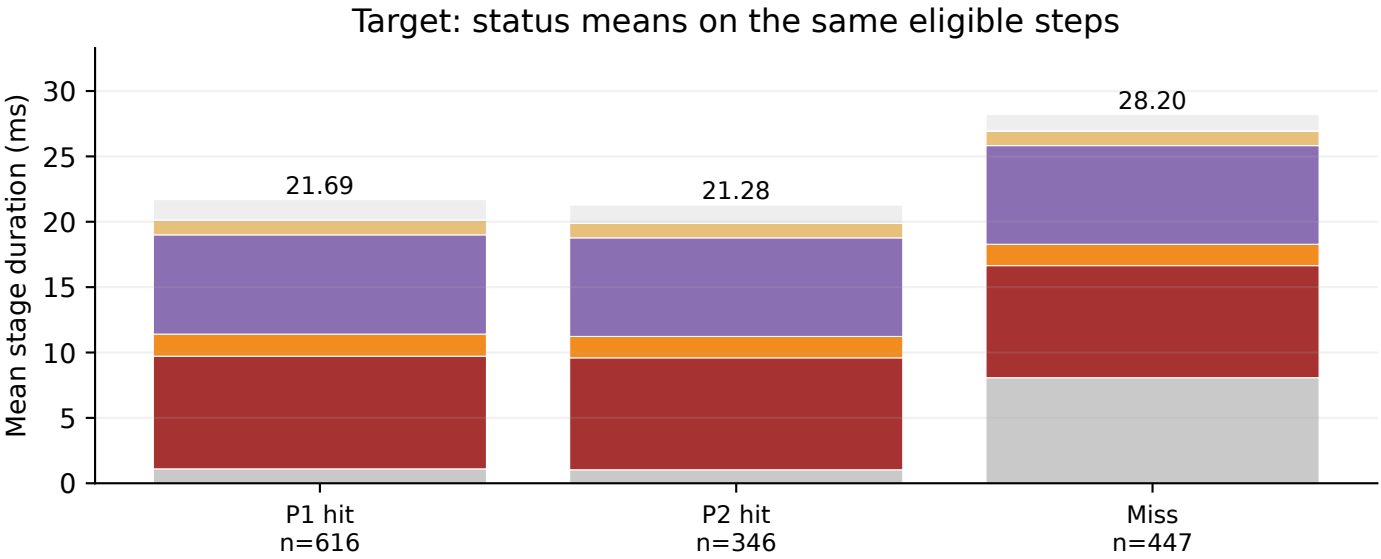
llama3_b1_tree_p2budget8
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=8, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1361 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

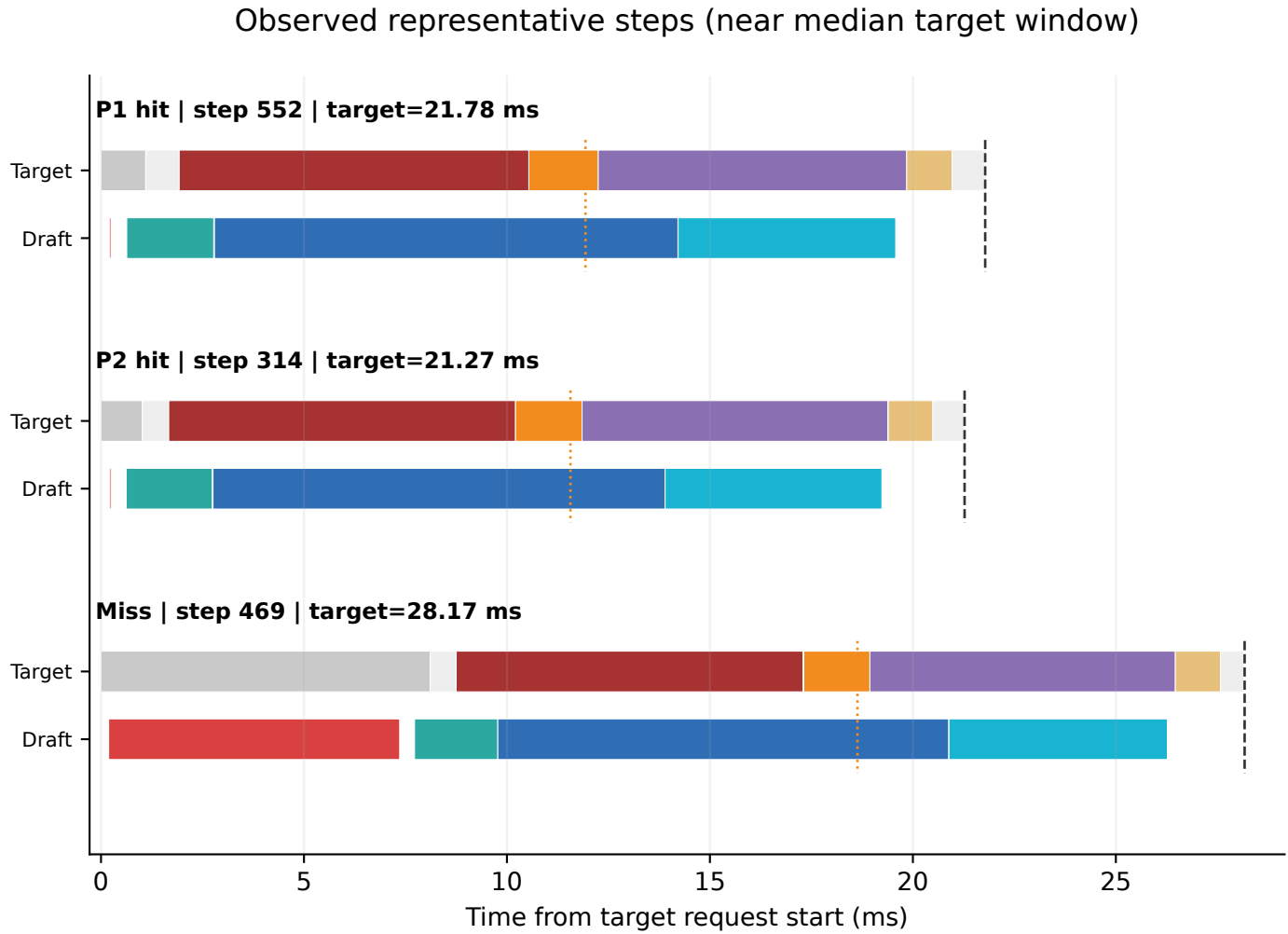
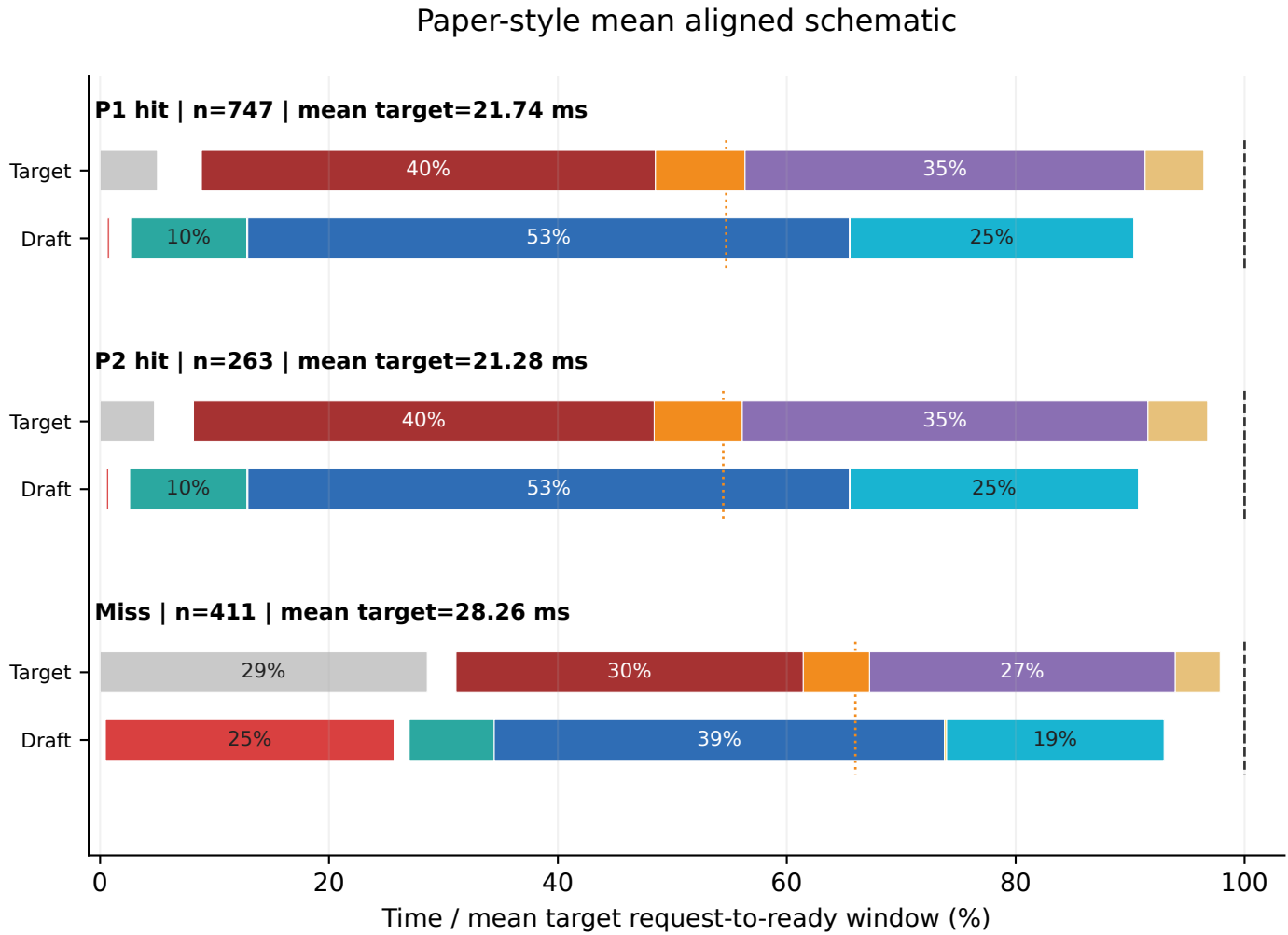
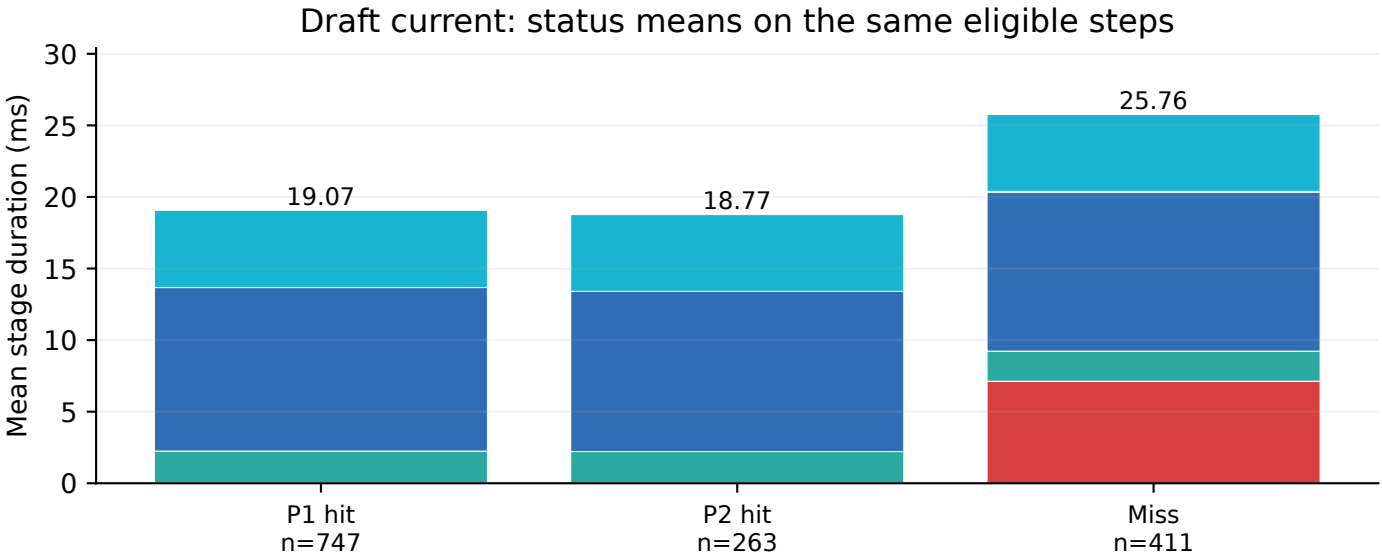
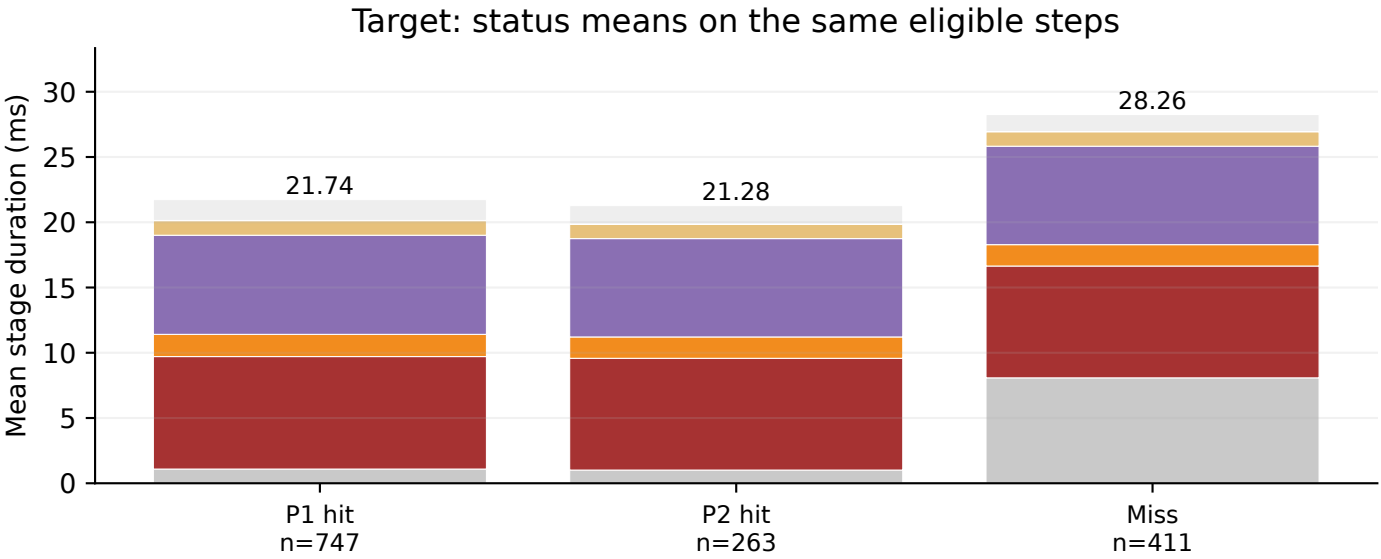


llama3_b1_tree_roots2_compute1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1409 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

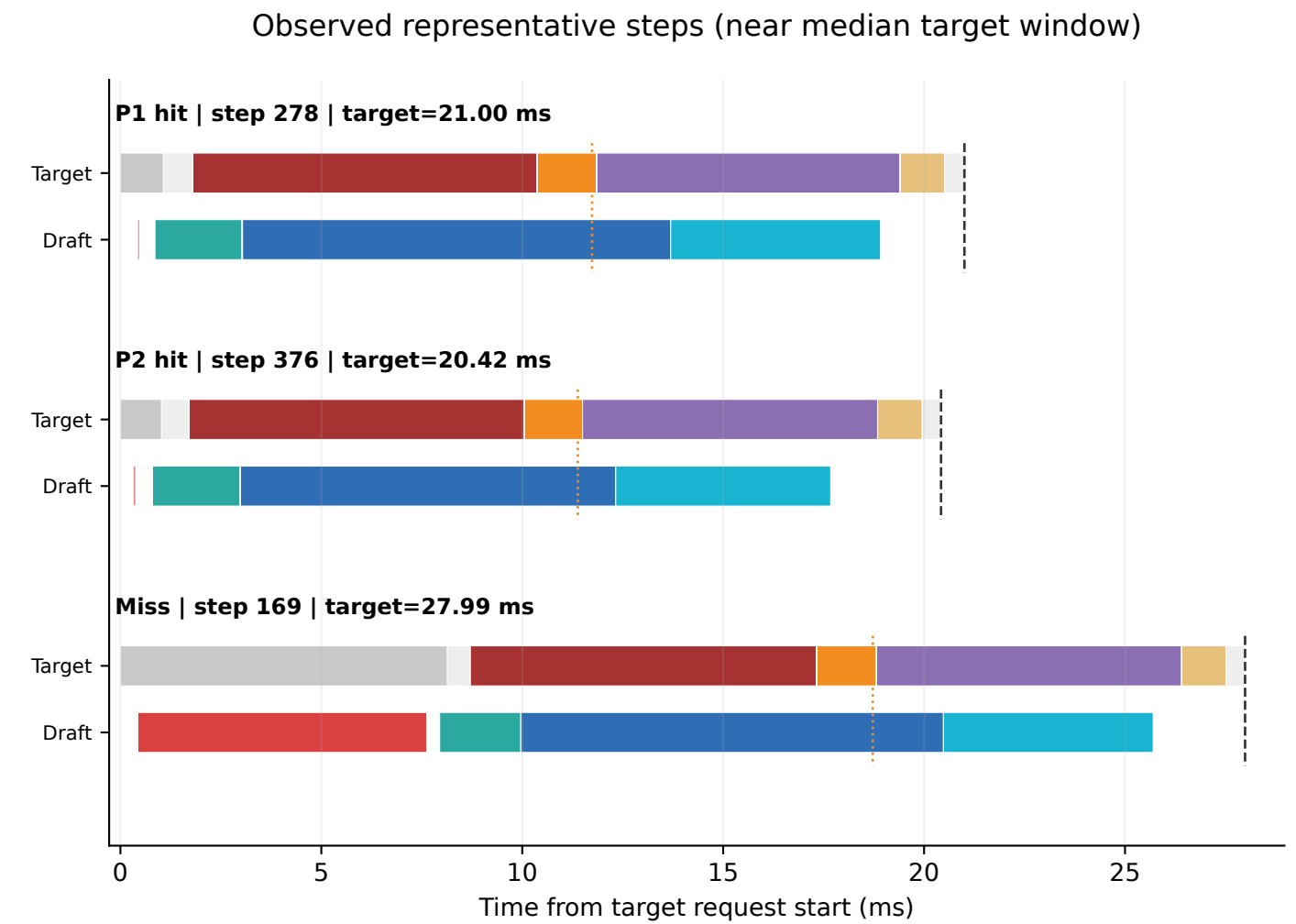
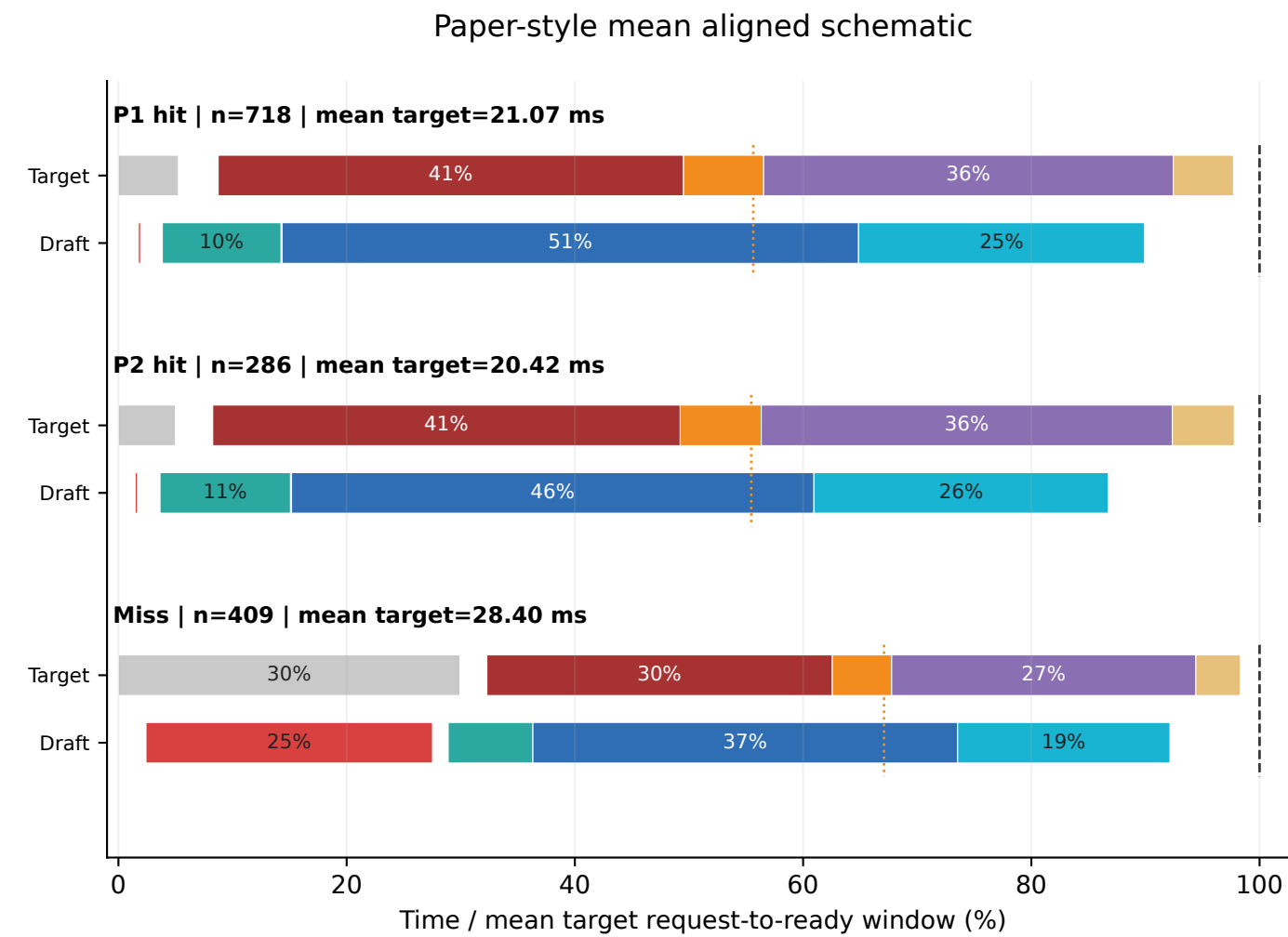
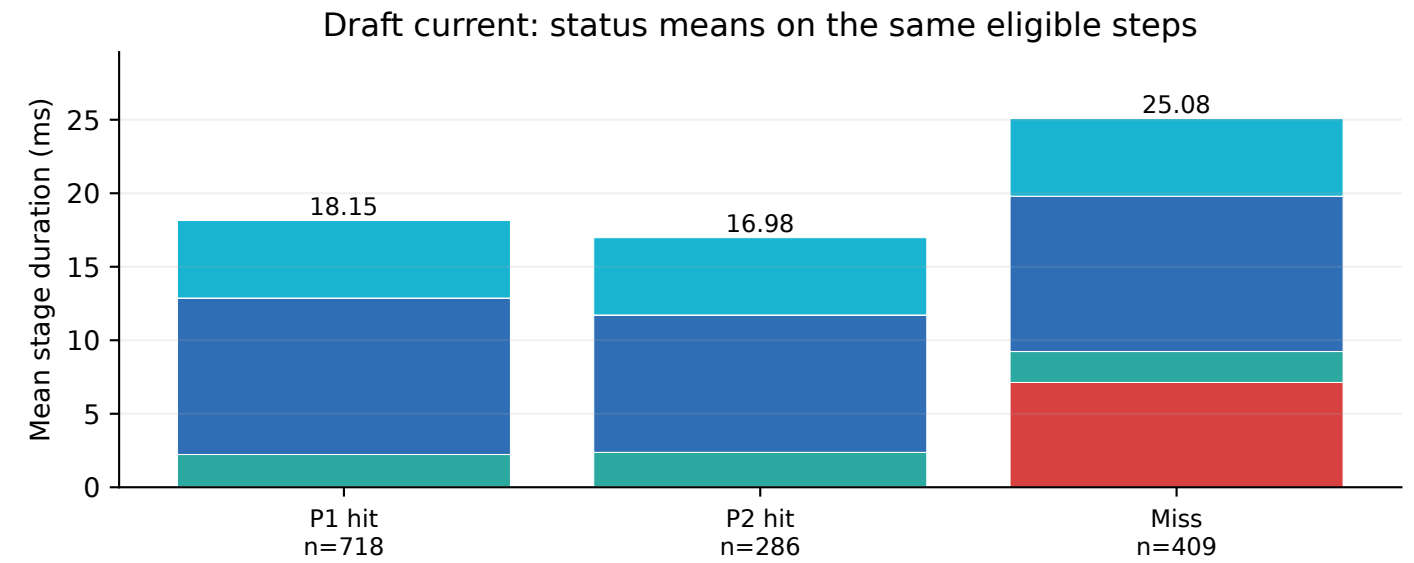
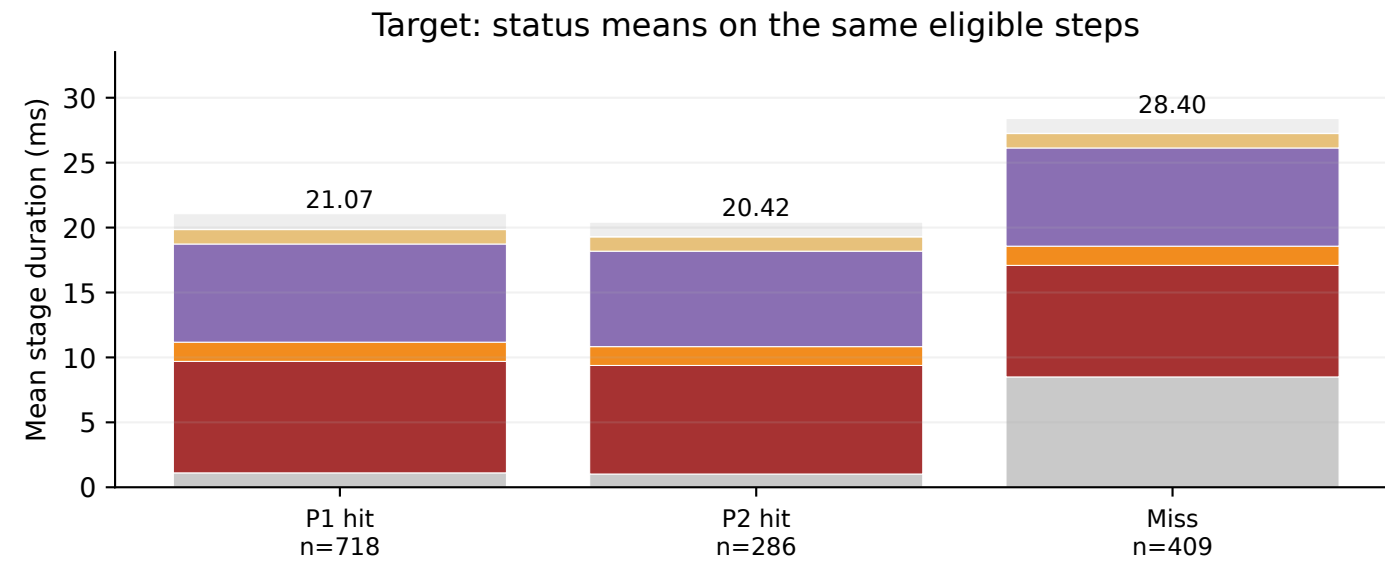
llama3_b1_tree_roots3
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=3, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1421 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

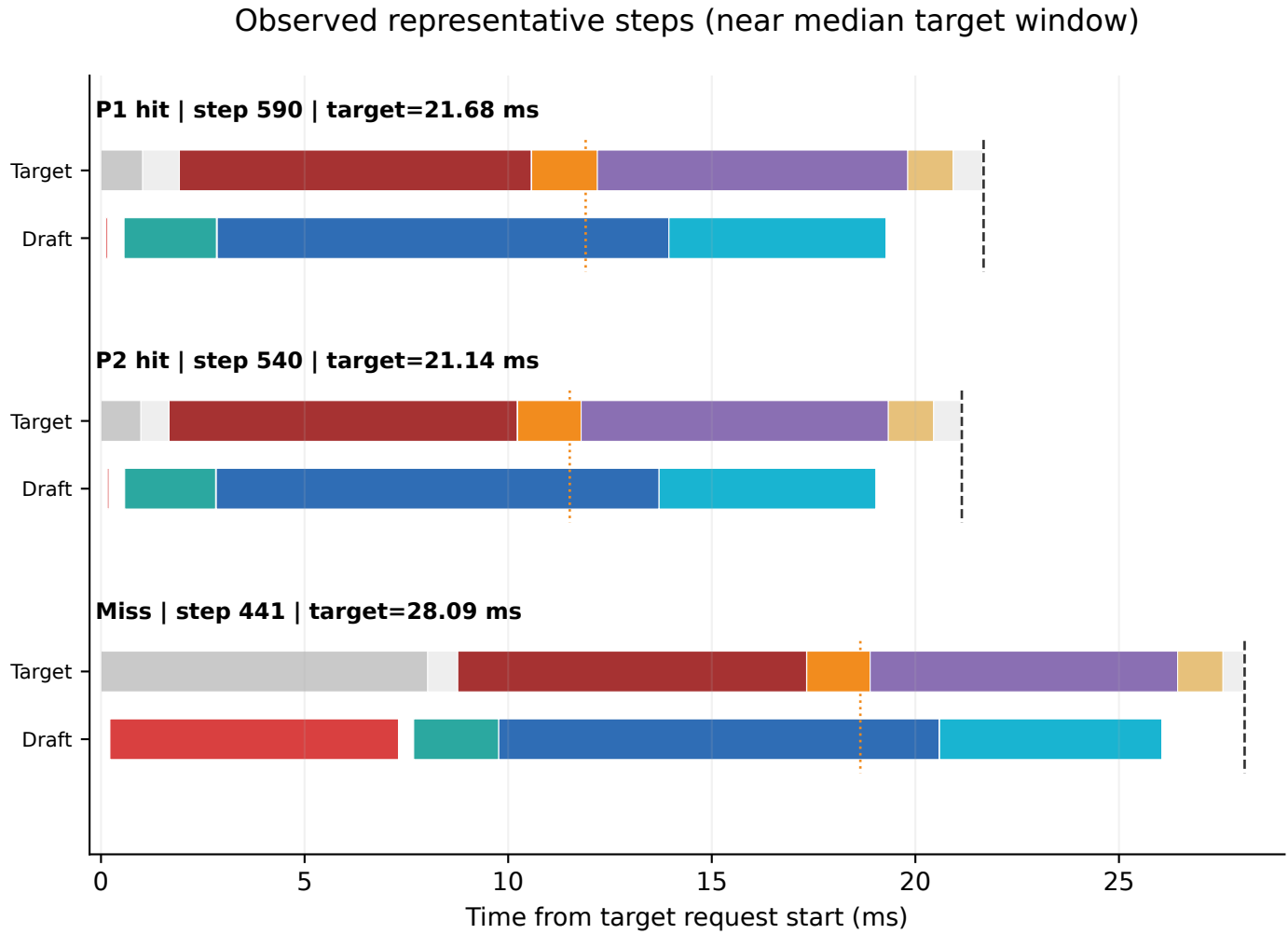
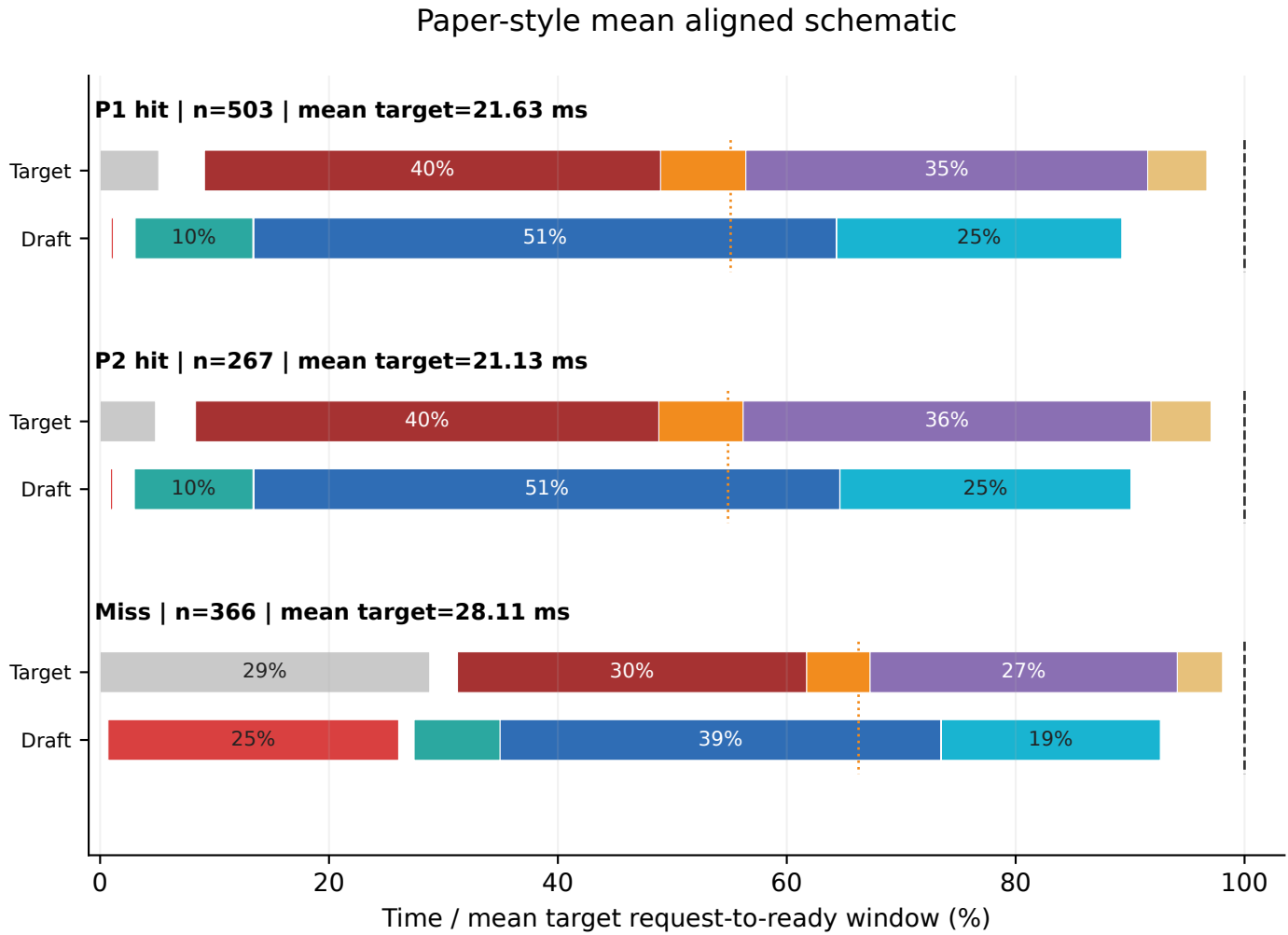
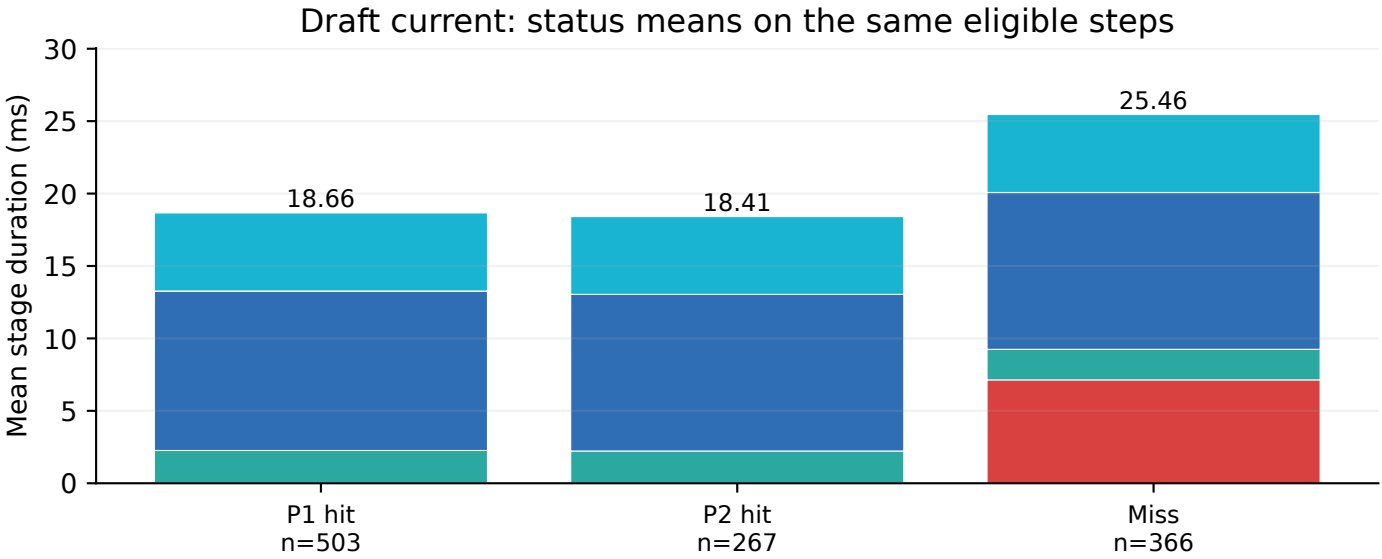
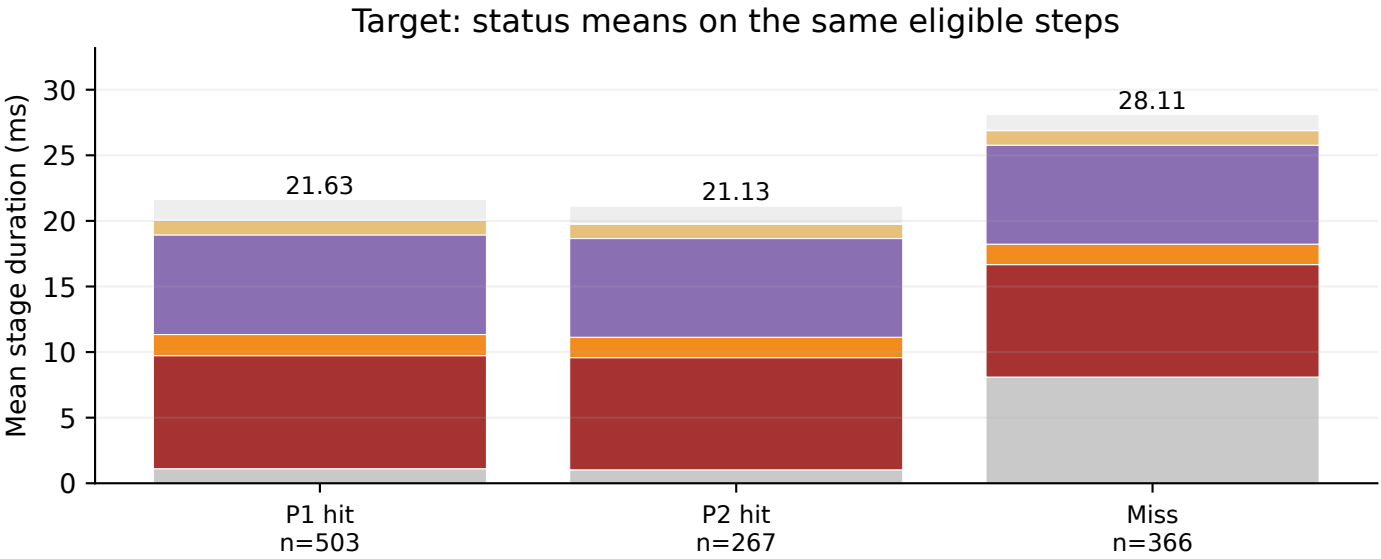
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_tree_width1
DUET: K1/K2=4/2, exit=16 (0-based), C=1, N1/N2=4/2, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1413 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

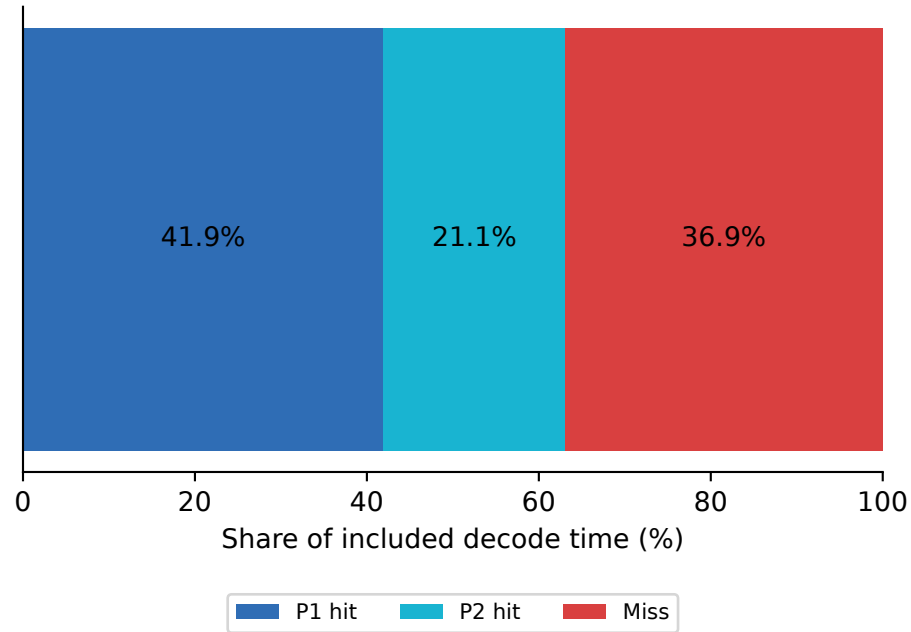
llama3_b1_tree_width2
DUET: K1/K2=4/2, exit=16 (0-based), C=2, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1136 eligible steps



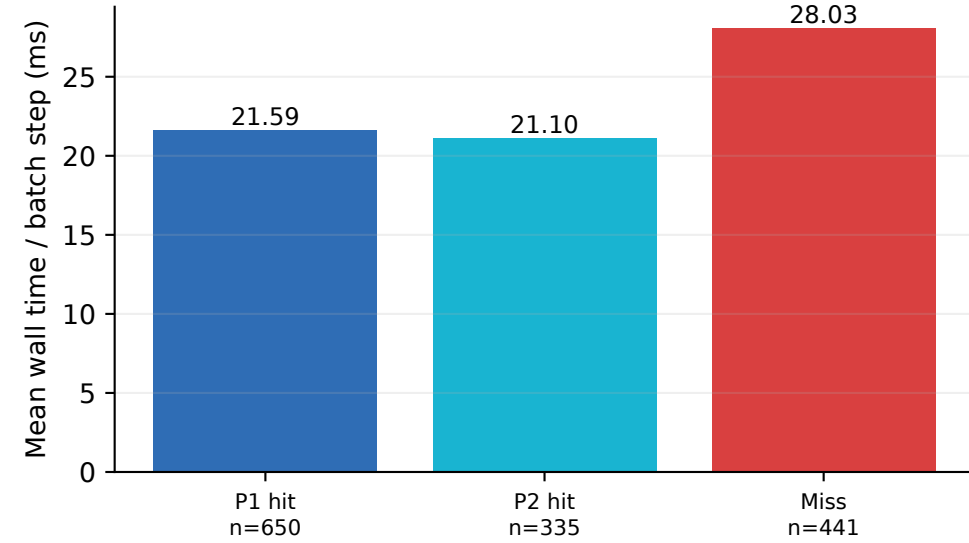
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_warm_duet_tree_1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=2.654, hit=69.0%, TPS*=113.10

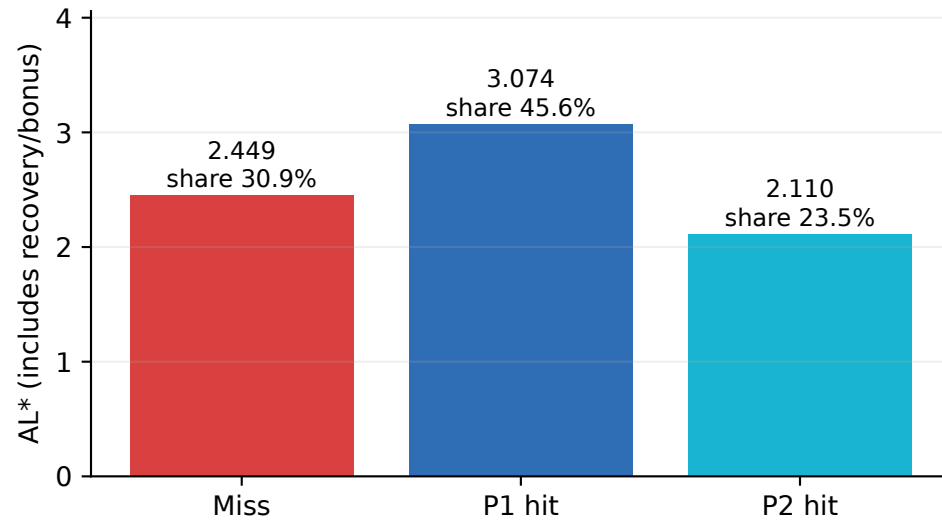
Measured wall time by batch cache status



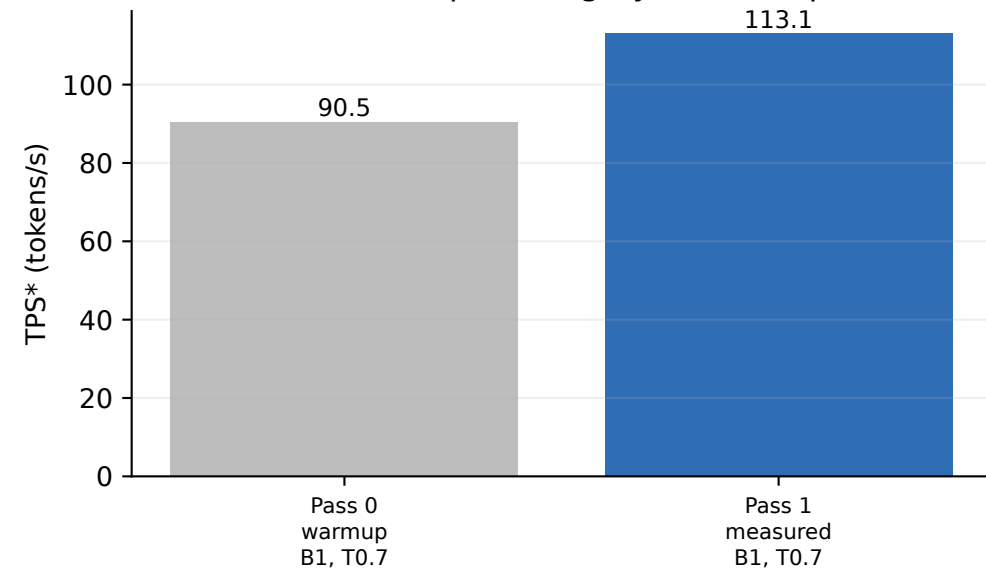
Boundary-free steps, including slow steps



Conditional AL* and source share



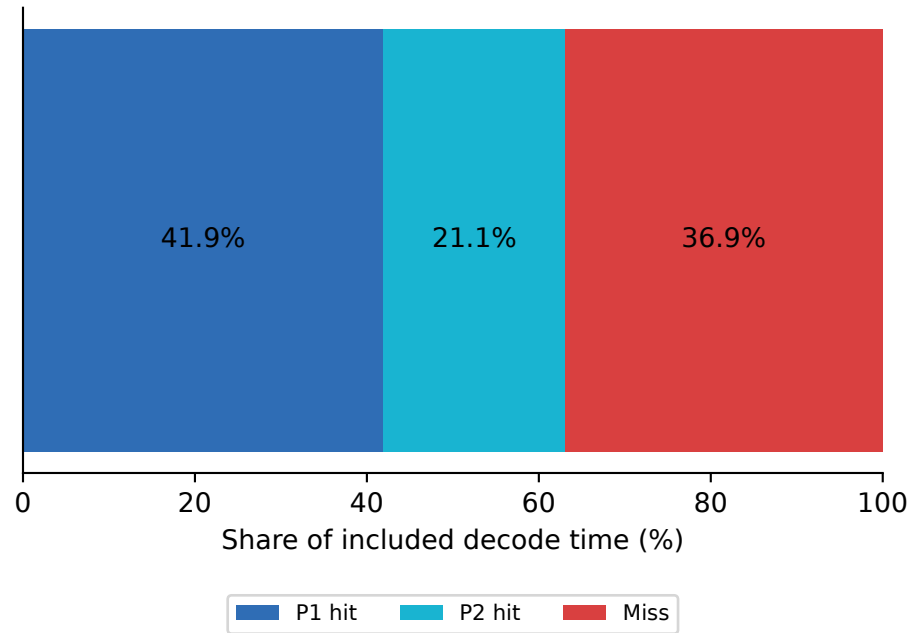
All saved passes; gray = warmup



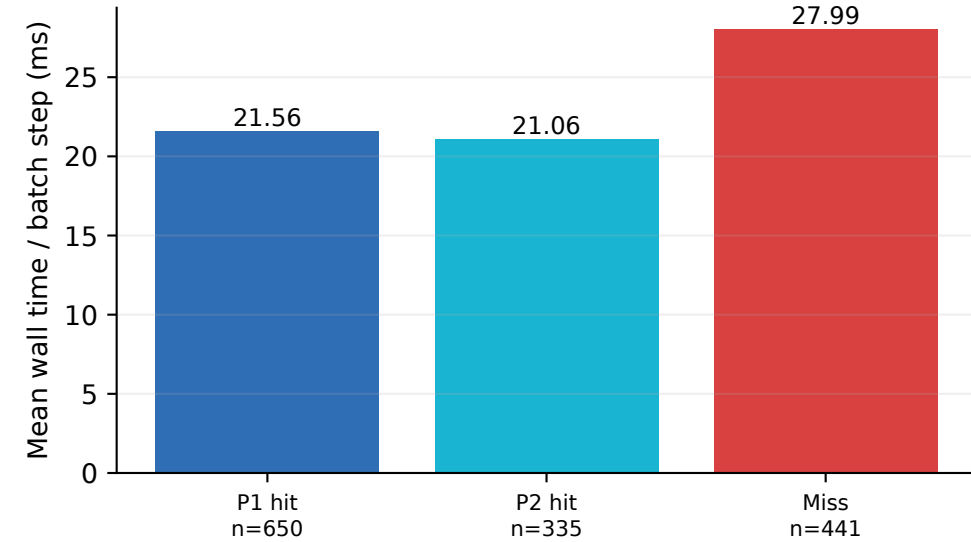
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_duet_tree_2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=2.654, hit=69.0%, TPS*=113.27

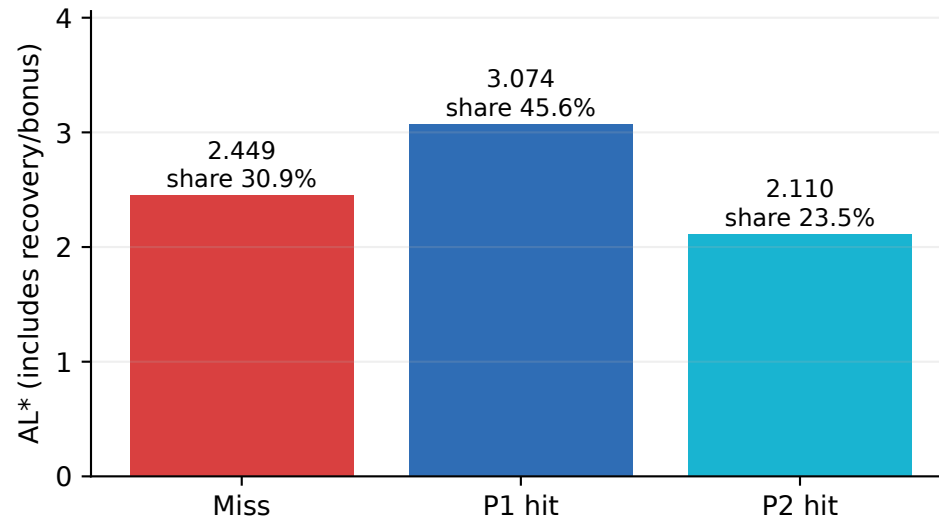
Measured wall time by batch cache status



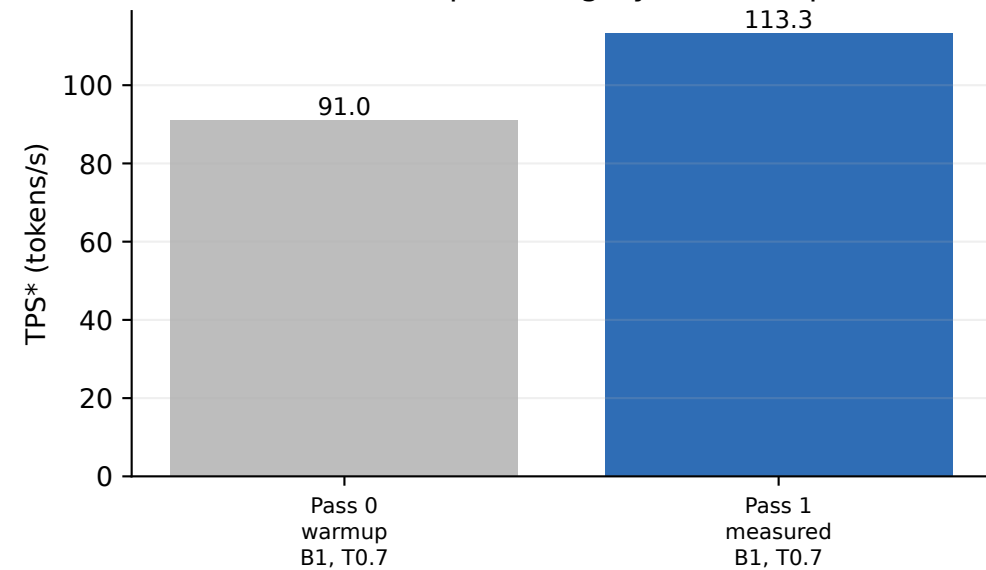
Boundary-free steps, including slow steps



Conditional AL* and source share



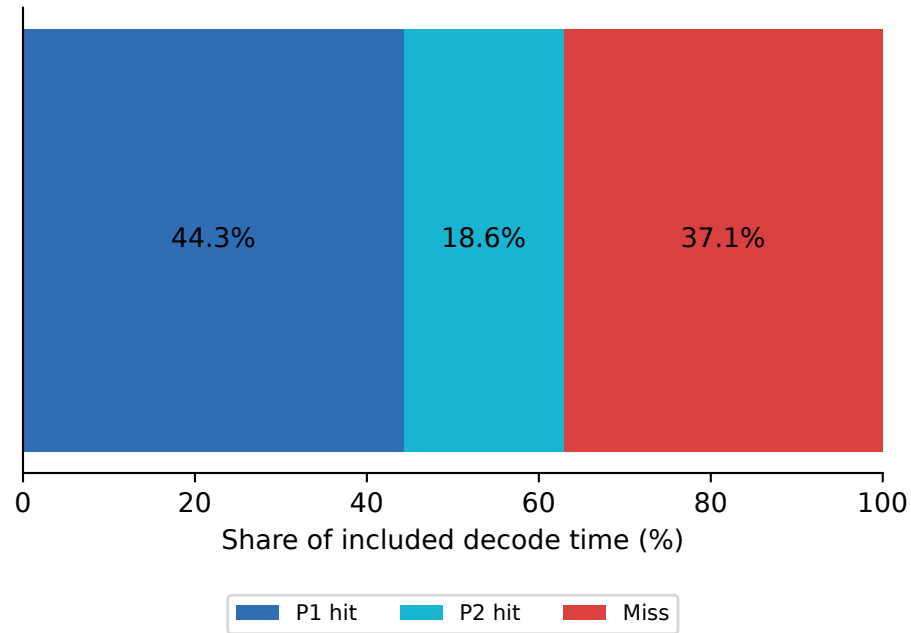
All saved passes; gray = warmup



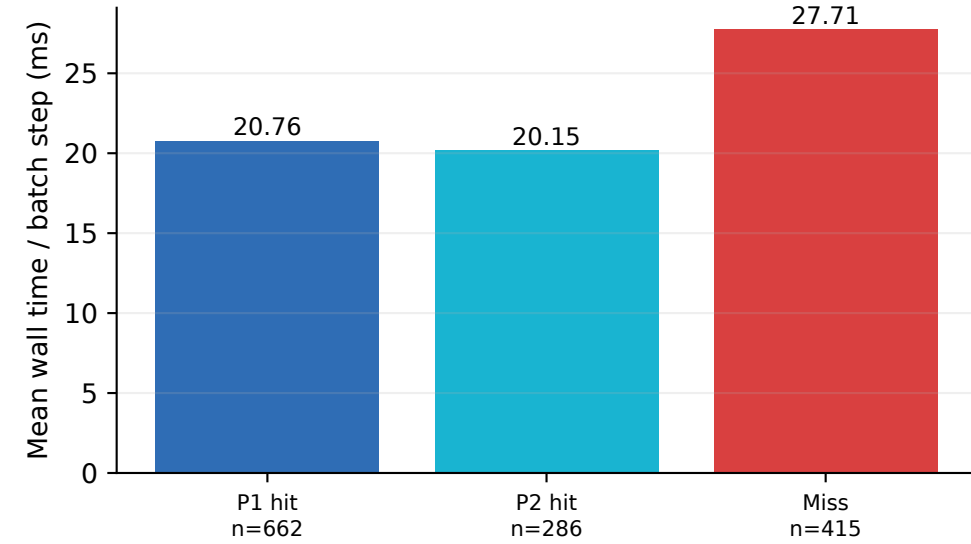
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_duet_tree_3
 DUET: K1/K2=4/2, exit=16 (0-based), C=1, N1/N2=4/2, U=2, W=5, trim/stream=0/0
 Last pass: B=1, T=0.7, N=48 | AL*=2.498, hit=69.3%, TPS*=109.83

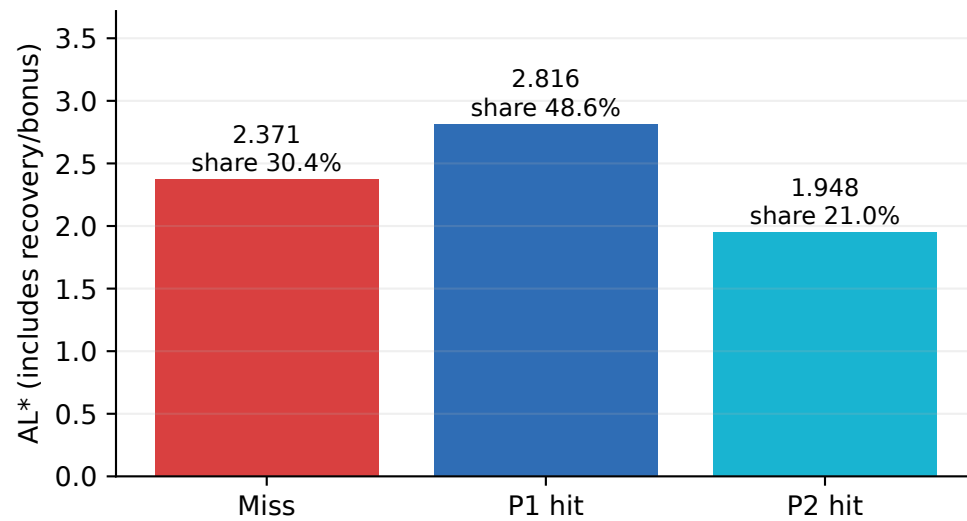
Measured wall time by batch cache status



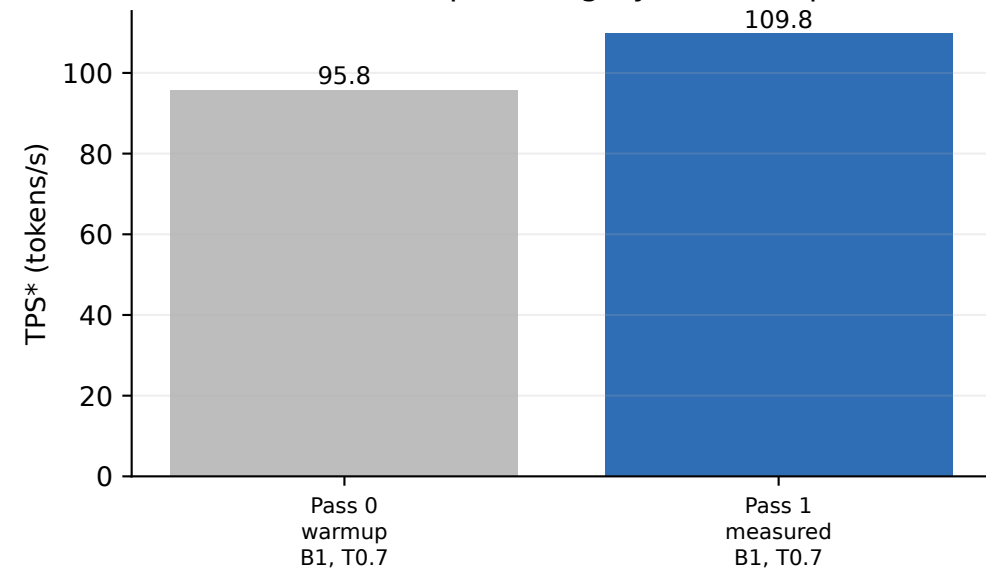
Boundary-free steps, including slow steps



Conditional AL* and source share



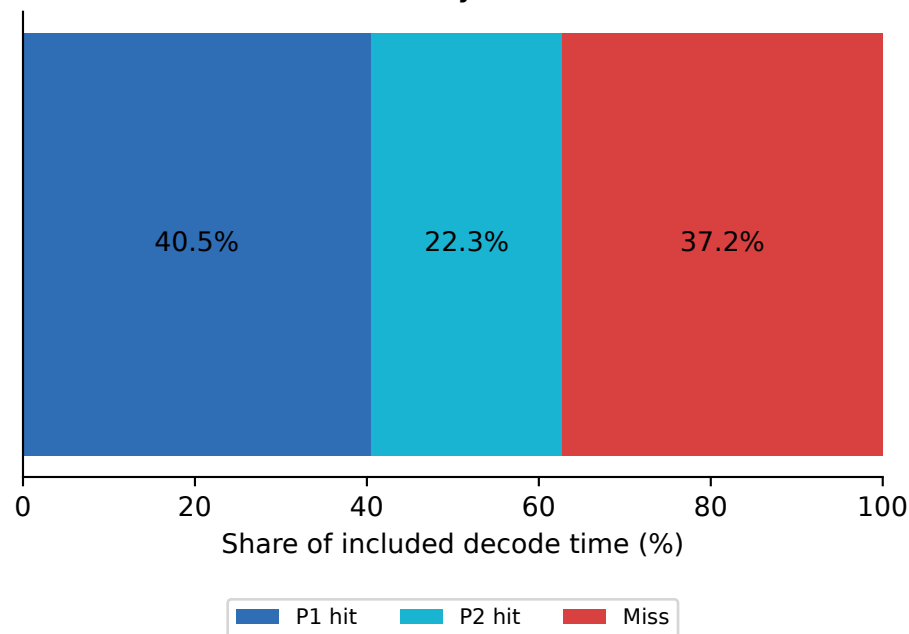
All saved passes; gray = warmup



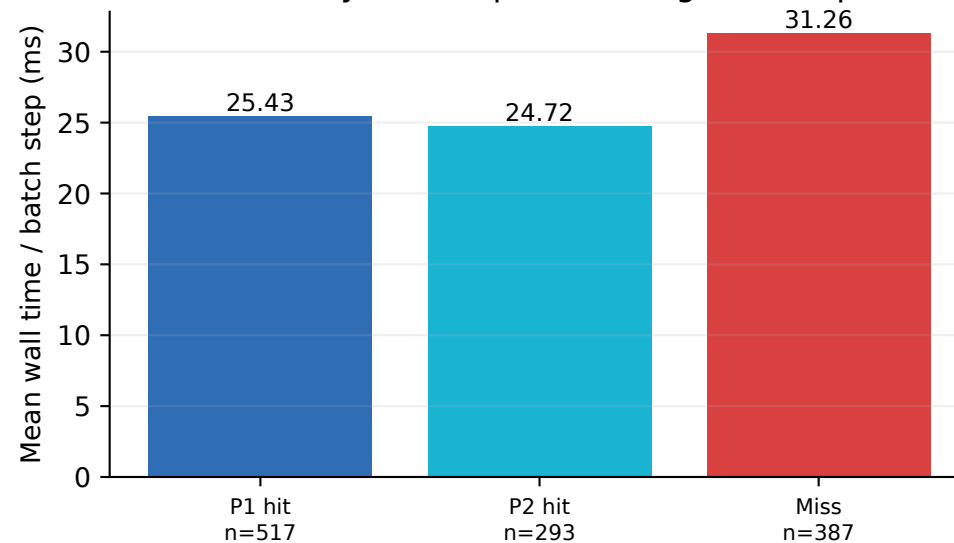
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_duet_tree_4
DUET: K1/K2=6/2, exit=16 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=2.828, hit=67.3%, TPS*=104.20

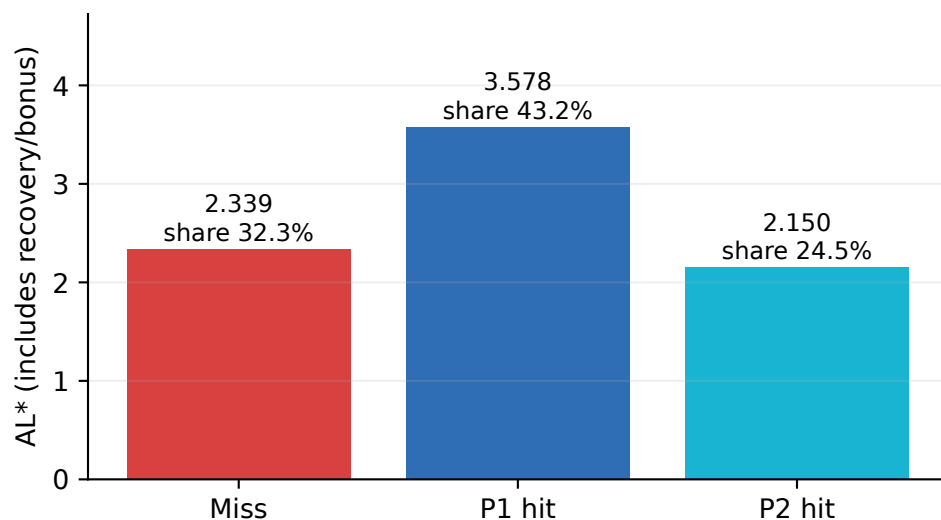
Measured wall time by batch cache status



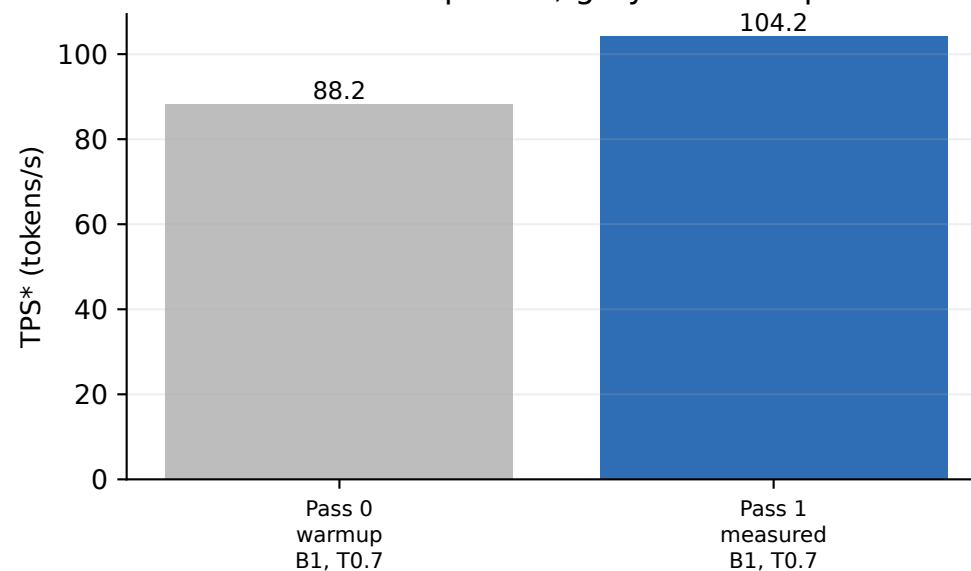
Boundary-free steps, including slow steps



Conditional AL* and source share



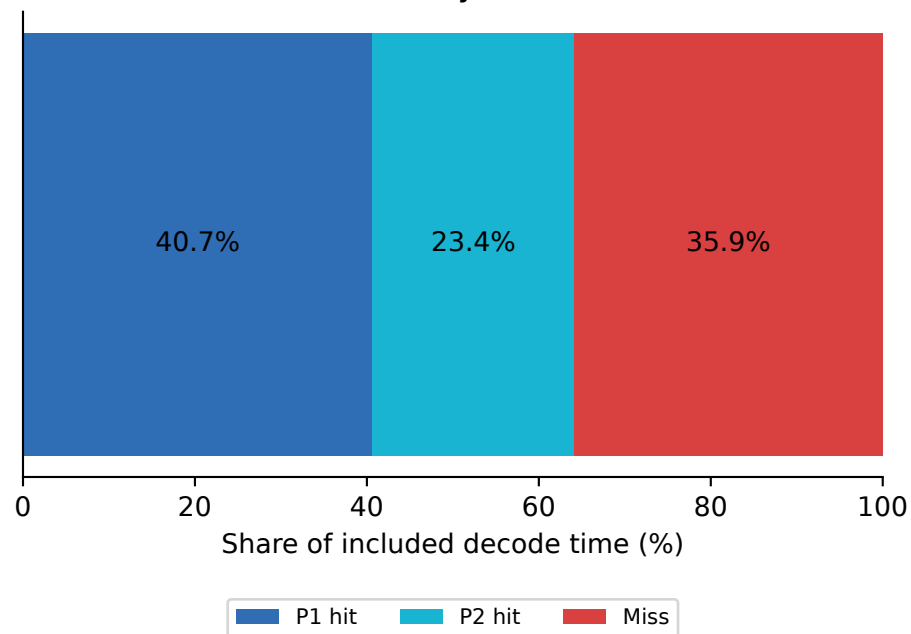
All saved passes; gray = warmup



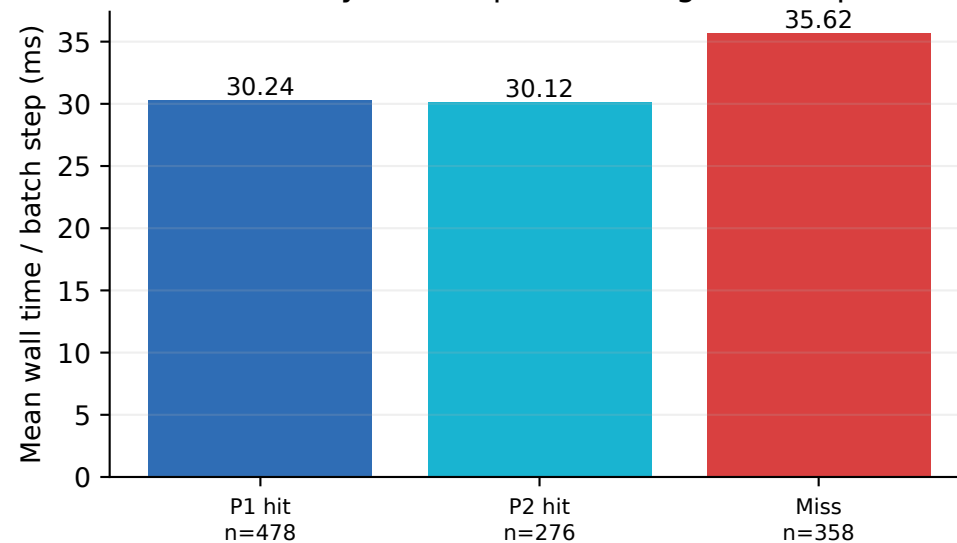
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_duet_tree_5
DUET: K1/K2=6/4, exit=16 (0-based), C=3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
Last pass: B=1, T=0.7, N=48 | AL*=3.028, hit=67.5%, TPS*=94.79

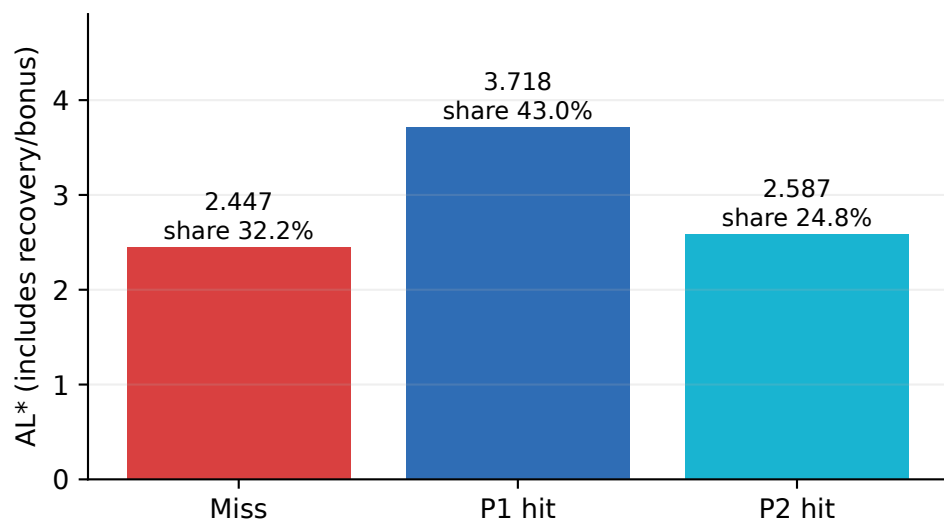
Measured wall time by batch cache status



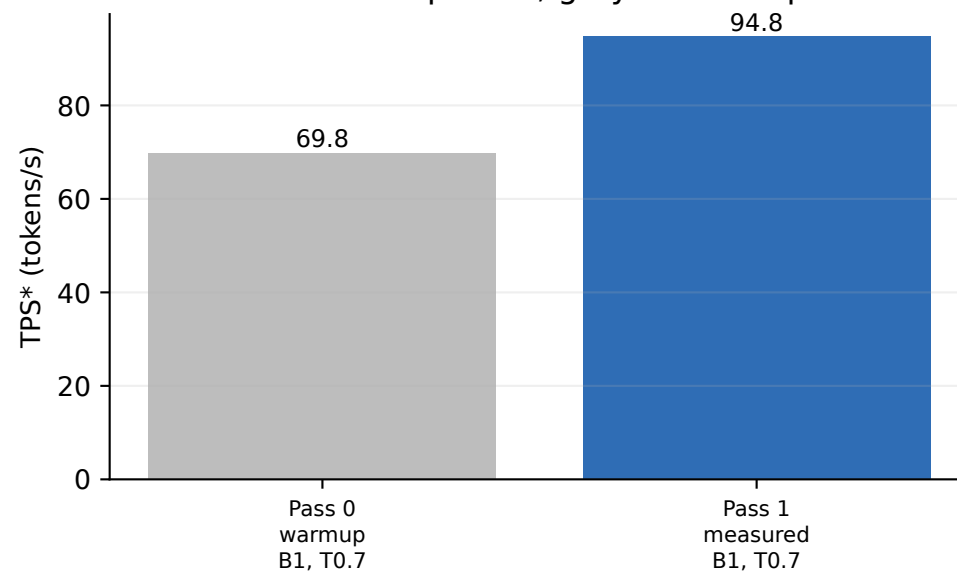
Boundary-free steps, including slow steps



Conditional AL* and source share



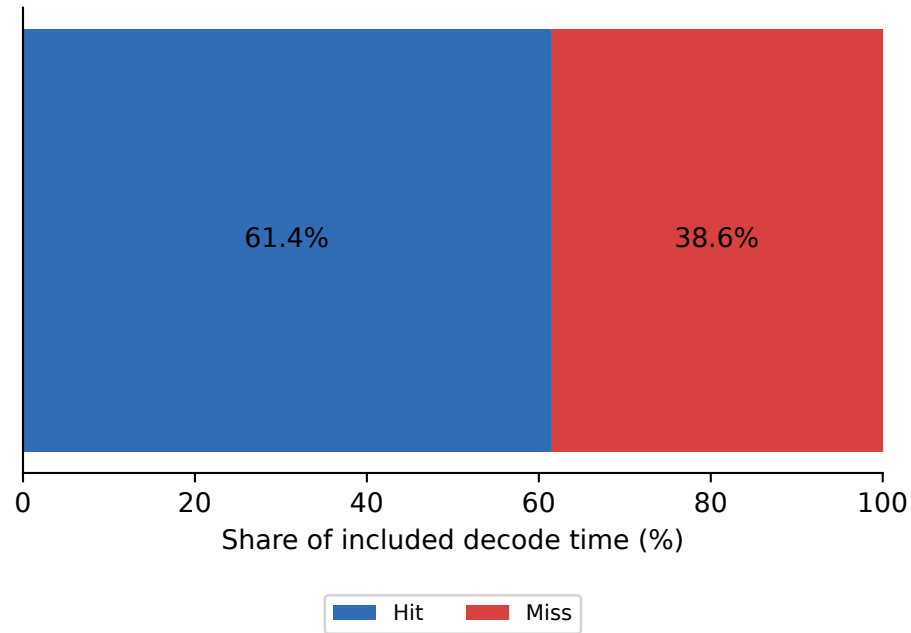
All saved passes; gray = warmup



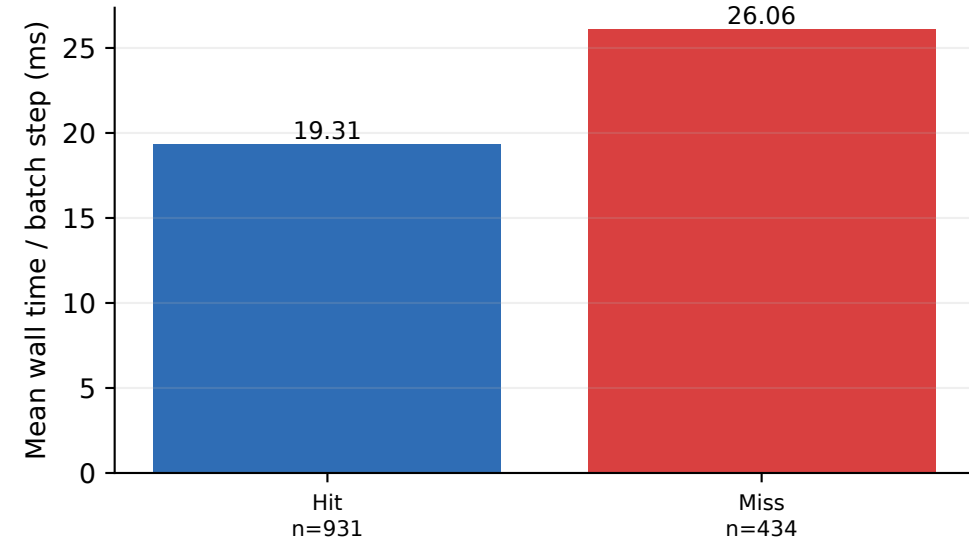
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_ssd_1
SSD: K=4, F=9
Last pass: B=1, T=0.7, N=48 | AL*=2.508, hit=67.9%, TPS*=116.85

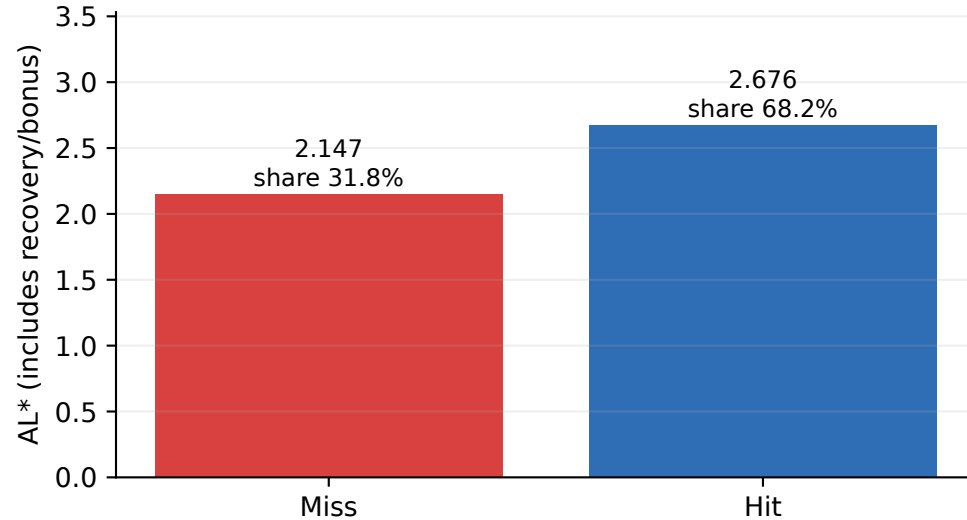
Measured wall time by batch cache status



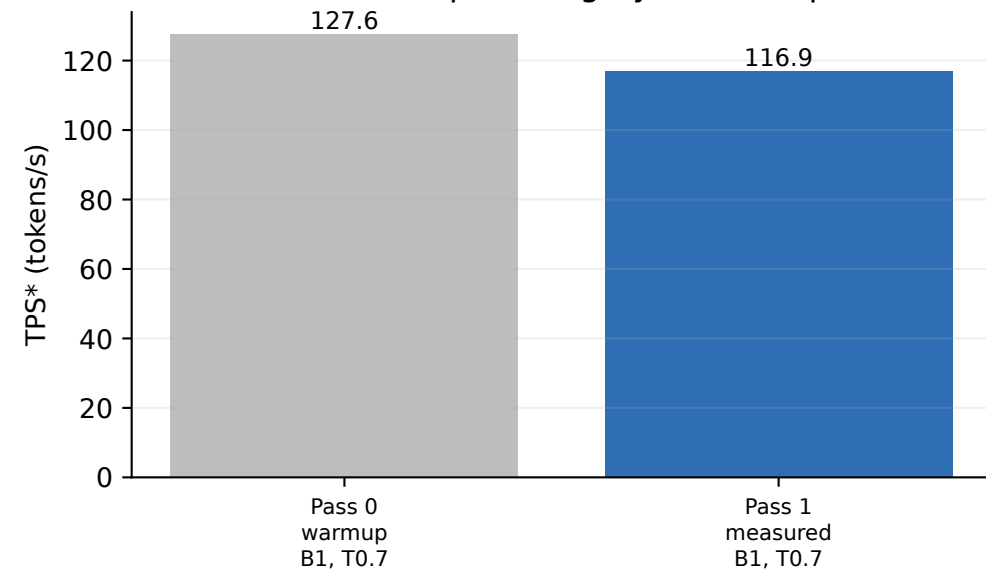
Boundary-free steps, including slow steps



Conditional AL* and source share



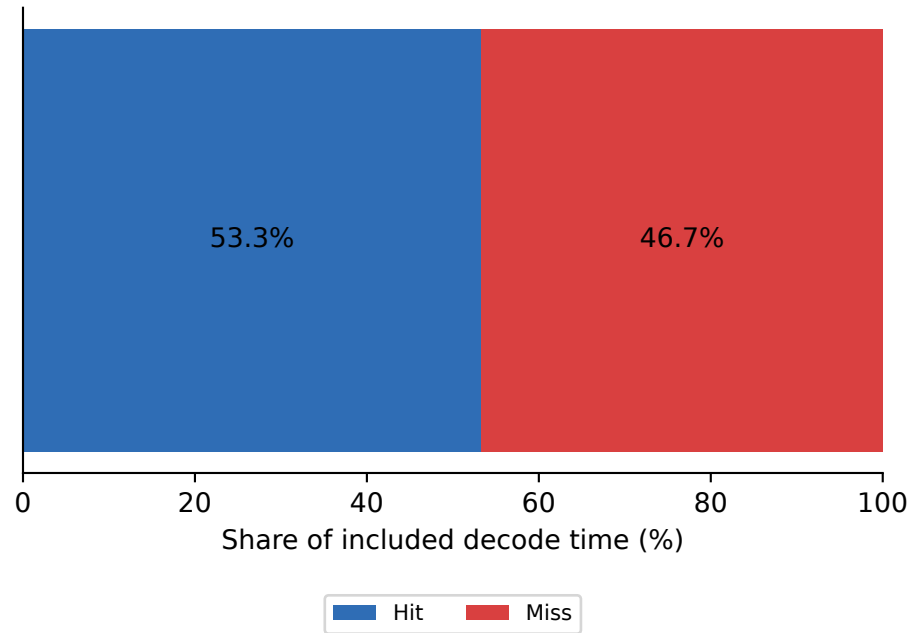
All saved passes; gray = warmup



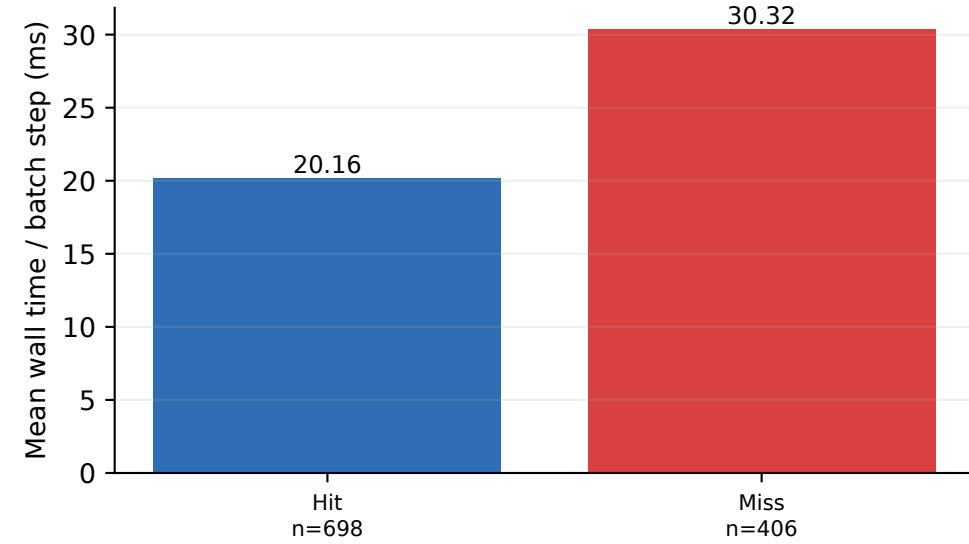
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_ssd_2
SSD: K=6, F=7
Last pass: B=1, T=0.7, N=48 | AL*=2.874, hit=63.1%, TPS*=120.27

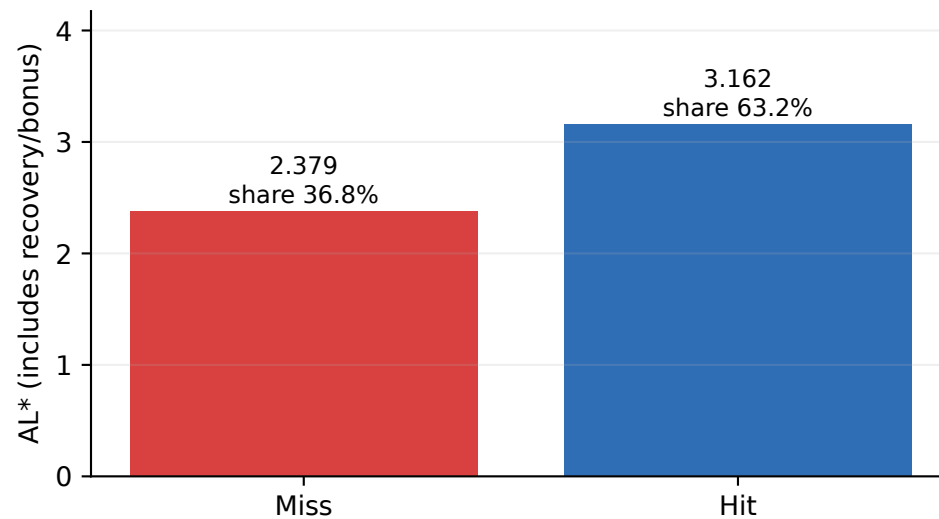
Measured wall time by batch cache status



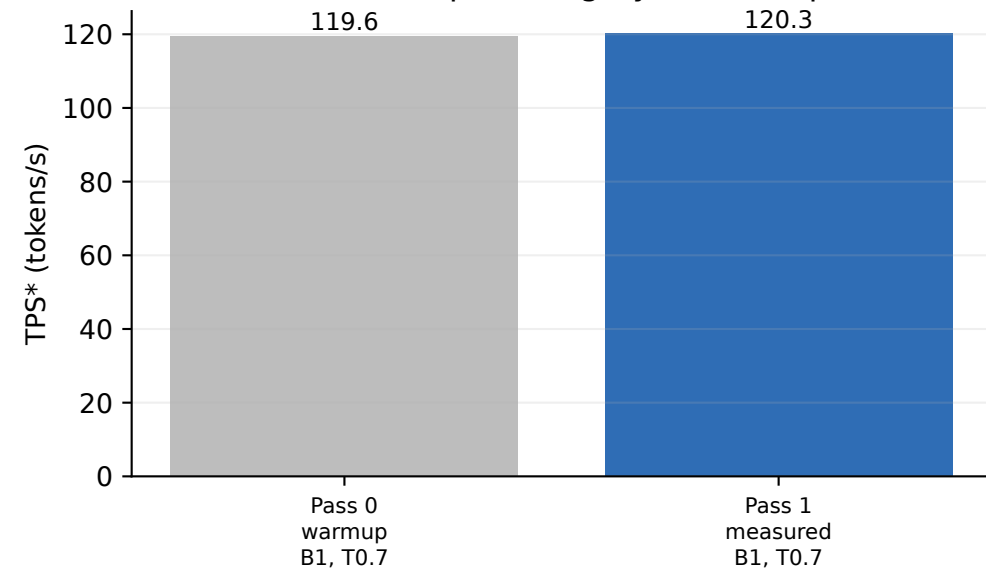
Boundary-free steps, including slow steps



Conditional AL* and source share



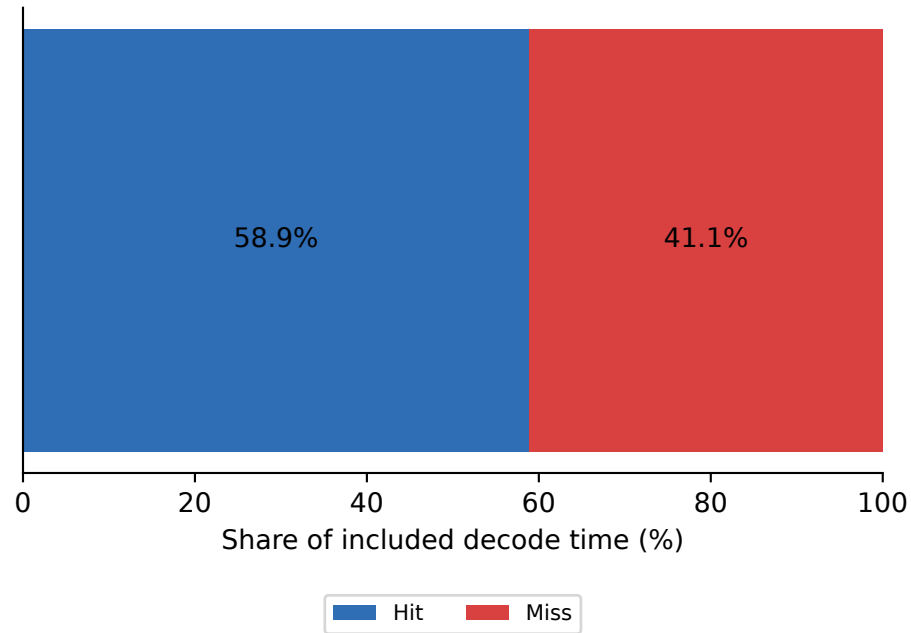
All saved passes; gray = warmup



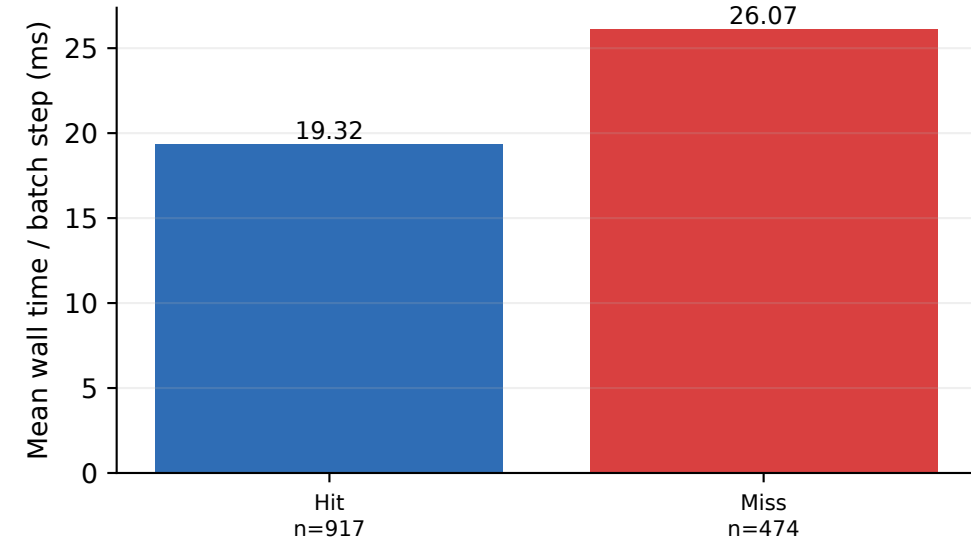
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_ssd_3
SSD: K=4, F=7
Last pass: B=1, T=0.7, N=48 | AL*=2.675, hit=65.7%, TPS*=123.73

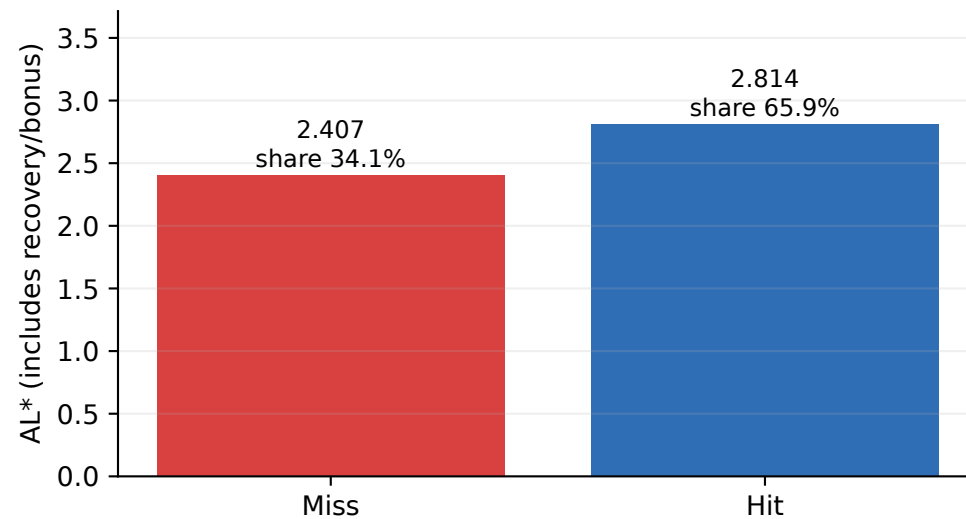
Measured wall time by batch cache status



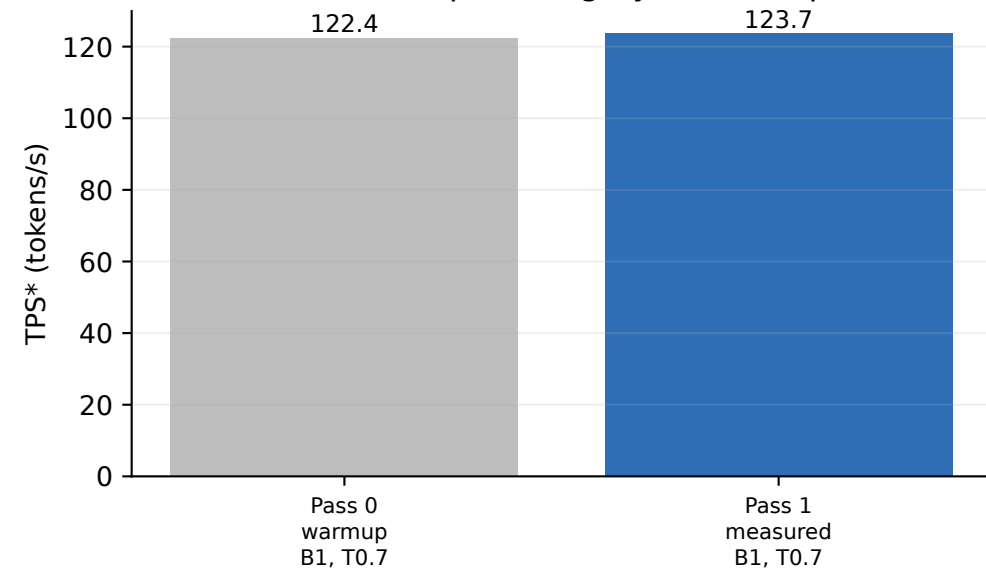
Boundary-free steps, including slow steps



Conditional AL* and source share

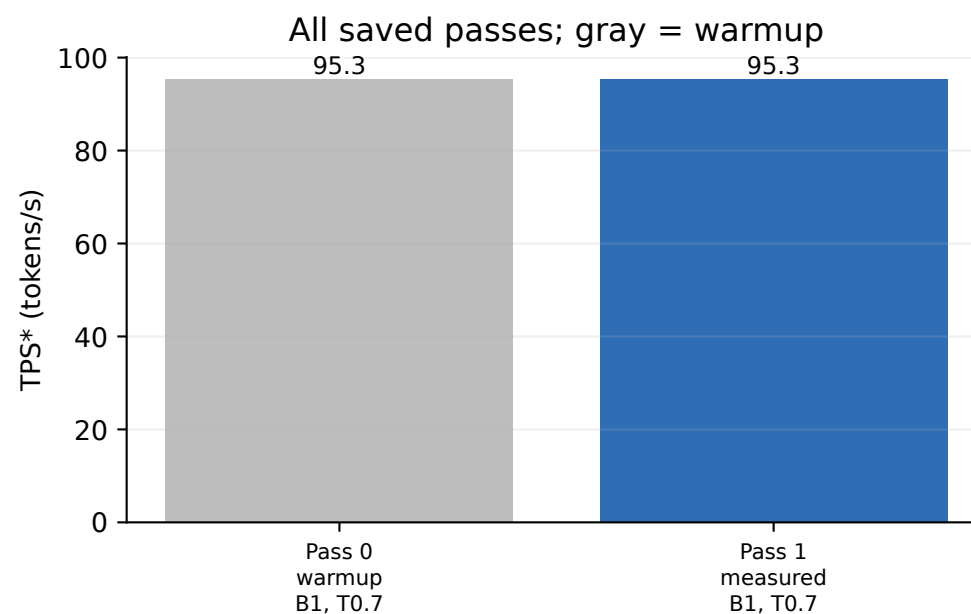
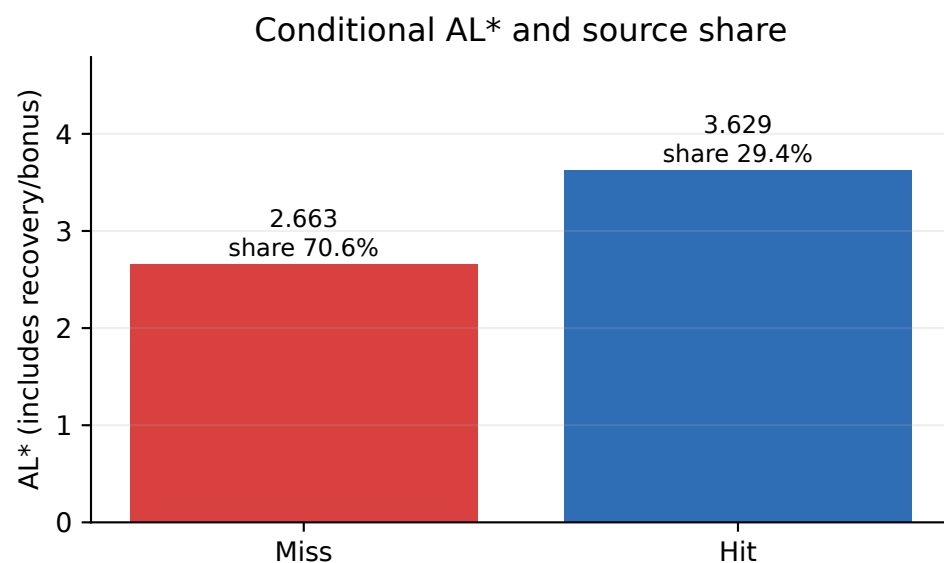
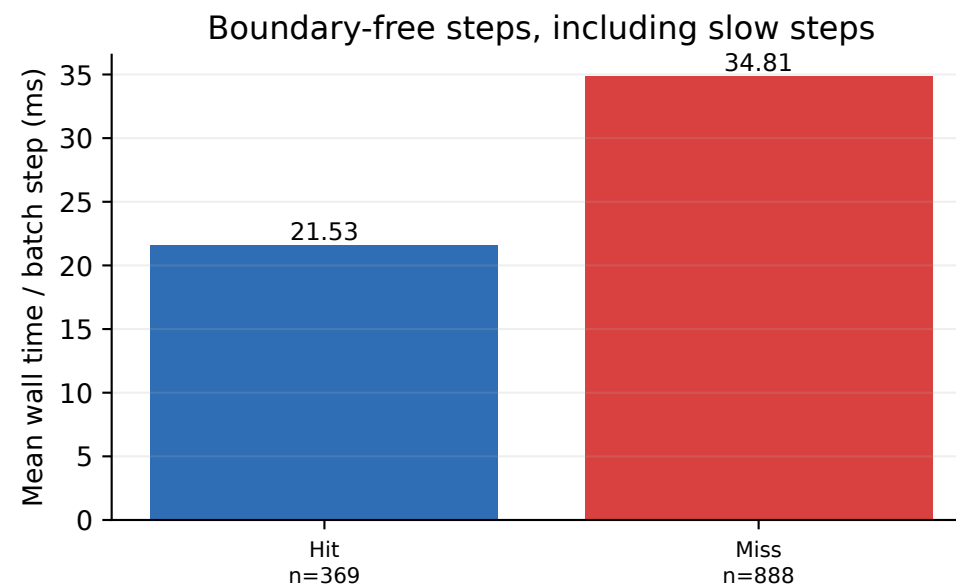
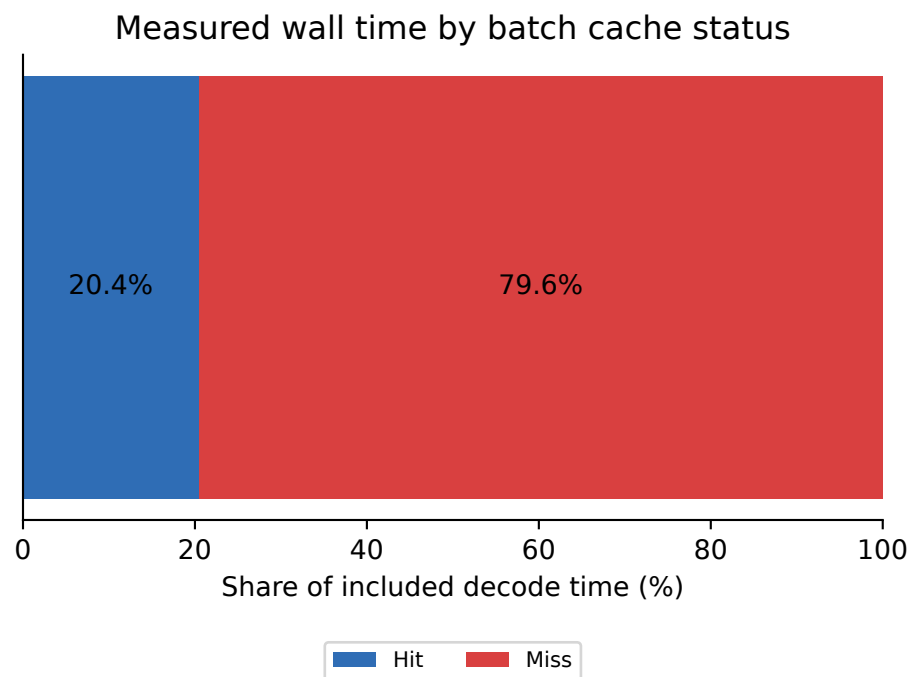


All saved passes; gray = warmup



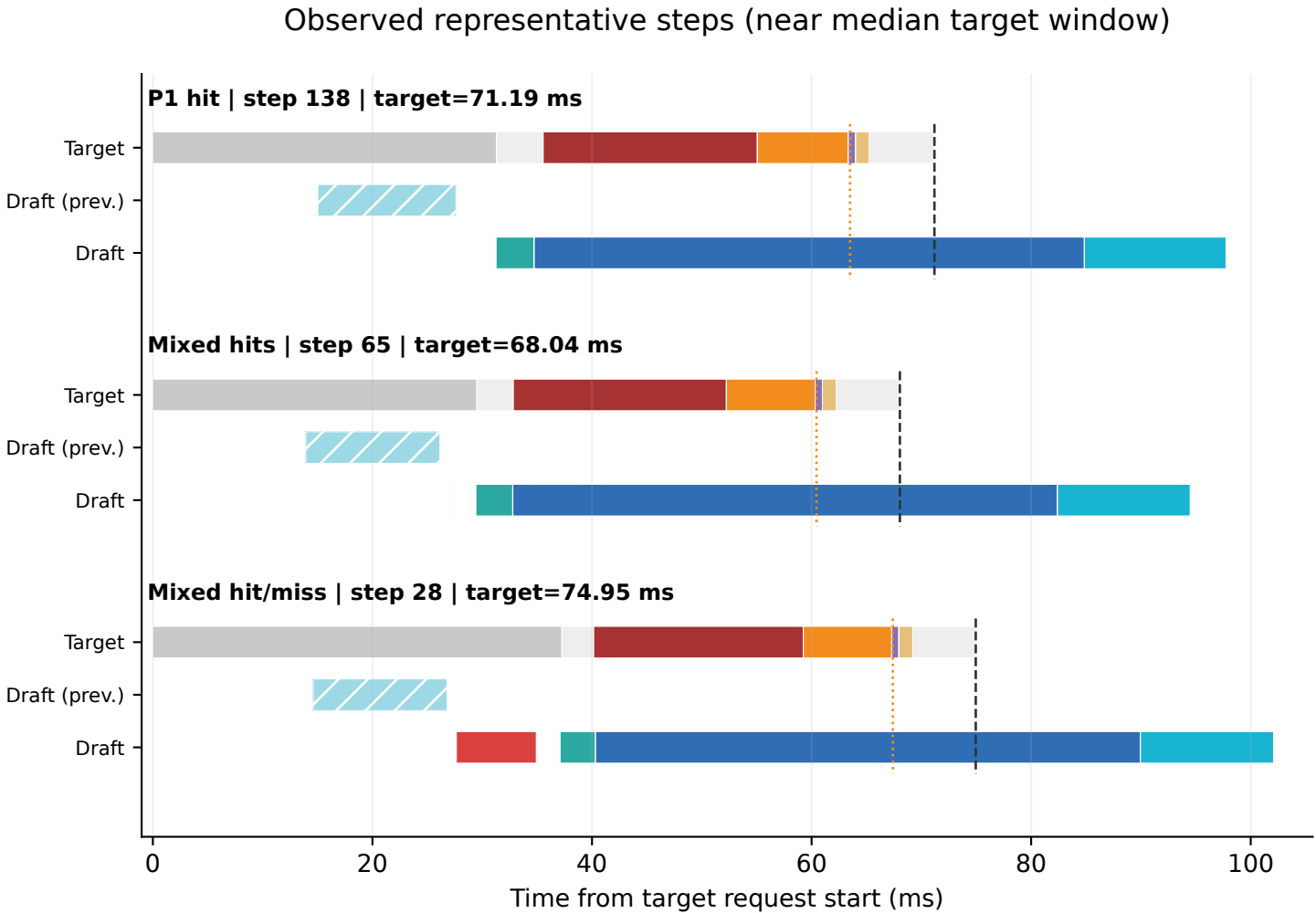
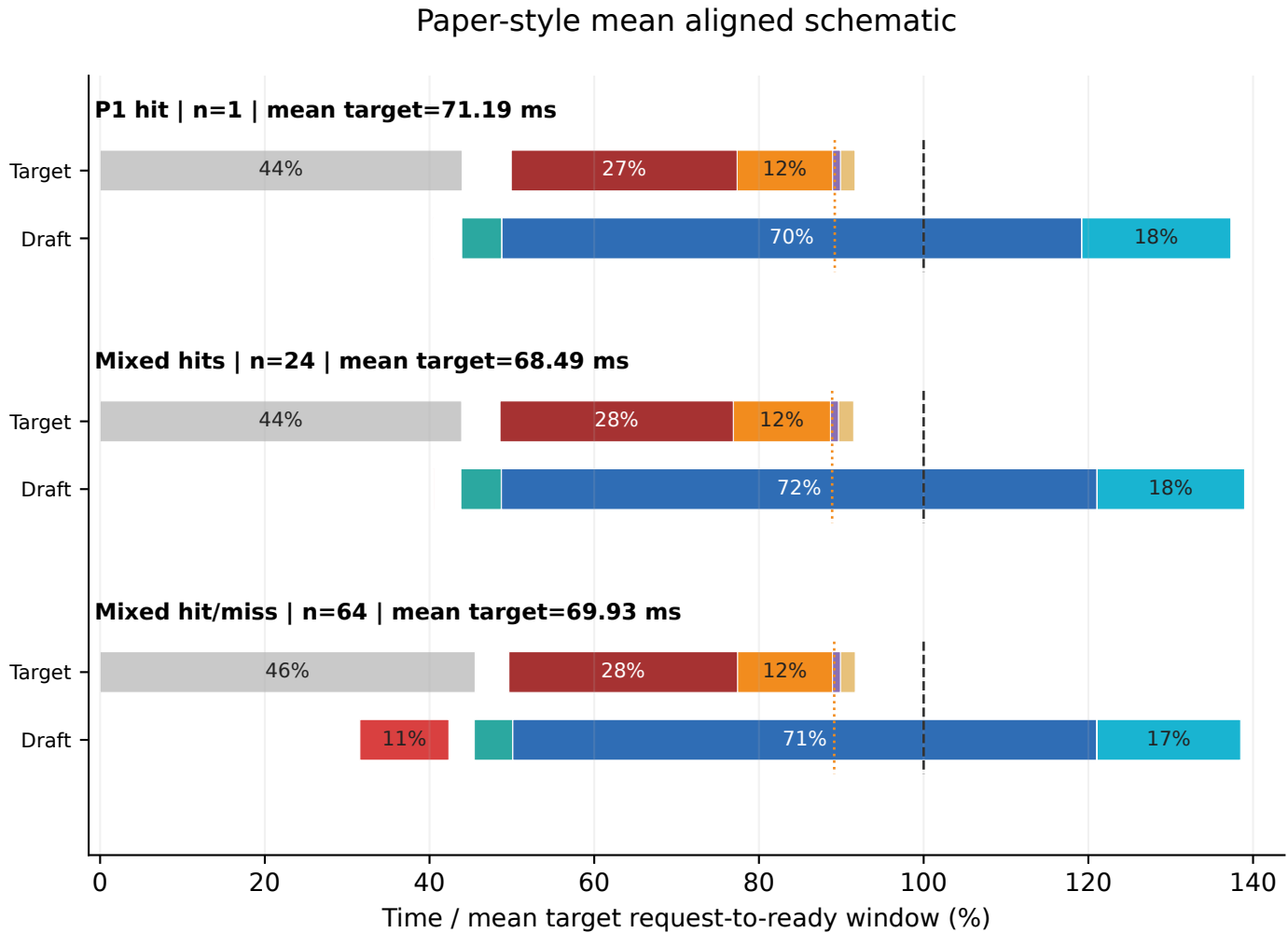
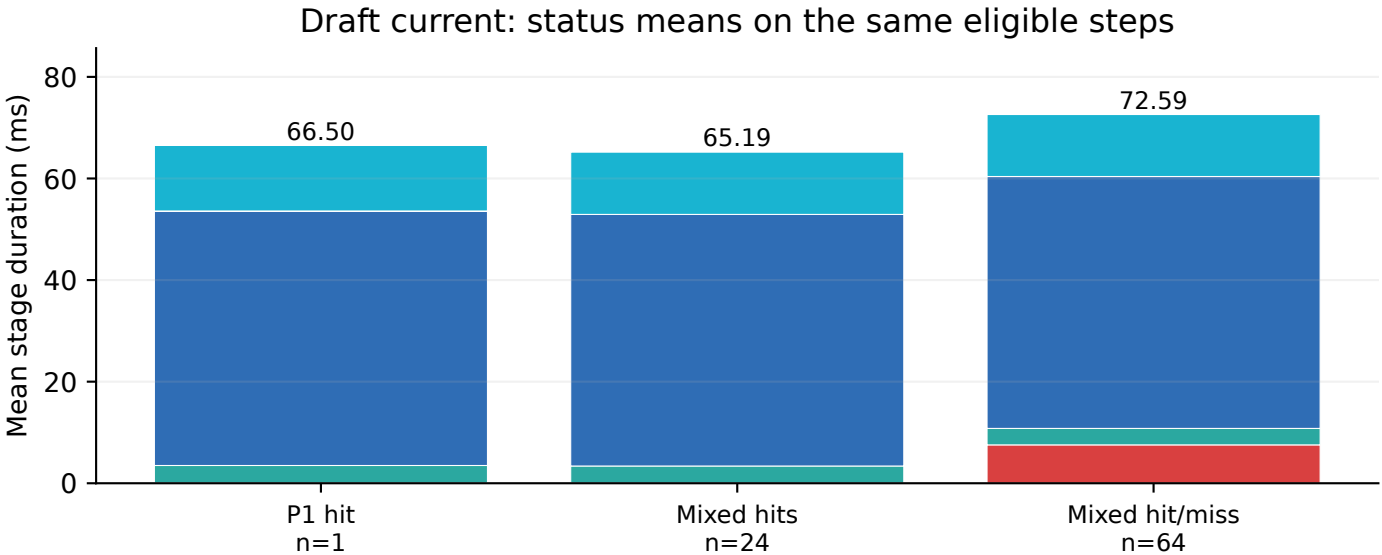
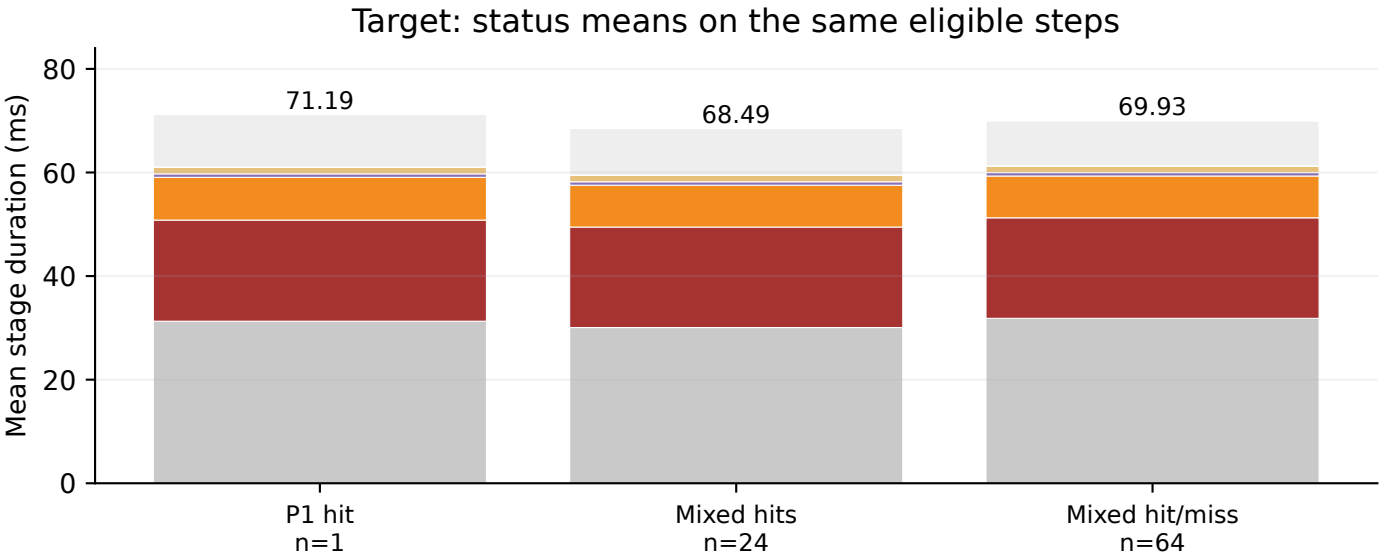
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_warm_ssd_4
SSD: K=8, F=1
Last pass: B=1, T=0.7, N=48 | AL*=2.947, hit=30.2%, TPS*=95.33



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

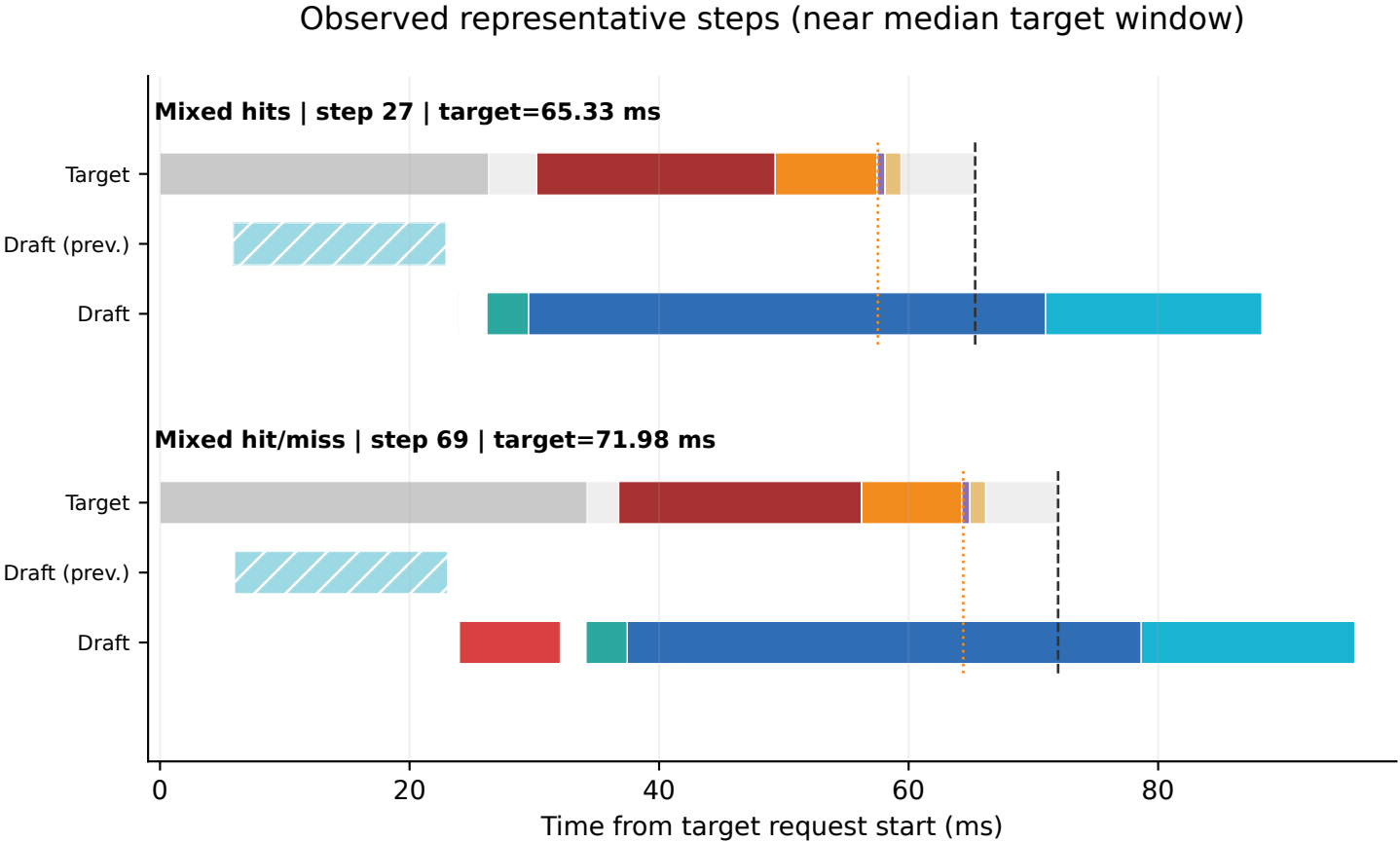
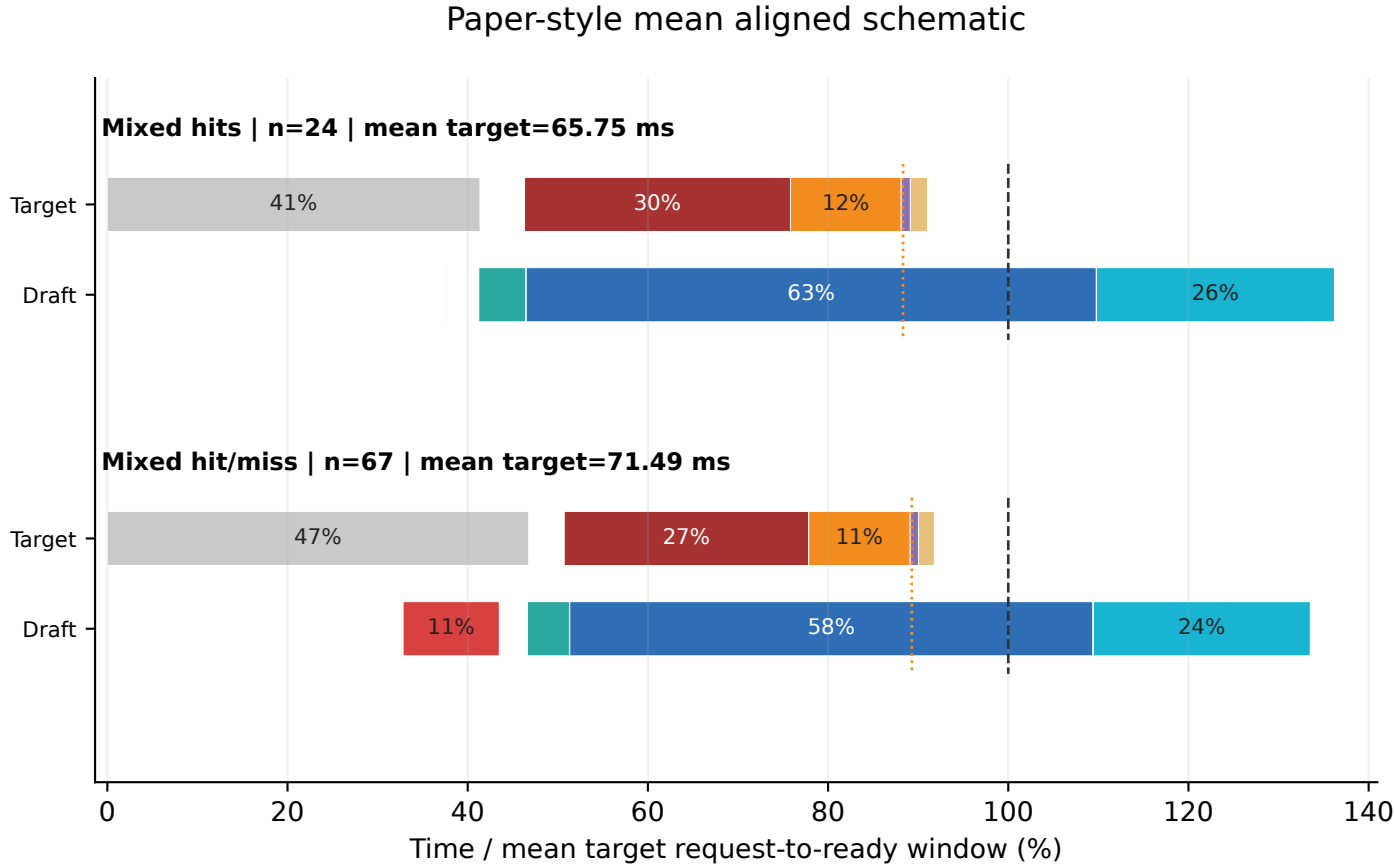
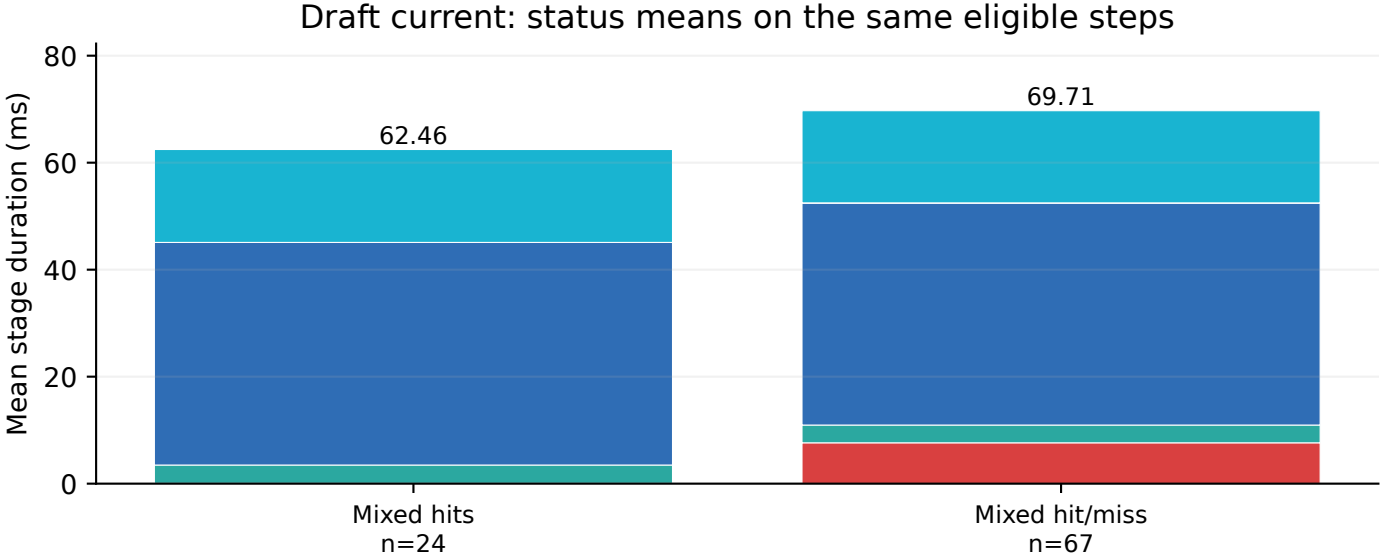
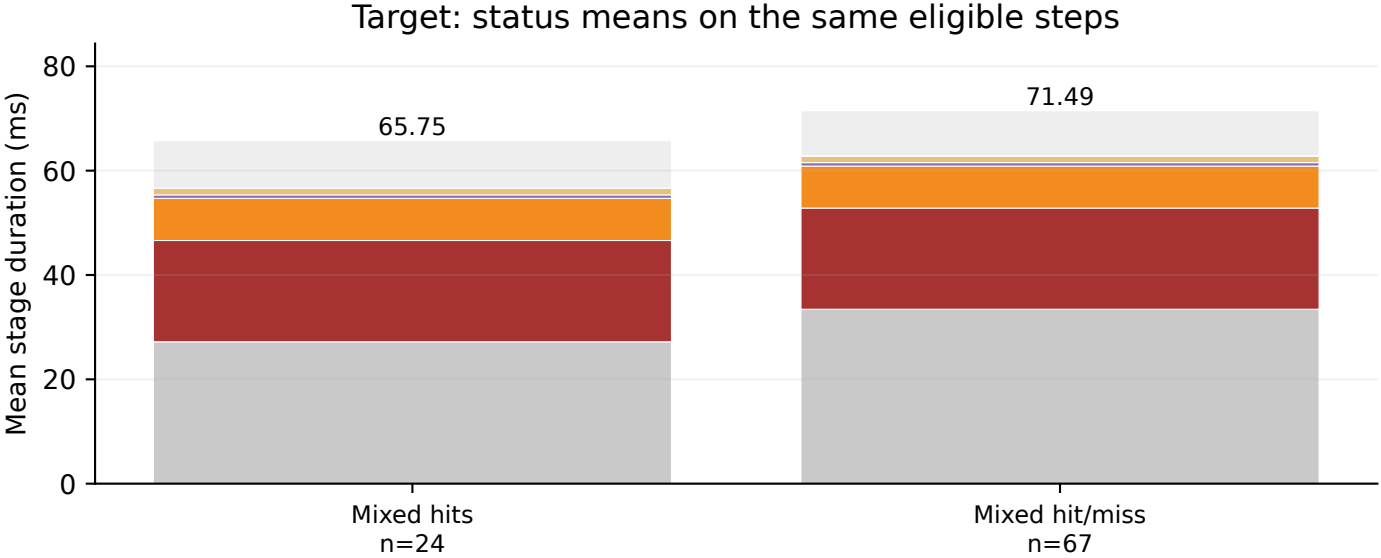
llama3_b8_combo_al_k1_7
DUET: K1/K2=7/2, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=8, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 89 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

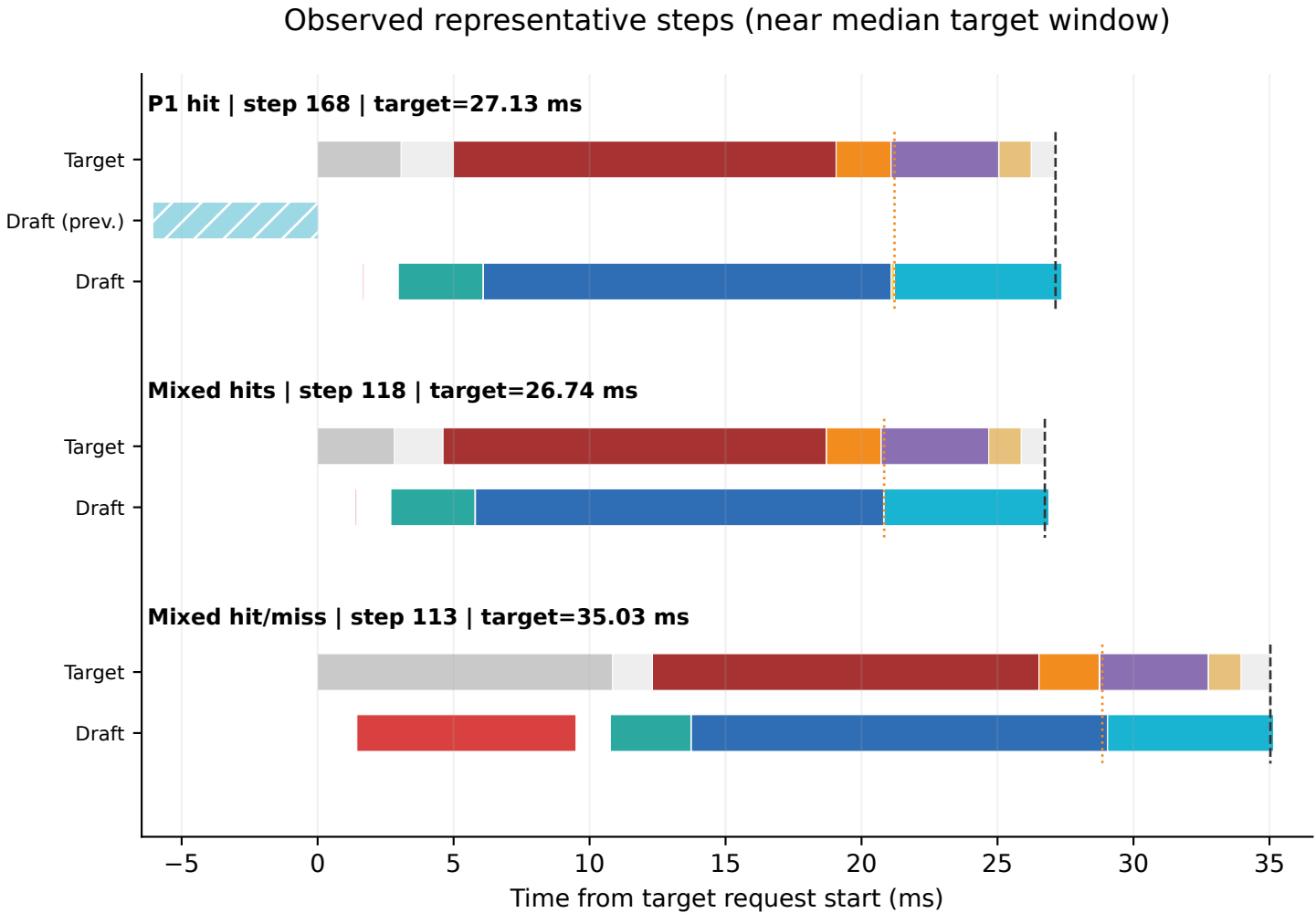
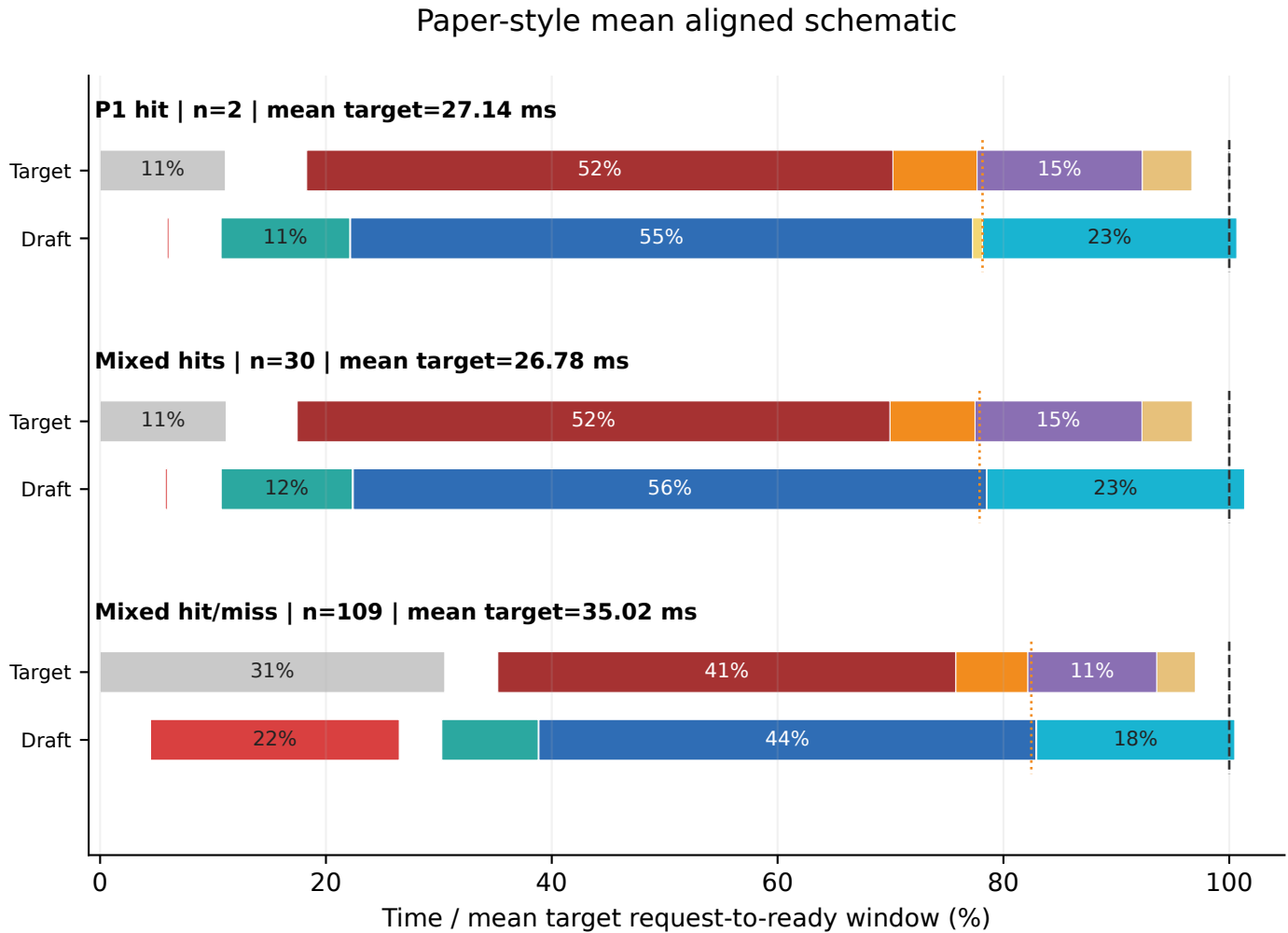
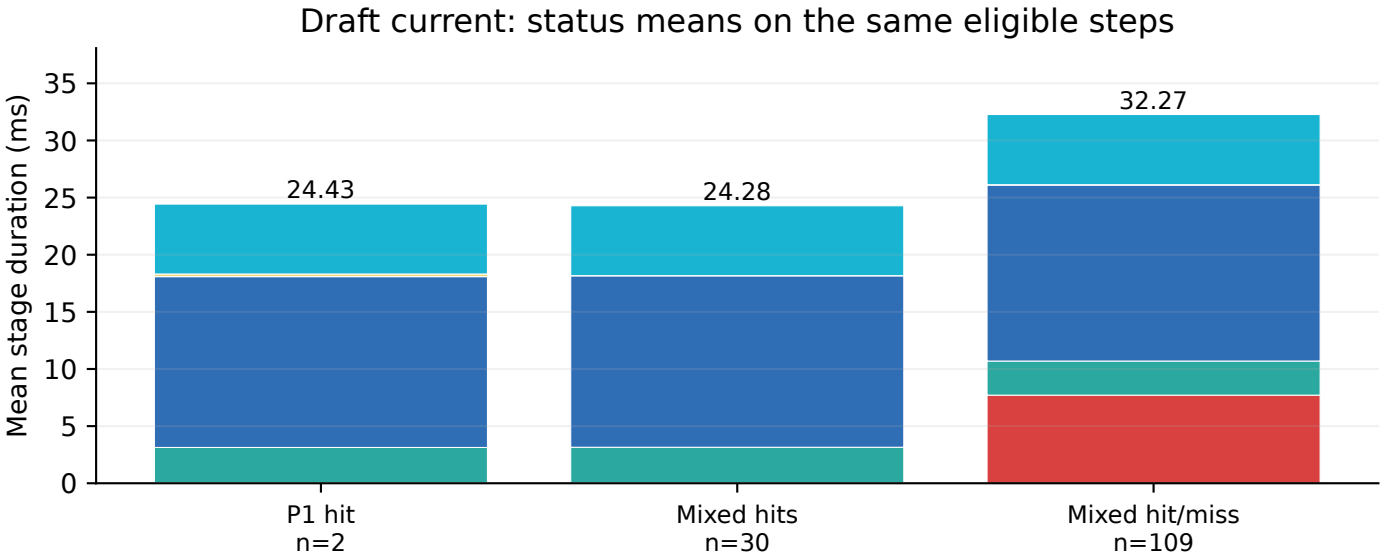
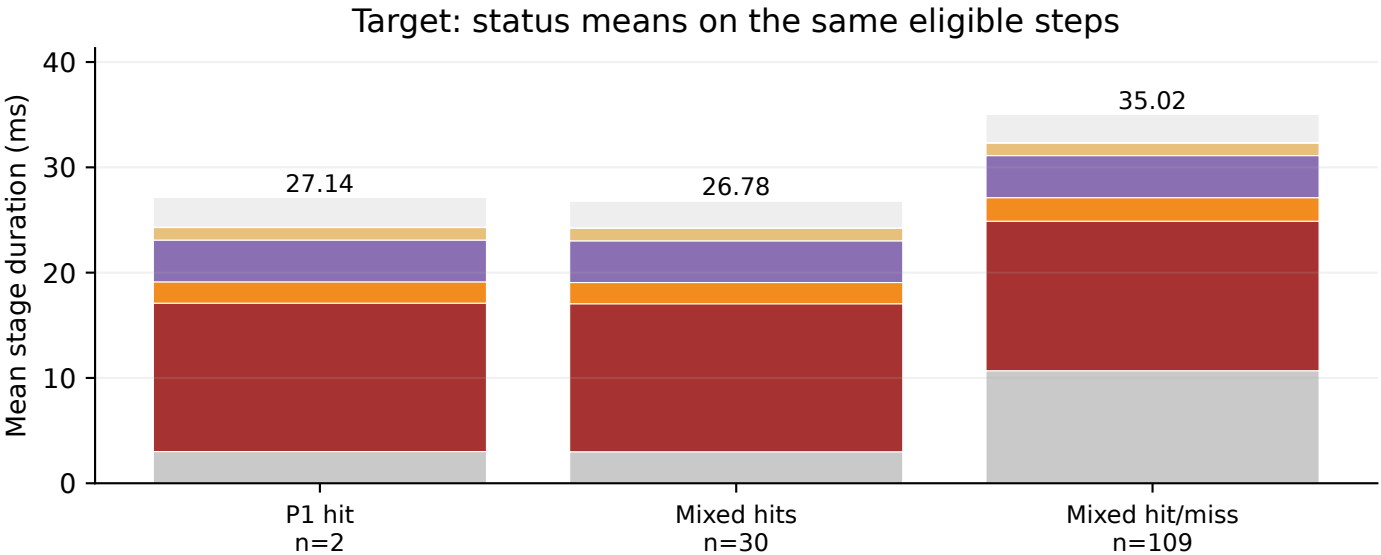
llama3_b8_combo_al_k2_3
DUET: K1/K2=6/3, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 91 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

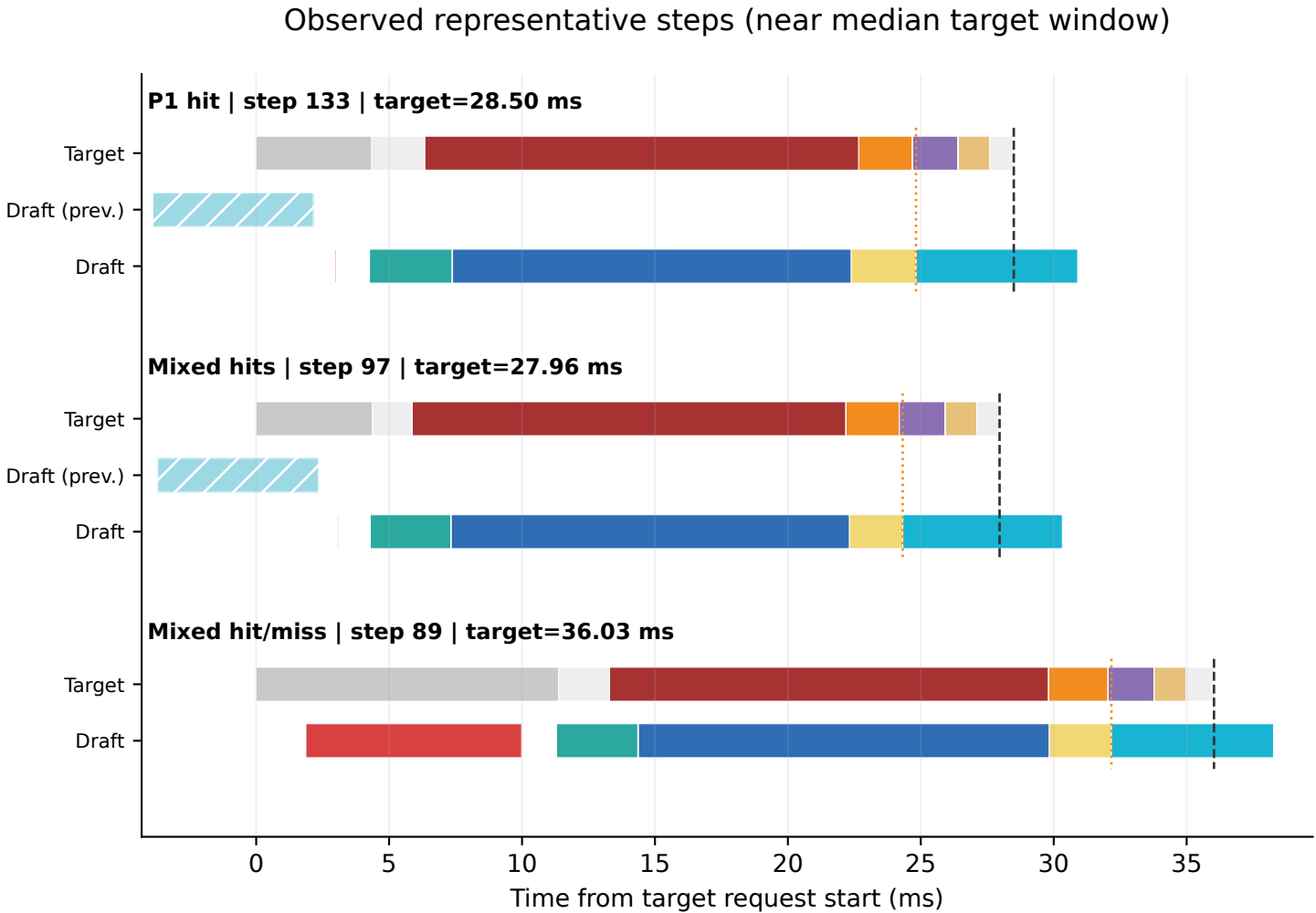
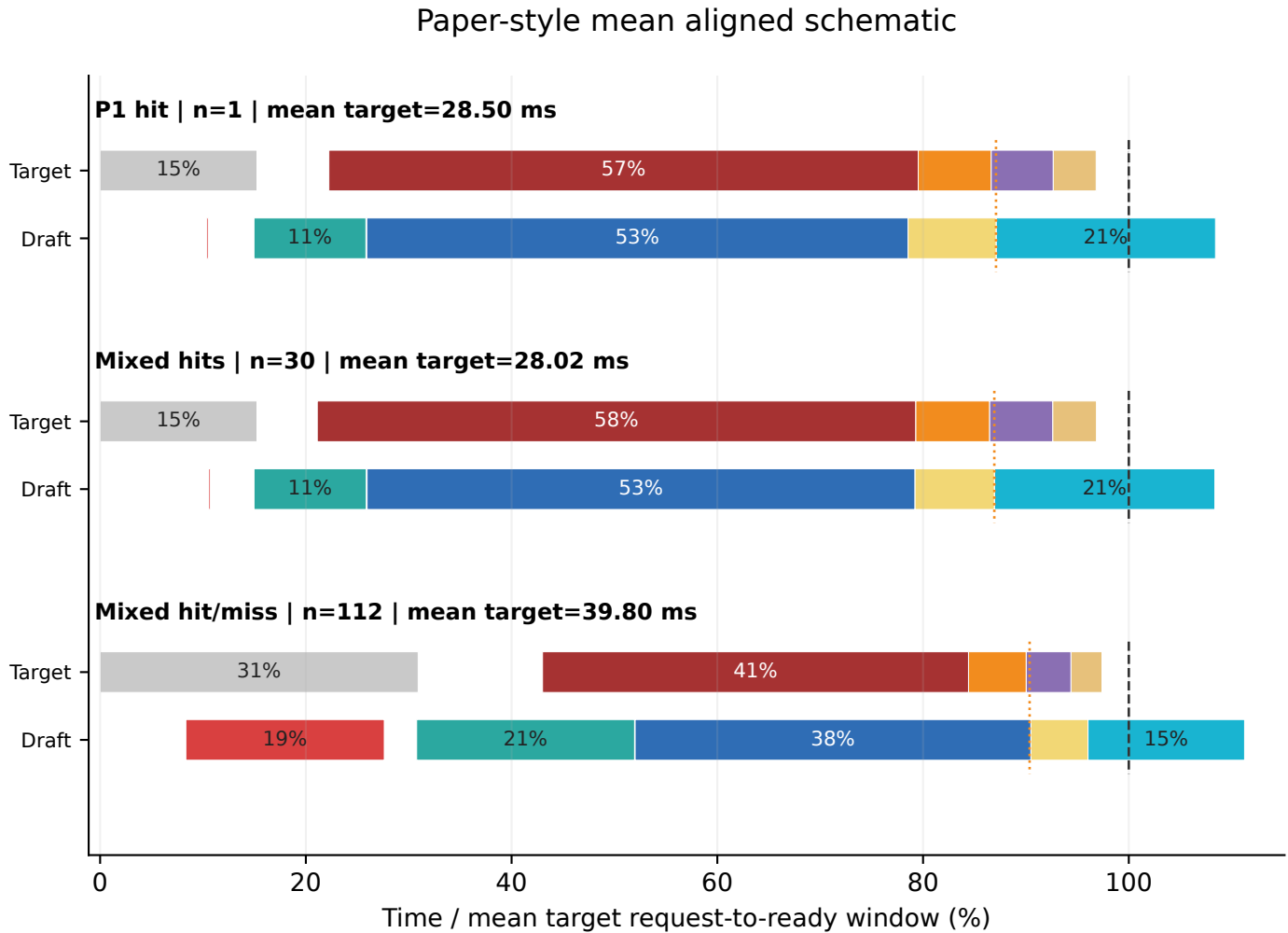
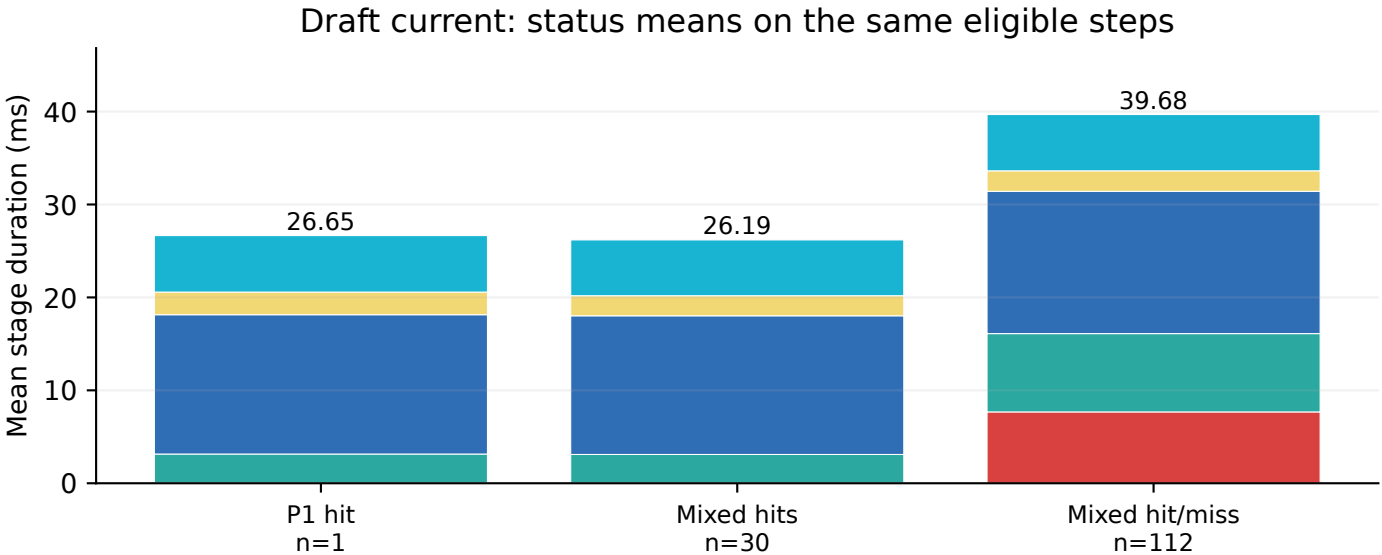
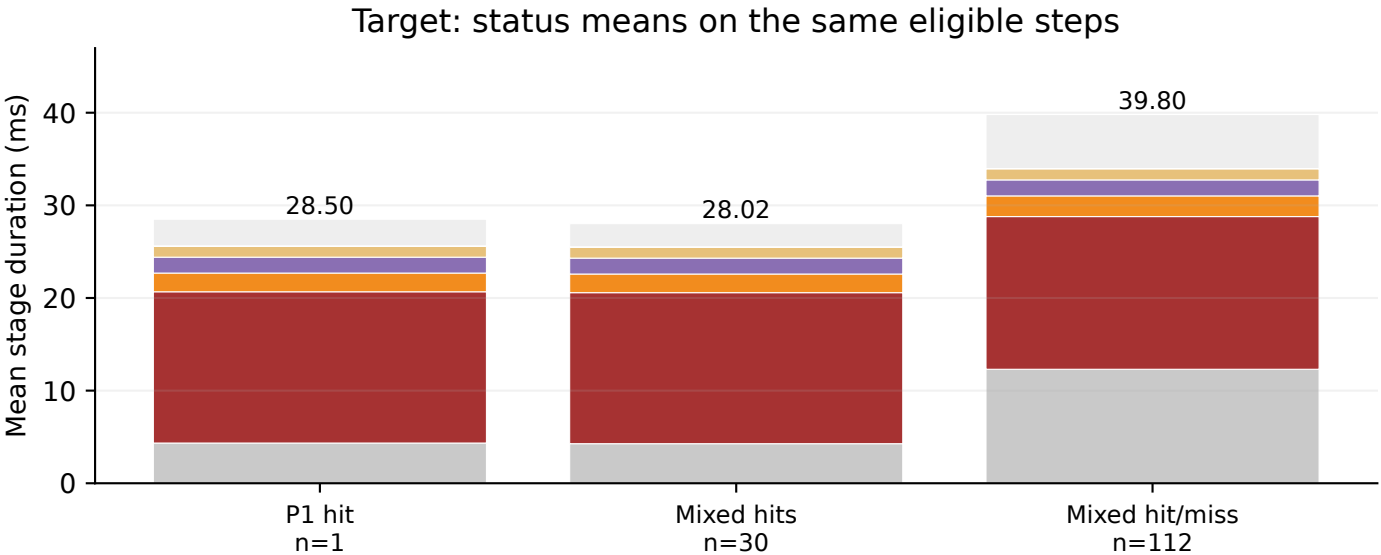
llama3_b8_combo_e24
DUET: K1/K2=3/1, exit=24 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 141 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

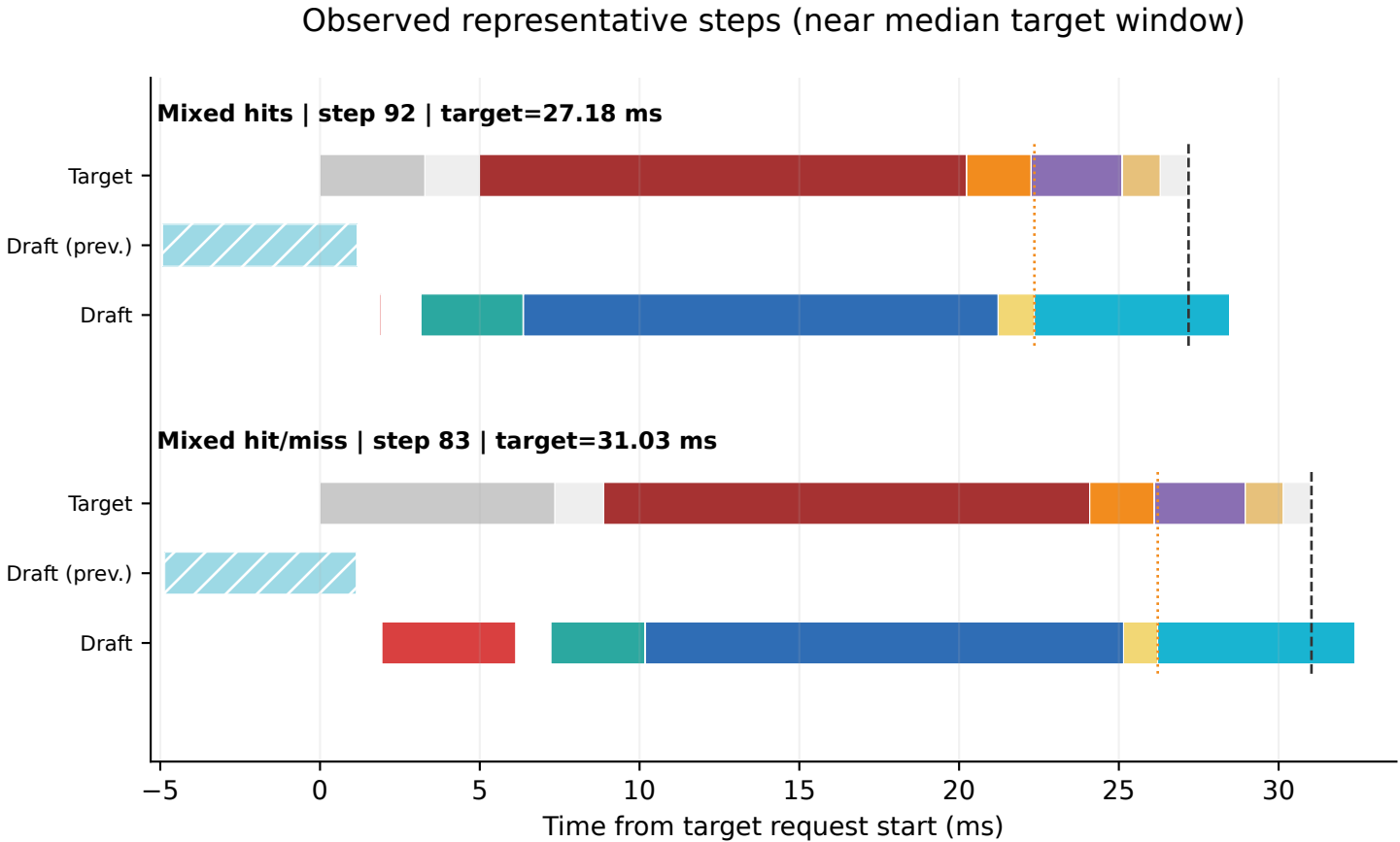
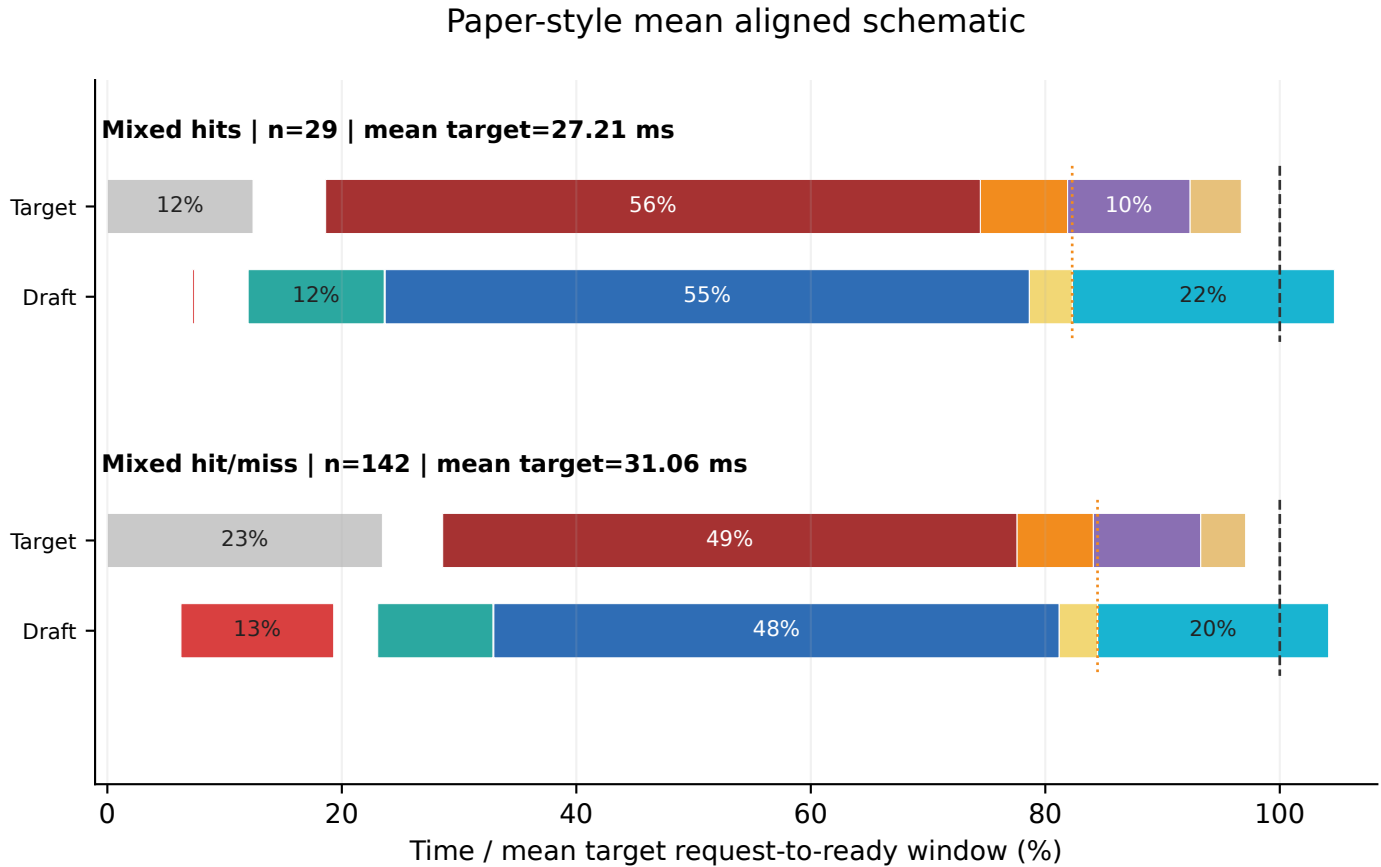
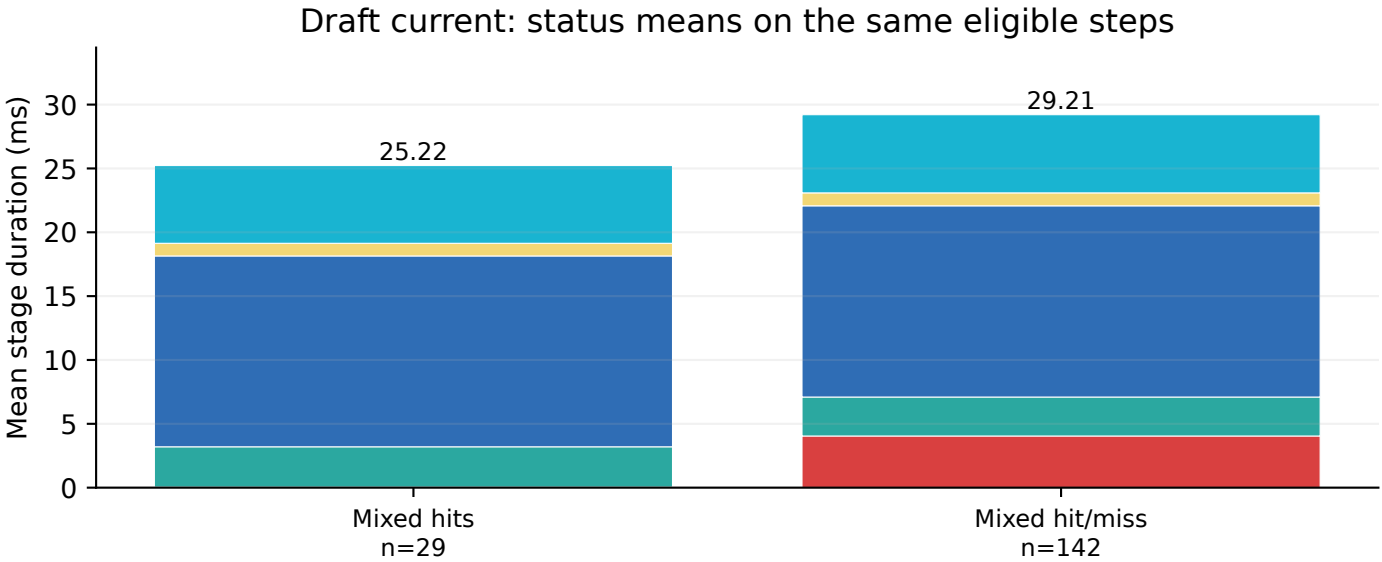
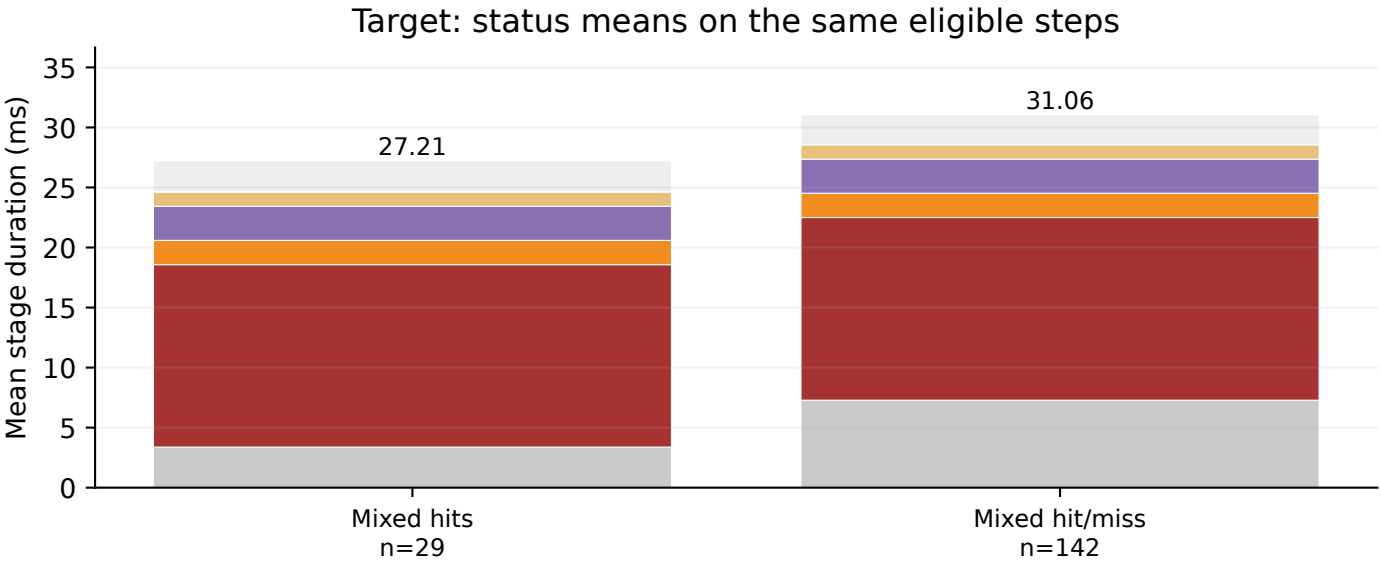
llama3_b8_combo_e28
DUET: K1/K2=3/1, exit=28 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 143 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

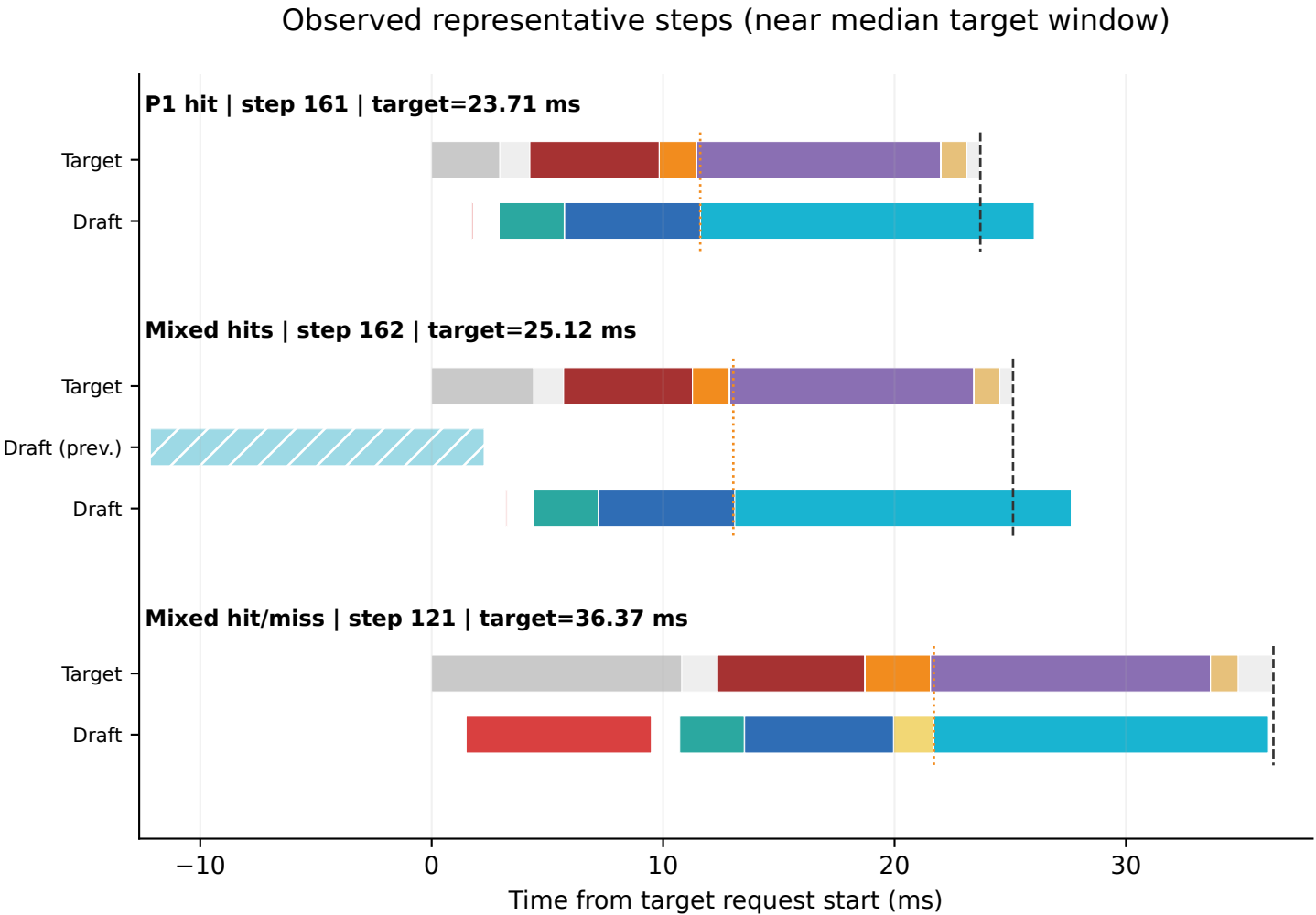
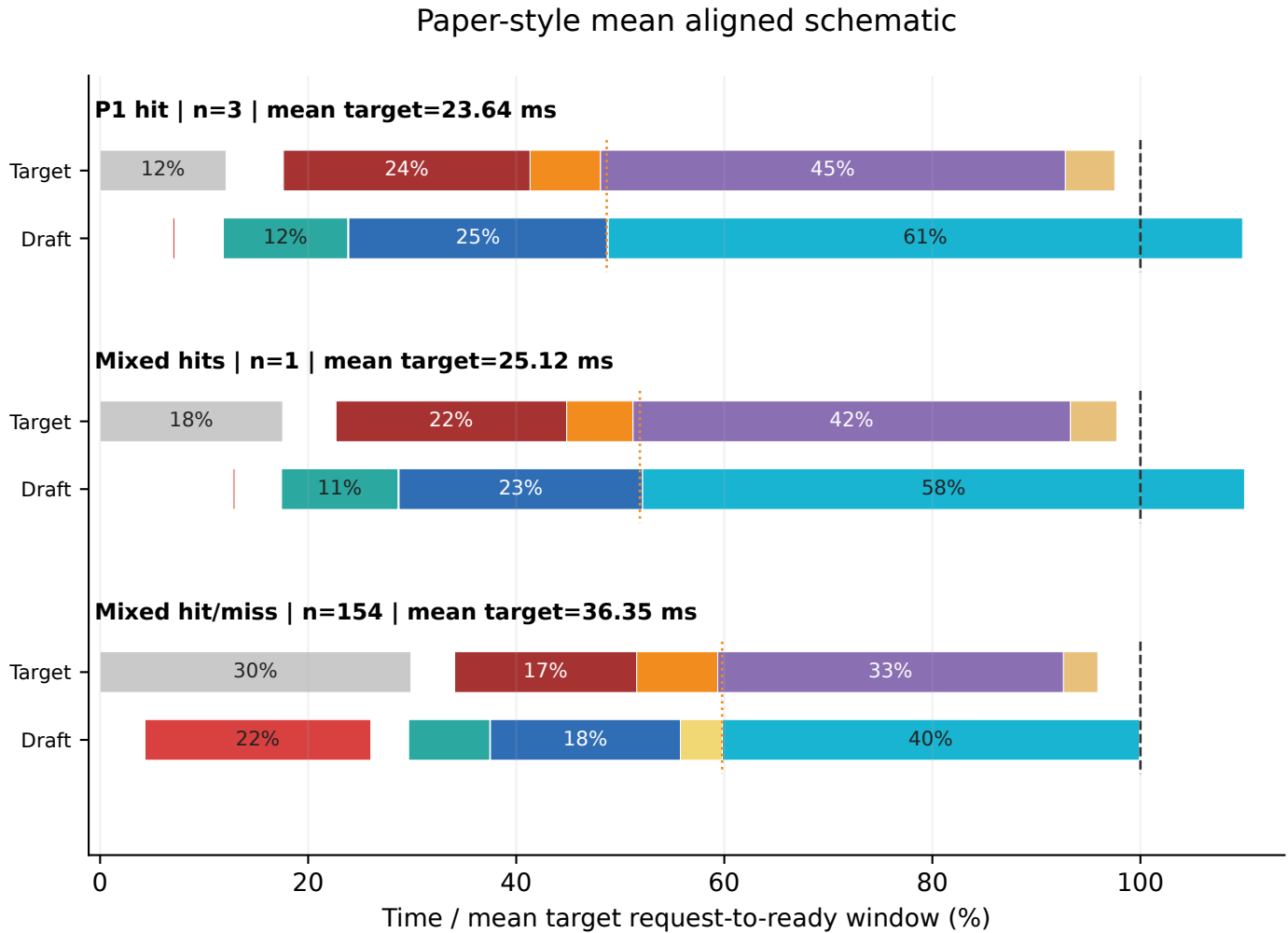
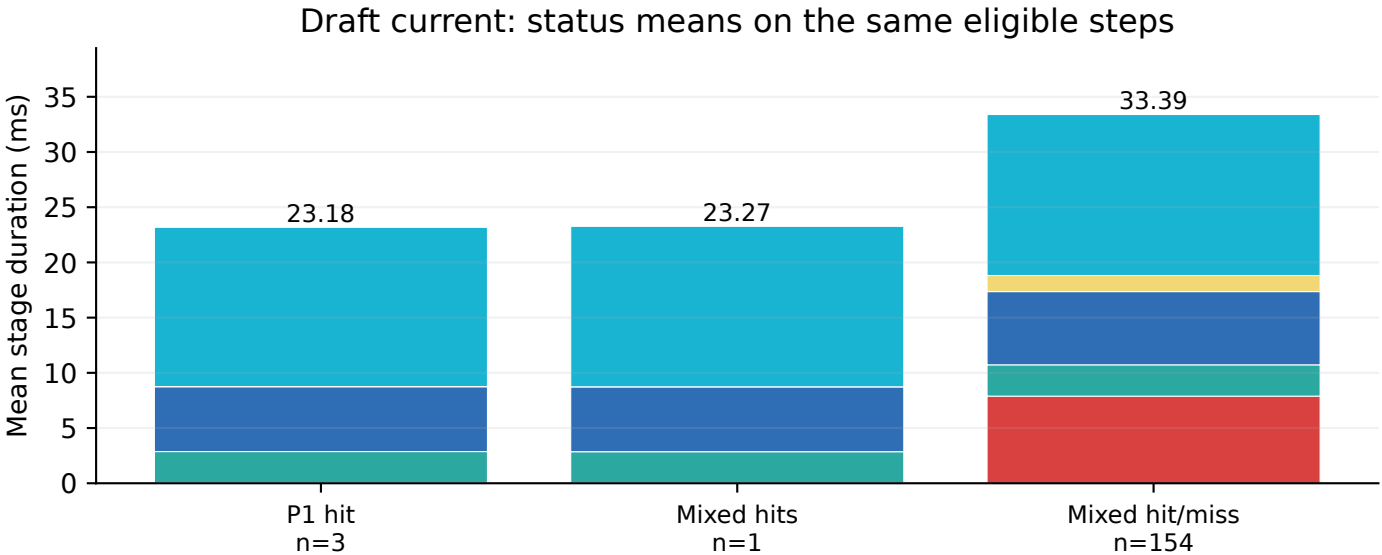
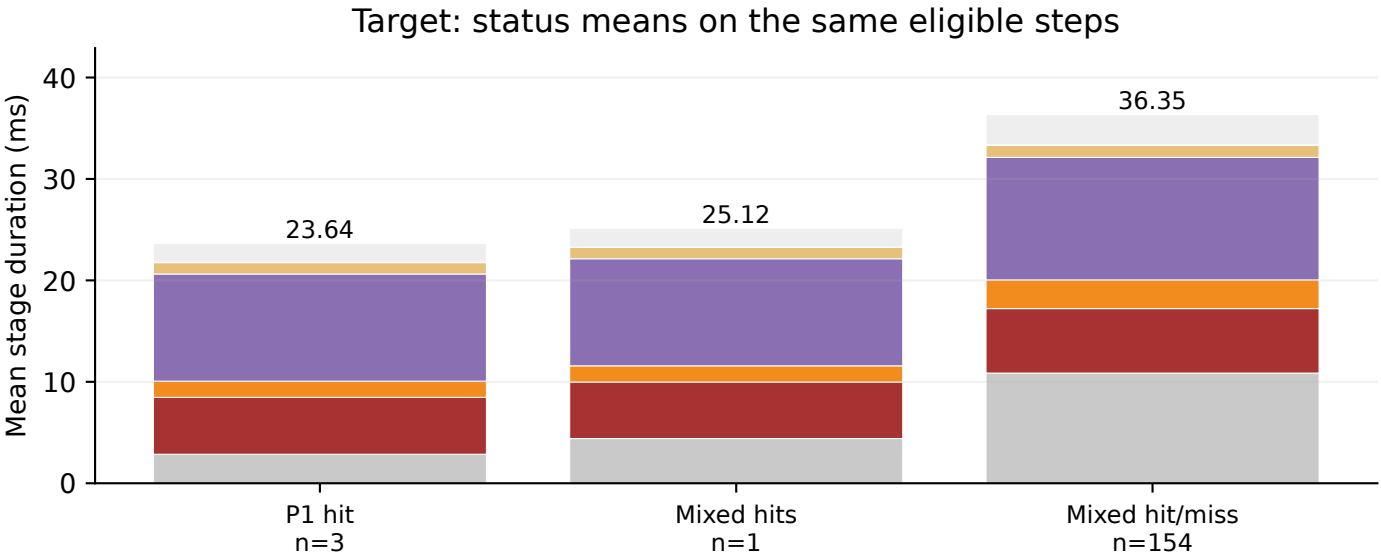
llama3_b8_combo_miss2
DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 171 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

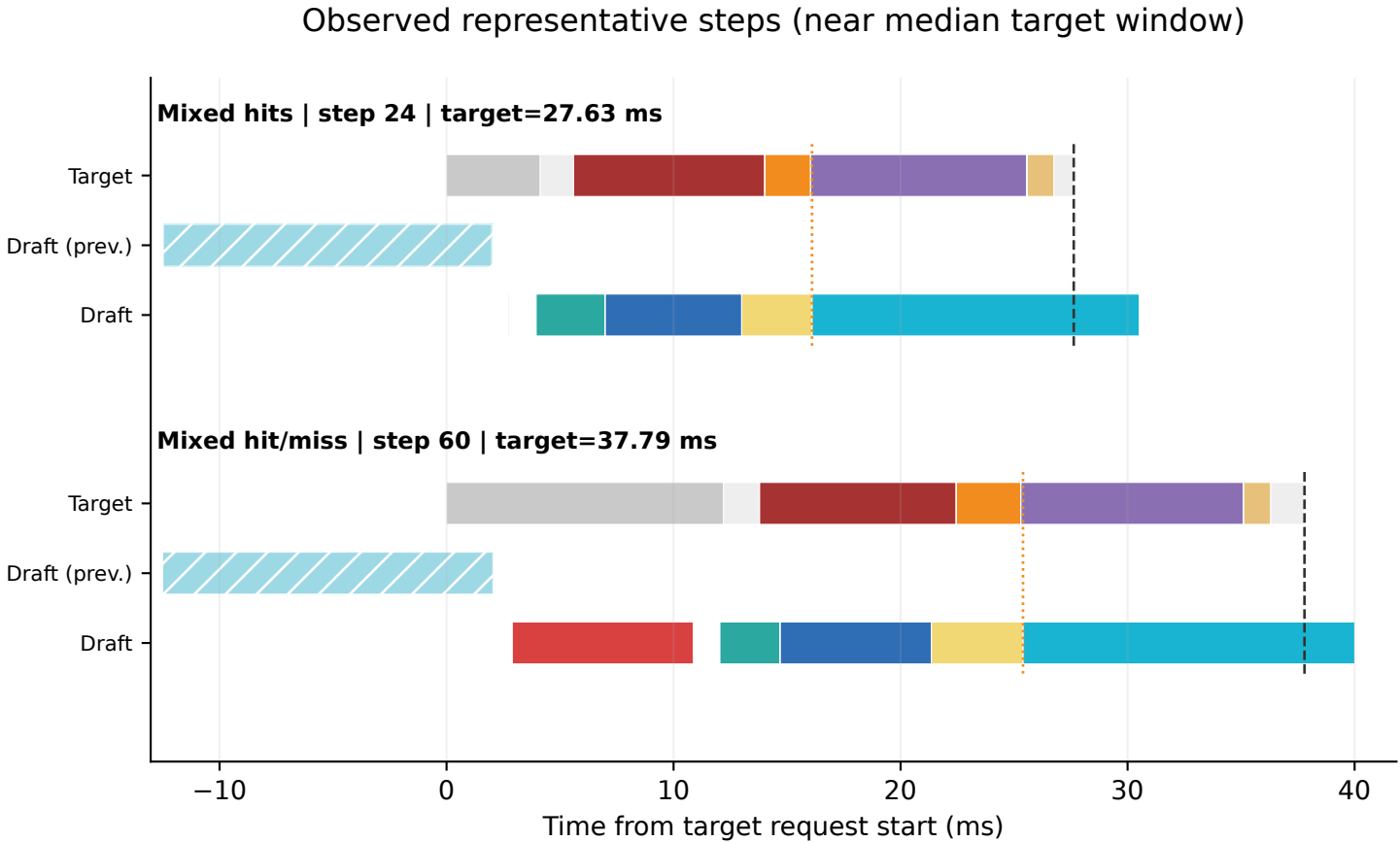
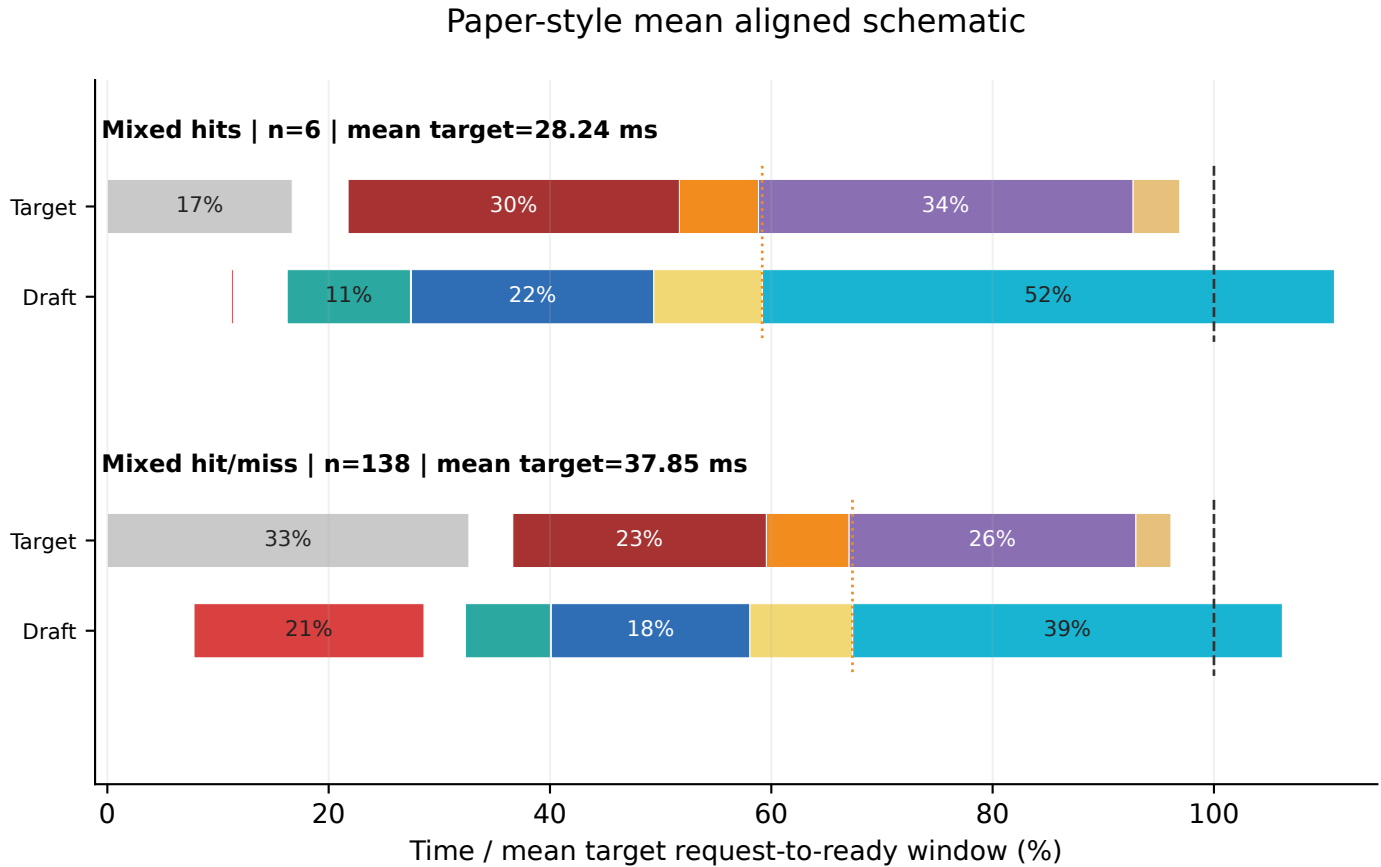
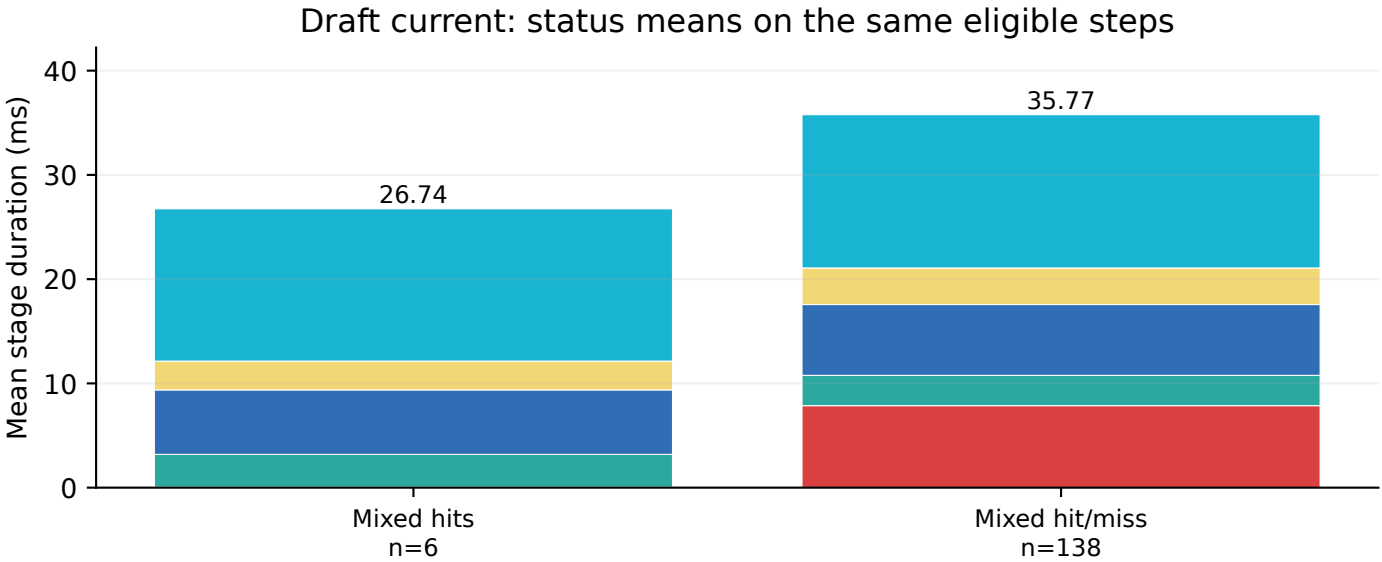
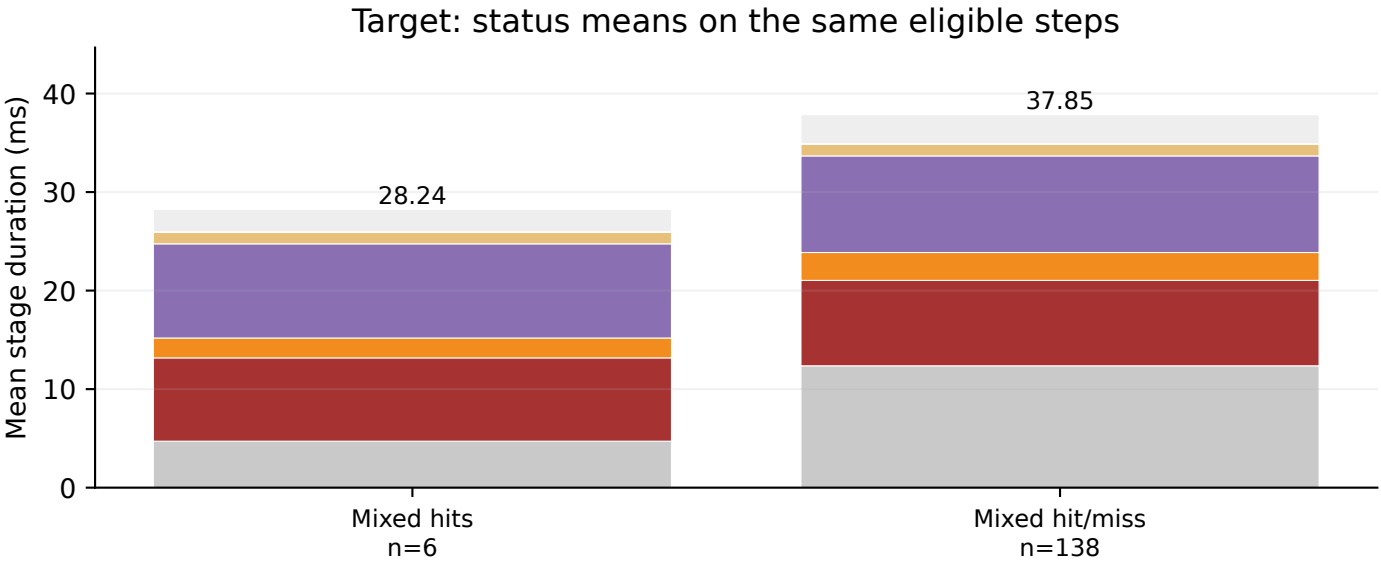
llama3_b8_combo_transfer_1_3_e10
DUET: K1/K2=1/3, exit=10 (0-based), C=1, N1/N2=2/6, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 158 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

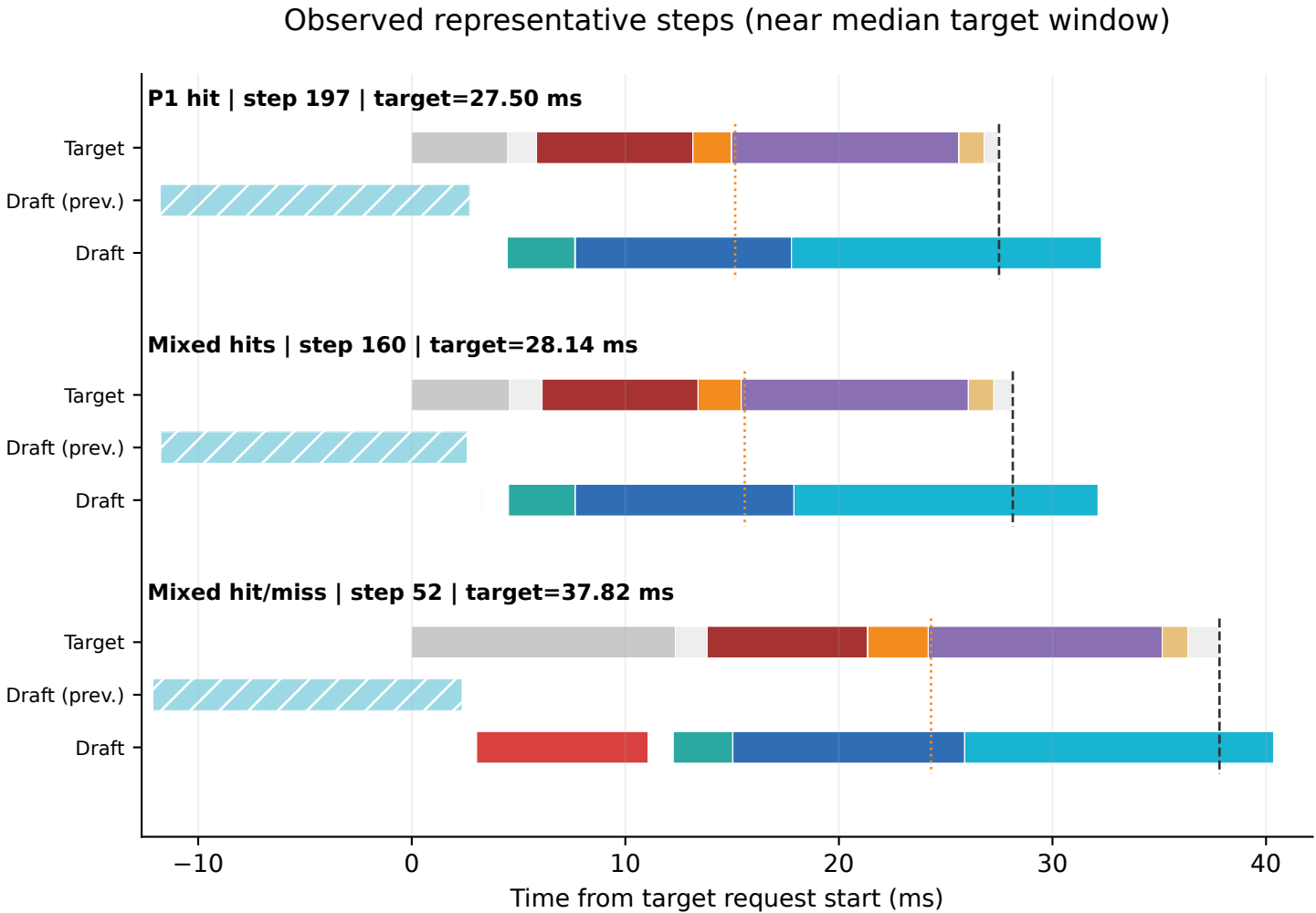
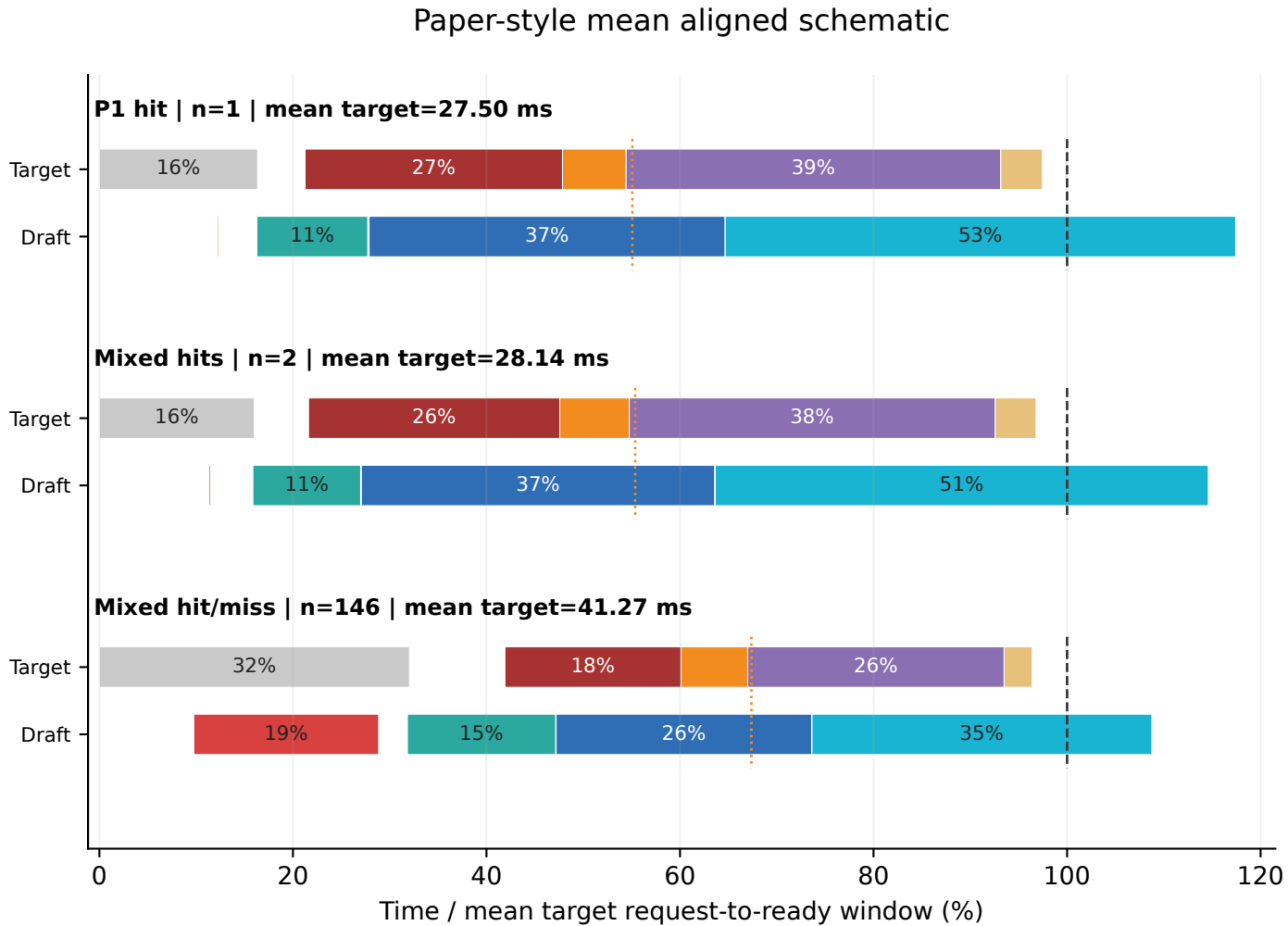
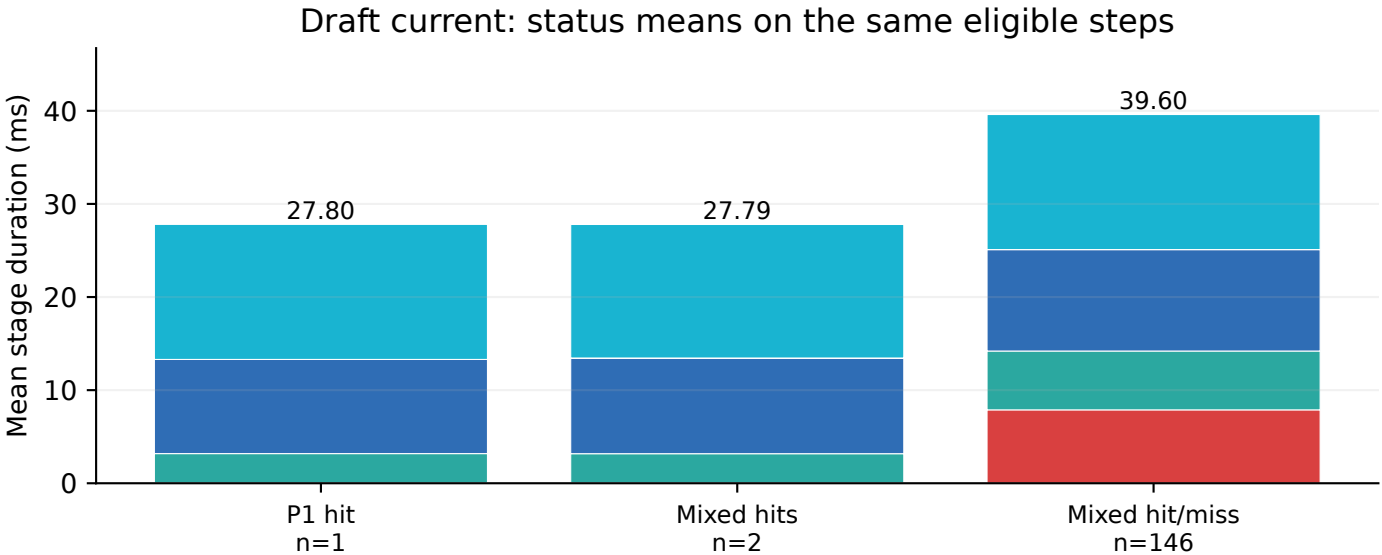
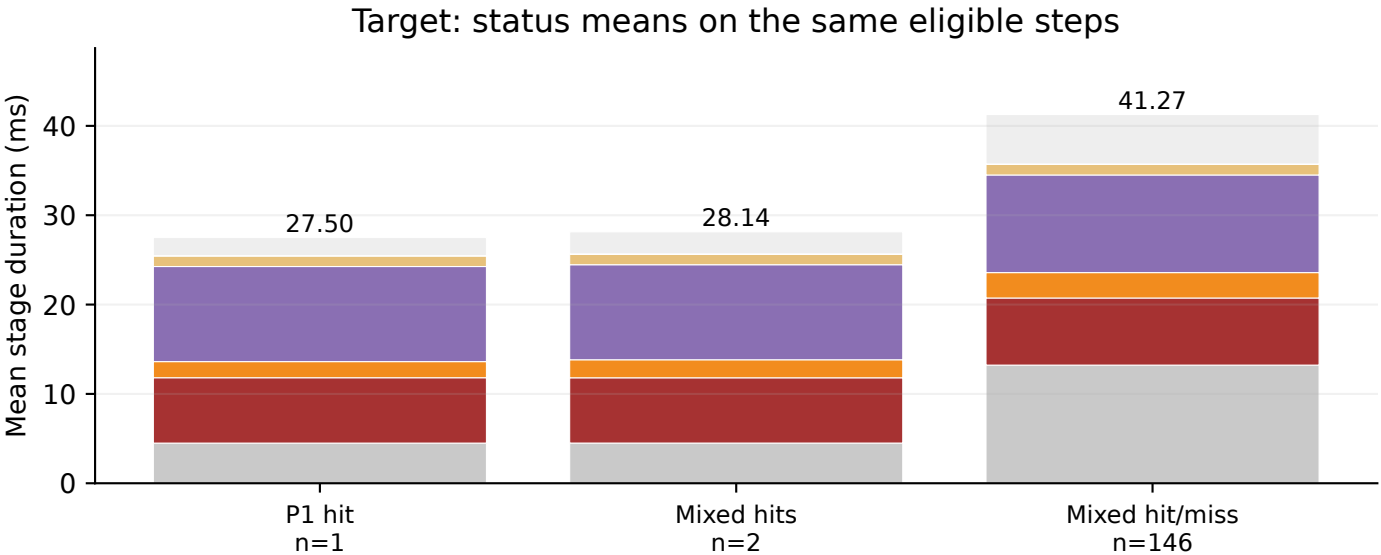
llama3_b8_combo_transfer_1_3_e14
DUET: K1/K2=1/3, exit=14 (0-based), C=1, N1/N2=2/6, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 144 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

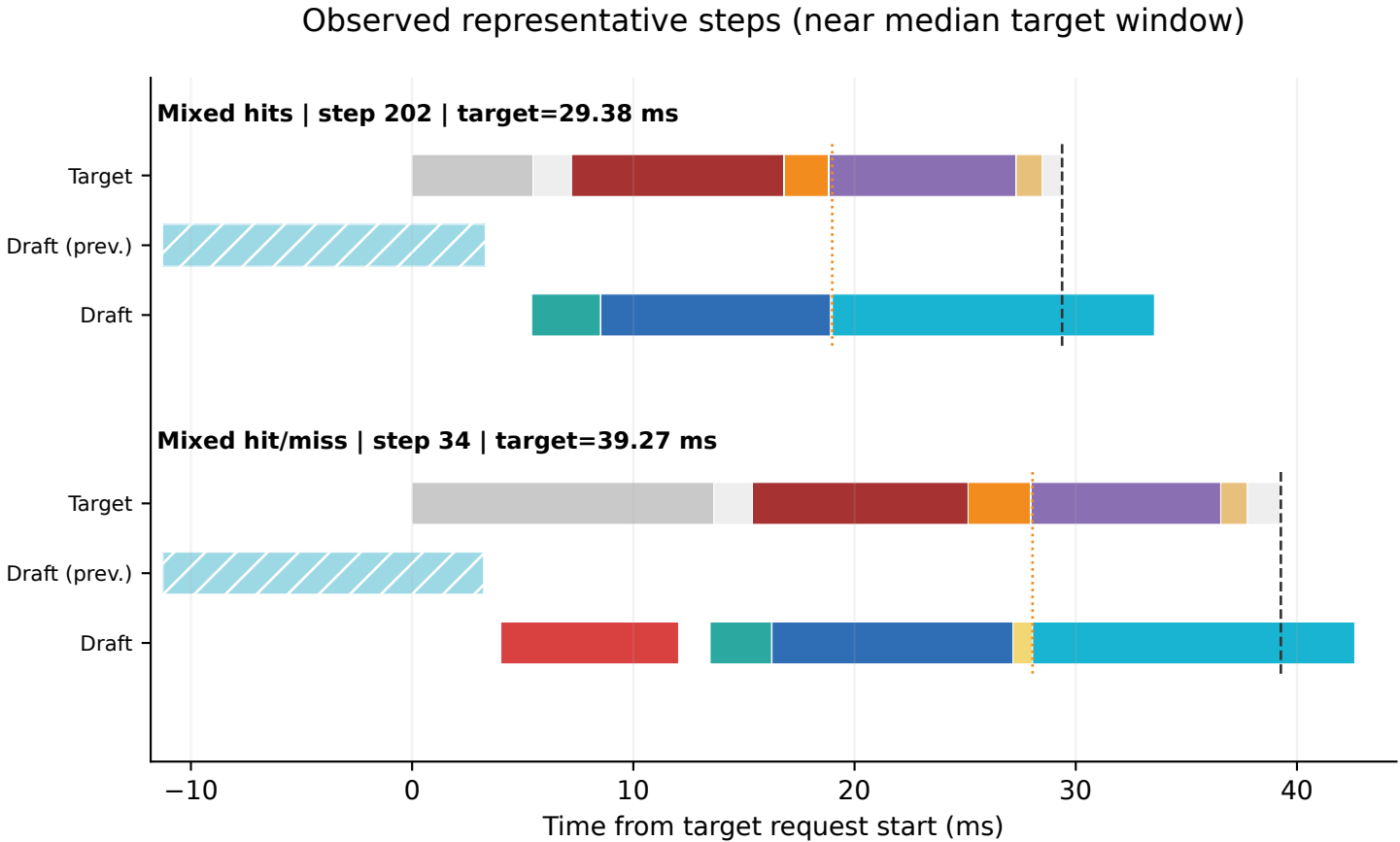
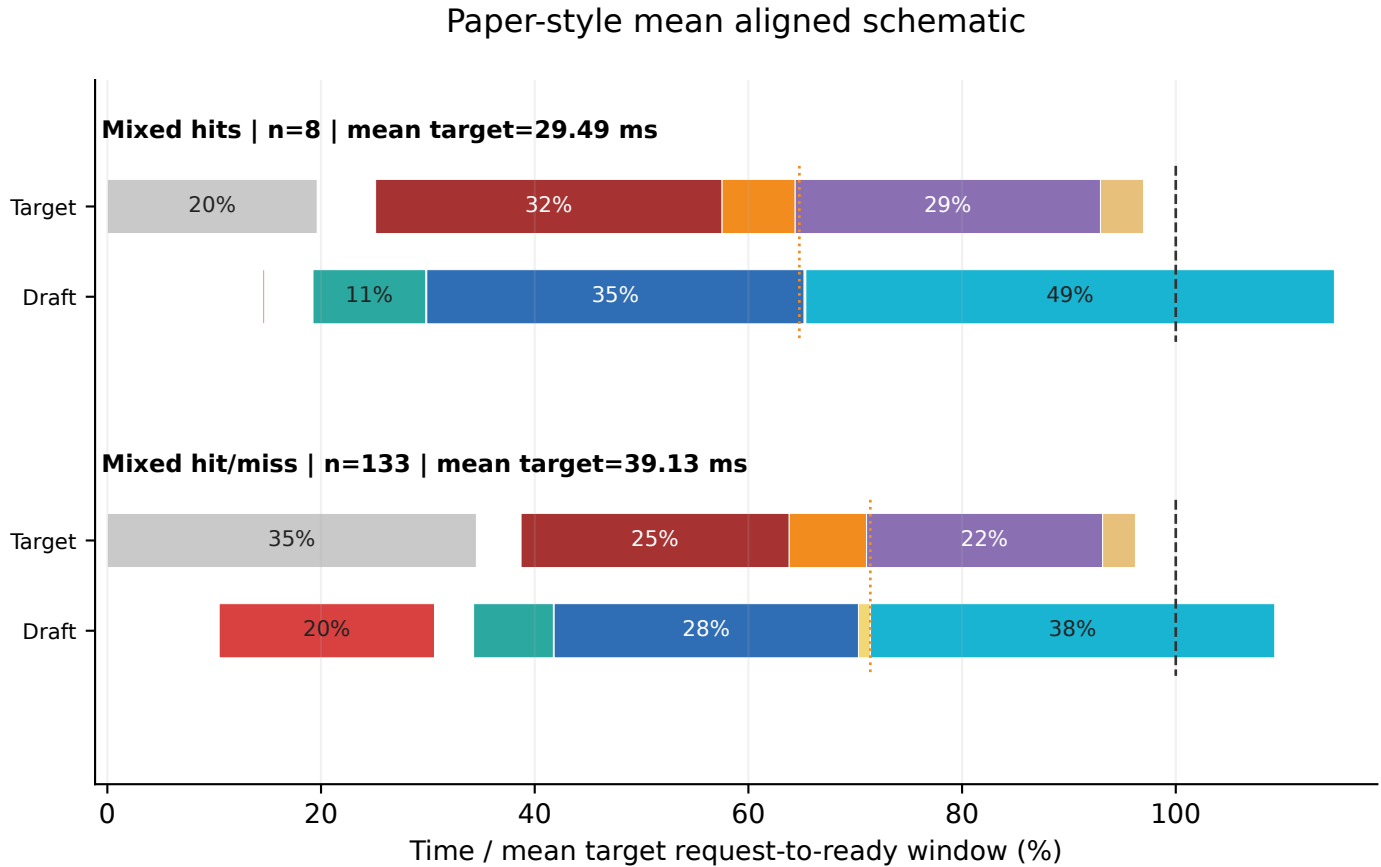
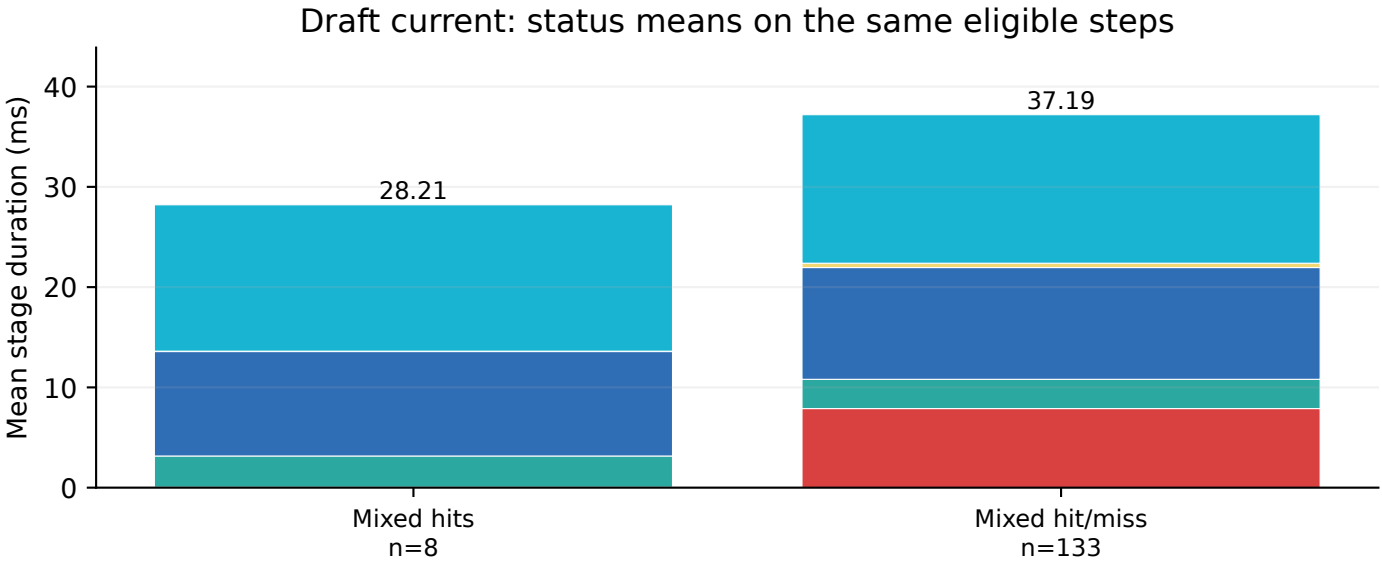
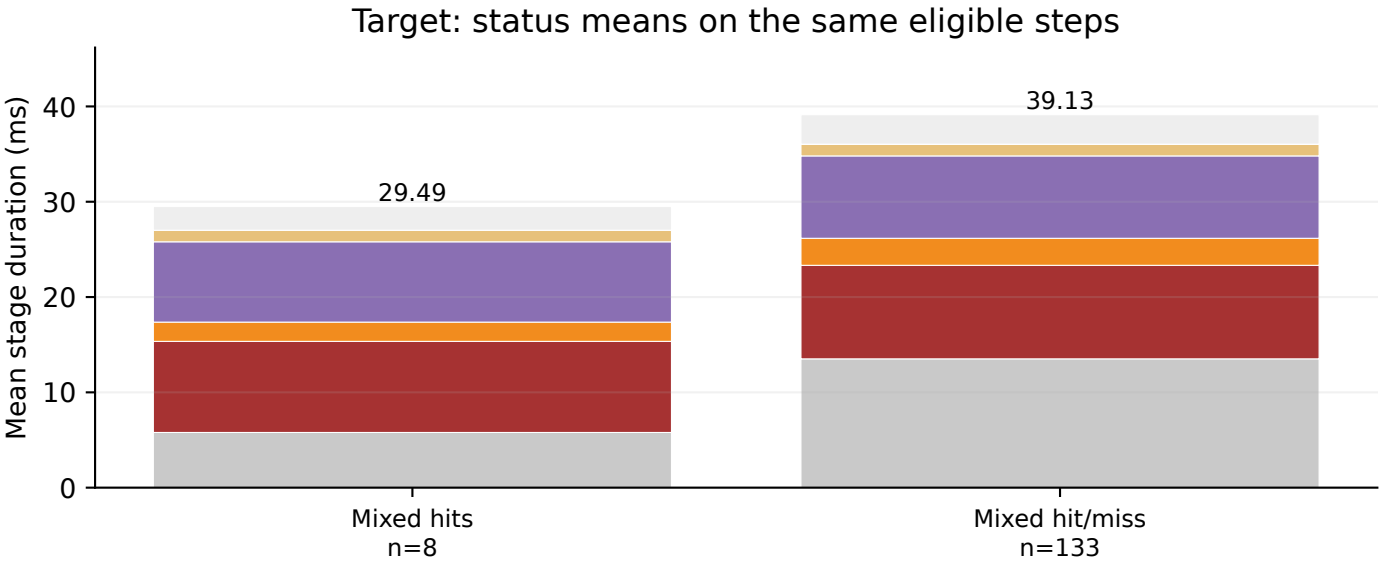
llama3_b8_combo_transfer_2_3_e12
DUET: K1/K2=2/3, exit=12 (0-based), C=1, N1/N2=4/6, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 149 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

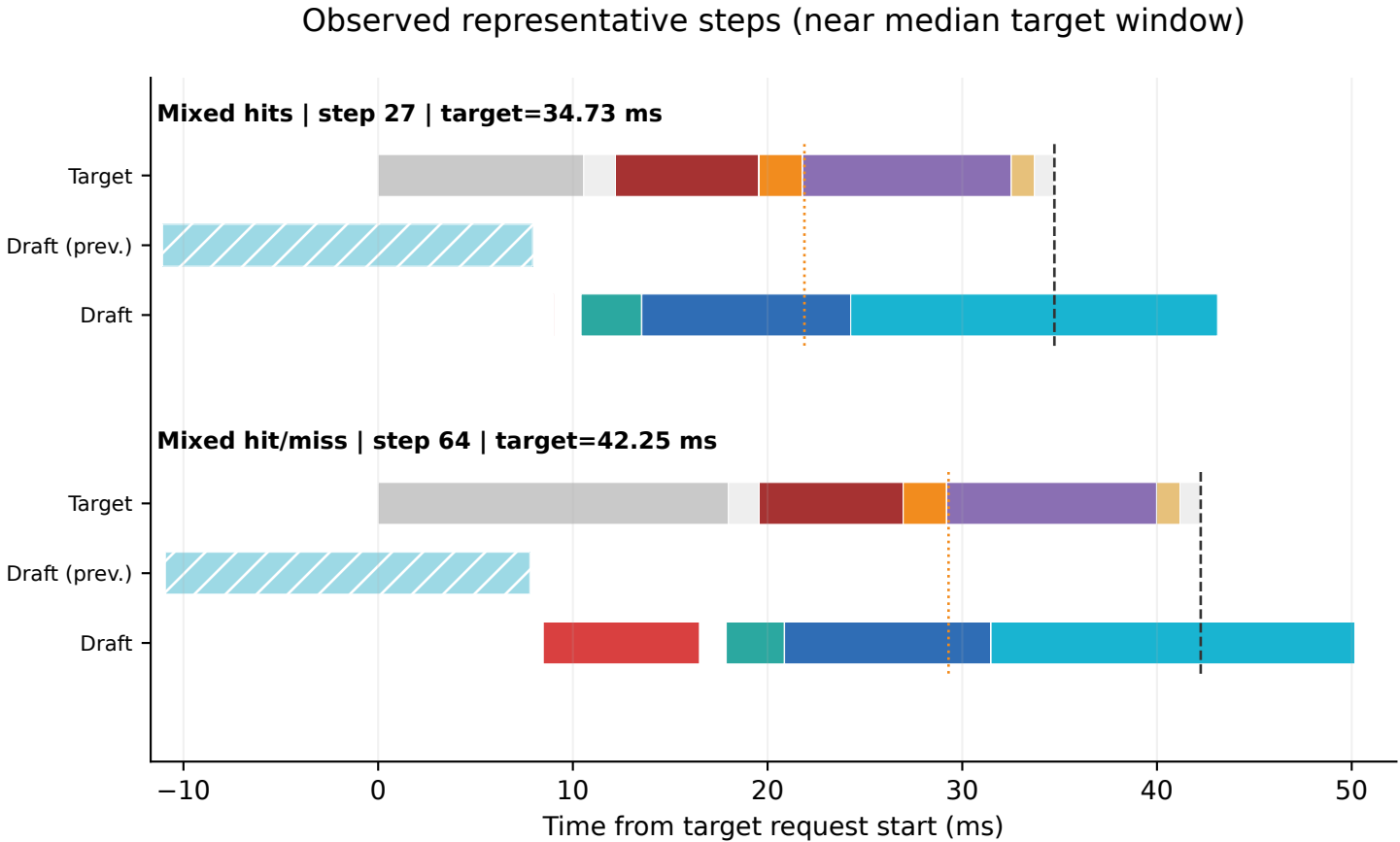
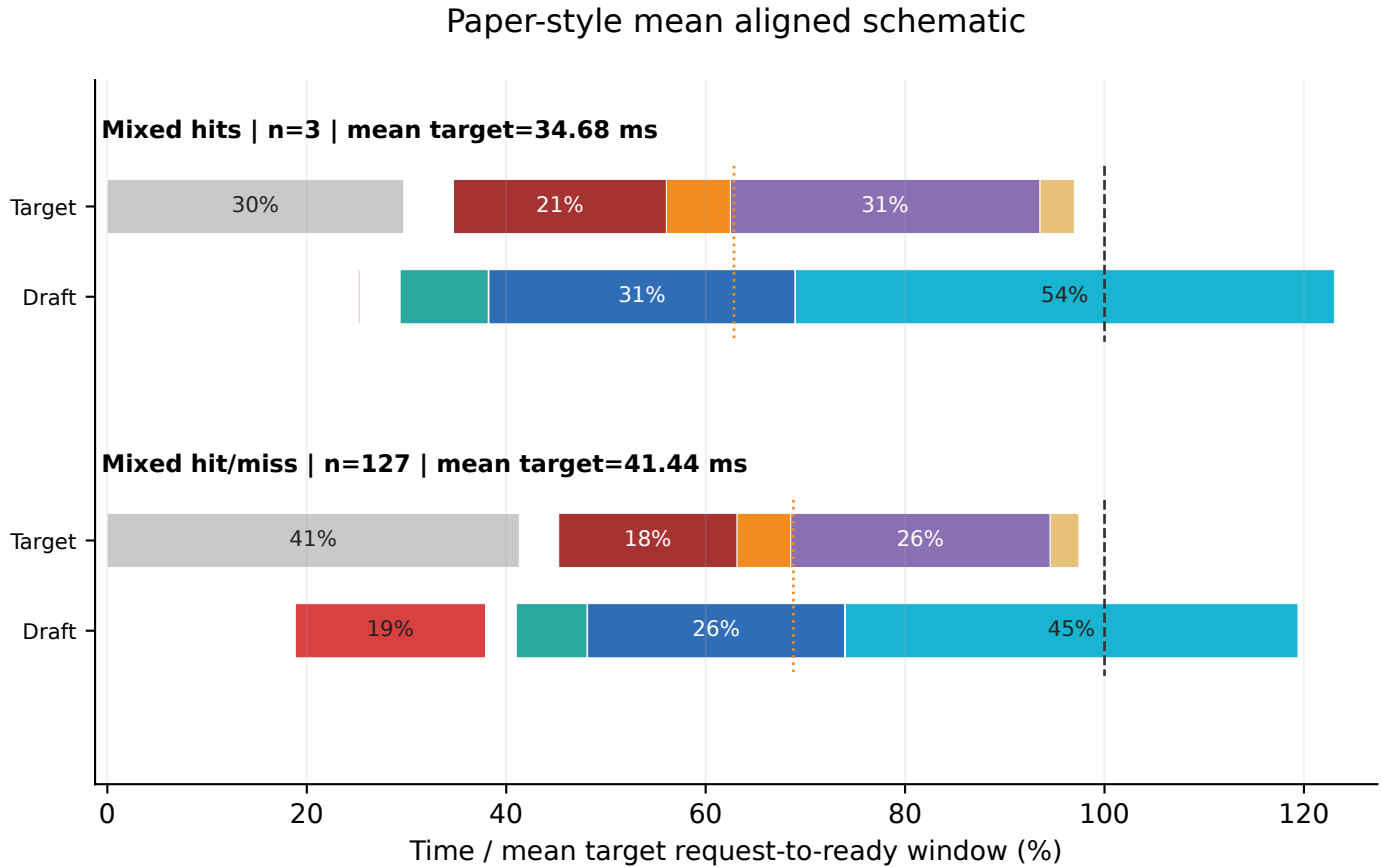
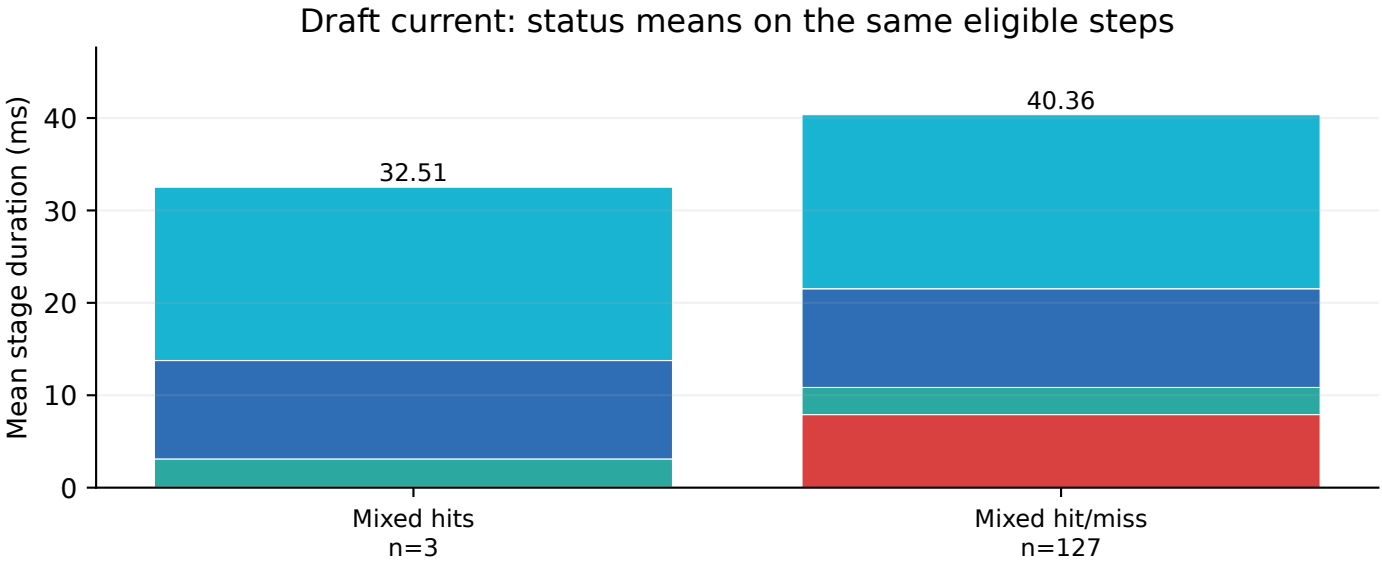
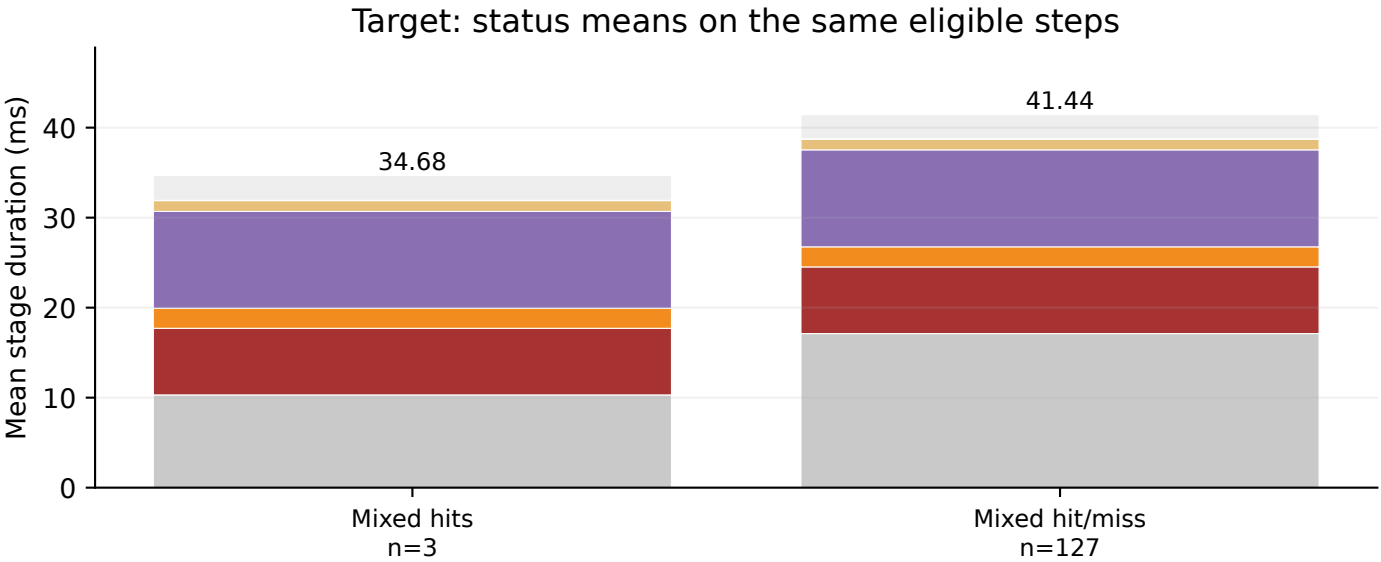
llama3_b8_combo_transfer_2_3_e16
DUET: K1/K2=2/3, exit=16 (0-based), C=1, N1/N2=4/6, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 141 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

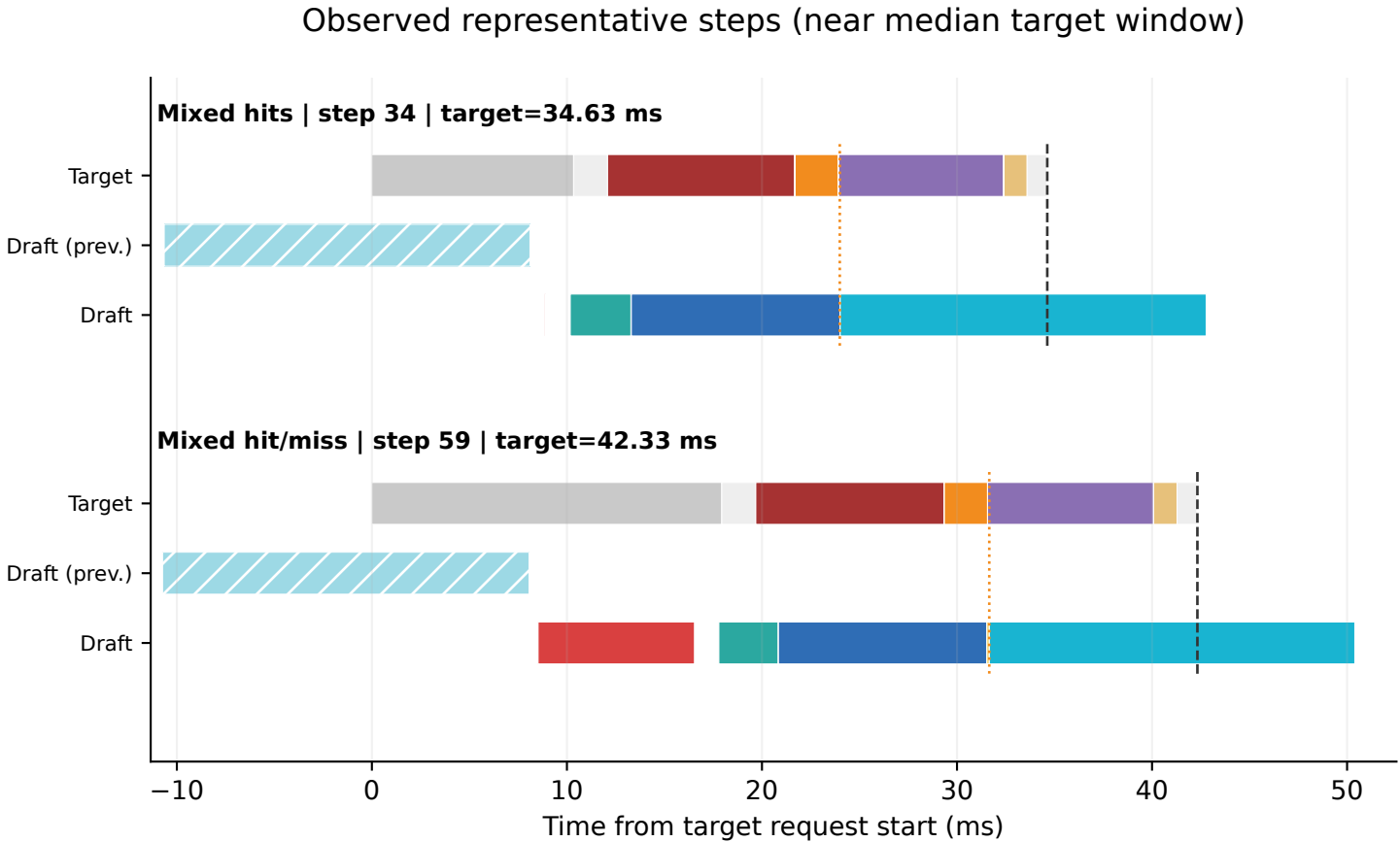
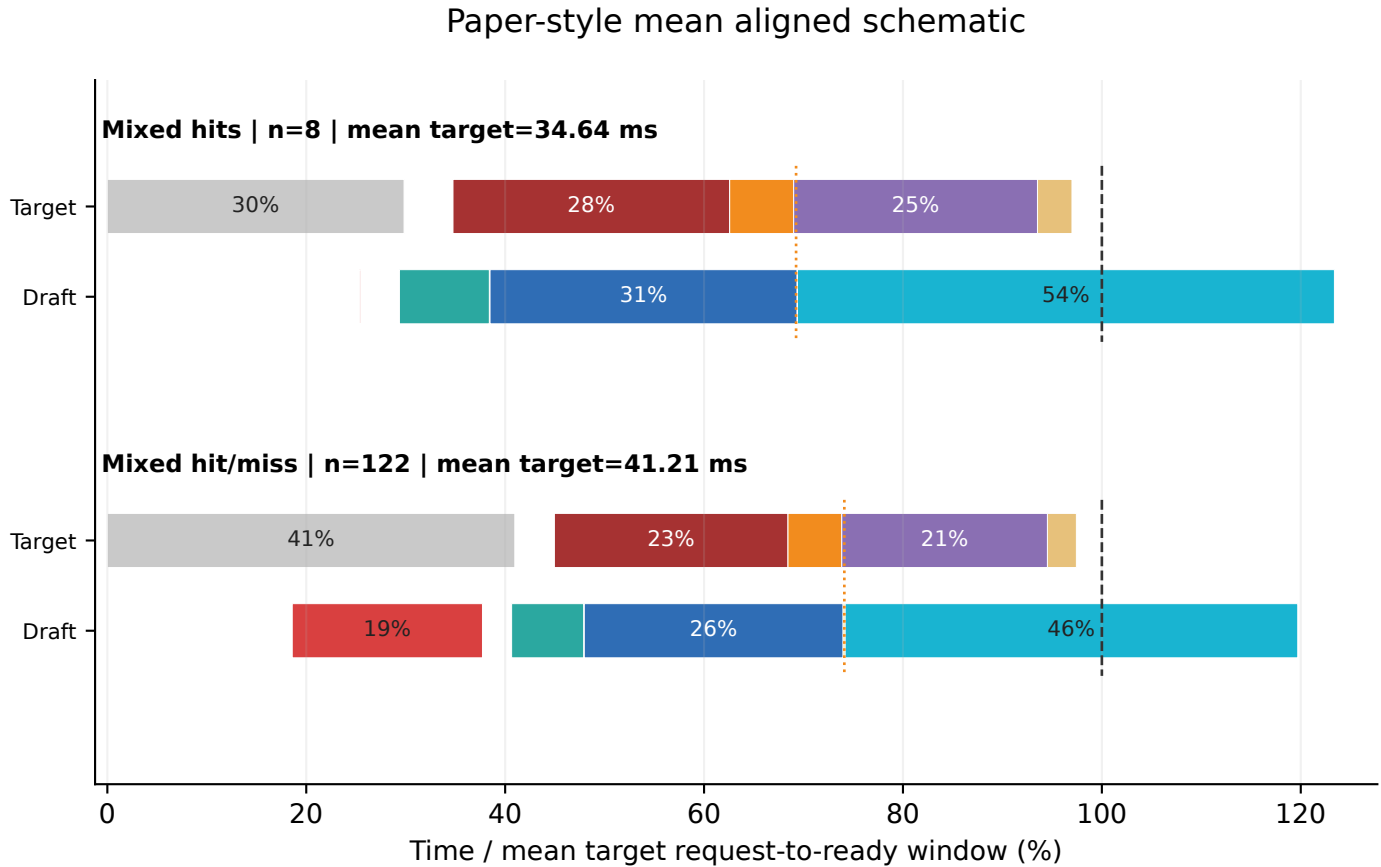
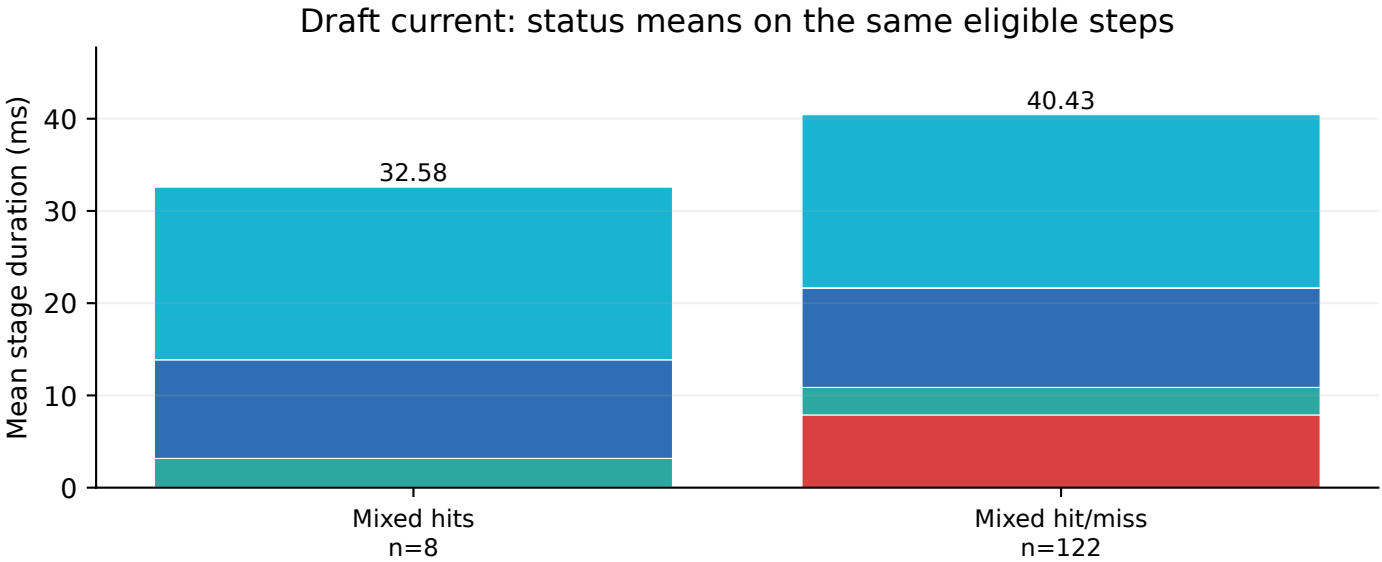
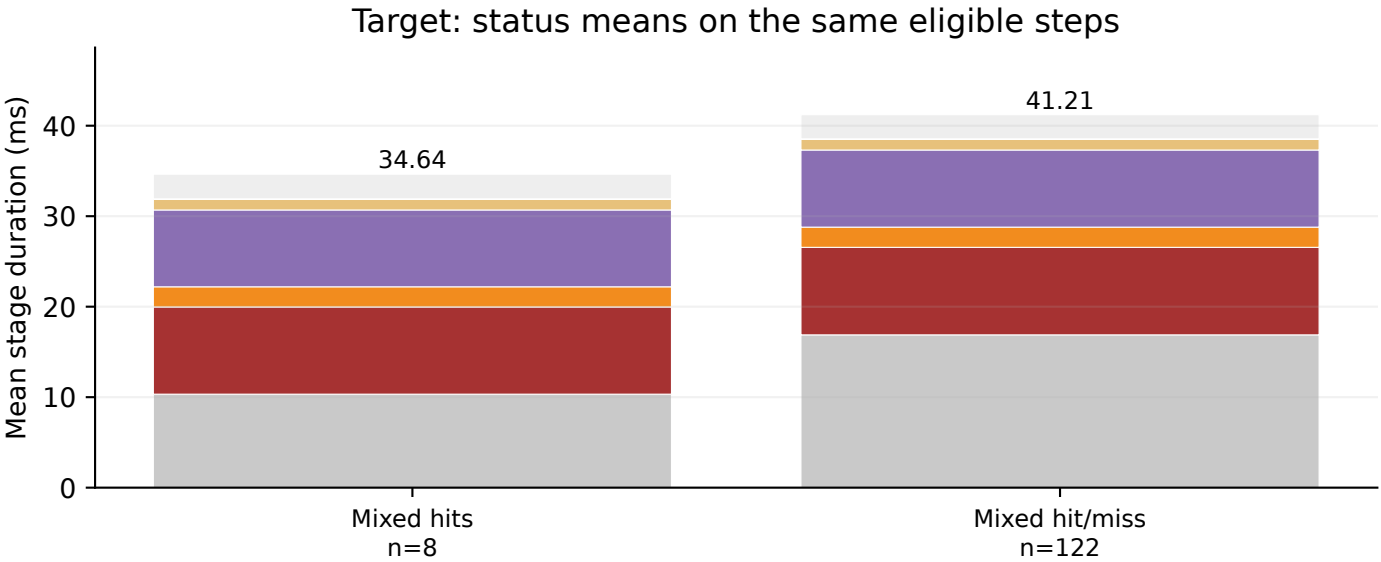
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_combo_transfer_2_4_e12
DUET: K1/K2=2/4, exit=12 (0-based), C=1, N1/N2=4/8, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 130 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

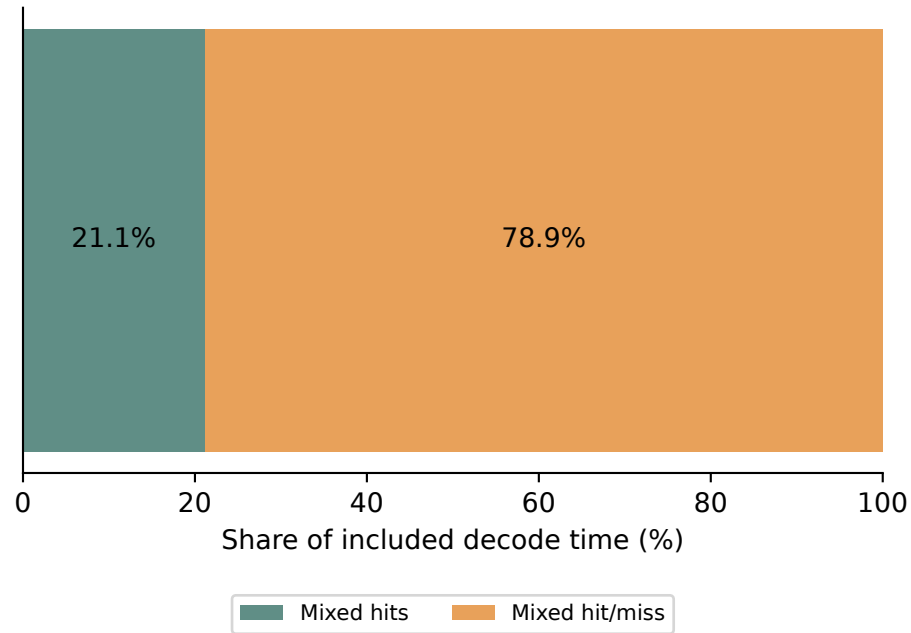


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

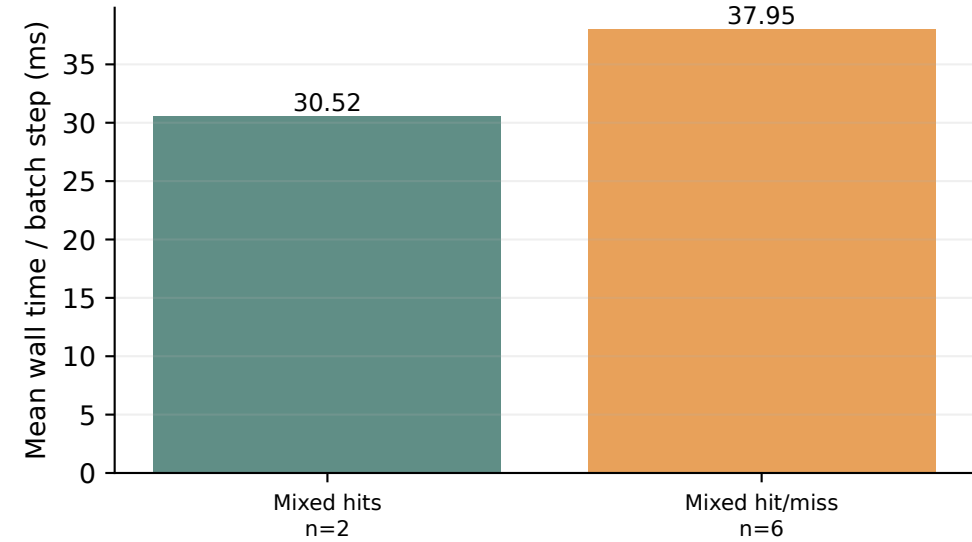
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_transfer_smoke
DUET: K1/K2=1/3, exit=12 (0-based), C=3, N1/N2=2/6, U=2, W=3, trim/stream=0/0
Last pass: B=8, T=0.7, N=8 | AL*=2.177, hit=59.4%, TPS*=346.33

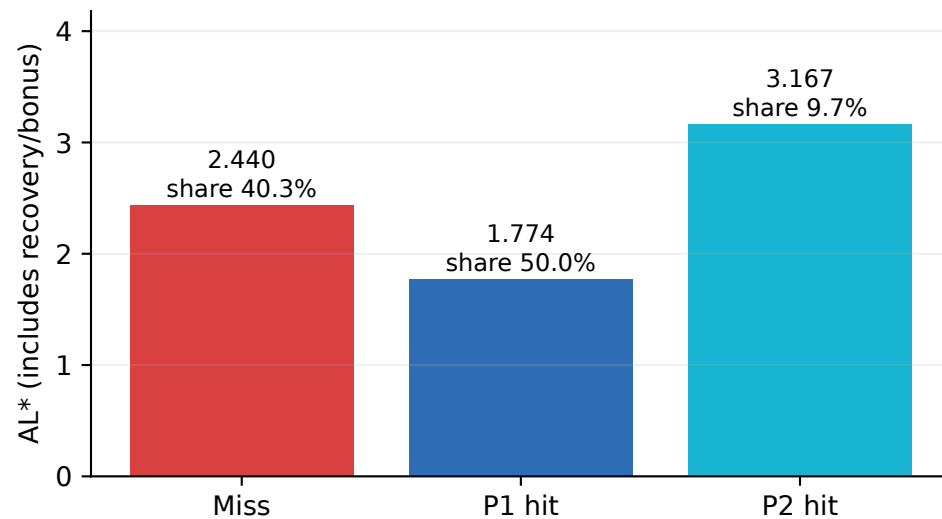
Measured wall time by batch cache status



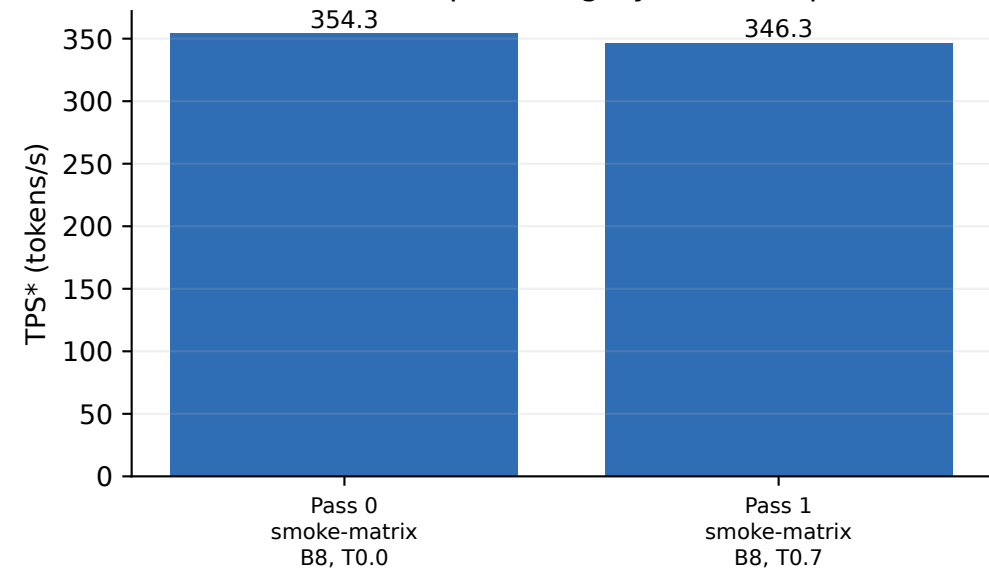
Boundary-free steps, including slow steps



Conditional AL* and source share

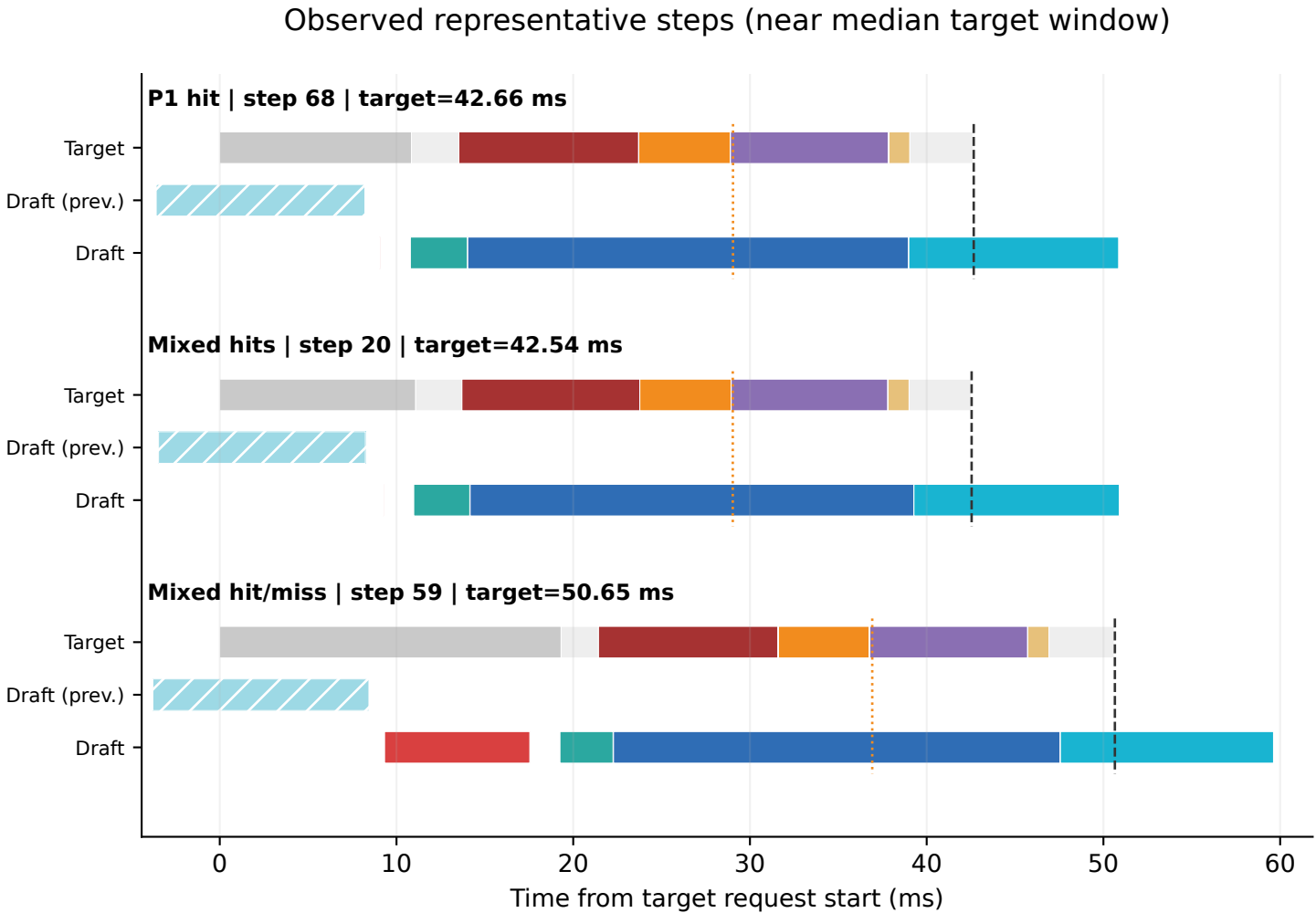
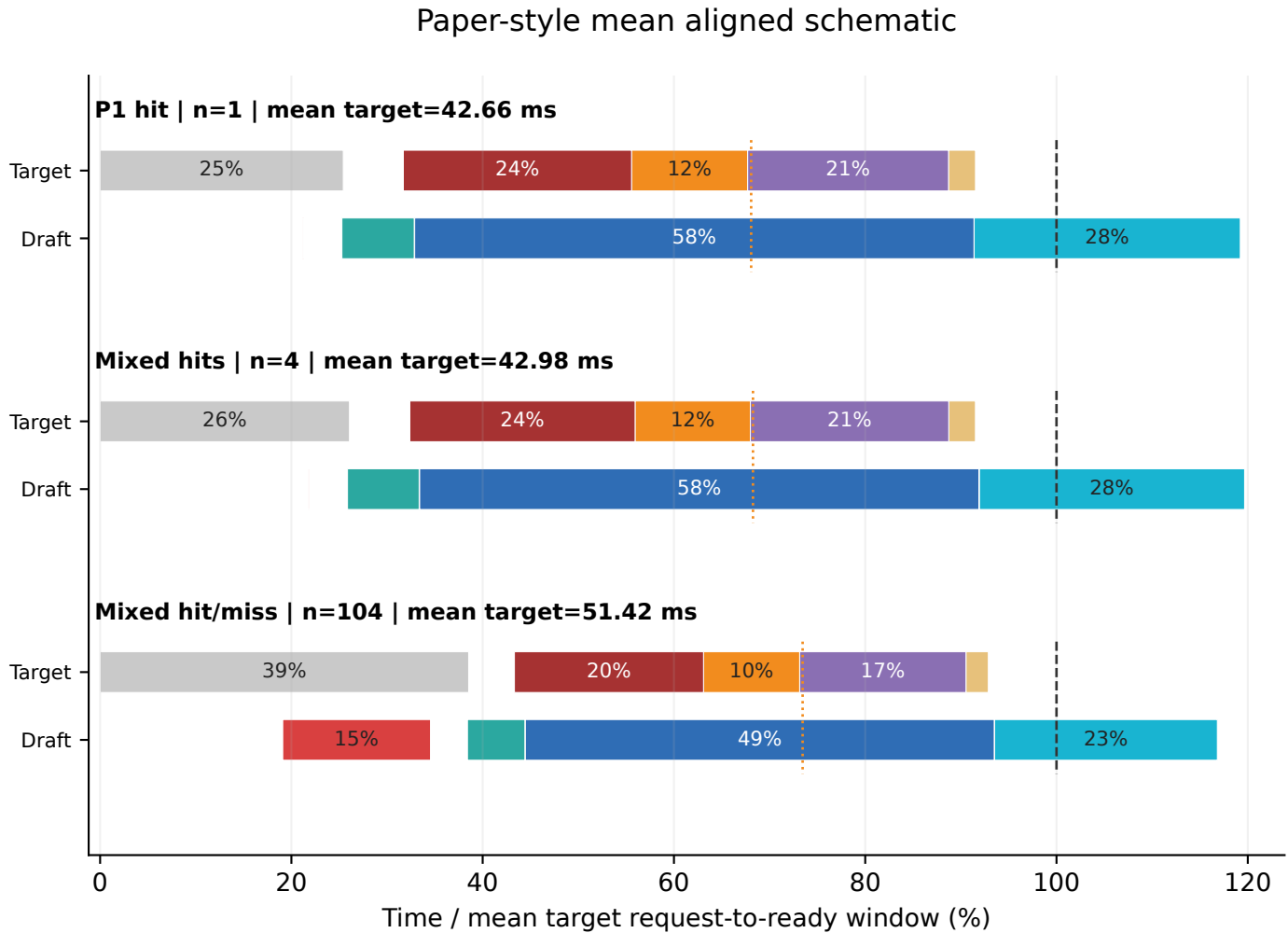
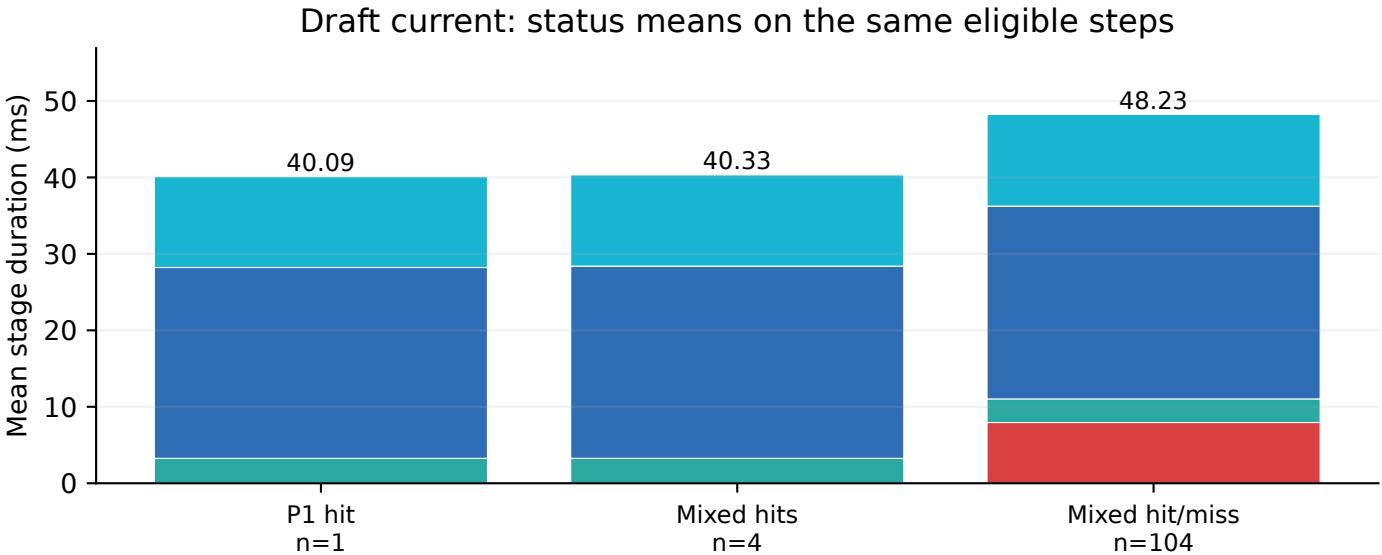
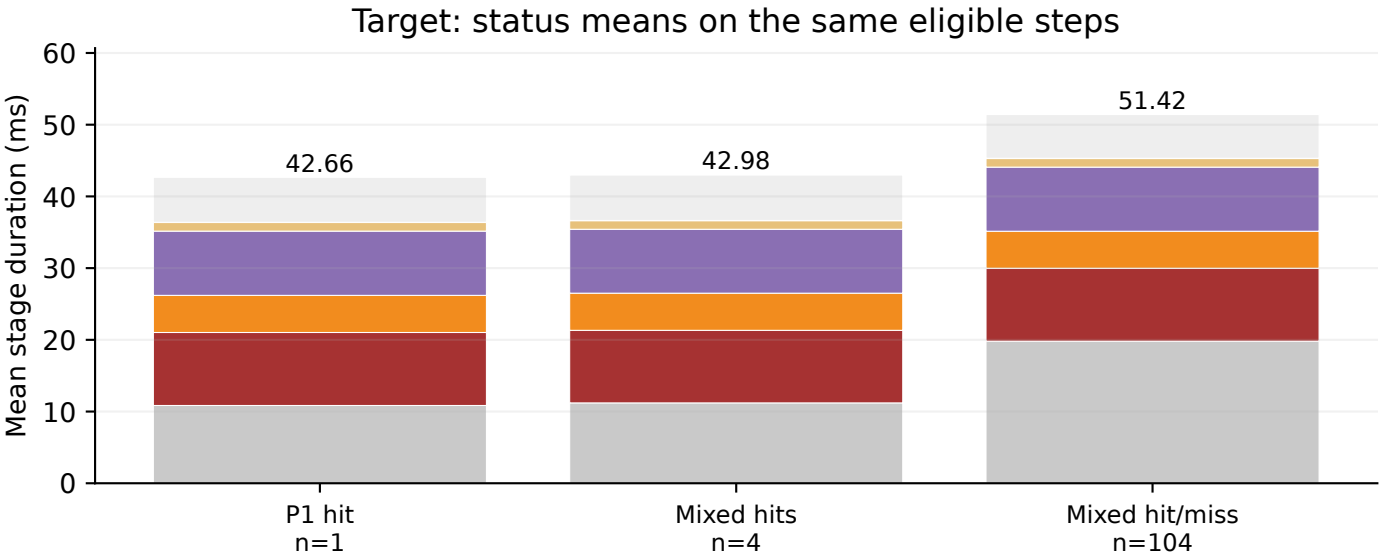


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

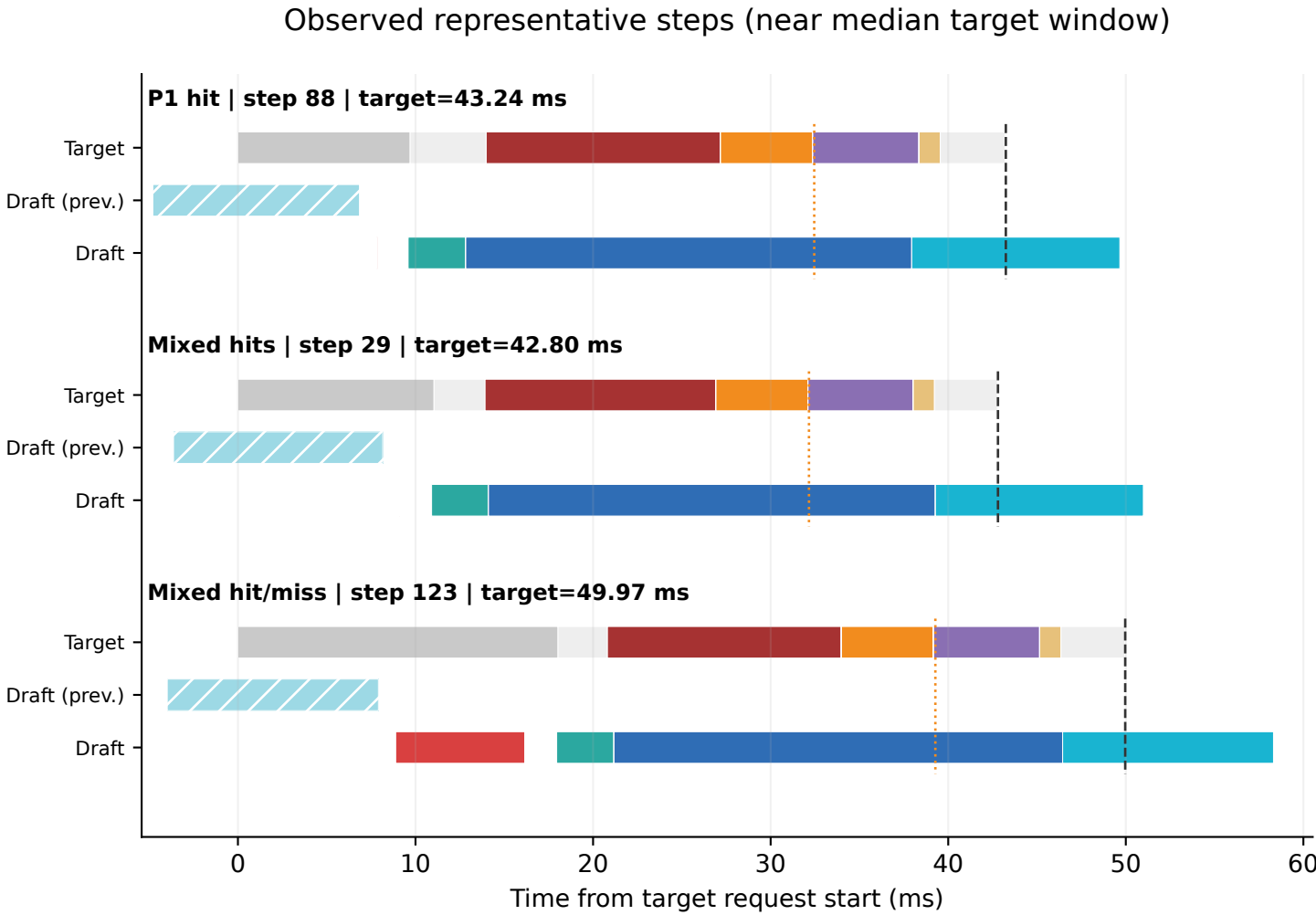
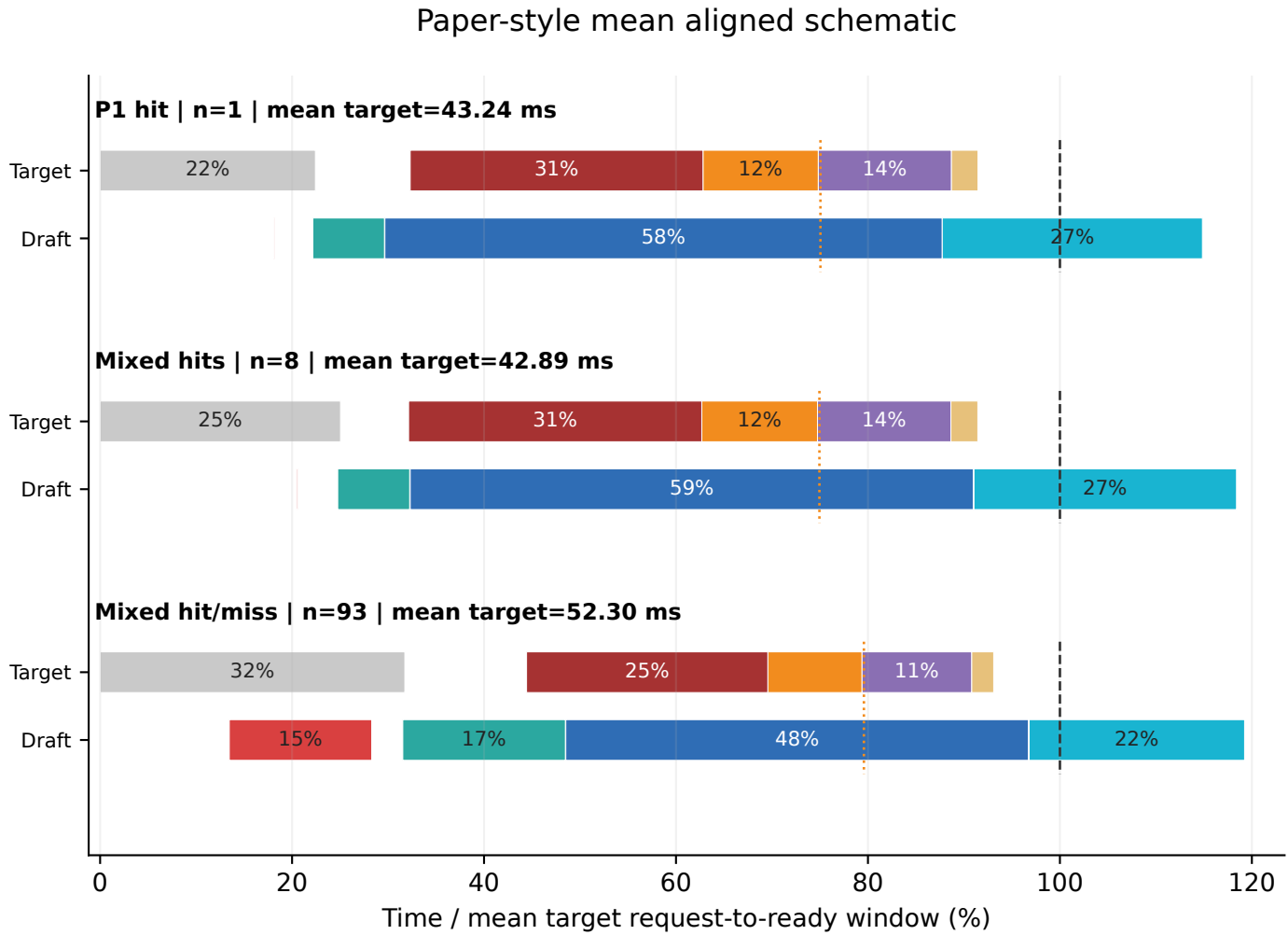
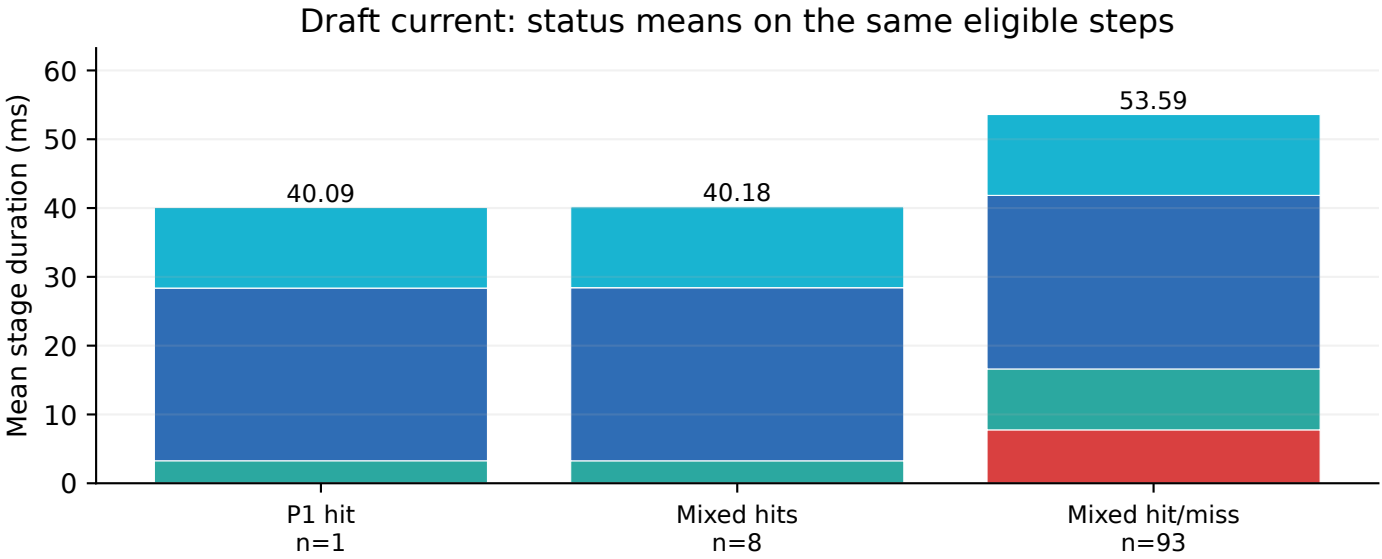
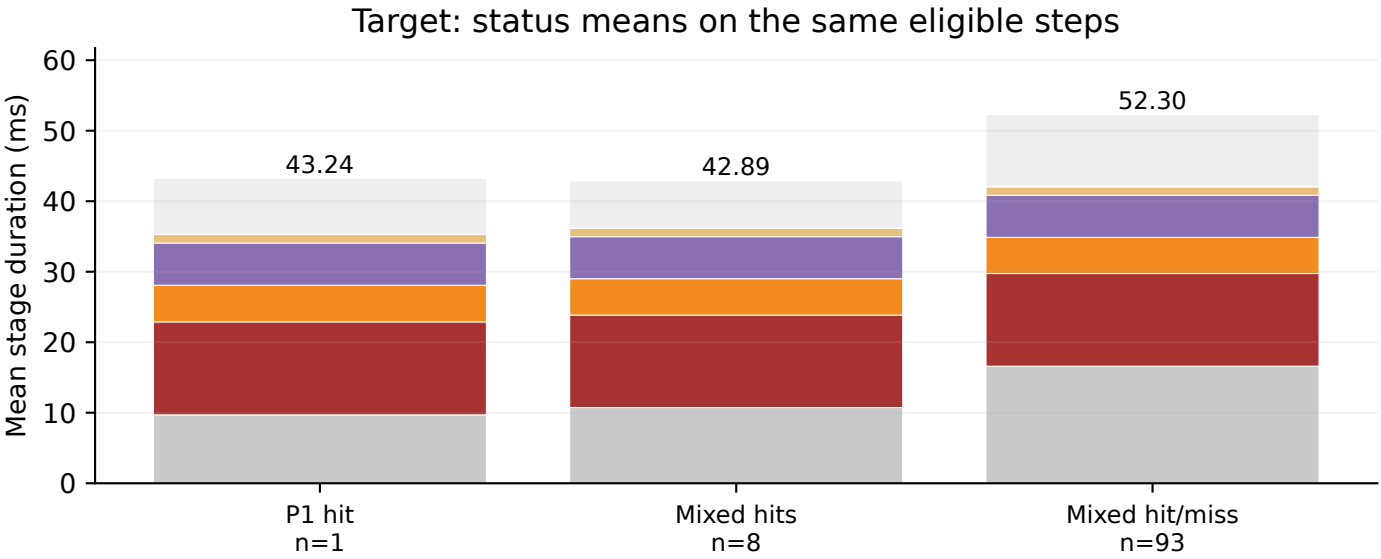
llama3_b8_duet_e16_k4_2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 109 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

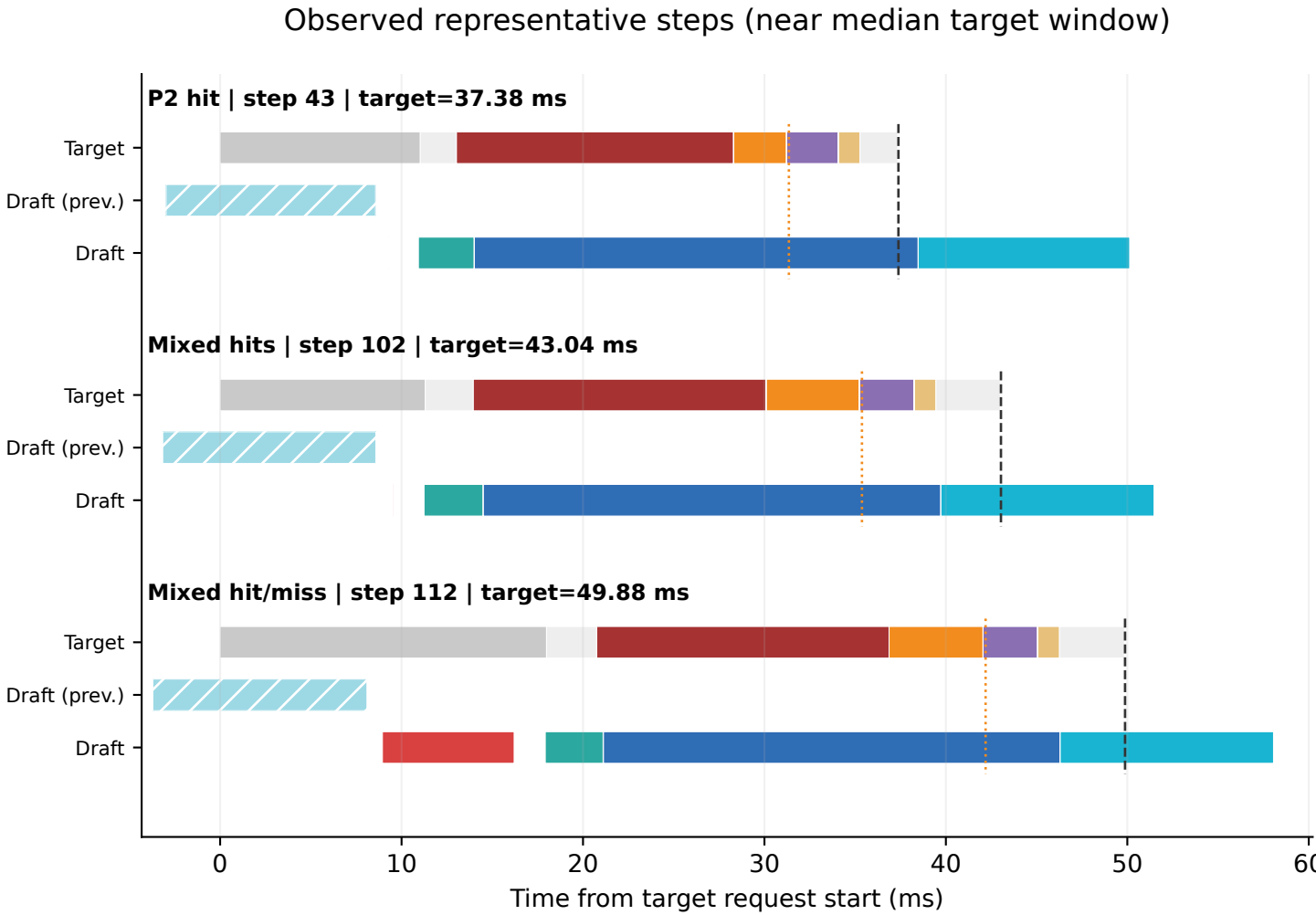
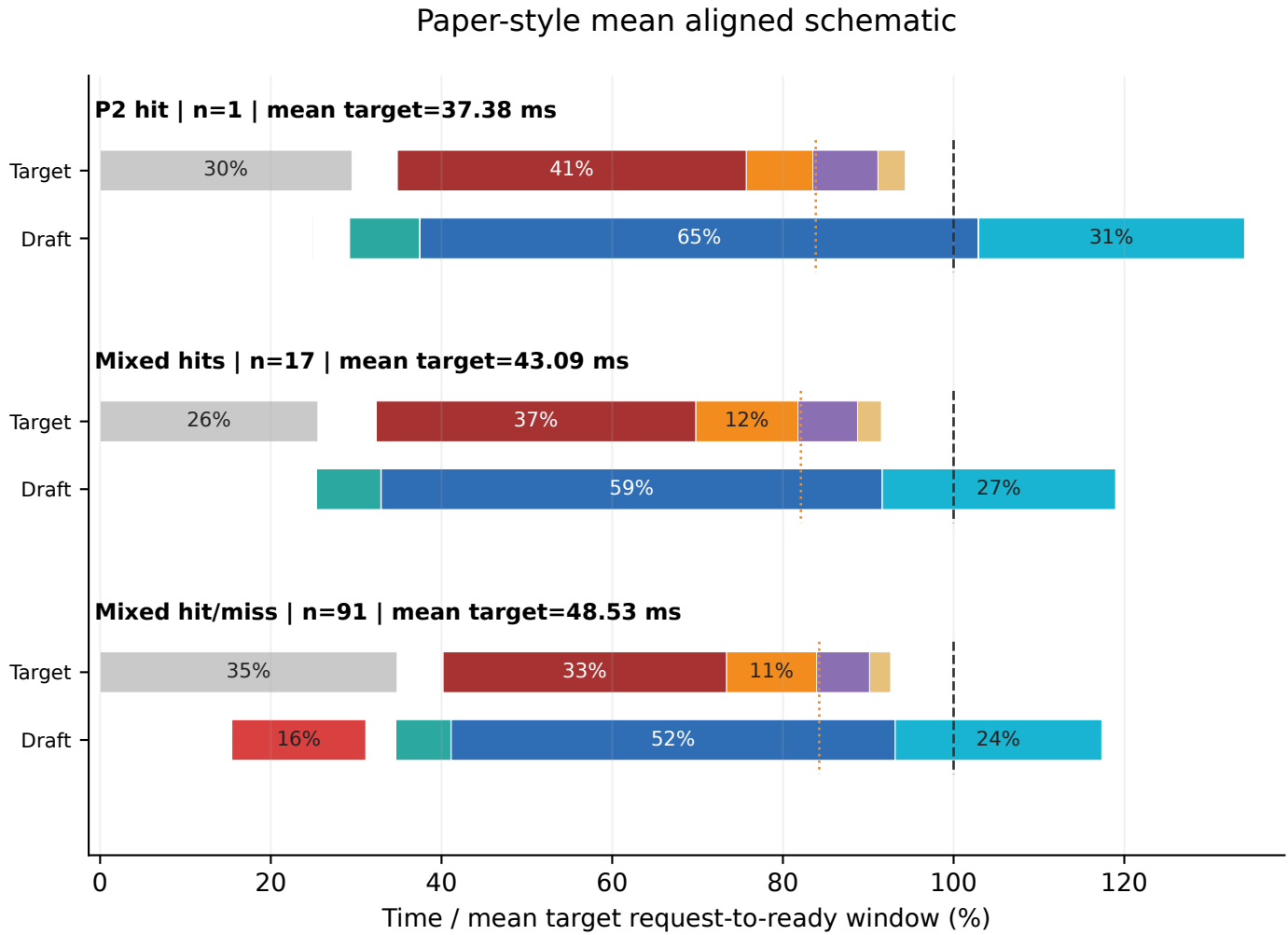
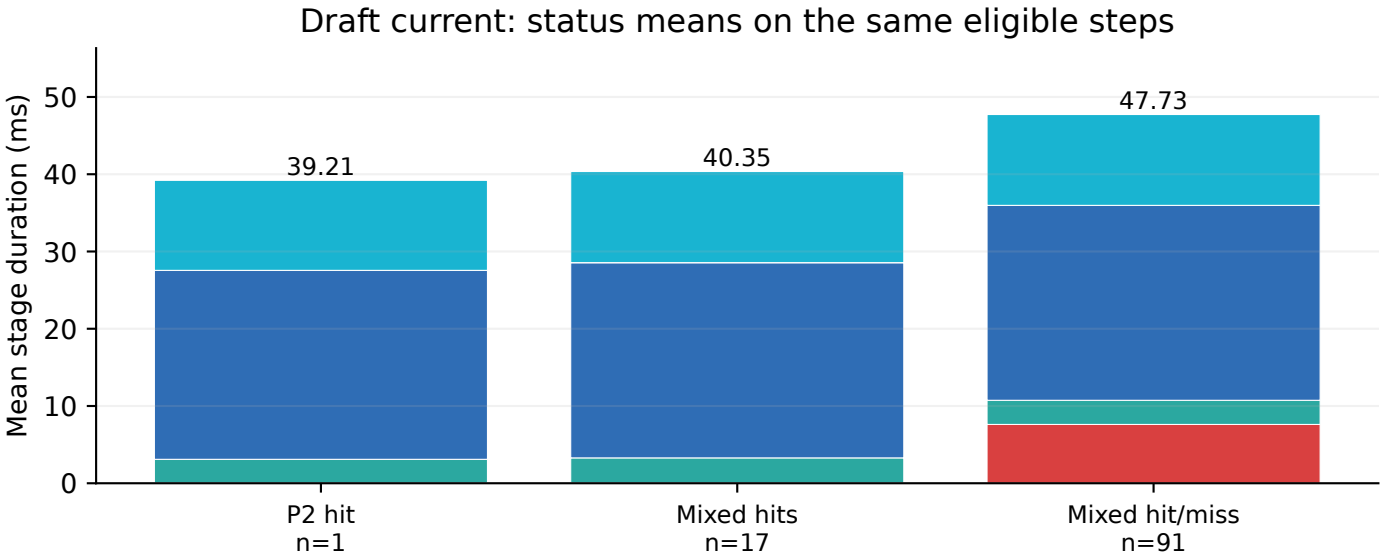
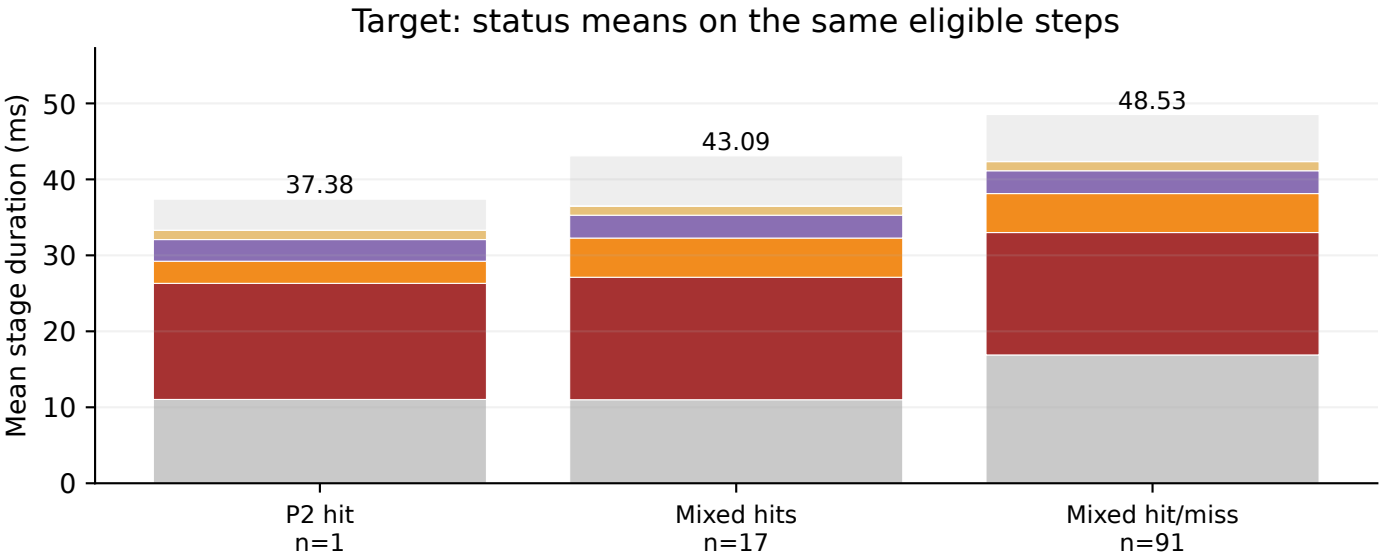
llama3_b8_duet_e21_k4_2
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 102 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

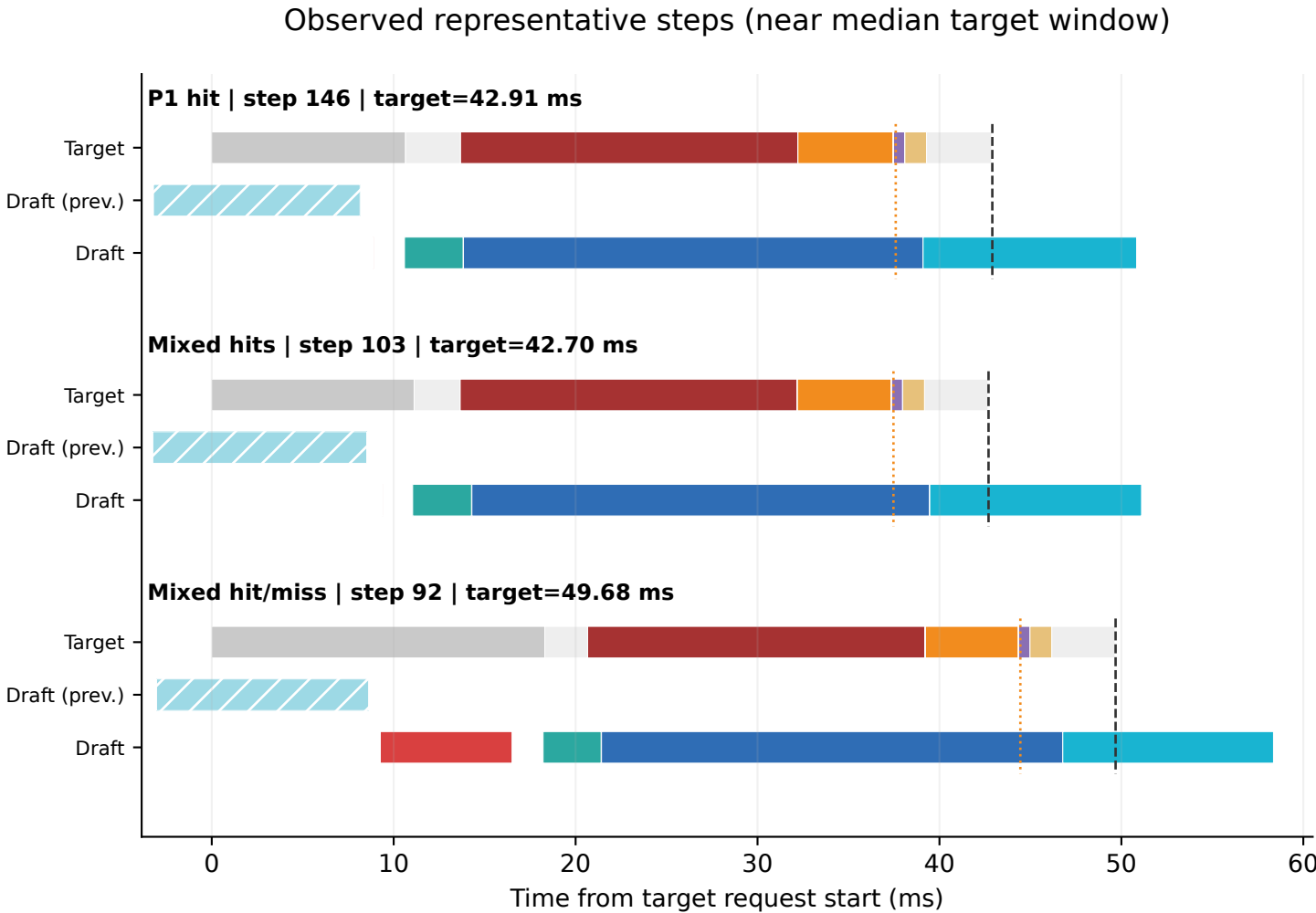
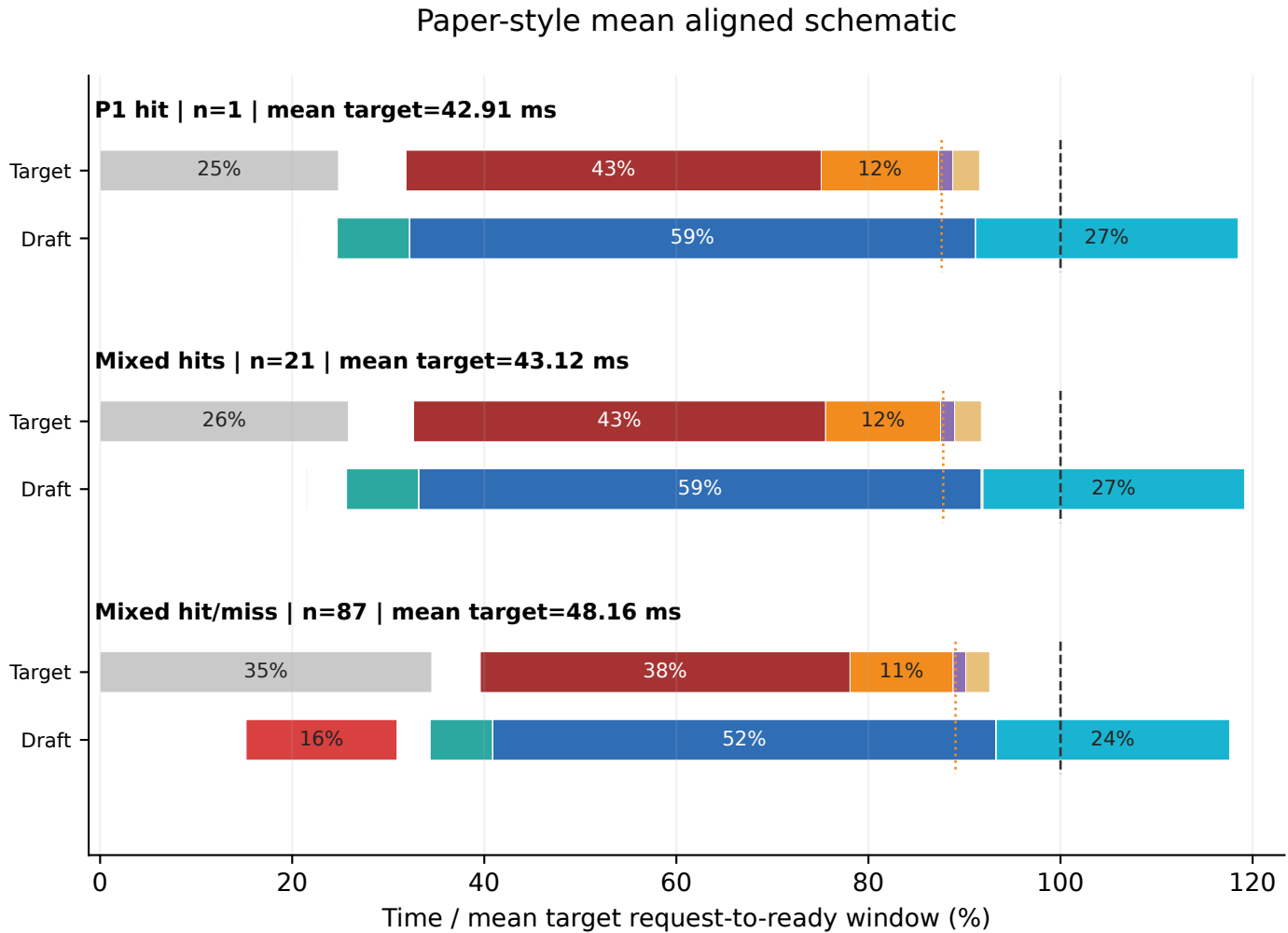
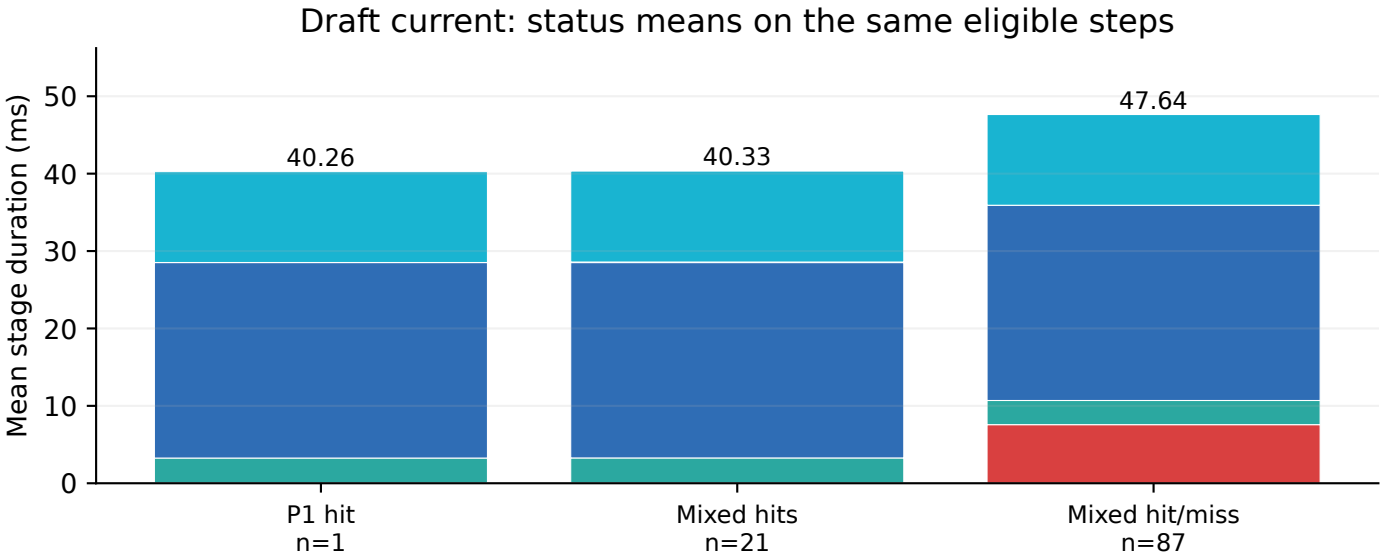
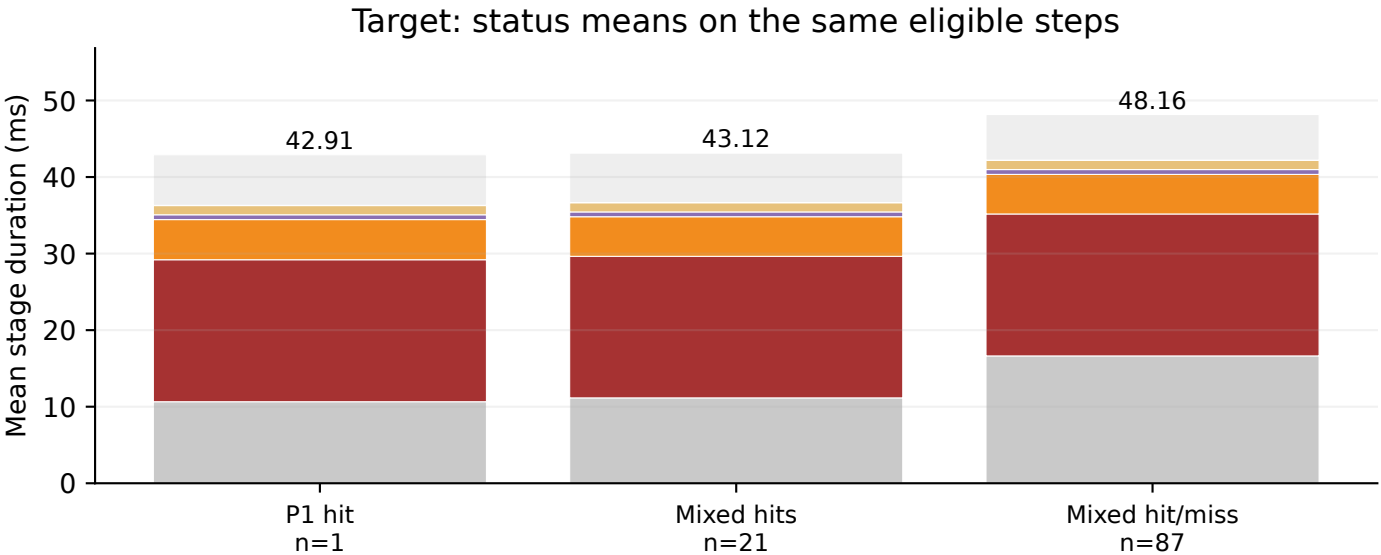
llama3_b8_duet_e26_k4_2
DUET: K1/K2=4/2, exit=26 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 109 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

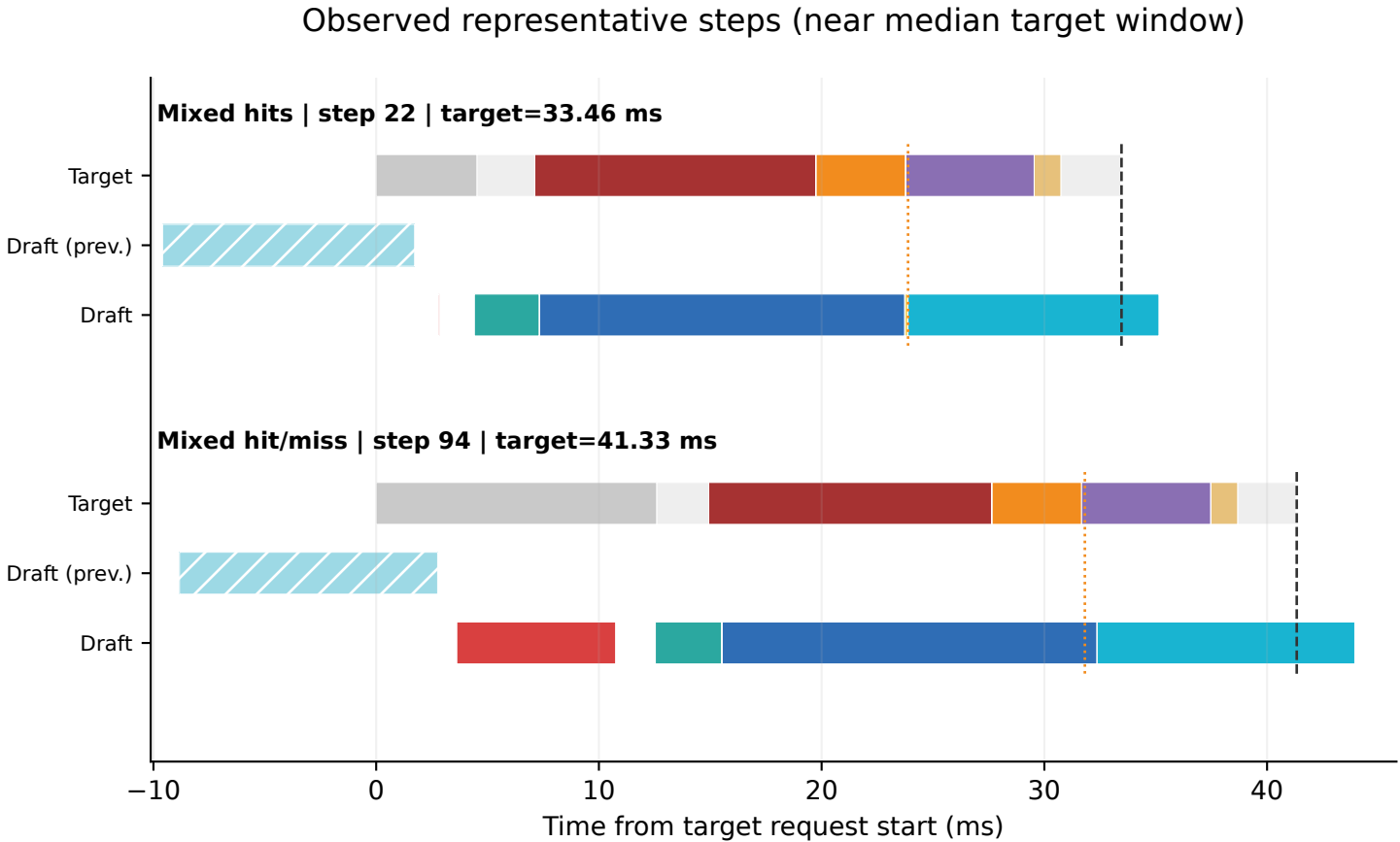
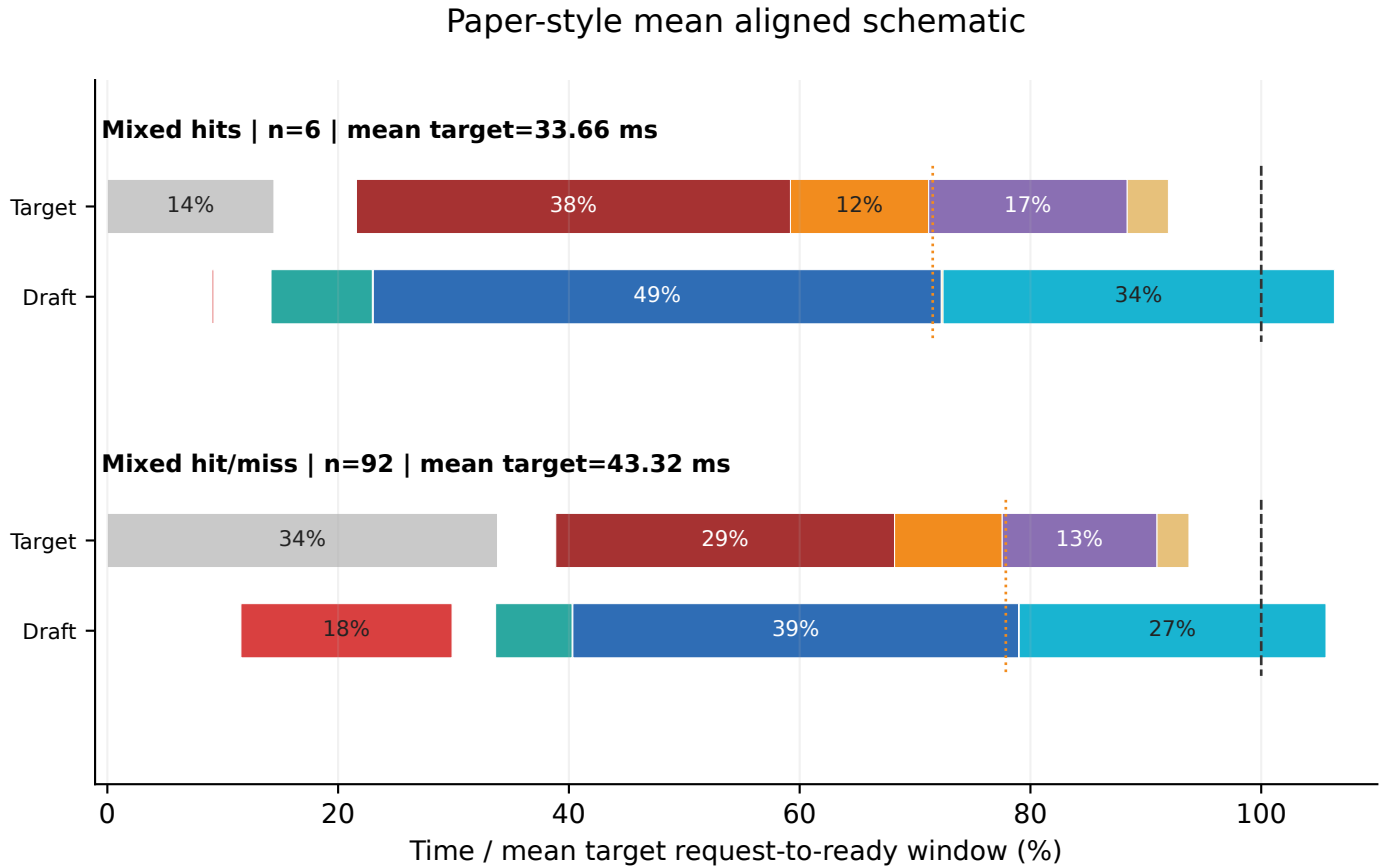
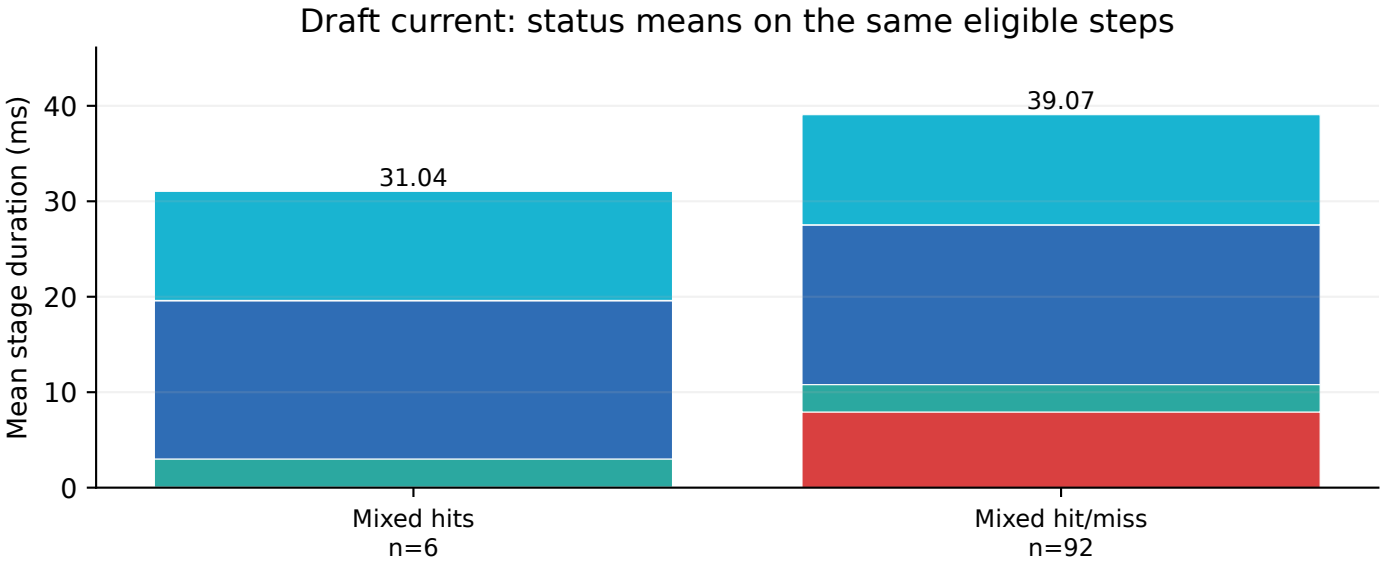
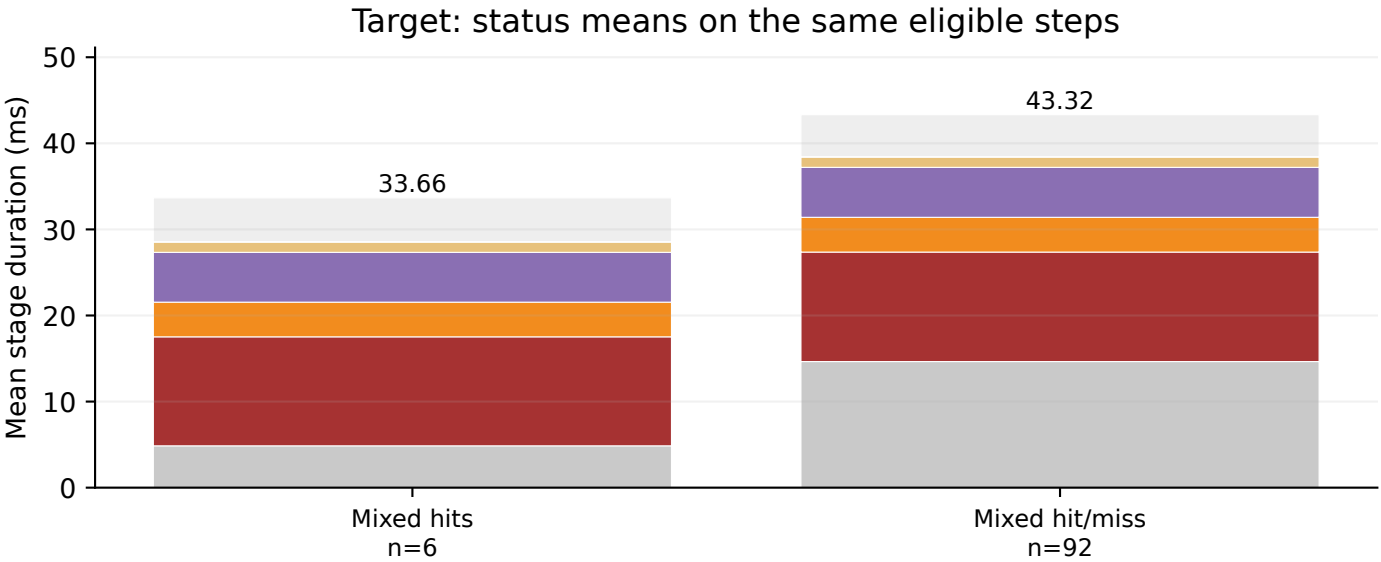
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_duet_e30_k4_2
DUET: K1/K2=4/2, exit=30 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 109 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

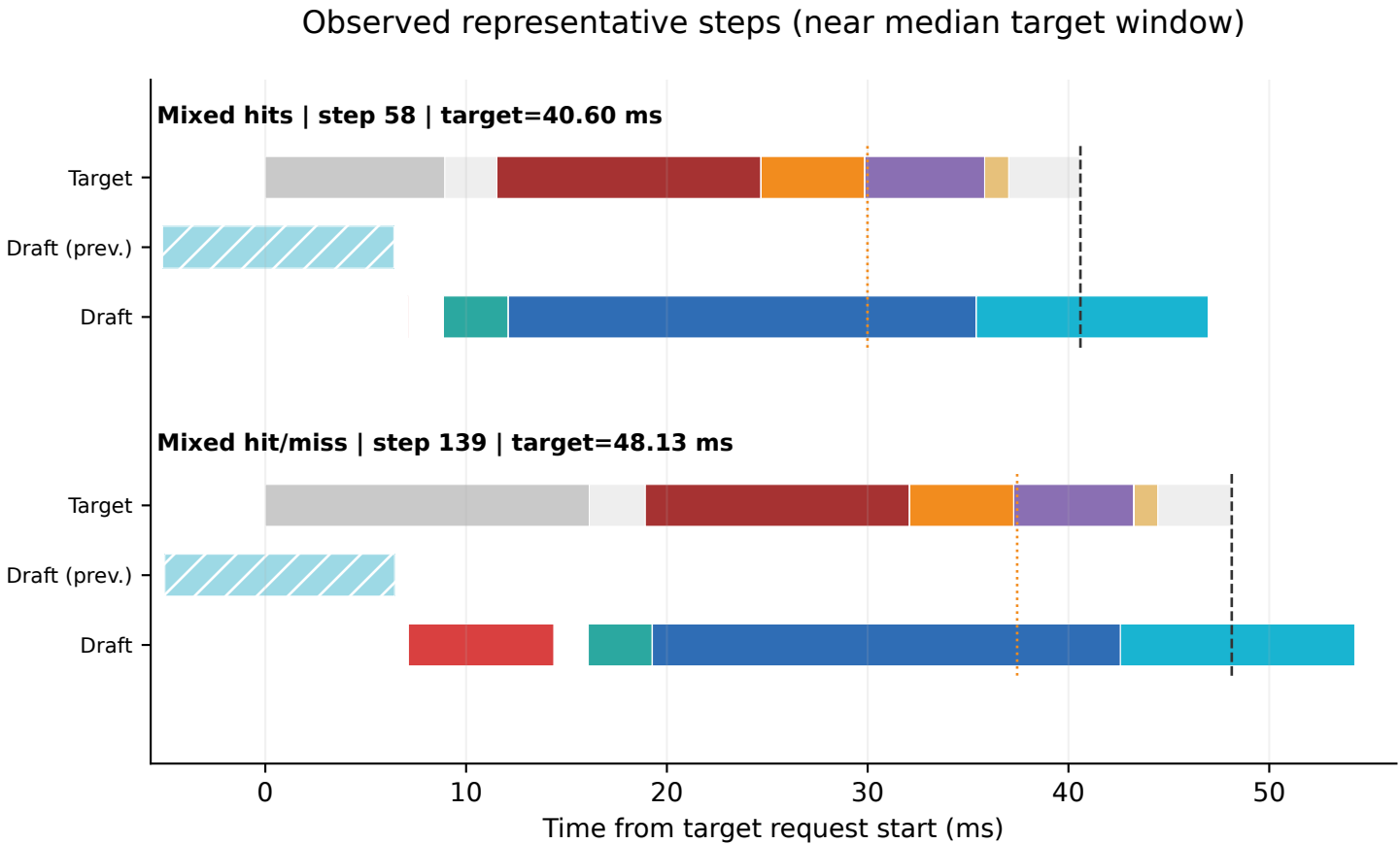
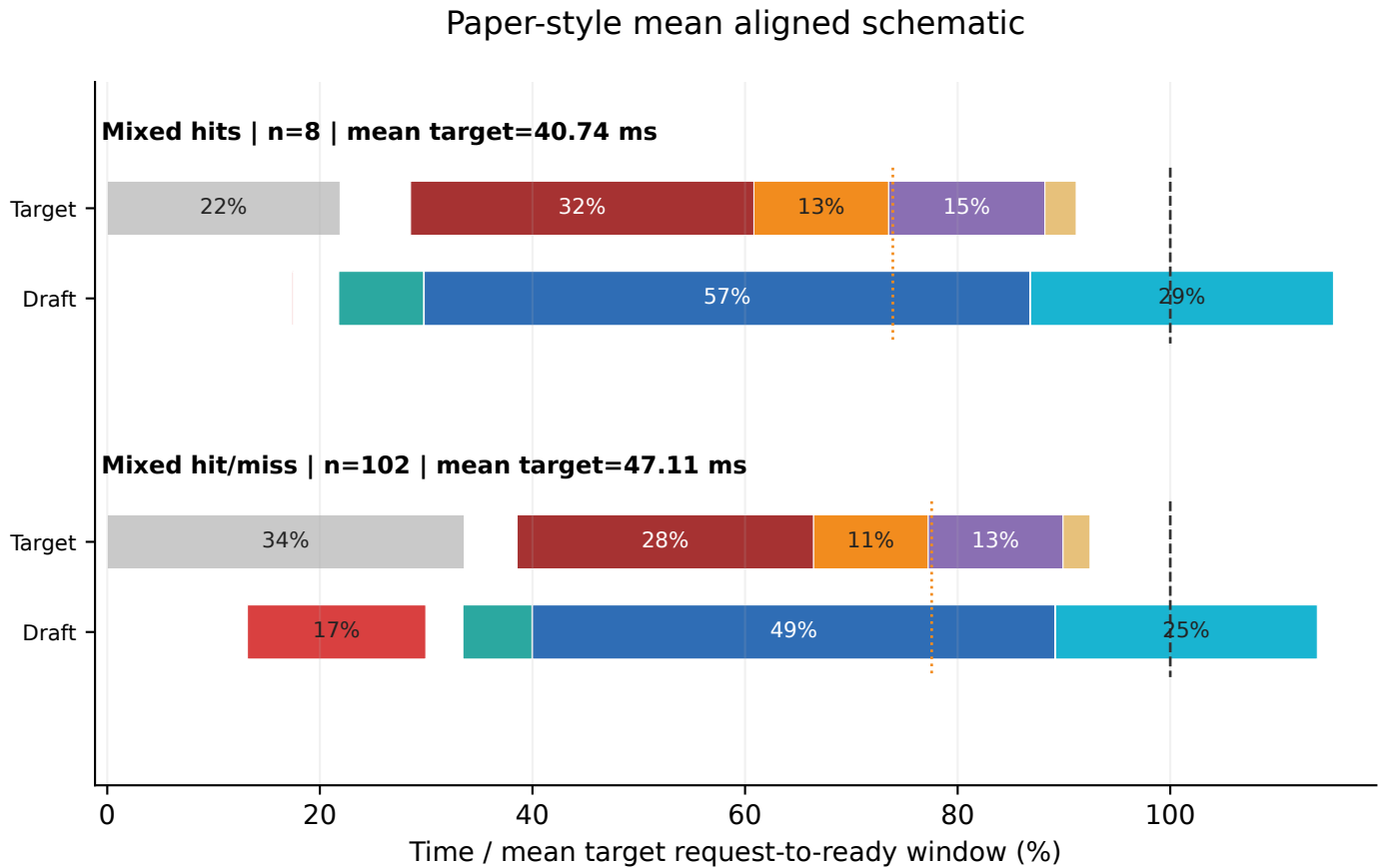
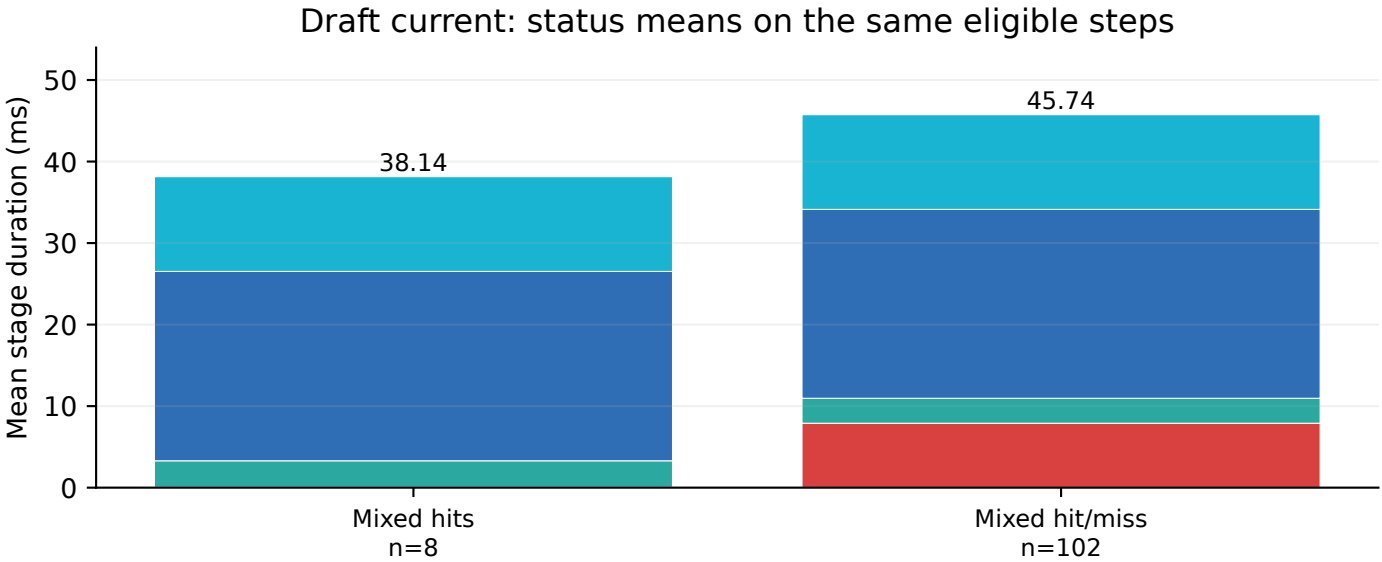
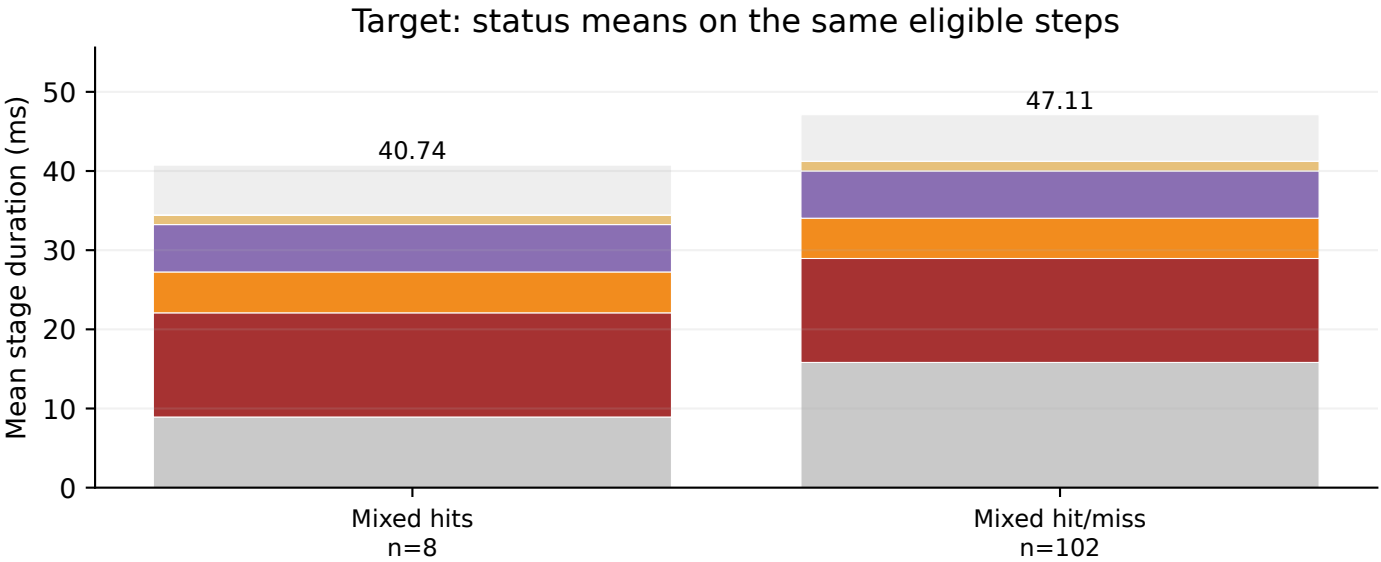
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

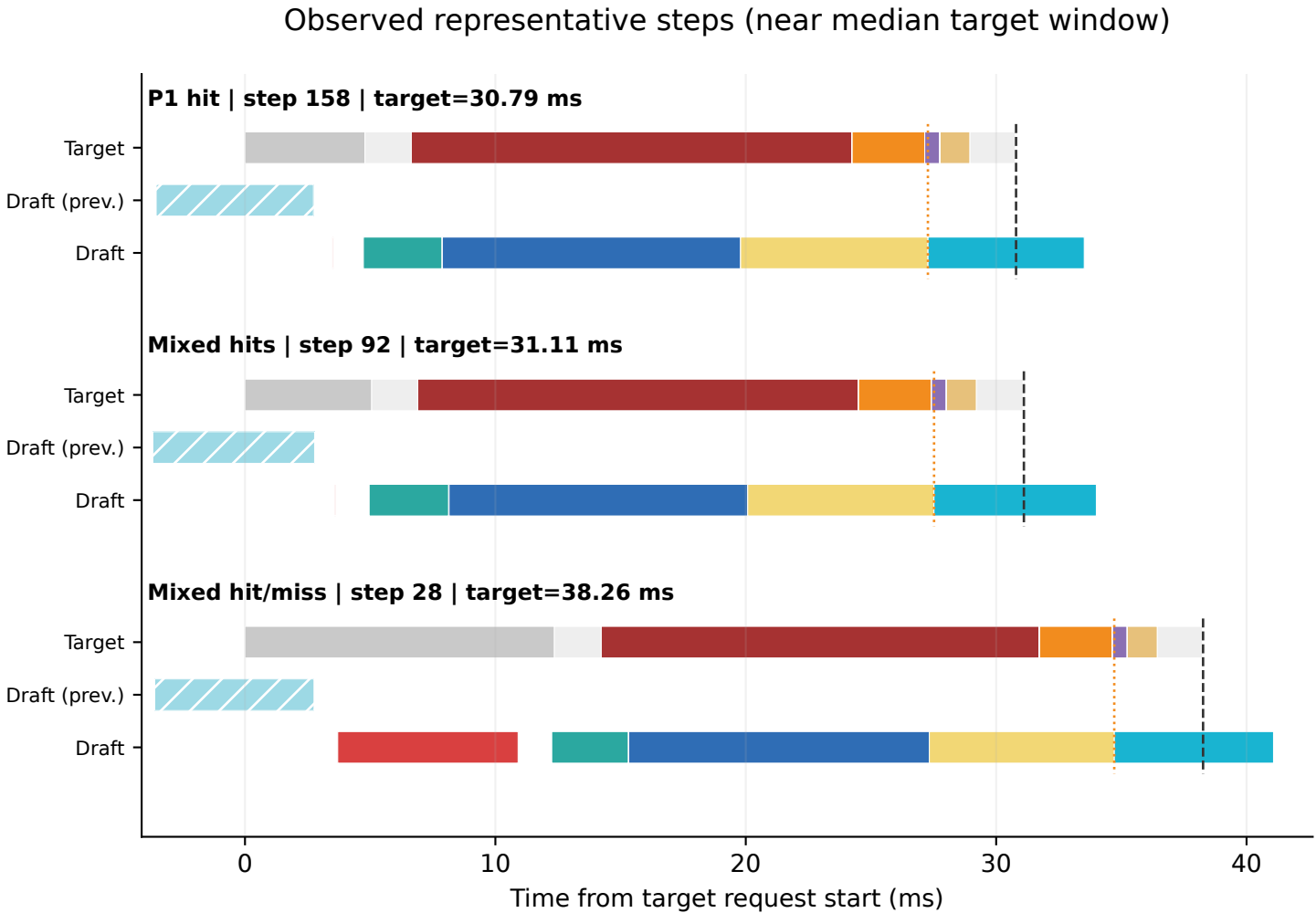
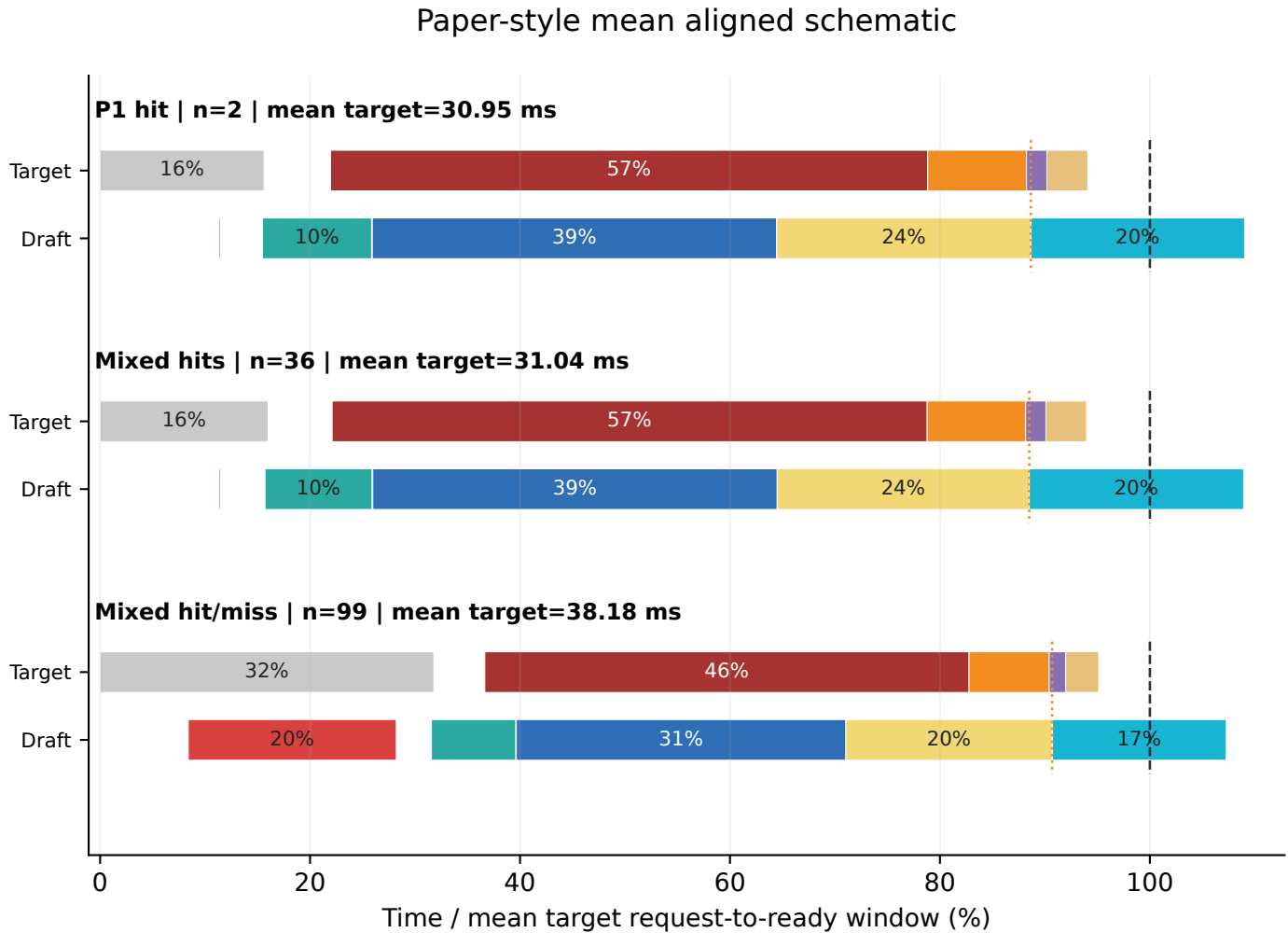
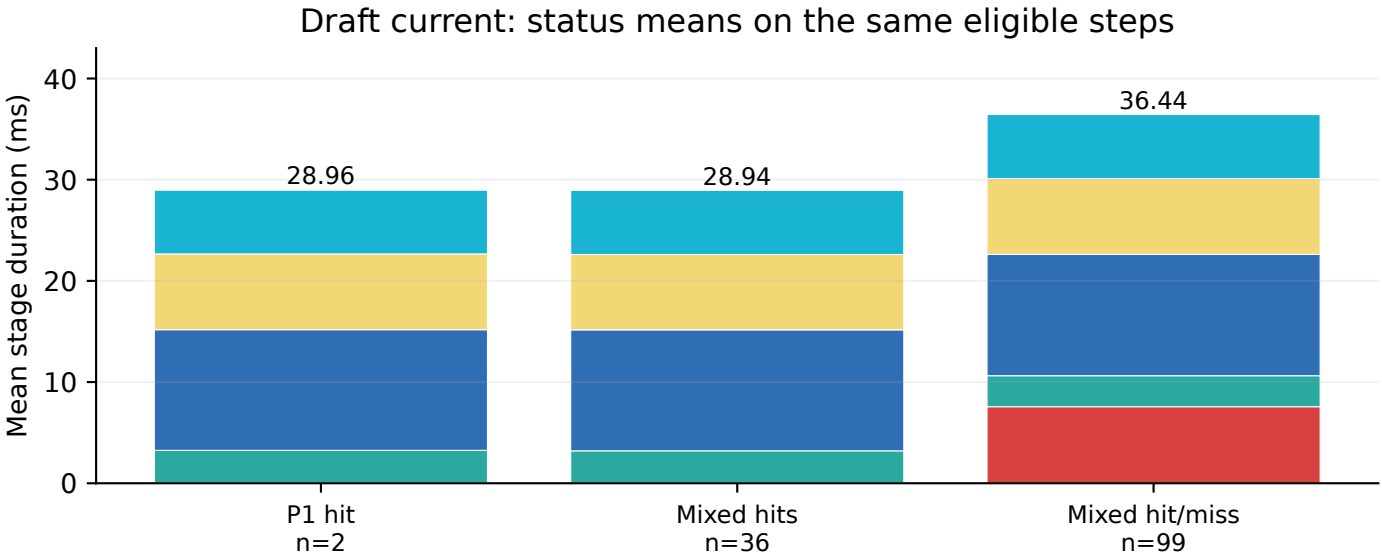
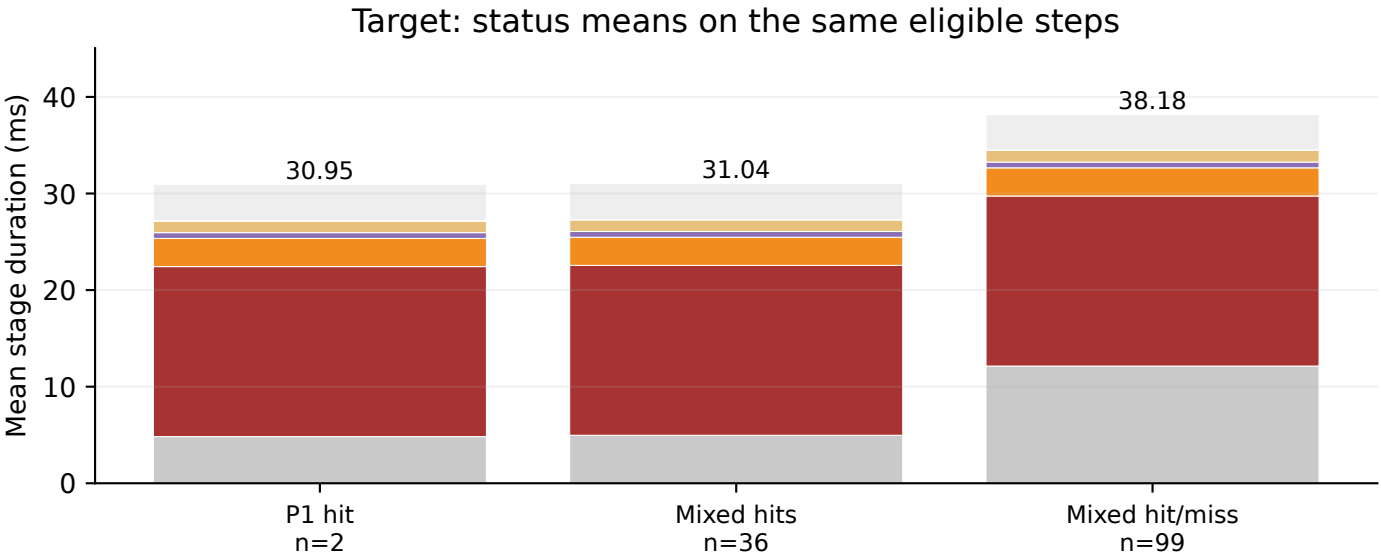
llama3_b8_refine_e21_k4_2_r1
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=1, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 110 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

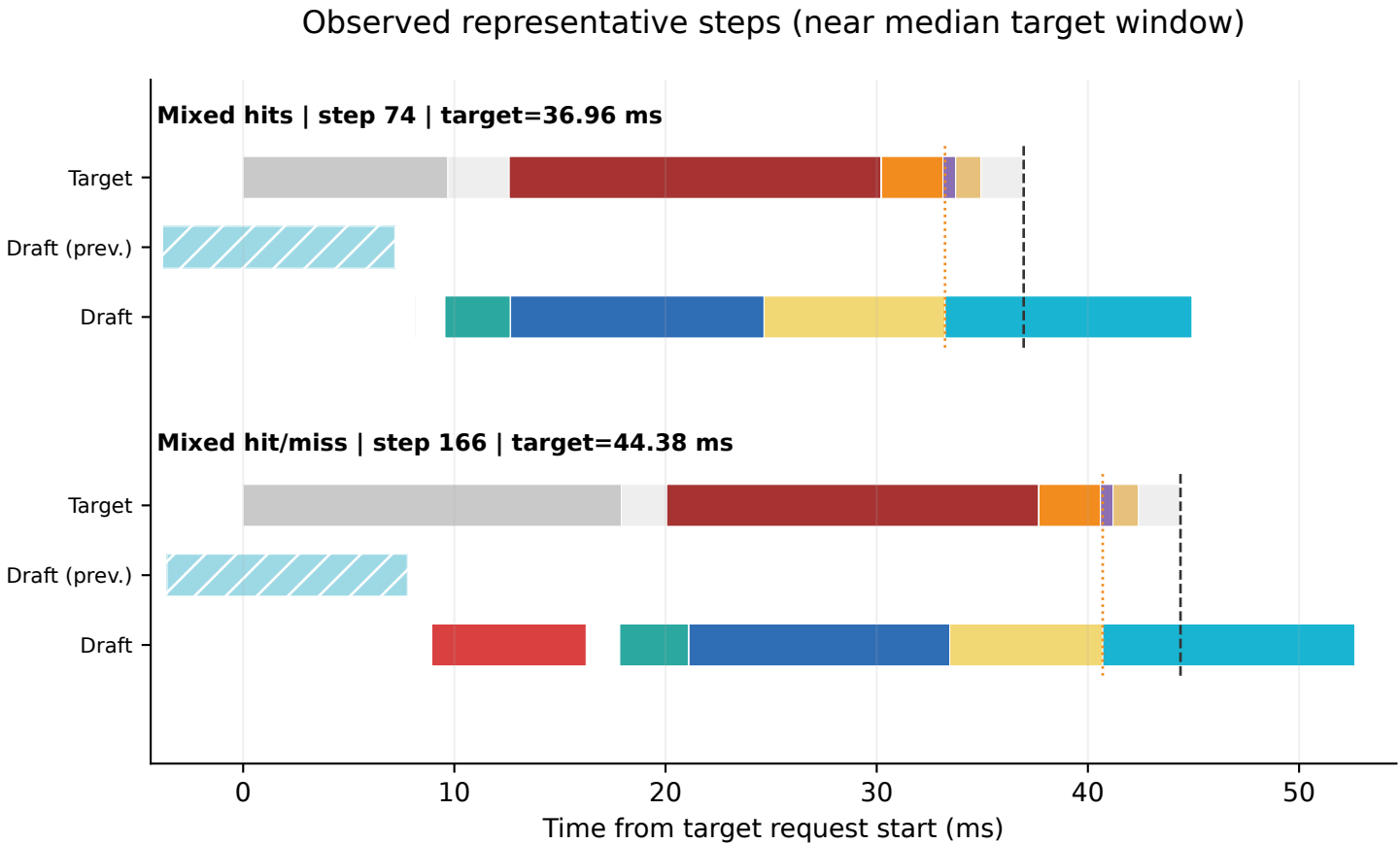
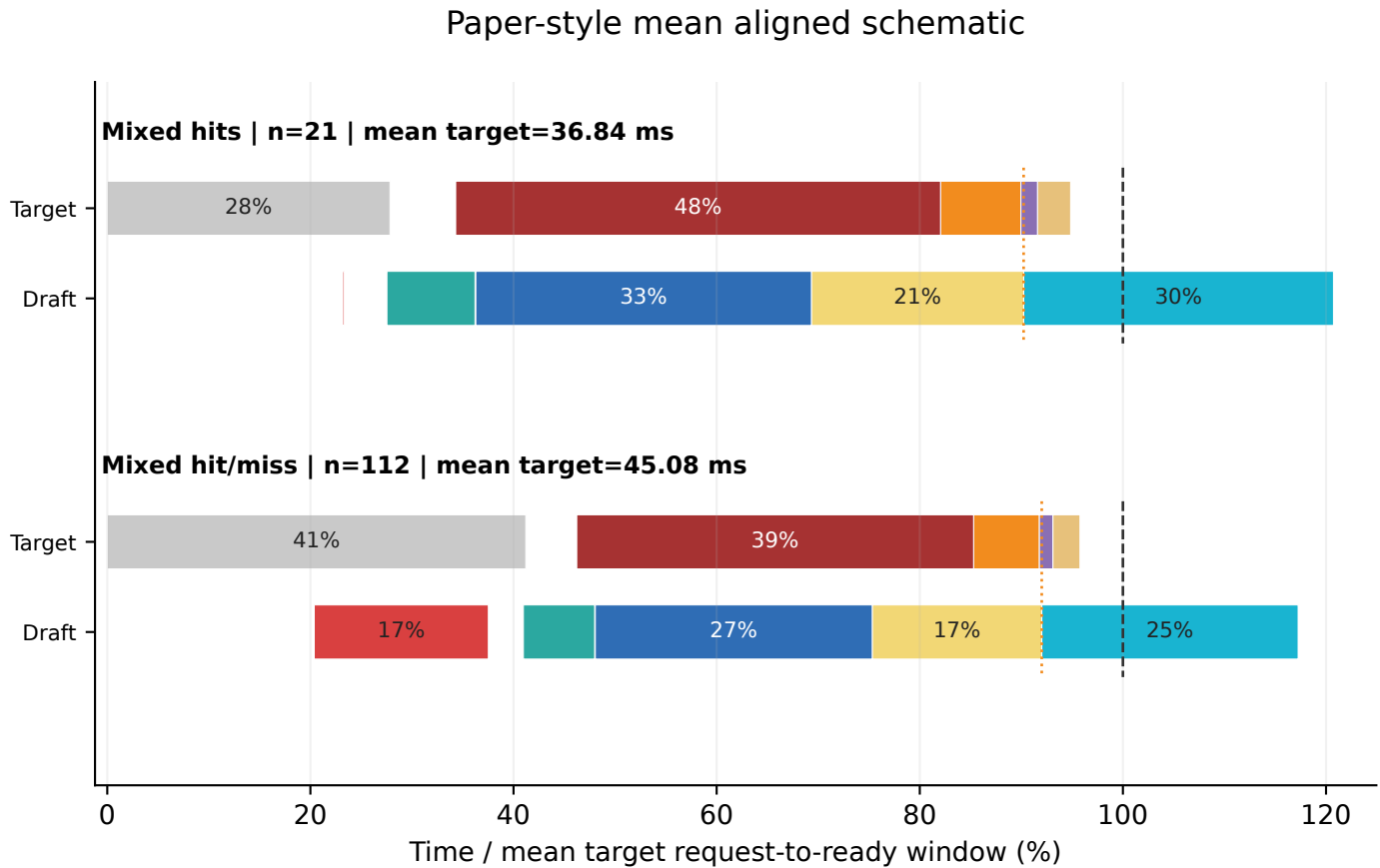
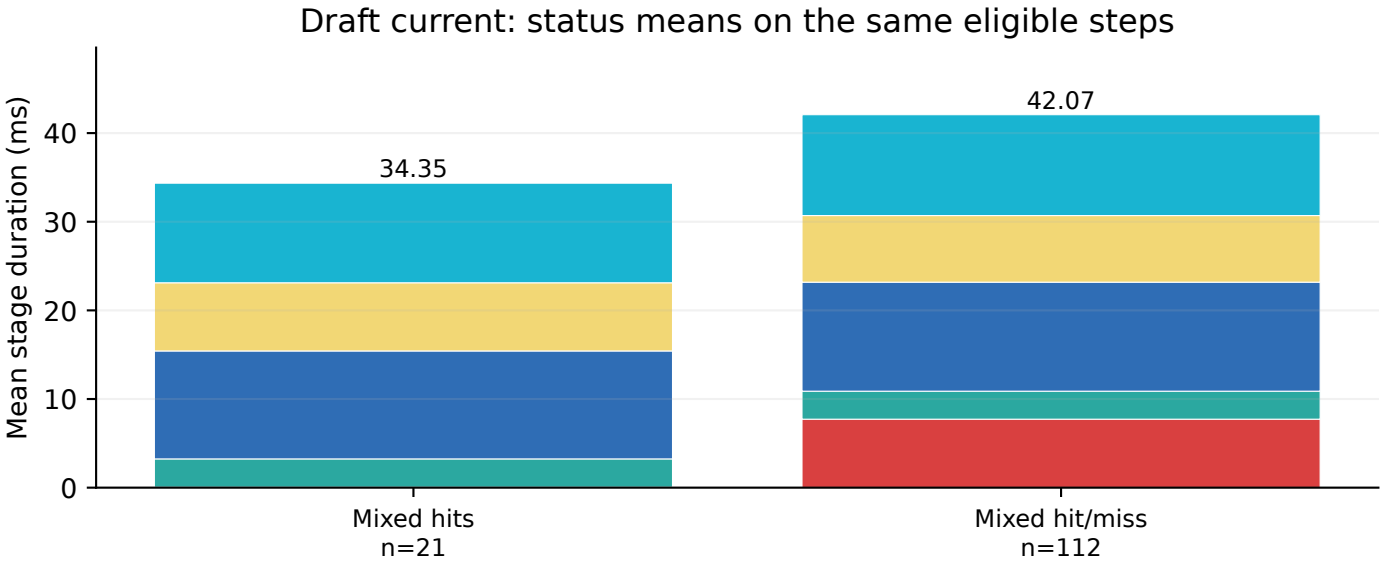
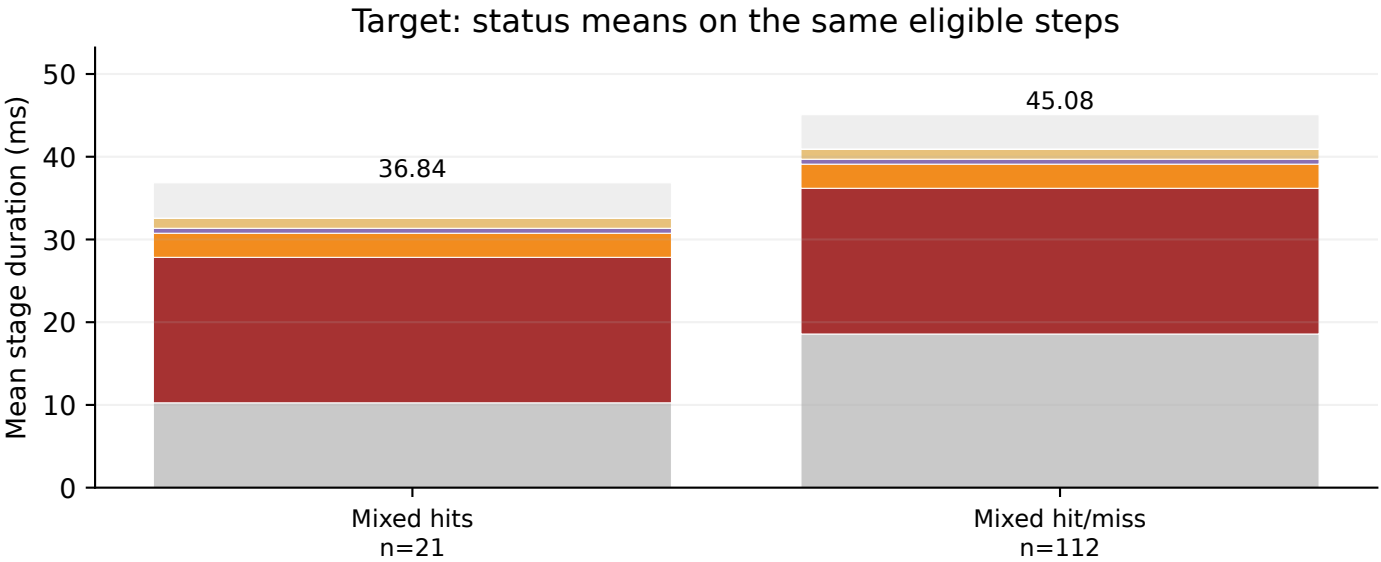
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_refine_e30_k2_1_r2
DUET: K1/K2=2/1, exit=30 (0-based), C=3, N1/N2=4/2, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 137 eligible steps



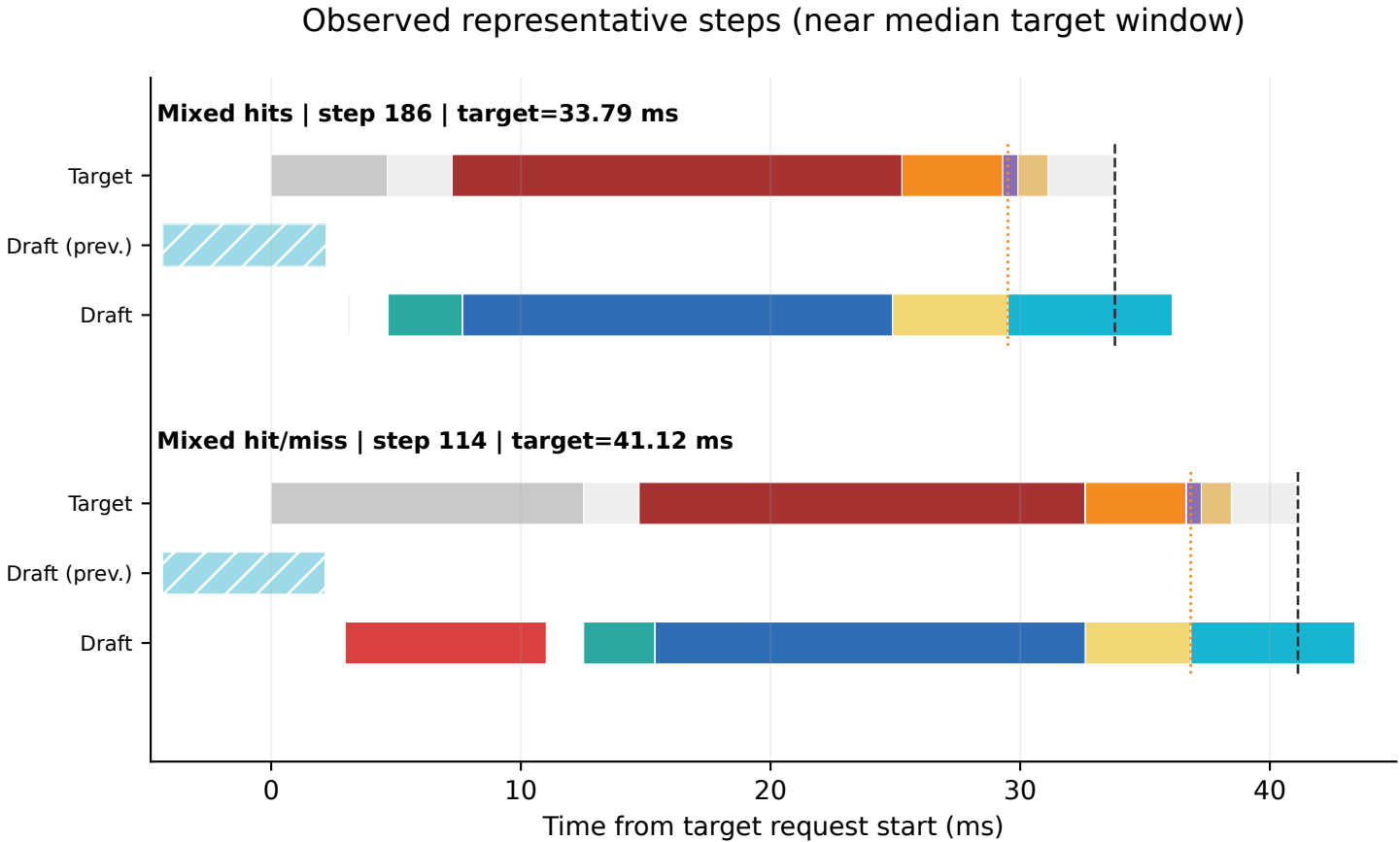
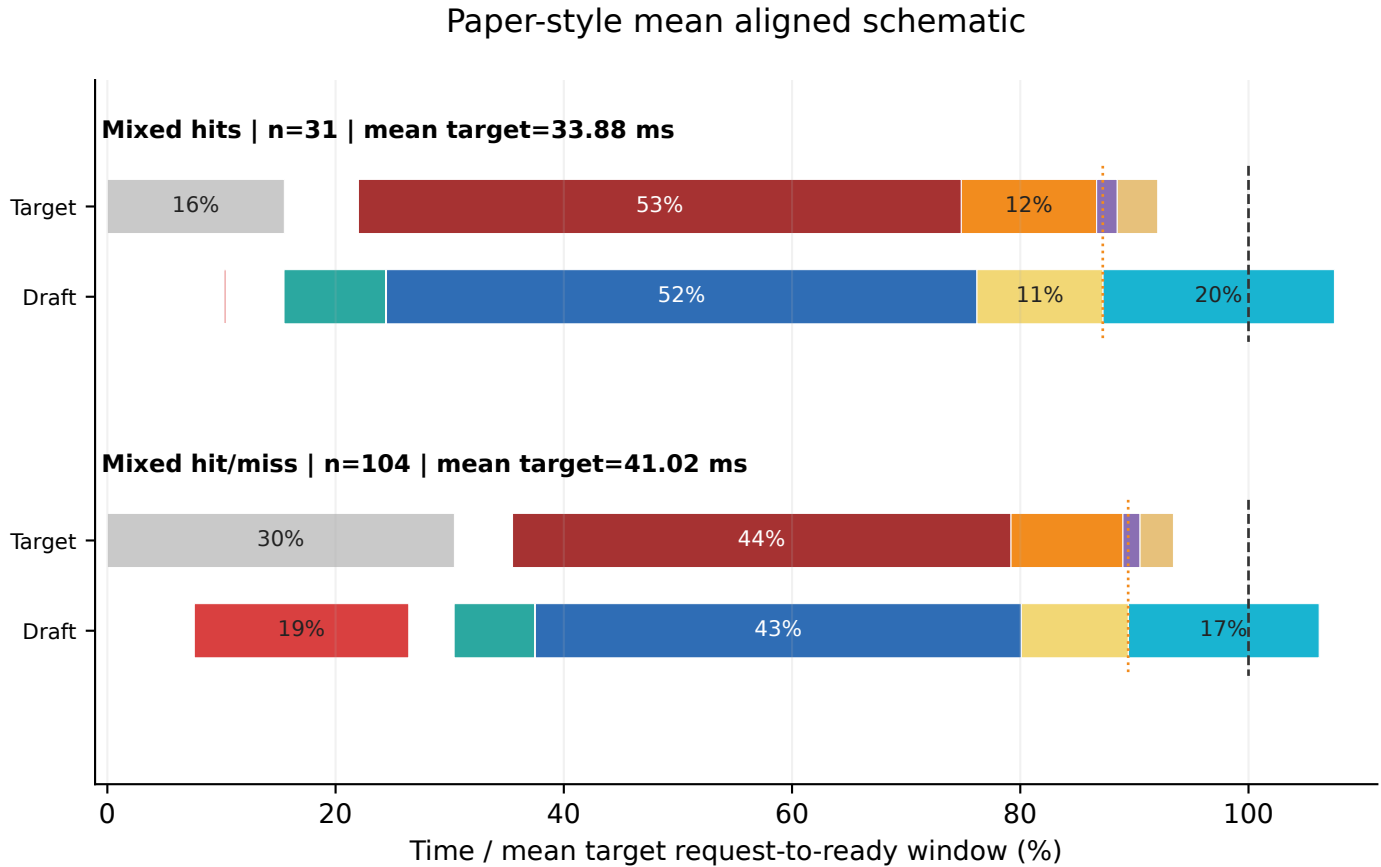
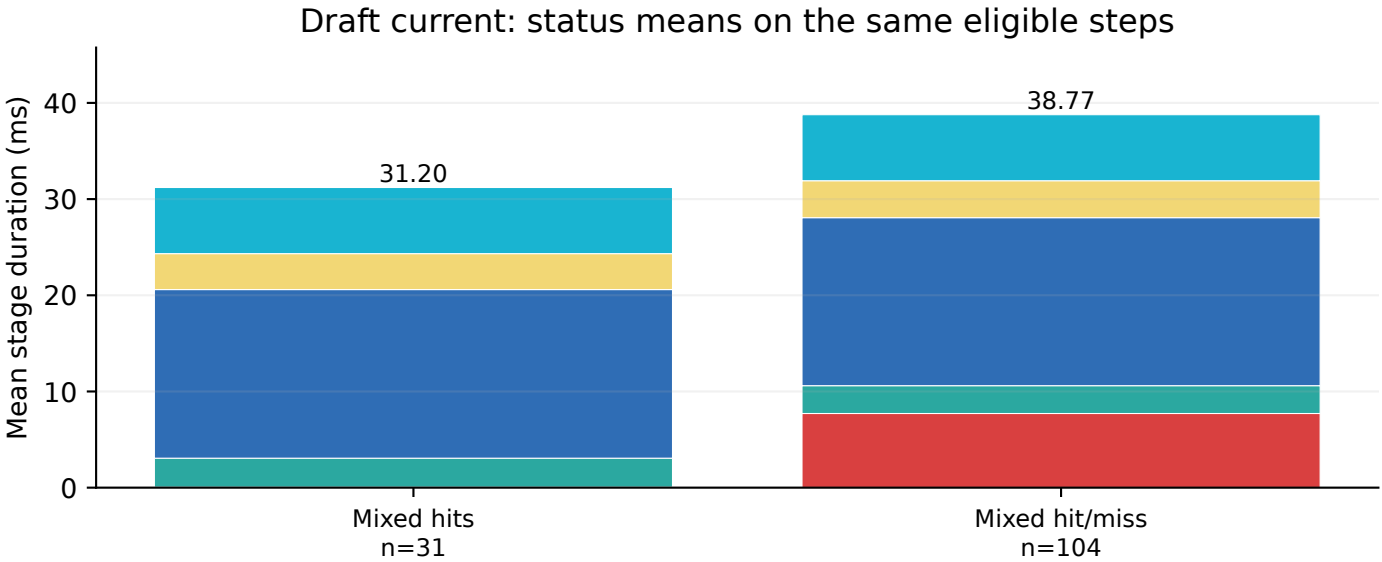
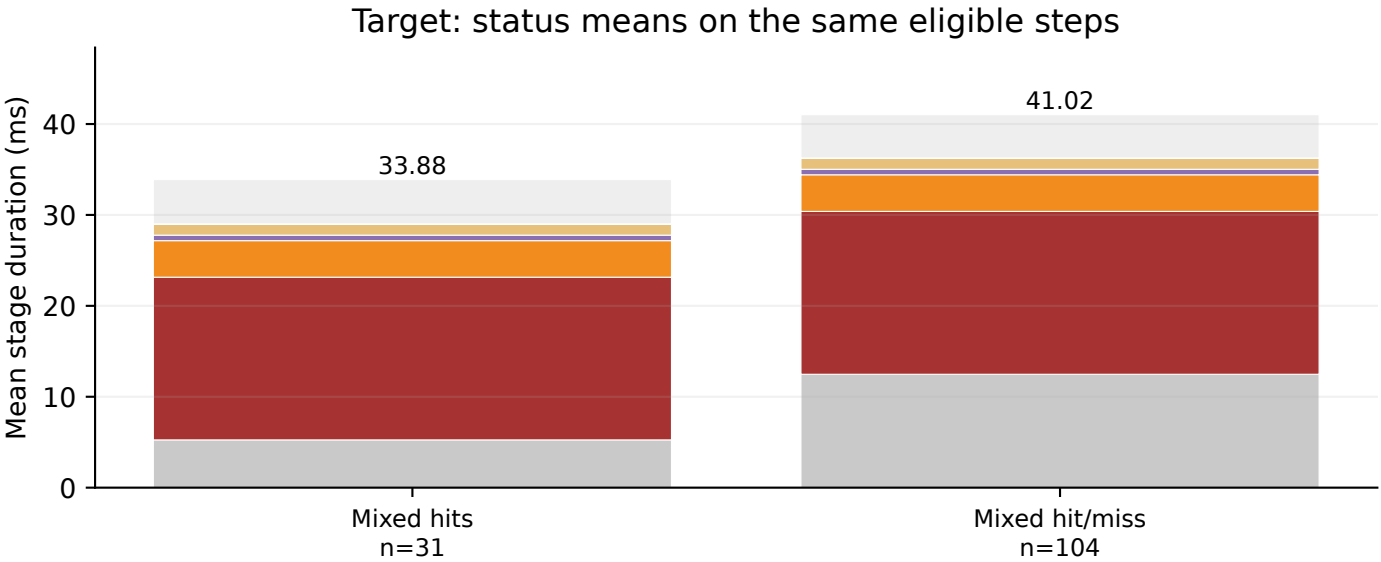
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



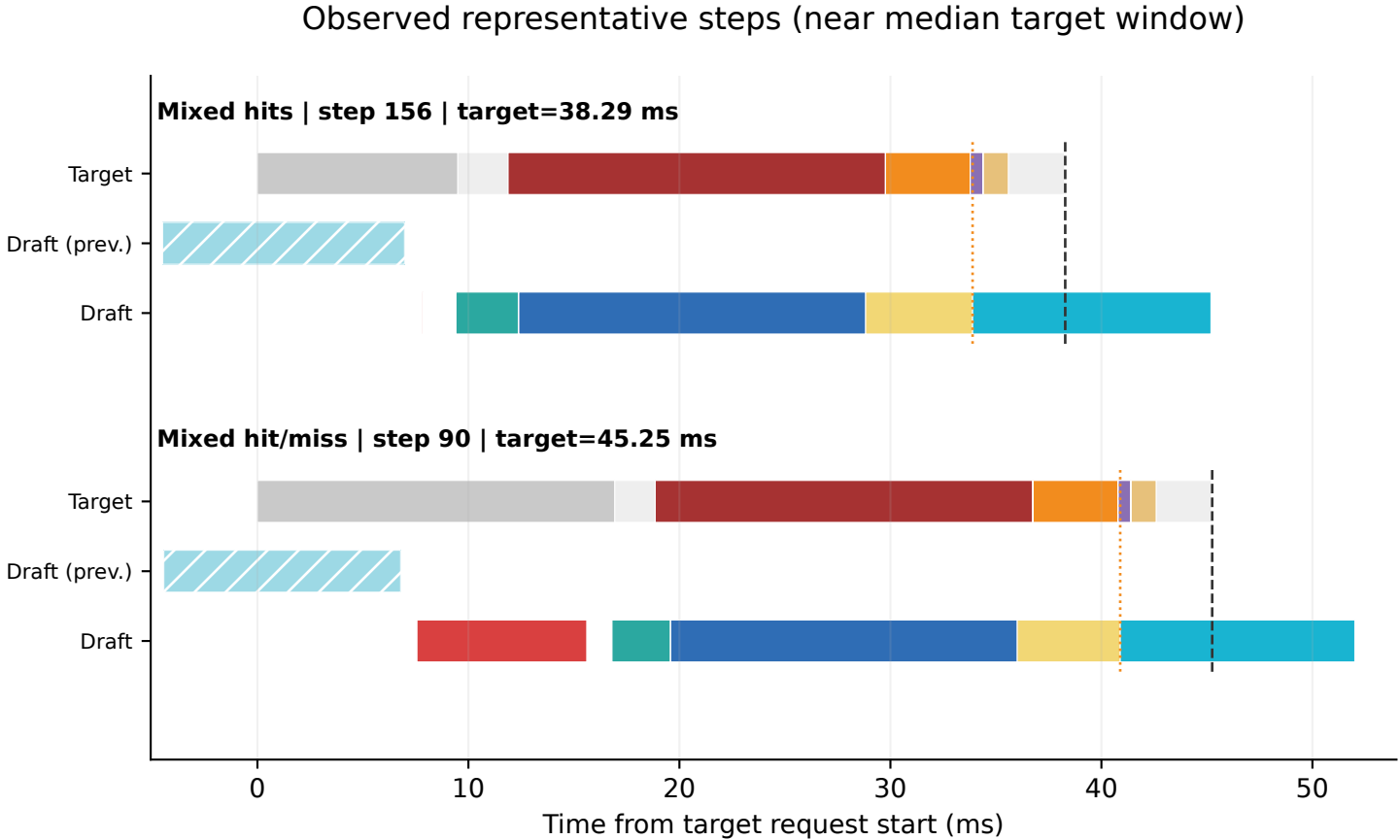
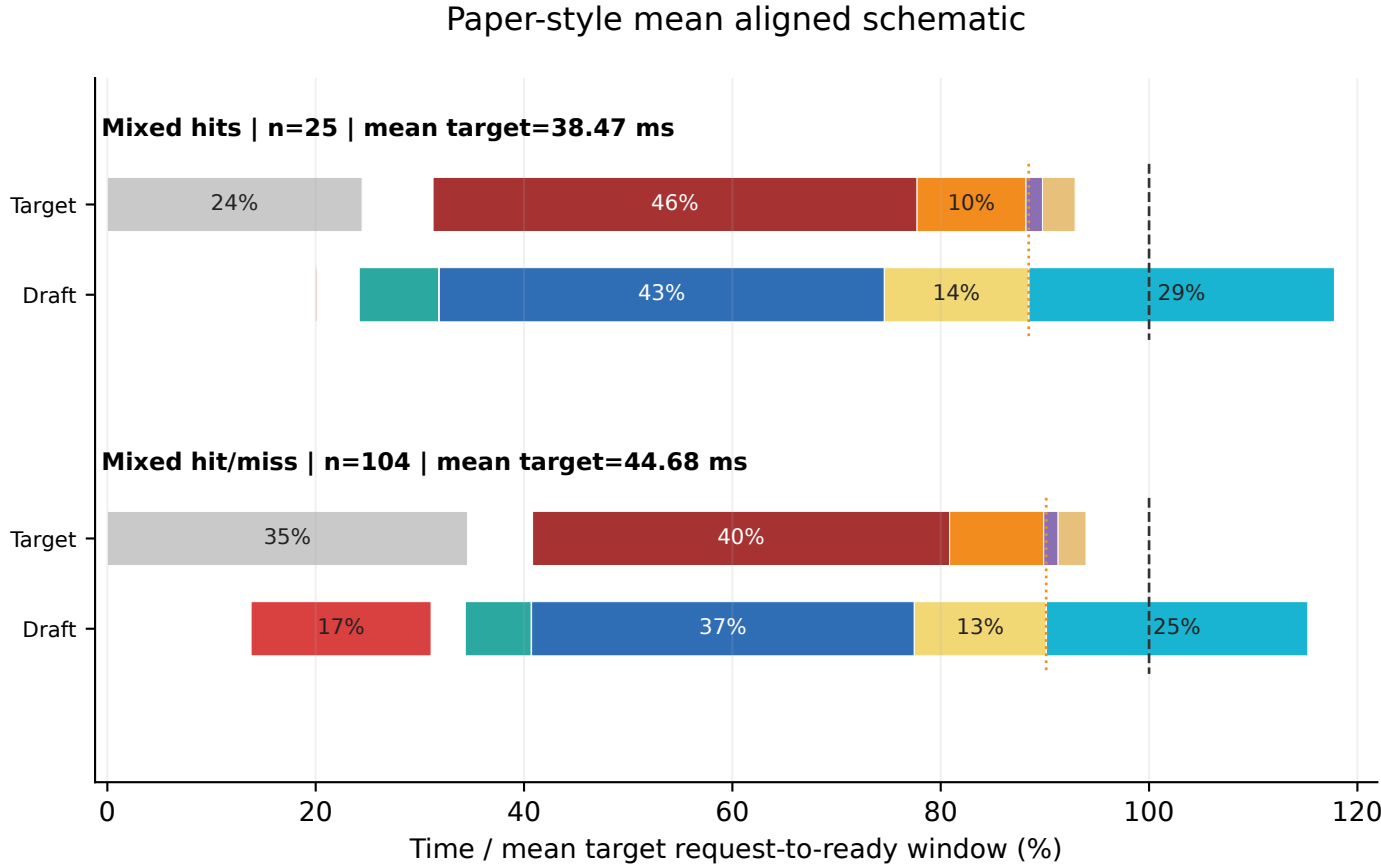
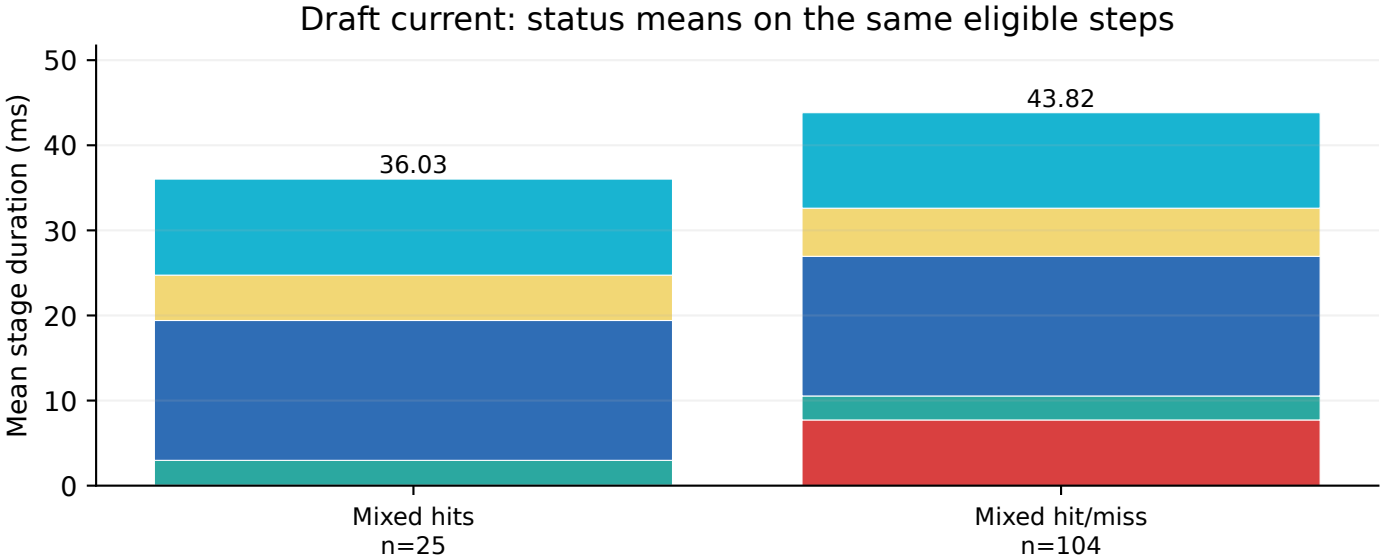
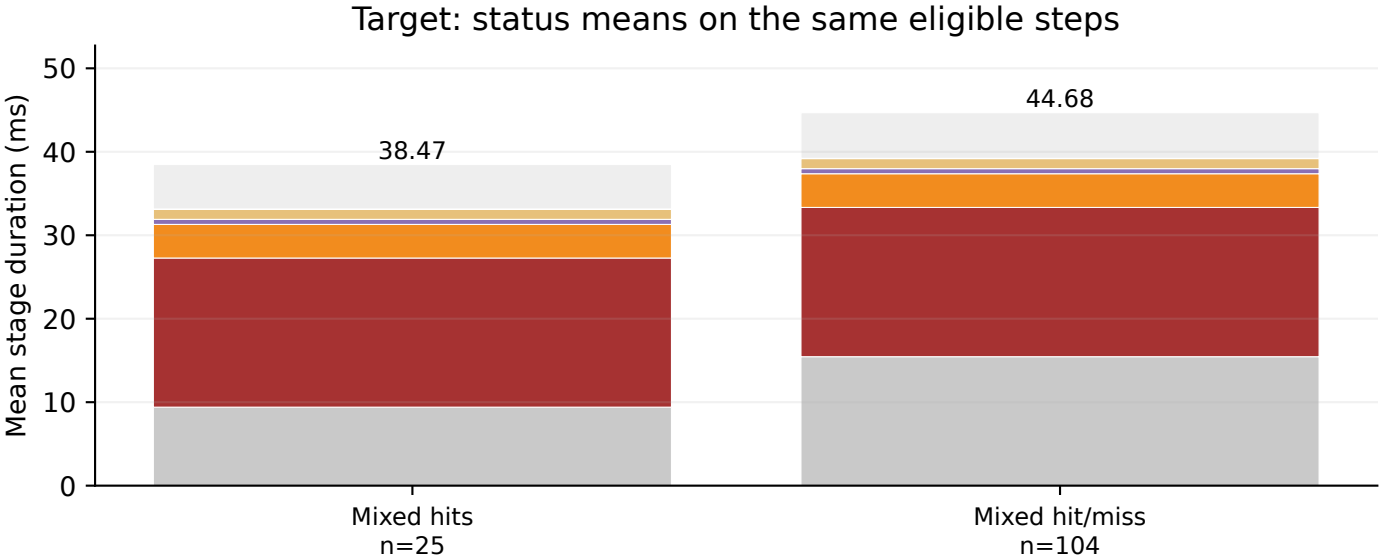
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



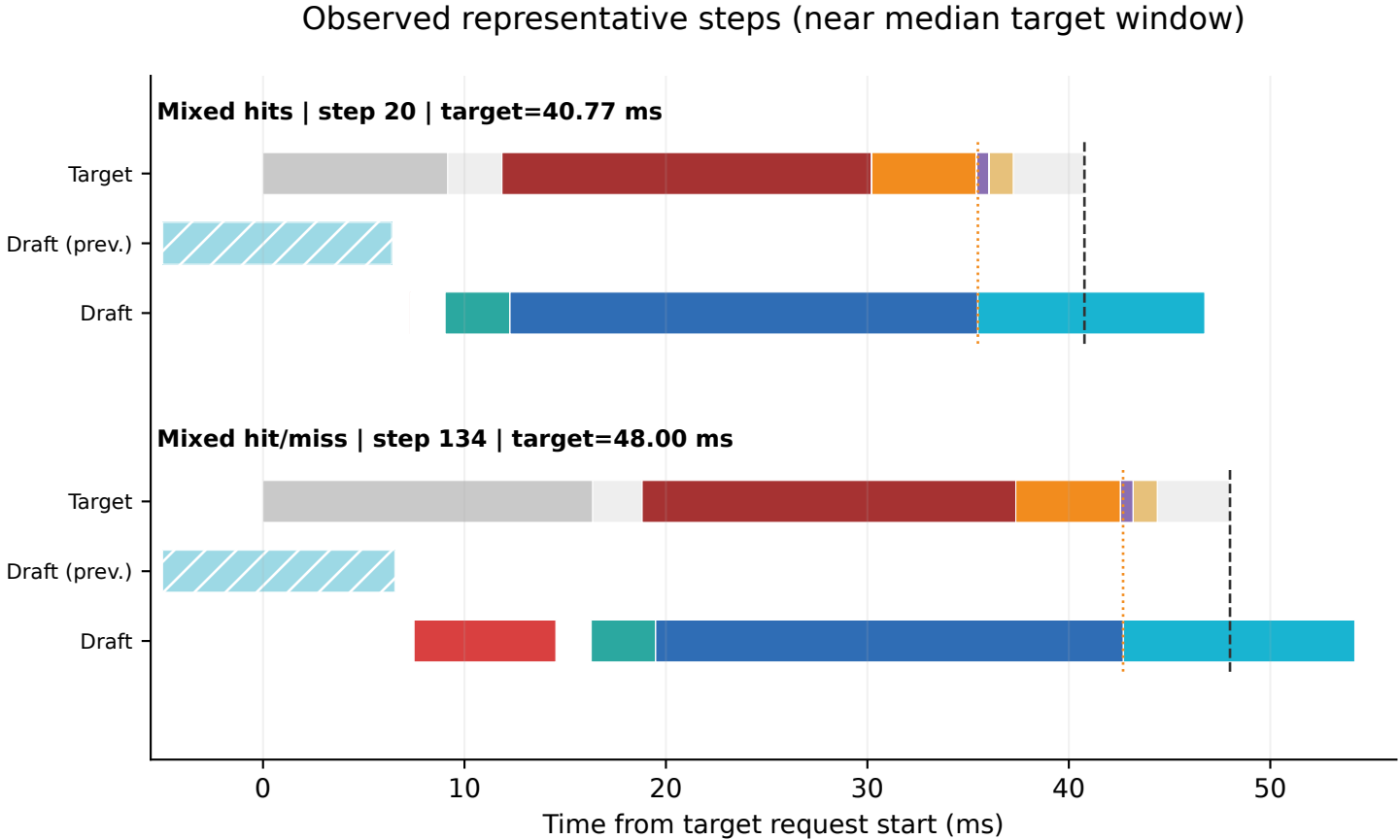
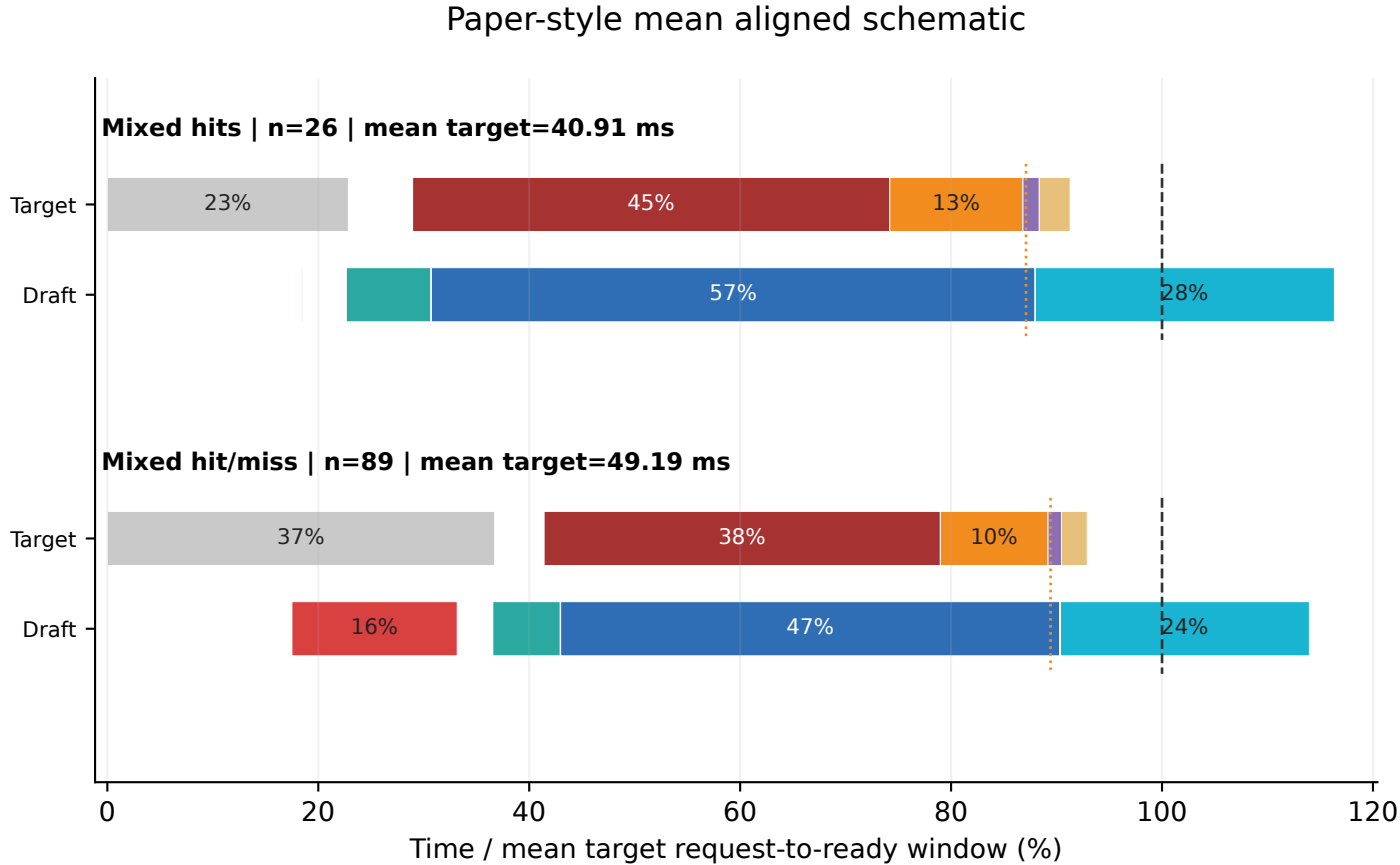
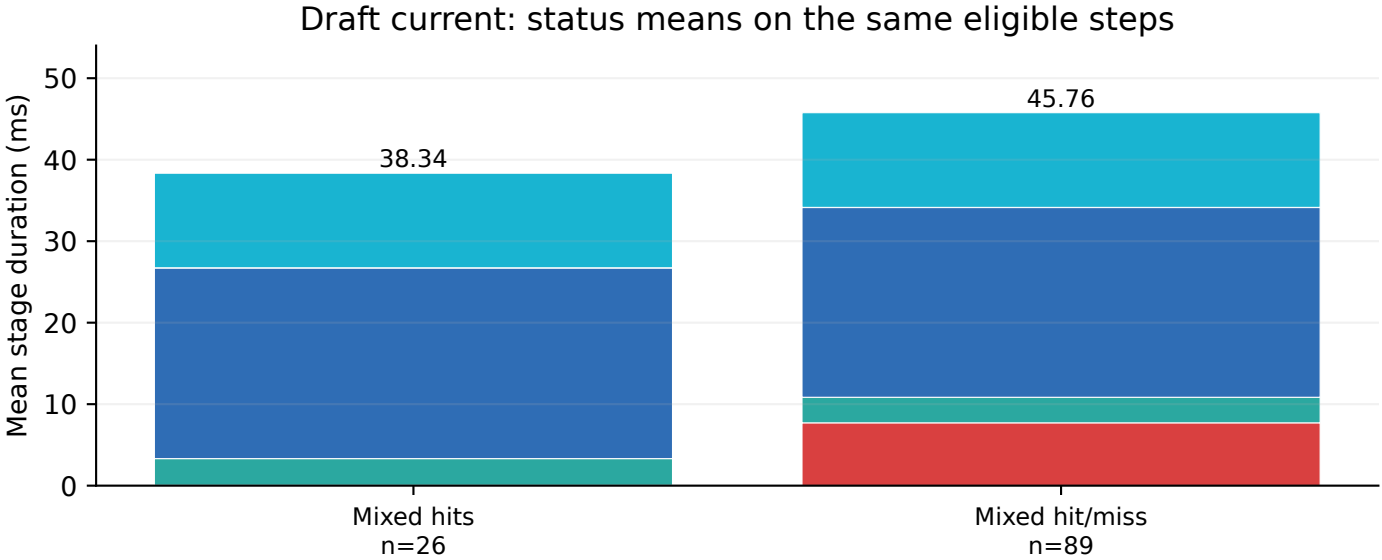
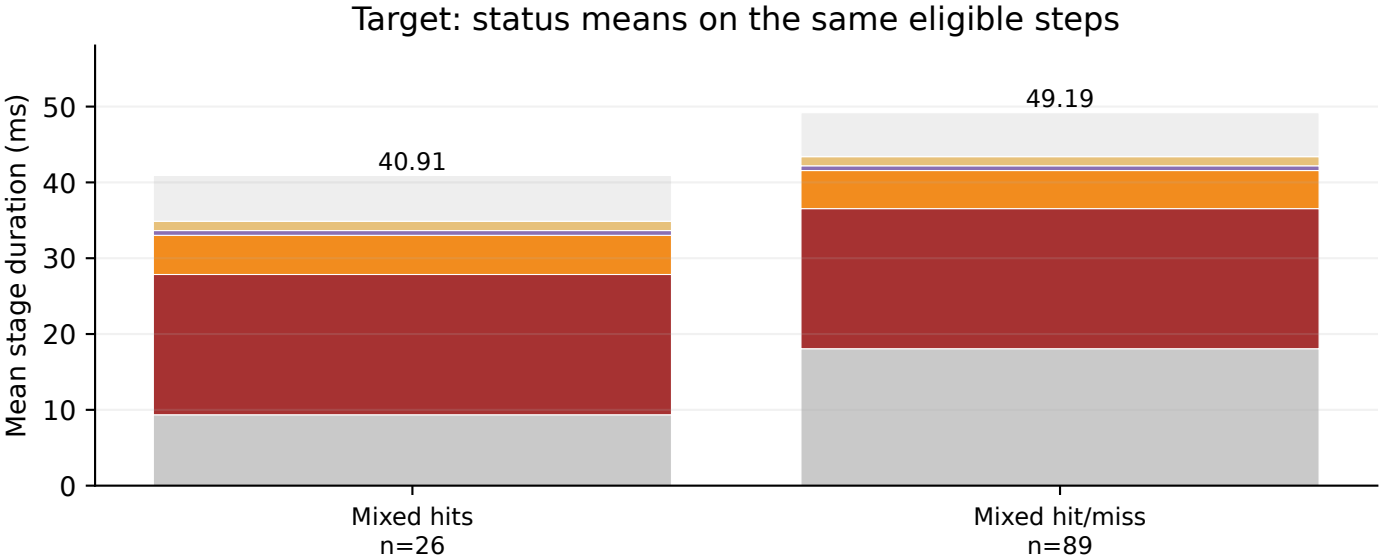
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

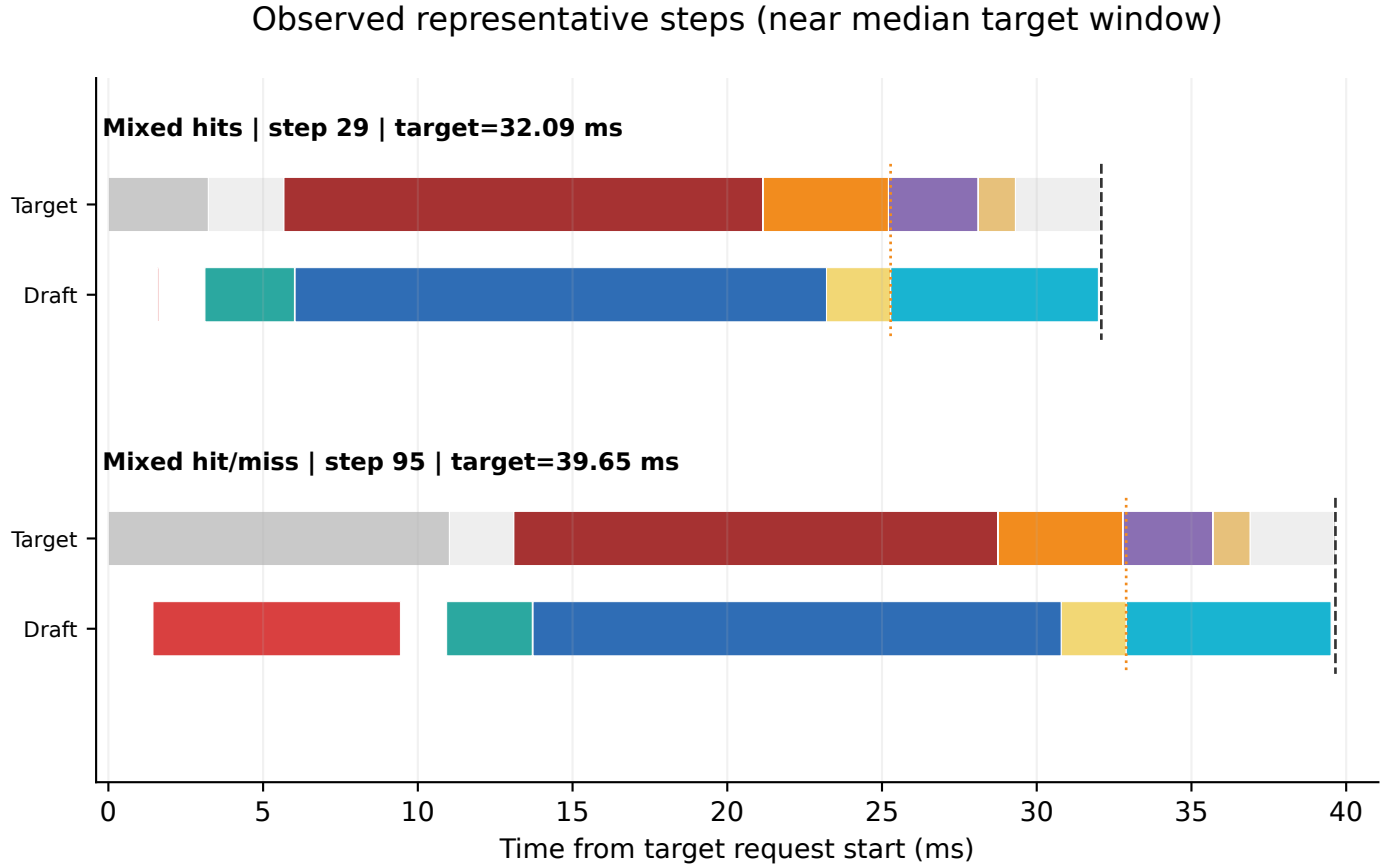
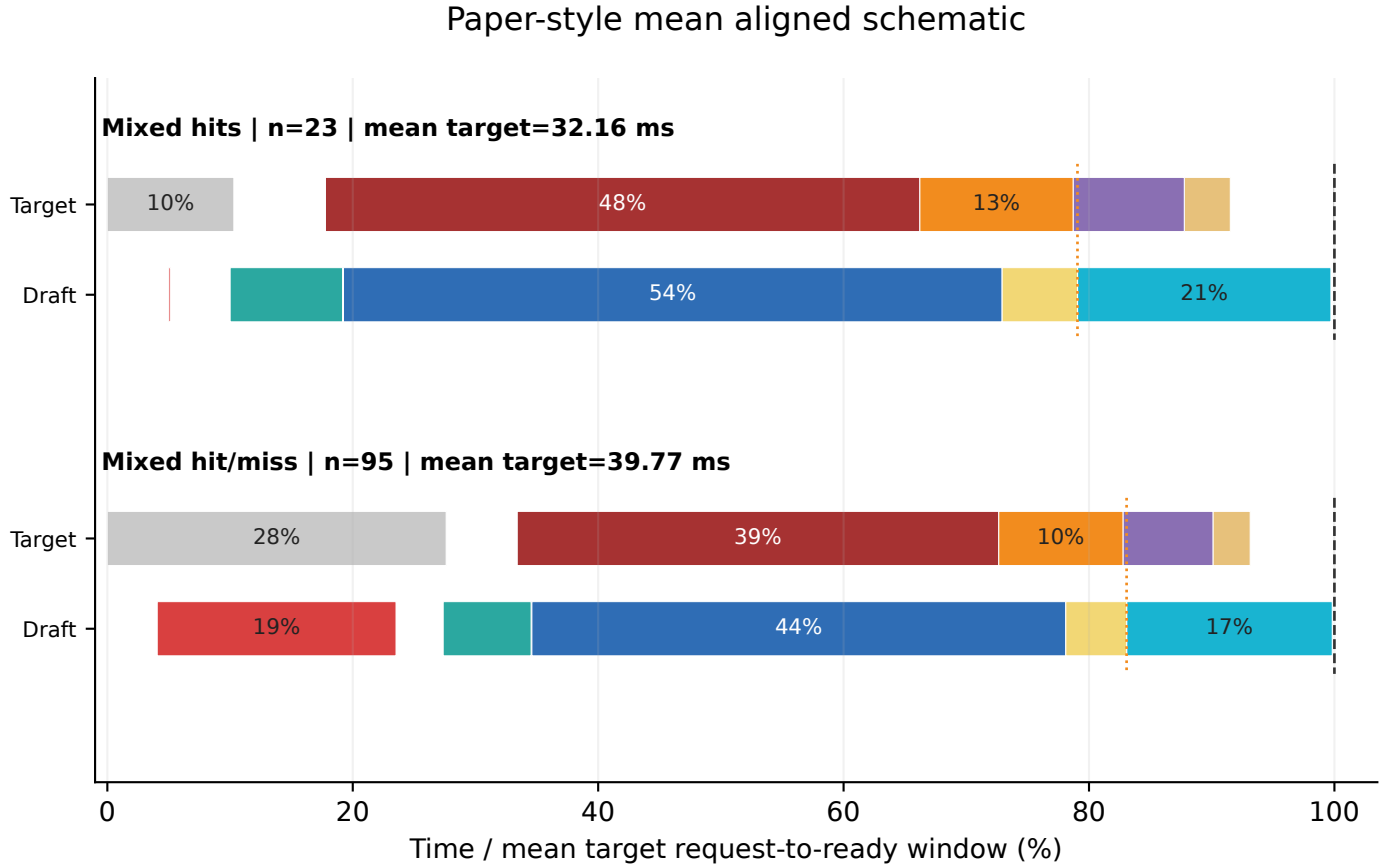
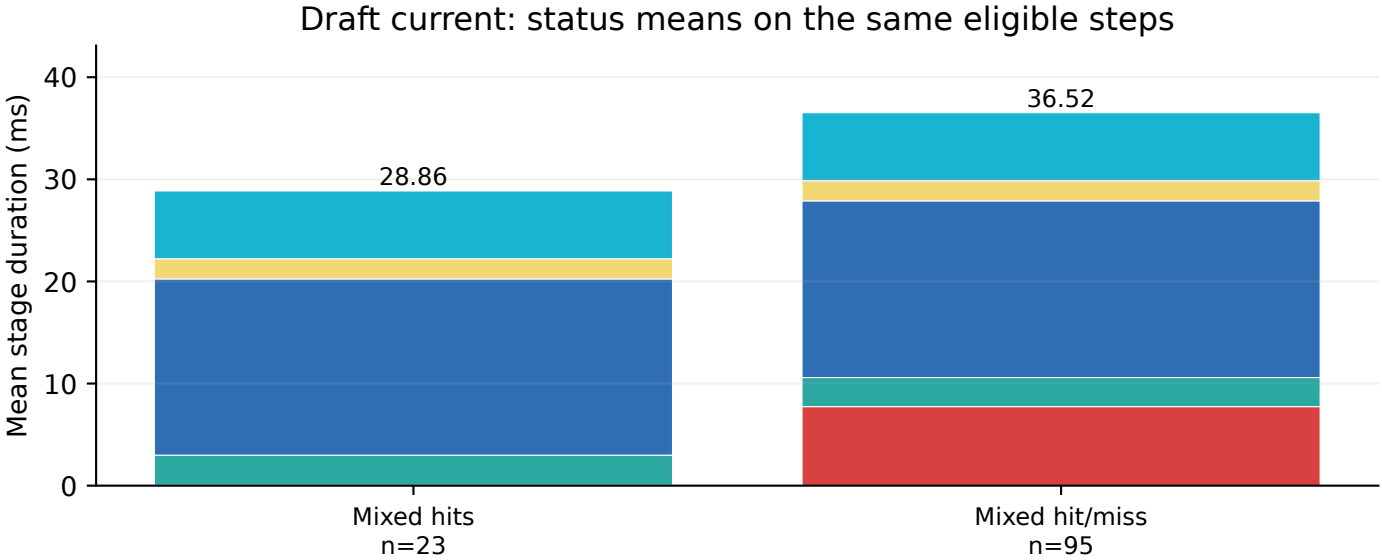
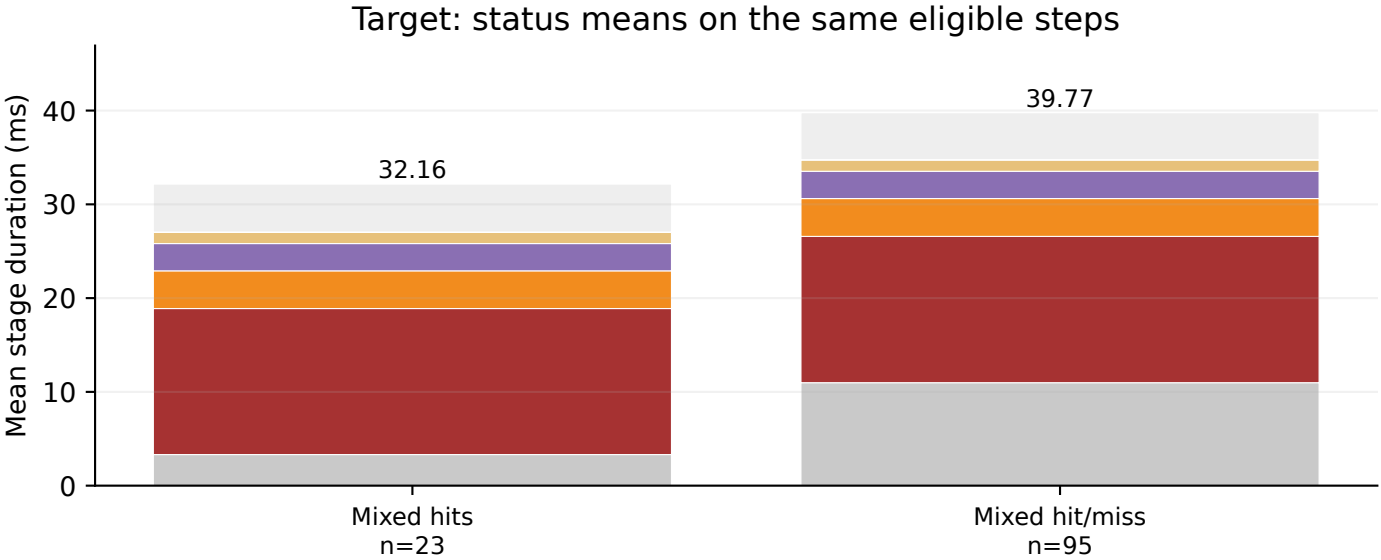
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

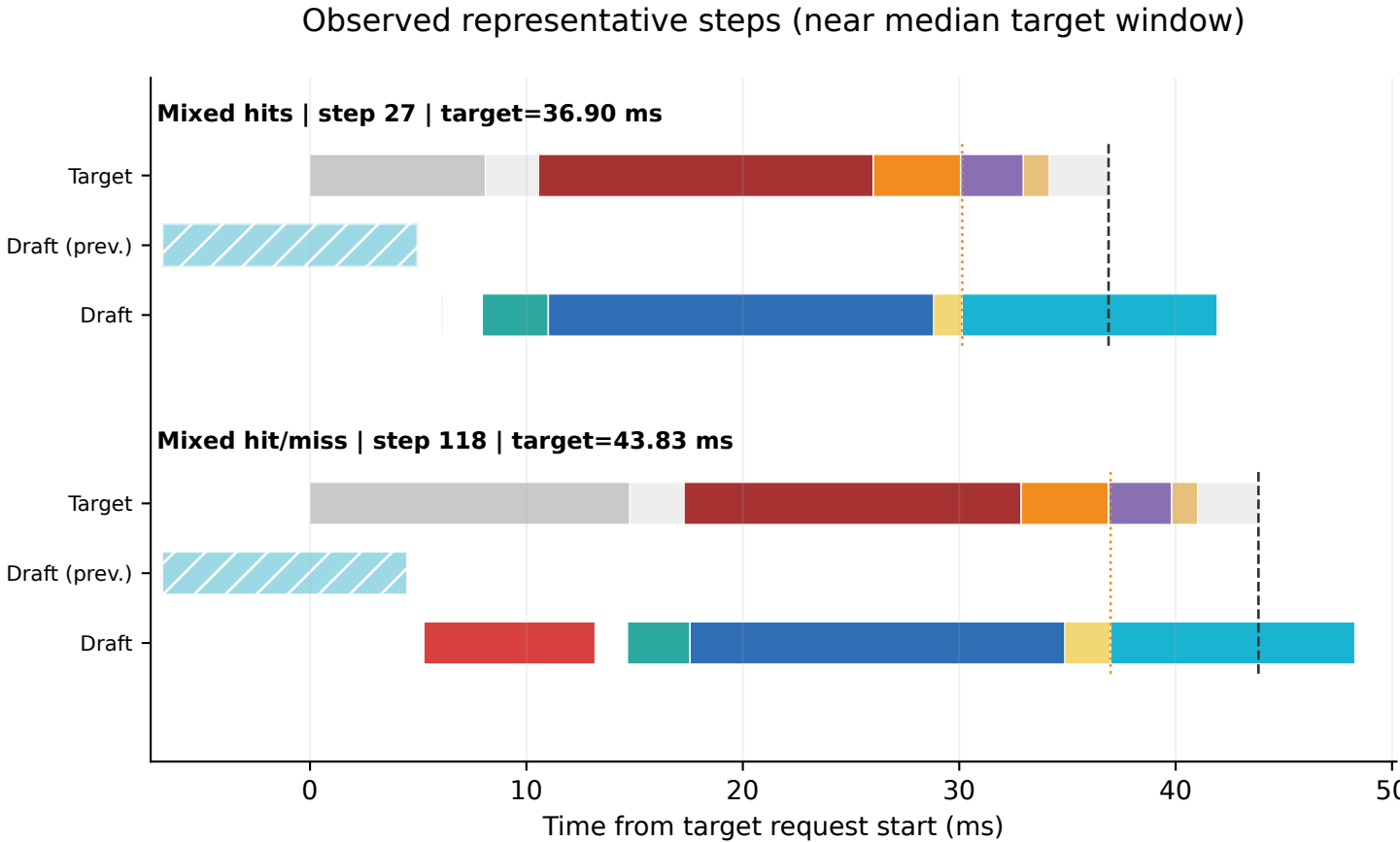
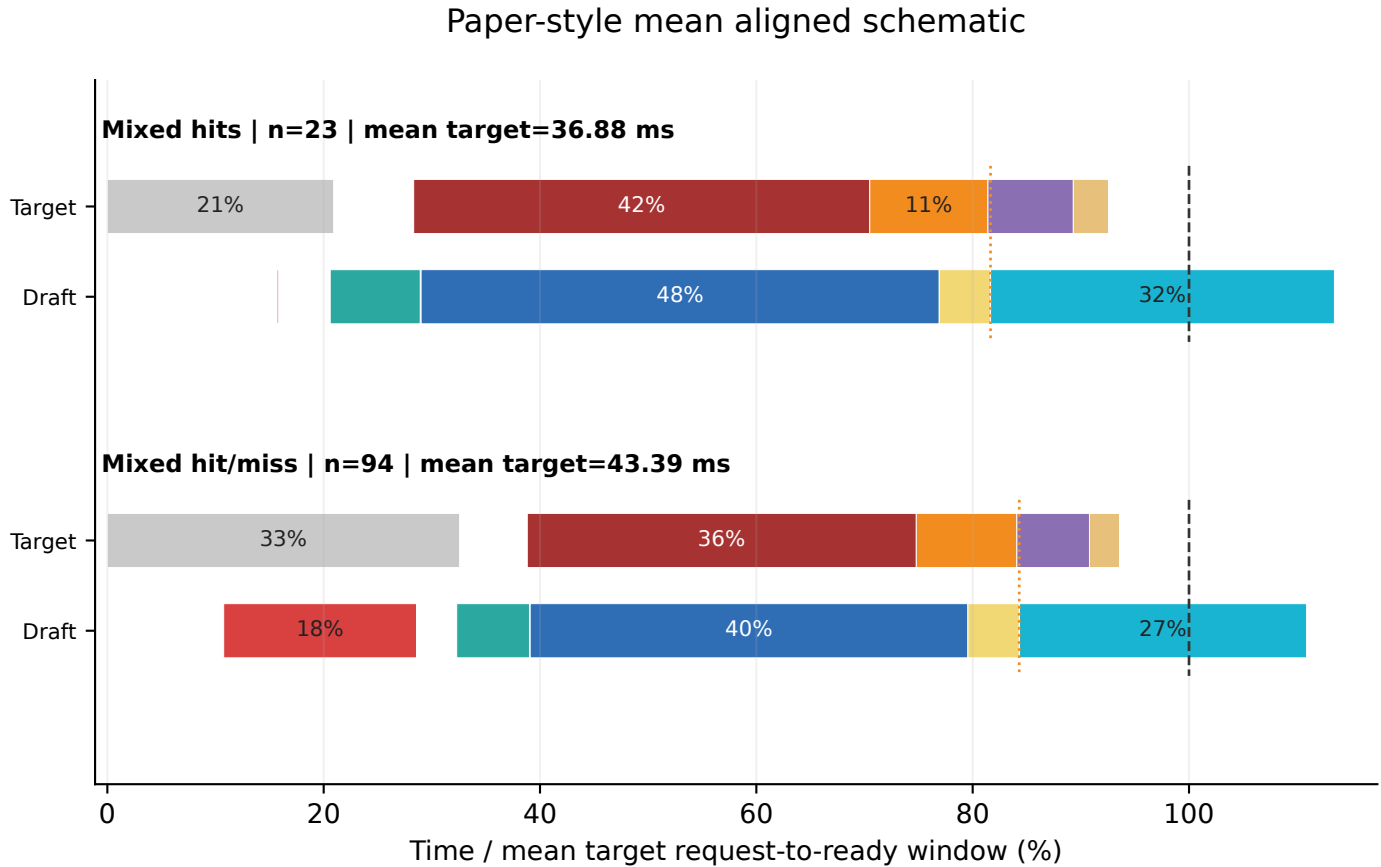
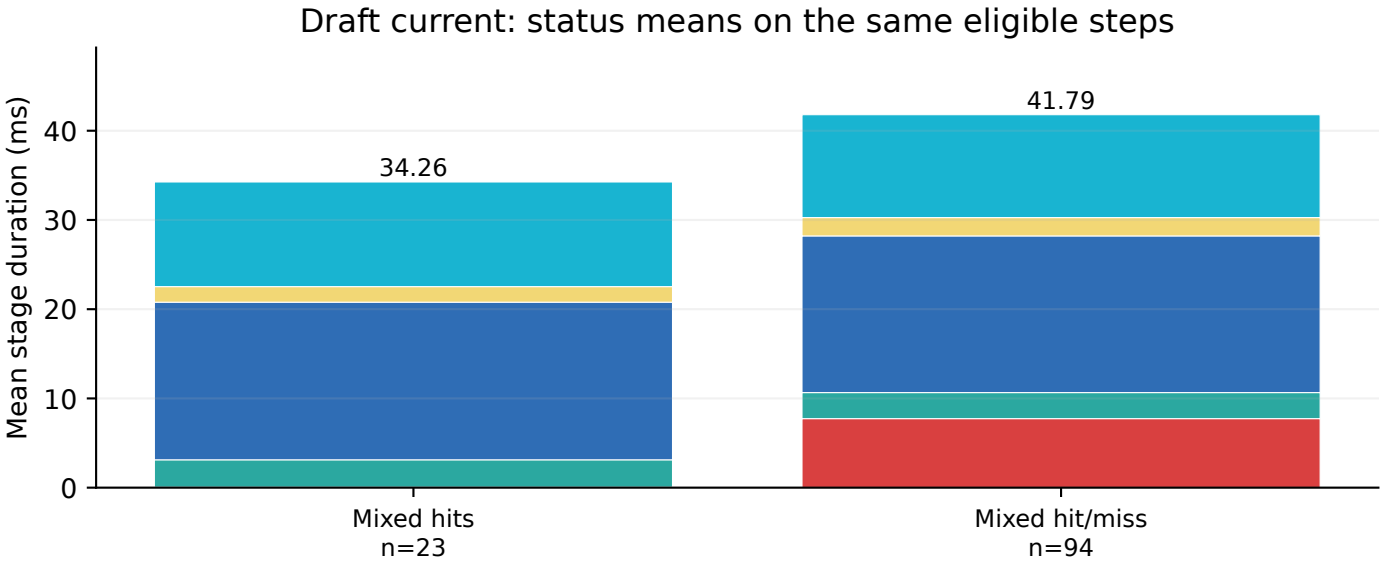
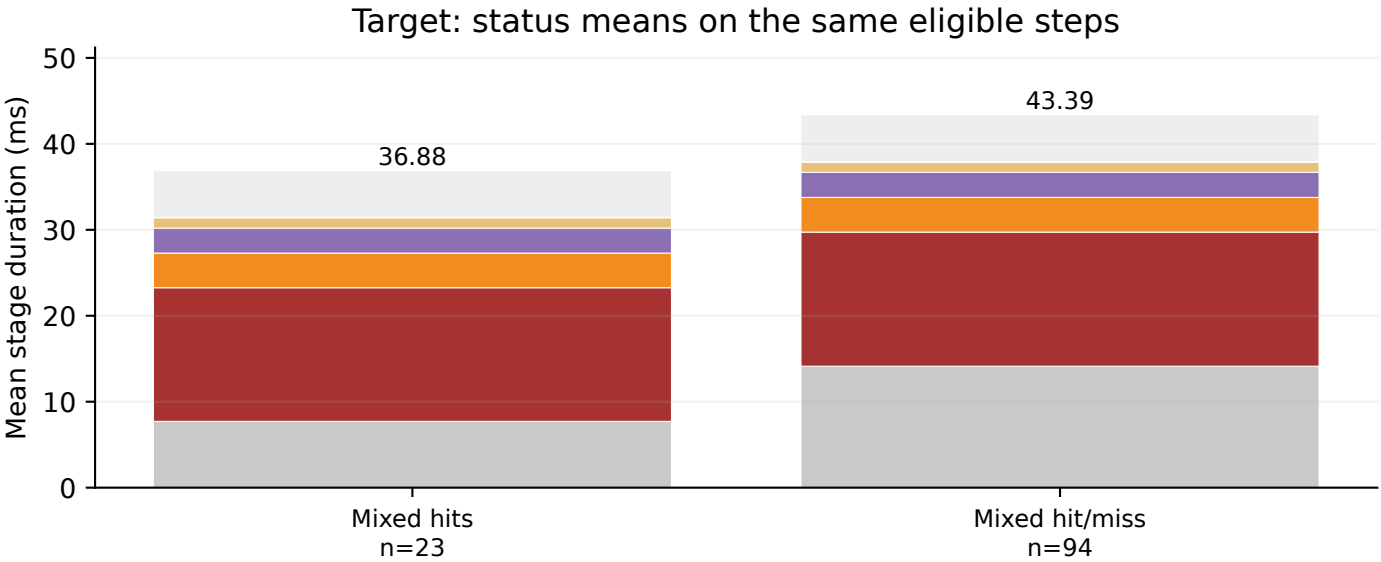
llama3_b8_refine_e26_k3_1_r2
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 118 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

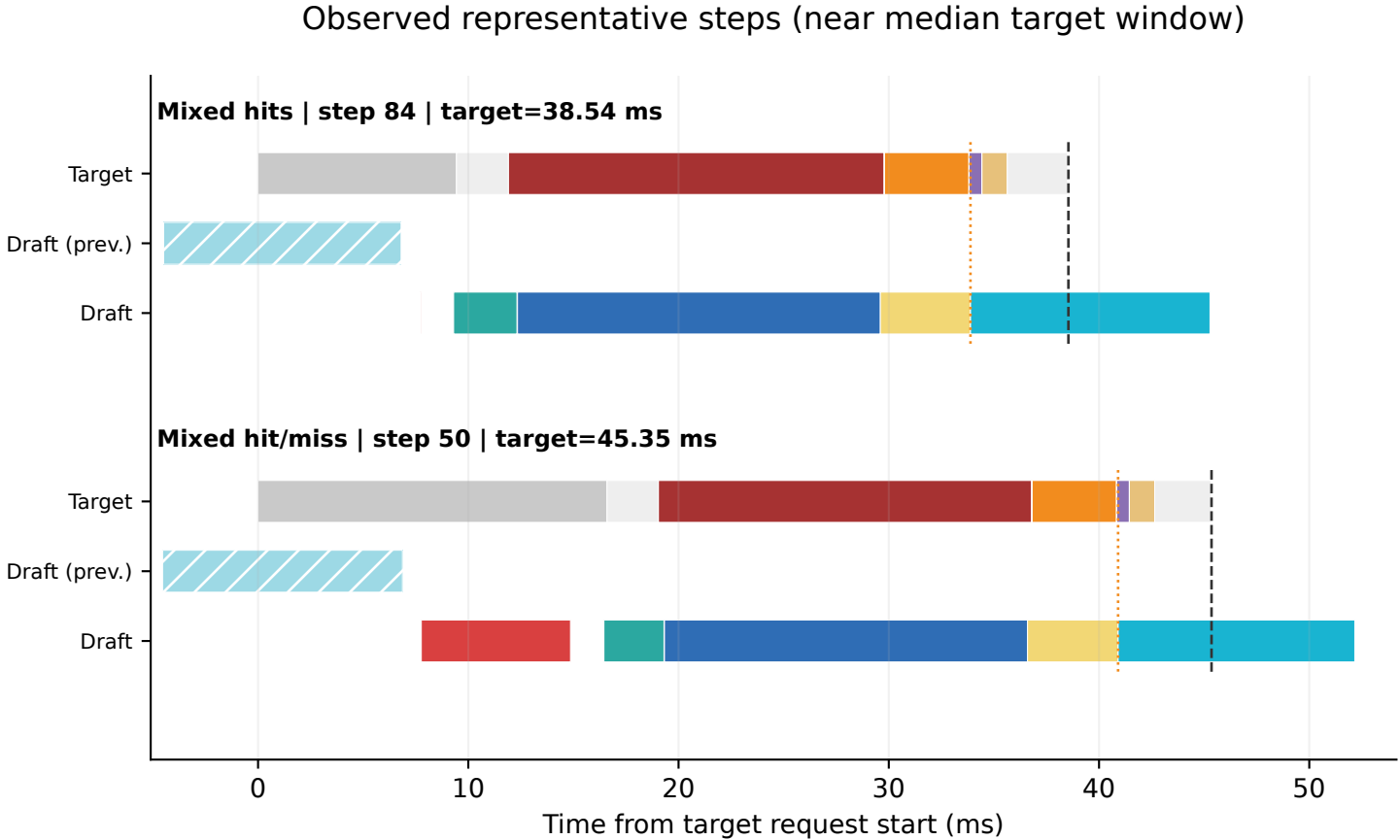
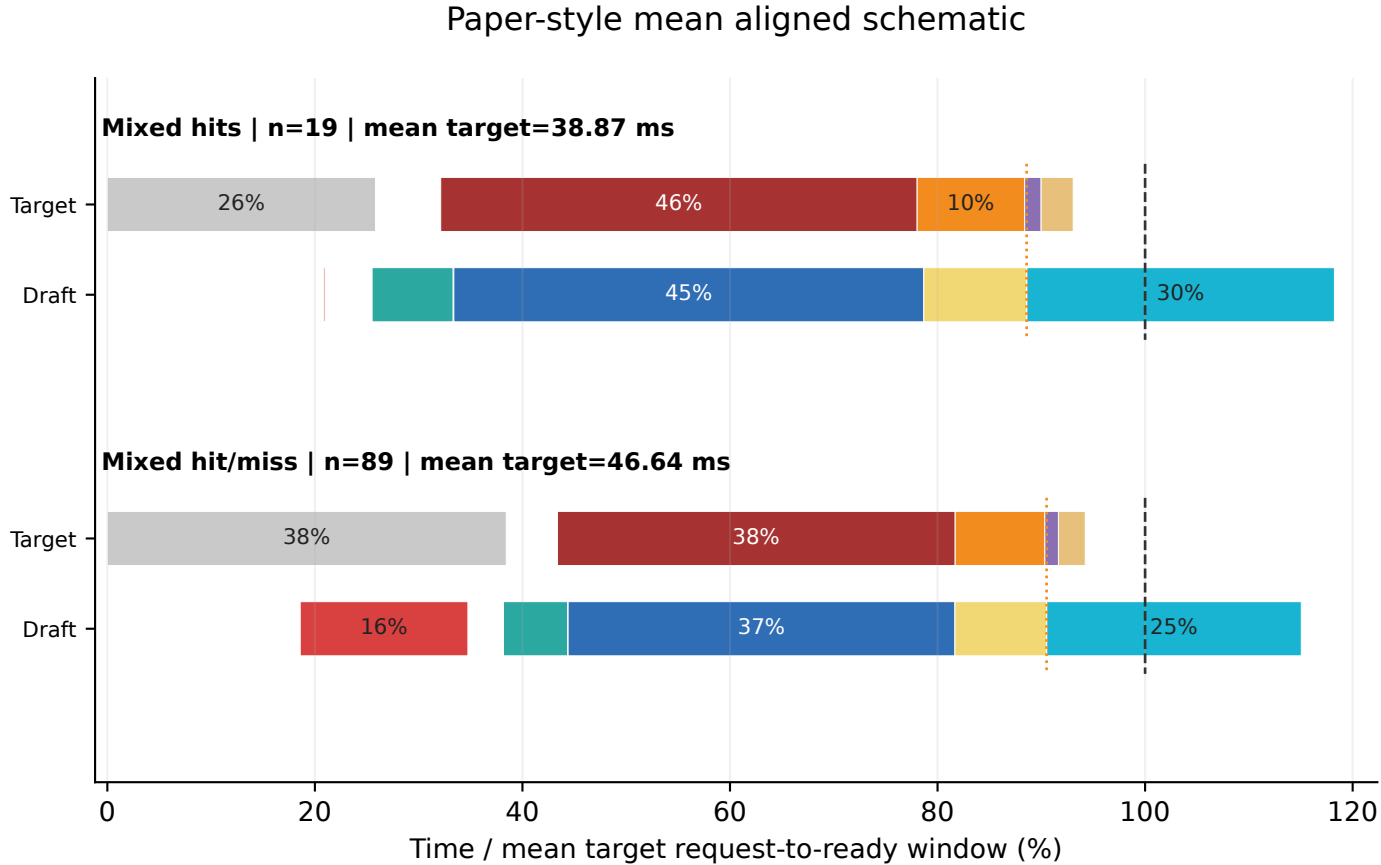
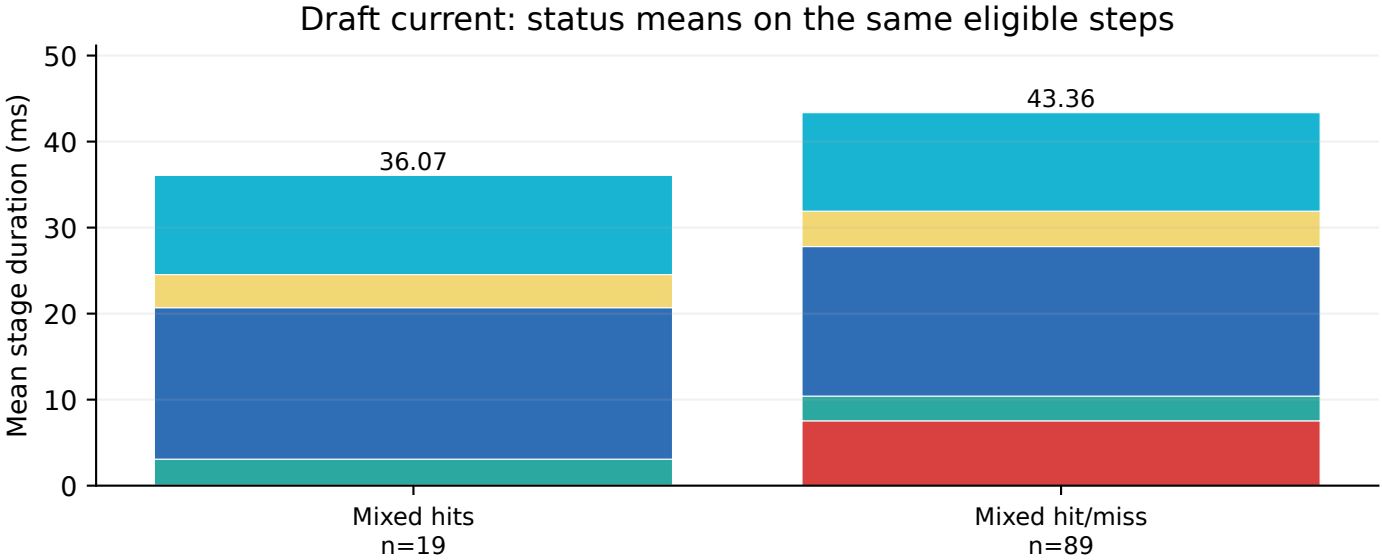
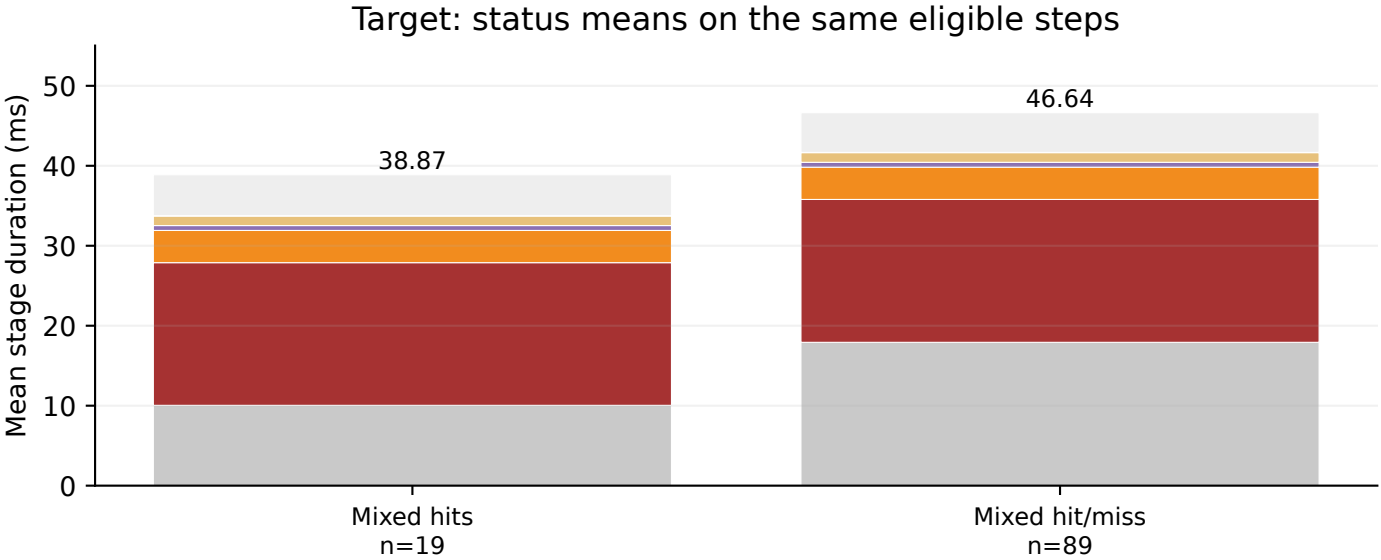
llama3_b8_refine_e26_k3_2_r2
DUET: K1/K2=3/2, exit=26 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 117 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

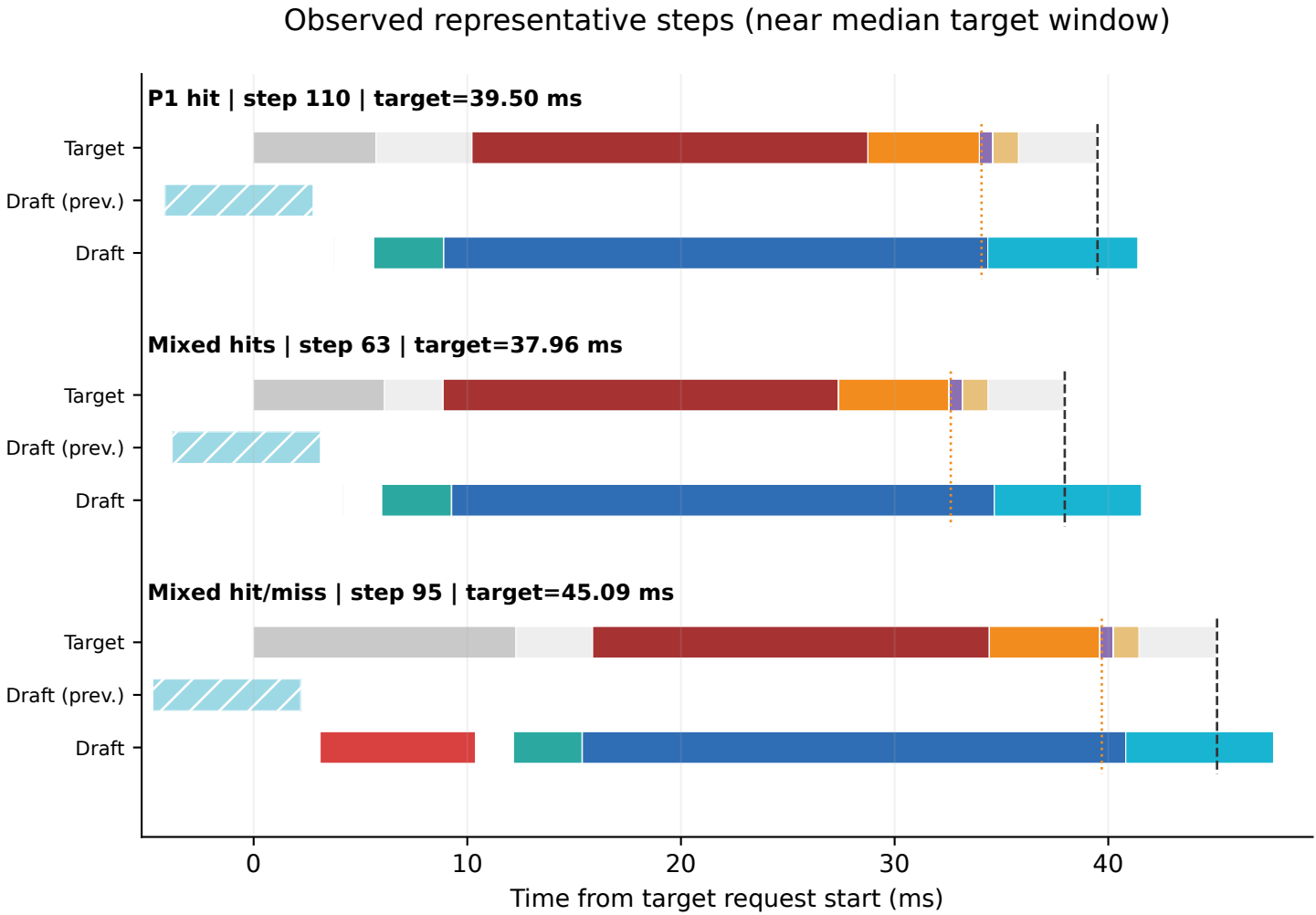
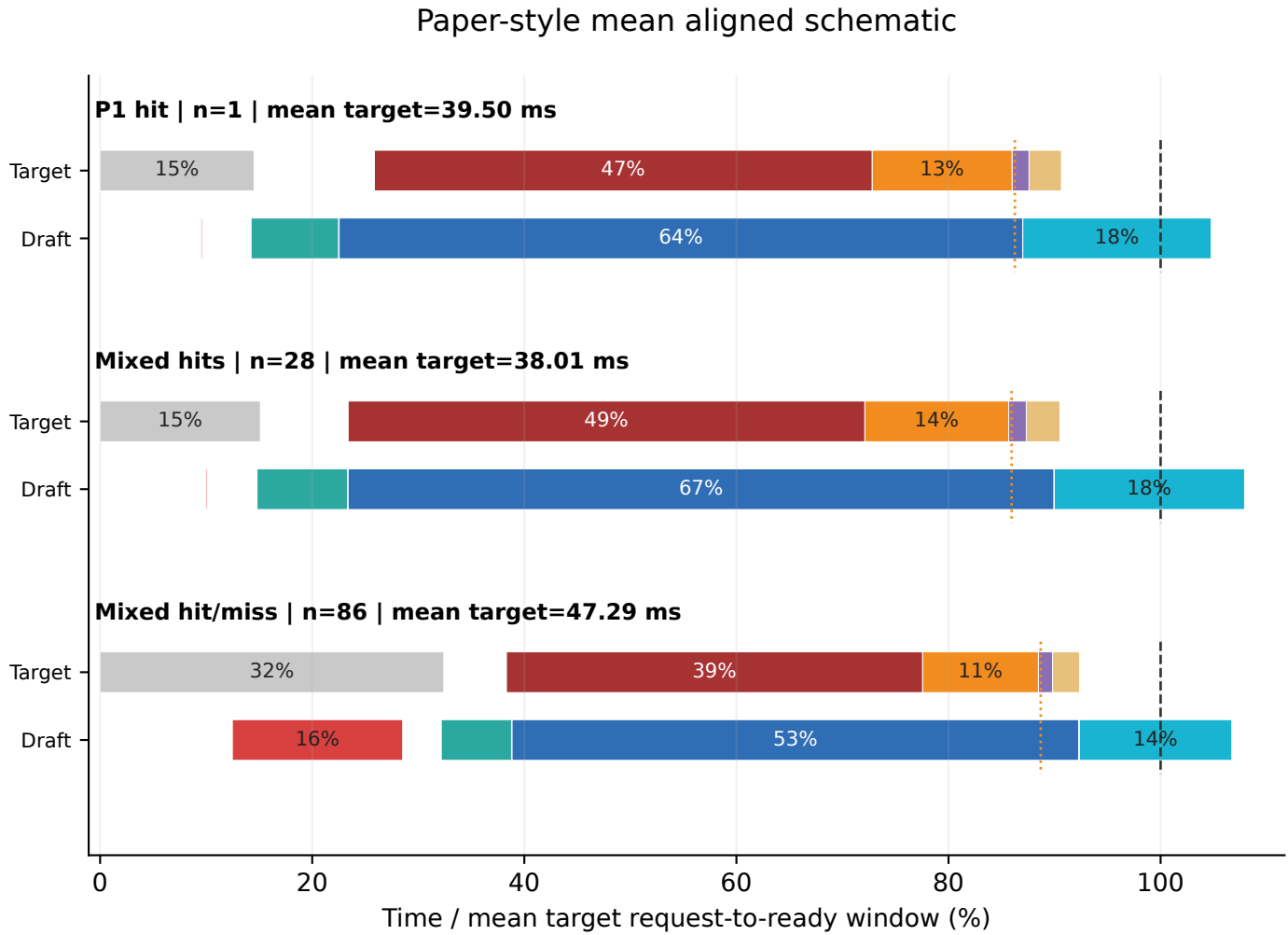
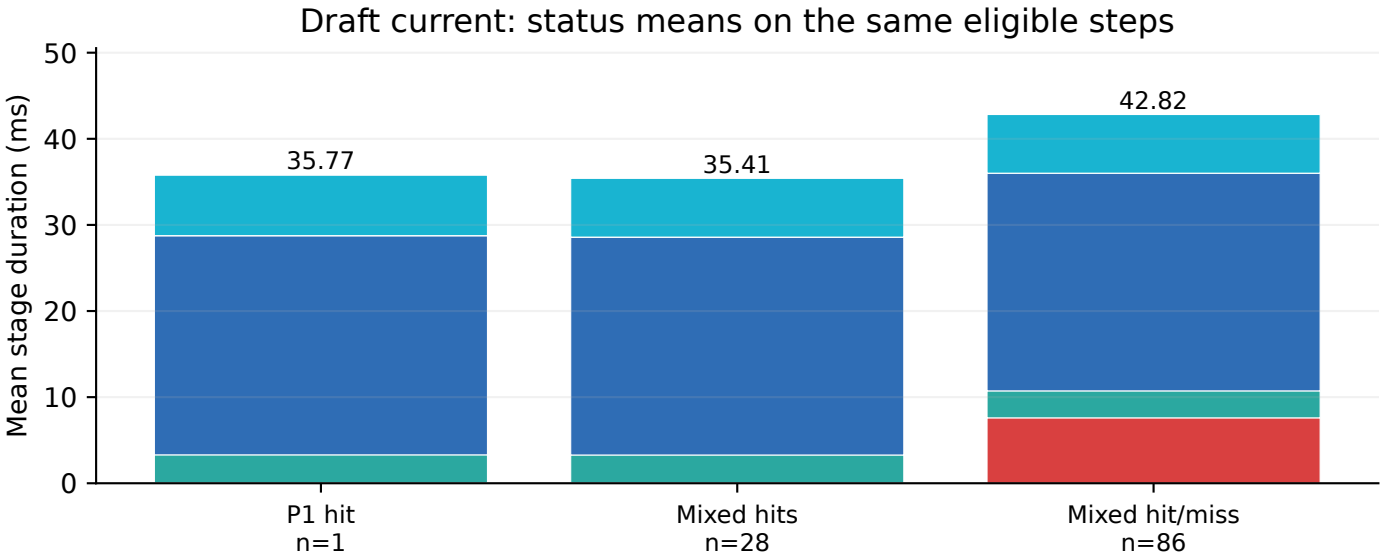
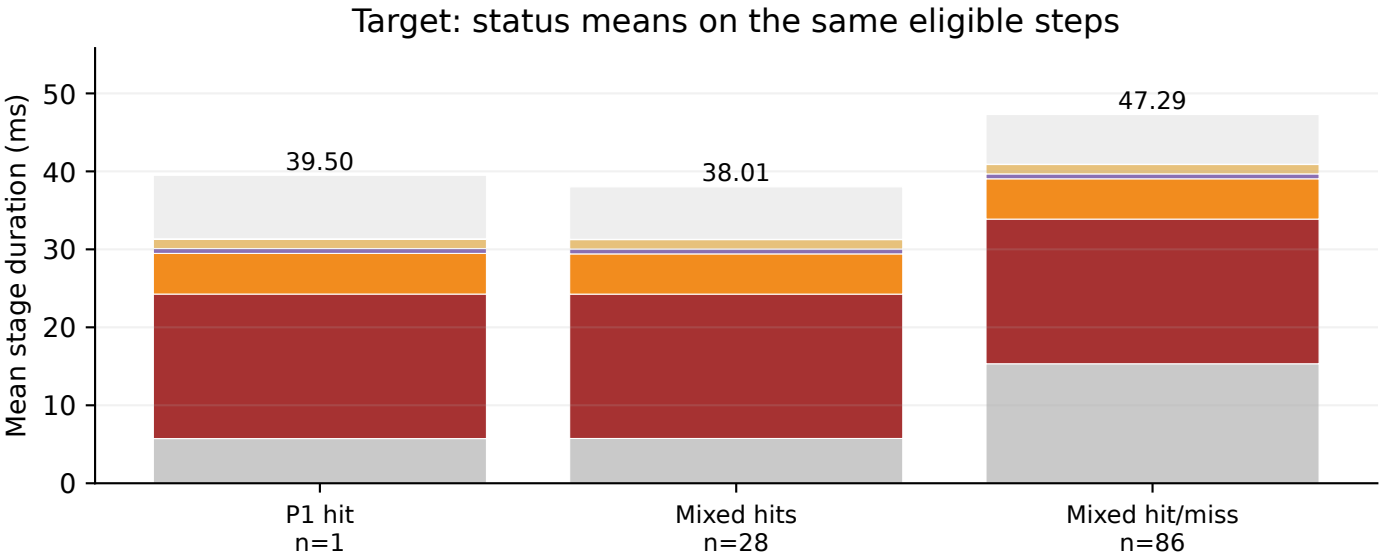
llama3_b8_refine_e30_k3_2_r2
DUET: K1/K2=3/2, exit=30 (0-based), C=3, N1/N2=6/4, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 108 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

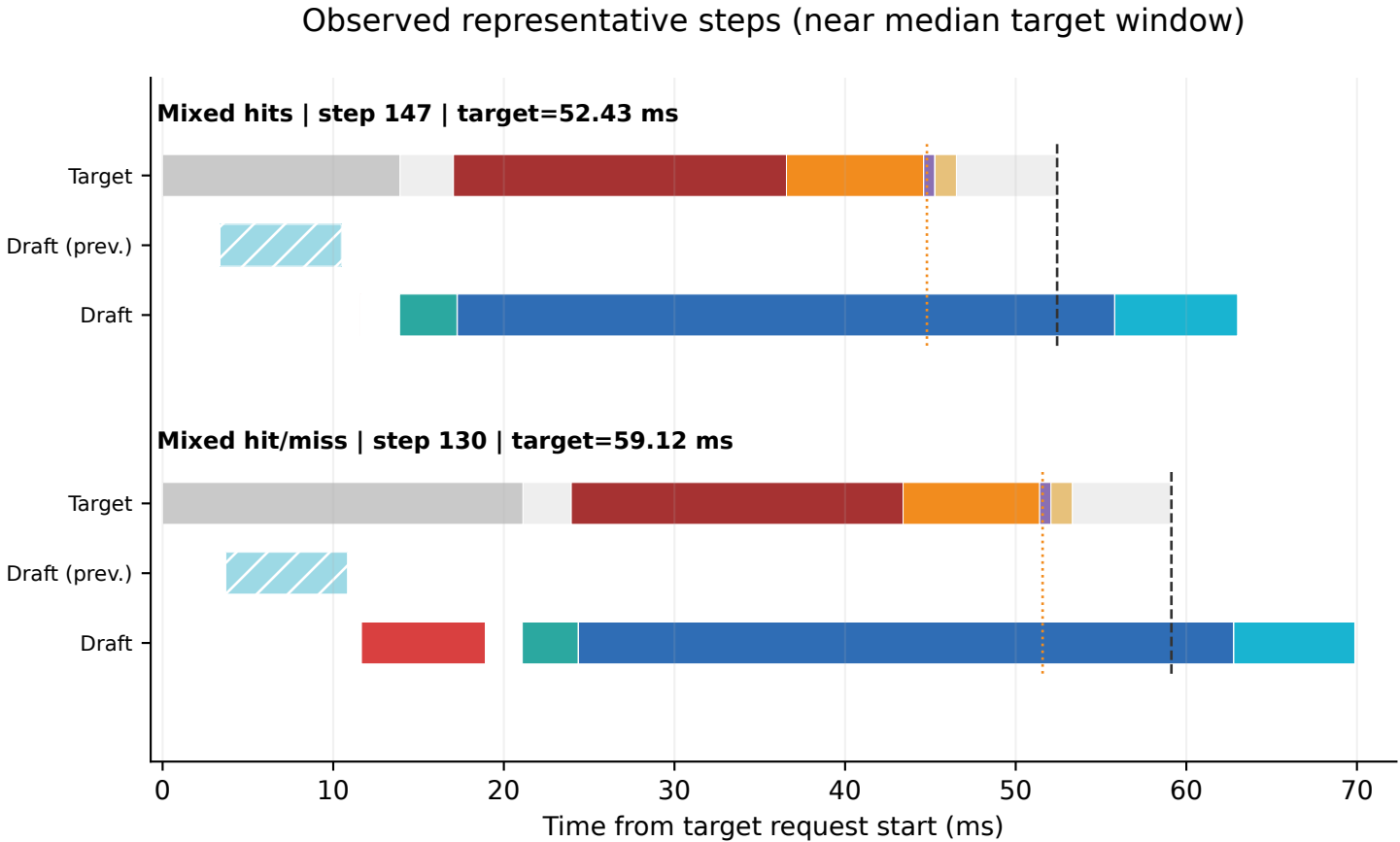
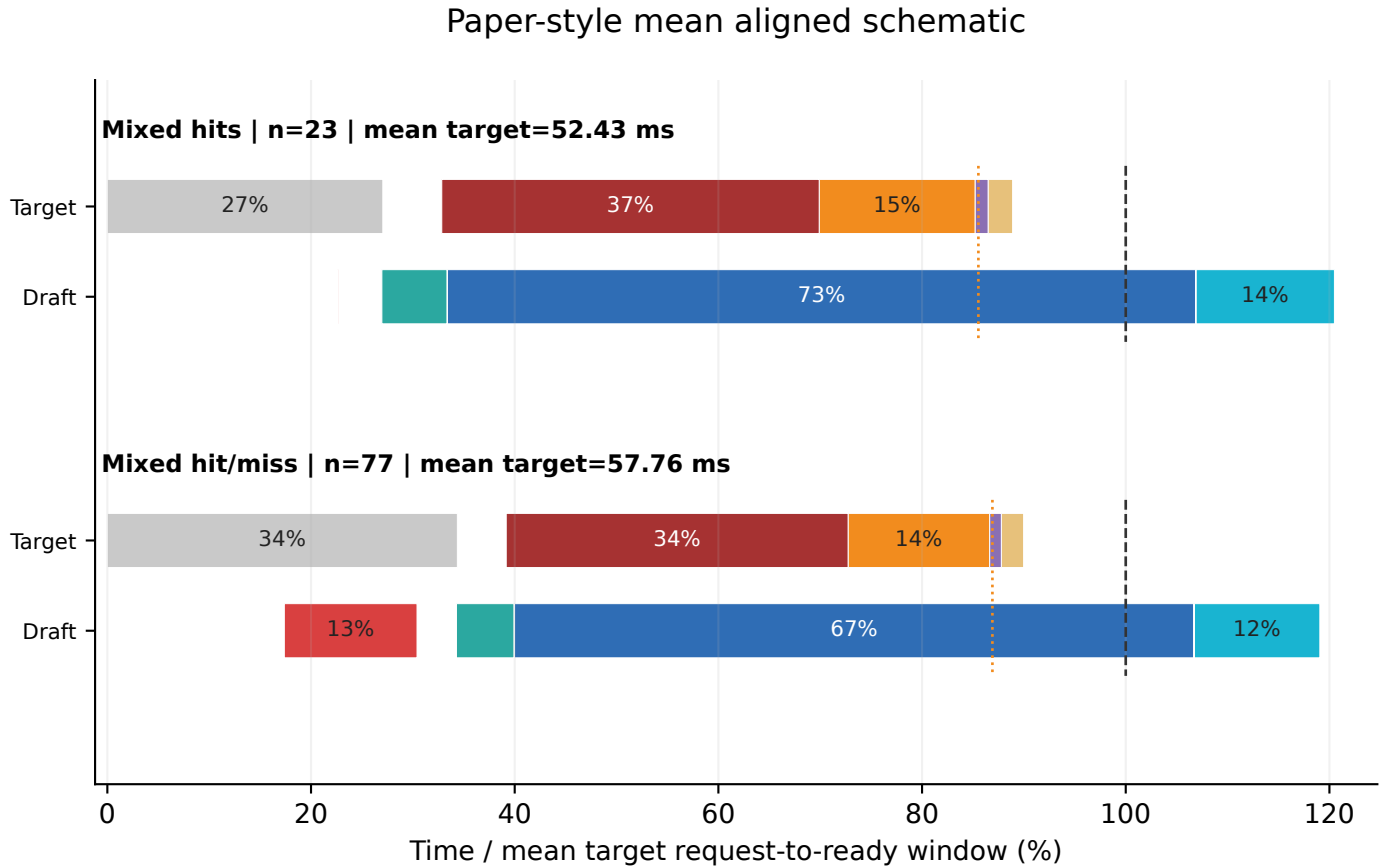
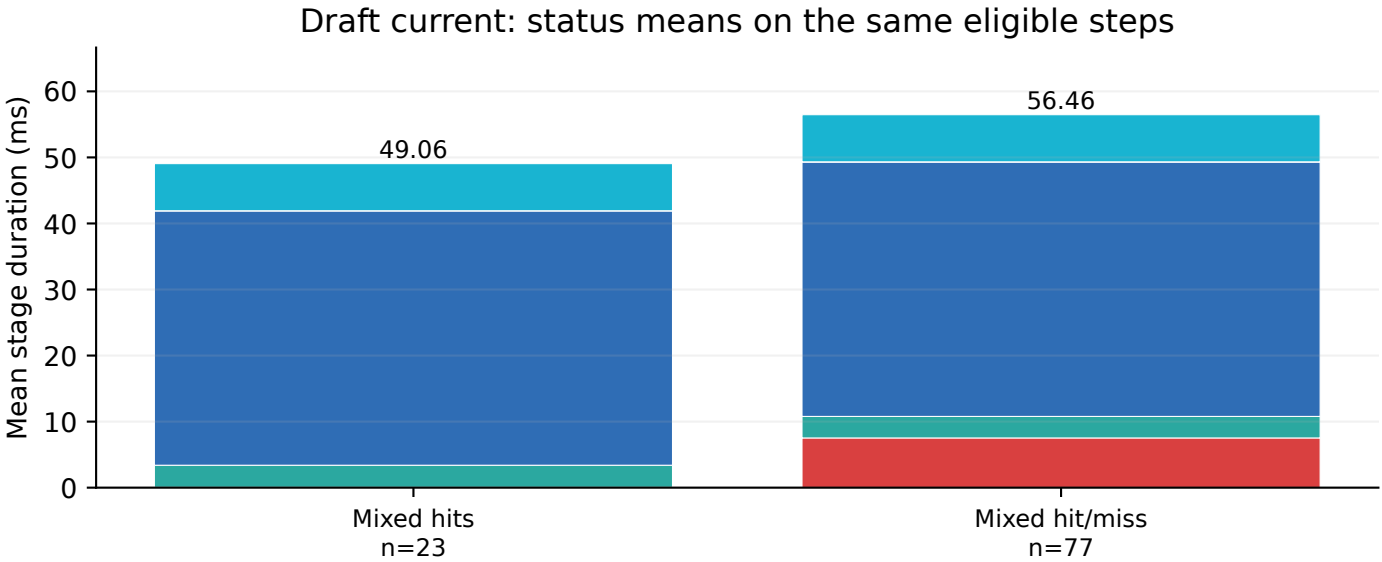
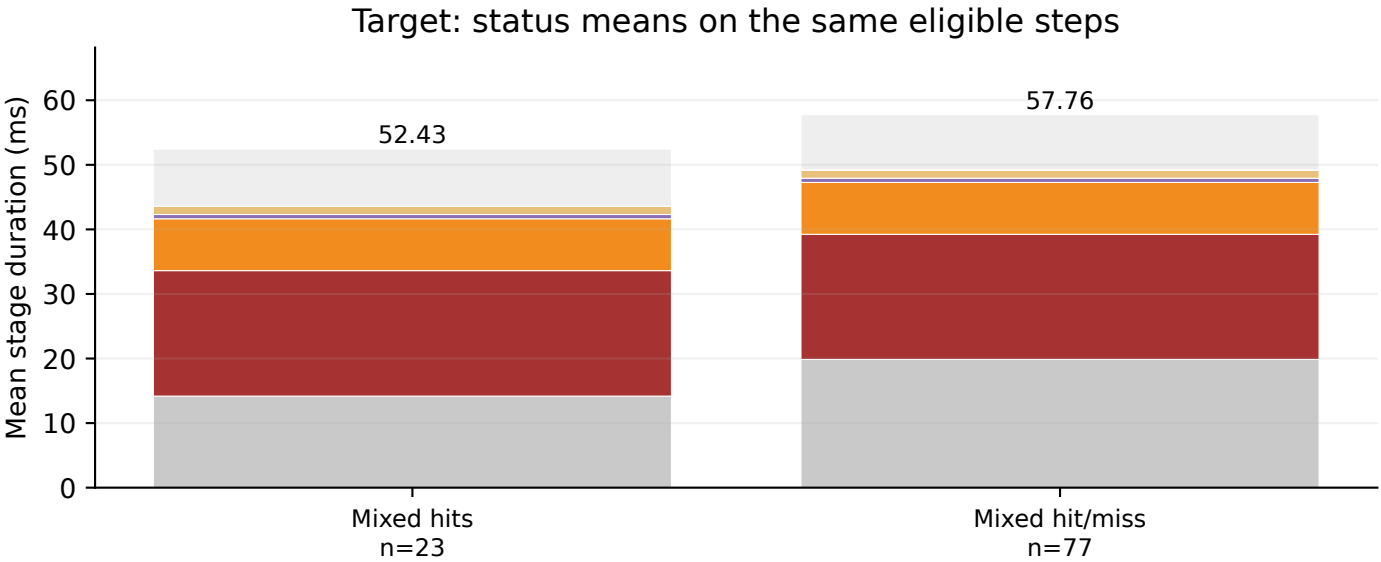
llama3_b8_refine_e30_k4_1_r2
DUET: K1/K2=4/1, exit=30 (0-based), C=3, N1/N2=8/2, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 115 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

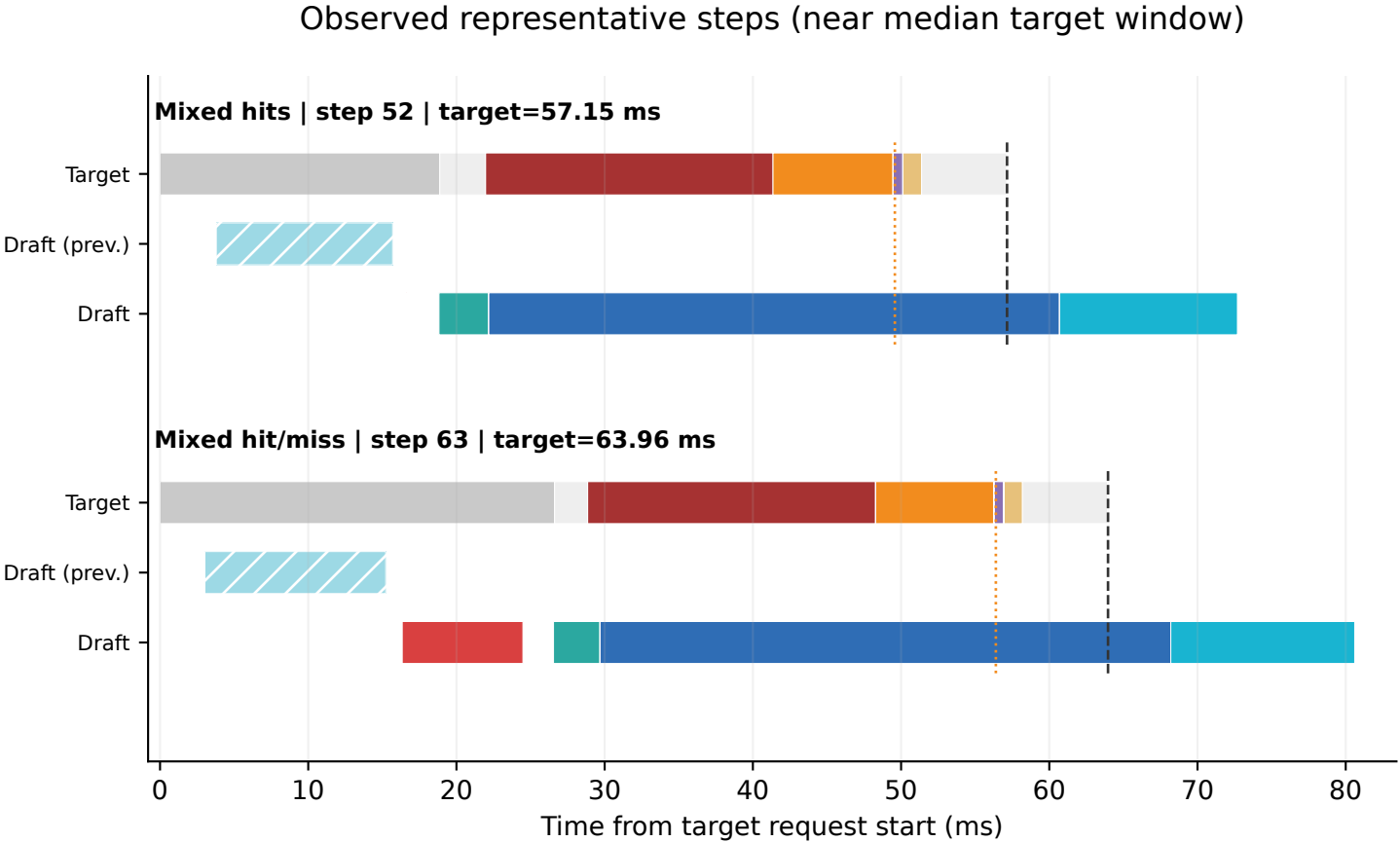
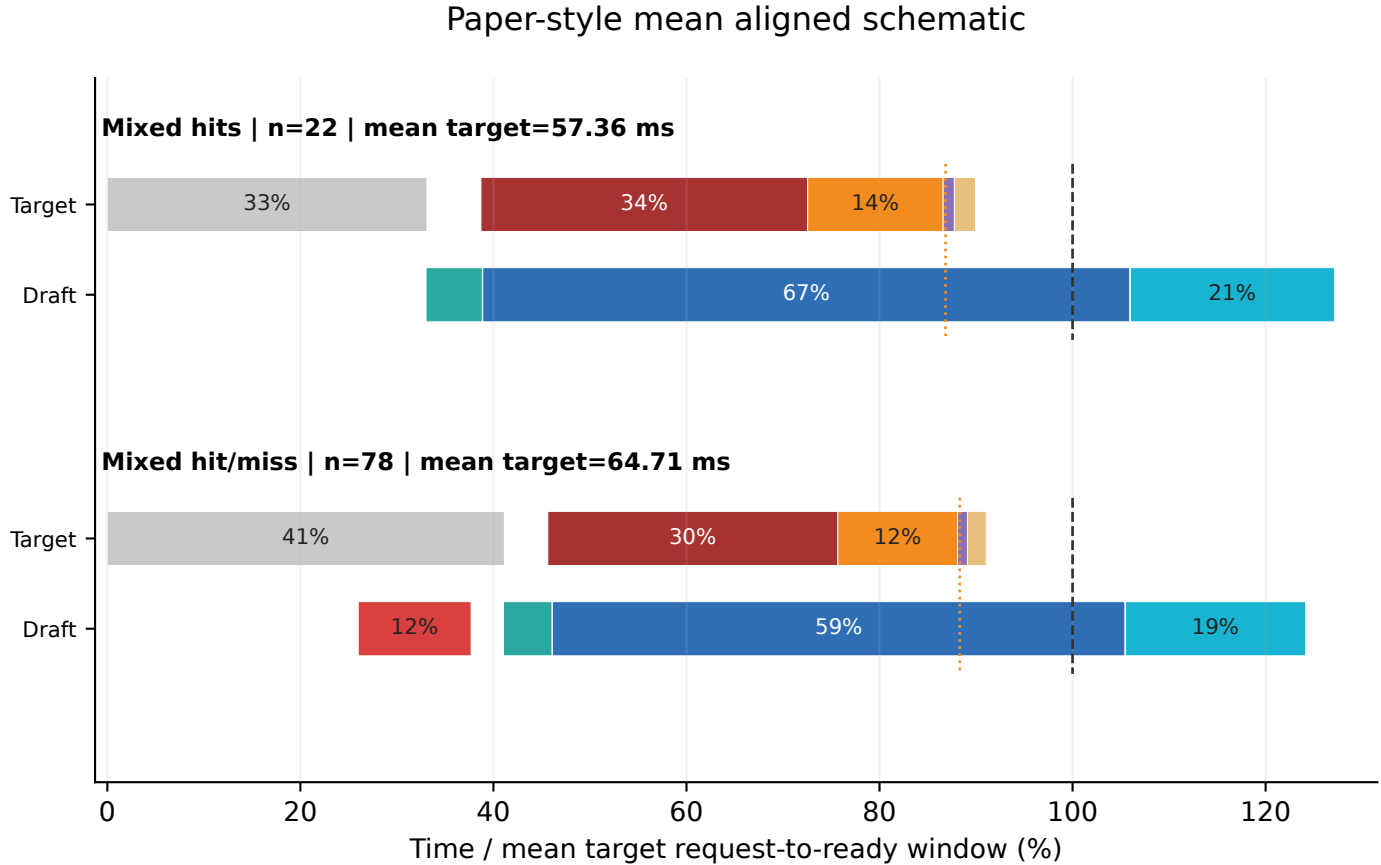
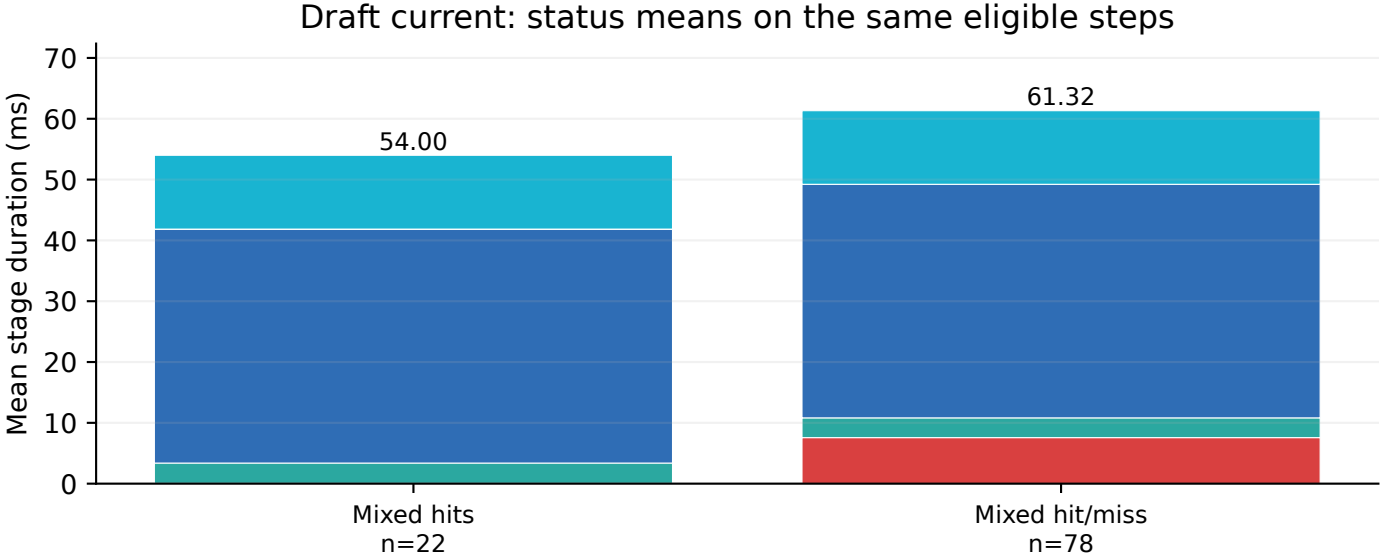
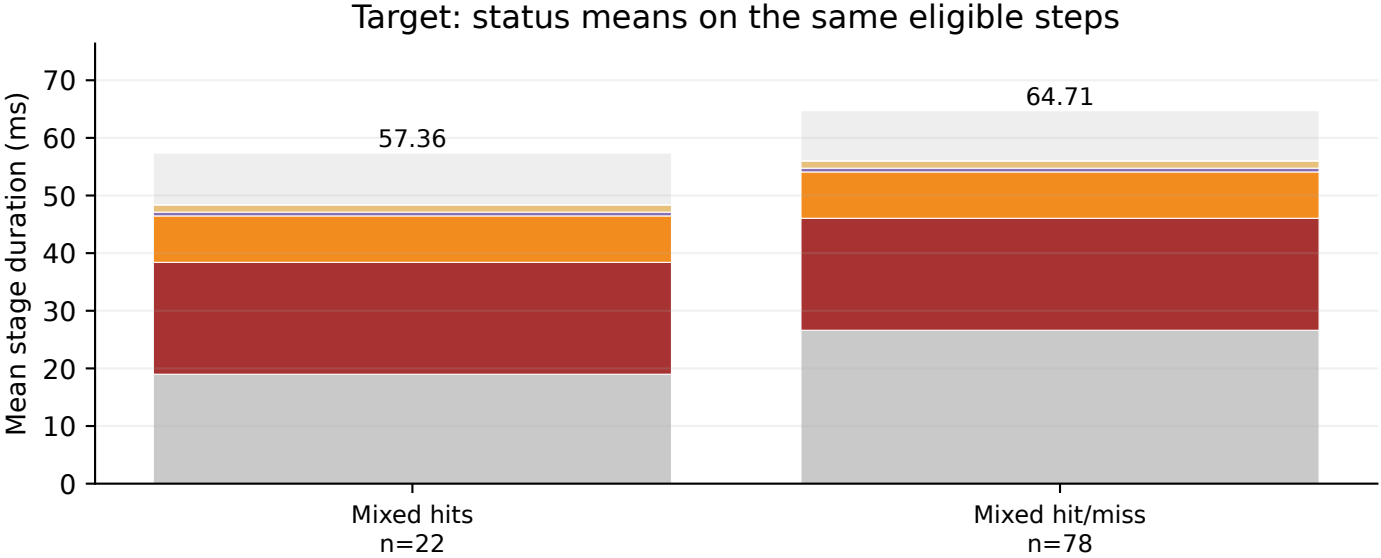
llama3_b8_refine_e30_k6_1_r2
DUET: K1/K2=6/1, exit=30 (0-based), C=3, N1/N2=12/2, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 100 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

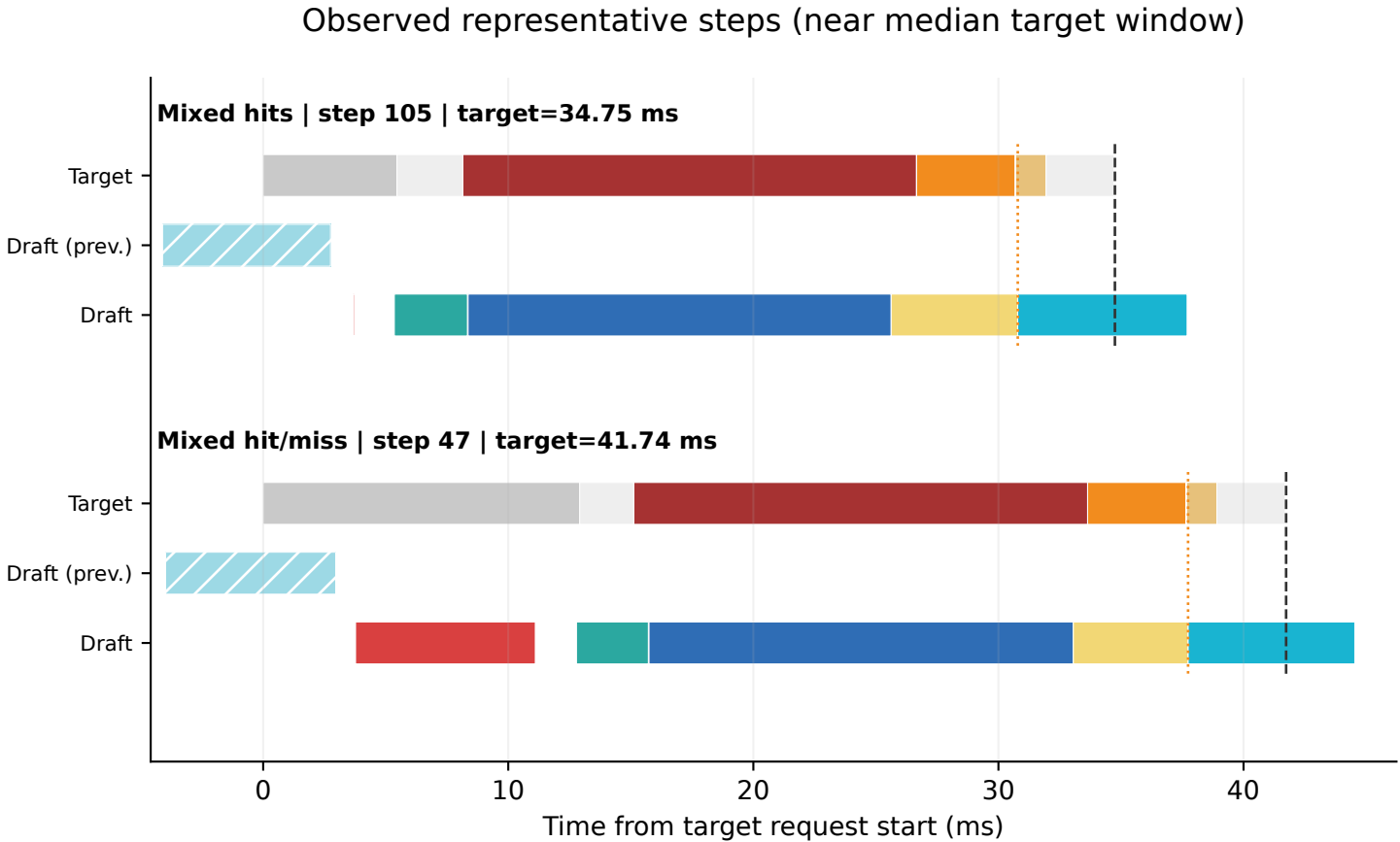
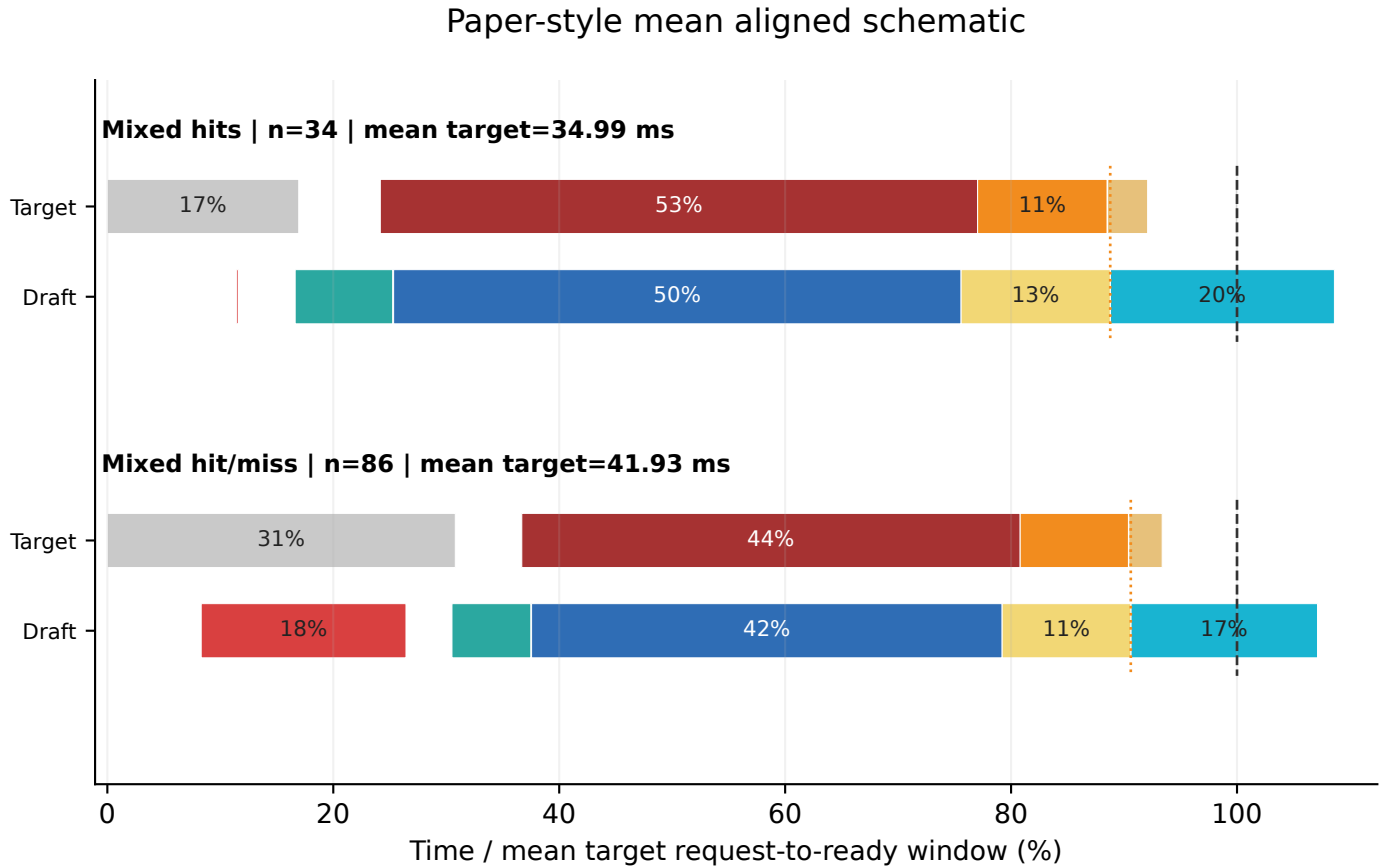
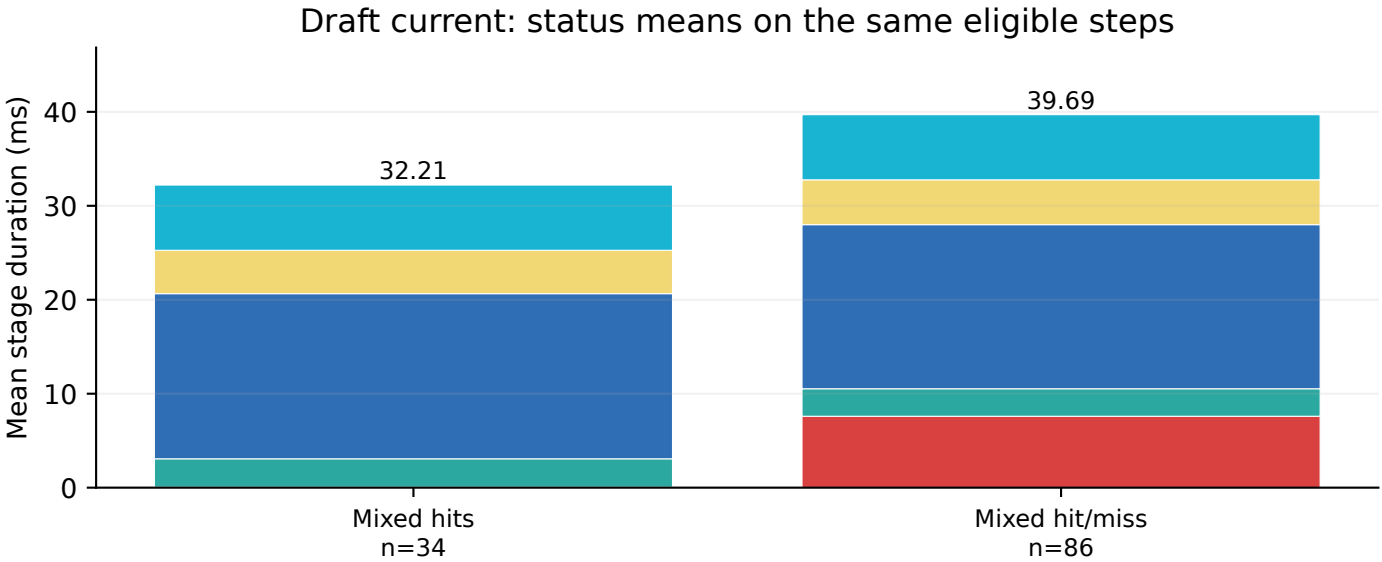
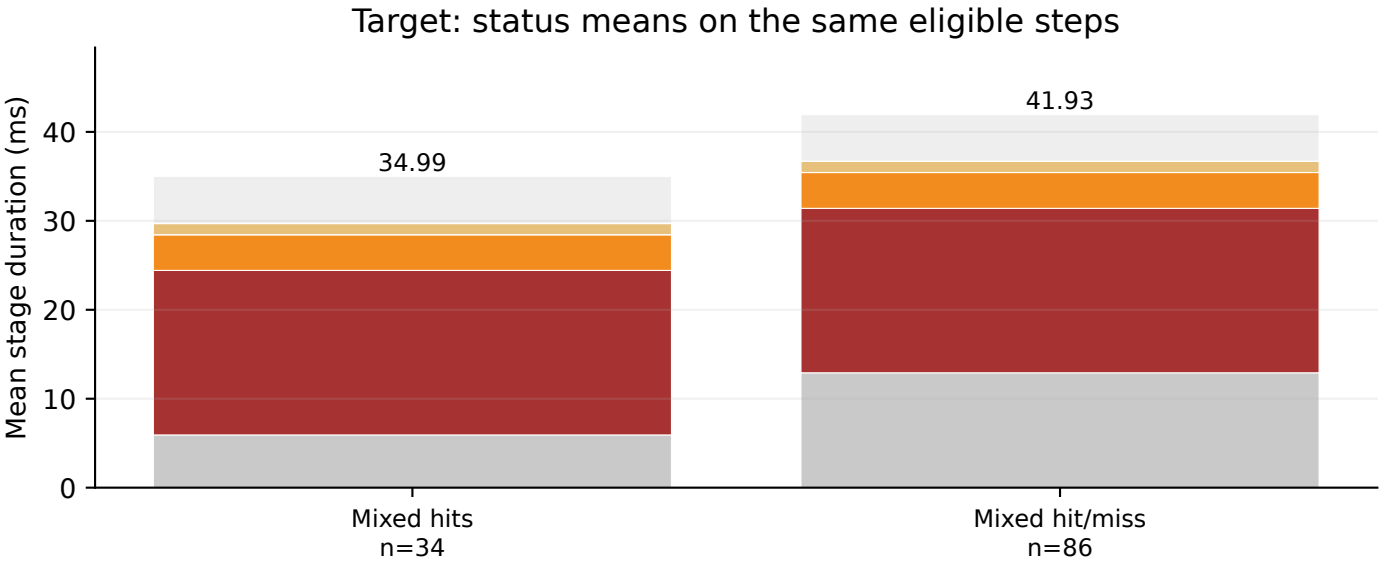
llama3_b8_refine_e30_k6_2_r2
DUET: K1/K2=6/2, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 100 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

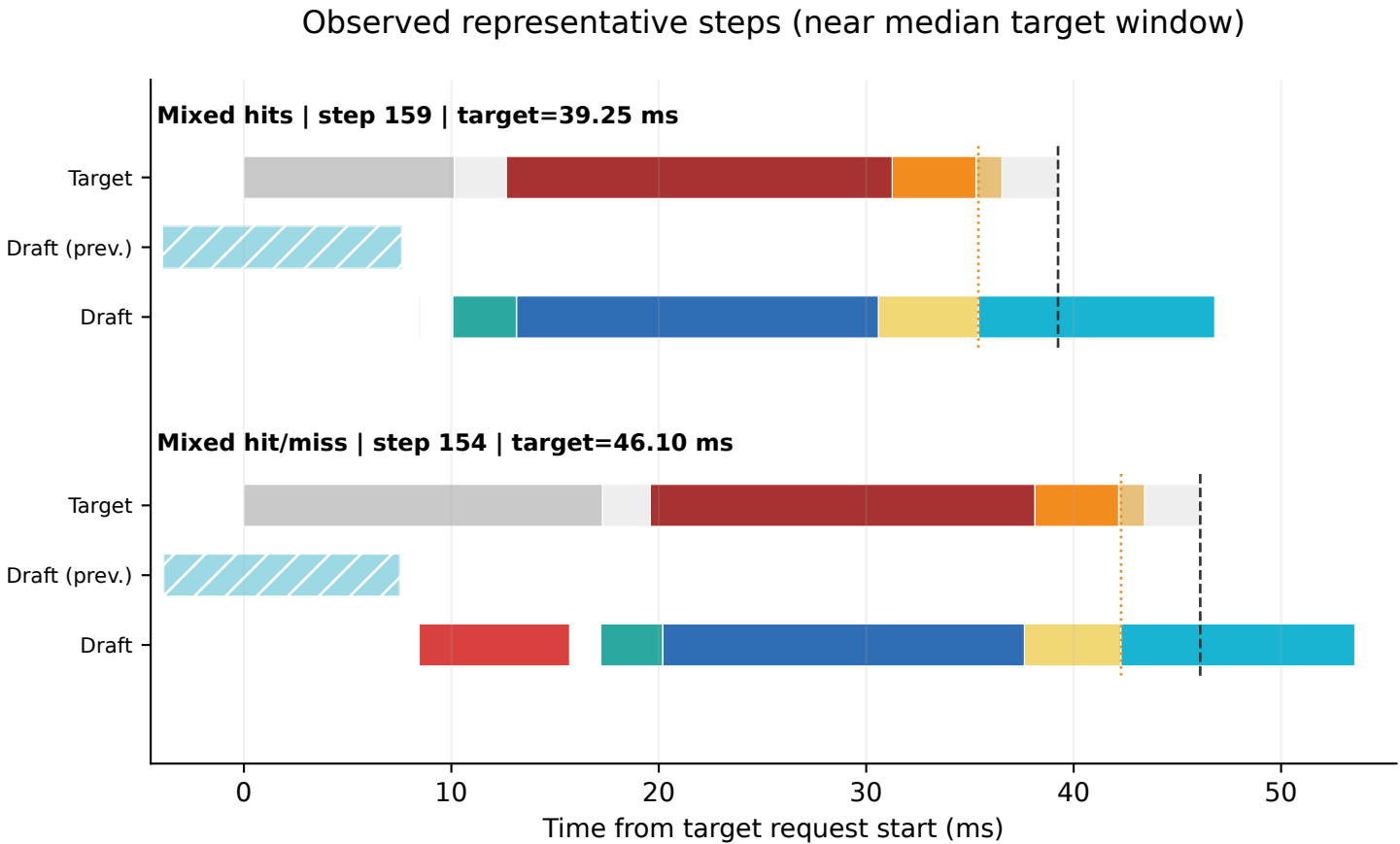
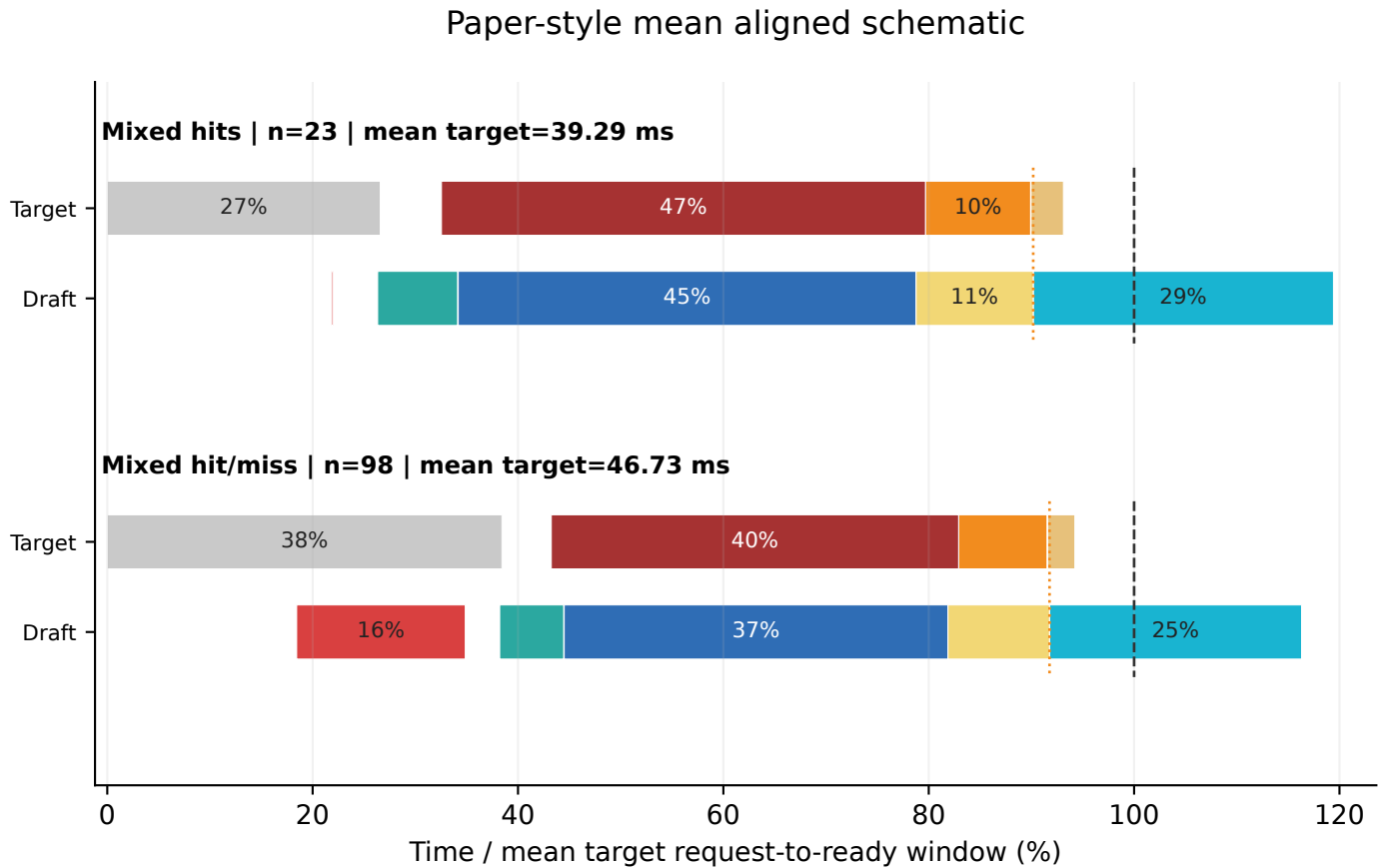
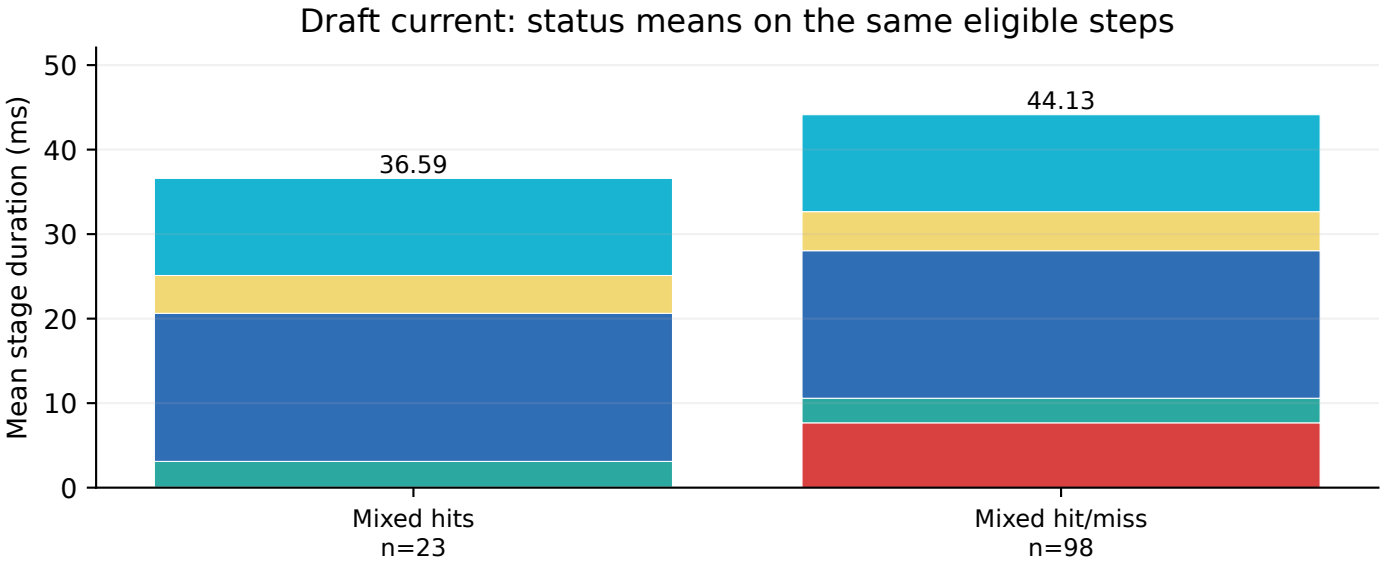
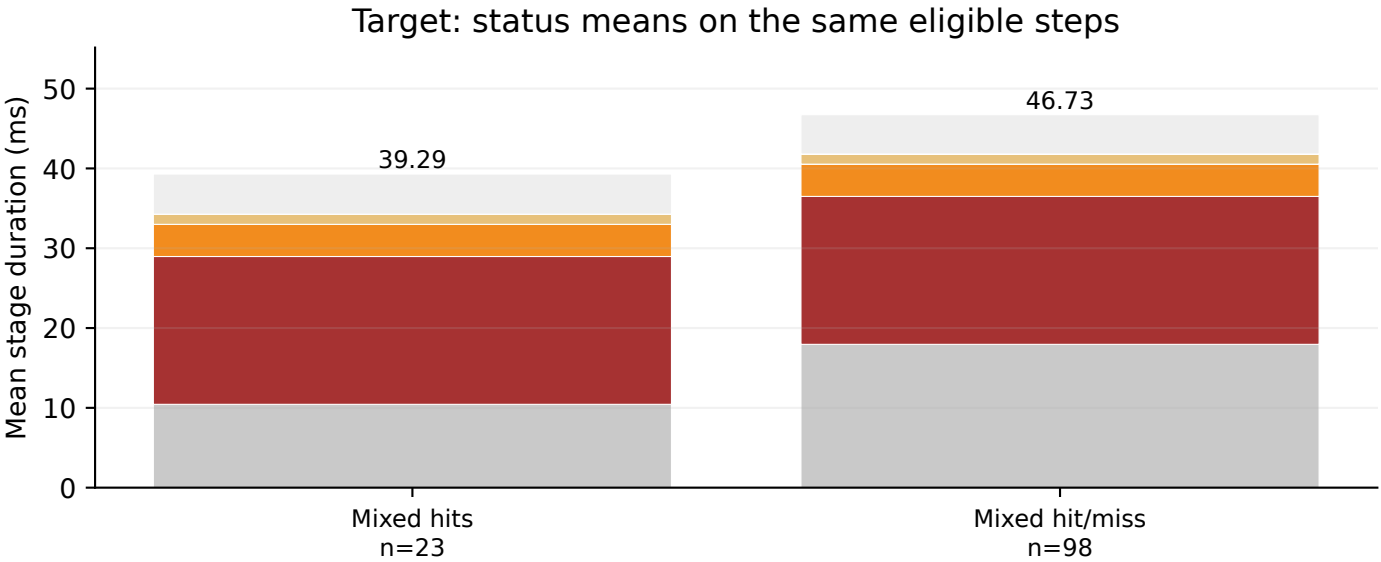
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_refine_e31_k3_1_r2
DUET: K1/K2=3/1, exit=31 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 120 eligible steps



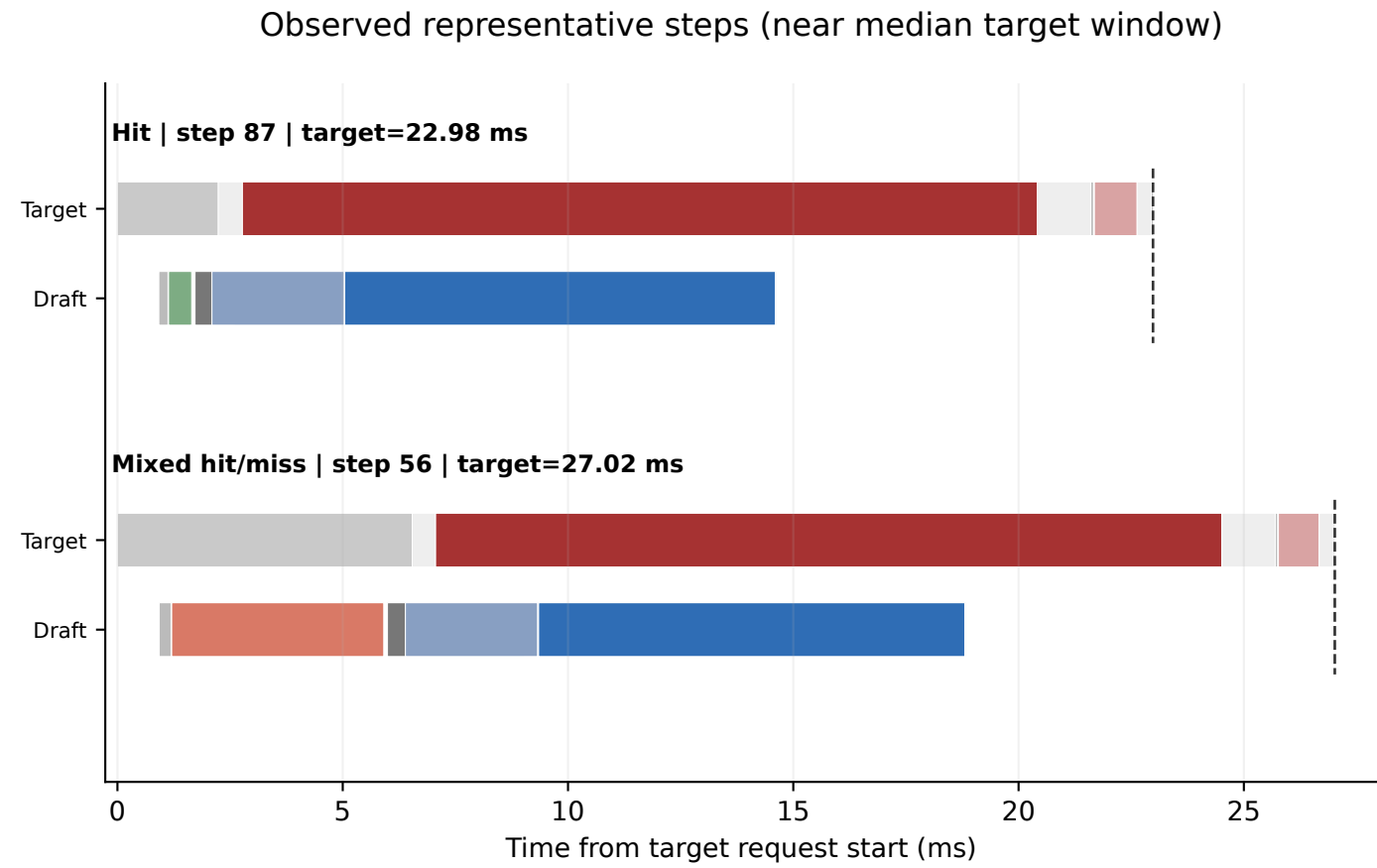
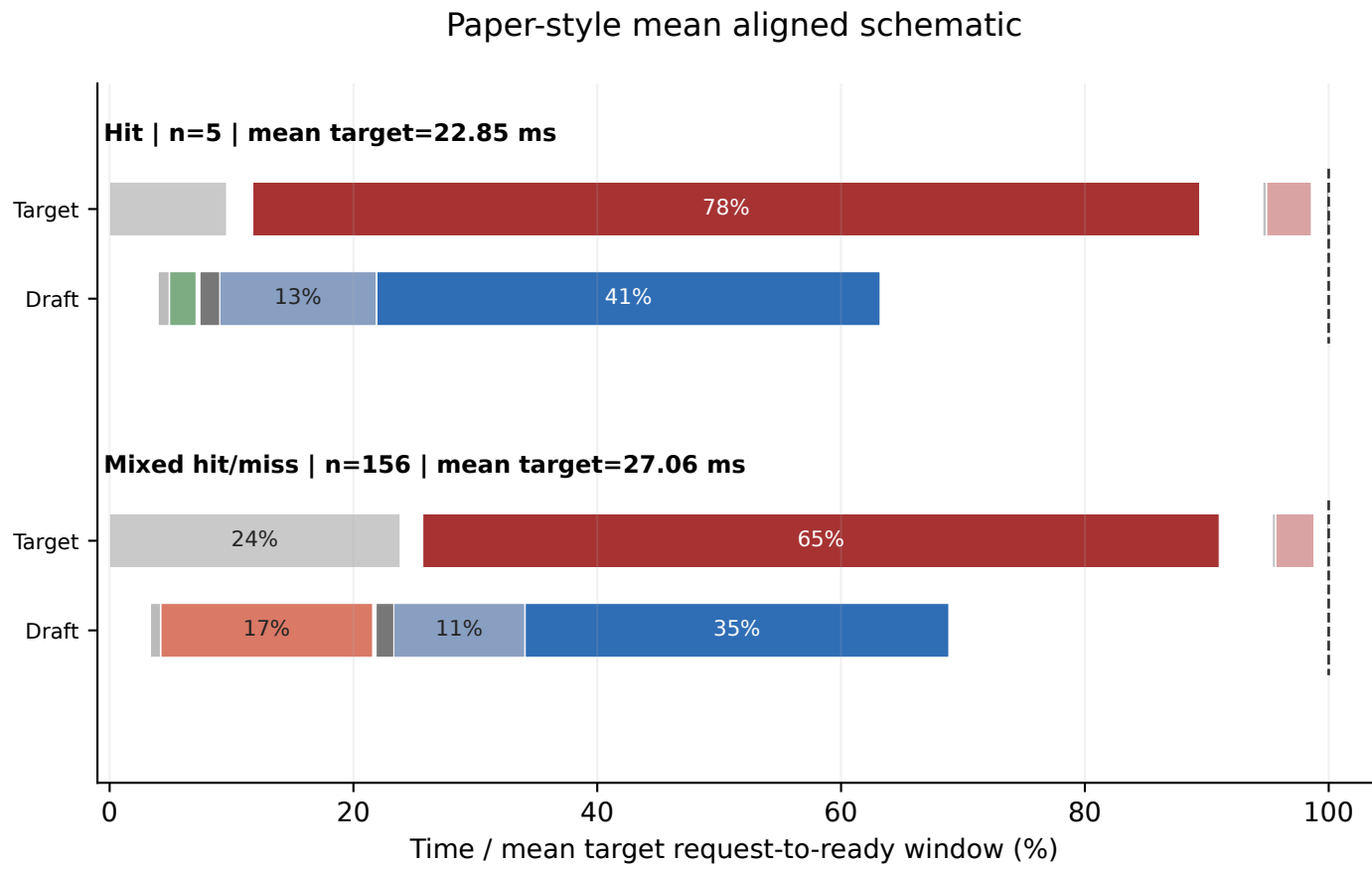
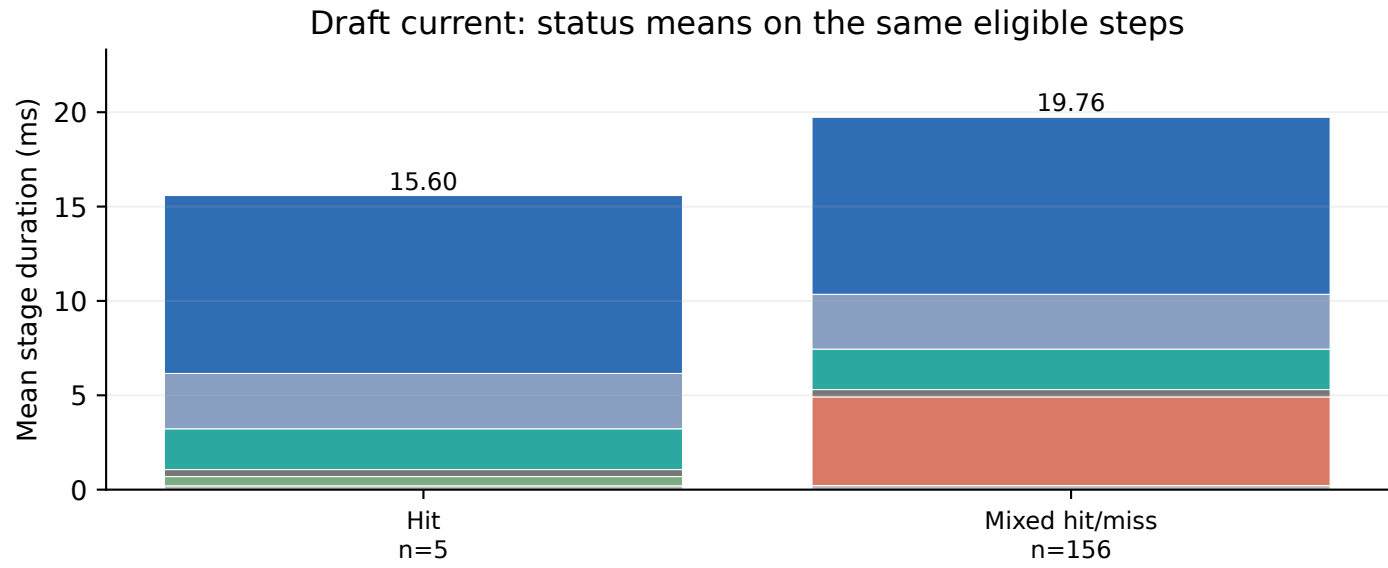
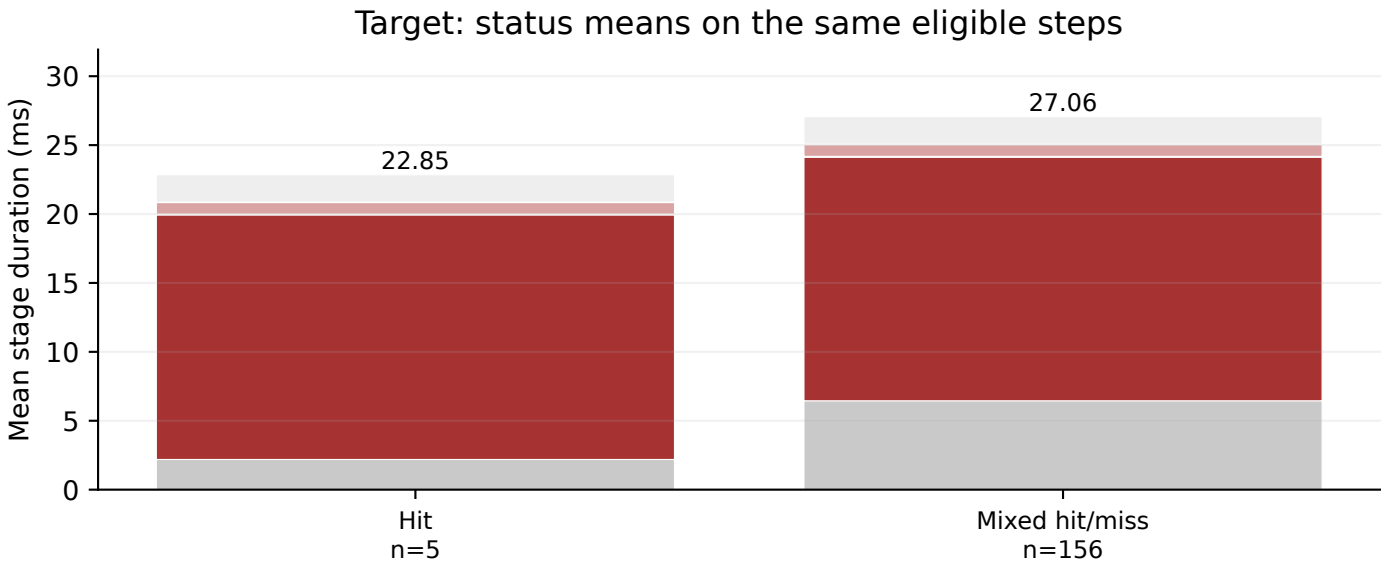
Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

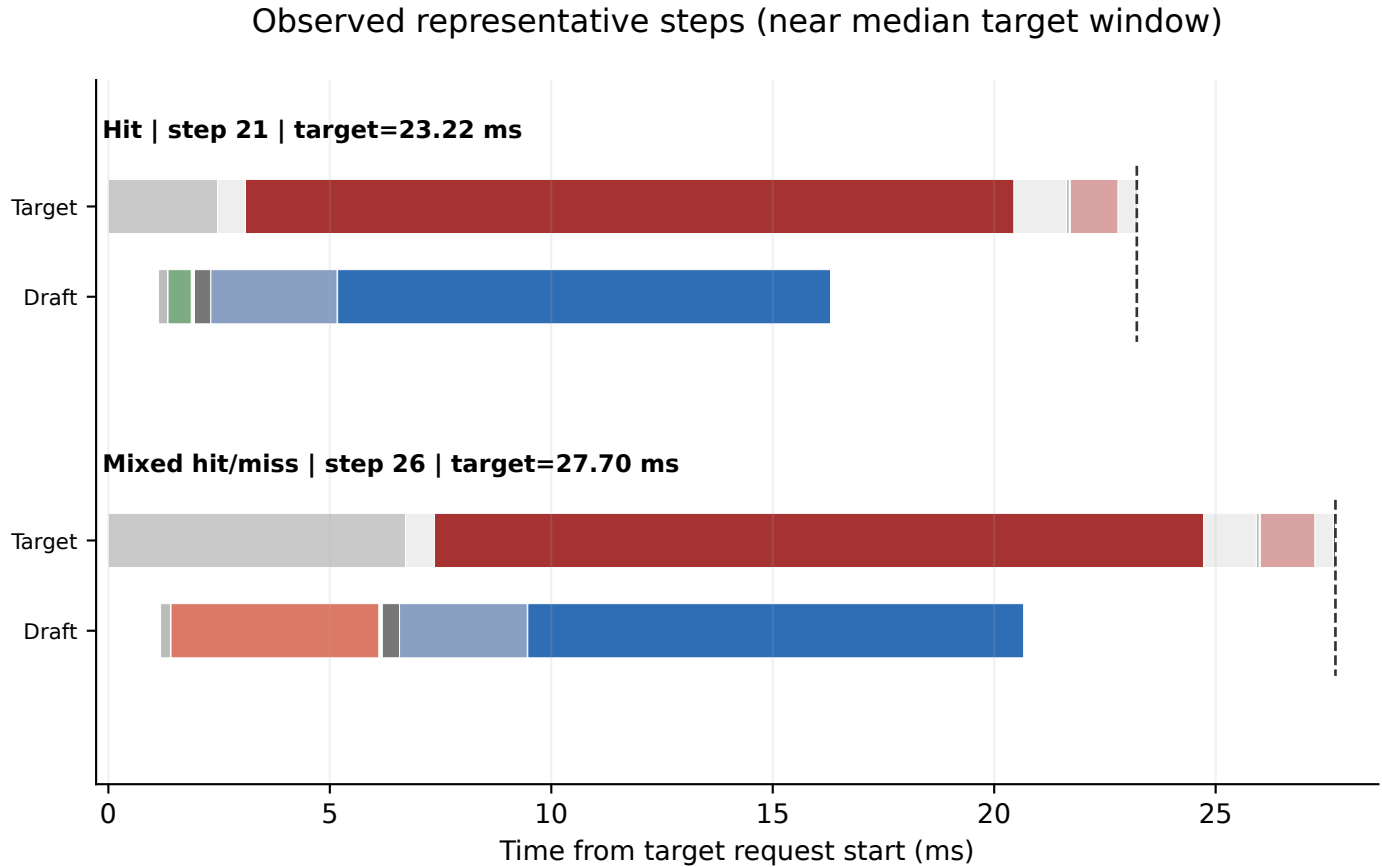
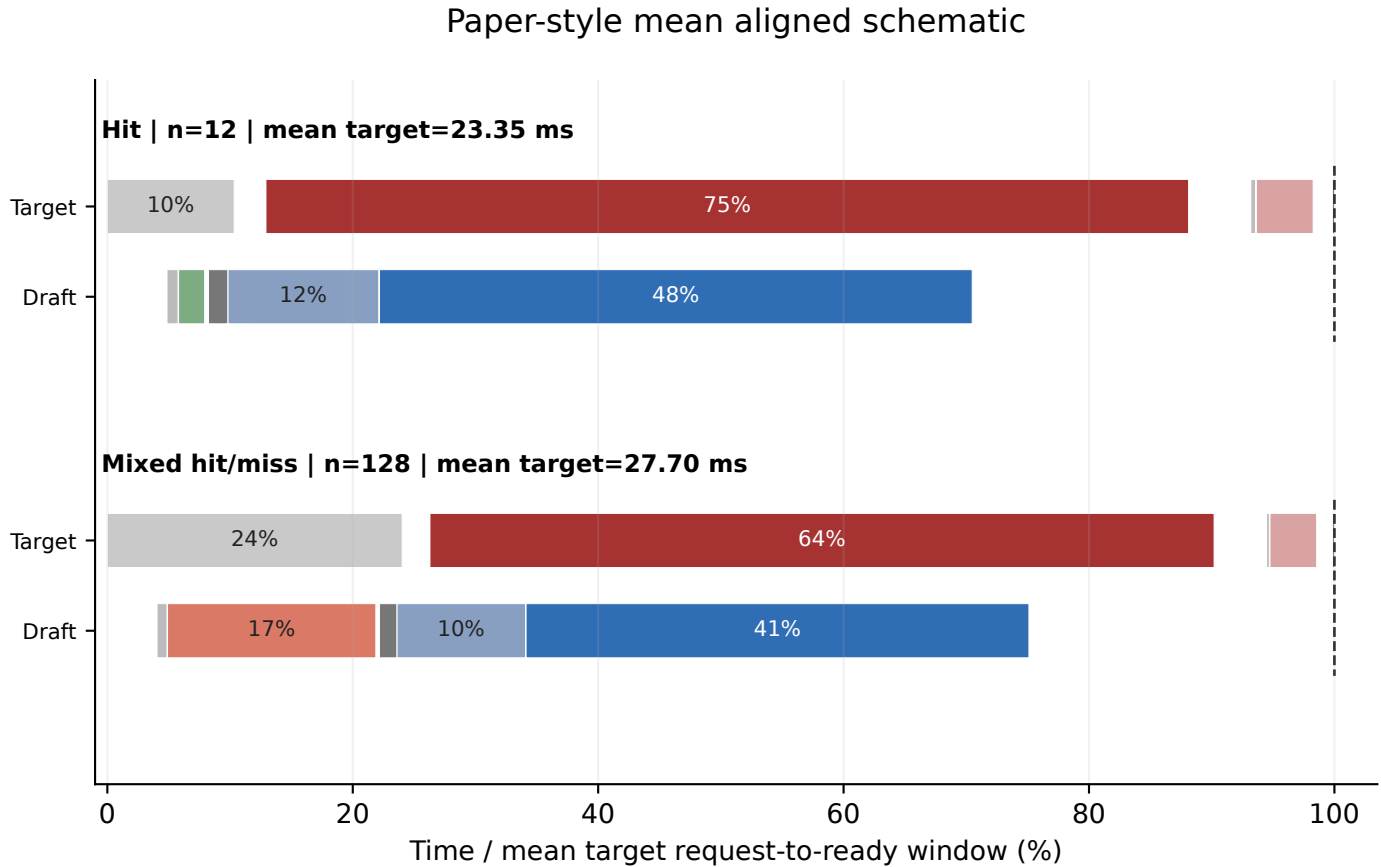
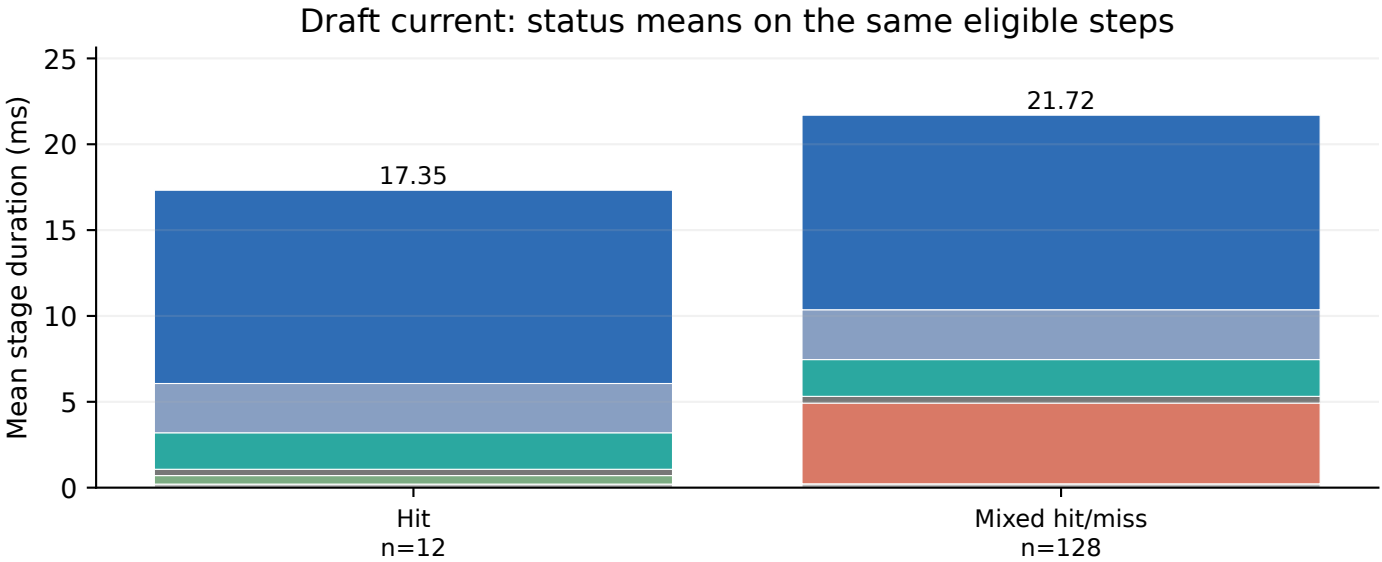
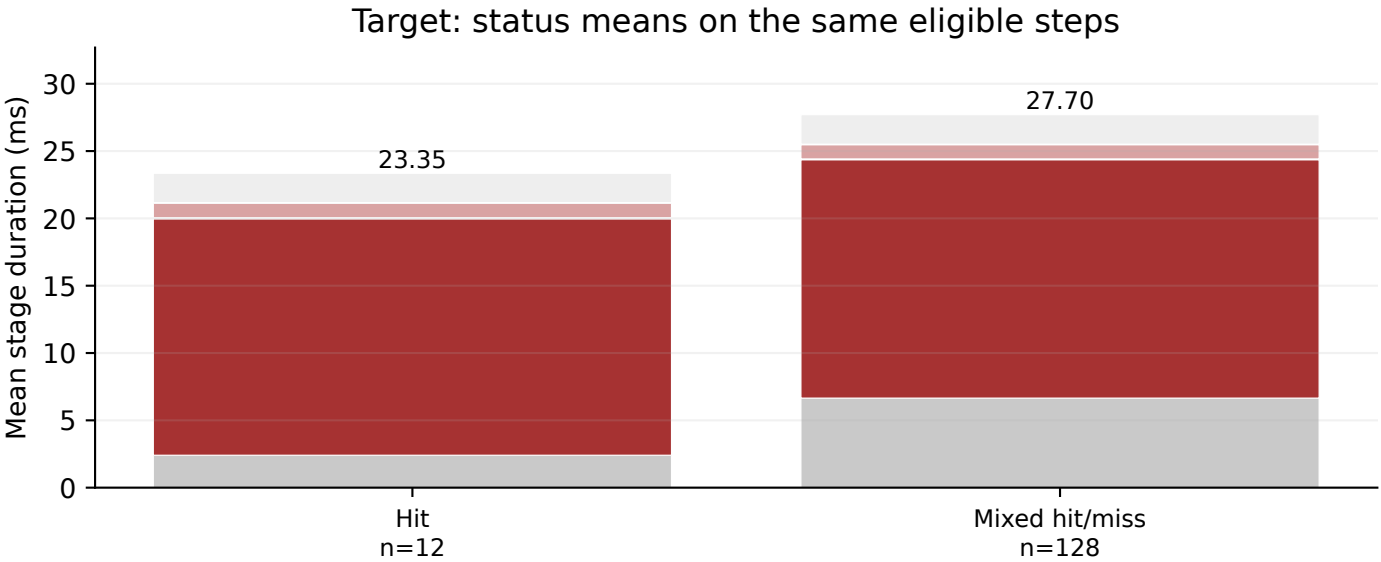
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

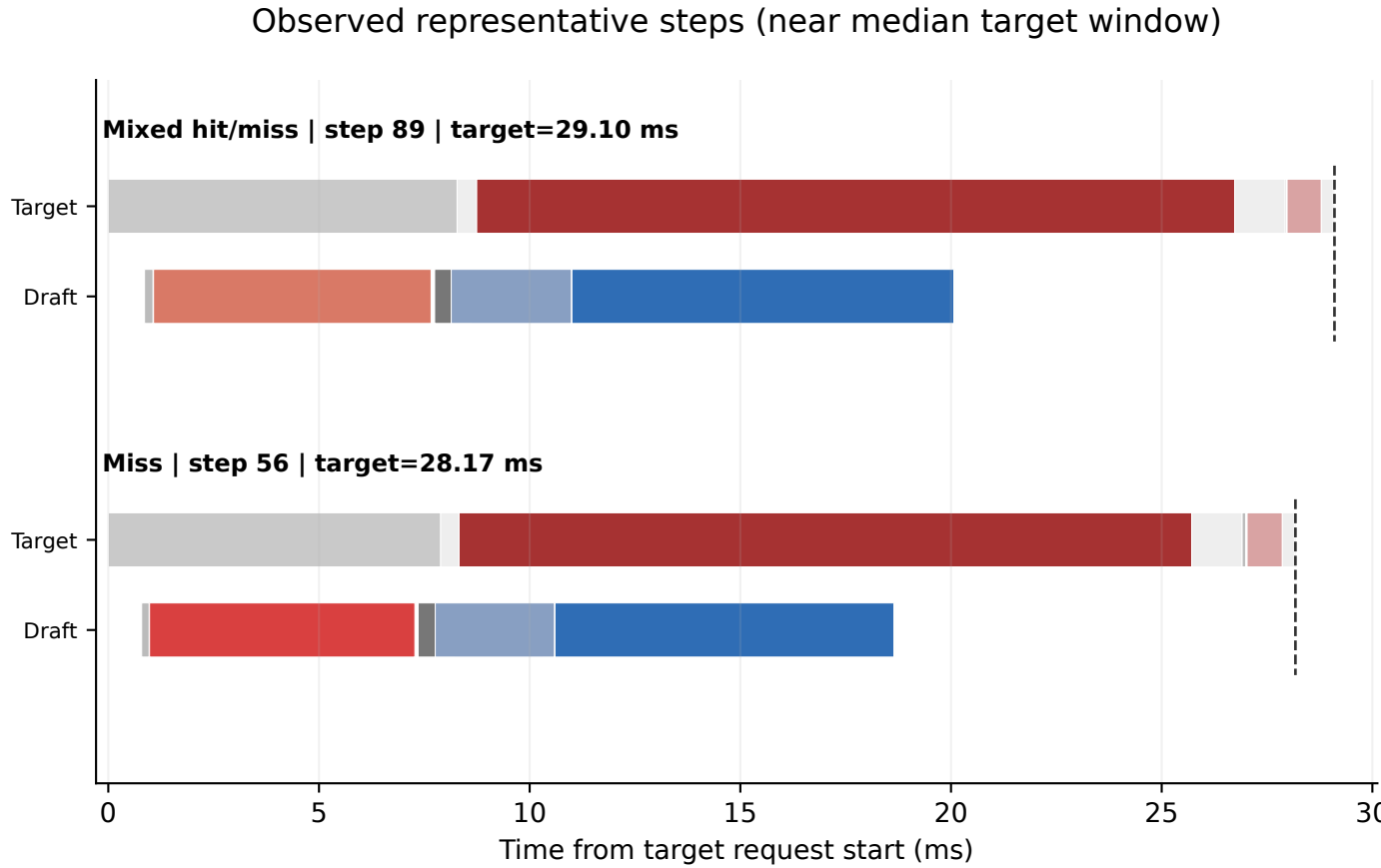
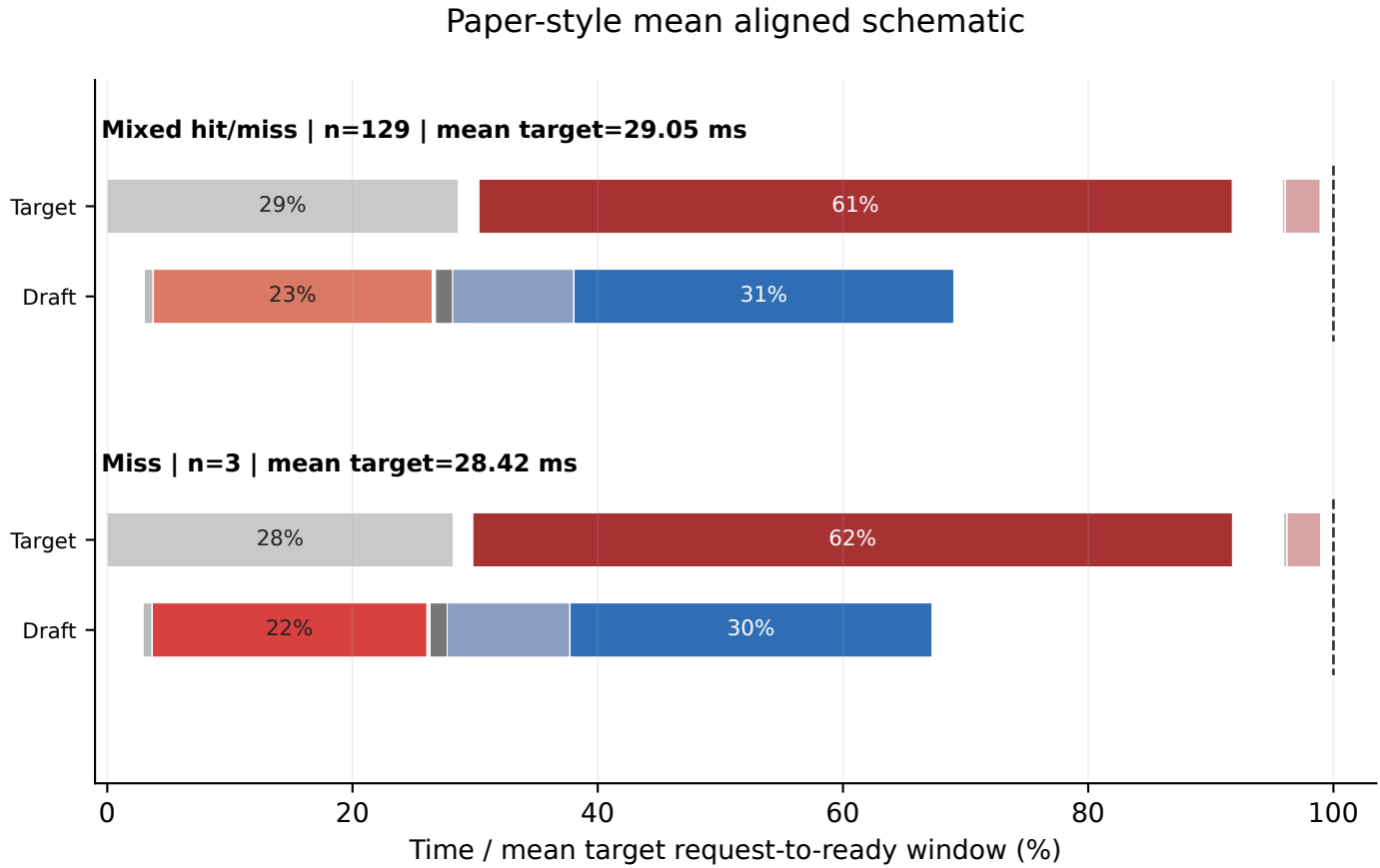
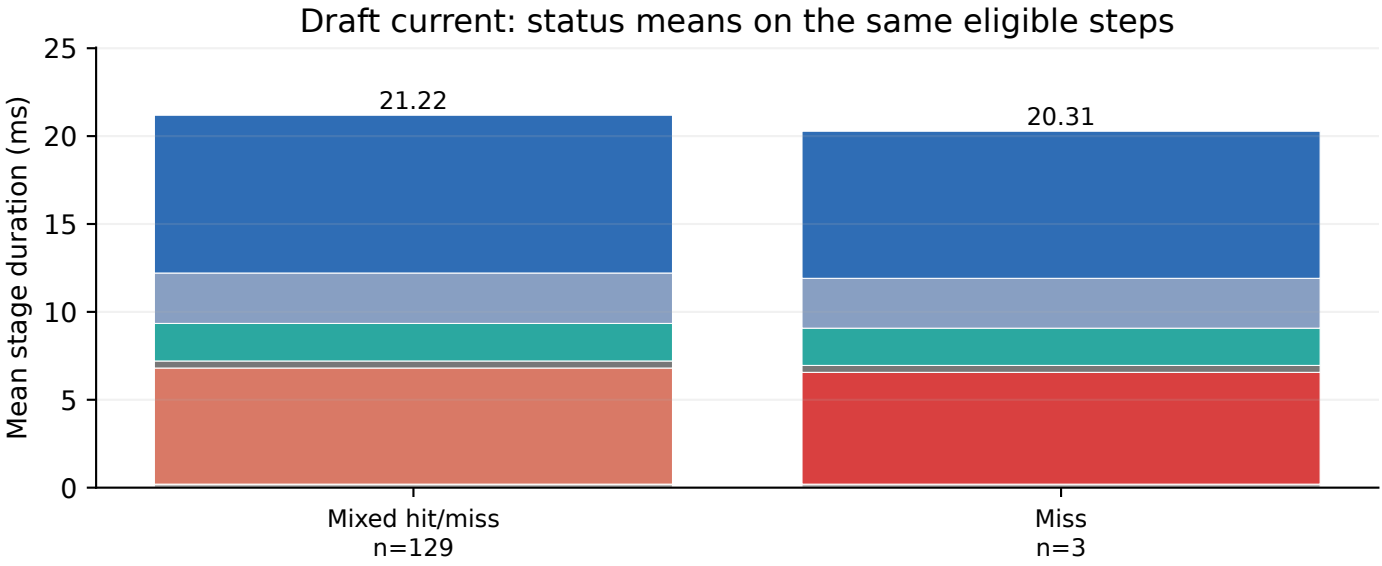
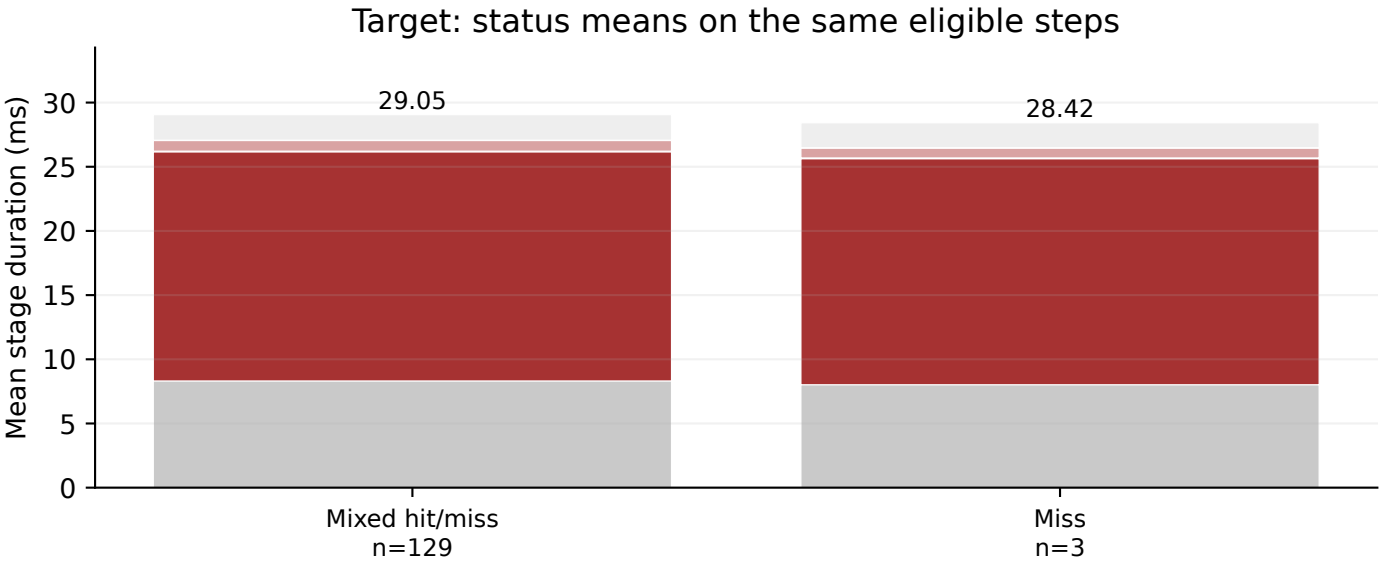
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.





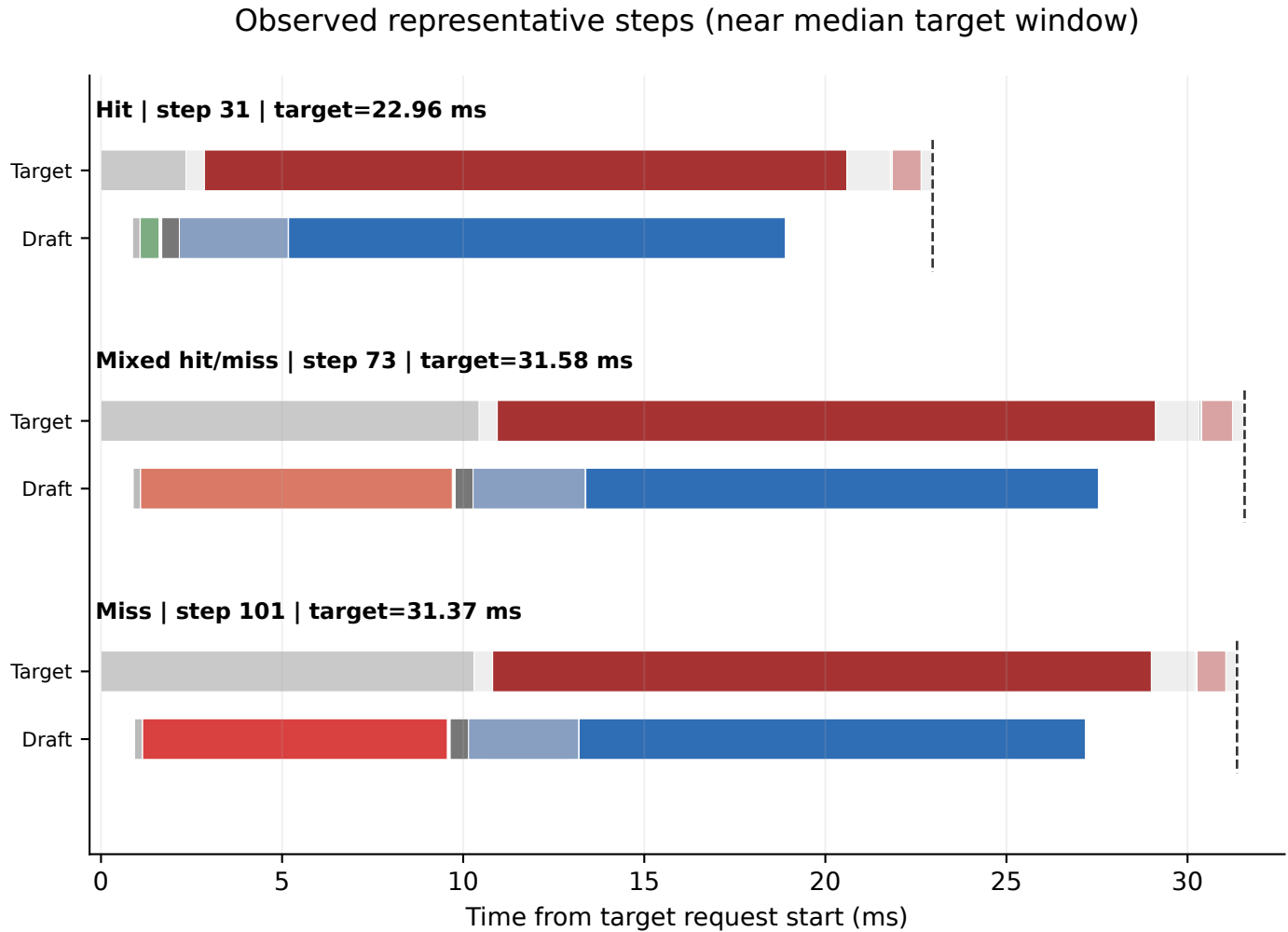
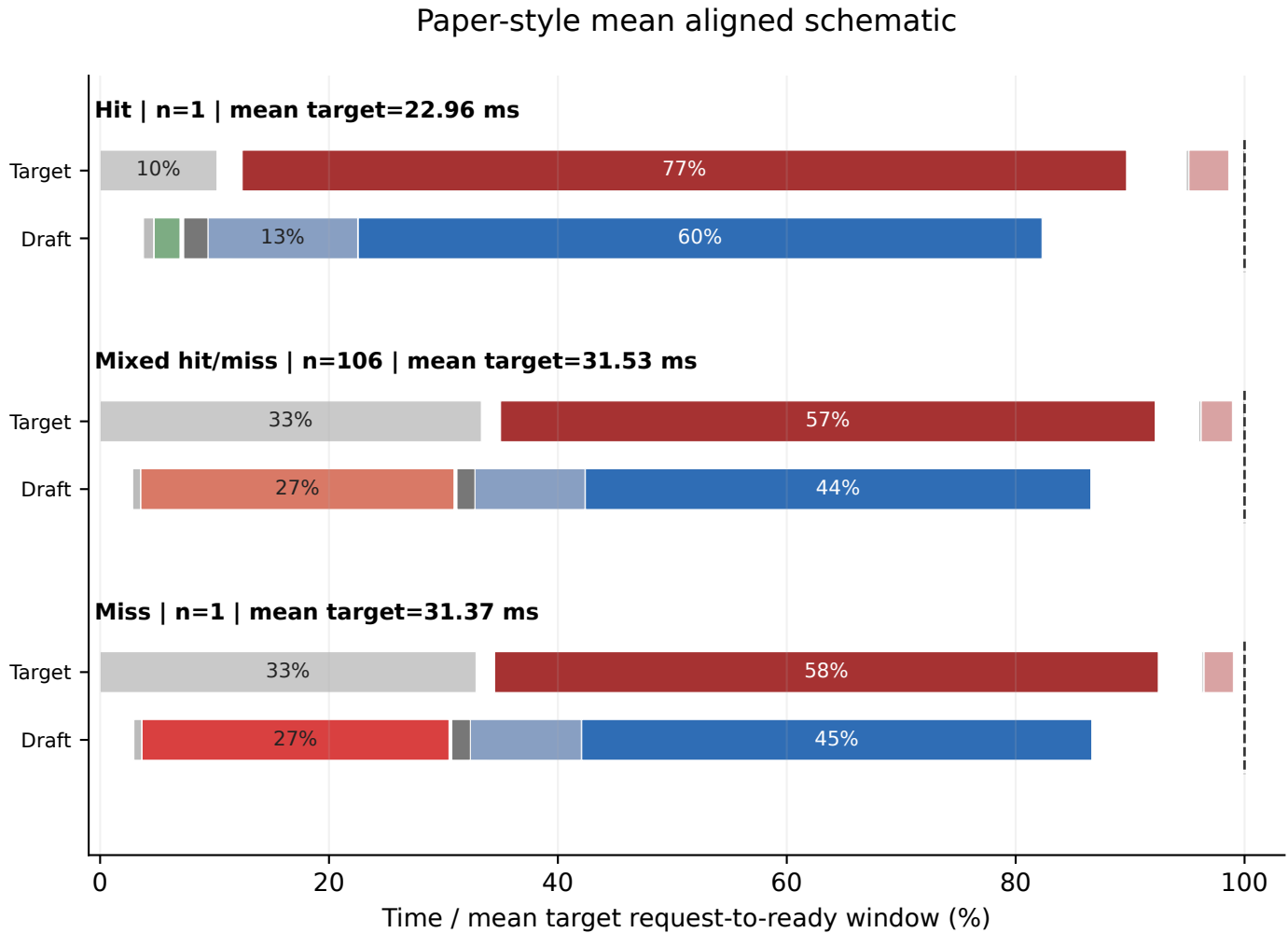
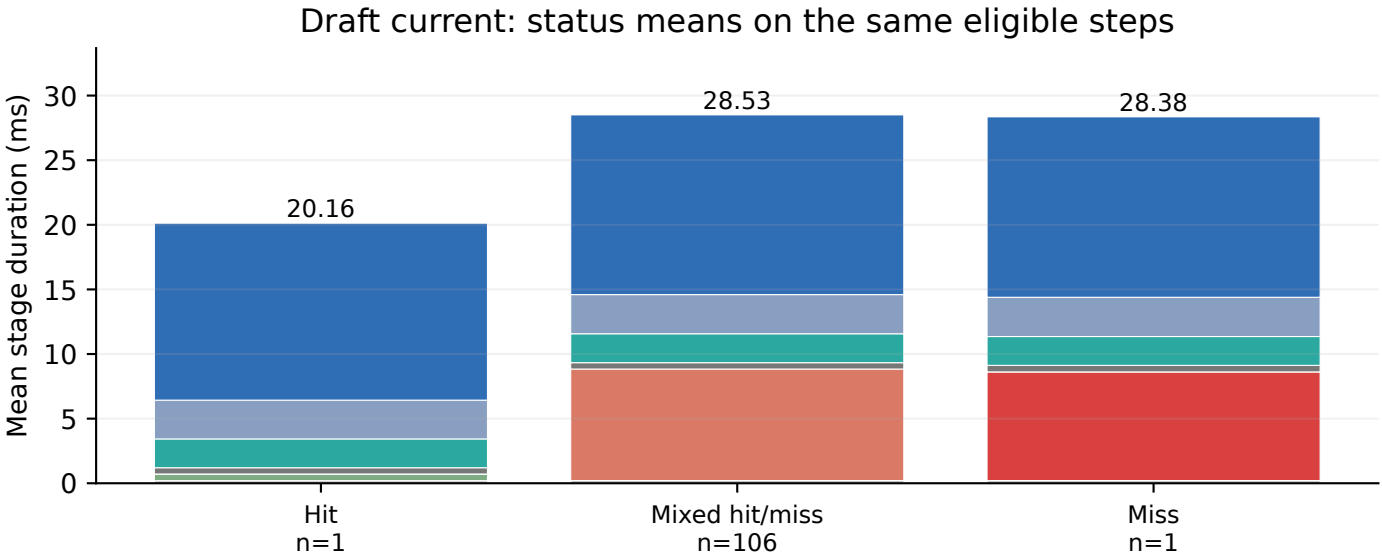
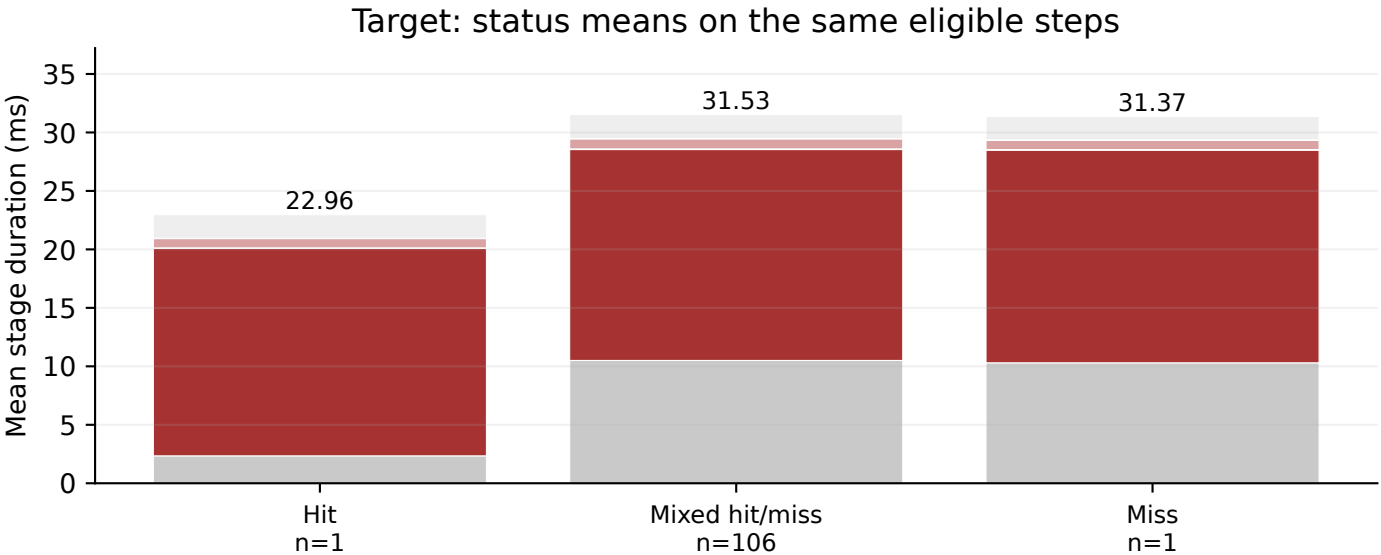
Wait / sync Accept prepare Mixed response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



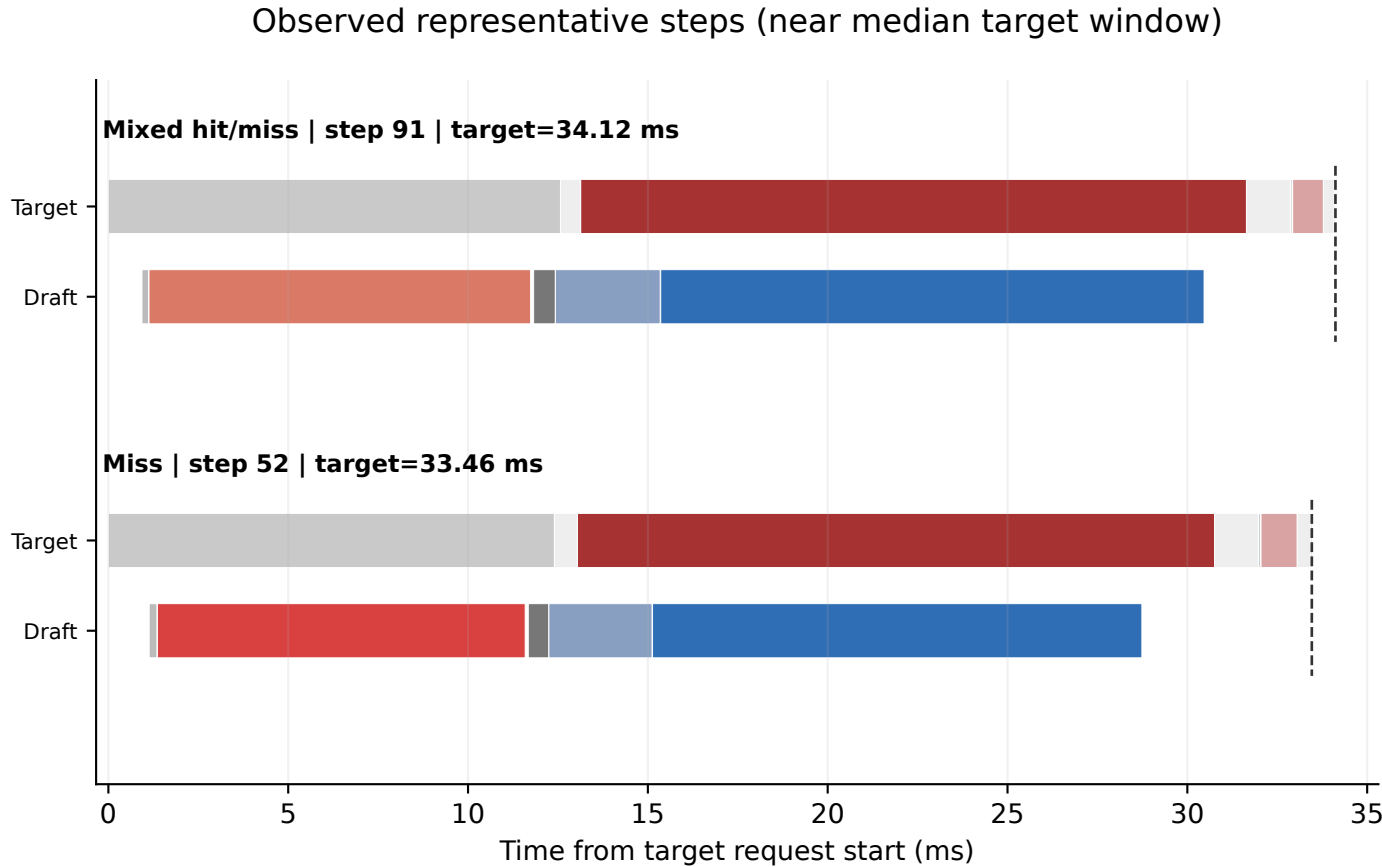
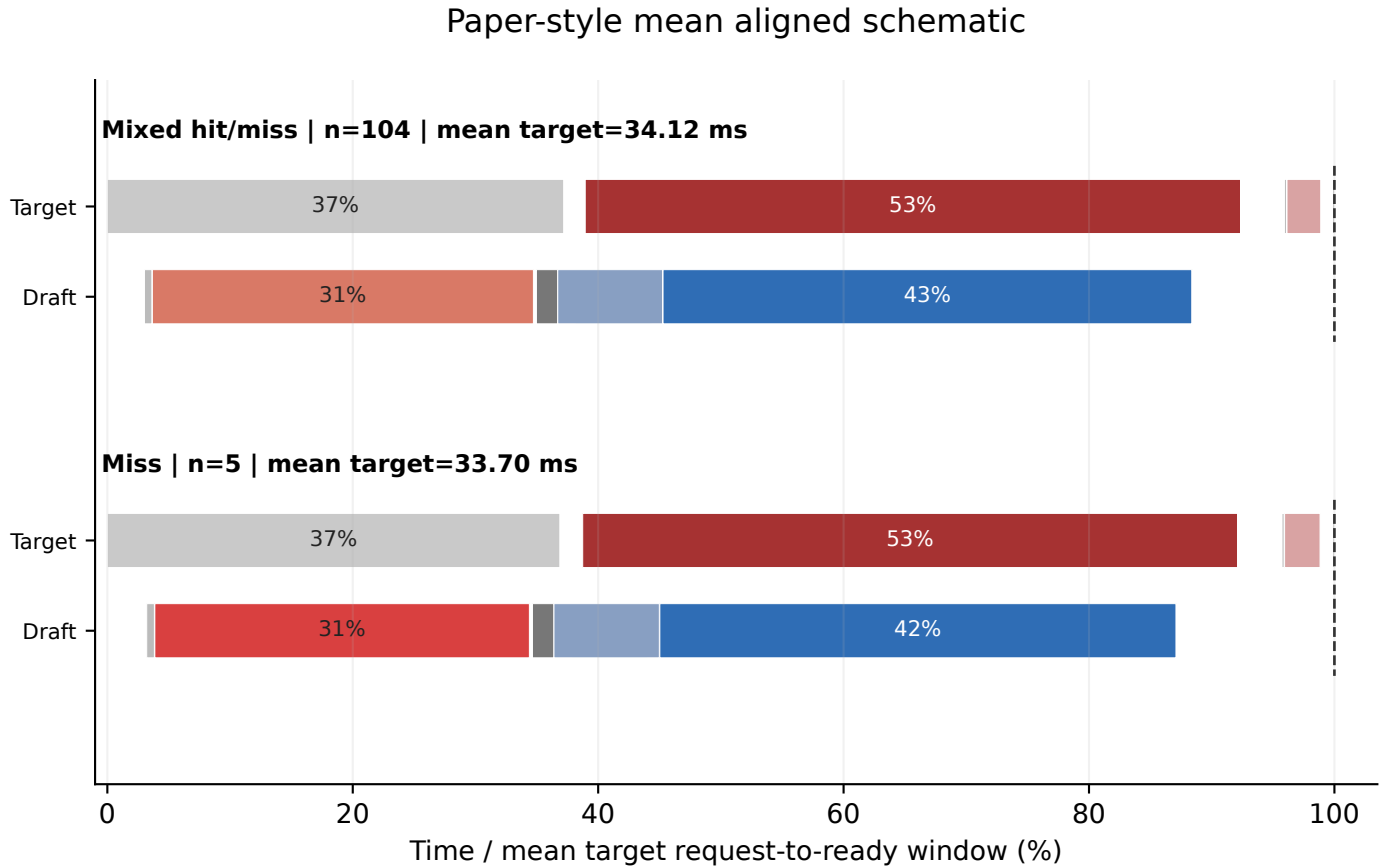
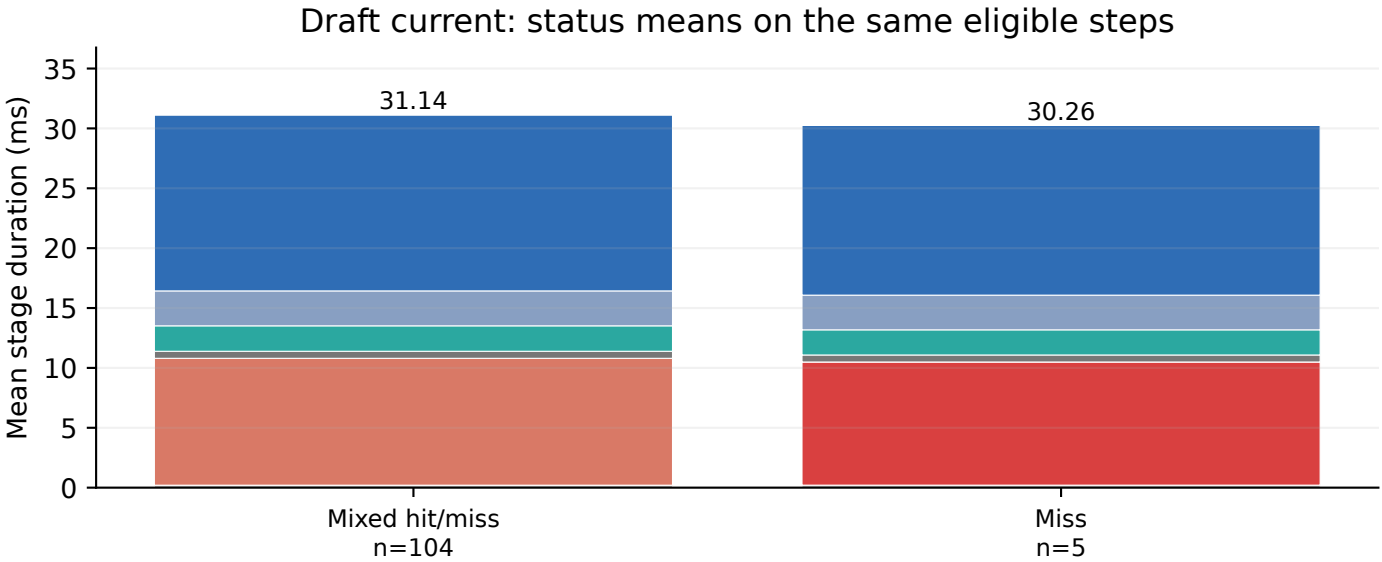
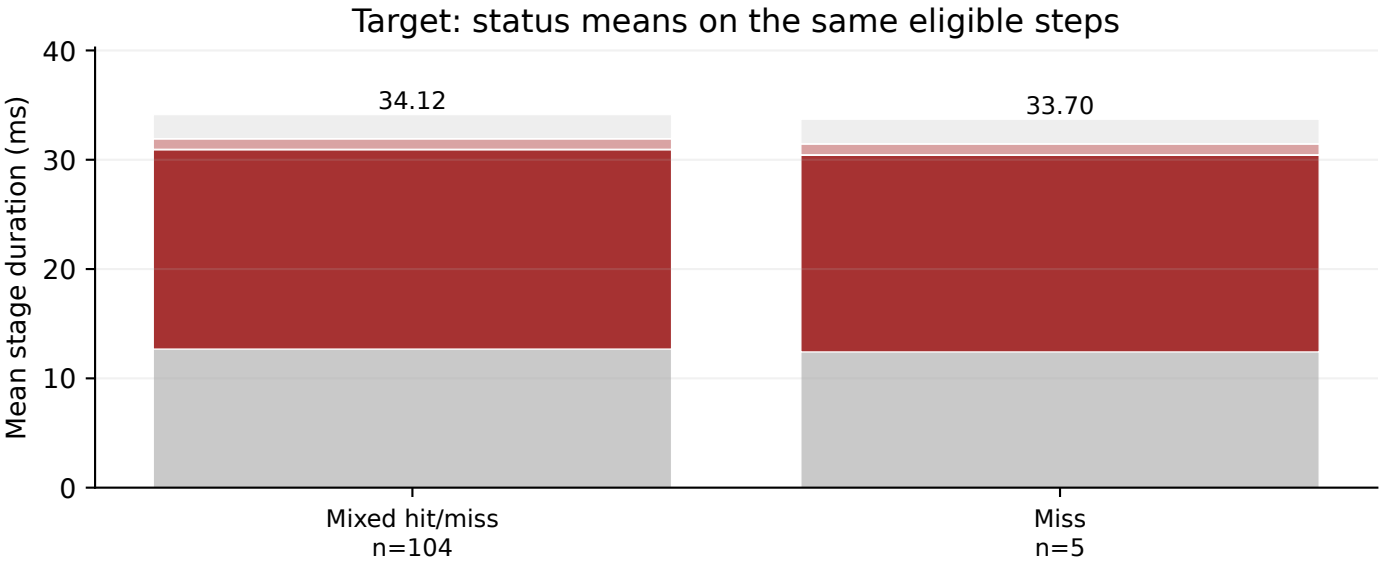
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



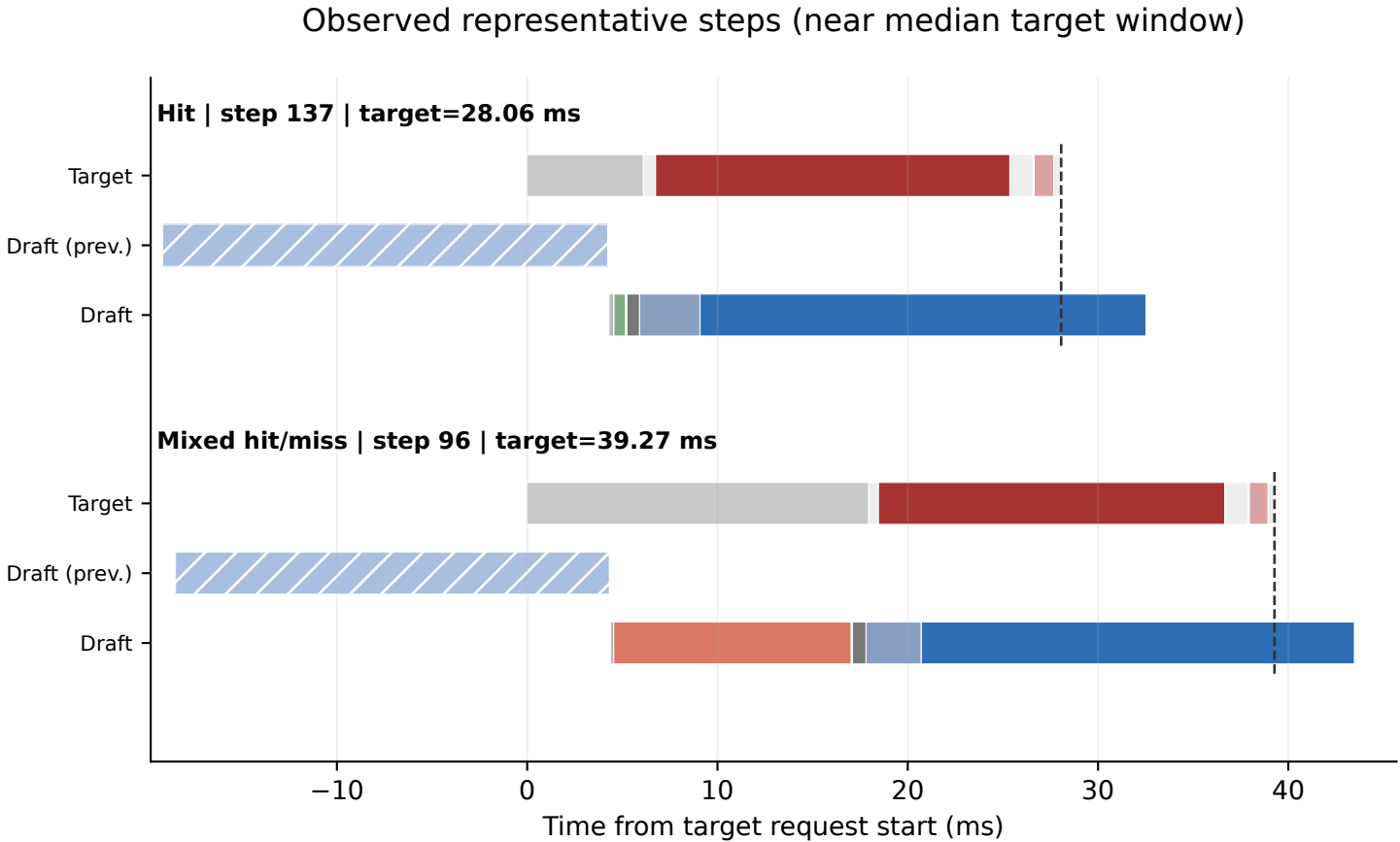
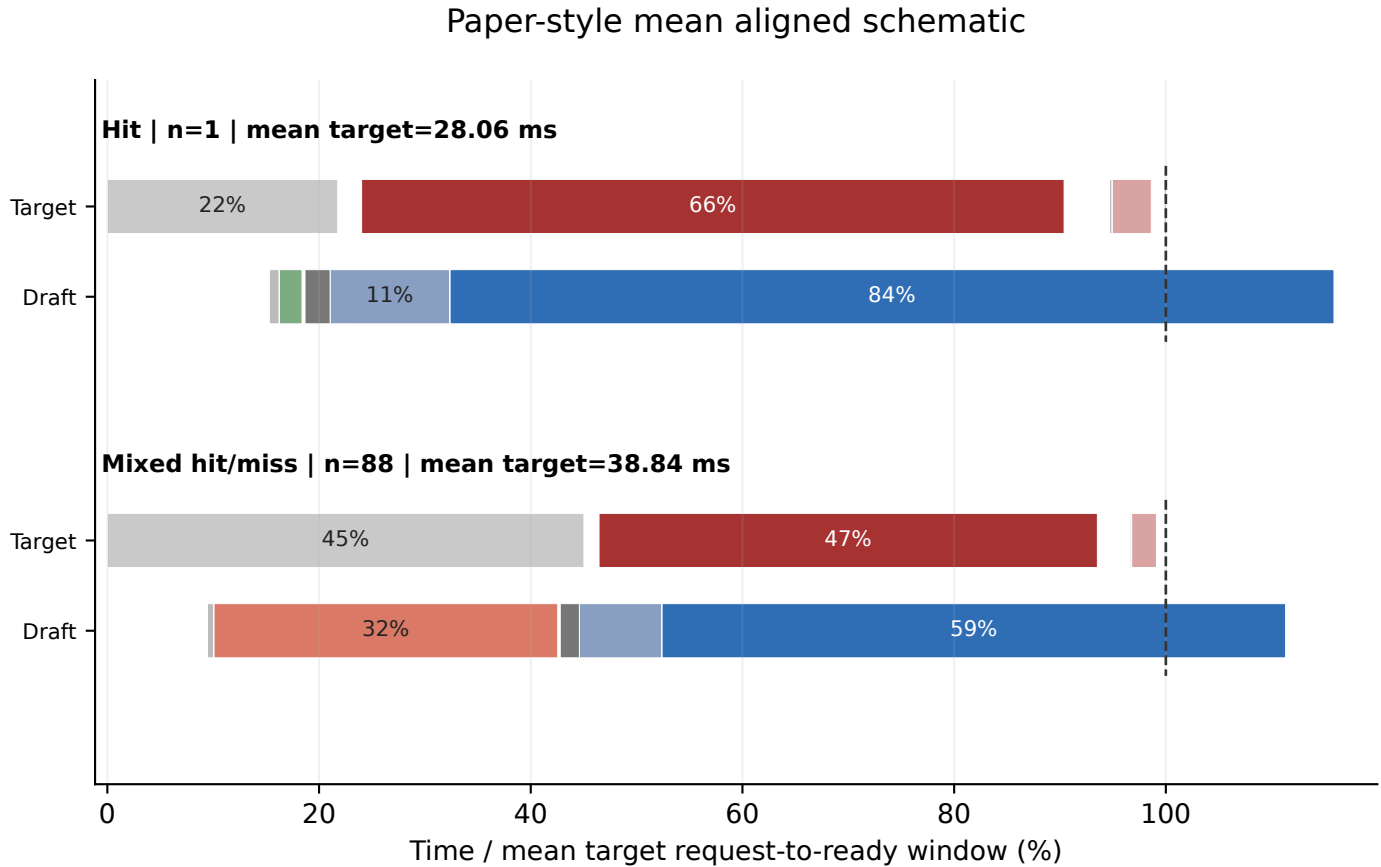
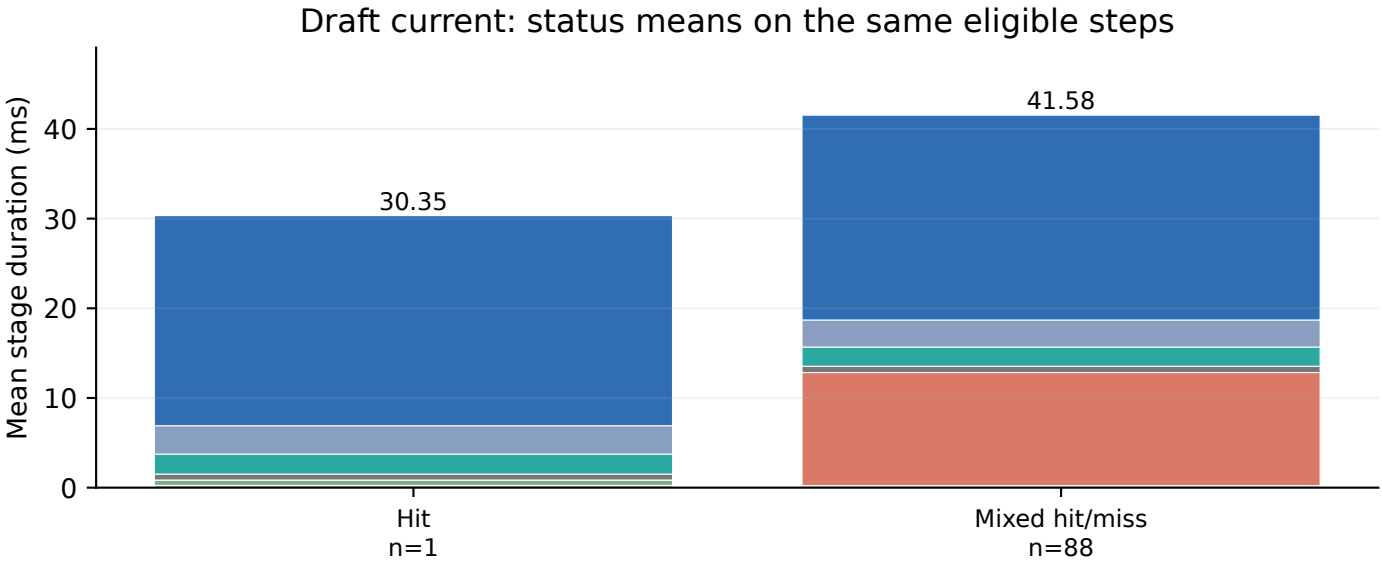
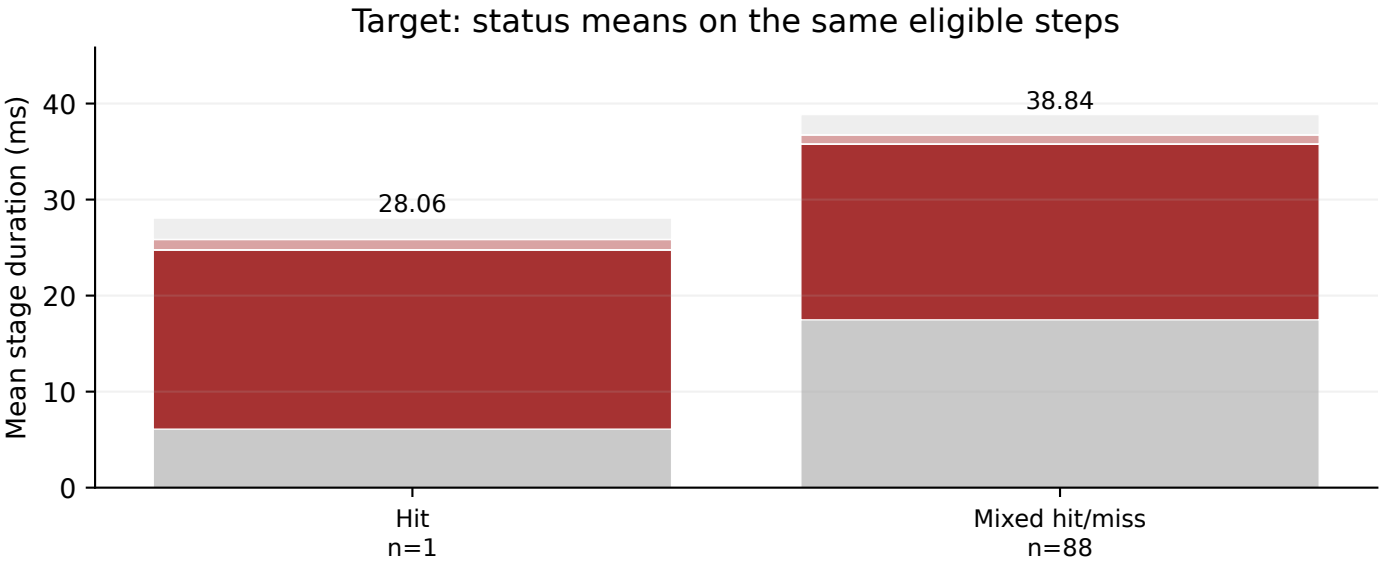
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



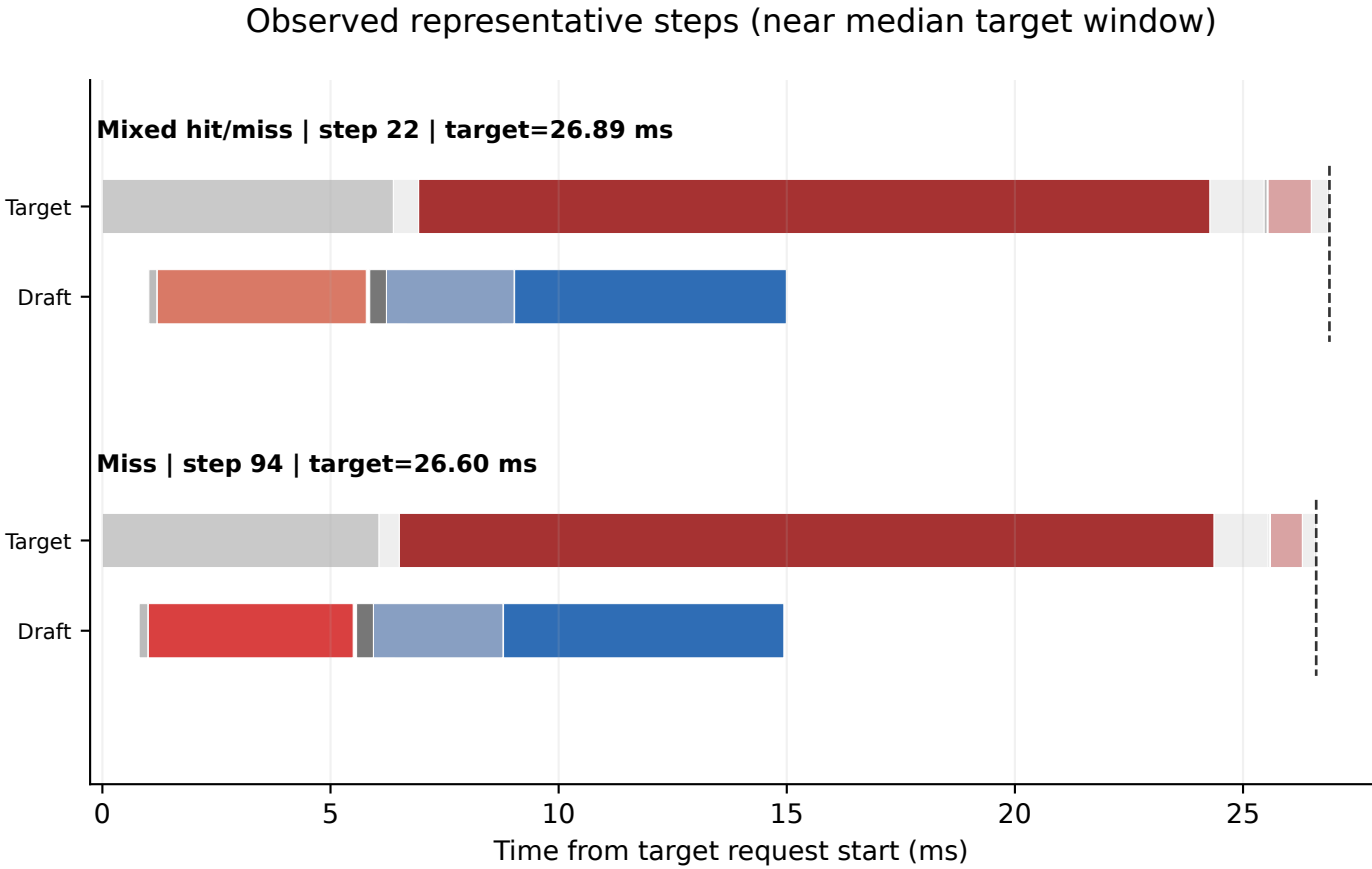
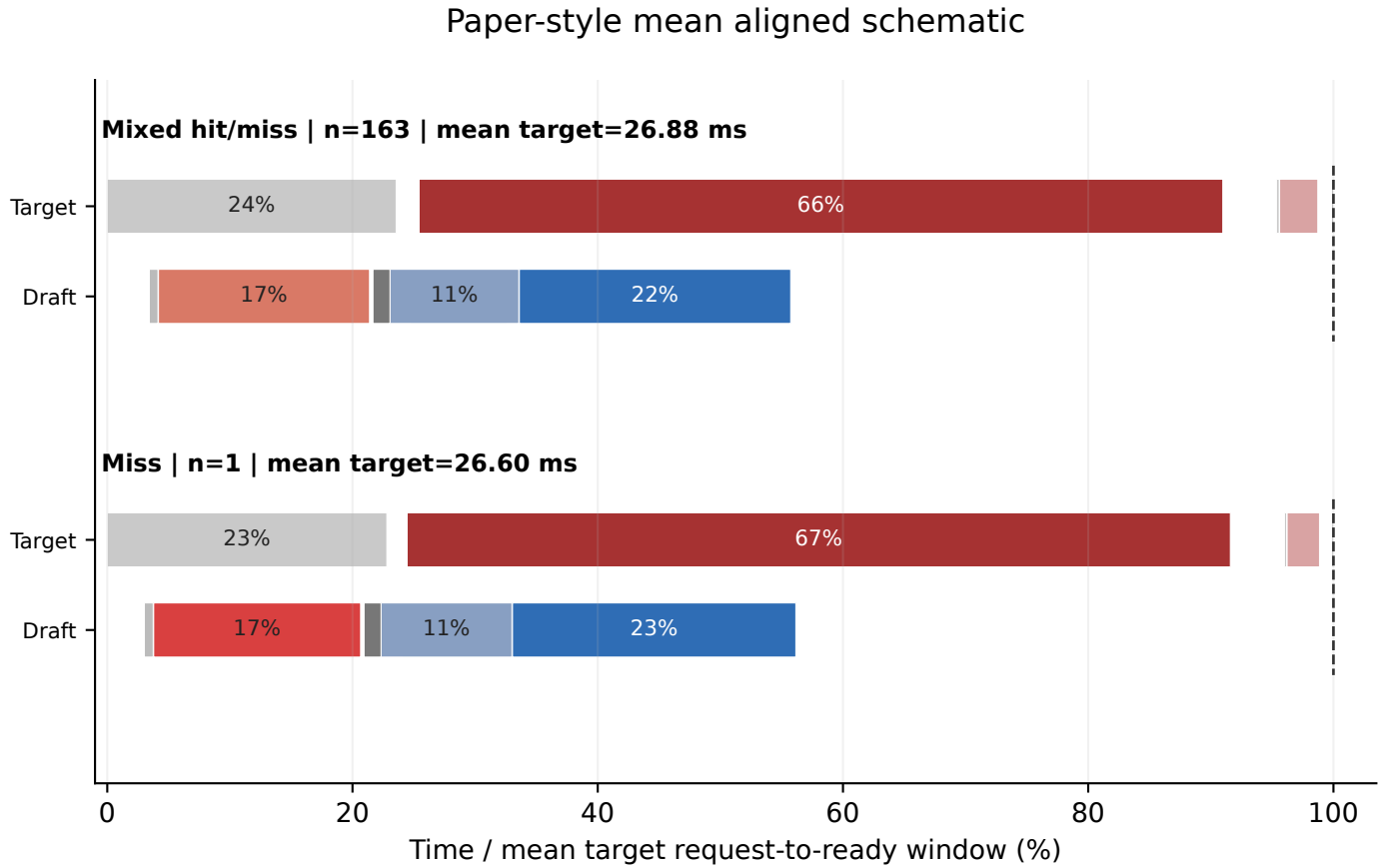
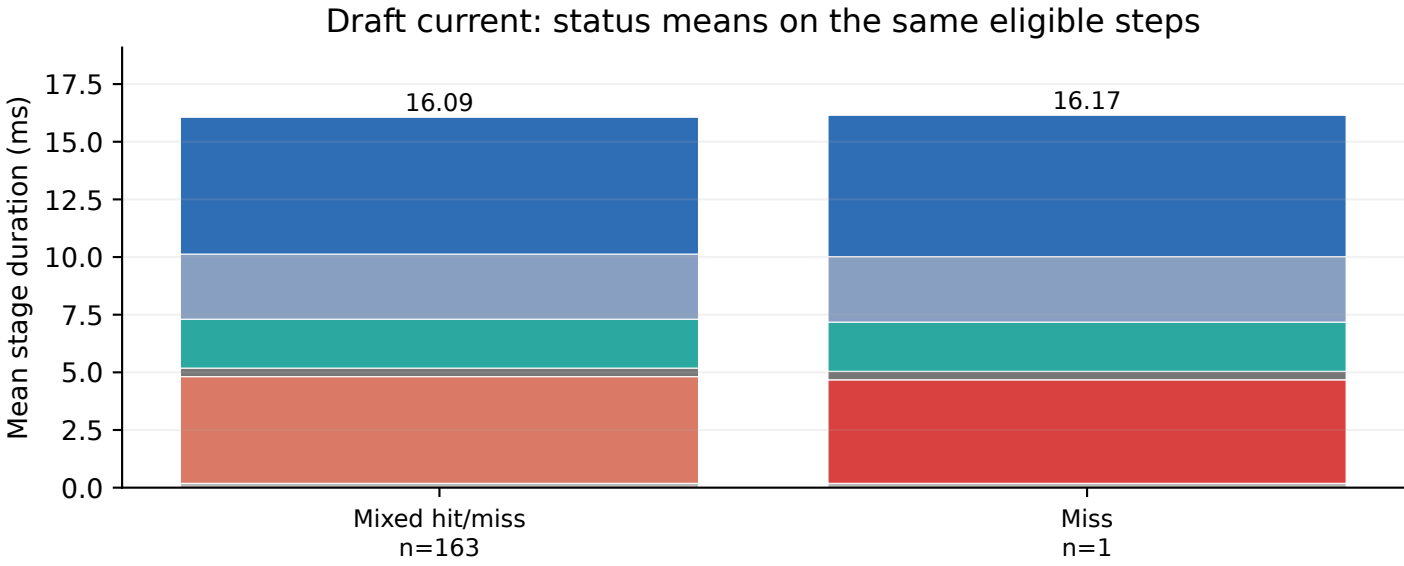
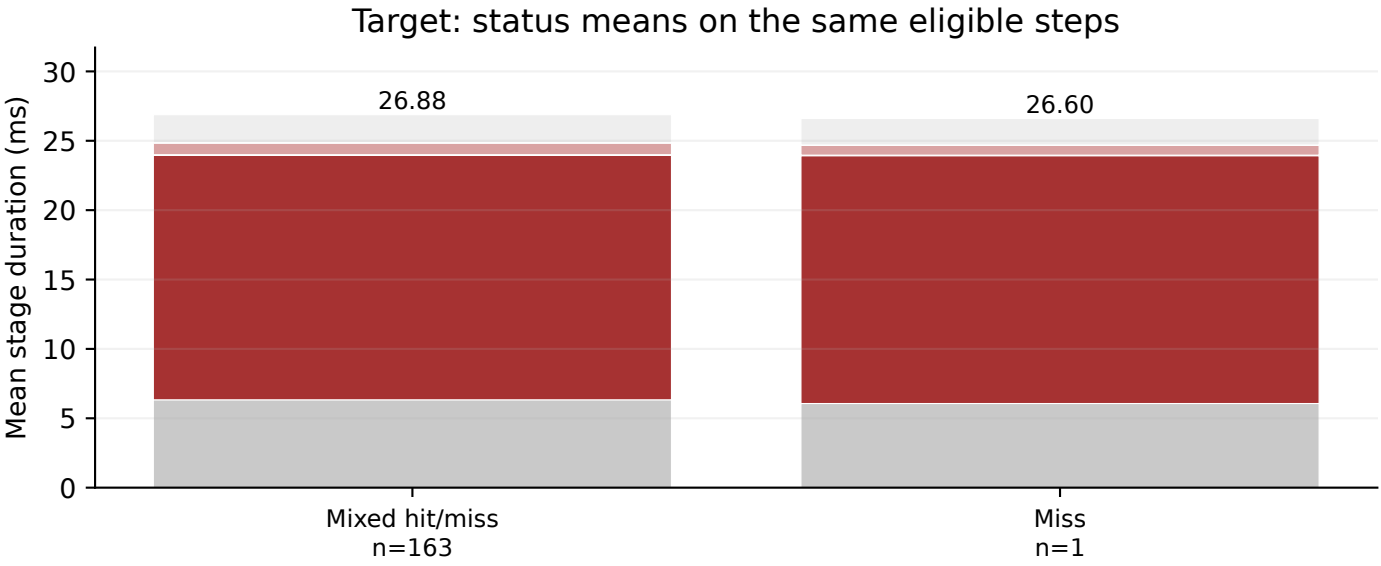
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



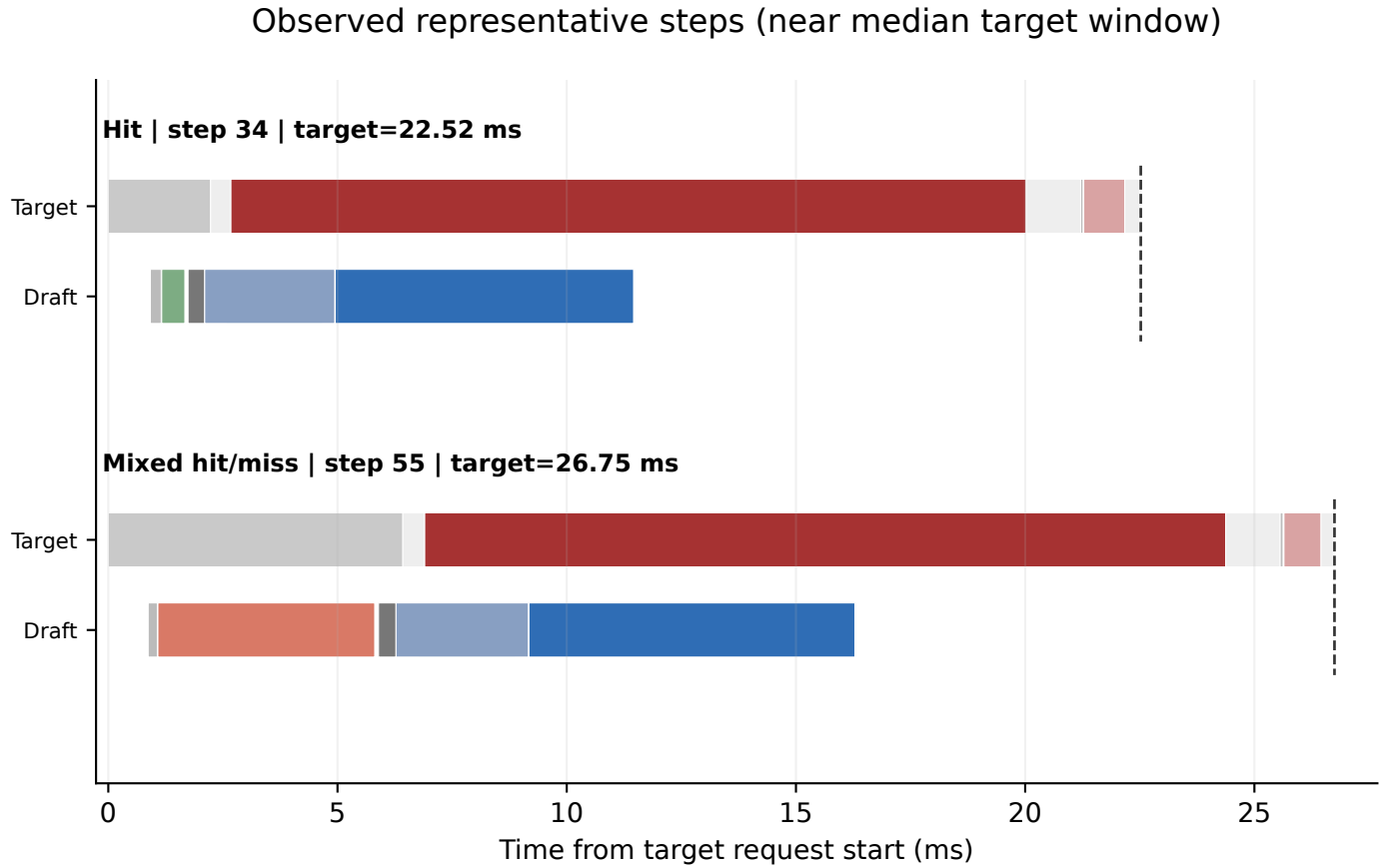
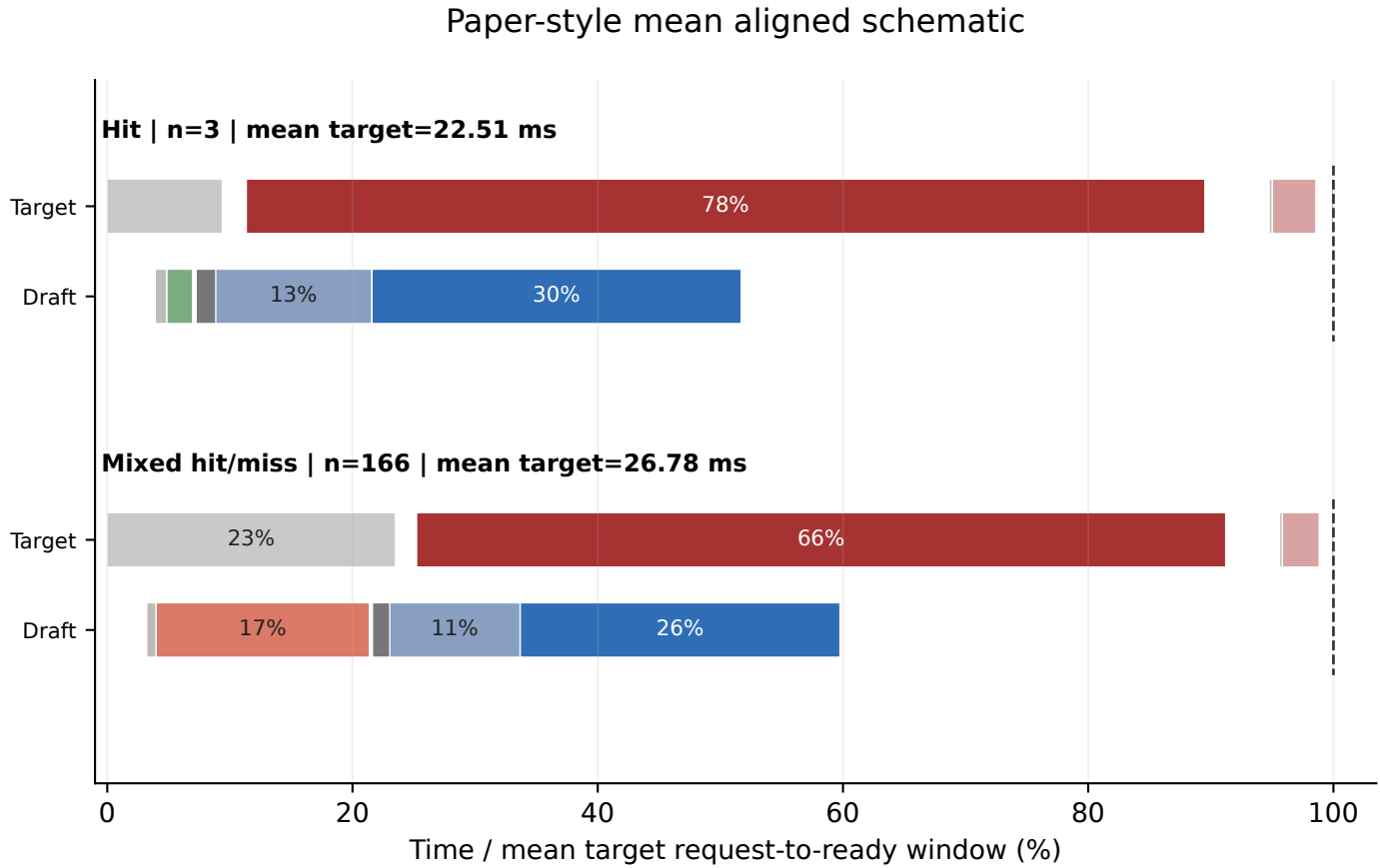
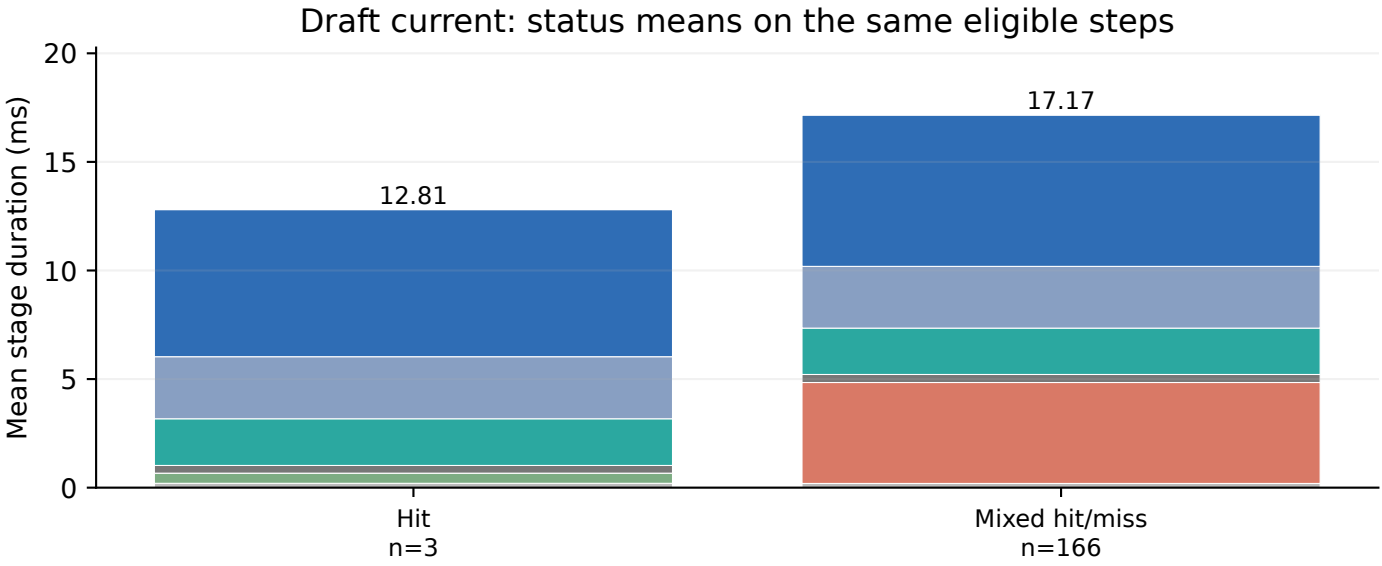
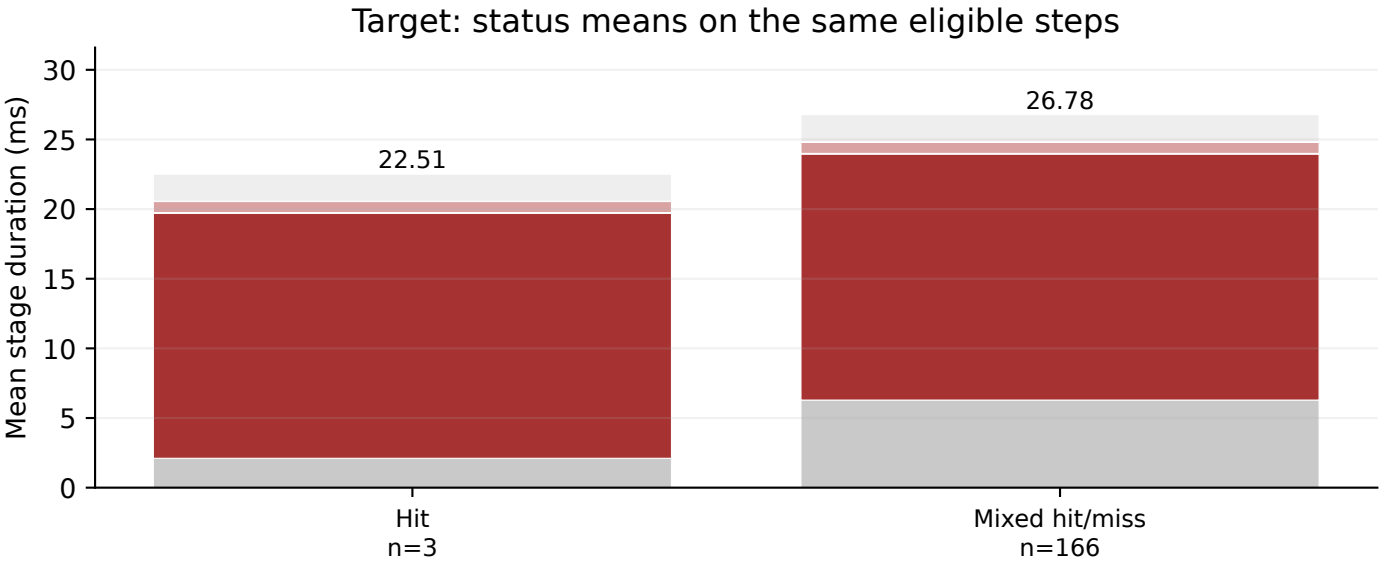
Wait / sync Accept prepare Mixed response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



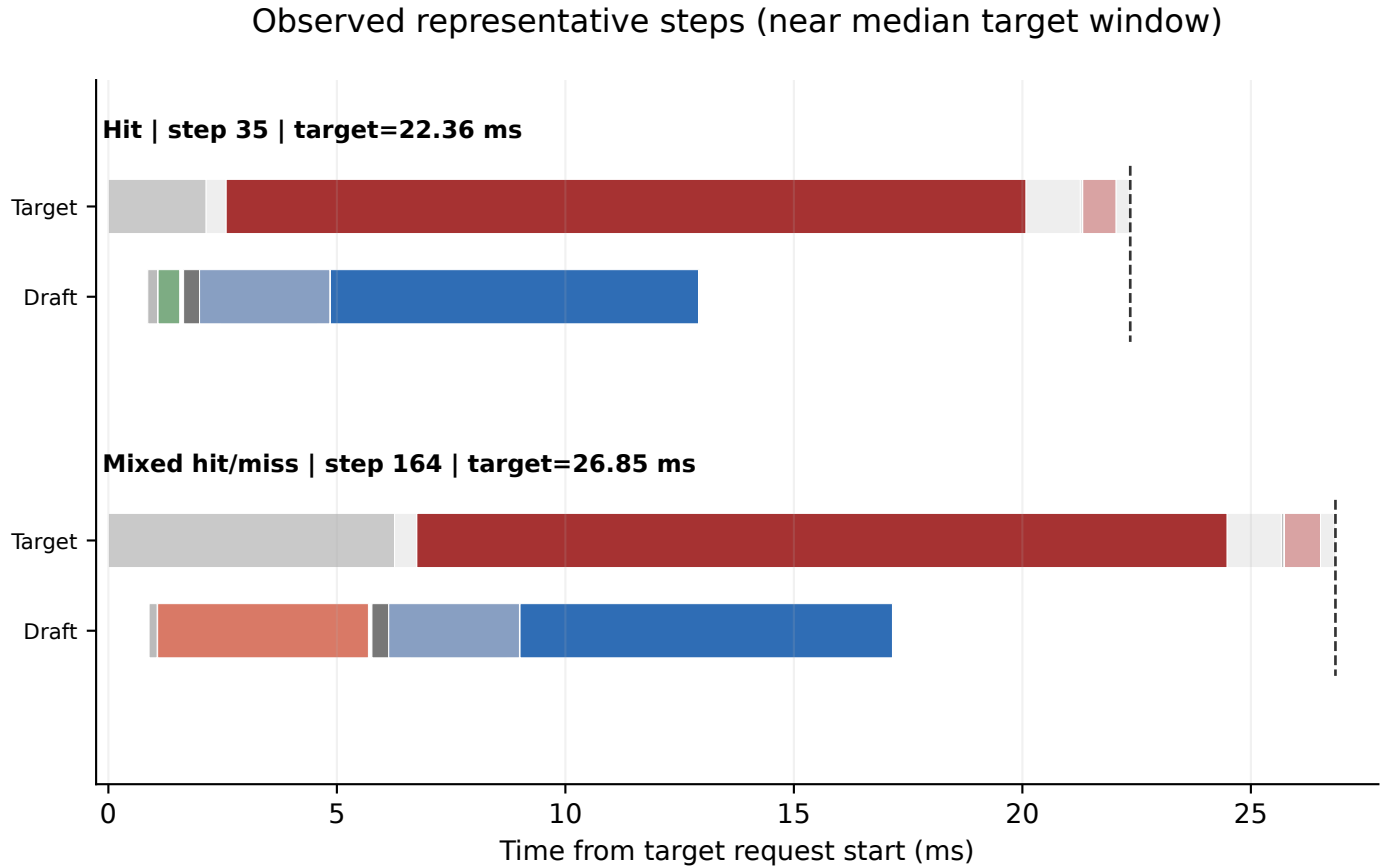
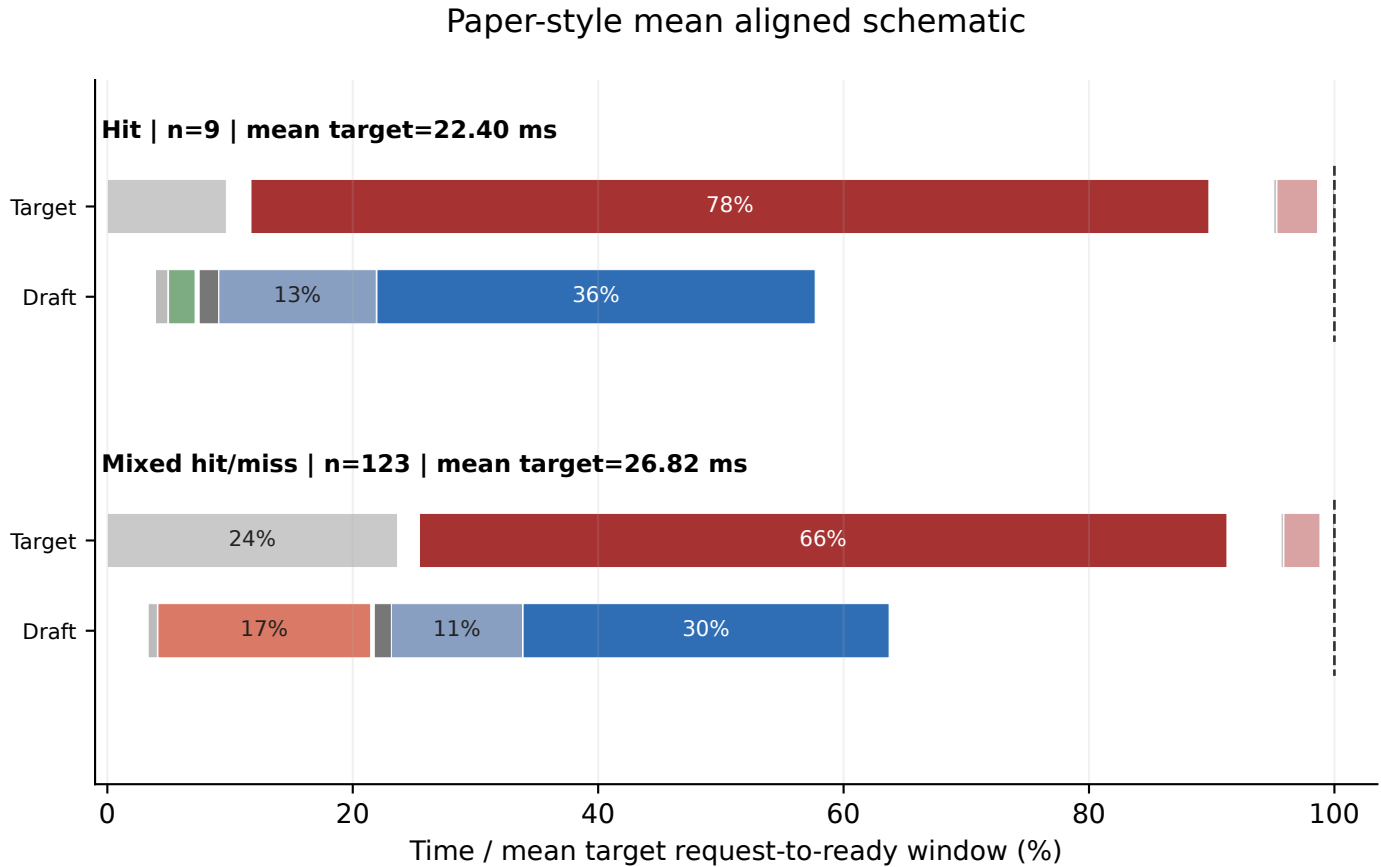
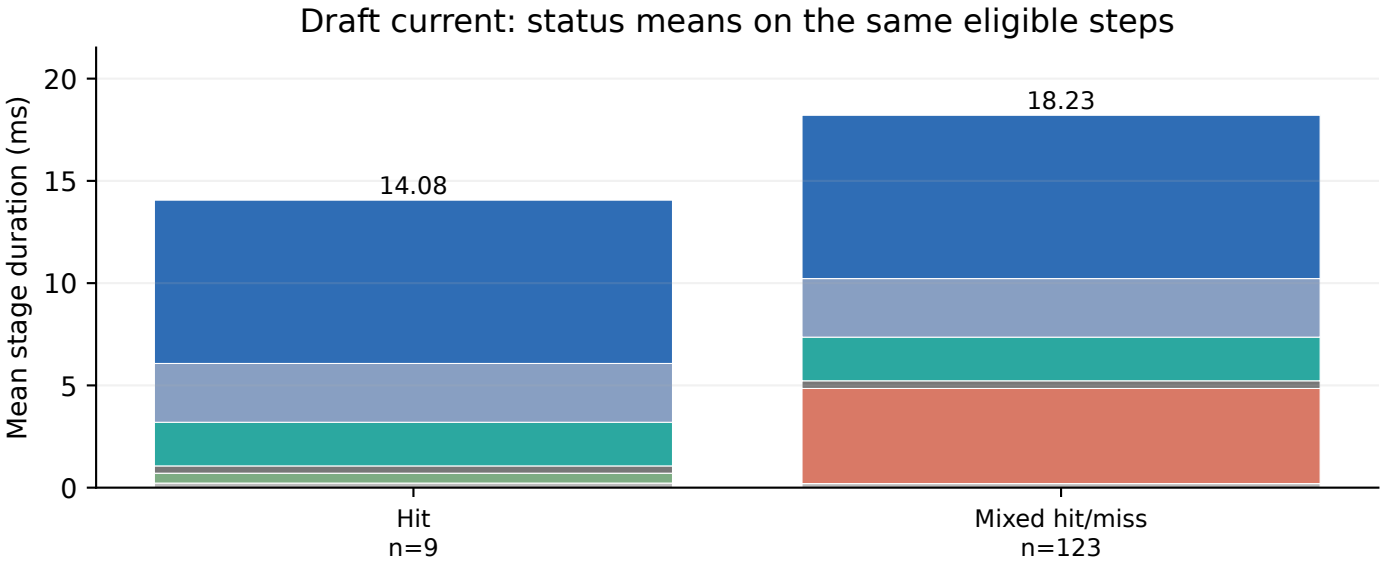
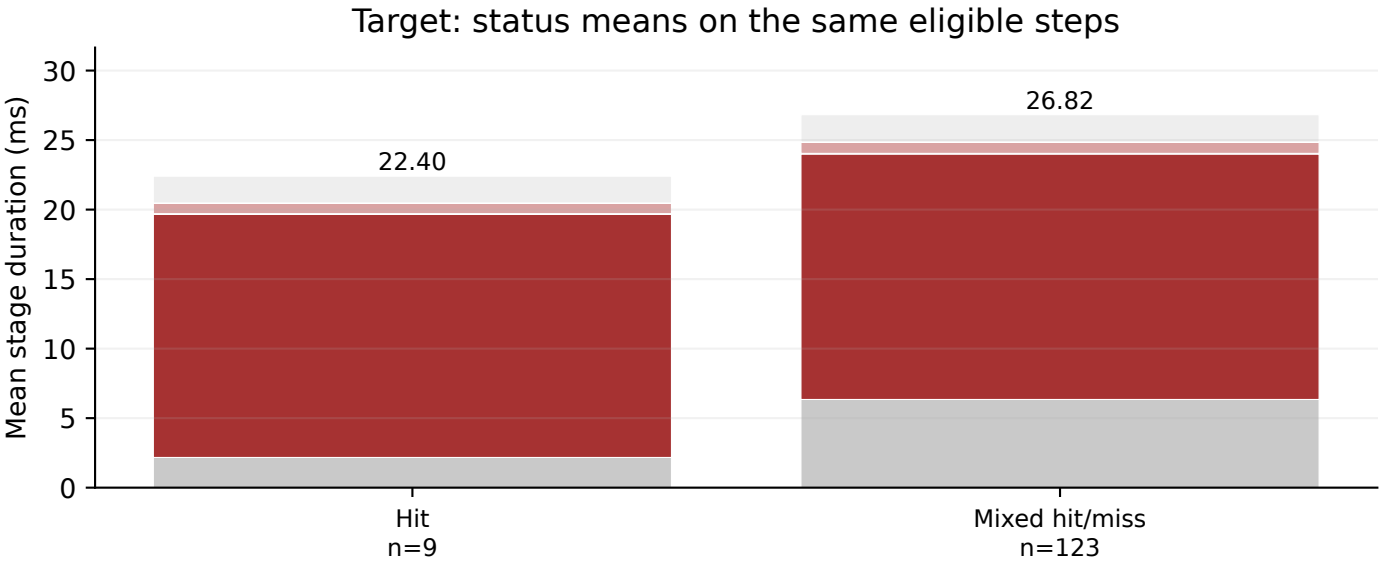
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



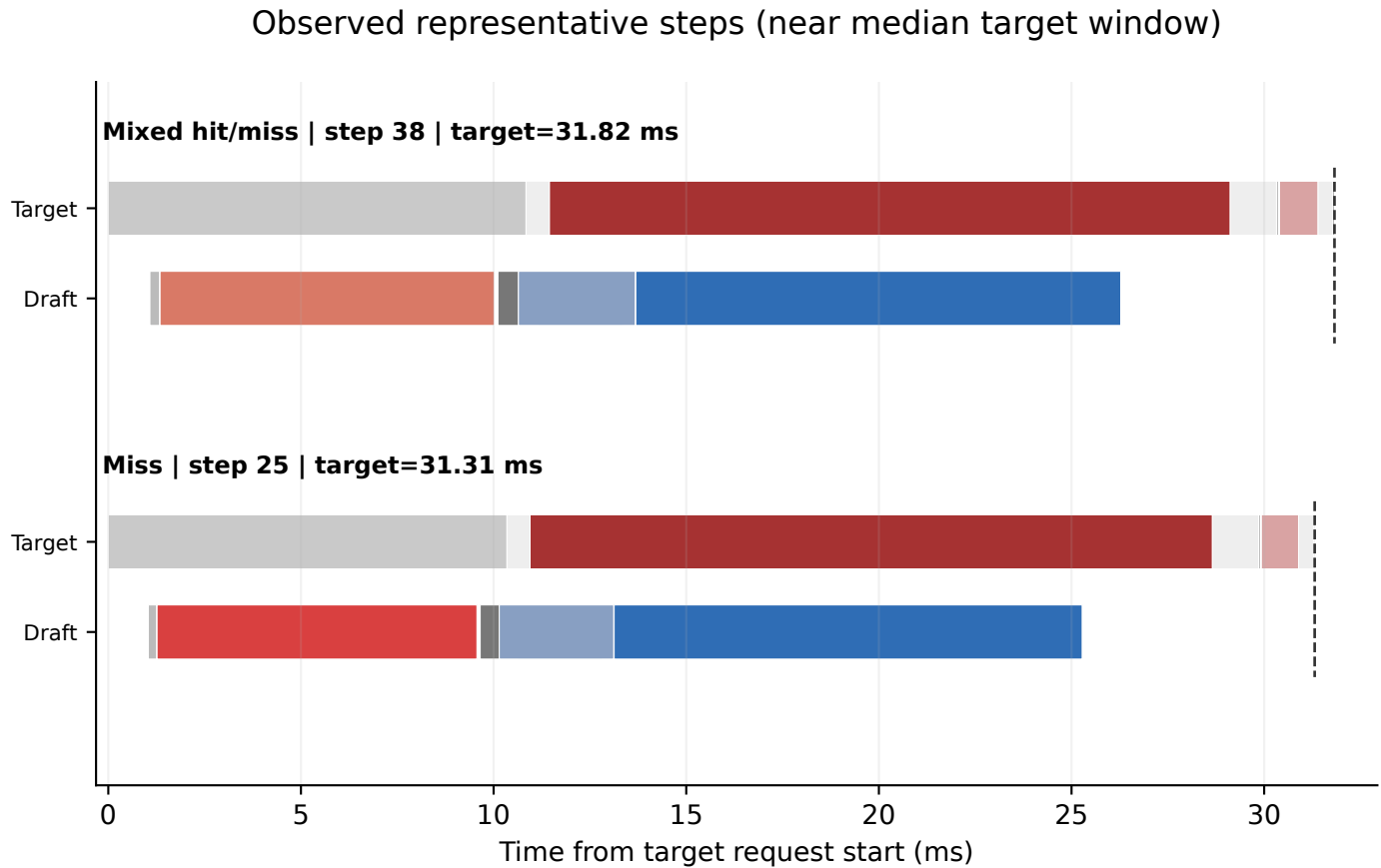
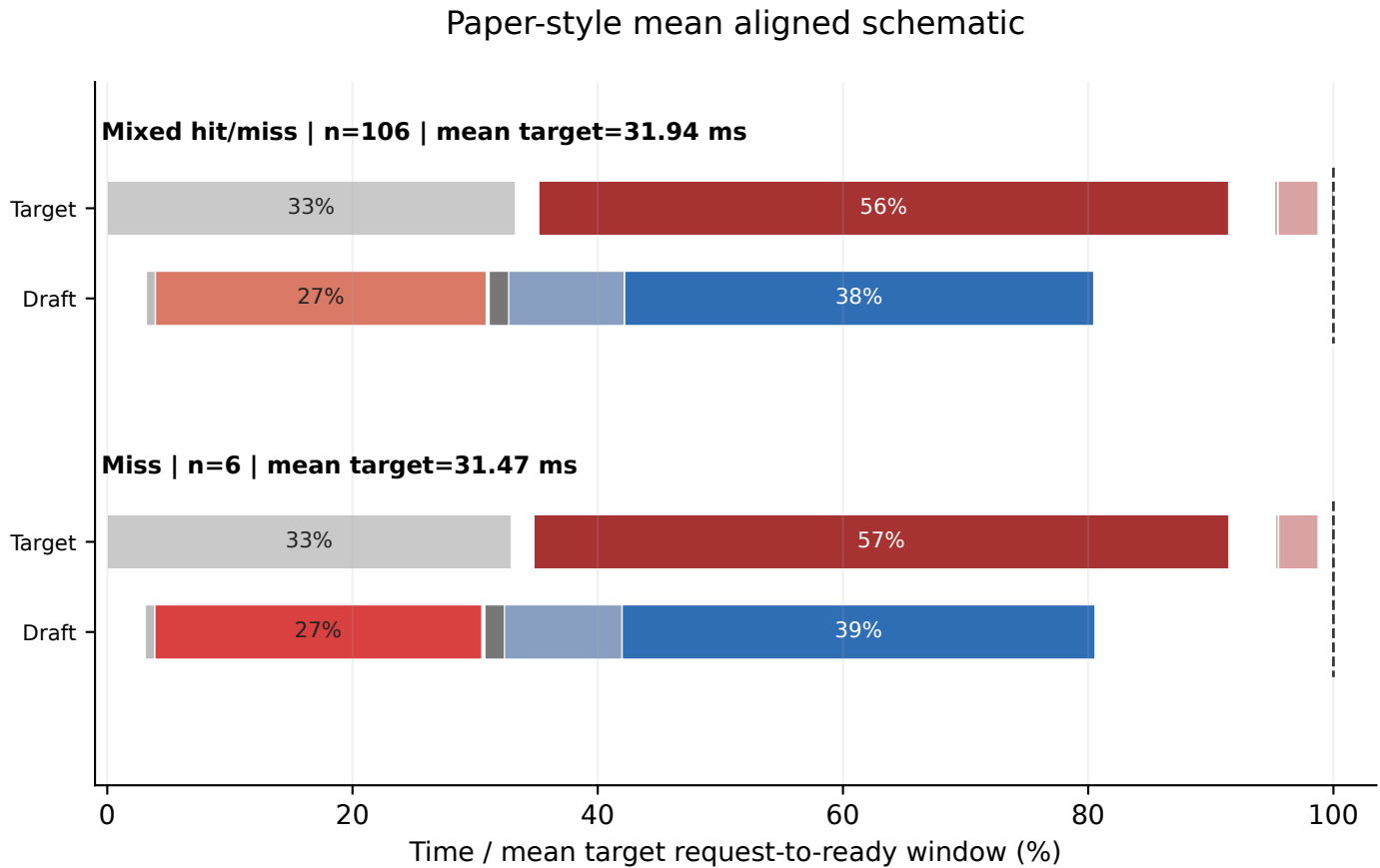
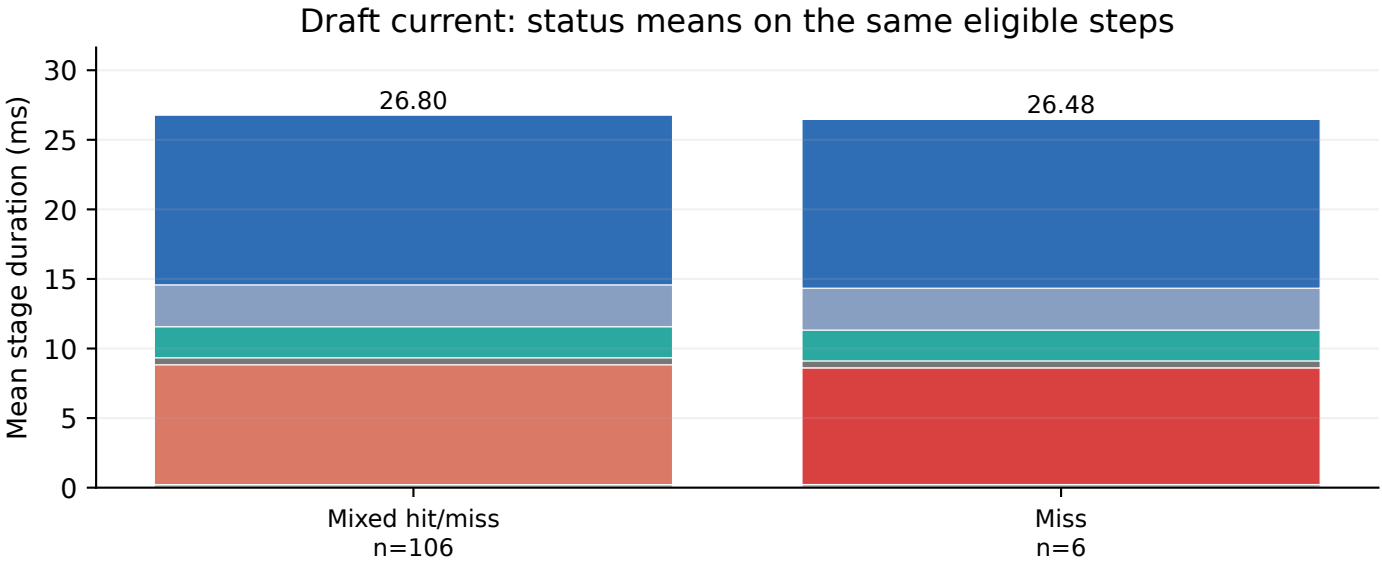
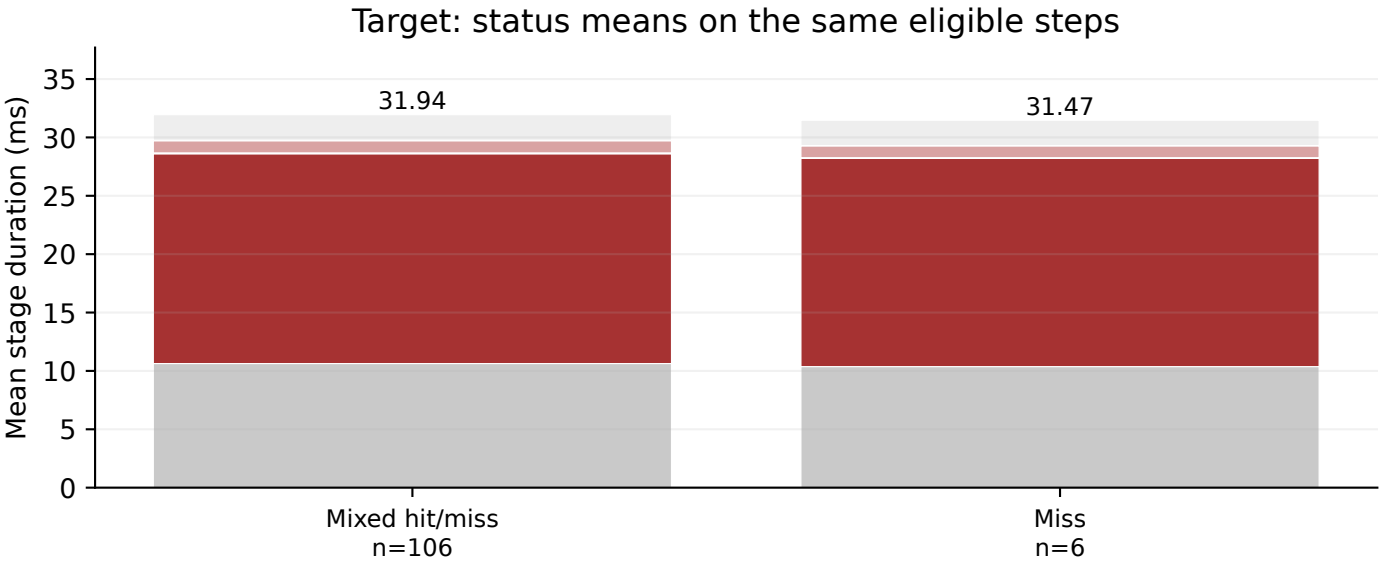
Wait / sync Accept prepare Mixed response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



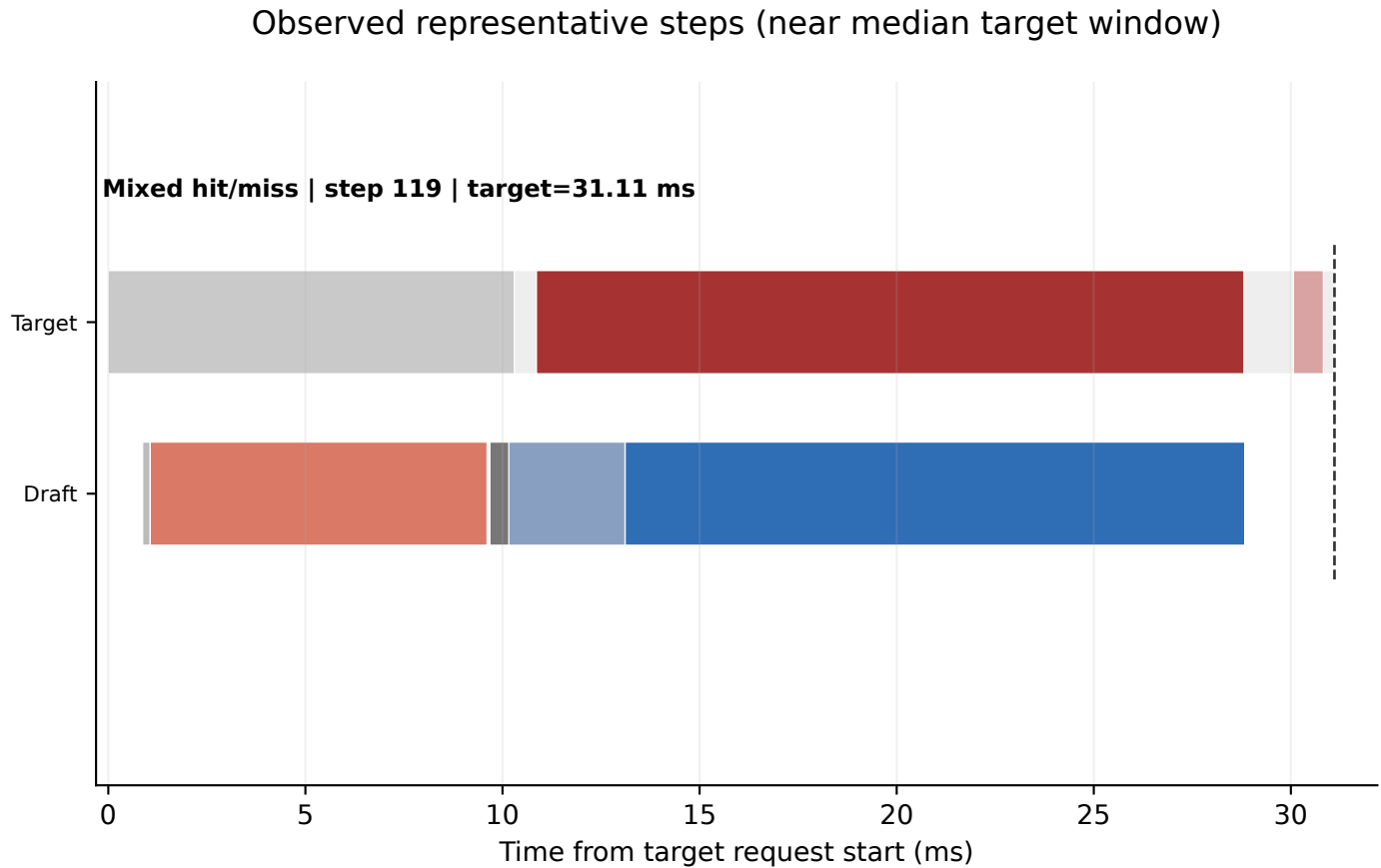
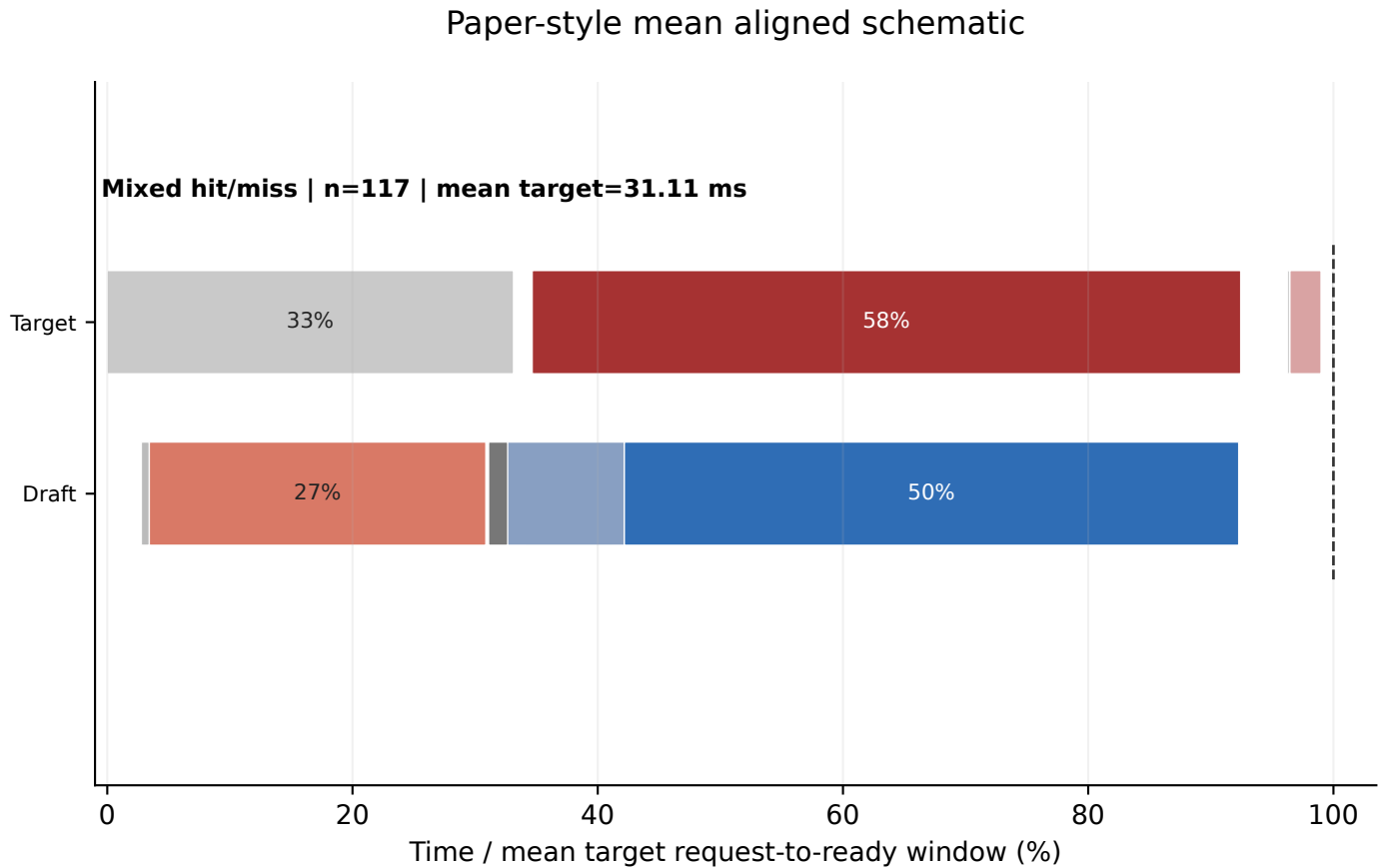
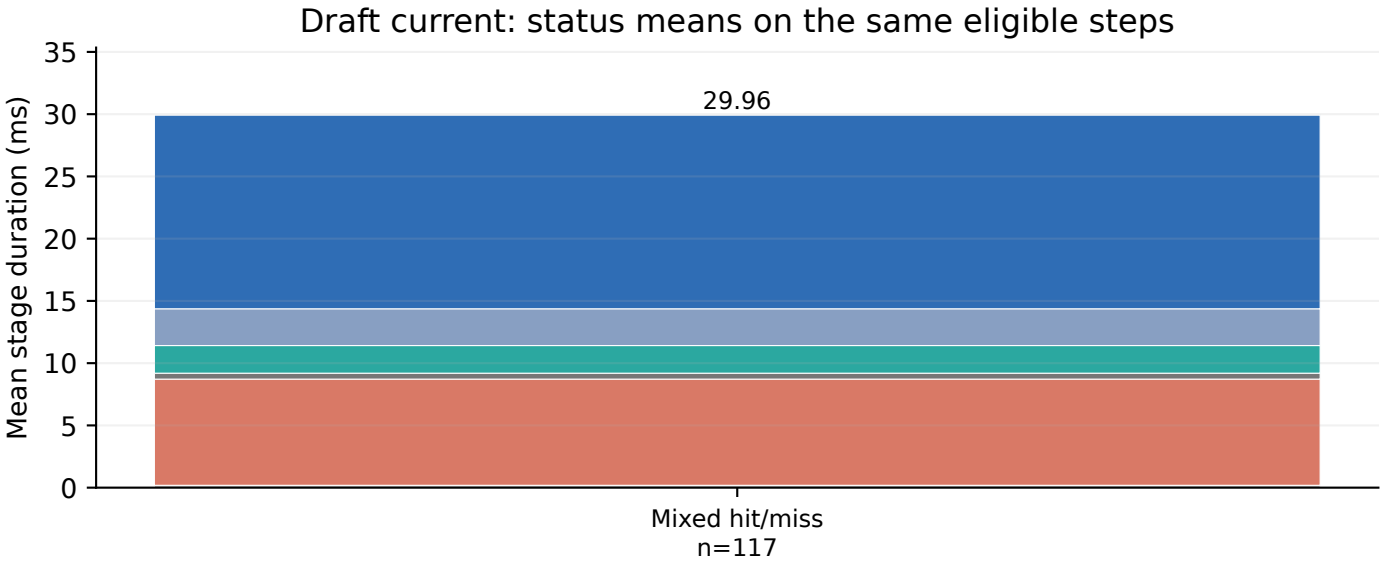
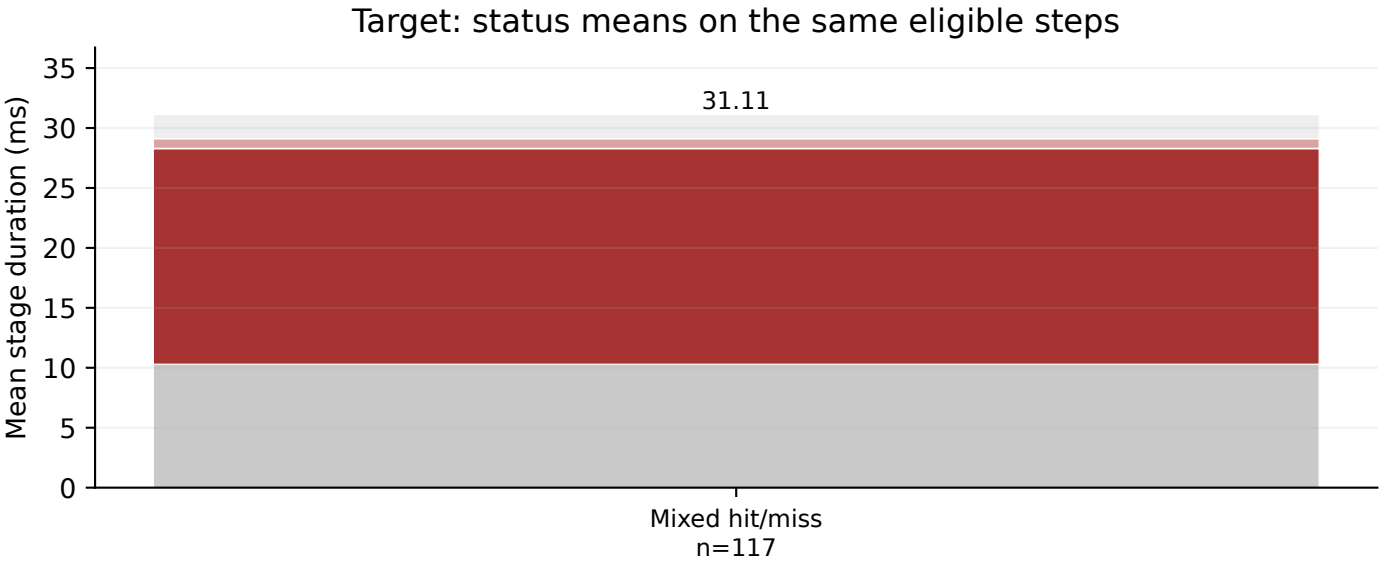
Wait / sync Accept prepare Mixed response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



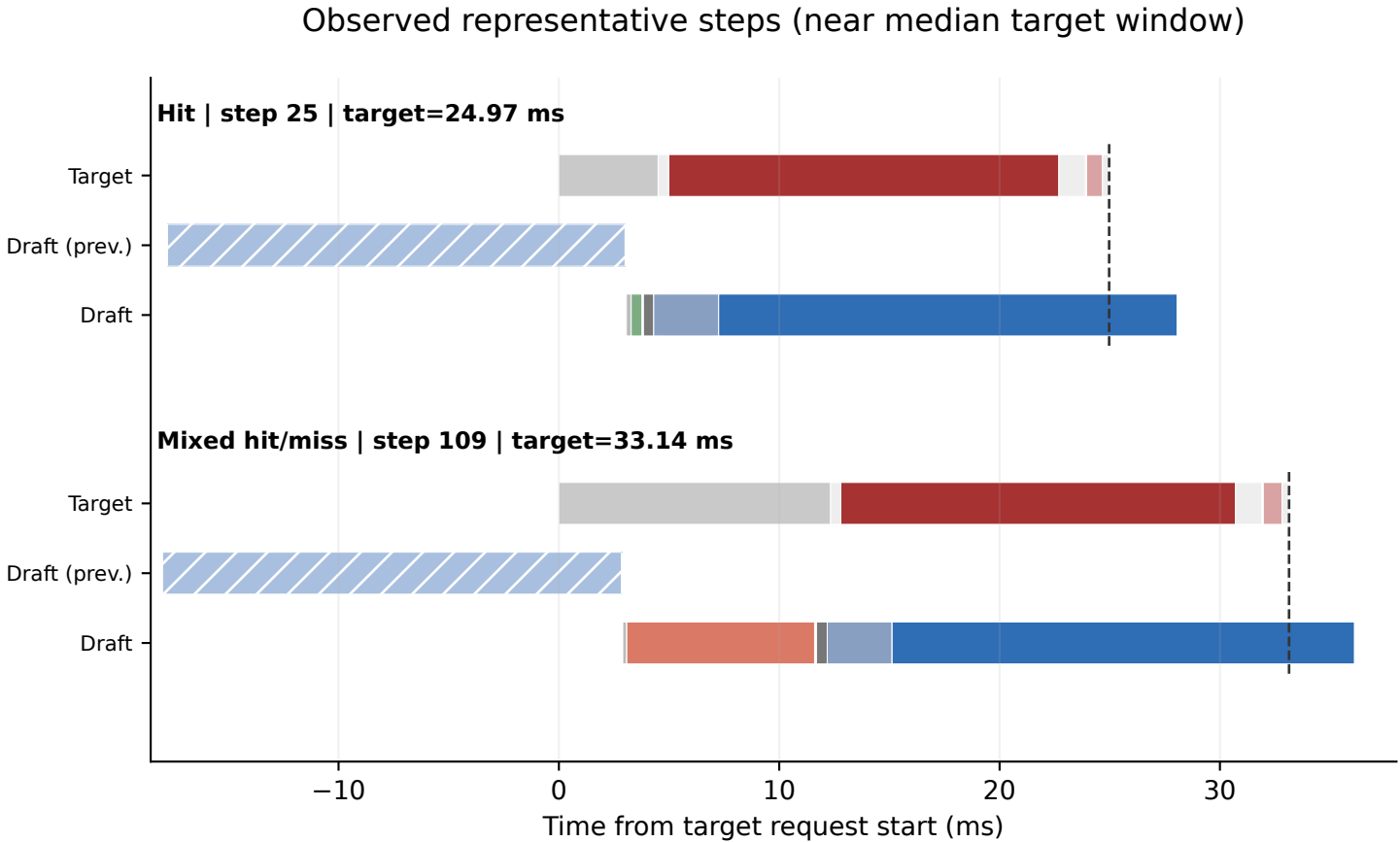
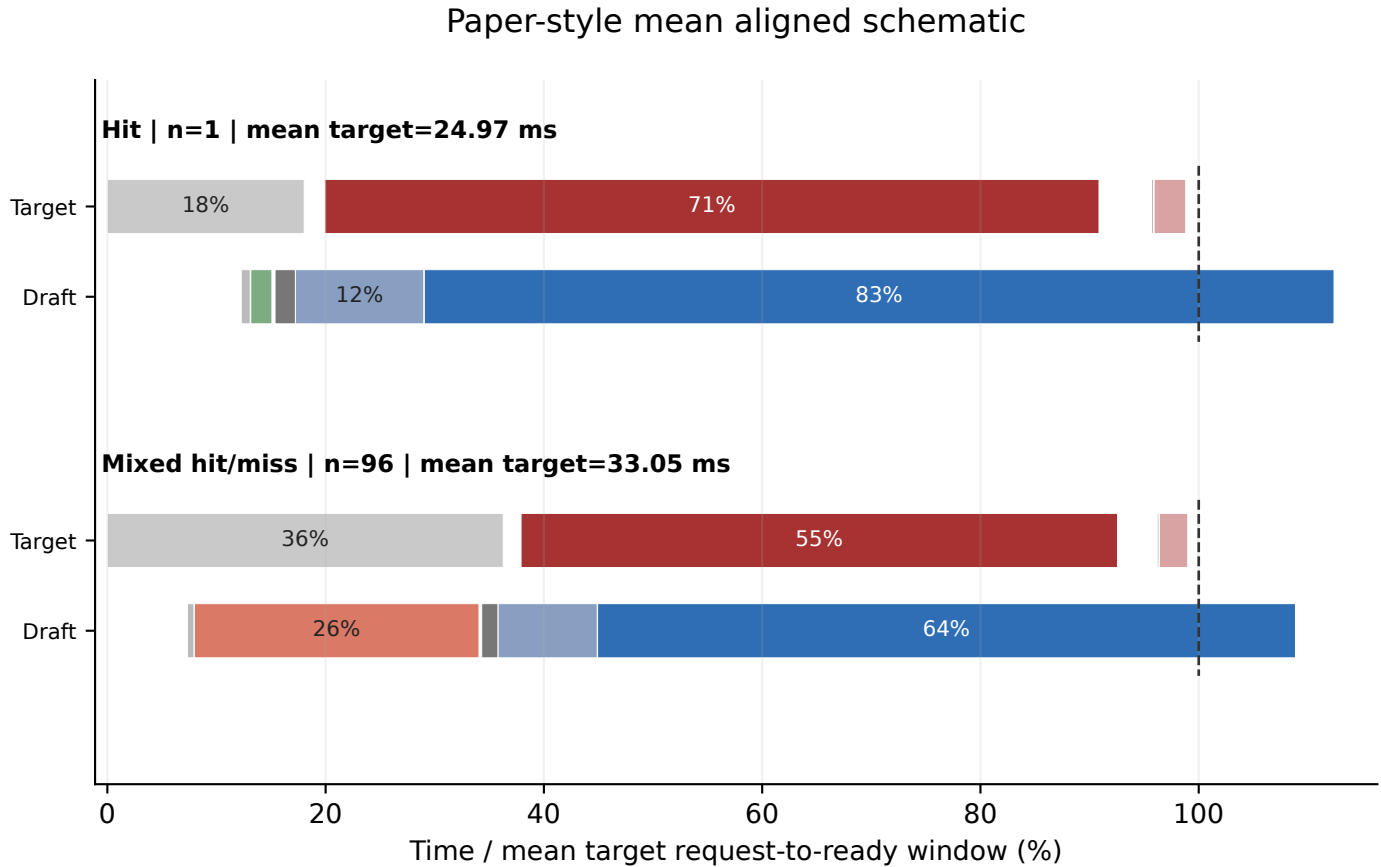
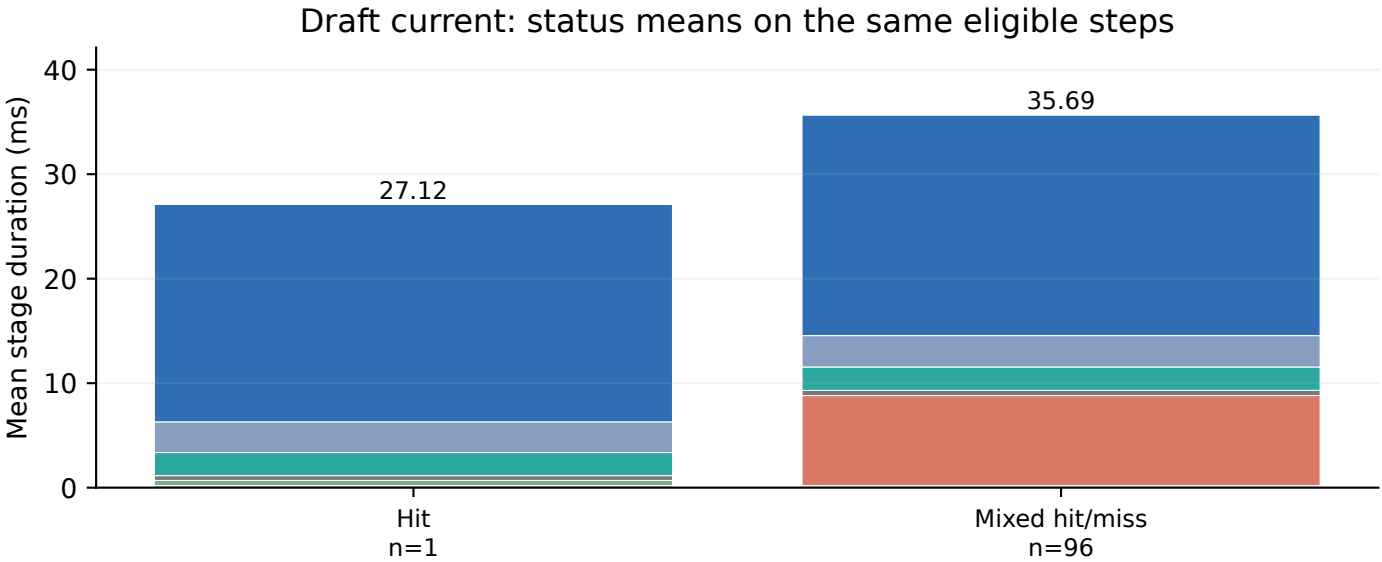
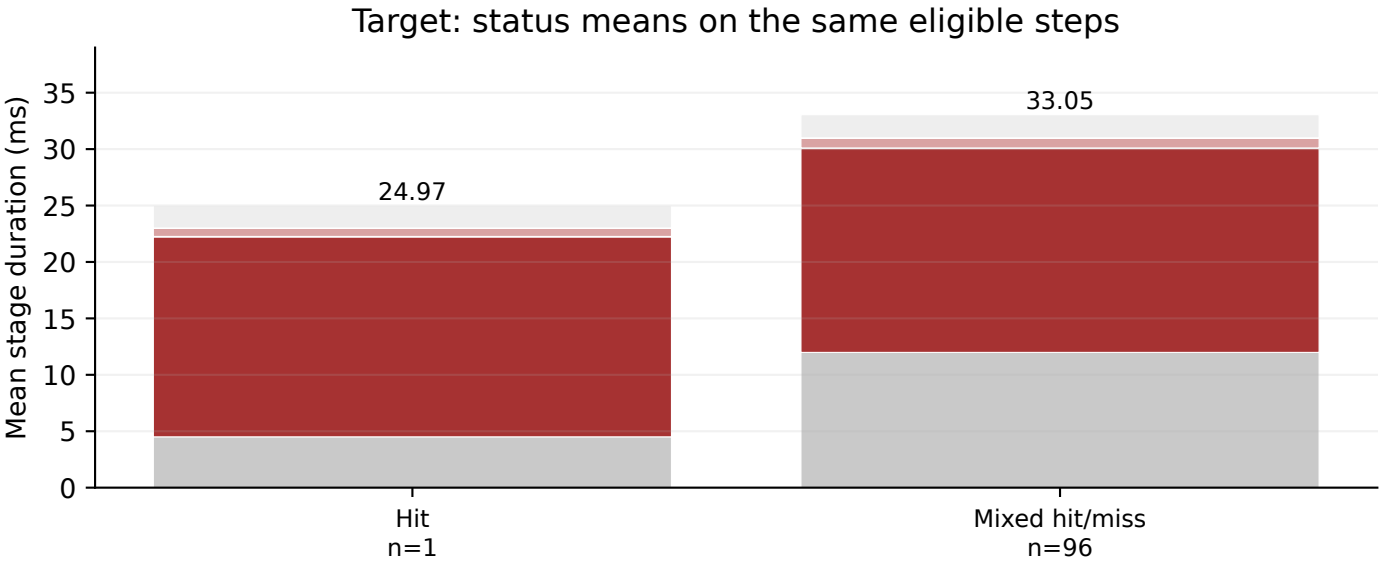
Wait / sync Accept prepare Miss response Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



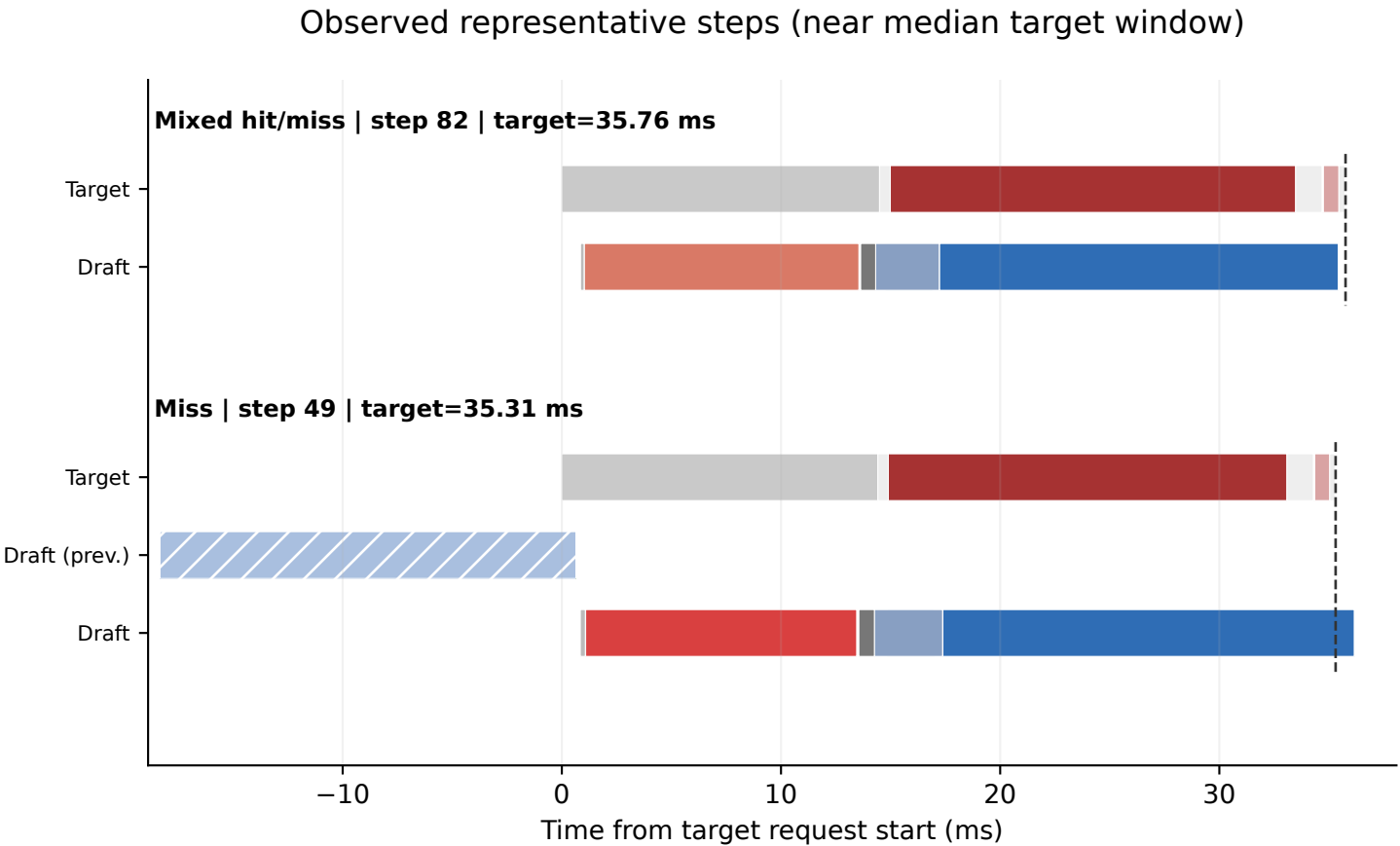
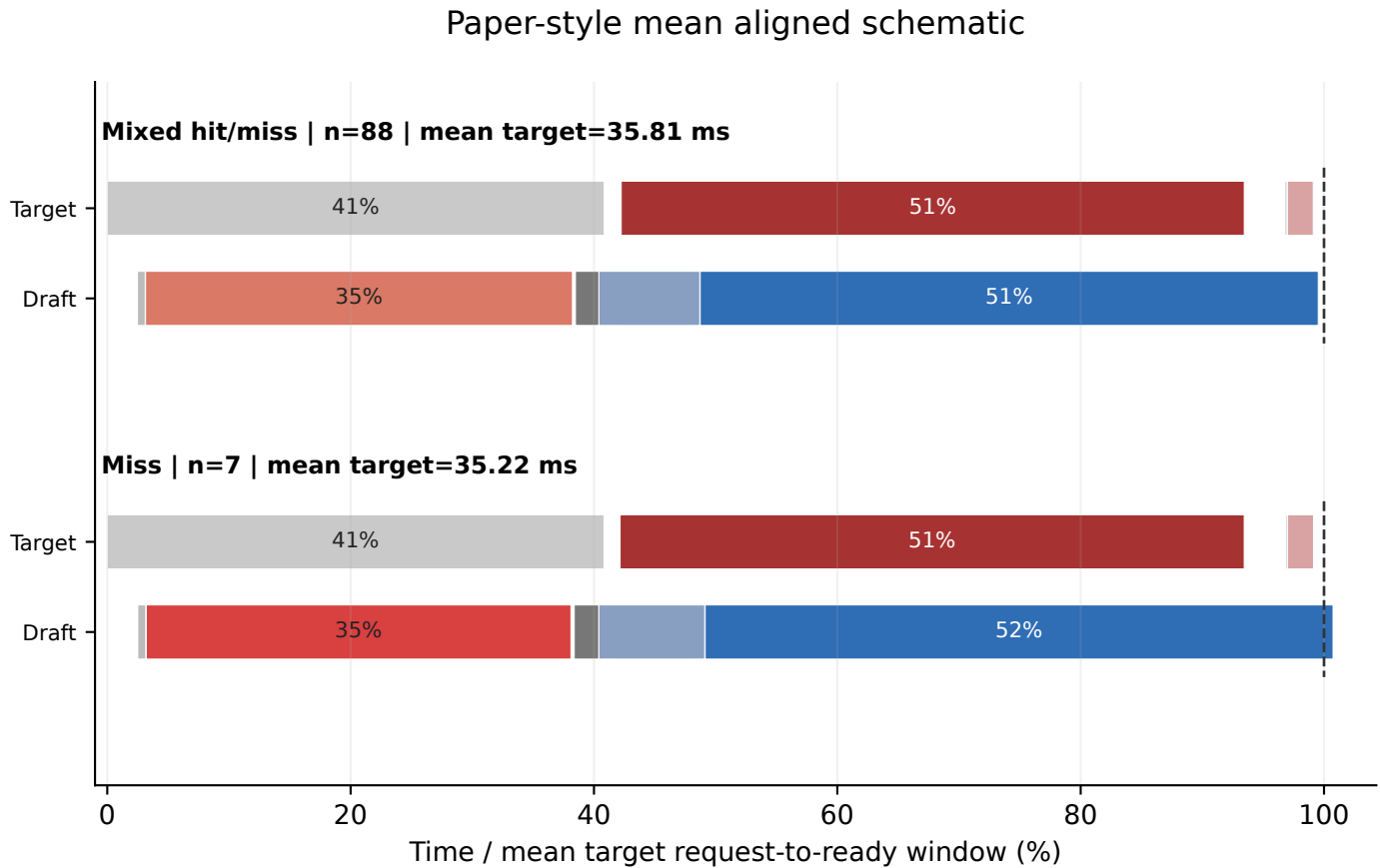
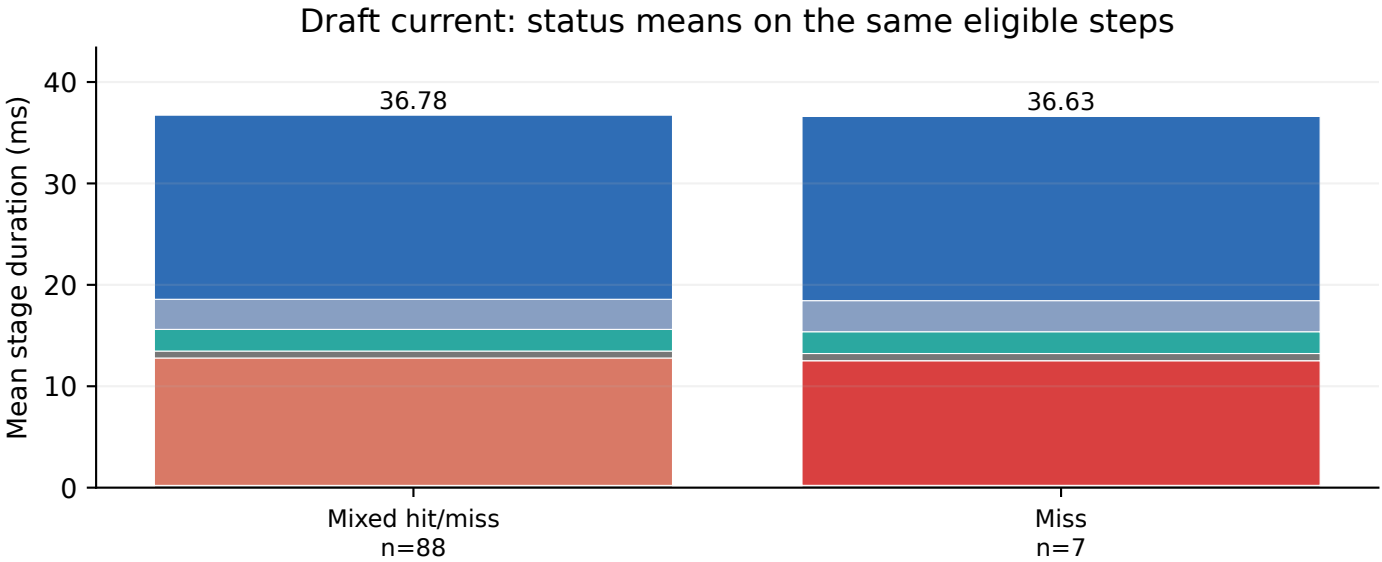
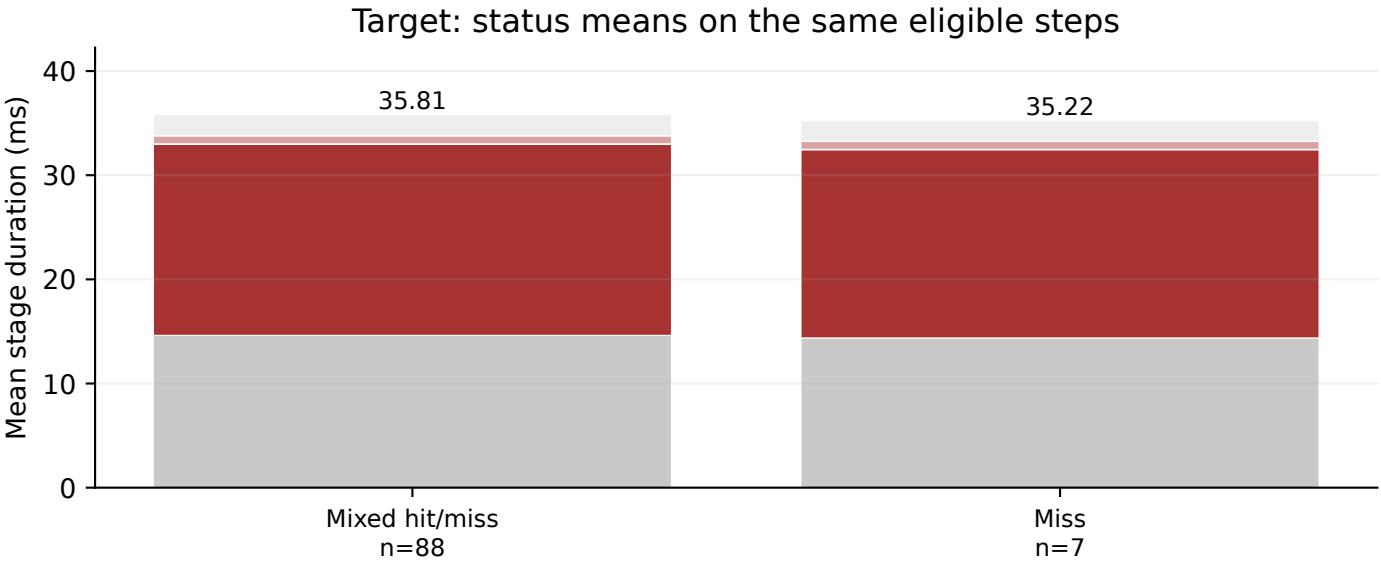
Wait / sync Accept prepare Unlabelled gap Context Send response SSD decode Target ready
Target verify Postprocess Mixed response Request SSD build Cache insert Proxy received
Accept

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



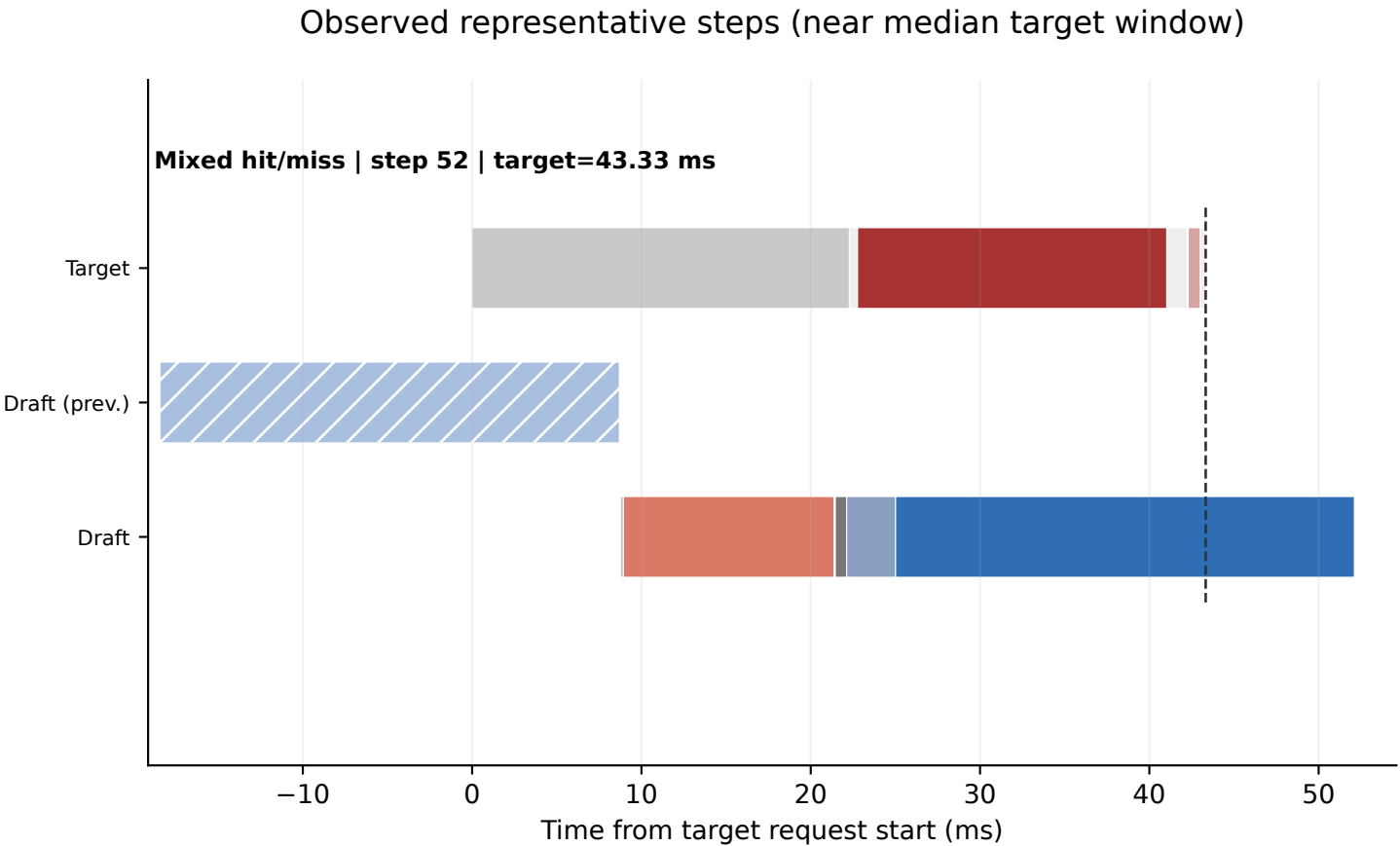
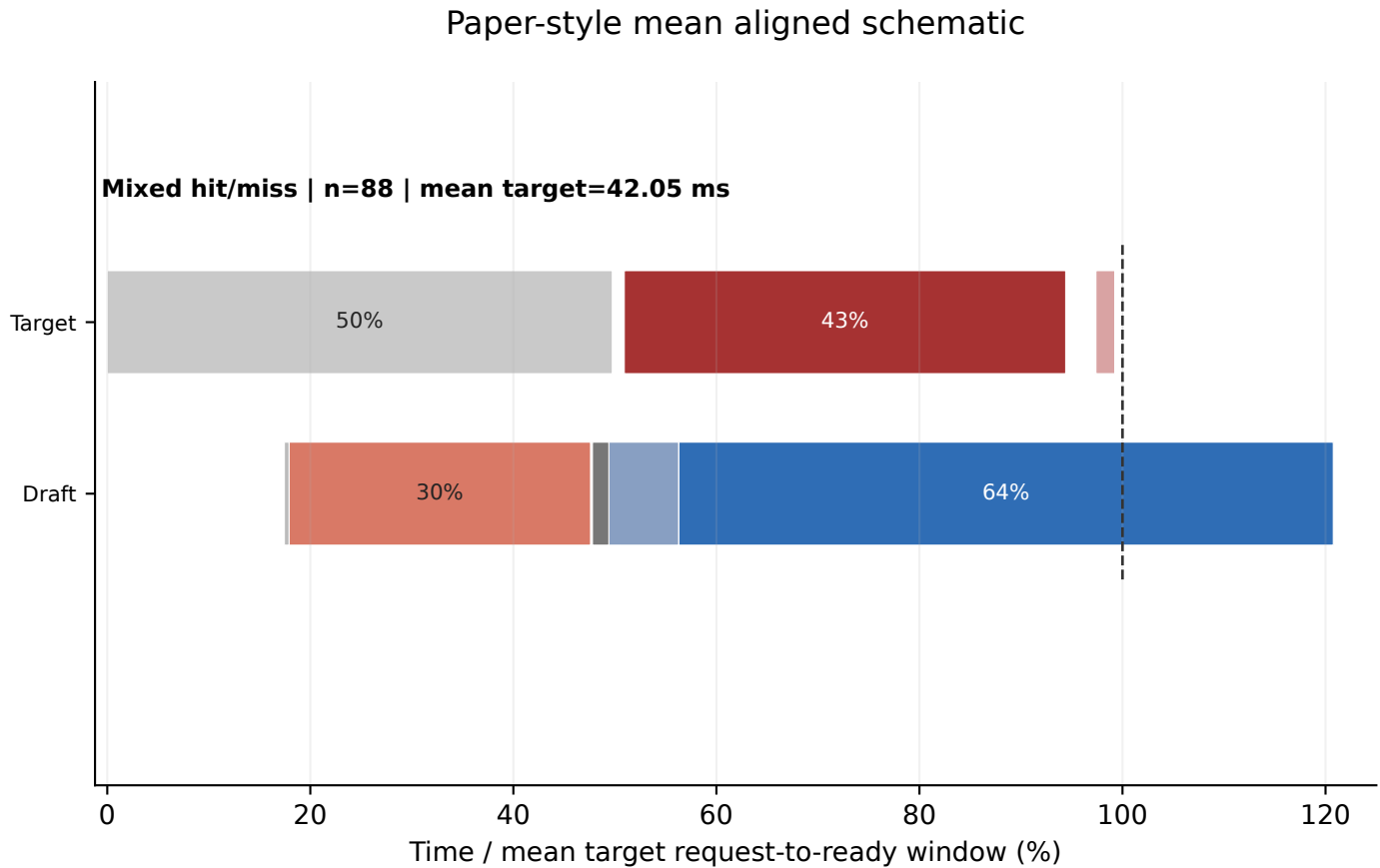
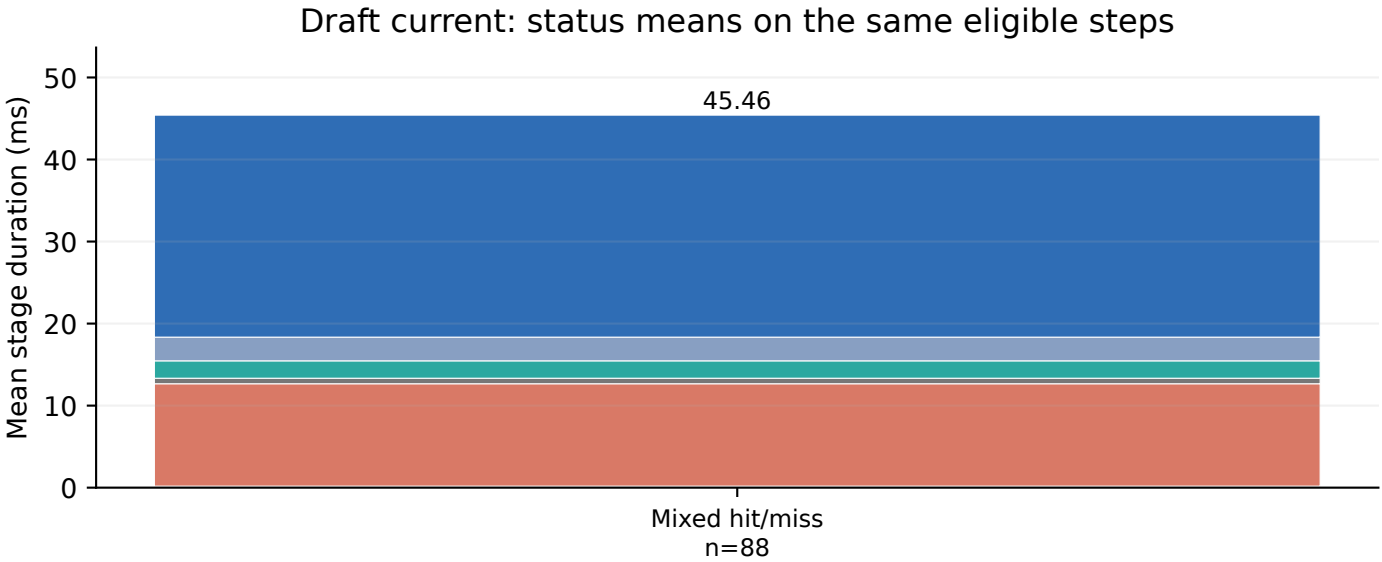
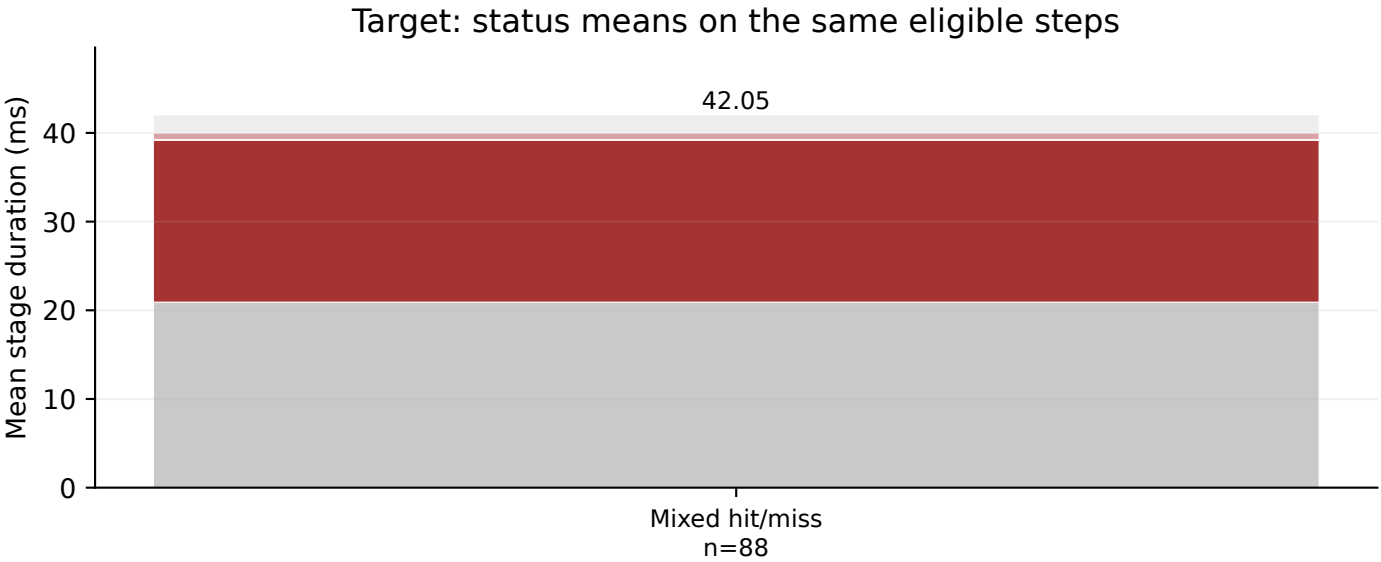
Wait / sync Accept prepare Mixed response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



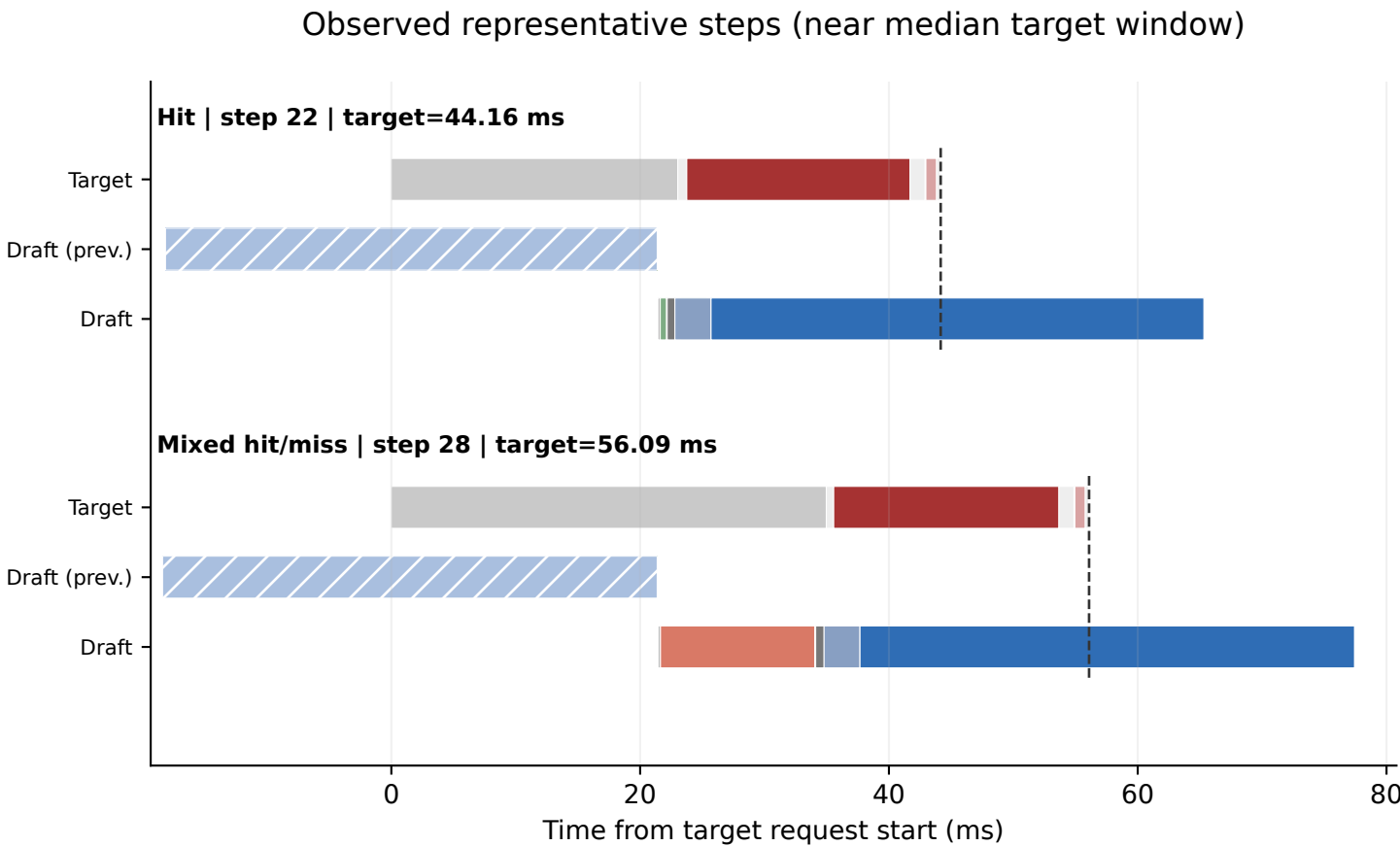
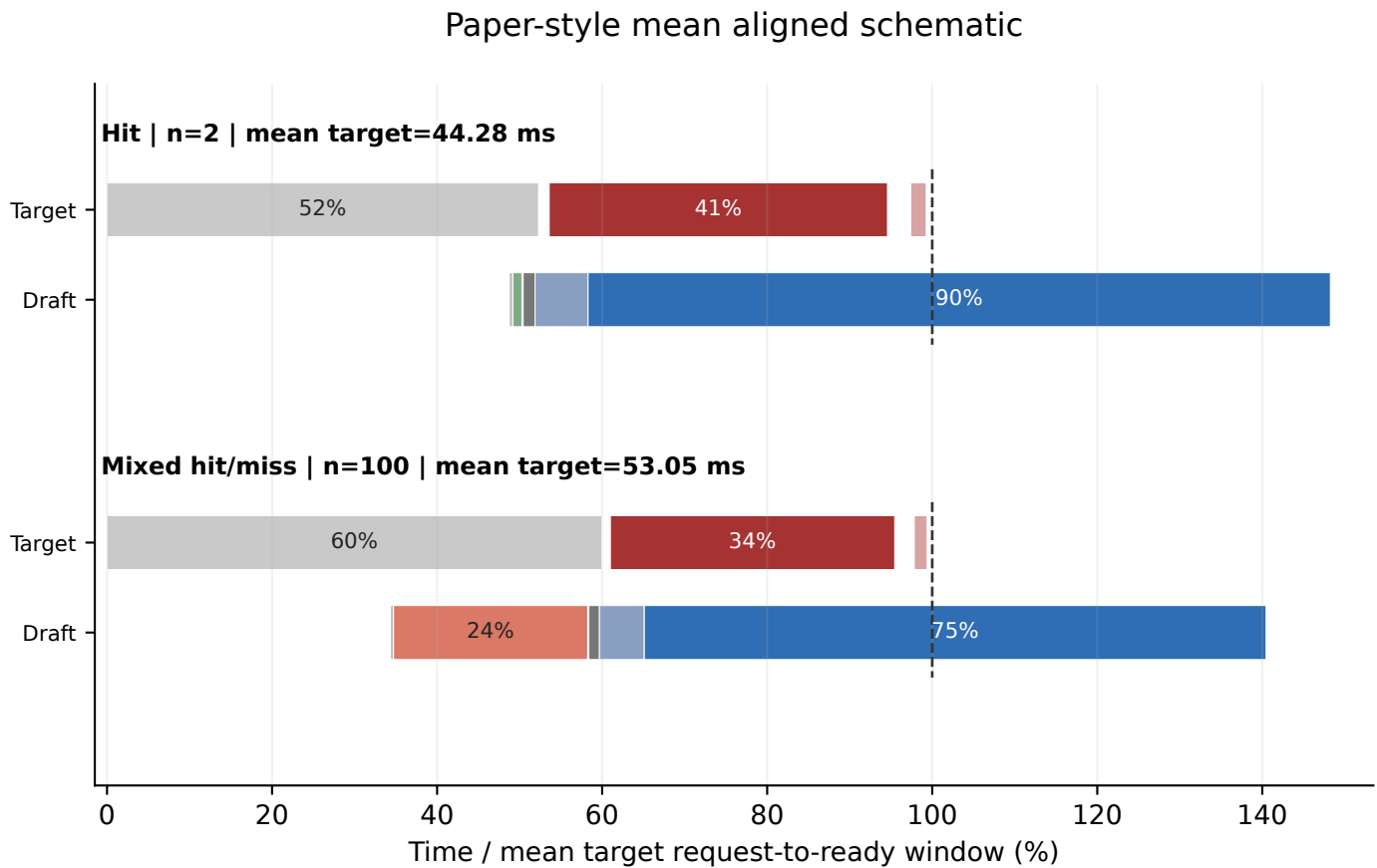
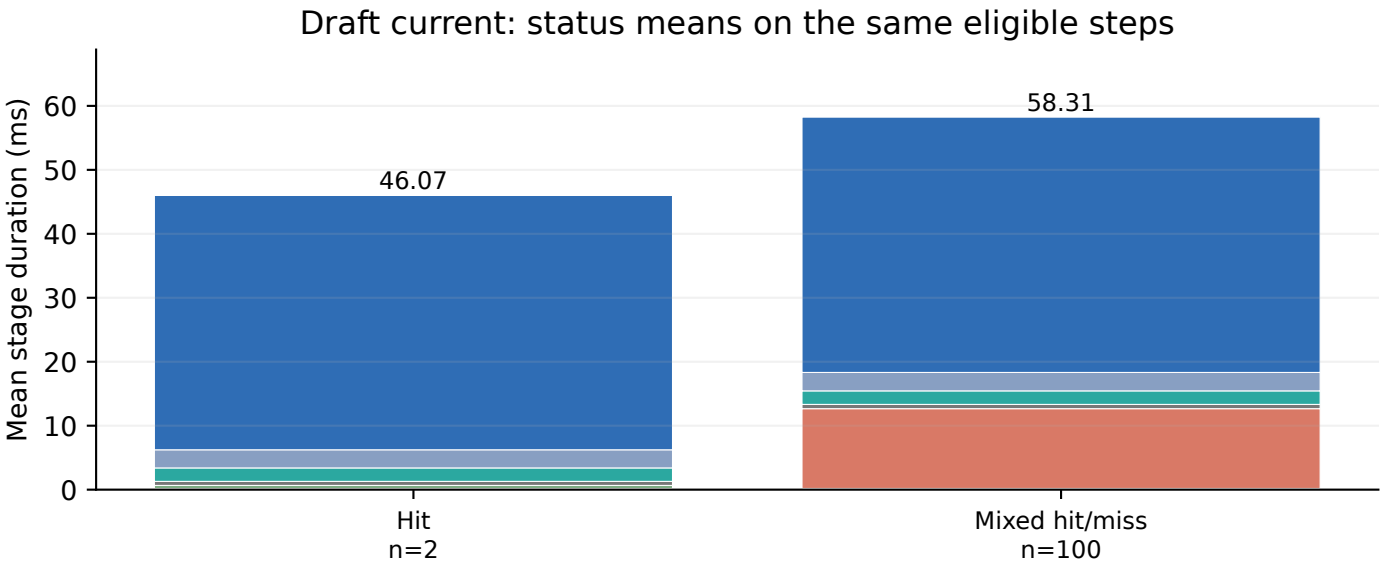
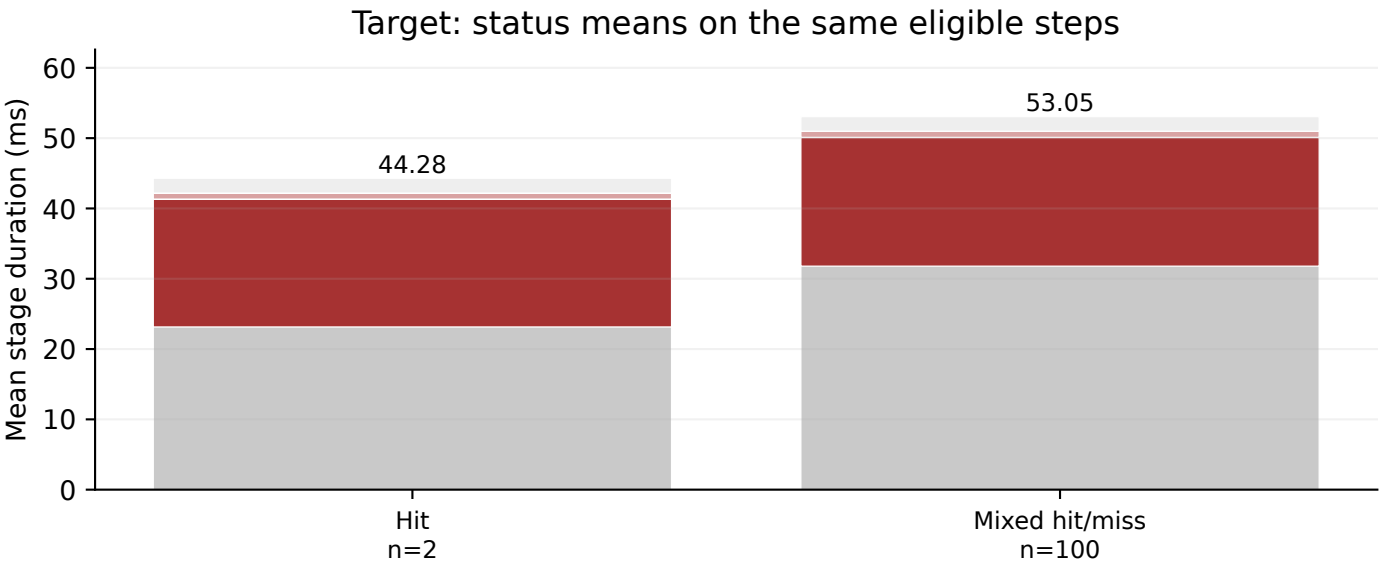
Wait / sync Accept prepare Miss response Request SSD decode Previous SSD decode Target ready
Target verify Postprocess Mixed response Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Context SSD build

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



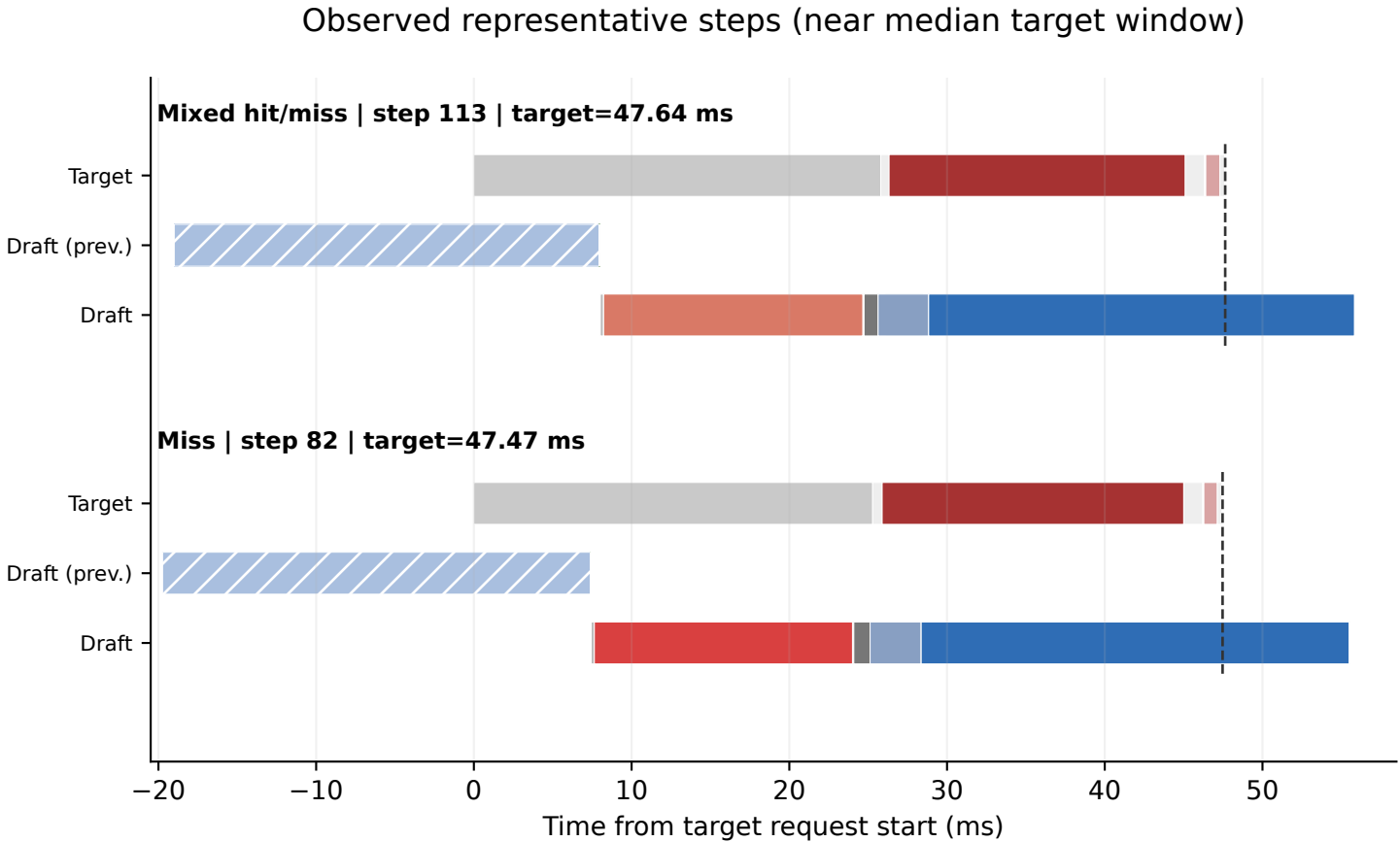
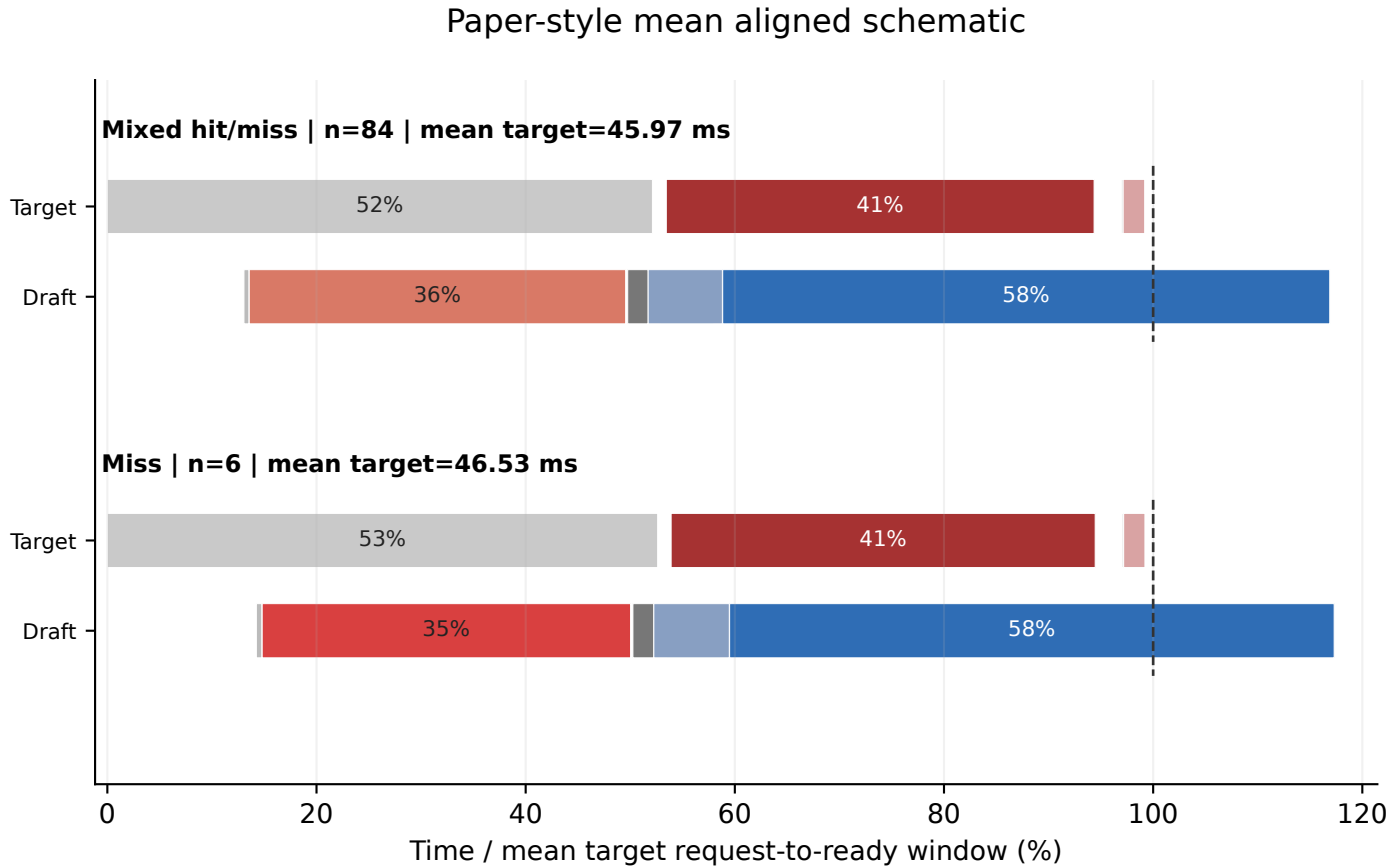
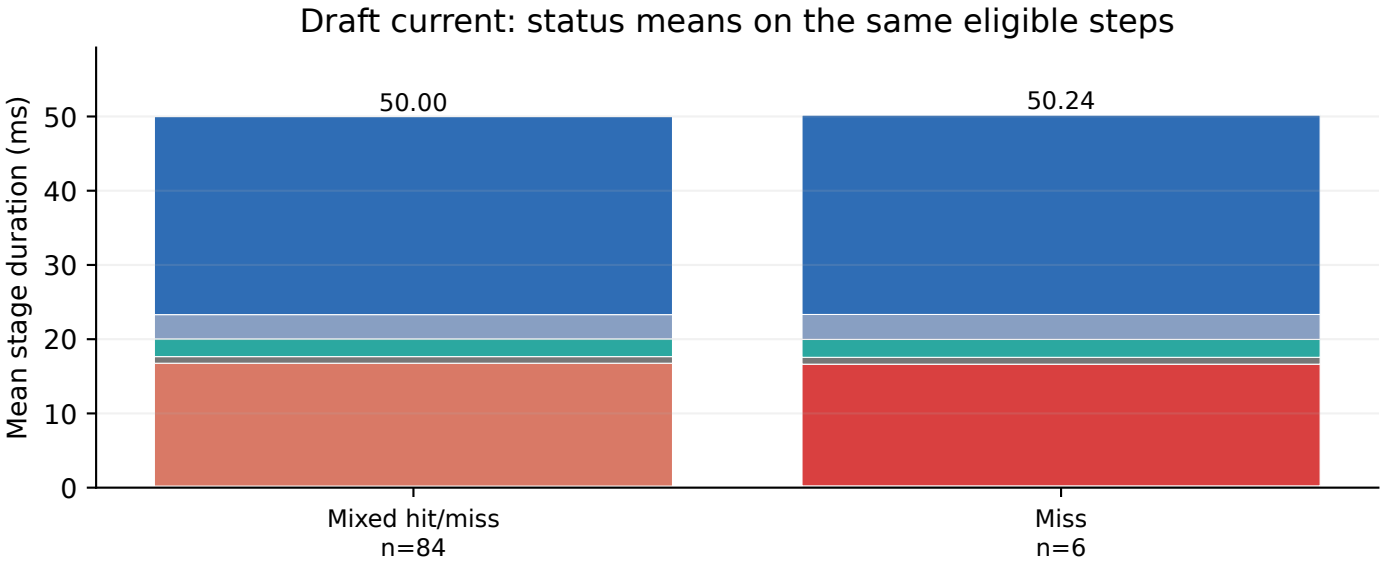
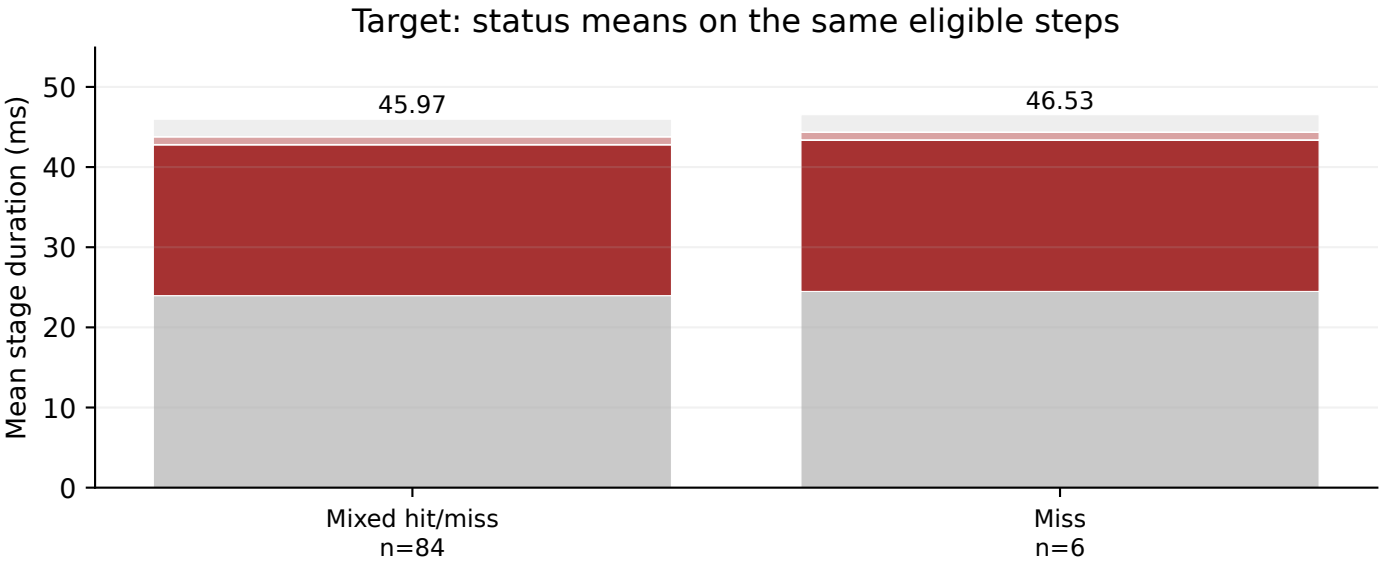
Wait / sync Accept prepare Mixed response Send response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context SSD build Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



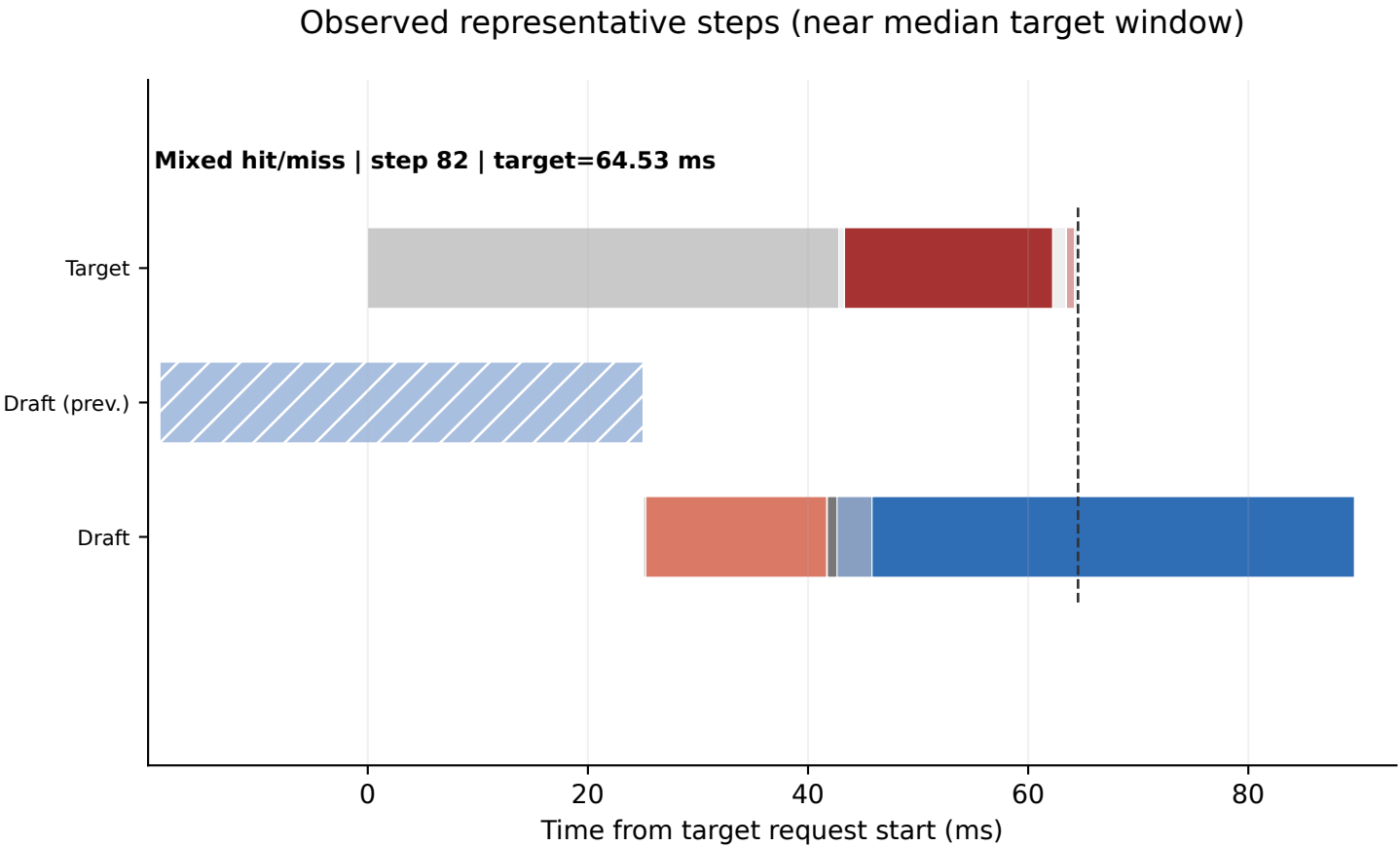
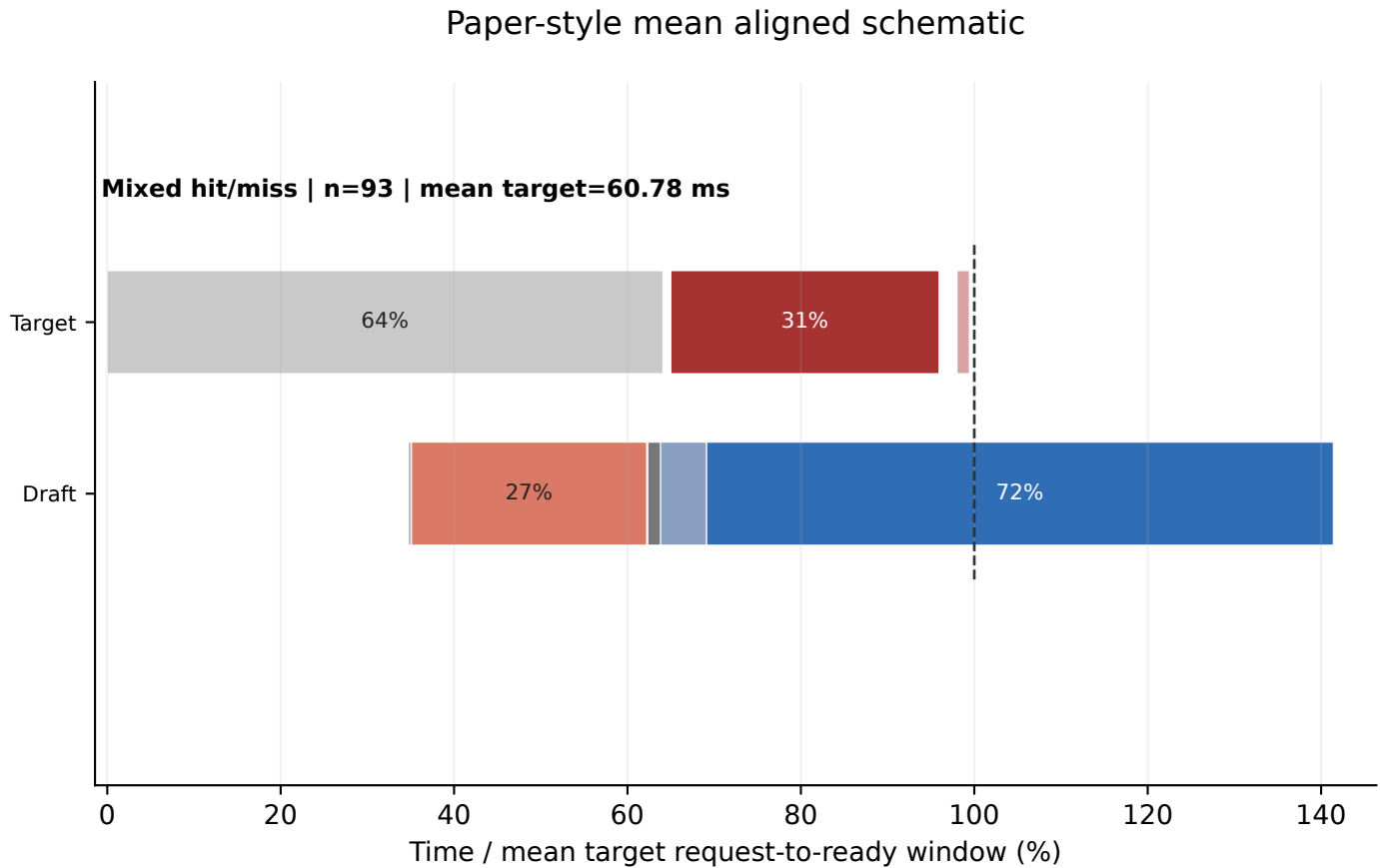
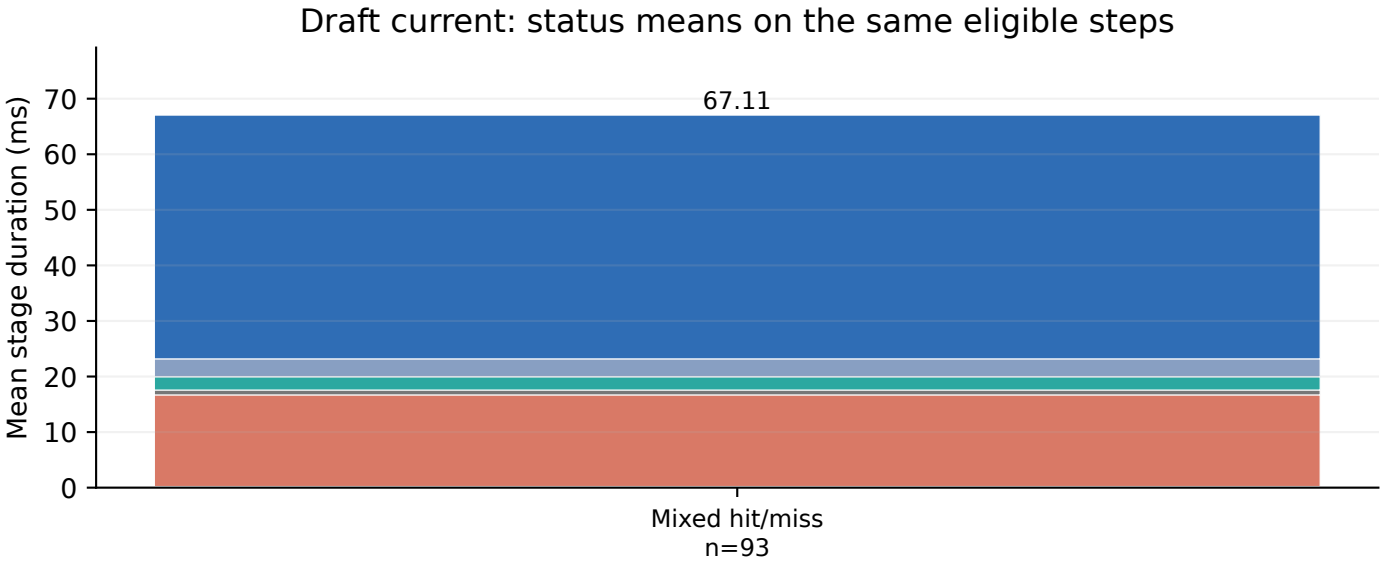
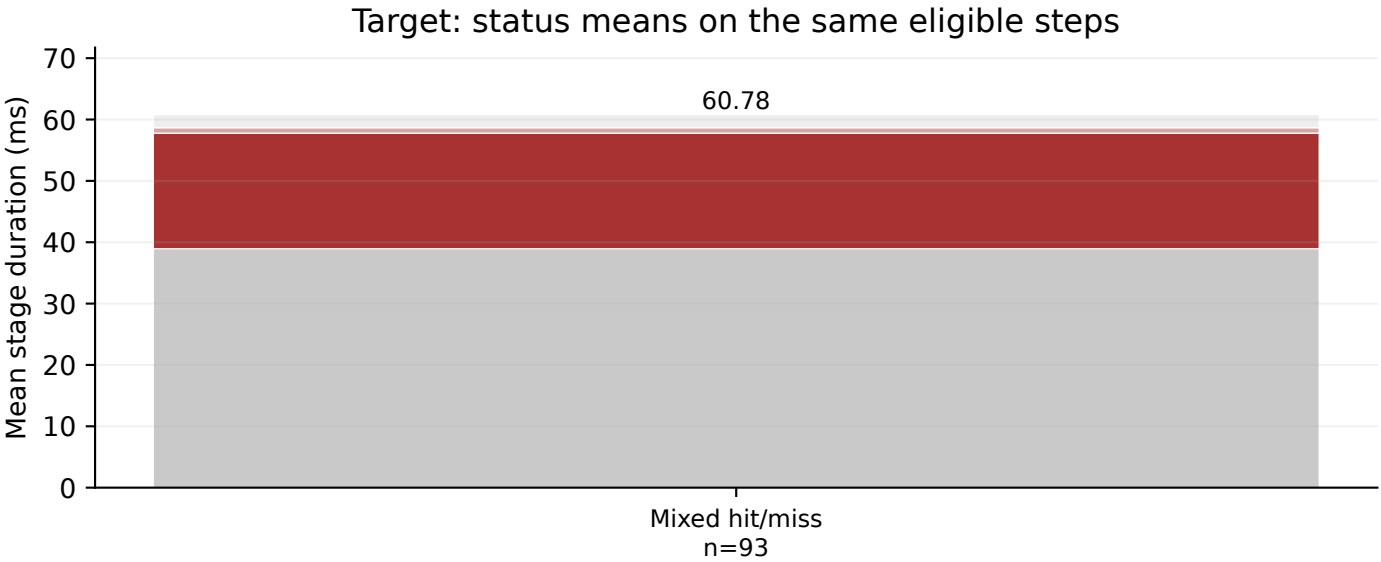
Wait / sync Accept prepare Mixed response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



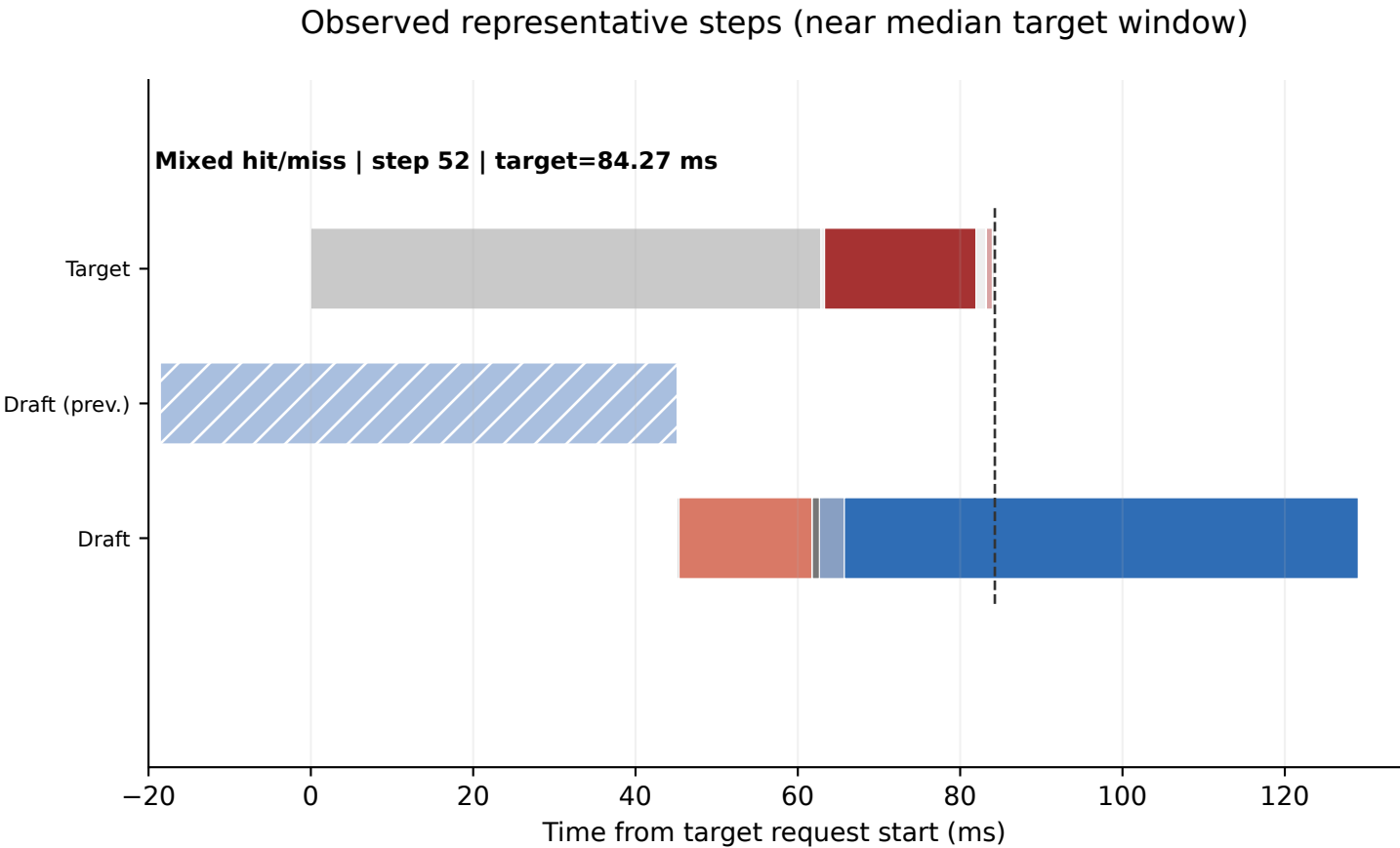
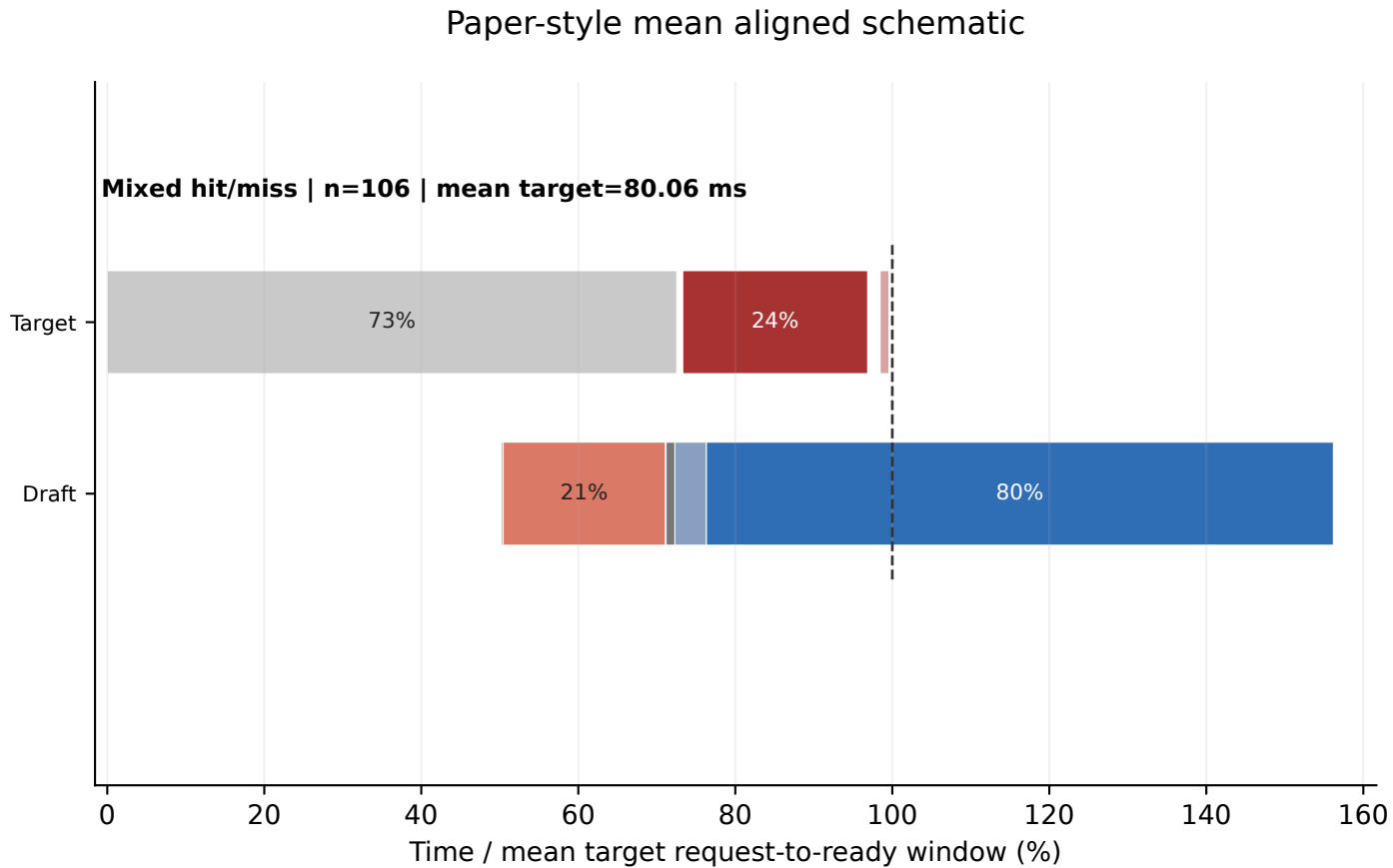
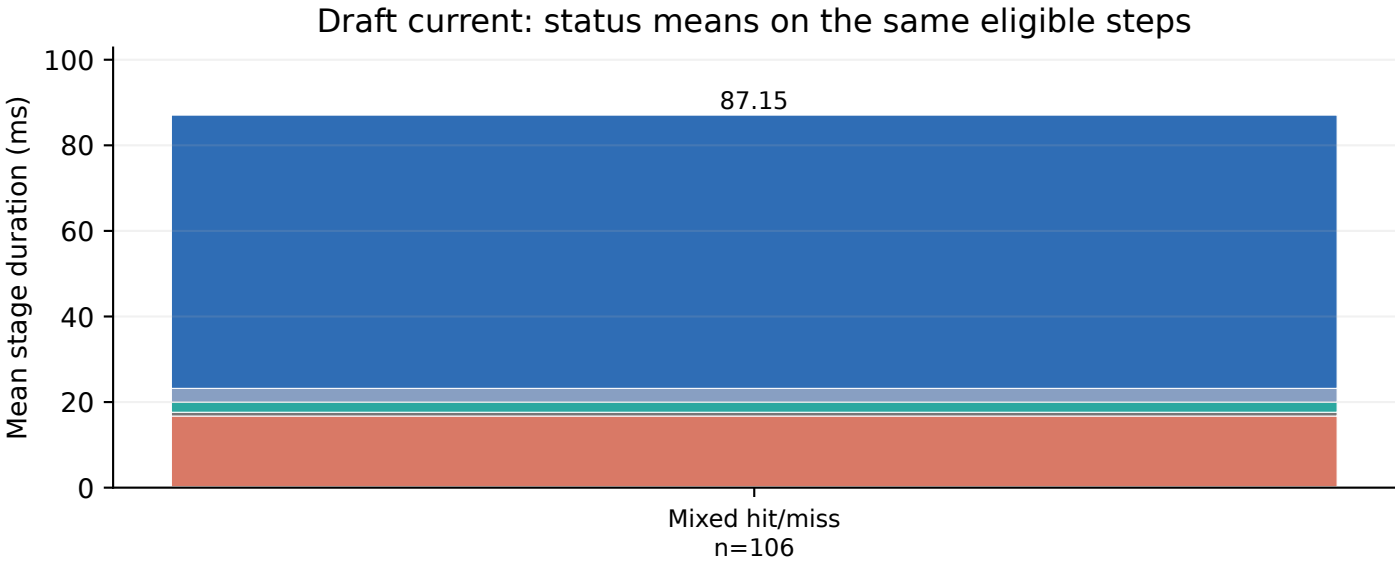
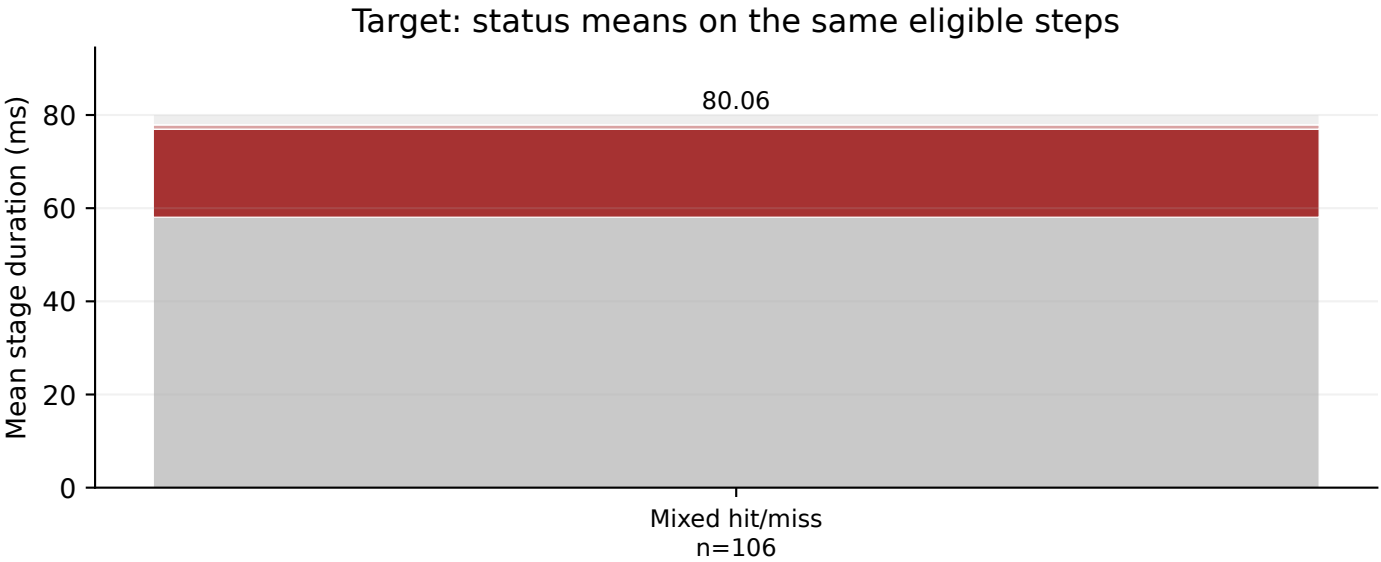
Wait / sync Accept prepare Miss response Request SSD decode Previous SSD decode Target ready
Target verify Postprocess Mixed response Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Context SSD build

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Mixed response Send response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context SSD build Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request

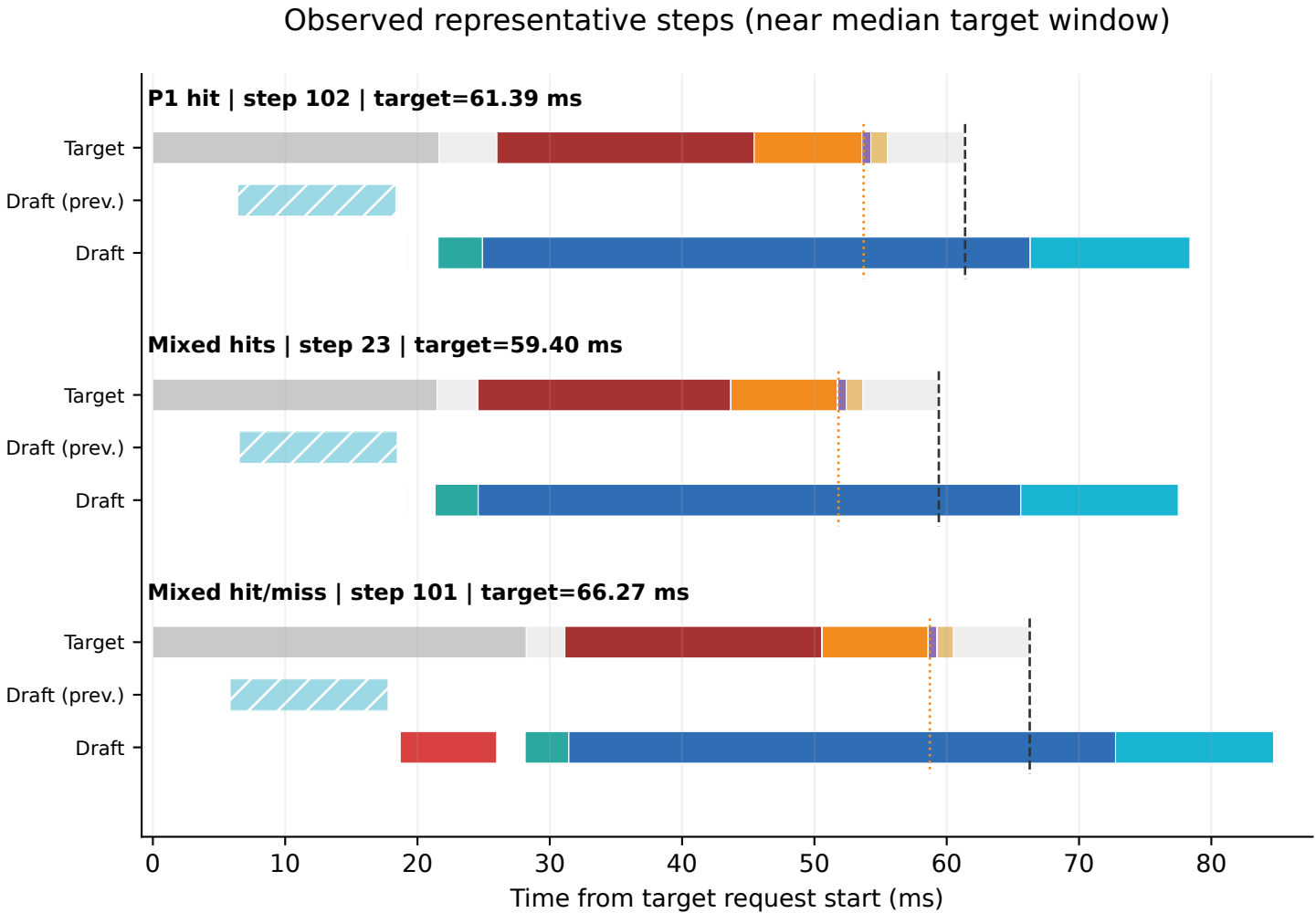
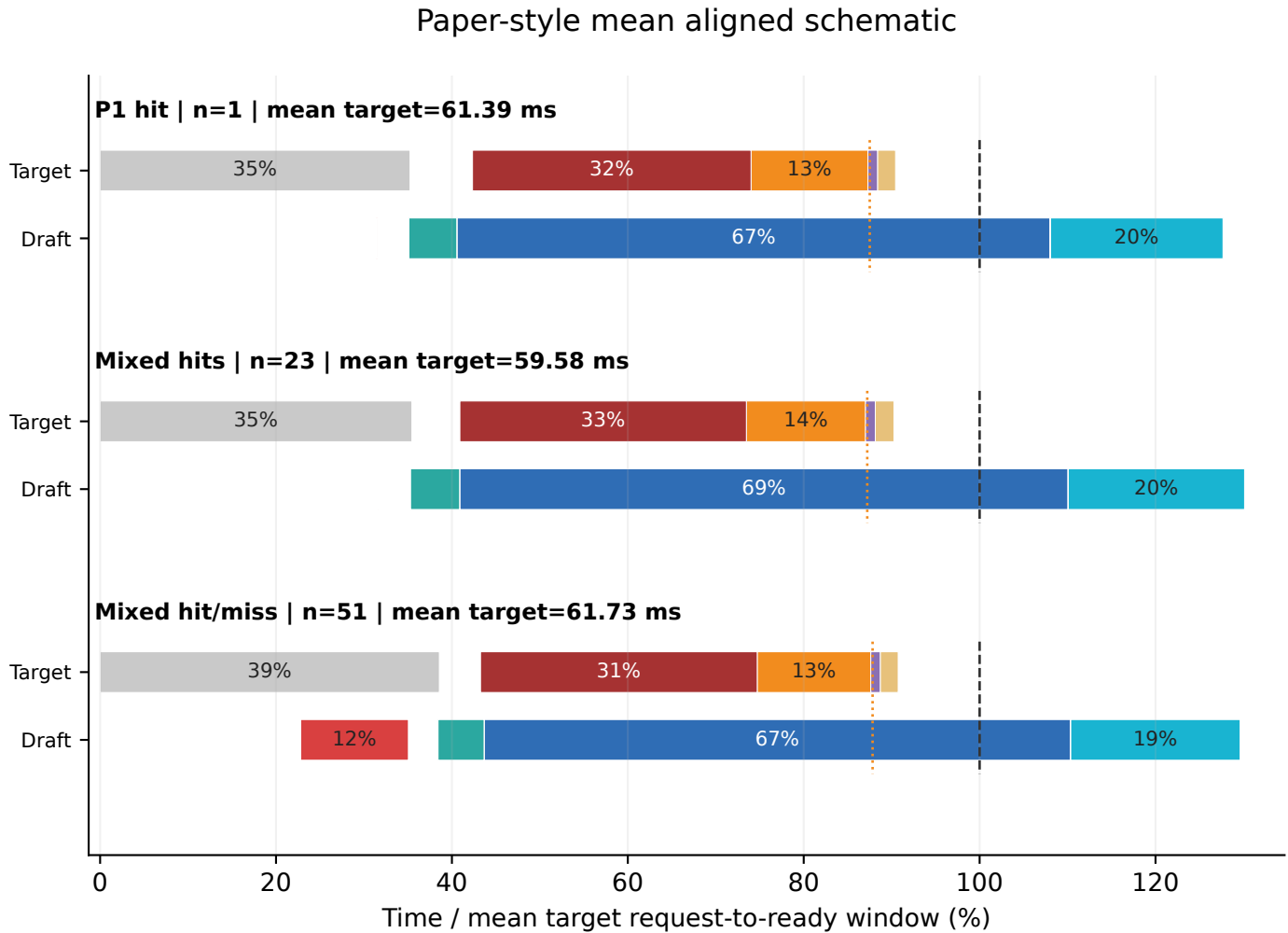
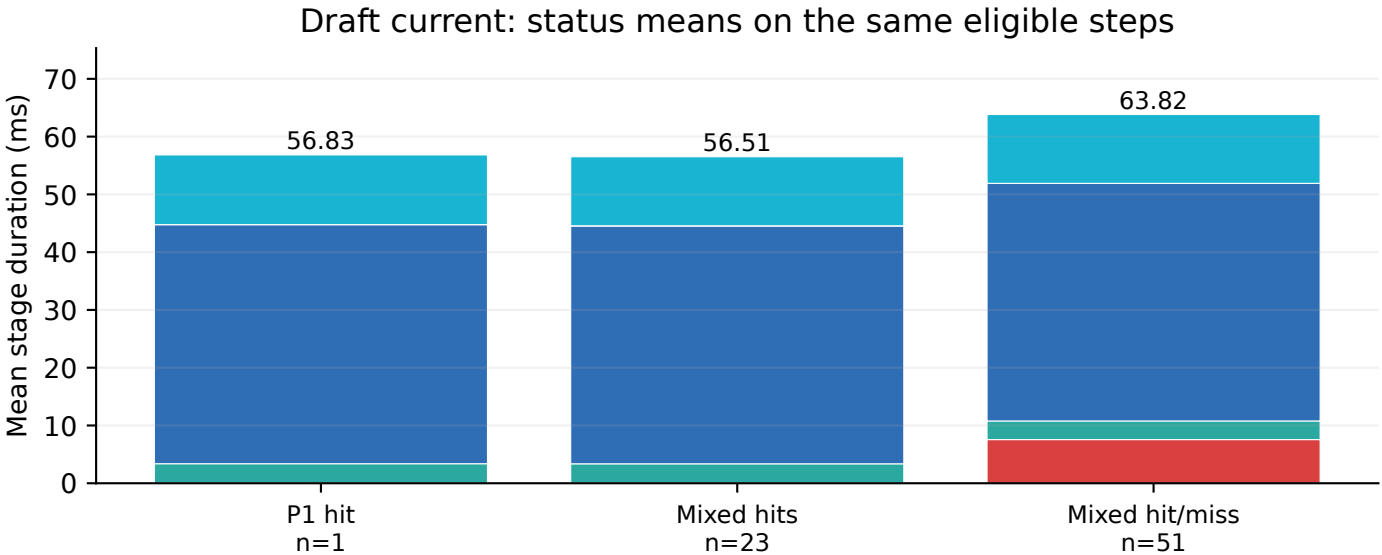
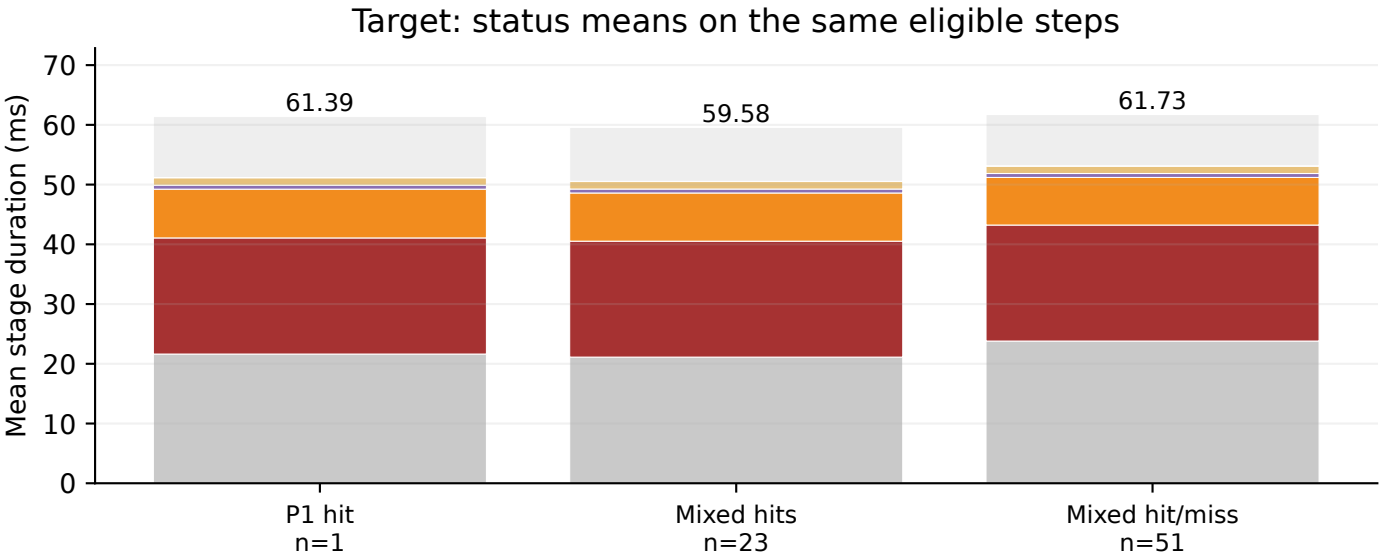
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Mixed response Send response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context SSD build Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

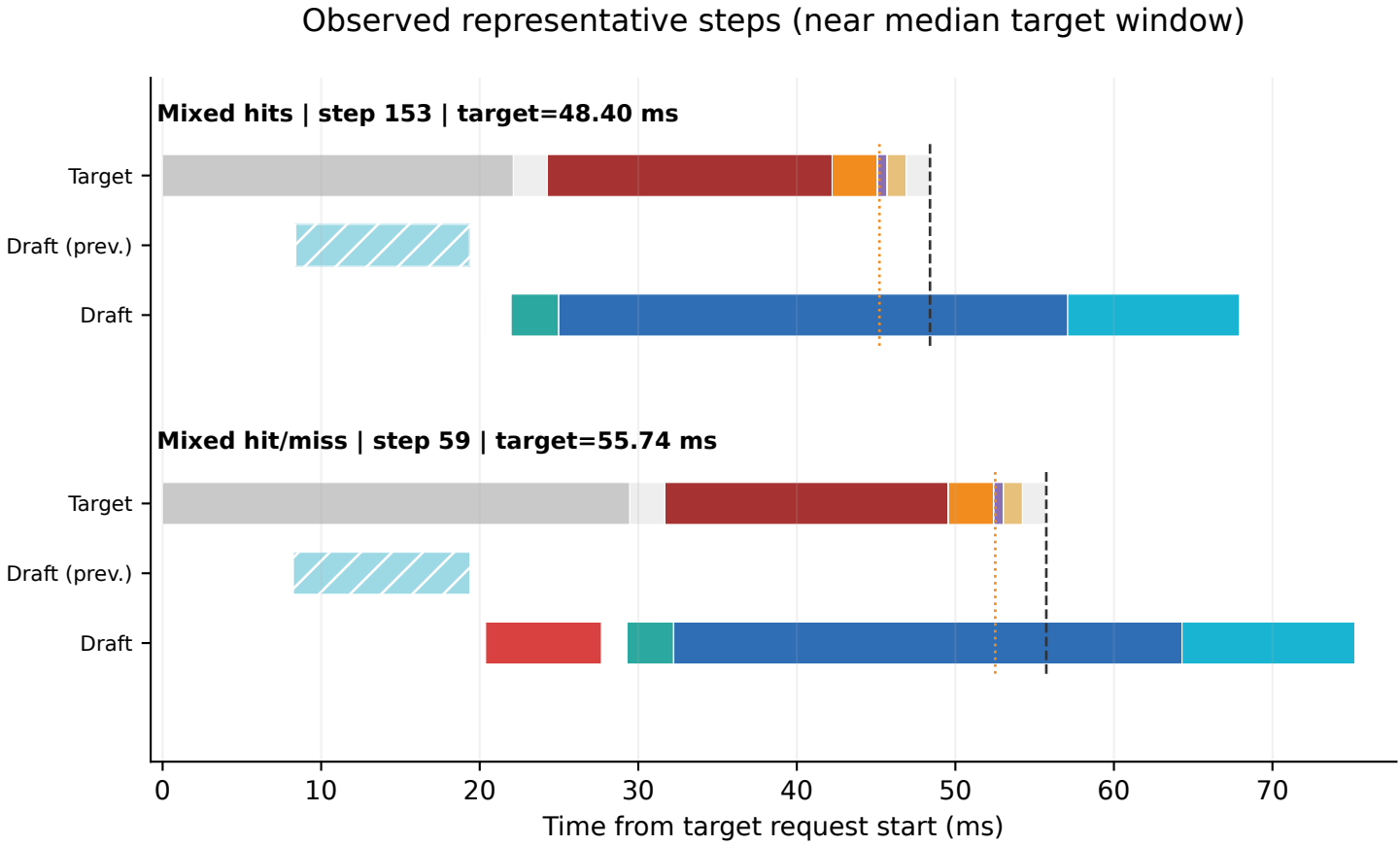
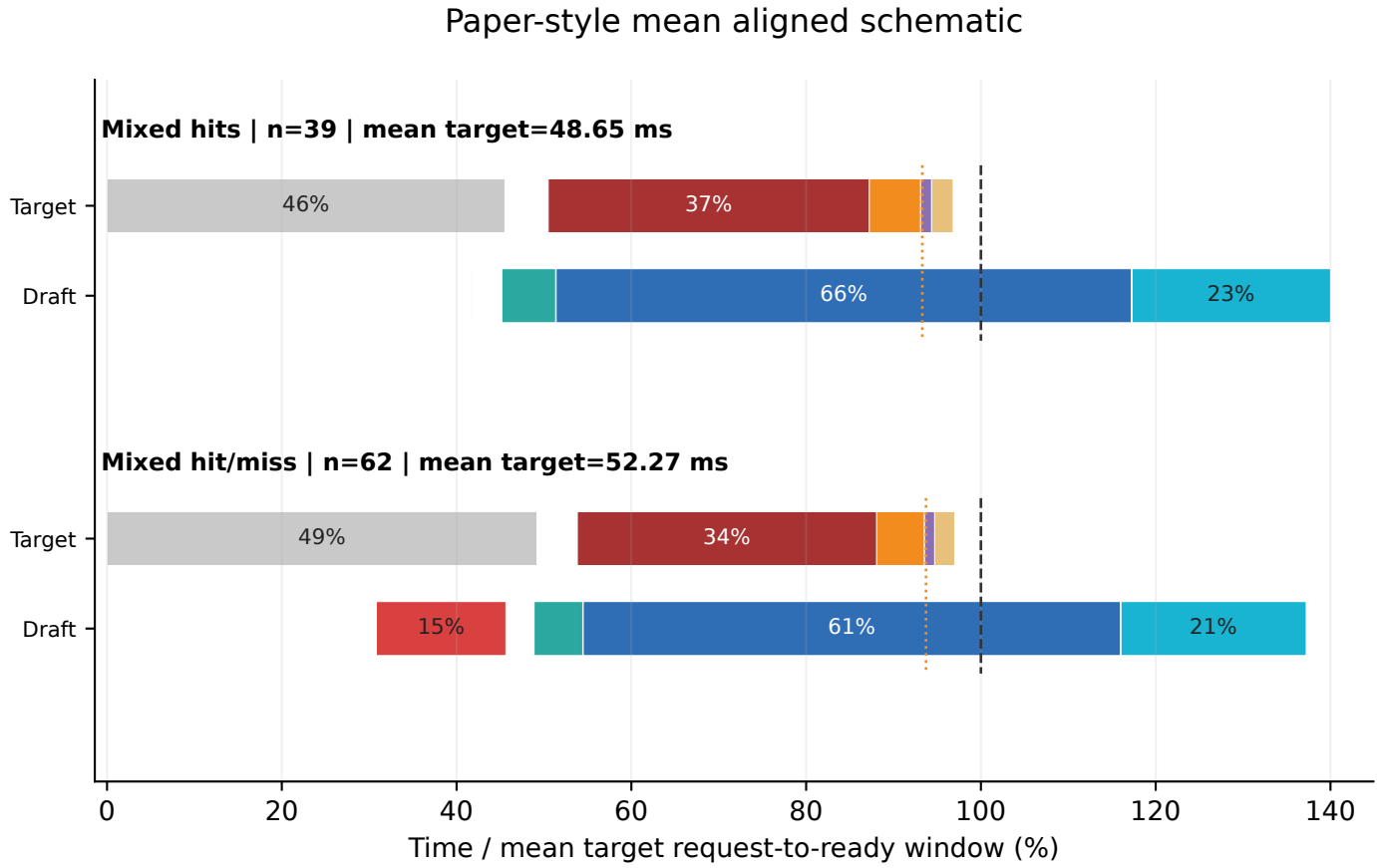
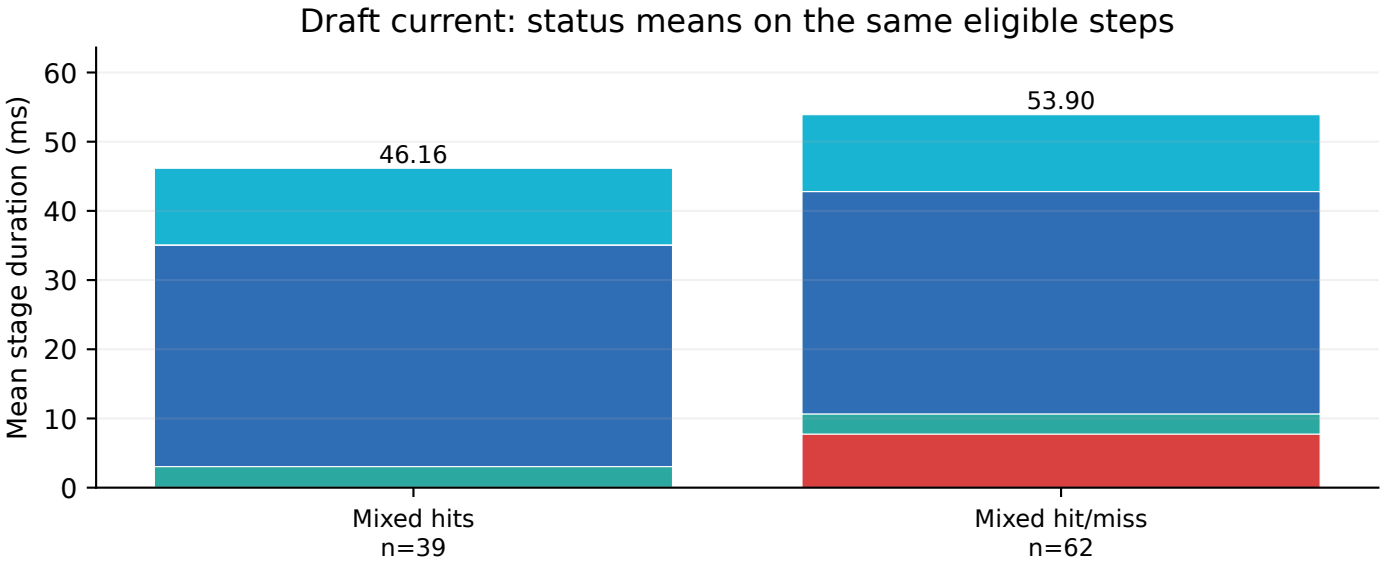
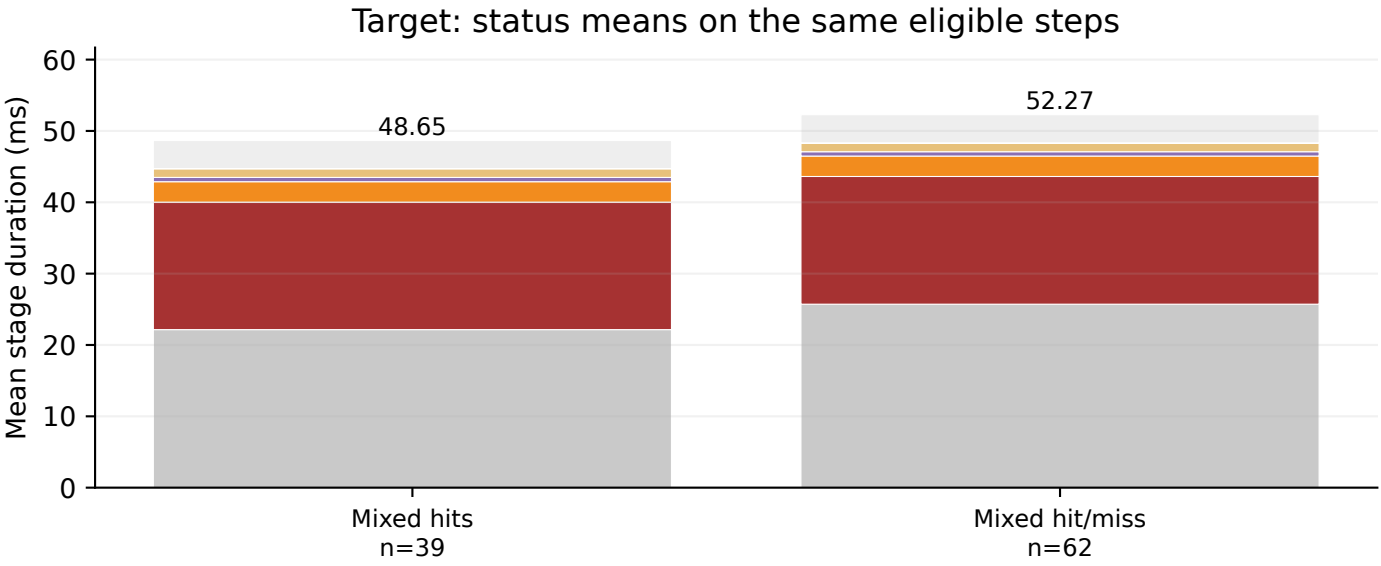
llama3_b8_tree_al_compute3
DUET: K1/K2=6/2, exit=30 (0-based), $\bar{C}$ =3, $\bar{N1}/\bar{N2}$ =12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 75 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

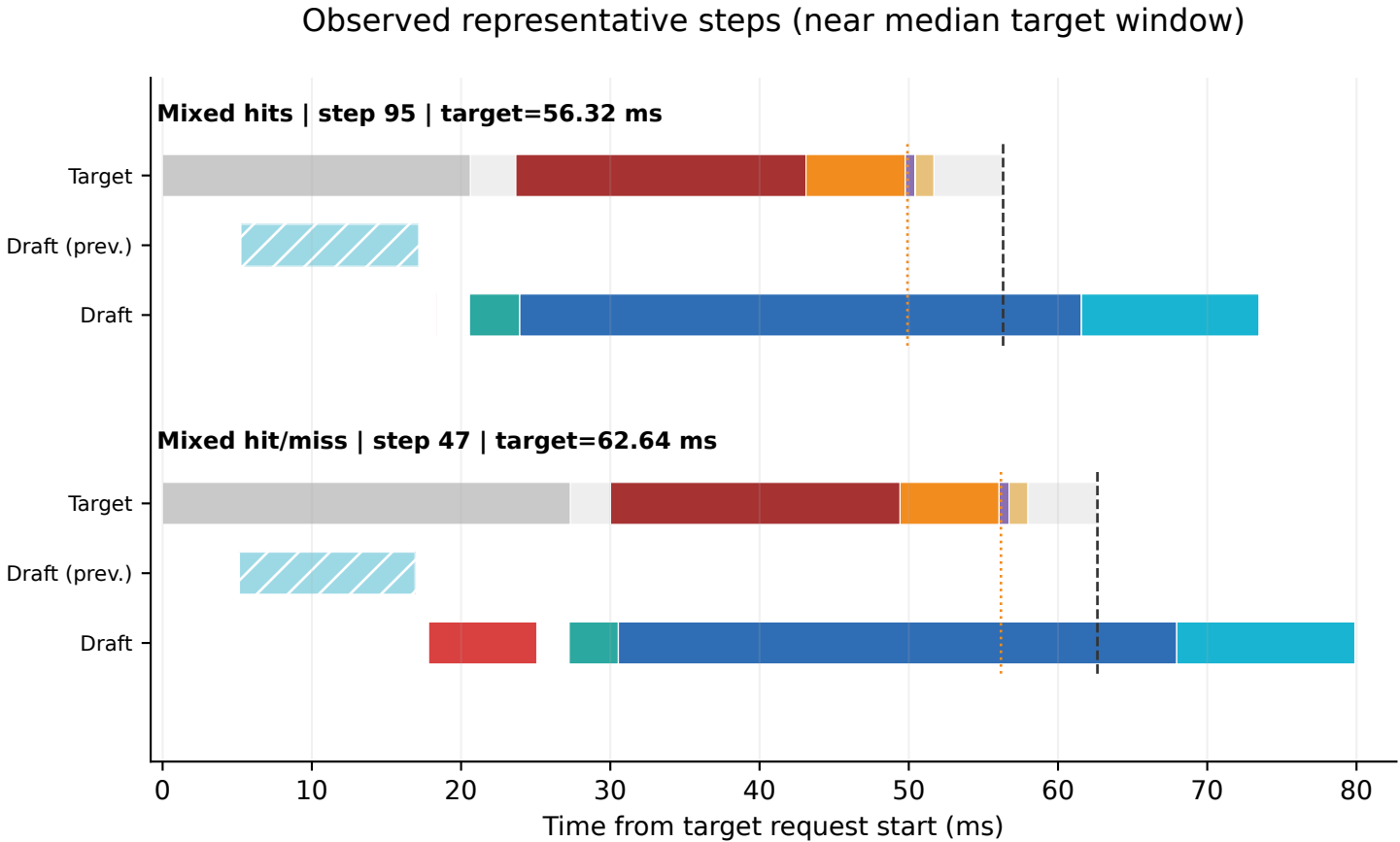
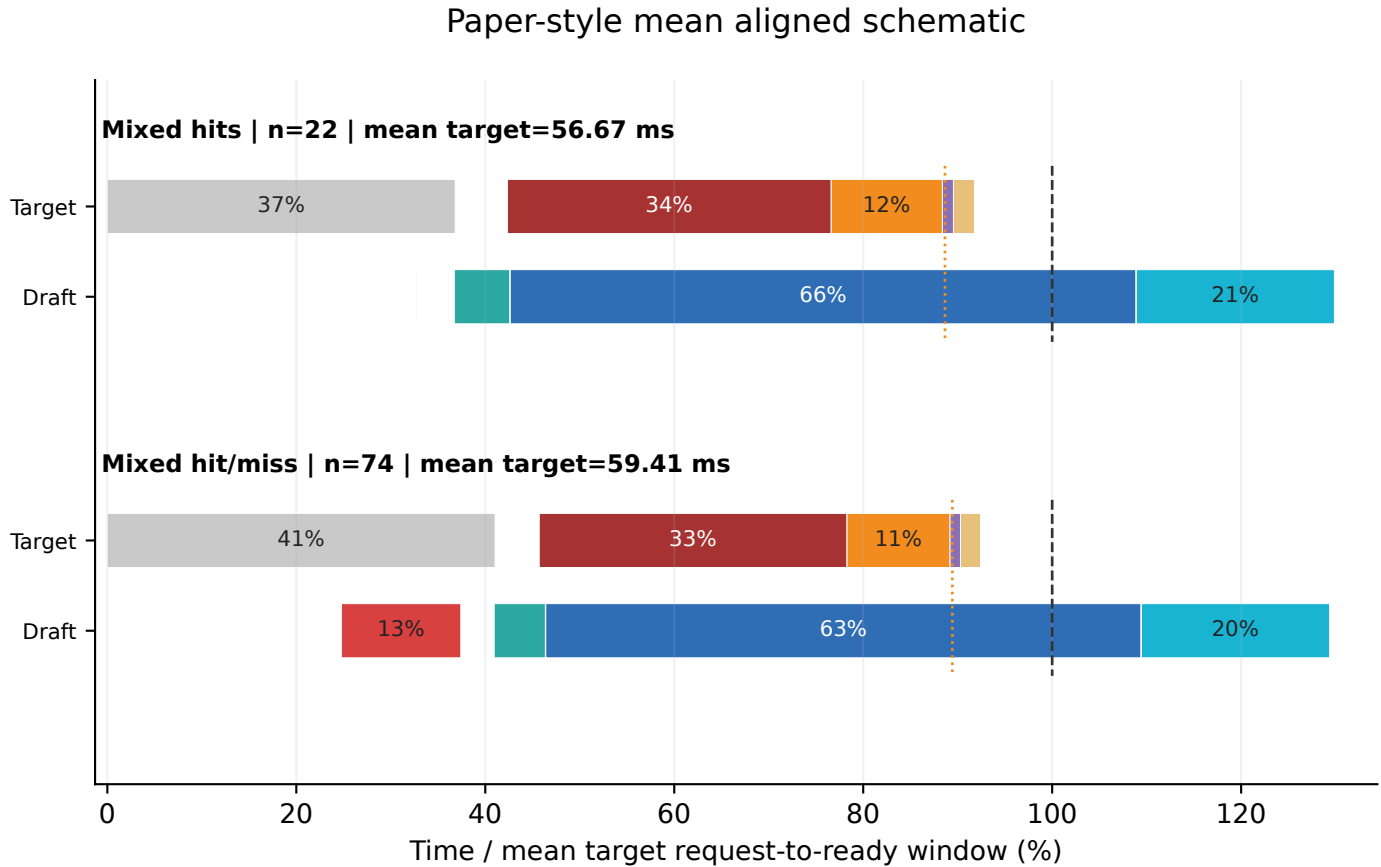
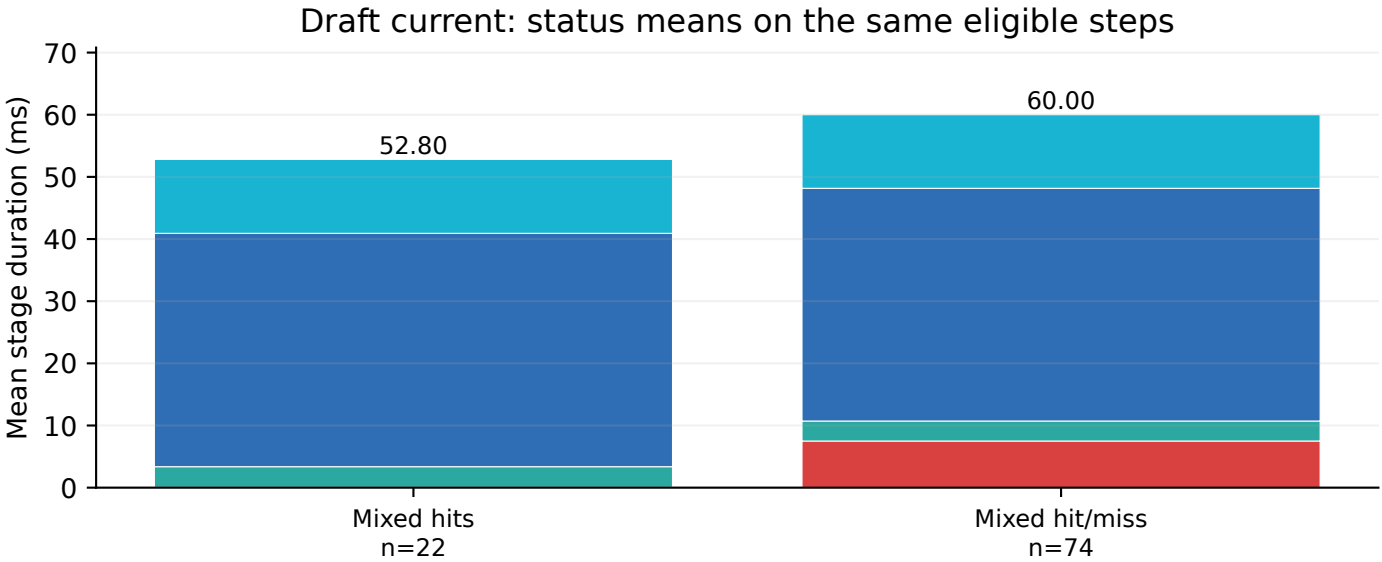
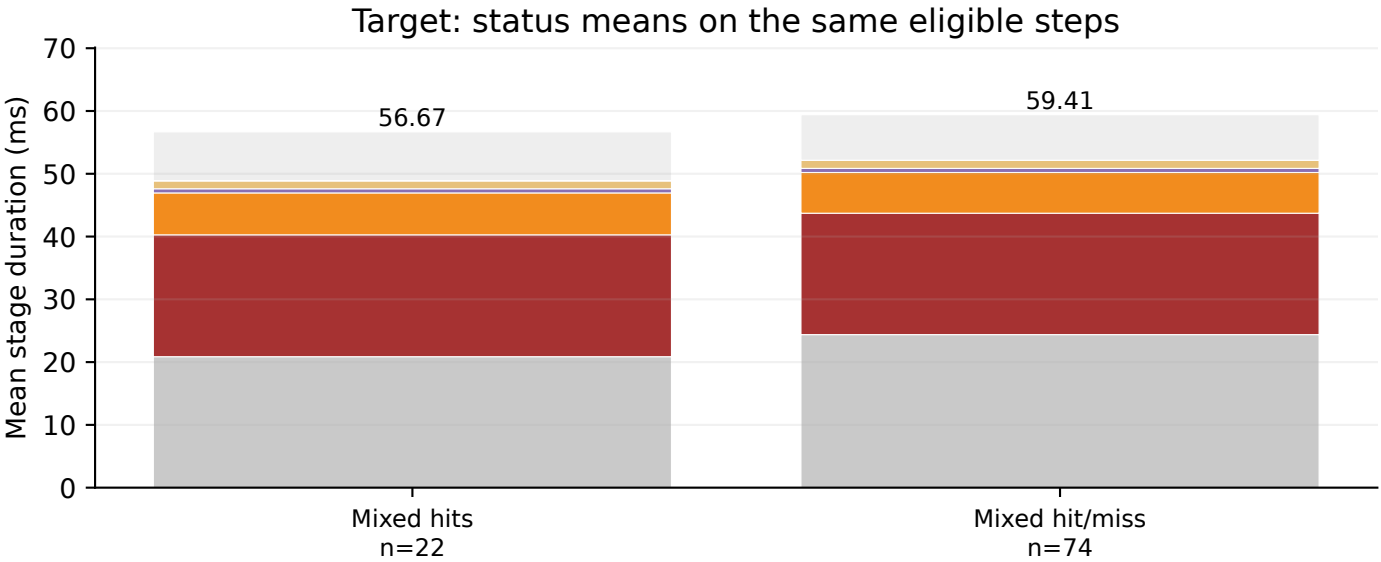
llama3_b8_tree_al_width1
DUET: K1/K2=6/2, exit=30 (0-based), C=1, N1/N2=6/2, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 101 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

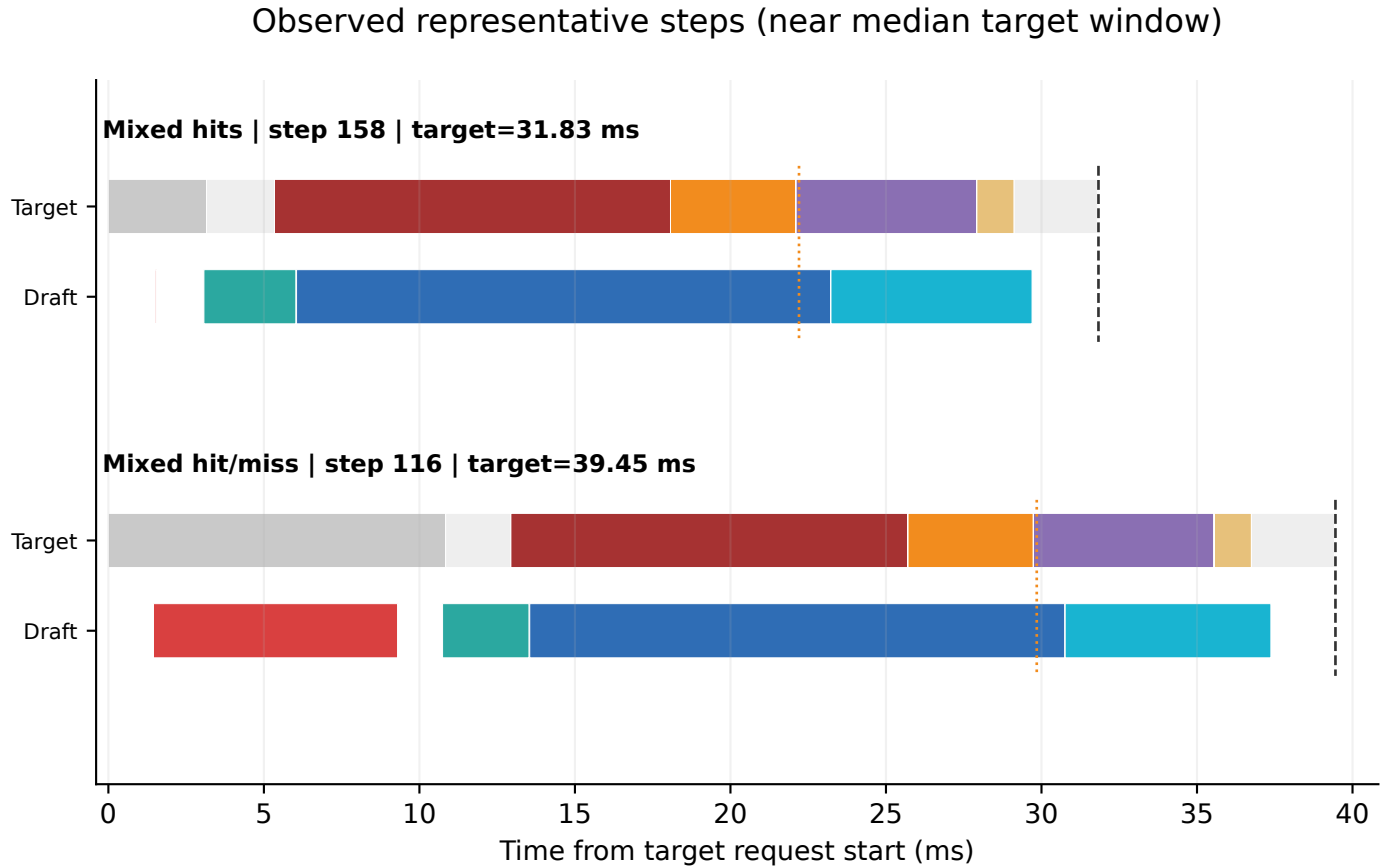
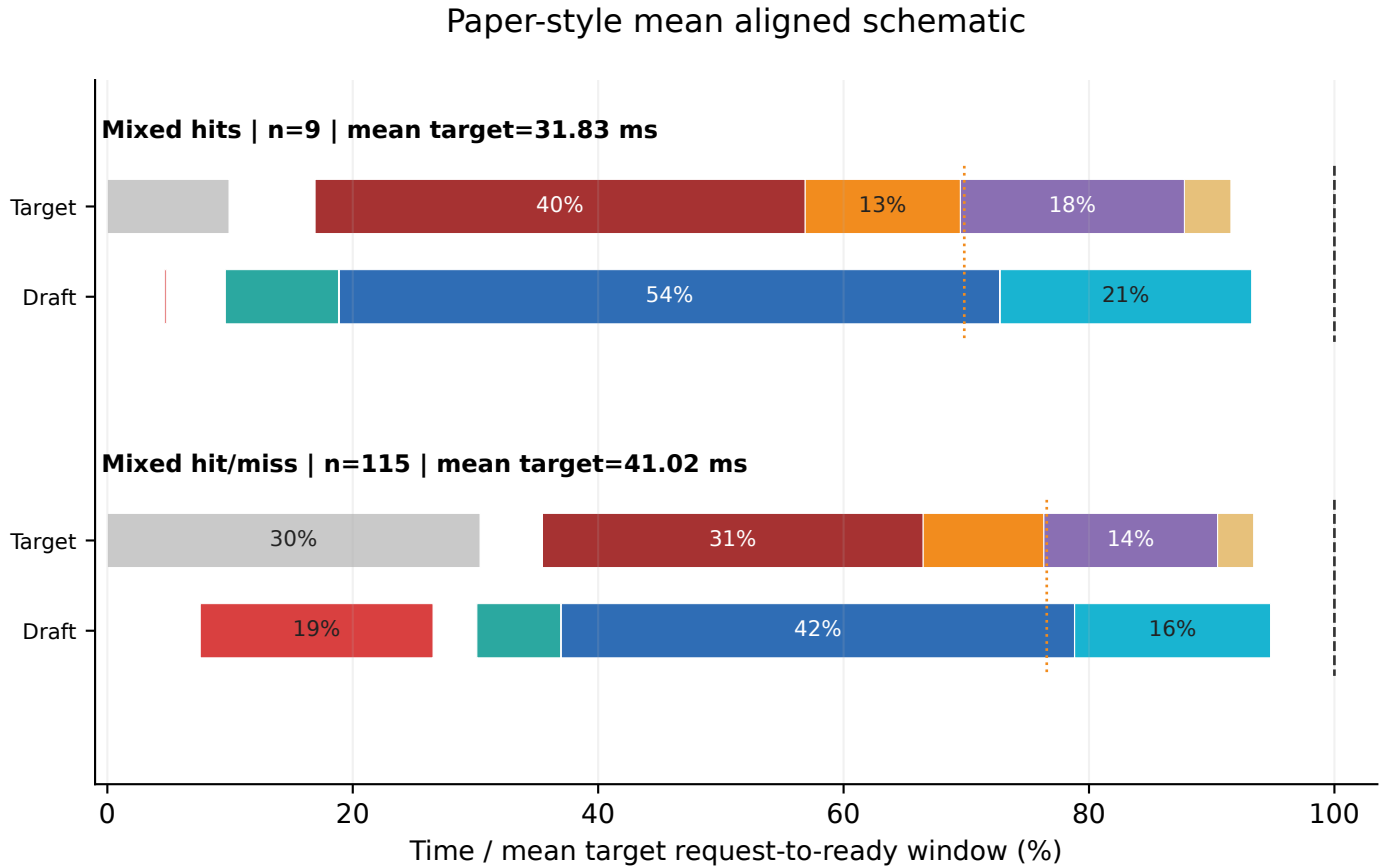
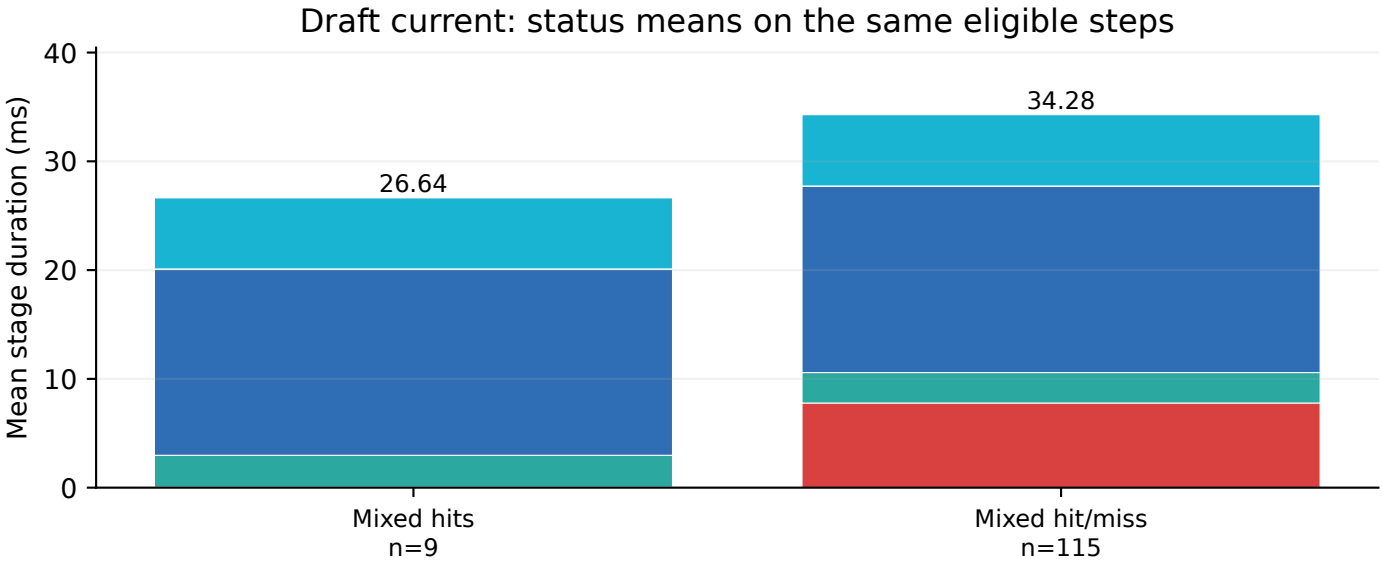
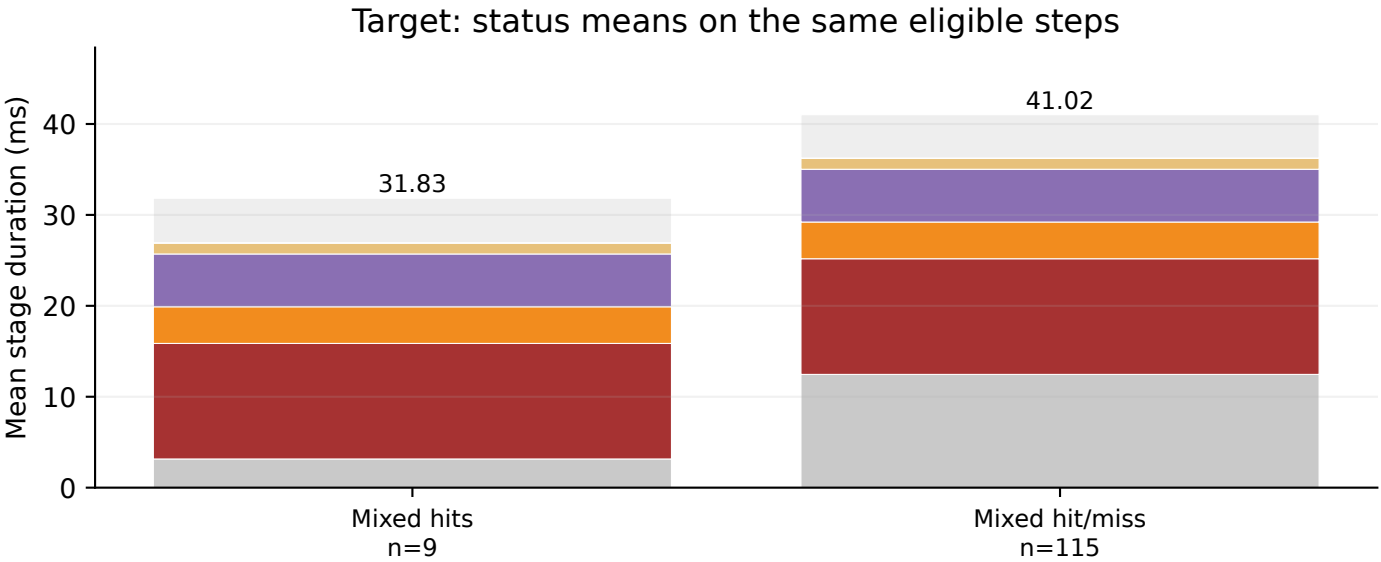
llama3_b8_tree_al_width2
DUET: K1/K2=6/2, exit=30 (0-based), C=2, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 96 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

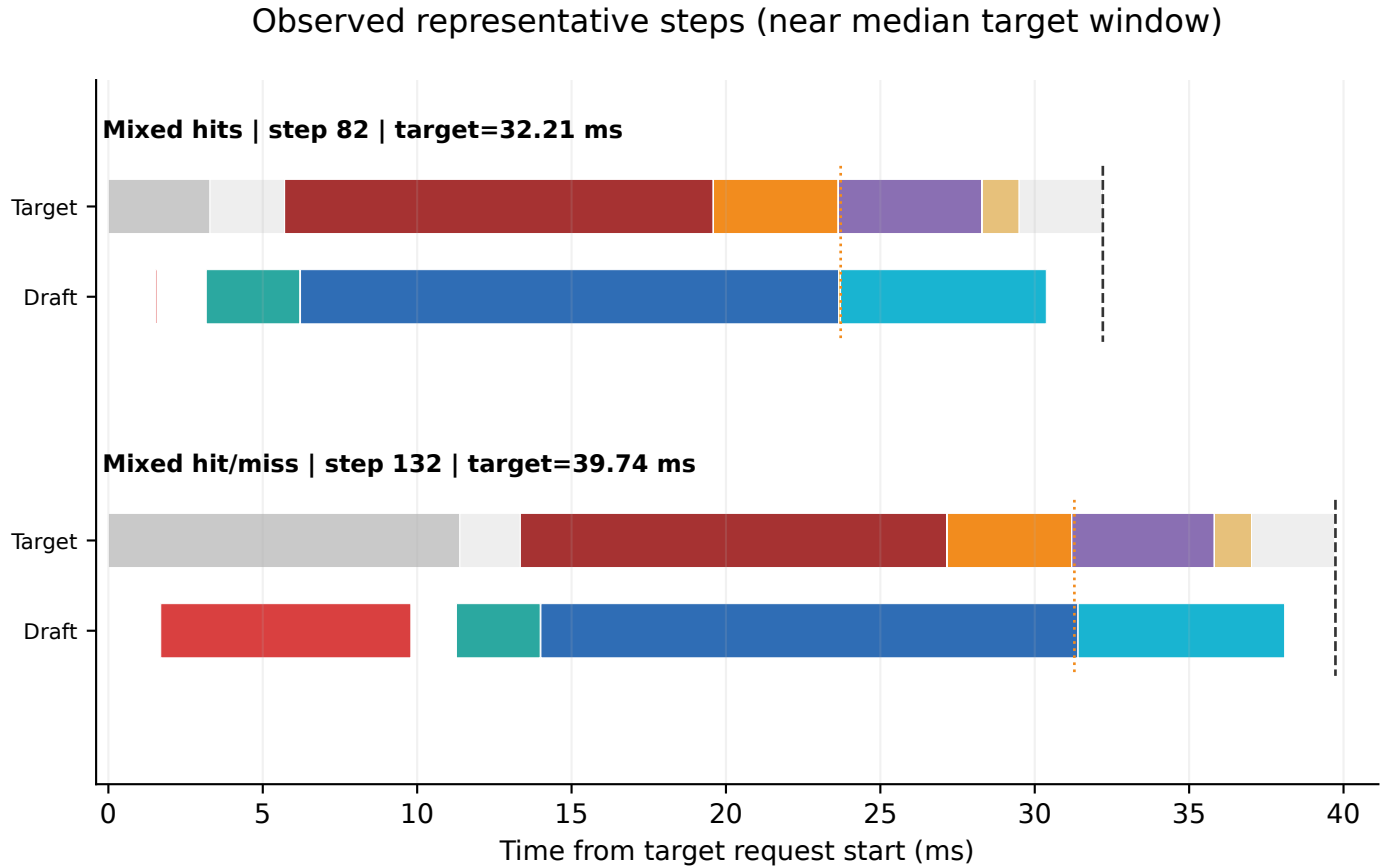
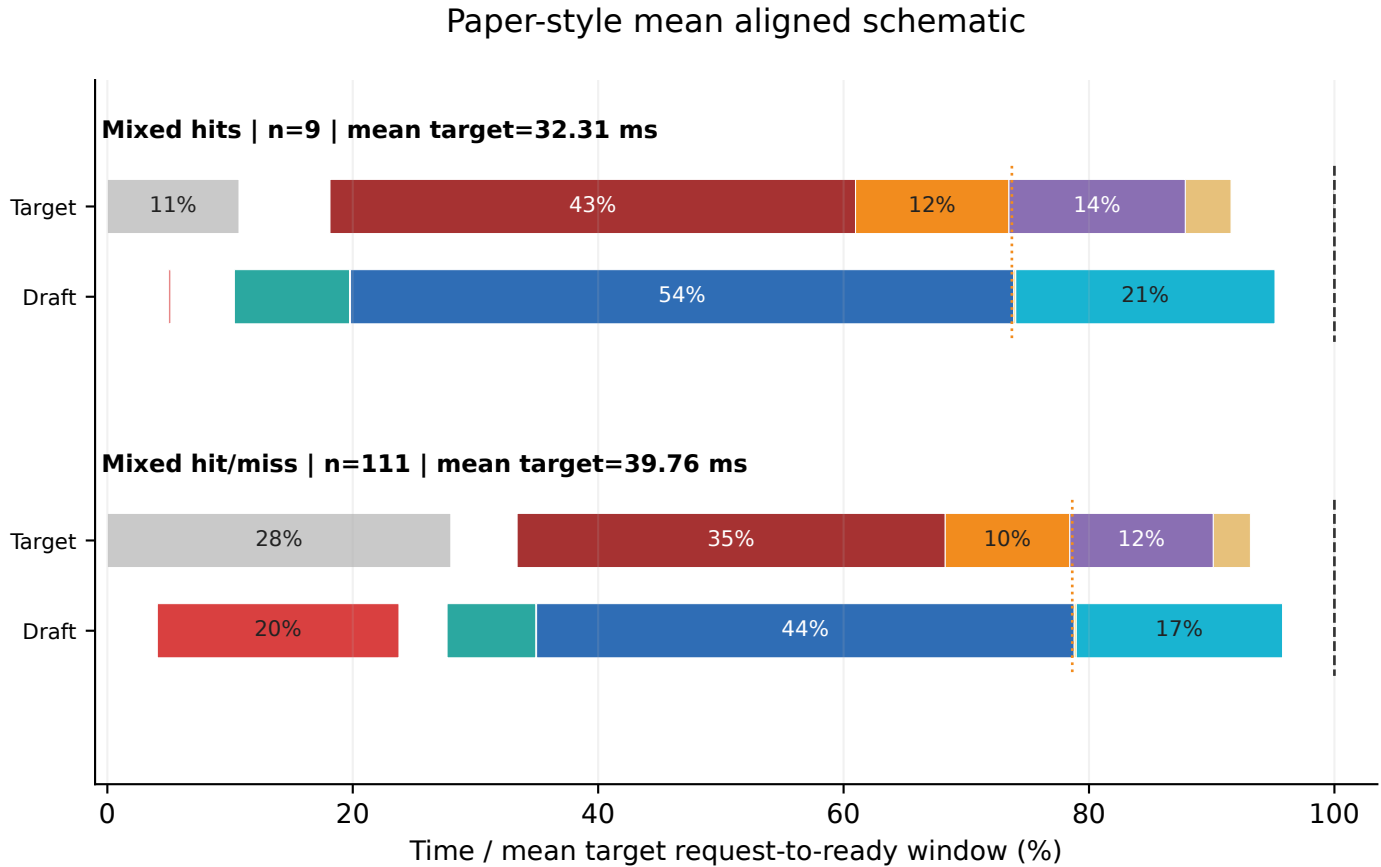
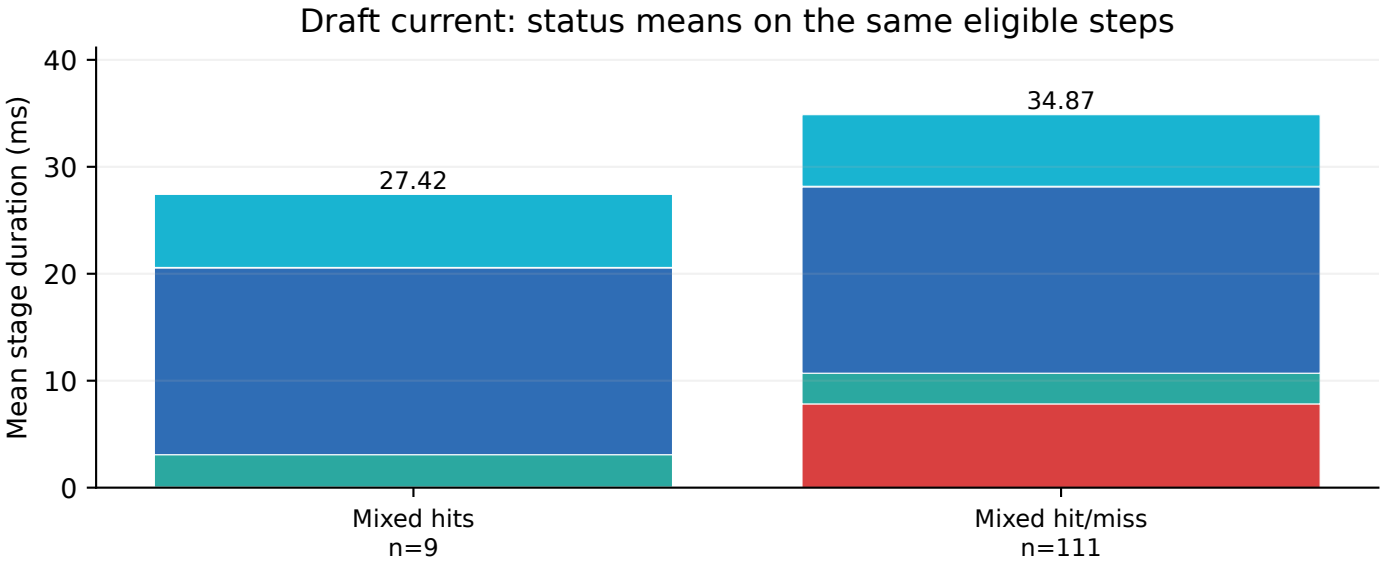
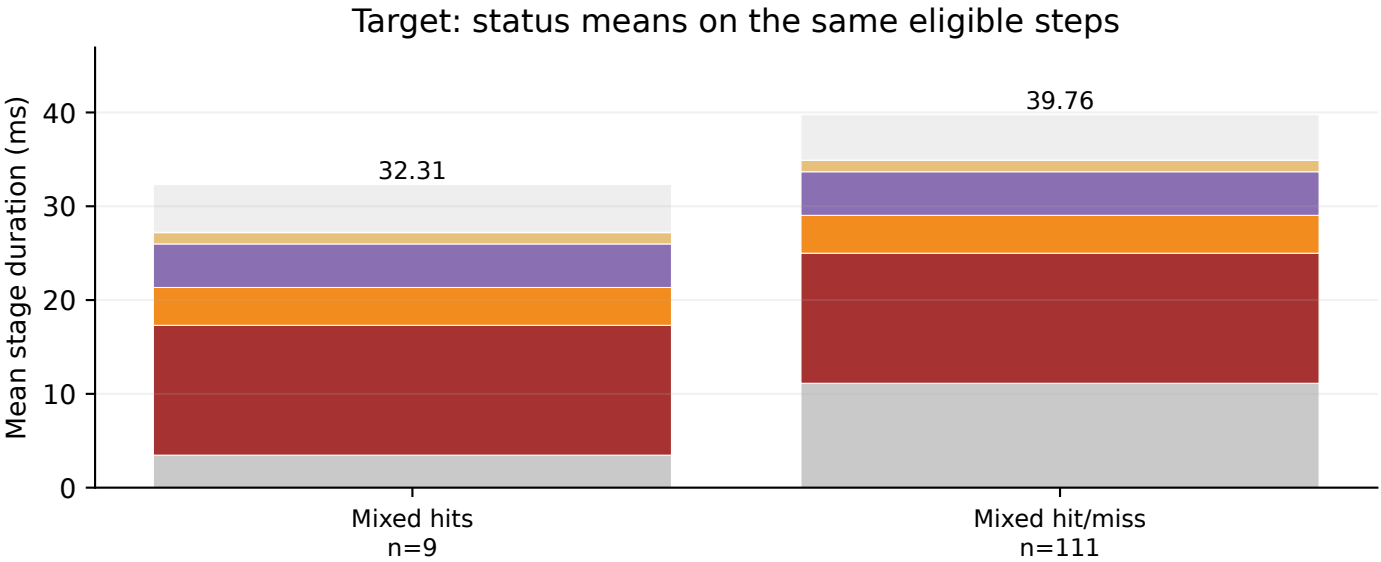
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_tree_align_exit21
DUET: K1/K2=3/1, exit=21 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 124 eligible steps



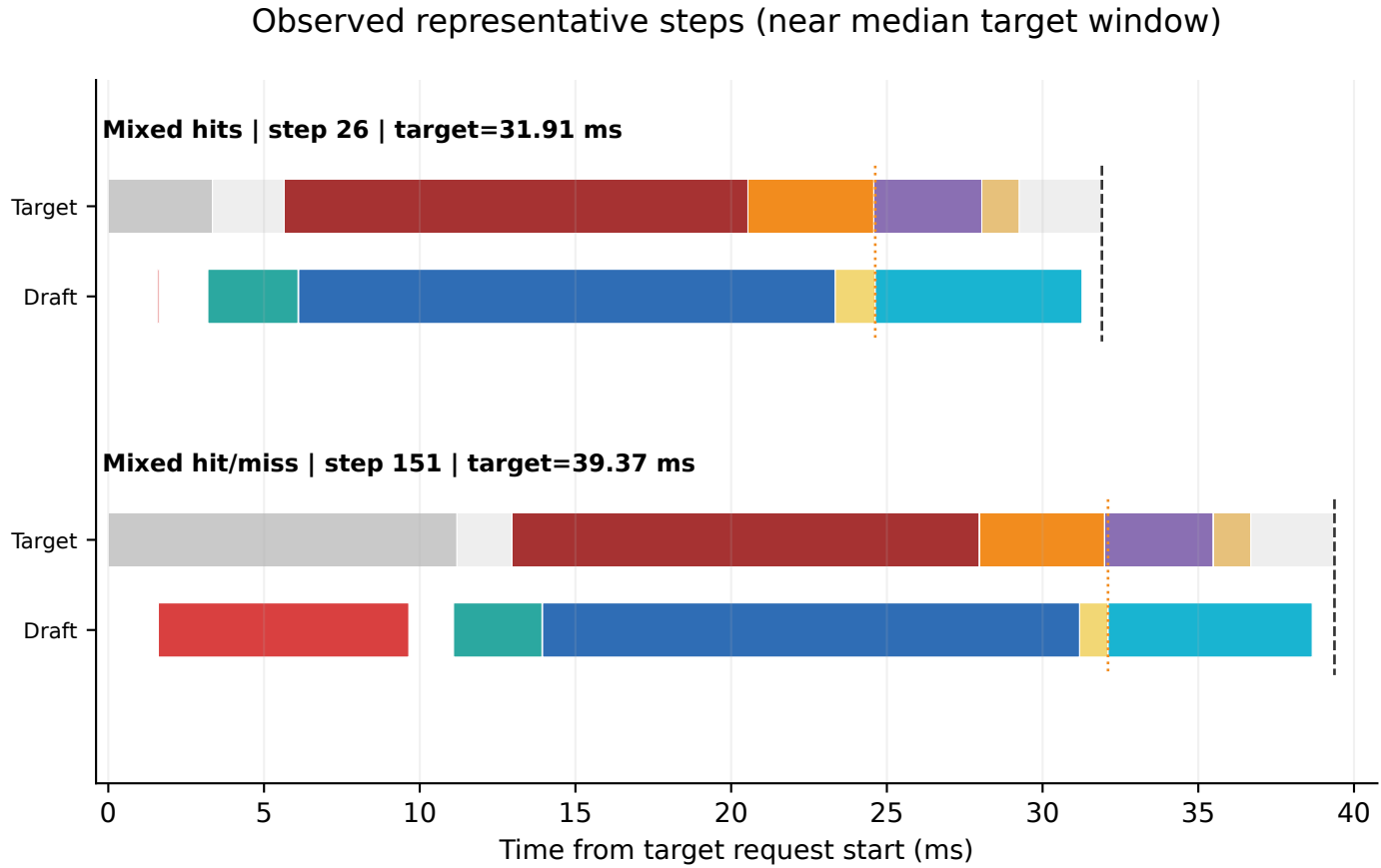
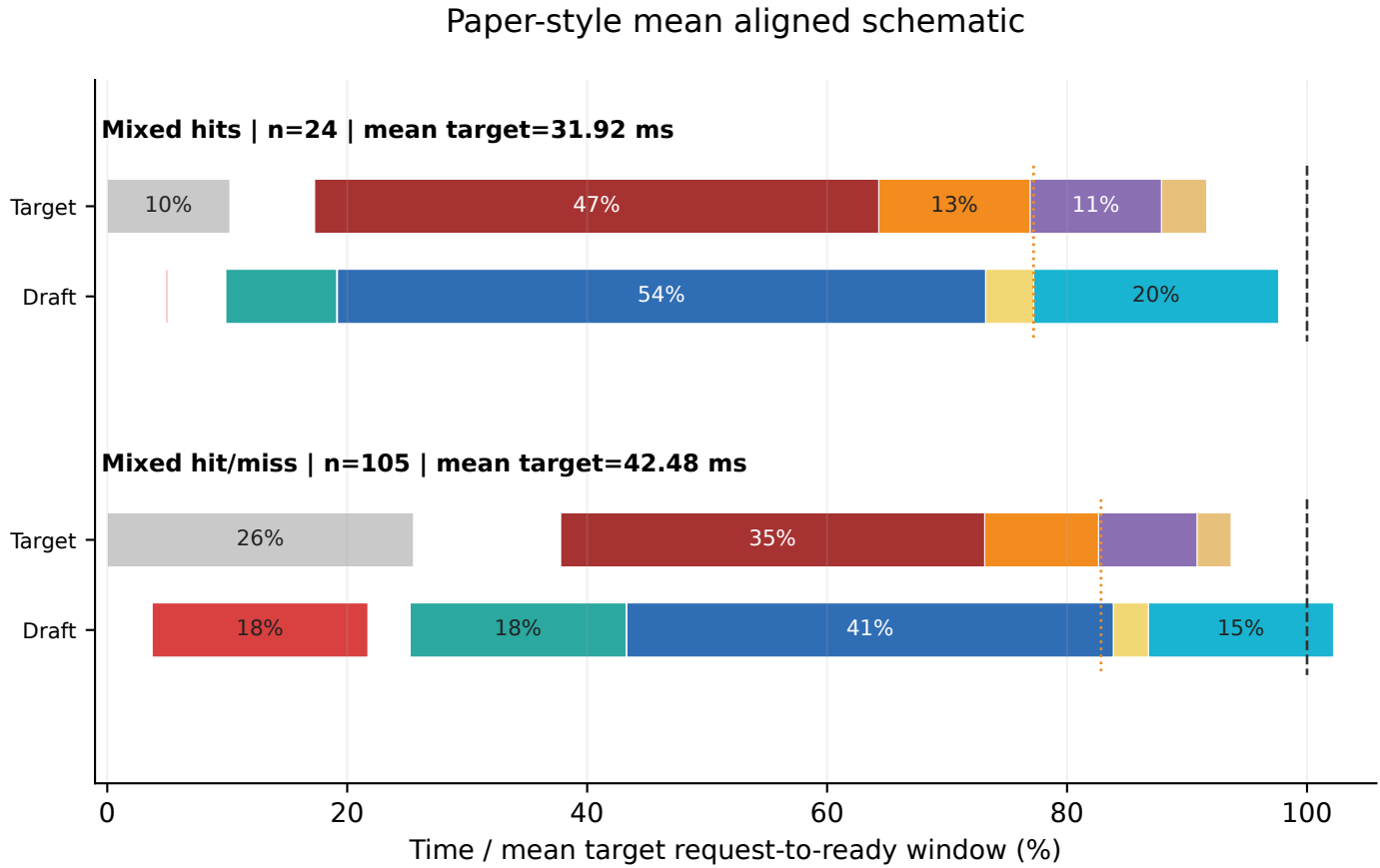
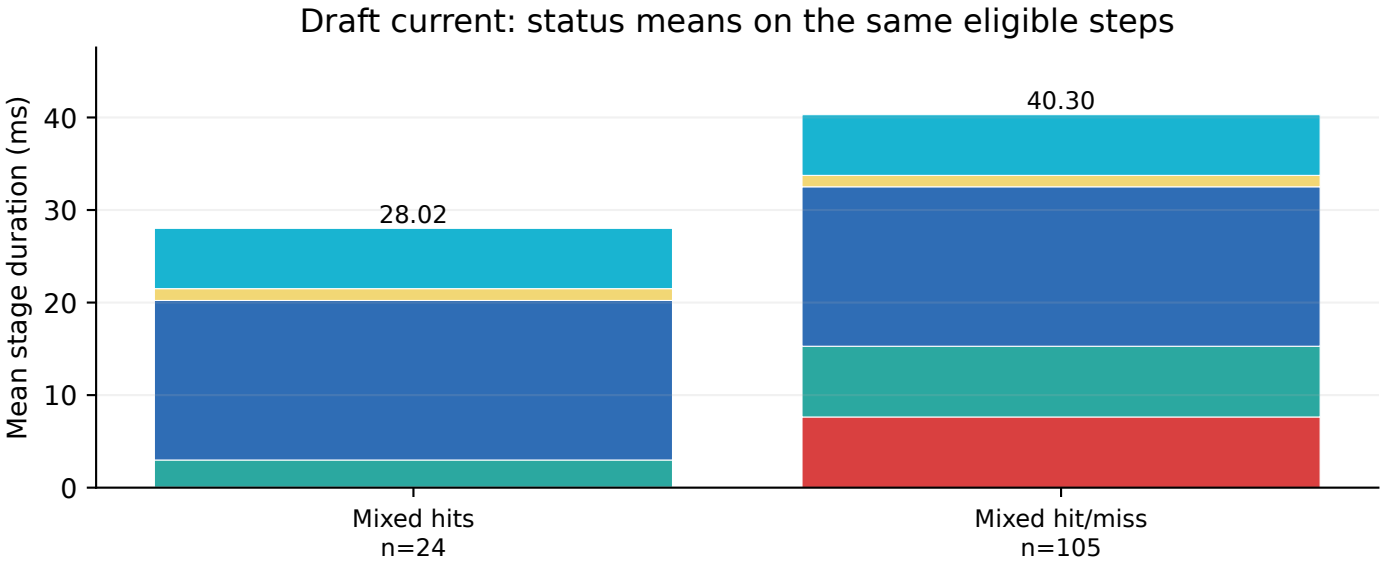
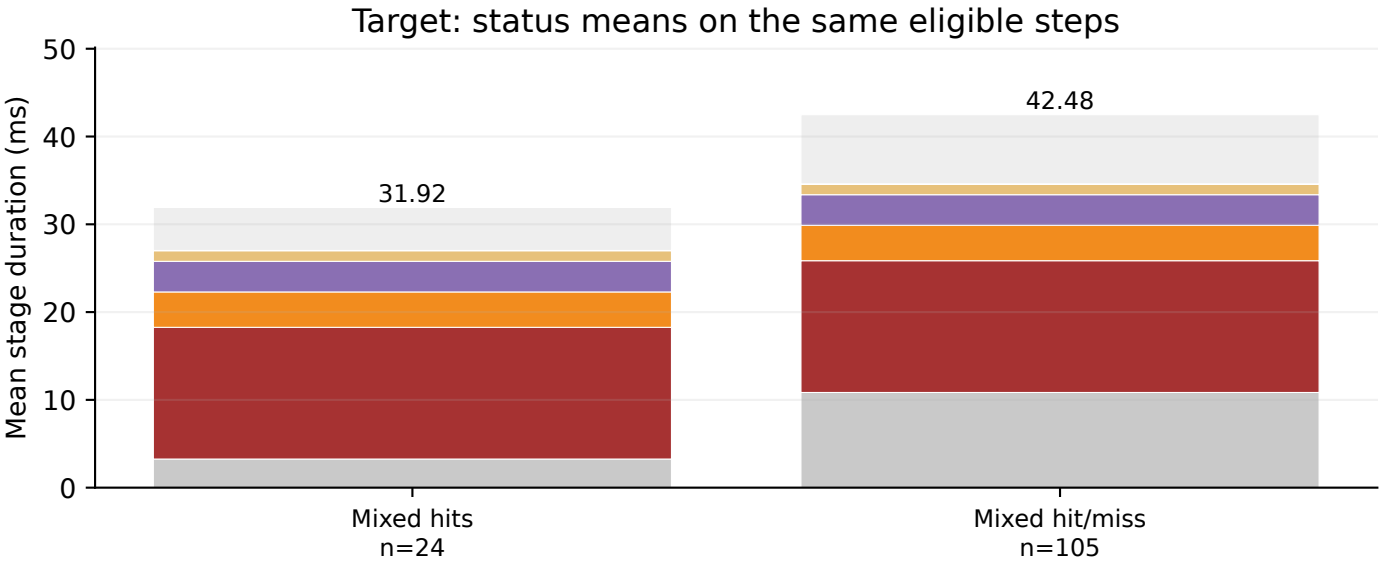
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_tree_align_exit23
DUET: K1/K2=3/1, exit=23 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 120 eligible steps



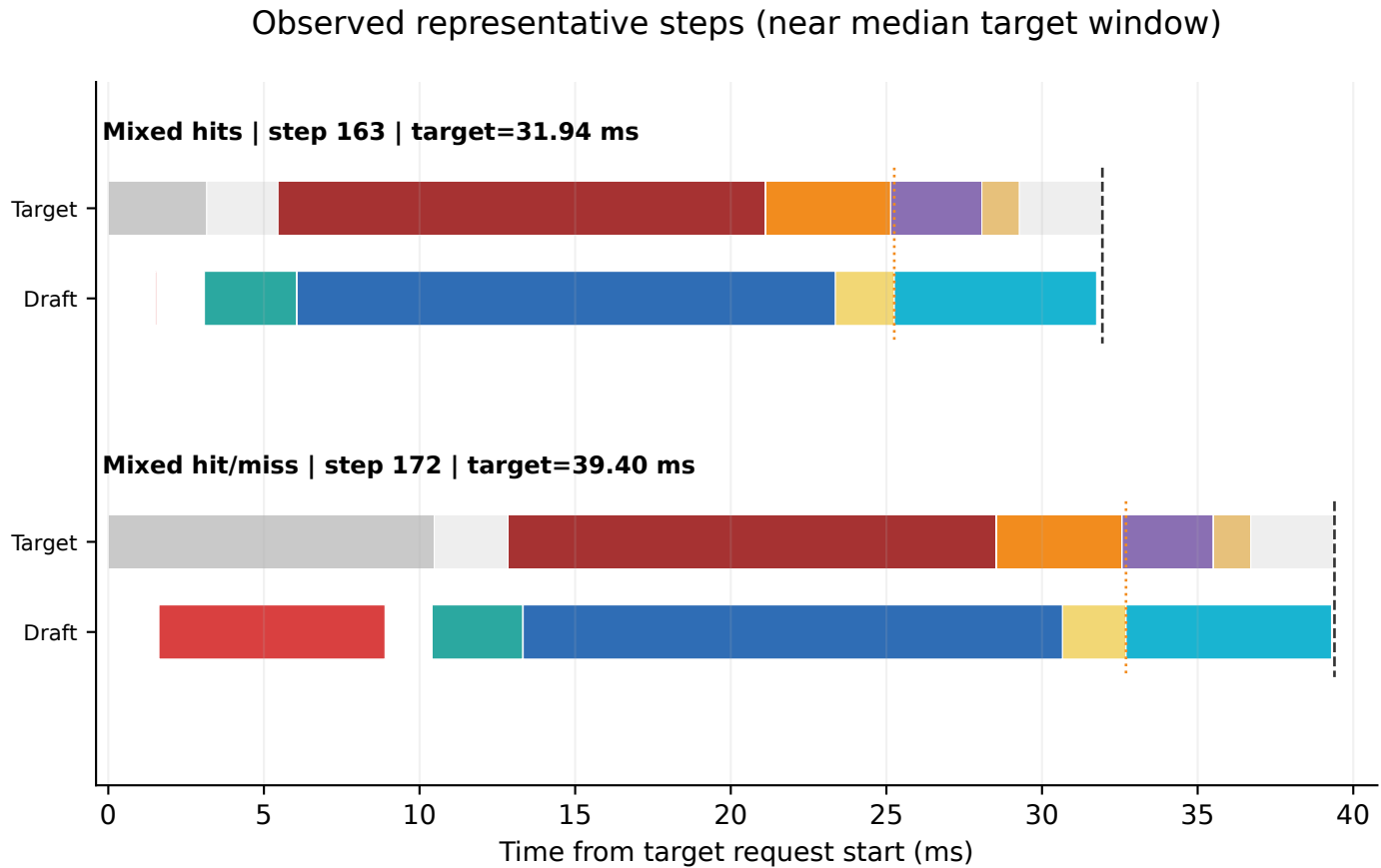
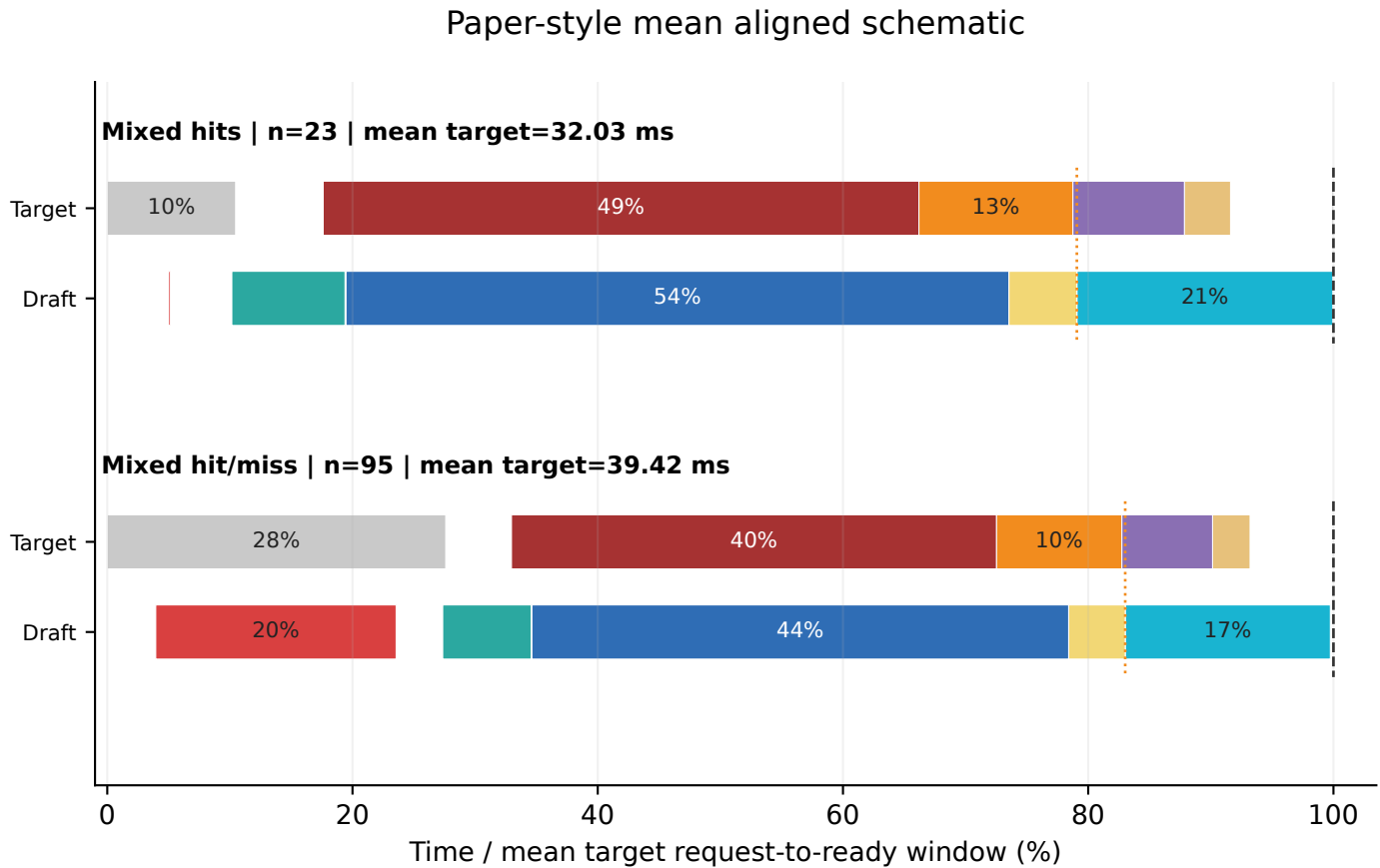
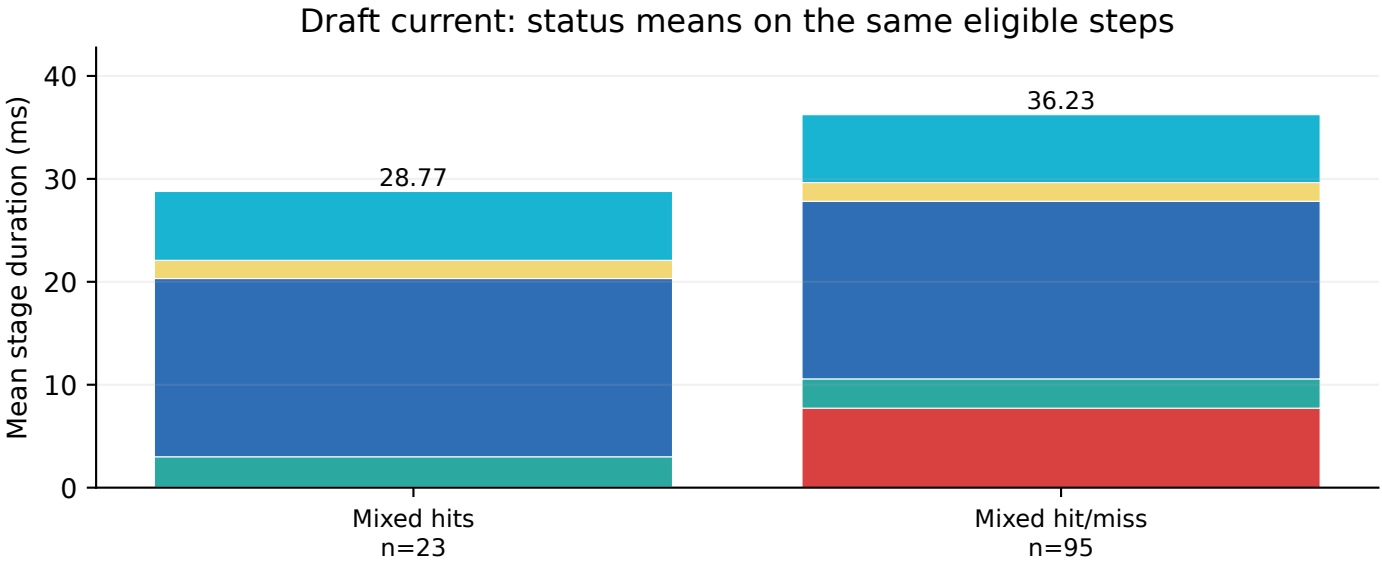
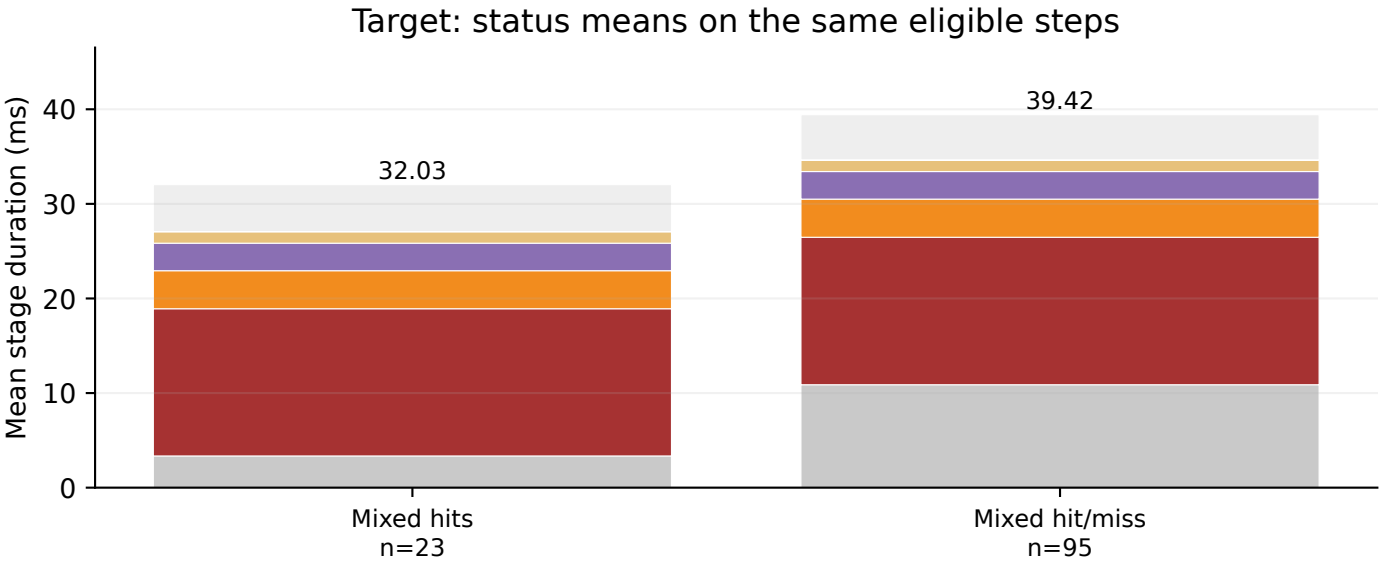
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_tree_align_exit25
DUET: K1/K2=3/1, exit=25 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 129 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

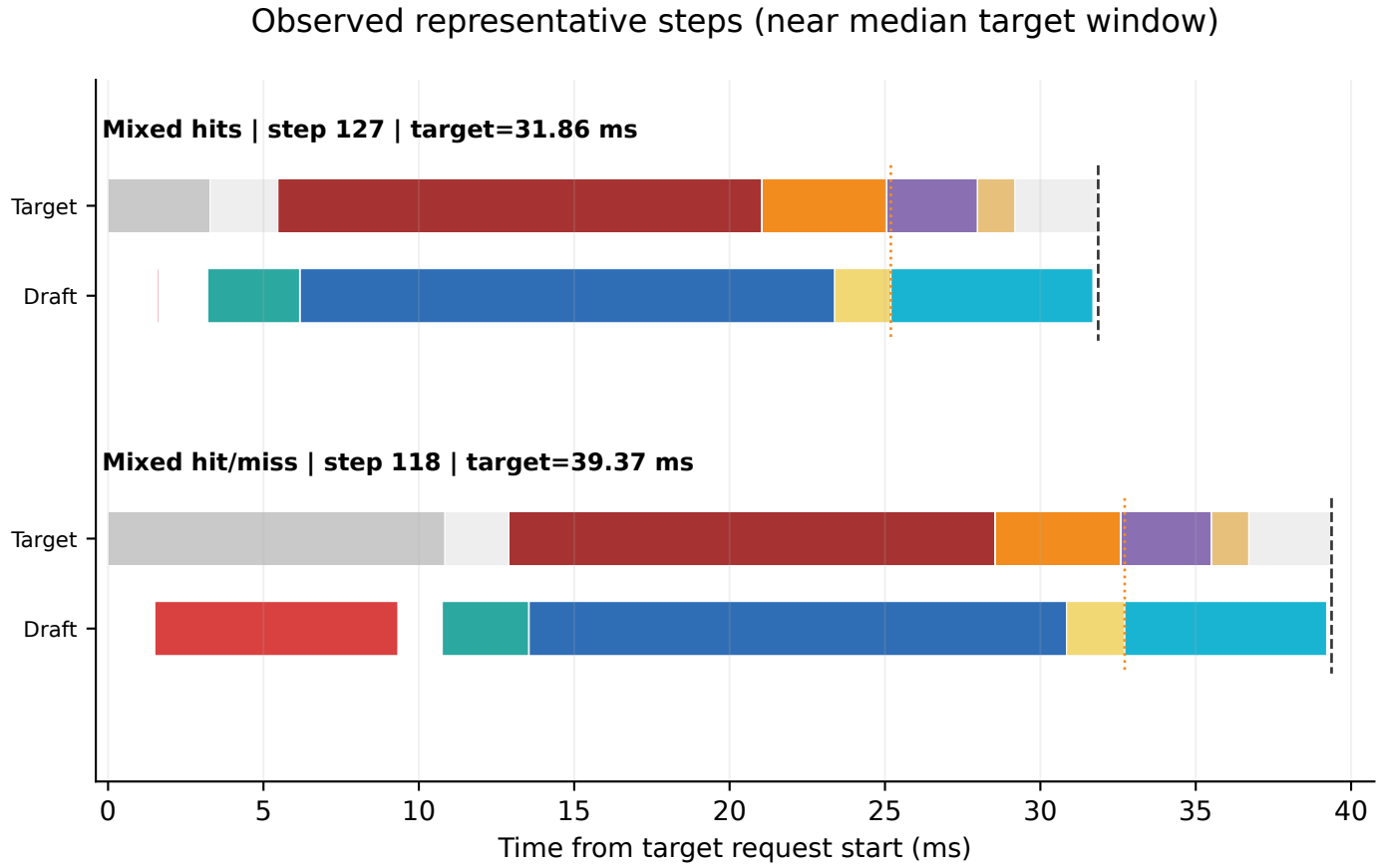
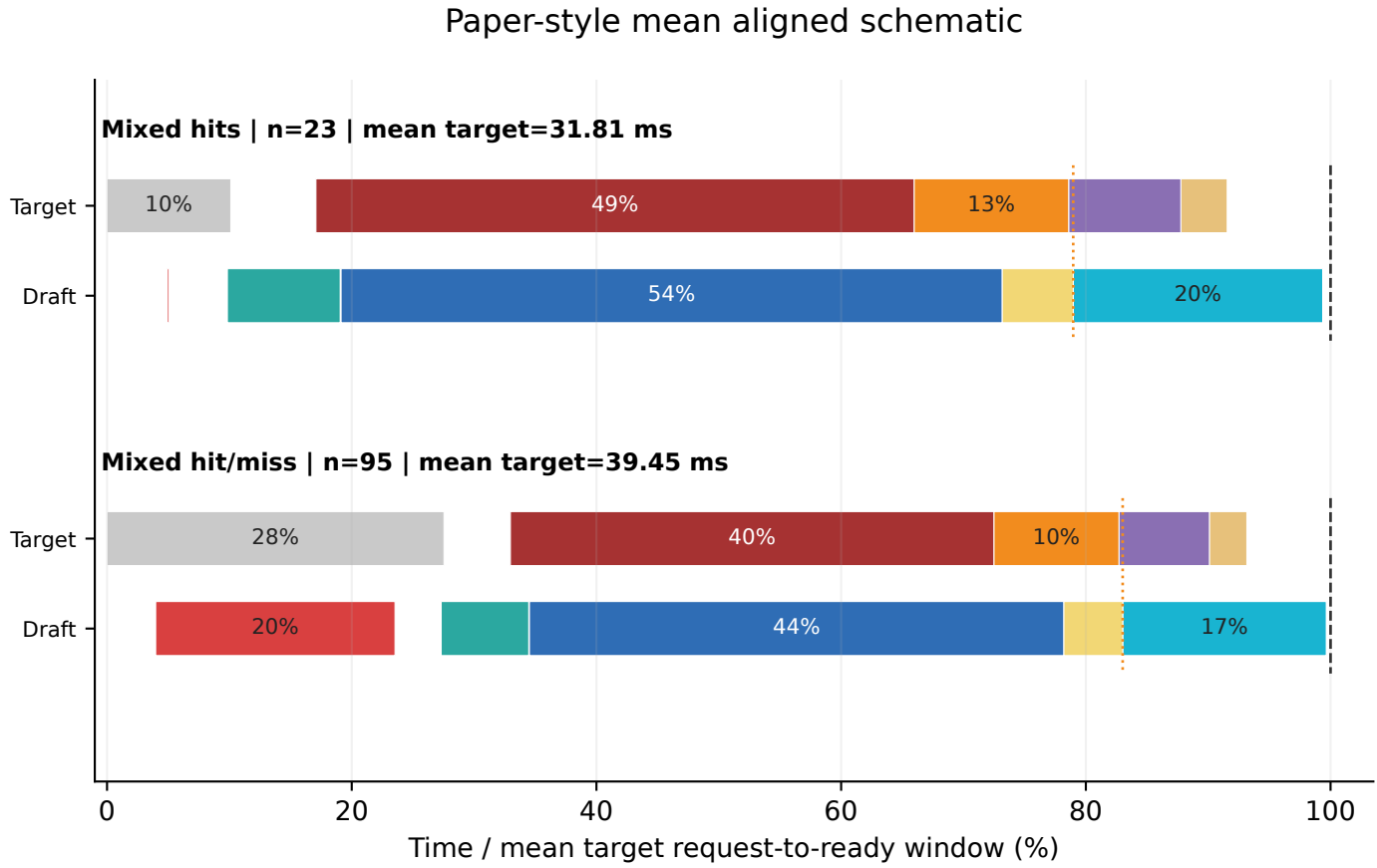
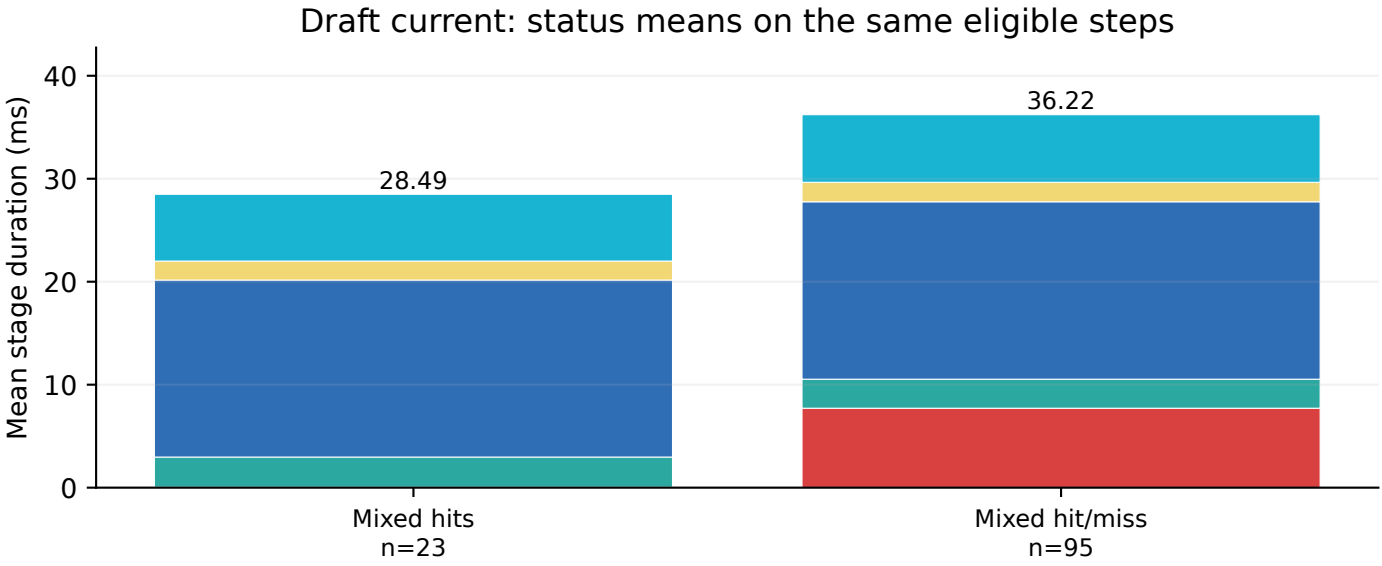
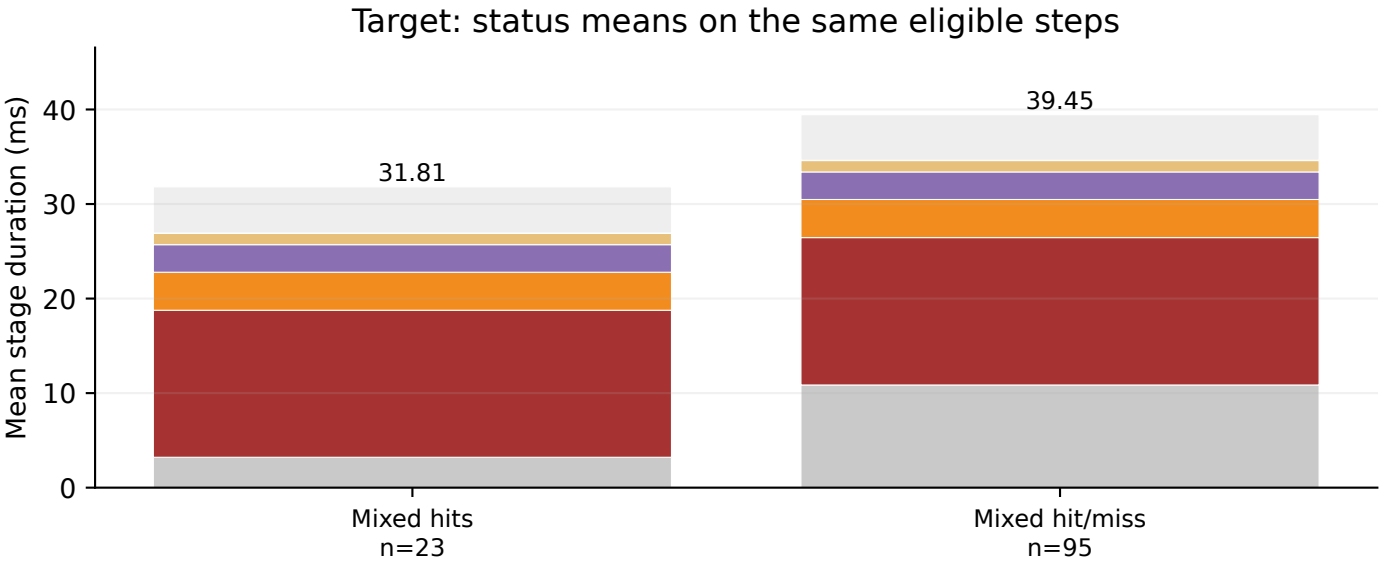
llama3_b8_tree_beta0
DUET: K1/K2=3/1, exit=26 (0-based), $\bar{C}$ =3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 118 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

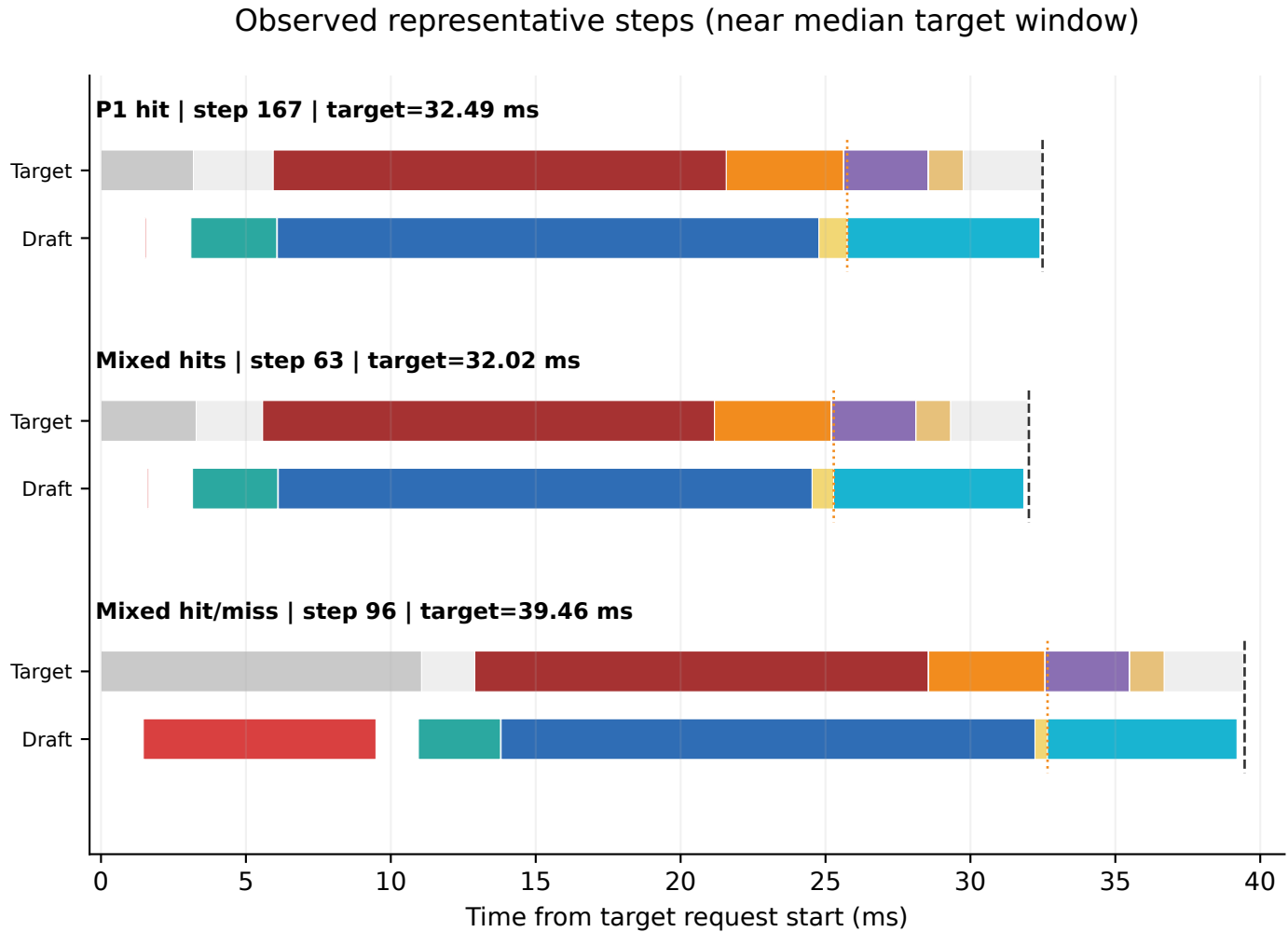
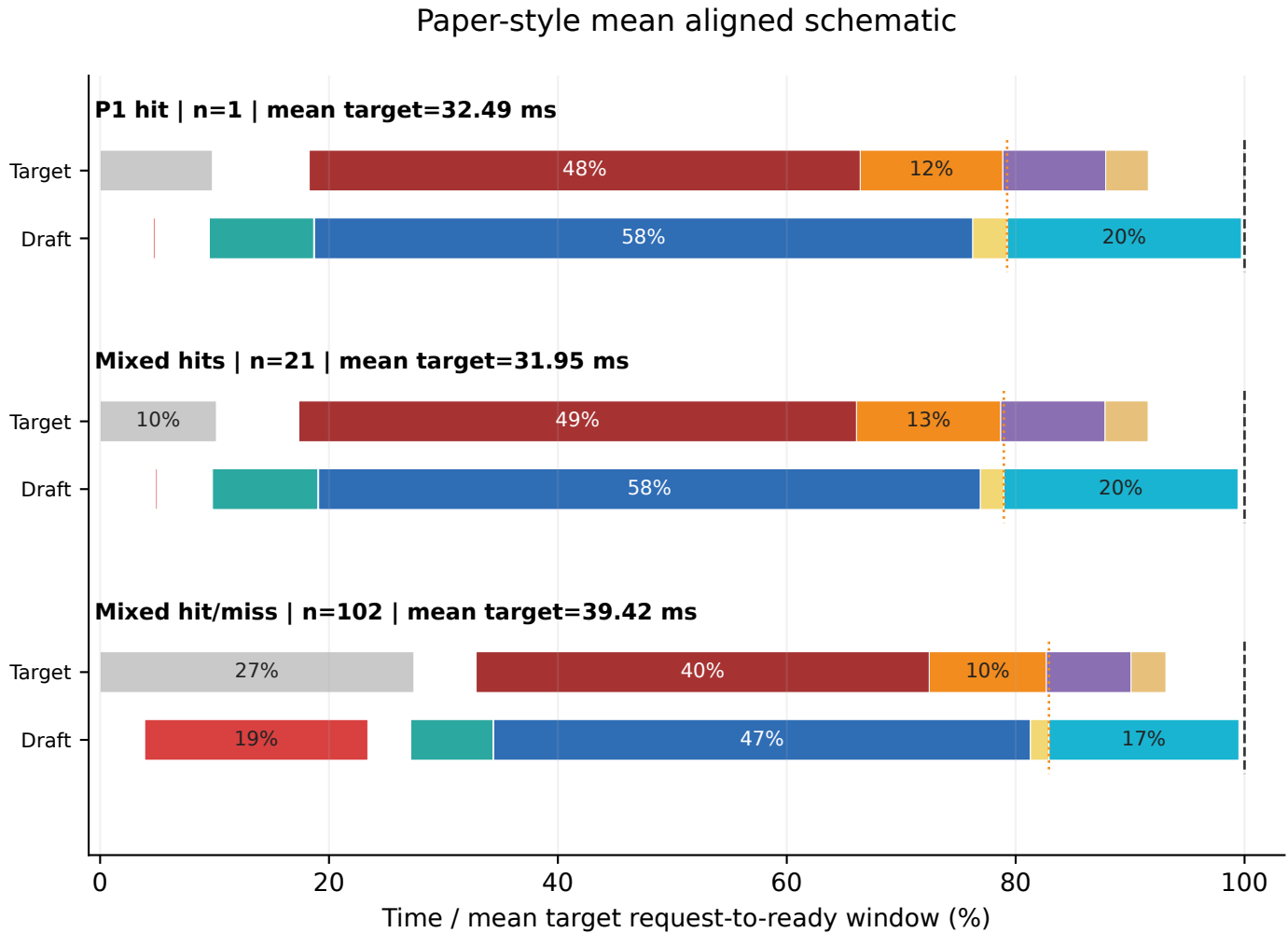
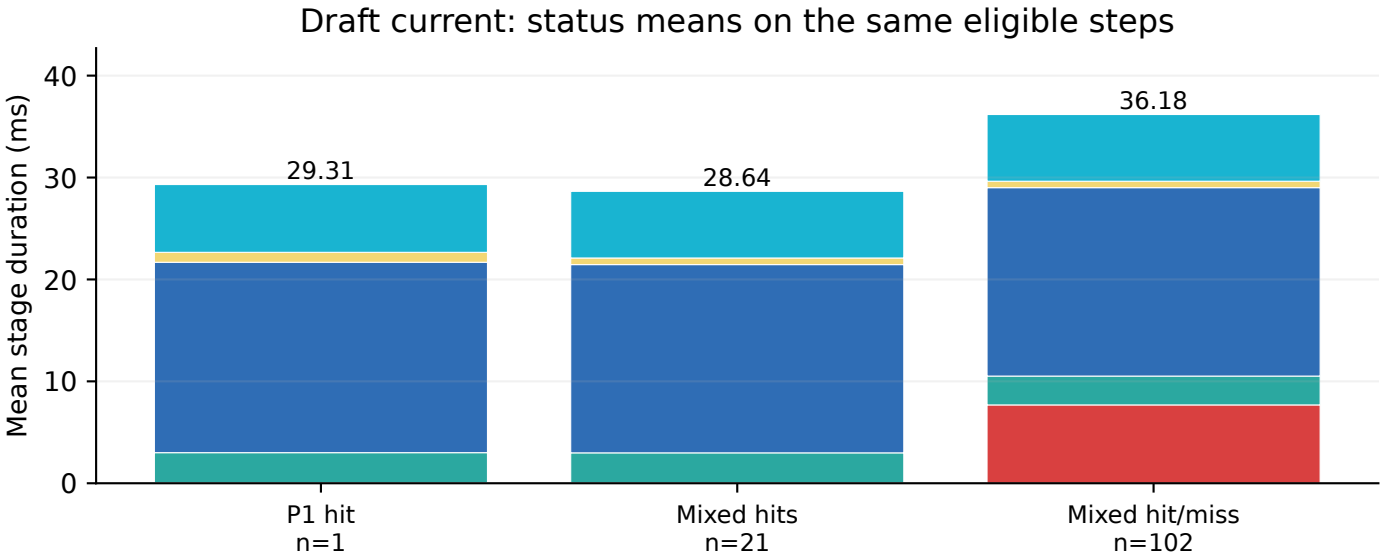
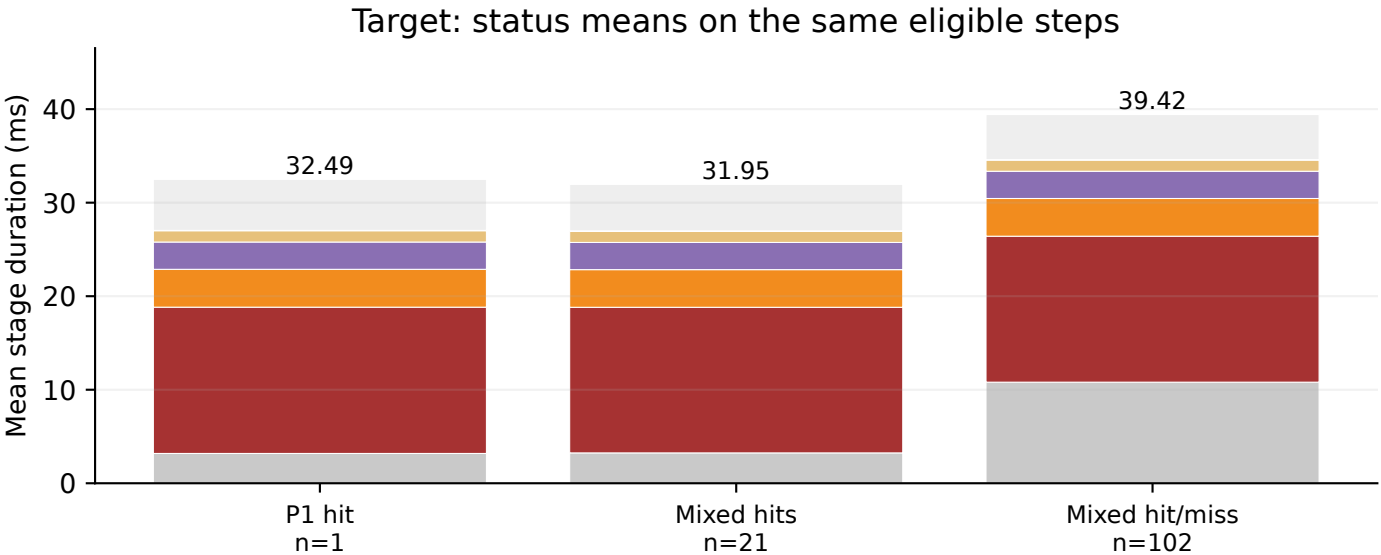
llama3_b8_tree_beta1
DUET: K1/K2=3/1, exit=26 (0-based), $\bar{C}=3$, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 118 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

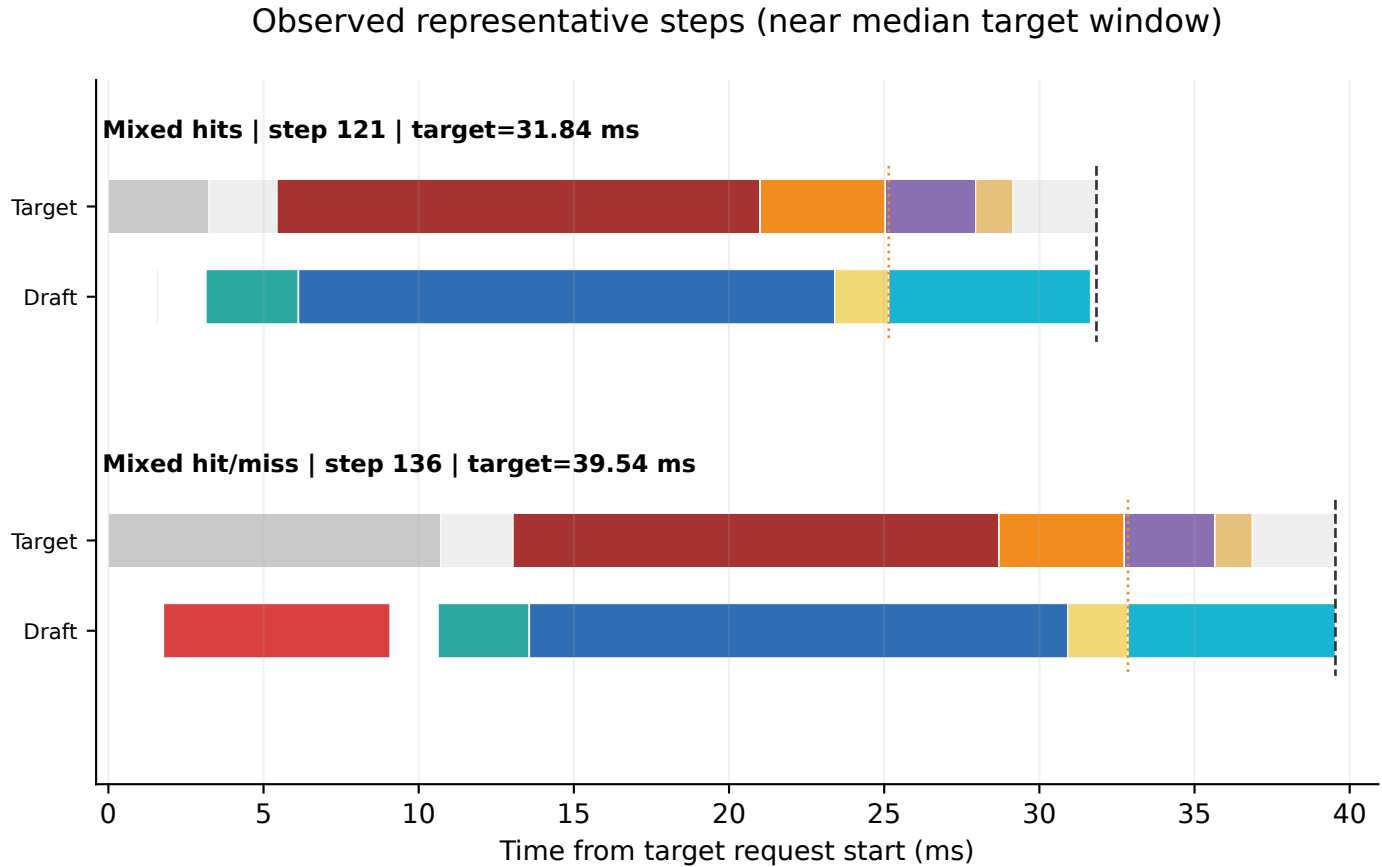
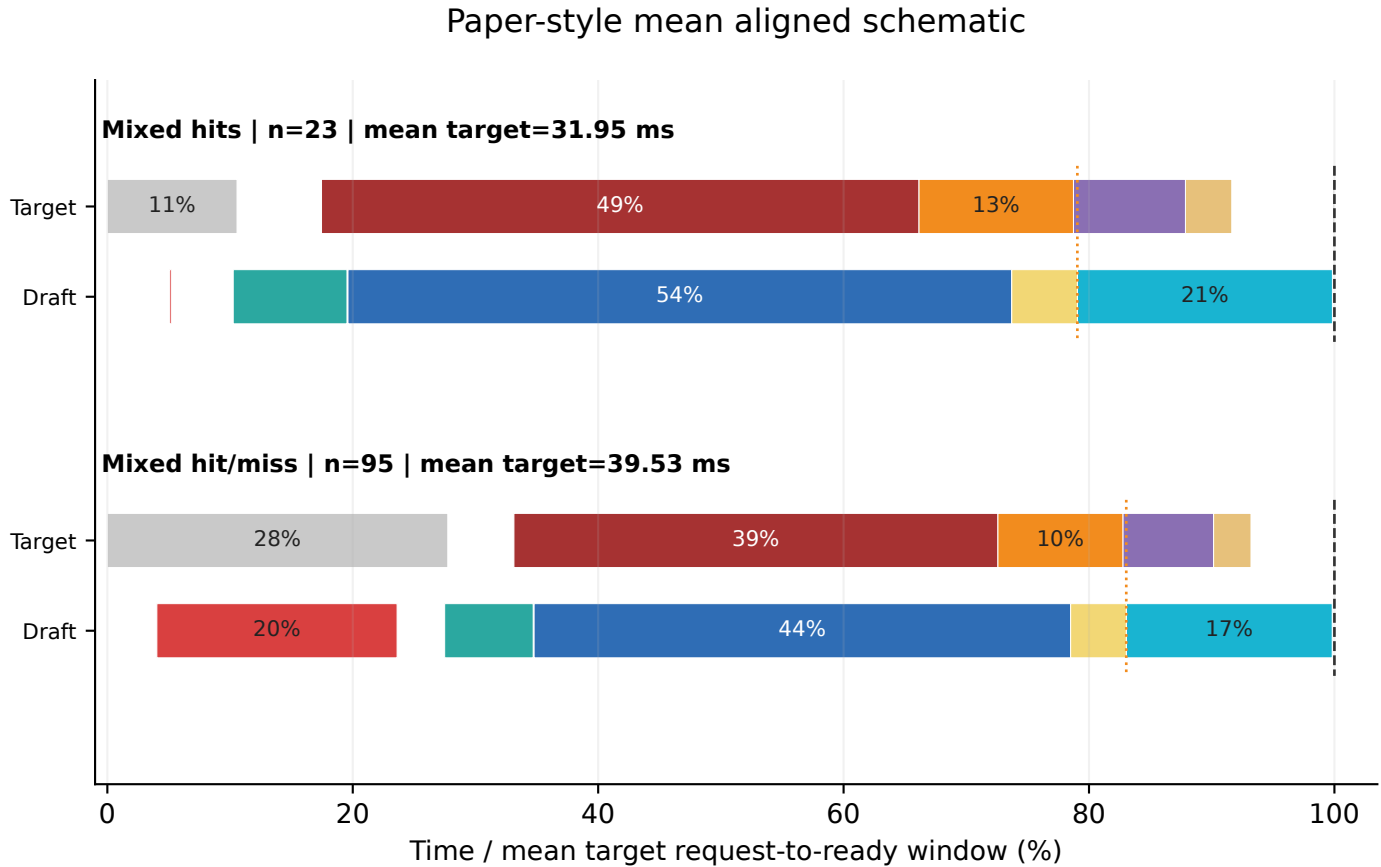
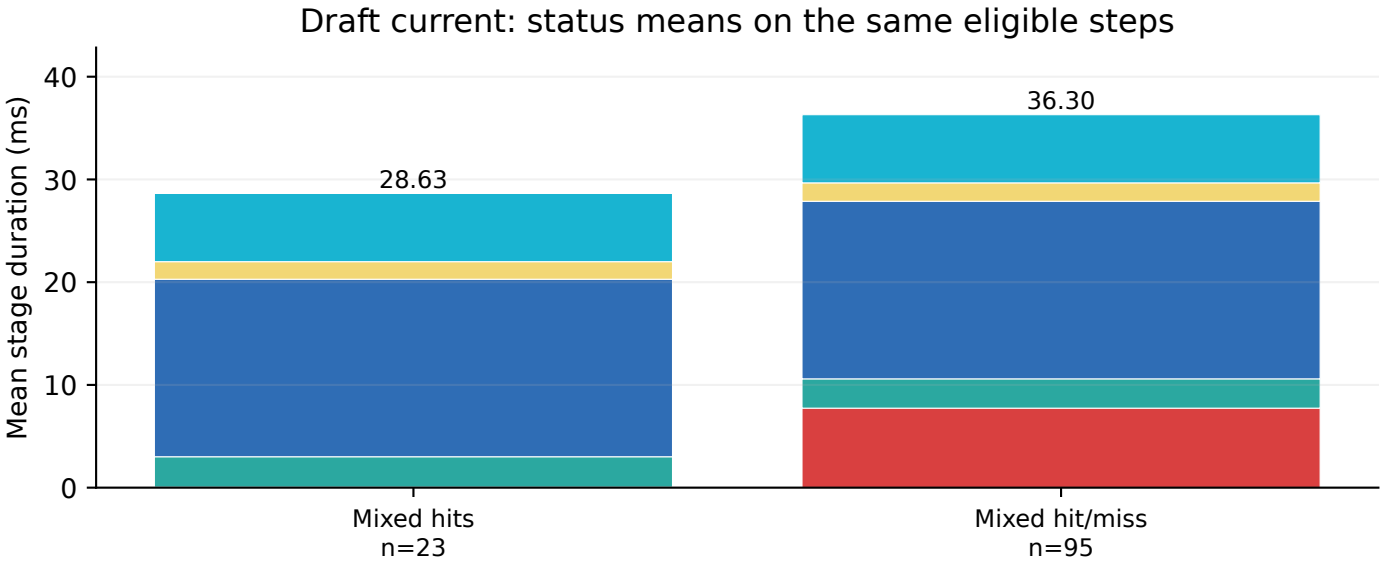
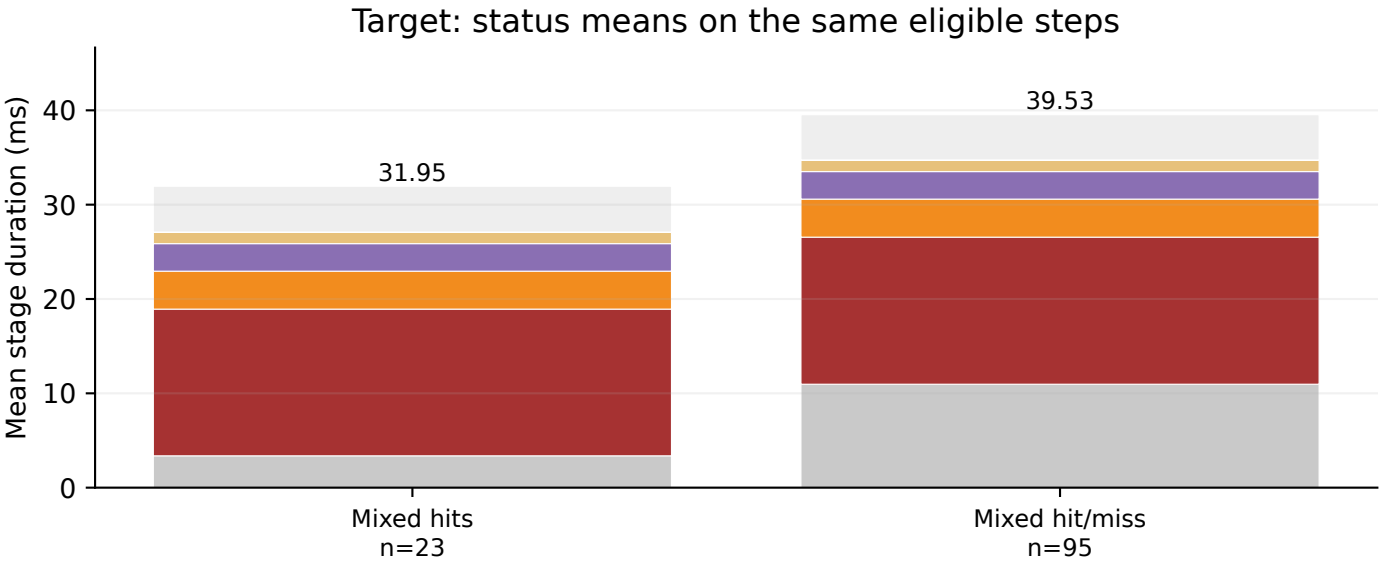
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_tree_compute3
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 124 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

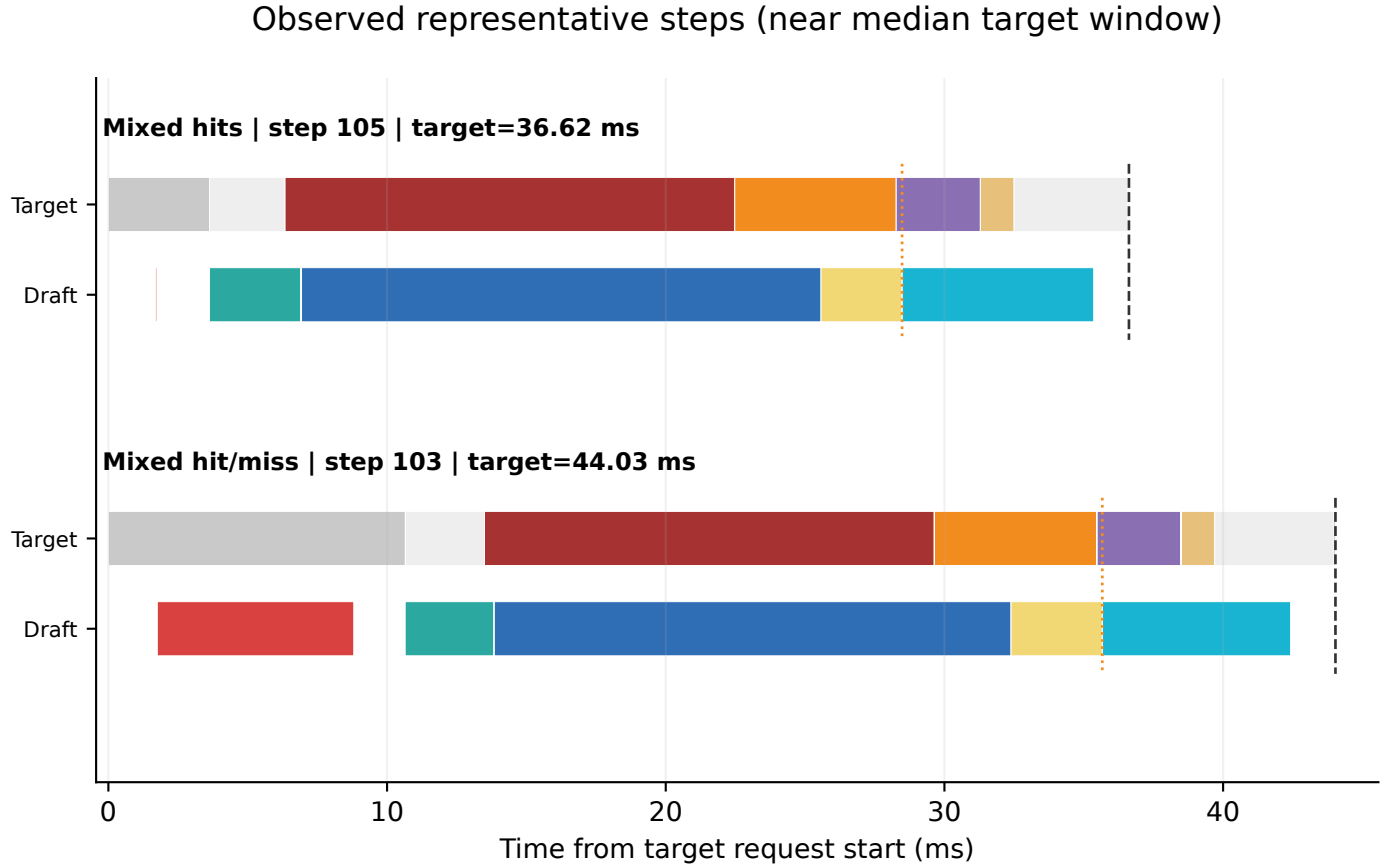
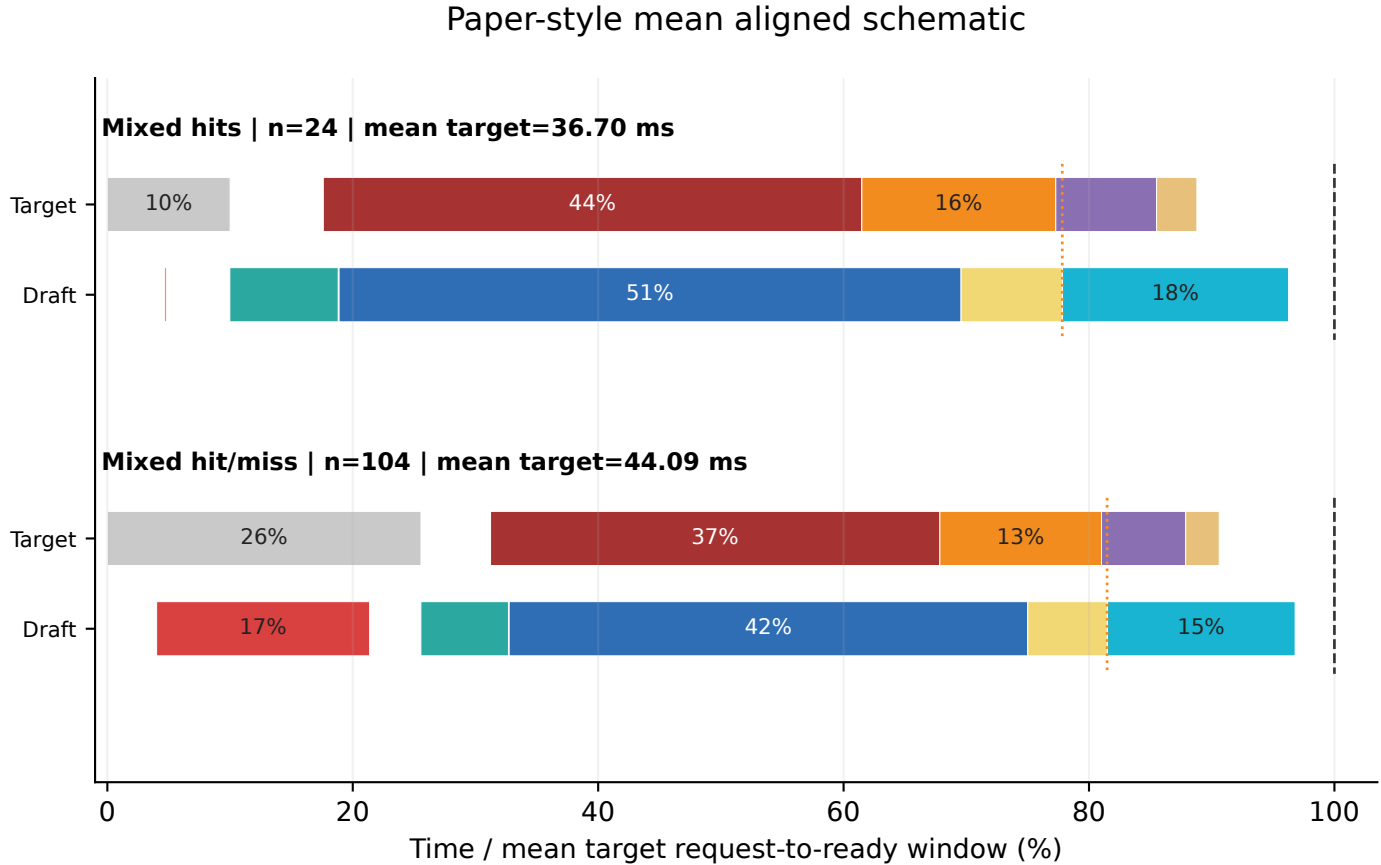
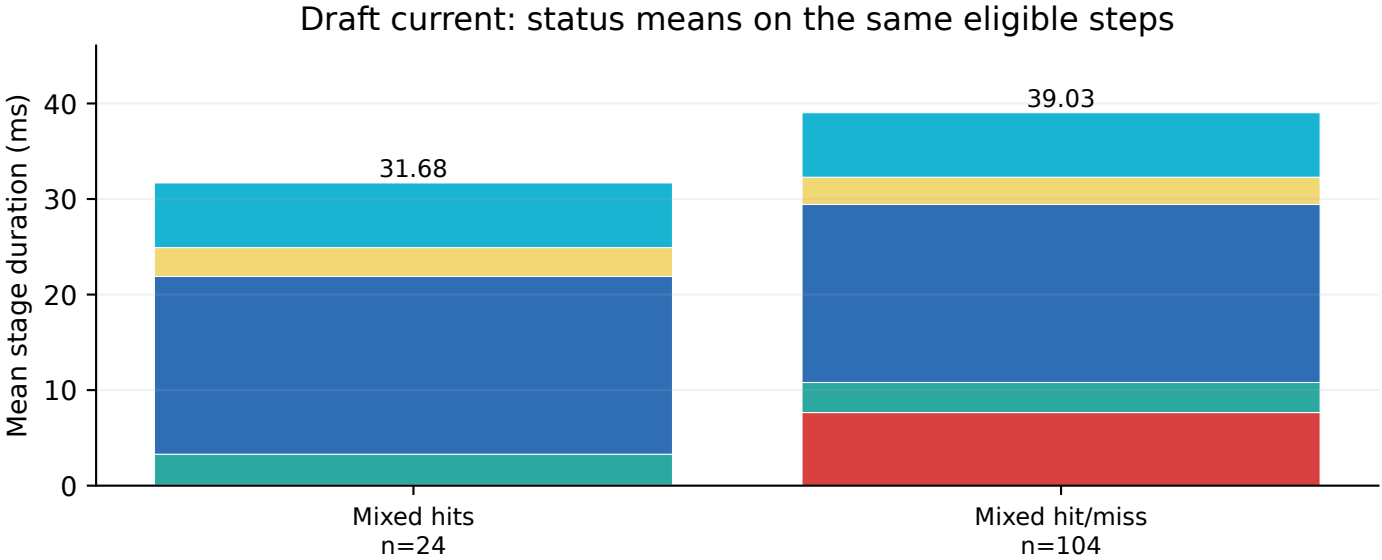
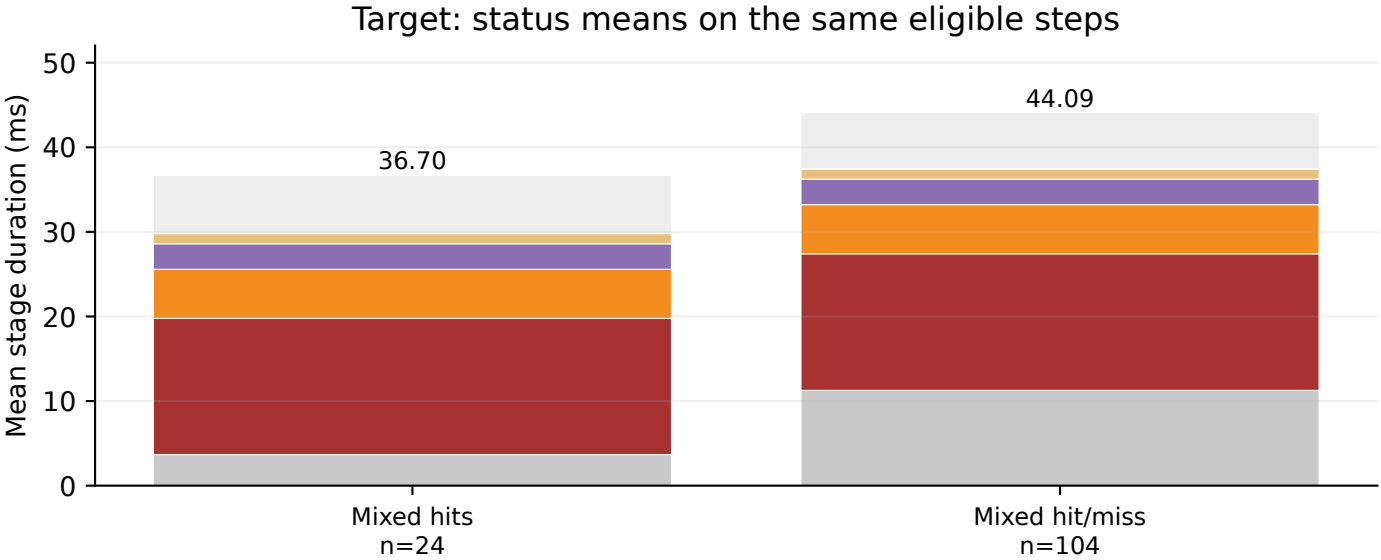
llama3_b8_tree_no_floors
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 118 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

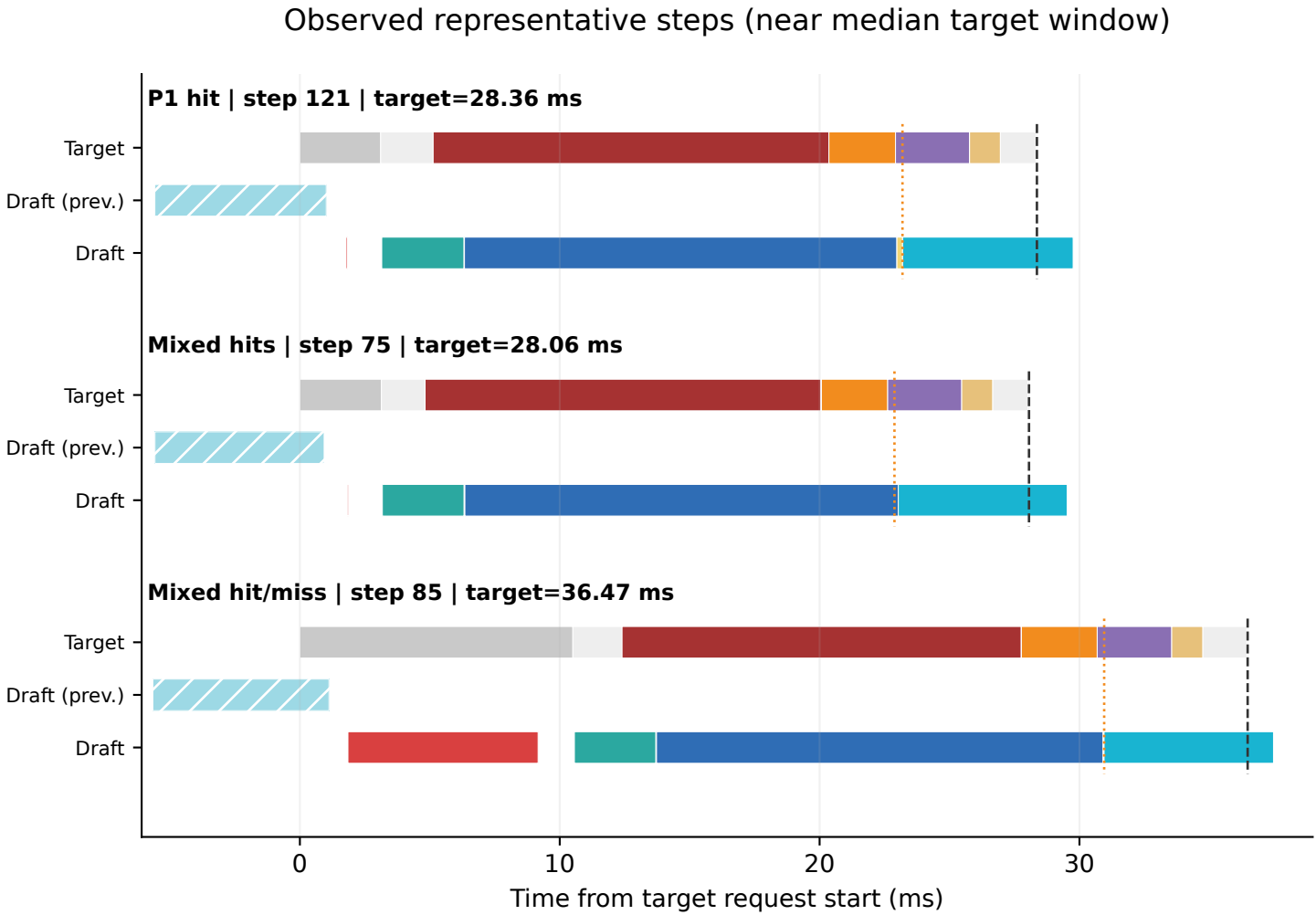
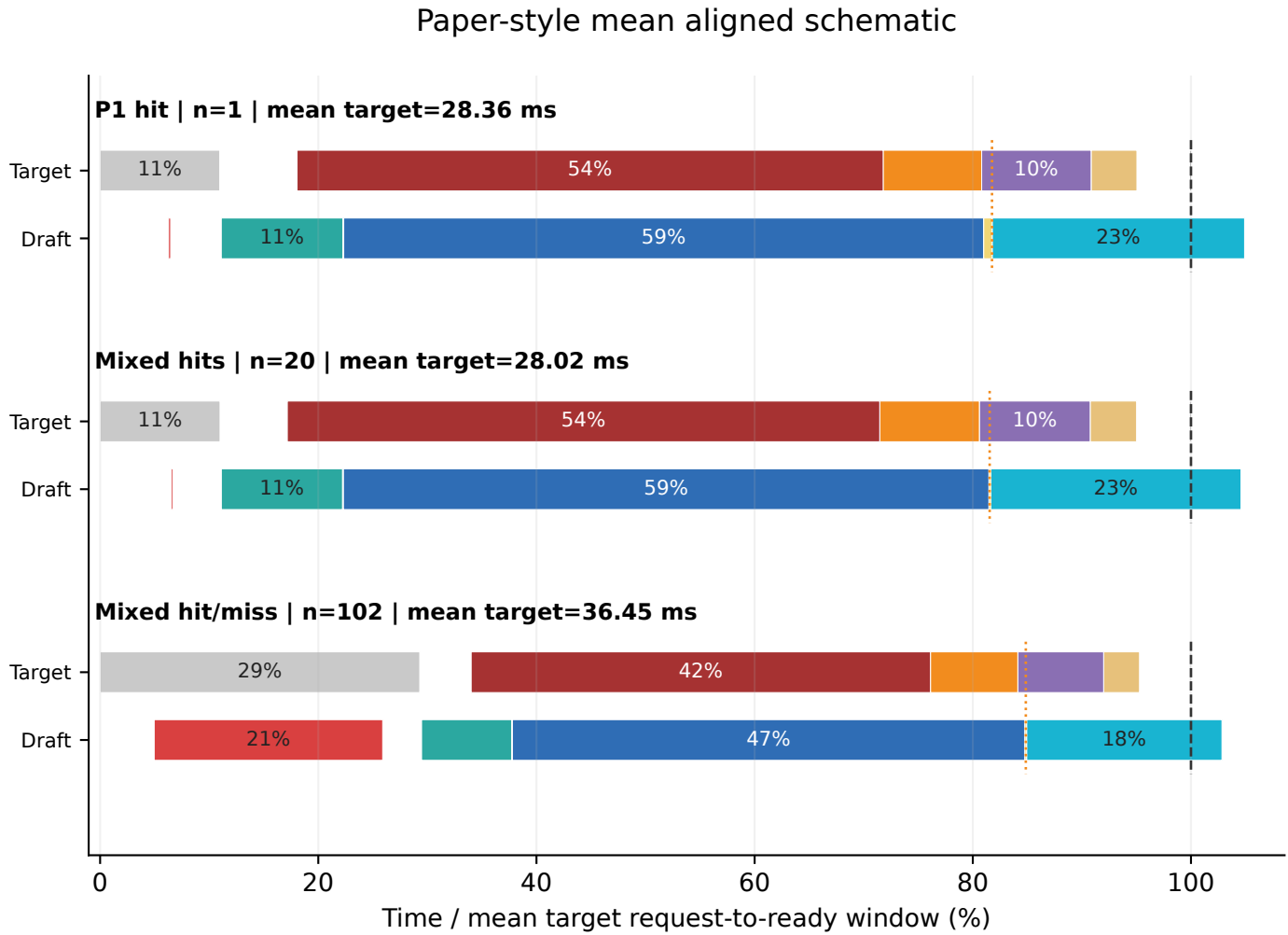
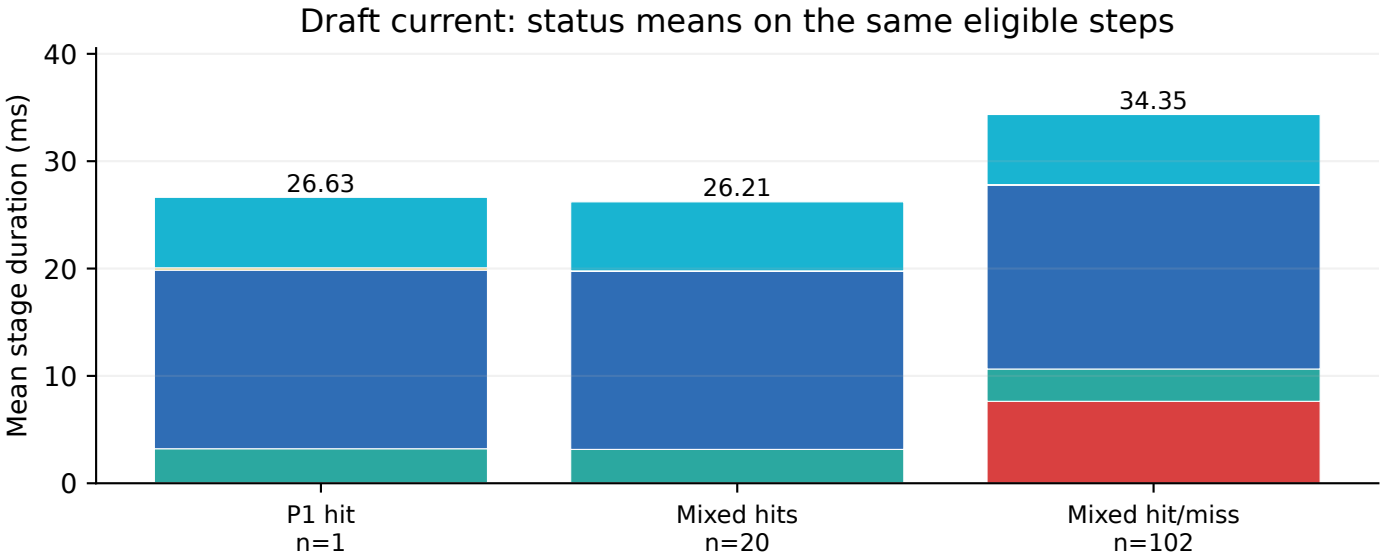
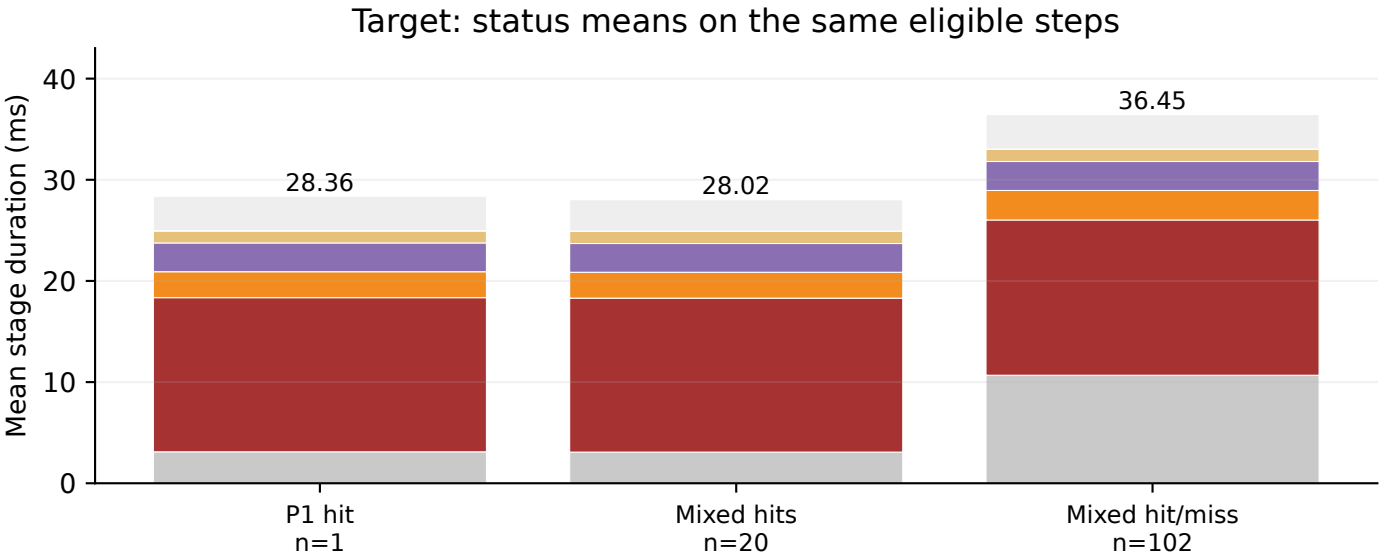
llama3_b8_tree_nodes_high
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=9/3, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 128 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

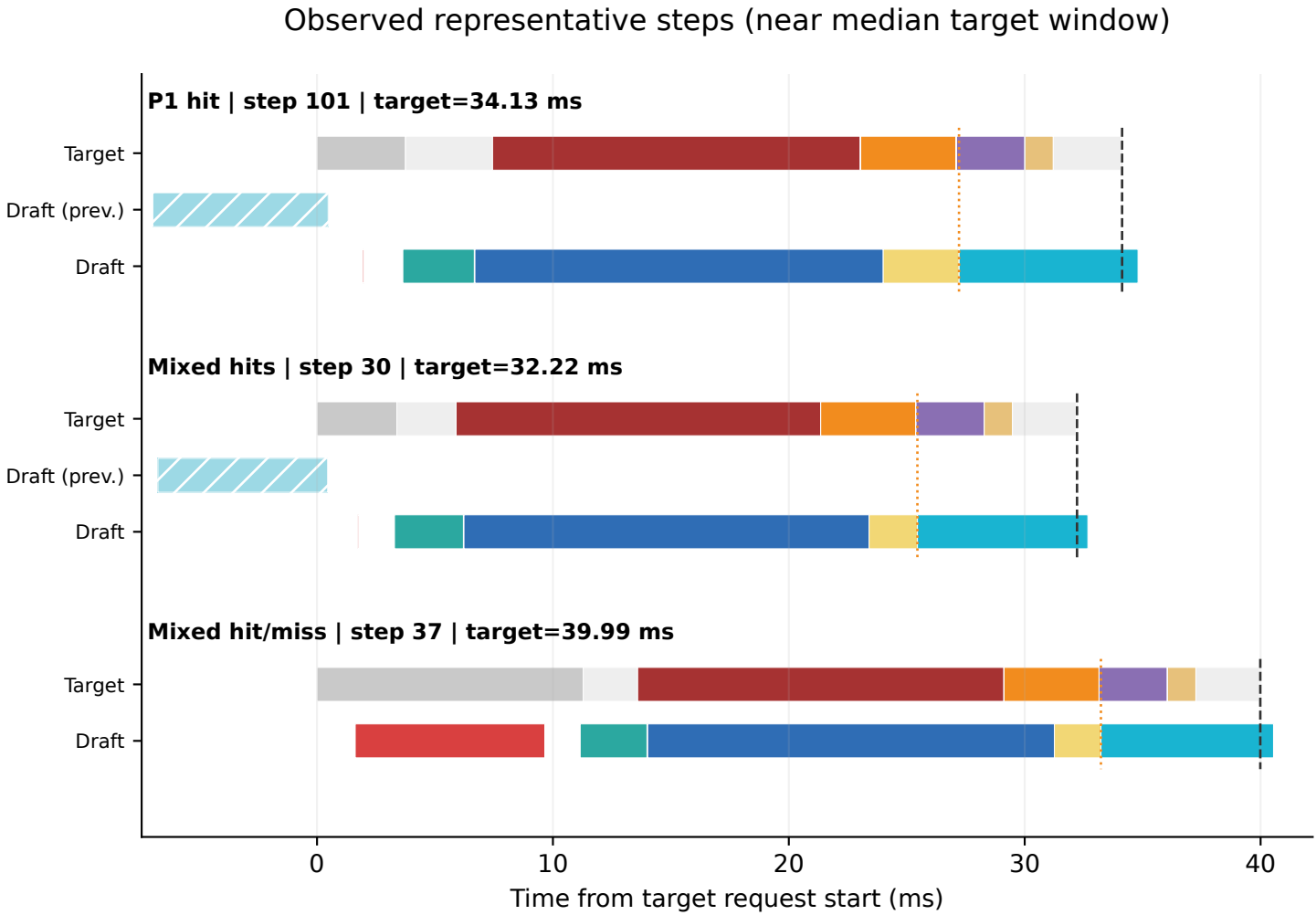
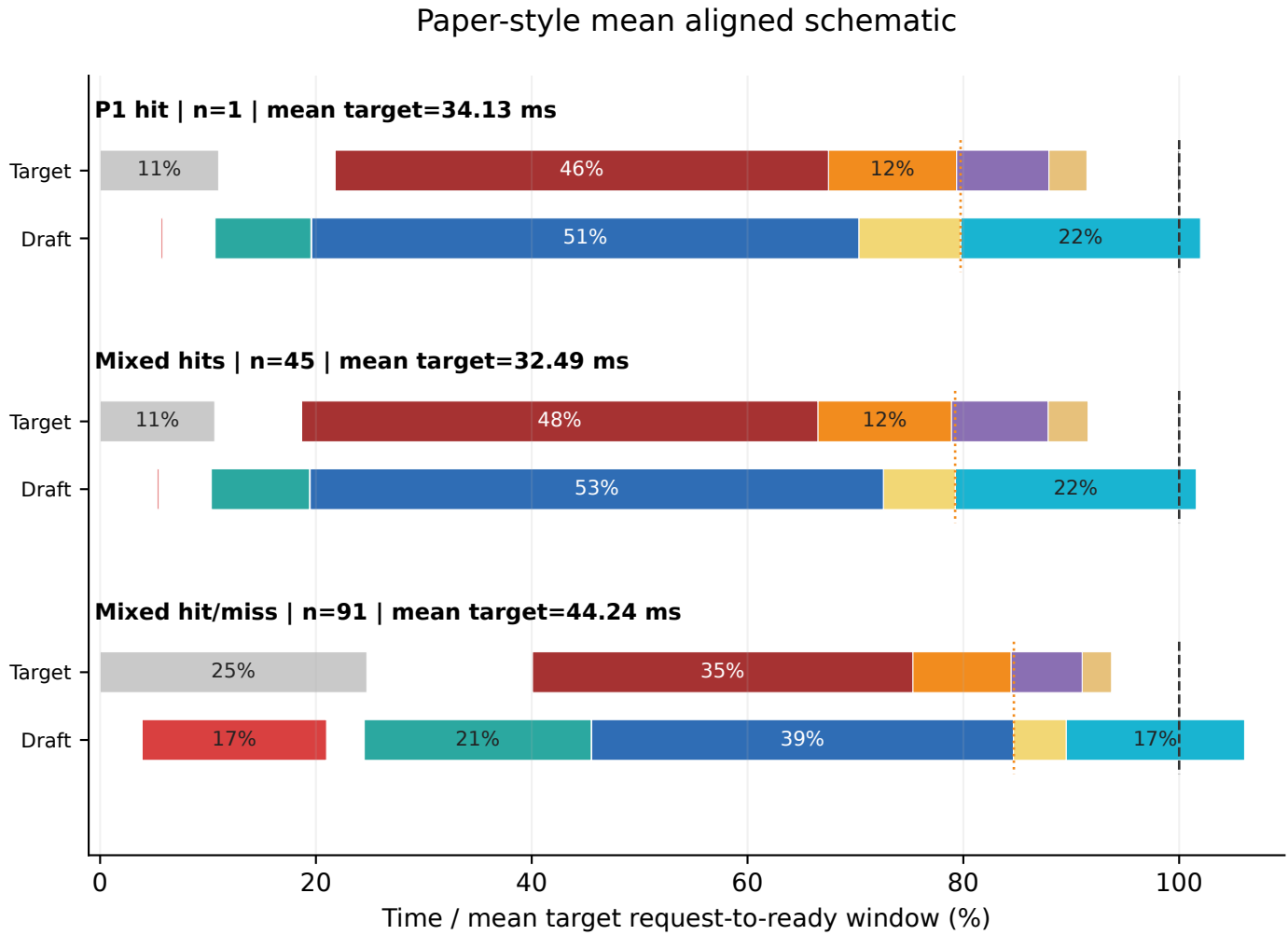
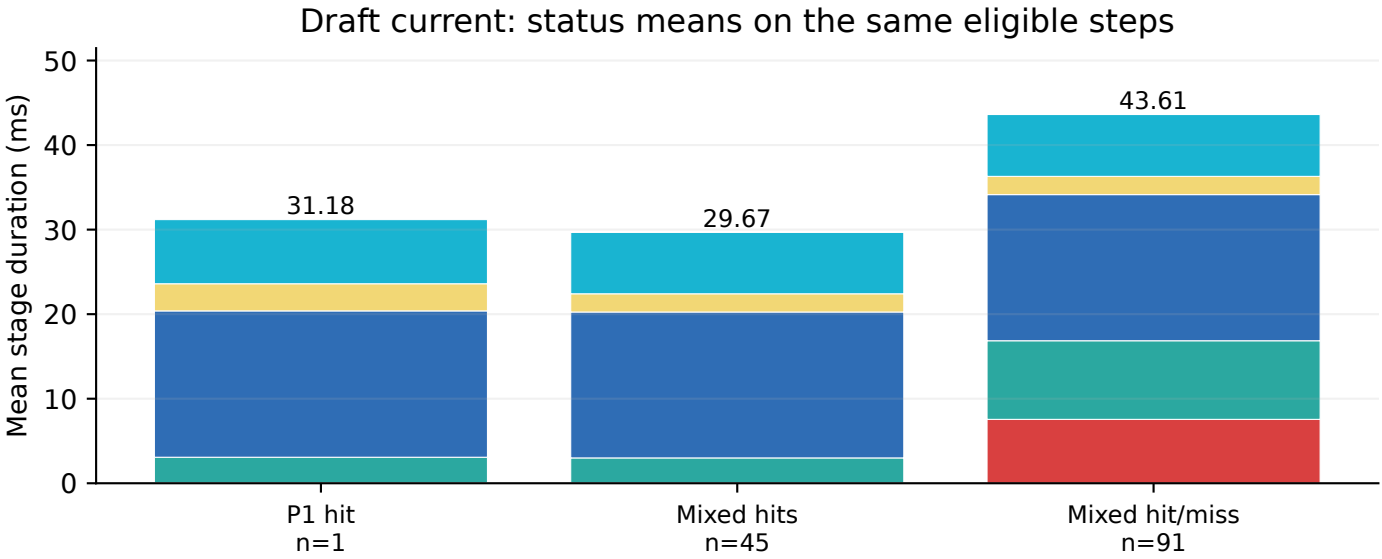
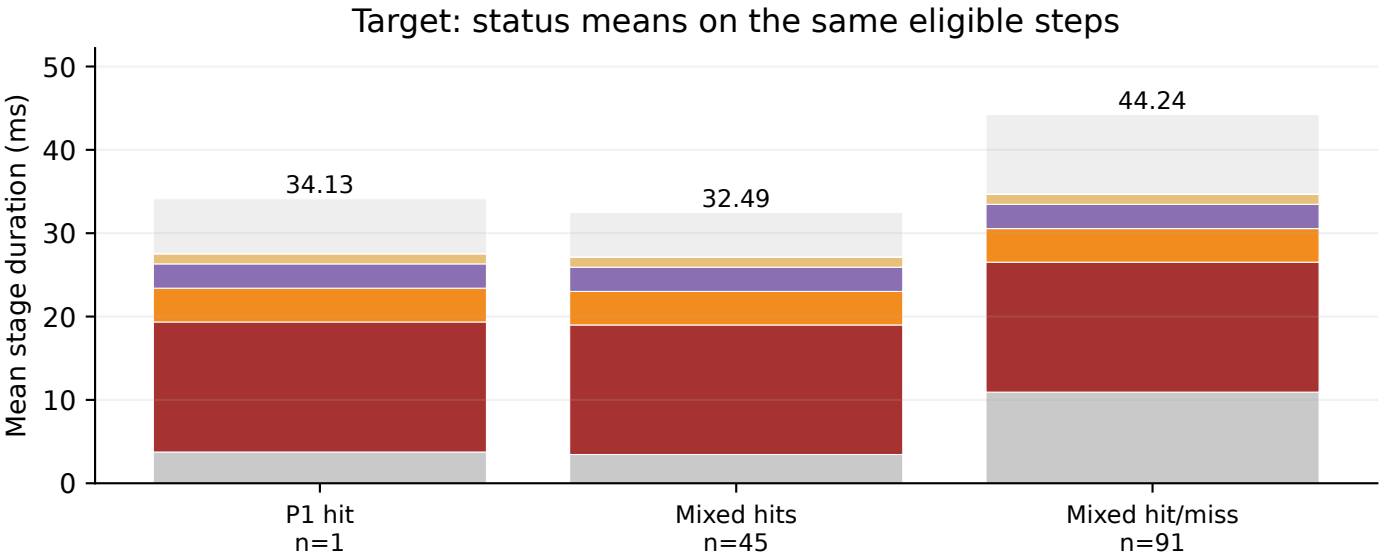
llama3_b8_tree_nodes_low
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=3/1, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 123 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

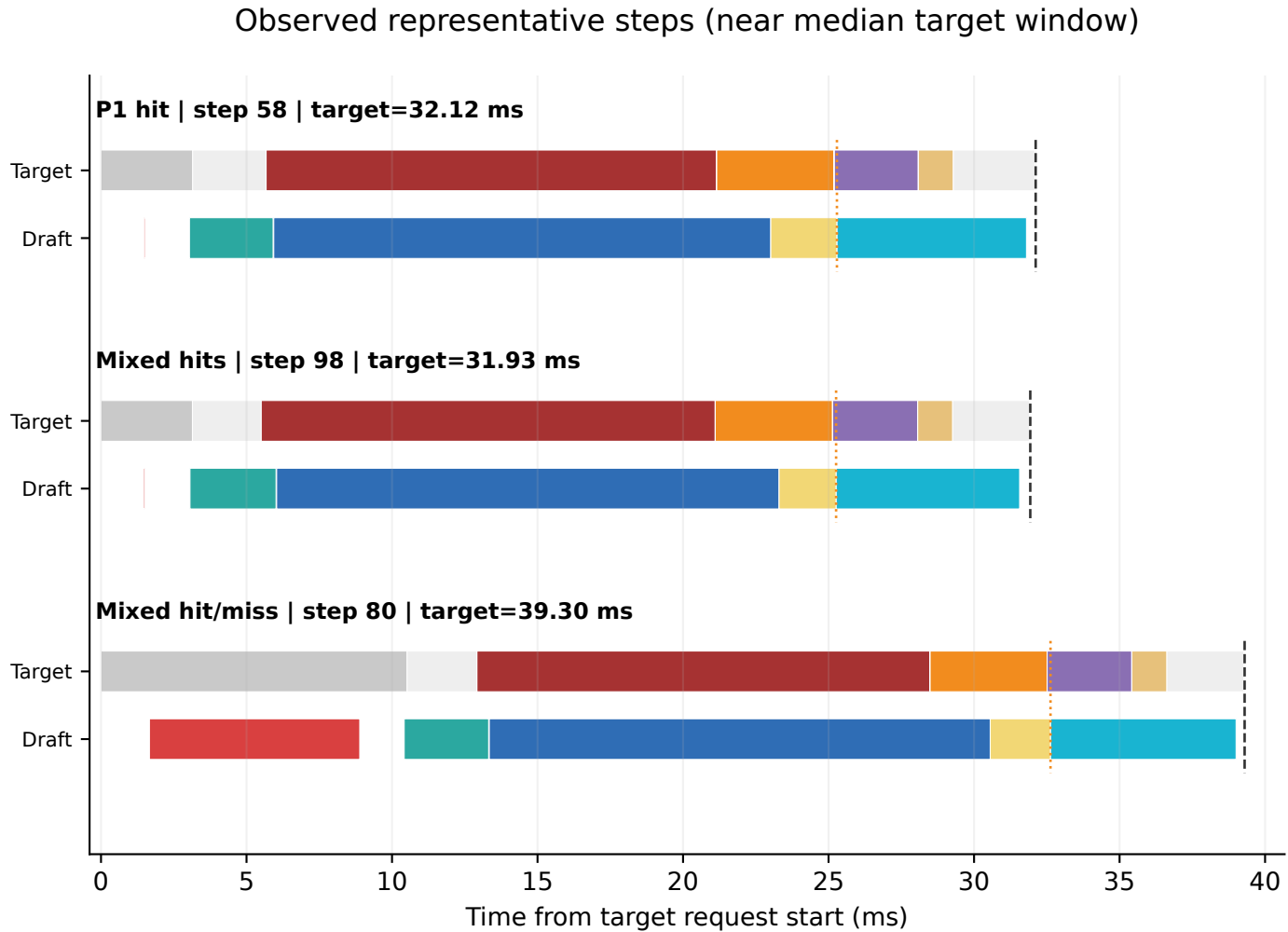
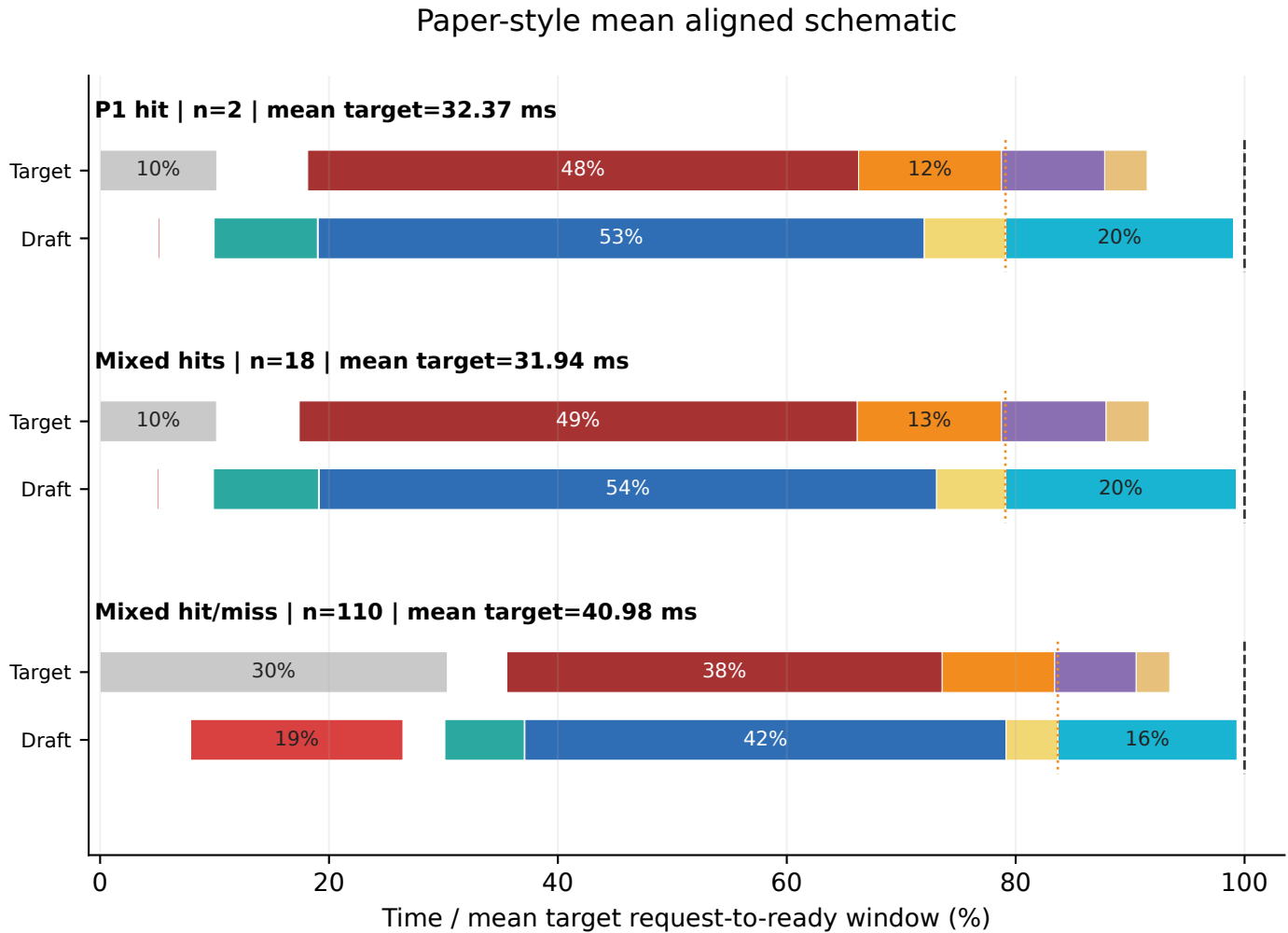
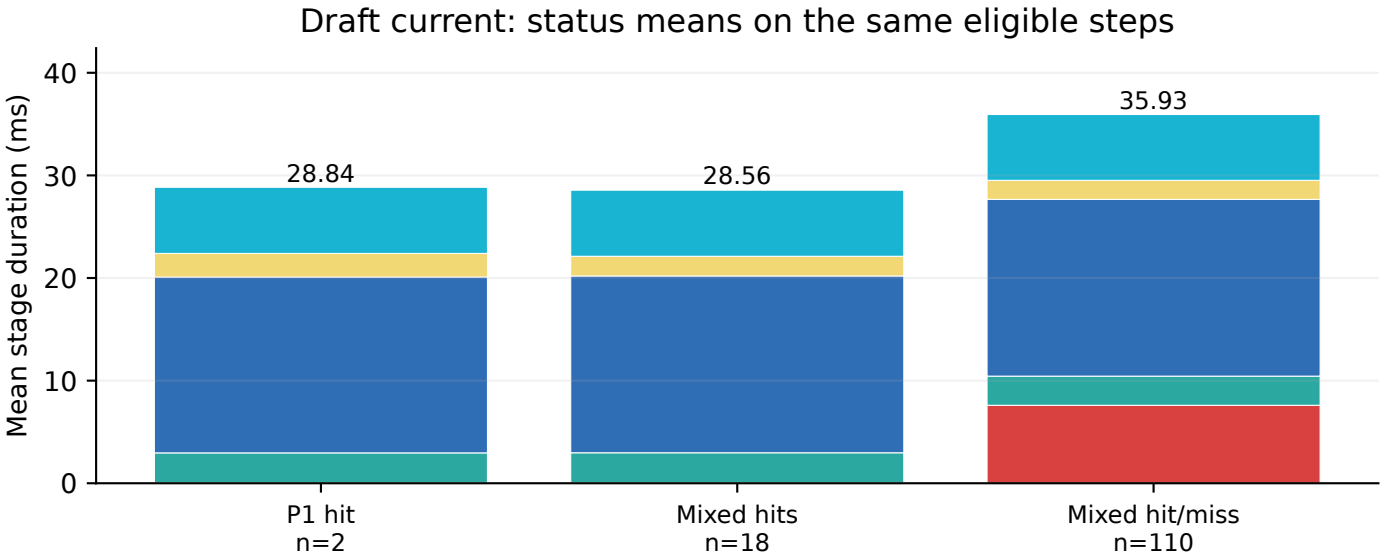
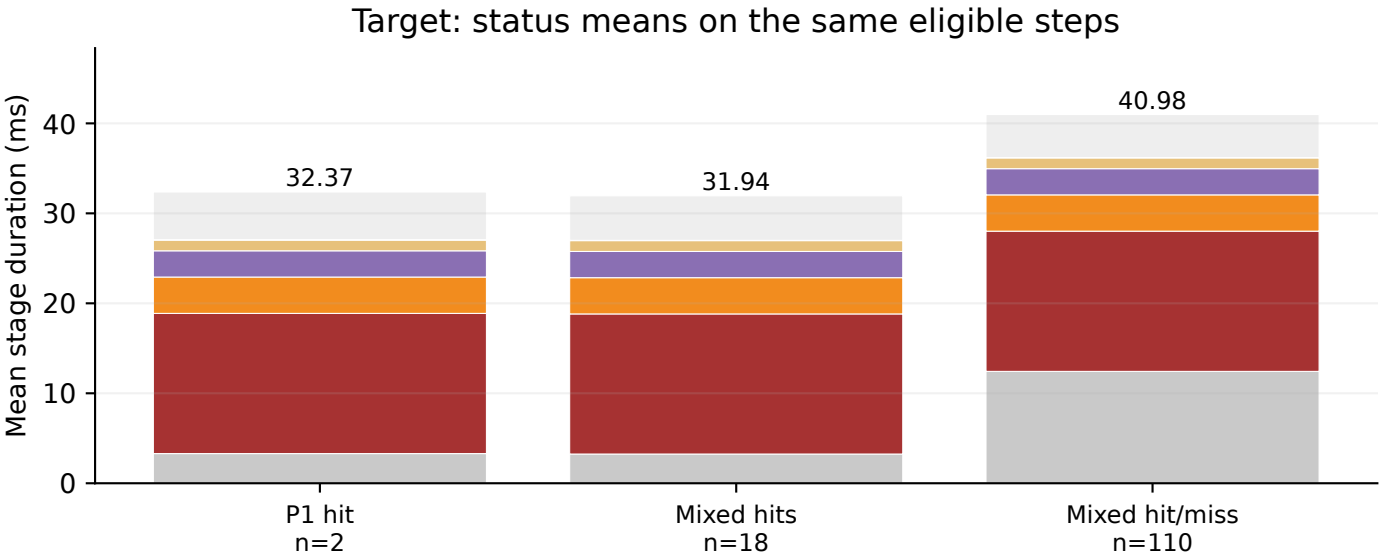
llama3_b8_tree_p2budget12
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=2, W=12, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 137 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

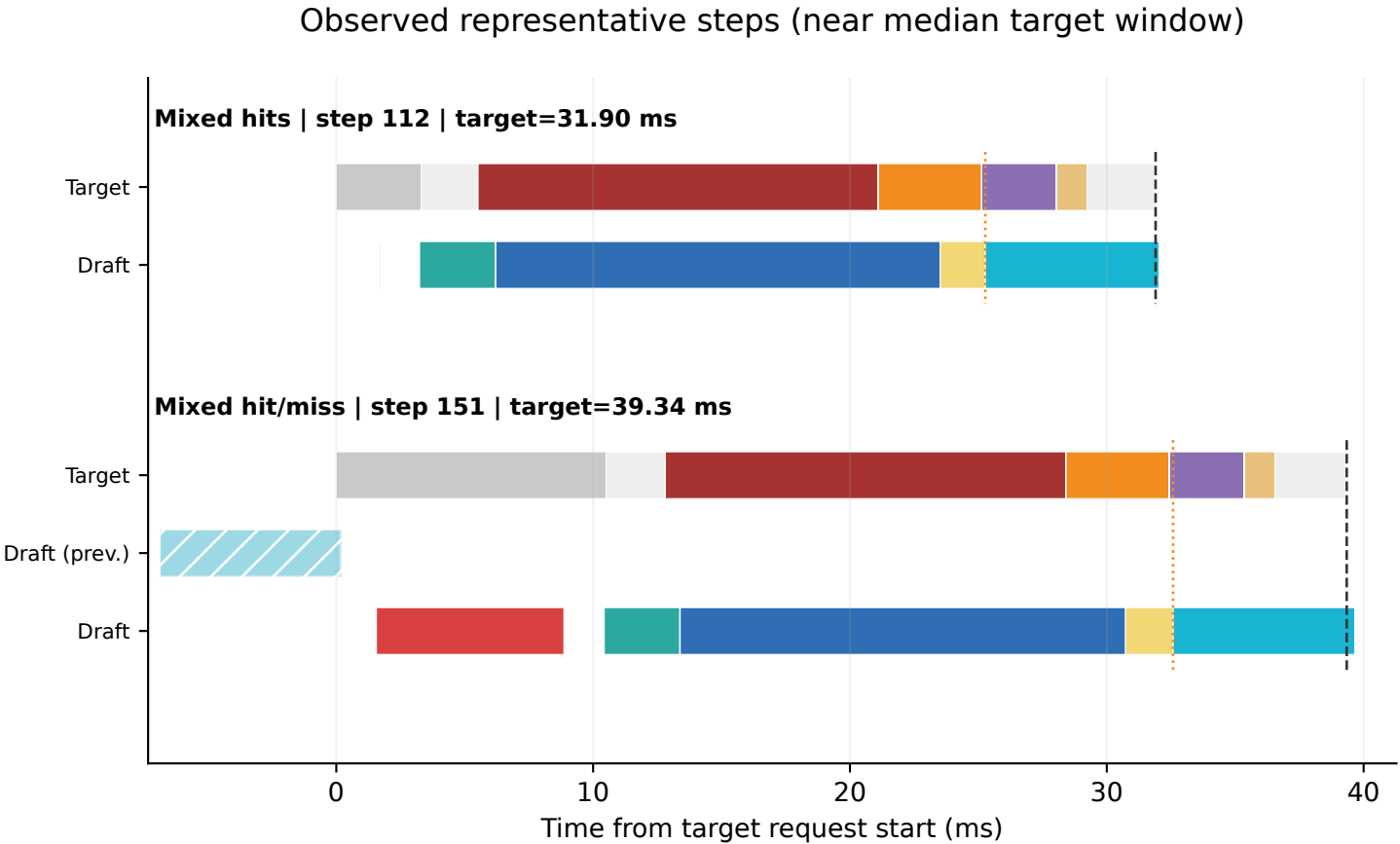
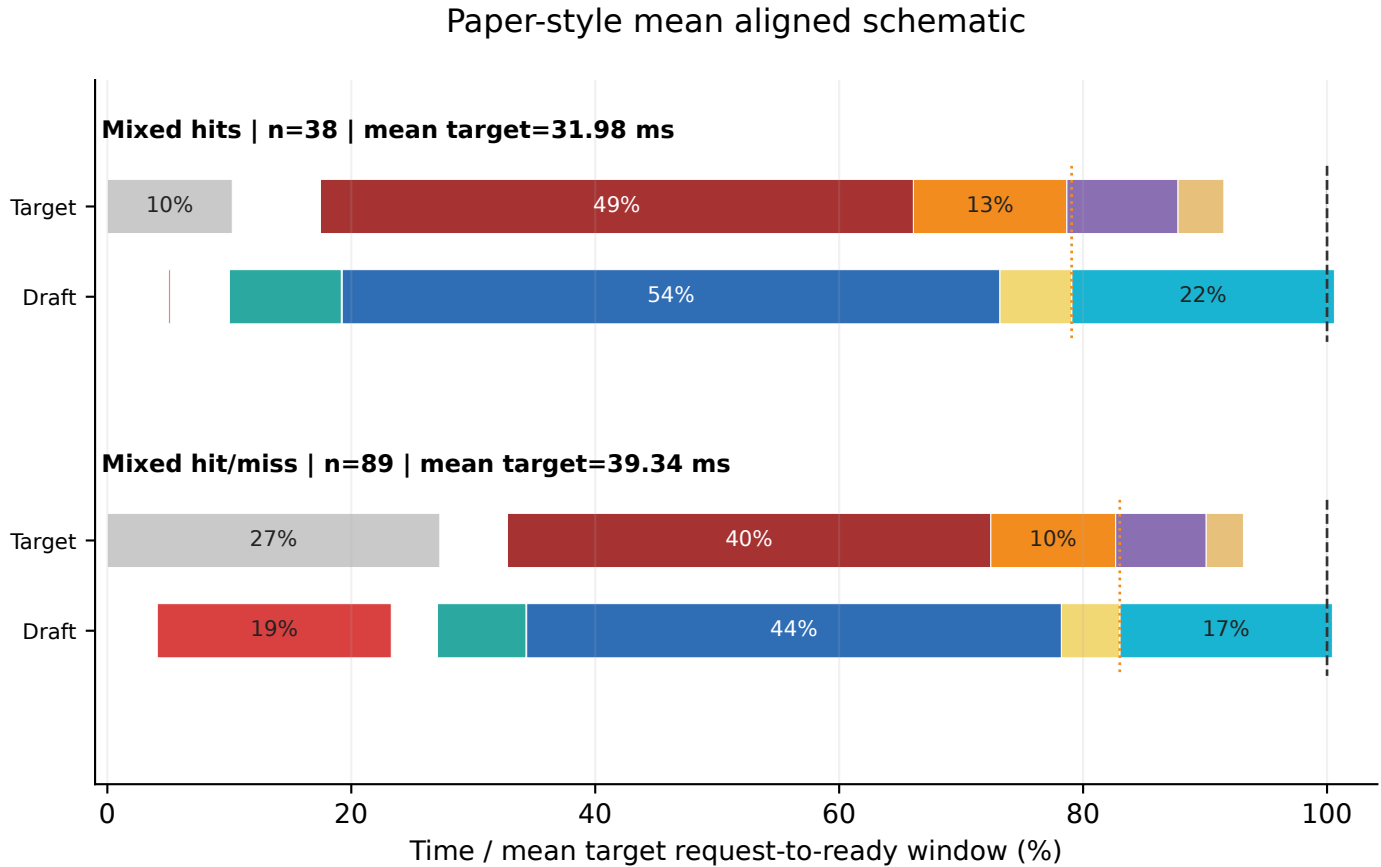
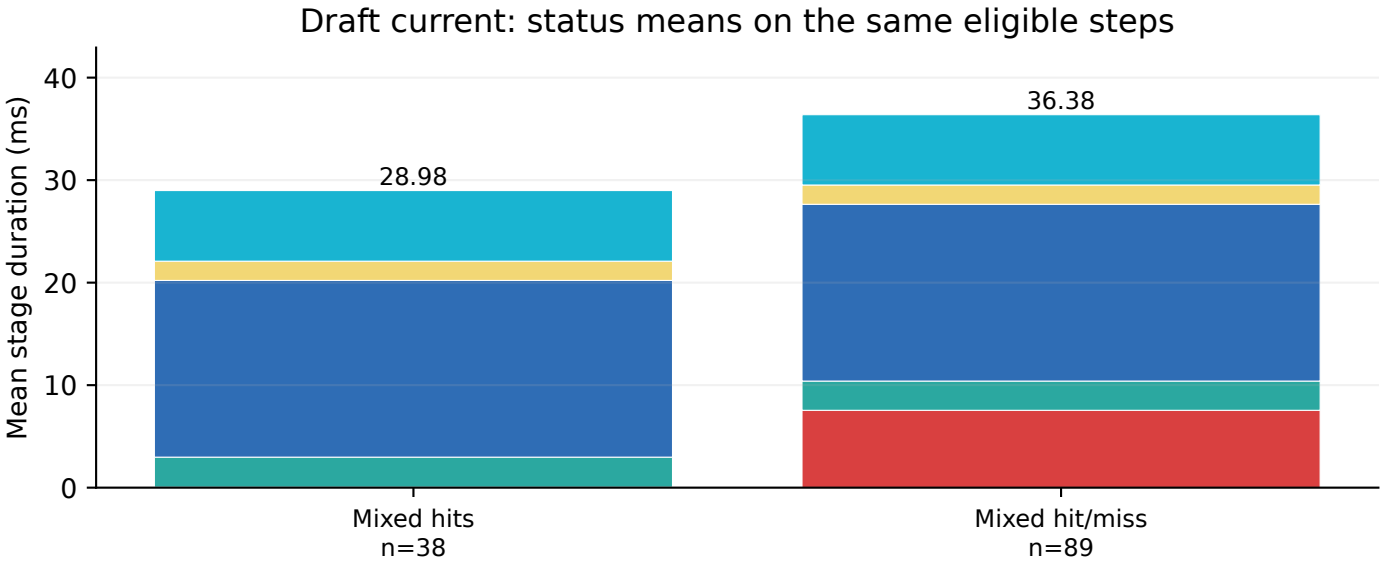
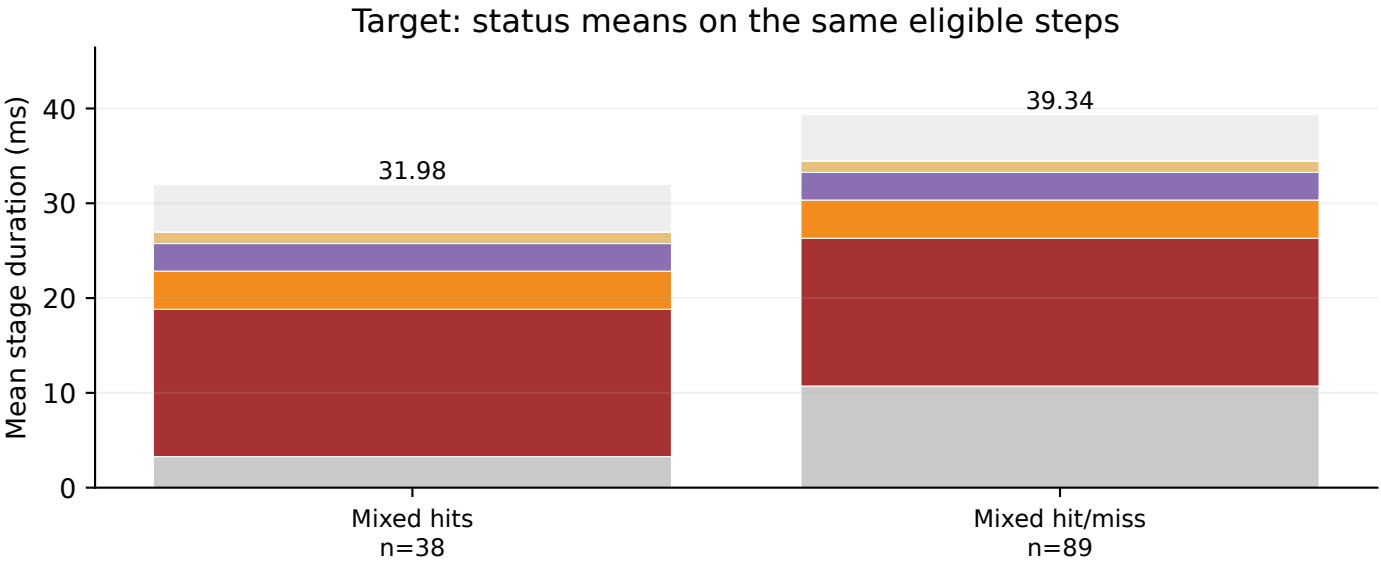
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_tree_p2budget3
DUET: K1/K2=3/1, exit=26 (0-based), $\bar{C}$ =3, N1/N2=6/2, U=2, W=3, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 130 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

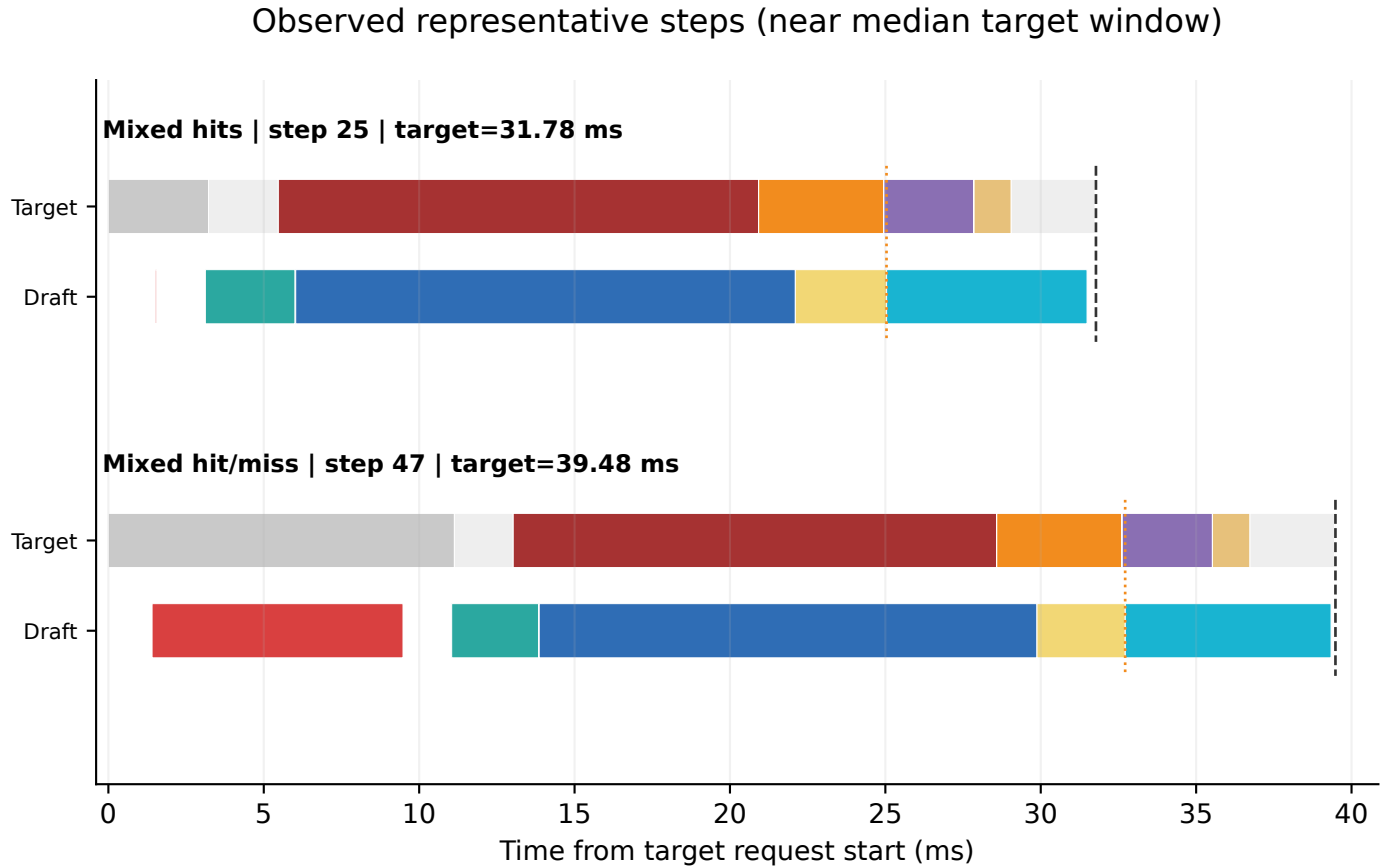
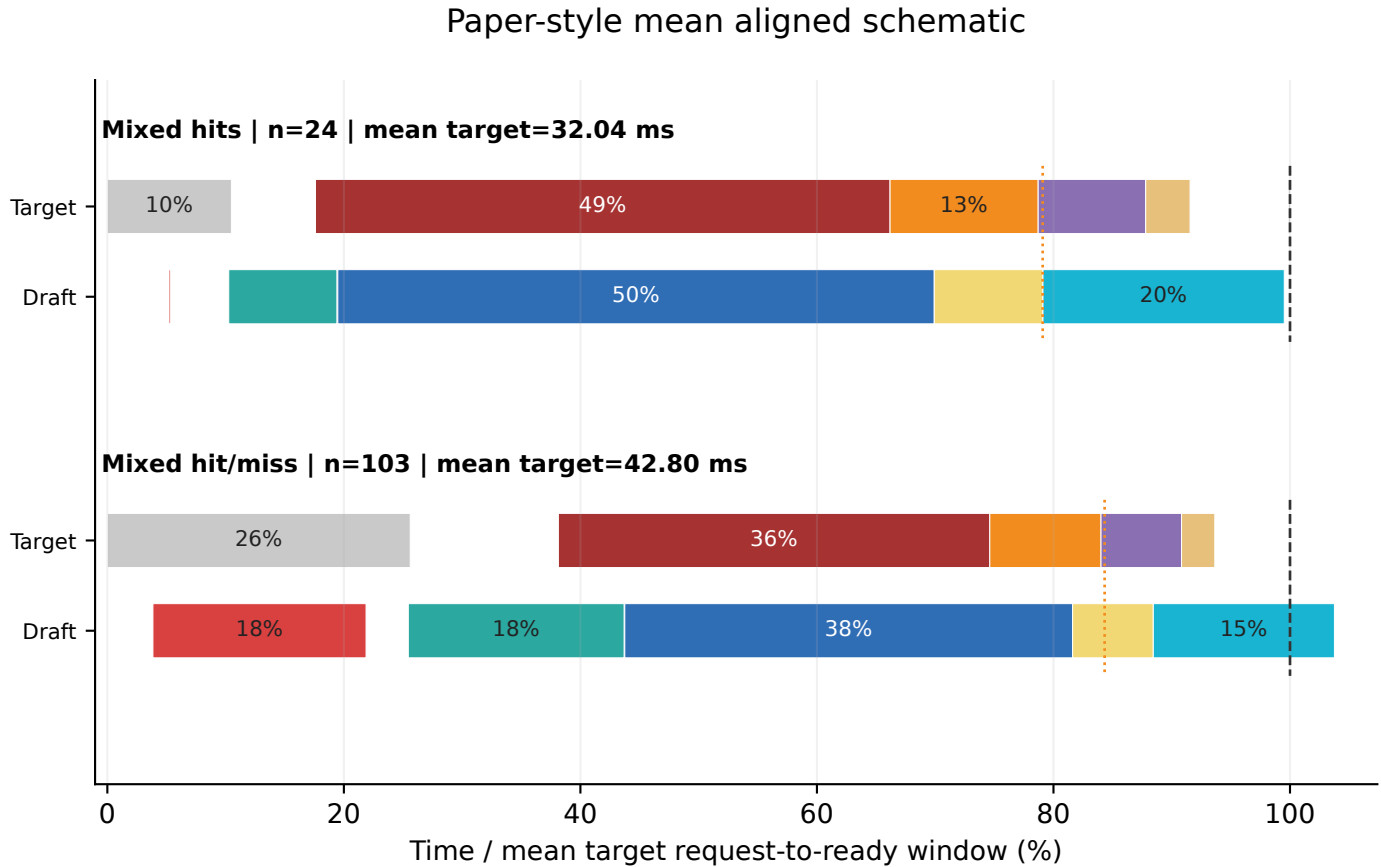
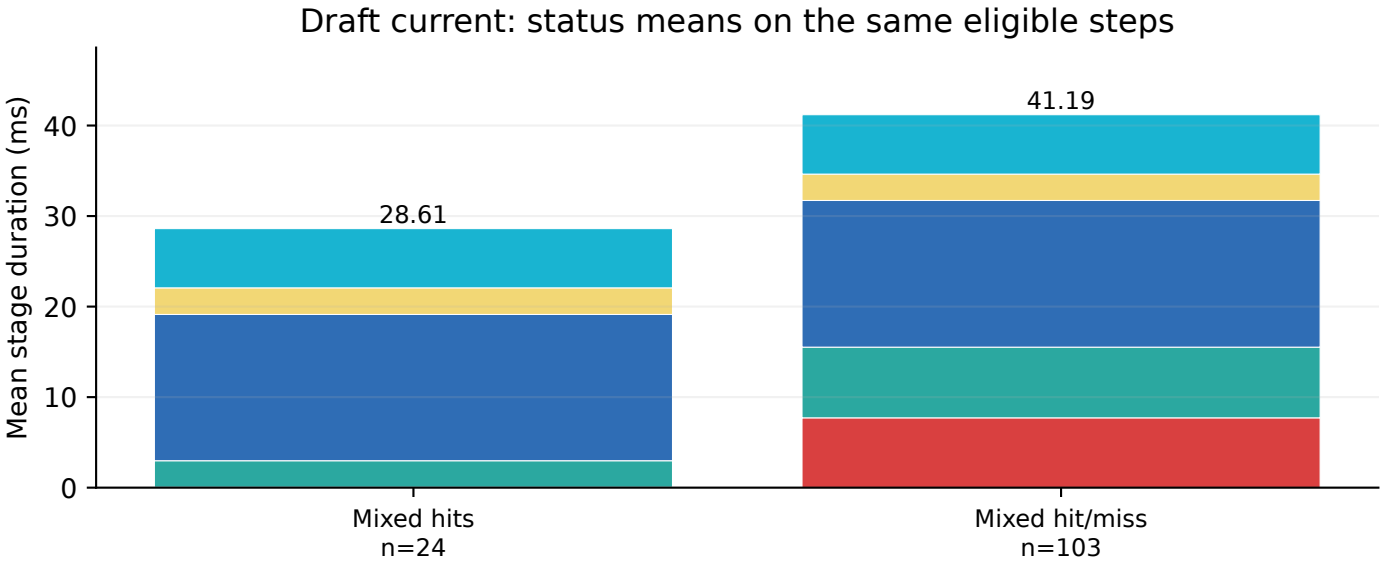
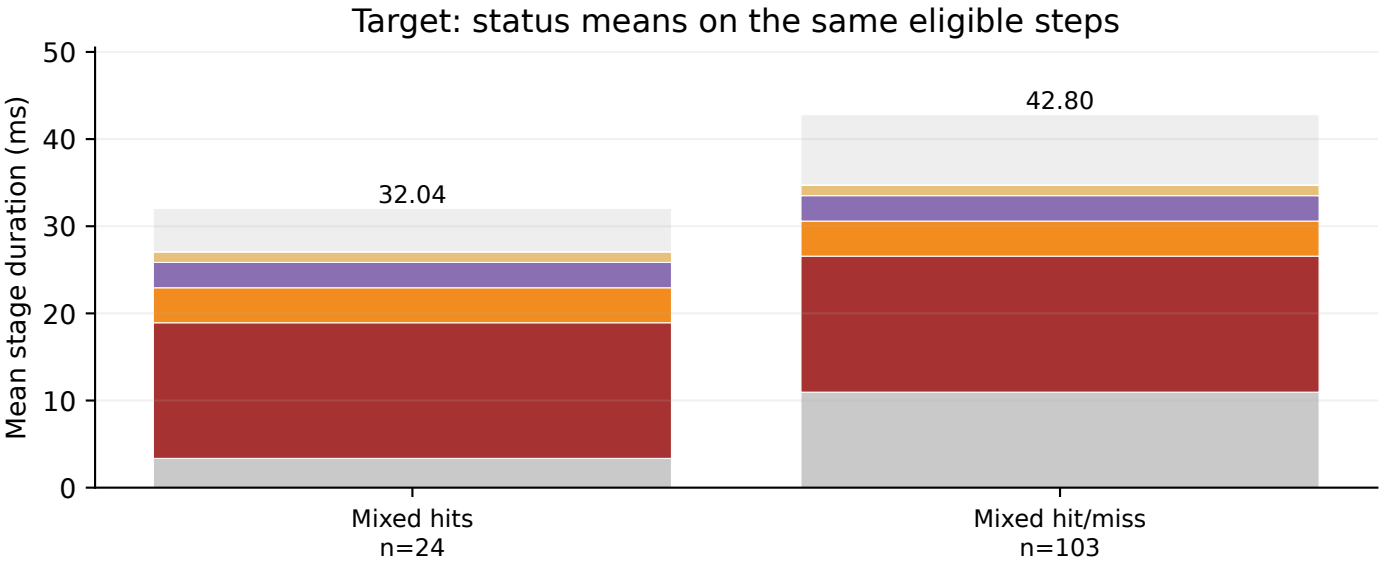
llama3_b8_tree_p2budget8
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=2, W=8, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 127 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

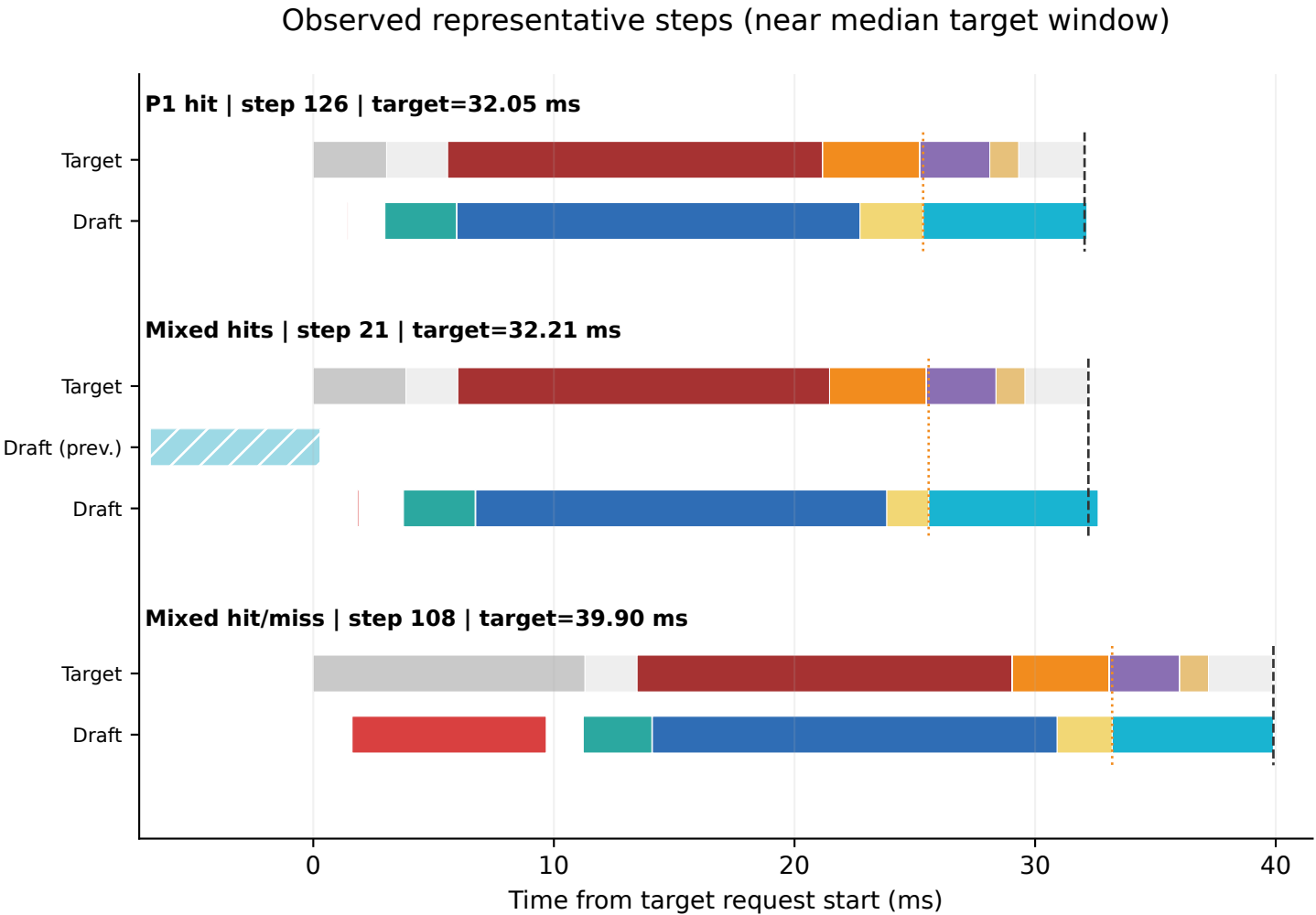
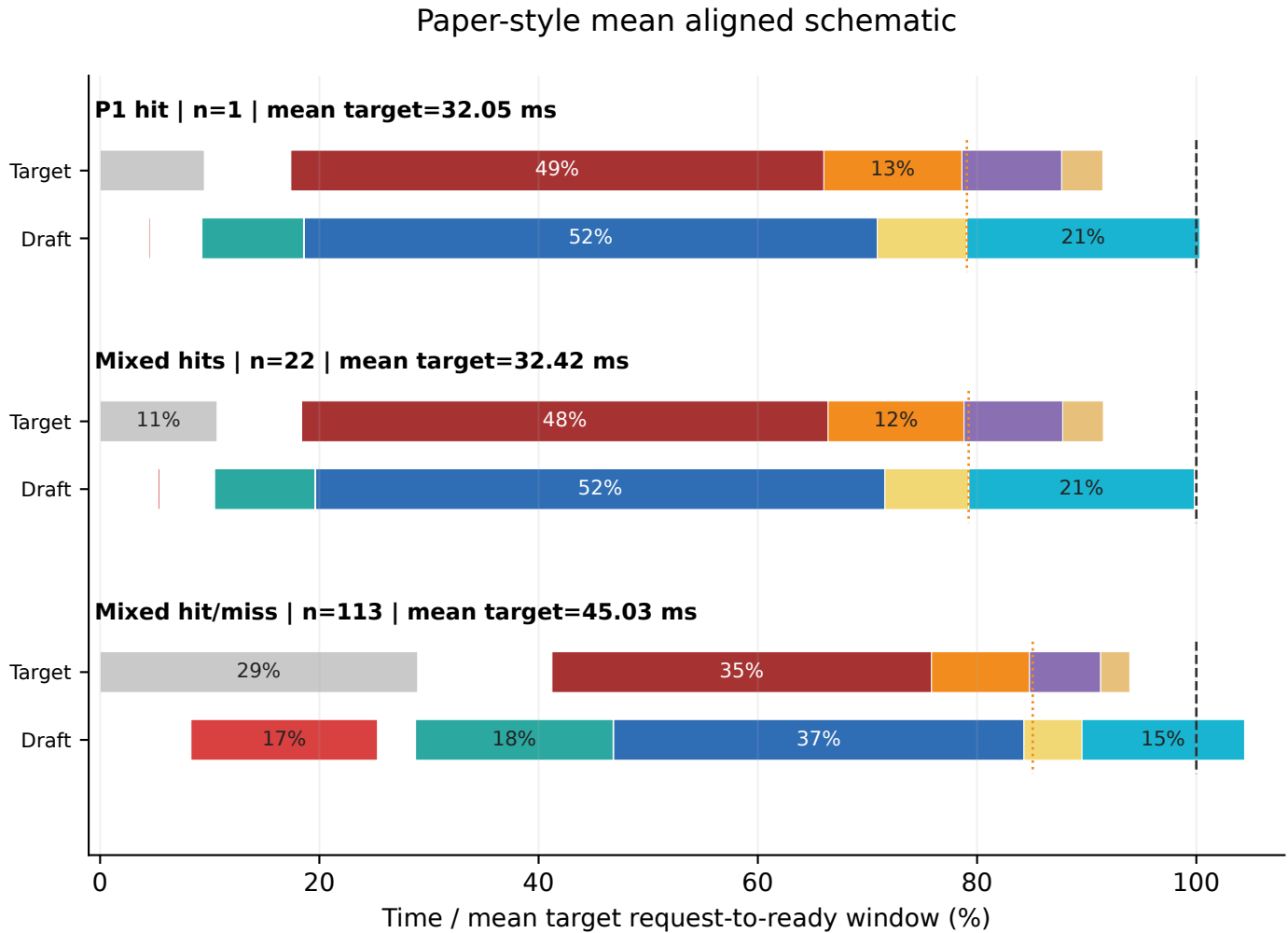
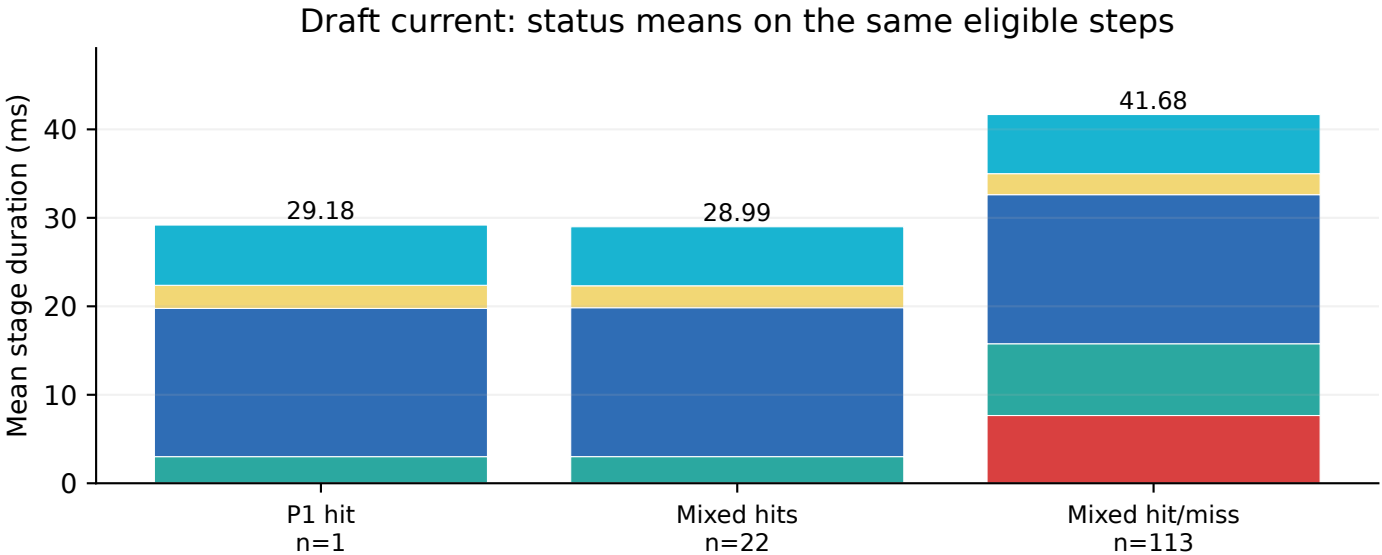
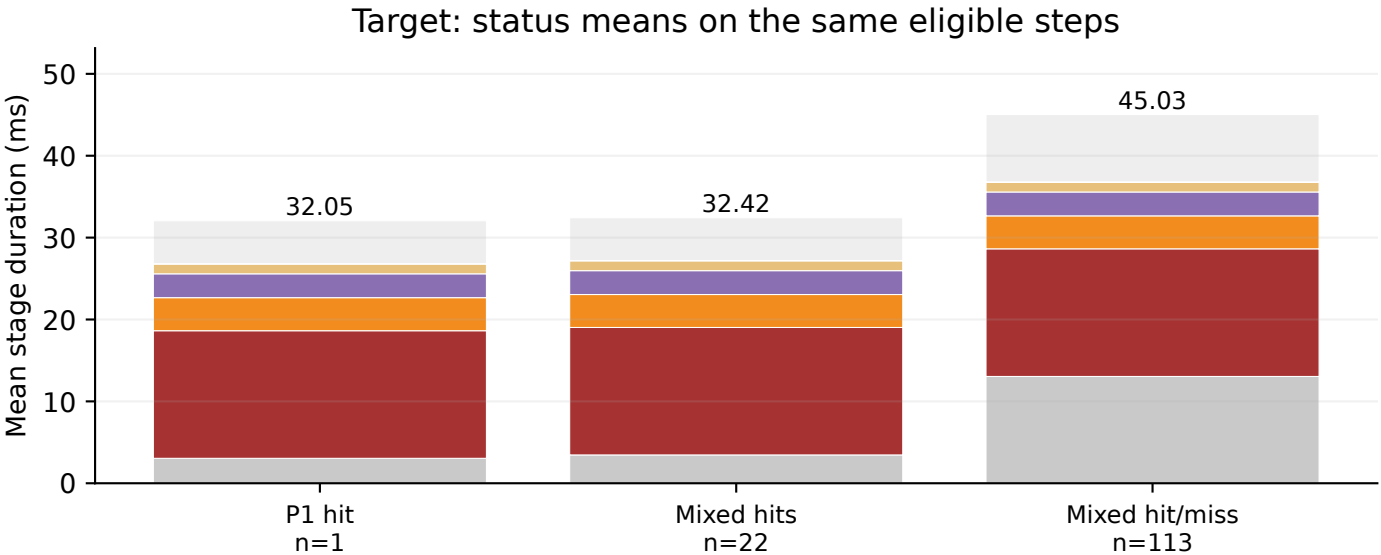
llama3_b8_tree_roots1_compute1
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=1, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 127 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

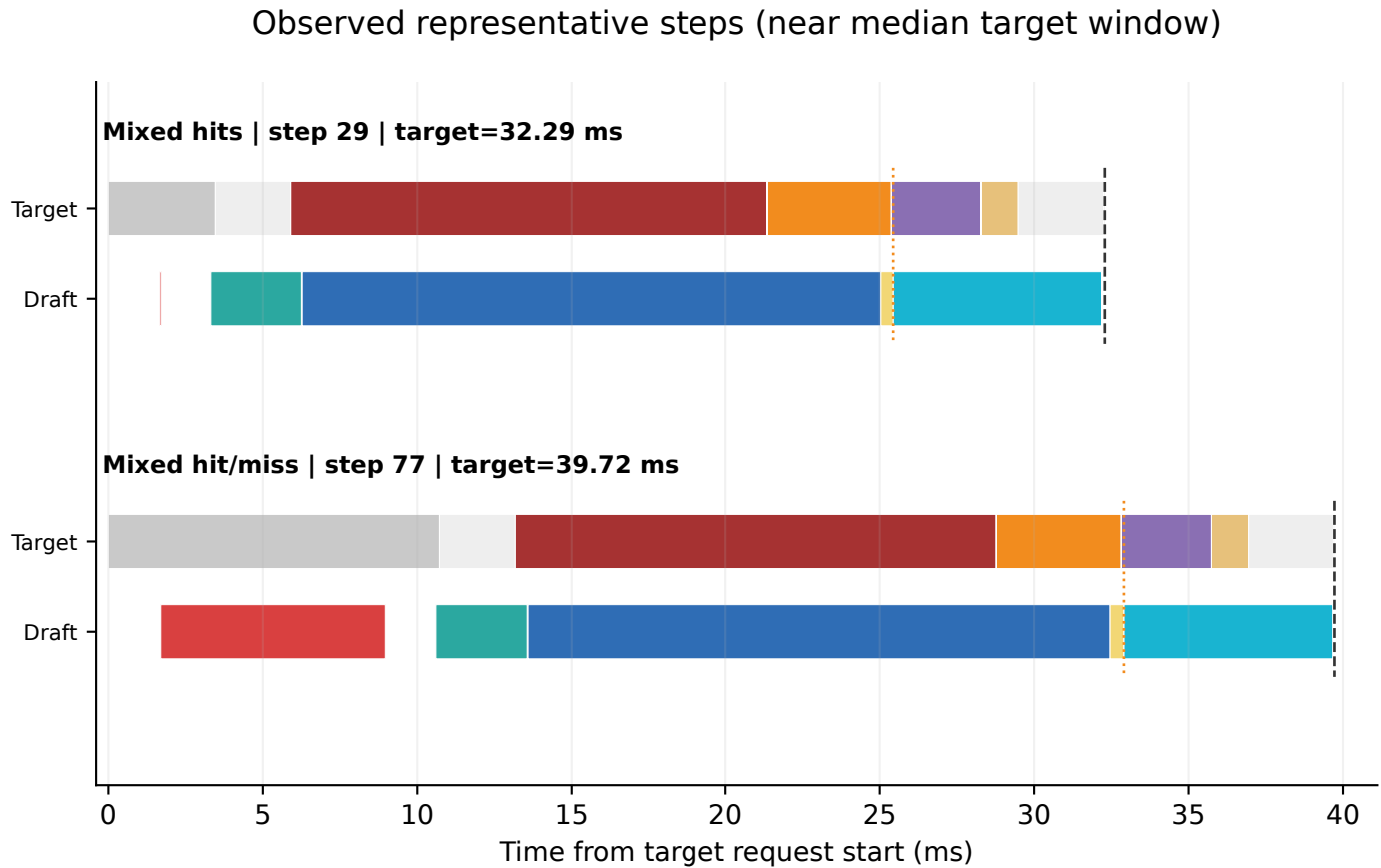
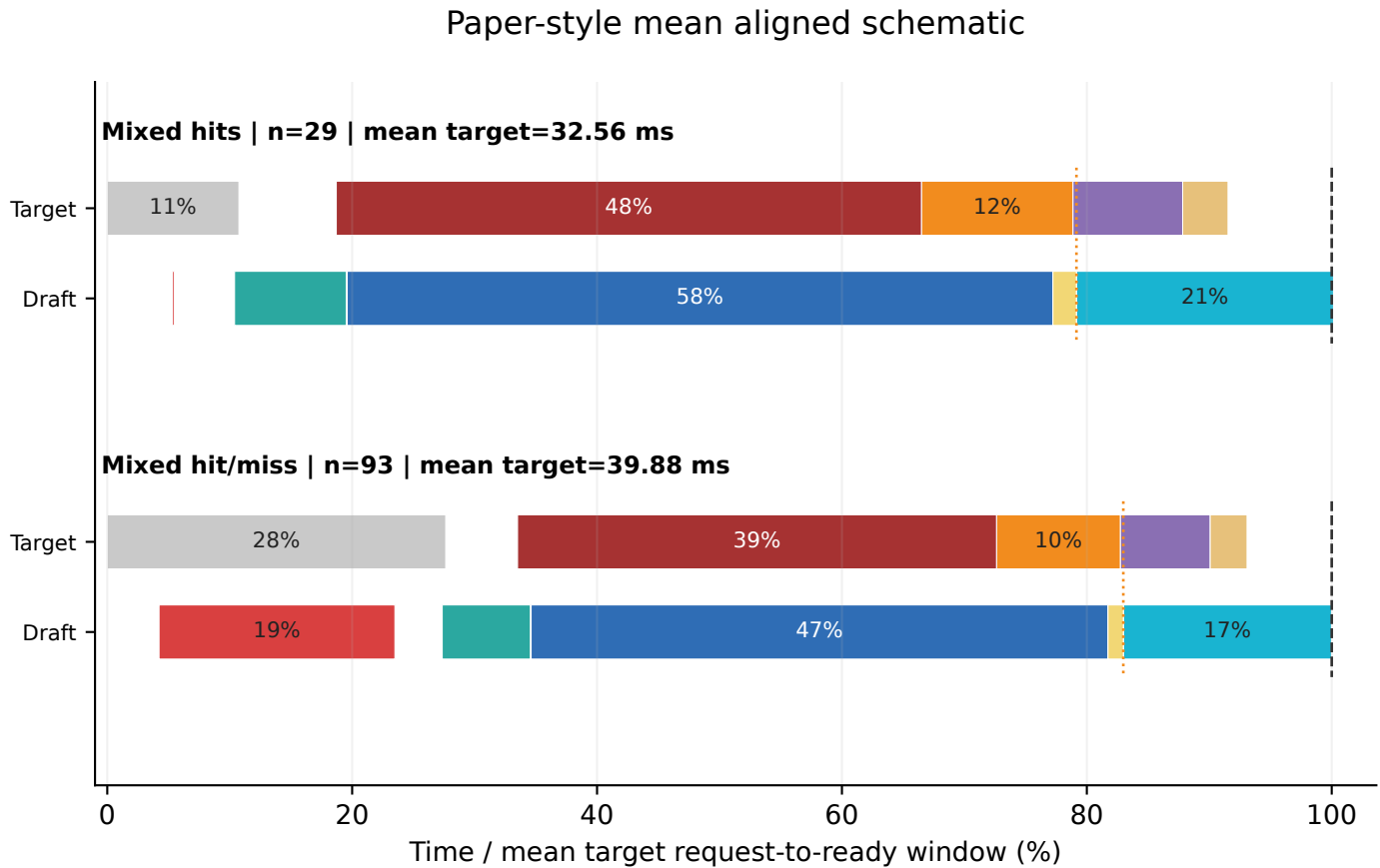
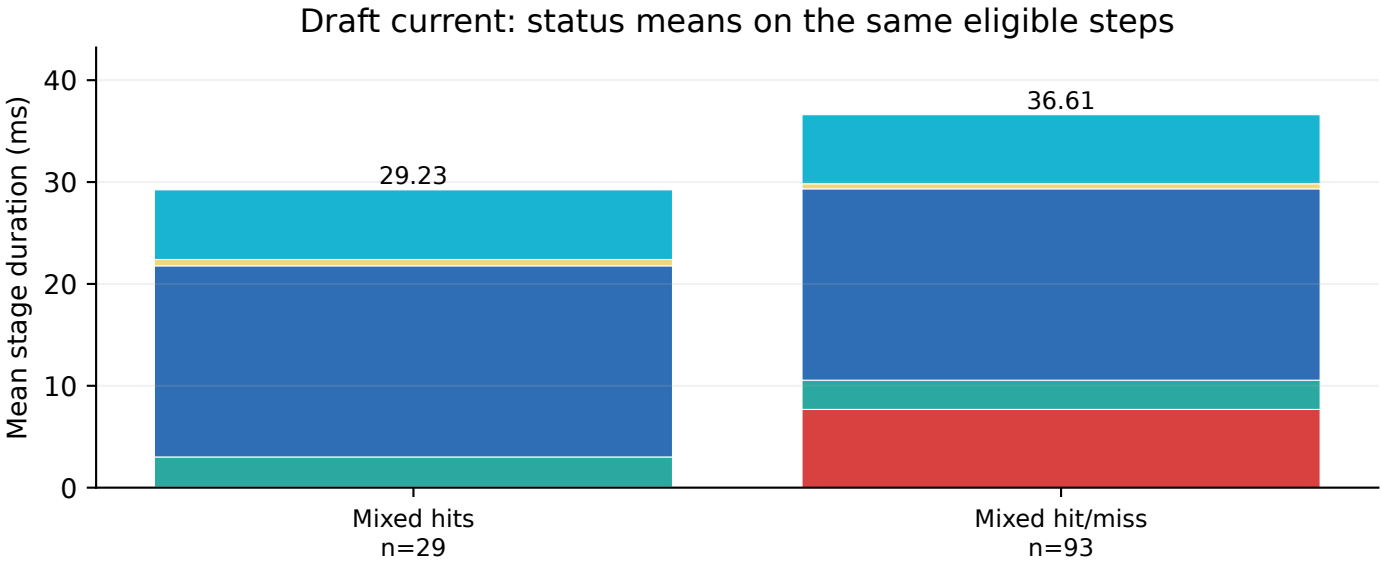
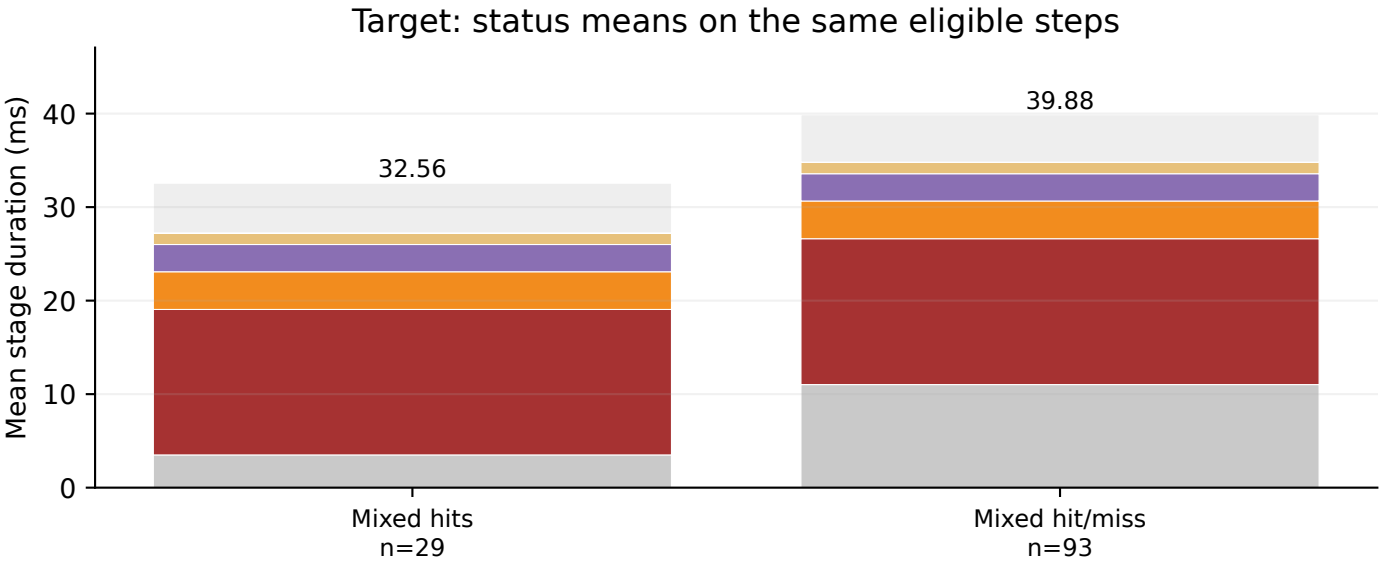
llama3_b8_tree_roots2_compute1
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 136 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

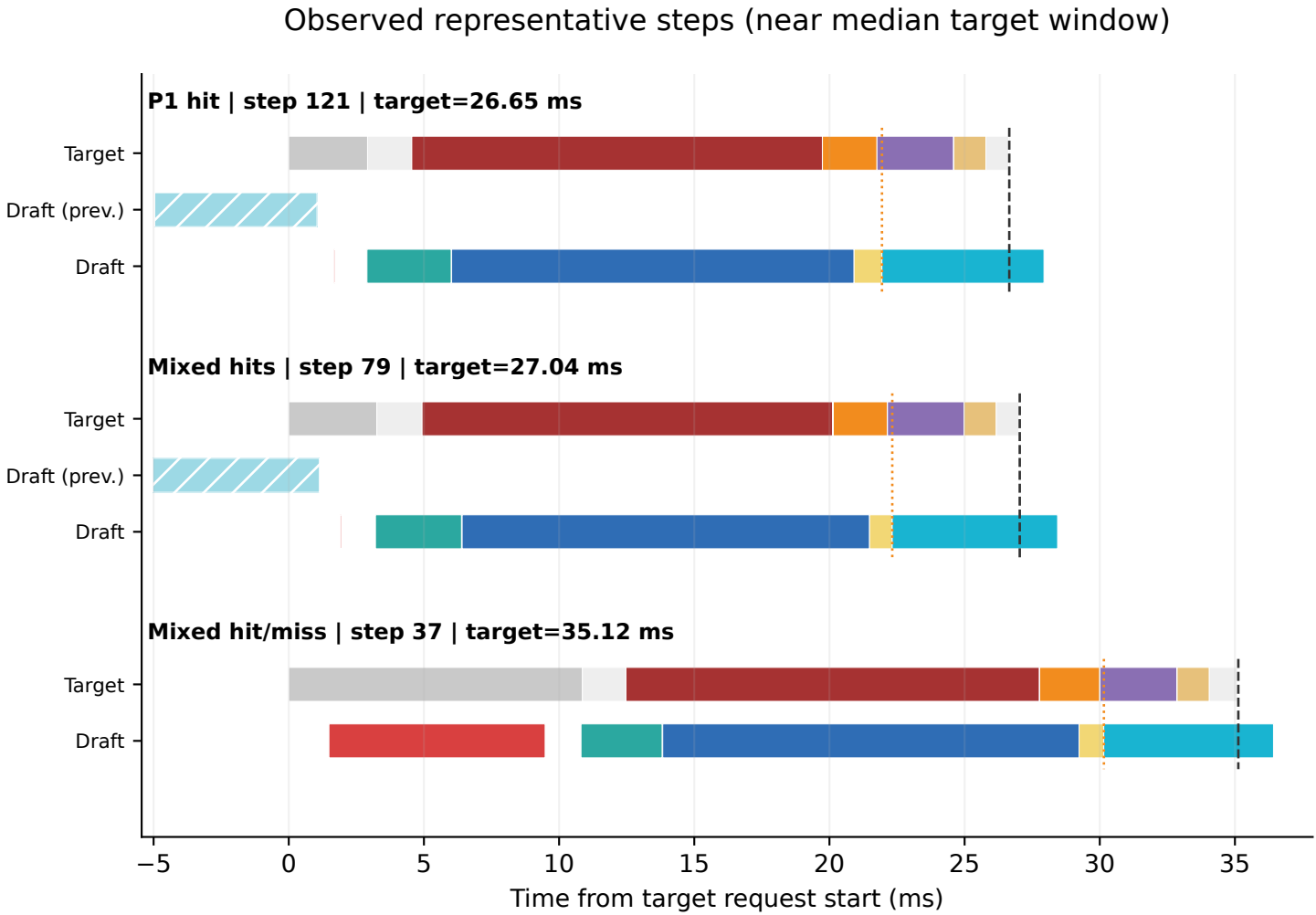
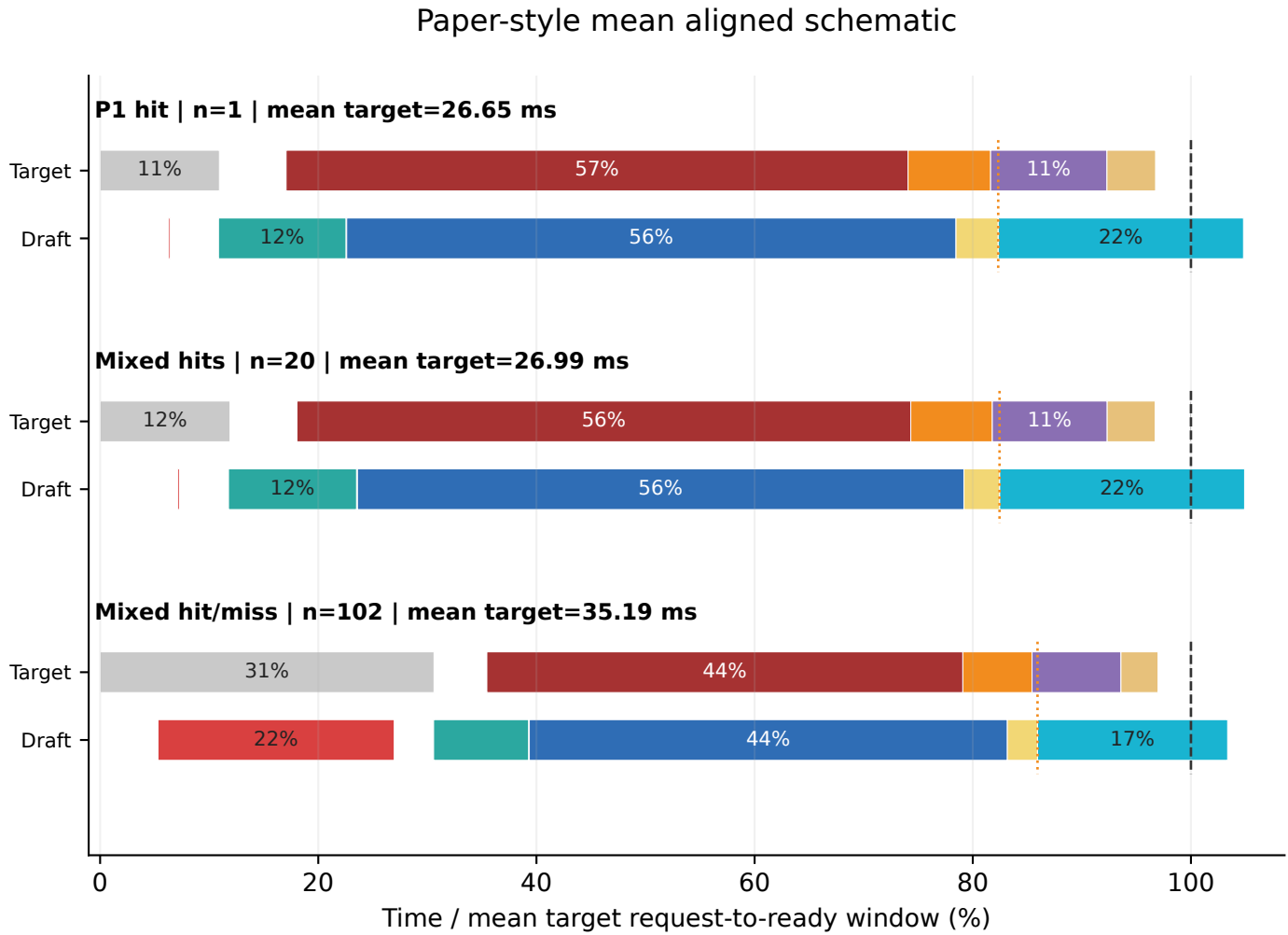
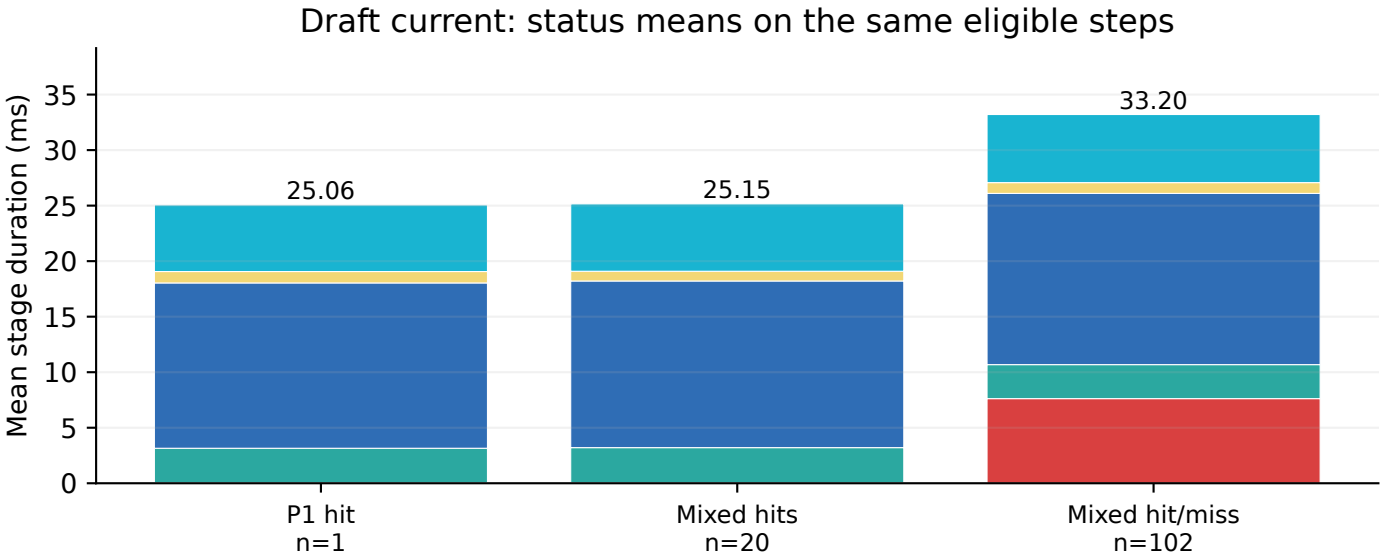
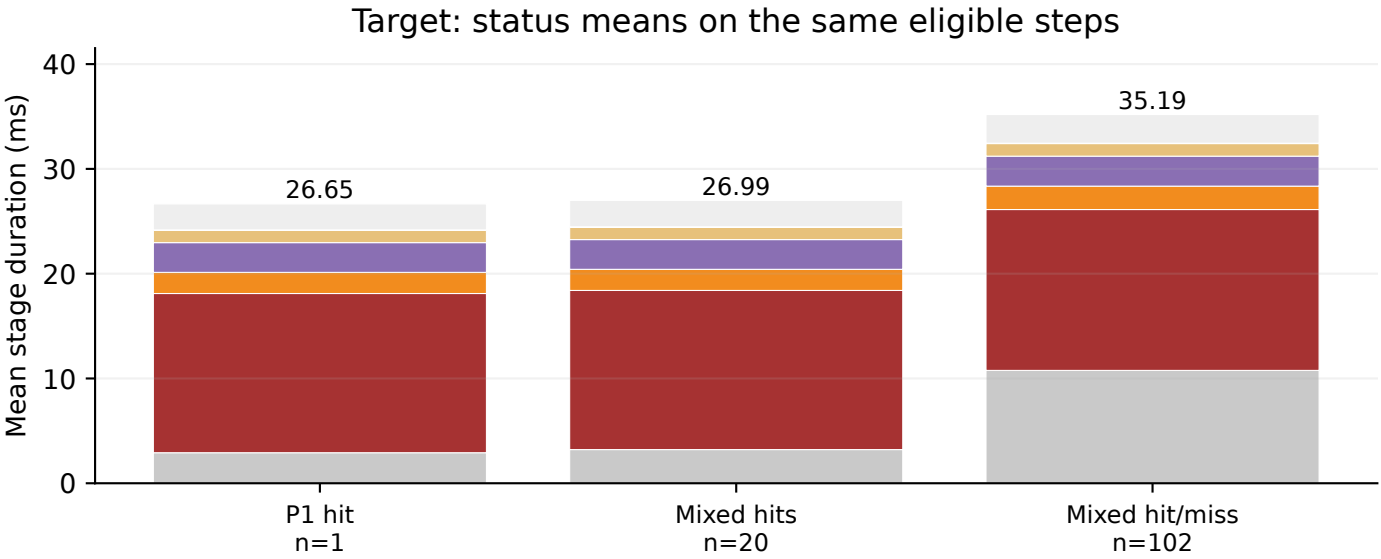
llama3_b8_tree_roots3
DUET: K1/K2=3/1, exit=26 (0-based), C=3, N1/N2=6/2, U=3, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 122 eligible steps



Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

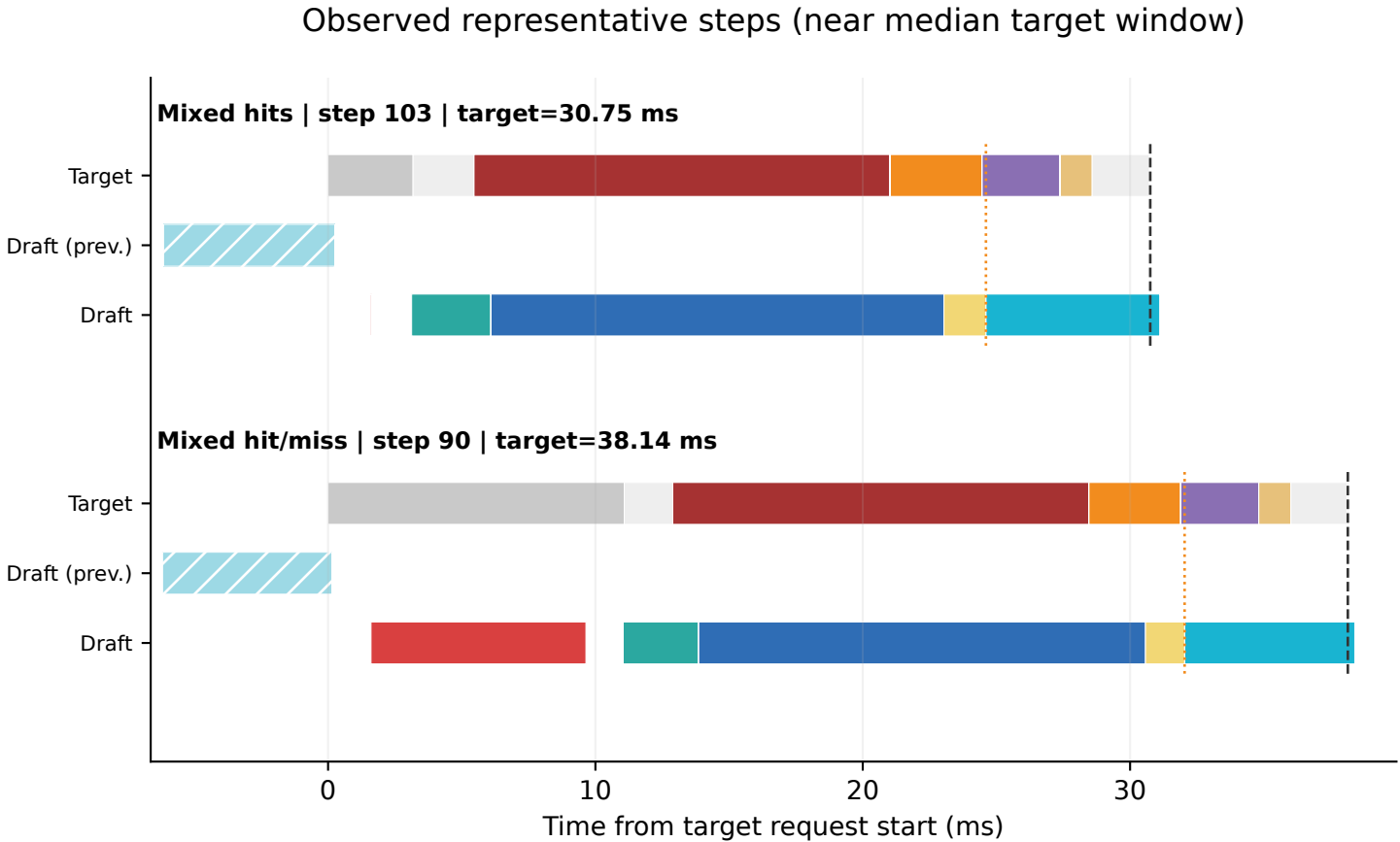
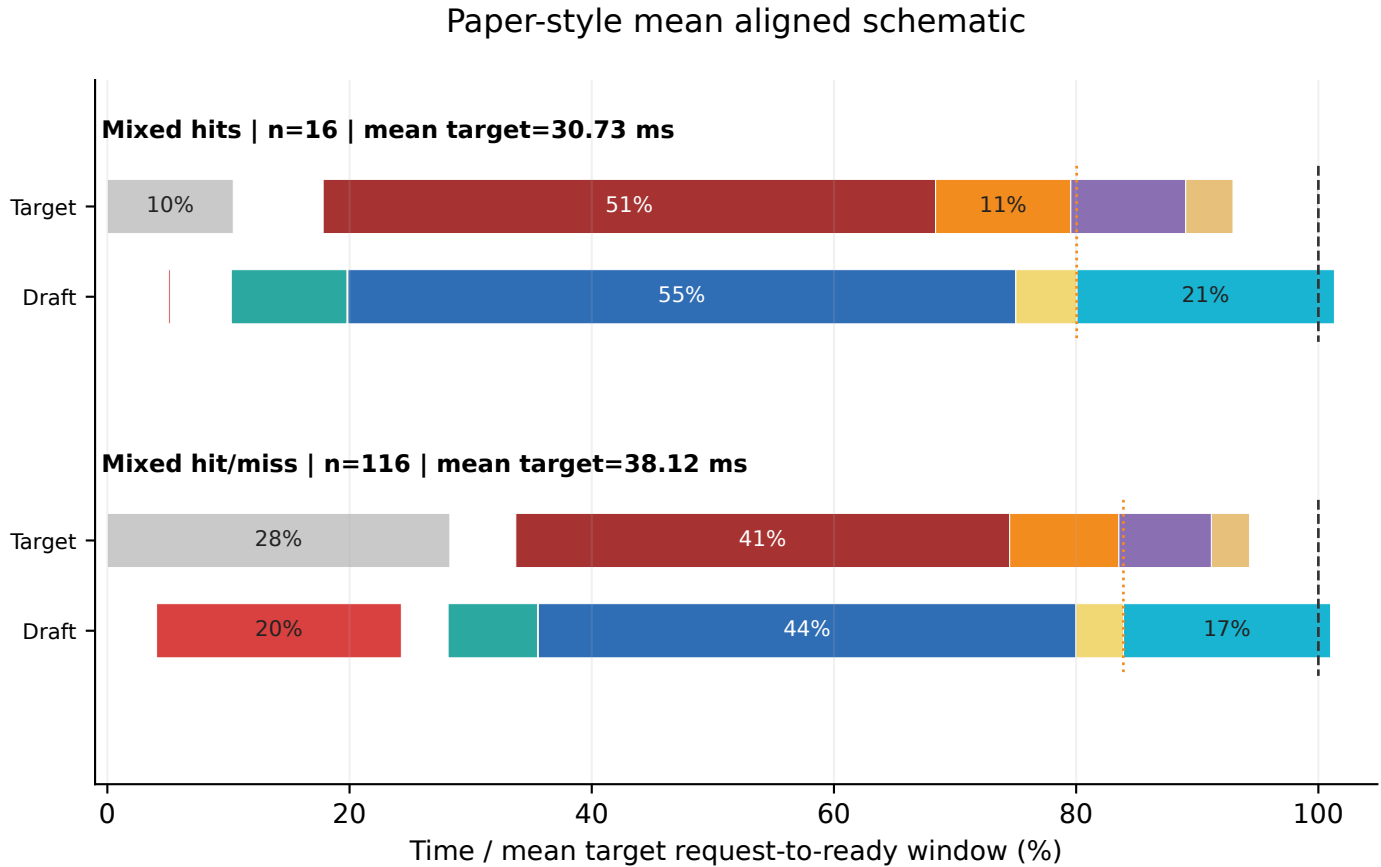
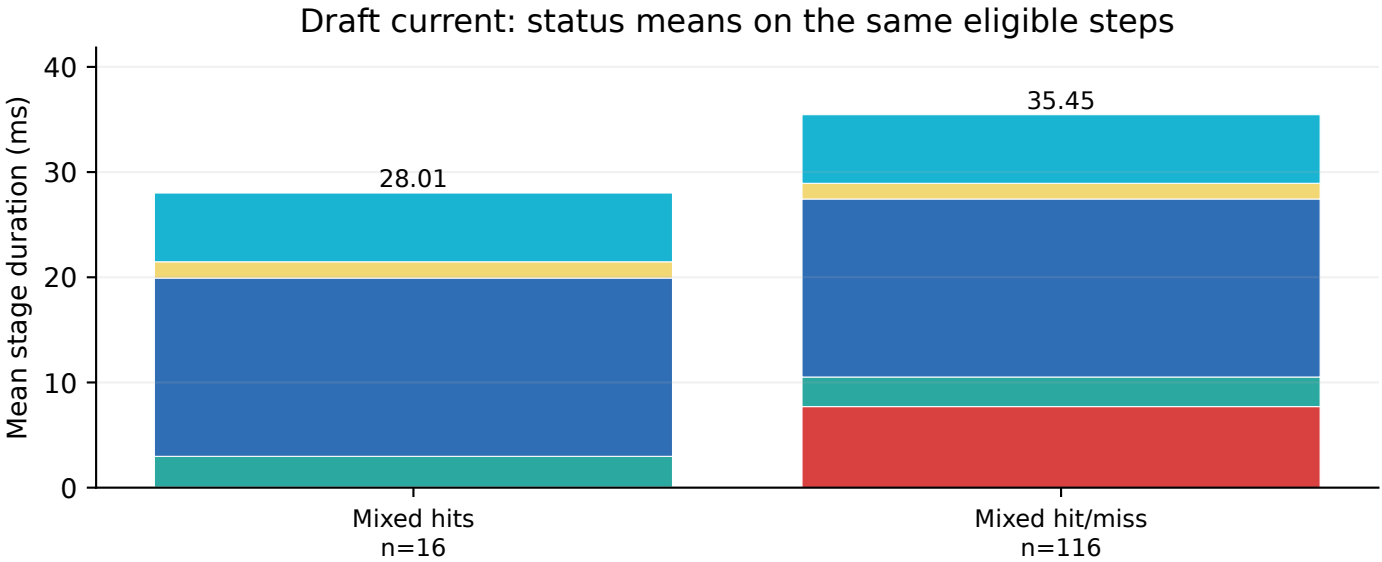
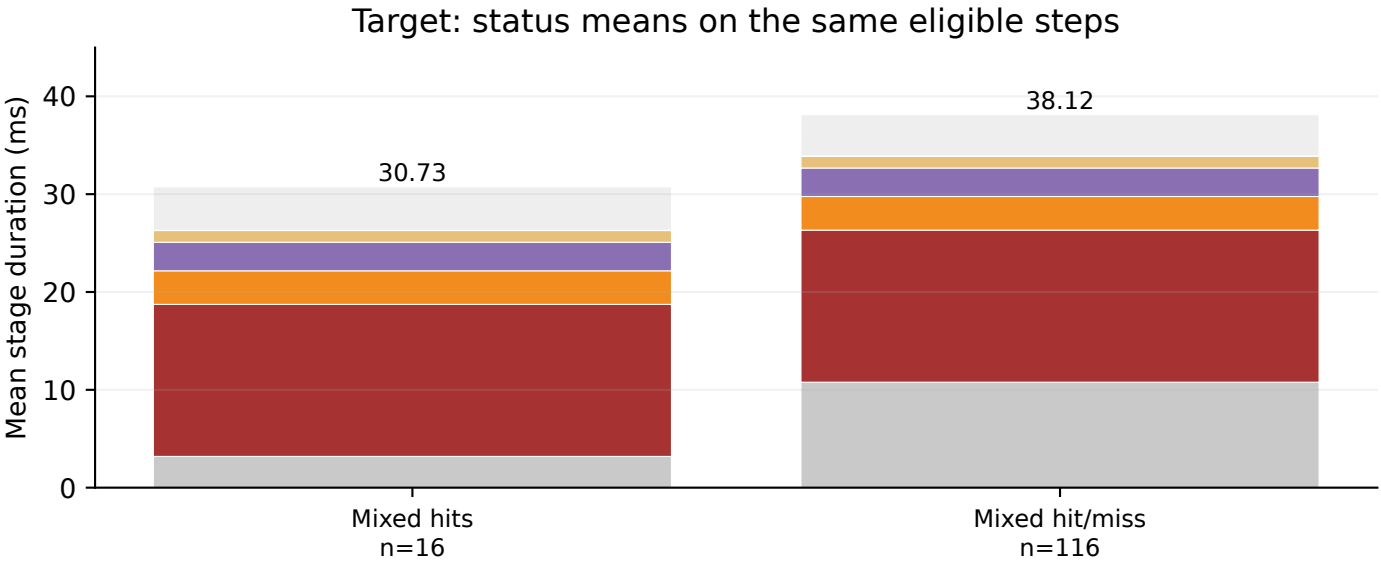
llama3_b8_tree_width1
DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 123 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_tree_width2
DUET: K1/K2=3/1, exit=26 (0-based), C=2, N1/N2=6/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 132 eligible steps

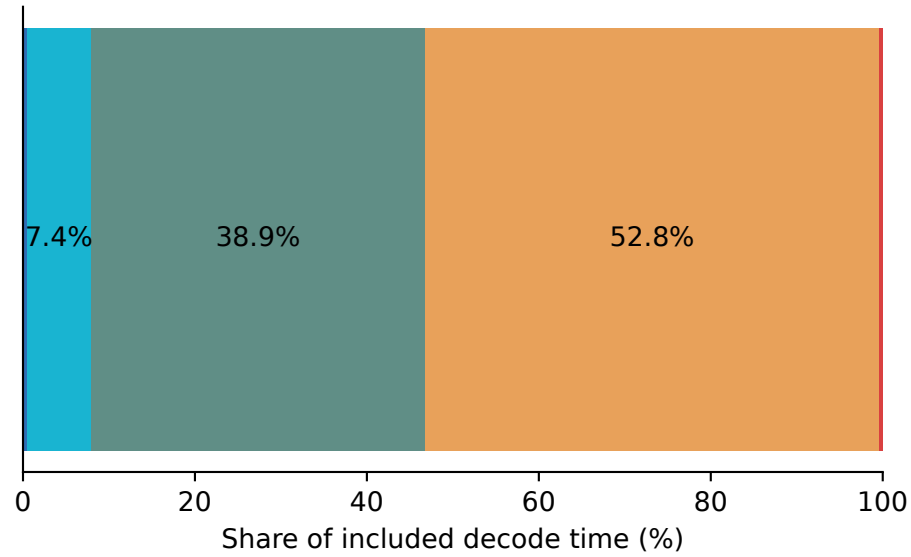


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

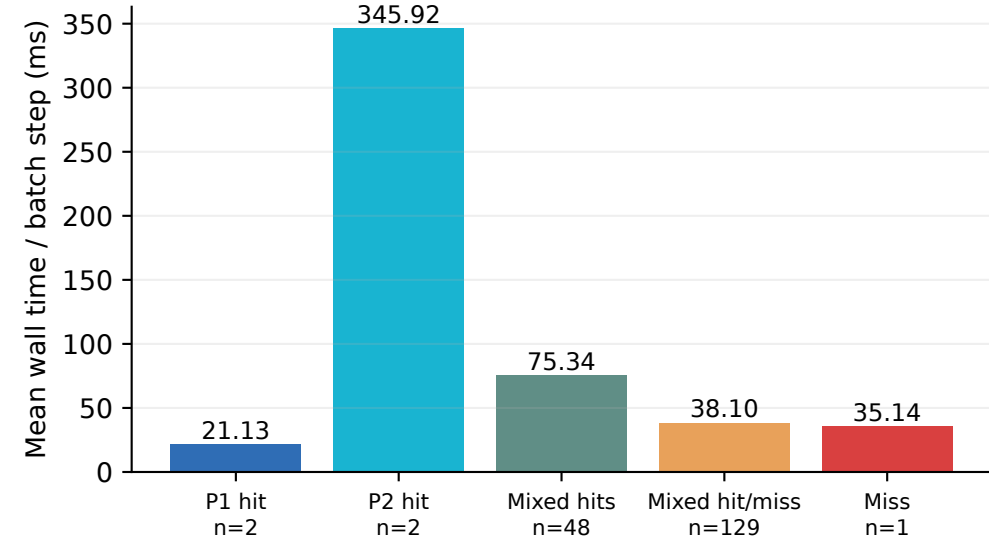
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_warm_confirm_grow_k1
 DUET: K1/K2=4/1, exit=26 (0-based), C=1, N1/N2=4/1, U=2, W=5, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.310, hit=81.7%, TPS*=335.48

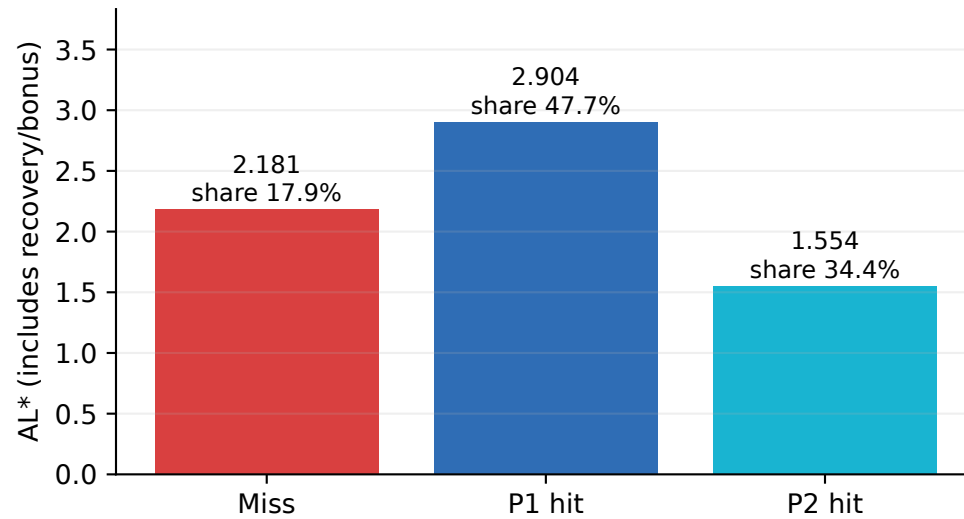
Measured wall time by batch cache status



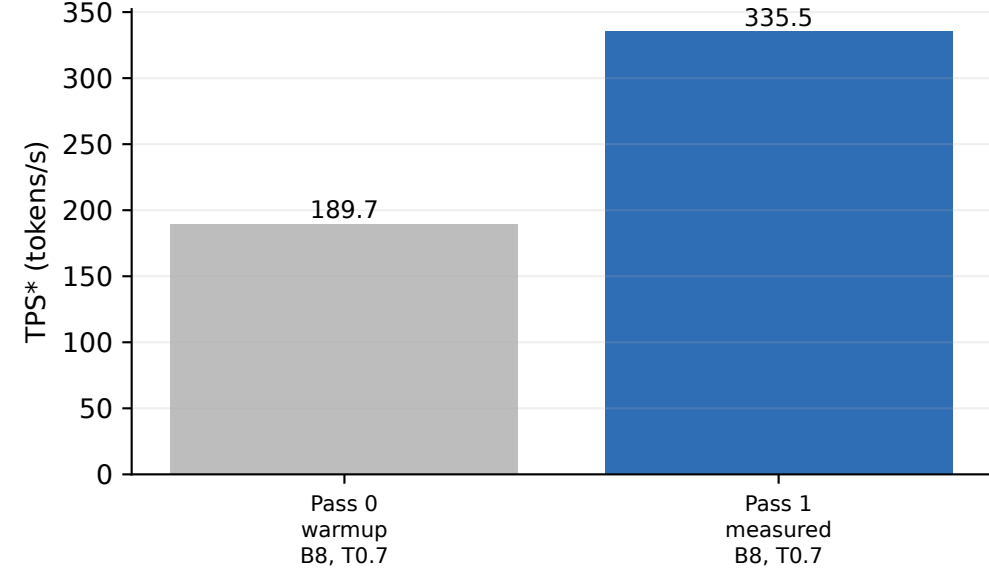
Boundary-free steps, including slow steps



Conditional AL* and source share



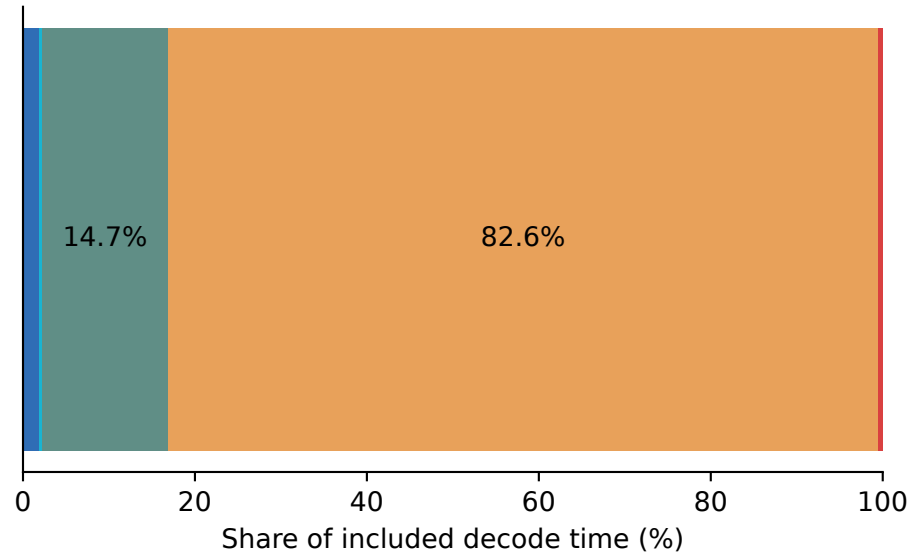
All saved passes; gray = warmup



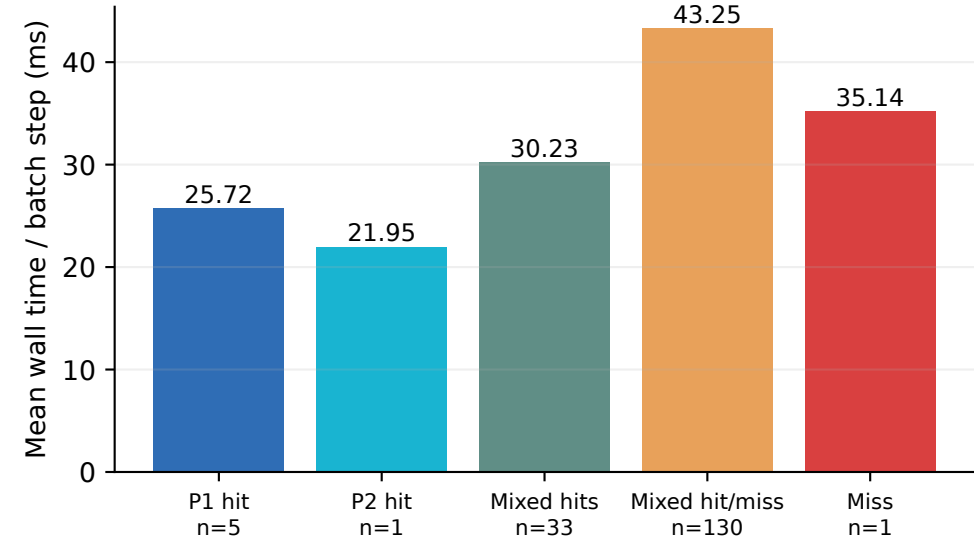
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_confirm_grow_k2
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.364, hit=80.8%, TPS*=451.58

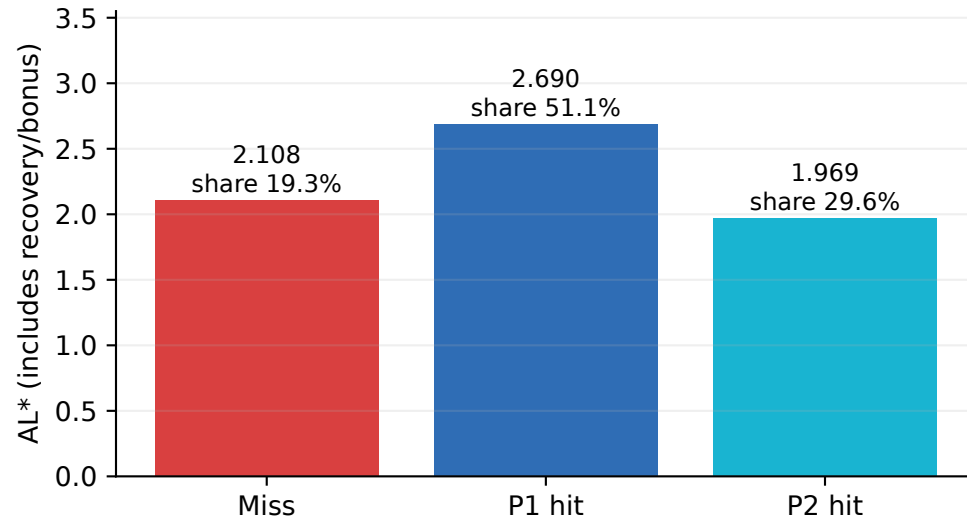
Measured wall time by batch cache status



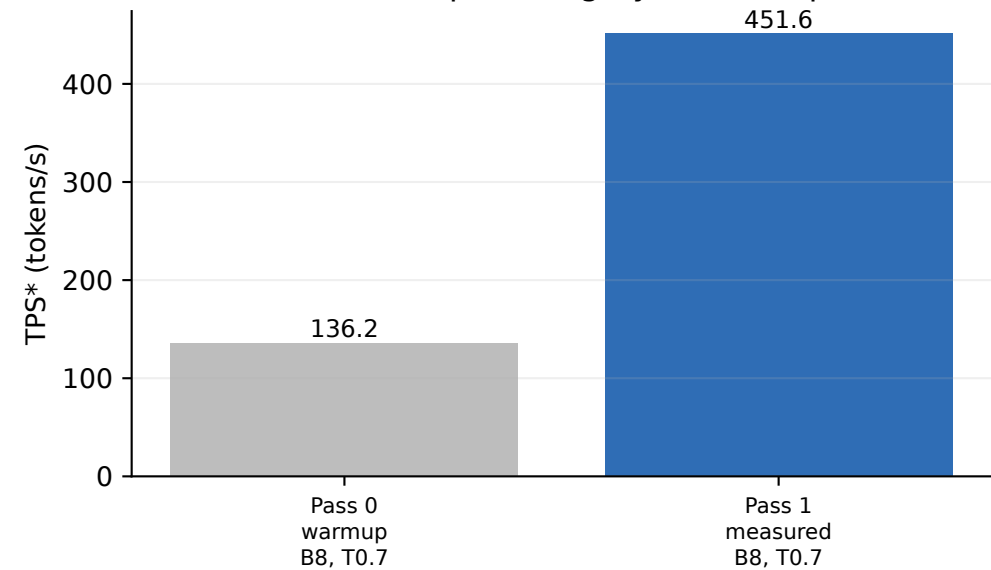
Boundary-free steps, including slow steps



Conditional AL* and source share

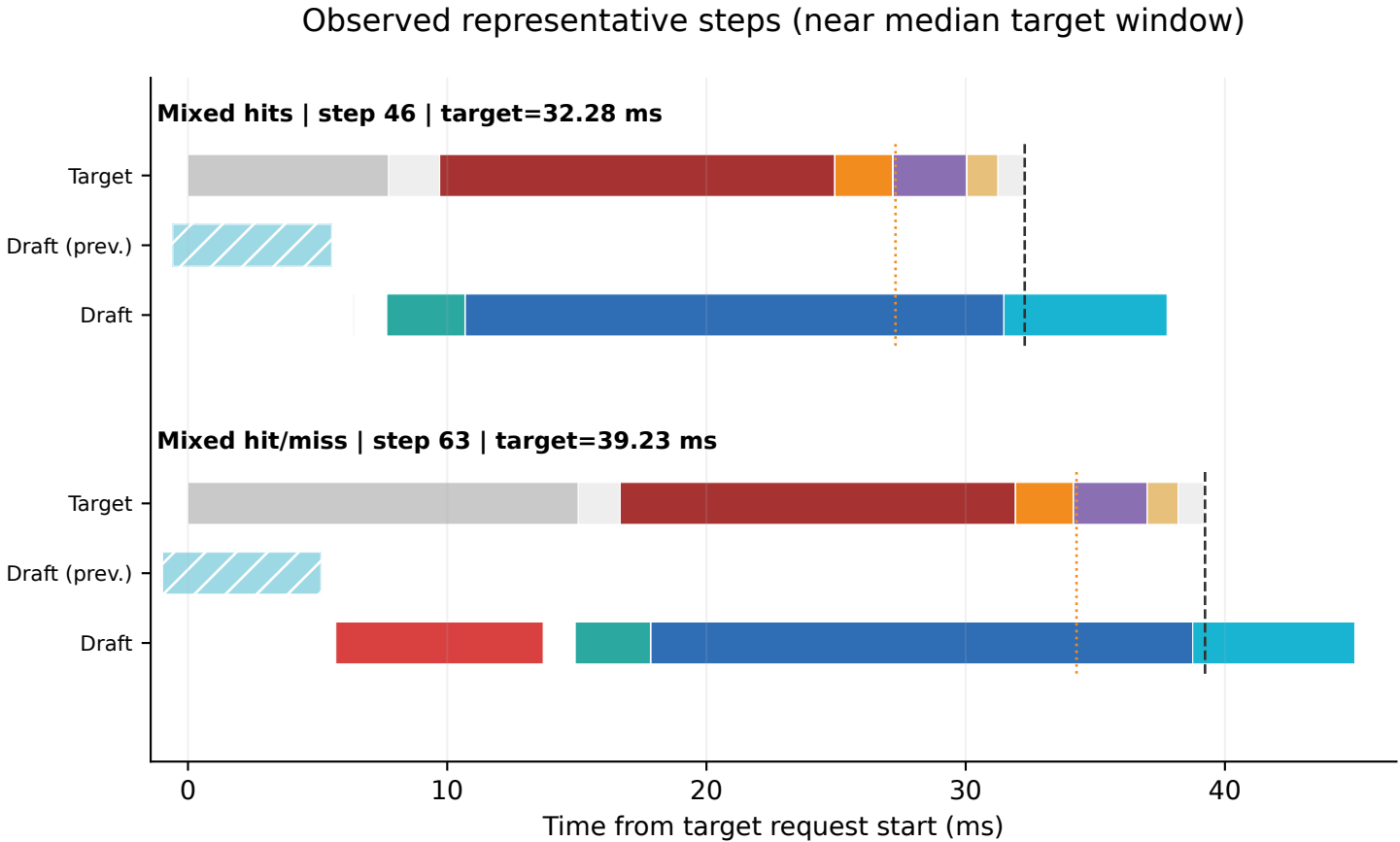
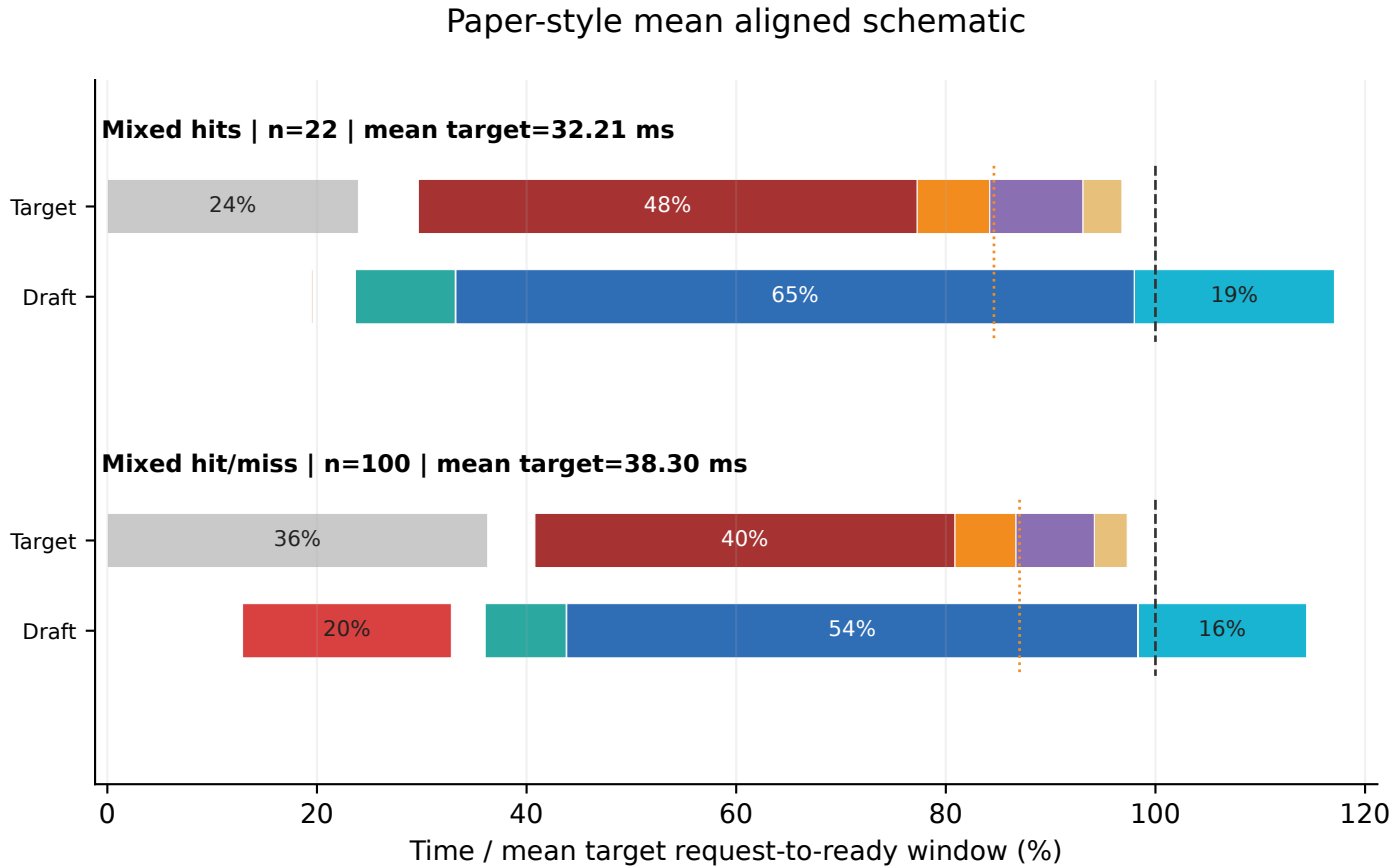
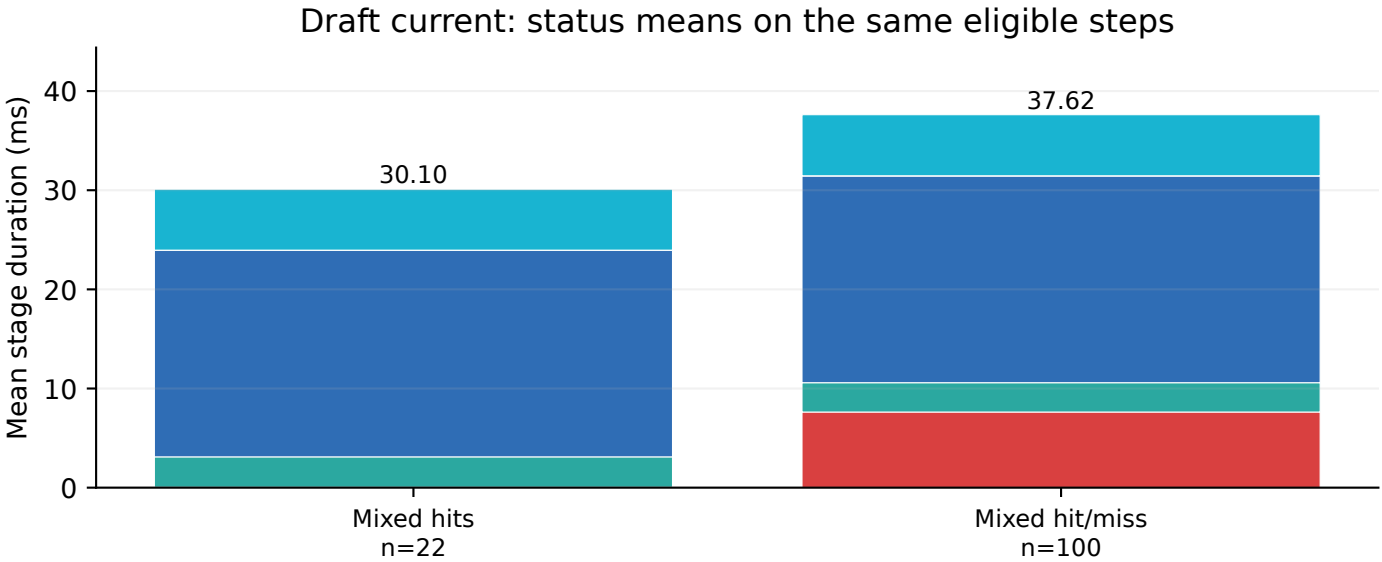
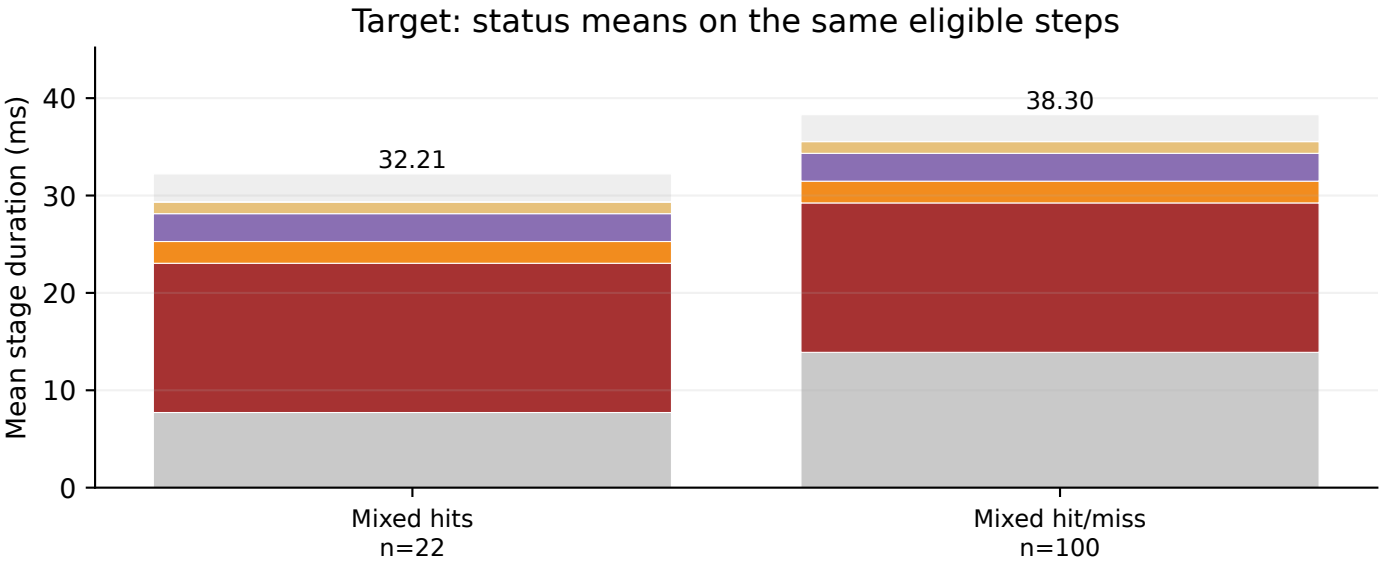


All saved passes; gray = warmup



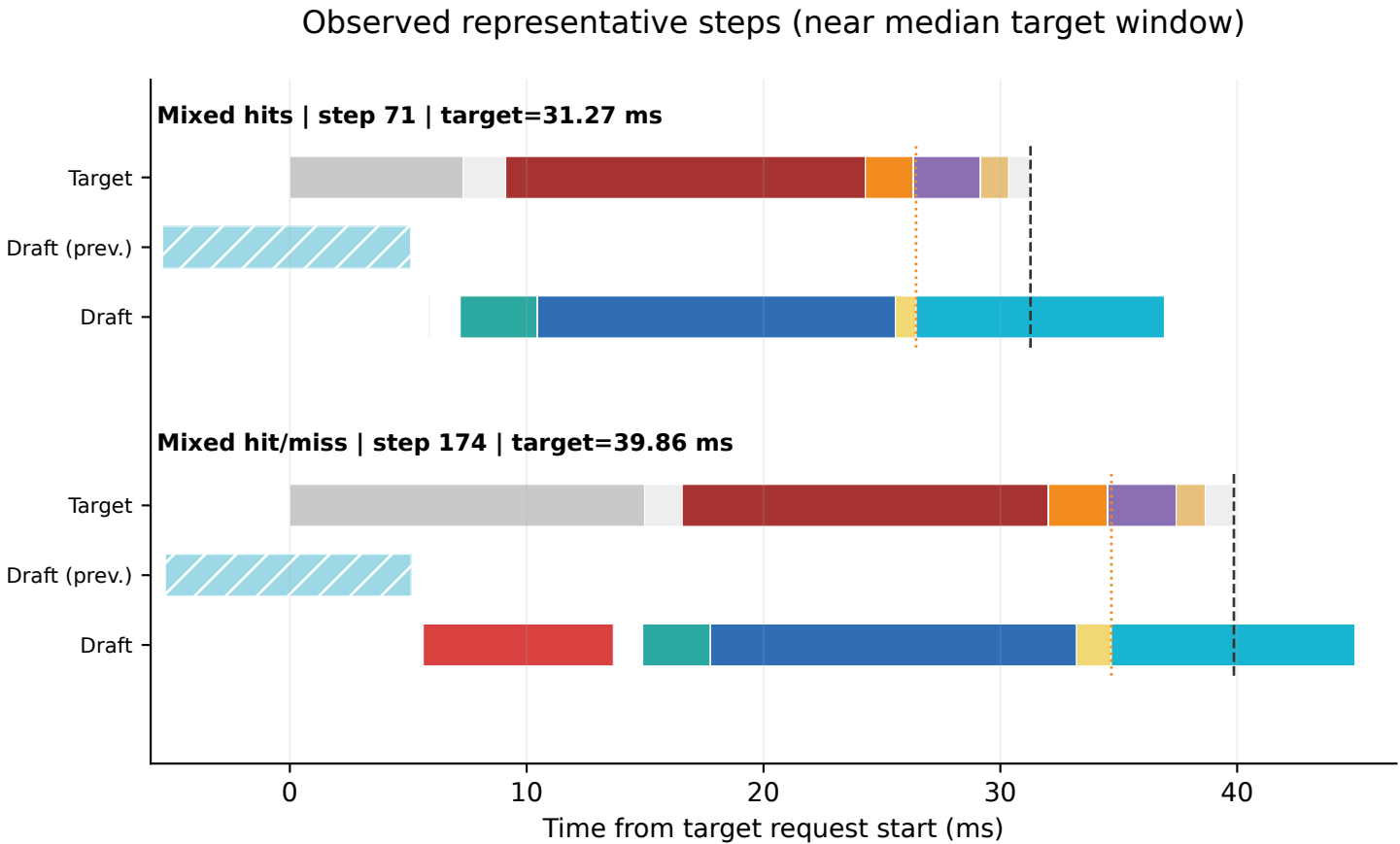
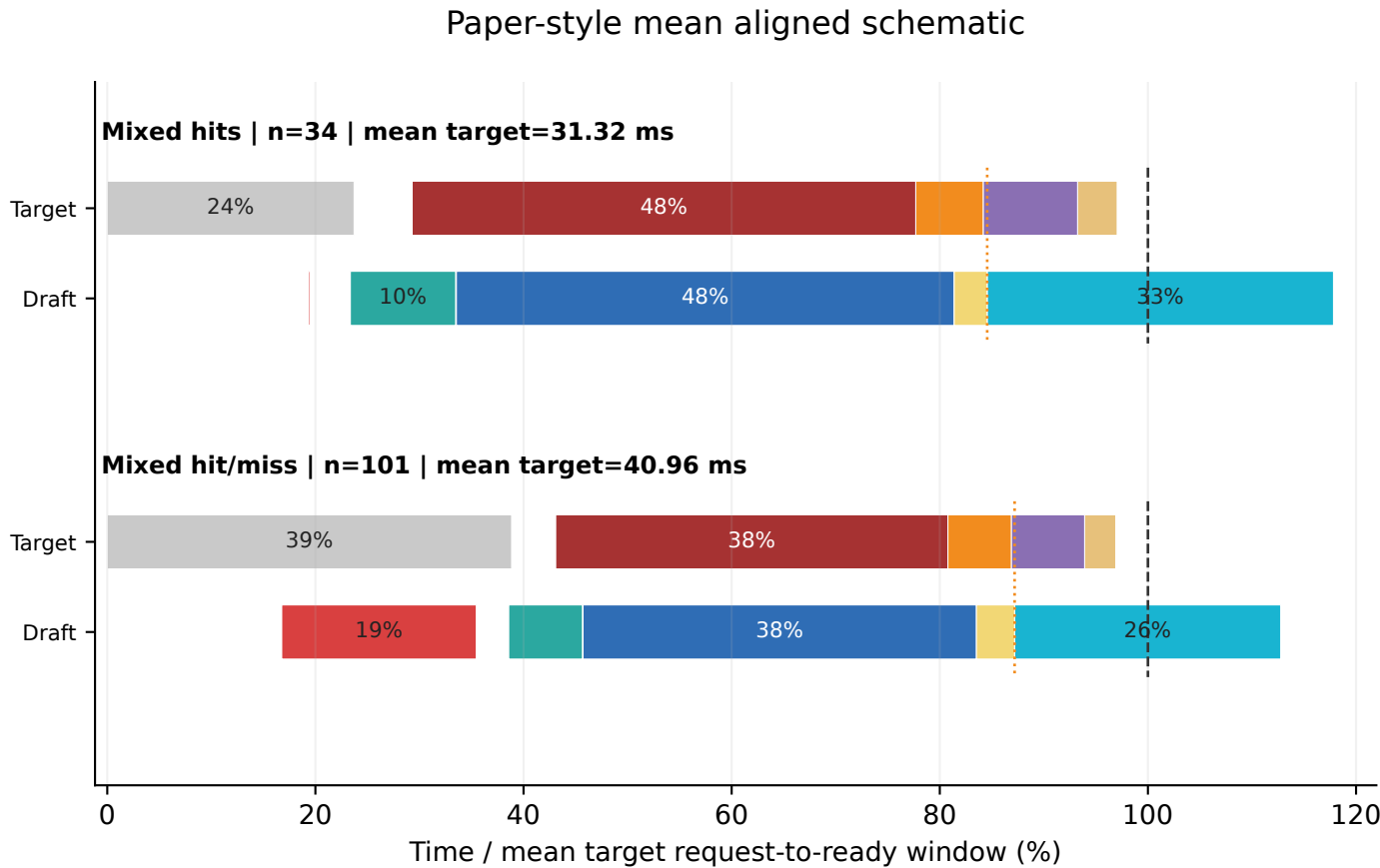
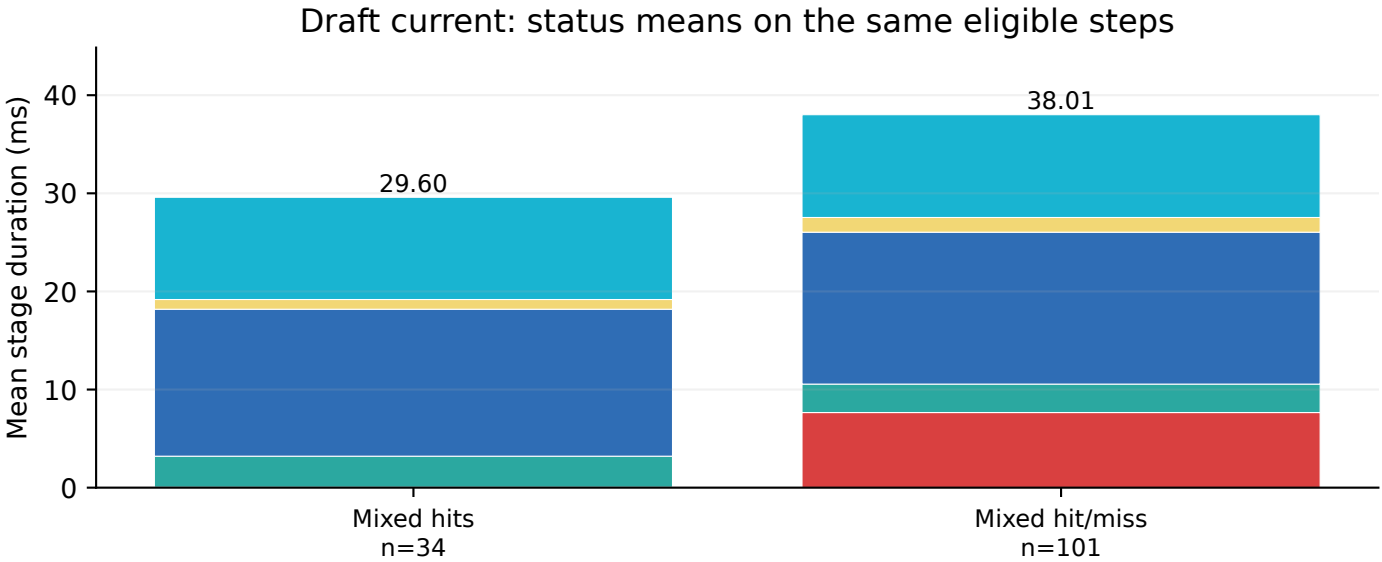
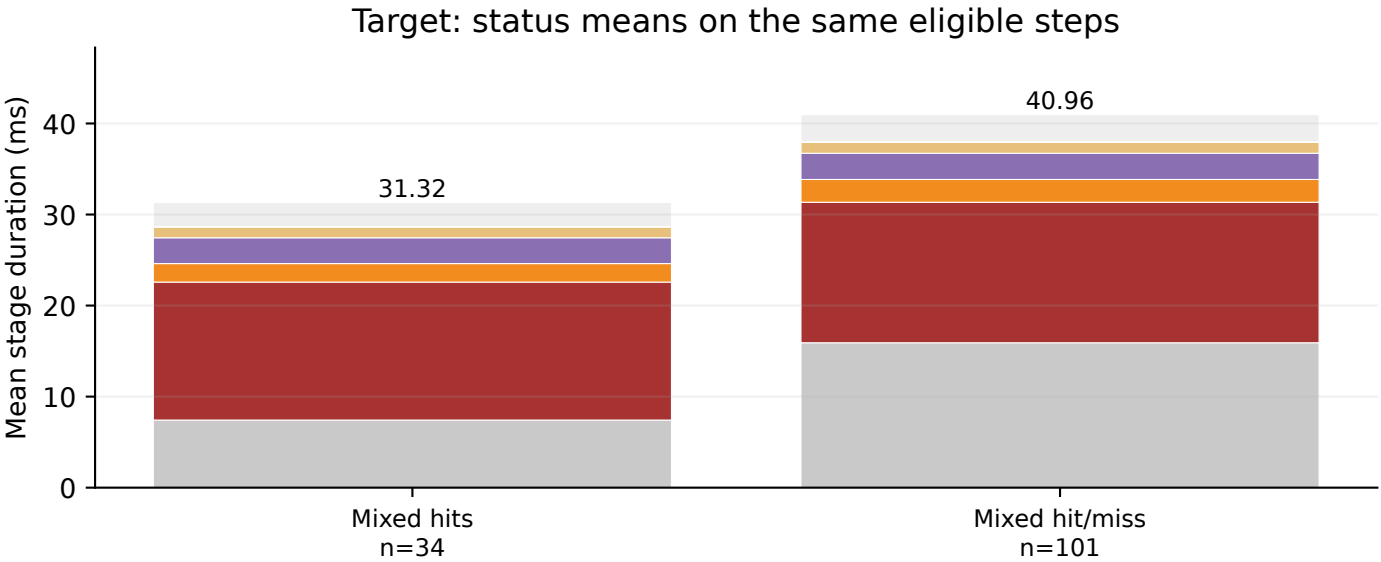
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_diagnostic_grow_k1
DUET: K1/K2=4/1, exit=26 (0-based), C=1, N1/N2=4/1, U=2, W=5, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 122 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

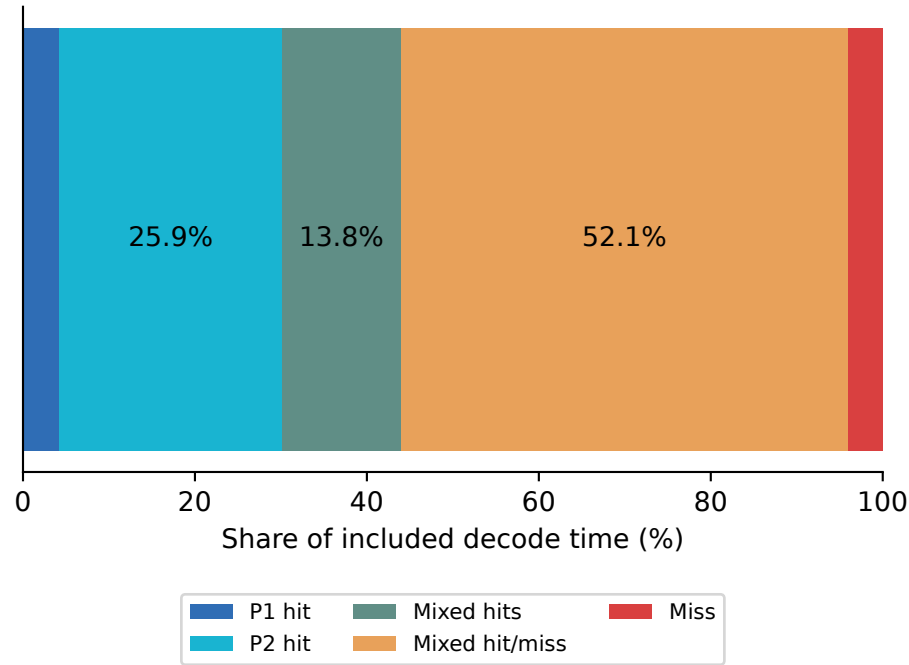


Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

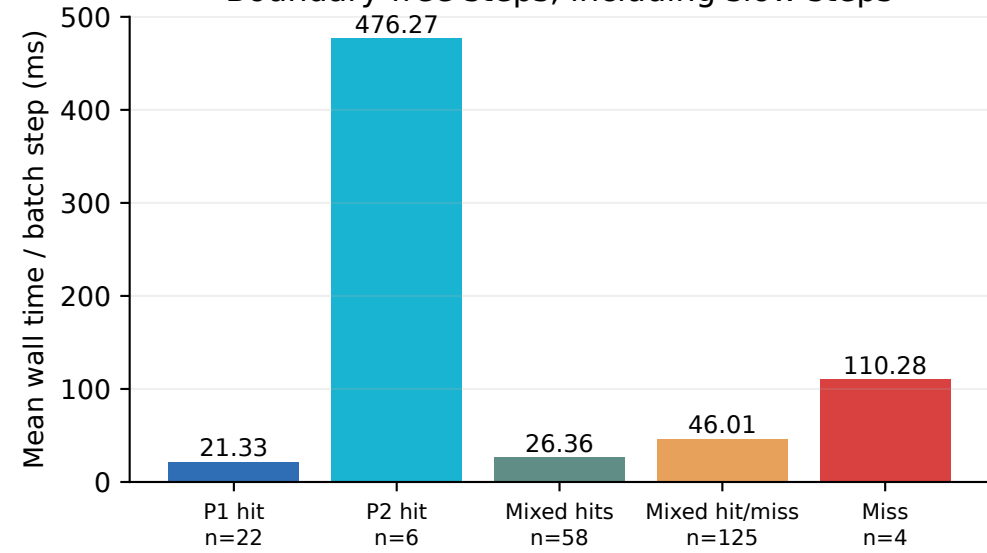
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_warm_duet_tree_1
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.314, hit=82.2%, TPS*=310.62

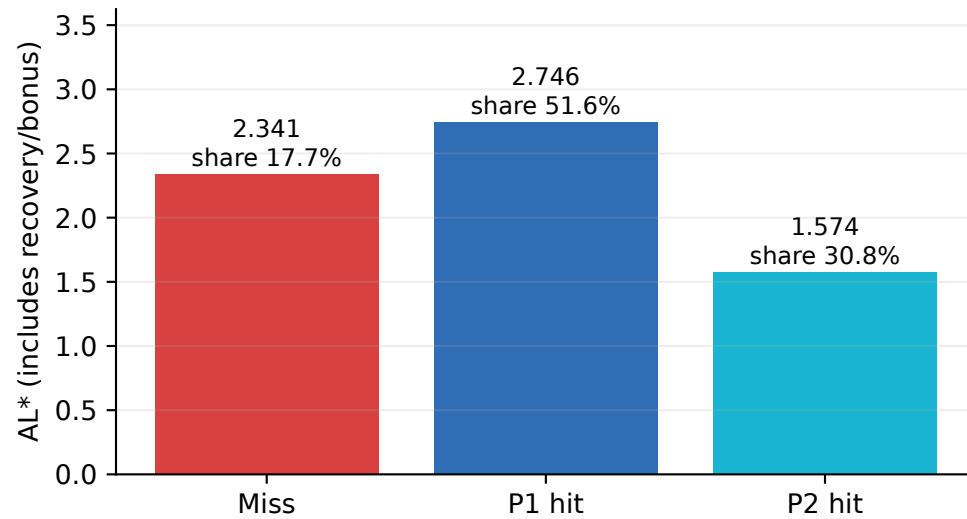
Measured wall time by batch cache status



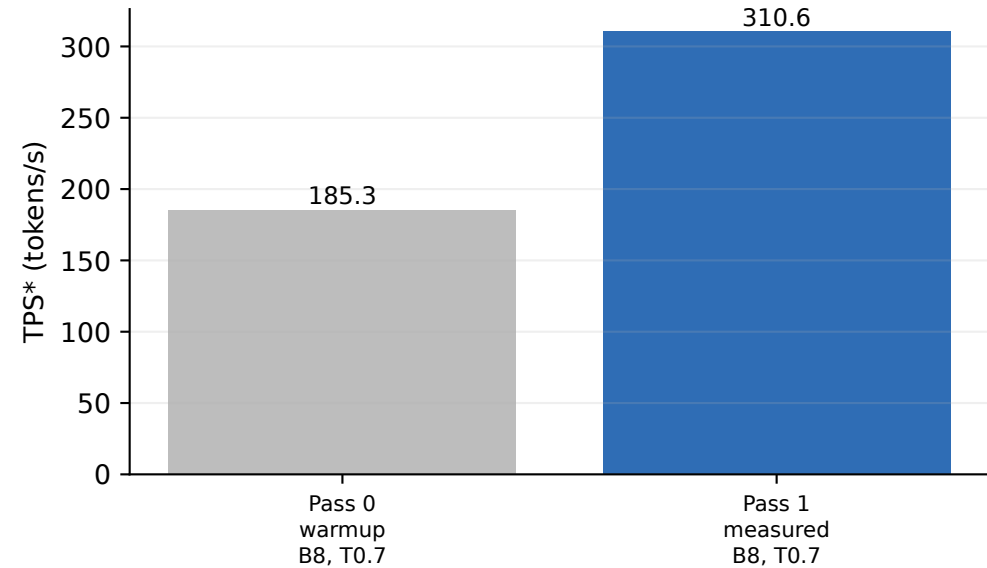
Boundary-free steps, including slow steps



Conditional AL* and source share



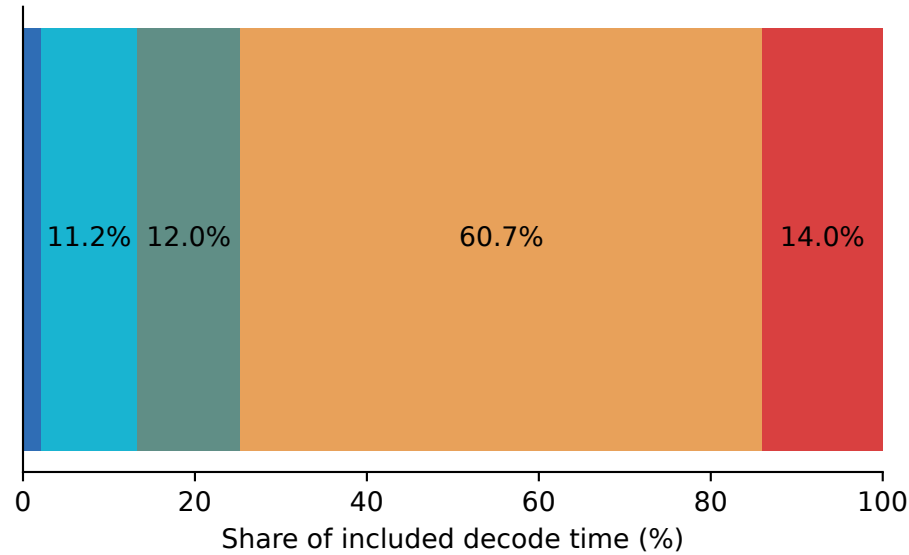
All saved passes; gray = warmup



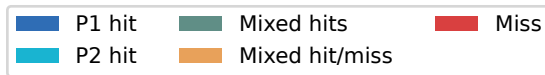
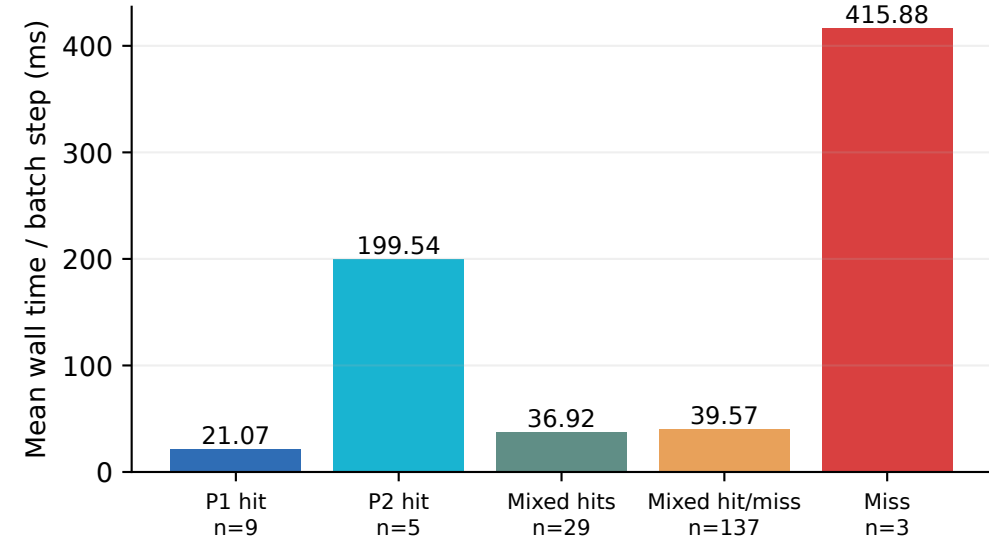
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_duet_tree_2
 DUET: K1/K2=3/1, exit=24 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.270, hit=78.7%, TPS*=337.87

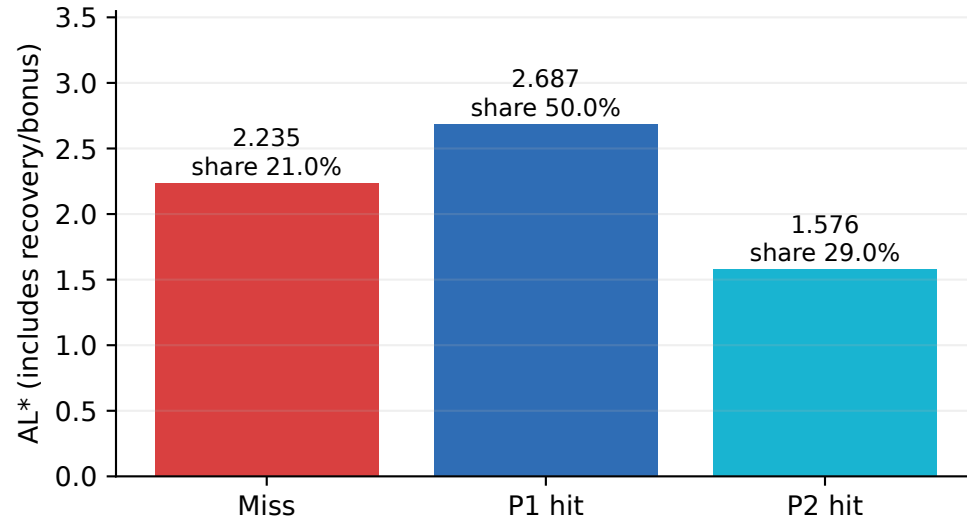
Measured wall time by batch cache status



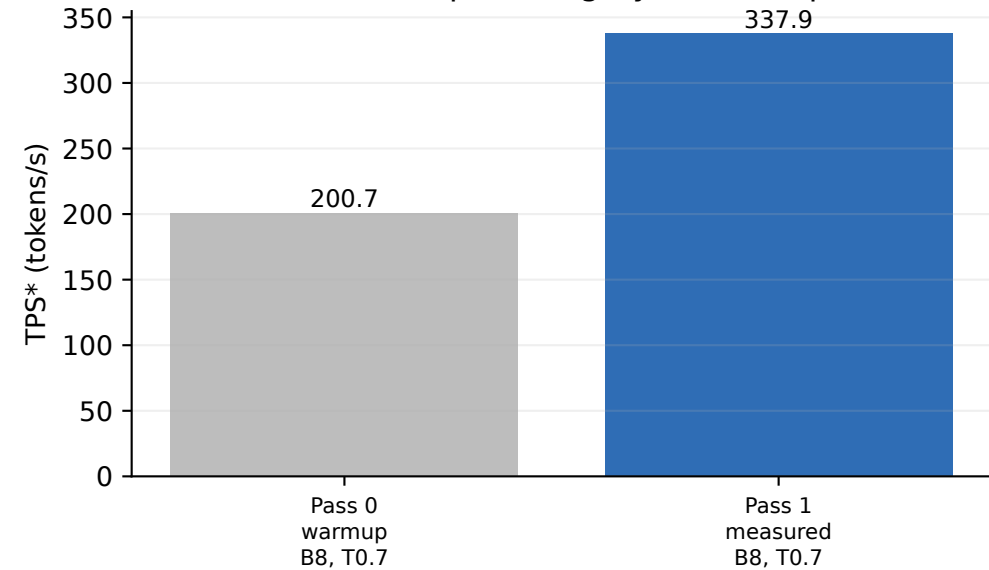
Boundary-free steps, including slow steps



Conditional AL* and source share



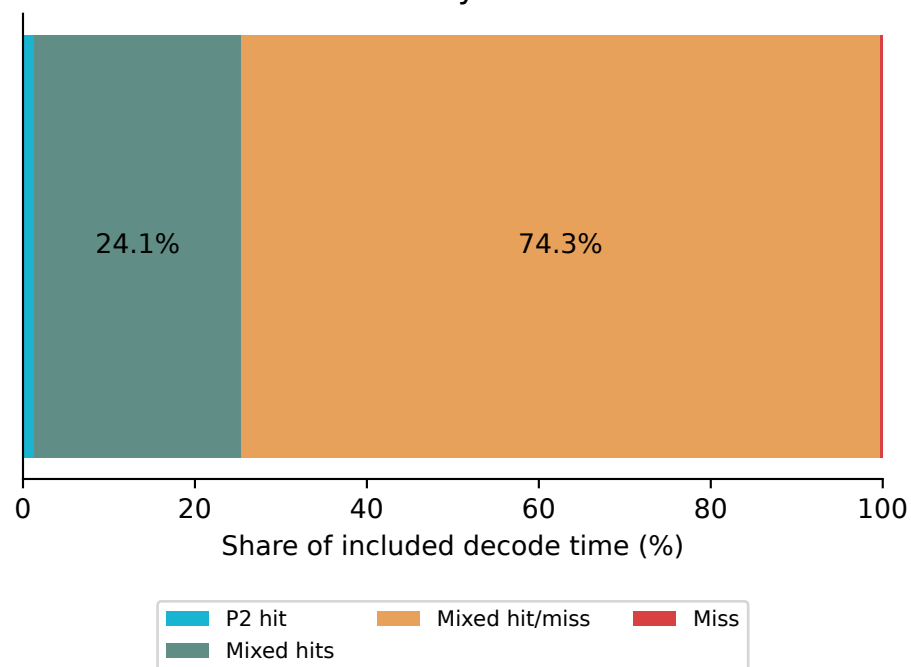
All saved passes; gray = warmup



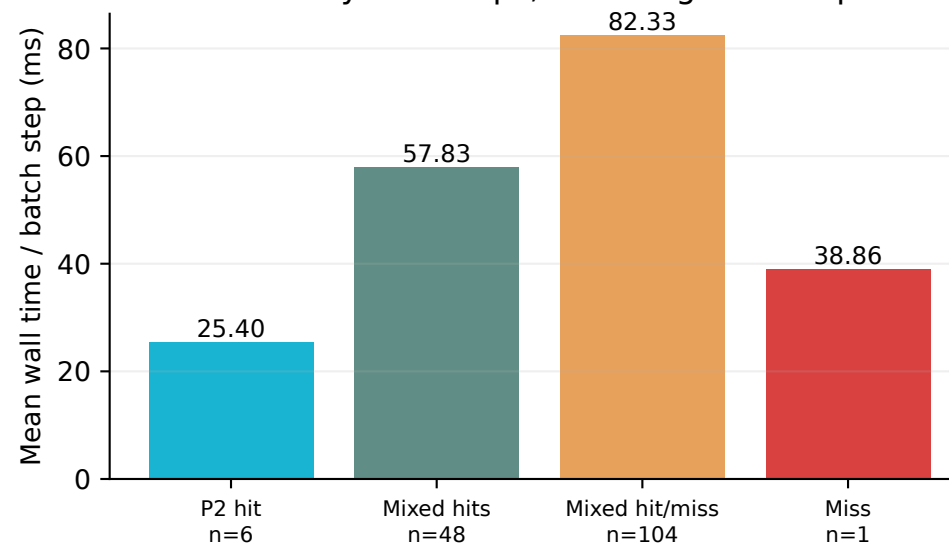
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_duet_tree_3
 DUET: K1/K2=6/2, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.753, hit=86.4%, TPS*=285.79

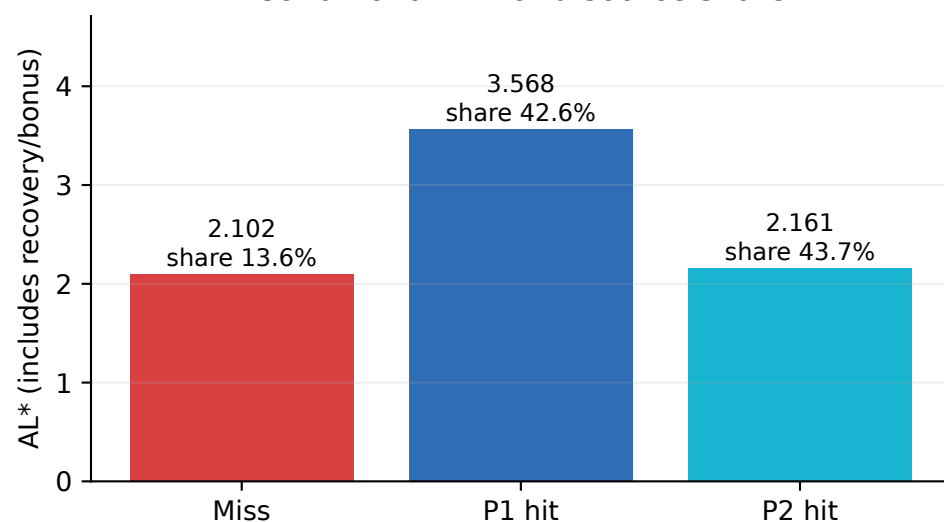
Measured wall time by batch cache status



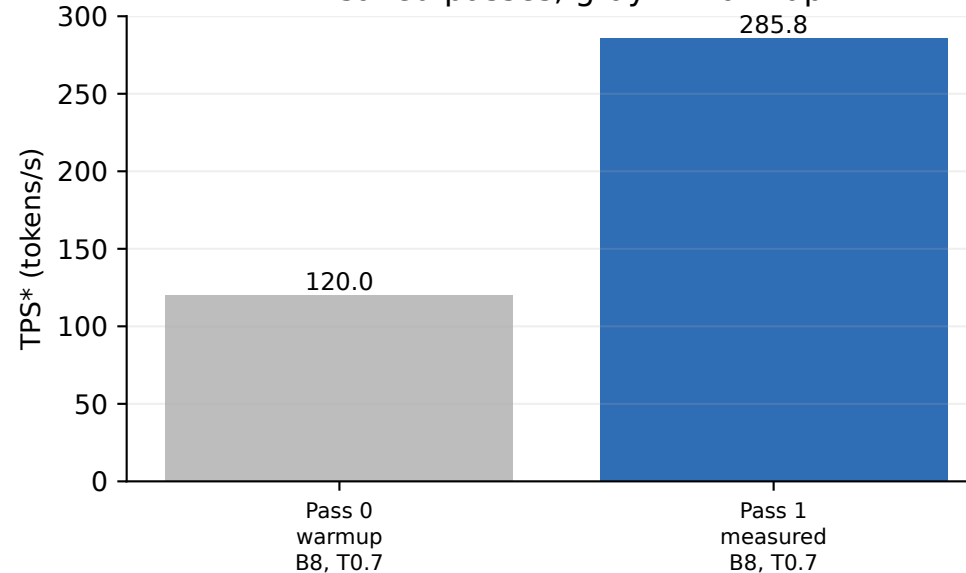
Boundary-free steps, including slow steps



Conditional AL* and source share



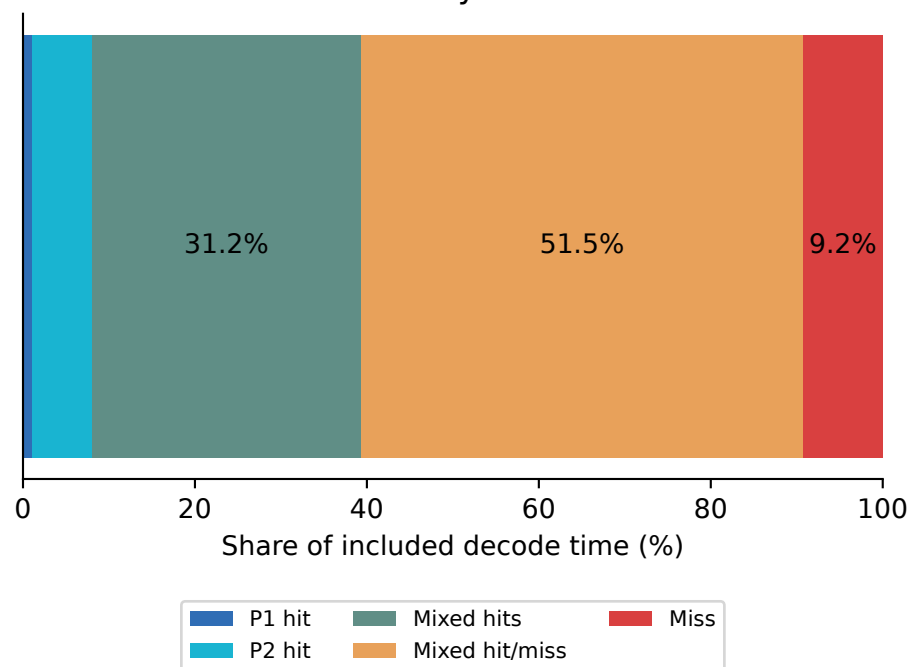
All saved passes; gray = warmup



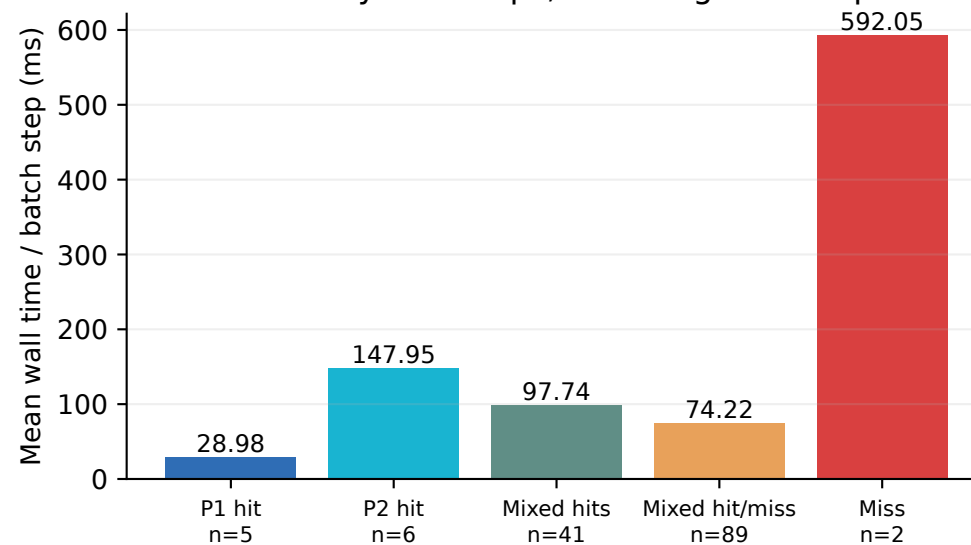
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_duet_tree_4
 DUET: K1/K2=6/3, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.843, hit=85.5%, TPS*=223.07

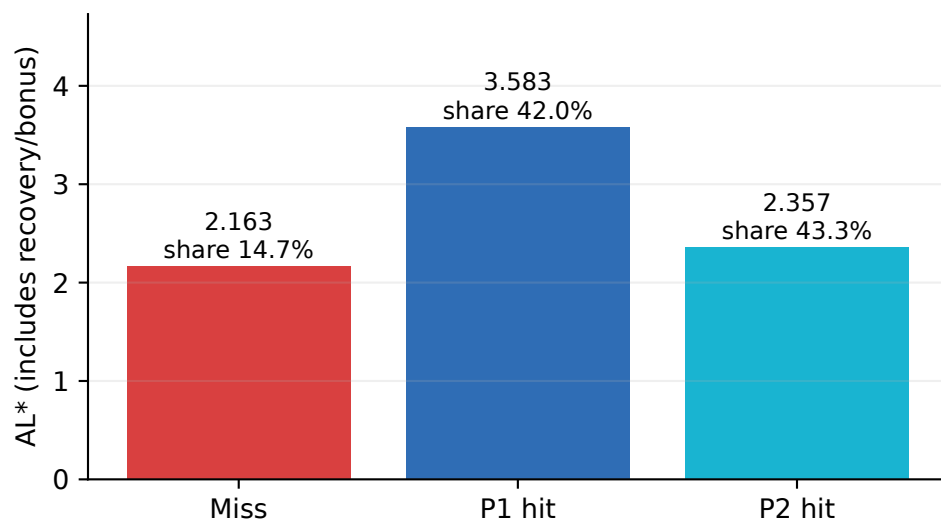
Measured wall time by batch cache status



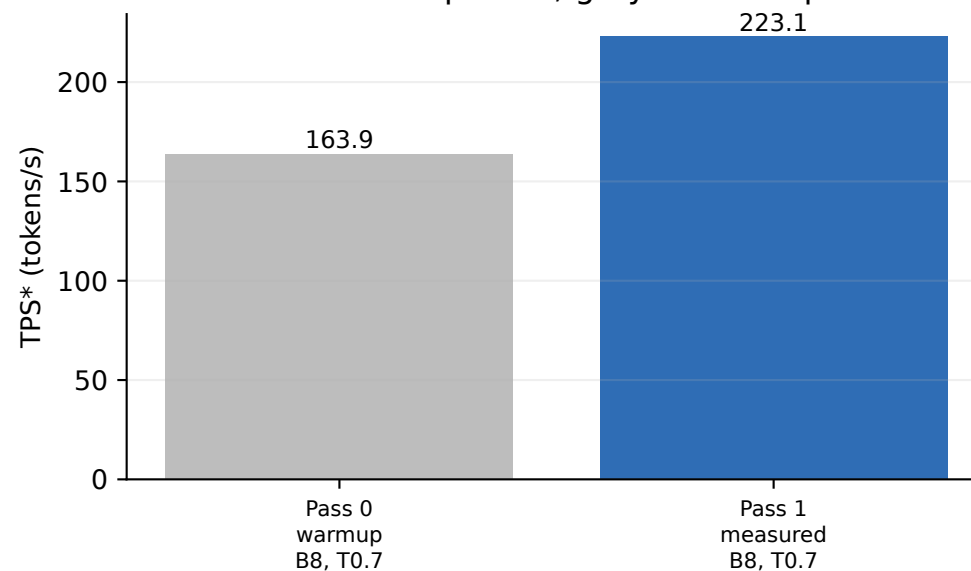
Boundary-free steps, including slow steps



Conditional AL* and source share



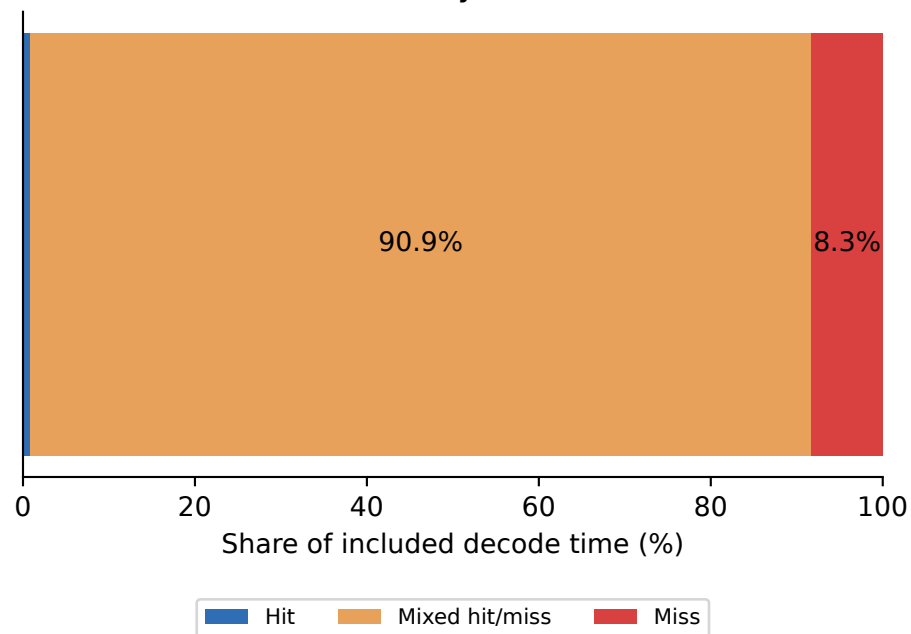
All saved passes; gray = warmup



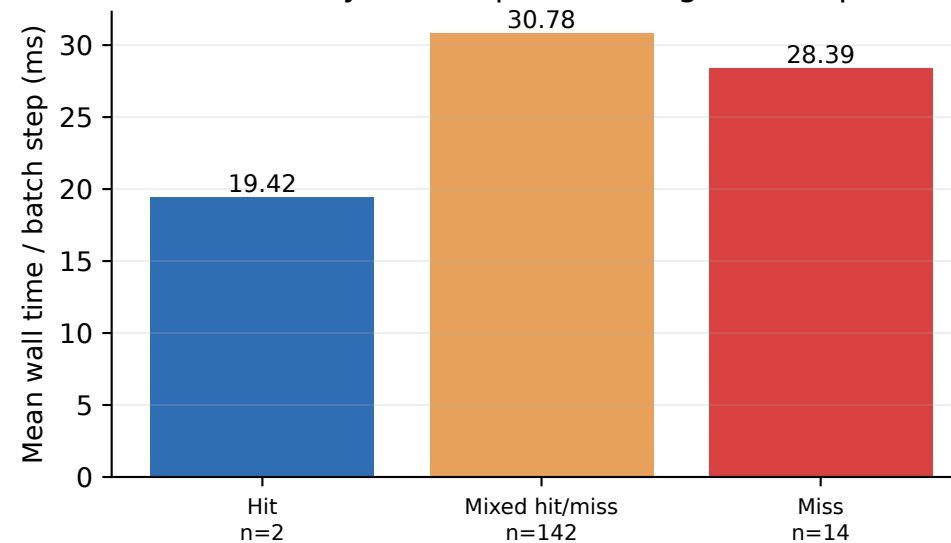
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_ssd_1
SSD: K=4, F=1
Last pass: B=8, T=0.7, N=48 | AL*=2.685, hit=34.5%, TPS*=649.05

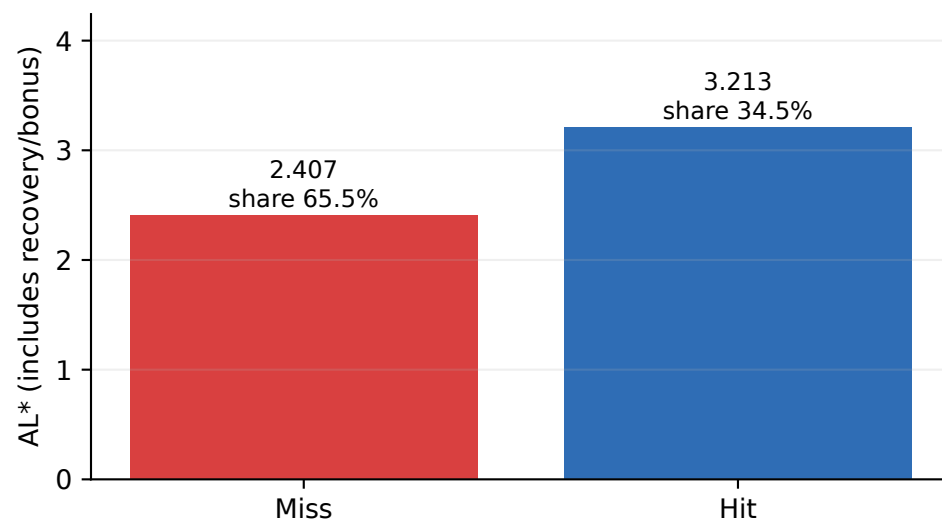
Measured wall time by batch cache status



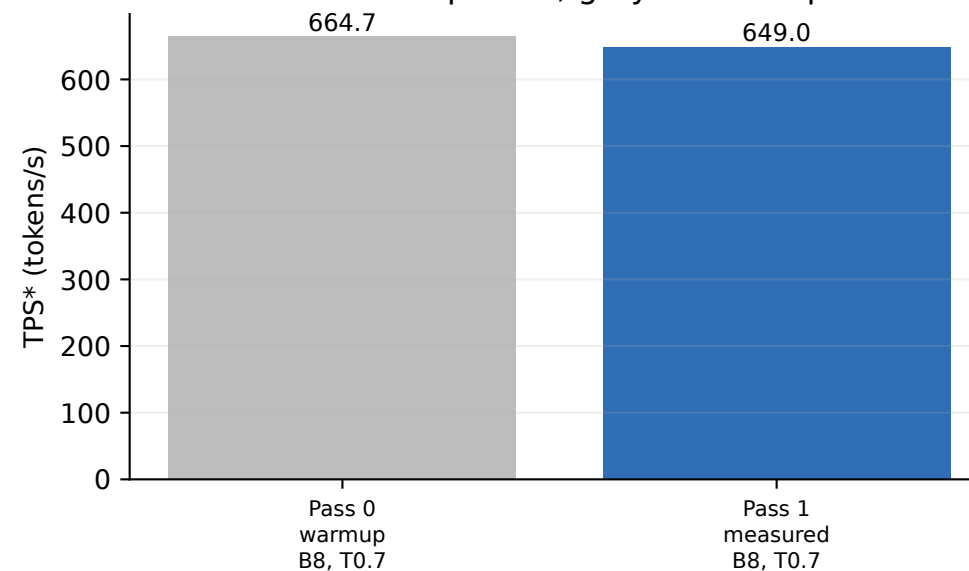
Boundary-free steps, including slow steps



Conditional AL* and source share



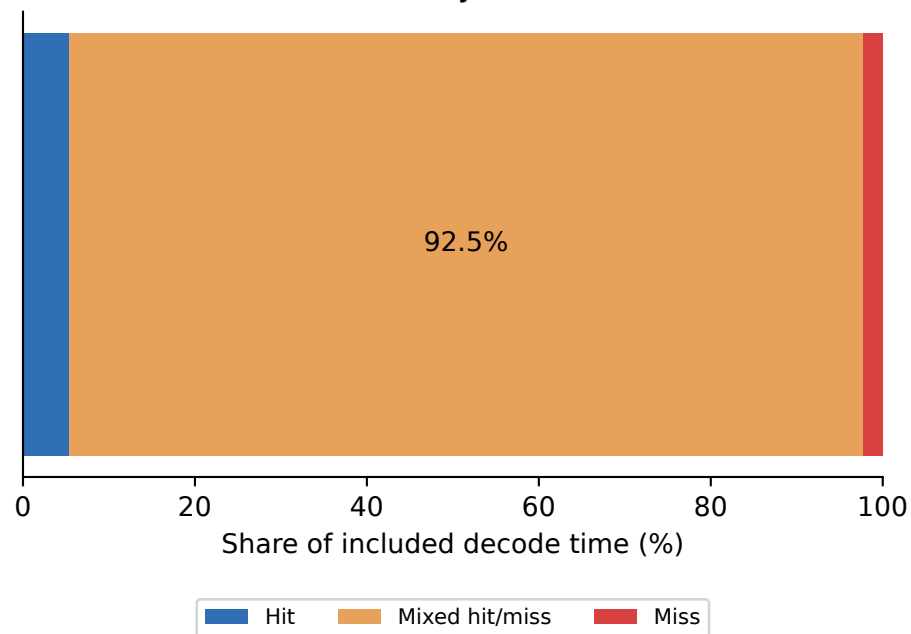
All saved passes; gray = warmup



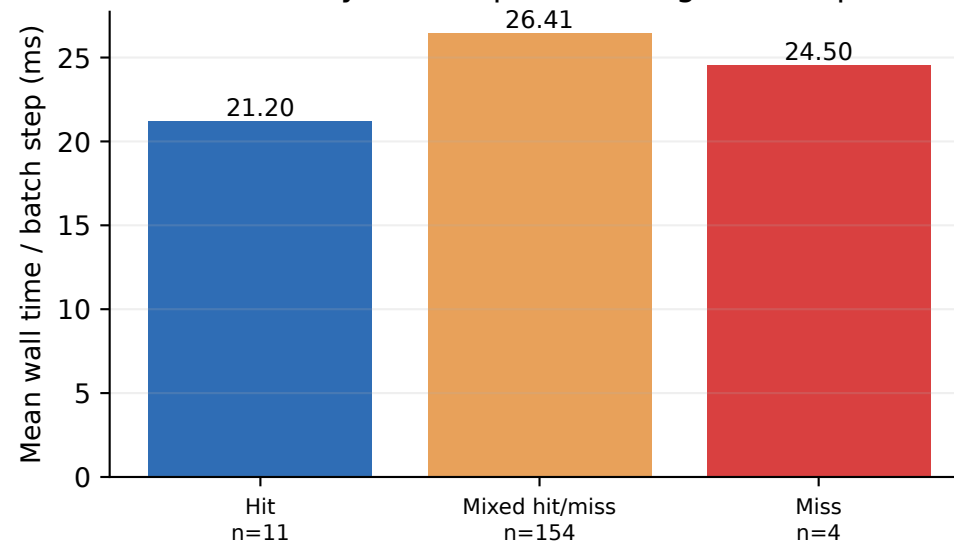
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_ssd_2
SSD: K=2, F=5
Last pass: B=8, T=0.7, N=48 | AL*=2.147, hit=67.4%, TPS*=617.32

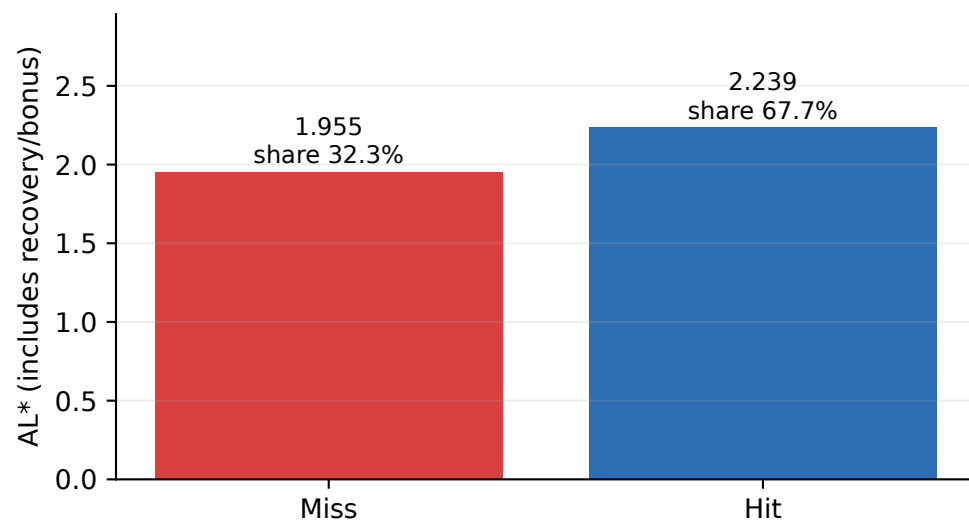
Measured wall time by batch cache status



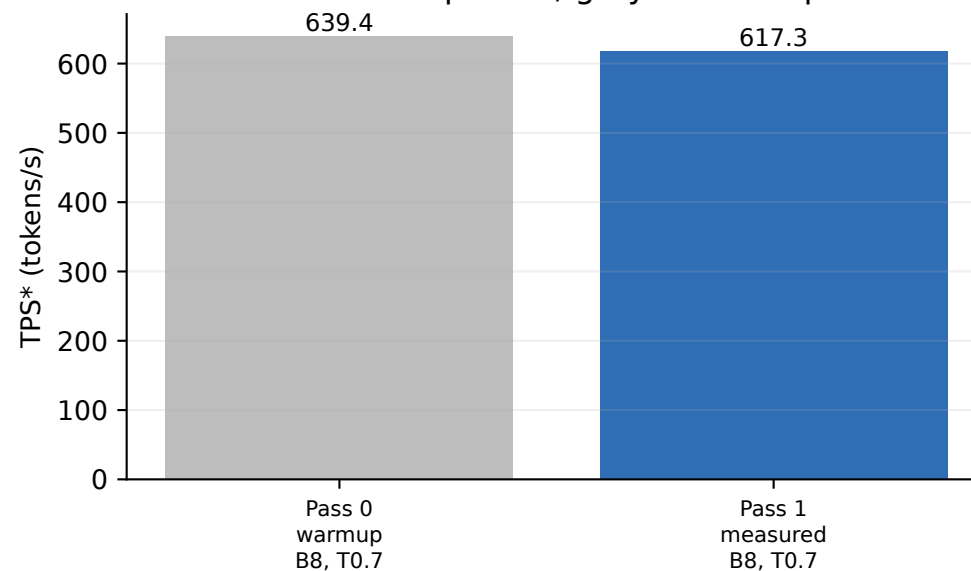
Boundary-free steps, including slow steps



Conditional AL* and source share



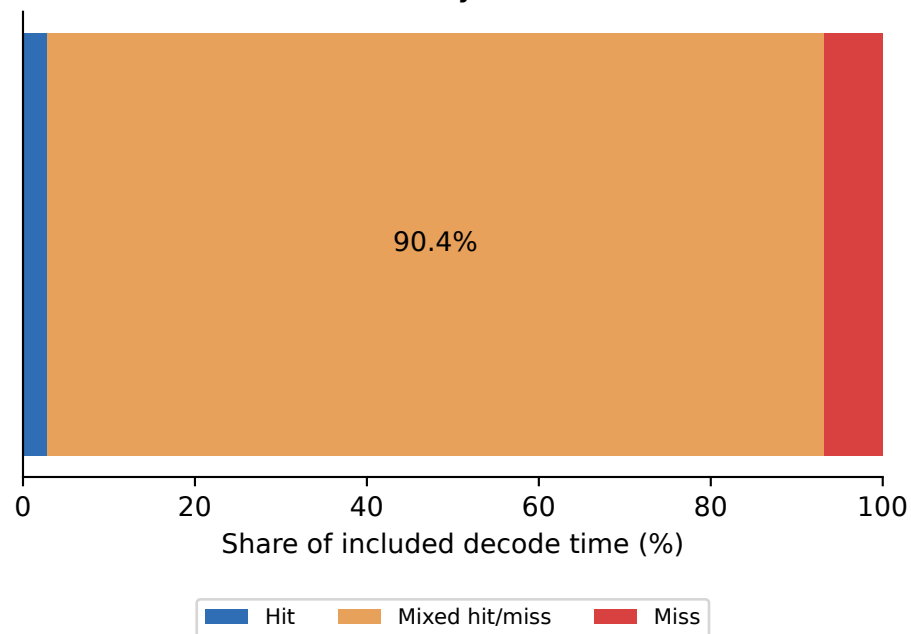
All saved passes; gray = warmup



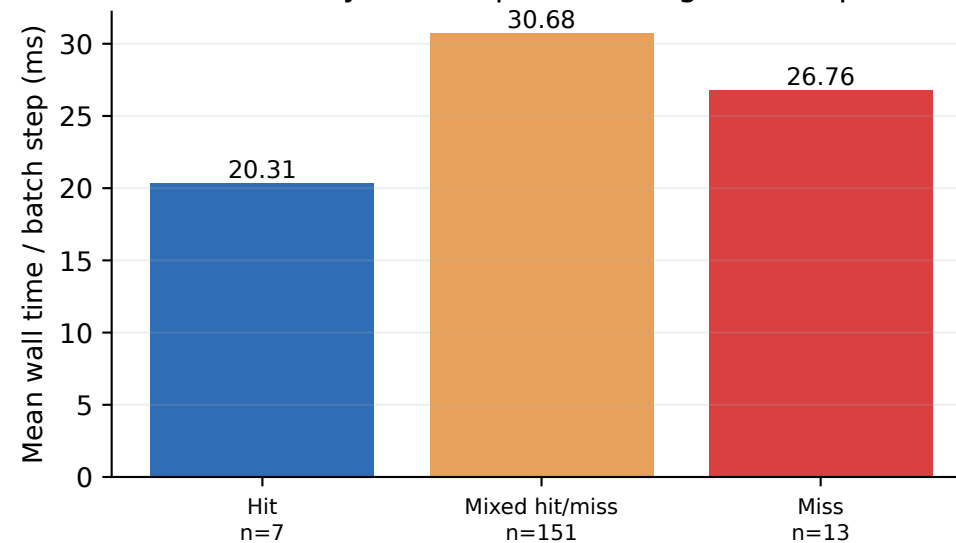
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_ssd_3
SSD: K=4, F=2
Last pass: B=8, T=0.7, N=48 | AL*=2.575, hit=46.5%, TPS*=611.74

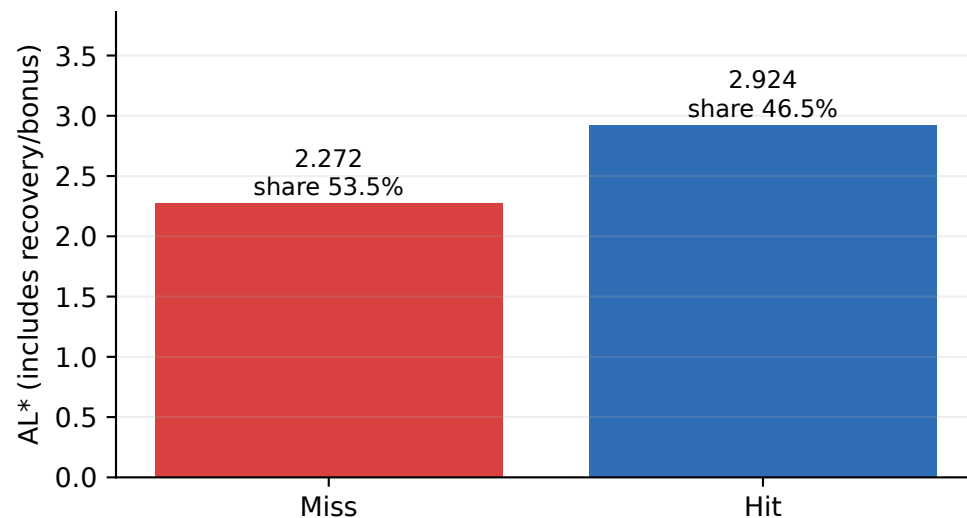
Measured wall time by batch cache status



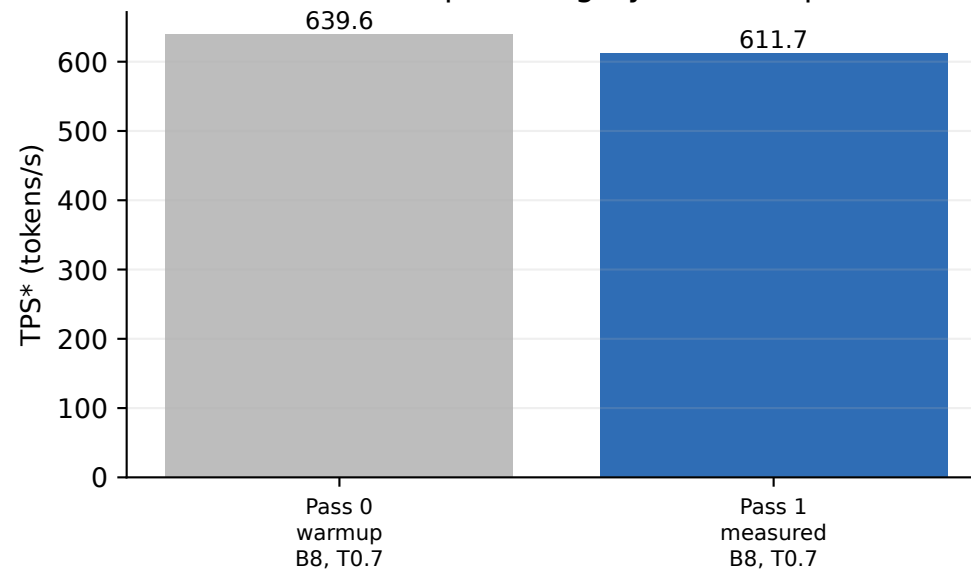
Boundary-free steps, including slow steps



Conditional AL* and source share



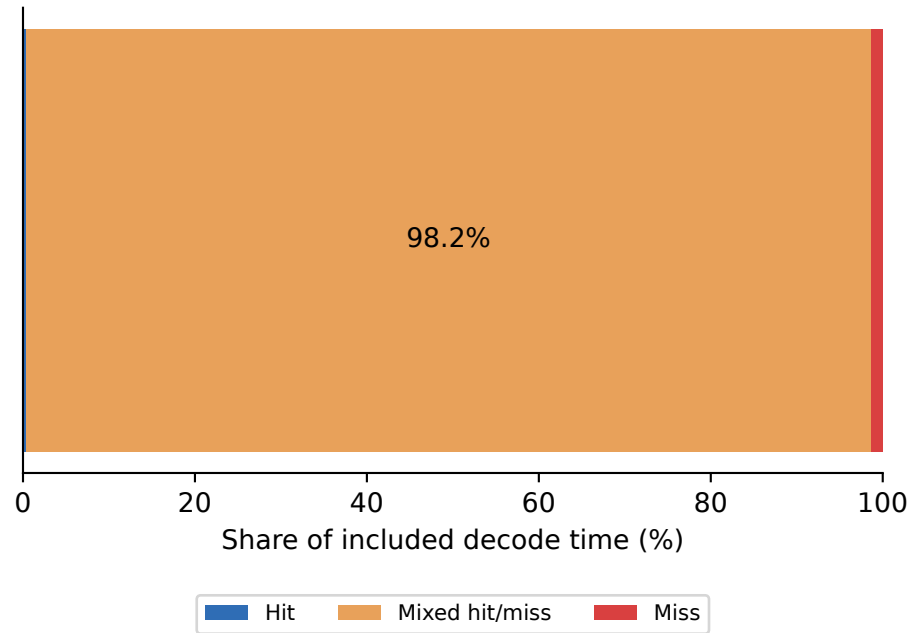
All saved passes; gray = warmup



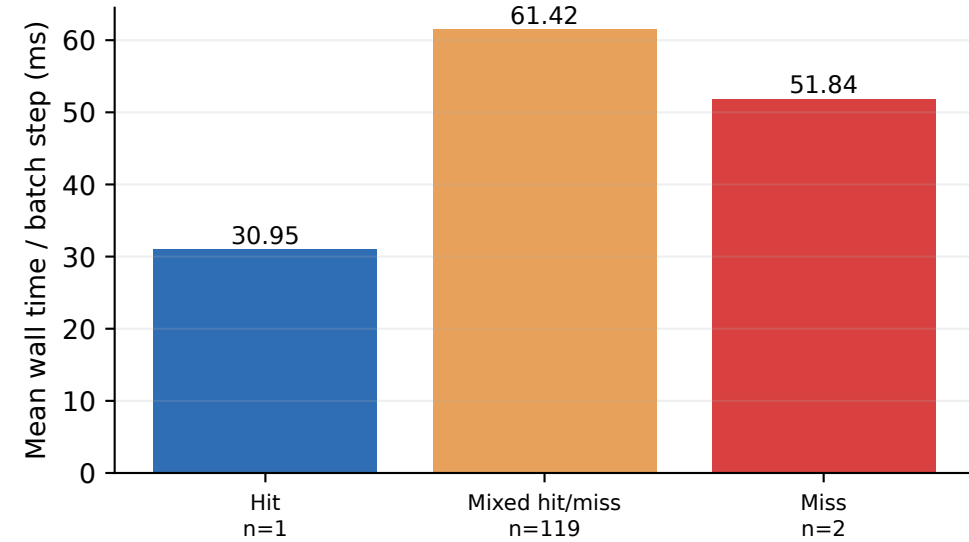
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_ssd_4
SSD: K=8, F=3
Last pass: B=8, T=0.7, N=48 | AL*=3.084, hit=46.9%, TPS*=386.61

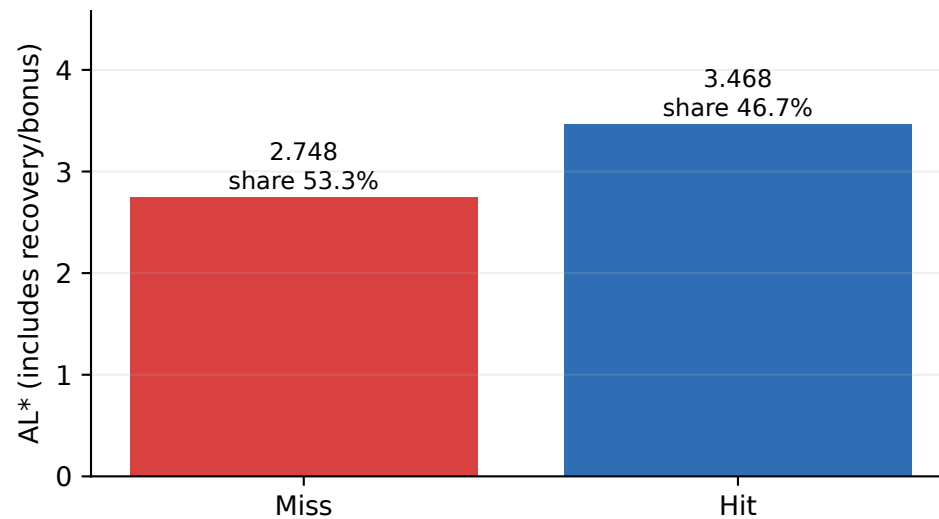
Measured wall time by batch cache status



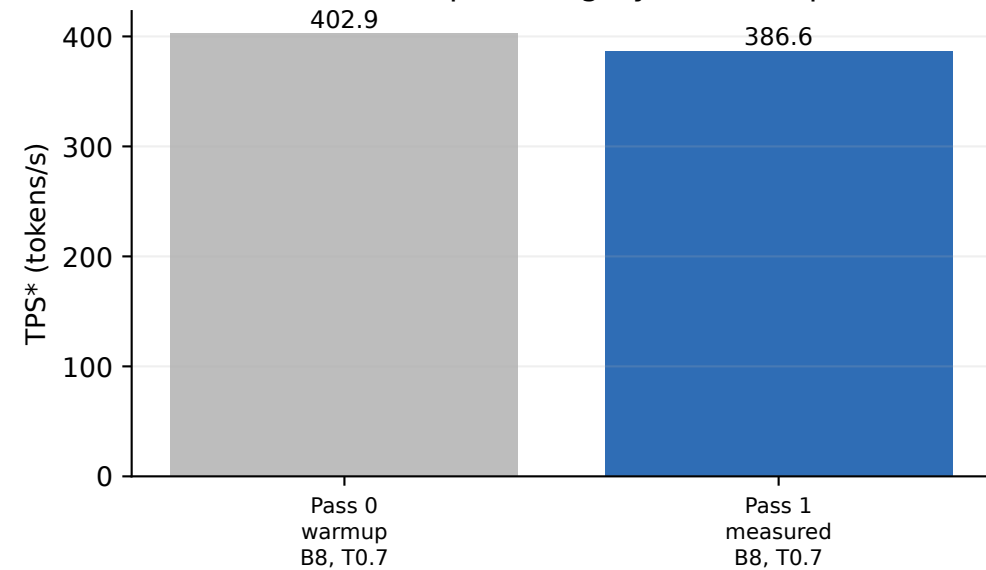
Boundary-free steps, including slow steps



Conditional AL* and source share



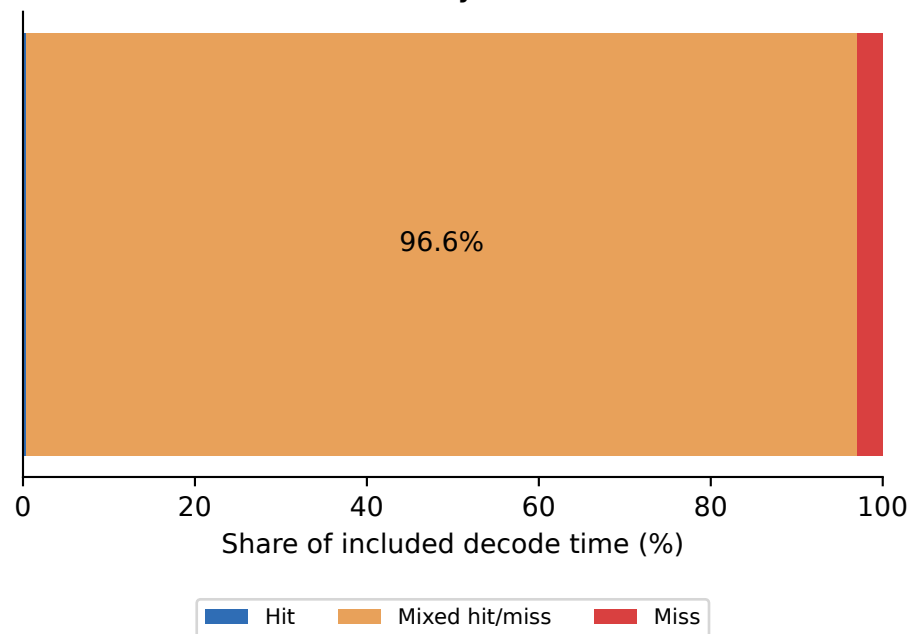
All saved passes; gray = warmup



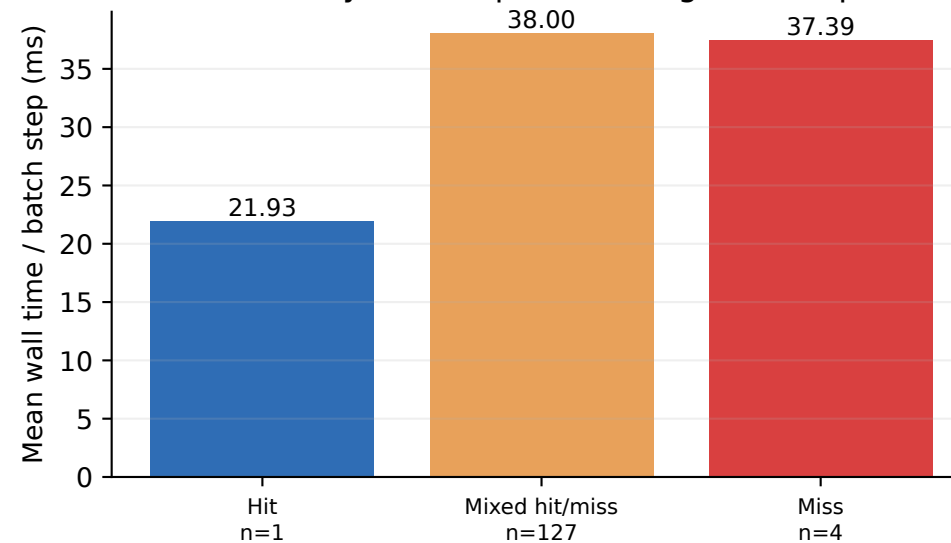
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_warm_ssd_5
SSD: K=6, F=2
Last pass: B=8, T=0.7, N=48 | AL*=2.787, hit=38.9%, TPS*=562.52

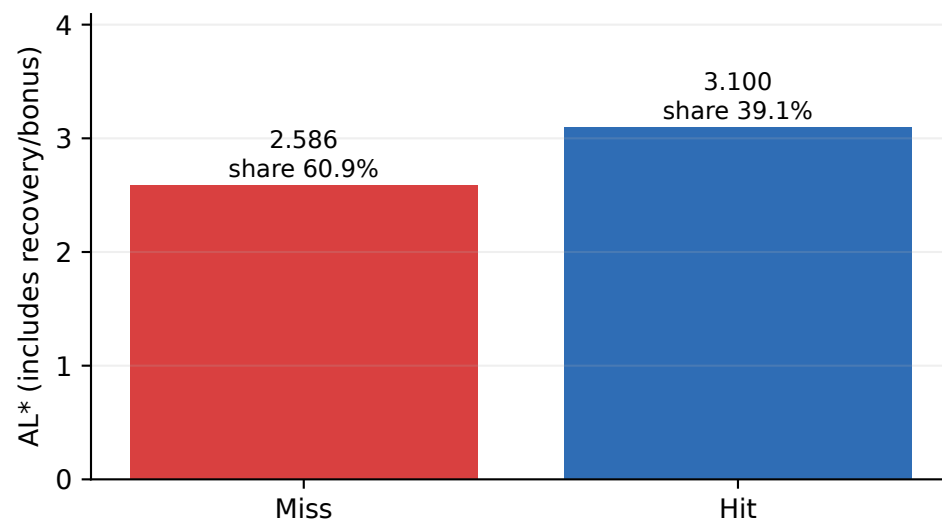
Measured wall time by batch cache status



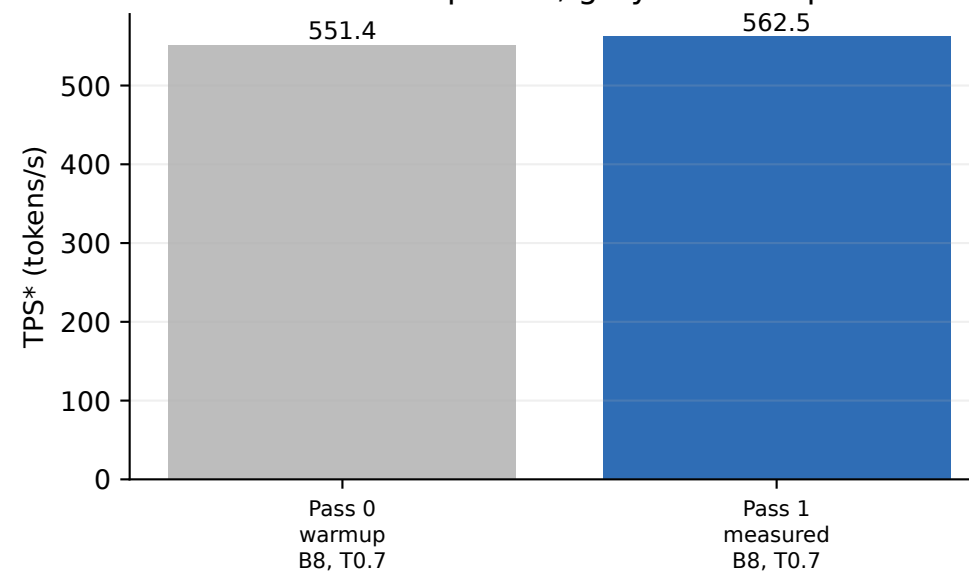
Boundary-free steps, including slow steps



Conditional AL* and source share



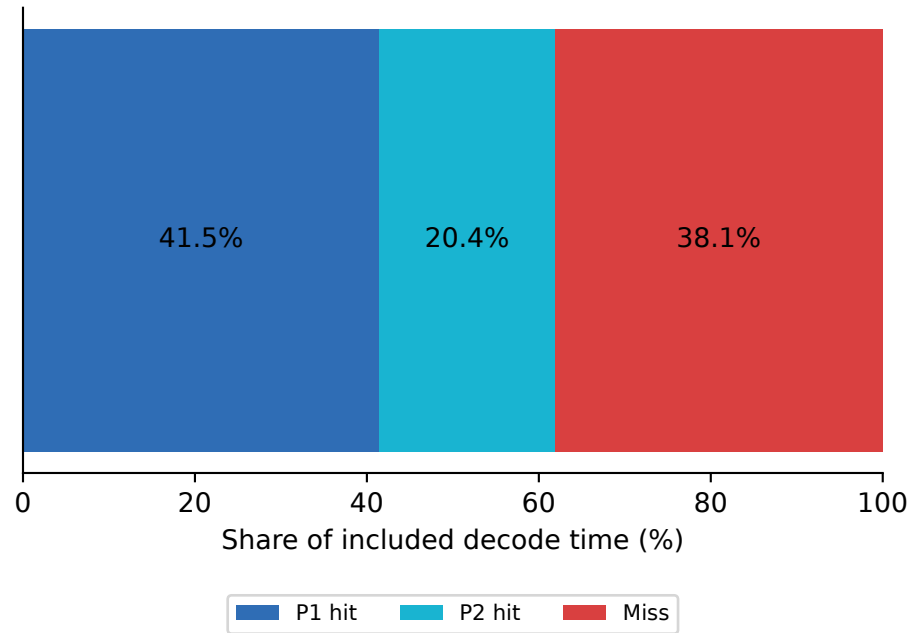
All saved passes; gray = warmup



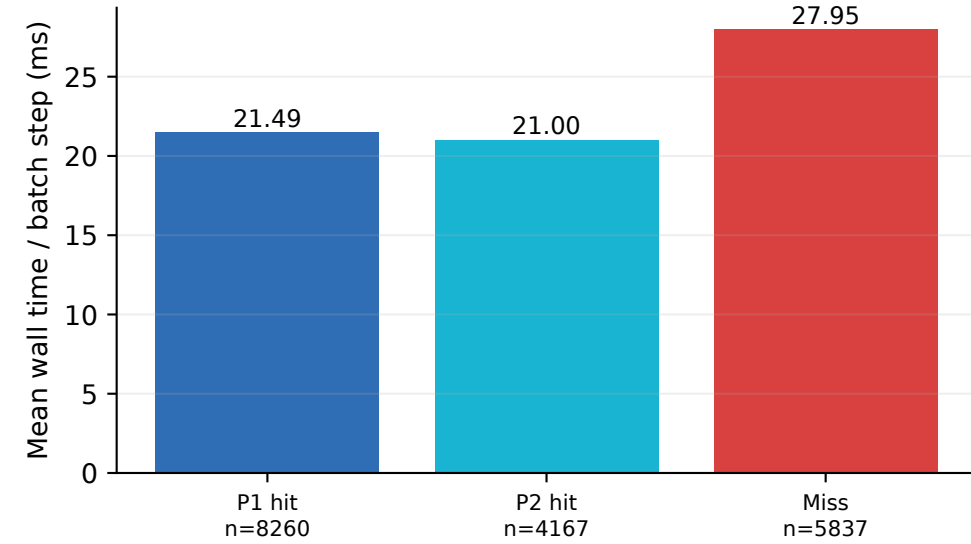
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_opt_fast_r0
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/0
Last pass: B=1, T=0.7, N=480 | AL*=2.679, hit=68.1%, TPS*=114.27

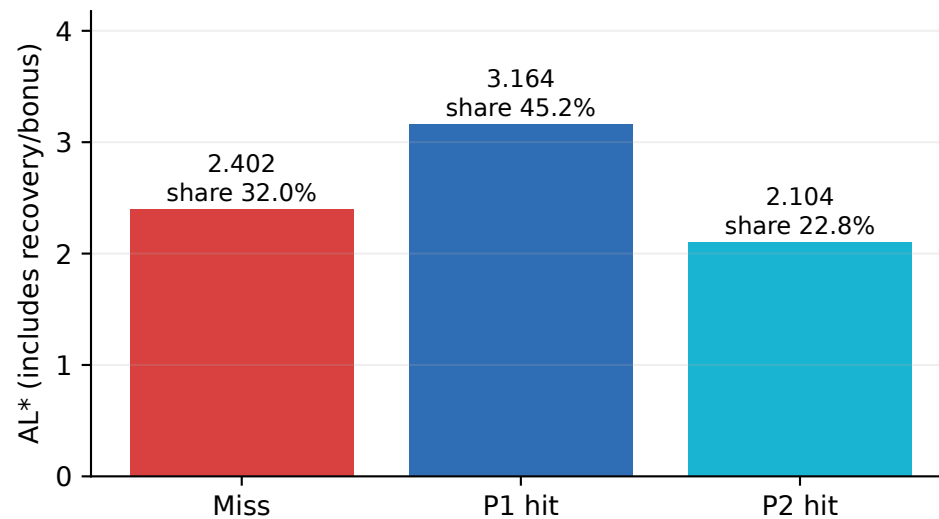
Measured wall time by batch cache status



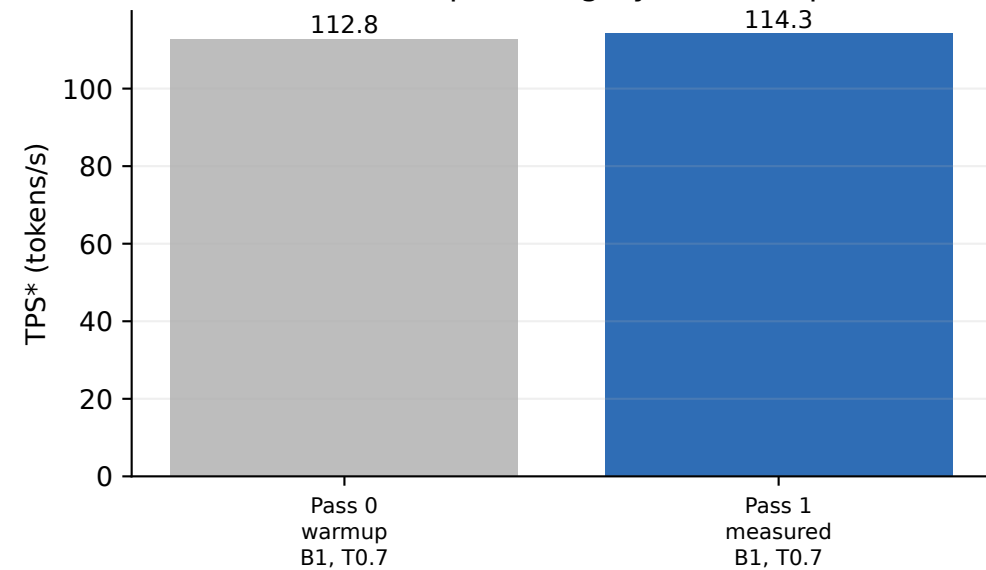
Boundary-free steps, including slow steps



Conditional AL* and source share



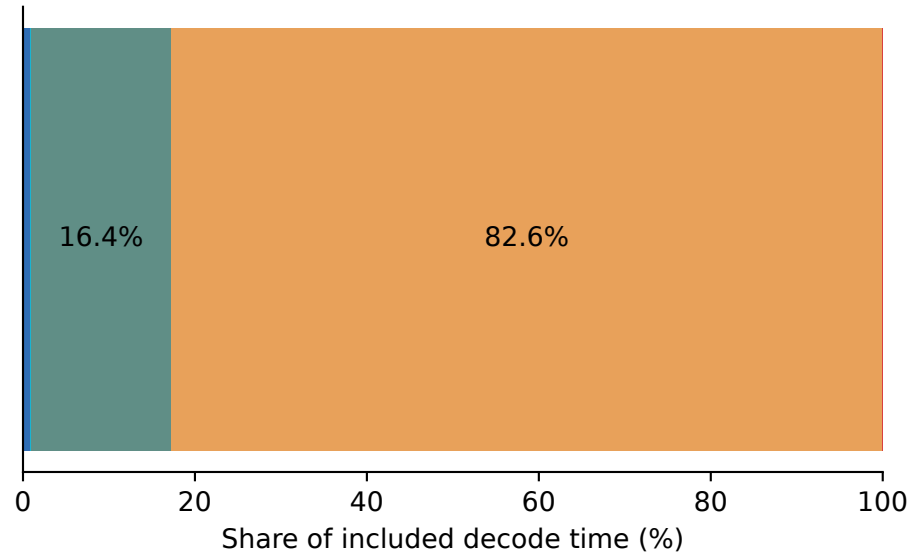
All saved passes; gray = warmup



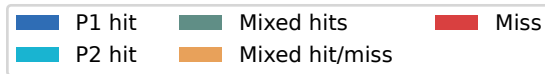
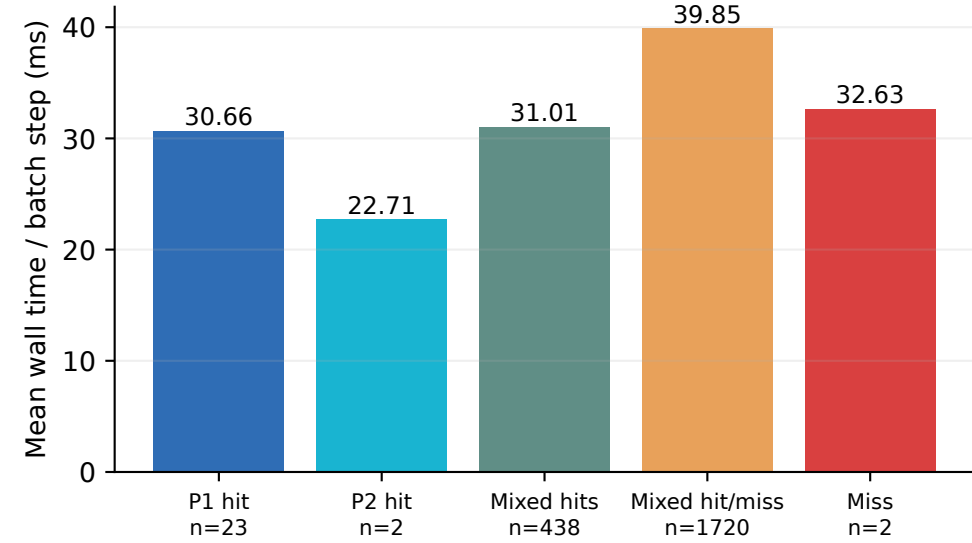
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_opt_fast_r1
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.386, hit=81.7%, TPS*=499.91

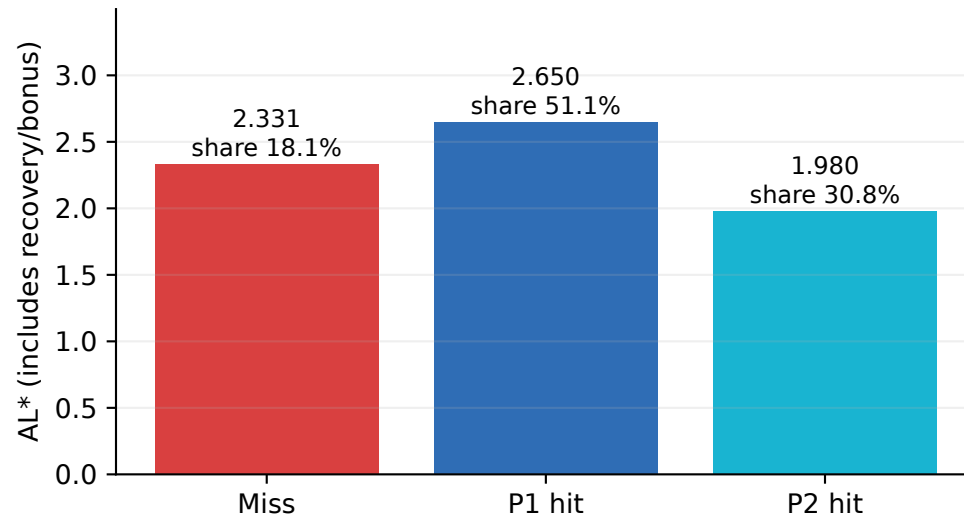
Measured wall time by batch cache status



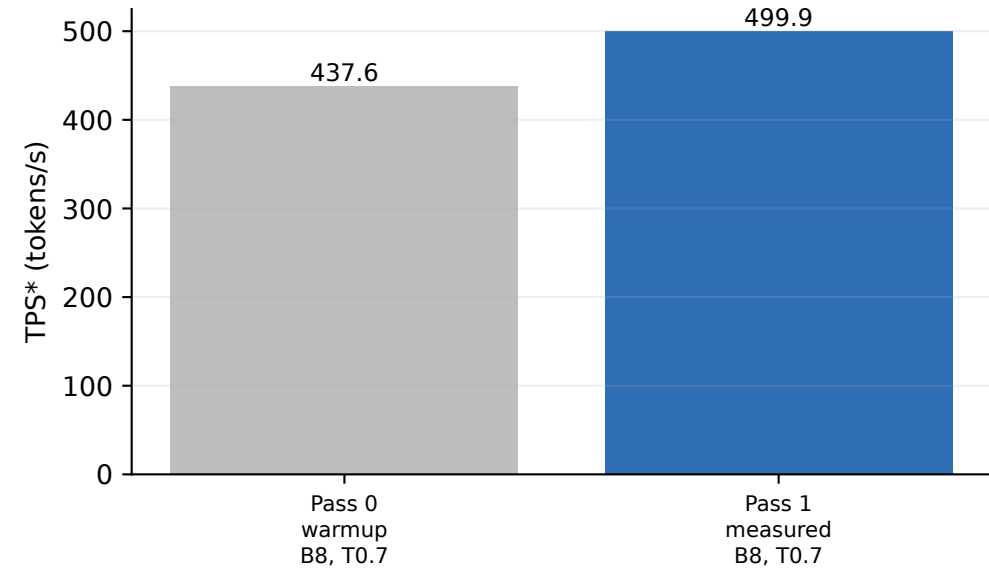
Boundary-free steps, including slow steps



Conditional AL* and source share

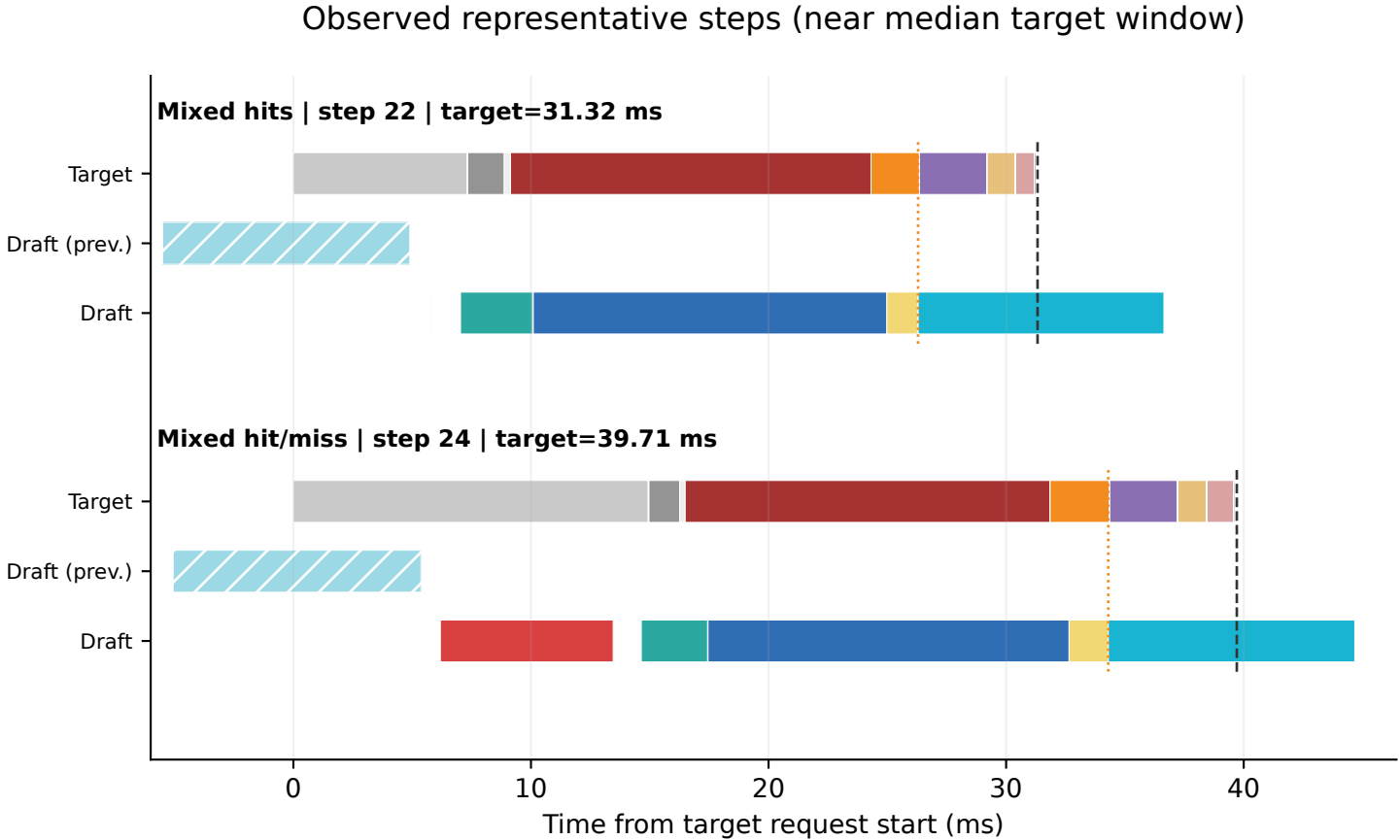
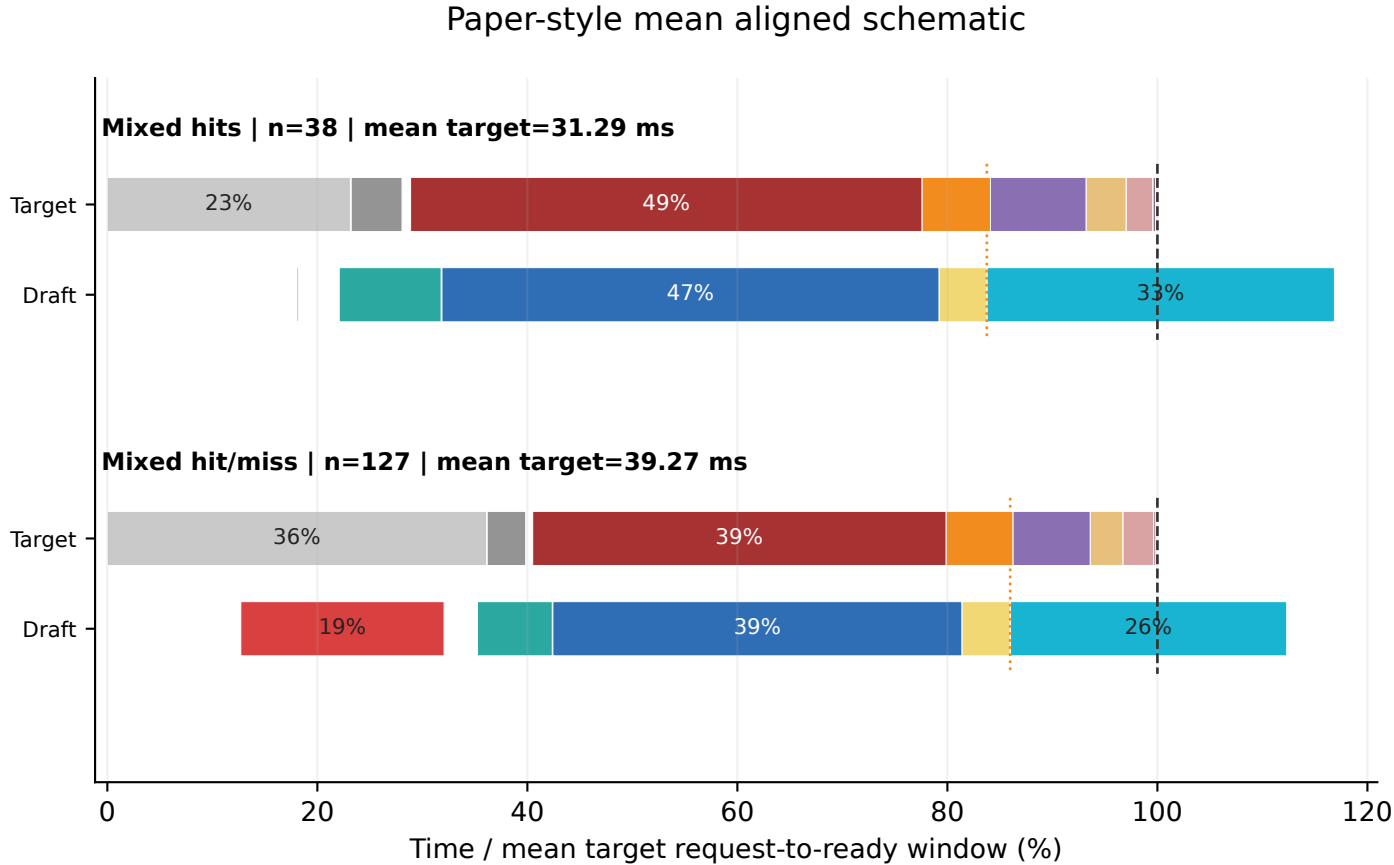
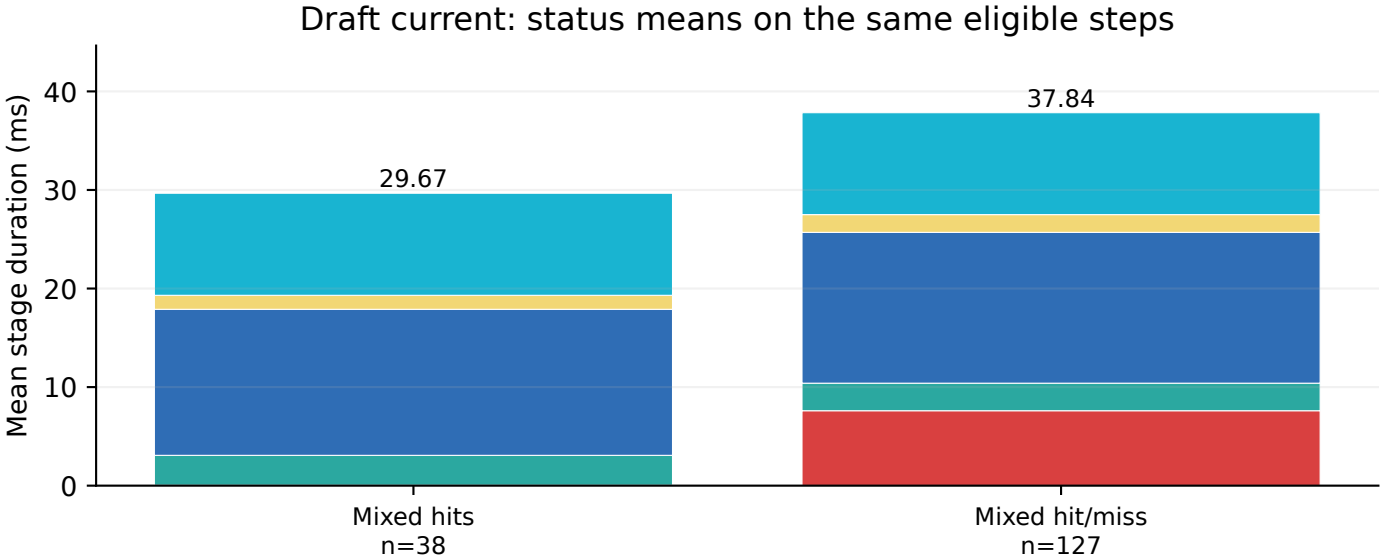
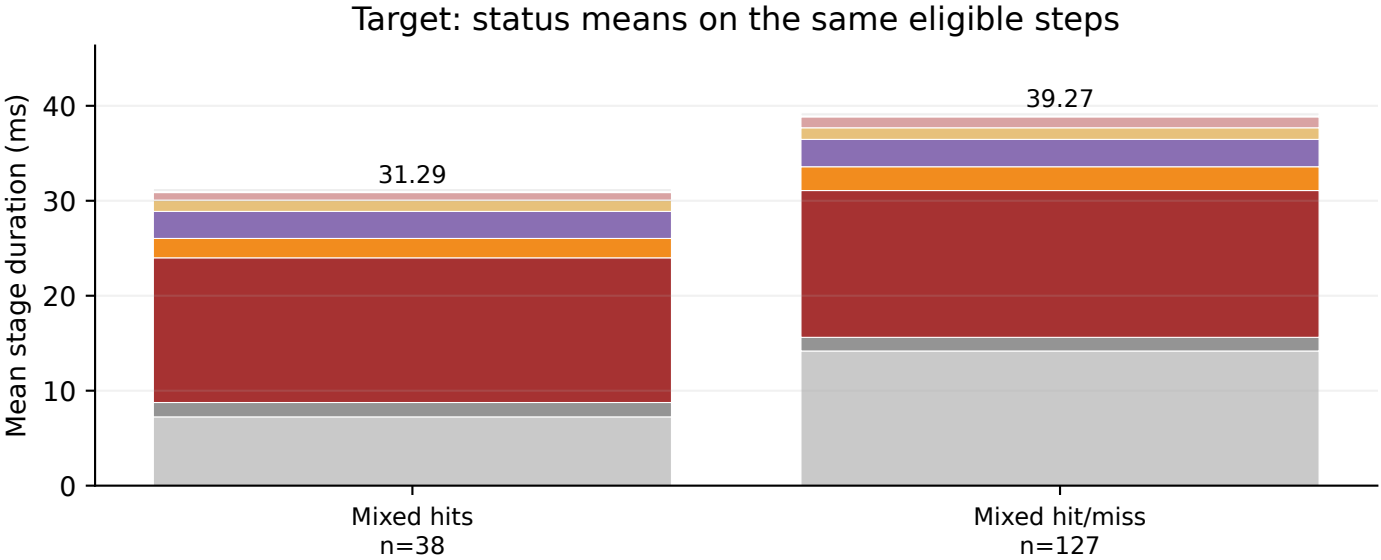


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

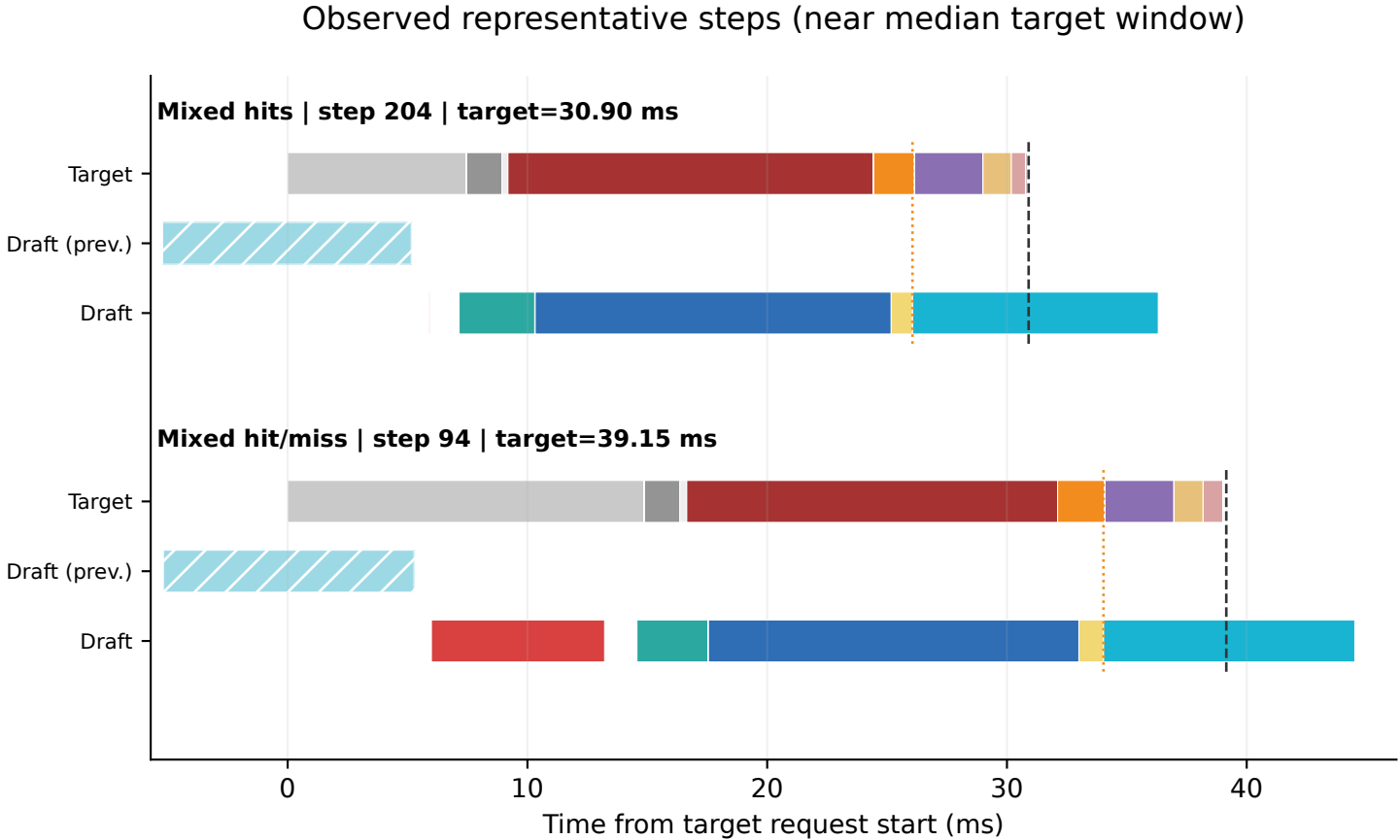
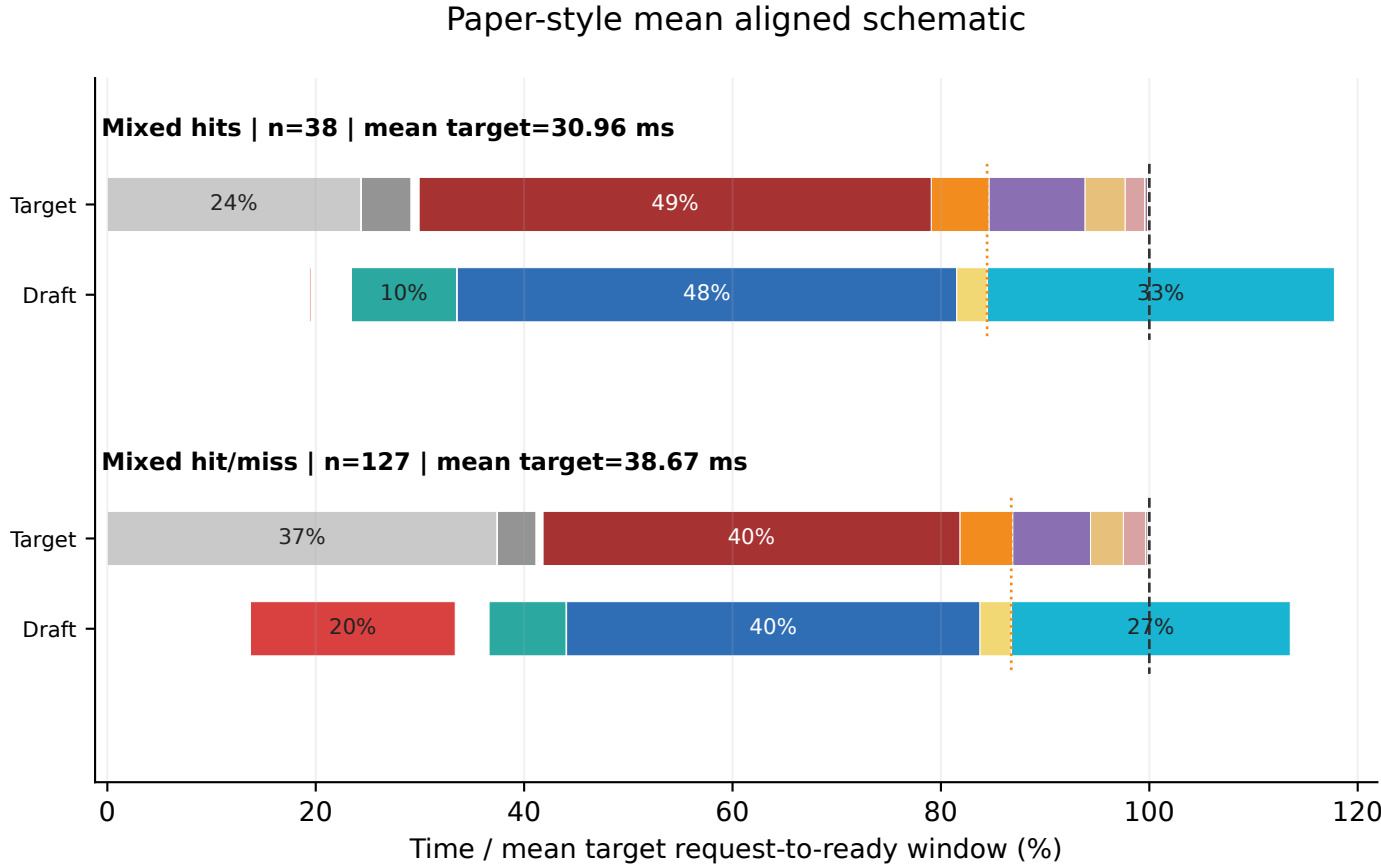
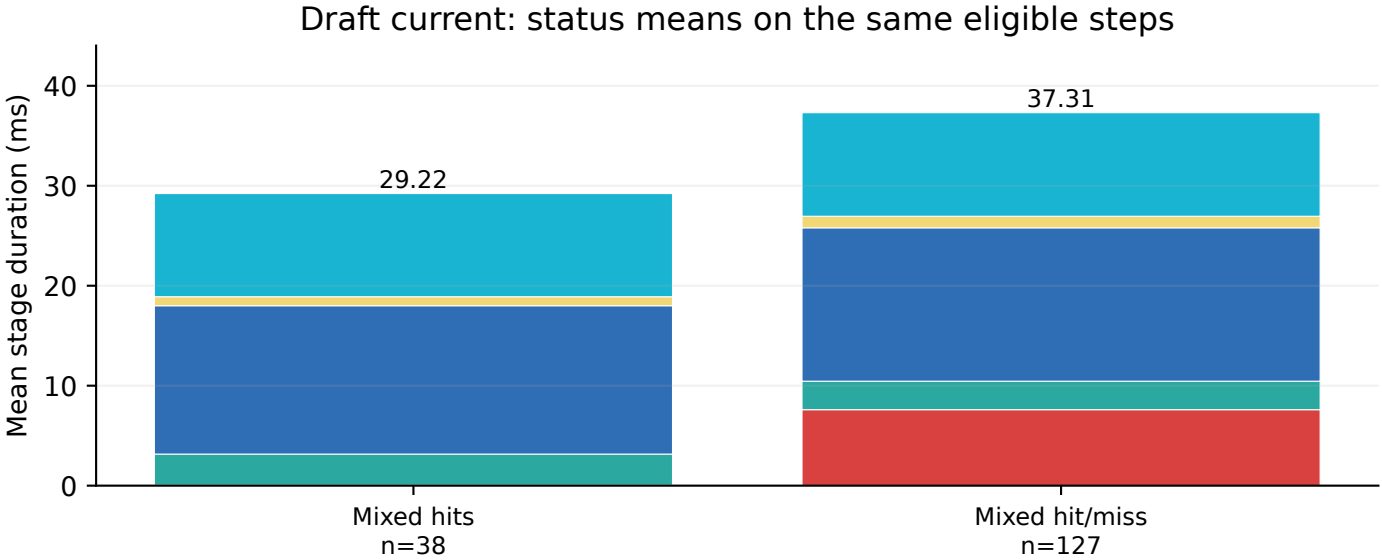
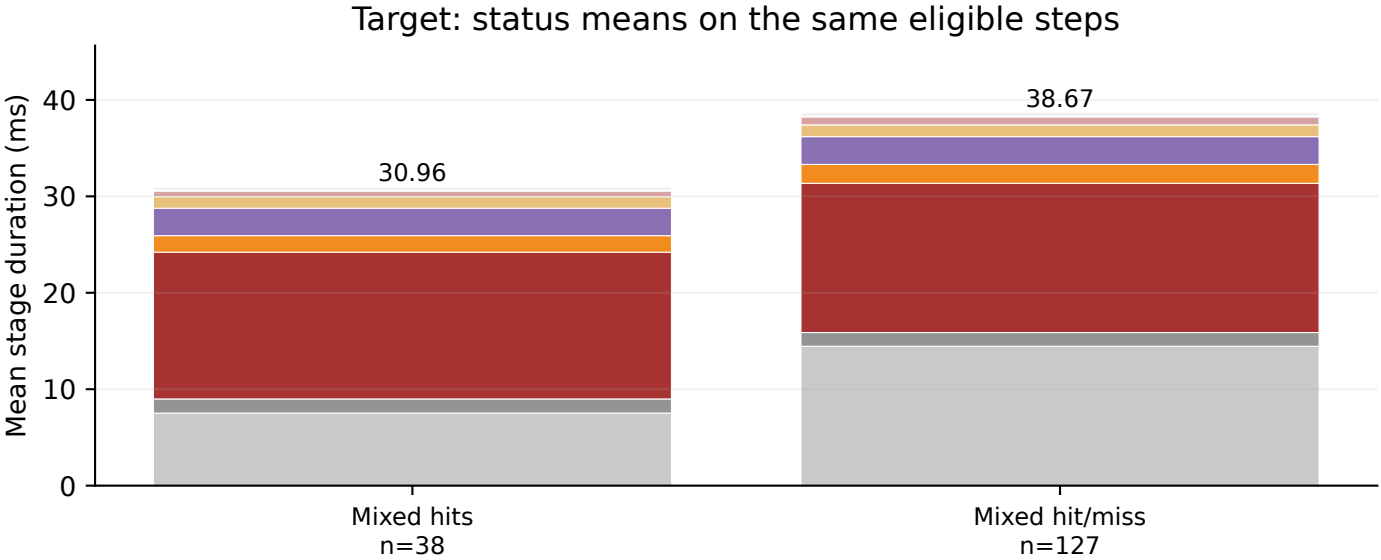
llama3_b8_opt_I0_profile_trim0
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 165 eligible steps



Wait / sync Prepare Target pre Target post Proxy / launch Final logits Accept Commit Postprocess Unlabelled gap P1 P2 Proxy wait Previous P2 Context Miss draft Target ready Proxy received

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_opt_I0_profile_trim1
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 165 eligible steps

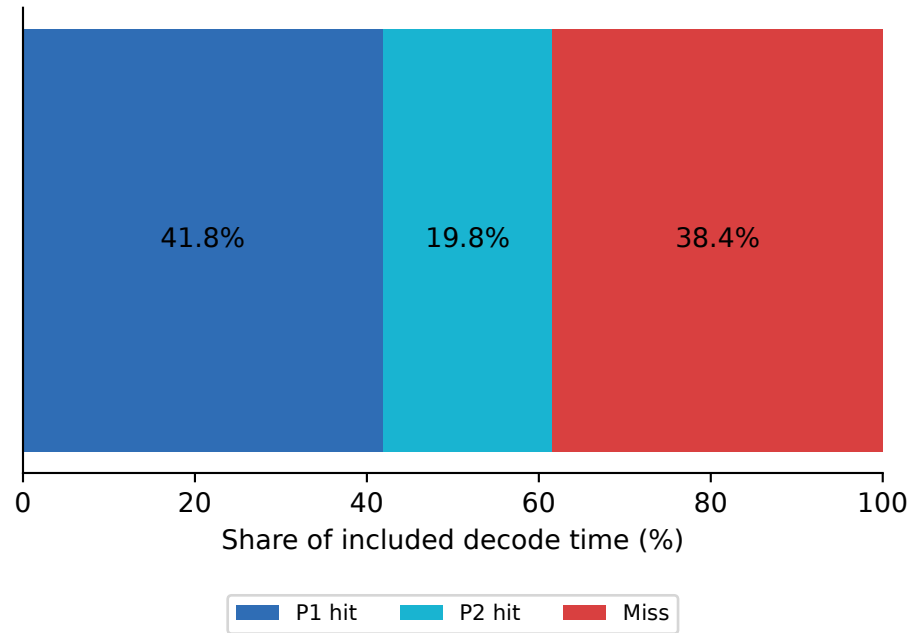


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

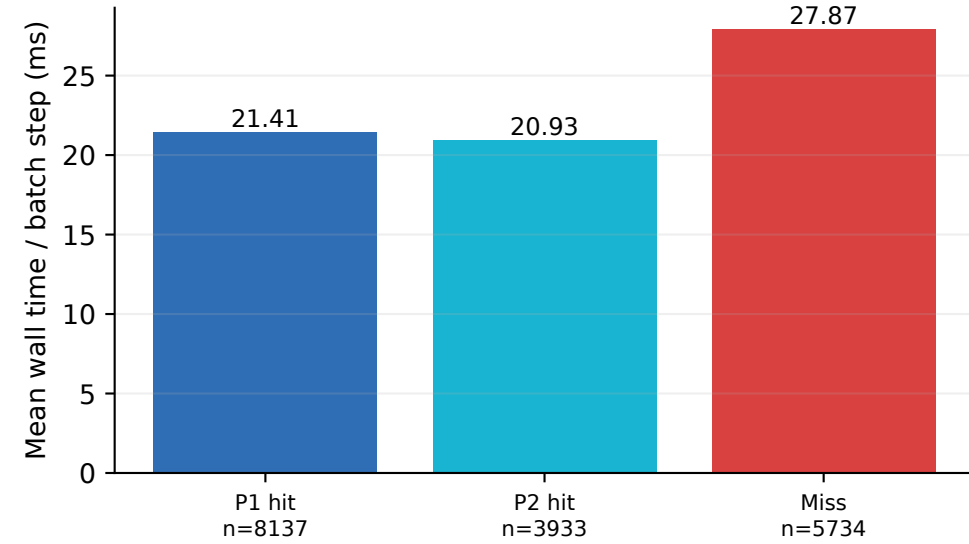
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_opt_fast_r1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/0
Last pass: B=1, T=0.7, N=480 | AL*=2.702, hit=67.7%, TPS*=115.53

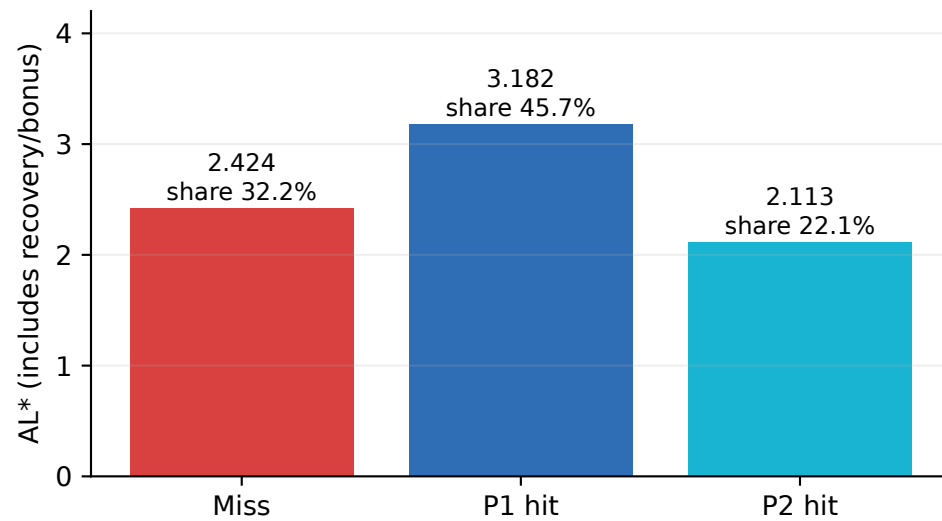
Measured wall time by batch cache status



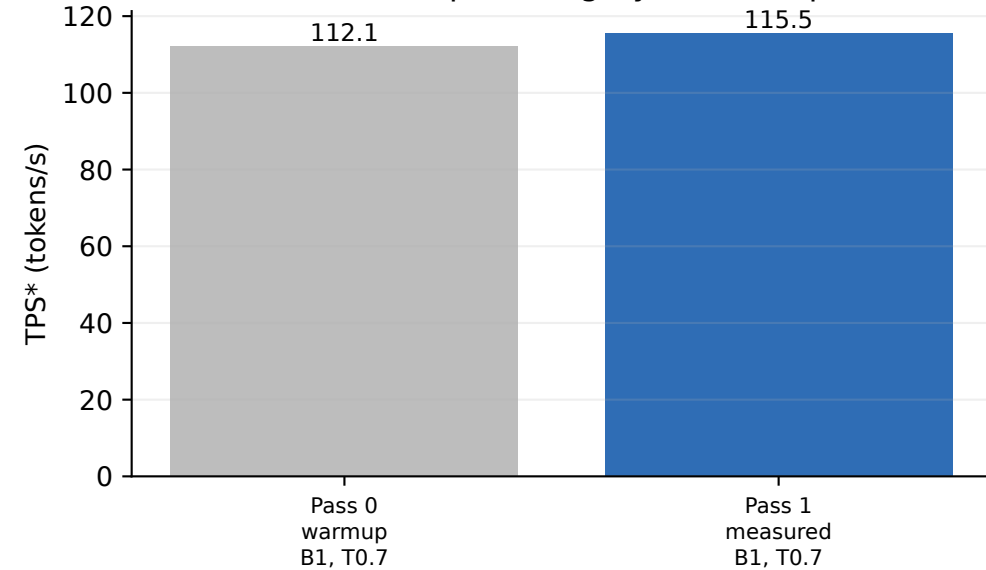
Boundary-free steps, including slow steps



Conditional AL* and source share



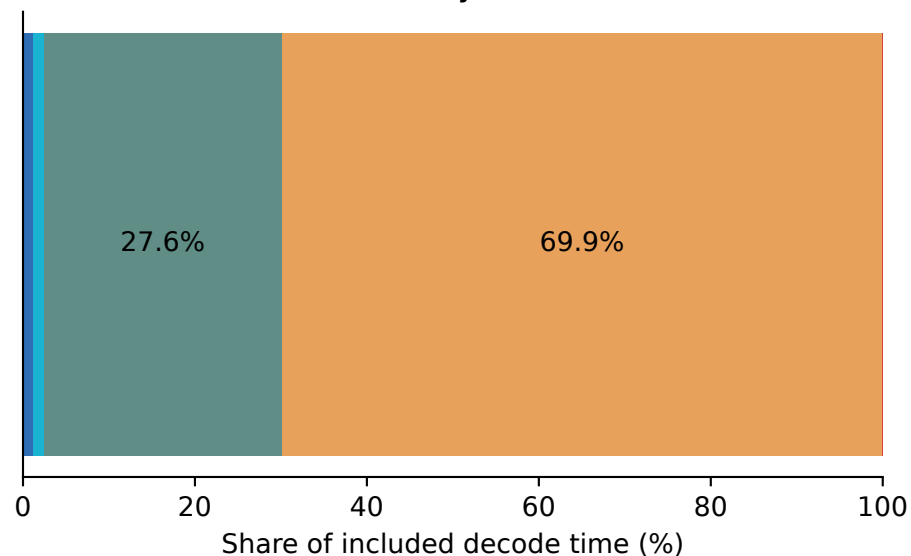
All saved passes; gray = warmup



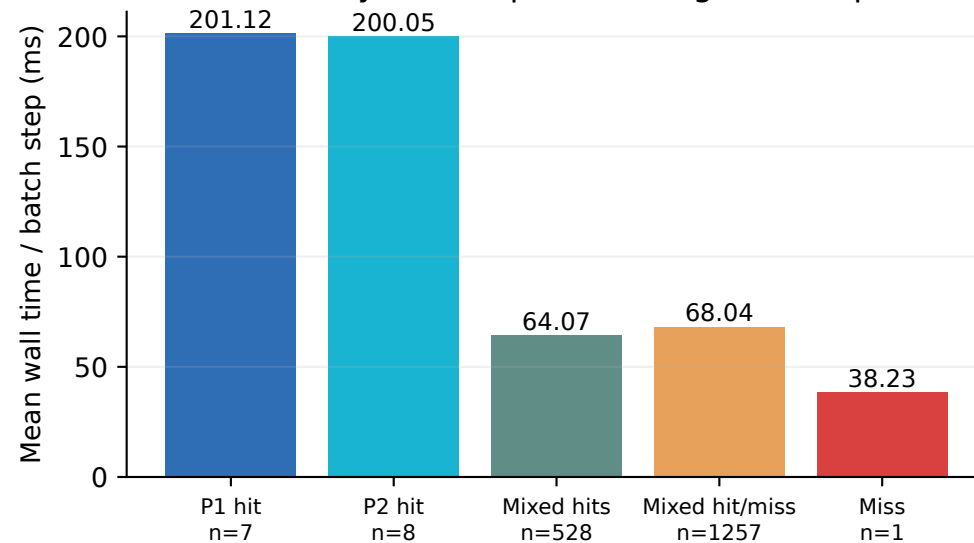
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_opt_al_r0
 DUET: K1/K2=6/3, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.903, hit=86.1%, TPS*=337.39

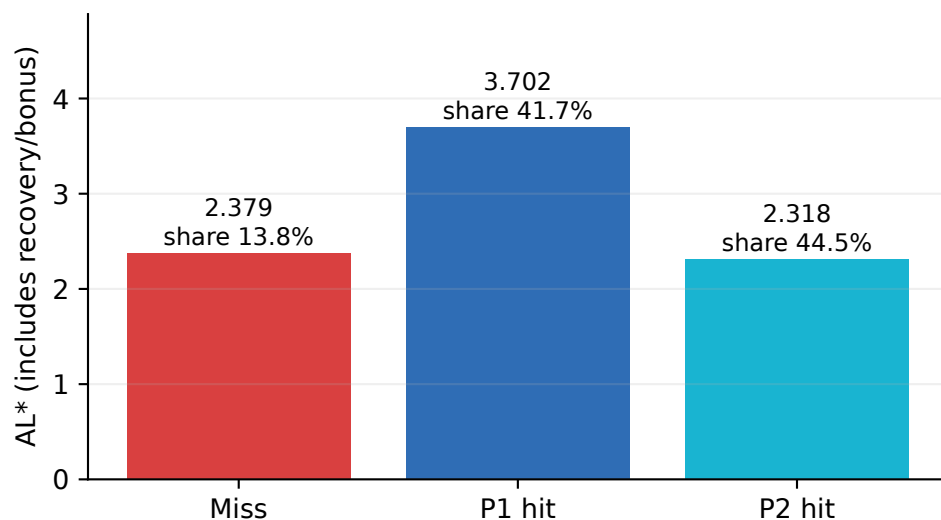
Measured wall time by batch cache status



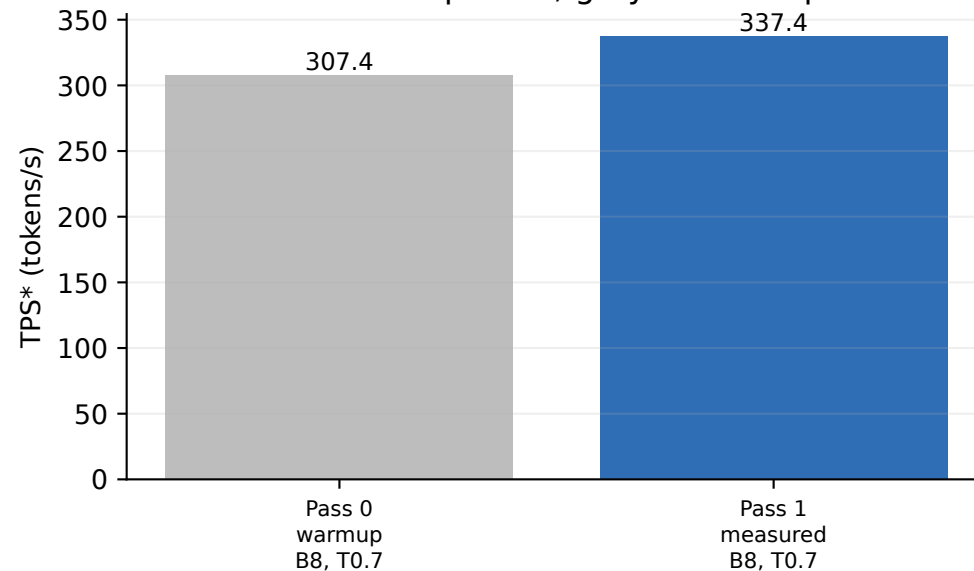
Boundary-free steps, including slow steps



Conditional AL* and source share



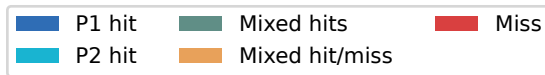
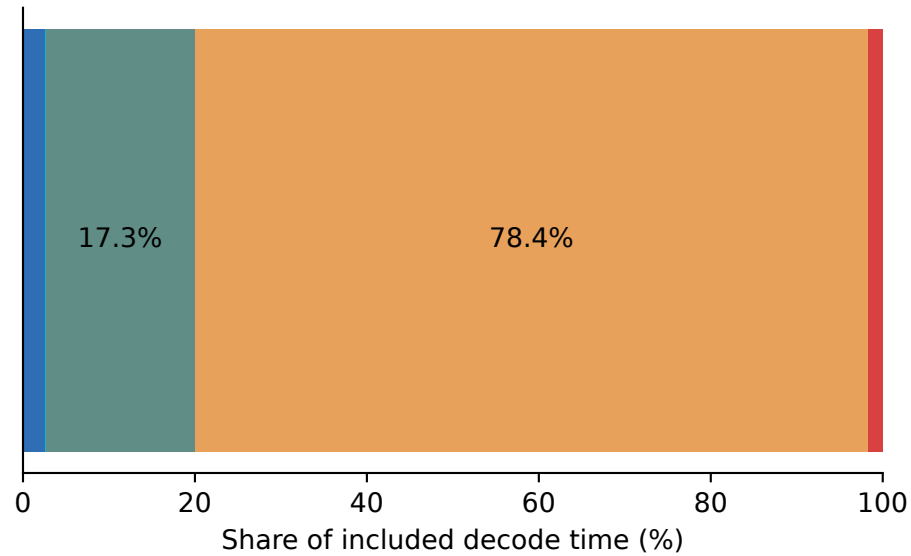
All saved passes; gray = warmup



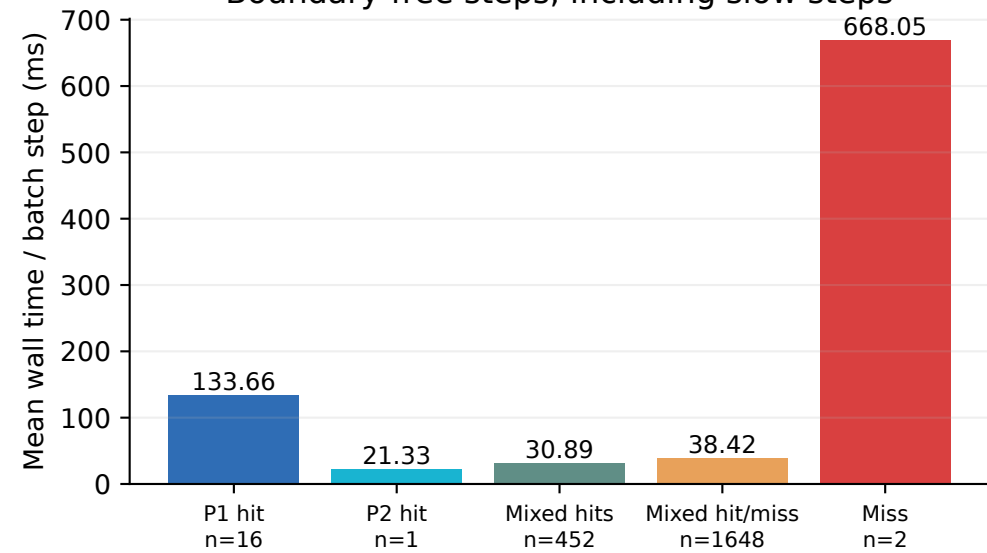
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_opt_fast_r0
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.429, hit=81.7%, TPS*=507.46

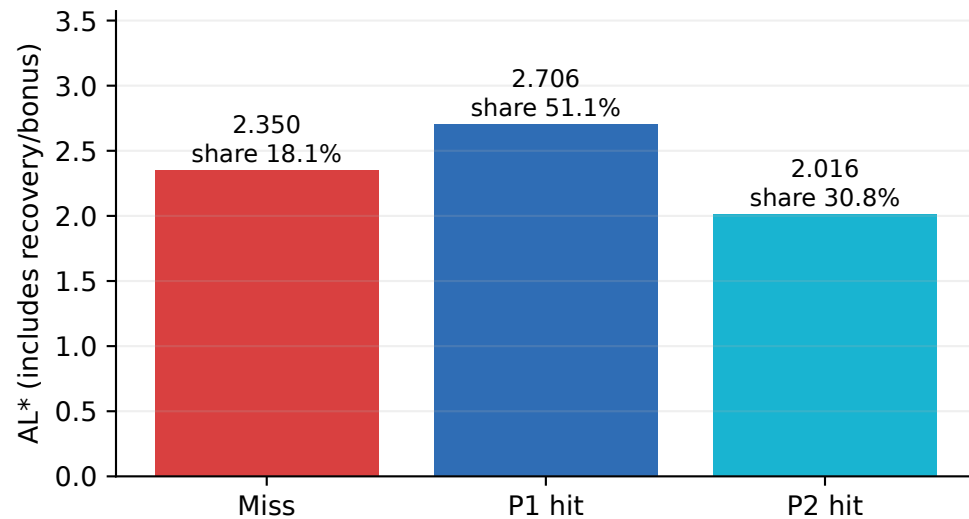
Measured wall time by batch cache status



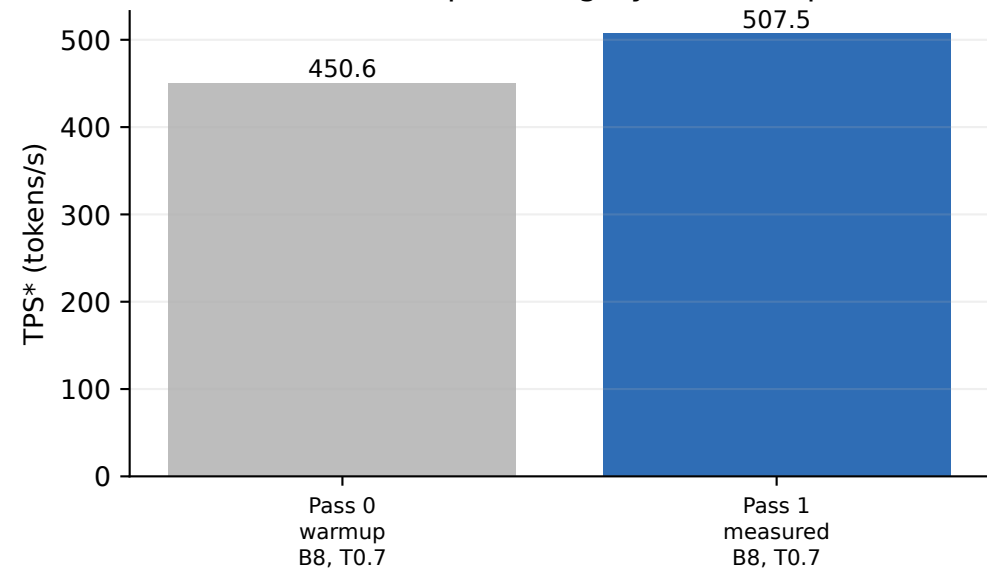
Boundary-free steps, including slow steps



Conditional AL* and source share

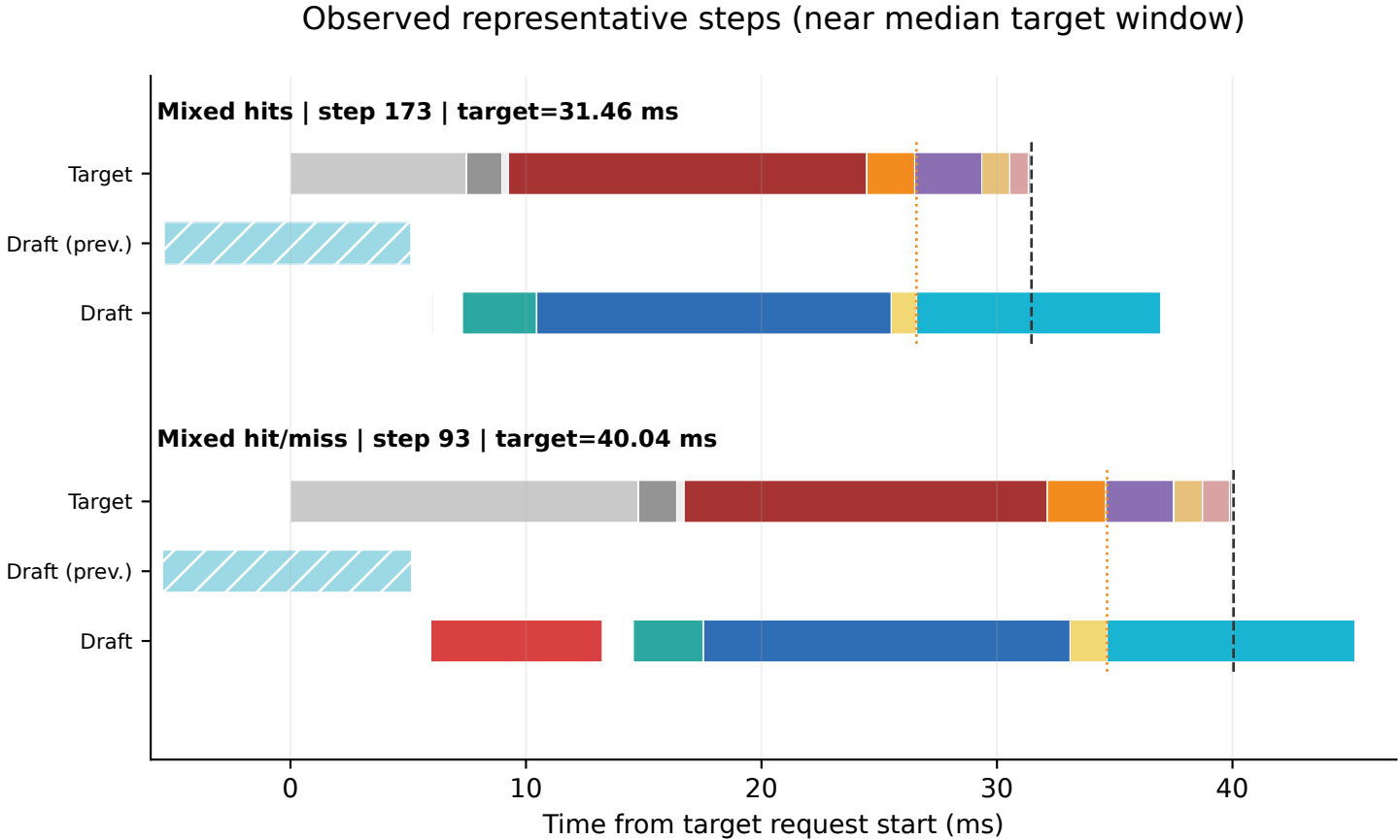
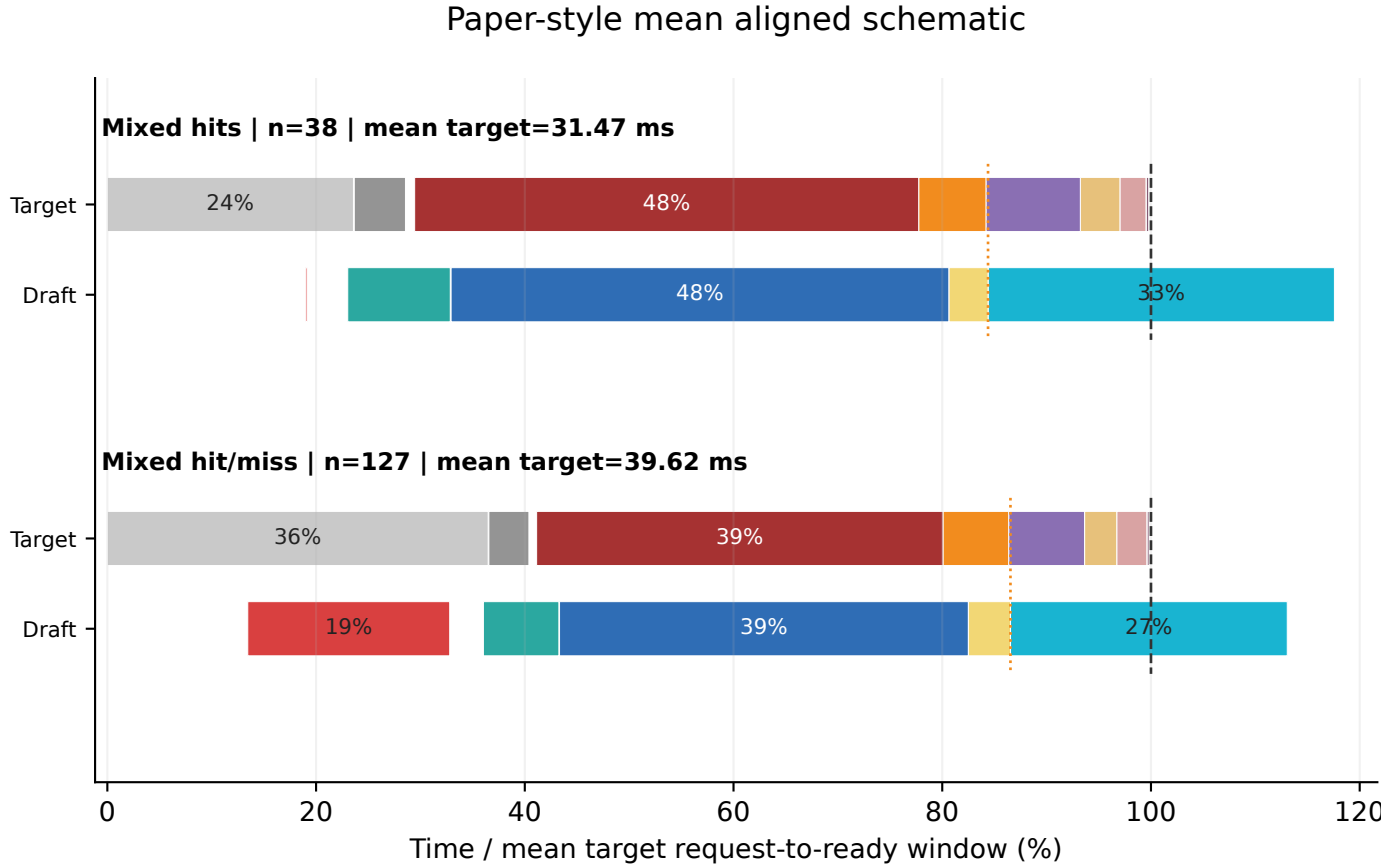
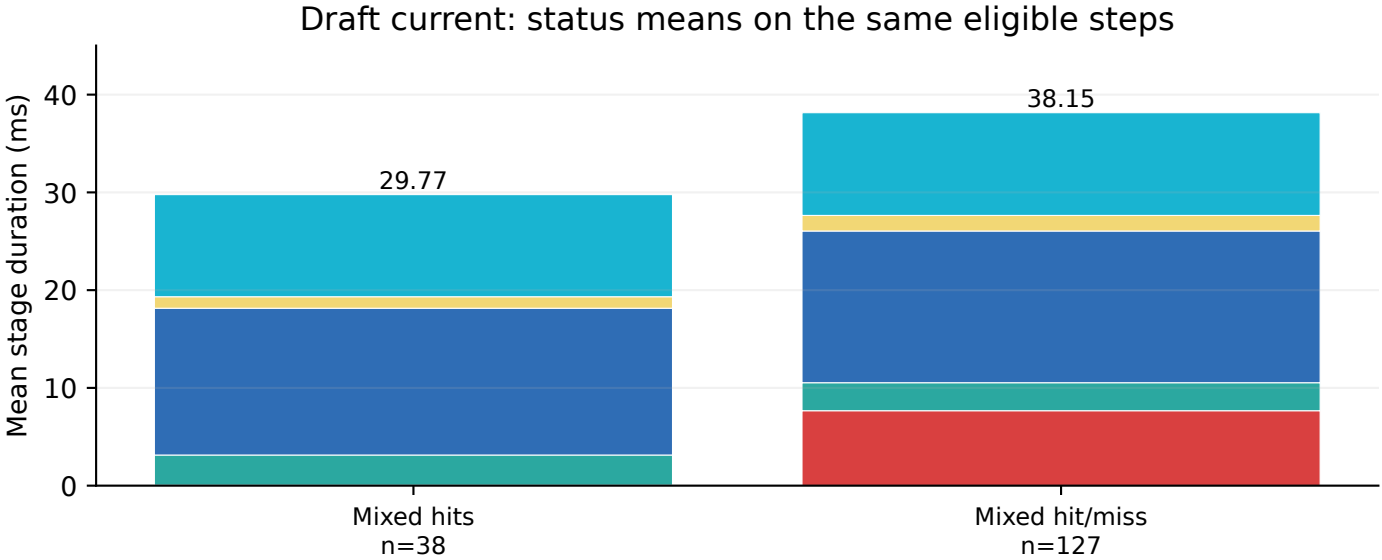
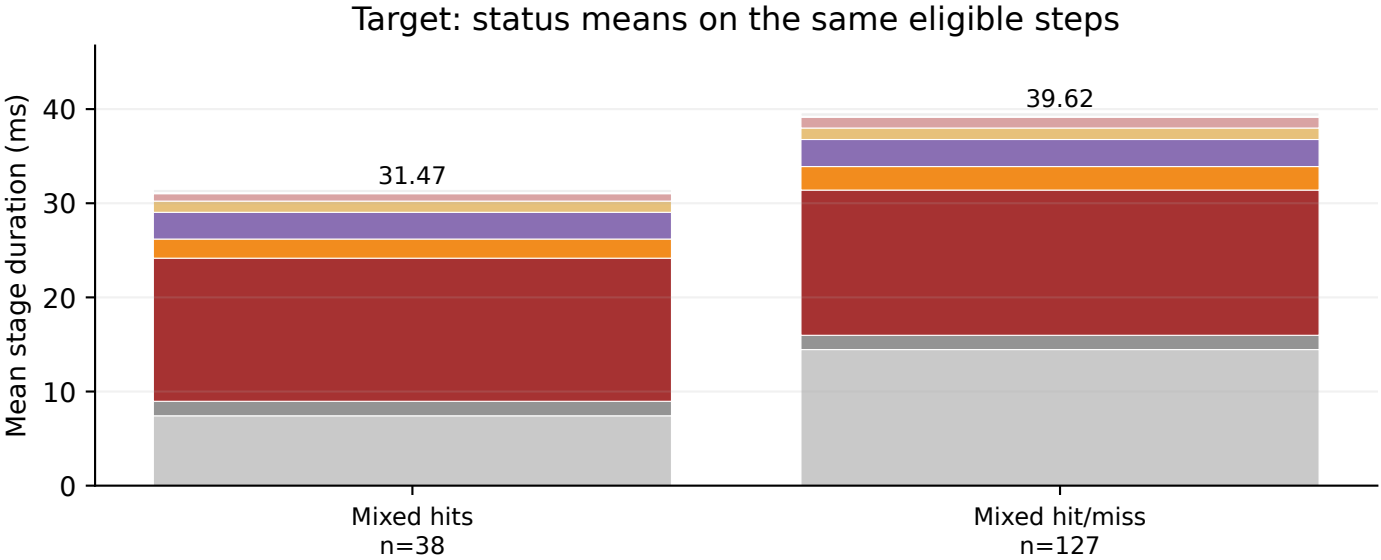


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

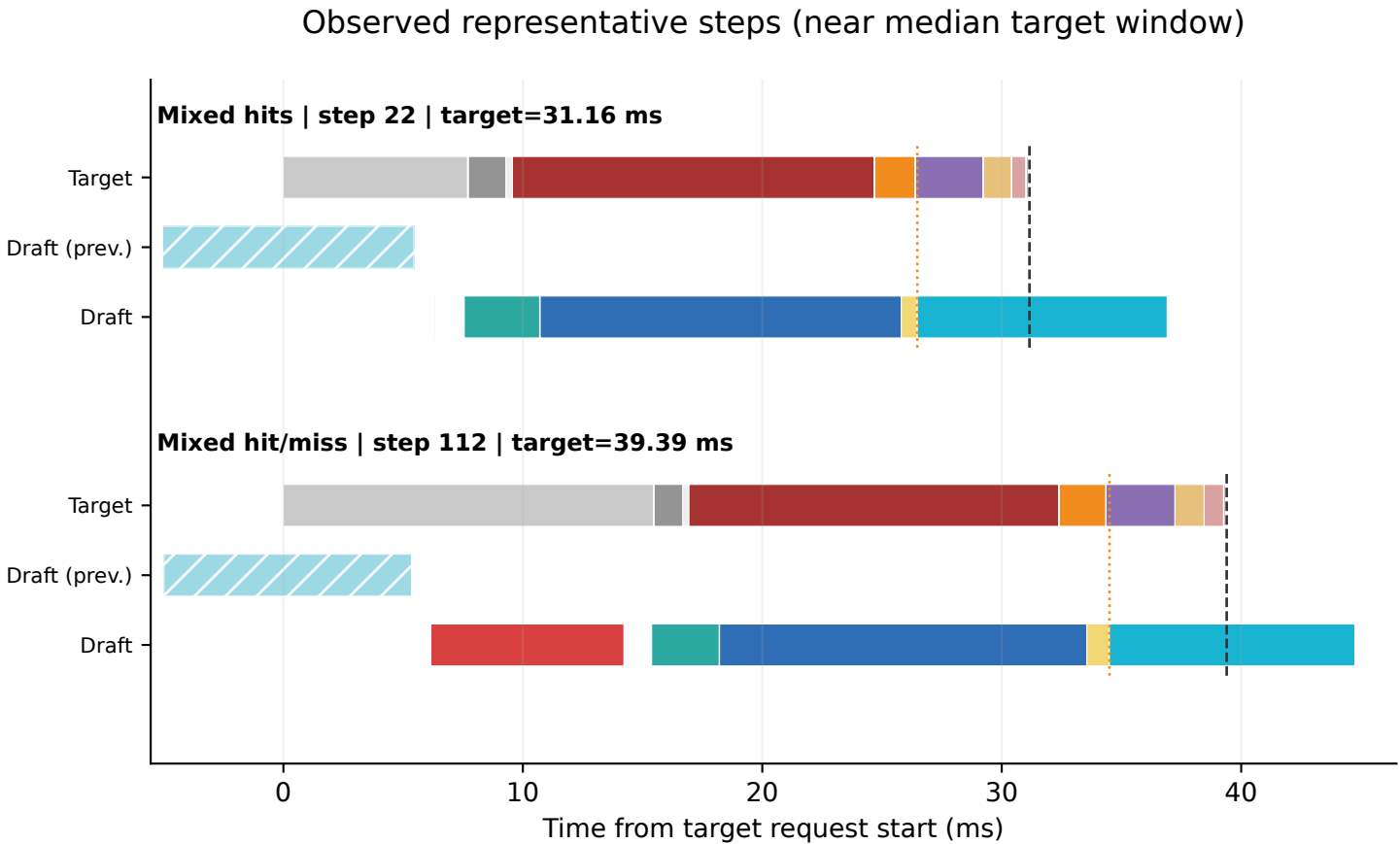
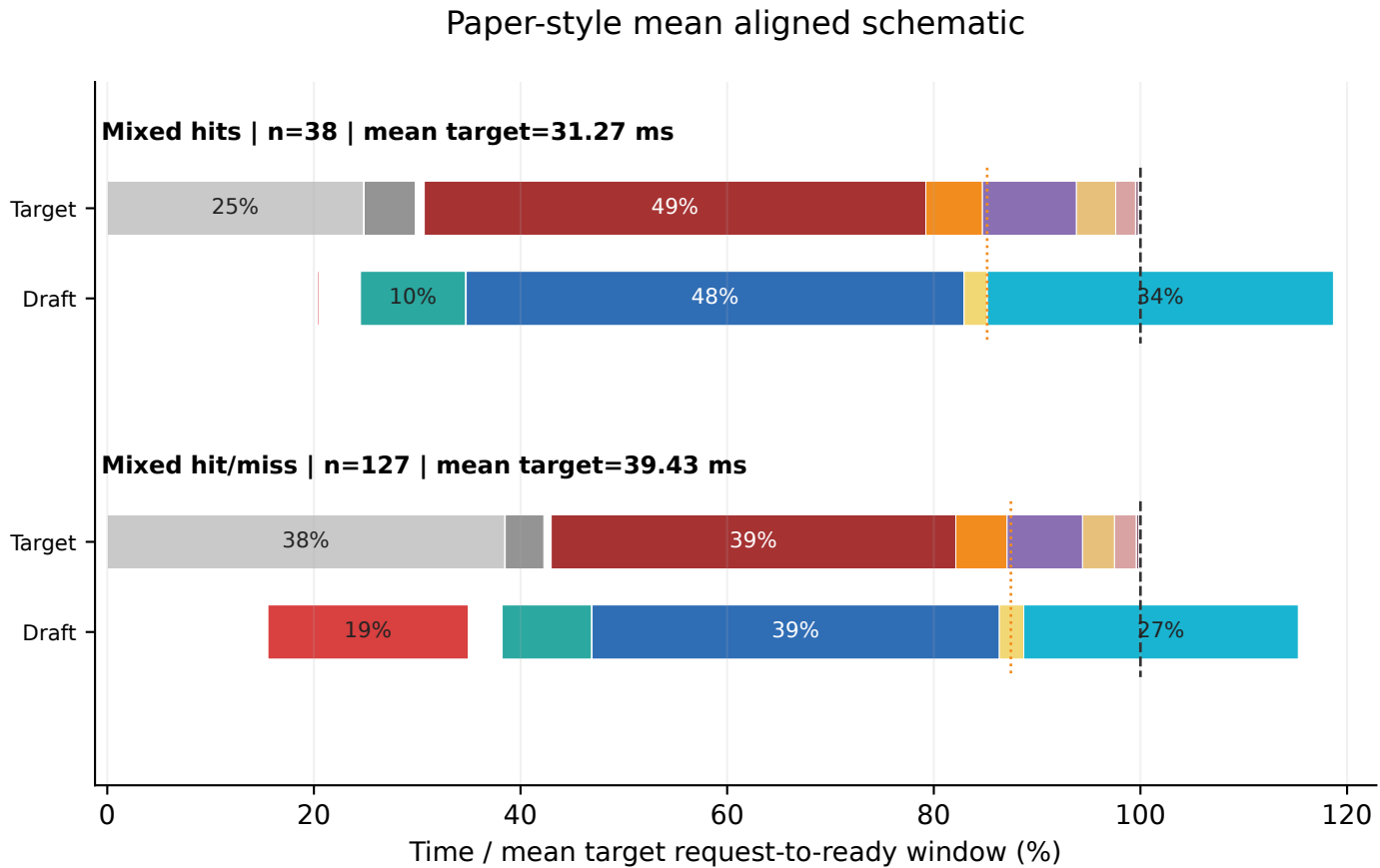
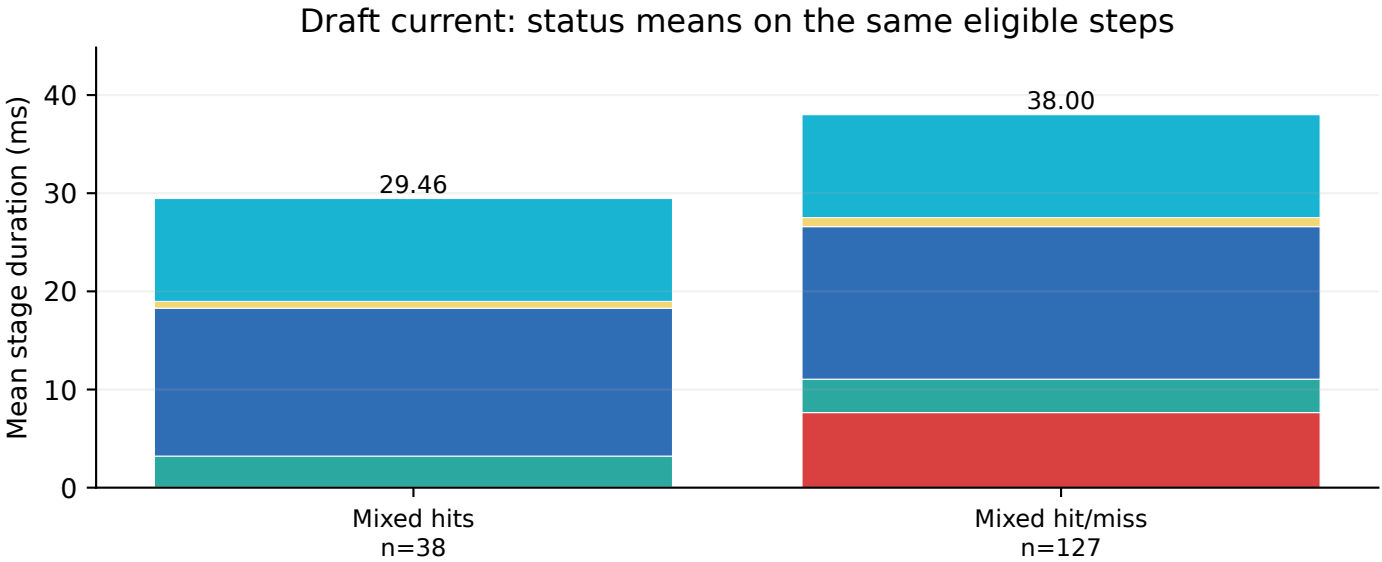
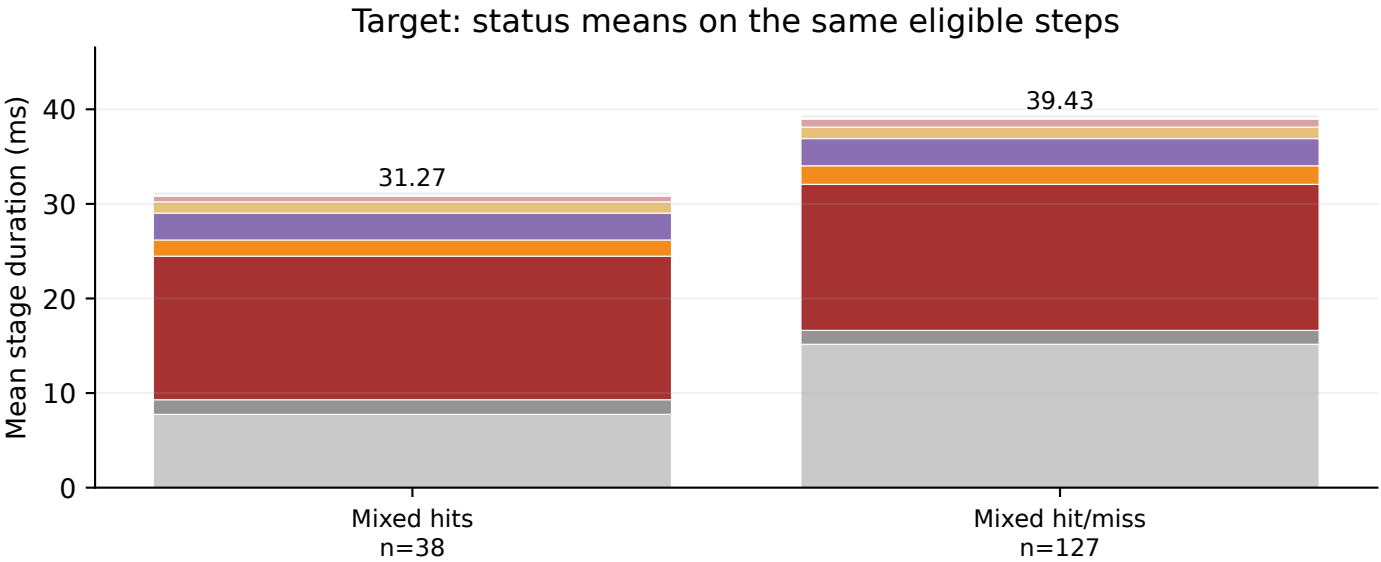
llama3_b8_opt_l1_profile_trim0
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 165 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_opt_l1_profile_trim1
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 165 eligible steps

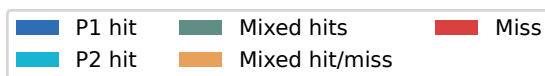
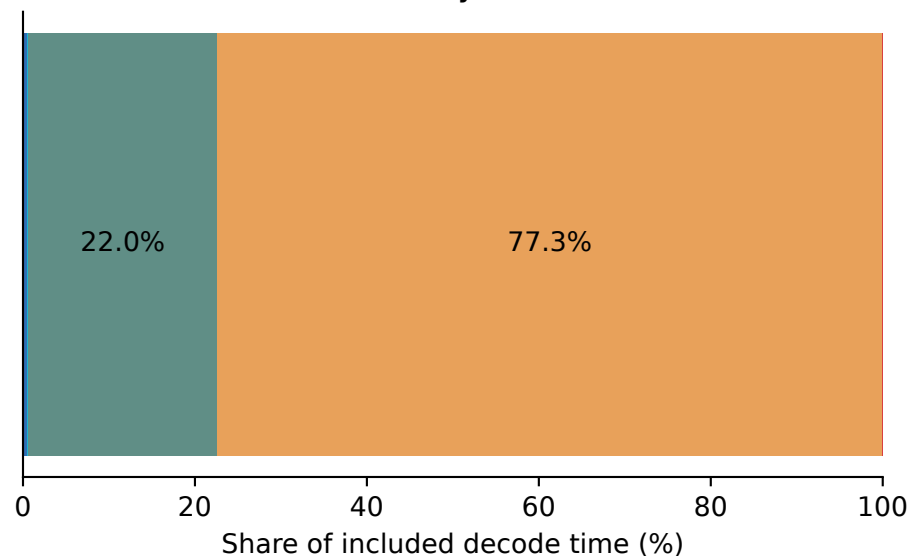


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

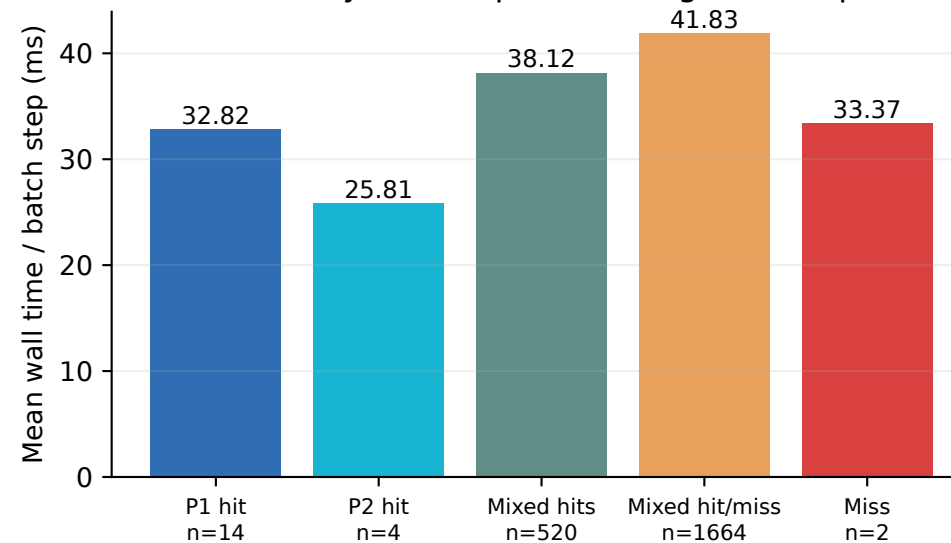
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_postopt_final_r1
 DUET: K1/K2=3/2, exit=28 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.366, hit=83.6%, TPS*=457.27

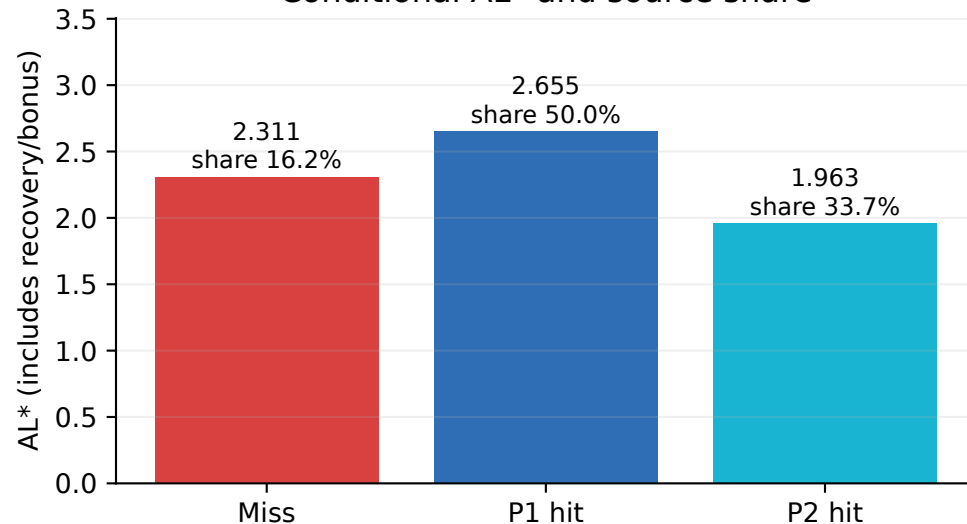
Measured wall time by batch cache status



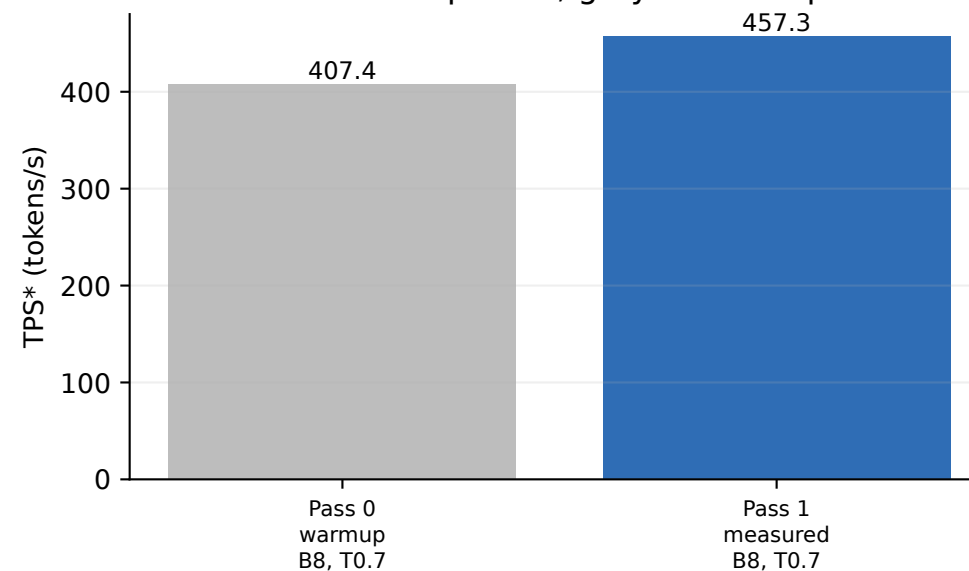
Boundary-free steps, including slow steps



Conditional AL* and source share



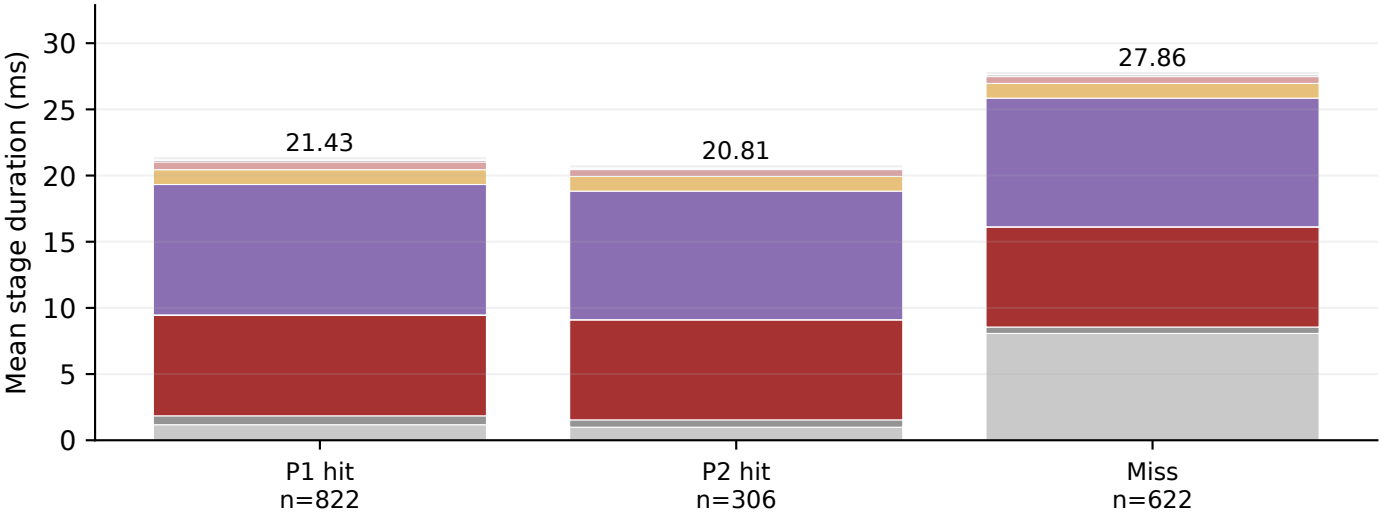
All saved passes; gray = warmup



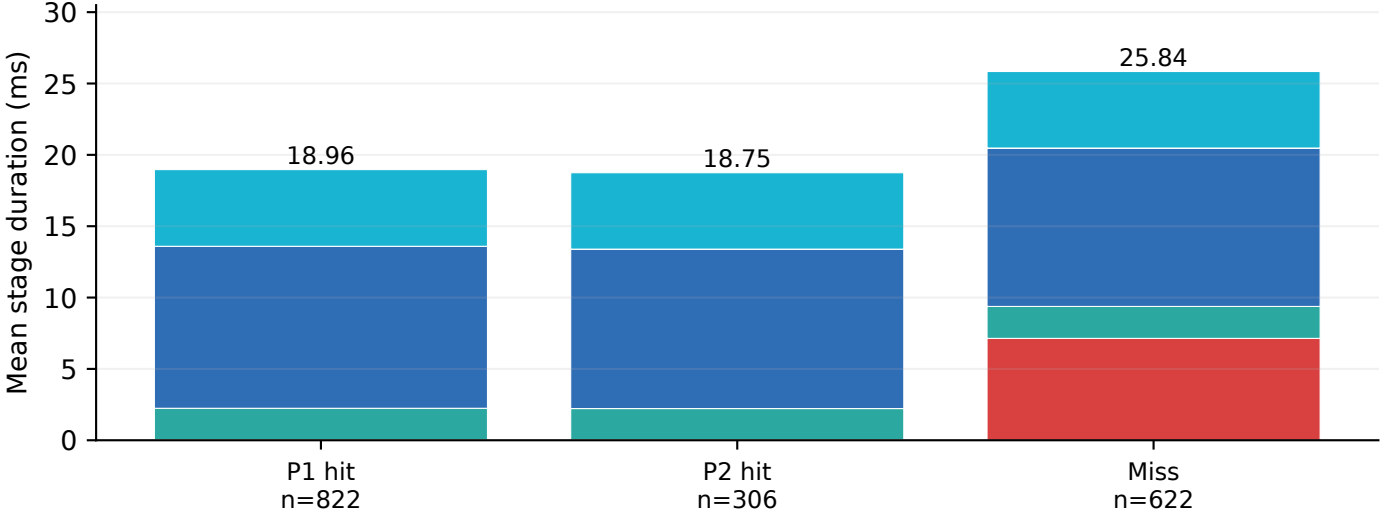
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_postopt_profile_exit_earlier
DUET: K1/K2=4/2, exit=14 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1750 eligible steps

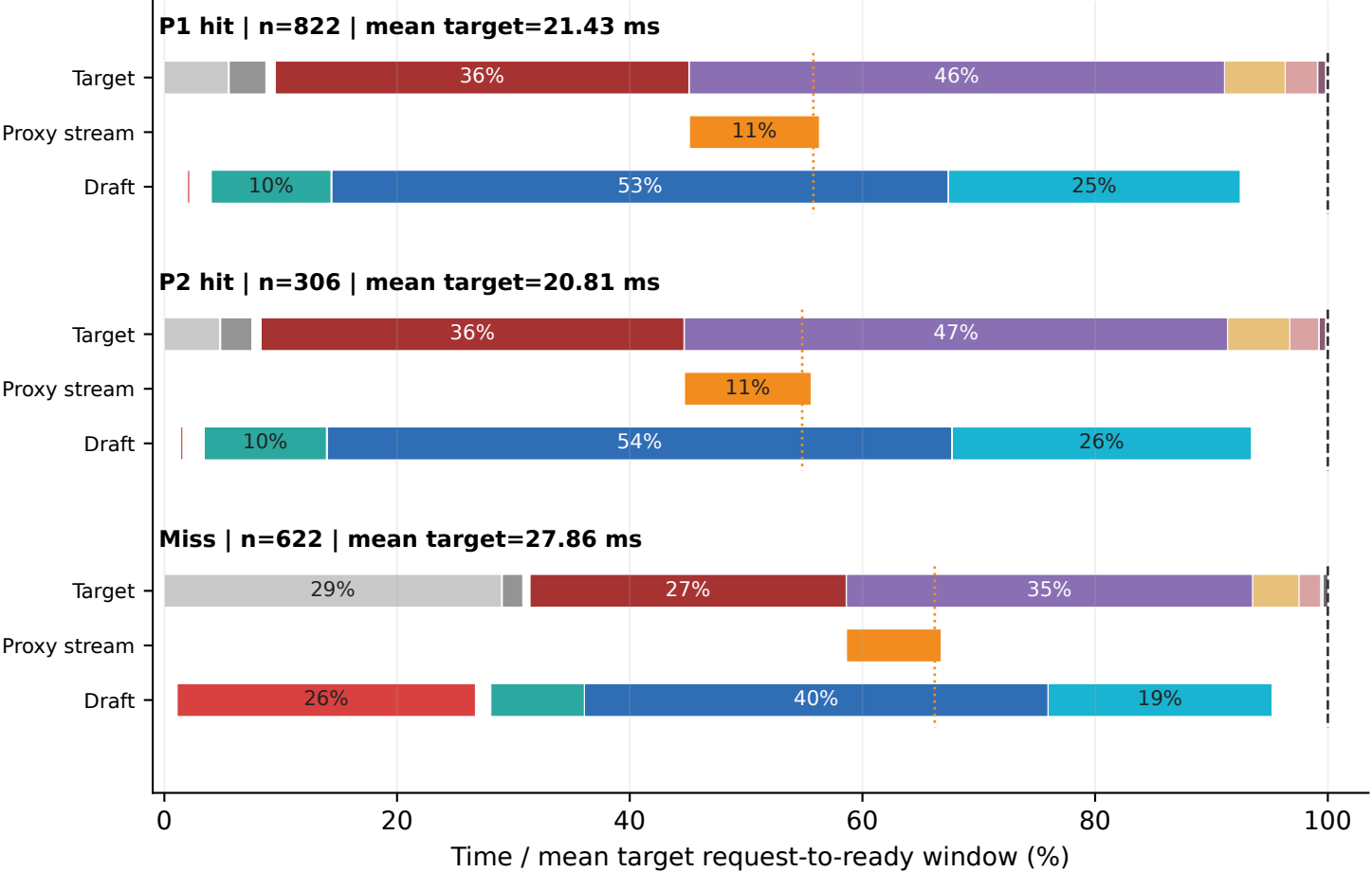
Target: status means on the same eligible steps



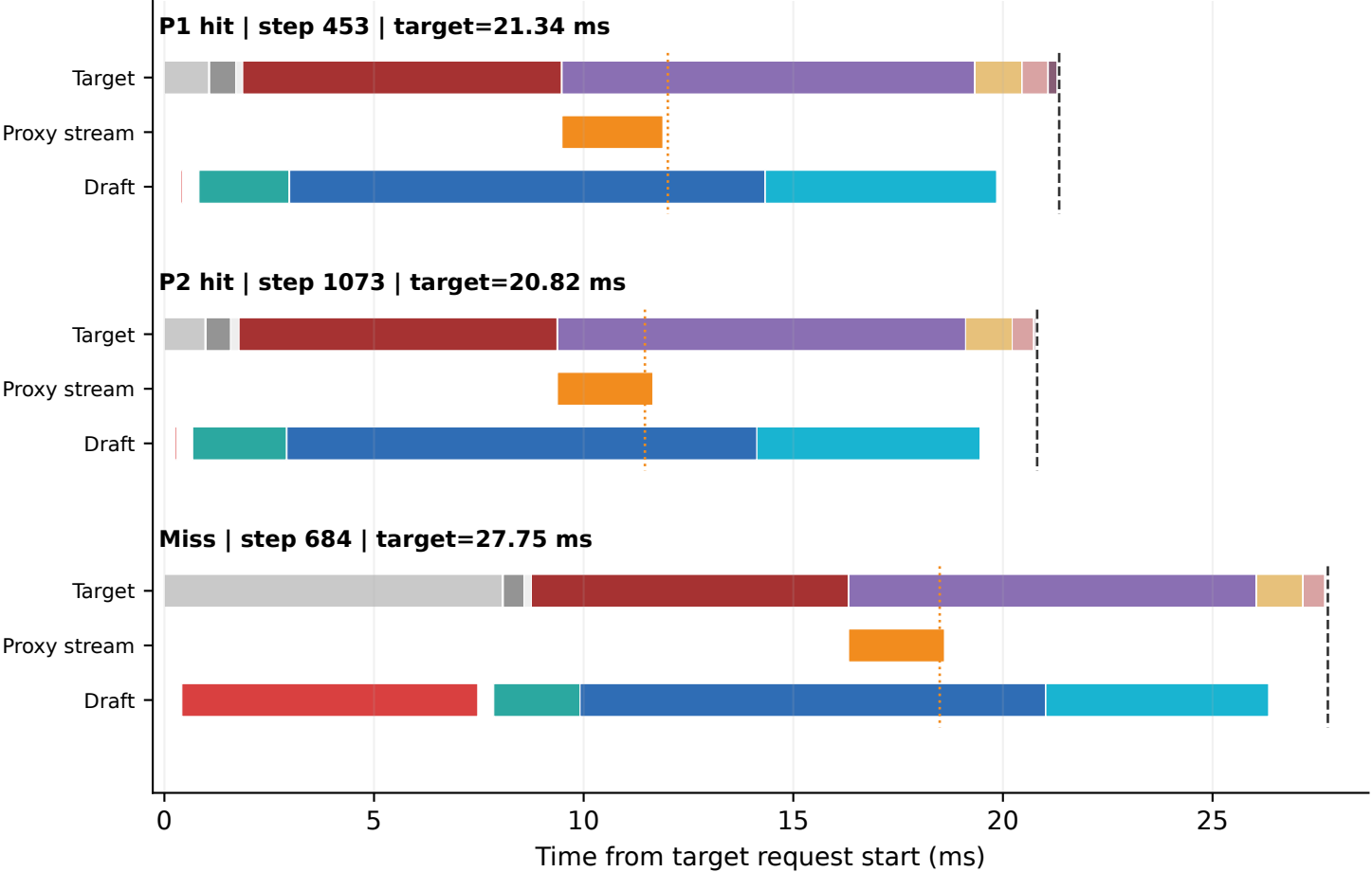
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

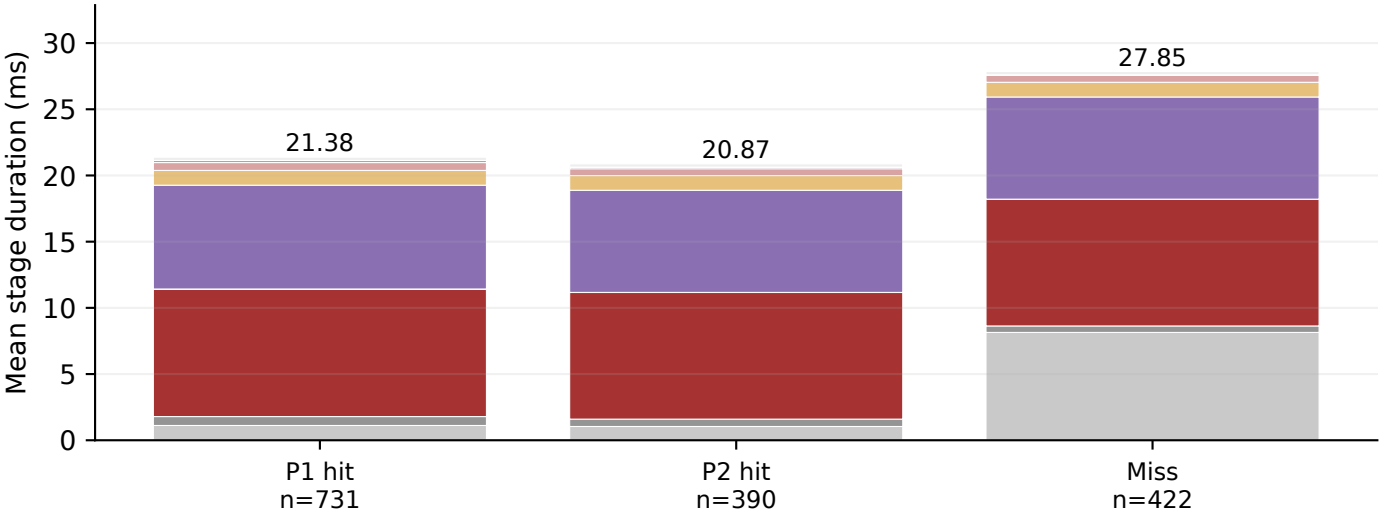


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

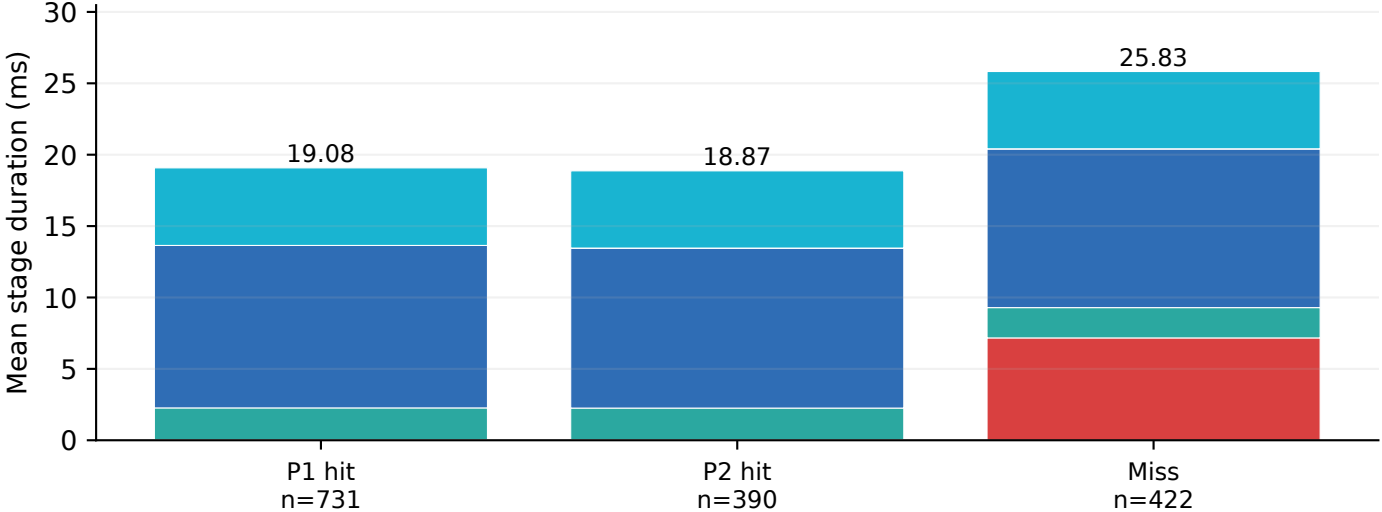
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_postopt_profile_exit_later
DUET: K1/K2=4/2, exit=18 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1543 eligible steps

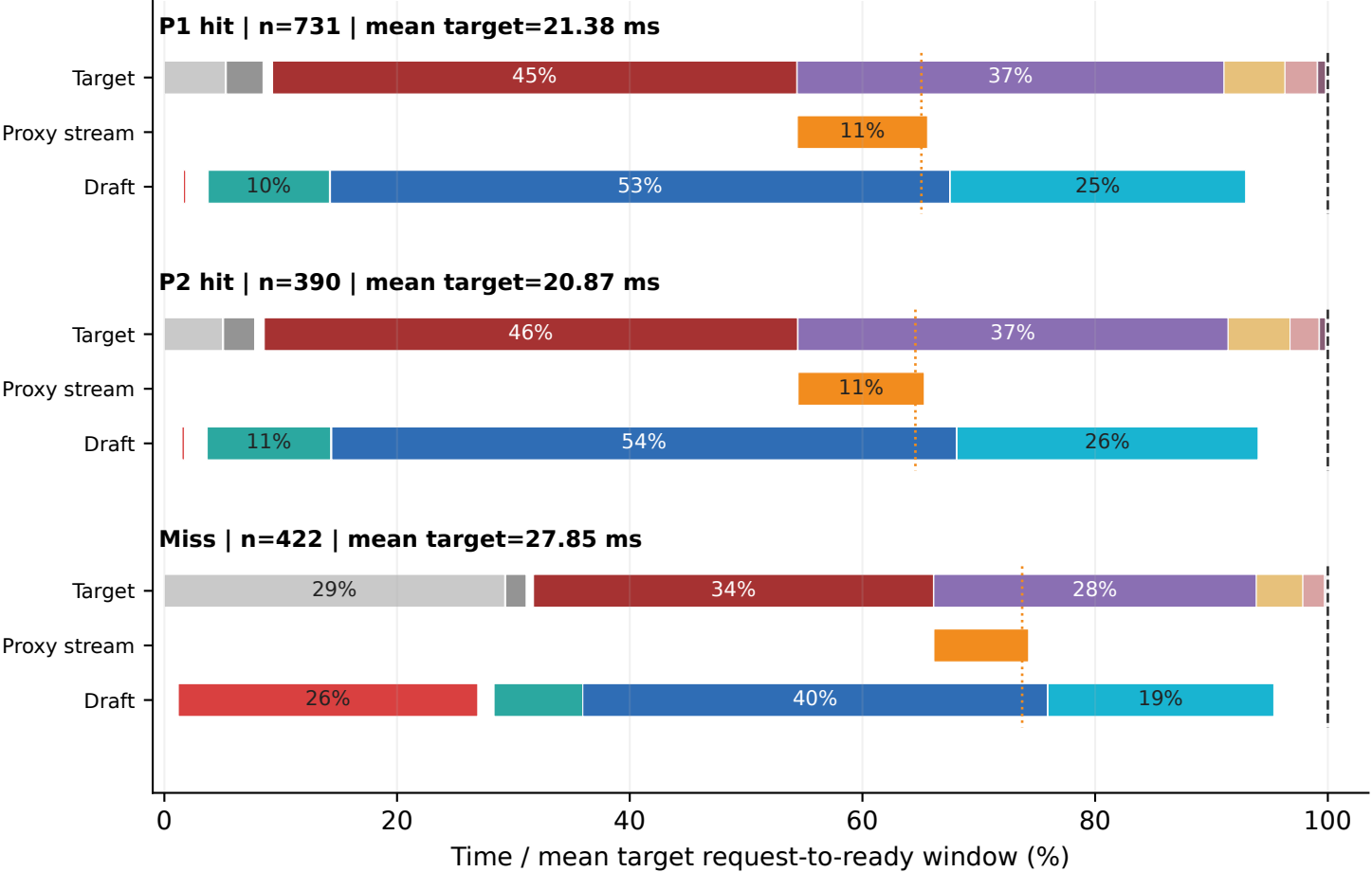
Target: status means on the same eligible steps



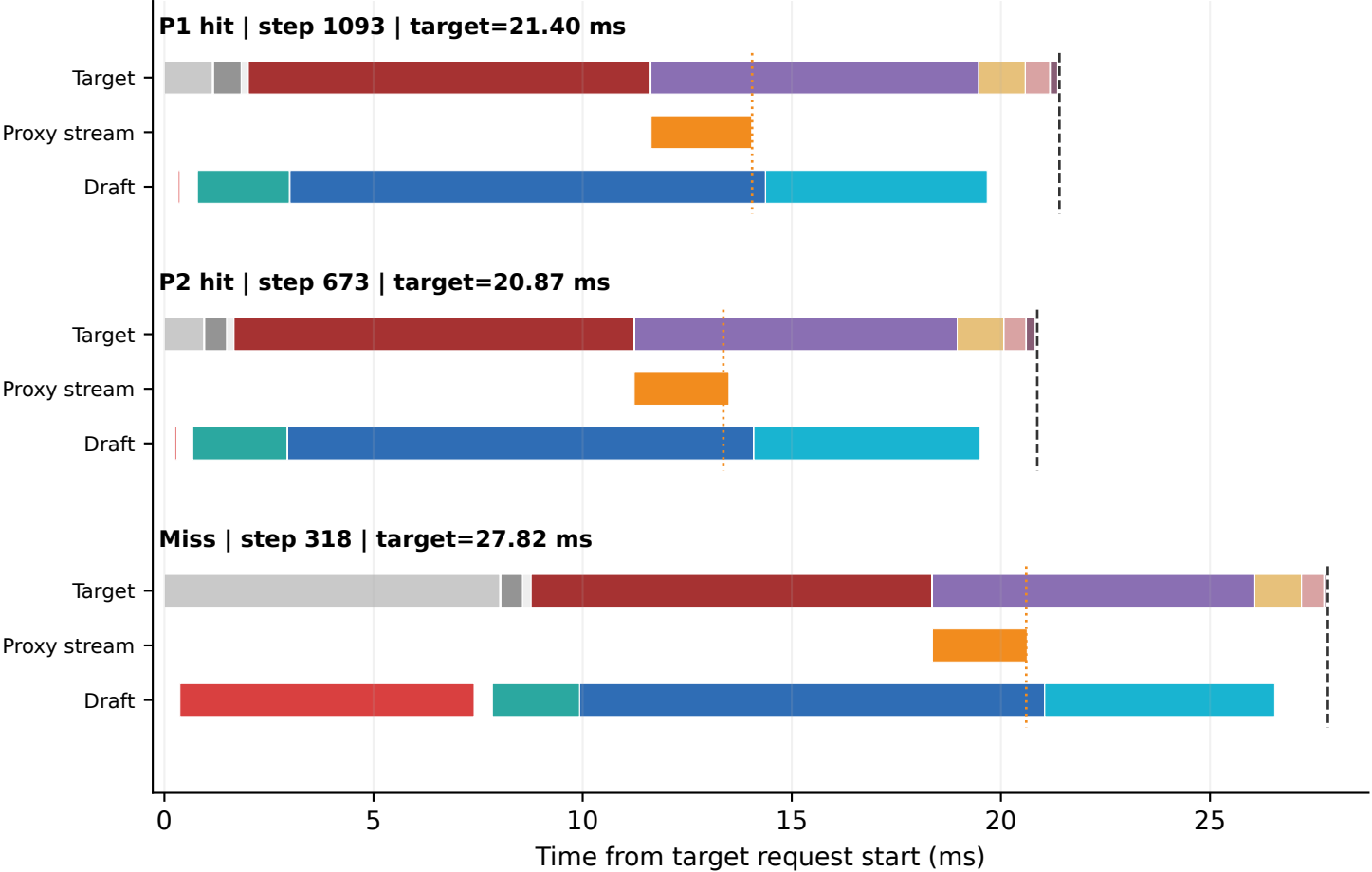
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

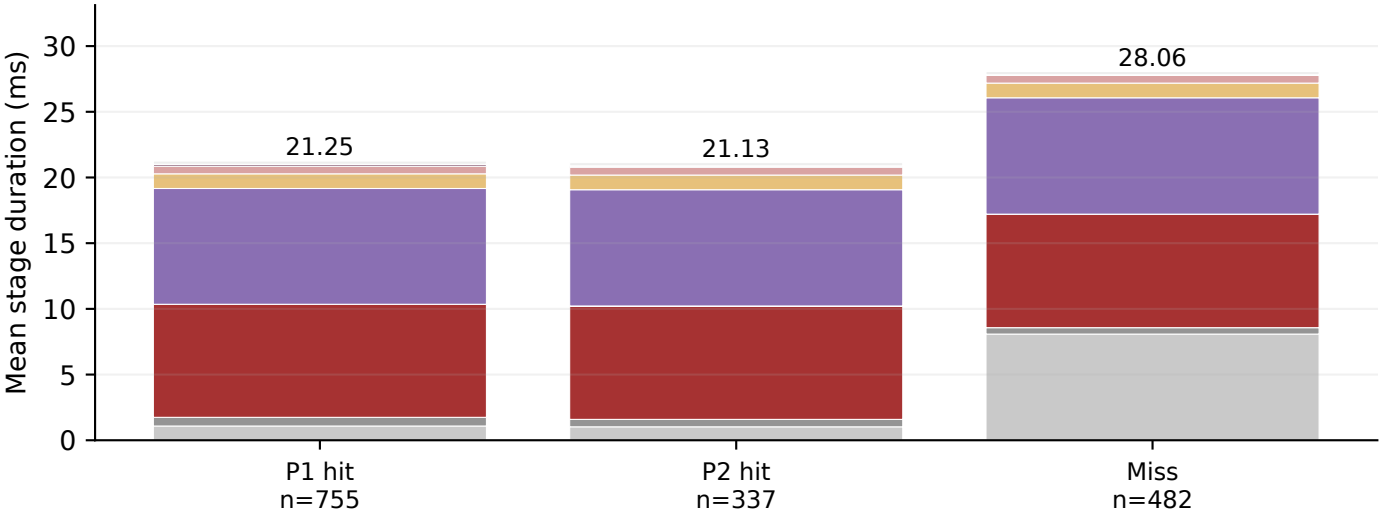


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

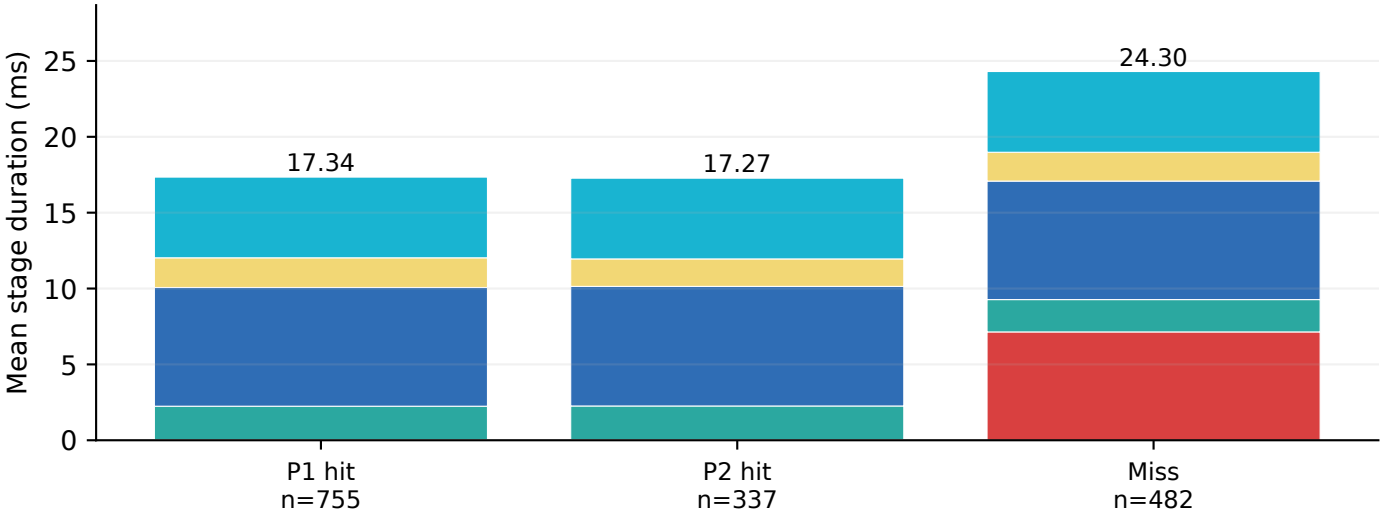
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_postopt_profile_k1_down
DUET: K1/K2=3/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=4, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1574 eligible steps

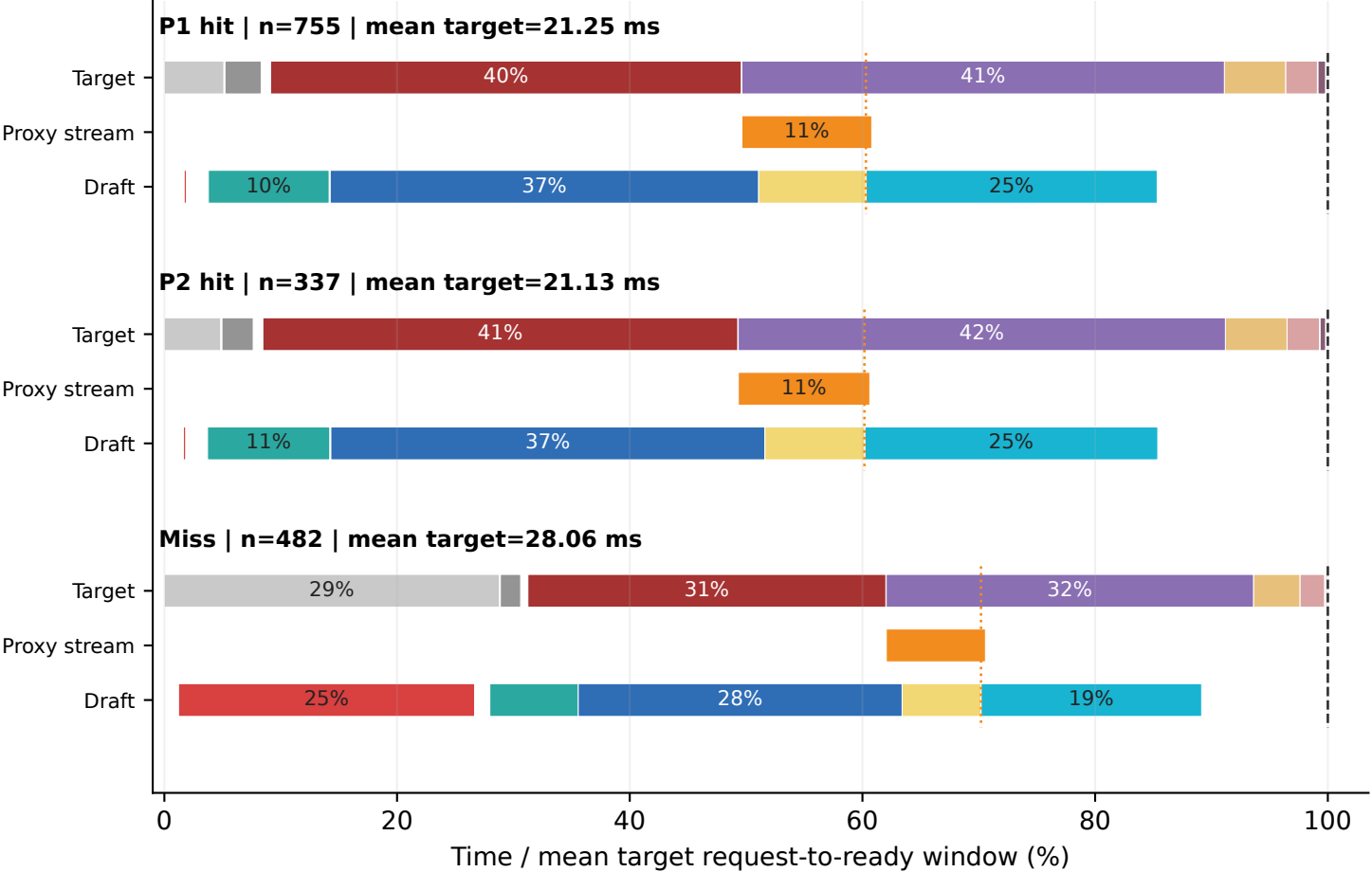
Target: status means on the same eligible steps



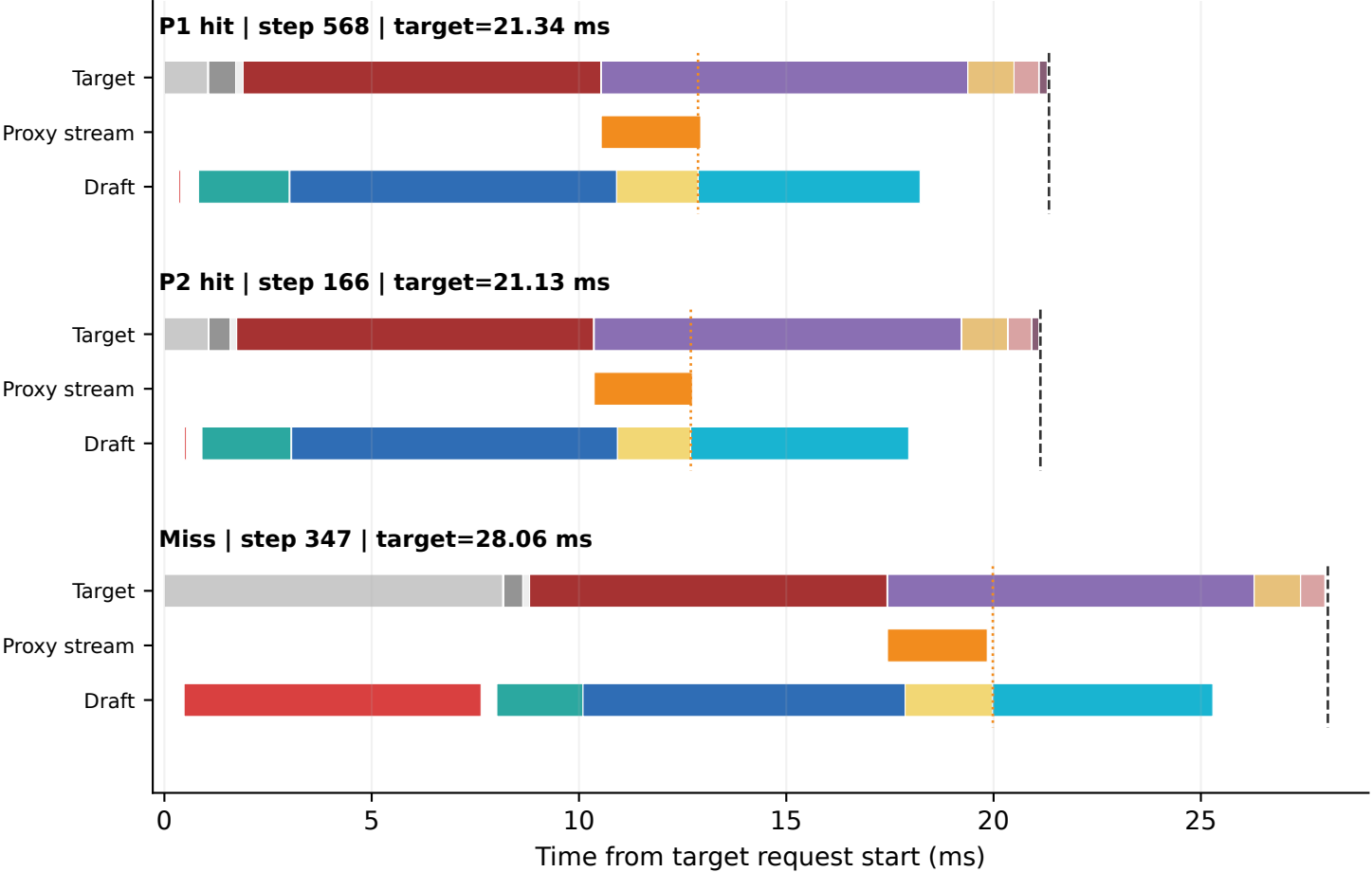
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



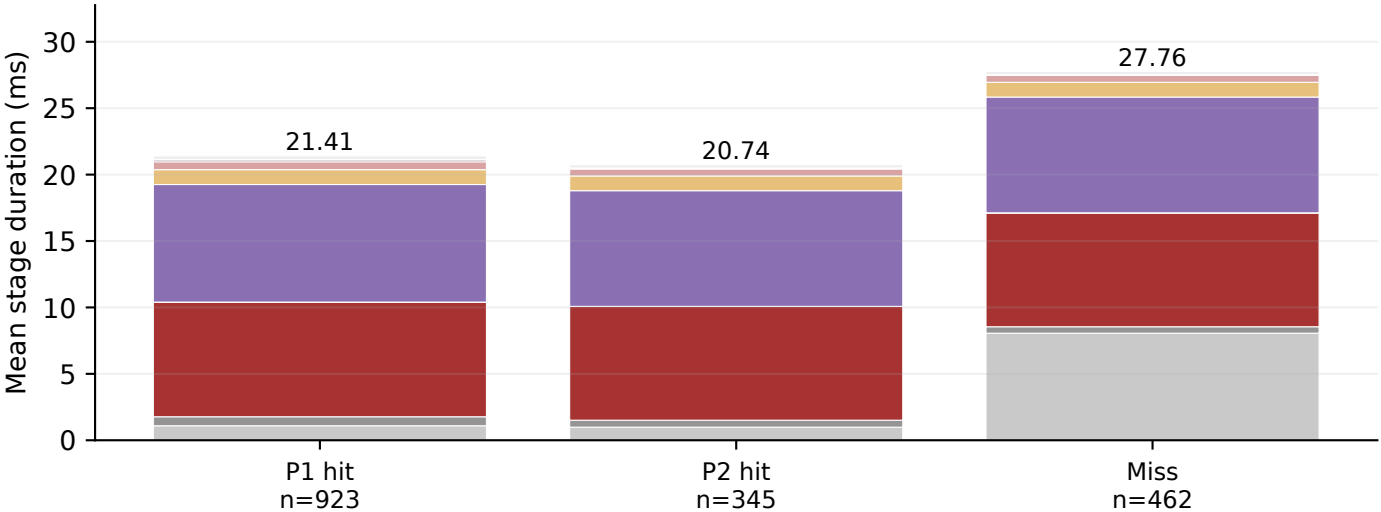
Observed representative steps (near median target window)



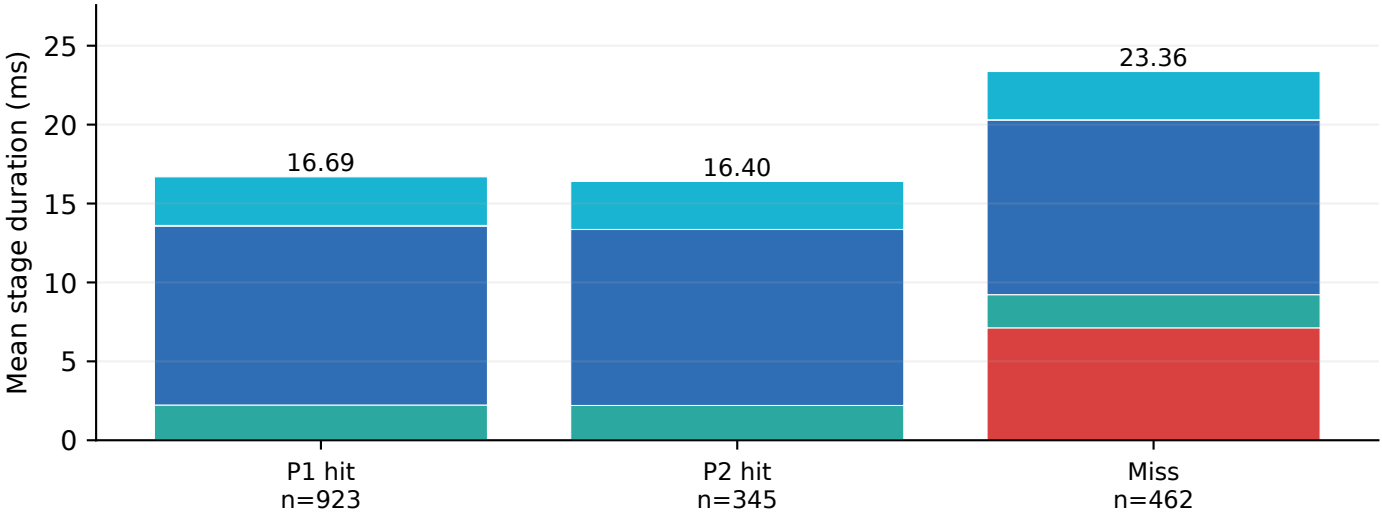
Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

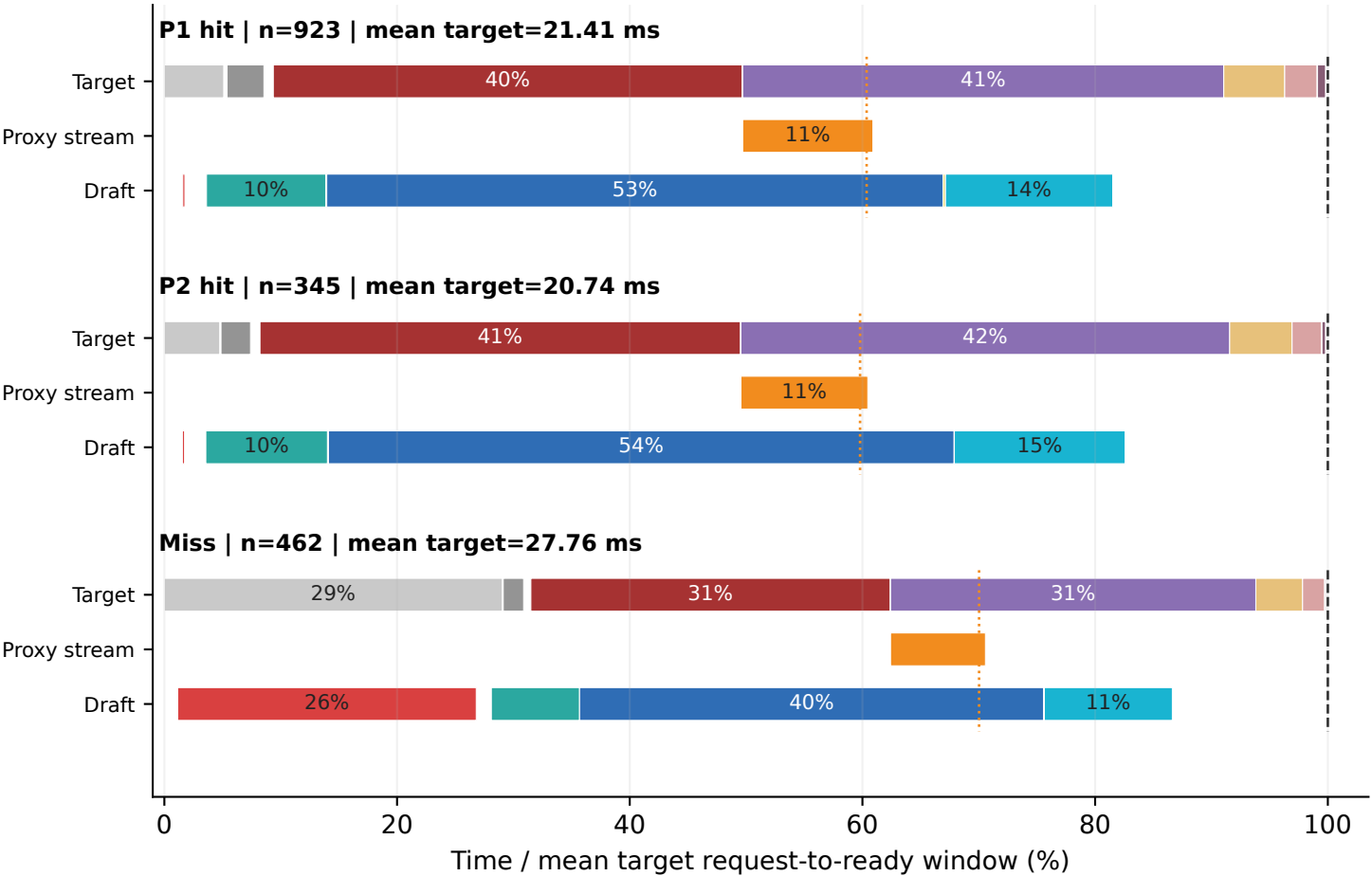
Target: status means on the same eligible steps



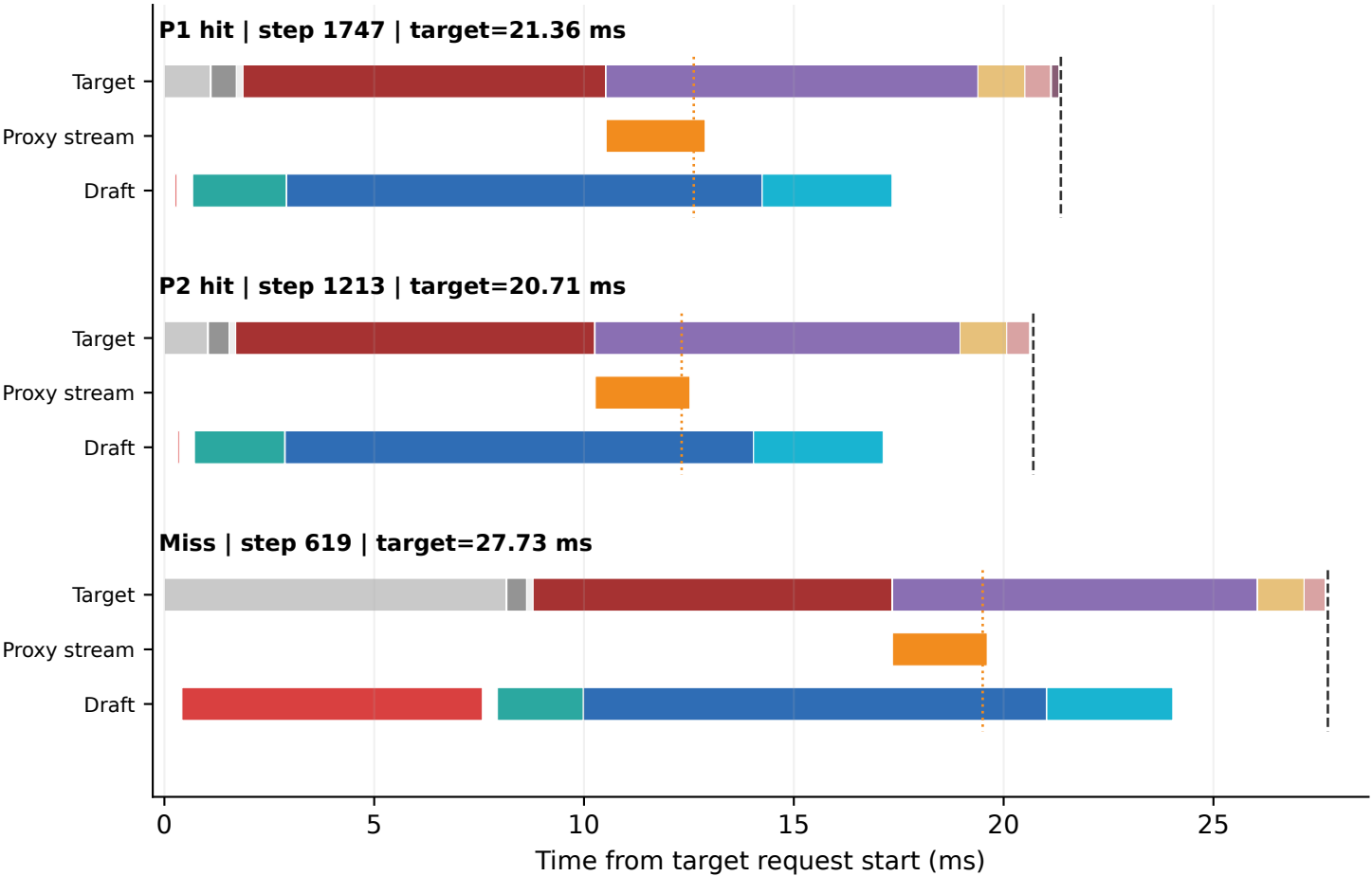
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

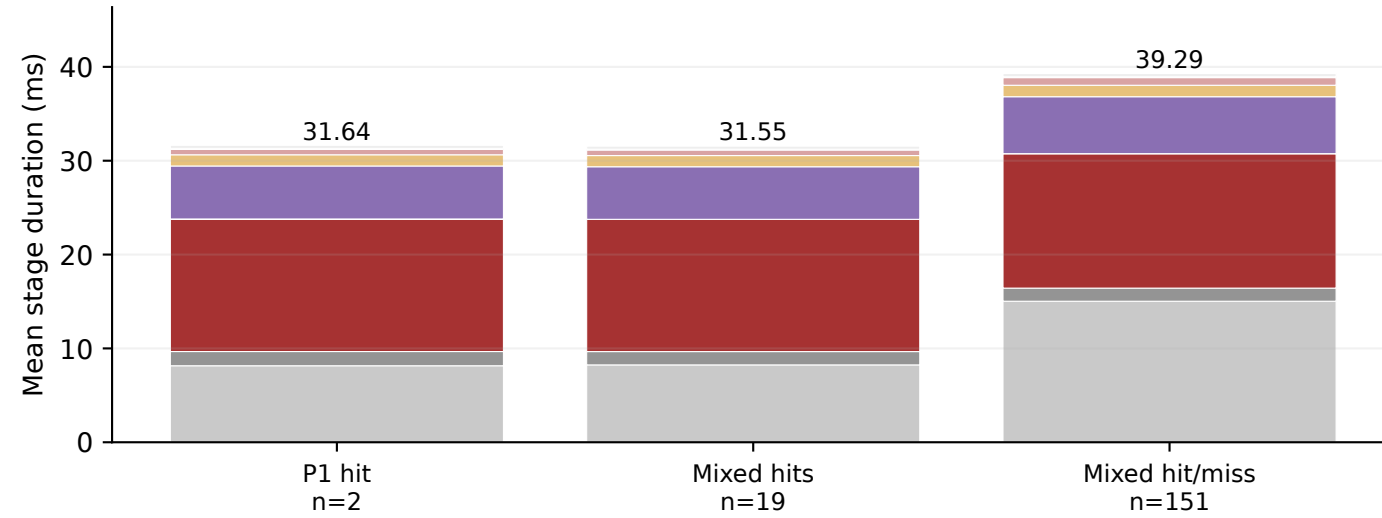


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received

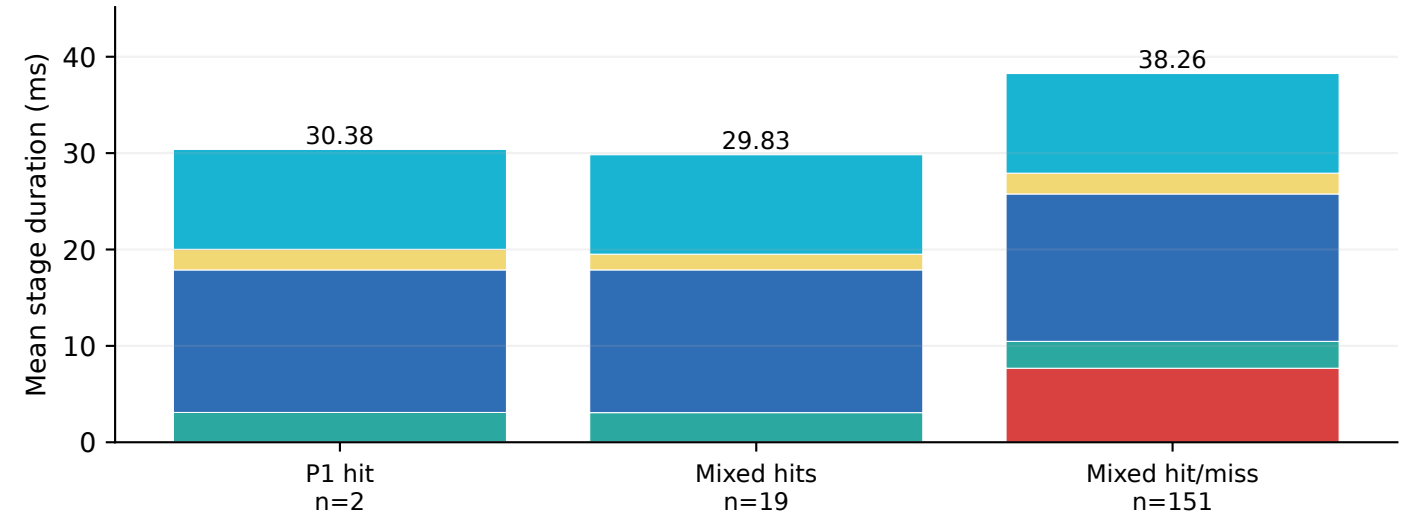
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_postopt_profile_exit_earlier
DUET: K1/K2=3/2, exit=24 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 172 eligible steps

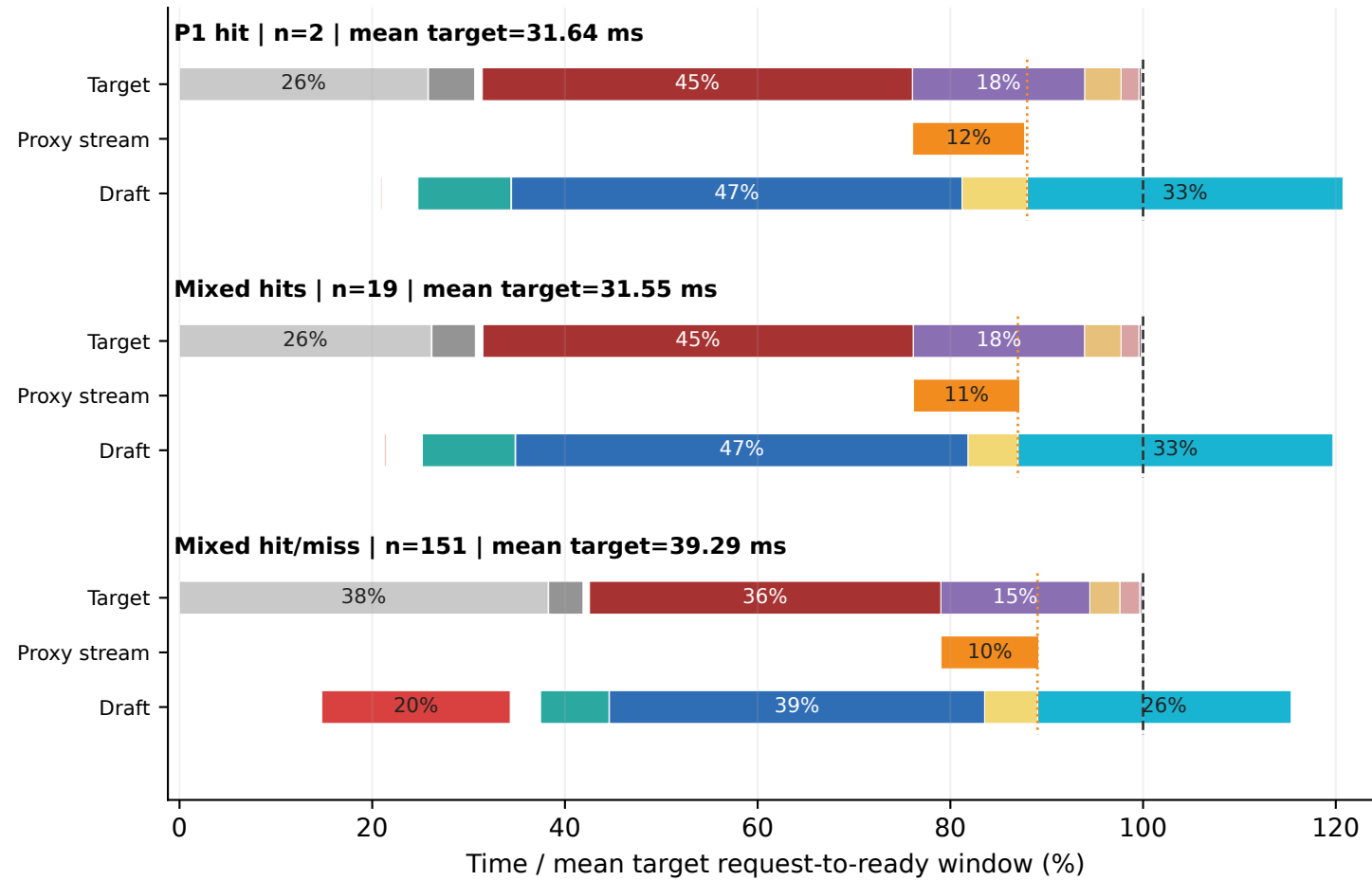
Target: status means on the same eligible steps



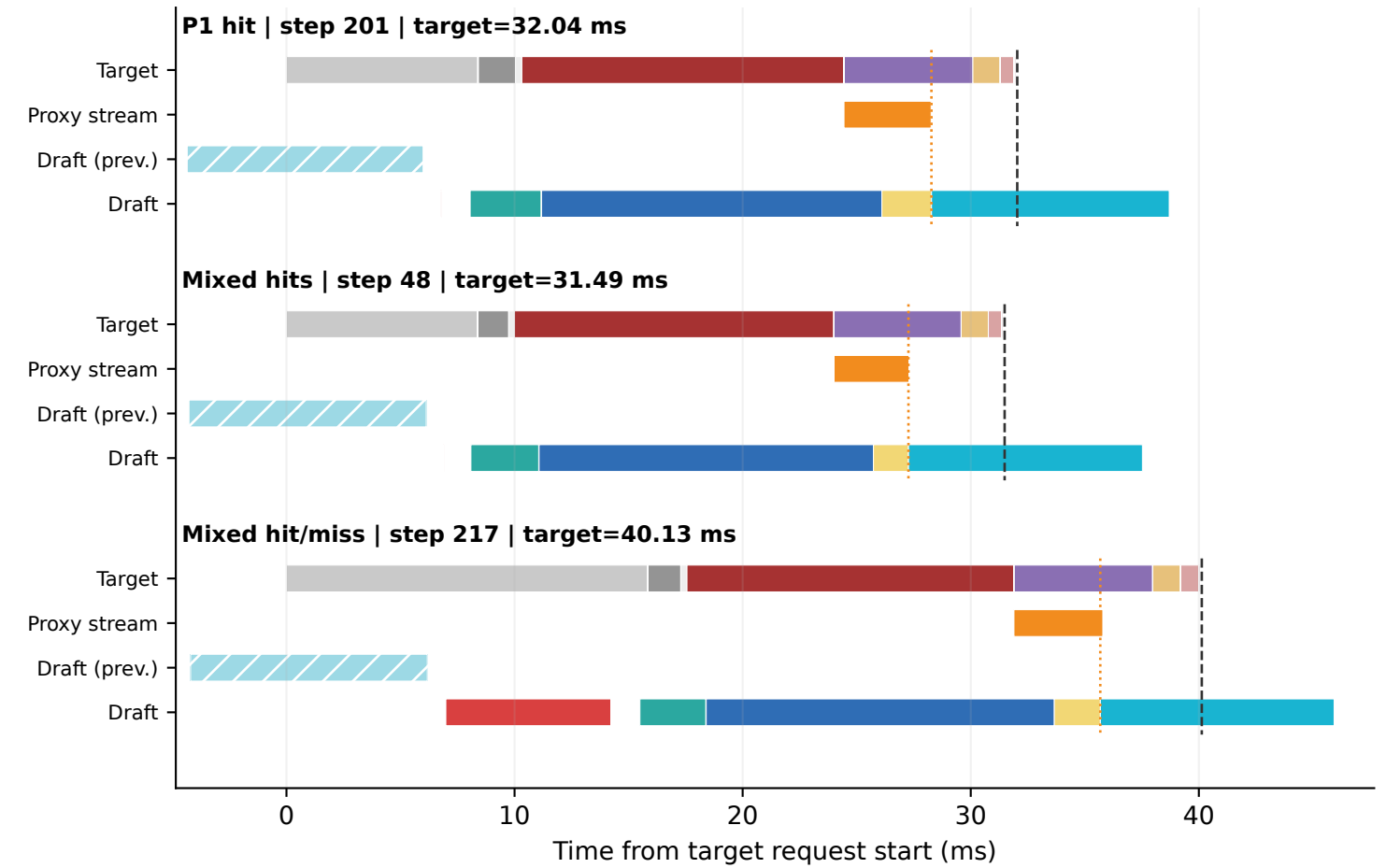
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

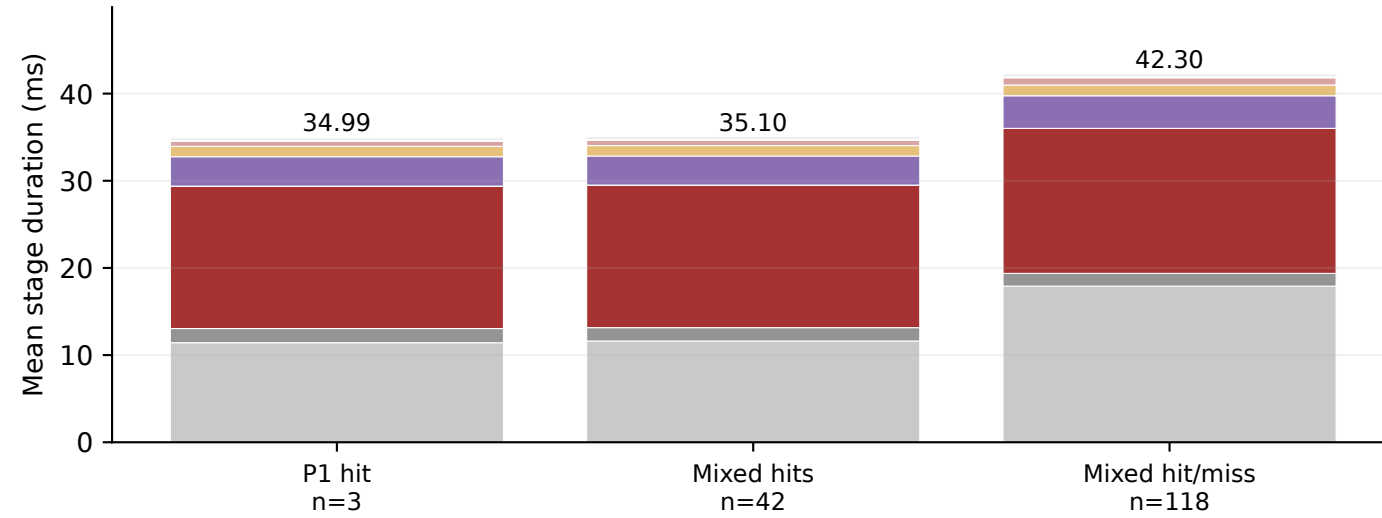


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

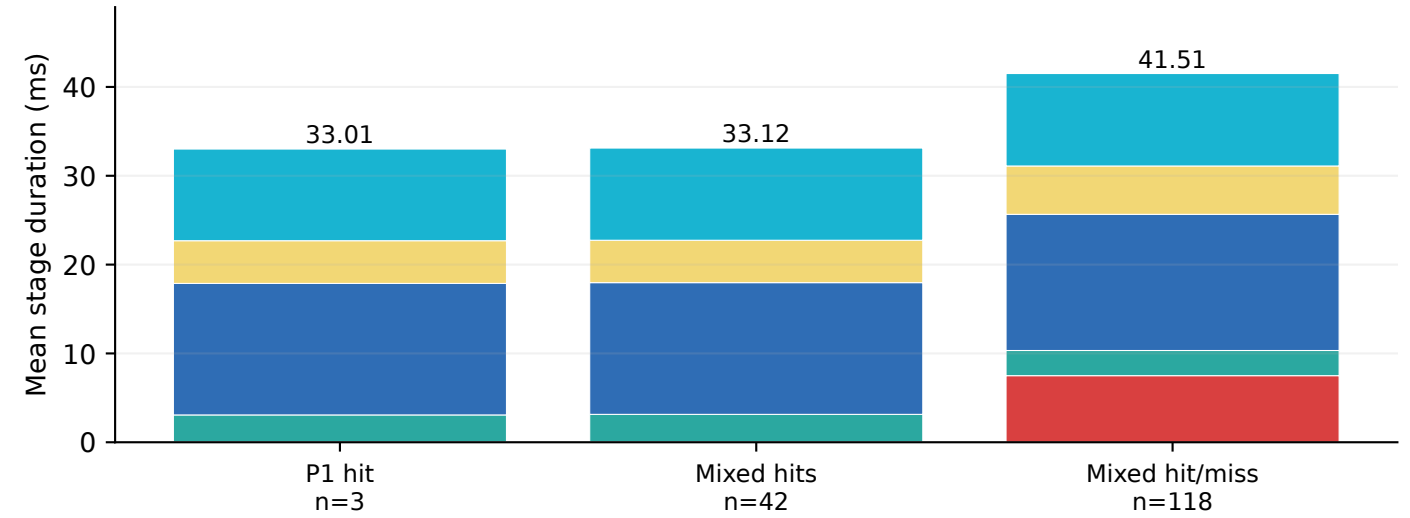
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_postopt_profile_exit_later
 DUET: K1/K2=3/2, exit=28 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 163 eligible steps

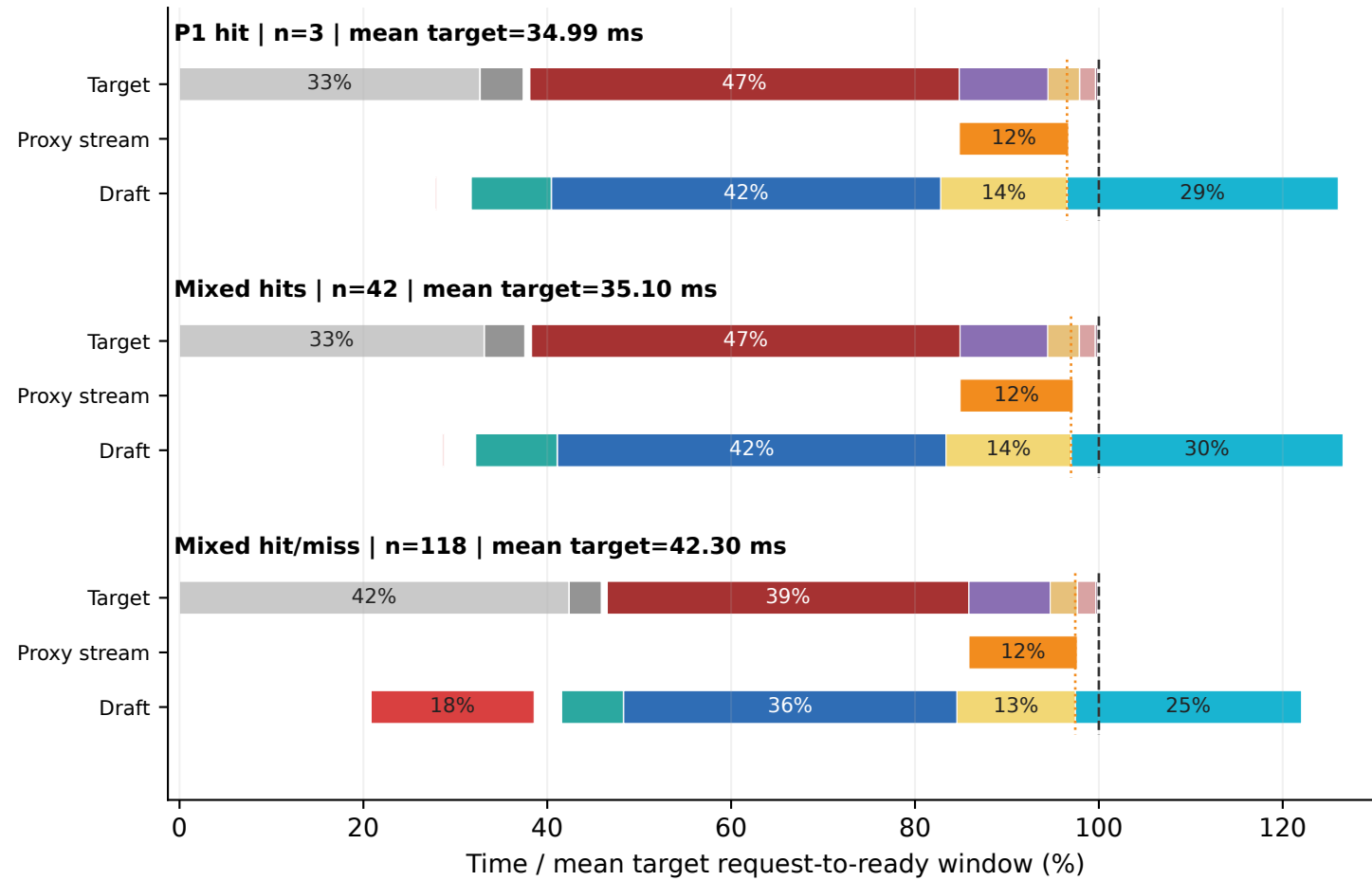
Target: status means on the same eligible steps



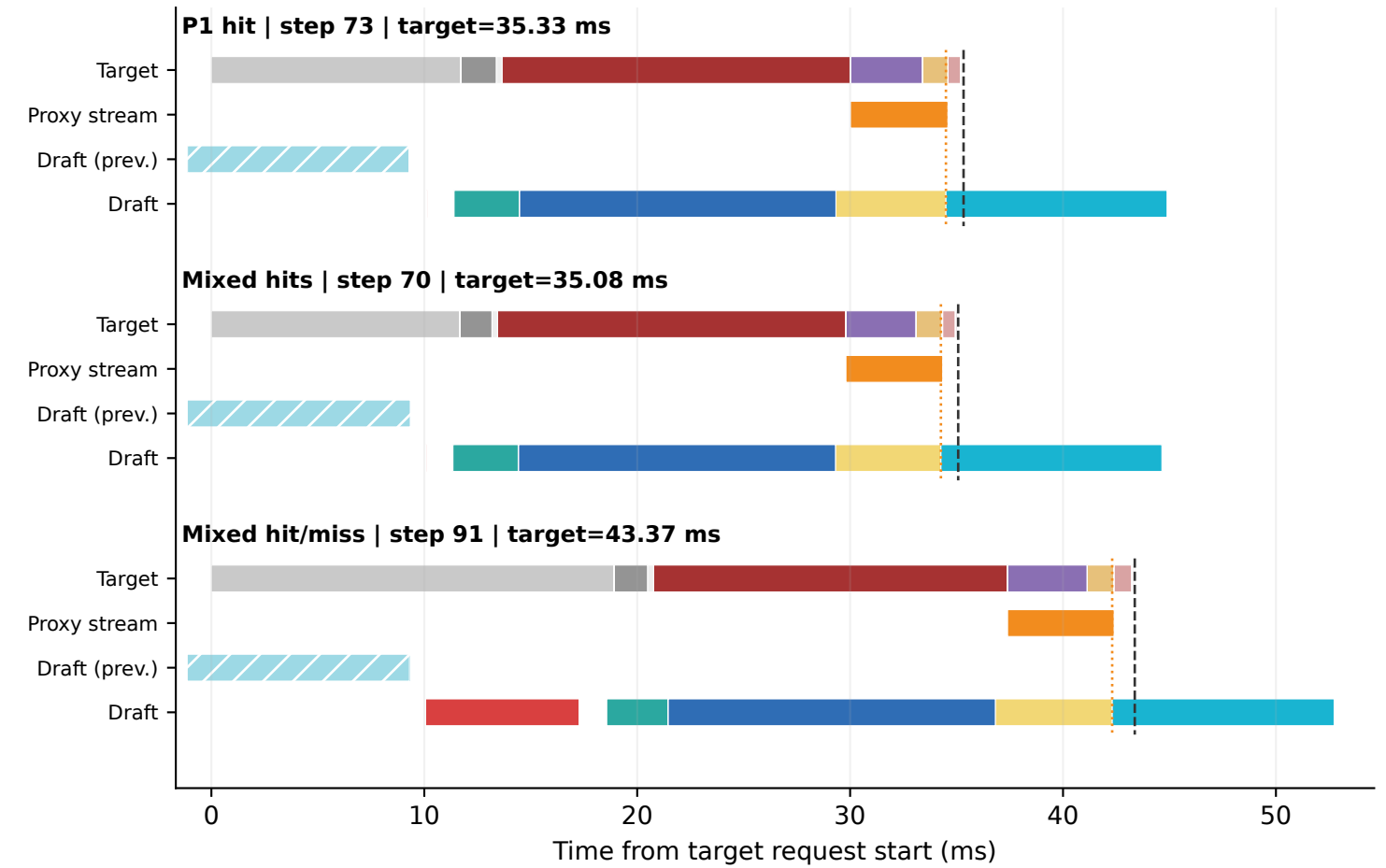
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



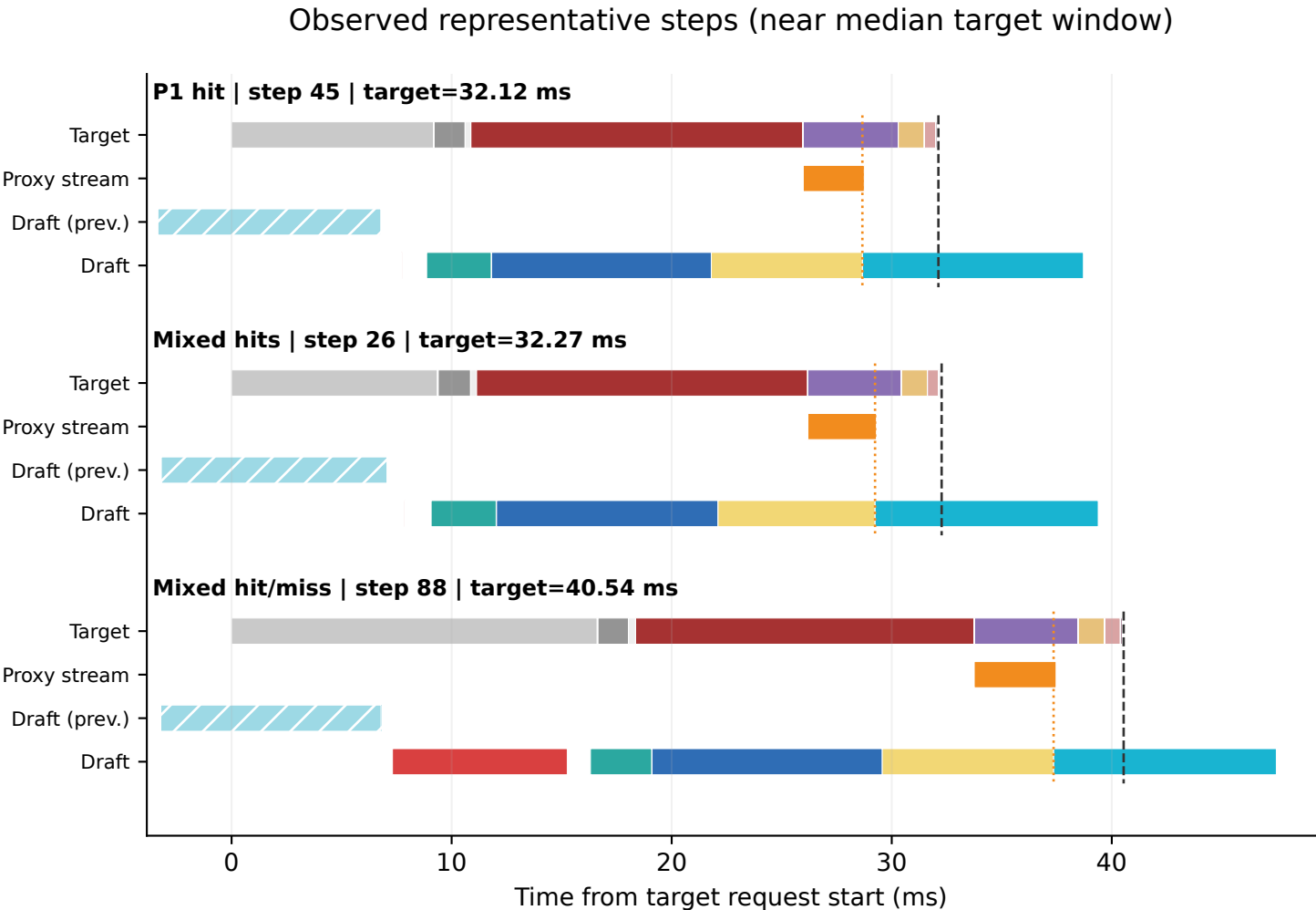
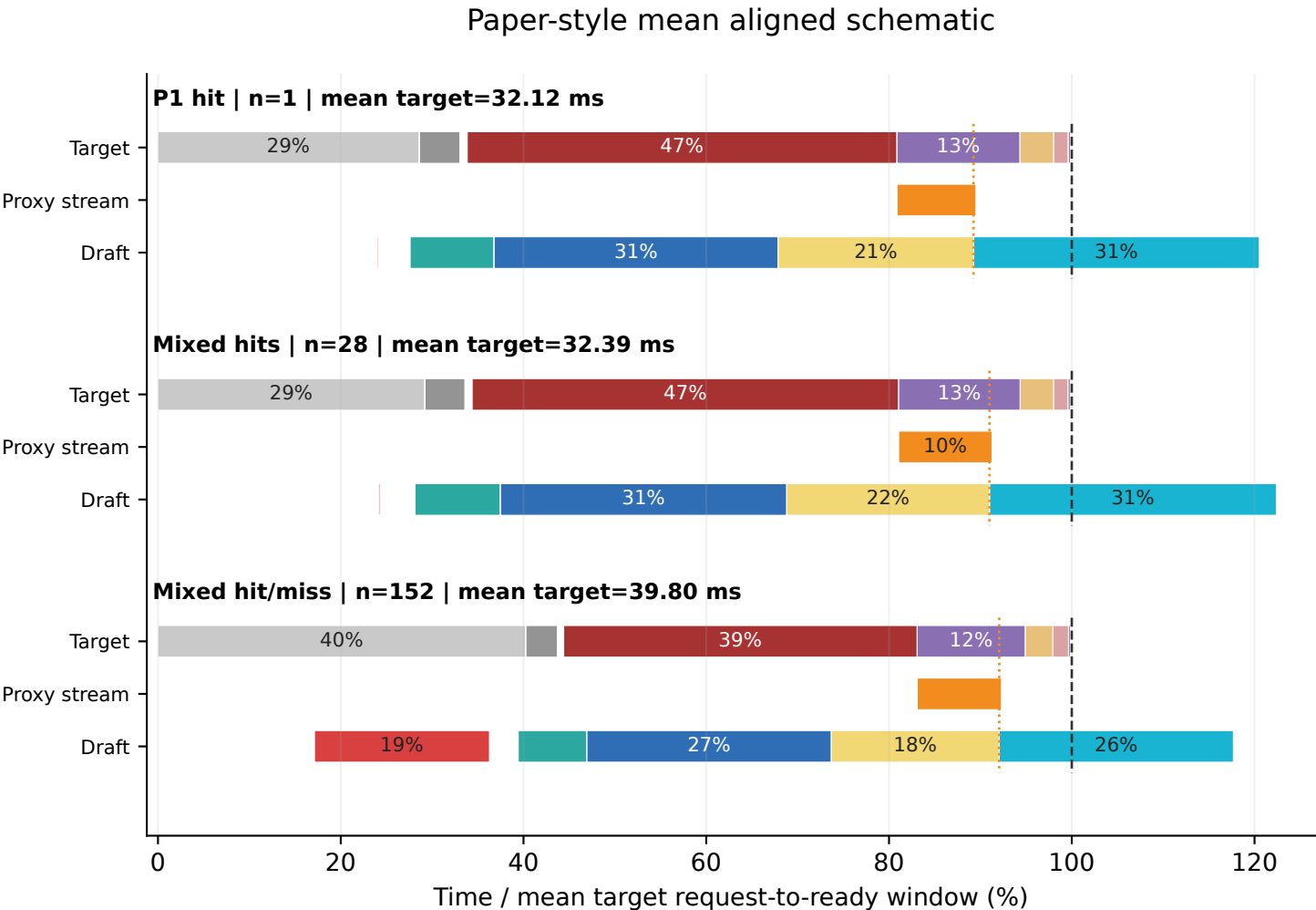
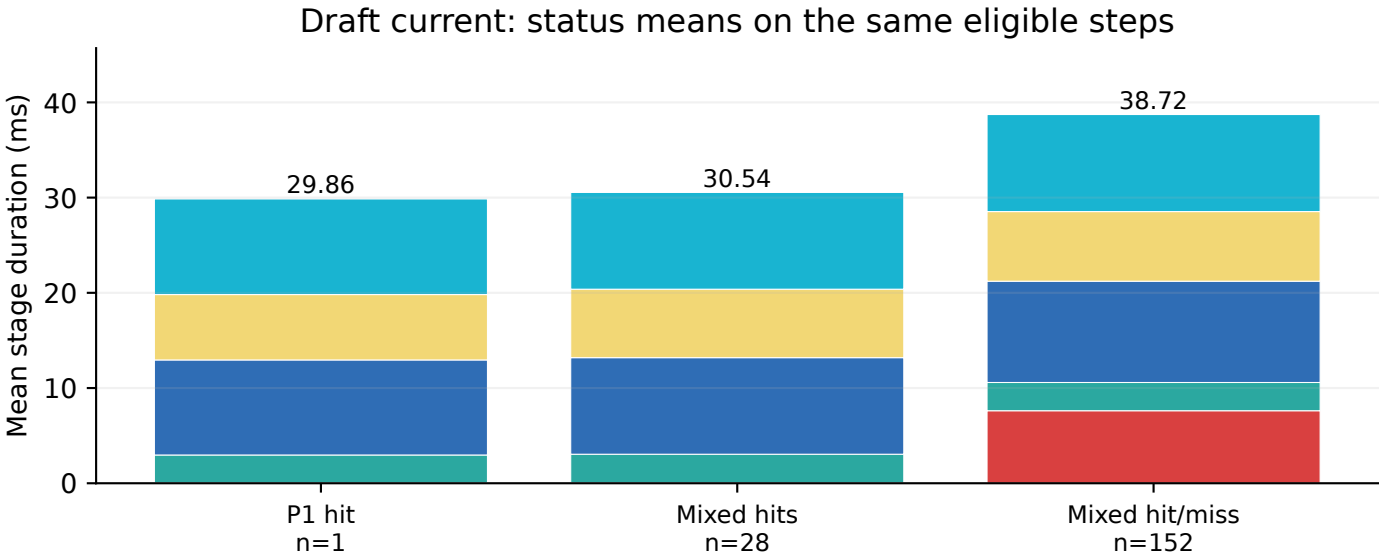
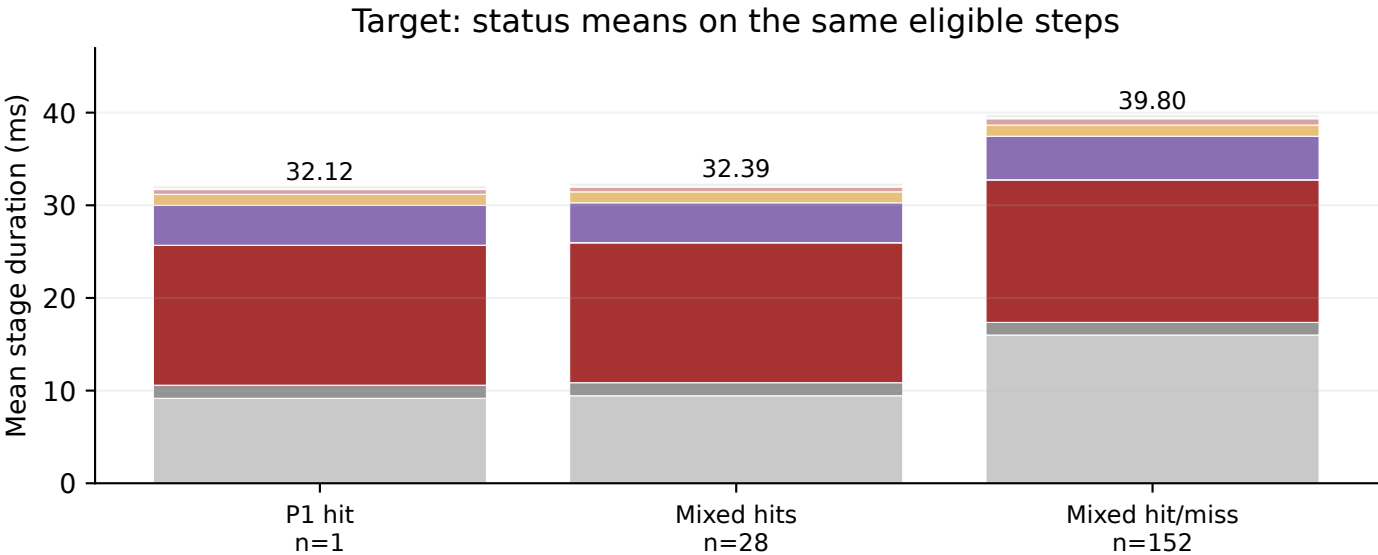
Observed representative steps (near median target window)



Wait / sync (grey), Prepare (dark grey), Target pre (red), Proxy / launch (orange), Target post (purple), Proxy compute (dark orange), Final logits (yellow), Commit (dark purple), Postprocess (dark grey), Unlabelled gap (light grey), Context (teal), P1 (blue), Previous P2 (light blue), Proxy wait (yellow), P2 (cyan), Target ready (dashed line), Proxy received (dotted line)

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
 Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

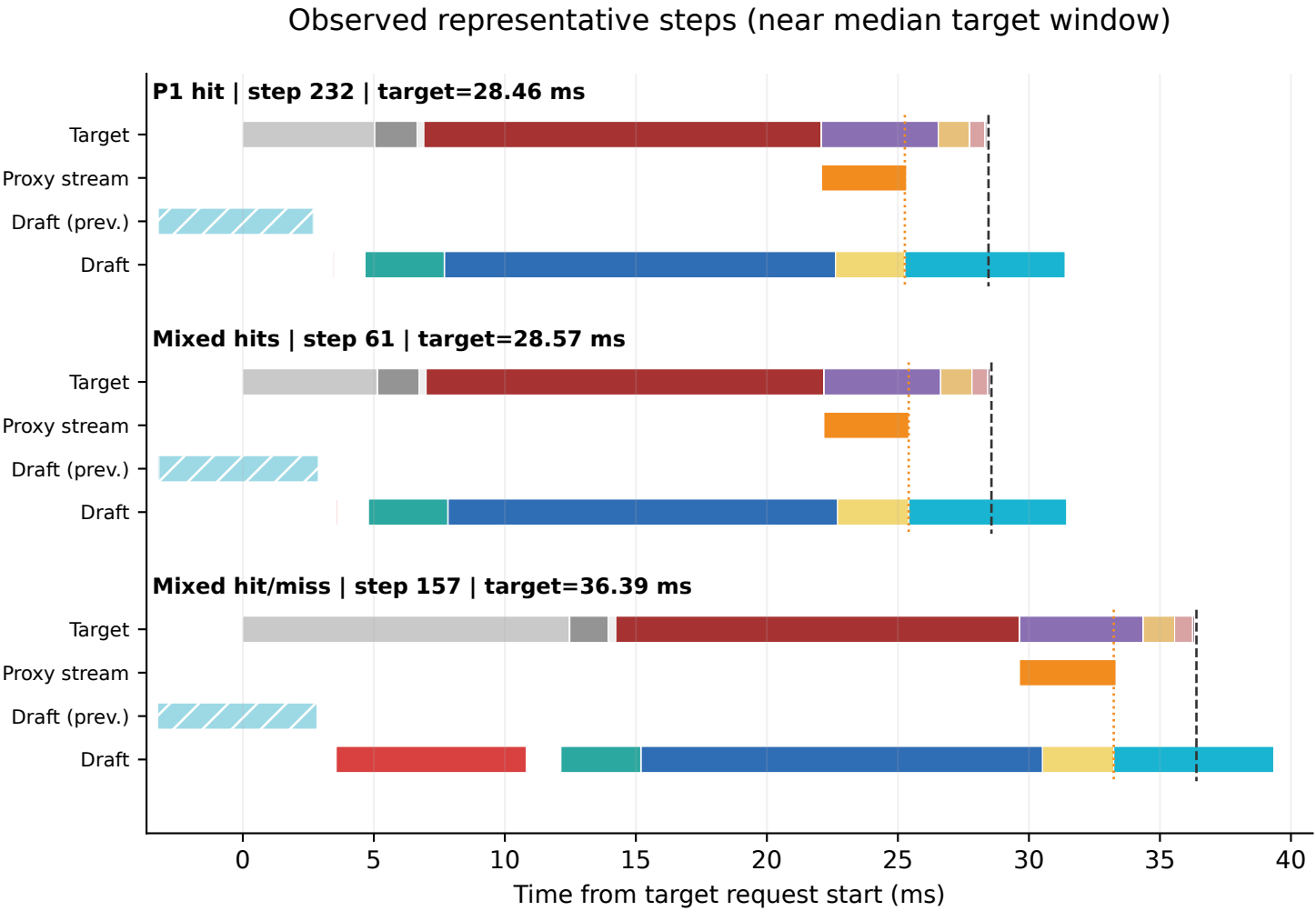
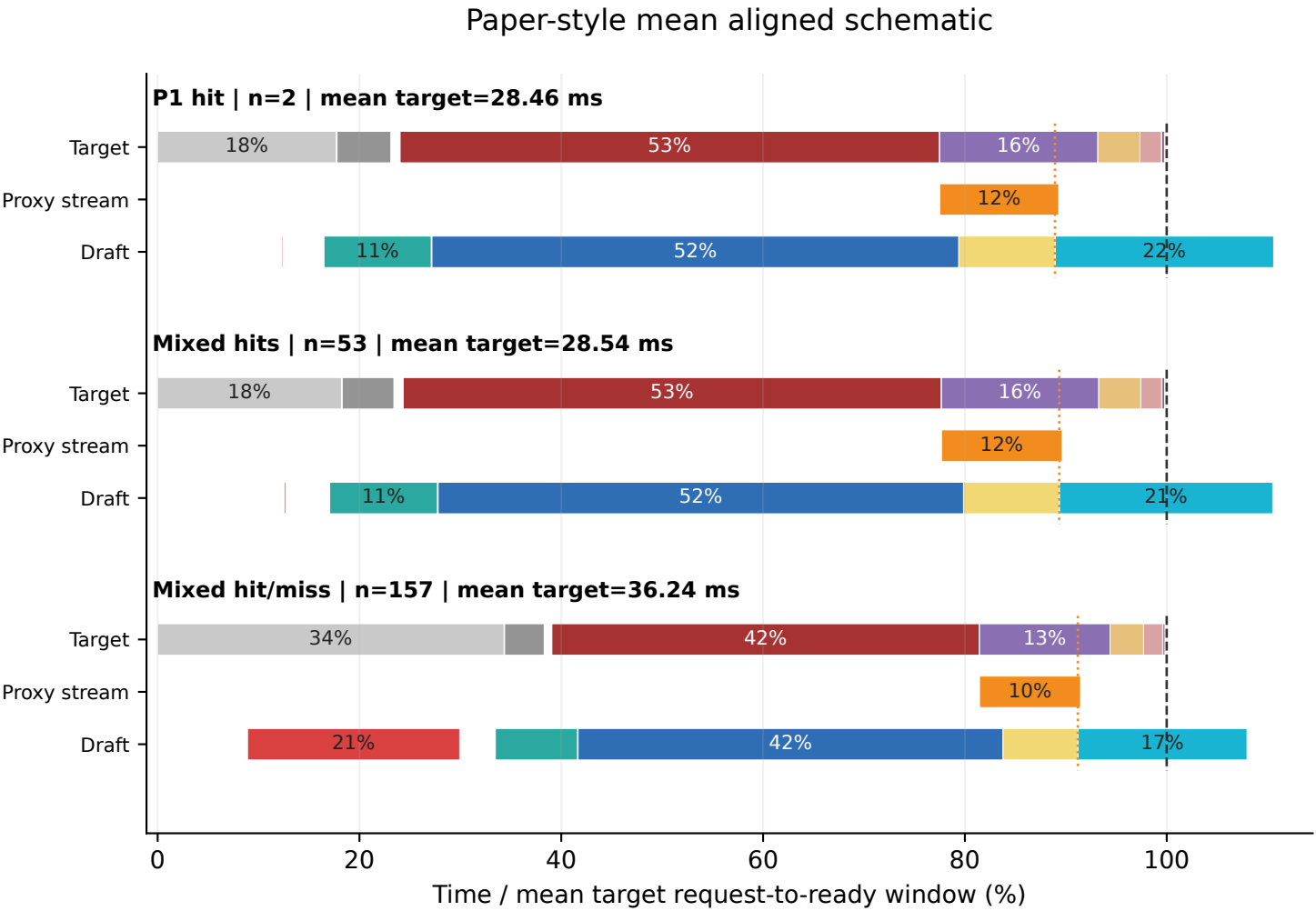
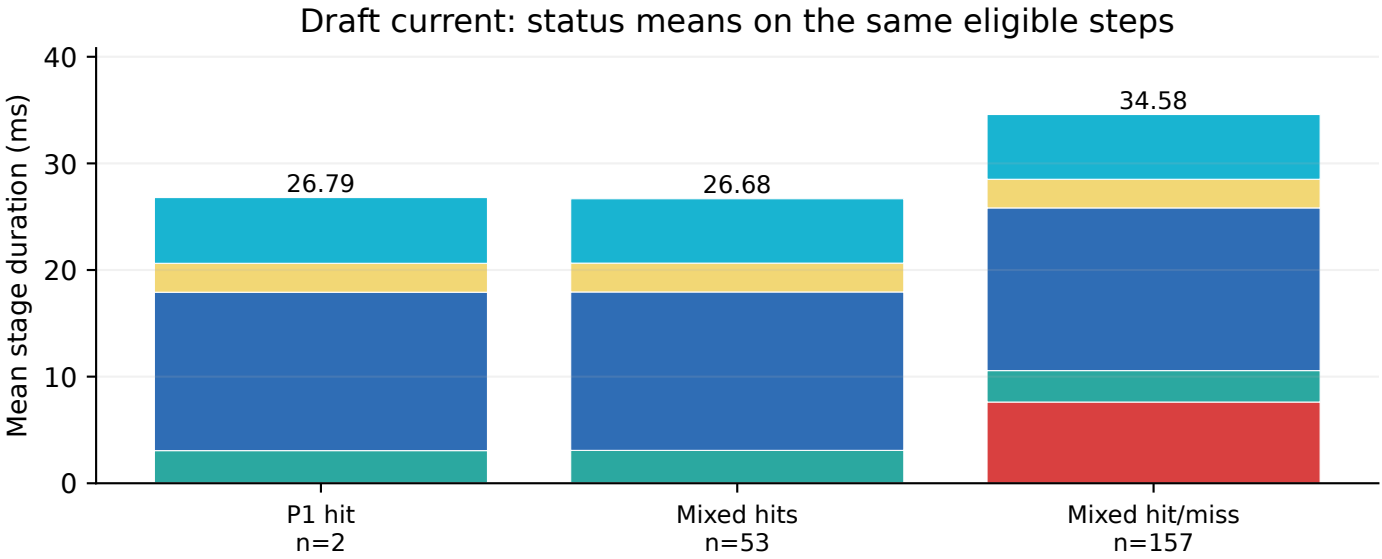
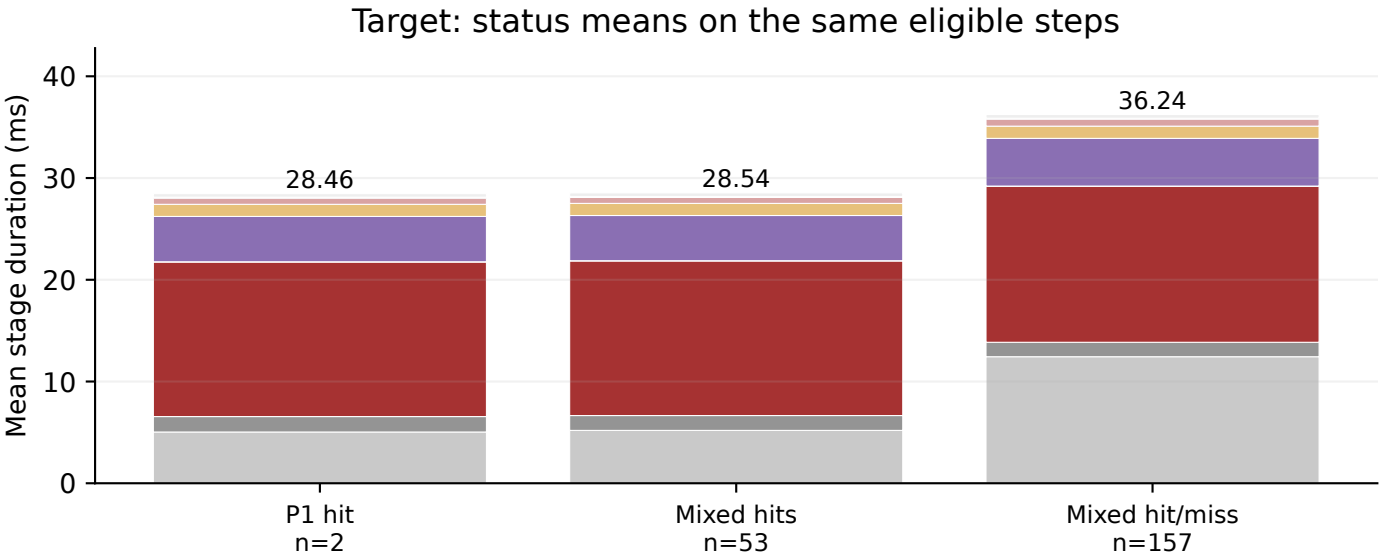
llama3_b8_postopt_profile_k1_down
DUET: K1/K2=2/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=3, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 181 eligible steps



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_postopt_profile_k2_down
DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 212 eligible steps

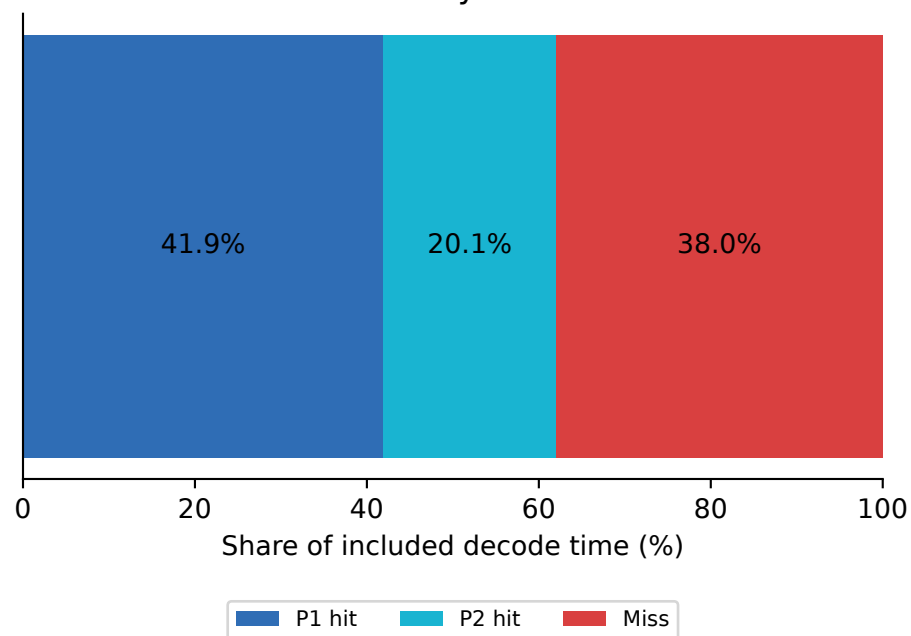


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

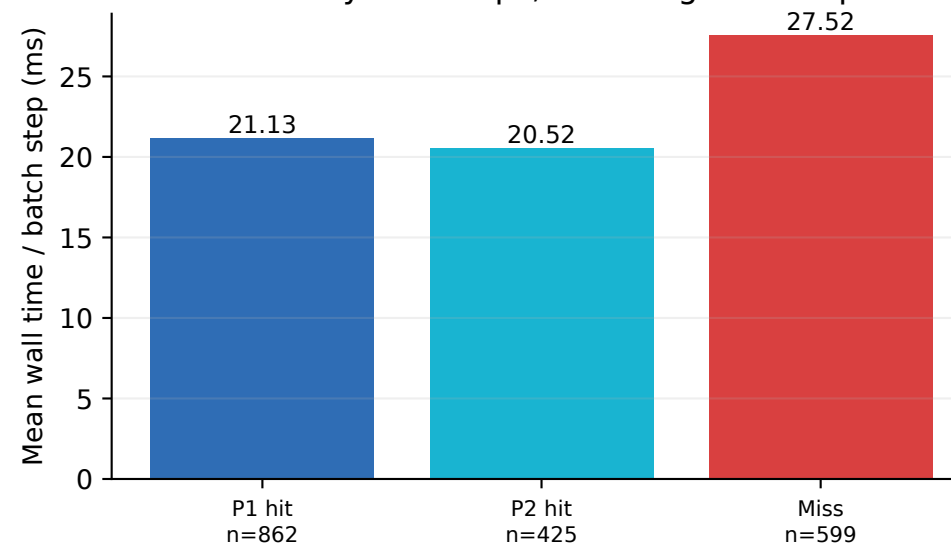
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_postopt_warm_0
DUET: K1/K2=4/1, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
Last pass: B=1, T=0.7, N=48 | AL*=2.601, hit=68.1%, TPS*=113.01

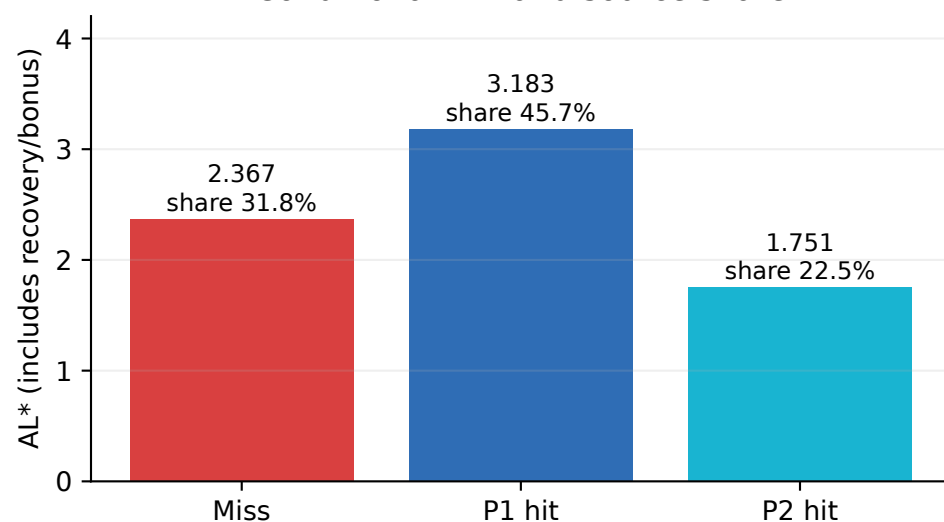
Measured wall time by batch cache status



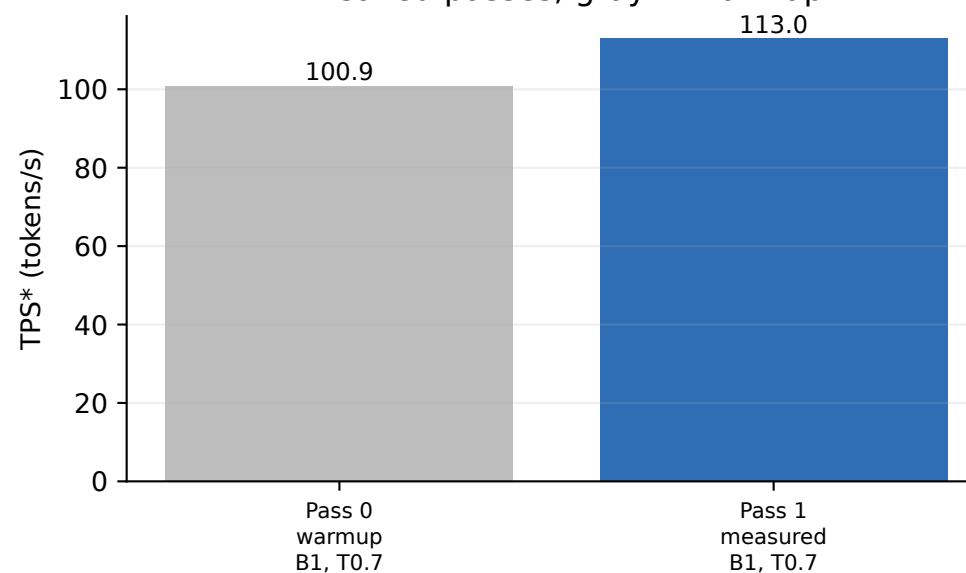
Boundary-free steps, including slow steps



Conditional AL* and source share



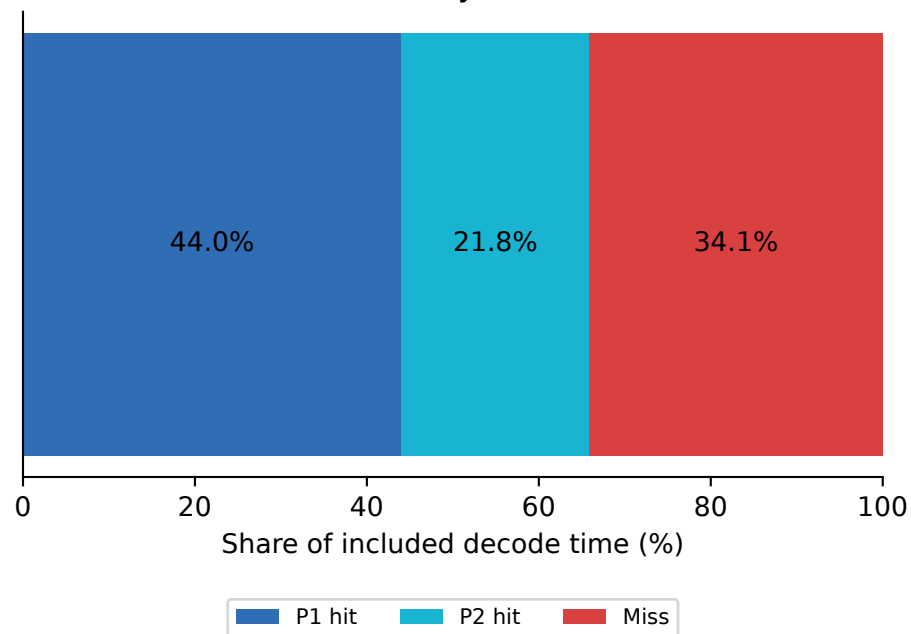
All saved passes; gray = warmup



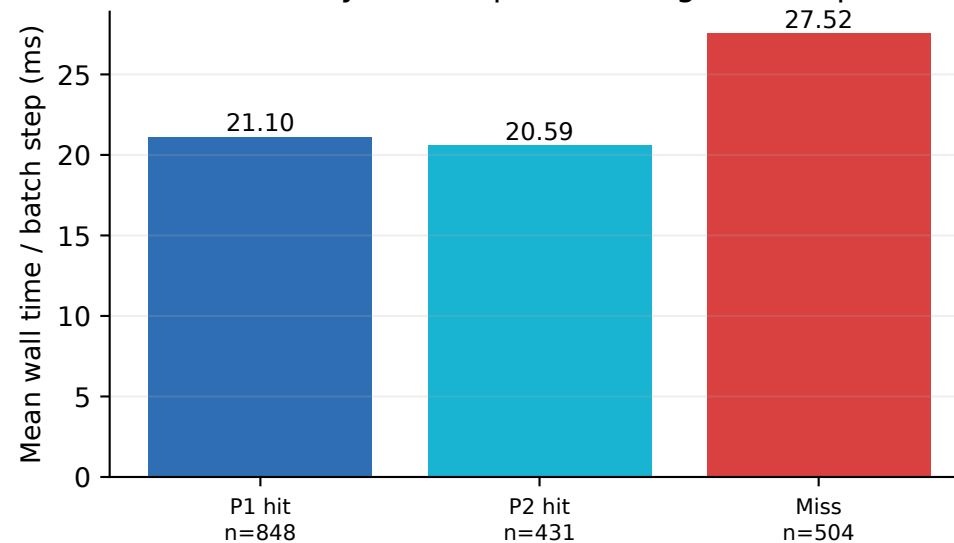
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_postopt_warm_1
DUET: K1/K2=4/2, exit=18 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
Last pass: B=1, T=0.7, N=48 | AL*=2.716, hit=71.9%, TPS*=119.15

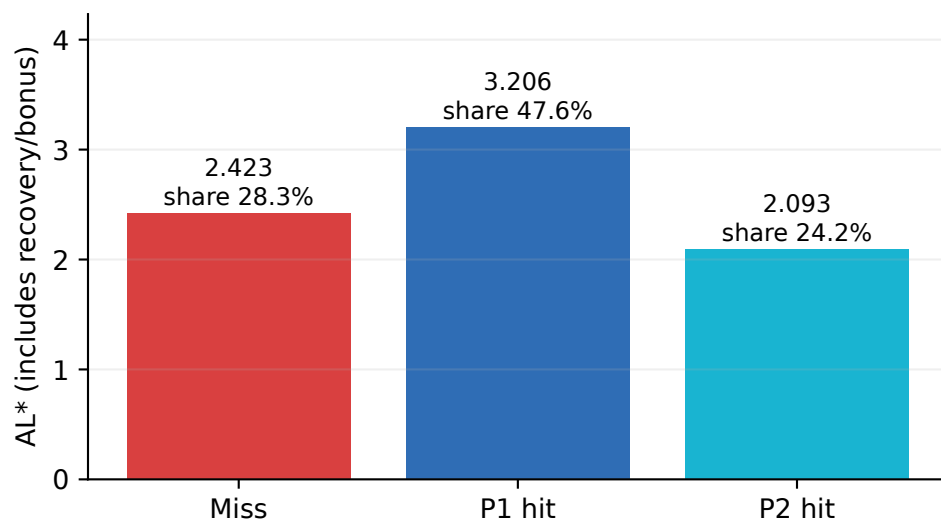
Measured wall time by batch cache status



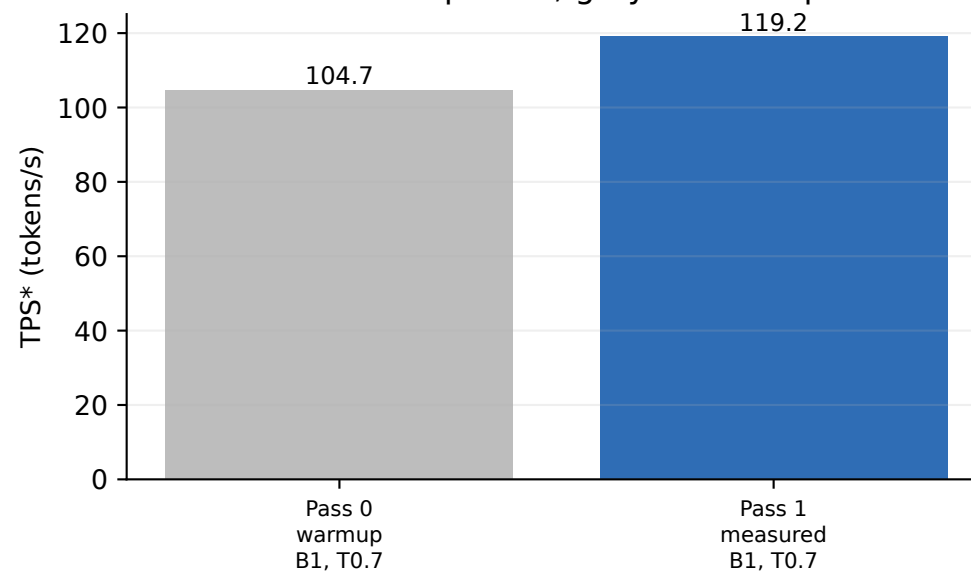
Boundary-free steps, including slow steps



Conditional AL* and source share



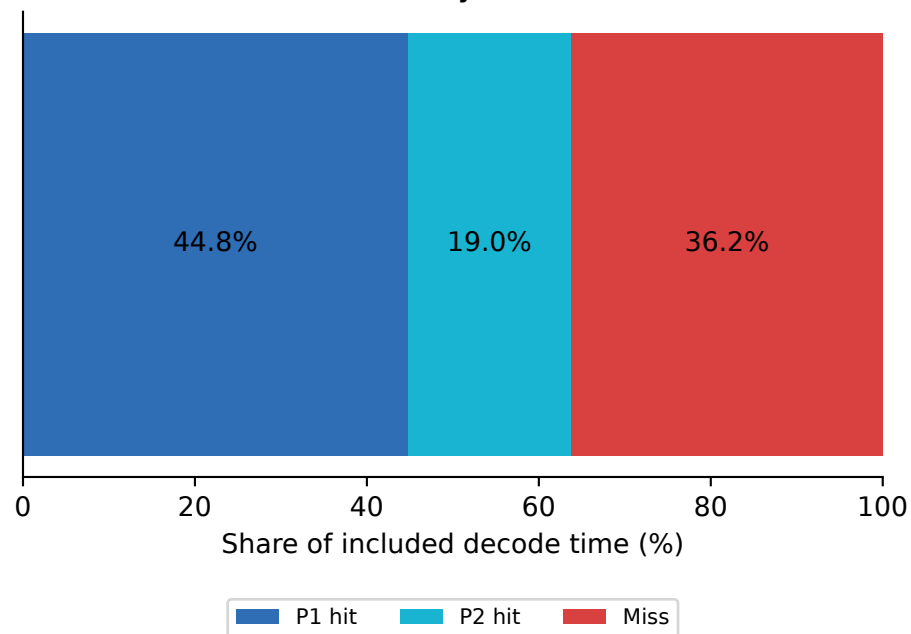
All saved passes; gray = warmup



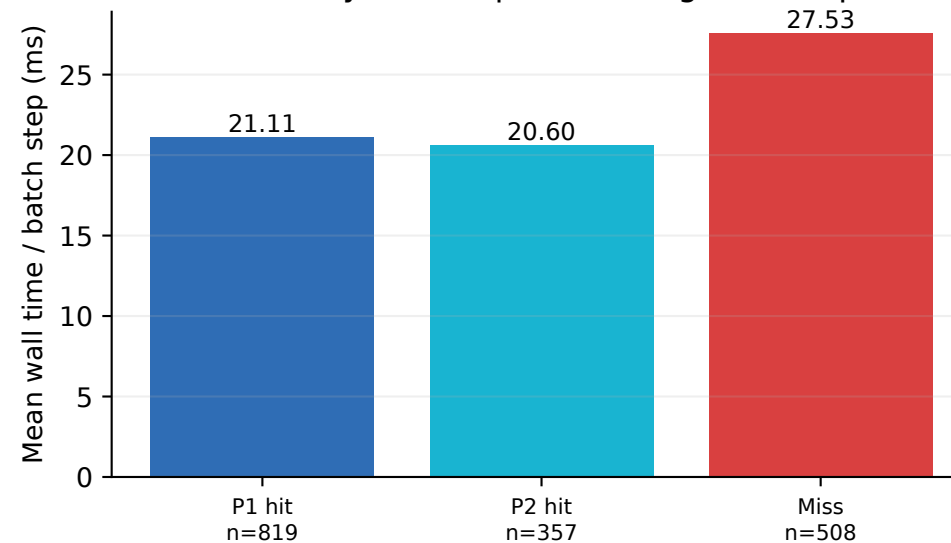
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_postopt_warm_2
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
Last pass: B=1, T=0.7, N=48 | AL*=2.794, hit=69.9%, TPS*=121.81

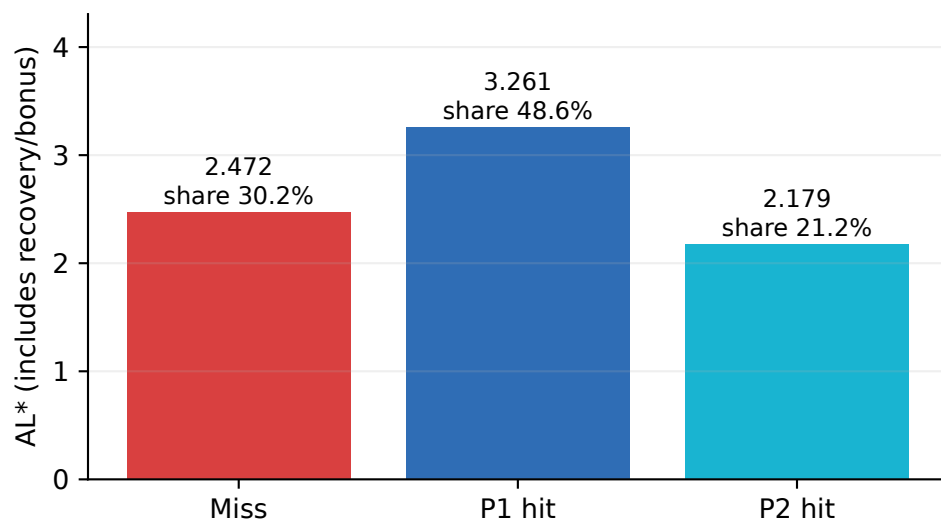
Measured wall time by batch cache status



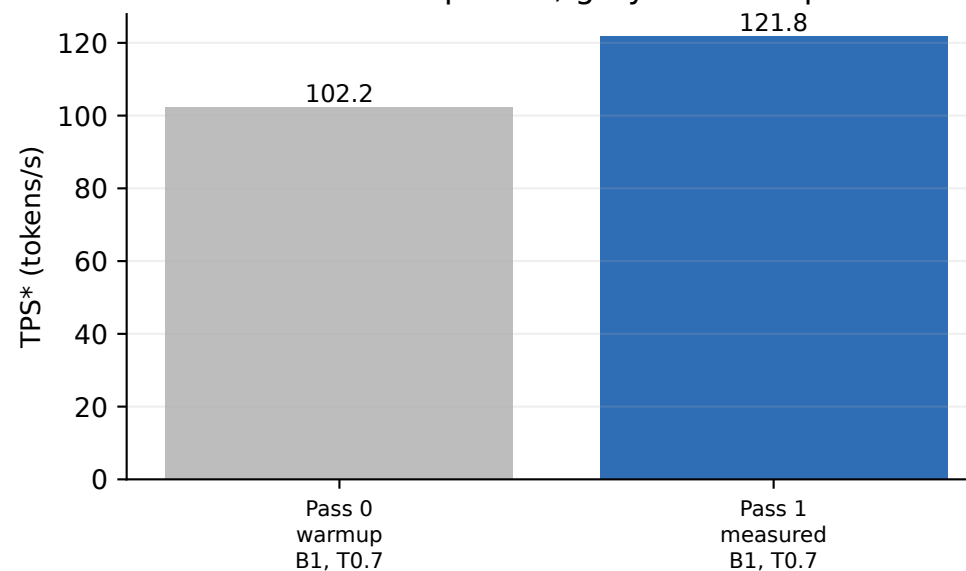
Boundary-free steps, including slow steps



Conditional AL* and source share



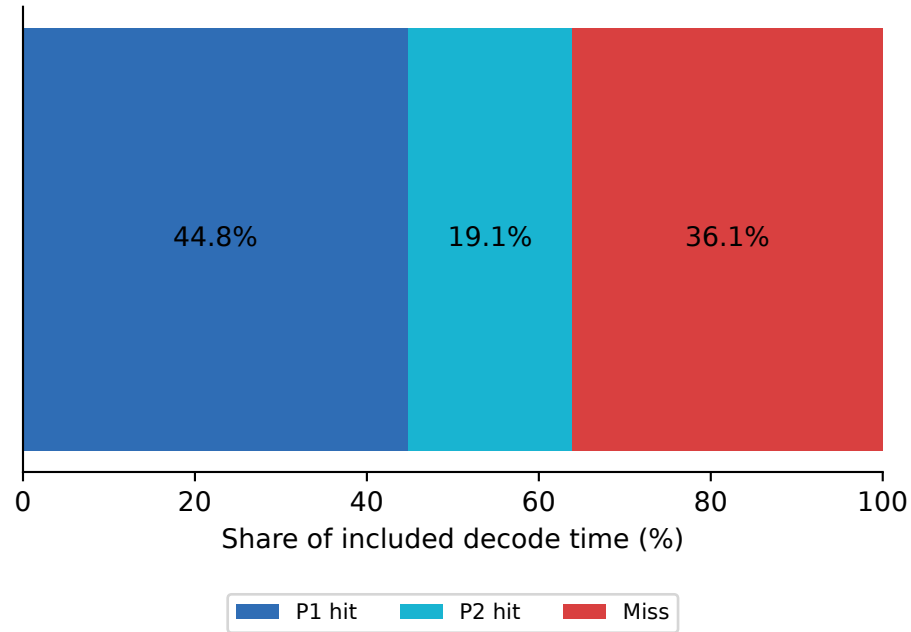
All saved passes; gray = warmup



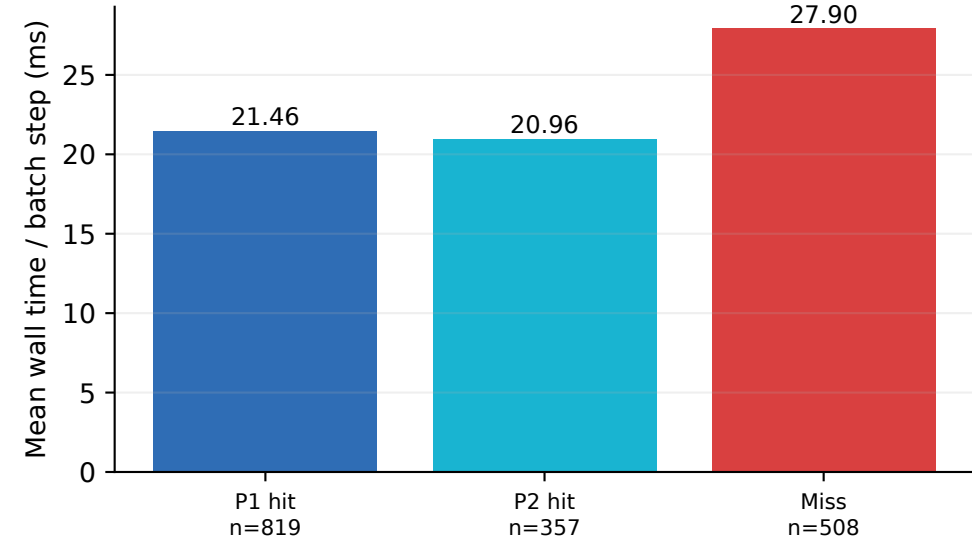
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b1_postopt_warm_3
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/0
Last pass: B=1, T=0.7, N=48 | AL*=2.794, hit=69.9%, TPS*=119.95

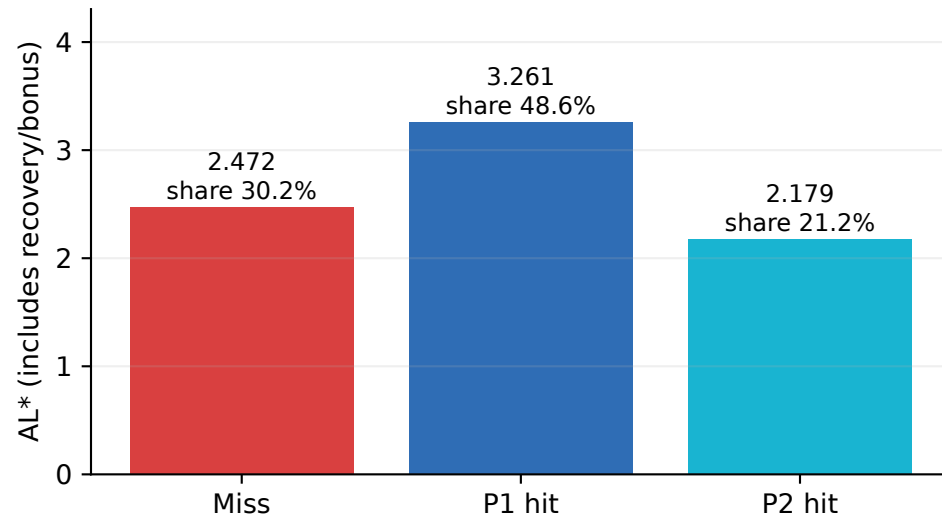
Measured wall time by batch cache status



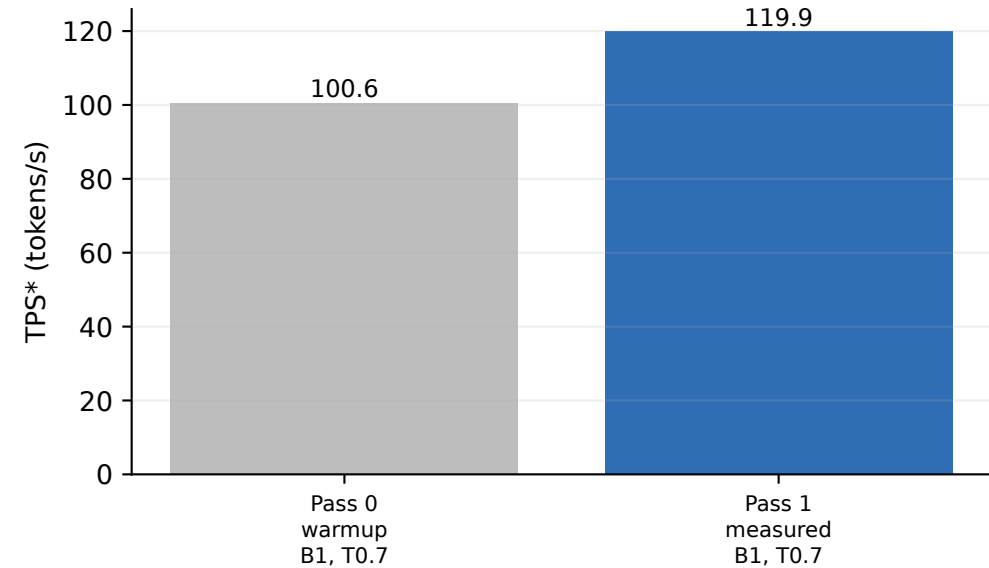
Boundary-free steps, including slow steps



Conditional AL* and source share



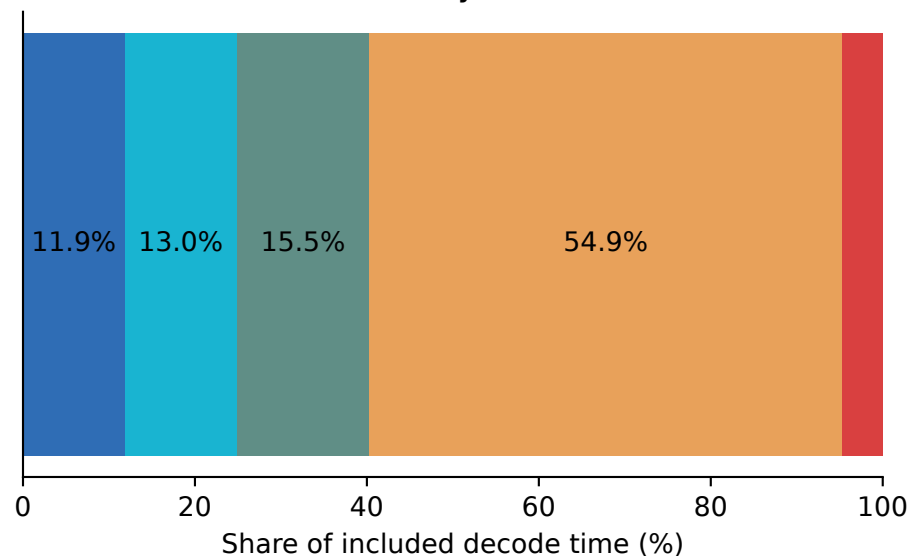
All saved passes; gray = warmup



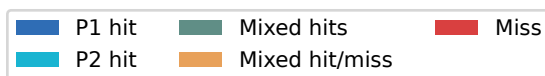
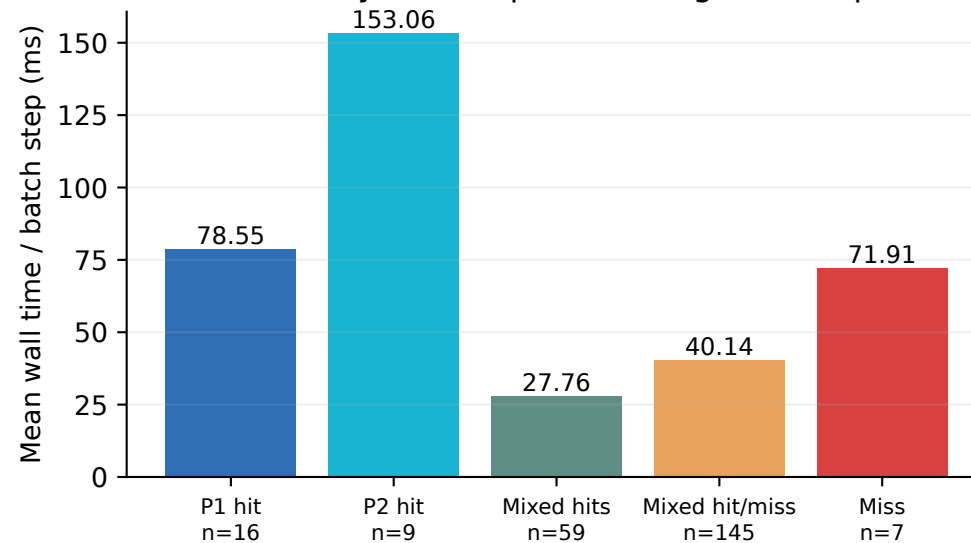
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_postopt_warm_0
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.408, hit=83.0%, TPS*=373.83

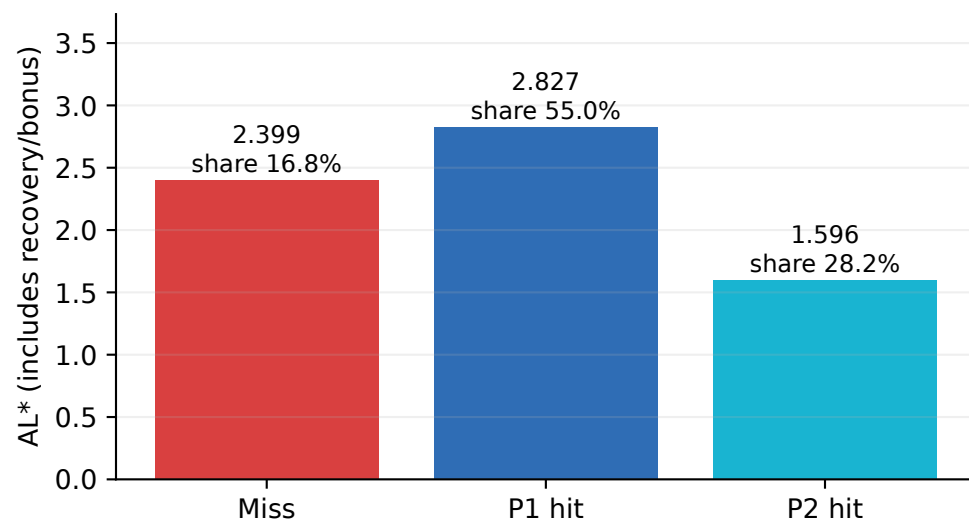
Measured wall time by batch cache status



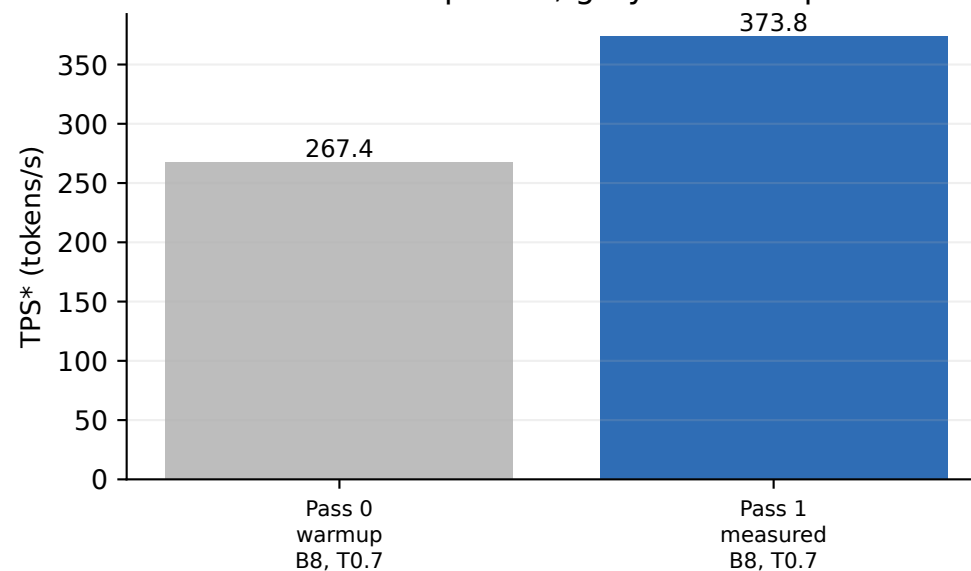
Boundary-free steps, including slow steps



Conditional AL* and source share



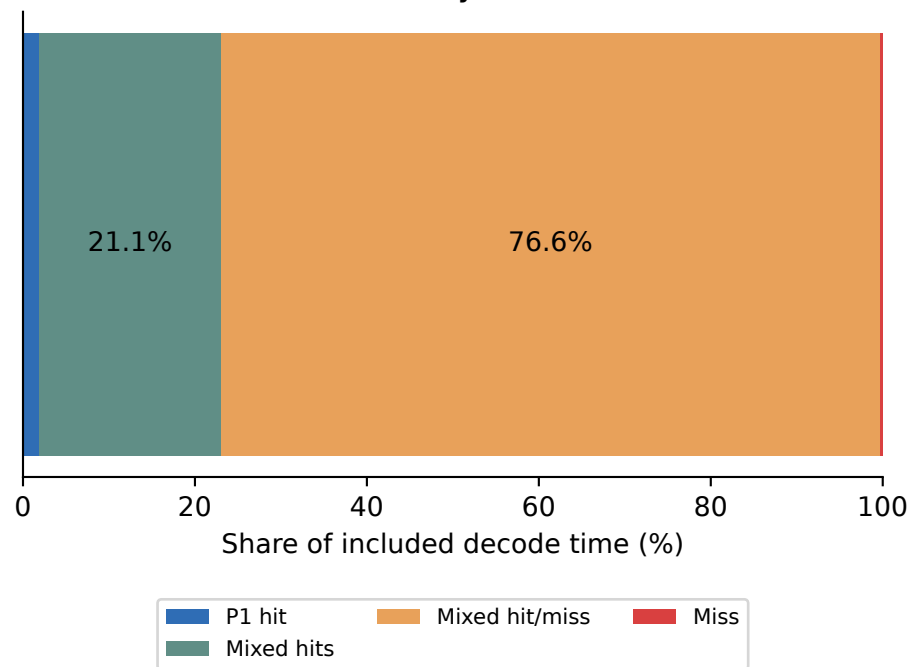
All saved passes; gray = warmup



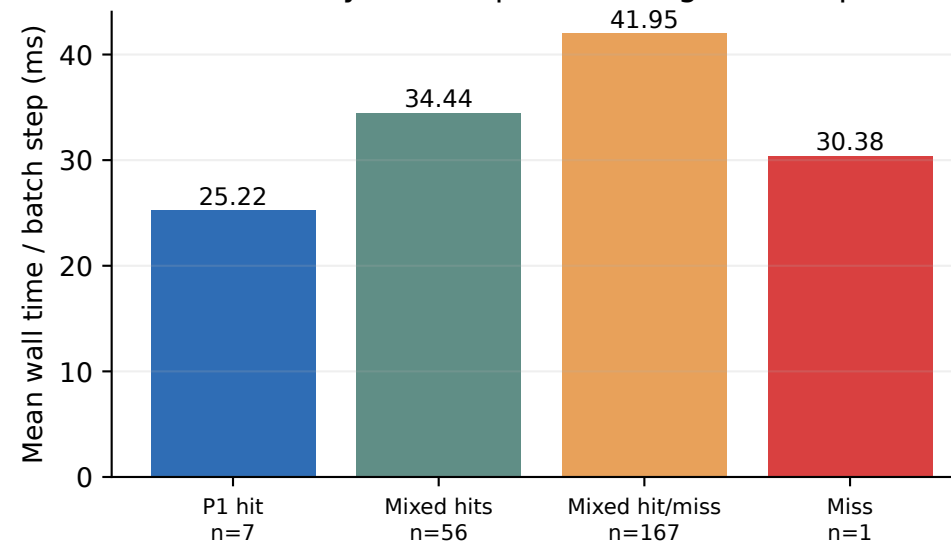
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_postopt_warm_1
 DUET: K1/K2=3/2, exit=28 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.316, hit=83.8%, TPS*=439.43

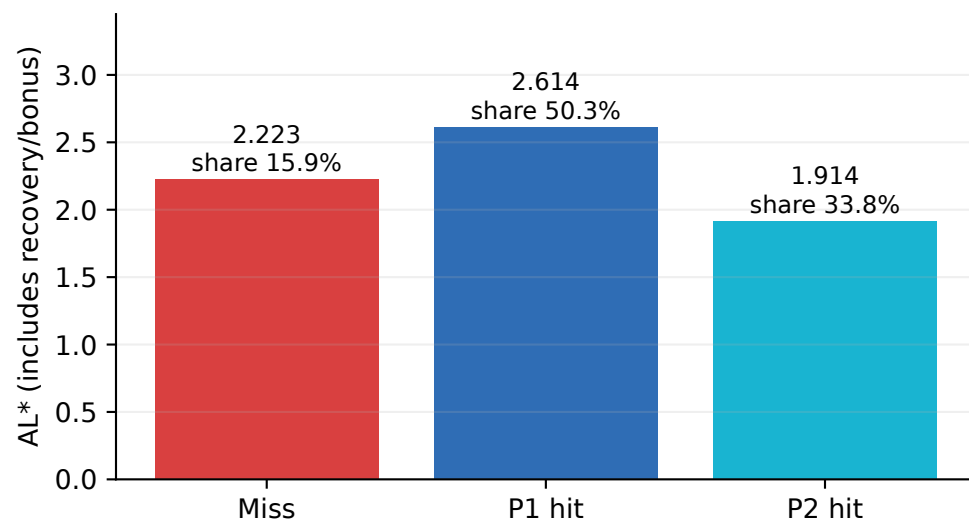
Measured wall time by batch cache status



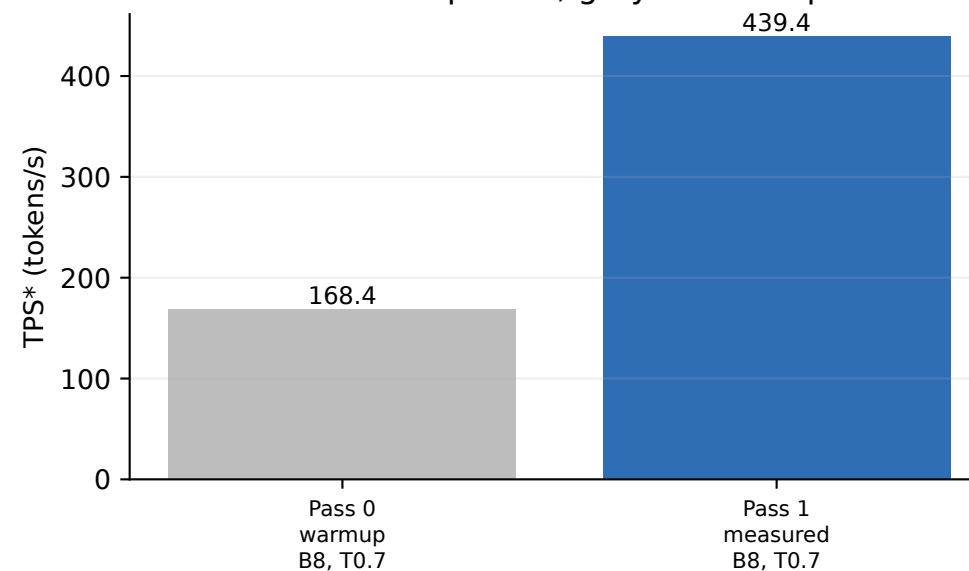
Boundary-free steps, including slow steps



Conditional AL* and source share



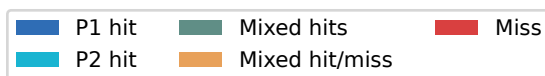
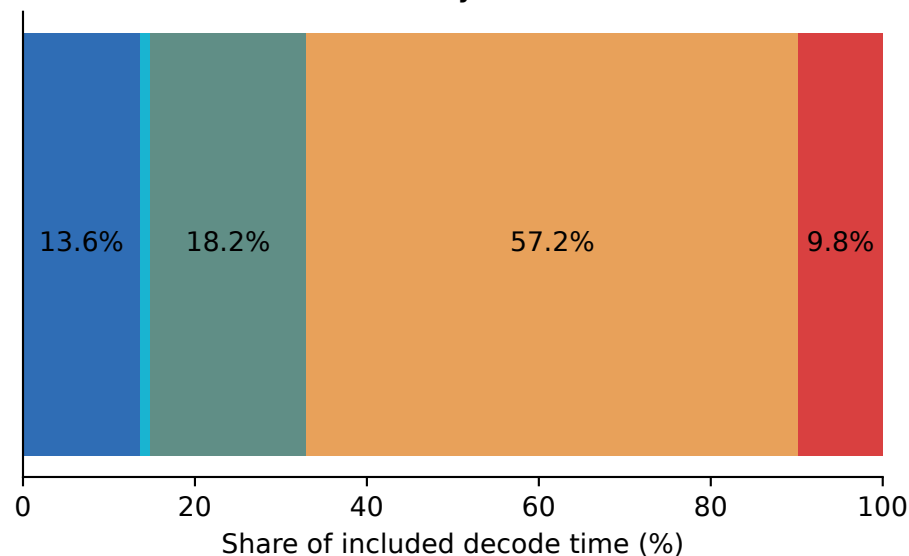
All saved passes; gray = warmup



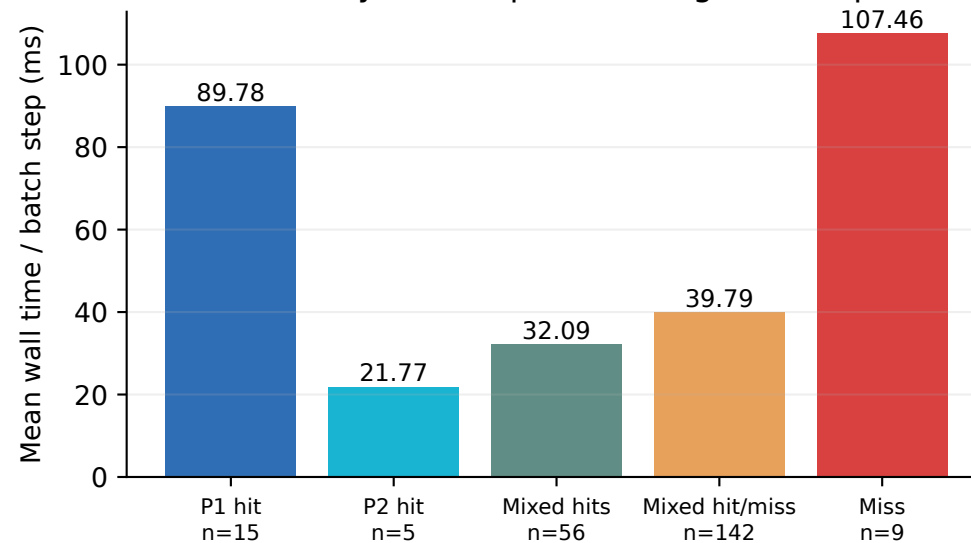
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_postopt_warm_2
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.406, hit=83.6%, TPS*=387.45

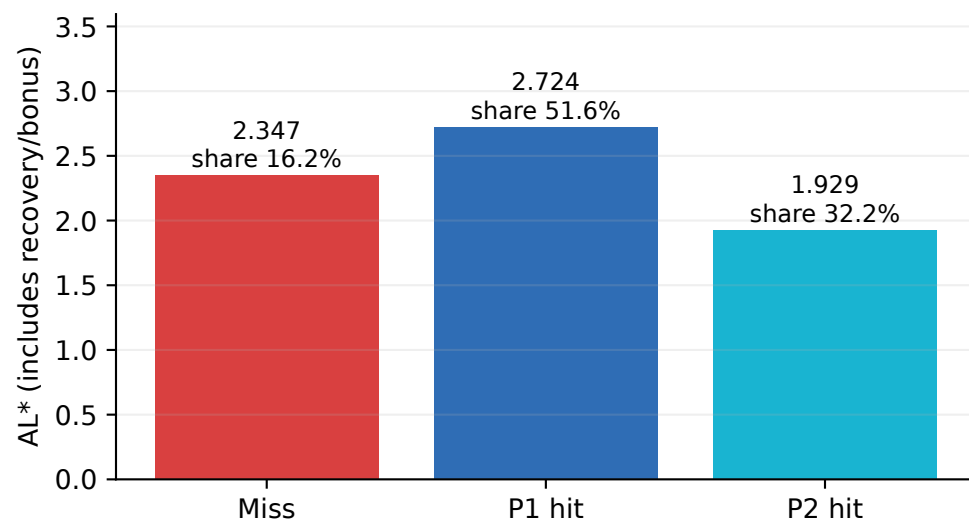
Measured wall time by batch cache status



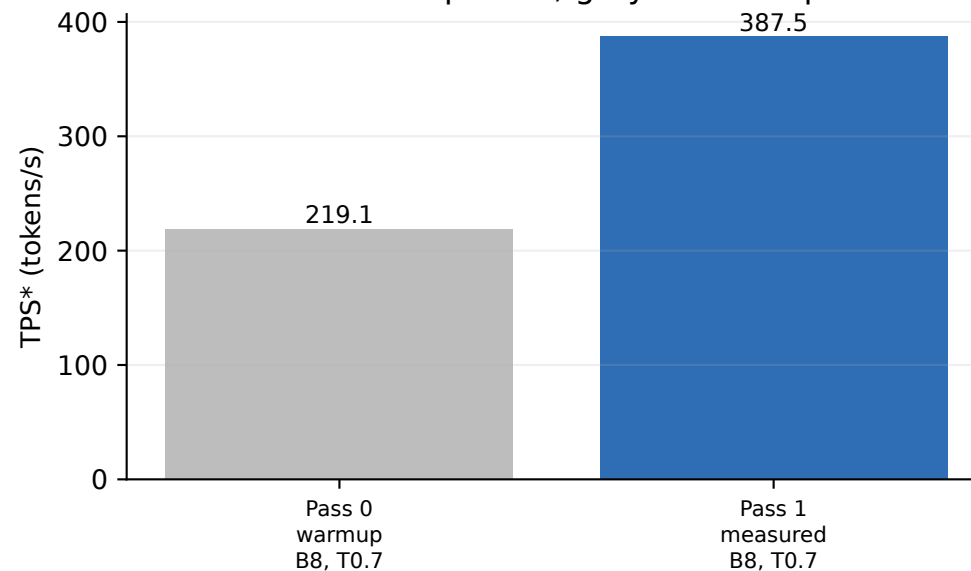
Boundary-free steps, including slow steps



Conditional AL* and source share



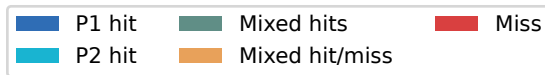
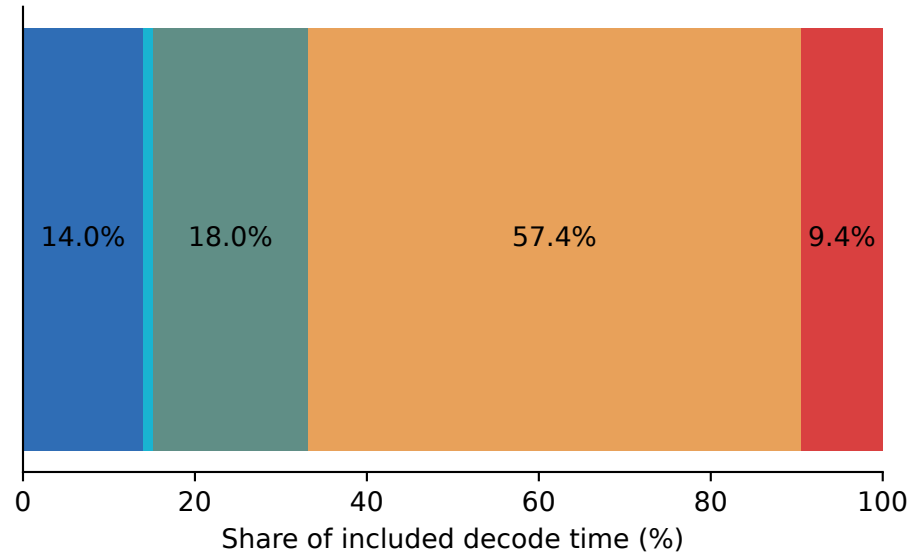
All saved passes; gray = warmup



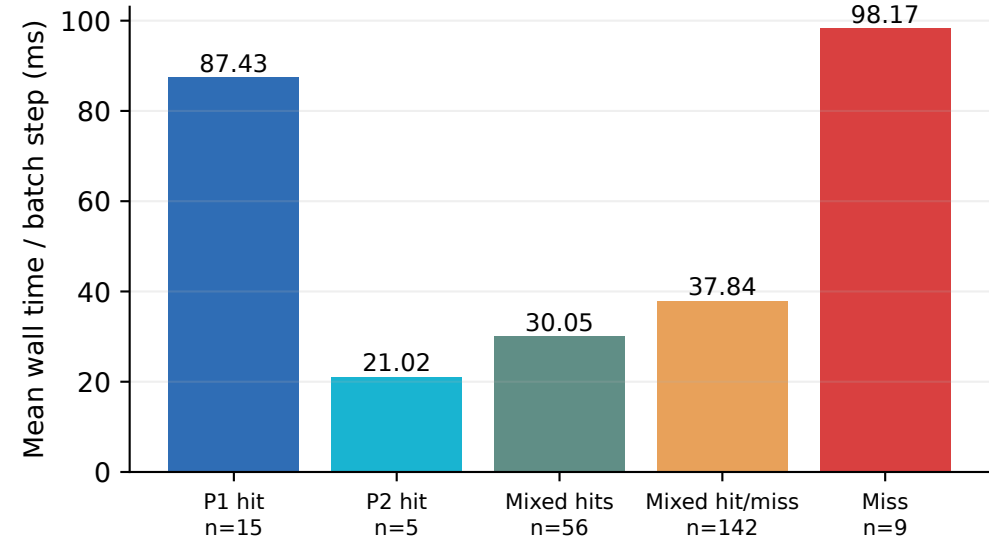
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_postopt_warm_3
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.406, hit=83.6%, TPS*=408.72

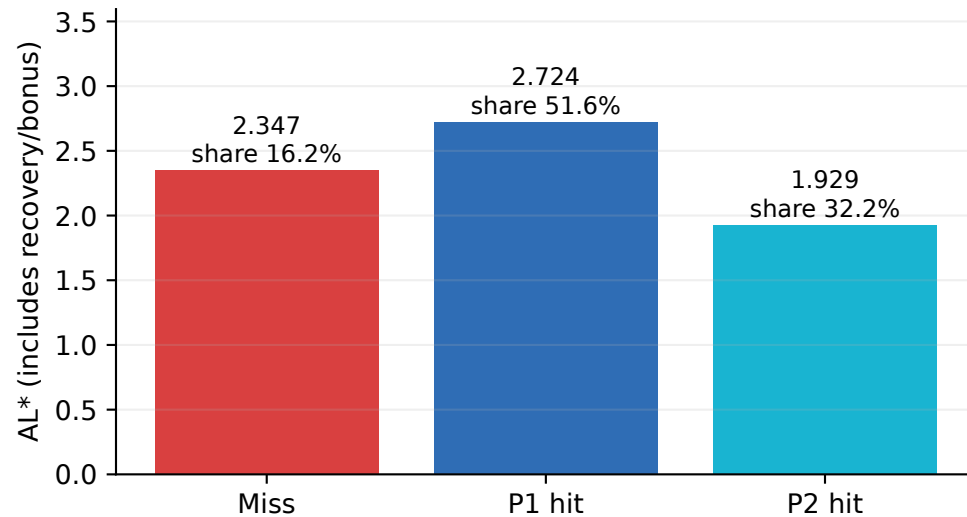
Measured wall time by batch cache status



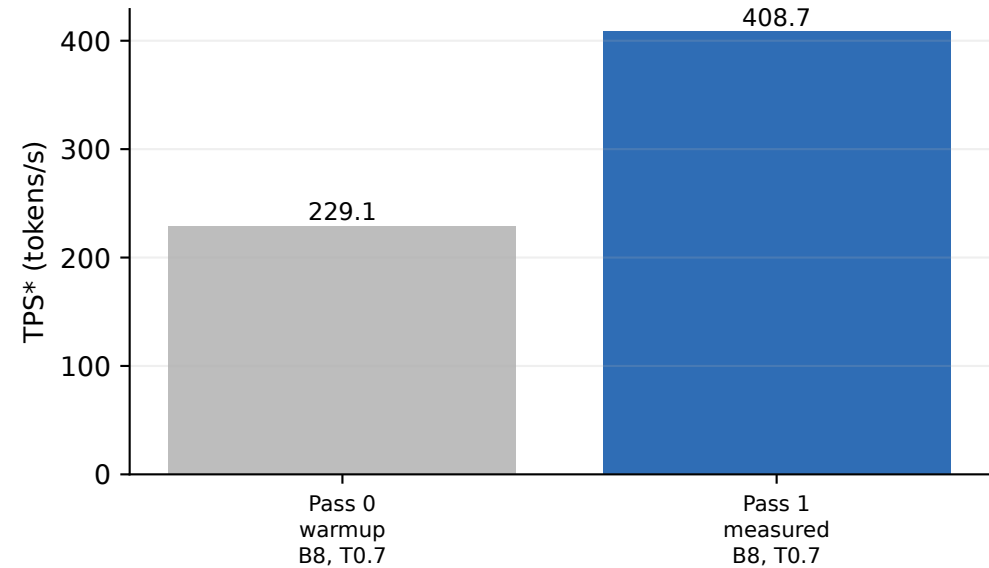
Boundary-free steps, including slow steps



Conditional AL* and source share



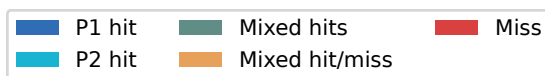
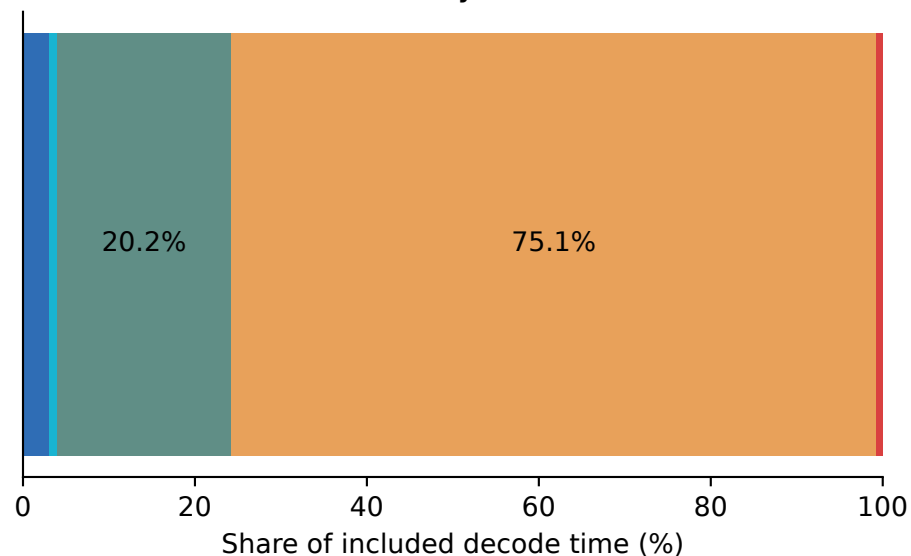
All saved passes; gray = warmup



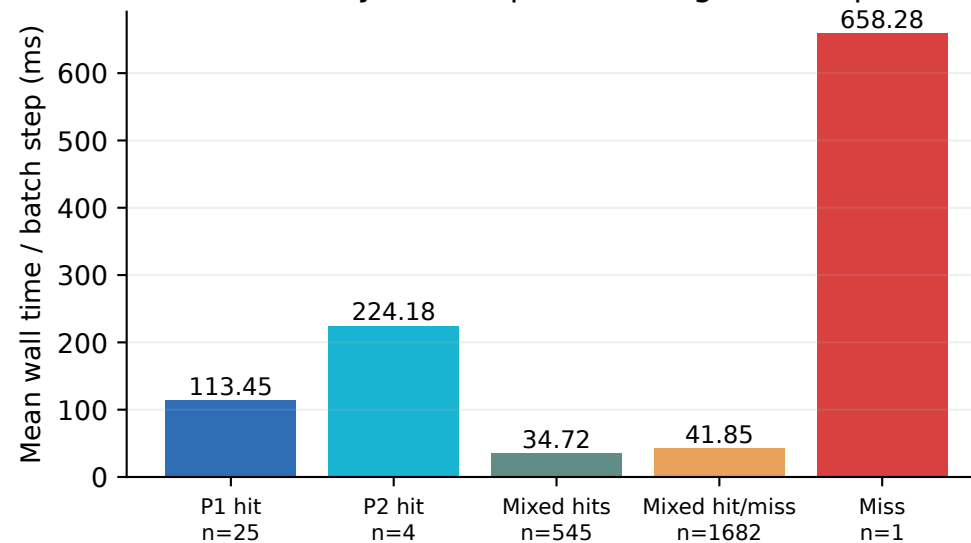
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_postopt_final_r0
 DUET: K1/K2=3/2, exit=28 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.381, hit=83.9%, TPS*=452.85

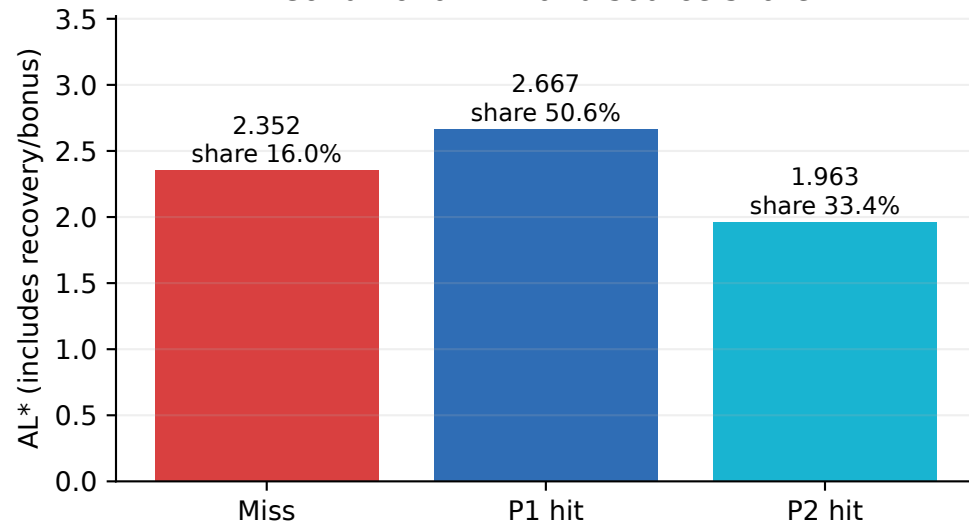
Measured wall time by batch cache status



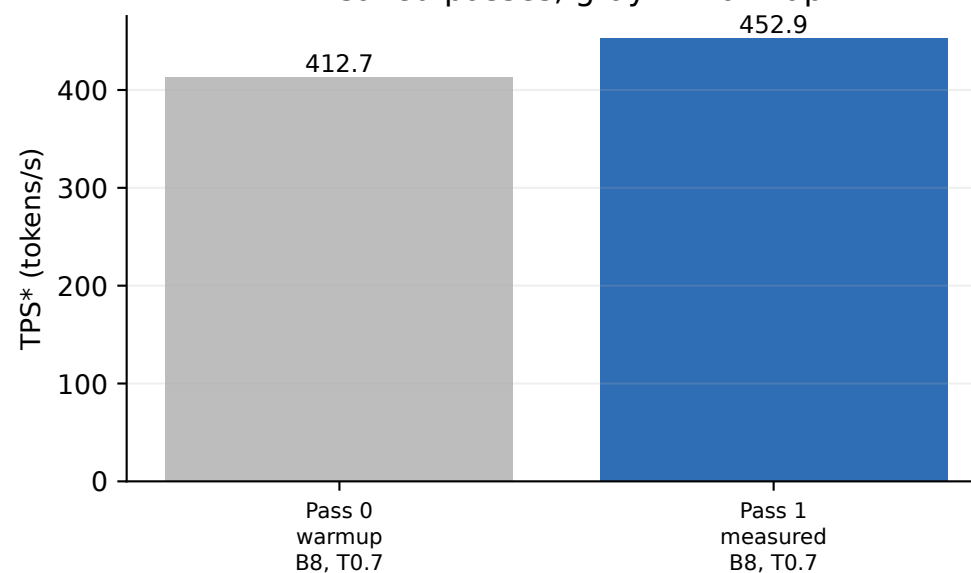
Boundary-free steps, including slow steps



Conditional AL* and source share



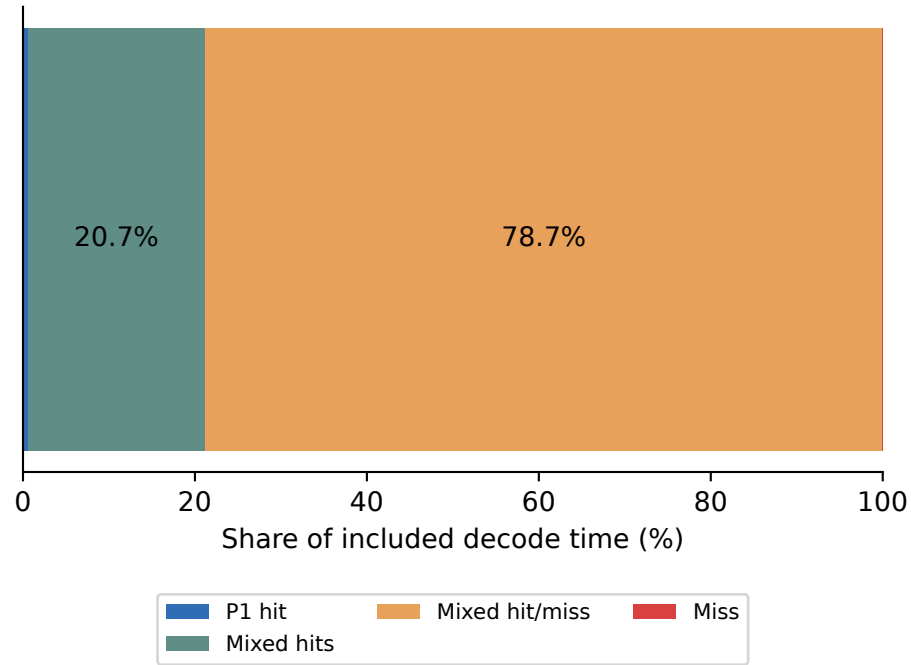
All saved passes; gray = warmup



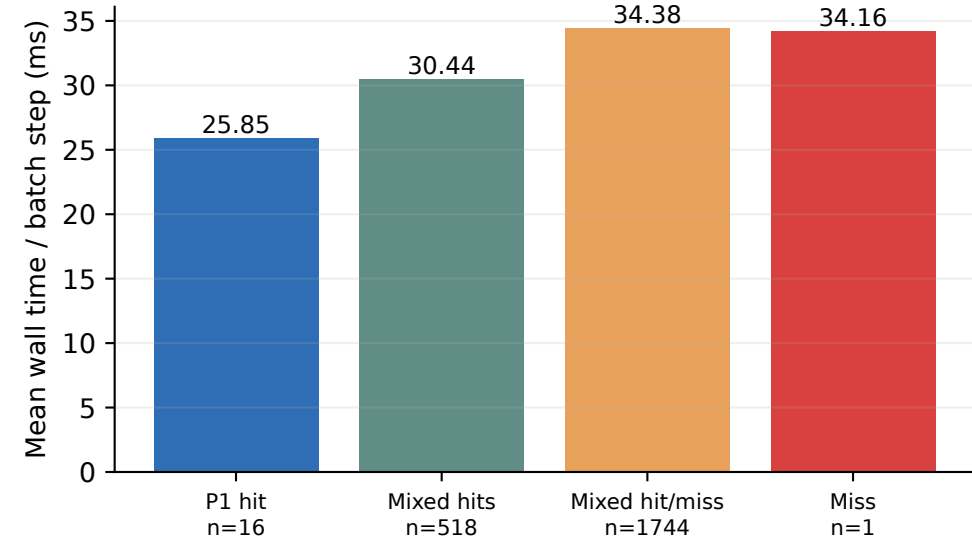
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_final_r1
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.279, hit=82.9%, TPS*=542.47

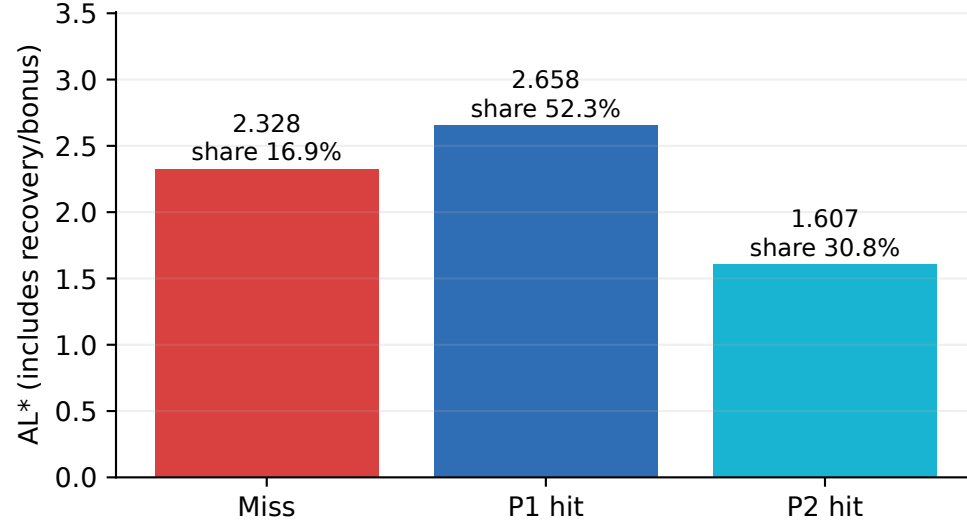
Measured wall time by batch cache status



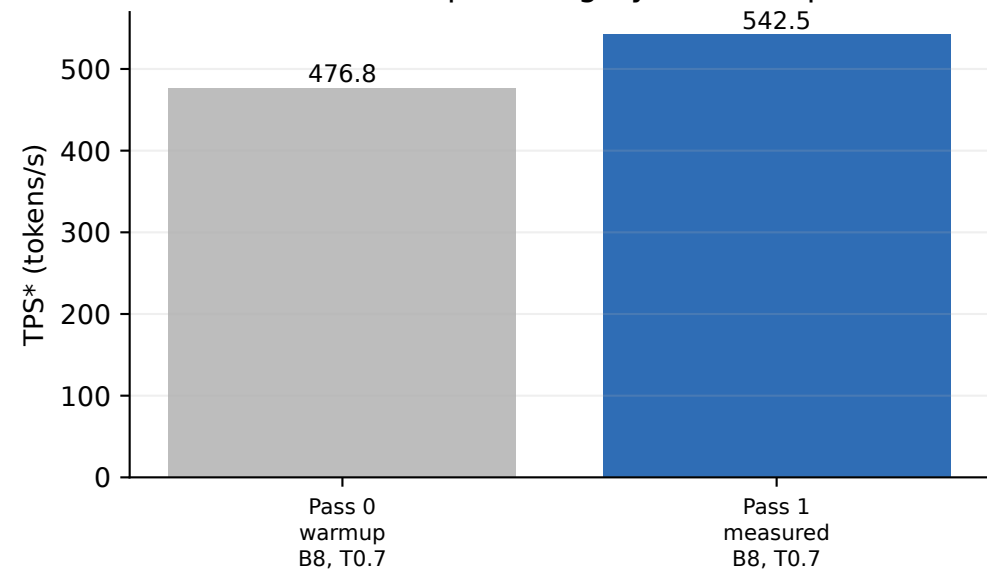
Boundary-free steps, including slow steps



Conditional AL* and source share



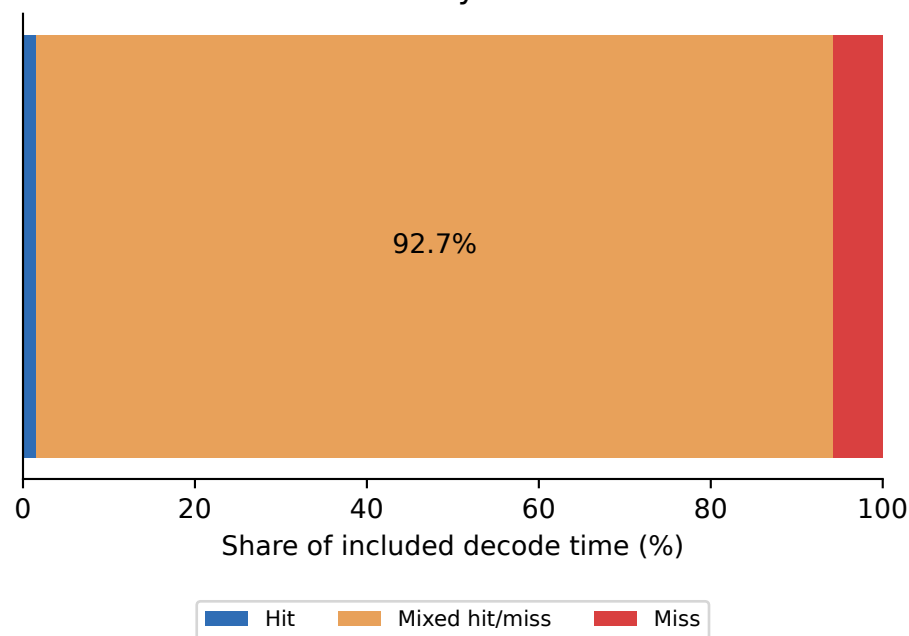
All saved passes; gray = warmup



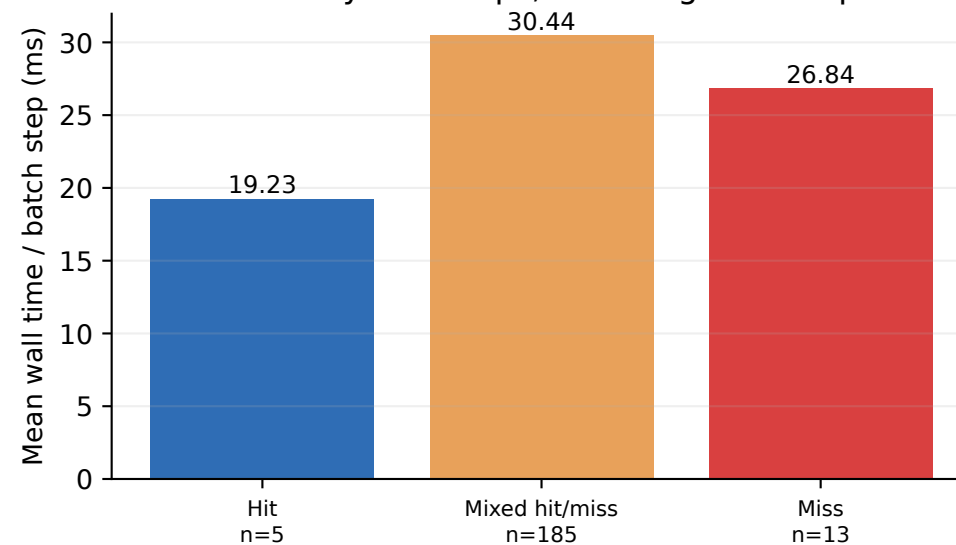
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_ssd_control
SSD: K=4, F=1
Last pass: B=8, T=0.7, N=48 | AL*=2.691, hit=36.7%, TPS*=639.44

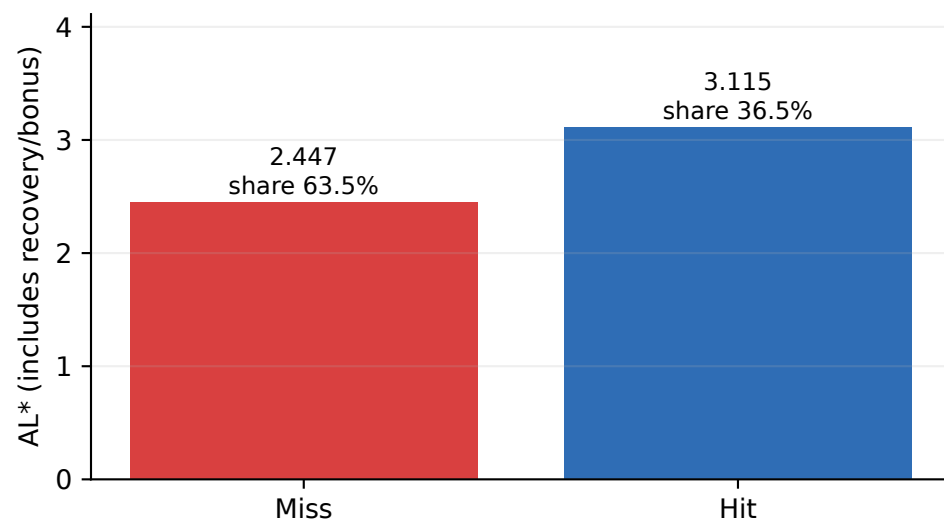
Measured wall time by batch cache status



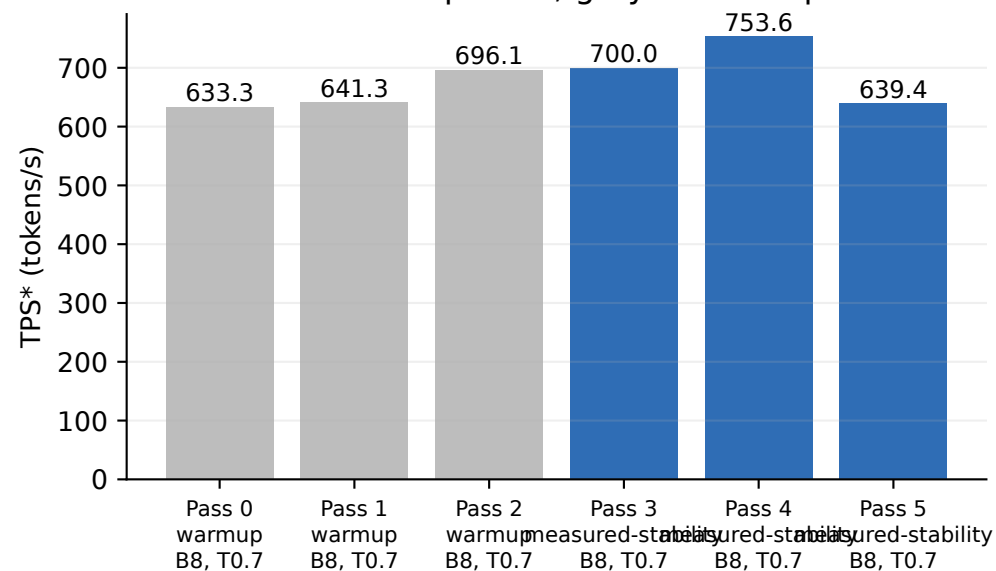
Boundary-free steps, including slow steps



Conditional AL* and source share



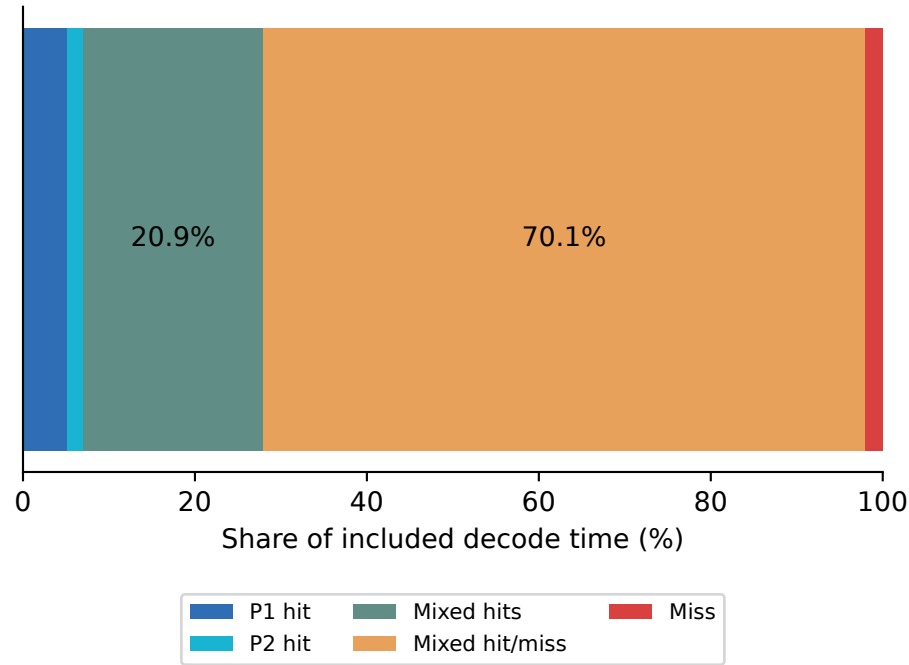
All saved passes; gray = warmup



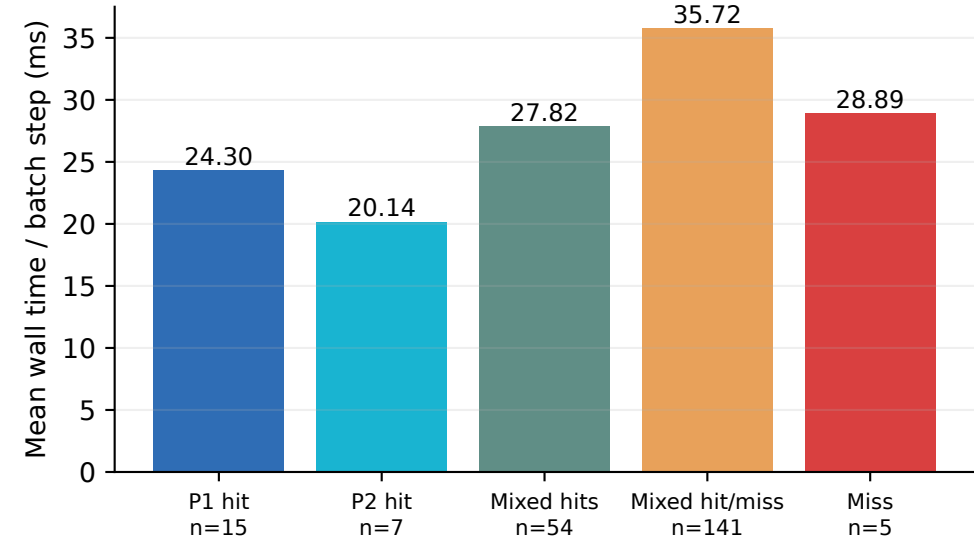
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_0
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.436, hit=85.0%, TPS*=529.43

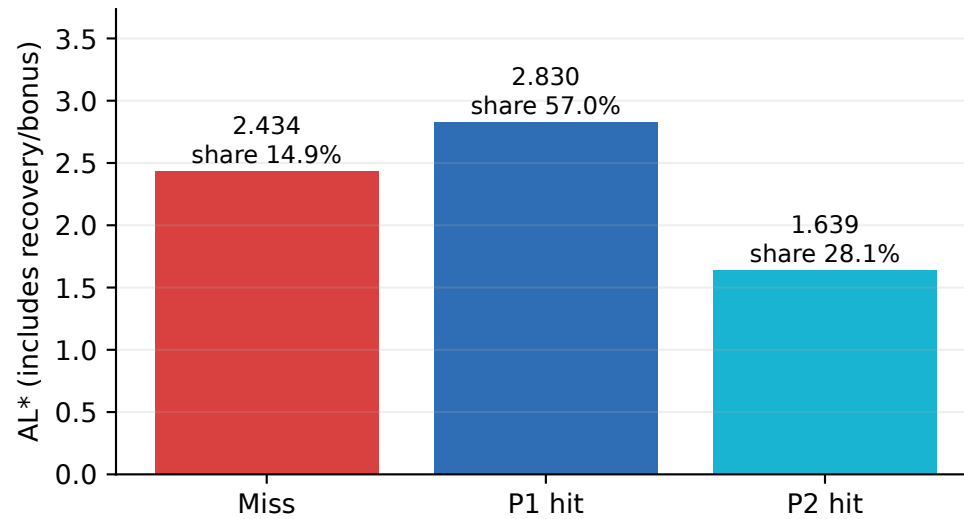
Measured wall time by batch cache status



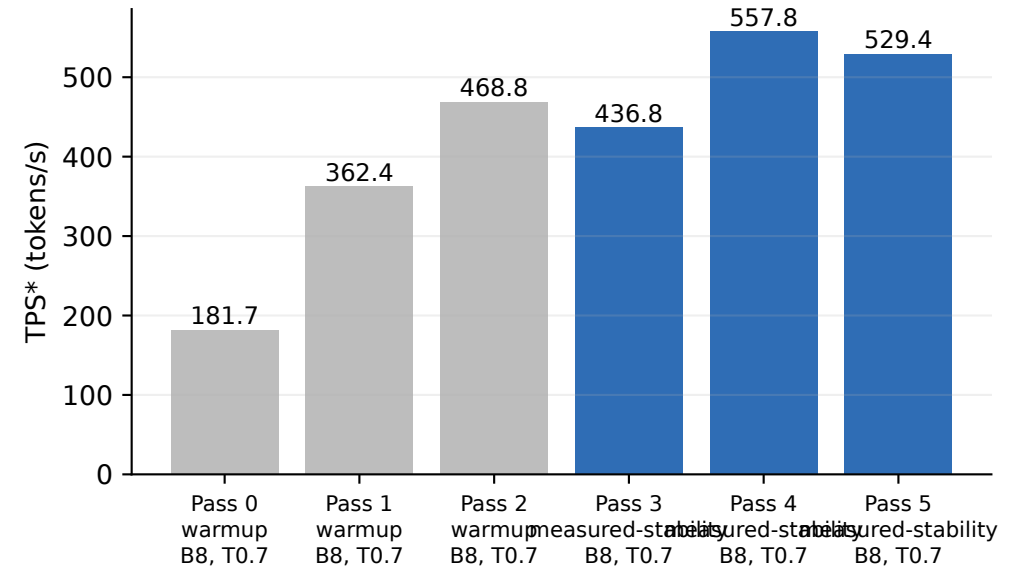
Boundary-free steps, including slow steps



Conditional AL* and source share



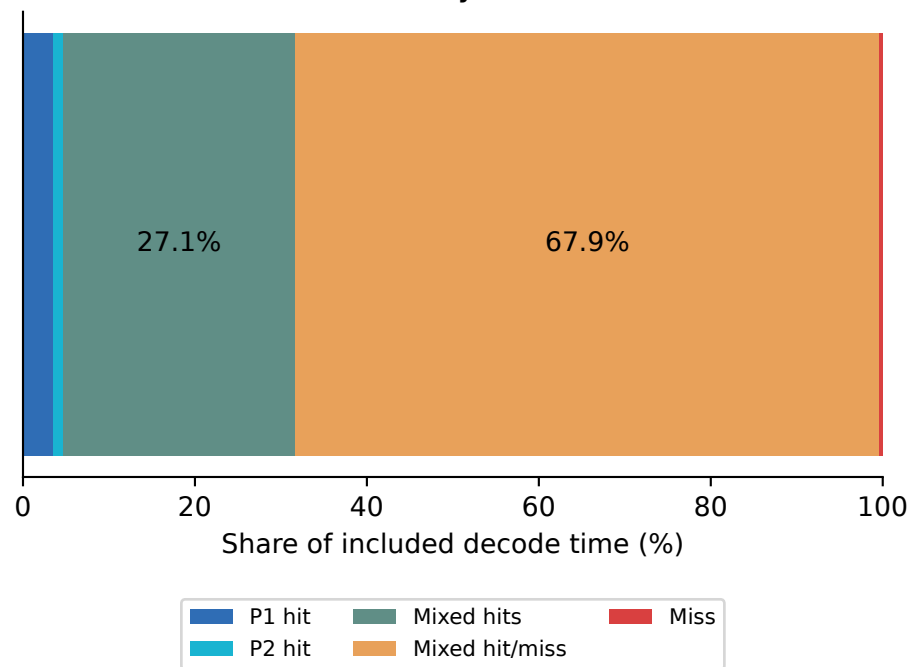
All saved passes; gray = warmup



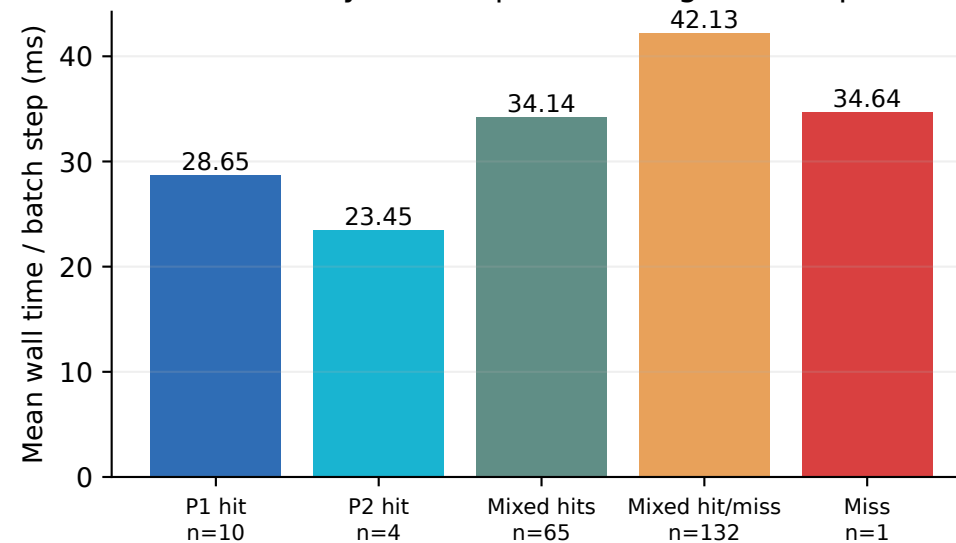
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_1
 DUET: K1/K2=3/2, exit=28 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.435, hit=83.7%, TPS*=467.61

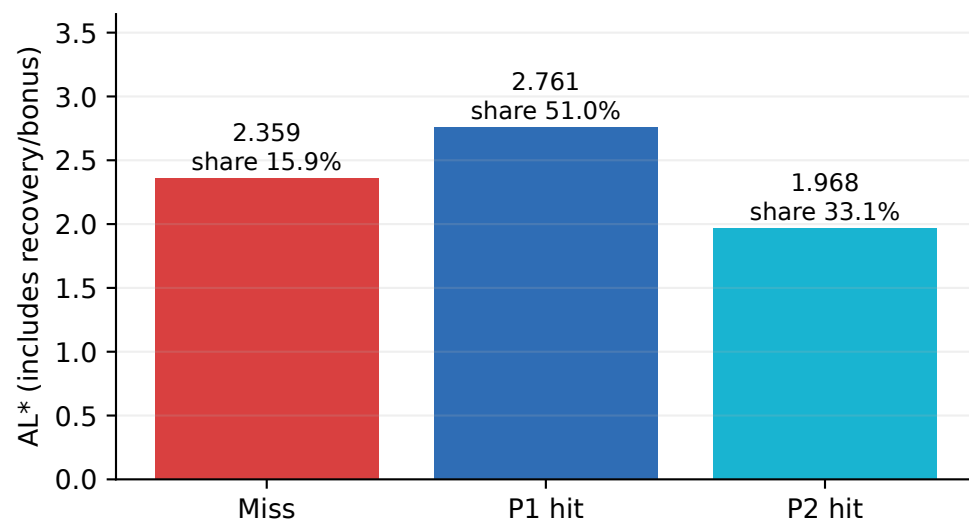
Measured wall time by batch cache status



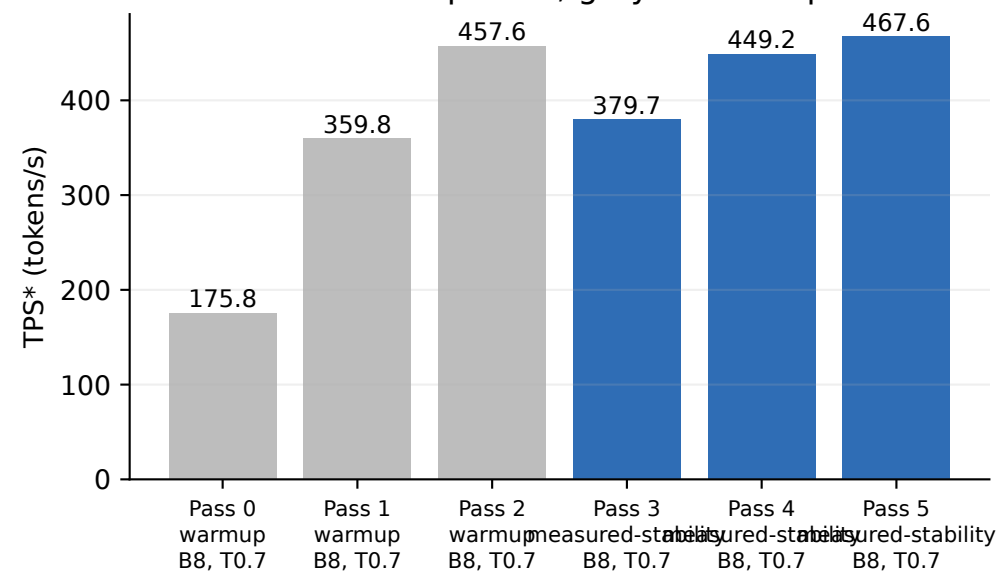
Boundary-free steps, including slow steps



Conditional AL* and source share



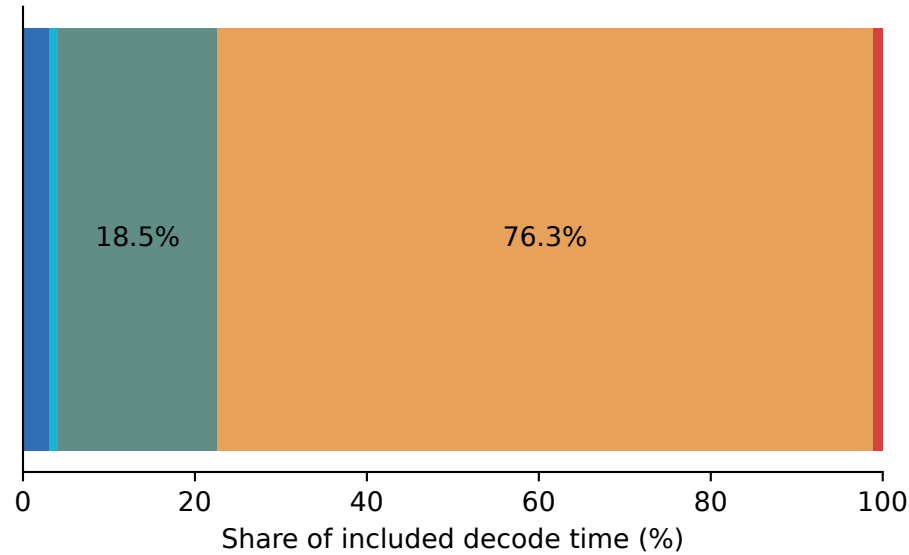
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

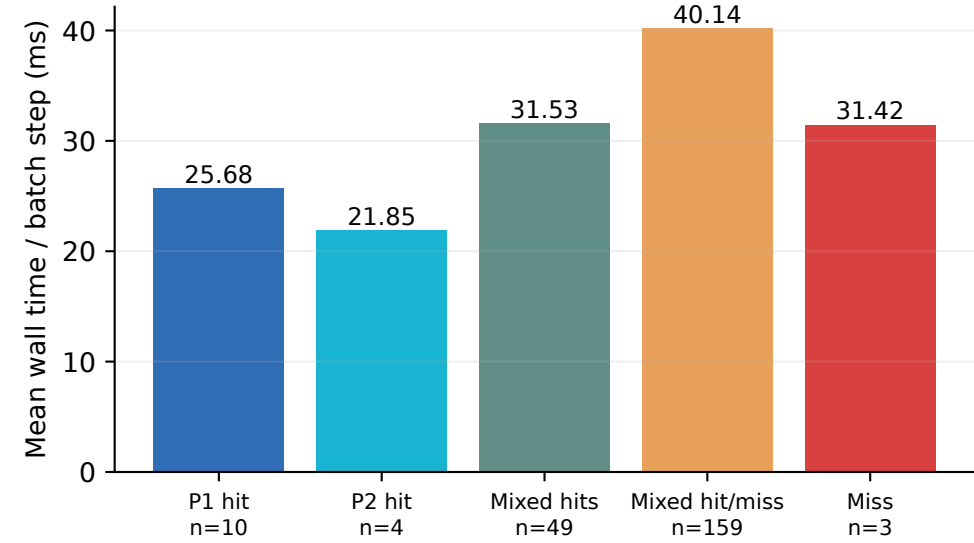
llama3_b8_stability_warm_2
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.438, hit=81.8%, TPS*=477.03

Measured wall time by batch cache status

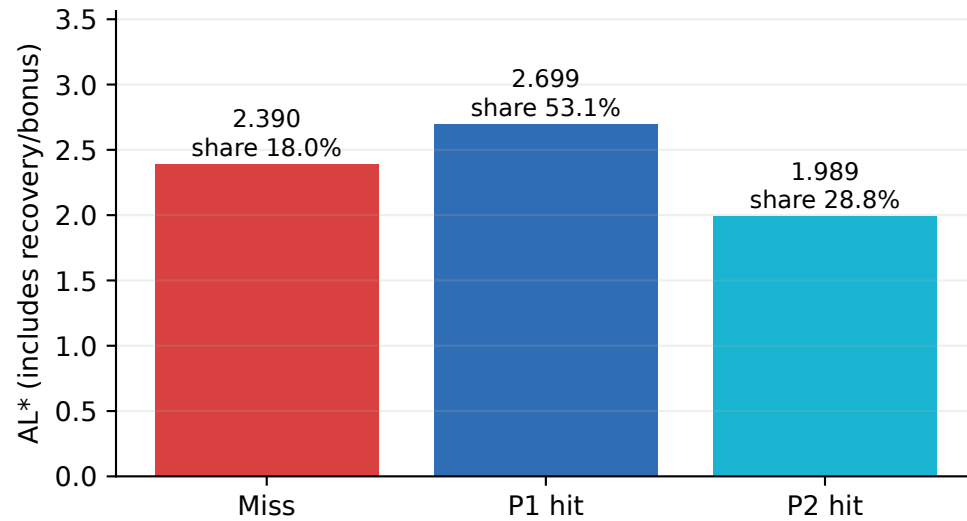


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

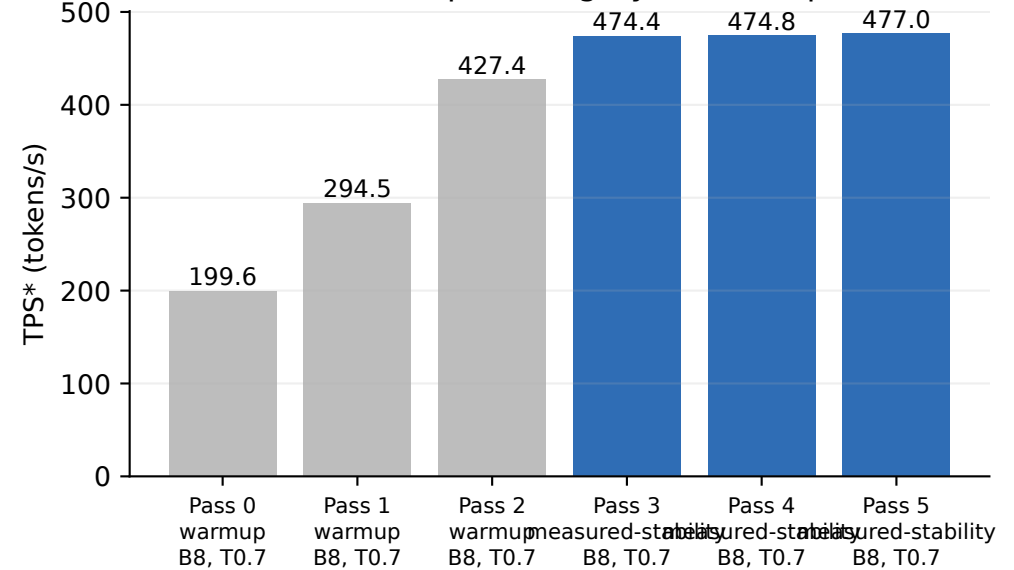
Boundary-free steps, including slow steps



Conditional AL* and source share



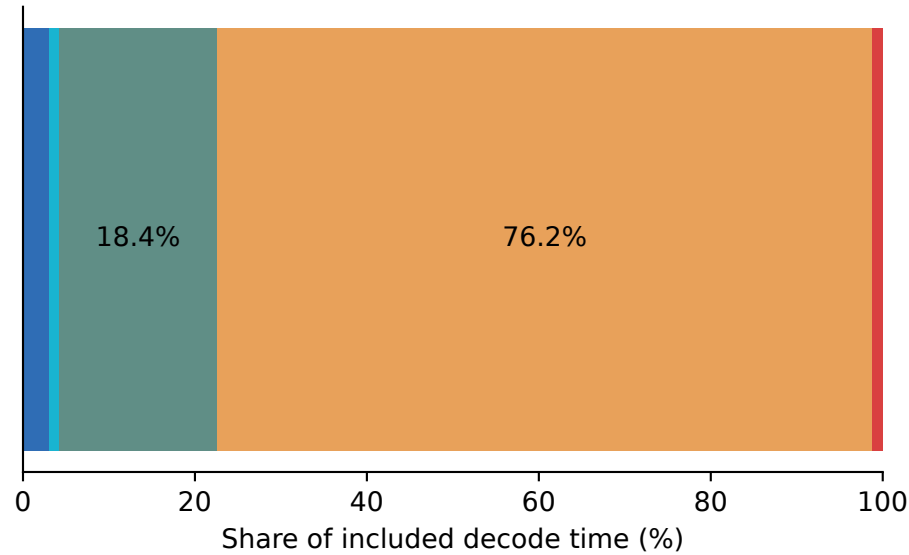
All saved passes; gray = warmup



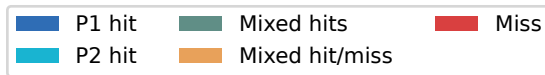
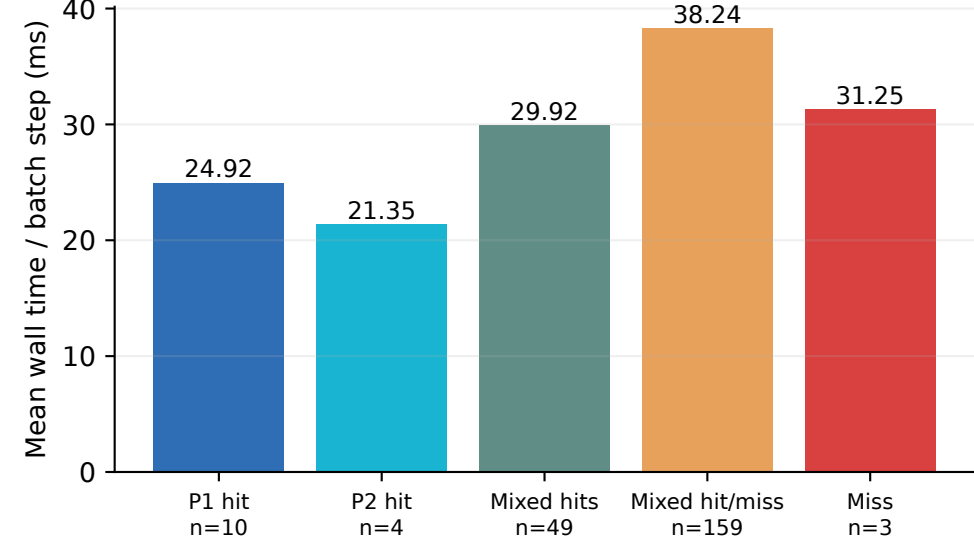
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_3
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.438, hit=81.8%, TPS*=500.44

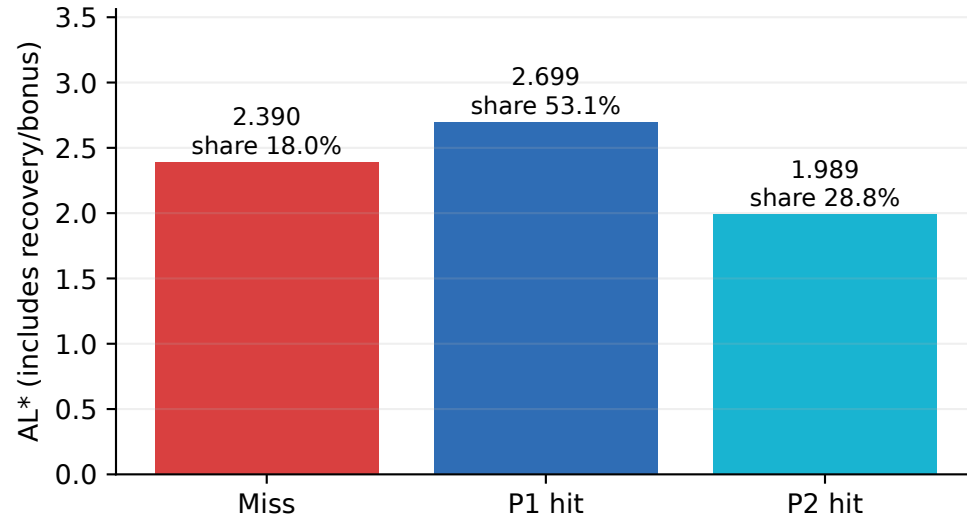
Measured wall time by batch cache status



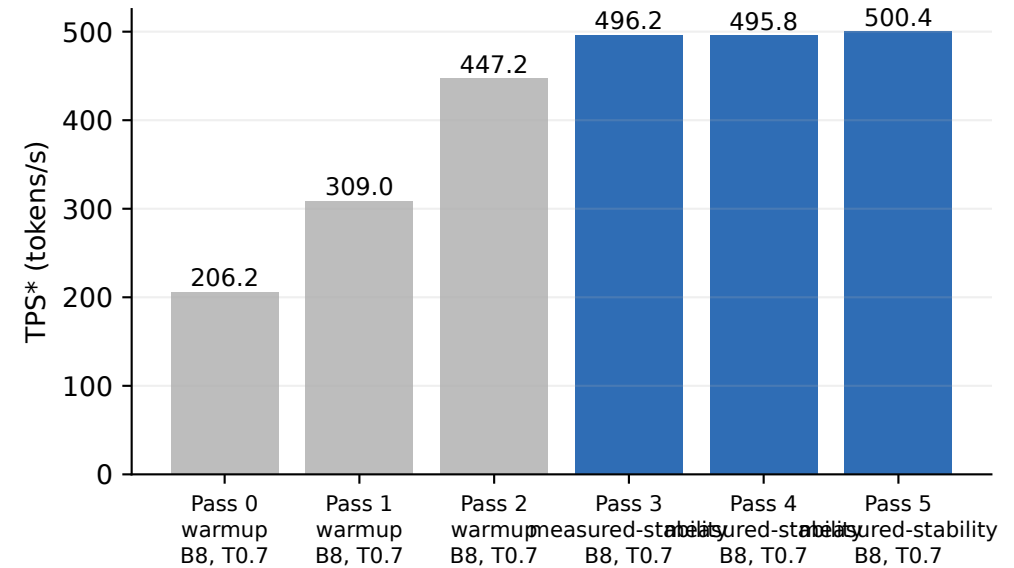
Boundary-free steps, including slow steps



Conditional AL* and source share



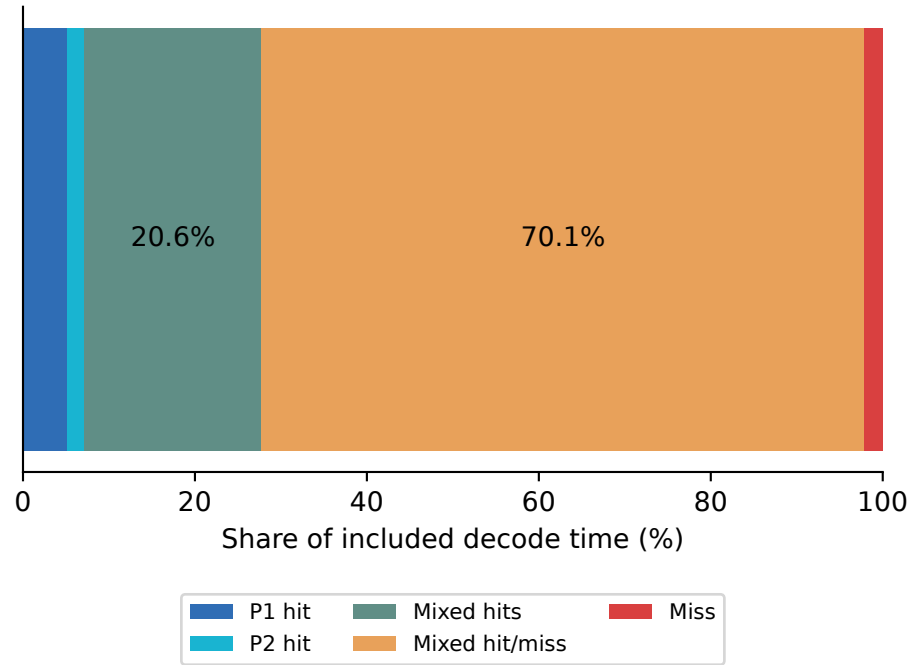
All saved passes; gray = warmup



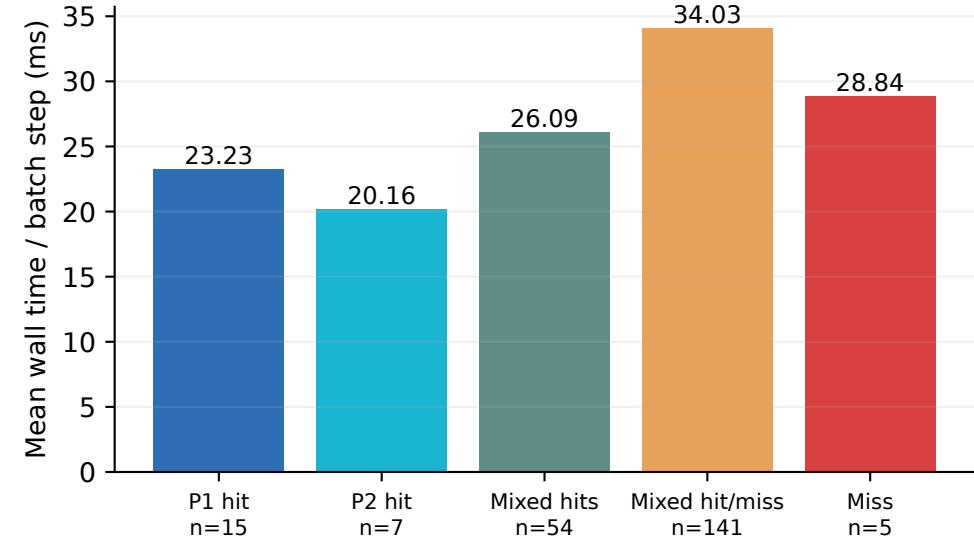
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_4
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.436, hit=85.0%, TPS*=556.40

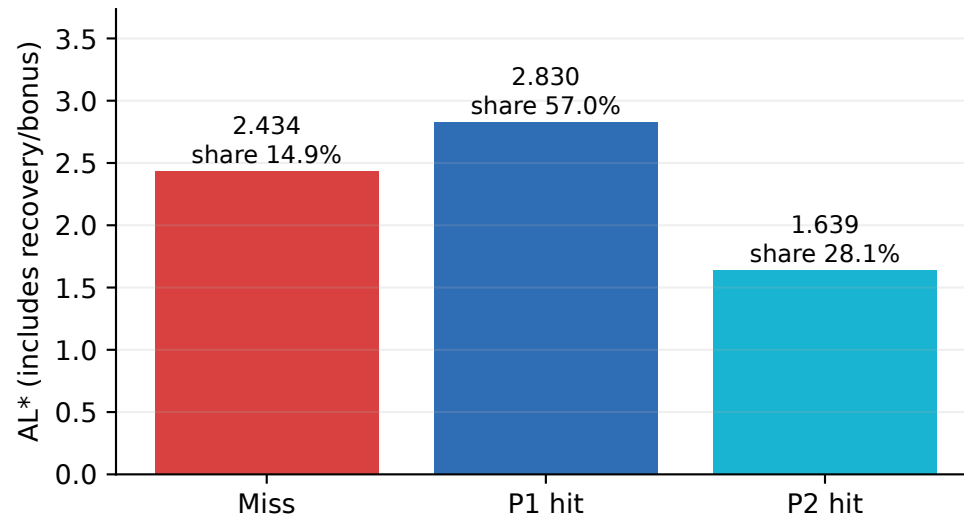
Measured wall time by batch cache status



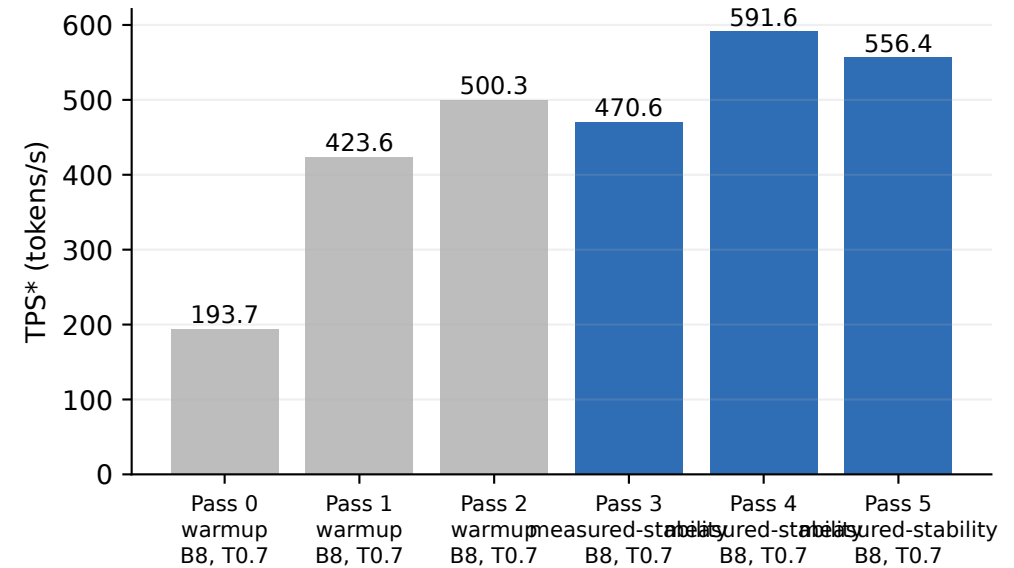
Boundary-free steps, including slow steps



Conditional AL* and source share



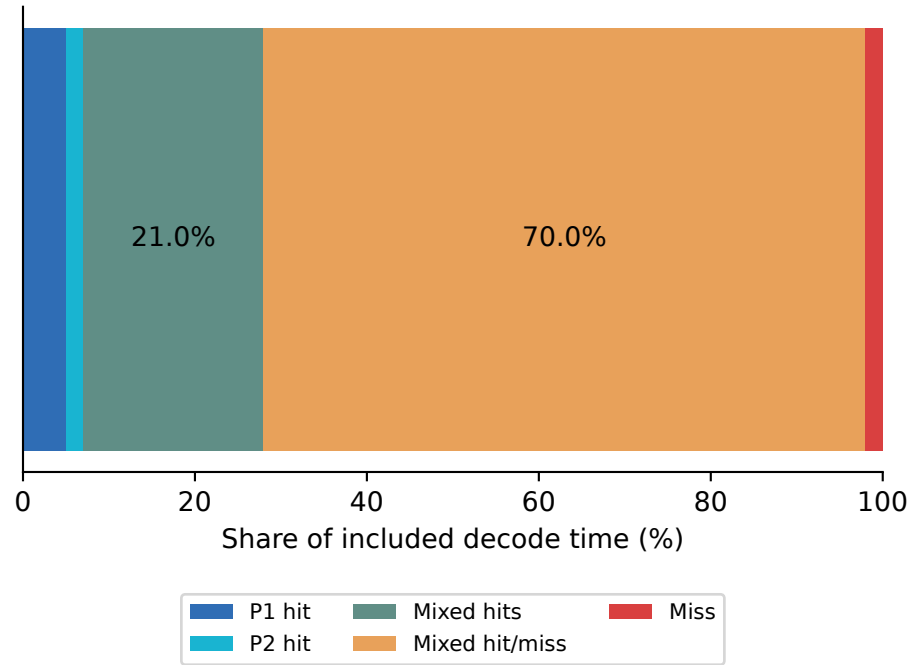
All saved passes; gray = warmup



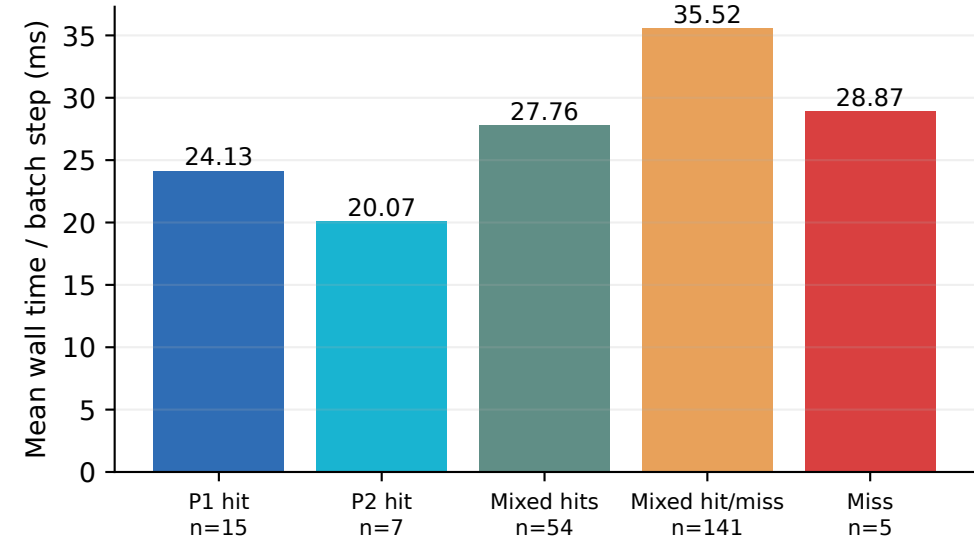
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_5
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.436, hit=85.0%, TPS*=532.01

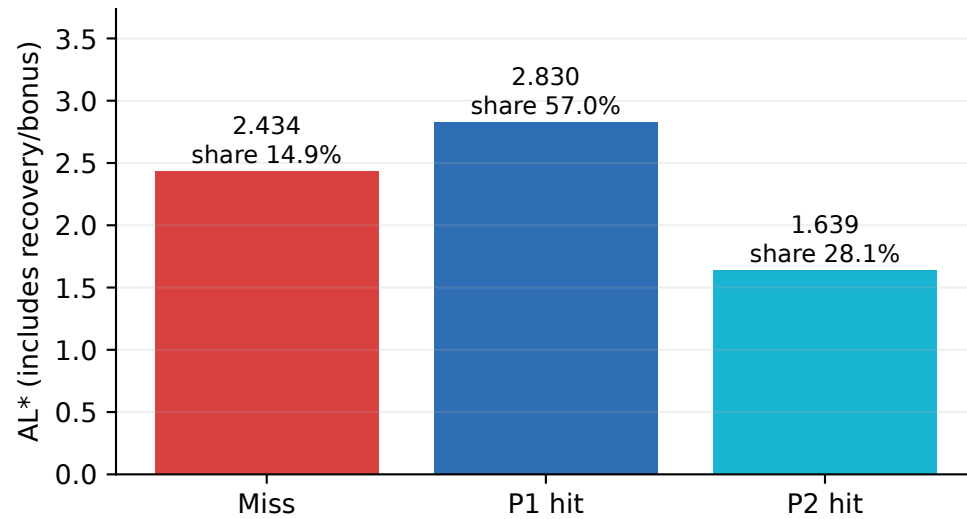
Measured wall time by batch cache status



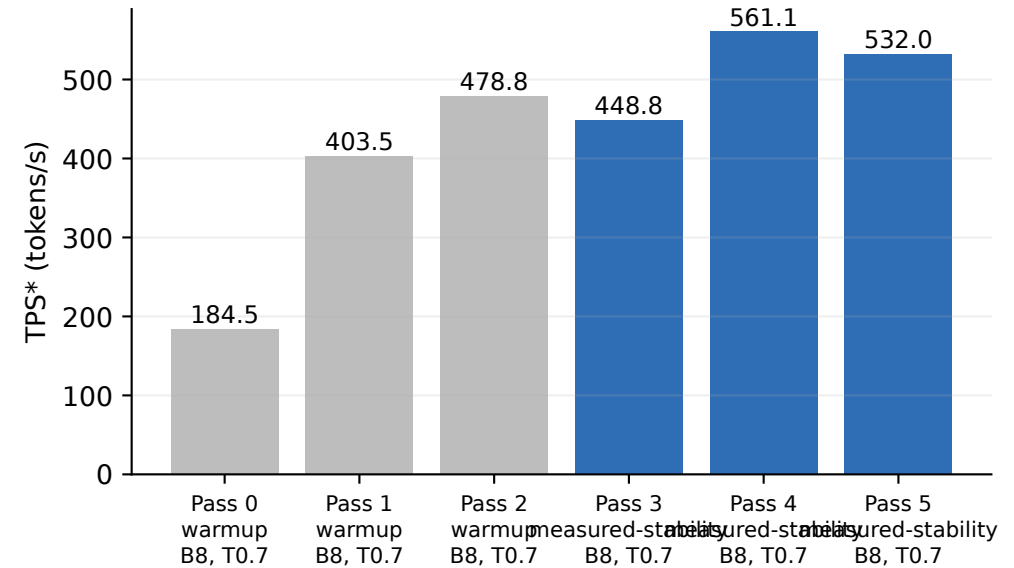
Boundary-free steps, including slow steps



Conditional AL* and source share



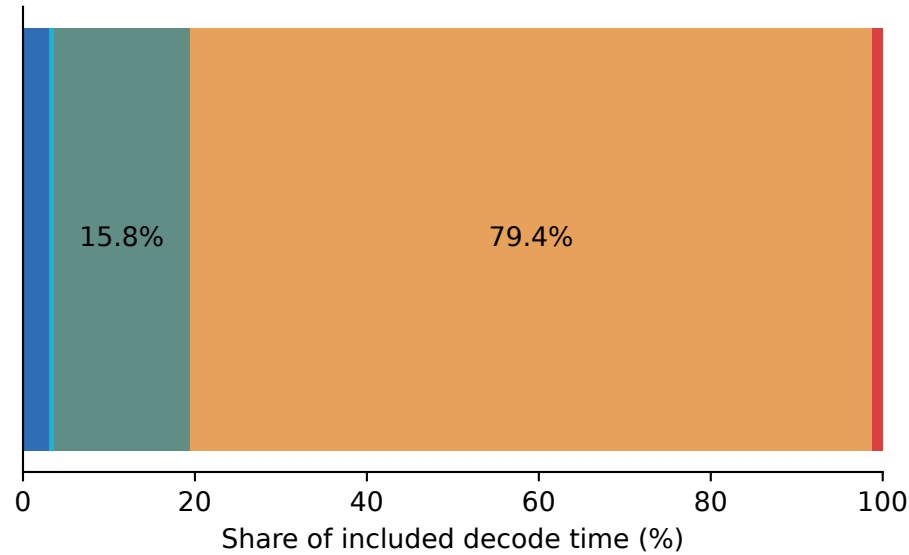
All saved passes; gray = warmup



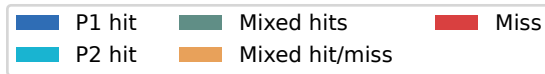
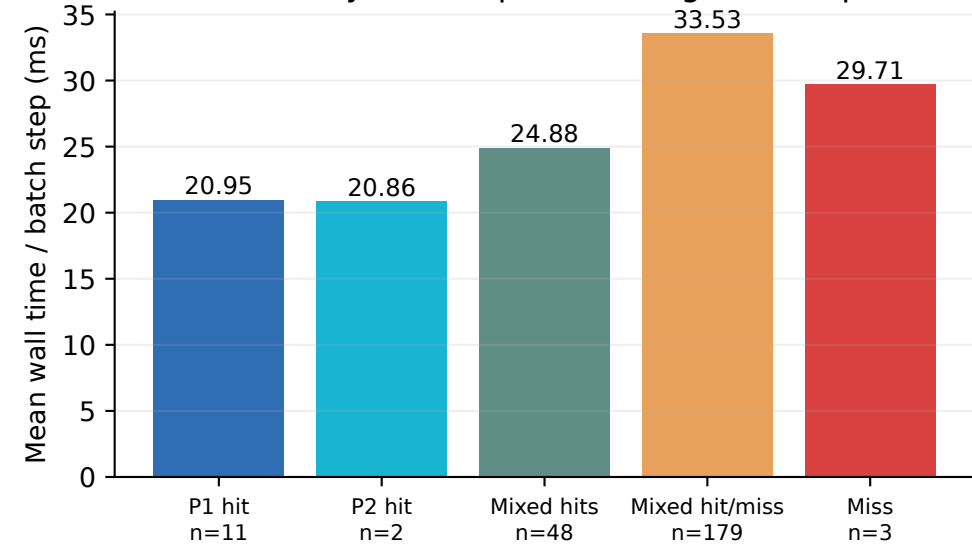
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_6
 DUET: K1/K2=3/1, exit=24 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=48 | AL*=2.352, hit=79.4%, TPS*=547.85

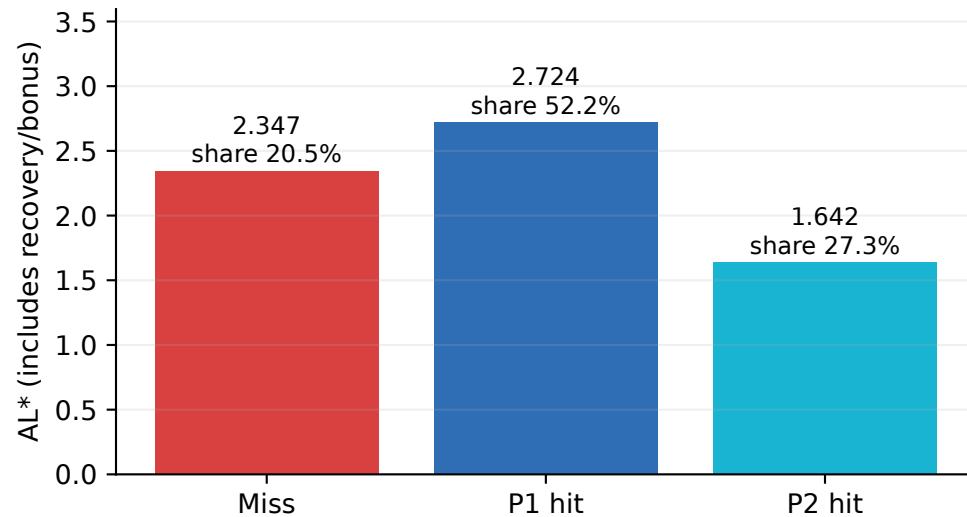
Measured wall time by batch cache status



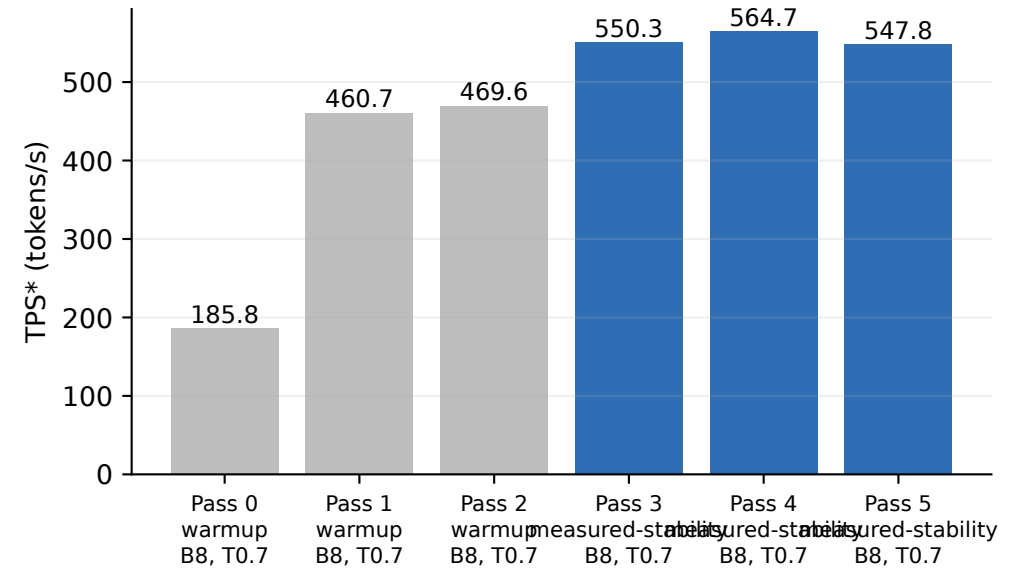
Boundary-free steps, including slow steps



Conditional AL* and source share



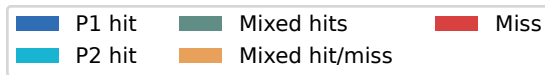
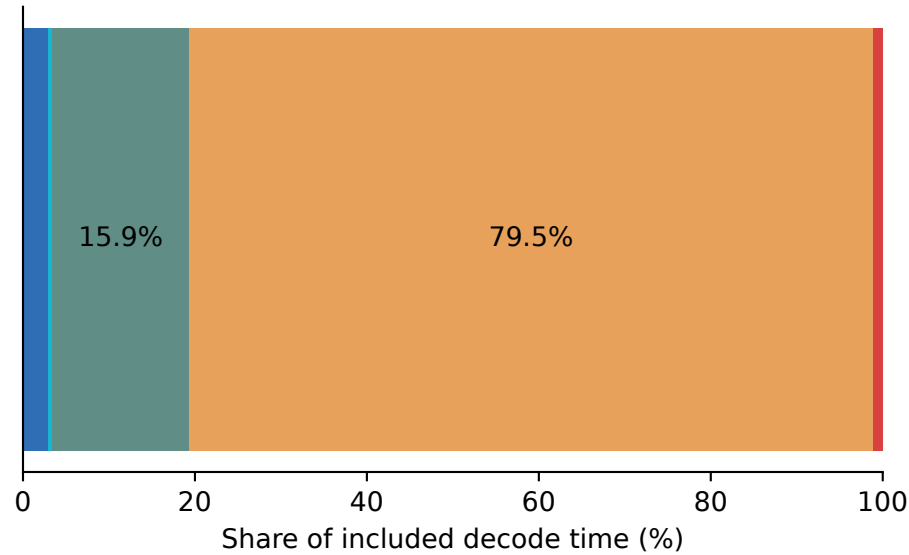
All saved passes; gray = warmup



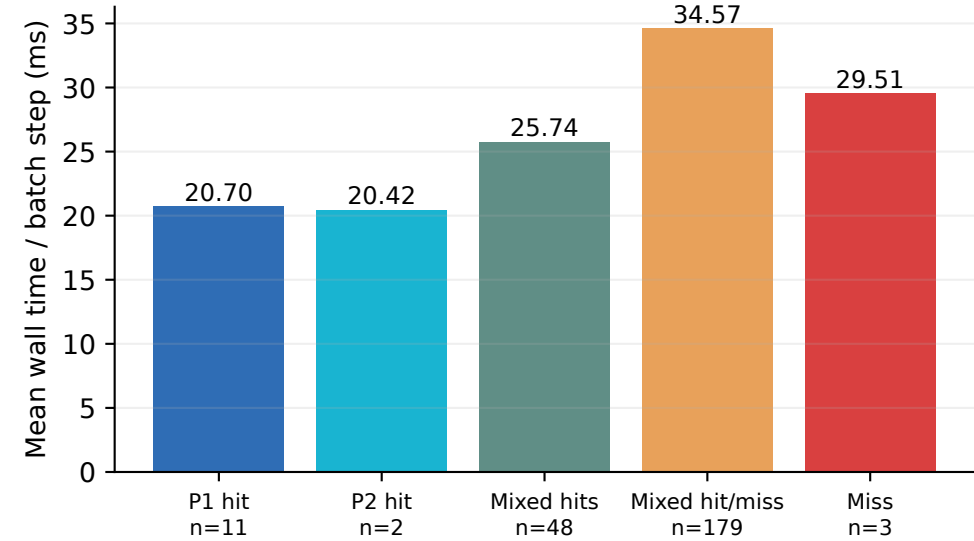
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_warm_7
 DUET: K1/K2=3/1, exit=24 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=48 | AL*=2.352, hit=79.4%, TPS*=532.08

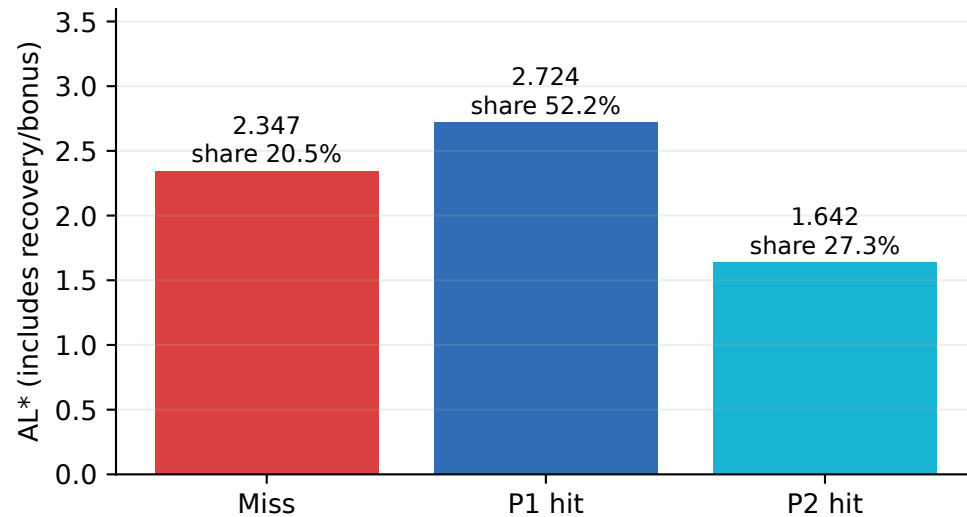
Measured wall time by batch cache status



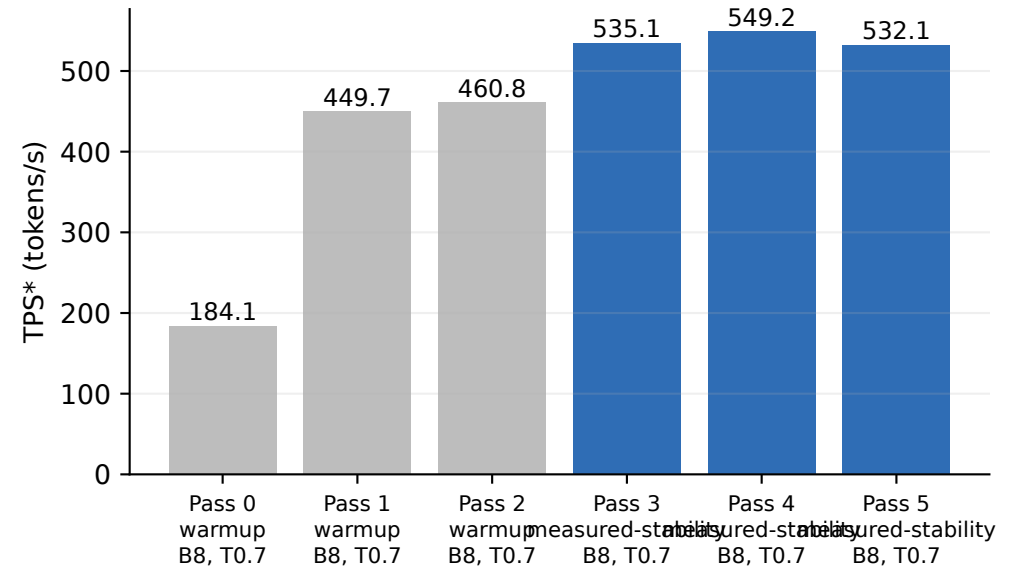
Boundary-free steps, including slow steps



Conditional AL* and source share



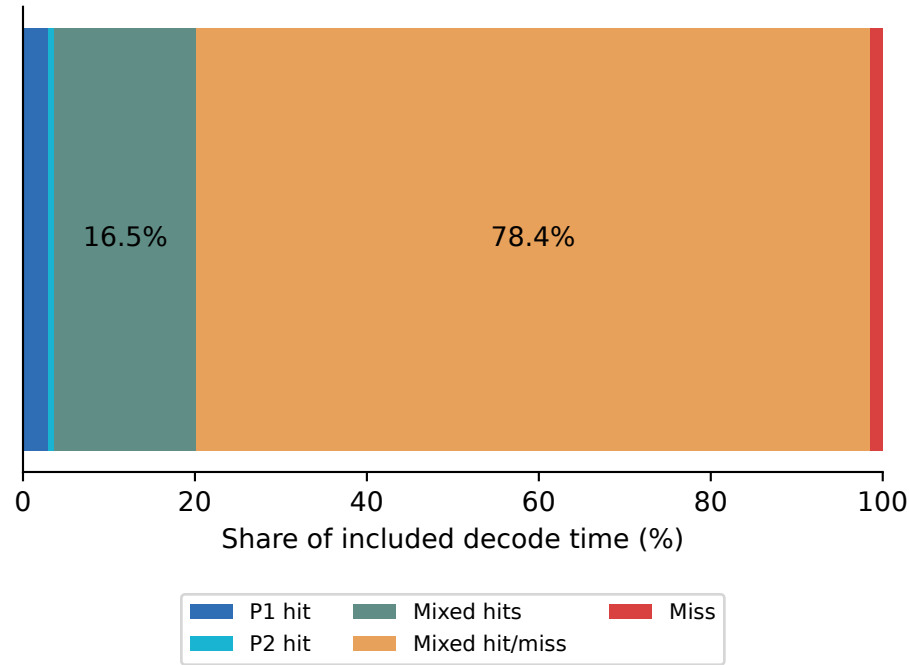
All saved passes; gray = warmup



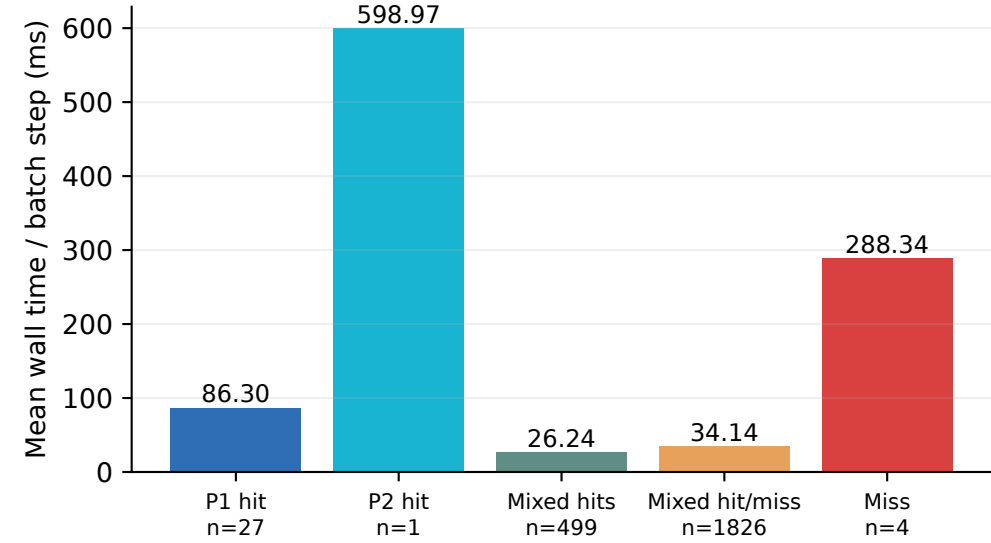
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stability_final_r0
 DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/0
 Last pass: B=8, T=0.7, N=480 | AL*=2.290, hit=82.4%, TPS*=537.68

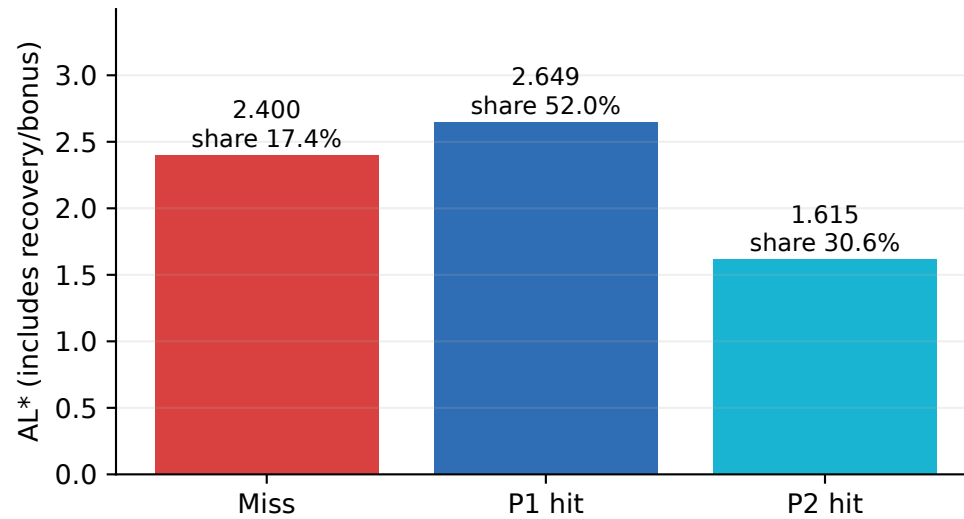
Measured wall time by batch cache status



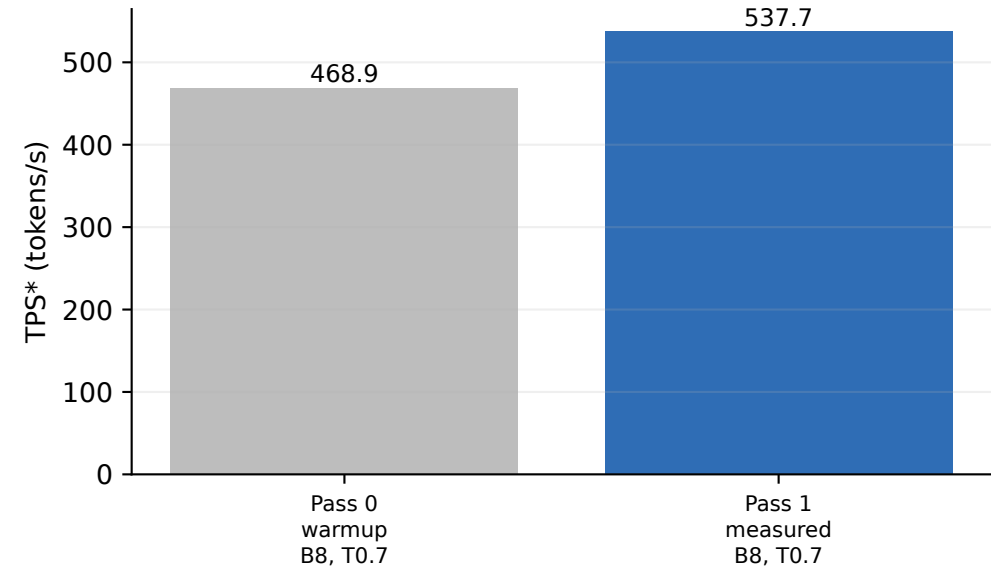
Boundary-free steps, including slow steps



Conditional AL* and source share

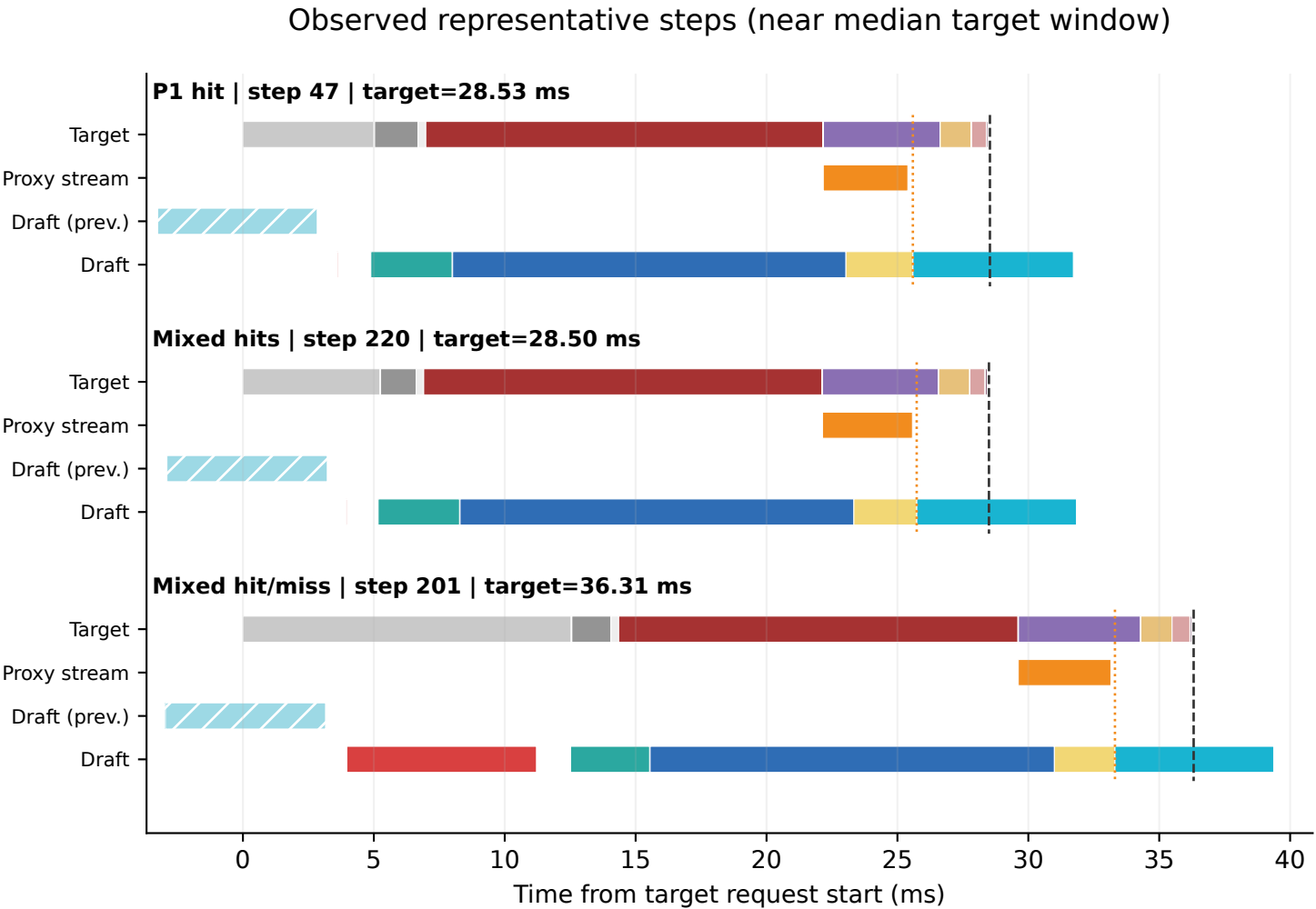
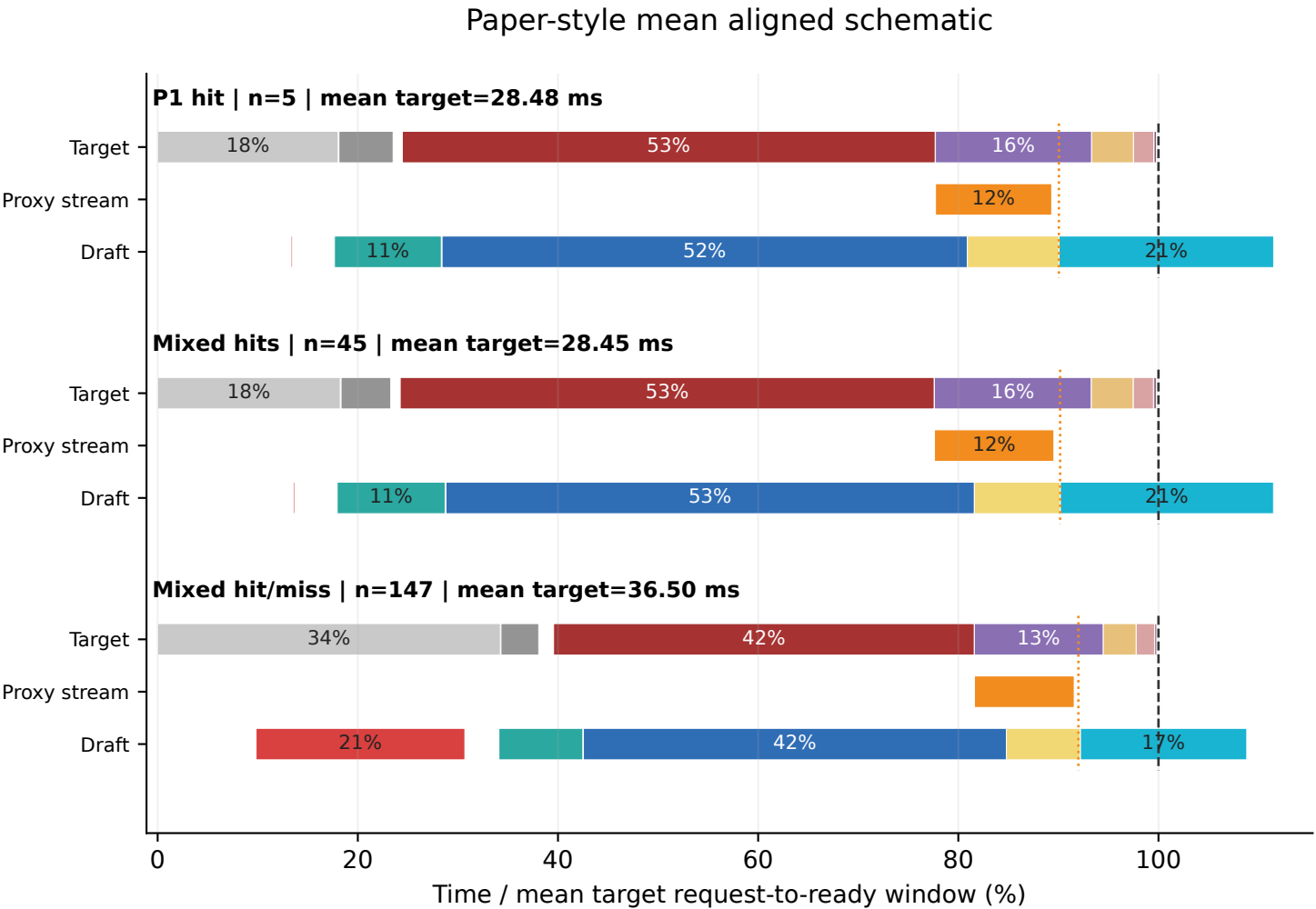
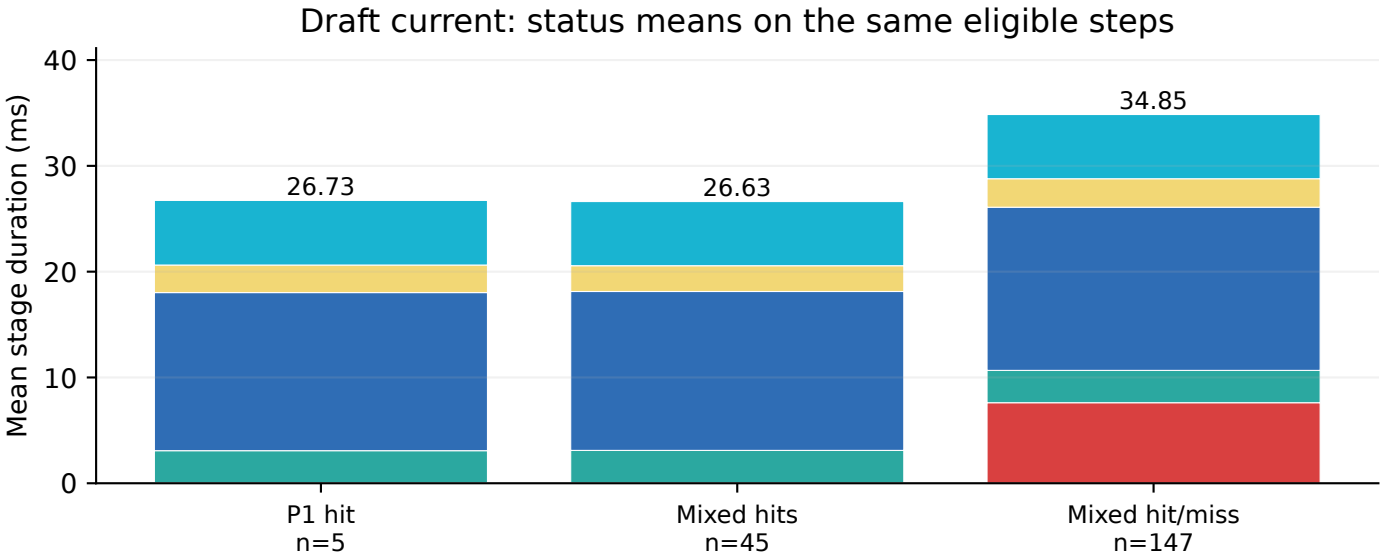
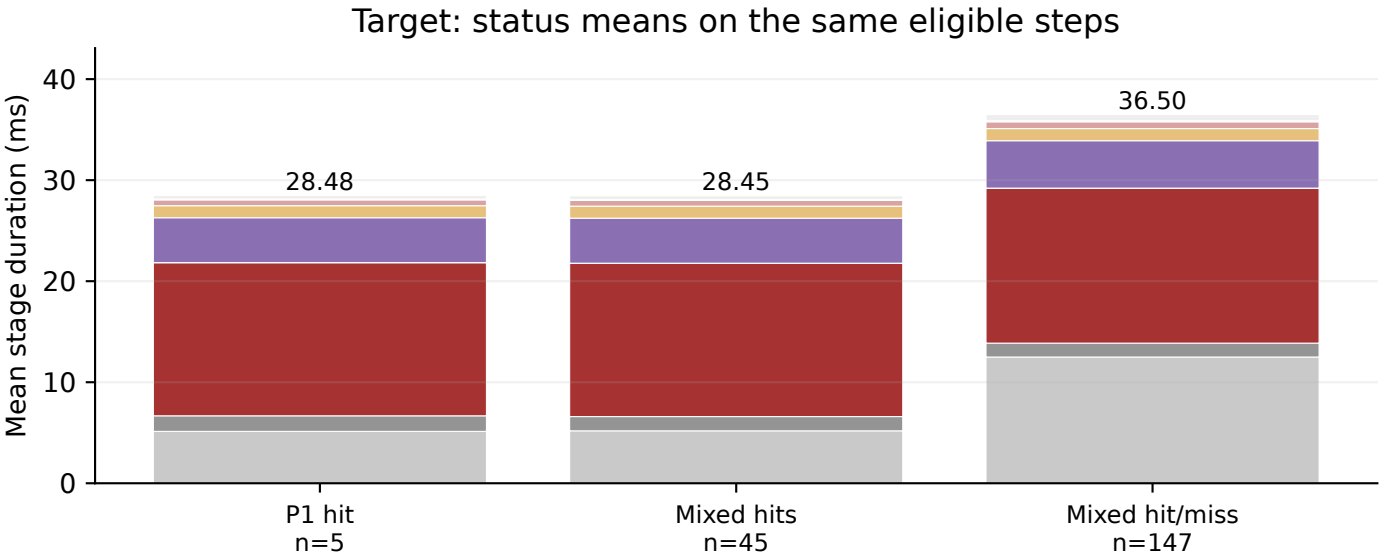


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

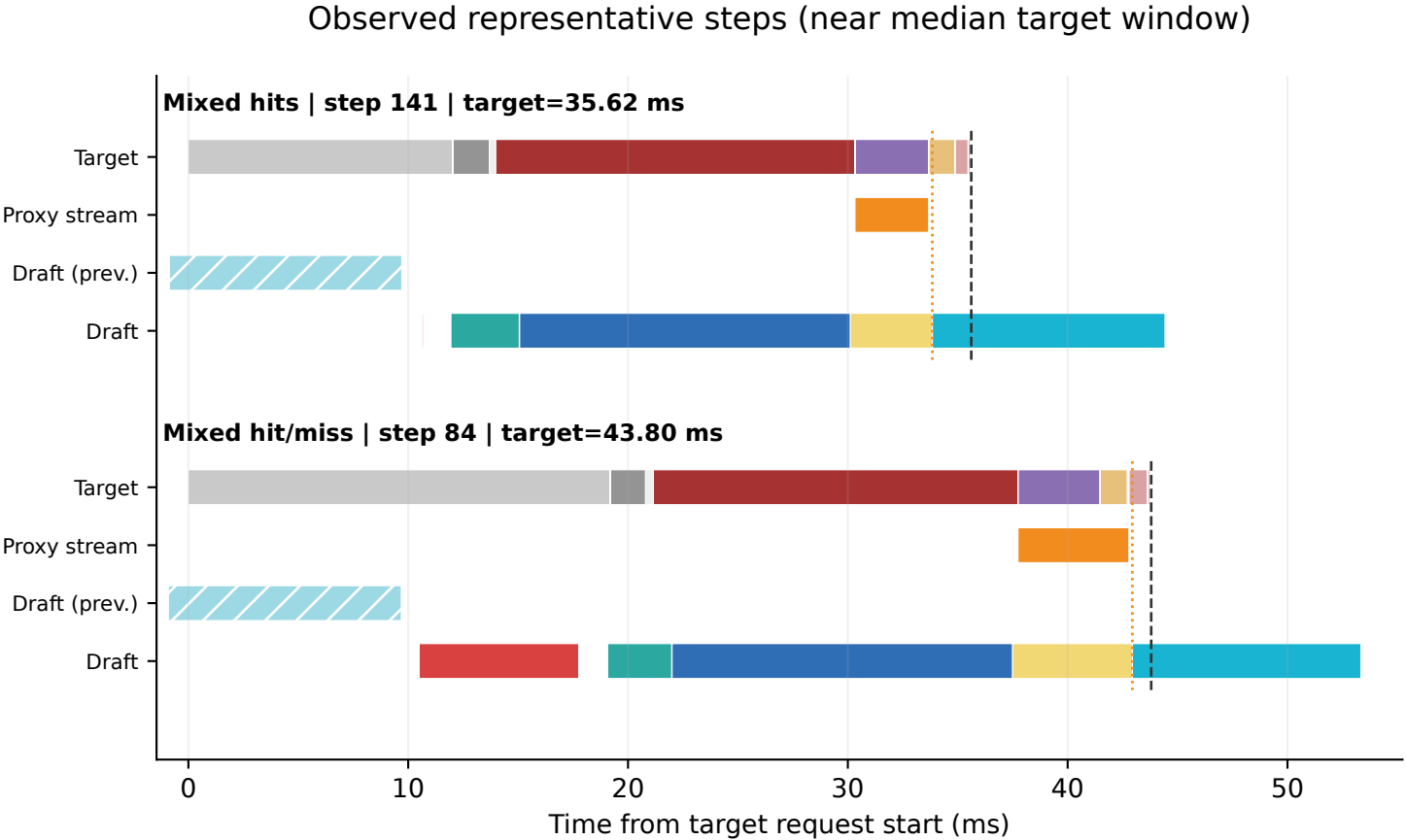
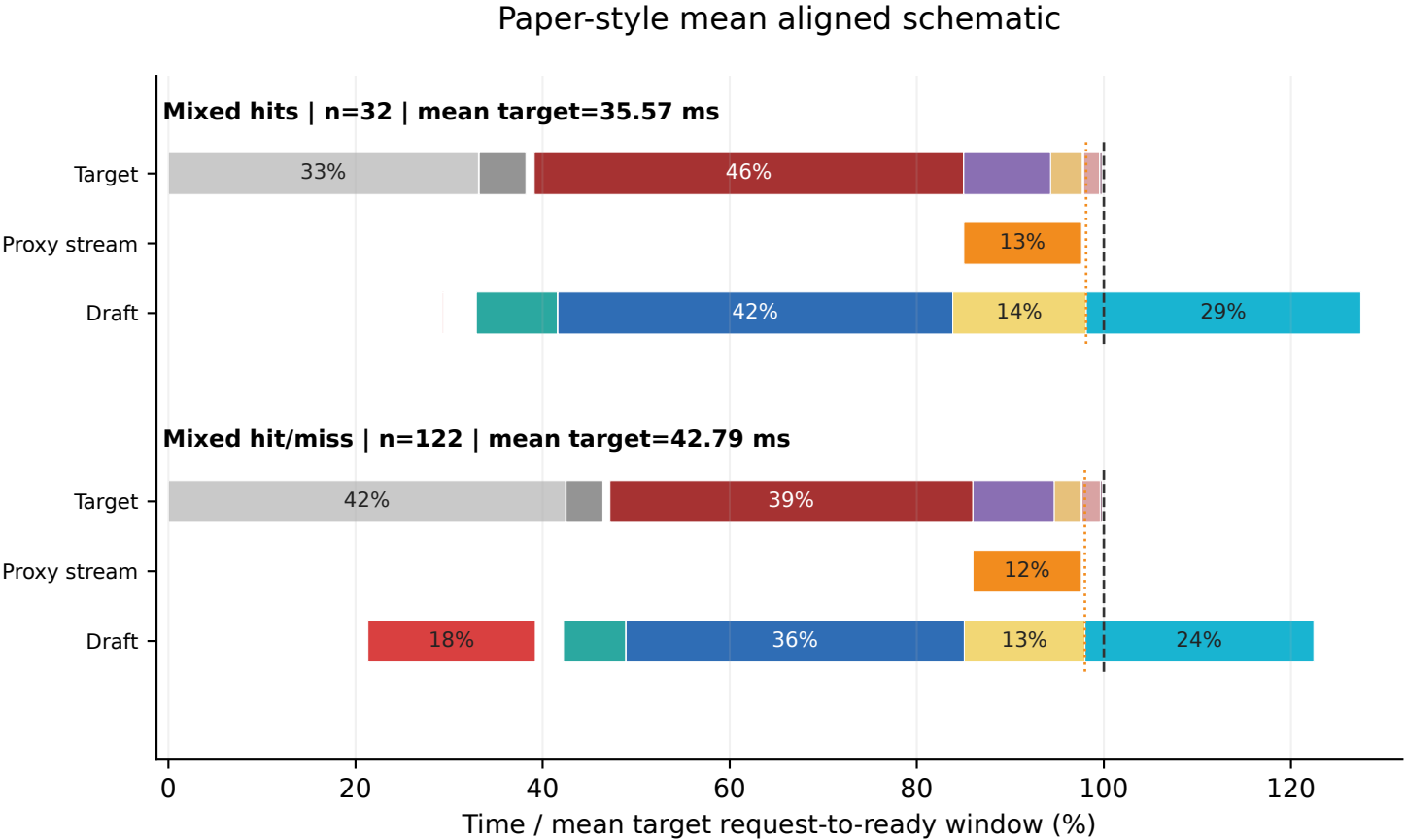
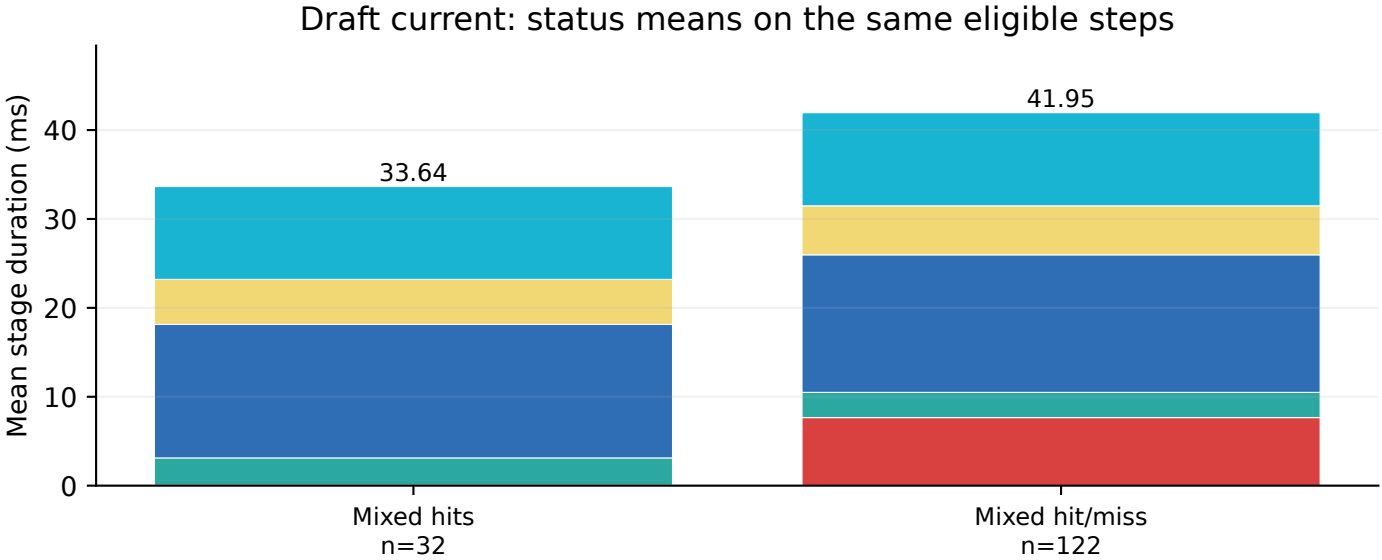
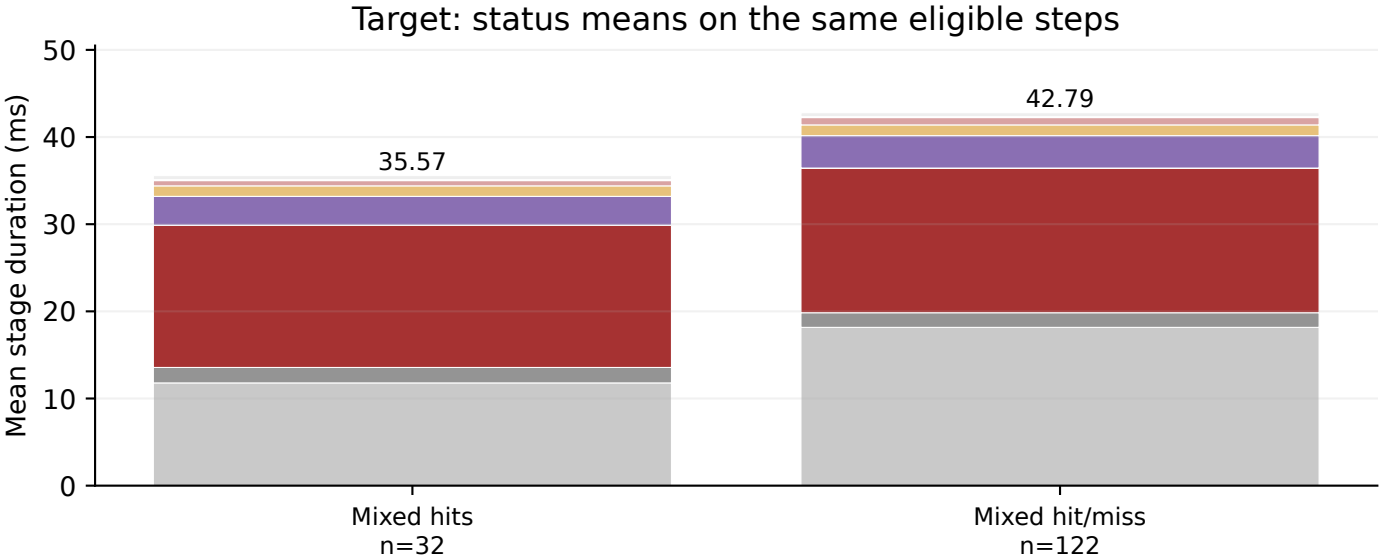
llama3_b8_stability_profile_0
DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 197 eligible steps



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

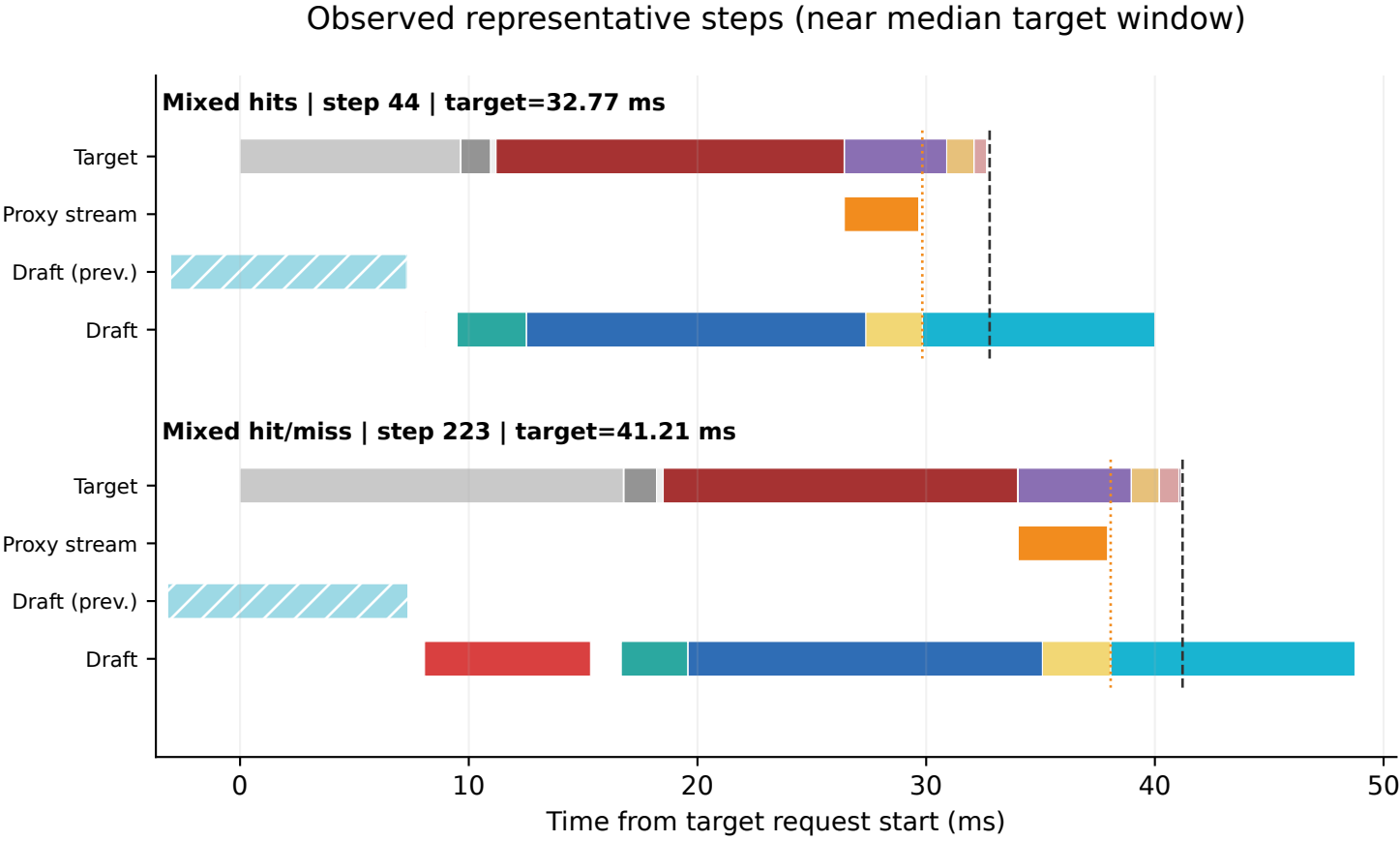
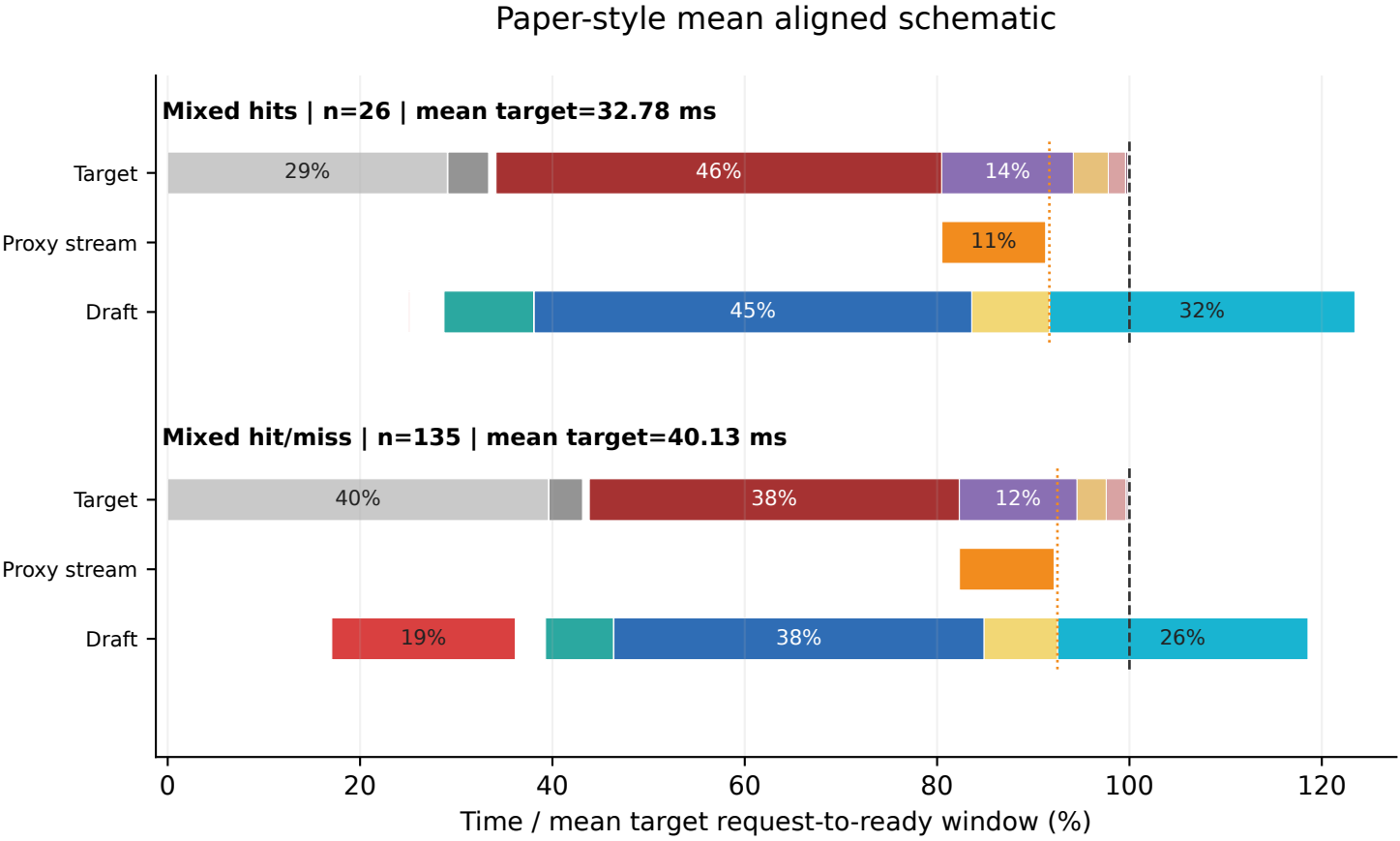
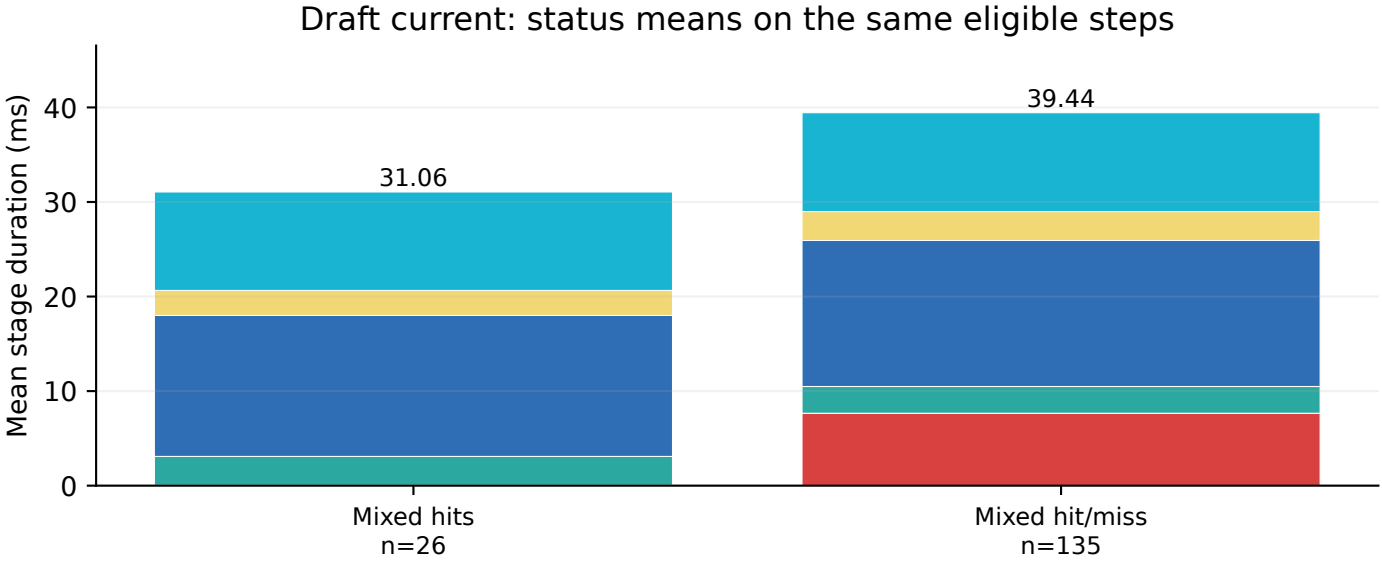
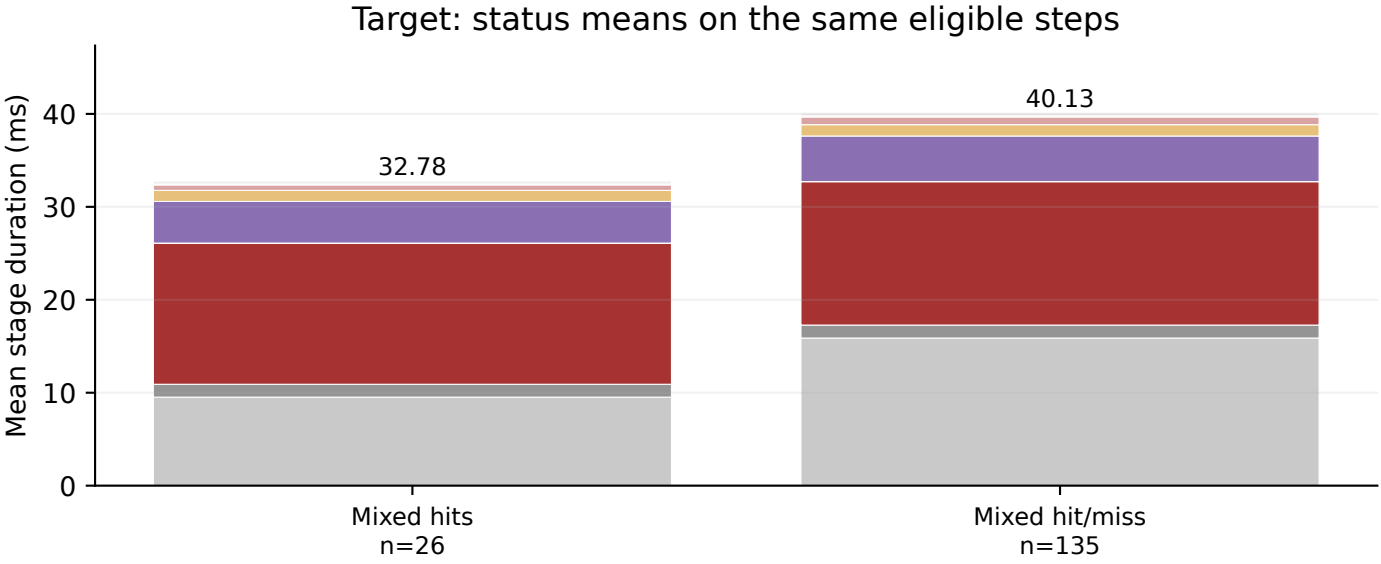
llama3_b8_stability_profile_1
DUET: K1/K2=3/2, exit=28 (0-based), $\bar{C}$ =1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 154 eligible steps



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

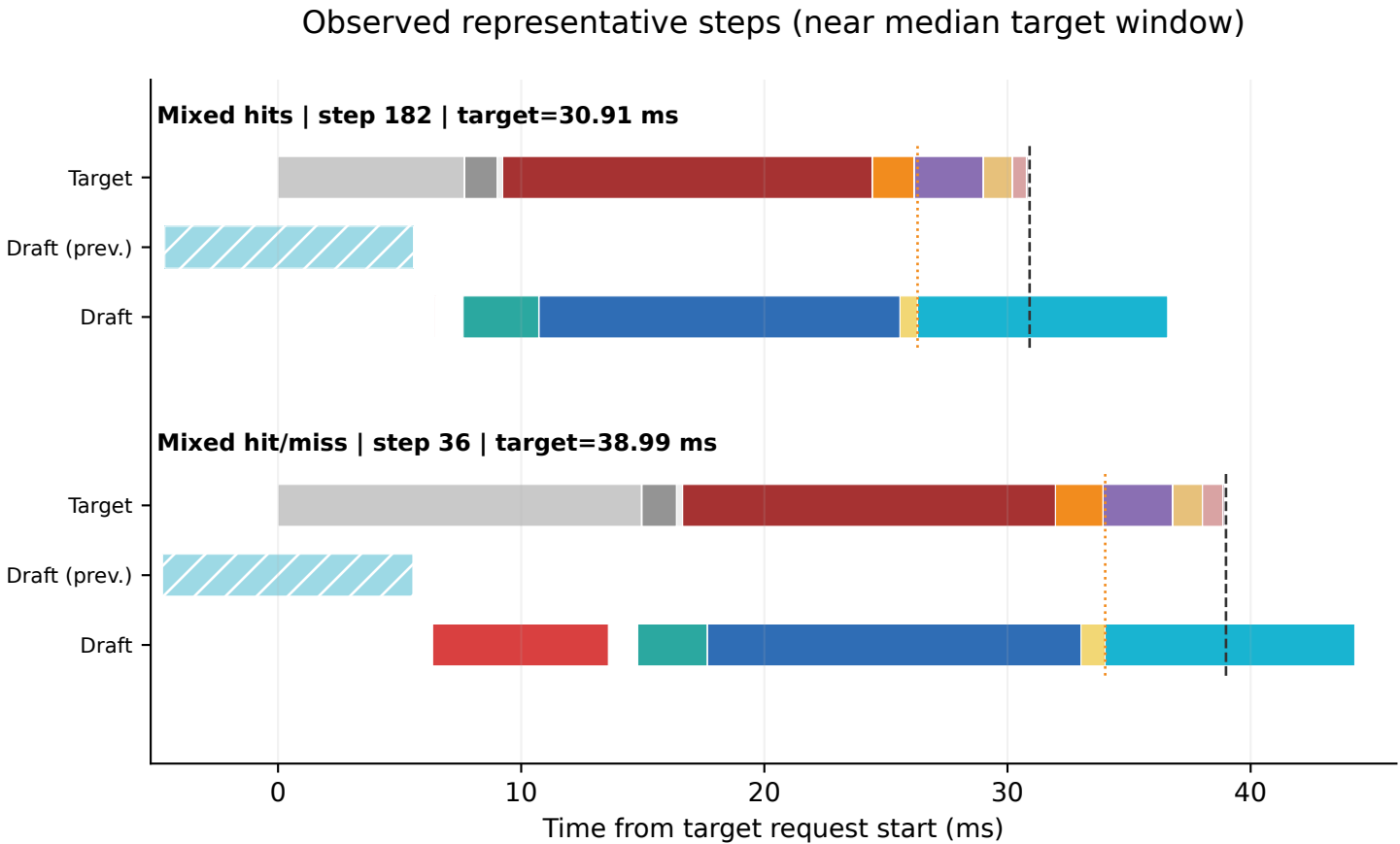
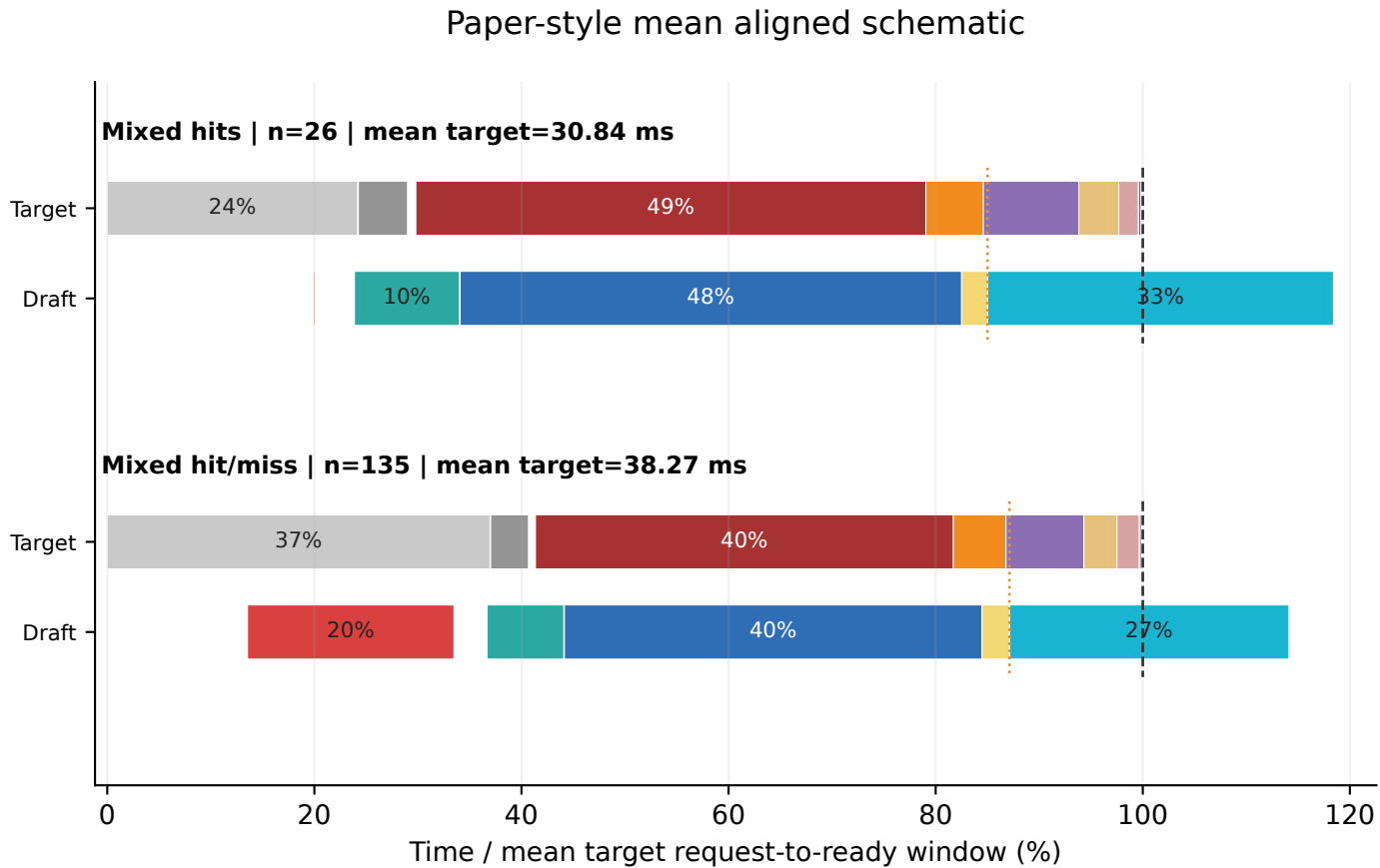
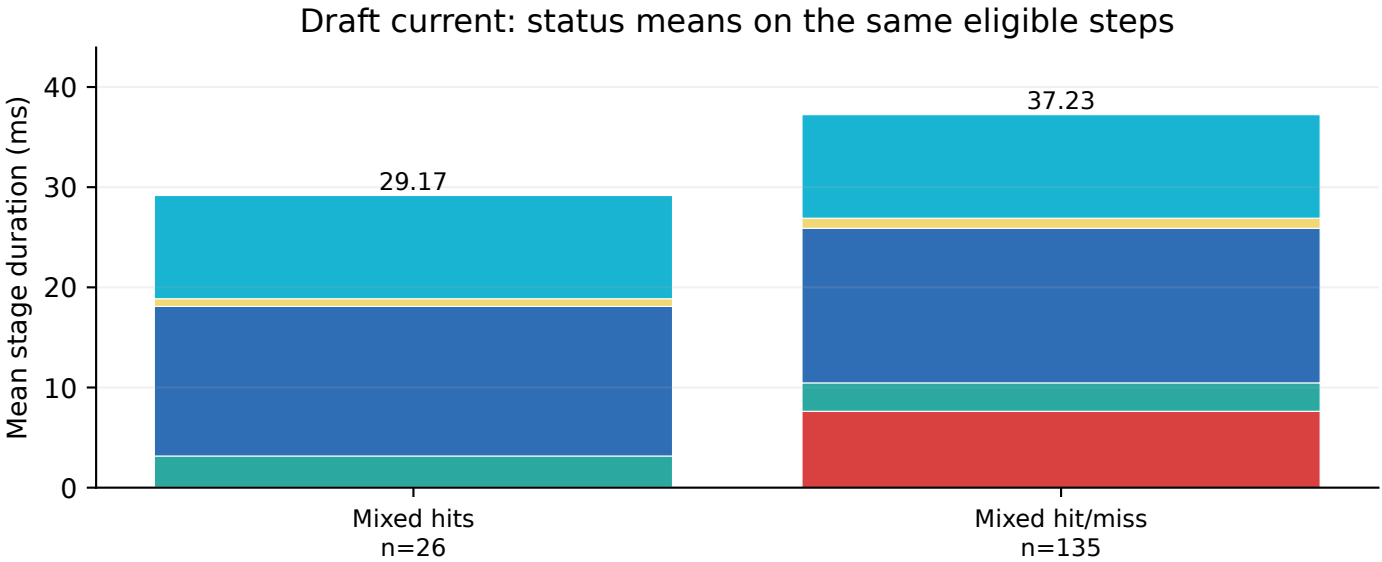
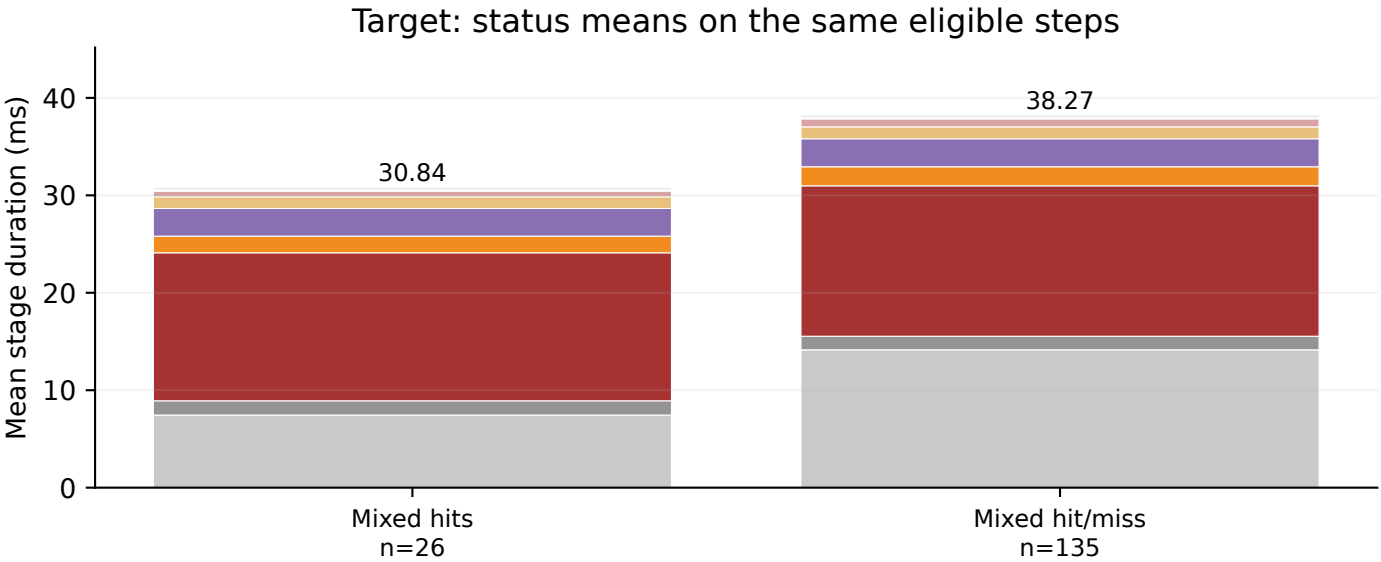
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stability_profile_2
DUET: K1/K2=3/2, exit=26 (0-based), $\bar{C}$ =1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 161 eligible steps



Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

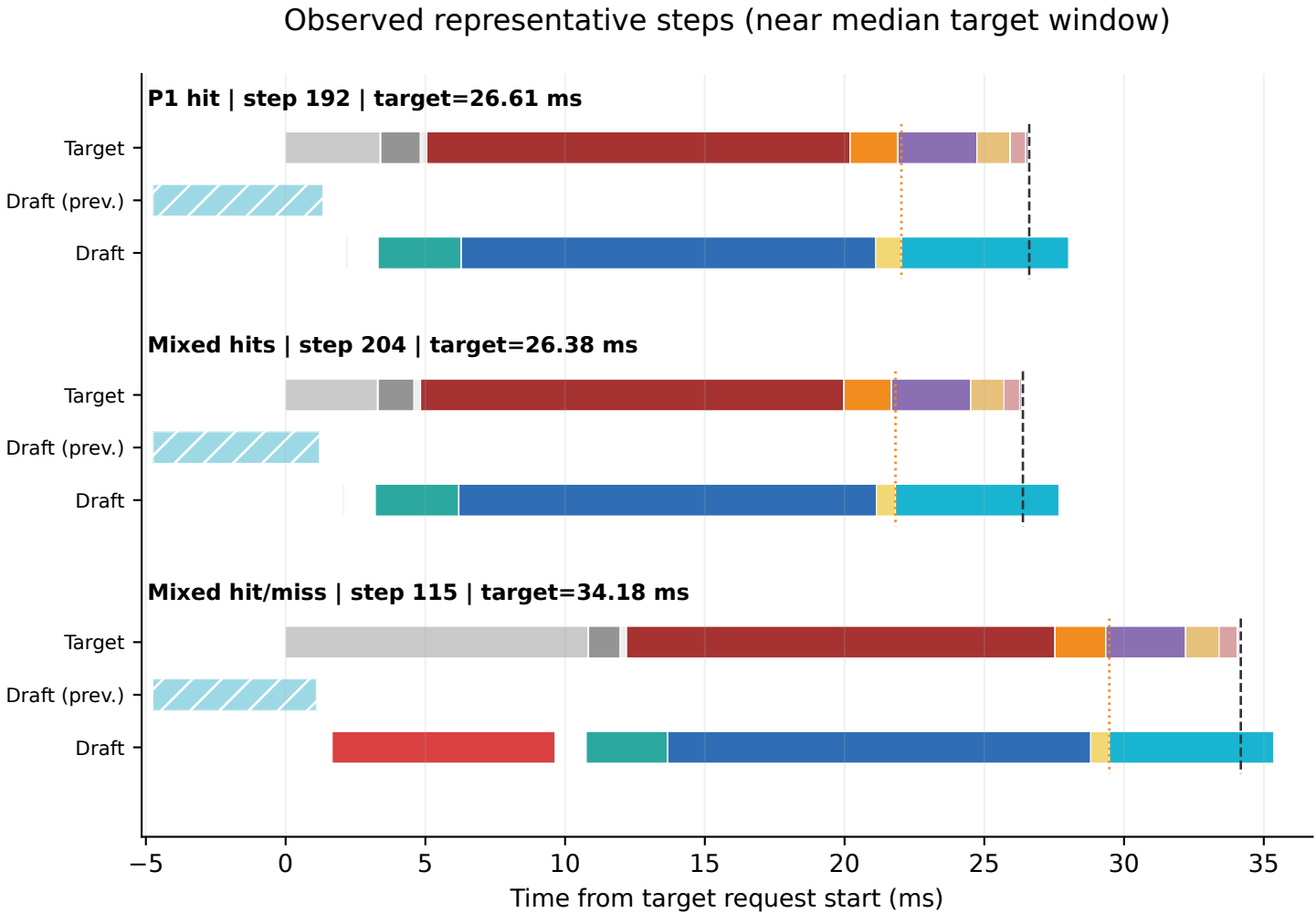
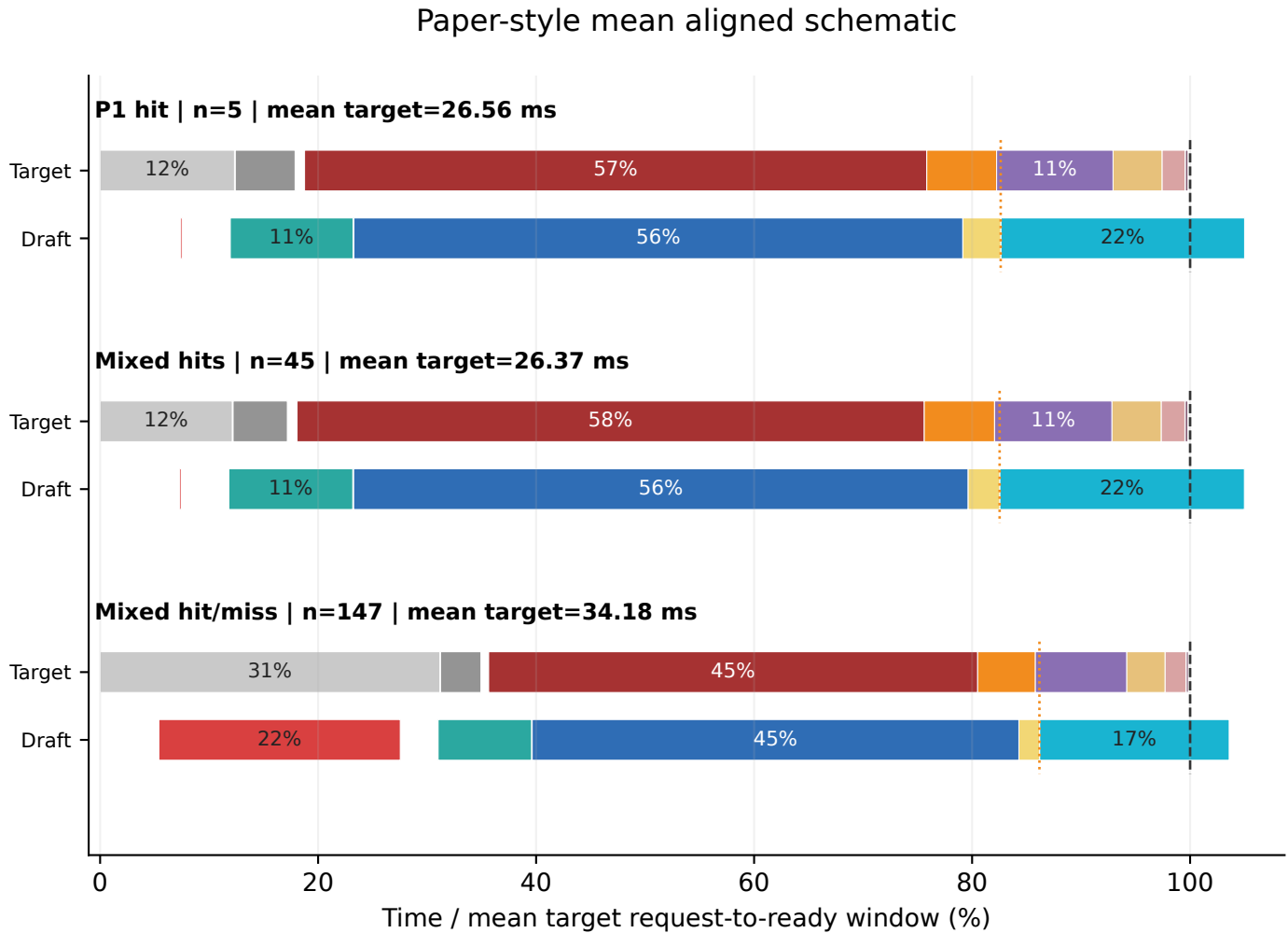
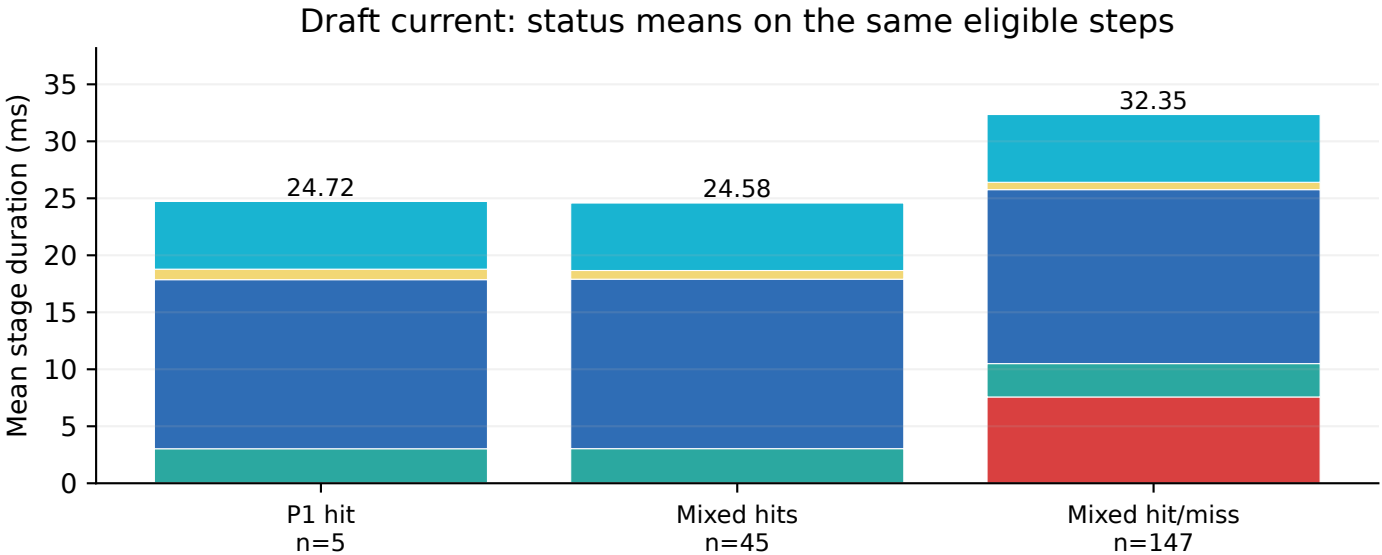
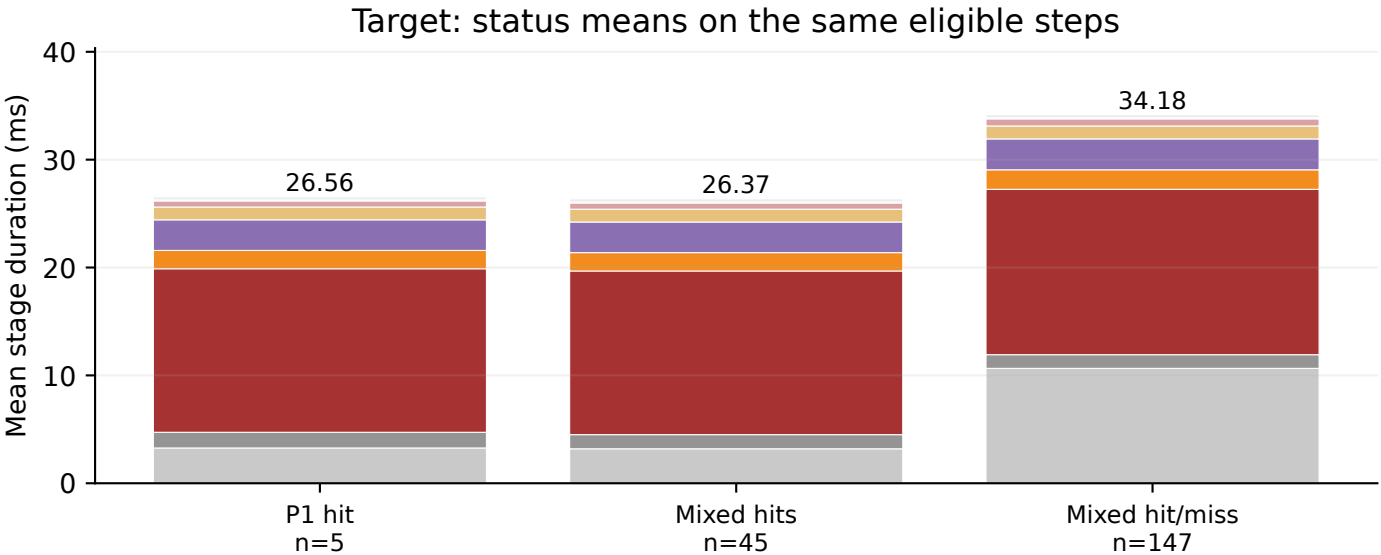
llama3_b8_stability_profile_3
DUET: K1/K2=3/2, exit=26 (0-based), $\bar{C}$ =1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 161 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stability_profile_4
DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 197 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

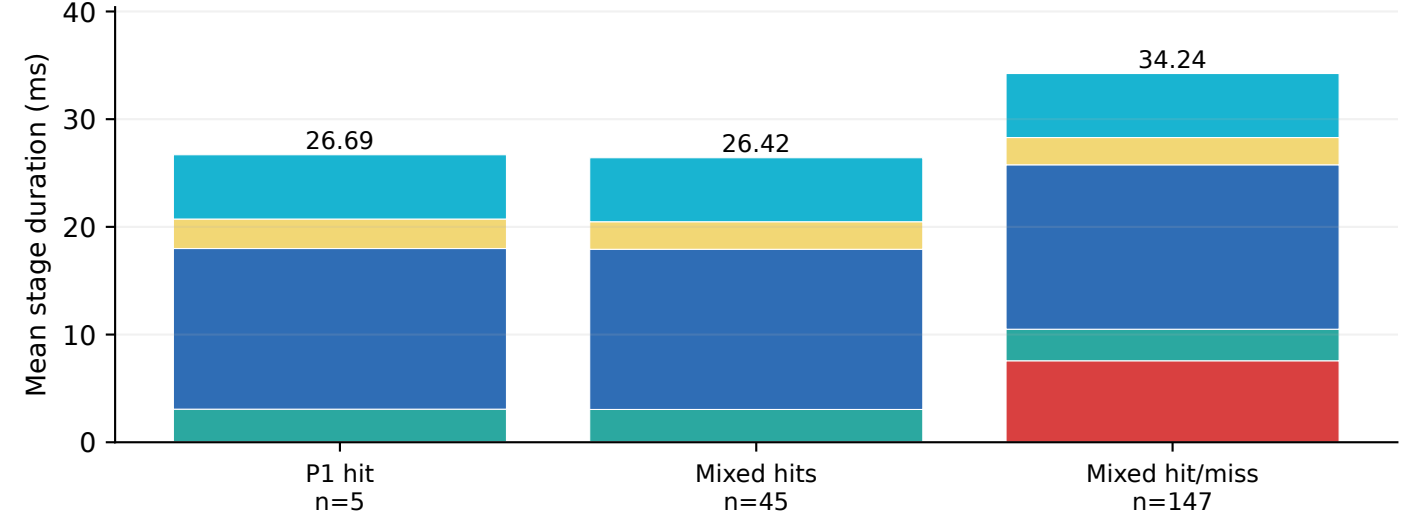
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stability_profile_5
DUET: K1/K2=3/1, exit=26 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 197 eligible steps

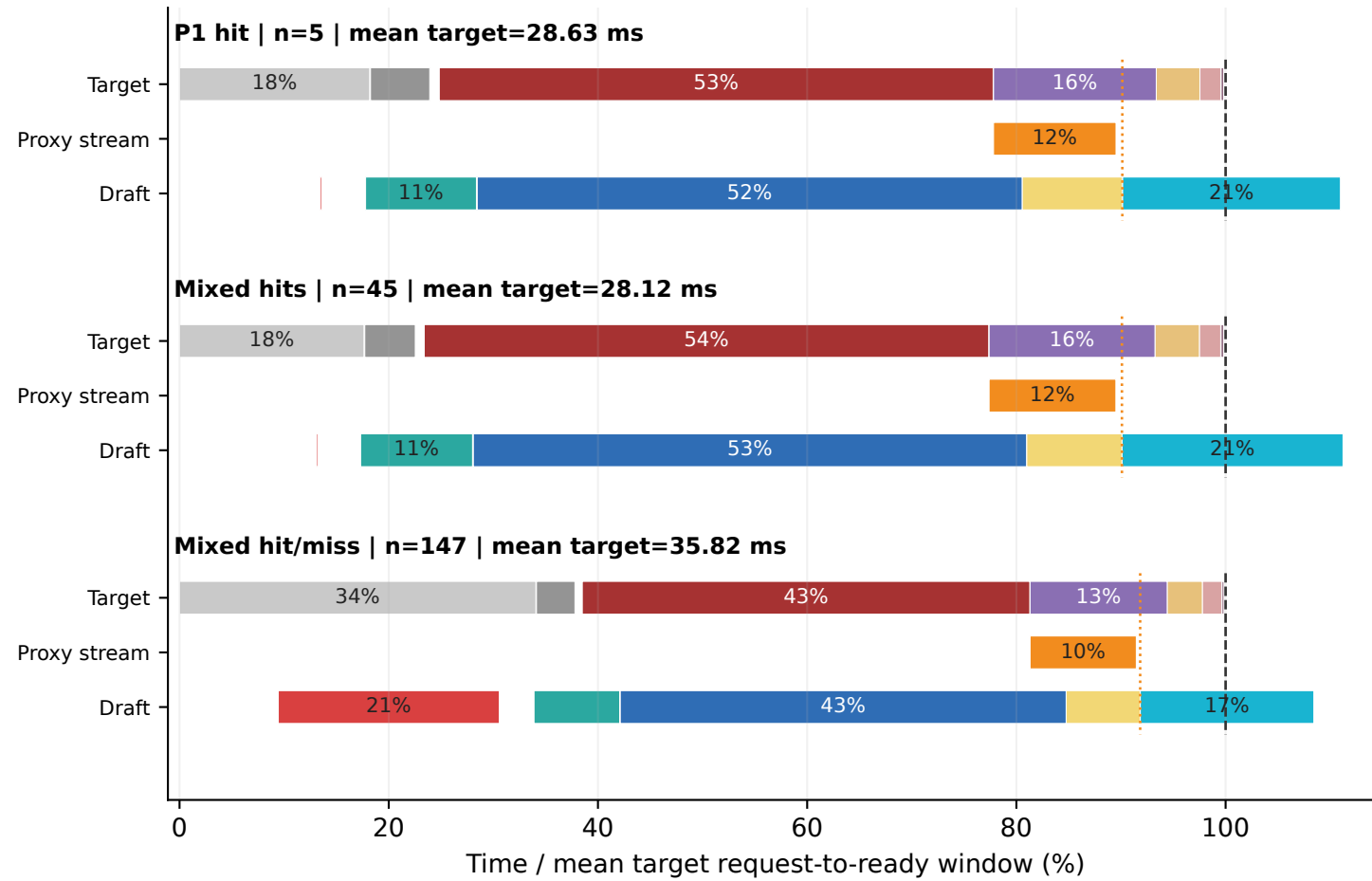
Target: status means on the same eligible steps



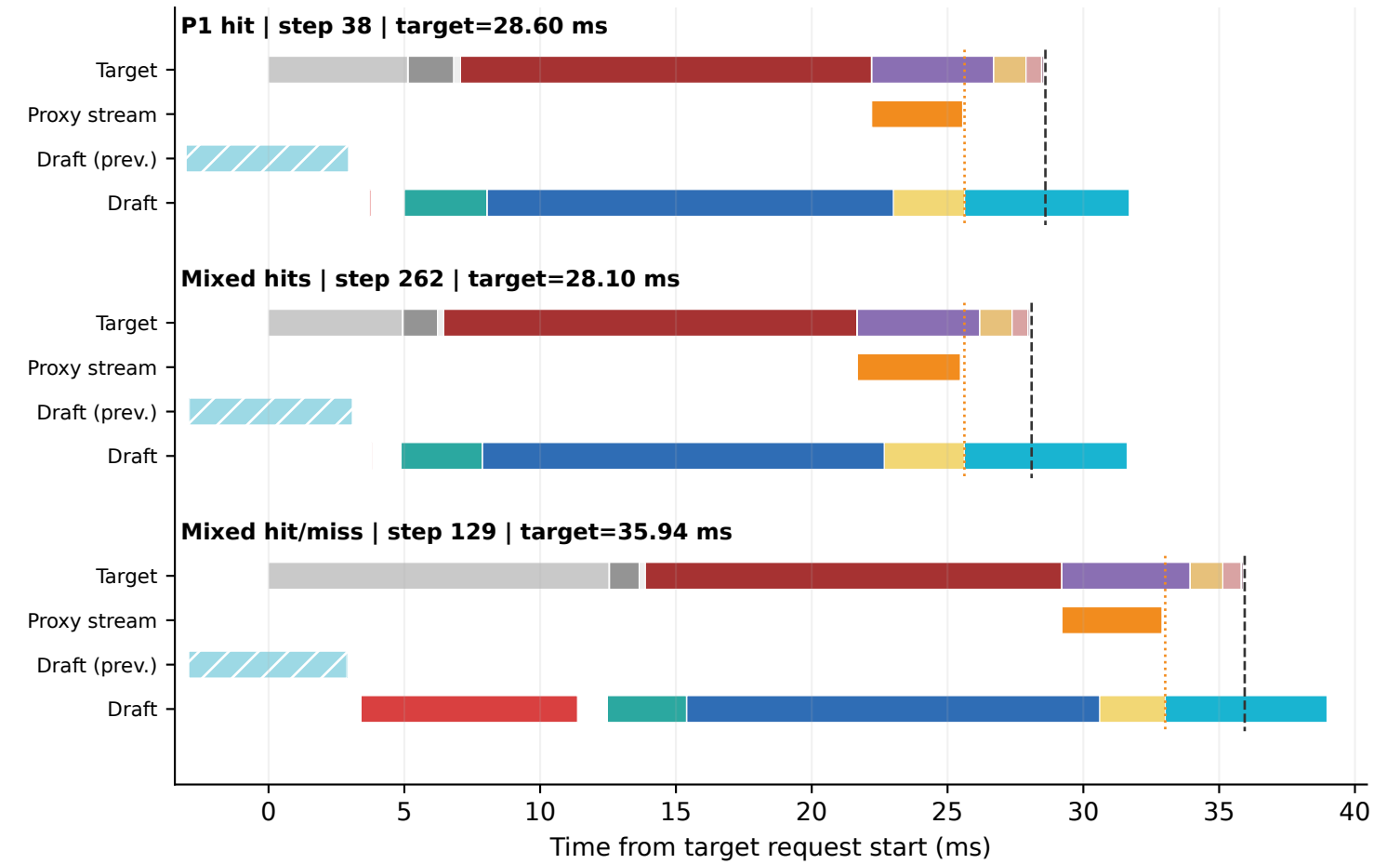
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



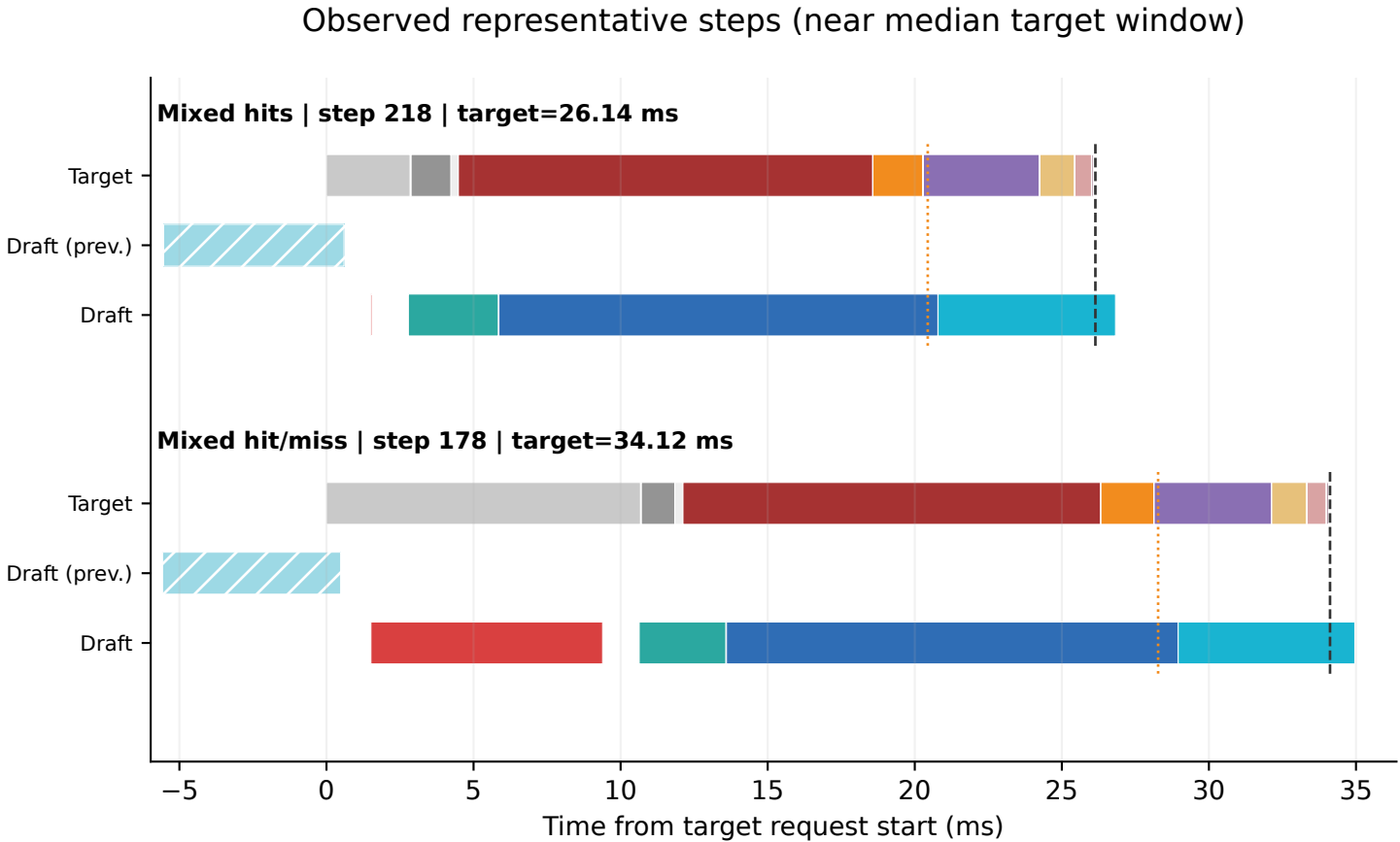
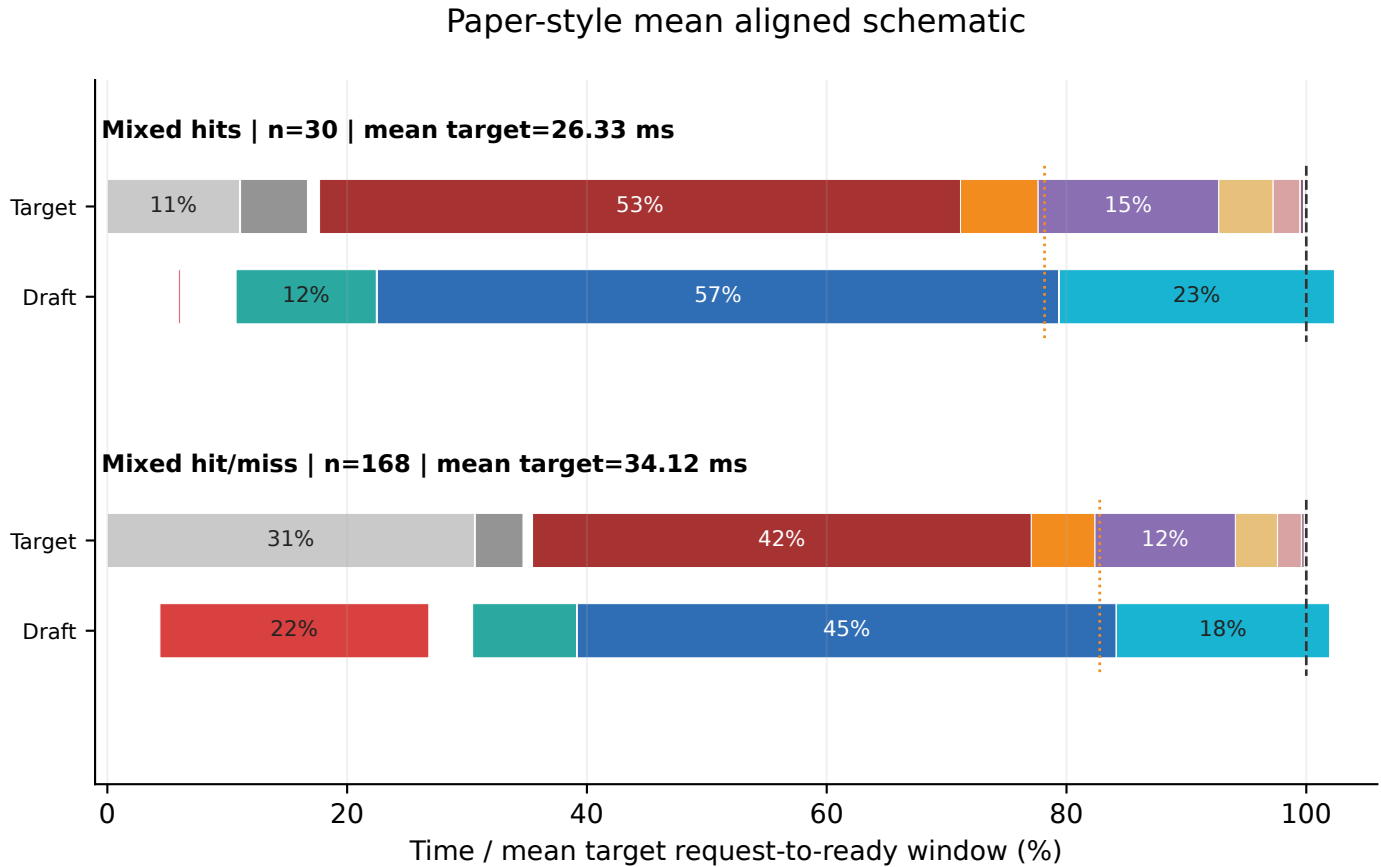
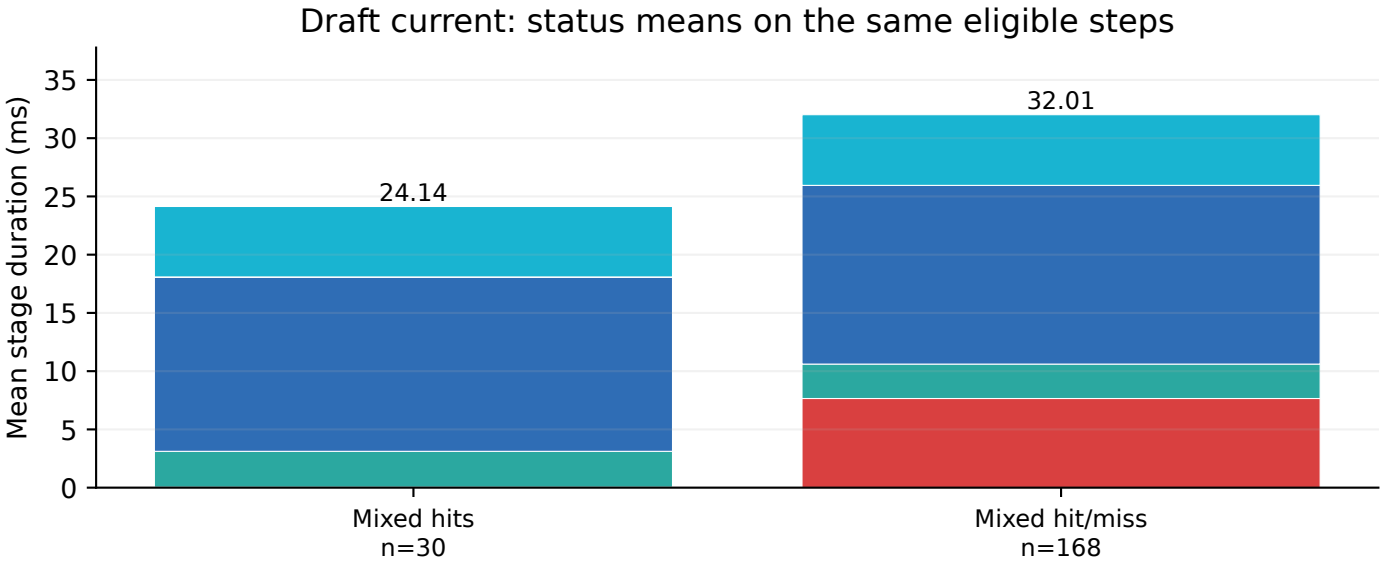
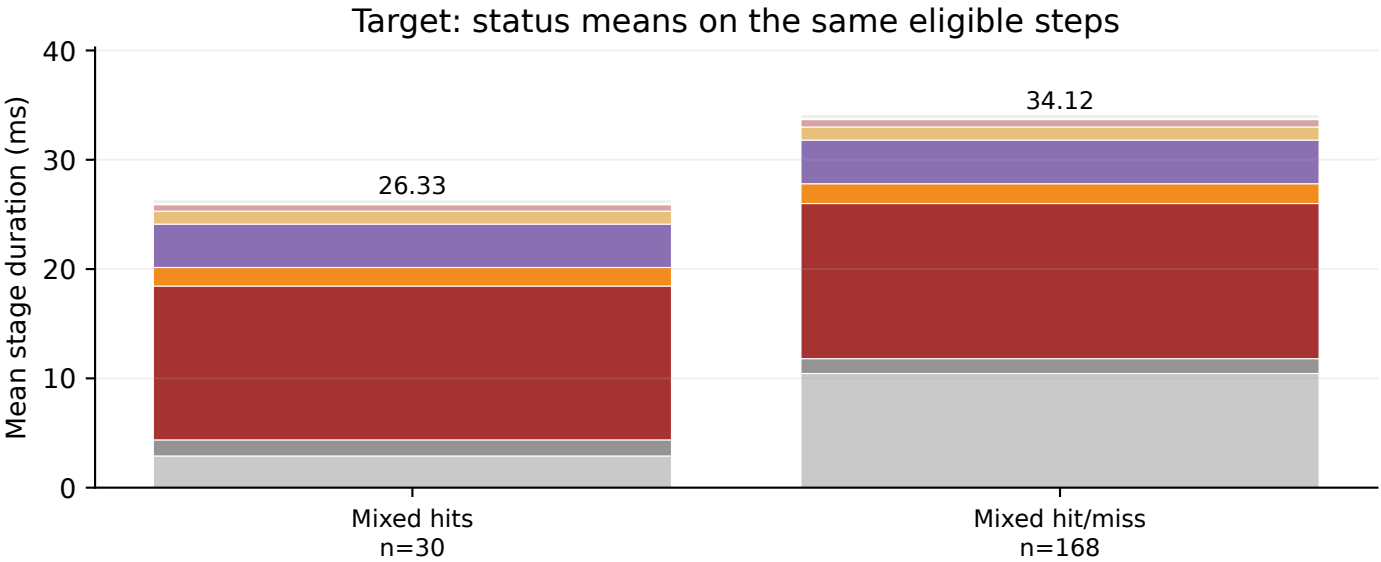
Observed representative steps (near median target window)



Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

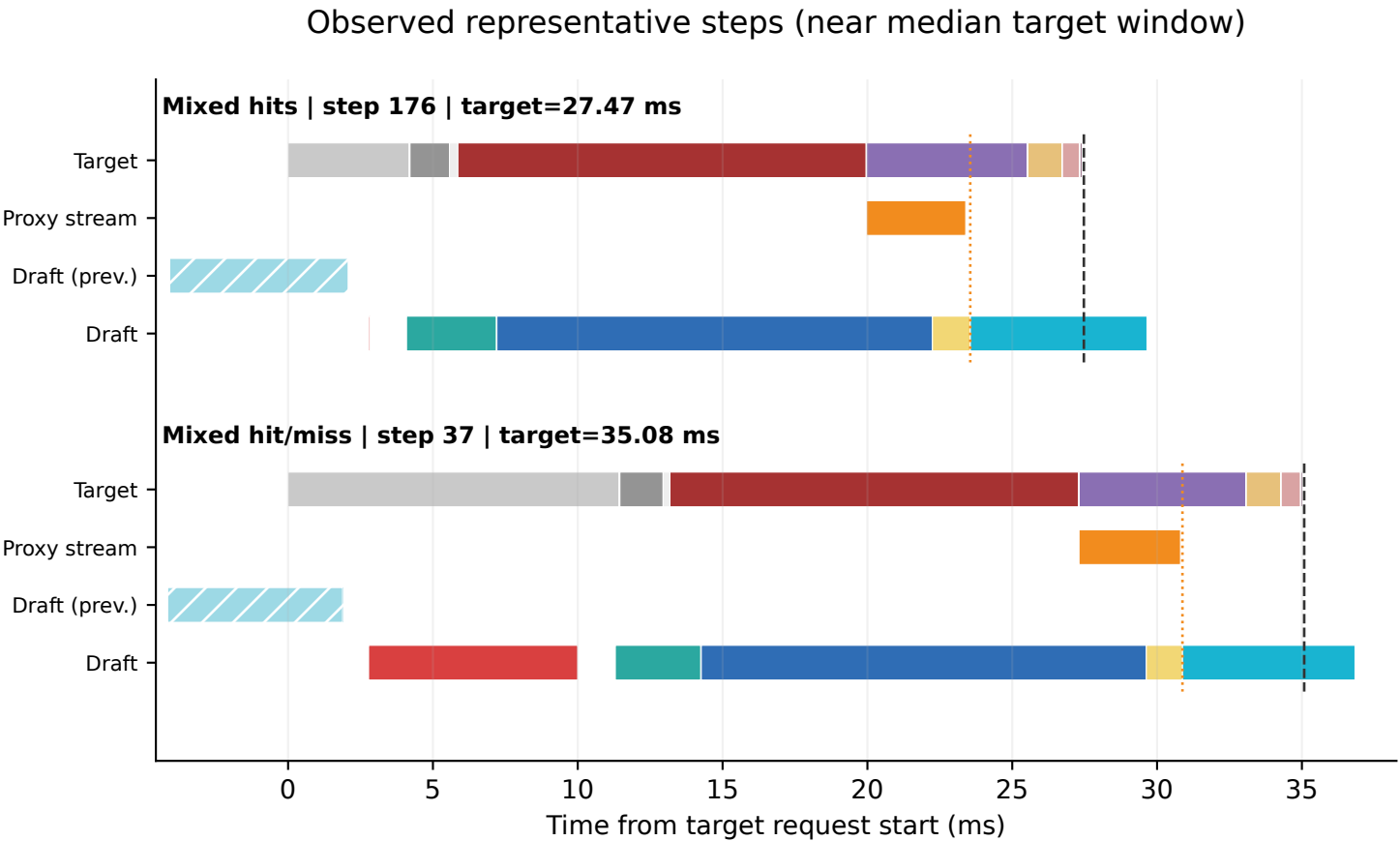
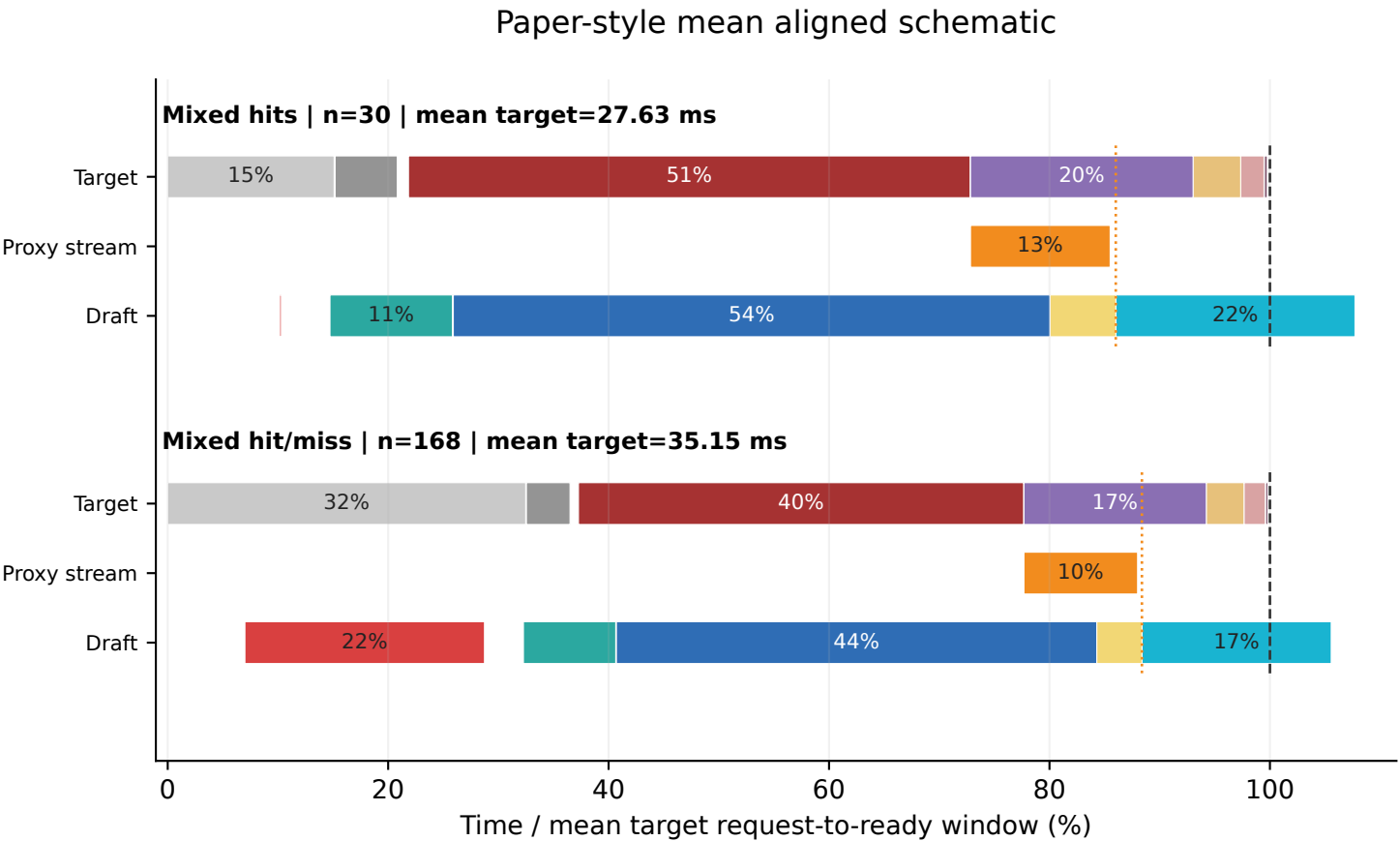
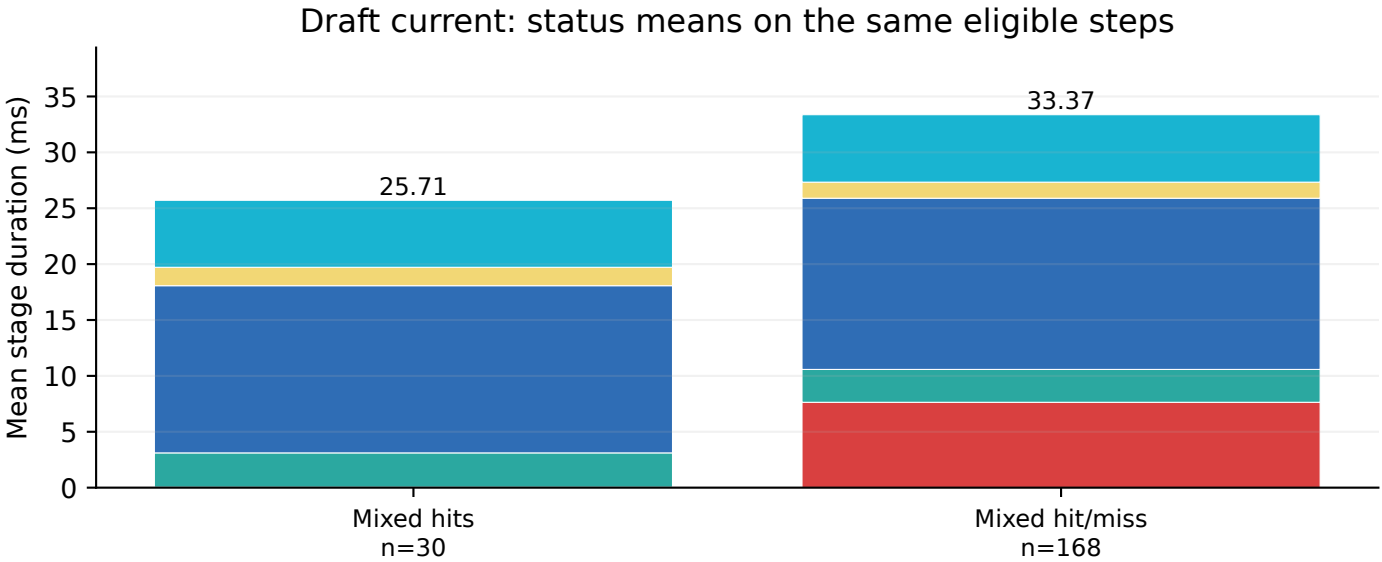
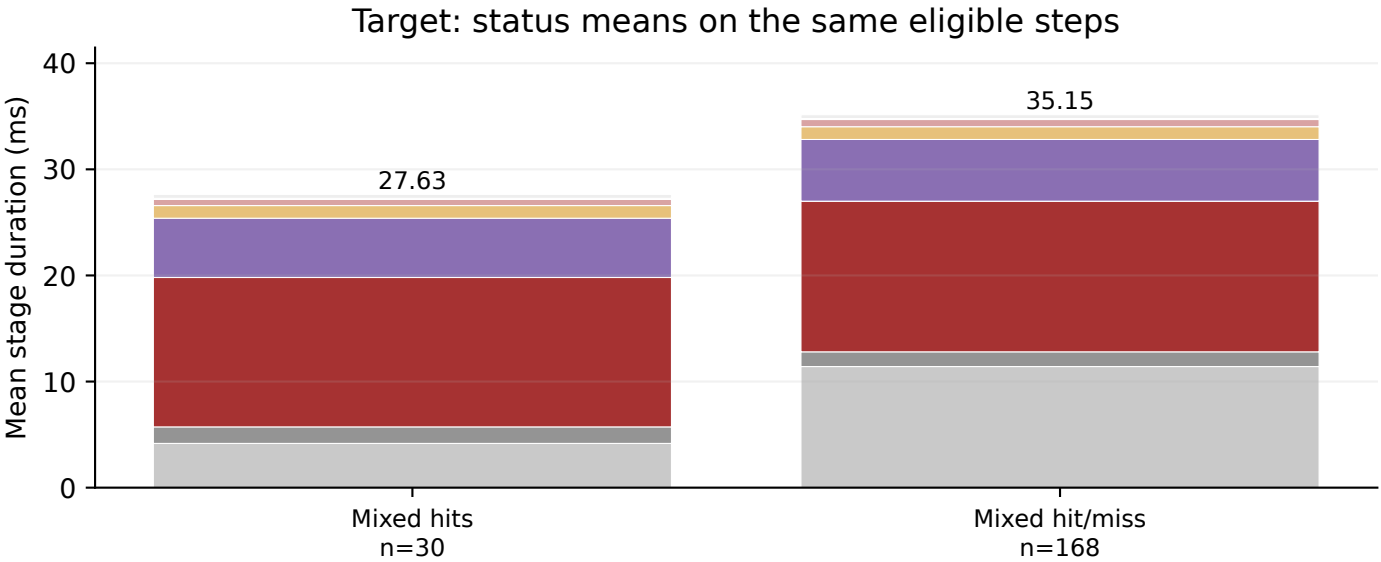
llama3_b8_stability_profile_6
DUET: K1/K2=3/1, exit=24 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 198 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stability_profile_7
DUET: K1/K2=3/1, exit=24 (0-based), C=1, N1/N2=3/1, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 198 eligible steps

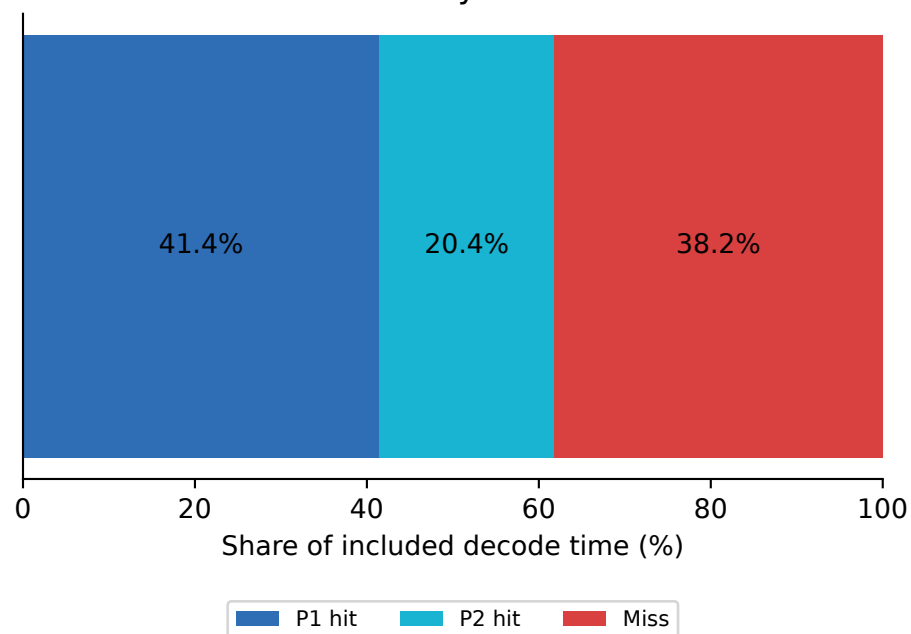


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

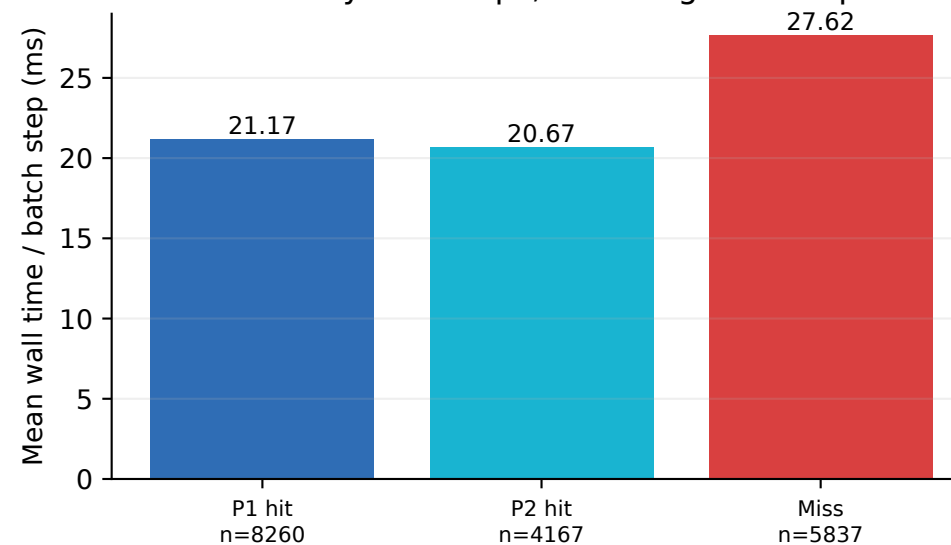
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_stream_fast_r0
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
Last pass: B=1, T=0.7, N=480 | AL*=2.679, hit=68.1%, TPS*=115.88

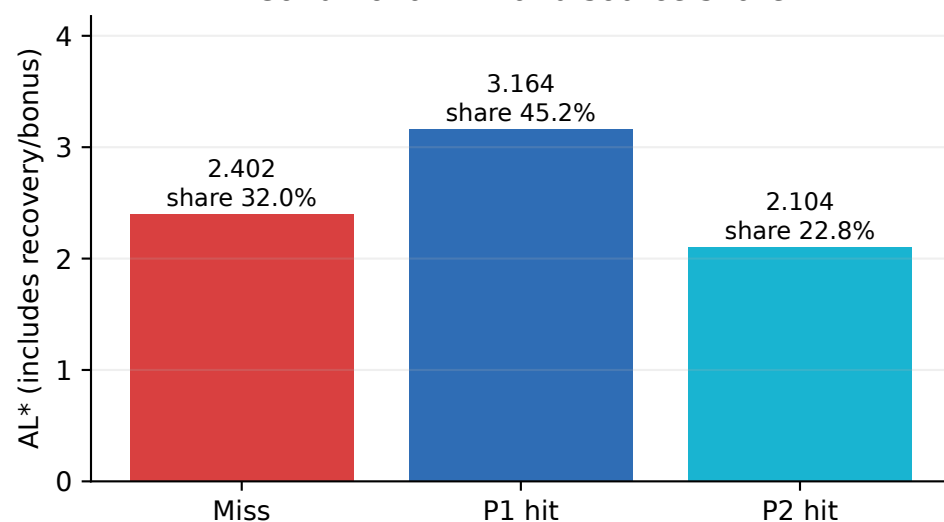
Measured wall time by batch cache status



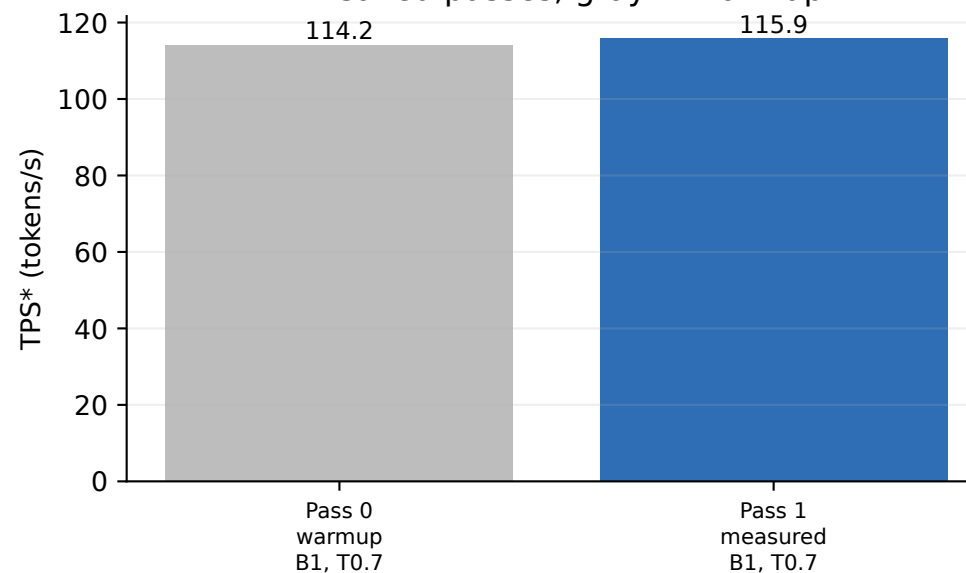
Boundary-free steps, including slow steps



Conditional AL* and source share



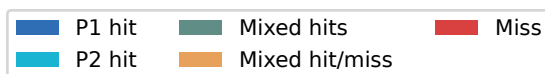
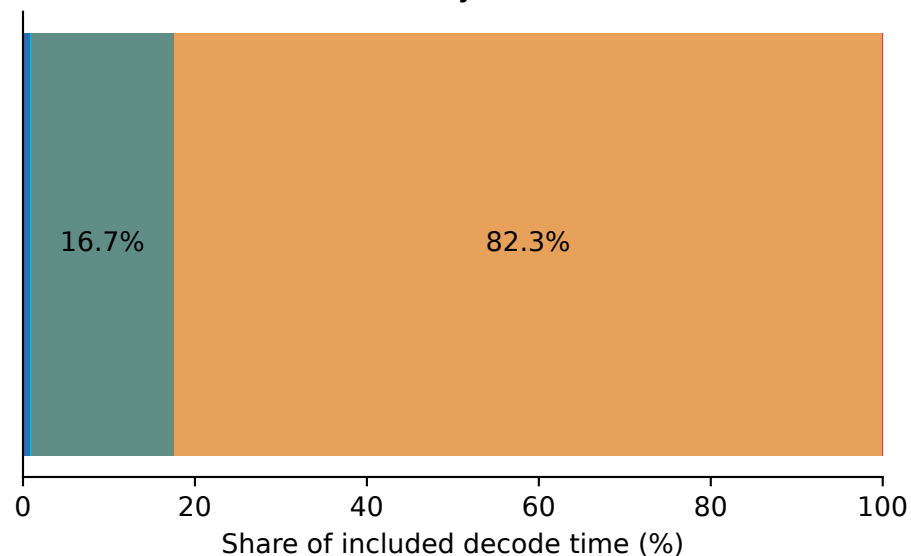
All saved passes; gray = warmup



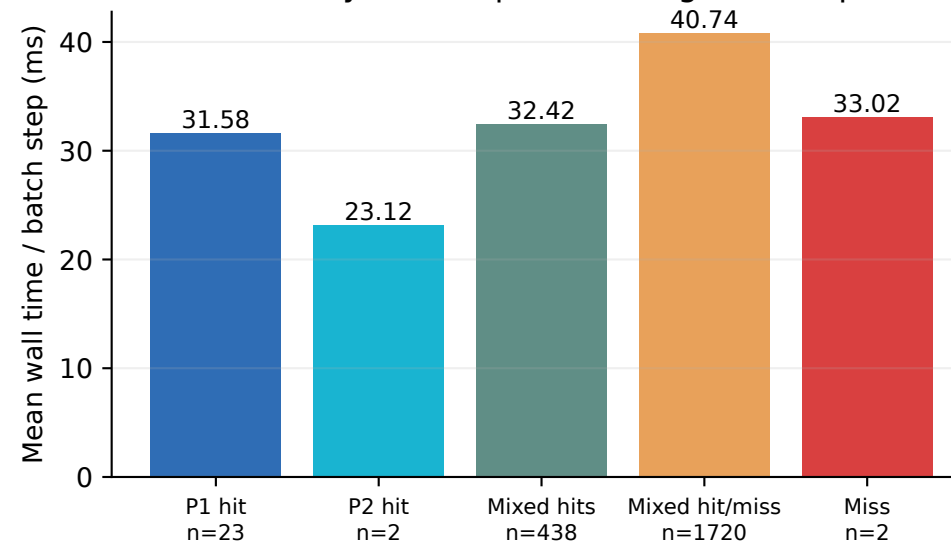
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stream_fast_r1
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.386, hit=81.7%, TPS*=487.12

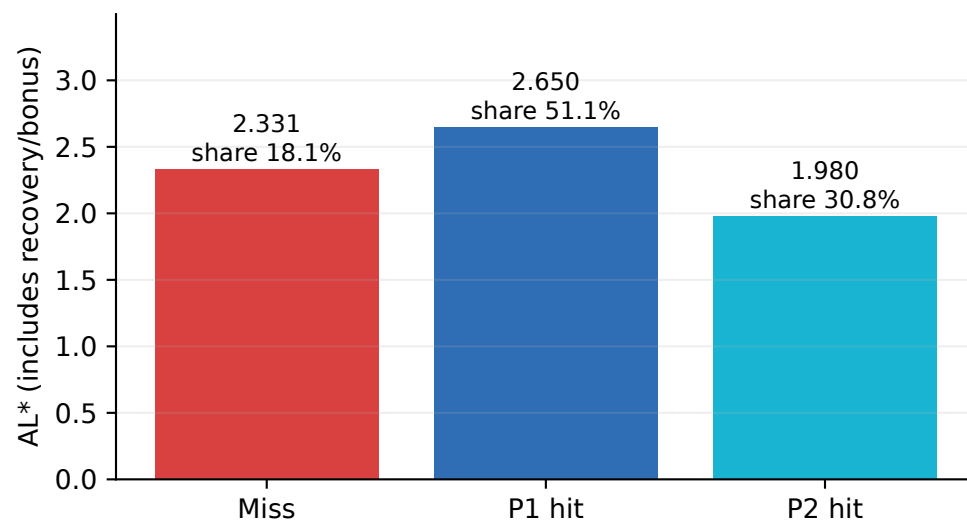
Measured wall time by batch cache status



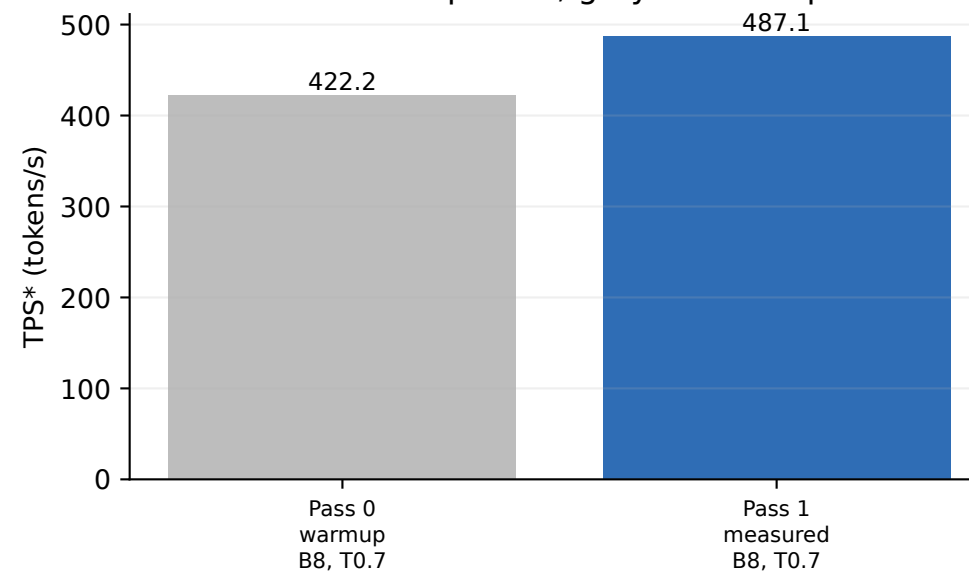
Boundary-free steps, including slow steps



Conditional AL* and source share

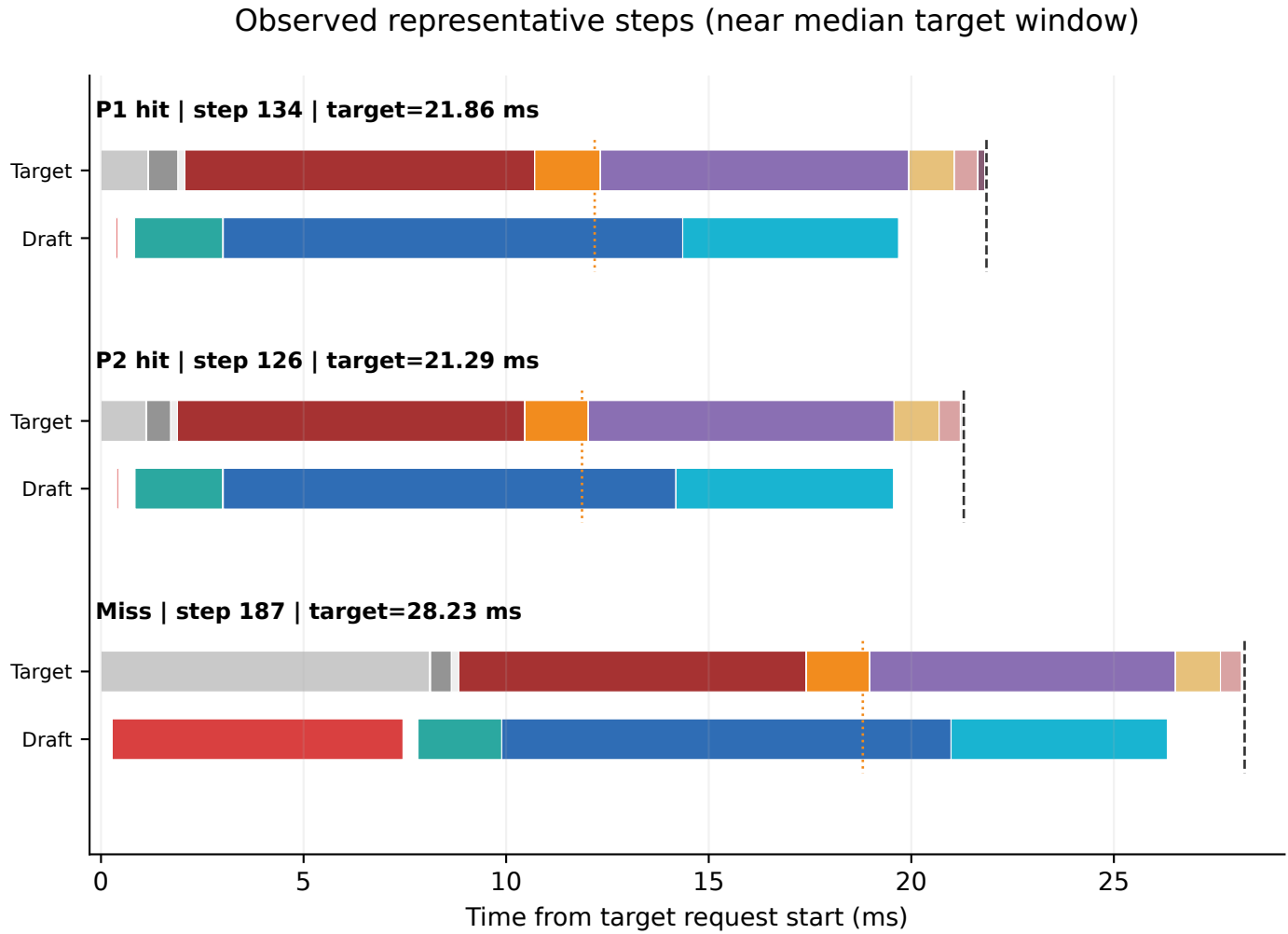
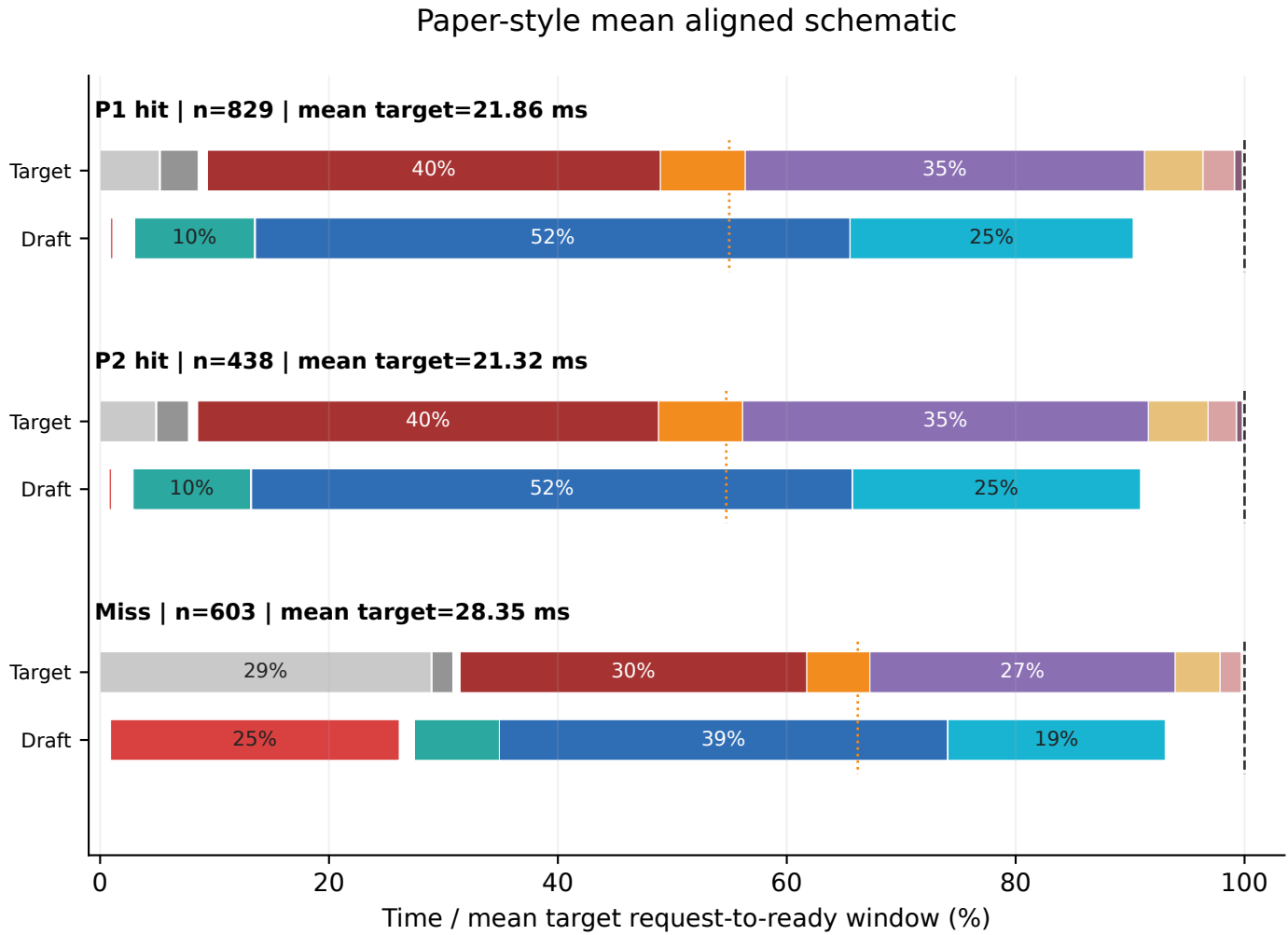
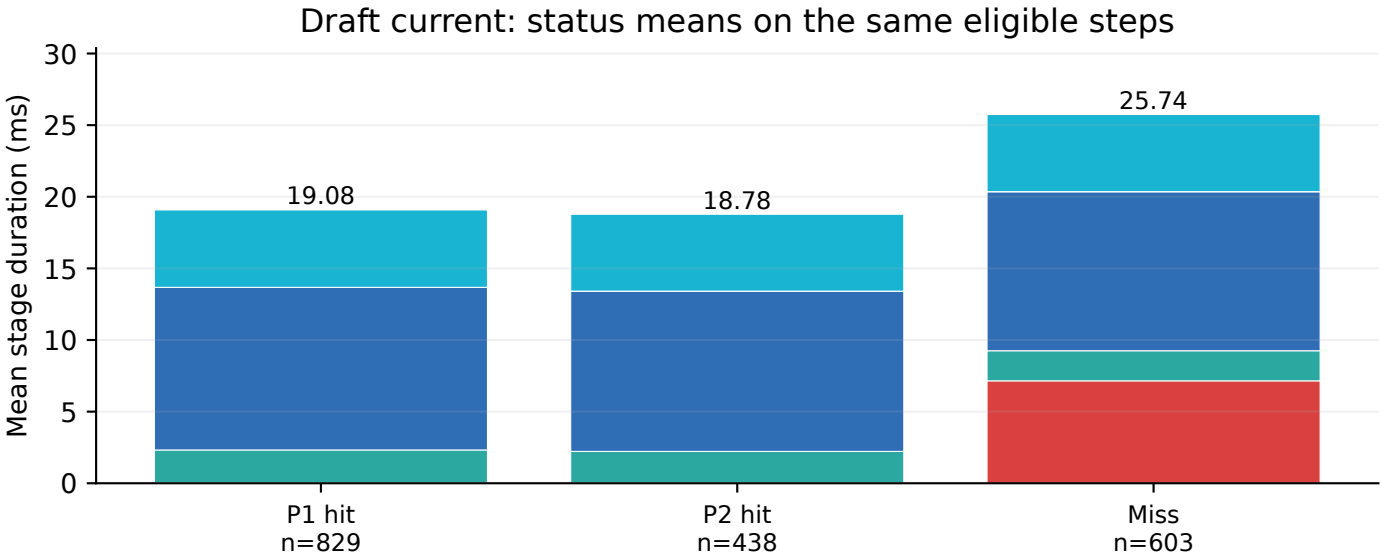
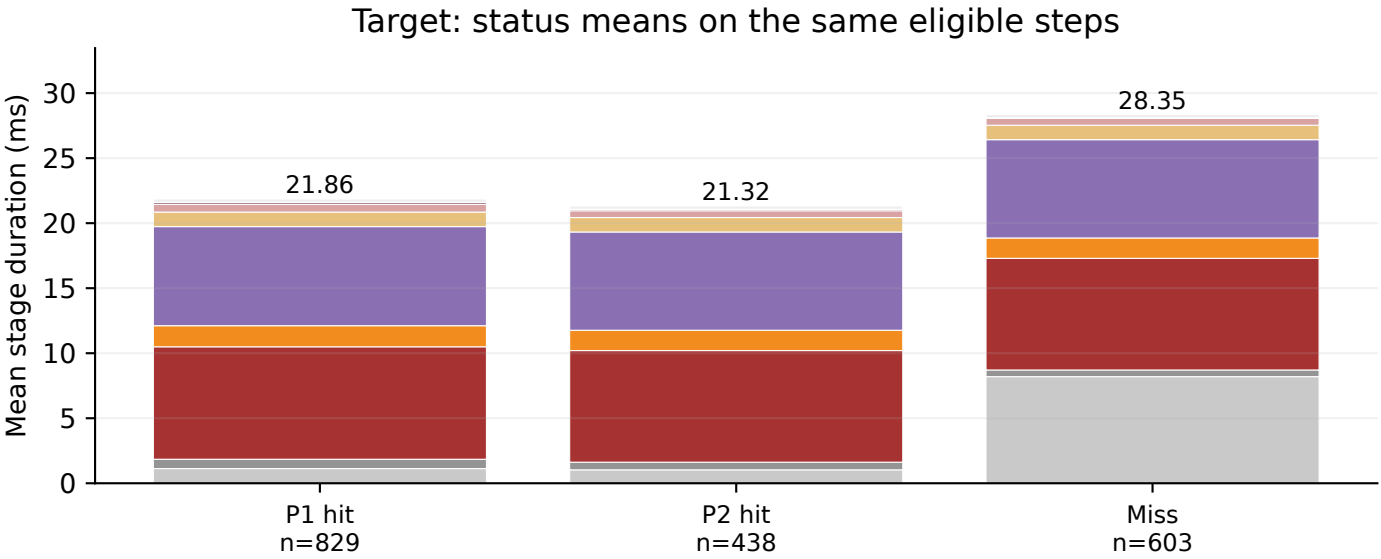


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

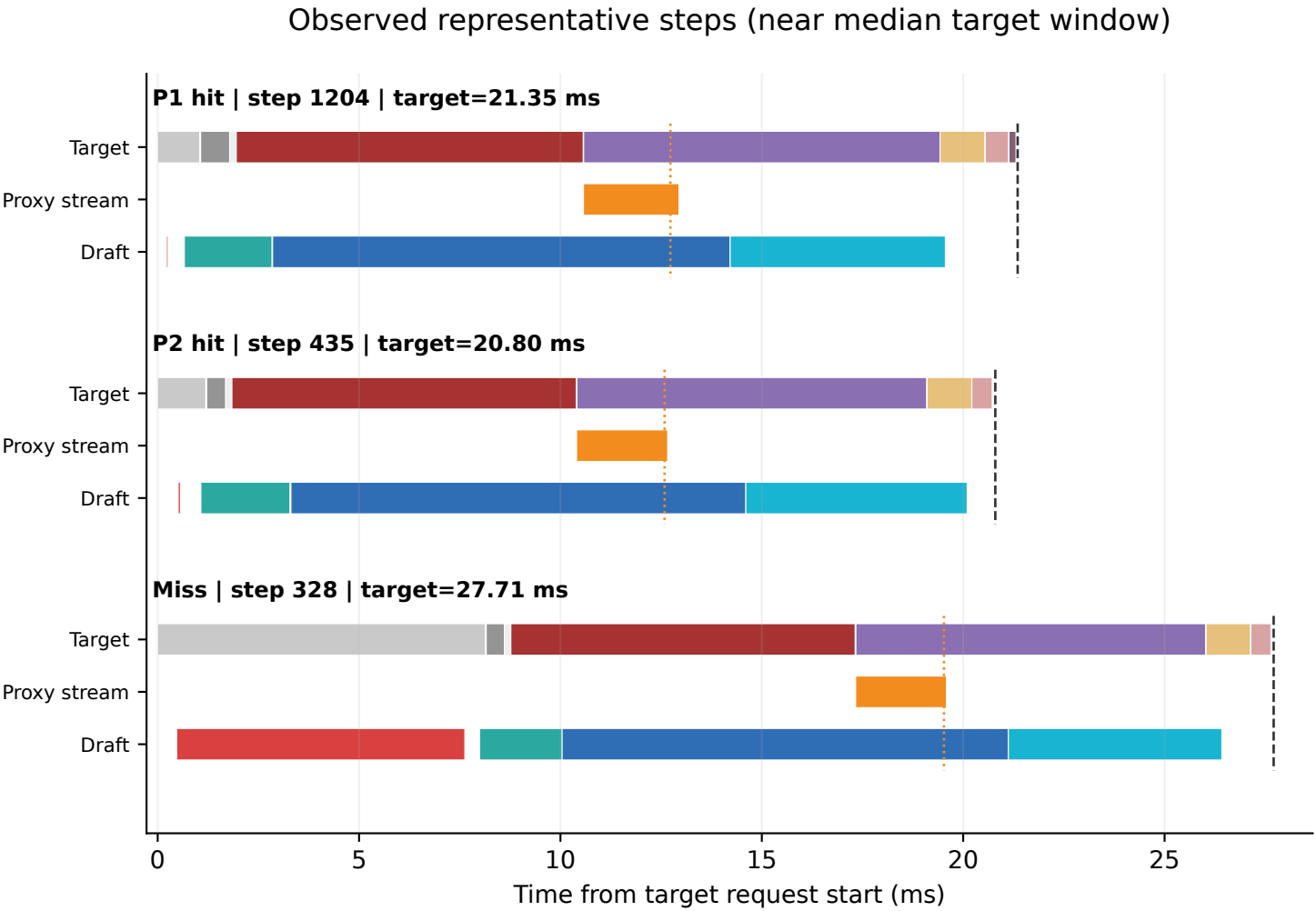
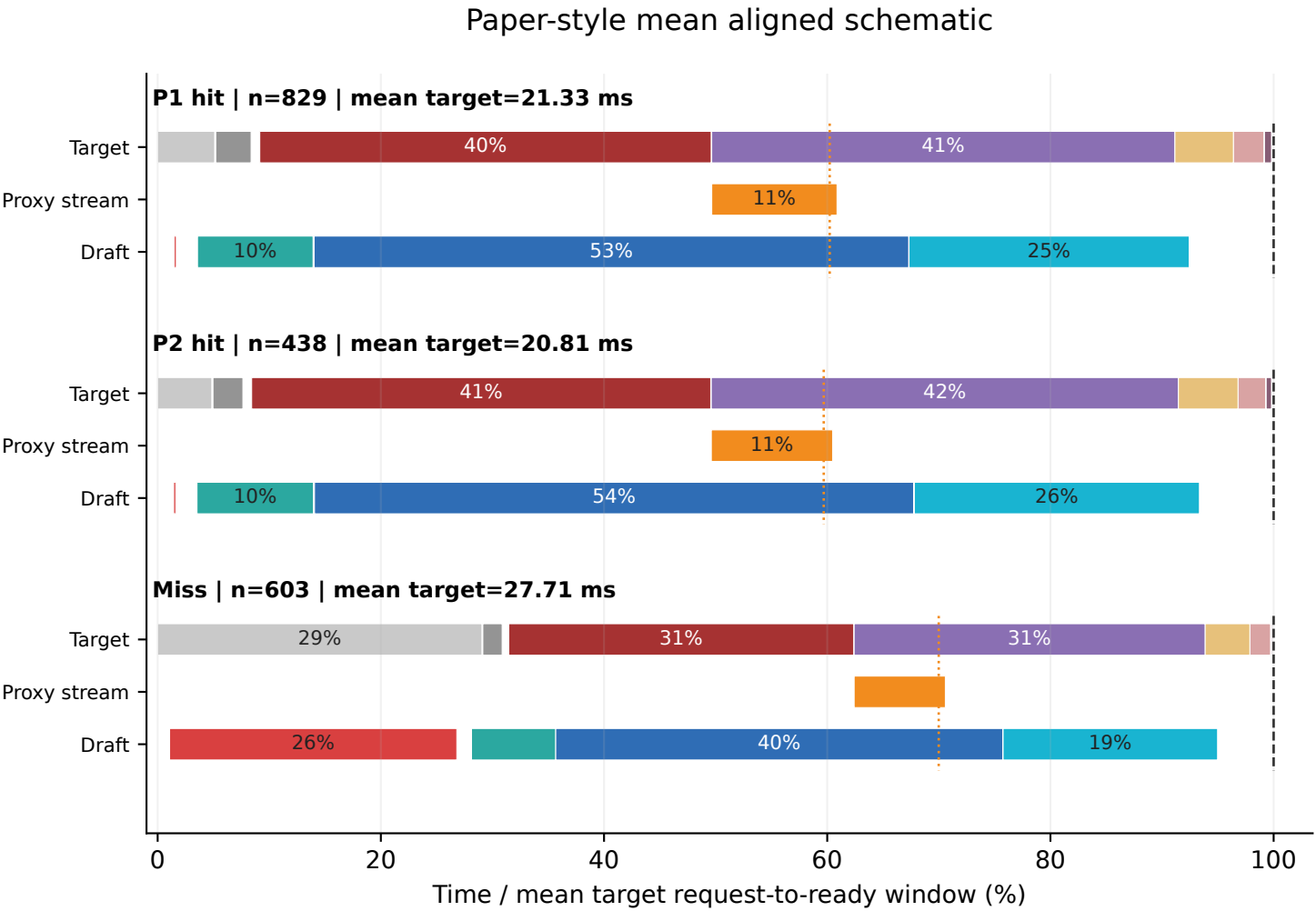
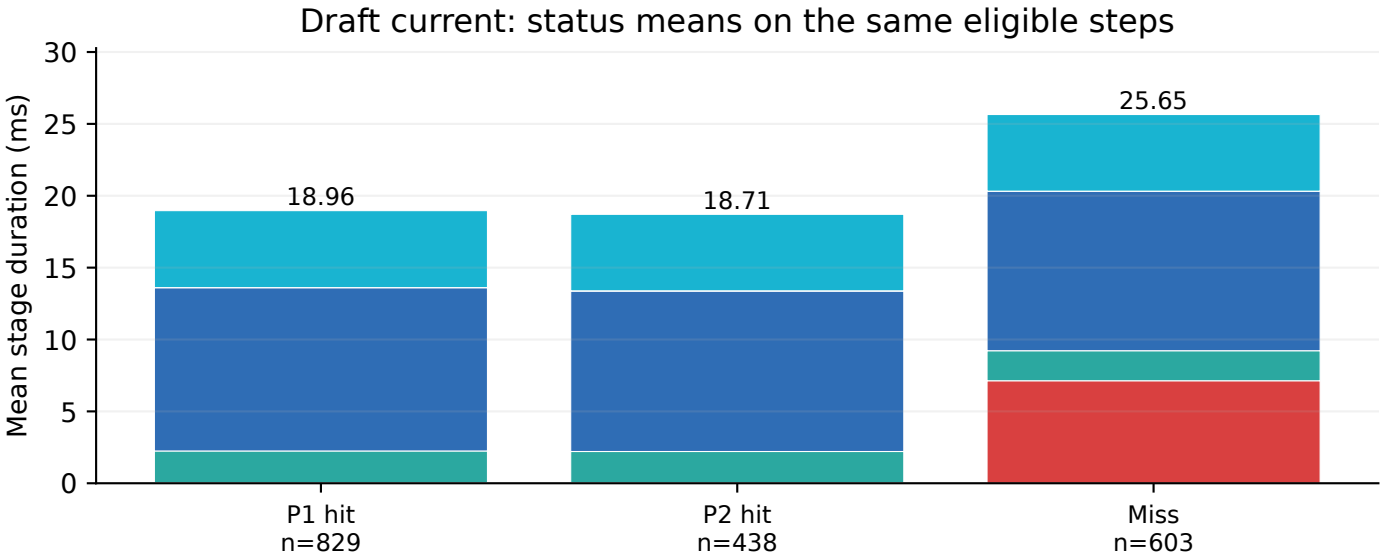
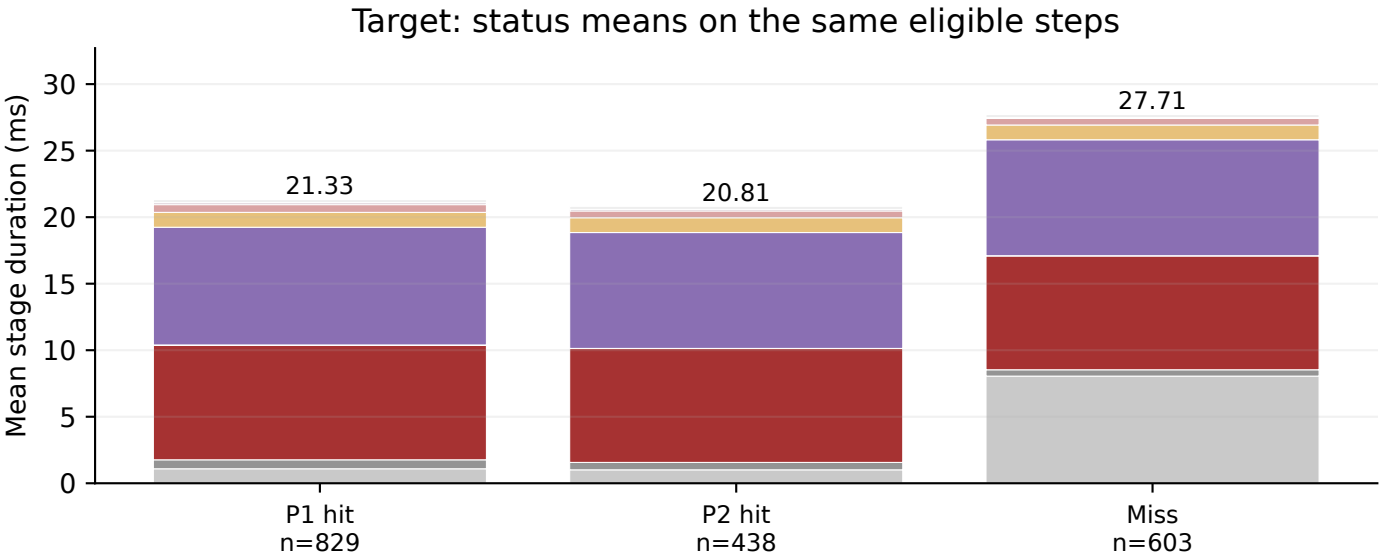
llama3_b1_stream_l0_profile_s0
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/0
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1870 eligible steps



Wait / sync Target post Accept Unlabelled gap Context P2 Target ready
Prepare Proxy / launch Commit Miss draft P1 Proxy wait Proxy received
Target pre Final logits Postprocess

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_stream_l0_profile_s1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1870 eligible steps



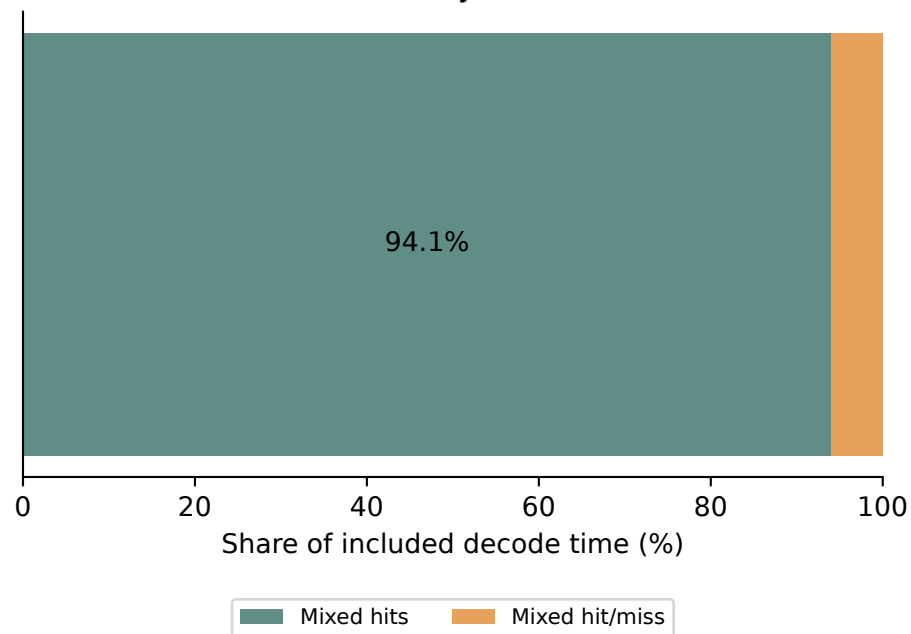
Legend:

- Wait / sync (grey)
- Prepare (dark grey)
- Target pre (red)
- Target post (purple)
- Proxy / launch (orange)
- Proxy compute (orange)
- Final logits (yellow)
- Accept (pink)
- Commit (dark purple)
- Miss draft (red)
- Postprocess (dark grey)
- Unlabelled gap (light grey)
- Context (teal)
- P1 (blue)
- P2 (cyan)
- Proxy wait (yellow)
- Target ready (dashed line)
- Proxy received (dotted line)

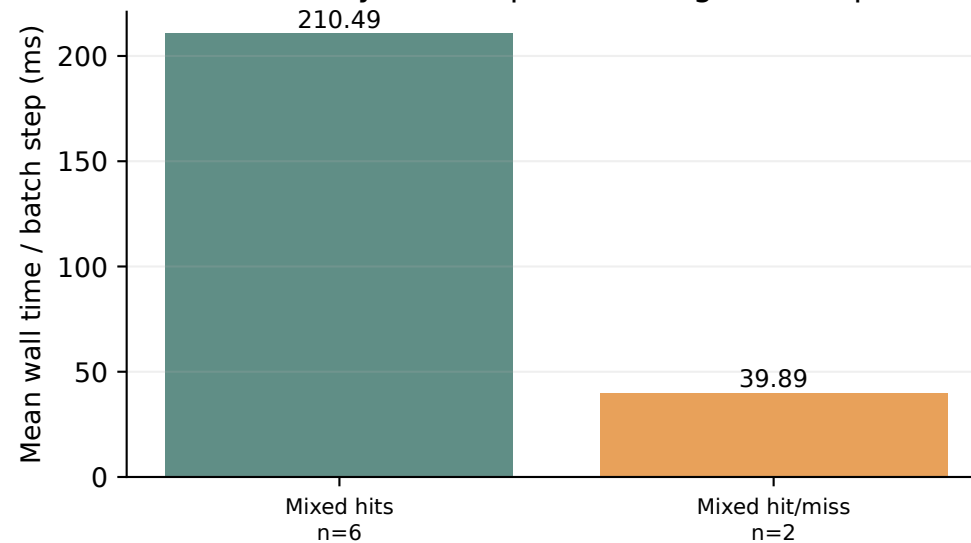
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stream_l0_greedy_smoke
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
Last pass: B=8, T=0.0, N=8 | AL*=2.197, hit=83.8%, TPS*=75.22

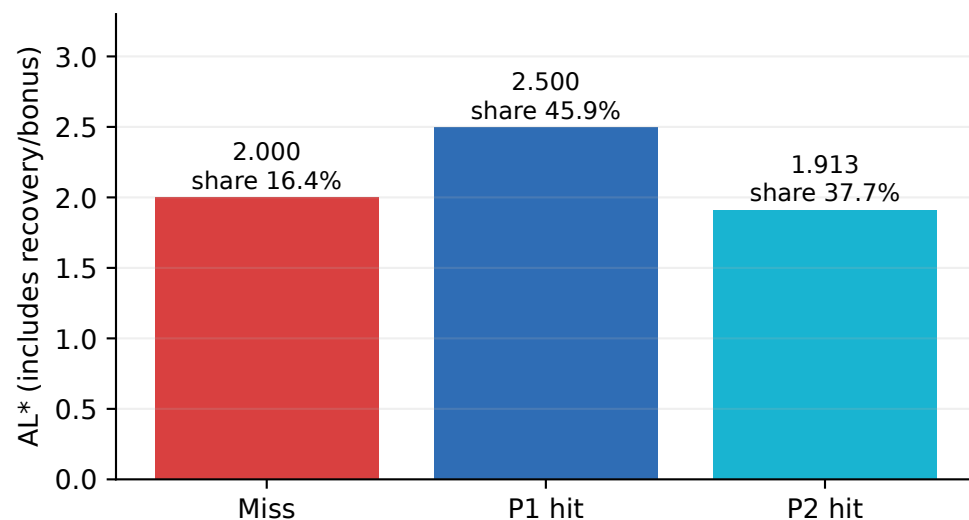
Measured wall time by batch cache status



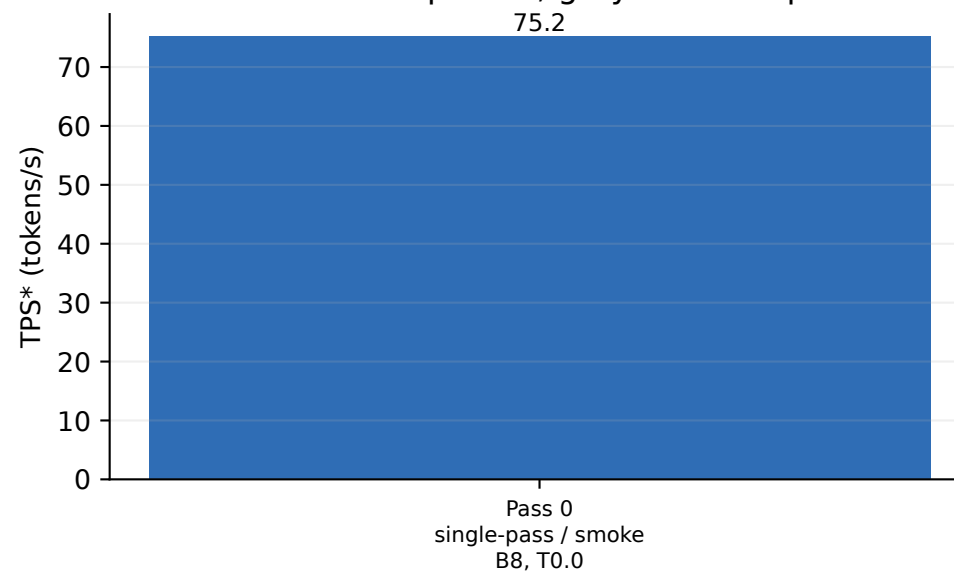
Boundary-free steps, including slow steps



Conditional AL* and source share

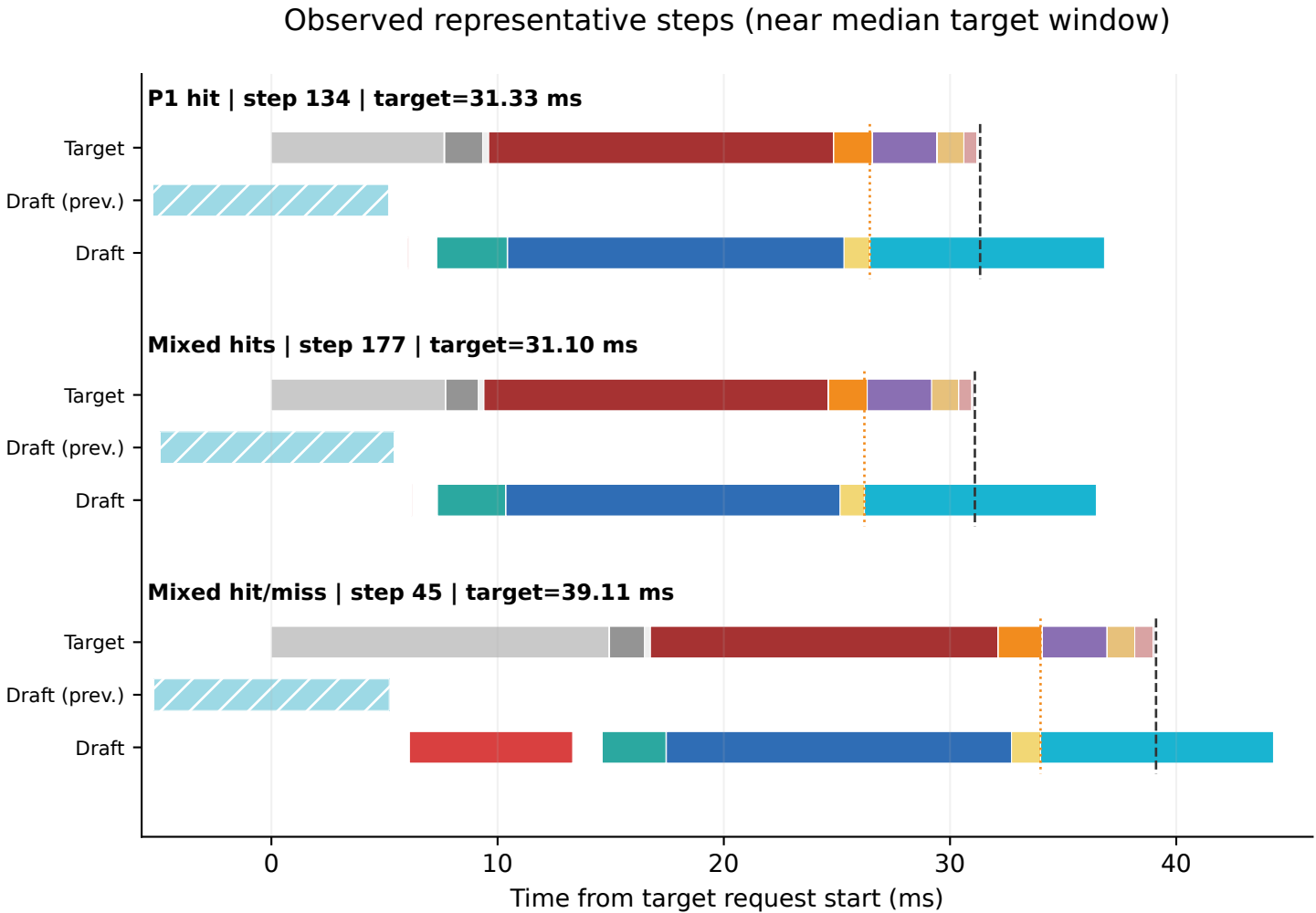
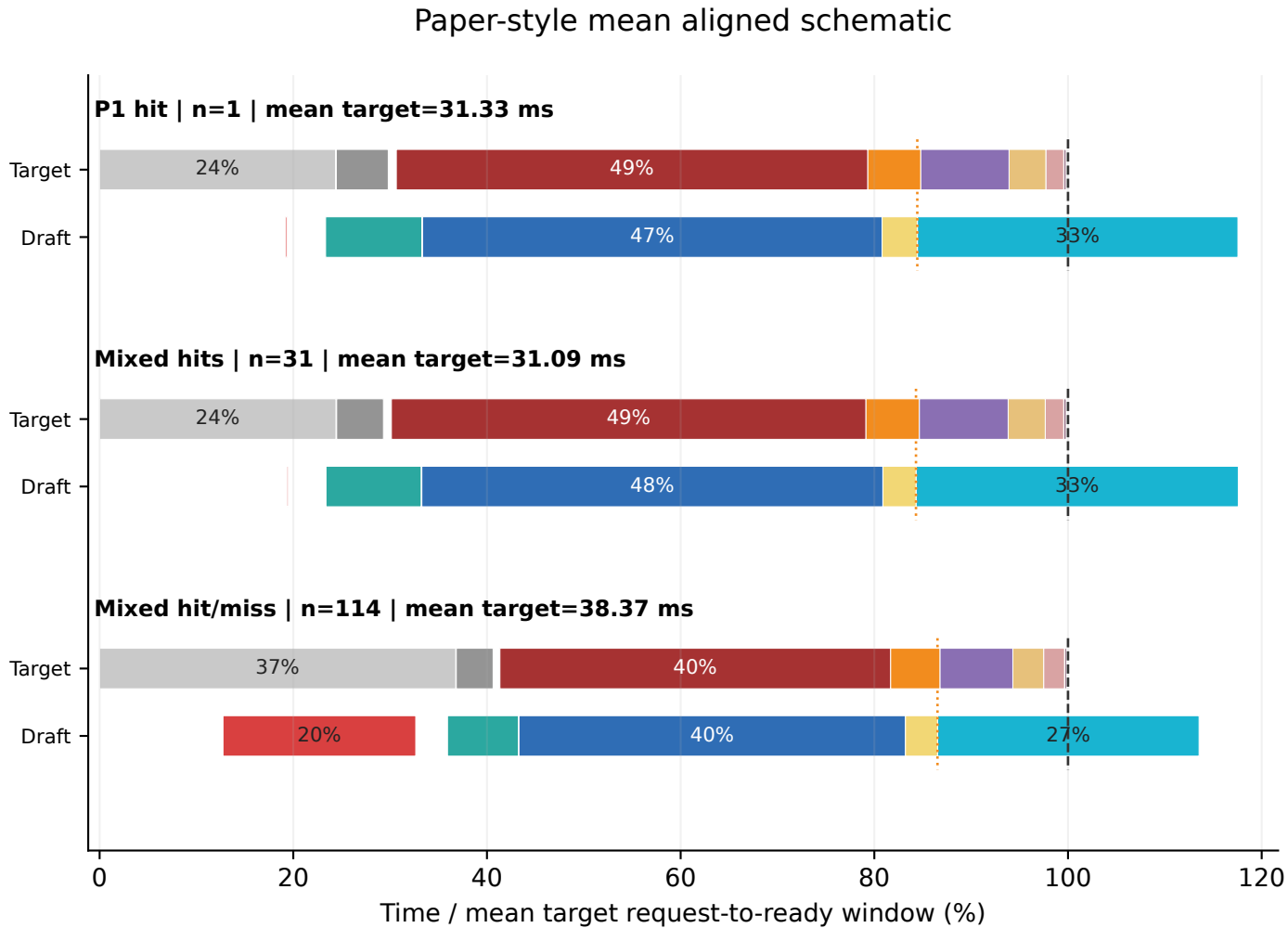
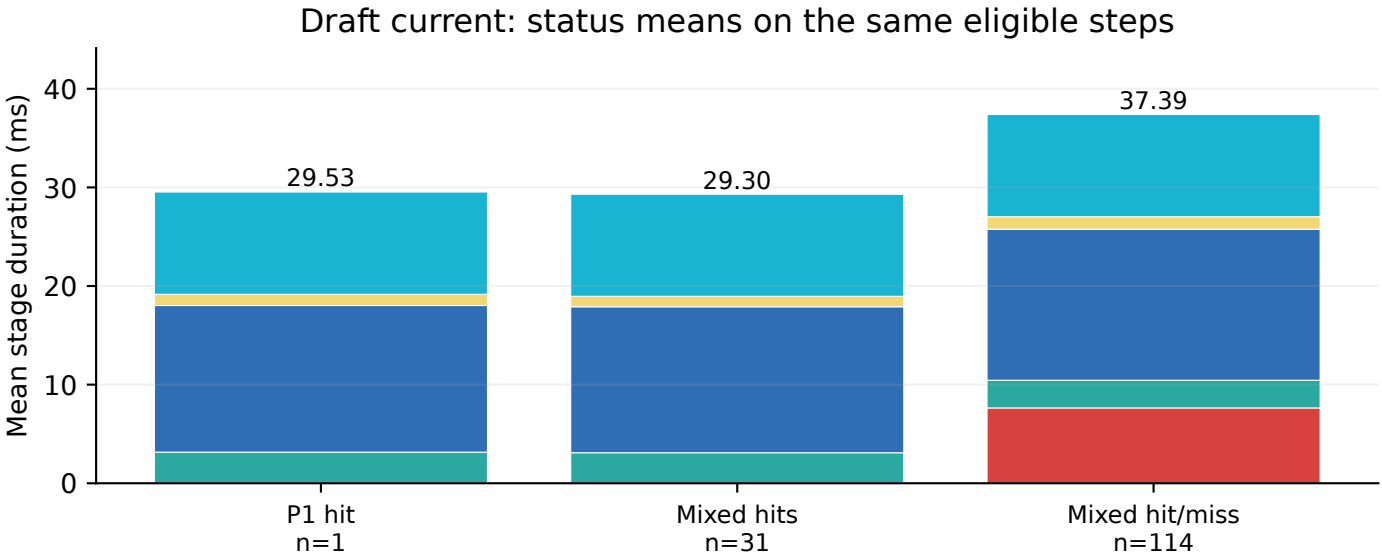
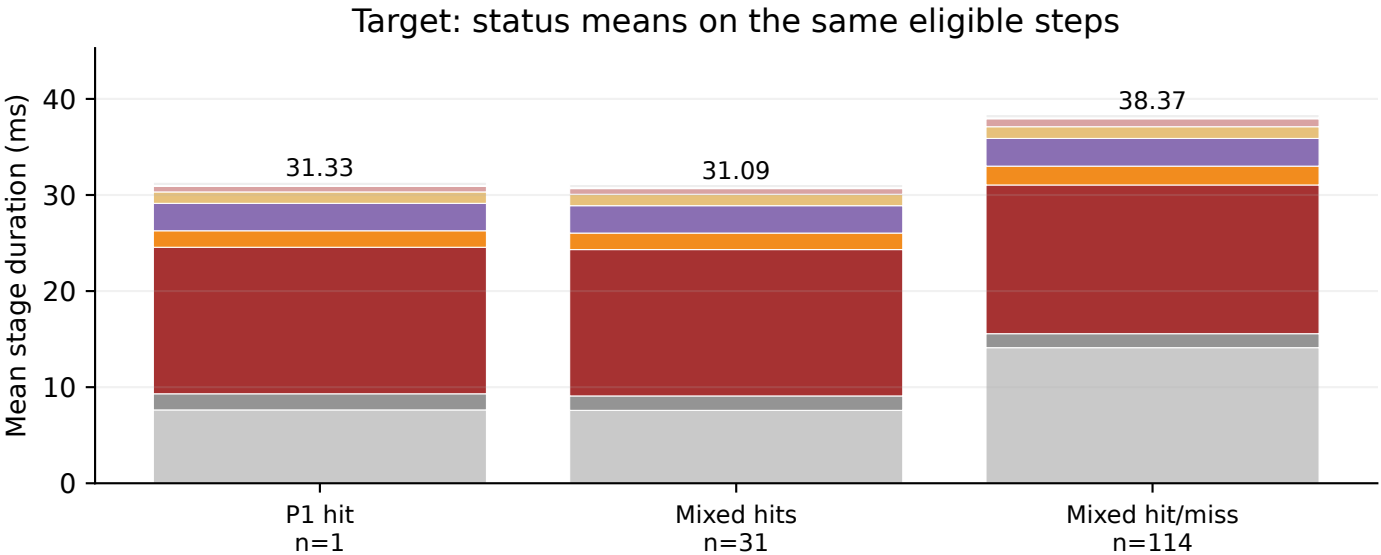


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stream_l0_profile_s0
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 146 eligible steps

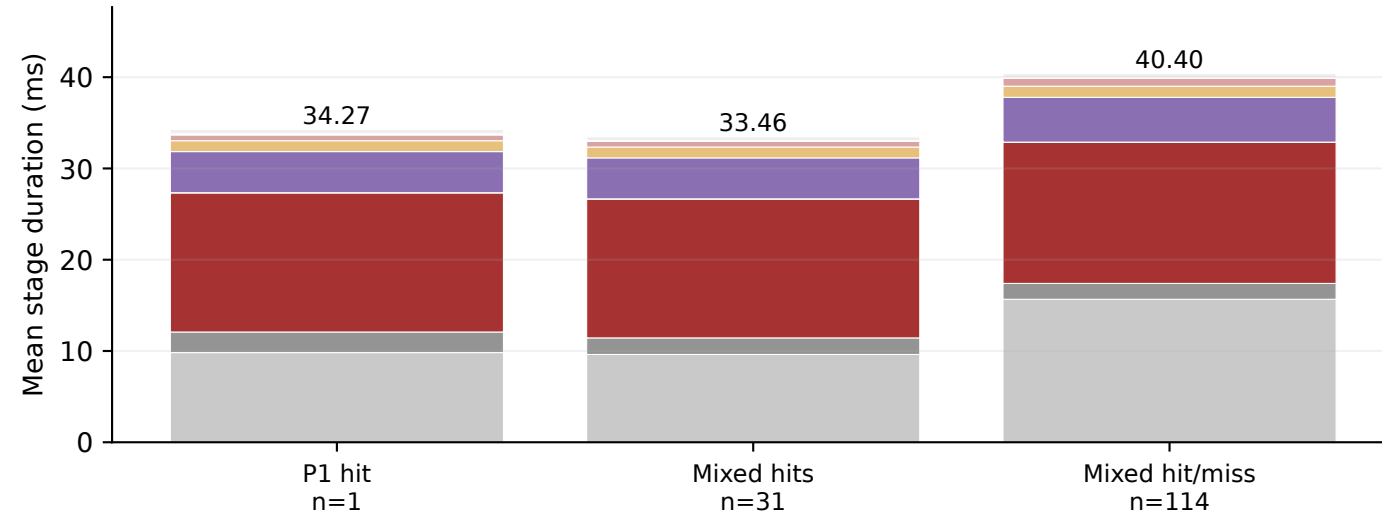


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

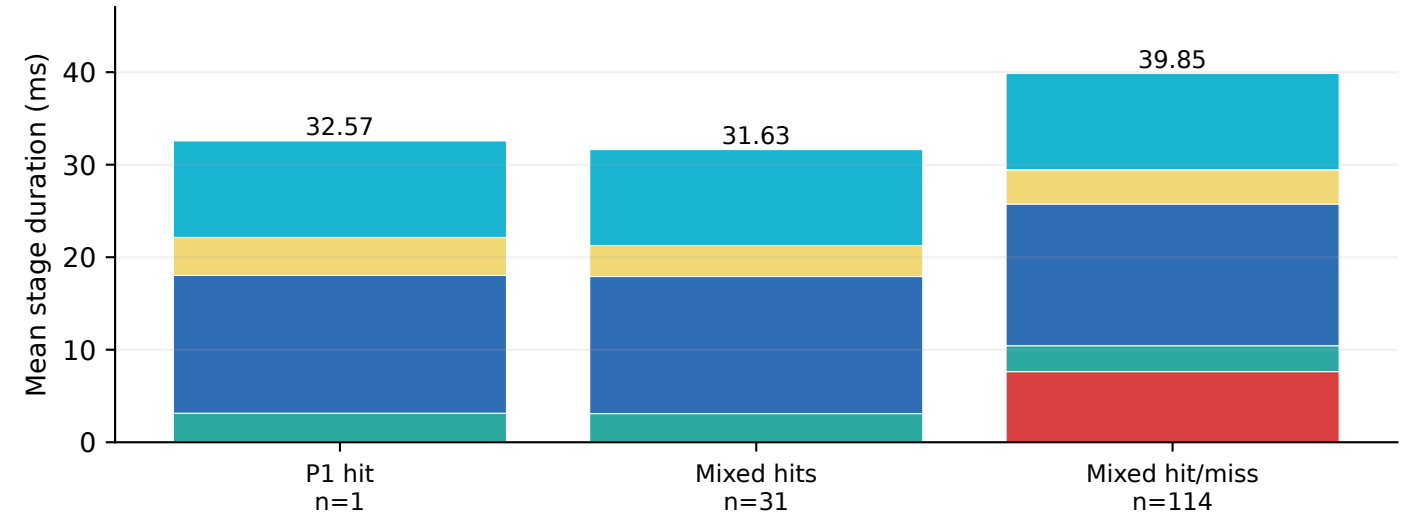
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stream_l0_profile_s1
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 146 eligible steps

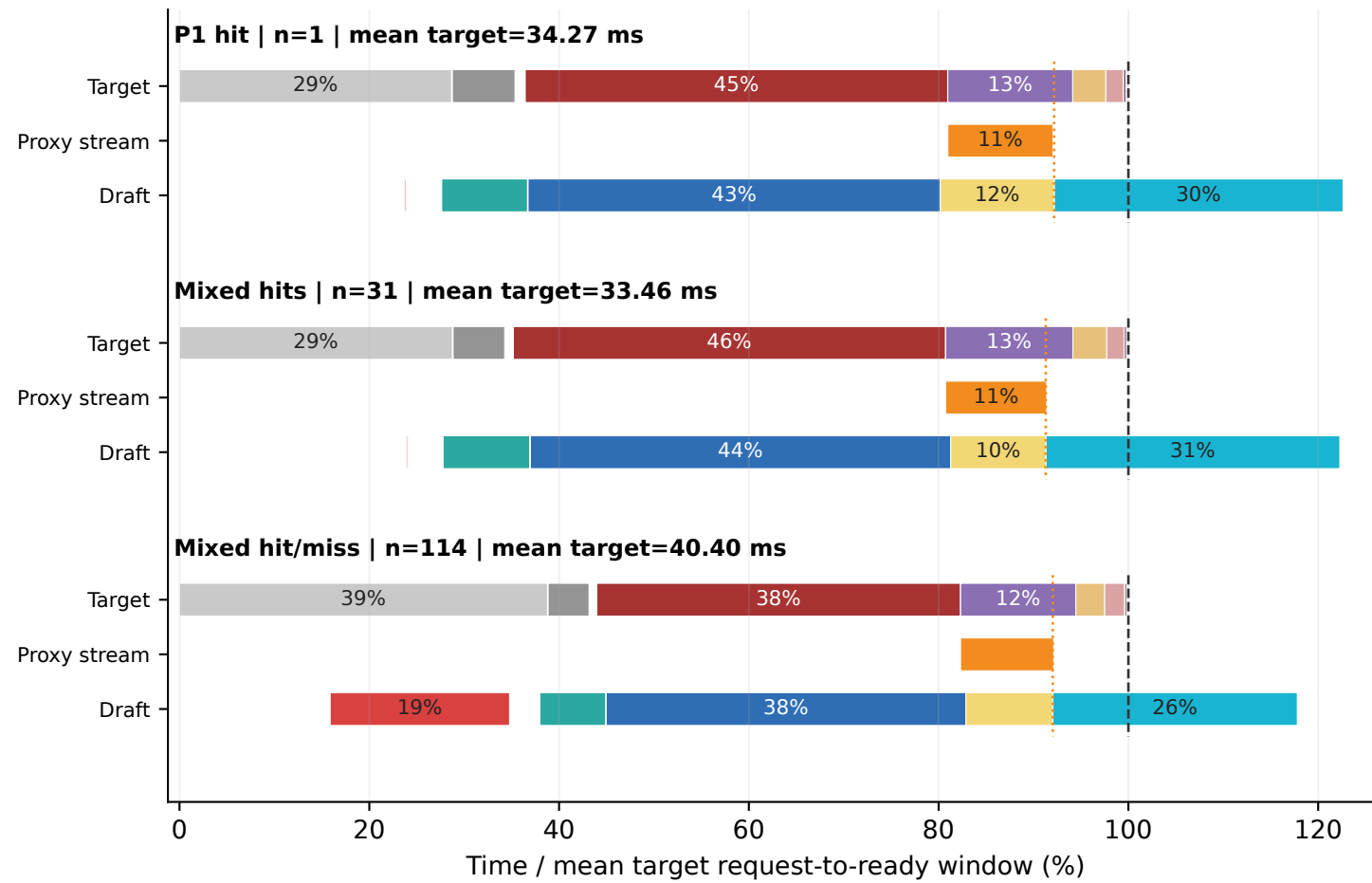
Target: status means on the same eligible steps



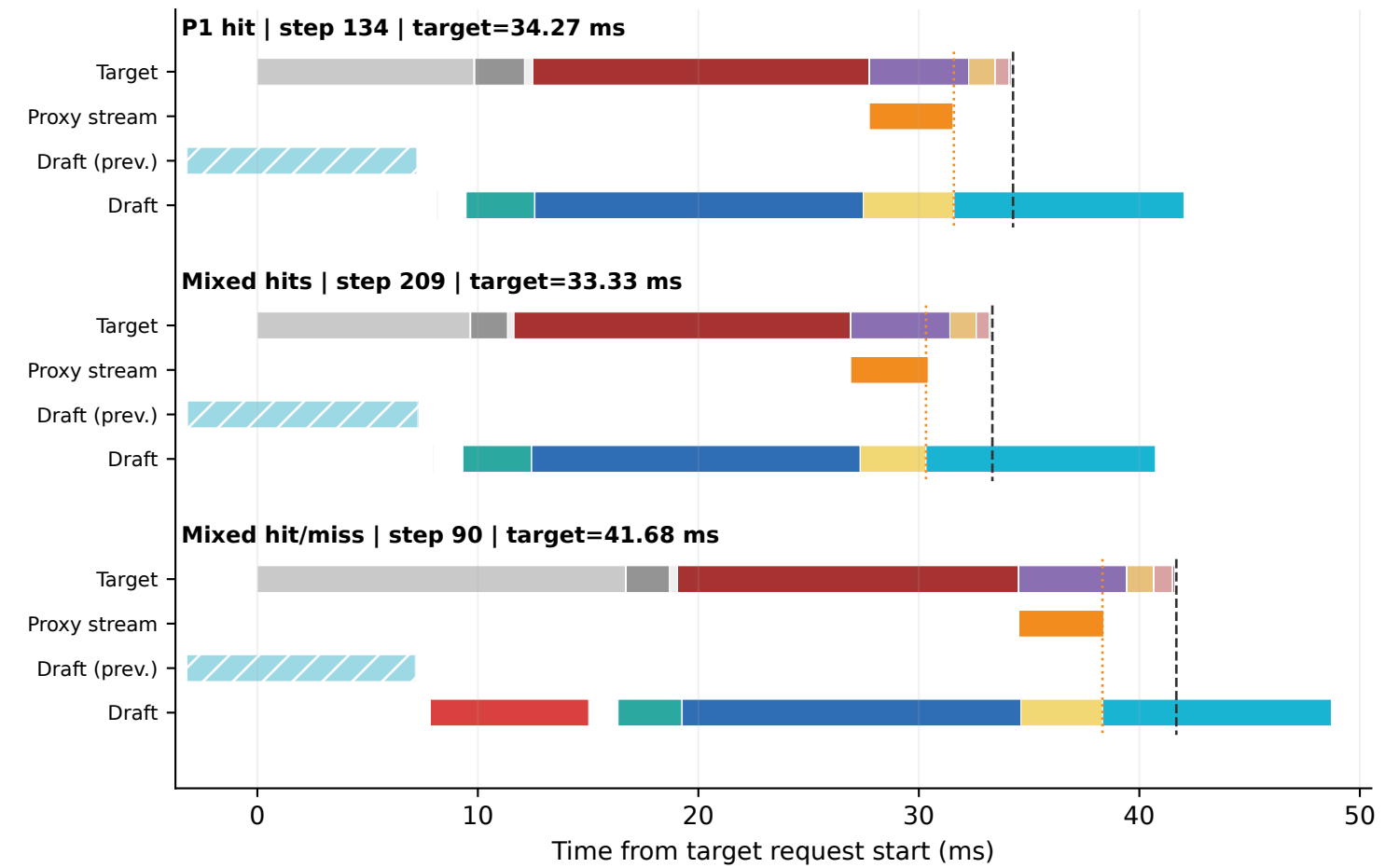
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

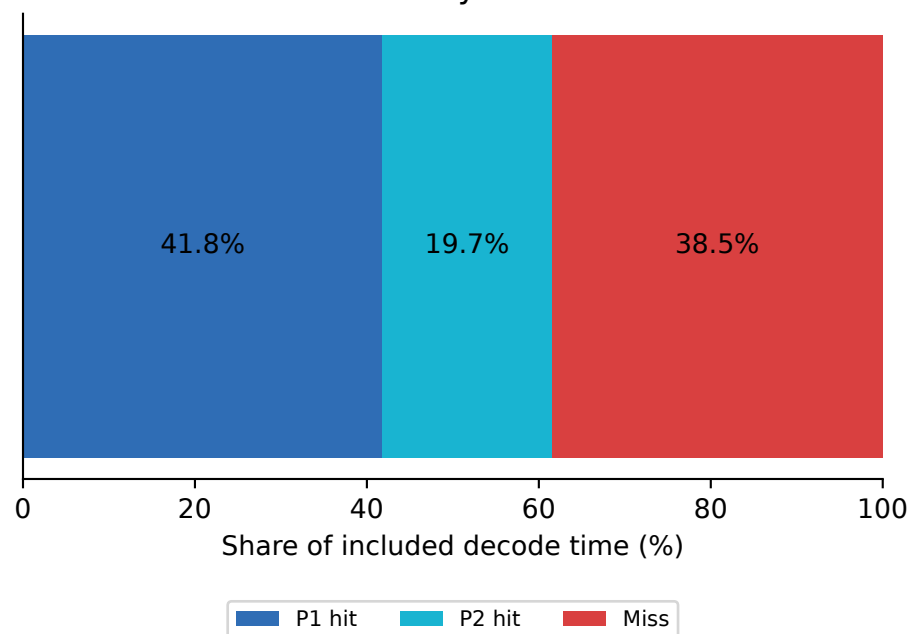


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

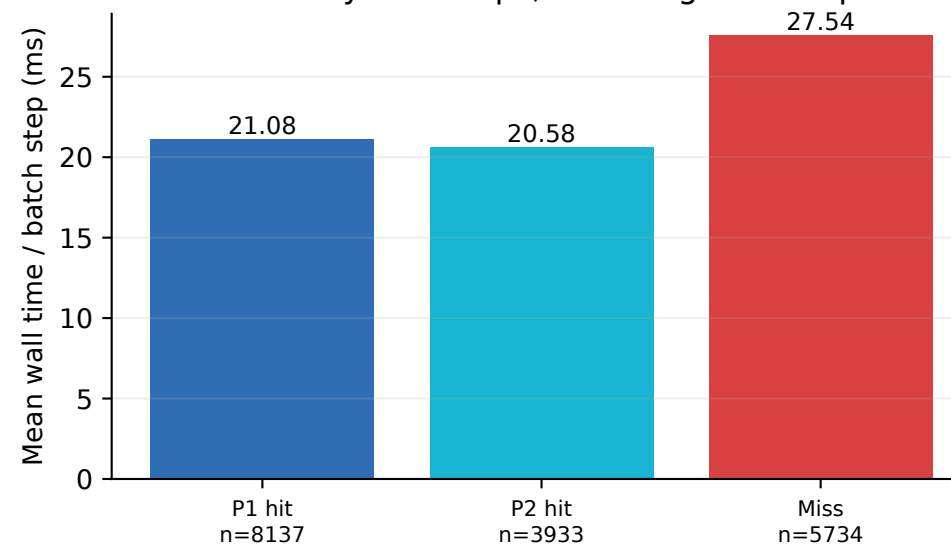
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_stream_fast_r1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
Last pass: B=1, T=0.7, N=480 | AL*=2.702, hit=67.7%, TPS*=117.21

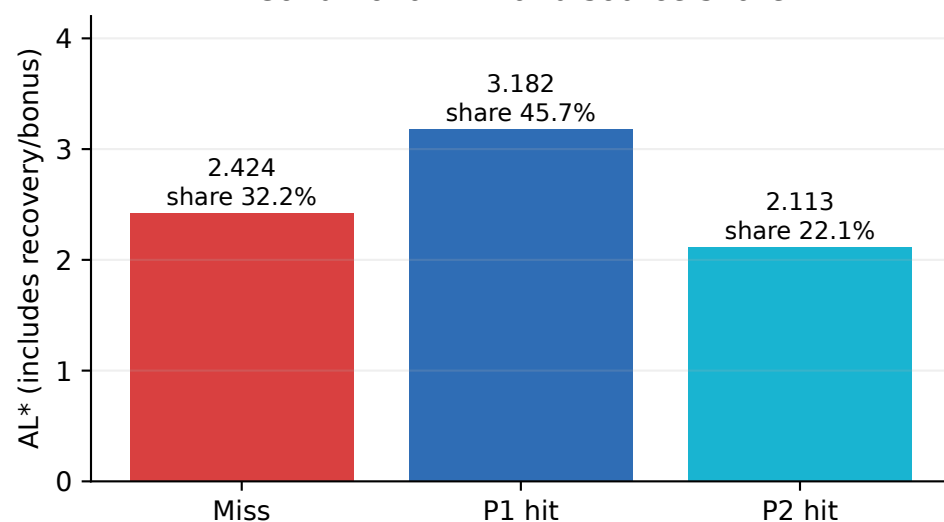
Measured wall time by batch cache status



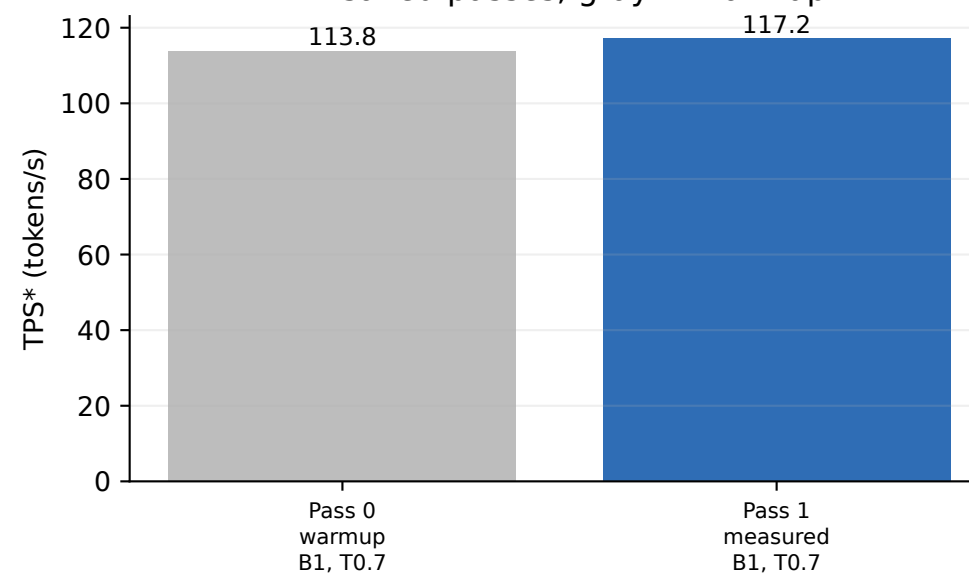
Boundary-free steps, including slow steps



Conditional AL* and source share



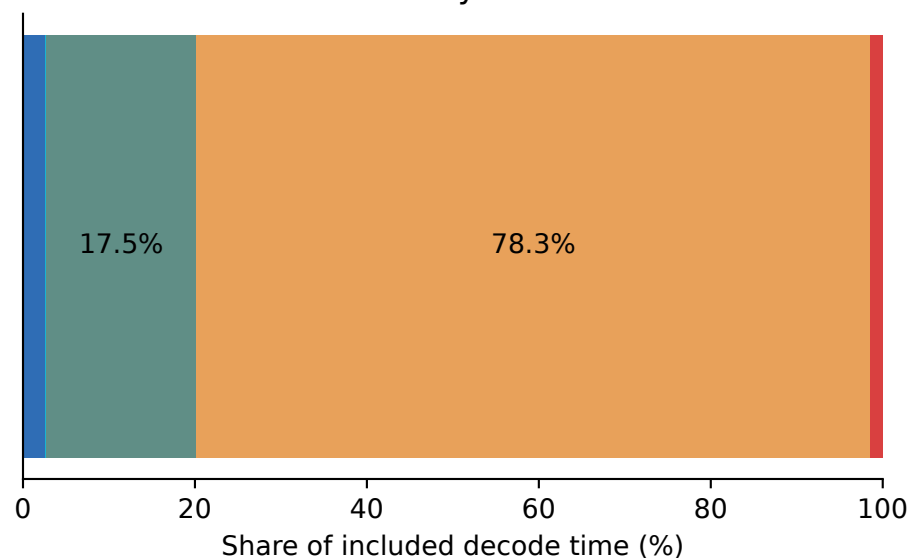
All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

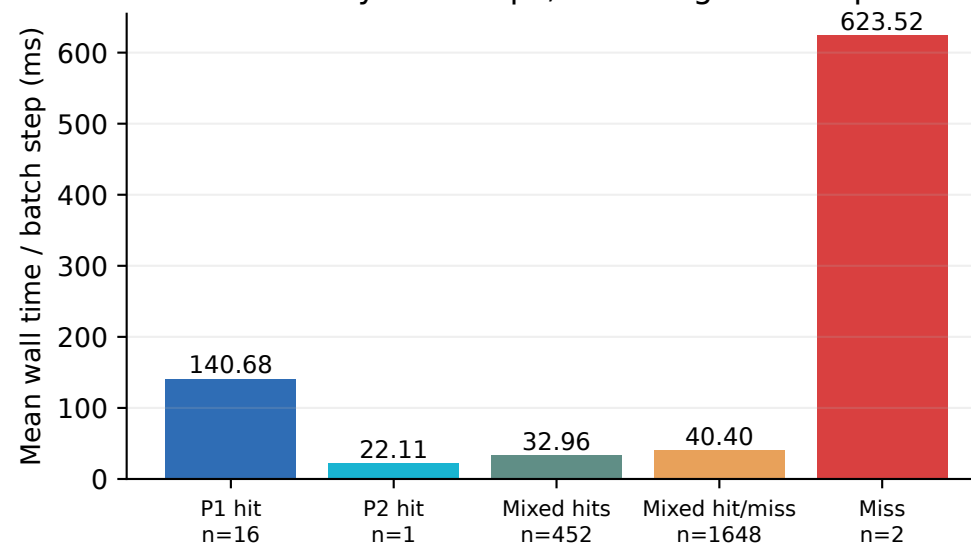
llama3_b8_stream_fast_r0
 DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
 Last pass: B=8, T=0.7, N=480 | AL*=2.429, hit=81.7%, TPS*=482.20

Measured wall time by batch cache status

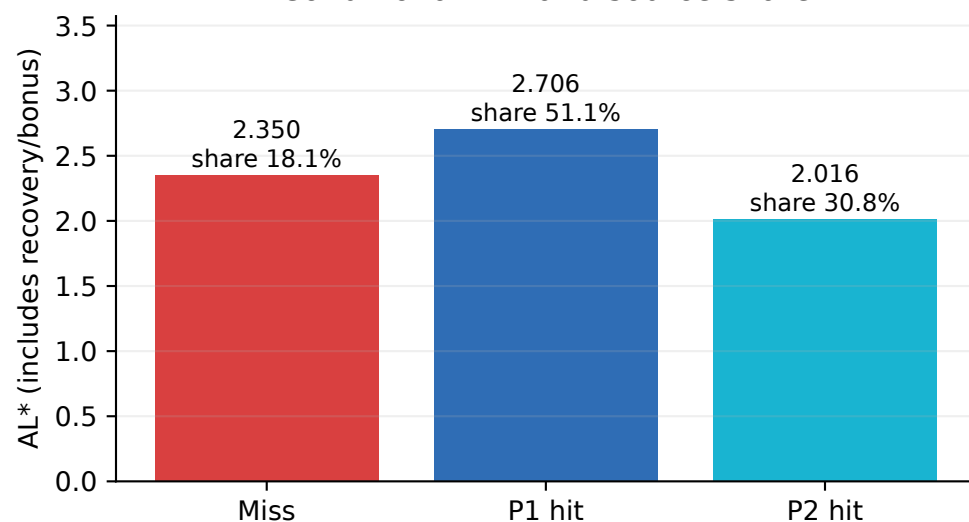


P1 hit Mixed hits Miss
 P2 hit Mixed hit/miss

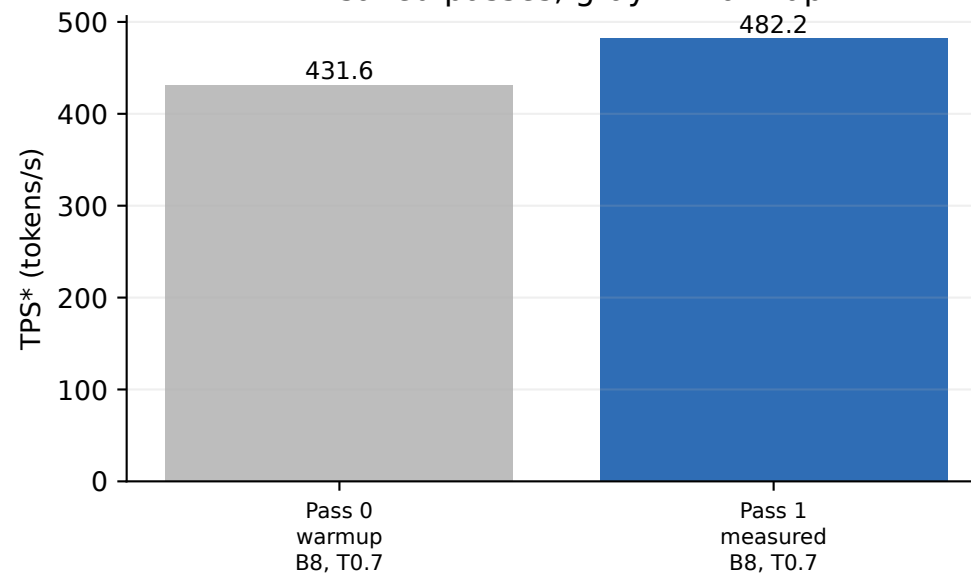
Boundary-free steps, including slow steps



Conditional AL* and source share

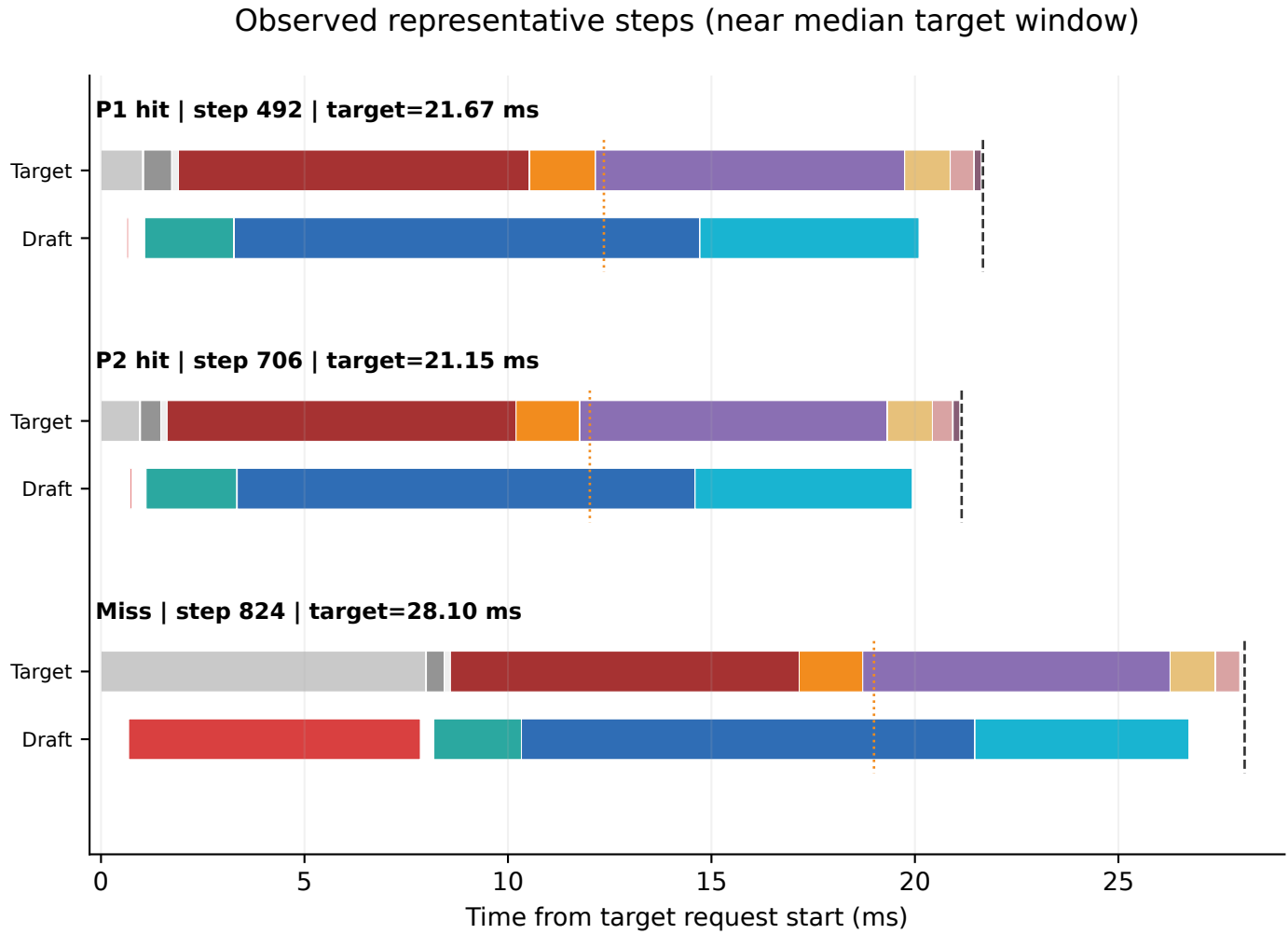
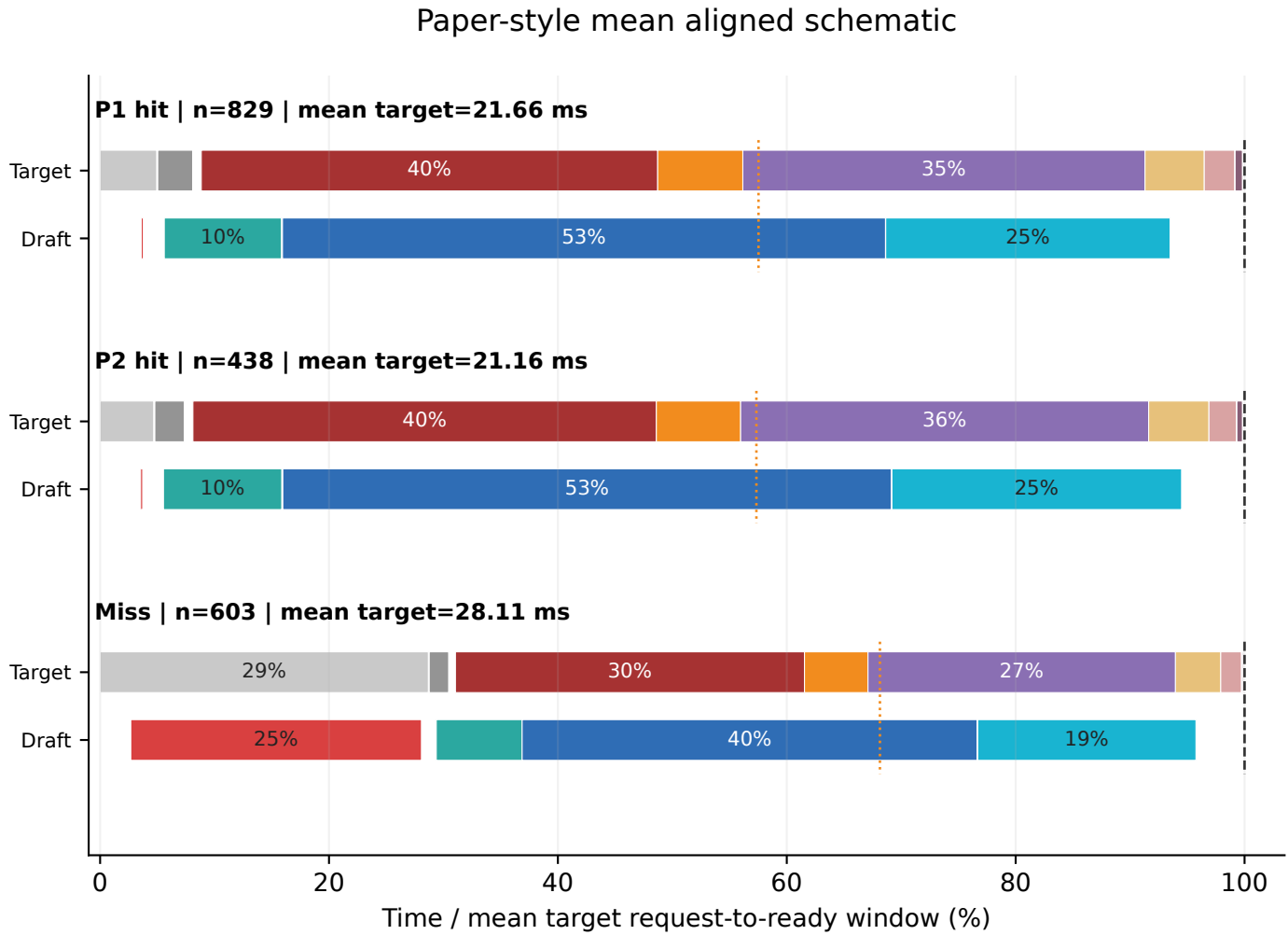
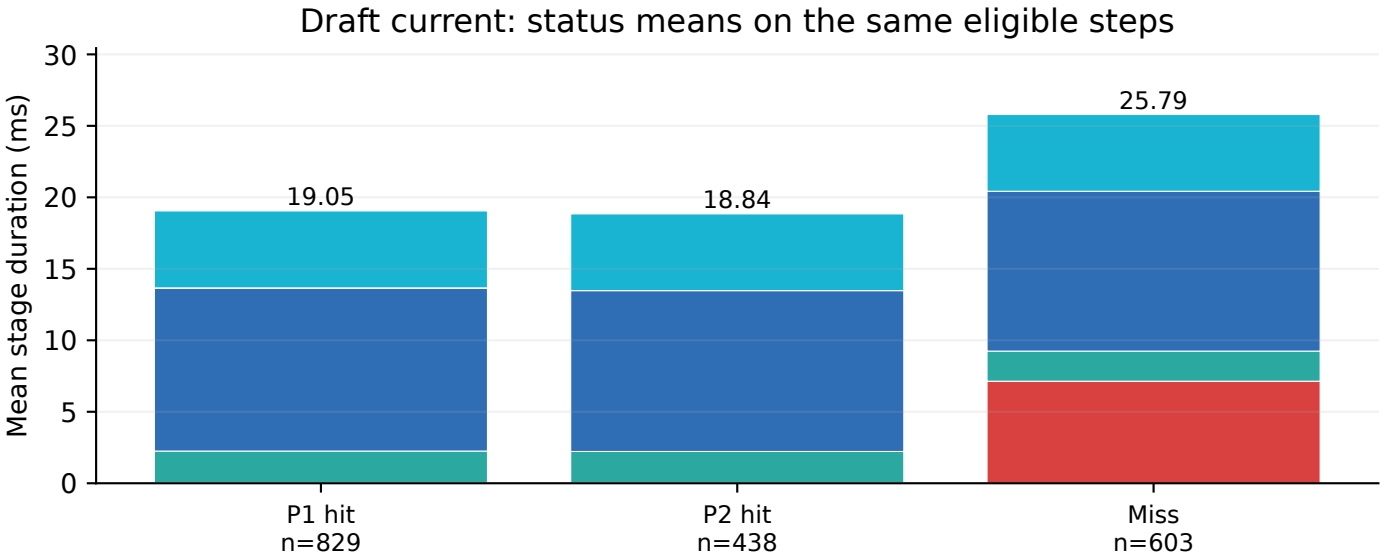
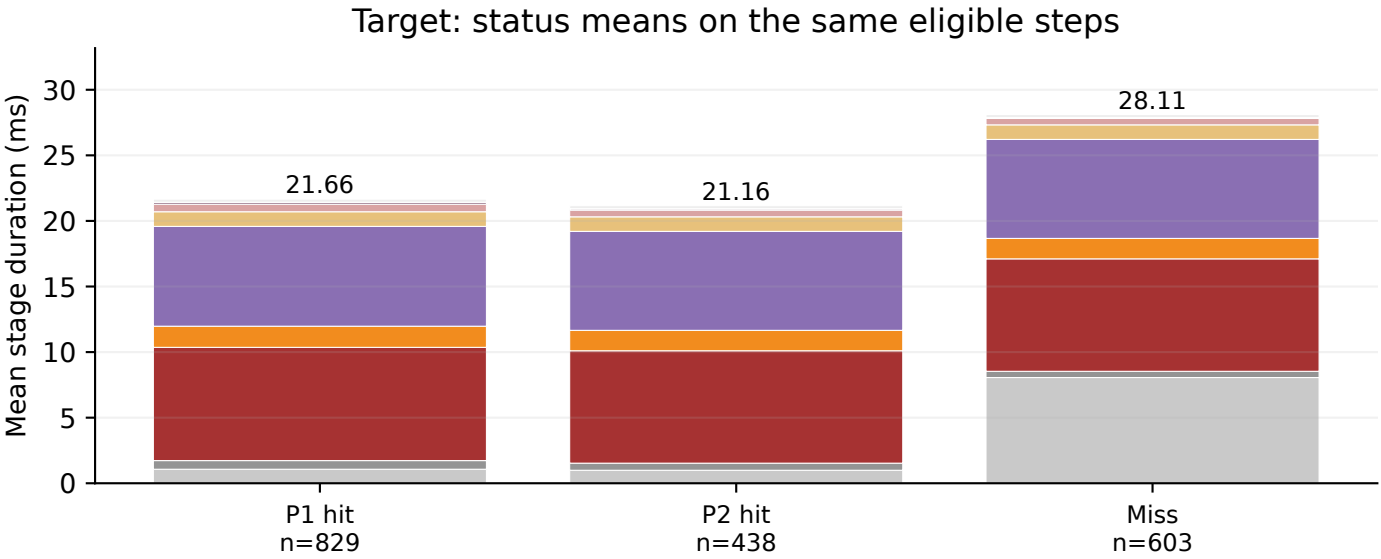


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

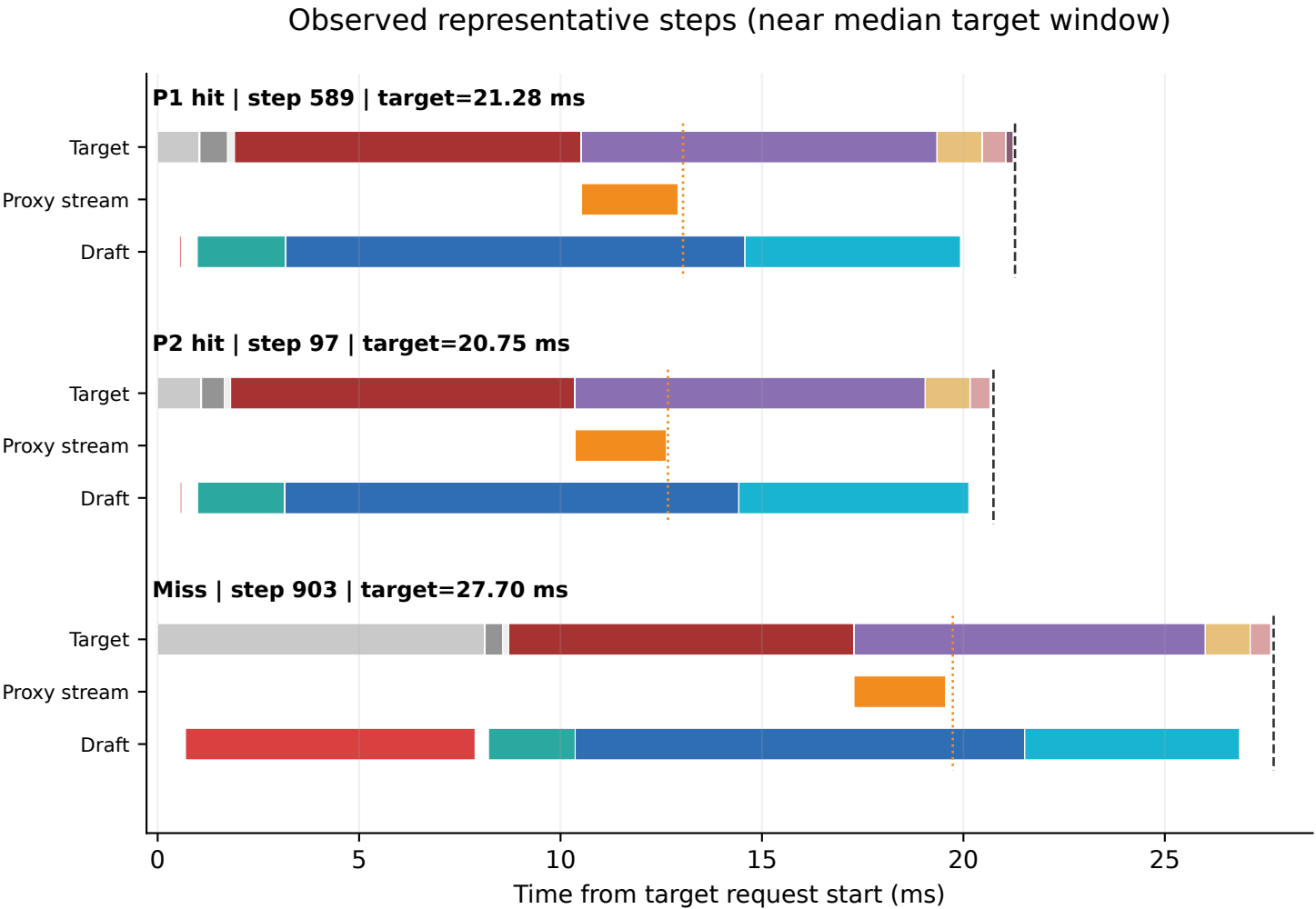
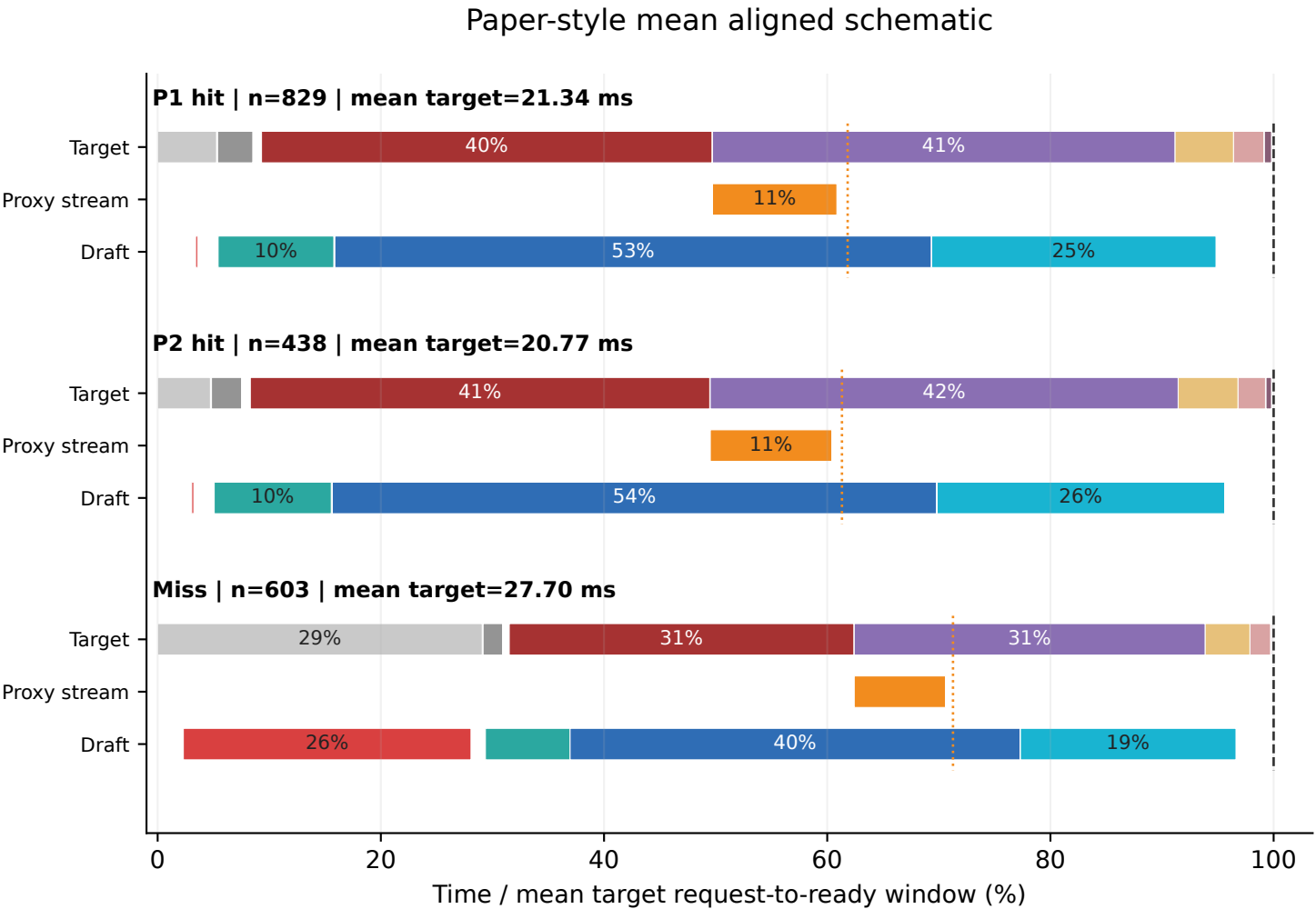
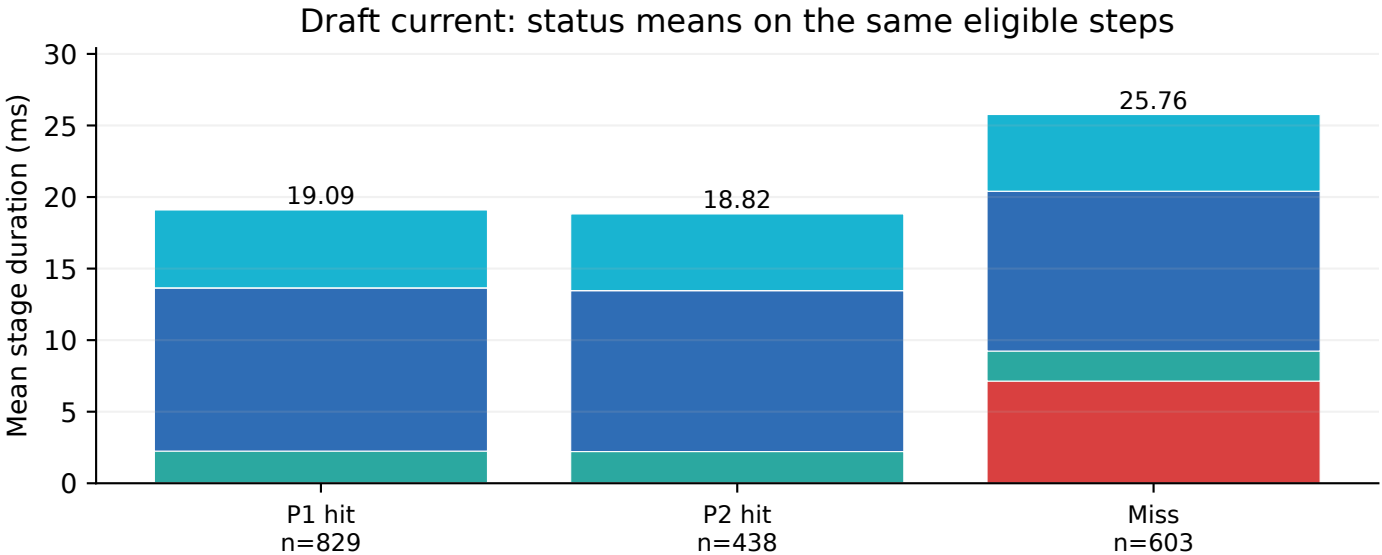
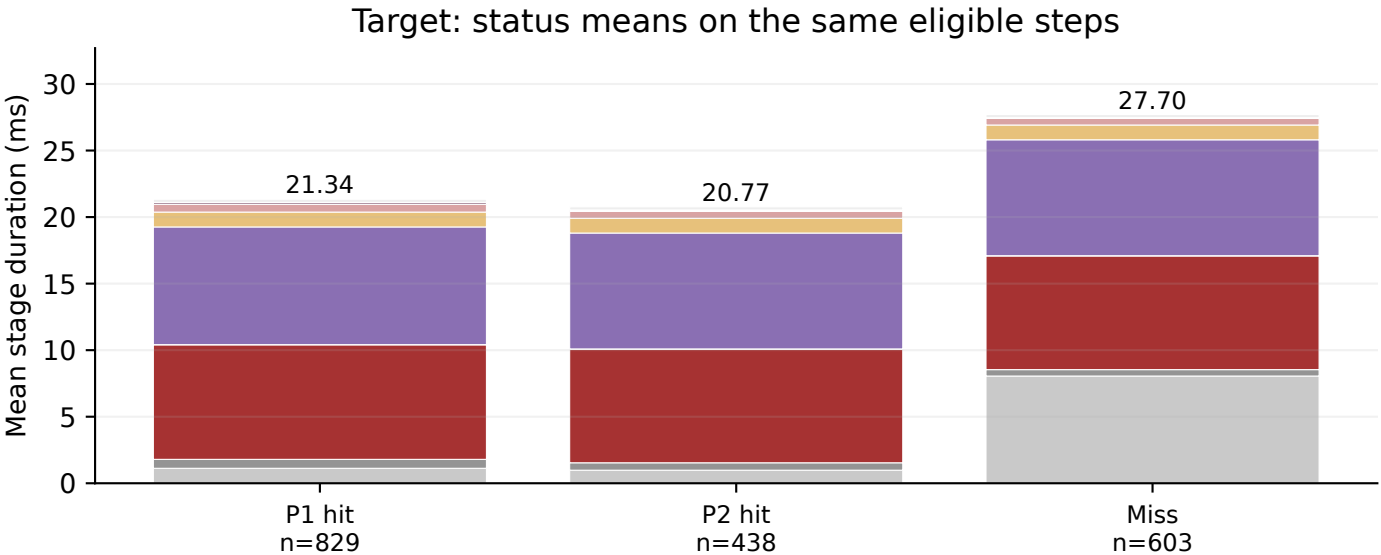
llama3_b1_stream_l1_profile_s0
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/0
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1870 eligible steps



Wait / sync Target post Accept Unlabelled gap Context P2 Target ready
Prepare Proxy / launch Commit Miss draft P1 Proxy wait Proxy received
Target pre Final logits Postprocess

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_stream_l1_profile_s1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1870 eligible steps

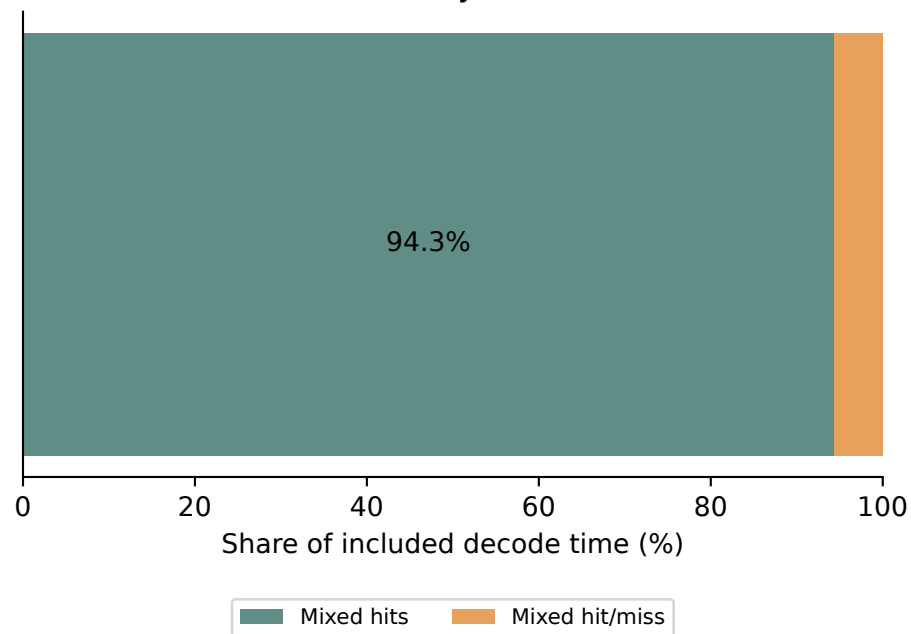


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

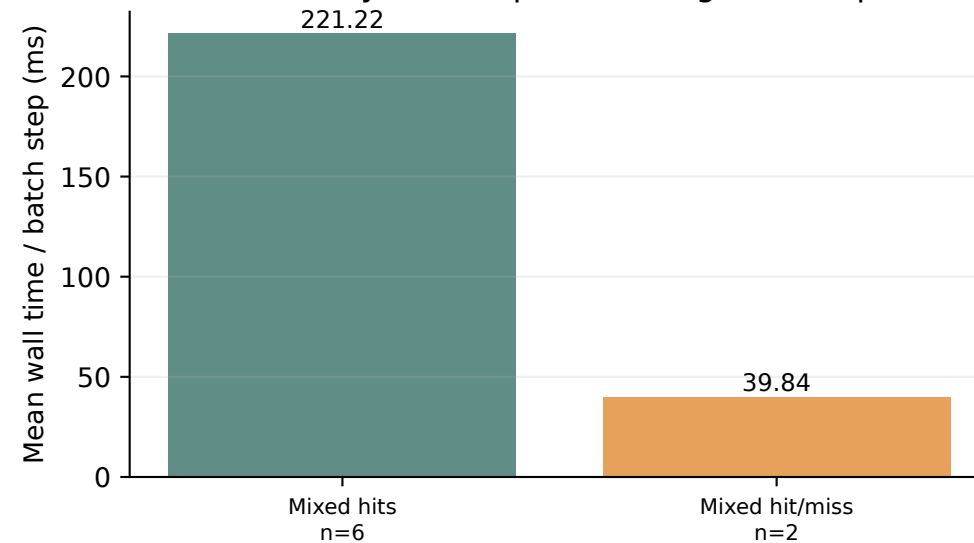
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stream_l1_greedy_smoke
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
Last pass: B=8, T=0.0, N=8 | AL*=2.197, hit=83.8%, TPS*=71.78

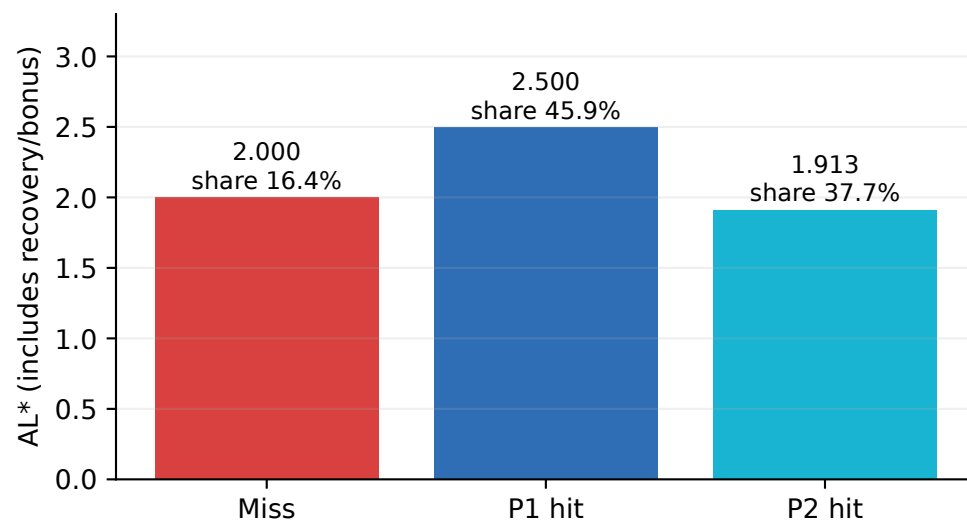
Measured wall time by batch cache status



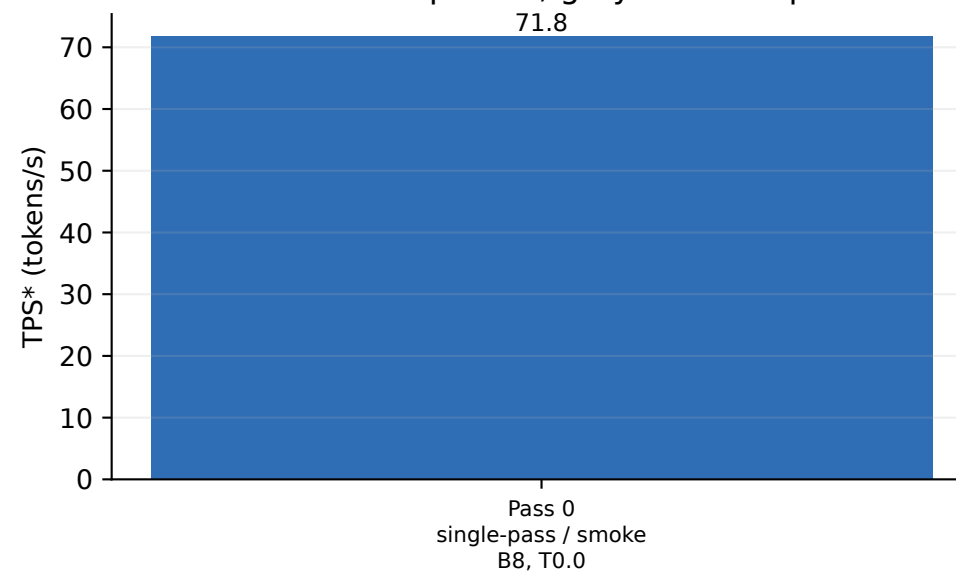
Boundary-free steps, including slow steps



Conditional AL* and source share

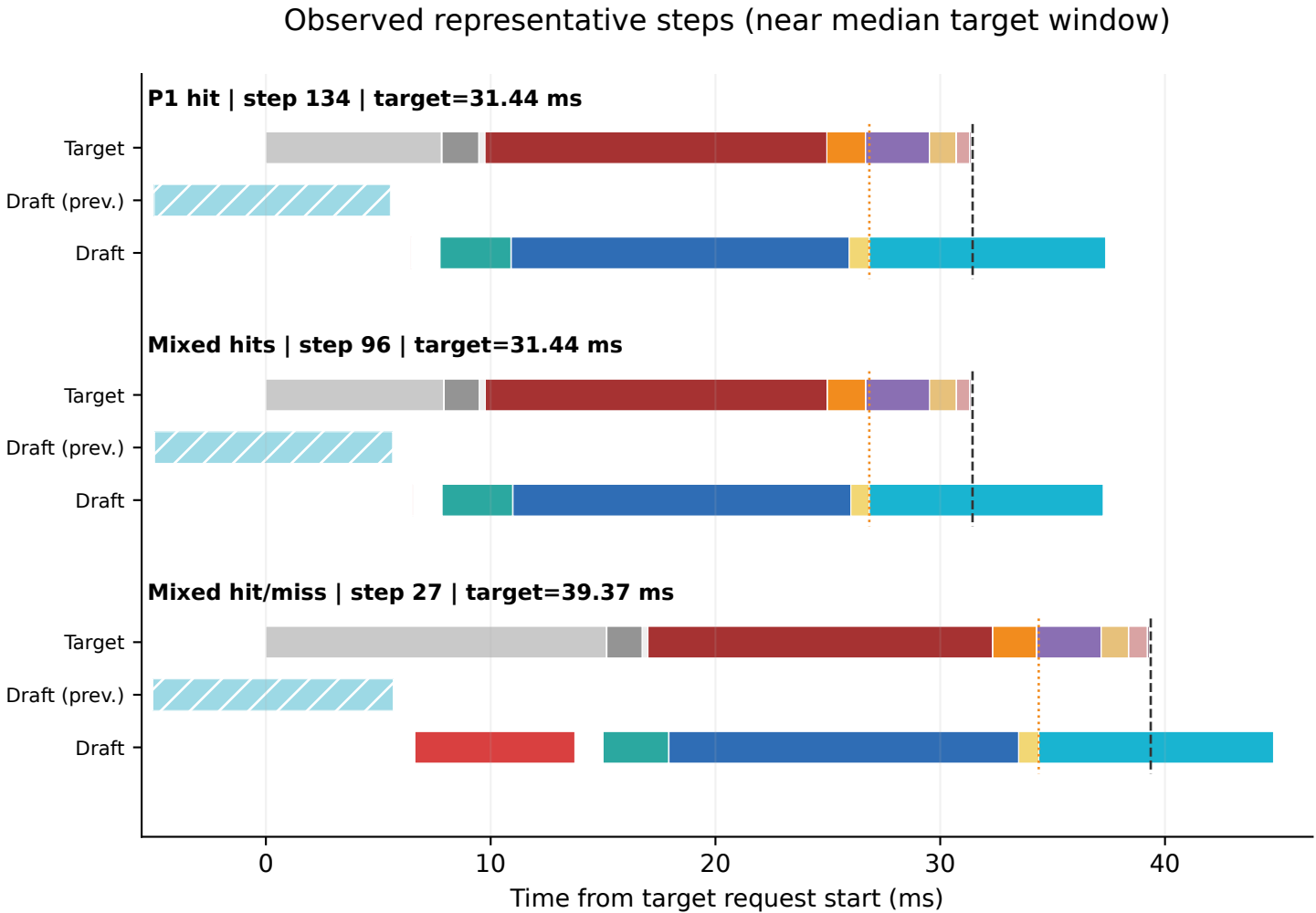
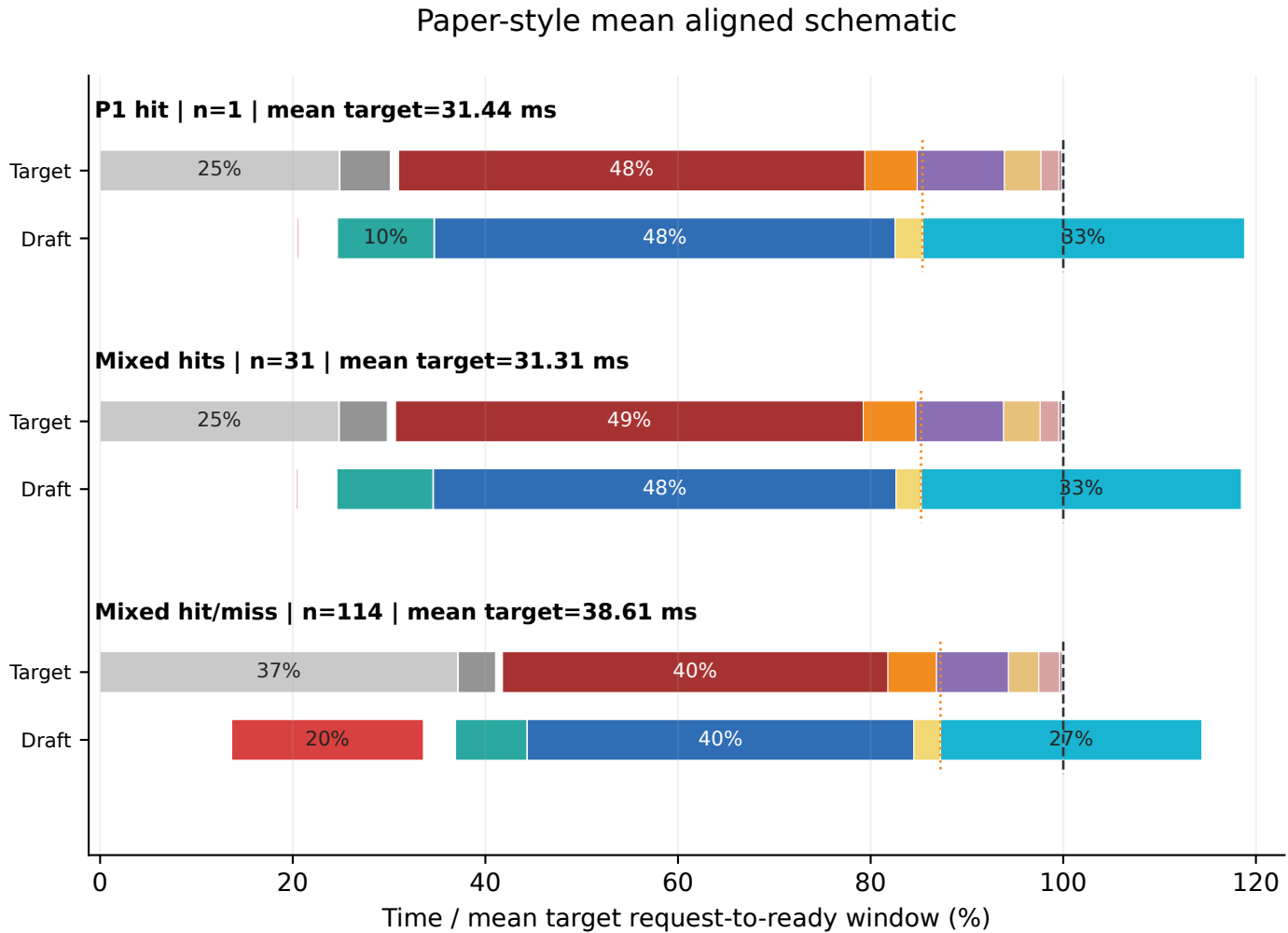
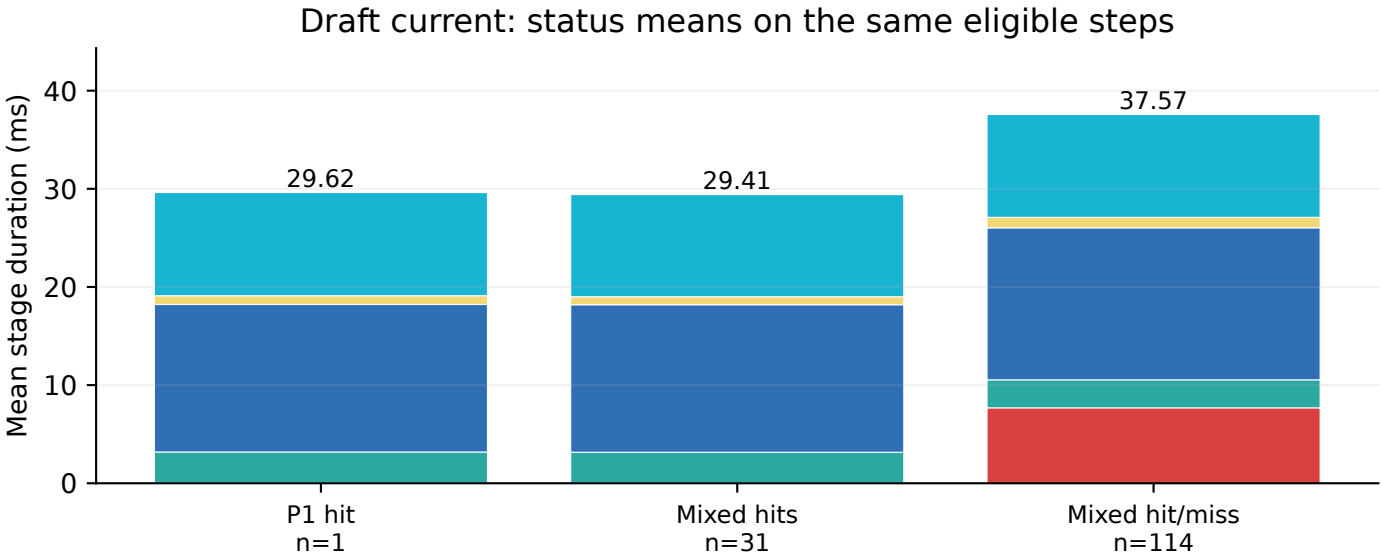
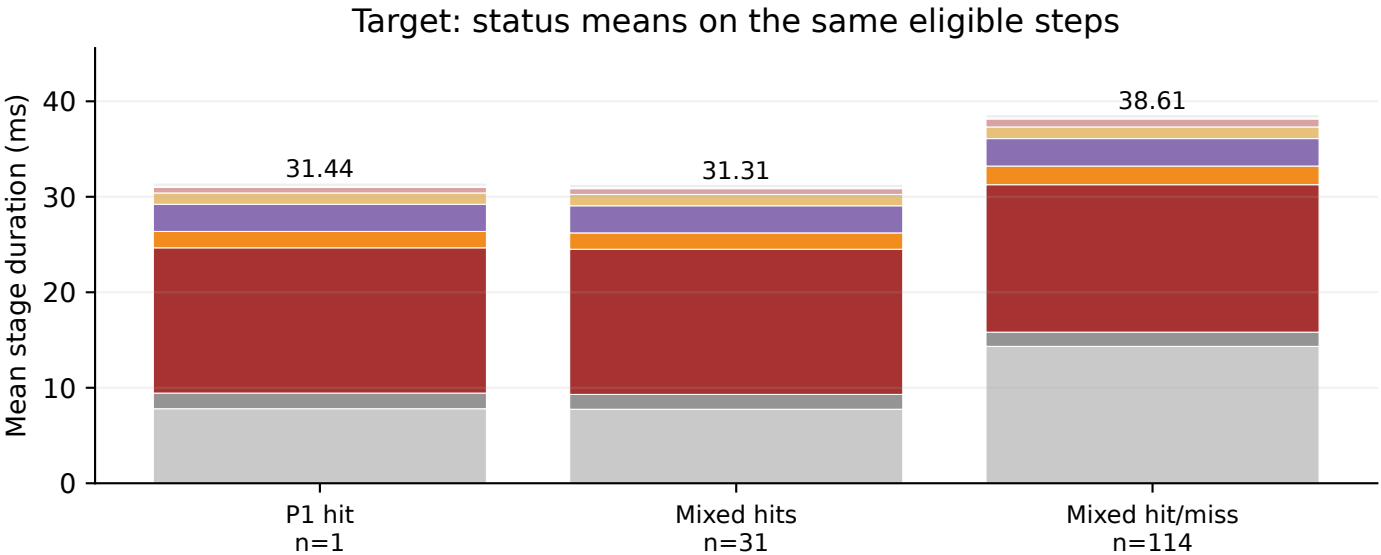


All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b8_stream_l1_profile_s0
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 146 eligible steps

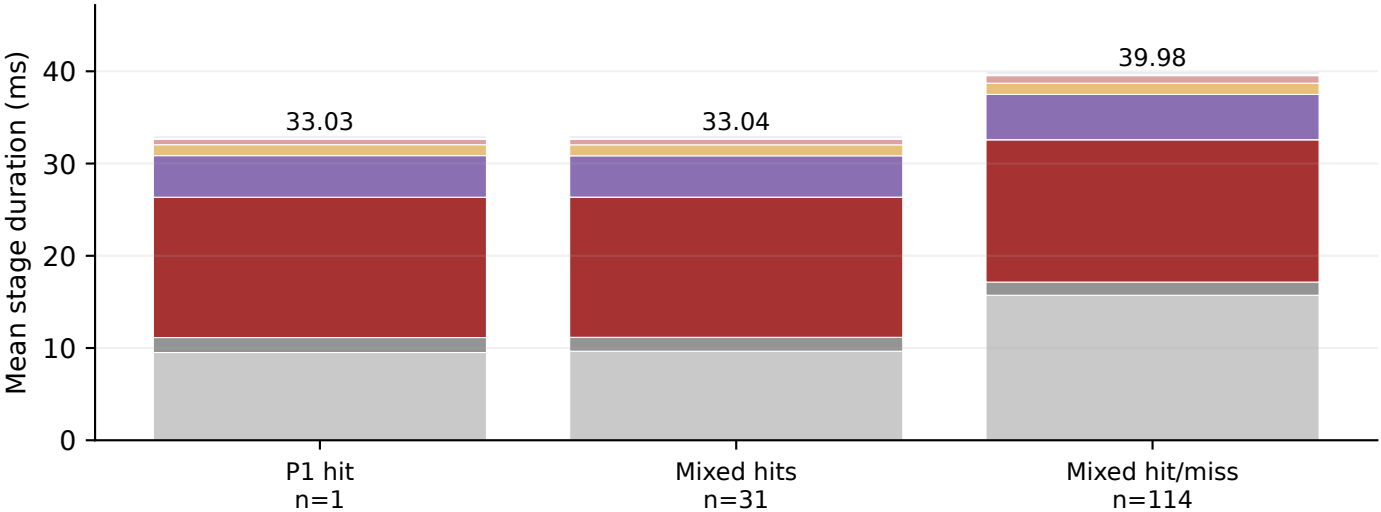


Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

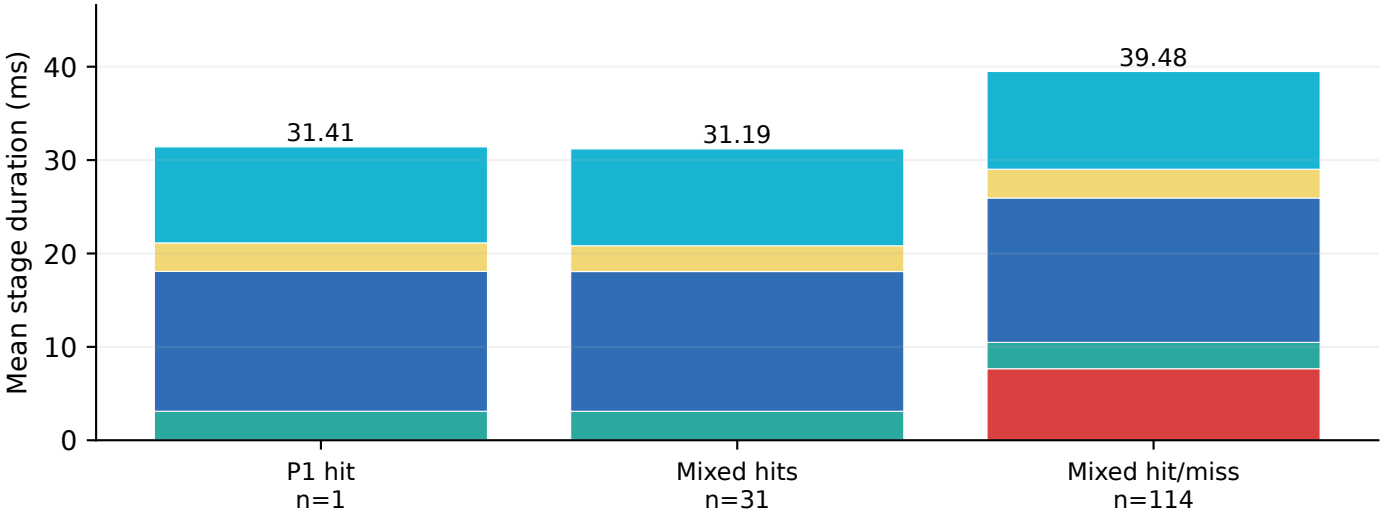
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_stream_l1_profile_s1
DUET: K1/K2=3/2, exit=26 (0-based), C=1, N1/N2=3/2, U=2, W=4, trim/stream=1/1
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 146 eligible steps

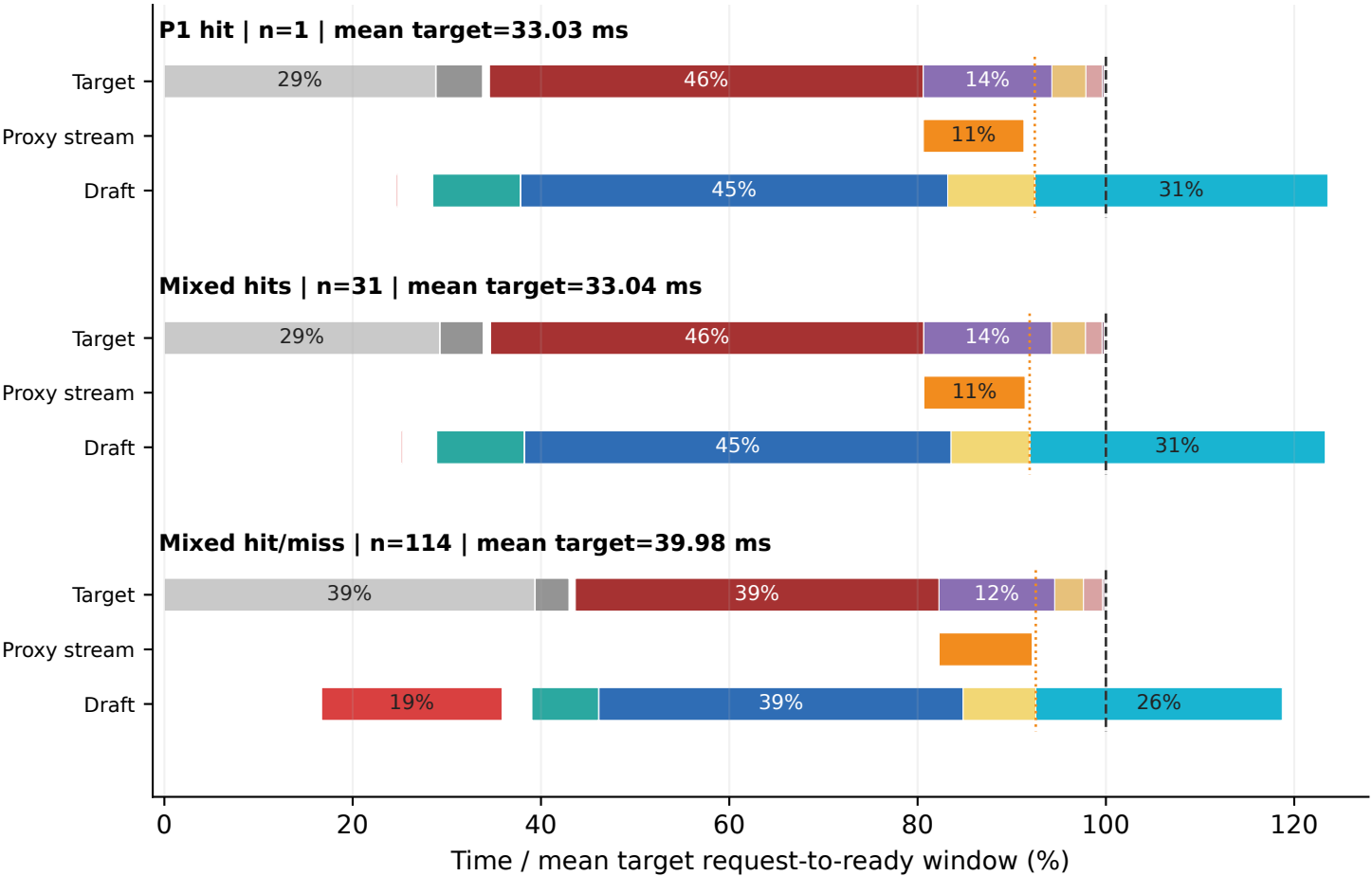
Target: status means on the same eligible steps



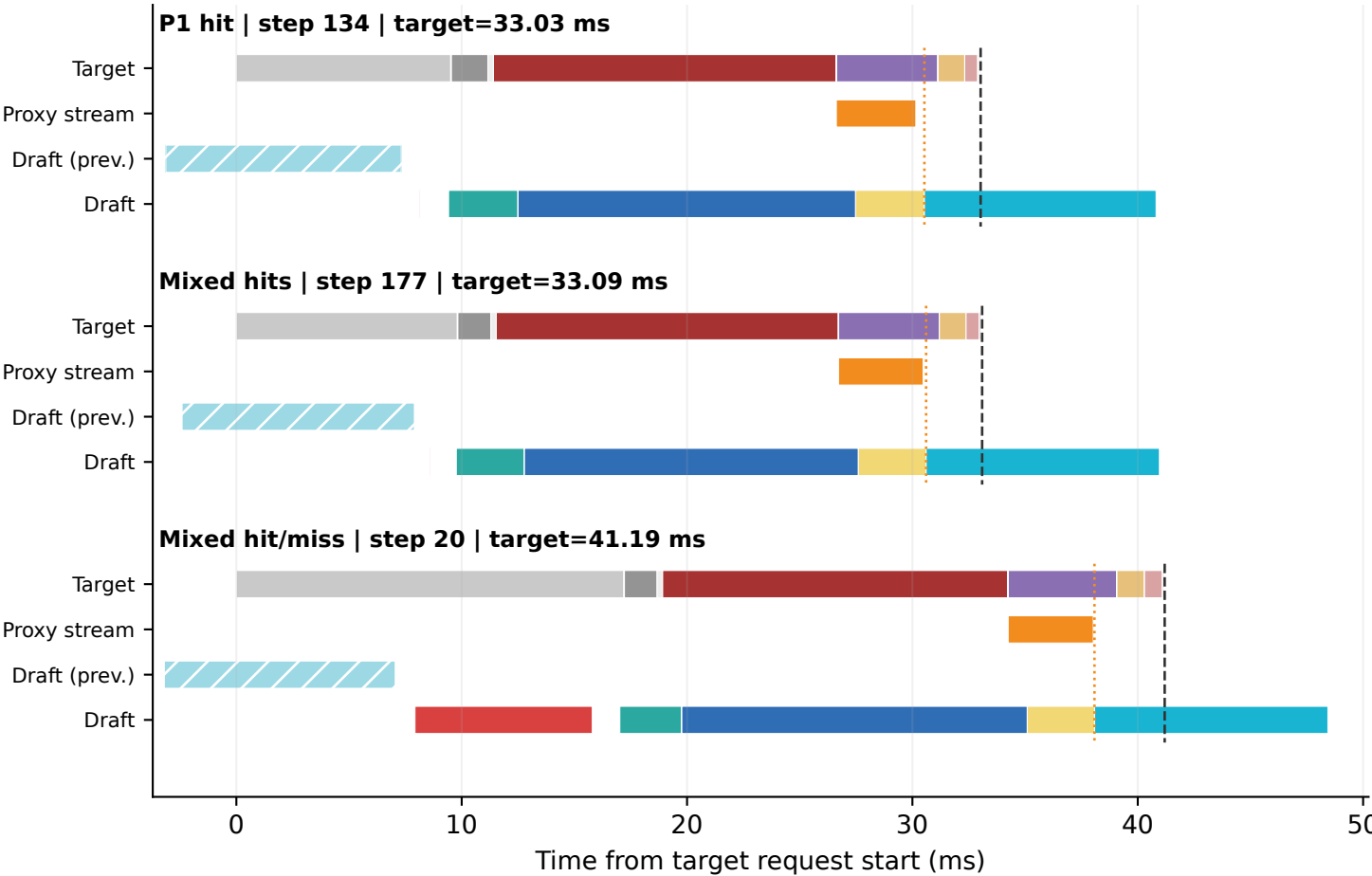
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic



Observed representative steps (near median target window)

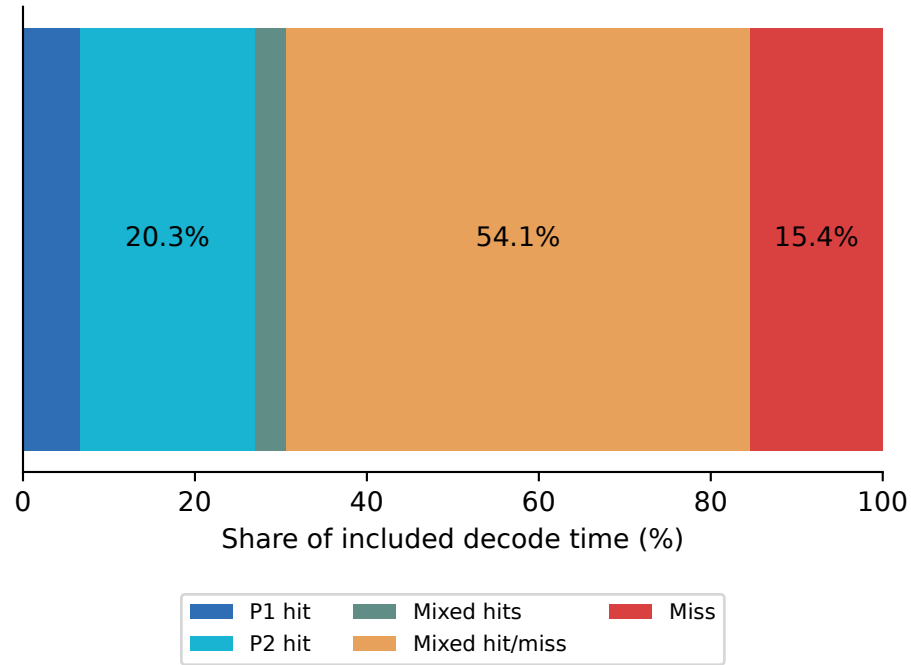


Wait / sync Target post Final logits Postprocess Context Proxy wait Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Previous P2 Proxy received
Target pre Proxy compute Commit Miss draft P2

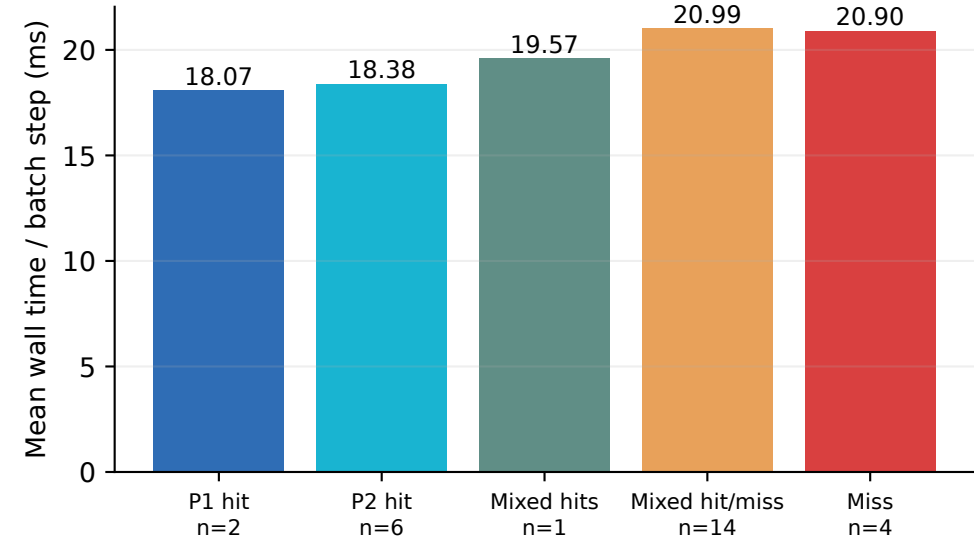
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b2_root_smoke
 DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
 Last pass: B=2, T=1.0, N=6 | AL*=1.352, hit=62.7%, TPS*=126.95

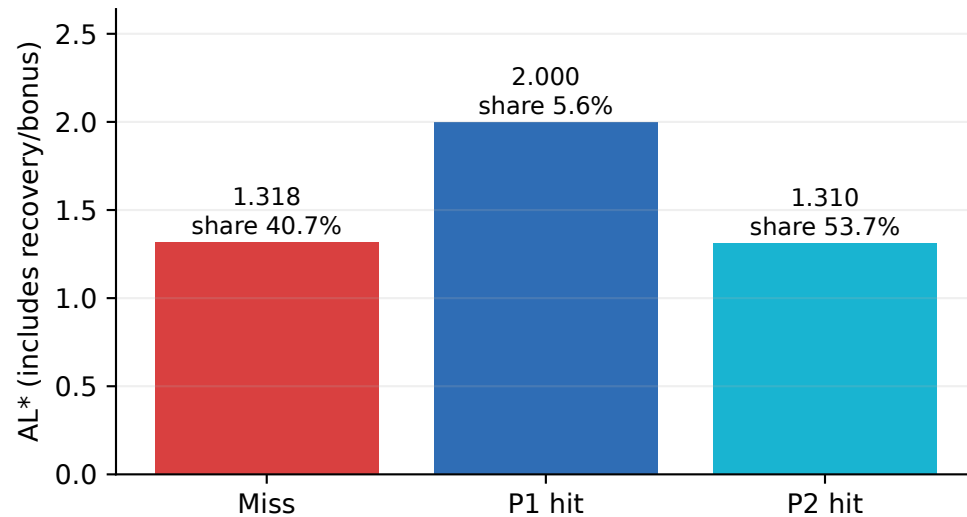
Measured wall time by batch cache status



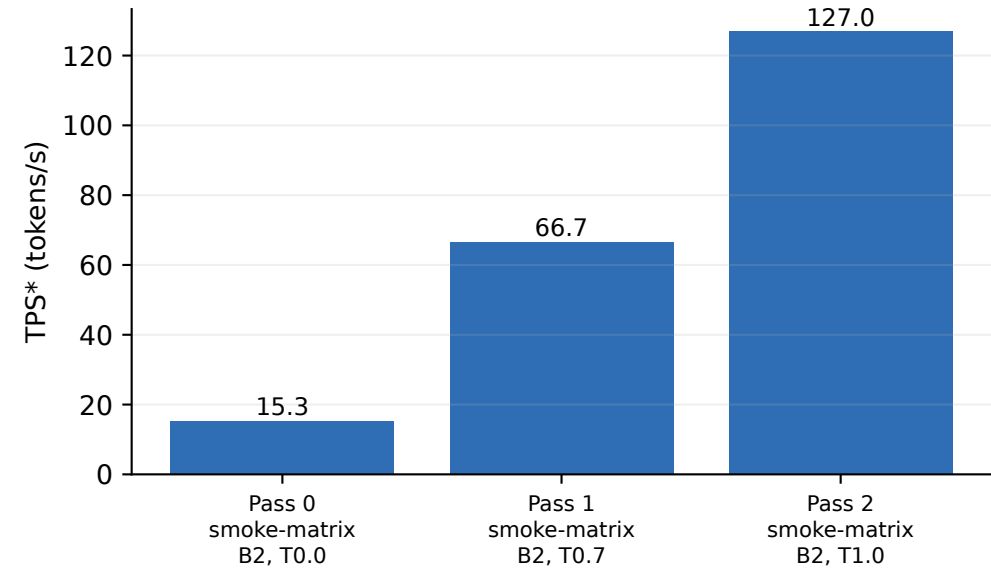
Boundary-free steps, including slow steps



Conditional AL* and source share



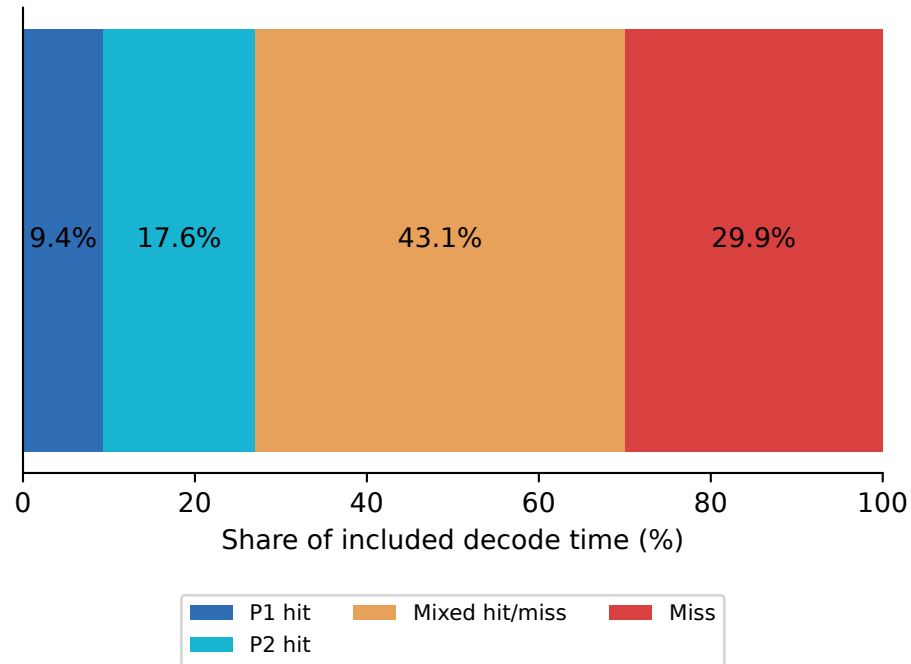
All saved passes; gray = warmup



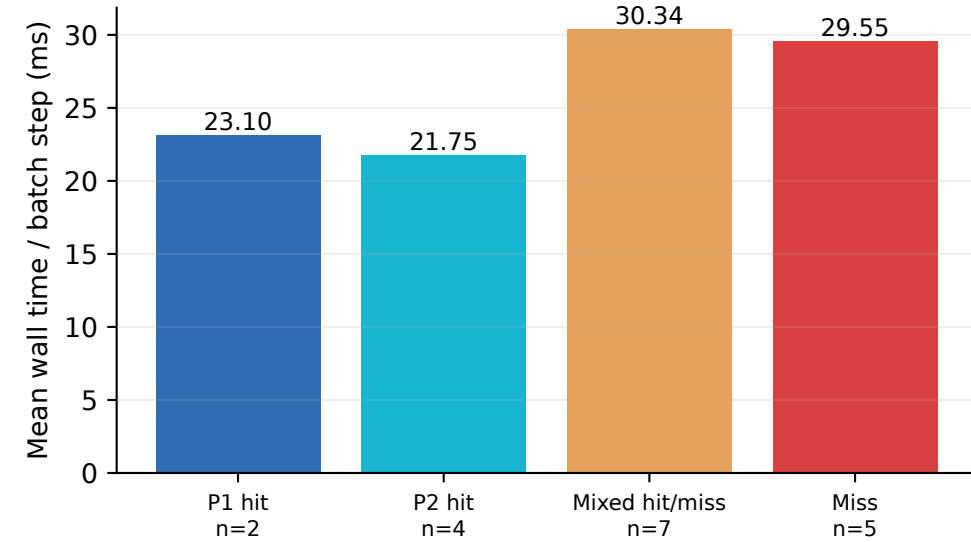
No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
 Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.

llama3_b2_root_smoke
DUET: K1/K2=4/2, exit=21 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=0/0
Last pass: B=2, T=1.0, N=6 | AL*=1.943, hit=45.0%, TPS*=117.57

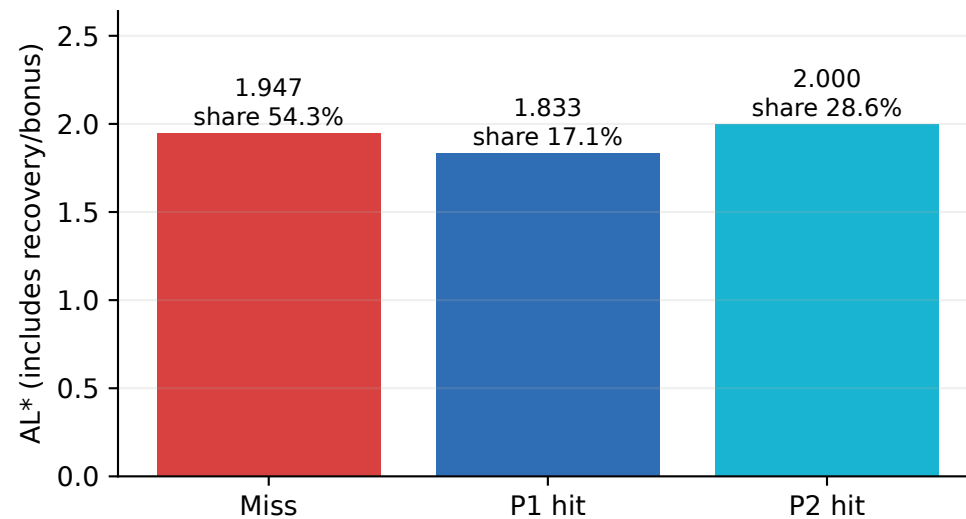
Measured wall time by batch cache status



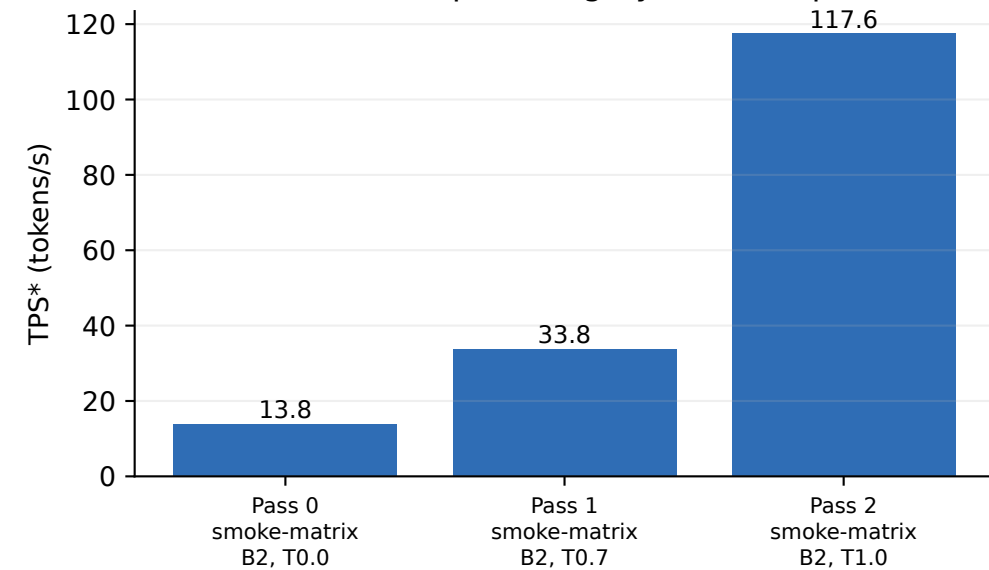
Boundary-free steps, including slow steps



Conditional AL* and source share



All saved passes; gray = warmup



No per-phase GPU trace was recorded for this run. These are observed whole-step/source breakdowns, not inferred phase times.
Top and lower-left panels use the last saved pass. AL* and TPS* use different boundary filters; partial batches remain in TPS*.